

Dual-View Stage 1 Output

Two views per country, presented as a paired Stage-1 reading:

View A – Section Plate · CoDa-Standard reading
XY: scatter of (h_i, h_j) for each carrier pair – plan view
XZ: bar of pairwise bearings θ_{ij} – magnitude index
YZ: bar of CLR coordinate per carrier – magnitude index
Generator: `stage1_plates_raw.py`
Answers: "what are the magnitudes at this timestep?"

View B – ILR-Helmert Orthogonal Triplet Plate · Orthonormal reading
Panel 1: scatter (ilr_1, ilr_2)
Panel 2: scatter (ilr_1, ilr_3)
Panel 3: scatter (ilr_2, ilr_3)
Trajectory drawn $\circ \rightarrow \blacksquare$; current year highlighted X on cine pages.
Generator: `ilr_triplet_plate.py`
Answers: "where is the composition in ILR space and where has it moved?"

Both views are Order-1 first-principles plates per Output Doctrine v1.0.
Both read from the same `stage1_output.json`. The pair is complementary:
· Section Plate gives the magnitude index reading at each timestep
· Triplet Plate gives the trajectory shape across all timesteps
· Together they answer "what" and "where it is going" in one folder.

Each country gets ~56 pages: View A divider + 27 Section plates + View B divider + 27 Triplet plates.

View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ AUS Australia

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

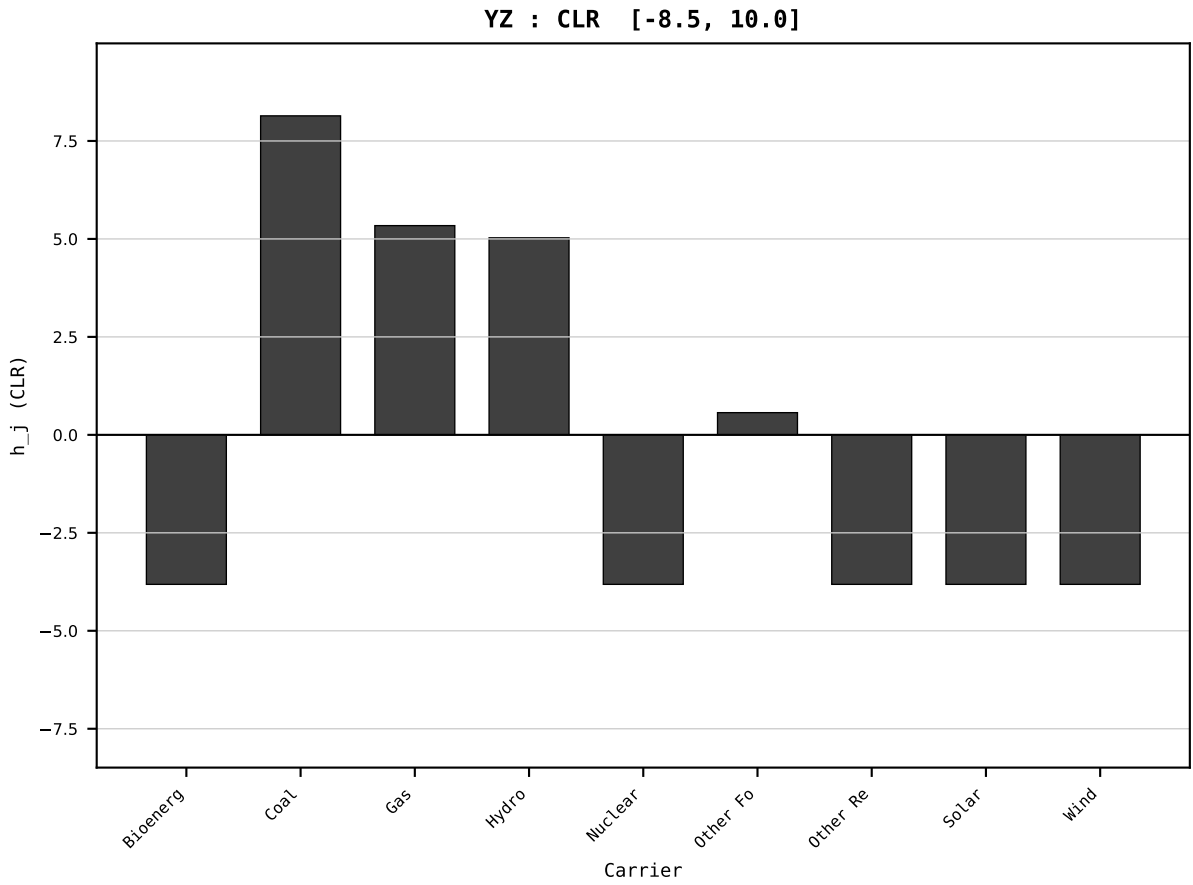
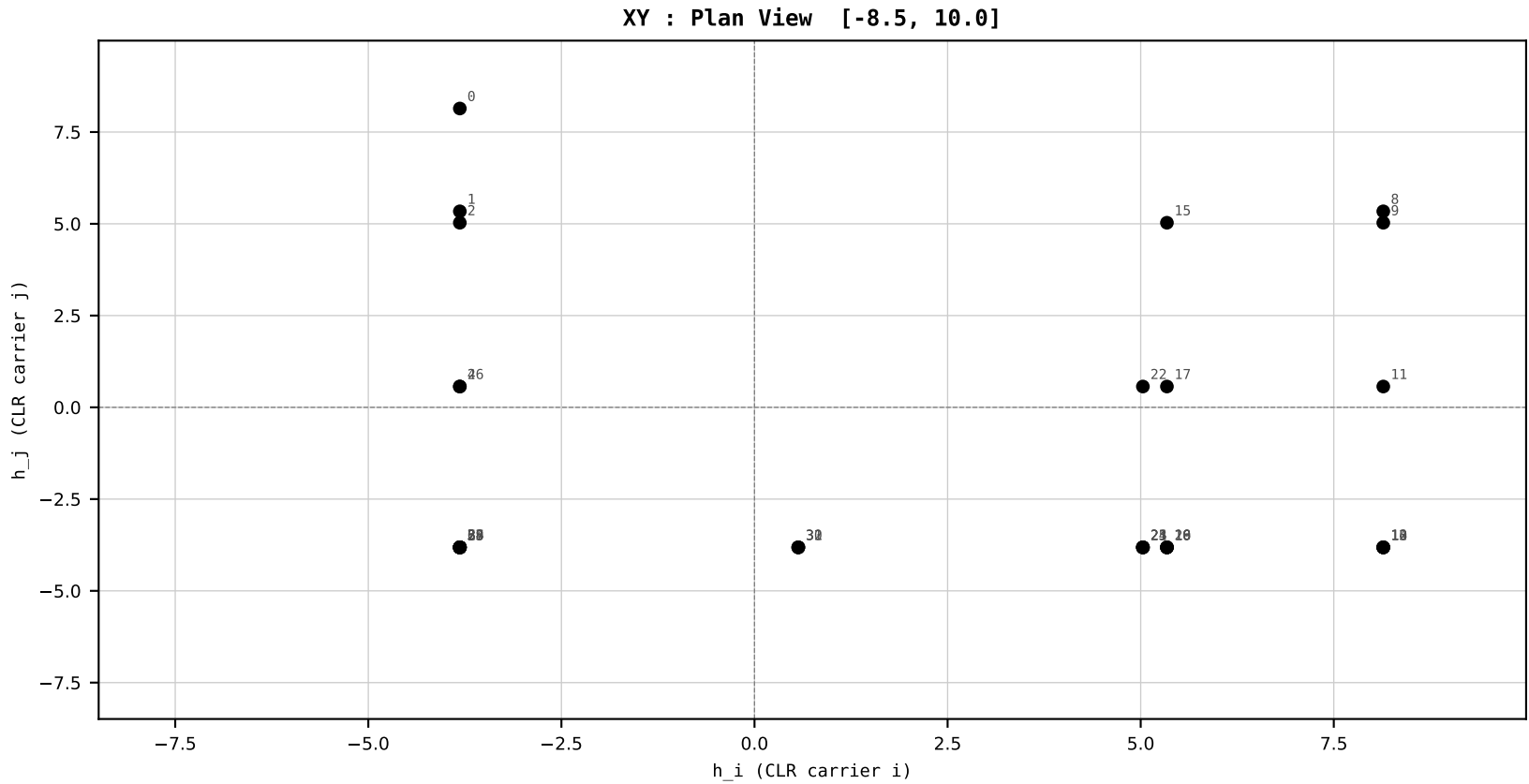
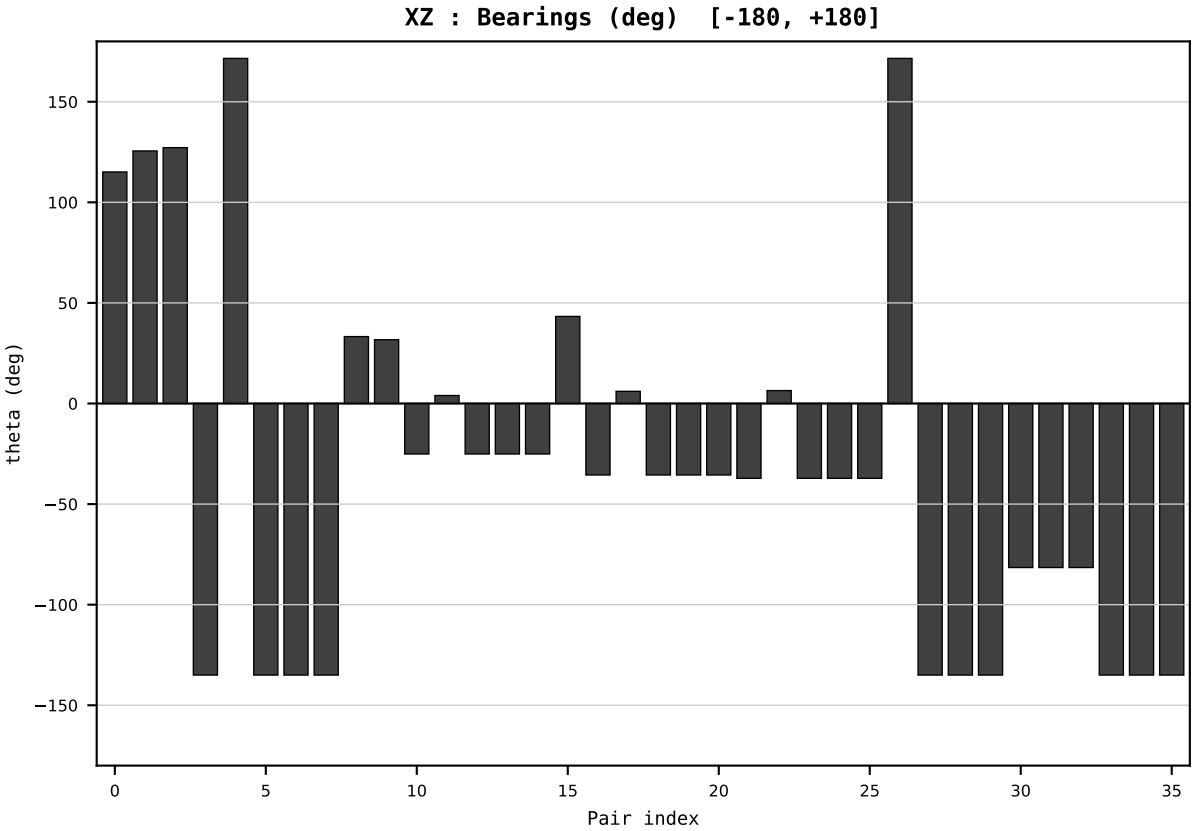
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.825462
Ring = Hs-5
E_metric= 193.1704
kappa_HS= 155760.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 11.956 HLR
DR ratio= 155760.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

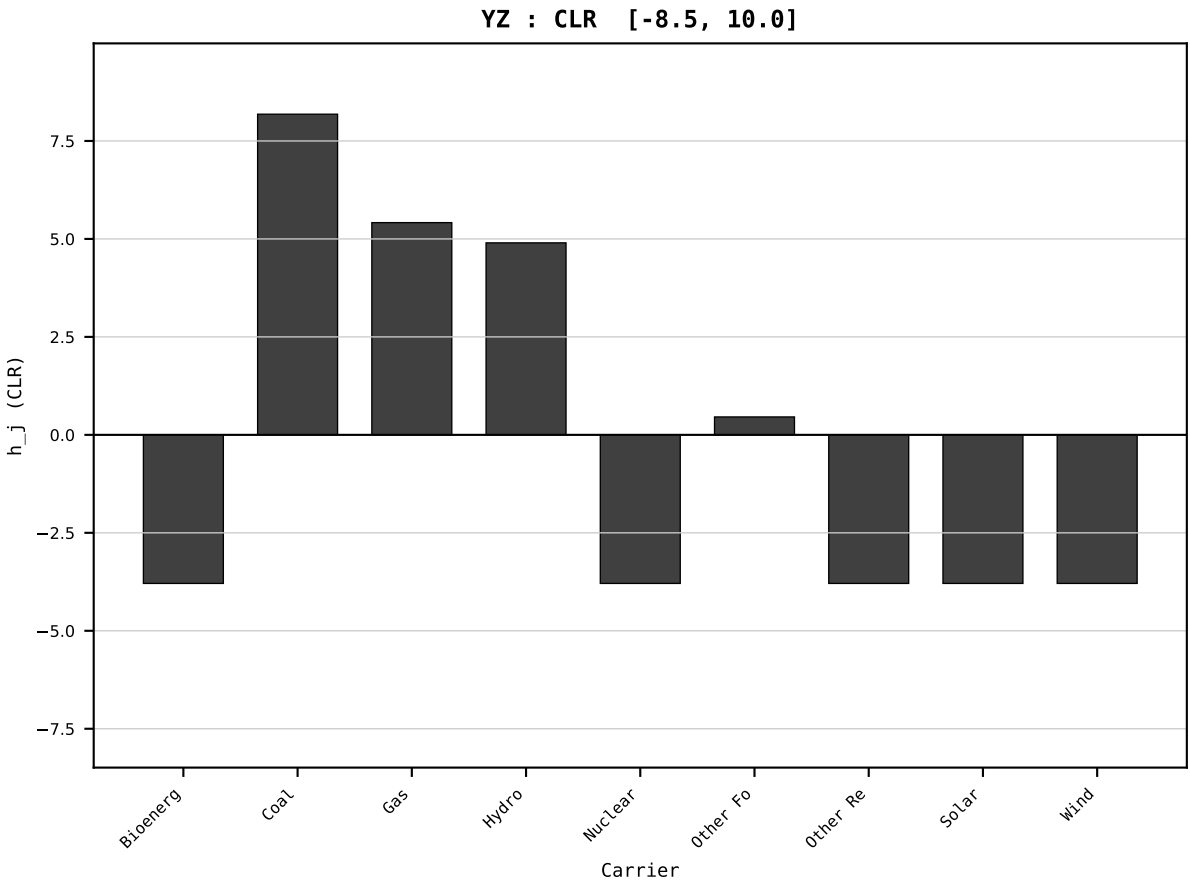
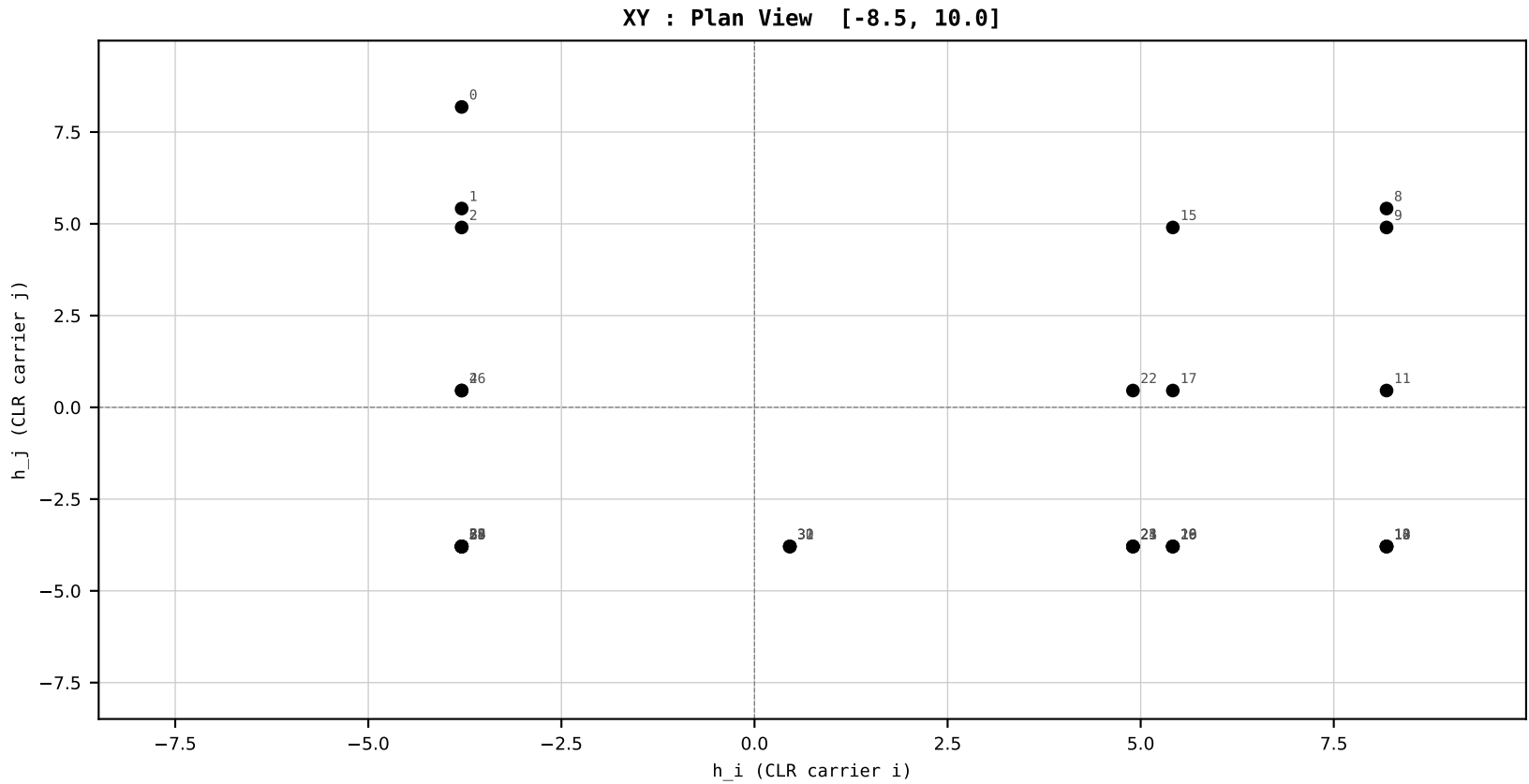
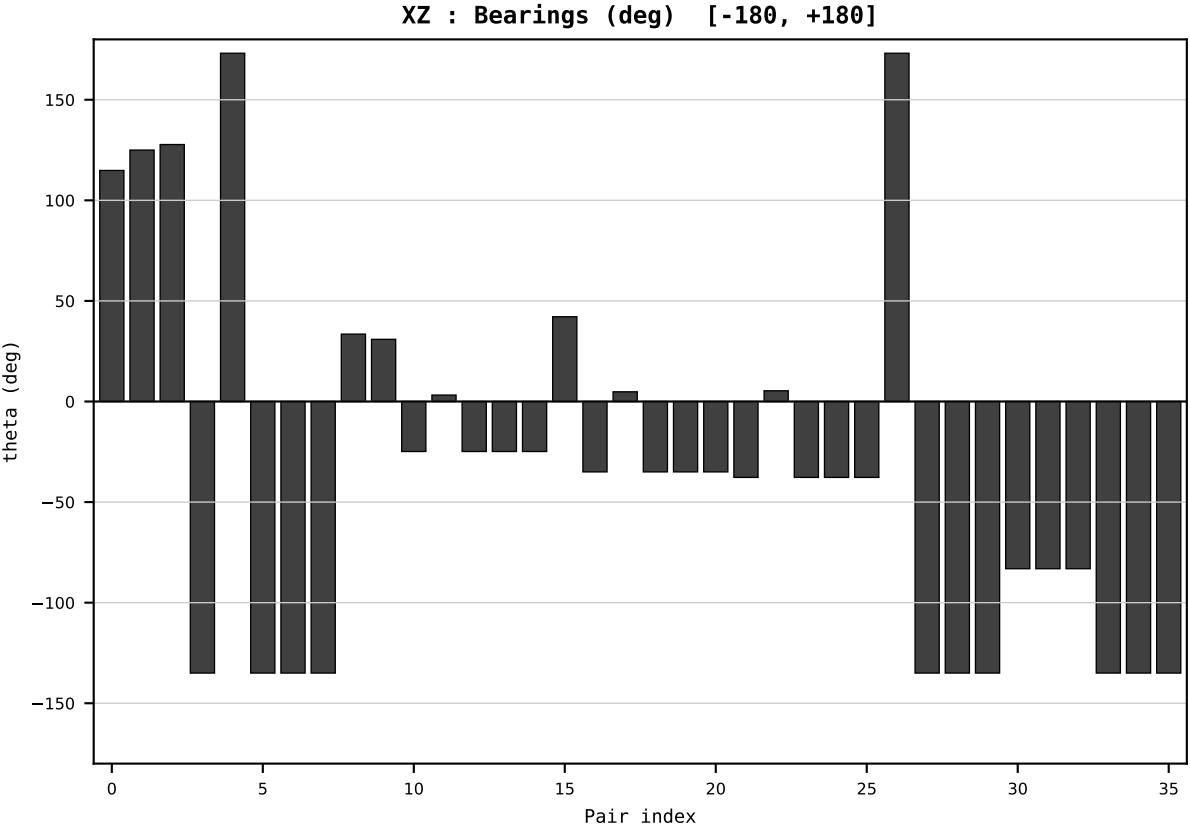
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.831929
Ring = Hs-5
E_metric= 192.3868
kappa_HS= 158670.00
omega = 0.8070 deg
d_A = 0.197590
Helm = Hydro
Helm d = -0.128582
DR = 11.975 HLR
DR ratio= 158670.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

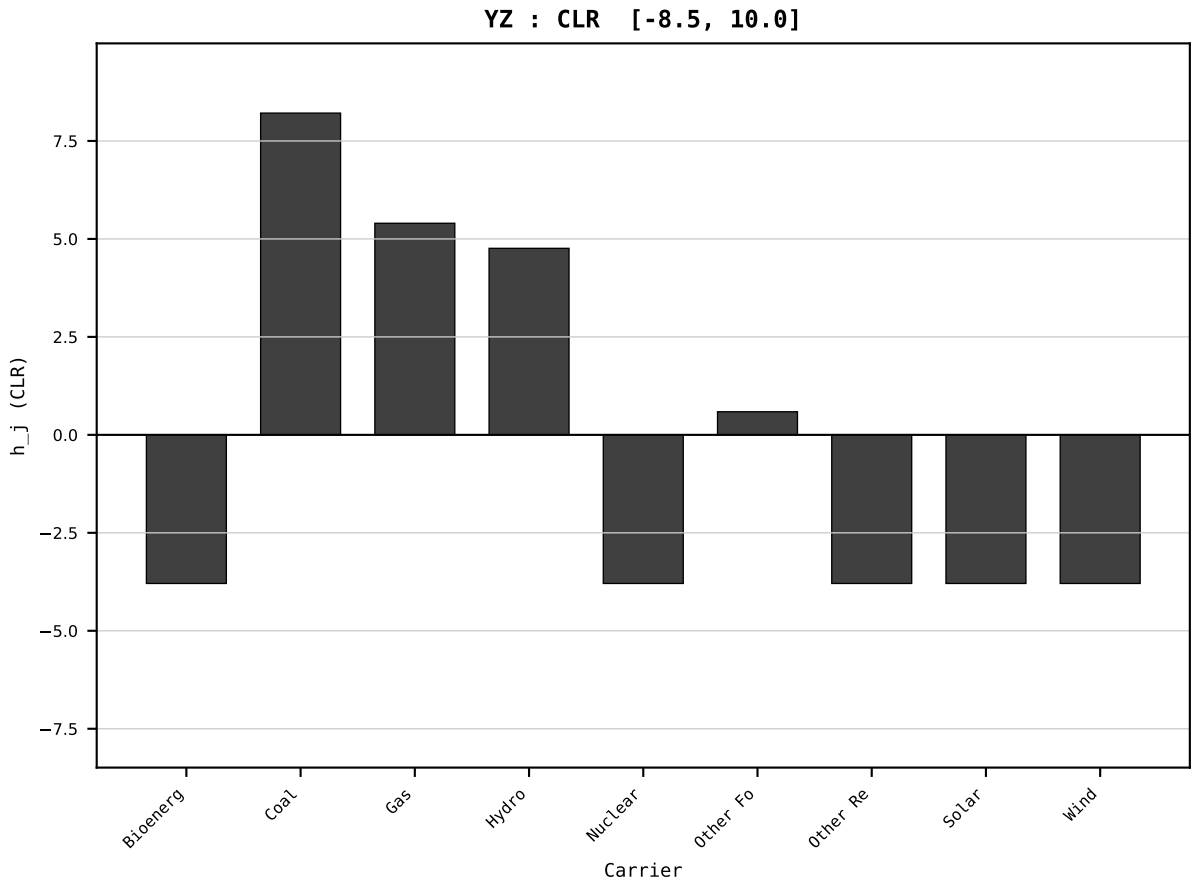
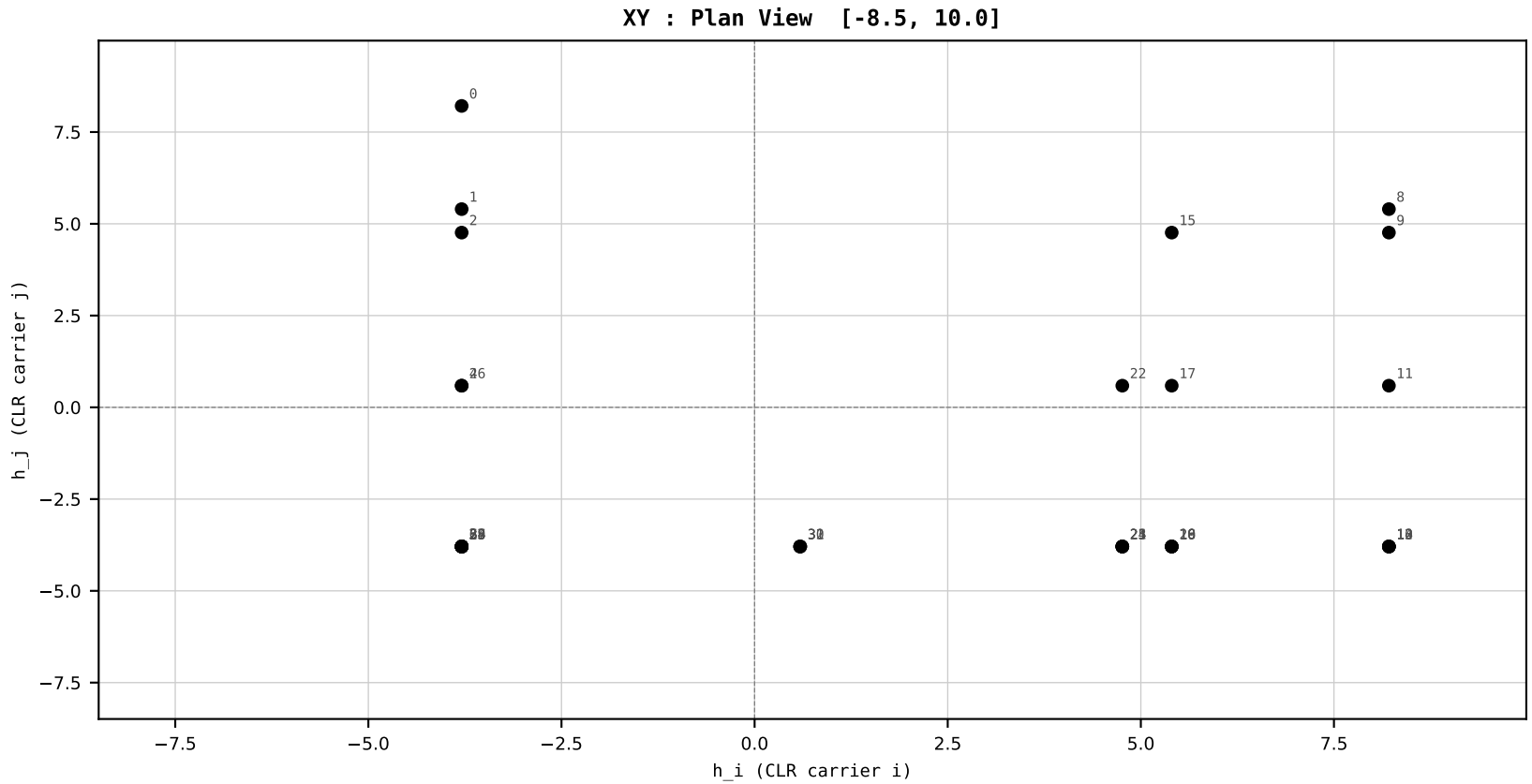
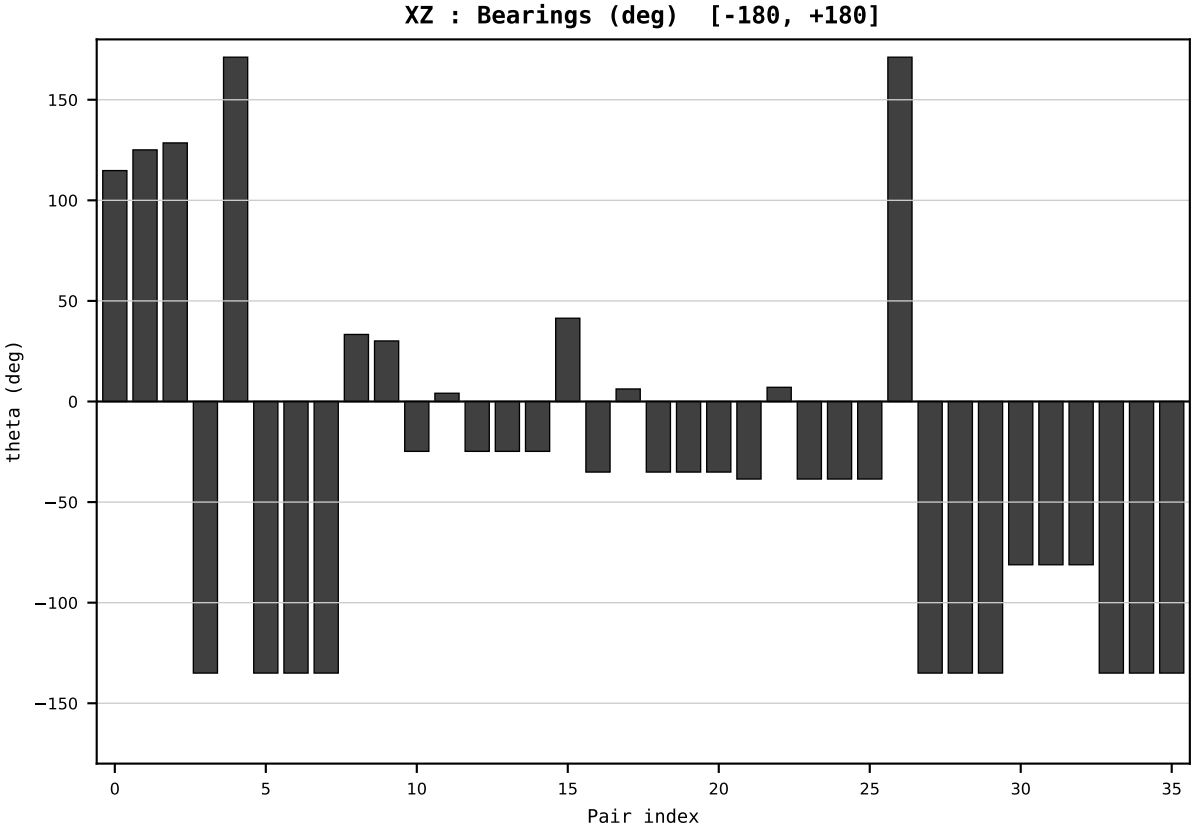
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.842056
Ring = Hs-5
E_metric= 191.4975
kappa_HS= 163320.00
omega = 0.7962 deg
d_A = 0.195173
Helm = Hydro
Helm d = -0.139550
DR = 12.003 HLR
DR ratio= 163320.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

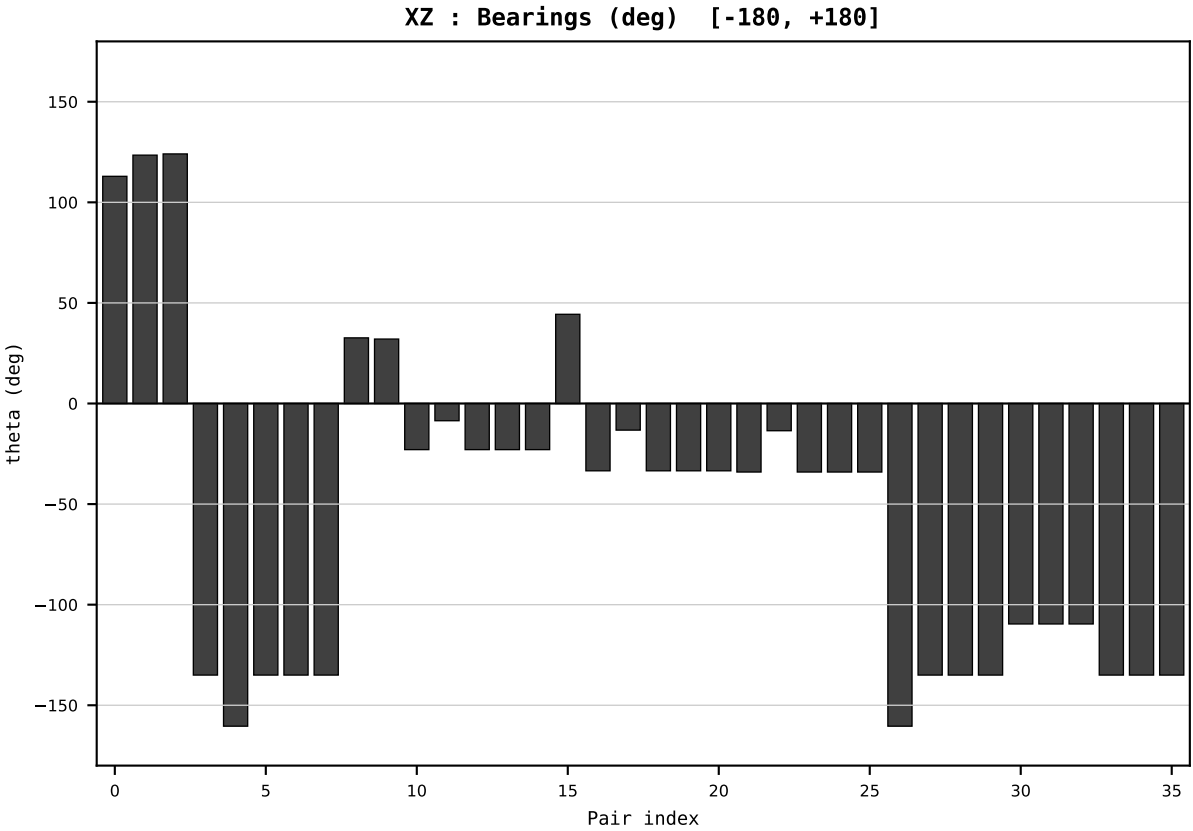
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.843732
Ring = Hs-5
E_metric= 194.1424
kappa_HS= 166860.00
omega = 8.2899 deg
d_A = 2.009584
Helm = Other Fossil
Helm d = -1.862126
DR = 12.025 HLR
DR ratio= 166860.0x

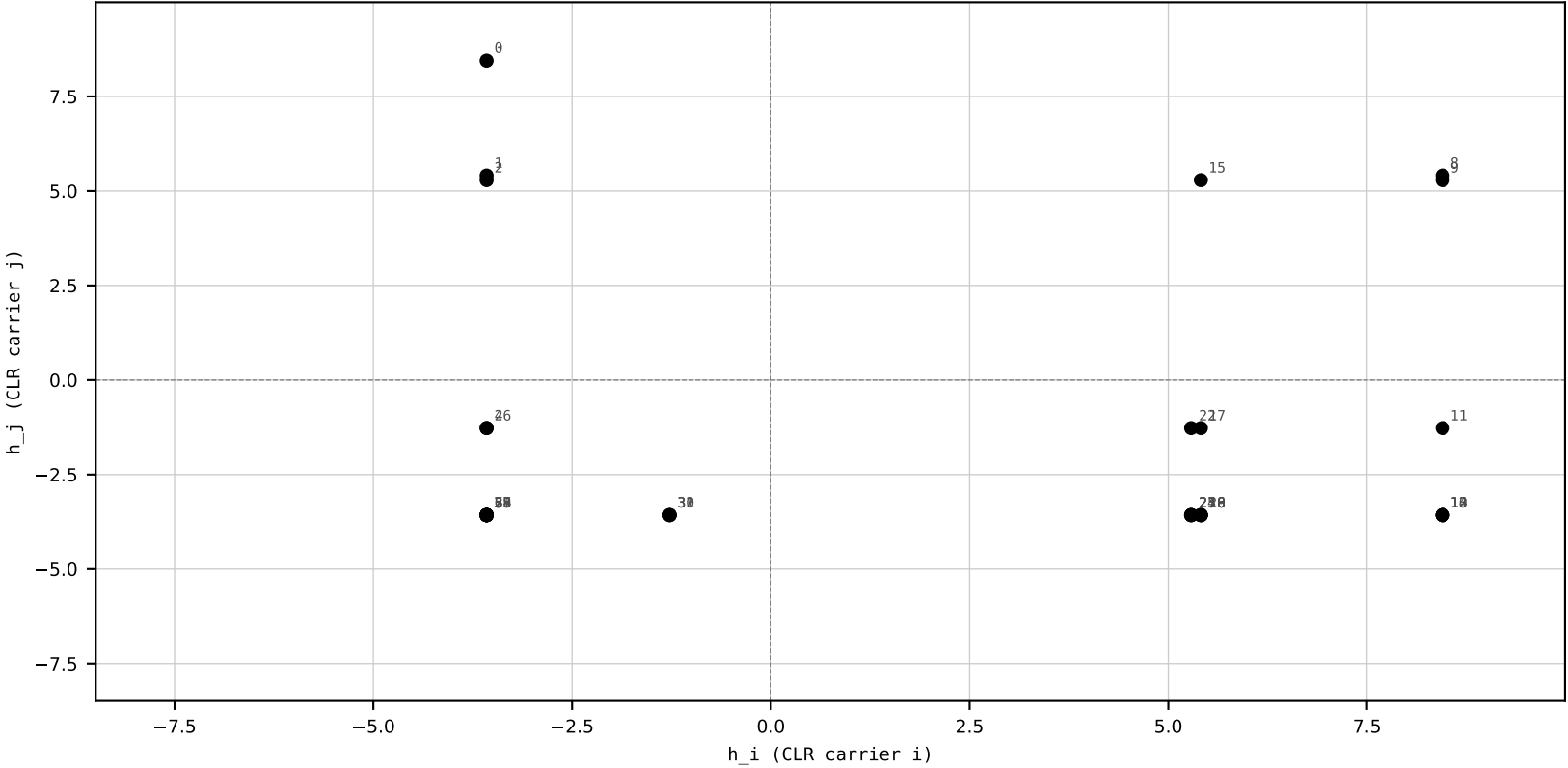
XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

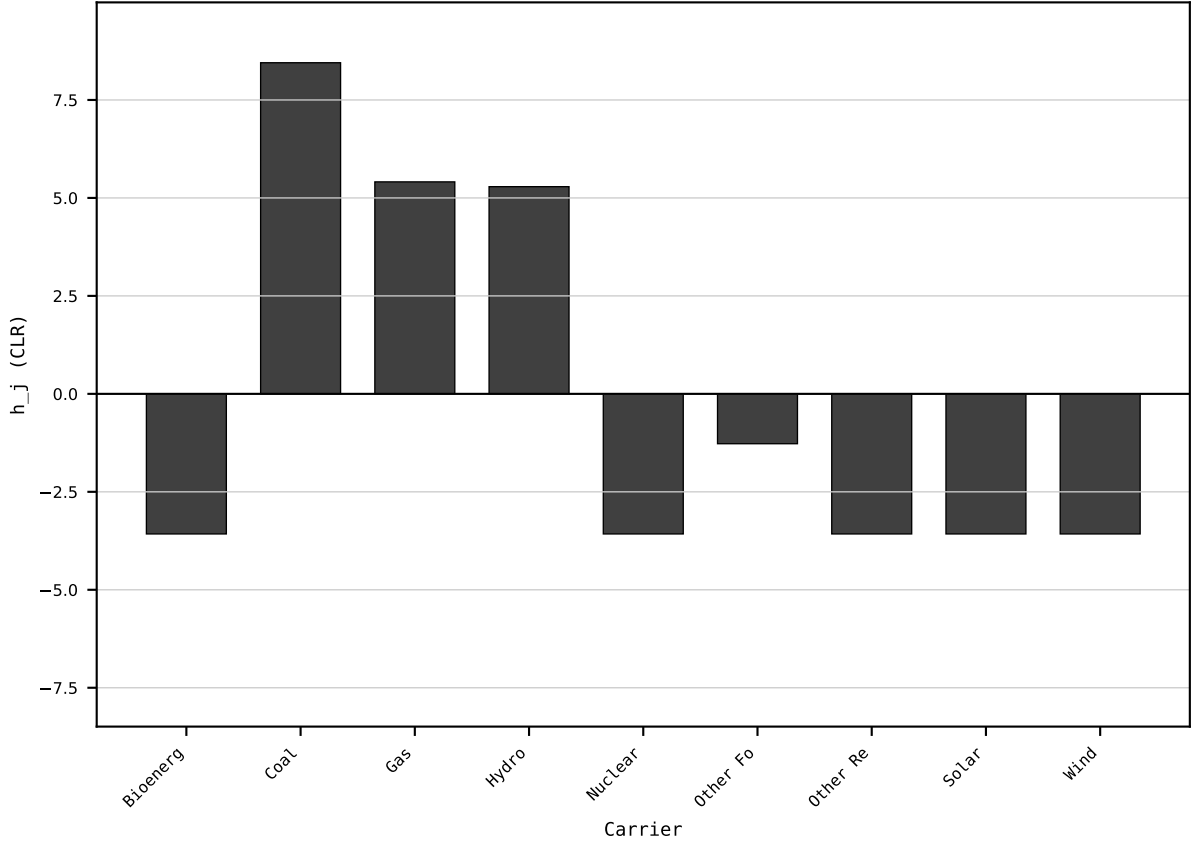
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



XY : Plan View [-8.5, 10.0]



YZ : CLR [-8.5, 10.0]



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

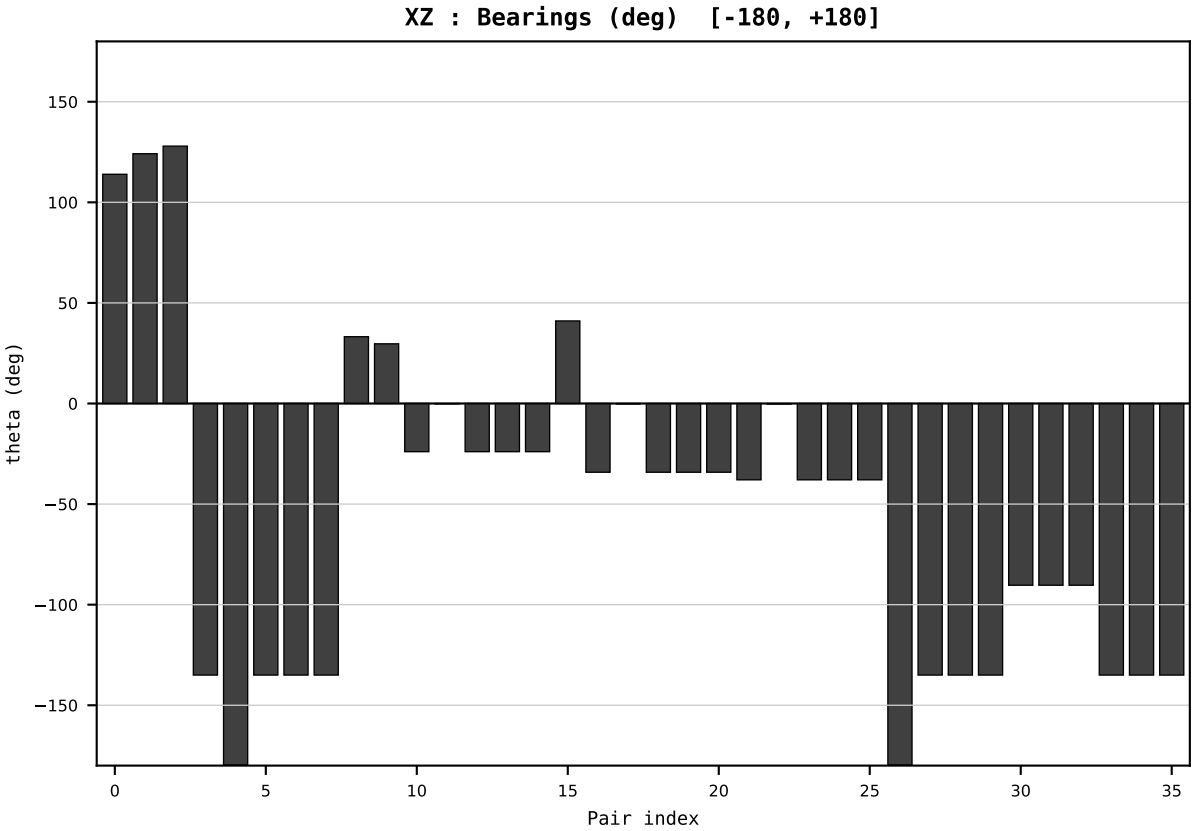
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.853390
Ring = Hs-5
E_metric= 191.1428
kappa_HS= 173390.00
omega = 5.7457 deg
d_A = 1.395449
Helm = Other Fossil
Helm d = +1.250220
DR = 12.063 HLR
DR ratio= 173390.0x

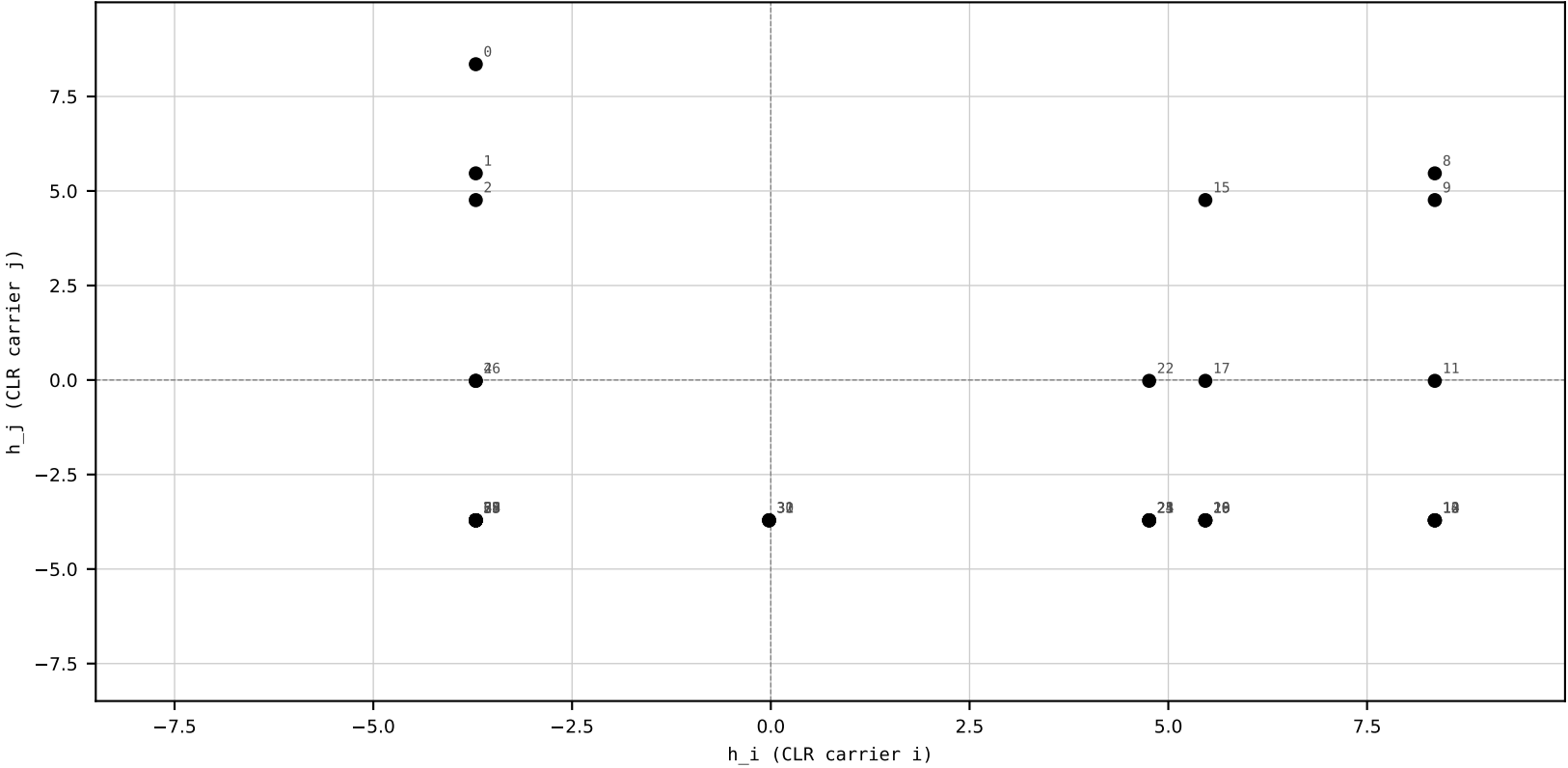
XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

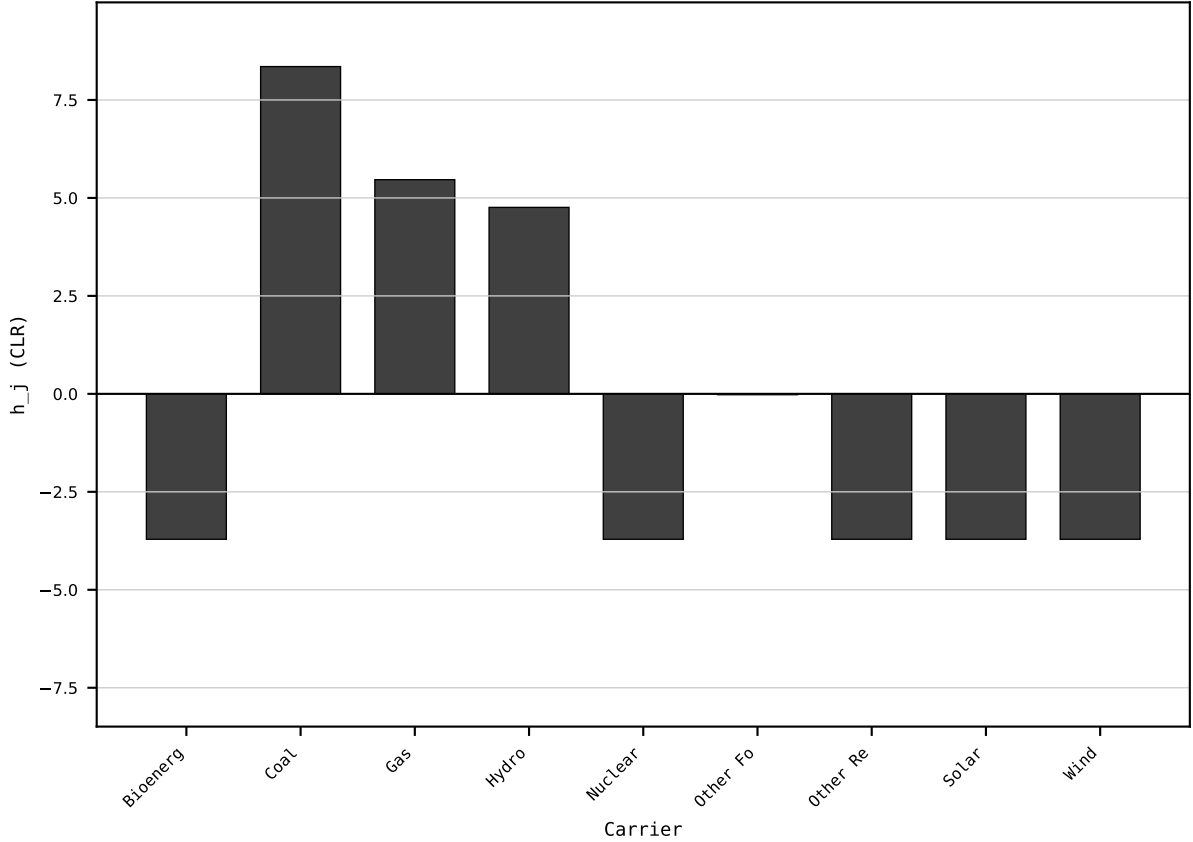
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



XY : Plan View [-8.5, 10.0]



YZ : CLR [-8.5, 10.0]



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

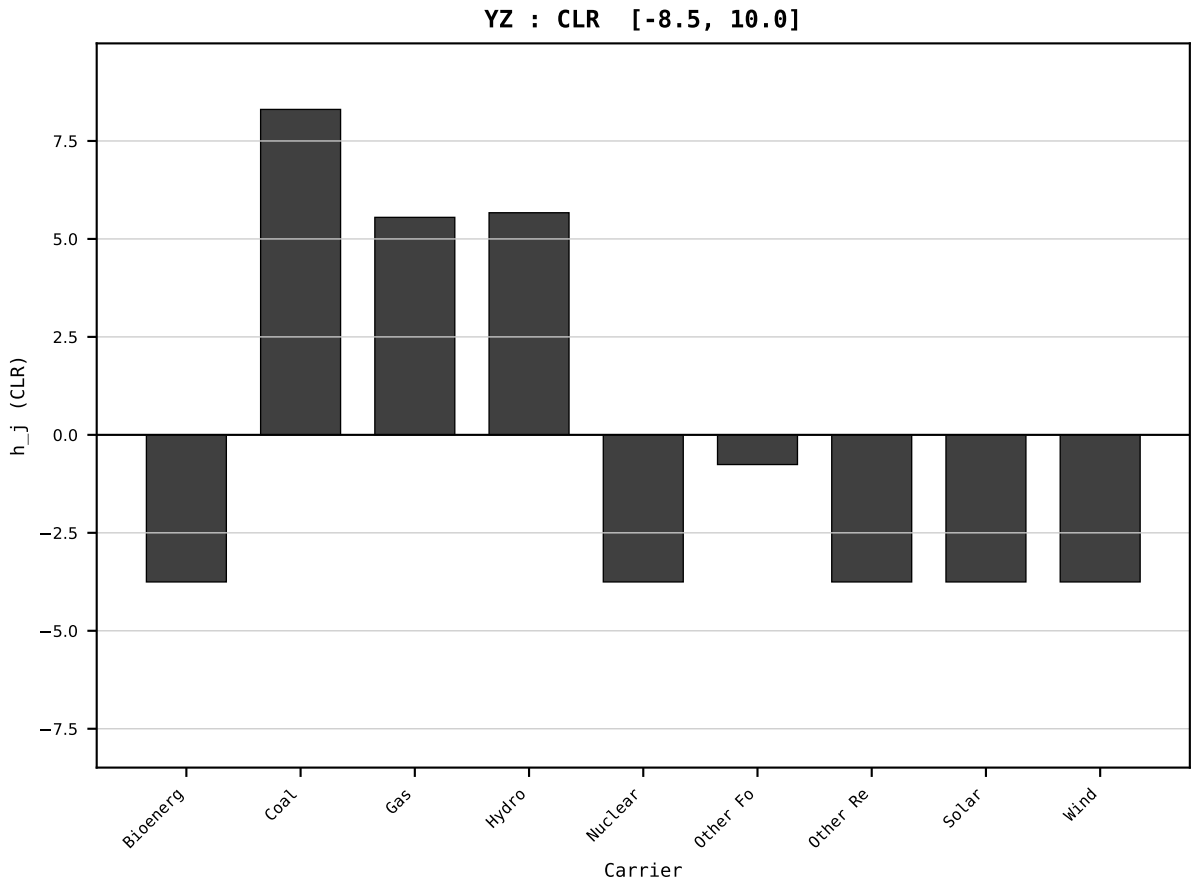
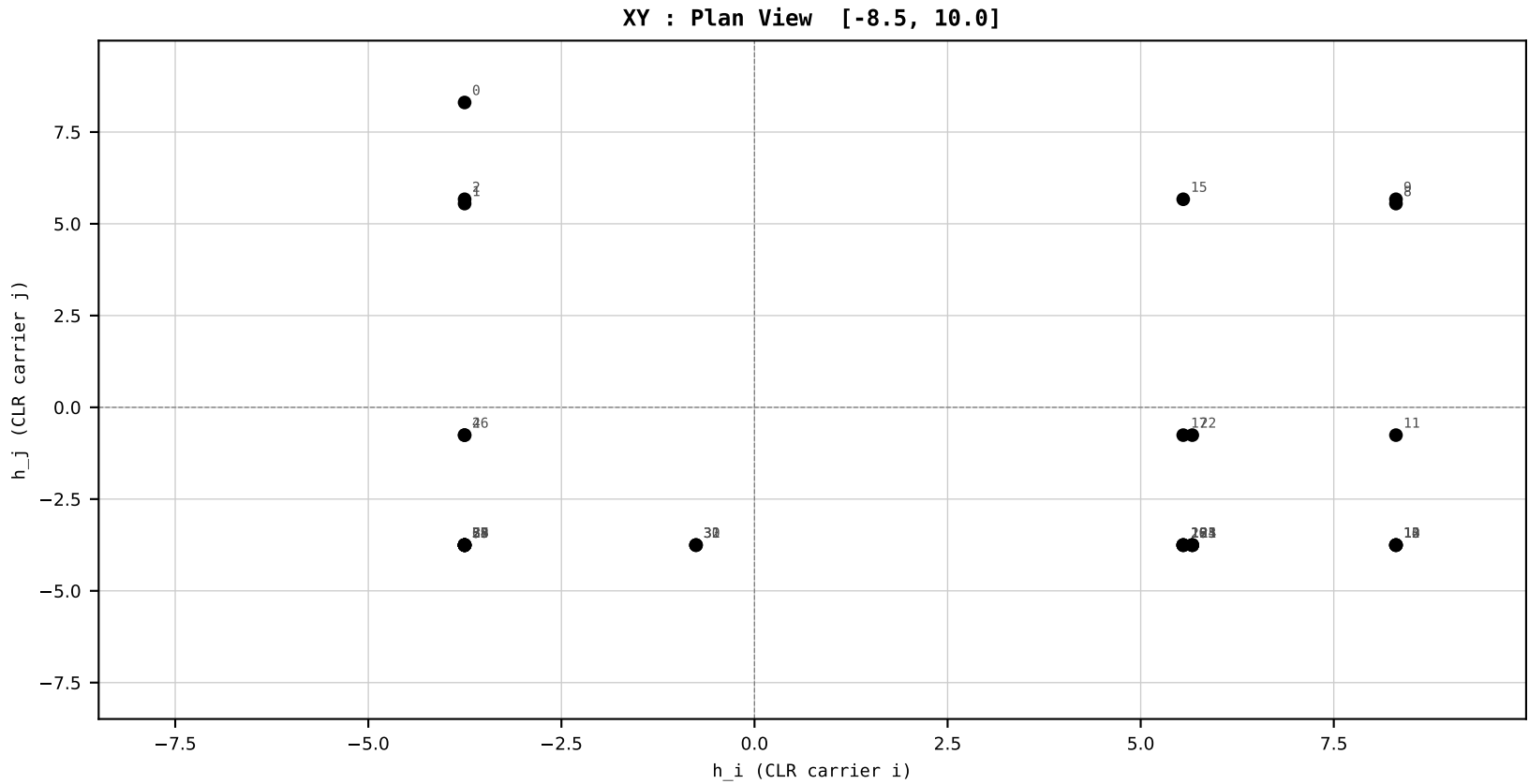
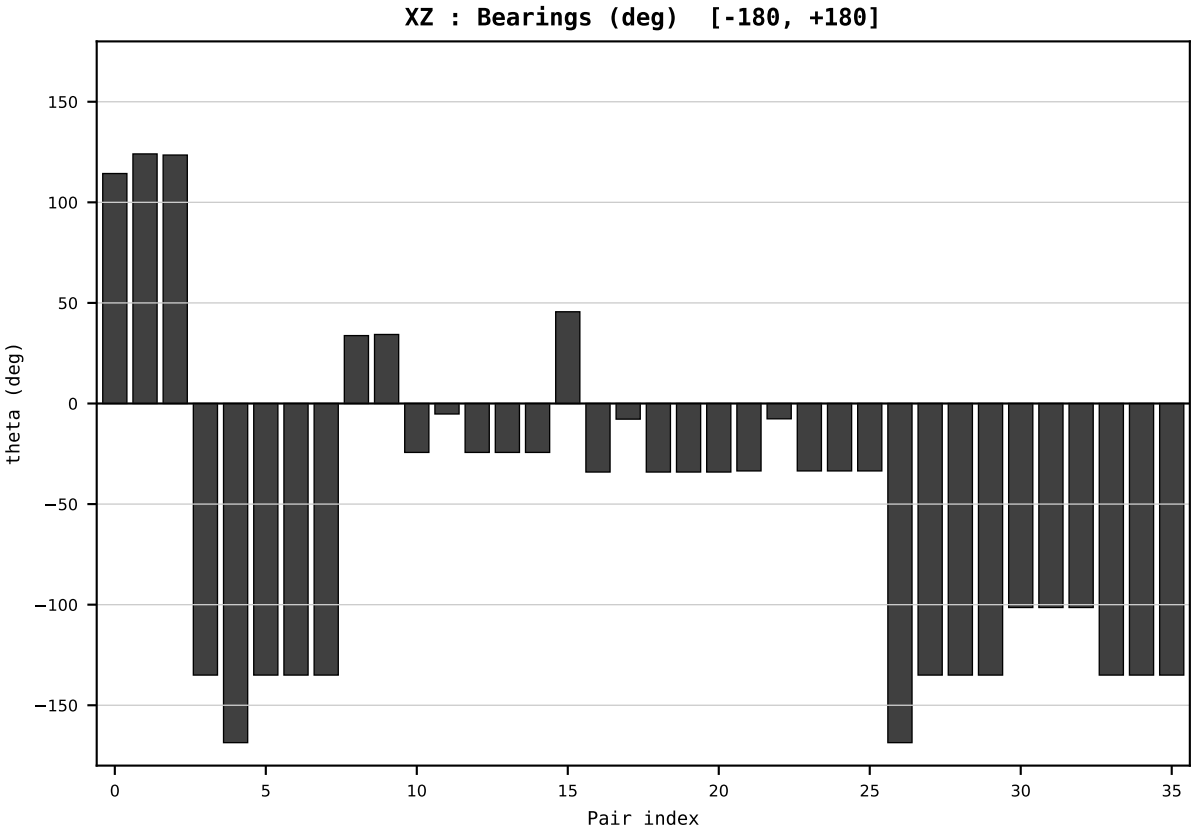
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.795717
Ring = Hs-5
E_metric= 202.9186
kappa_HS= 172520.00
omega = 4.4914 deg
d_A = 1.177094
Helm = Hydro
Helm d = +0.909068
DR = 12.058 HLR
DR ratio= 172520.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

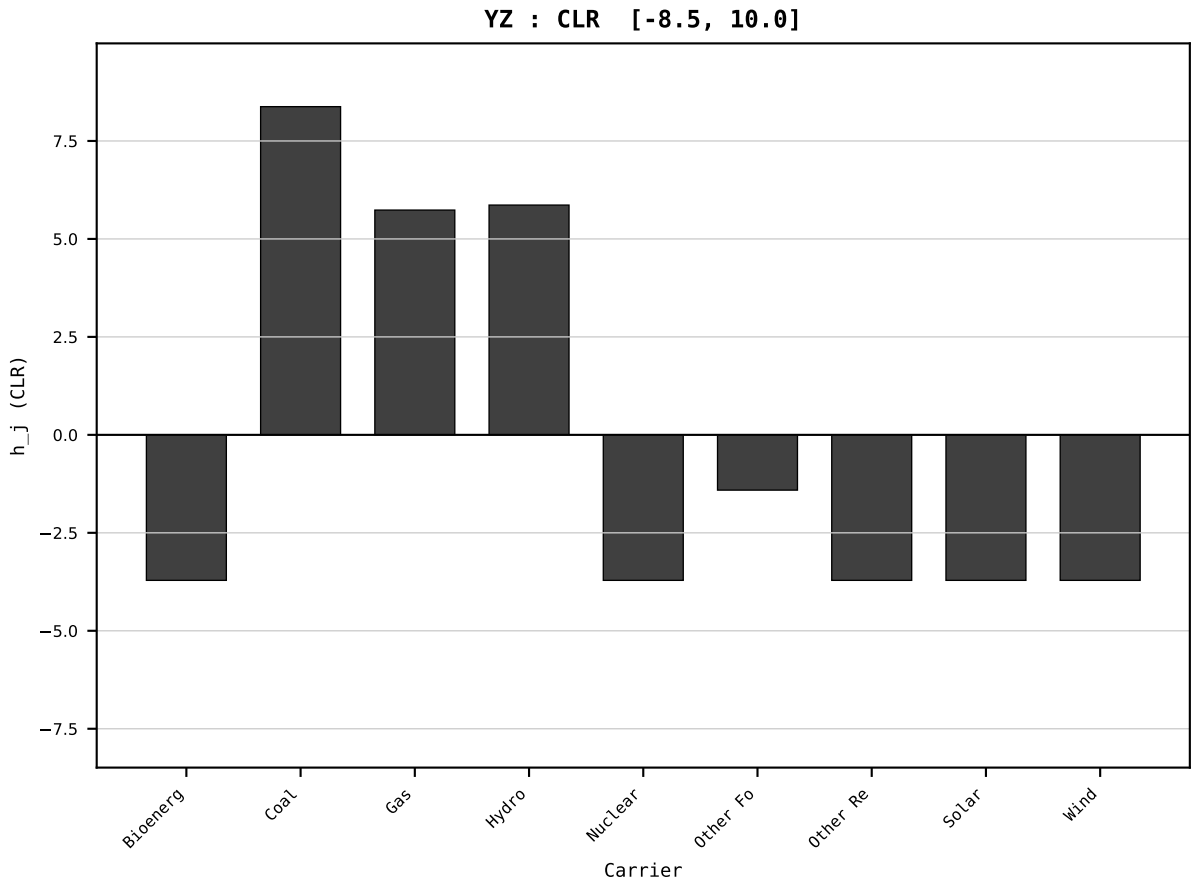
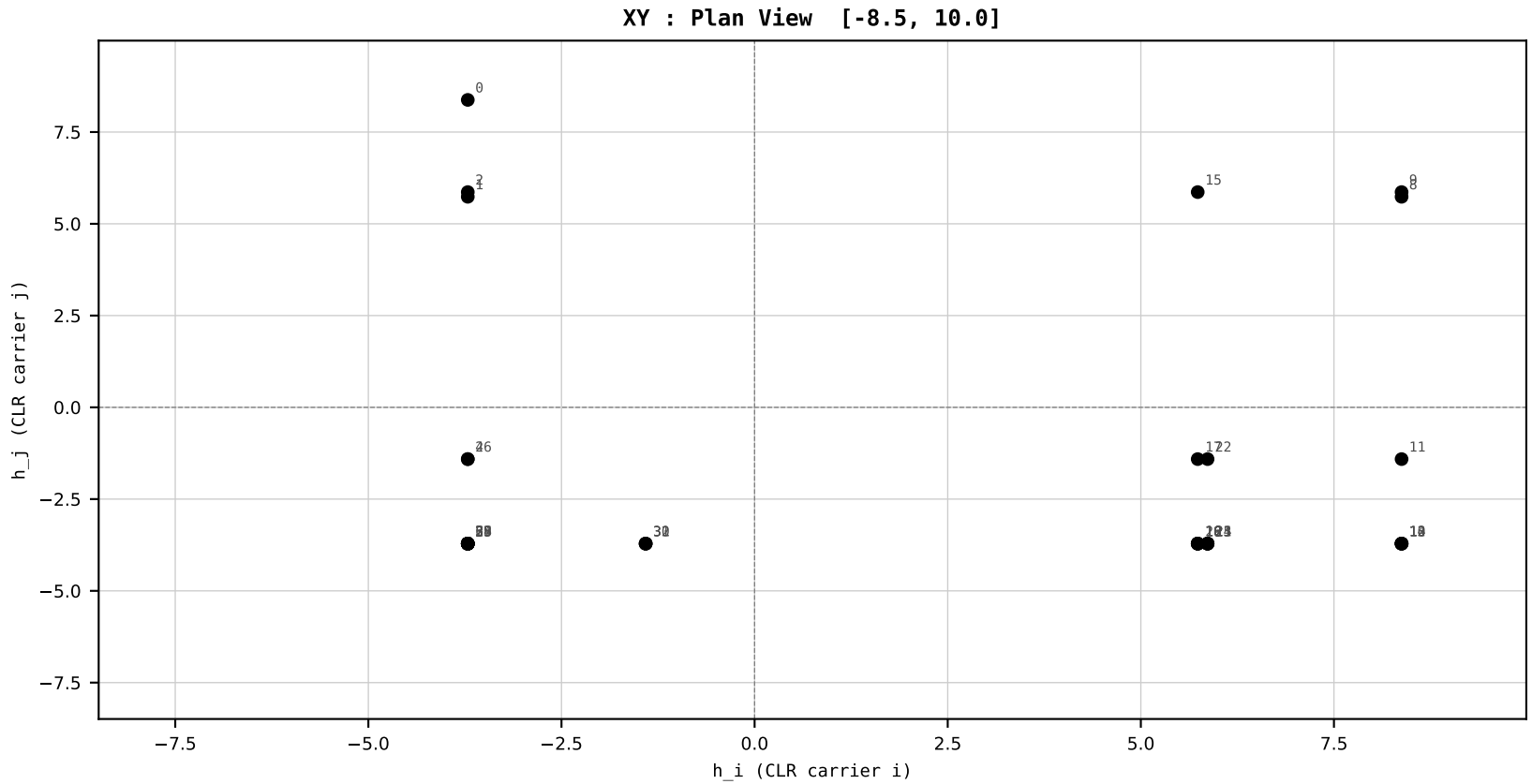
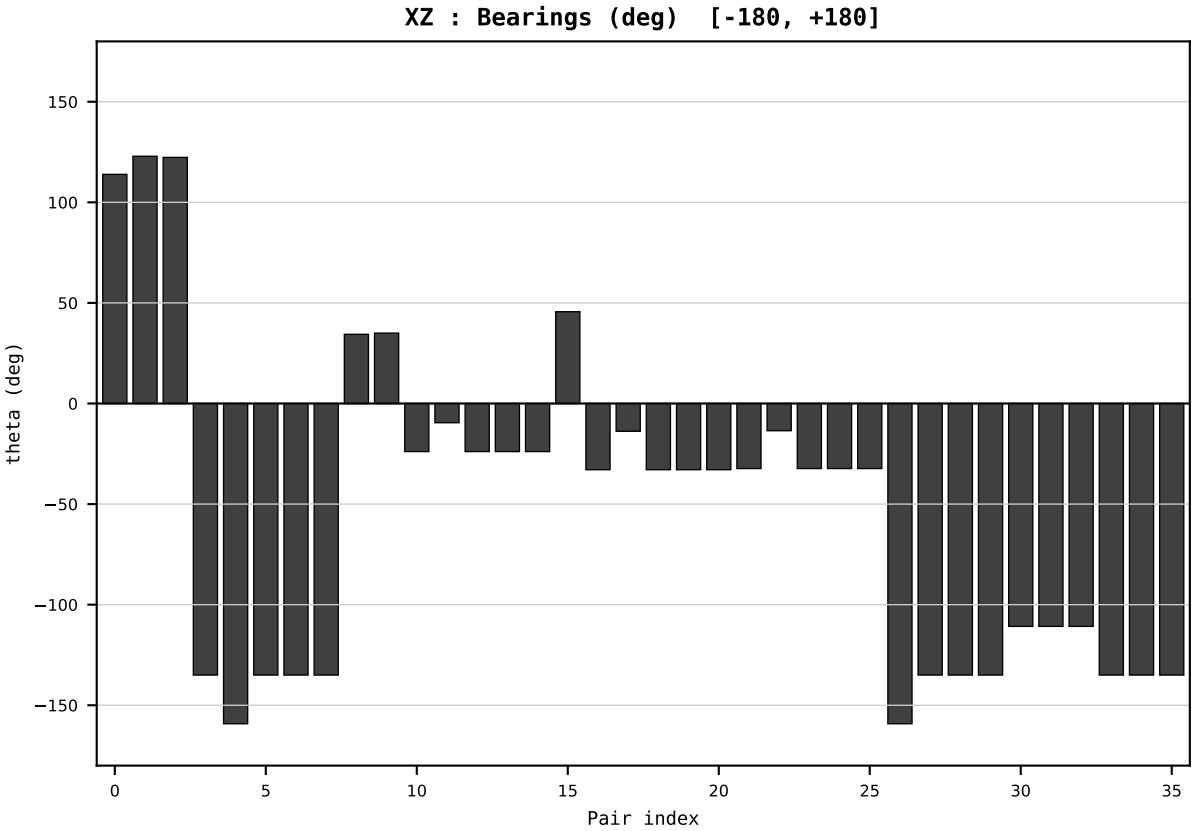
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.780110
Ring = Hs-5
E_metric= 208.3655
kappa_HS= 177840.00
omega = 2.7575 deg
d_A = 0.715719
Helm = Other Fossil
Helm d = -0.652970
DR = 12.089 HLR
DR ratio= 177840.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

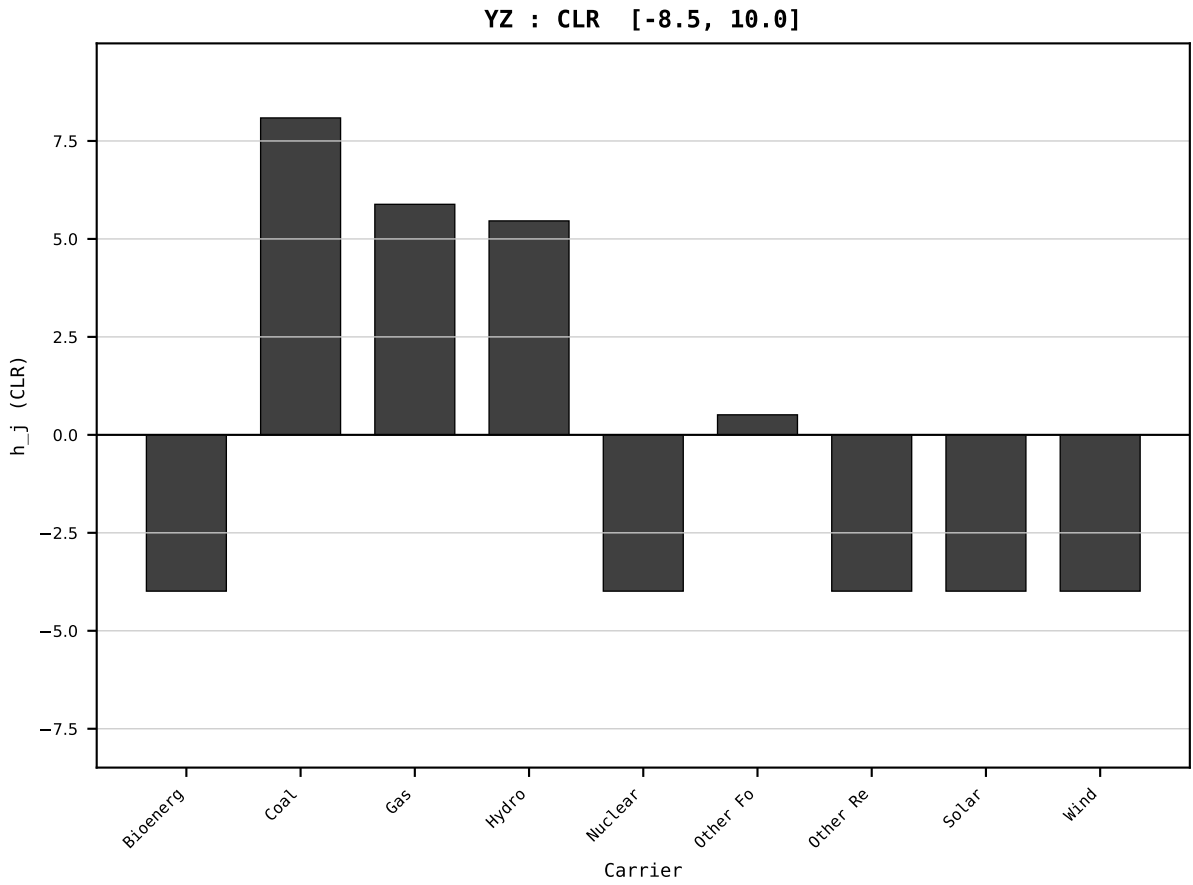
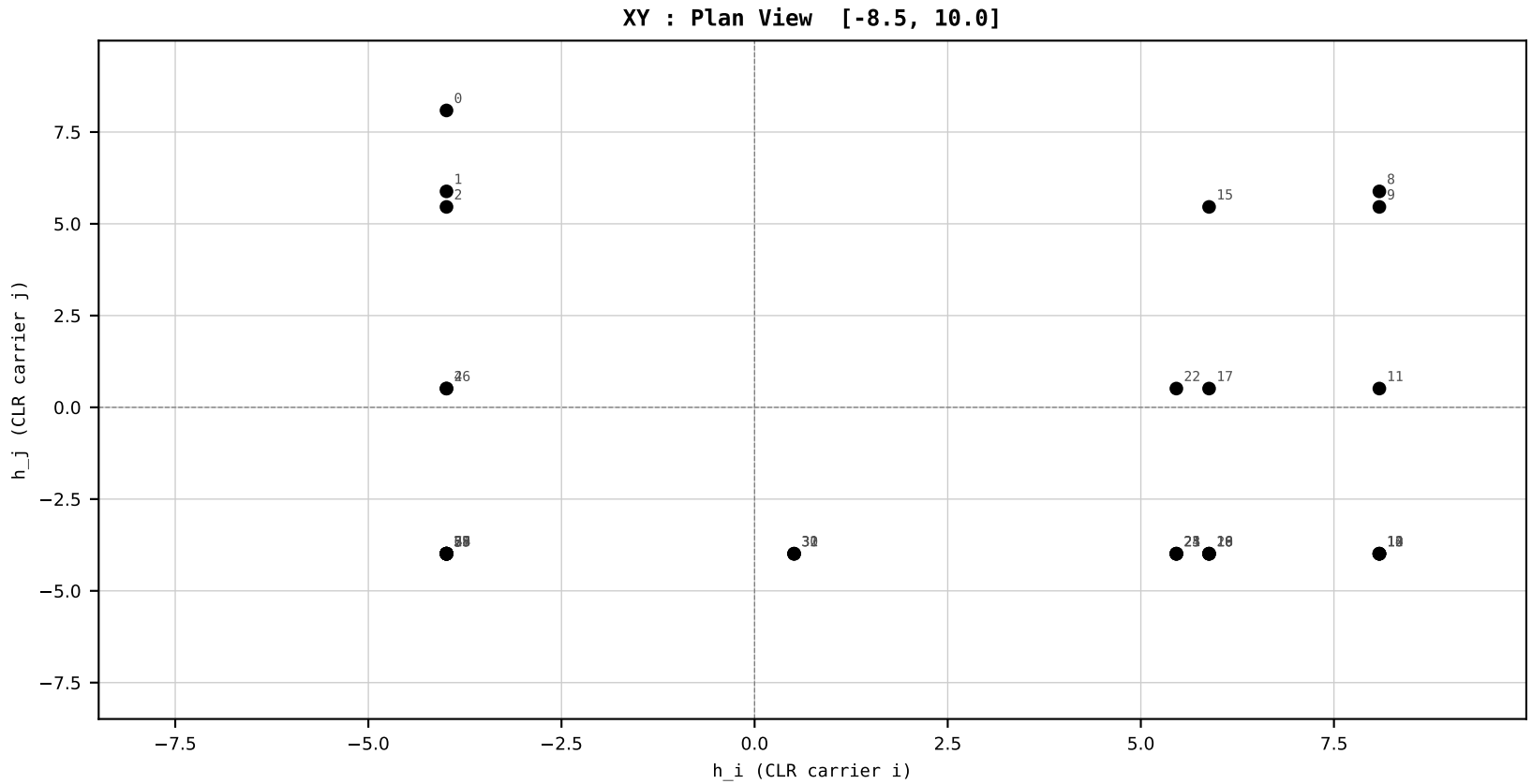
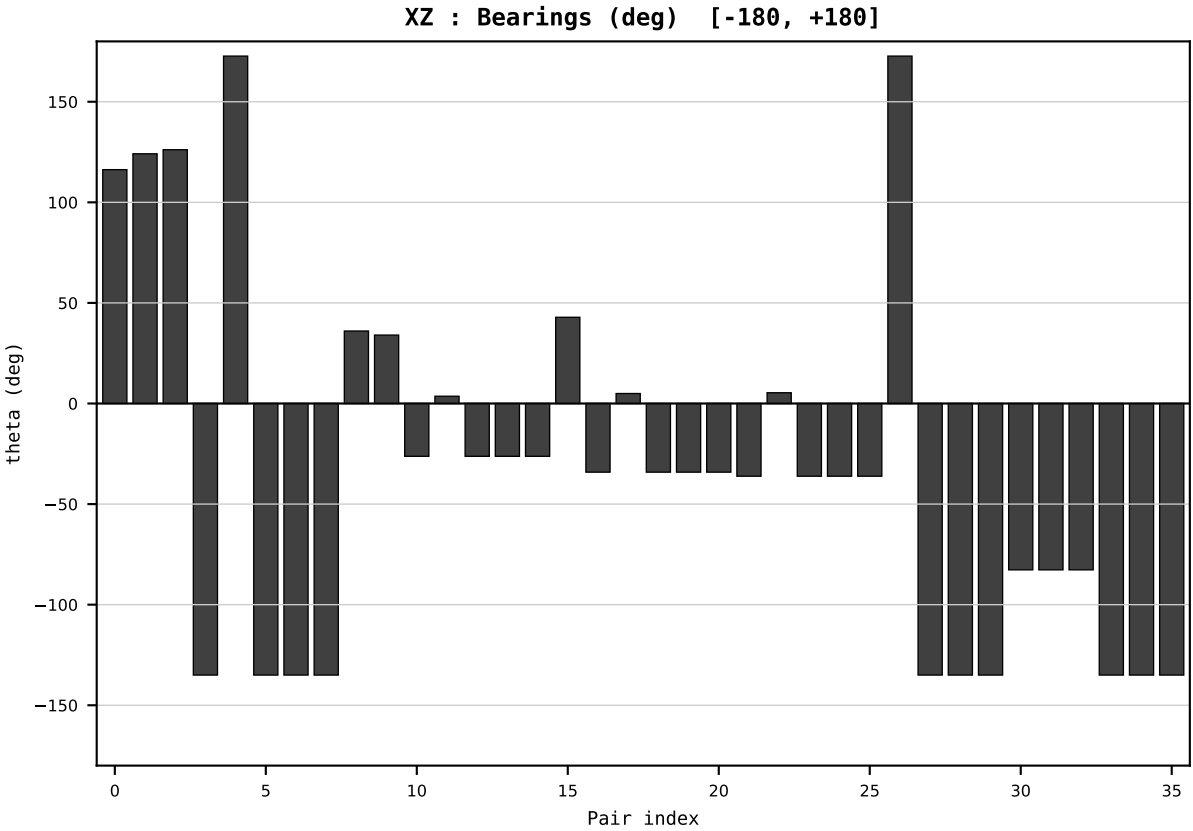
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.755279
Ring = Hs-5
E_metric= 209.6435
kappa_HS= 175670.00
omega = 8.2619 deg
d_A = 2.083316
Helm = Other Fossil
Helm d = +1.921770
DR = 12.076 HLR
DR ratio= 175670.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

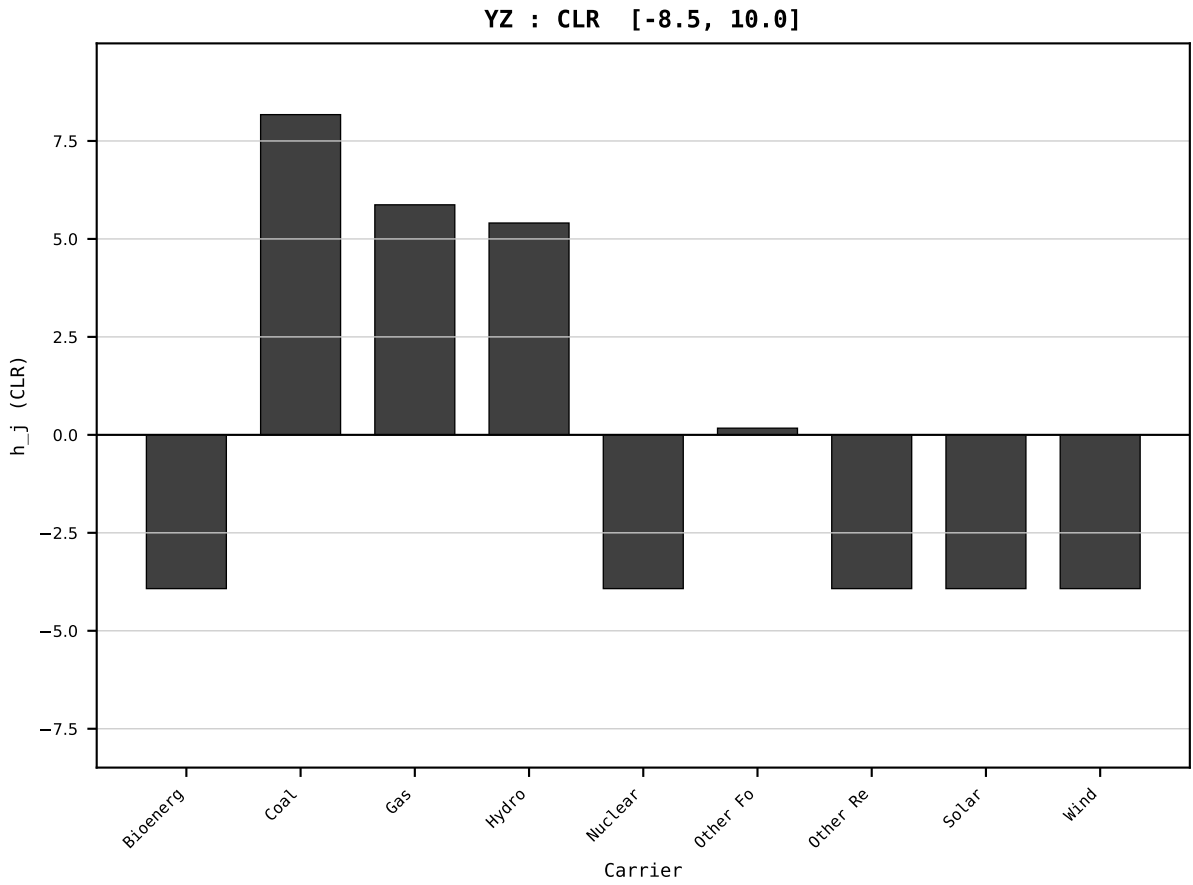
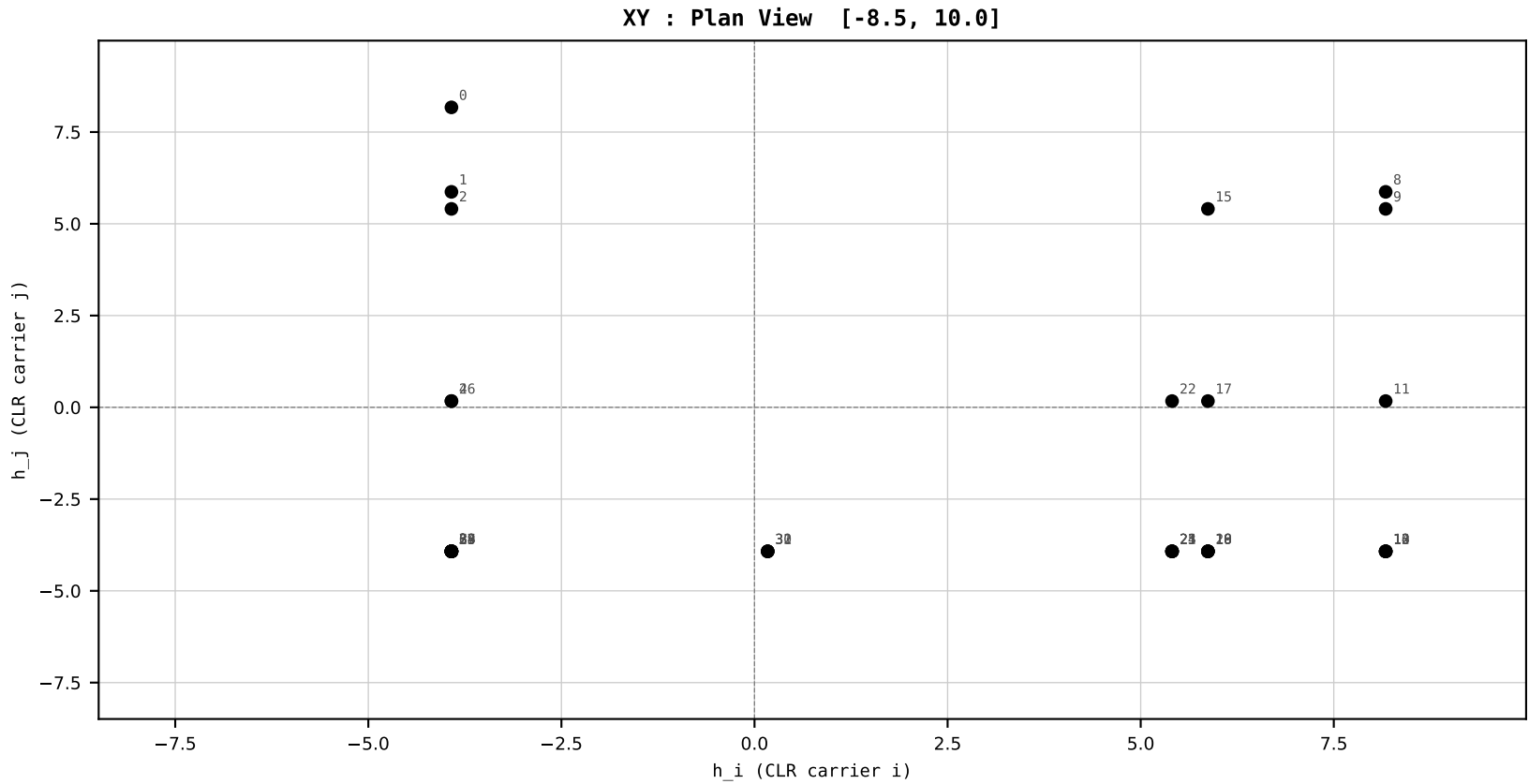
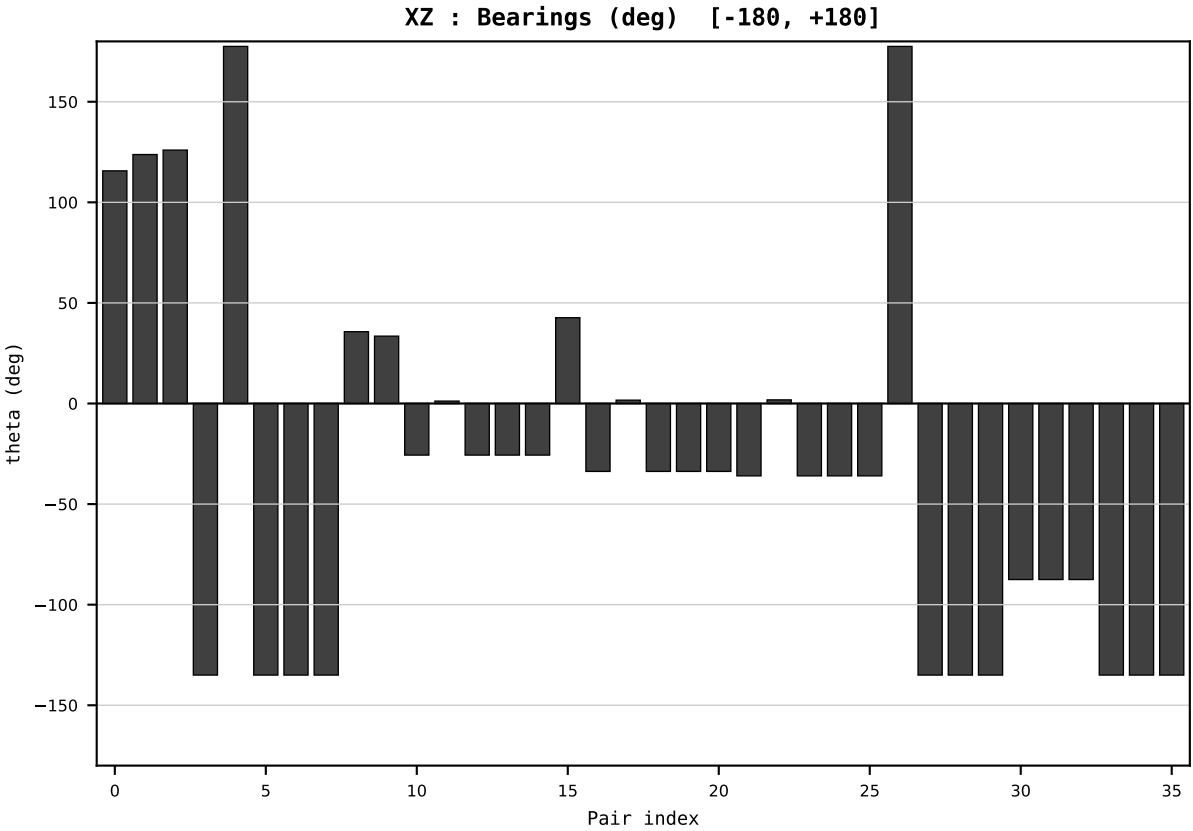
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.771736
Ring = Hs-5
E_metric= 207.4798
kappa_HS= 179120.00
omega = 1.4930 deg
d_A = 0.383687
Helm = Other Fossil
Helm d = -0.340748
DR = 12.096 HLR
DR ratio= 179120.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

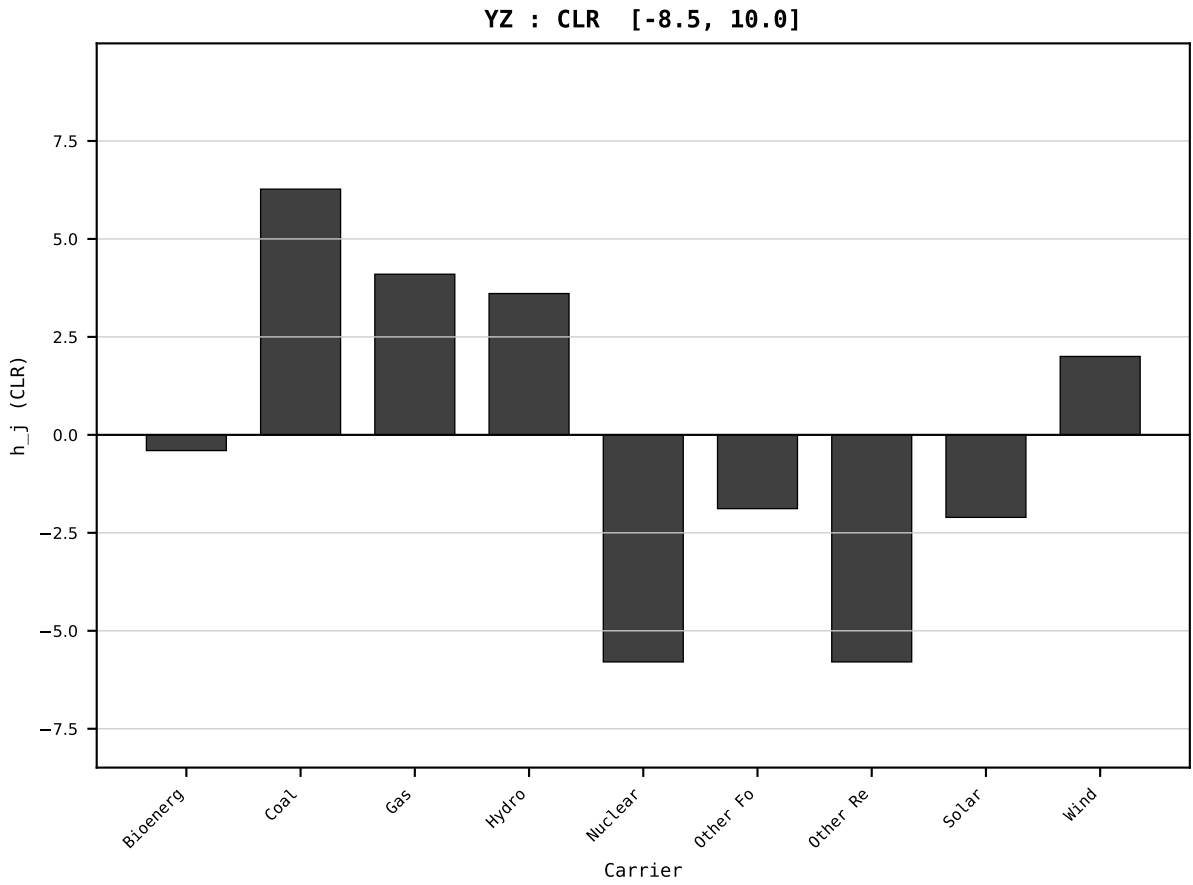
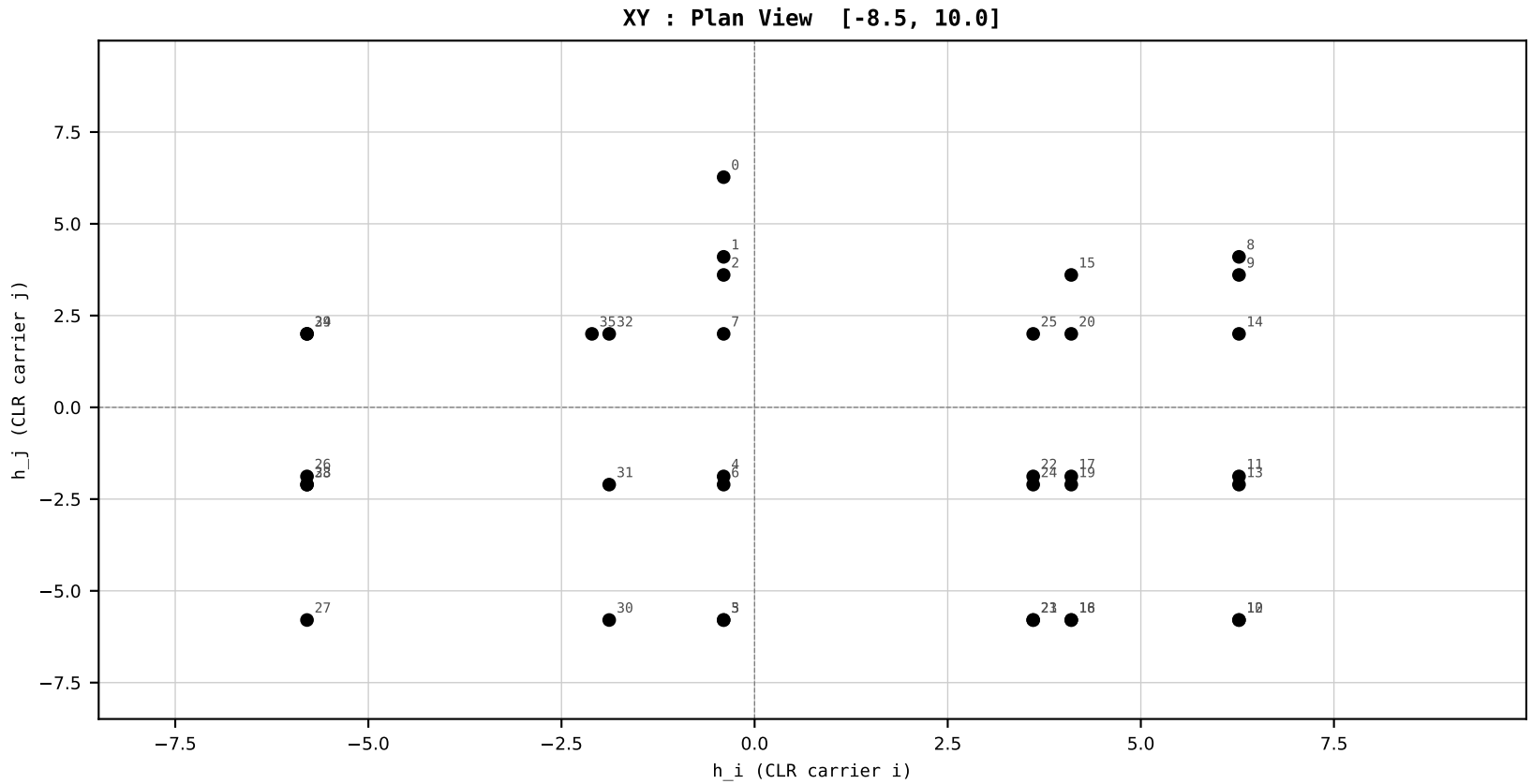
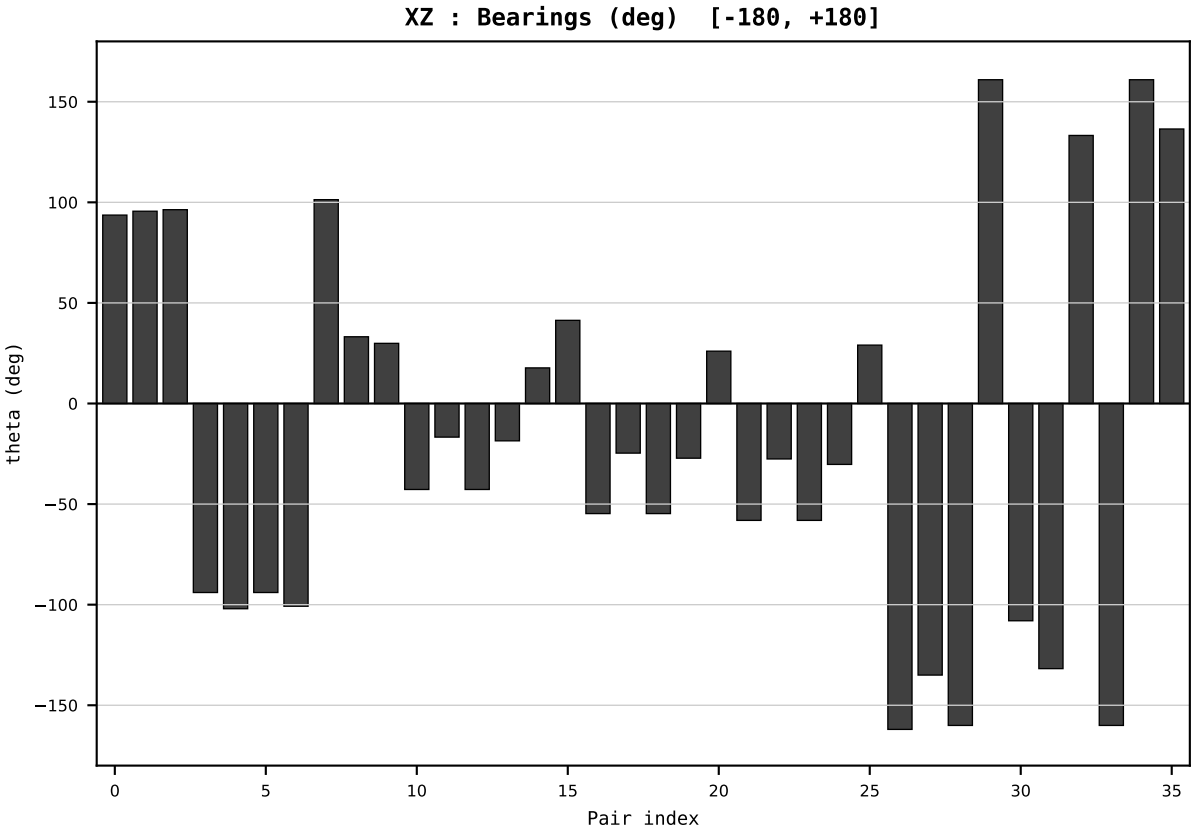
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.725332
Ring = Hs-4
E_metric= 148.4295
kappa_HS= 173730.00
omega = 36.0173 deg
d_A = 8.486789
Helm = Wind
Helm d = +5.924974
DR = 12.065 HLR
DR ratio= 173730.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

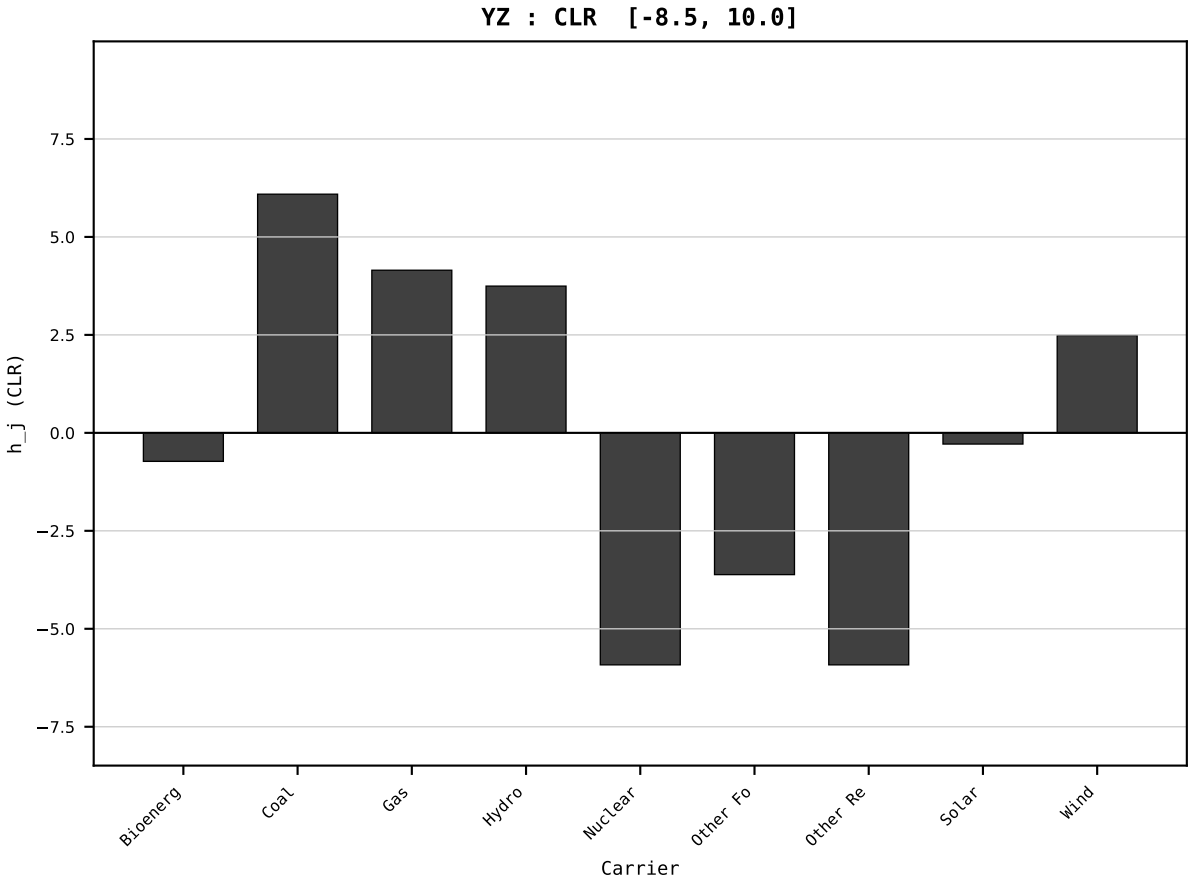
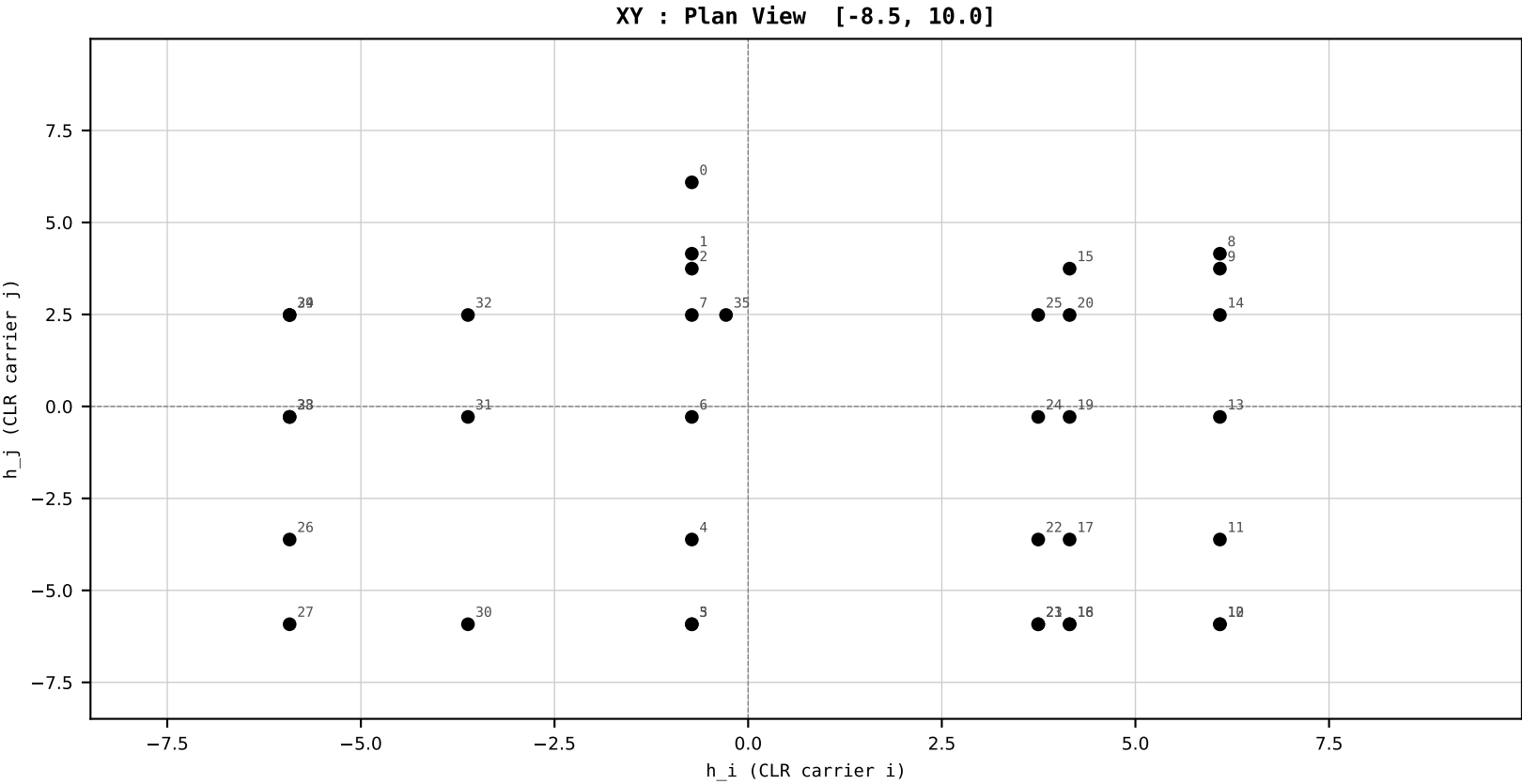
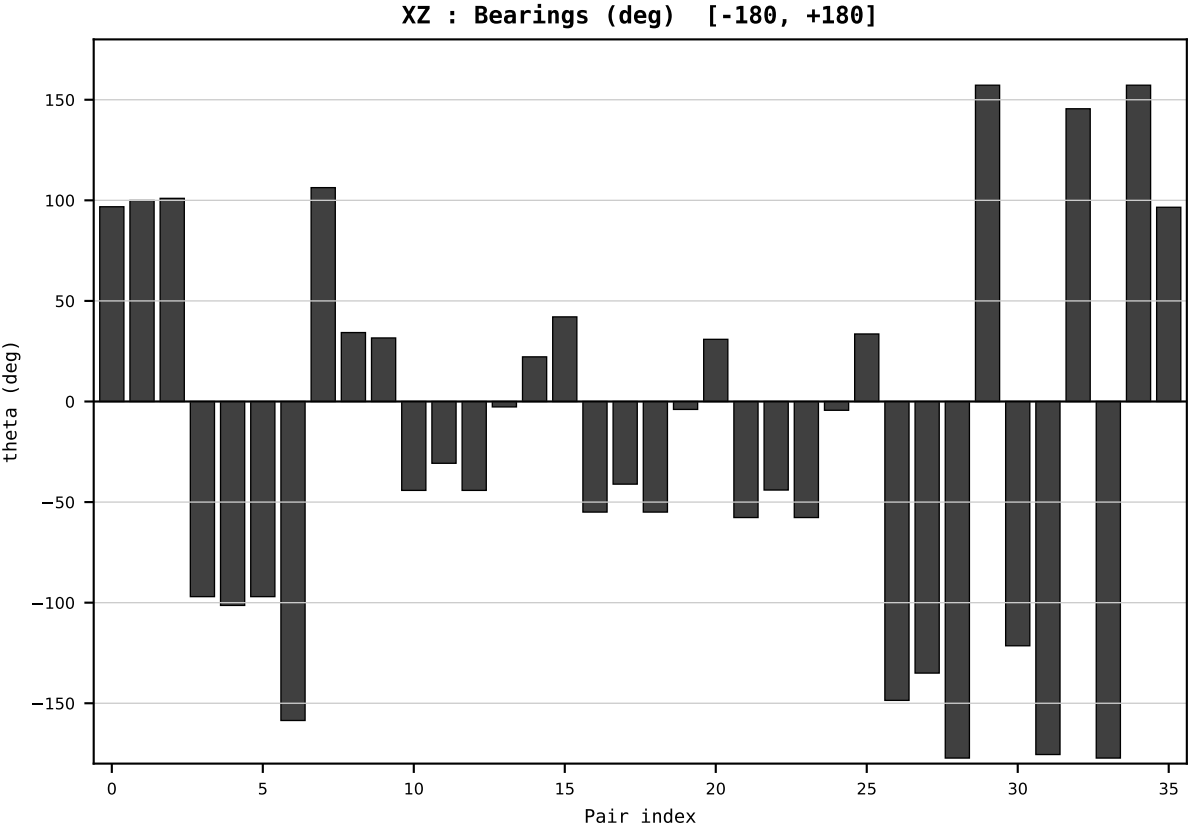
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.669015
Ring = Hs-4
E_metric= 158.3215
kappa_HS= 164580.00
omega = 11.9015 deg
d_A = 2.598112
Helm = Solar
Helm d = +1.820108
DR = 12.011 HLR
DR ratio= 164580.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

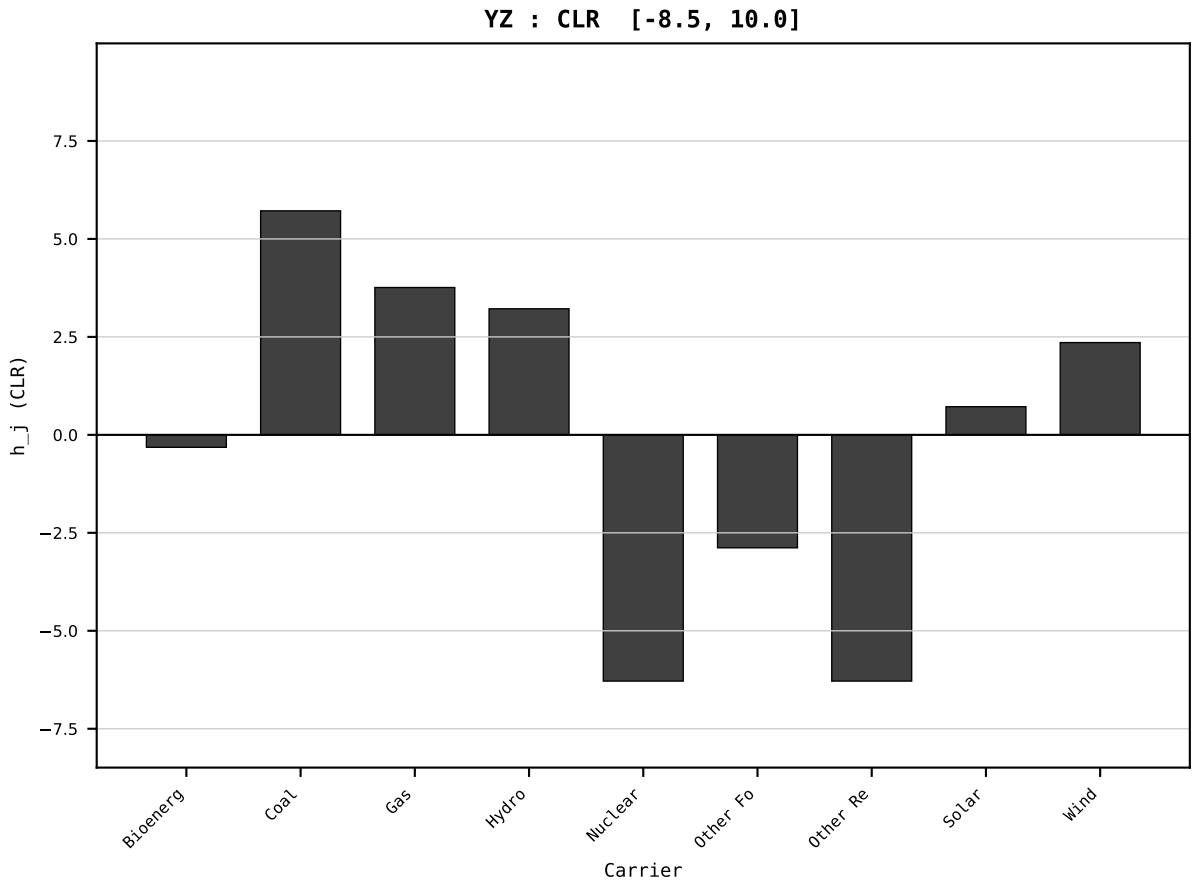
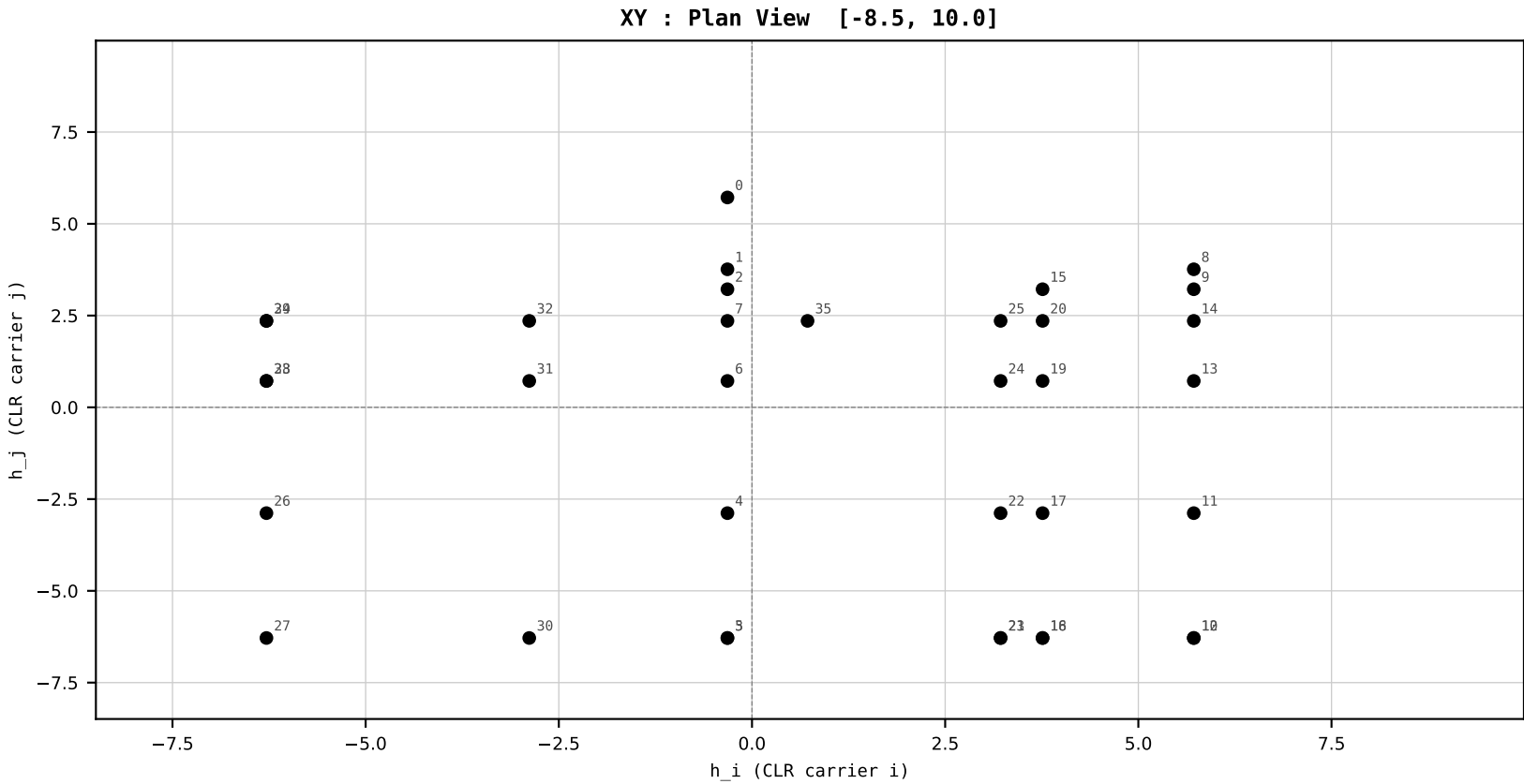
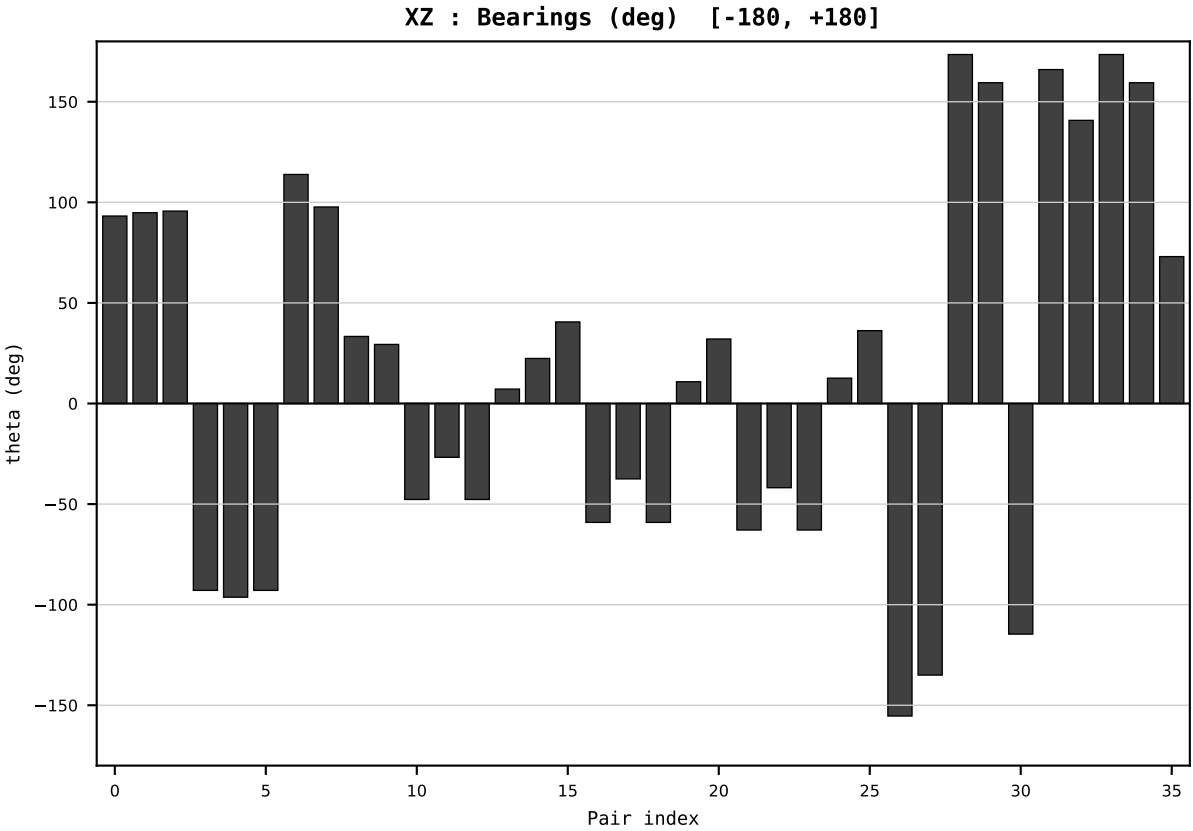
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.659385
Ring = Hs-4
E_metric= 150.6365
kappa_HS= 162980.00
omega = 7.2543 deg
d_A = 1.602461
Helm = Solar
Helm d = +1.004418
DR = 12.001 HLR
DR ratio= 162980.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



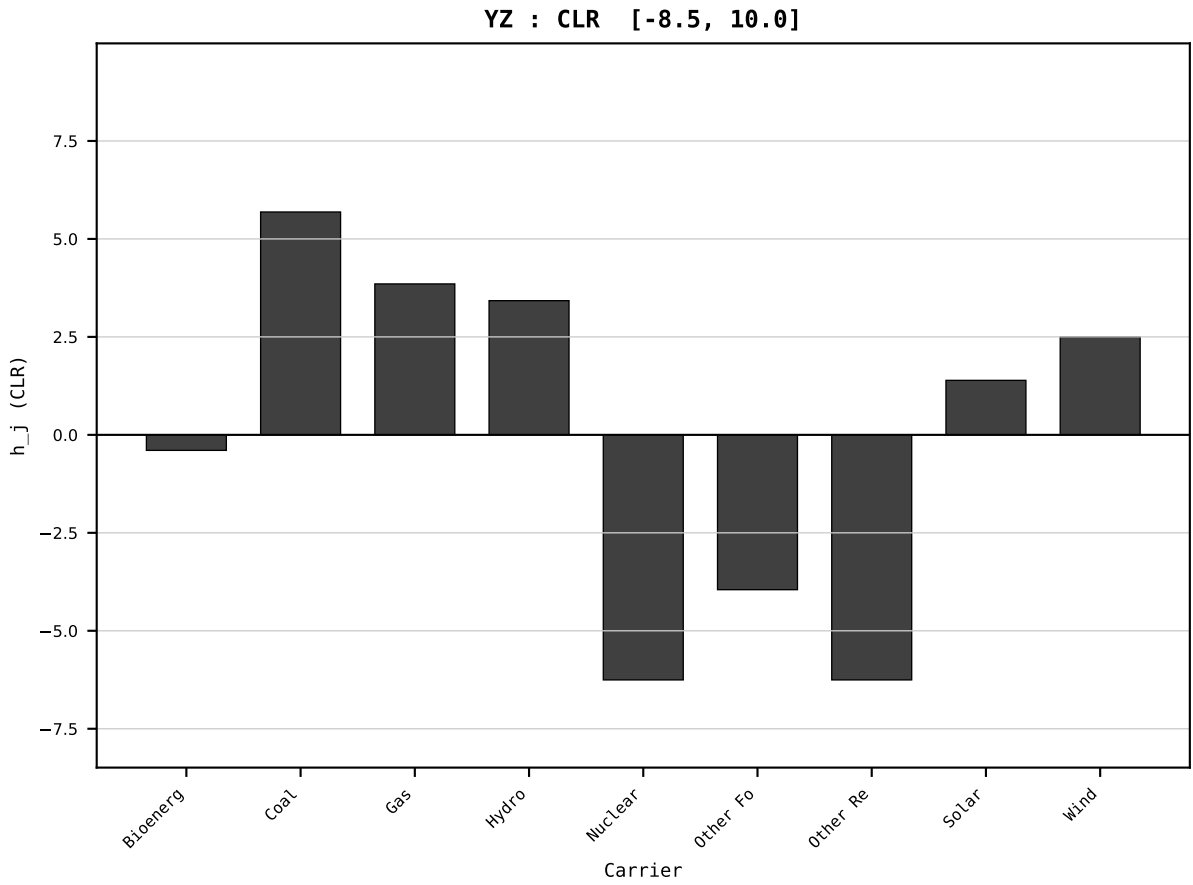
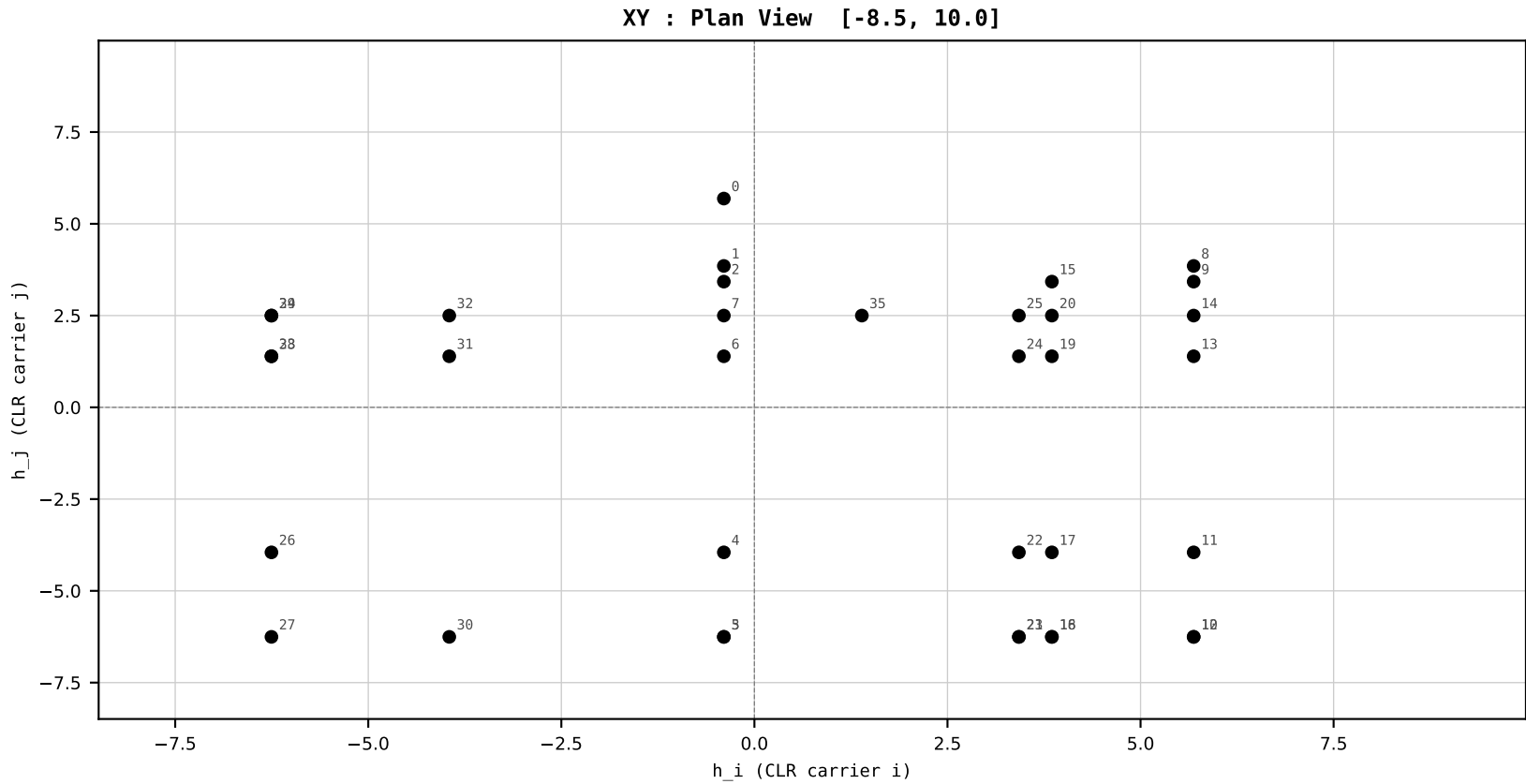
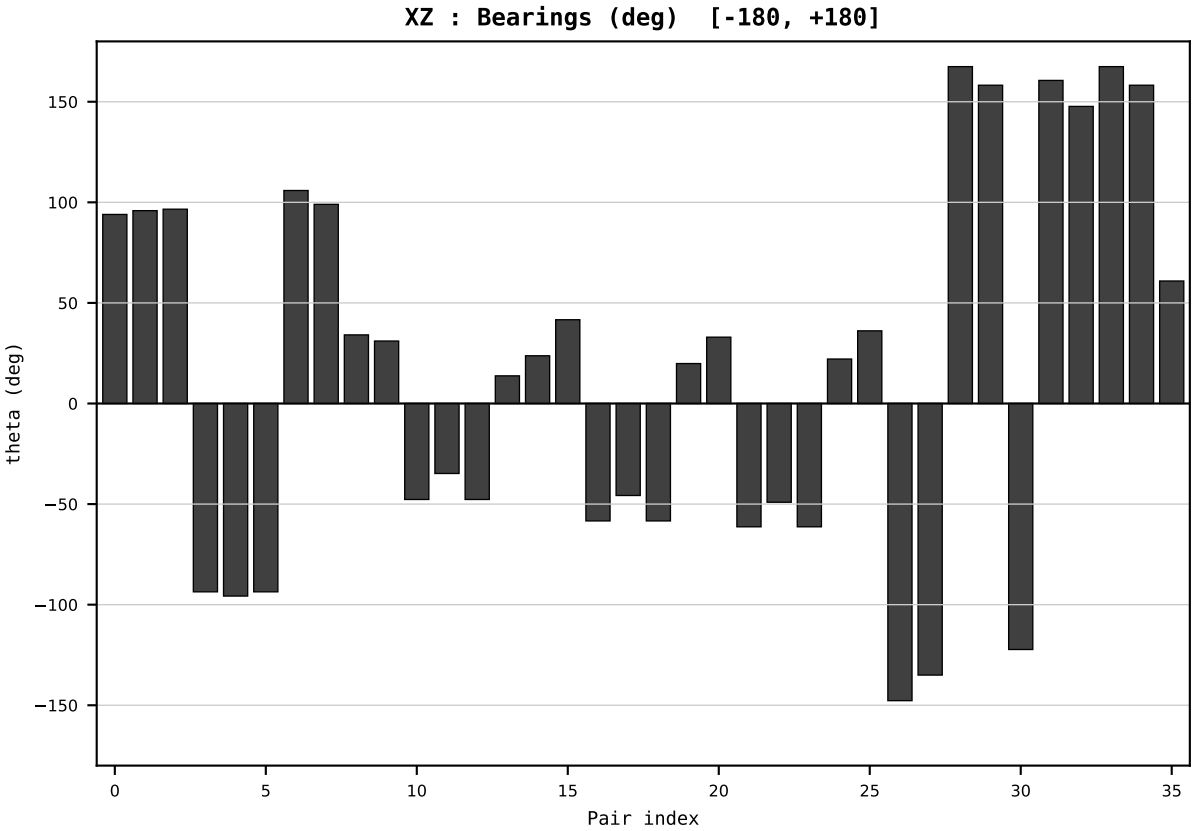
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.620841
Ring = Hs-4
E_metric= 161.0838
kappa_HS= 153490.00
omega = 5.6224 deg
d_A = 1.293789
Helm = Other Fossil
Helm d = -1.068245
DR = 11.941 HLR
DR ratio= 153490.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

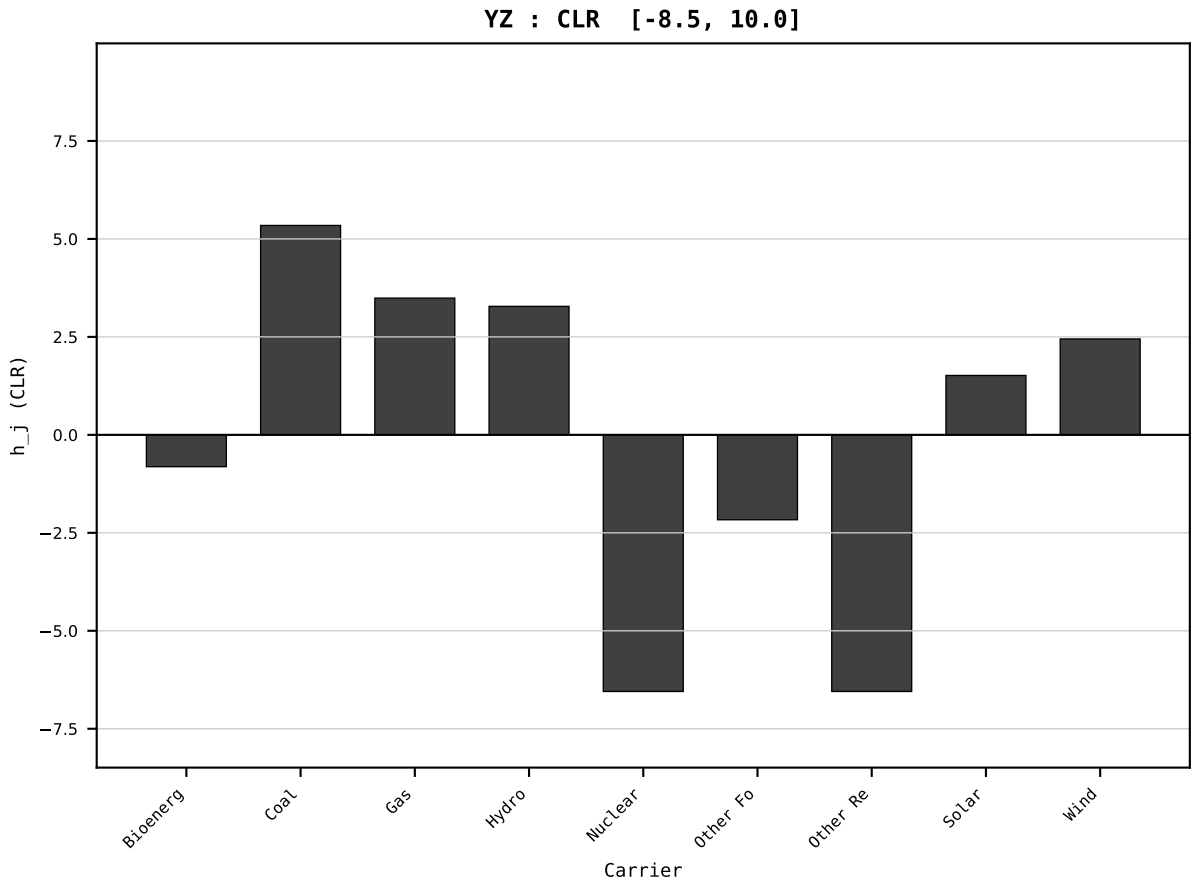
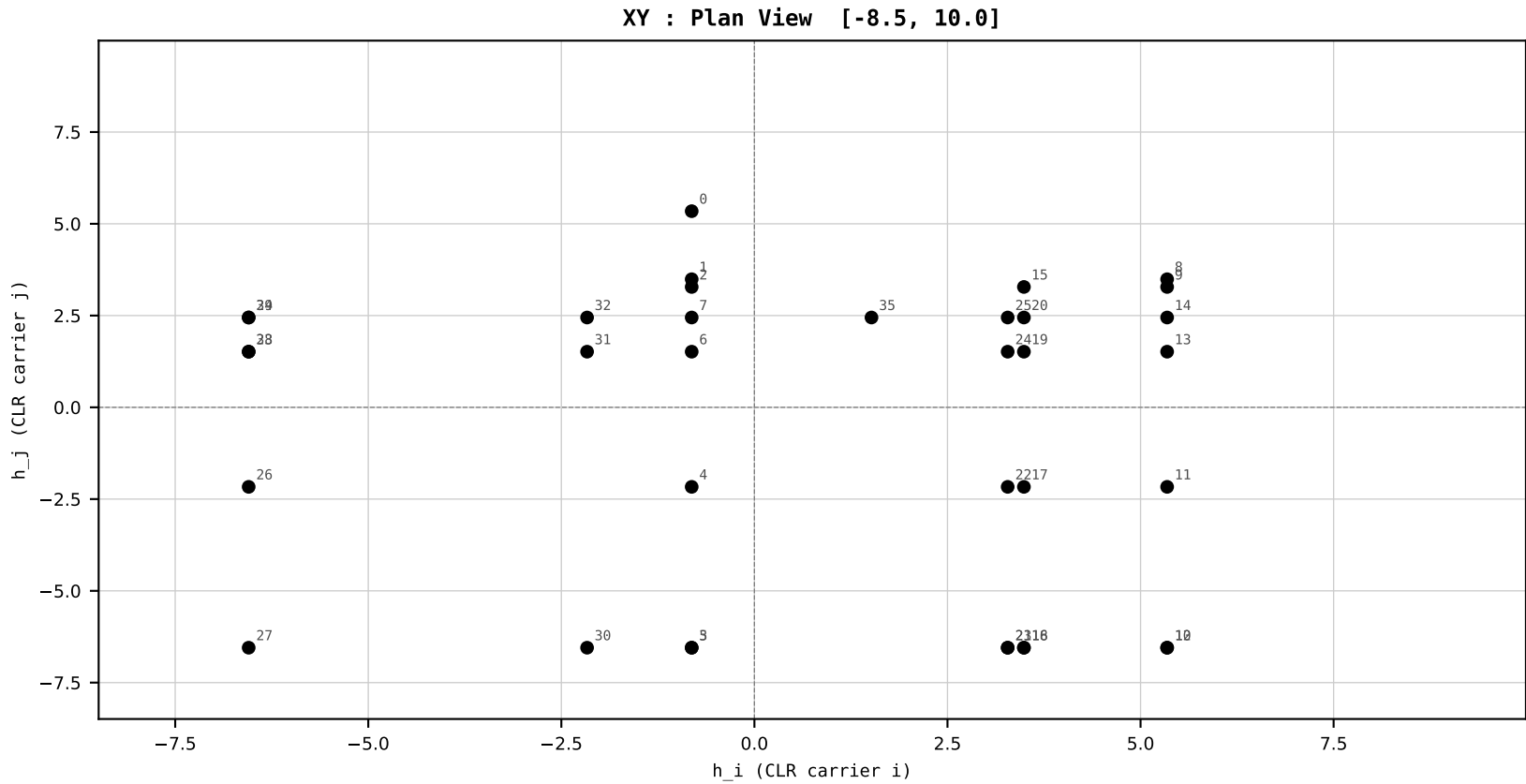
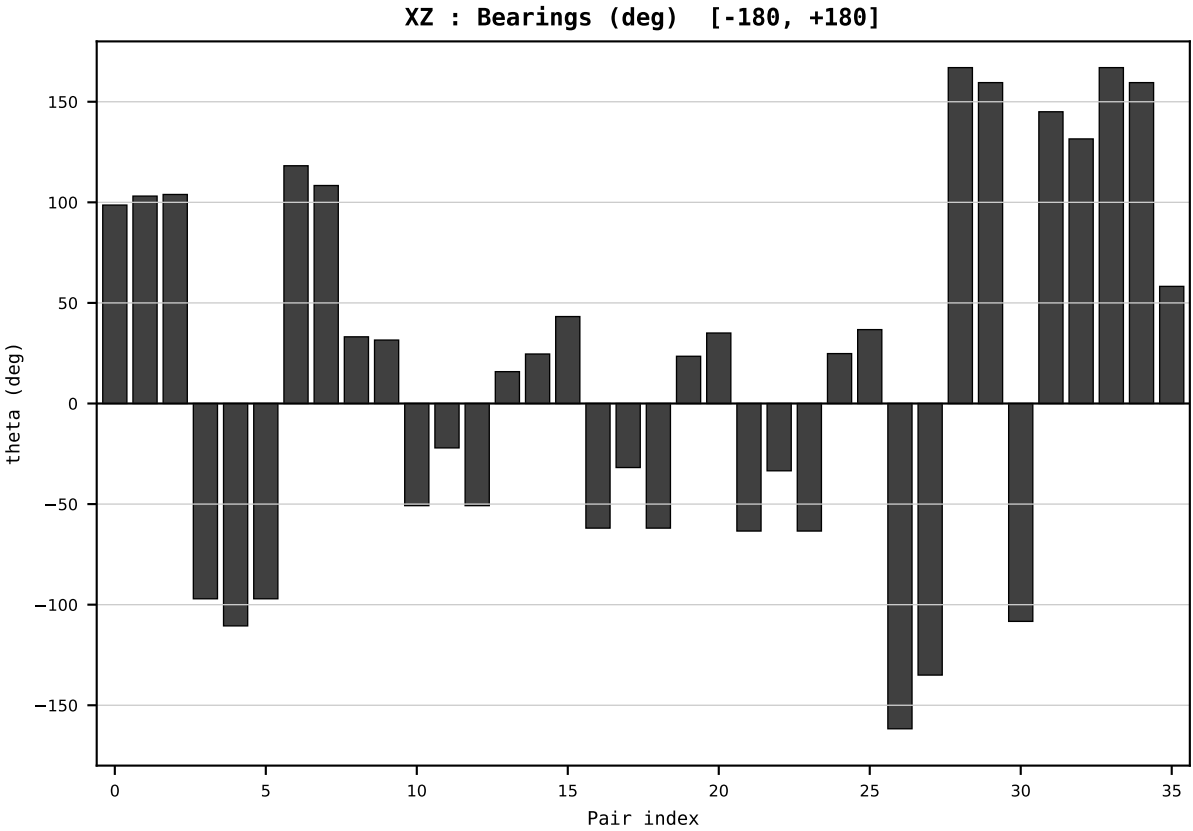
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.587671
Ring = Hs-4
E_metric= 150.9049
kappa_HS= 146210.00
omega = 8.7801 deg
d_A = 1.954538
Helm = Other Fossil
Helm d = +1.784479
DR = 11.893 HLR
DR ratio= 146210.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

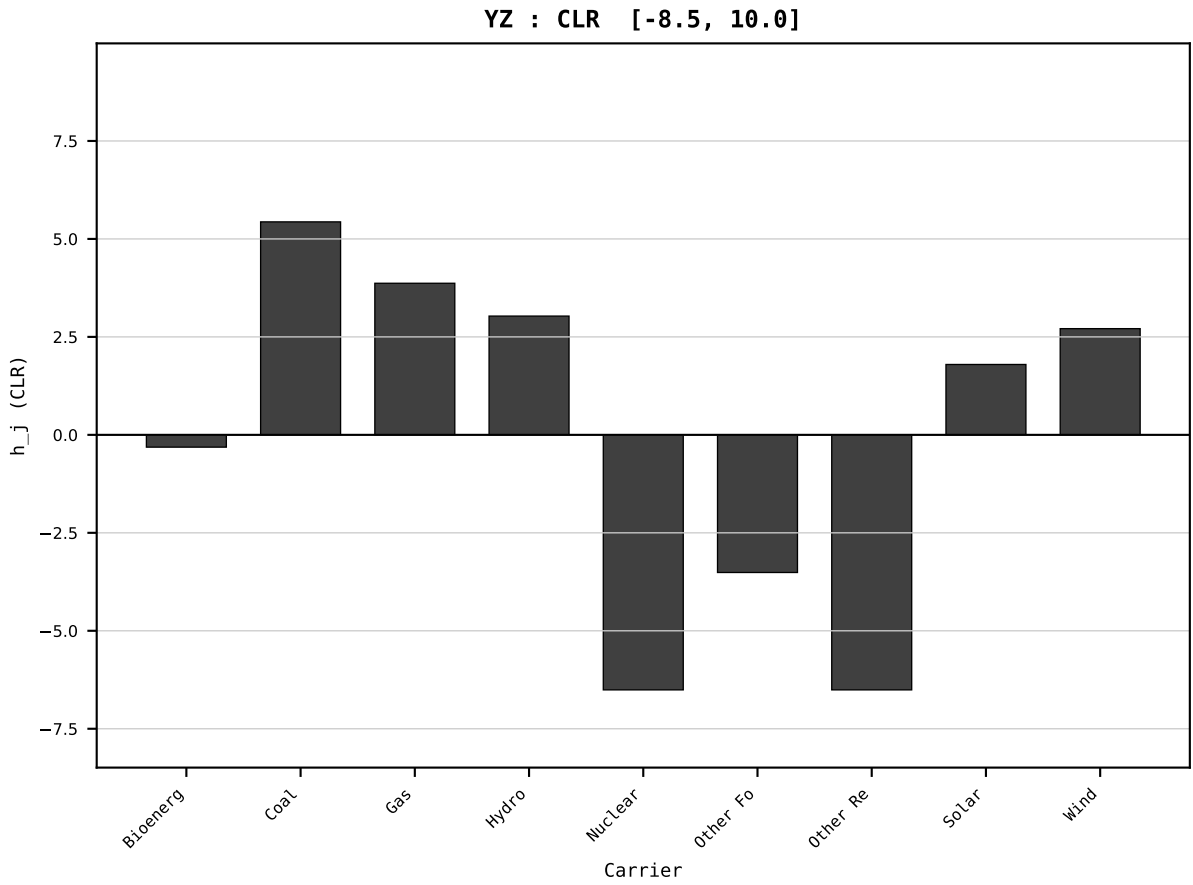
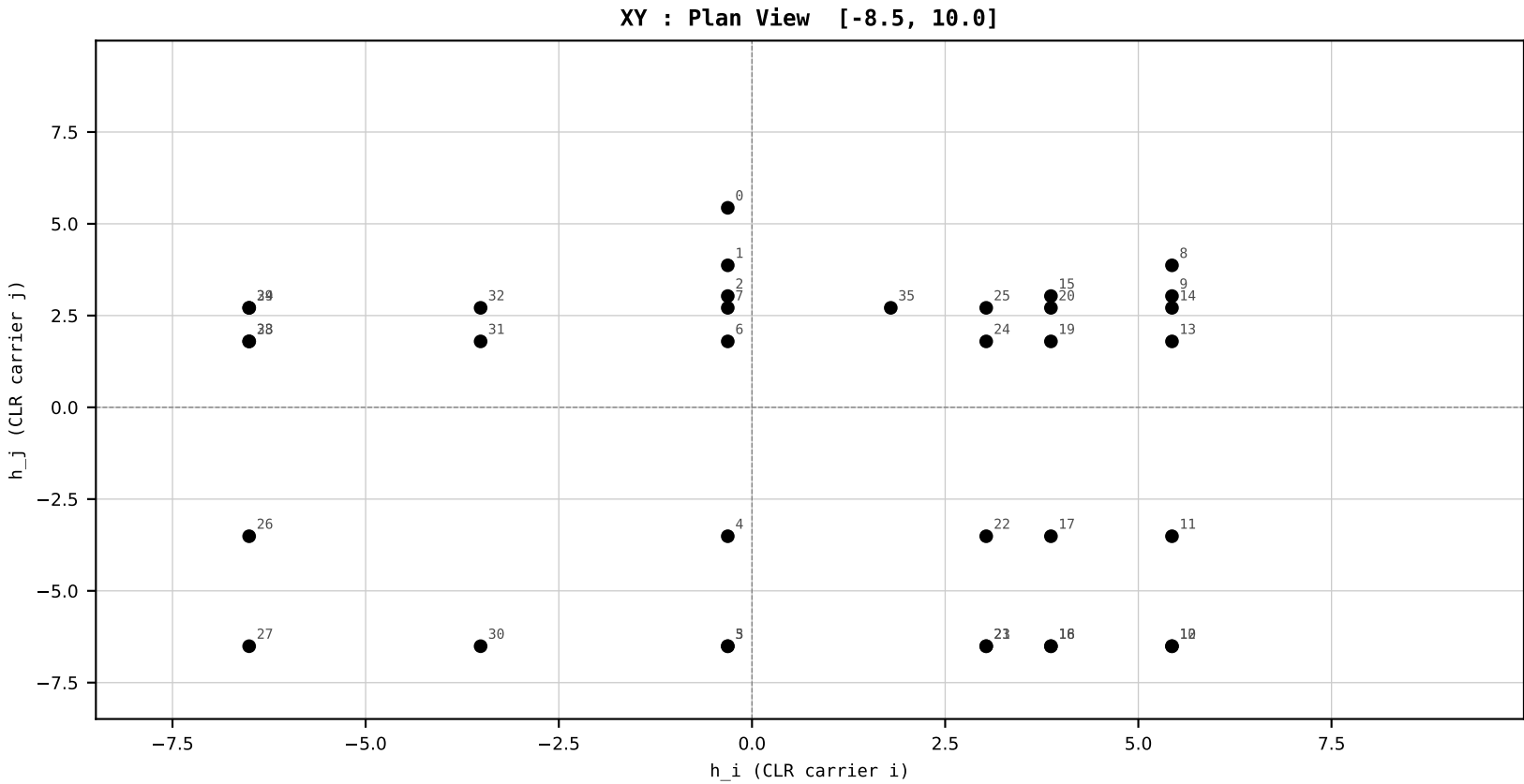
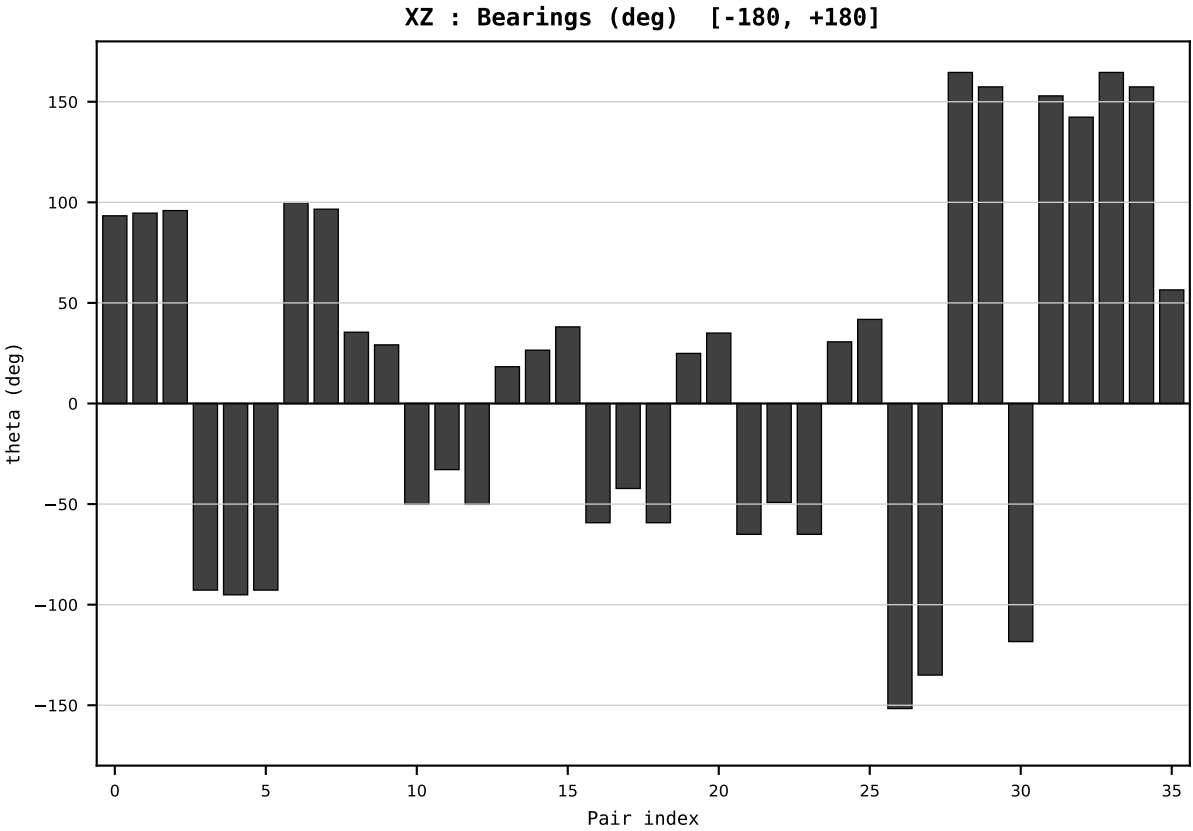
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.575153
Ring = Hs-4
E_metric= 161.4074
kappa_HS= 153760.00
omega = 6.8792 deg
d_A = 1.556841
Helm = Other Fossil
Helm d = -1.345462
DR = 11.943 HLR
DR ratio= 153760.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

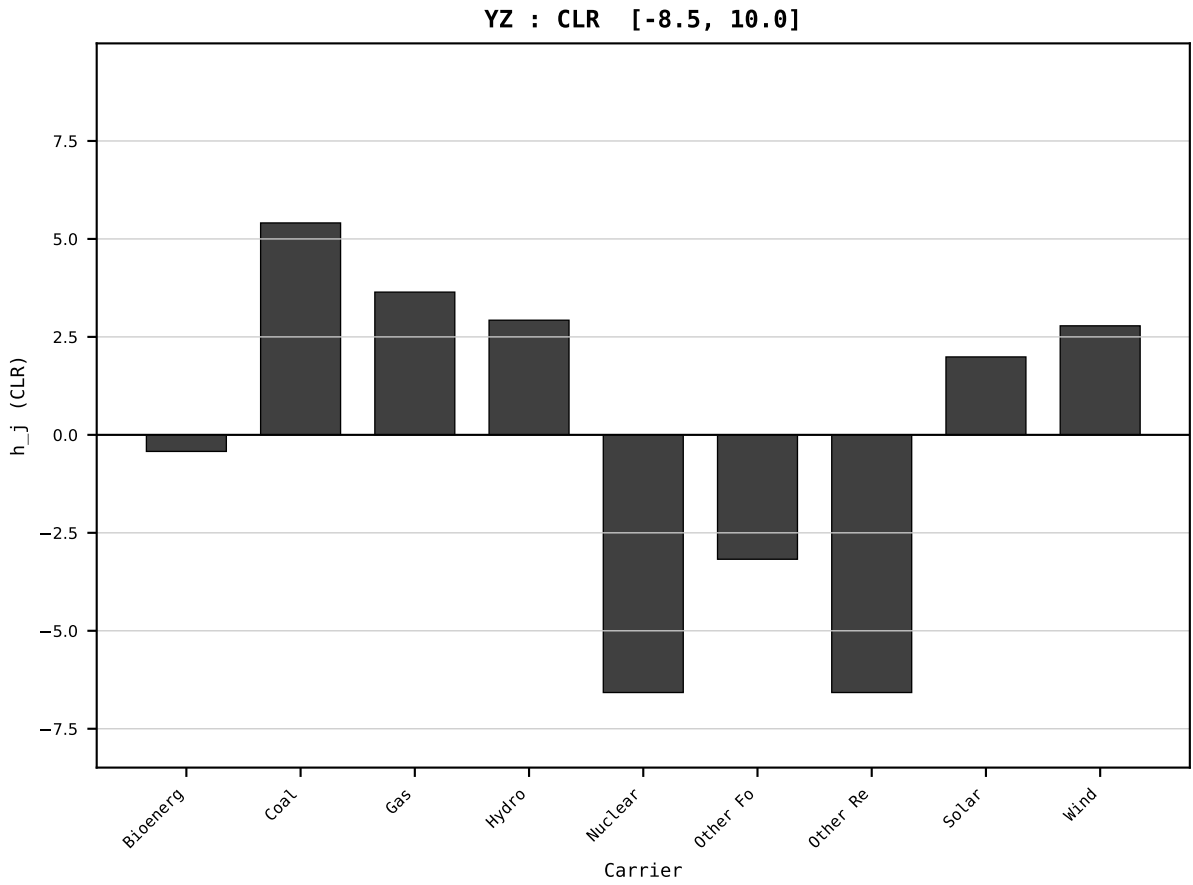
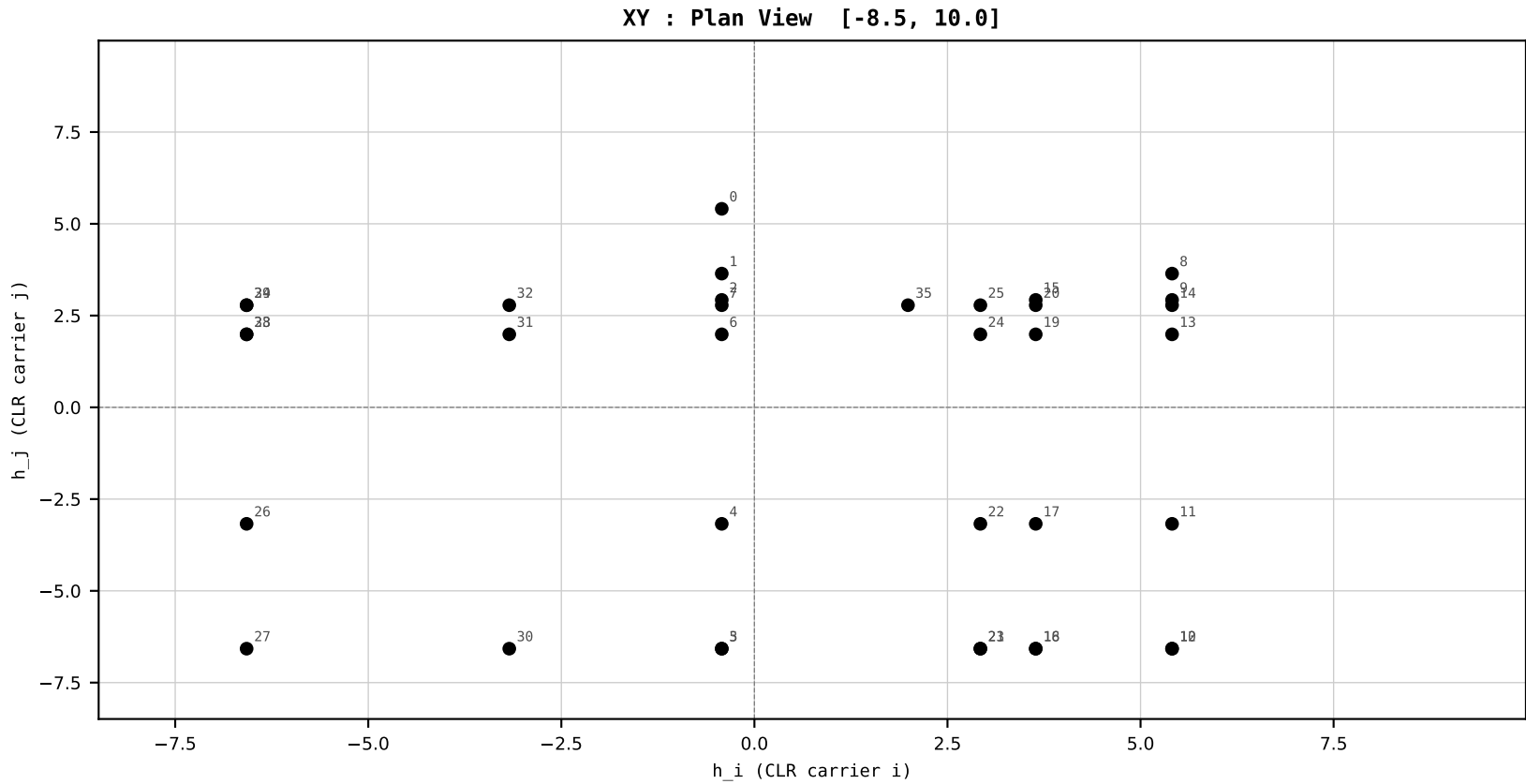
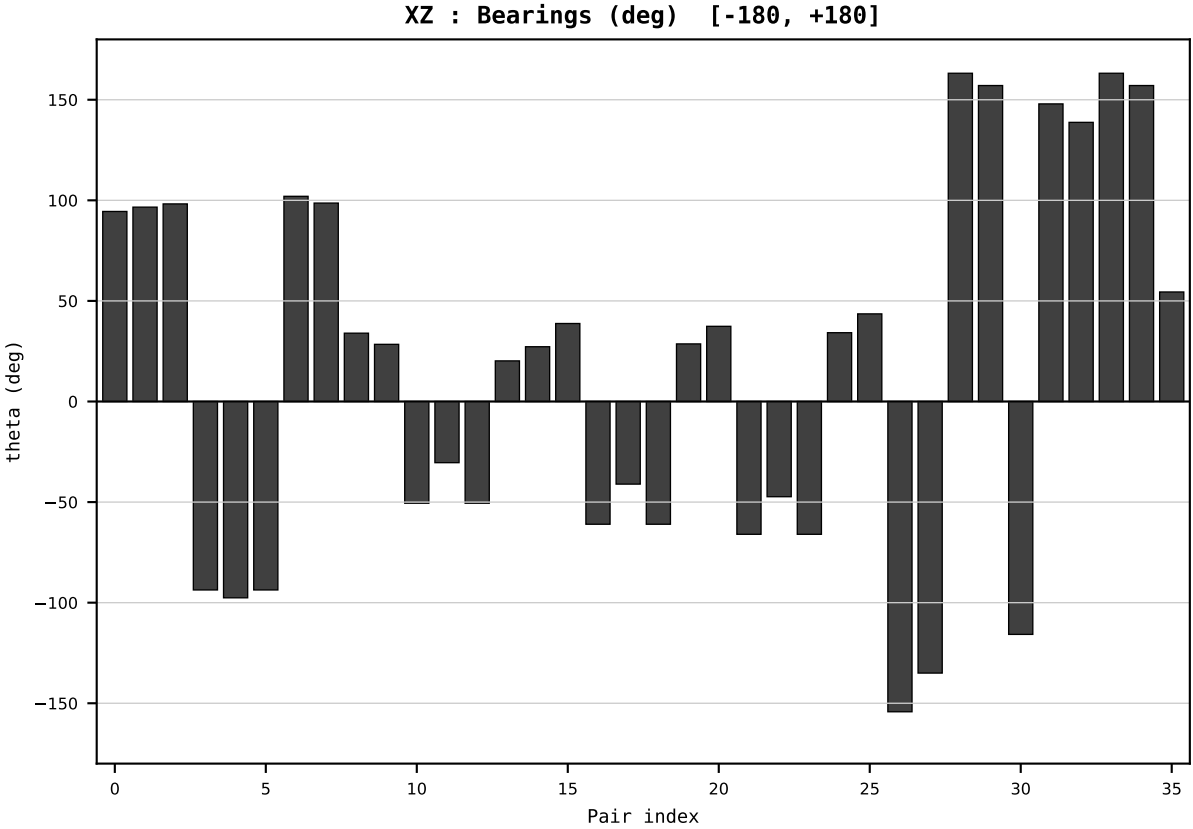
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.581737
Ring = Hs-4
E_metric= 159.5021
kappa_HS= 160070.00
omega = 2.1911 deg
d_A = 0.490179
Helm = Other Fossil
Helm d = +0.337914
DR = 11.983 HLR
DR ratio= 160070.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



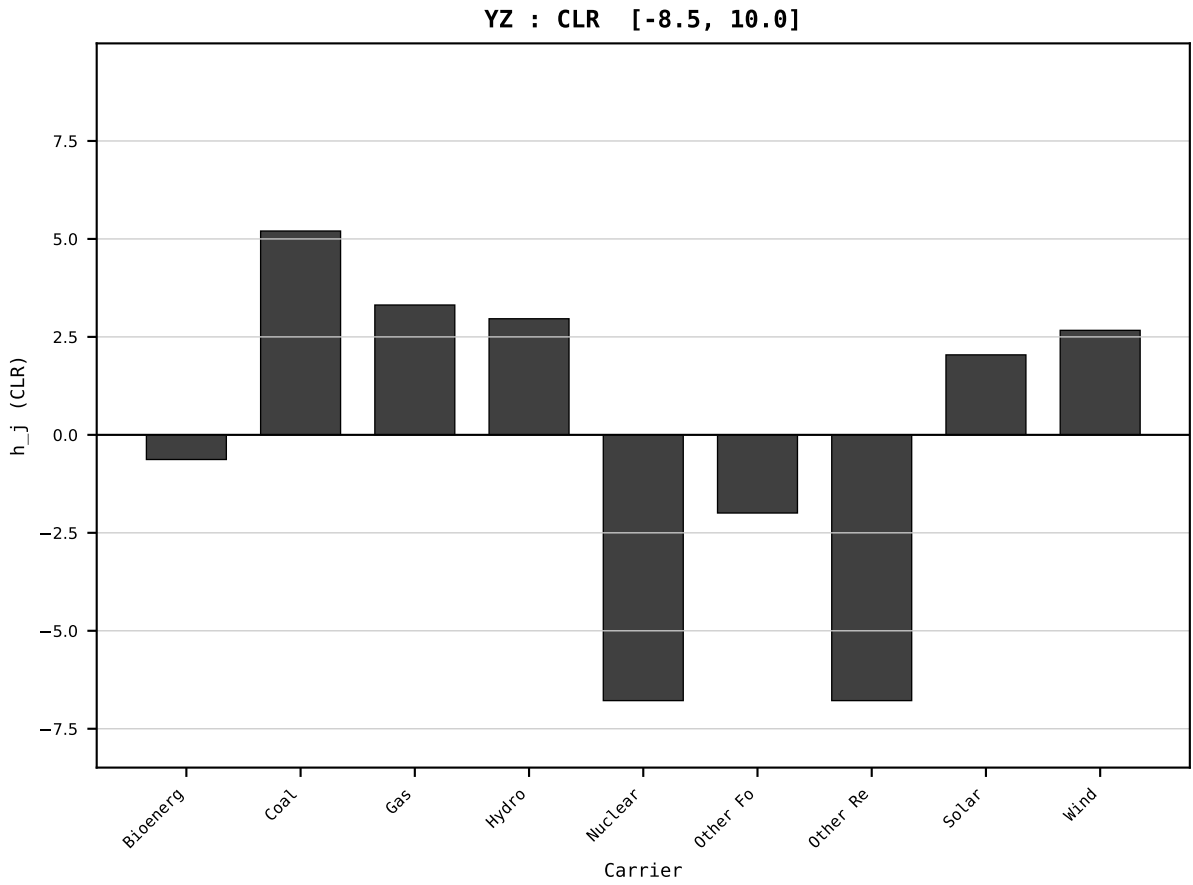
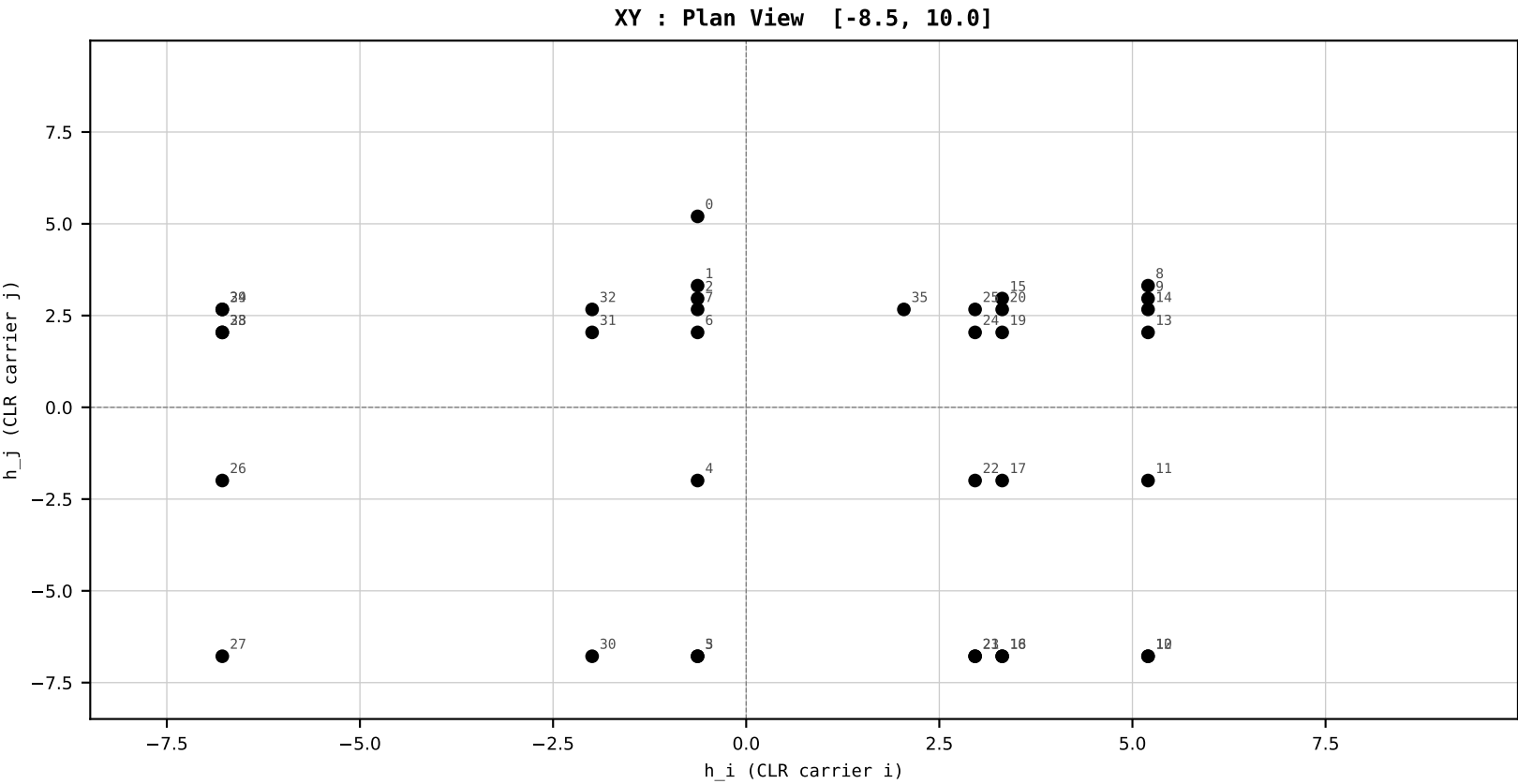
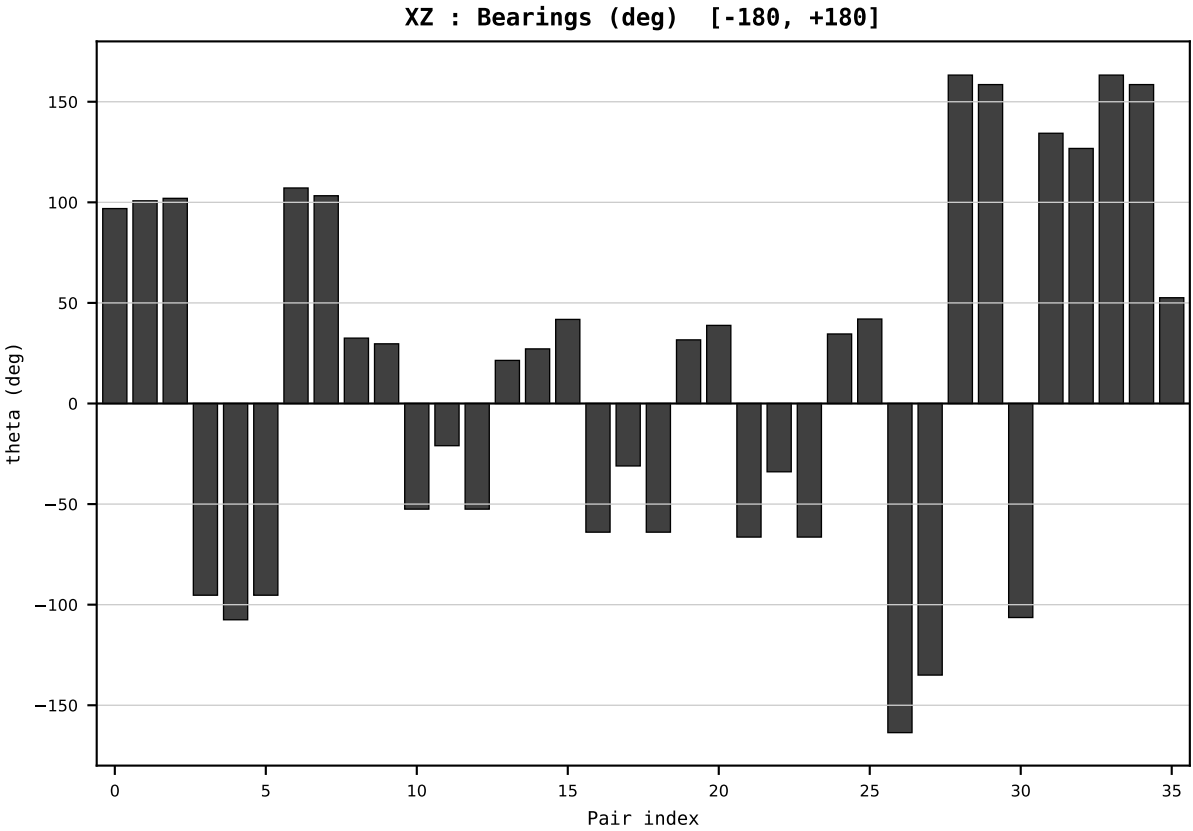
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.562143
Ring = Hs-4
E_metric= 154.4679
kappa_HS= 160170.00
omega = 5.8725 deg
d_A = 1.299181
Helm = Other Fossil
Helm d = +1.179791
DR = 11.984 HLR
DR ratio= 160170.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



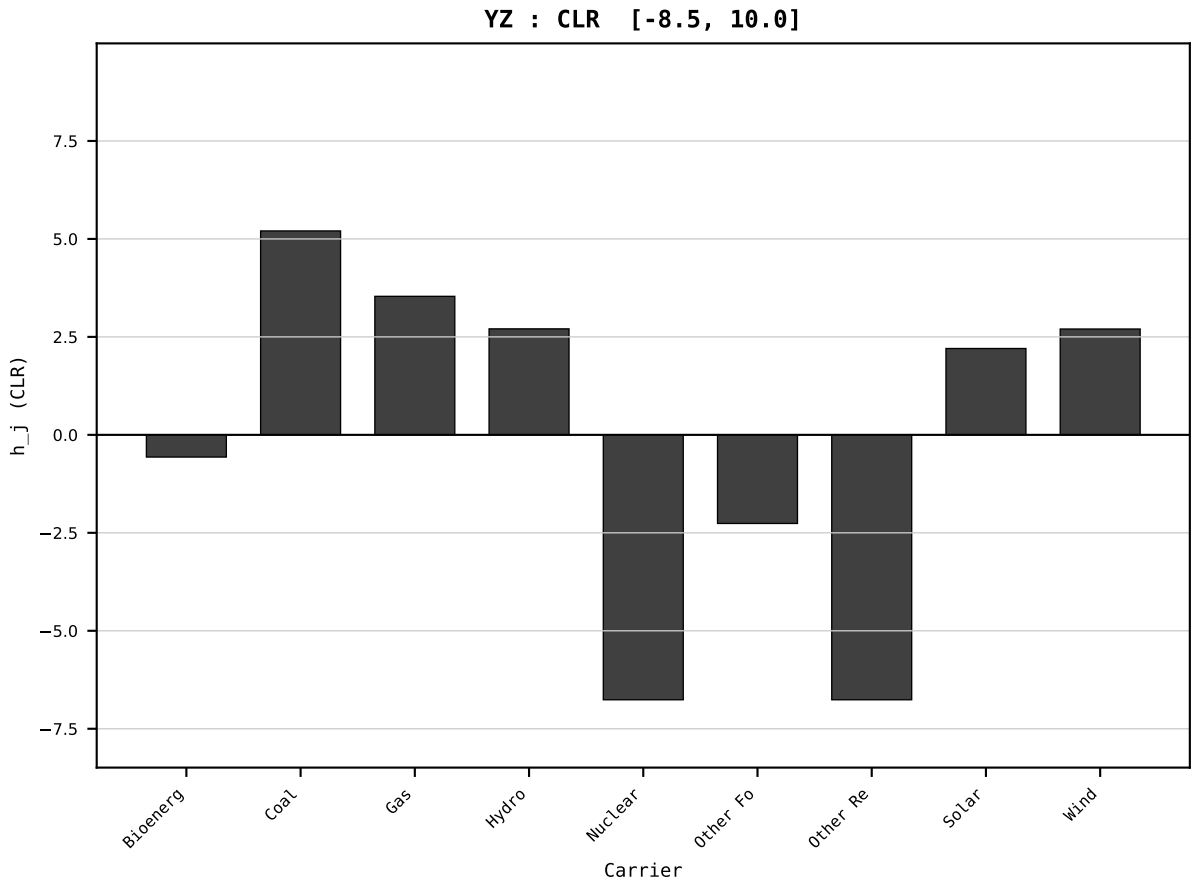
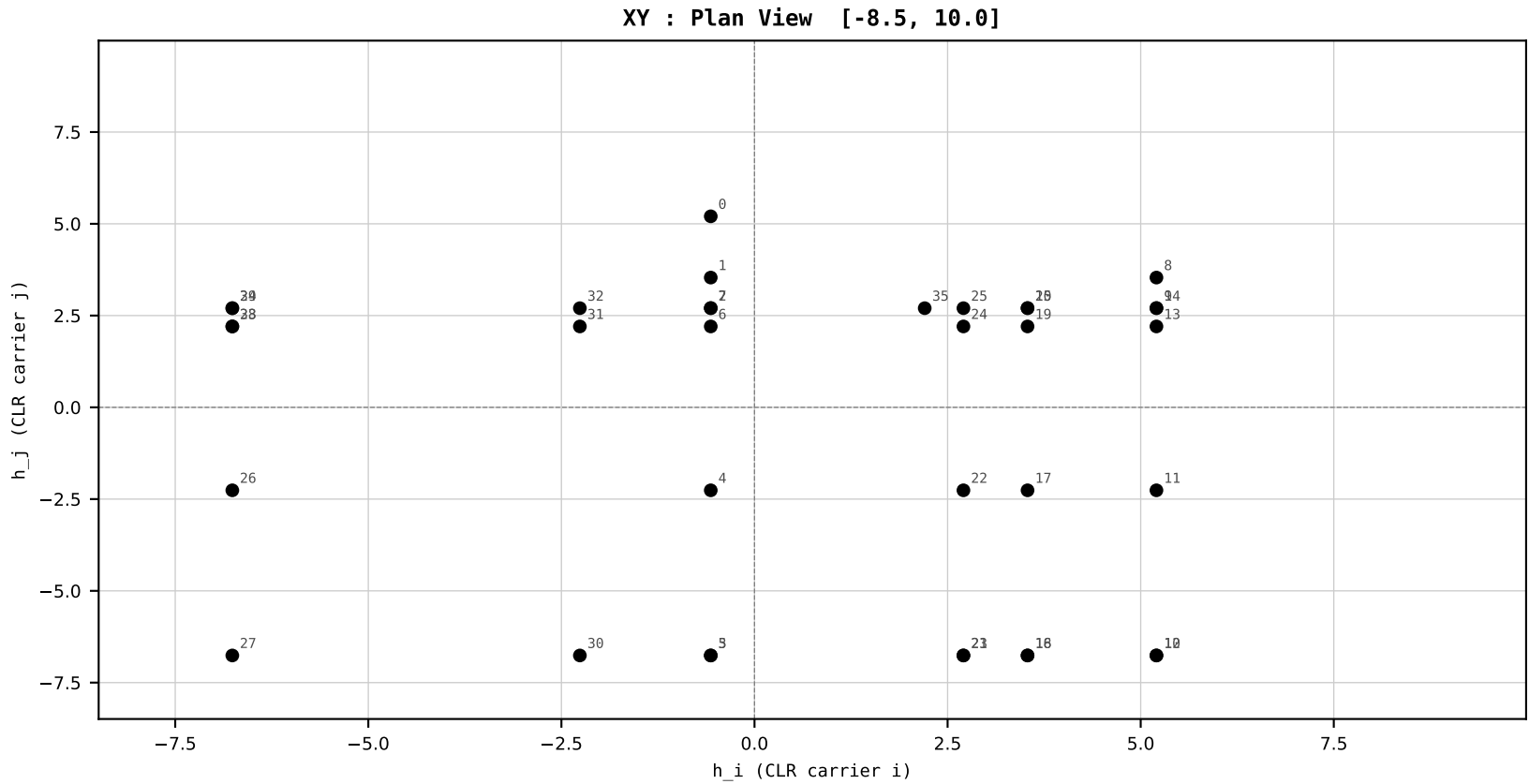
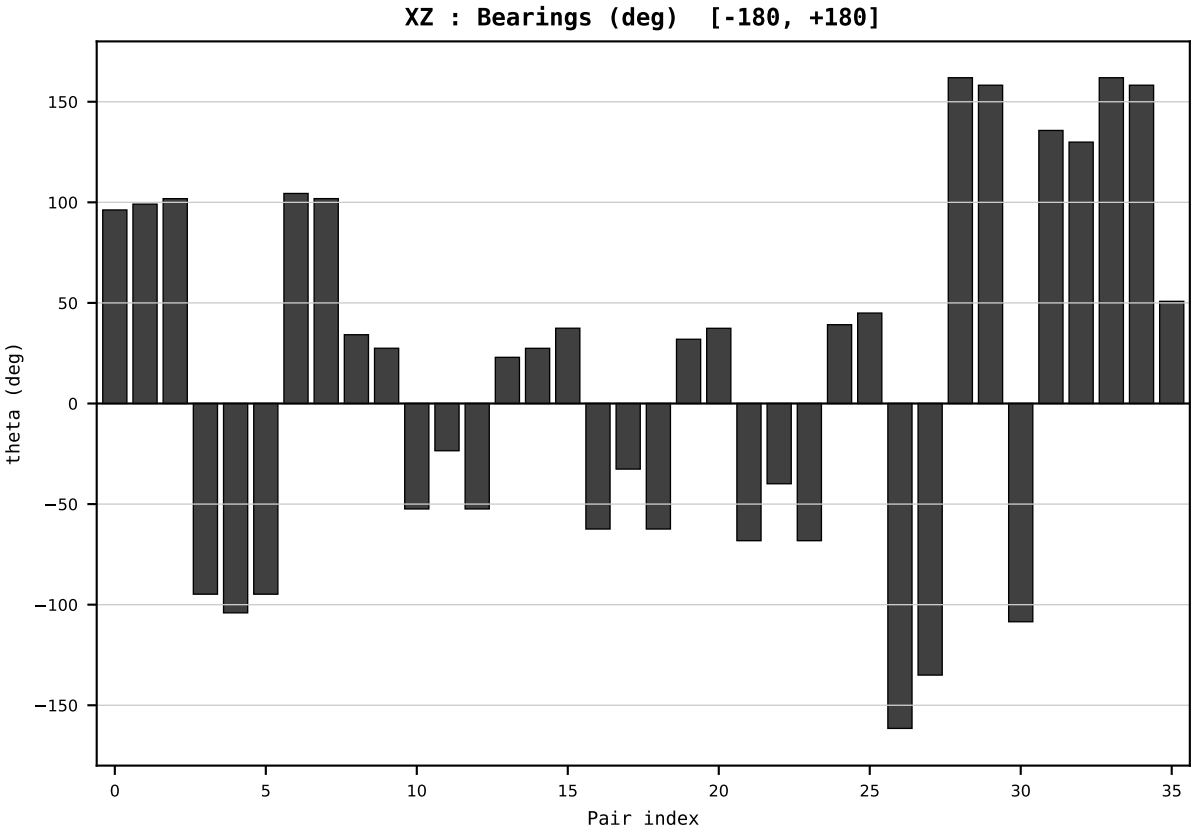
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.554731
Ring = Hs-4
E_metric= 155.8956
kappa_HS= 157110.00
omega = 2.1363 deg
d_A = 0.467967
Helm = Other Fossil
Helm d = -0.266487
DR = 11.965 HLR
DR ratio= 157110.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



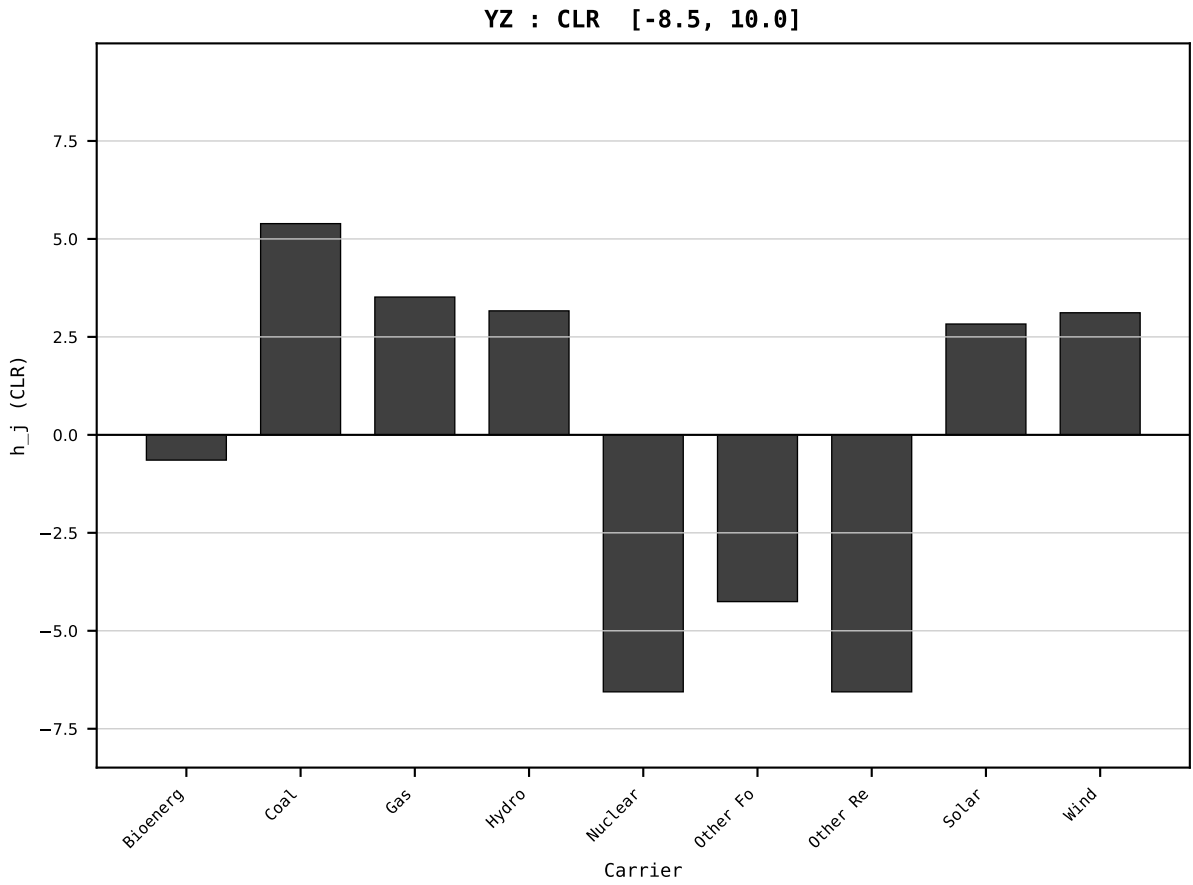
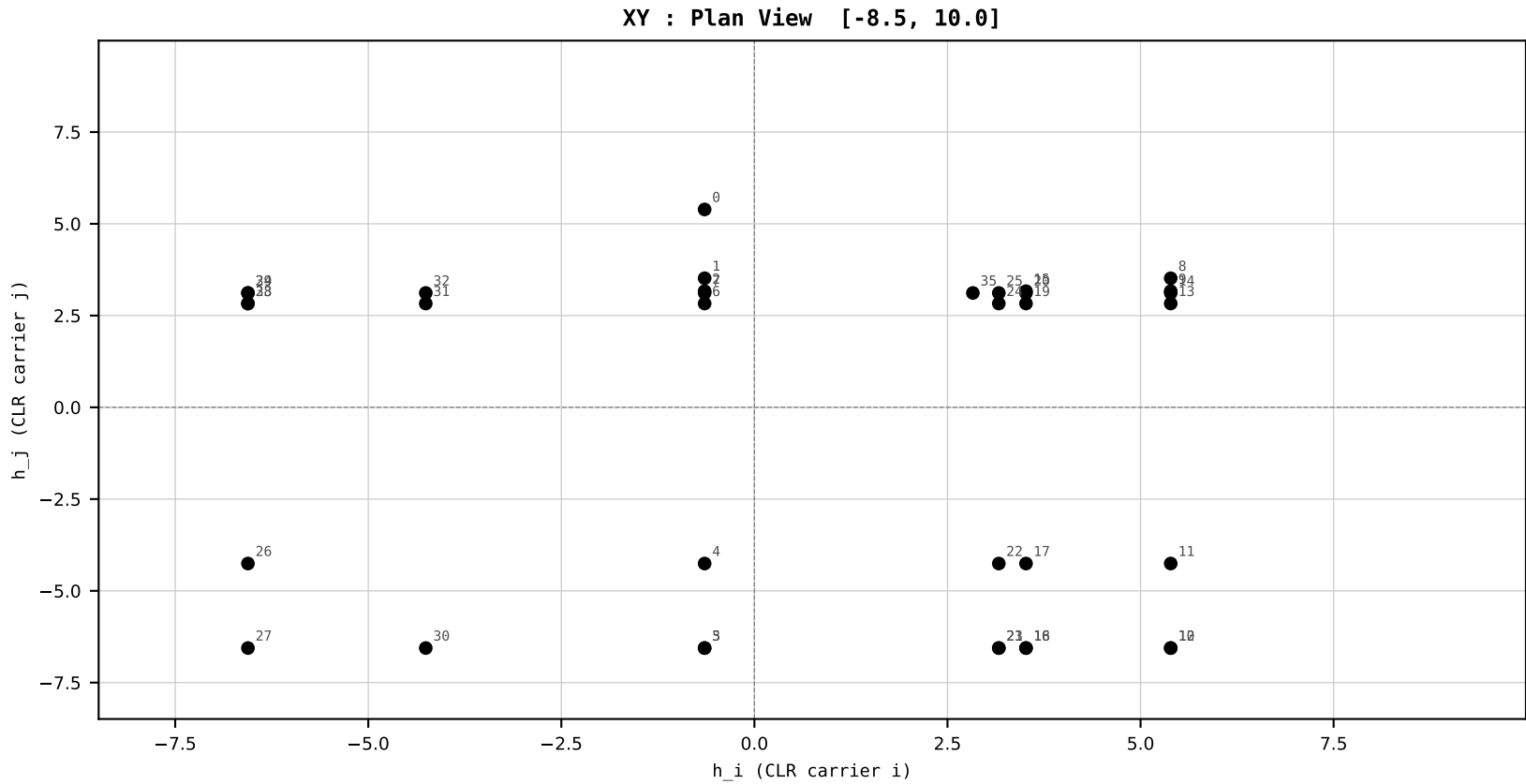
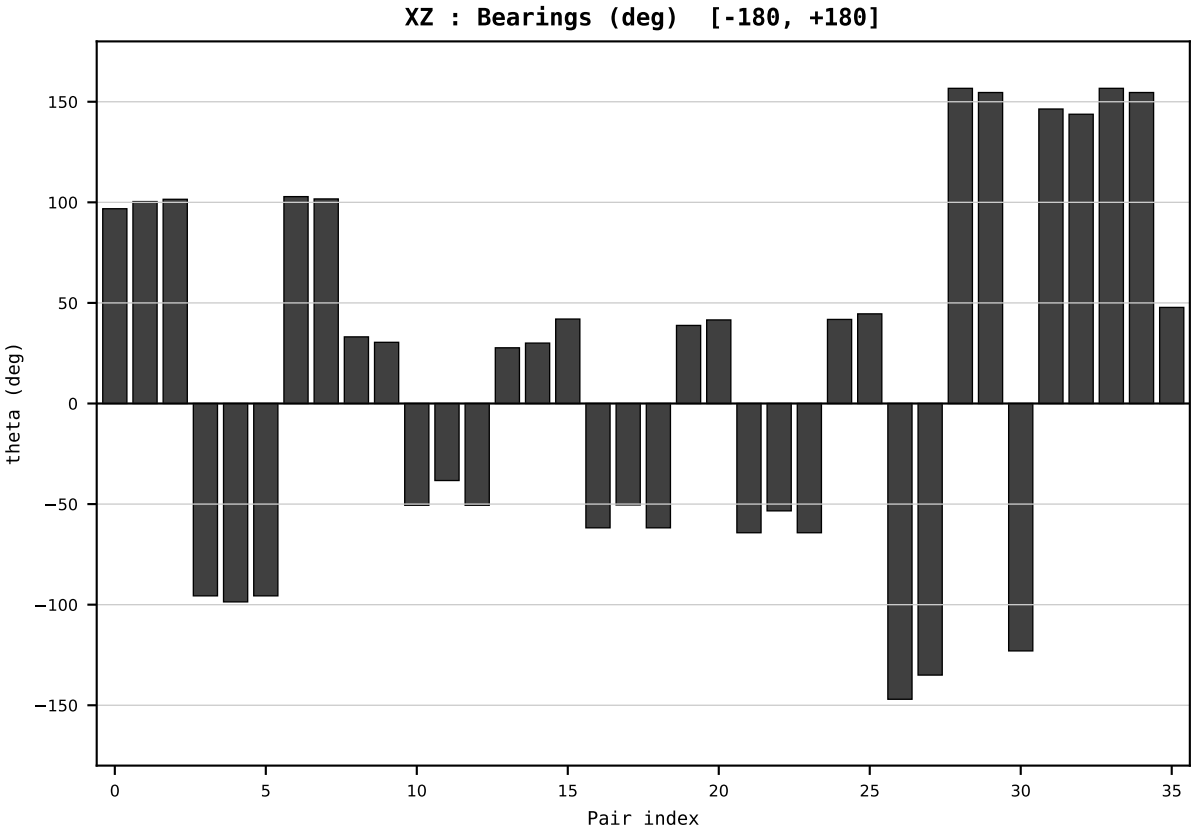
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.525552
Ring = Hs-4
E_metric= 173.6939
kappa_HS= 154620.00
omega = 9.3738 deg
d_A = 2.208080
Helm = Other Fossil
Helm d = -1.994594
DR = 11.949 HLR
DR ratio= 154620.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



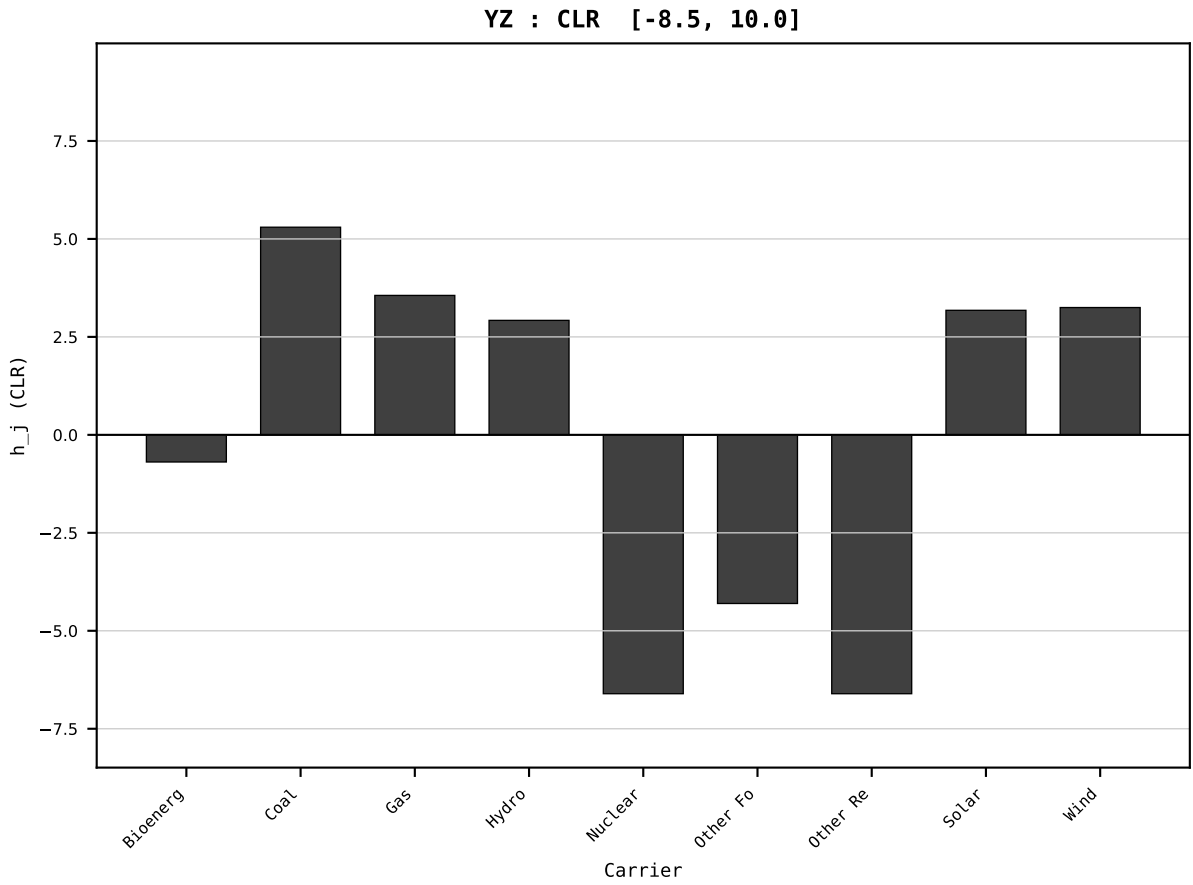
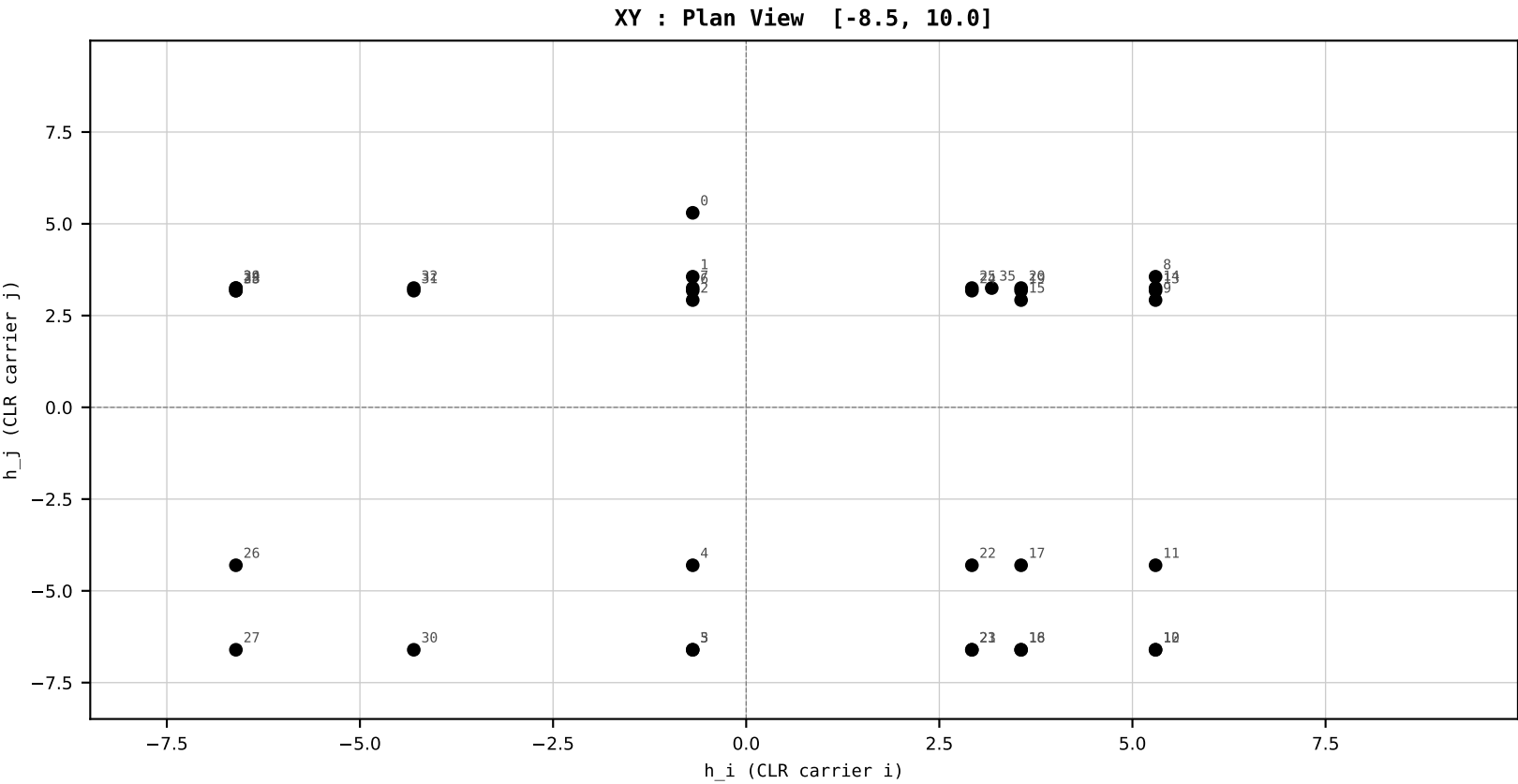
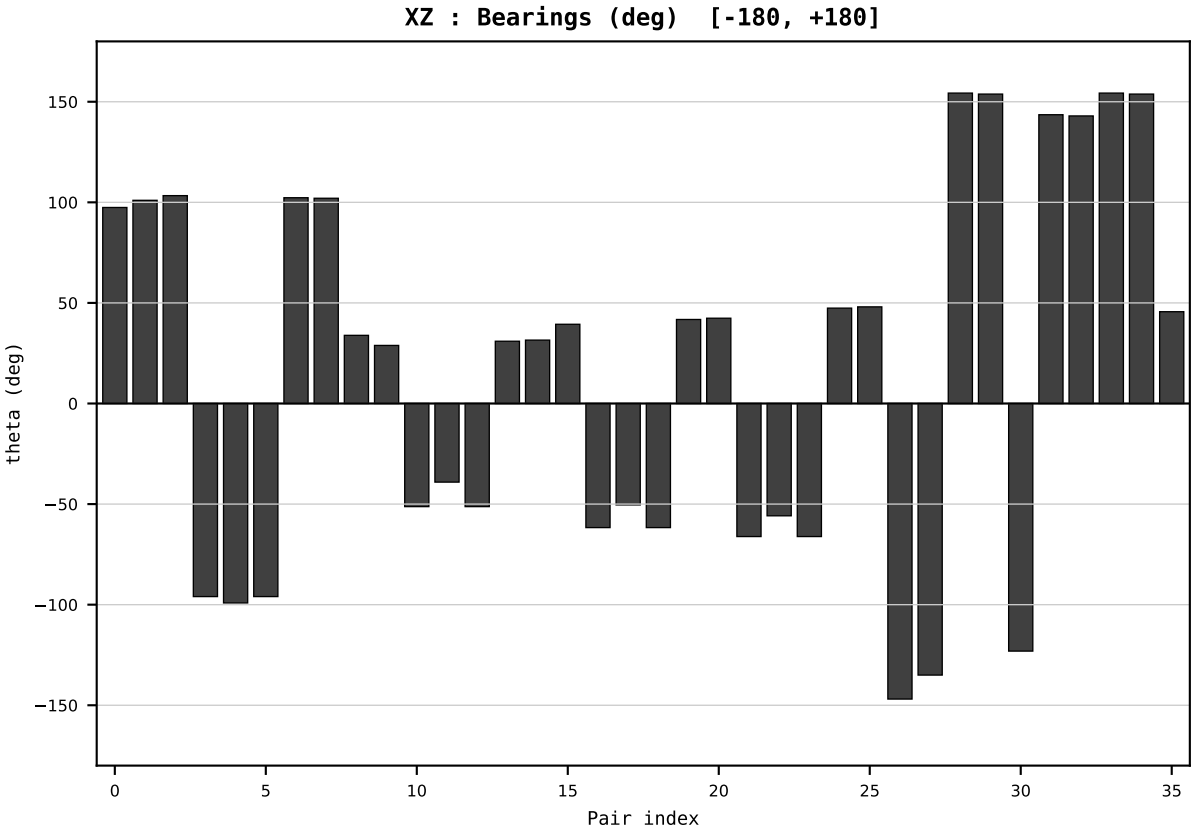
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.492113
Ring = Hs-3
E_metric= 176.2070
kappa_HS= 147990.00
omega = 1.9812 deg
d_A = 0.467089
Helm = Solar
Helm d = +0.349727
DR = 11.905 HLR
DR ratio= 147990.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

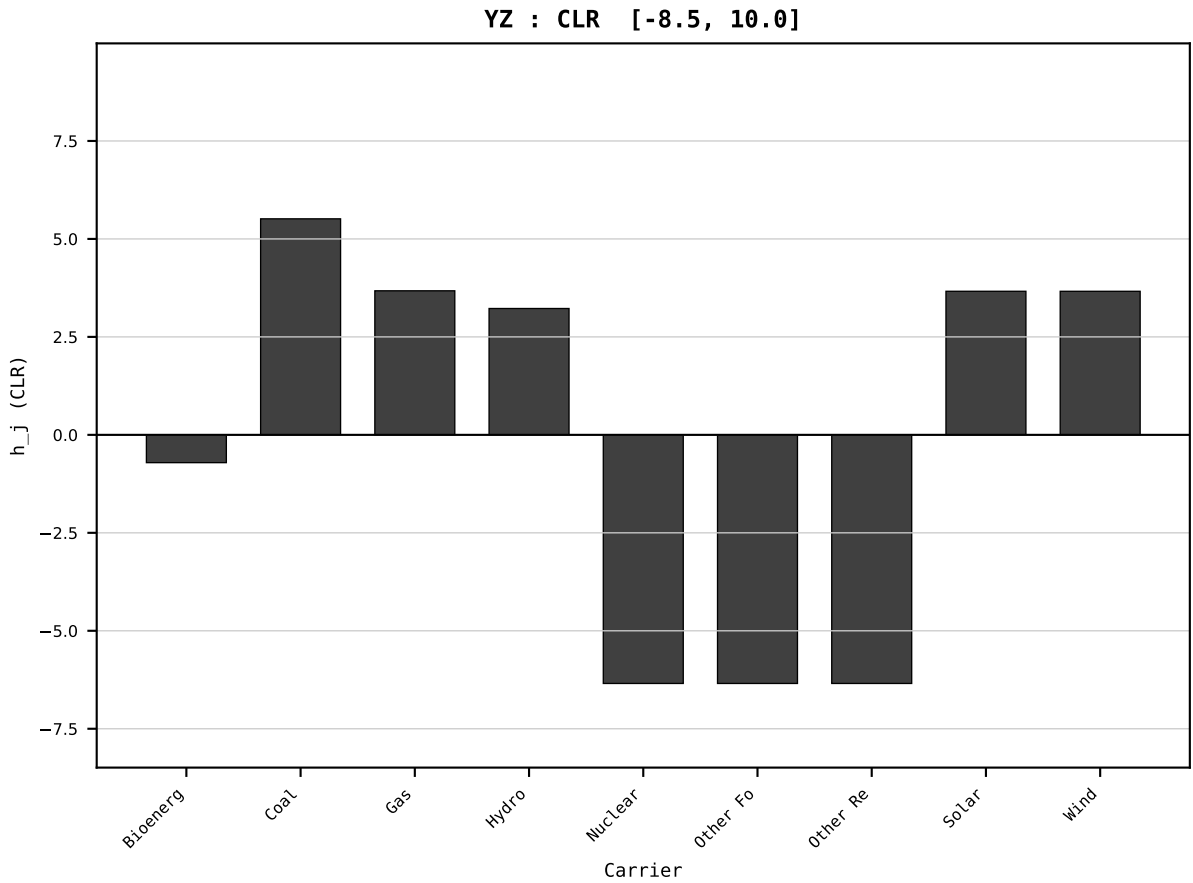
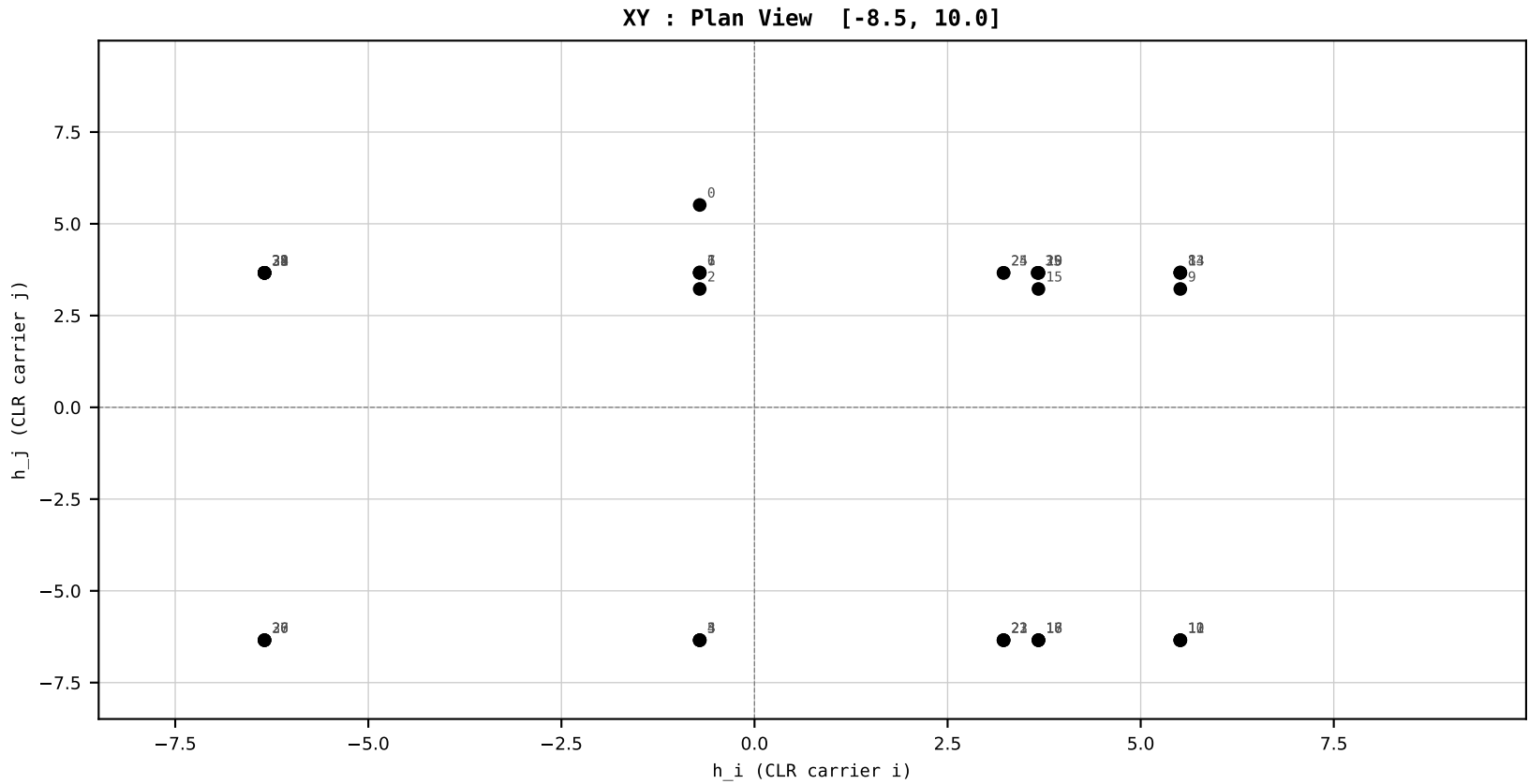
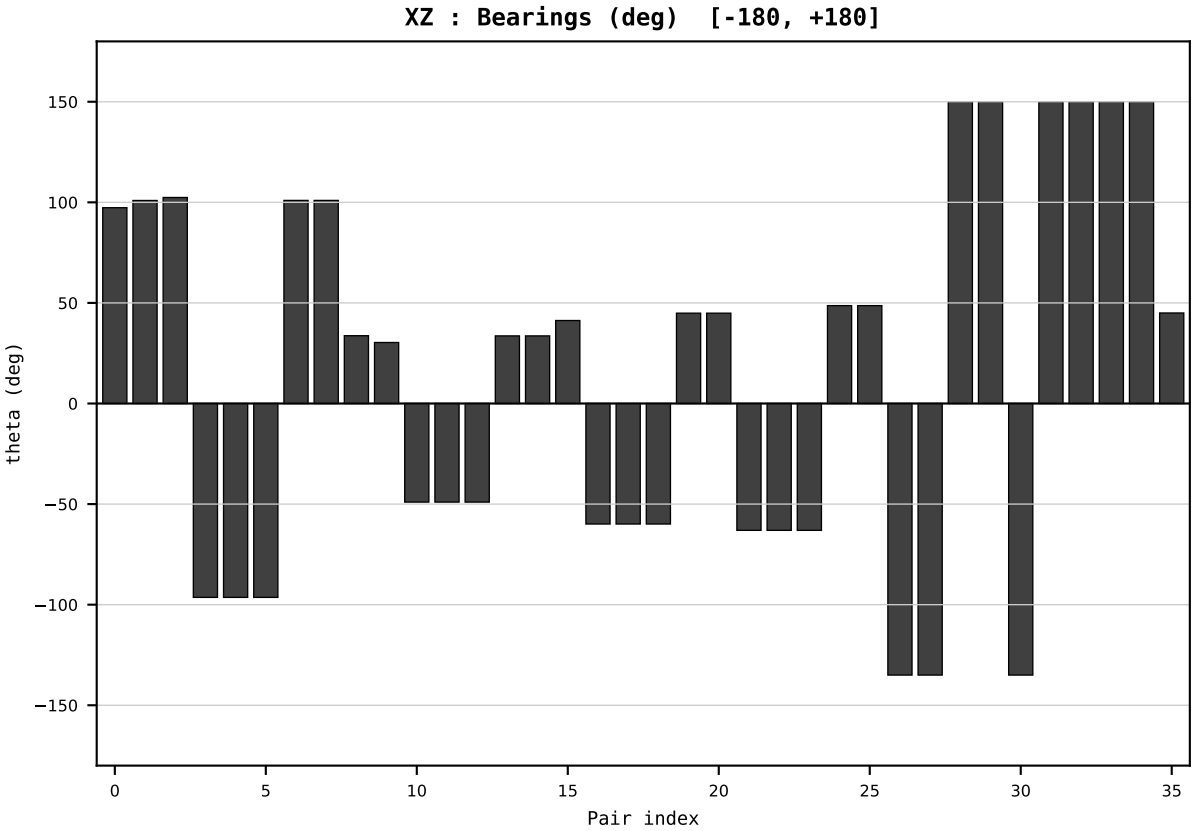
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.469557
Ring = Hs-3
E_metric= 202.4007
kappa_HS= 141070.00
omega = 8.2986 deg
d_A = 2.204987
Helm = Other Fossil
Helm d = -2.040967
DR = 11.857 HLR
DR ratio= 141070.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

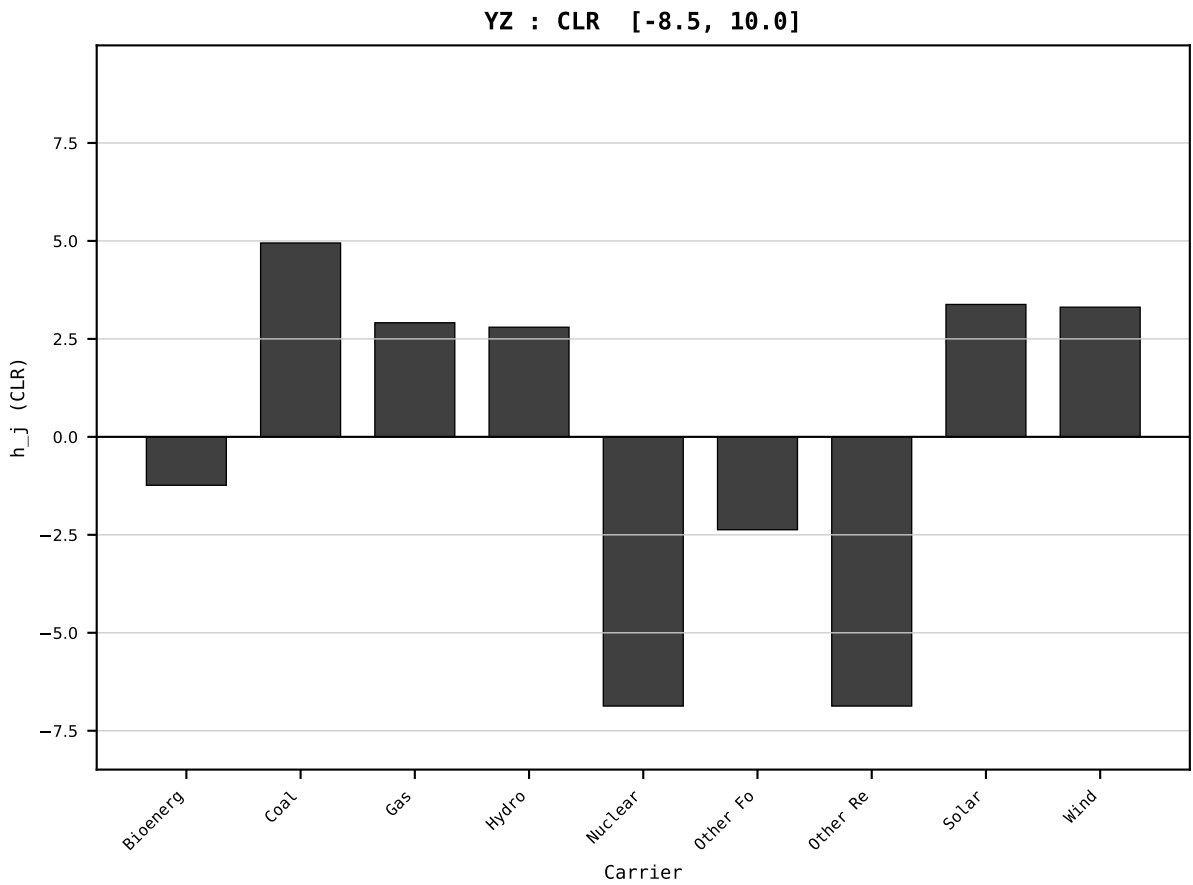
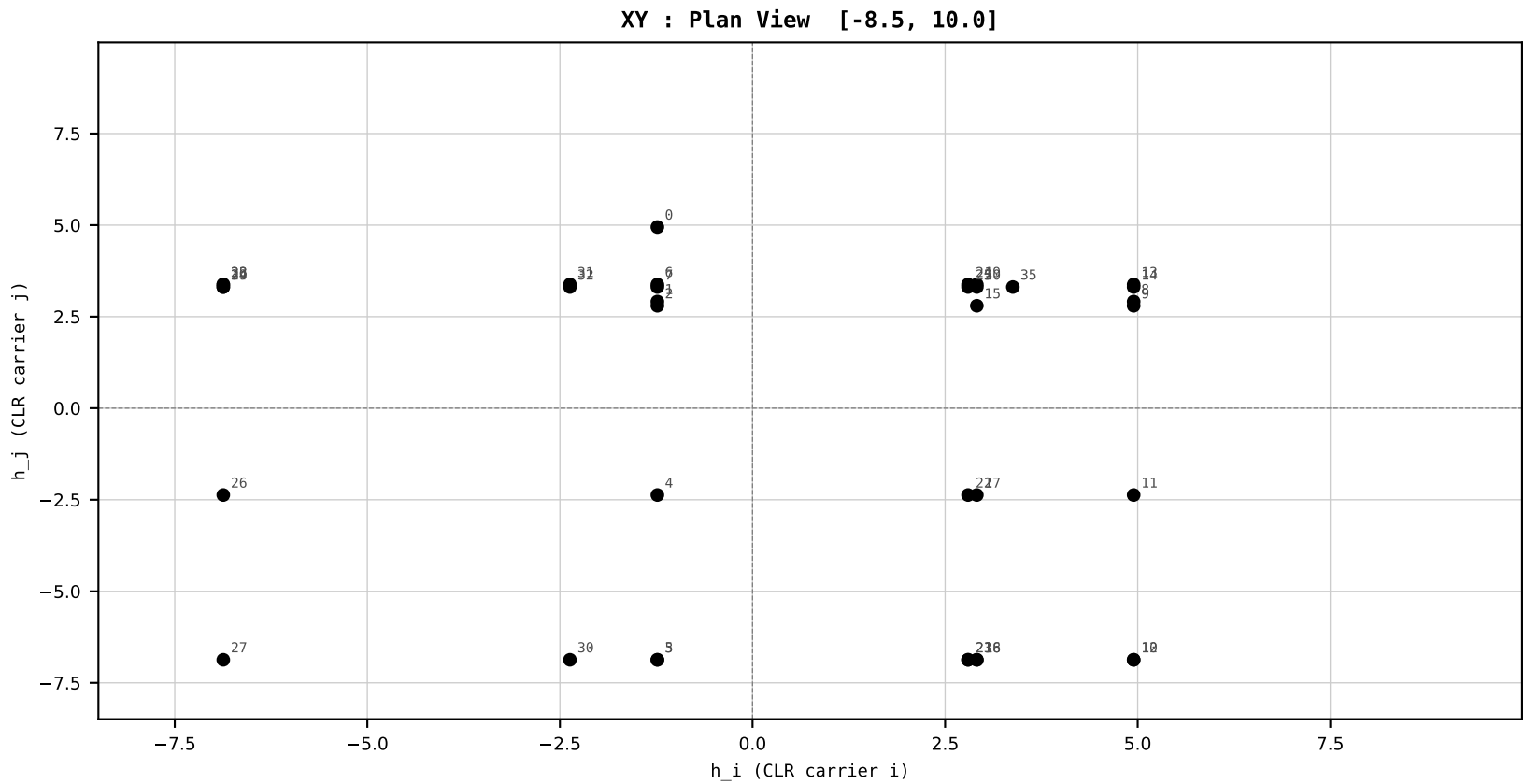
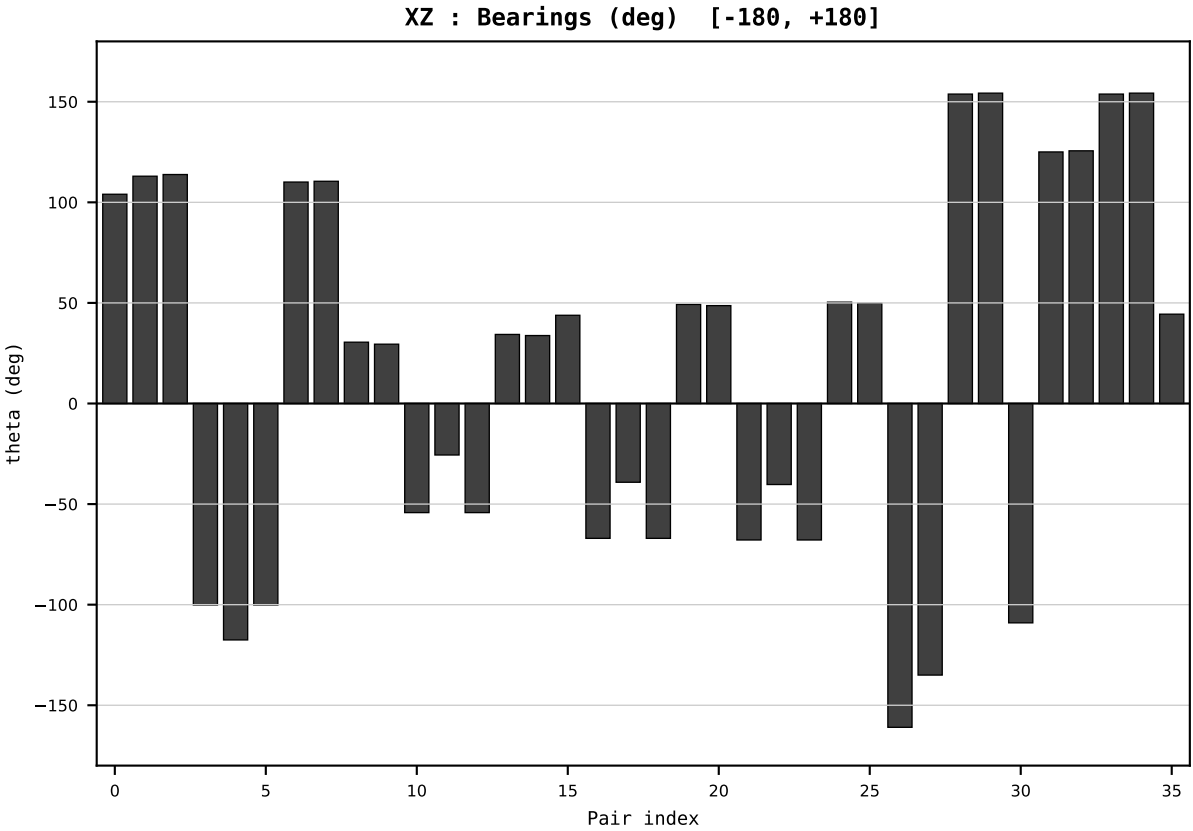
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.446631
Ring = Hs-3
E_metric= 164.6967
kappa_HS= 135640.00
omega = 17.0092 deg
d_A = 4.232518
Helm = Other Fossil
Helm d = +3.973835
DR = 11.818 HLR
DR ratio= 135640.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

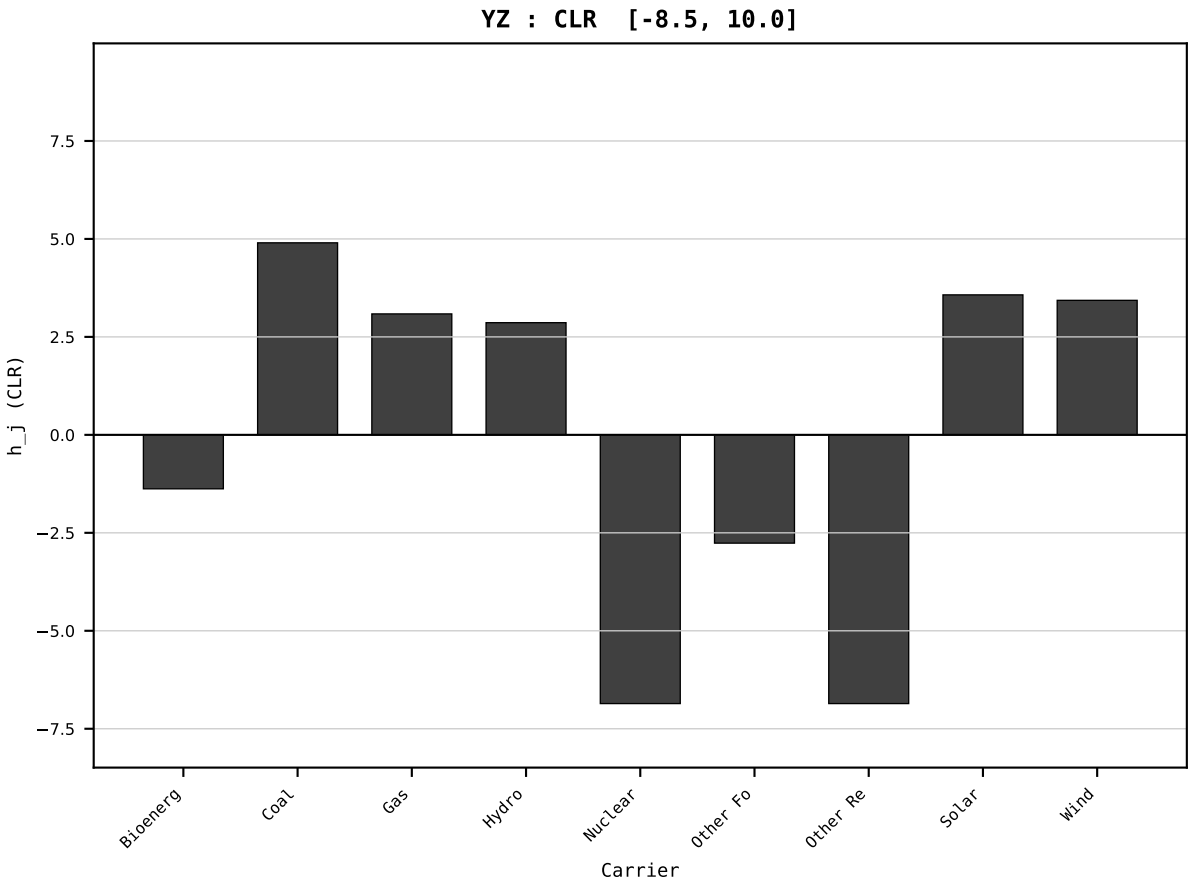
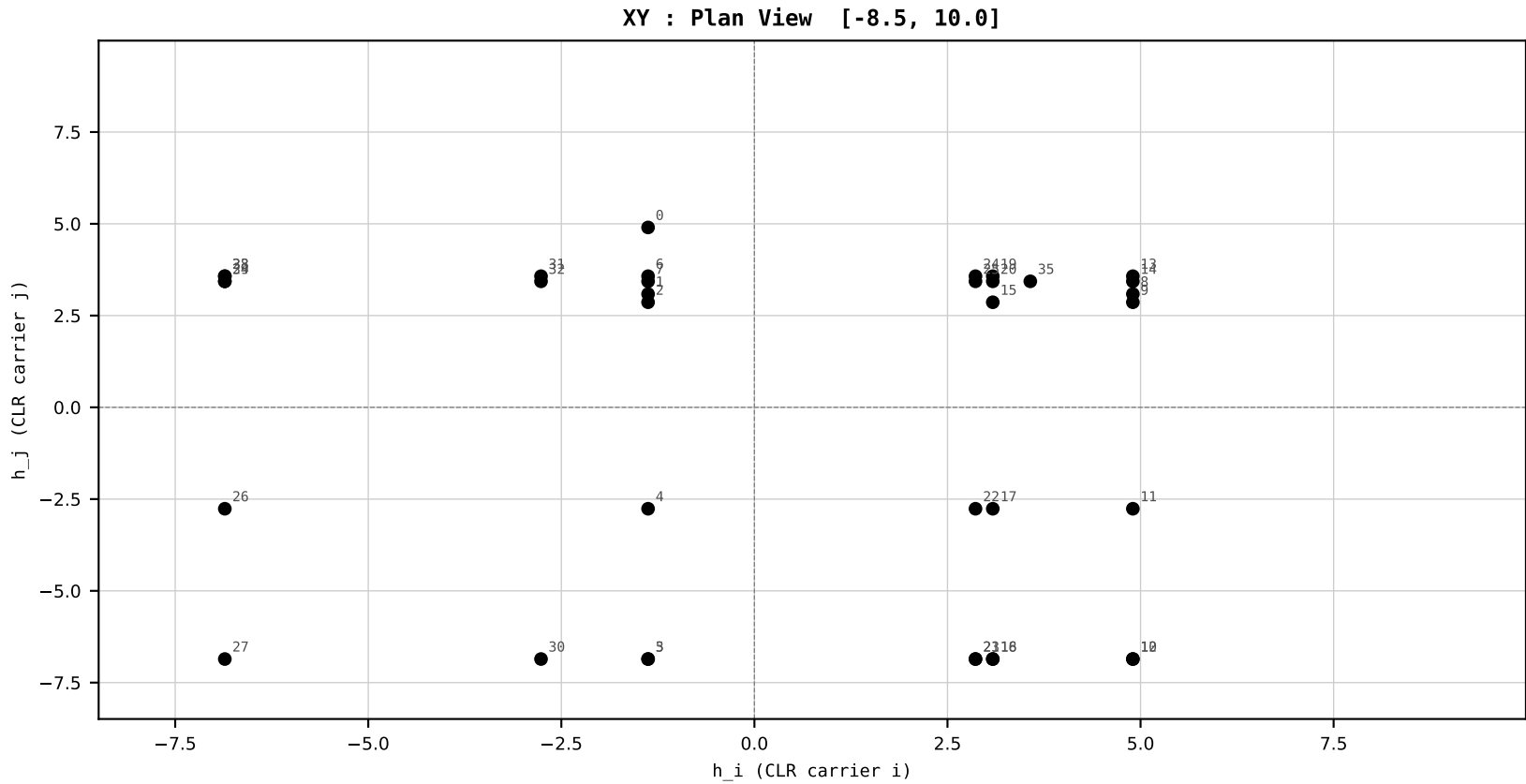
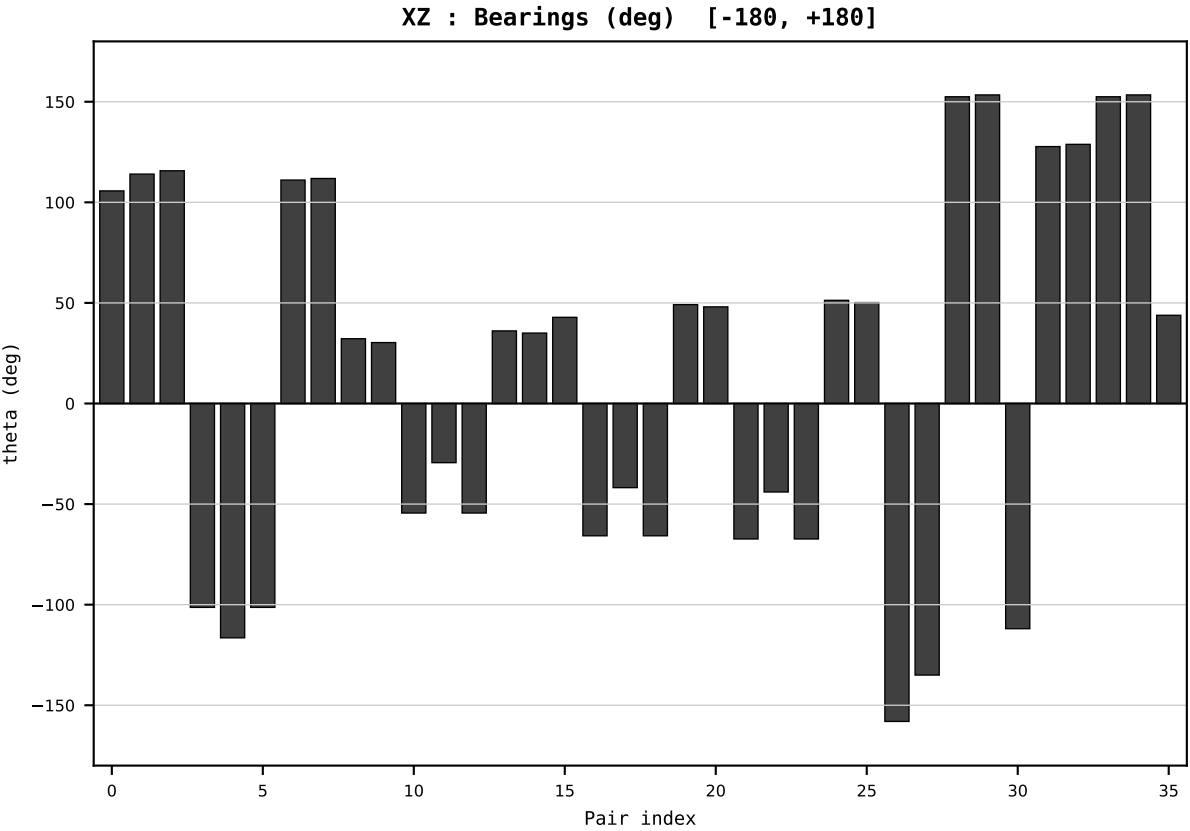
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.412813
Ring = Hs-3
E_metric= 169.8678
kappa_HS= 127790.00
omega = 2.0987 deg
d_A = 0.514148
Helm = Other Fossil
Helm d = -0.392917
DR = 11.758 HLR
DR ratio= 127790.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



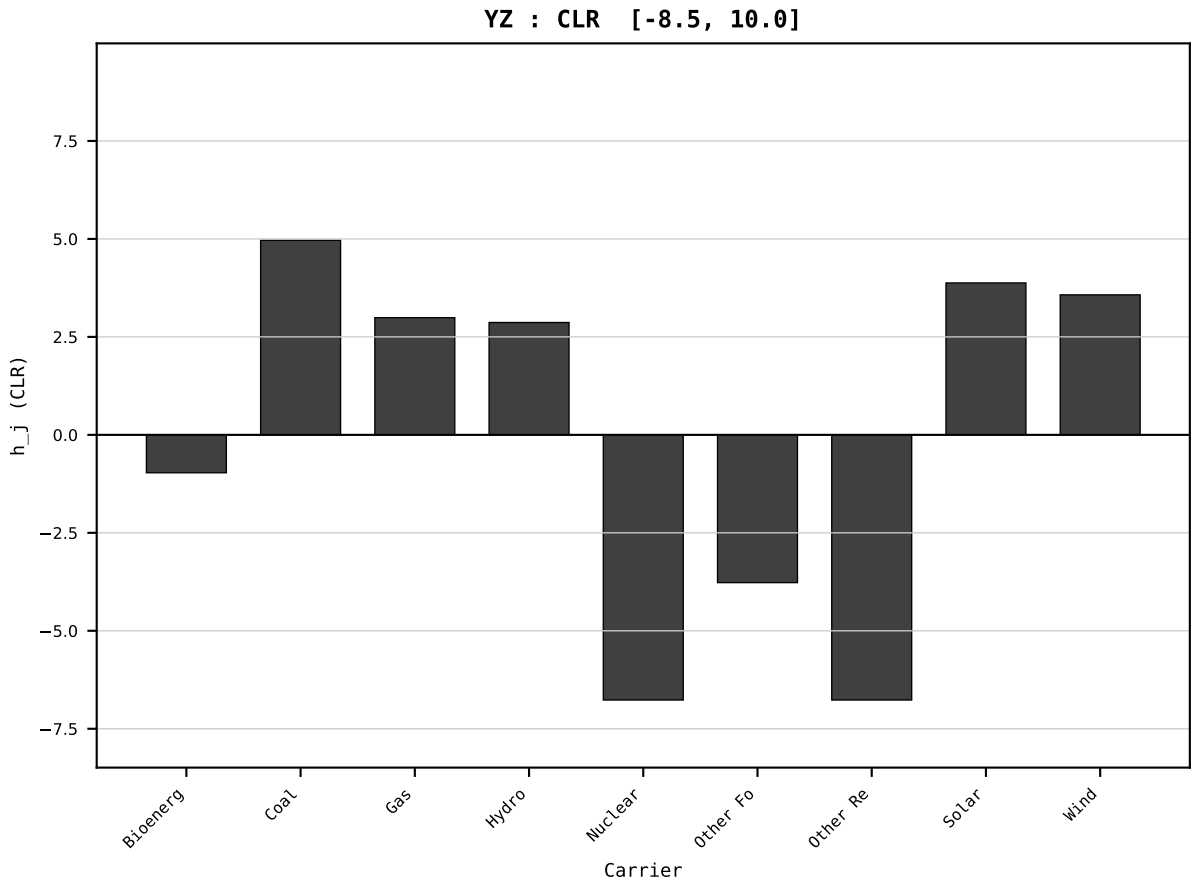
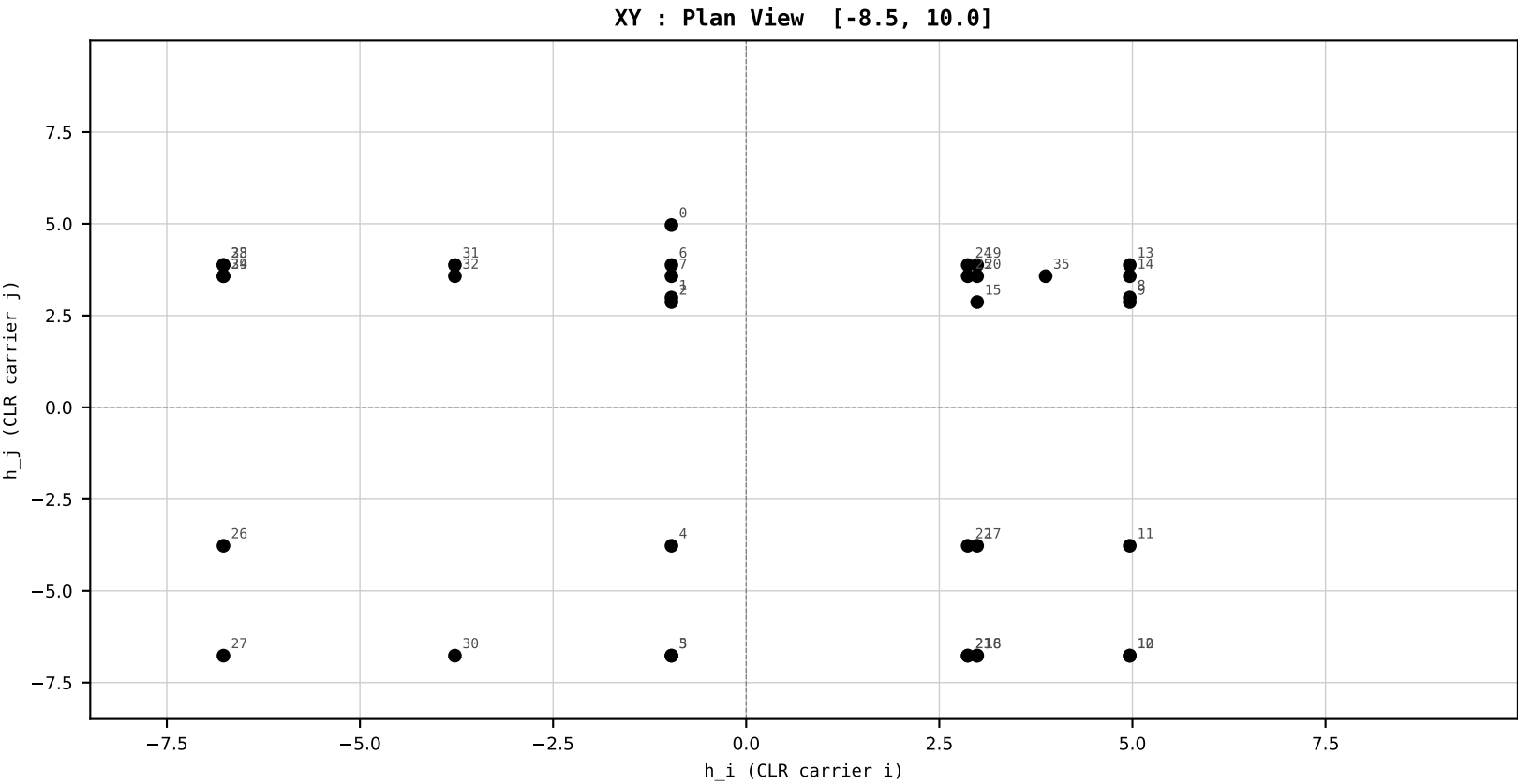
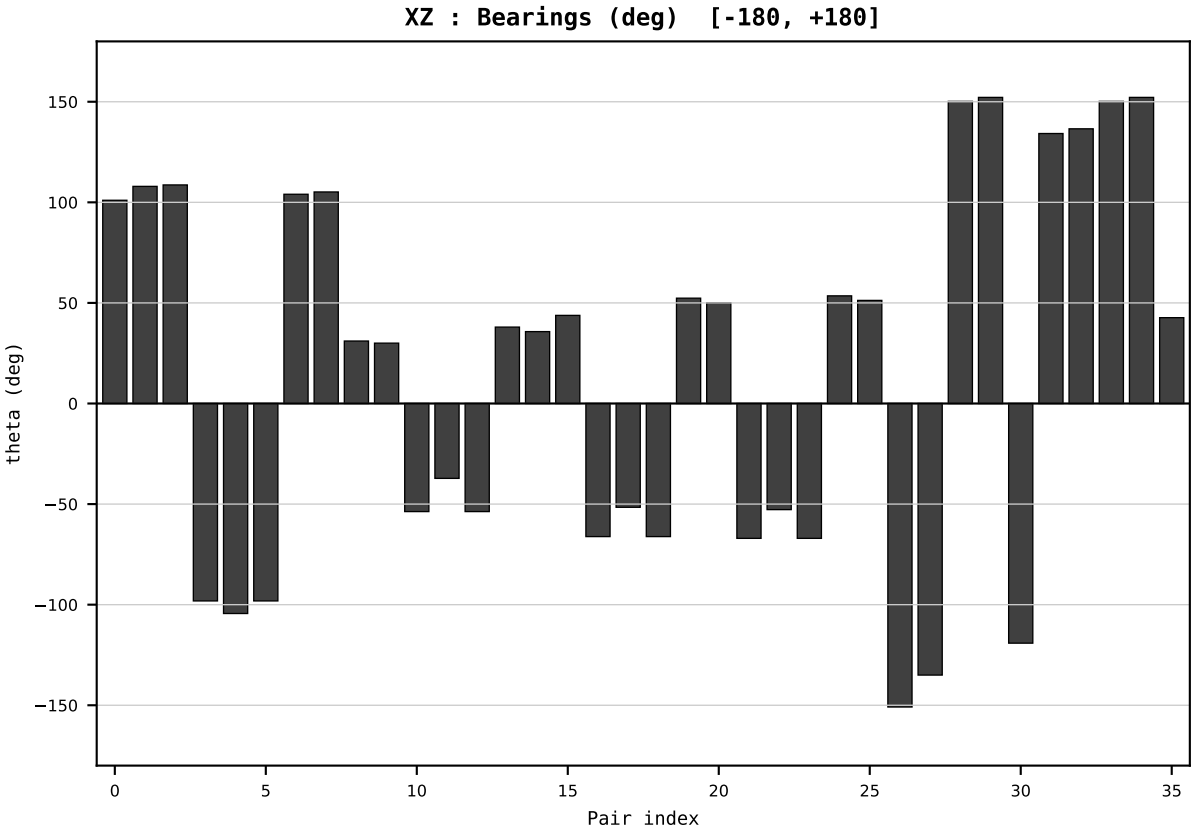
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409750
Ring = Hs-3
E_metric= 176.3471
kappa_HS= 124490.00
omega = 4.8995 deg
d_A = 1.151287
Helm = Other Fossil
Helm d = -1.007865
DR = 11.732 HLR
DR ratio= 124490.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

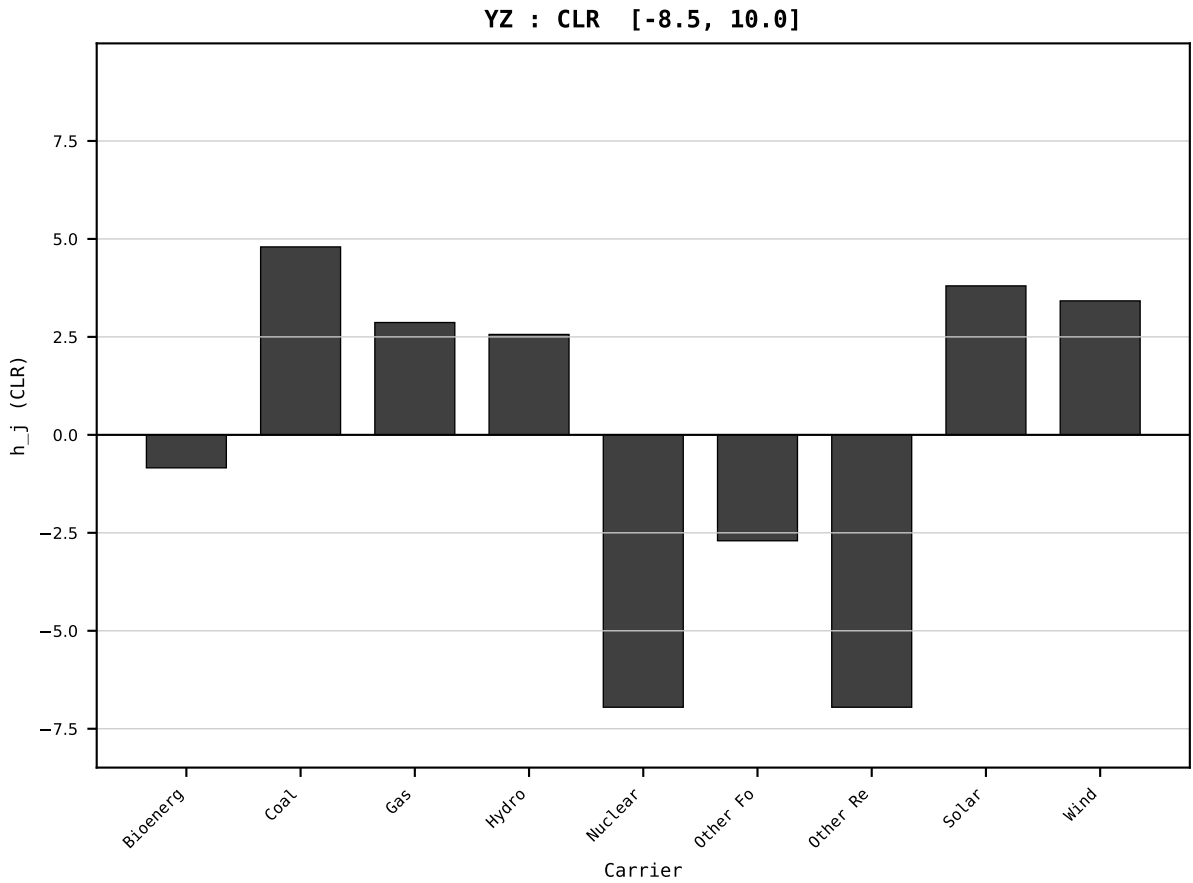
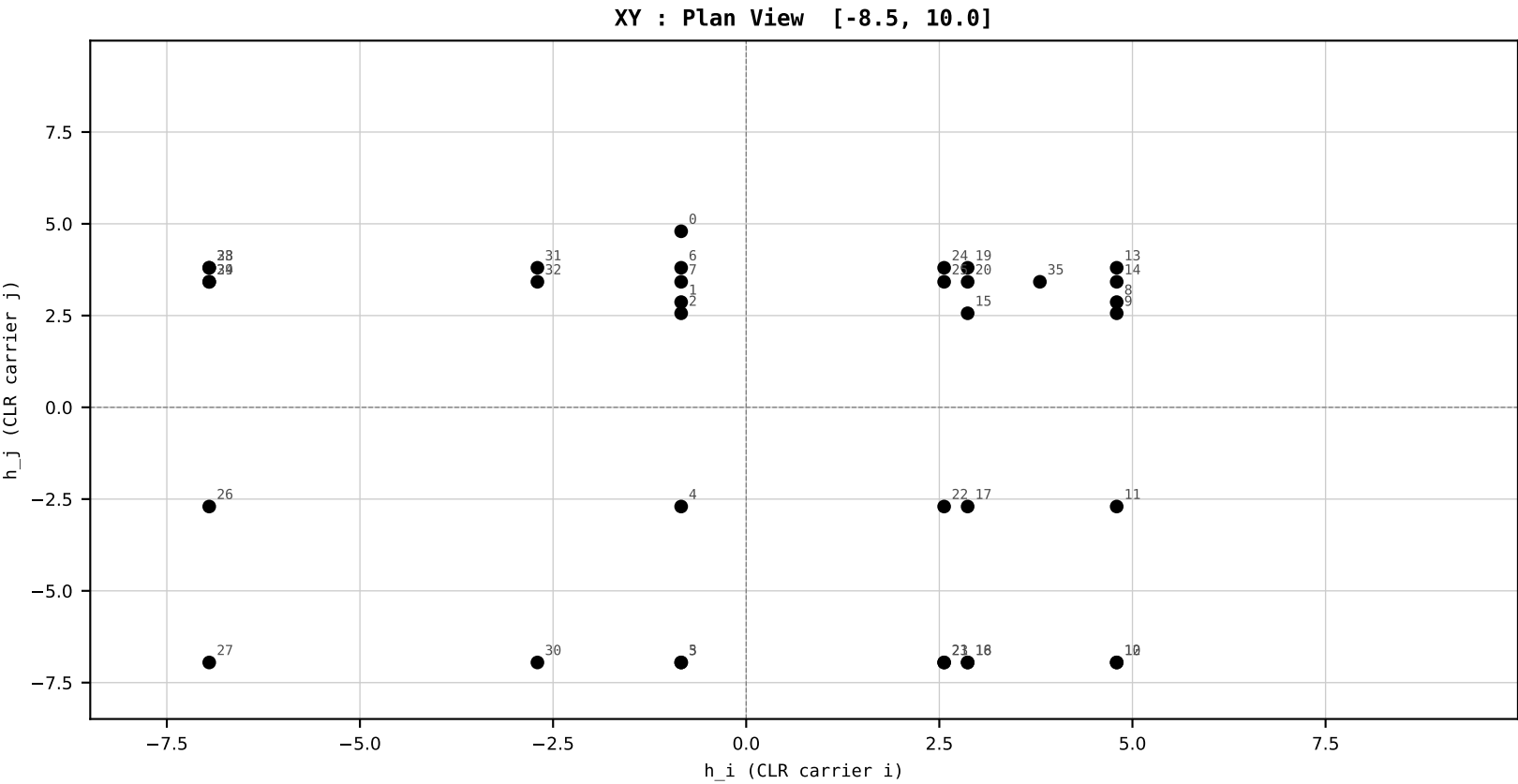
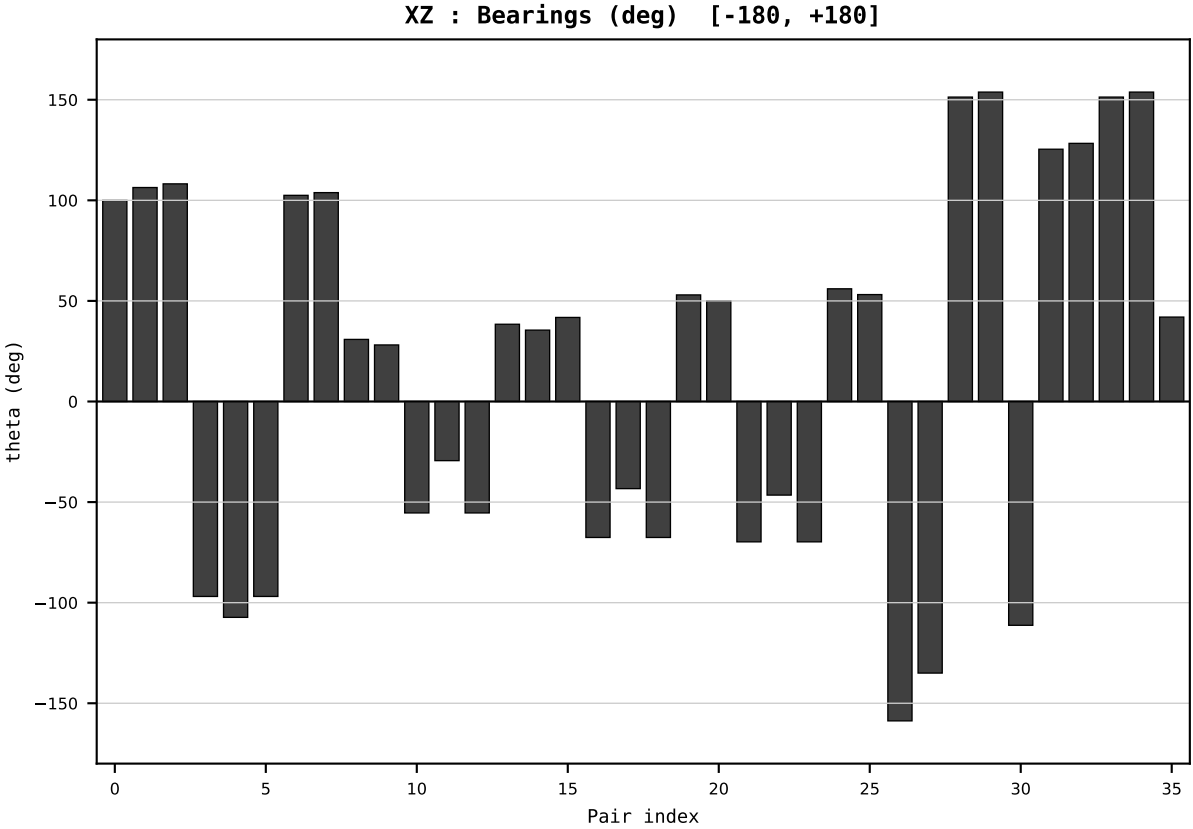
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.407970
Ring = Hs-3
E_metric= 168.5832
kappa_HS= 126430.00
omega = 4.9844 deg
d_A = 1.179587
Helm = Other Fossil
Helm d = +1.068460
DR = 11.747 HLR
DR ratio= 126430.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

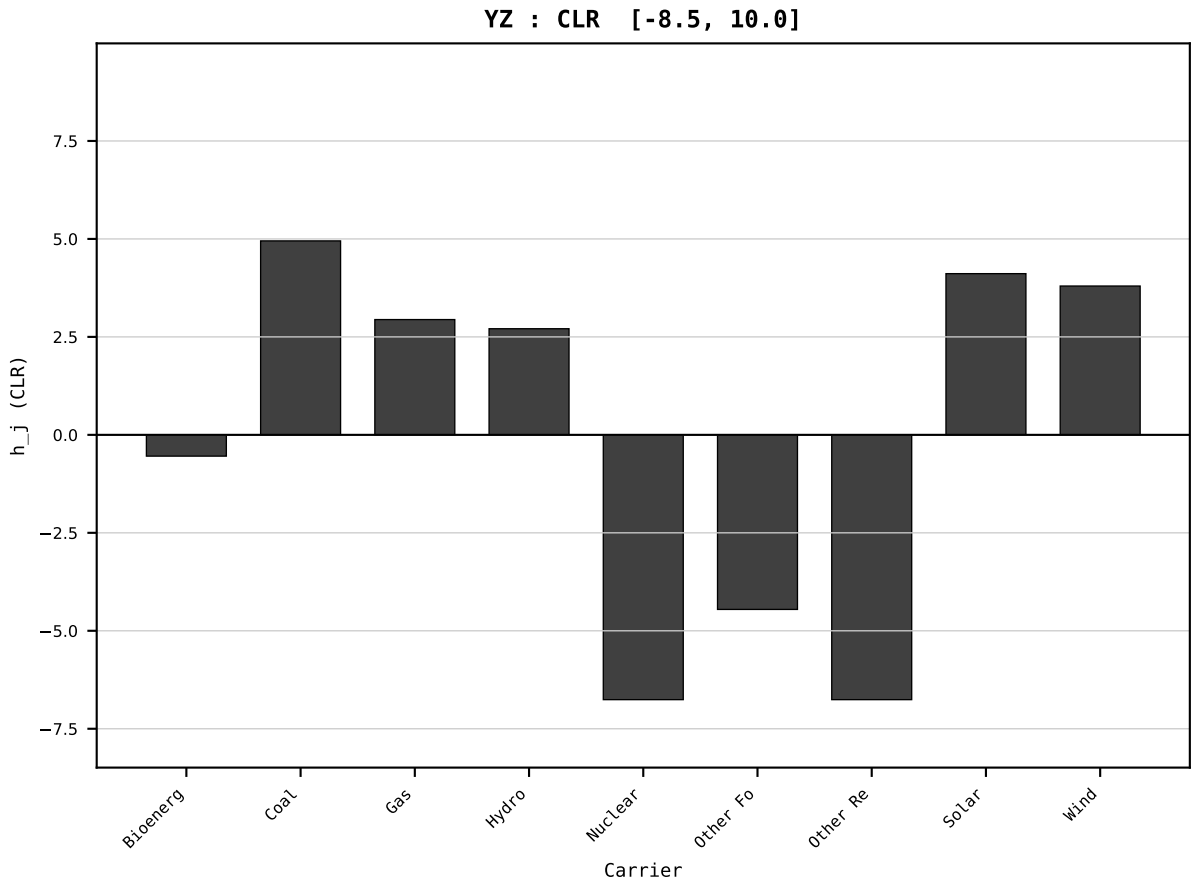
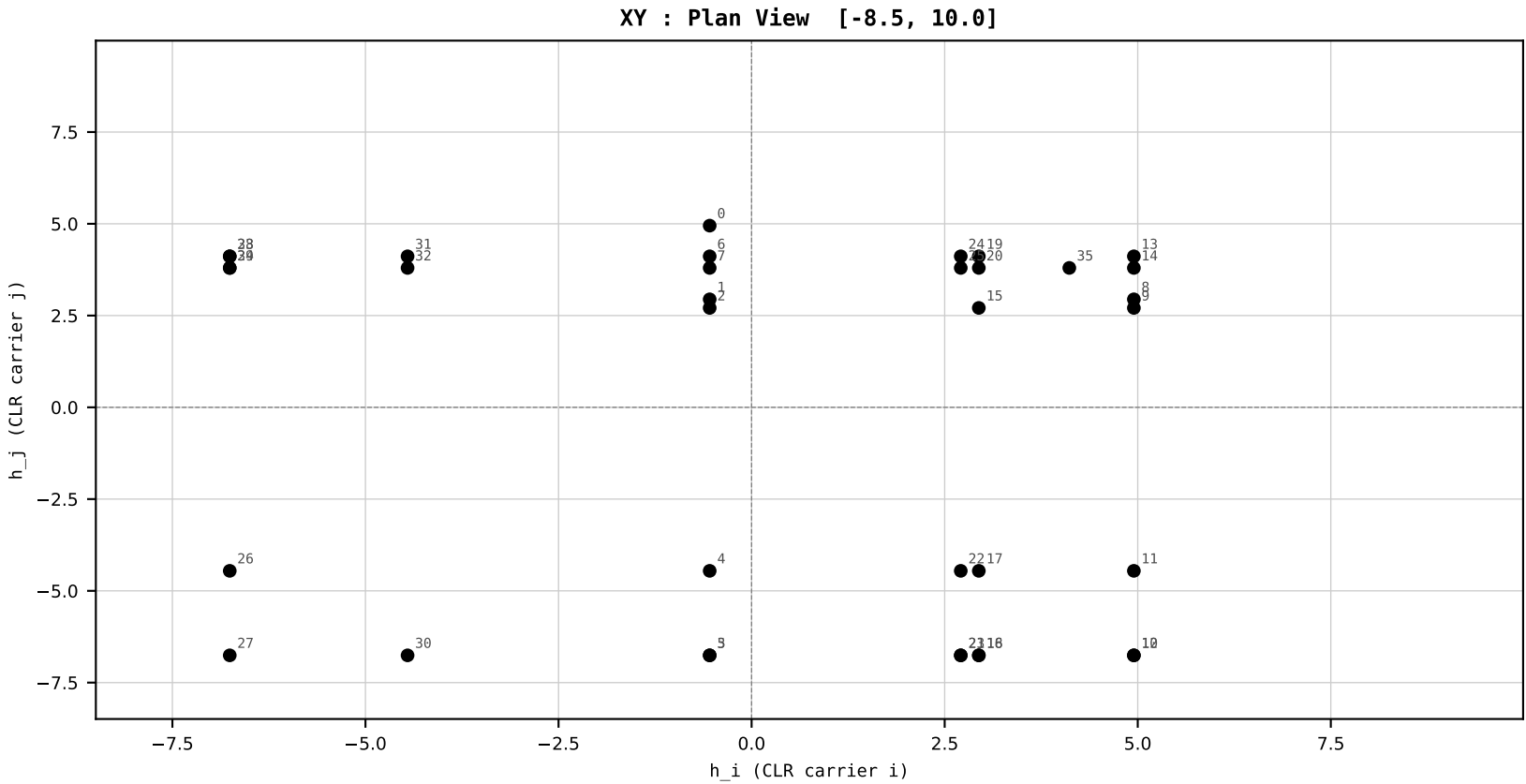
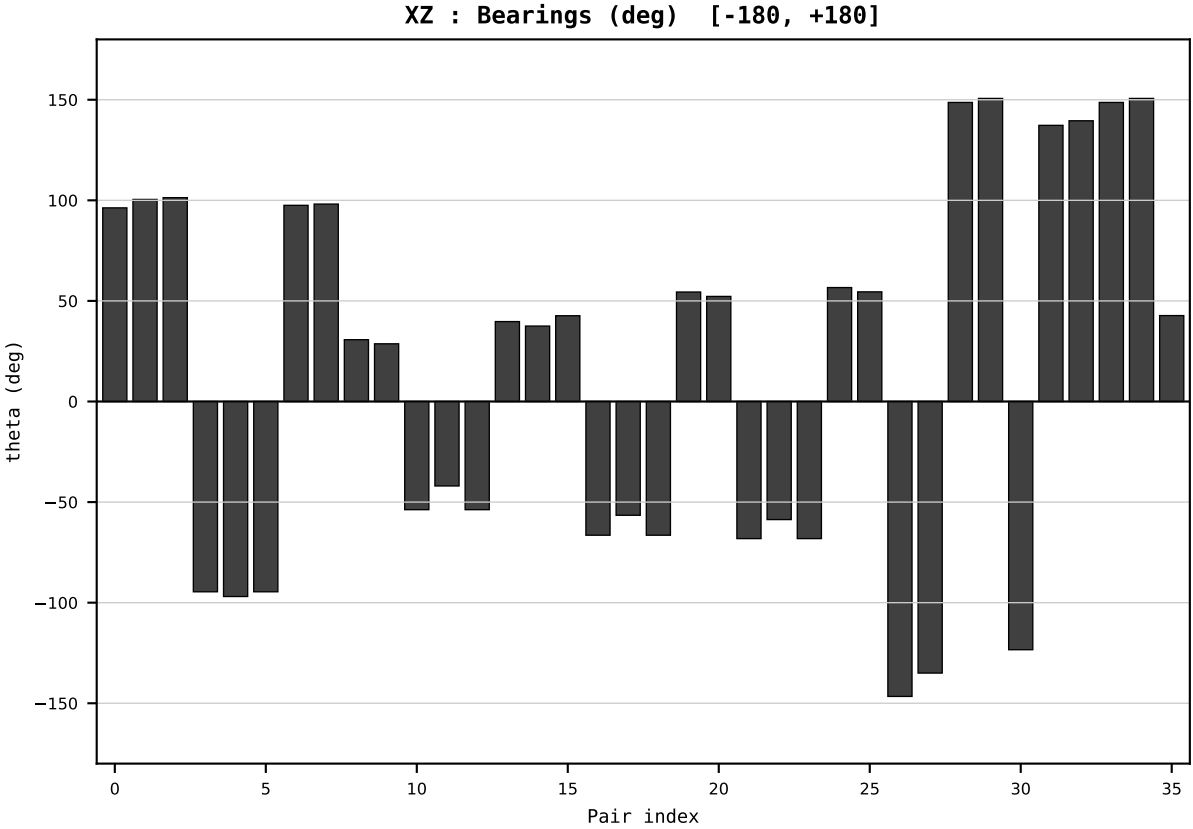
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

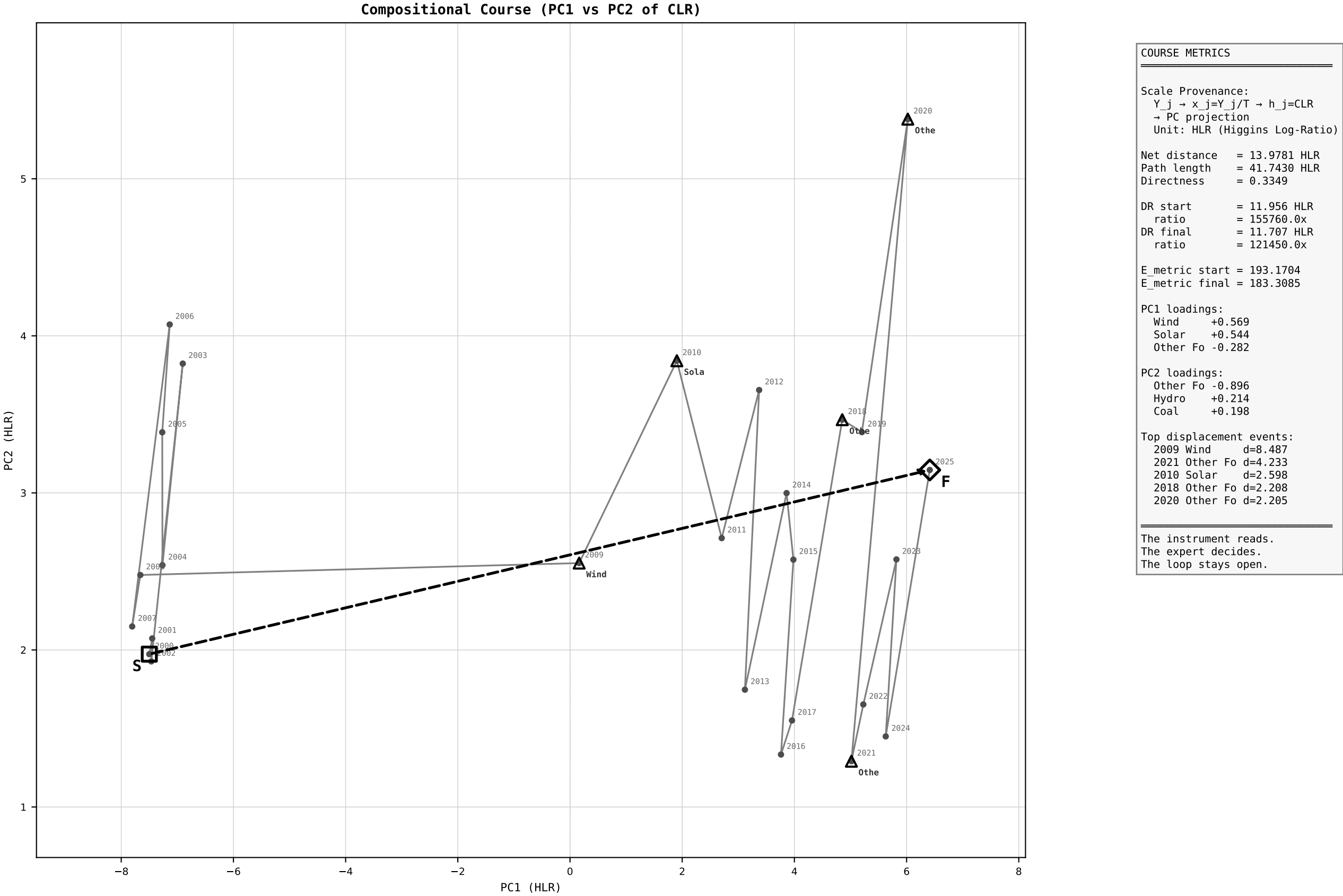
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.398765
Ring = Hs-3
E_metric= 183.3085
kappa_HS= 121450.00
omega = 7.7580 deg
d_A = 1.877835
Helm = Other Fossil
Helm d = -1.752225
DR = 11.707 HLR
DR ratio= 121450.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





View B

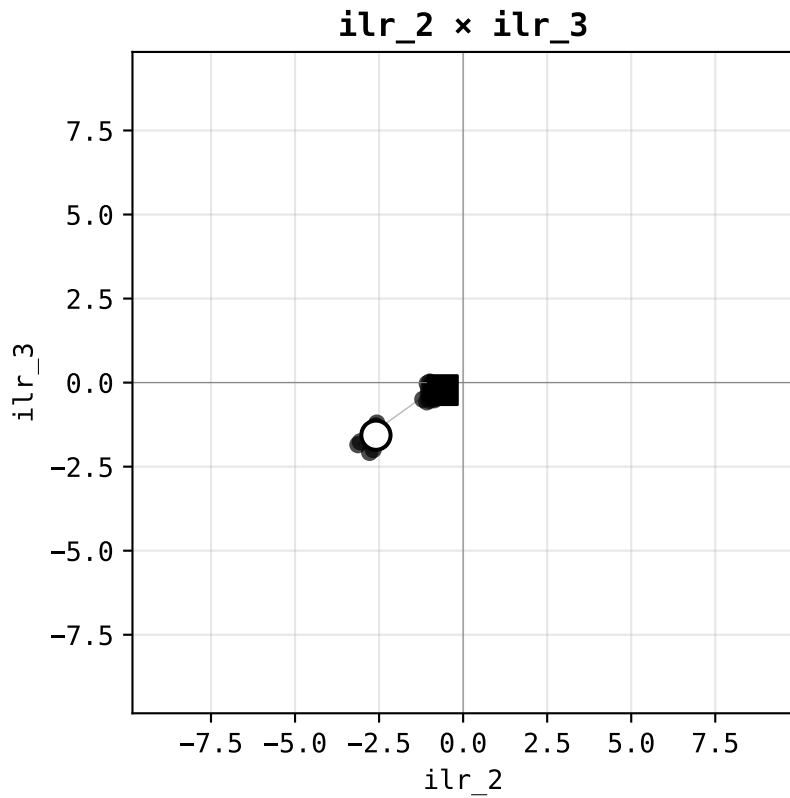
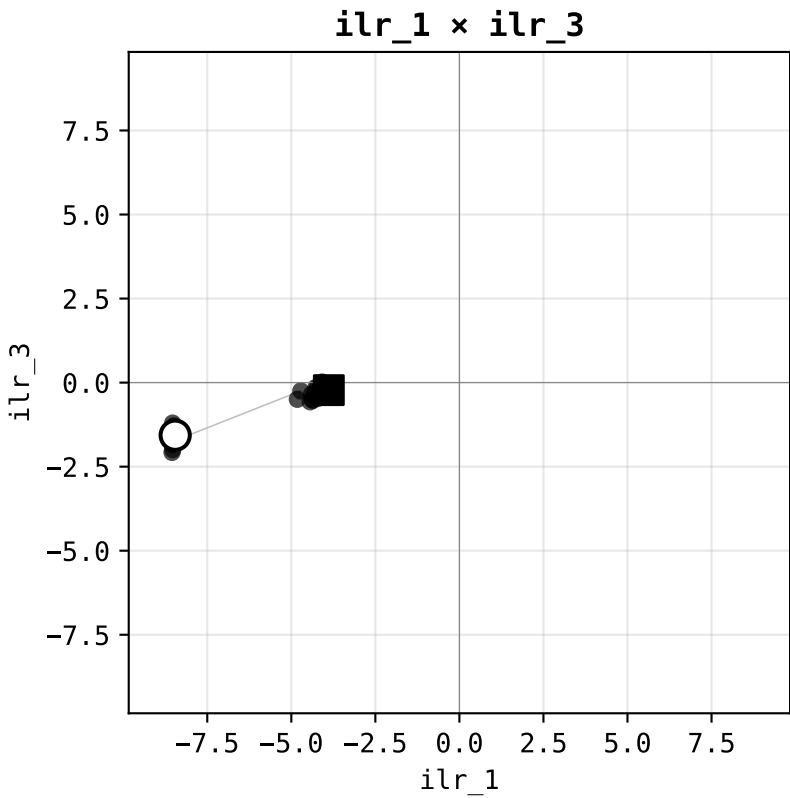
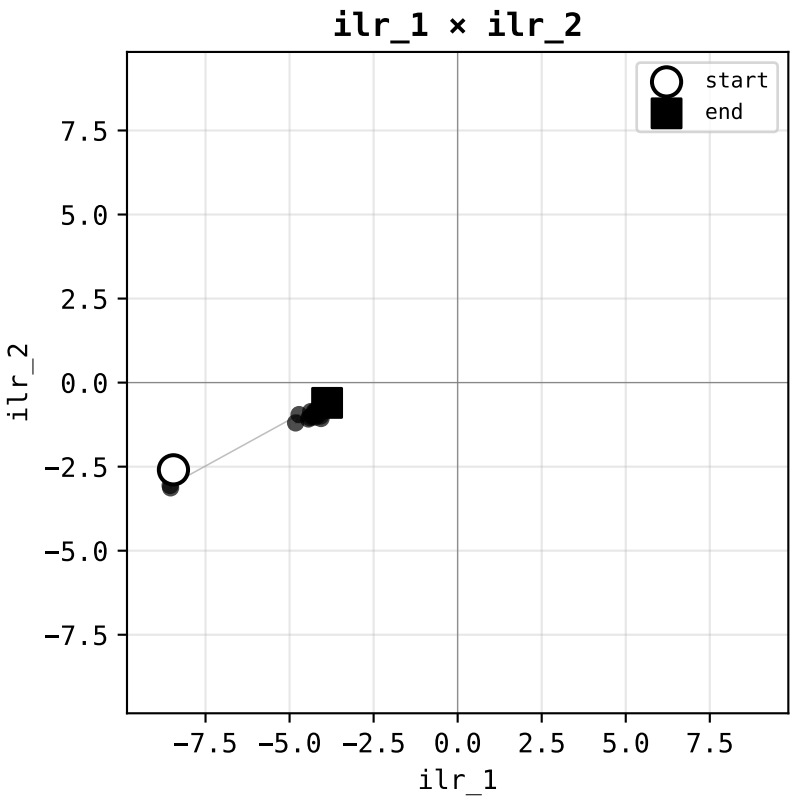
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ AUS Australia

ILR-Helmert Orthogonal Triplet – AUS Australia · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=26 years · D=9 carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:

start year: 2000
end year: 2025
N readings: 26

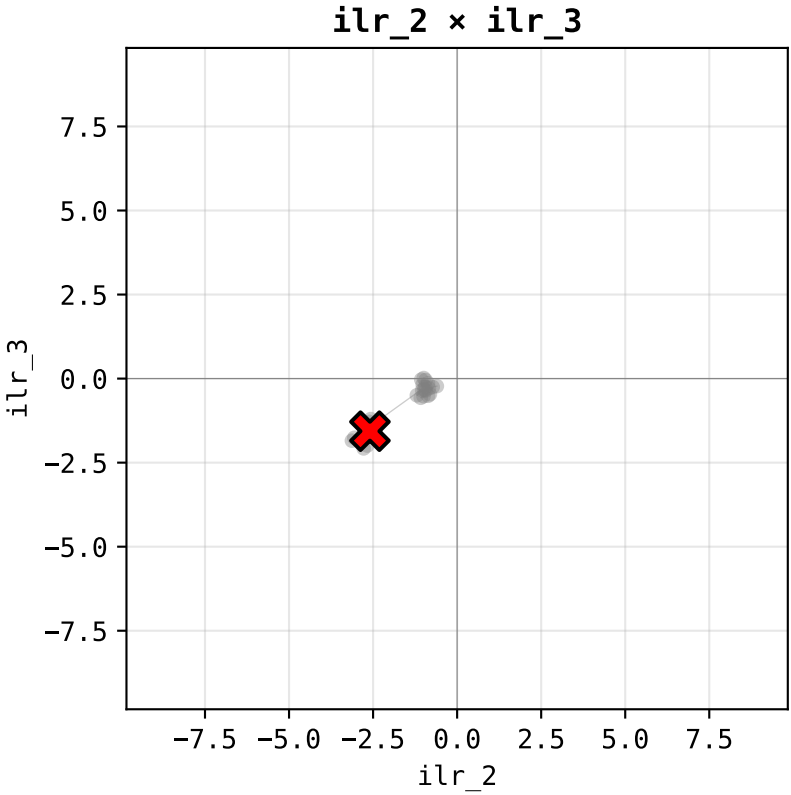
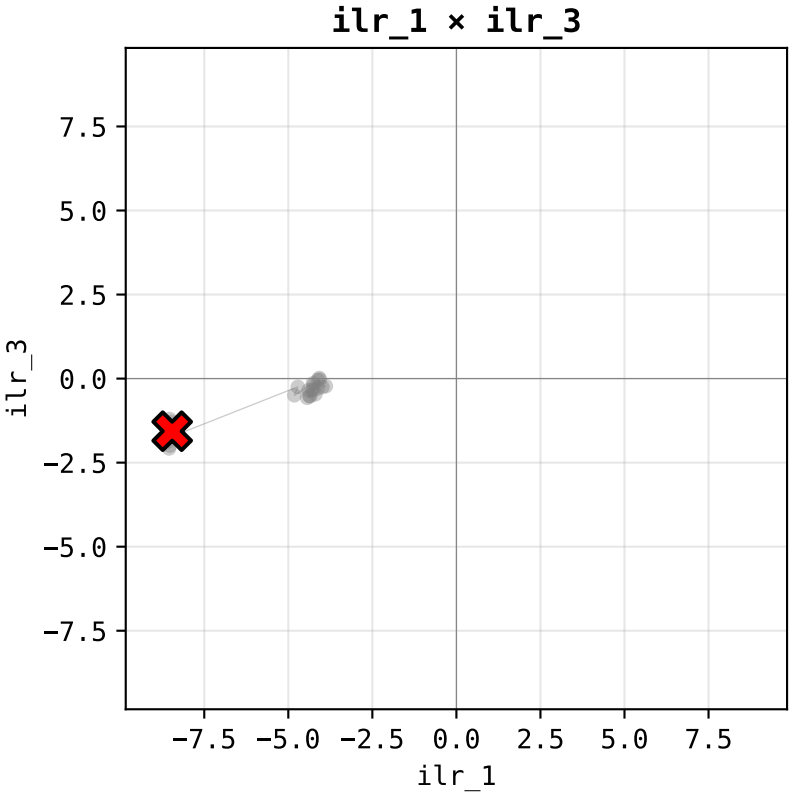
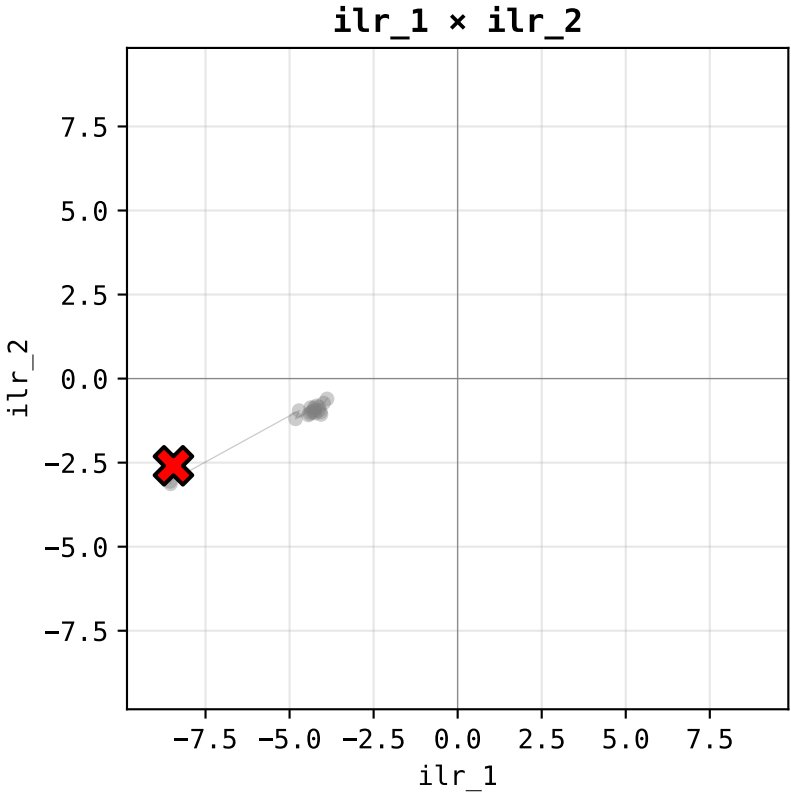
Display:

fixed scale ±9.84 ILR
Order 1 (first principles)
per Output Doctrine v1.0

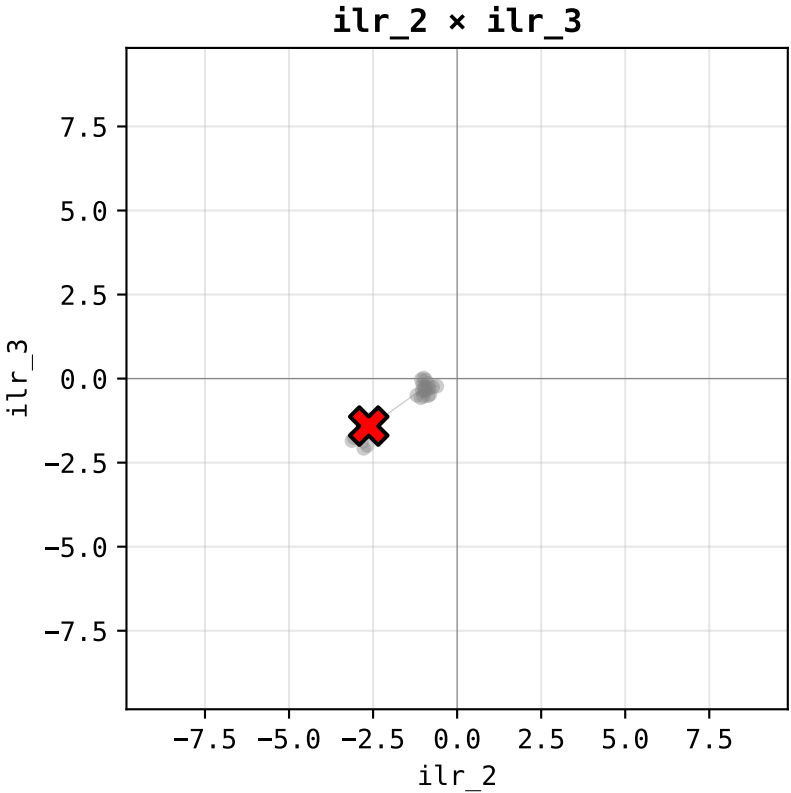
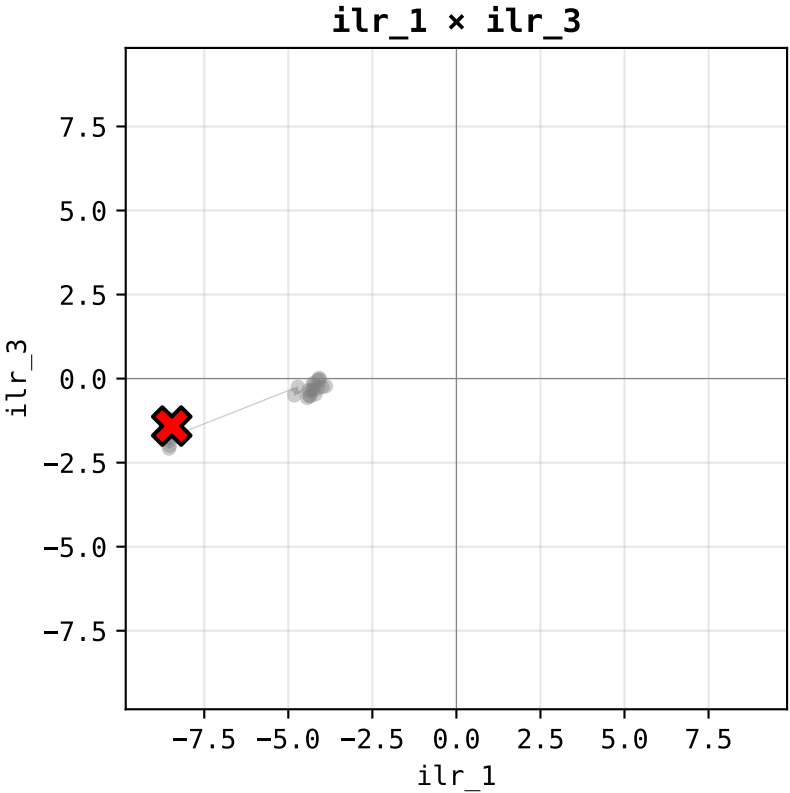
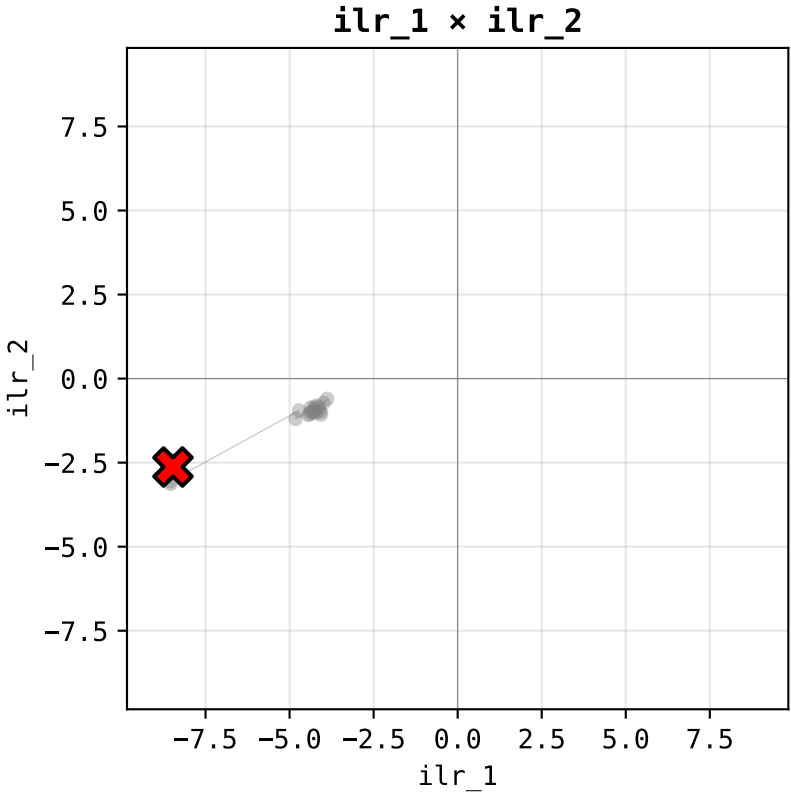
Symbology:

○ start · ■ end
· intermediate years

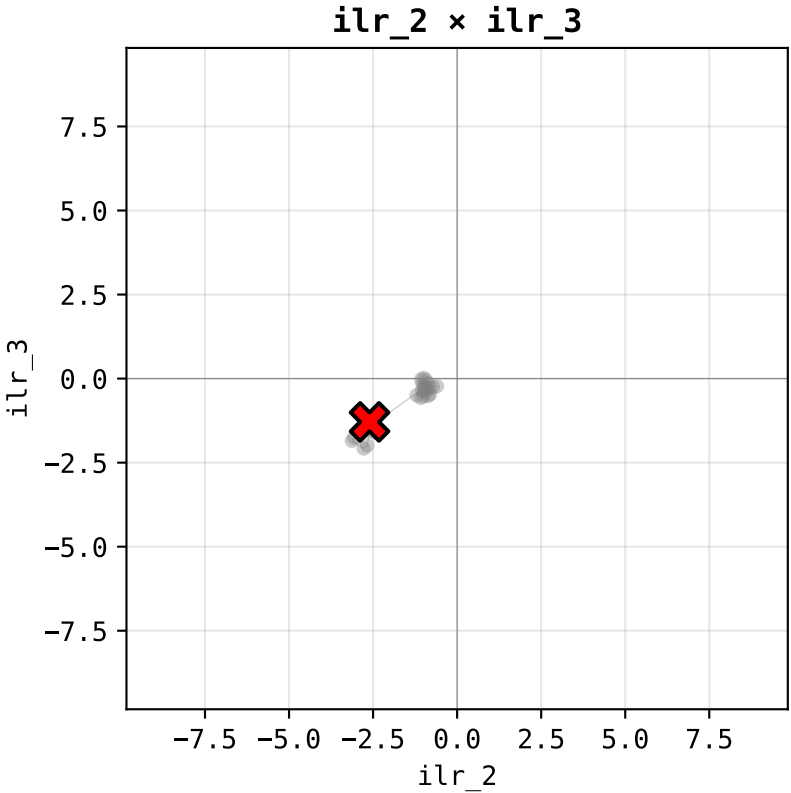
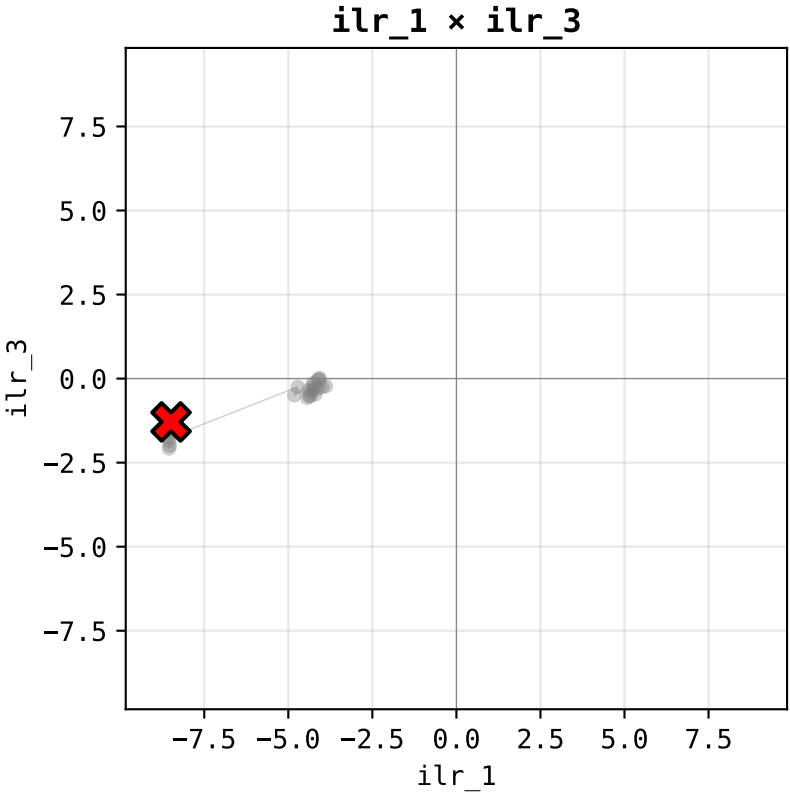
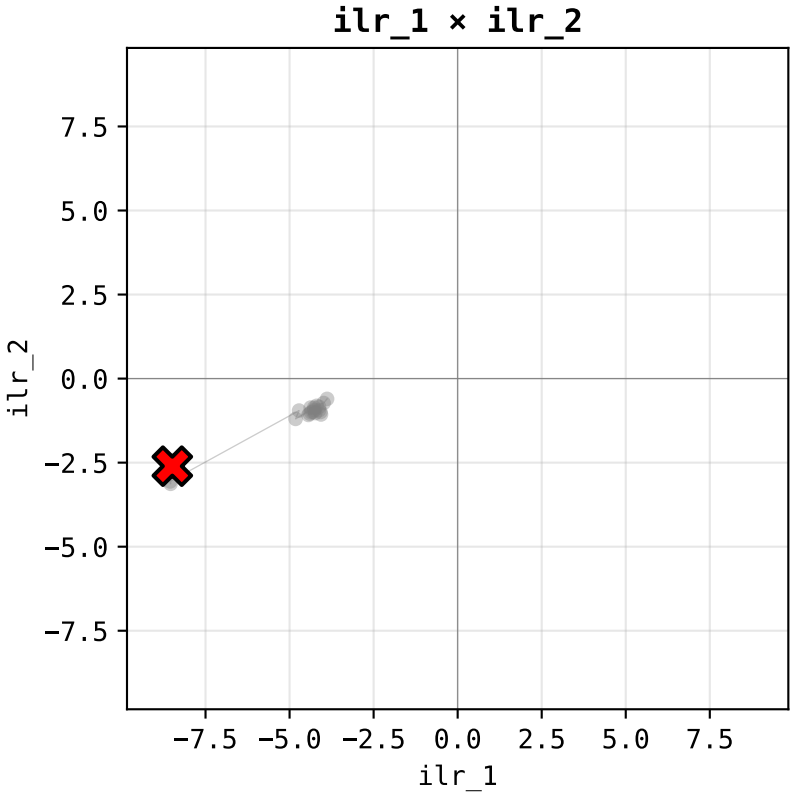
ILR Triplet – AUS Australia · year 2000 (1/26)



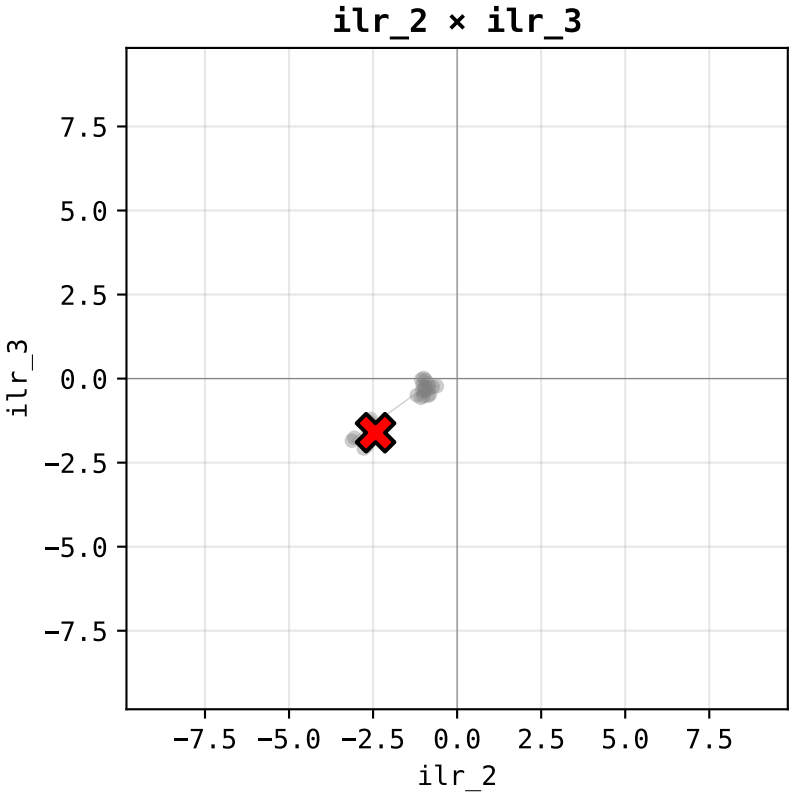
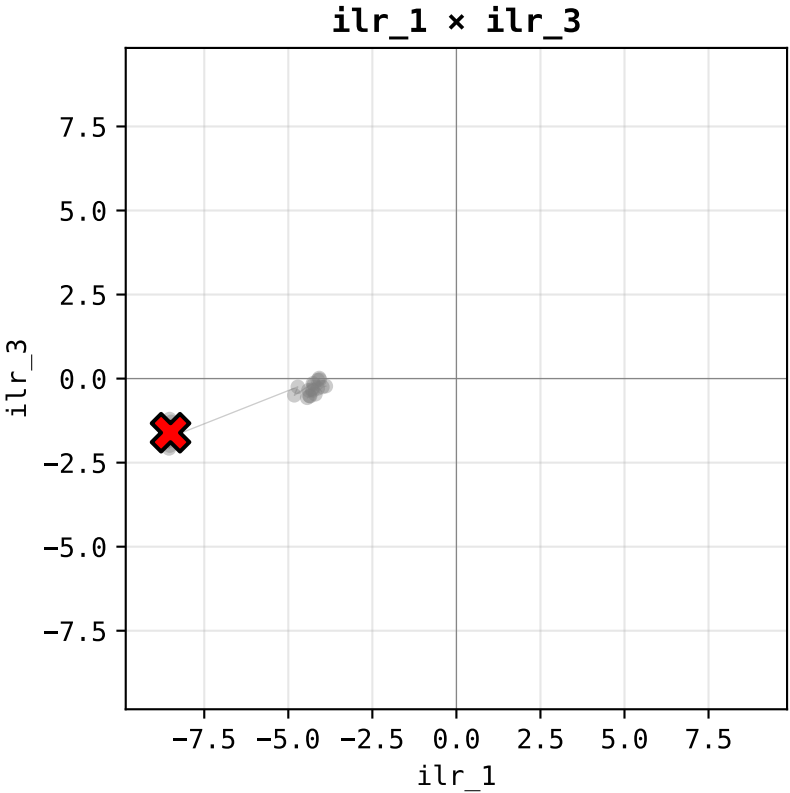
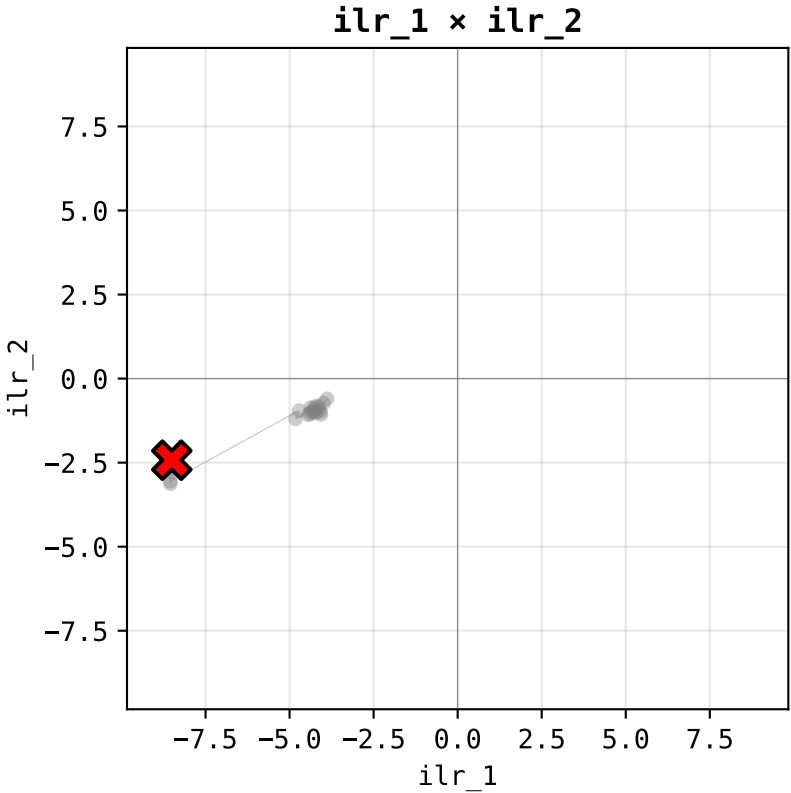
ILR Triplet – AUS Australia · year 2001 (2/26)



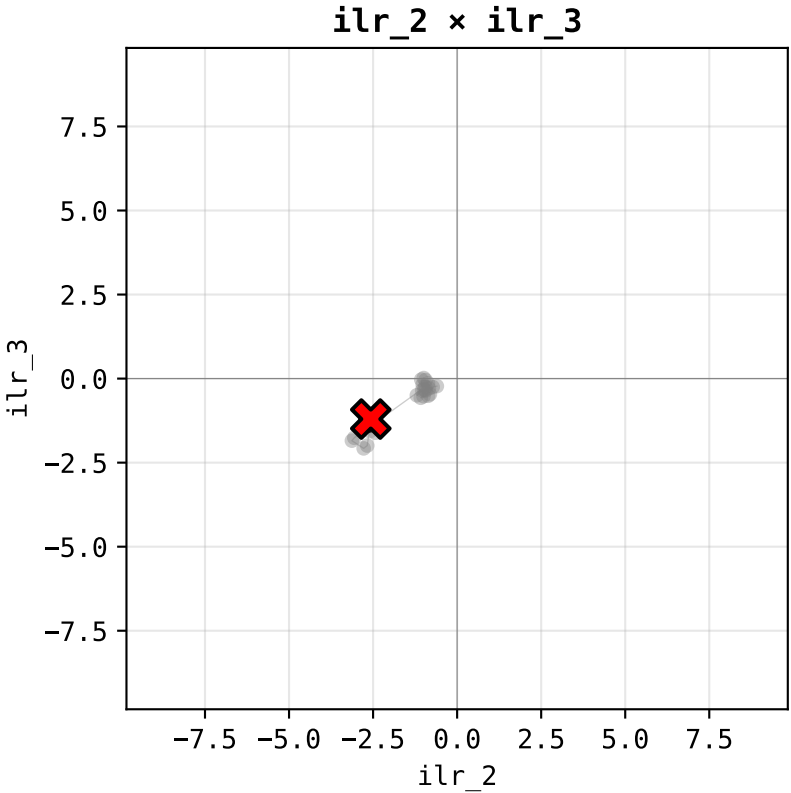
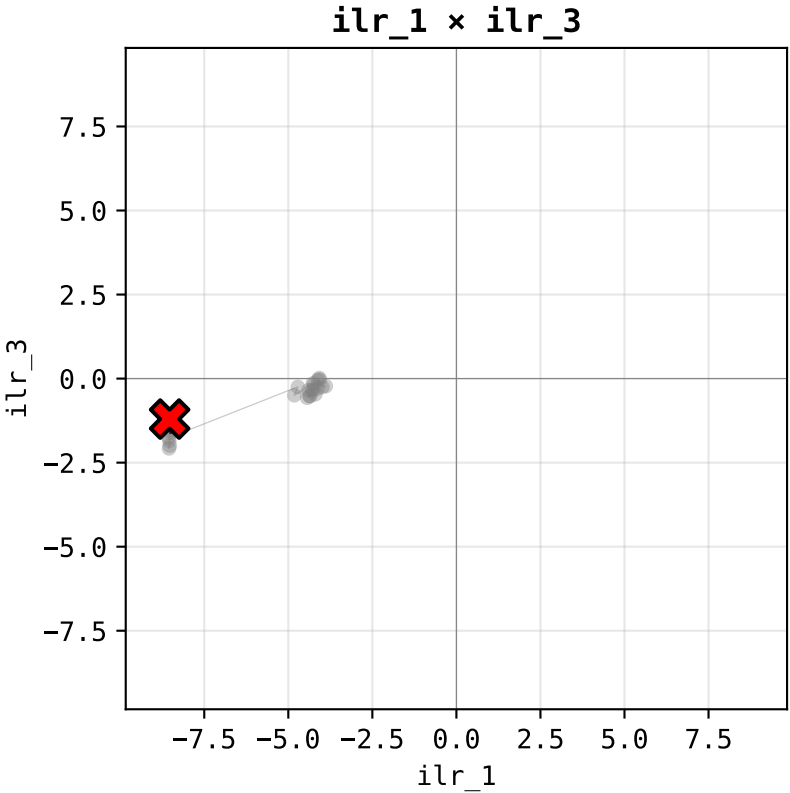
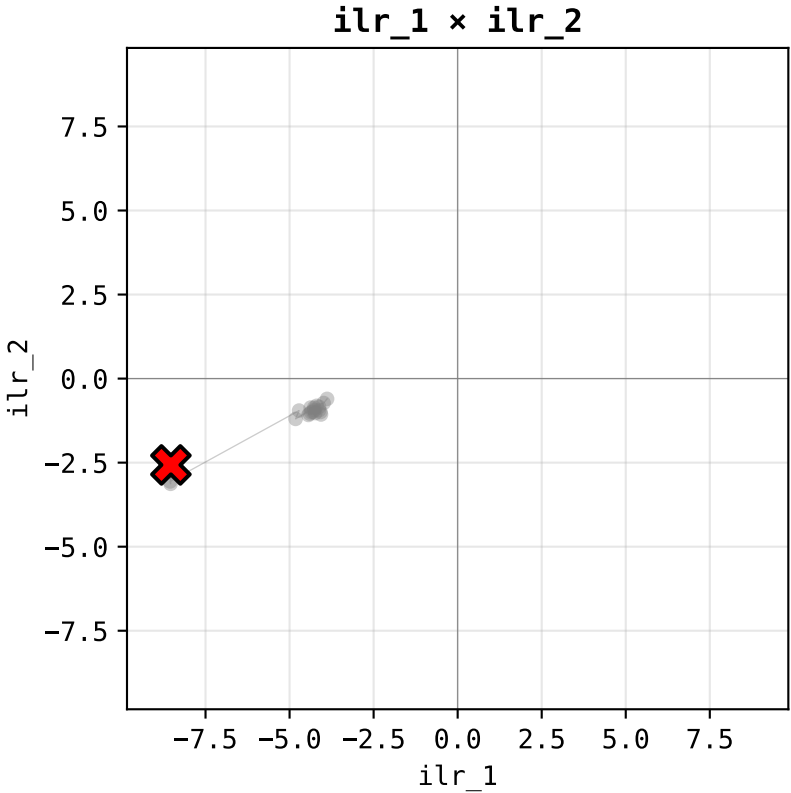
ILR Triplet – AUS Australia · year 2002 (3/26)



ILR Triplet – AUS Australia · year 2003 (4/26)

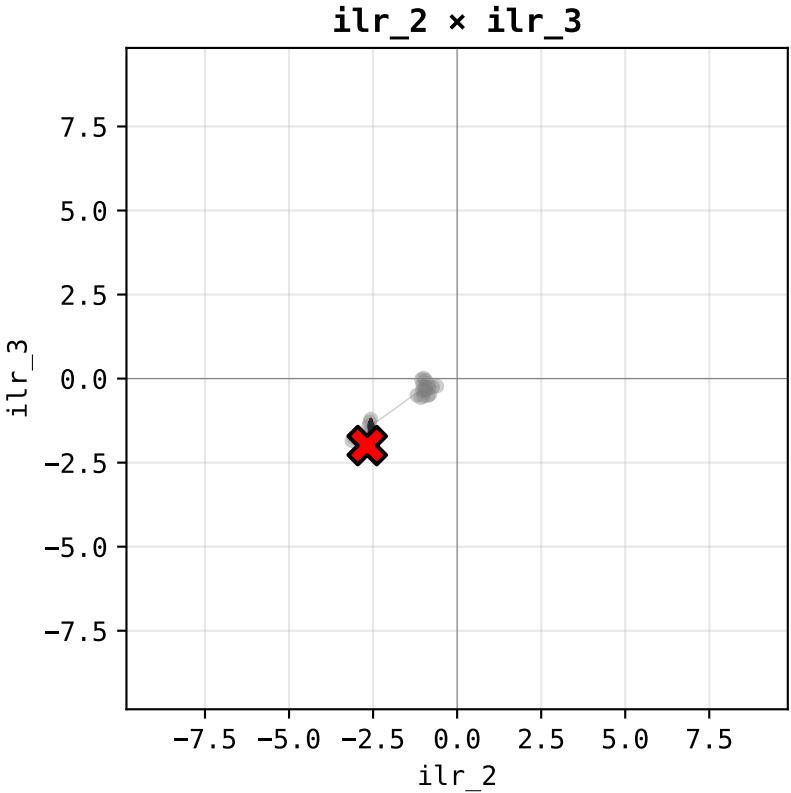
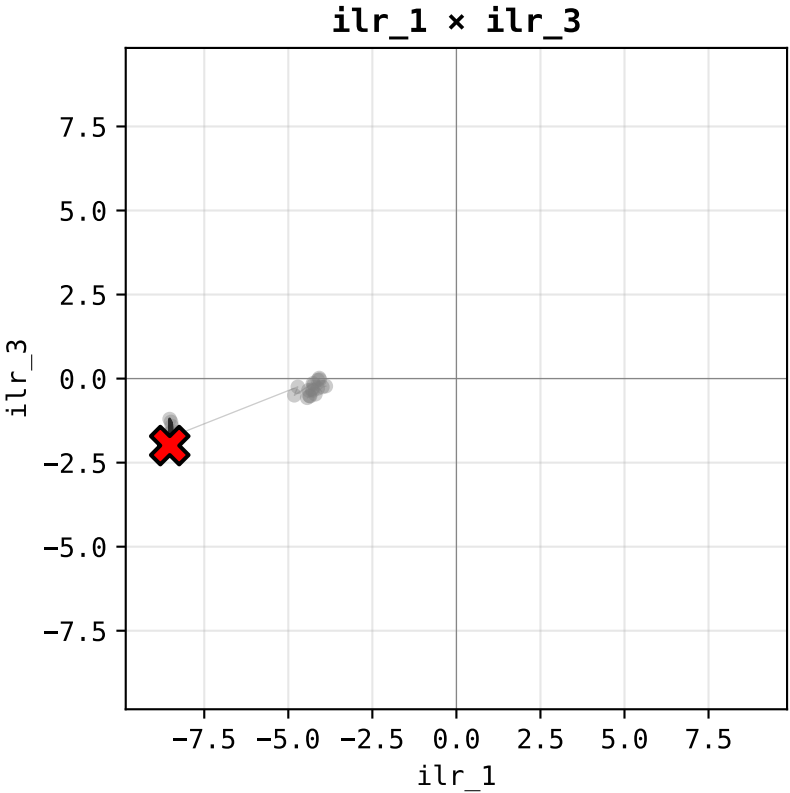
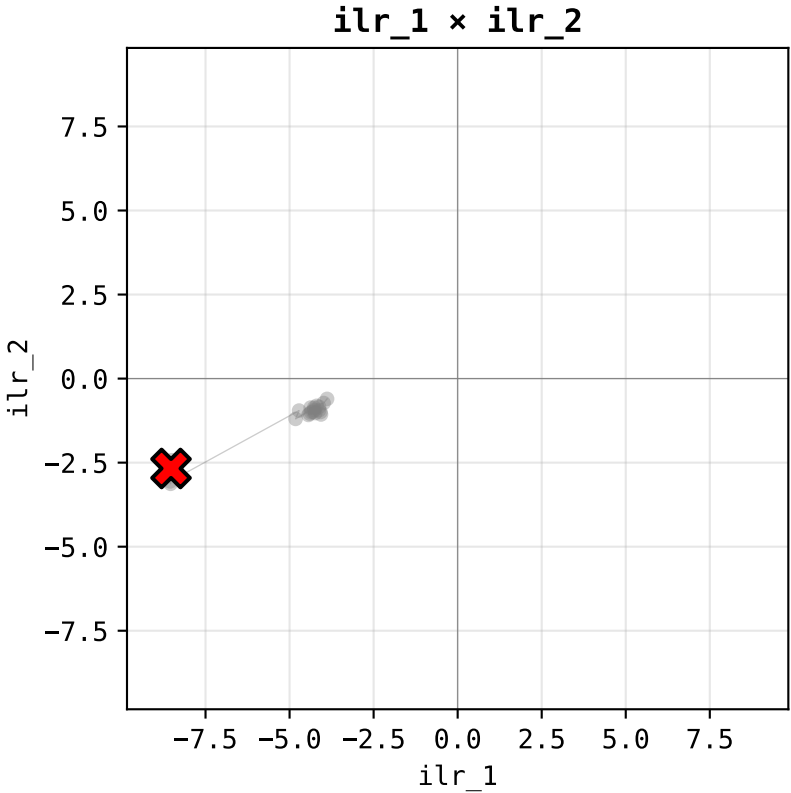


ILR Triplet – AUS Australia · year 2004 (5/26)

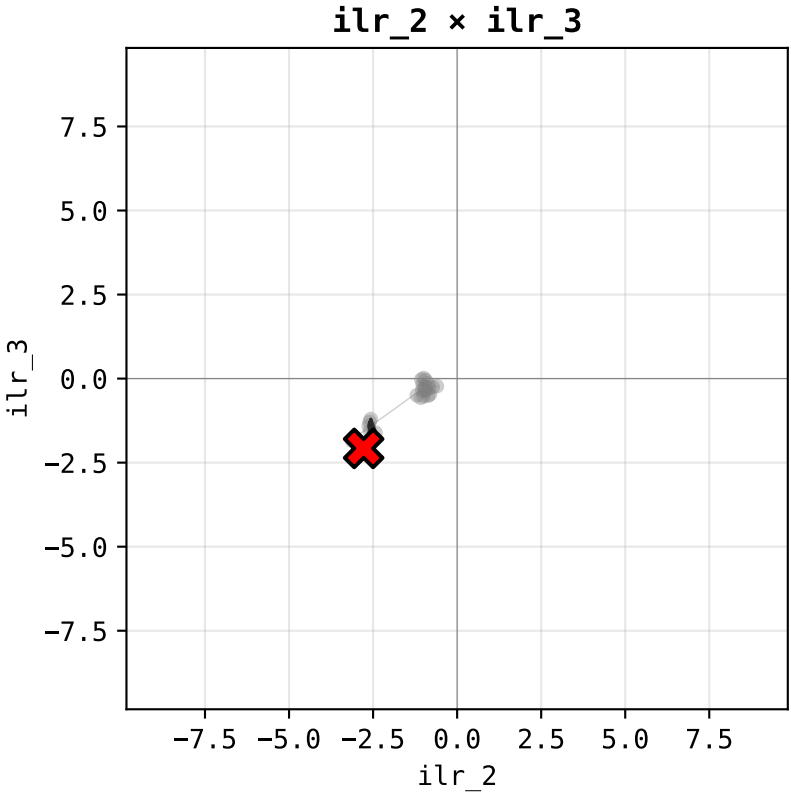
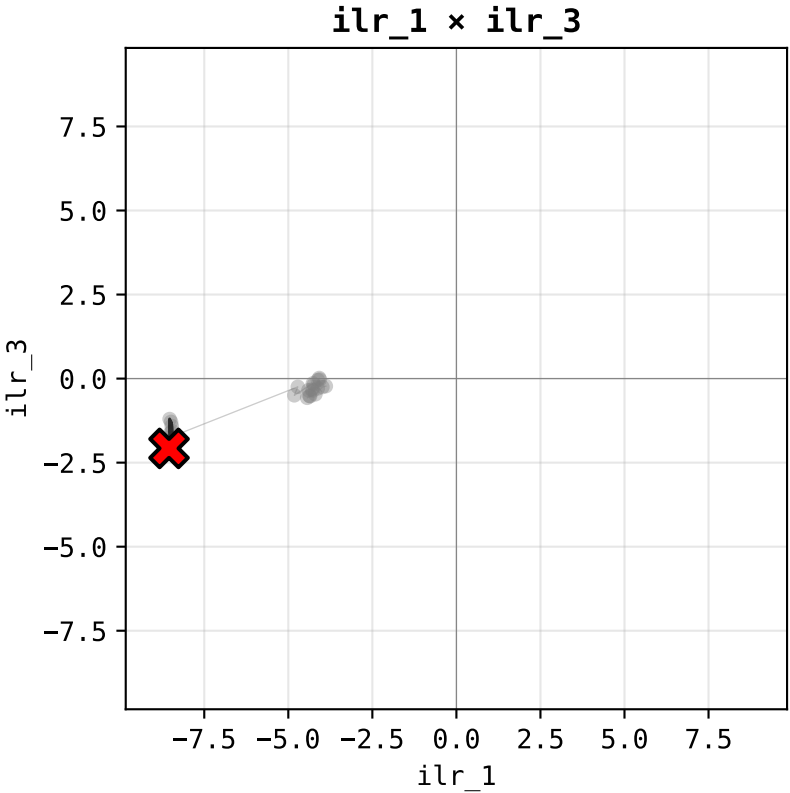
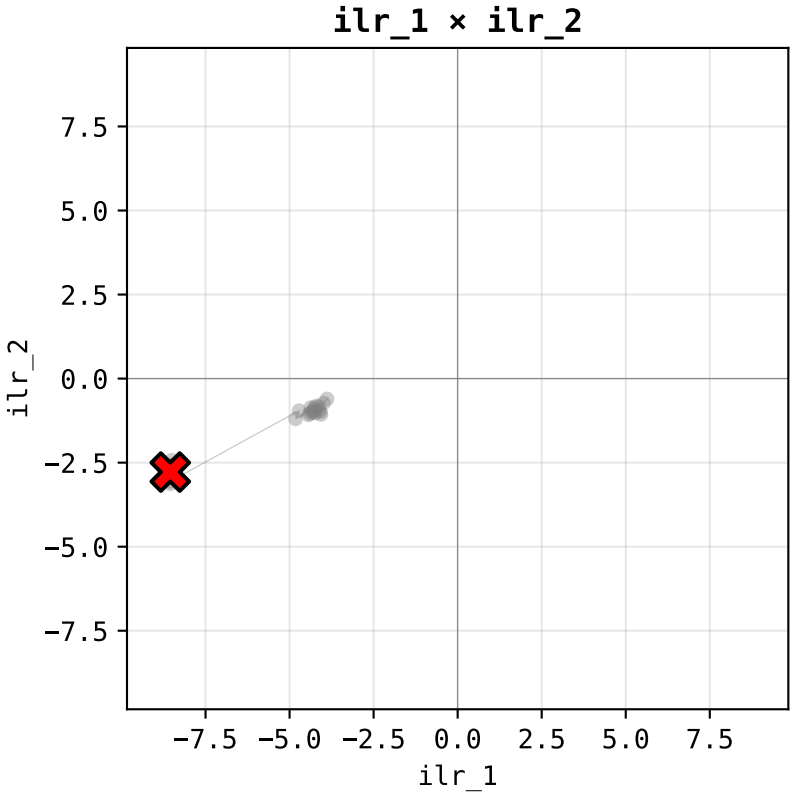


X = current year (2004) · black = past trajectory · gray = future · ilr range ±9.84

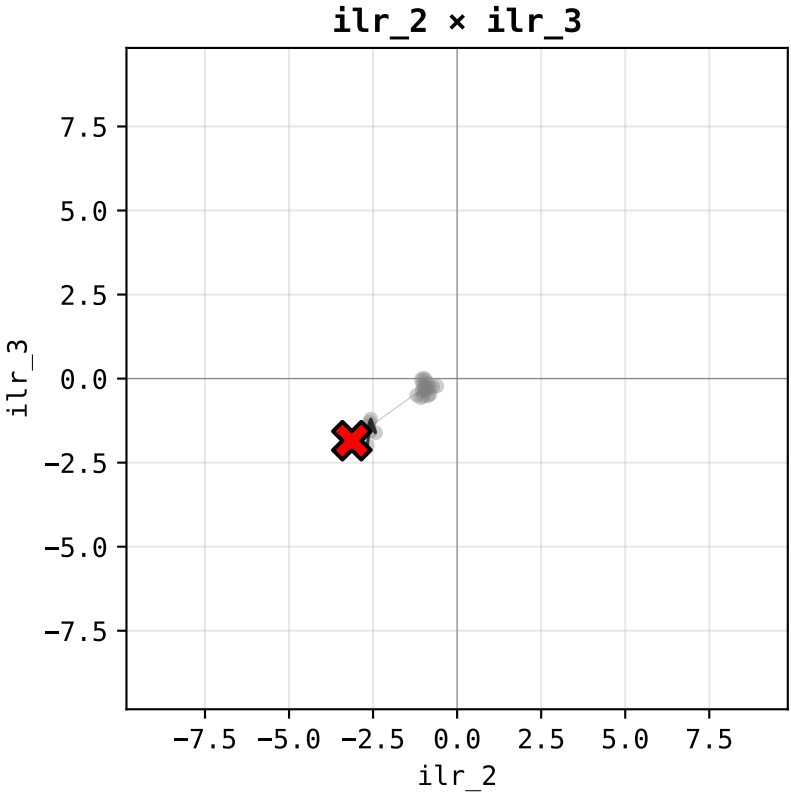
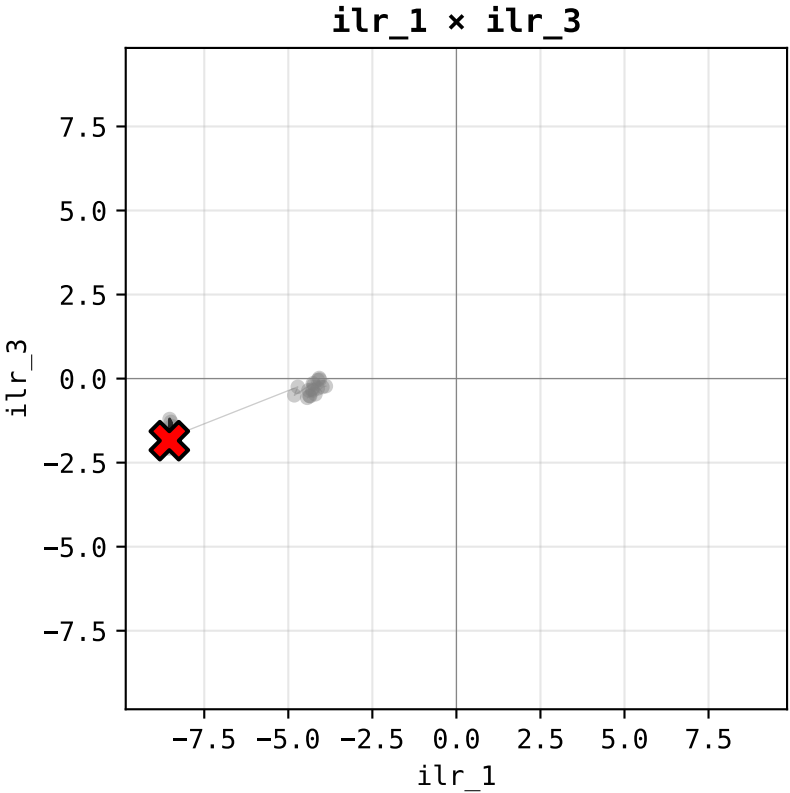
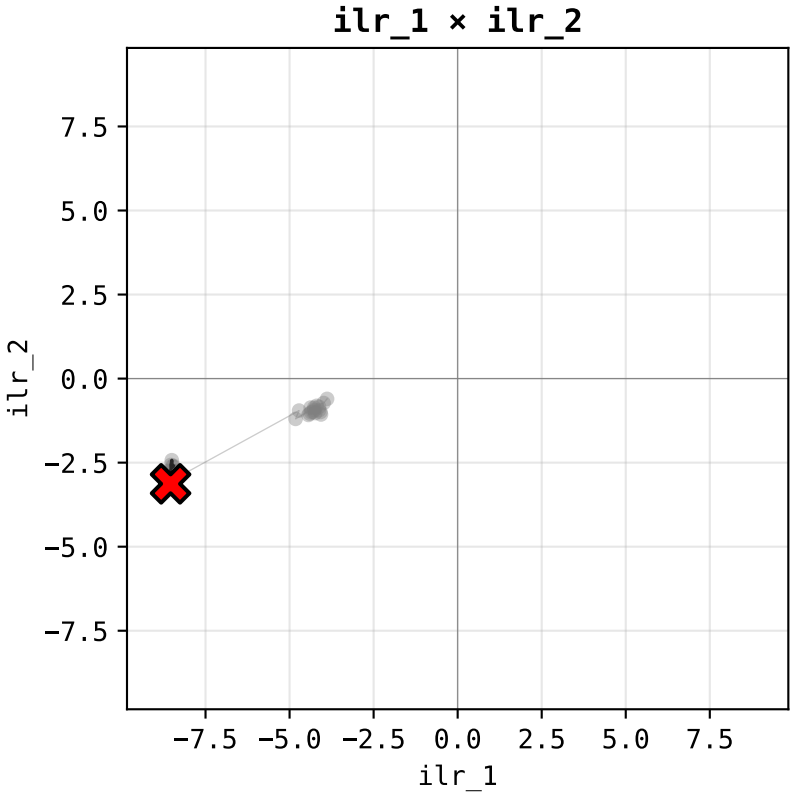
ILR Triplet – AUS Australia · year 2005 (6/26)



ILR Triplet – AUS Australia · year 2006 (7/26)

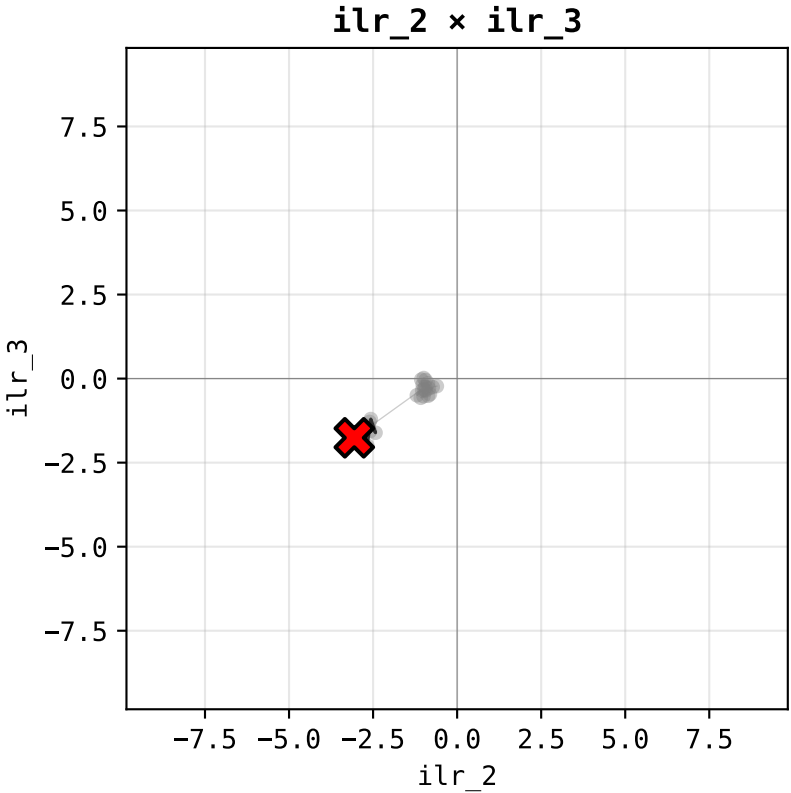
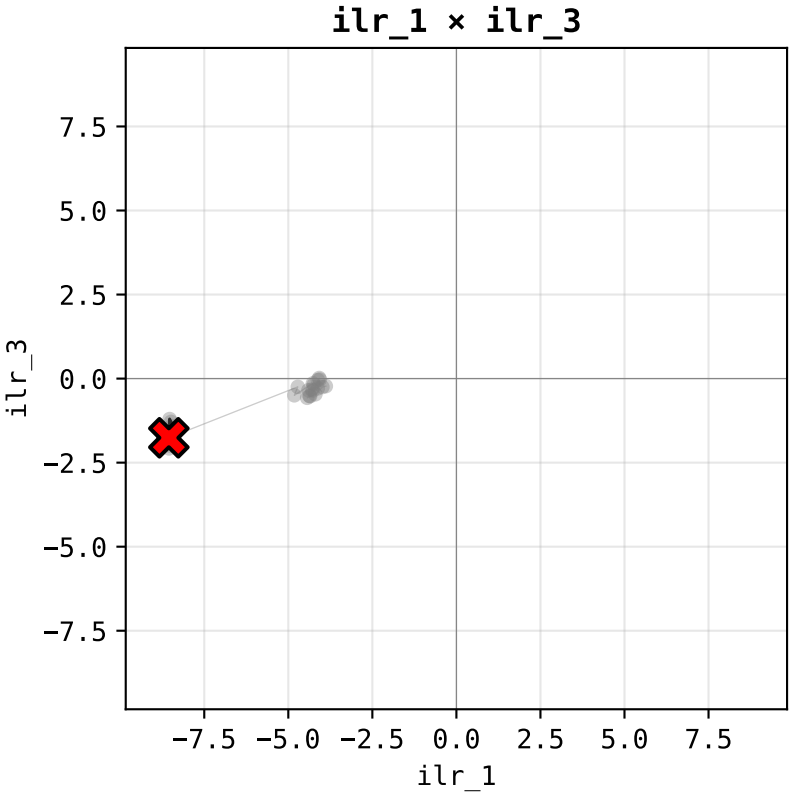
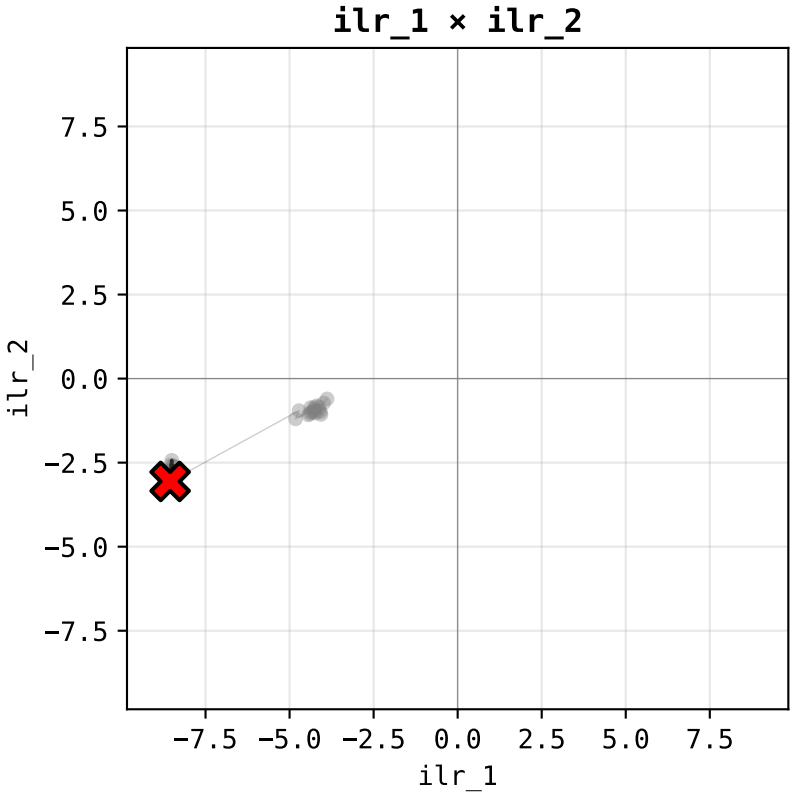


ILR Triplet – AUS Australia · year 2007 (8/26)



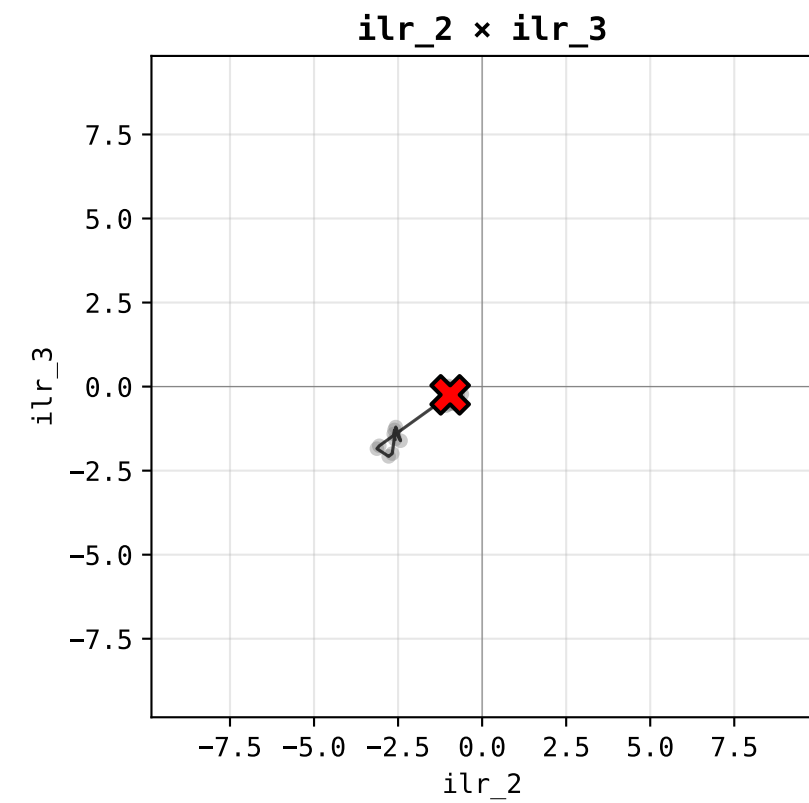
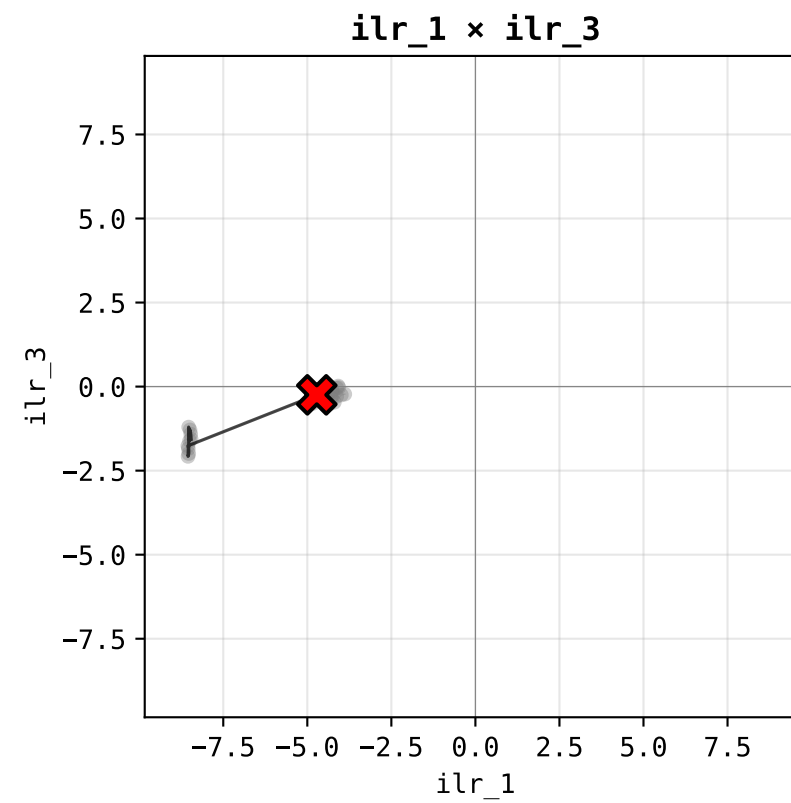
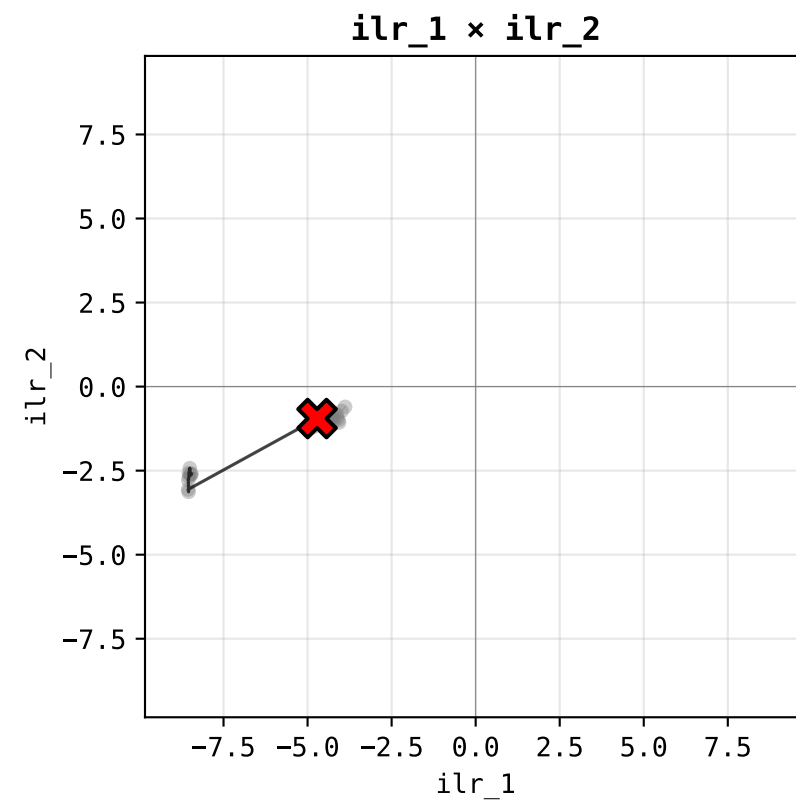
X = current year (2007) · black = past trajectory · gray = future · ilr range ±9.84

ILR Triplet – AUS Australia · year 2008 (9/26)



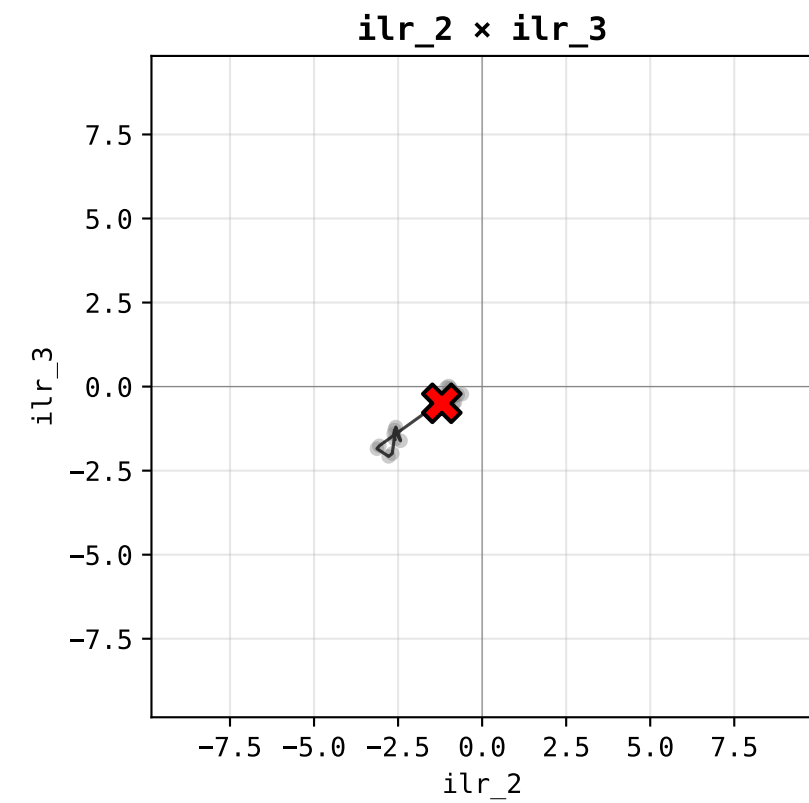
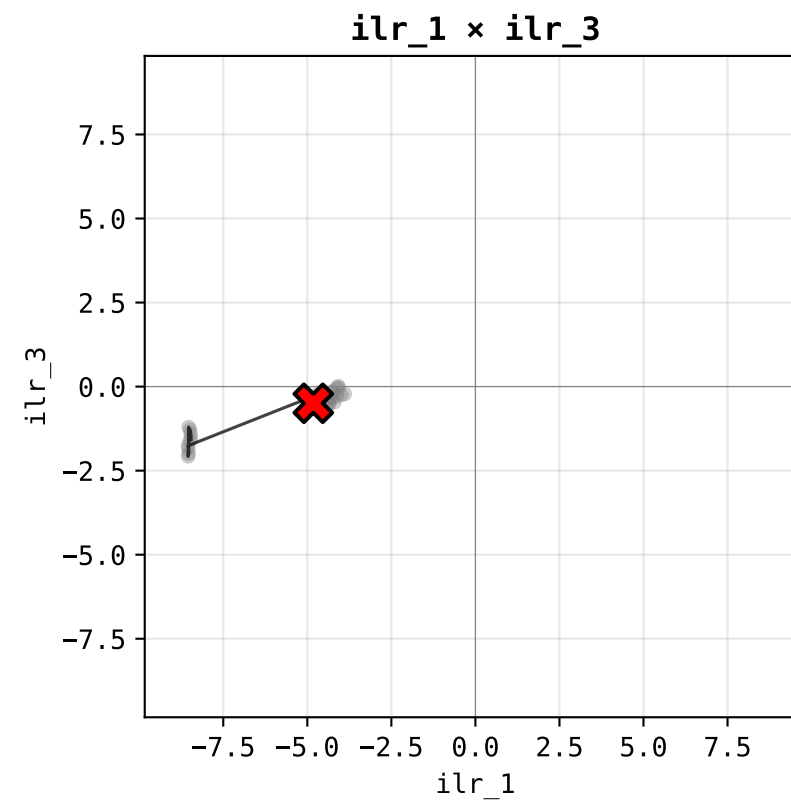
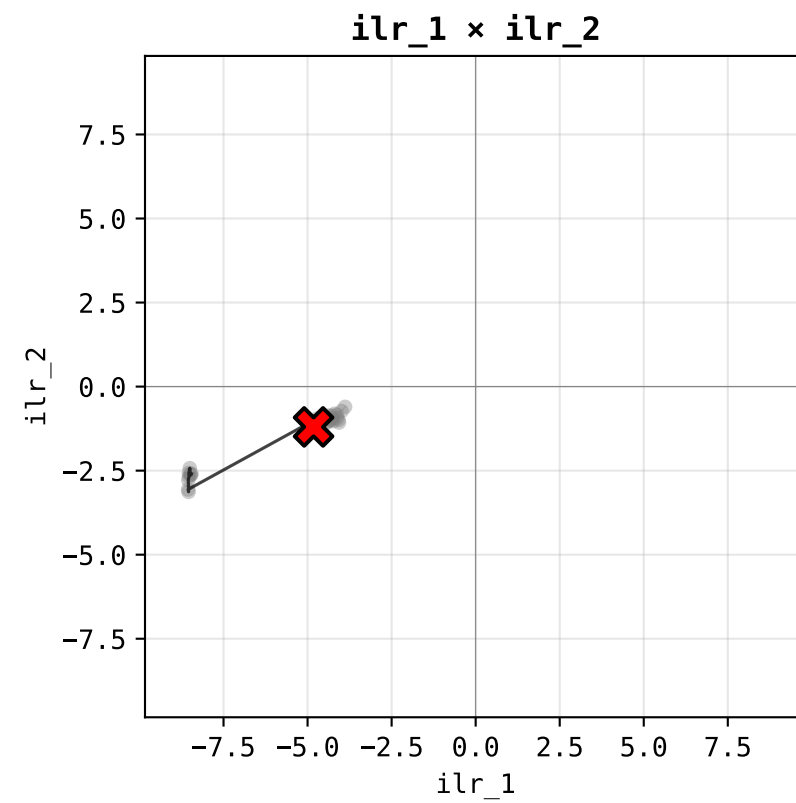
X = current year (2008) · black = past trajectory · gray = future · ilr range ±9.84

ILR Triplet – AUS Australia · year 2009 (10/26)



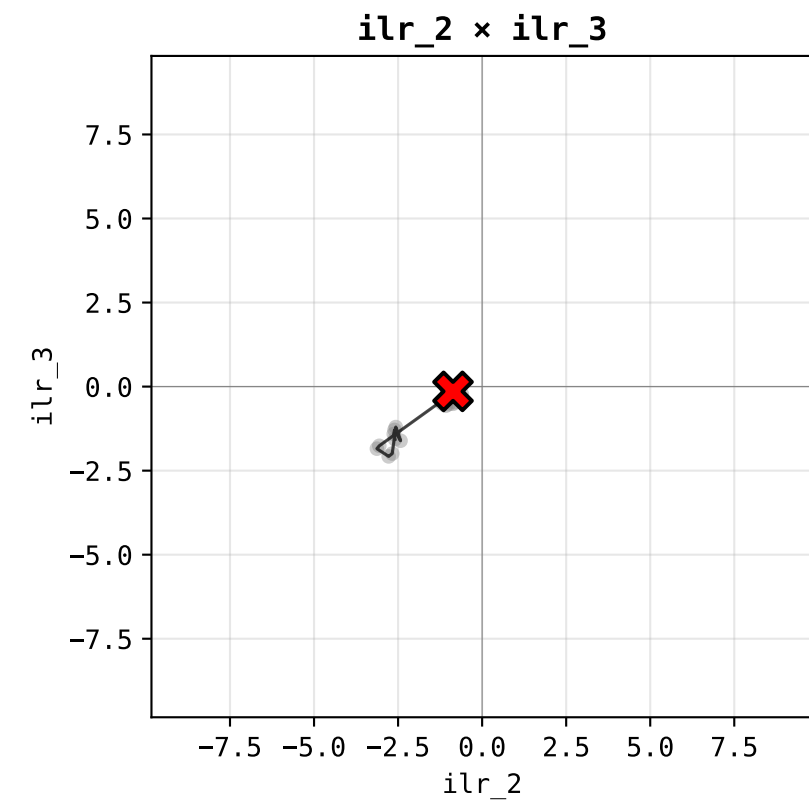
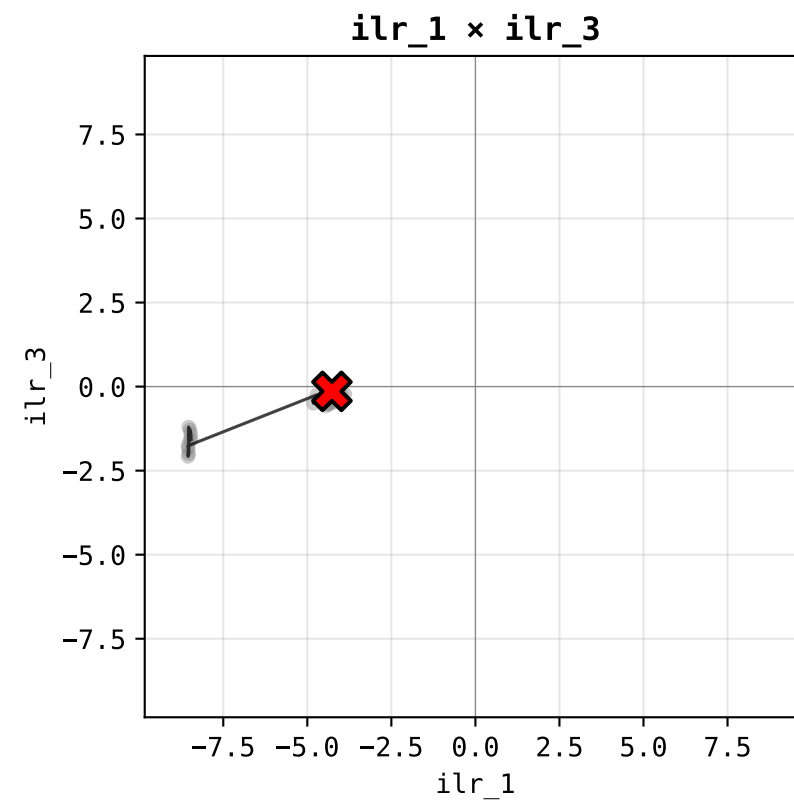
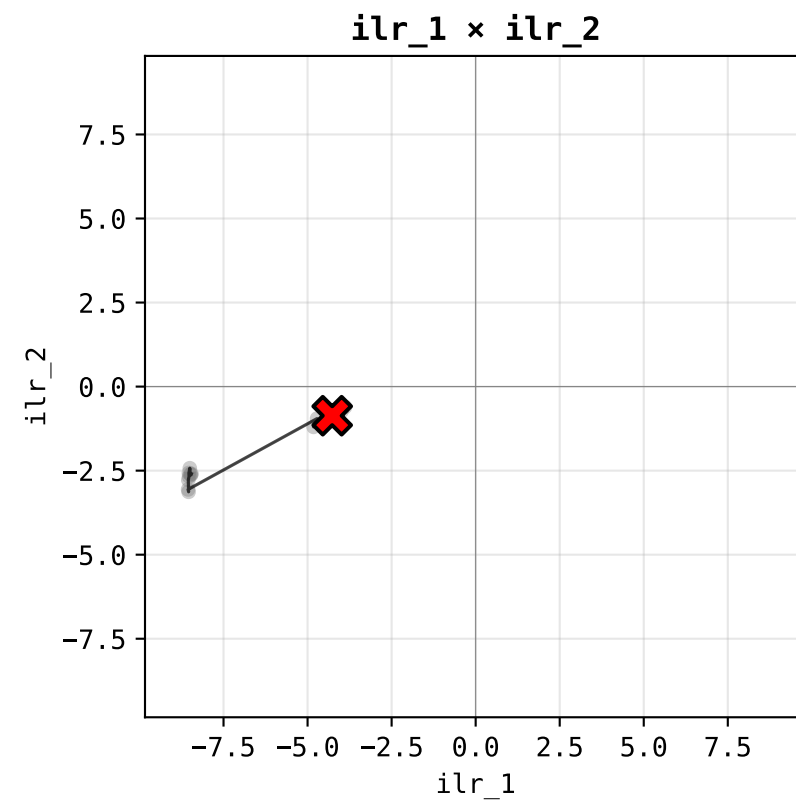
X = current year (2009) · black = past trajectory · gray = future · ilr range ±9.84

ILR Triplet – AUS Australia · year 2010 (11/26)



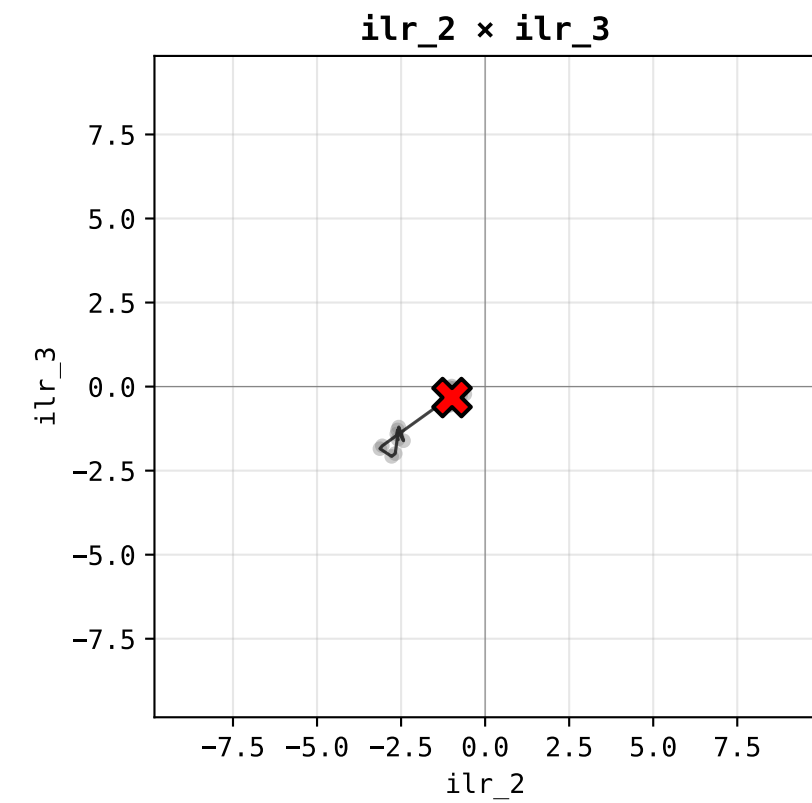
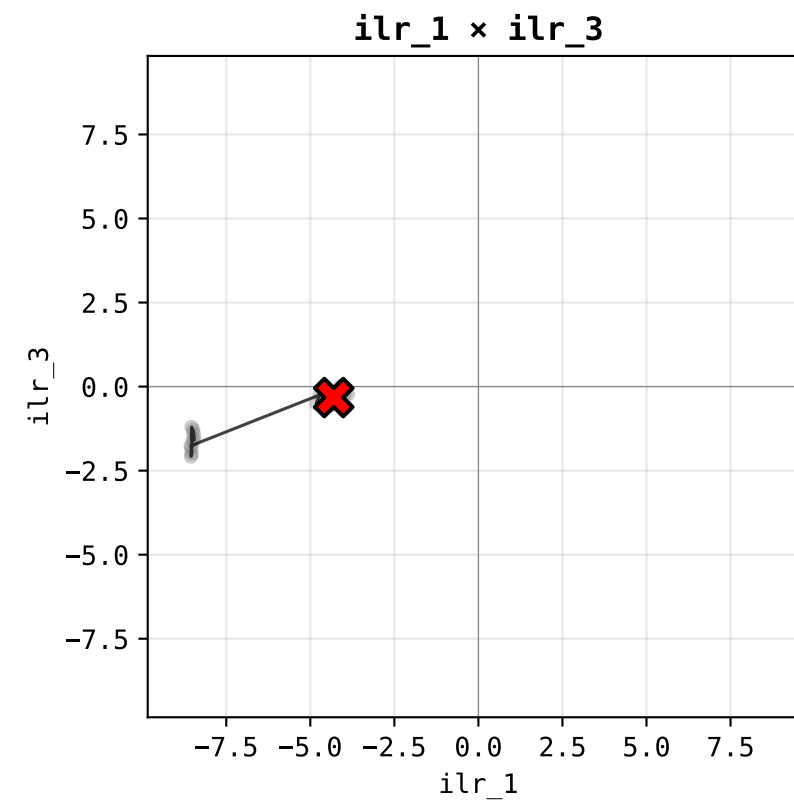
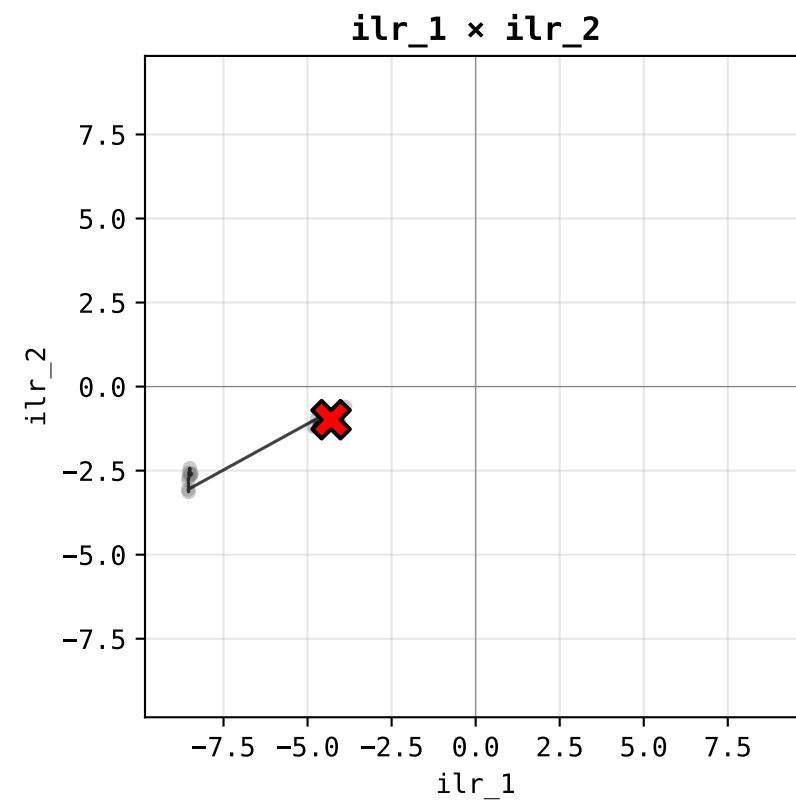
X = current year (2010) · black = past trajectory · gray = future · ilr range ±9.84

ILR Triplet – AUS Australia · year 2011 (12/26)

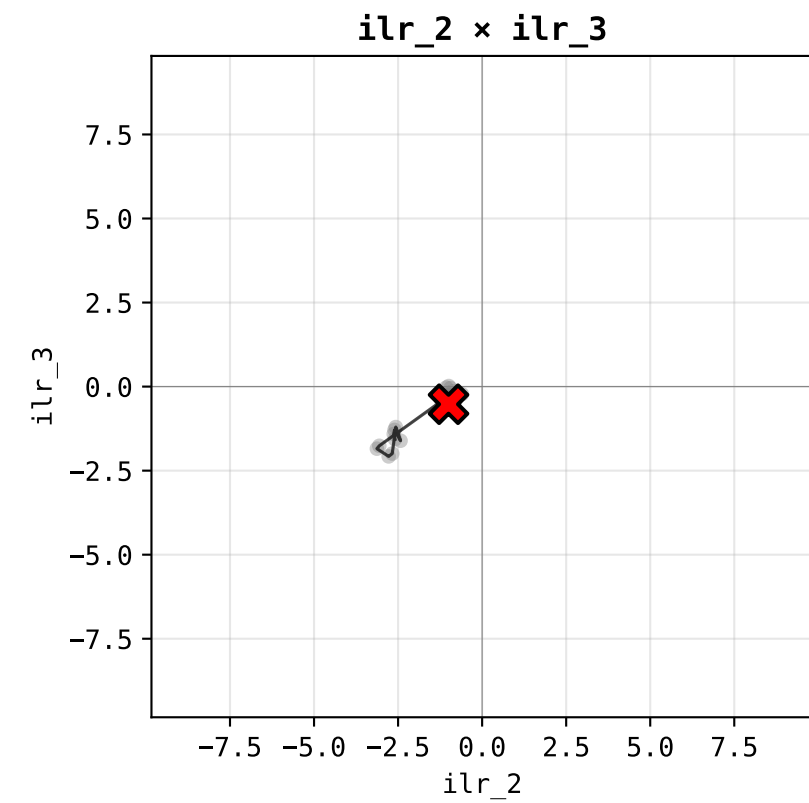
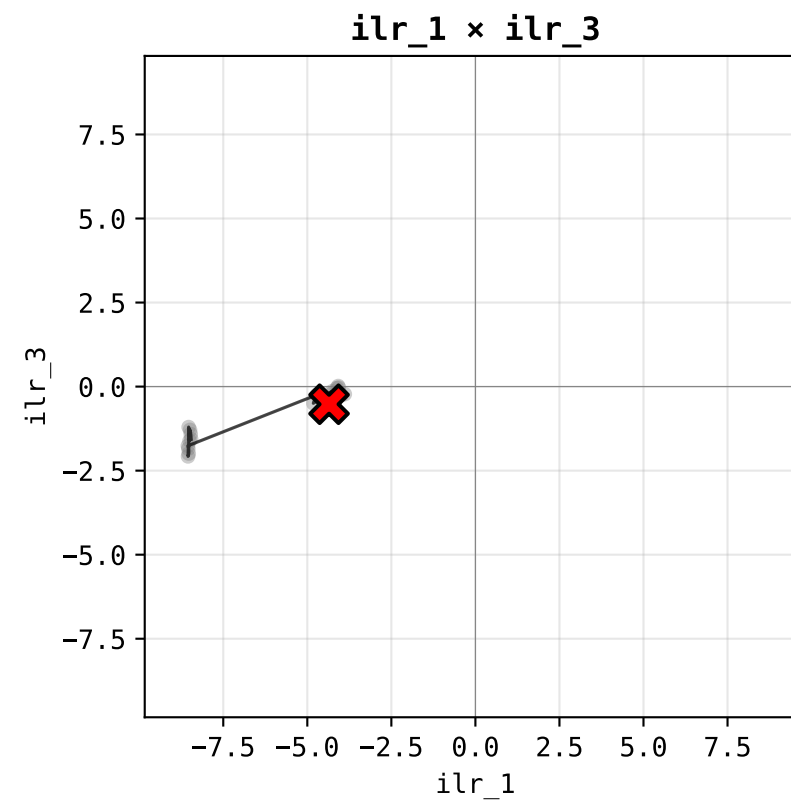
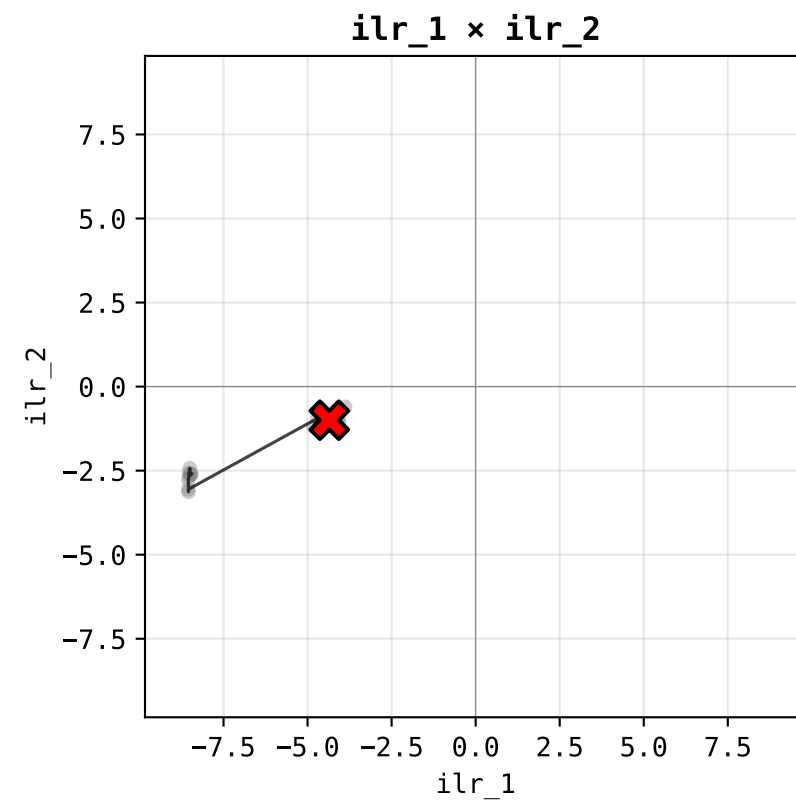


X = current year (2011) · black = past trajectory · gray = future · ilr range ±9.84

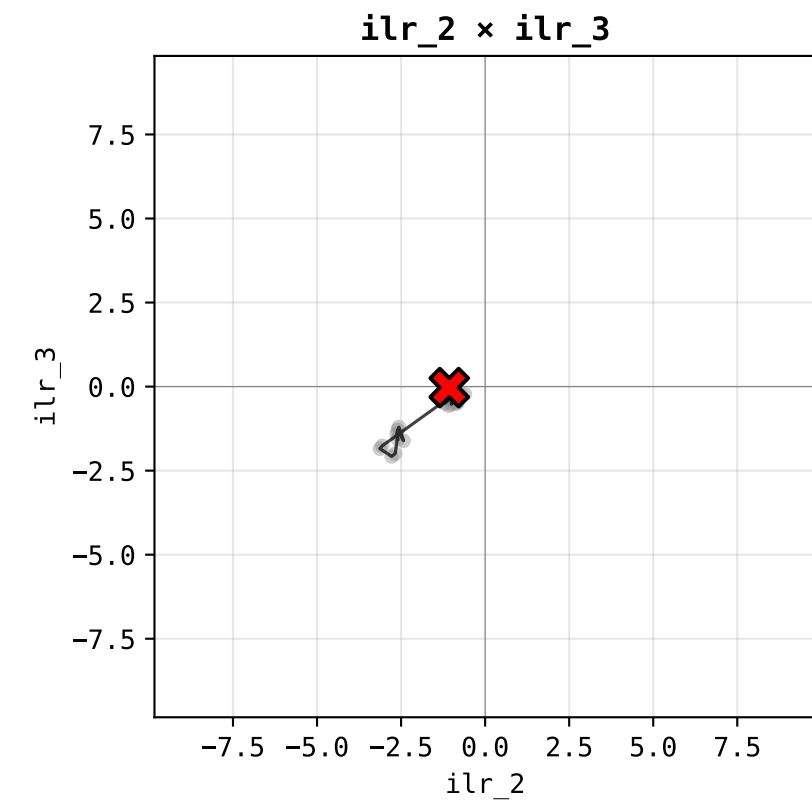
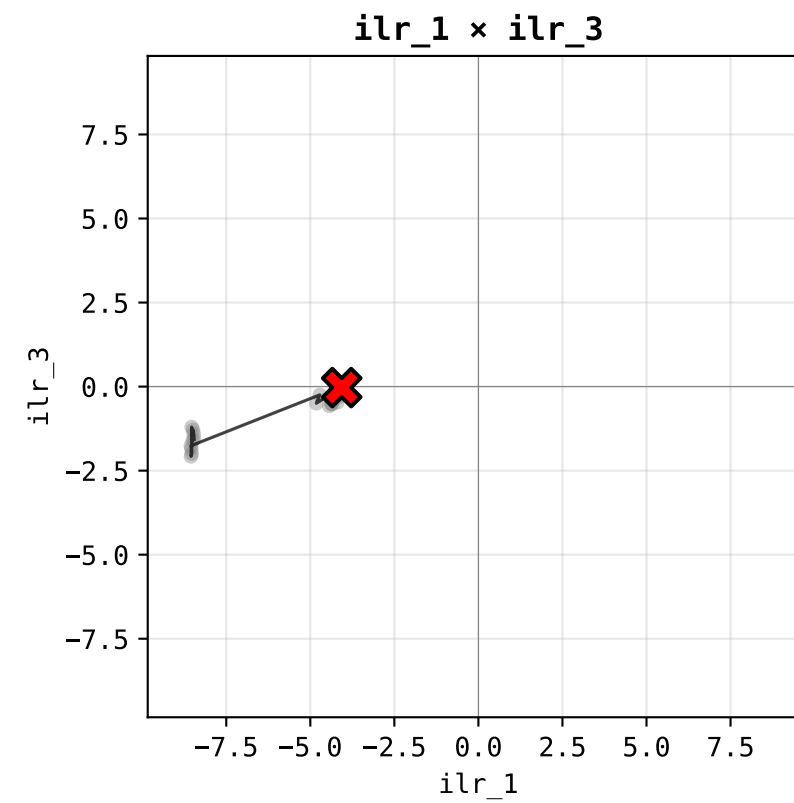
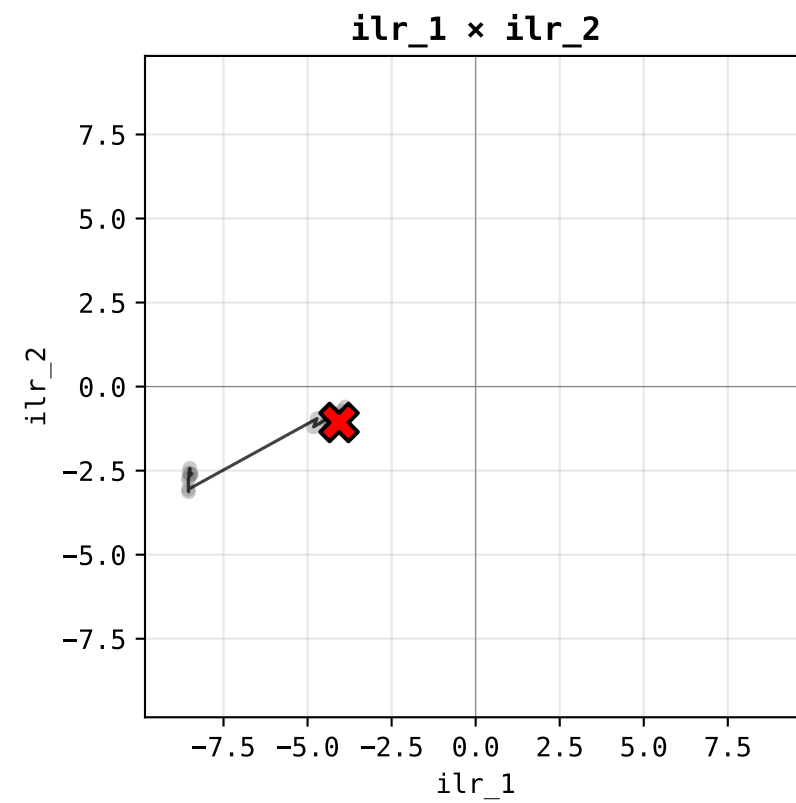
ILR Triplet – AUS Australia · year 2012 (13/26)



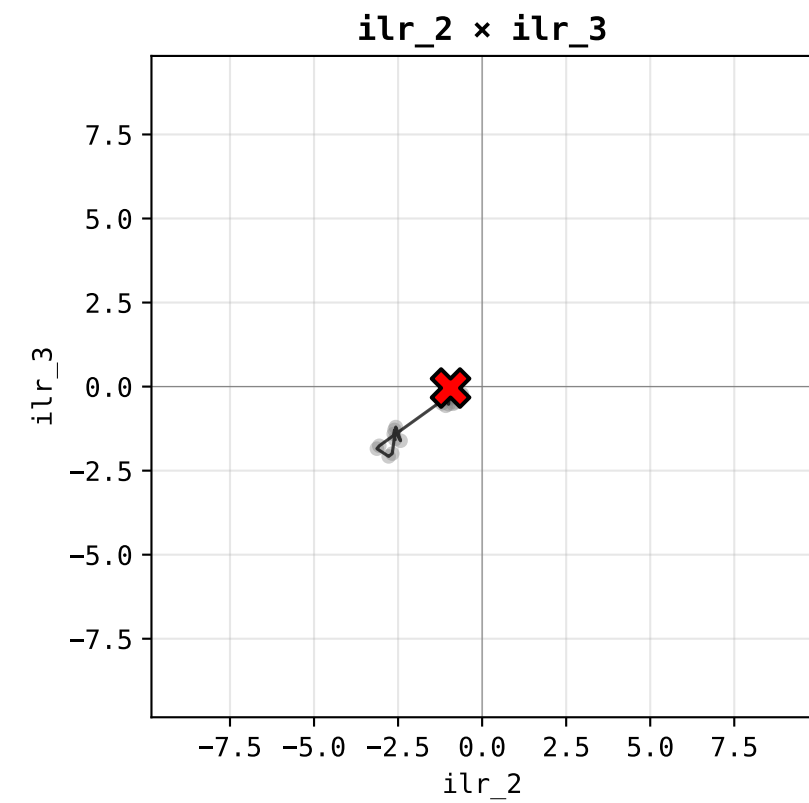
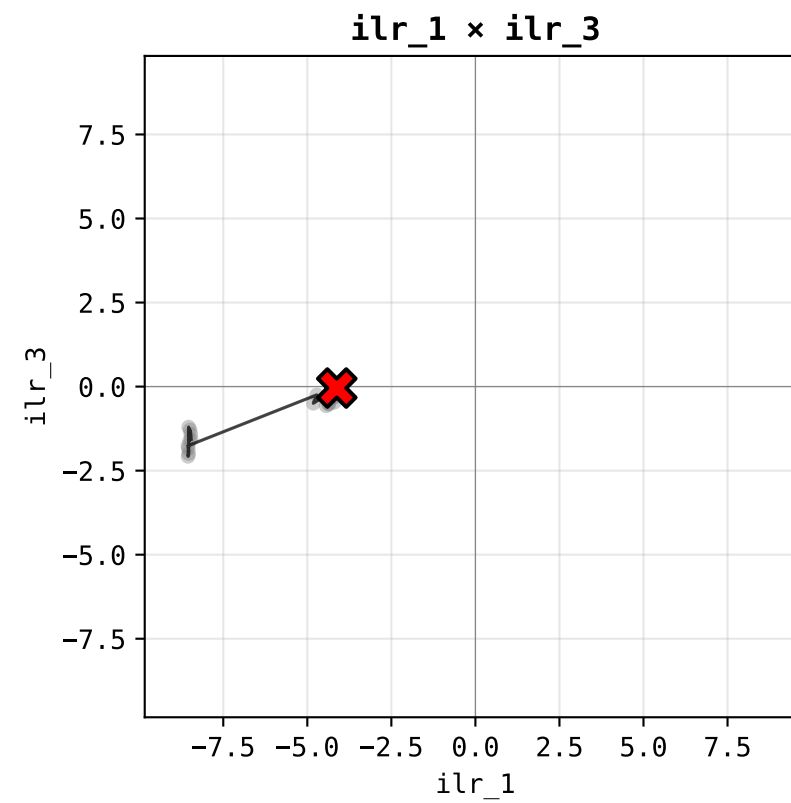
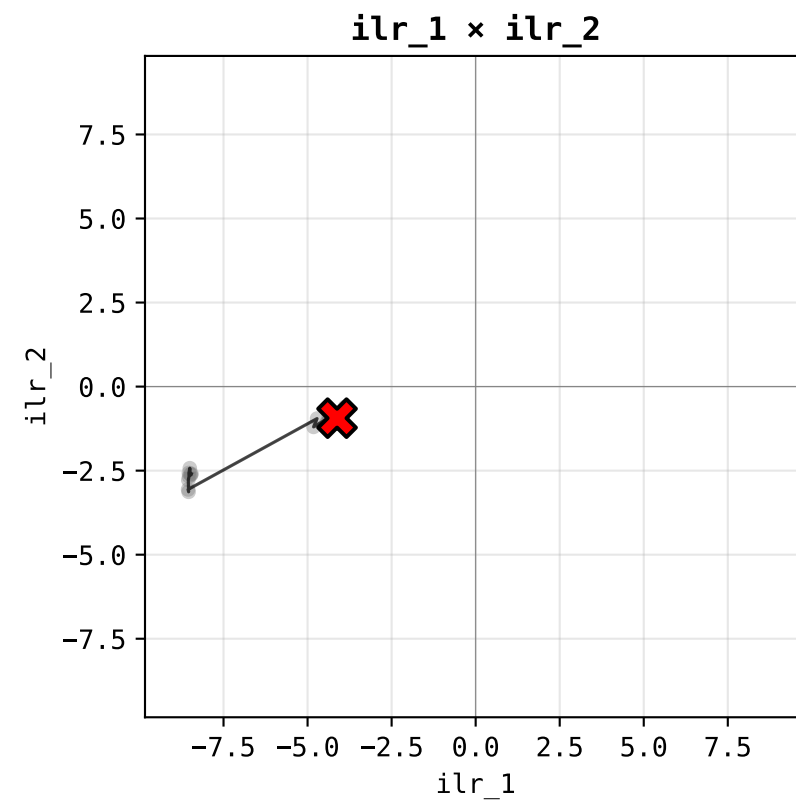
ILR Triplet – AUS Australia · year 2013 (14/26)



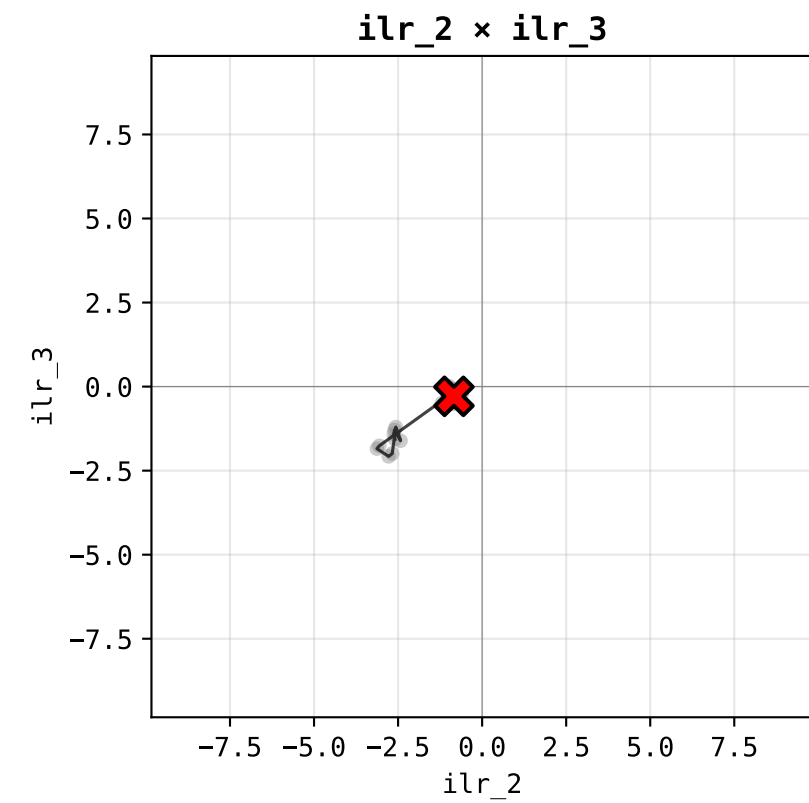
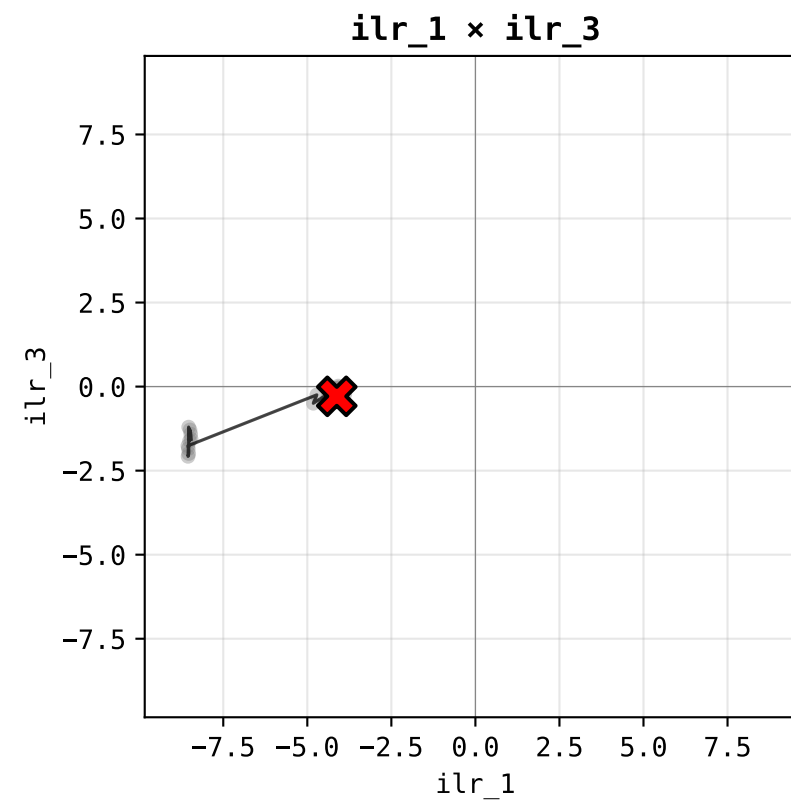
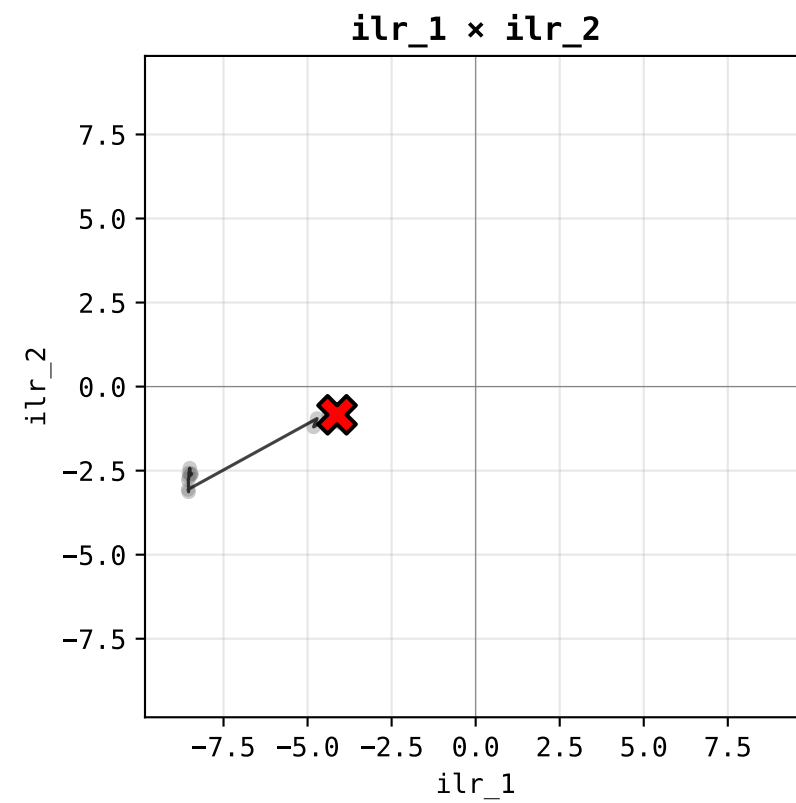
ILR Triplet – AUS Australia · year 2014 (15/26)

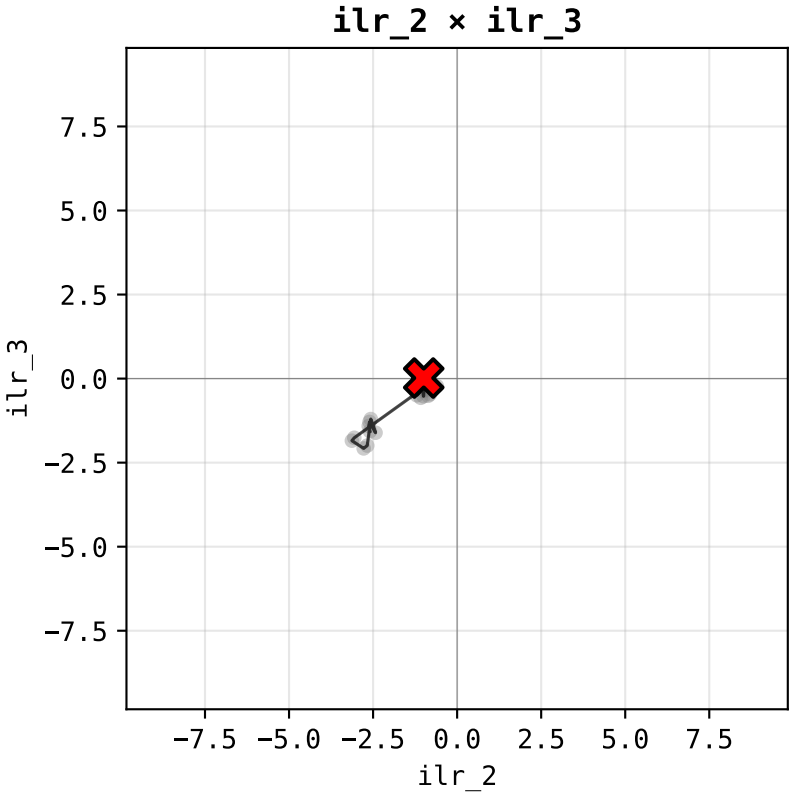
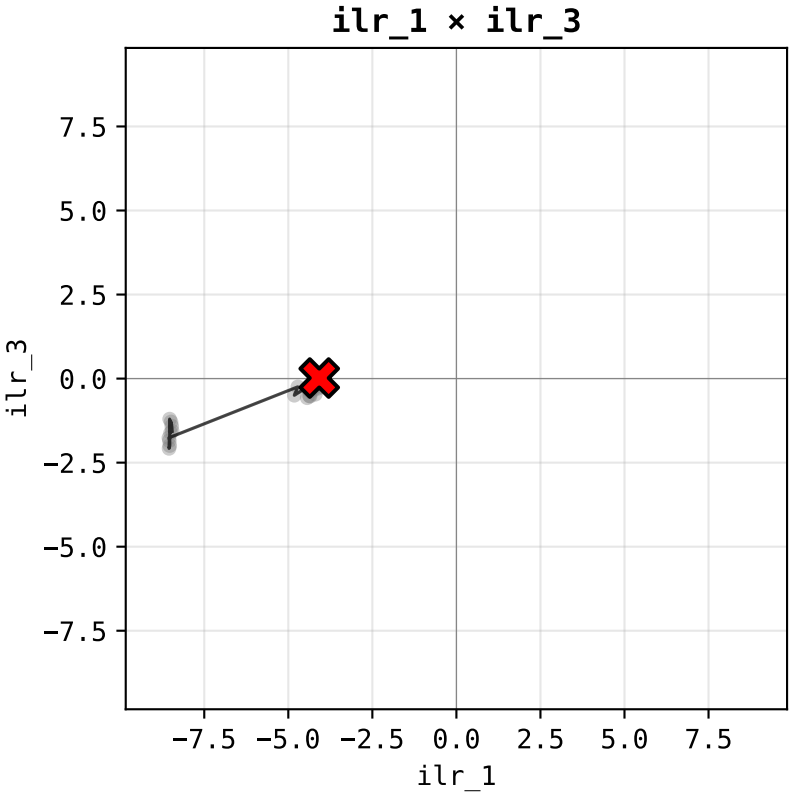
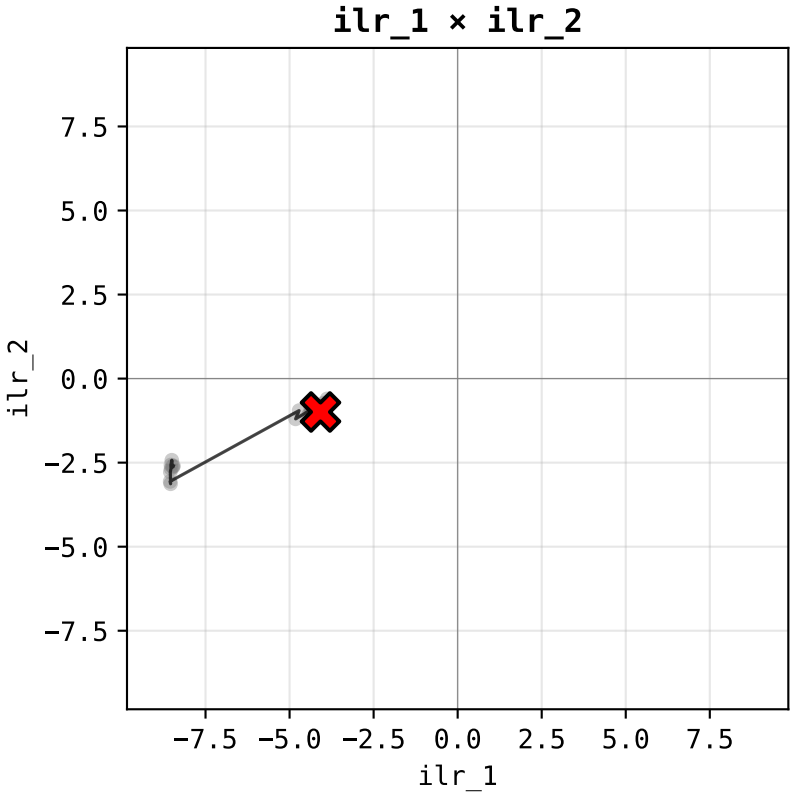


ILR Triplet – AUS Australia · year 2015 (16/26)

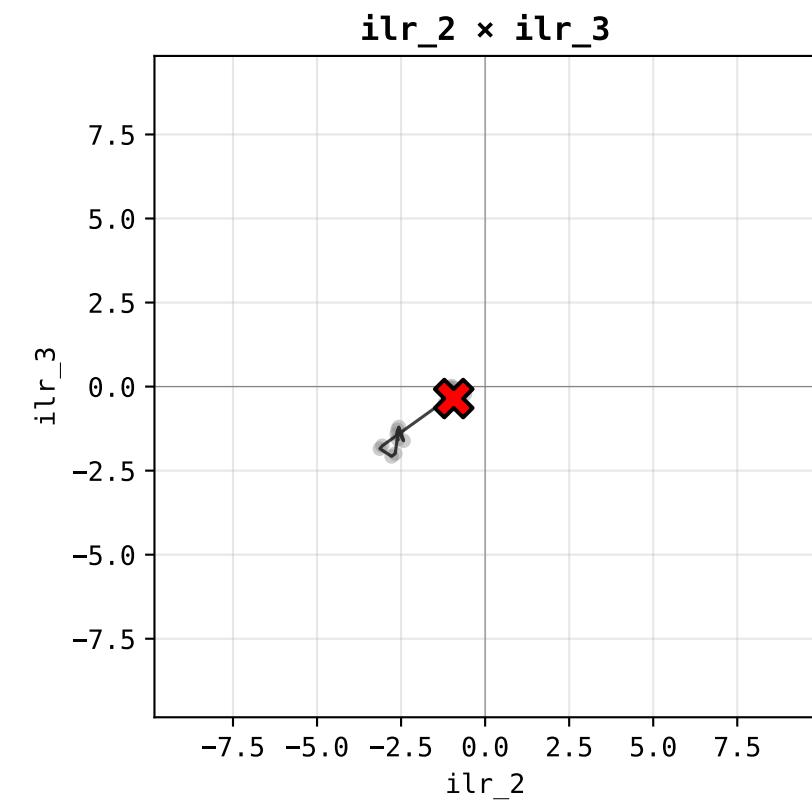
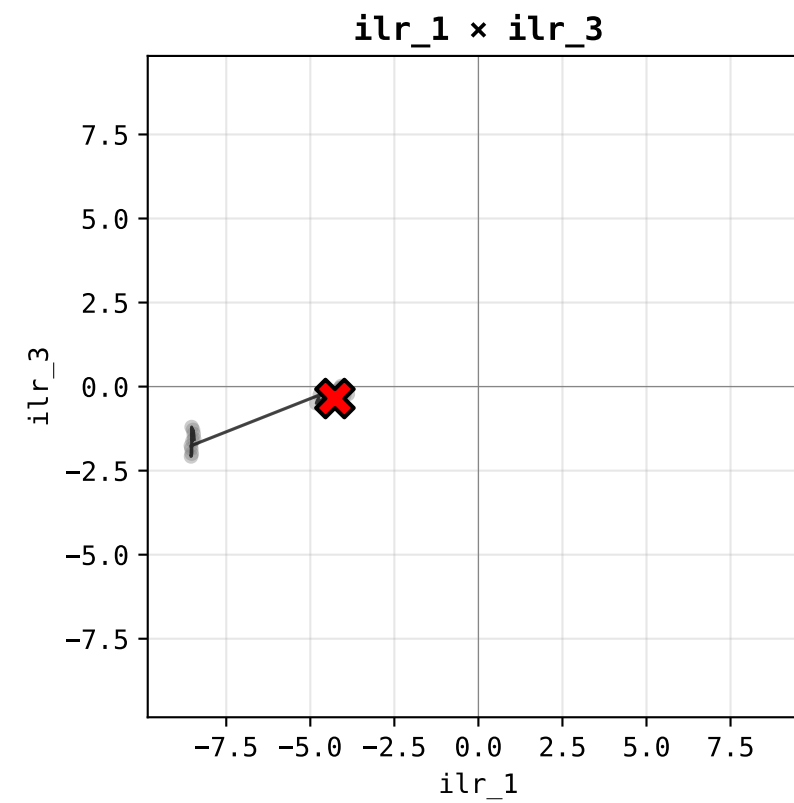
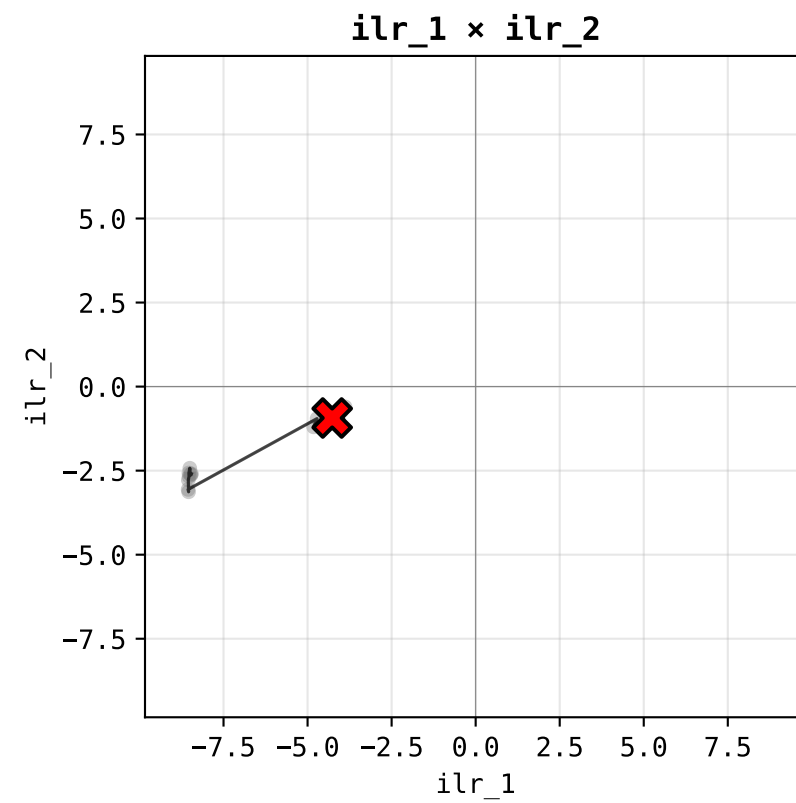


ILR Triplet – AUS Australia · year 2016 (17/26)



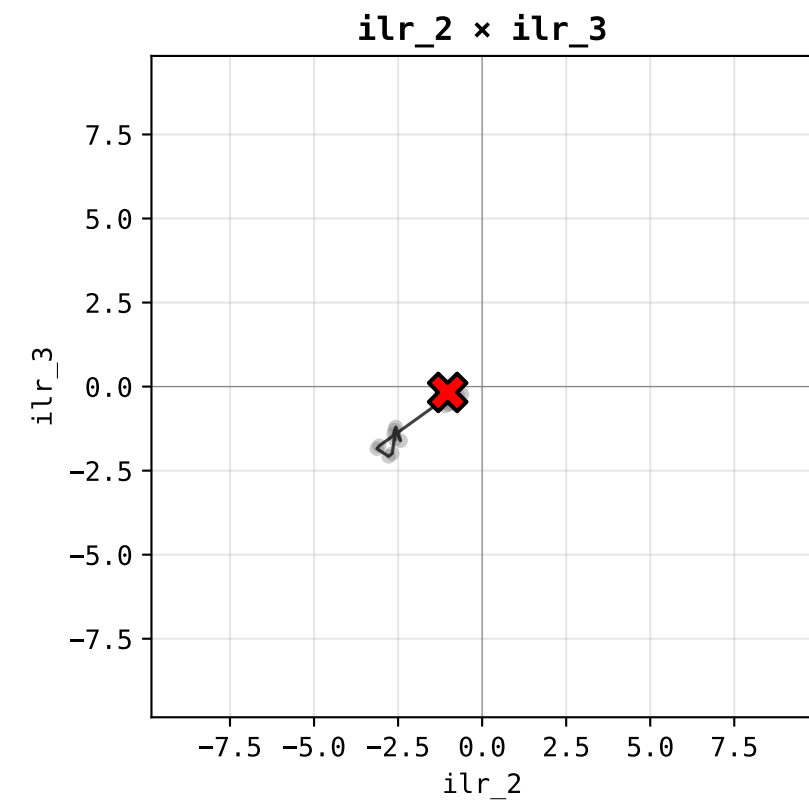
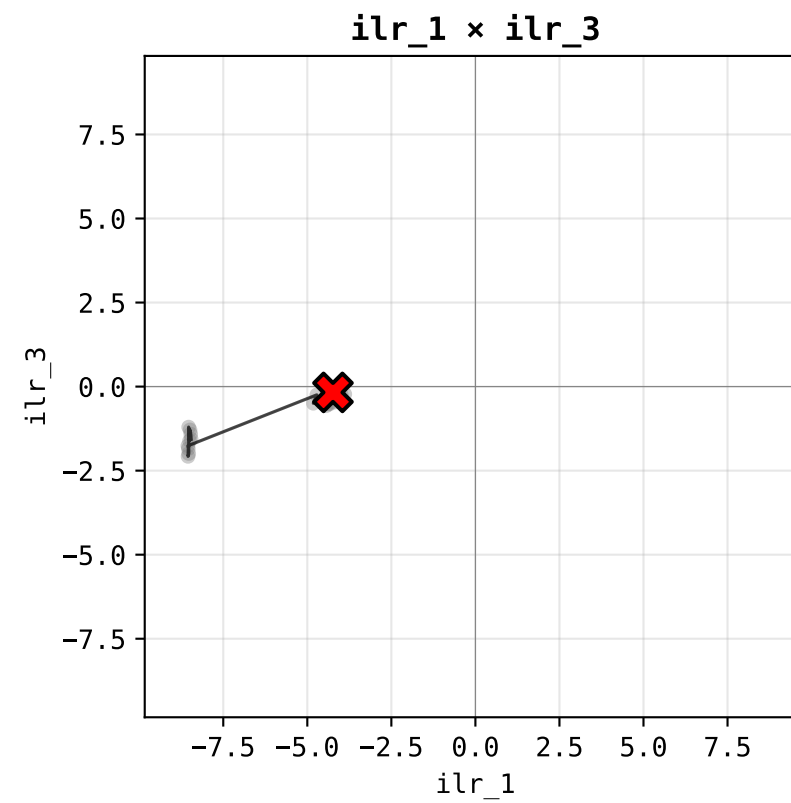
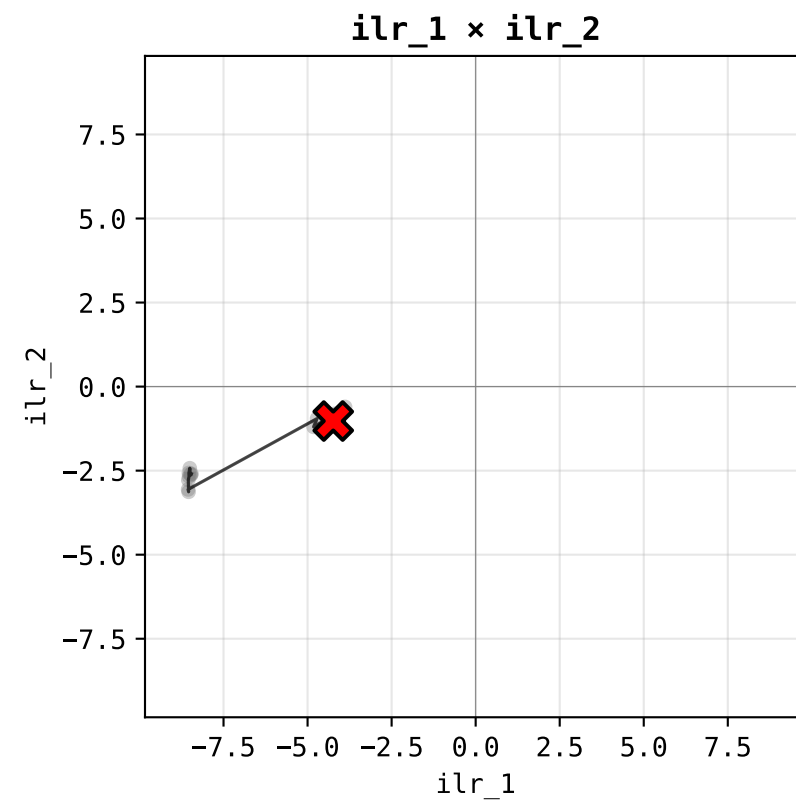


ILR Triplet – AUS Australia · year 2018 (19/26)

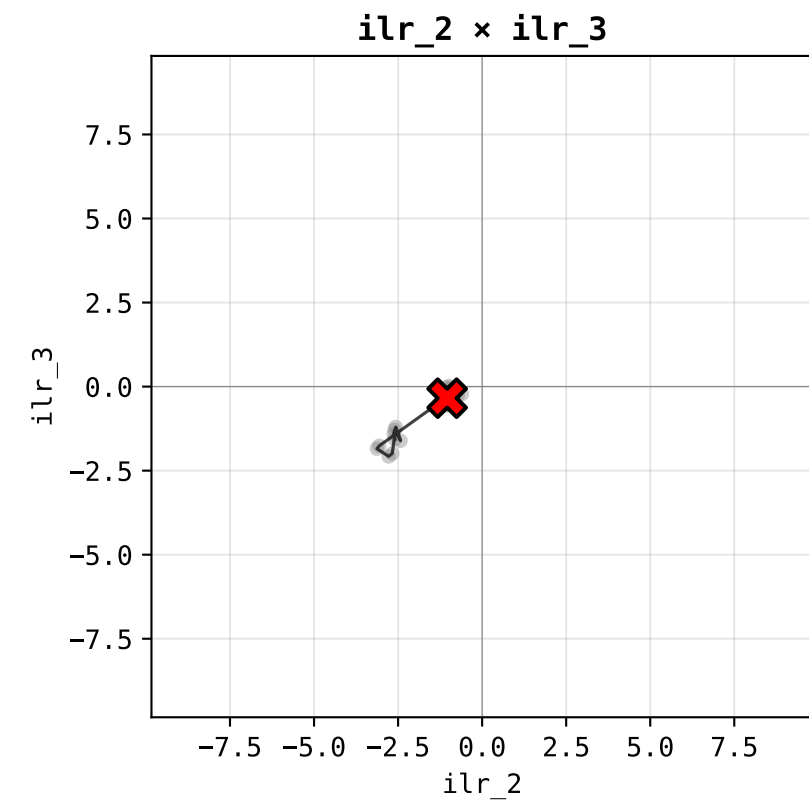
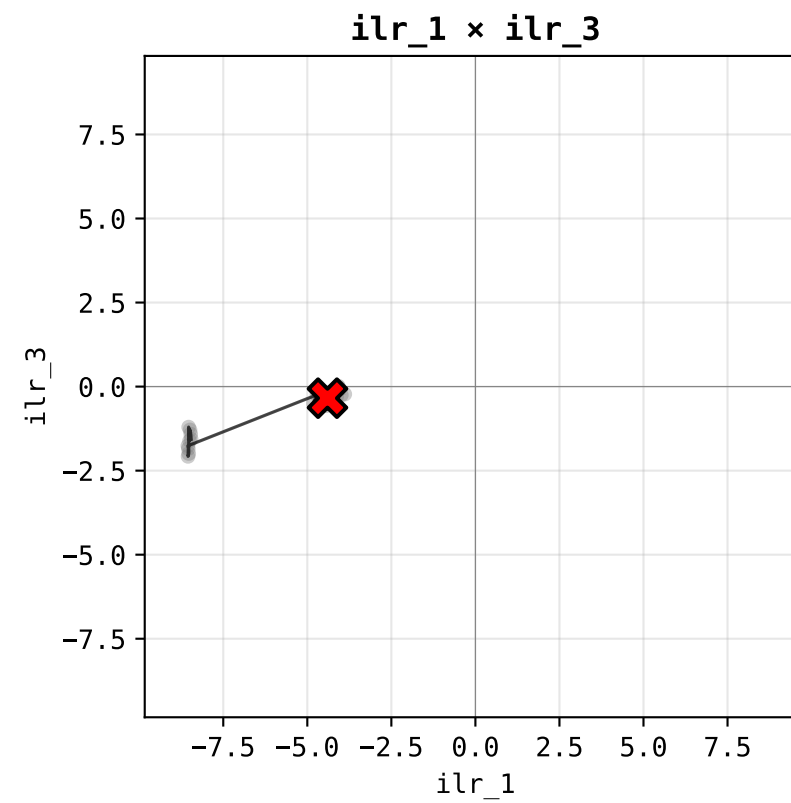
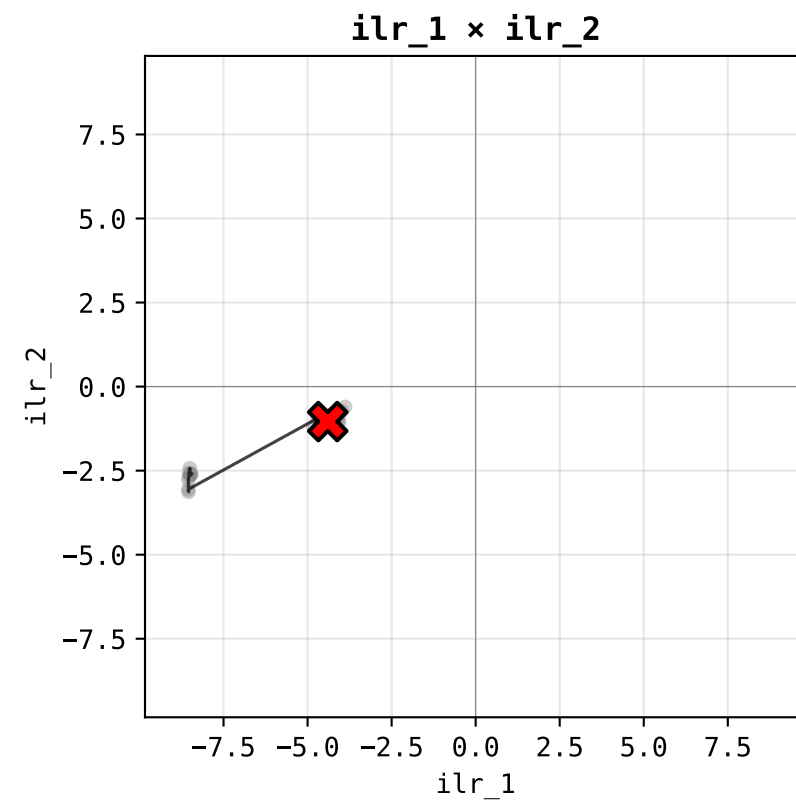


X = current year (2018) · black = past trajectory · gray = future · ilr range ±9.84

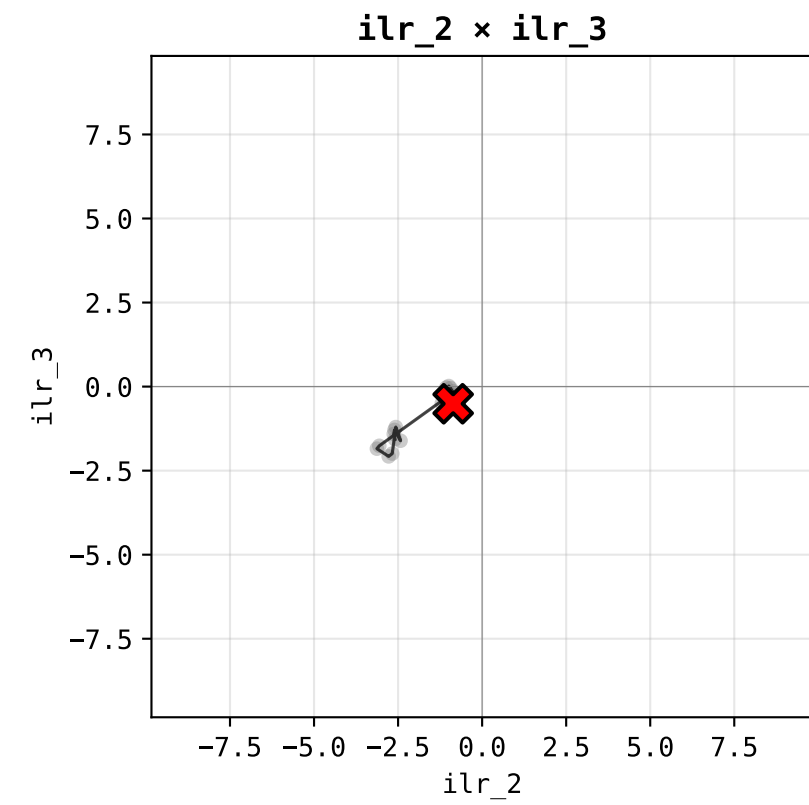
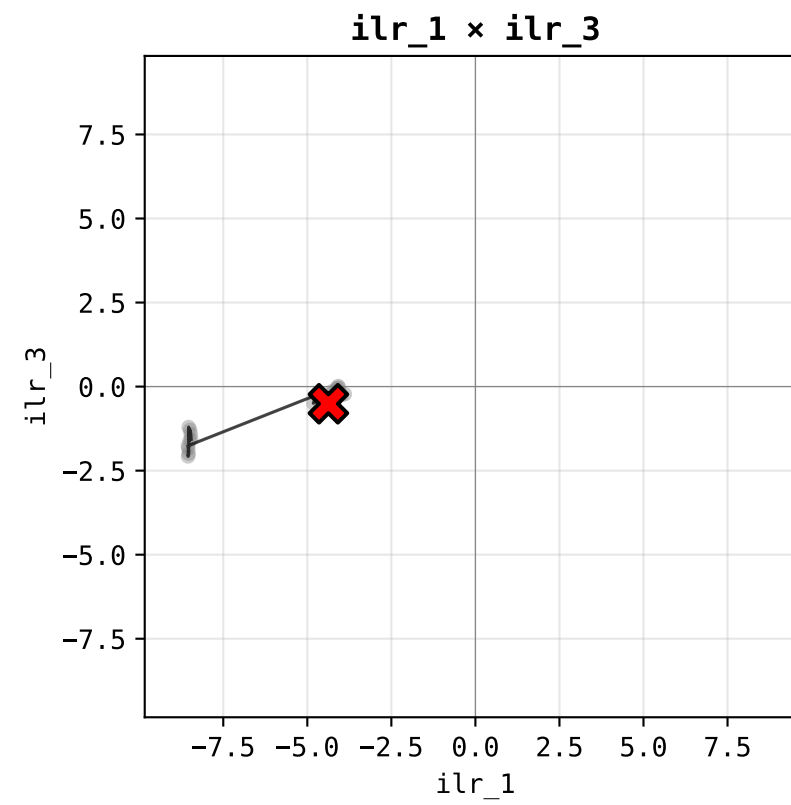
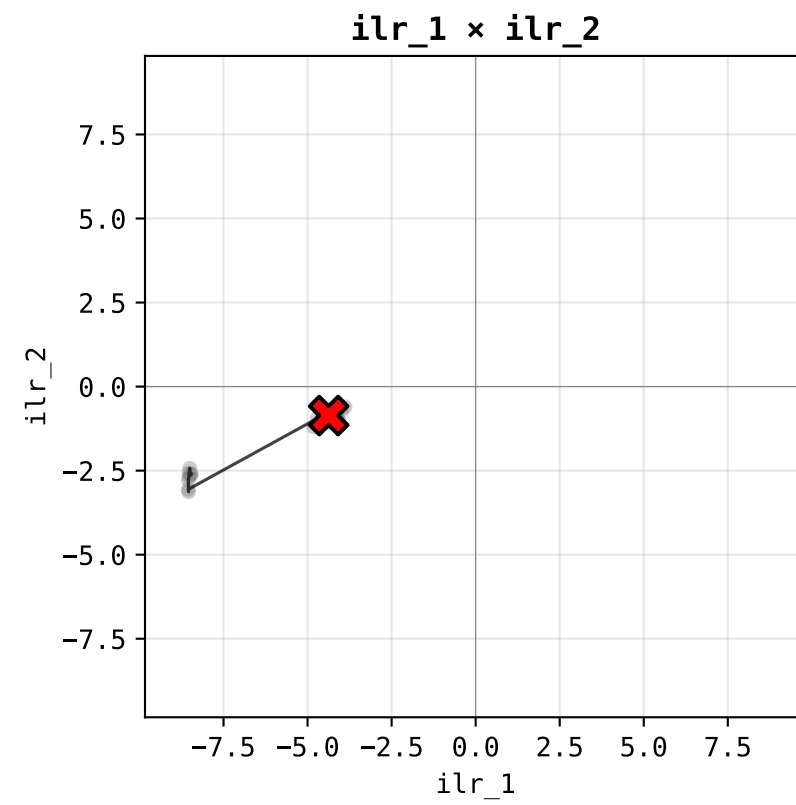
ILR Triplet – AUS Australia · year 2019 (20/26)



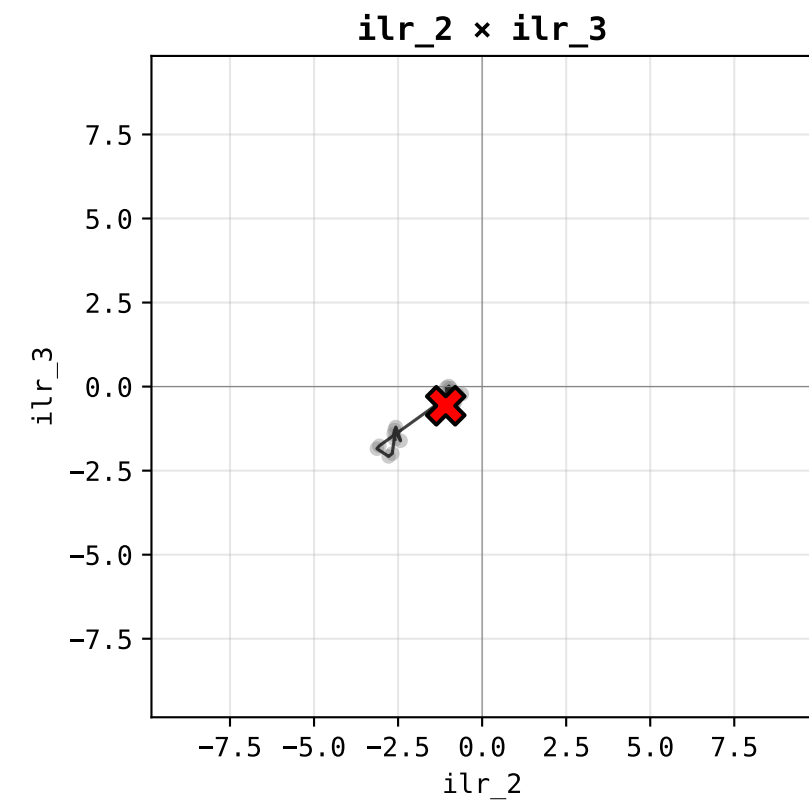
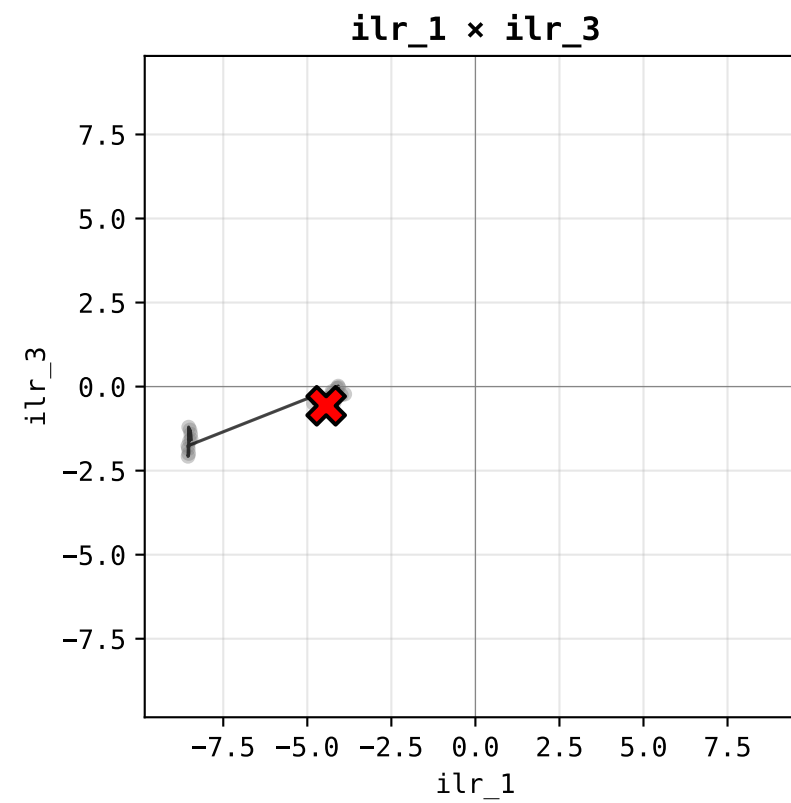
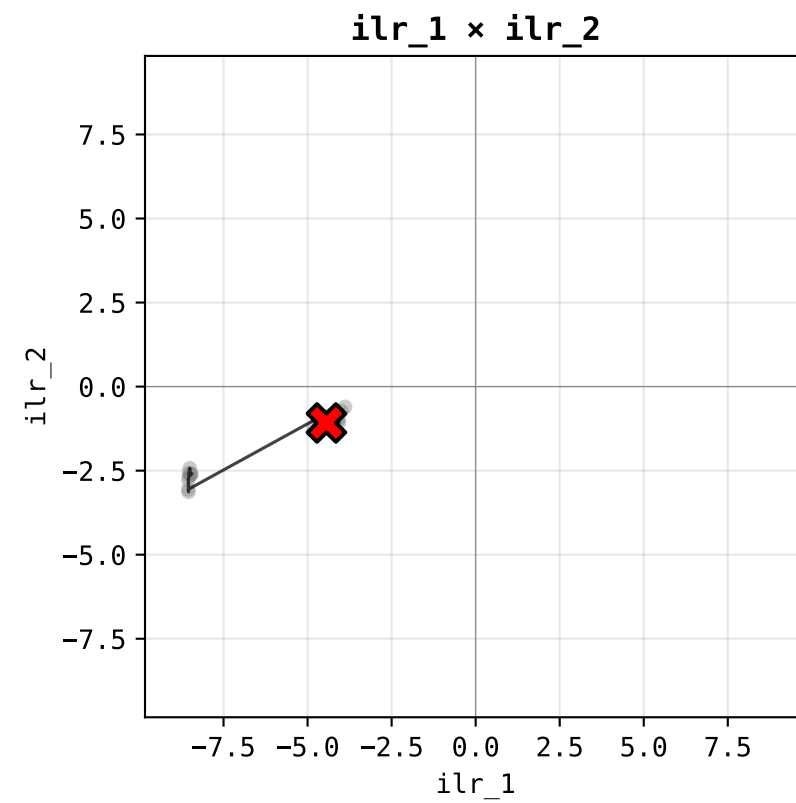
ILR Triplet – AUS Australia · year 2020 (21/26)



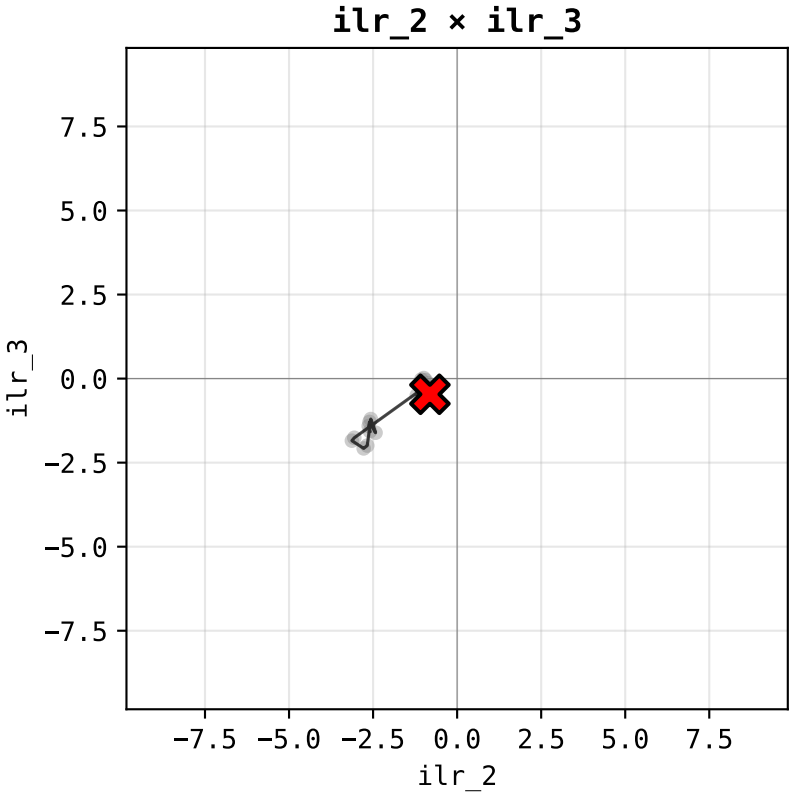
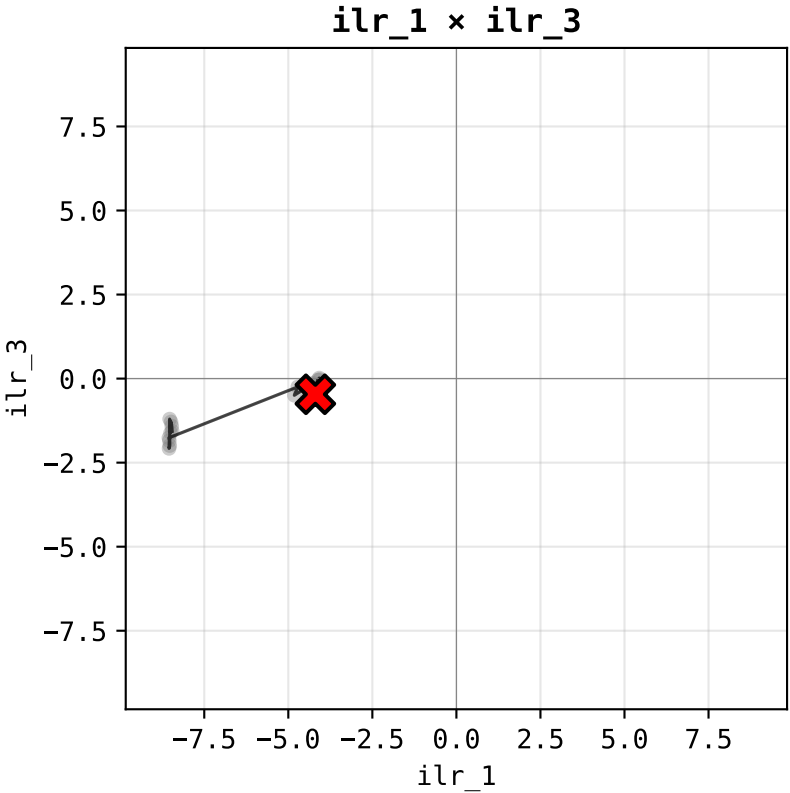
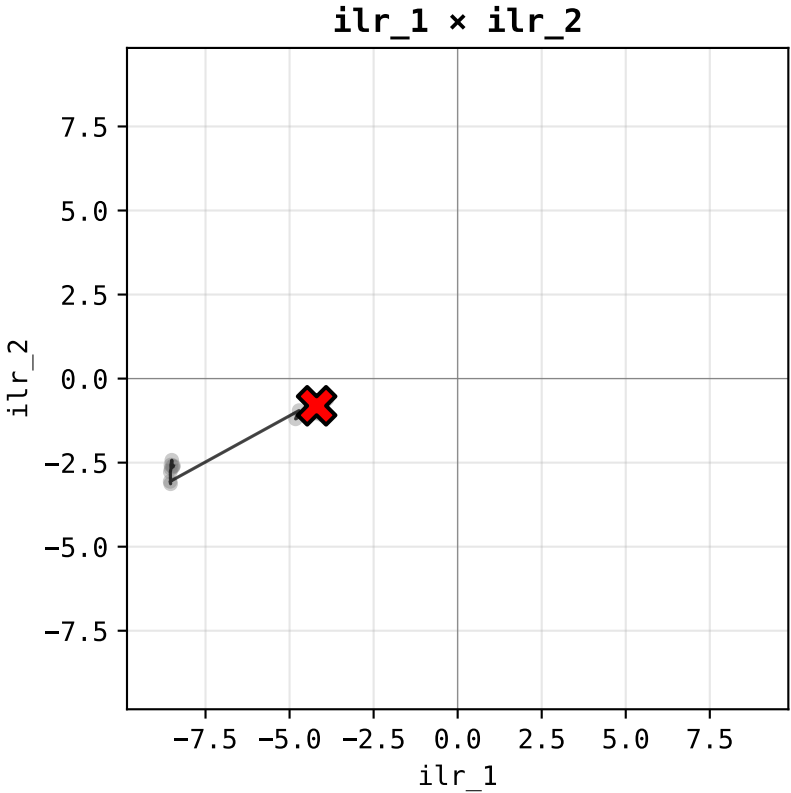
ILR Triplet – AUS Australia · year 2021 (22/26)



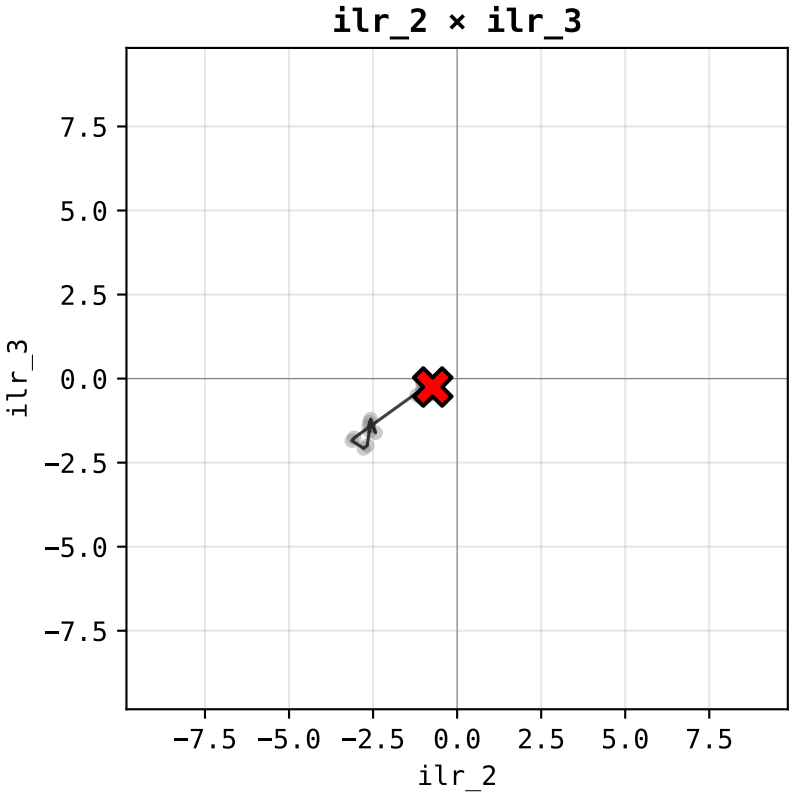
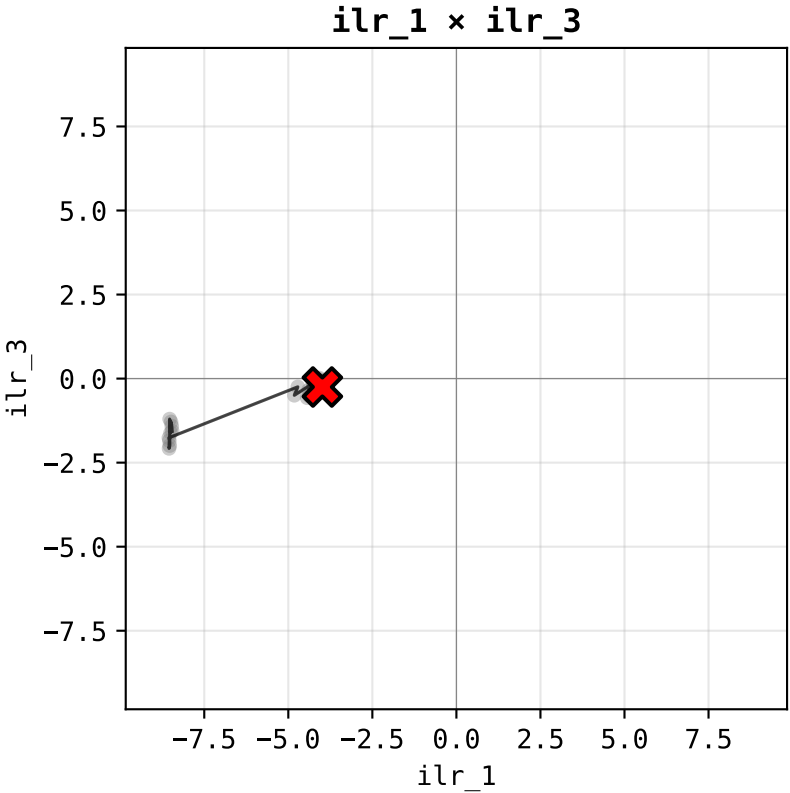
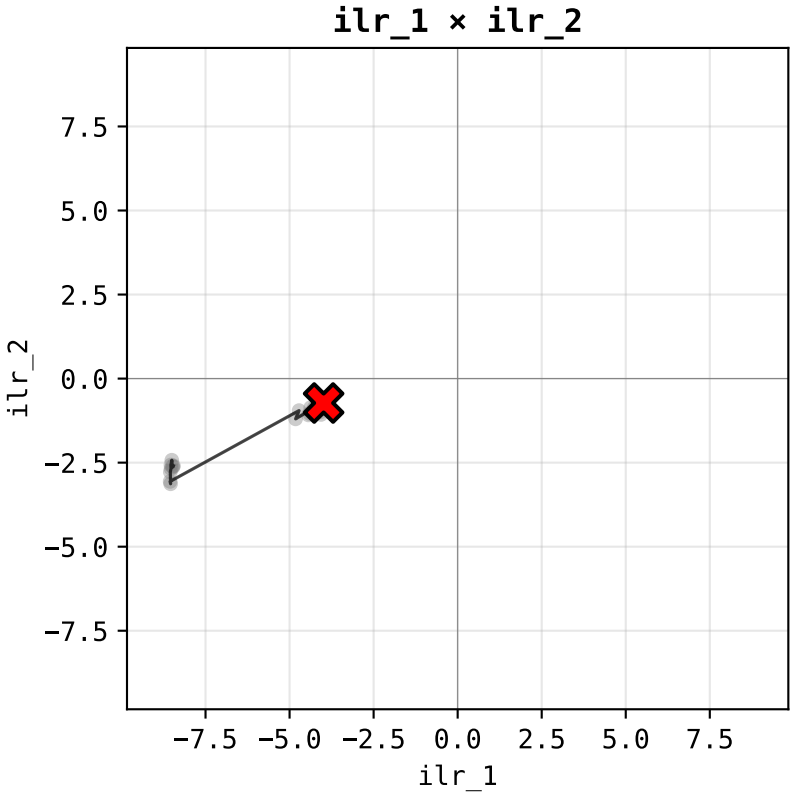
ILR Triplet – AUS Australia · year 2022 (23/26)

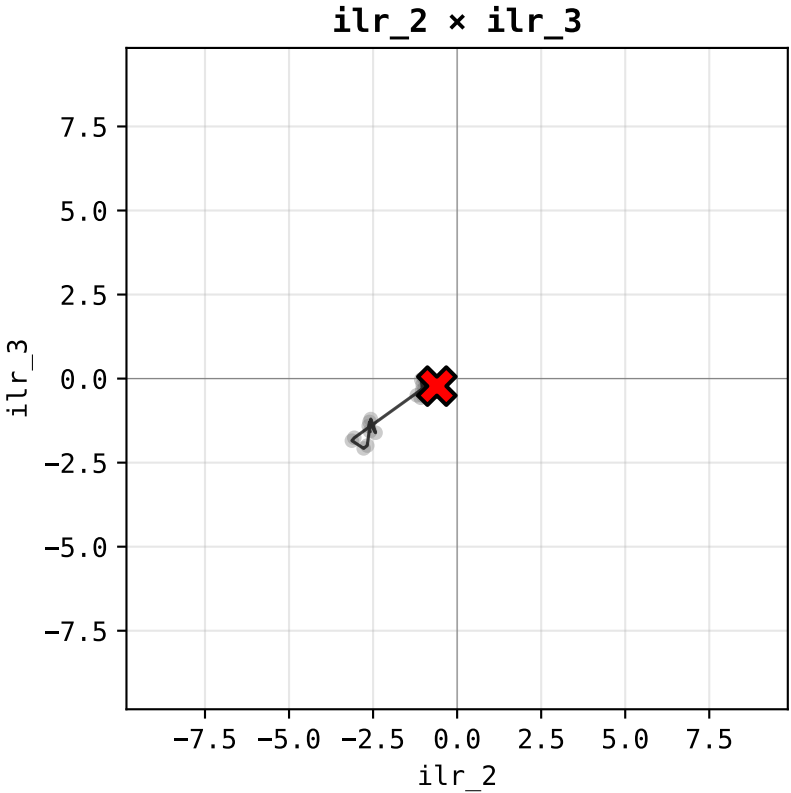
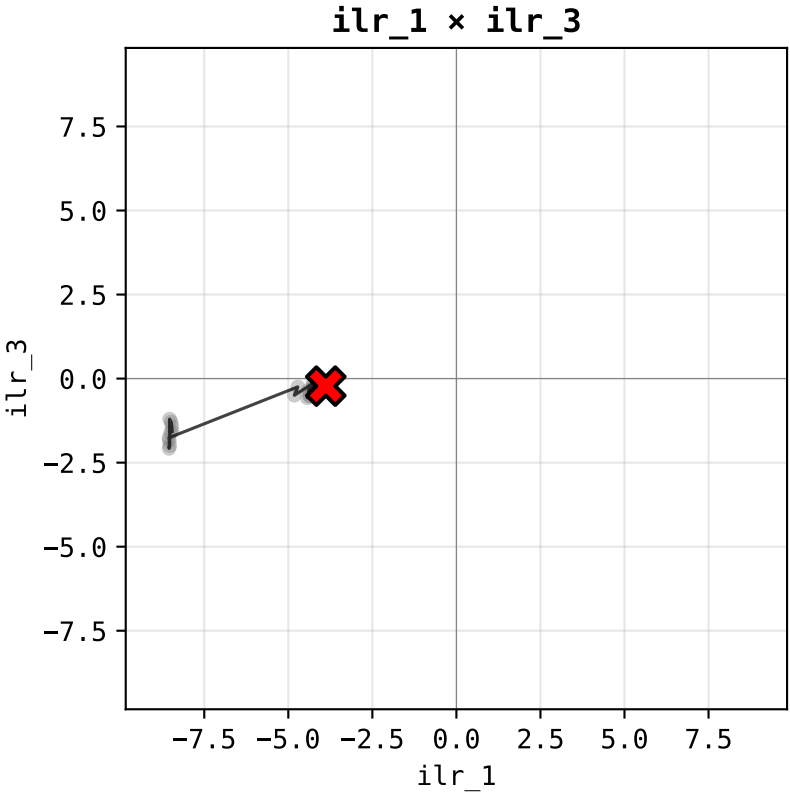
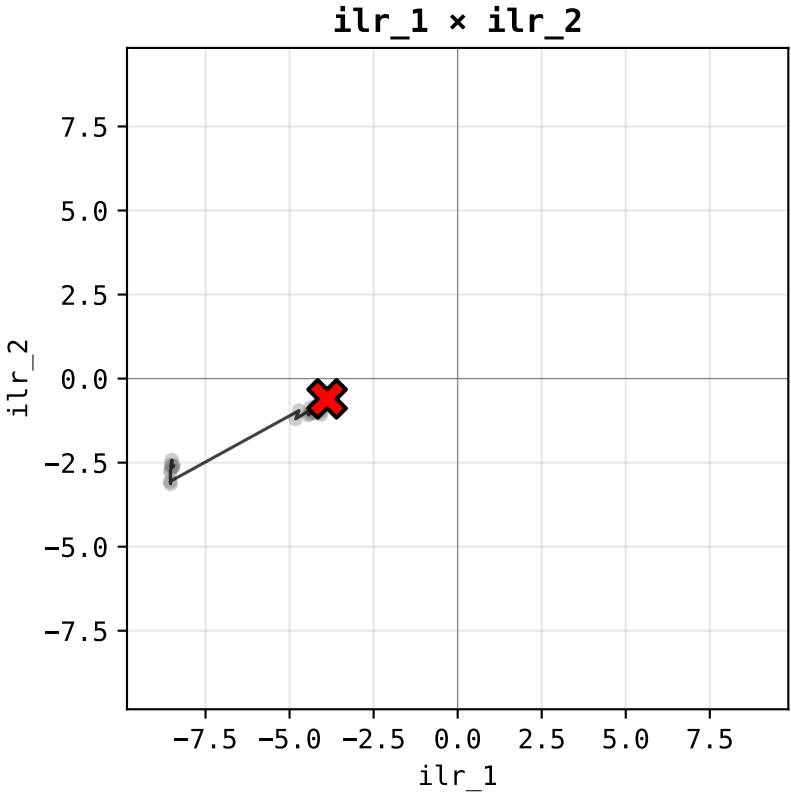


ILR Triplet – AUS Australia · year 2023 (24/26)



ILR Triplet – AUS Australia · year 2024 (25/26)





View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ CHN China

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

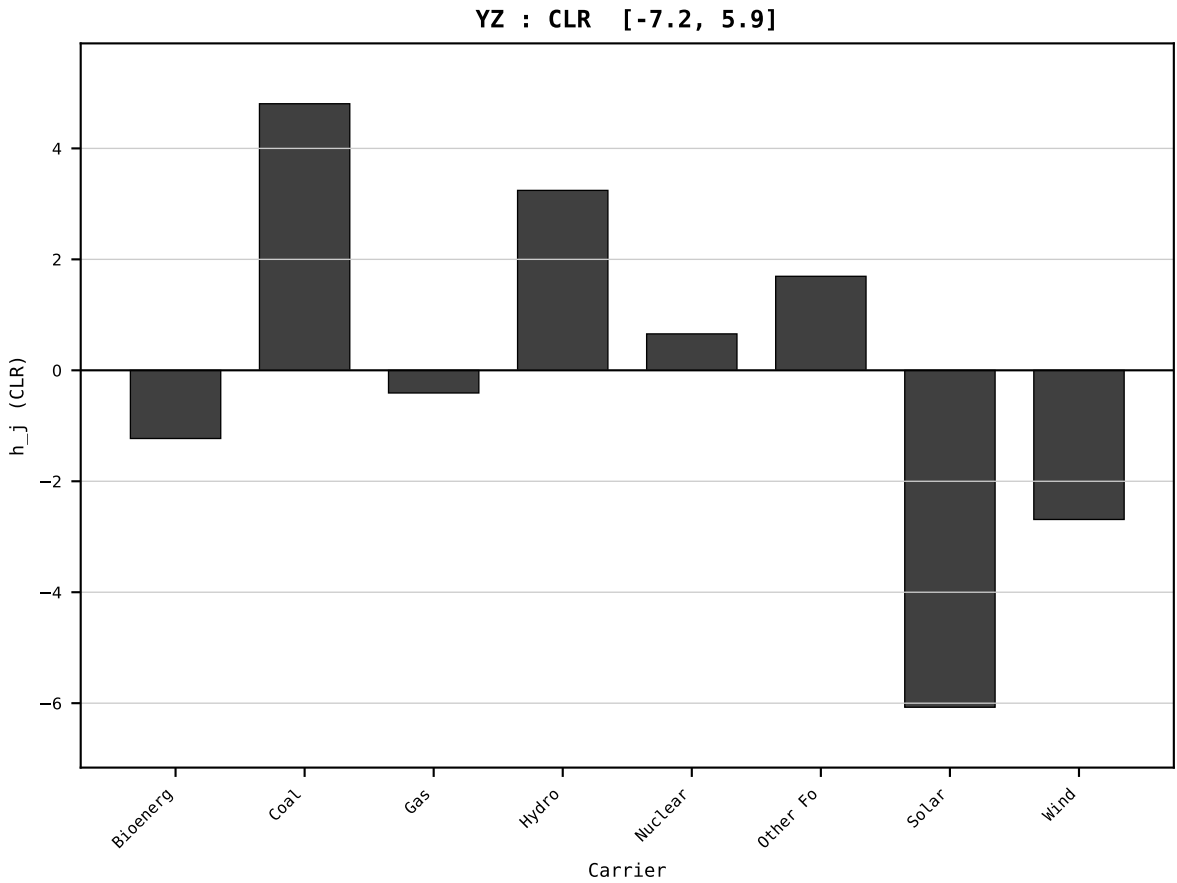
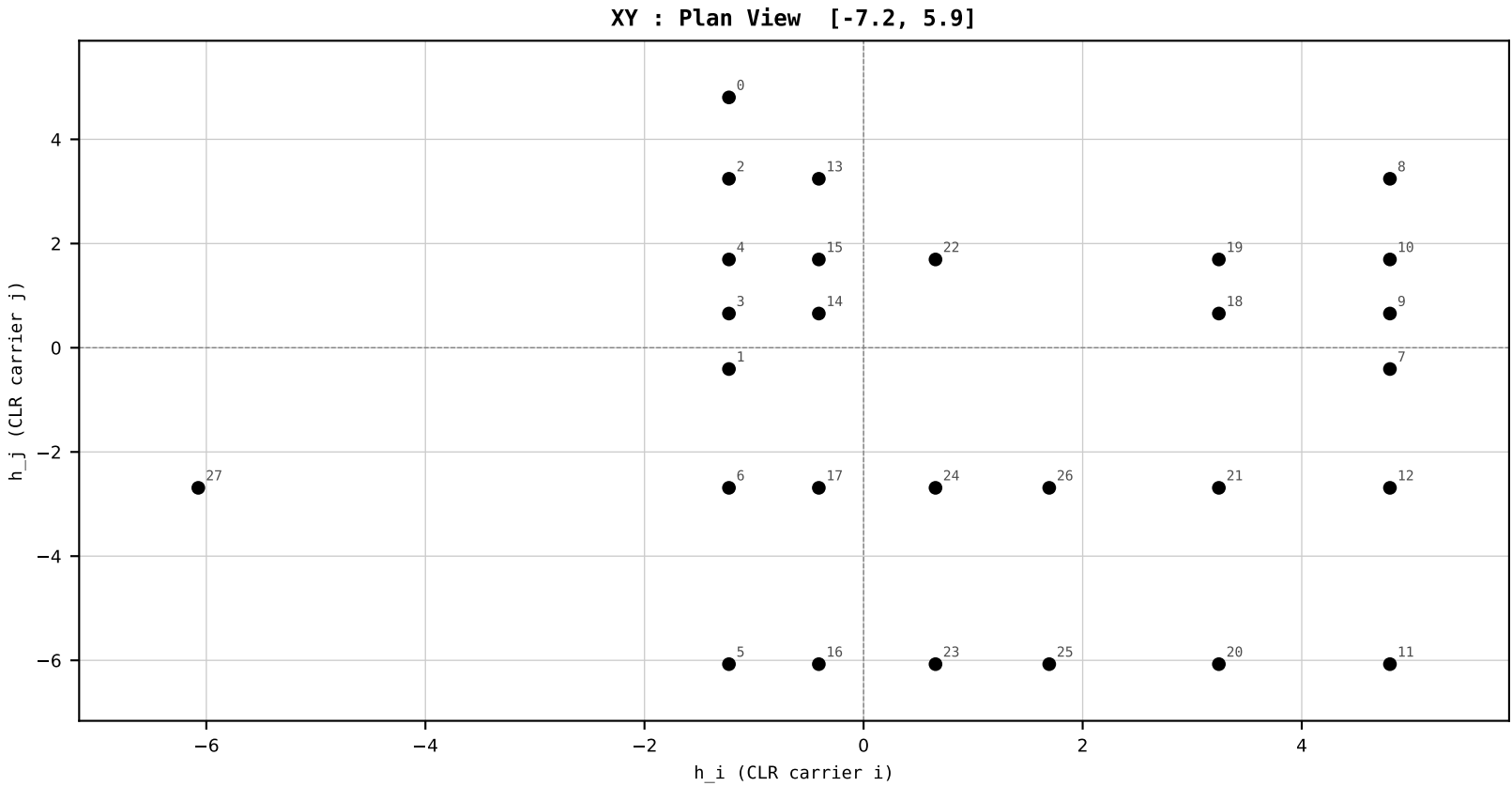
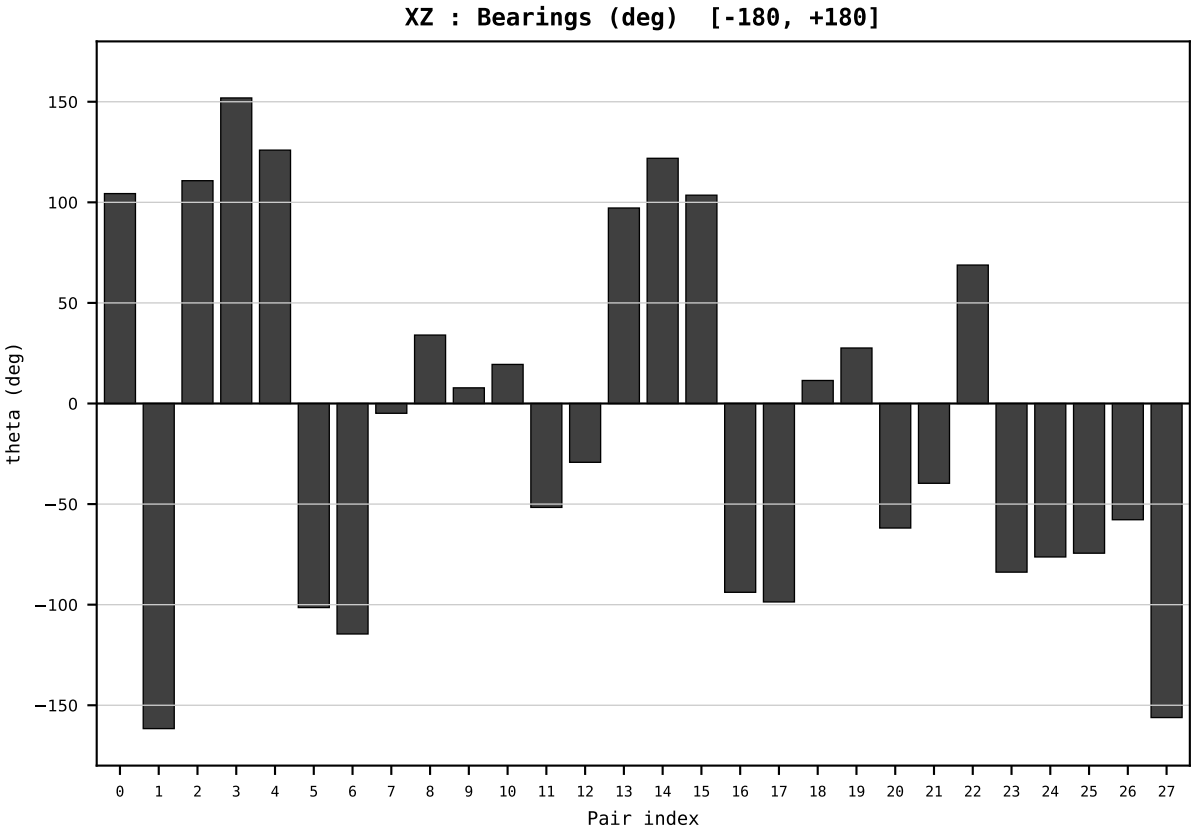
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.664059
Ring = Hs-4
E_metric= 82.7022
kappa_HS= 53013.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.878 HLR
DR ratio= 53013.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

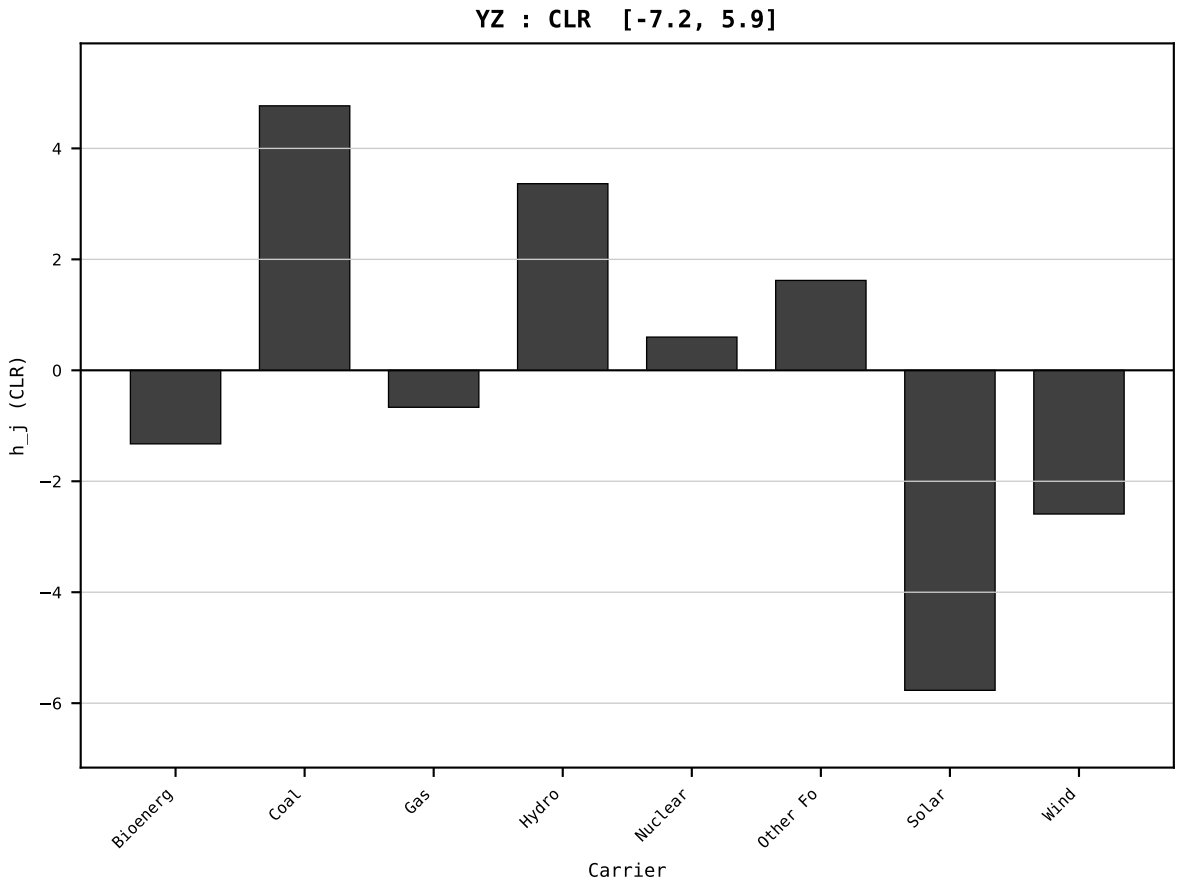
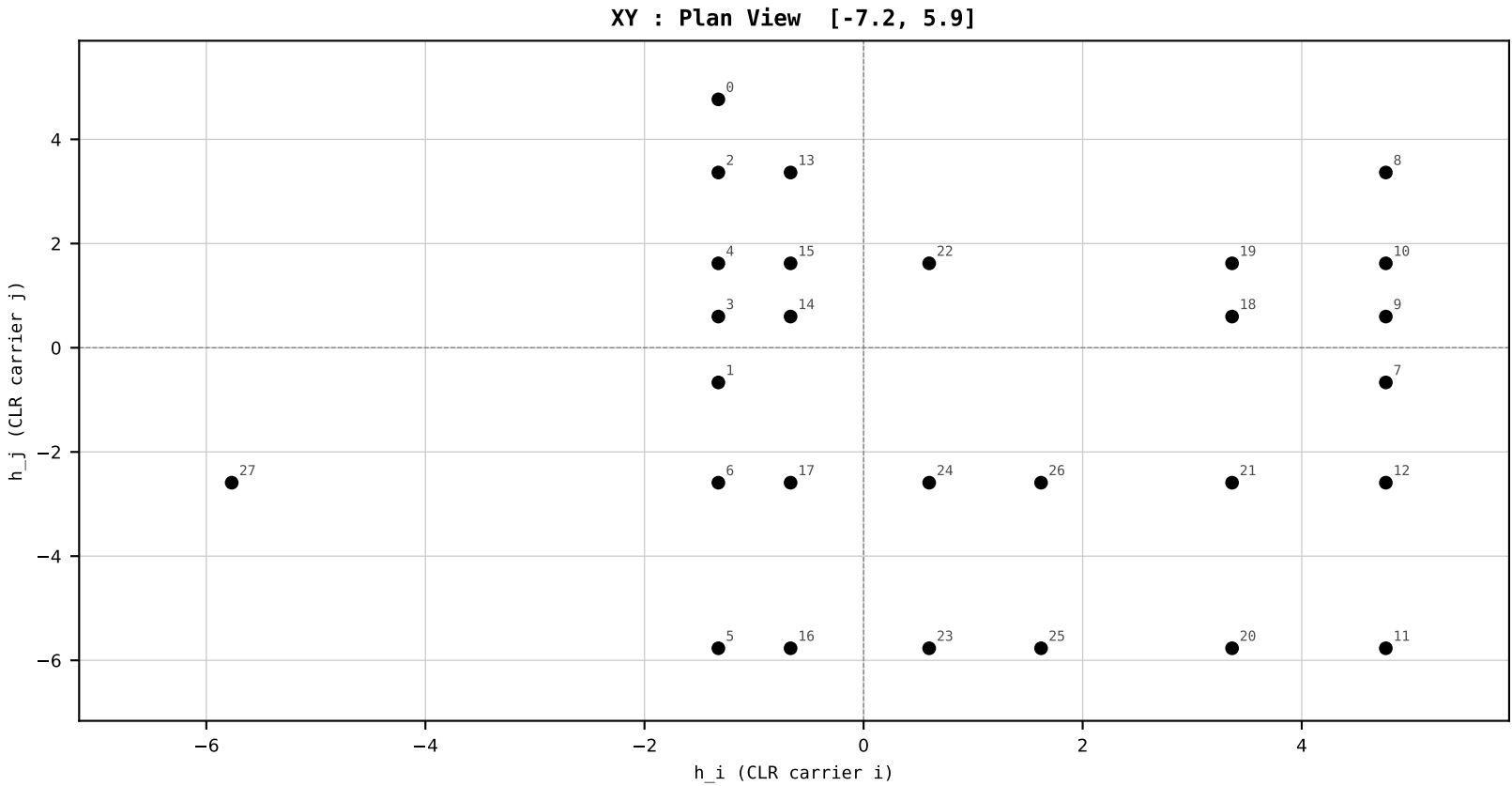
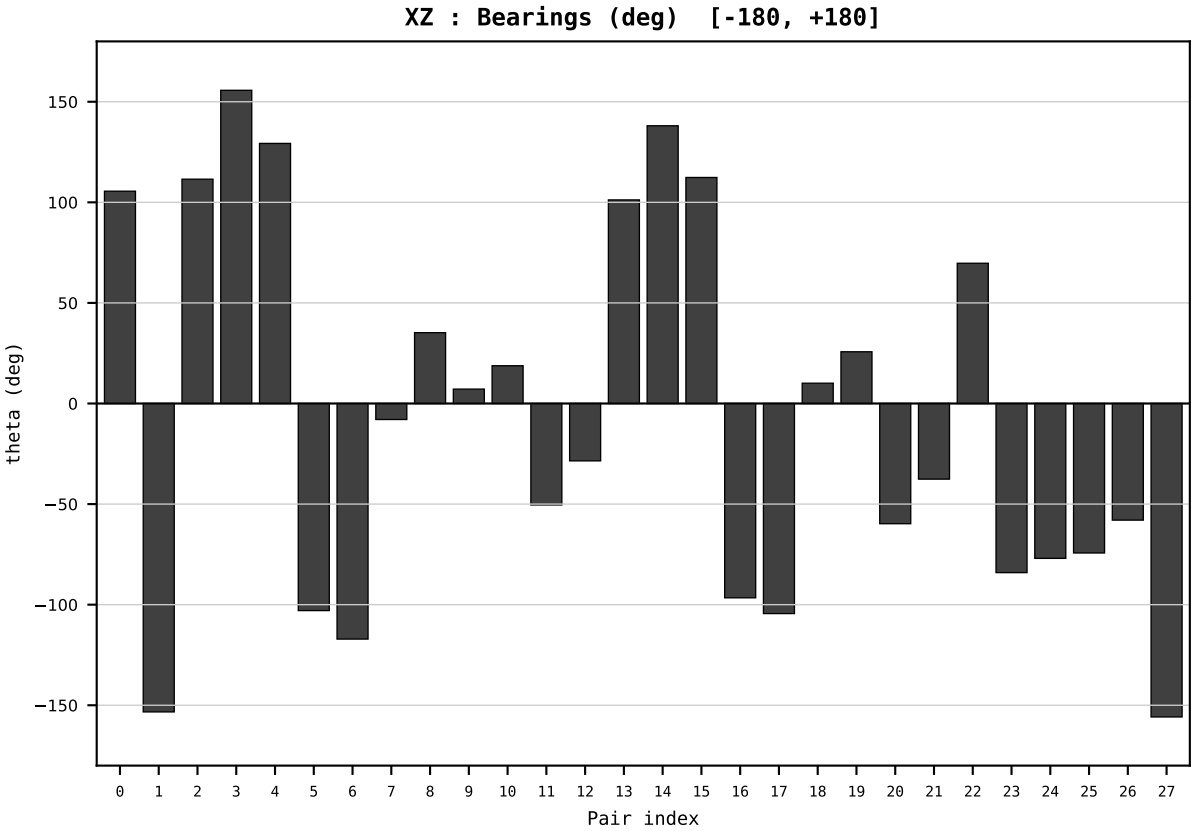
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.654342
Ring = Hs-4
E_metric= 79.2118
kappa_HS= 37638.00
omega = 2.5929 deg
d_A = 0.450949
Helm = Solar
Helm d = +0.304970
DR = 10.536 HLR
DR ratio= 37638.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

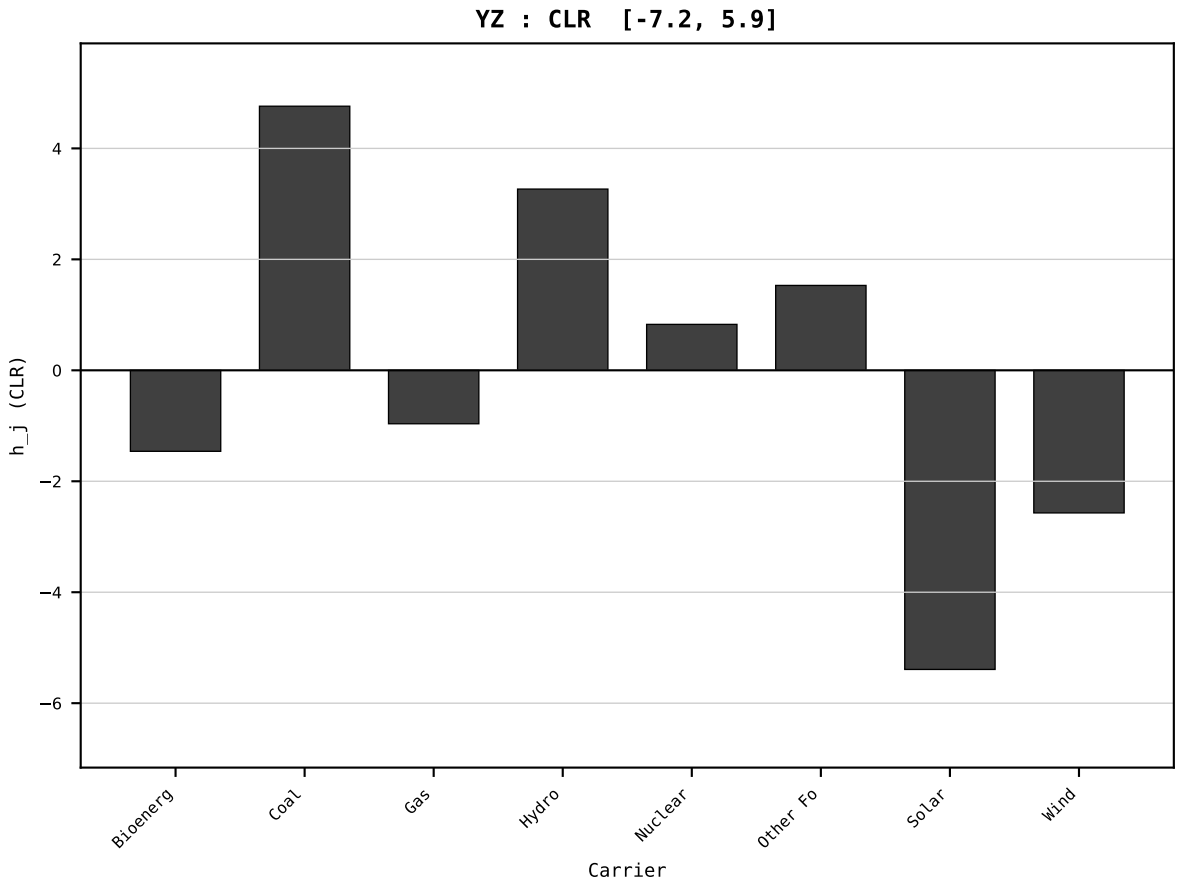
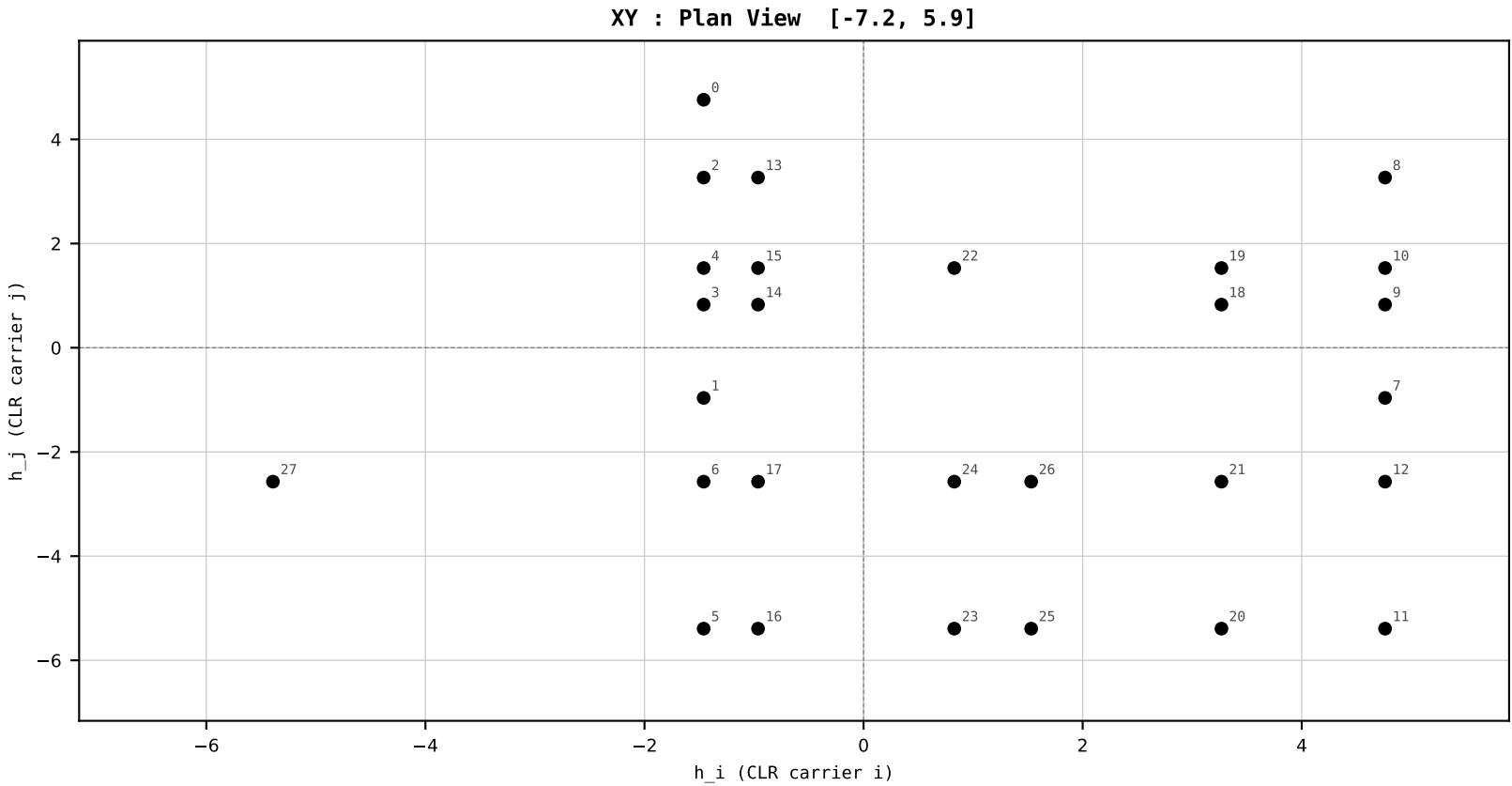
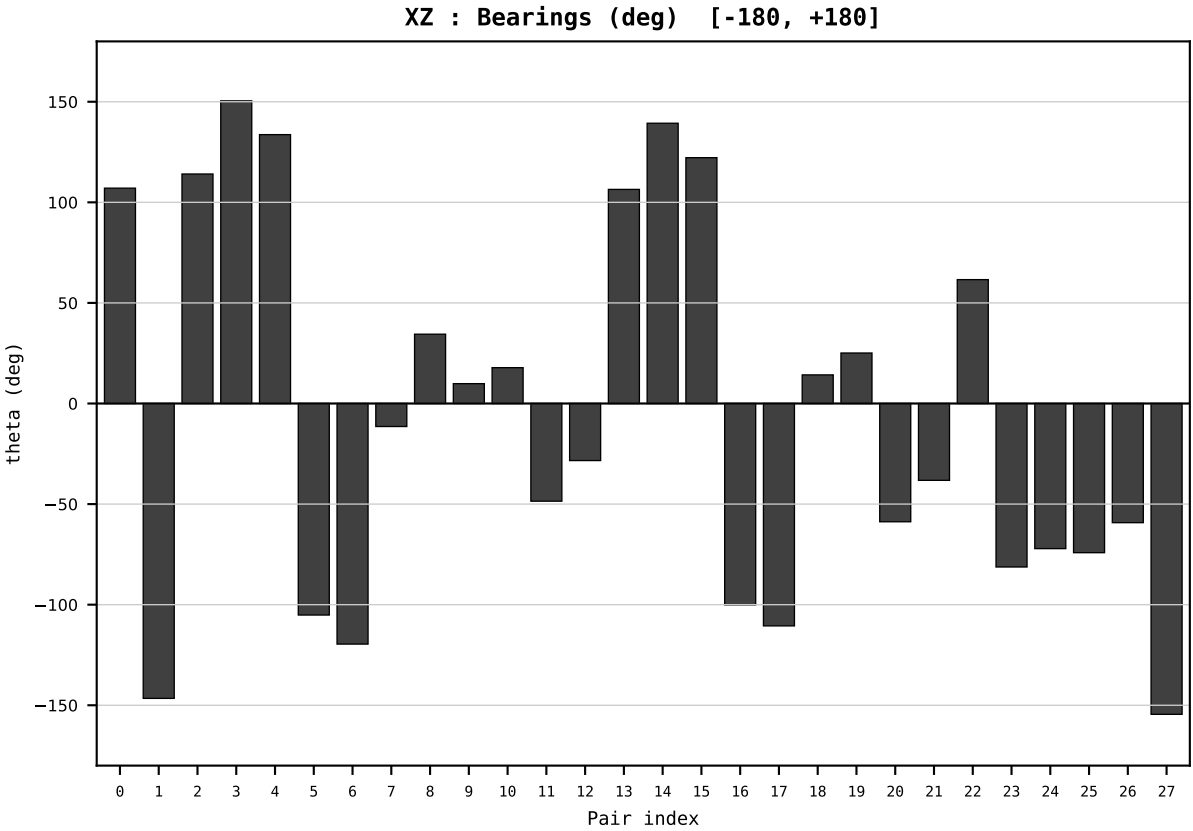
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.662736
Ring = Hs-4
E_metric= 75.0996
kappa_HS= 25651.40
omega = 3.3508 deg
d_A = 0.564384
Helm = Solar
Helm d = +0.376481
DR = 10.152 HLR
DR ratio= 25651.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

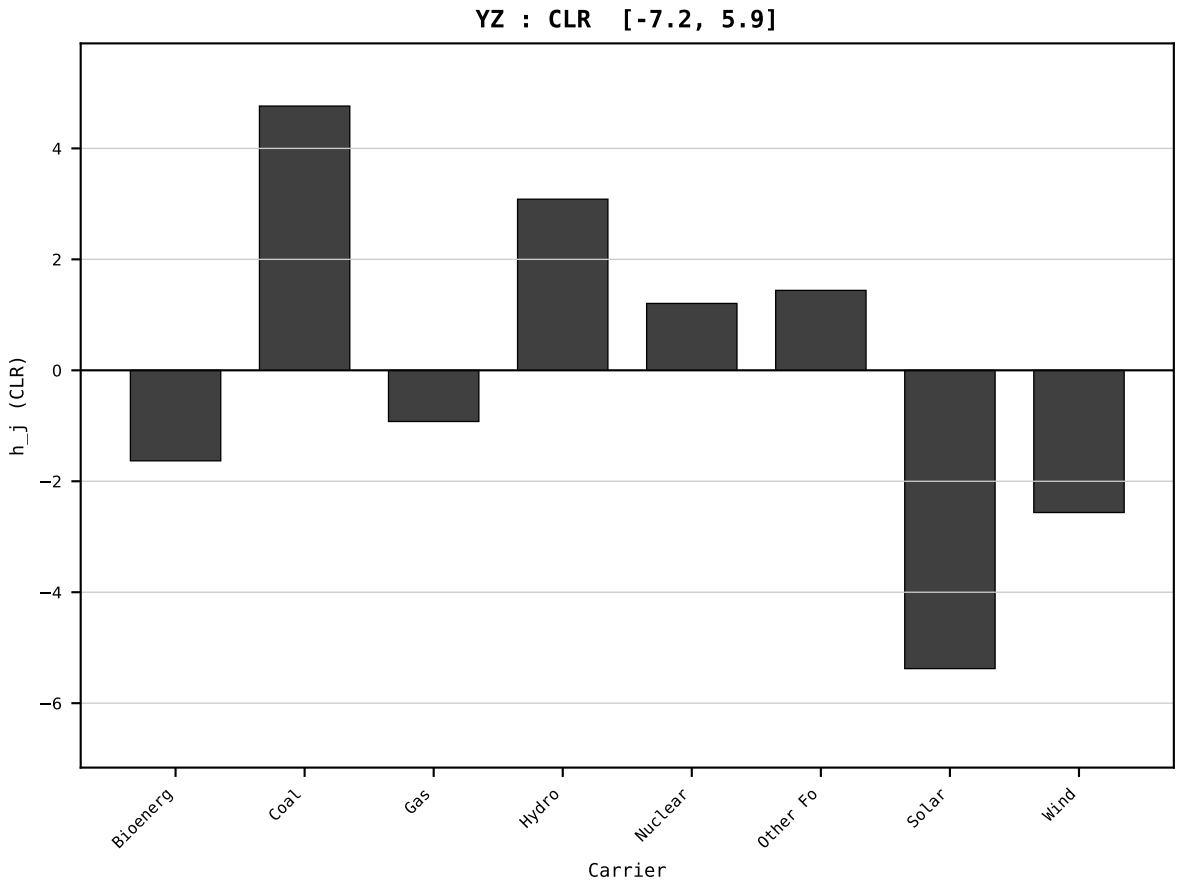
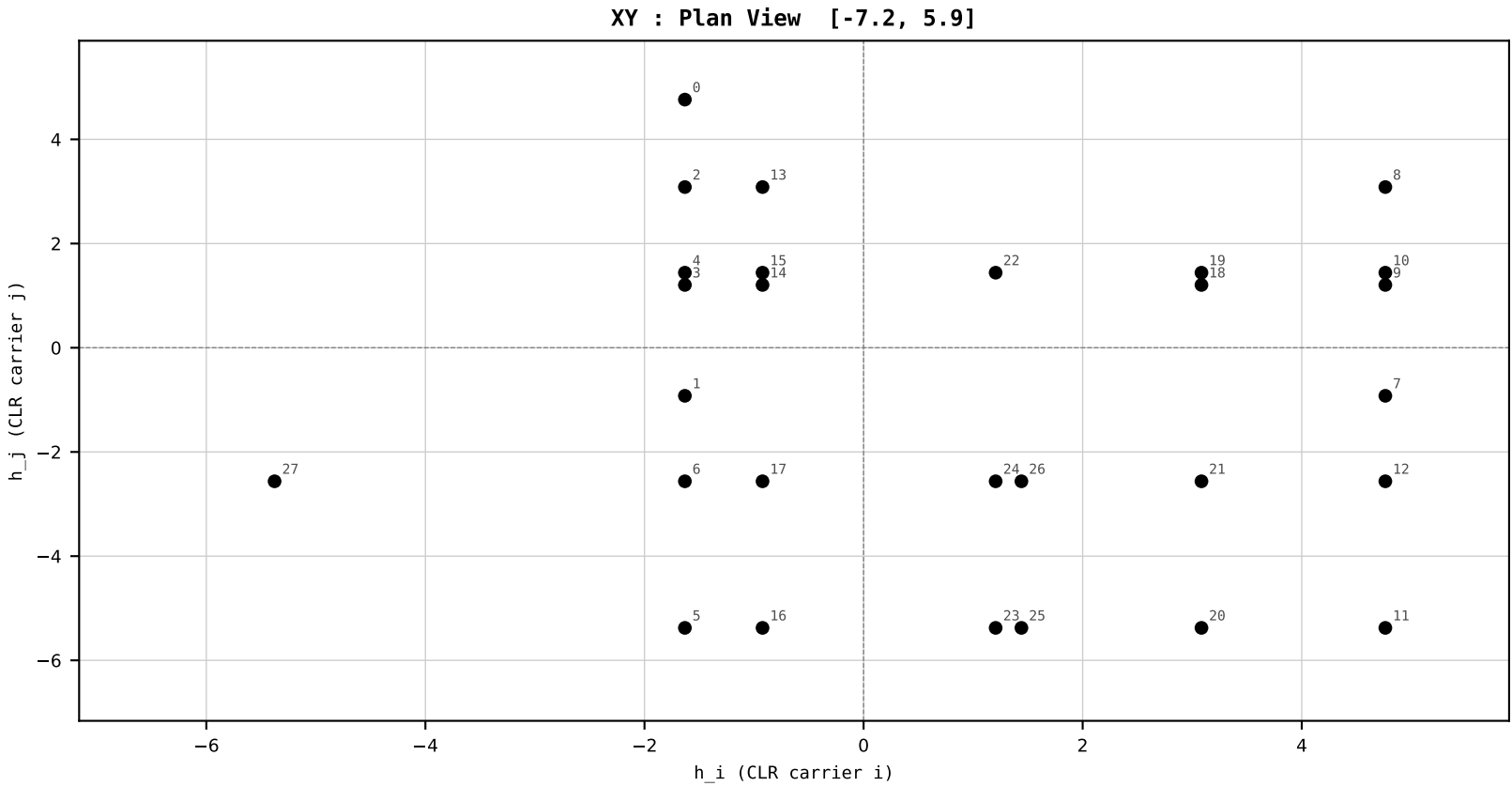
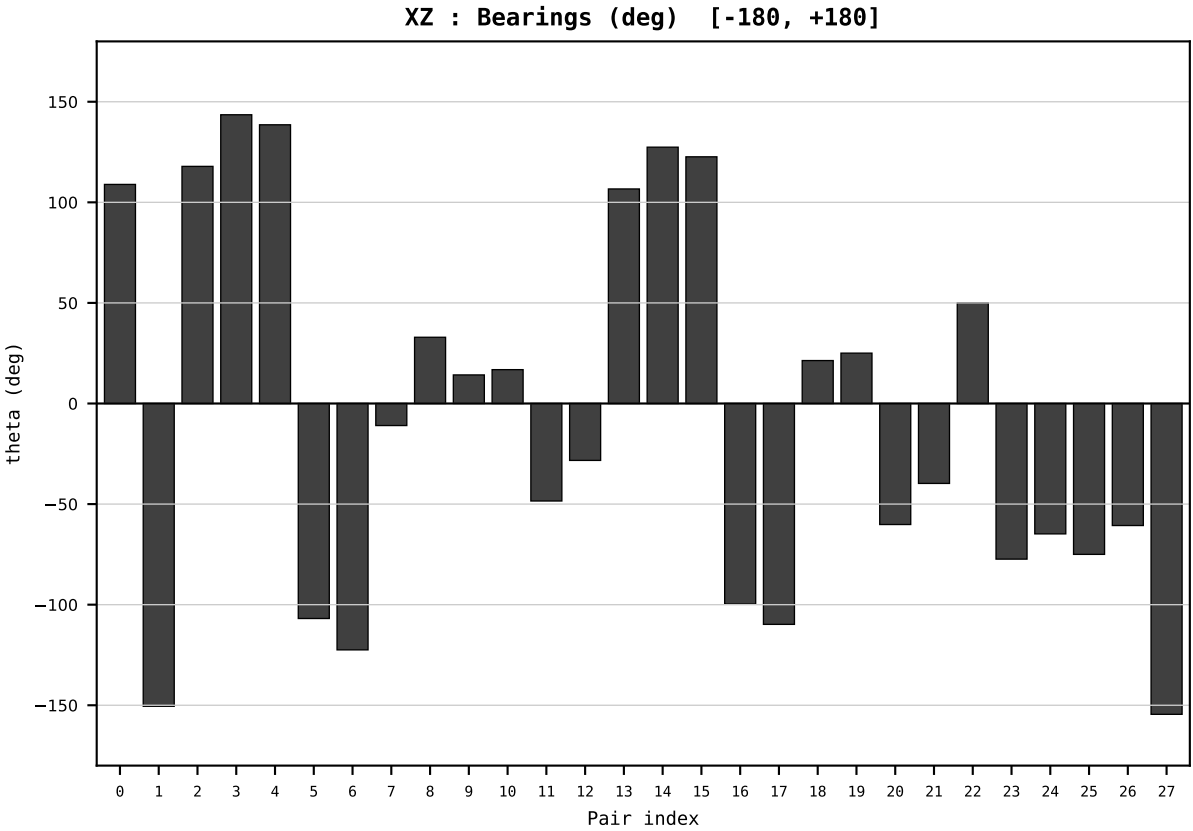
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.672028
Ring = Hs-4
E_metric= 74.7231
kappa_HS= 25332.83
omega = 3.0675 deg
d_A = 0.463827
Helm = Nuclear
Helm d = +0.377648
DR = 10.140 HLR
DR ratio= 25332.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

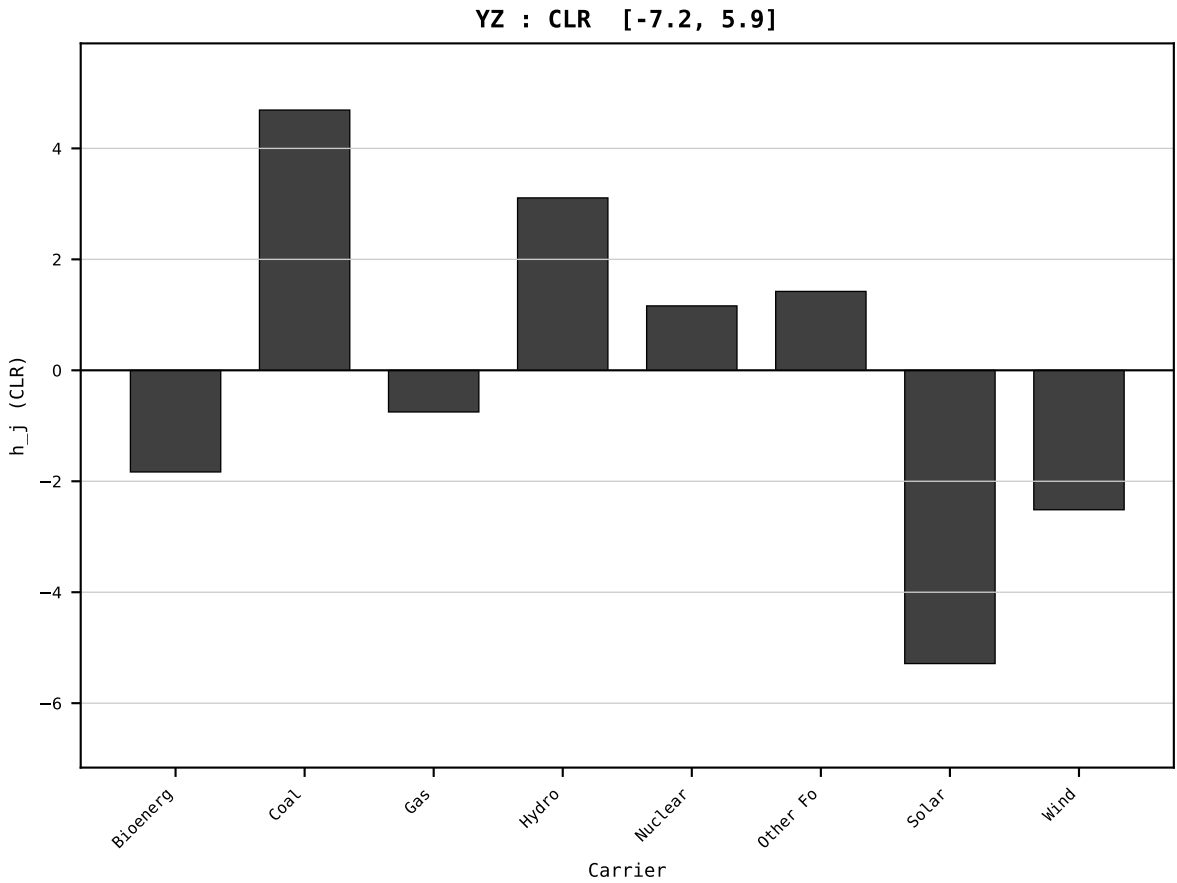
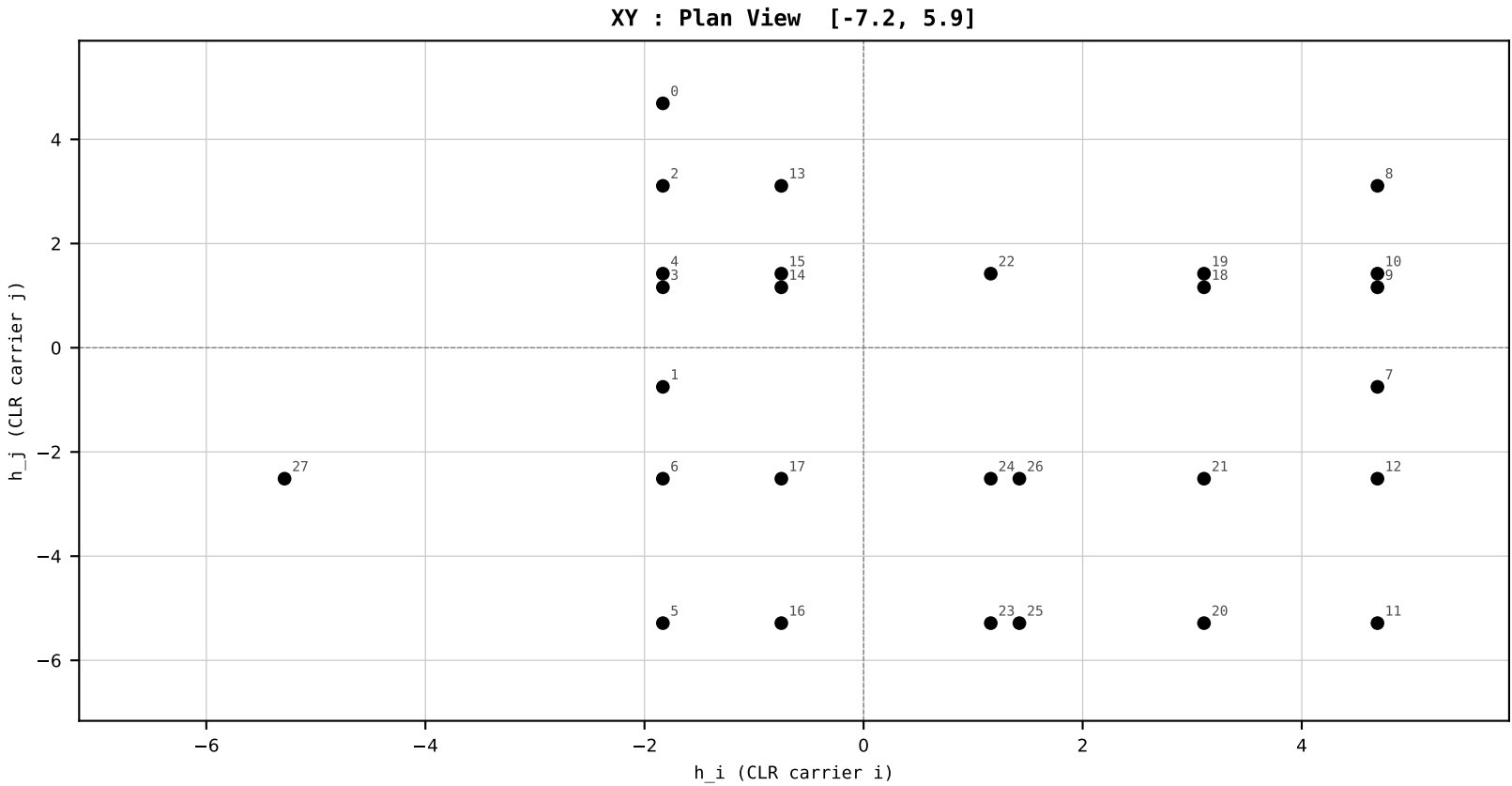
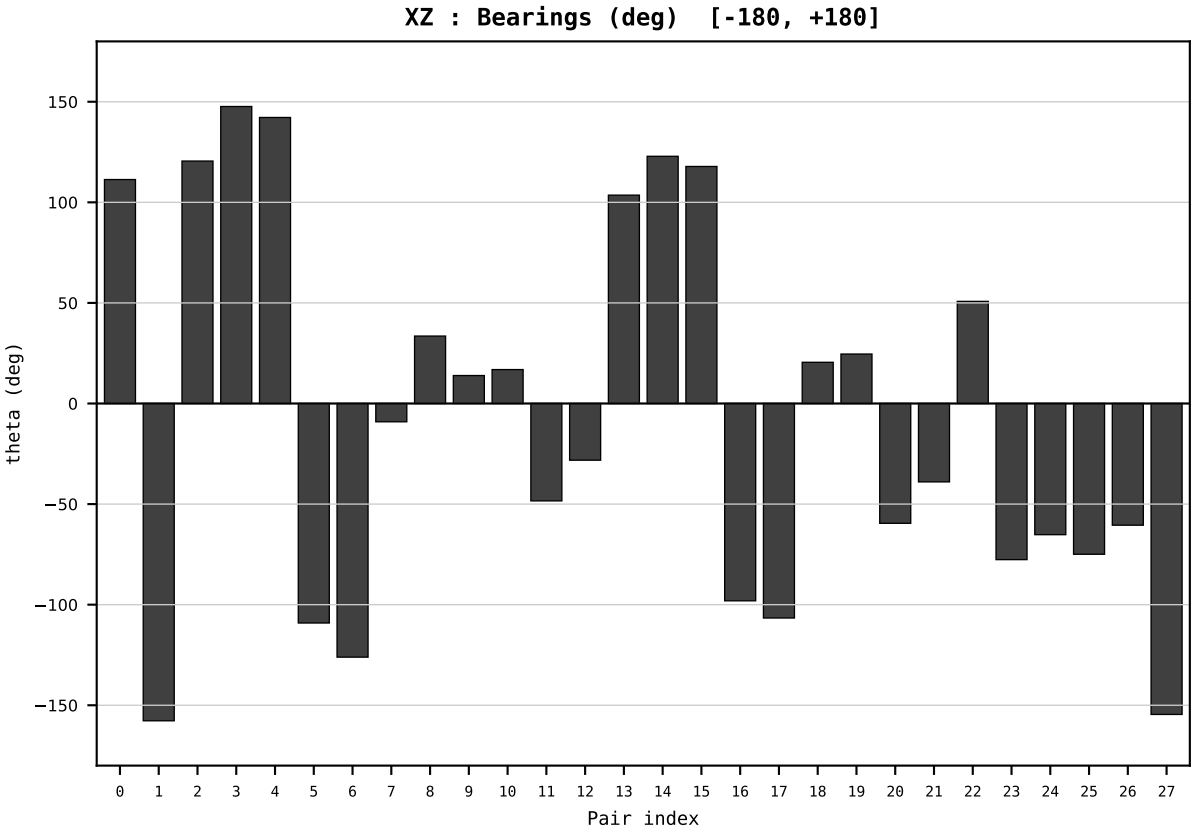
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.659108
Ring = Hs-4
E_metric= 73.2154
kappa_HS= 21530.25
omega = 1.8952 deg
d_A = 0.297664
Helm = Bioenergy
Helm d = -0.200817
DR = 9.977 HLR
DR ratio= 21530.2x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

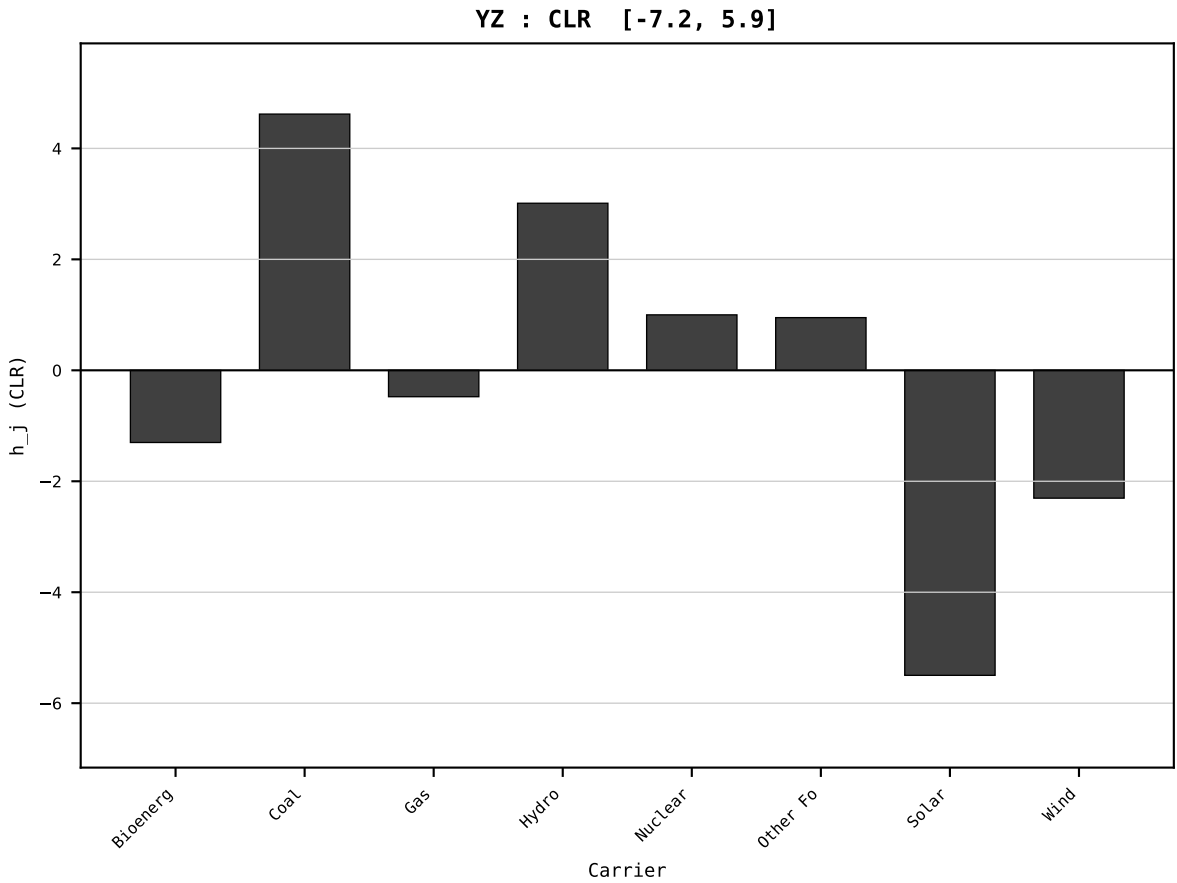
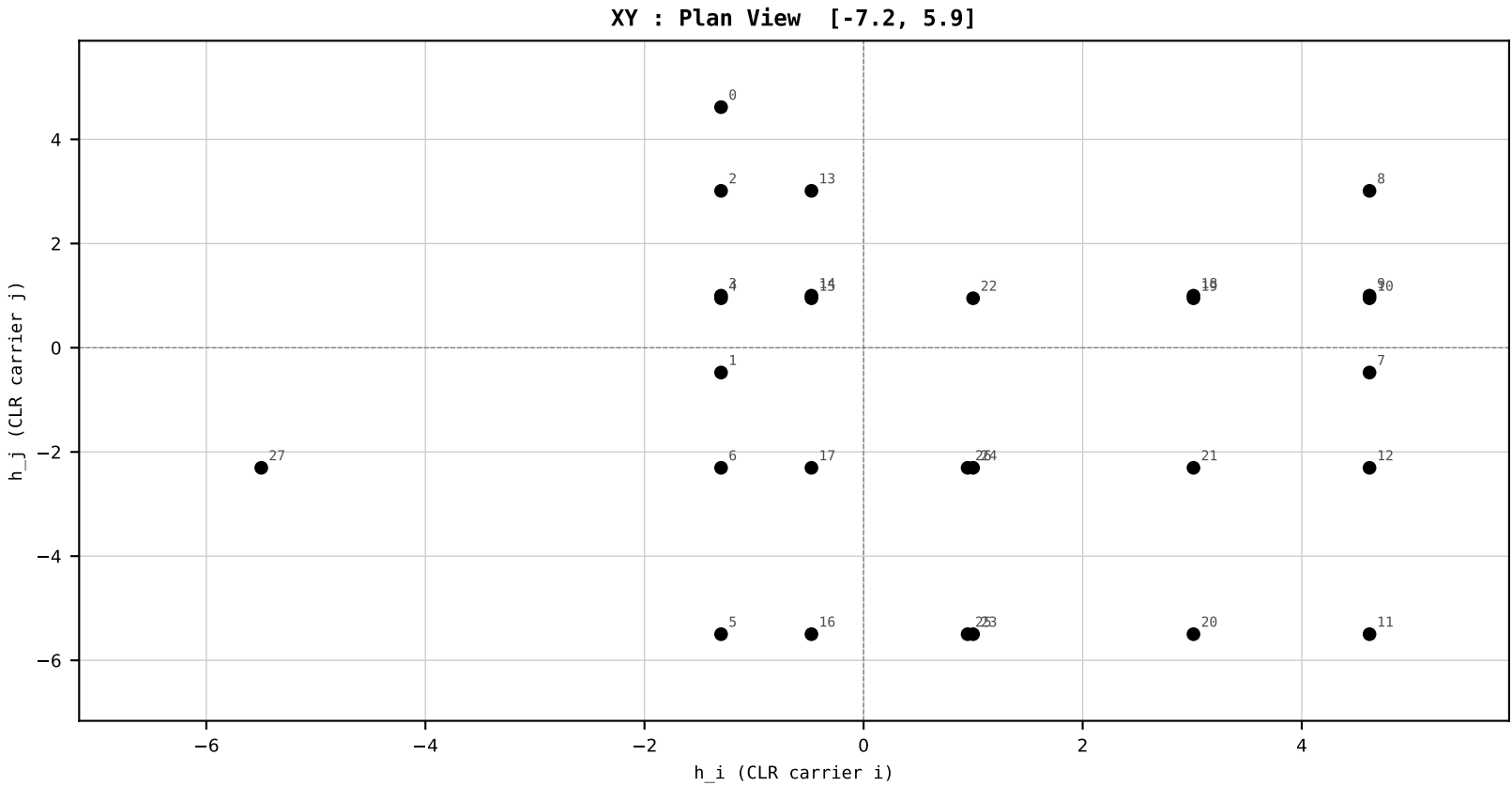
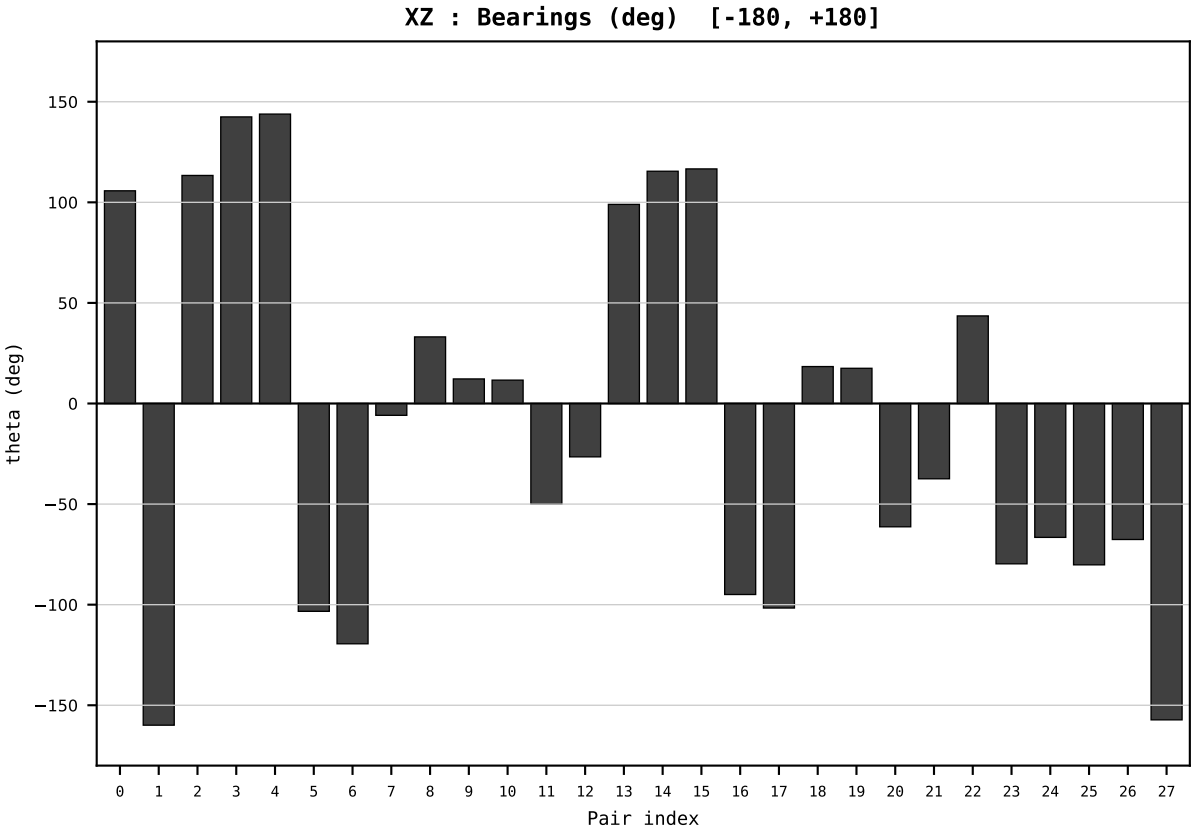
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.671833
Ring = Hs-4
E_metric= 69.7610
kappa_HS= 24751.88
omega = 5.5429 deg
d_A = 0.842658
Helm = Bioenergy
Helm d = +0.531144
DR = 10.117 HLR
DR ratio= 24751.9x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

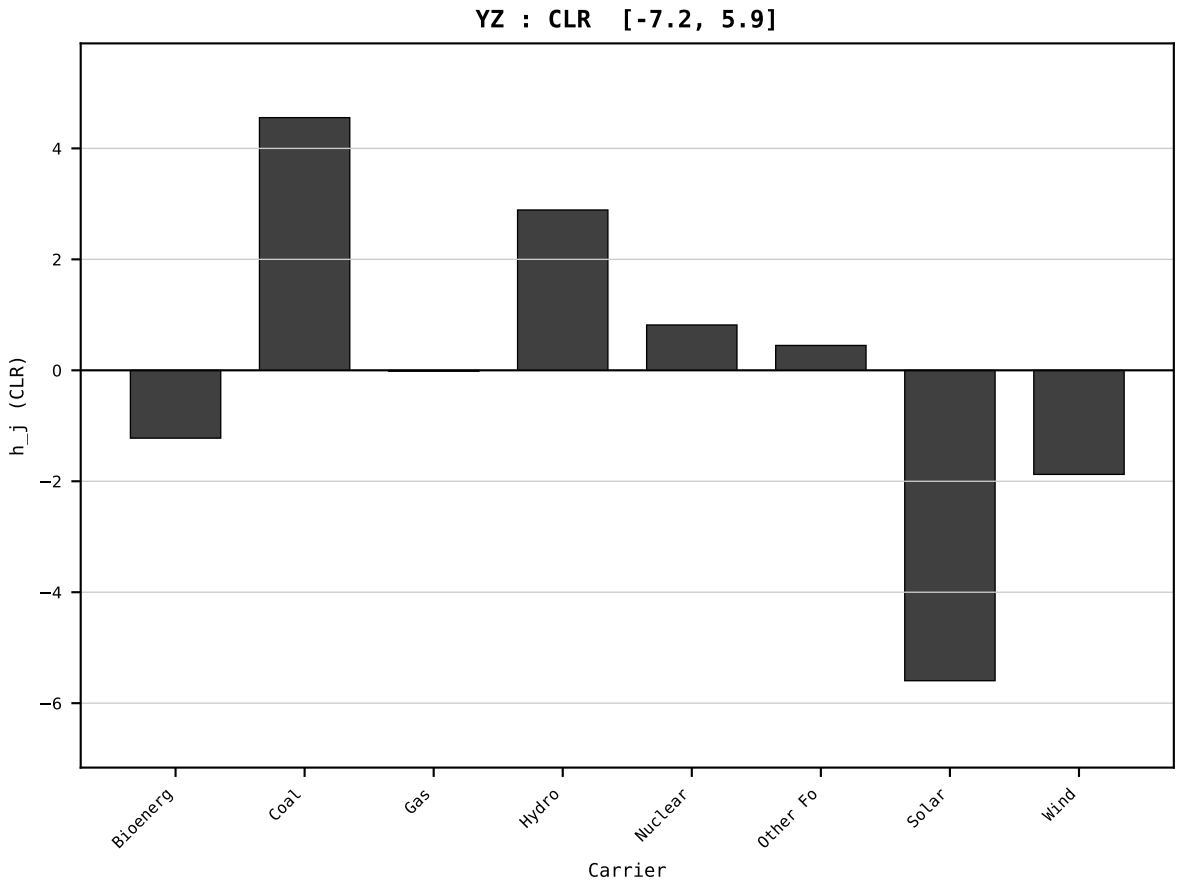
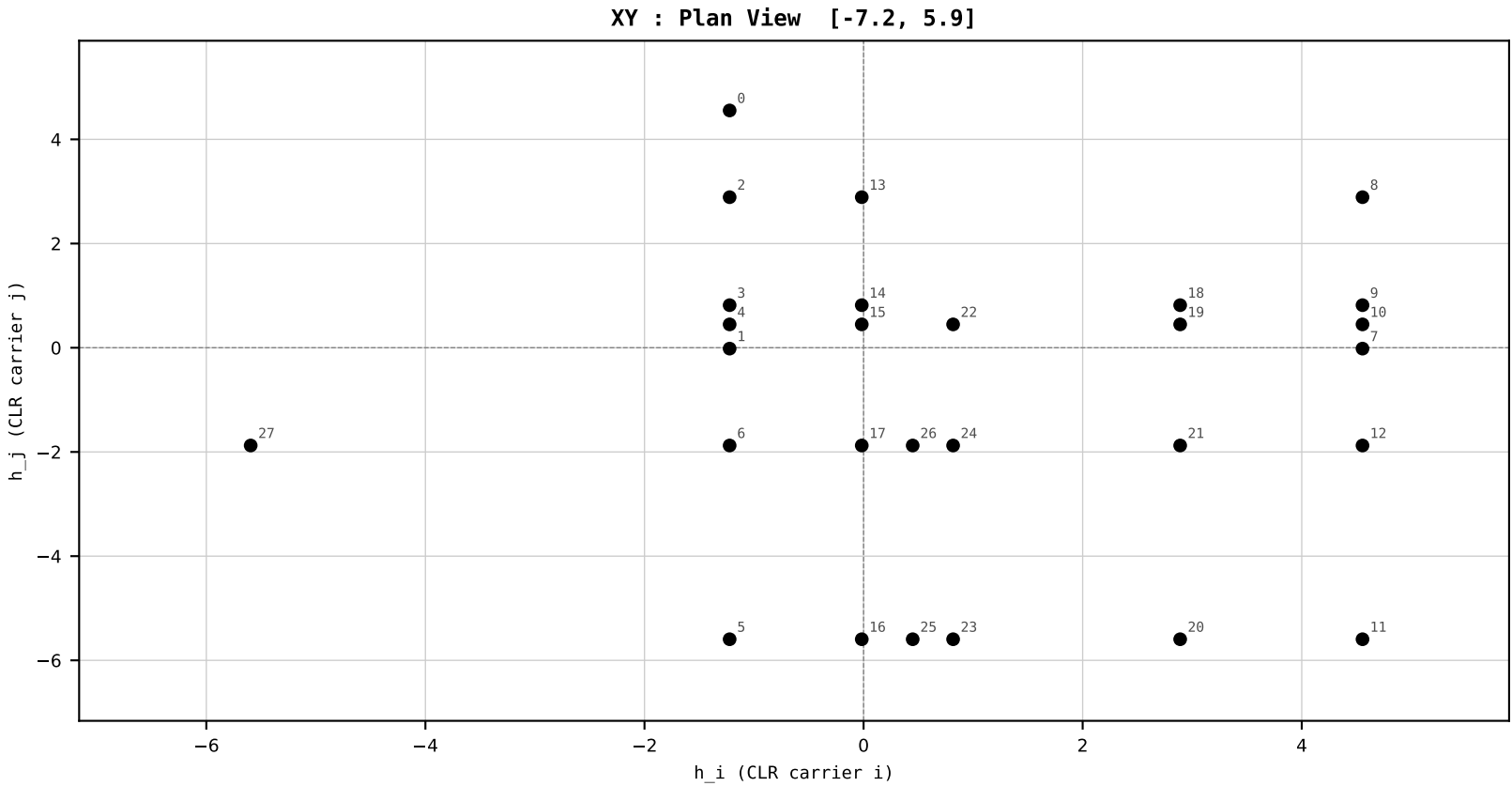
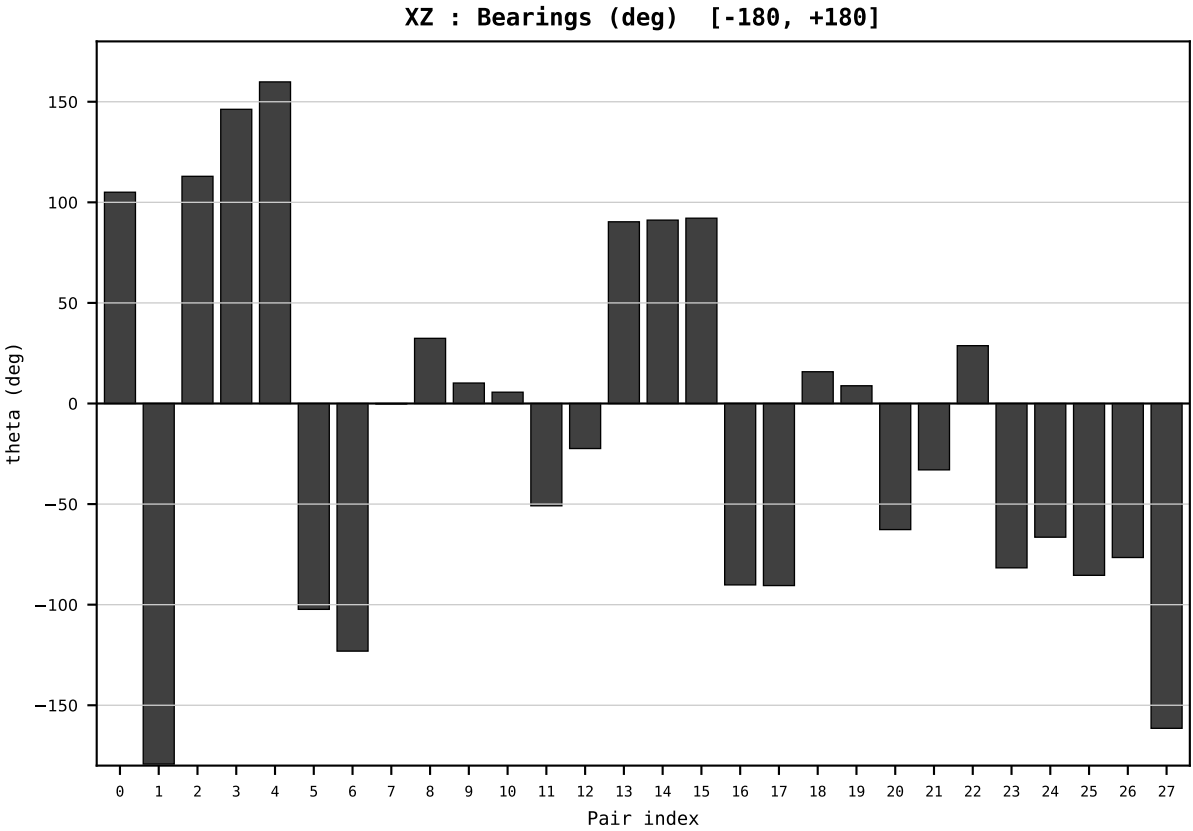
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.683157
Ring = Hs-4
E_metric= 66.2857
kappa_HS= 25582.44
omega = 5.6860 deg
d_A = 0.844723
Helm = Other Fossil
Helm d = -0.501386
DR = 10.150 HLR
DR ratio= 25582.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

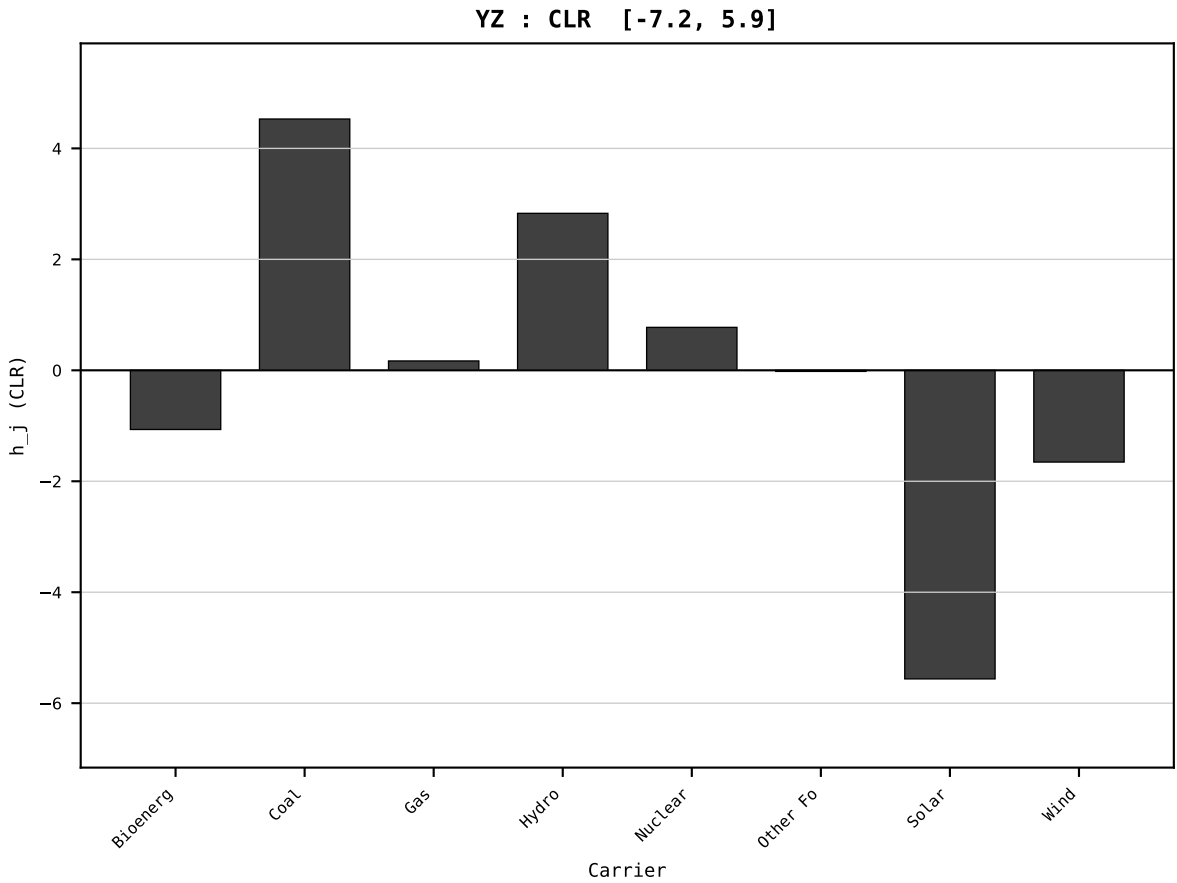
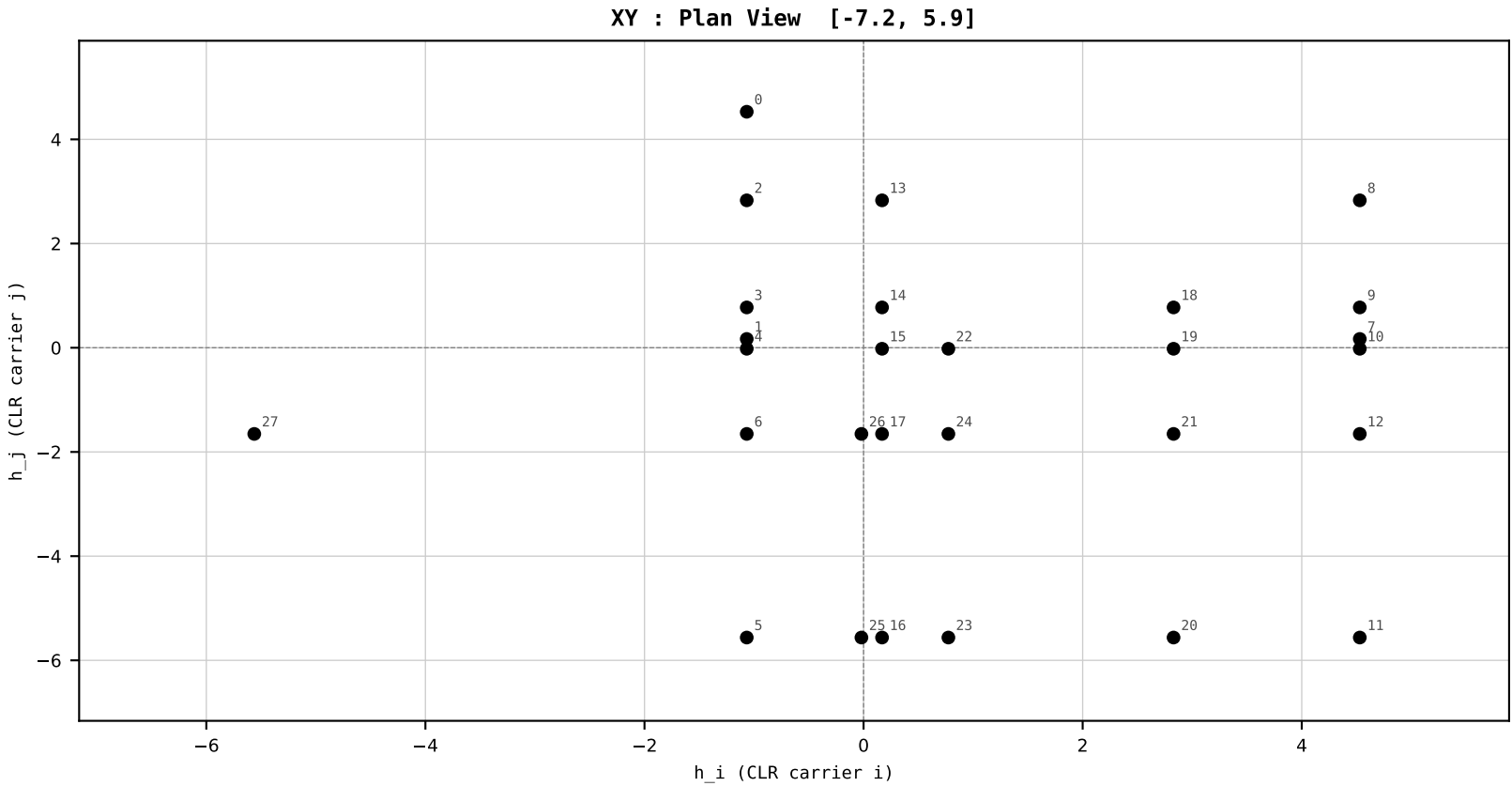
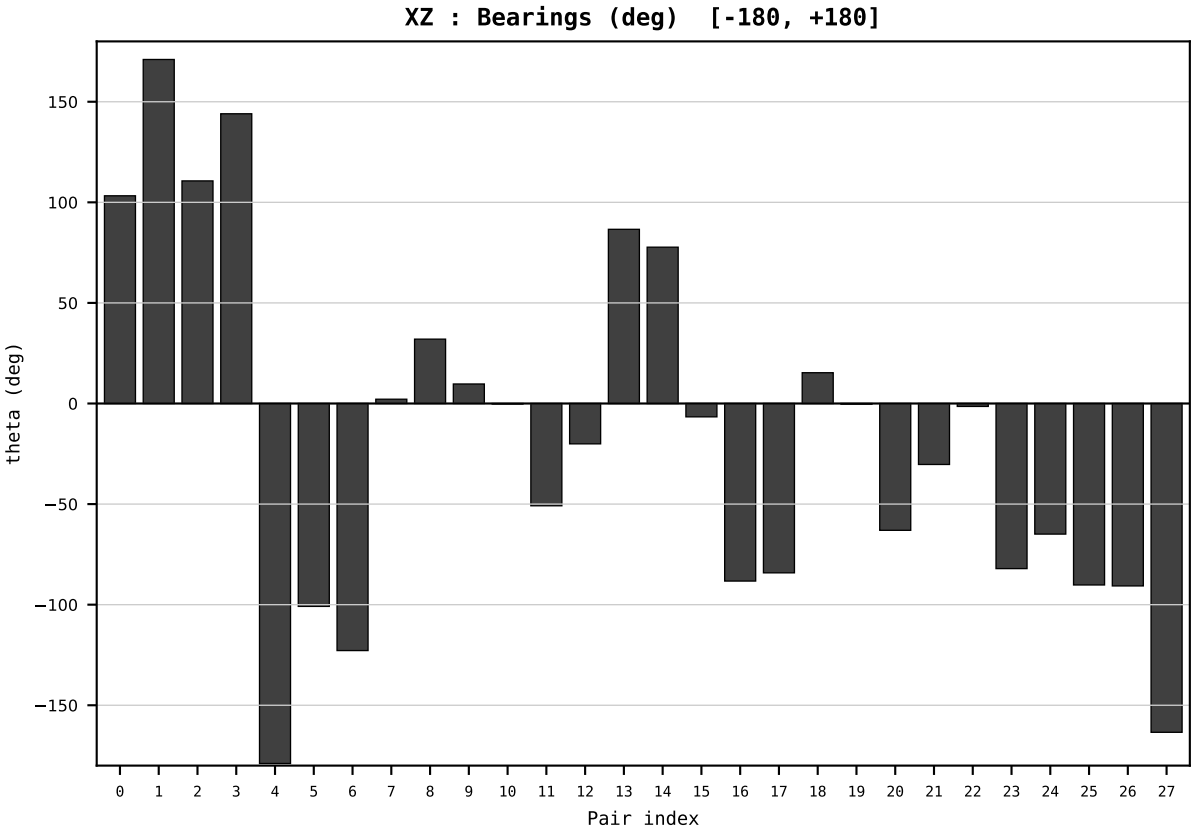
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.689708
Ring = Hs-4
E_metric= 63.9666
kappa_HS= 24151.82
omega = 3.9773 deg
d_A = 0.578182
Helm = Other Fossil
Helm d = -0.468002
DR = 10.092 HLR
DR ratio= 24151.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

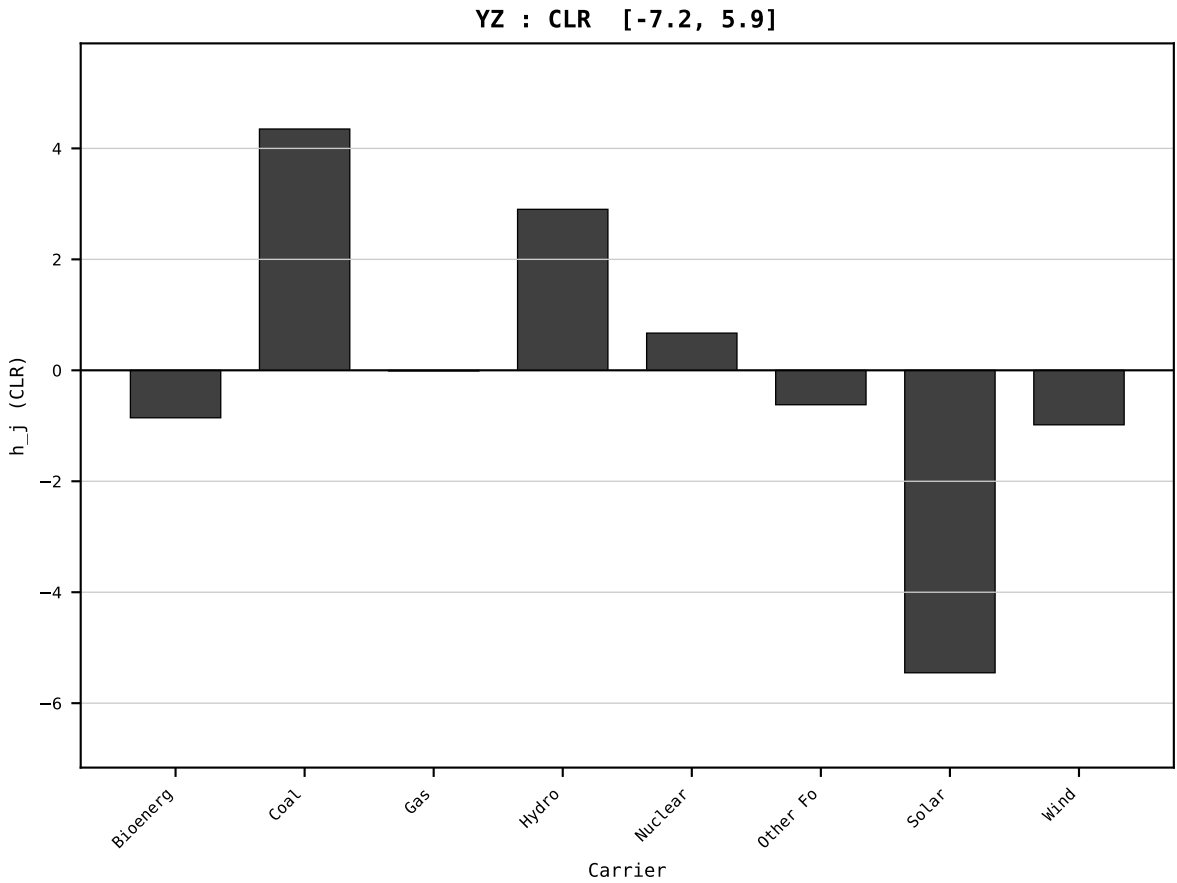
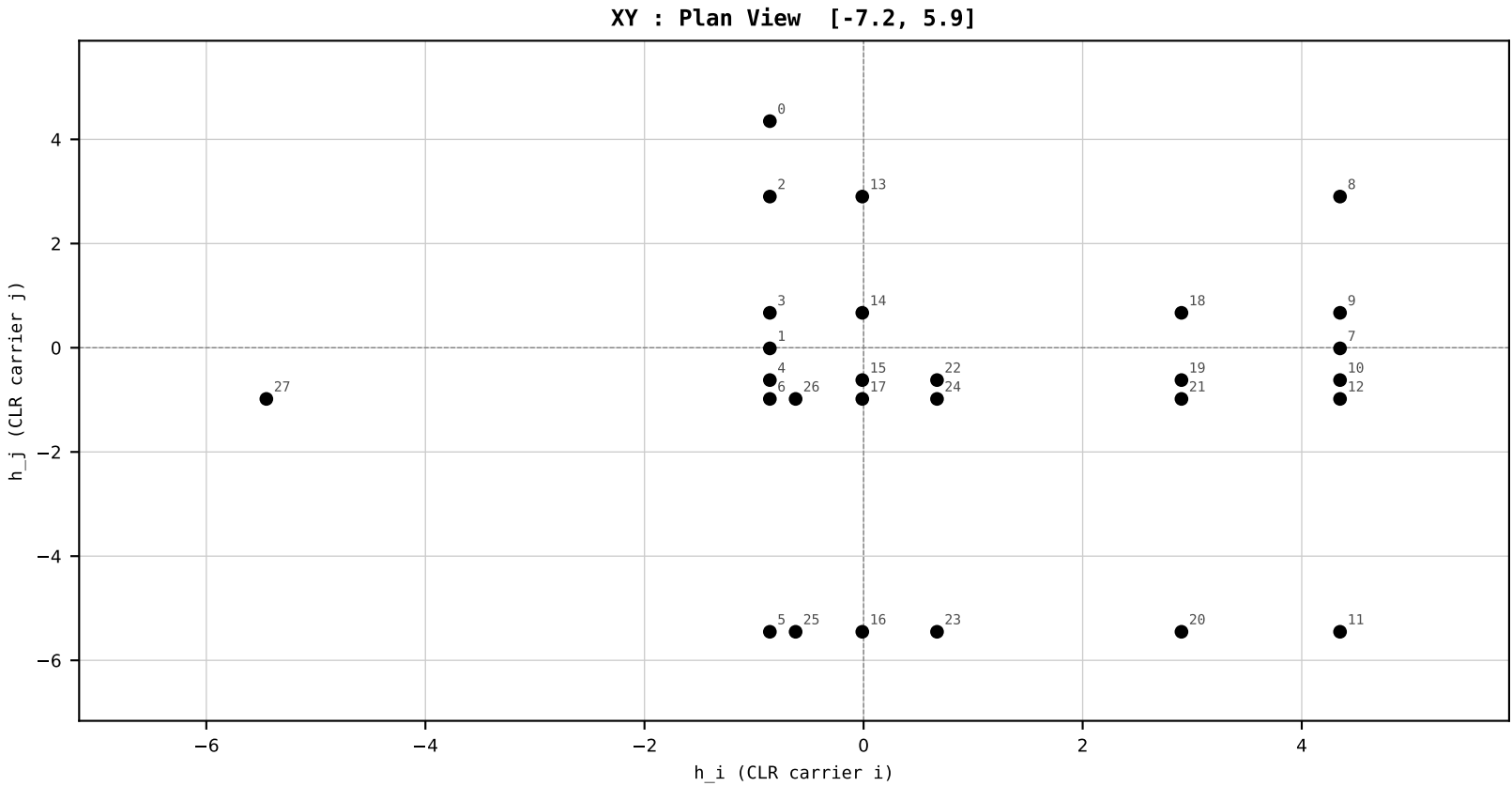
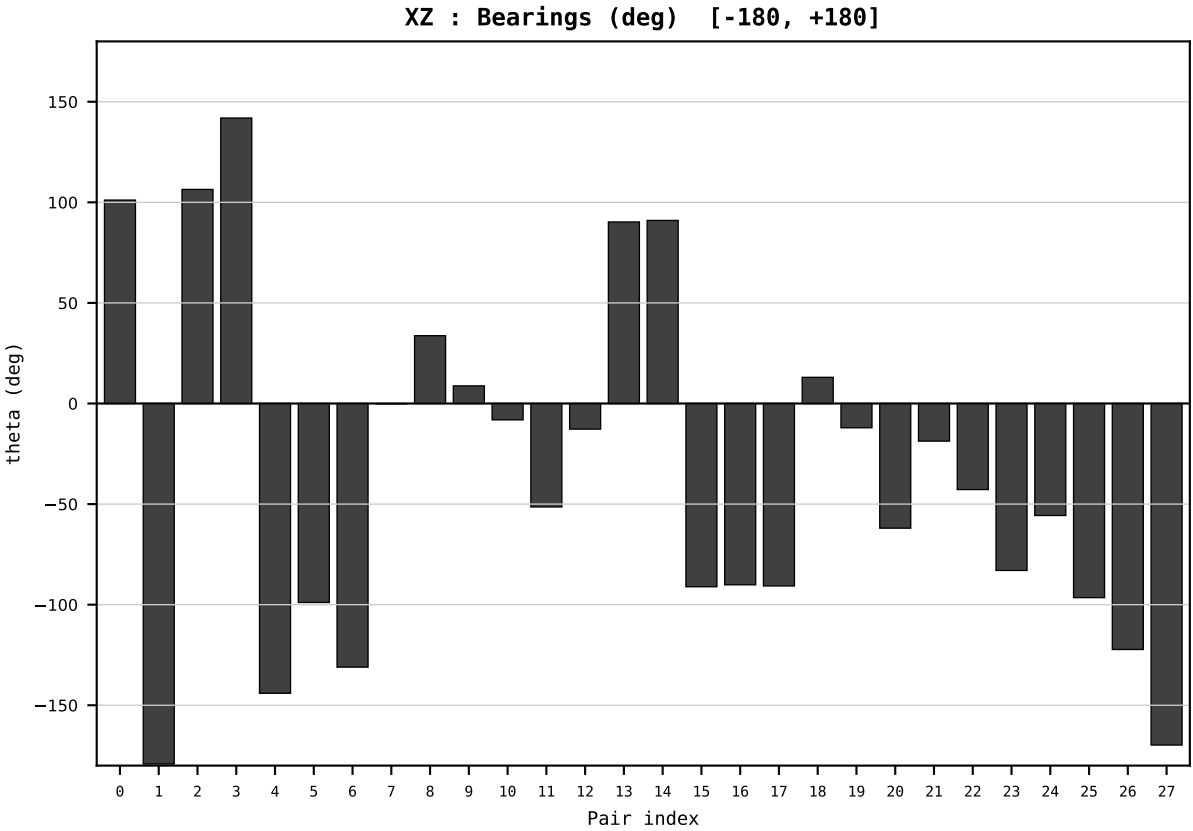
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.661852
Ring = Hs-4
E_metric= 59.5937
kappa_HS= 18059.40
omega = 6.8155 deg
d_A = 0.974683
Helm = Wind
Helm d = +0.671725
DR = 9.801 HLR
DR ratio= 18059.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=8 N=26 pairs=28

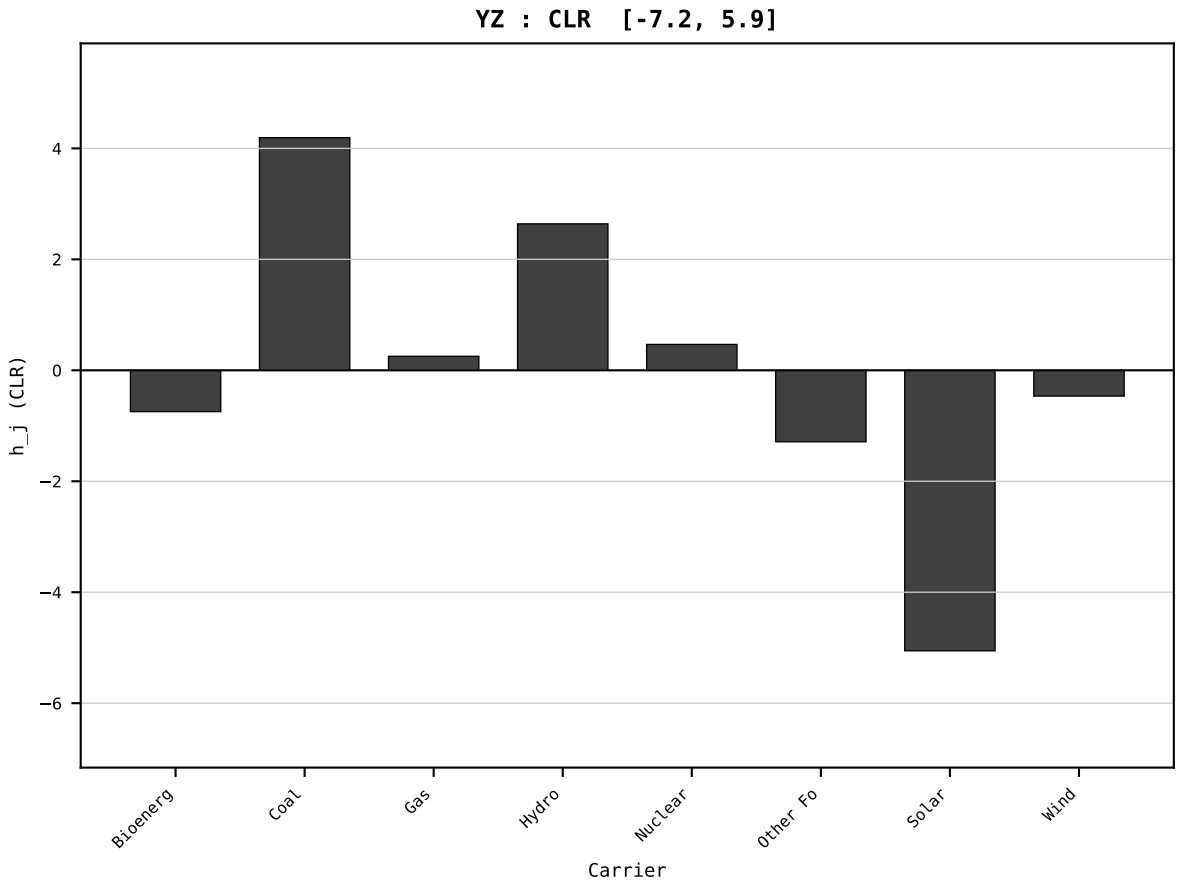
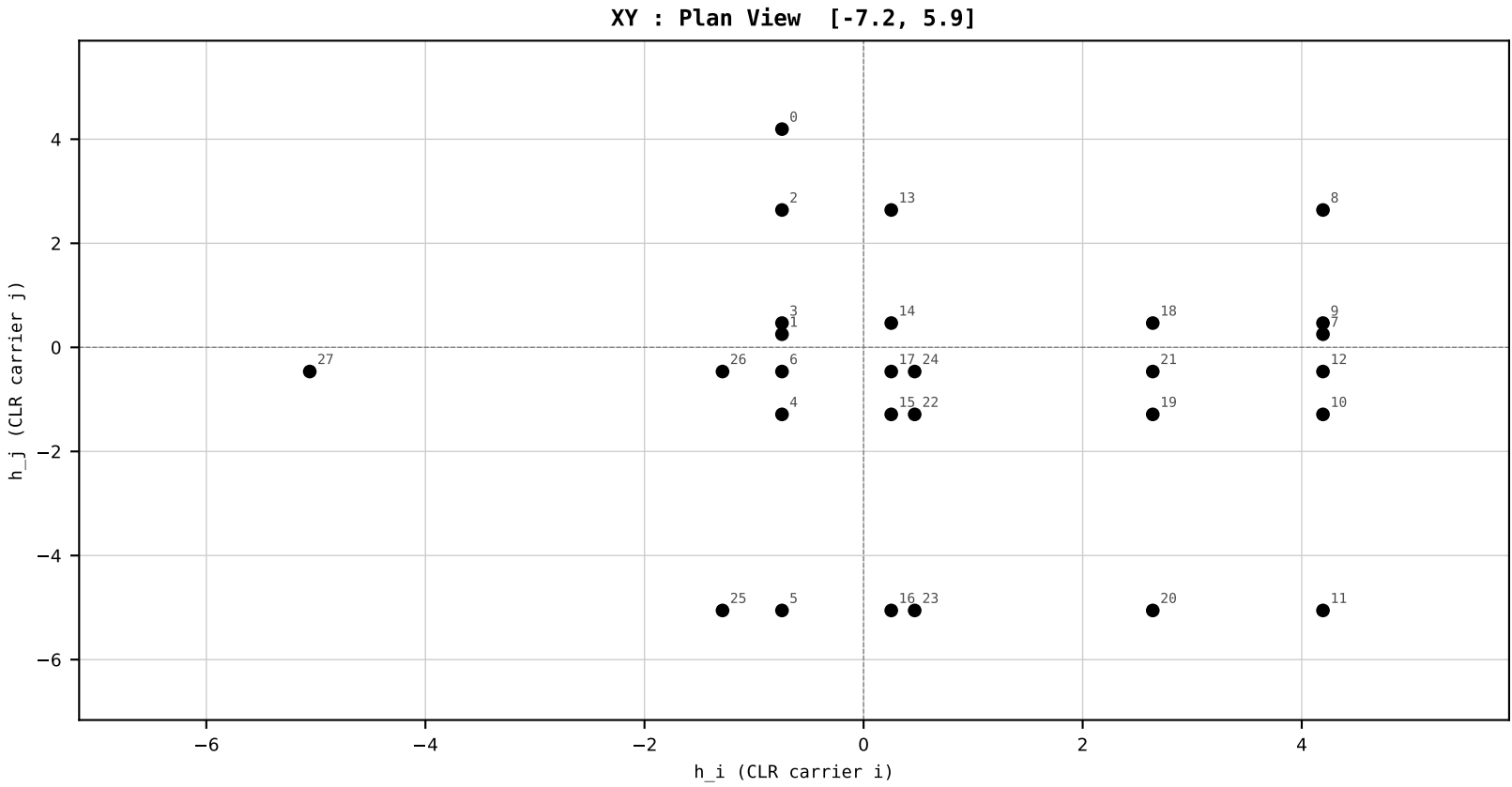
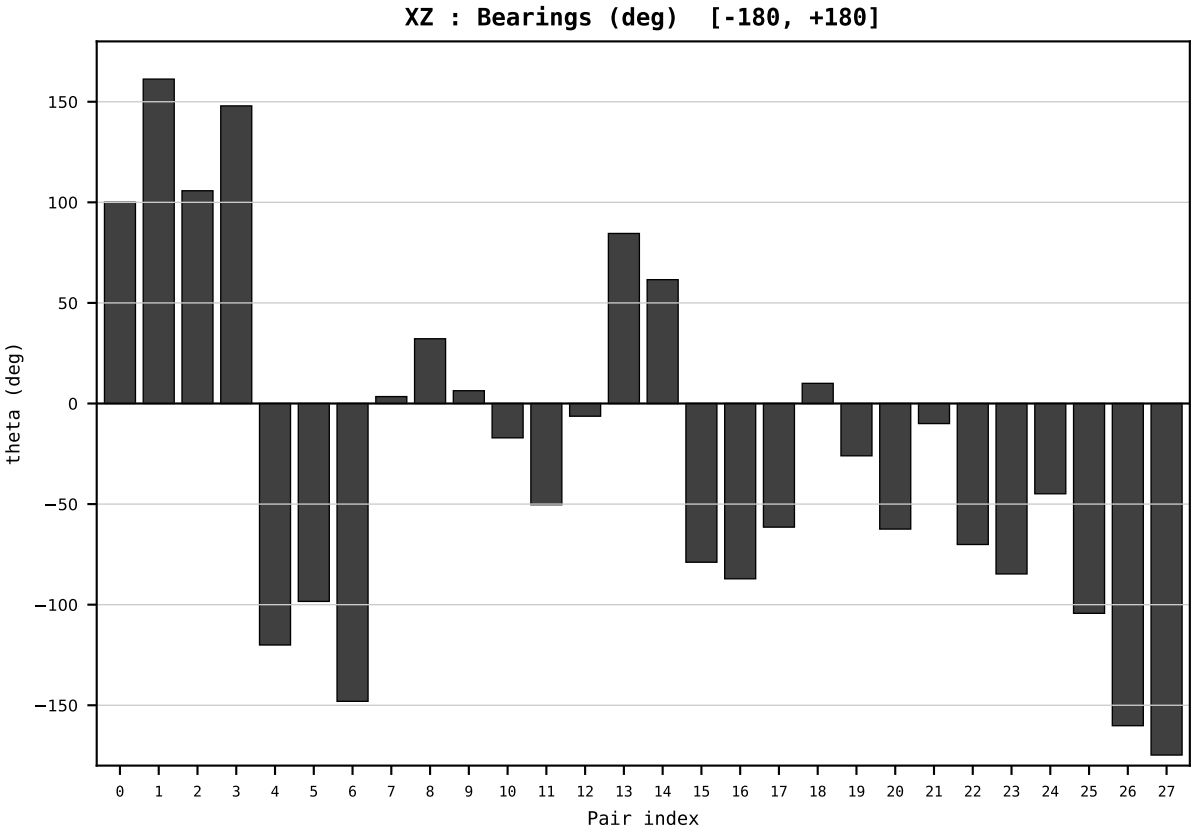
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.657431
Ring = Hs-4
E_metric= 52.8271
kappa_HS= 10398.29
omega = 7.1934 deg
d_A = 1.042625
Helm = Other Fossil
Helm d = -0.667414
DR = 9.249 HLR
DR ratio= 10398.3x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

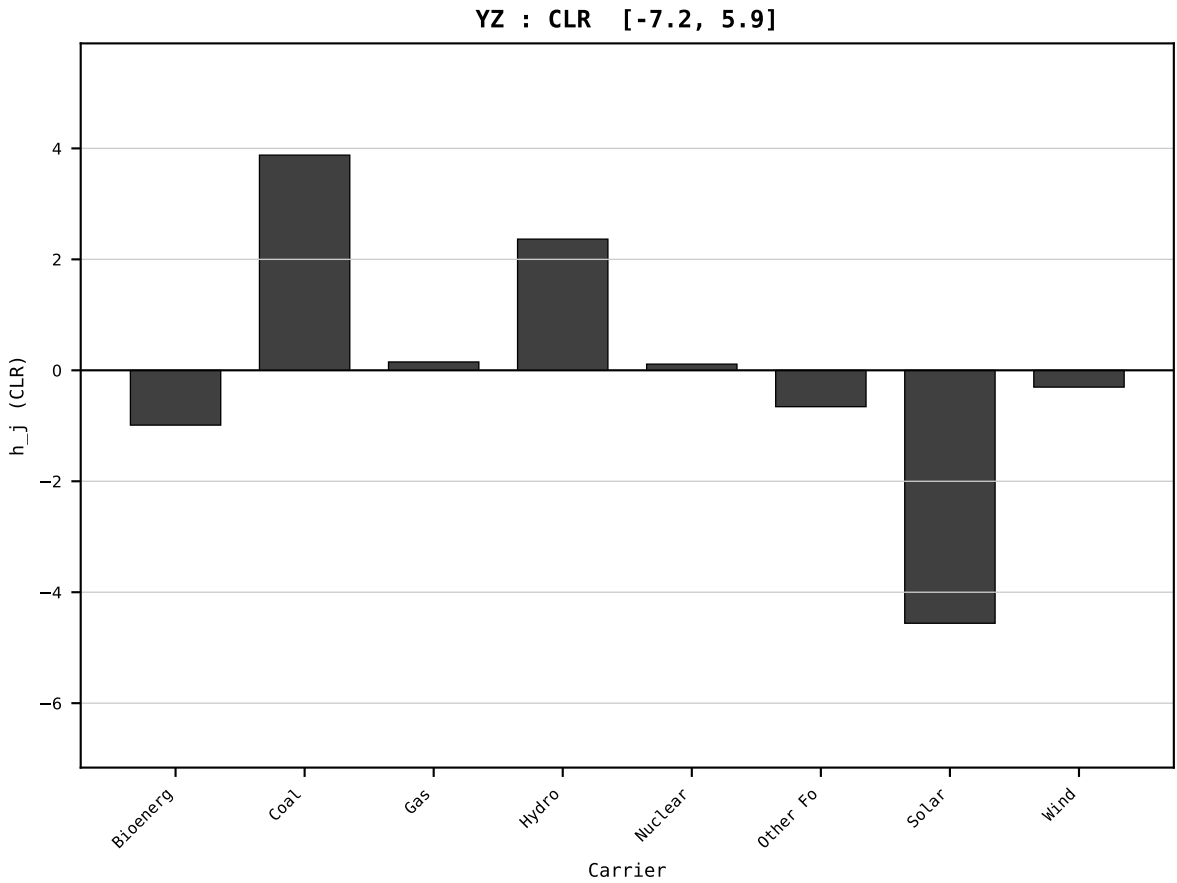
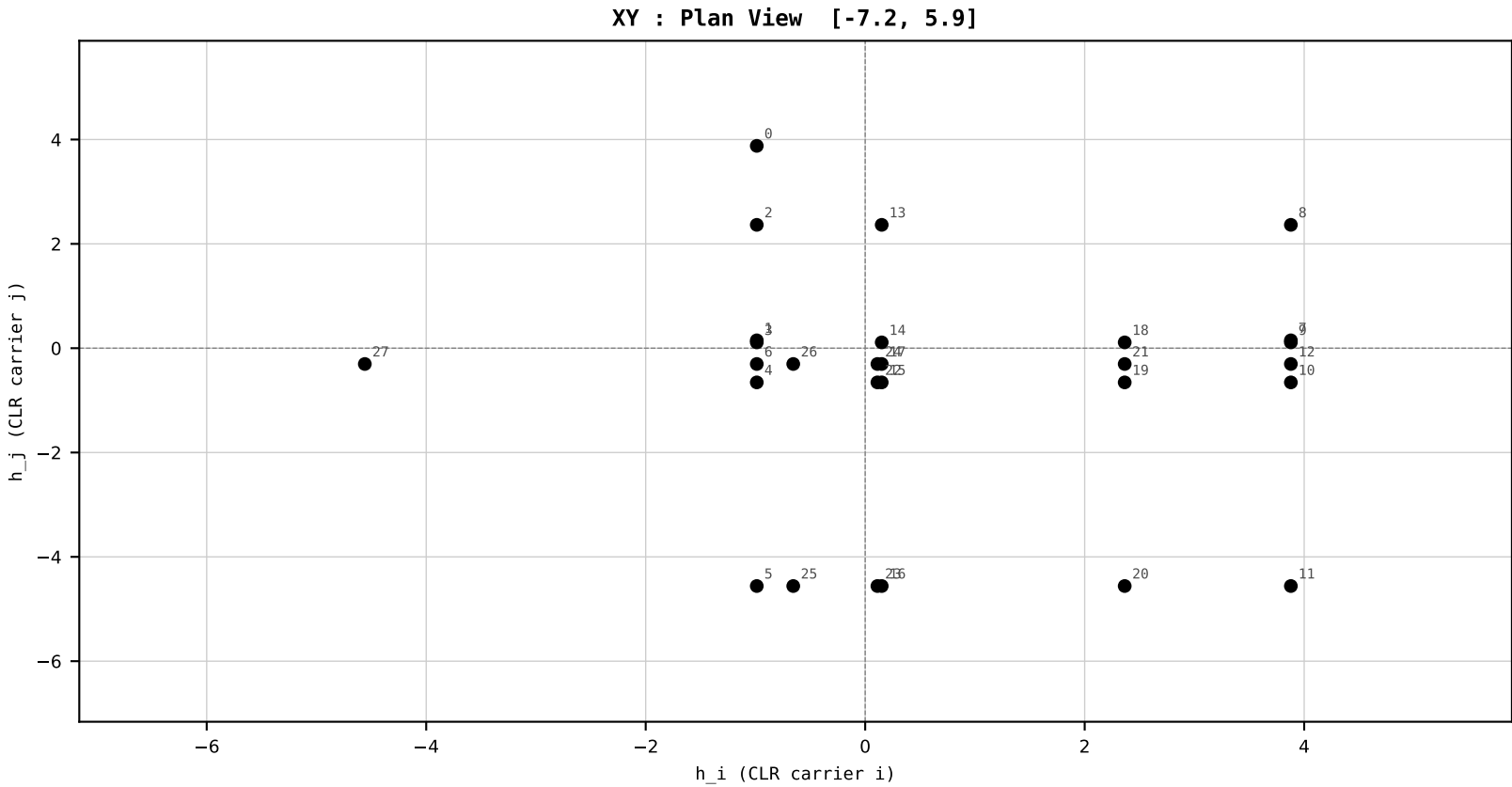
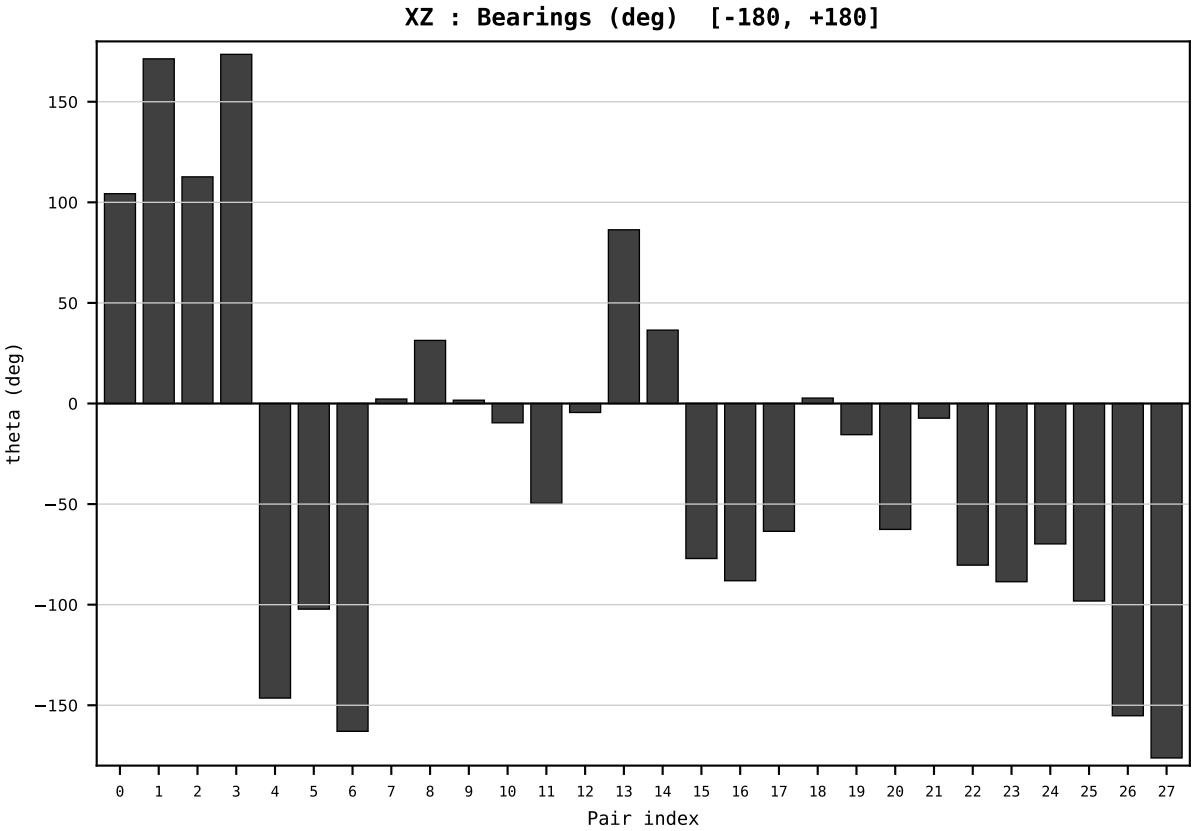
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.628874
Ring = Hs-4
E_metric= 42.9552
kappa_HS= 4619.43
omega = 6.0622 deg
d_A = 1.021213
Helm = Other Fossil
Helm d = +0.632236
DR = 8.438 HLR
DR ratio= 4619.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

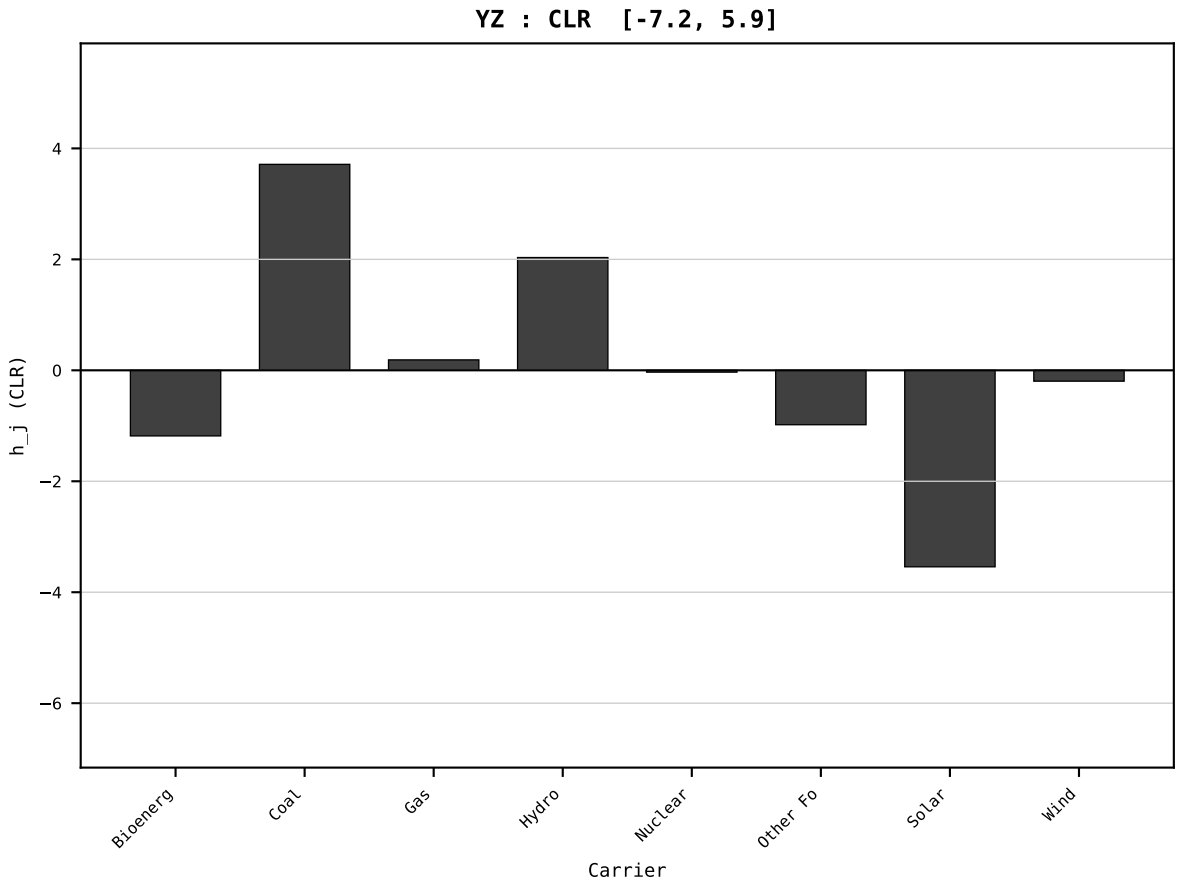
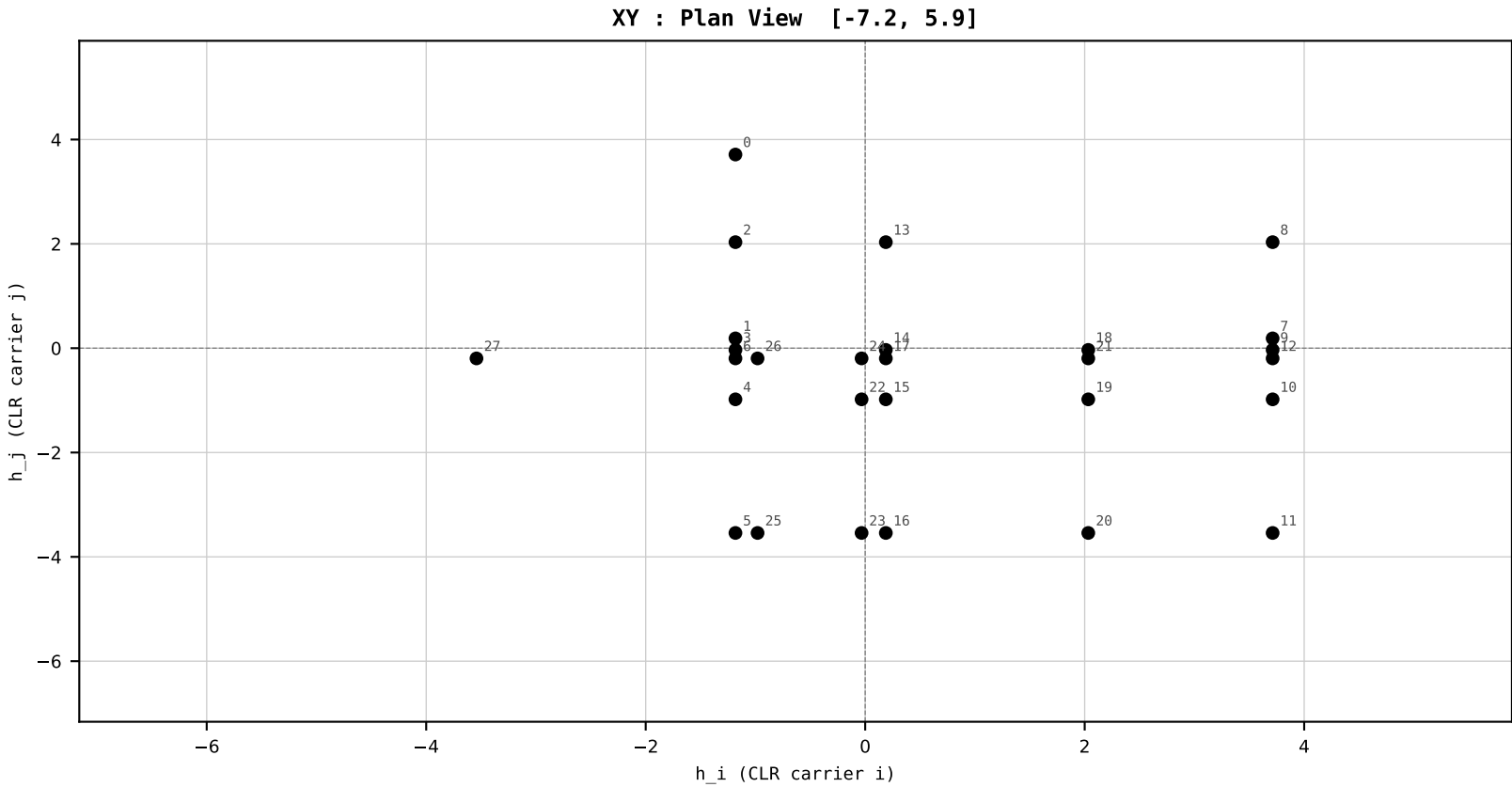
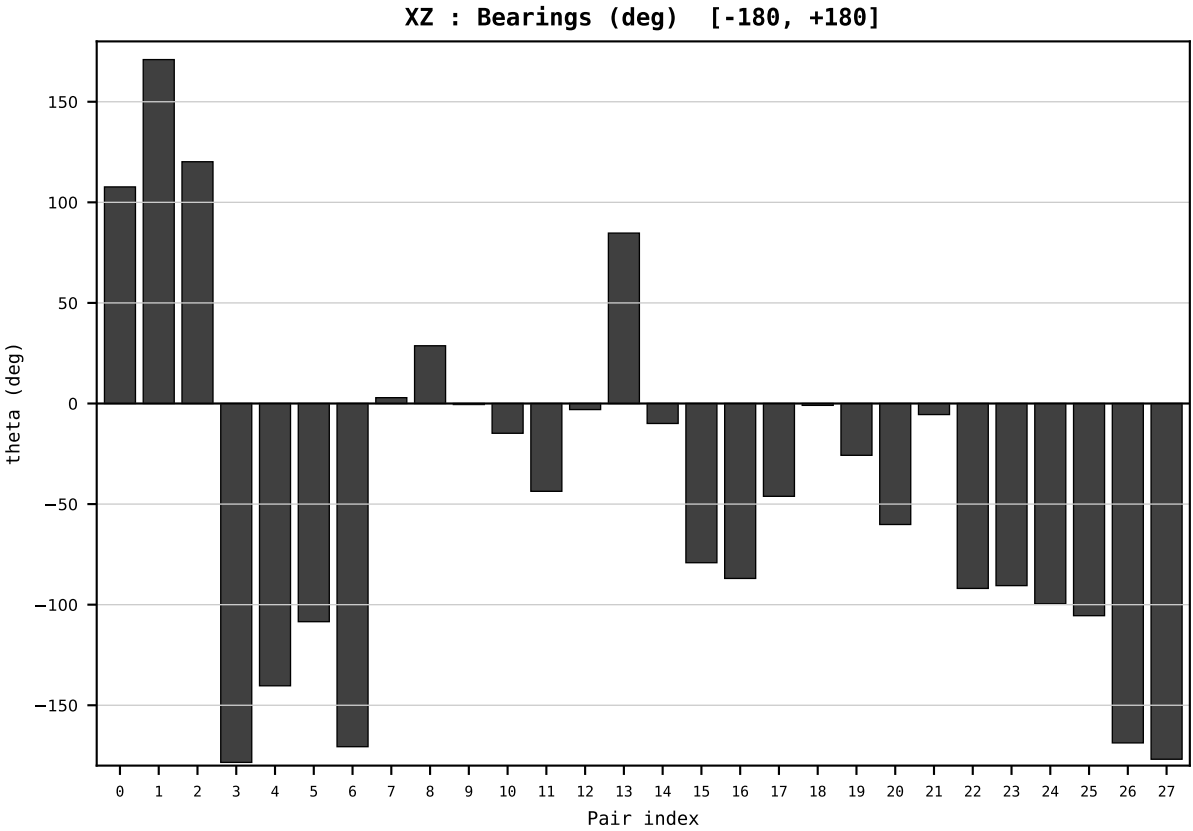
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.630601
Ring = Hs-4
E_metric= 32.8929
kappa_HS= 1414.13
omega = 7.7114 deg
d_A = 1.162016
Helm = Solar
Helm d = +1.017519
DR = 7.254 HLR
DR ratio= 1414.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

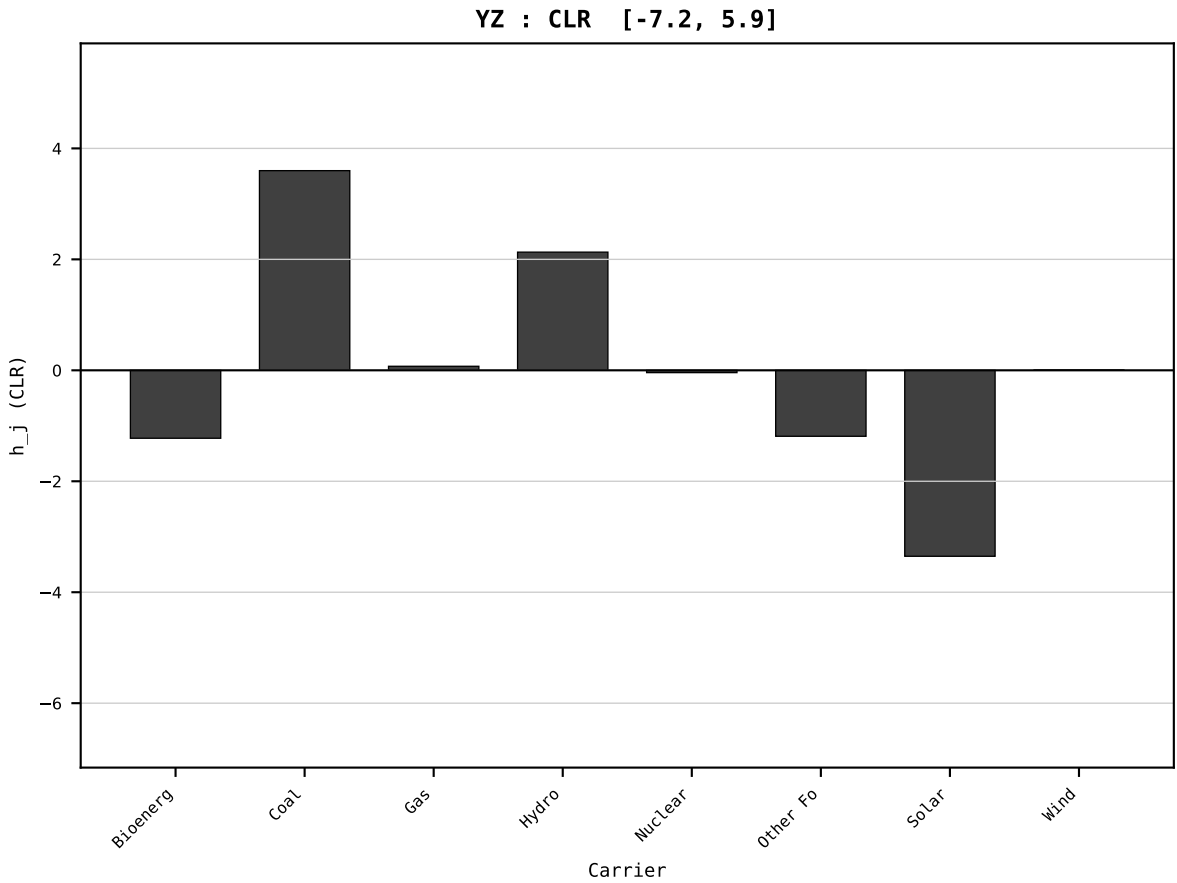
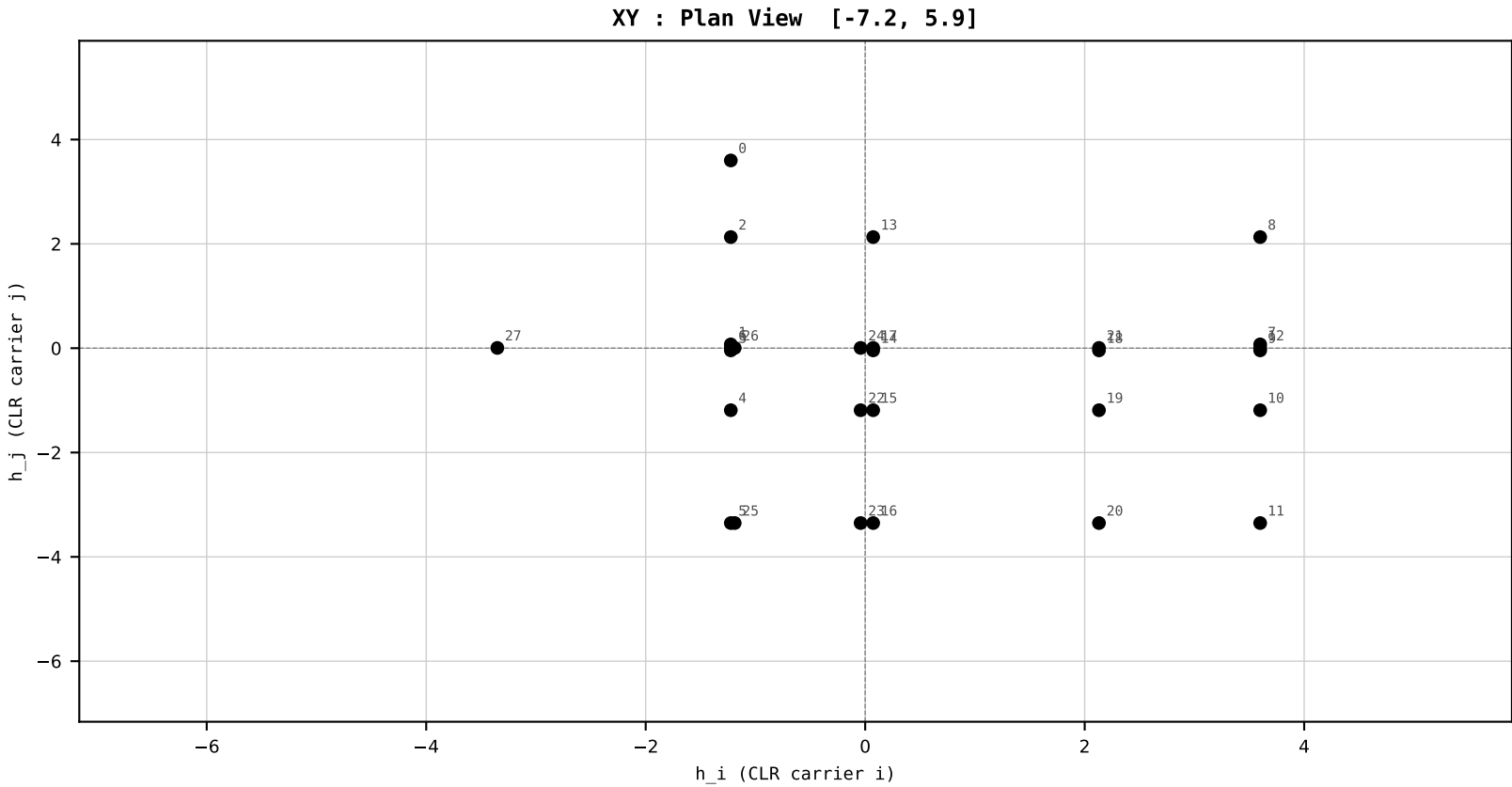
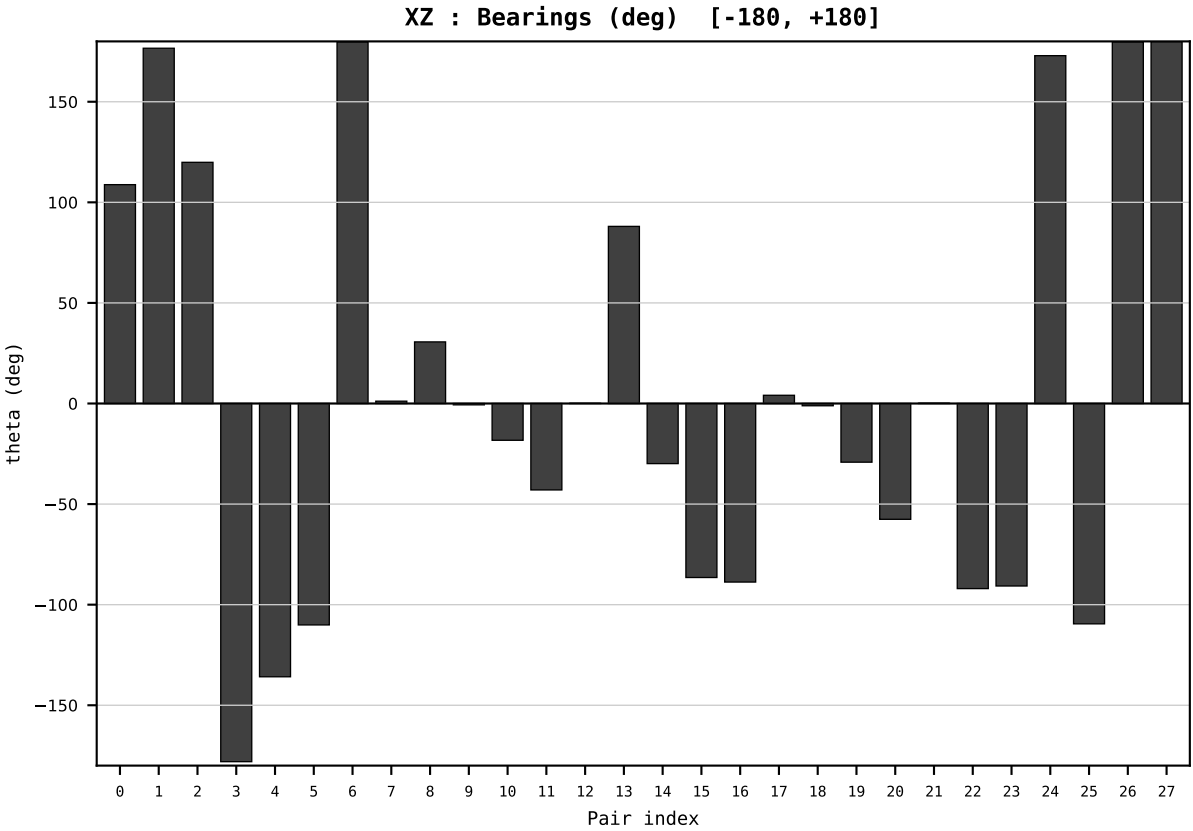
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.601887
Ring = Hs-4
E_metric= 31.6455
kappa_HS= 1044.06
omega = 3.8459 deg
d_A = 0.396692
Helm = Other Fossil
Helm d = -0.208288
DR = 6.951 HLR
DR ratio= 1044.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=8 N=26 pairs=28

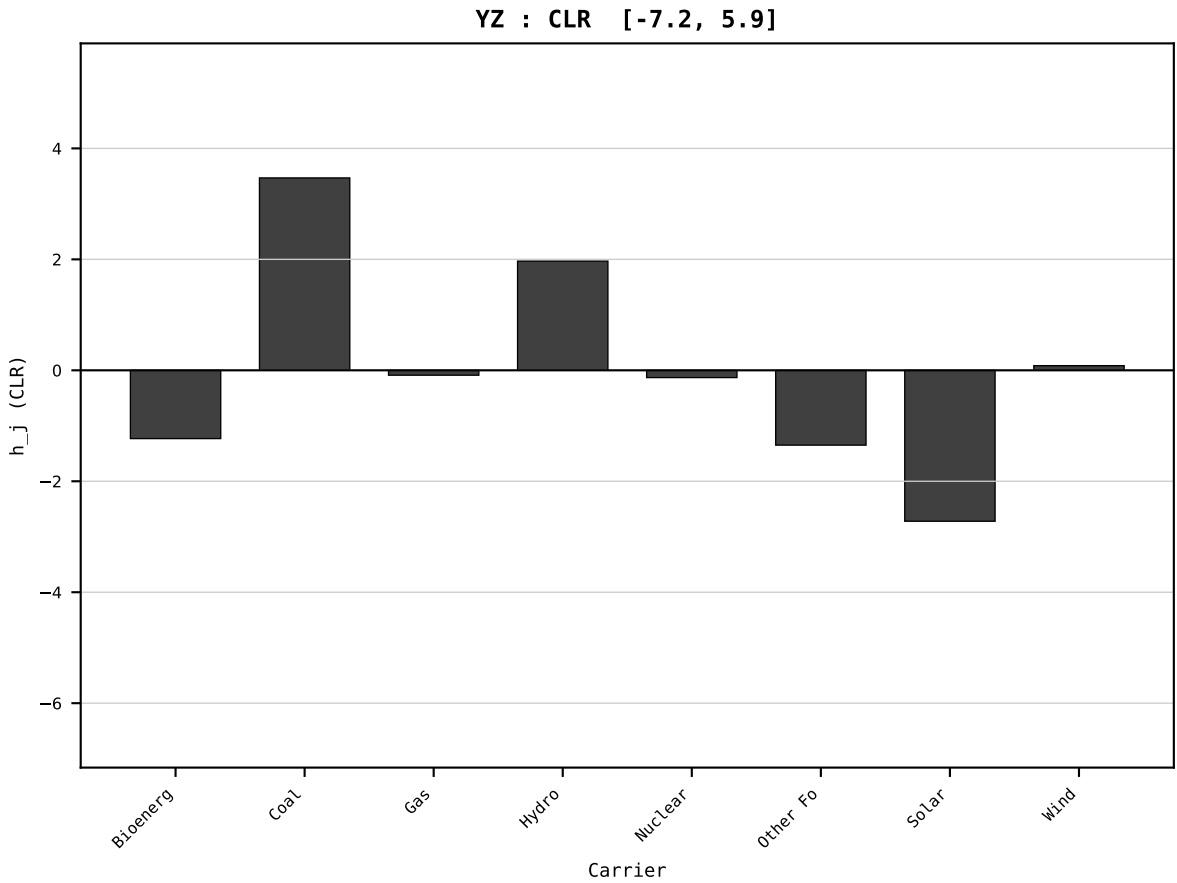
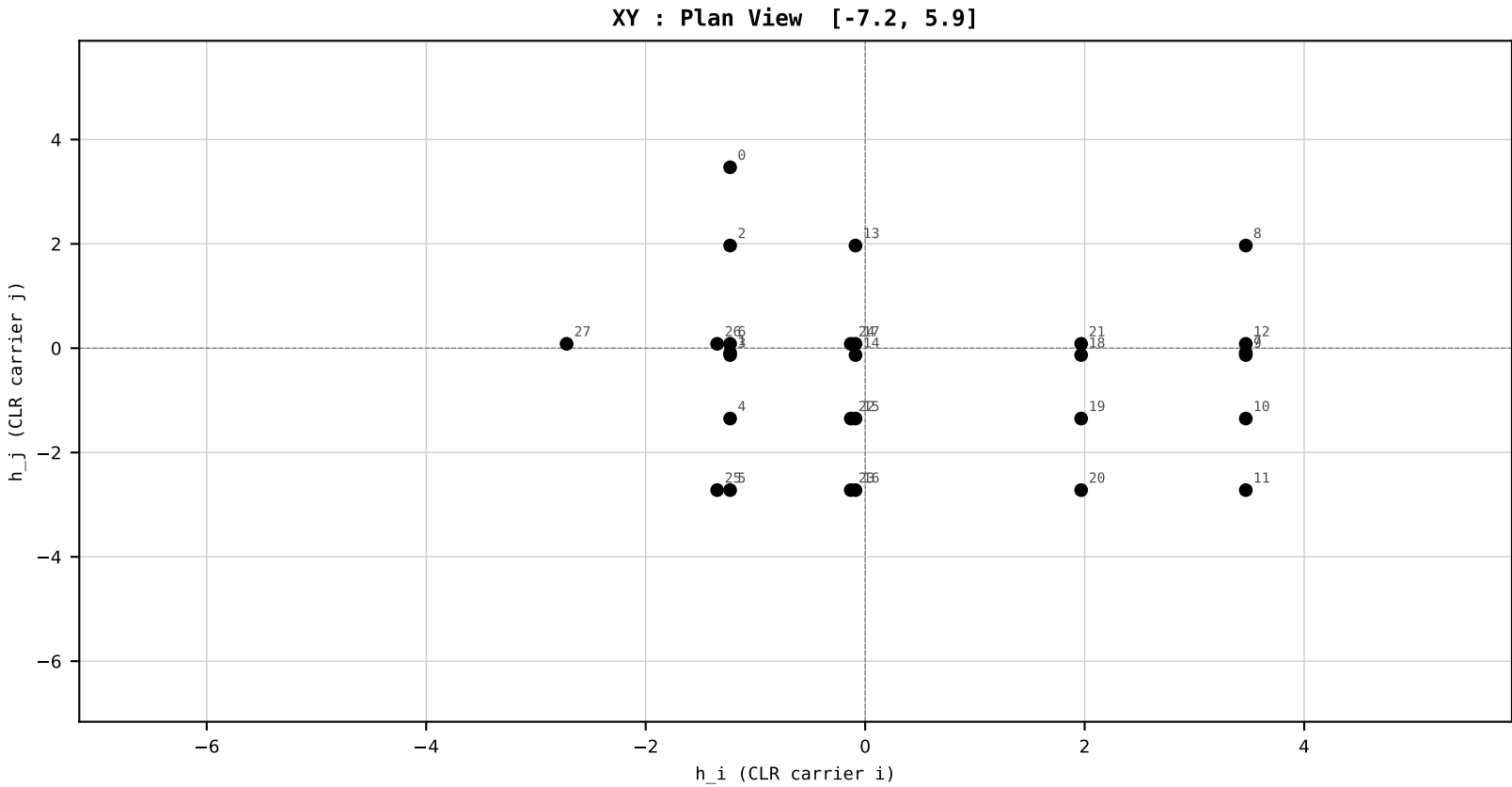
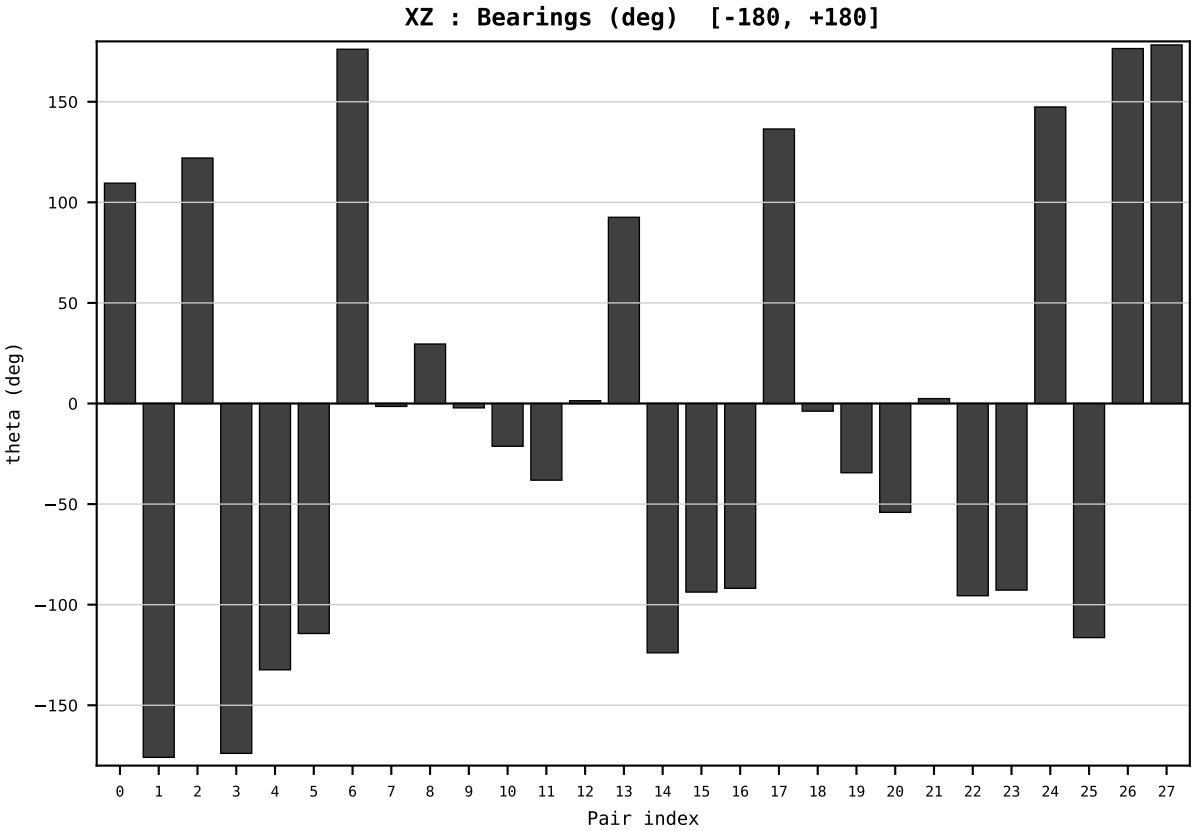
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.593561
Ring = Hs-4
E_metric= 26.6666
kappa_HS= 487.14
omega = 5.7742 deg
d_A = 0.712559
Helm = Solar
Helm d = +0.631278
DR = 6.189 HLR
DR ratio= 487.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

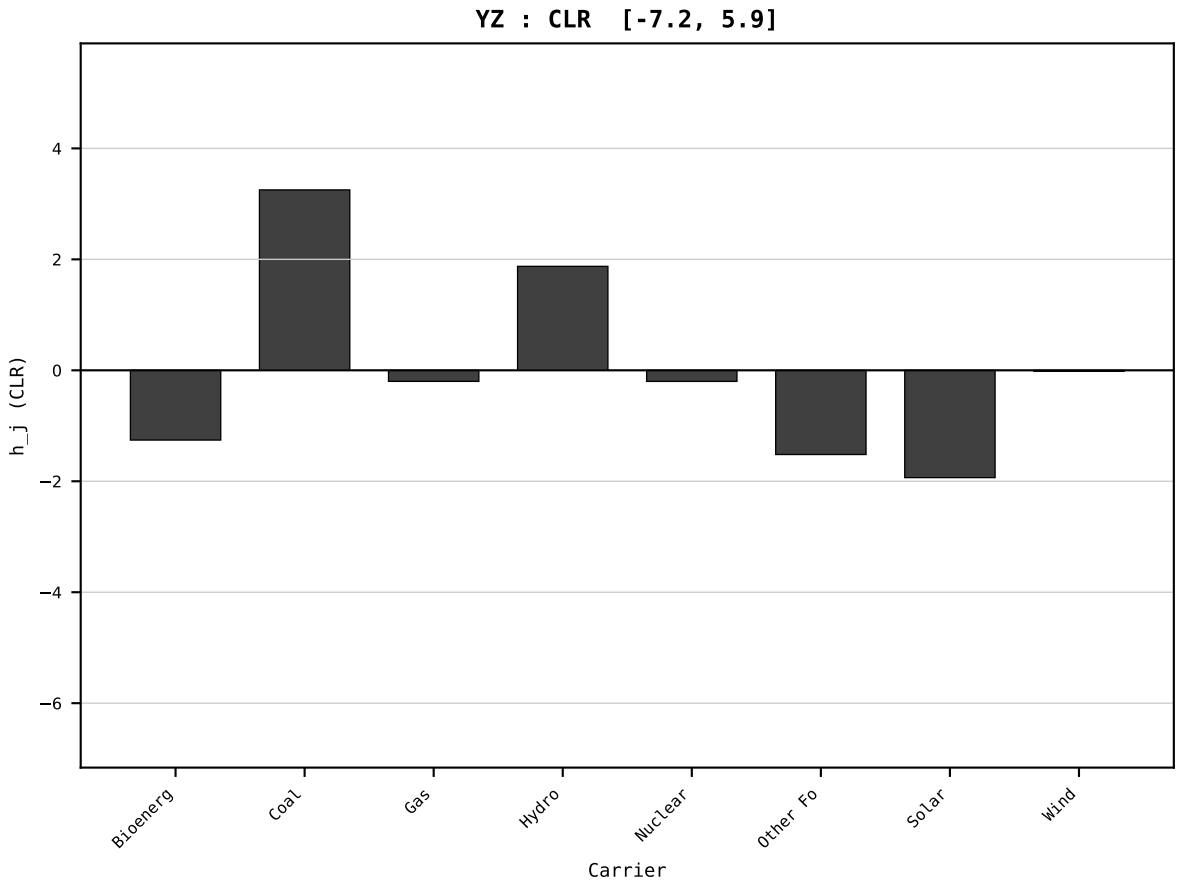
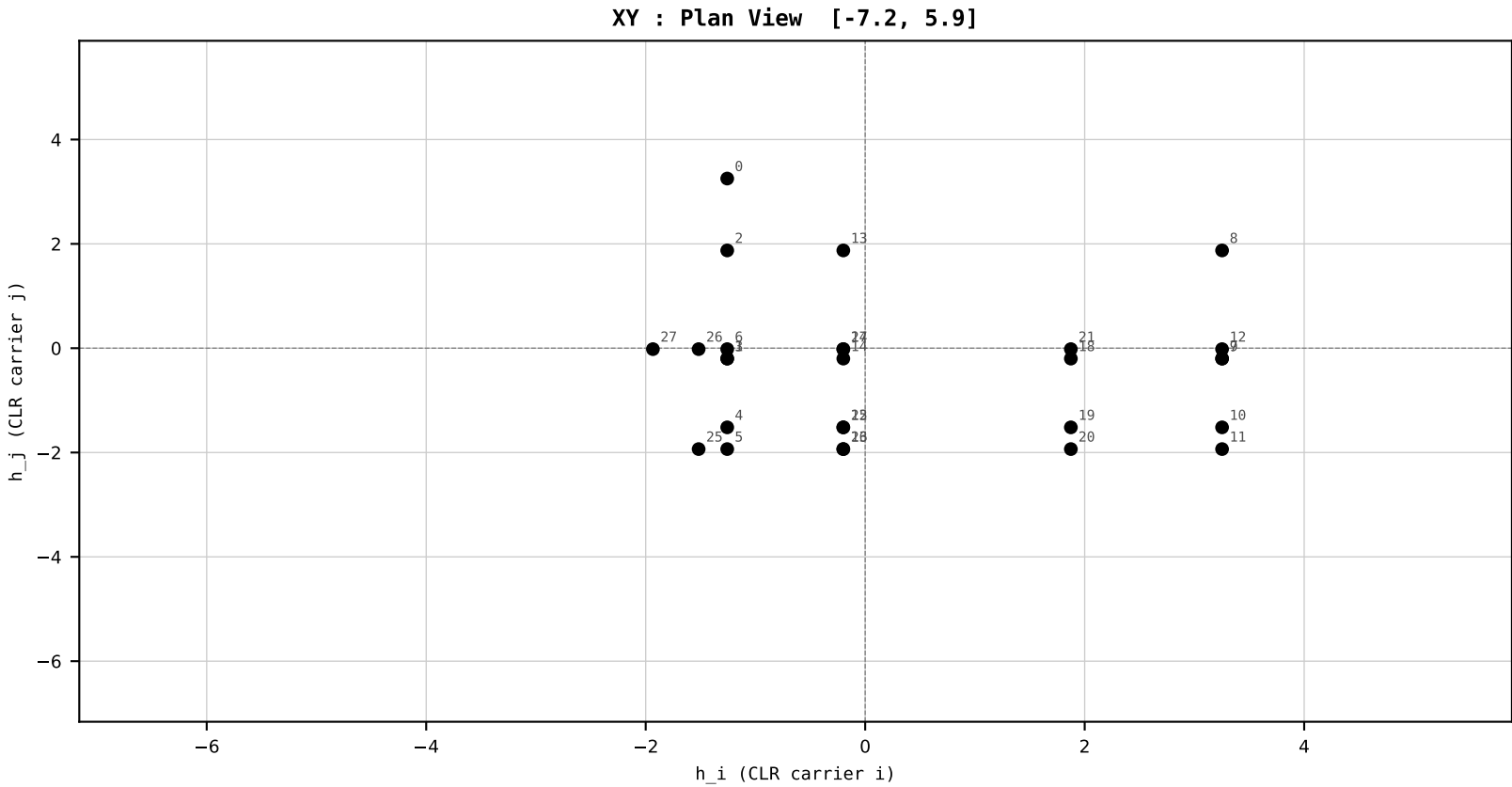
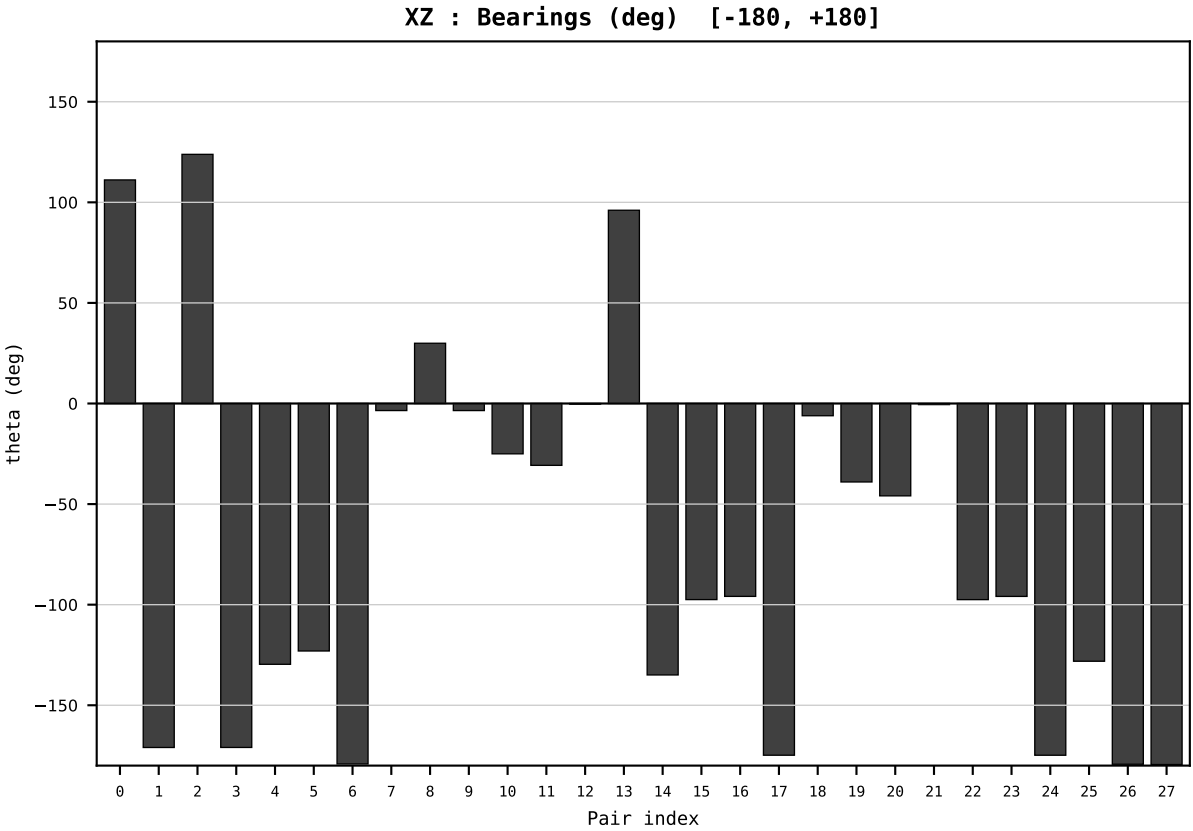
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.563173
Ring = Hs-4
E_metric= 21.7919
kappa_HS= 178.78
omega = 8.1303 deg
d_A = 0.854634
Helm = Solar
Helm d = +0.786340
DR = 5.186 HLR
DR ratio= 178.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

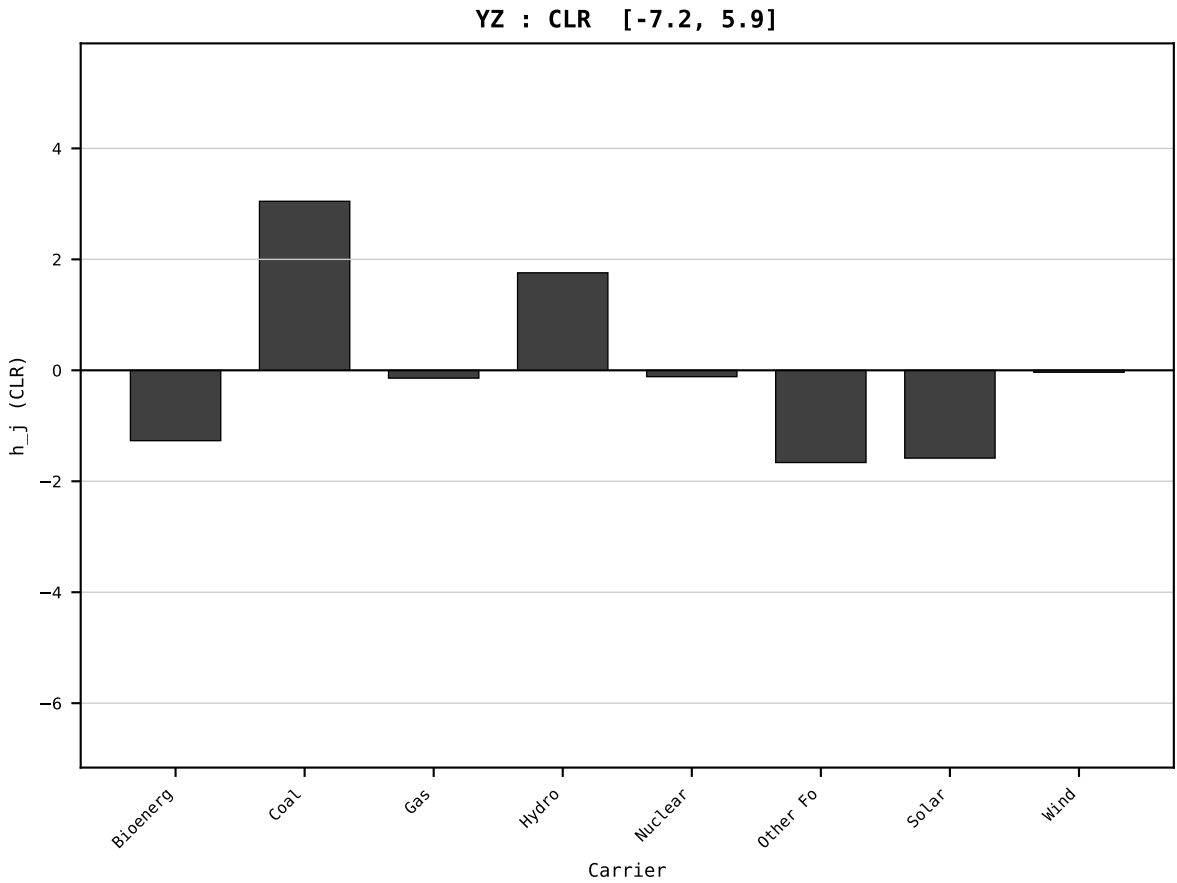
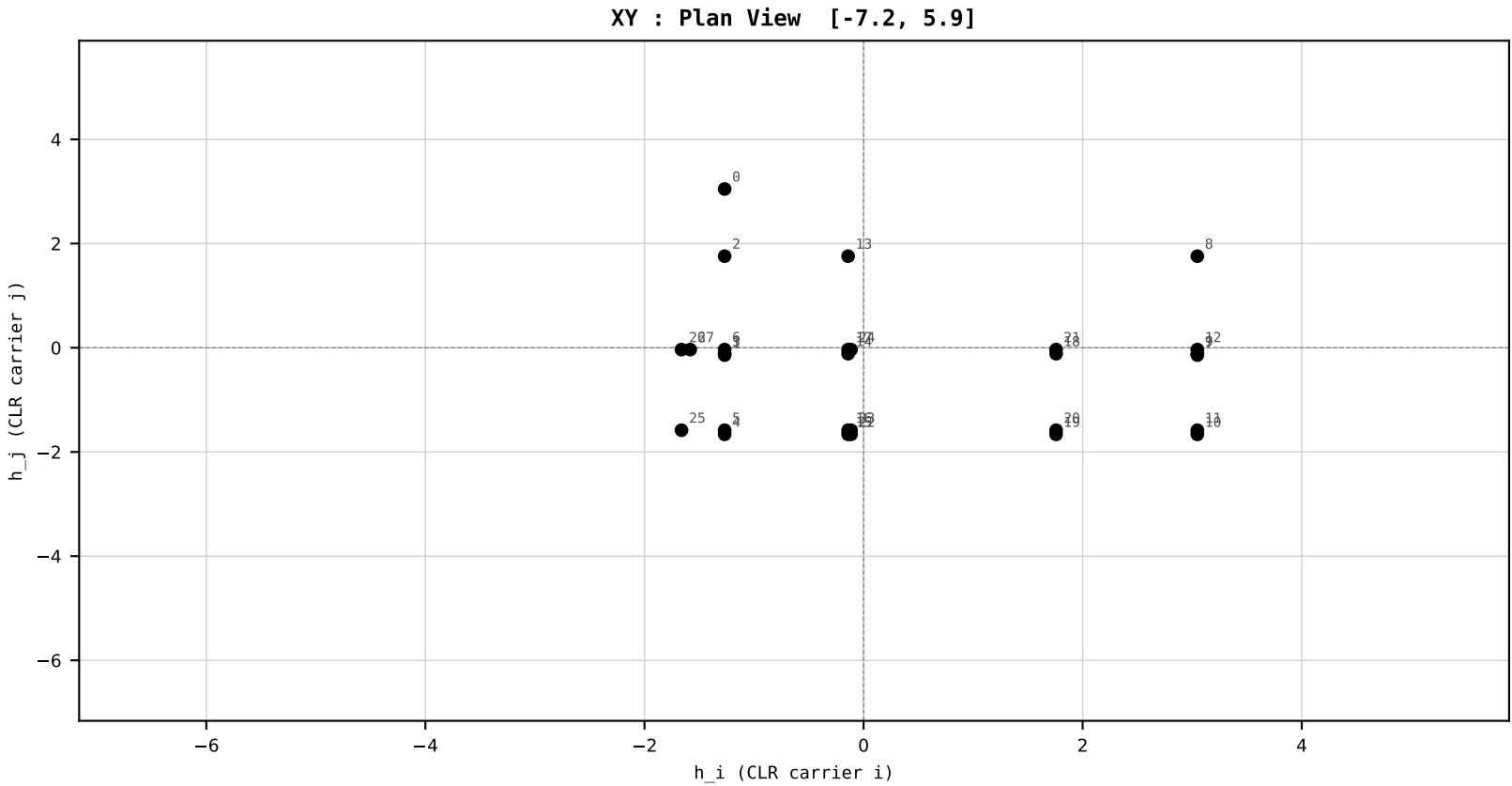
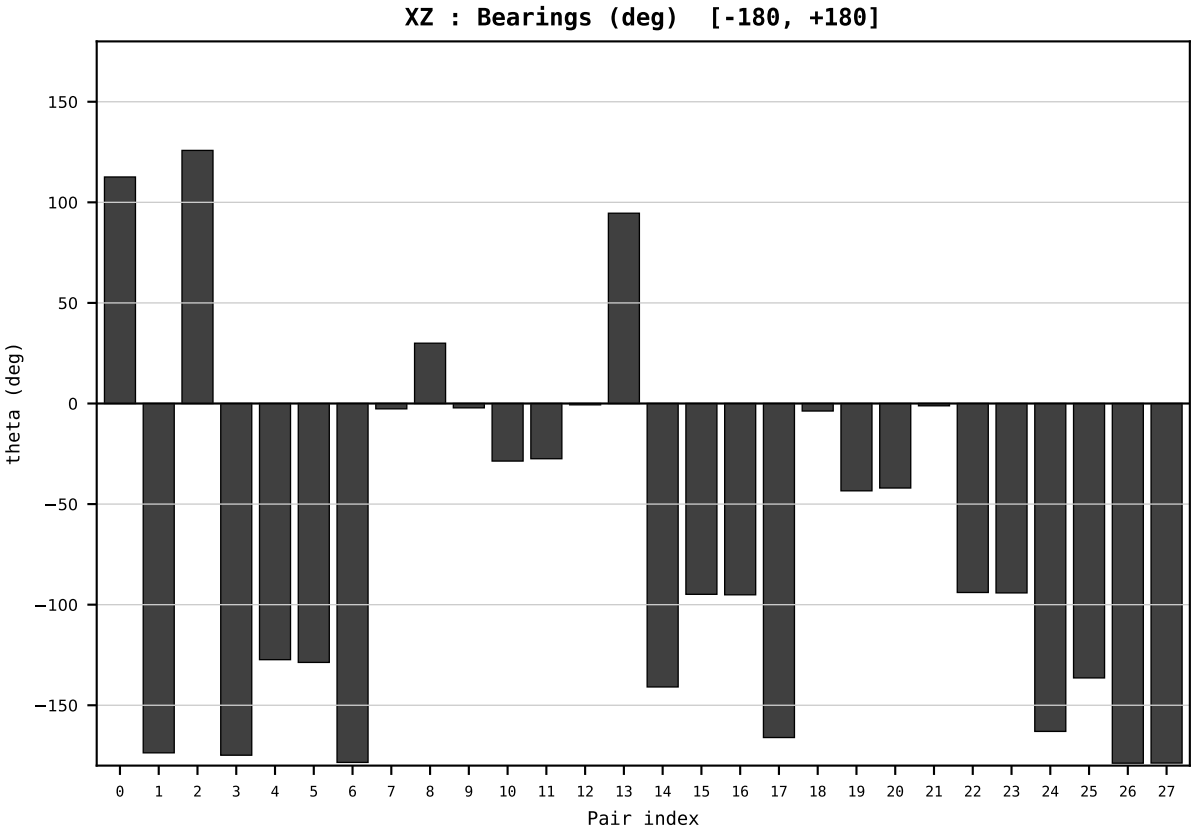
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.522035
Ring = Hs-4
E_metric= 19.2846
kappa_HS= 110.98
omega = 4.6414 deg
d_A = 0.459397
Helm = Solar
Helm d = +0.351431
DR = 4.709 HLR
DR ratio= 111.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



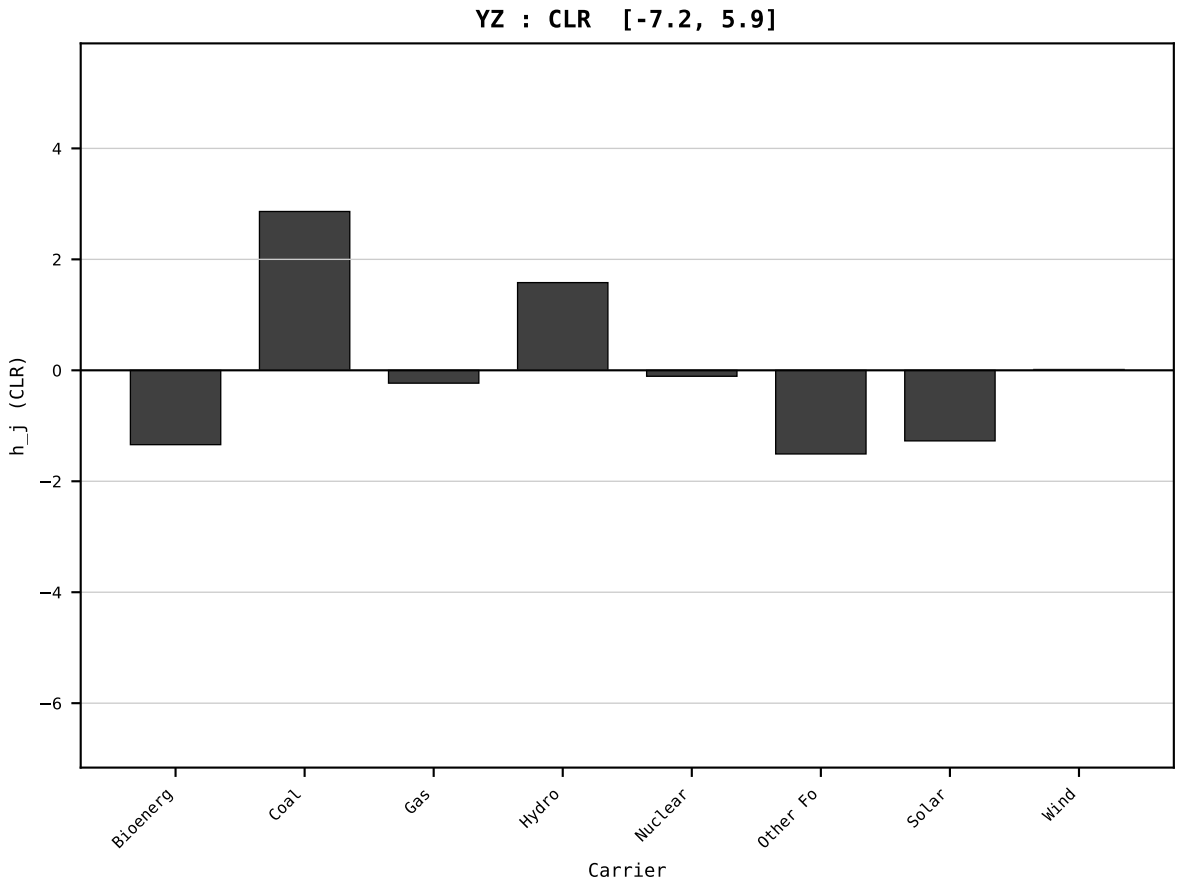
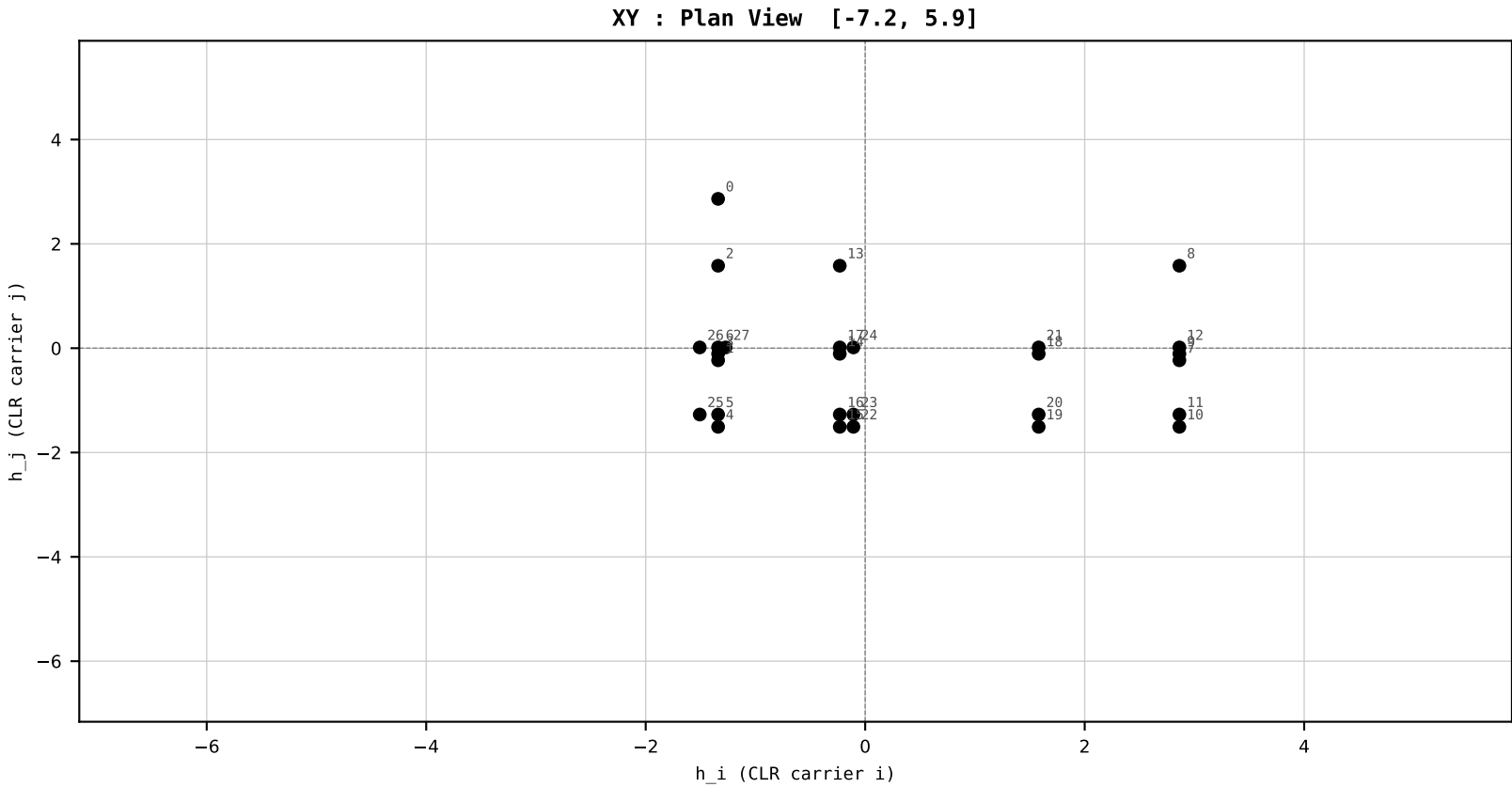
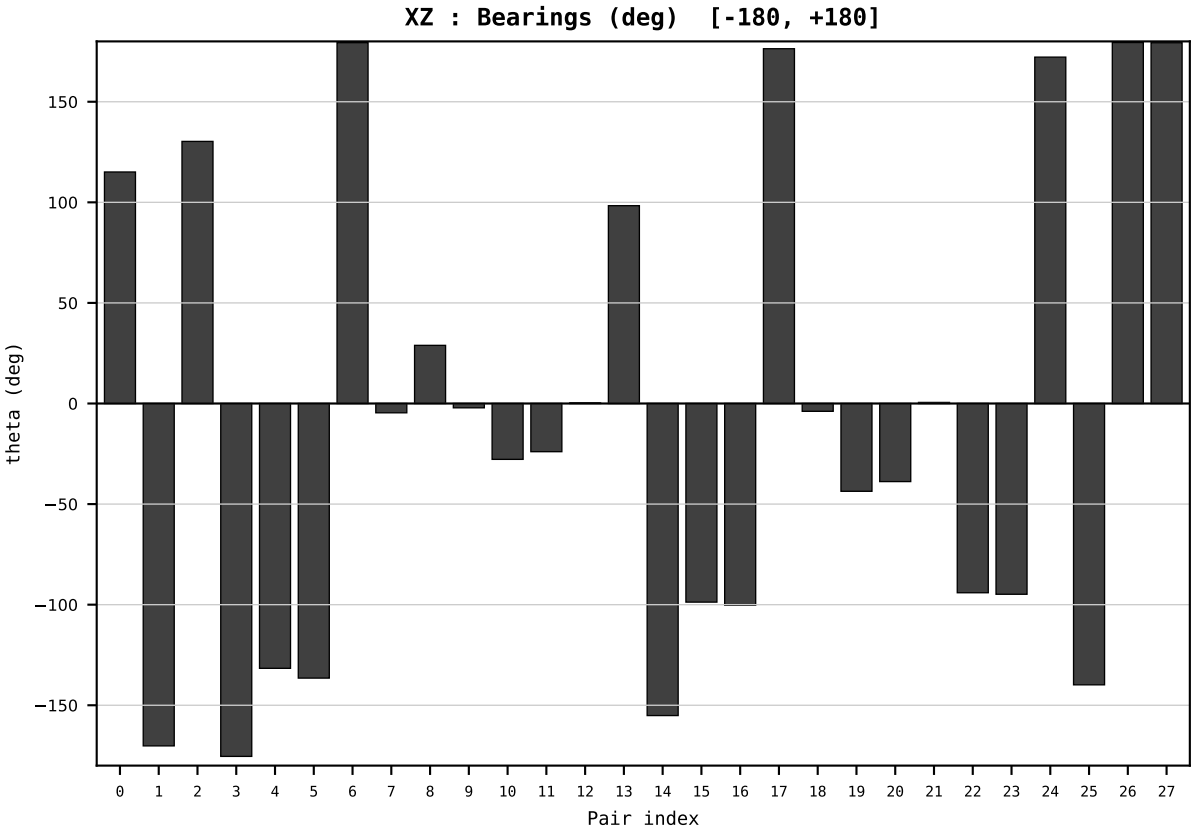
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.487785
Ring = Hs-3
E_metric= 16.4472
kappa_HS= 79.08
omega = 4.0472 deg
d_A = 0.449060
Helm = Solar
Helm d = +0.310984
DR = 4.370 HLR
DR ratio= 79.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

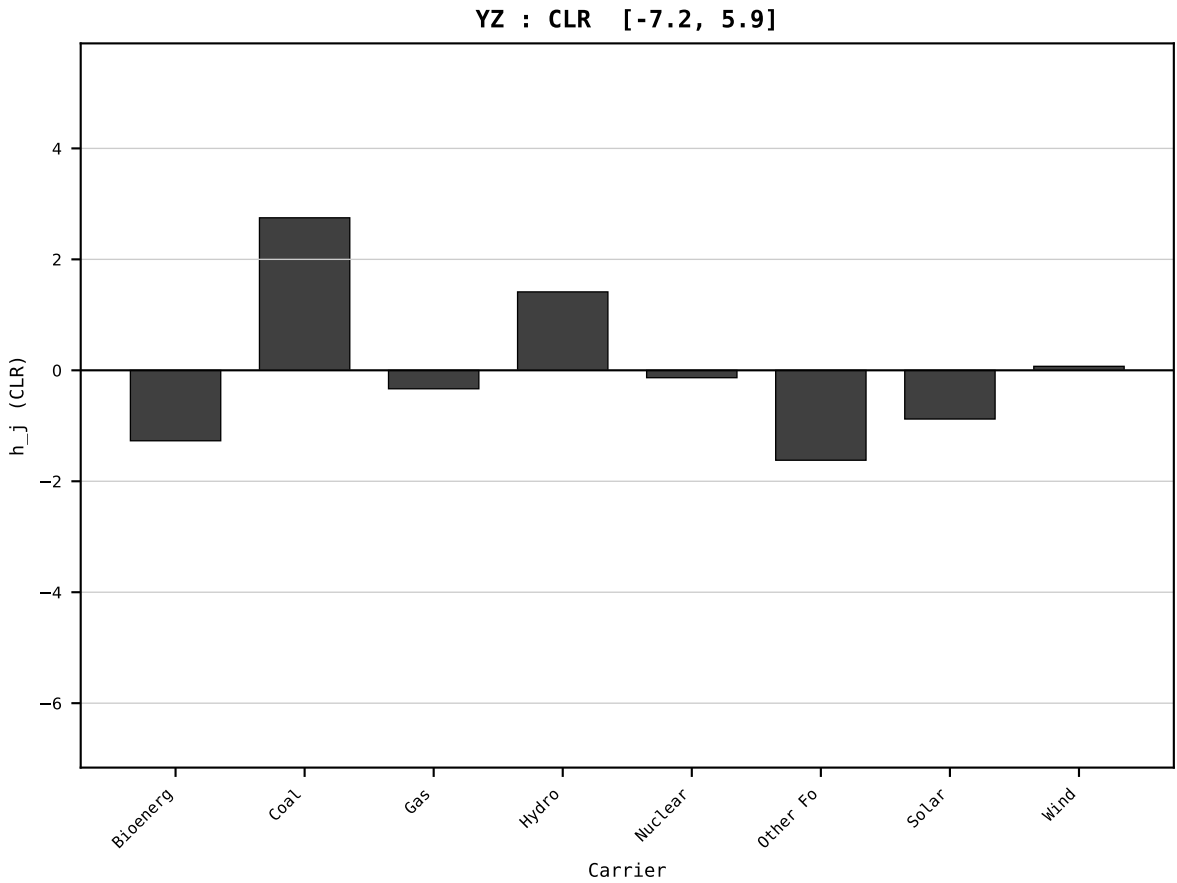
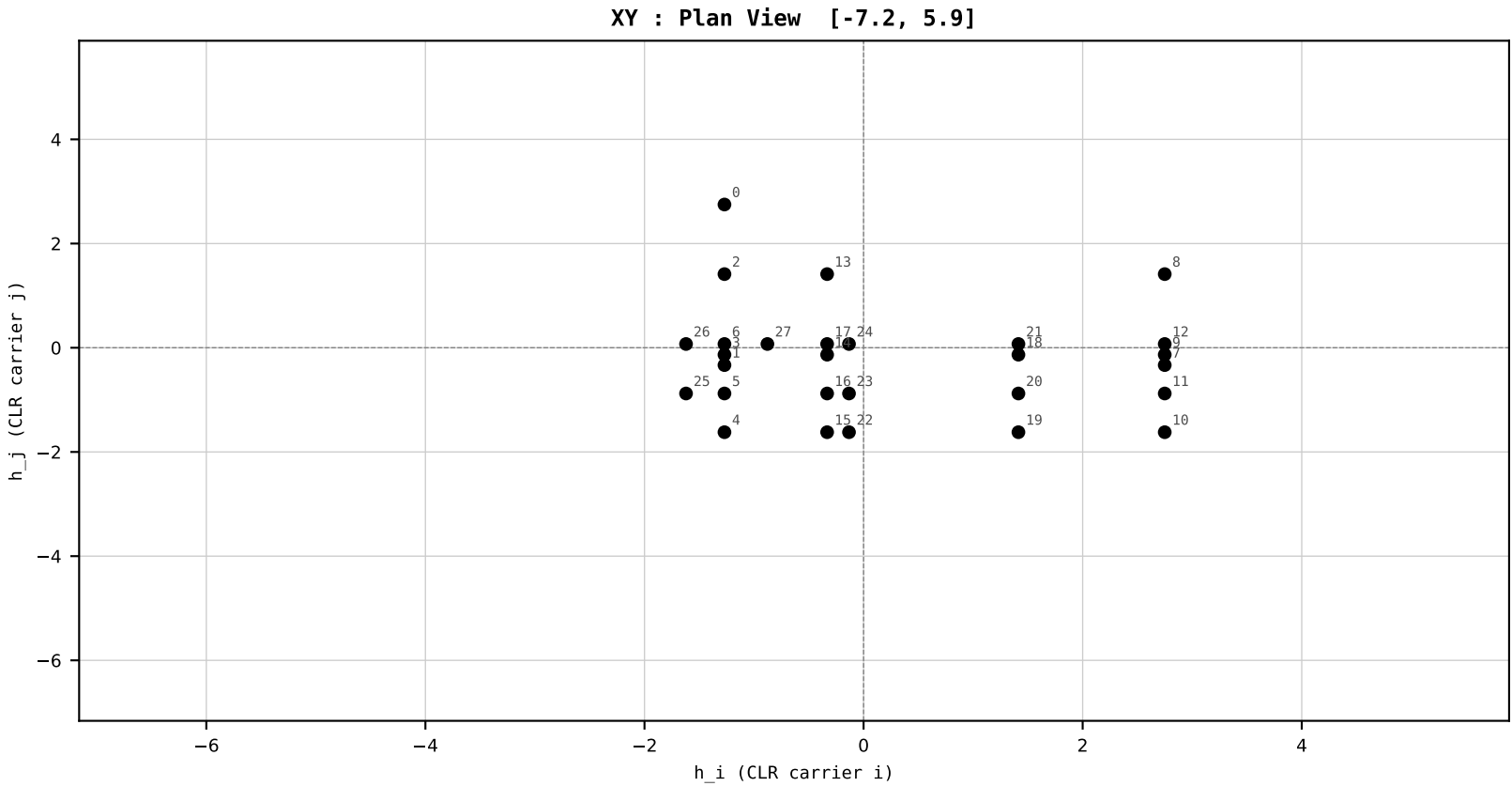
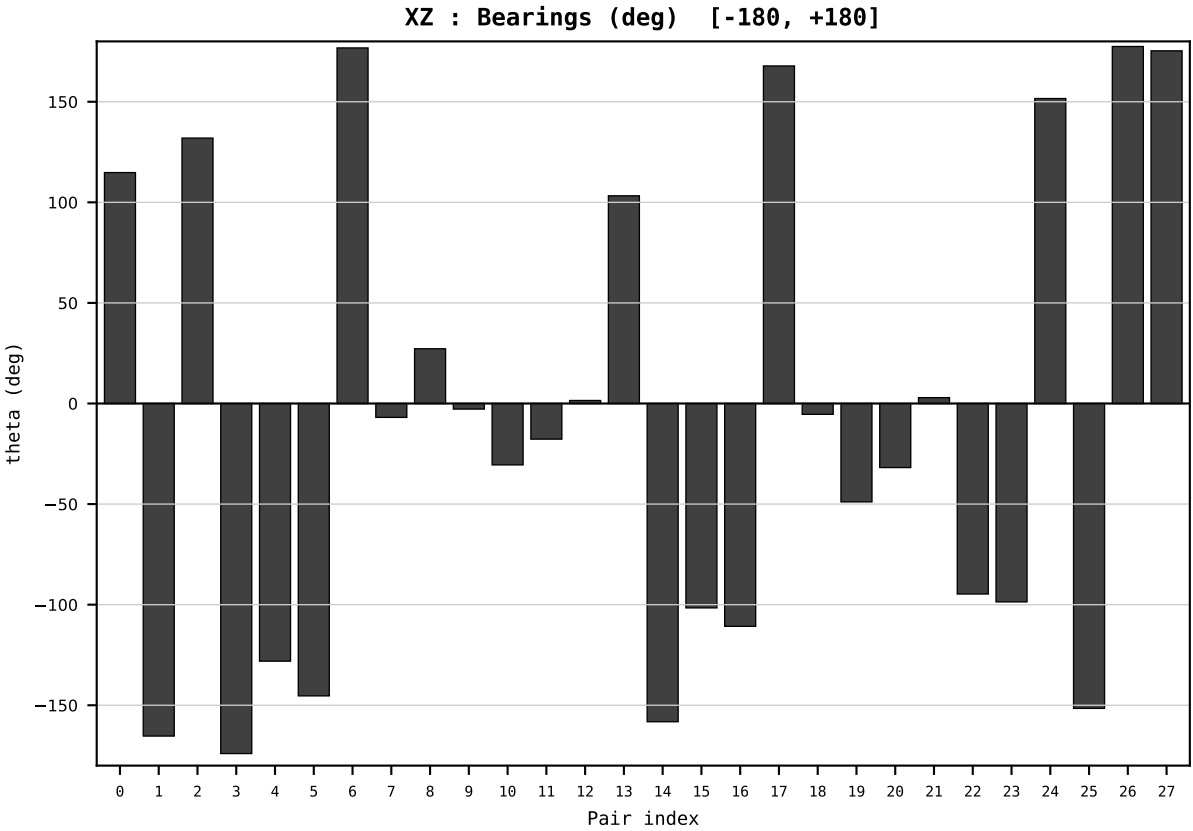
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.465364
Ring = Hs-3
E_metric= 14.7002
kappa_HS= 79.04
omega = 6.1498 deg
d_A = 0.477487
Helm = Solar
Helm d = +0.393824
DR = 4.370 HLR
DR ratio= 79.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

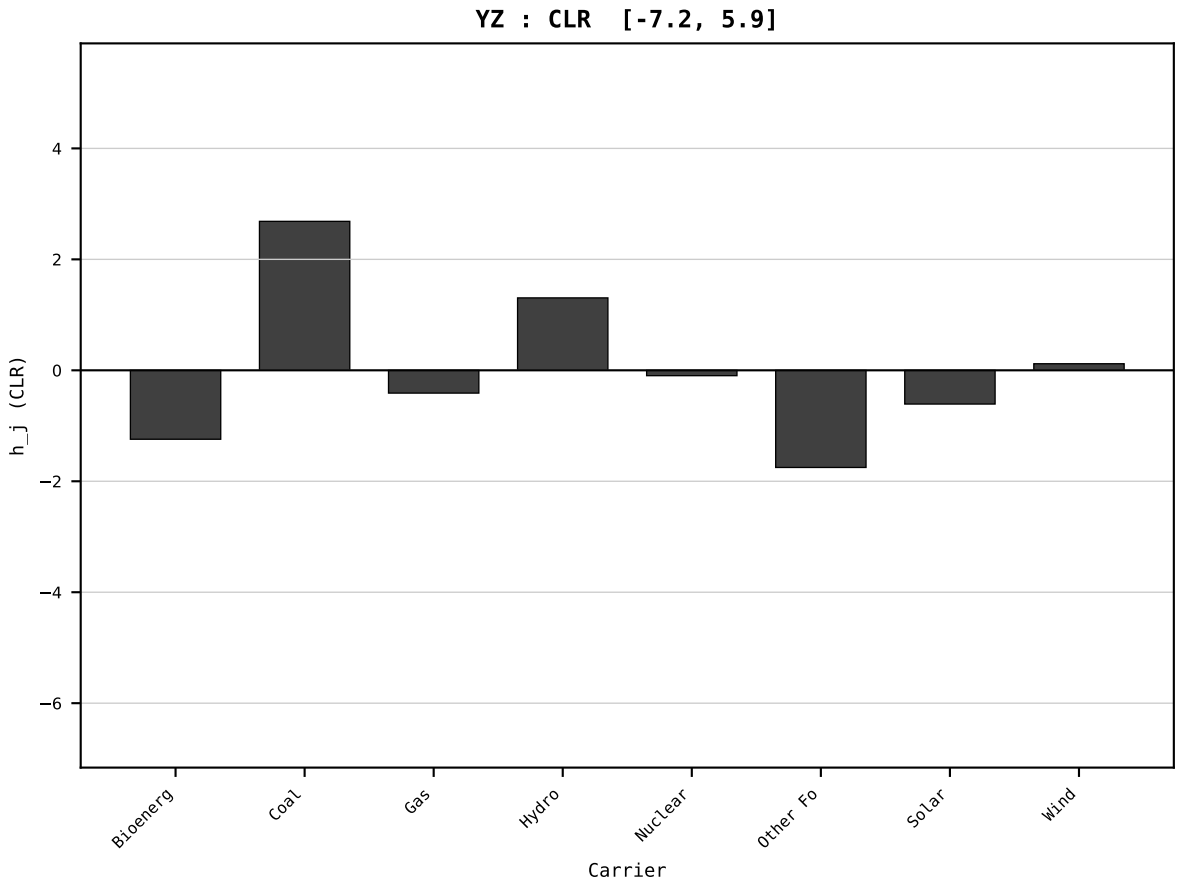
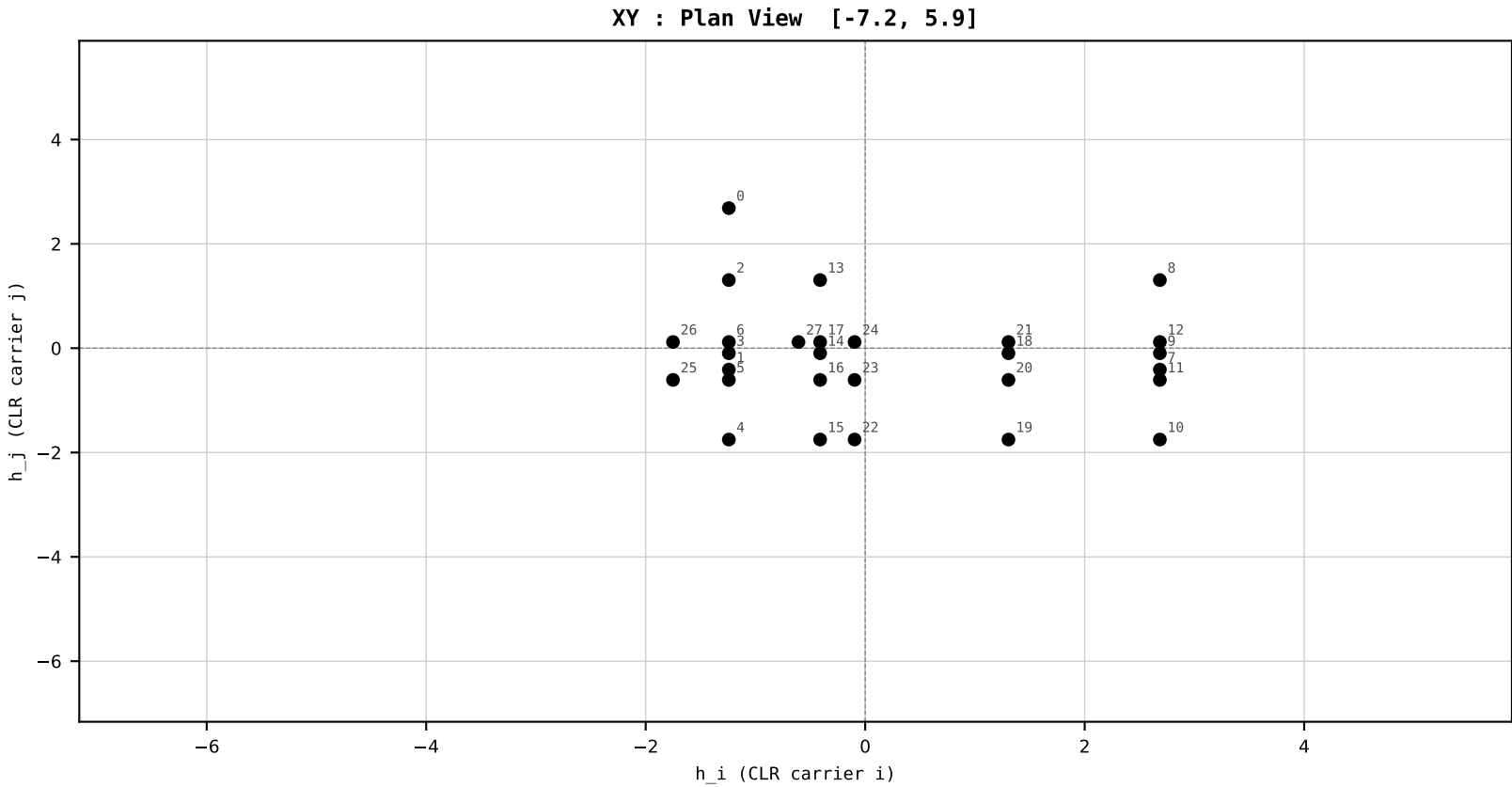
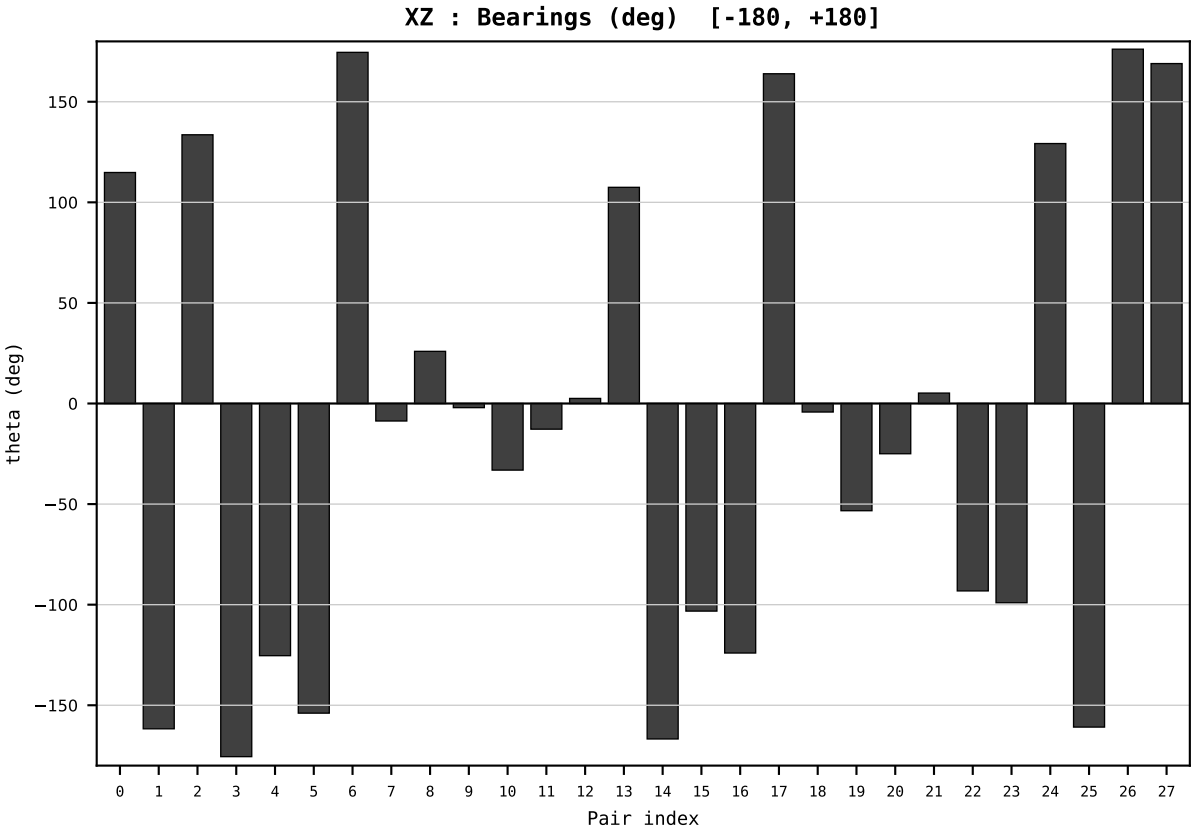
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.449187
Ring = Hs-3
E_metric= 14.0866
kappa_HS= 84.47
omega = 4.9977 deg
d_A = 0.340525
Helm = Solar
Helm d = +0.269941
DR = 4.436 HLR
DR ratio= 84.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=8 N=26 pairs=28

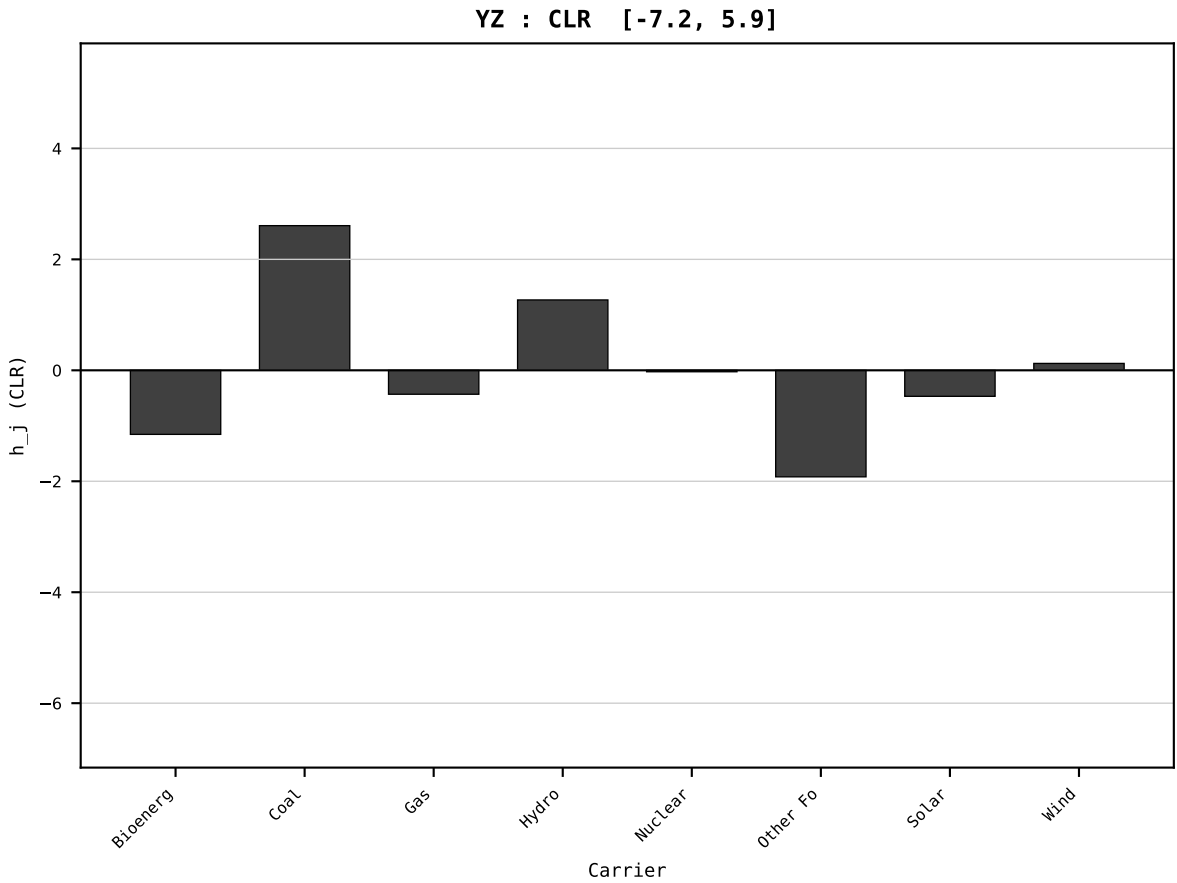
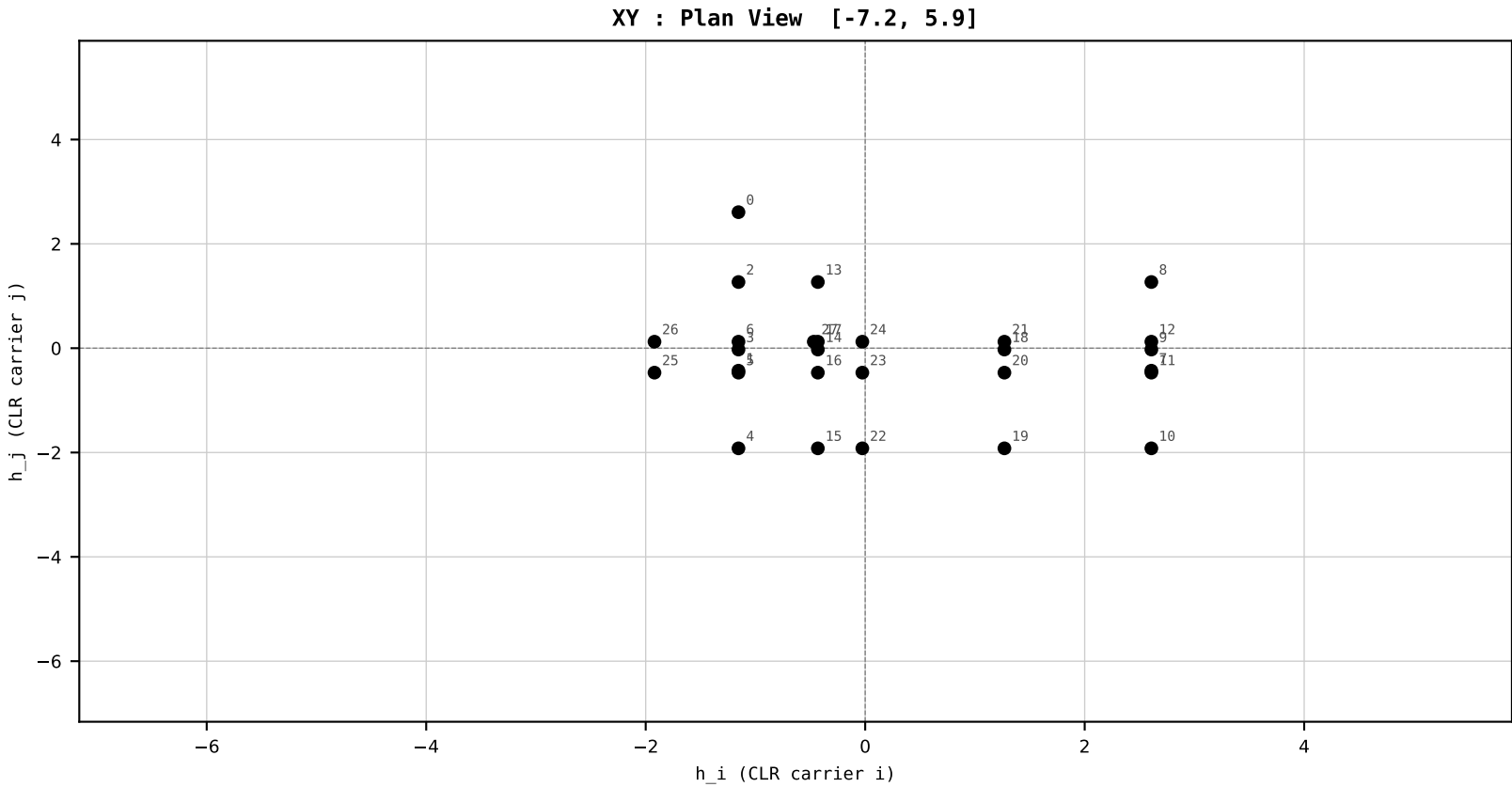
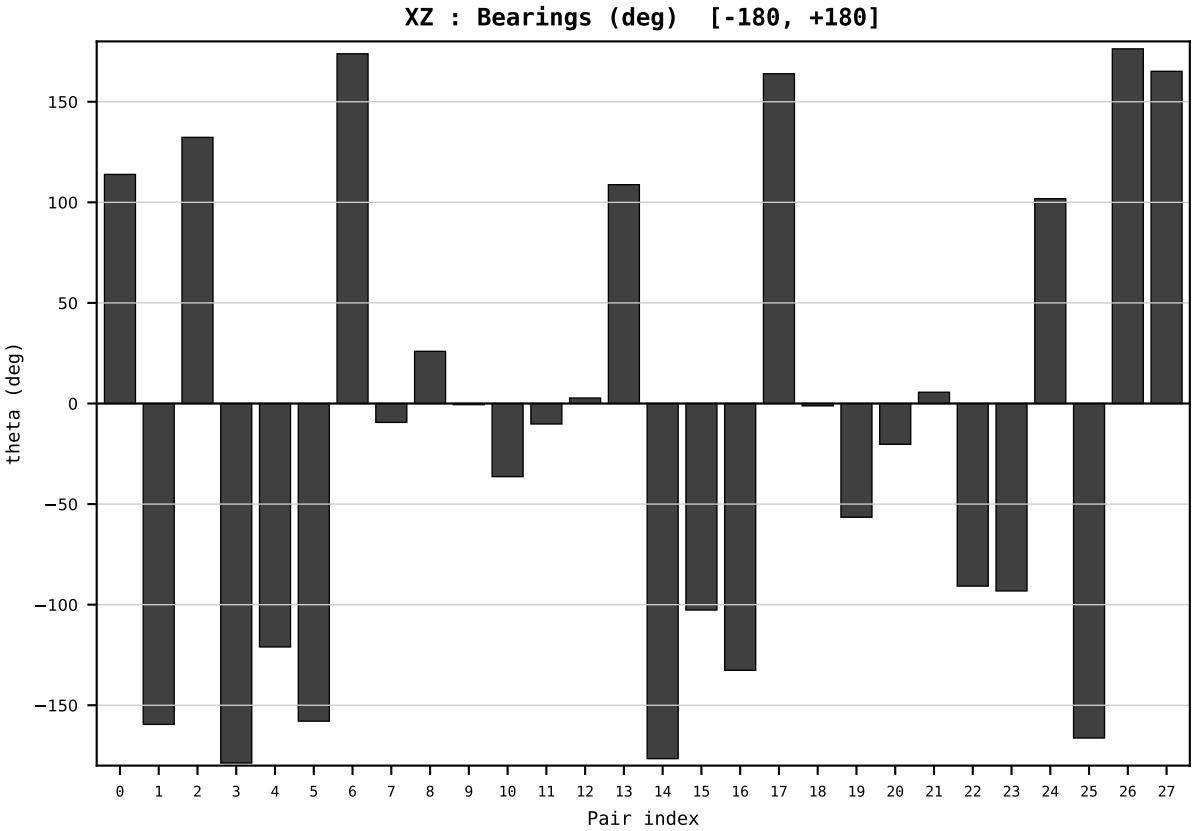
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.426218
Ring = Hs-3
E_metric= 13.8498
kappa_HS= 92.53
omega = 3.9842 deg
d_A = 0.261754
Helm = Other Fossil
Helm d = -0.168801
DR = 4.528 HLR
DR ratio= 92.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

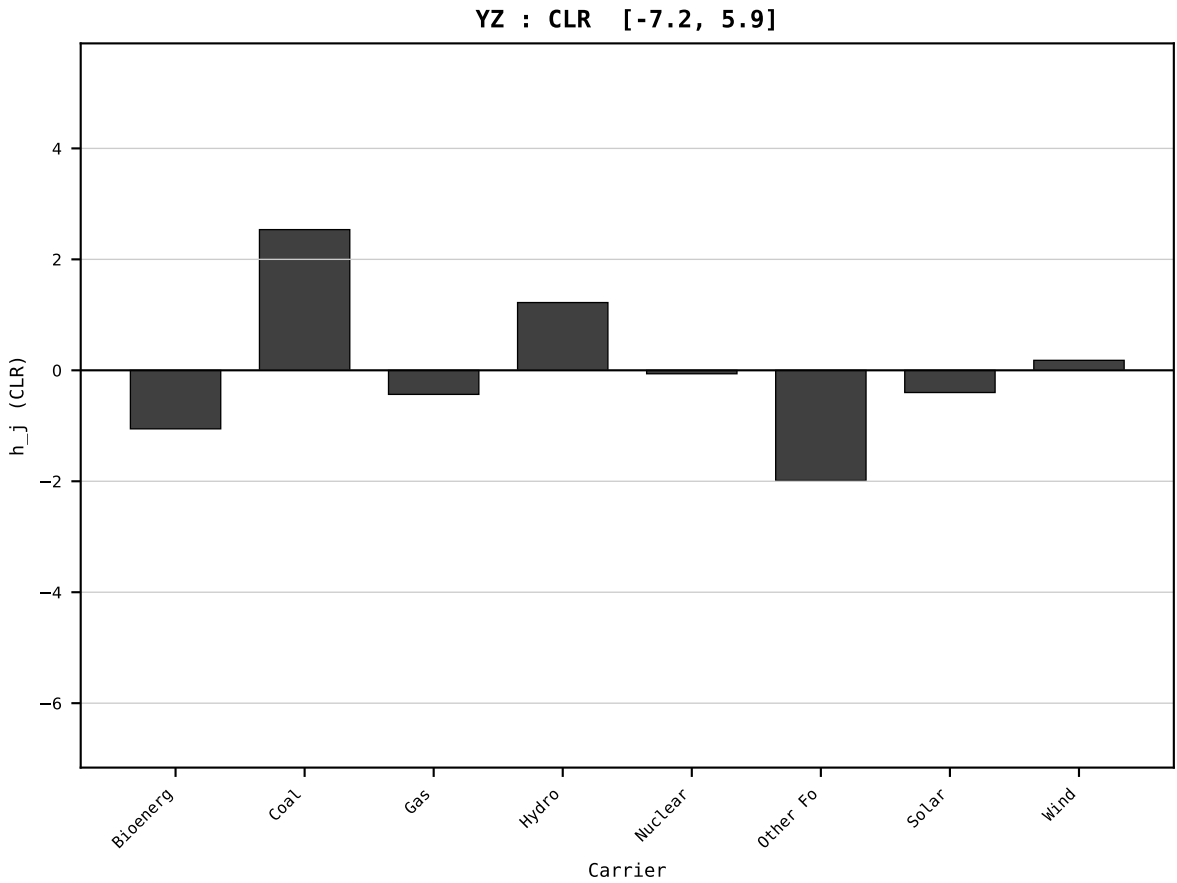
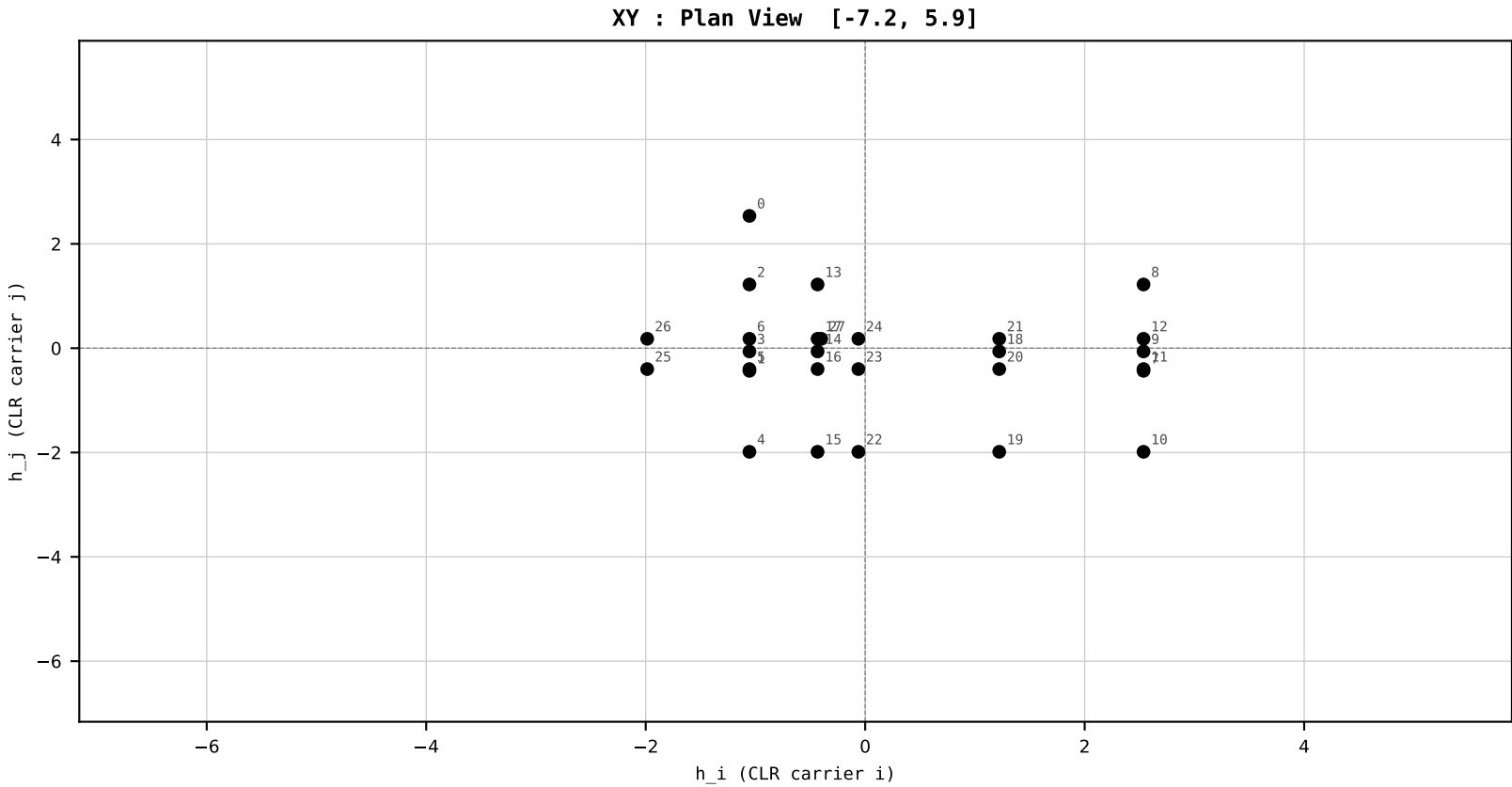
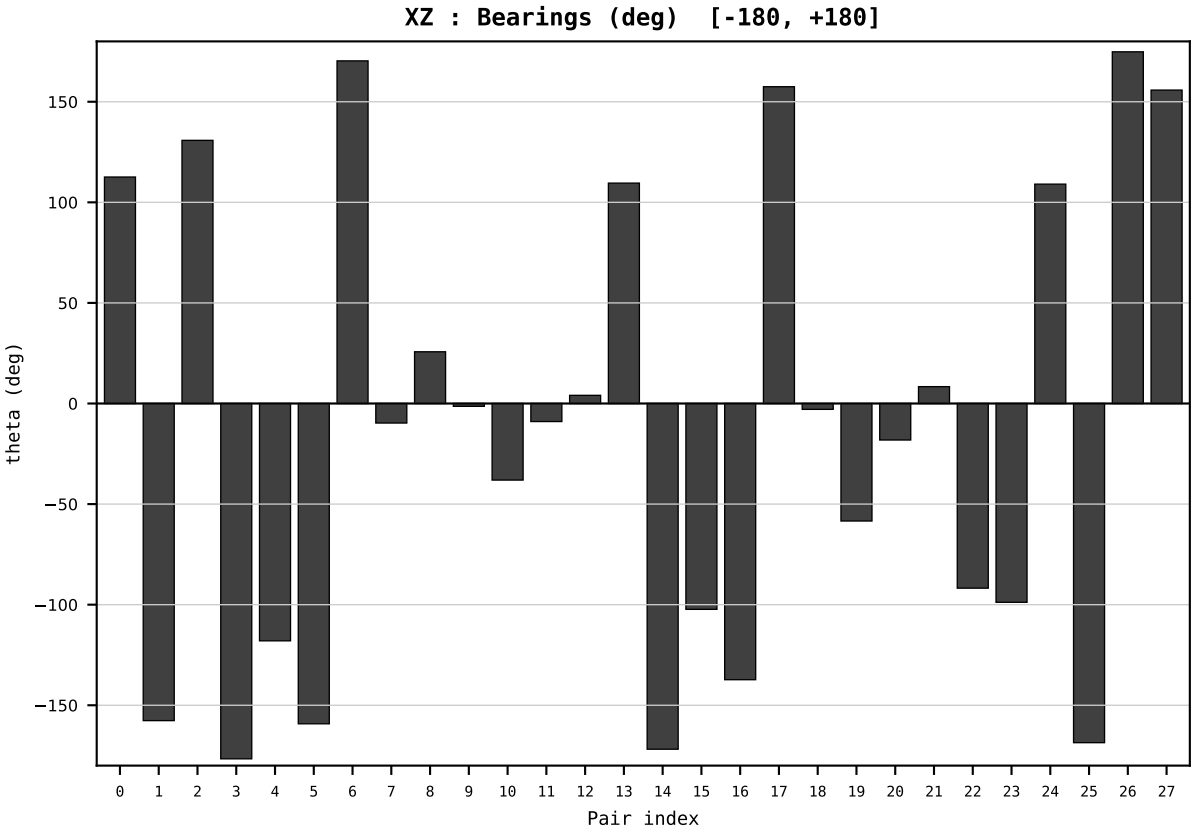
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.406843
Ring = Hs-3
E_metric= 13.3663
kappa_HS= 92.05
omega = 2.5244 deg
d_A = 0.175223
Helm = Bioenergy
Helm d = +0.099748
DR = 4.522 HLR
DR ratio= 92.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

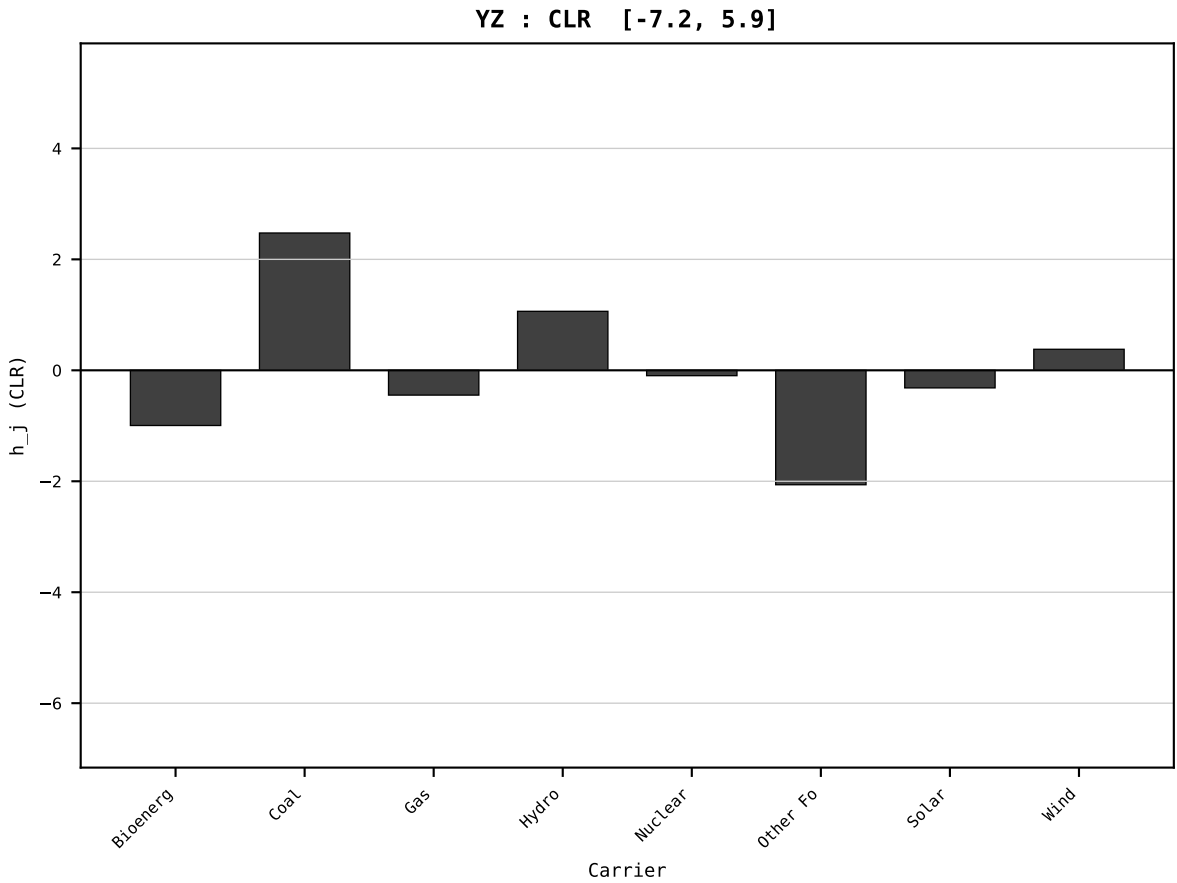
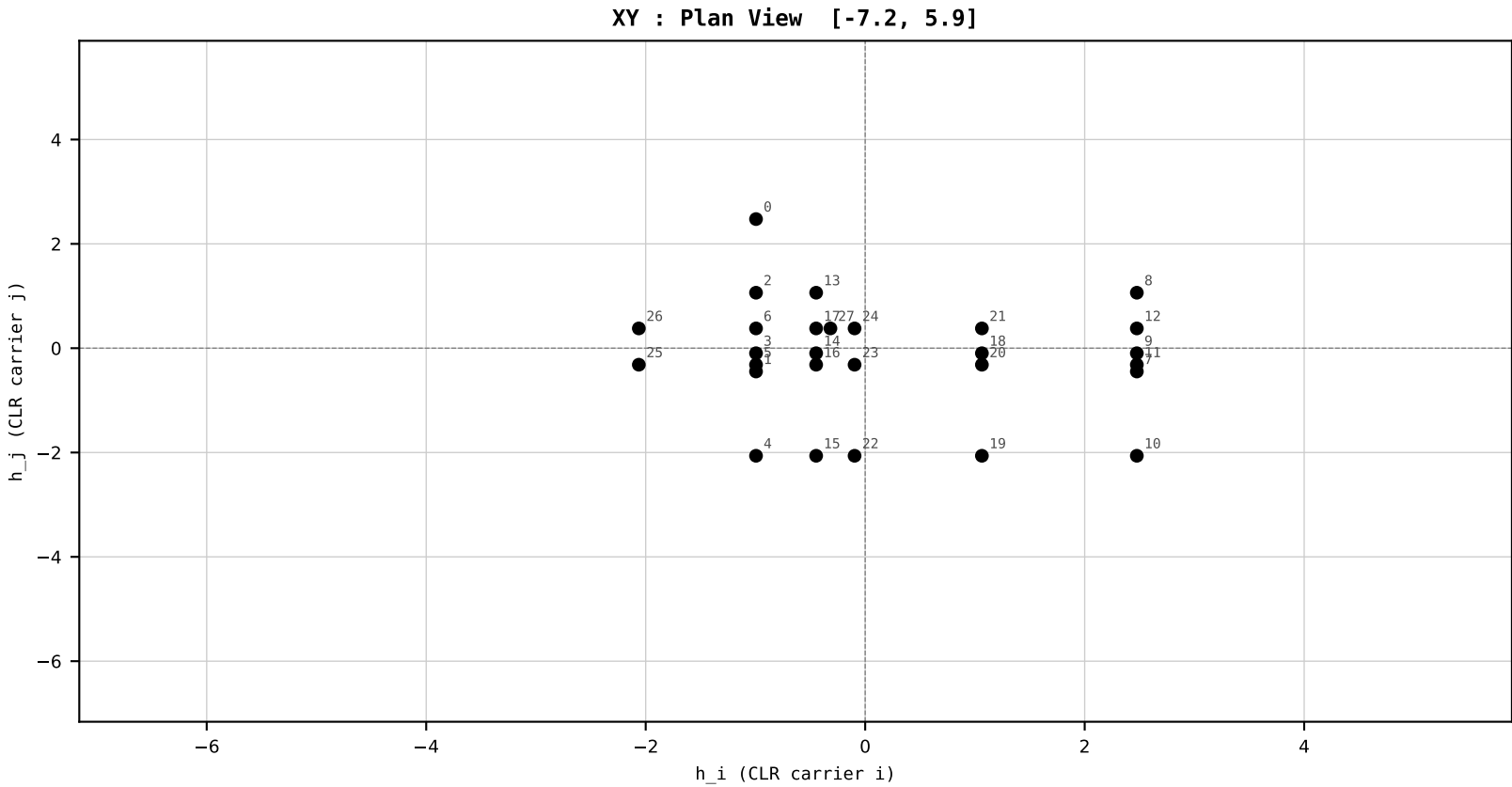
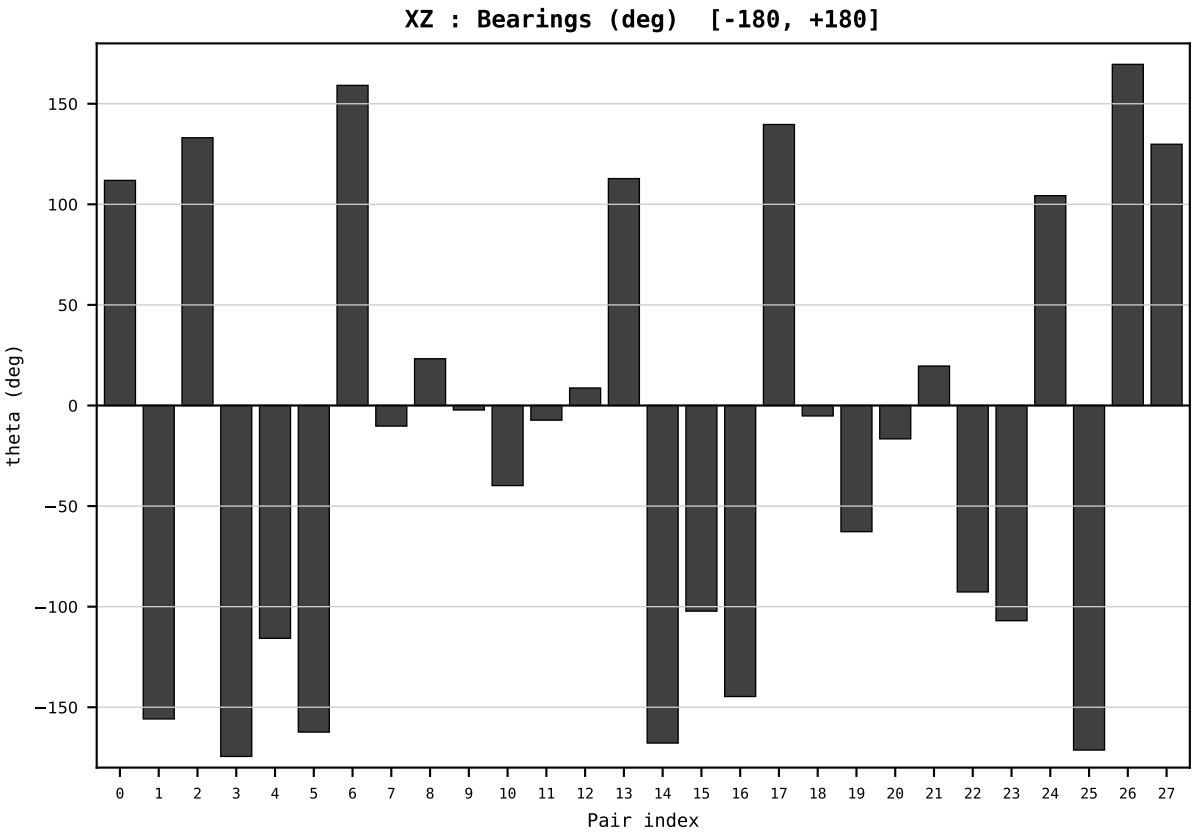
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.388284
Ring = Hs-3
E_metric= 12.9540
kappa_HS= 93.48
omega = 4.5501 deg
d_A = 0.293552
Helm = Wind
Helm d = +0.199189
DR = 4.538 HLR
DR ratio= 93.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

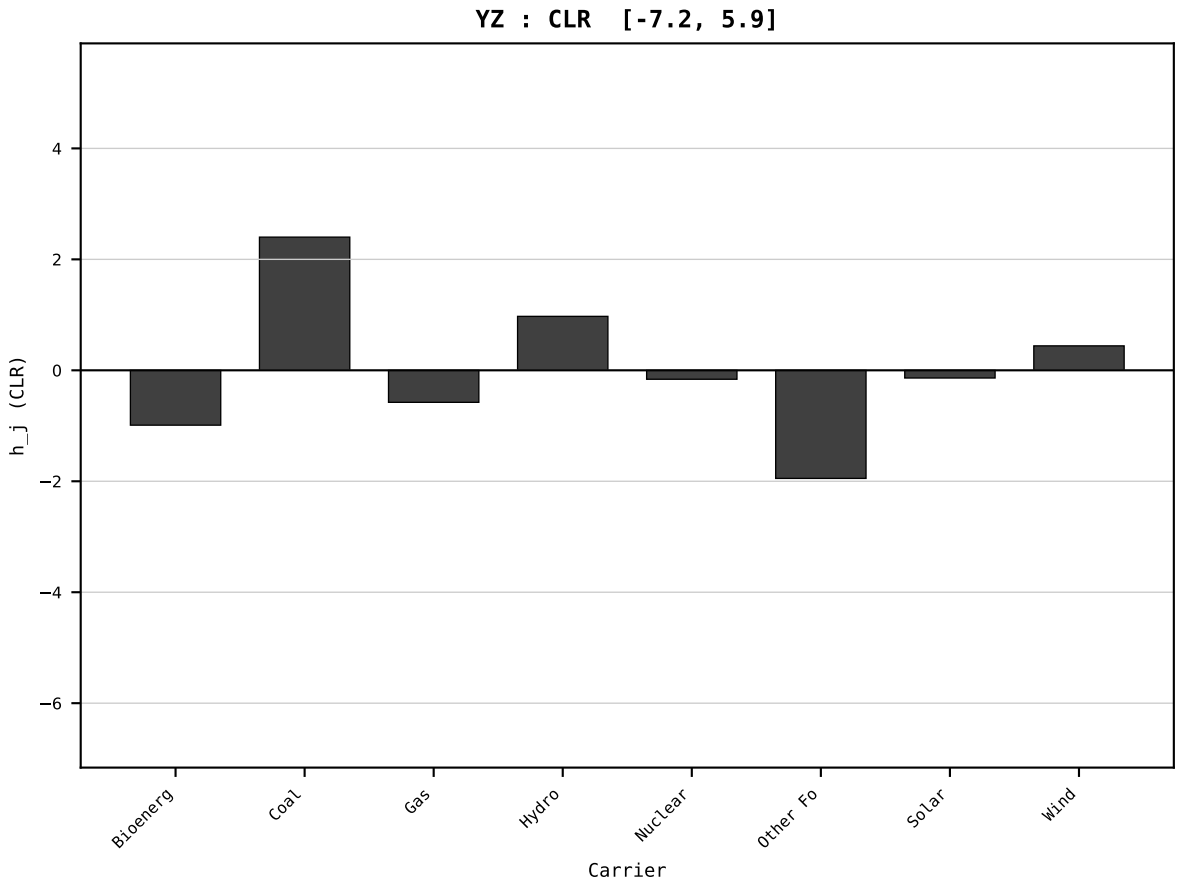
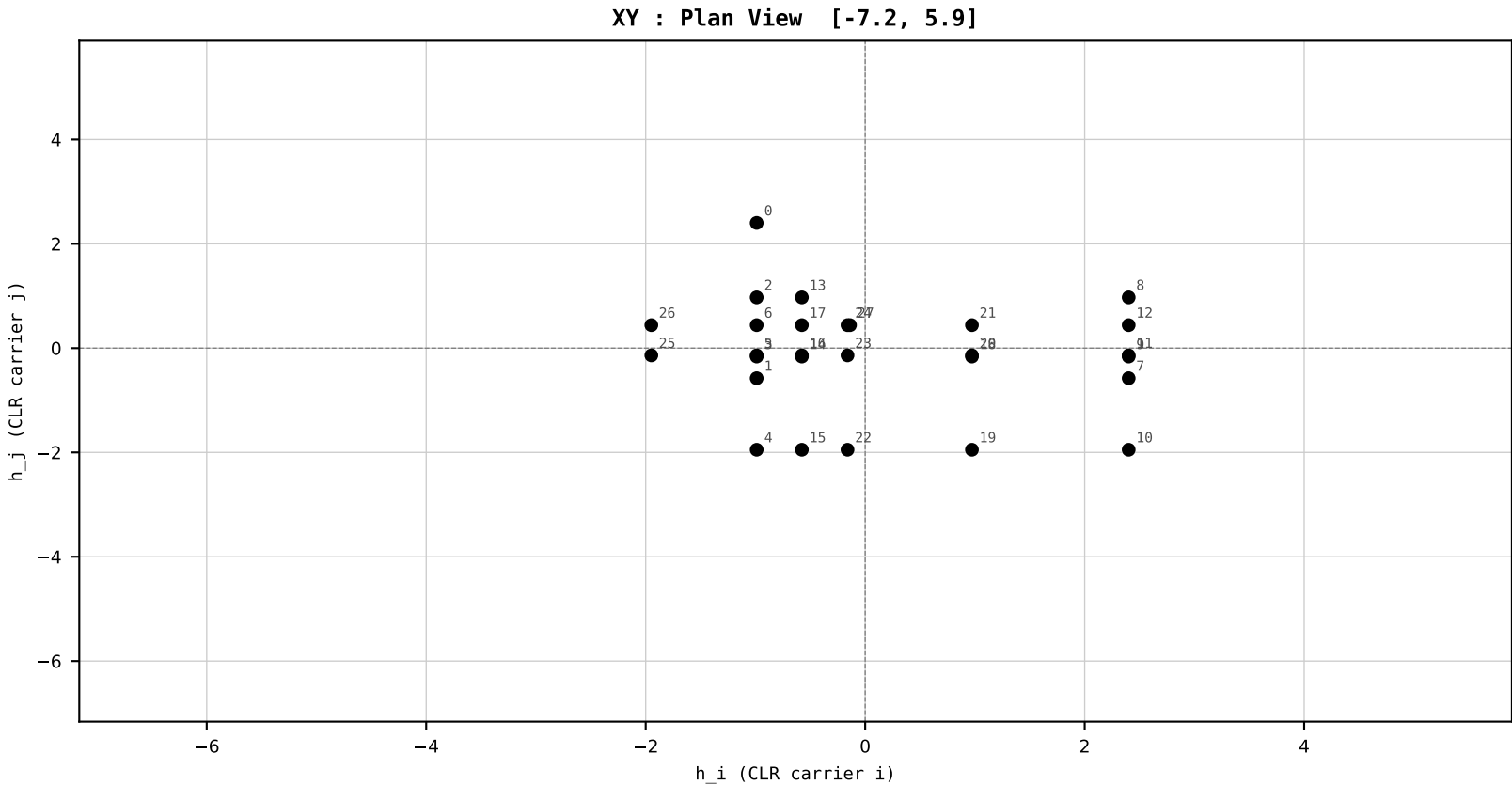
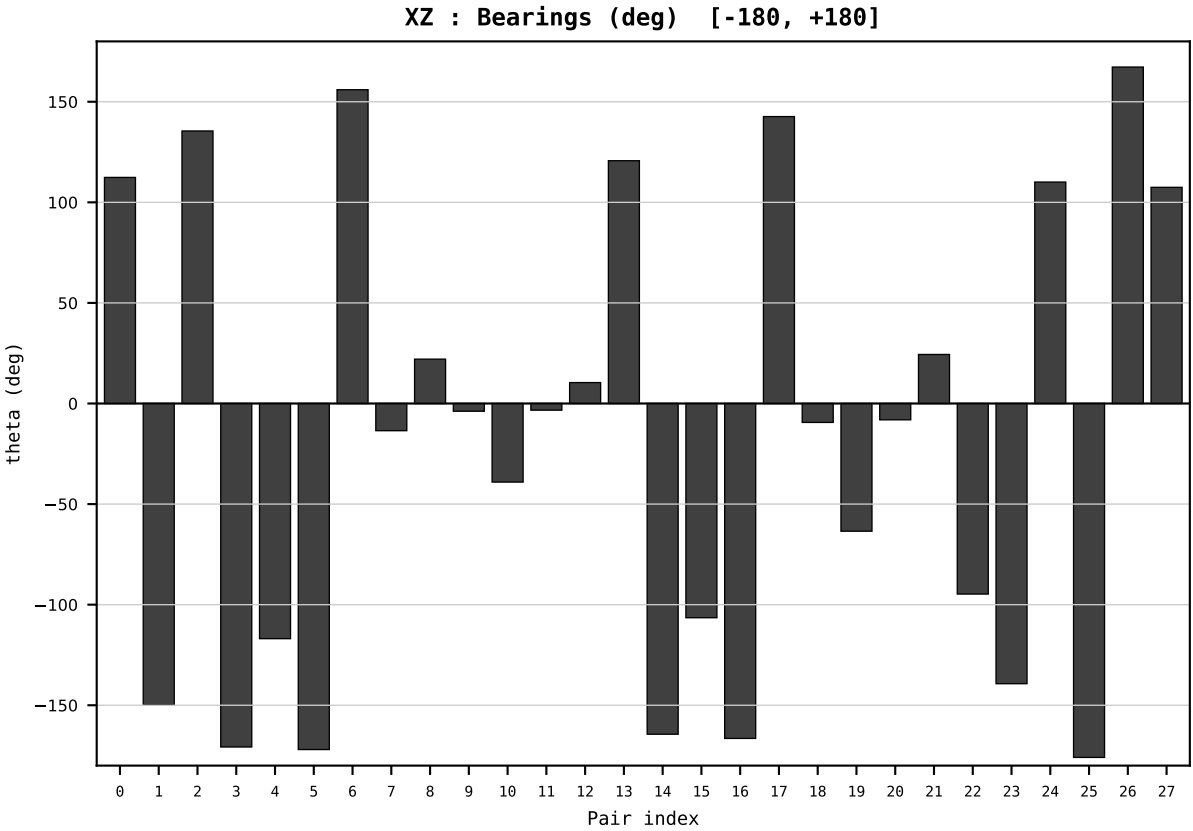
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.369899
Ring = Hs-3
E_metric= 12.0569
kappa_HS= 77.43
omega = 4.2062 deg
d_A = 0.288819
Helm = Solar
Helm d = +0.178100
DR = 4.349 HLR
DR ratio= 77.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

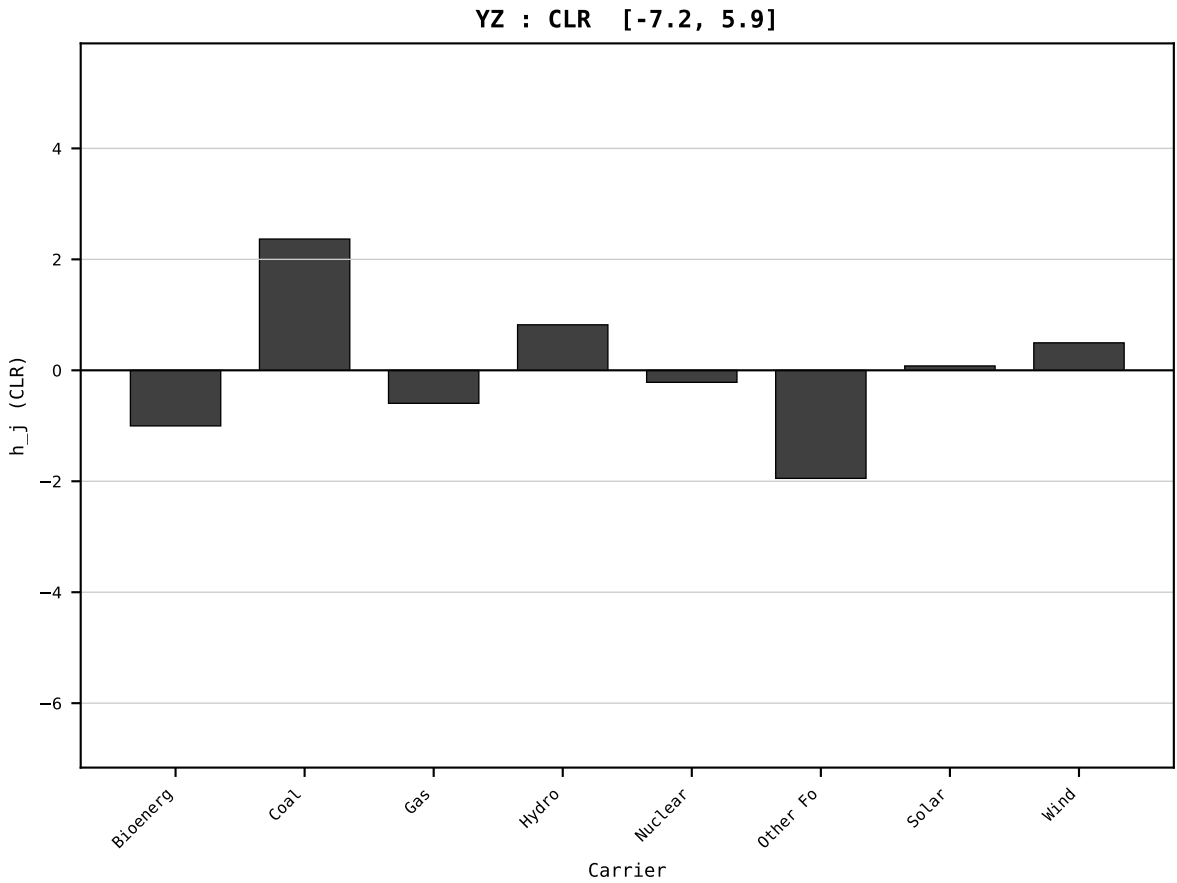
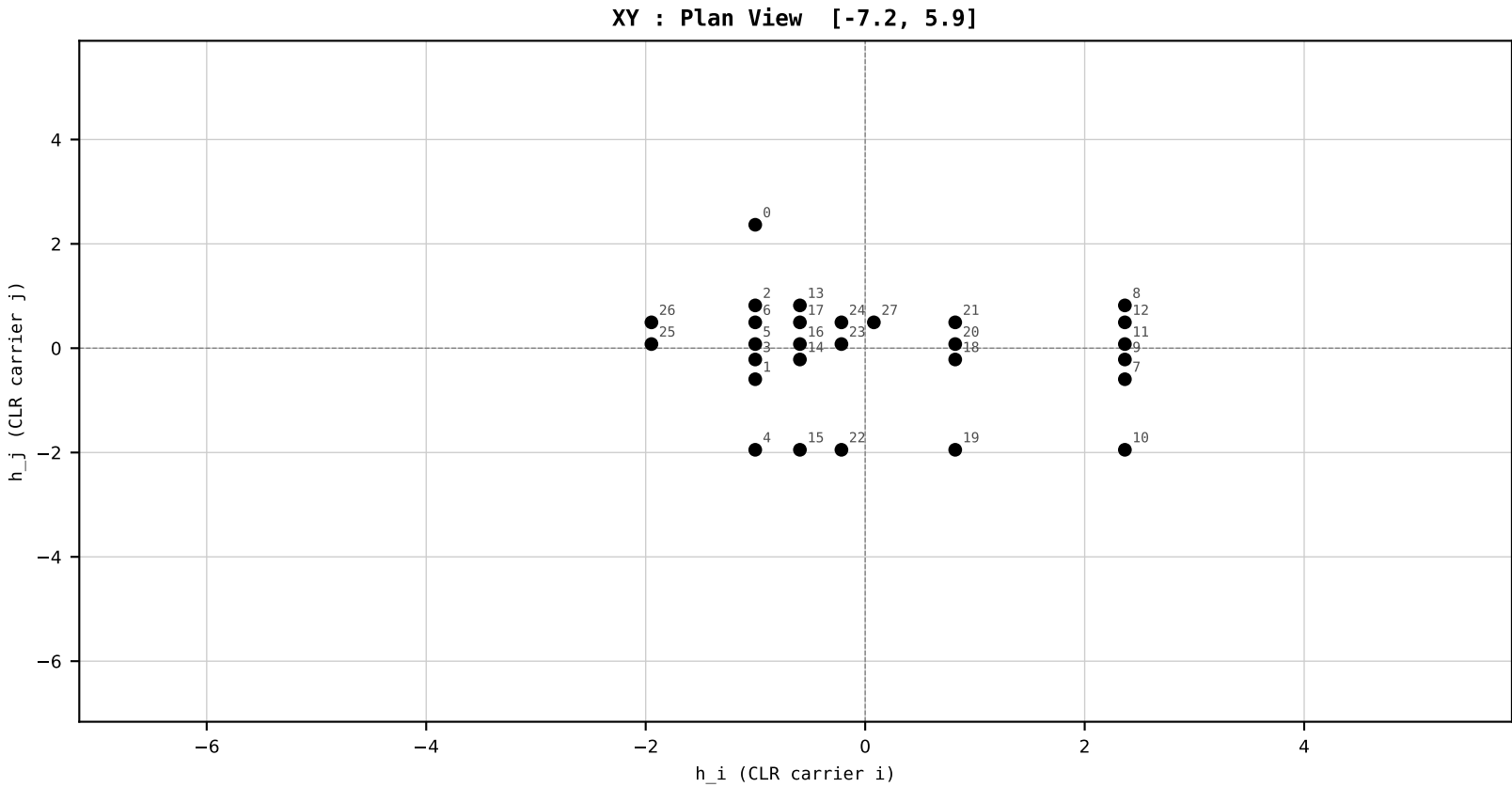
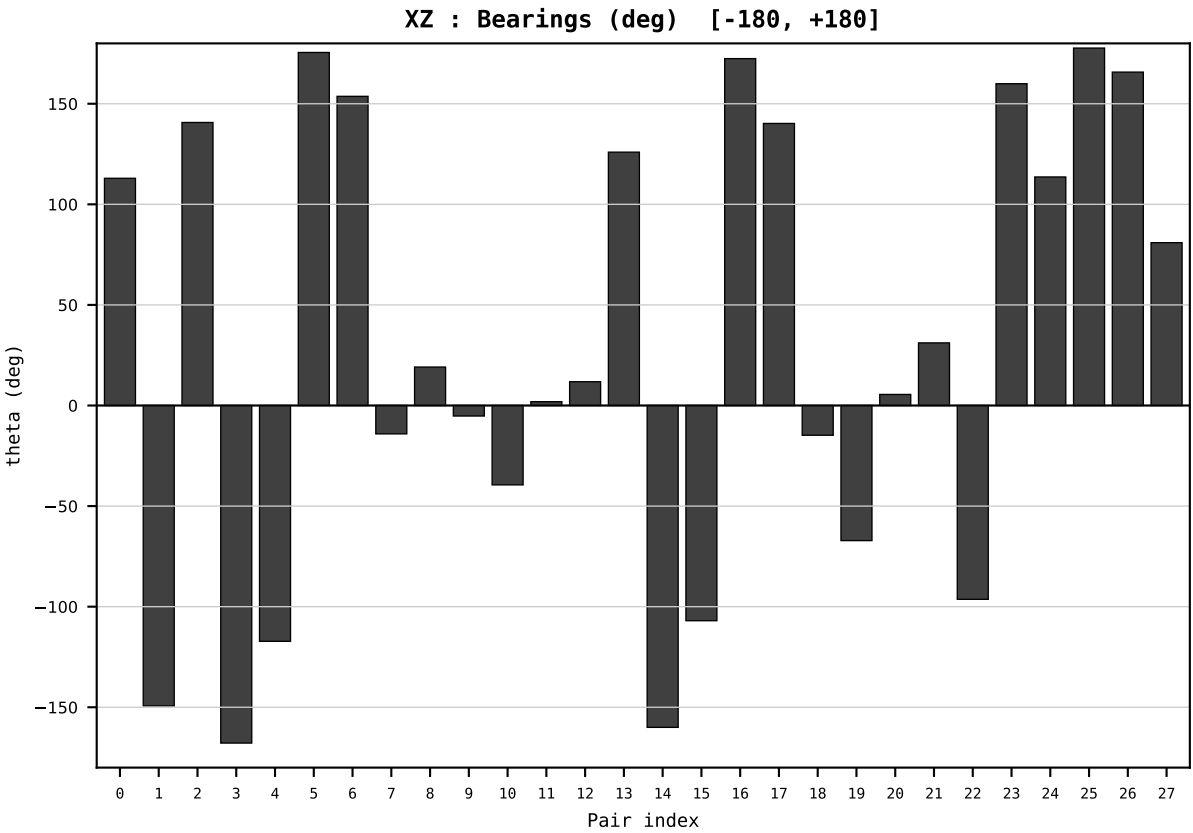
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.359597
Ring = Hs-3
E_metric= 11.7219
kappa_HS= 74.74
omega = 4.5800 deg
d_A = 0.279787
Helm = Solar
Helm d = +0.217626
DR = 4.314 HLR
DR ratio= 74.7x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

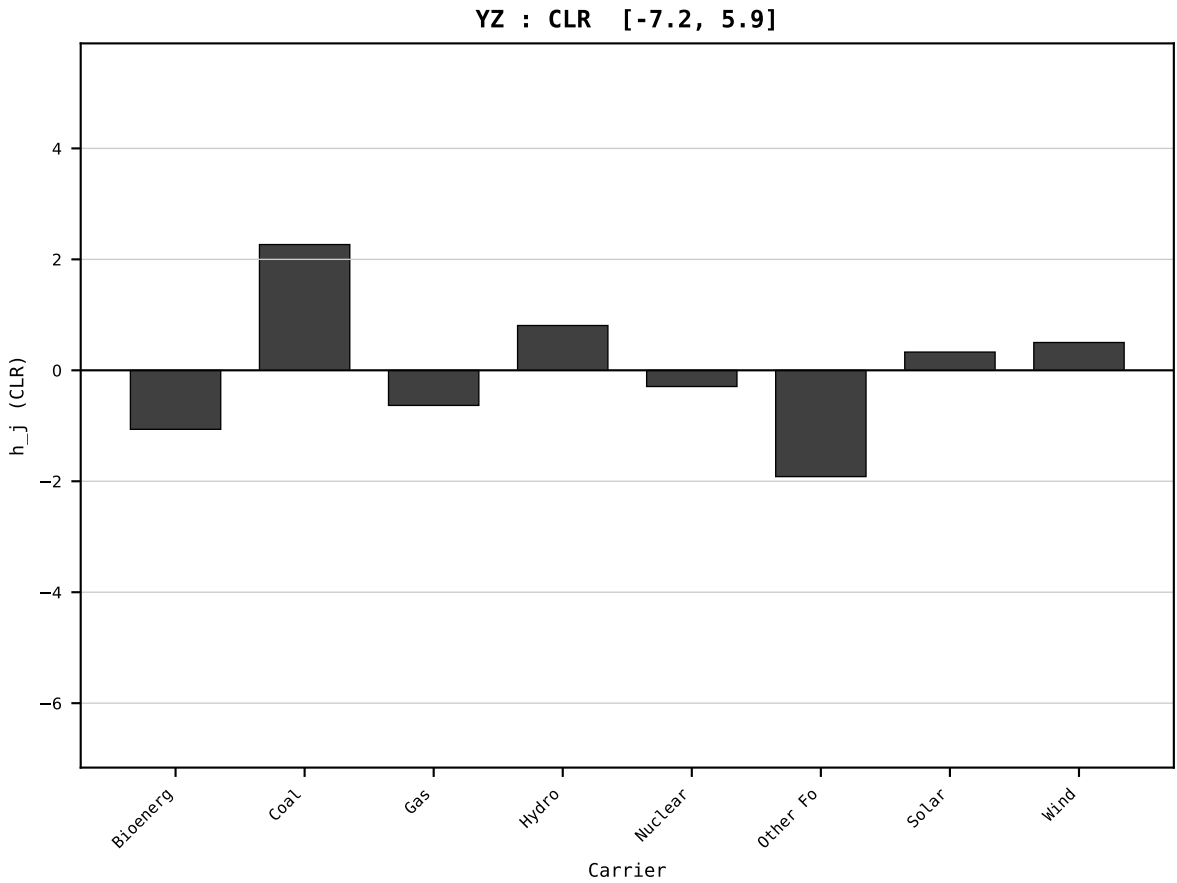
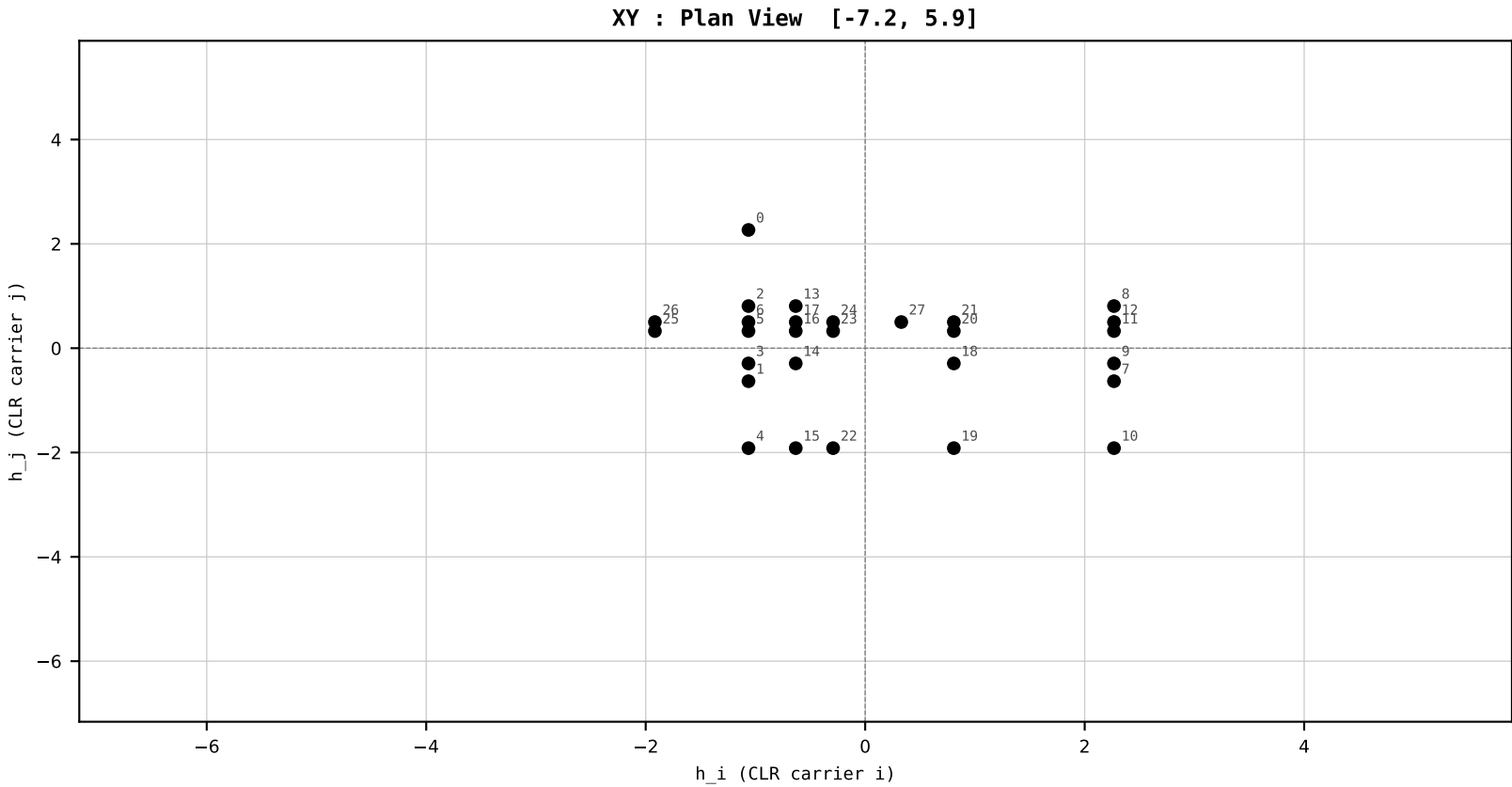
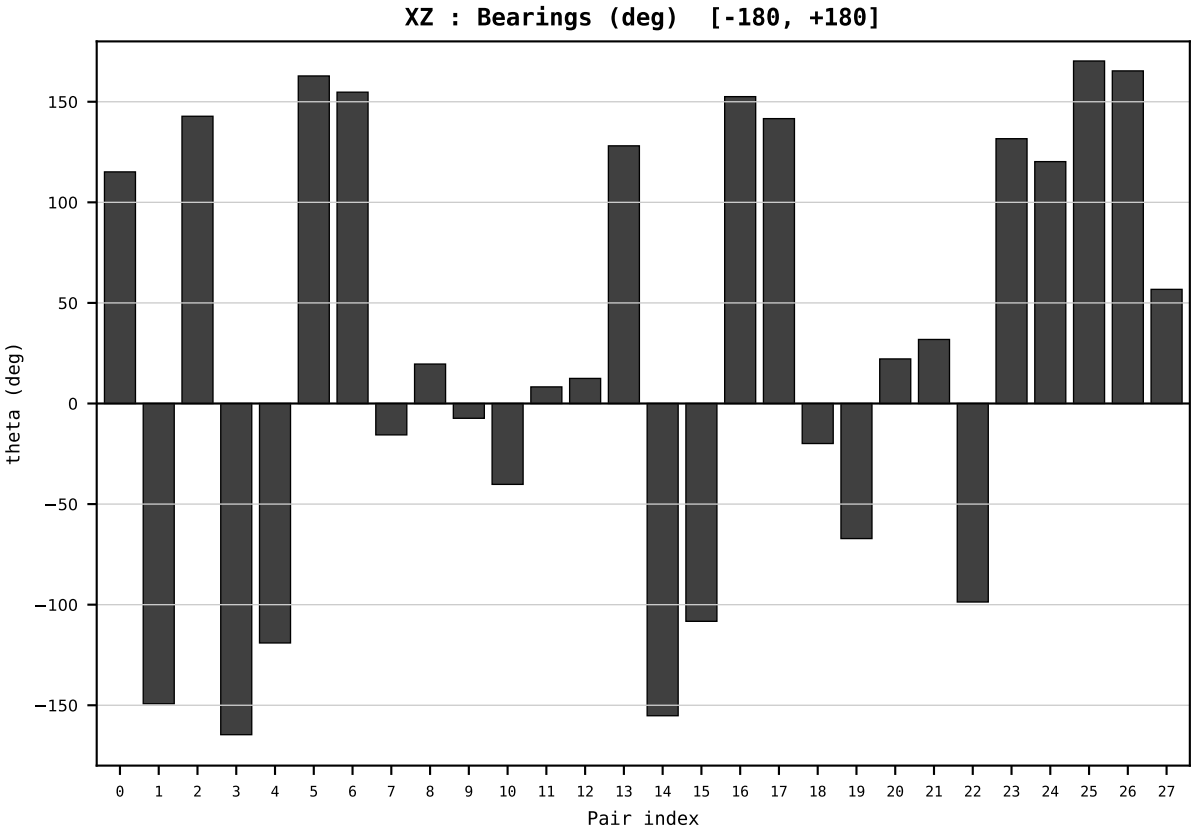
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.330309
Ring = Hs-3
E_metric= 11.4392
kappa_HS= 65.57
omega = 4.8451 deg
d_A = 0.290657
Helm = Solar
Helm d = +0.249869
DR = 4.183 HLR
DR ratio= 65.6x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

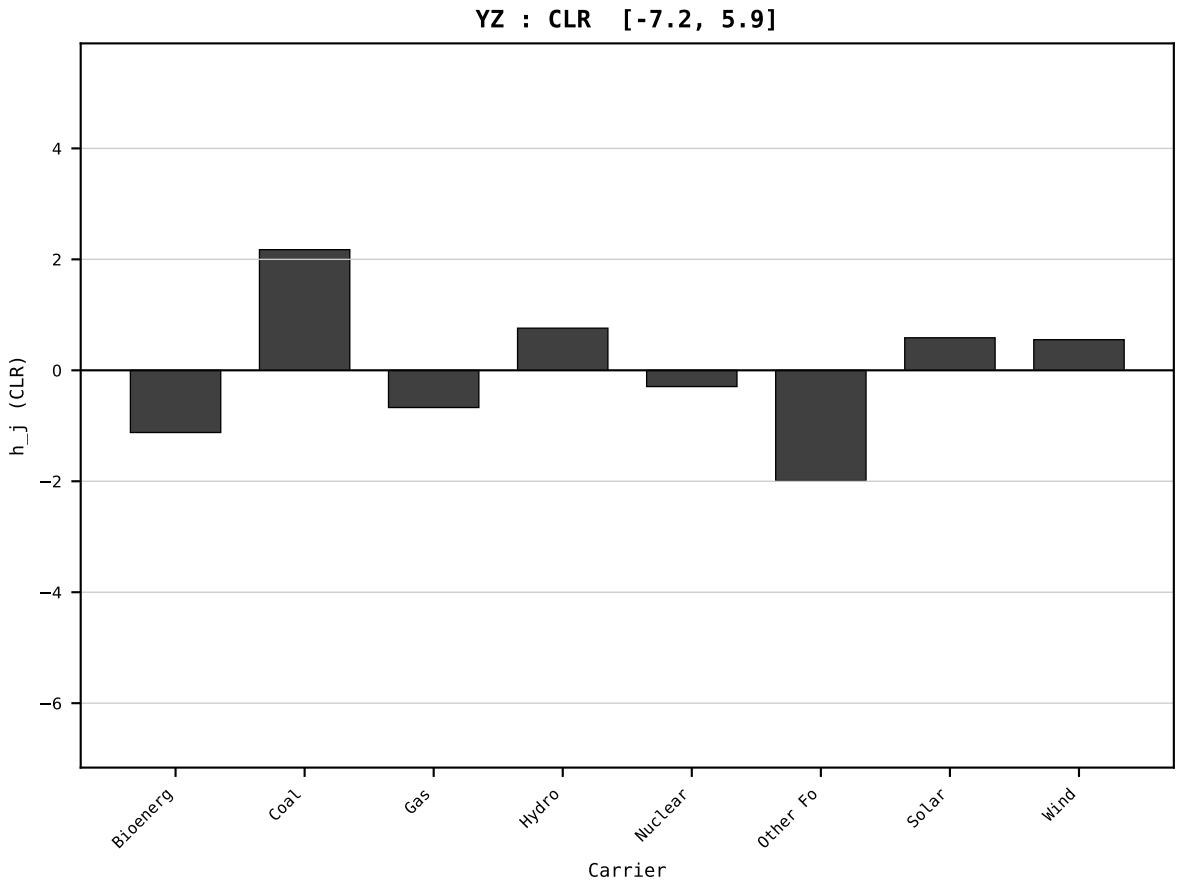
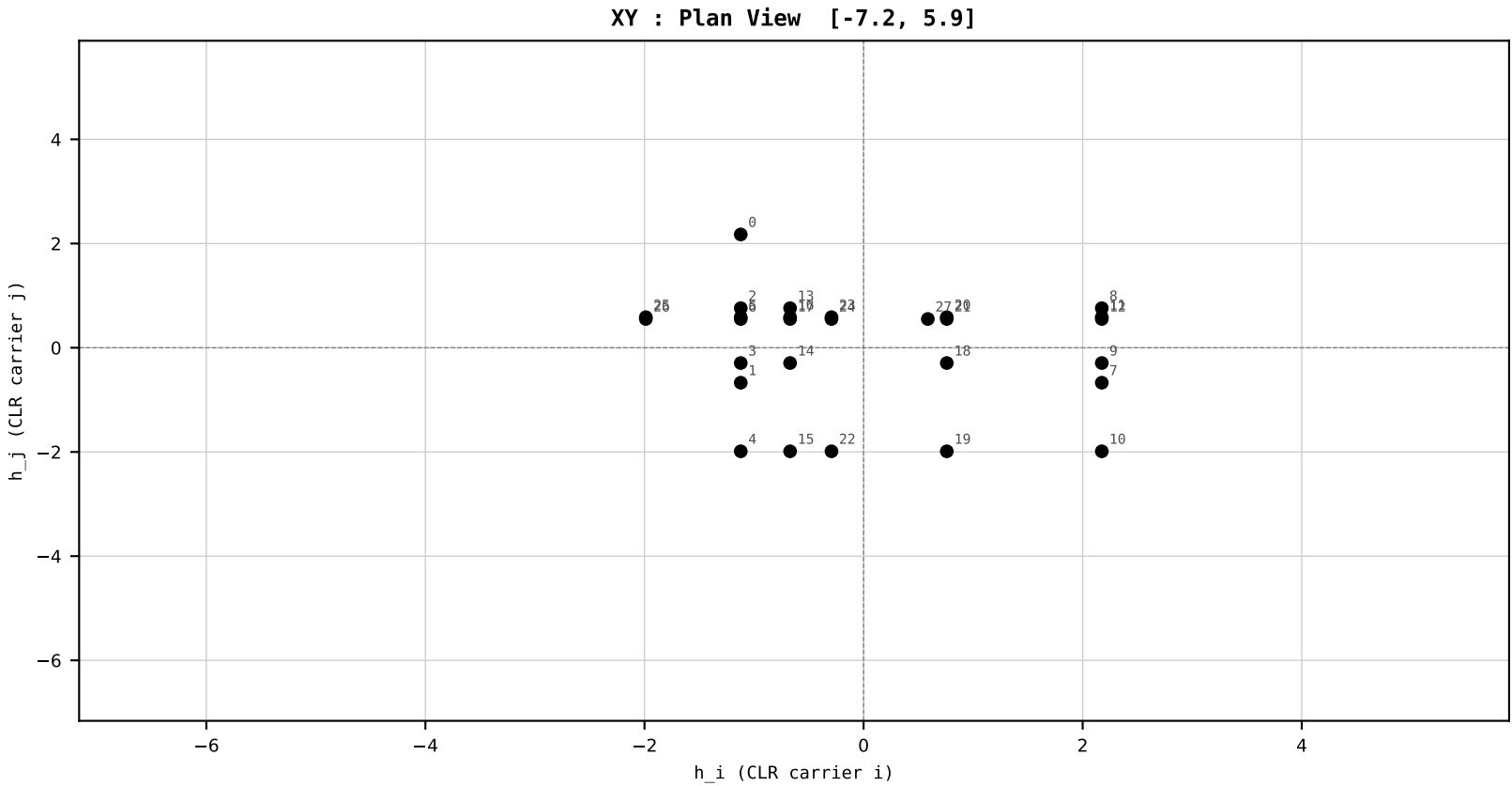
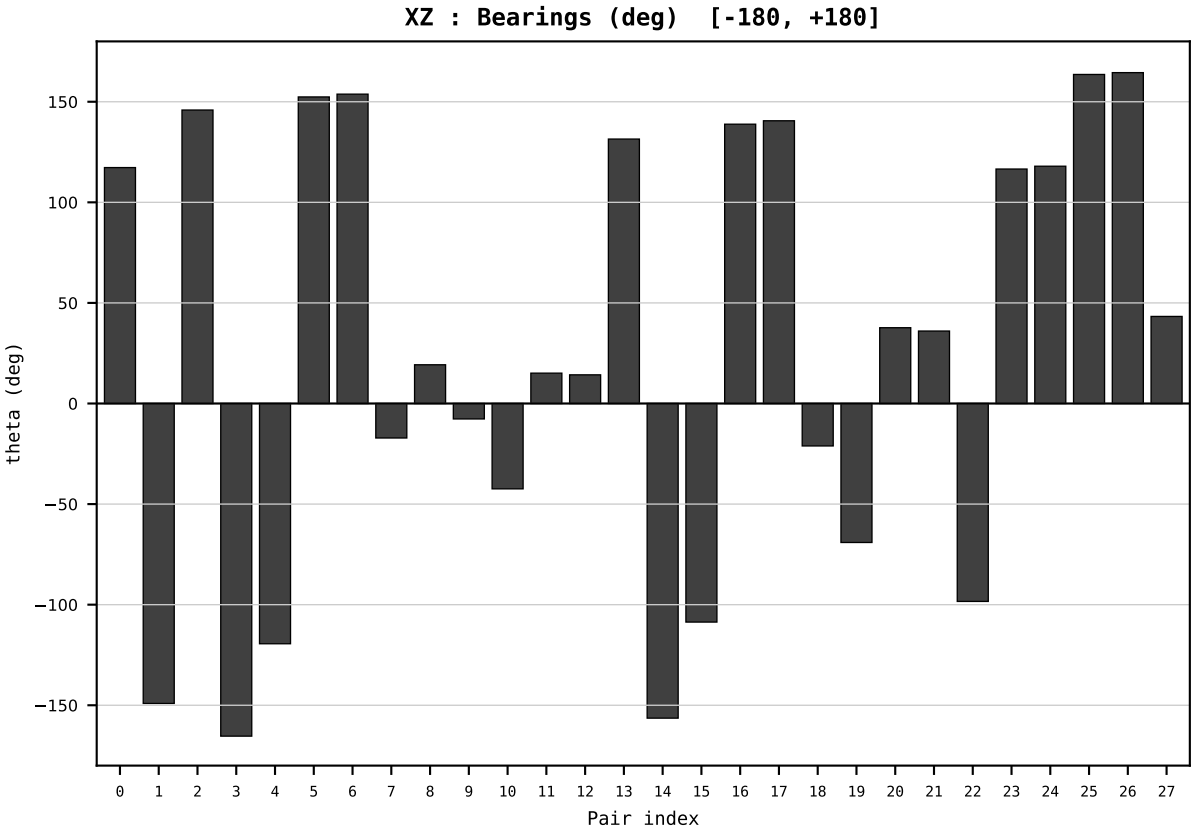
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

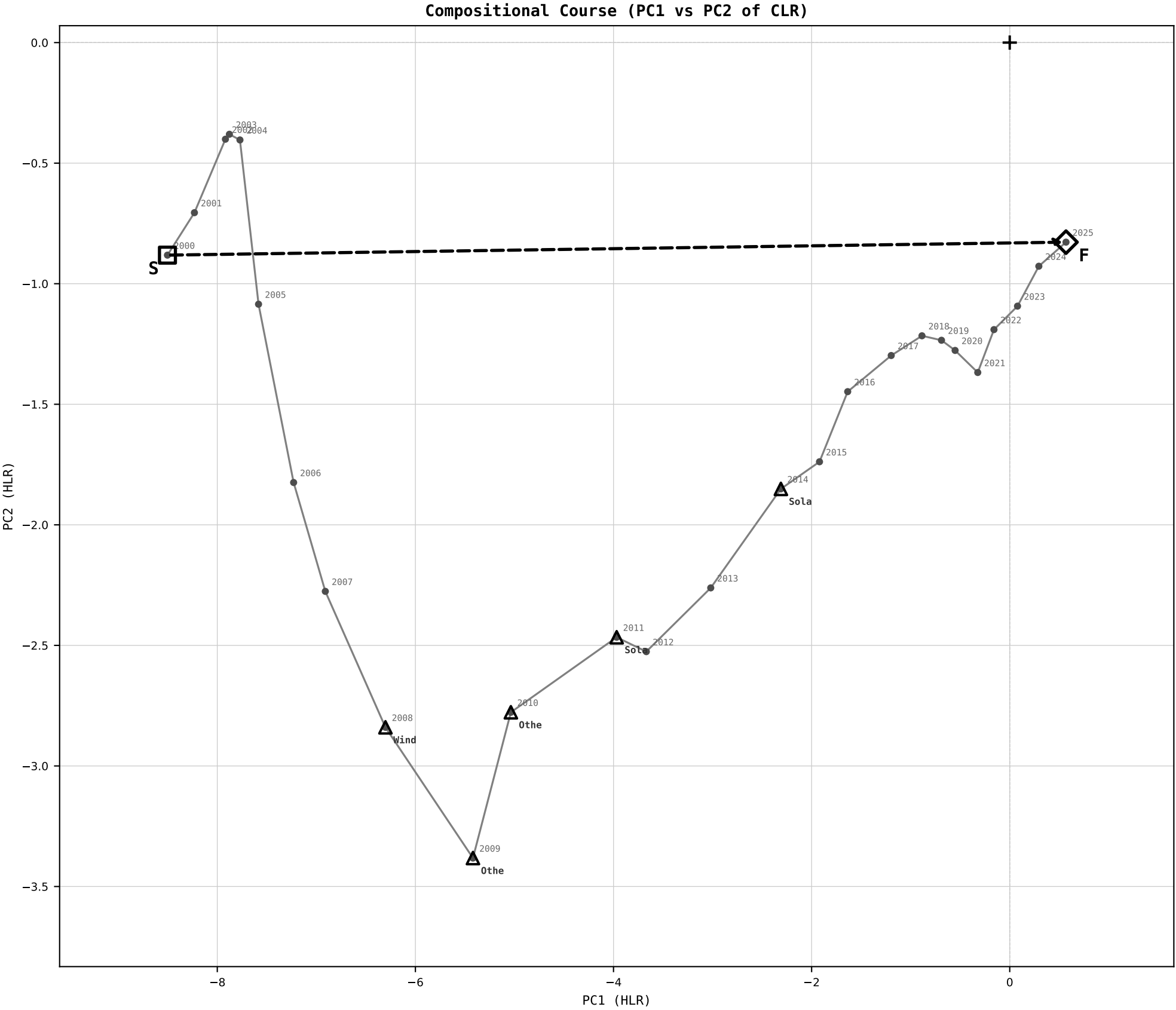
Dataset : ember_CHN_China_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.301686
Ring = Hs-3
E_metric= 11.6993
kappa_HS= 64.23
omega = 5.0047 deg
d_A = 0.299453
Helm = Solar
Helm d = +0.257596
DR = 4.162 HLR
DR ratio= 64.2x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 9.0822 HLR
Path length = 13.8225 HLR
Directness = 0.6571

DR start = 10.878 HLR
ratio = 53013.0x
DR final = 4.162 HLR
ratio = 64.2x

E_{metric} start = 82.7022
E_{metric} final = 11.6993

PC1 loadings:
Solar +0.735
Other Fo -0.396
Wind +0.347

PC2 loadings:
Other Fo +0.557
Solar +0.502
Wind -0.472

Top displacement events:
2011 Solar d=1.162
2009 Other Fo d=1.043
2010 Other Fo d=1.021
2008 Wind d=0.975
2014 Solar d=0.855

The instrument reads.
The expert decides.
The loop stays open.

View B

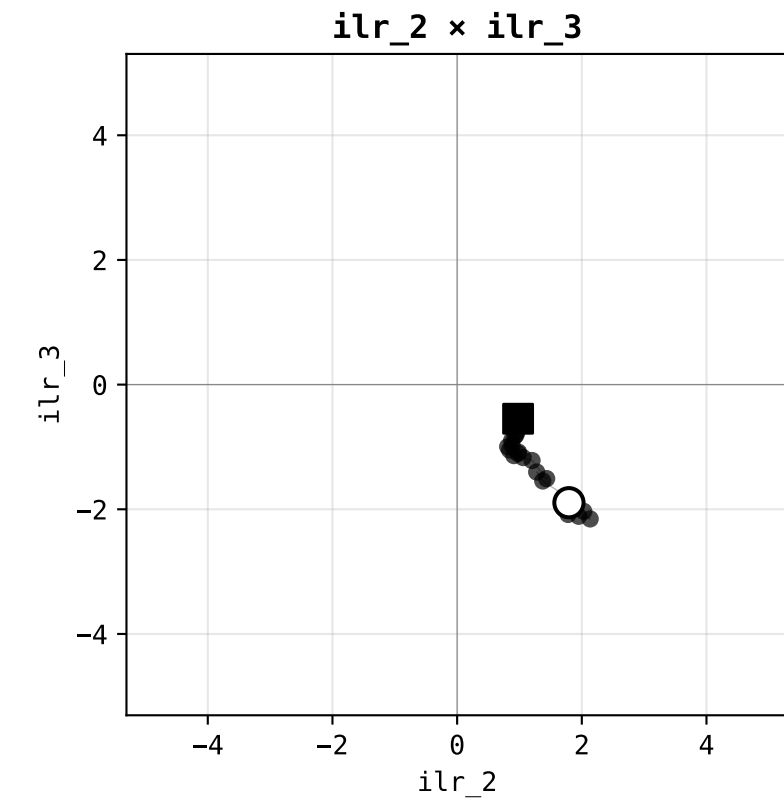
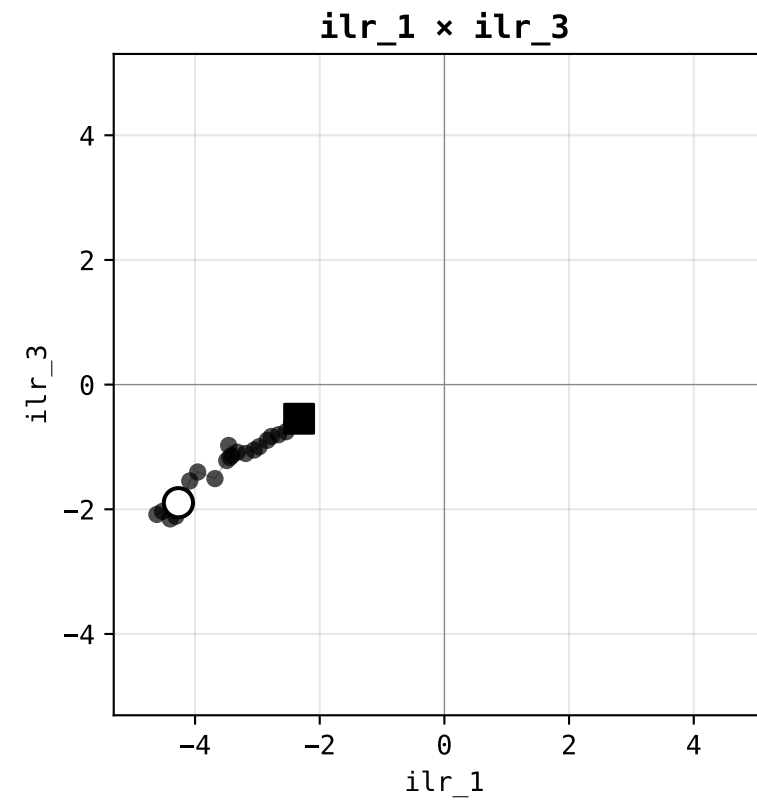
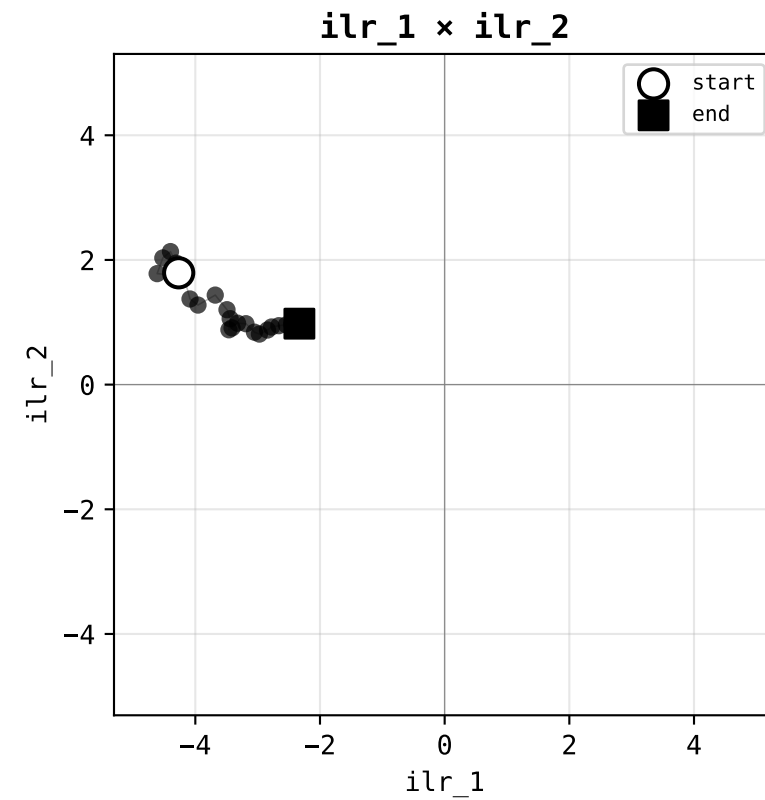
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ CHN China

ILR-Helmert Orthogonal Triplet – CHN China : Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · $N=26$ years · $D=8$ carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:

start year: 2000
end year: 2025
N readings: 26

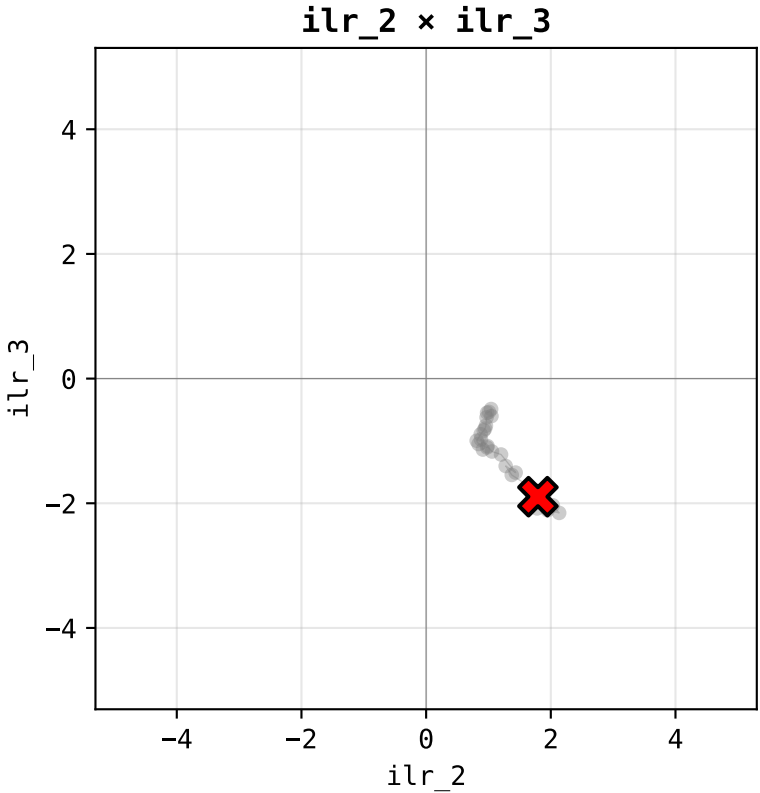
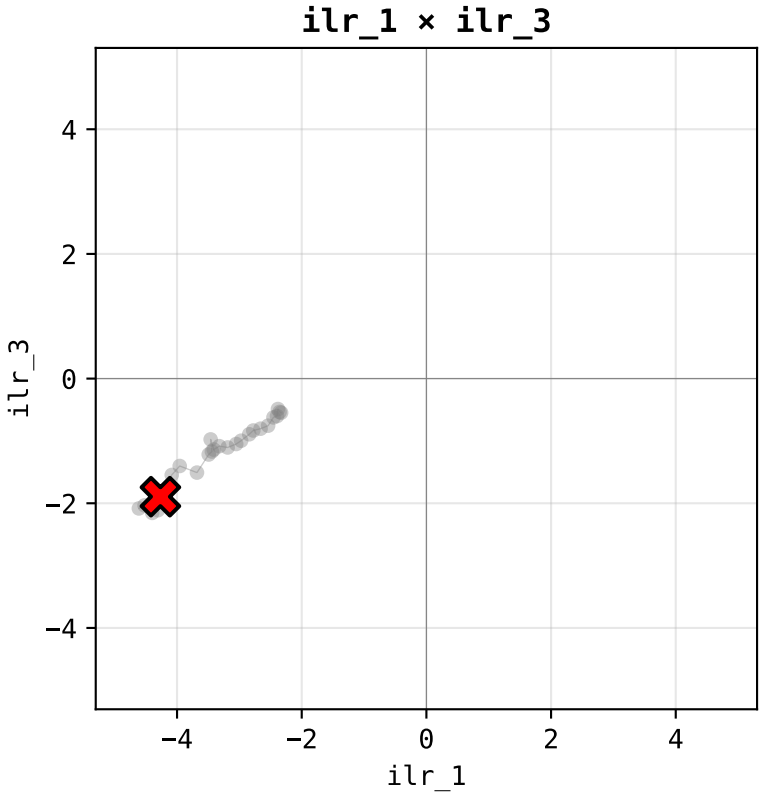
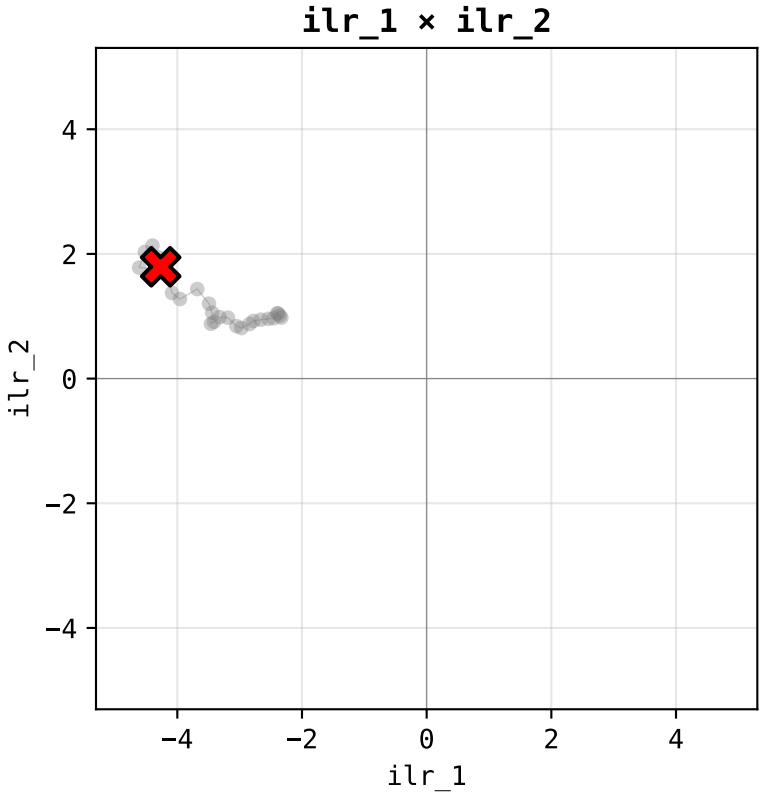
Display:

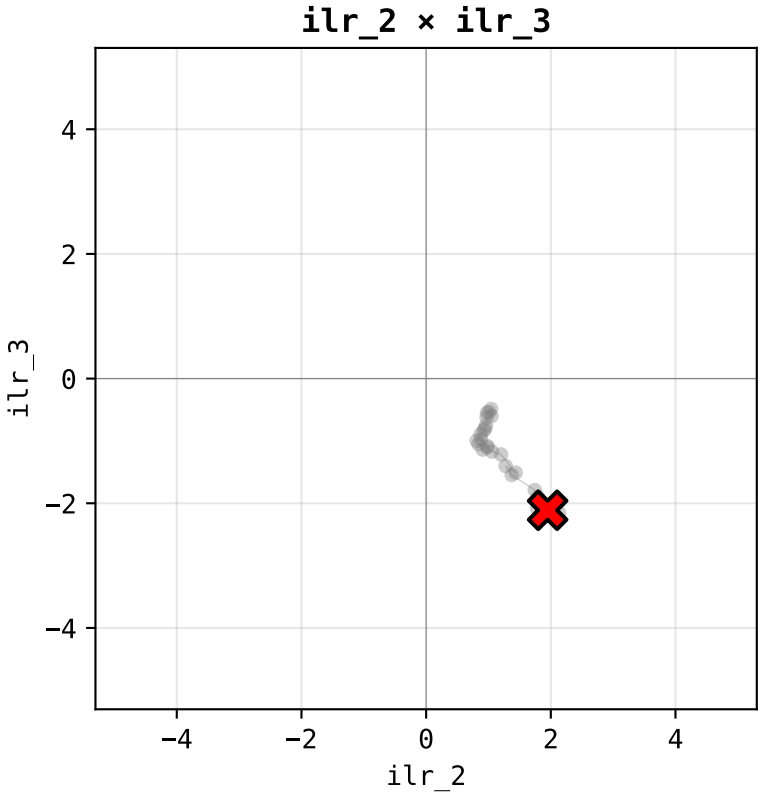
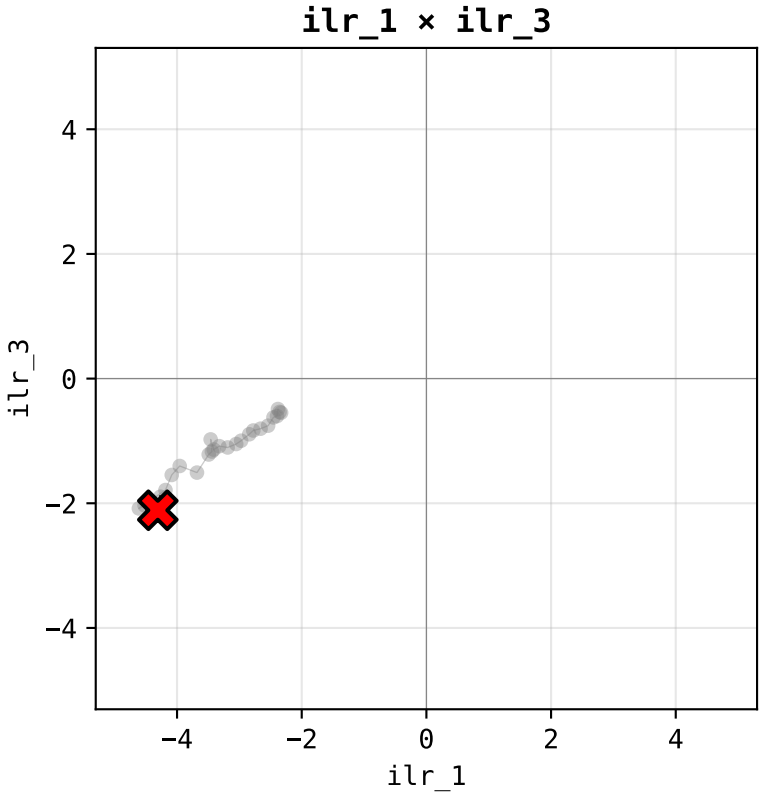
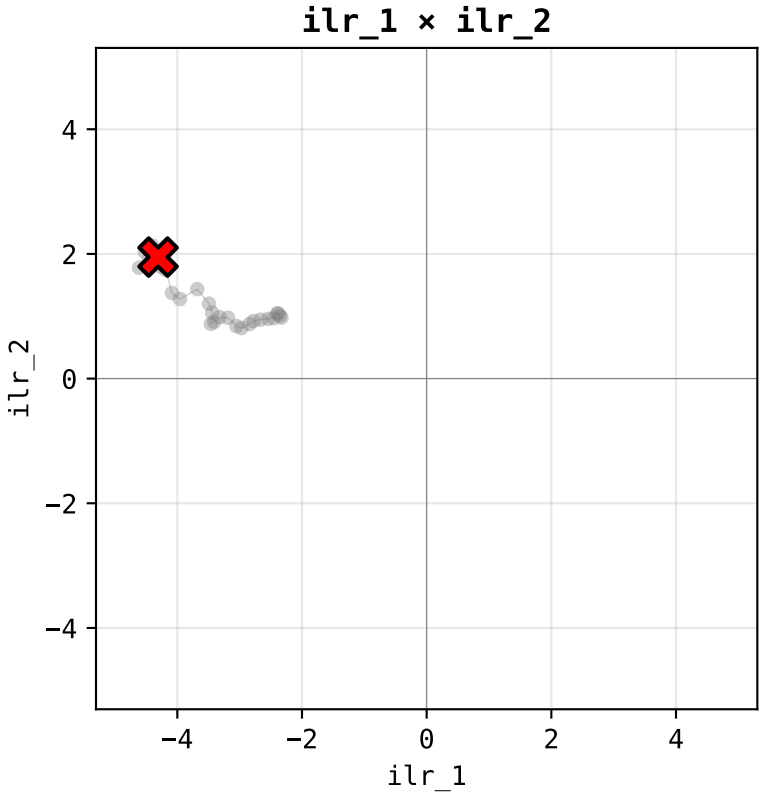
fixed scale ± 5.30 ILR
Order 1 (first principles)
per Output Doctrine v1.0

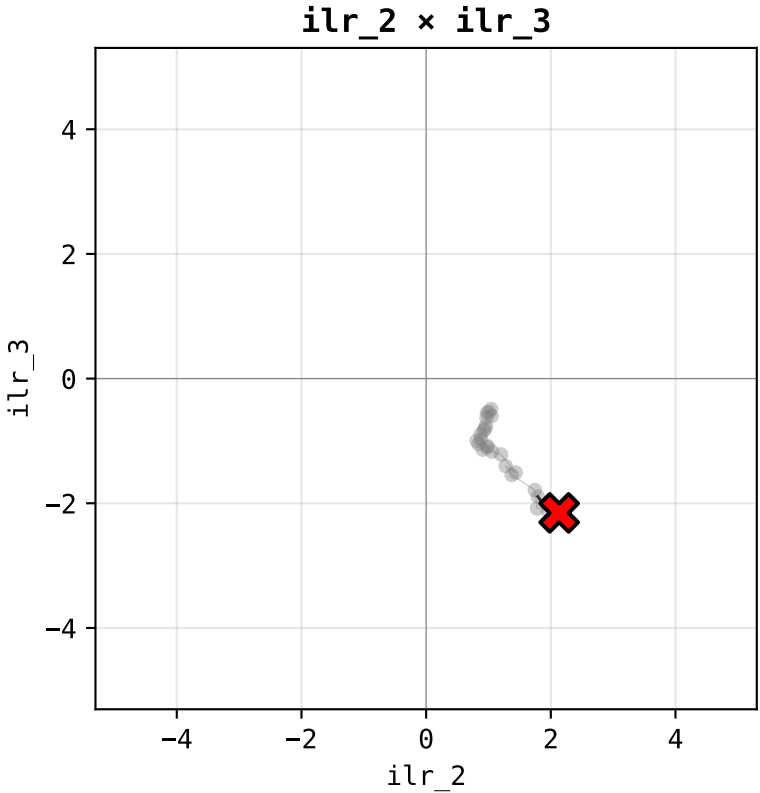
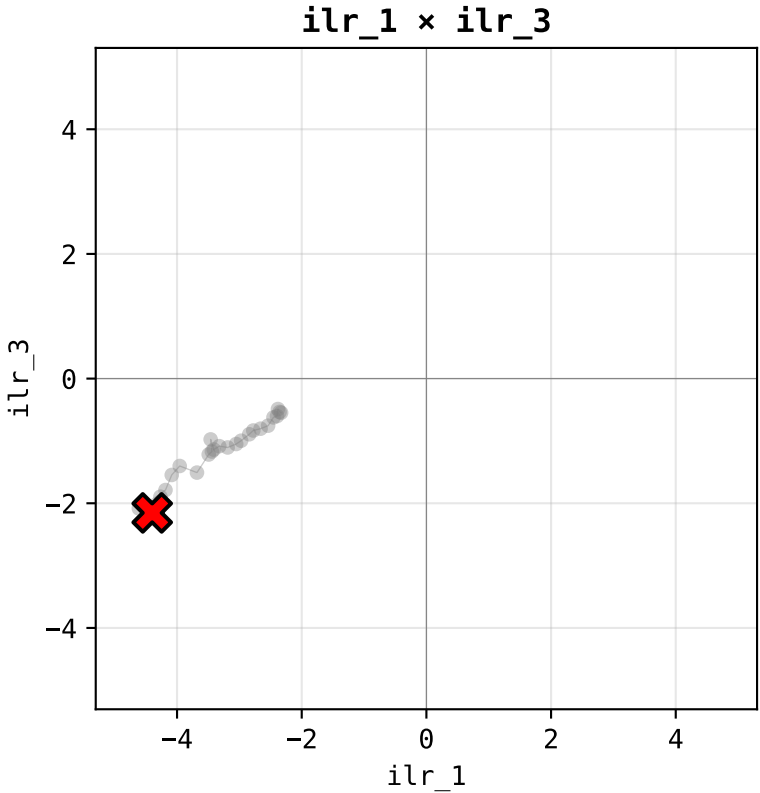
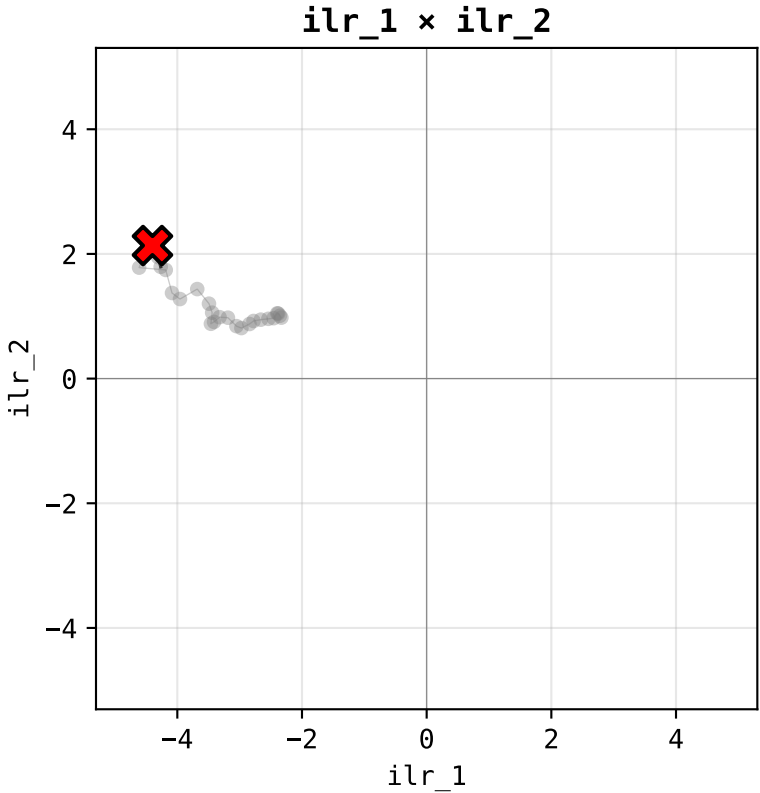
Symbology:

○ start · ■ end
· intermediate years

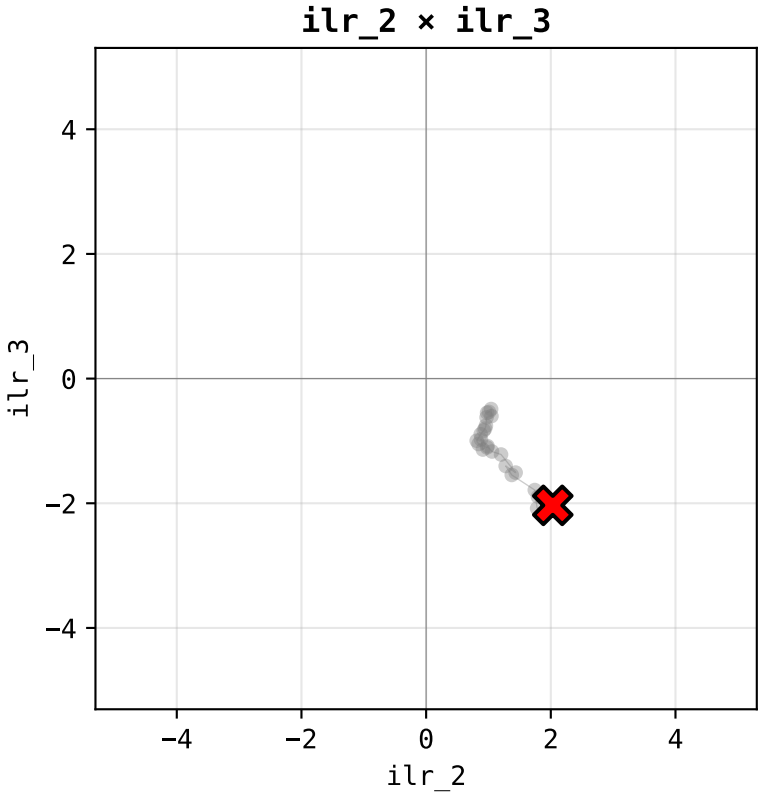
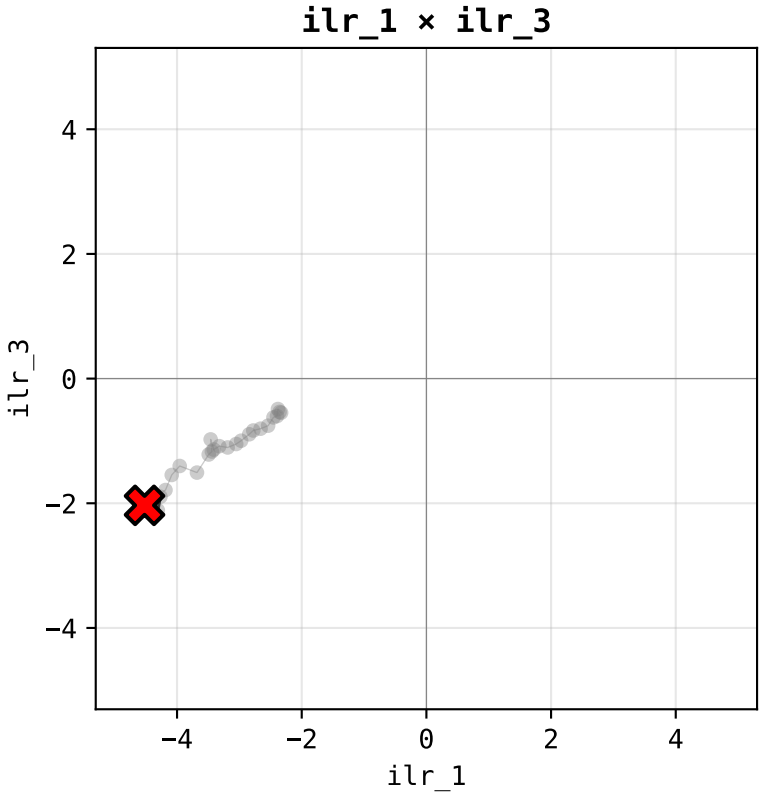
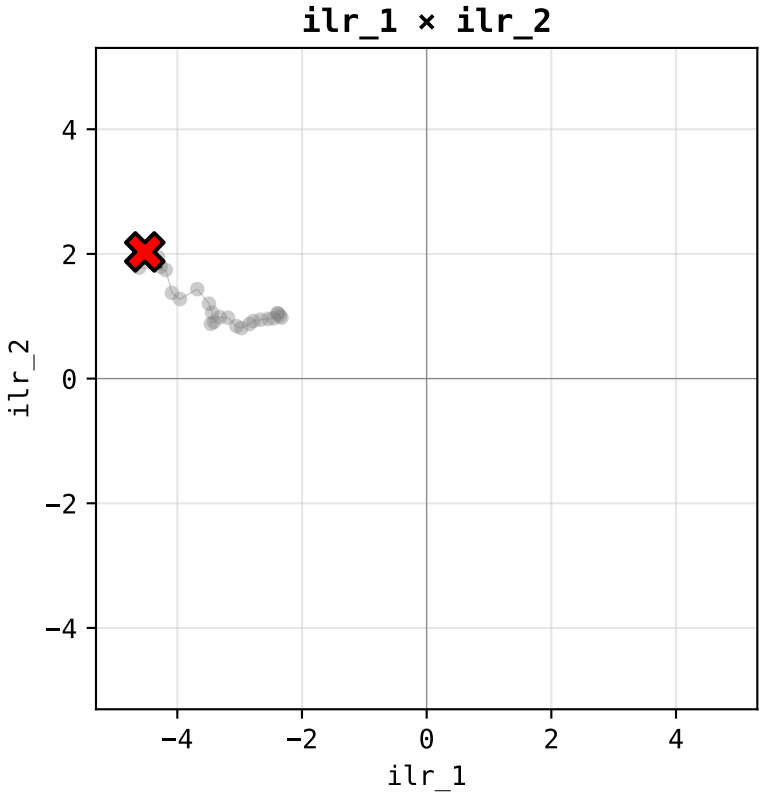
ILR Triplet – CHN China · year 2000 (1/26)

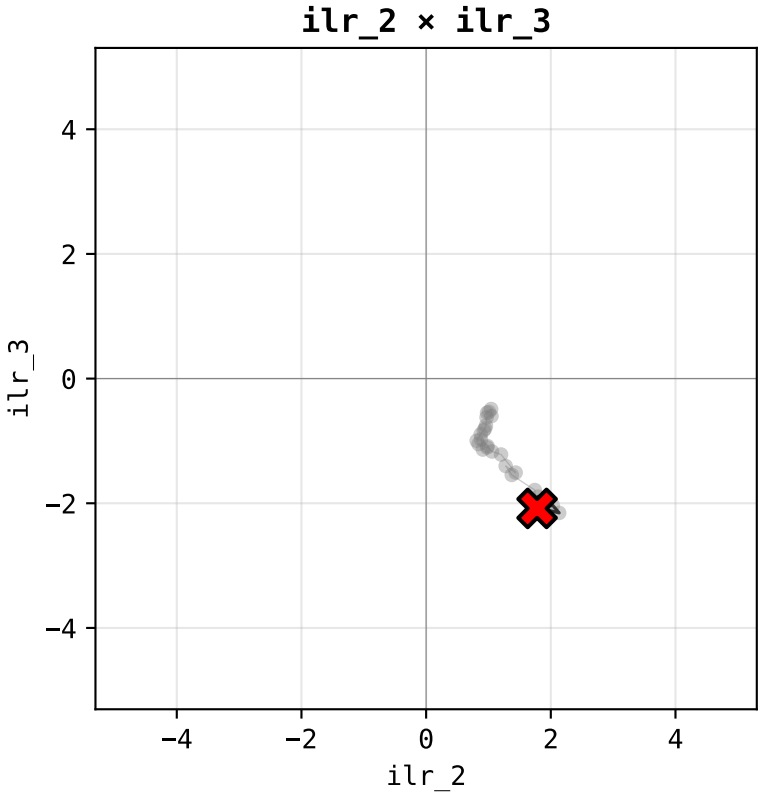
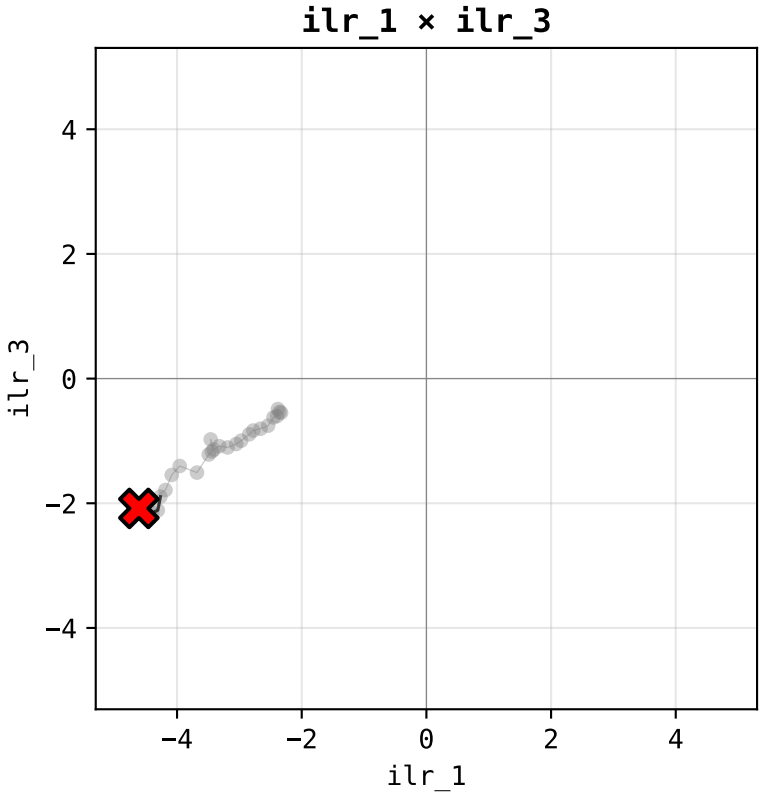
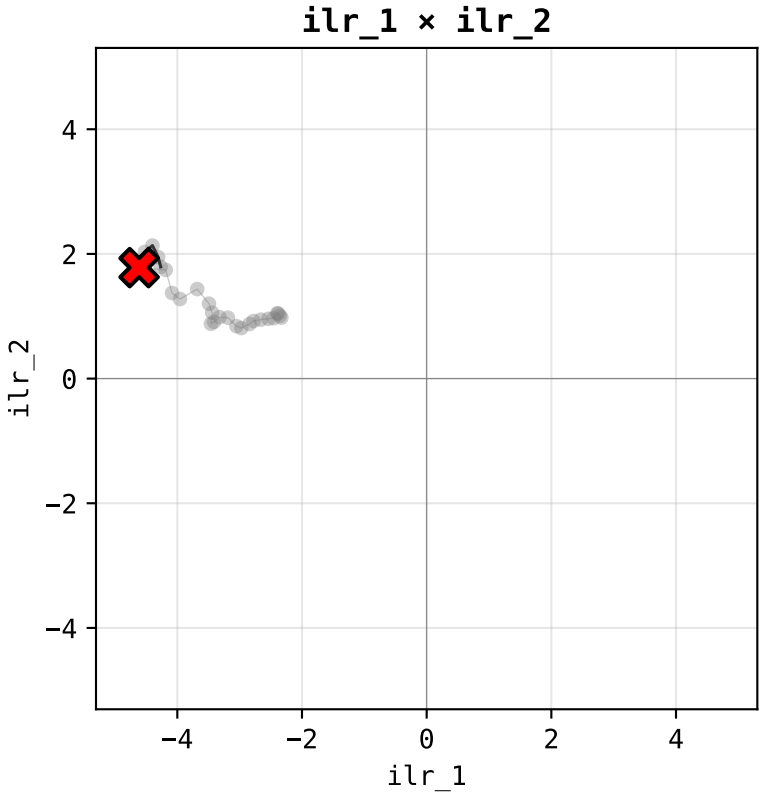


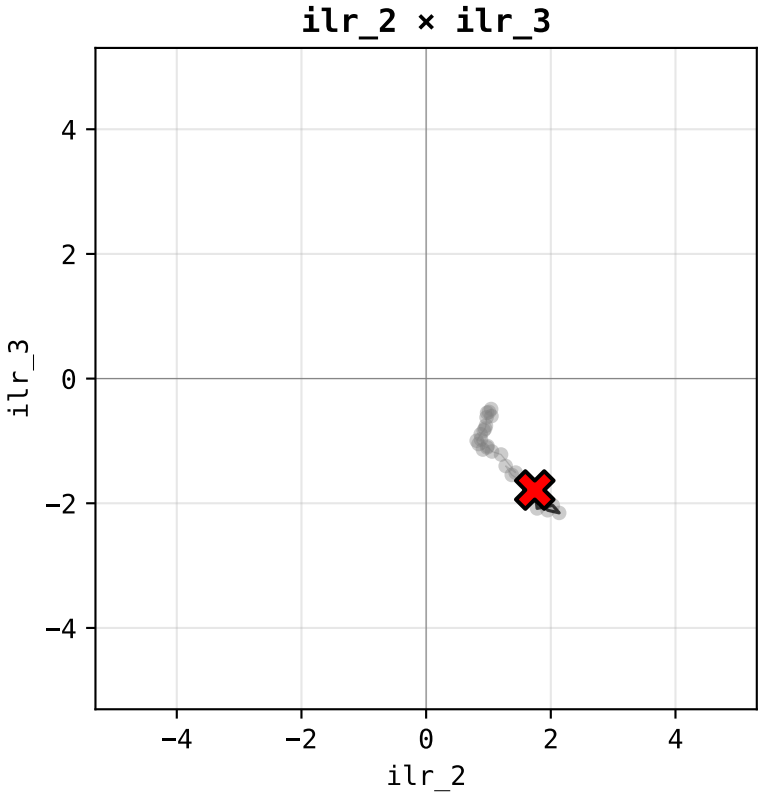
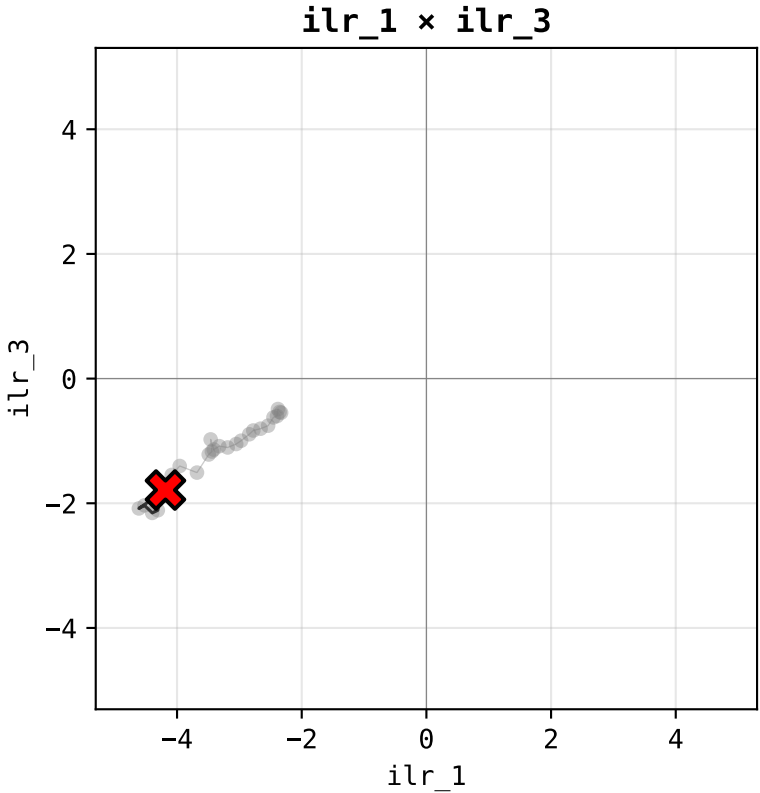
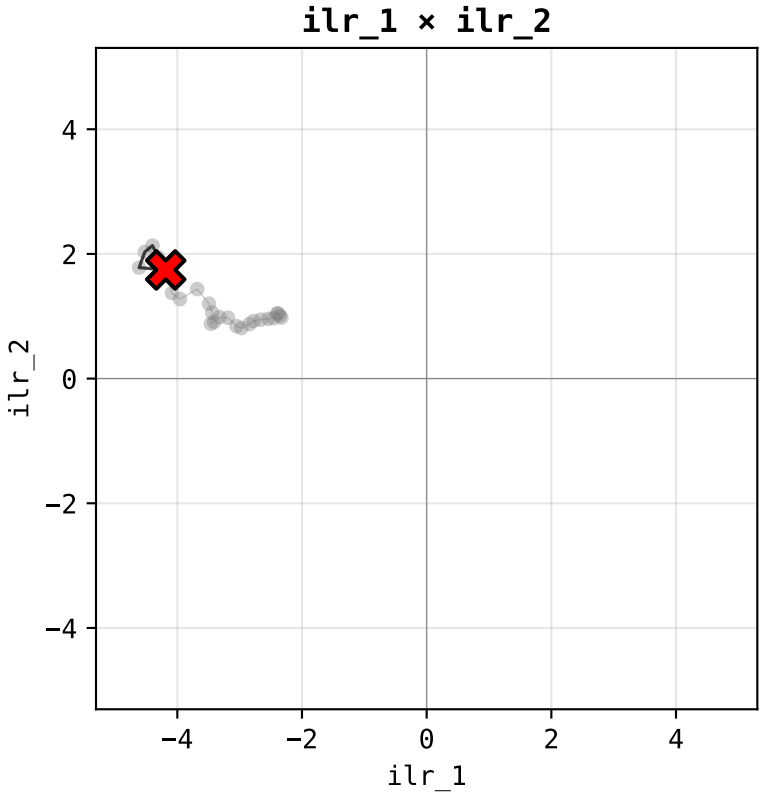




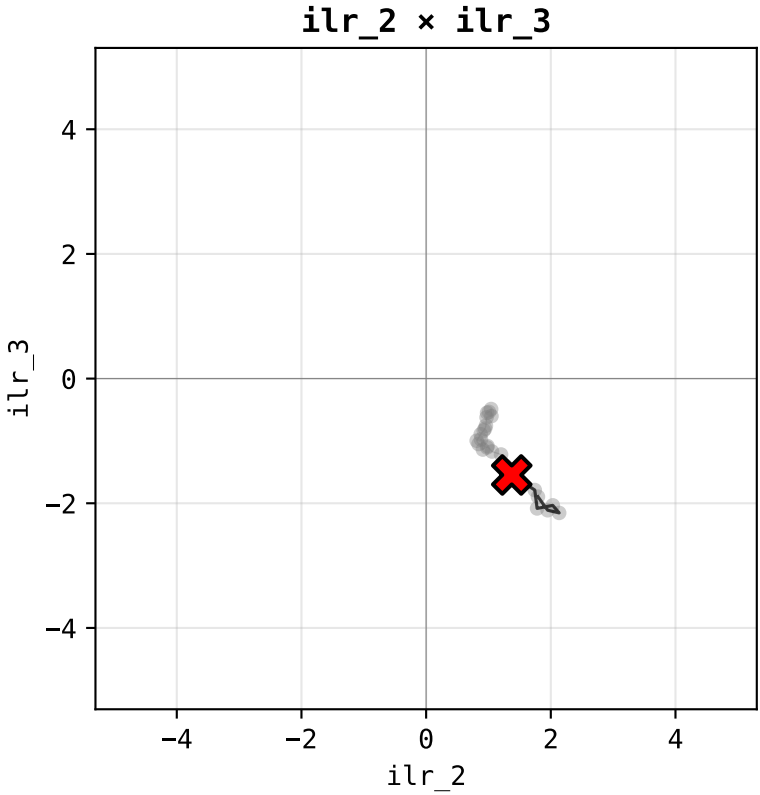
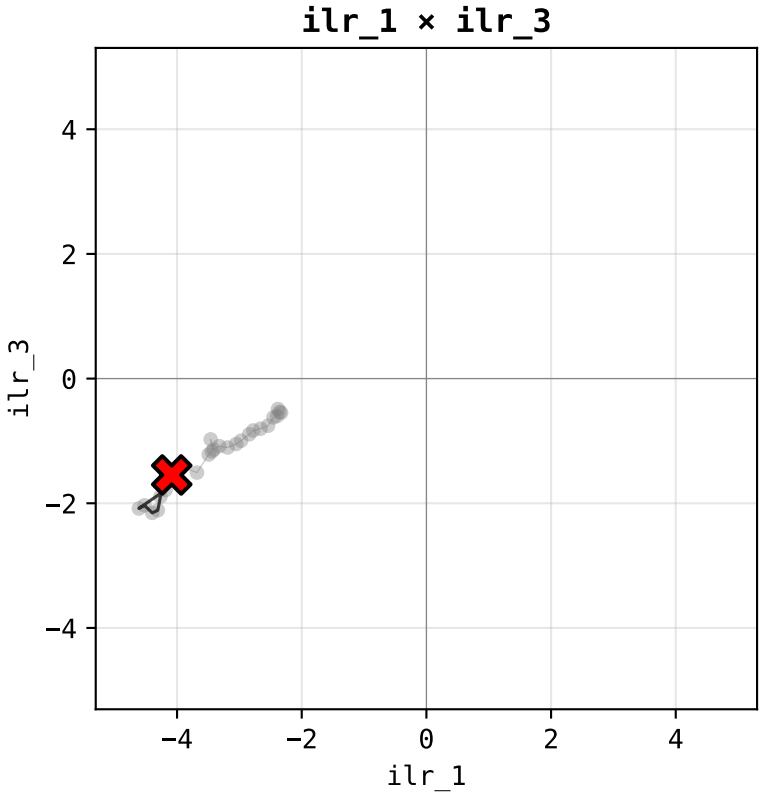
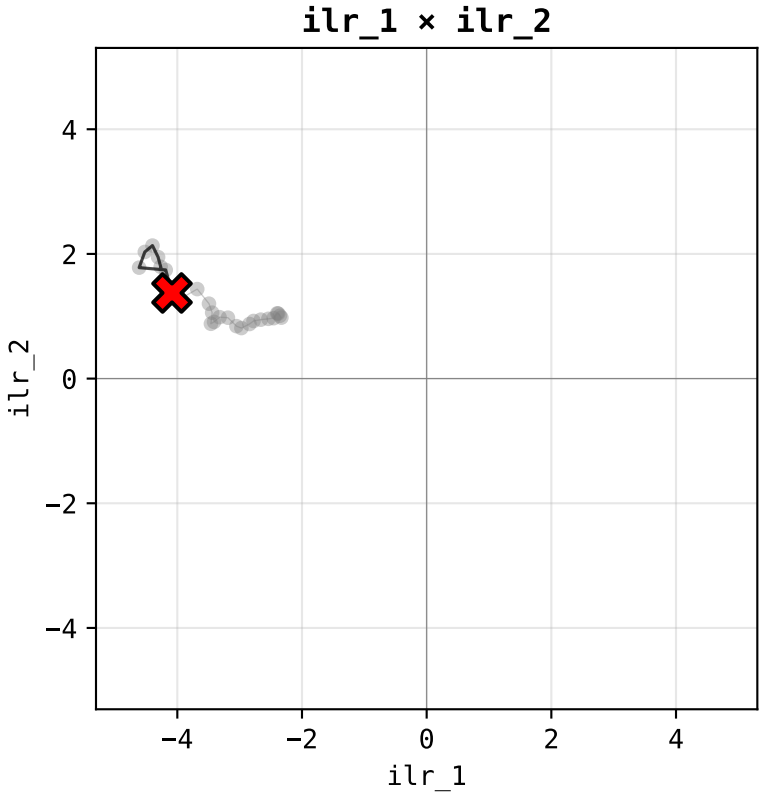
ILR Triplet – CHN China · year 2003 (4/26)



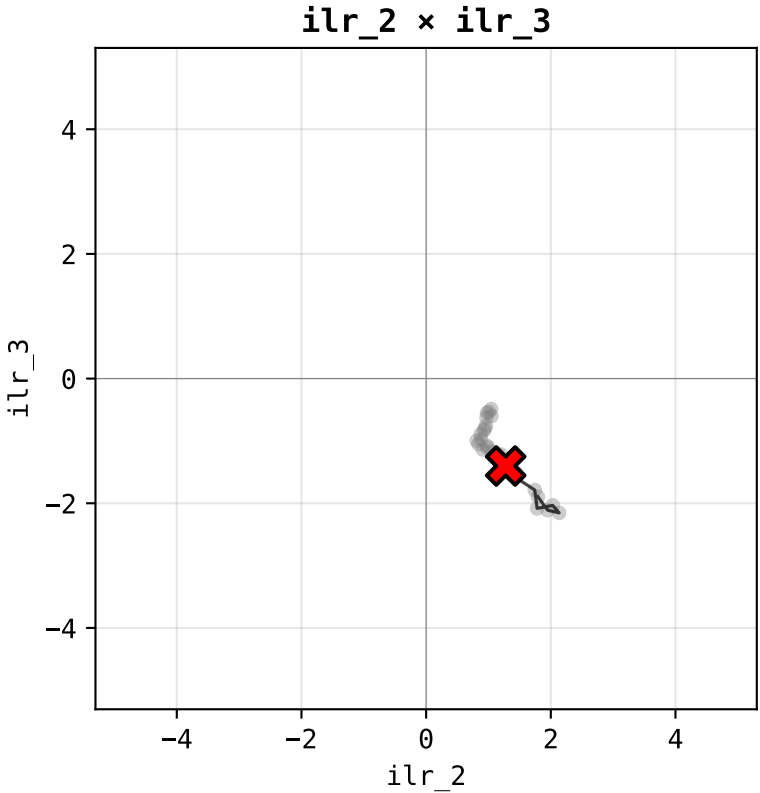
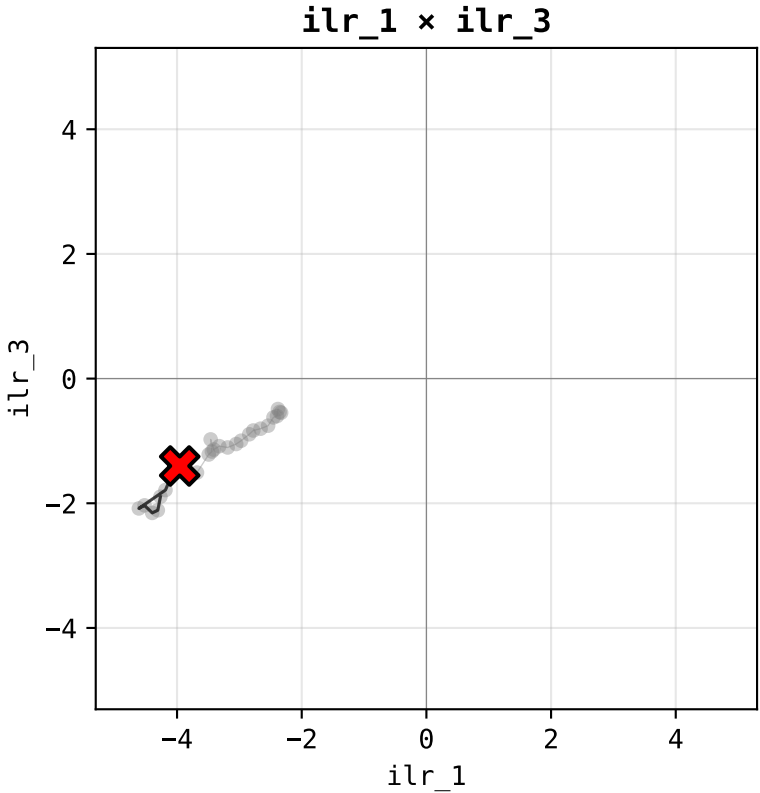
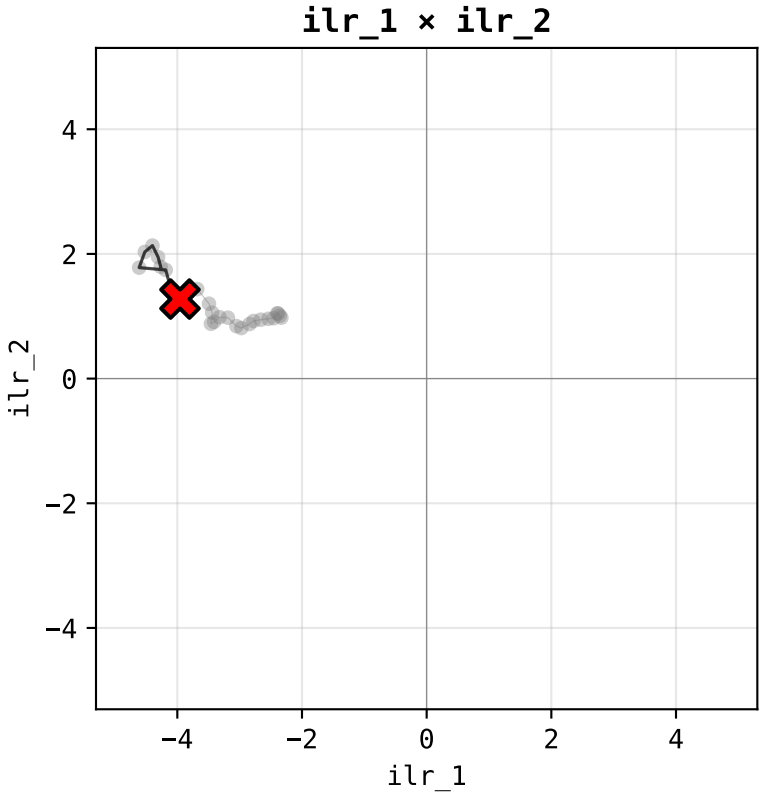




ILR Triplet – CHN China · year 2006 (7/26)

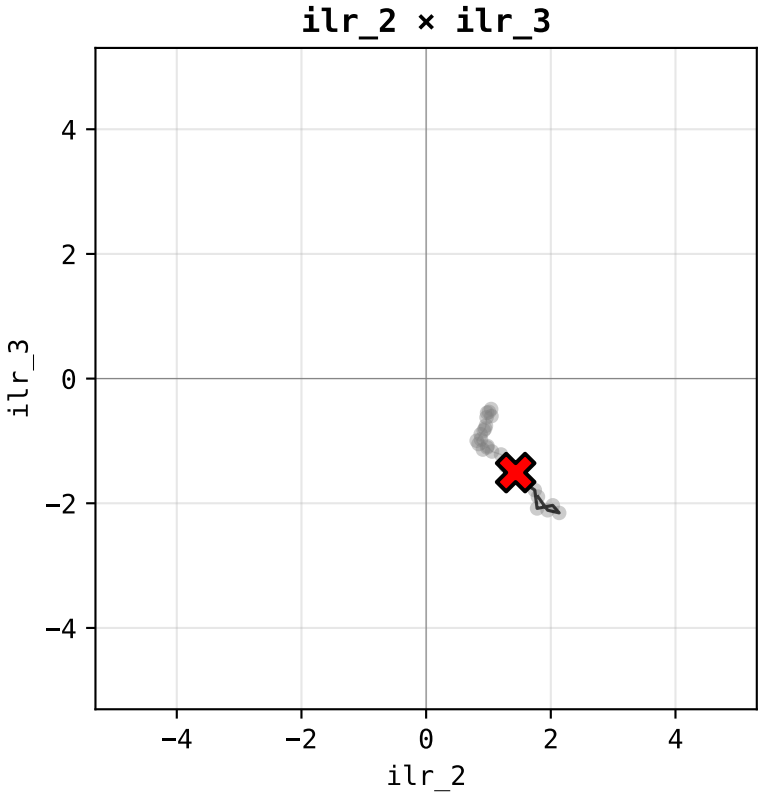
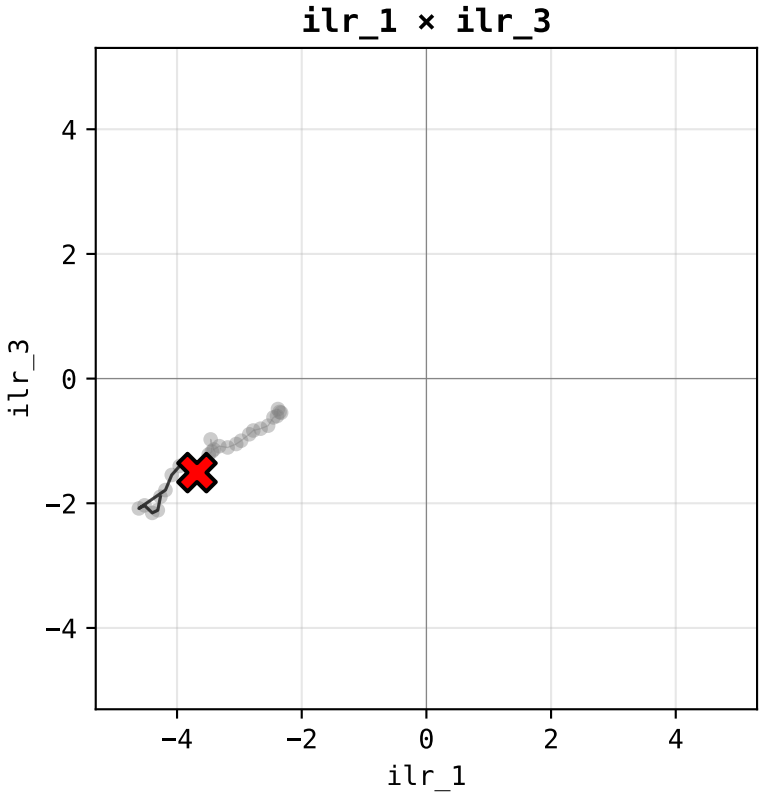
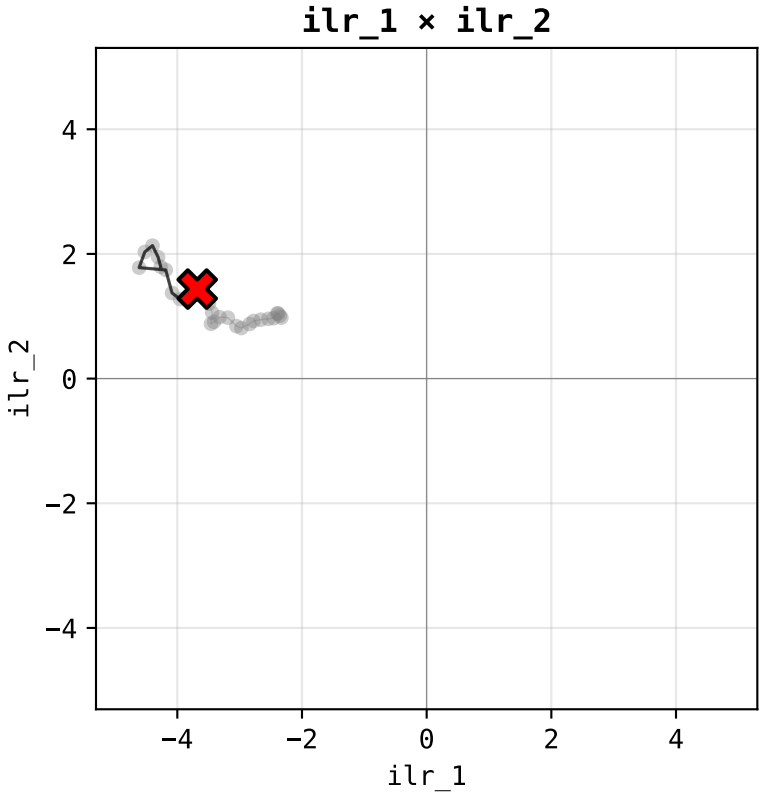


ILR Triplet – CHN China · year 2007 (8/26)

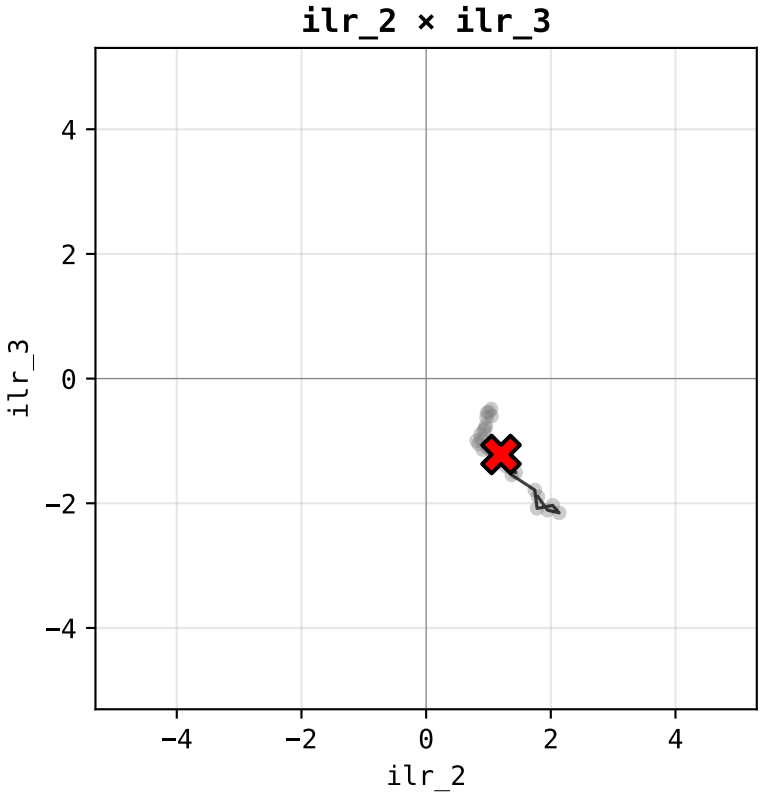
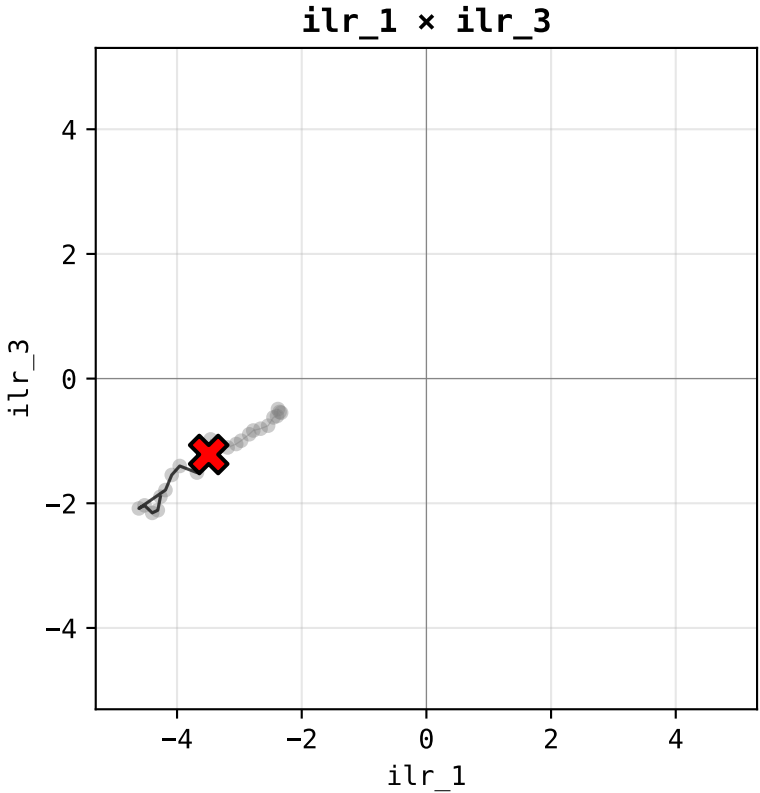
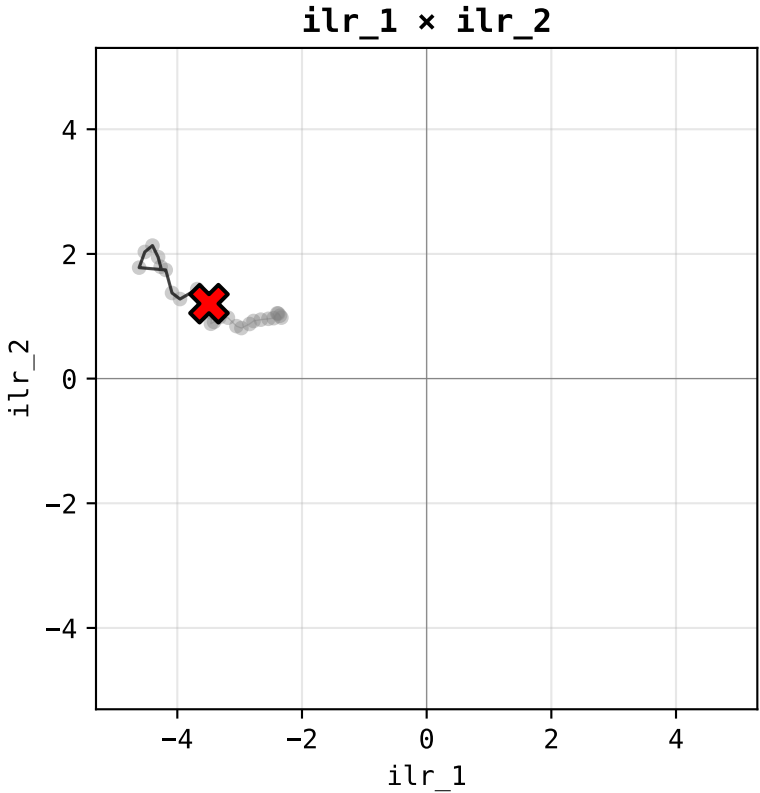


X = current year (2007) · black = past trajectory · gray = future · ilr range ±5.30

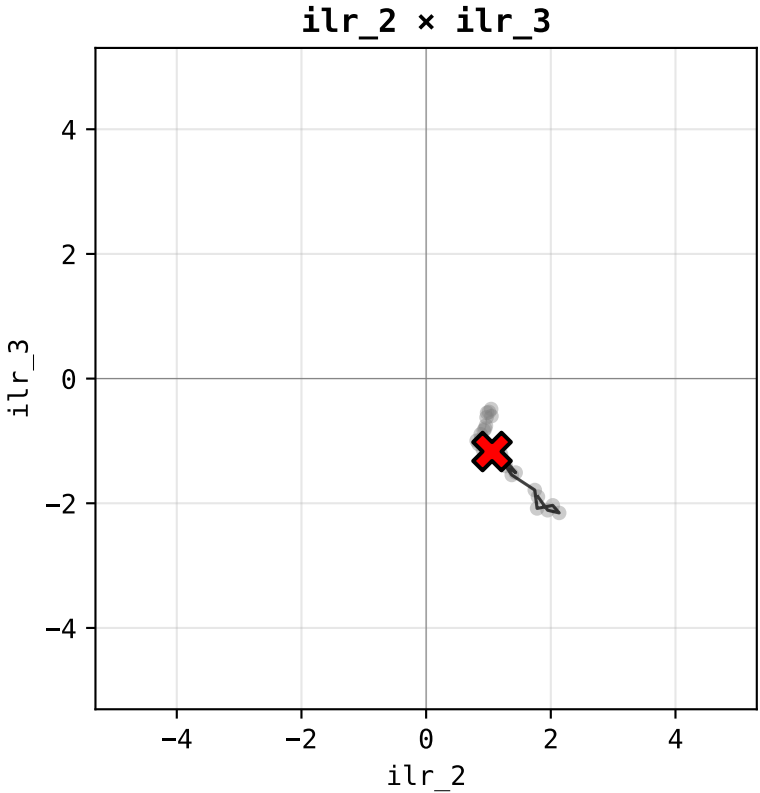
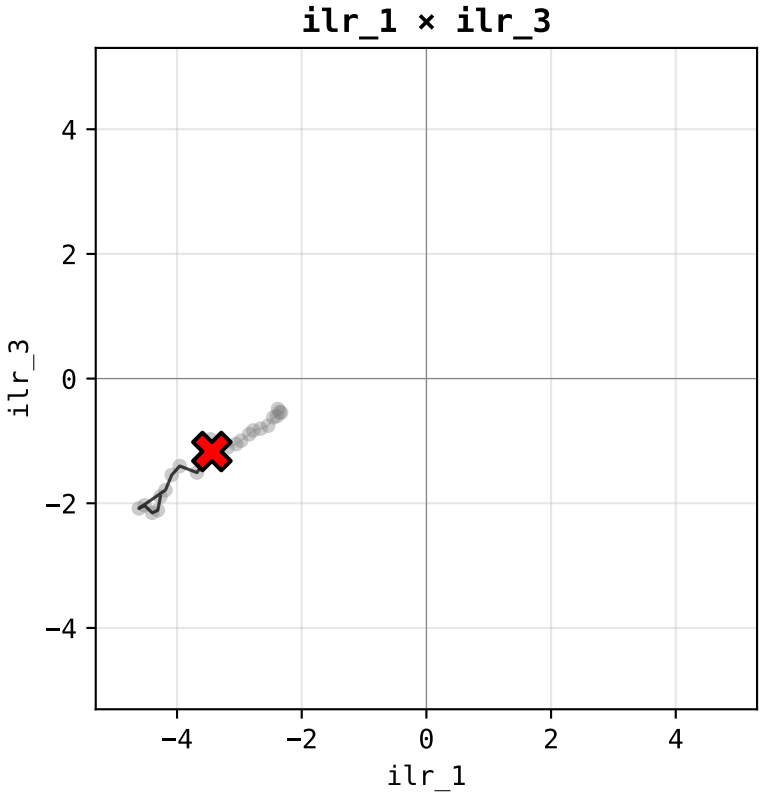
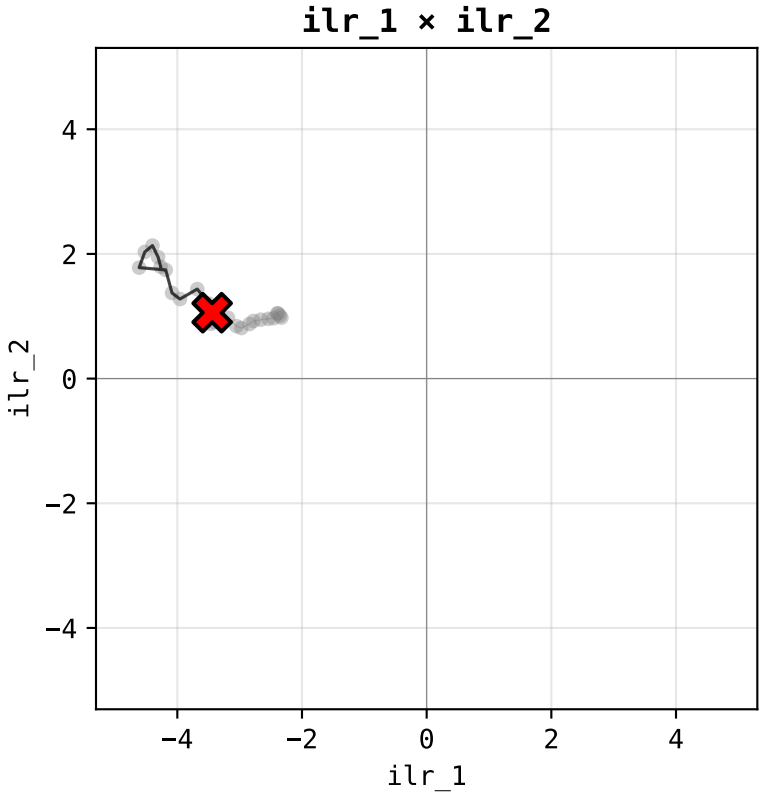
ILR Triplet – CHN China · year 2008 (9/26)



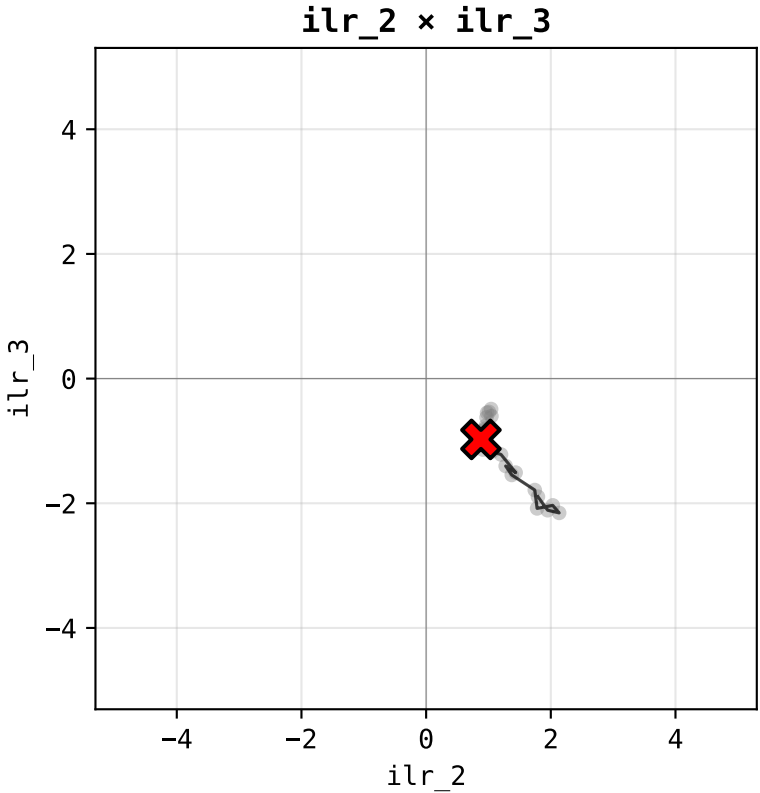
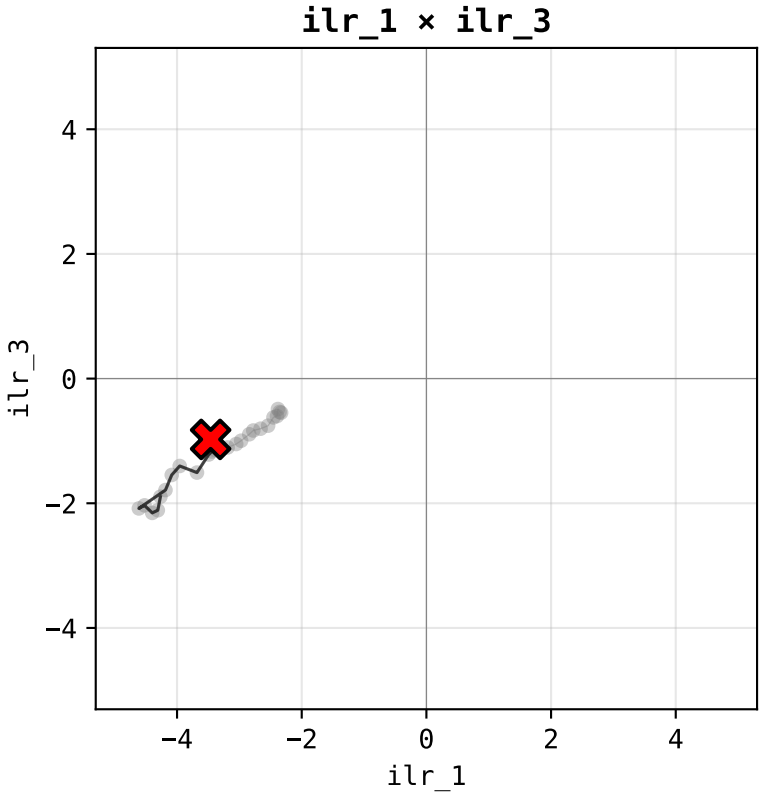
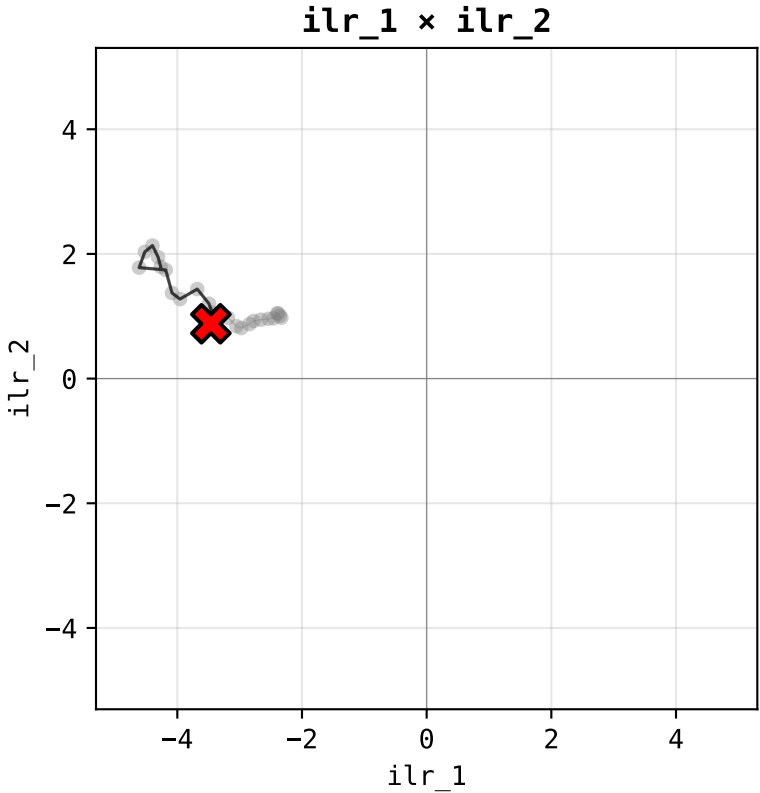
ILR Triplet – CHN China · year 2009 (10/26)

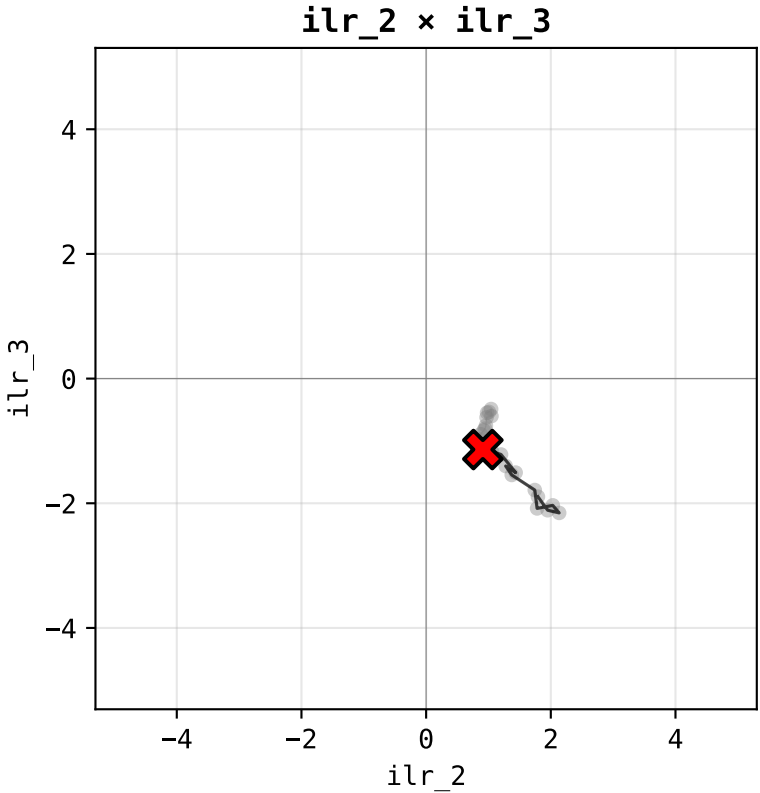
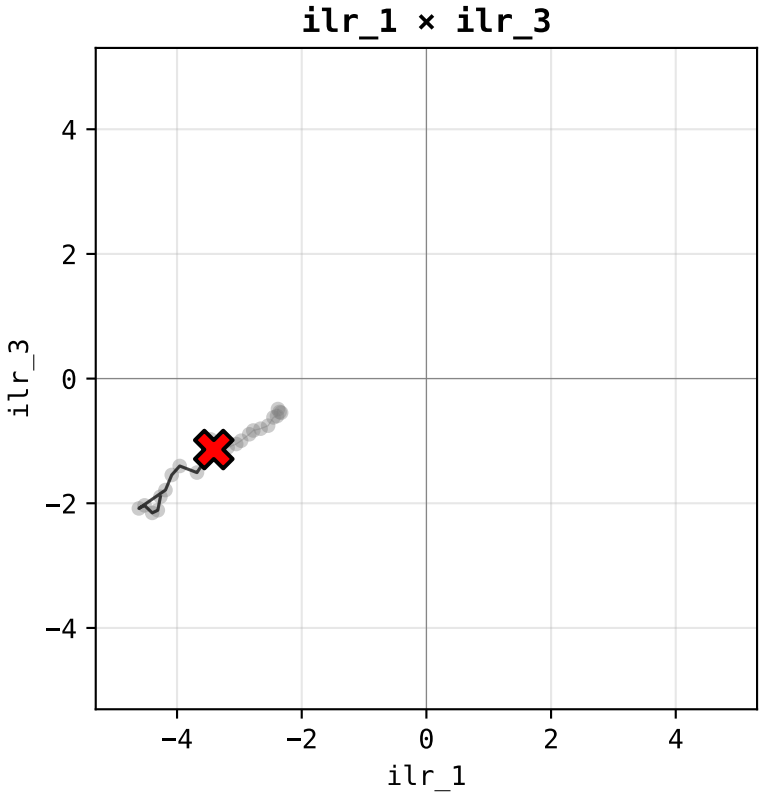
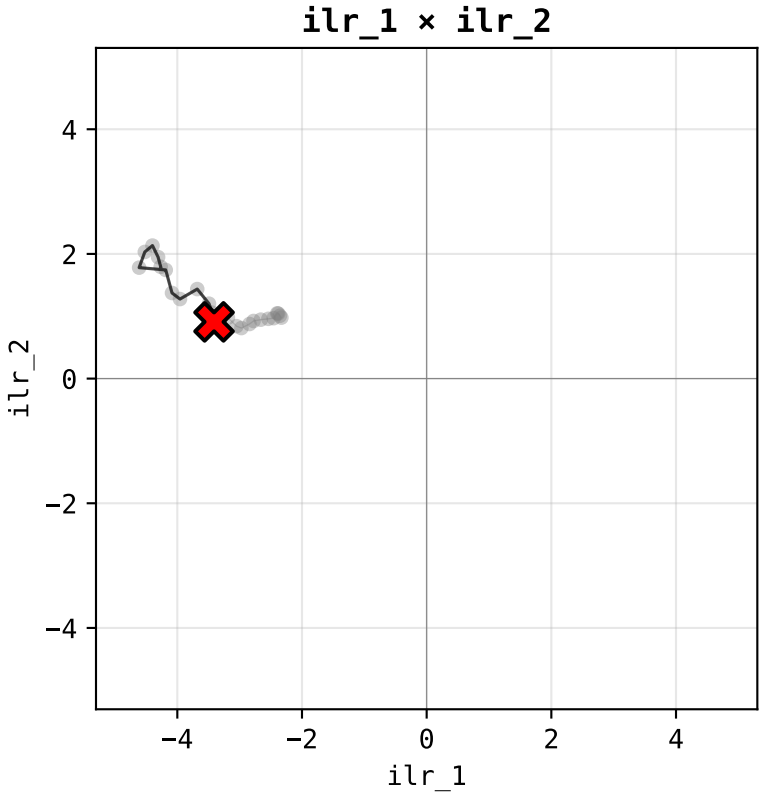


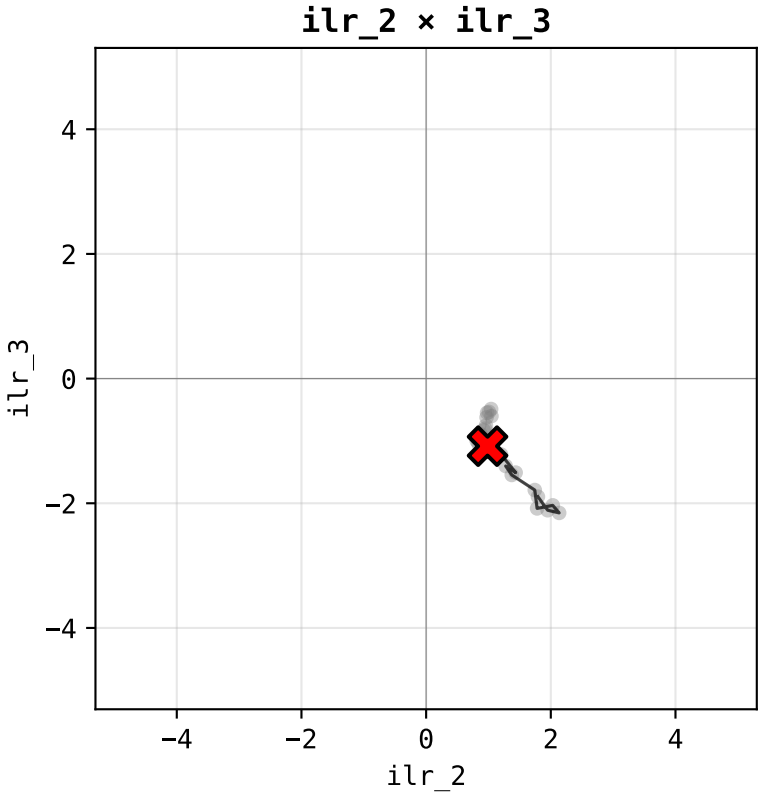
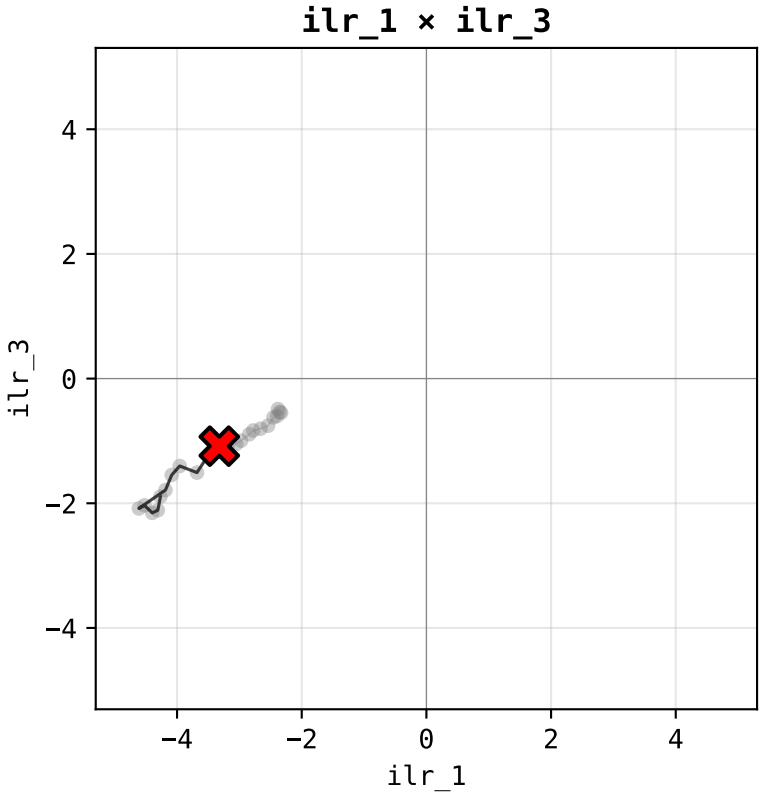
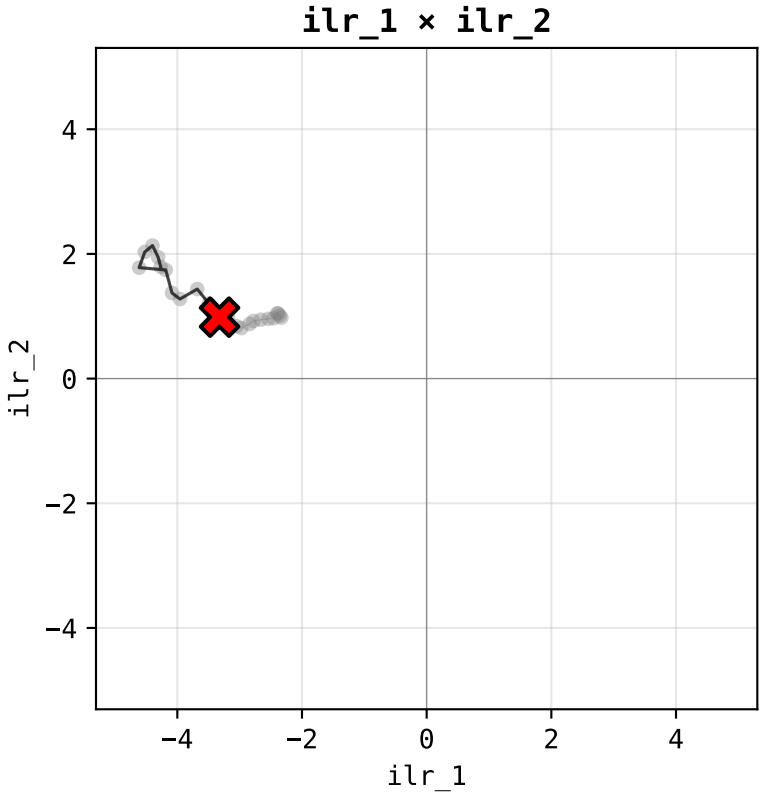
ILR Triplet – CHN China · year 2010 (11/26)

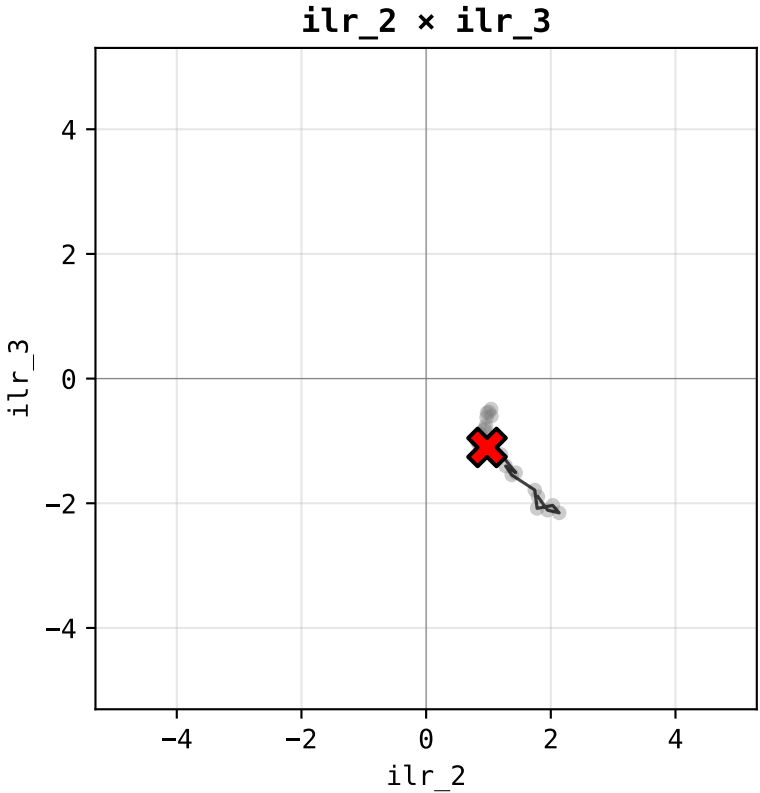
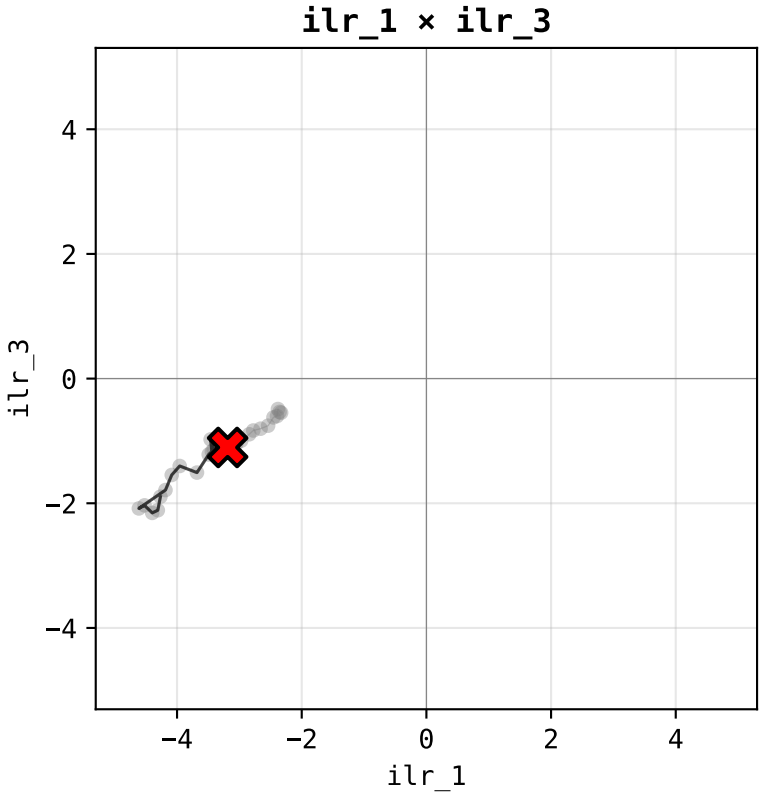
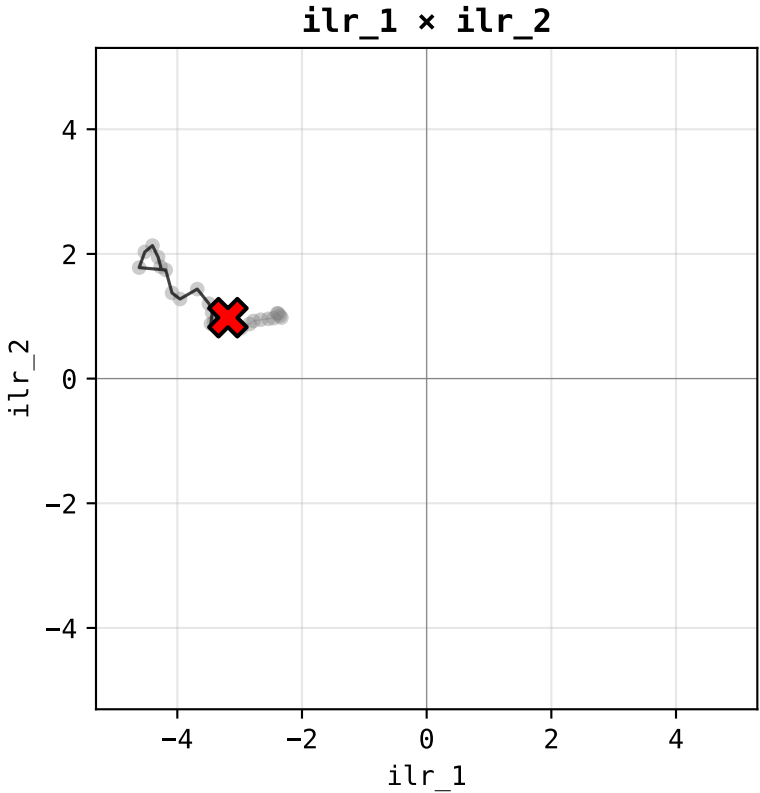


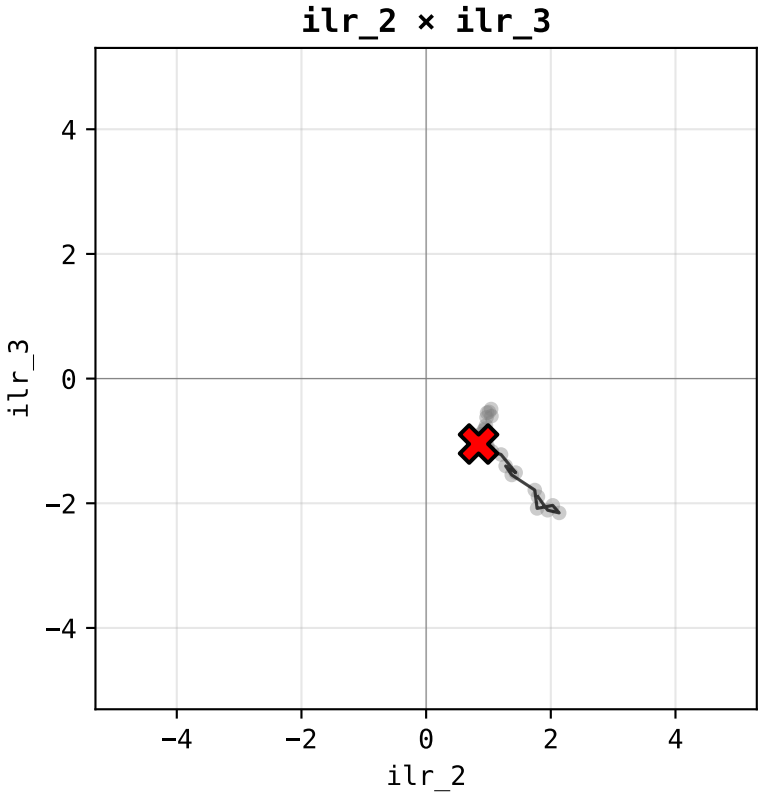
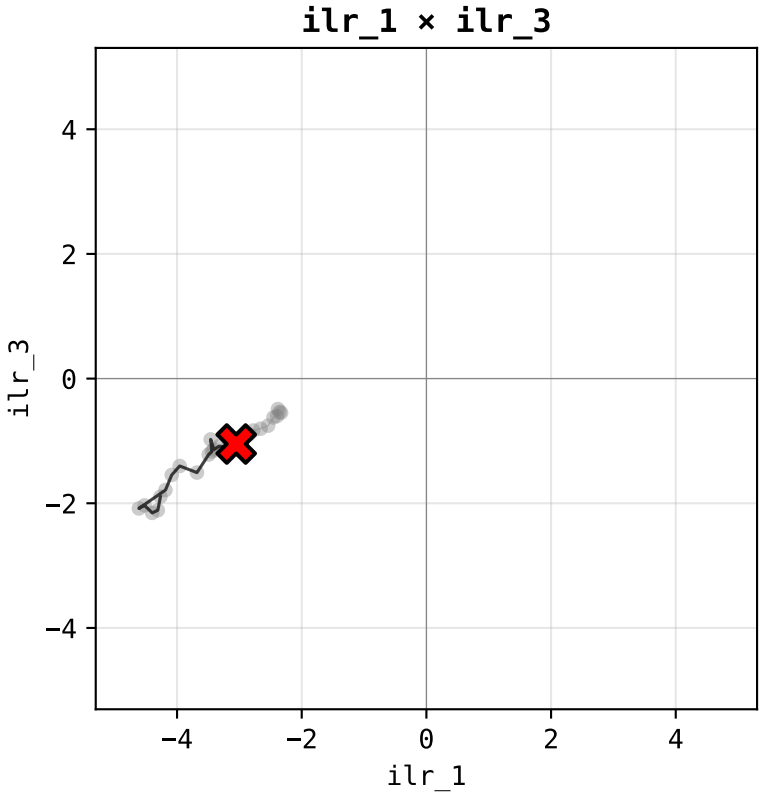
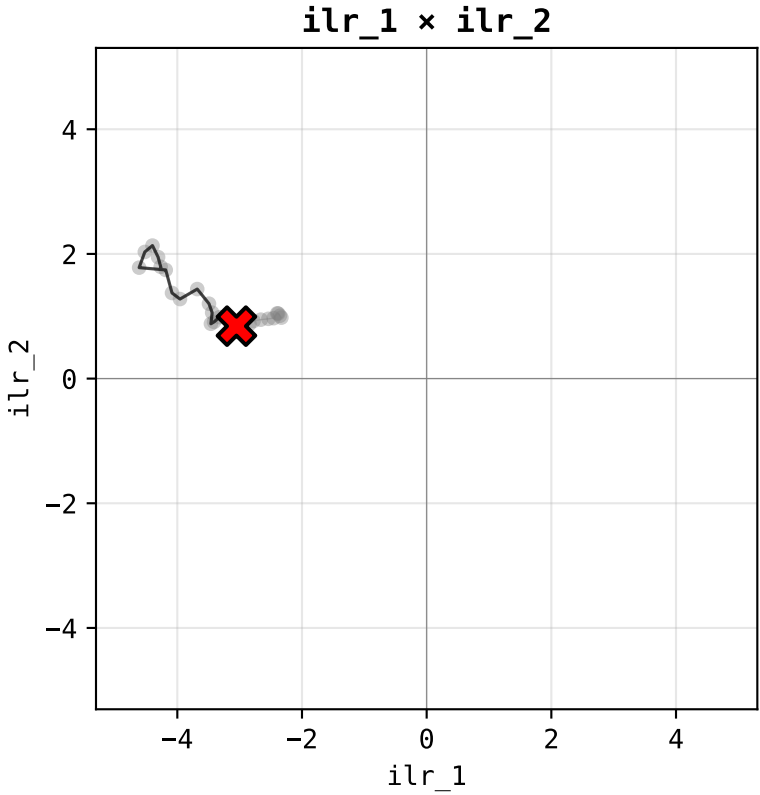
ILR Triplet – CHN China · year 2011 (12/26)

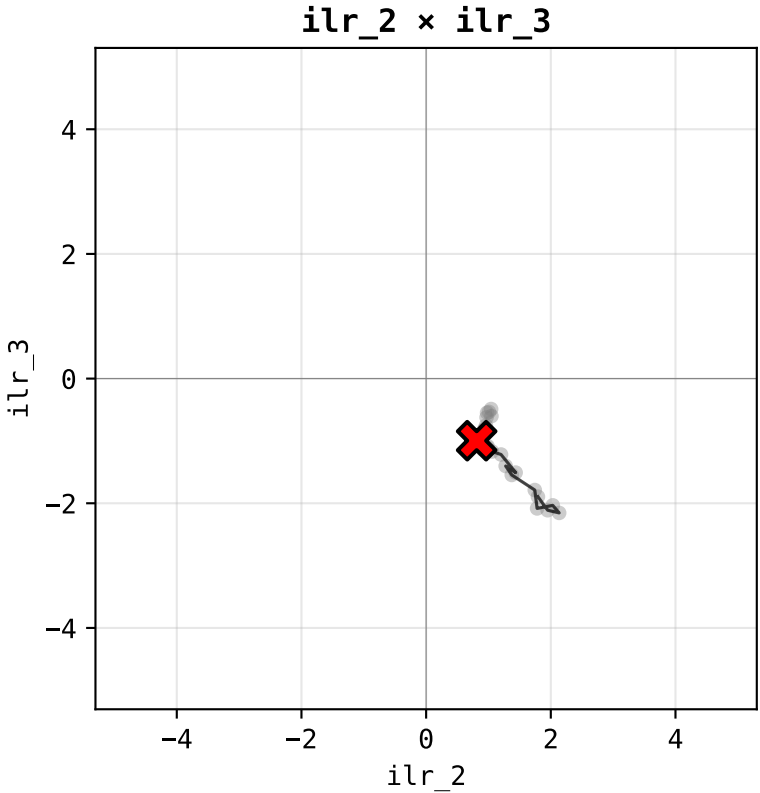
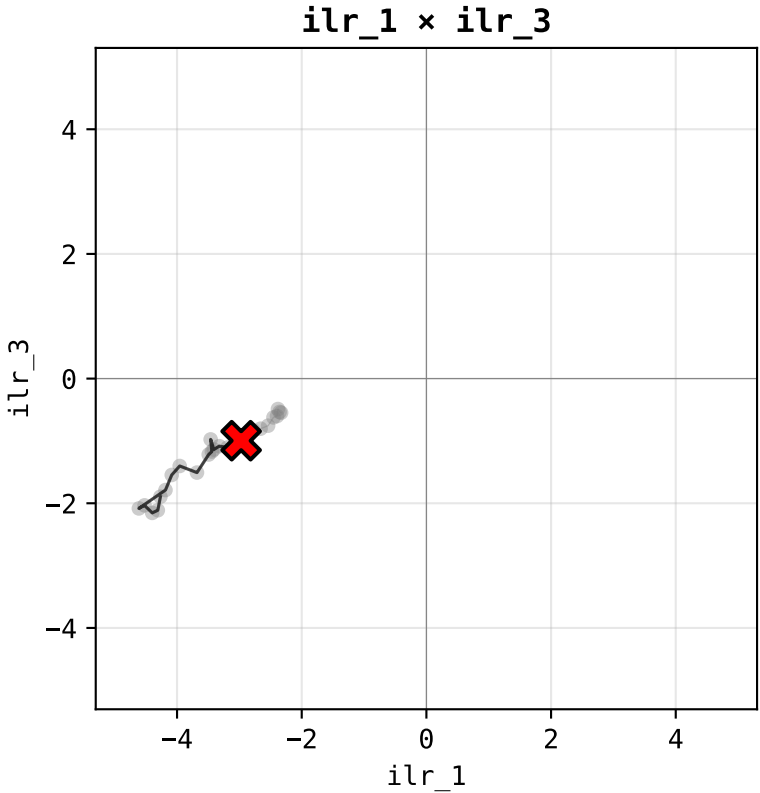
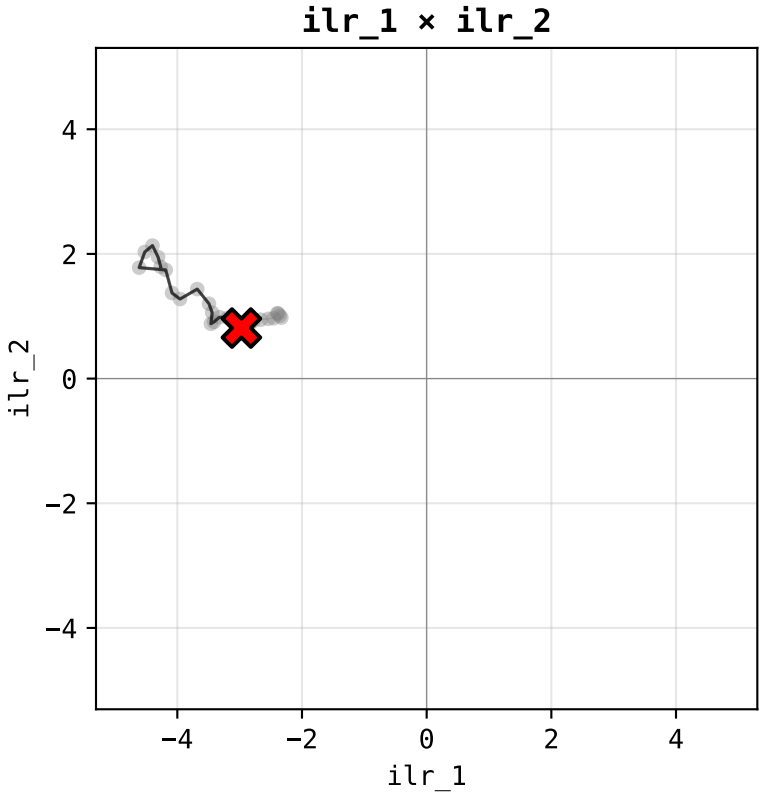


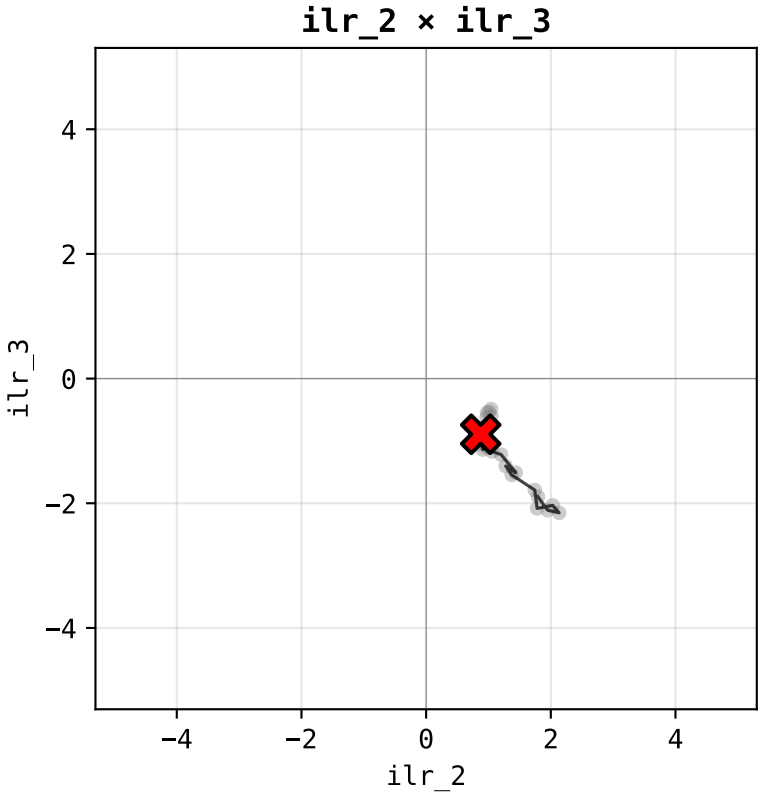
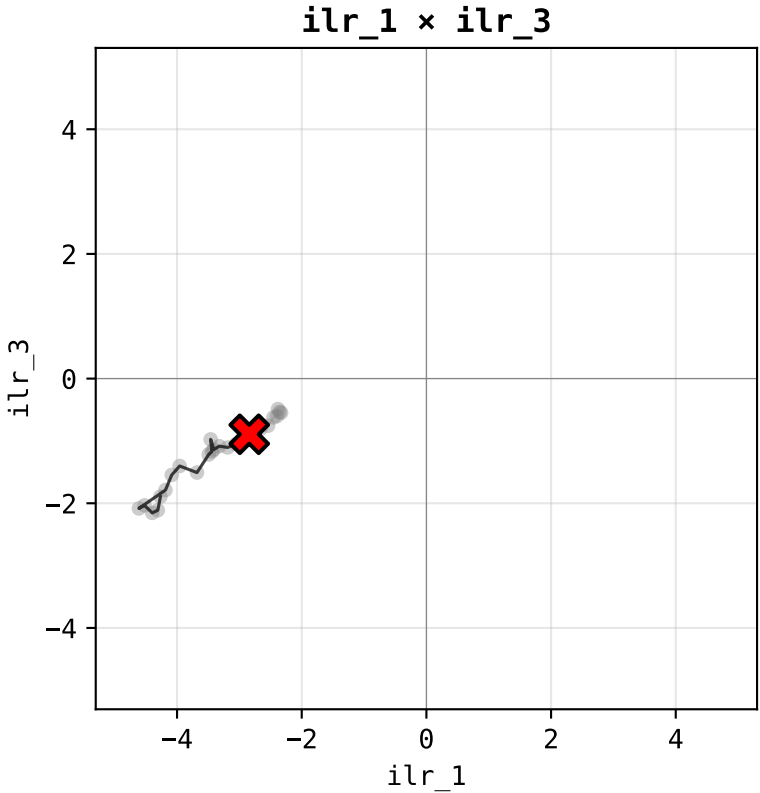
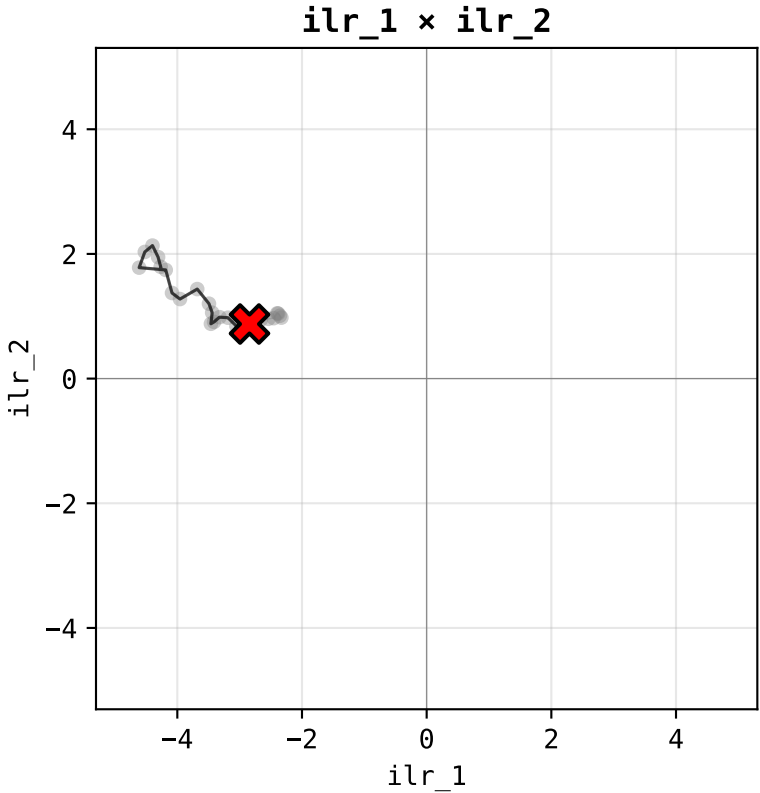


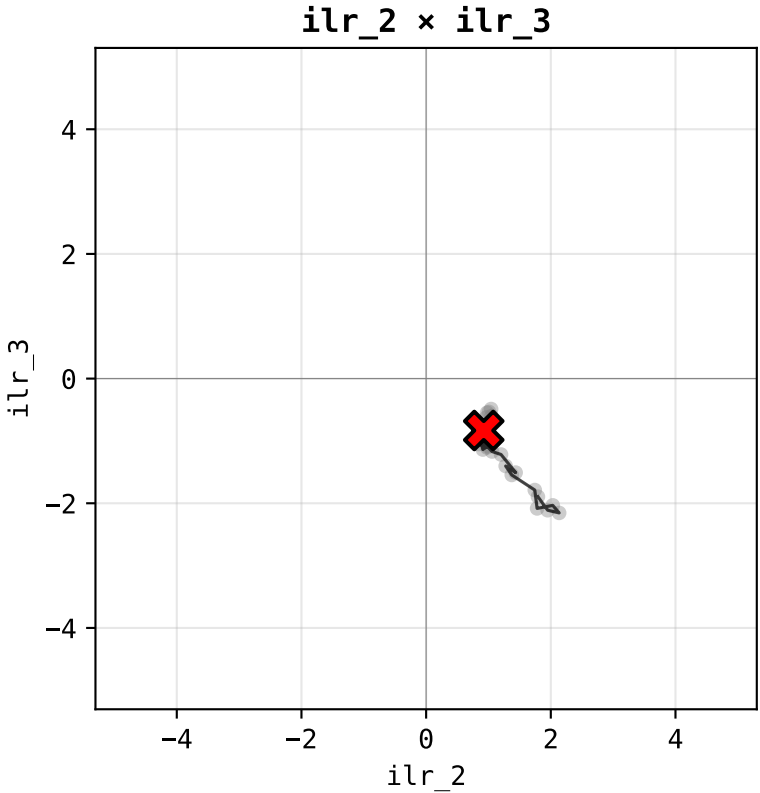
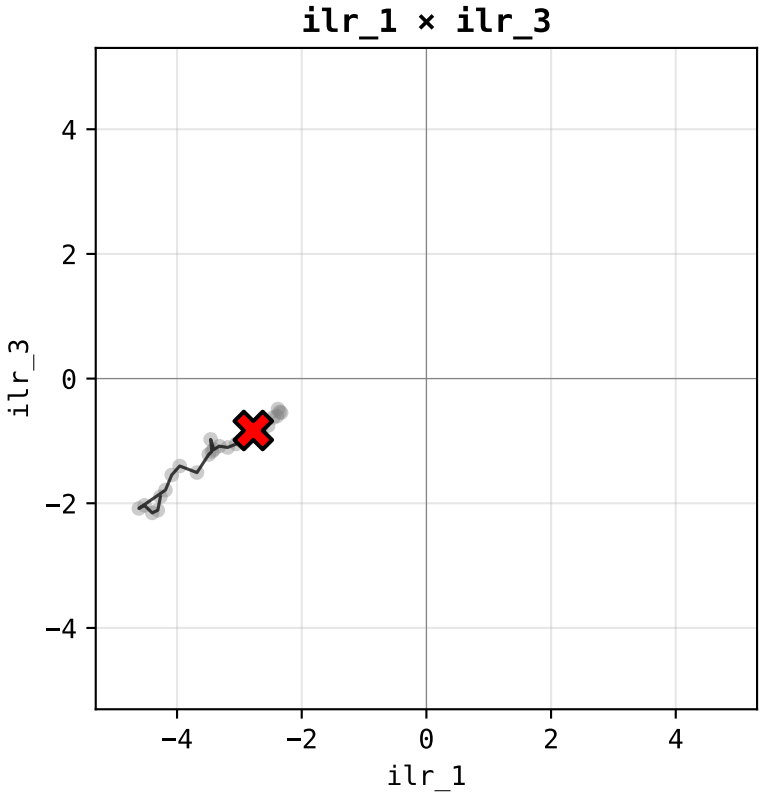
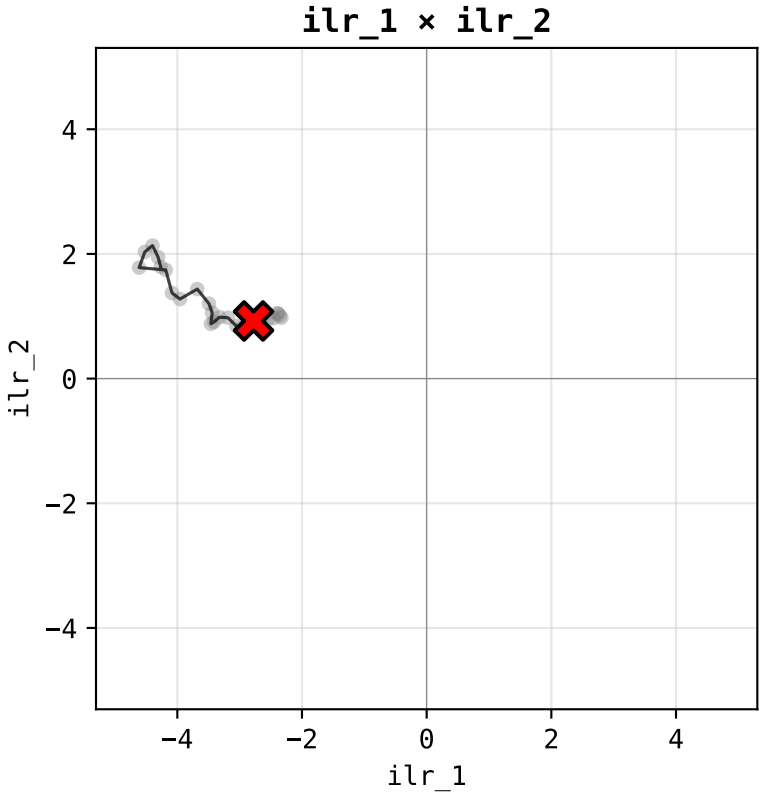


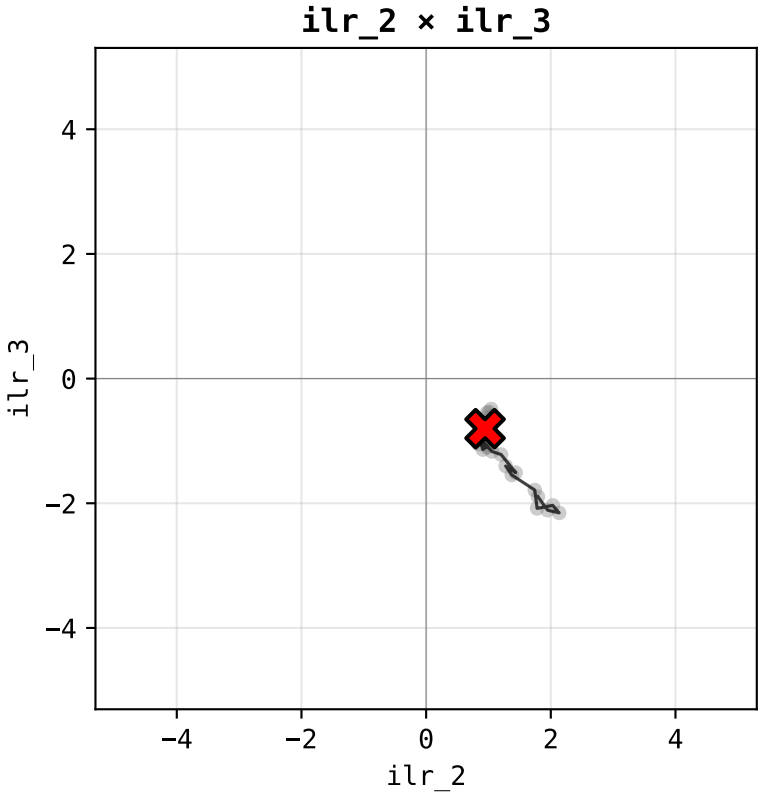
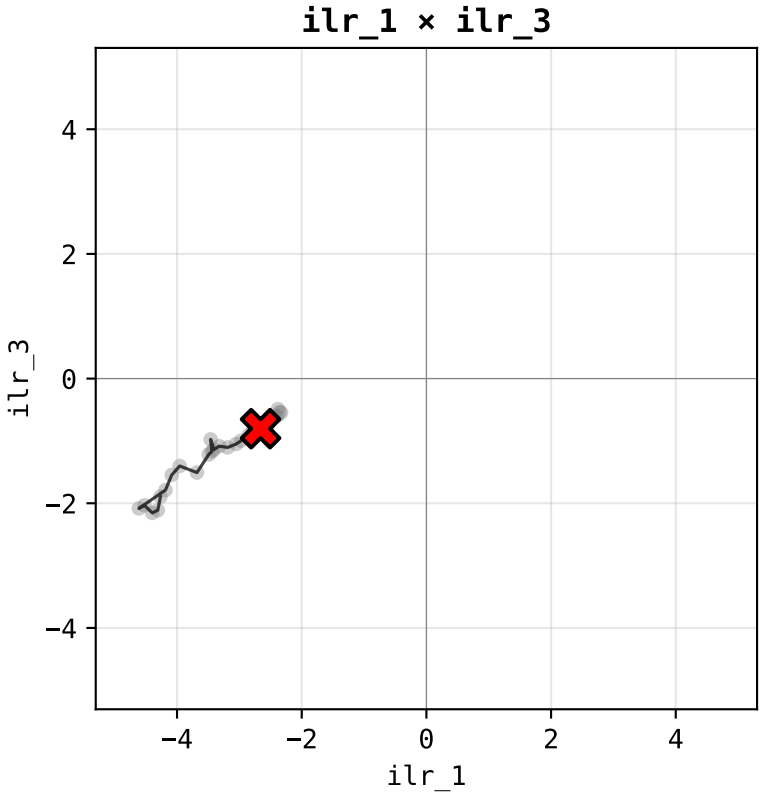
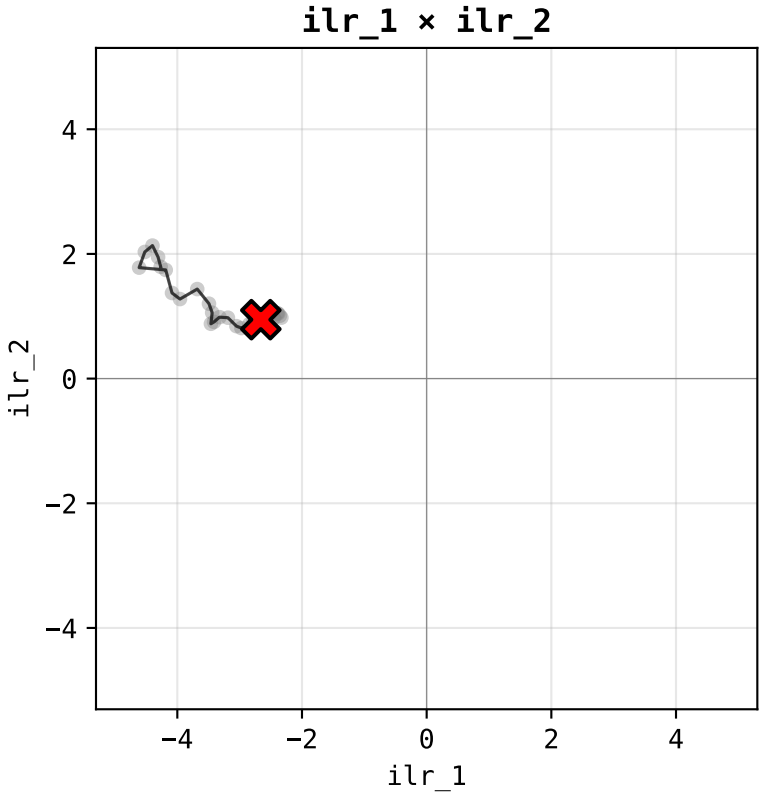


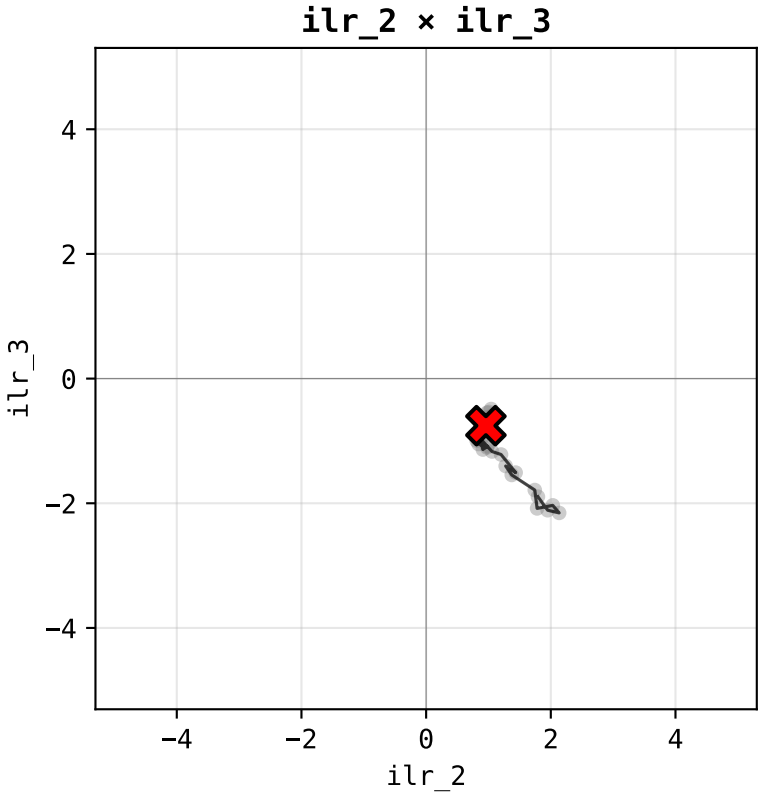
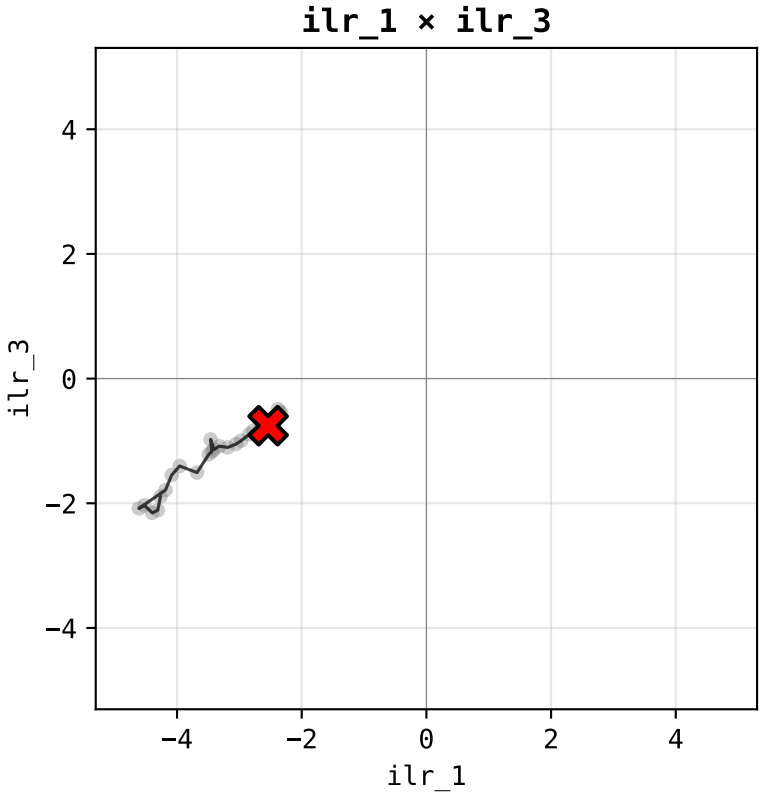
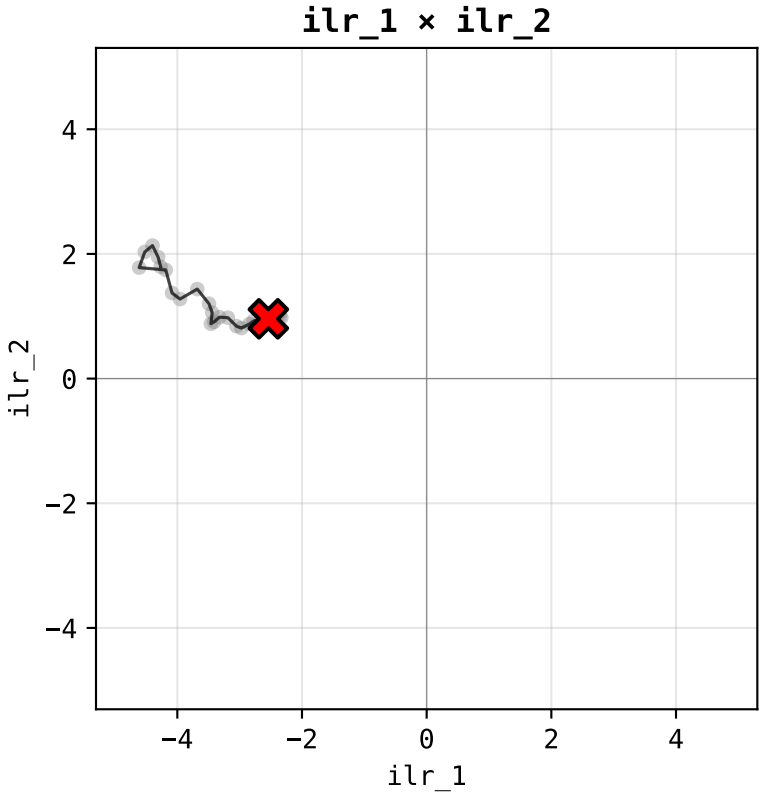


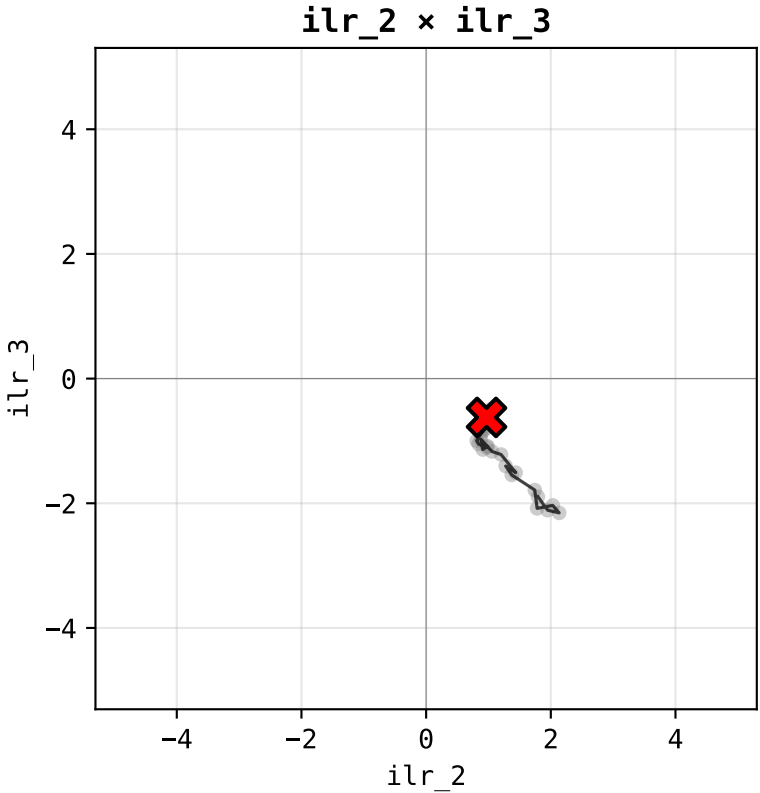
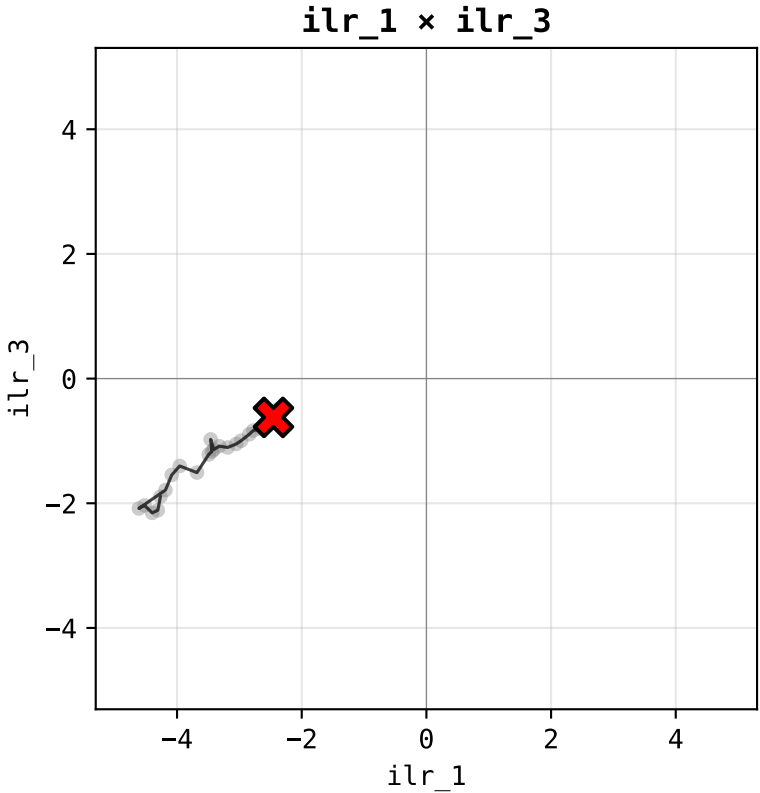
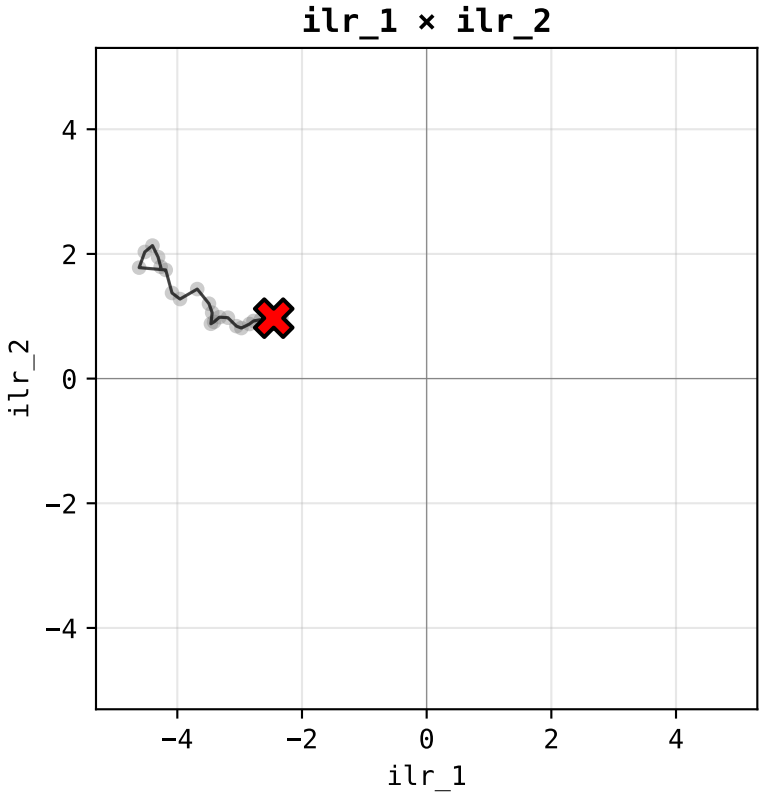


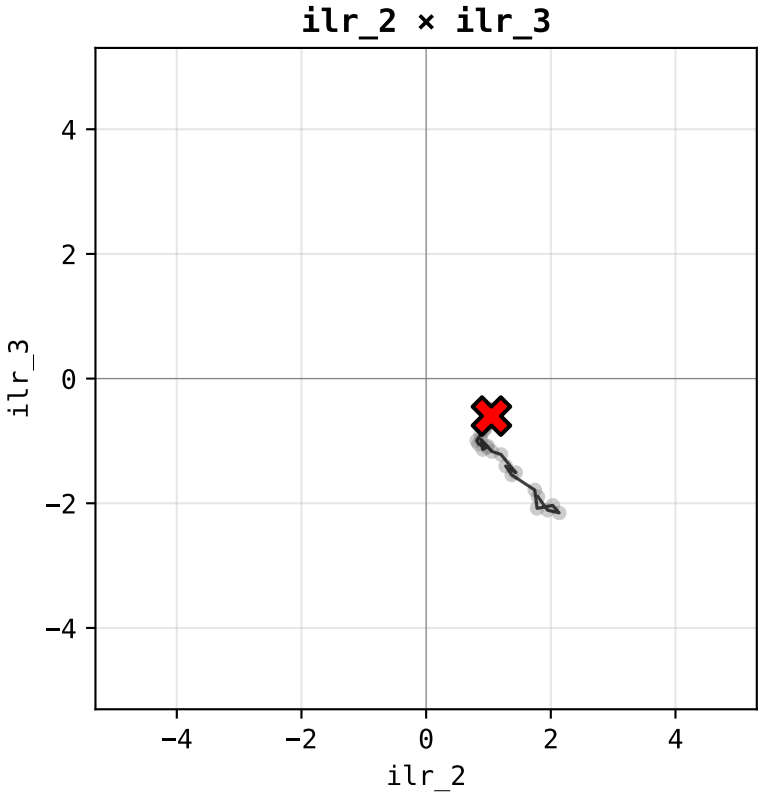
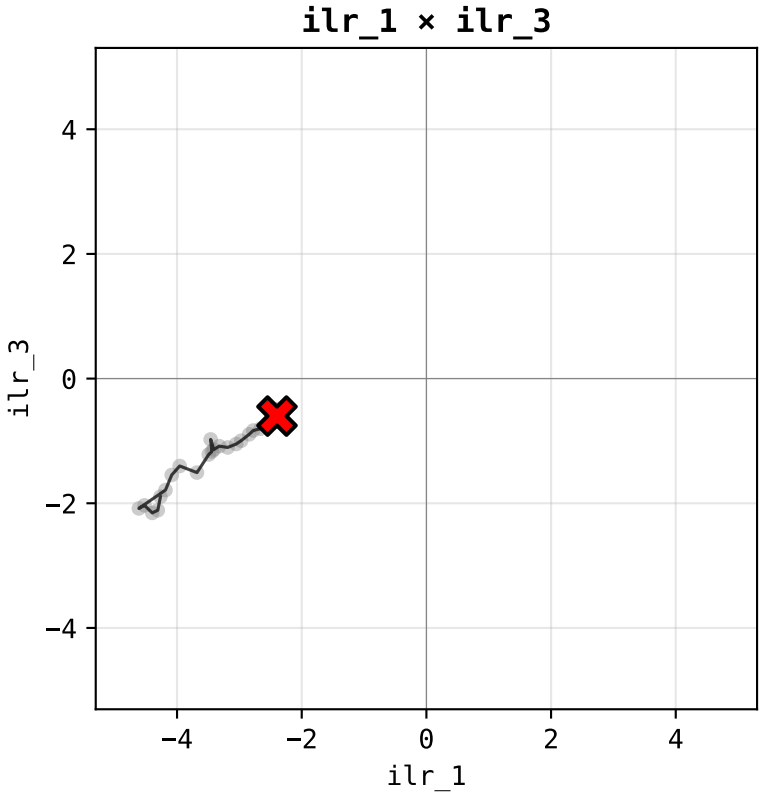
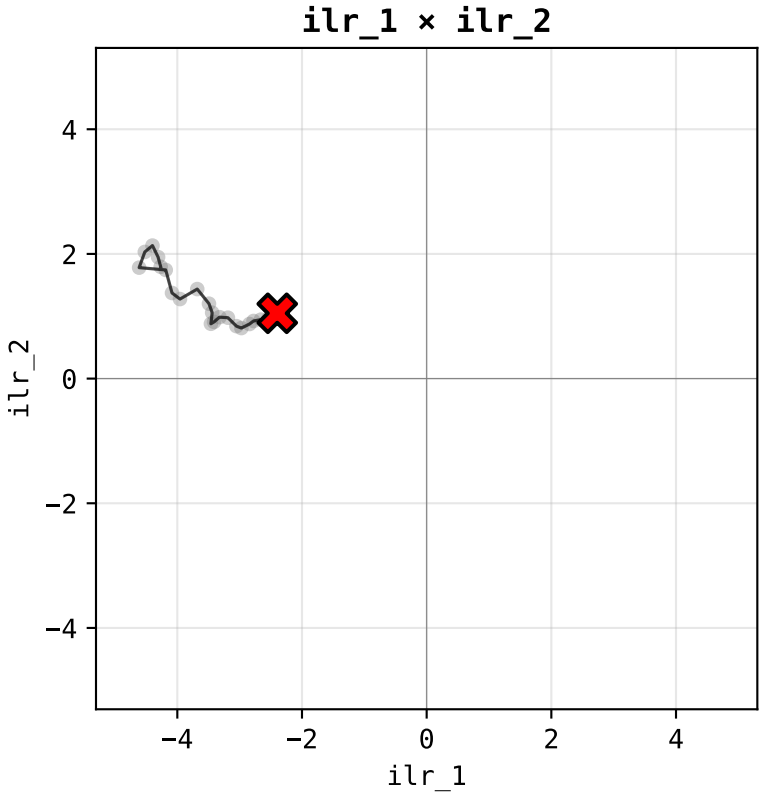


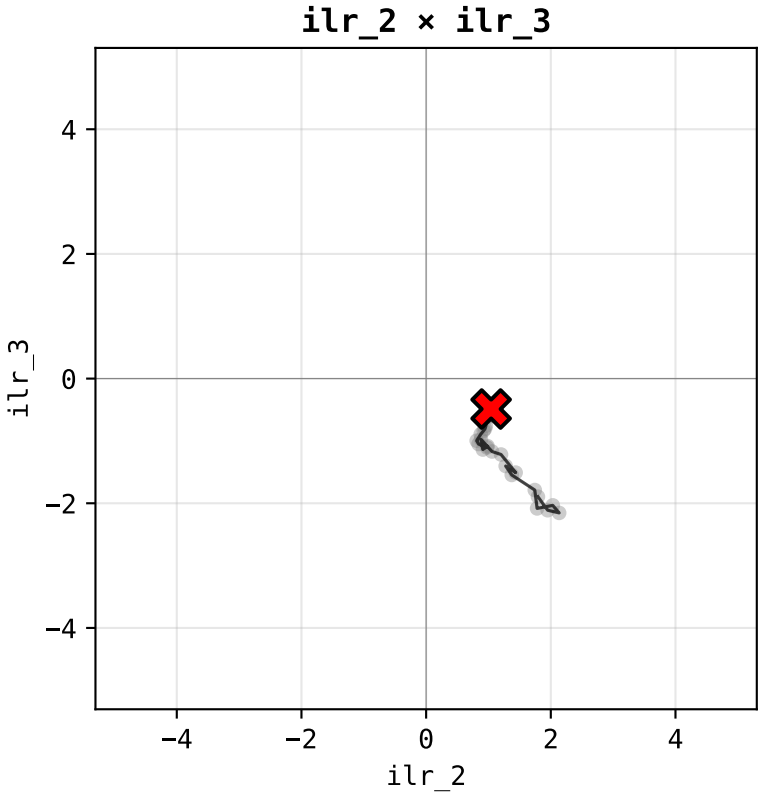
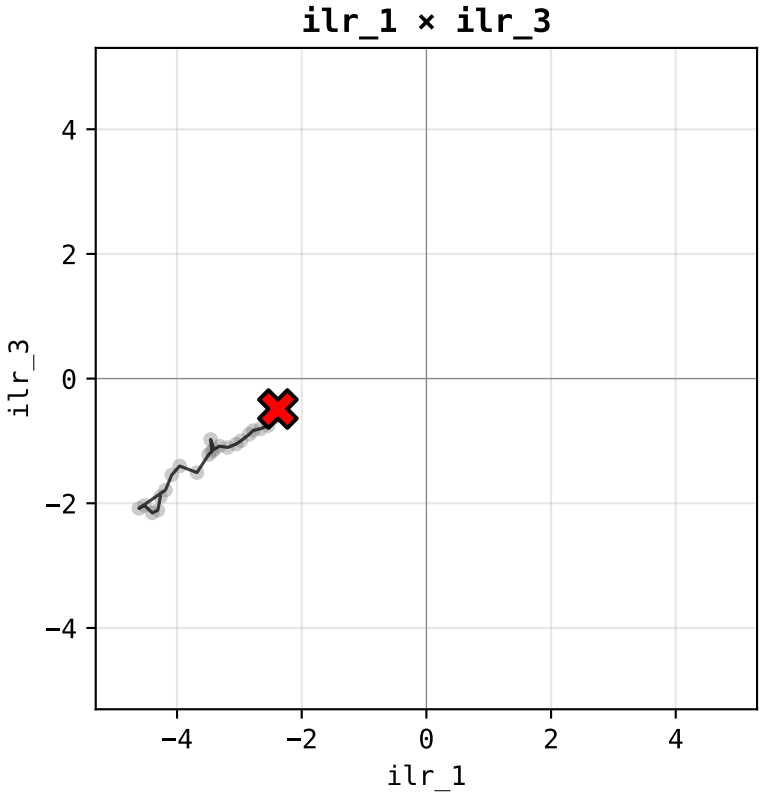
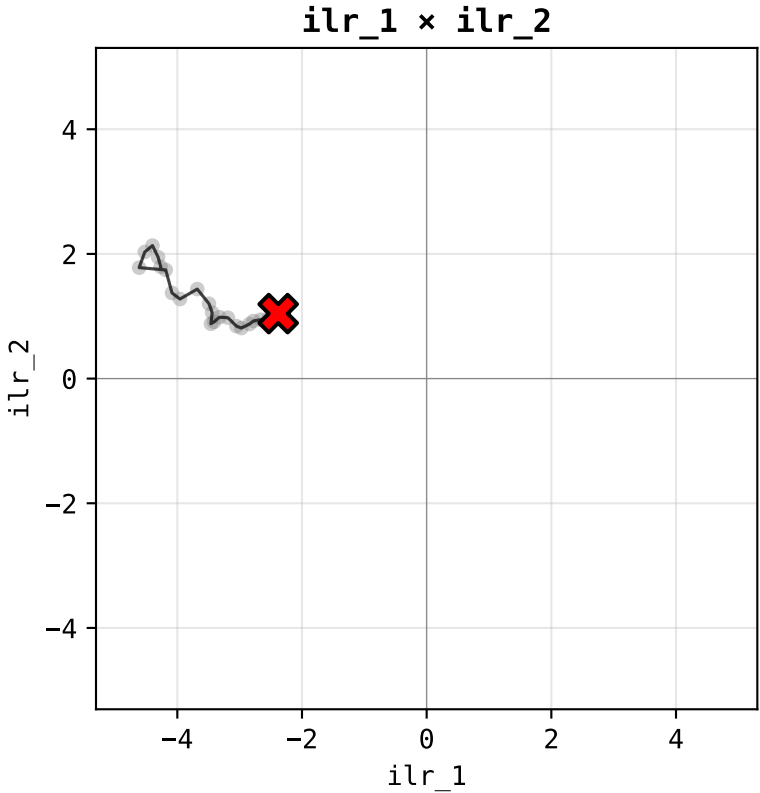


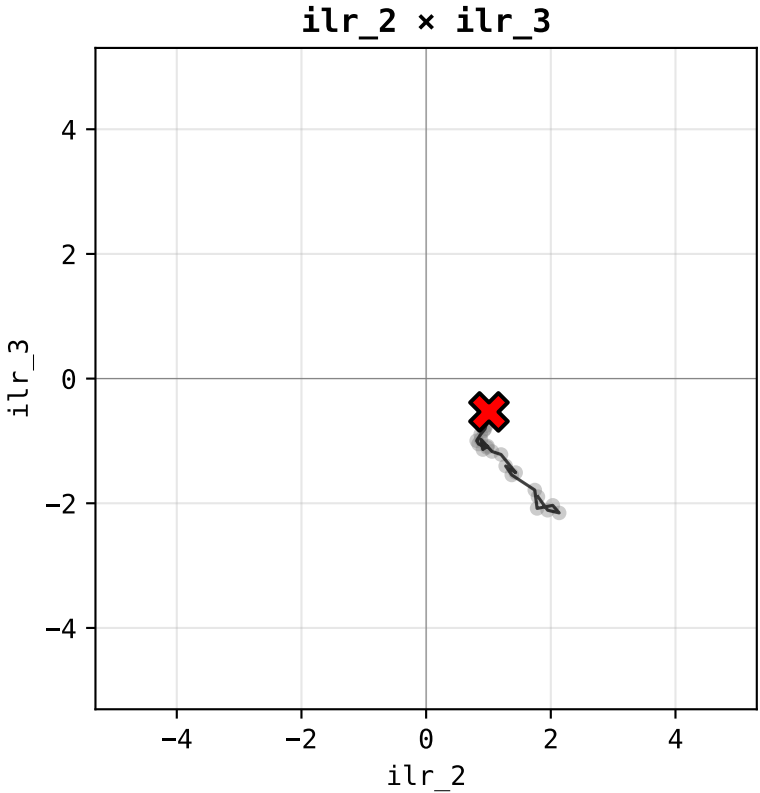
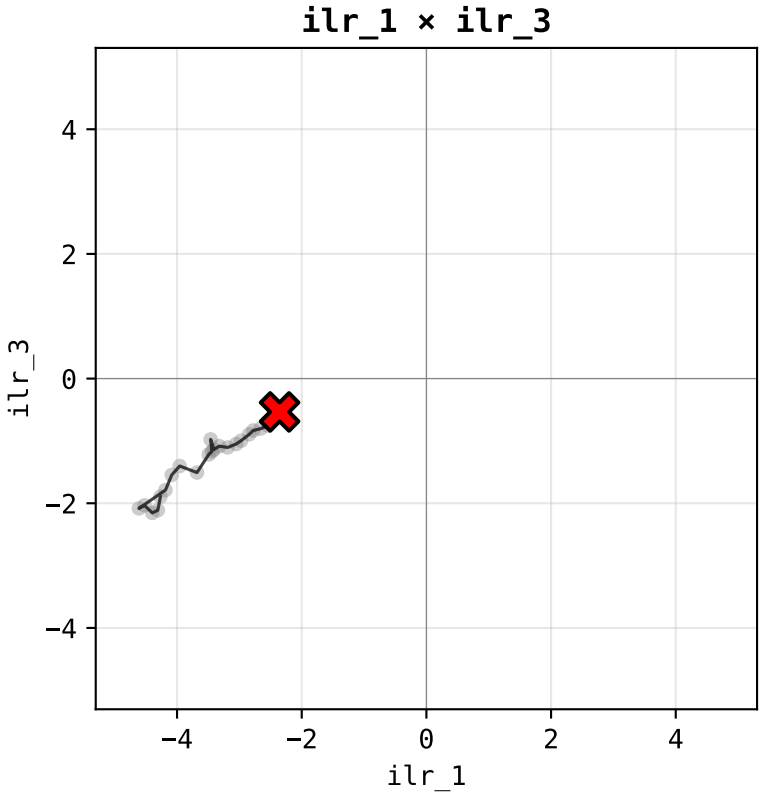
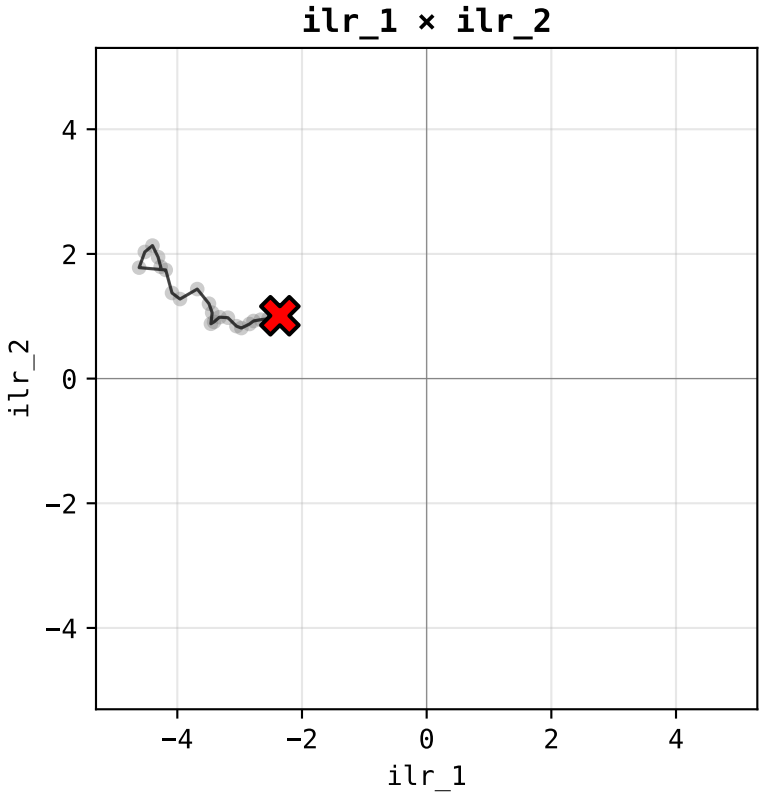


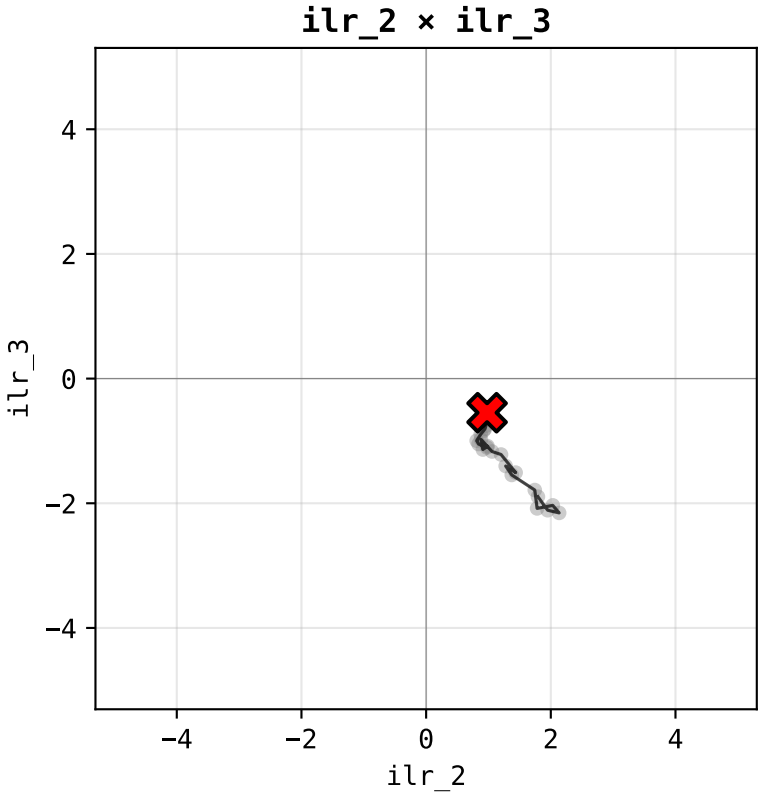
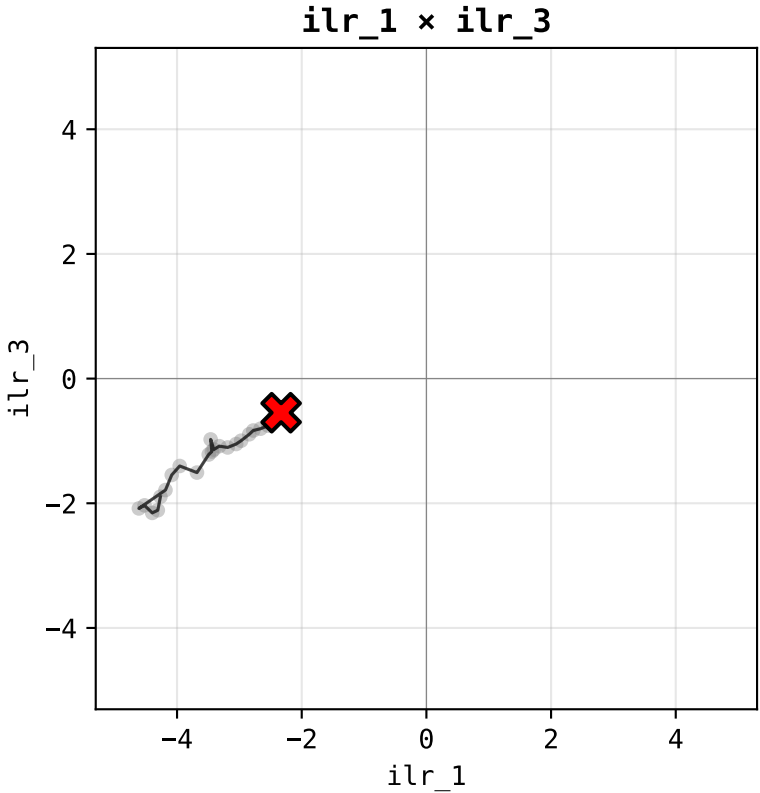
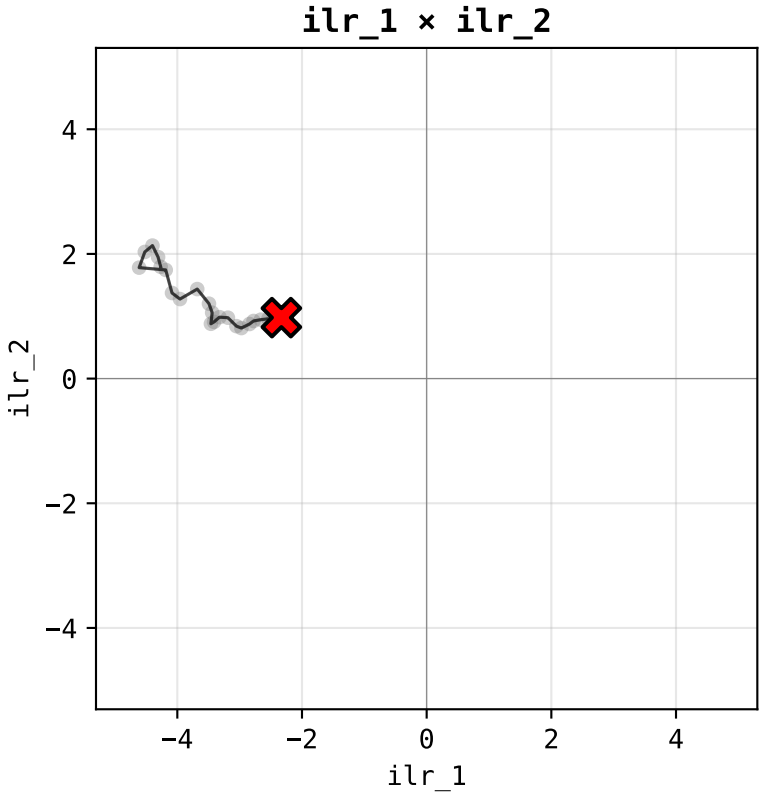












View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ DEU Germany

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

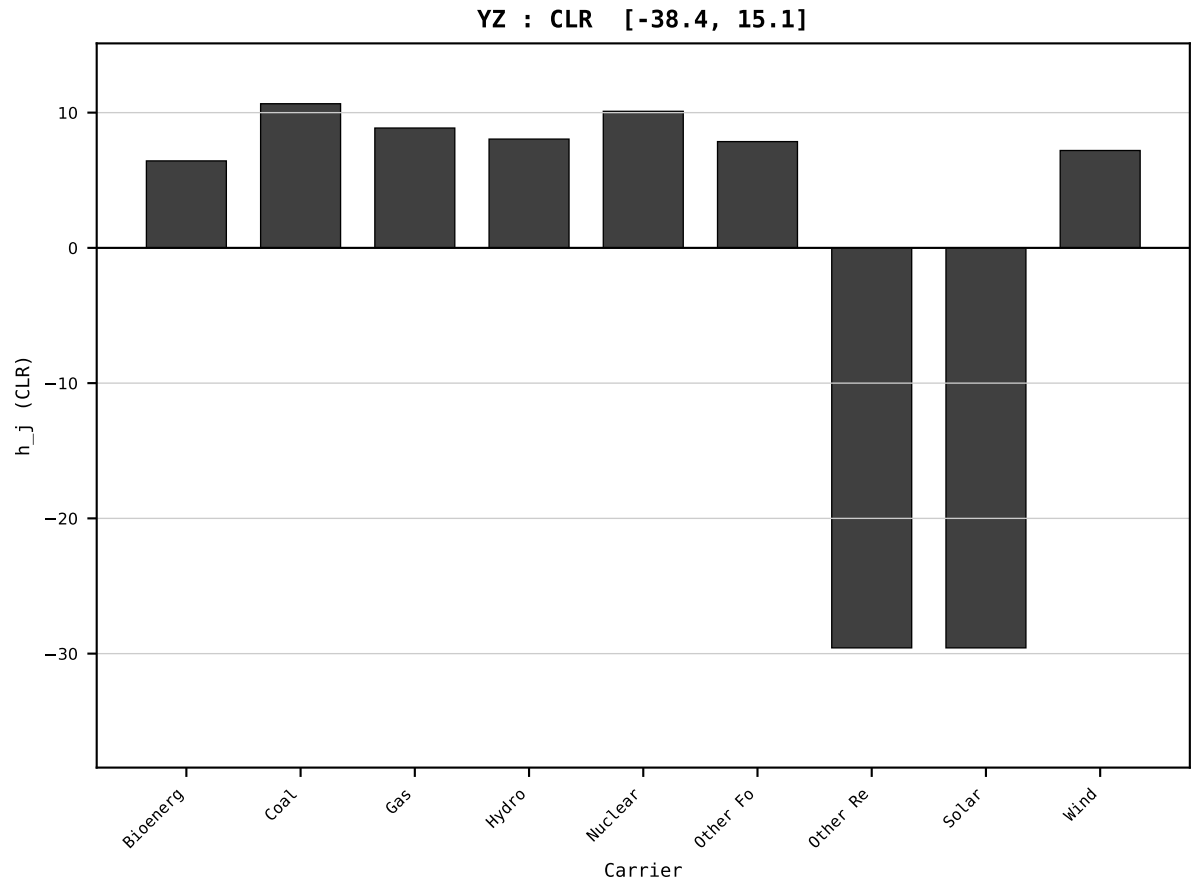
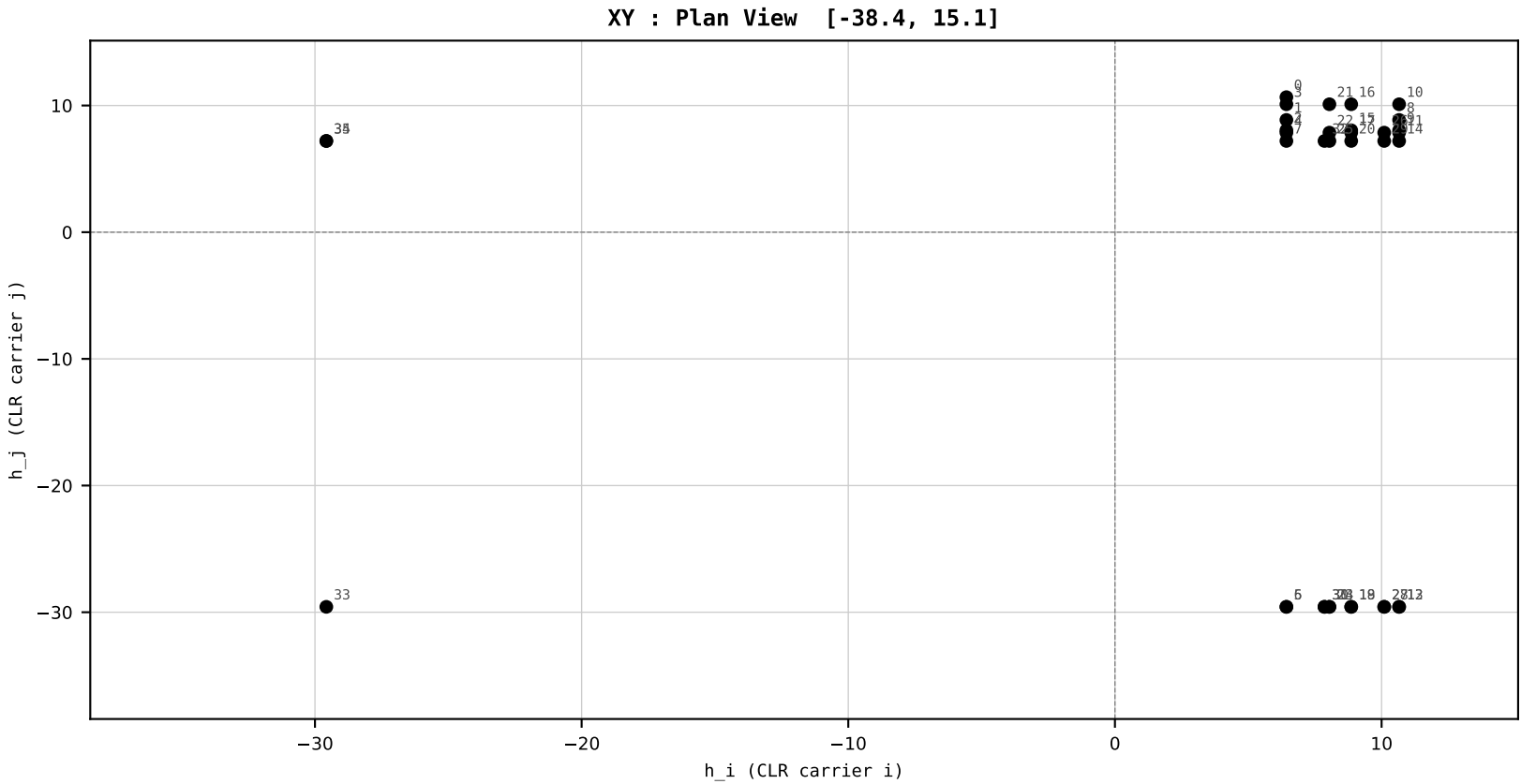
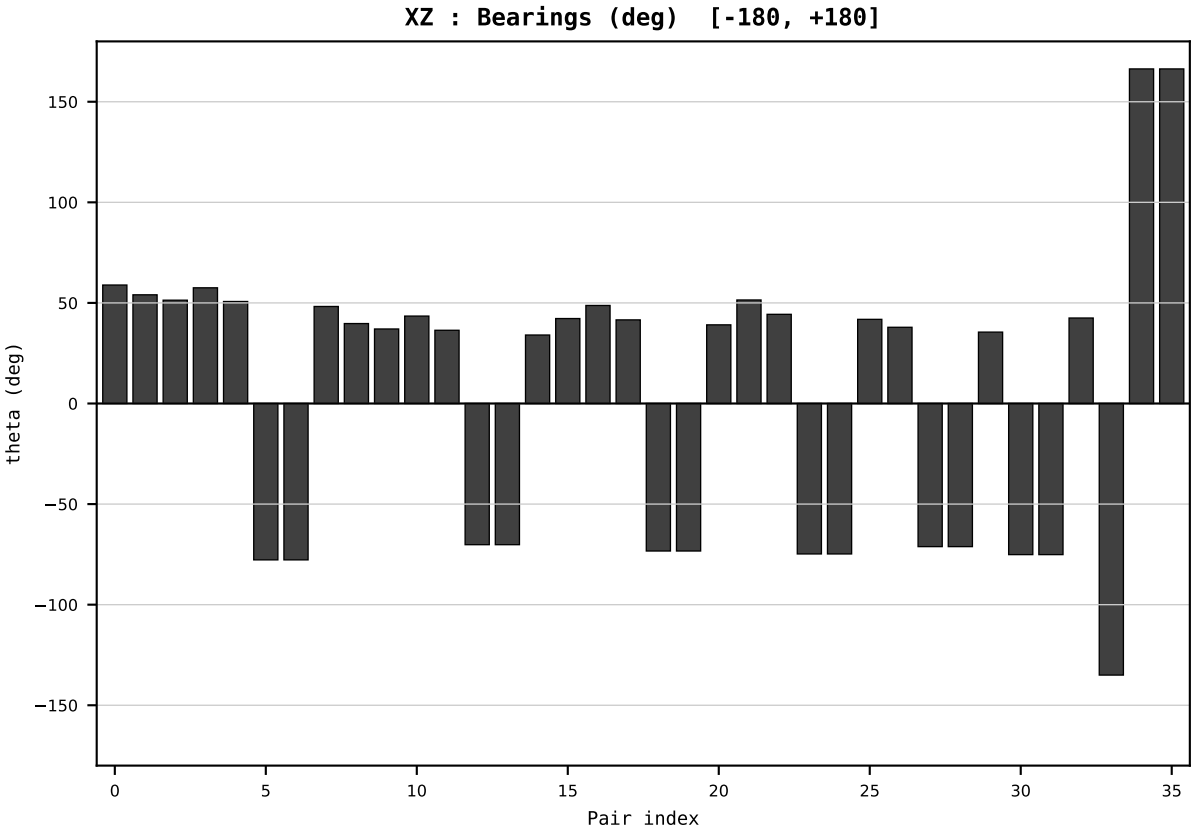
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.430719
Ring = Hs-3
E_metric= 1575.1016
kappa_HS= 296680000000000000.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 40.231 HLR
DR ratio= 296679999472192320.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

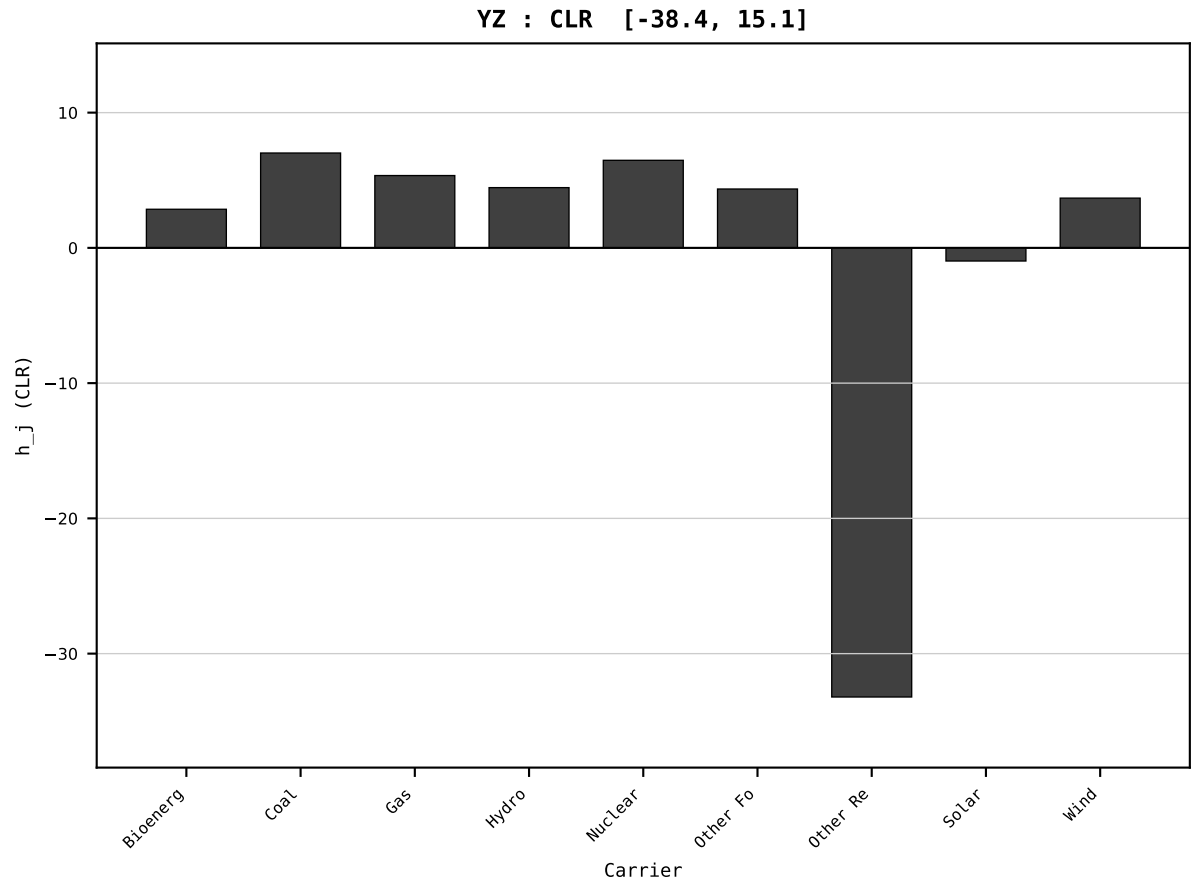
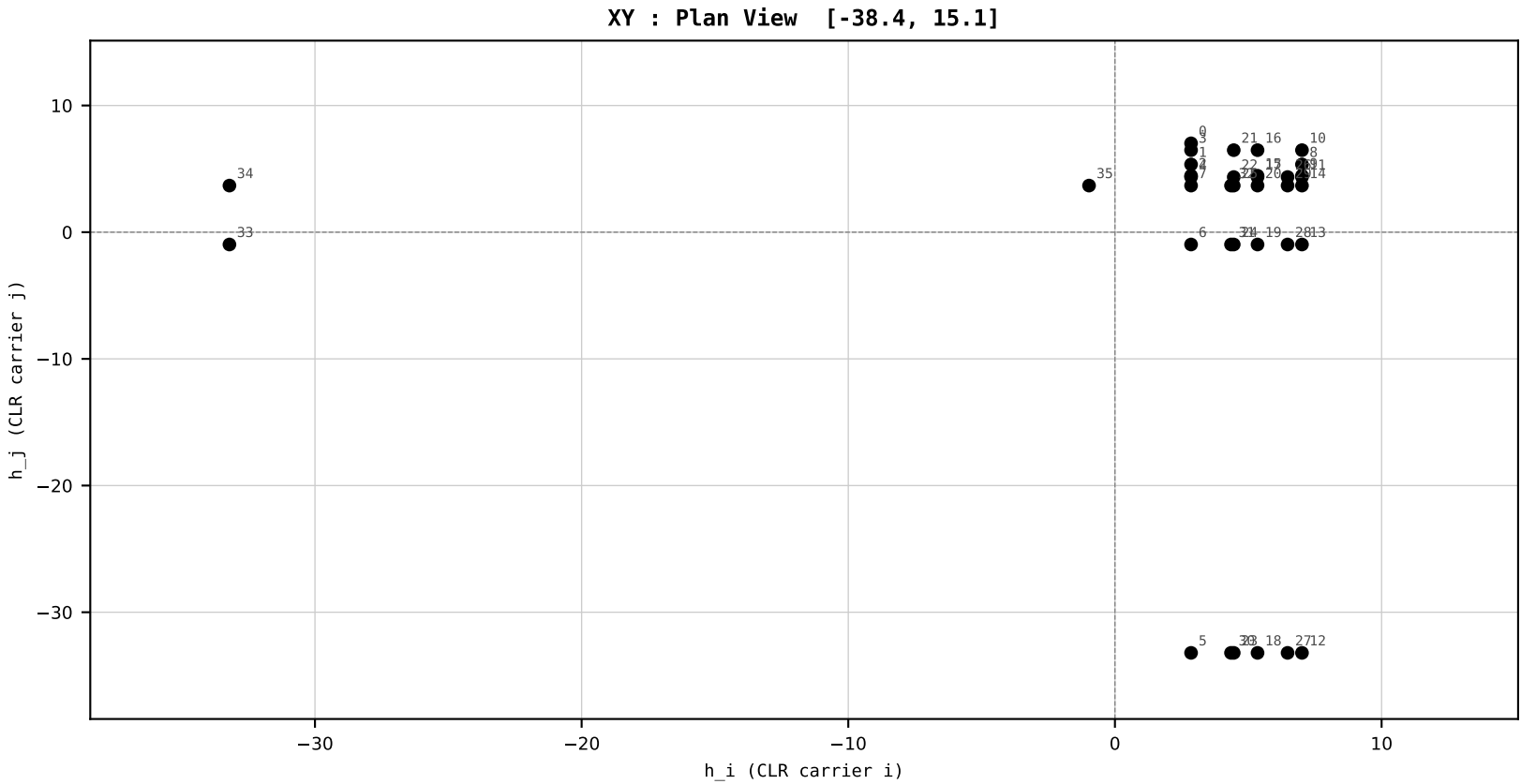
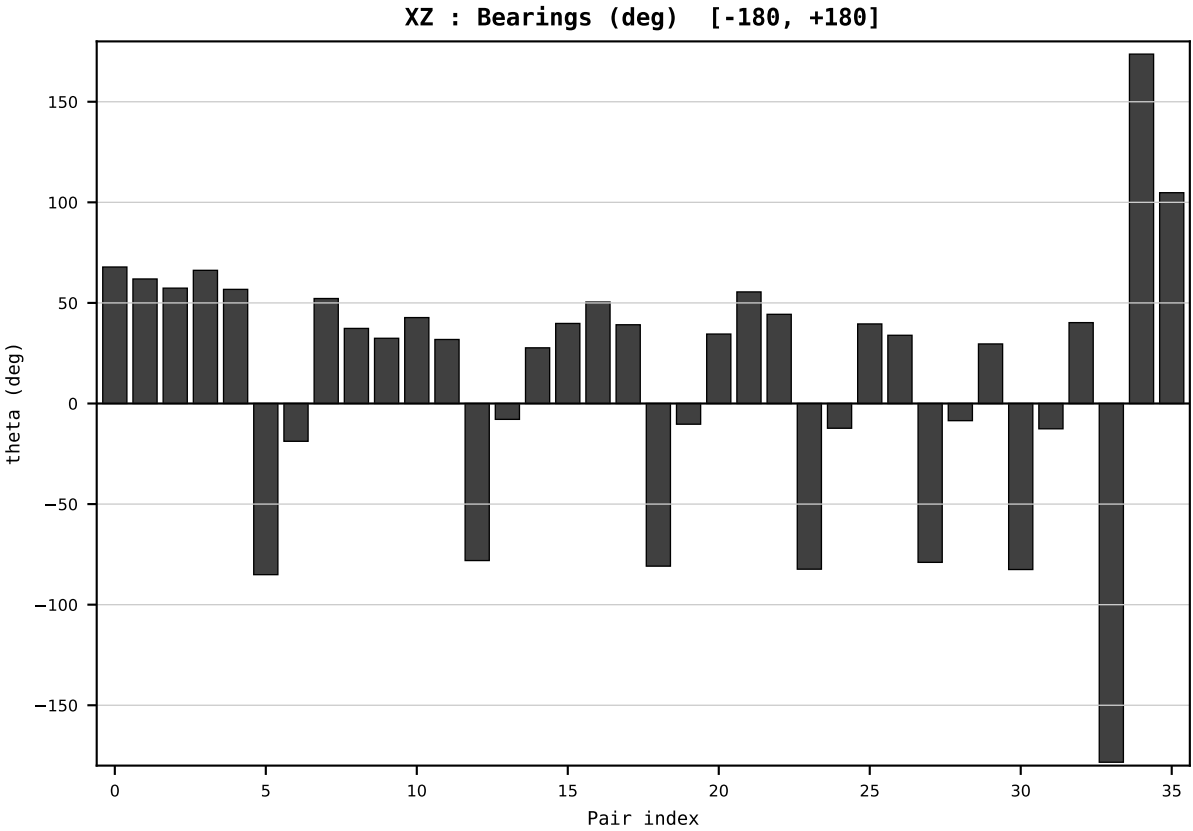
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.414163
Ring = Hs-3
E_metric= 897.3656
kappa_HS= 29374000000000000.00
omega = 39.6073 deg
d_A = 30.338234
Helm = Solar
Helm d = +28.602816
DR = 40.221 HLR
DR ratio= 293740000811779456.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

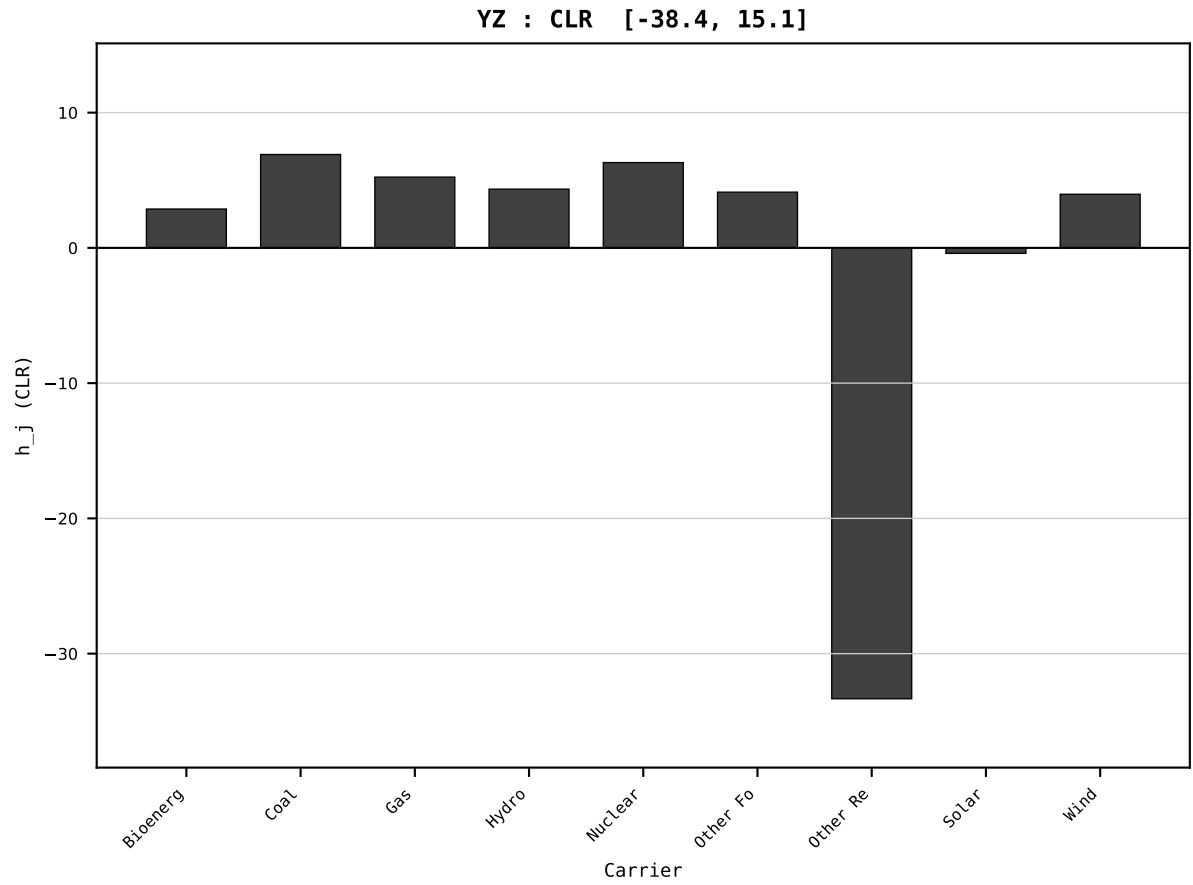
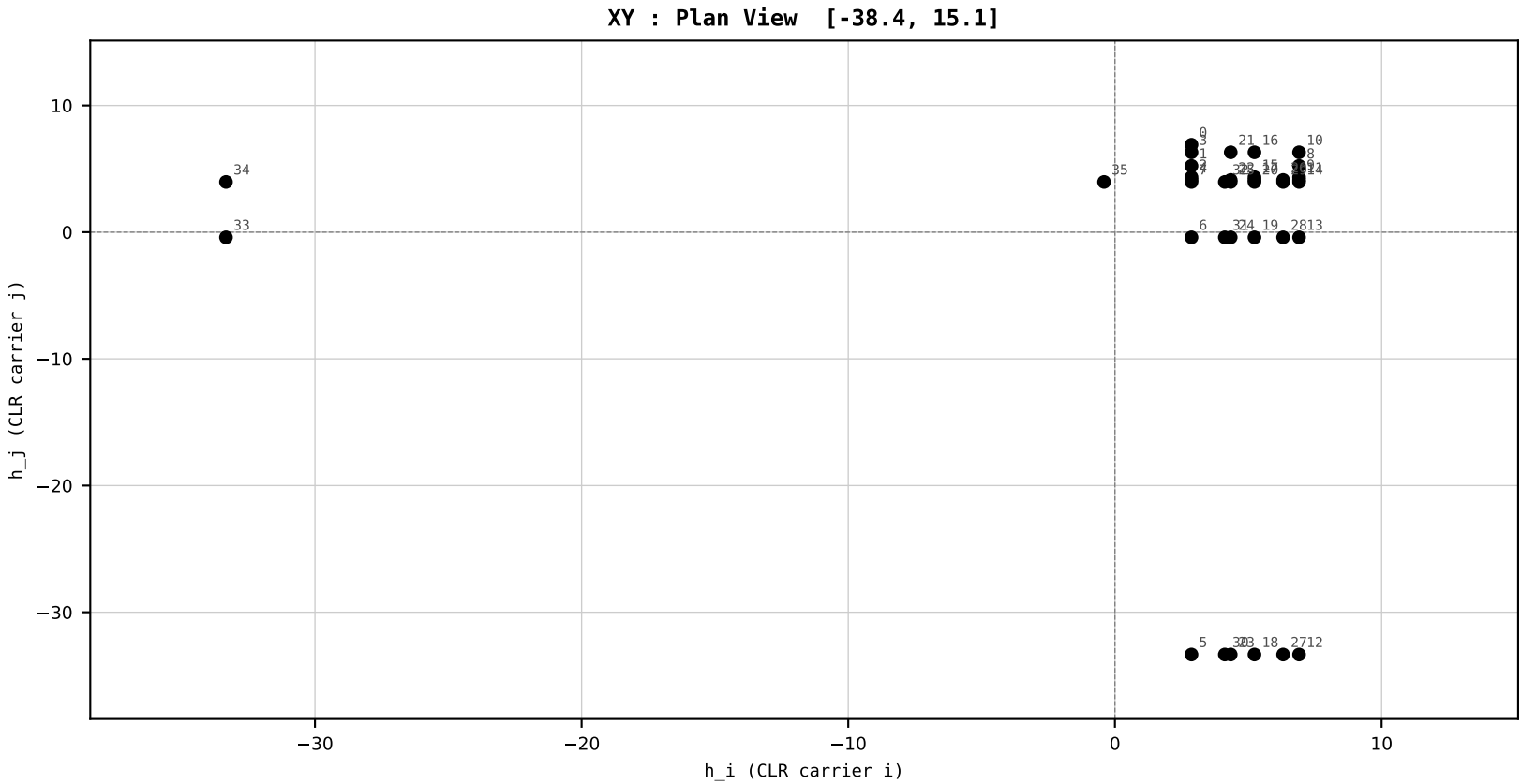
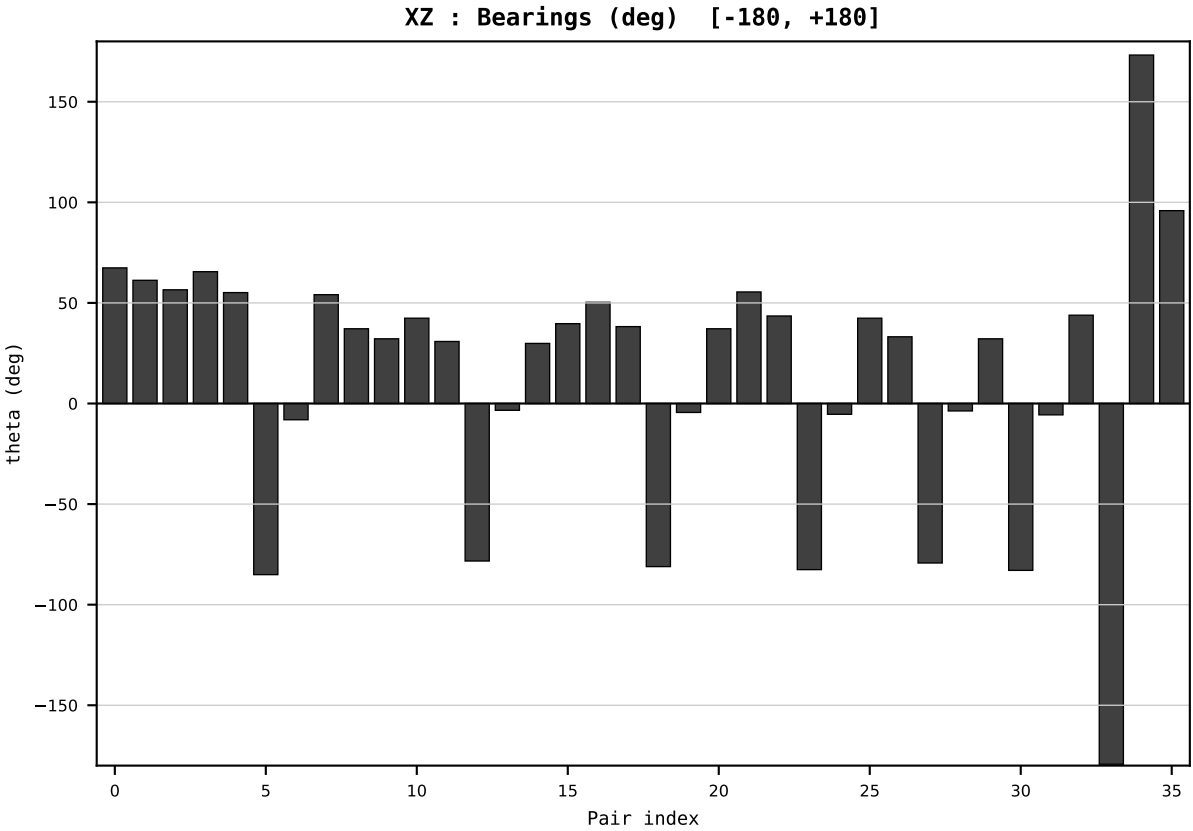
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.405699
Ring = Hs-3
E_metric= 897.5397
kappa_HS= 299600000000000000.00
omega = 1.1699 deg
d_A = 0.732611
Helm = Solar
Helm d = +0.563622
DR = 40.241 HLR
DR ratio= 299600001032496448.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

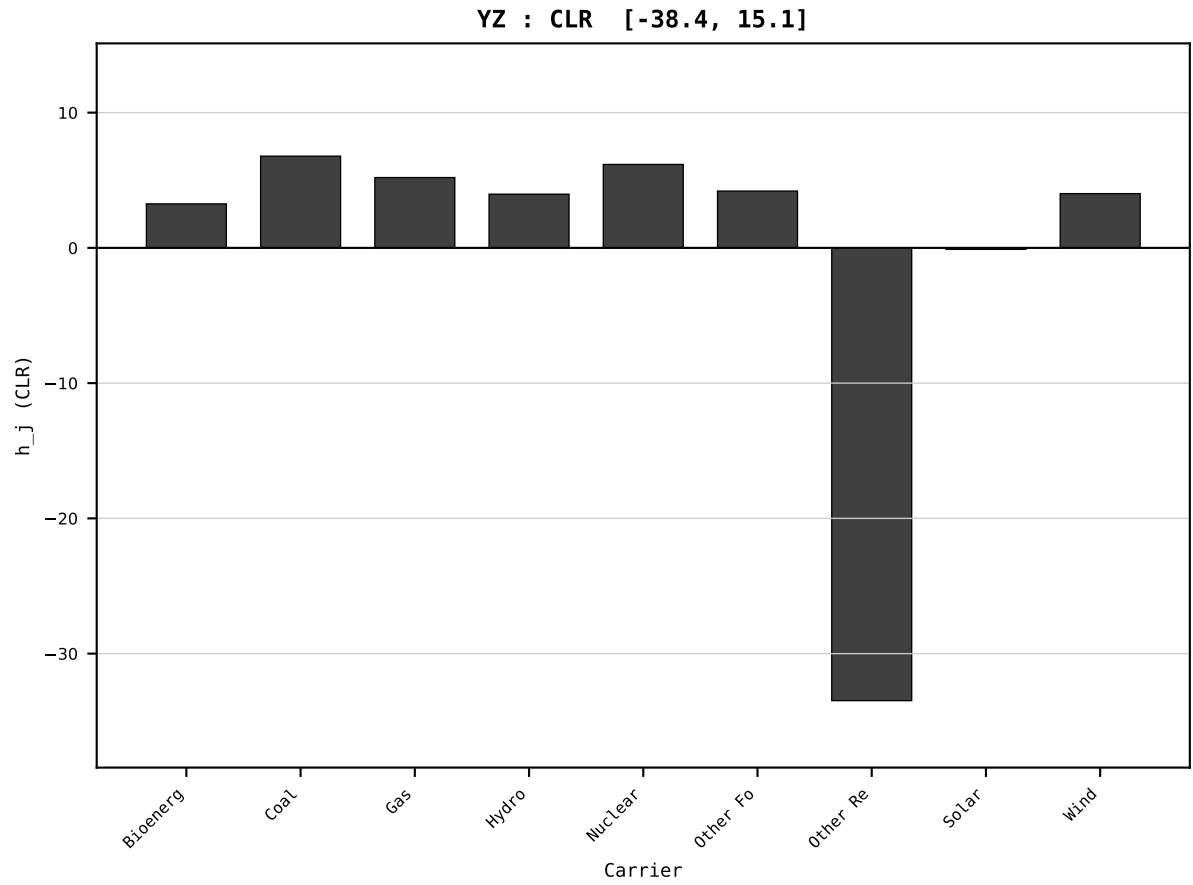
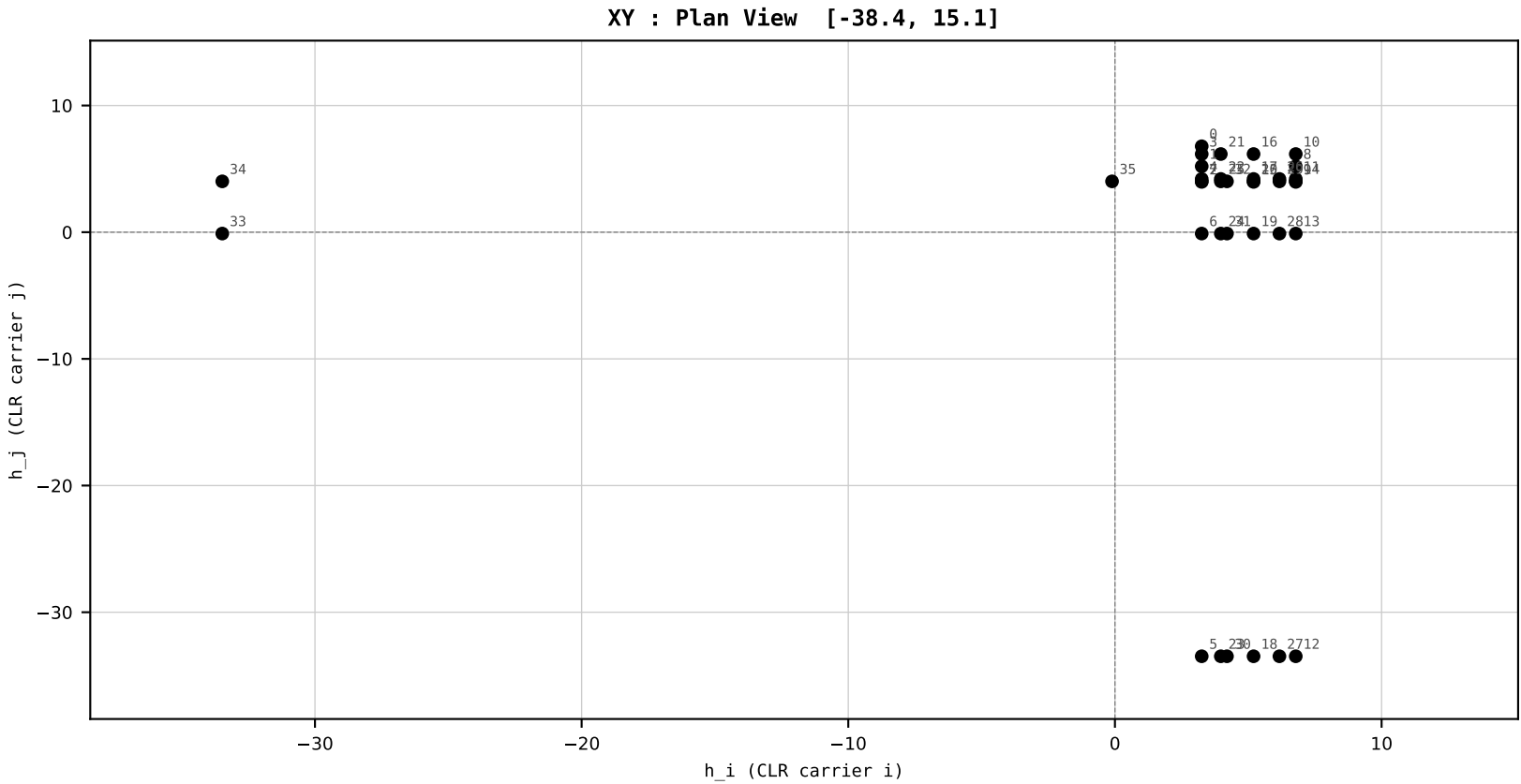
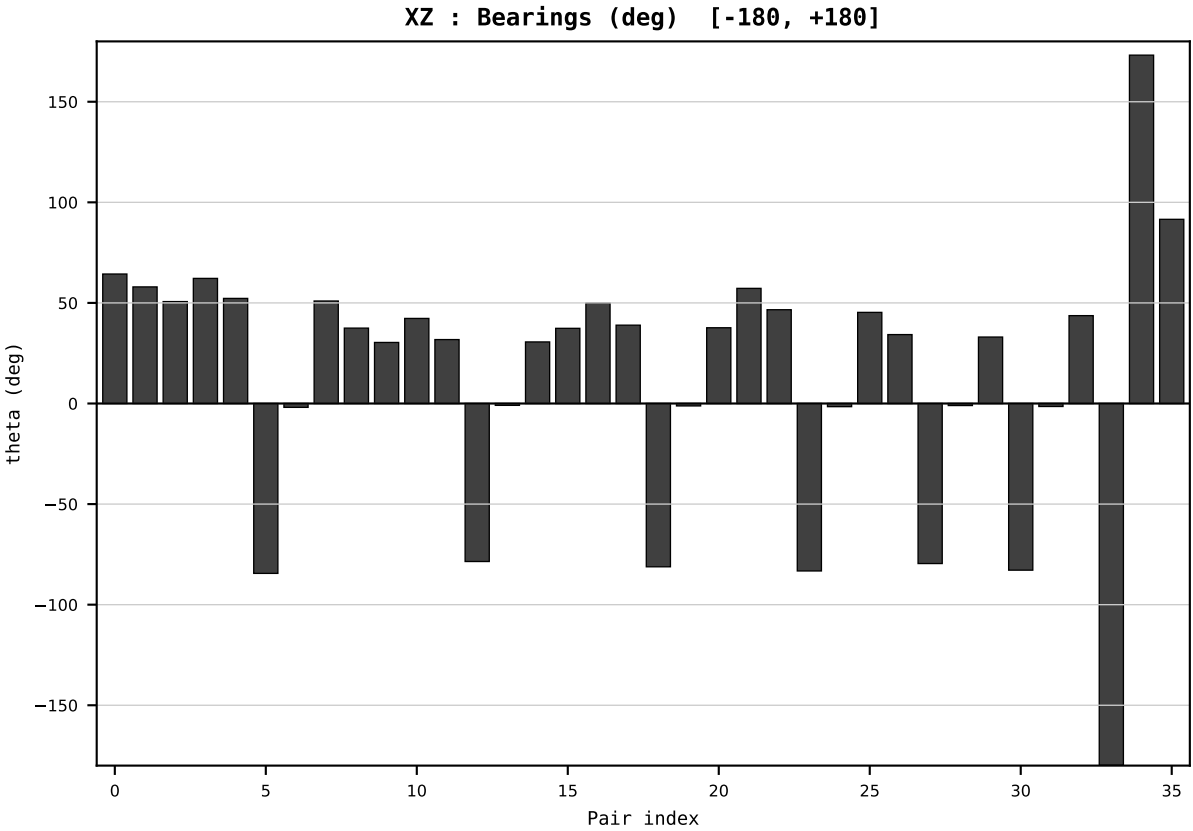
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.389220
Ring = Hs-3
E_metric= 899.7401
kappa_HS= 304629999999999936.00
omega = 1.0479 deg
d_A = 0.661385
Helm = Bioenergy
Helm d = +0.382457
DR = 40.258 HLR
DR ratio= 304629999895005696.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

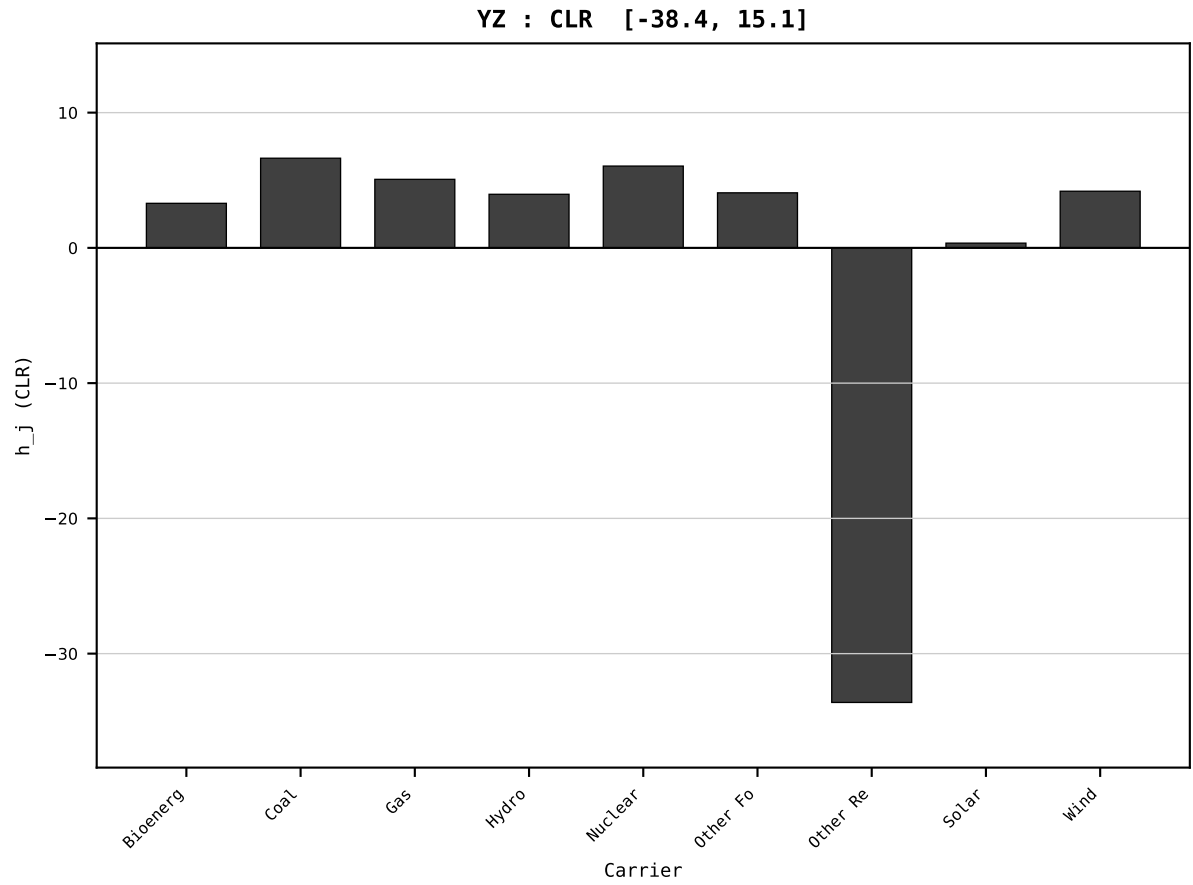
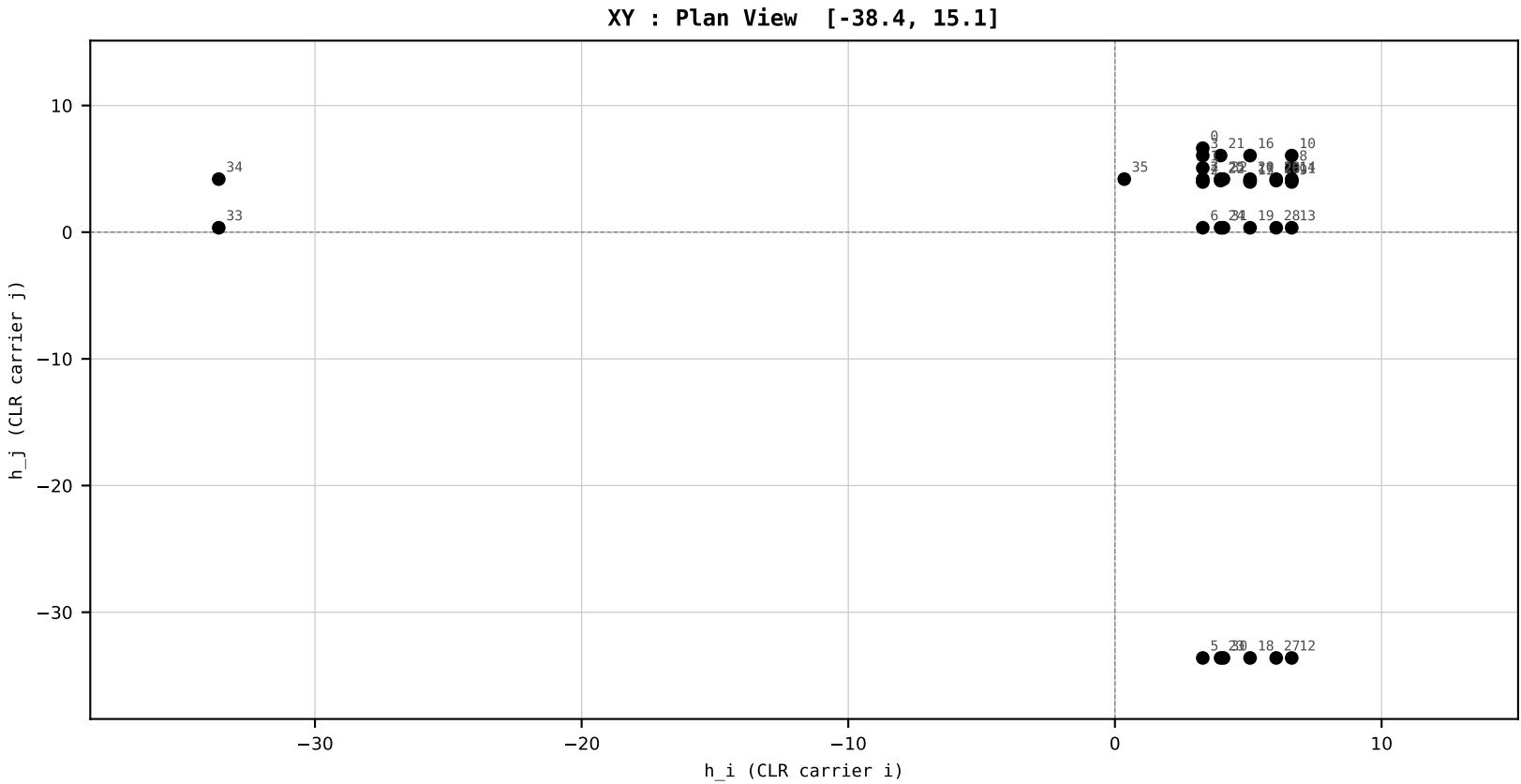
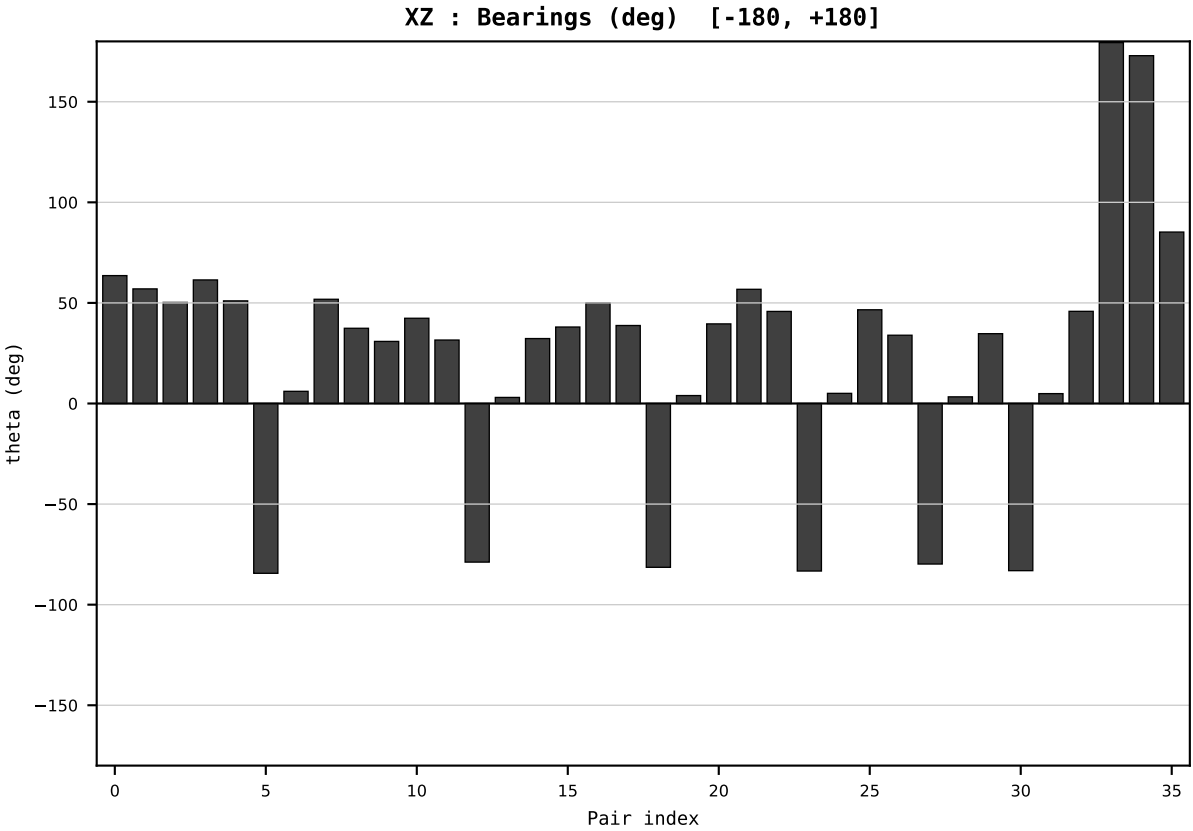
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.367842
Ring = Hs-3
E_metric= 902.0612
kappa_HS= 29876999999999936.00
omega = 0.9133 deg
d_A = 0.577156
Helm = Solar
Helm d = +0.458667
DR = 40.238 HLR
DR ratio= 298769999545169920.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

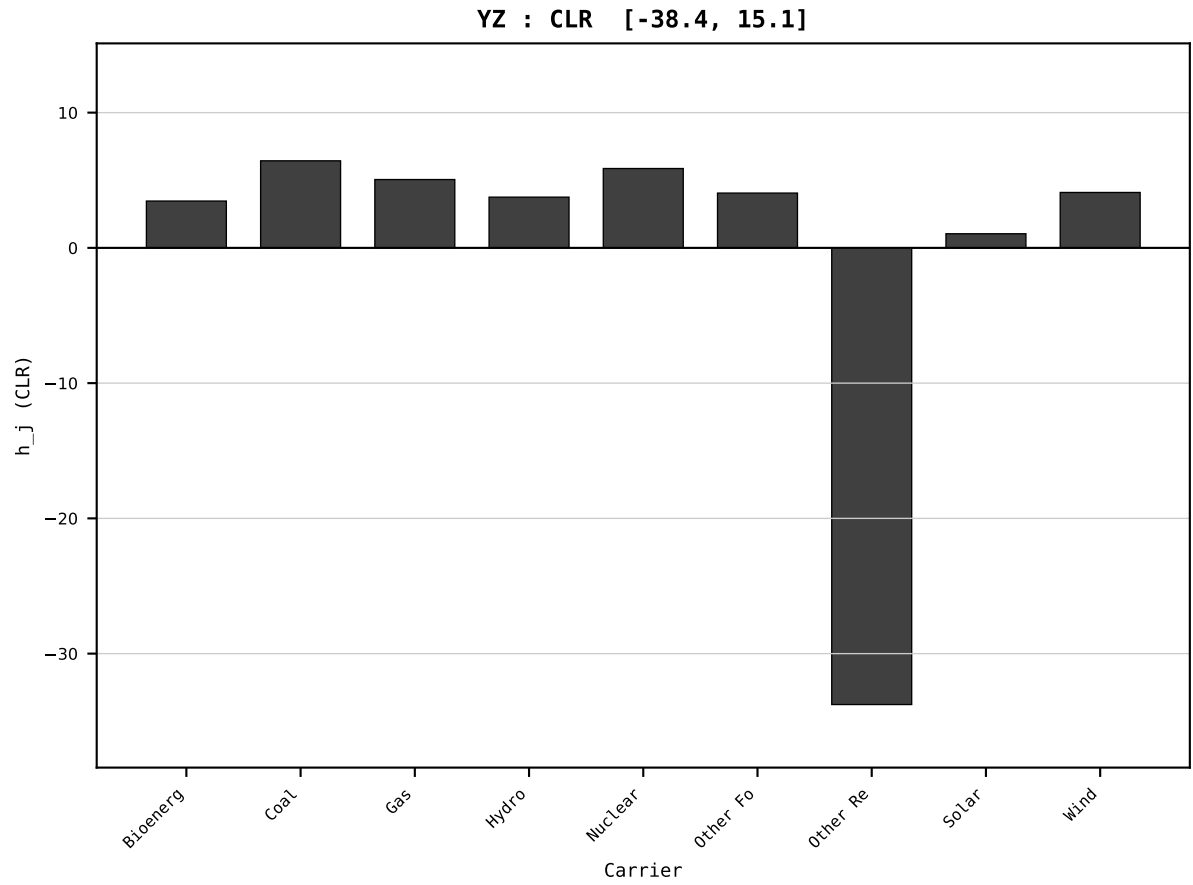
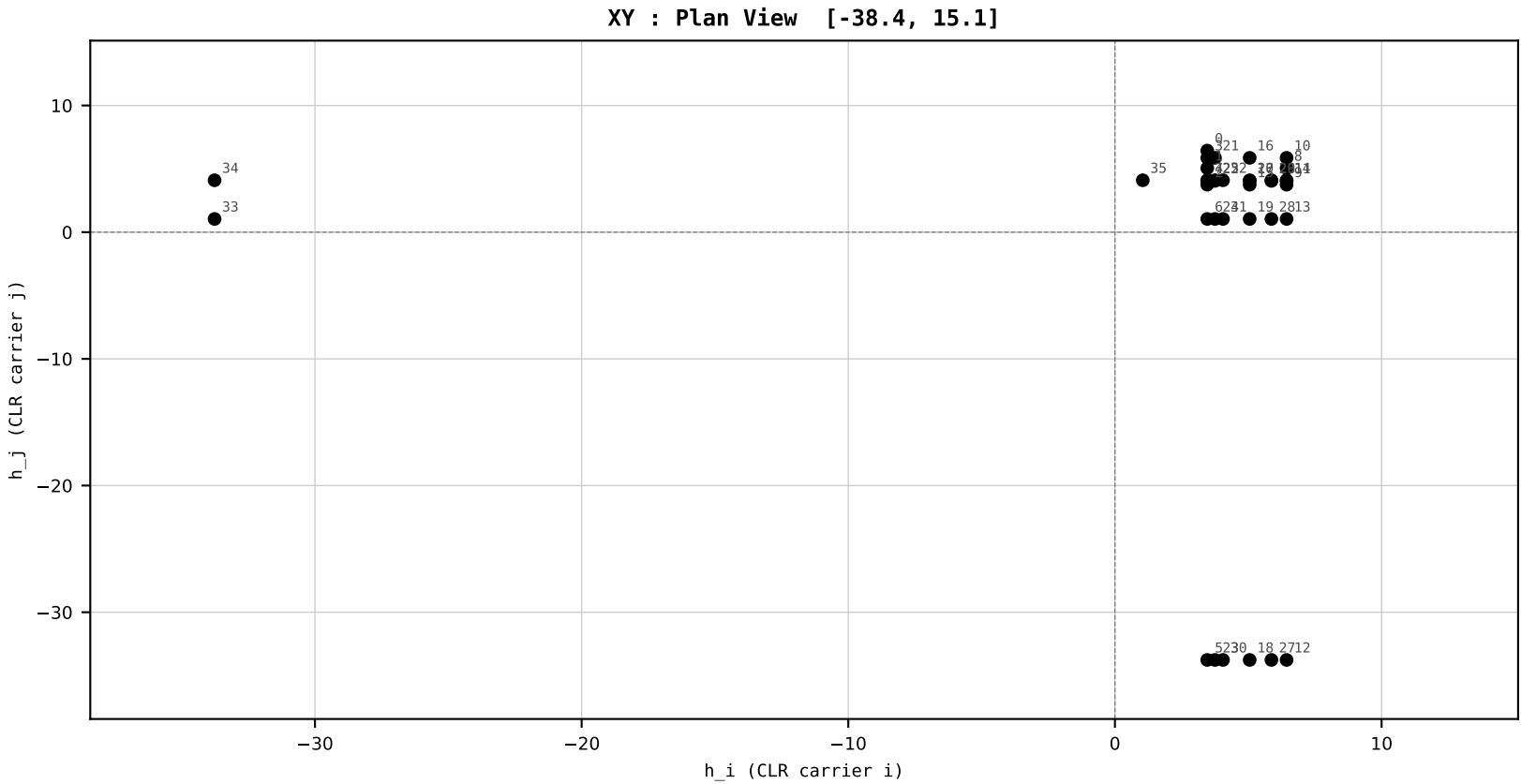
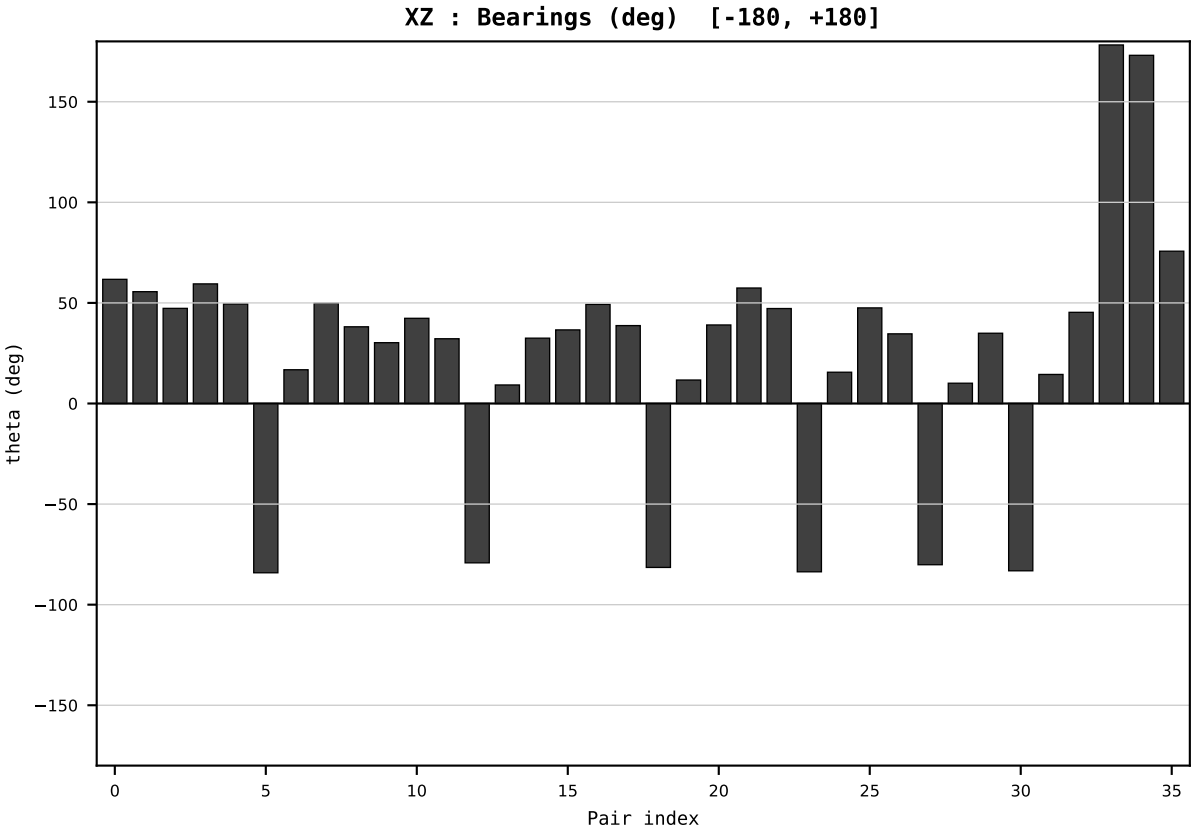
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.341114
Ring = Hs-3
E_metric= 904.9926
kappa_HS= 28813999999999968.00
omega = 1.2830 deg
d_A = 0.810406
Helm = Solar
Helm d = +0.694037
DR = 40.202 HLR
DR ratio= 288140000563598304.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

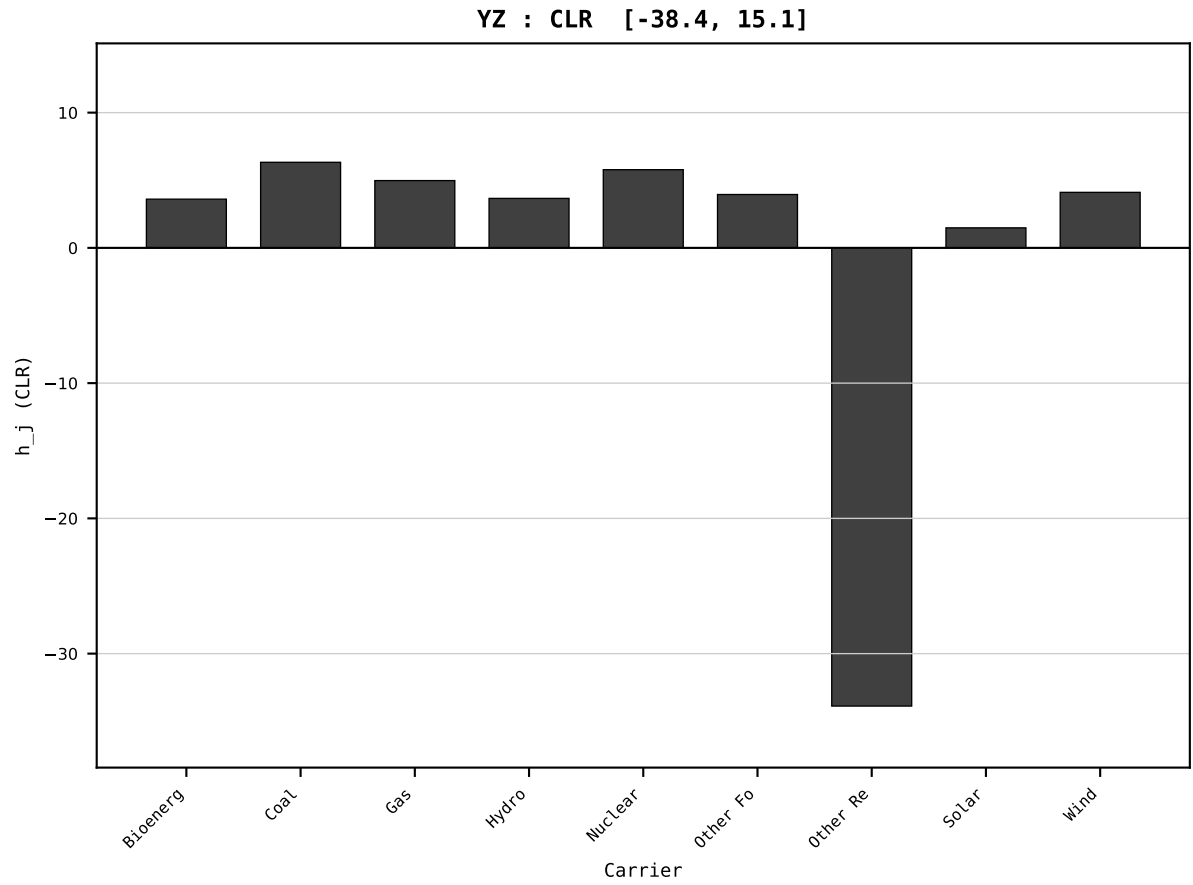
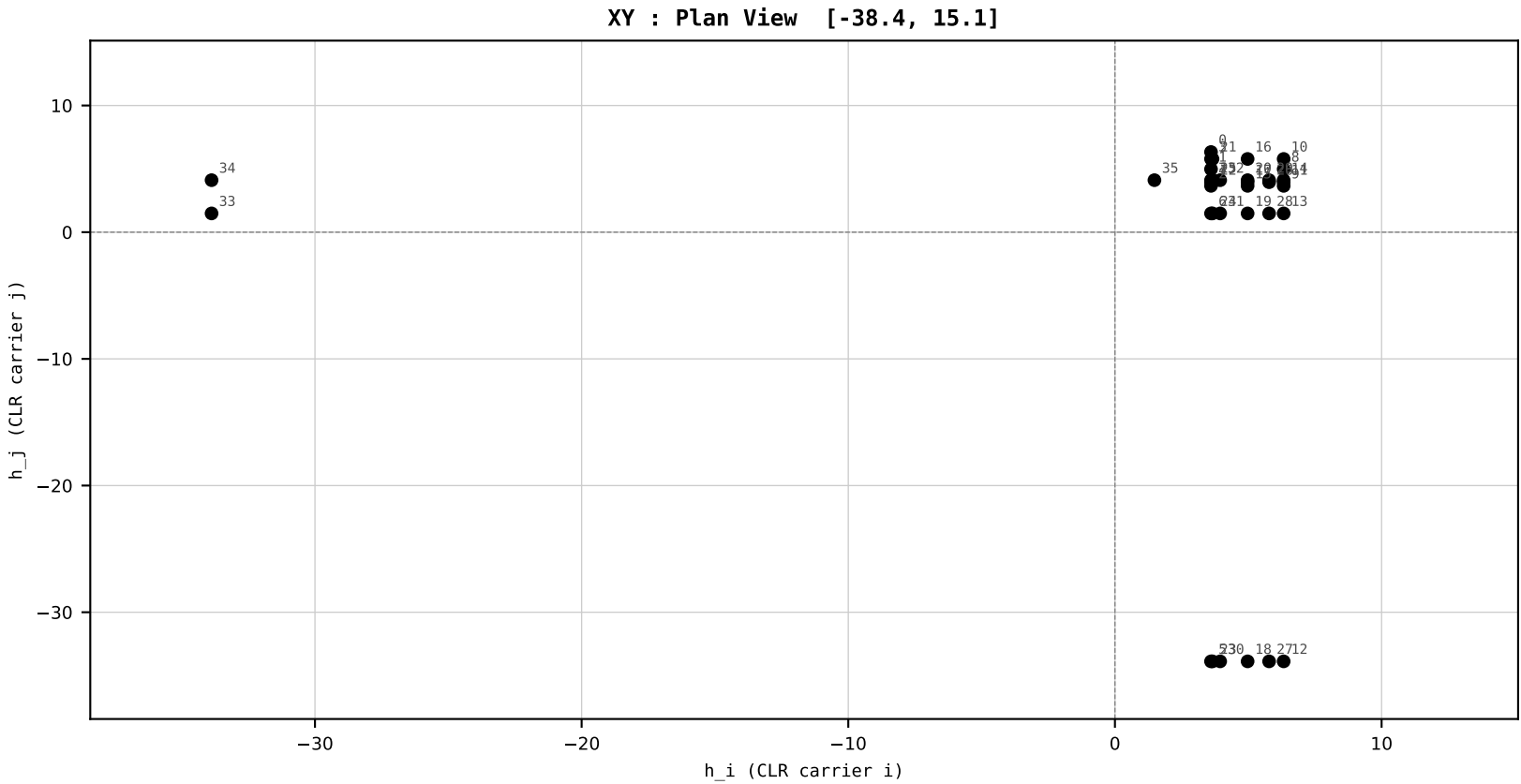
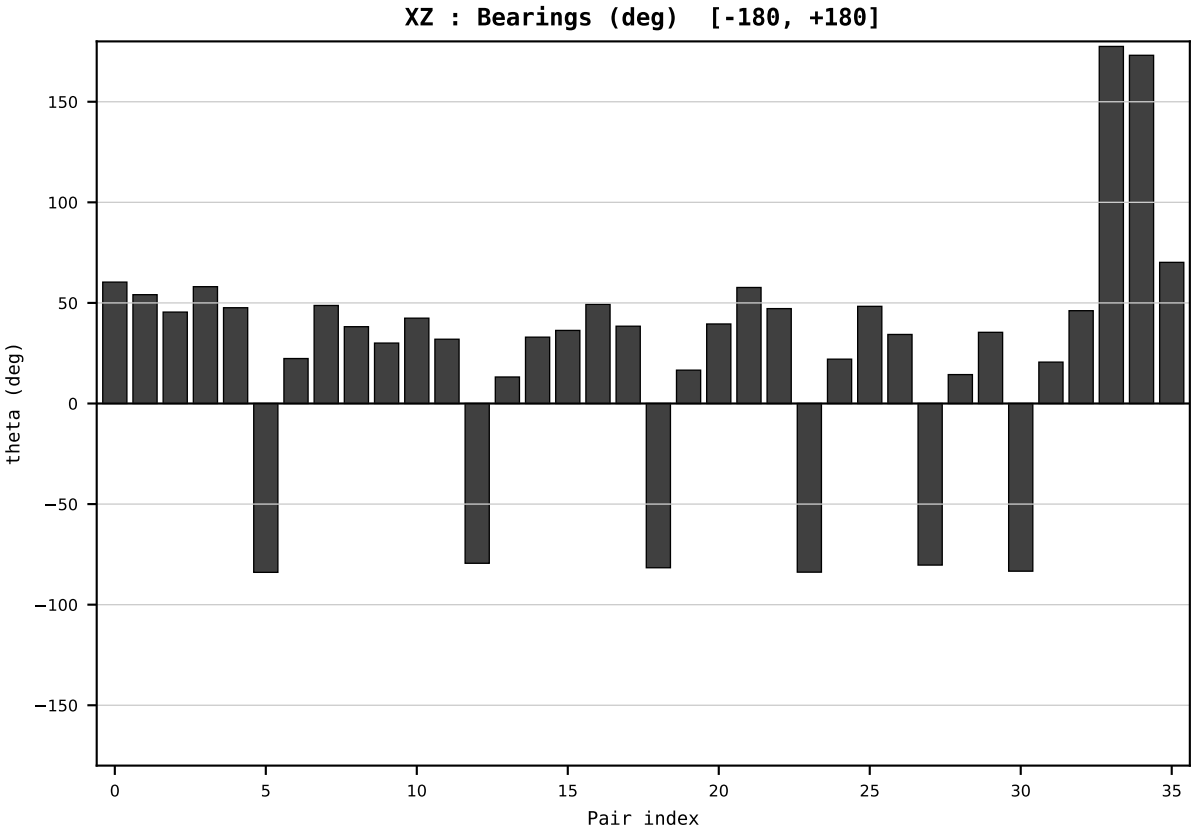
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.326053
Ring = Hs-3
E_metric= 907.0614
kappa_HS= 288930000000000000.00
omega = 0.8154 deg
d_A = 0.518705
Helm = Solar
Helm d = +0.437297
DR = 40.205 HLR
DR ratio= 288930000245136960.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

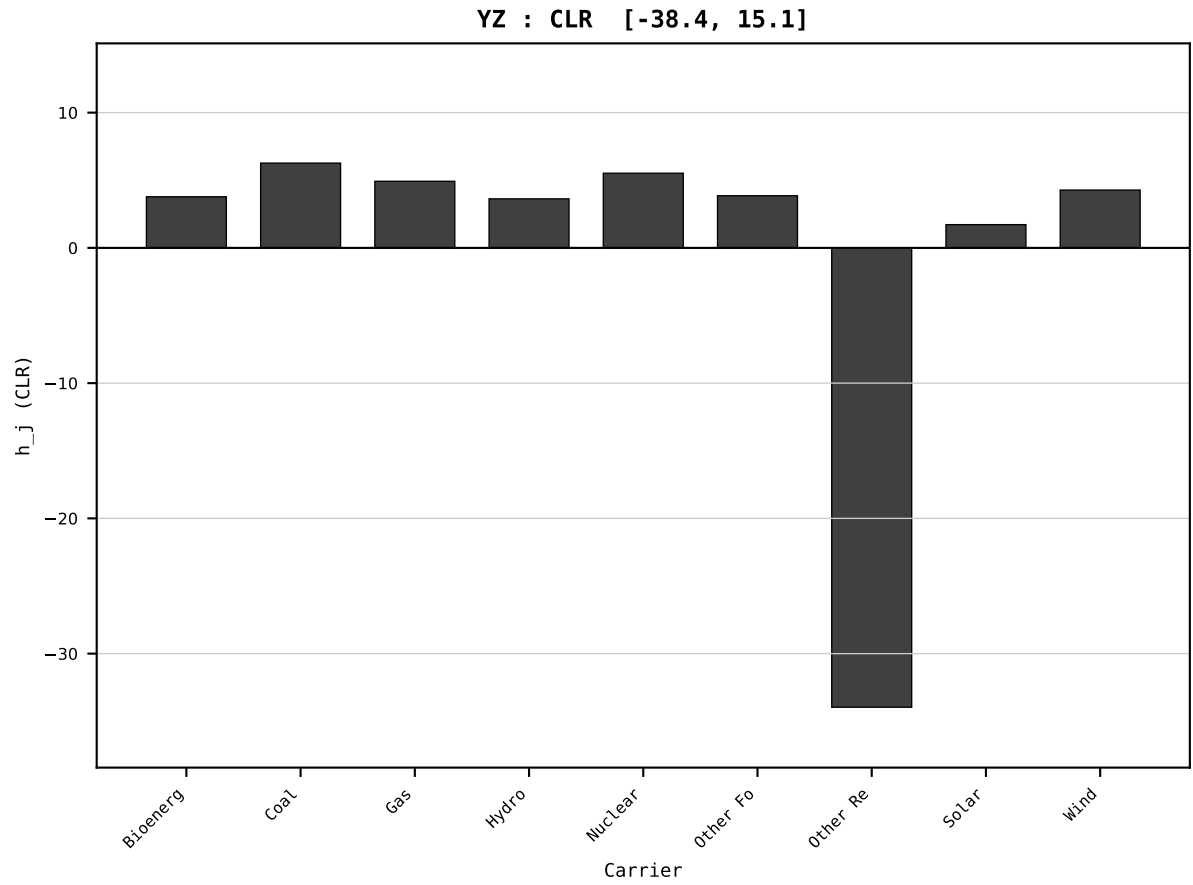
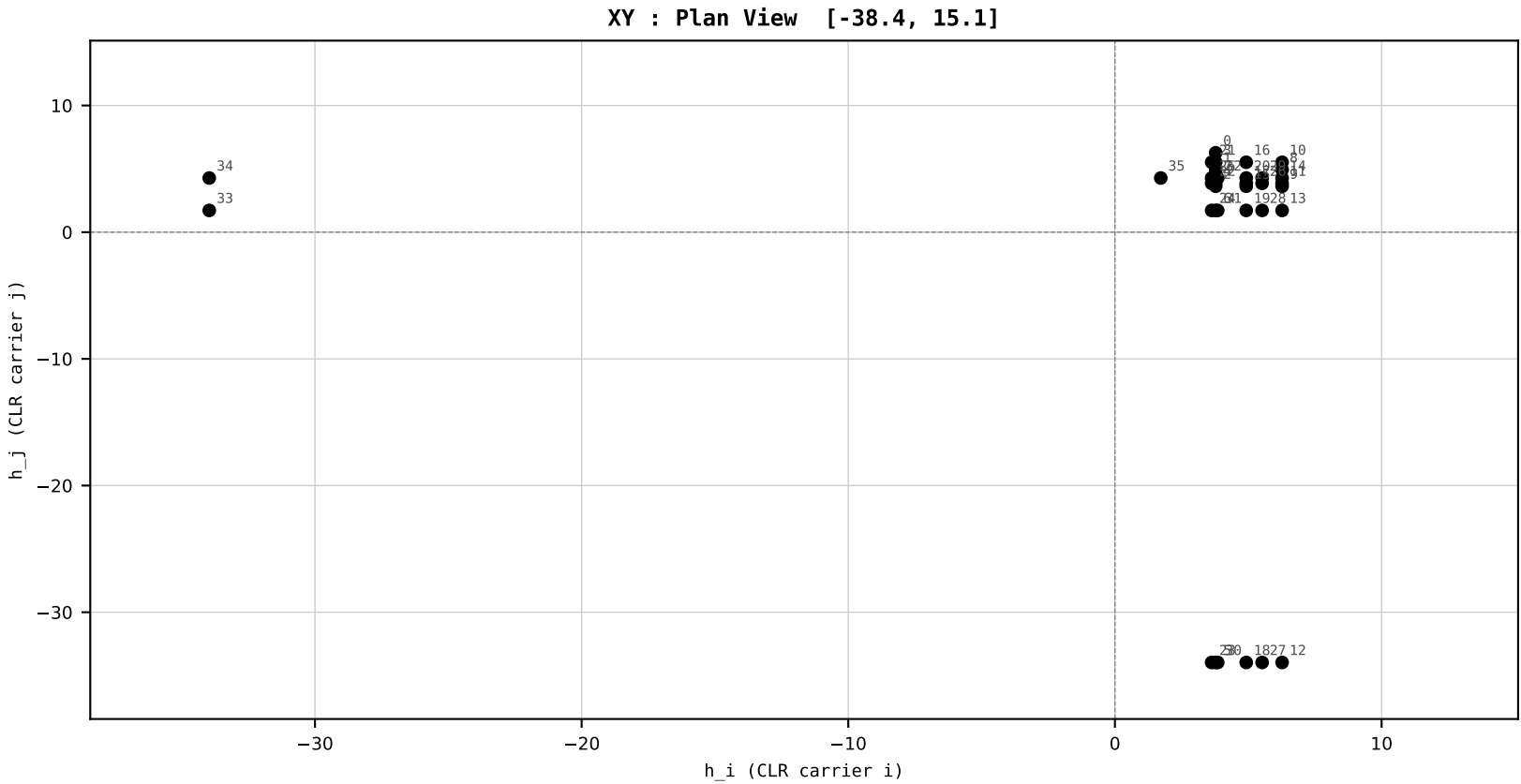
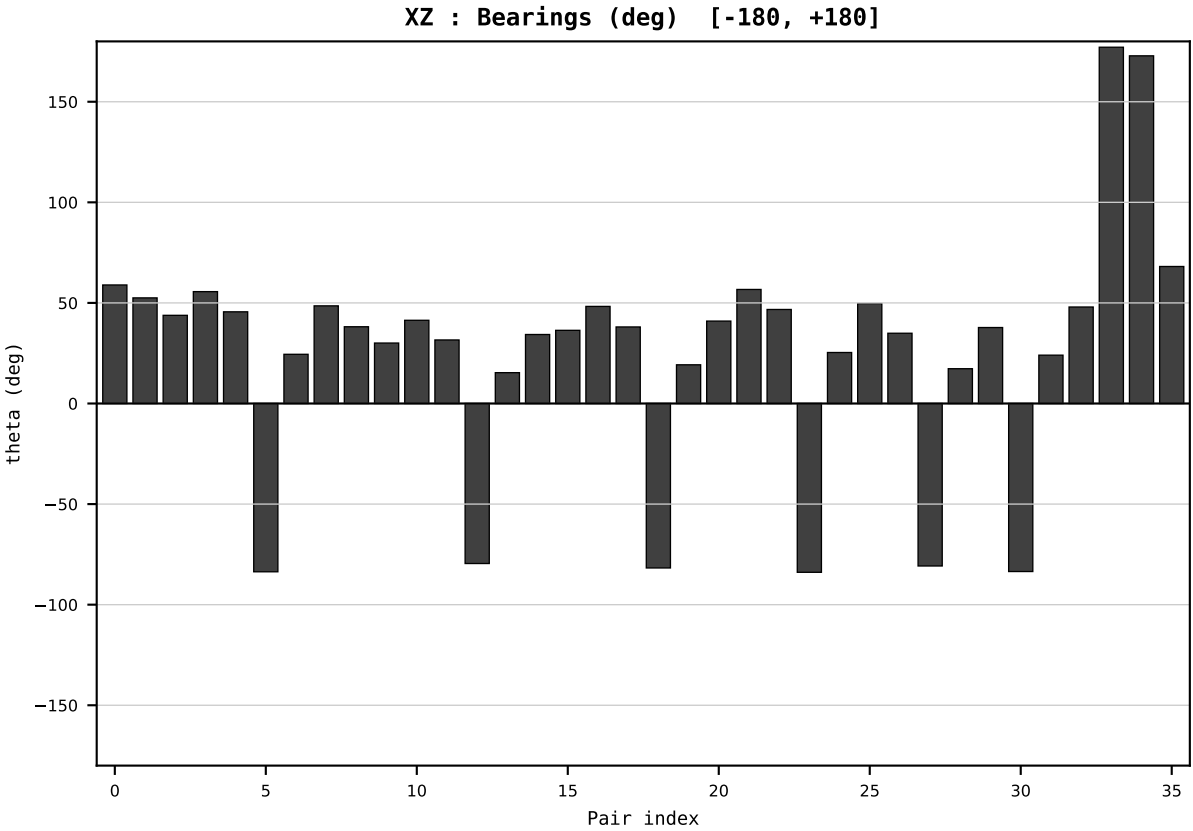
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.306825
Ring = Hs-3
E_metric= 910.0899
kappa_HS= 297100000000000000.00
omega = 0.7153 deg
d_A = 0.455473
Helm = Nuclear
Helm d = -0.261293
DR = 40.233 HLR
DR ratio= 297099997807627008.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

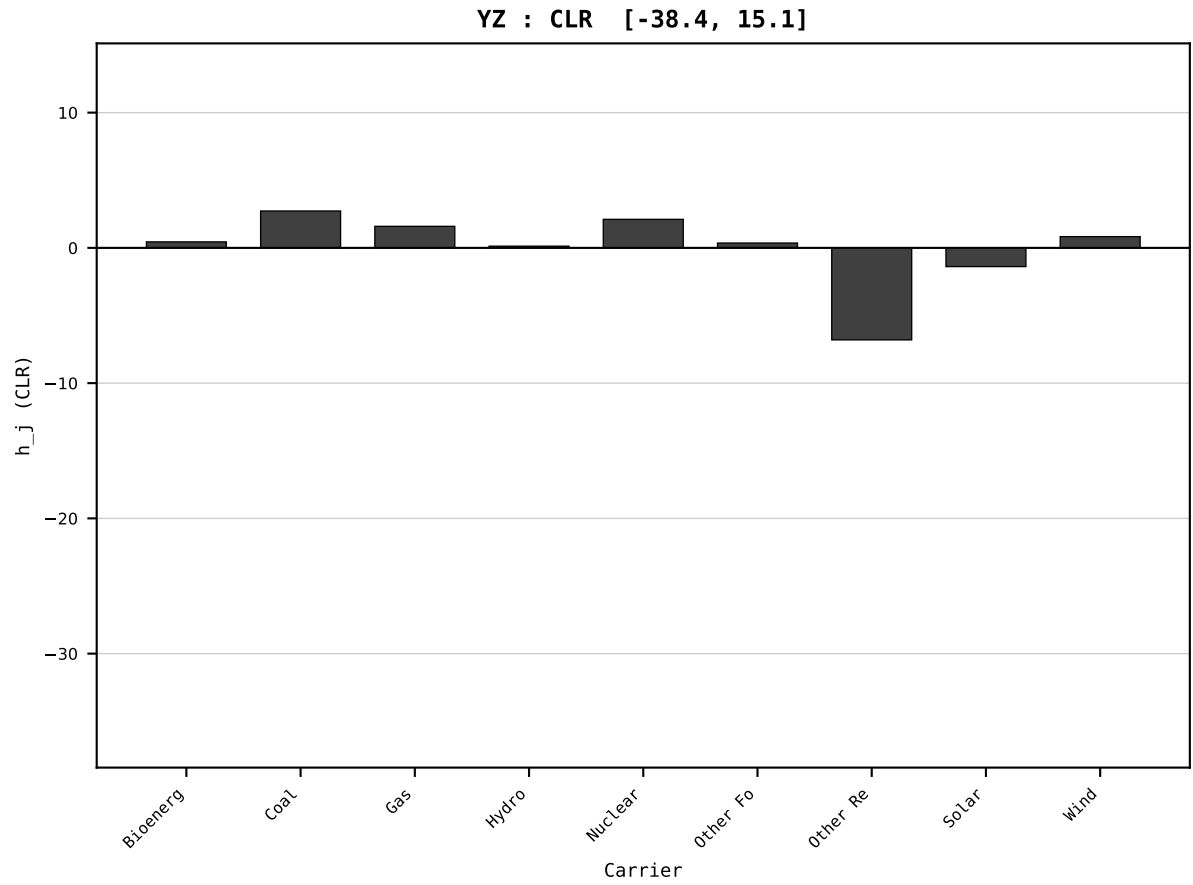
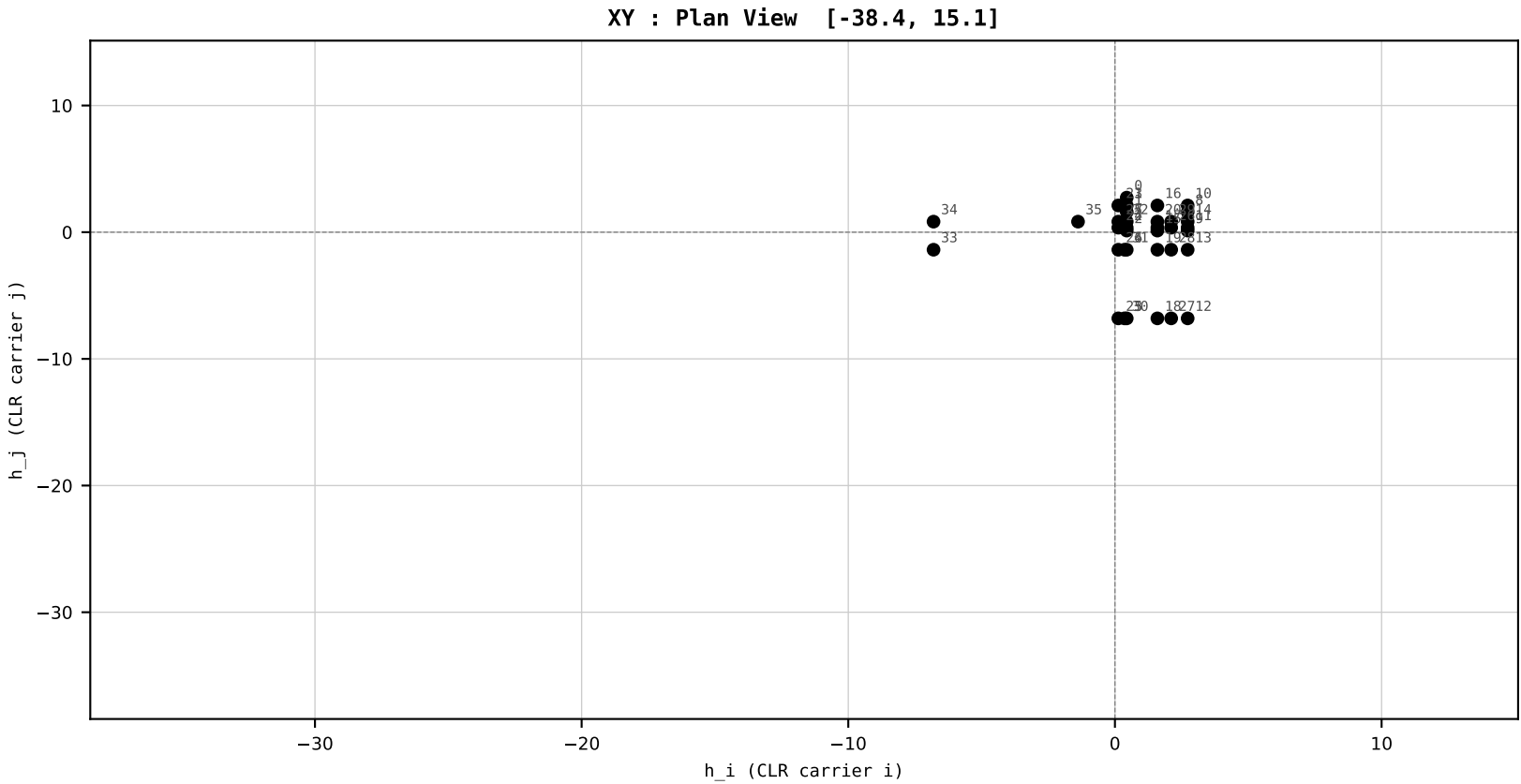
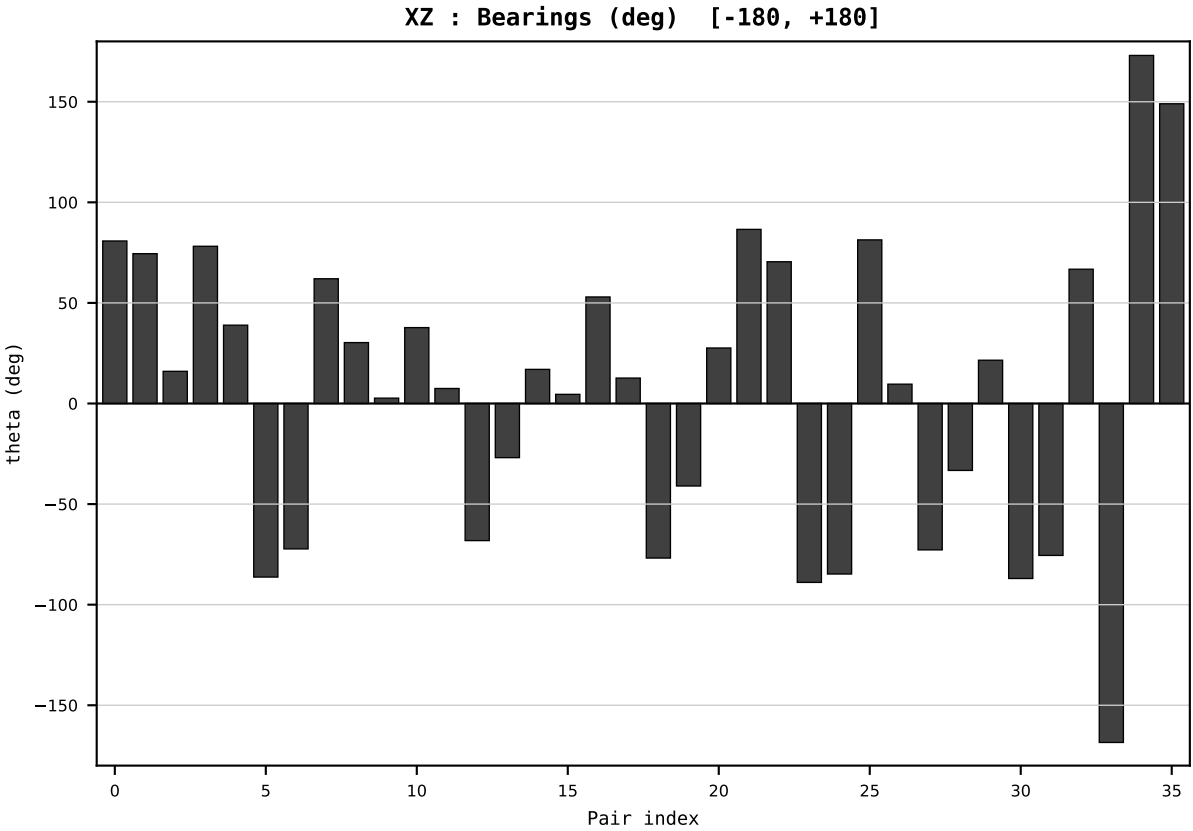
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.285057
Ring = Hs-2
E_metric= 63.6470
kappa_HS= 13760.50
omega = 19.4937 deg
d_A = 28.811916
Helm = Other Renewables
Helm d = +27.161871
DR = 9.530 HLR
DR ratio= 13760.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



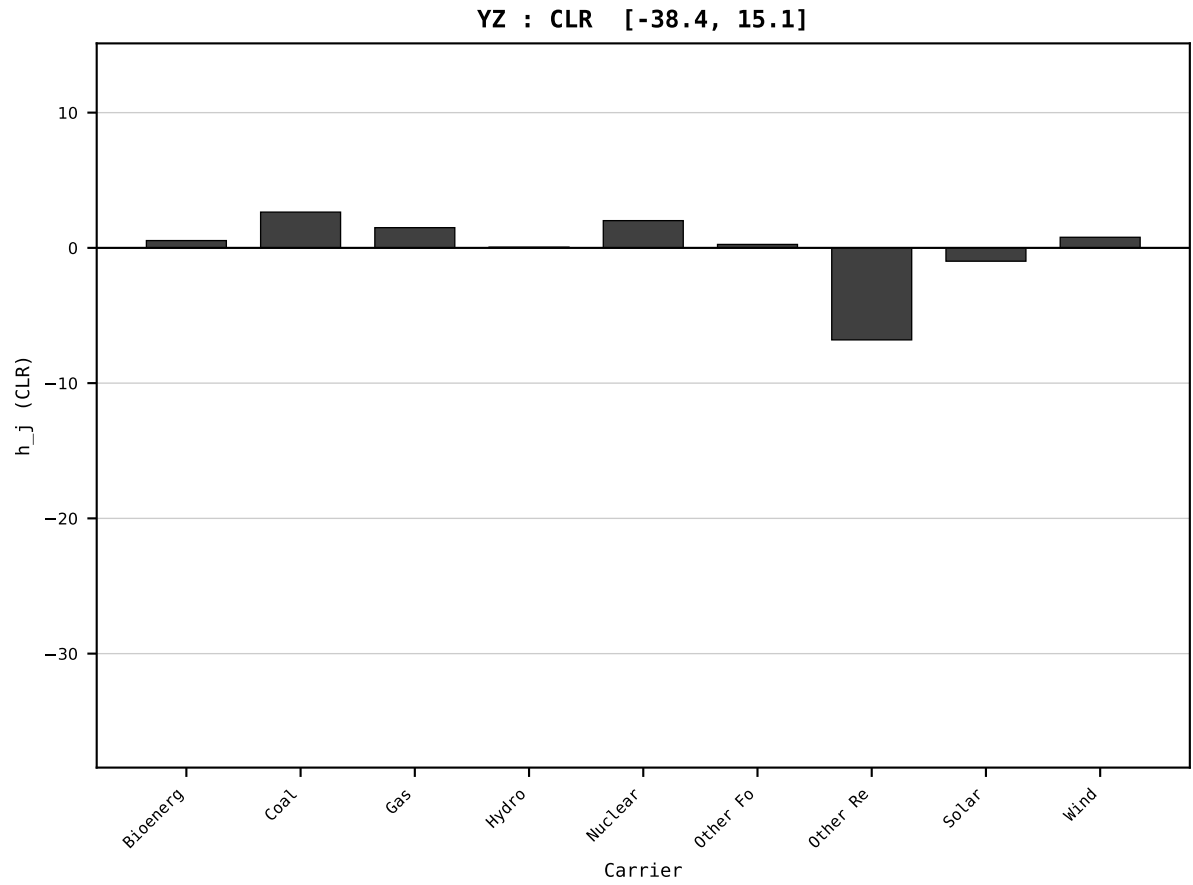
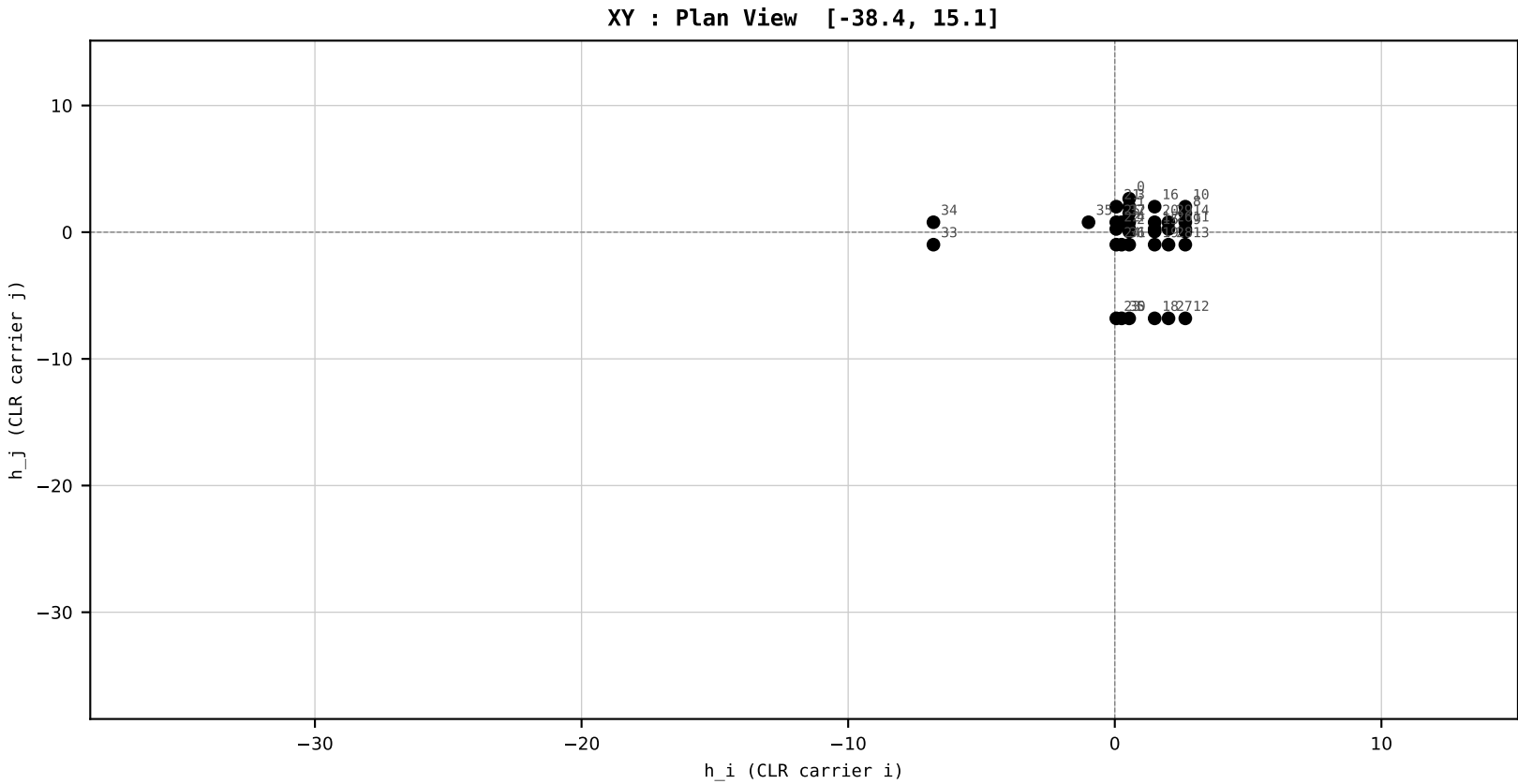
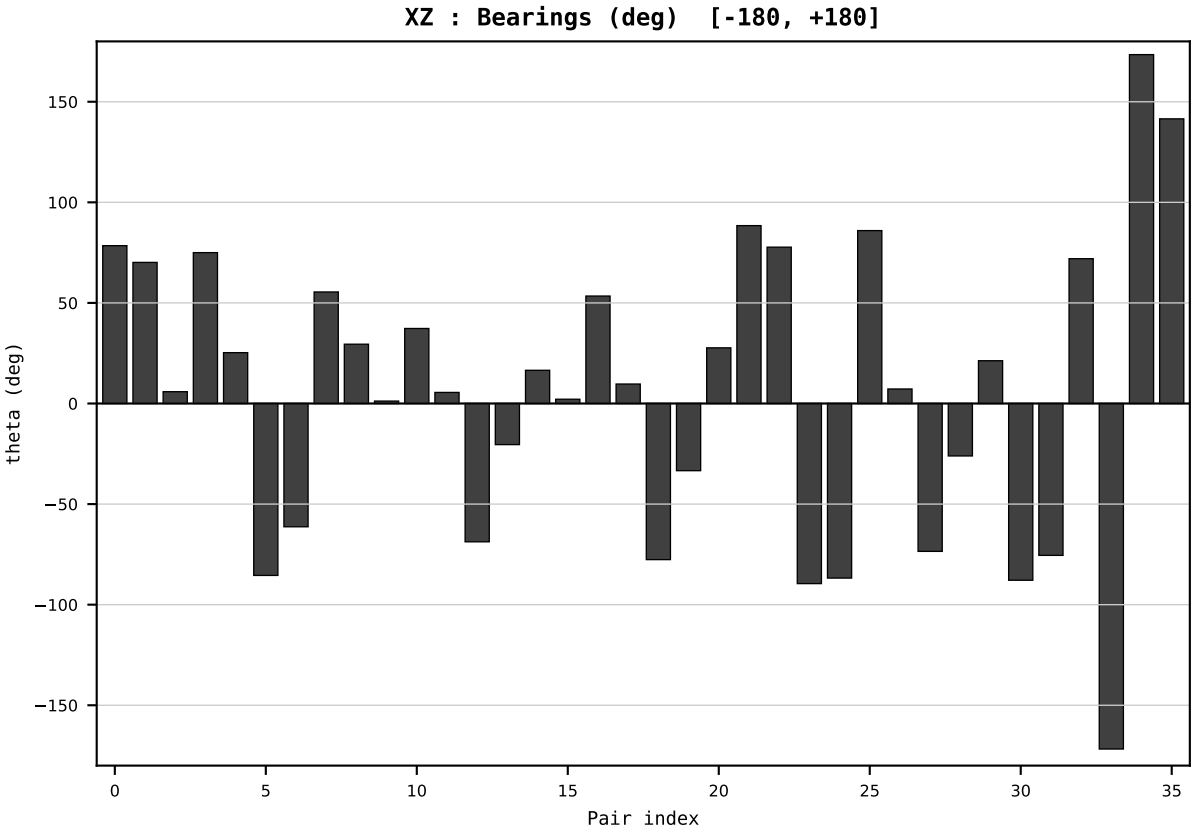
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.272489
Ring = Hs-2
E_metric= 61.5055
kappa_HS= 12672.50
omega = 3.1880 deg
d_A = 0.460403
Helm = Solar
Helm d = +0.398992
DR = 9.447 HLR
DR ratio= 12672.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

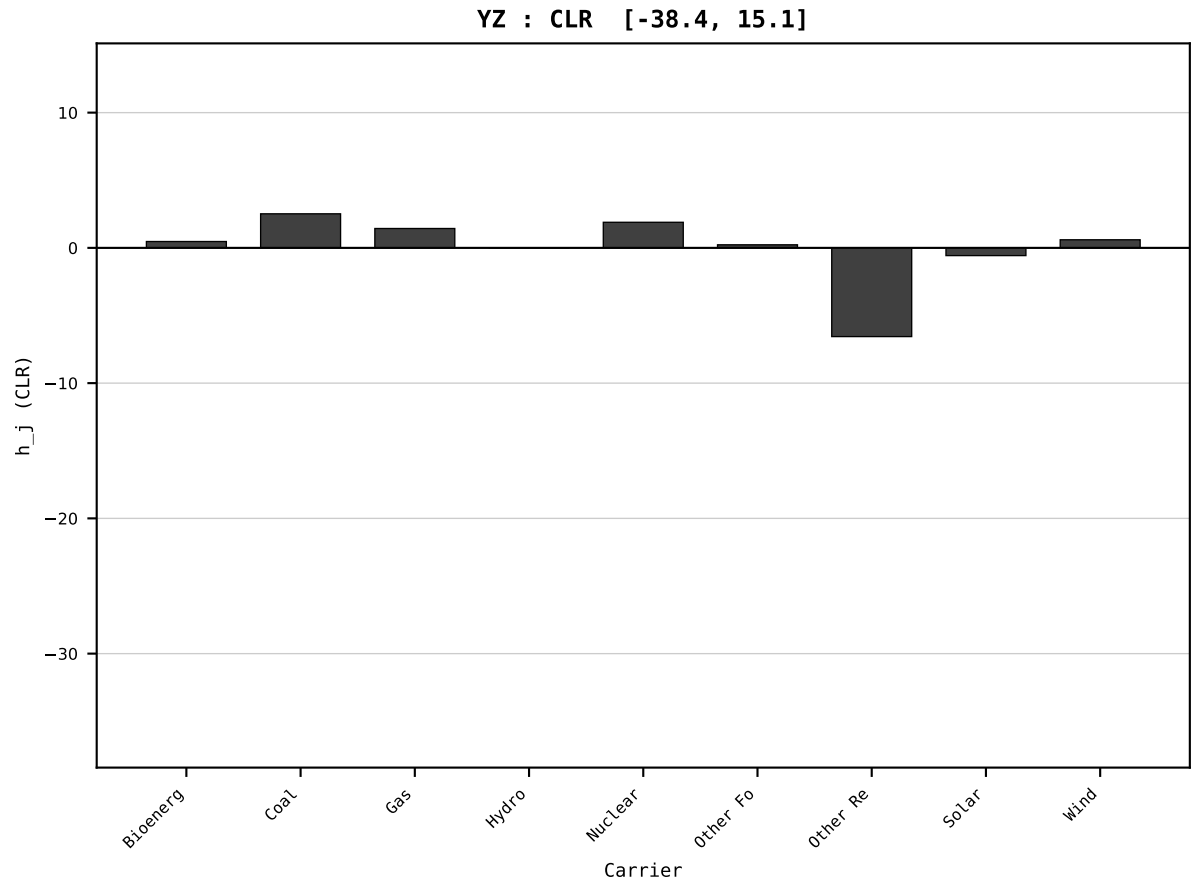
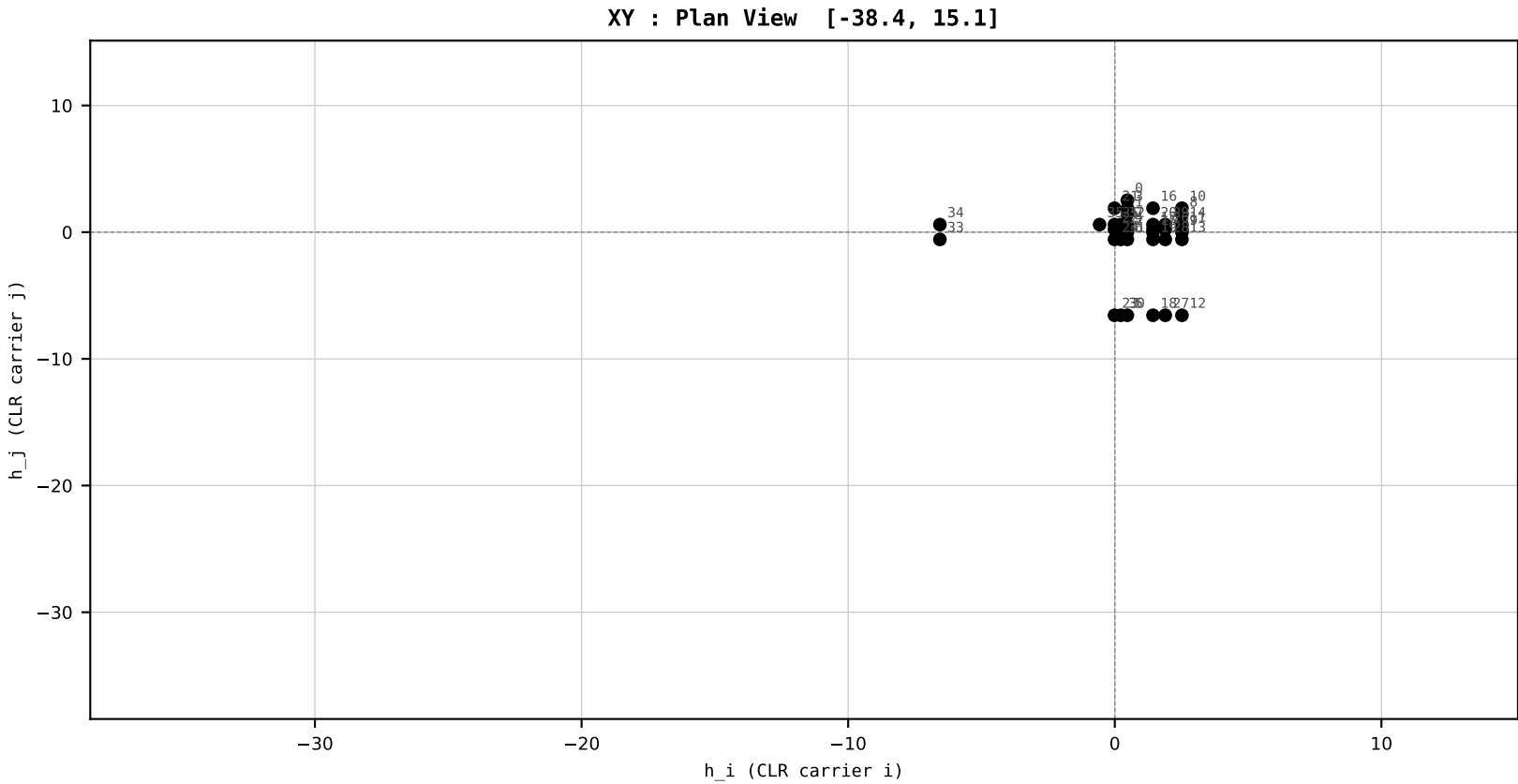
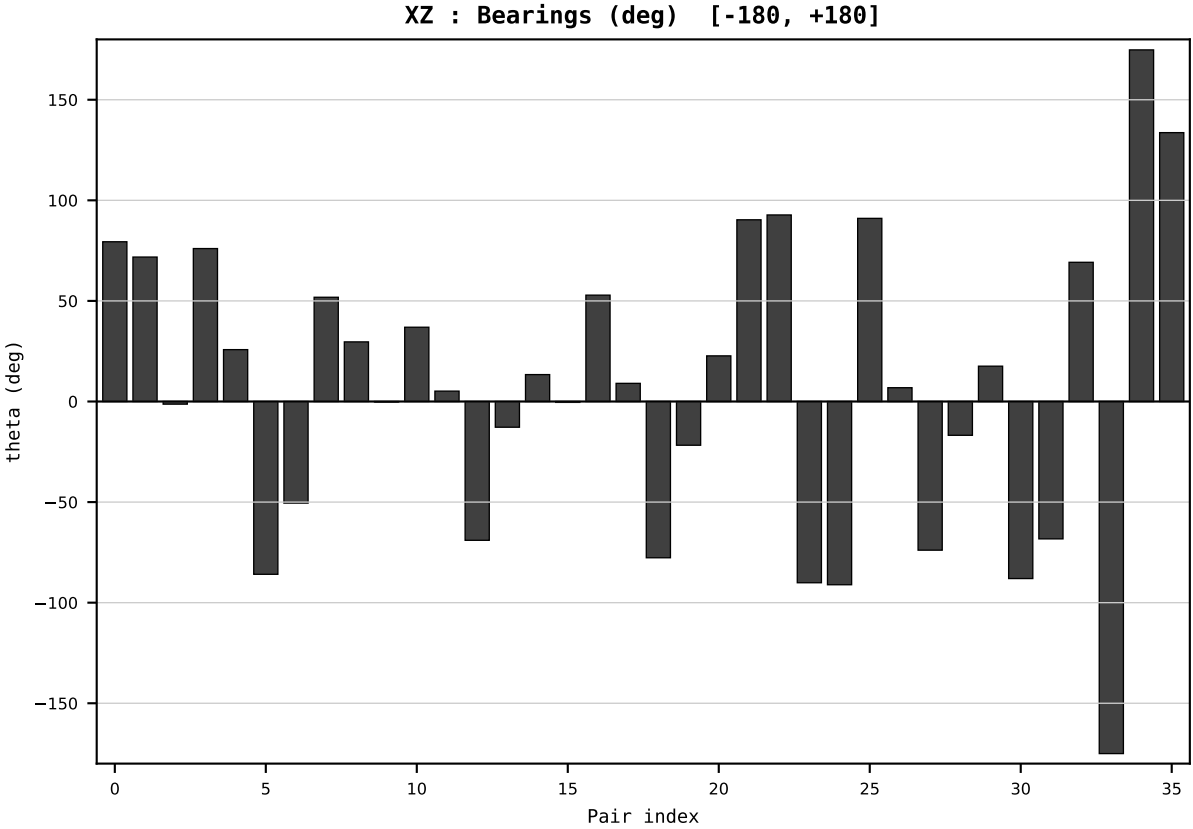
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.257193
Ring = Hs-2
E_metric= 55.9662
kappa_HS= 8763.00
omega = 3.1589 deg
d_A = 0.555847
Helm = Solar
Helm d = +0.413991
DR = 9.078 HLR
DR ratio= 8763.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

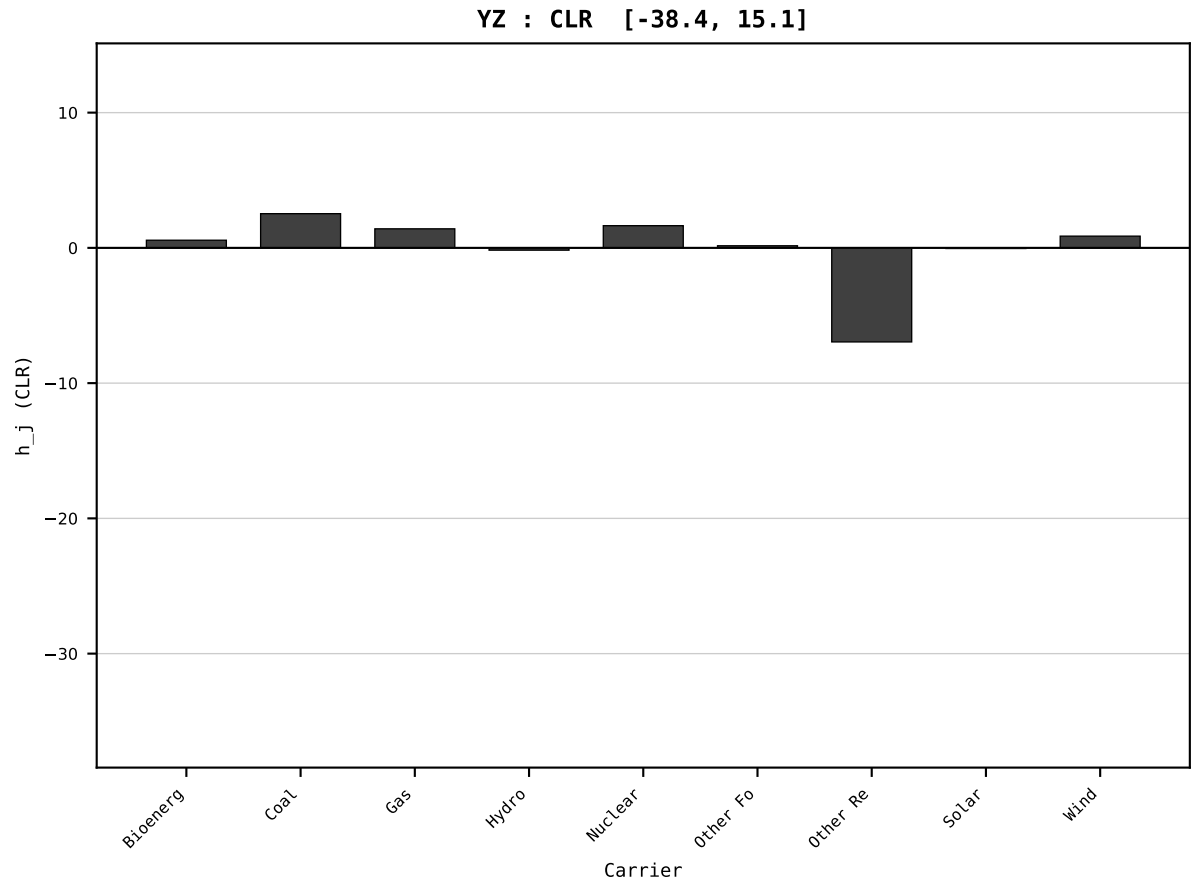
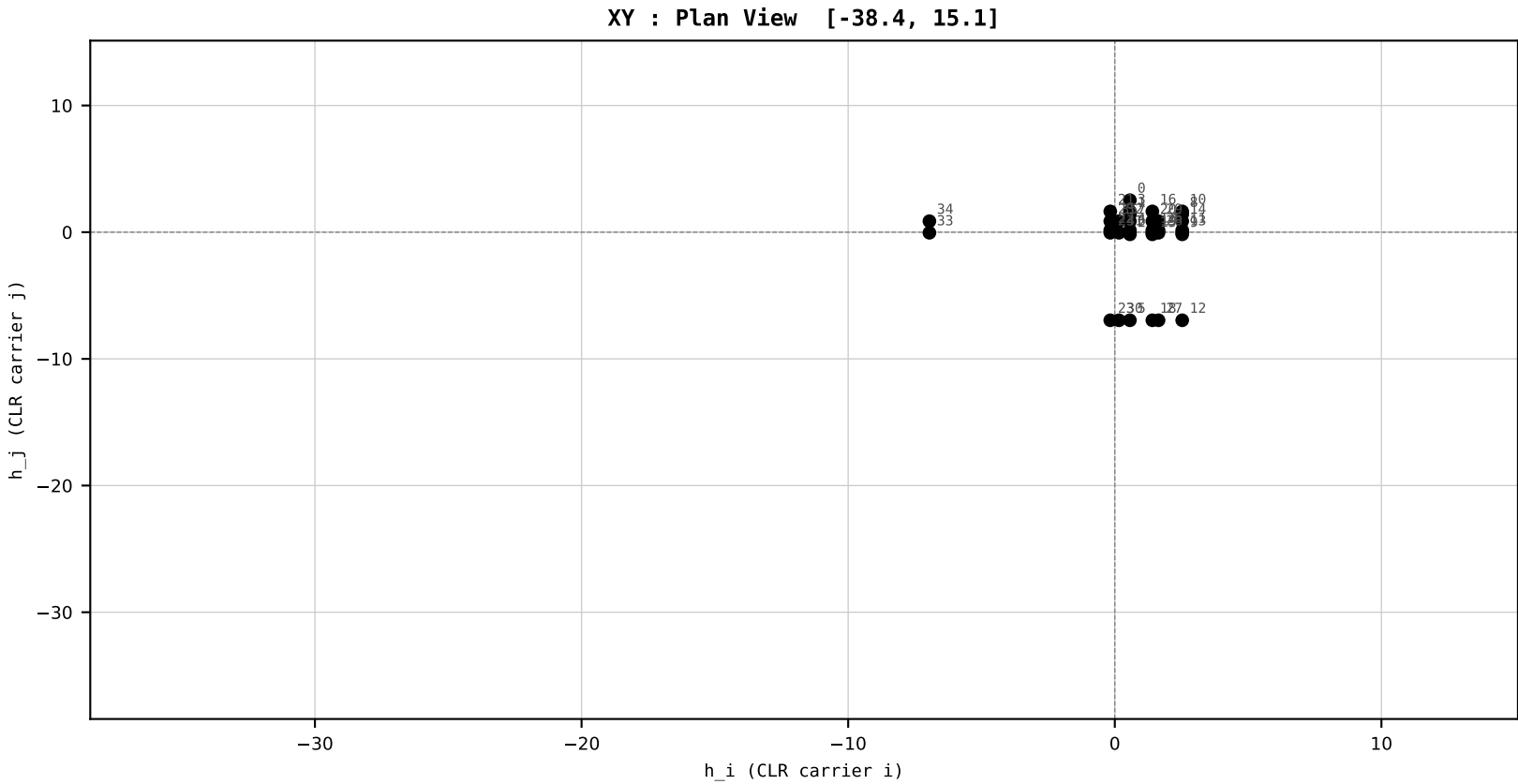
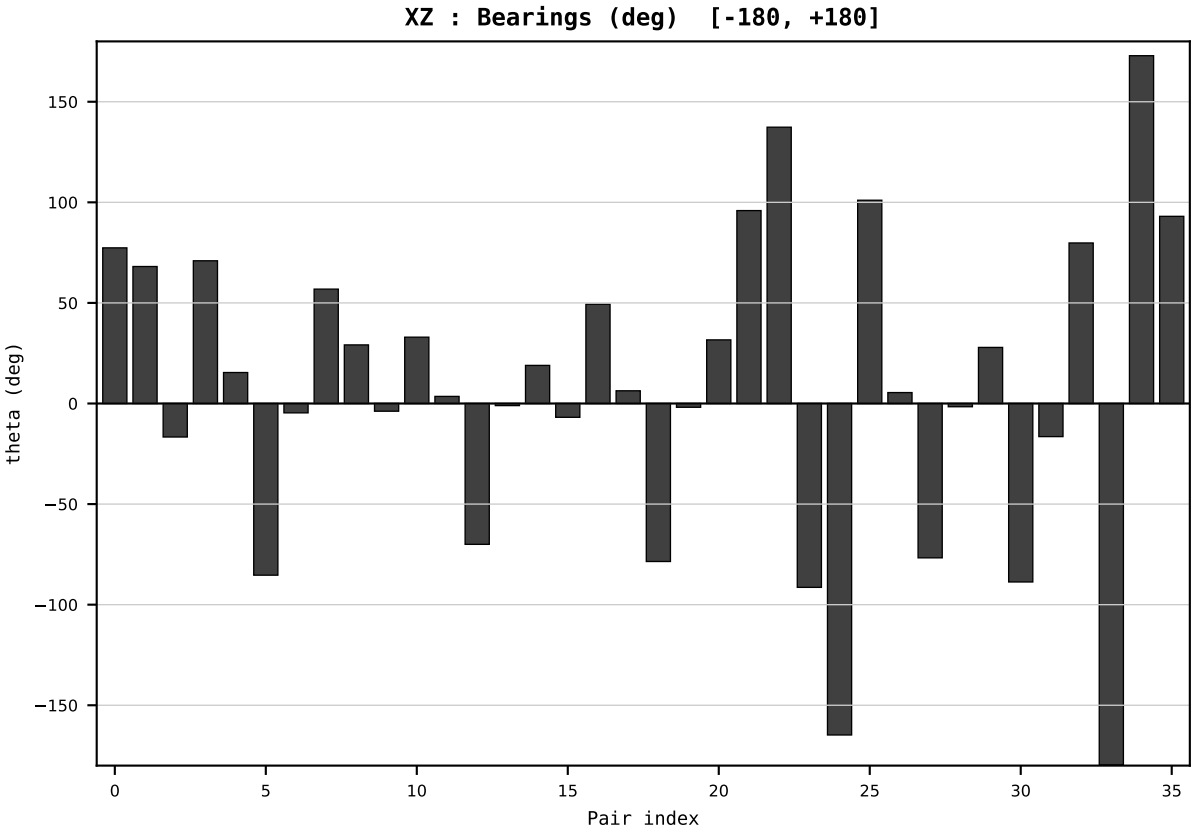
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.240465
Ring = Hs-2
E_metric= 60.5504
kappa_HS= 13123.00
omega = 5.4004 deg
d_A = 0.779100
Helm = Solar
Helm d = +0.525310
DR = 9.482 HLR
DR ratio= 13123.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

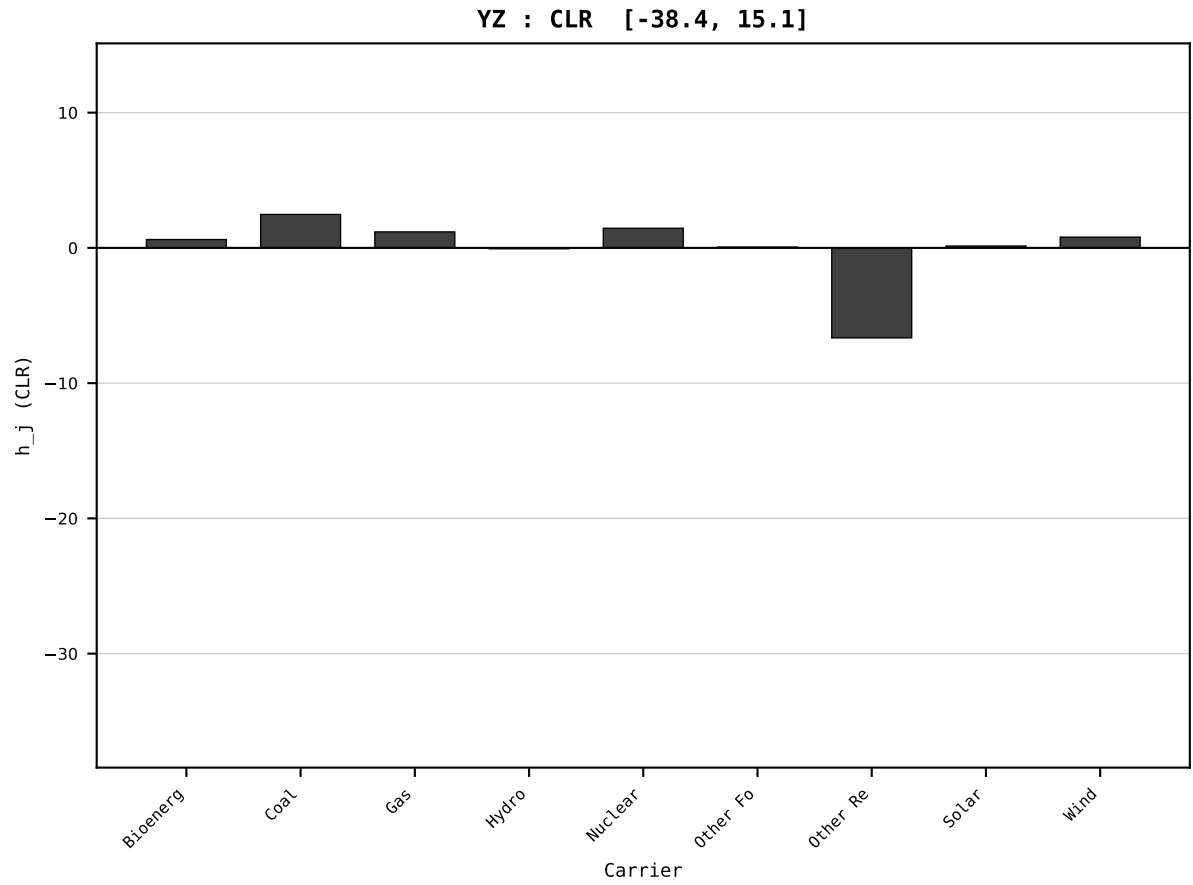
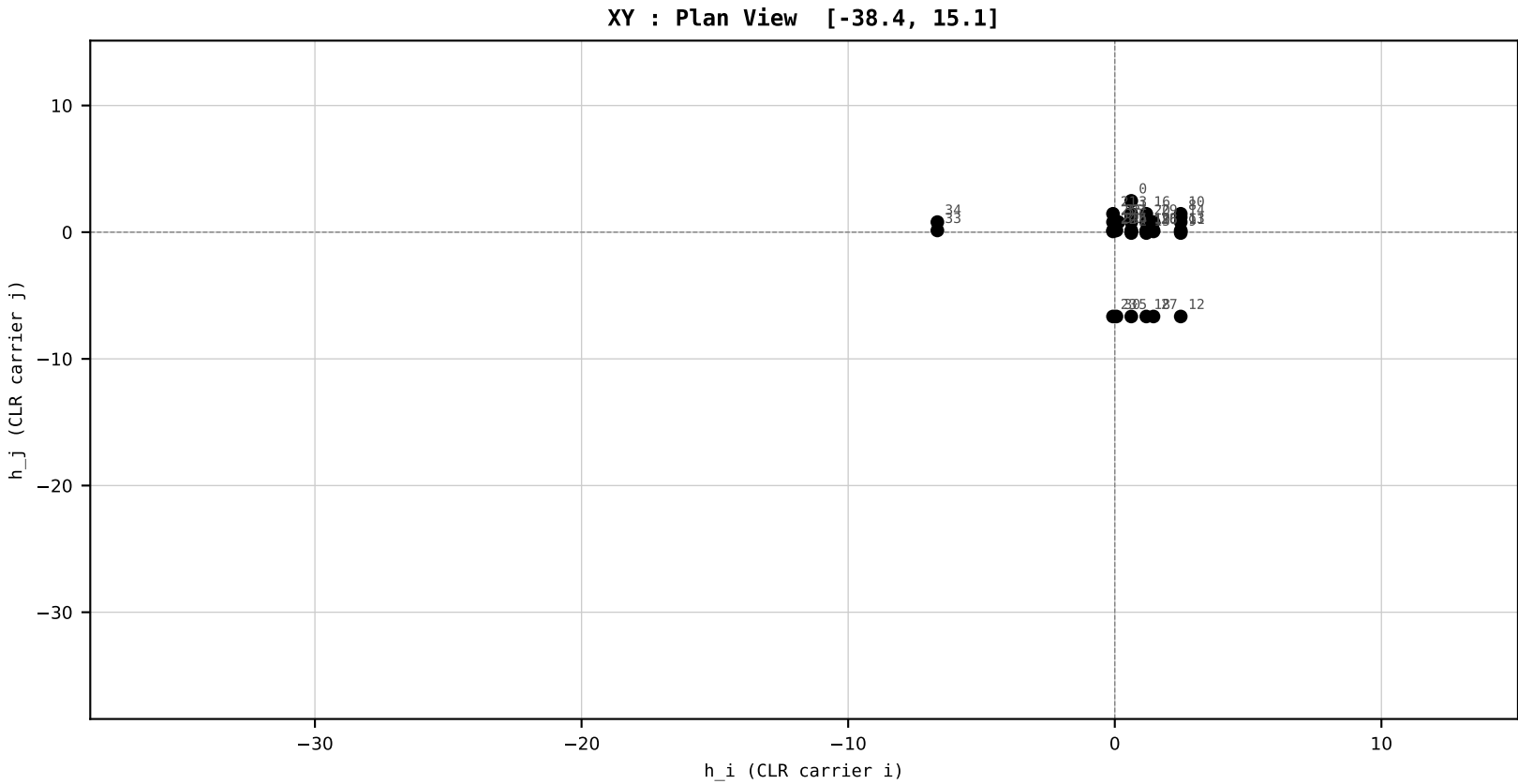
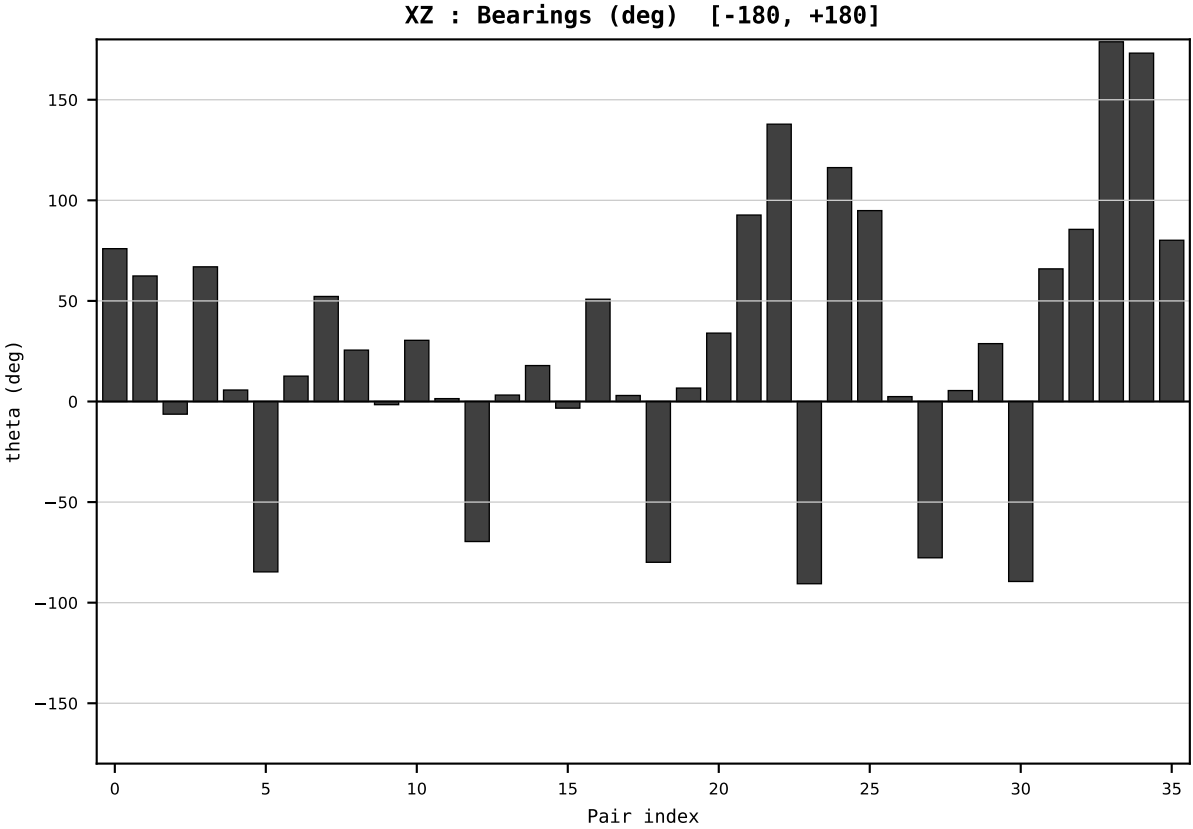
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.232607
Ring = Hs-2
E_metric= 54.9495
kappa_HS= 9207.67
omega = 2.4380 deg
d_A = 0.490208
Helm = Other Renewables
Helm d = +0.298944
DR = 9.128 HLR
DR ratio= 9207.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

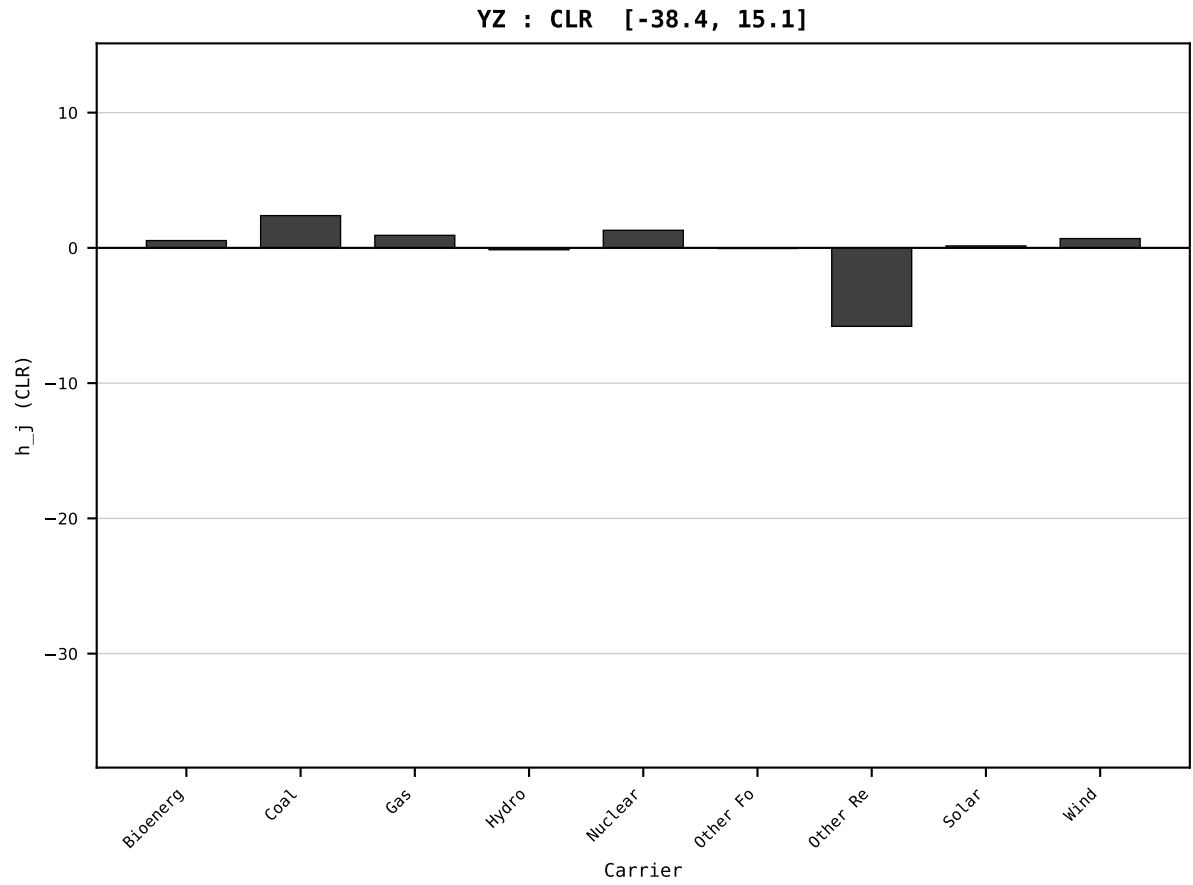
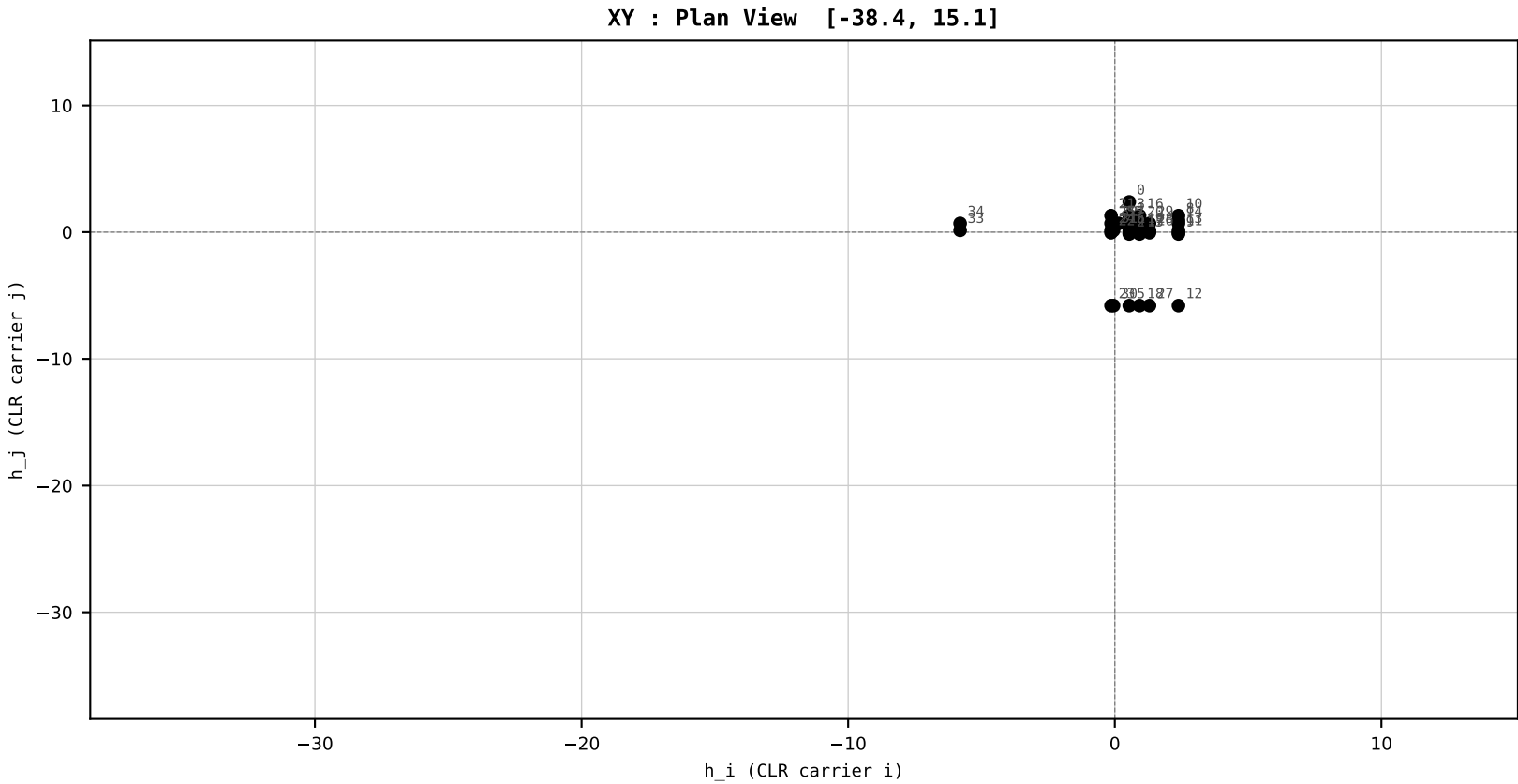
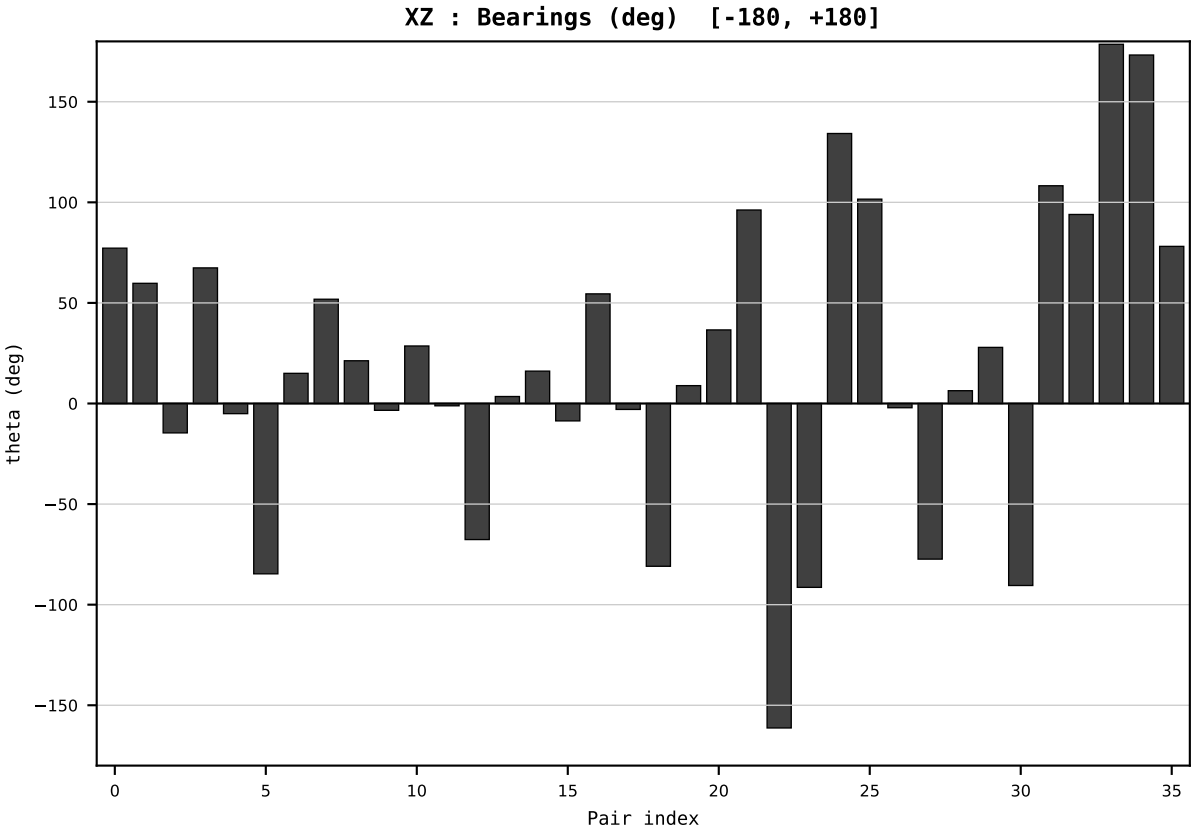
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.235275
Ring = Hs-2
E_metric= 42.7299
kappa_HS= 3602.50
omega = 2.4484 deg
d_A = 0.925102
Helm = Other Renewables
Helm d = +0.852077
DR = 8.189 HLR
DR ratio= 3602.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

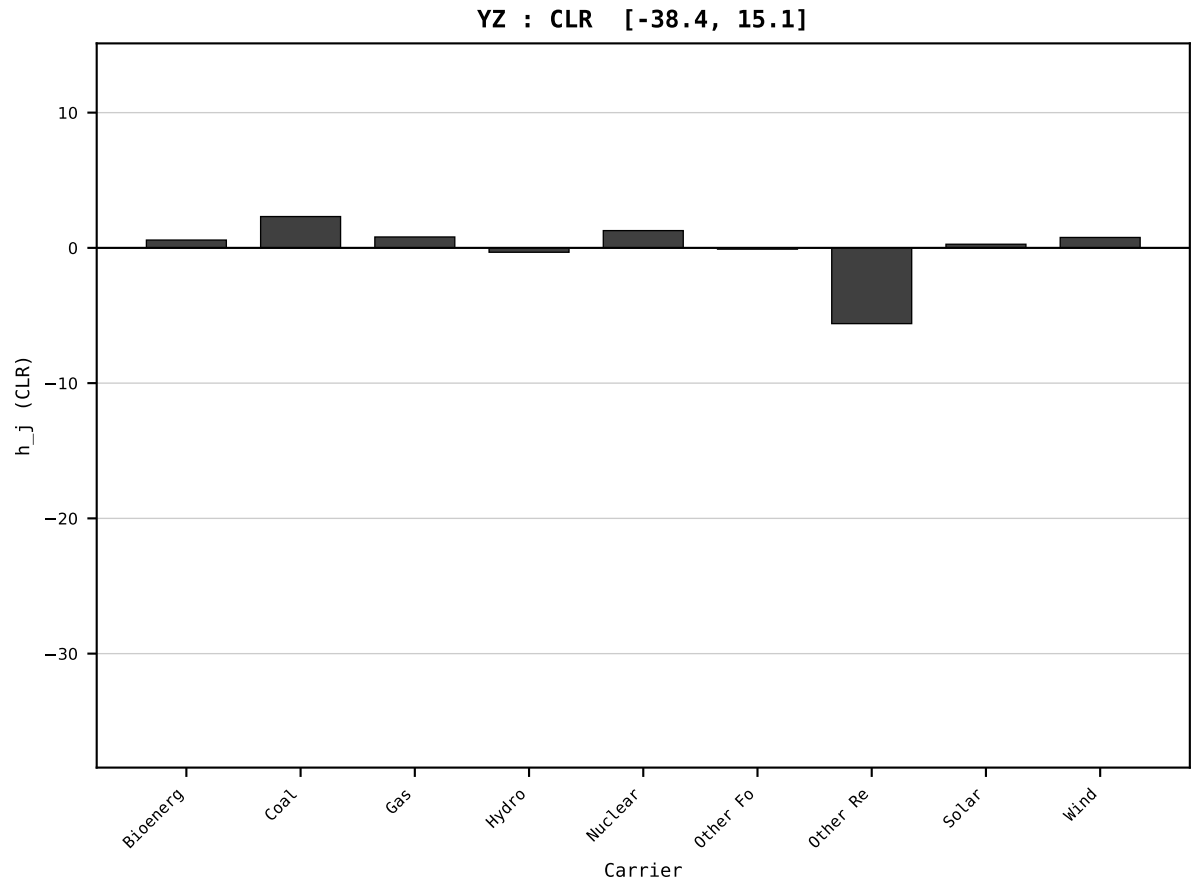
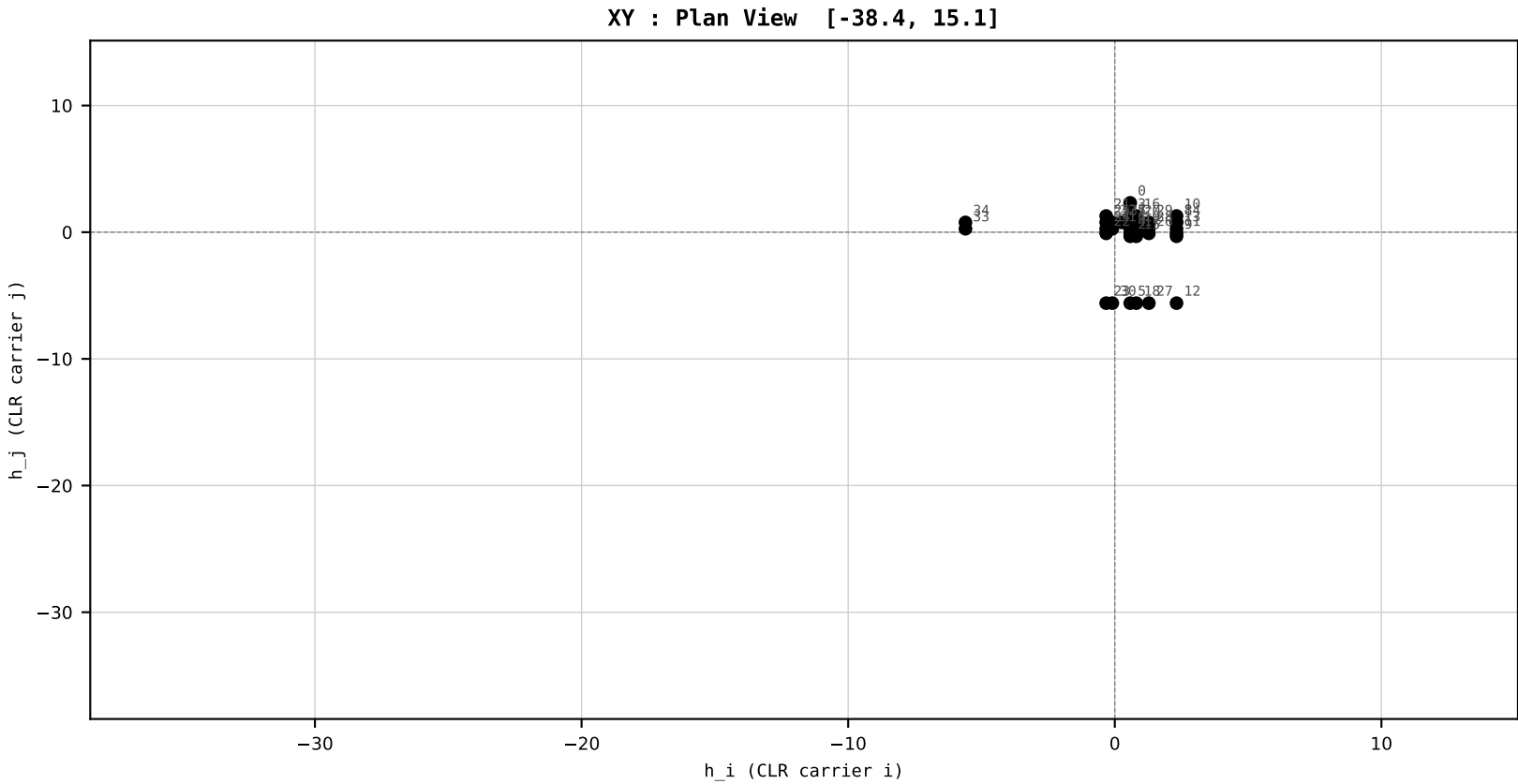
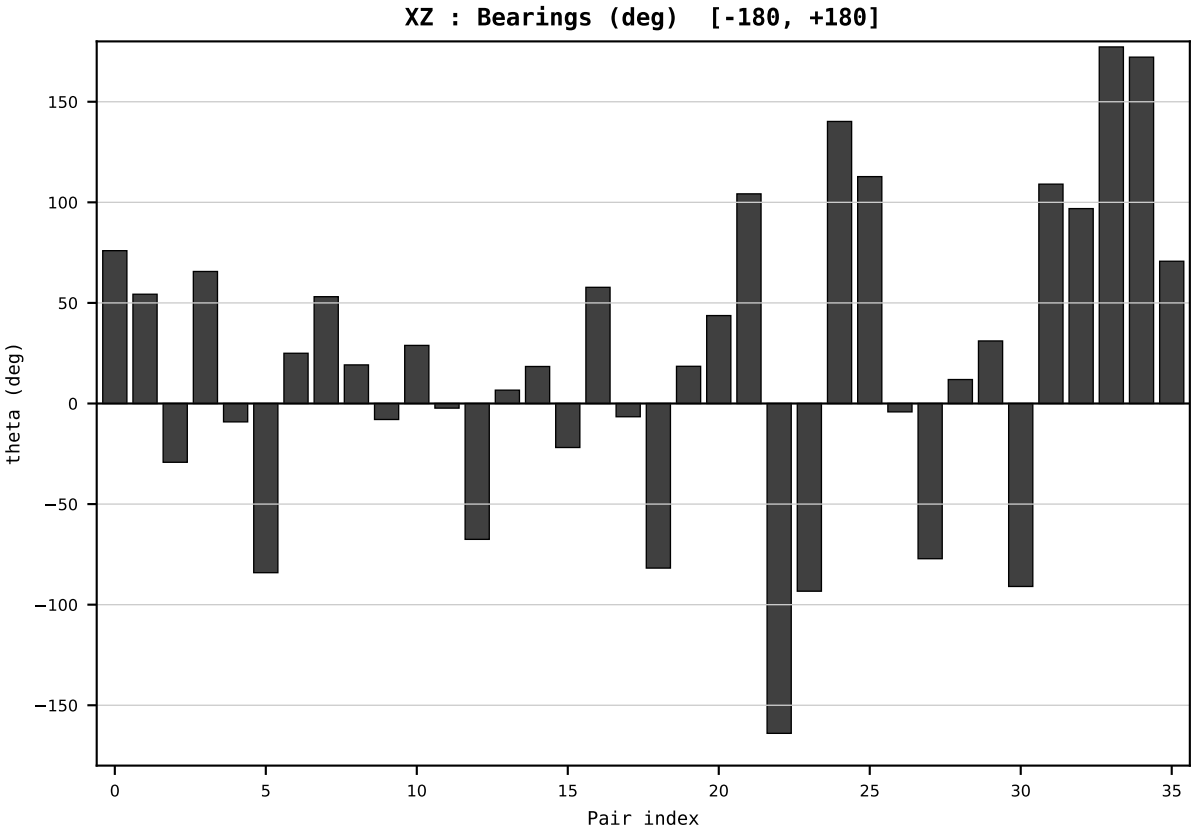
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.224515
Ring = Hs-2
E_metric= 40.1329
kappa_HS= 2744.10
omega = 2.5057 deg
d_A = 0.346263
Helm = Other Renewables
Helm d = +0.201151
DR = 7.917 HLR
DR ratio= 2744.1x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

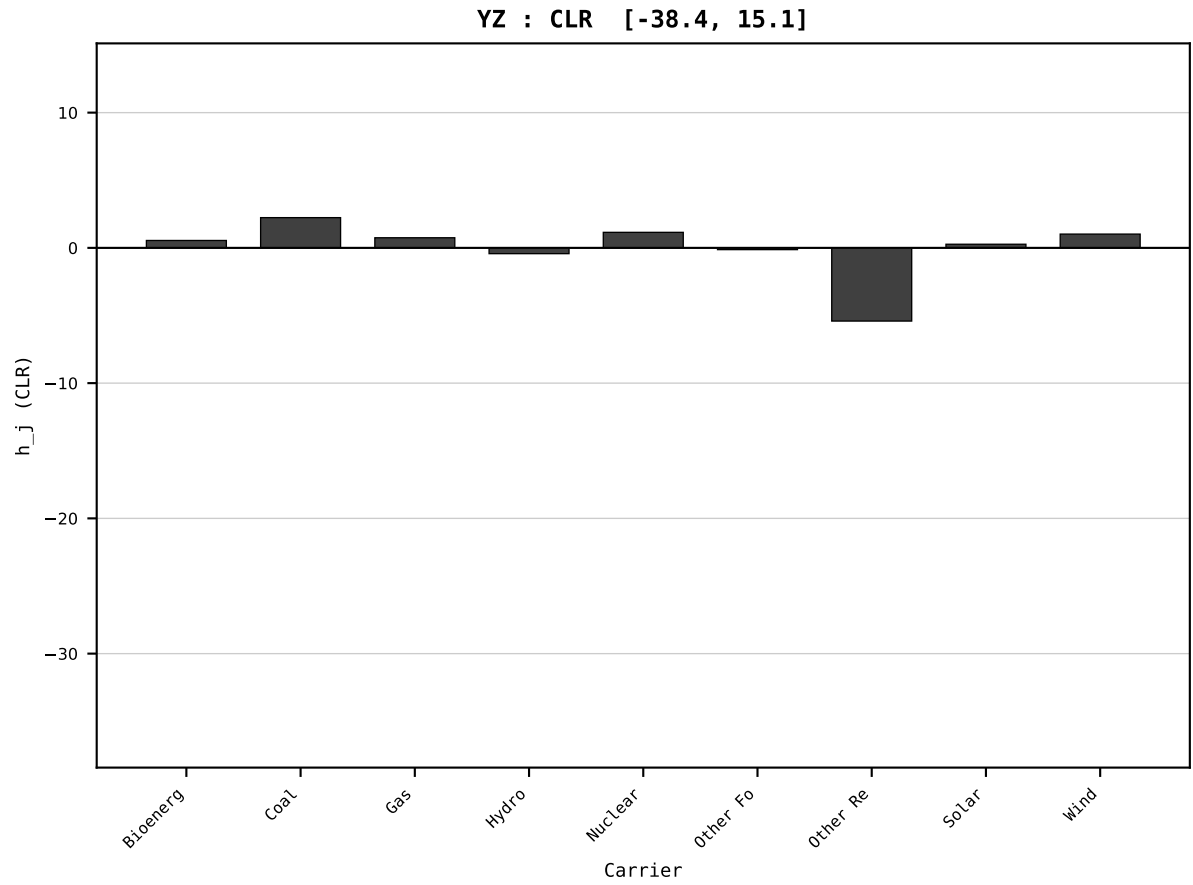
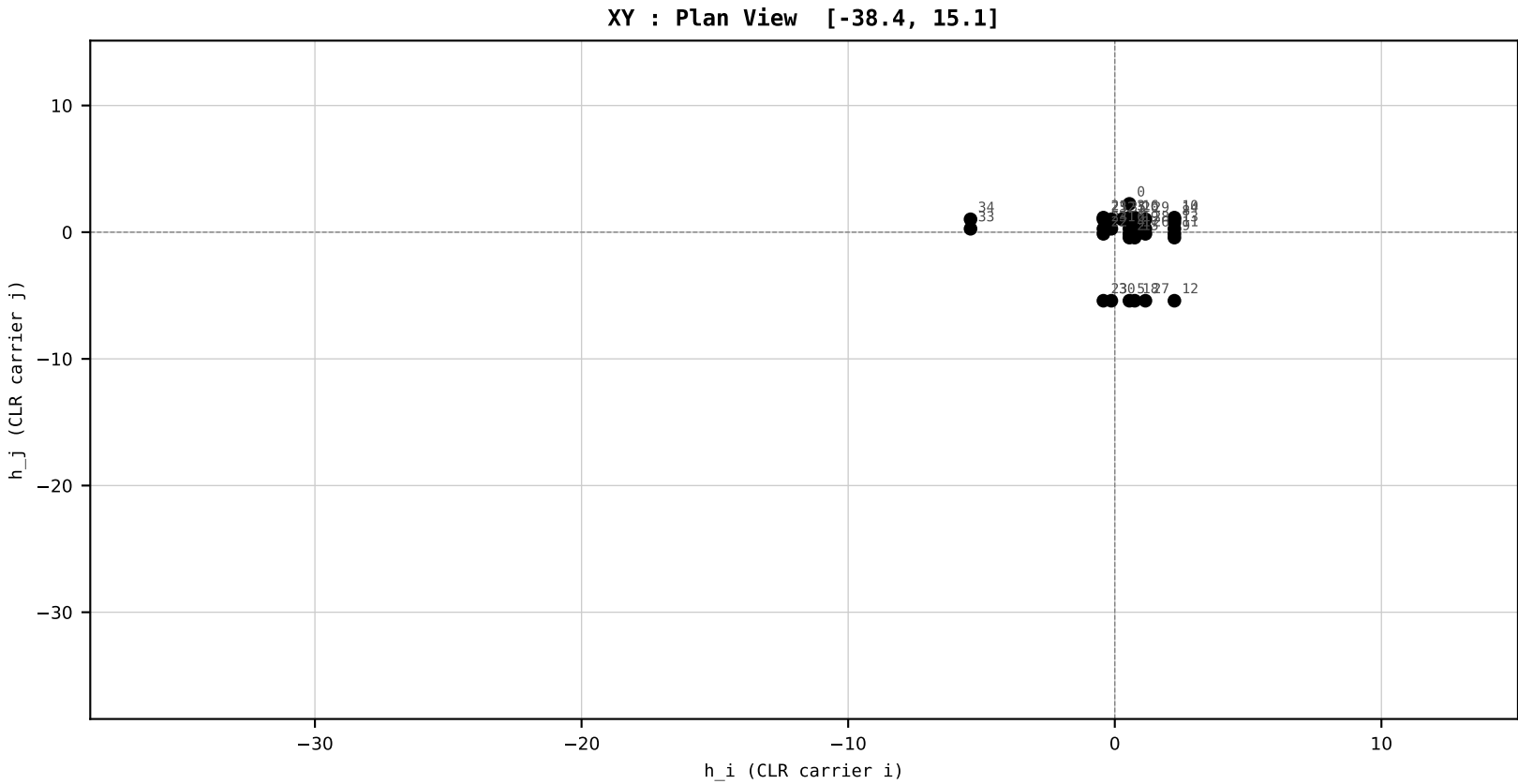
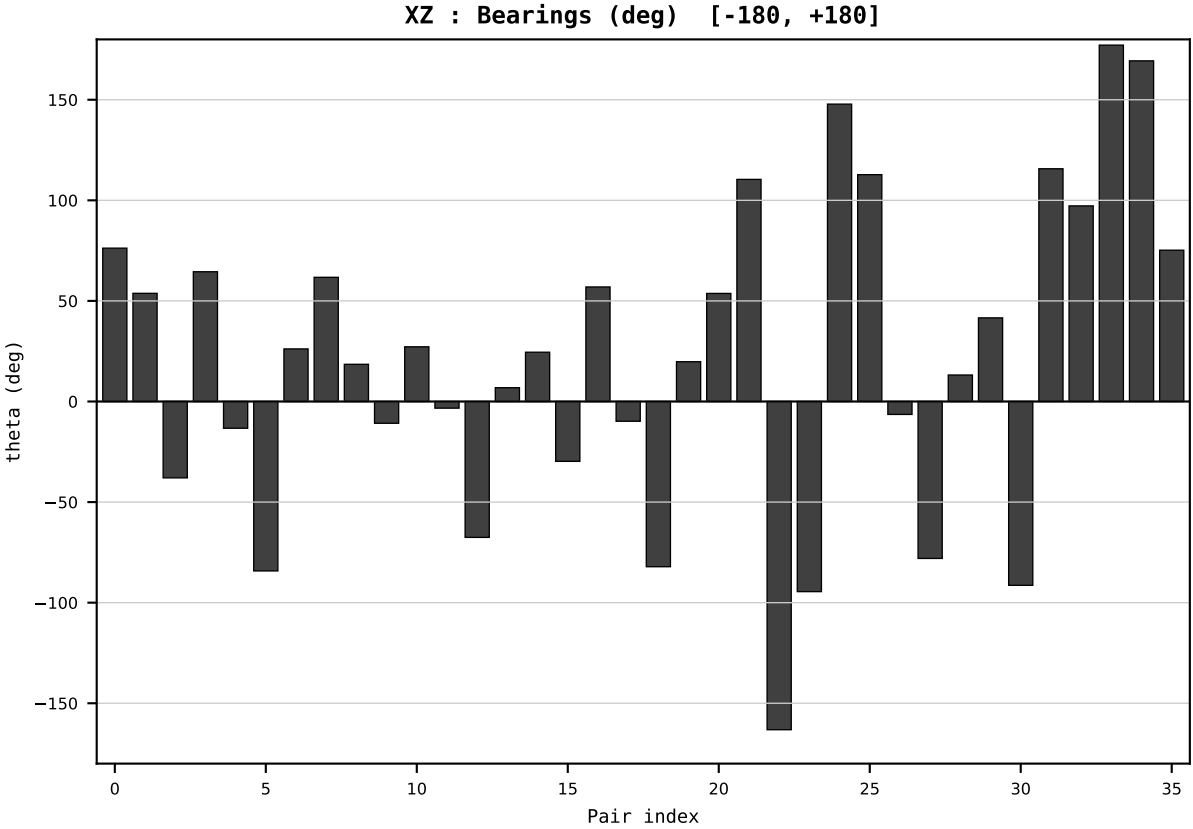
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.211885
Ring = Hs-2
E_metric= 37.7664
kappa_HS= 2093.85
omega = 2.9247 deg
d_A = 0.370645
Helm = Wind
Helm d = +0.248533
DR = 7.647 HLR
DR ratio= 2093.8x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

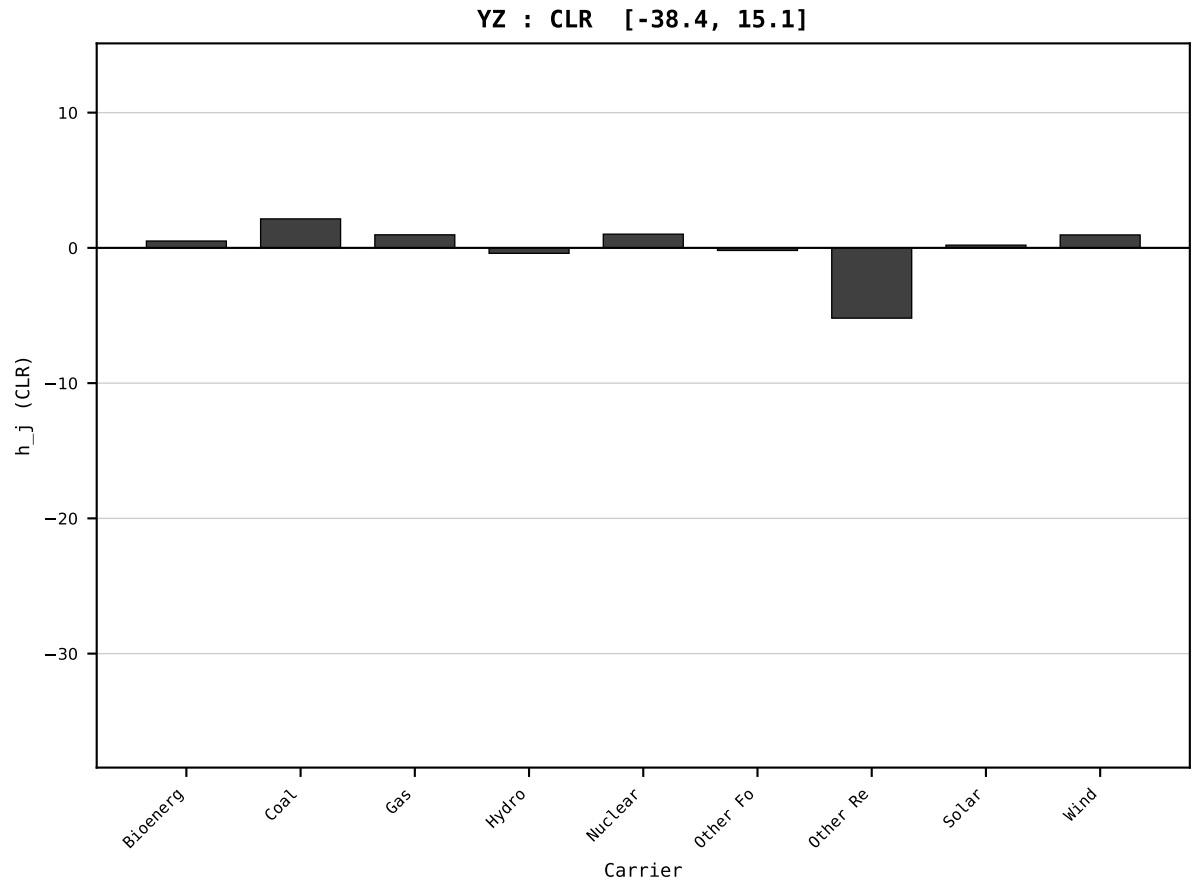
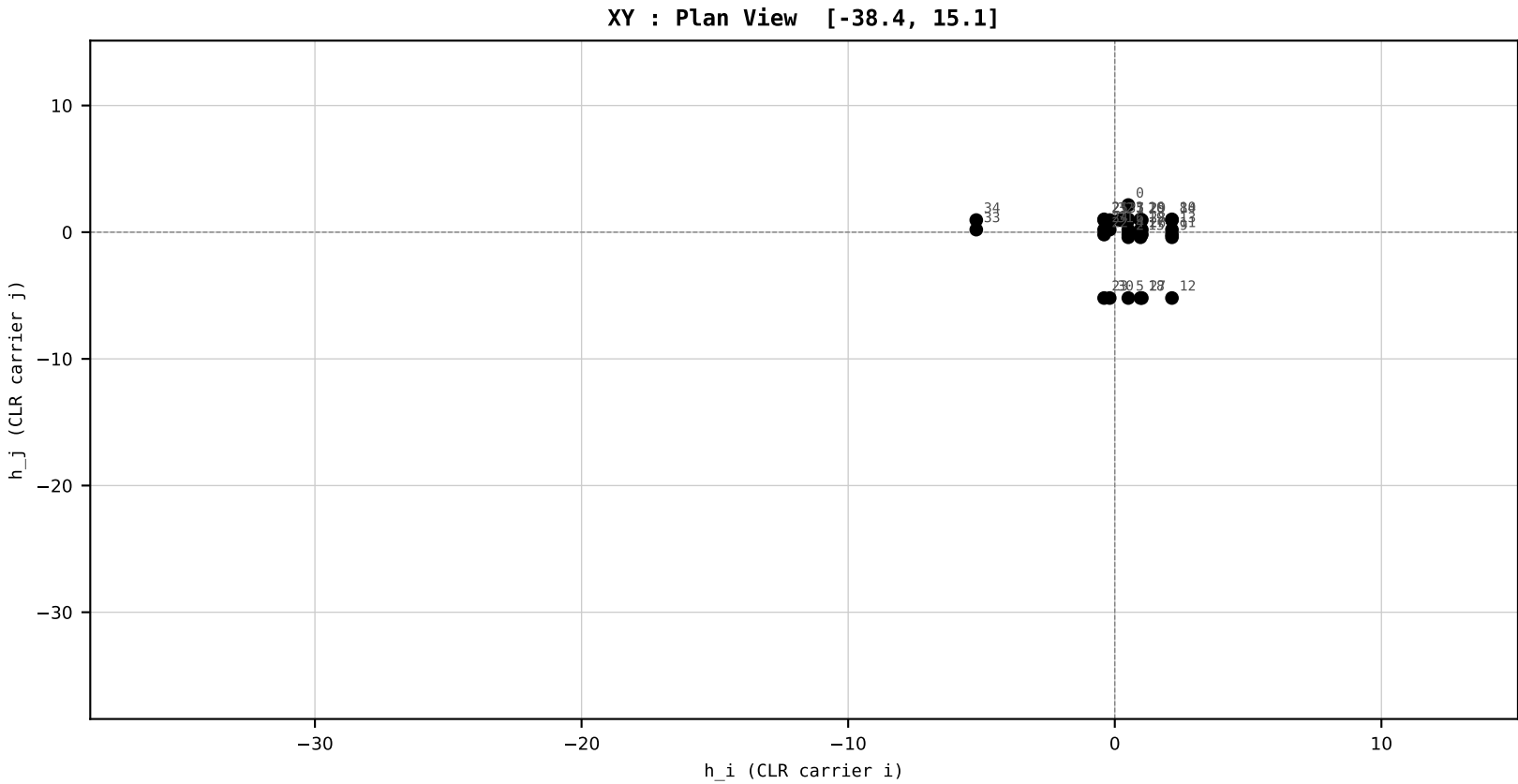
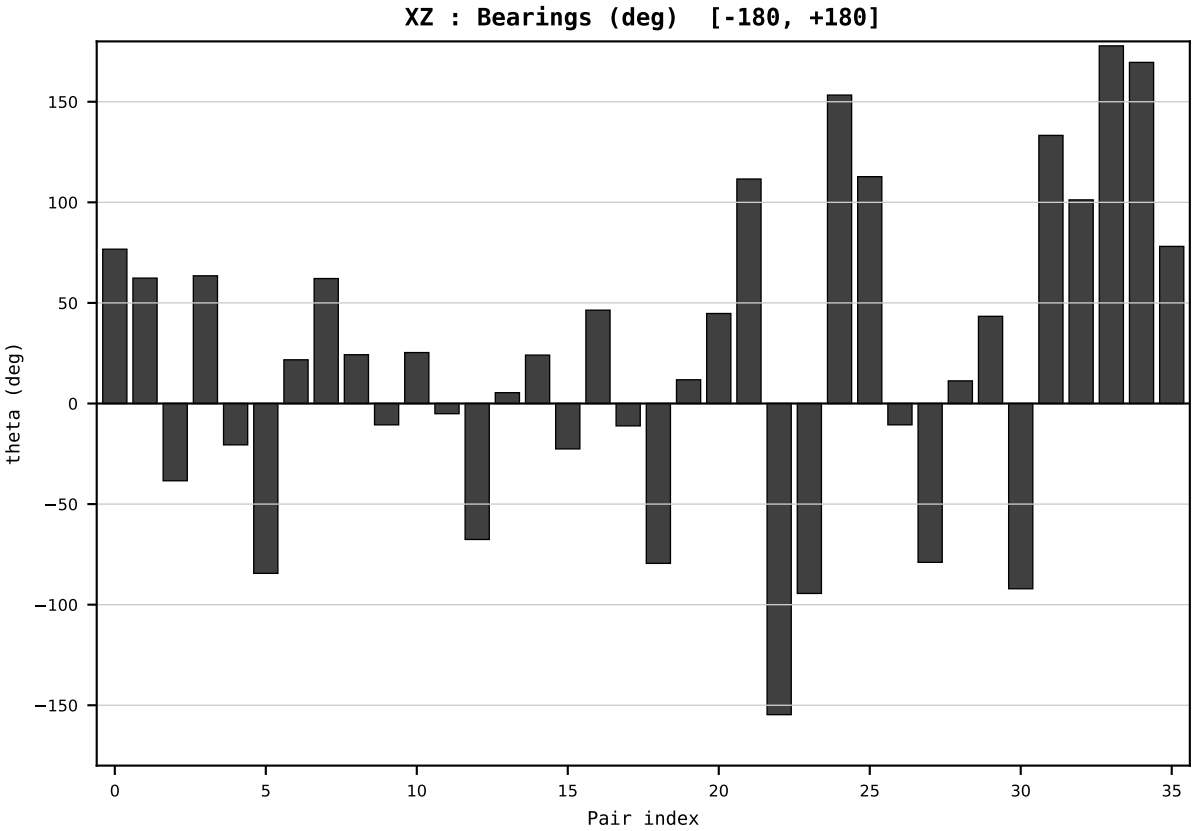
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.198712
Ring = Hs-2
E_metric= 34.9636
kappa_HS= 1539.71
omega = 2.7002 deg
d_A = 0.367040
Helm = Gas
Helm d = +0.217765
DR = 7.339 HLR
DR ratio= 1539.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

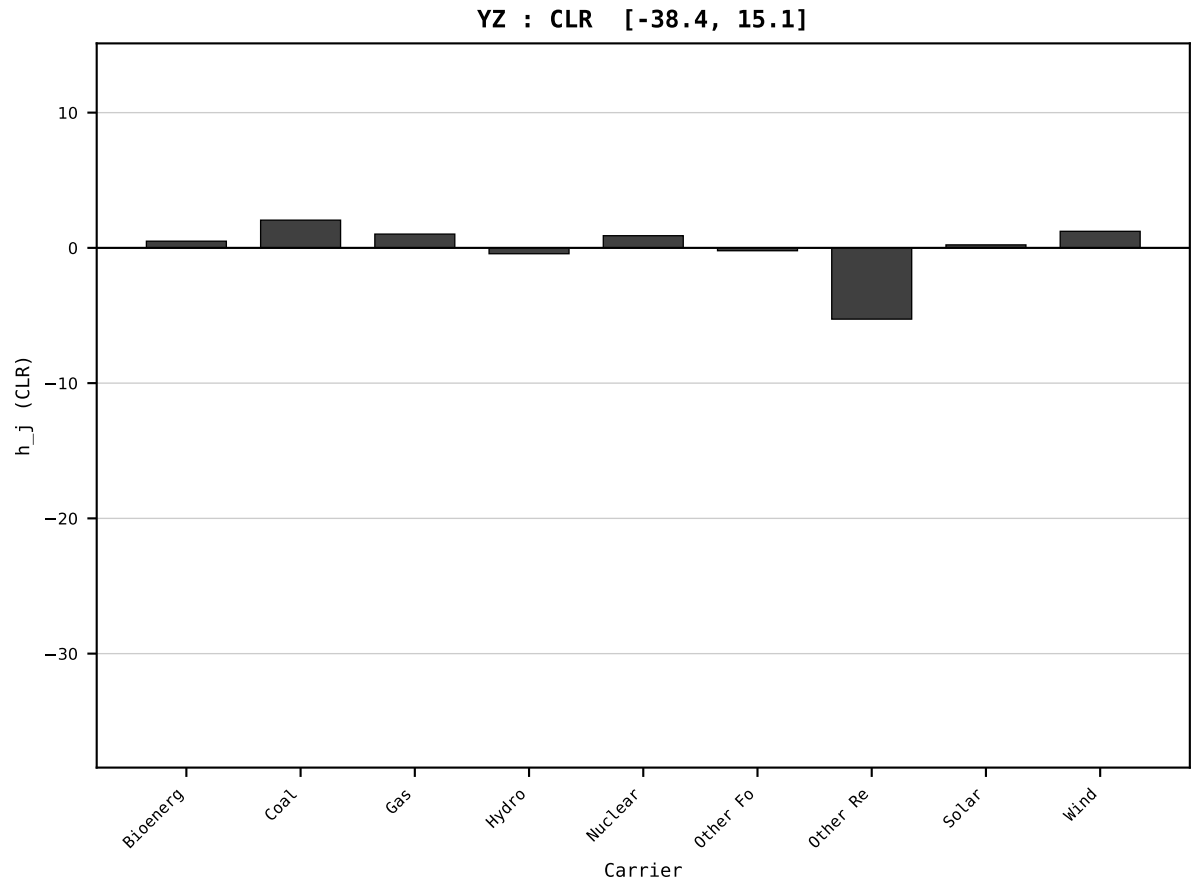
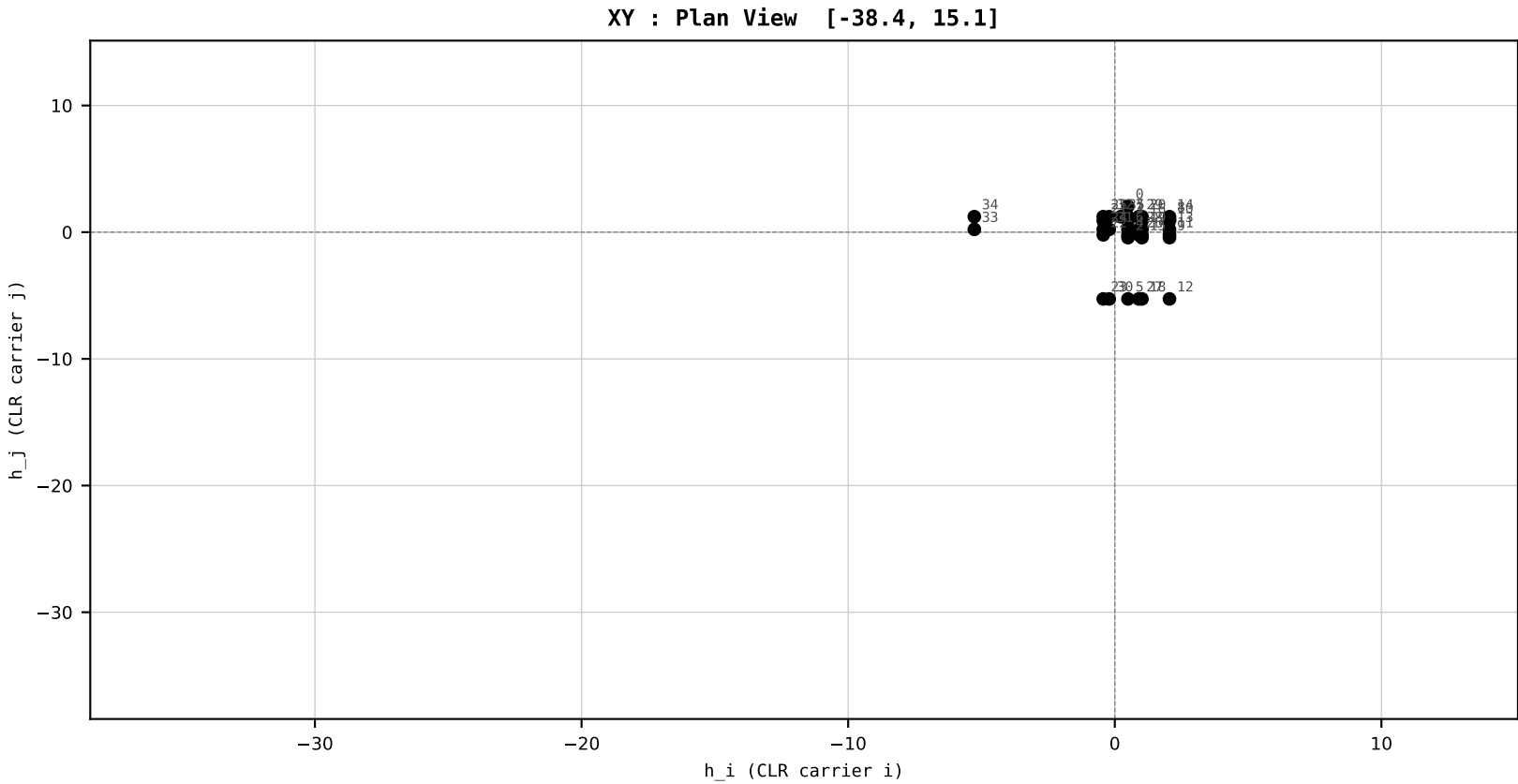
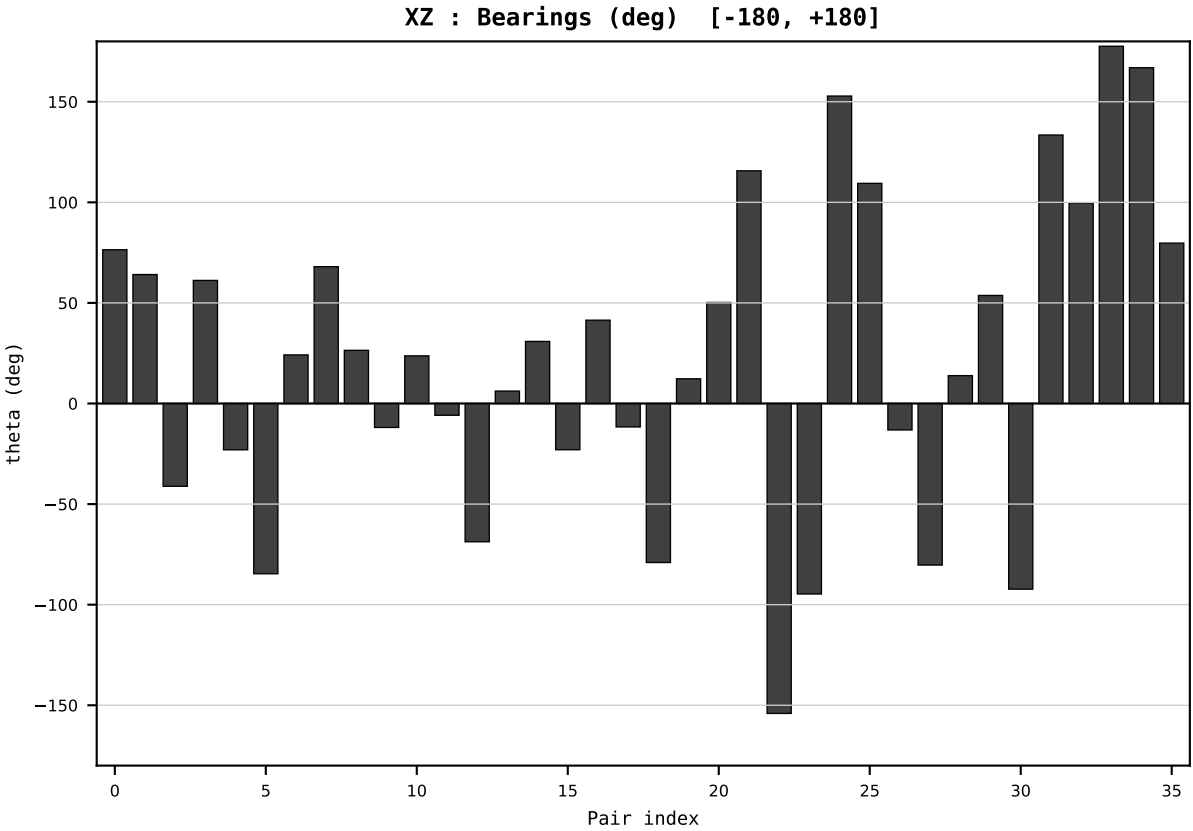
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.184230
Ring = Hs-2
E_metric= 35.8305
kappa_HS= 1507.88
omega = 3.0206 deg
d_A = 0.321962
Helm = Wind
Helm d = +0.268201
DR = 7.318 HLR
DR ratio= 1507.9x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

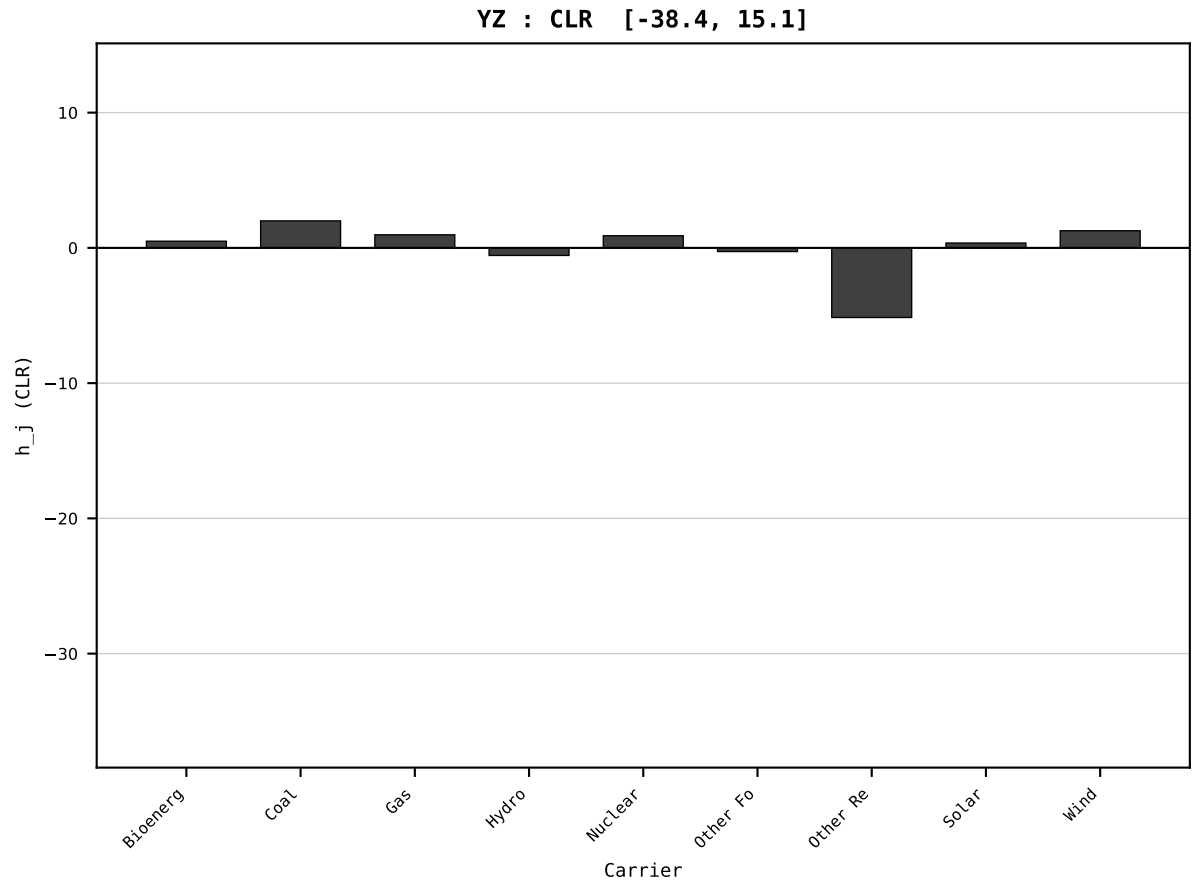
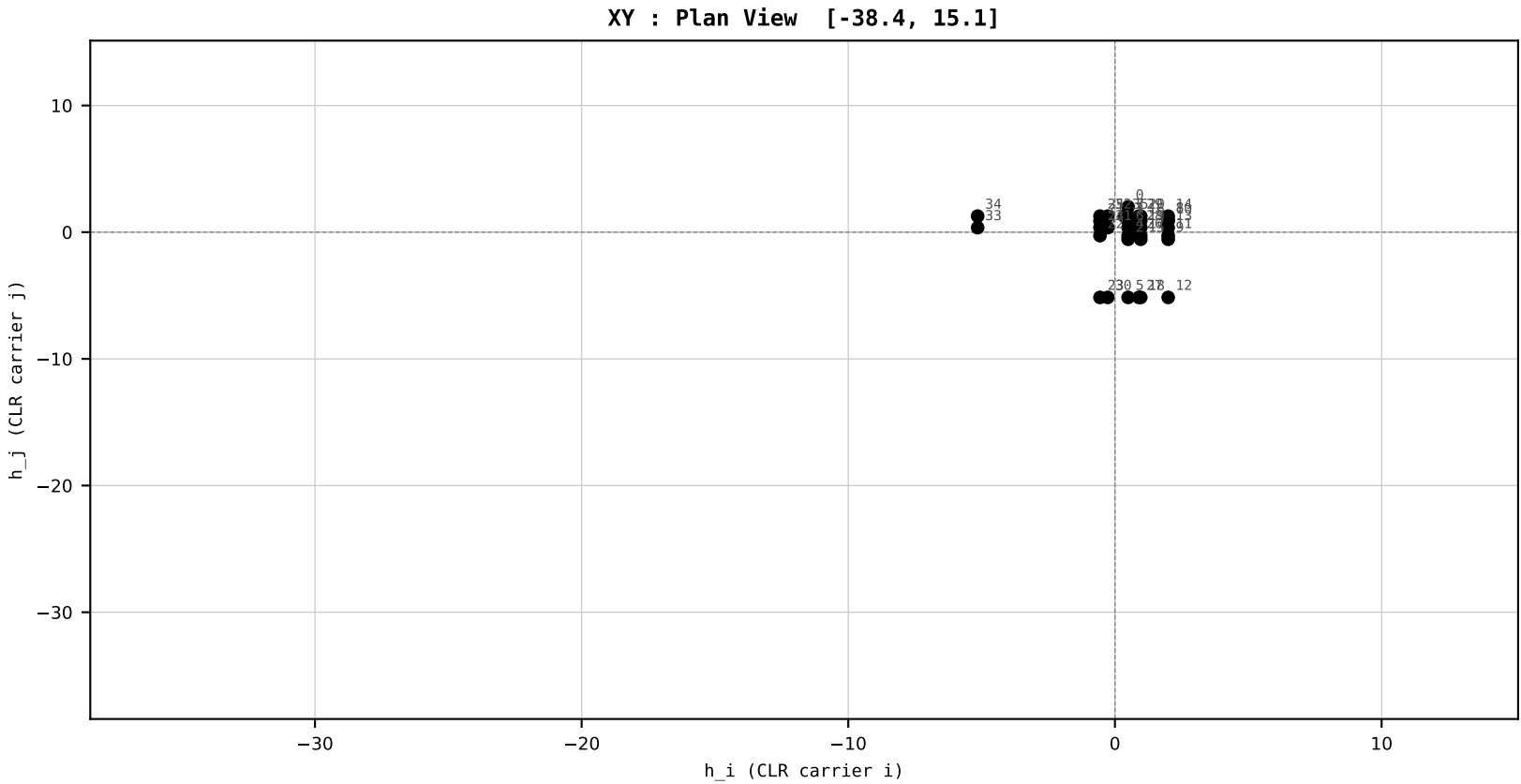
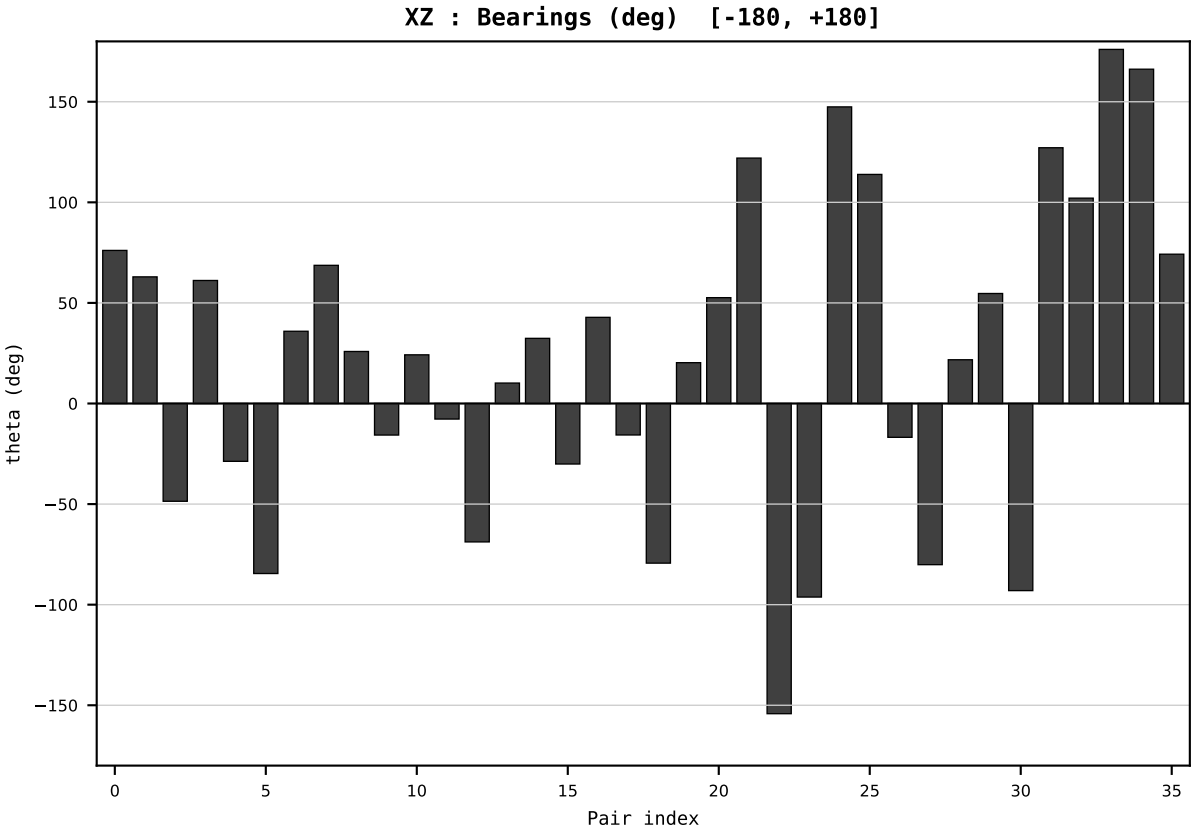
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.178543
Ring = Hs-2
E_metric= 34.5969
kappa_HS= 1267.56
omega = 2.1475 deg
d_A = 0.245475
Helm = Solar
Helm d = +0.136009
DR = 7.145 HLR
DR ratio= 1267.6x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



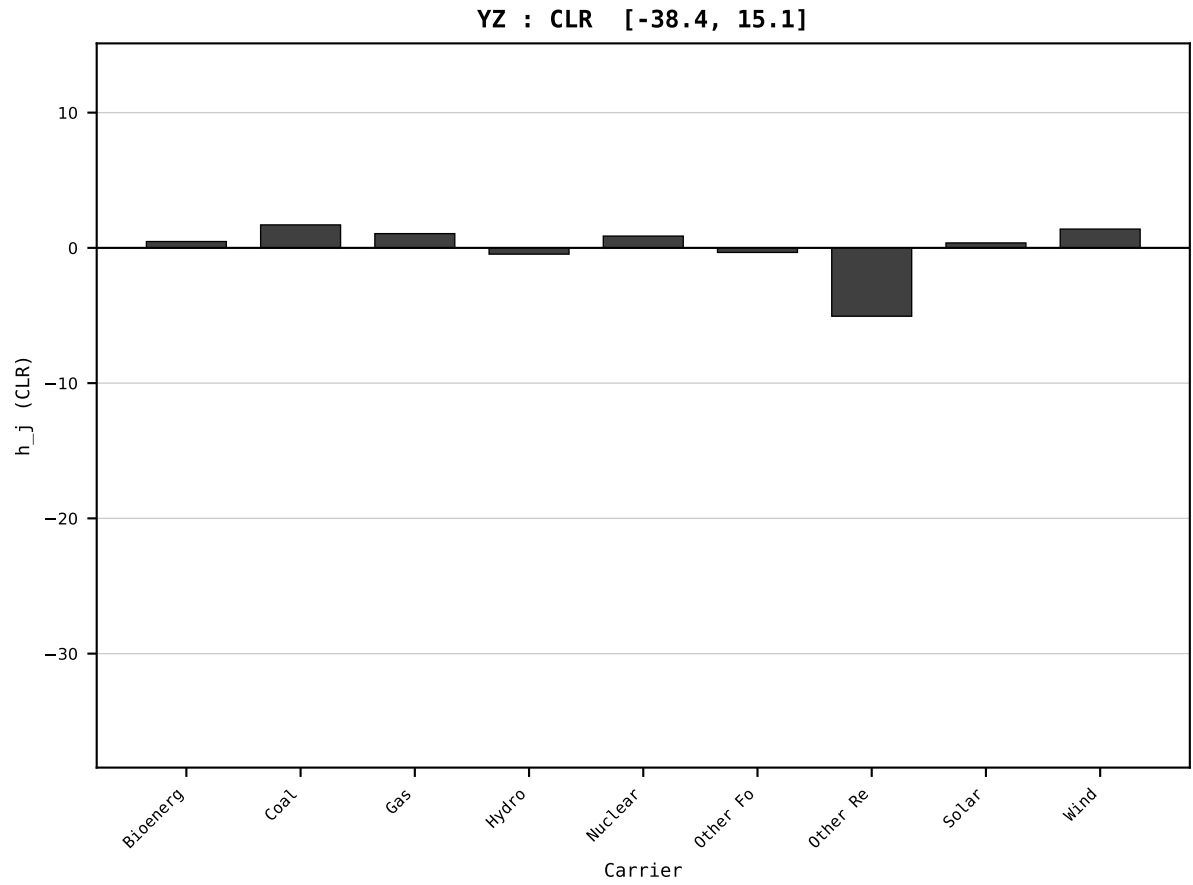
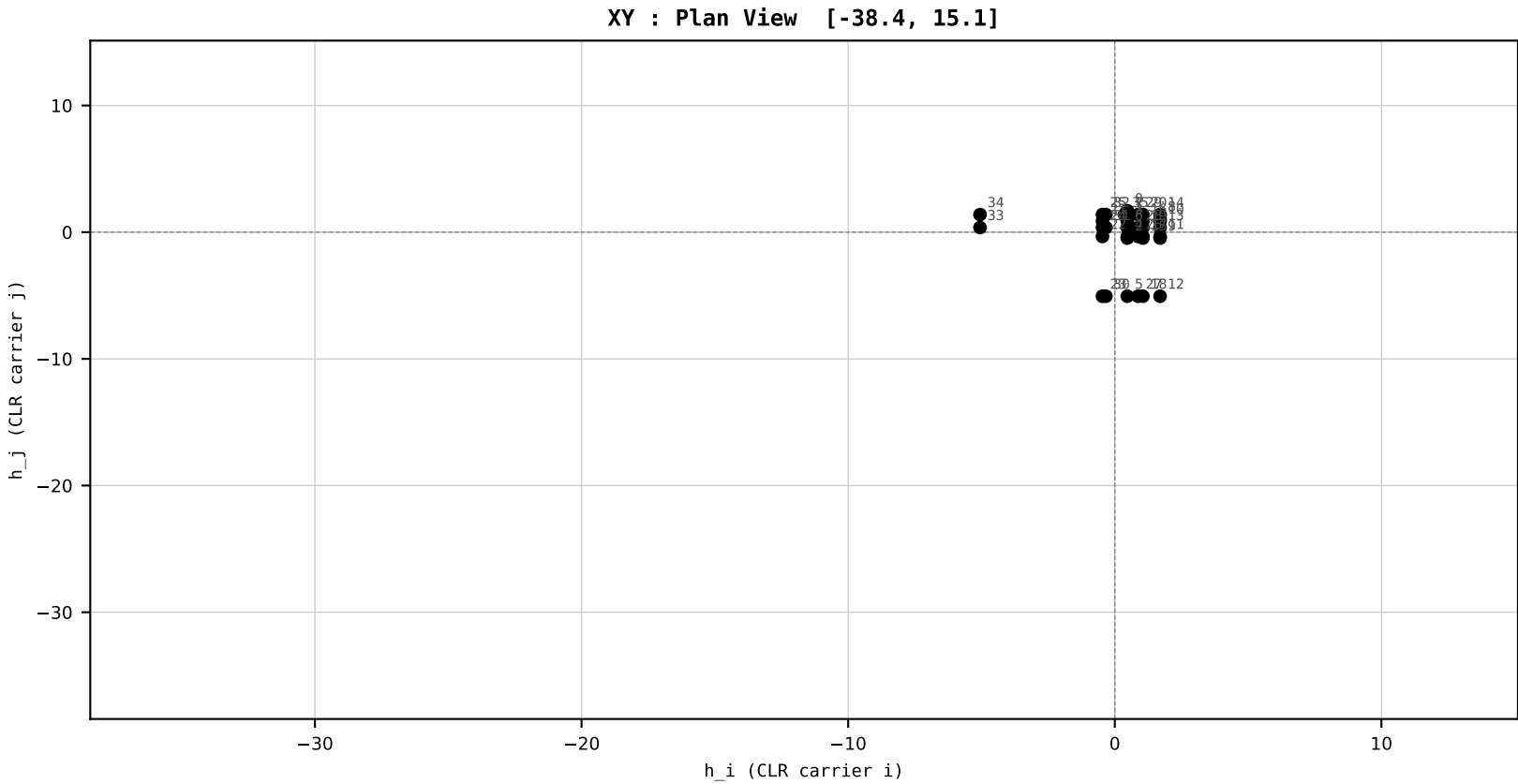
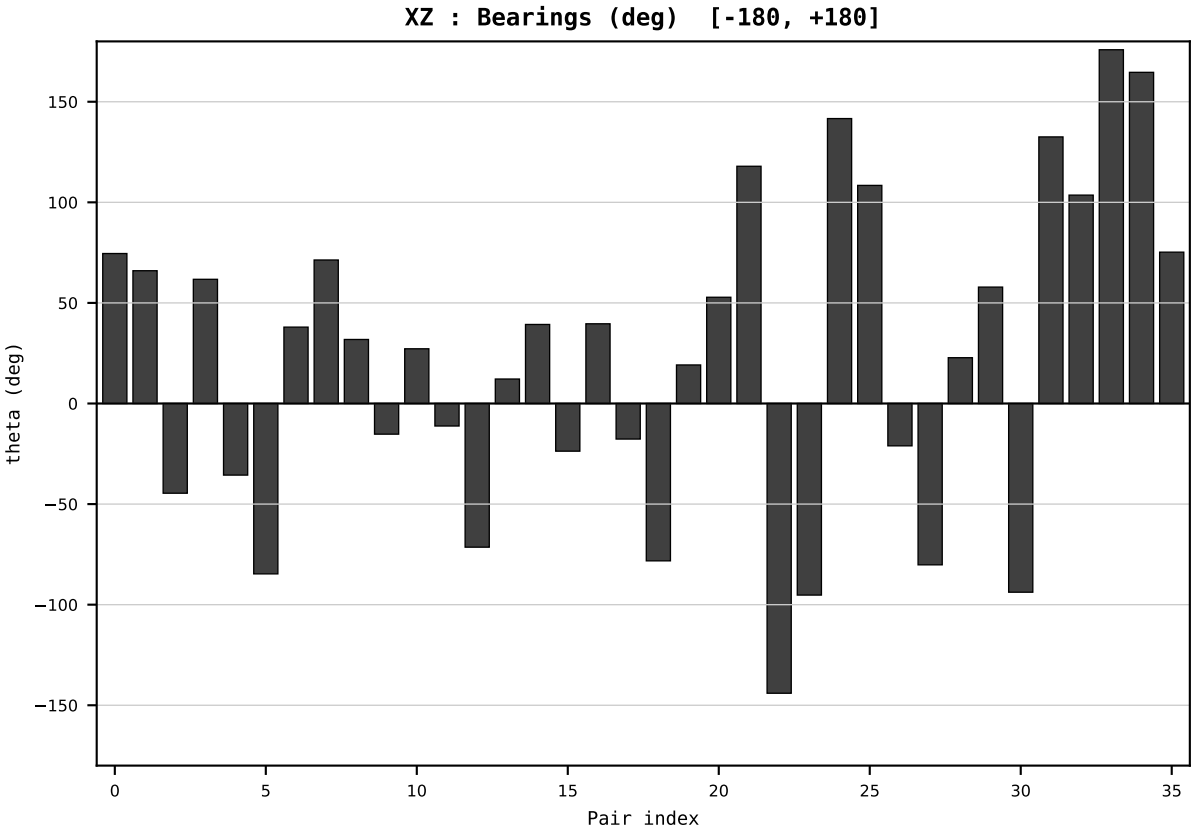
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.148916
Ring = Hs-1
E_metric= 32.9256
kappa_HS= 857.25
omega = 3.3361 deg
d_A = 0.367532
Helm = Coal
Helm d = -0.297295
DR = 6.754 HLR
DR ratio= 857.2x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

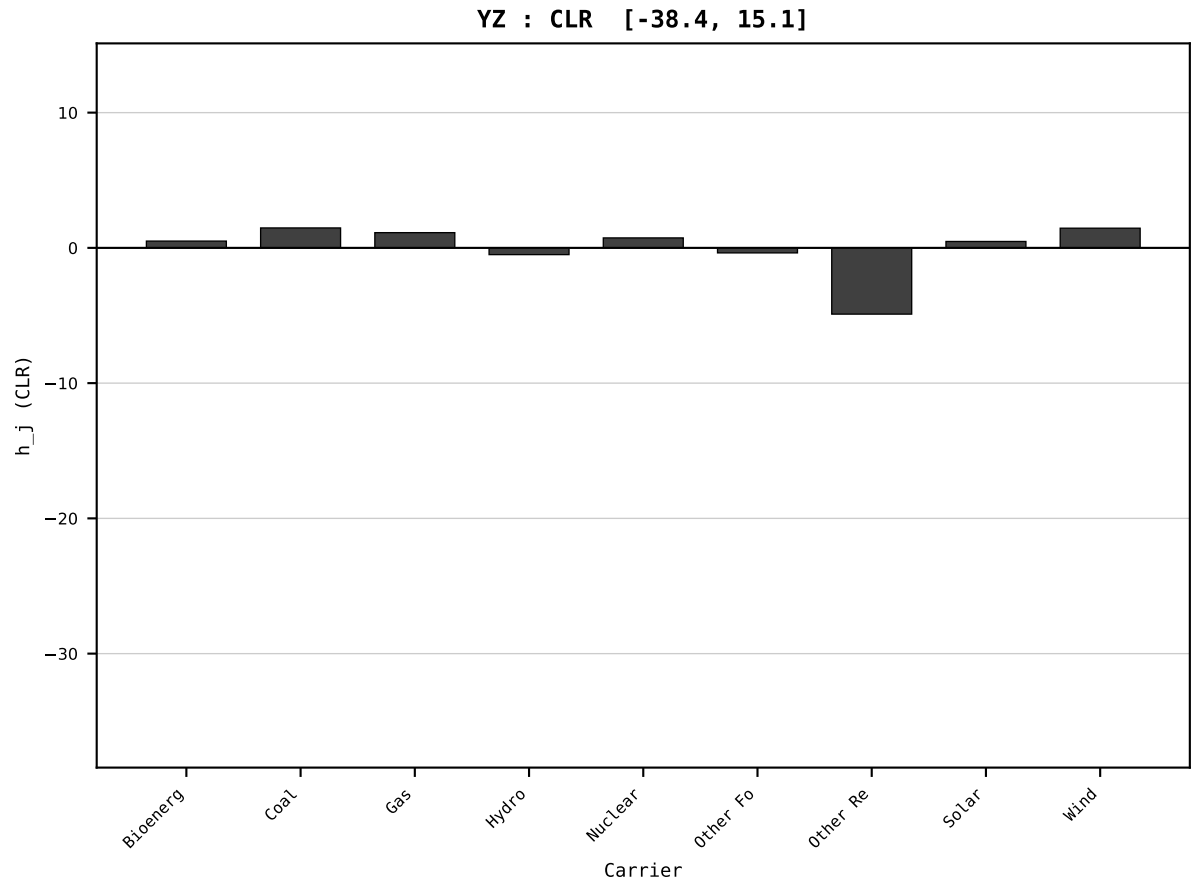
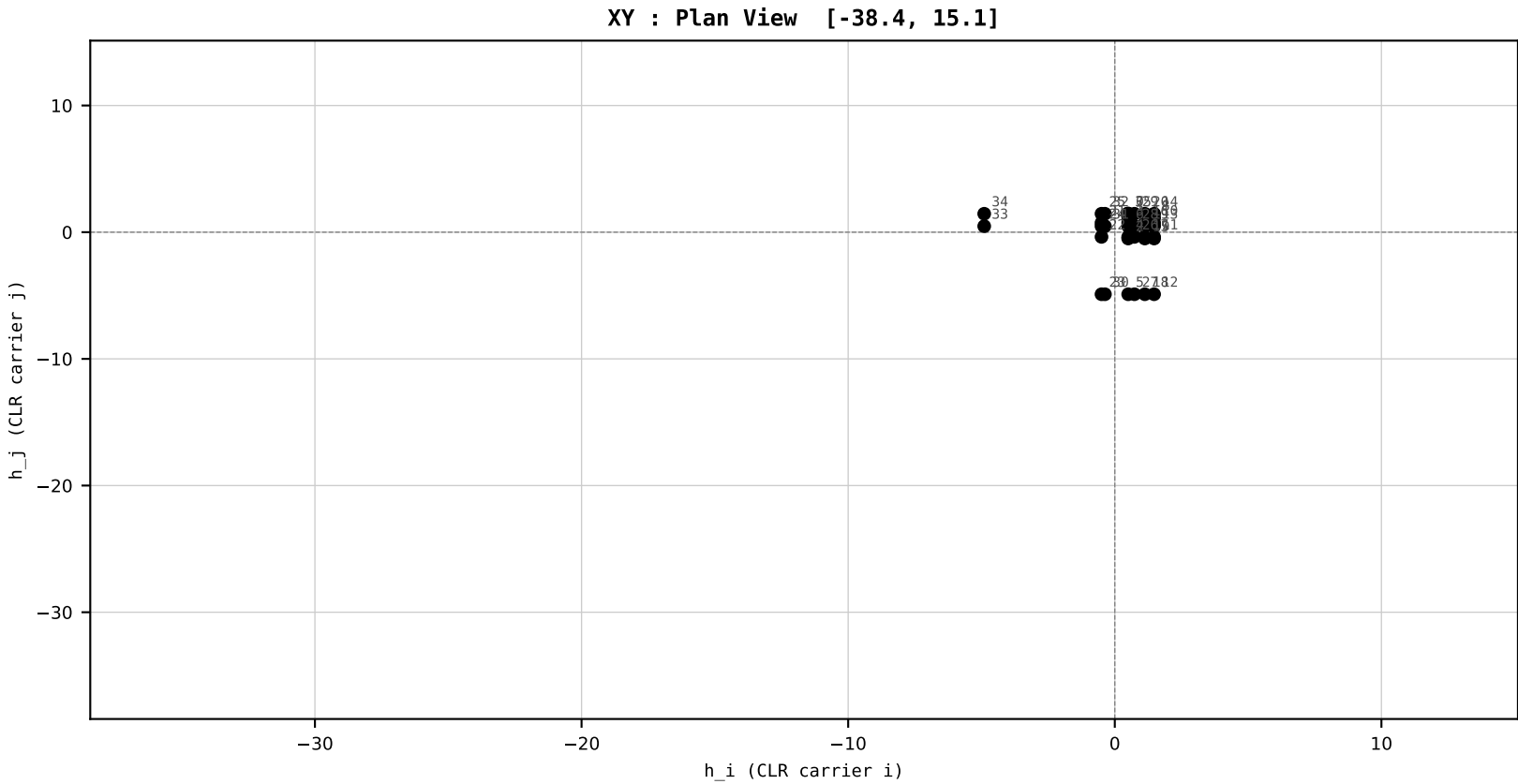
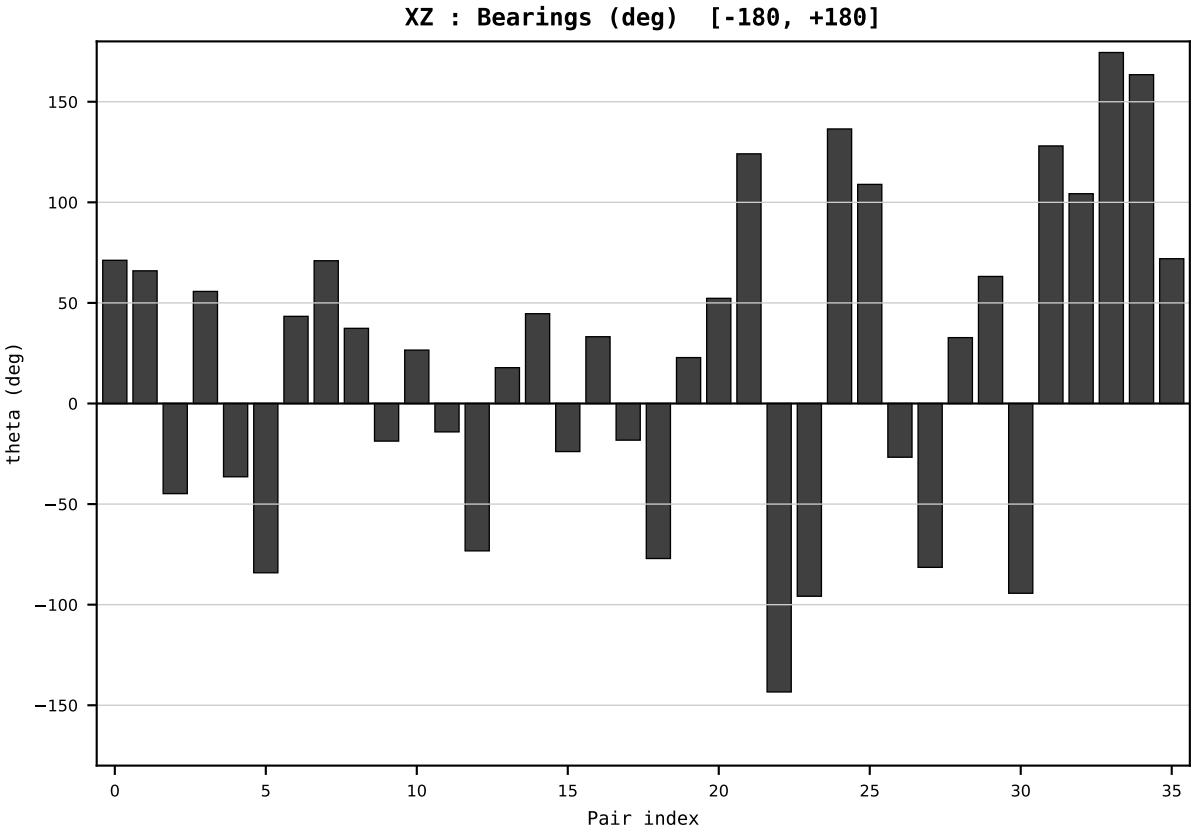
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.137161
Ring = Hs-1
E_metric= 30.9528
kappa_HS= 585.22
omega = 3.0042 deg
d_A = 0.343833
Helm = Coal
Helm d = -0.225193
DR = 6.372 HLR
DR ratio= 585.2x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

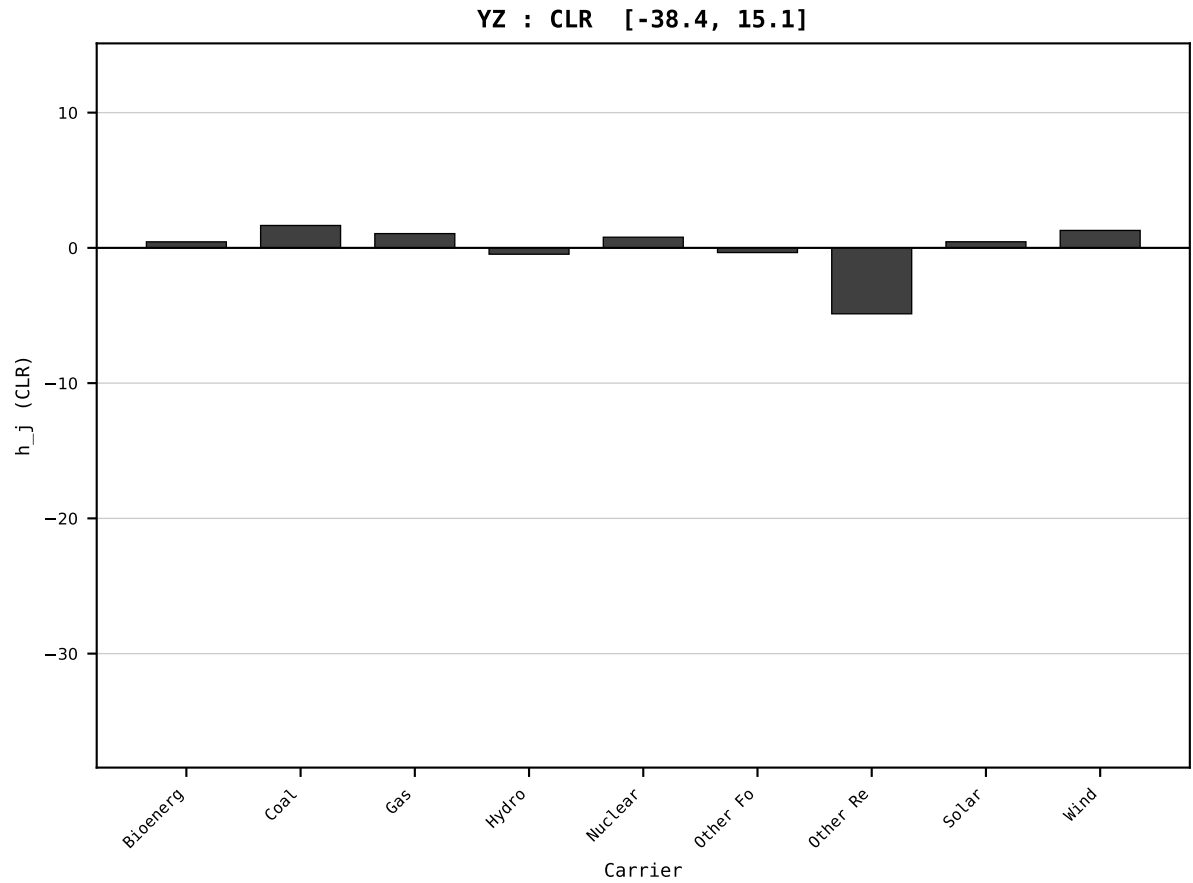
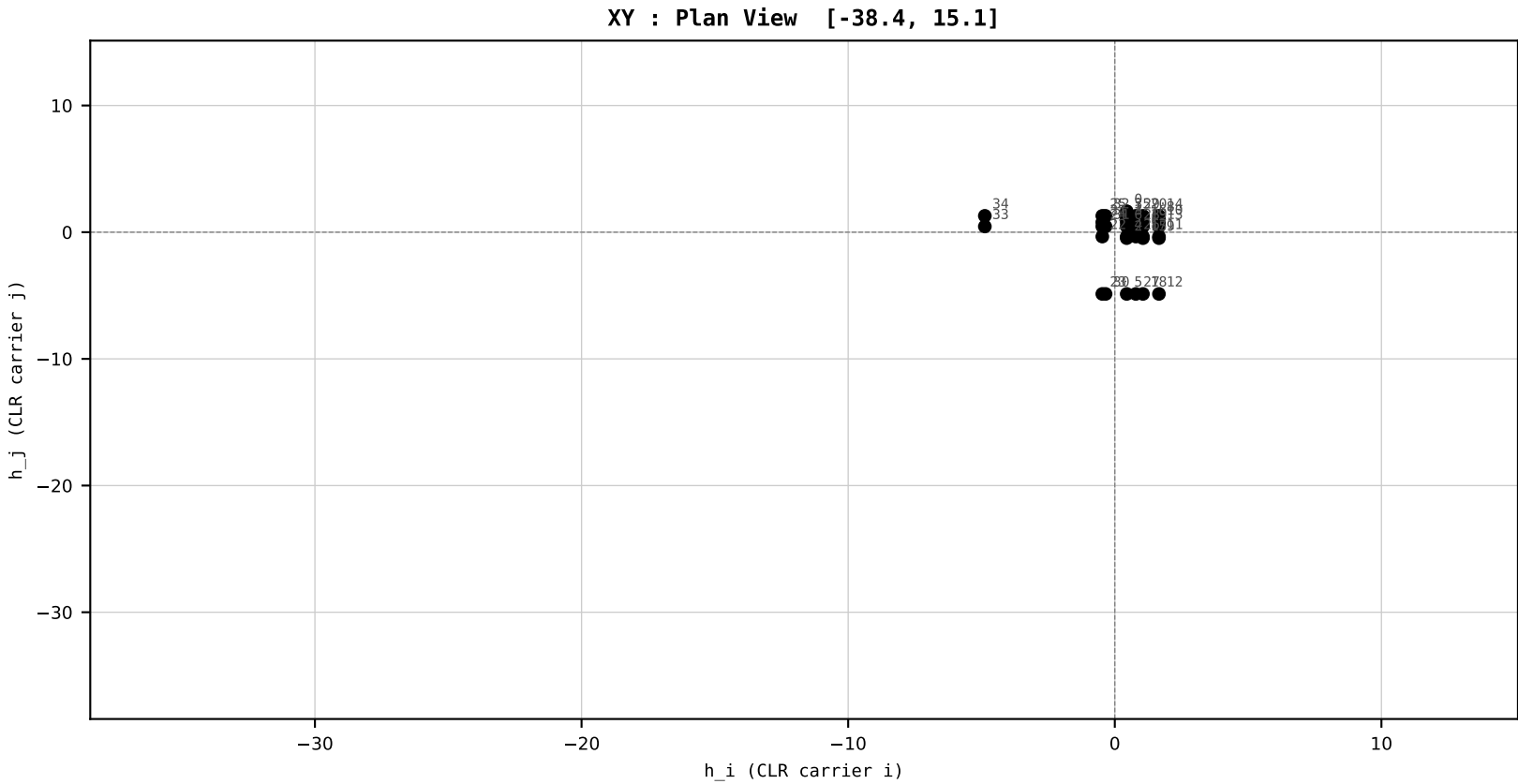
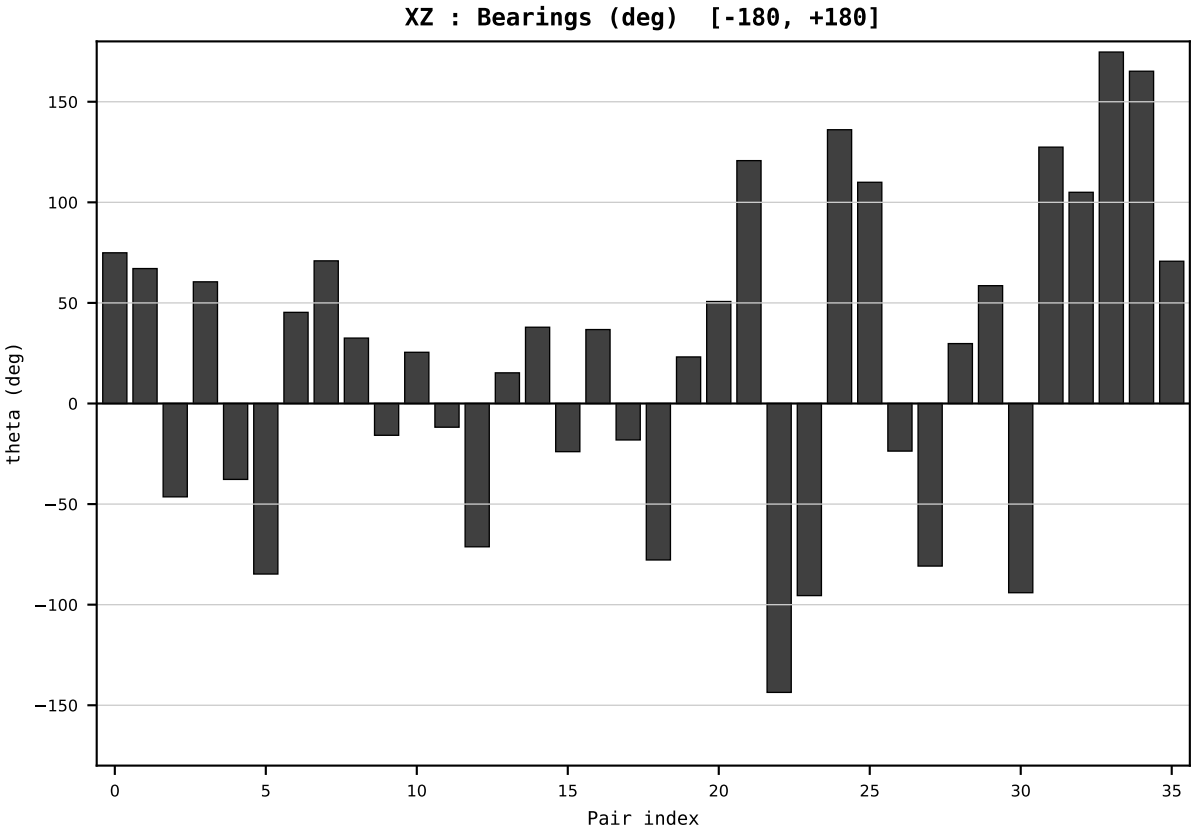
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.142585
Ring = Hs-1
E_metric= 30.6481
kappa_HS= 686.04
omega = 2.7904 deg
d_A = 0.271644
Helm = Coal
Helm d = +0.182416
DR = 6.531 HLR
DR ratio= 686.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

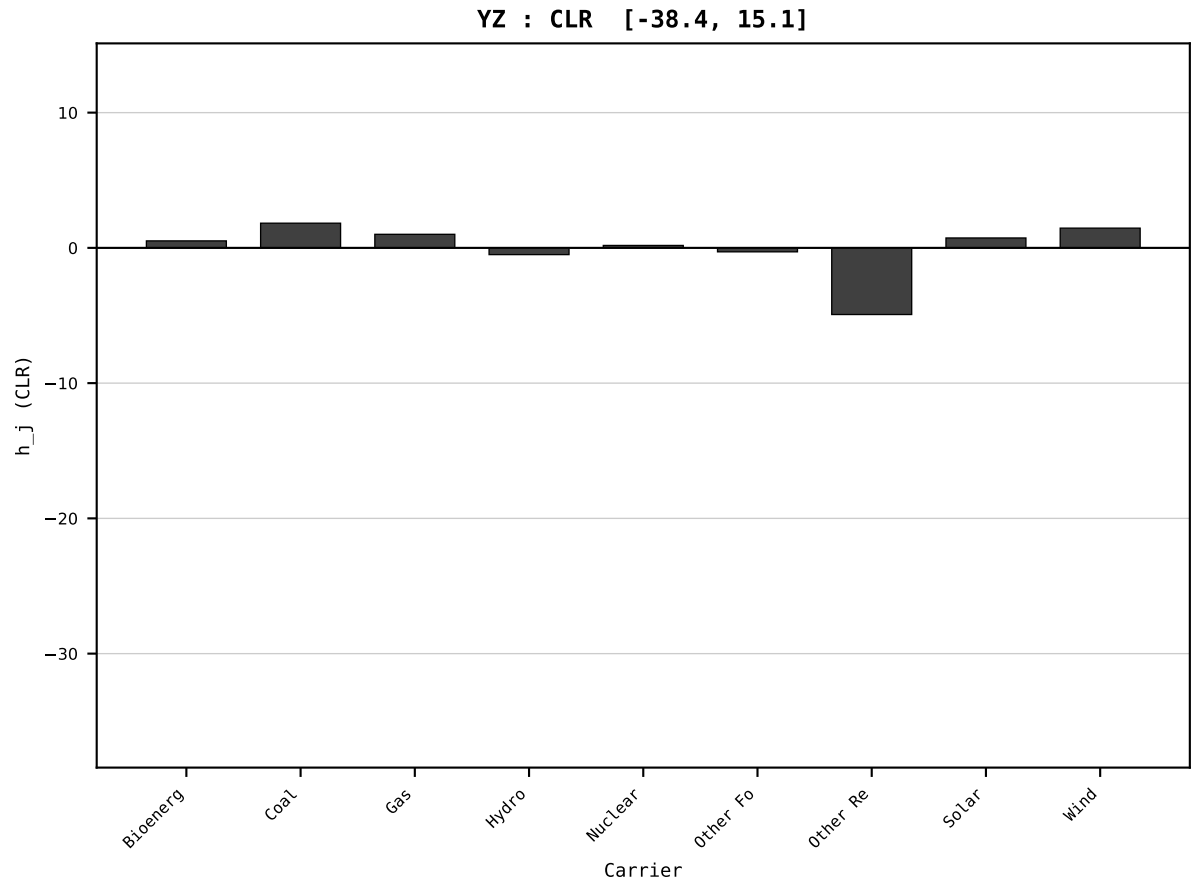
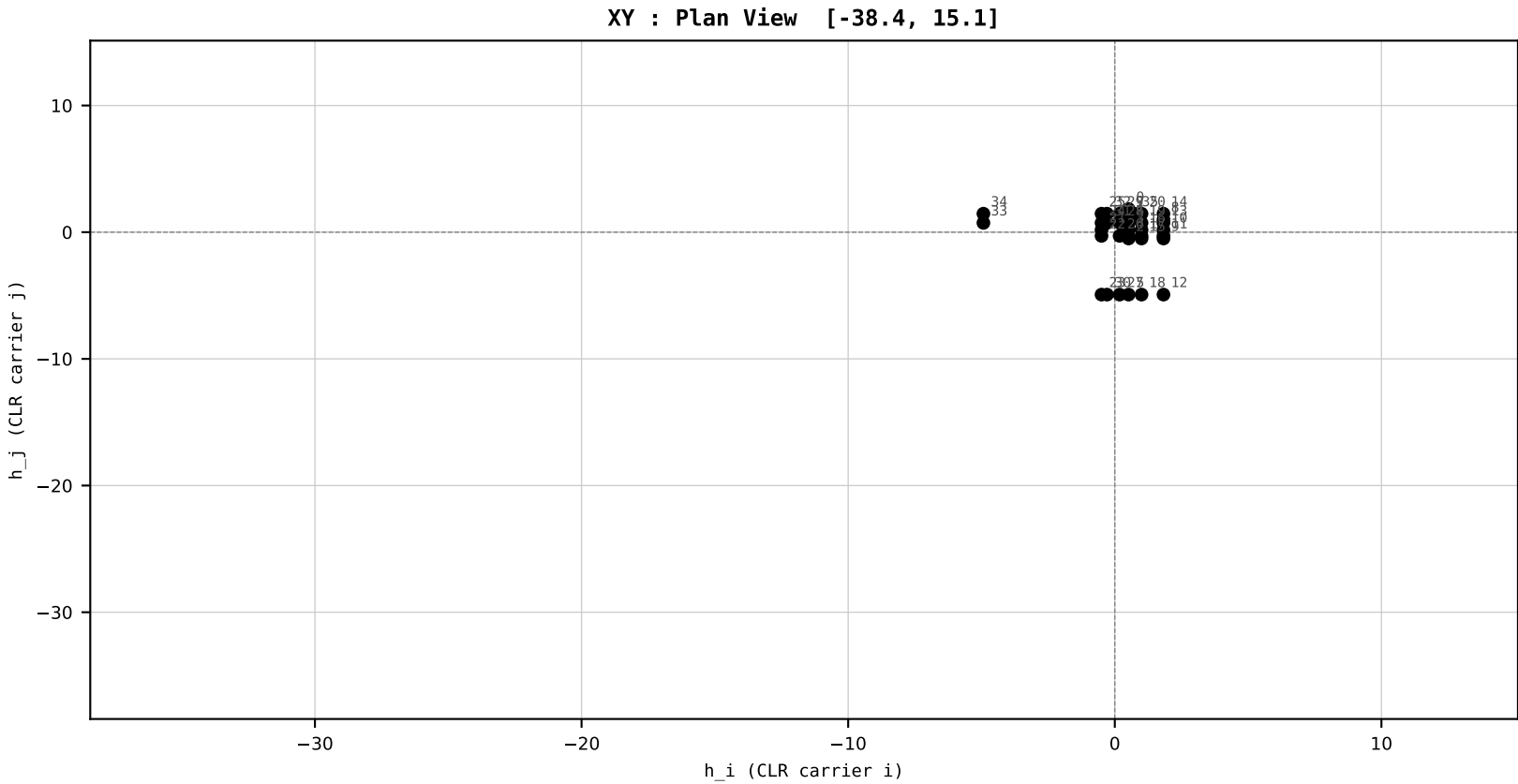
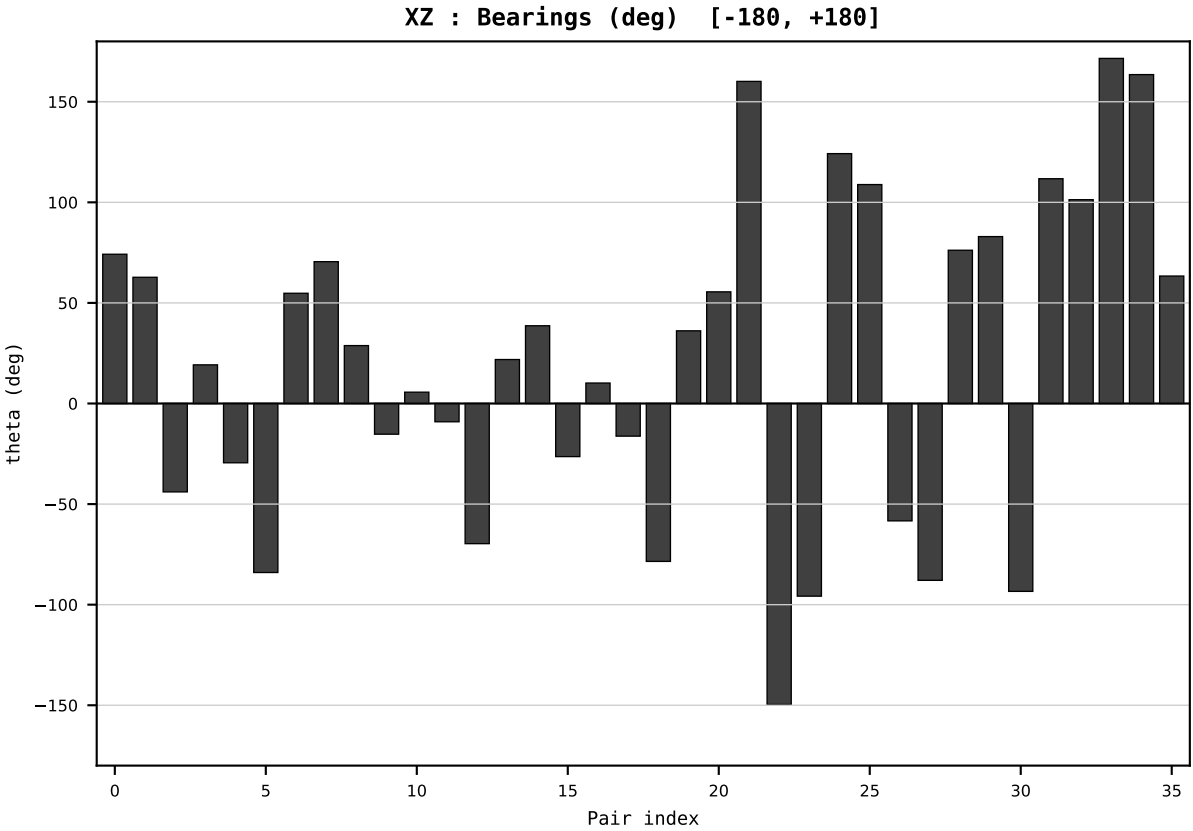
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.168193
Ring = Hs-2
E_metric= 31.9204
kappa_HS= 856.81
omega = 7.3067 deg
d_A = 0.721749
Helm = Nuclear
Helm d = -0.608539
DR = 6.753 HLR
DR ratio= 856.8x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



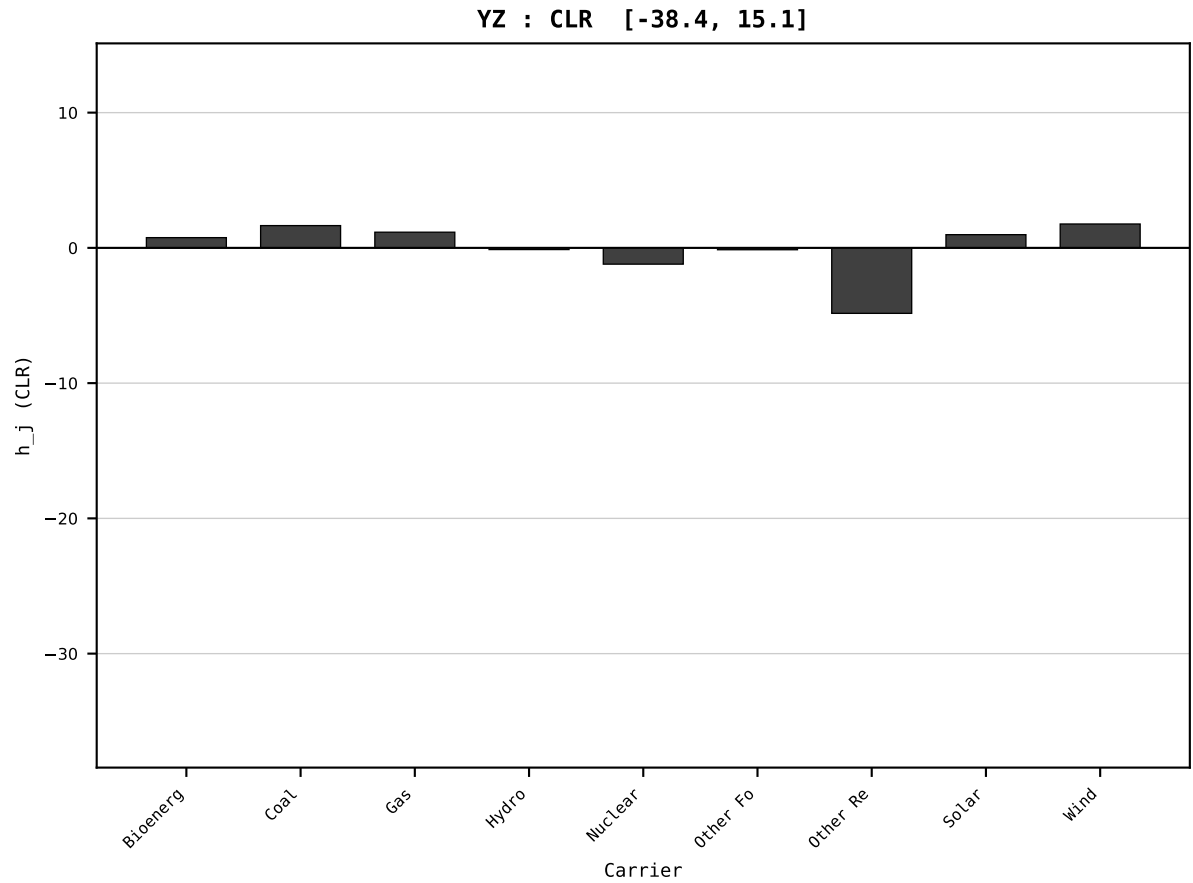
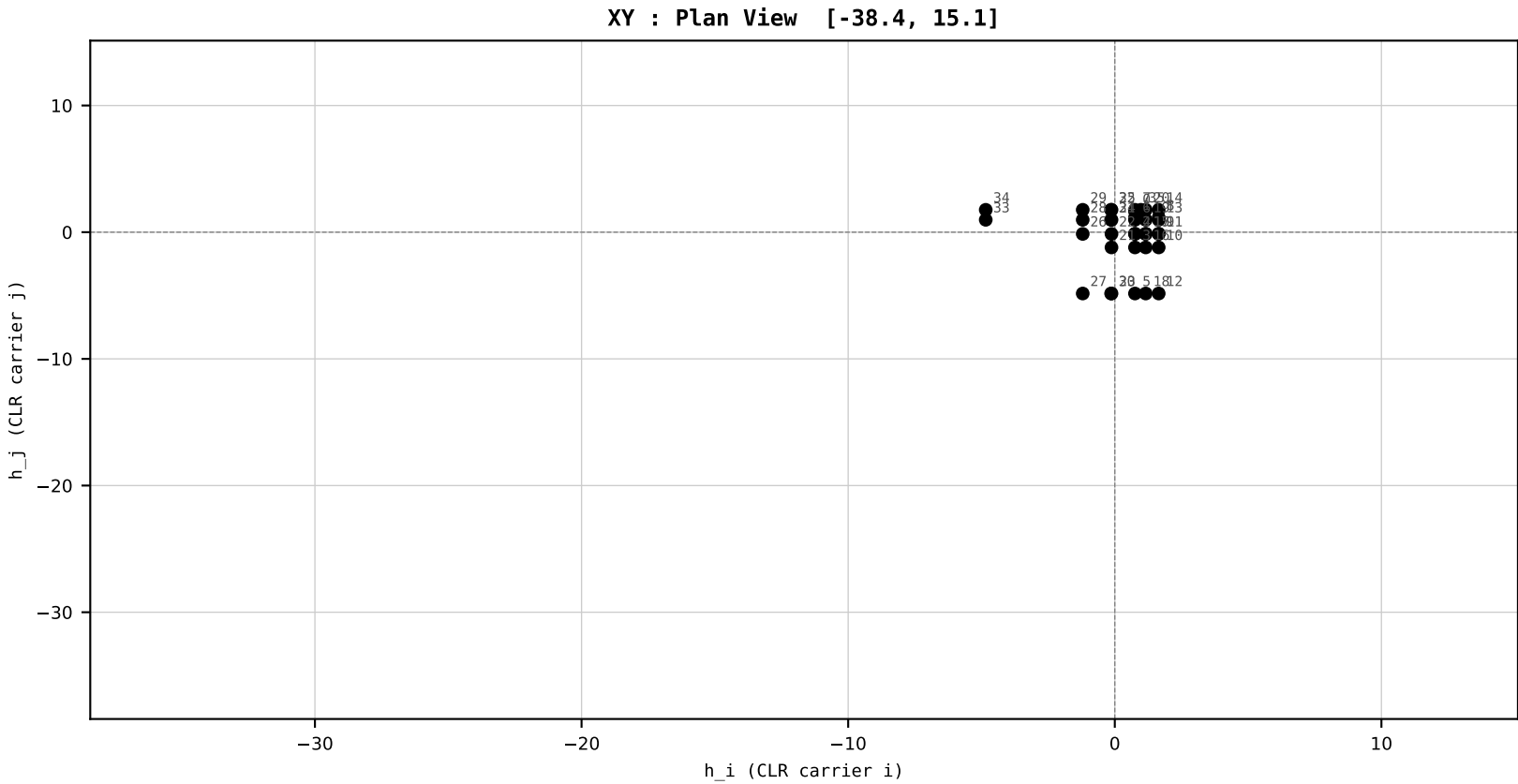
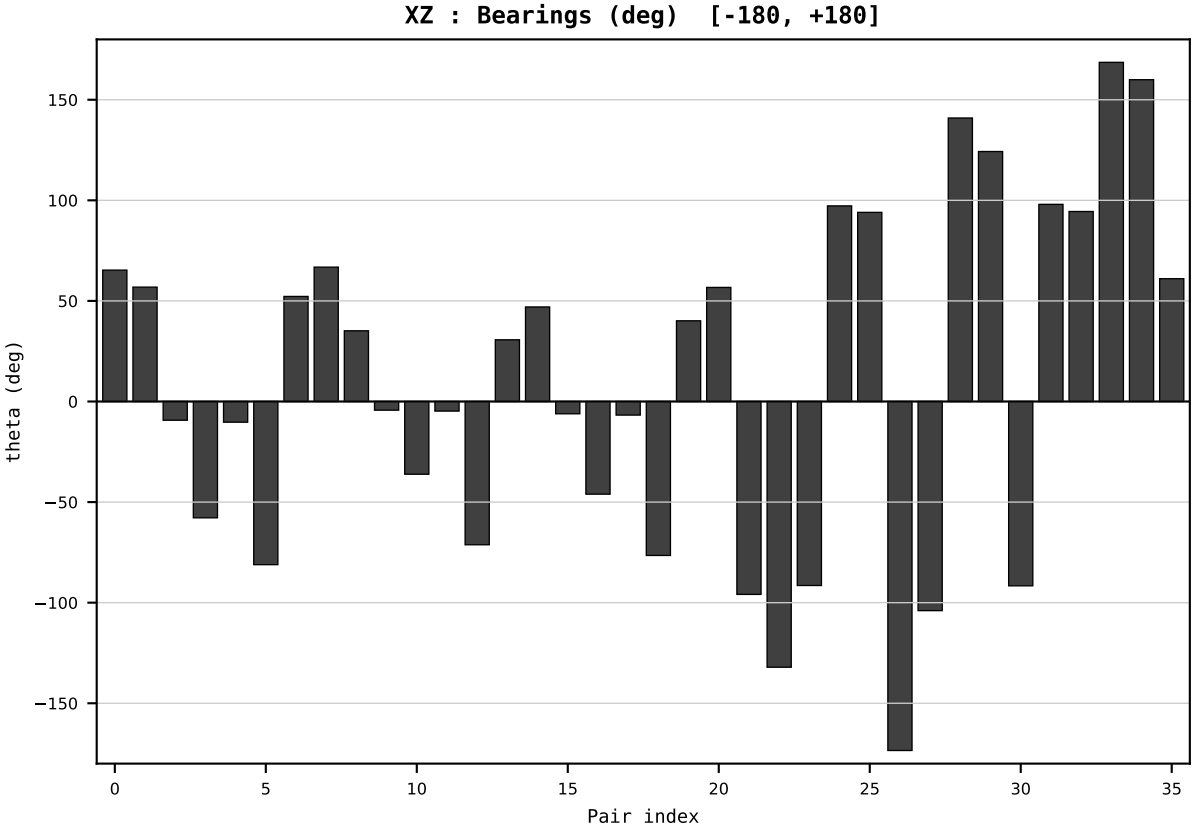
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.177336
Ring = Hs-2
E_metric= 33.6139
kappa_HS= 739.68
omega = 15.3266 deg
d_A = 1.533573
Helm = Nuclear
Helm d = -1.383172
DR = 6.606 HLR
DR ratio= 739.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

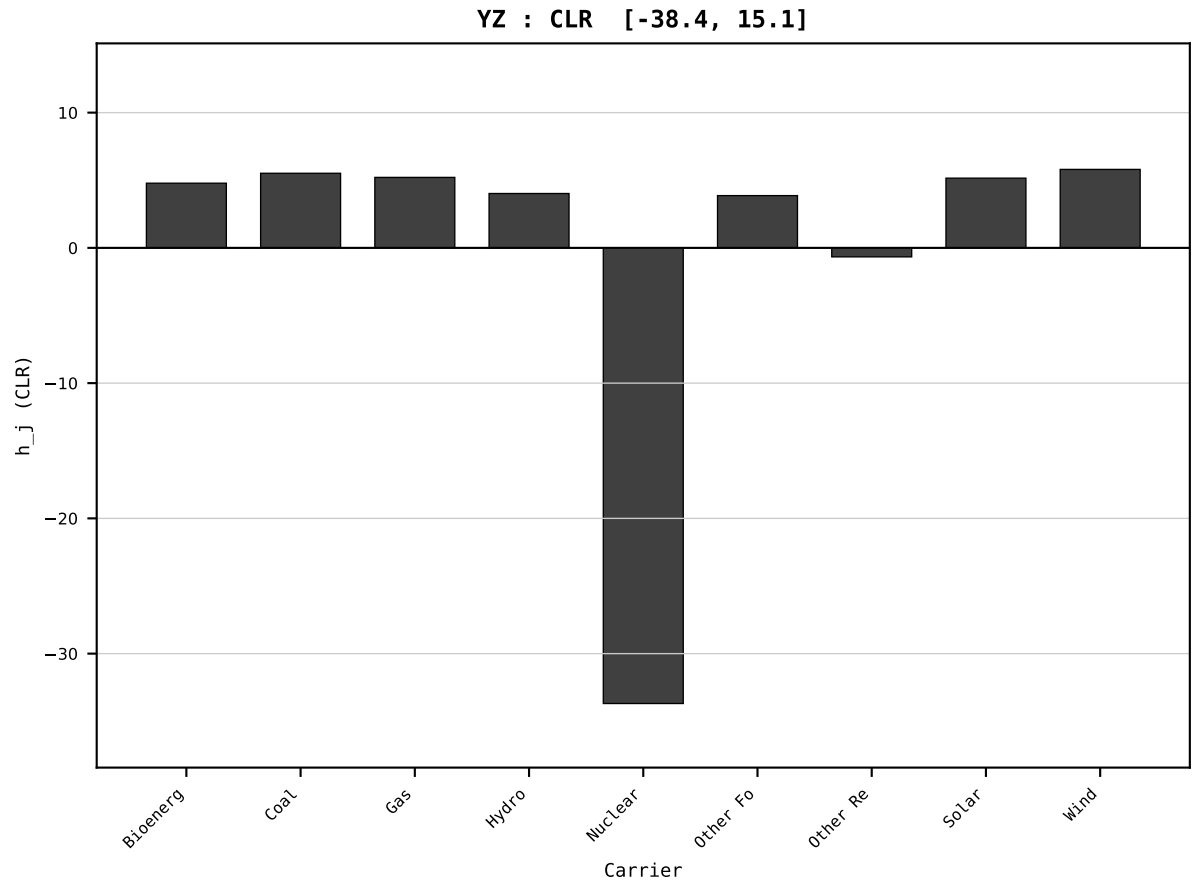
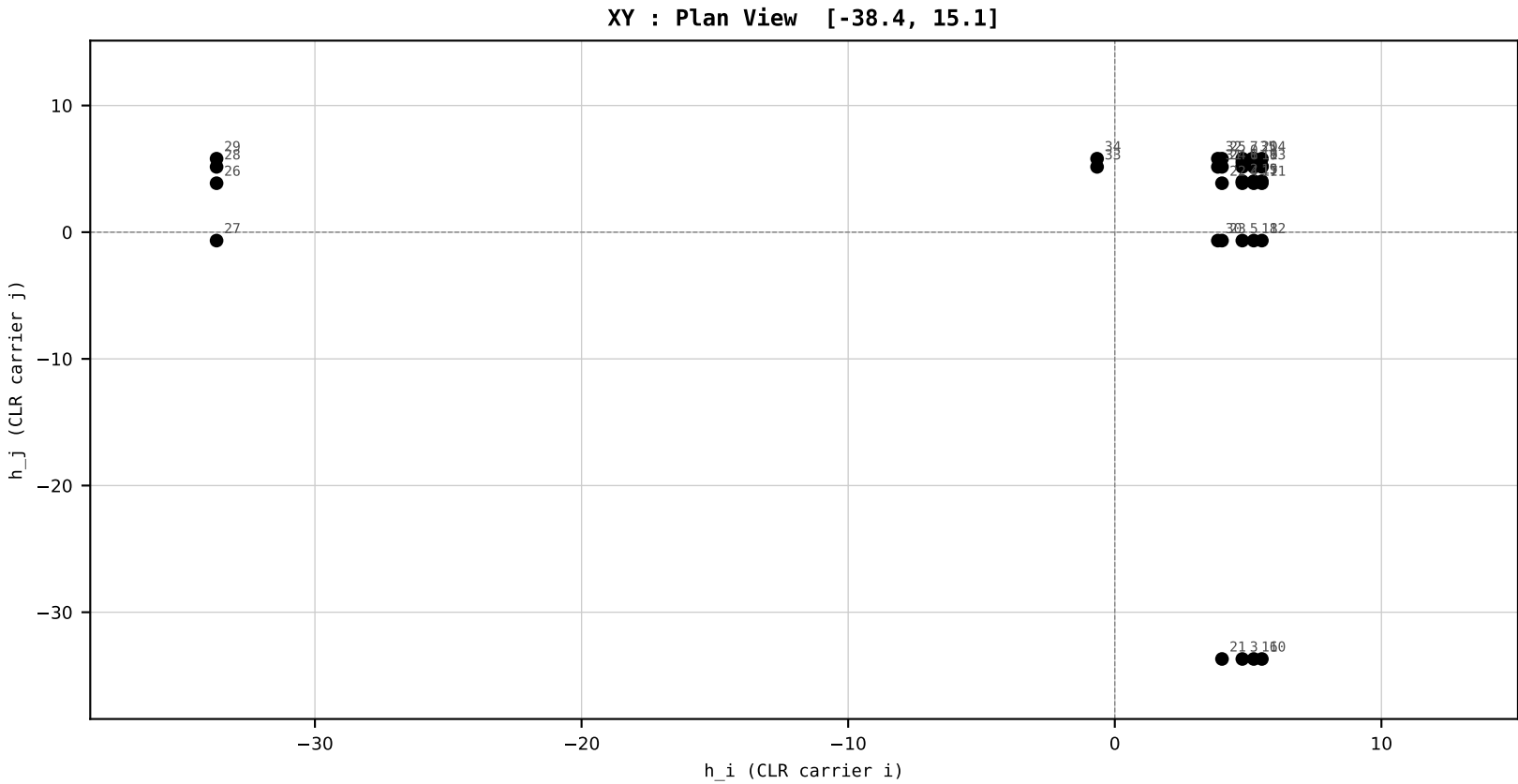
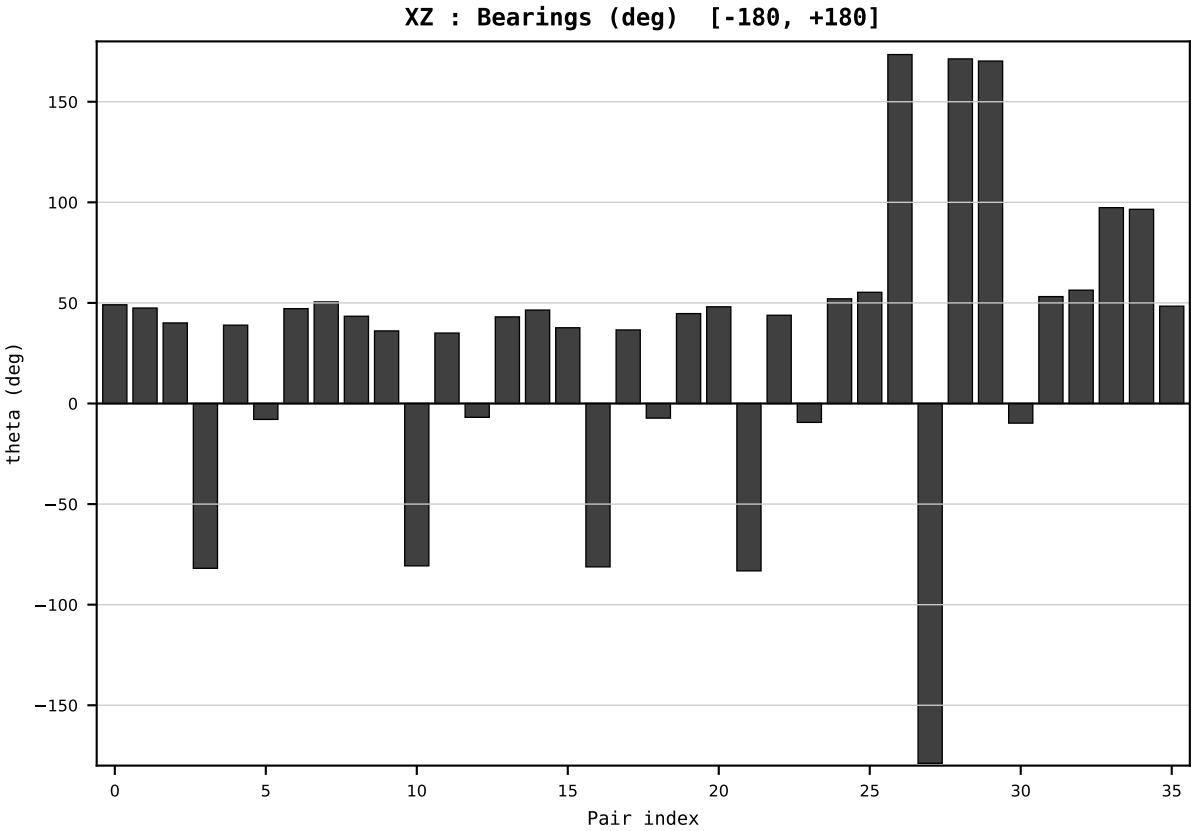
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.190666
Ring = Hs-2
E_metric= 923.2696
kappa_HS= 14159999999999984.00
omega = 68.5237 deg
d_A = 34.457790
Helm = Nuclear
Helm d = -32.486075
DR = 39.492 HLR
DR ratio= 141599999126284592.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

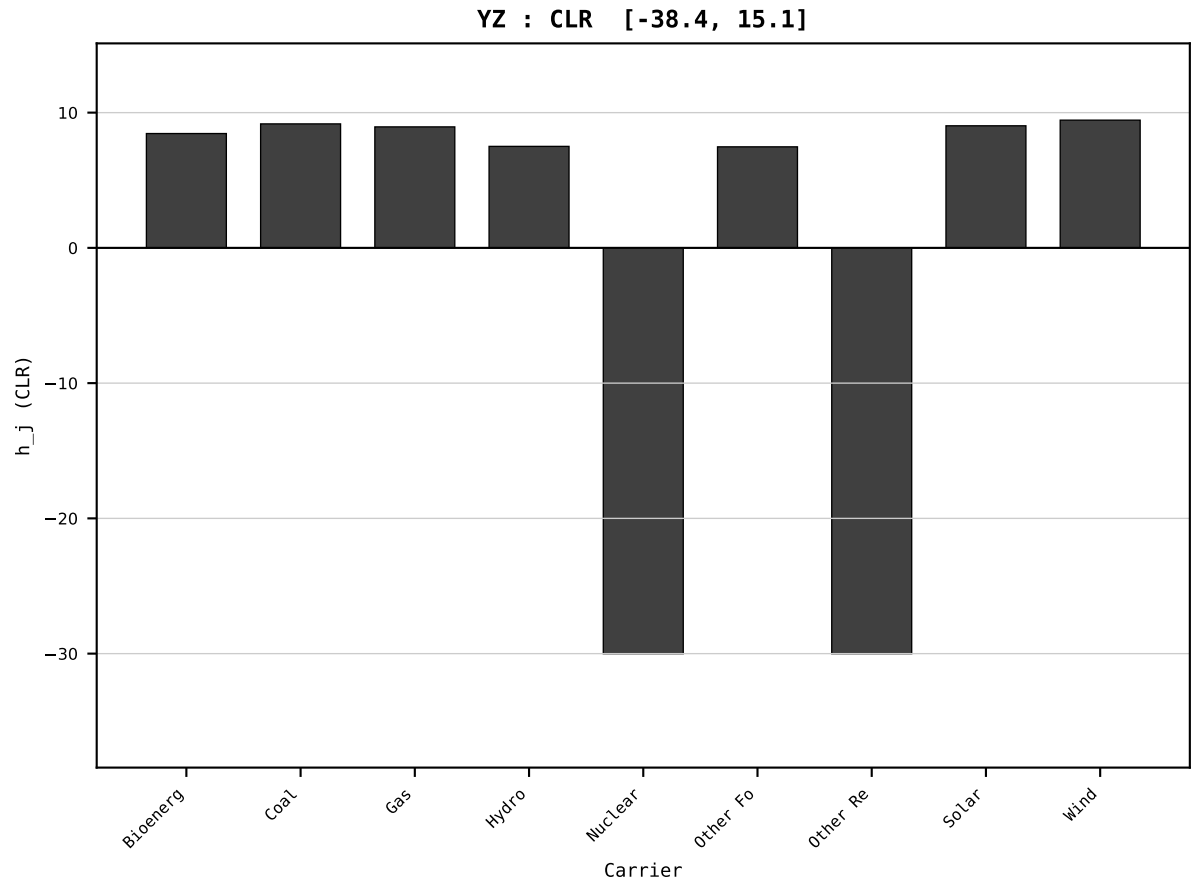
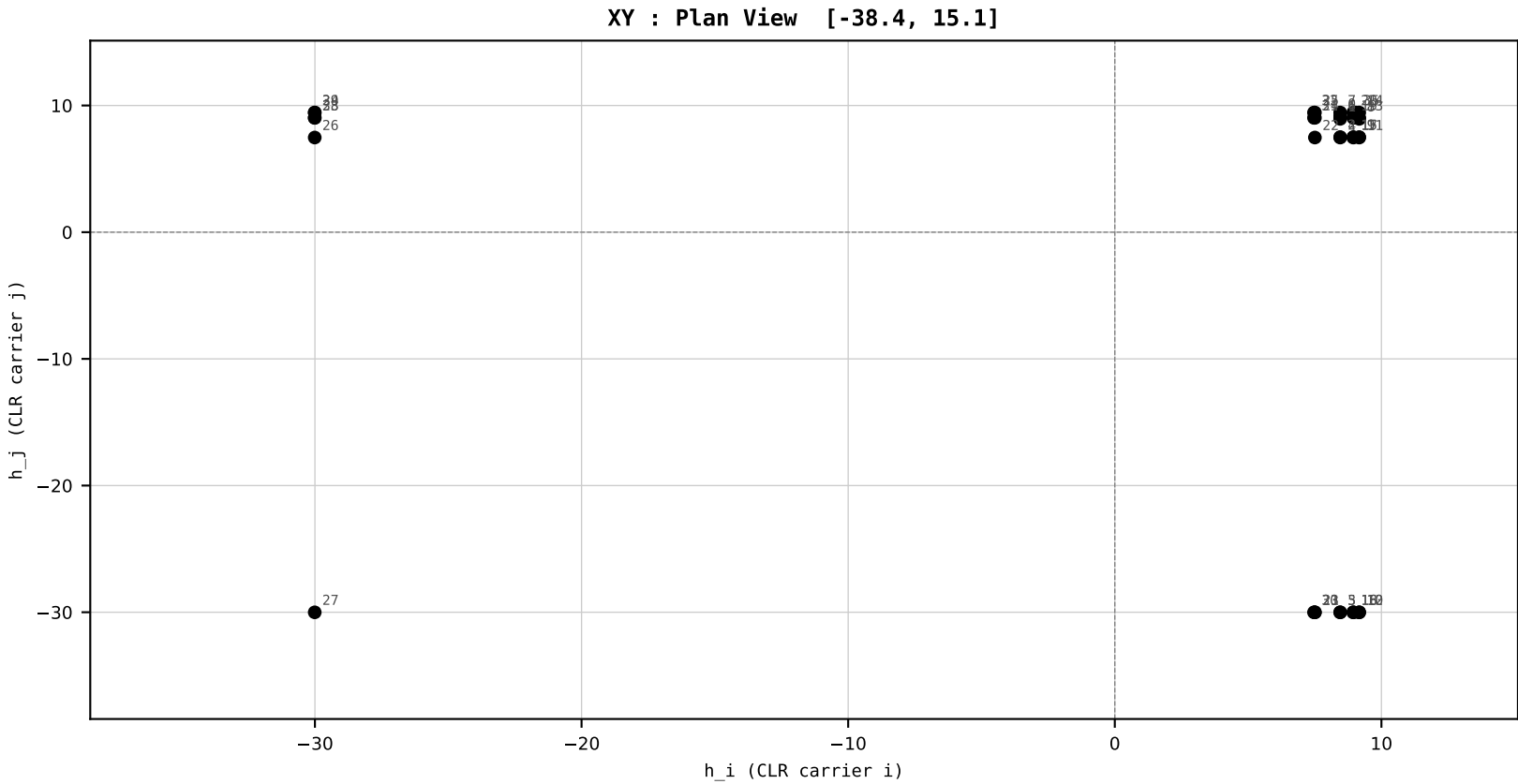
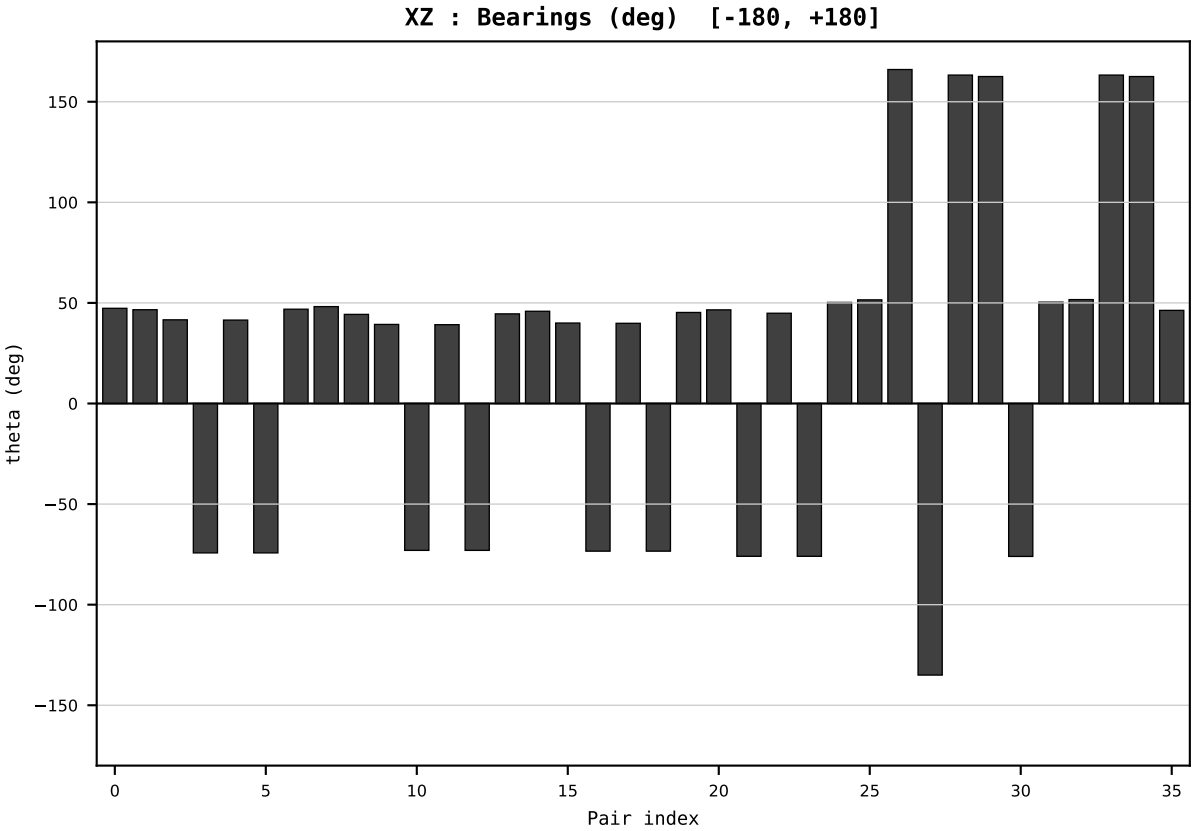
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

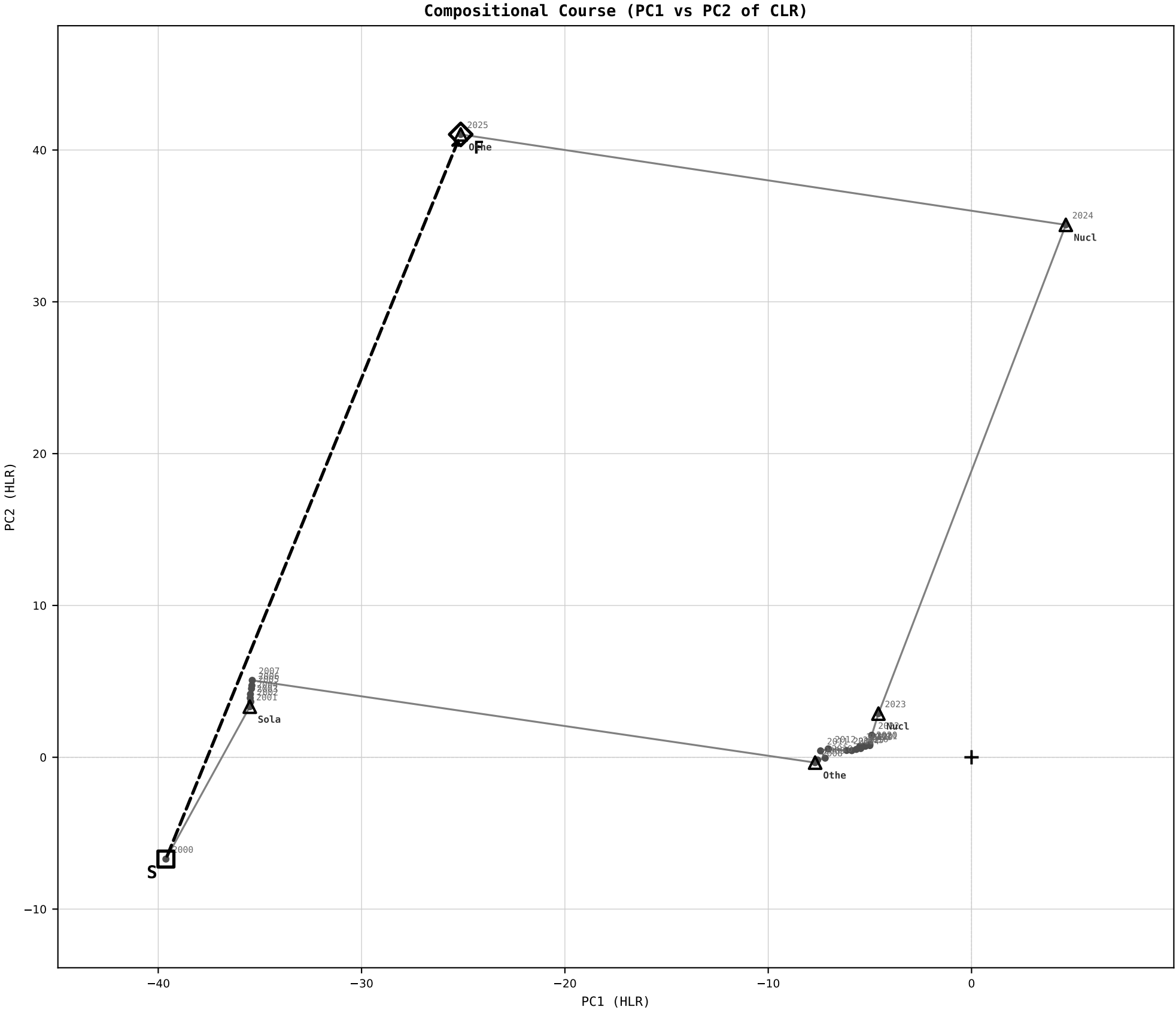
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.195789
Ring = Hs-2
E_metric= 1633.3021
kappa_HS= 13602999999999984.00
omega = 40.2530 deg
d_A = 31.124159
Helm = Other Renewables
Helm d = -29.342858
DR = 39.452 HLR
DR ratio= 136030000740543728.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow \text{PC projection}$
Unit: HLR (Higgins Log-Ratio)

Net distance = 55.7717 HLR
Path length = 136.5882 HLR
Directness = 0.4083

DR start = 40.231 HLR
ratio = 296679999472192320.0x
DR final = 39.452 HLR
ratio = 136030000740543728.0x

E_metric start = 2262.9463
E_metric final = 2319.2562

PC1 loadings:
Other Re +0.903
Nuclear -0.248
Coal -0.157

PC2 loadings:
Nuclear -0.881
Solar +0.310
Other Re -0.180

Top displacement events:
2024 Nuclear d=34.458
2025 Other Re d=31.124
2001 Solar d=30.338
2008 Other Re d=28.812
2023 Nuclear d=1.534

The instrument reads.
The expert decides.
The loop stays open.

View B

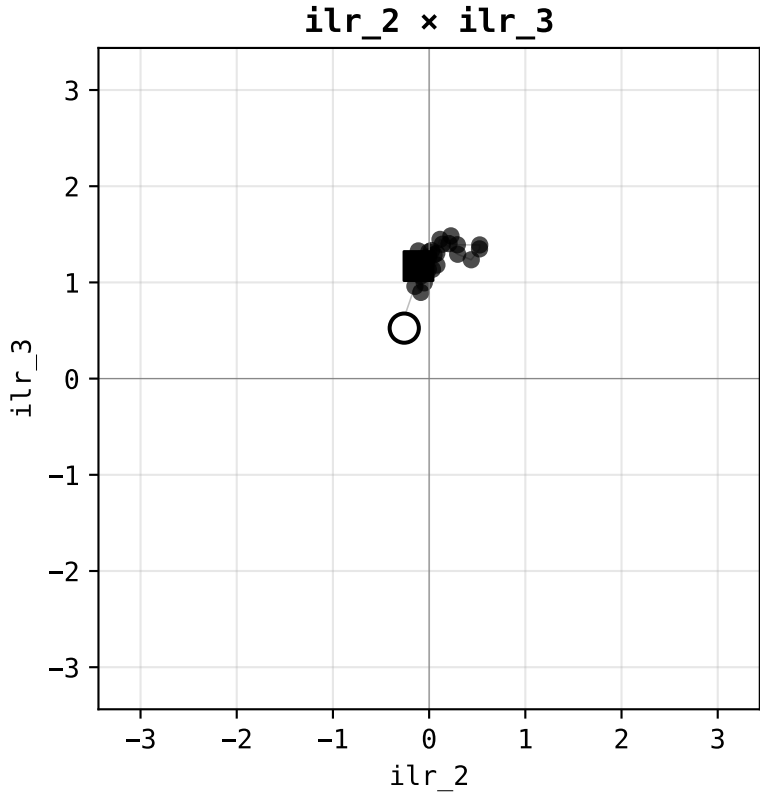
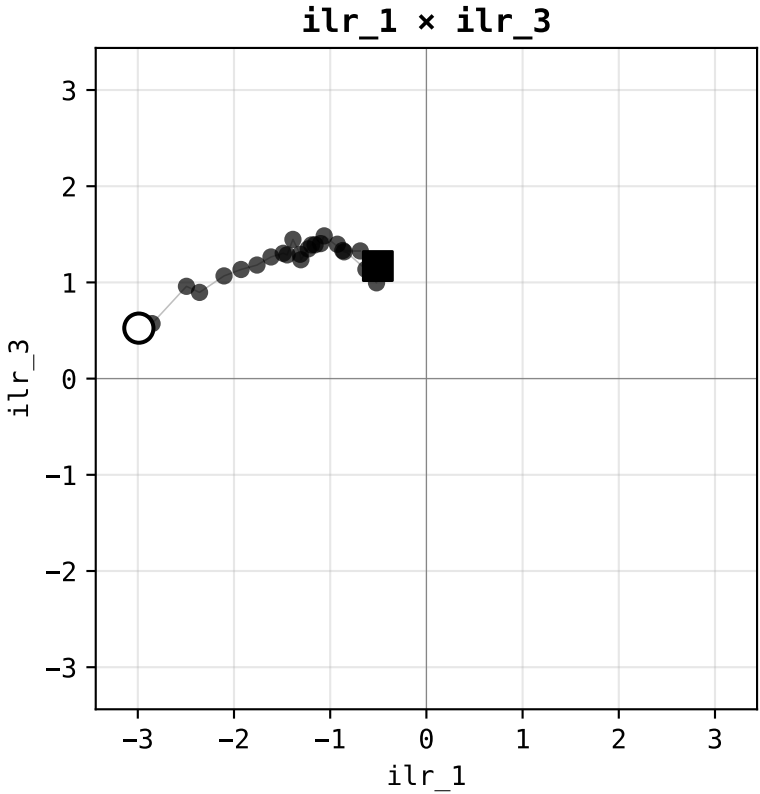
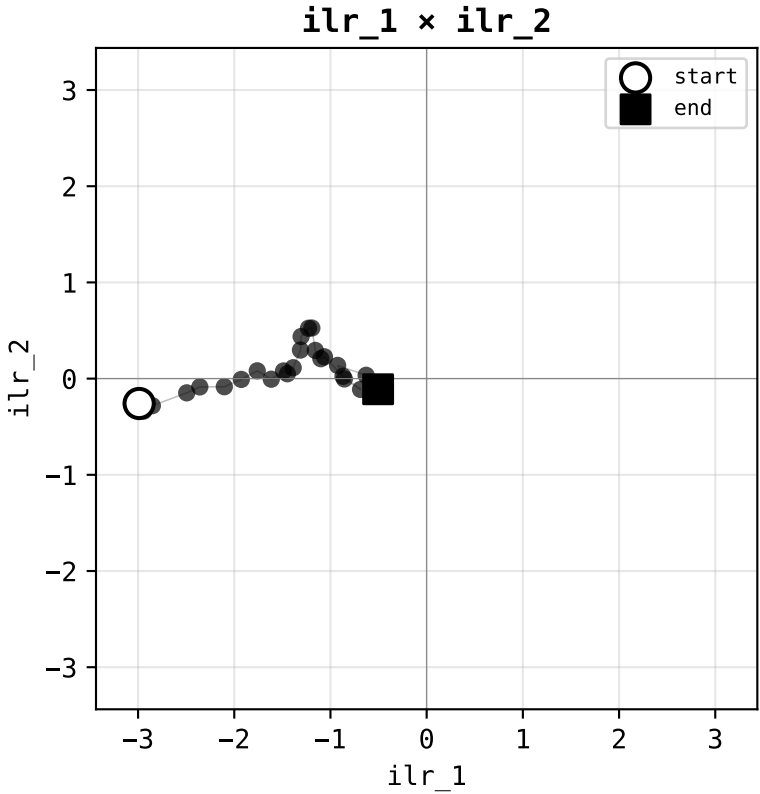
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ DEU Germany

ILR-Helmert Orthogonal Triplet – DEU Germany · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=26 years · D=9 carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:

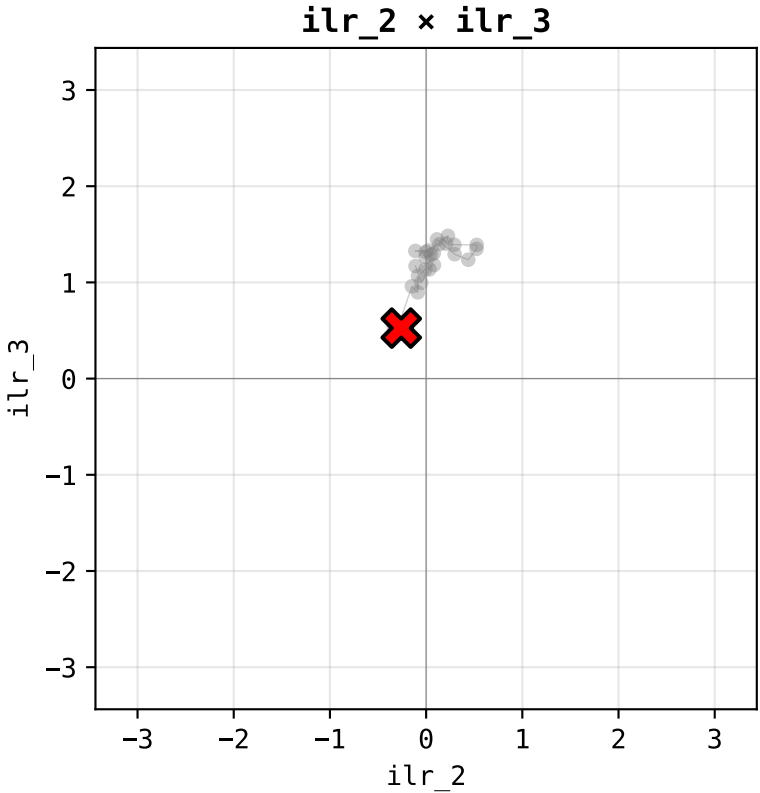
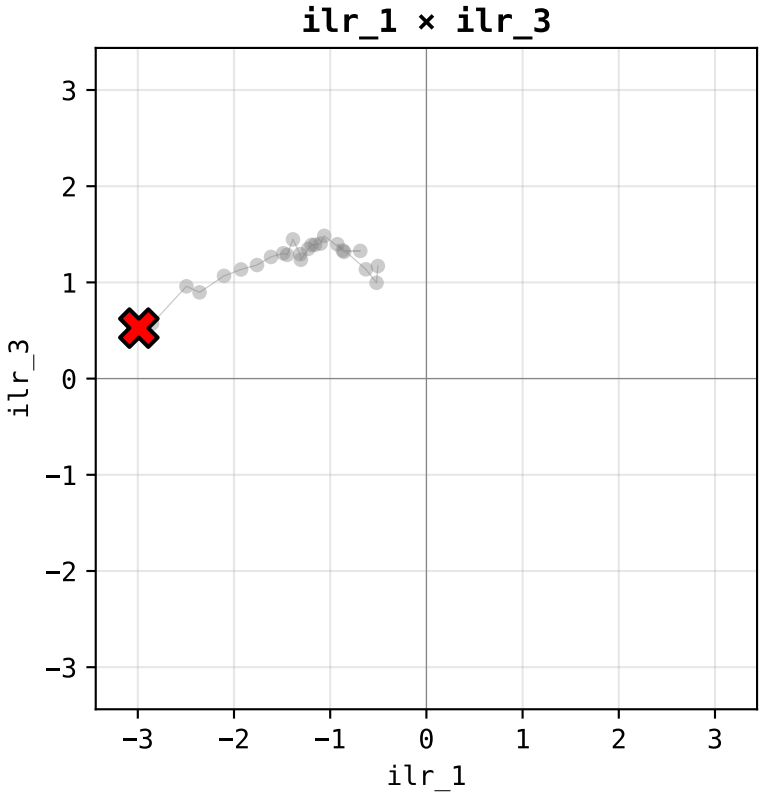
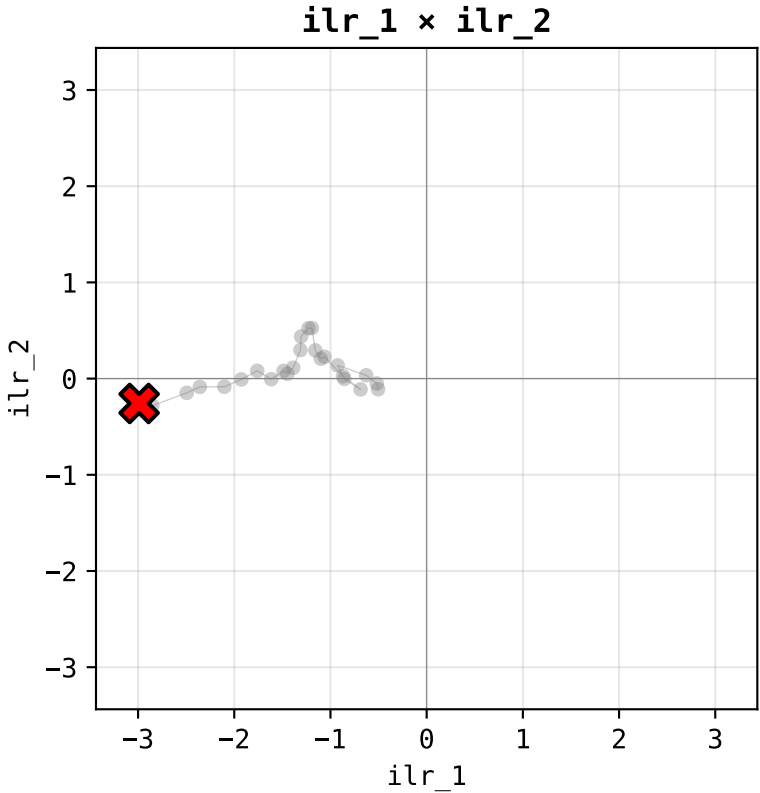
start year: 2000
end year: 2025
N readings: 26

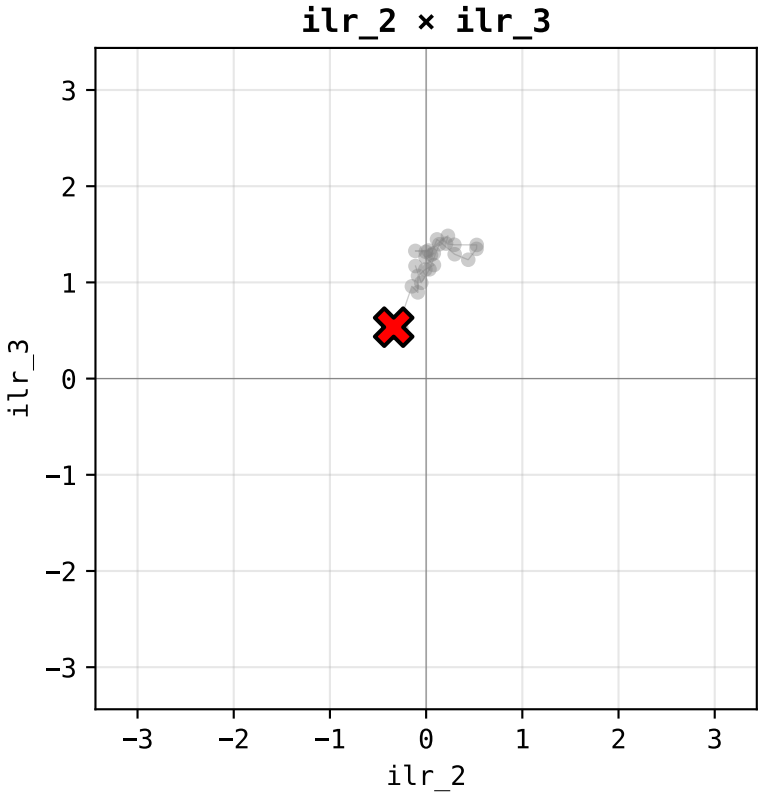
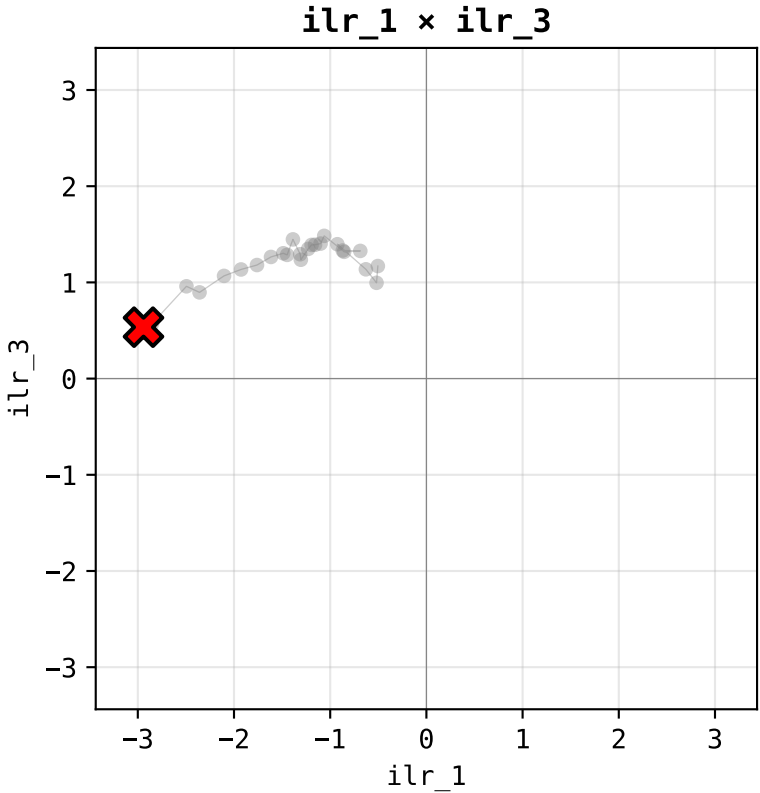
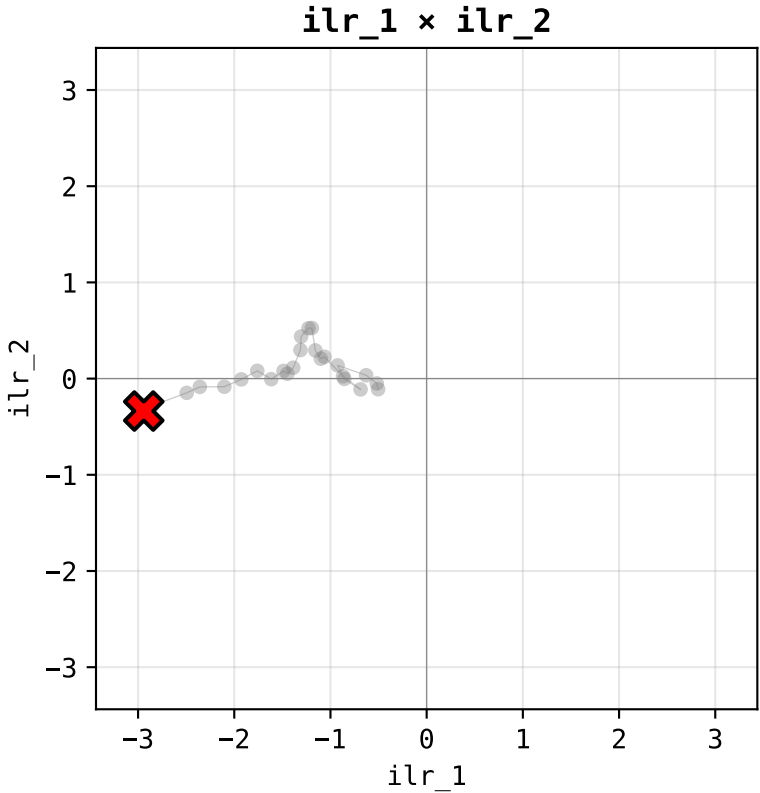
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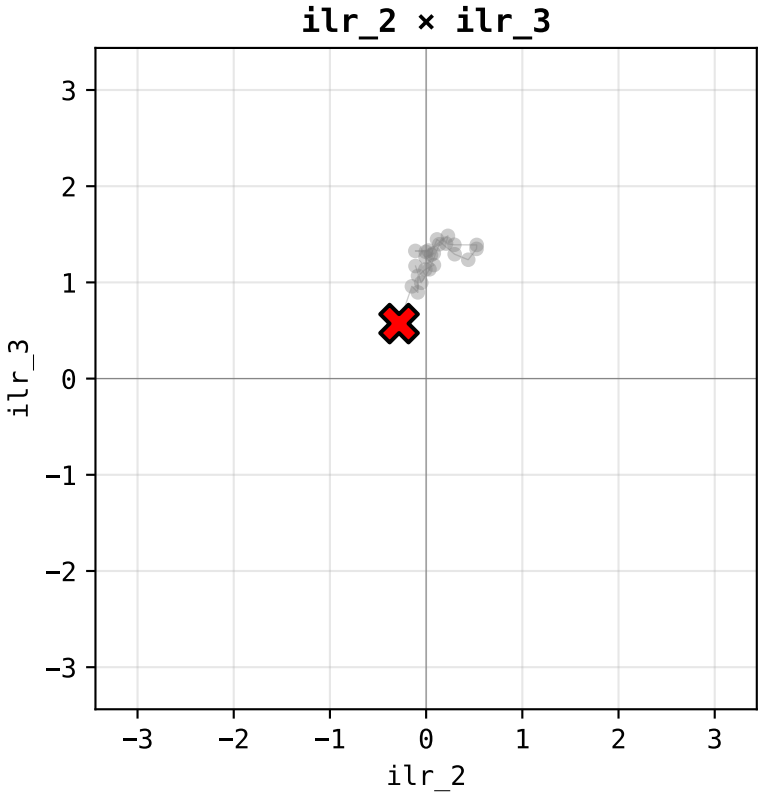
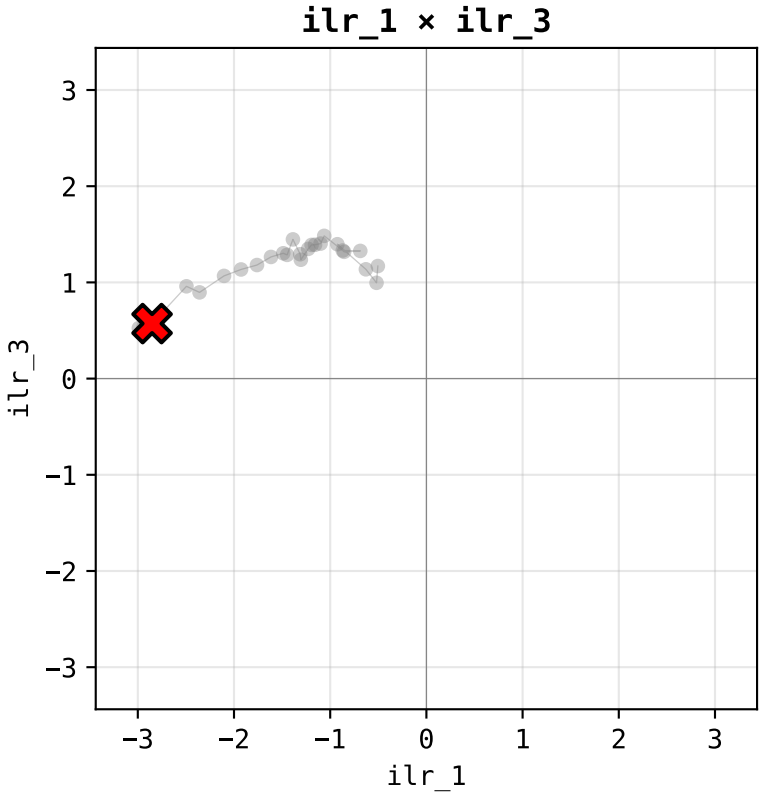
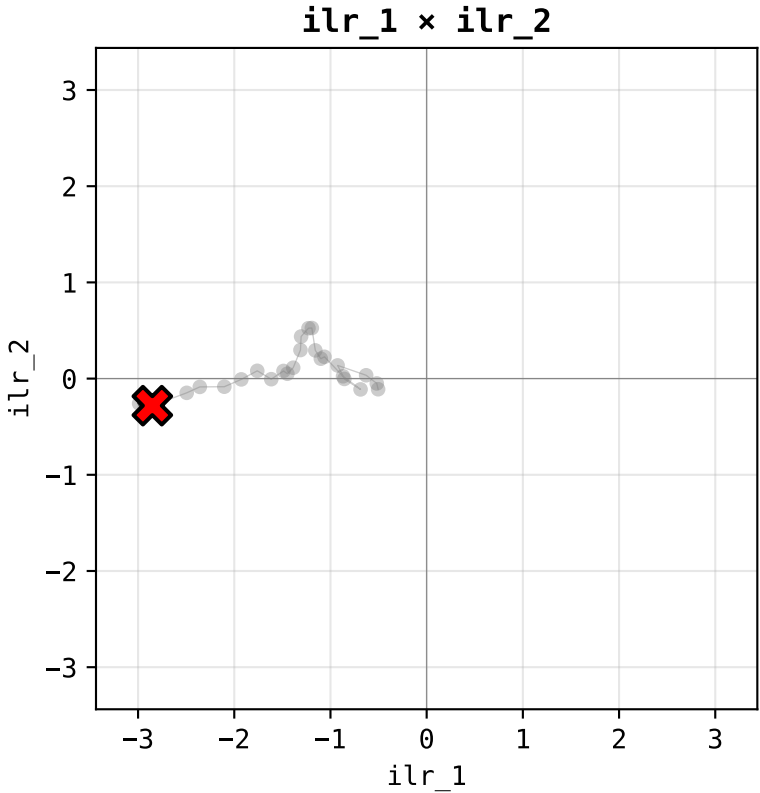
fixed scale ± 3.44 ILR
Order 1 (first principles)
per Output Doctrine v1.0

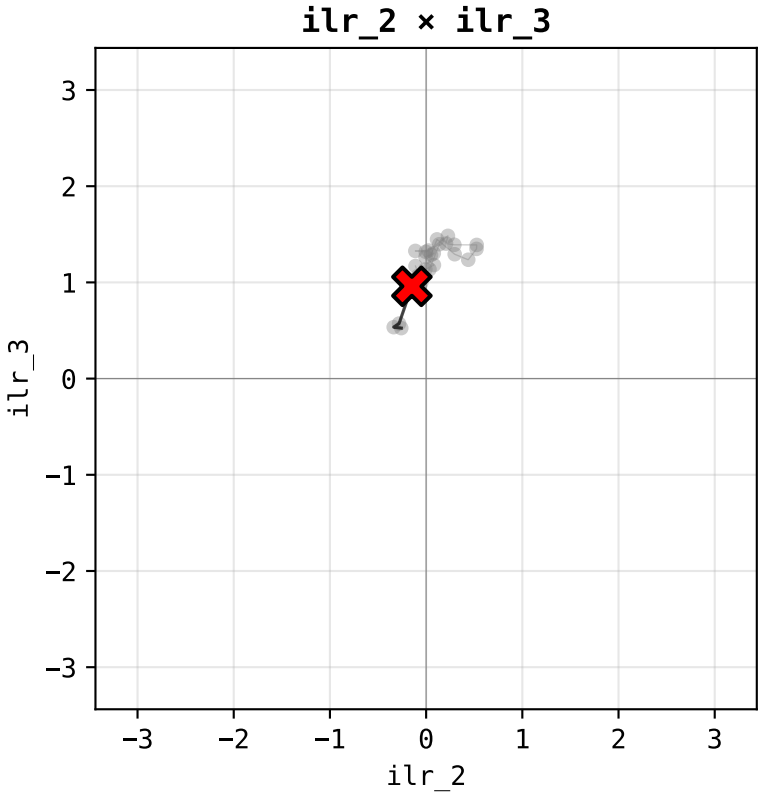
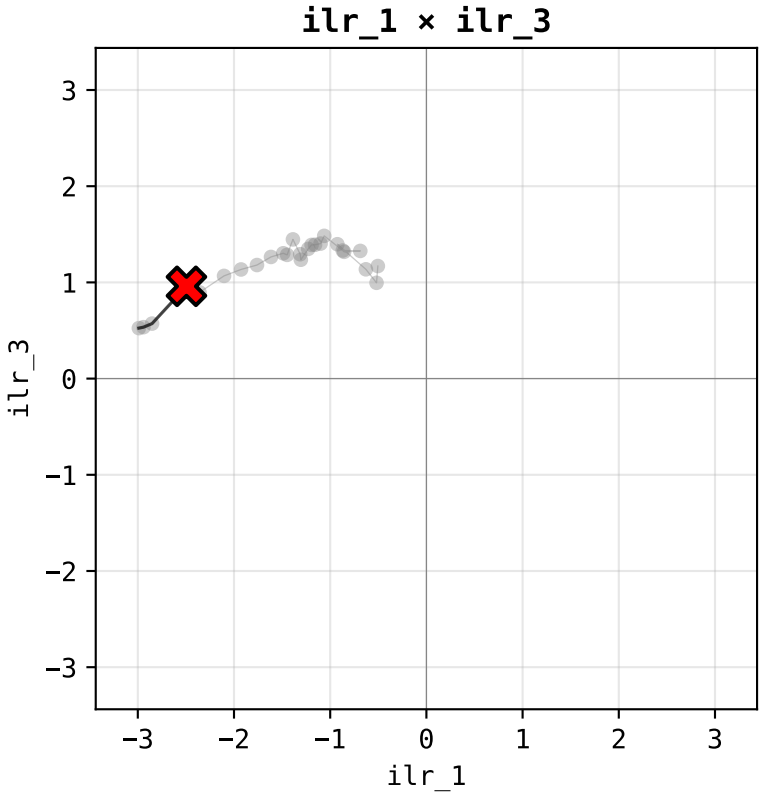
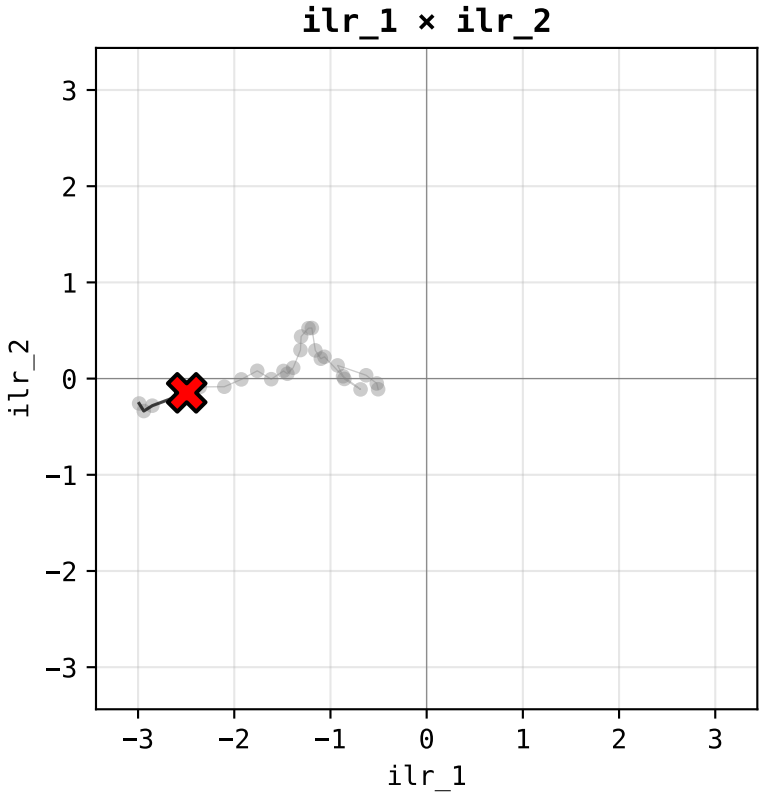
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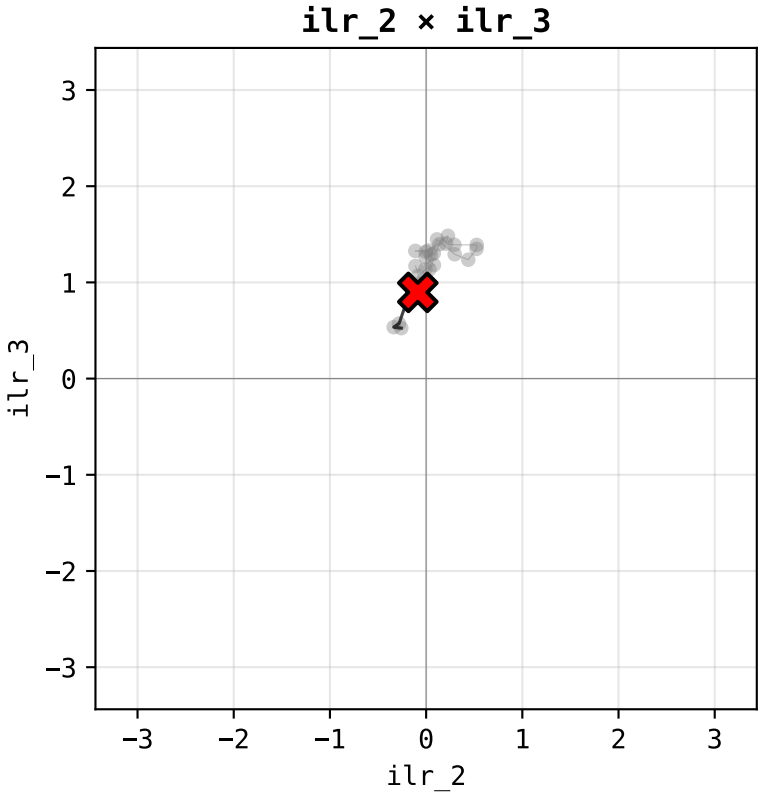
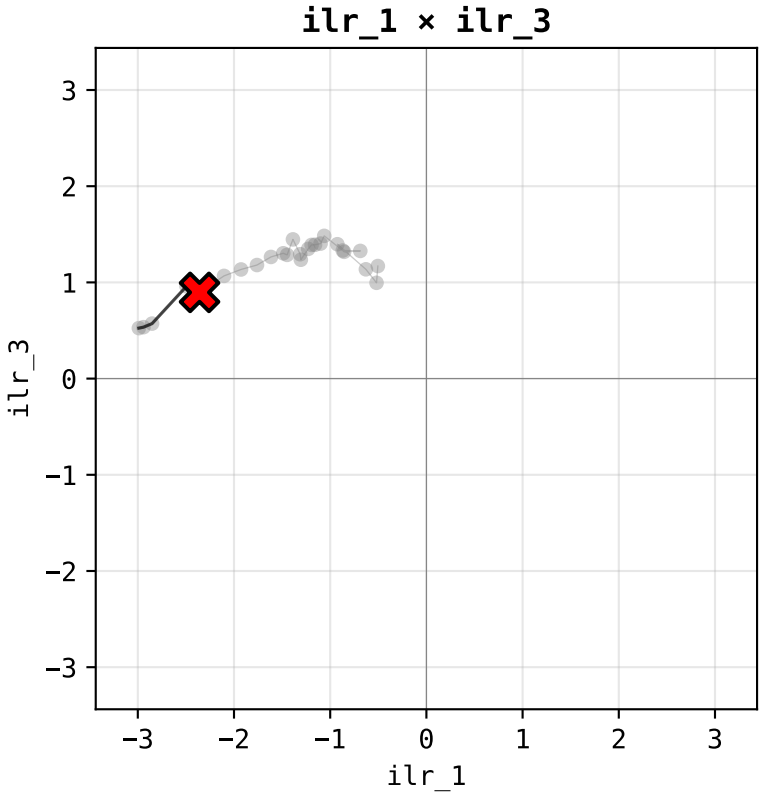
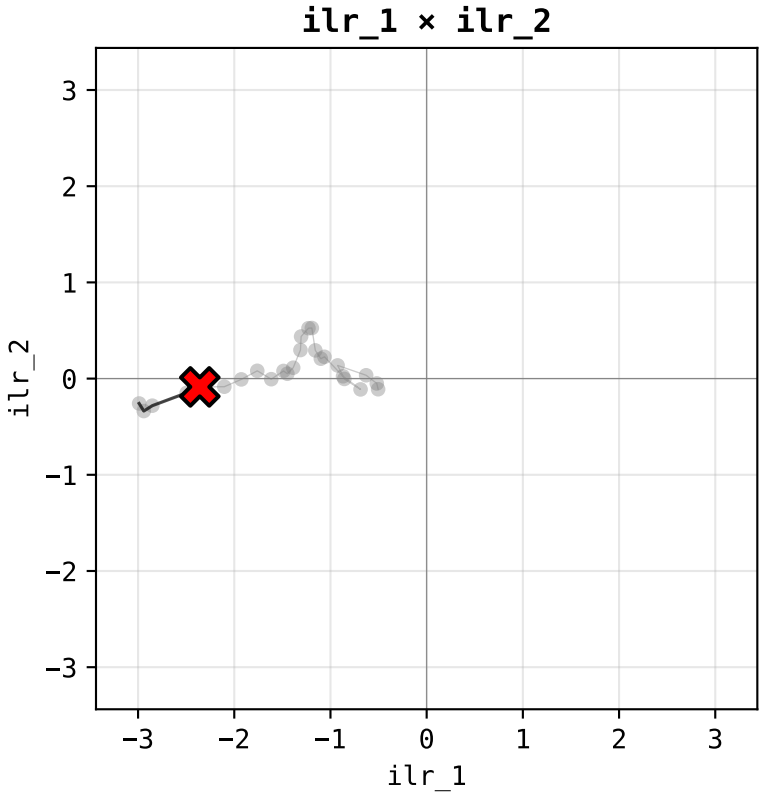
○ start · ■ end
· intermediate years

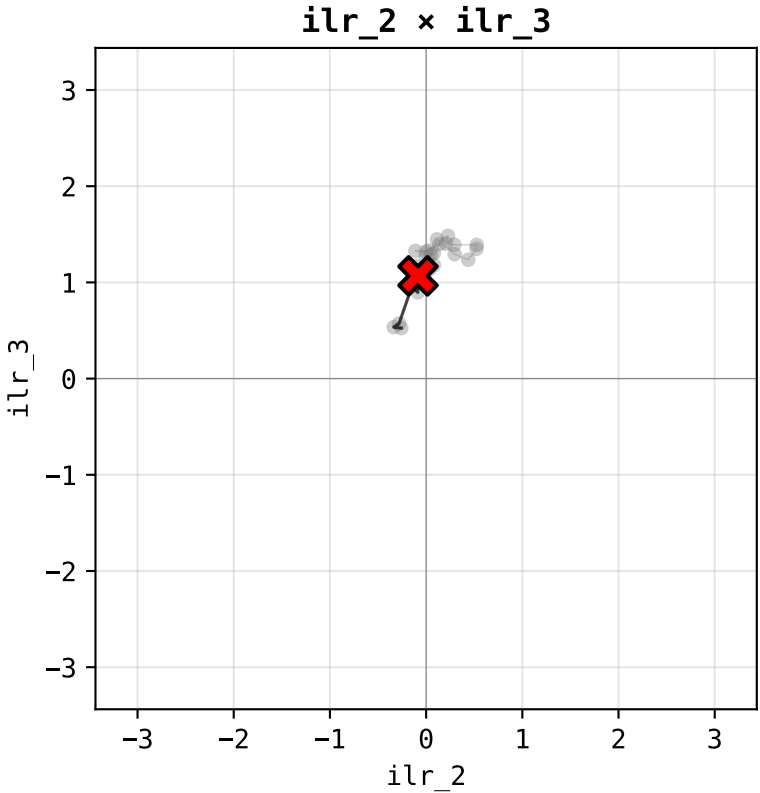
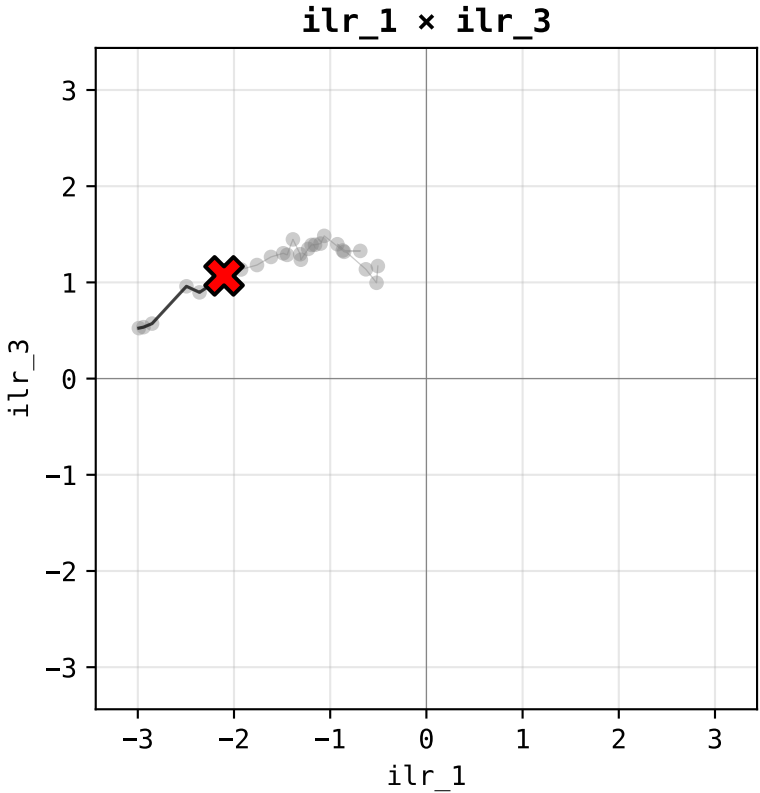
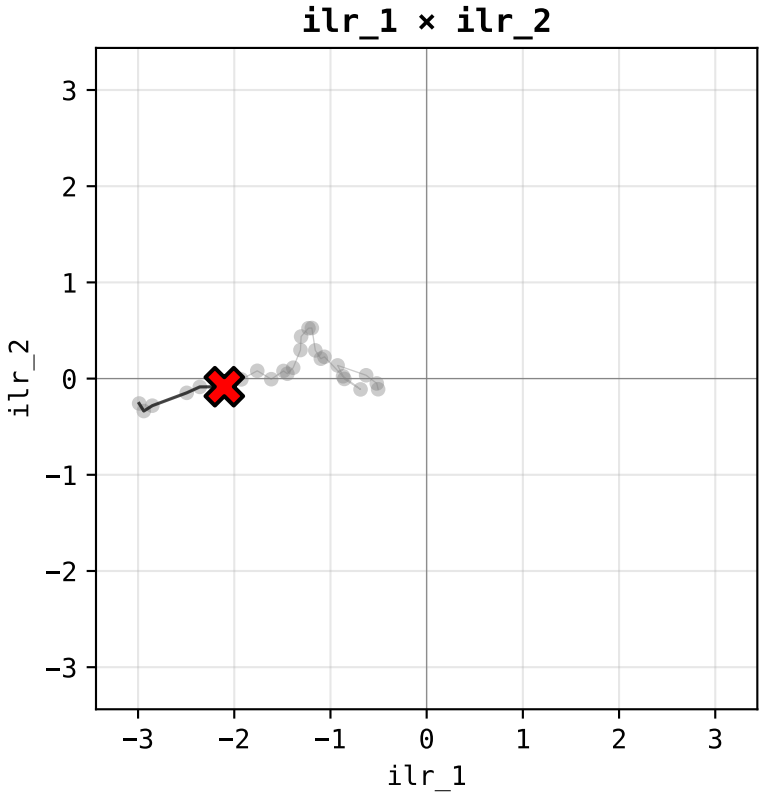


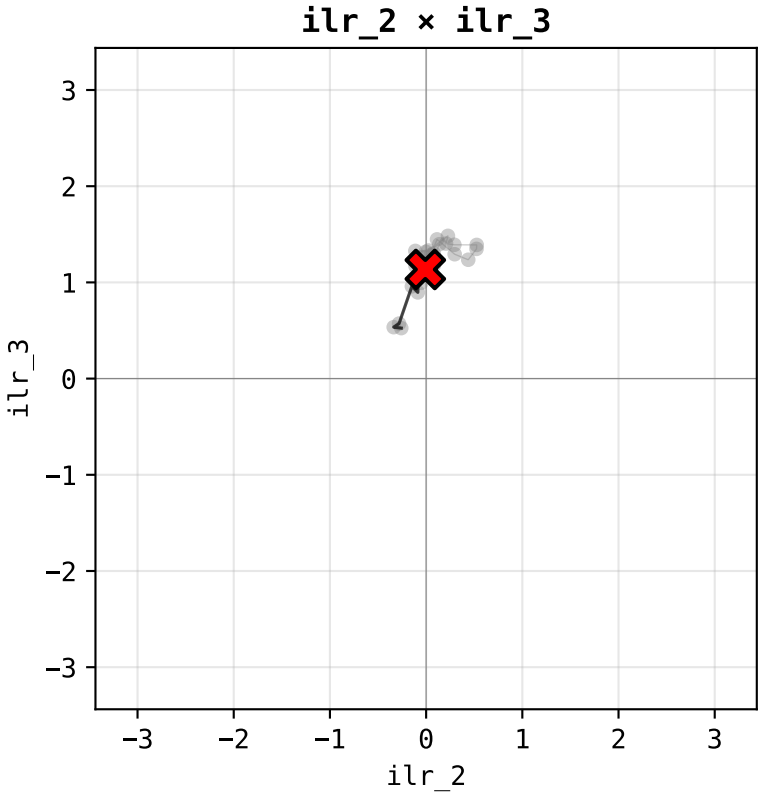
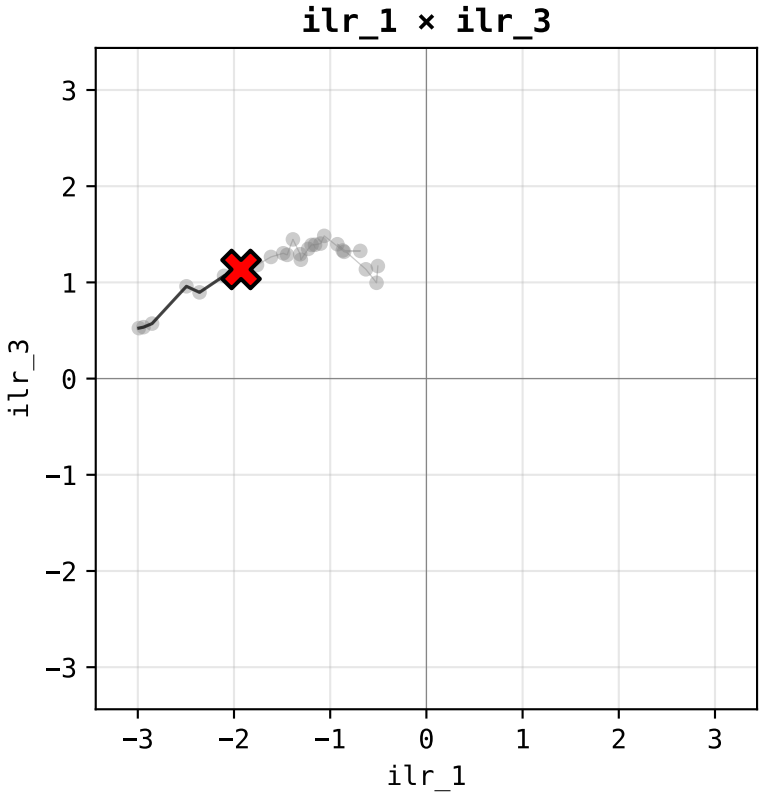
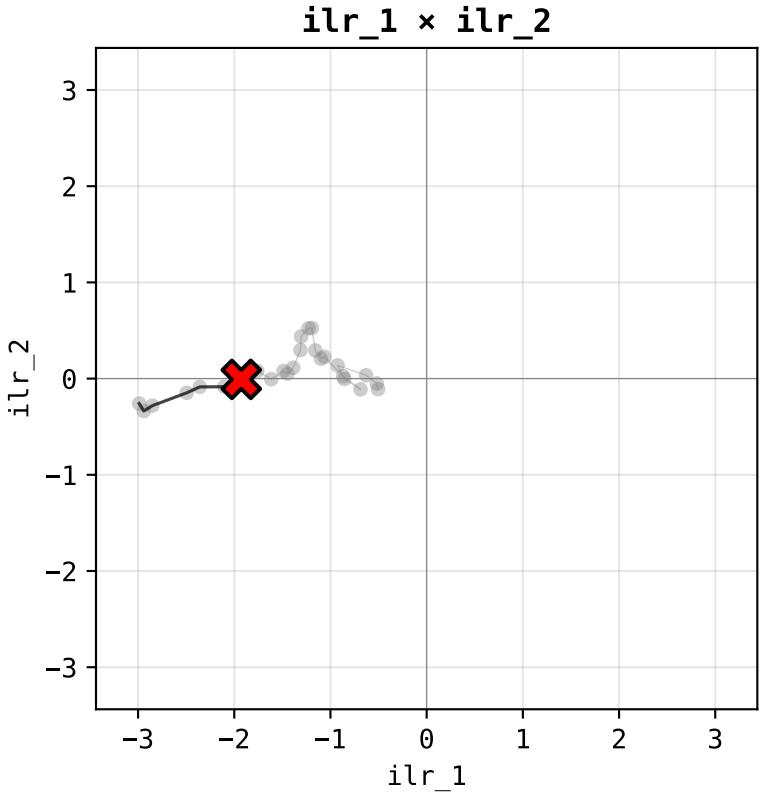


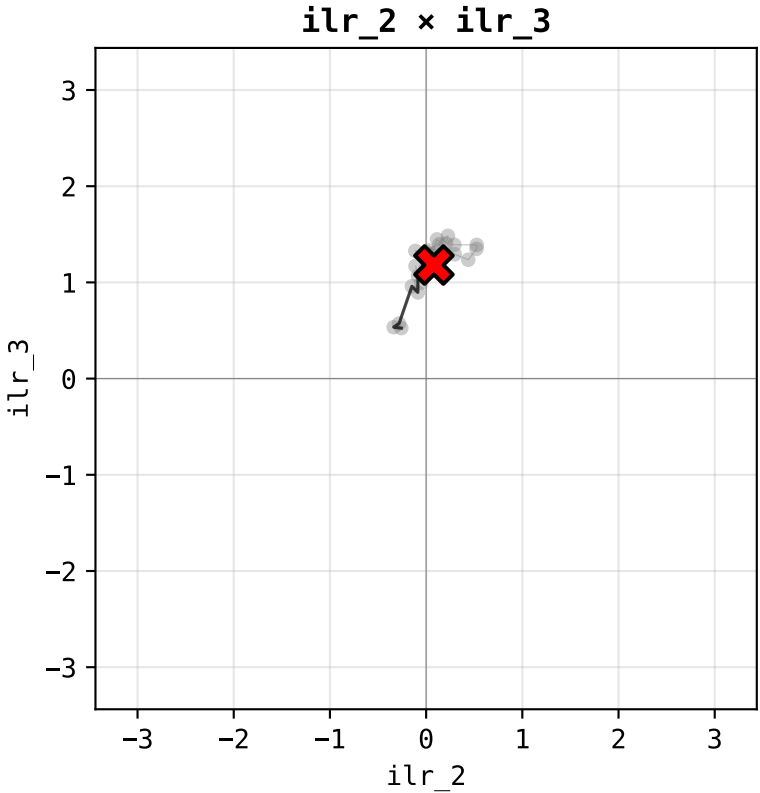
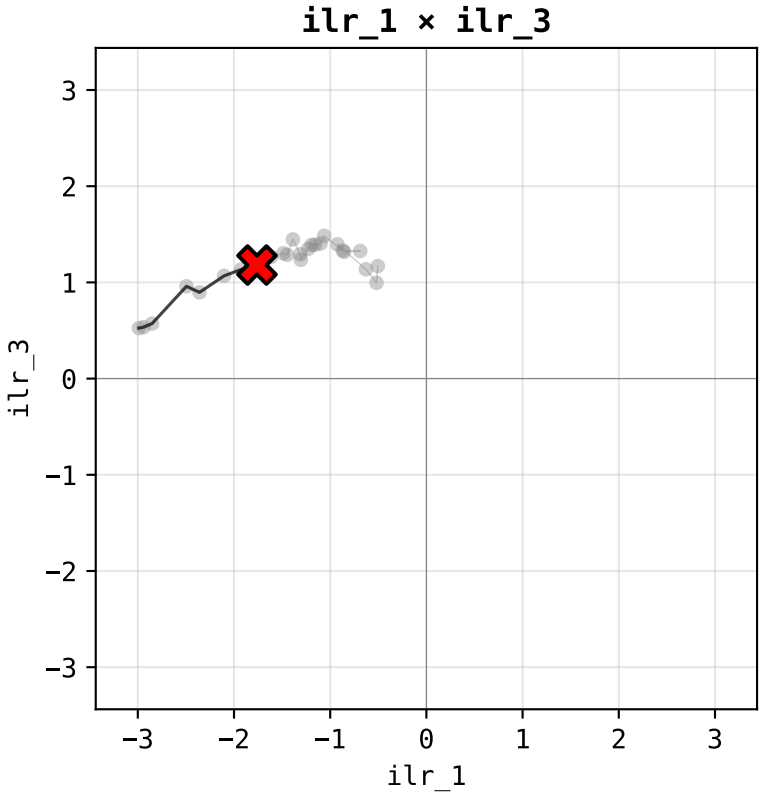
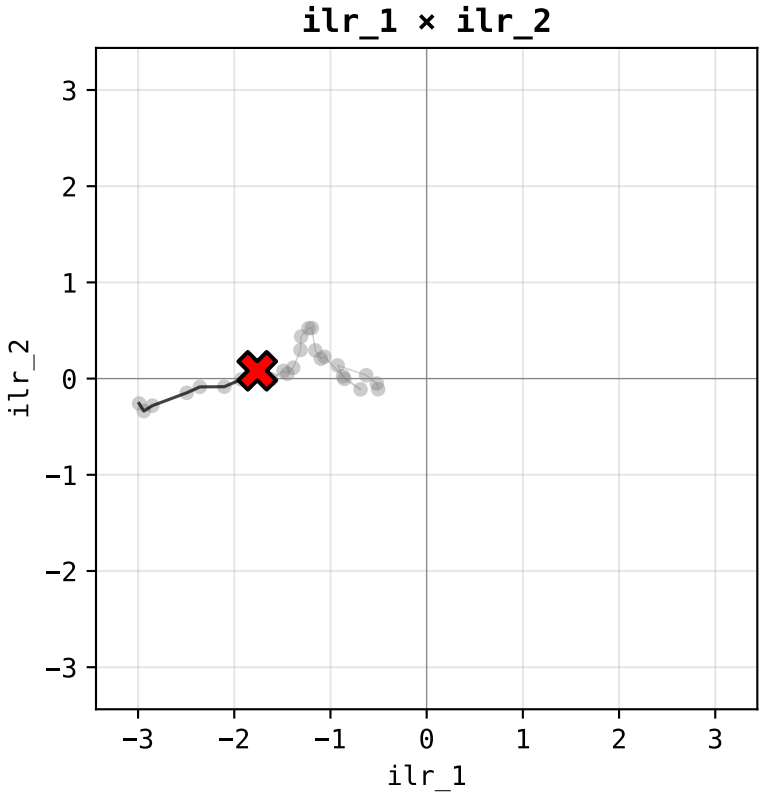


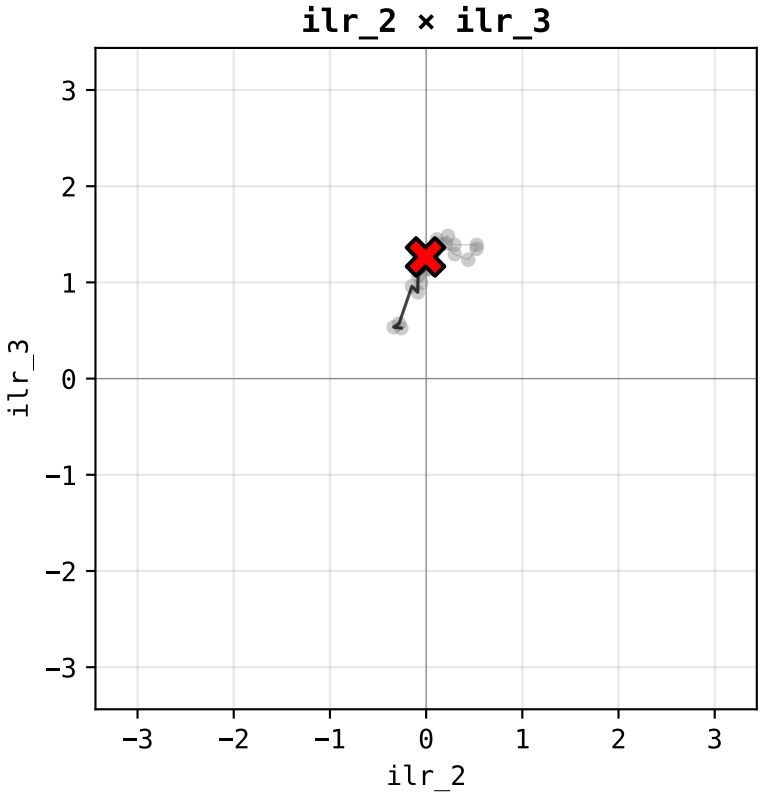
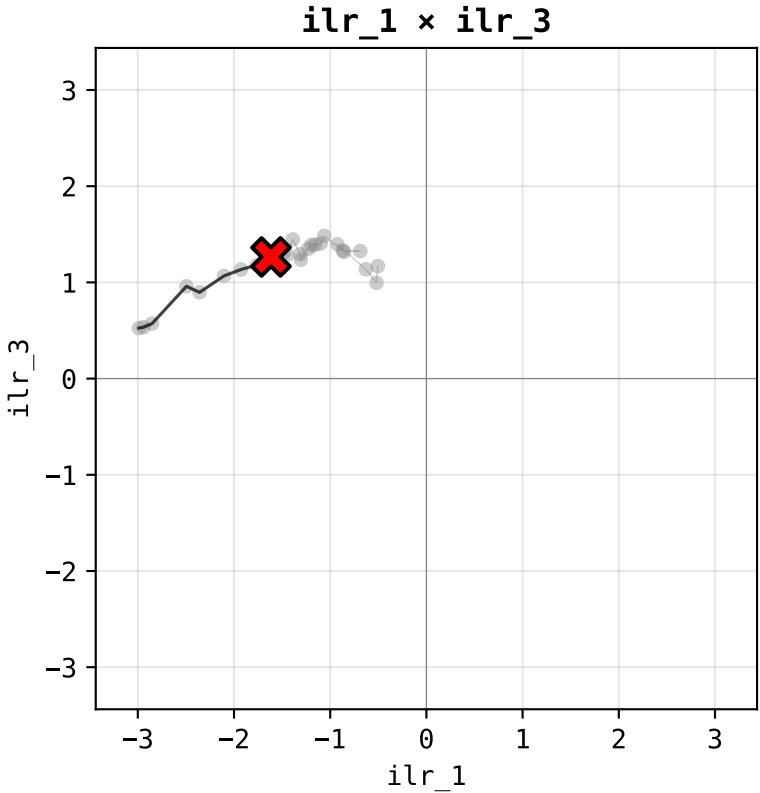
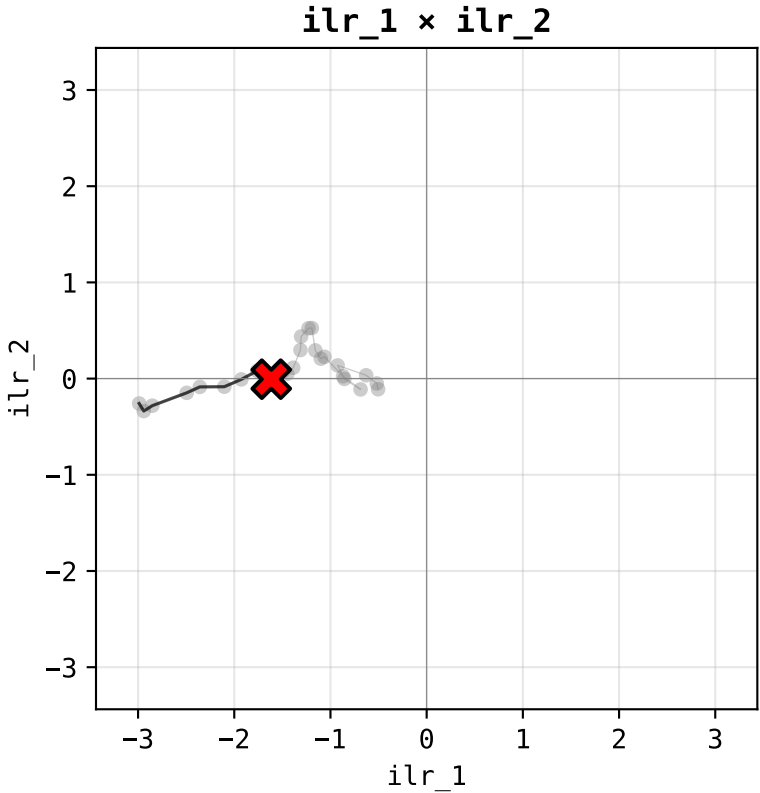


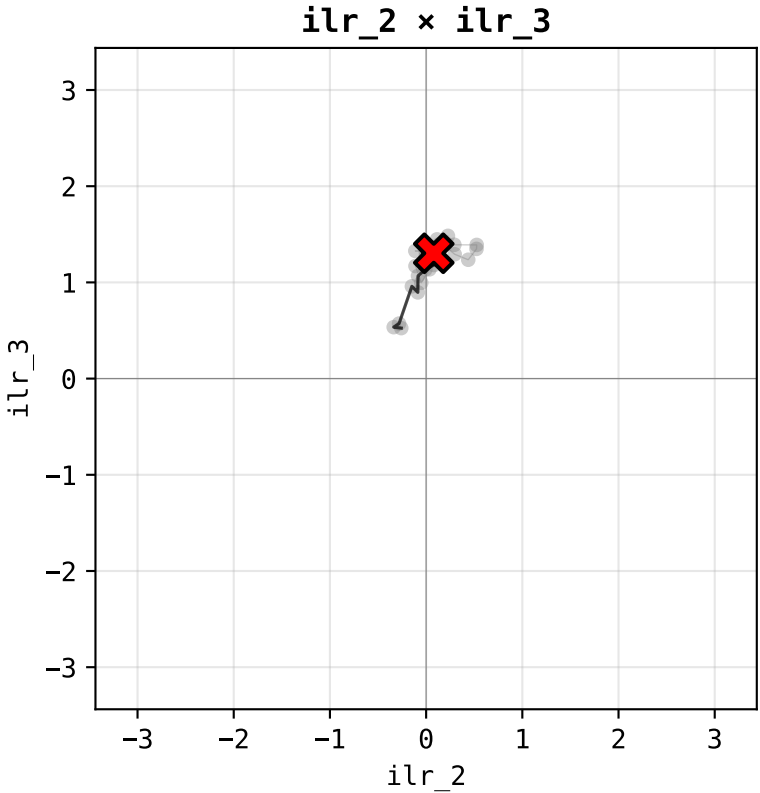
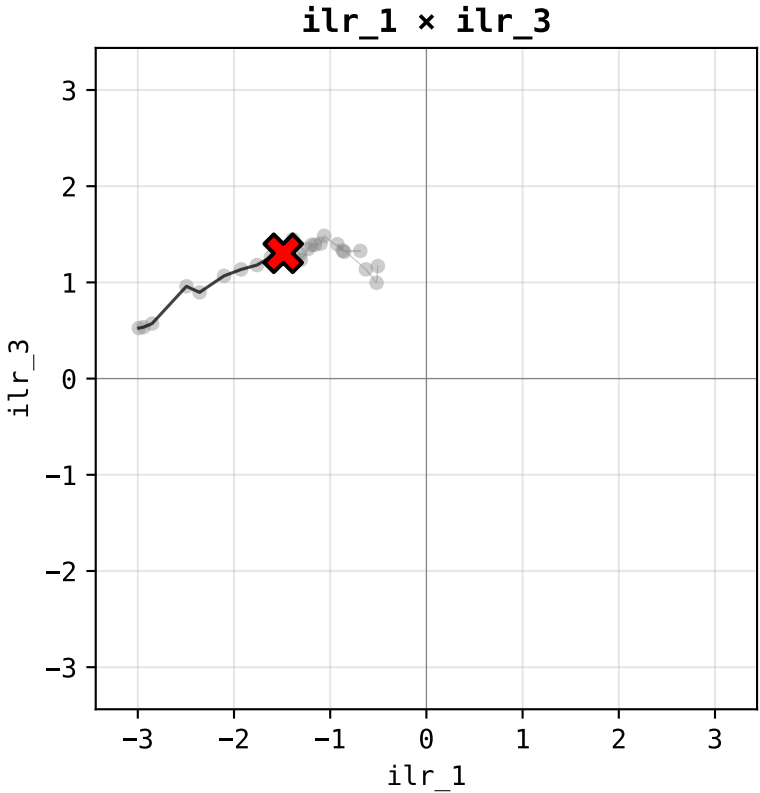
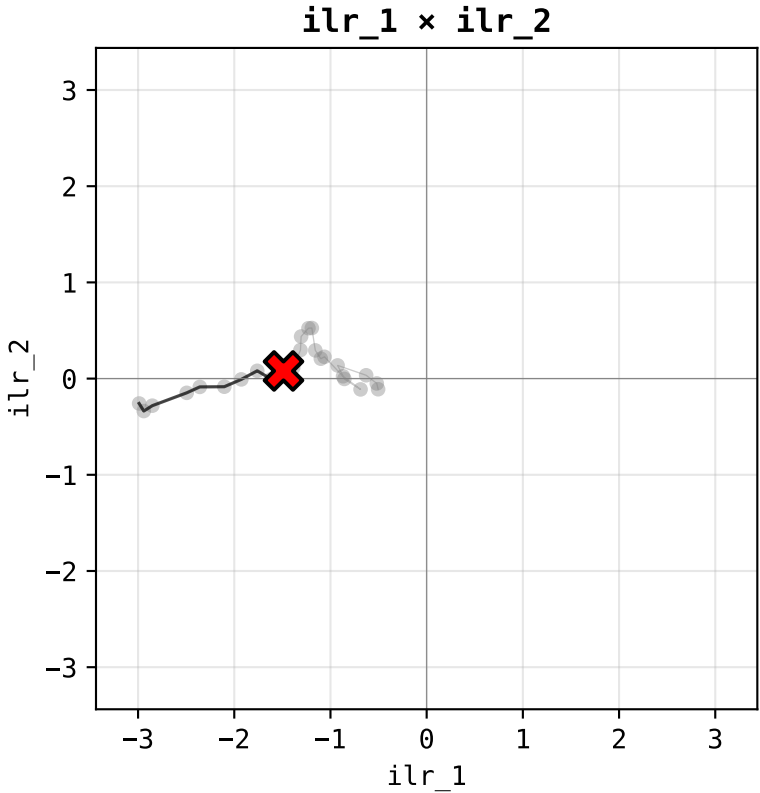




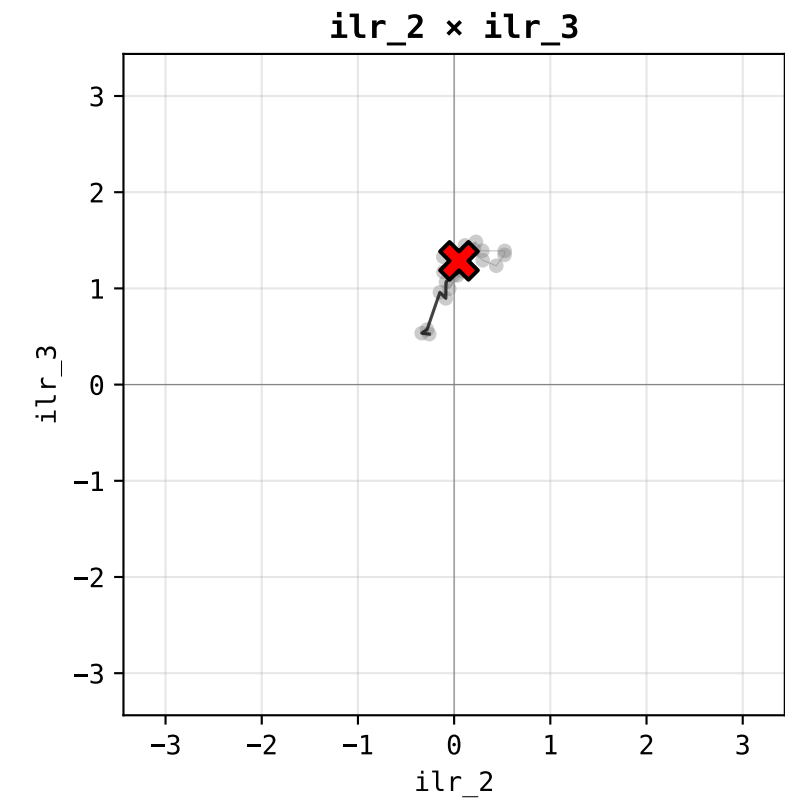
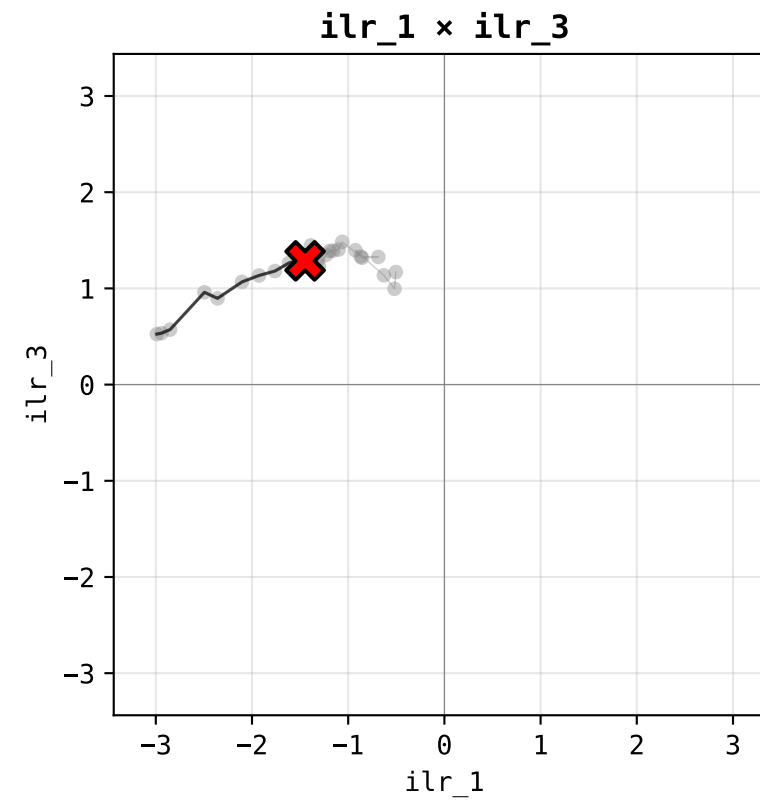
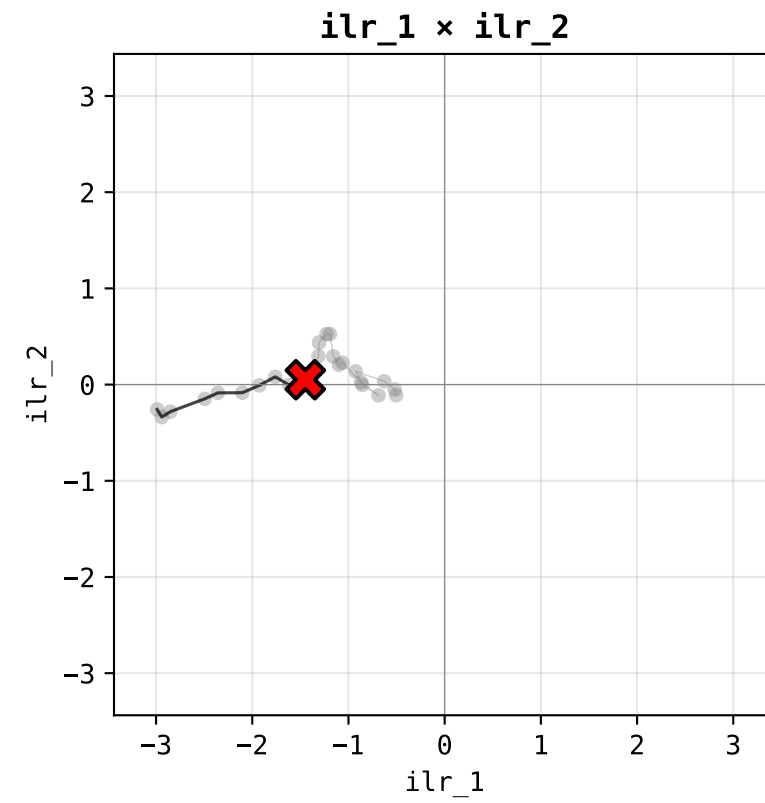






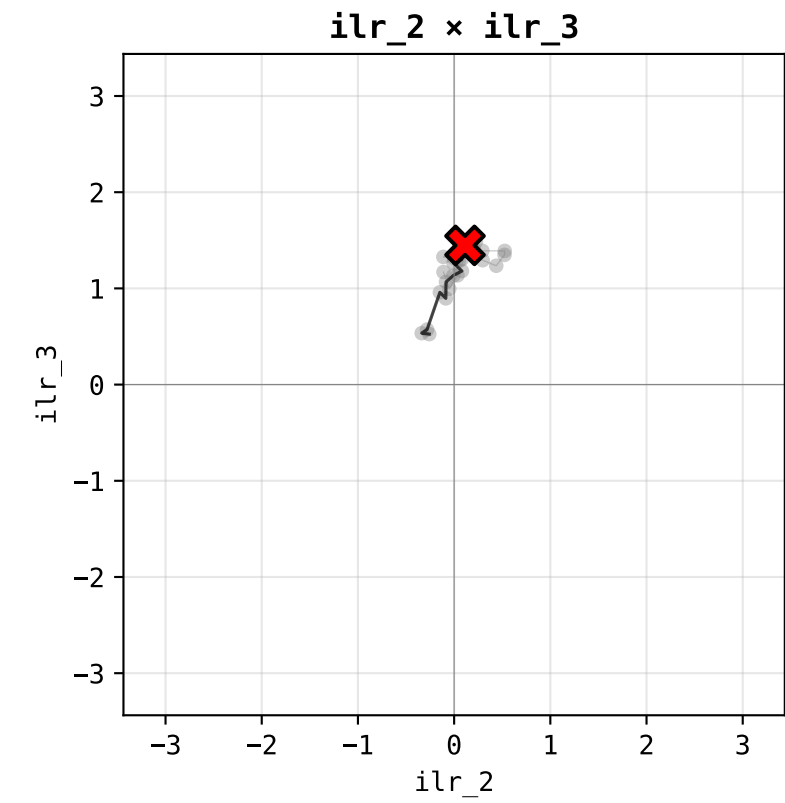
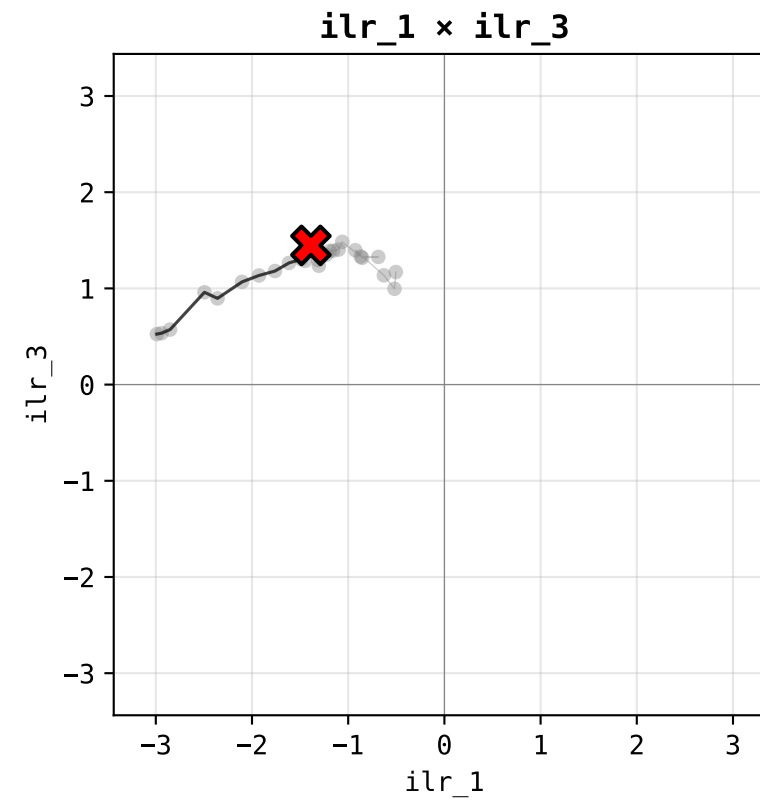
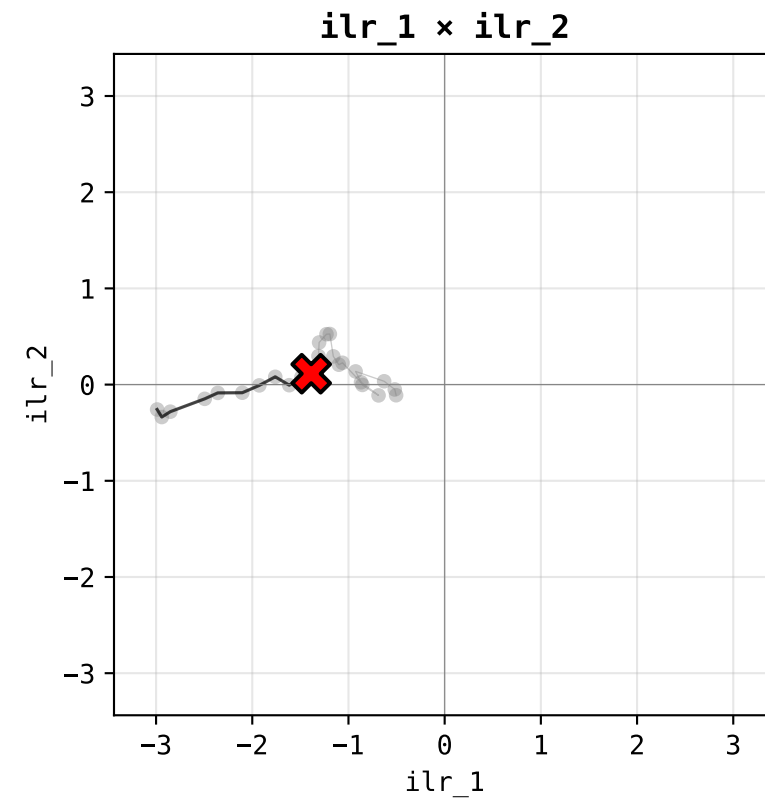


ILR Triplet – DEU Germany · year 2010 (11/26)

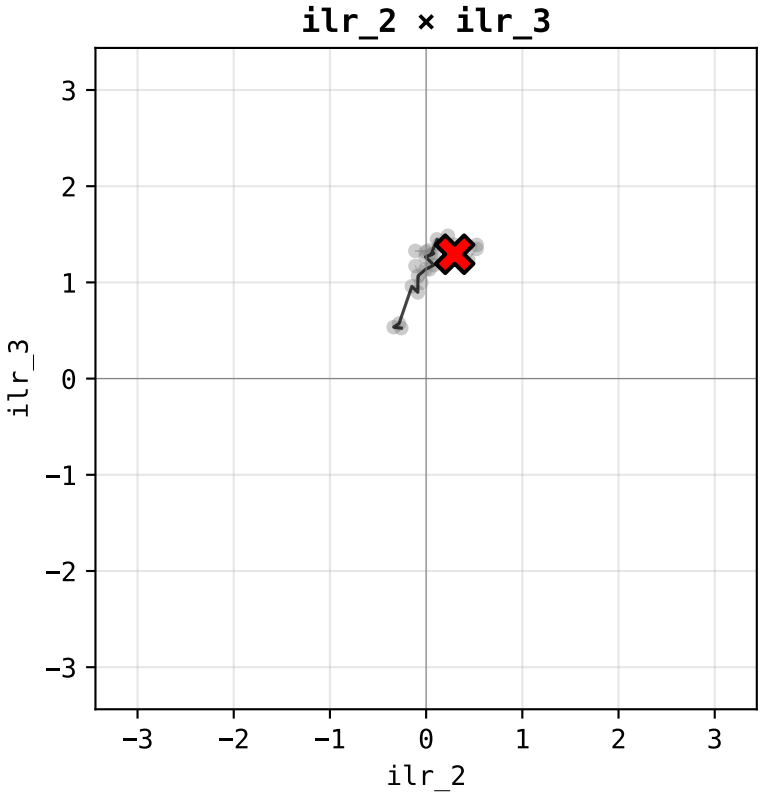
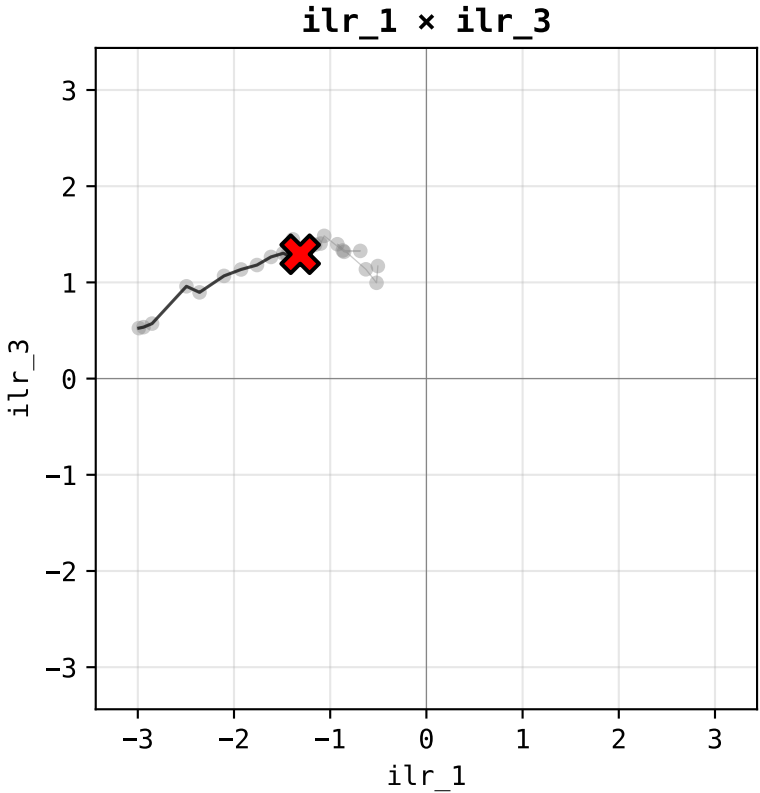
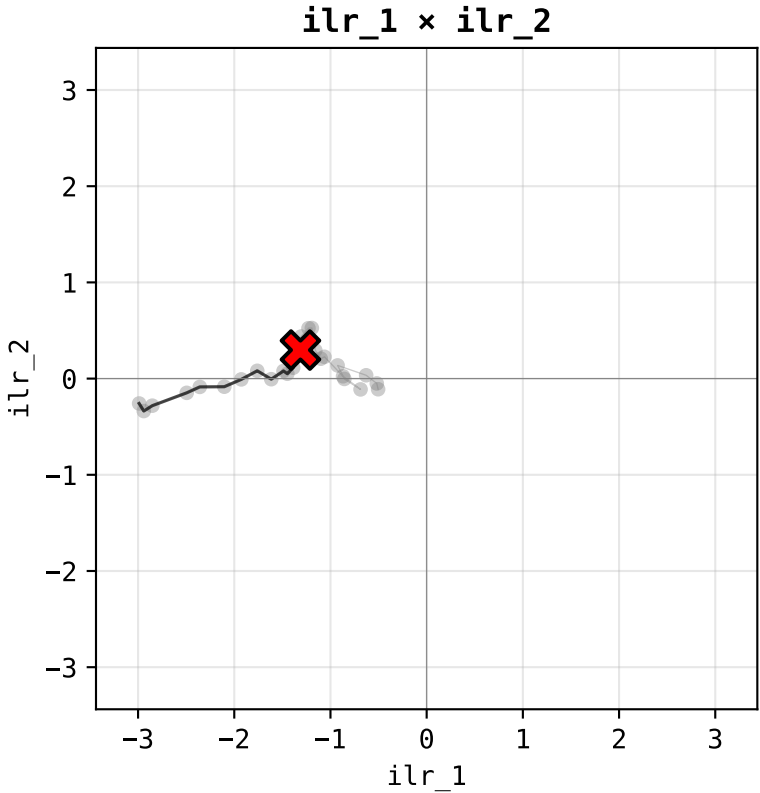


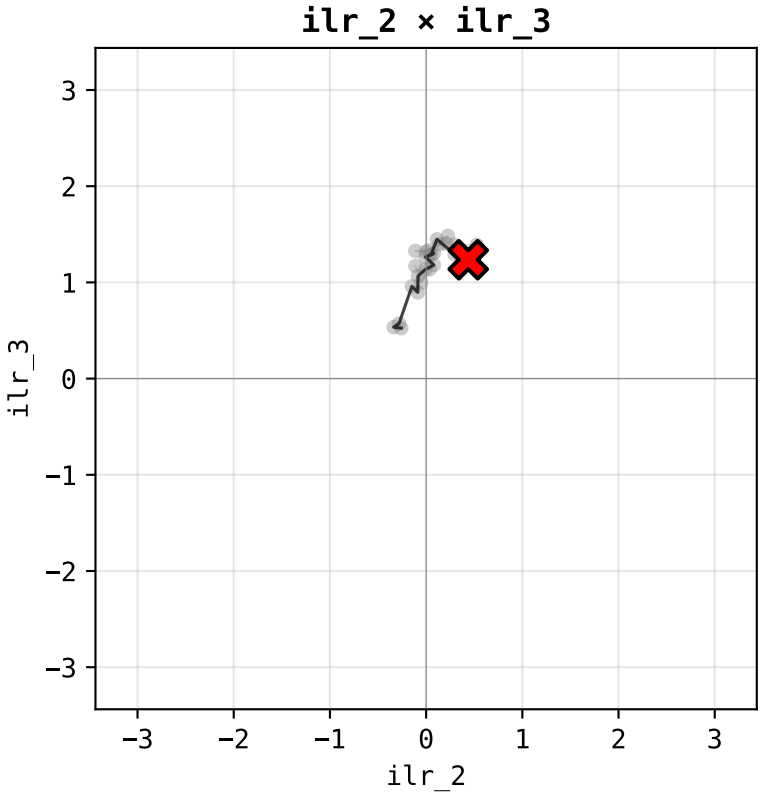
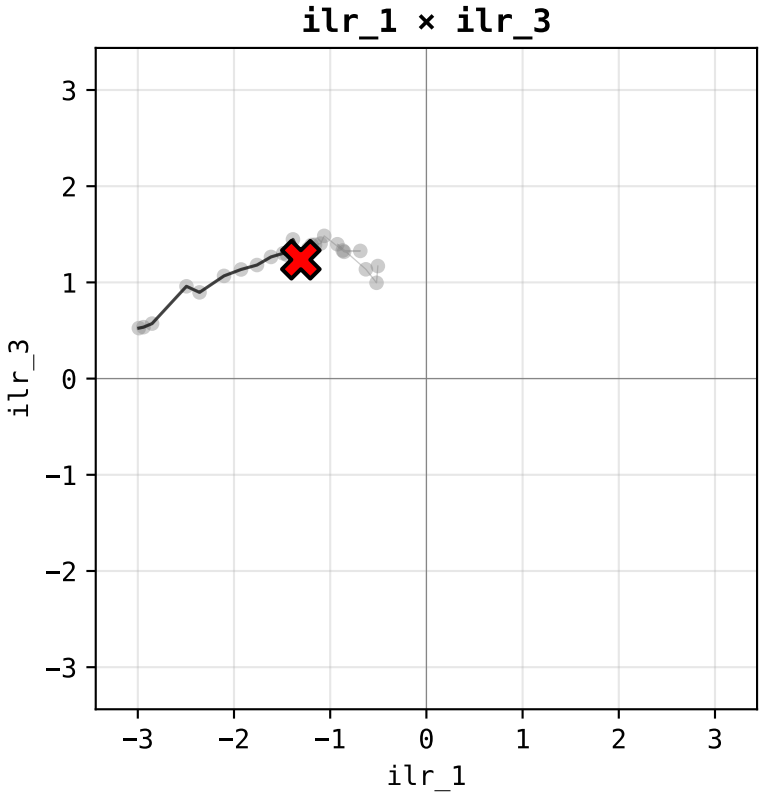
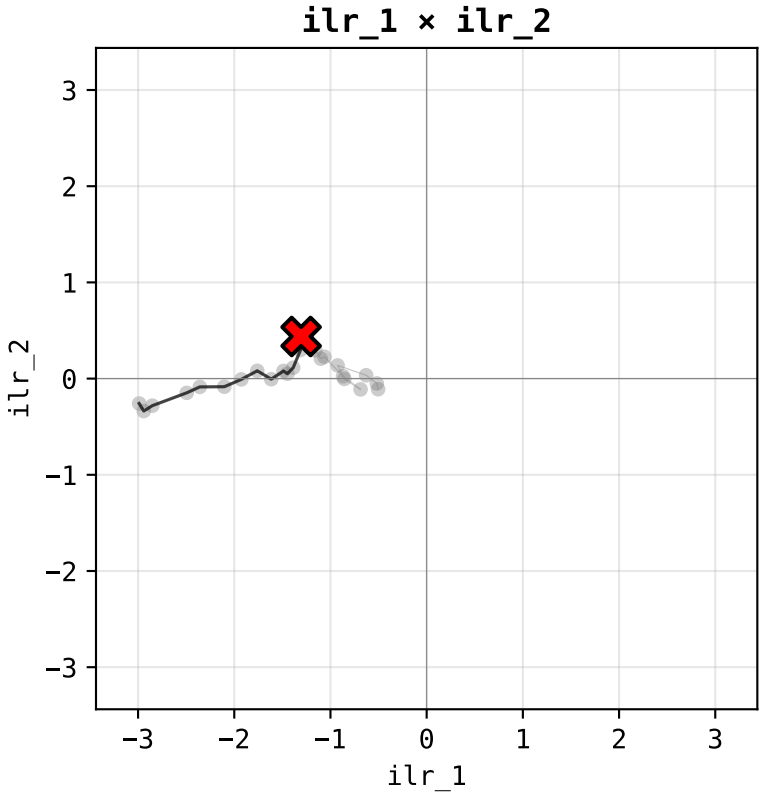
X = current year (2010) · black = past trajectory · gray = future · ilr range ±3.44

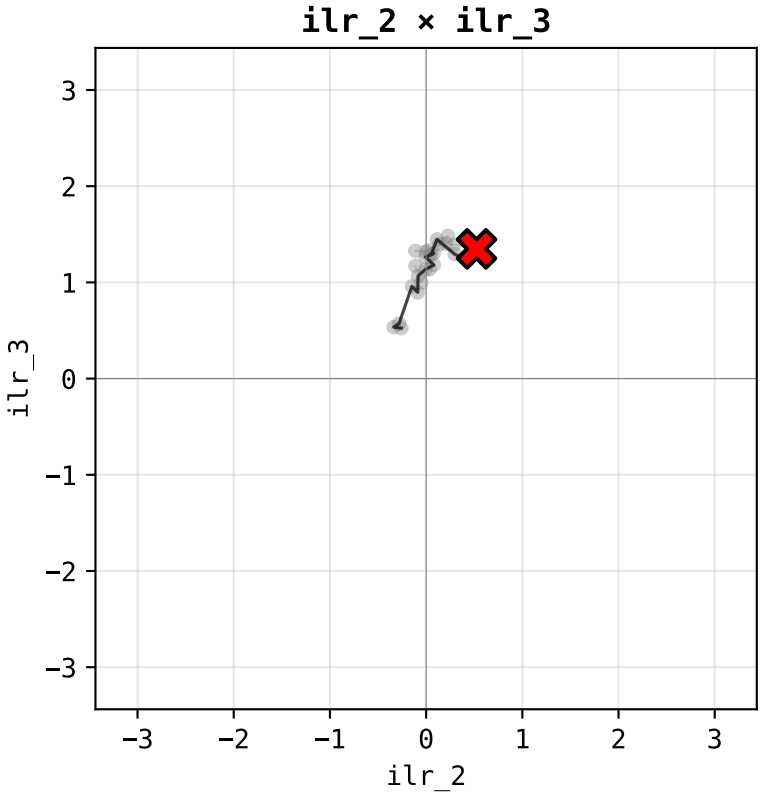
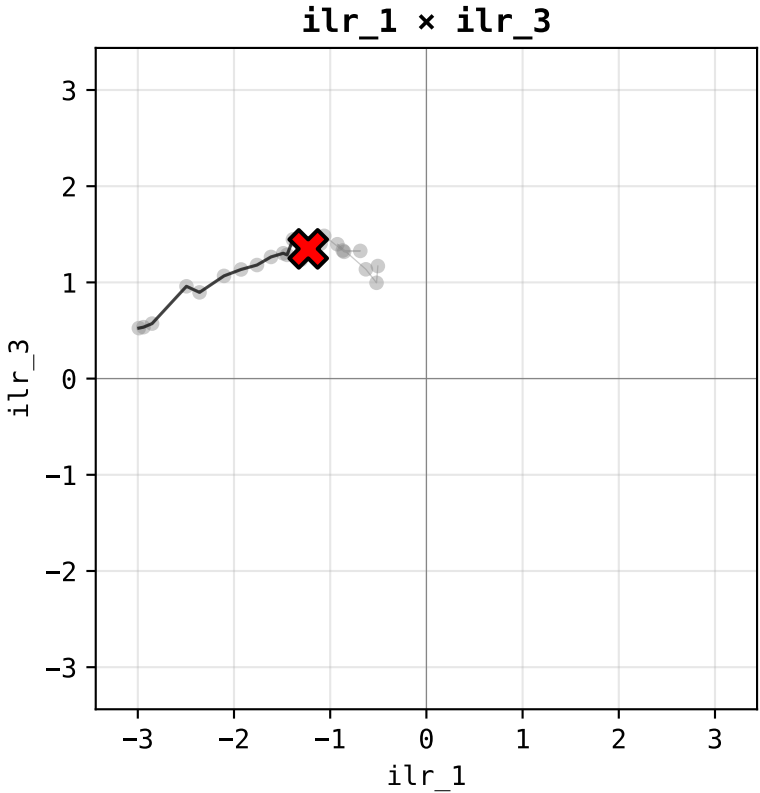
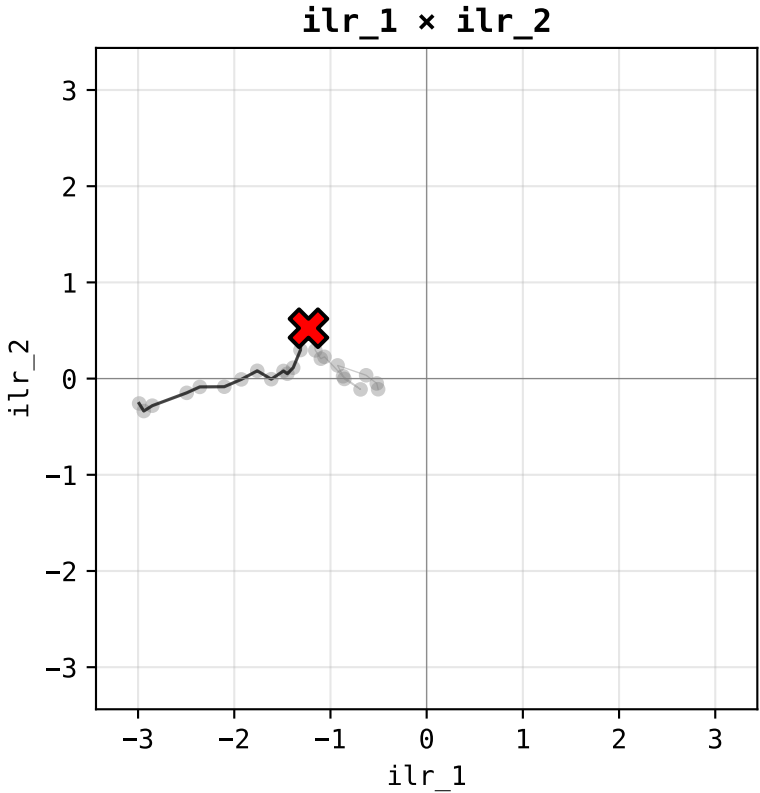
ILR Triplet – DEU Germany · year 2011 (12/26)

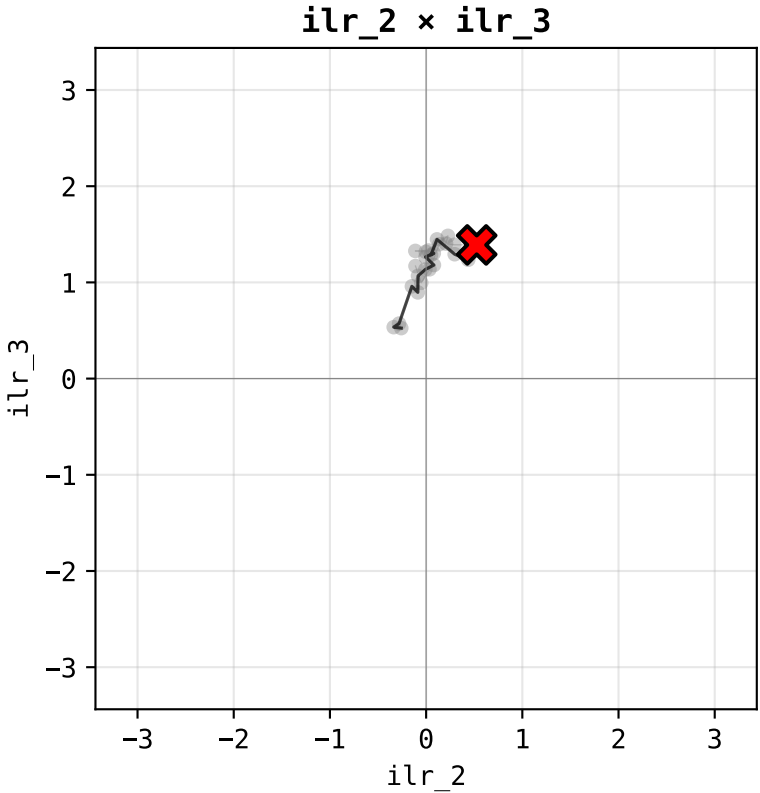
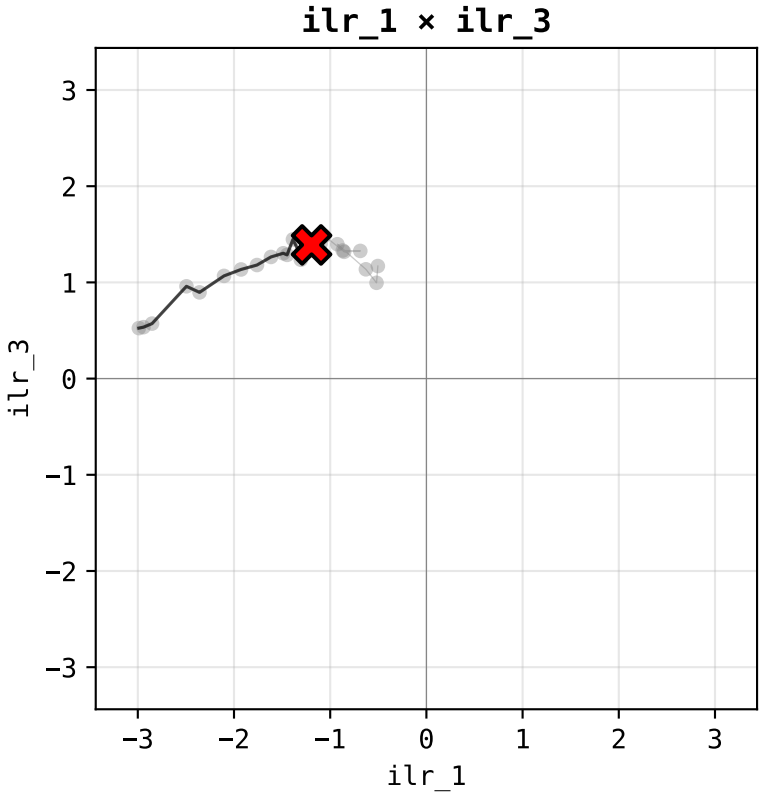
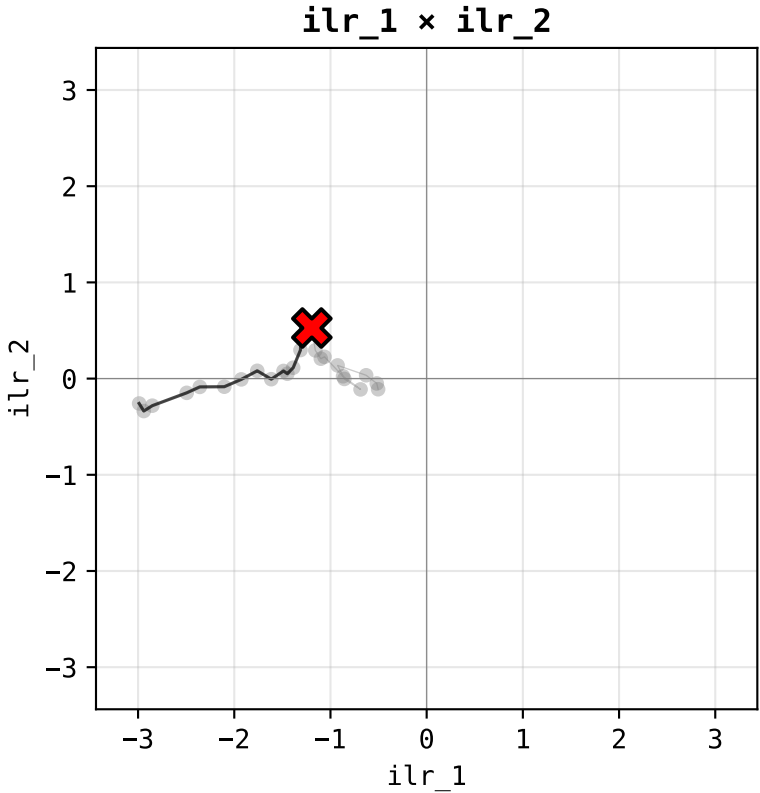


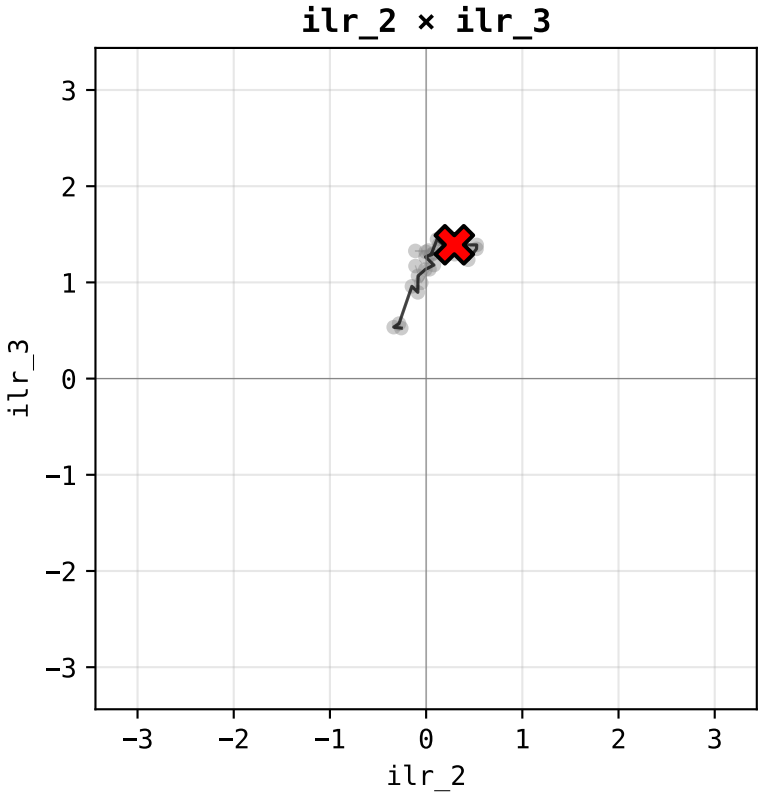
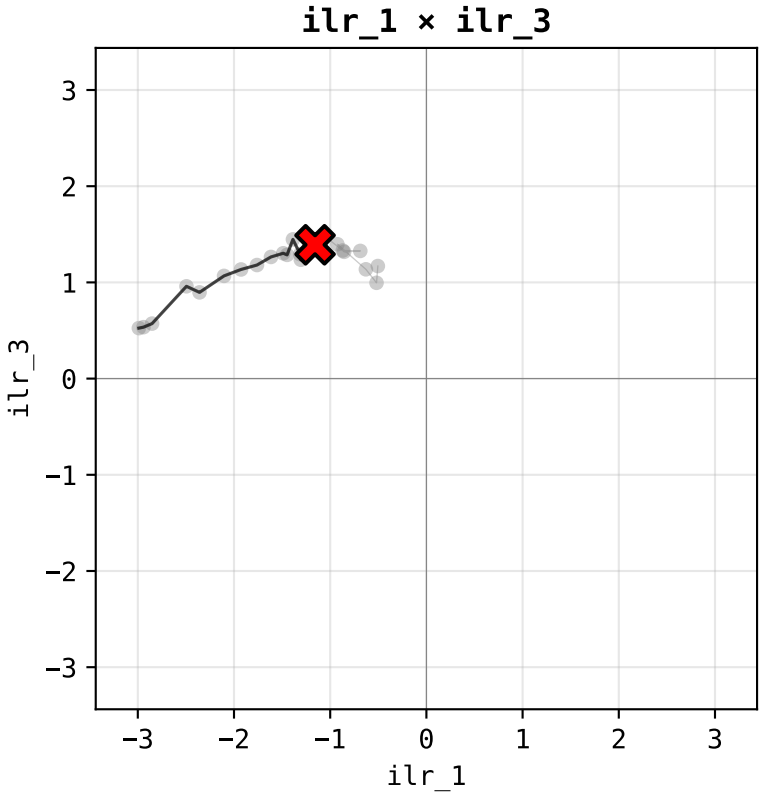
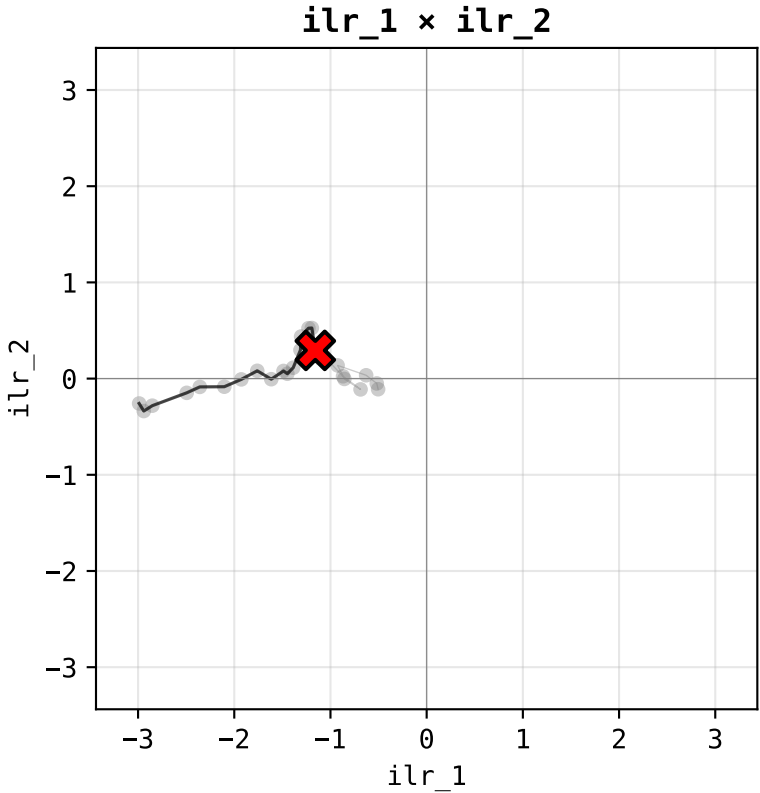
X = current year (2011) · black = past trajectory · gray = future · ilr range ±3.44

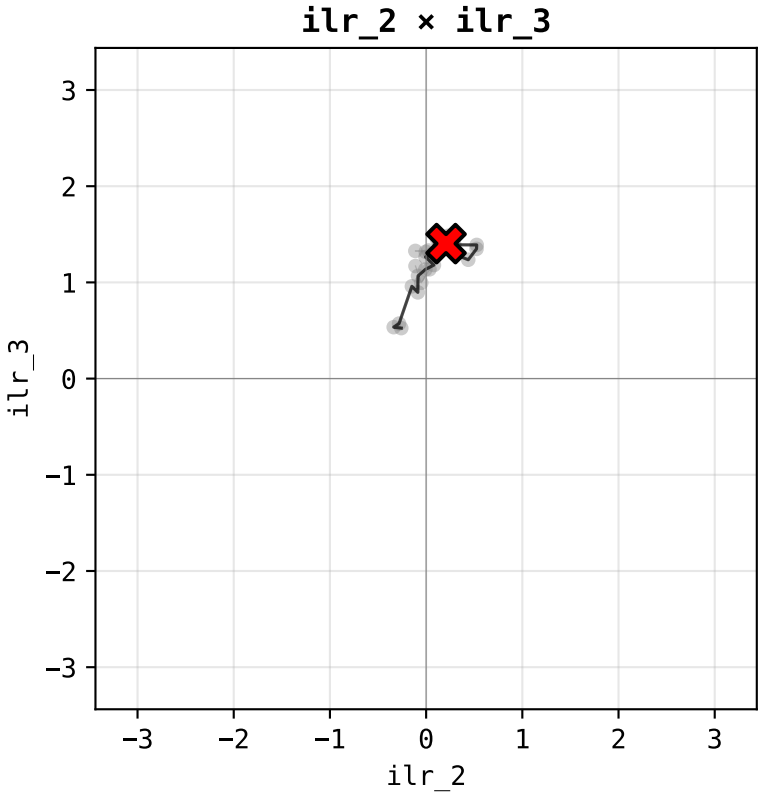
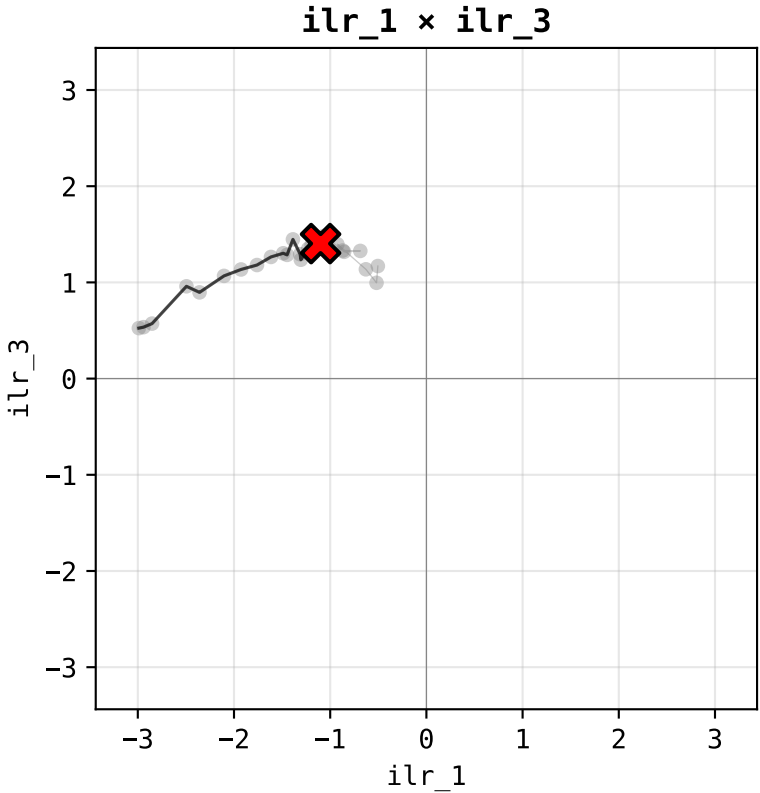
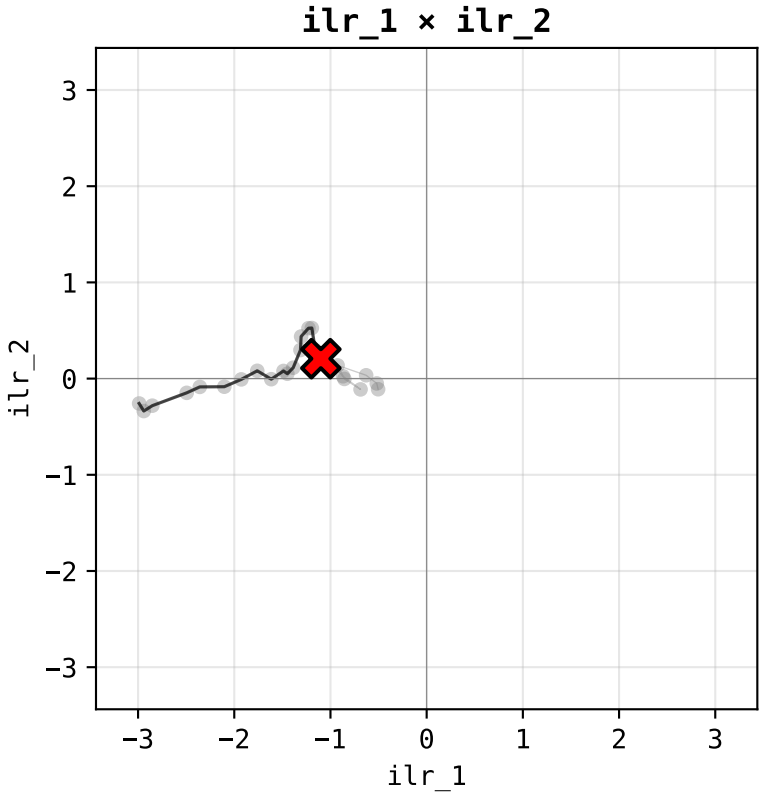


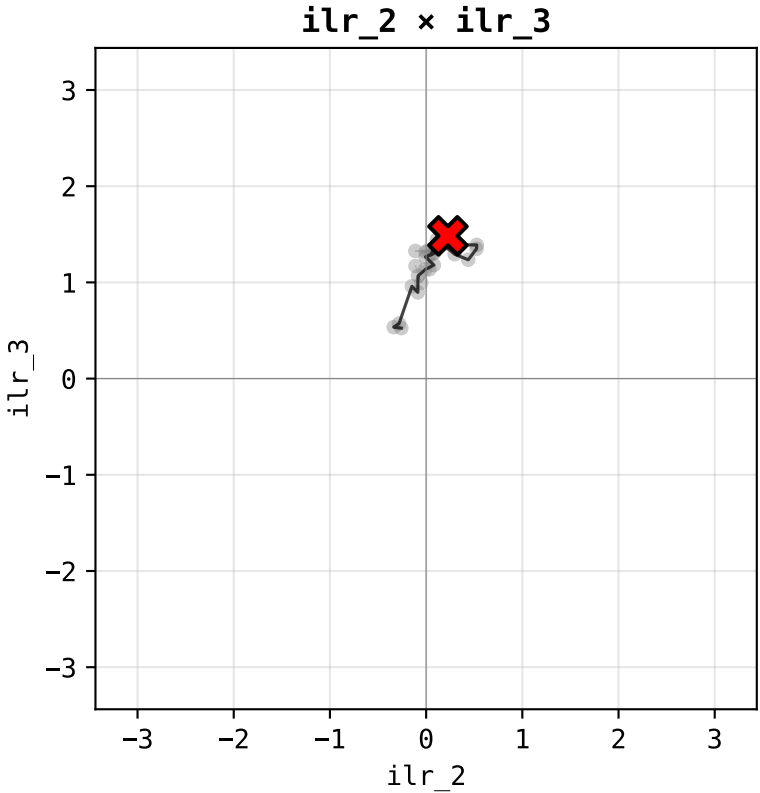
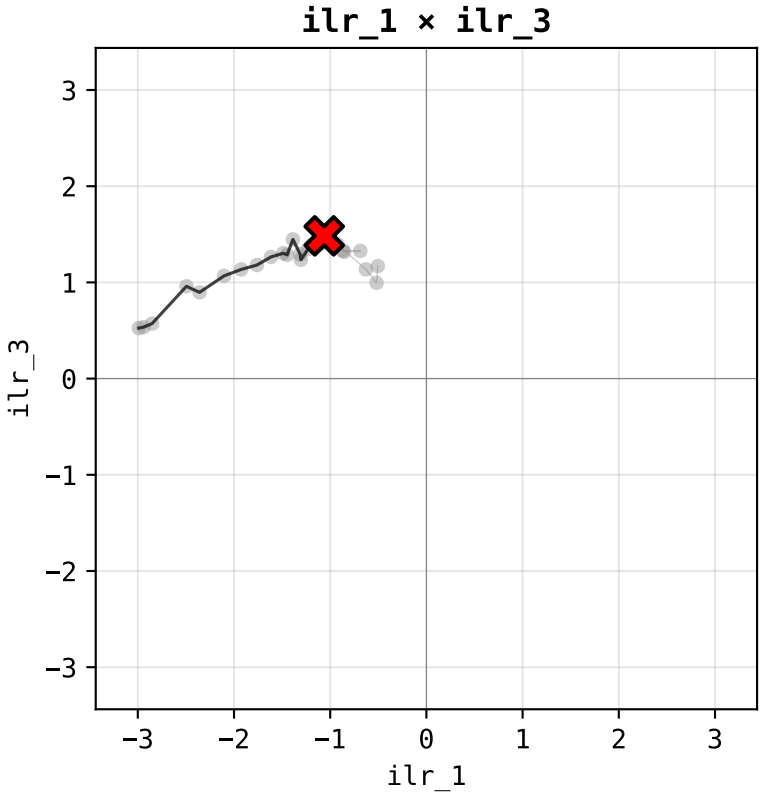
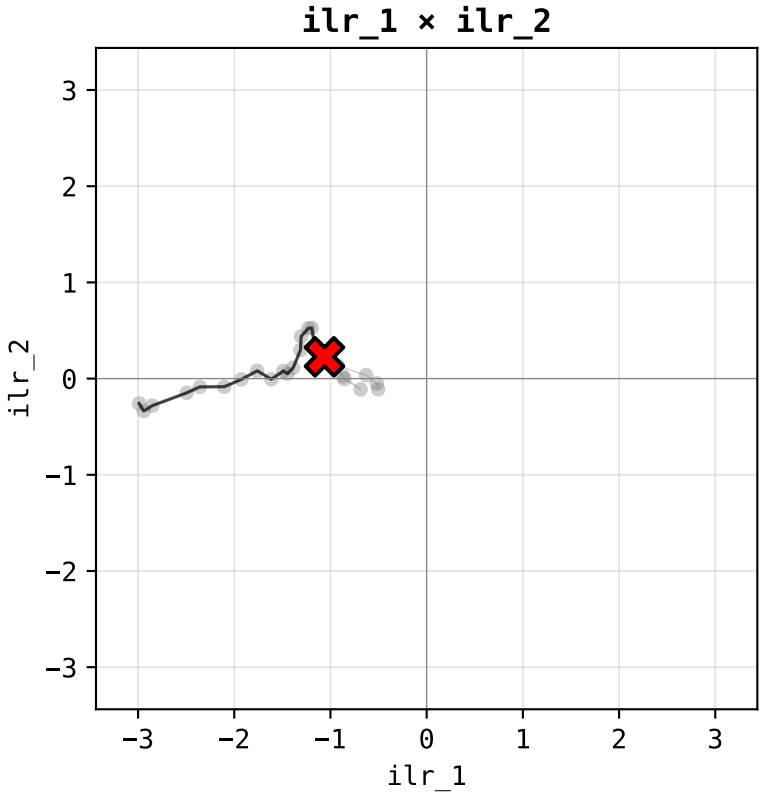


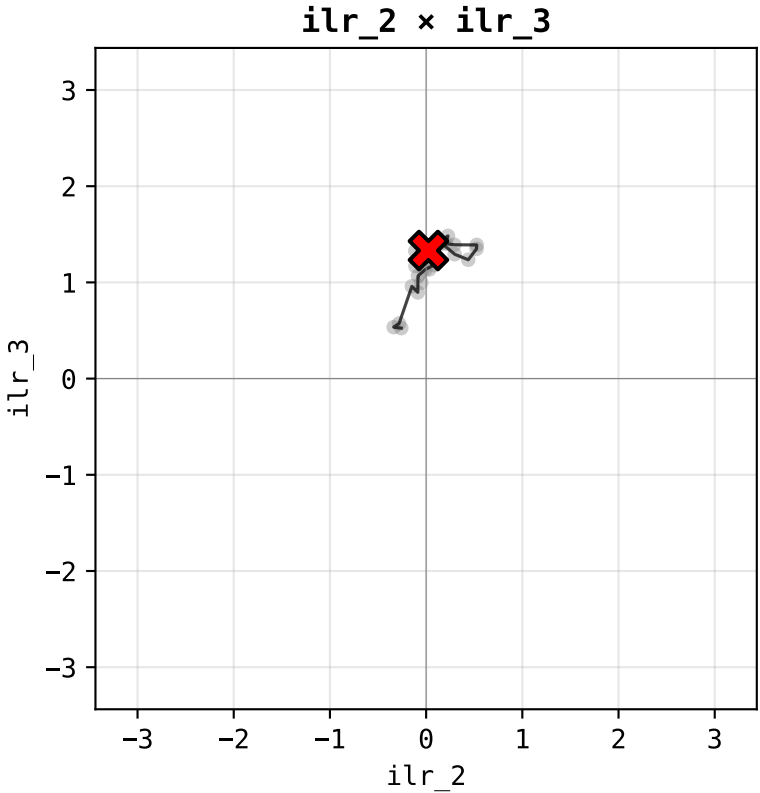
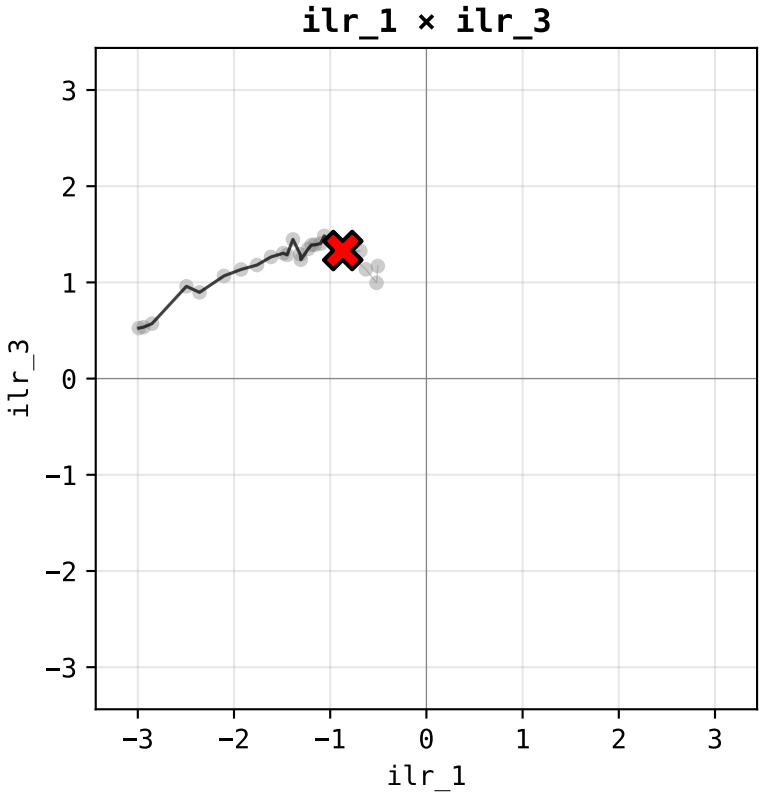
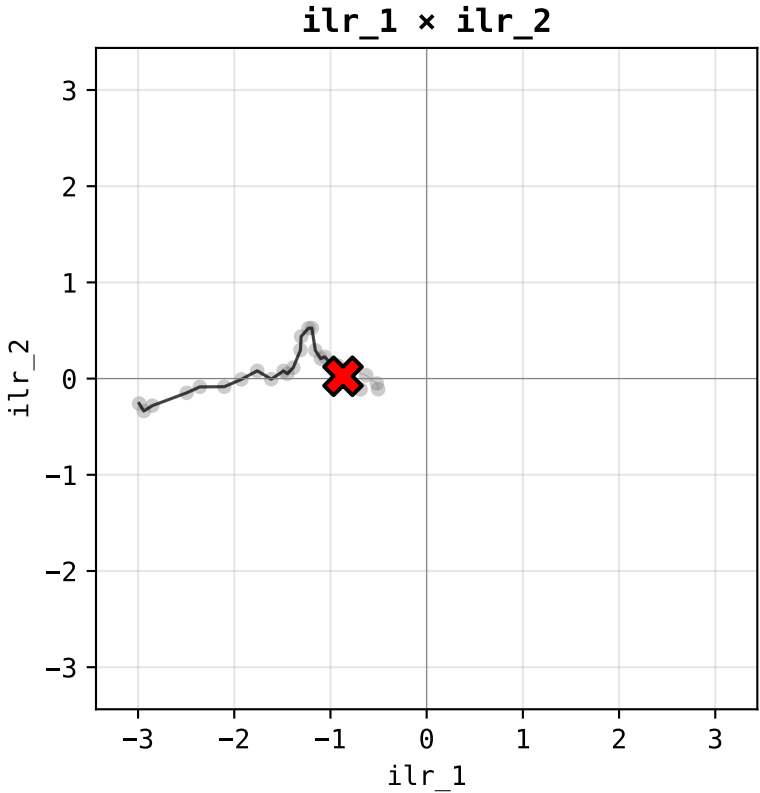


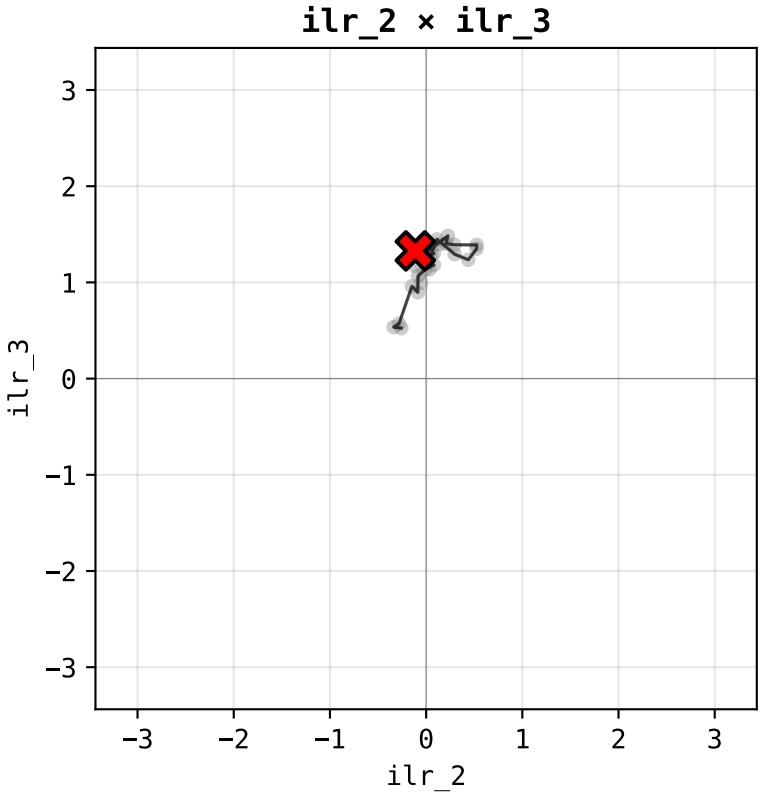
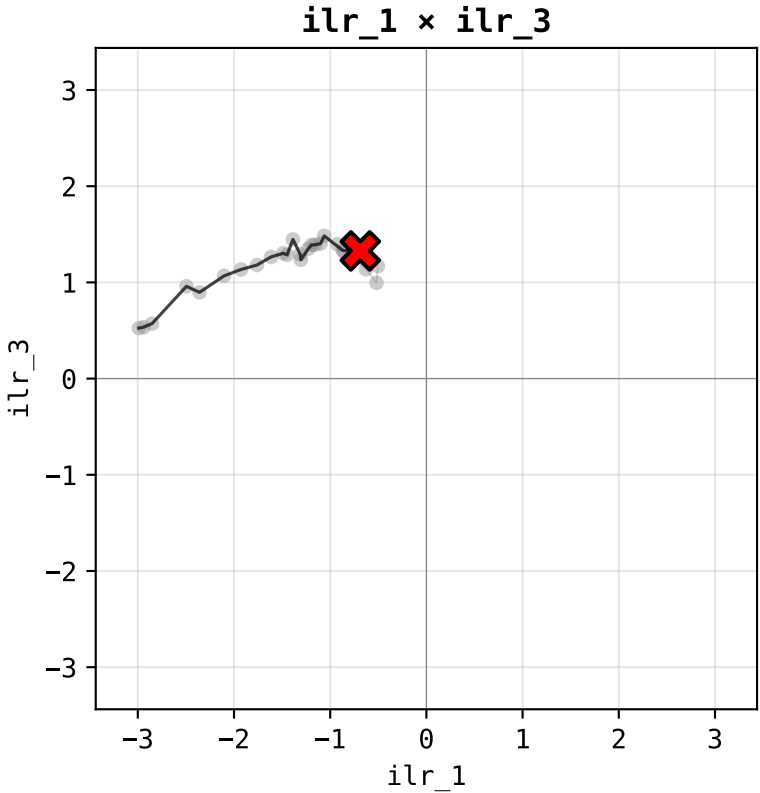
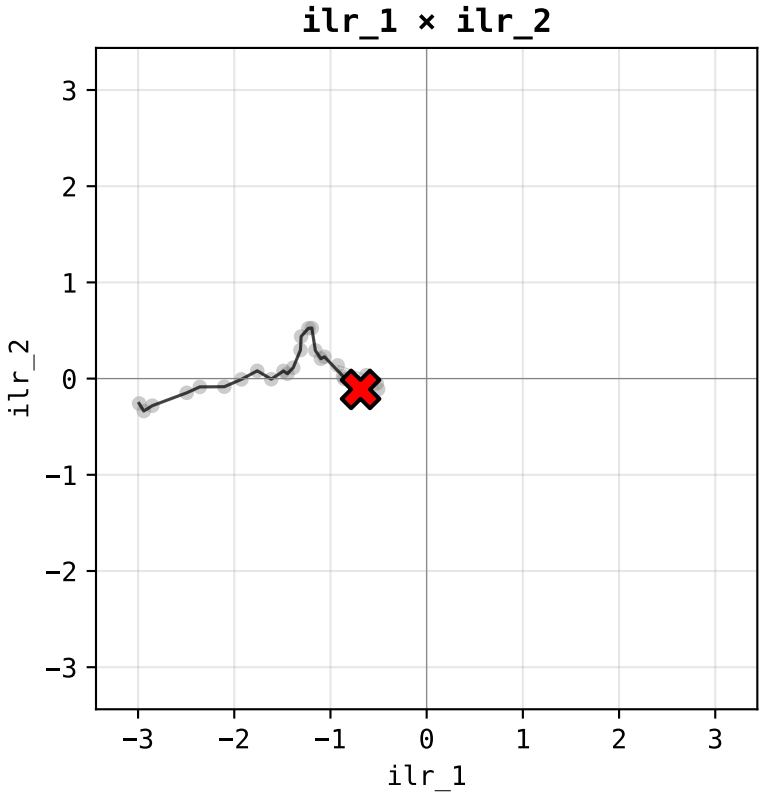


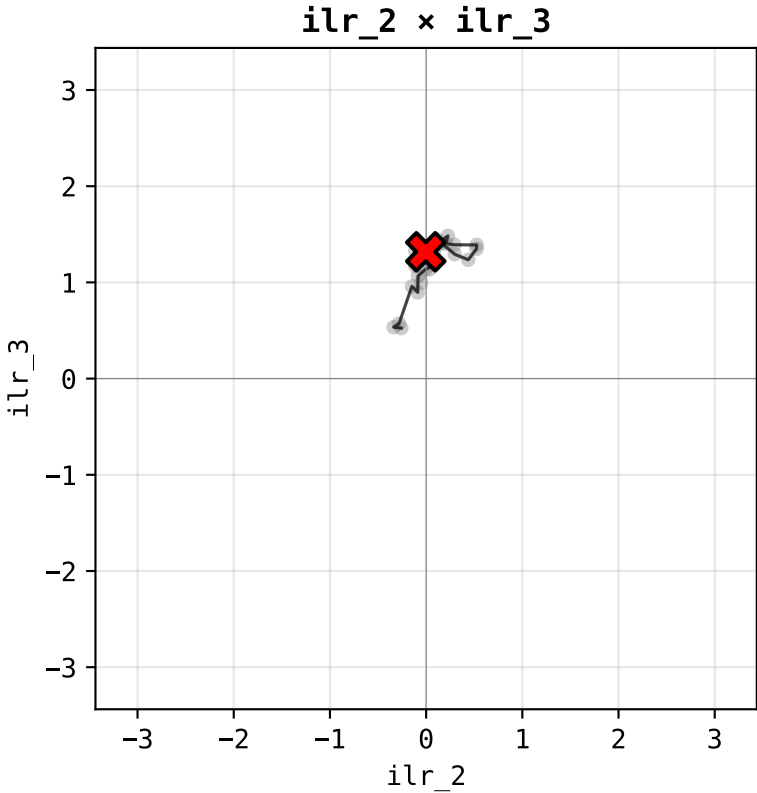
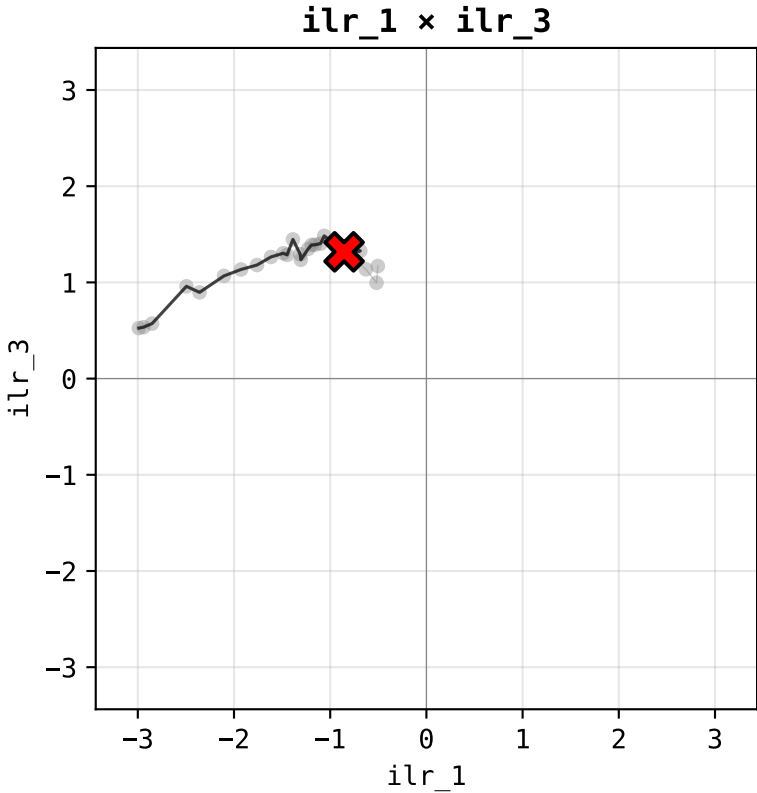
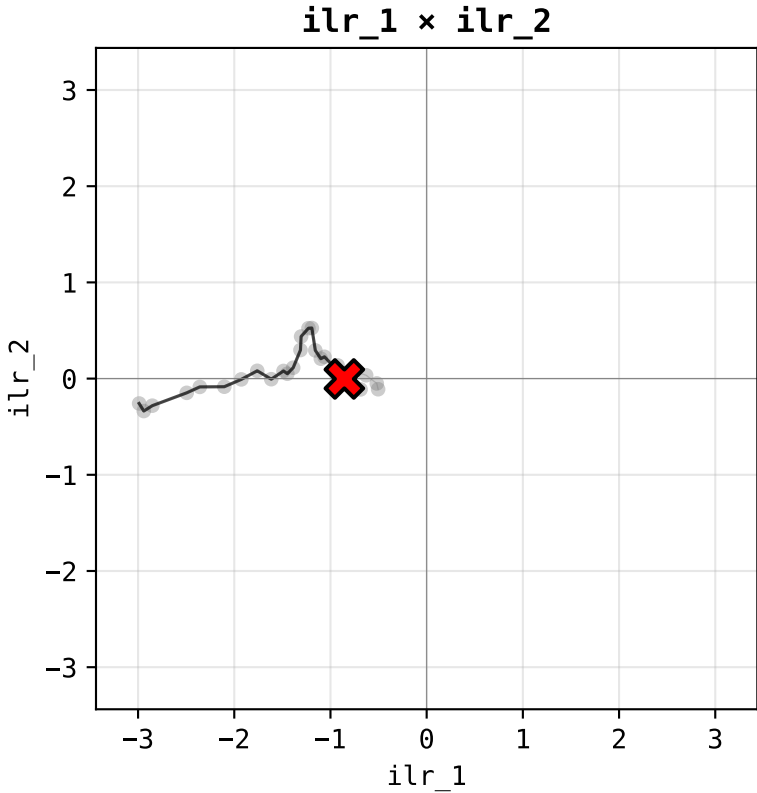


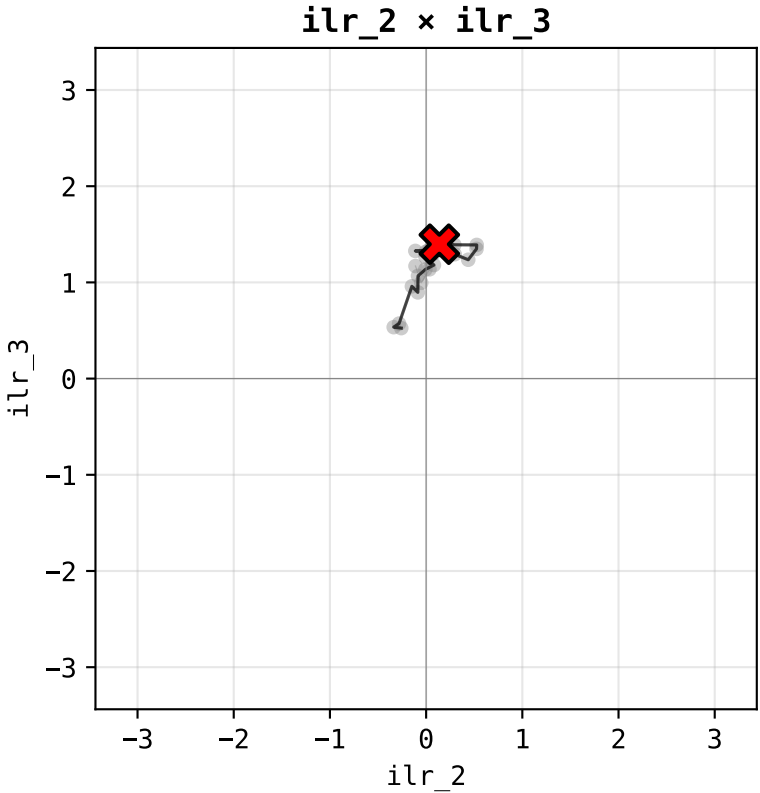
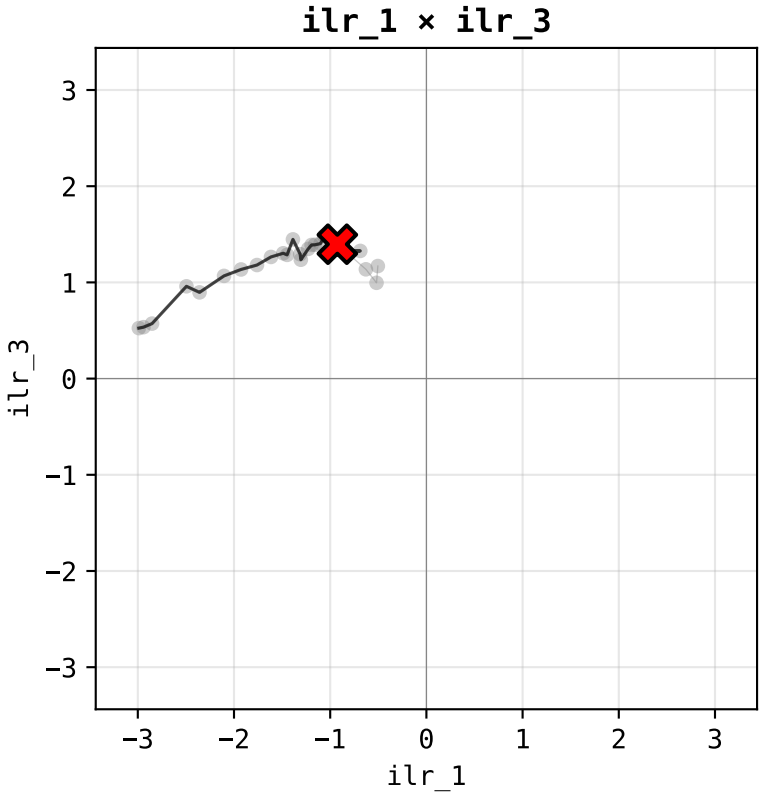
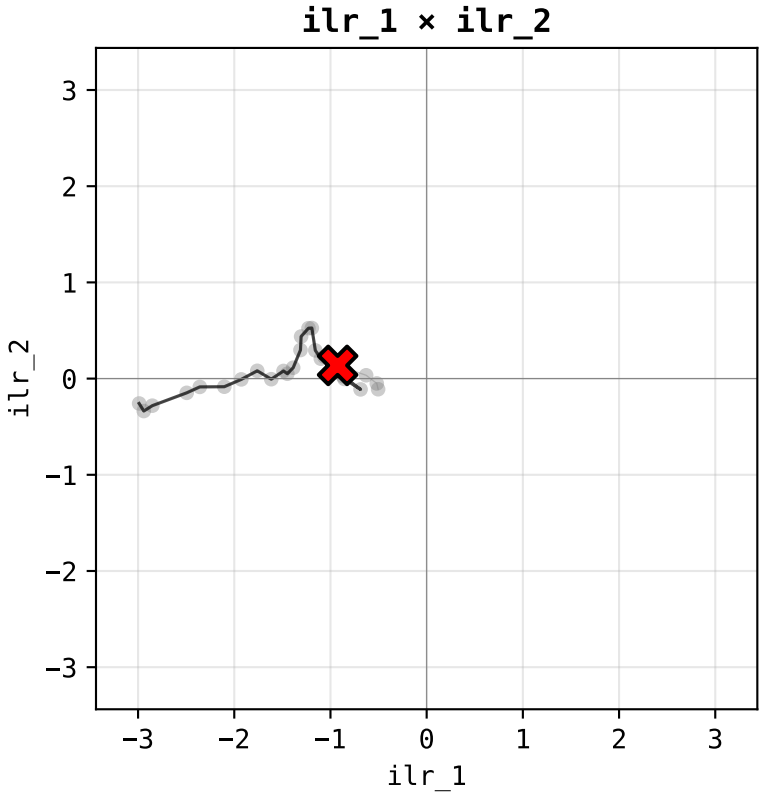


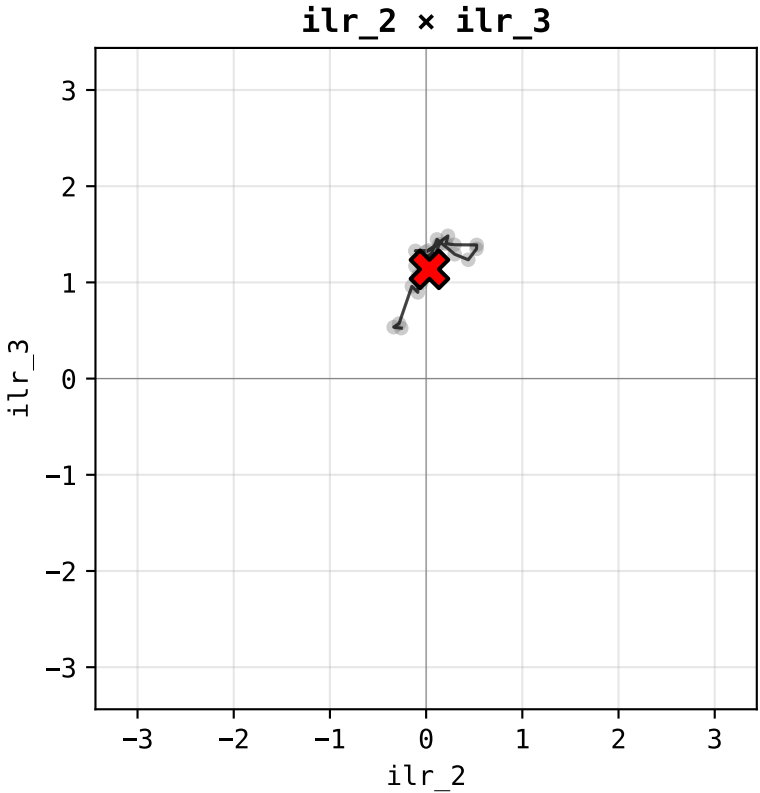
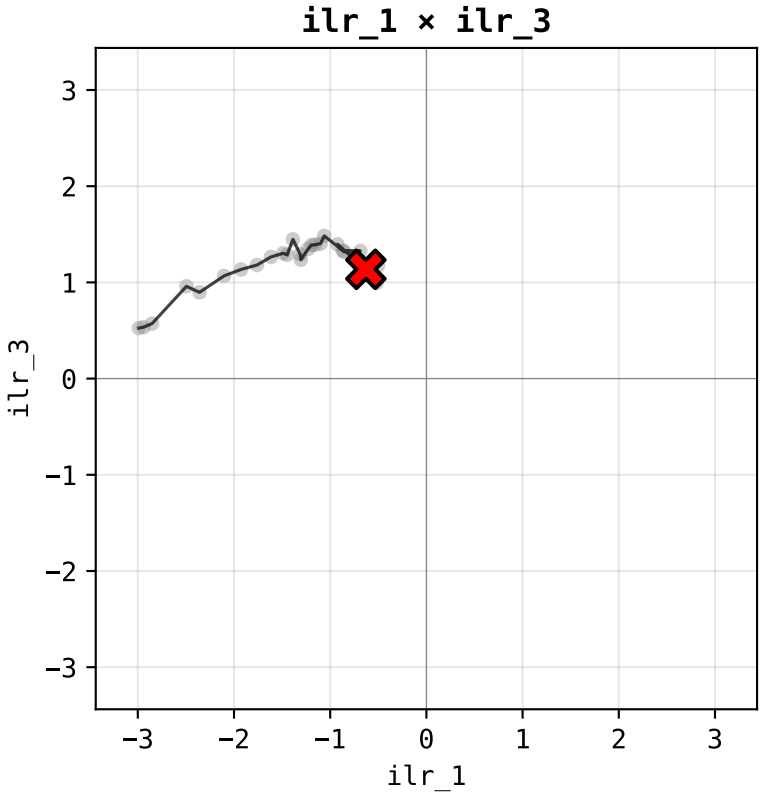
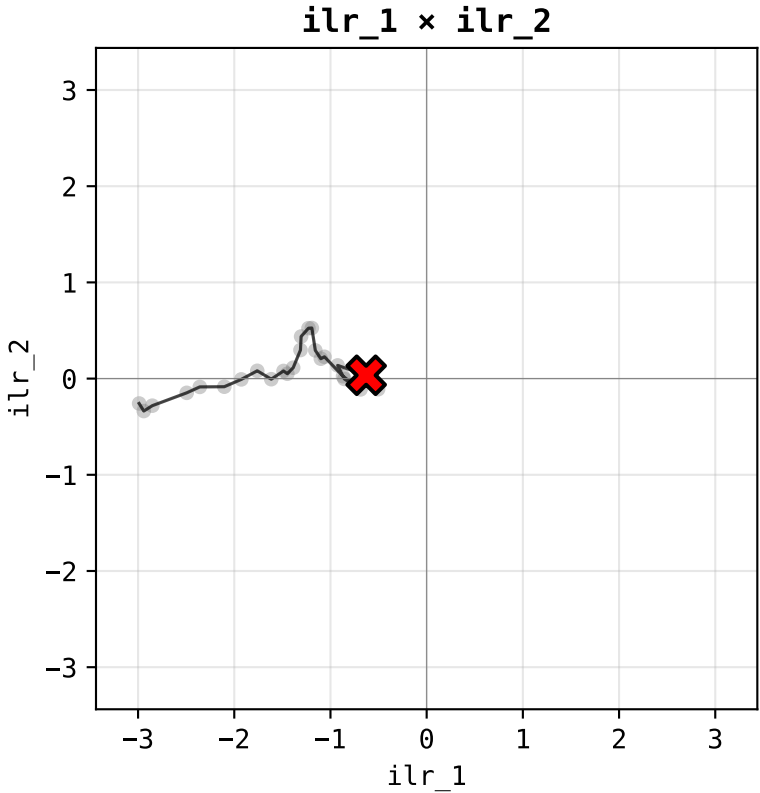


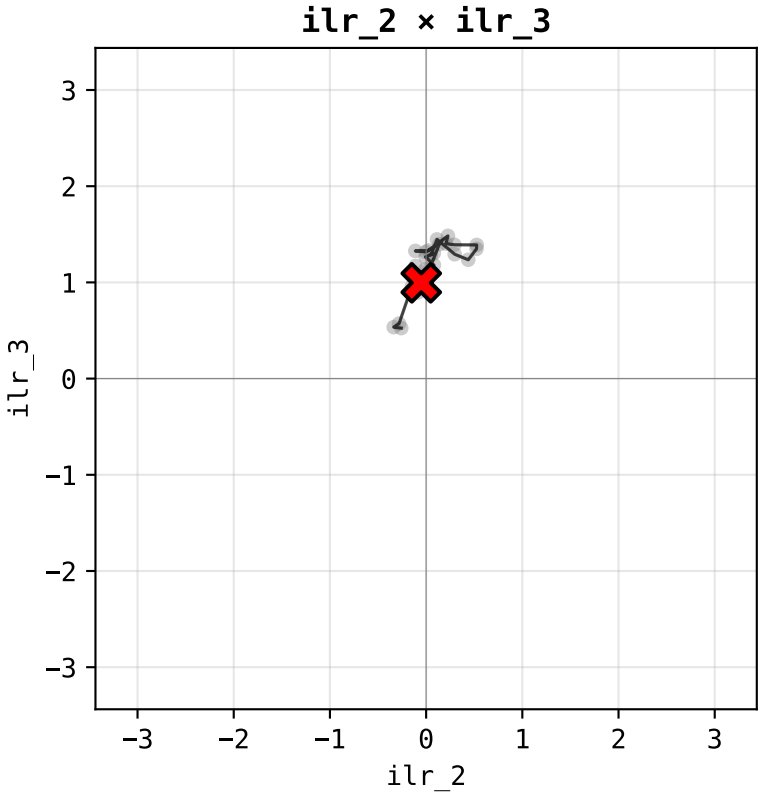
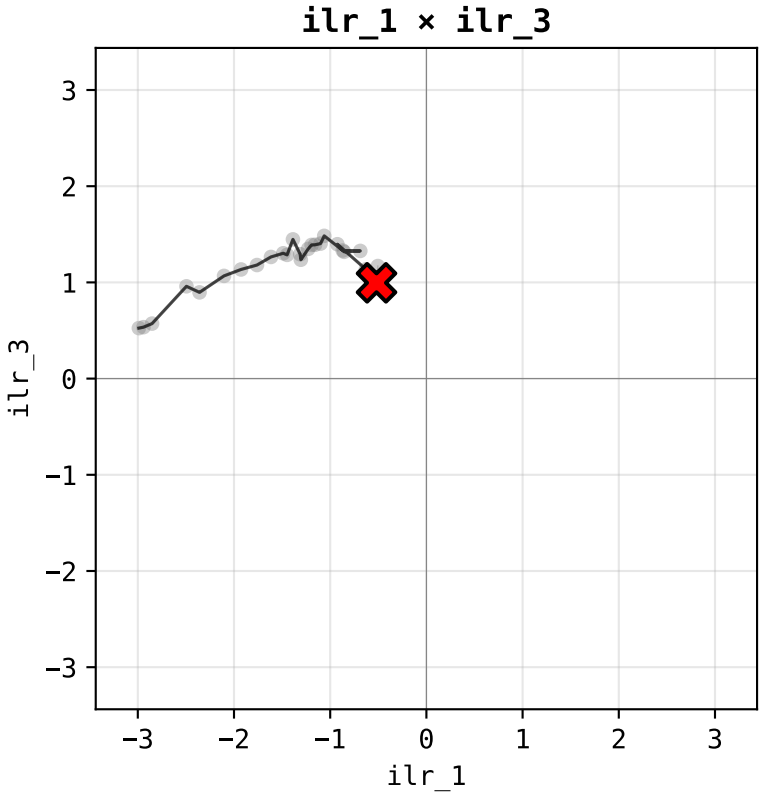
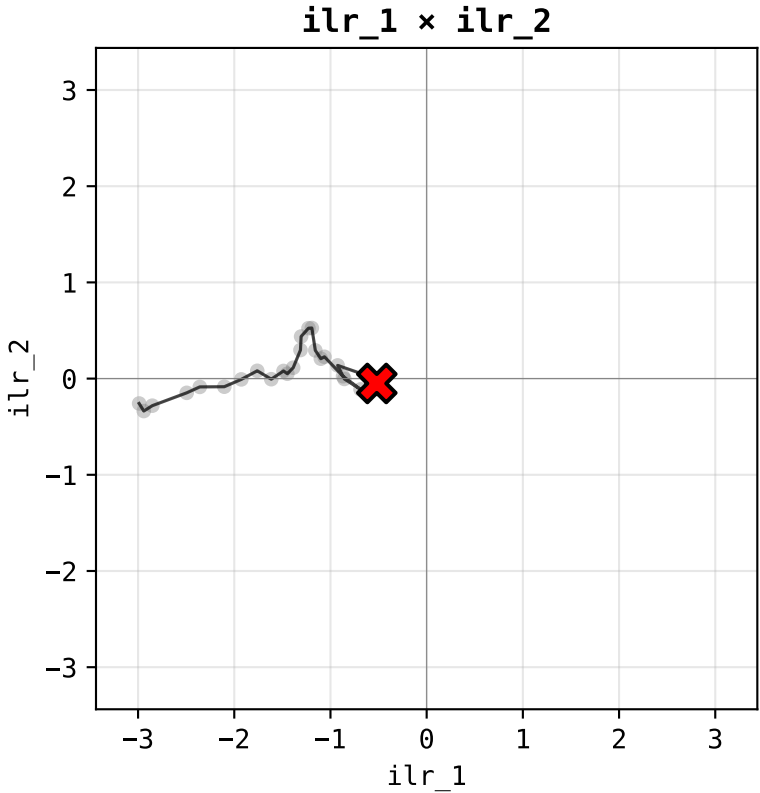


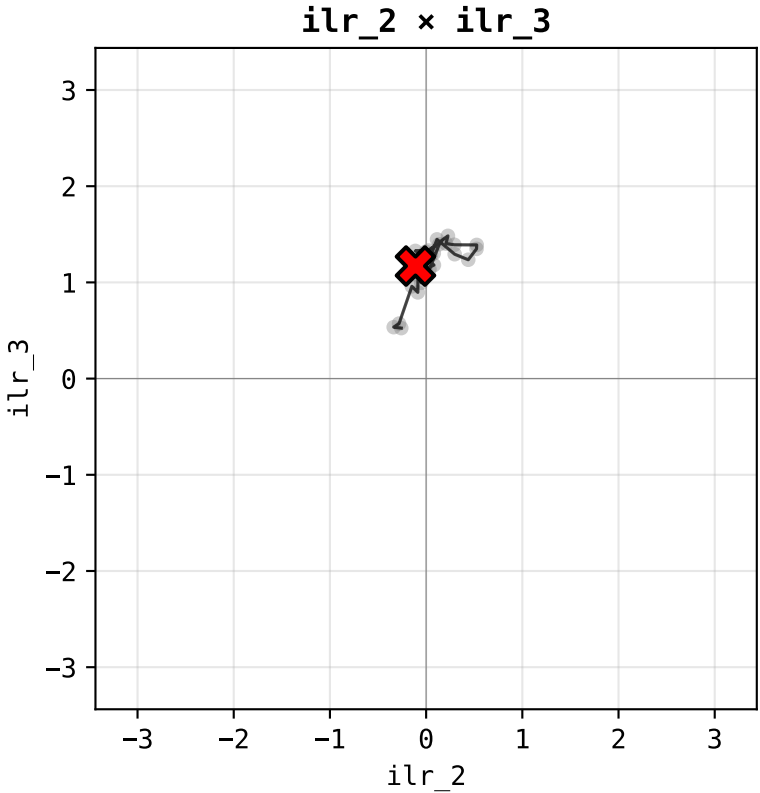
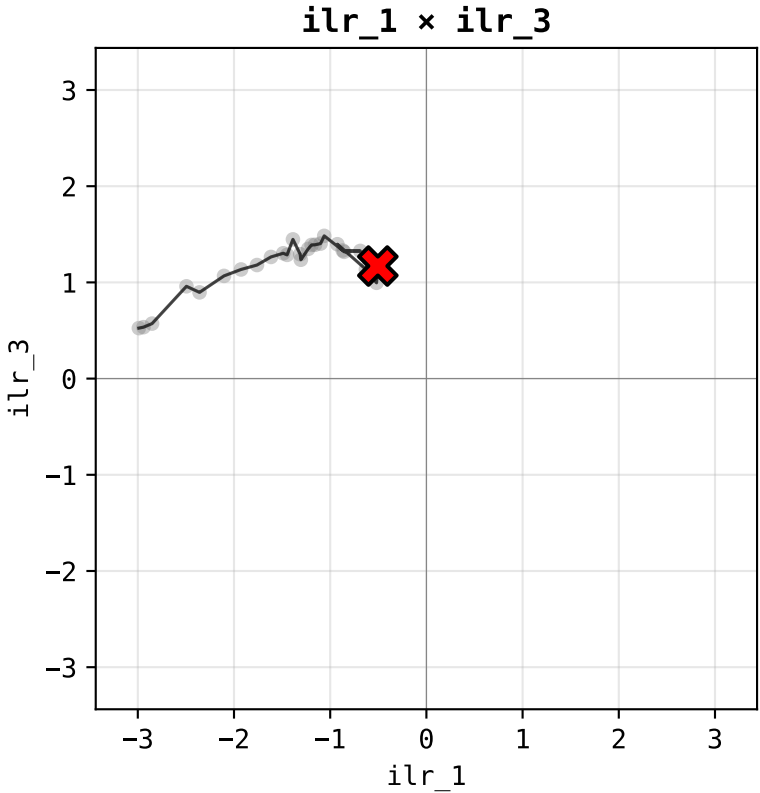
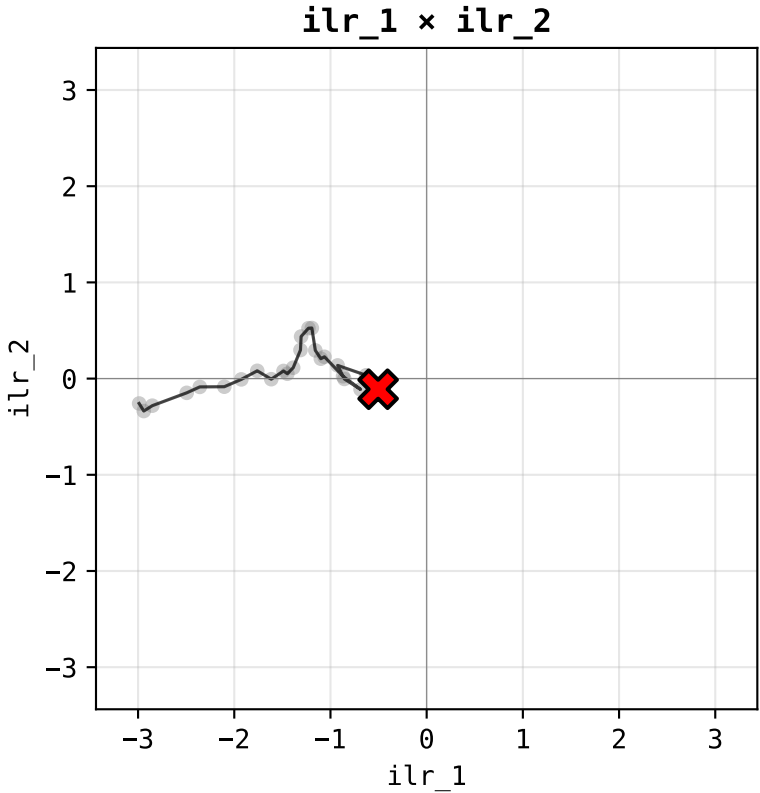












View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ FRA France

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

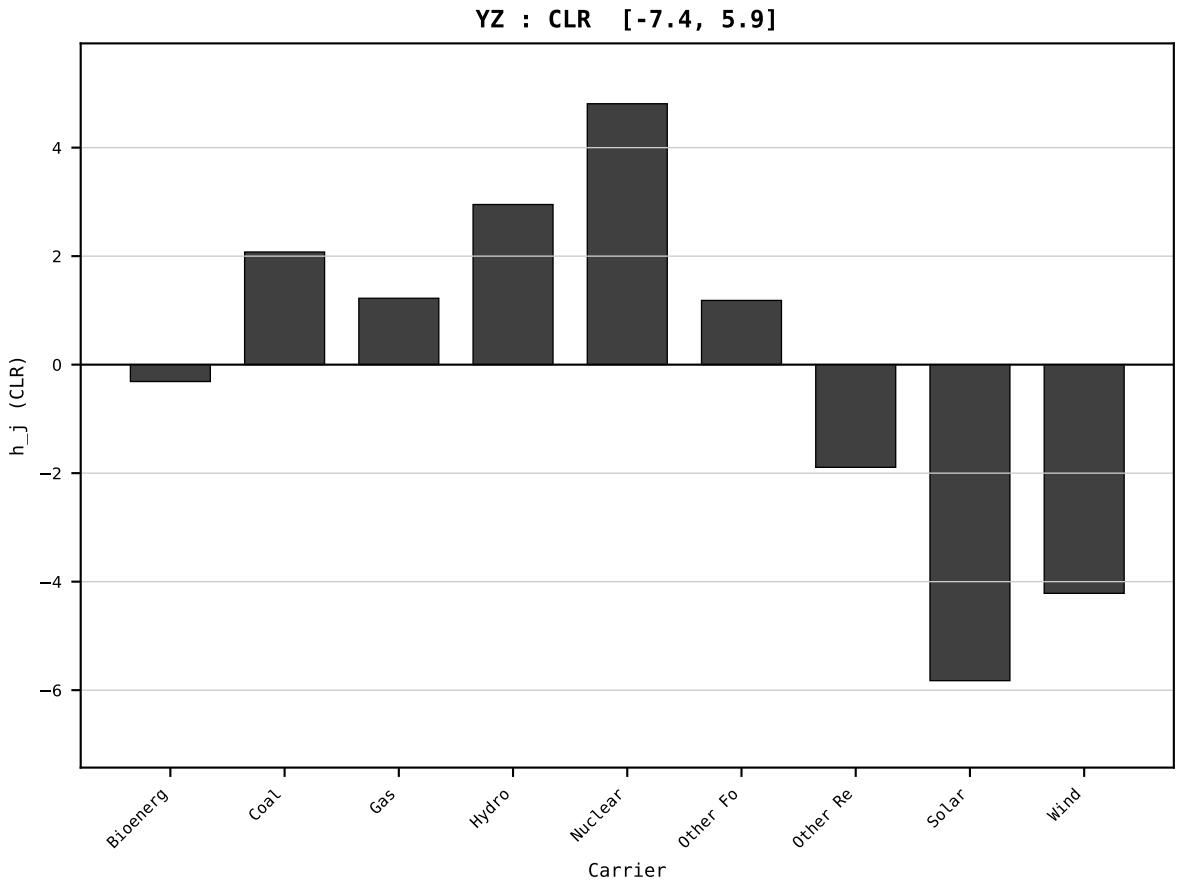
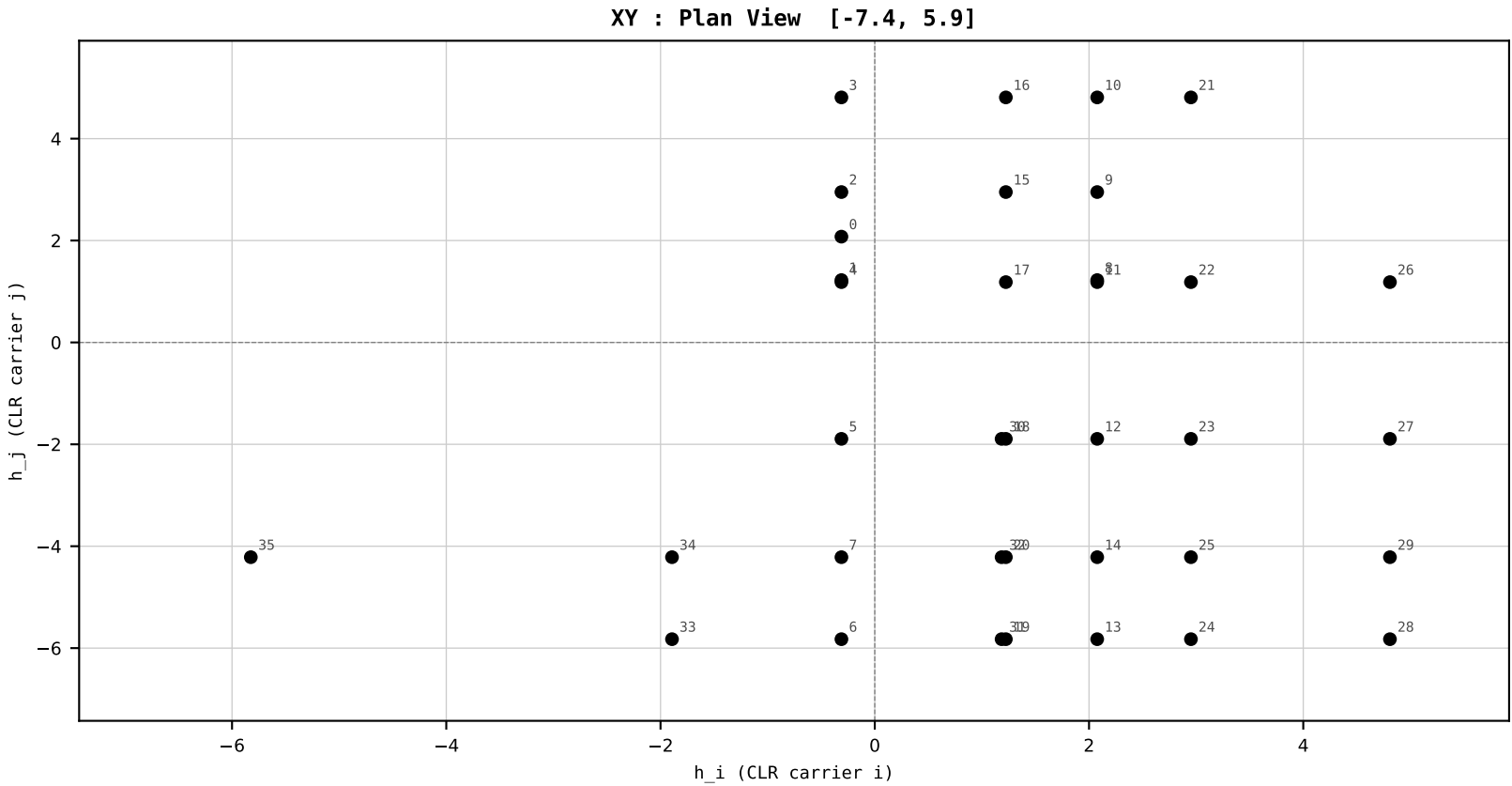
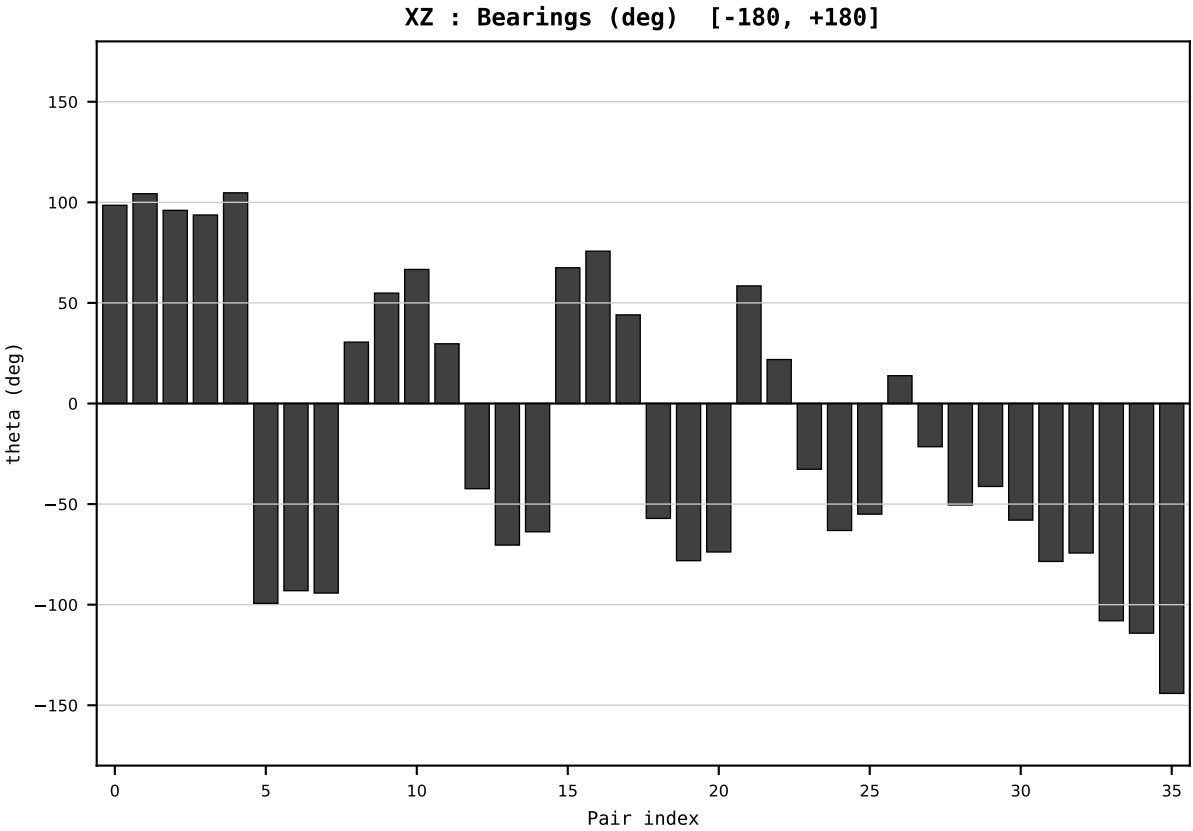
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636980
Ring = Hs-4
E_metric= 94.4242
kappa_HS= 41516.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.634 HLR
DR ratio= 41516.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

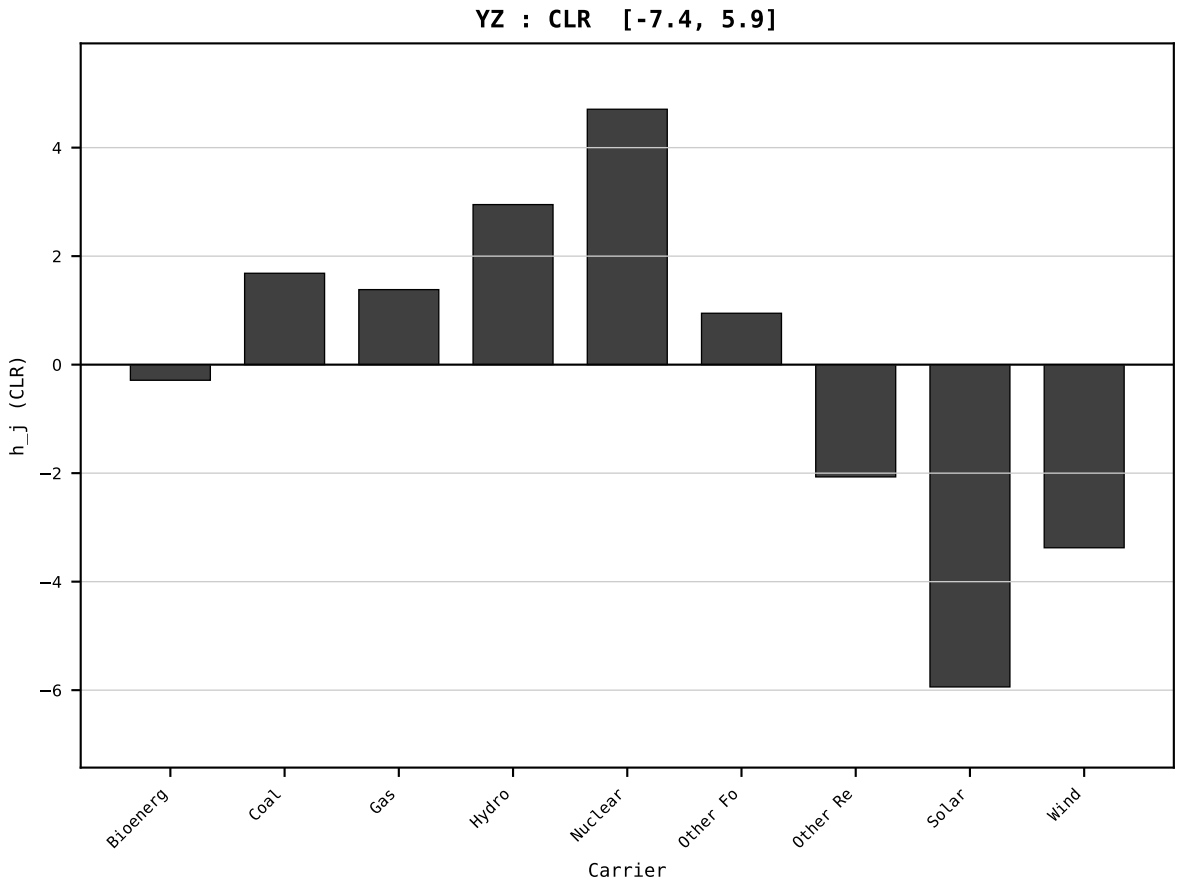
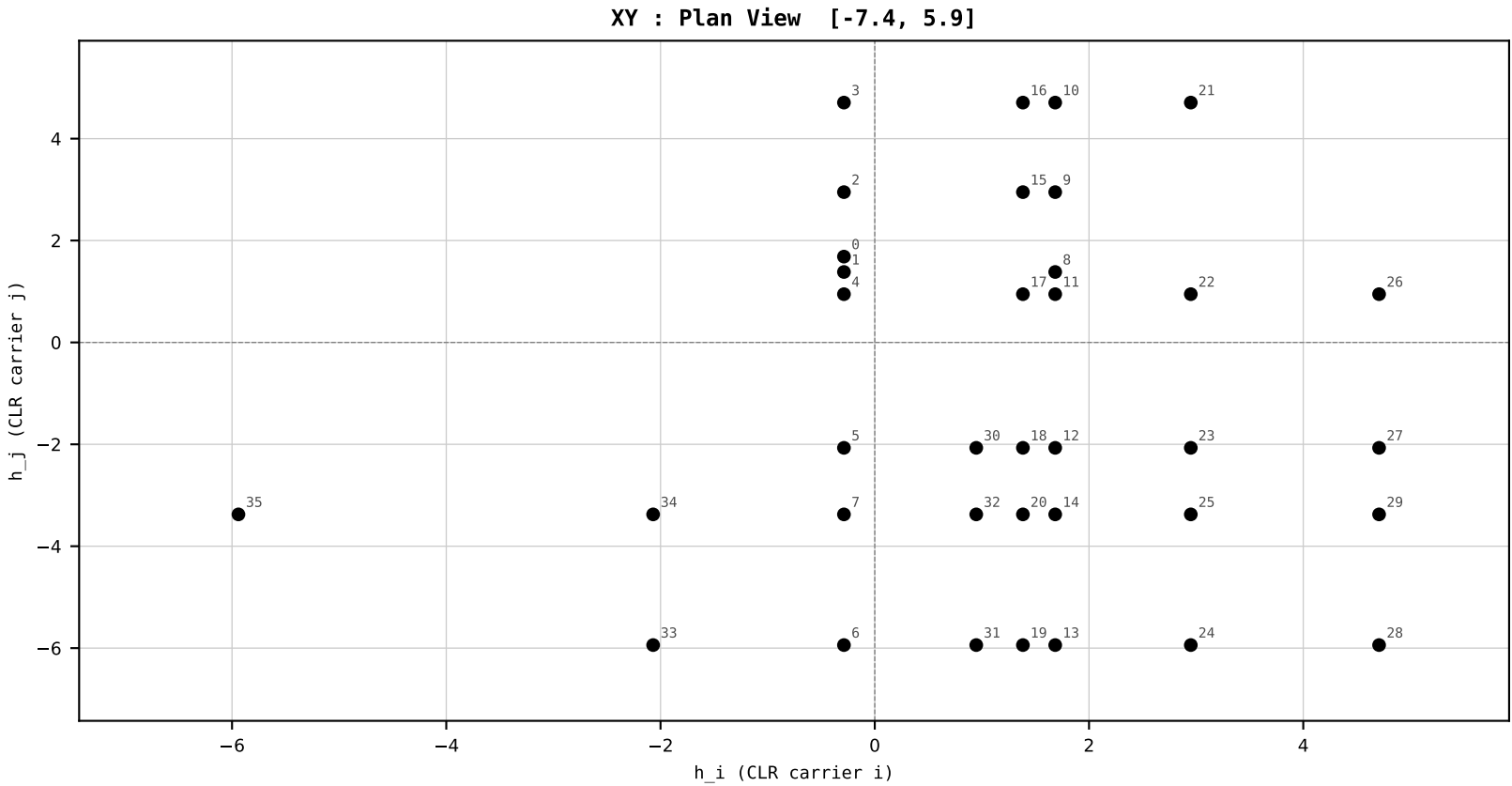
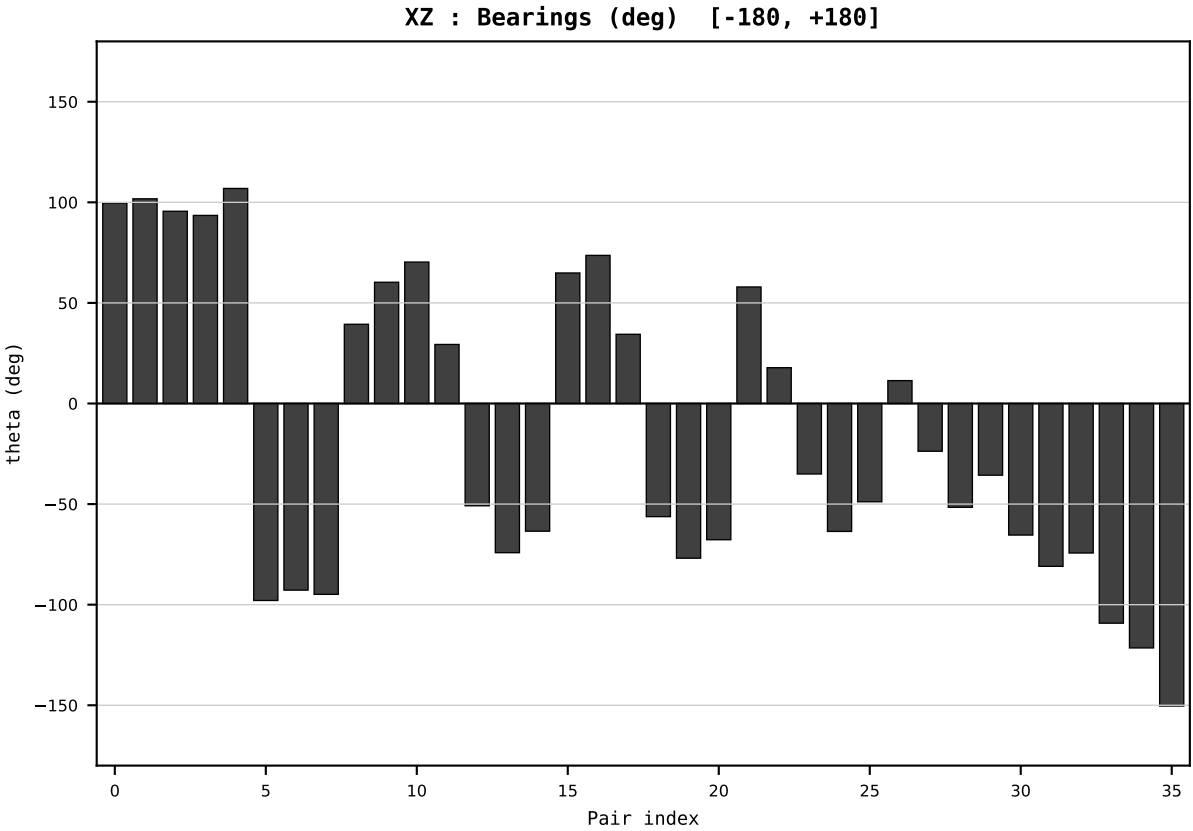
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636809
Ring = Hs-4
E_metric= 87.5562
kappa_HS= 42108.00
omega = 5.5939 deg
d_A = 0.997829
Helm = Wind
Helm d = +0.839979
DR = 10.648 HLR
DR ratio= 42108.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

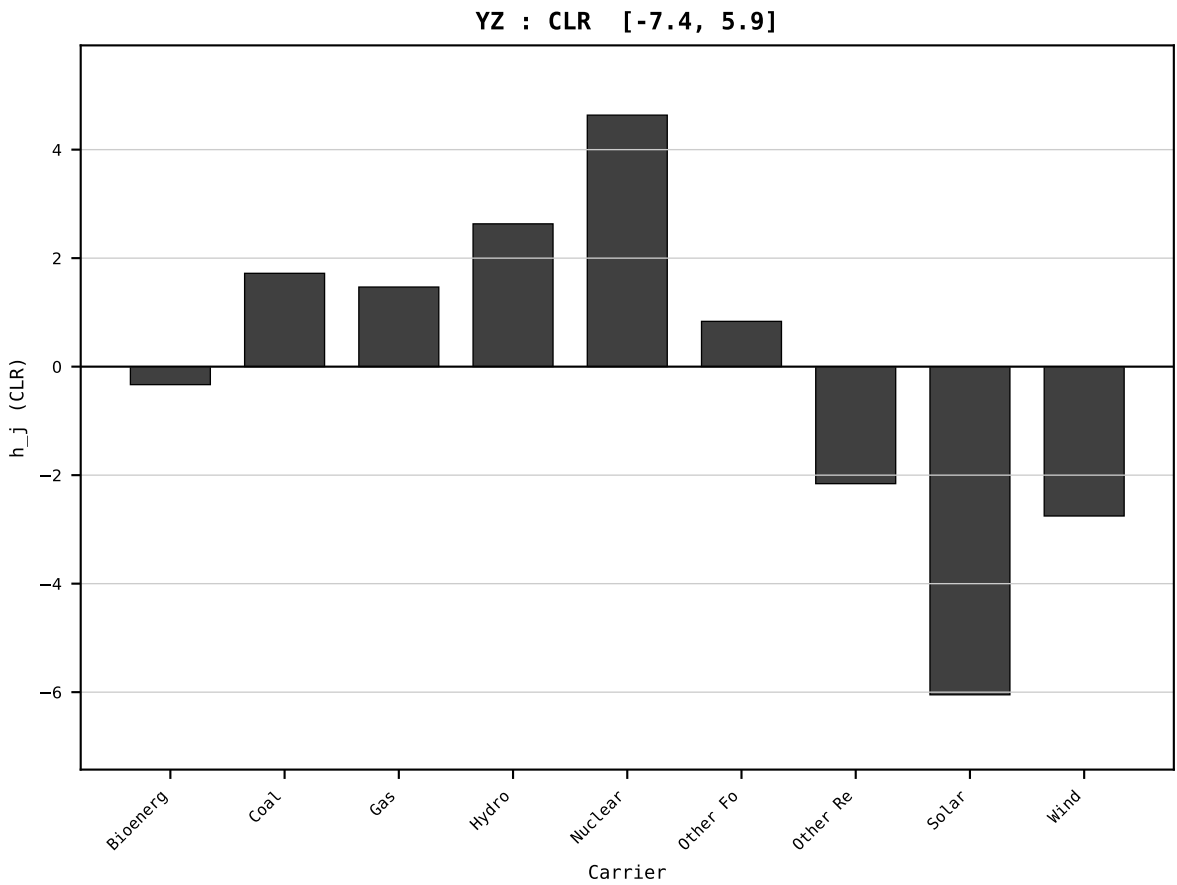
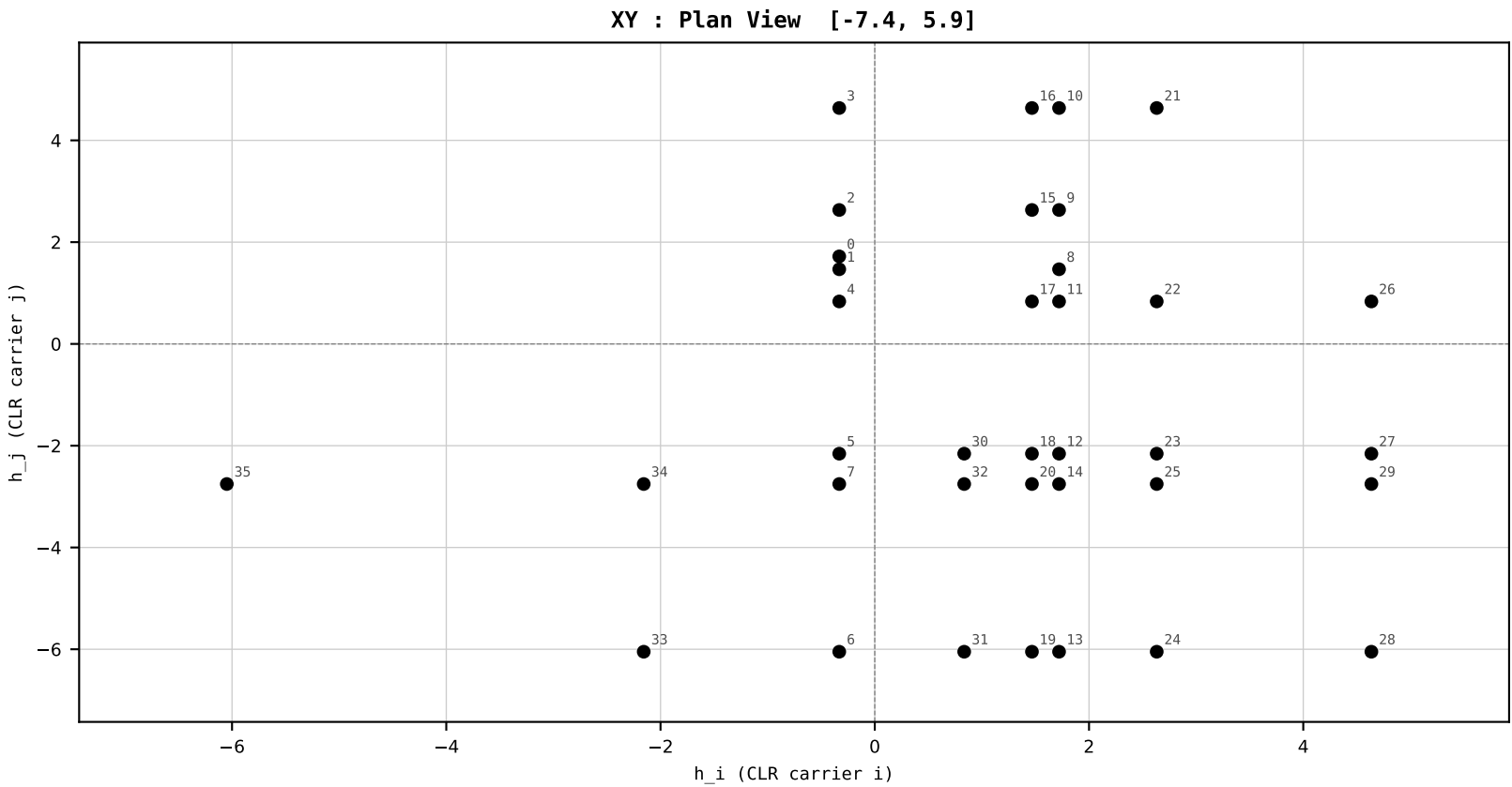
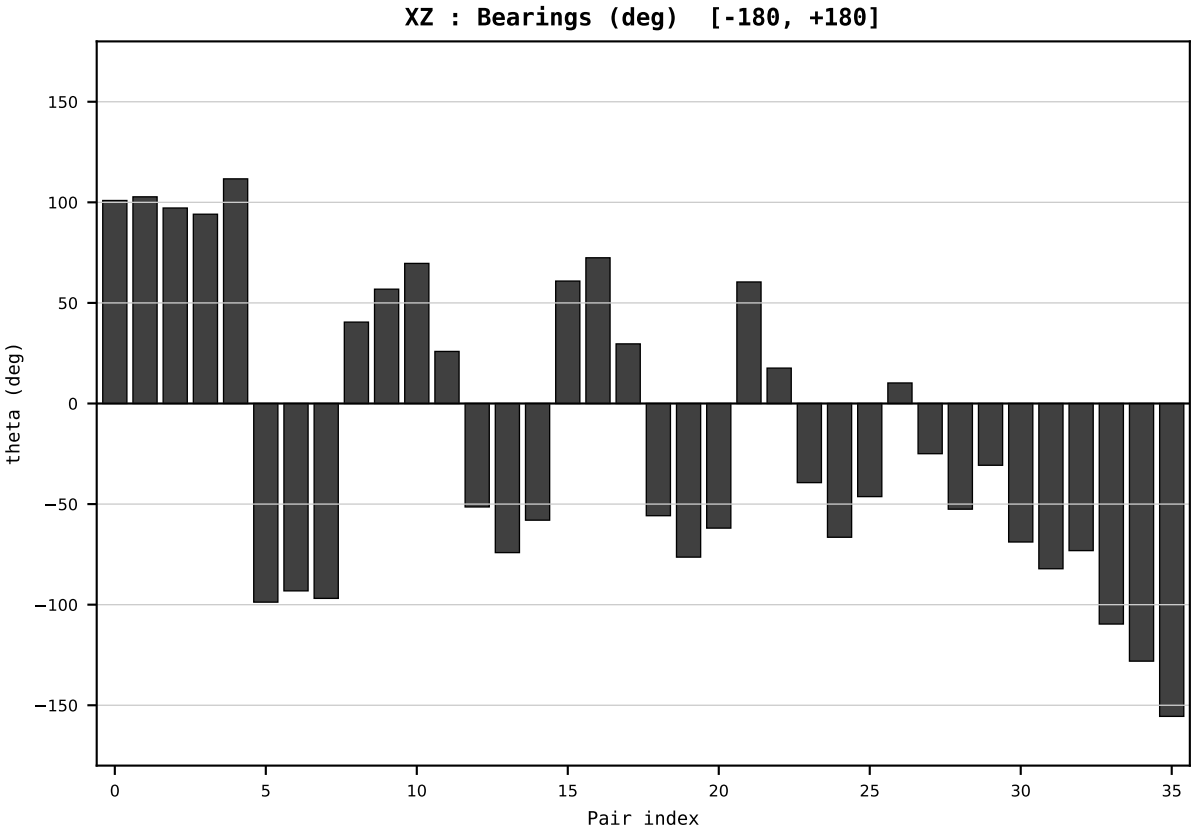
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.644046
Ring = Hs-4
E_metric= 83.1543
kappa_HS= 43676.00
omega = 4.2963 deg
d_A = 0.732324
Helm = Wind
Helm d = +0.622462
DR = 10.685 HLR
DR ratio= 43676.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

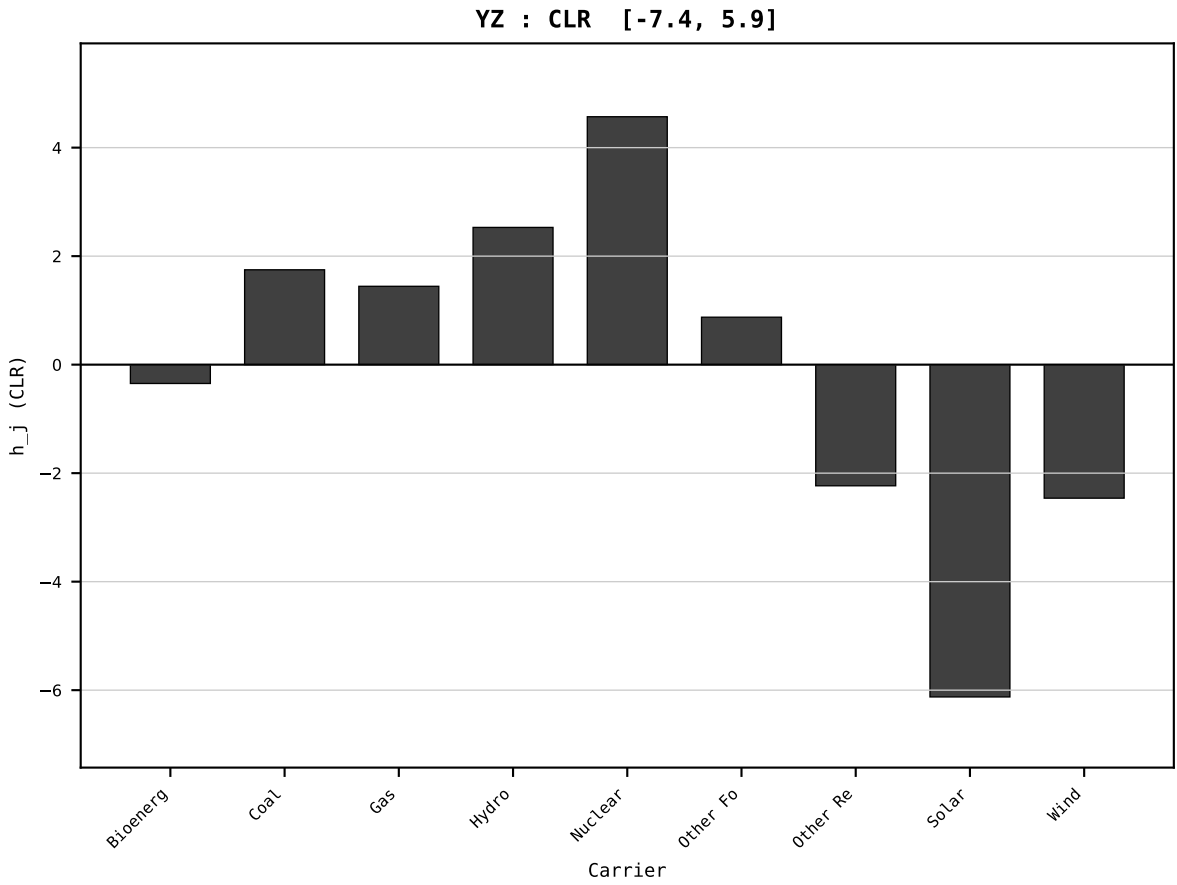
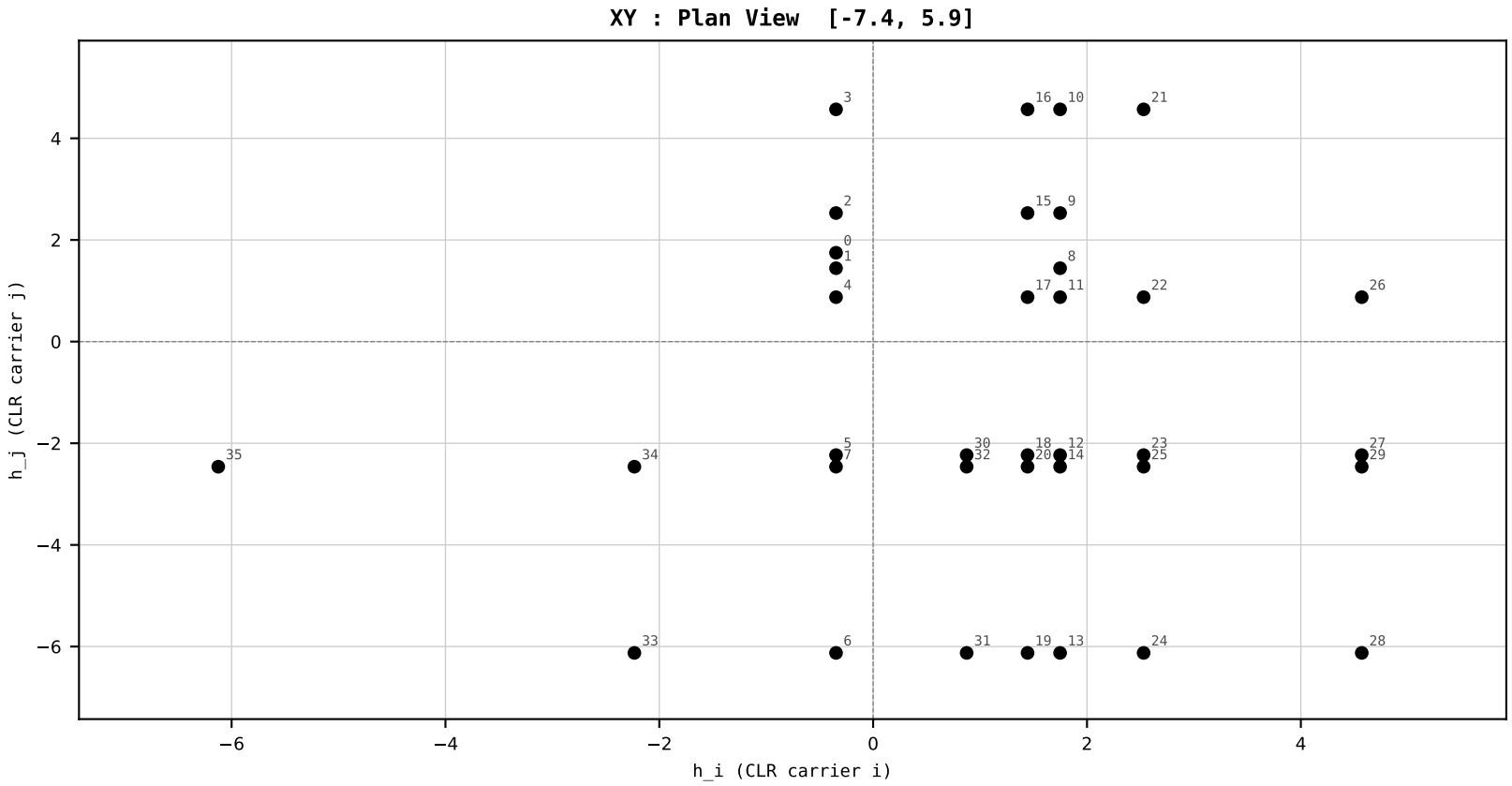
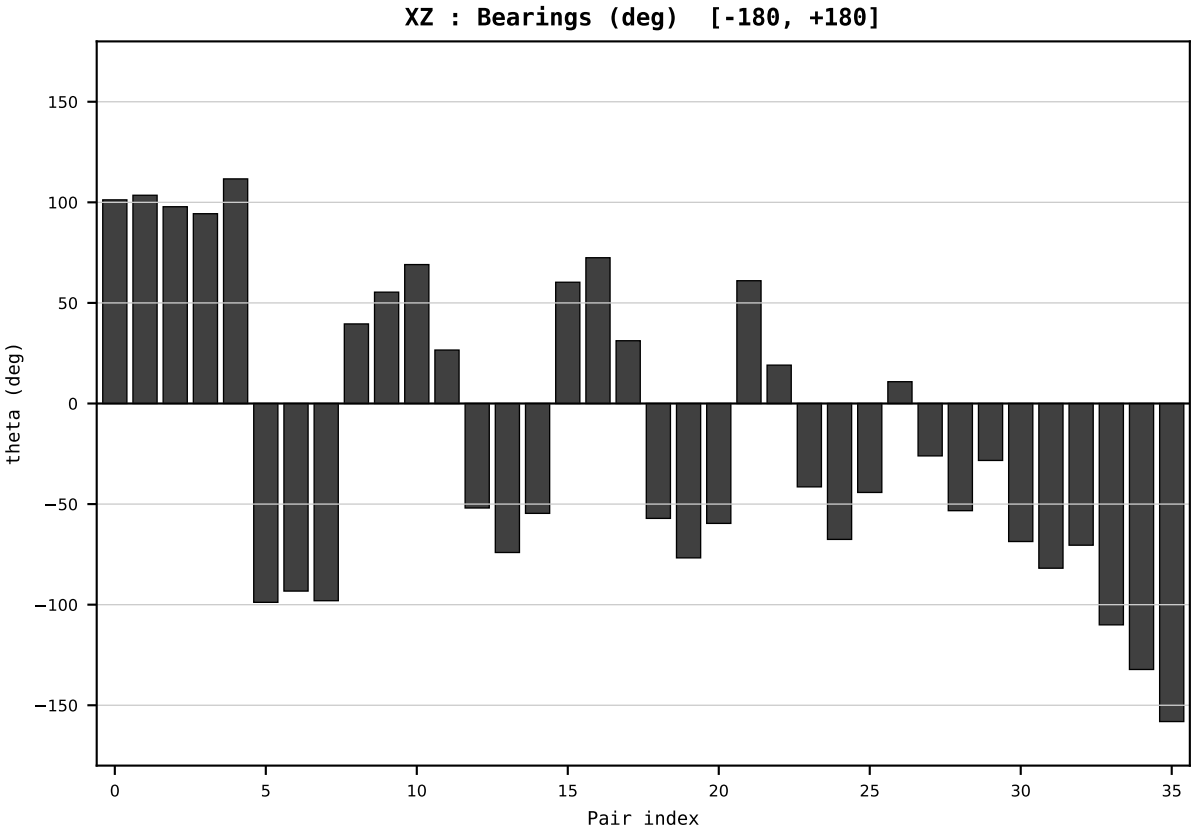
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636320
Ring = Hs-4
E_metric= 81.8645
kappa_HS= 44107.00
omega = 2.0886 deg
d_A = 0.338625
Helm = Wind
Helm d = +0.291606
DR = 10.694 HLR
DR ratio= 44107.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

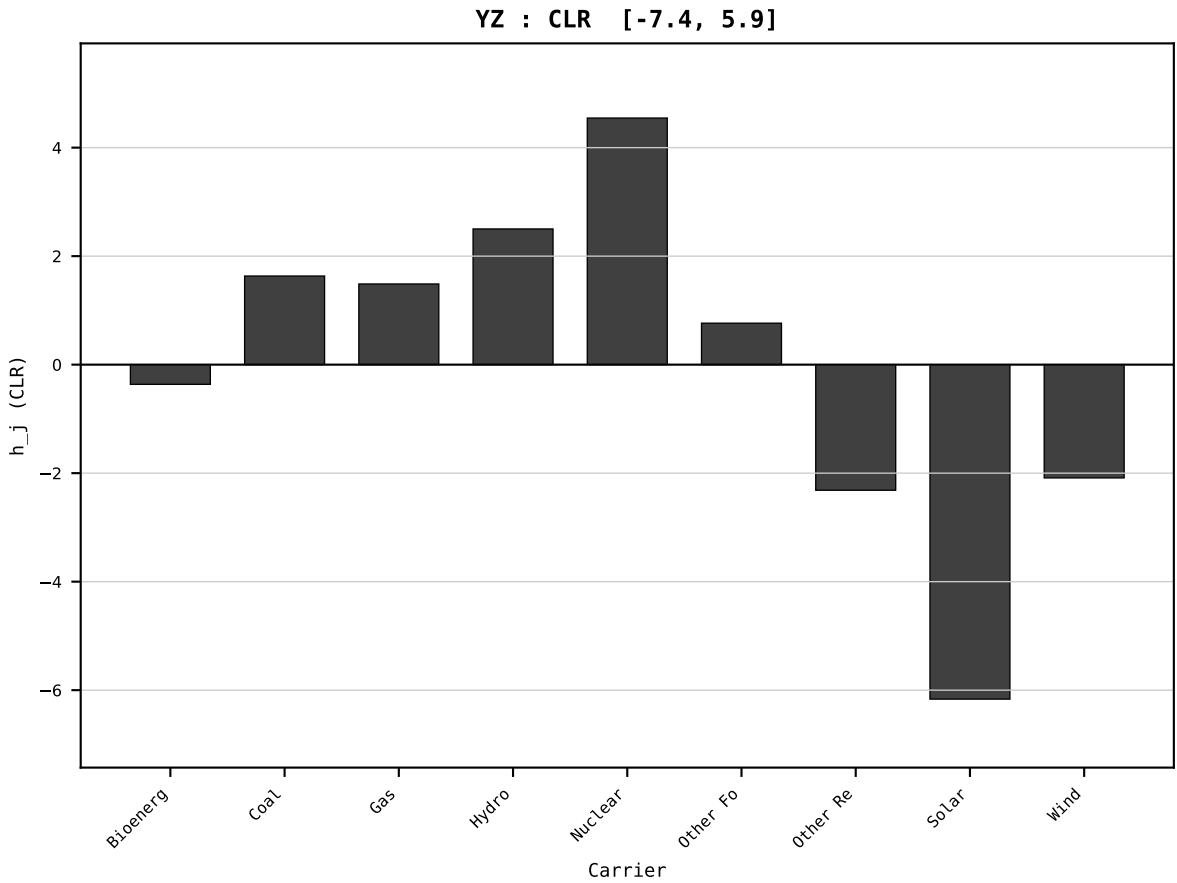
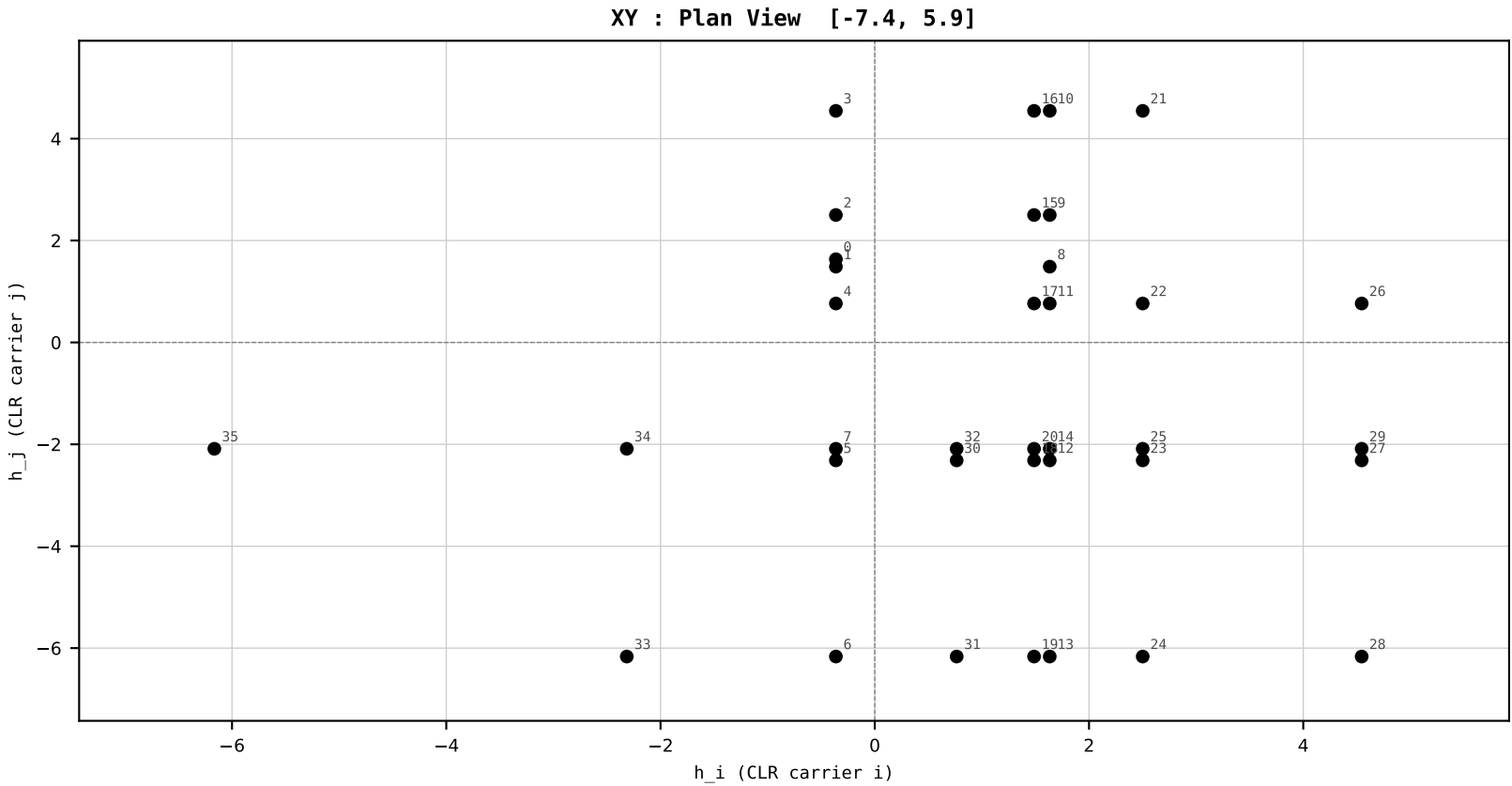
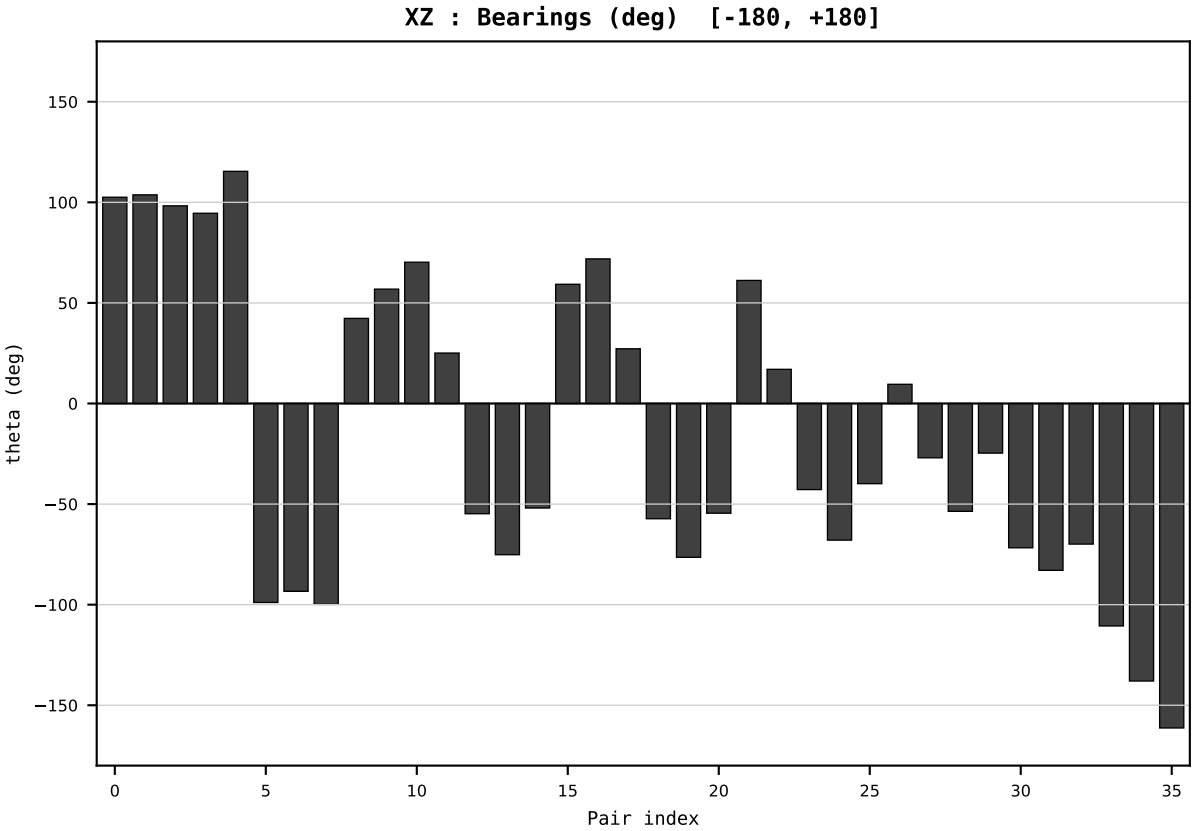
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.639536
Ring = Hs-4
E_metric= 80.2373
kappa_HS= 44824.00
omega = 2.6139 deg
d_A = 0.420503
Helm = Wind
Helm d = +0.373595
DR = 10.710 HLR
DR ratio= 44824.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

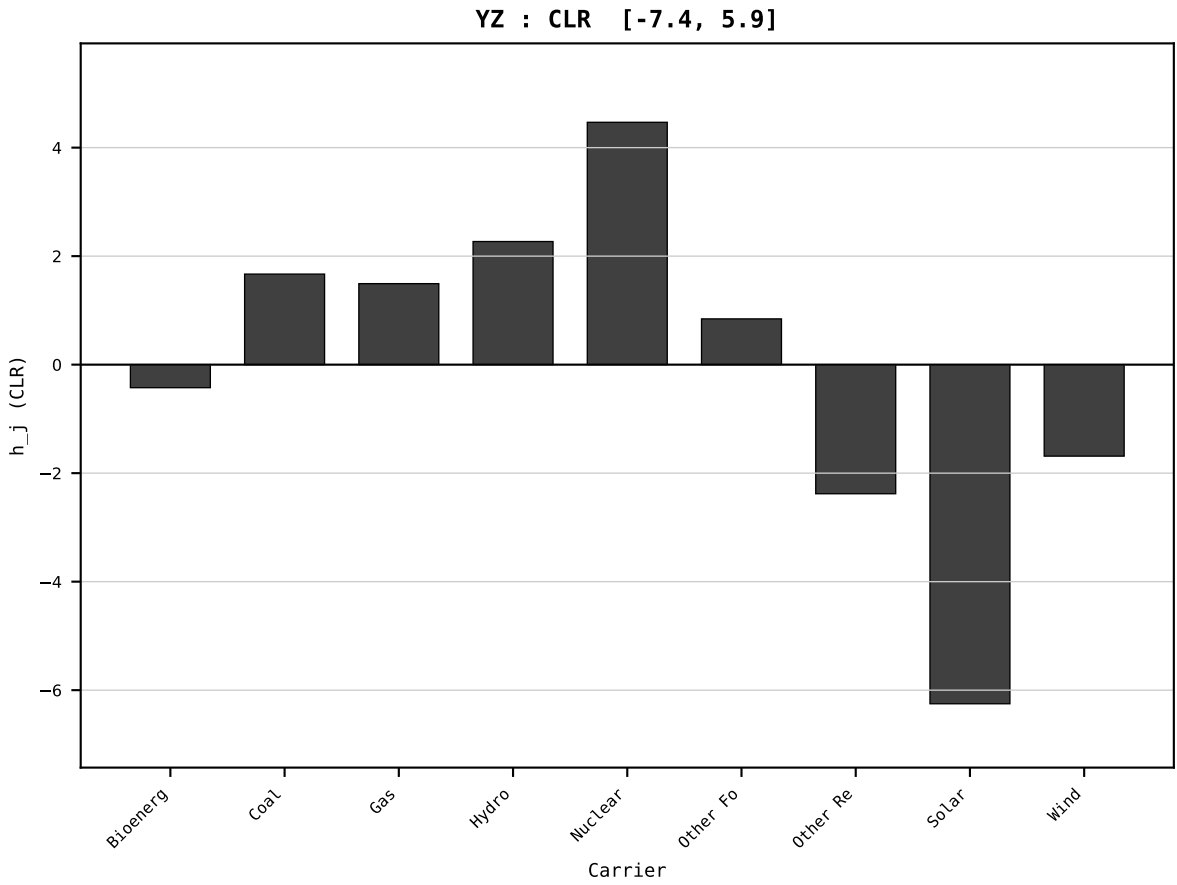
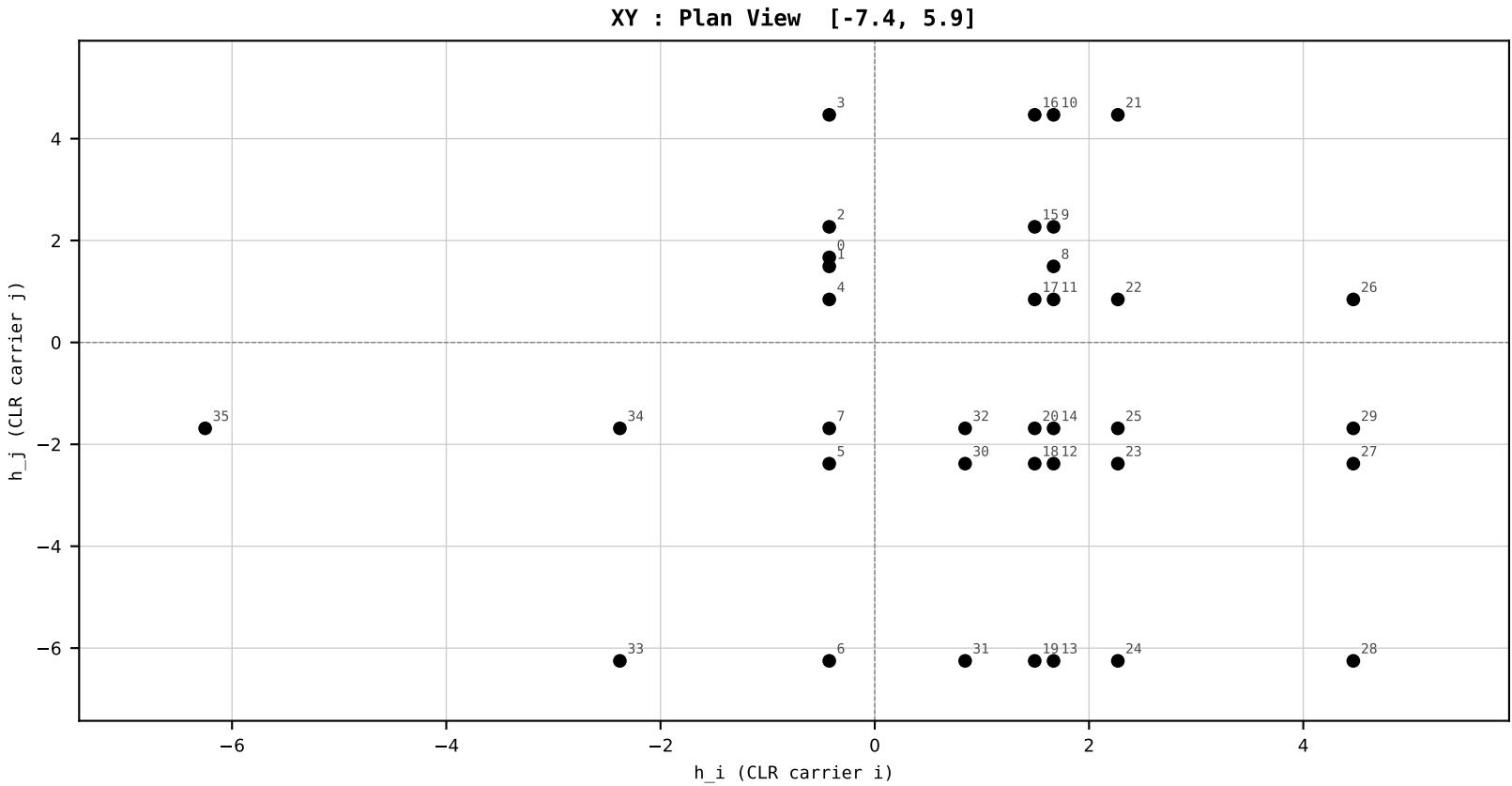
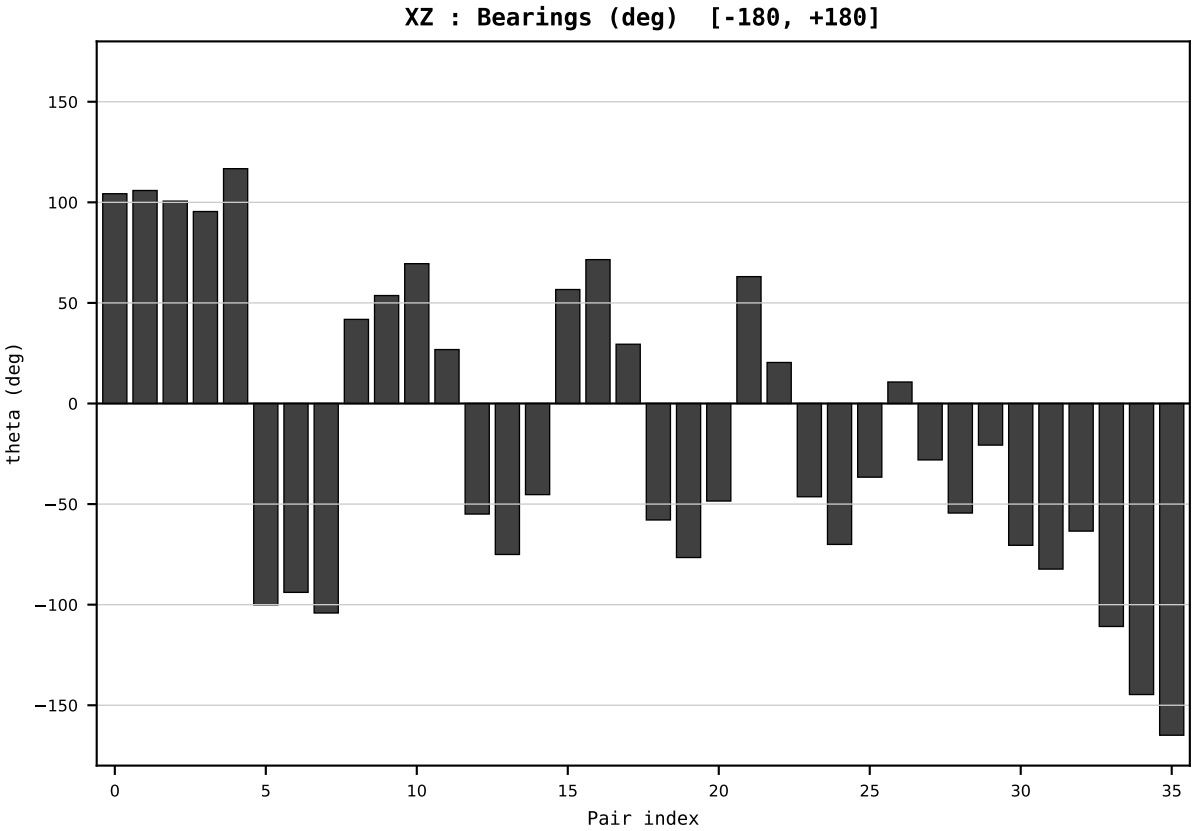
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.634489
Ring = Hs-4
E_metric= 78.5884
kappa_HS= 45153.00
omega = 3.1185 deg
d_A = 0.493709
Helm = Wind
Helm d = +0.401223
DR = 10.718 HLR
DR ratio= 45153.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

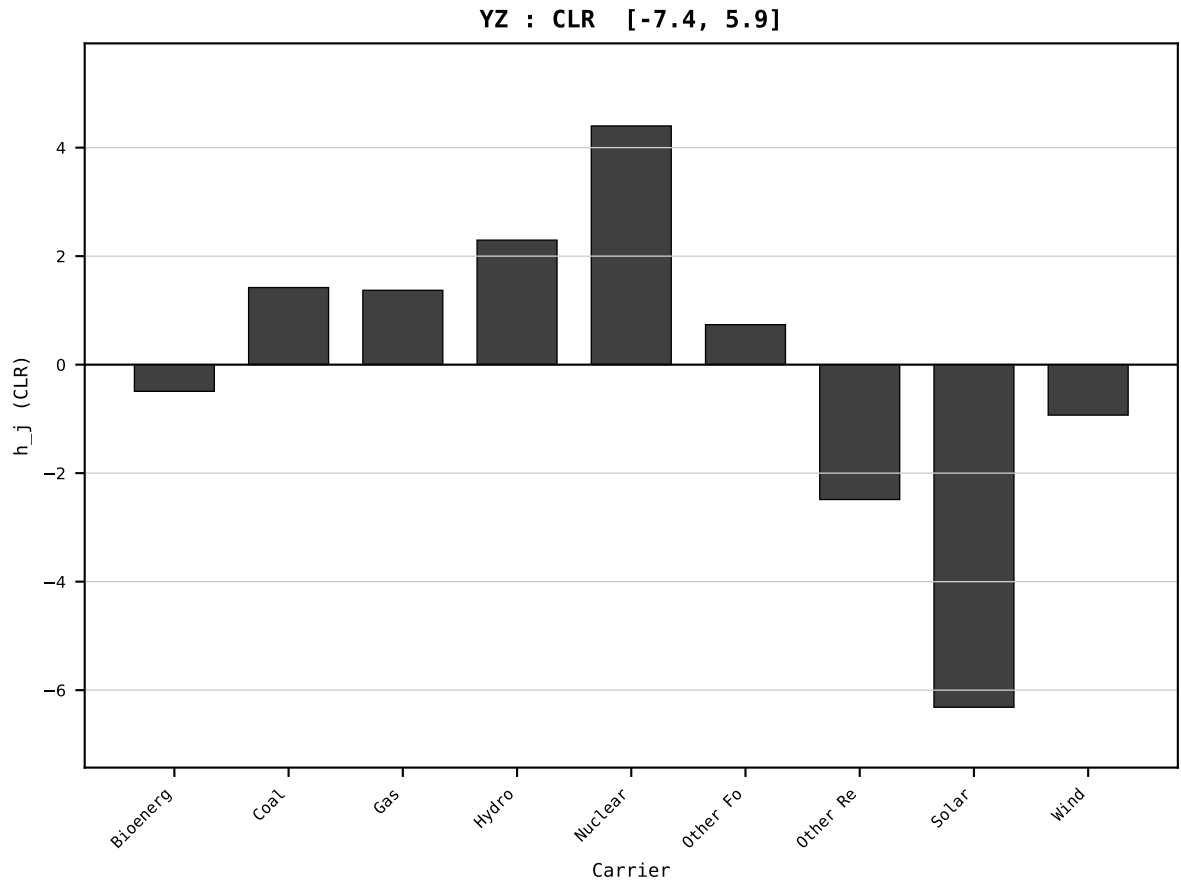
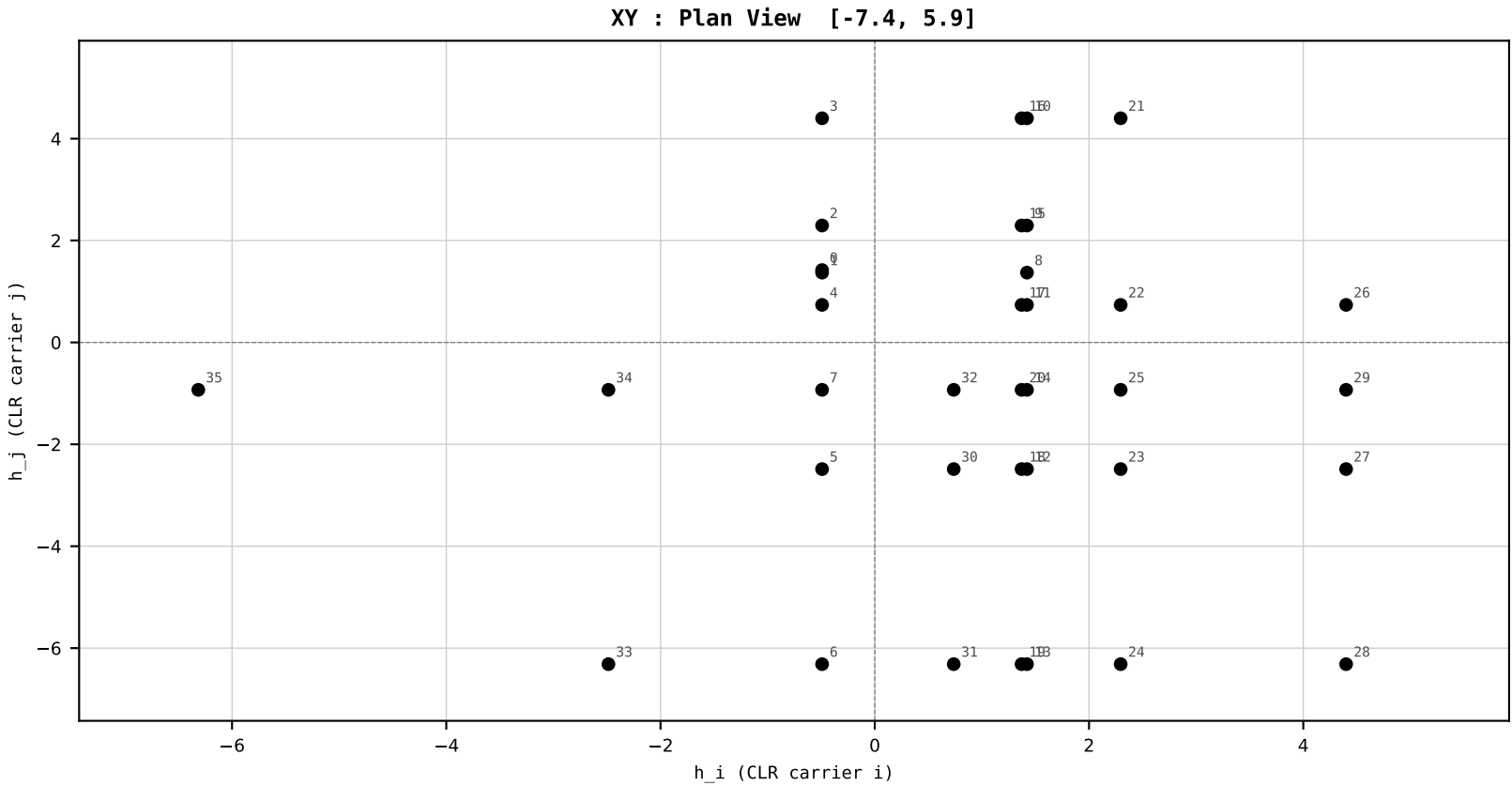
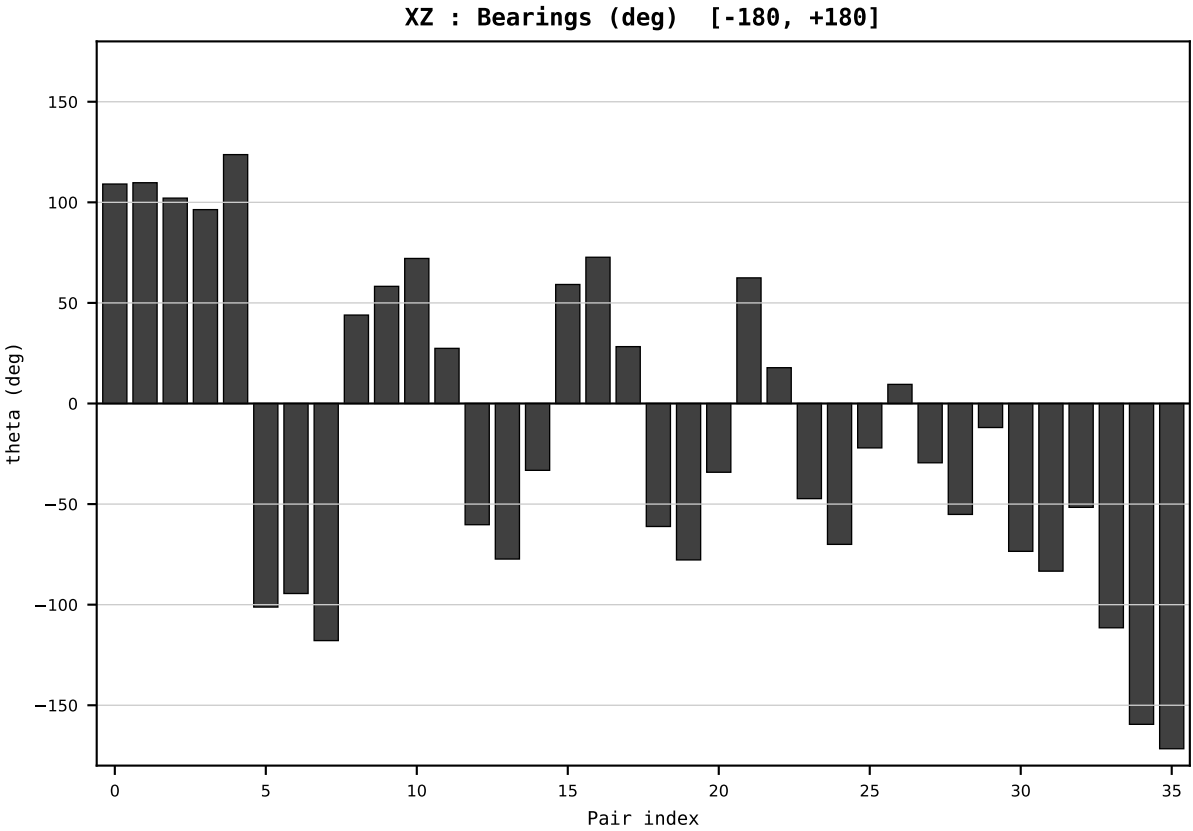
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.635362
Ring = Hs-4
E_metric= 76.2355
kappa_HS= 45019.00
omega = 5.3196 deg
d_A = 0.827413
Helm = Wind
Helm d = +0.756002
DR = 10.715 HLR
DR ratio= 45019.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

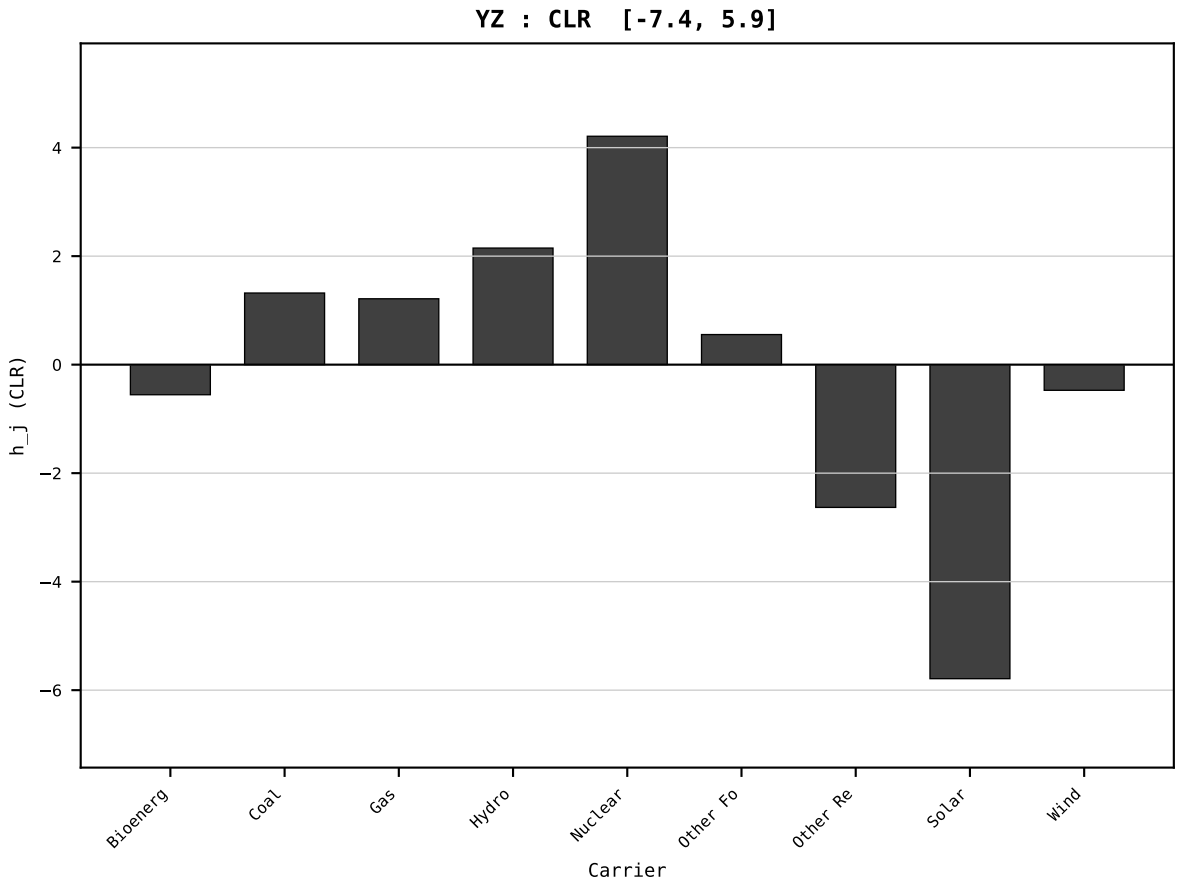
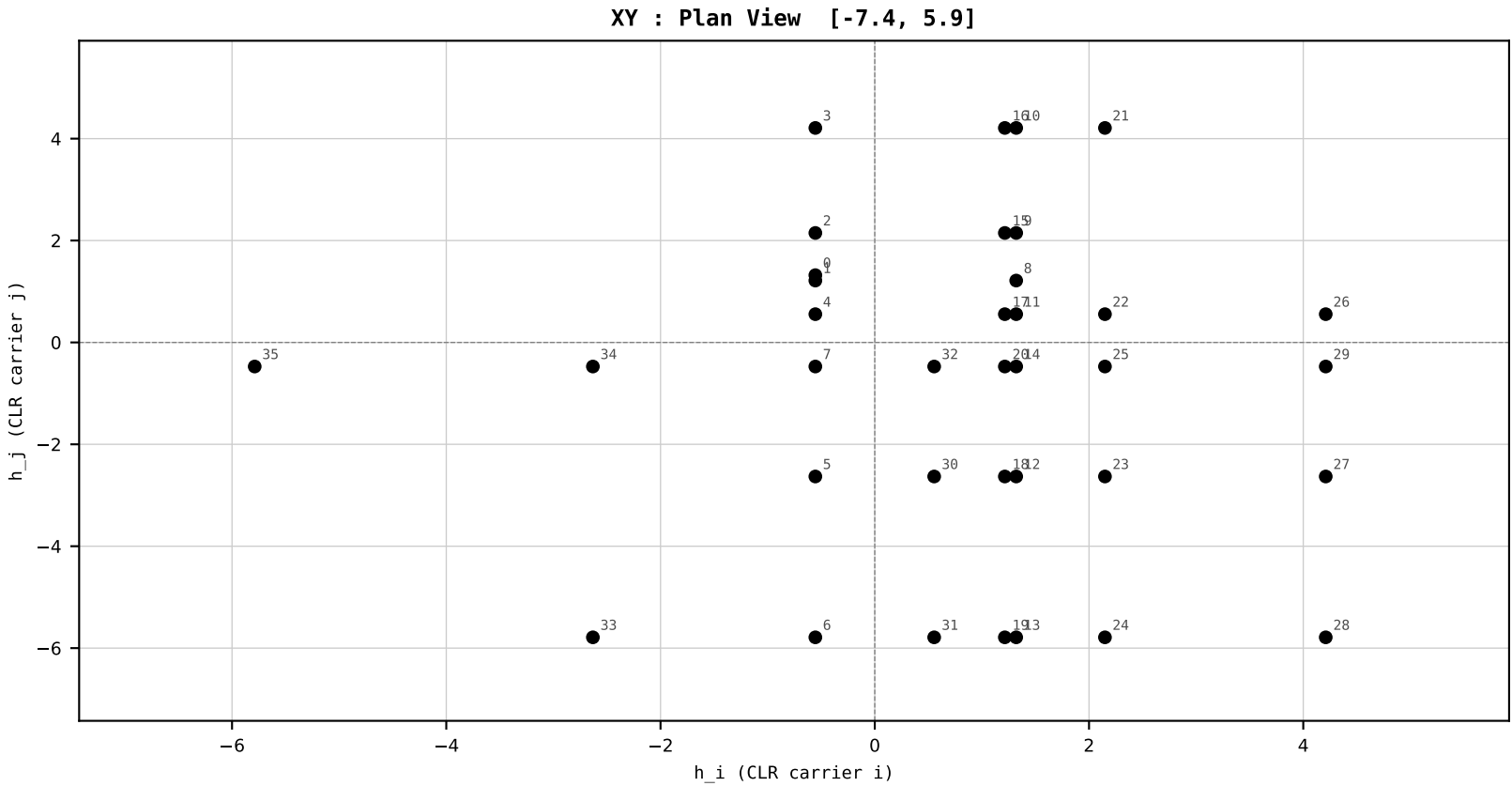
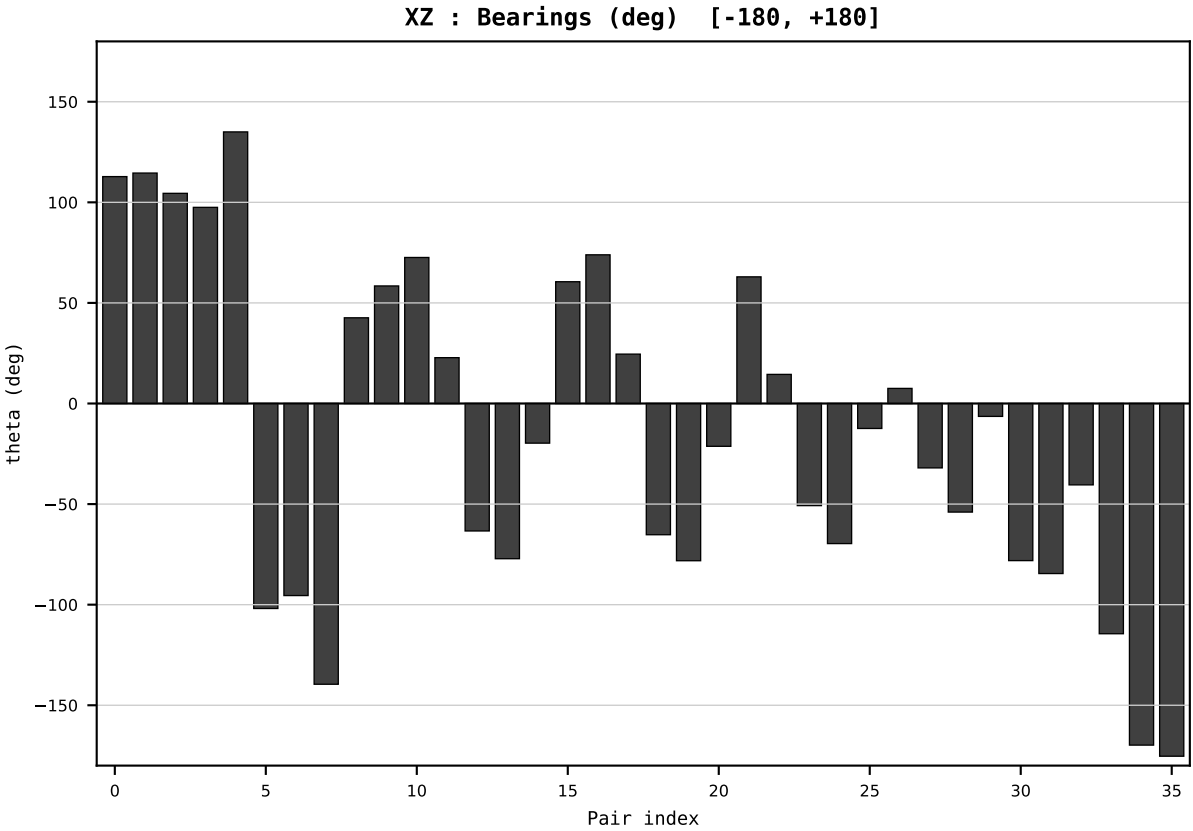
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.618062
Ring = Hs-4
E_metric= 66.8255
kappa_HS= 21986.50
omega = 3.8806 deg
d_A = 0.798187
Helm = Solar
Helm d = +0.526469
DR = 9.998 HLR
DR ratio= 21986.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

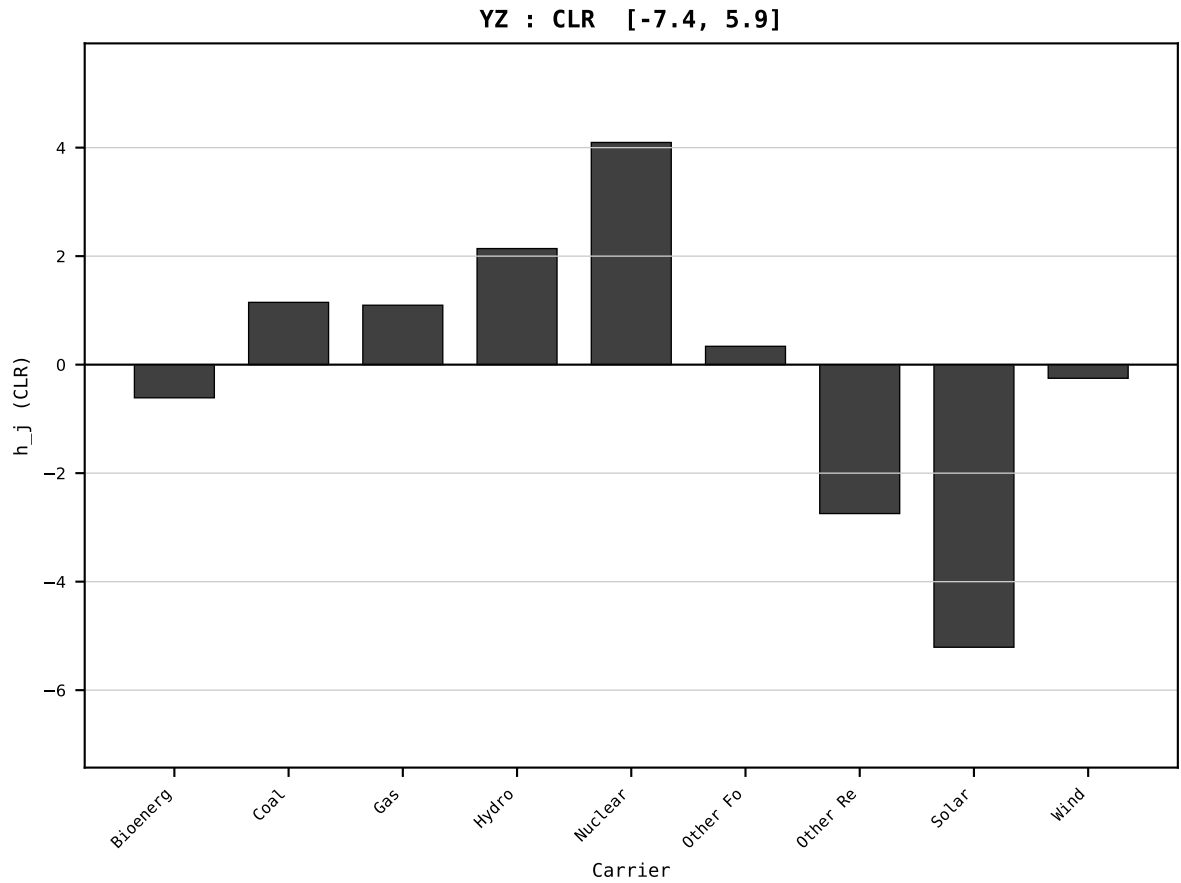
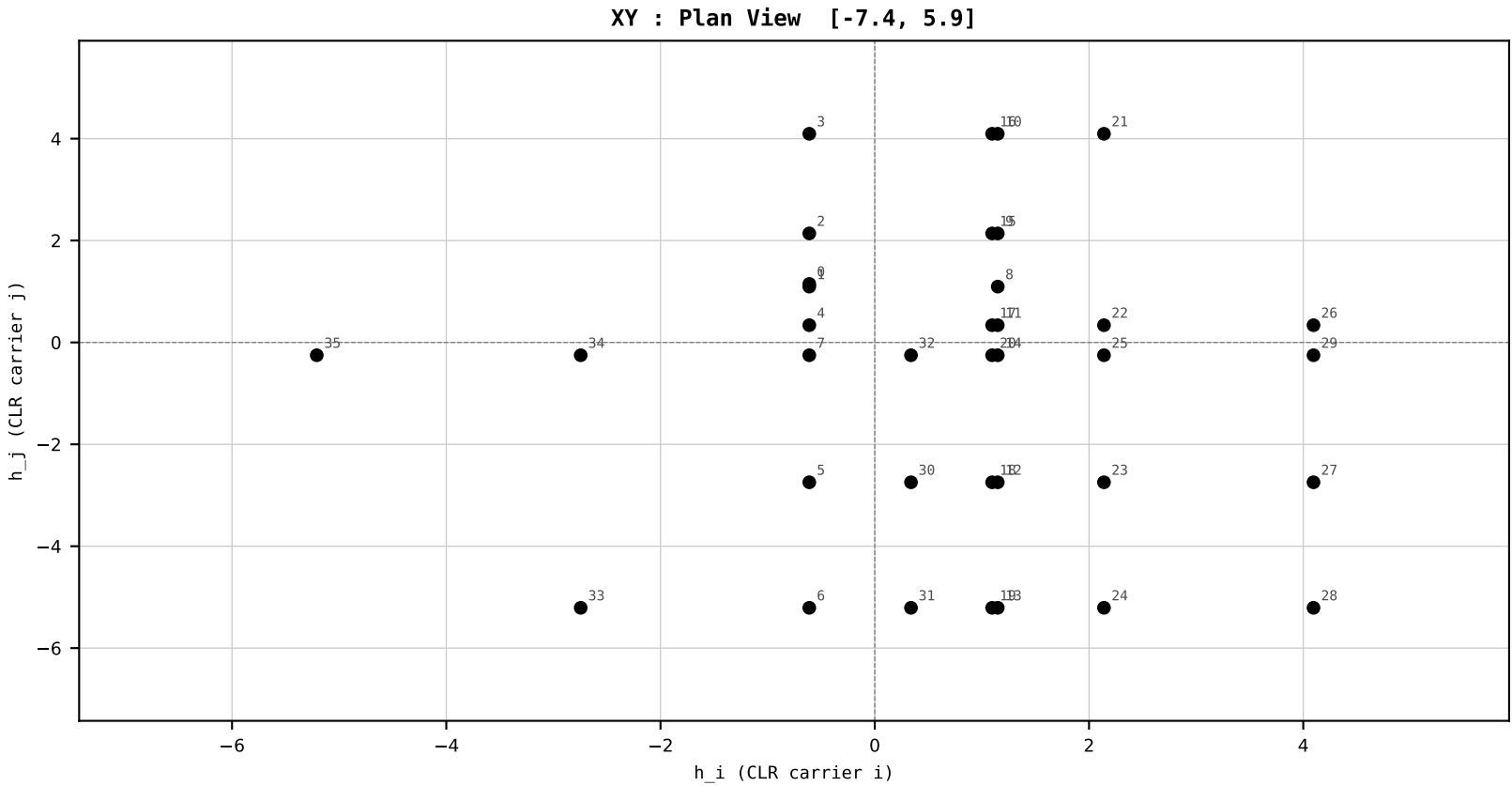
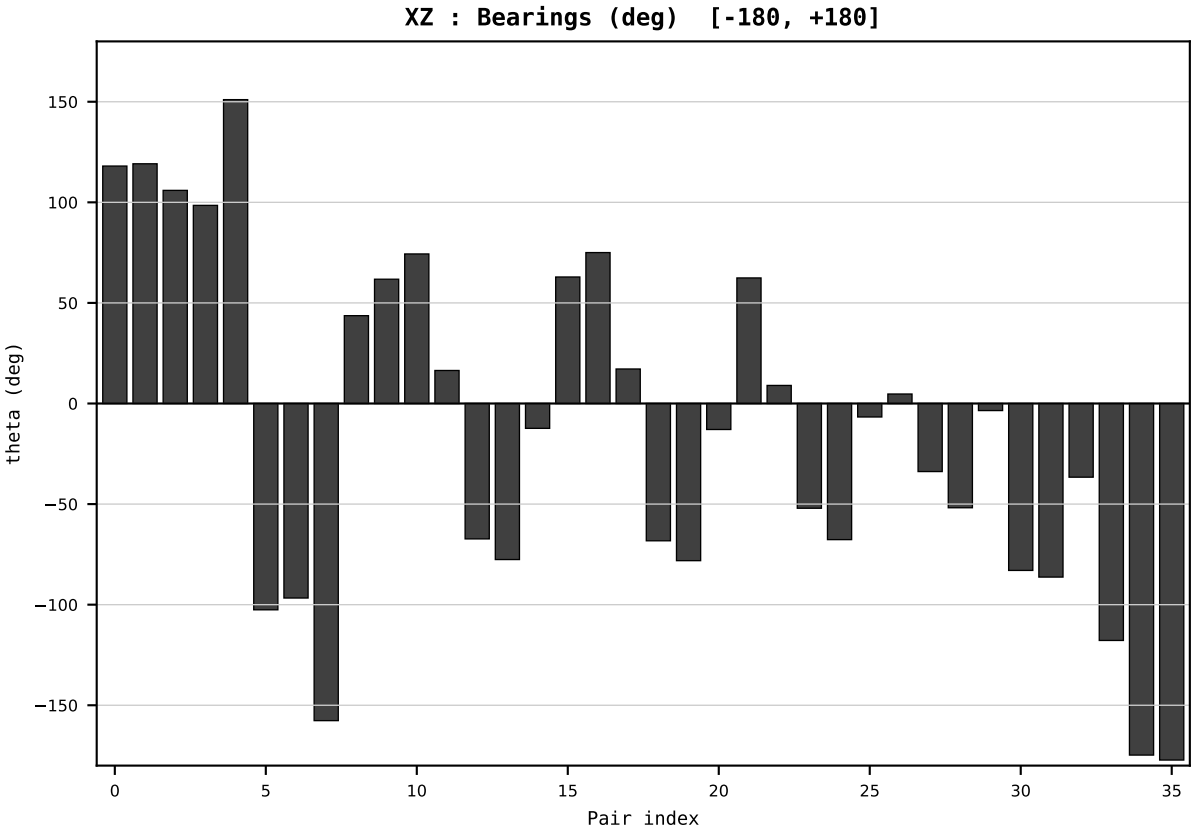
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.610524
Ring = Hs-4
E_metric= 59.0888
kappa_HS= 10986.25
omega = 3.7337 deg
d_A = 0.710392
Helm = Solar
Helm d = +0.579497
DR = 9.304 HLR
DR ratio= 10986.3x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=9 N=26 pairs=36

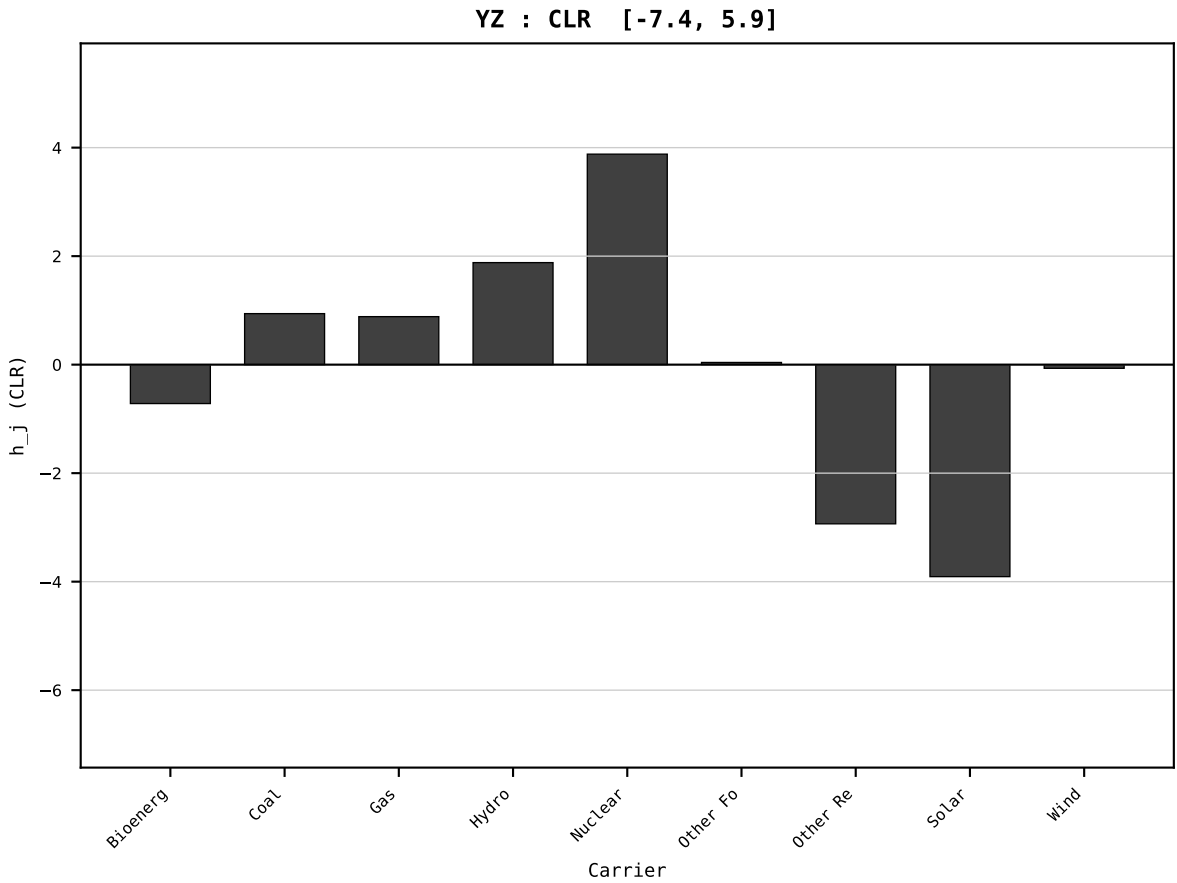
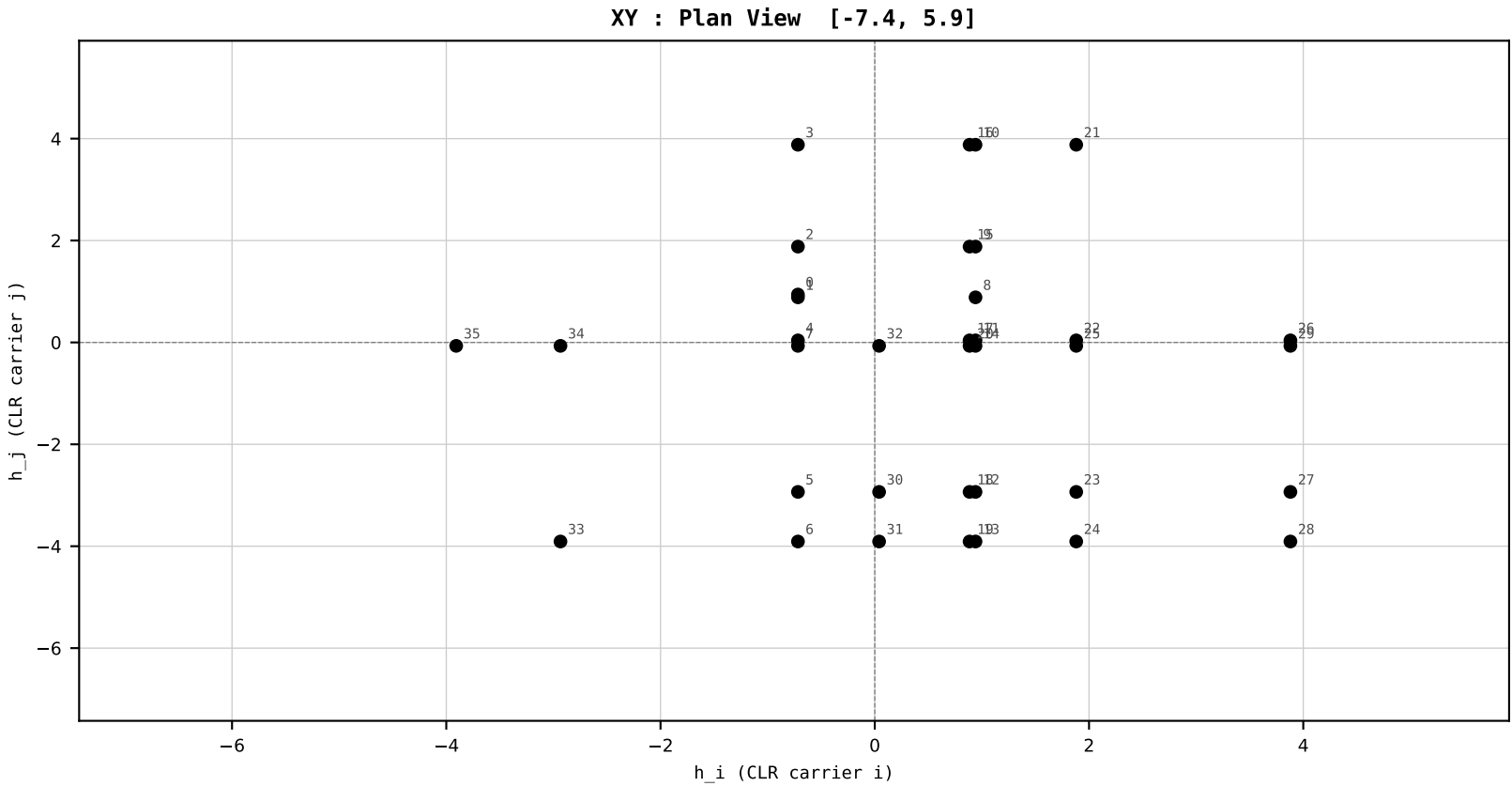
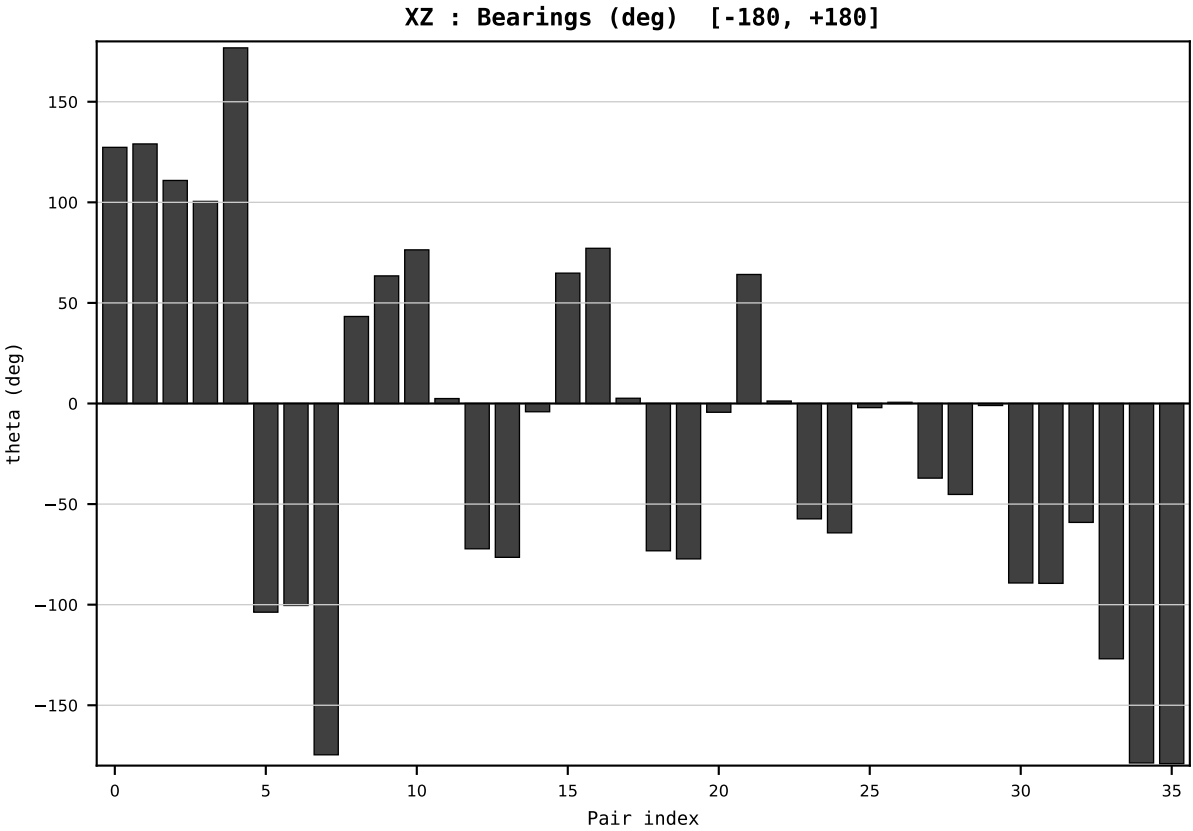
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.604621
Ring = Hs-4
E_metric= 44.6563
kappa_HS= 2410.24
omega = 8.2198 deg
d_A = 1.436738
Helm = Solar
Helm d = +1.301456
DR = 7.787 HLR
DR ratio= 2410.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



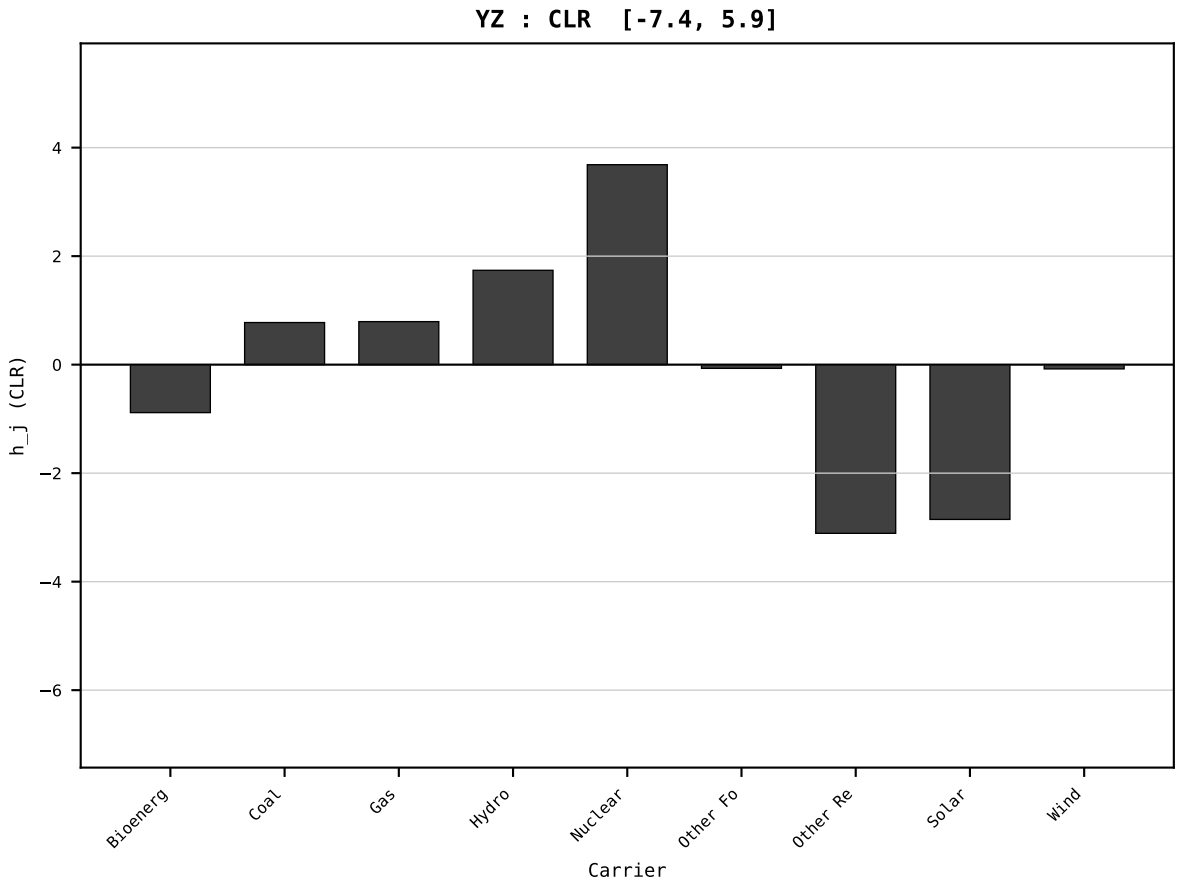
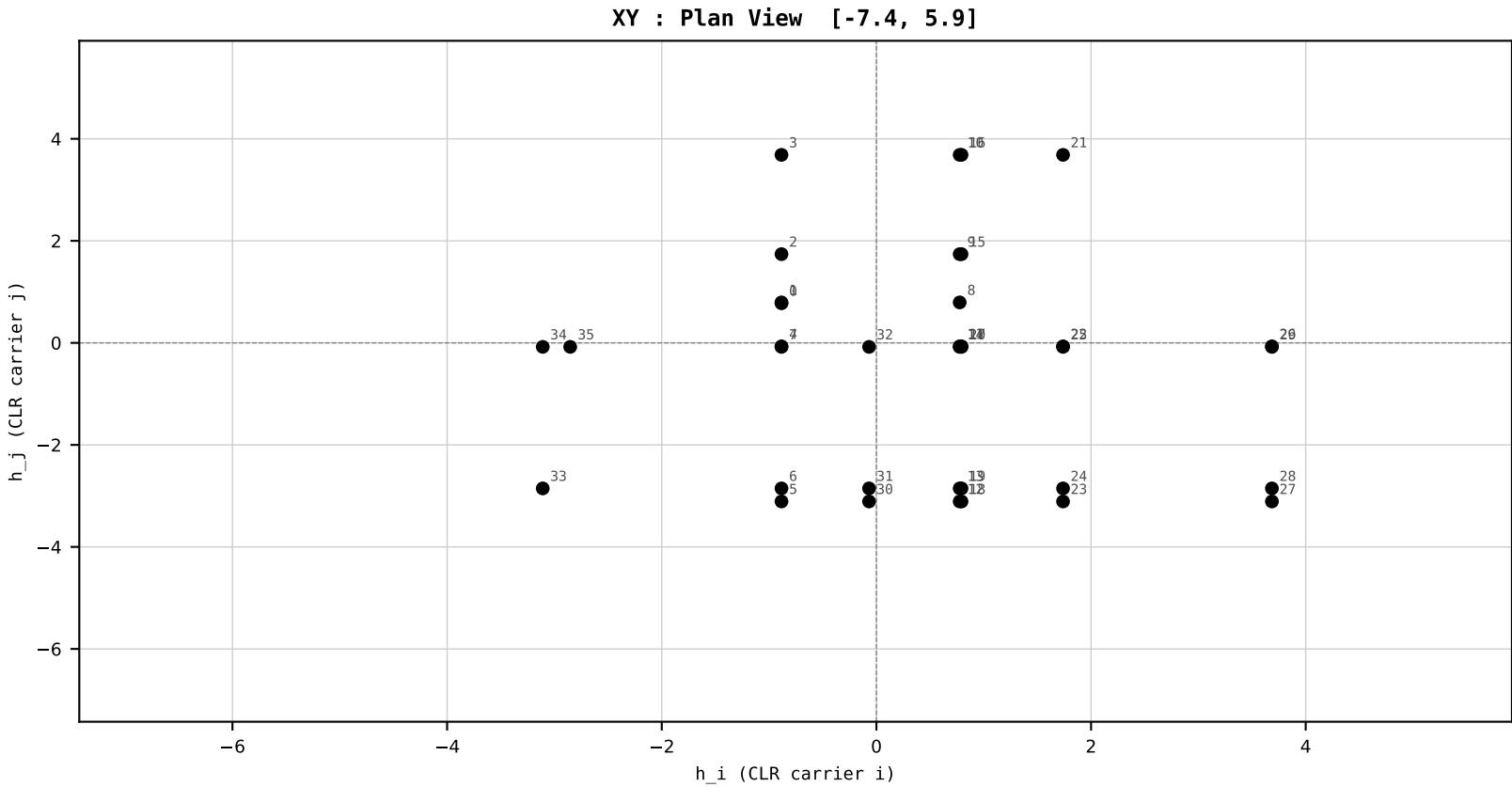
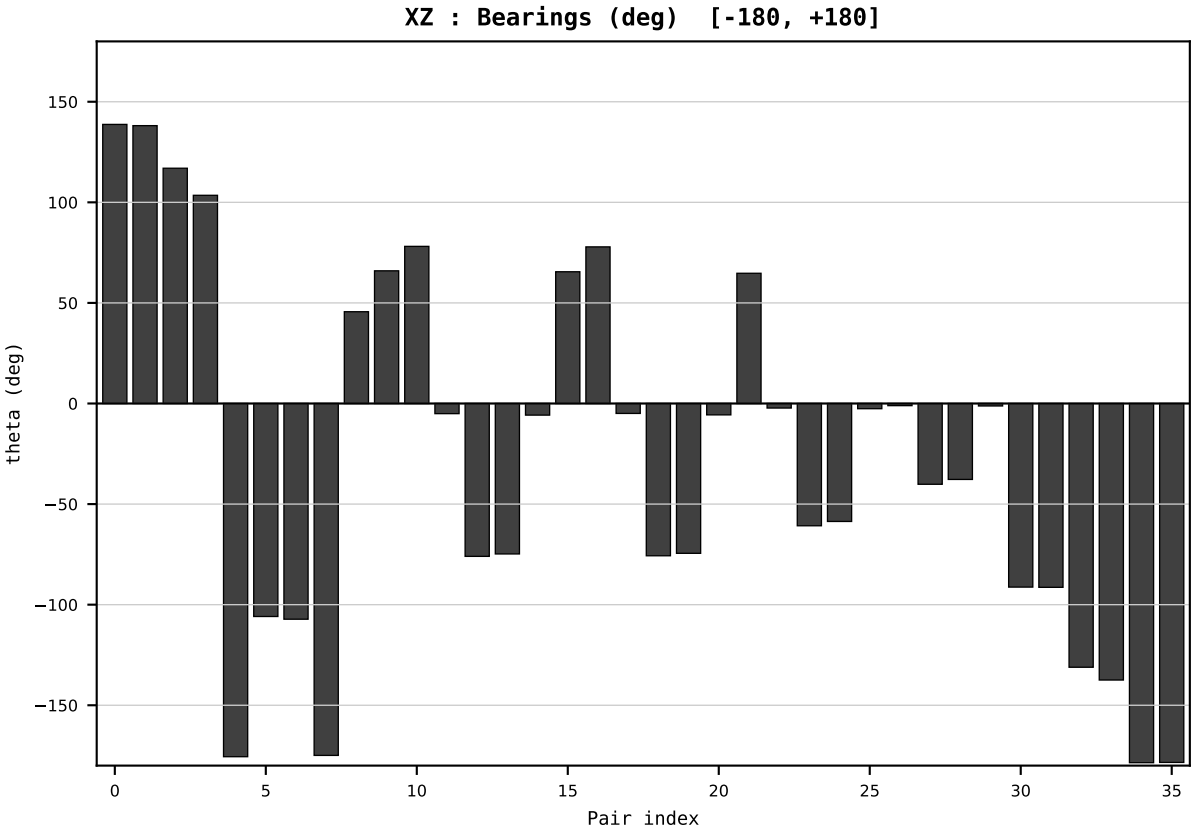
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.586166
Ring = Hs-4
E_metric= 36.4358
kappa_HS= 892.75
omega = 8.3636 deg
d_A = 1.129480
Helm = Solar
Helm d = +1.054432
DR = 6.794 HLR
DR ratio= 892.7x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

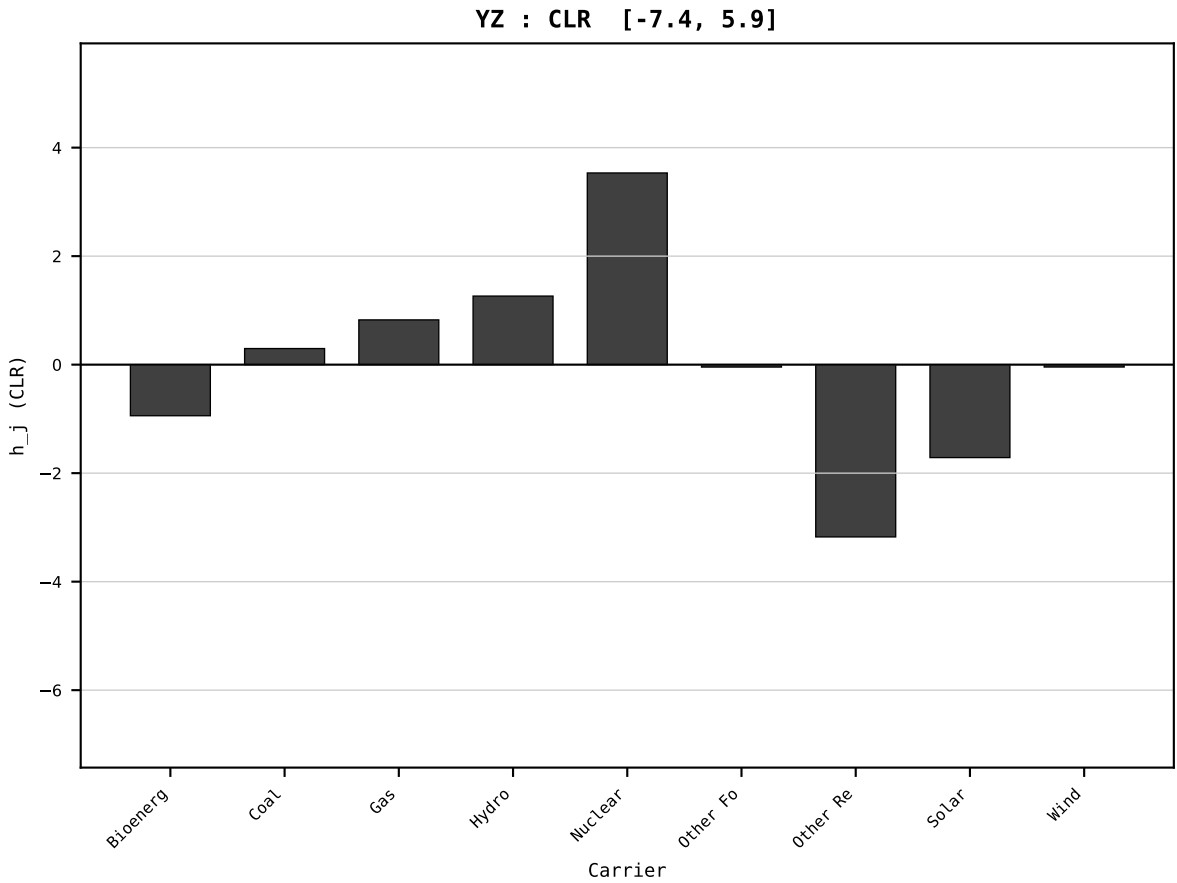
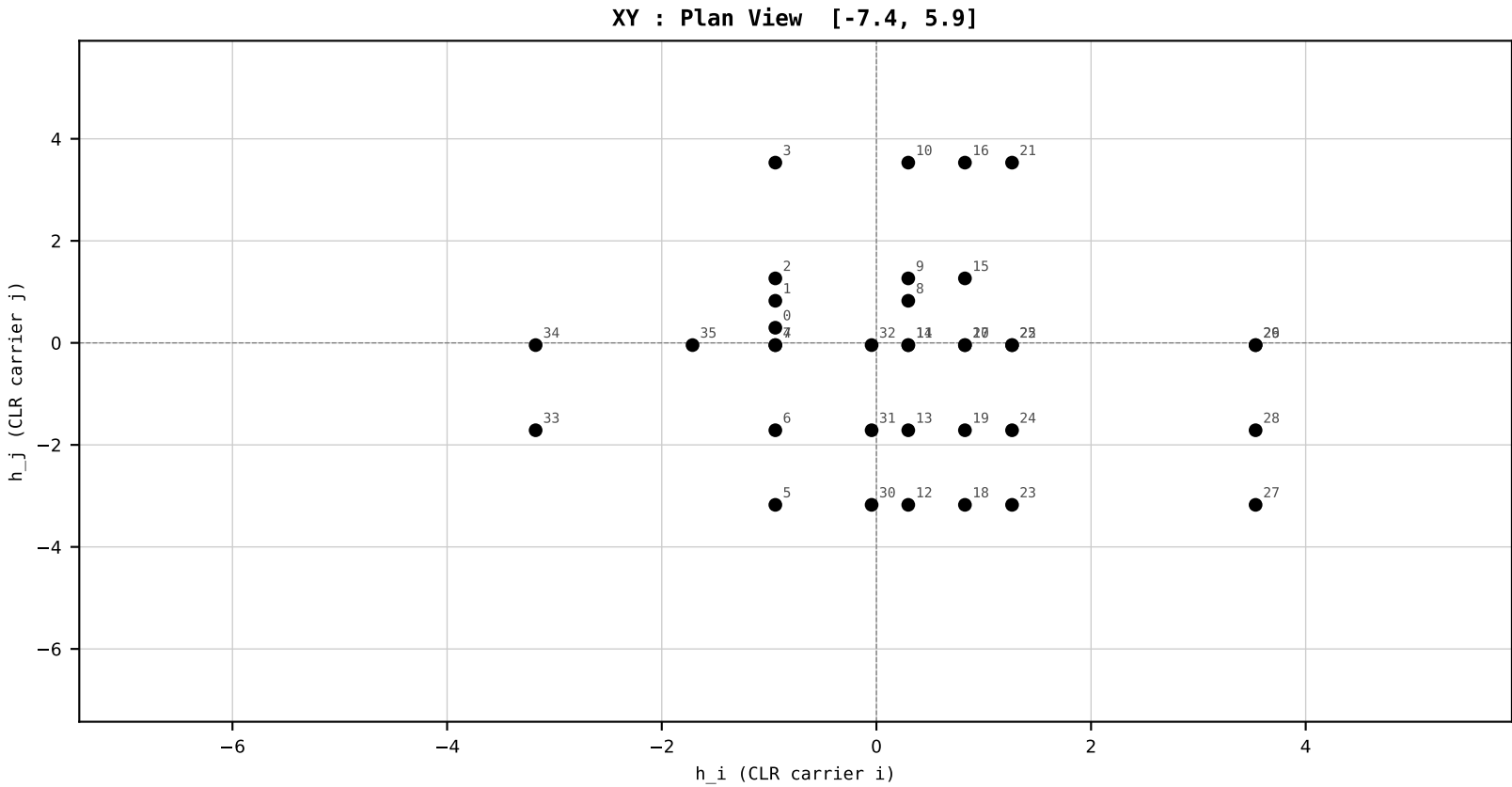
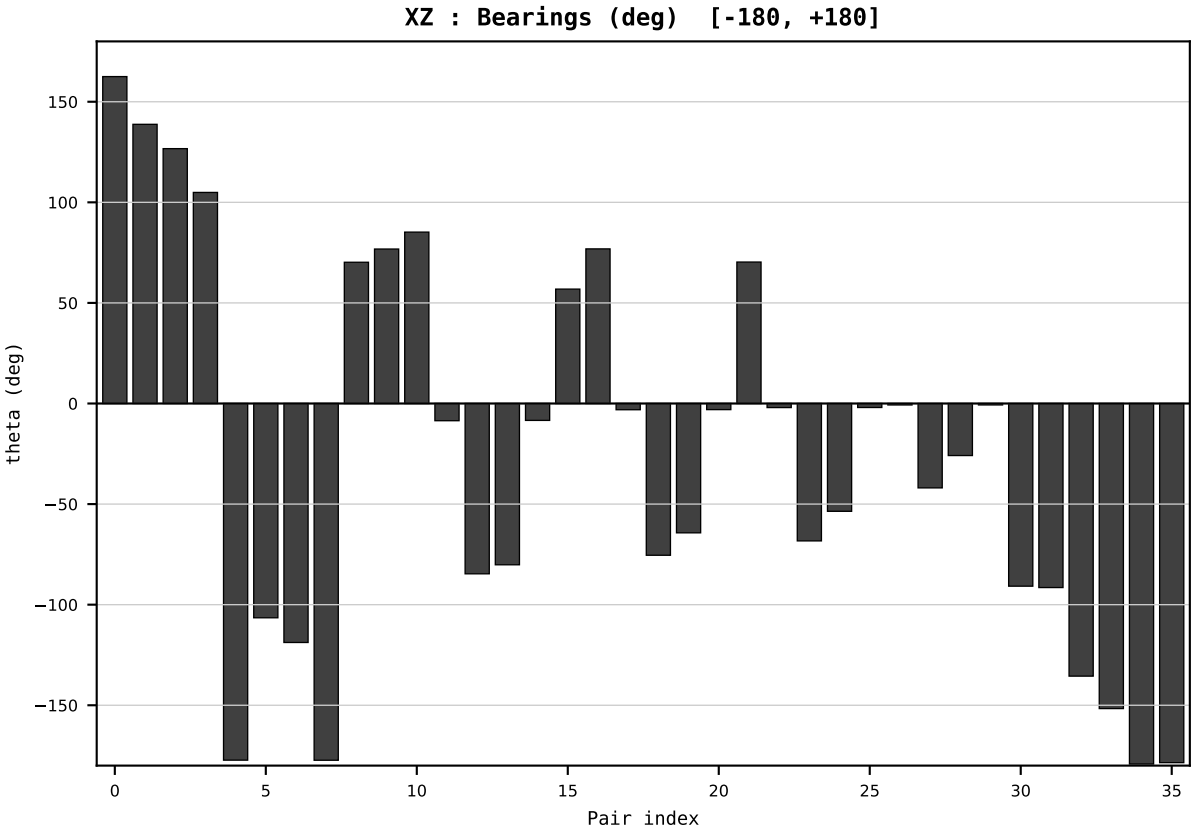
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.592444
Ring = Hs-4
E_metric= 28.7571
kappa_HS= 819.24
omega = 11.6539 deg
d_A = 1.337292
Helm = Solar
Helm d = +1.139834
DR = 6.708 HLR
DR ratio= 819.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



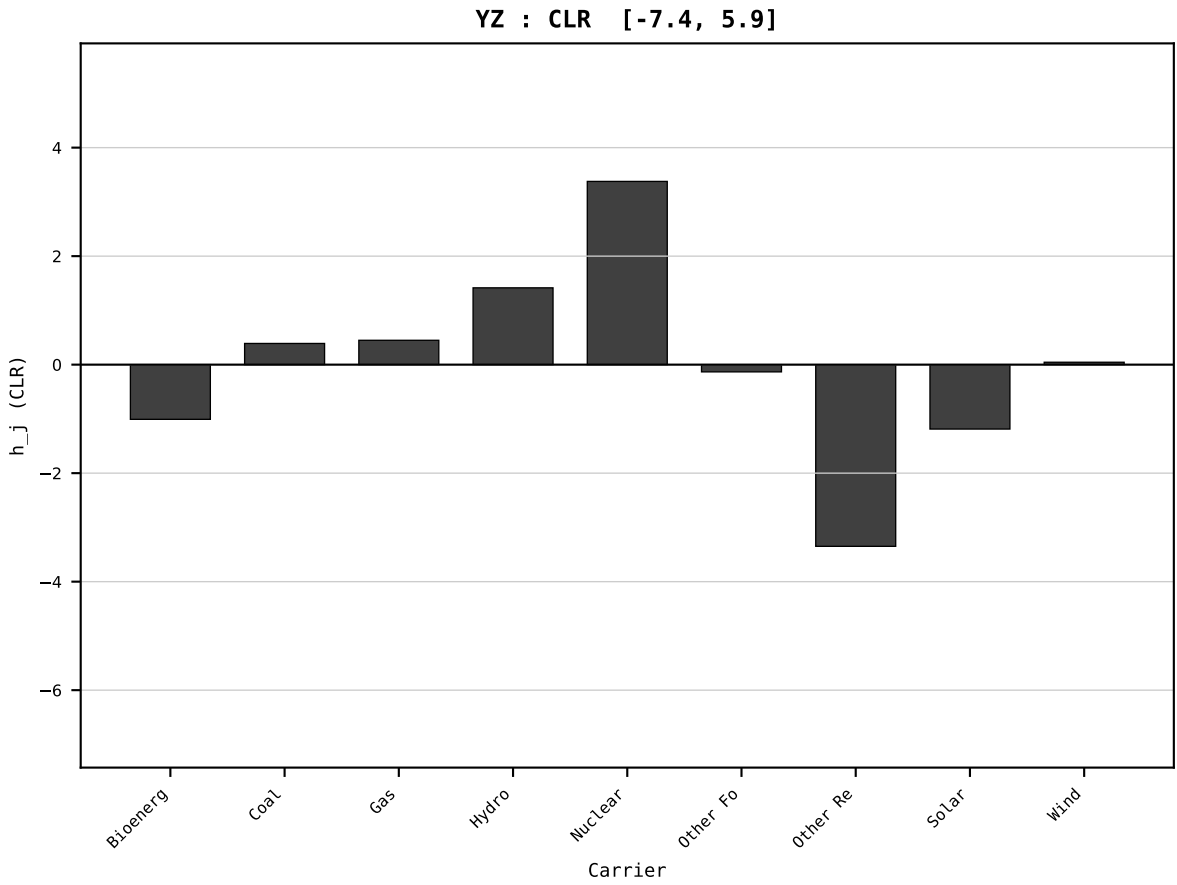
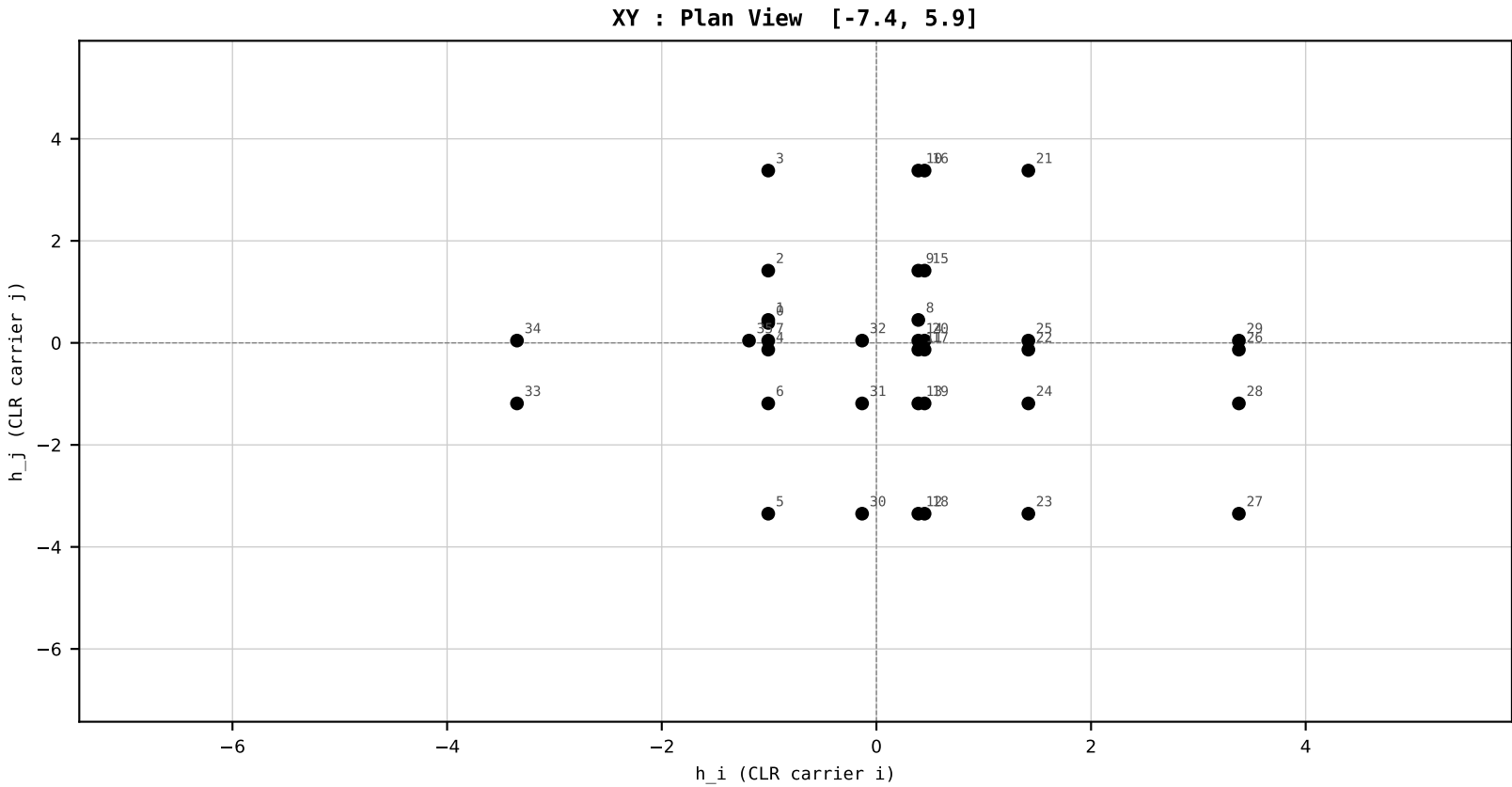
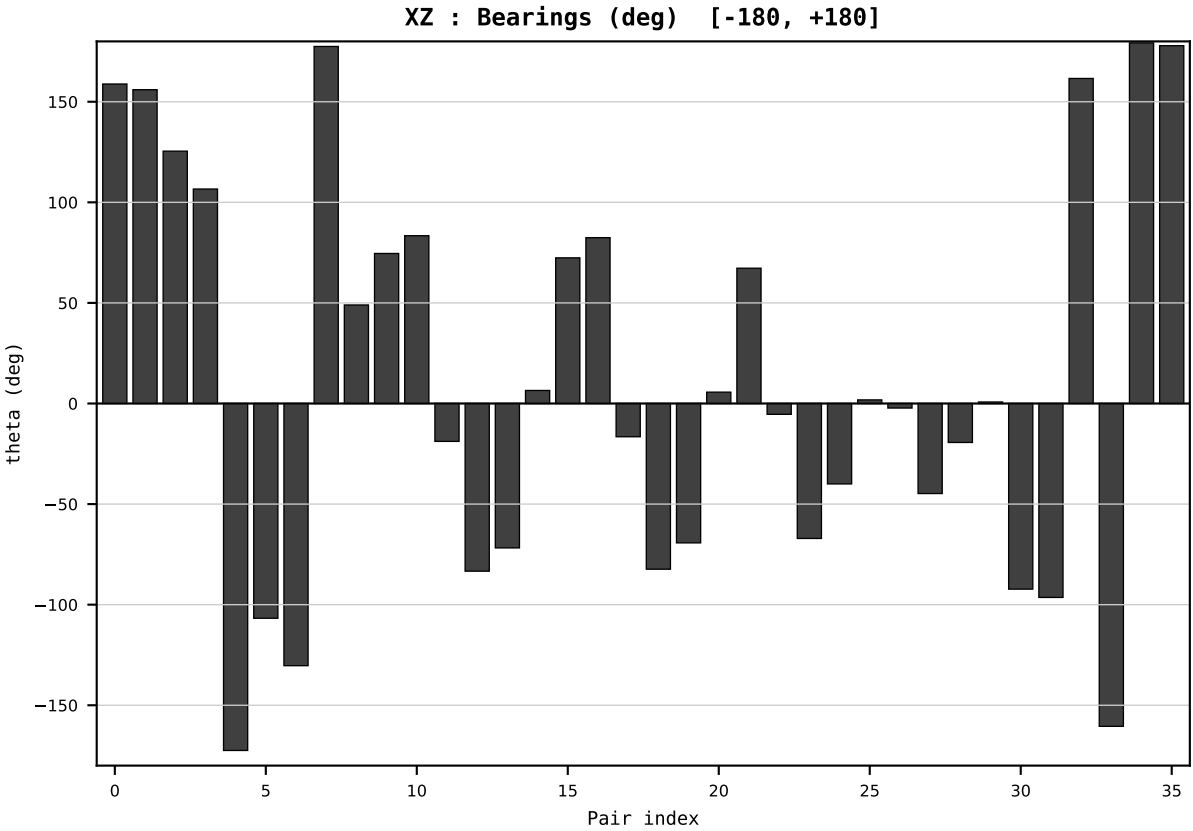
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.555897
Ring = Hs-4
E_metric= 27.4266
kappa_HS= 834.14
omega = 7.7165 deg
d_A = 0.724138
Helm = Solar
Helm d = +0.526031
DR = 6.726 HLR
DR ratio= 834.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

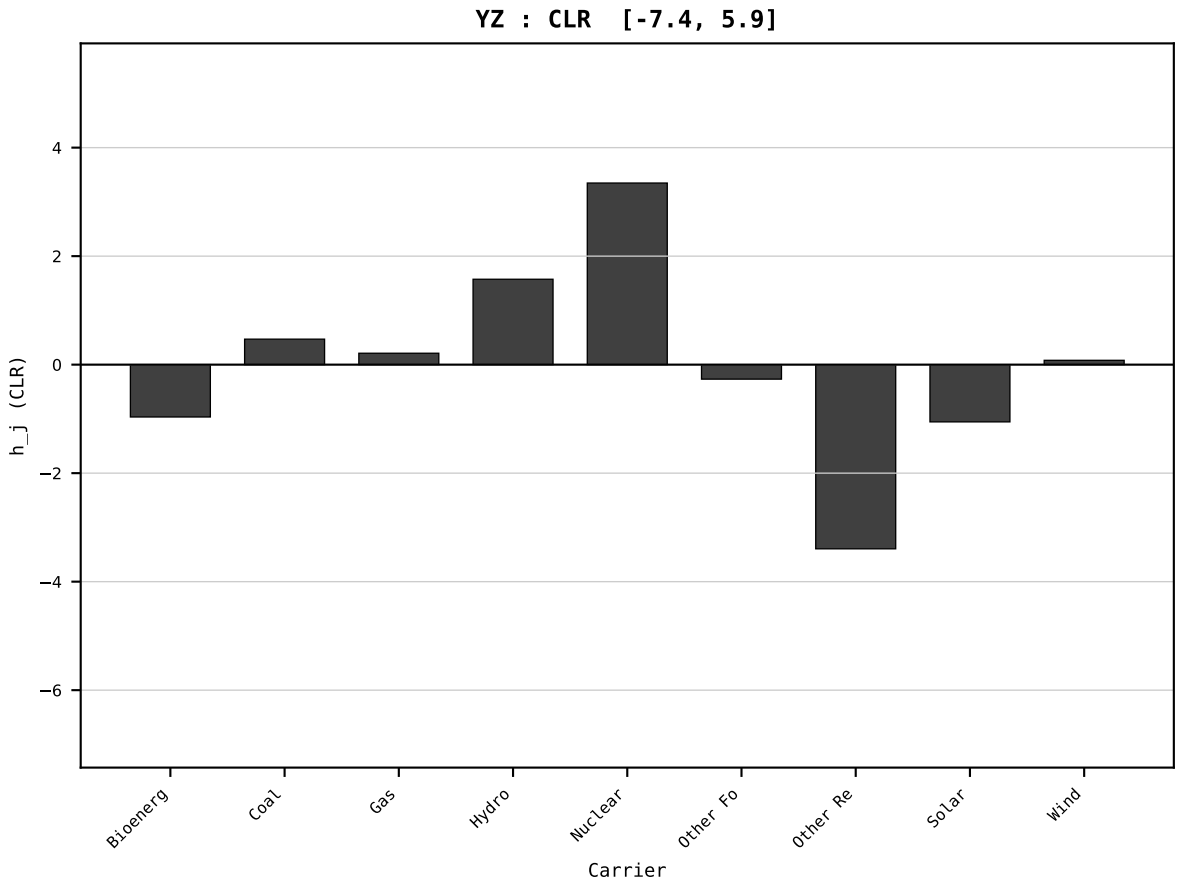
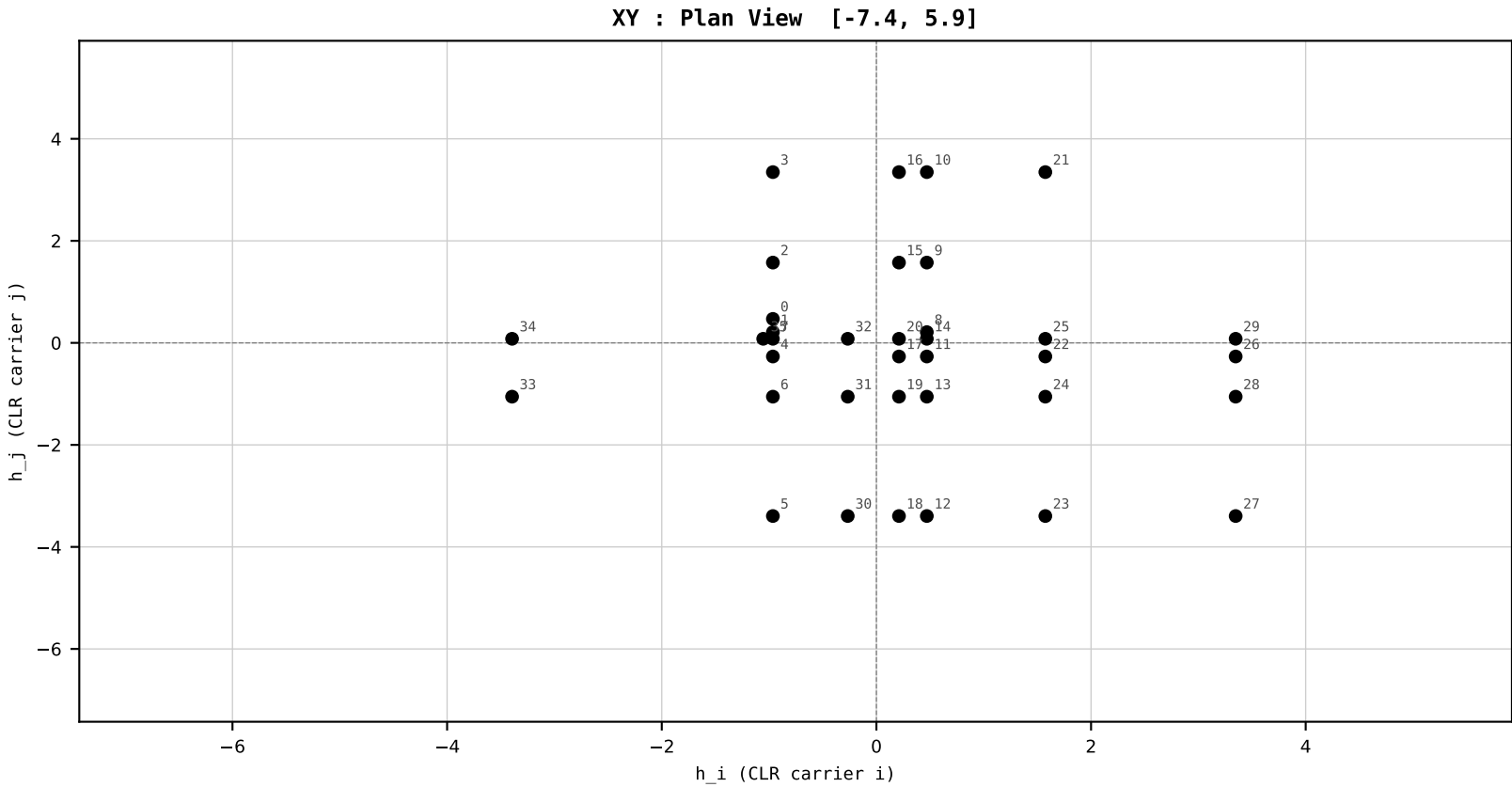
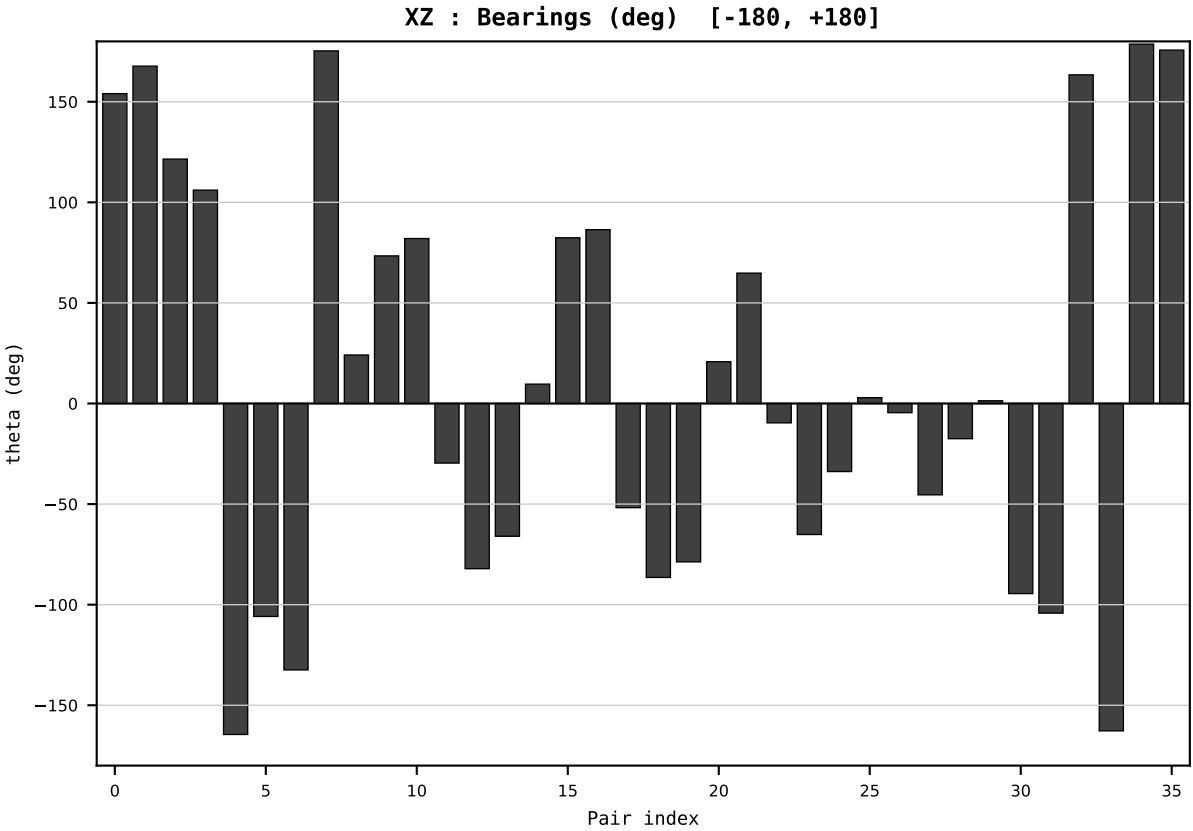
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.545141
Ring = Hs-4
E_metric= 27.5924
kappa_HS= 847.36
omega = 3.9329 deg
d_A = 0.360297
Helm = Gas
Helm d = -0.238712
DR = 6.742 HLR
DR ratio= 847.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

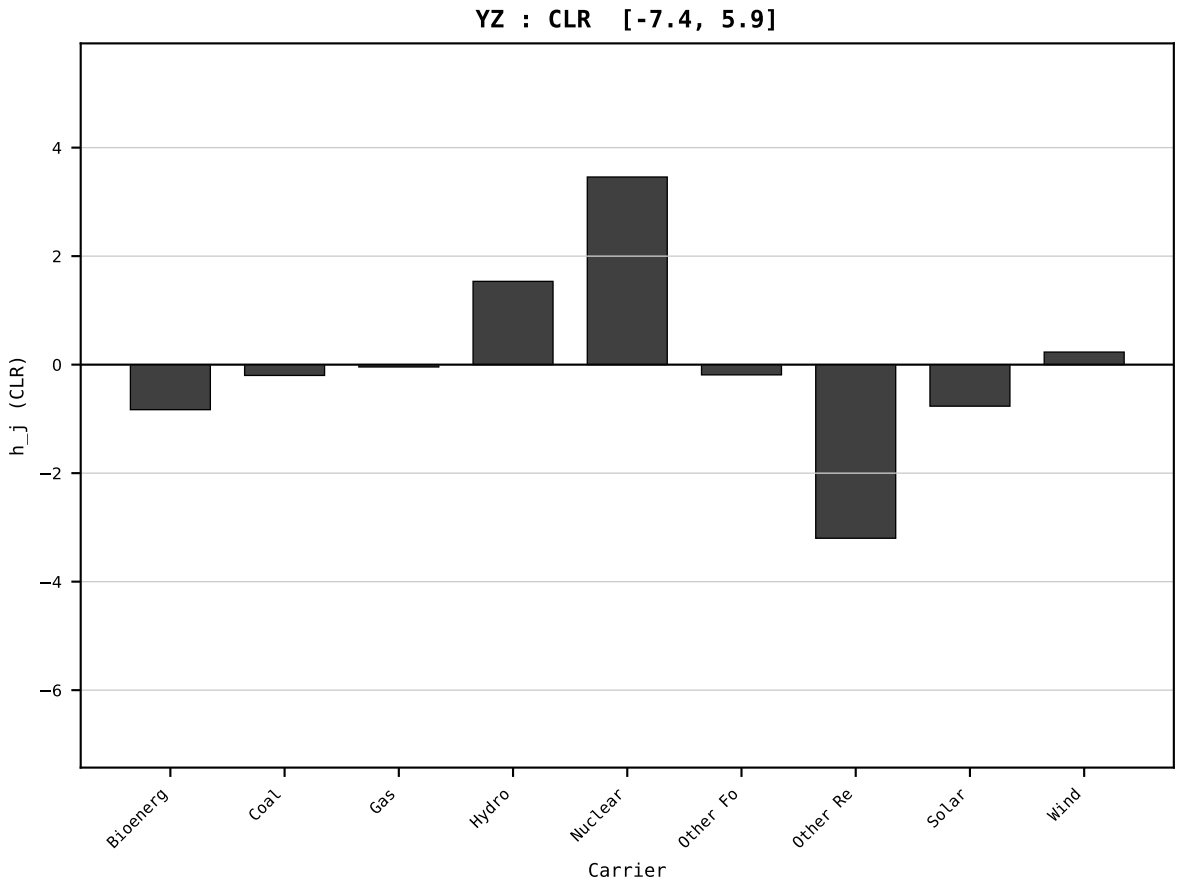
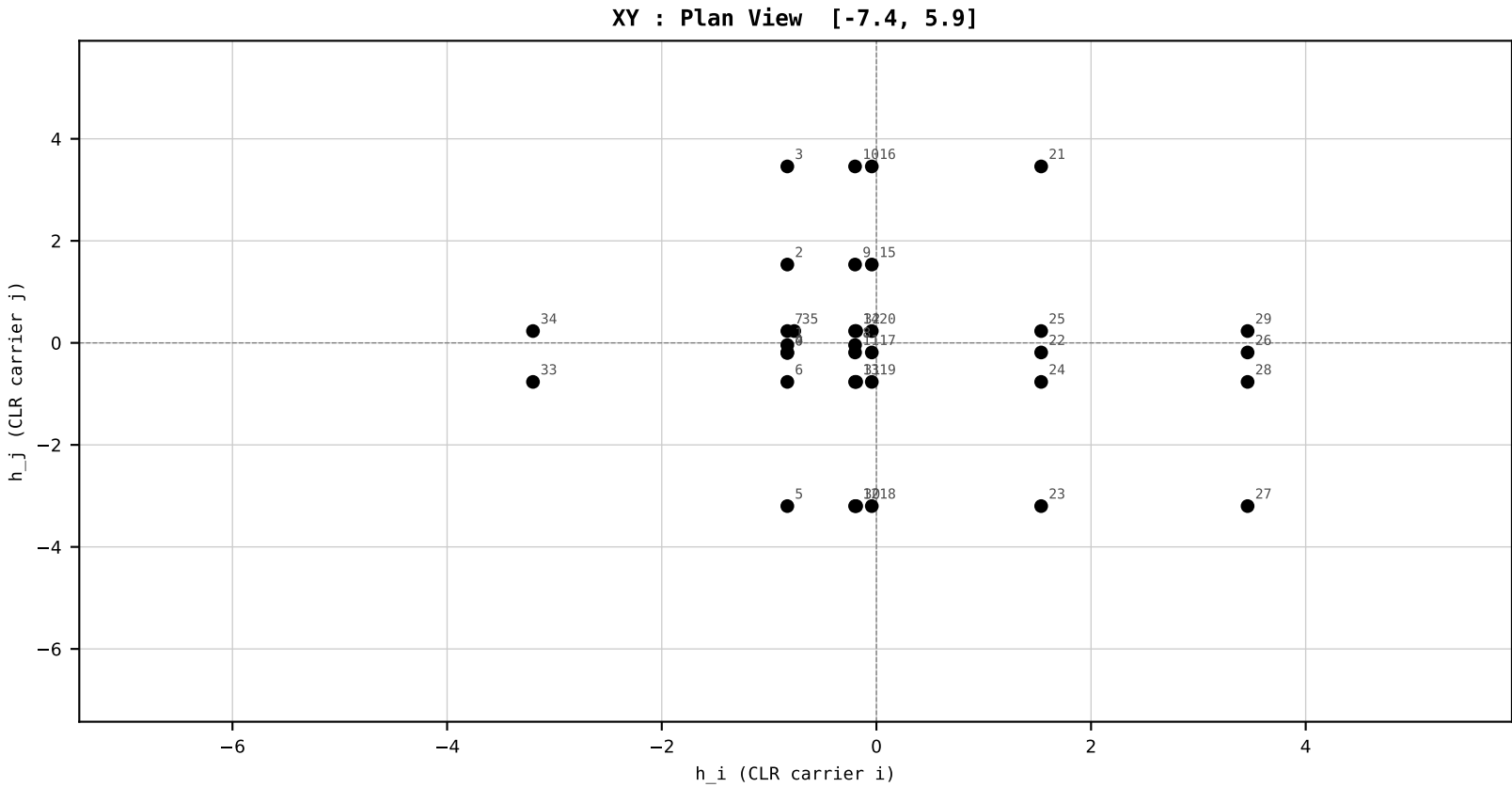
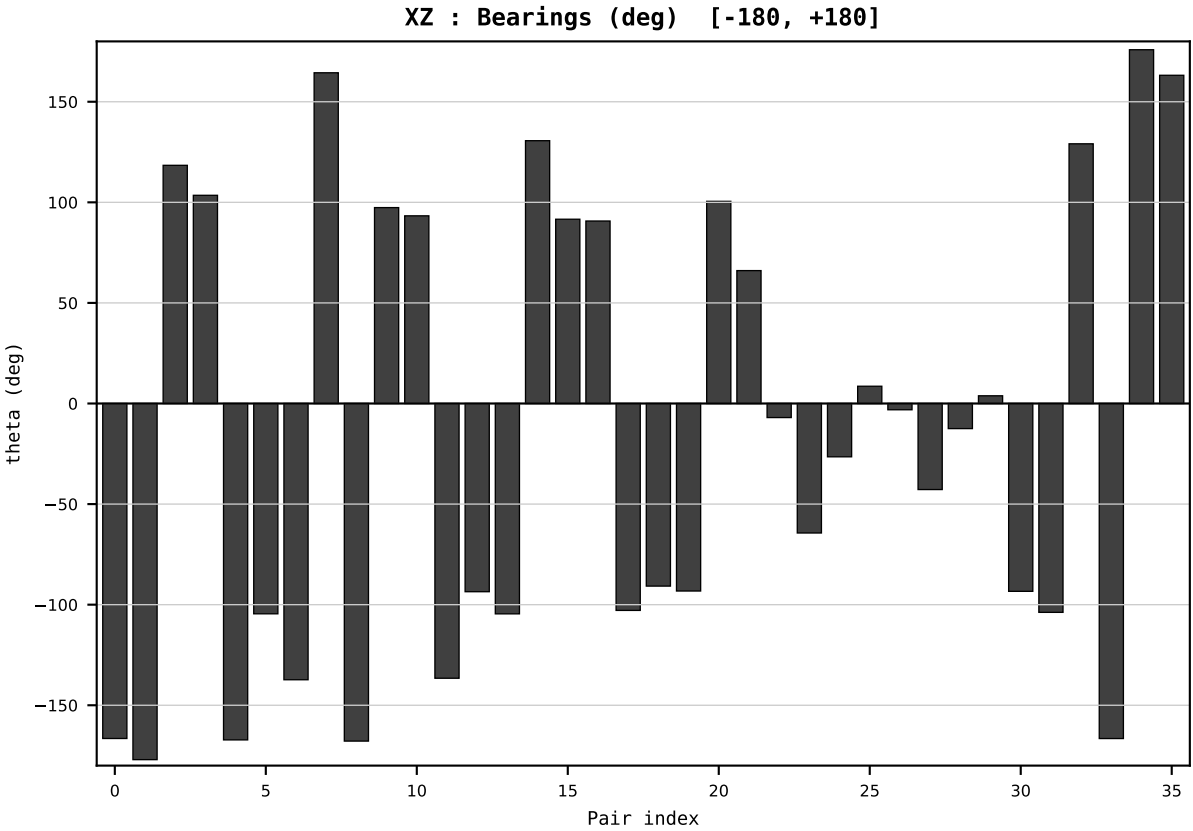
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.589178
Ring = Hs-4
E_metric= 25.9637
kappa_HS= 779.43
omega = 9.0709 deg
d_A = 0.833206
Helm = Coal
Helm d = -0.668681
DR = 6.659 HLR
DR ratio= 779.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

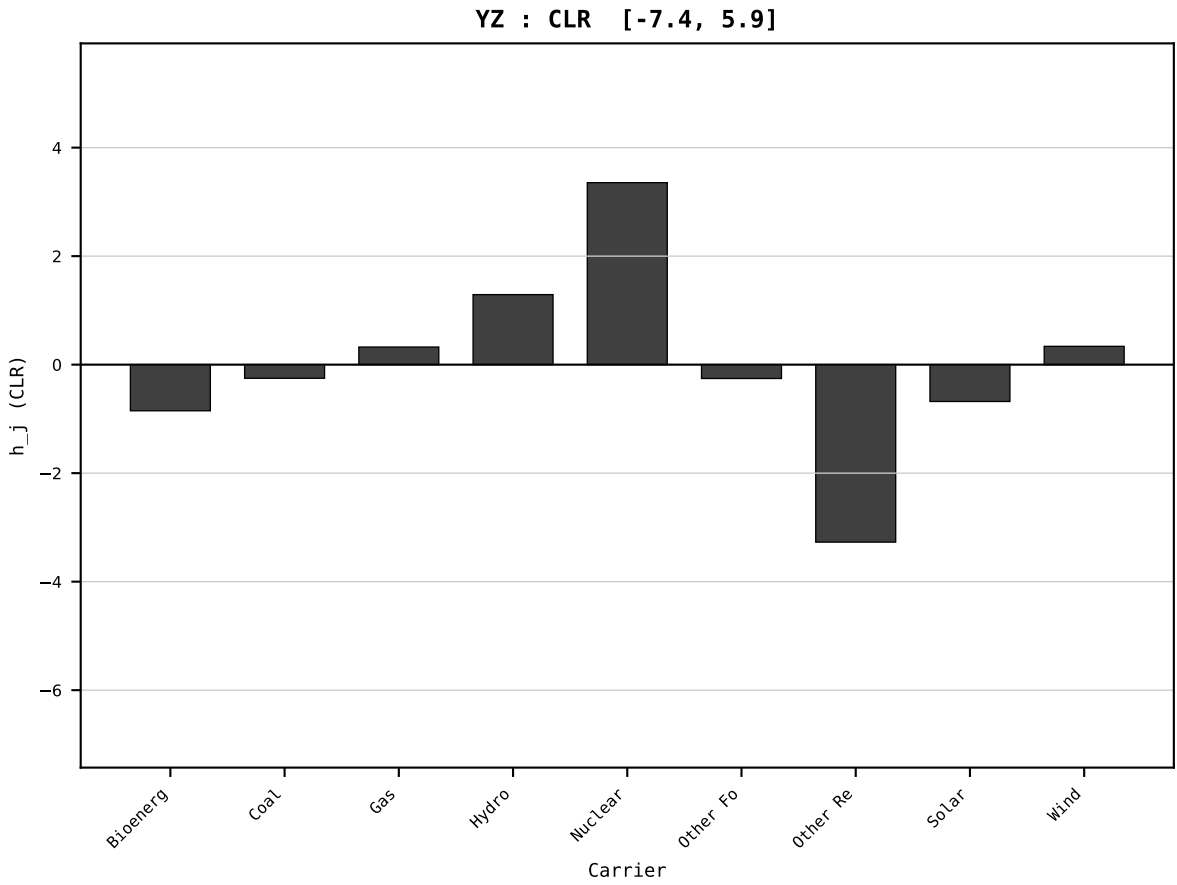
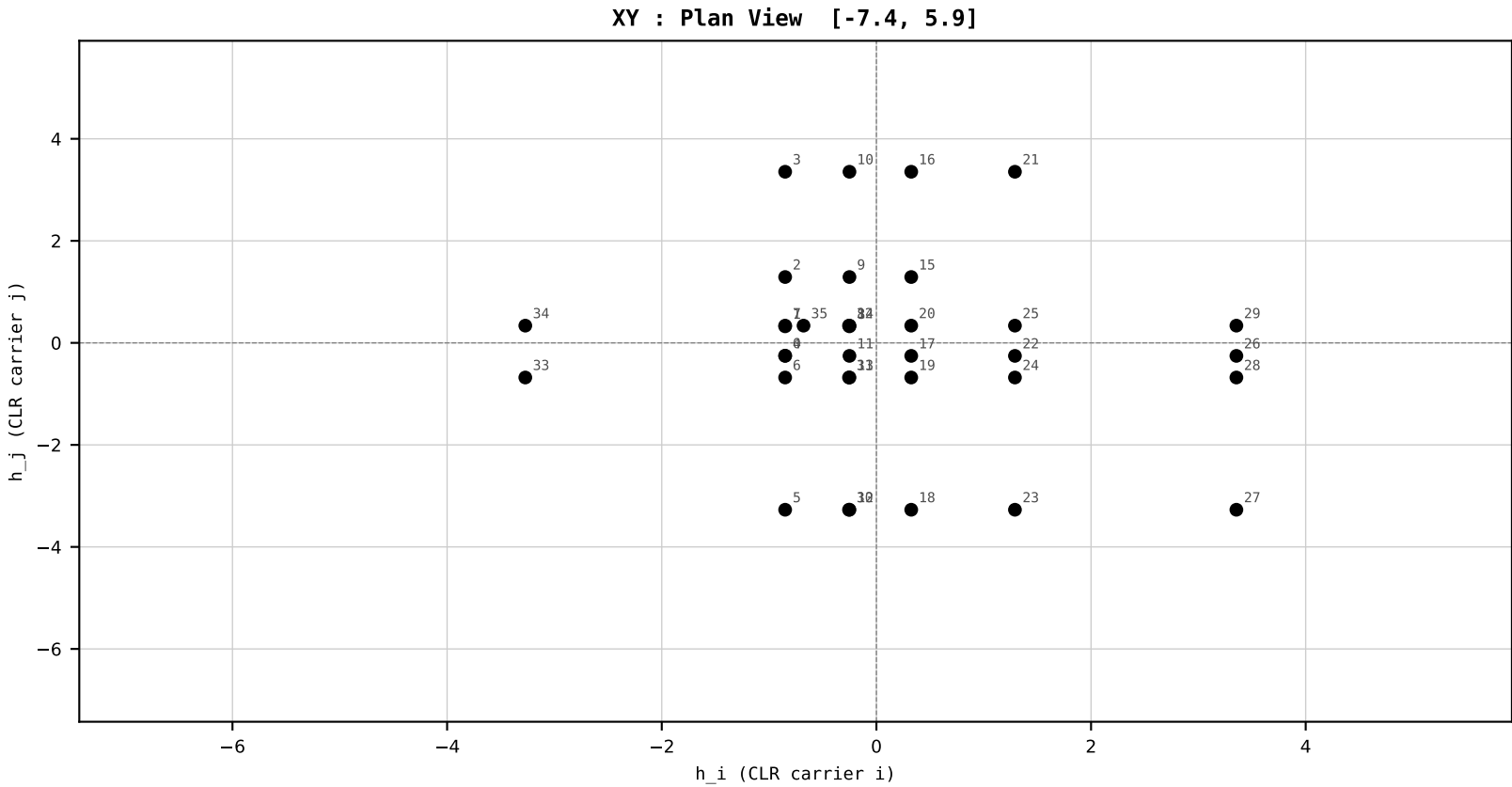
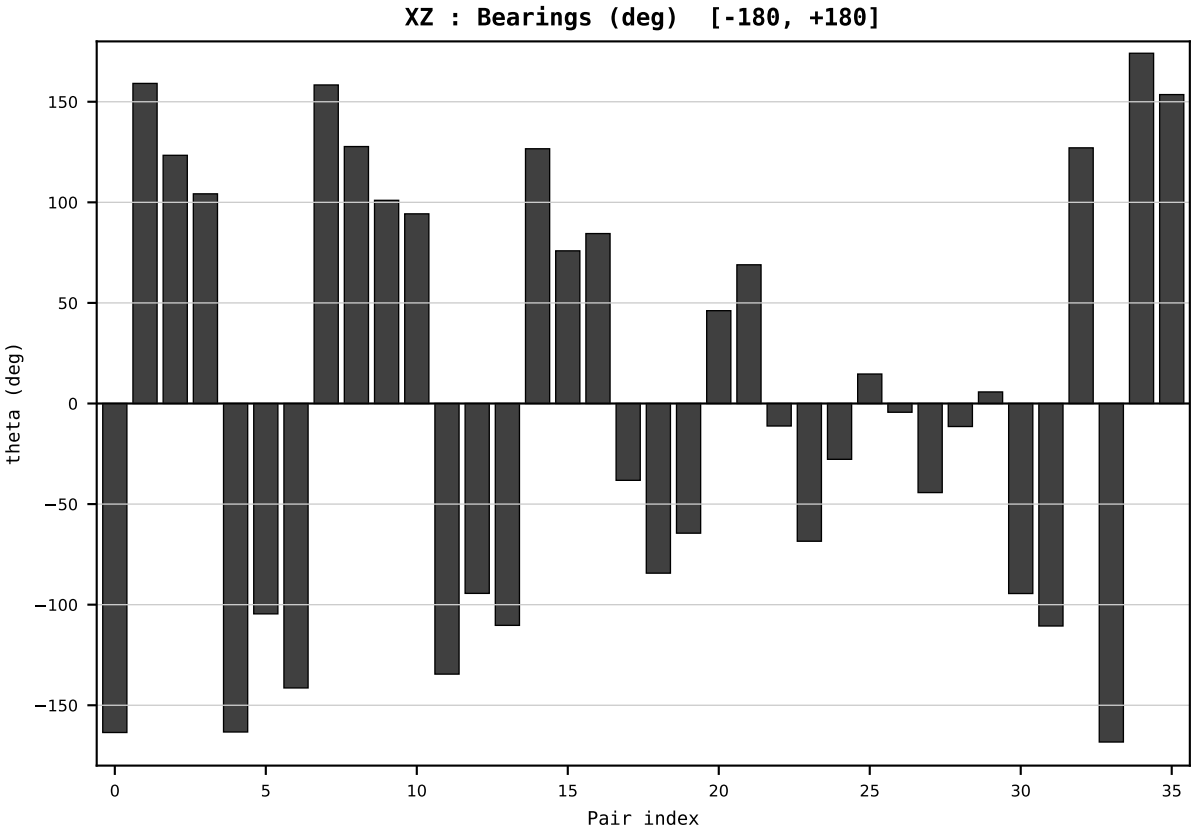
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.565912
Ring = Hs-4
E_metric= 25.1505
kappa_HS= 754.19
omega = 5.4442 deg
d_A = 0.486840
Helm = Gas
Helm d = +0.367403
DR = 6.626 HLR
DR ratio= 754.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

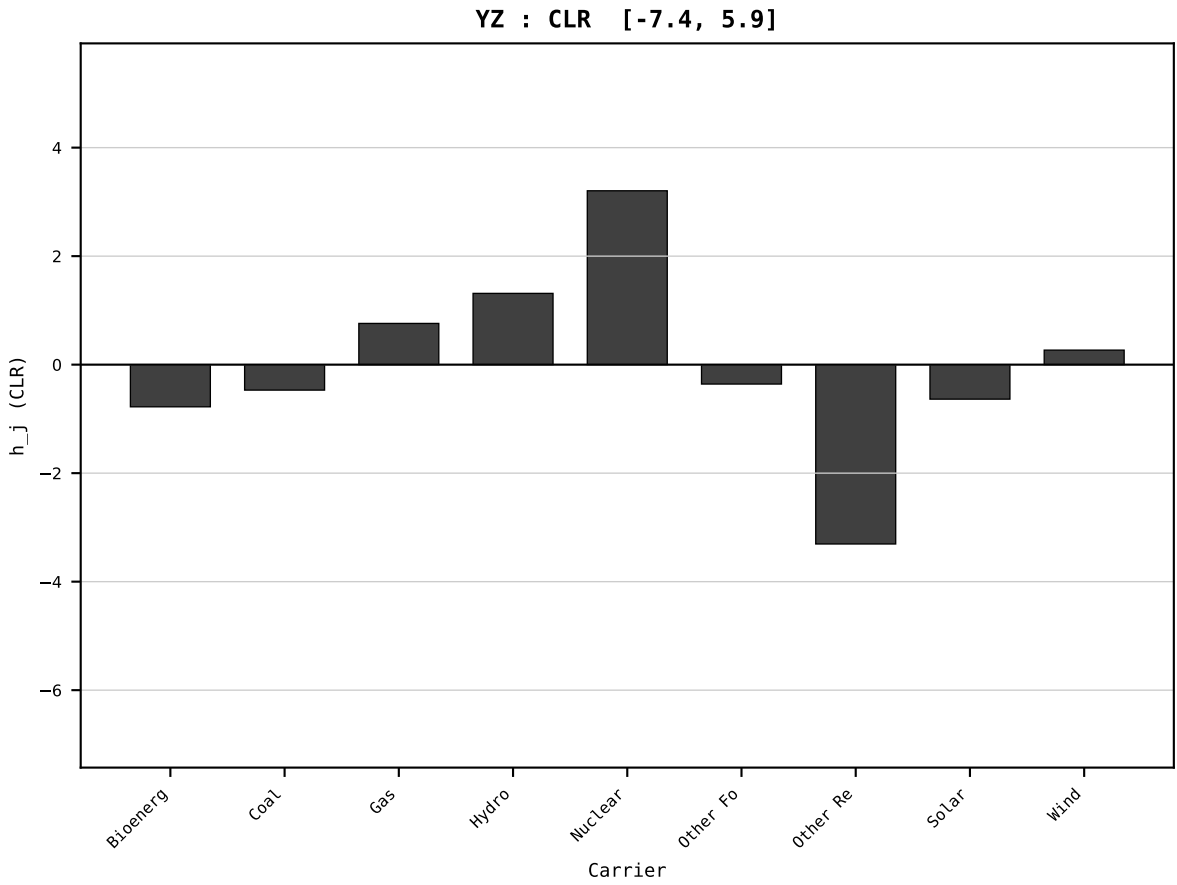
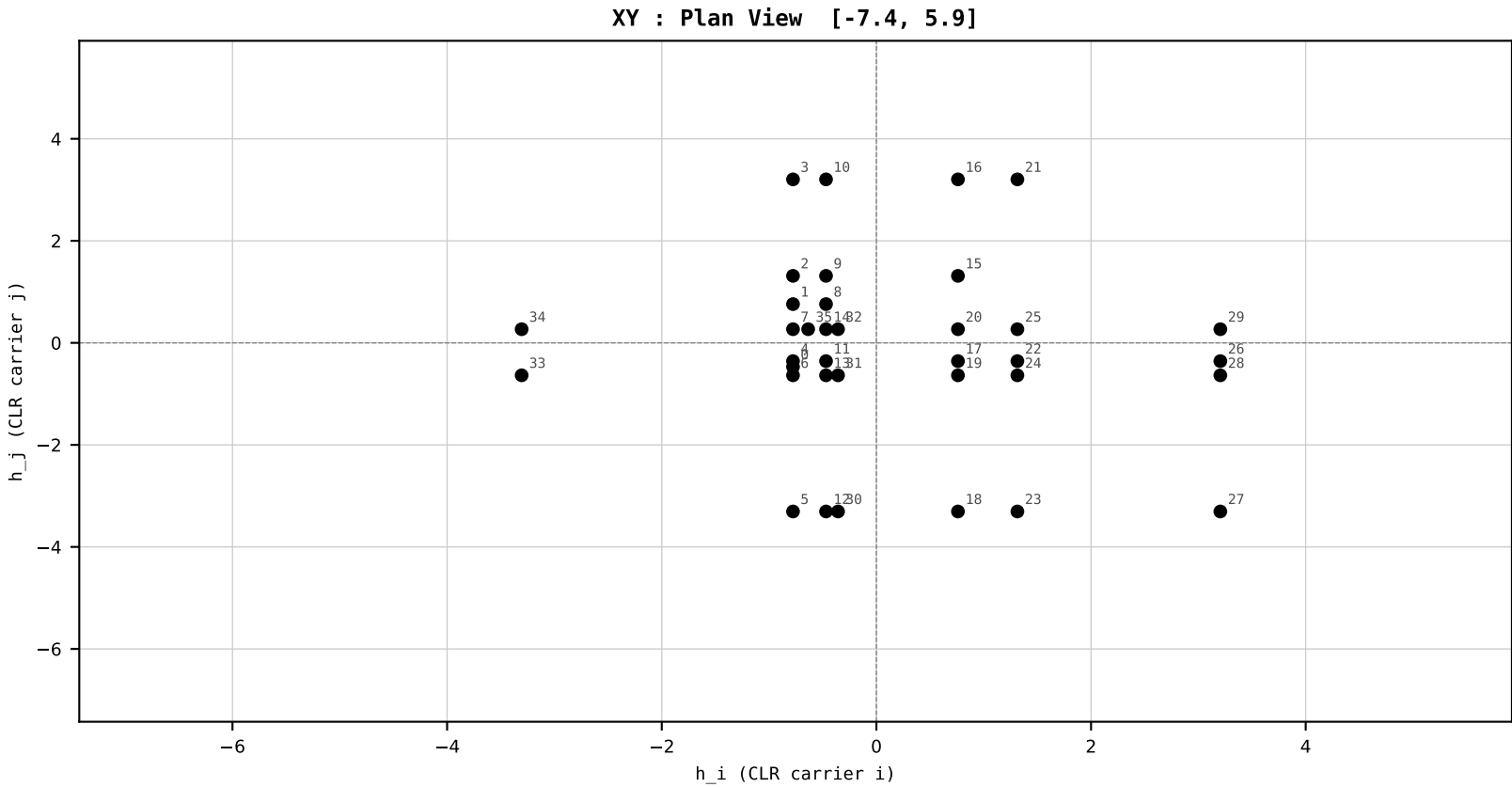
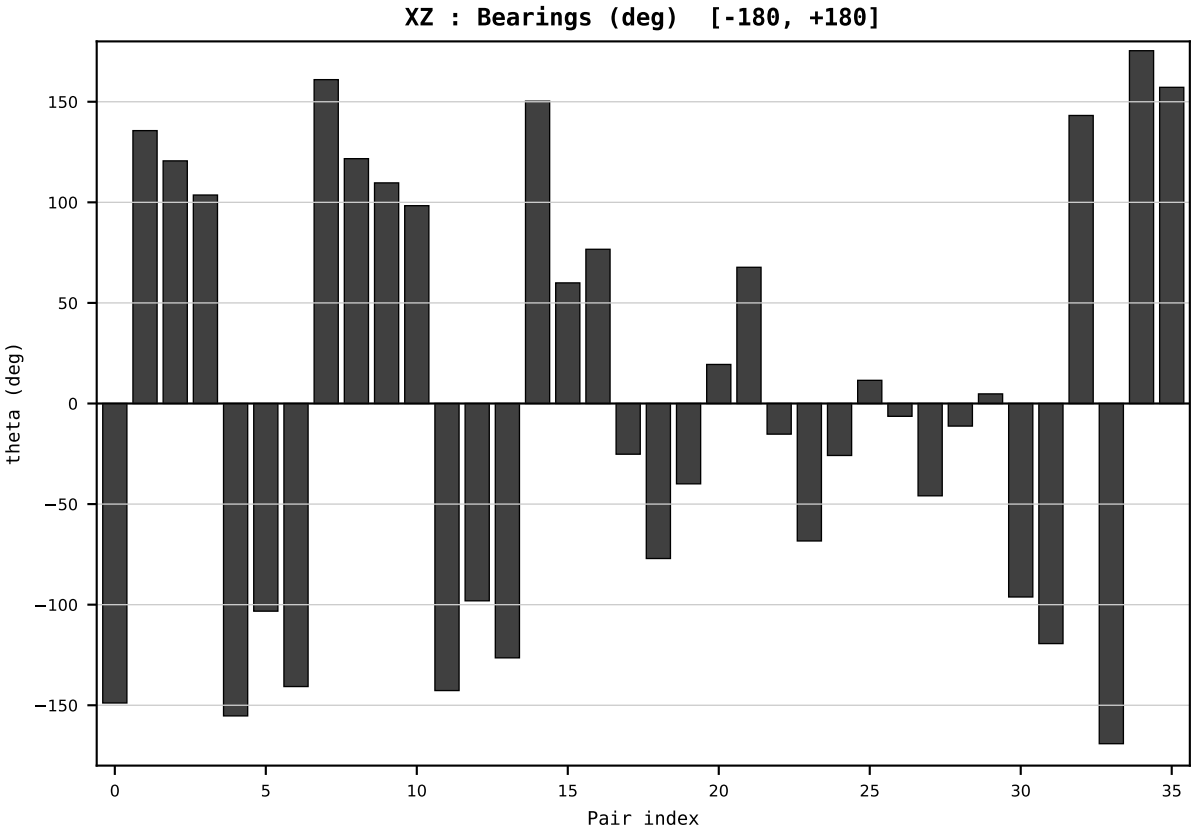
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.518507
Ring = Hs-4
E_metric= 24.9278
kappa_HS= 672.00
omega = 6.0973 deg
d_A = 0.532714
Helm = Gas
Helm d = +0.435313
DR = 6.510 HLR
DR ratio= 672.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

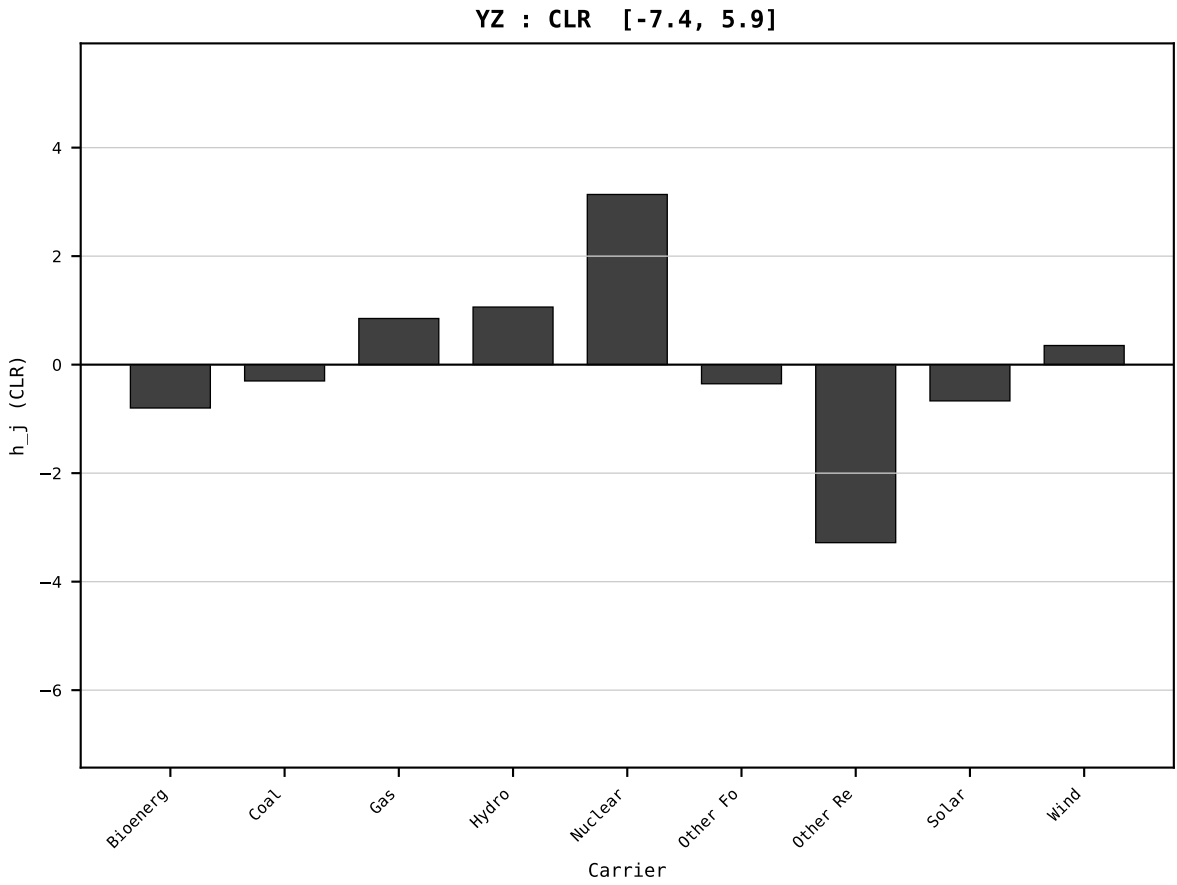
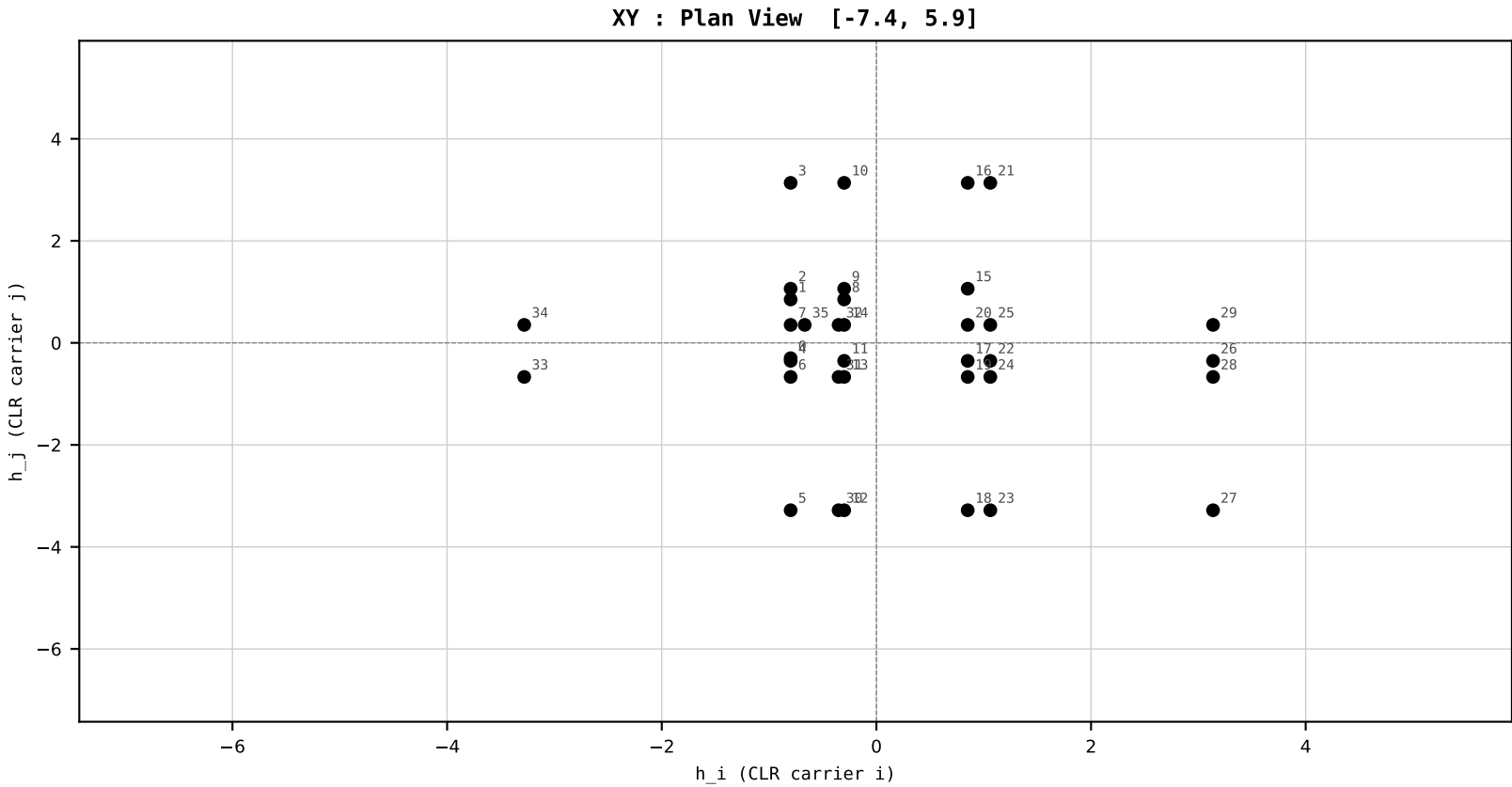
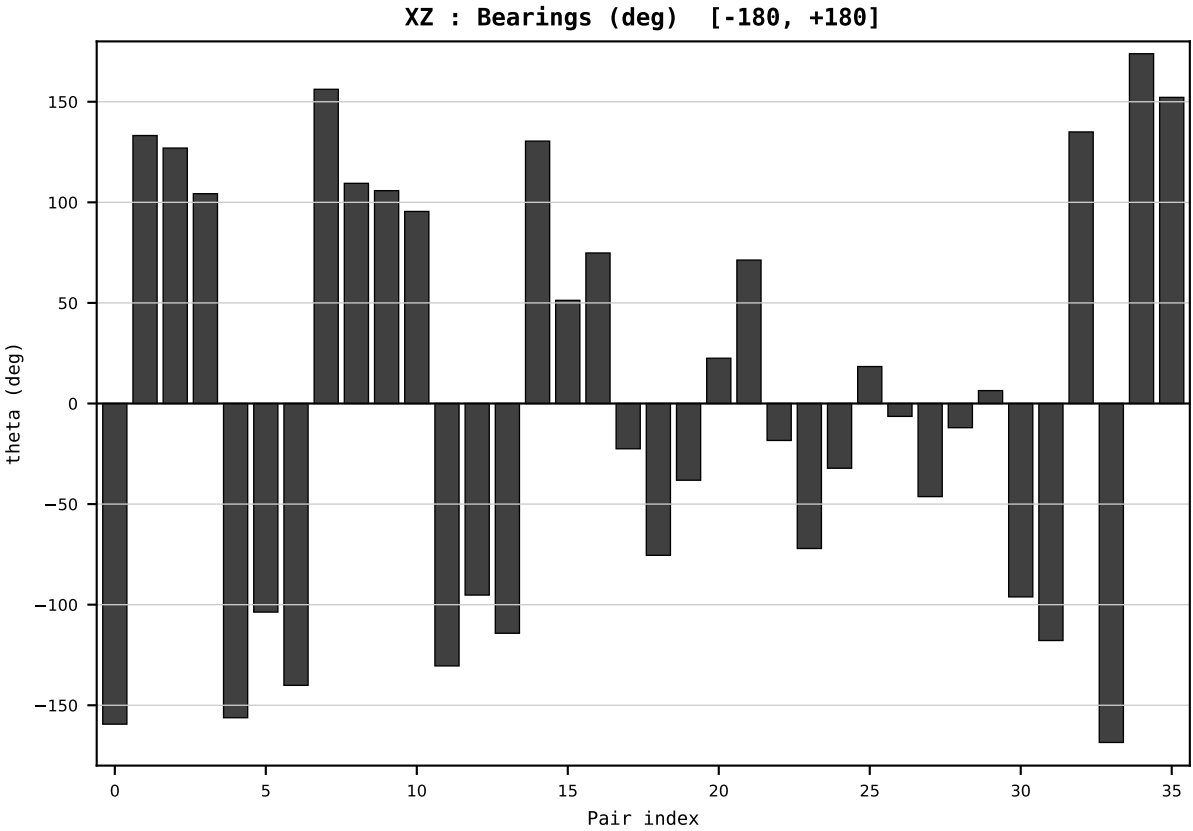
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.504621
Ring = Hs-4
E_metric= 23.8804
kappa_HS= 612.86
omega = 3.7247 deg
d_A = 0.338103
Helm = Hydro
Helm d = -0.252129
DR = 6.418 HLR
DR ratio= 612.9x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



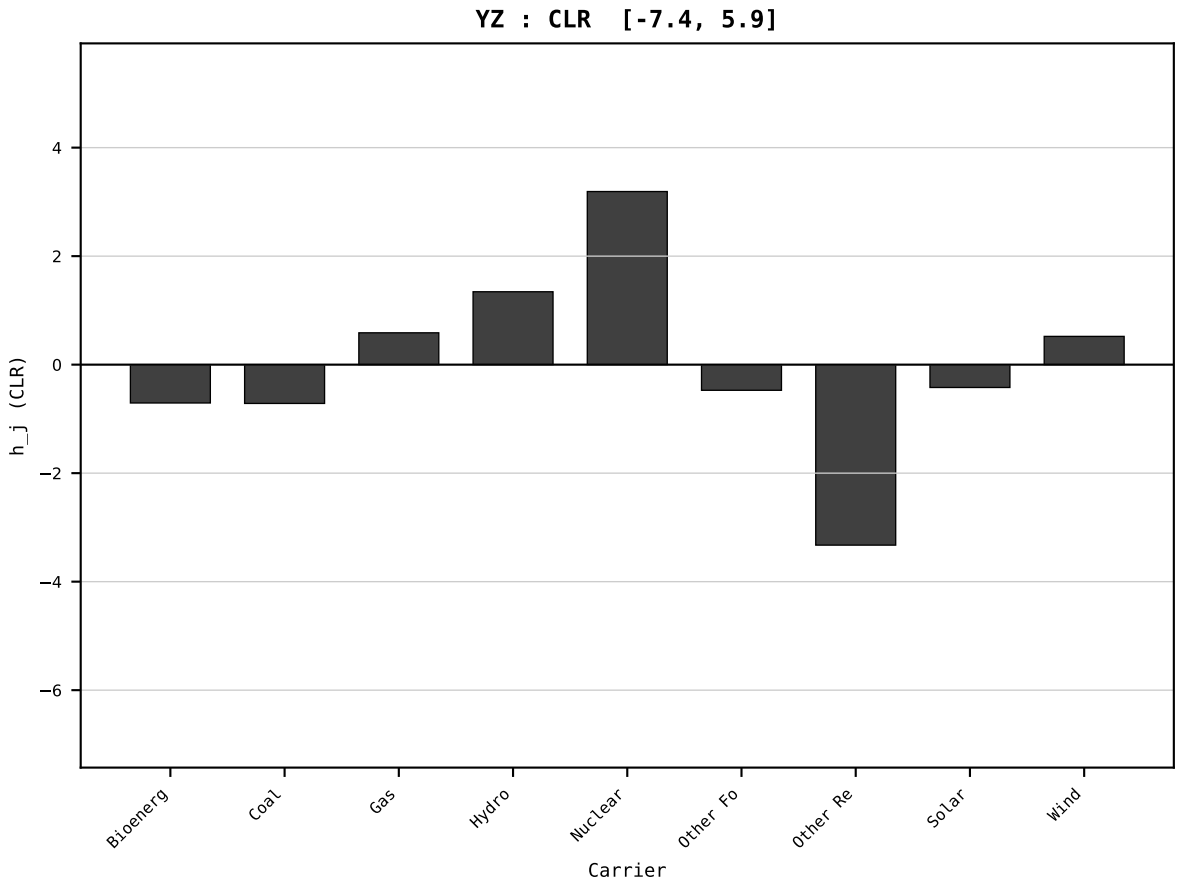
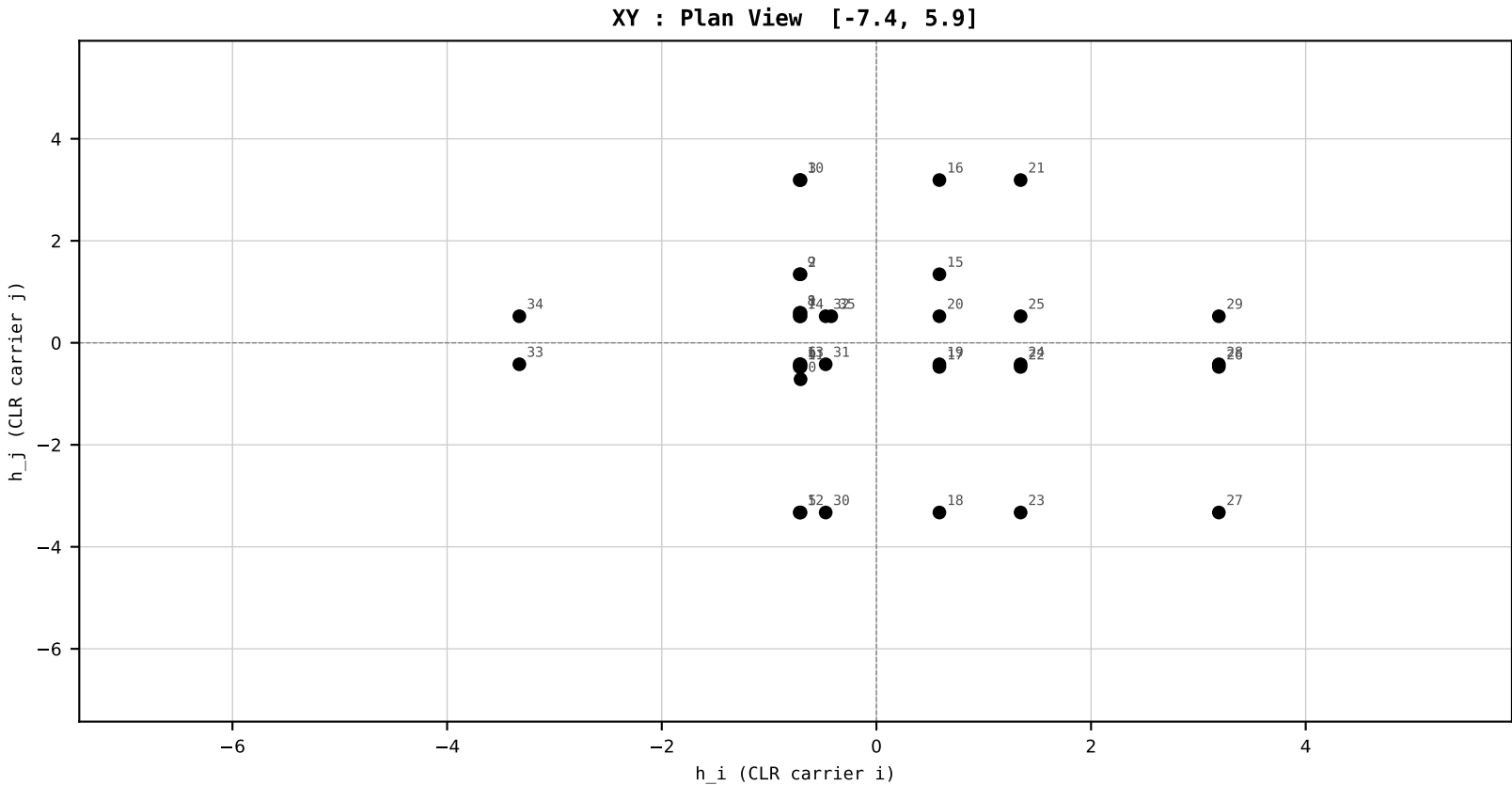
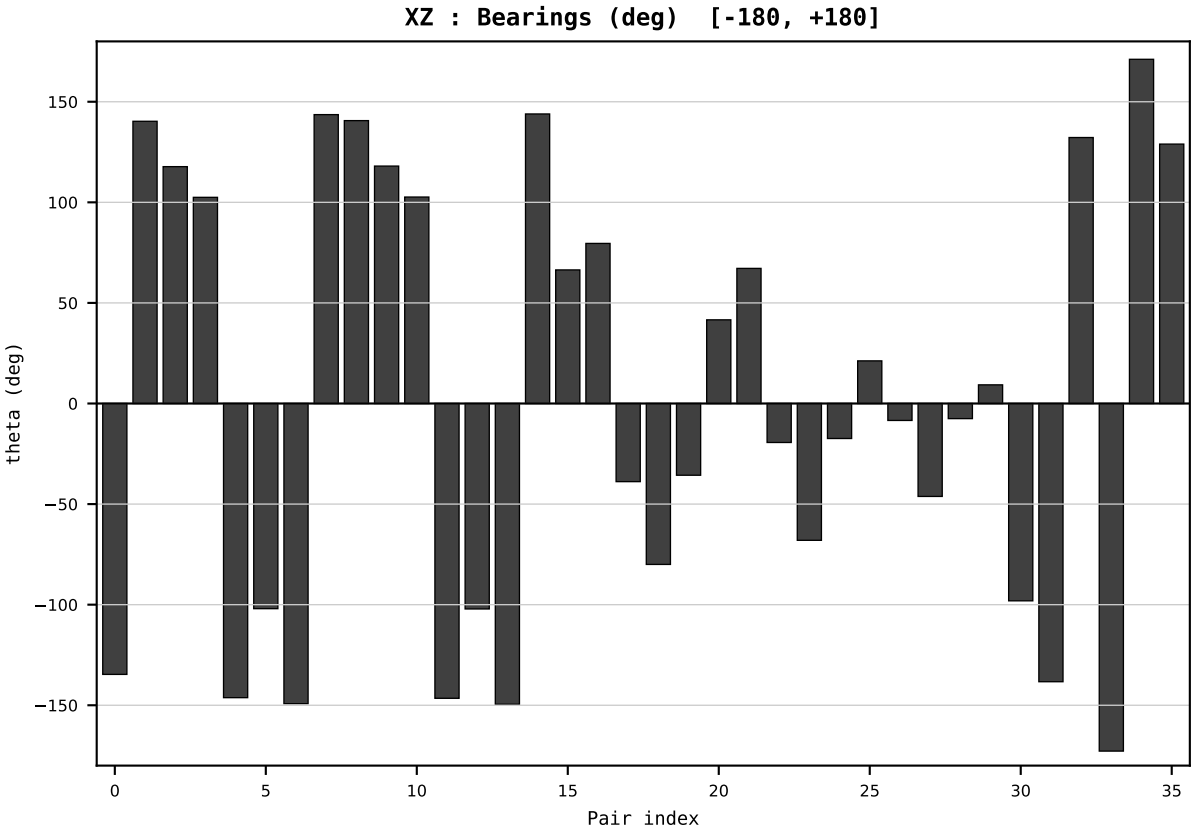
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.513194
Ring = Hs-4
E_metric= 25.0810
kappa_HS= 676.95
omega = 7.5442 deg
d_A = 0.662124
Helm = Coal
Helm d = -0.414639
DR = 6.518 HLR
DR ratio= 677.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



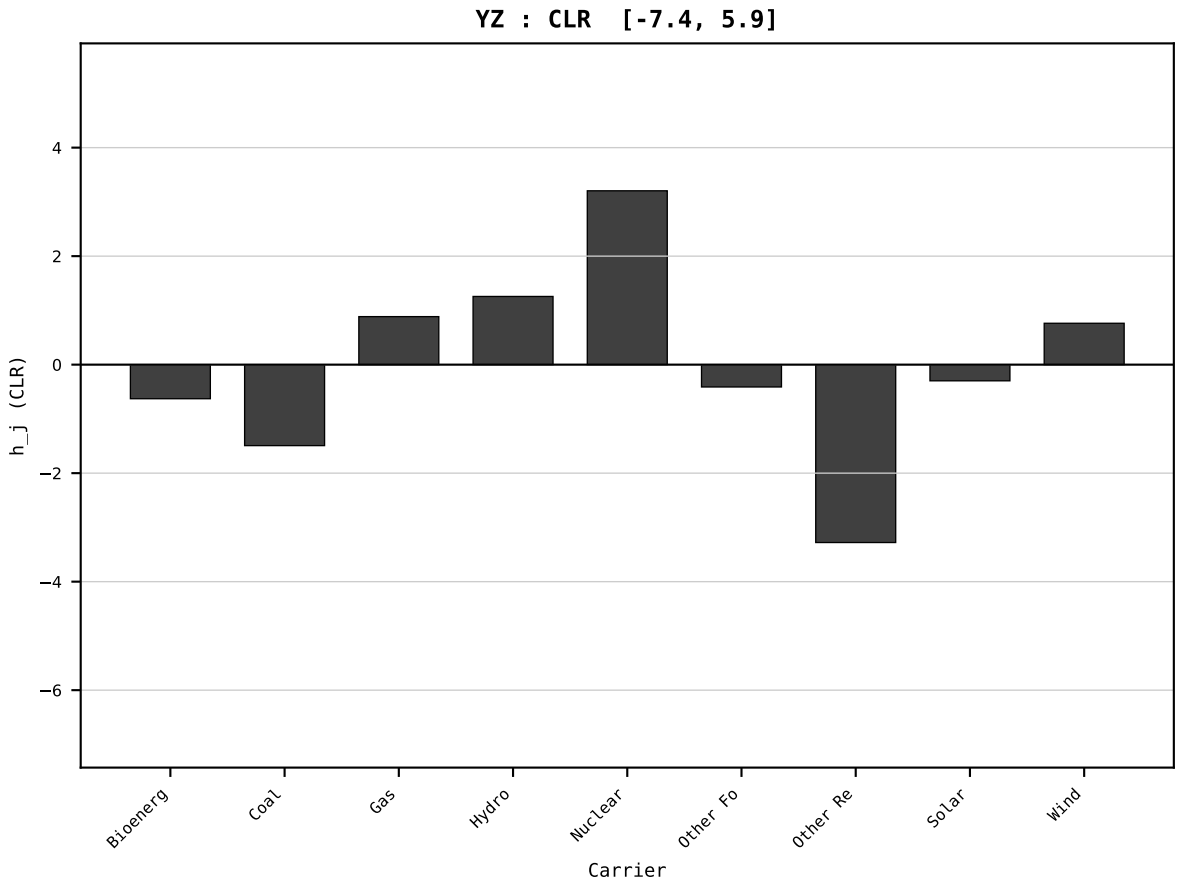
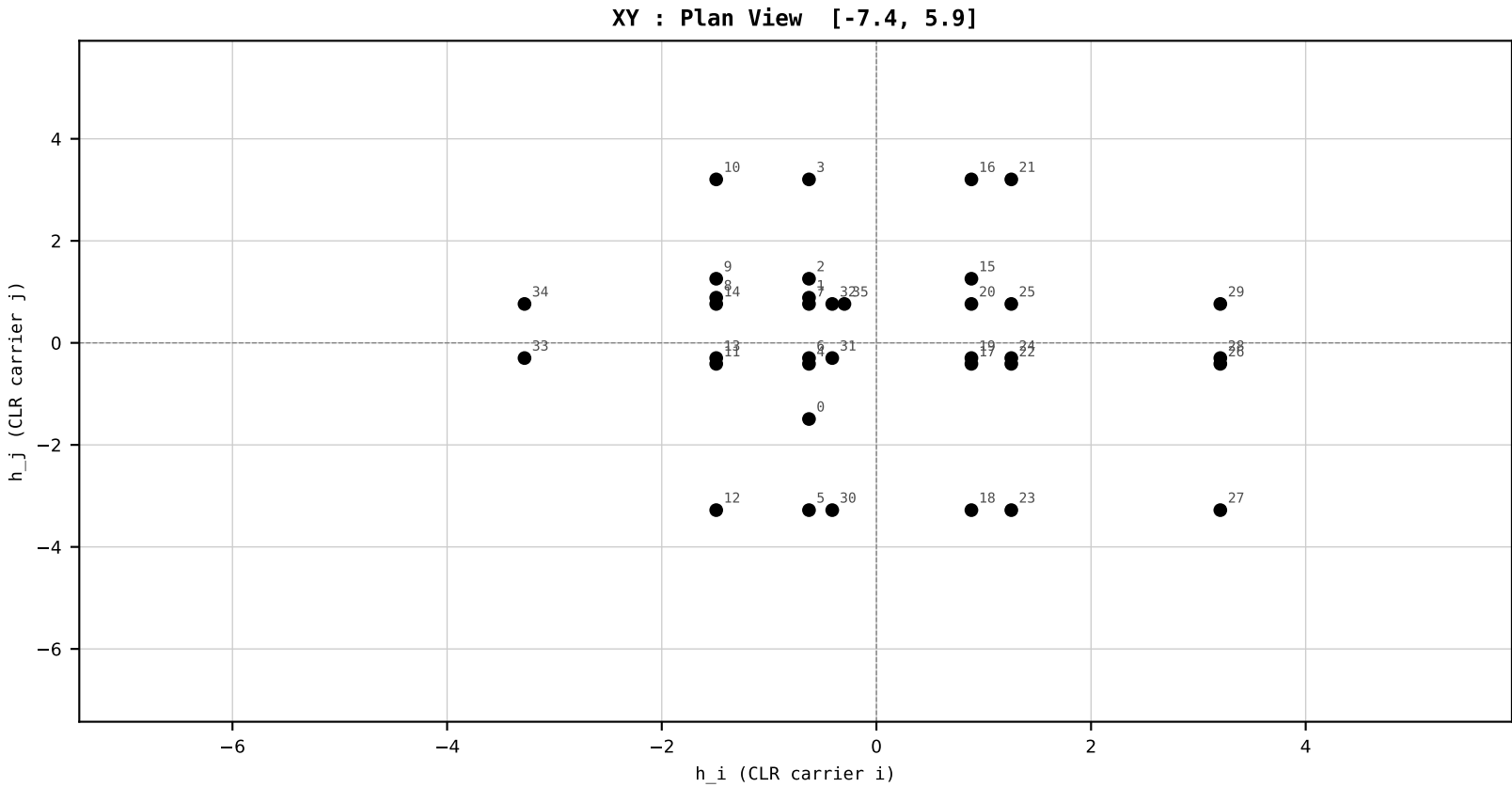
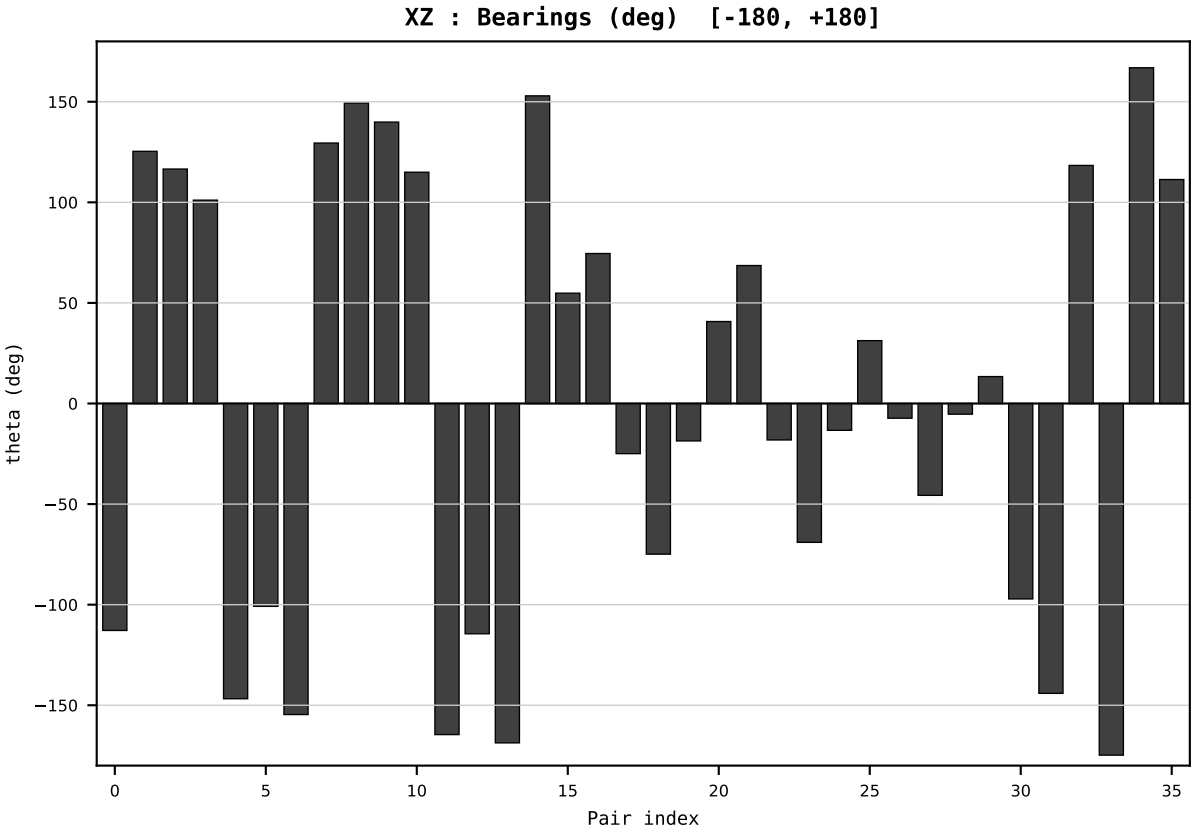
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.501913
Ring = Hs-4
E_metric= 26.8448
kappa_HS= 654.11
omega = 9.7988 deg
d_A = 0.887160
Helm = Coal
Helm d = -0.777550
DR = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

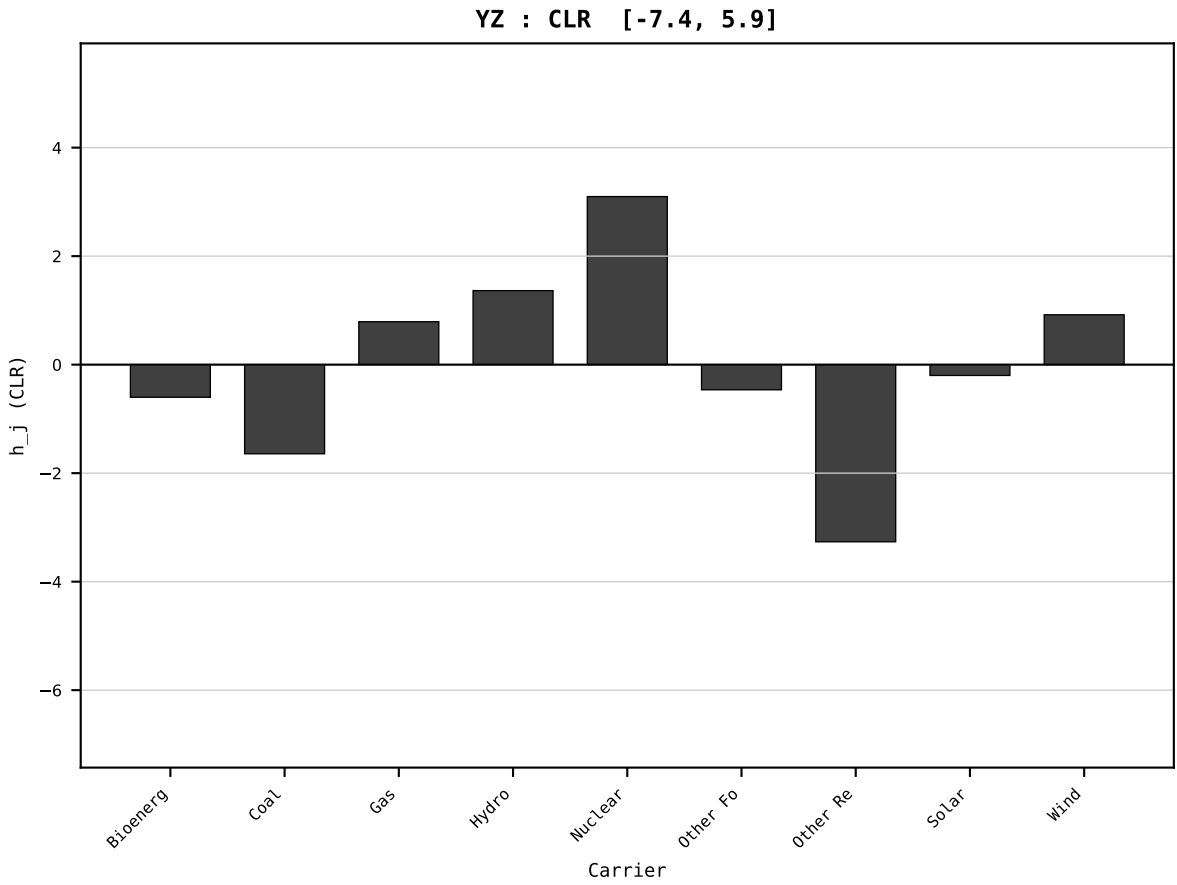
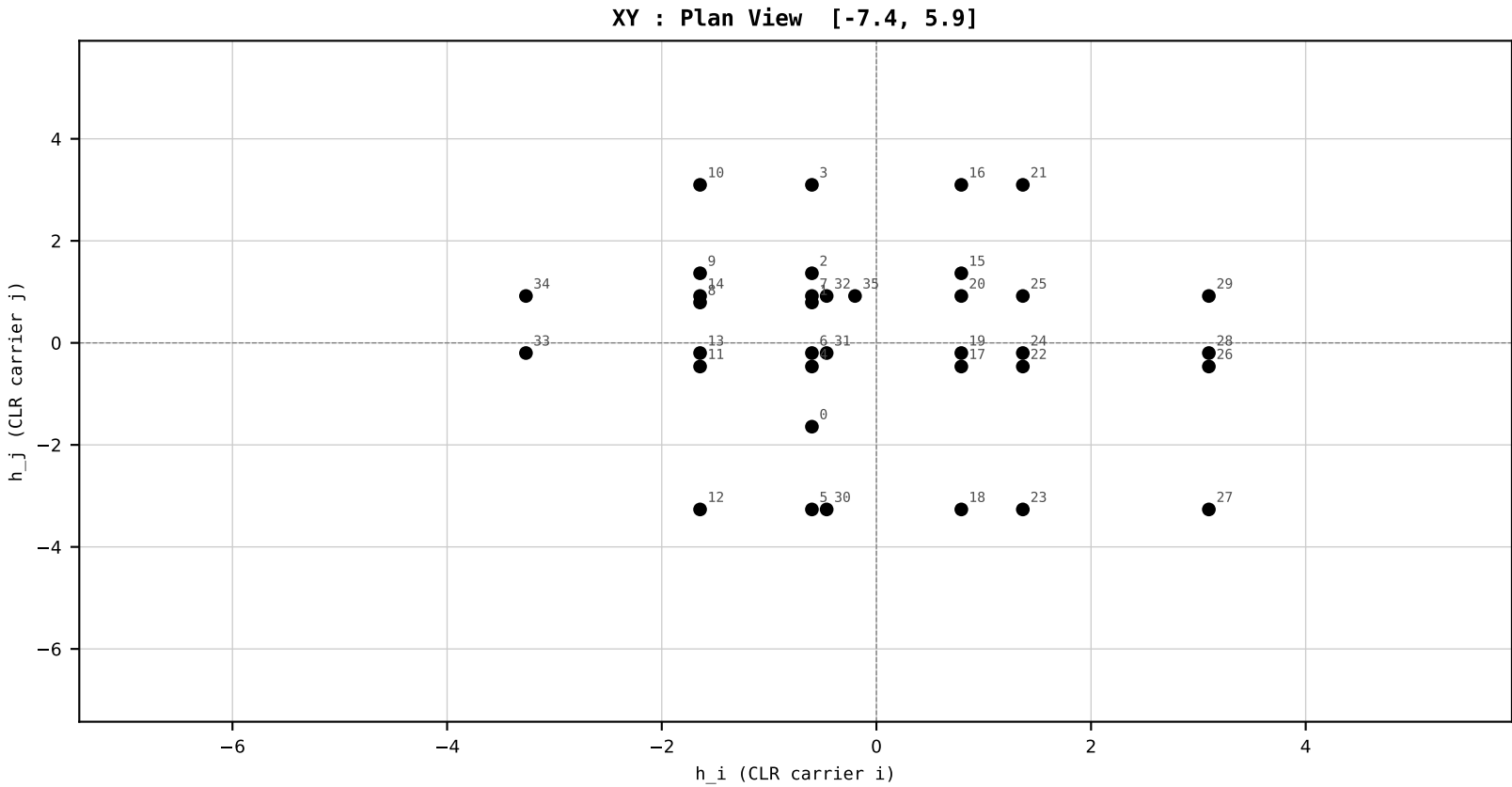
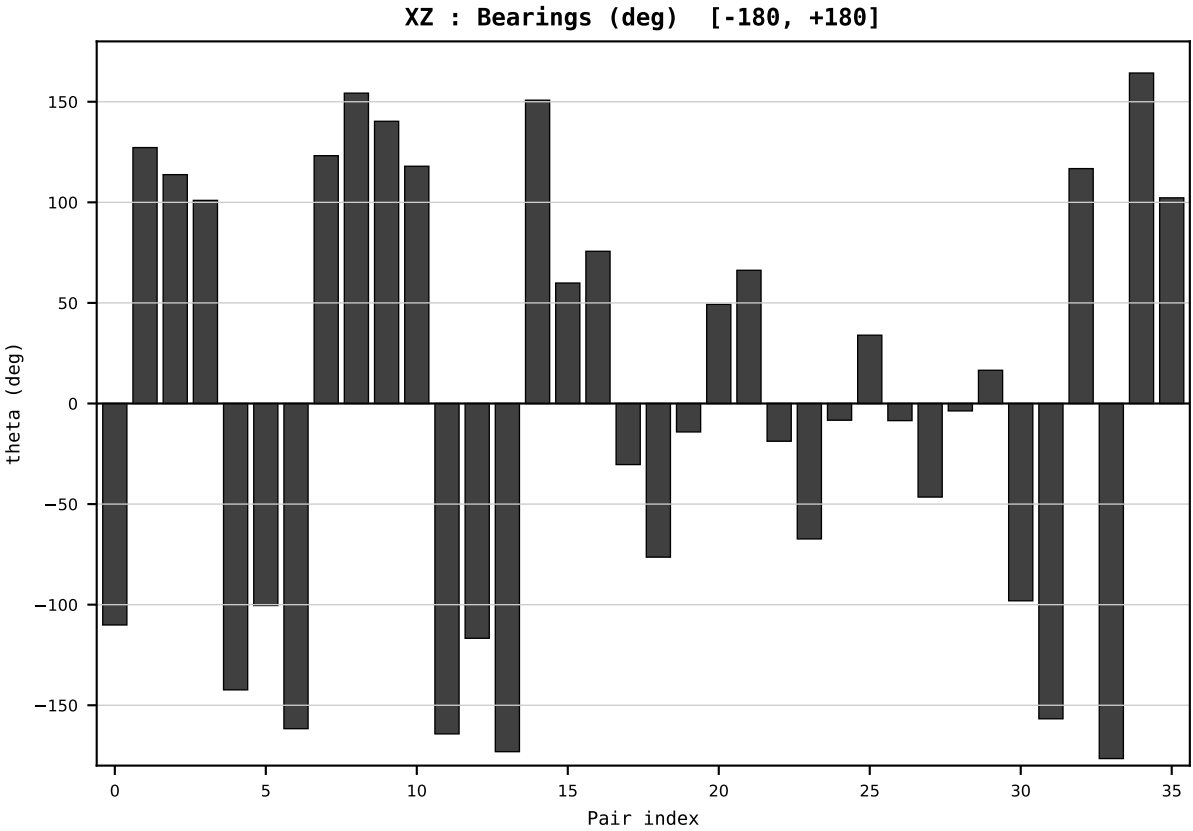
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.467323
Ring = Hs-3
E_metric= 26.9059
kappa_HS= 580.05
omega = 3.3546 deg
d_A = 0.303539
Helm = Wind
Helm d = +0.156233
DR = 6.363 HLR
DR ratio= 580.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

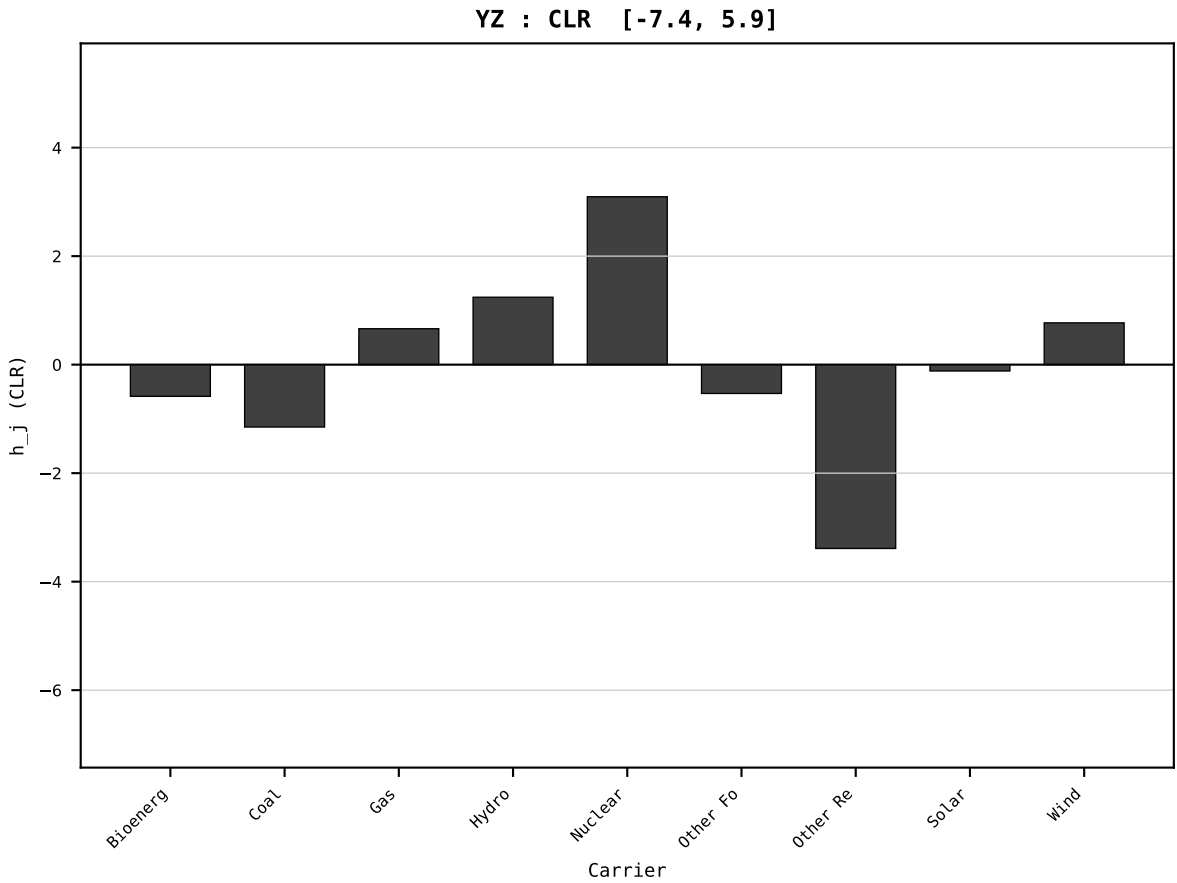
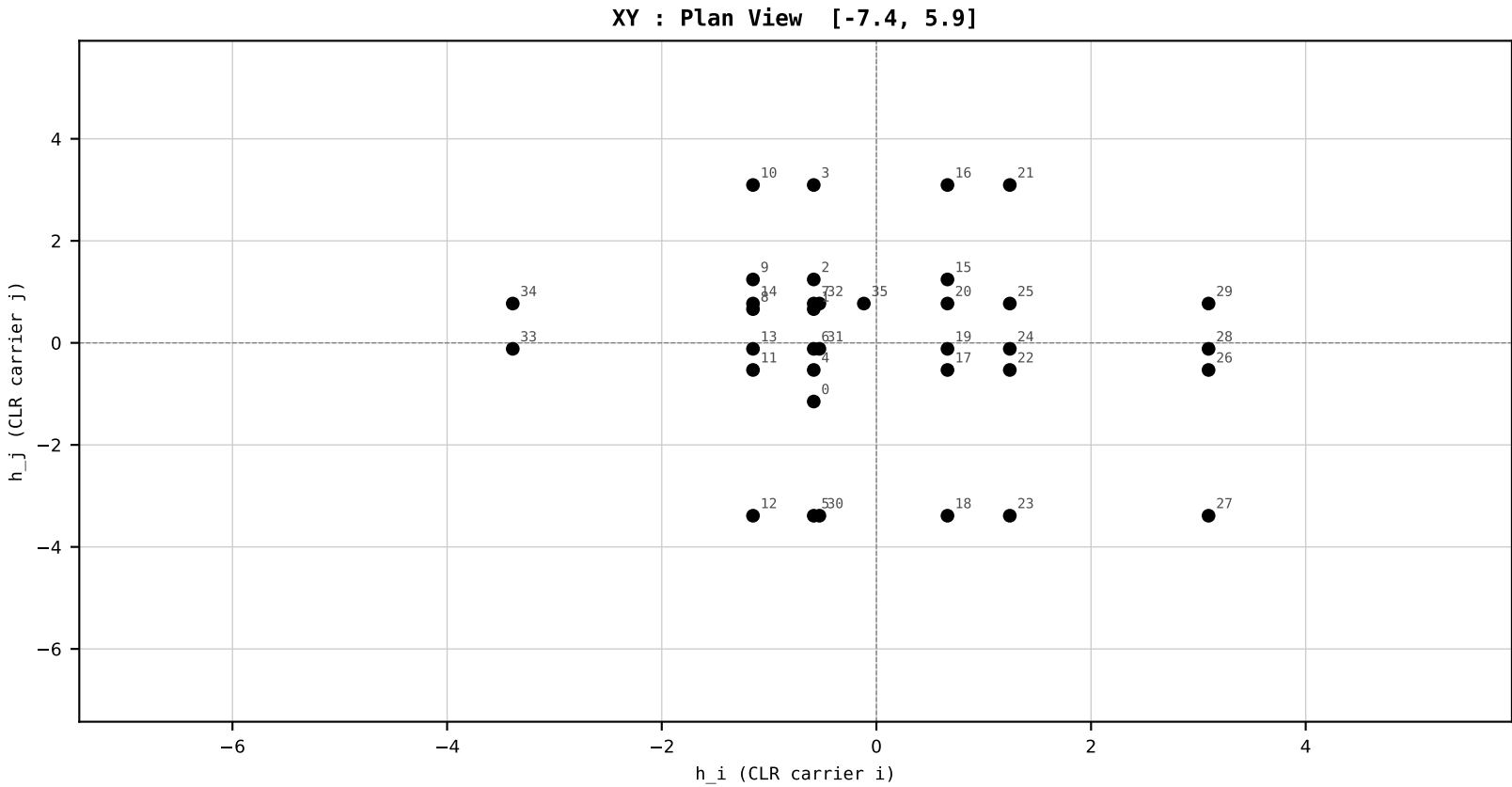
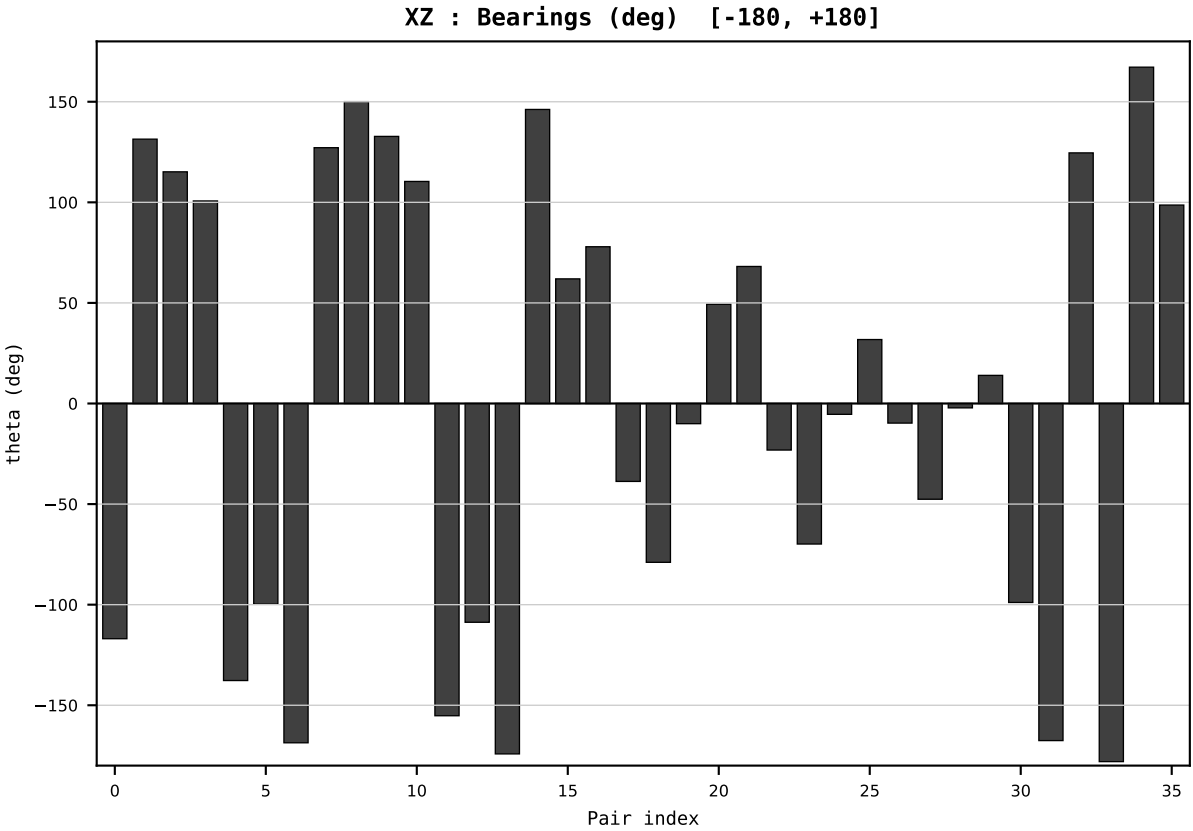
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.478912
Ring = Hs-3
E_metric= 25.5954
kappa_HS= 654.07
omega = 6.1997 deg
d_A = 0.568610
Helm = Coal
Helm d = +0.493100
DR = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

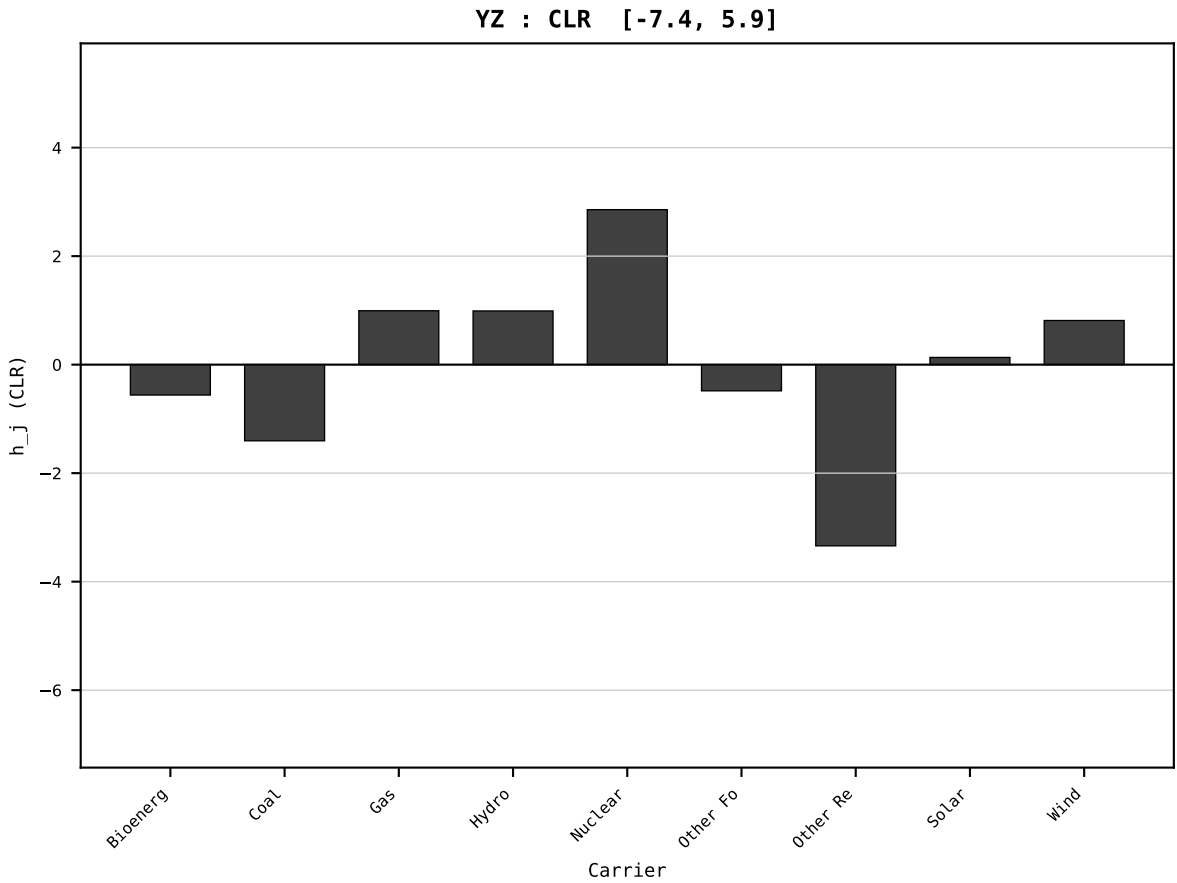
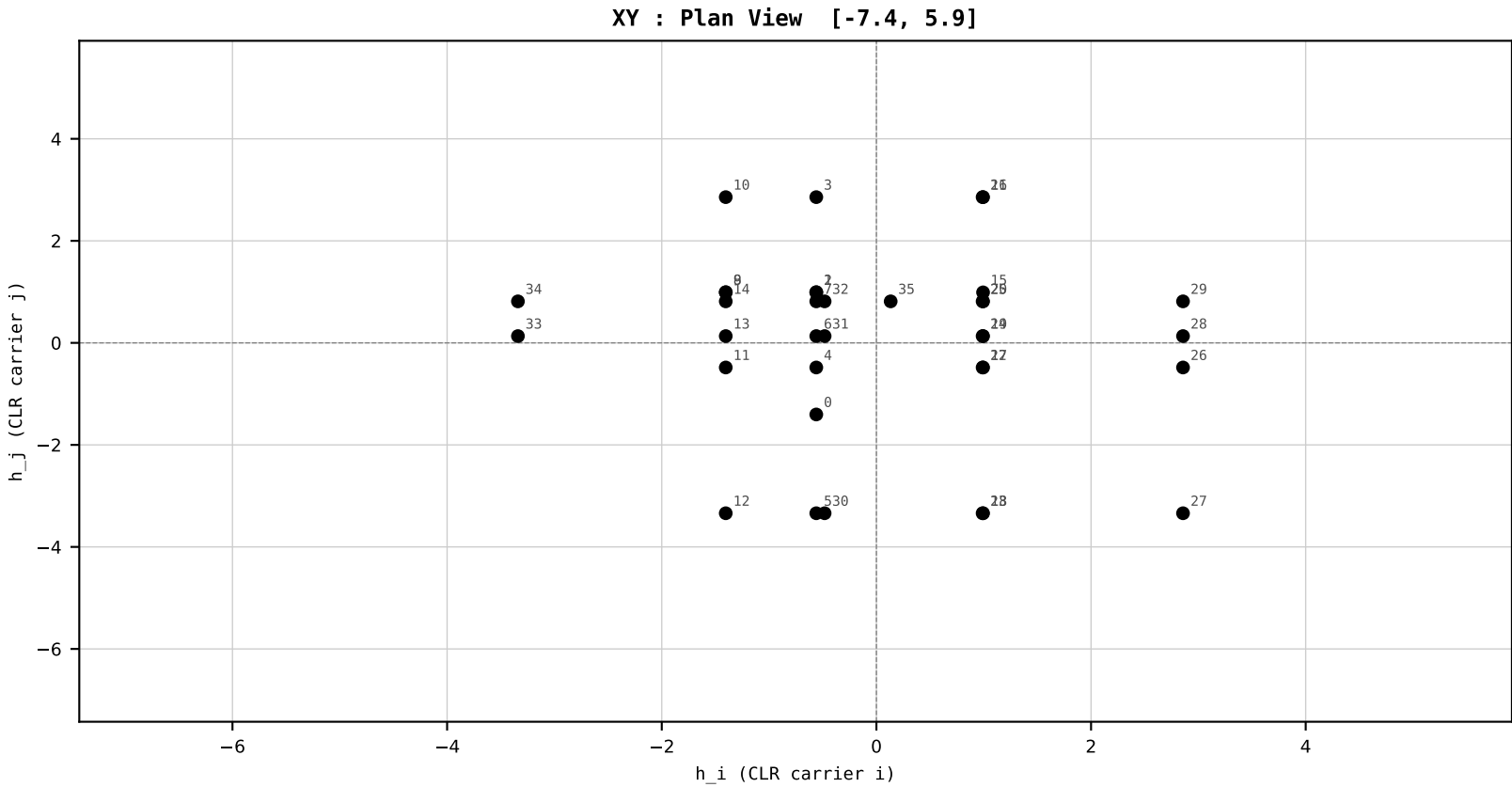
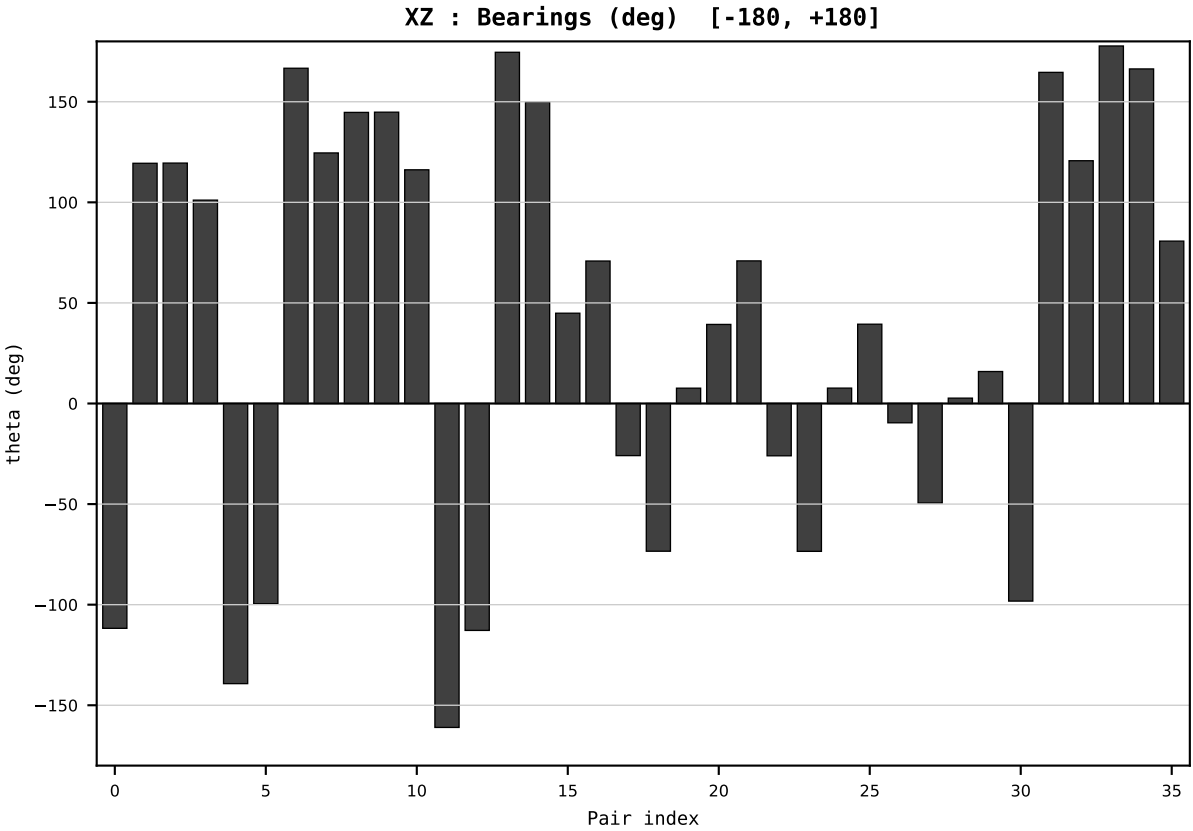
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409828
Ring = Hs-3
E_metric= 24.4815
kappa_HS= 491.22
omega = 6.8017 deg
d_A = 0.603939
Helm = Gas
Helm d = +0.331511
DR = 6.197 HLR
DR ratio= 491.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



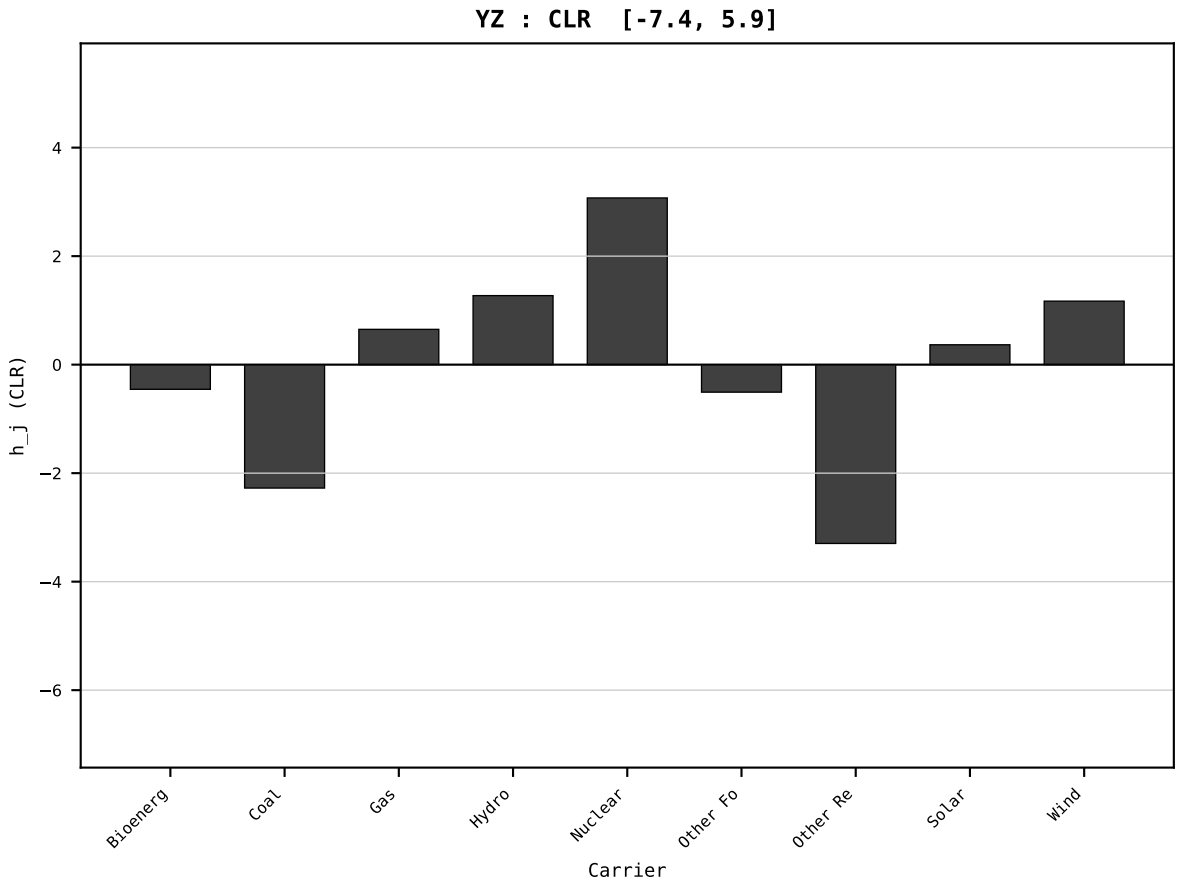
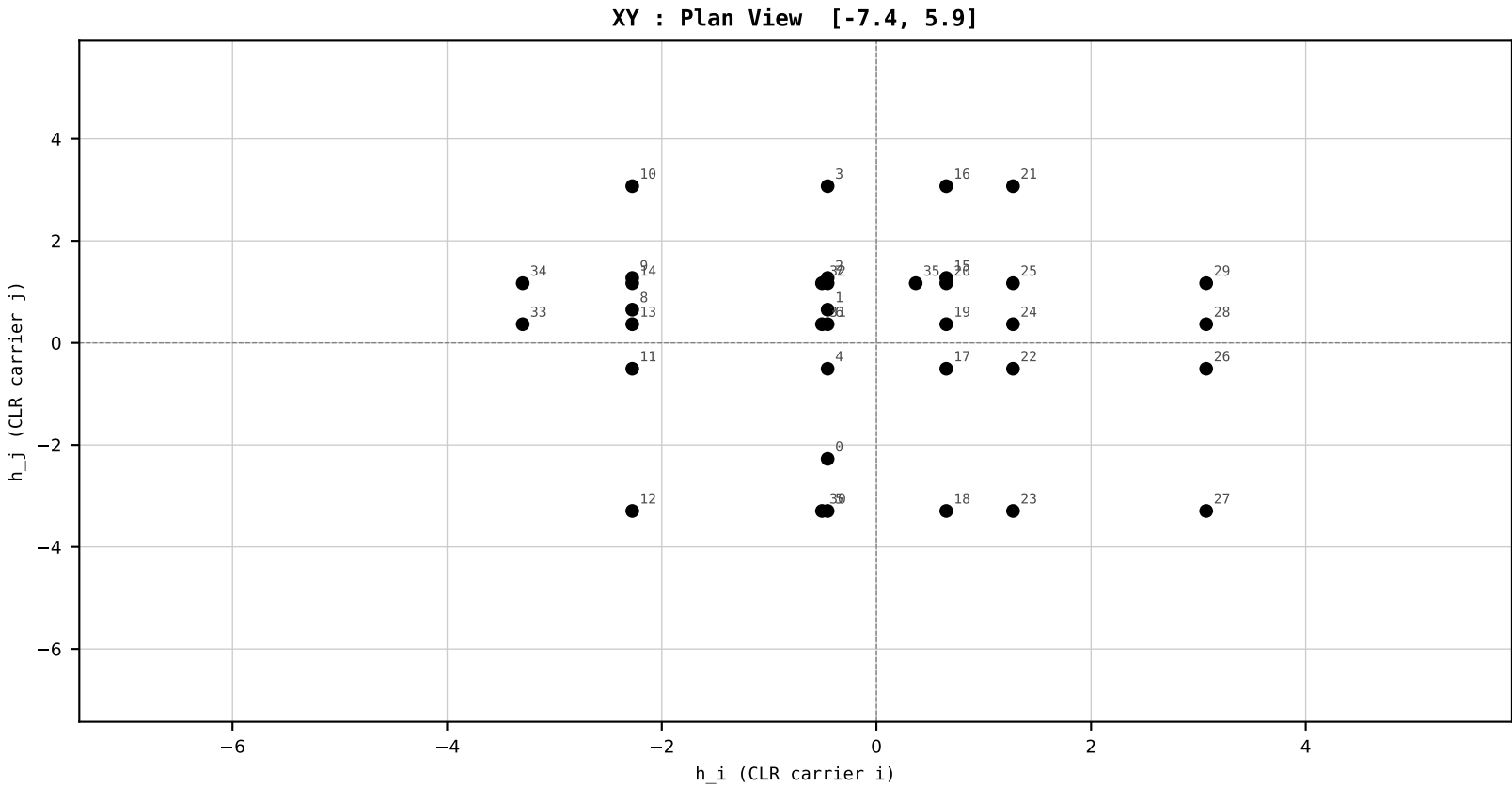
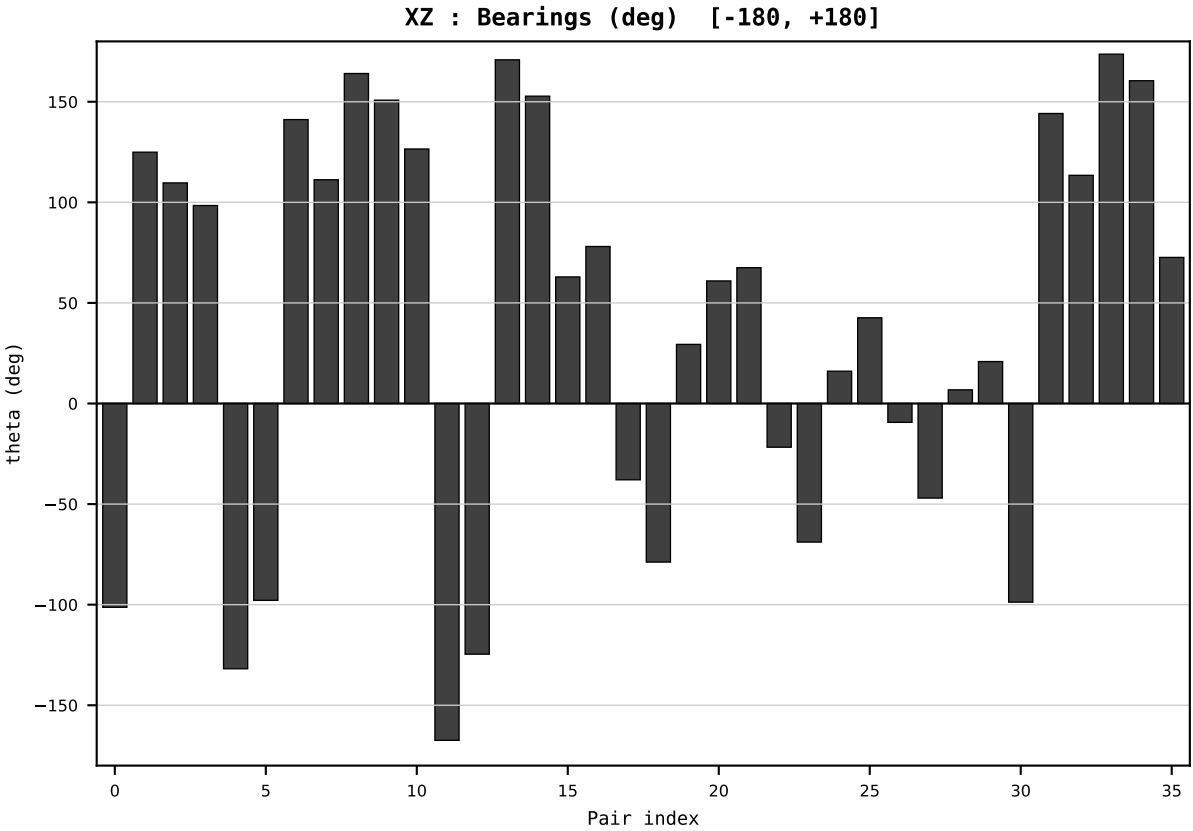
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.444201
Ring = Hs-3
E_metric= 29.4888
kappa_HS= 583.10
omega = 10.8790 deg
d_A = 1.094786
Helm = Coal
Helm d = -0.871139
DR = 6.368 HLR
DR ratio= 583.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



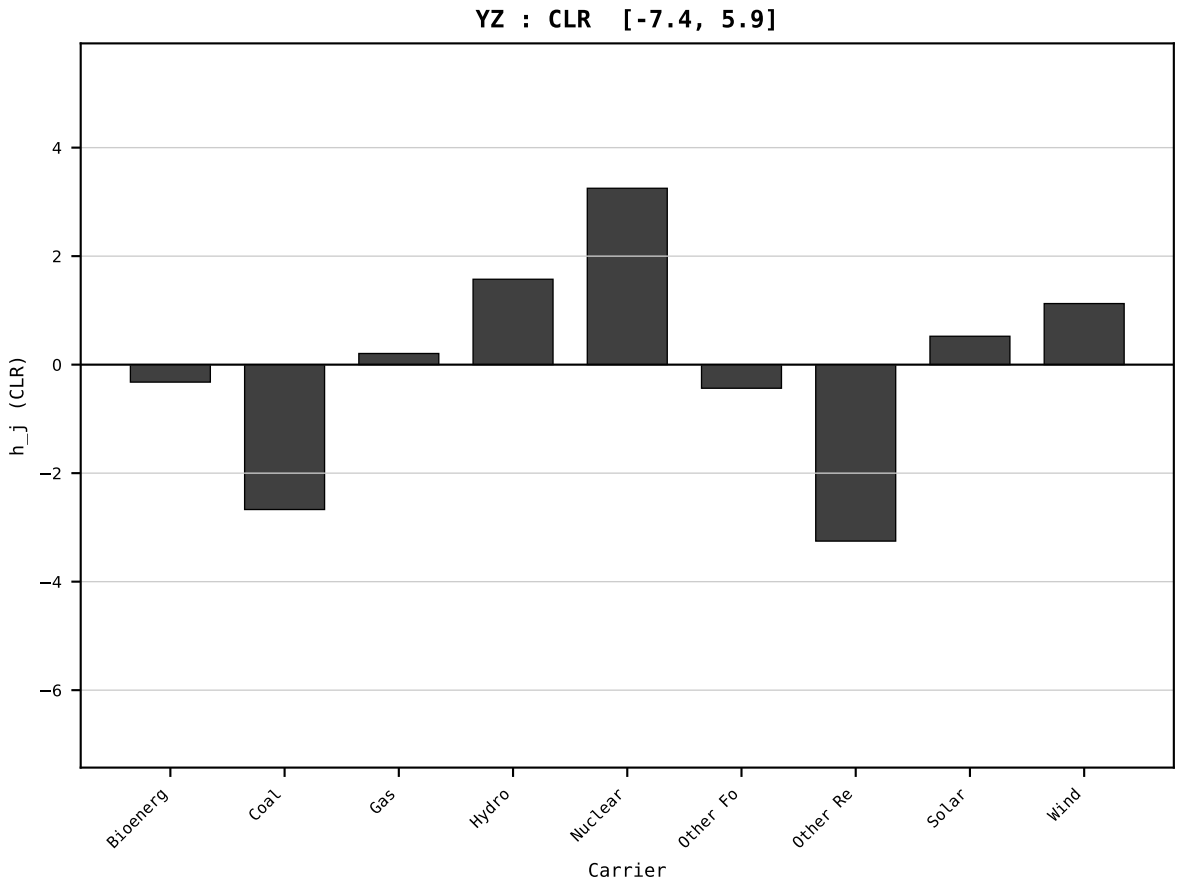
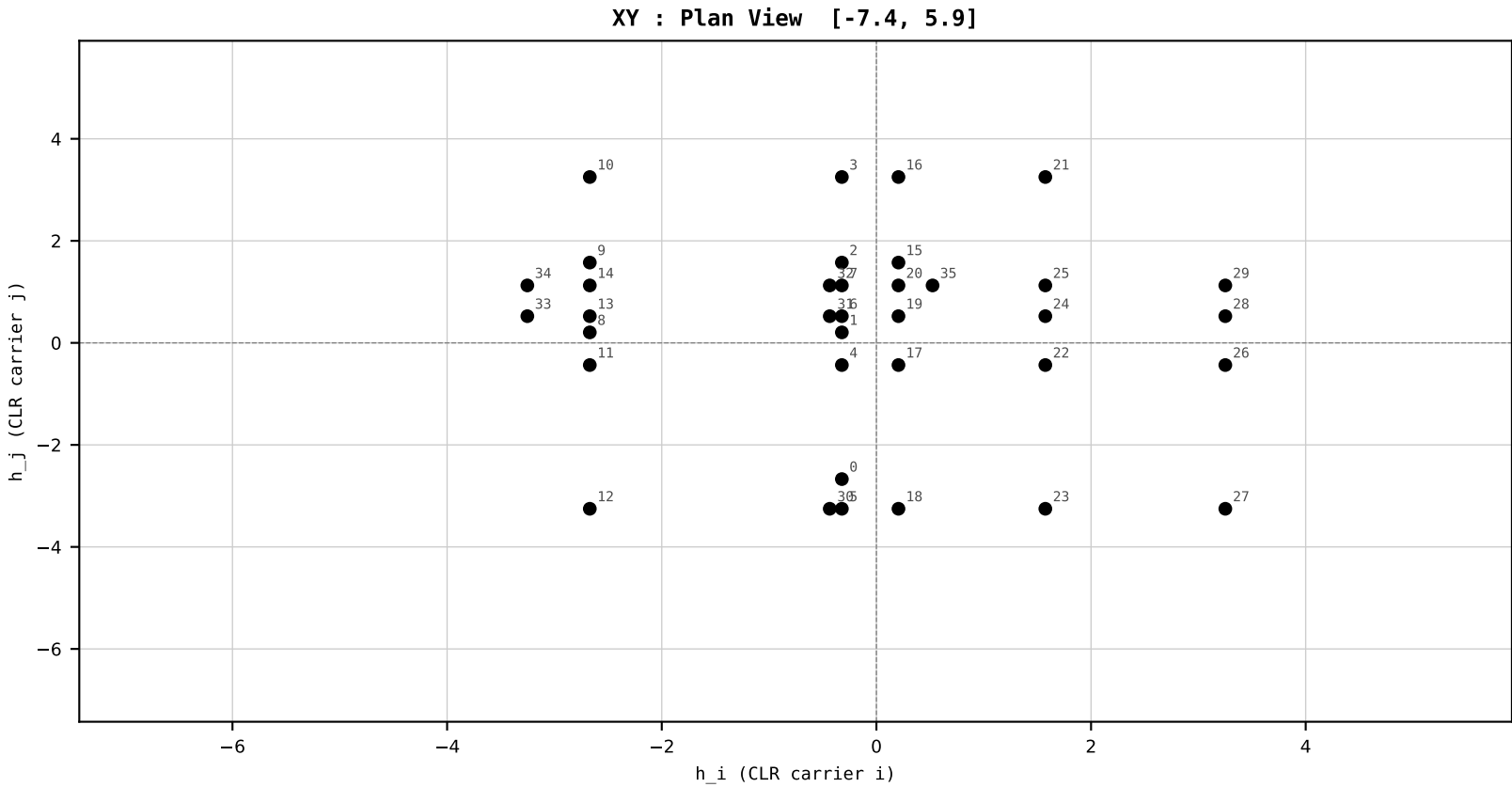
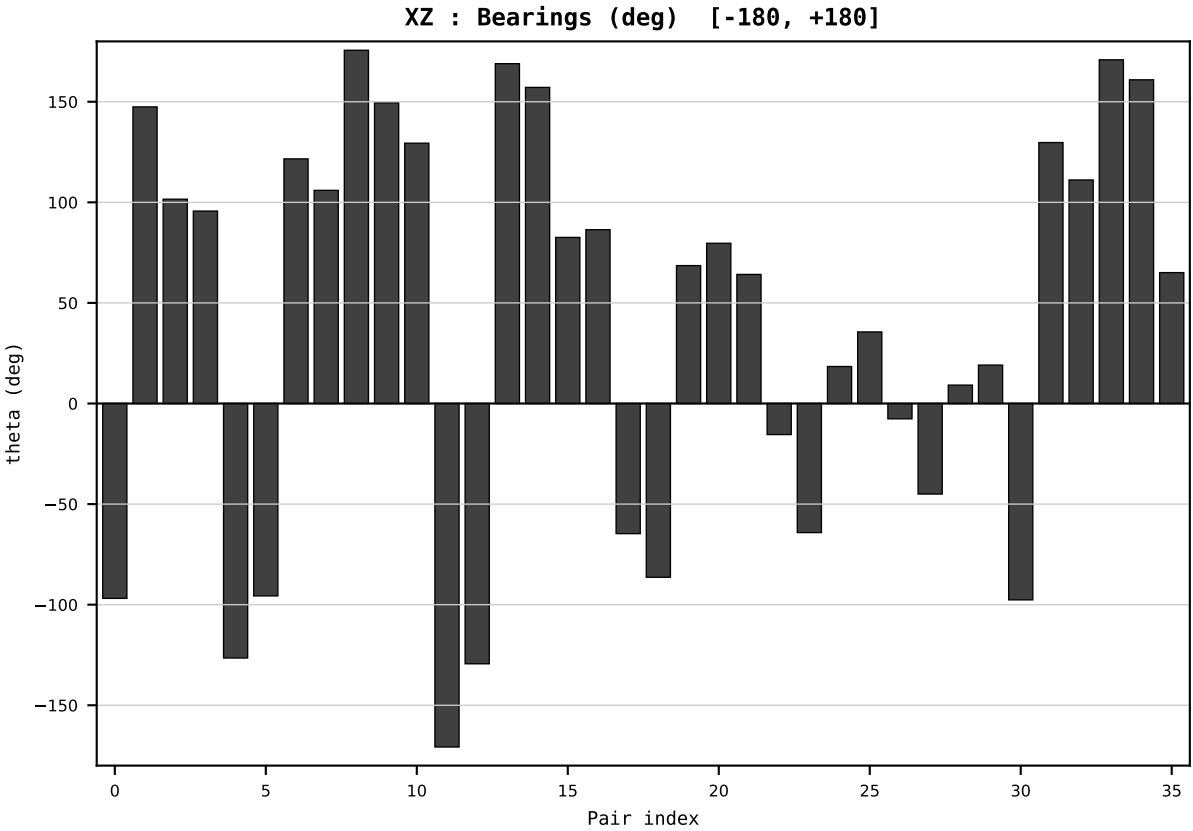
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.480863
Ring = Hs-3
E_metric= 32.6333
kappa_HS= 667.46
omega = 6.8989 deg
d_A = 0.727222
Helm = Gas
Helm d = -0.445163
DR = 6.503 HLR
DR ratio= 667.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

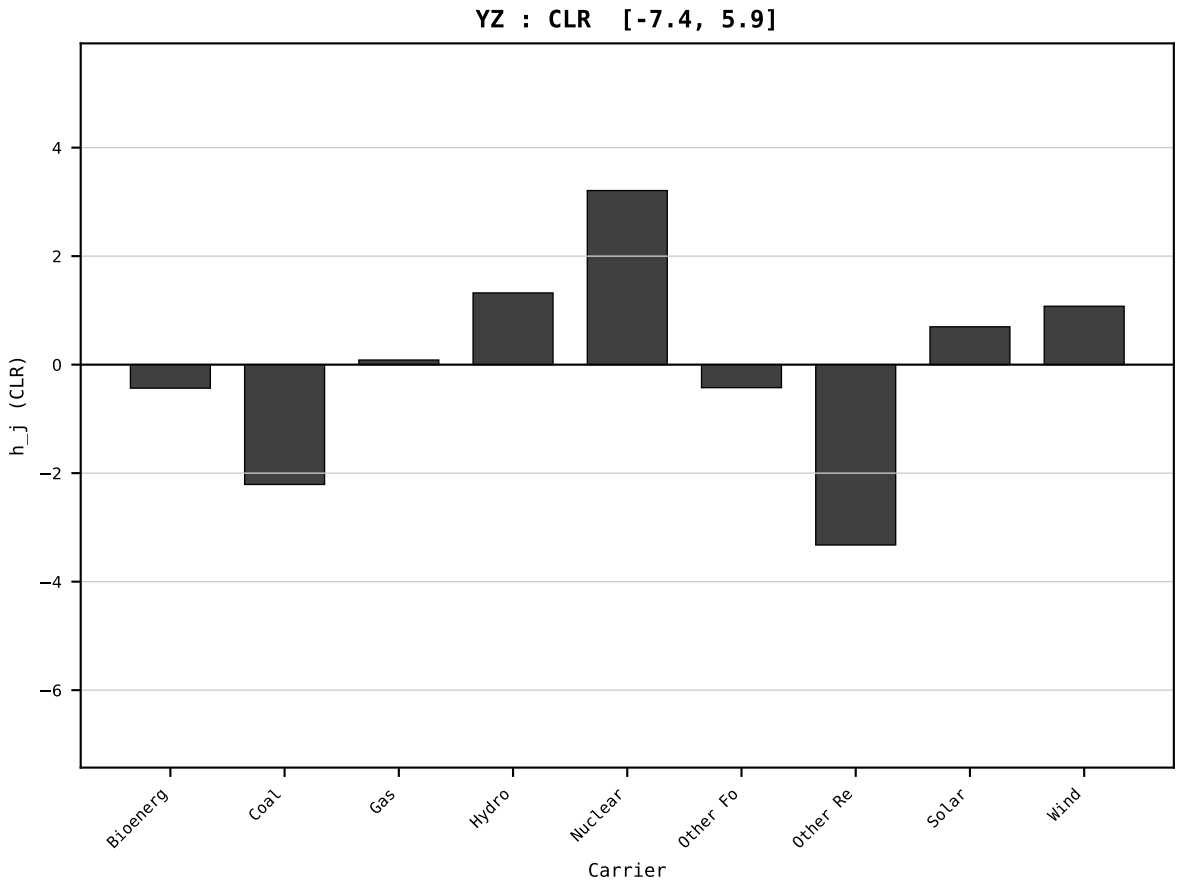
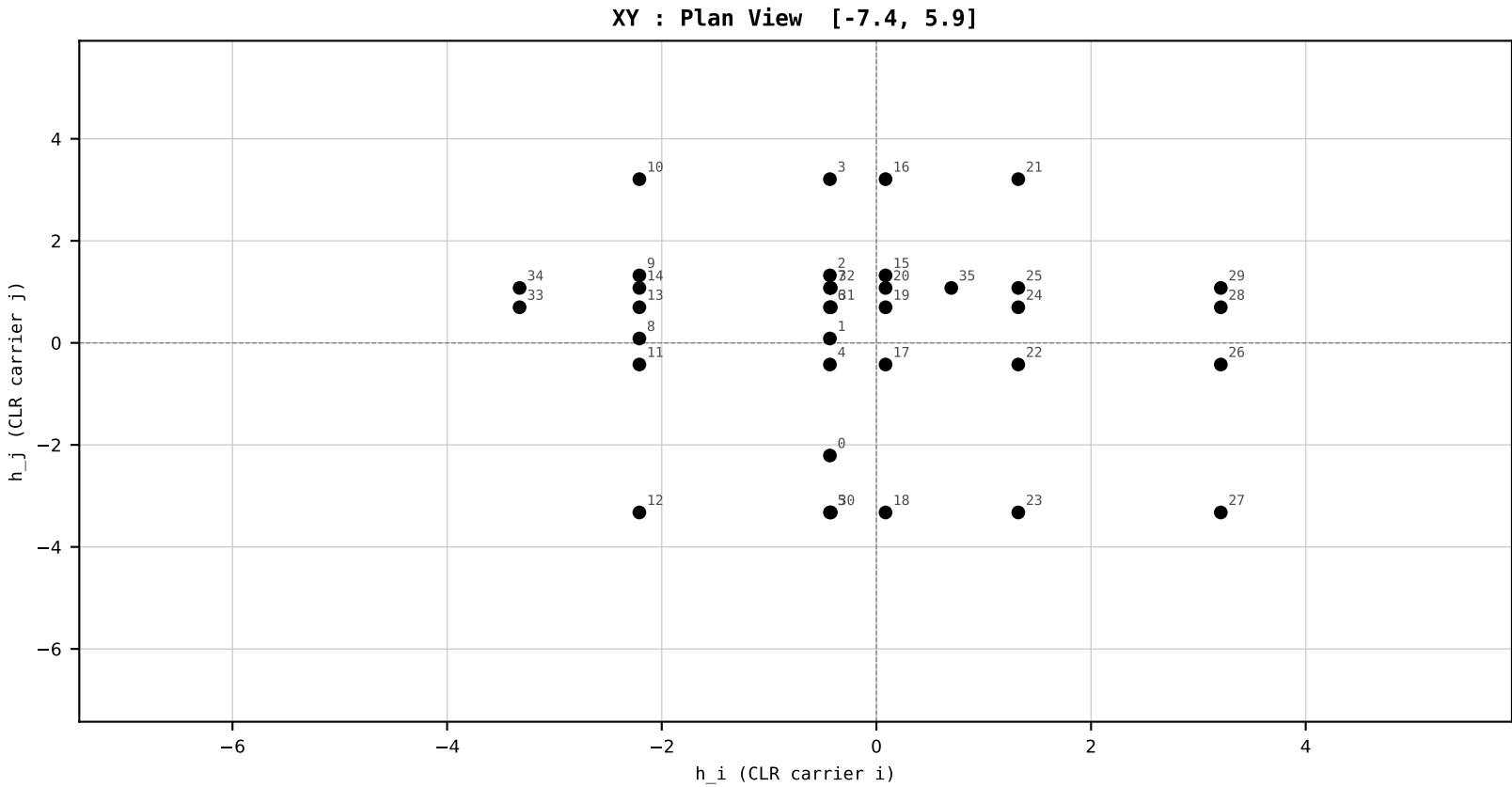
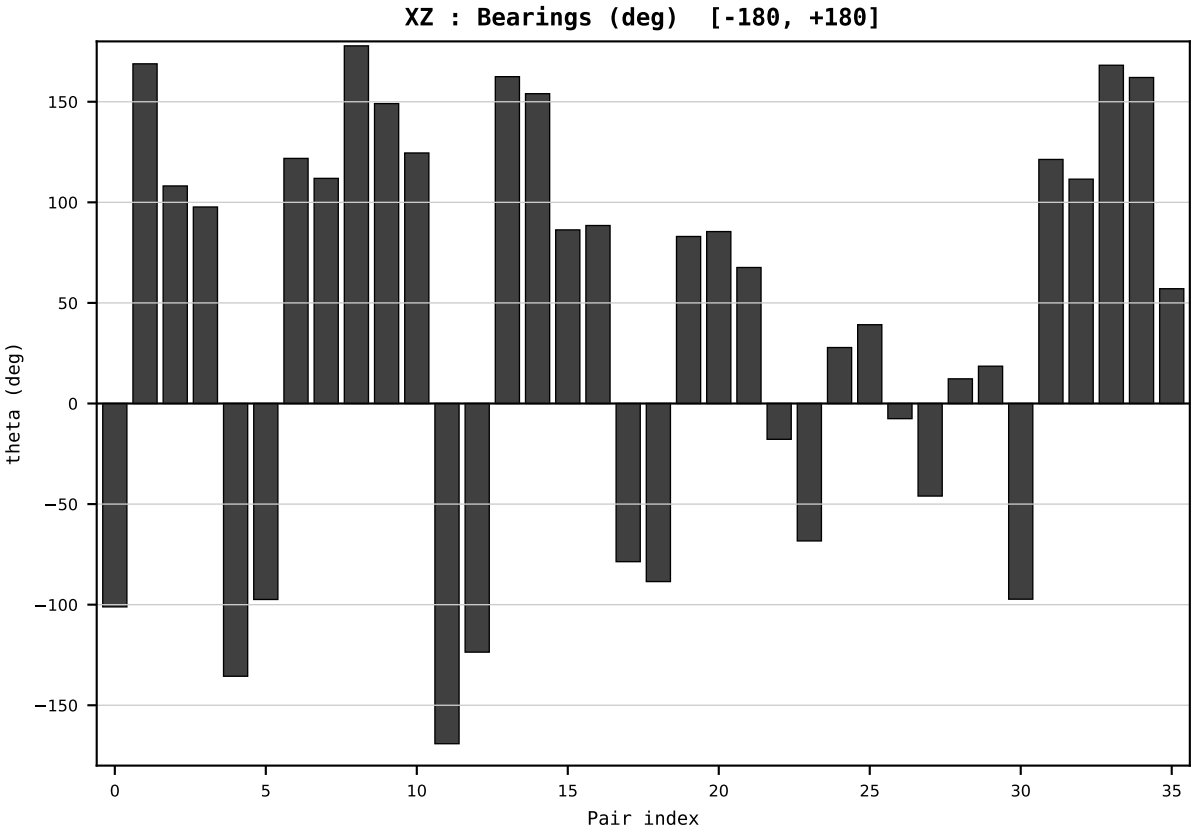
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

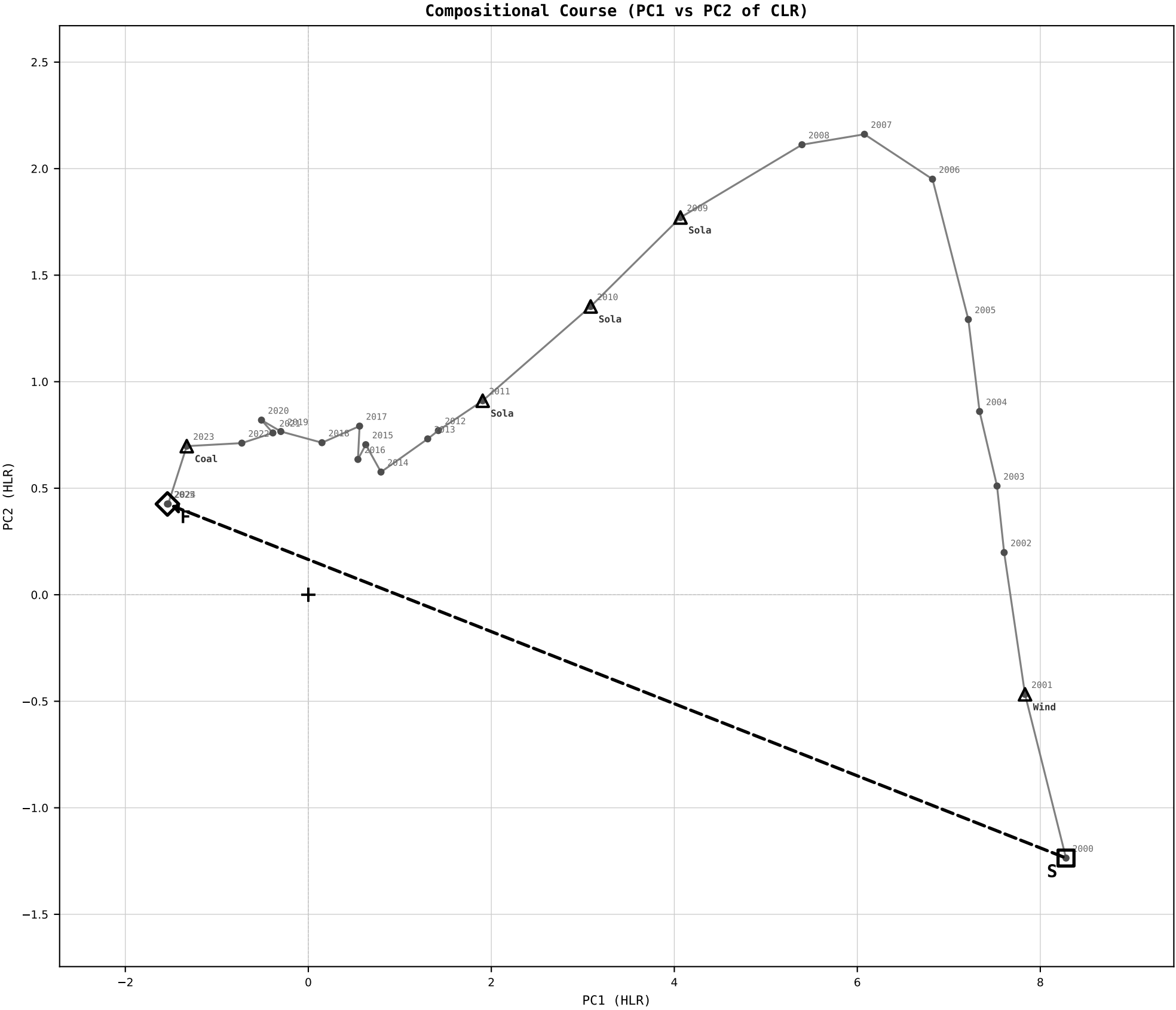
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.483790
Ring = Hs-3
E_metric= 29.9972
kappa_HS= 687.84
omega = 5.5004 deg
d_A = 0.586196
Helm = Coal
Helm d = +0.461974
DR = 6.534 HLR
DR ratio= 687.8x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 10.0035 HLR
Path length = 17.9314 HLR
Directness = 0.5579

DR start = 10.634 HLR
ratio = 41516.0x
DR final = 6.534 HLR
ratio = 687.8x

E_{metric} start = 94.4242
E_{metric} final = 29.9972

PC1 loadings:
Solar -0.768
Coal +0.387
Wind -0.385

PC2 loadings:
Wind +0.841
Solar -0.448
Hydro -0.160

Top displacement events:
2009 Solar d=1.437
2011 Solar d=1.337
2010 Solar d=1.129
2023 Coal d=1.095
2001 Wind d=0.998

The instrument reads.
The expert decides.
The loop stays open.

View B

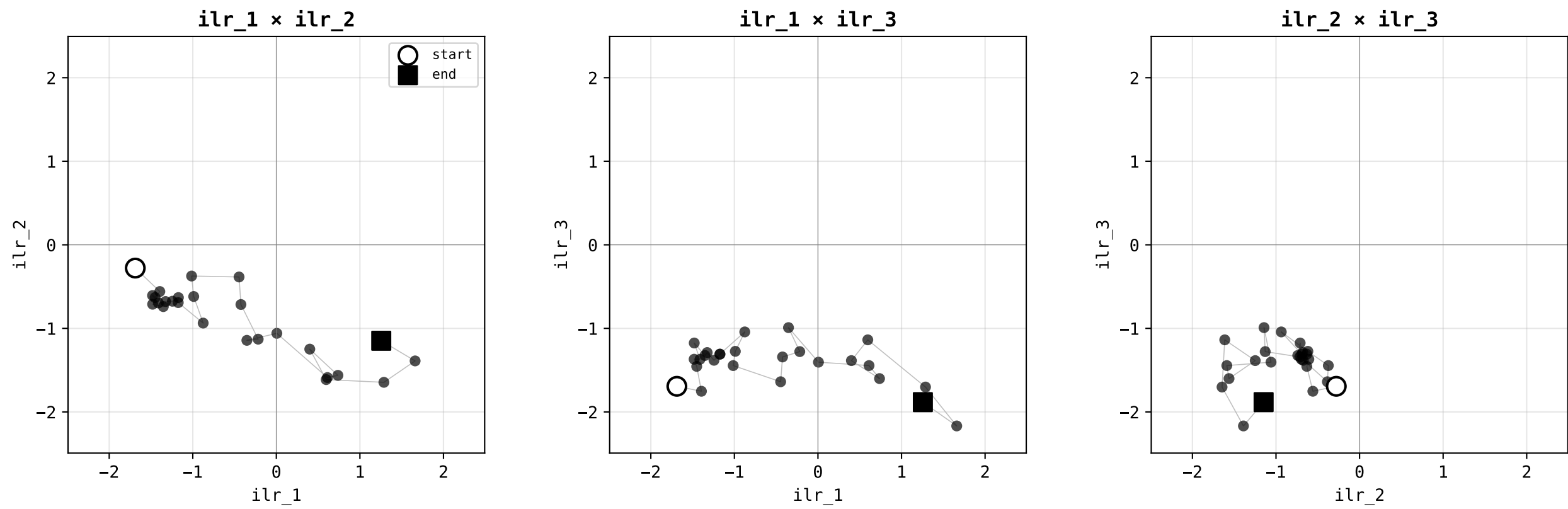
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ FRA France

ILR-Helmert Orthogonal Triplet – FRA France · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=26 years · D=9 carriers



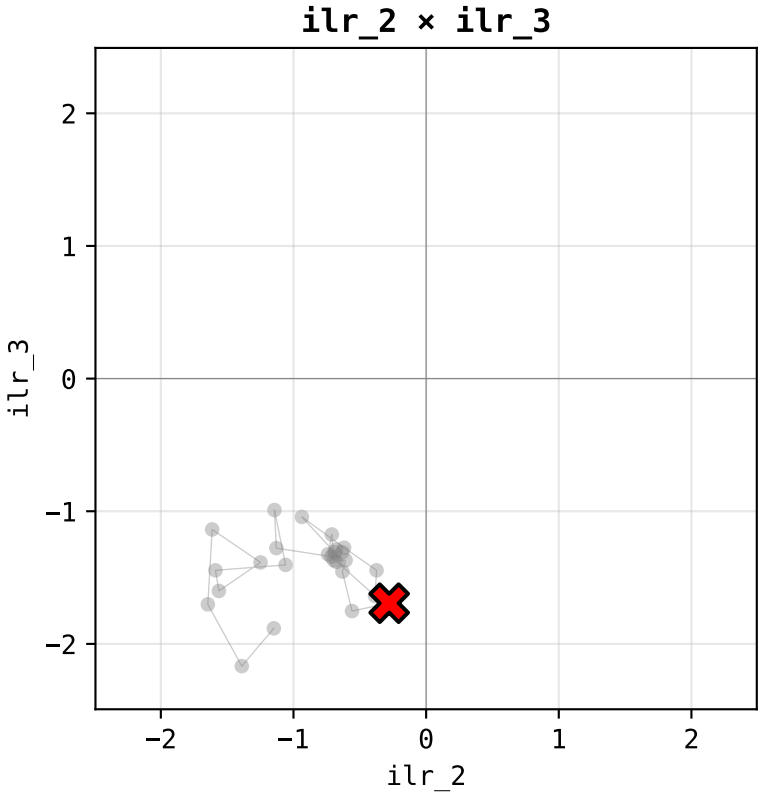
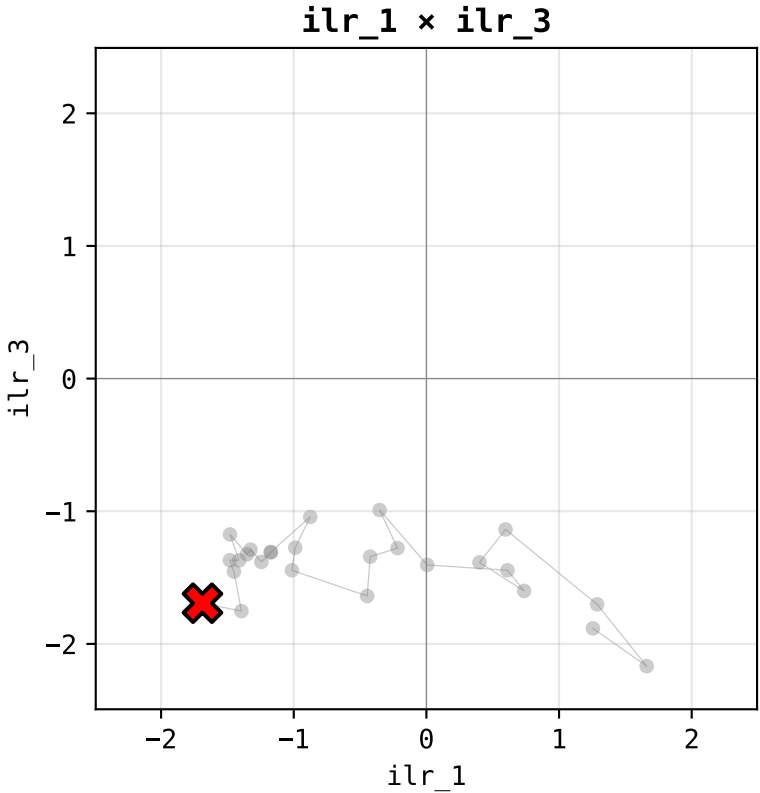
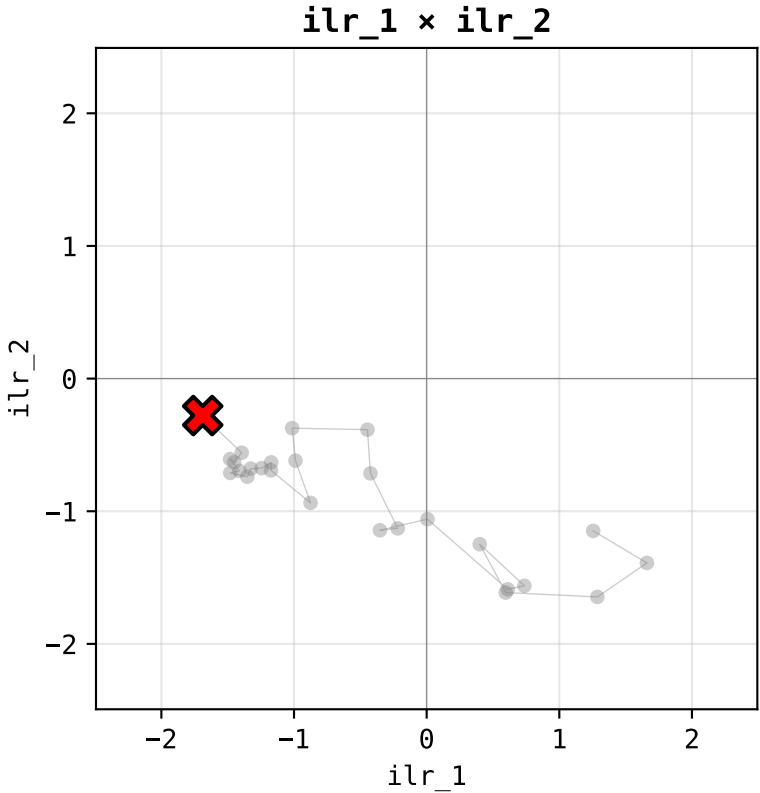
Helmert basis loadings (which carriers each ILR axis contrasts):

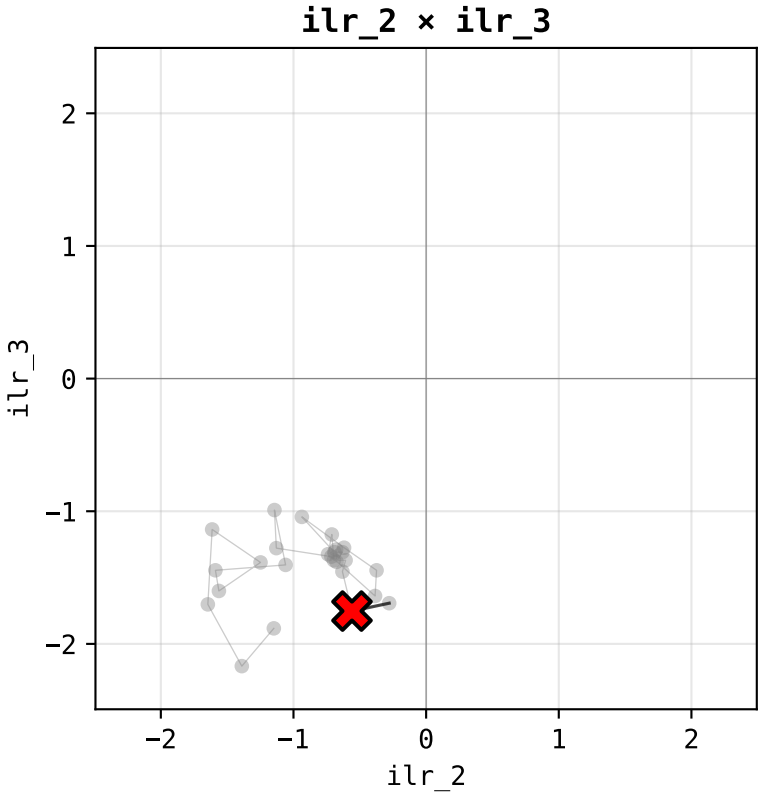
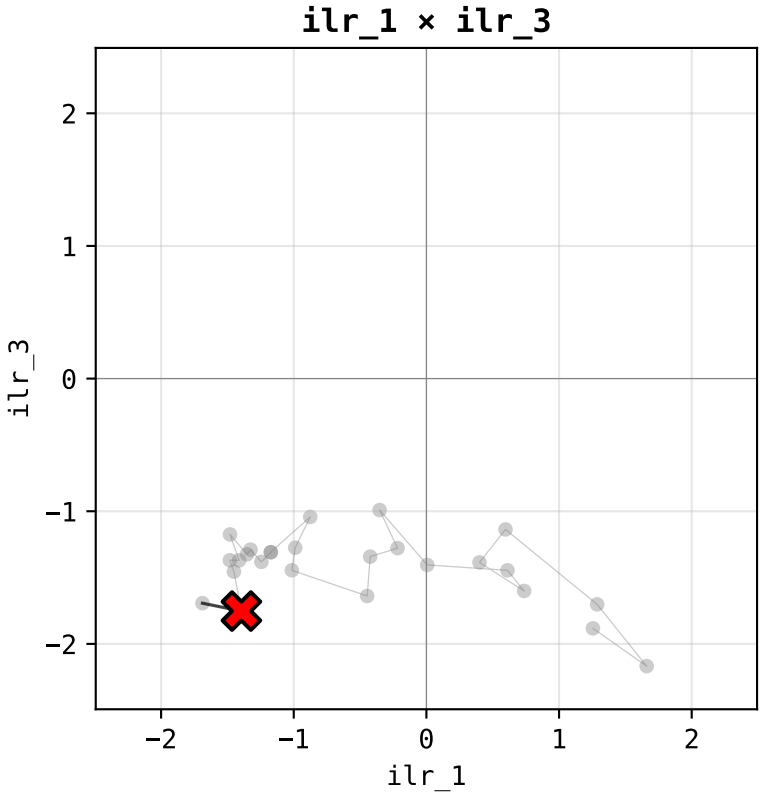
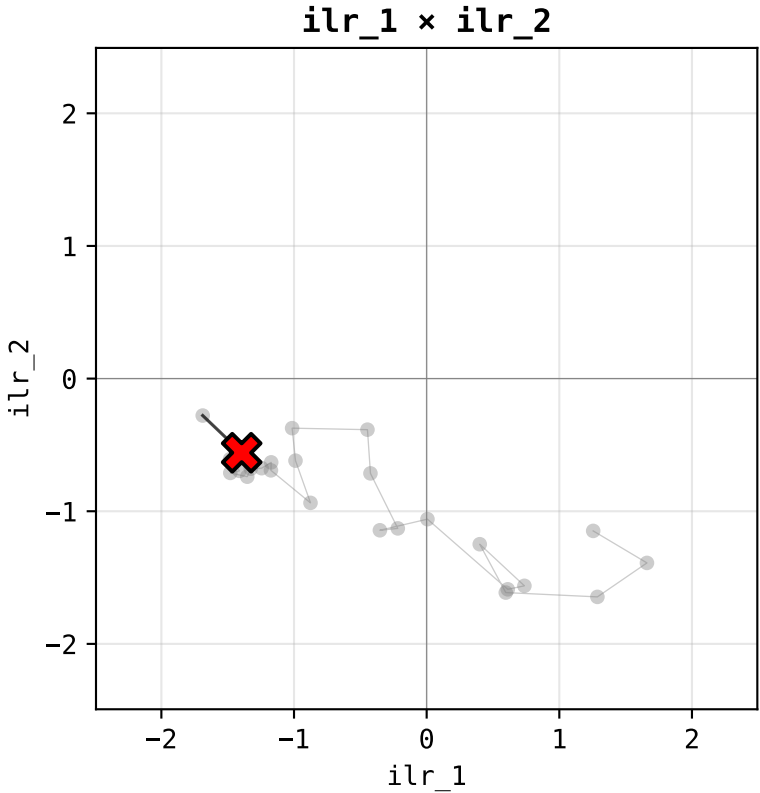
- ilr_1: (Bioenergy) vs Coal
- ilr_2: (Bioenergy + Coal) vs Gas
- ilr_3: (Bioenergy + Coal + Gas) vs Hydro

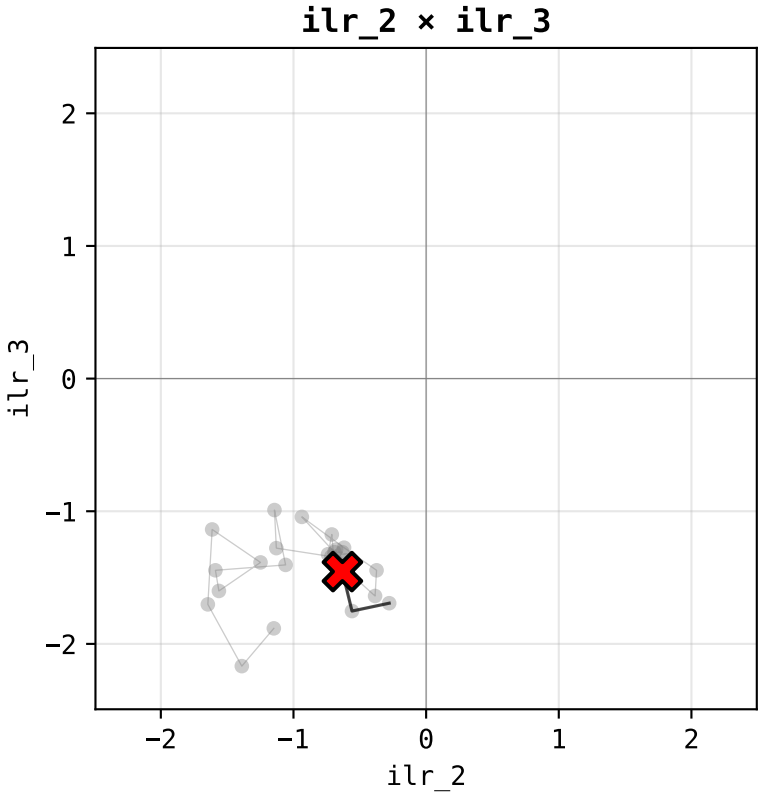
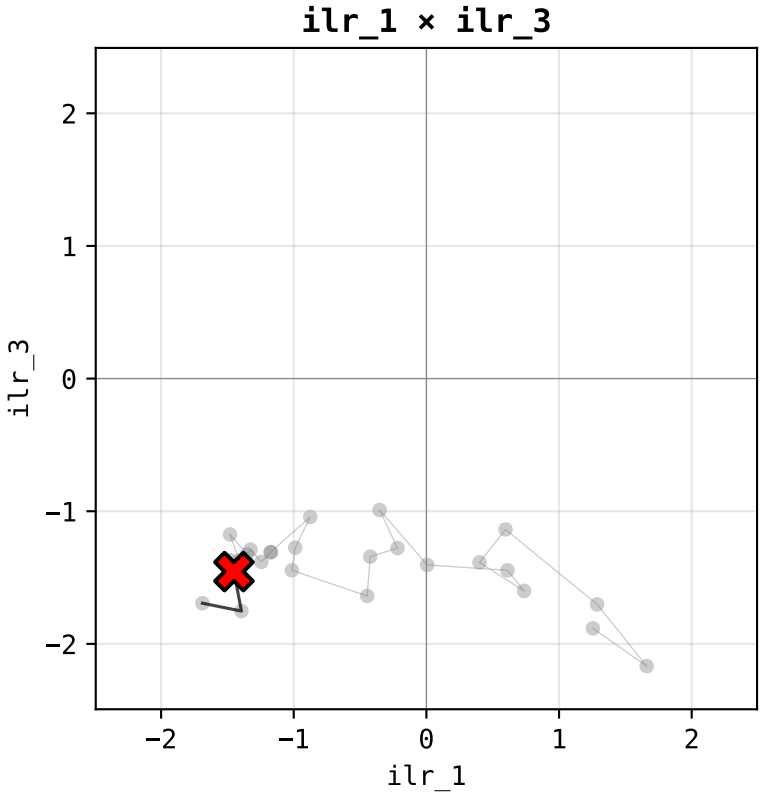
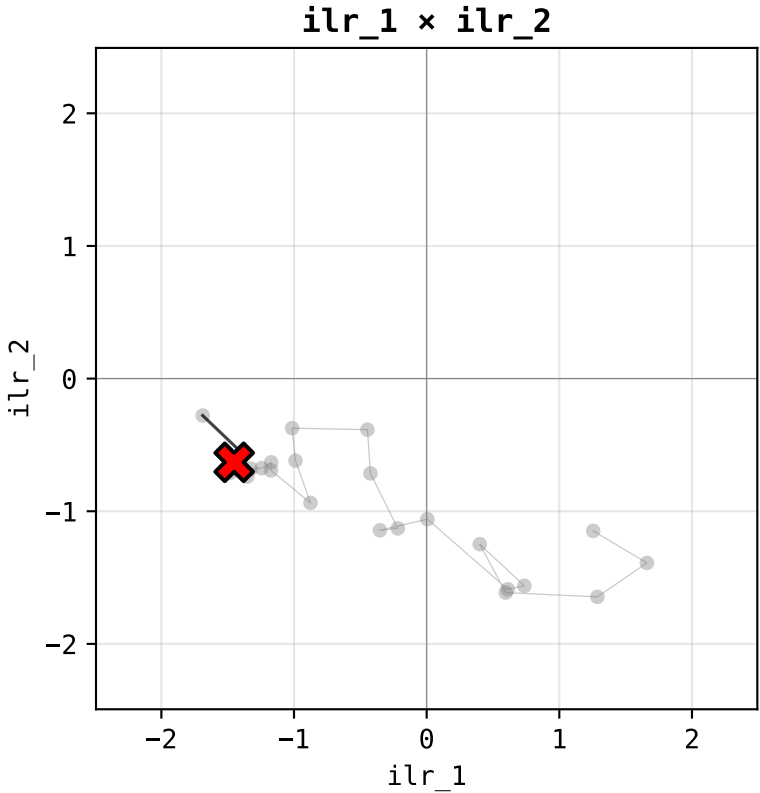
Trajectory:
start year: 2000
end year: 2025
N readings: 26

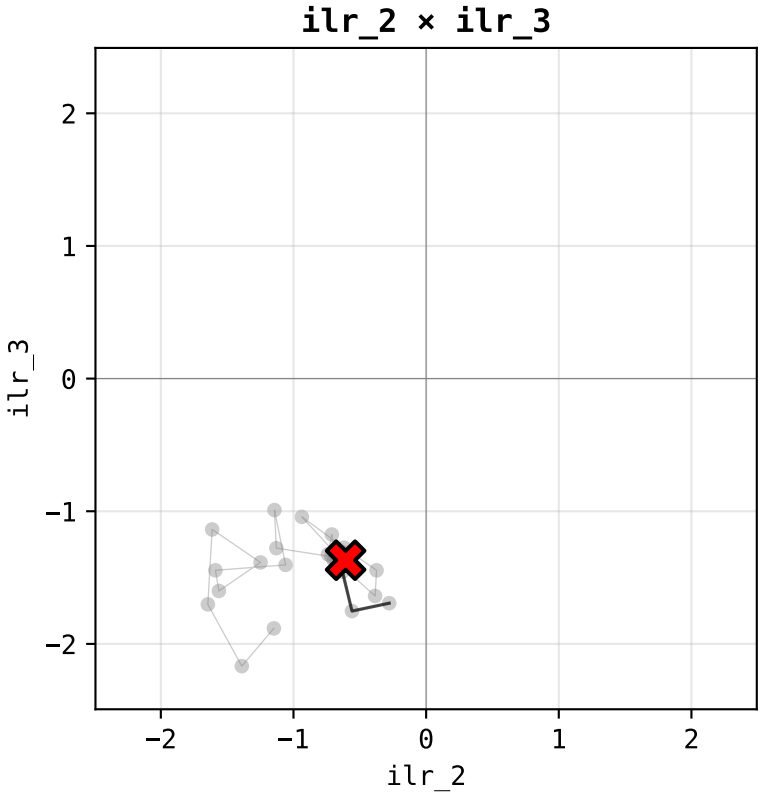
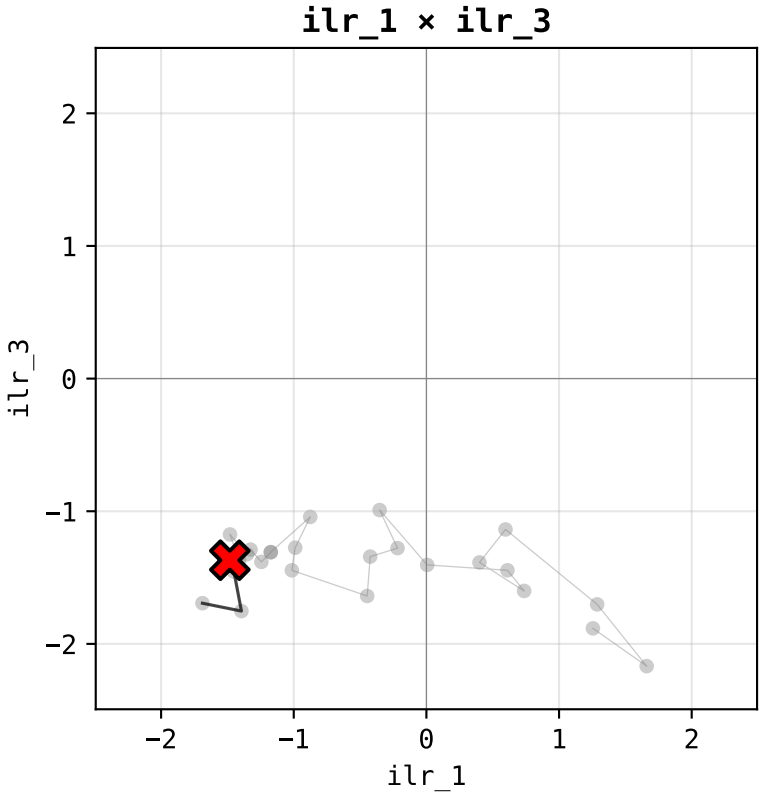
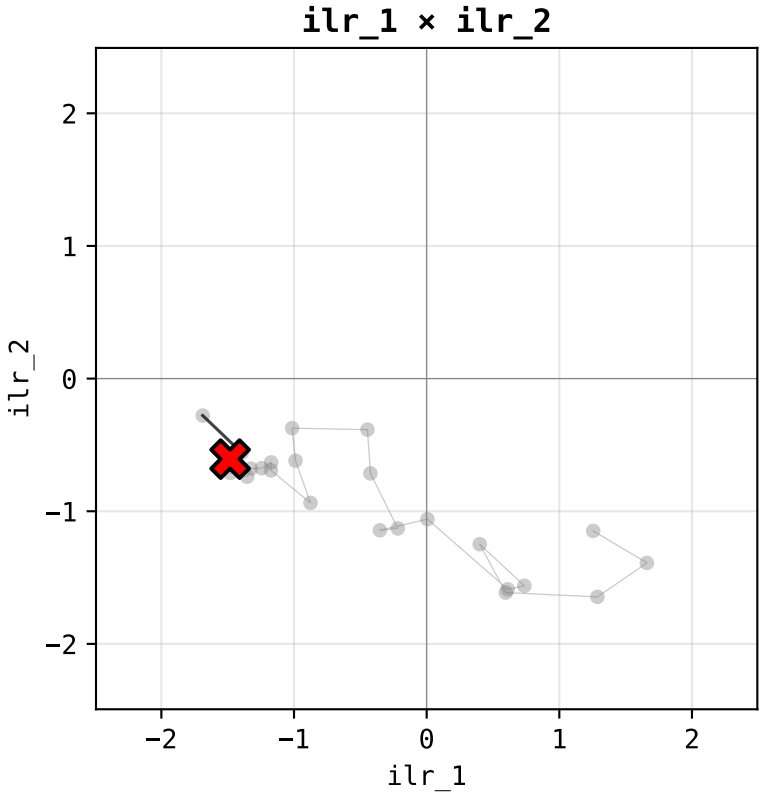
Display:
fixed scale ± 2.49 ILR
Order 1 (first principles)
per Output Doctrine v1.0

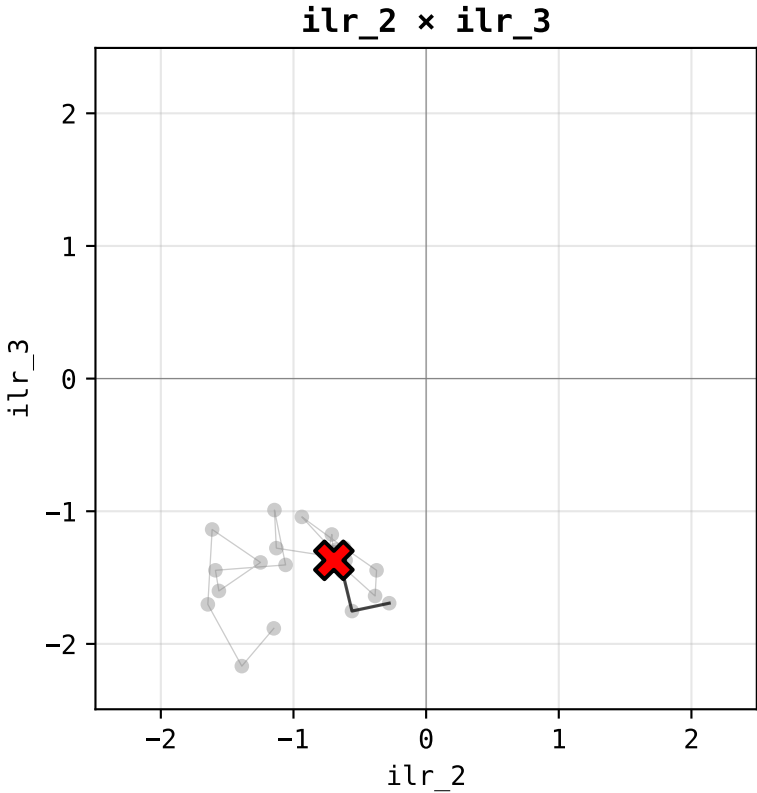
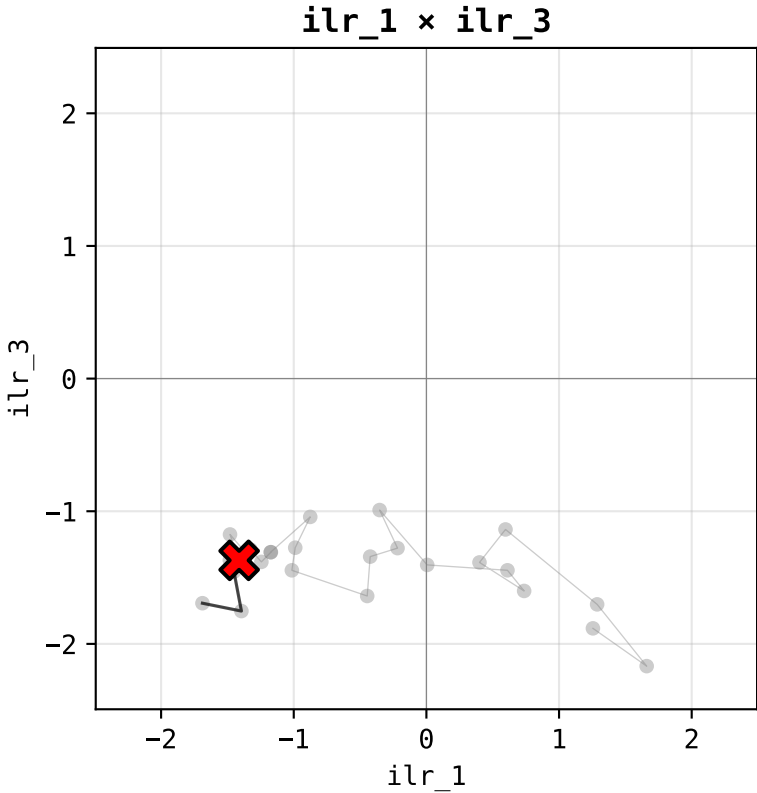
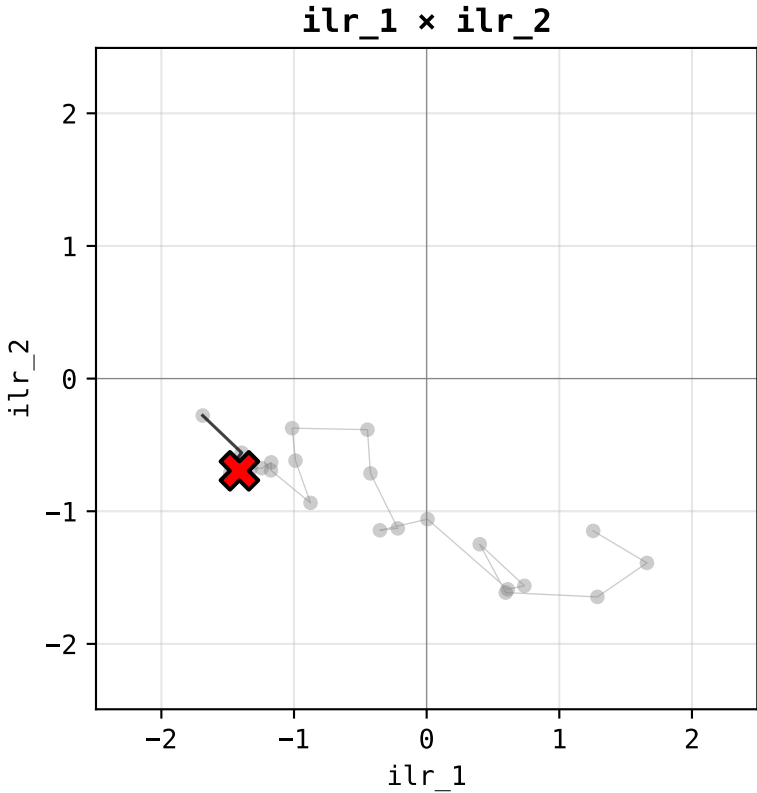
Symbology:
○ start · ■ end
· intermediate years

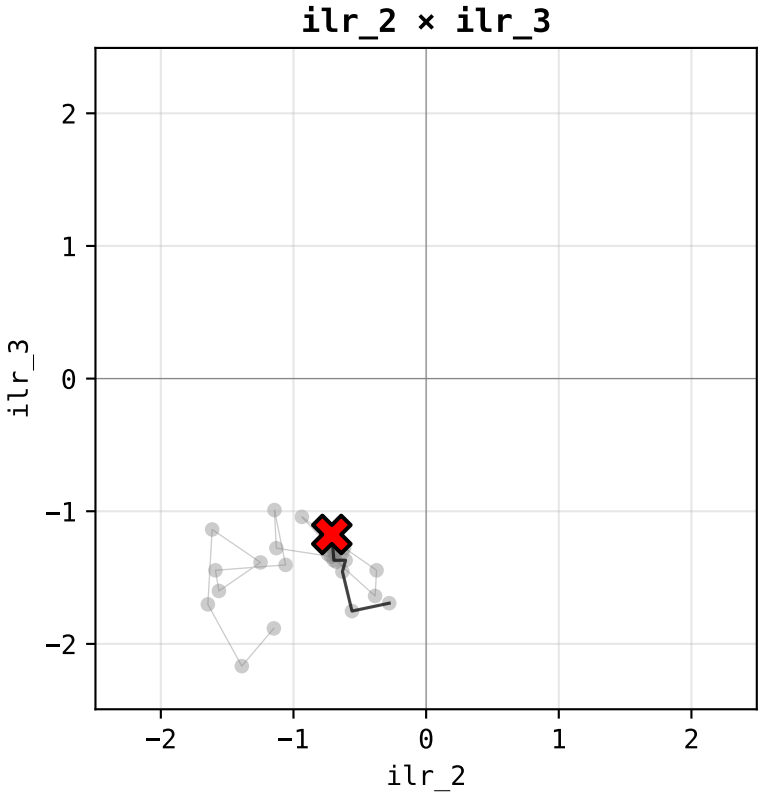
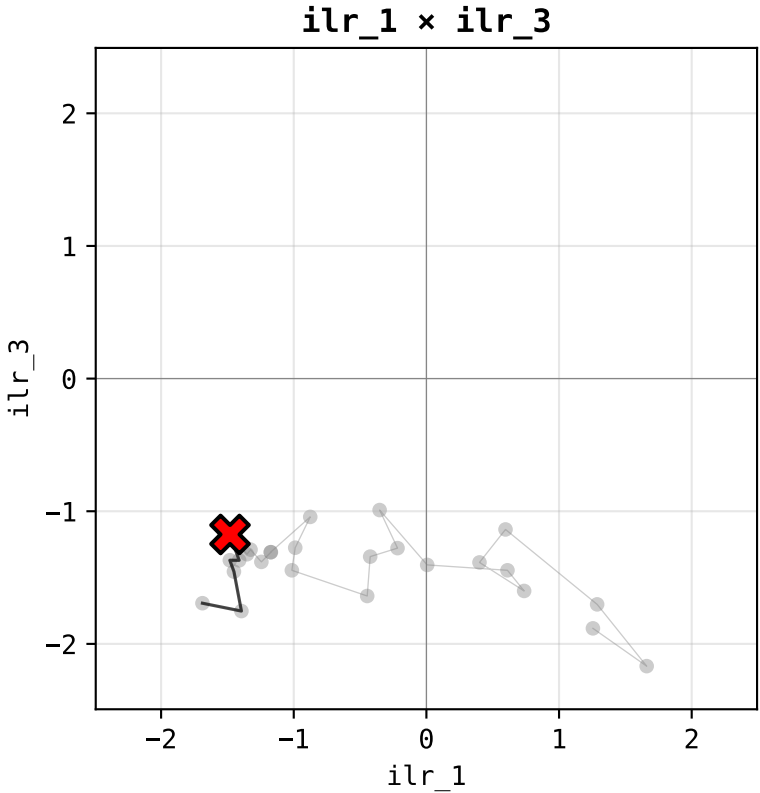
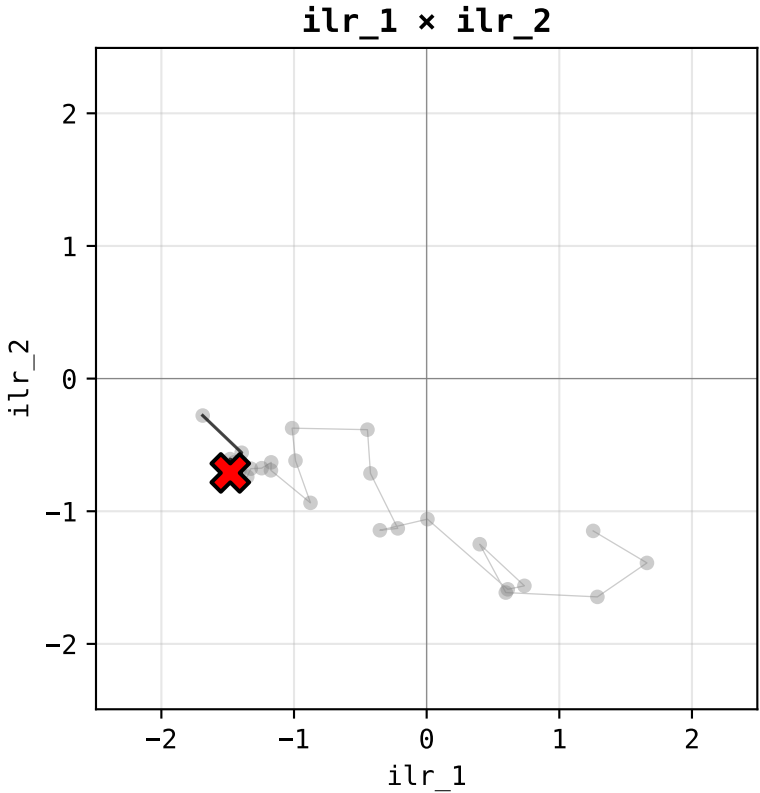


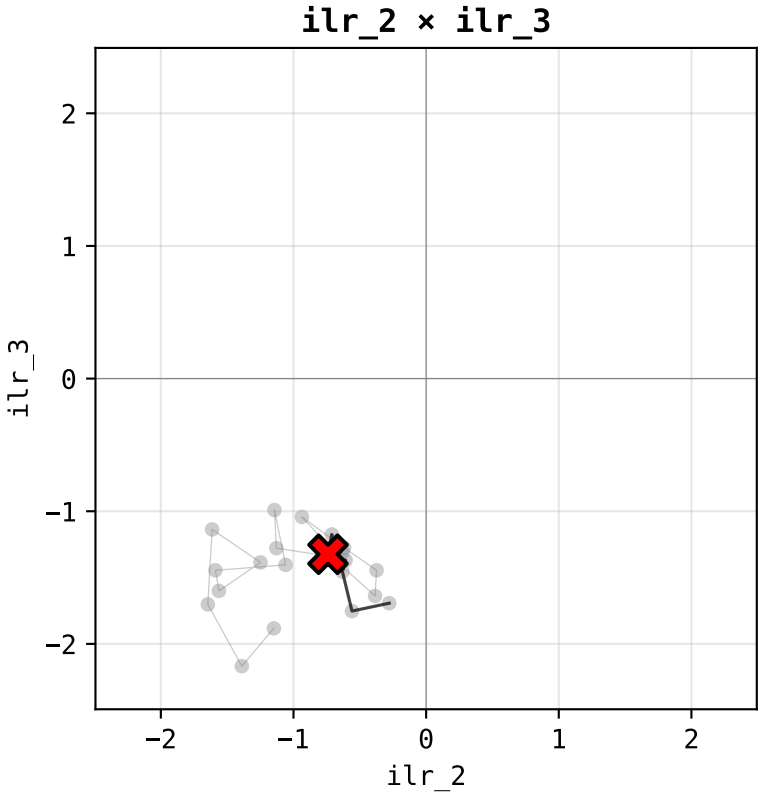
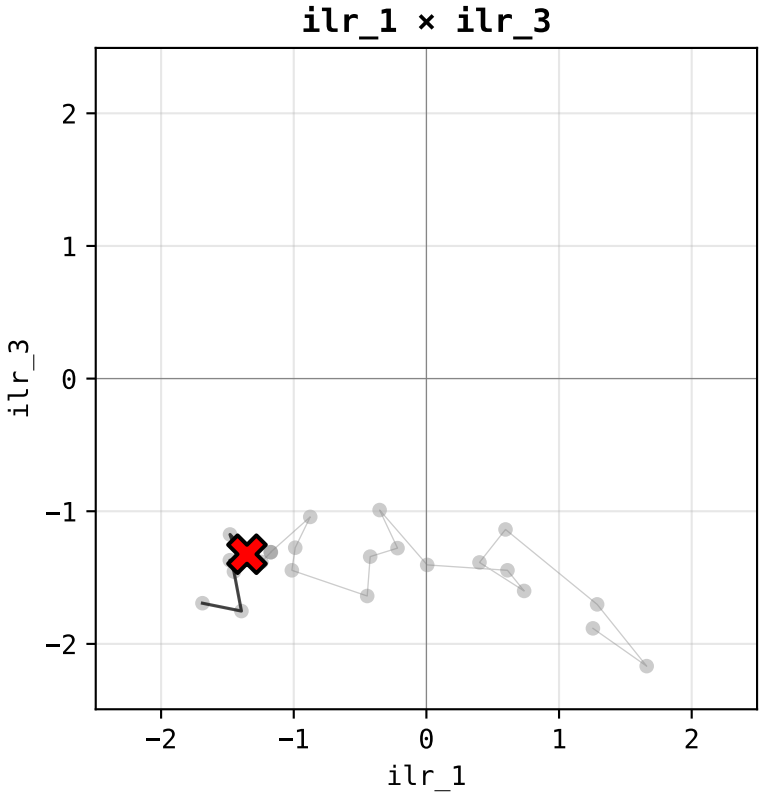
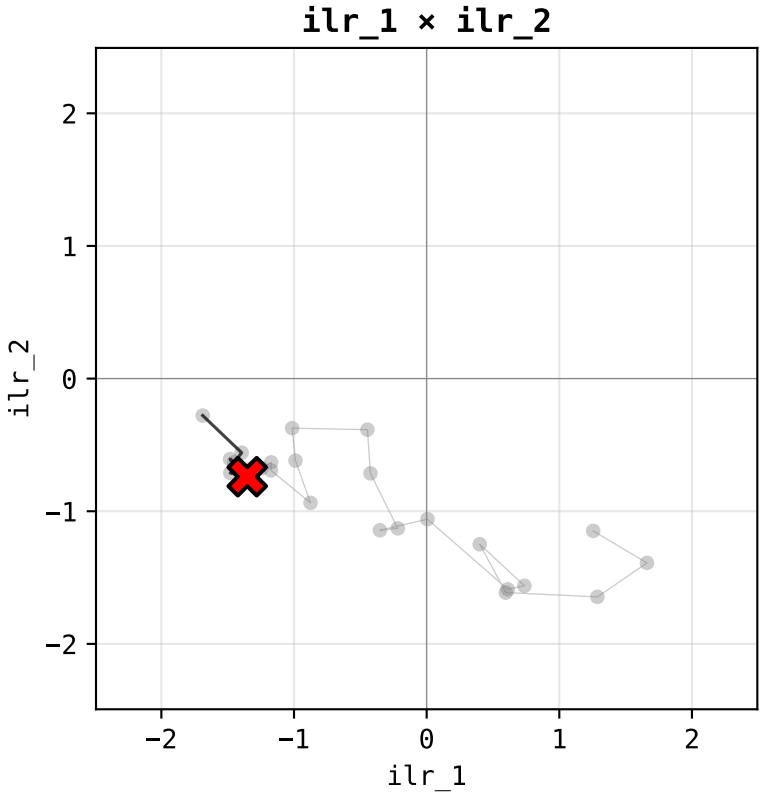


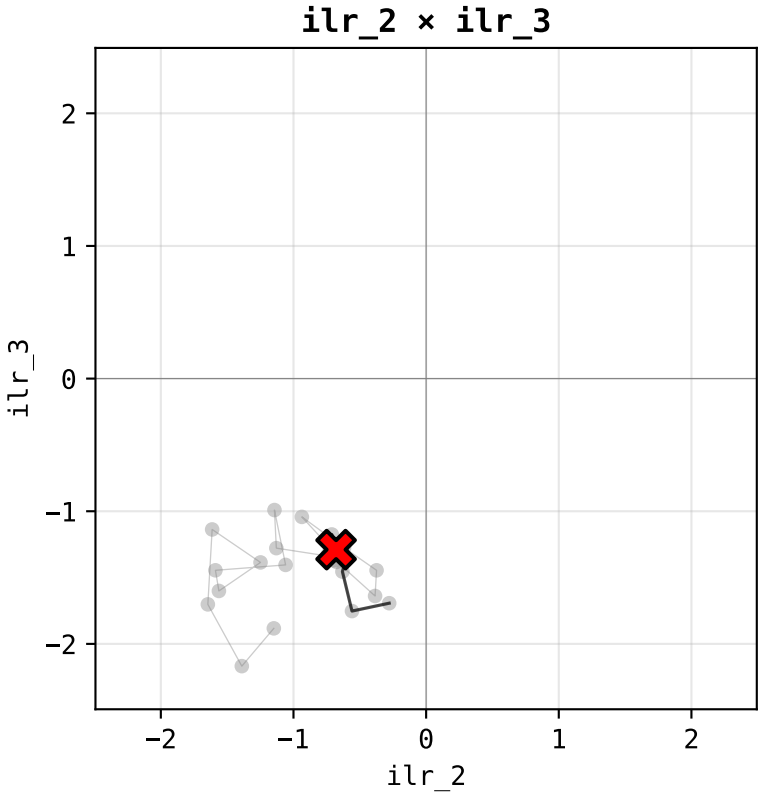
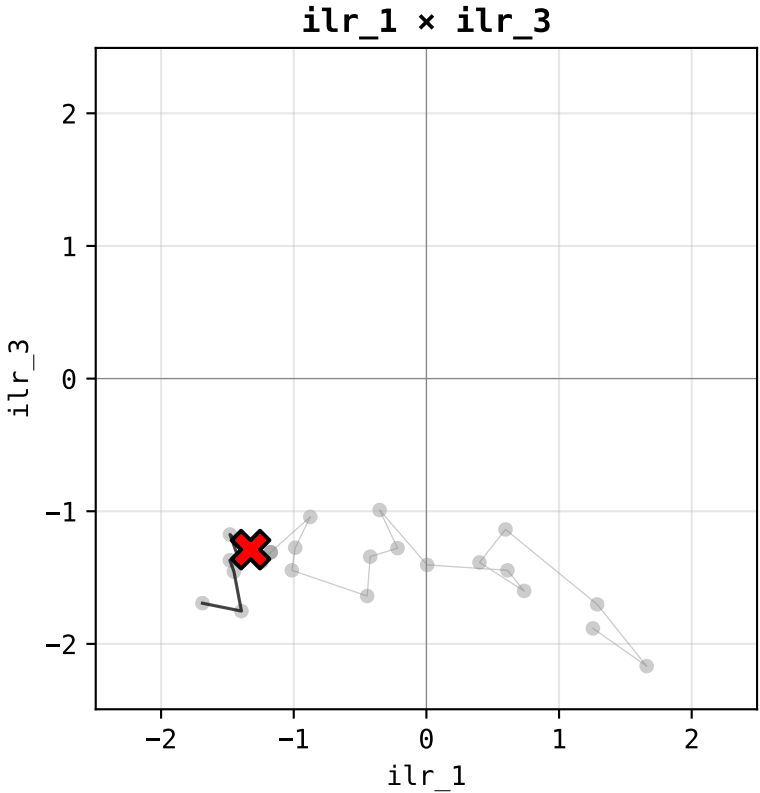
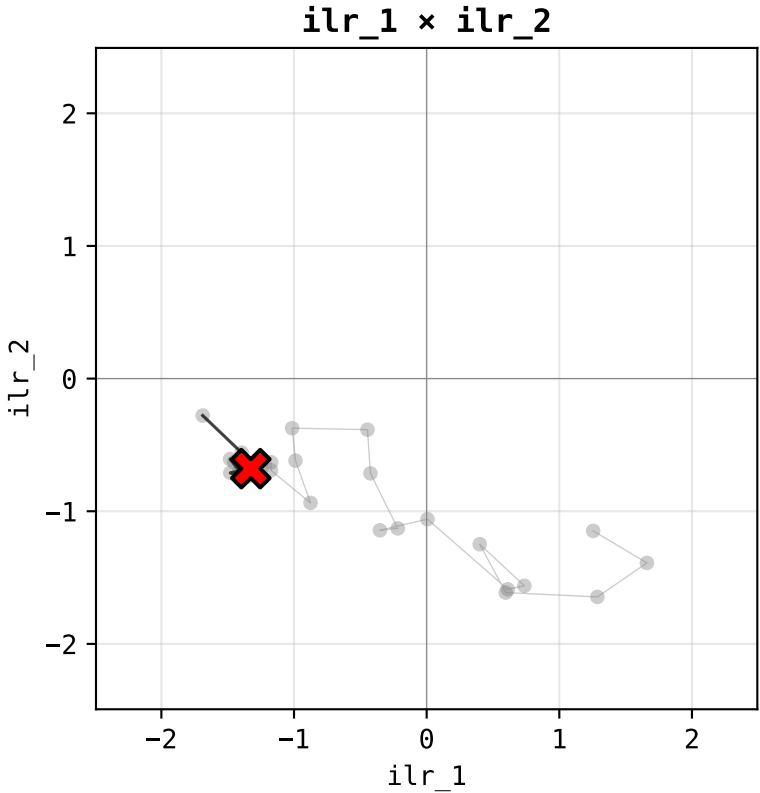




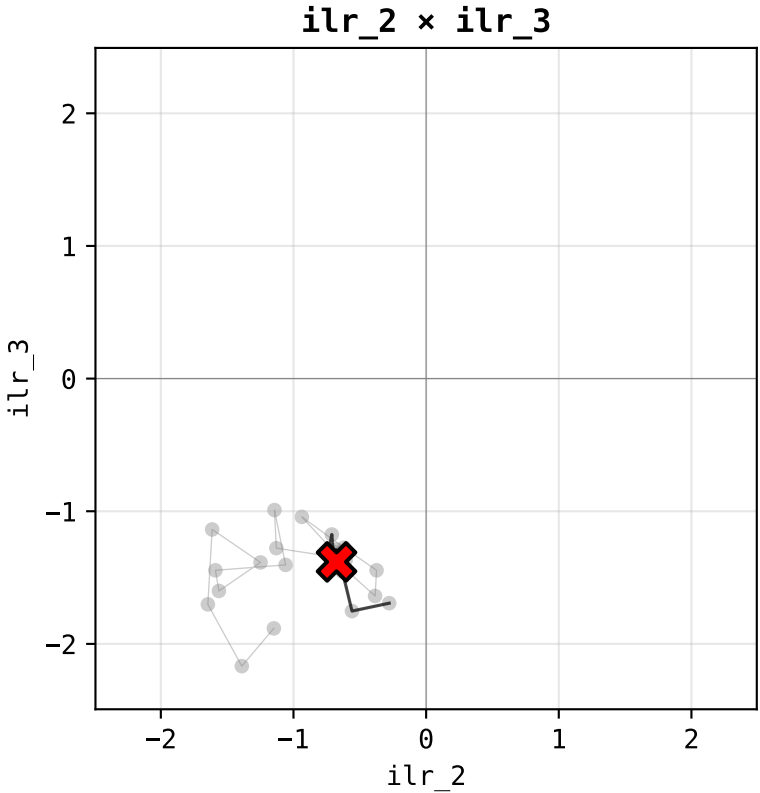
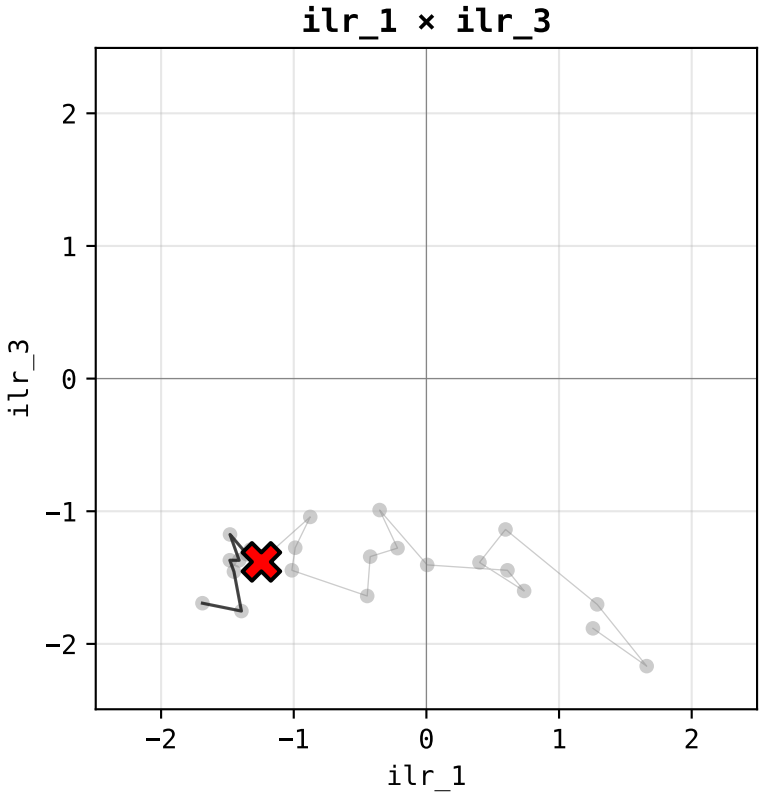
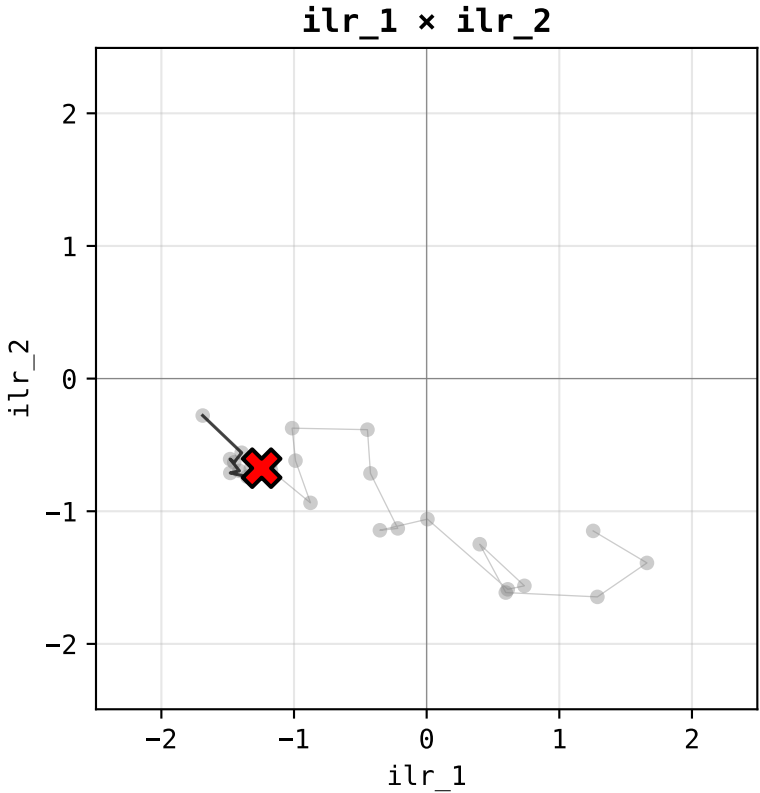




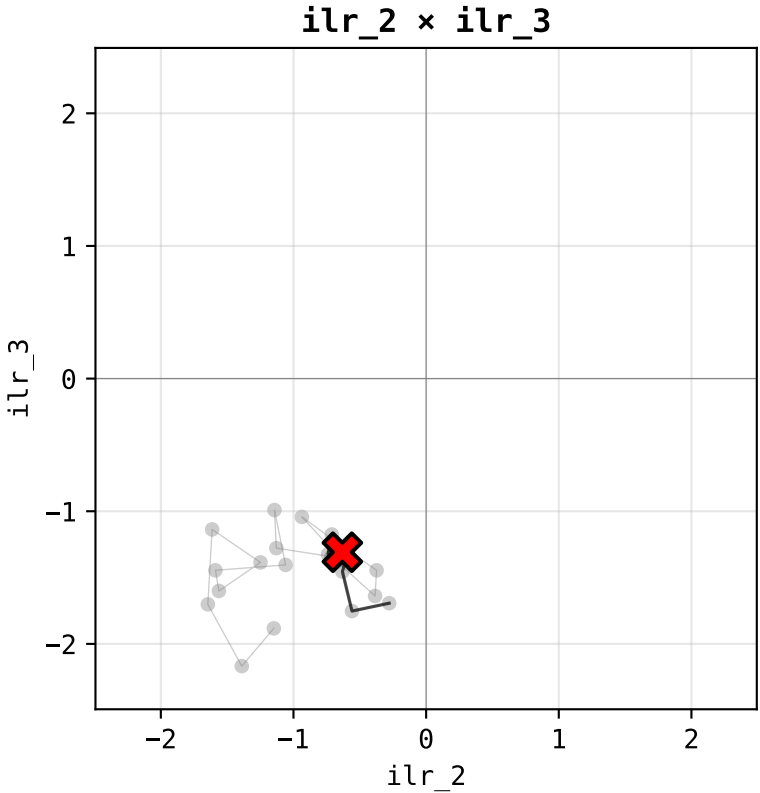
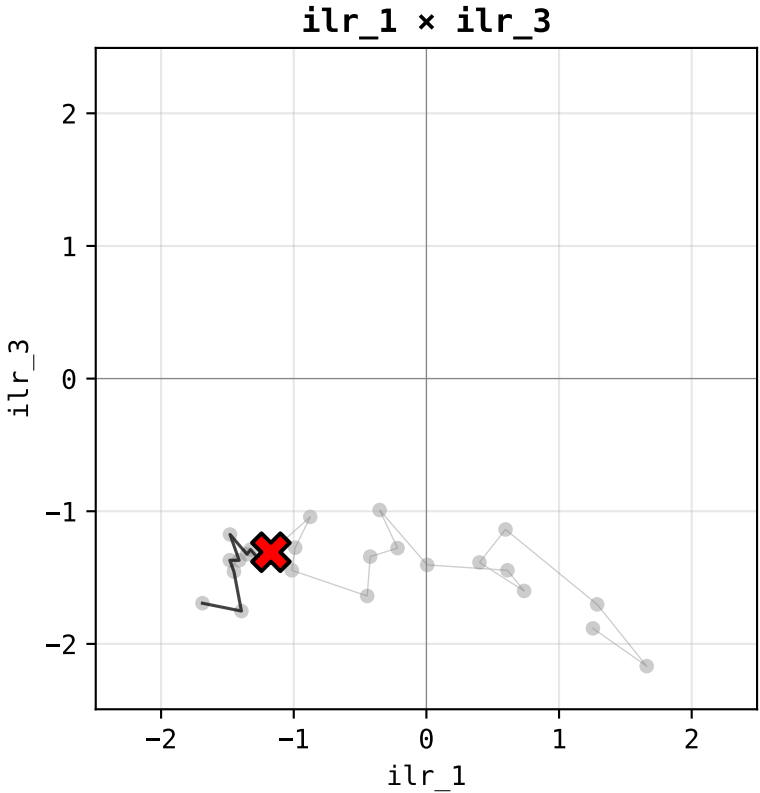
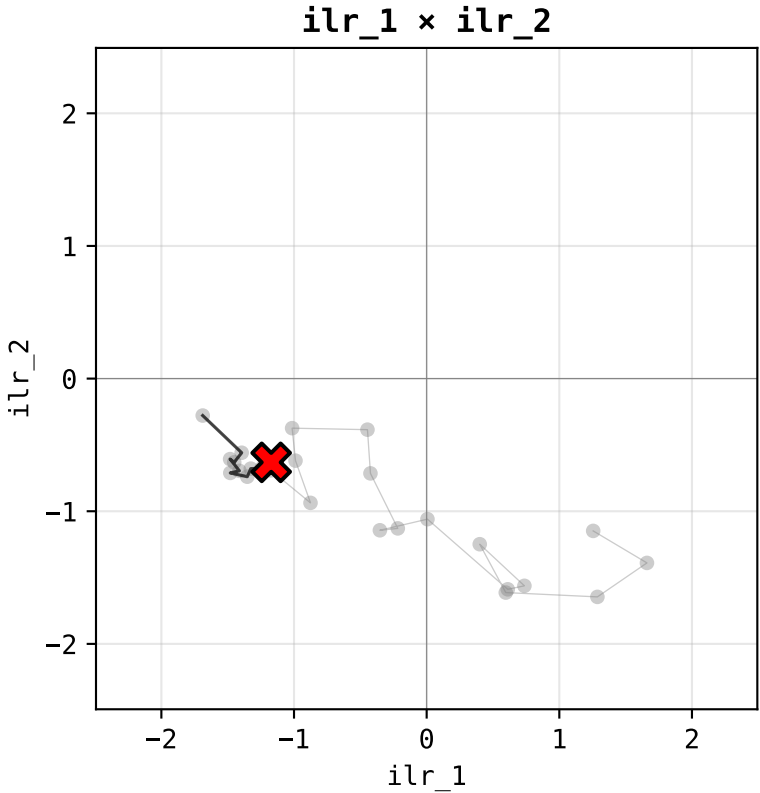




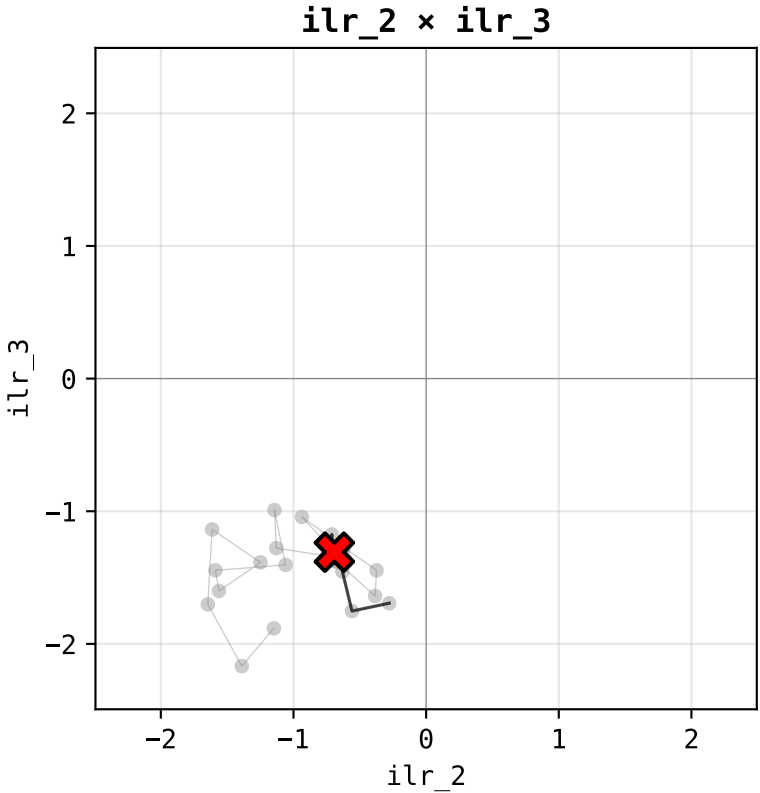
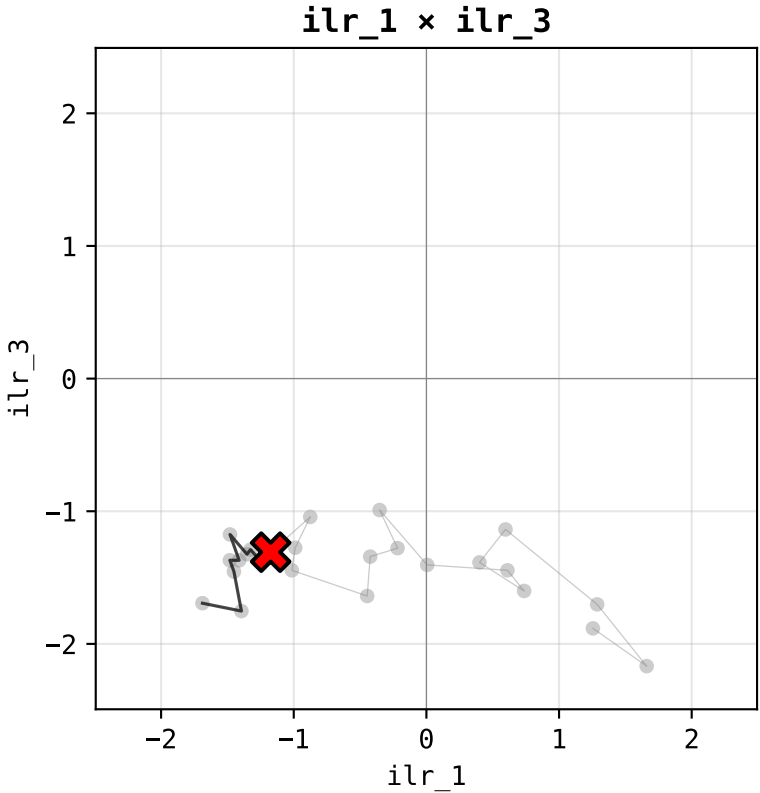
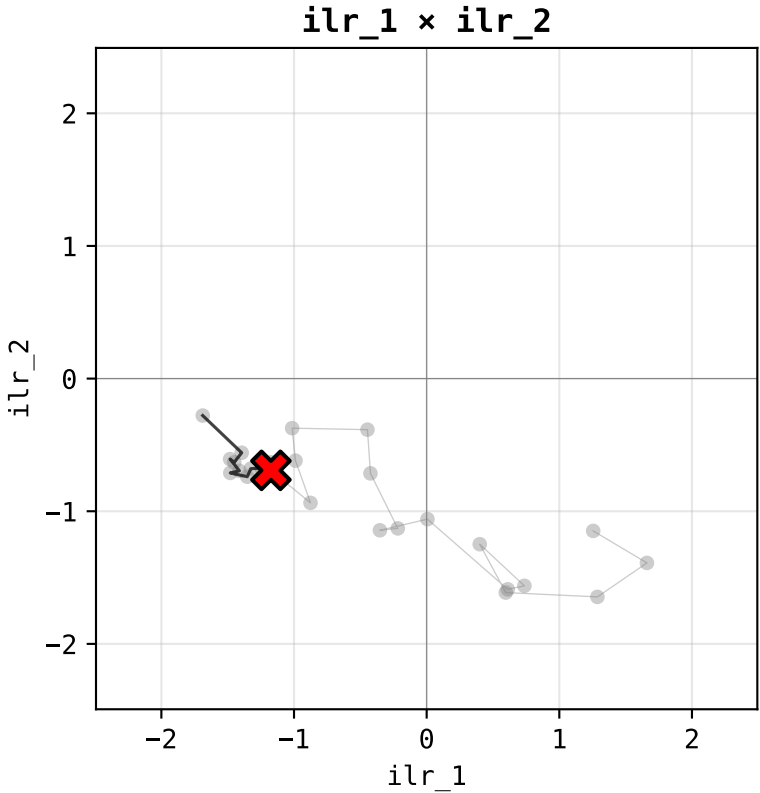
ILR Triplet – FRA France · year 2008 (9/26)

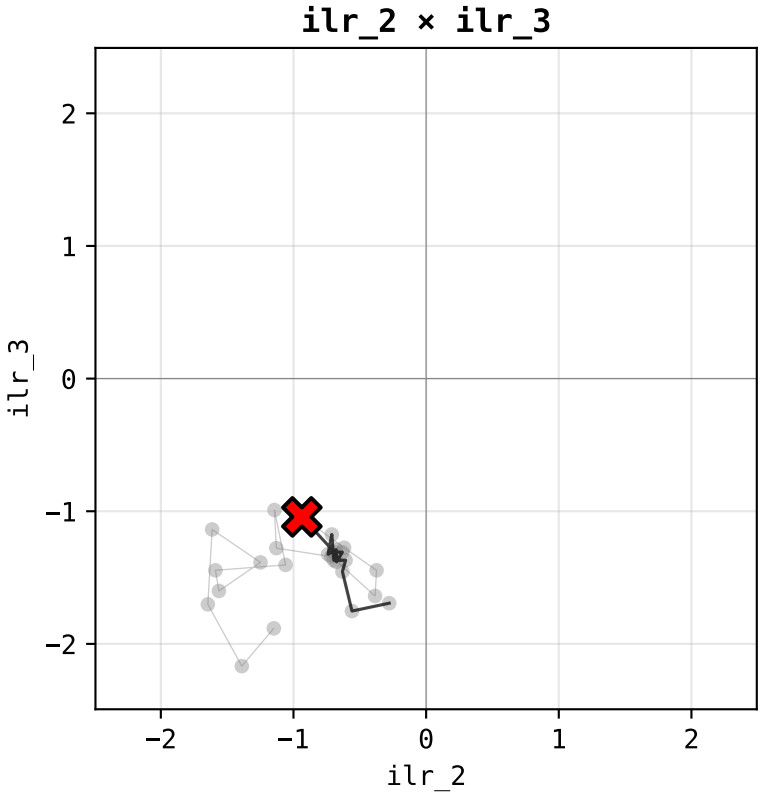
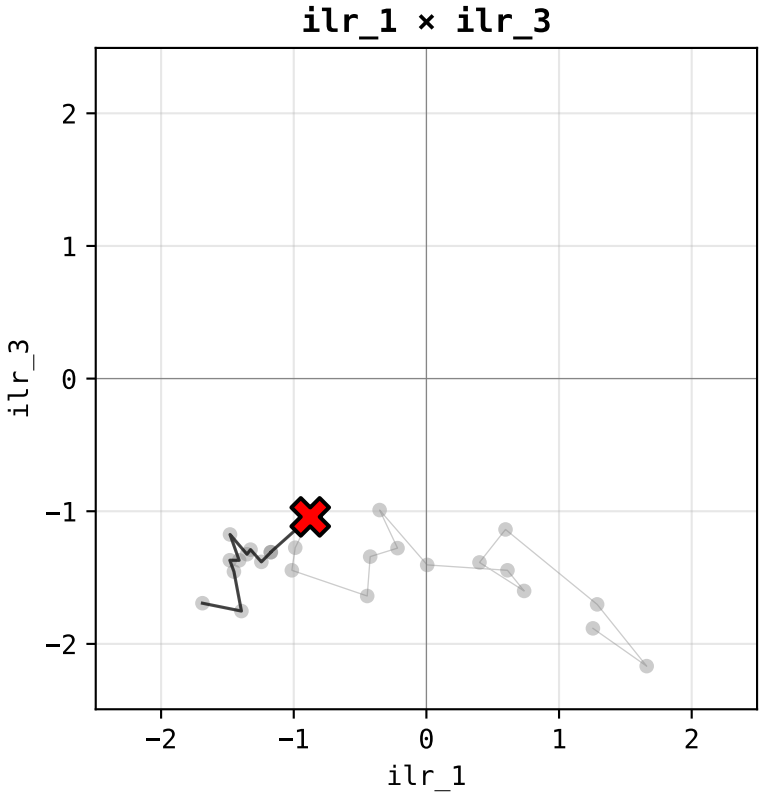
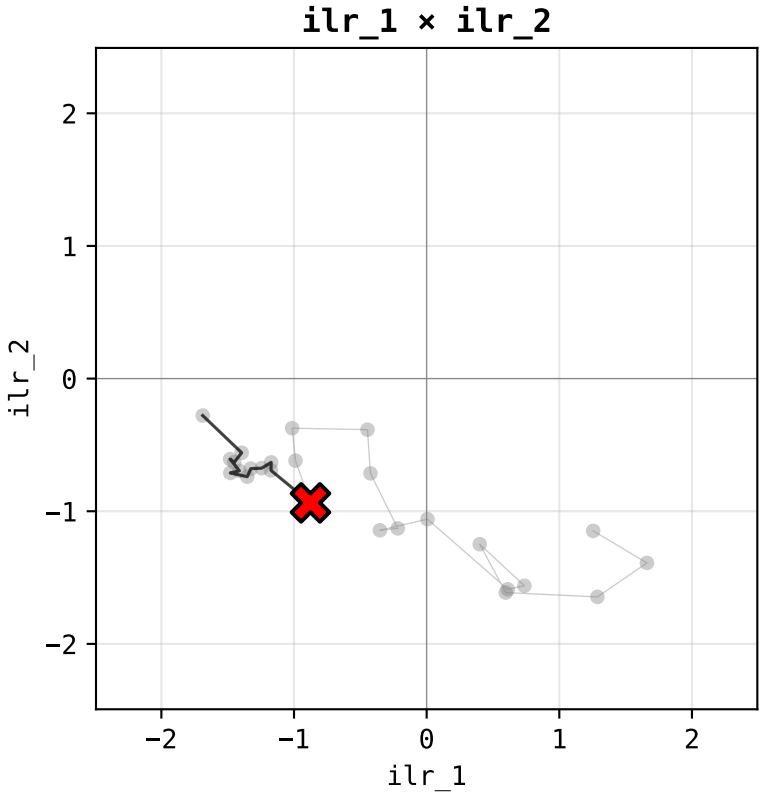


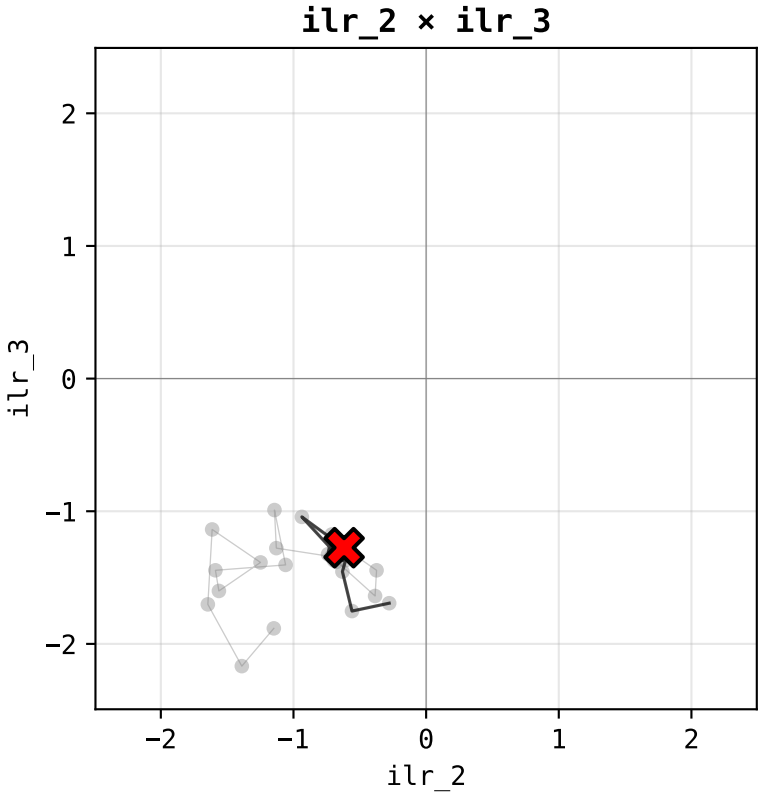
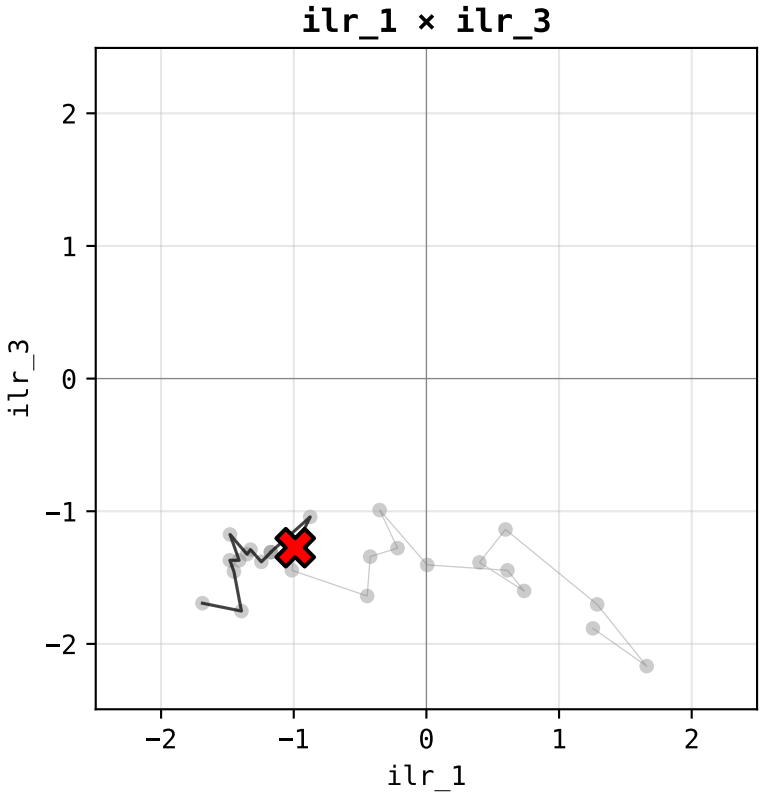
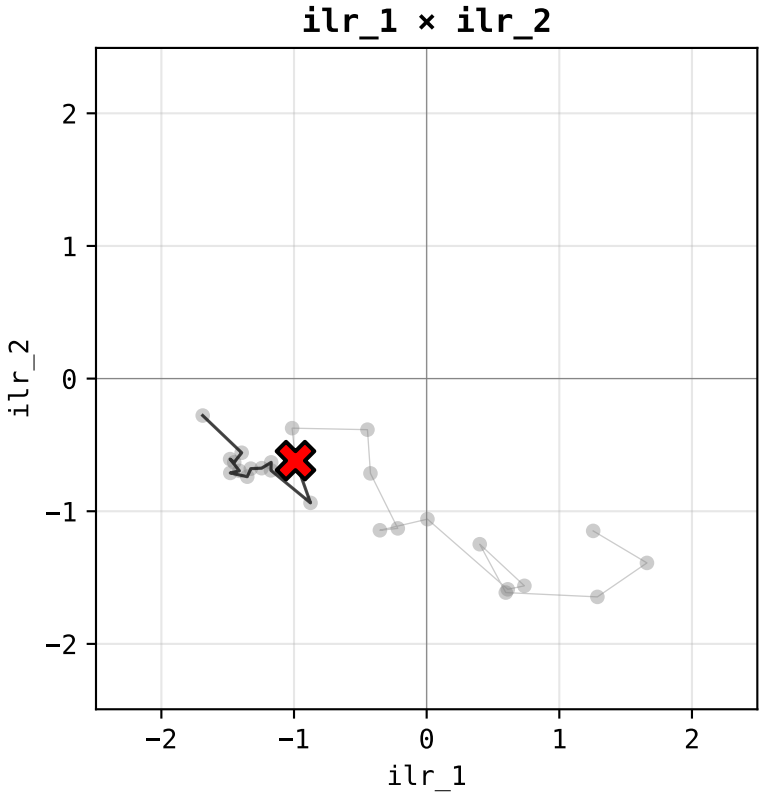
ILR Triplet – FRA France · year 2009 (10/26)

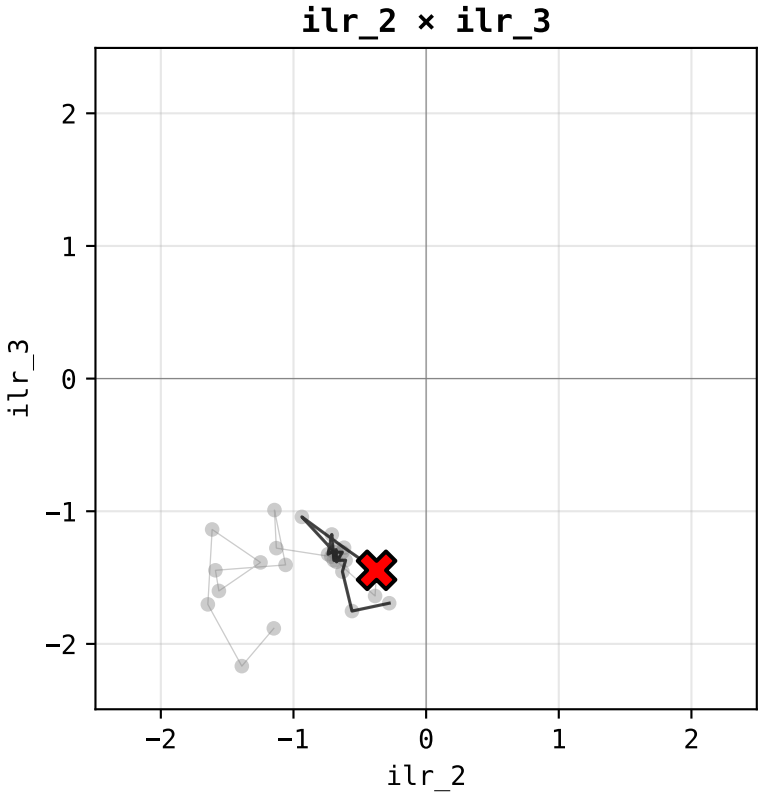
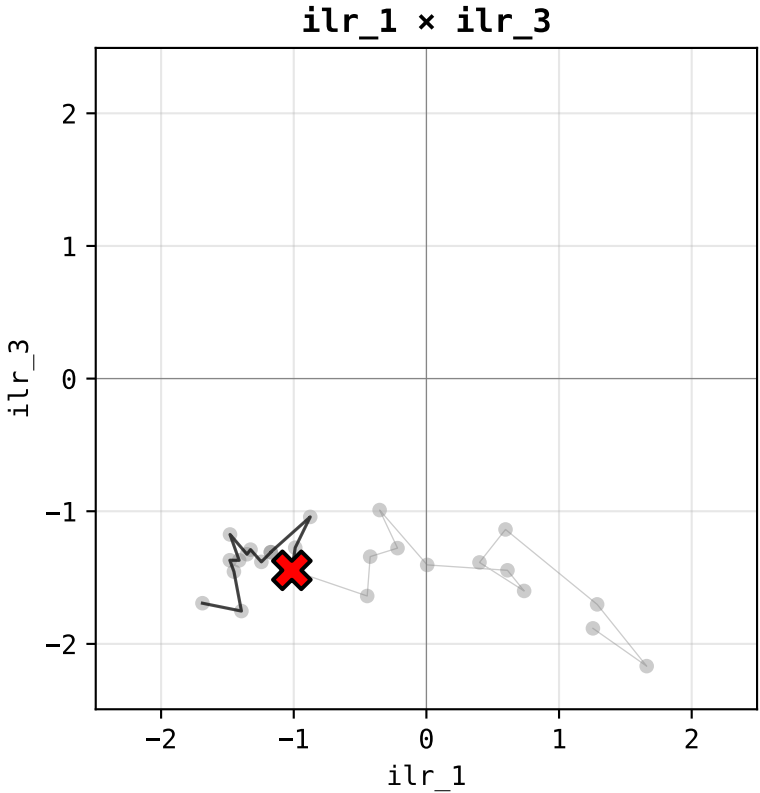
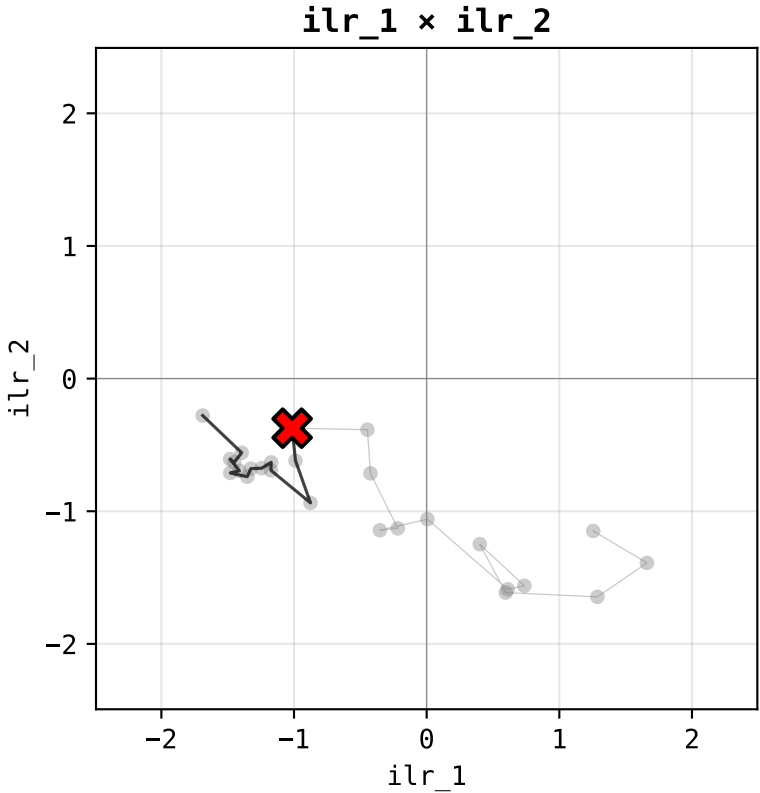


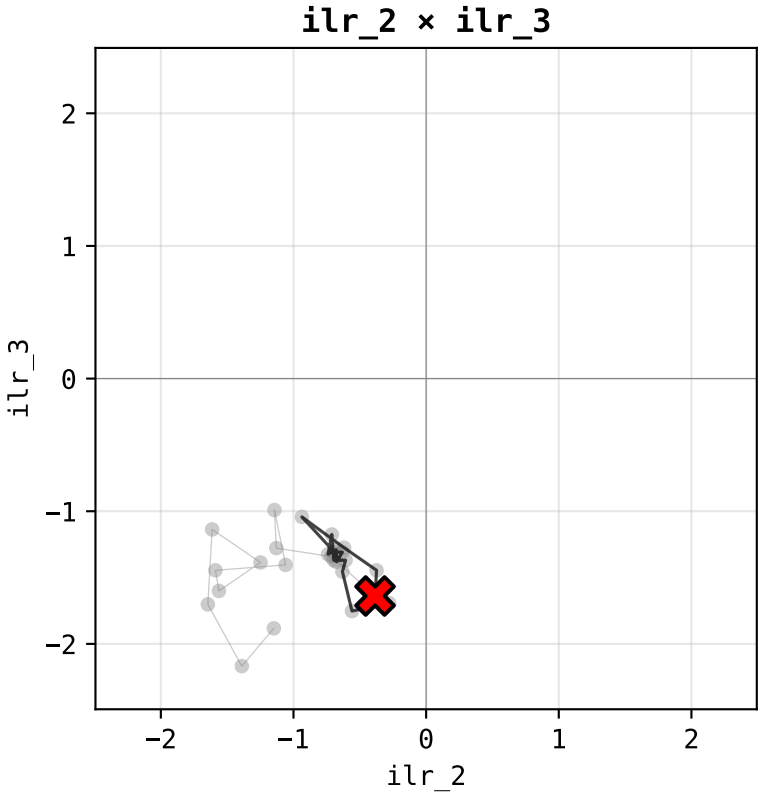
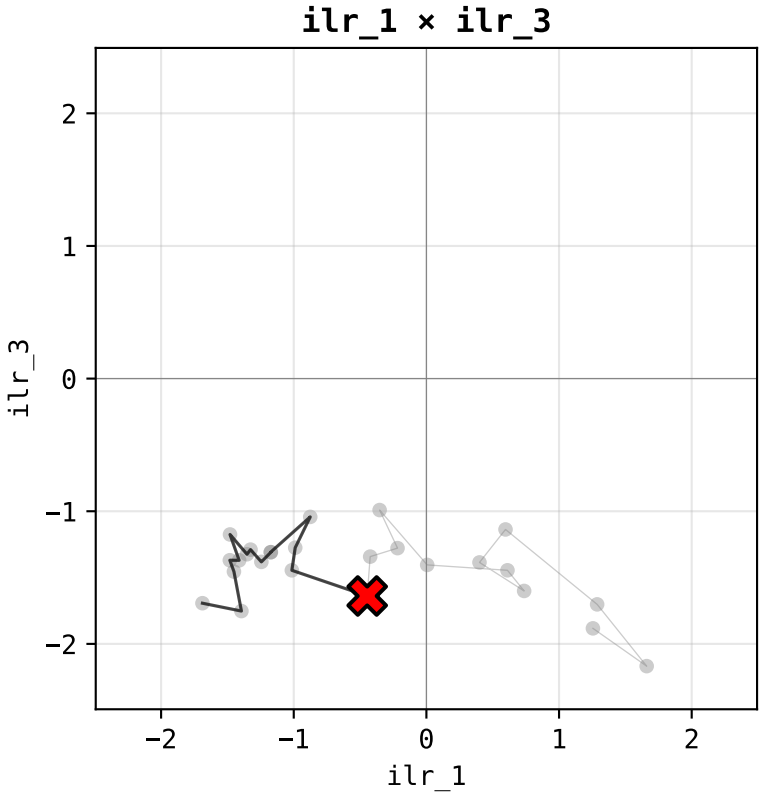
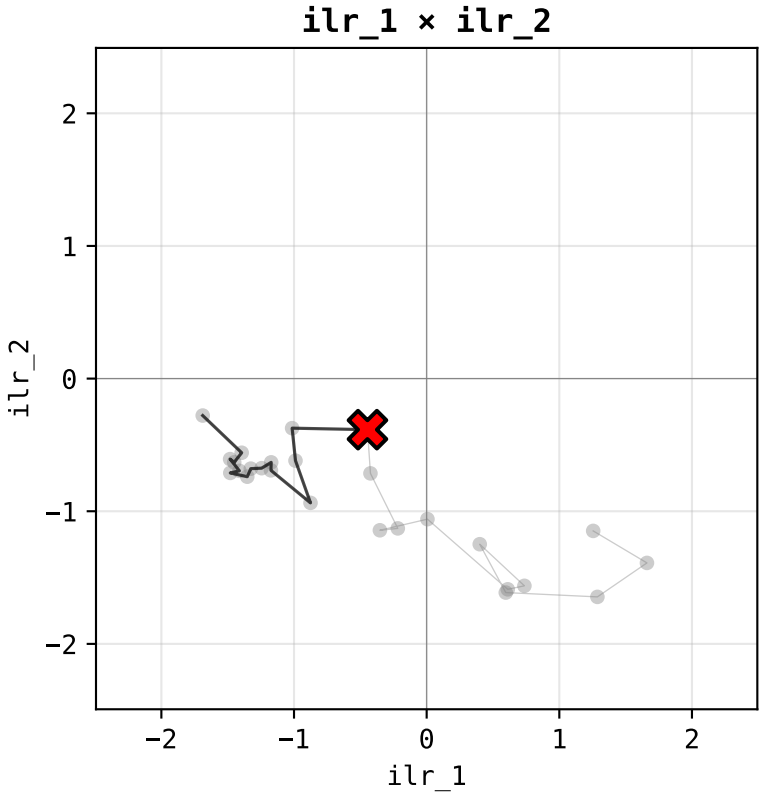
ILR Triplet – FRA France · year 2010 (11/26)

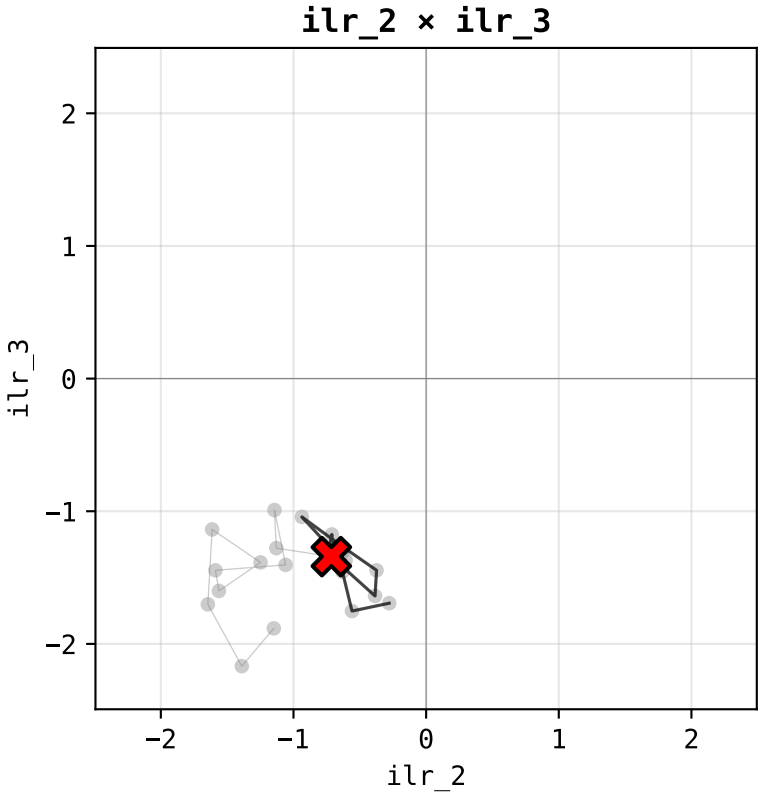
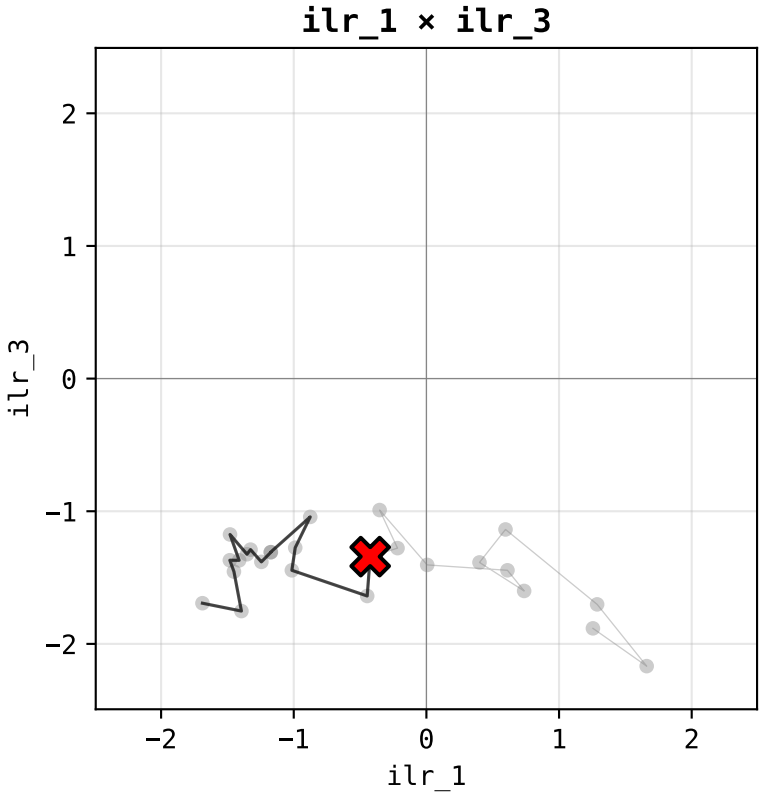
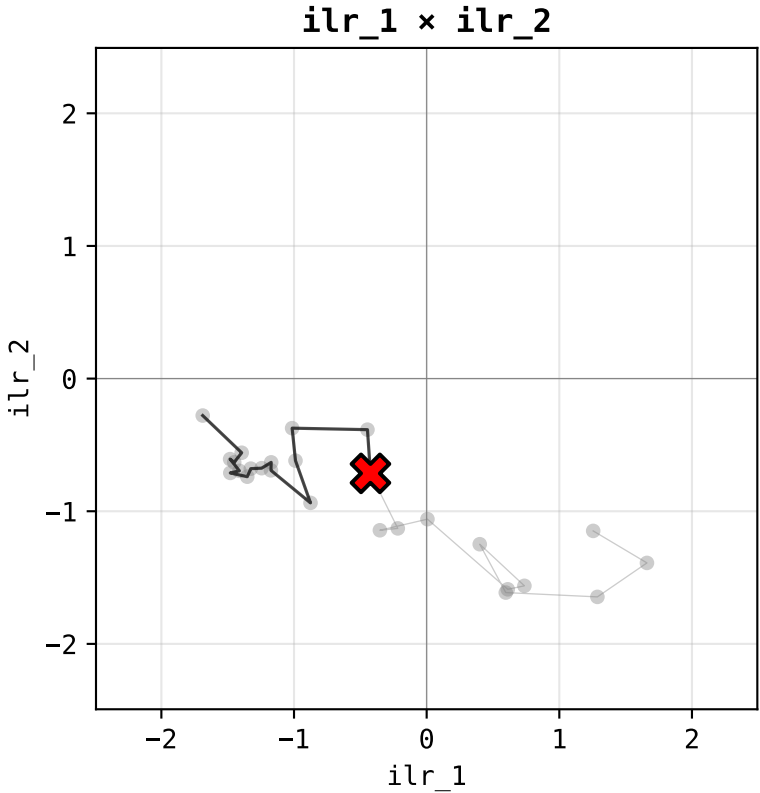


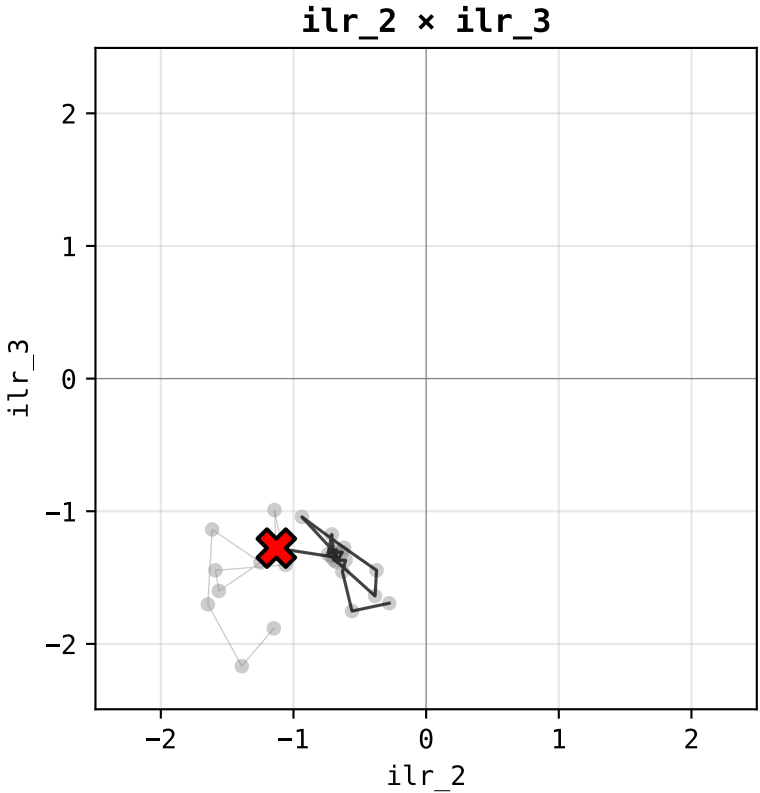
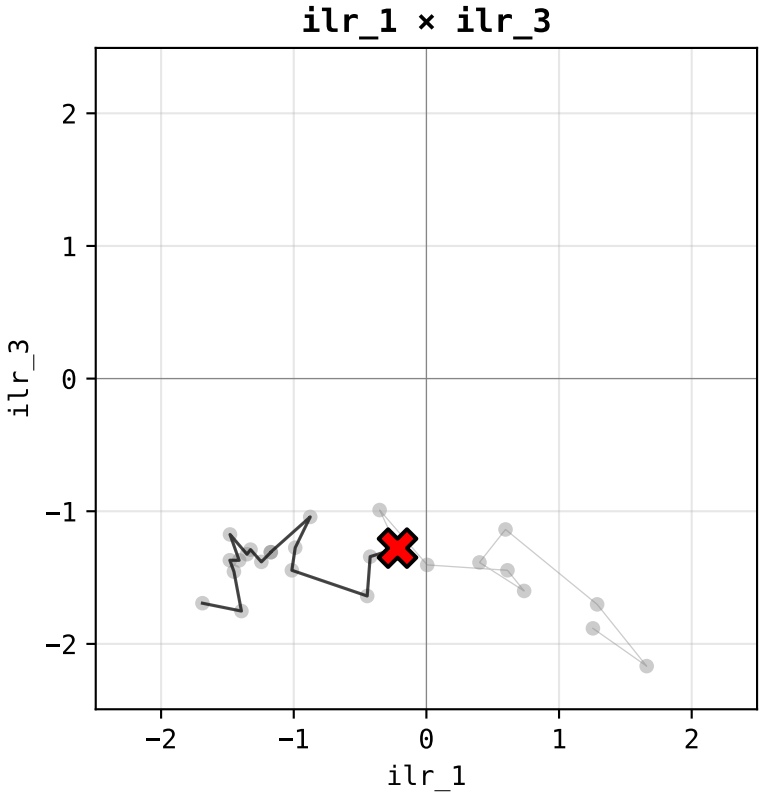
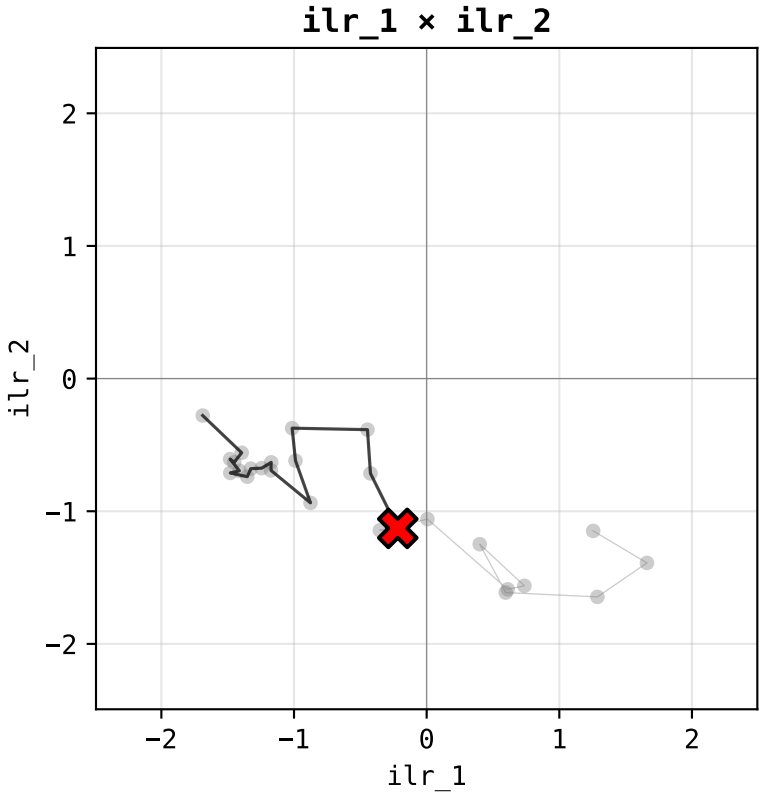


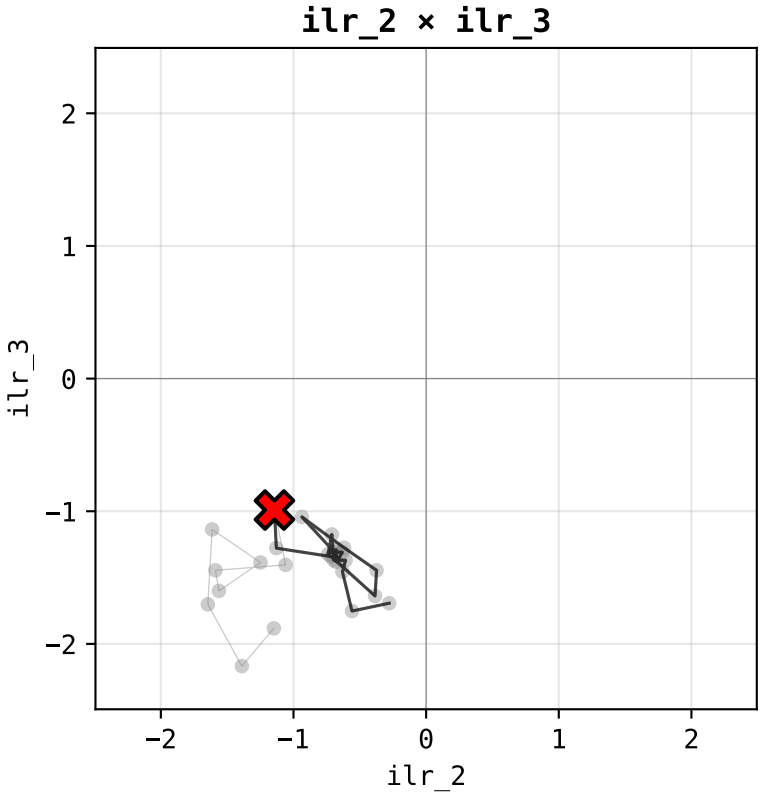
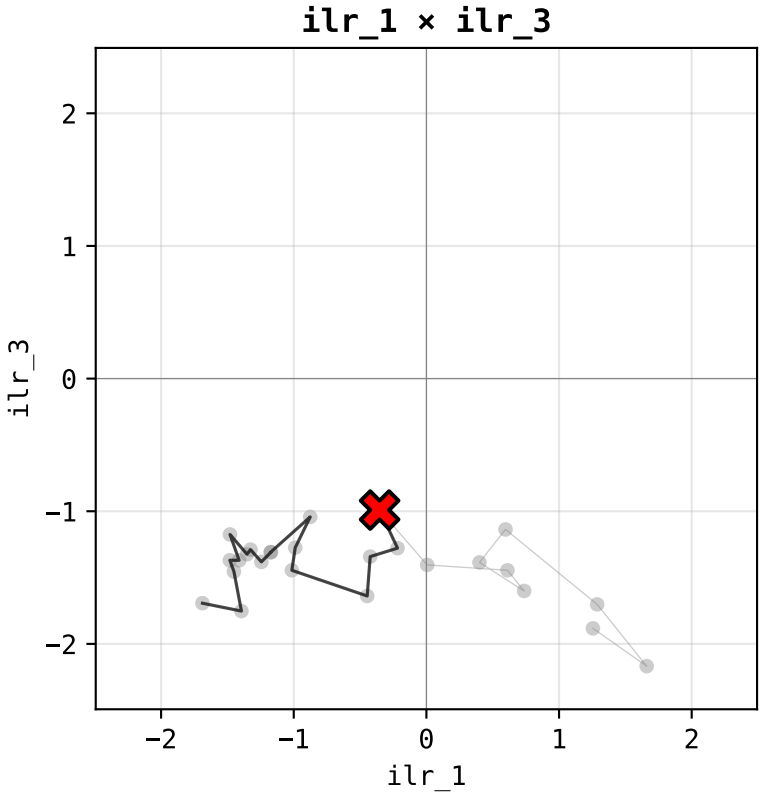
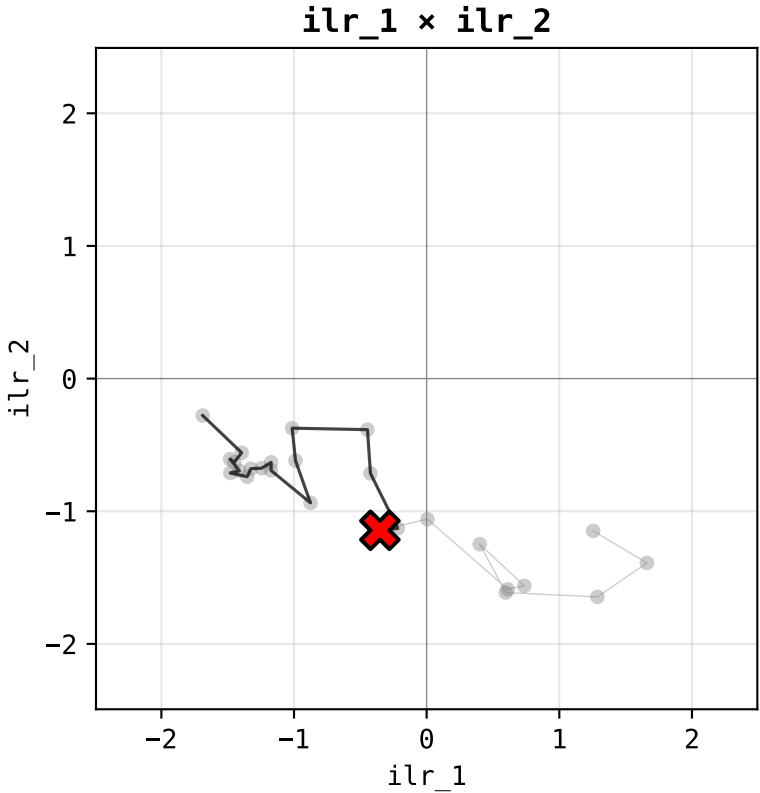


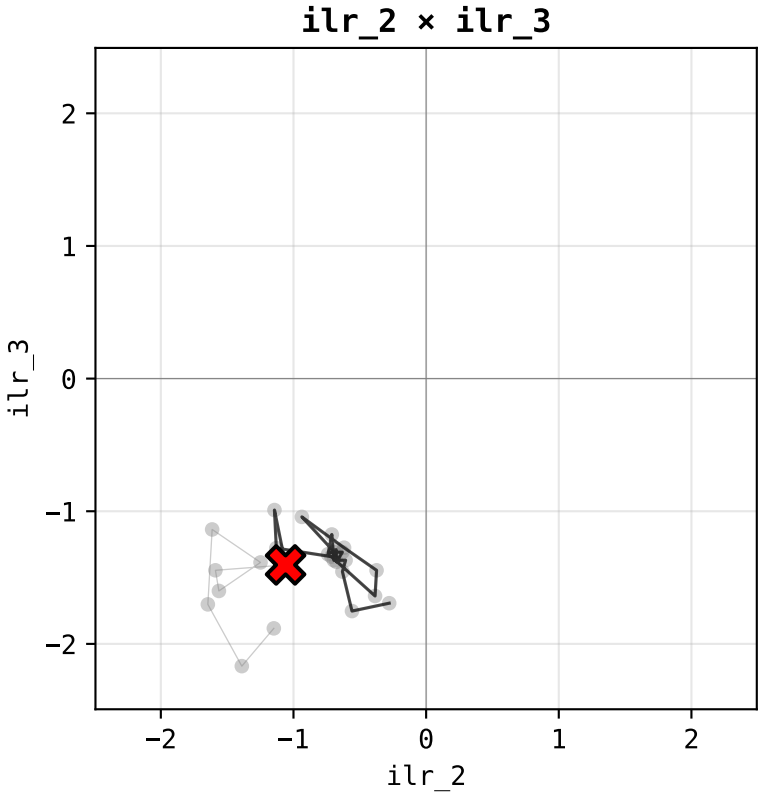
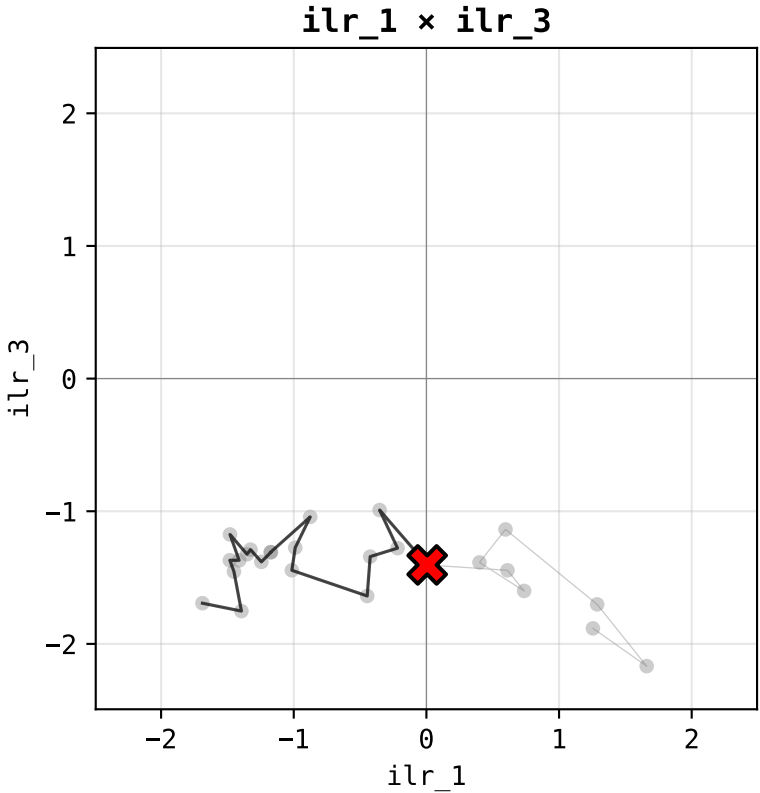
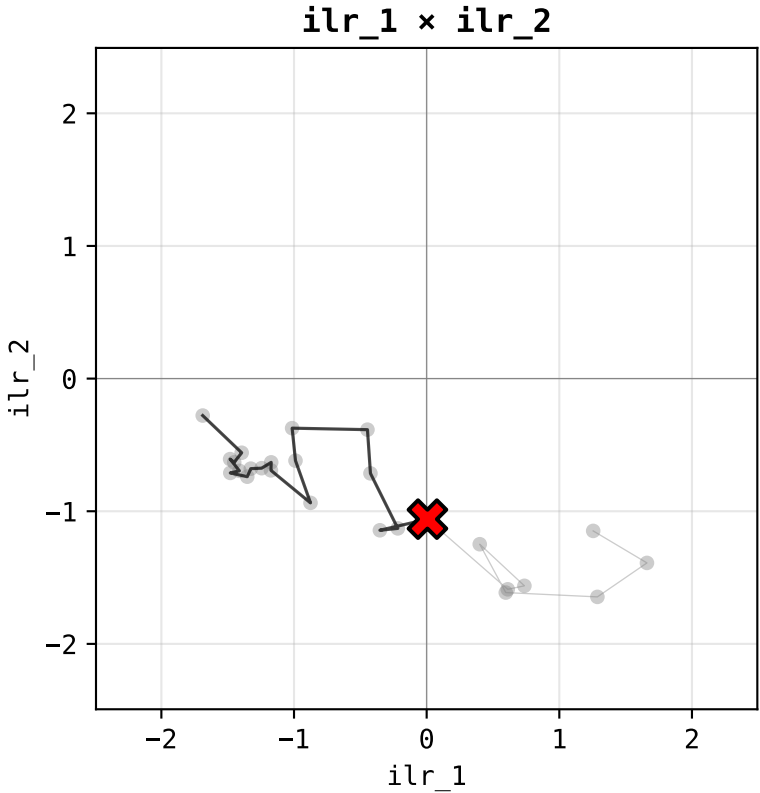


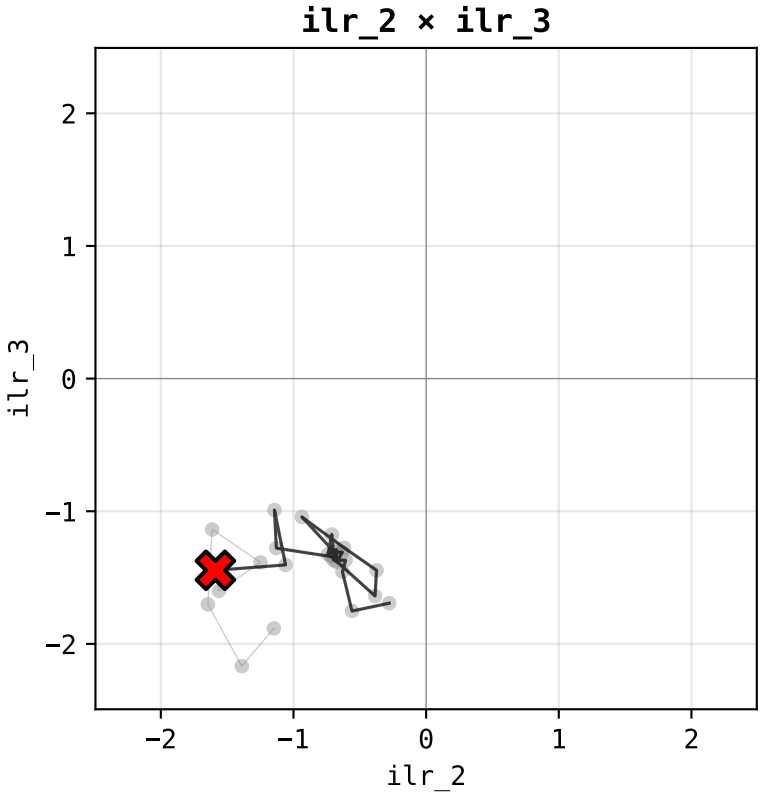
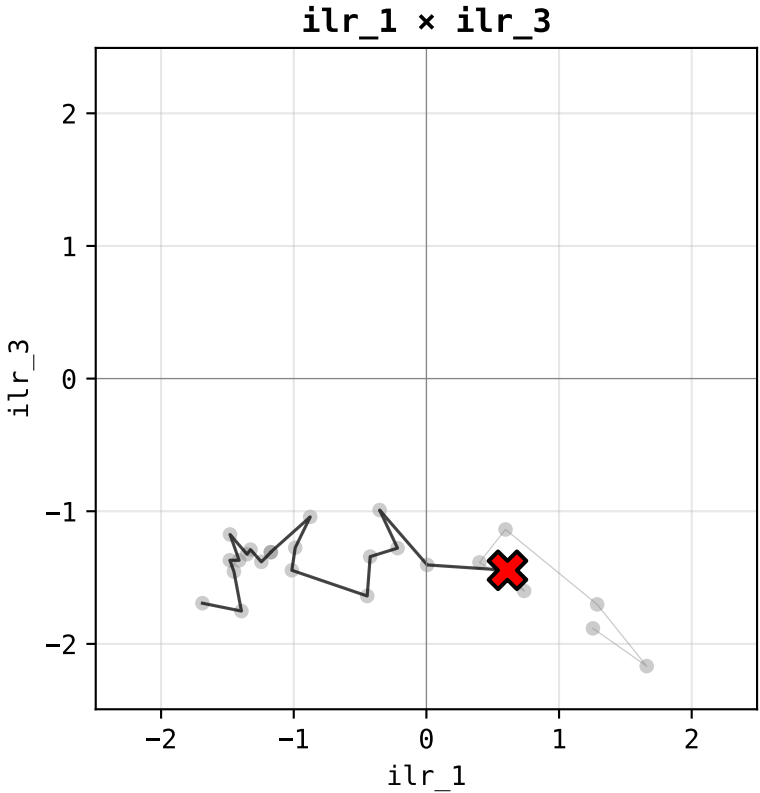
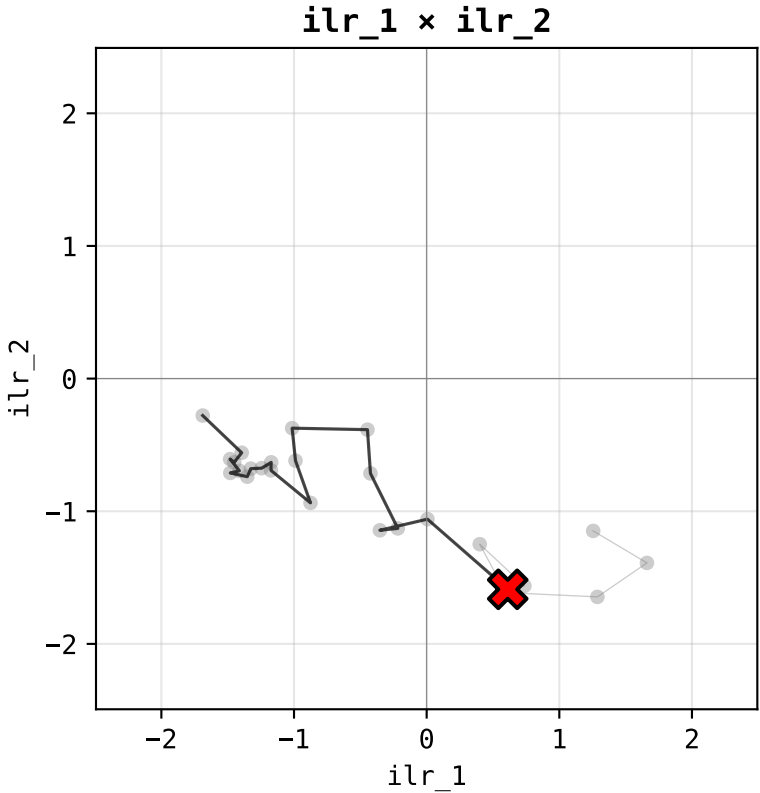


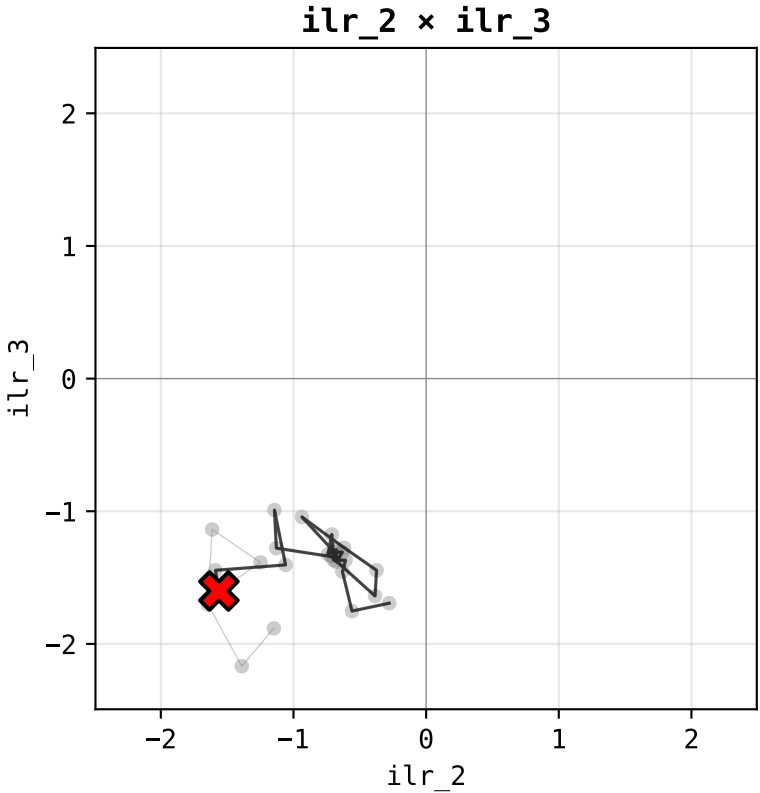
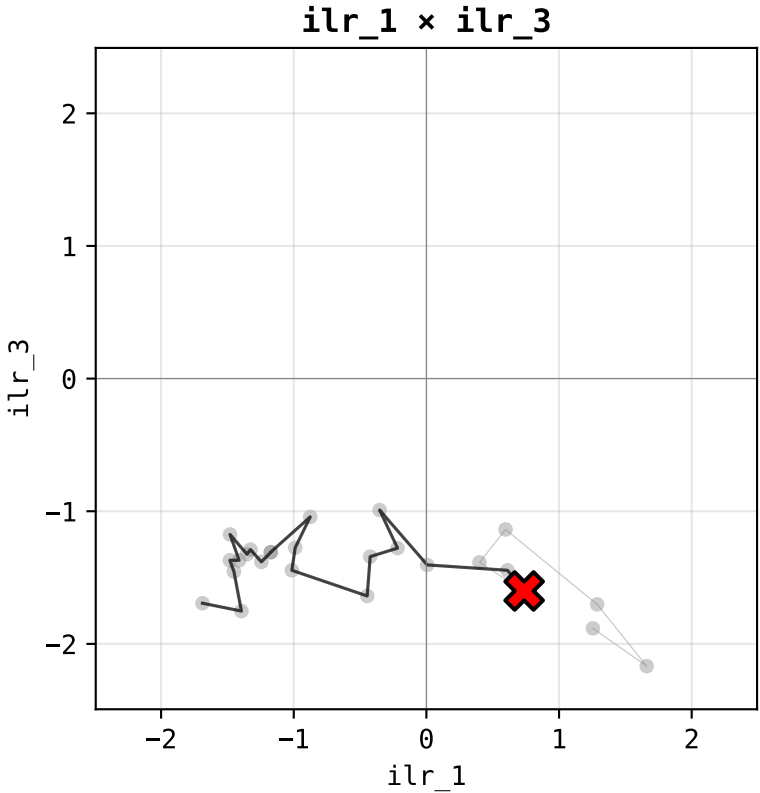
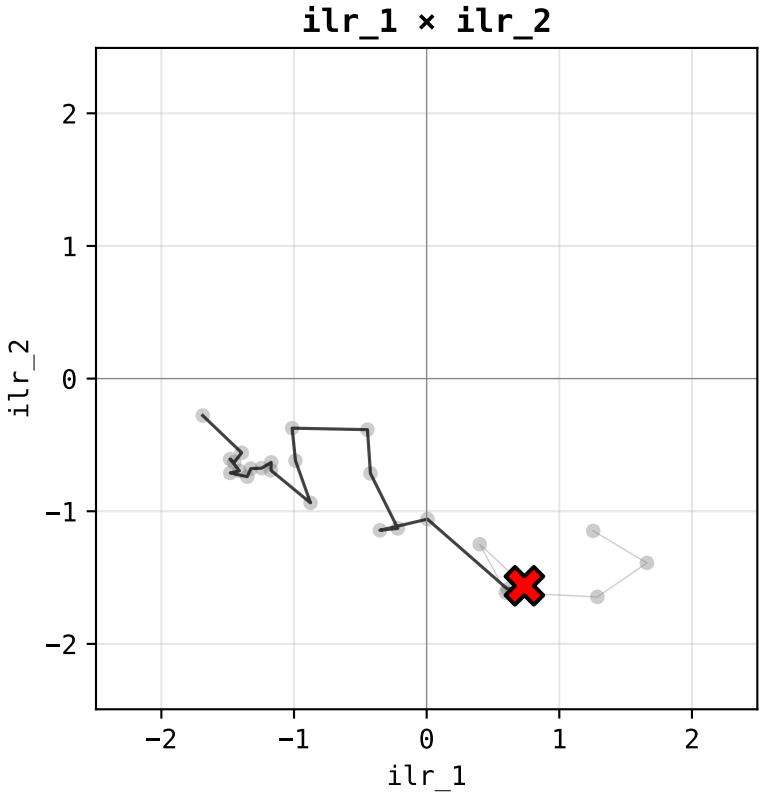


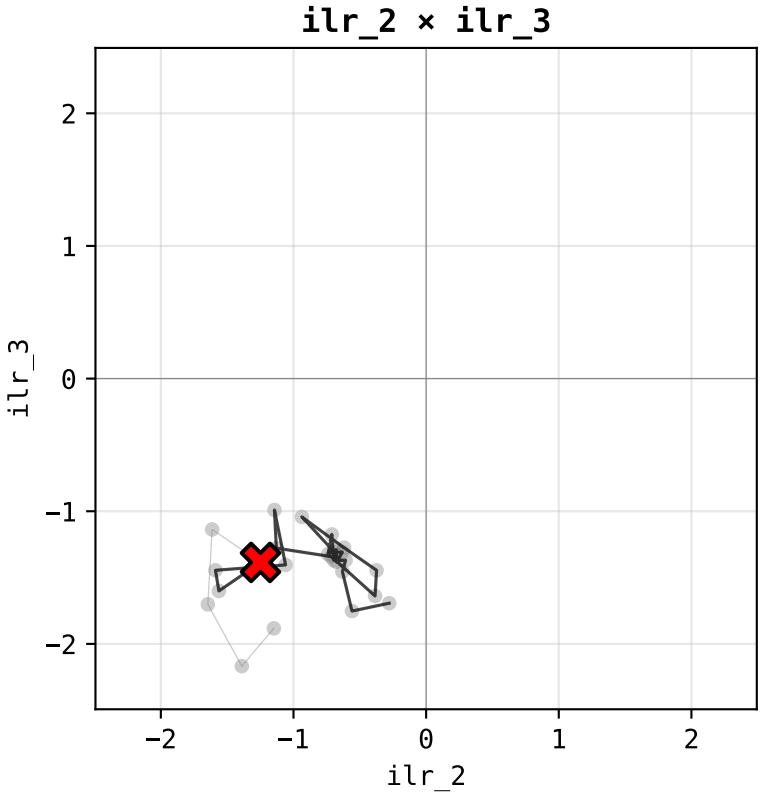
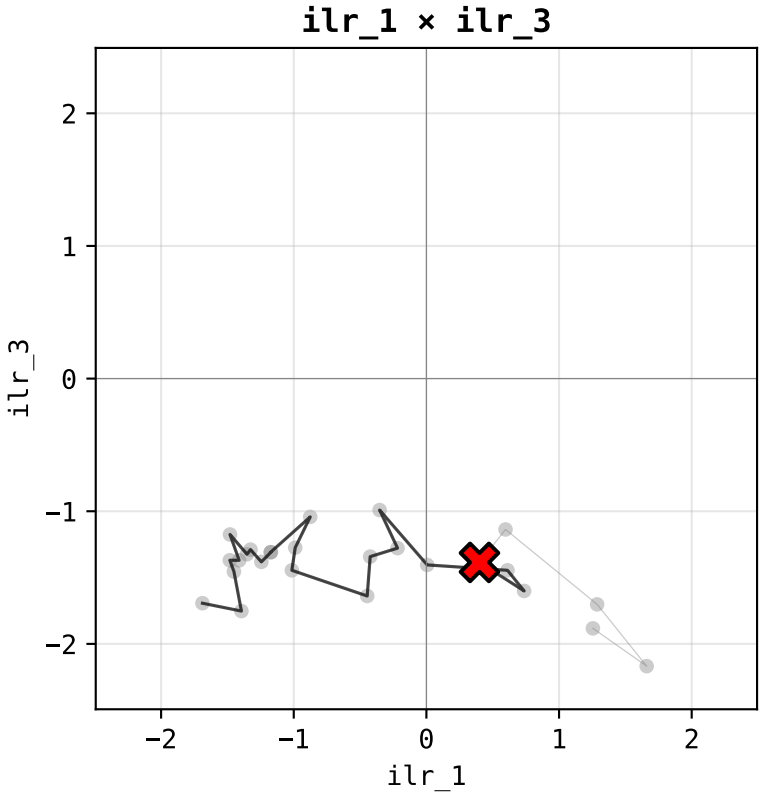
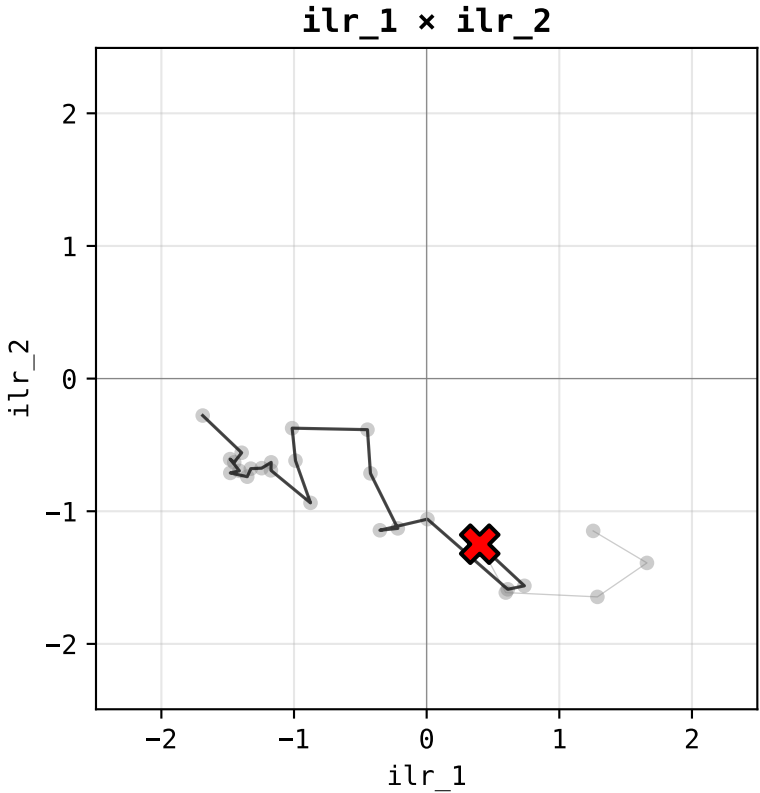


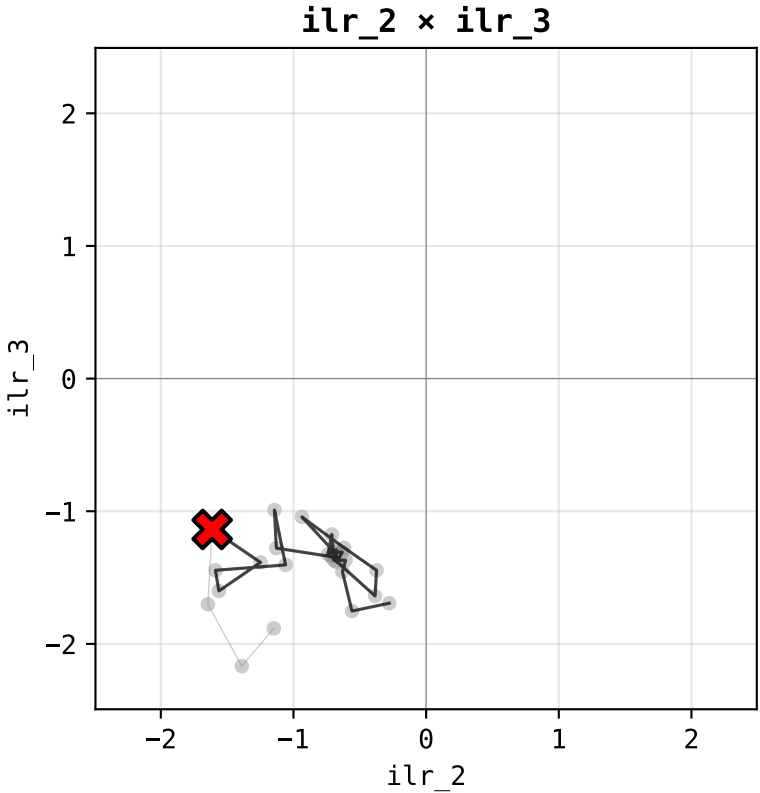
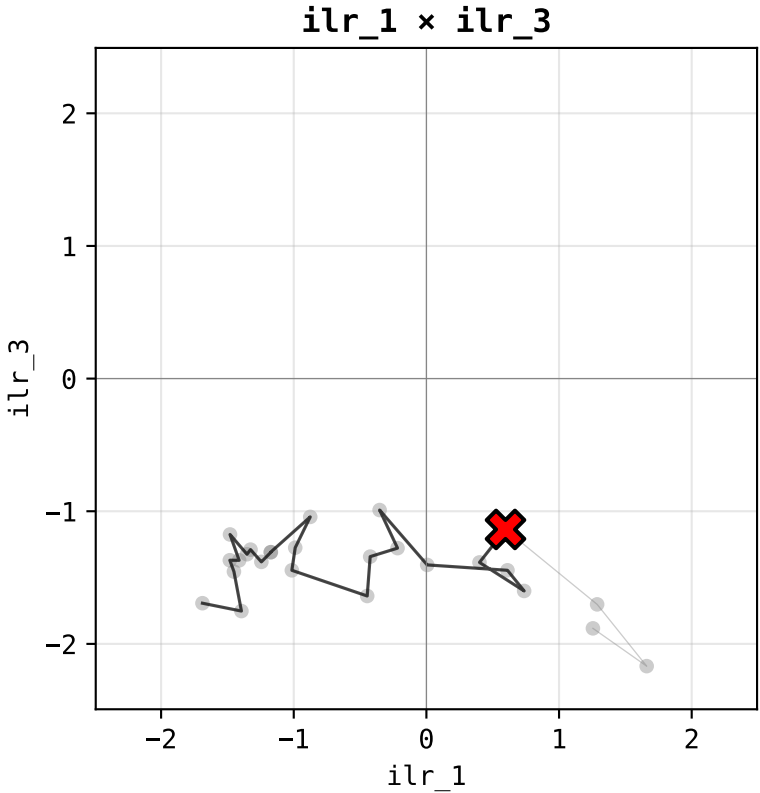
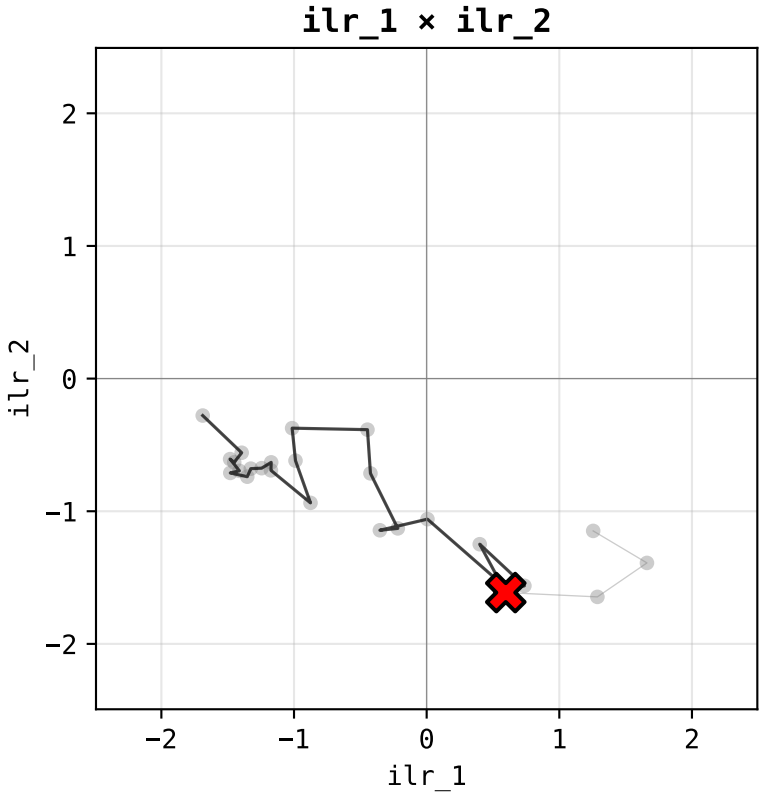


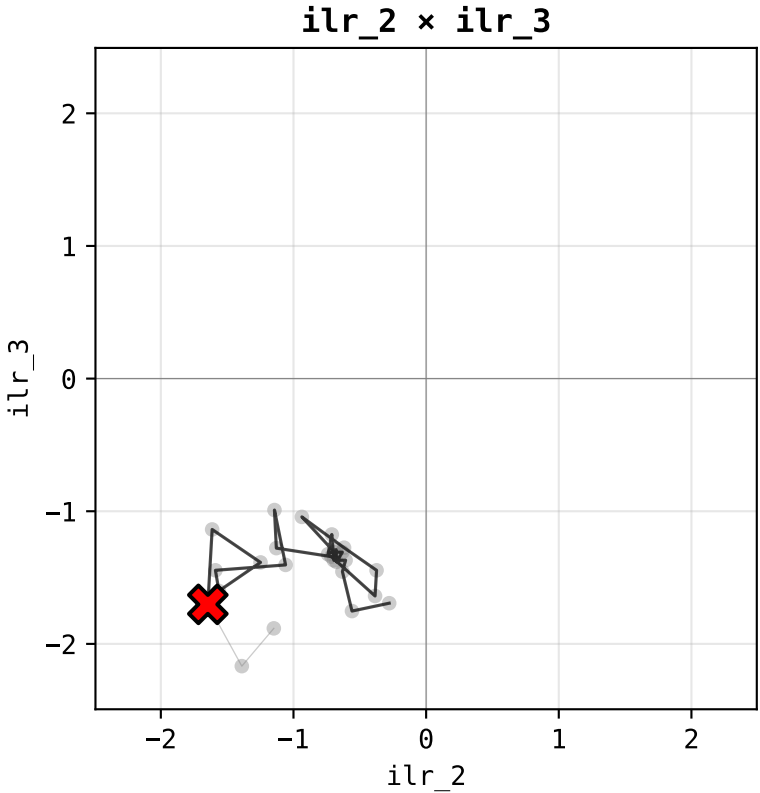
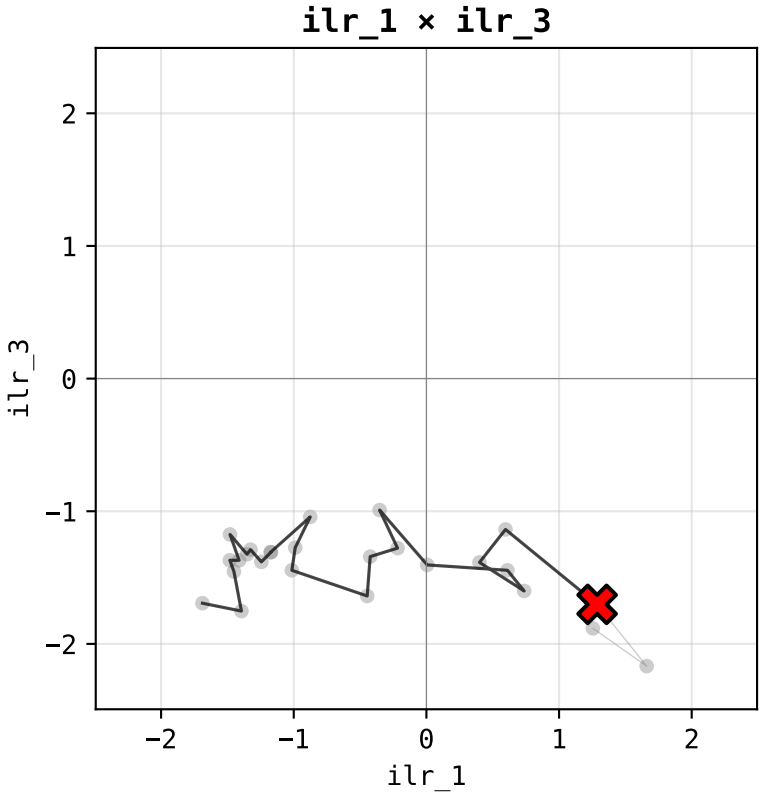
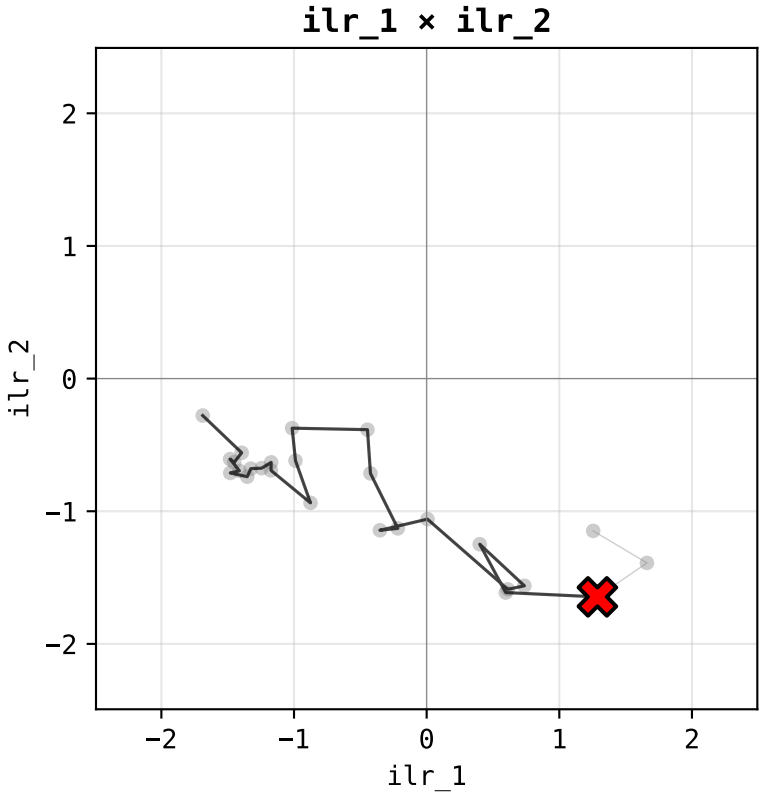


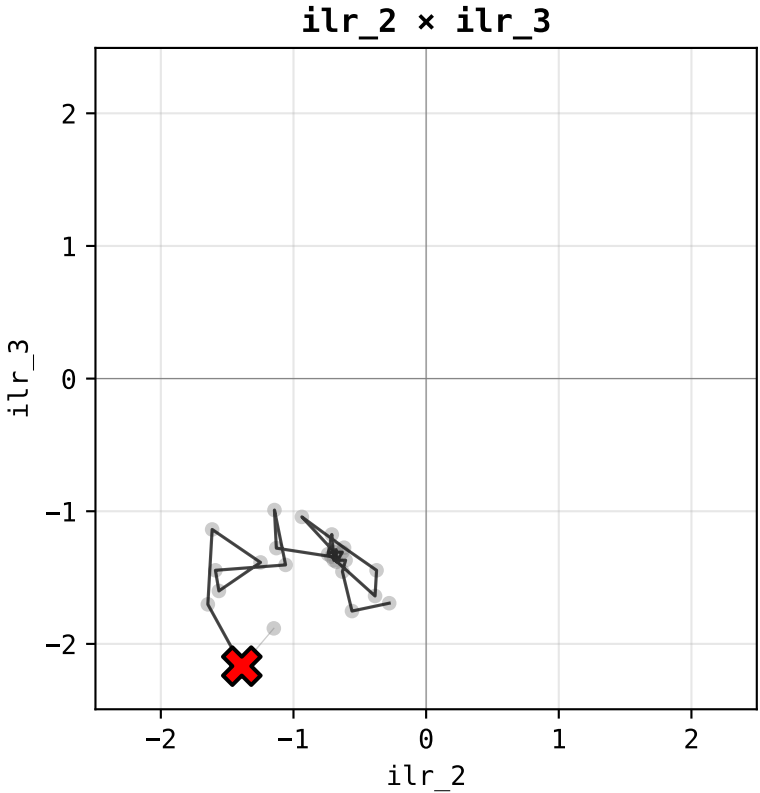
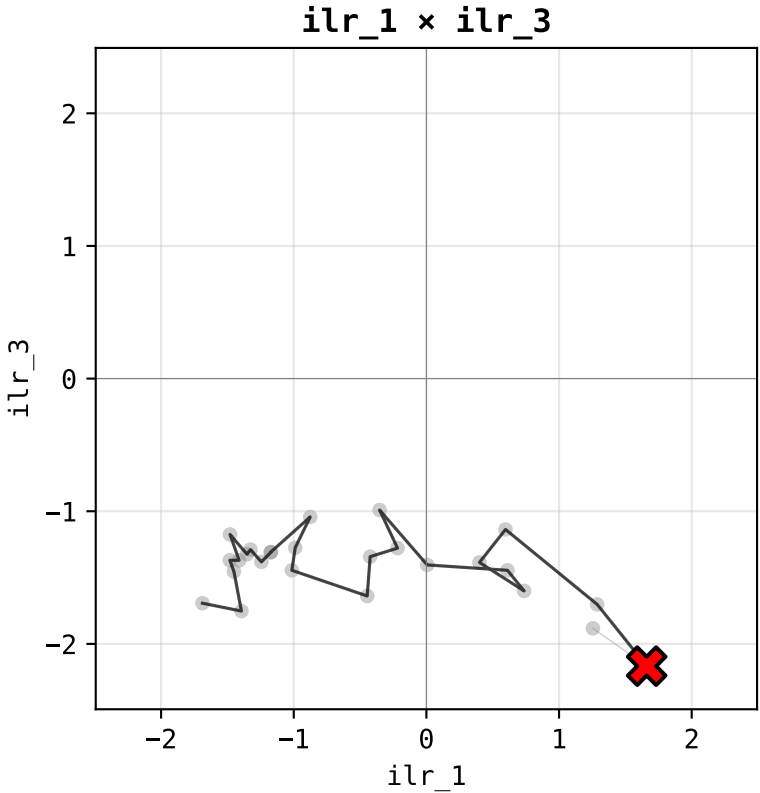
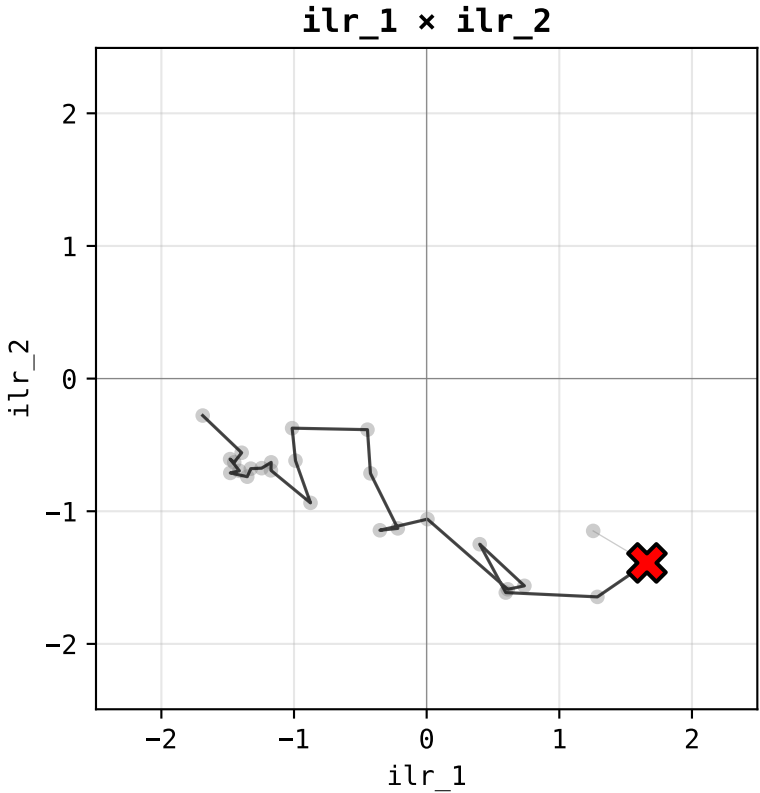


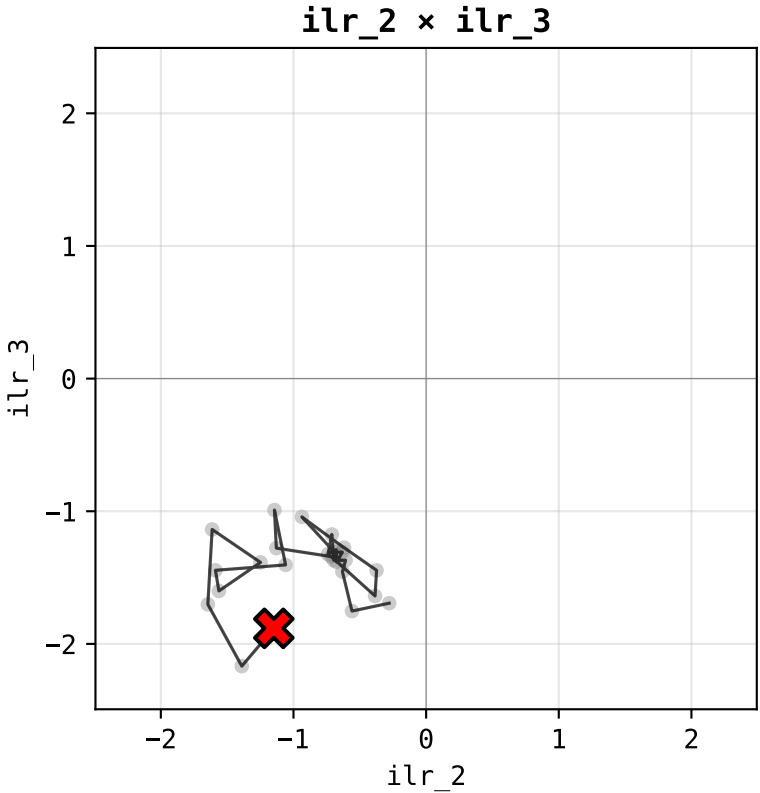
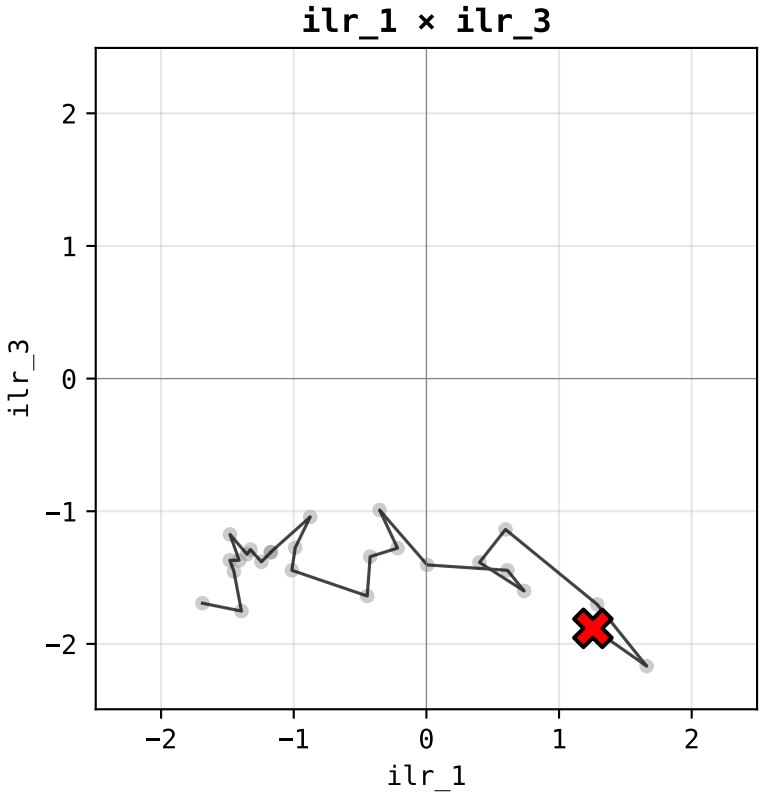
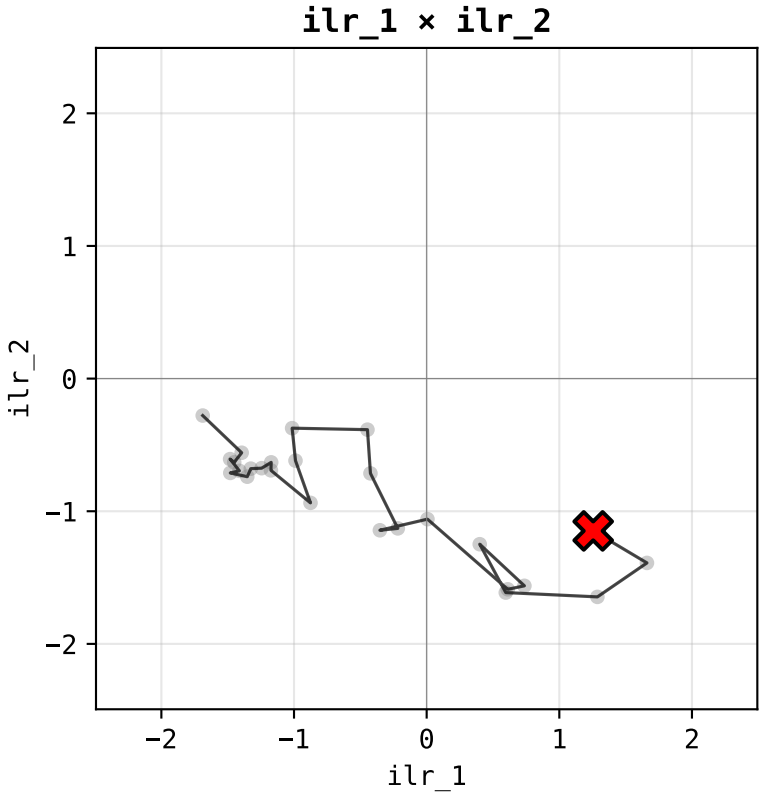












View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ GBR United Kingdom

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

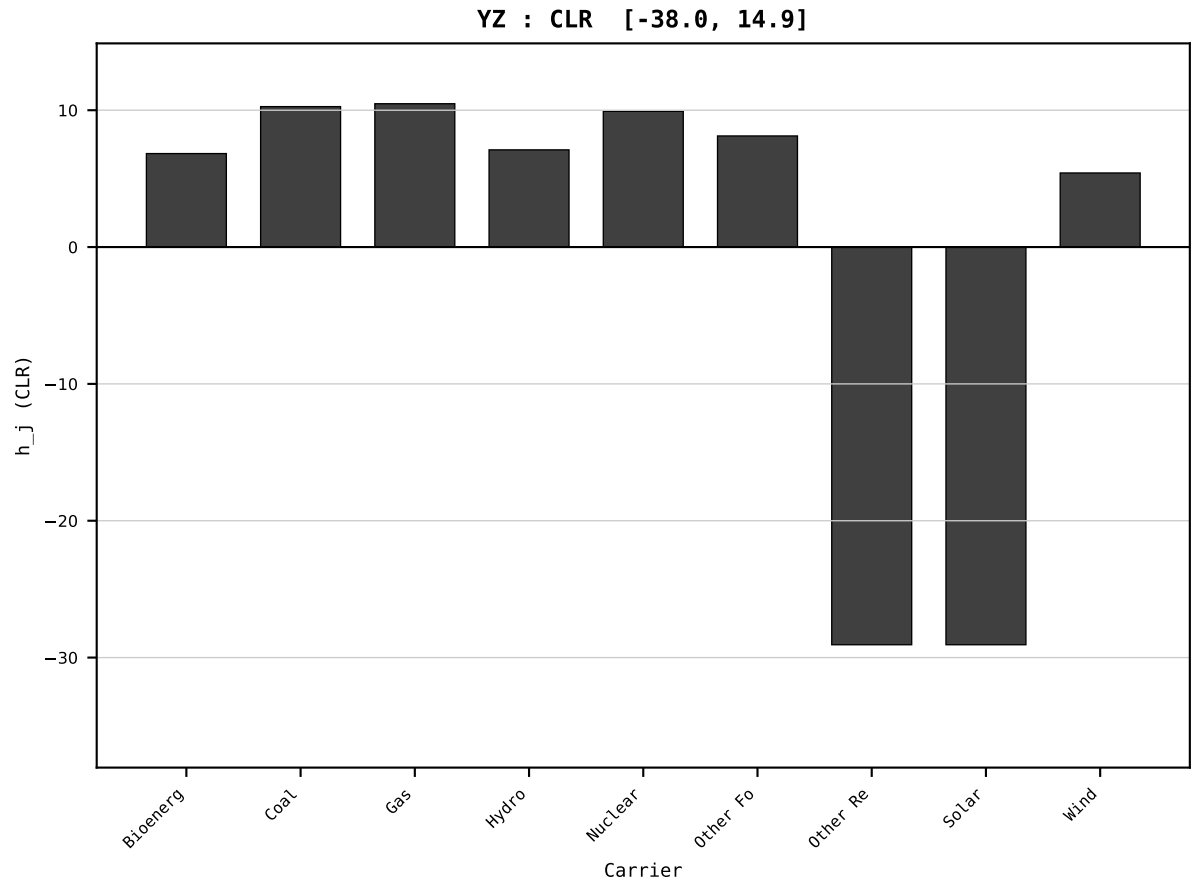
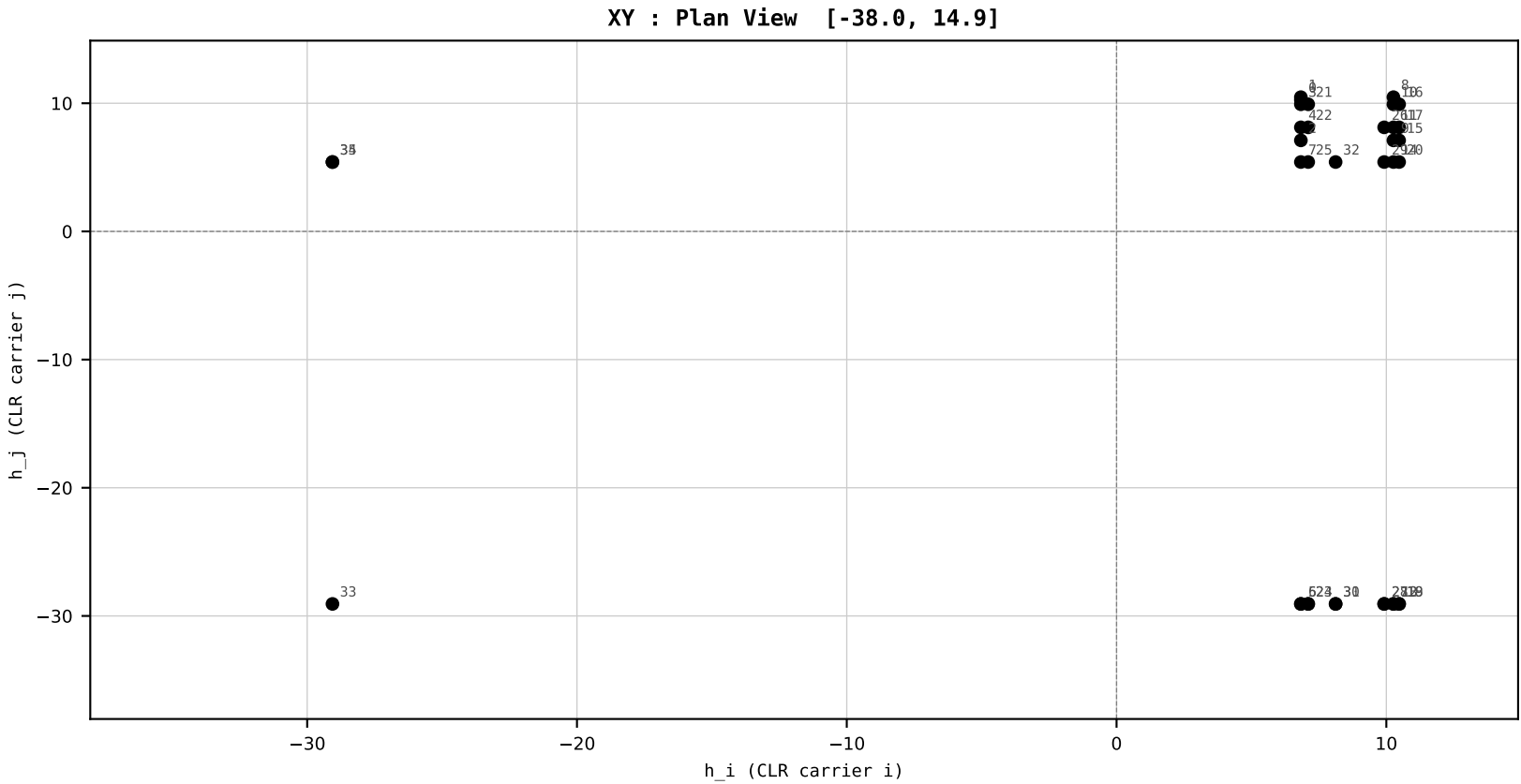
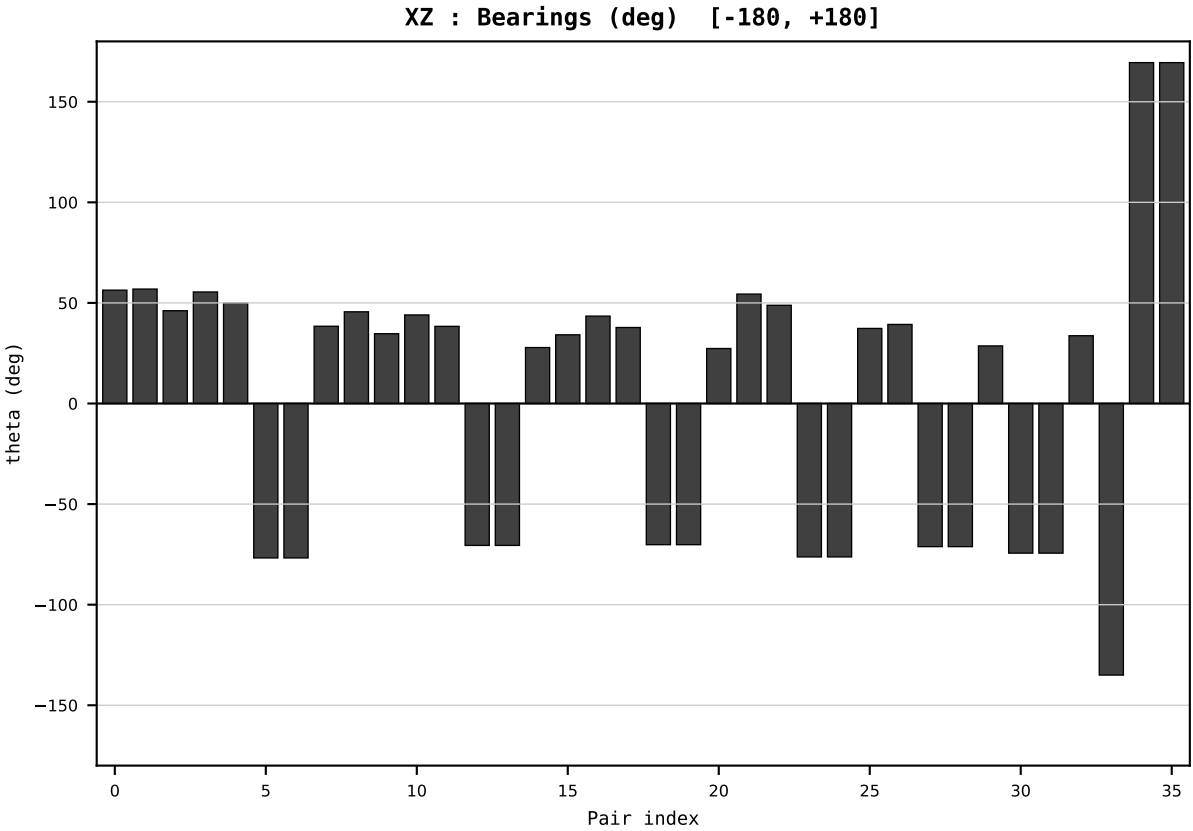
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.403740
Ring = Hs-3
E_metric= 1560.1795
kappa_HS= 148080000000000032.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 39.537 HLR
DR ratio= 148079999529708608.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

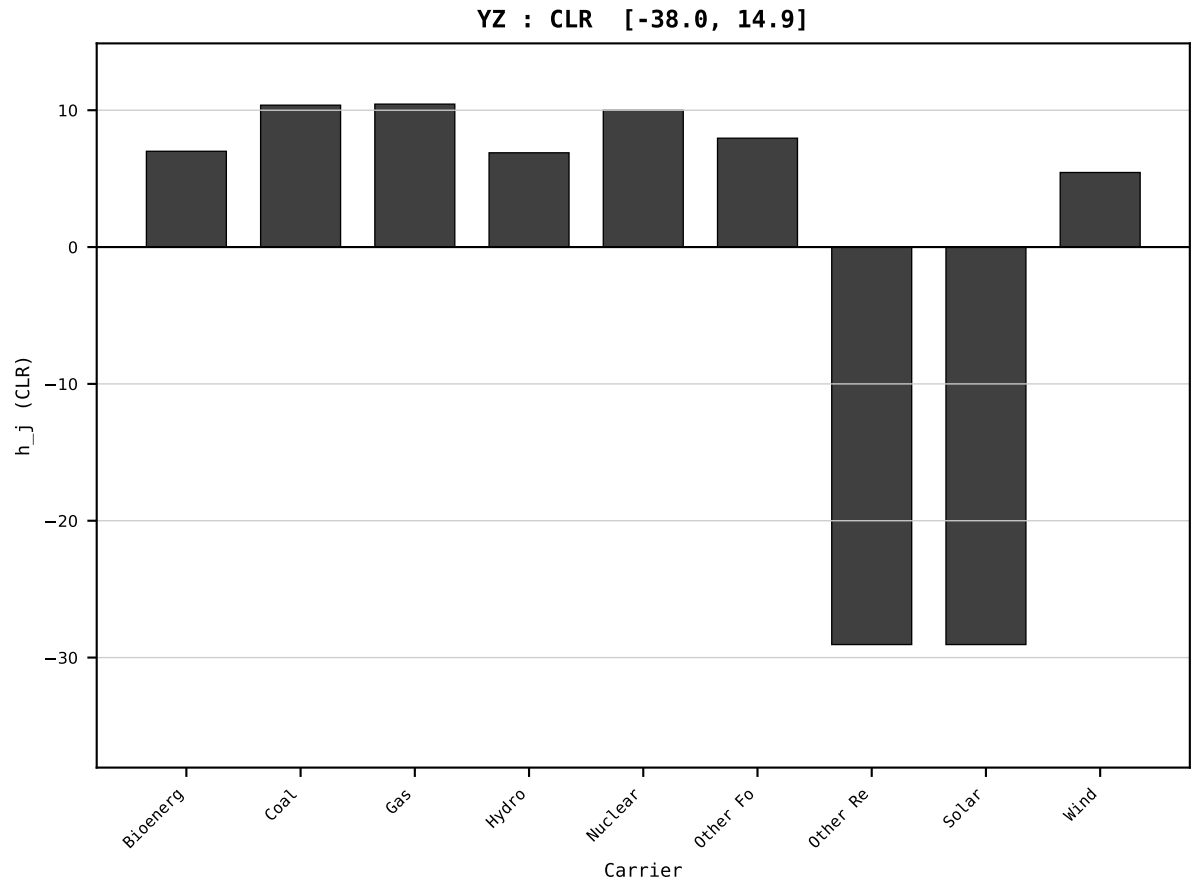
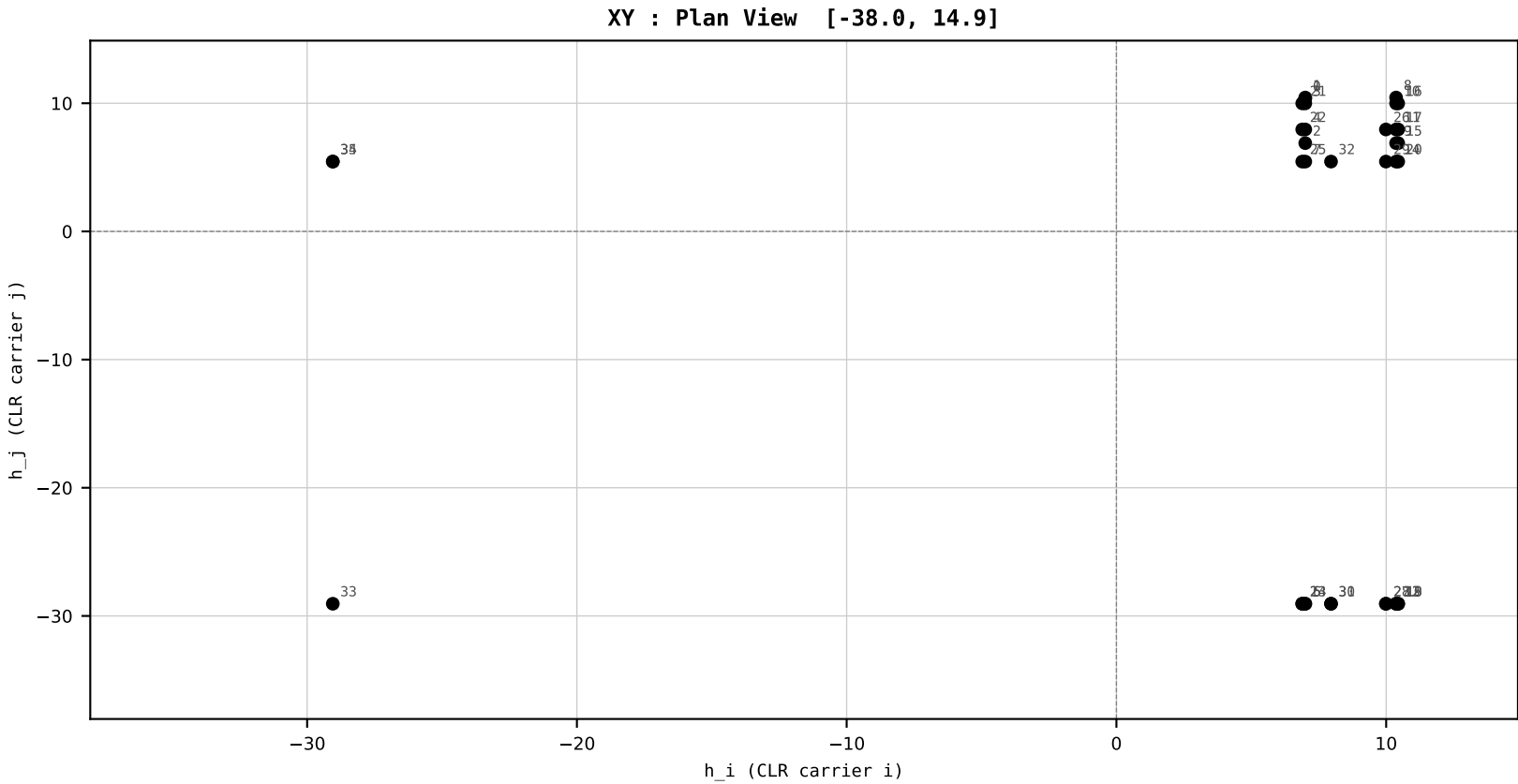
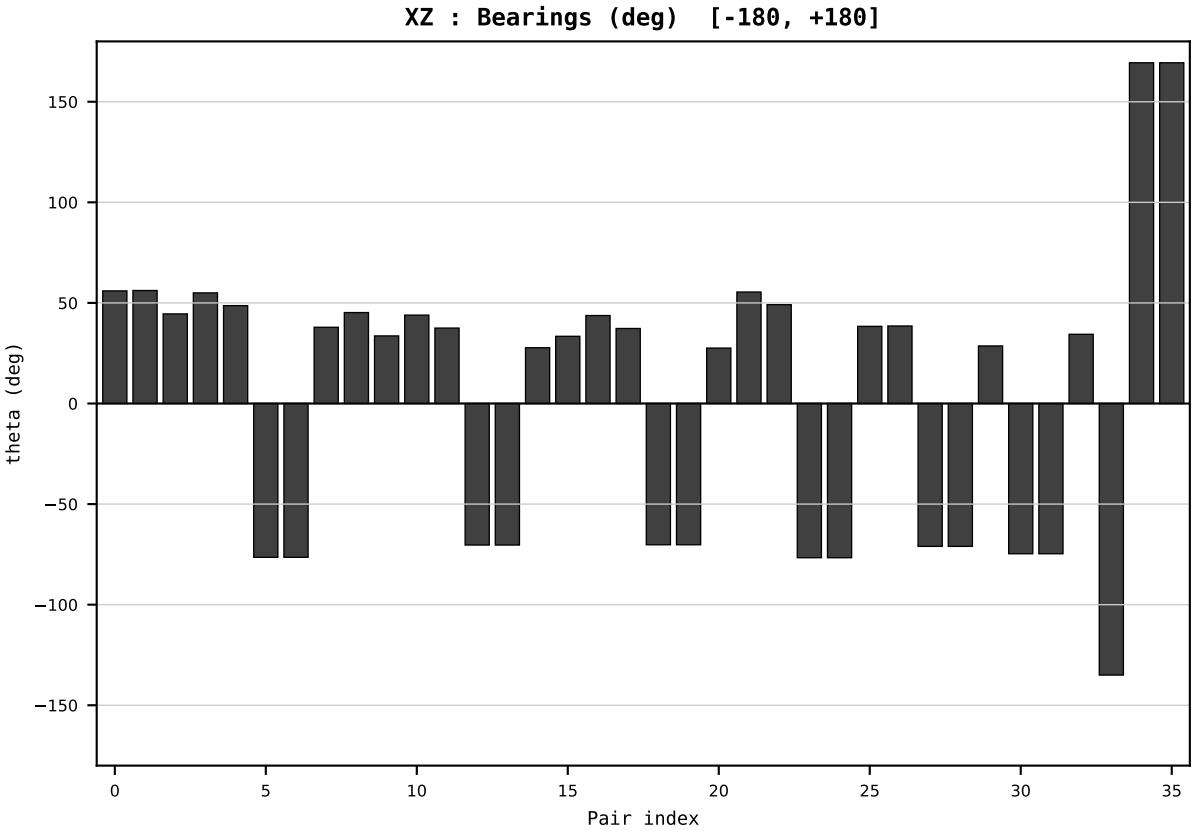
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409899
Ring = Hs-3
E_metric= 1556.9878
kappa_HS= 141909999999999984.00
omega = 0.4234 deg
d_A = 0.346528
Helm = Hydro
Helm d = -0.214796
DR = 39.494 HLR
DR ratio= 141910000176375712.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

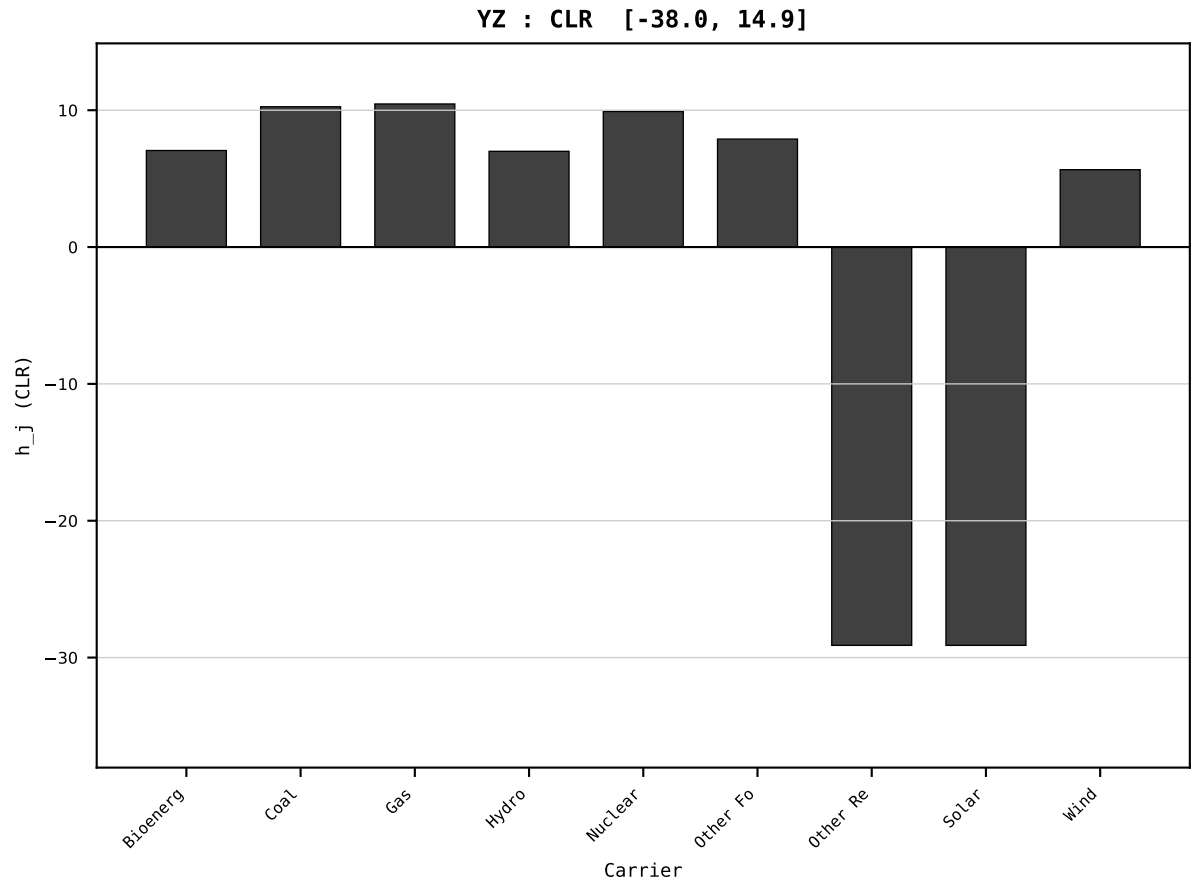
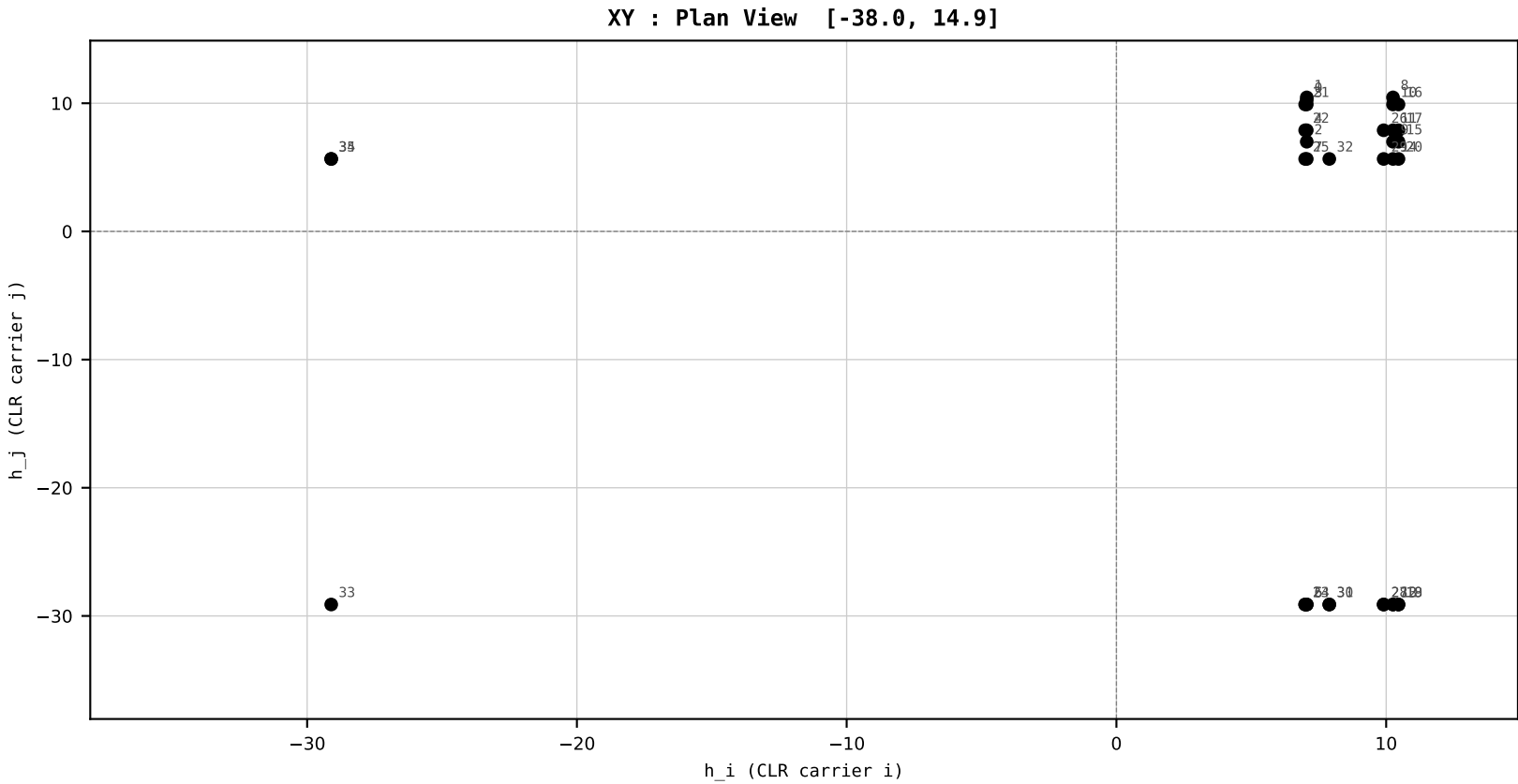
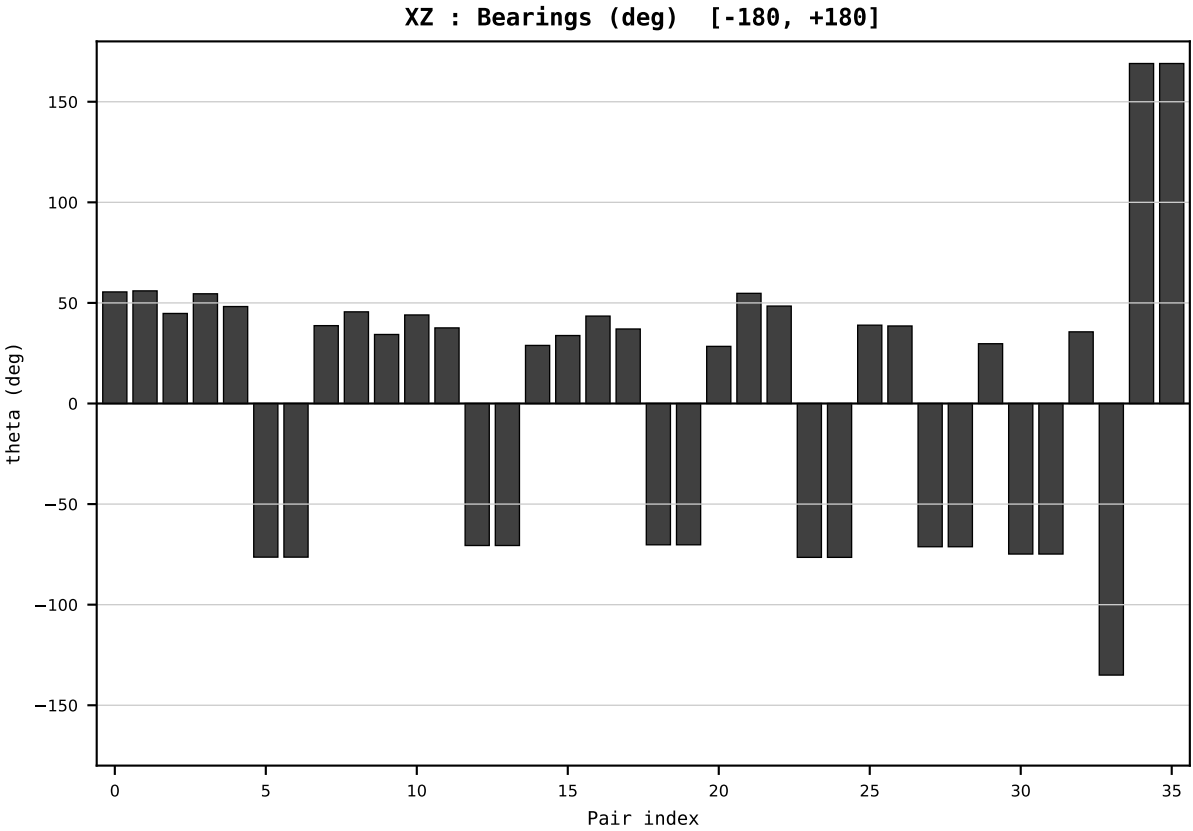
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.406615
Ring = Hs-3
E_metric= 1561.5638
kappa_HS= 15228000000000000.00
omega = 0.3533 deg
d_A = 0.297482
Helm = Wind
Helm d = +0.205028
DR = 39.564 HLR
DR ratio= 152279999021603904.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



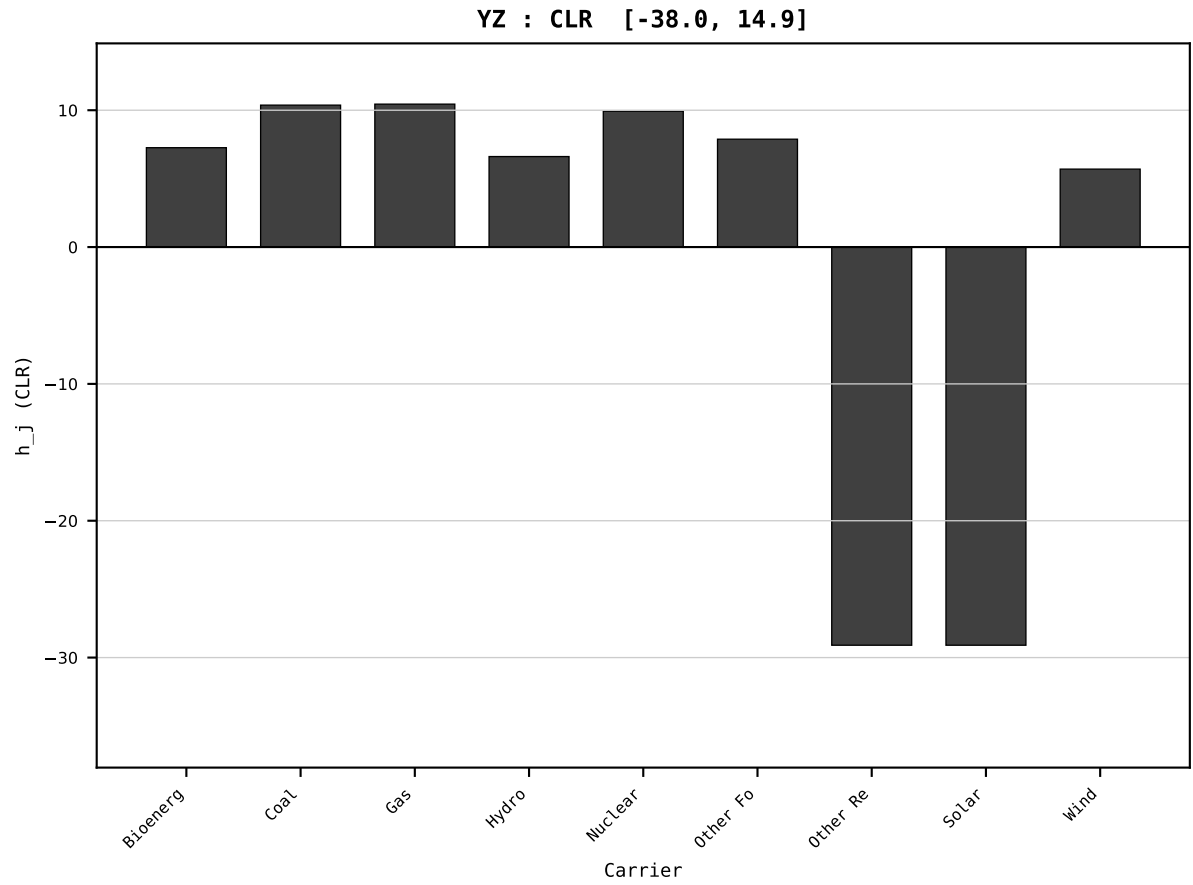
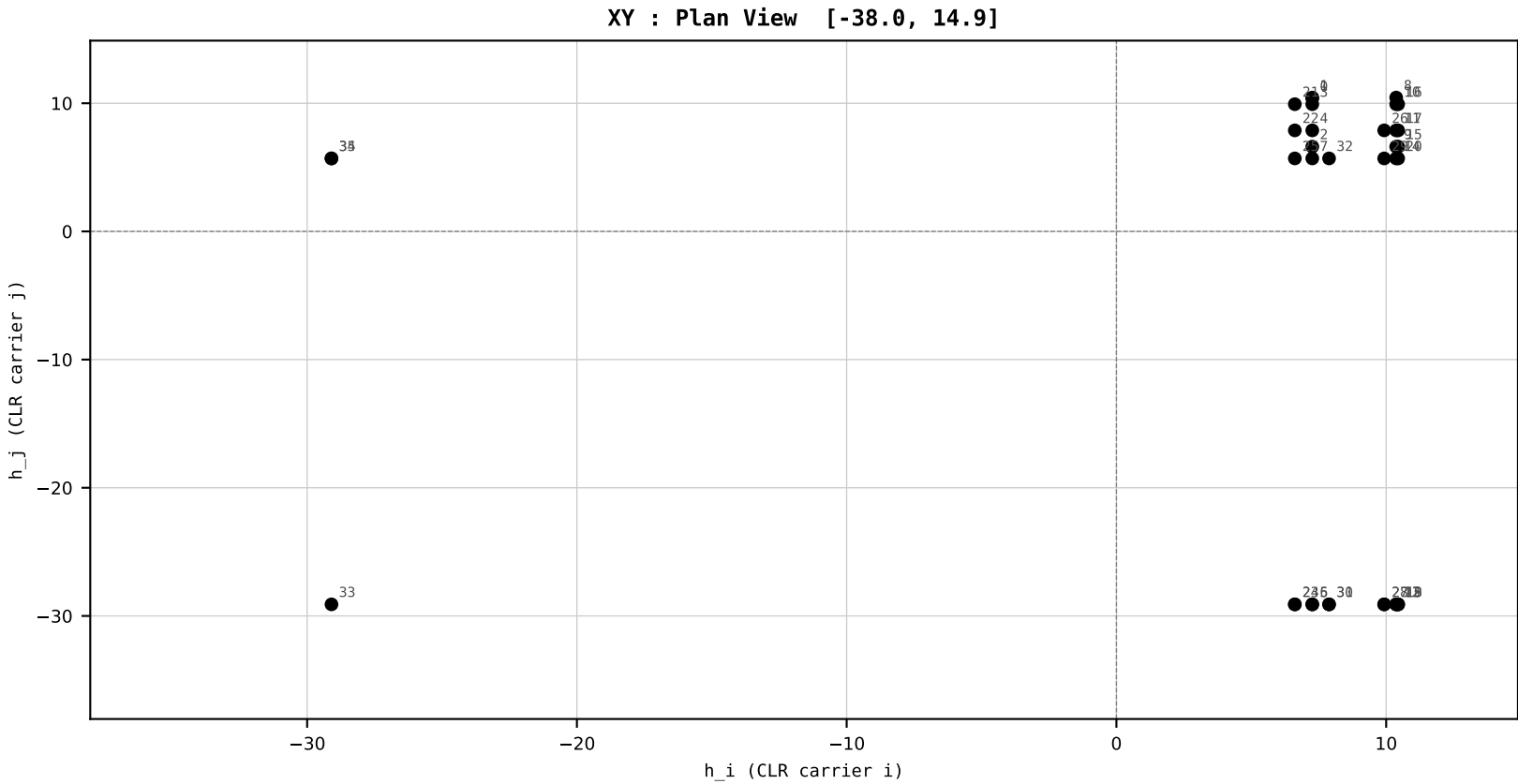
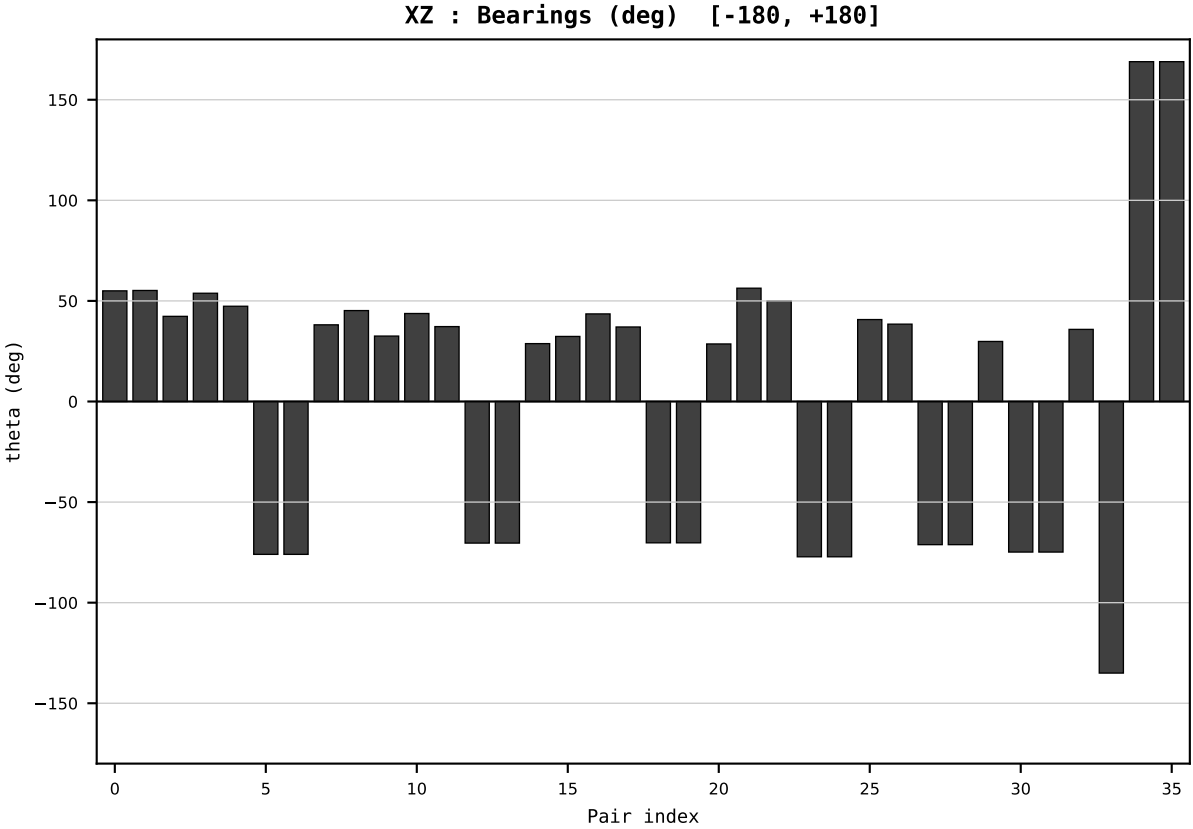
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.411016
Ring = Hs-3
E_metric= 1558.4988
kappa_HS= 14888000000000000.00
omega = 0.5541 deg
d_A = 0.453589
Helm = Hydro
Helm d = -0.383810
DR = 39.542 HLR
DR ratio= 148880000410211776.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

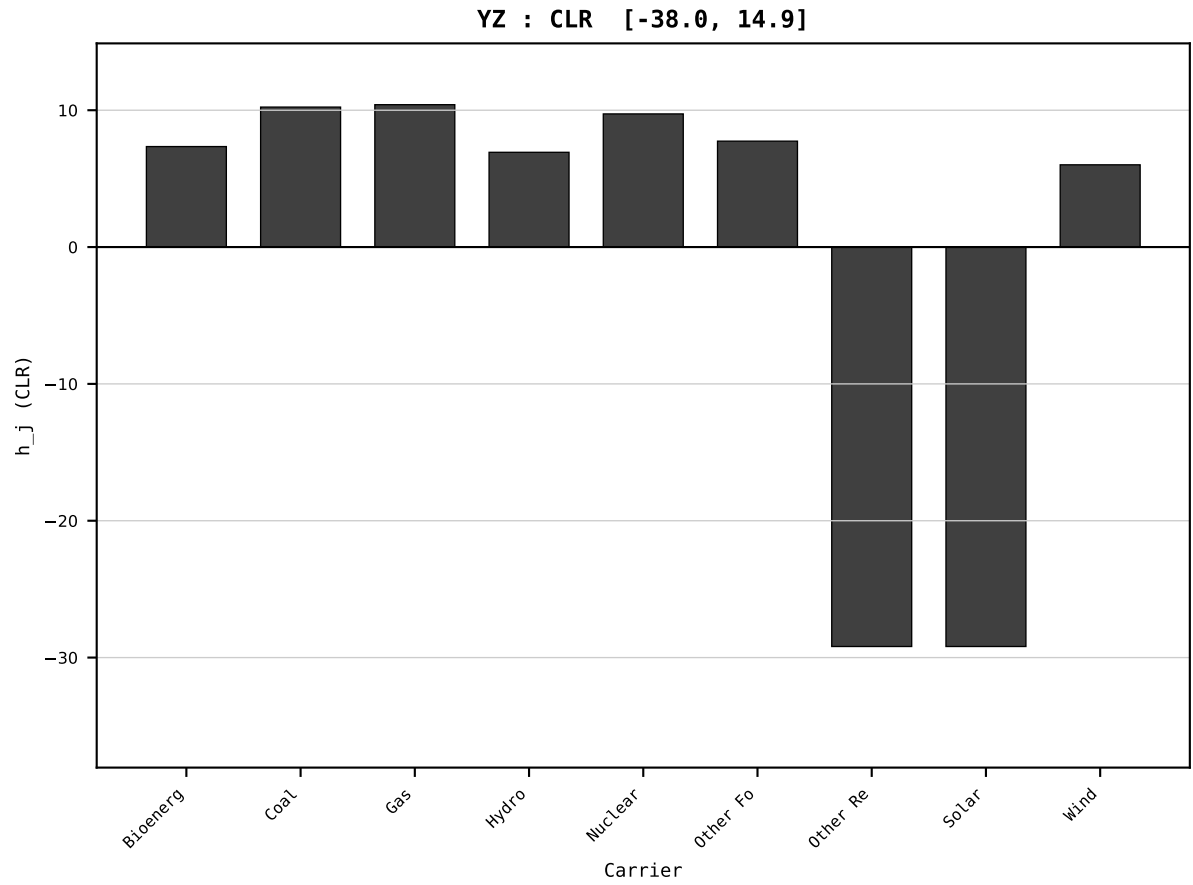
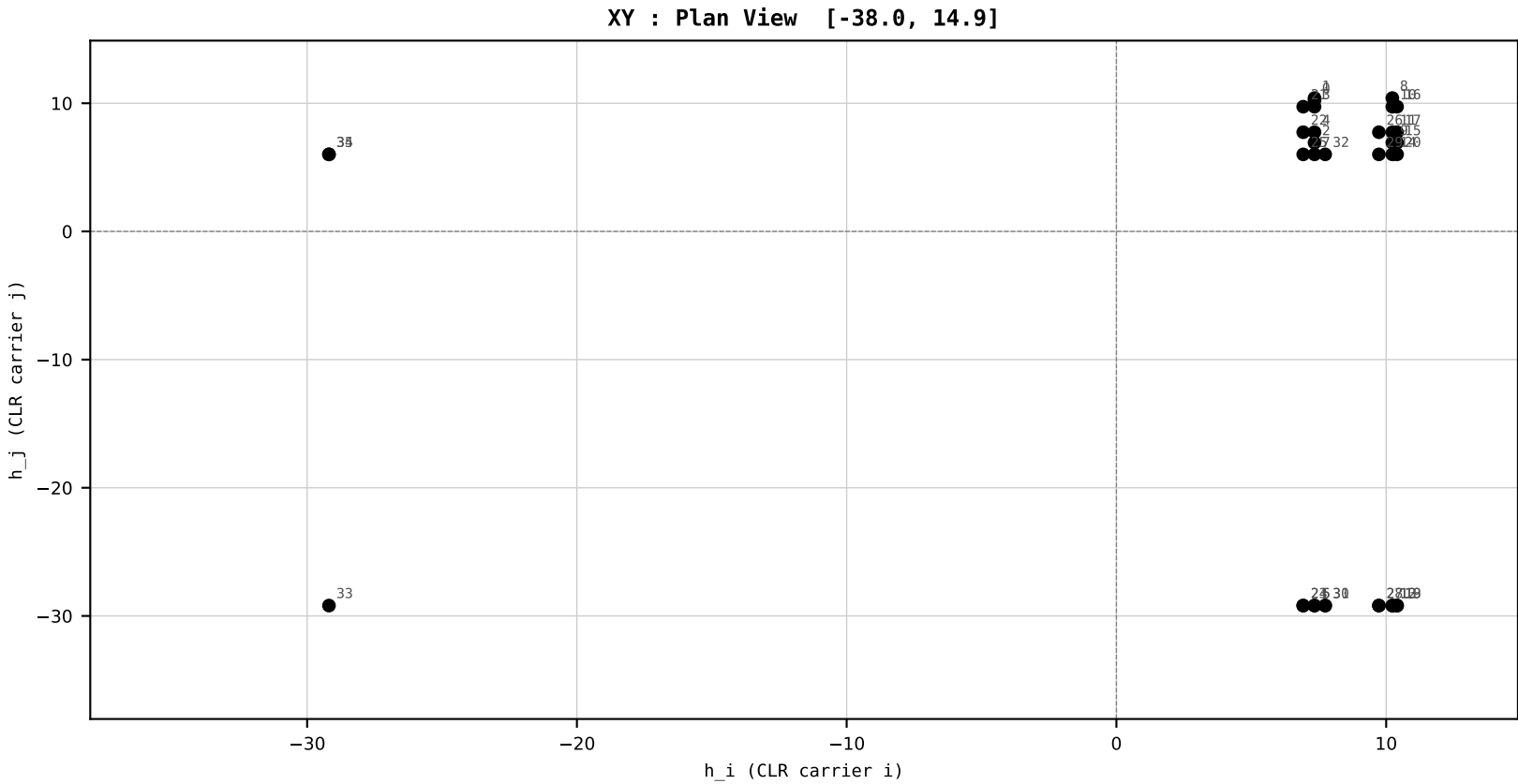
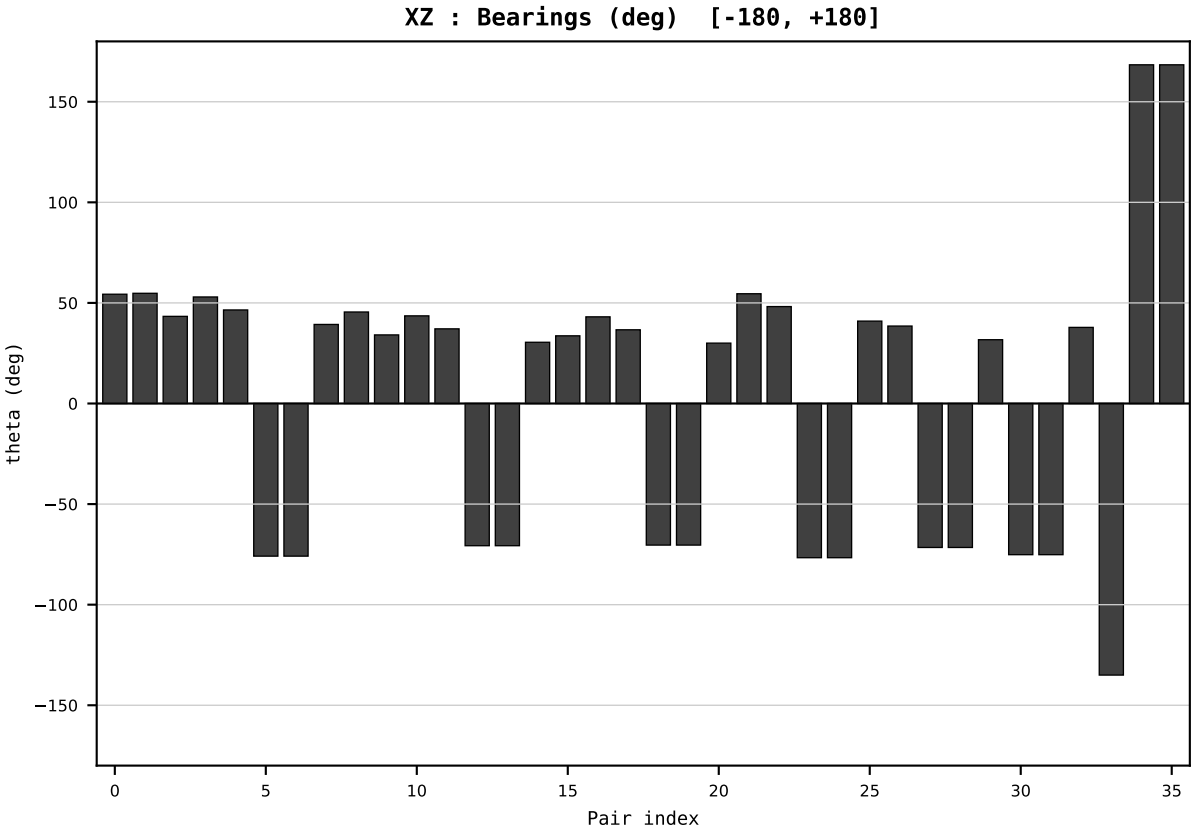
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.403458
Ring = Hs-3
E_metric= 1567.4455
kappa_HS= 15706000000000000.00
omega = 0.6567 deg
d_A = 0.548838
Helm = Wind
Helm d = +0.314727
DR = 39.595 HLR
DR ratio= 157059999551214208.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

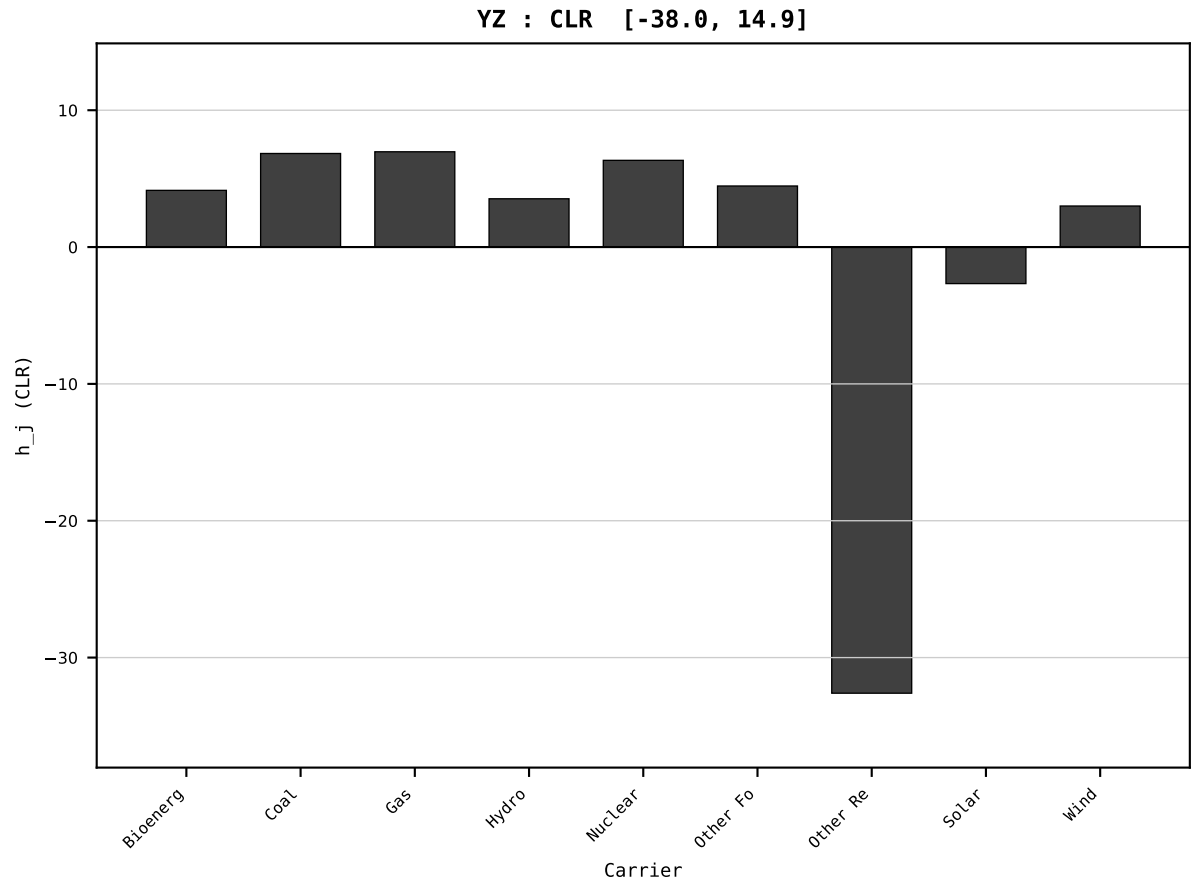
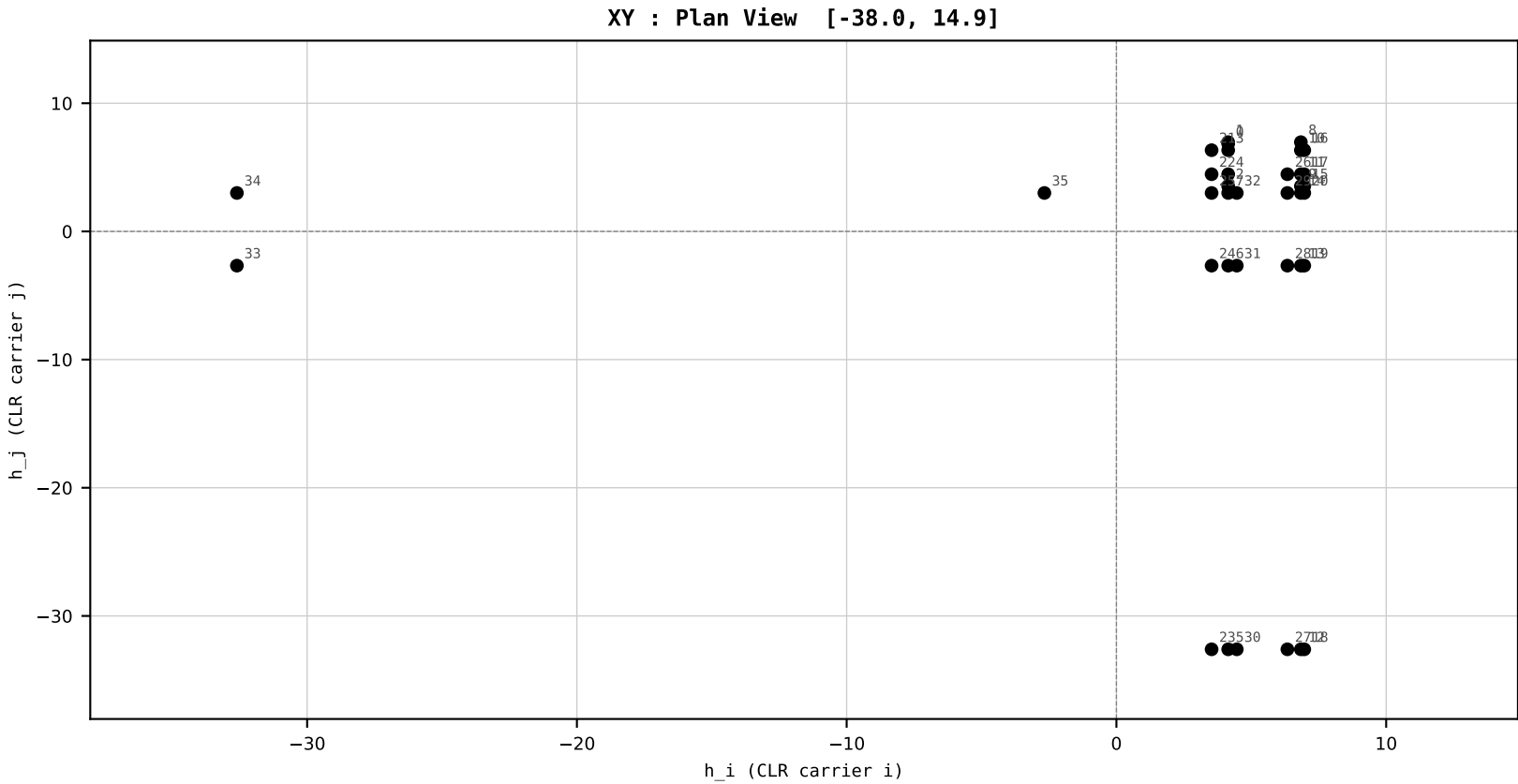
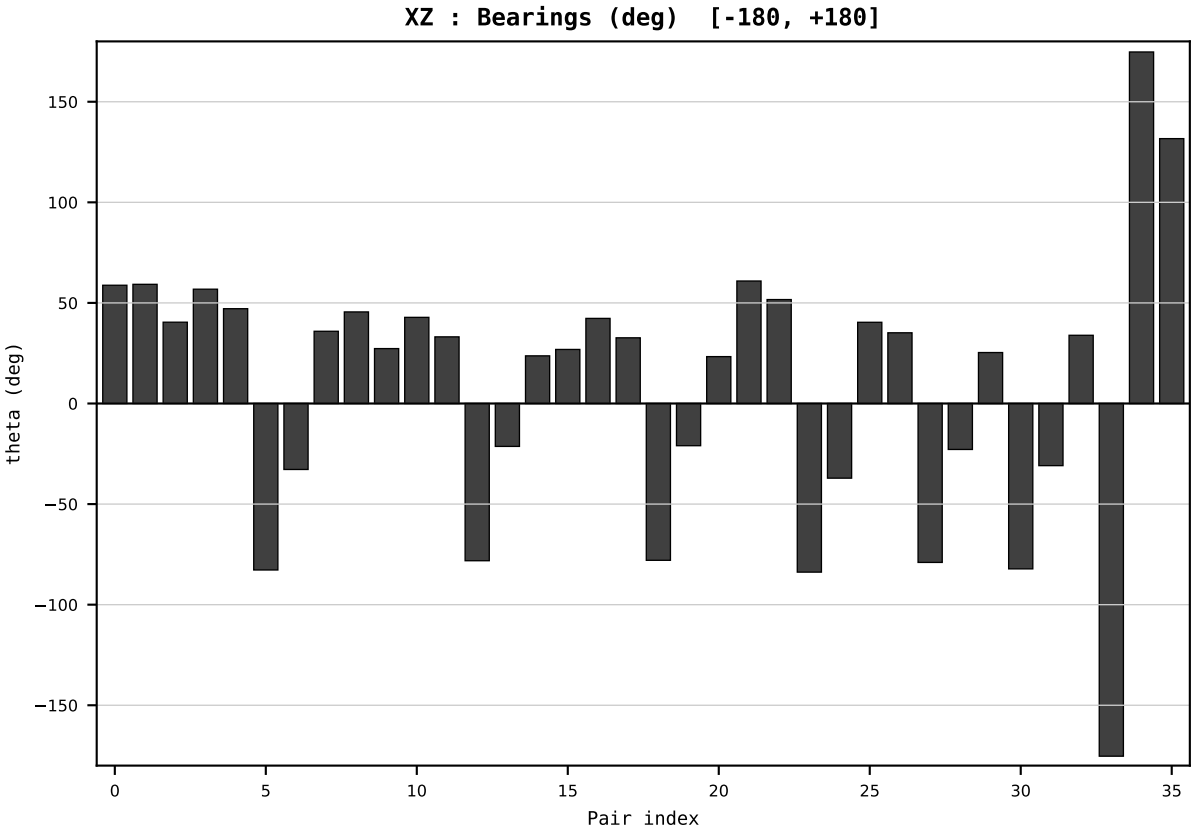
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.388111
Ring = Hs-3
E_metric= 905.5074
kappa_HS= 15264000000000000.00
omega = 36.6332 deg
d_A = 28.132514
Helm = Solar
Helm d = +26.521056
DR = 39.567 HLR
DR ratio= 152640001127908160.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

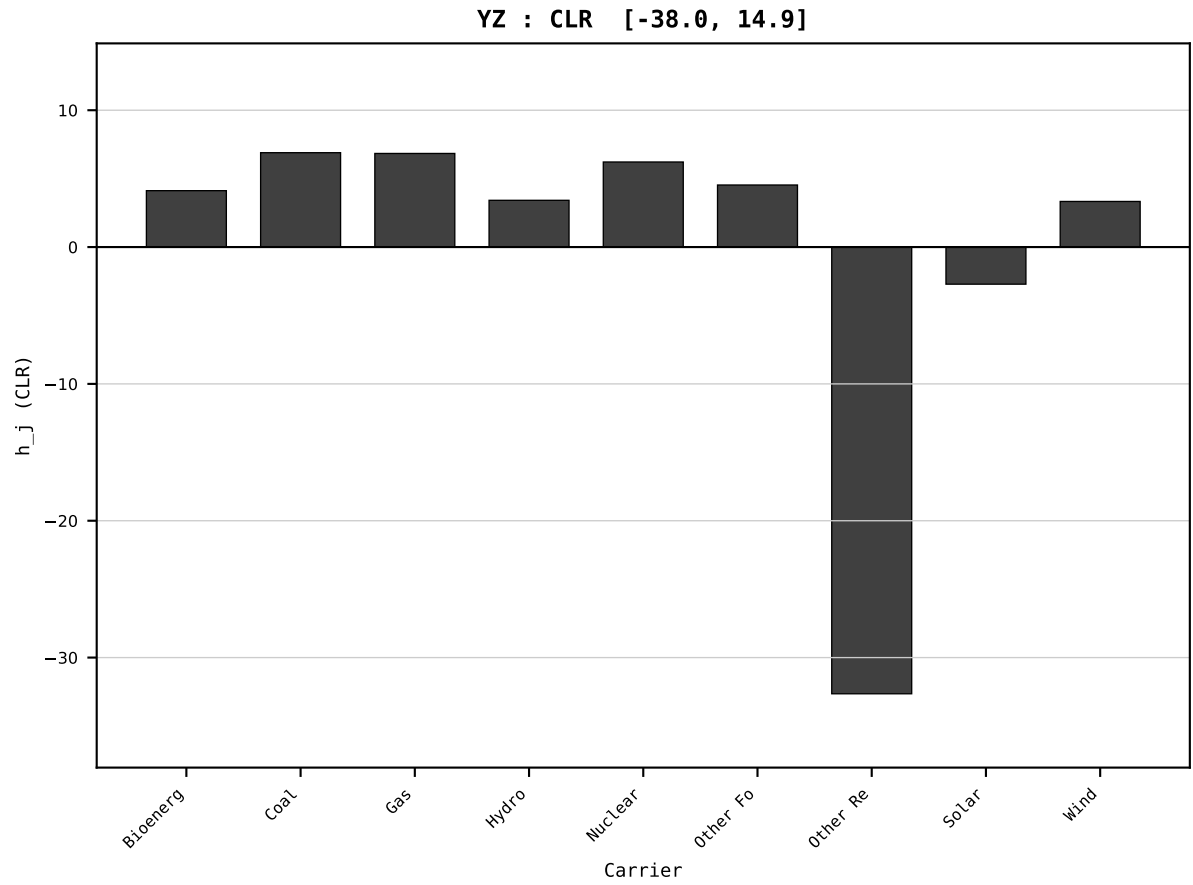
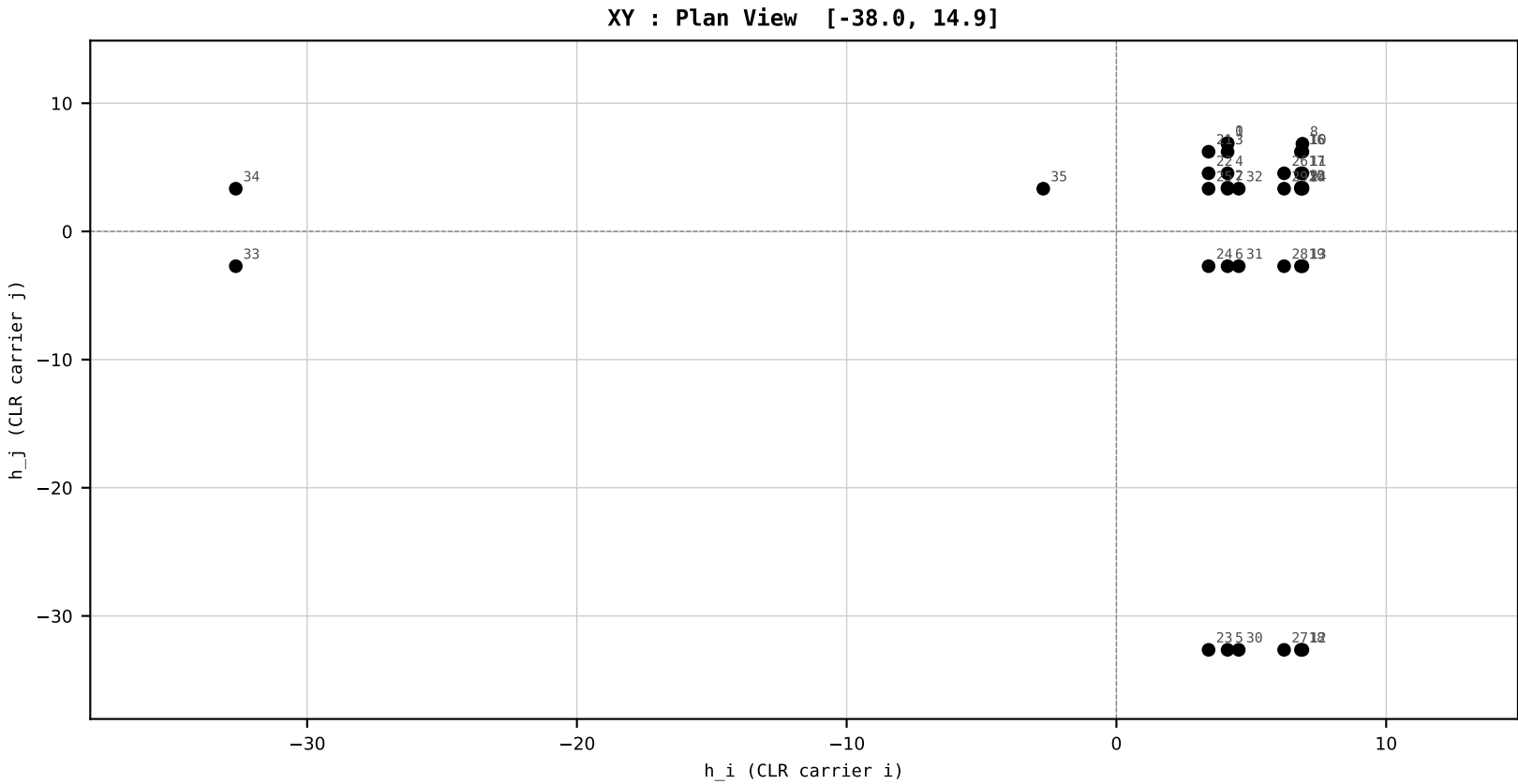
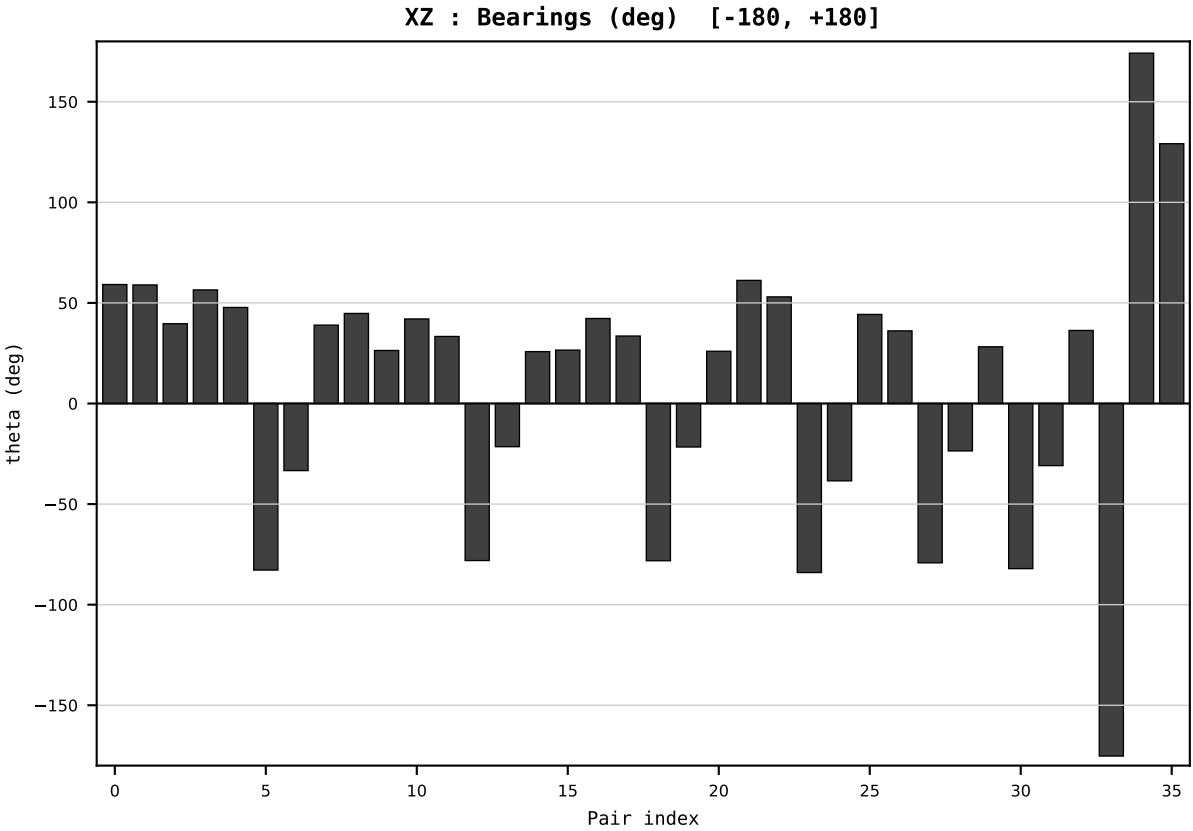
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.382444
Ring = Hs-3
E_metric= 907.5146
kappa_HS= 14885000000000000.00
omega = 0.6534 deg
d_A = 0.406998
Helm = Wind
Helm d = +0.332784
DR = 39.542 HLR
DR ratio= 148849999646857824.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

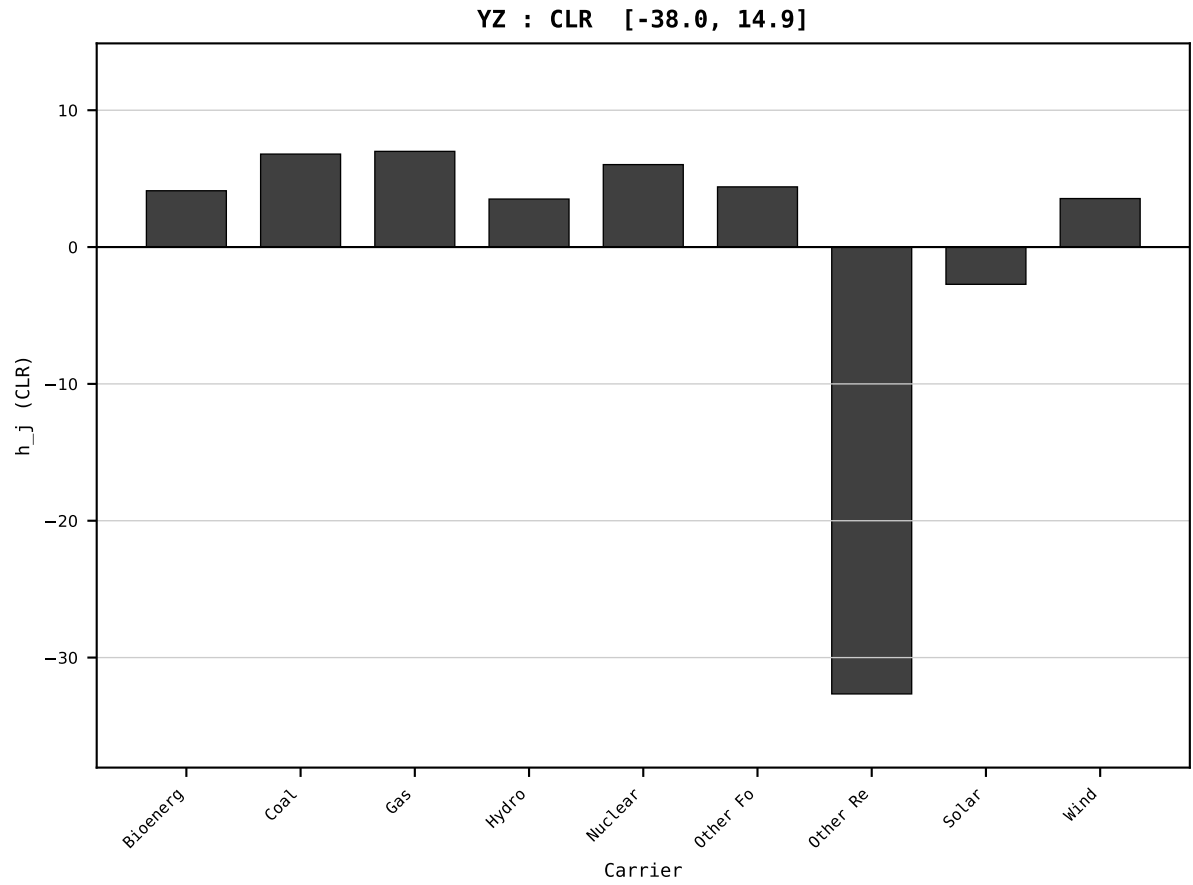
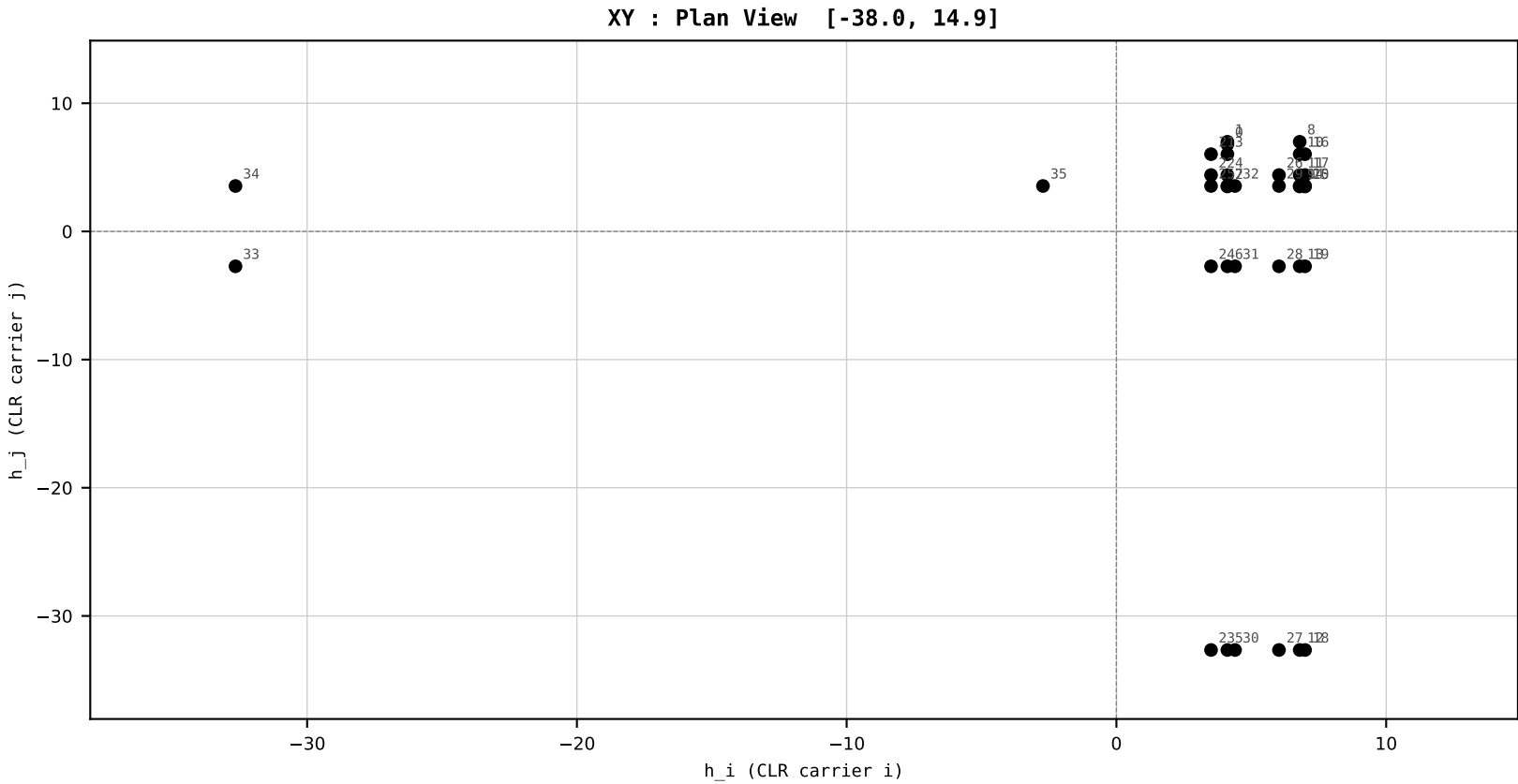
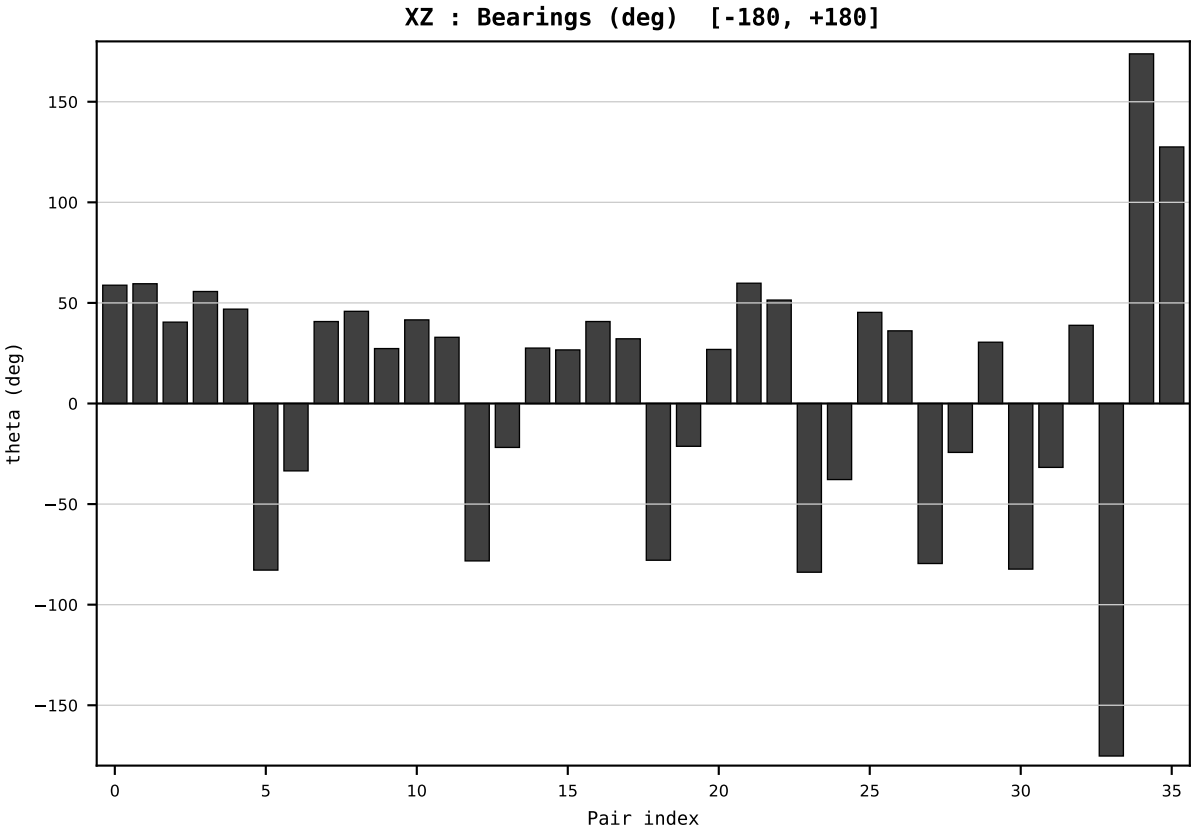
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.393078
Ring = Hs-3
E_metric= 907.4010
kappa_HS= 16578999999999968.00
omega = 0.6066 deg
d_A = 0.376749
Helm = Wind
Helm d = +0.211765
DR = 39.649 HLR
DR ratio= 165789999641552096.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

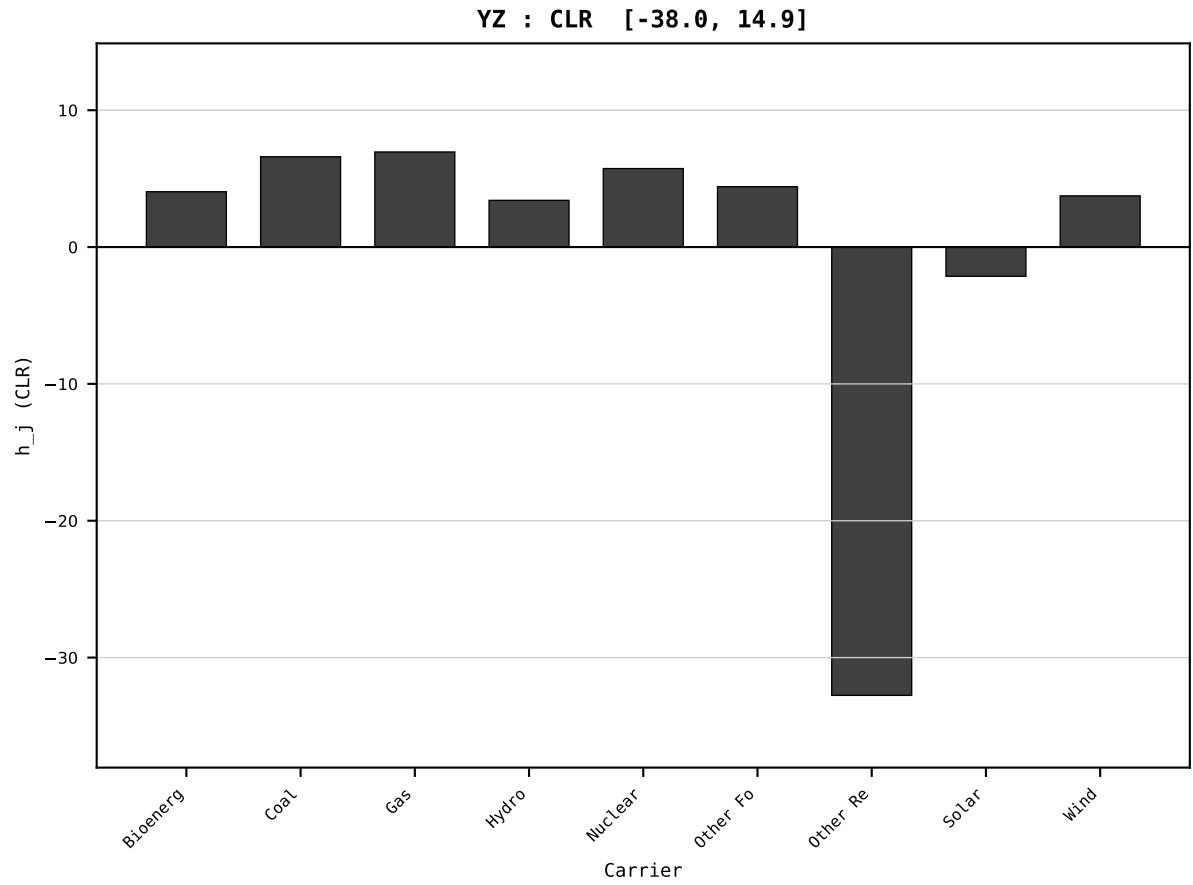
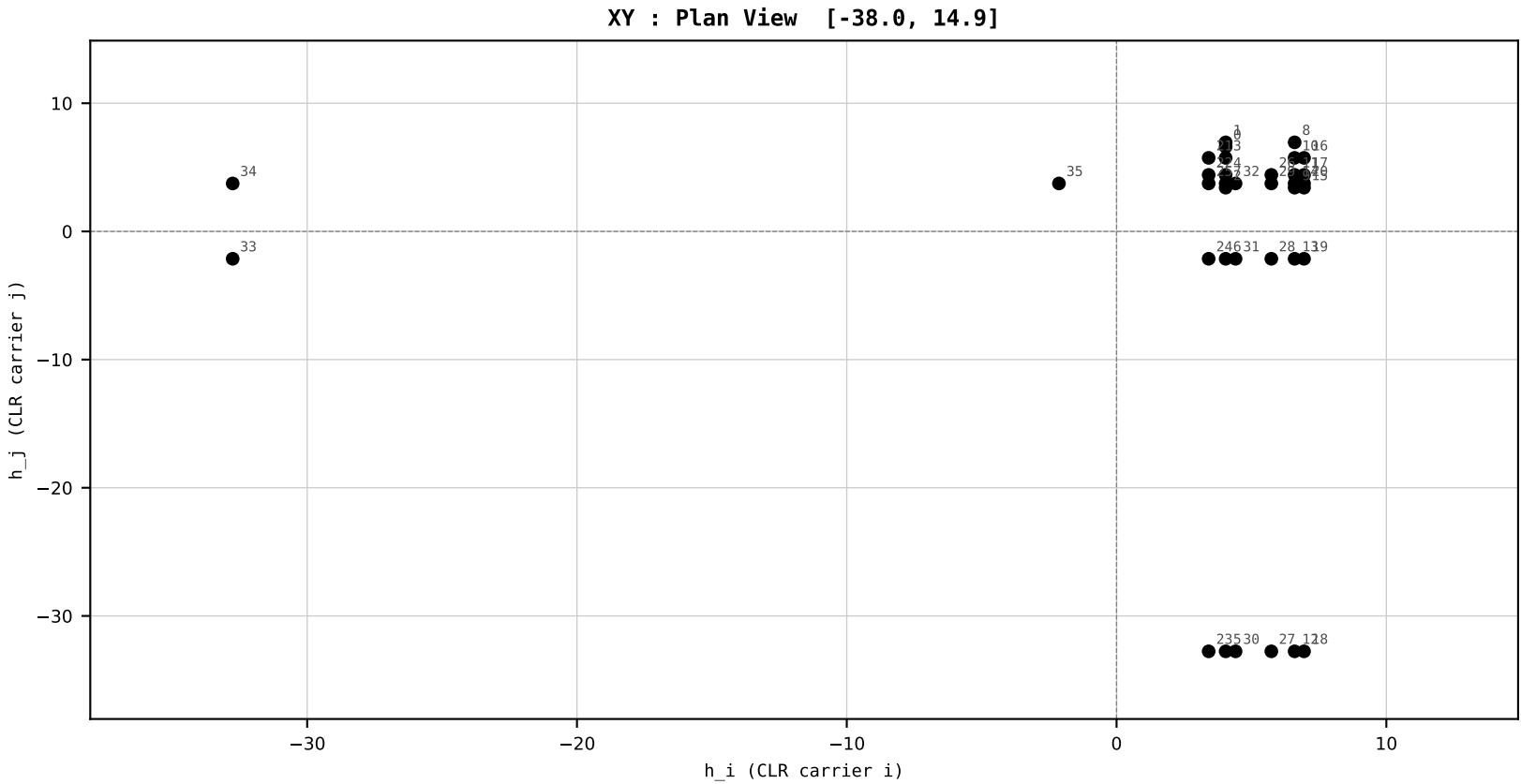
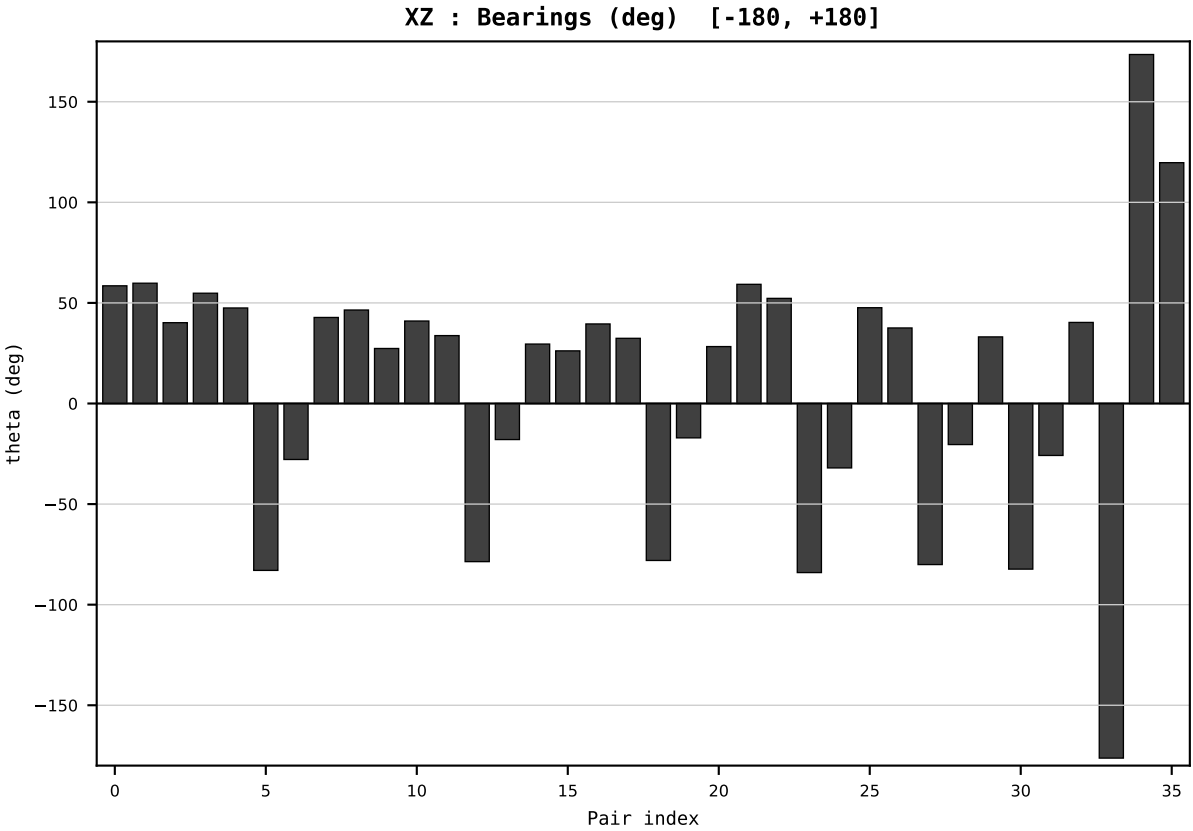
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.392183
Ring = Hs-3
E_metric= 904.9892
kappa_HS= 17622000000000000.00
omega = 1.1739 deg
d_A = 0.729475
Helm = Solar
Helm d = +0.587604
DR = 39.711 HLR
DR ratio= 176220000114672288.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

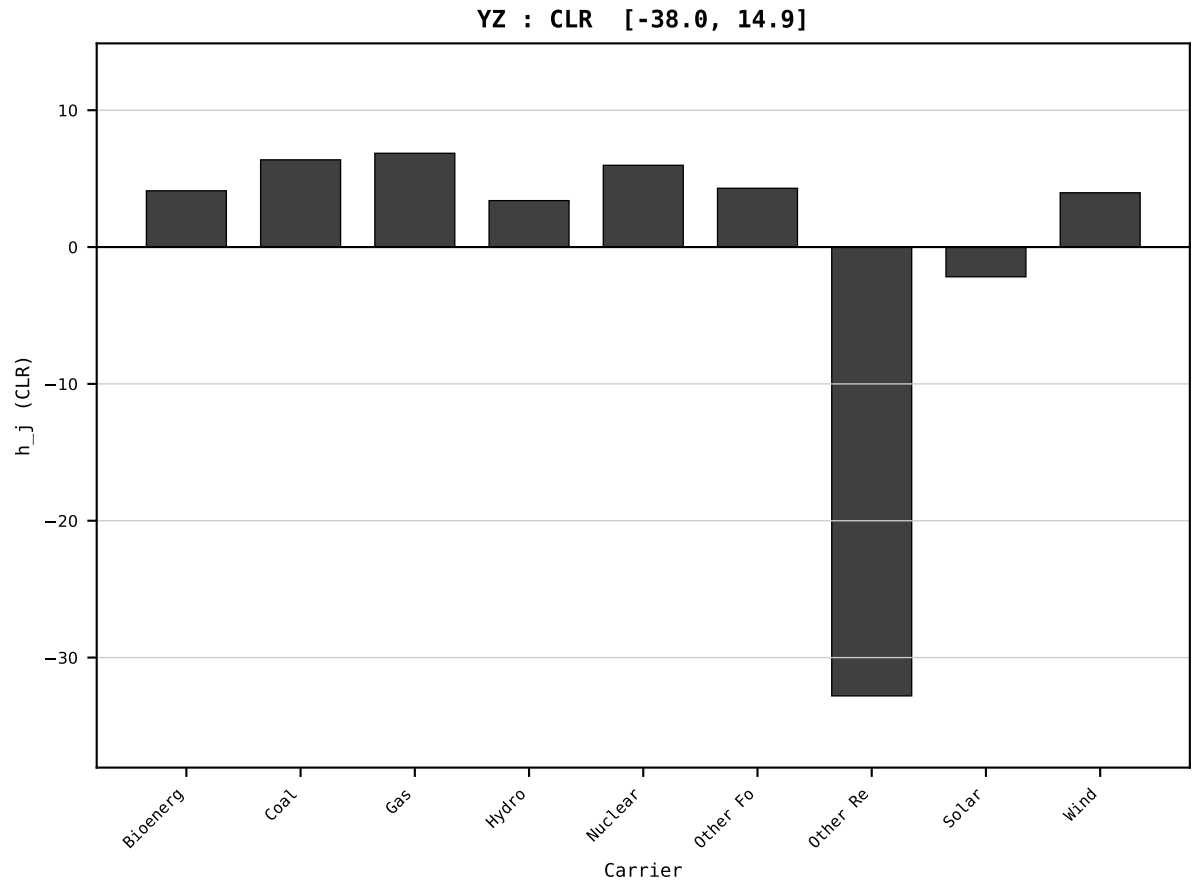
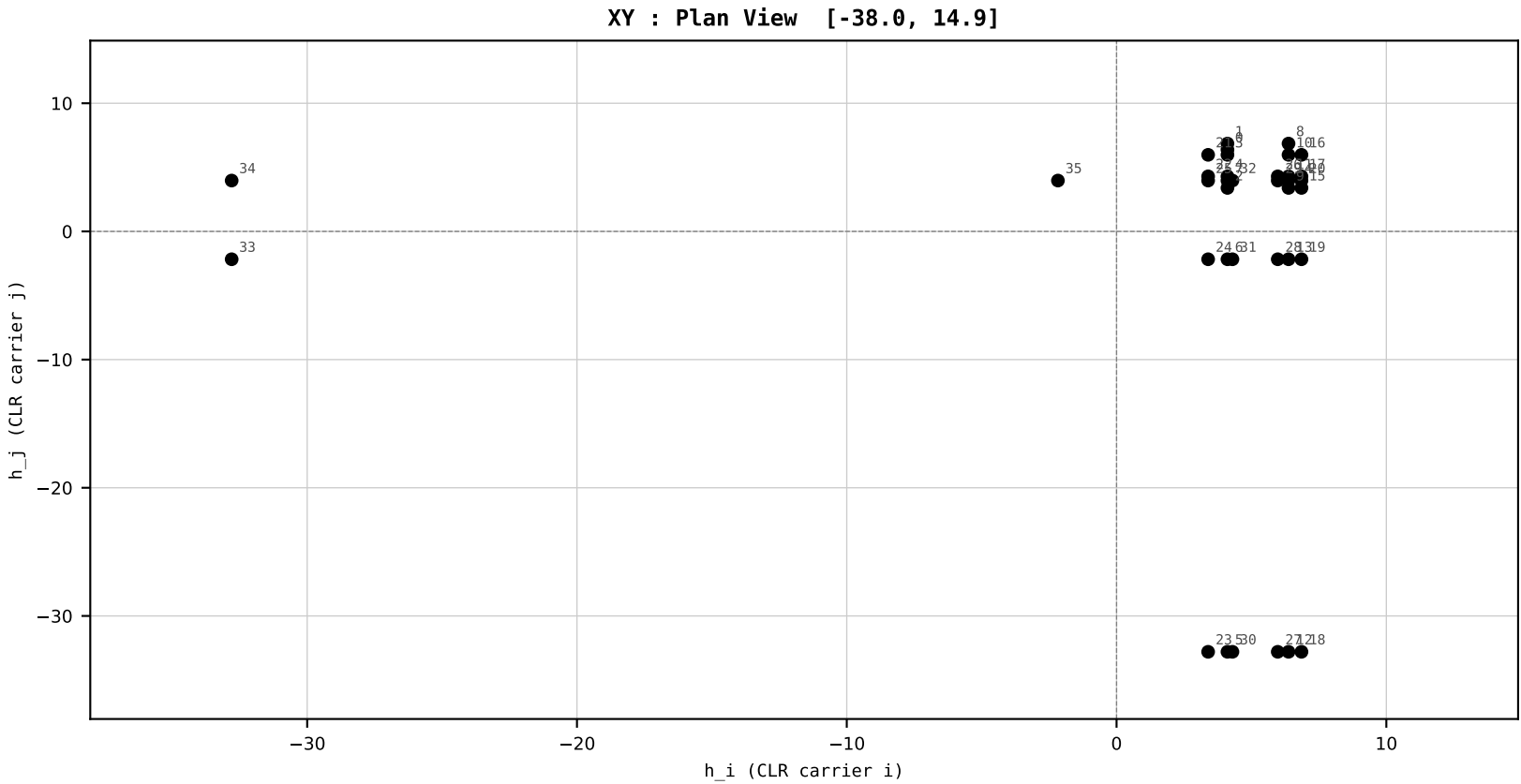
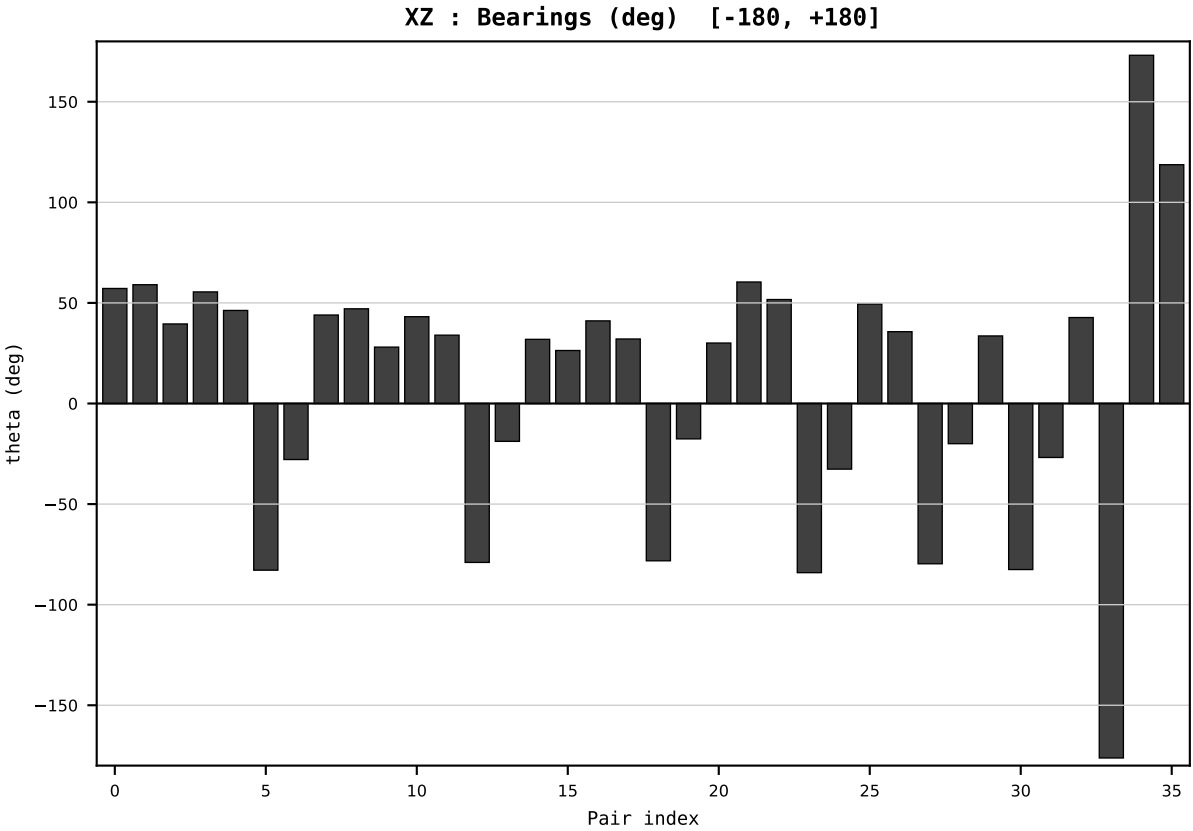
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.365440
Ring = Hs-3
E_metric= 908.6087
kappa_HS= 16650000000000000.00
omega = 0.6972 deg
d_A = 0.434051
Helm = Nuclear
Helm d = +0.237518
DR = 39.654 HLR
DR ratio= 166499999278857920.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

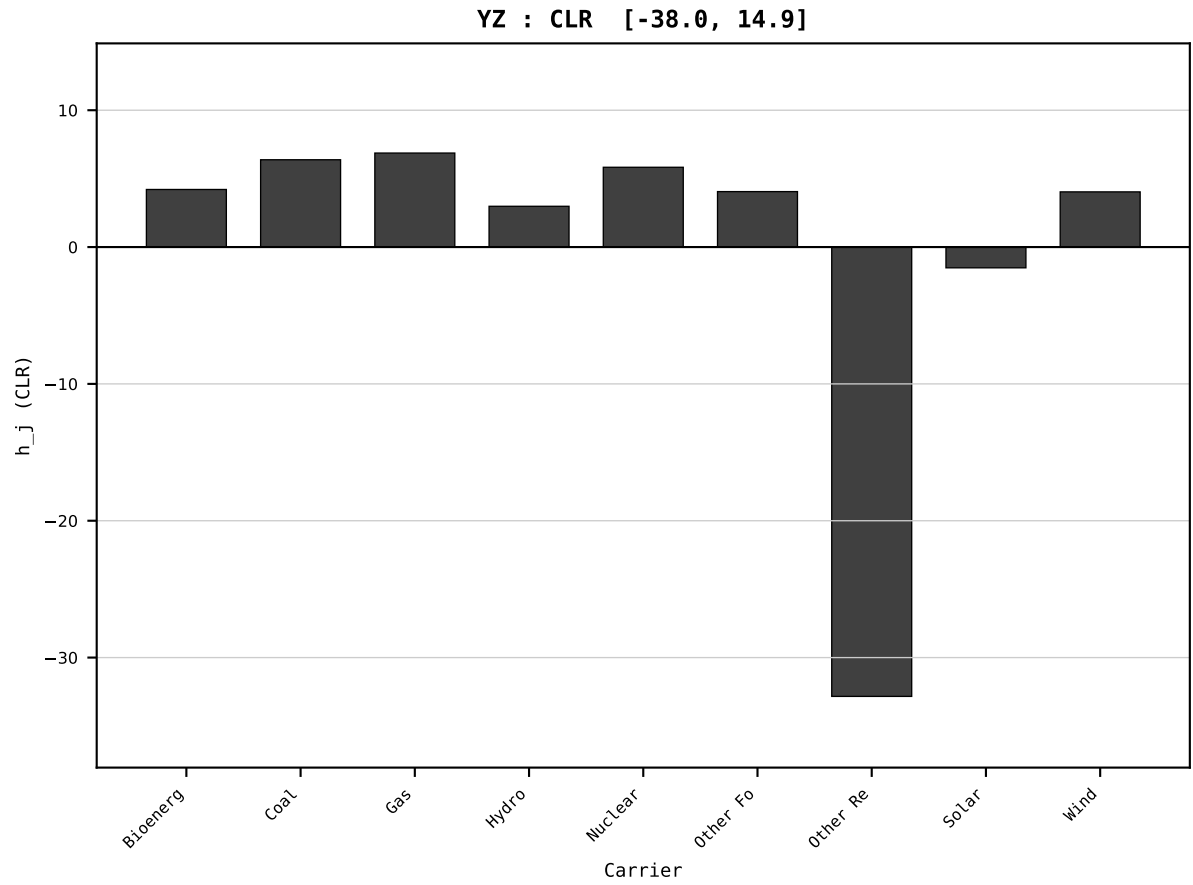
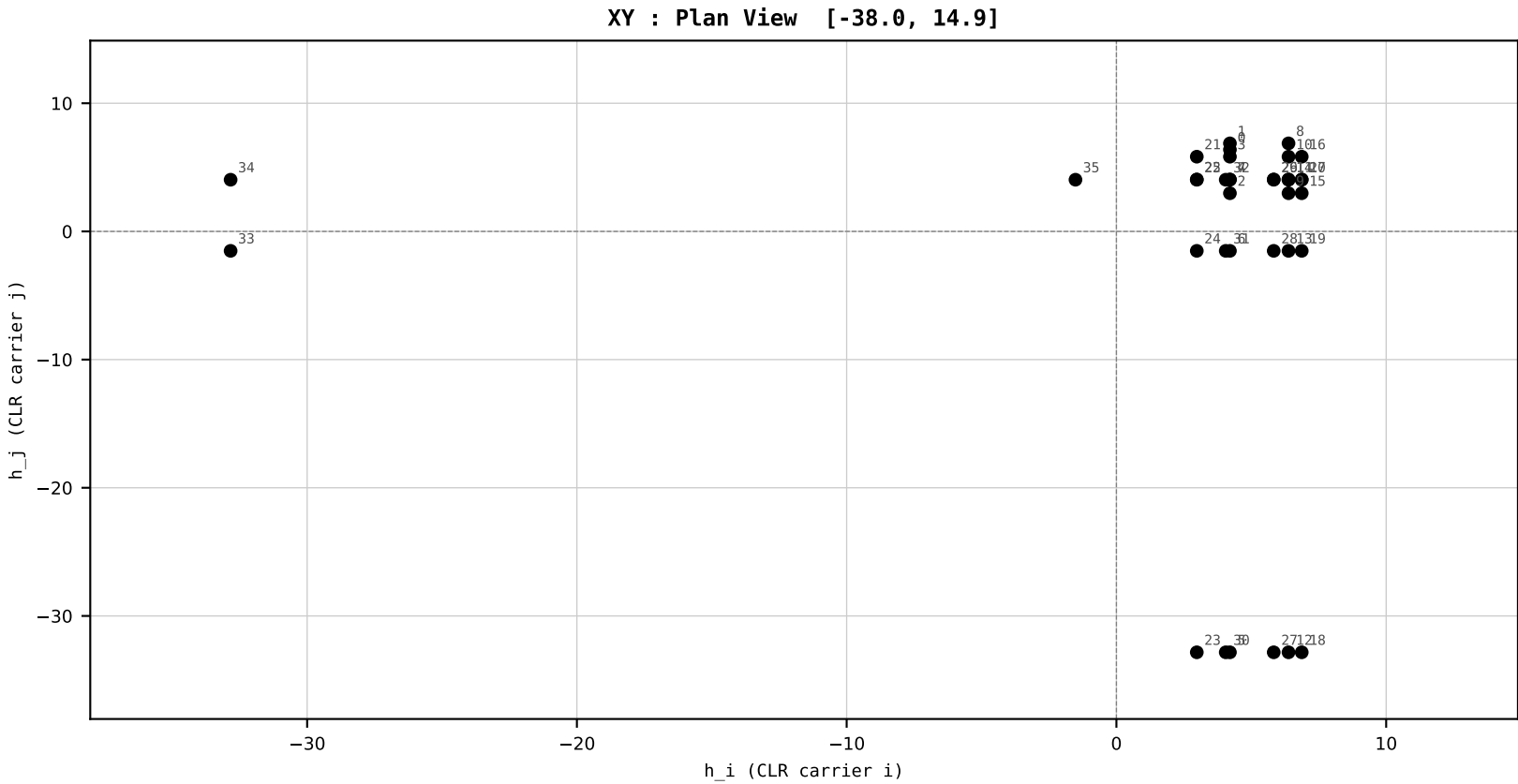
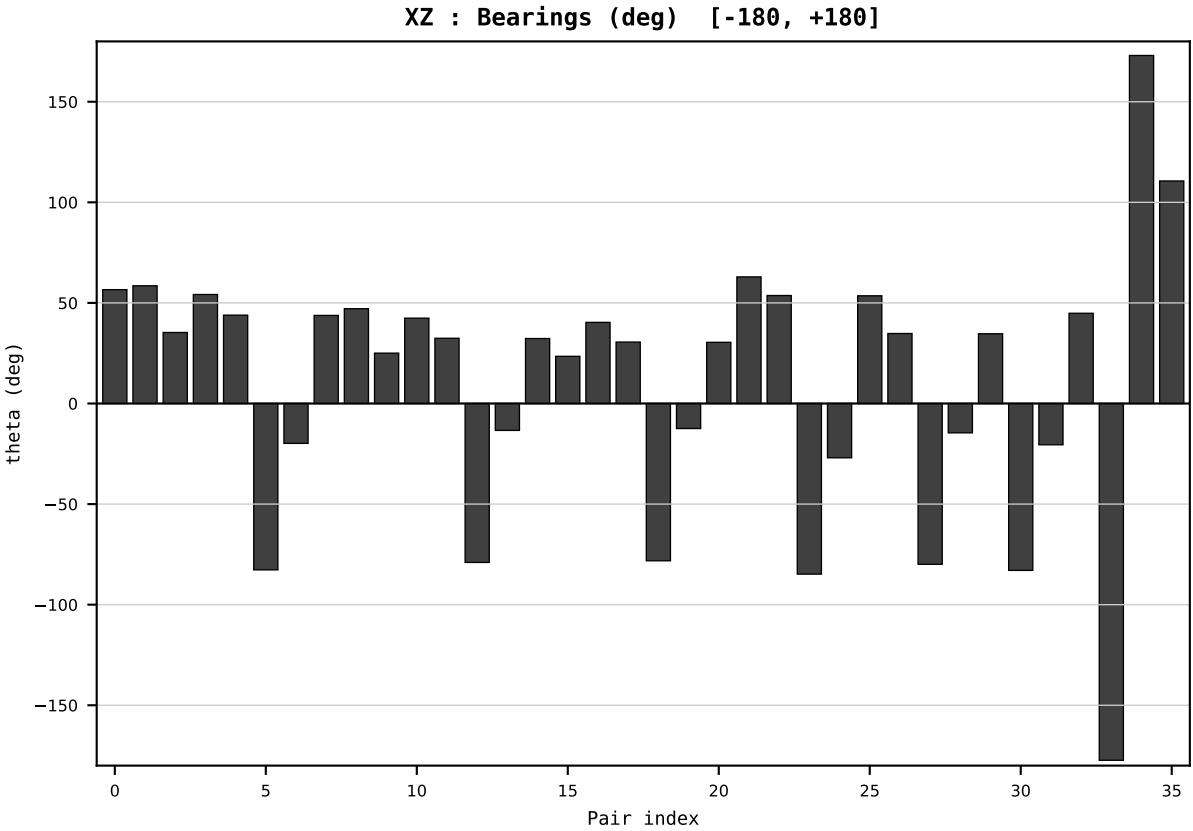
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.380664
Ring = Hs-3
E_metric= 902.7101
kappa_HS= 17565000000000000.00
omega = 1.3400 deg
d_A = 0.834110
Helm = Solar
Helm d = +0.655191
DR = 39.707 HLR
DR ratio= 175649999363704736.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

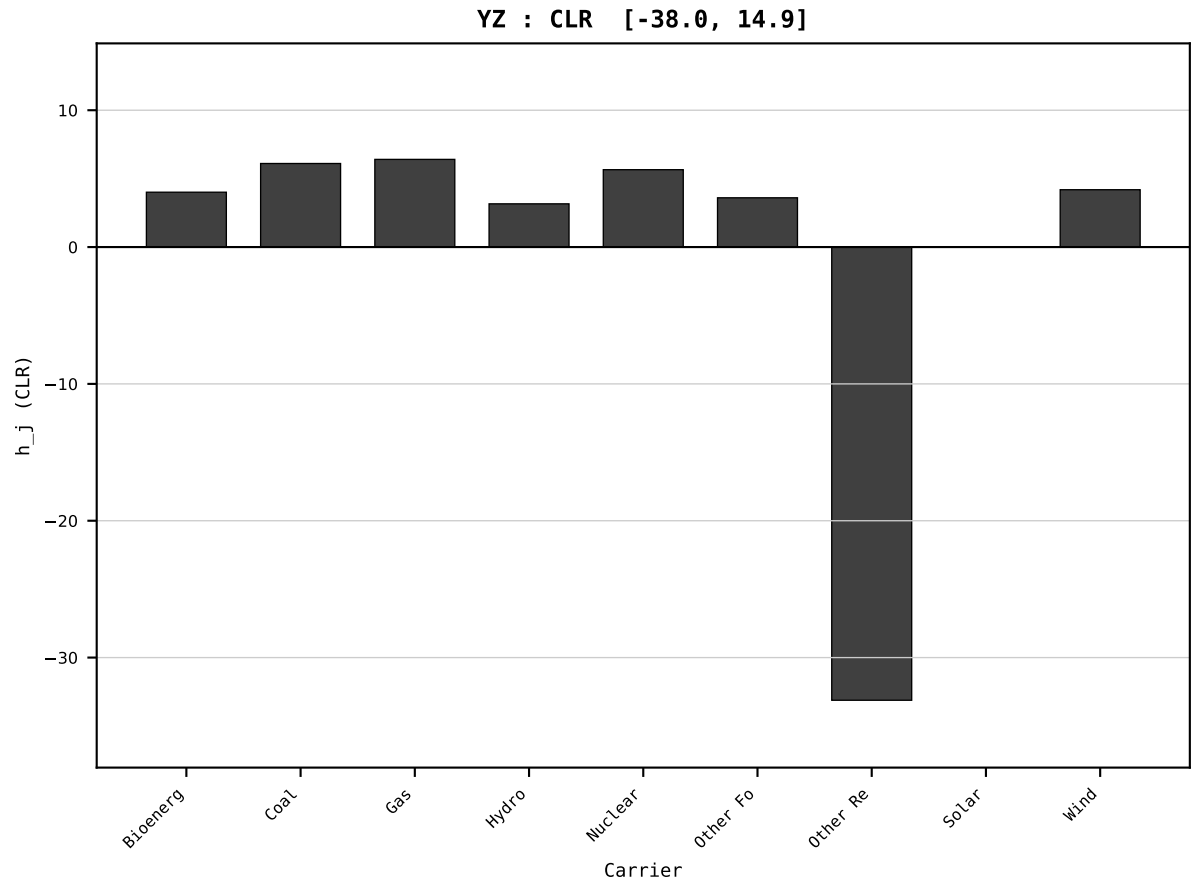
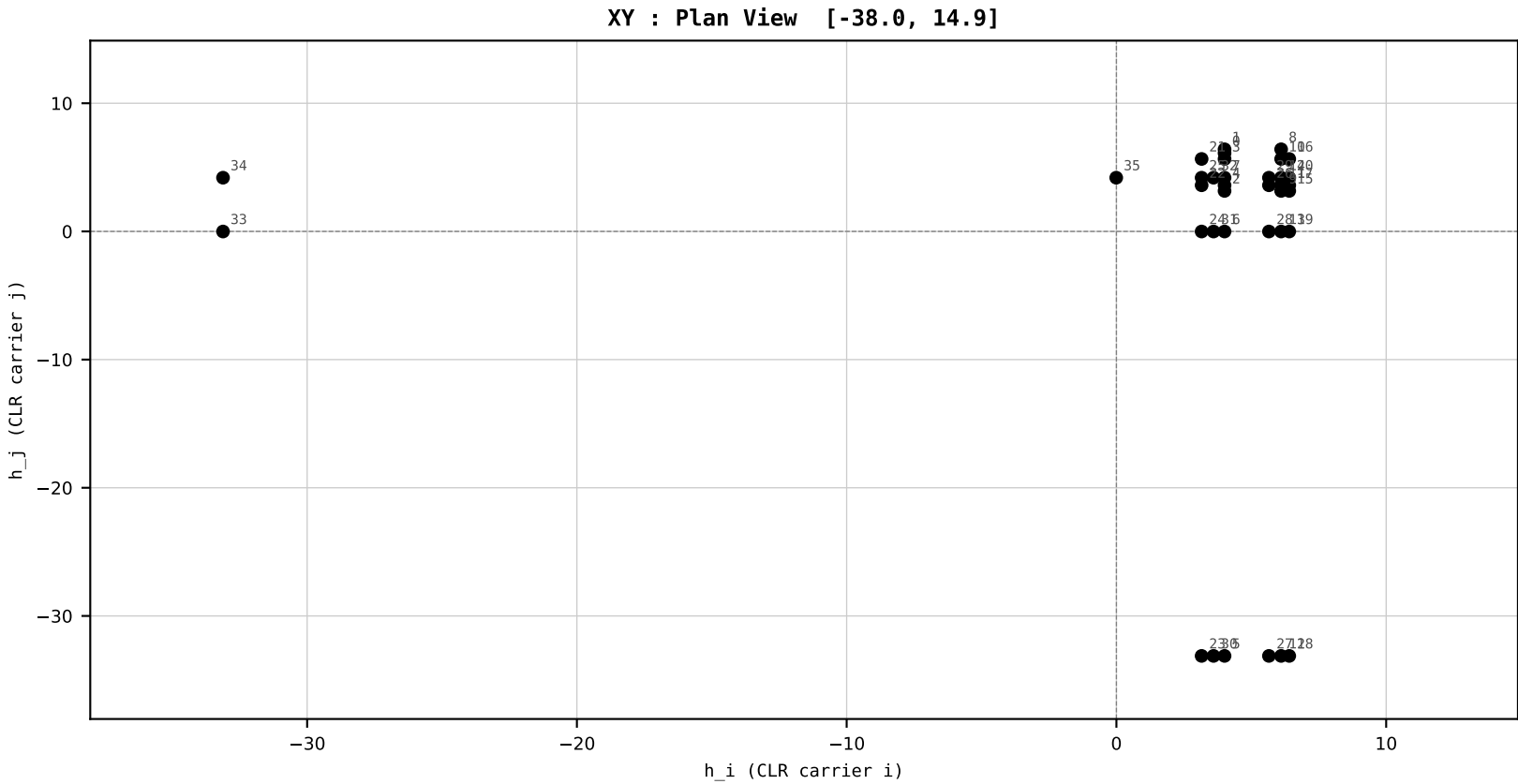
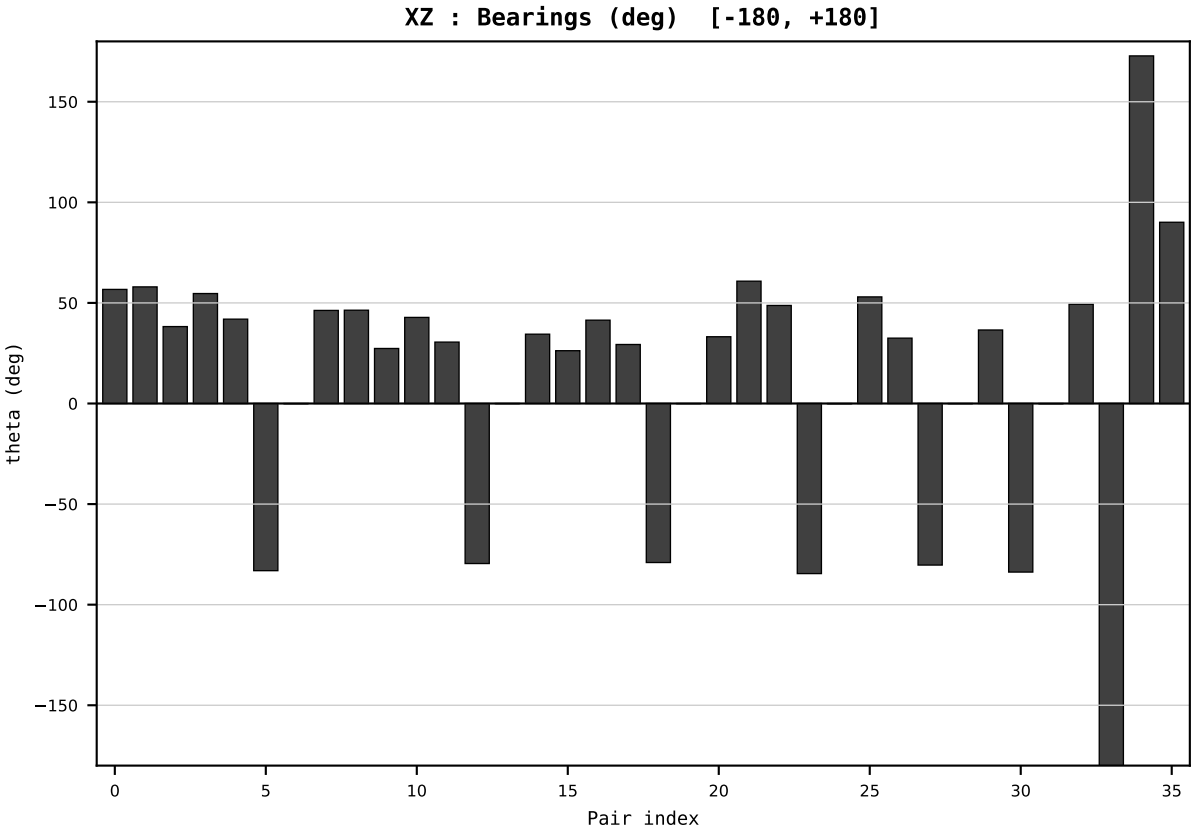
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.337503
Ring = Hs-3
E_metric= 903.4163
kappa_HS= 14649999999999968.00
omega = 2.7842 deg
d_A = 1.726842
Helm = Solar
Helm d = +1.510285
DR = 39.526 HLR
DR ratio= 146499999506615968.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

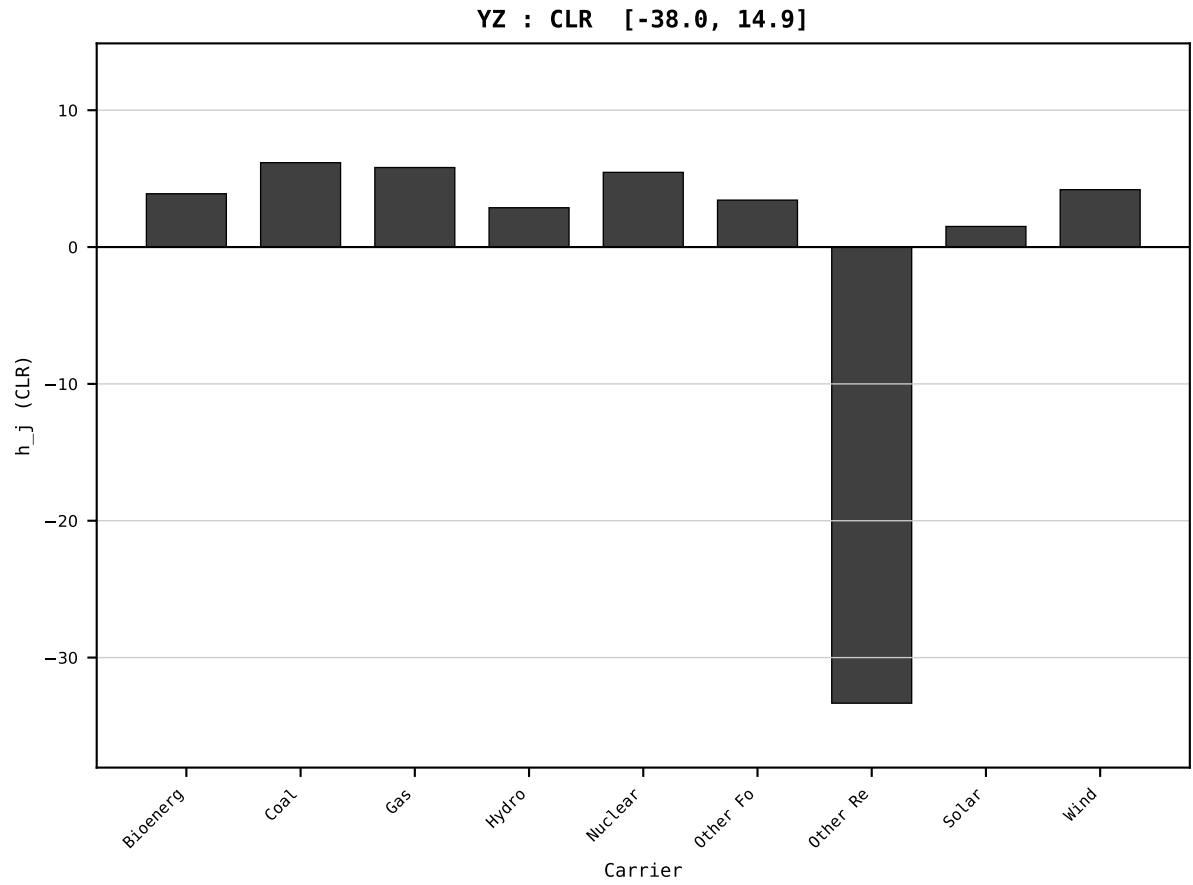
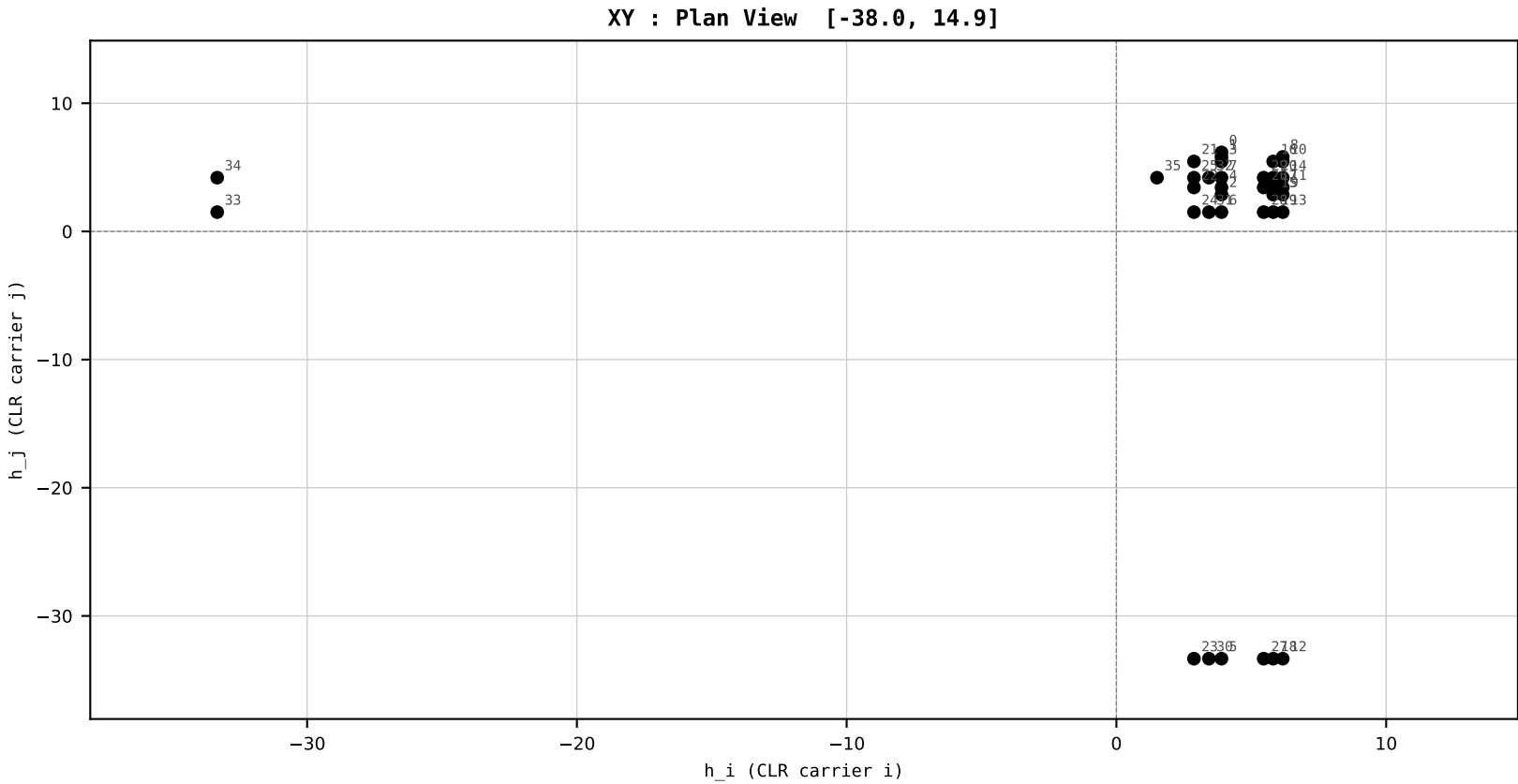
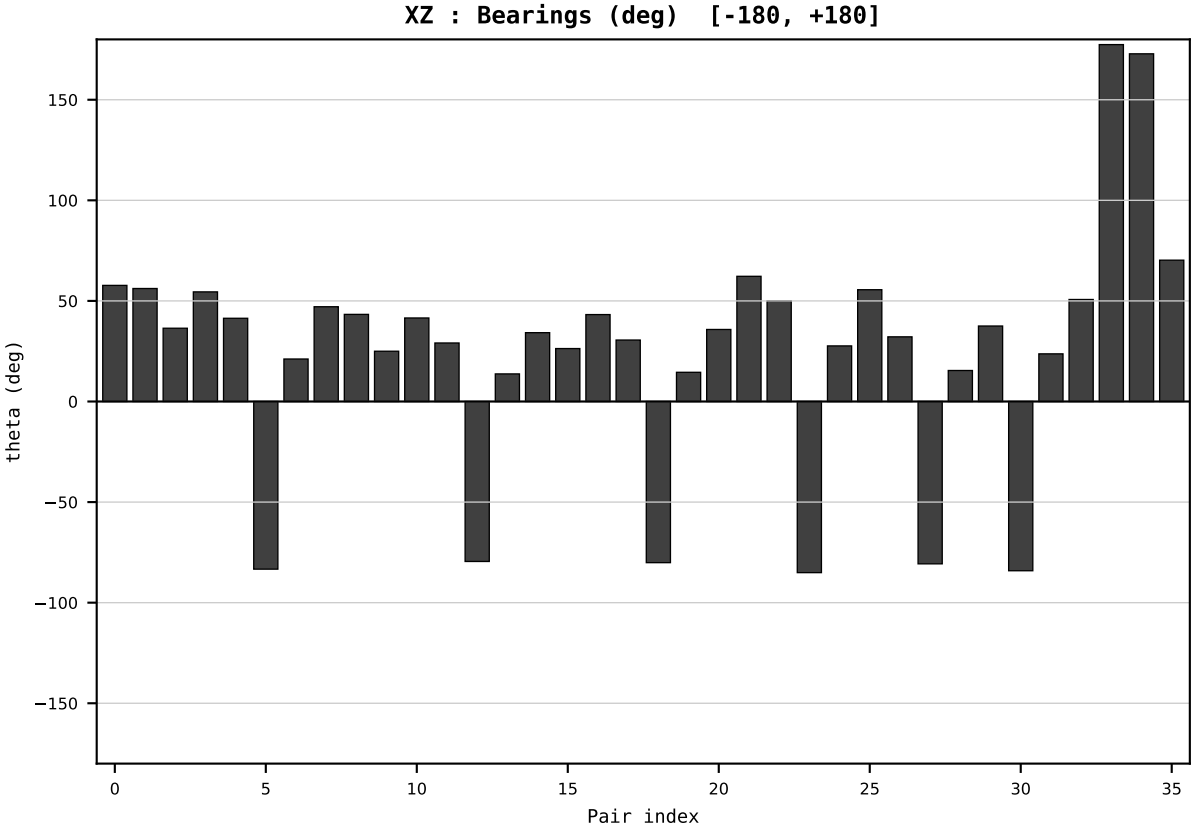
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.315326
Ring = Hs-3
E_metric= 905.5799
kappa_HS= 142789999999999984.00
omega = 2.7179 deg
d_A = 1.688485
Helm = Solar
Helm d = +1.511822
DR = 39.500 HLR
DR ratio= 142789999378451088.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

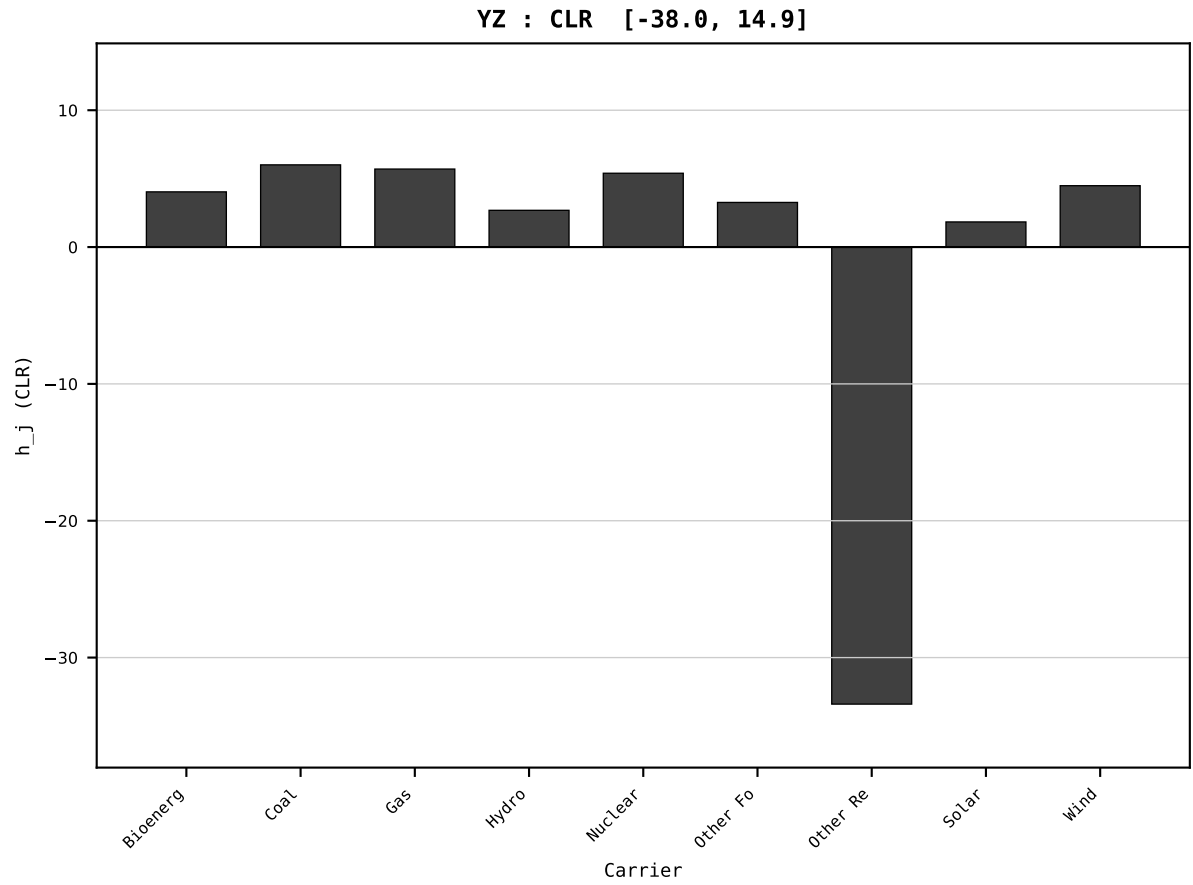
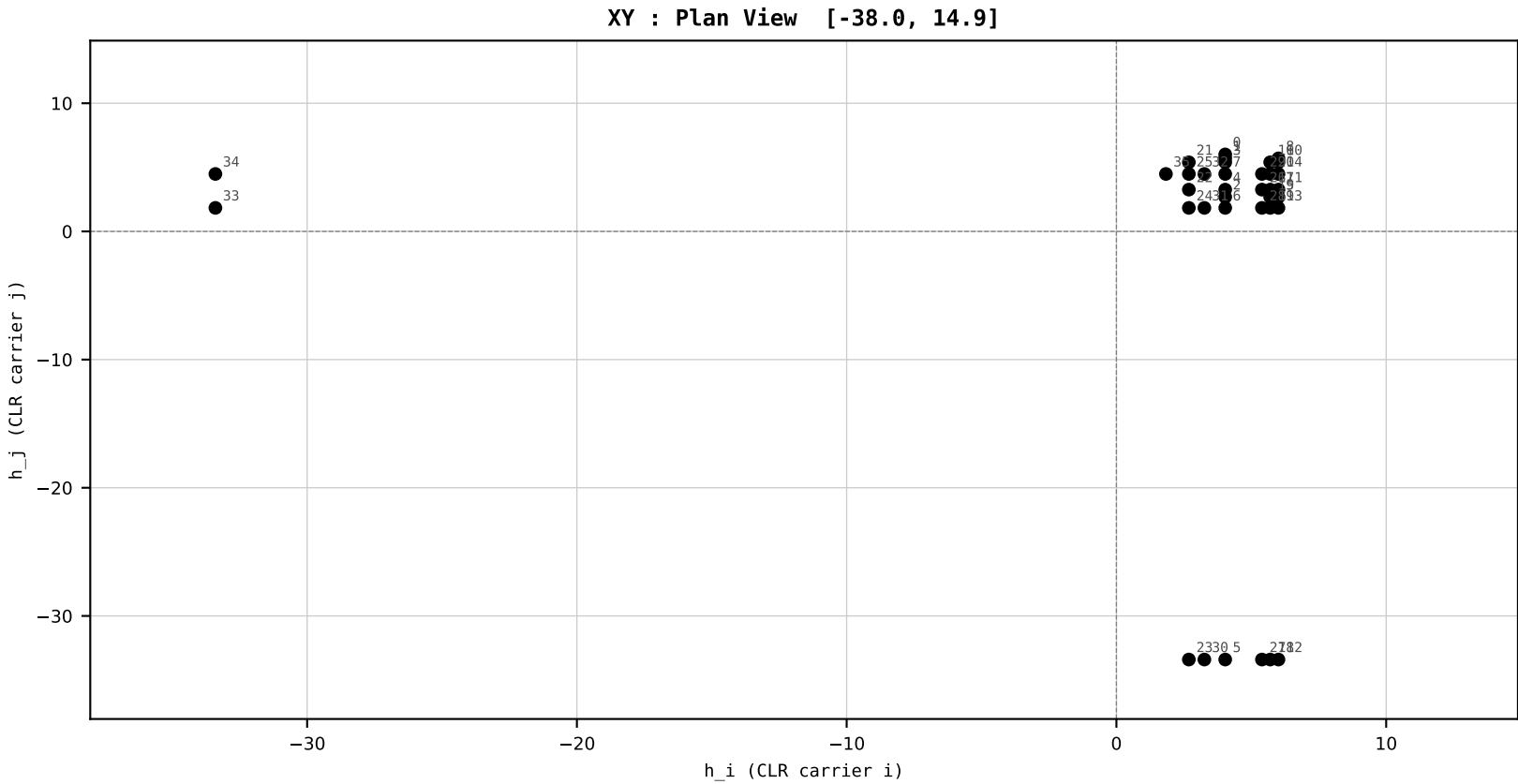
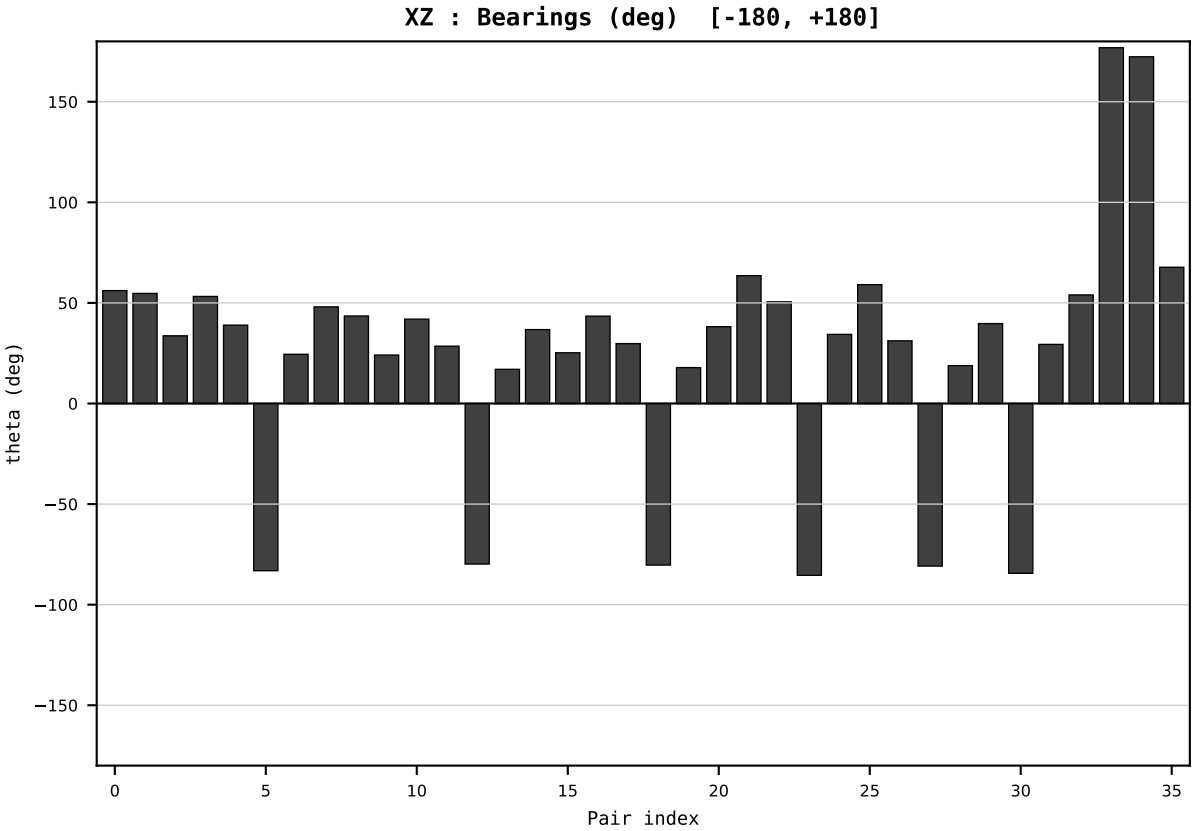
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287261
Ring = Hs-2
E_metric= 908.7712
kappa_HS= 130259999999999984.00
omega = 0.9132 deg
d_A = 0.569543
Helm = Solar
Helm d = +0.330853
DR = 39.408 HLR
DR ratio= 130260000256749840.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

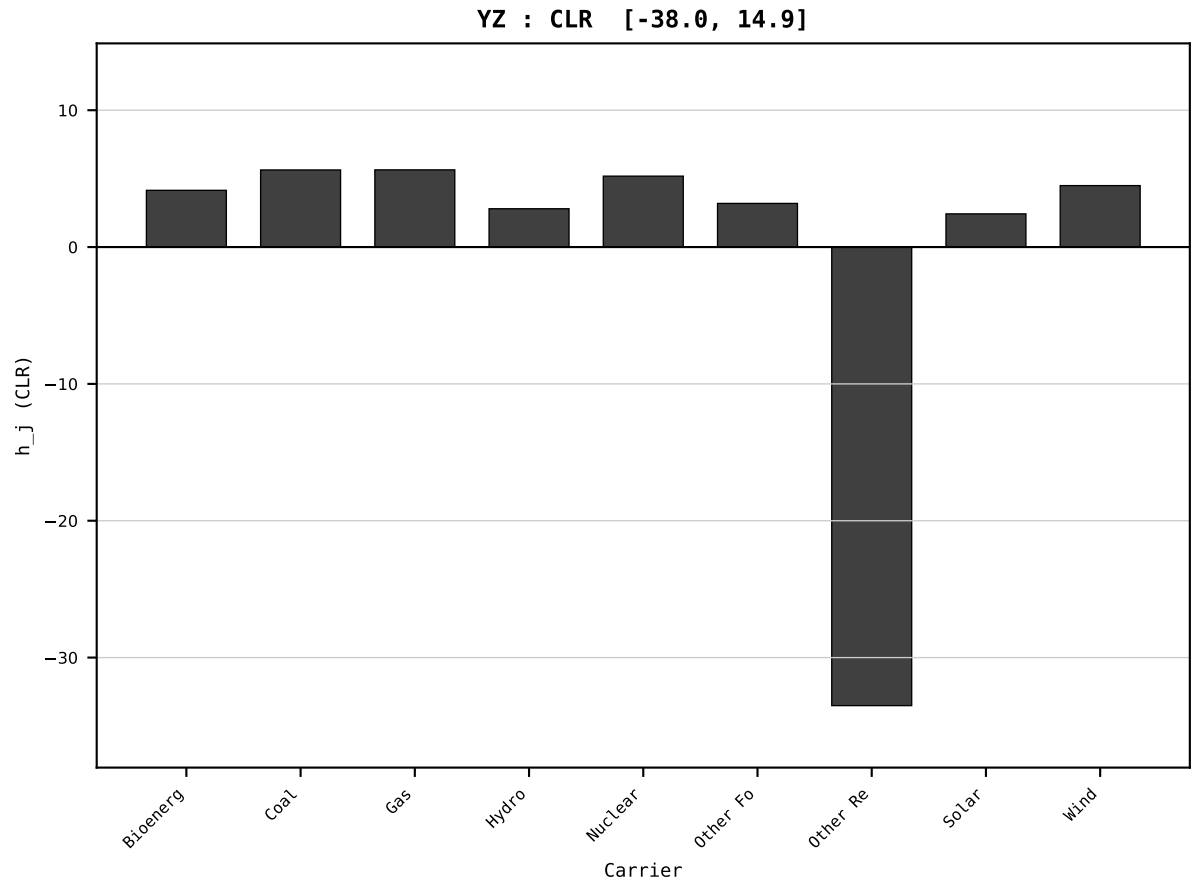
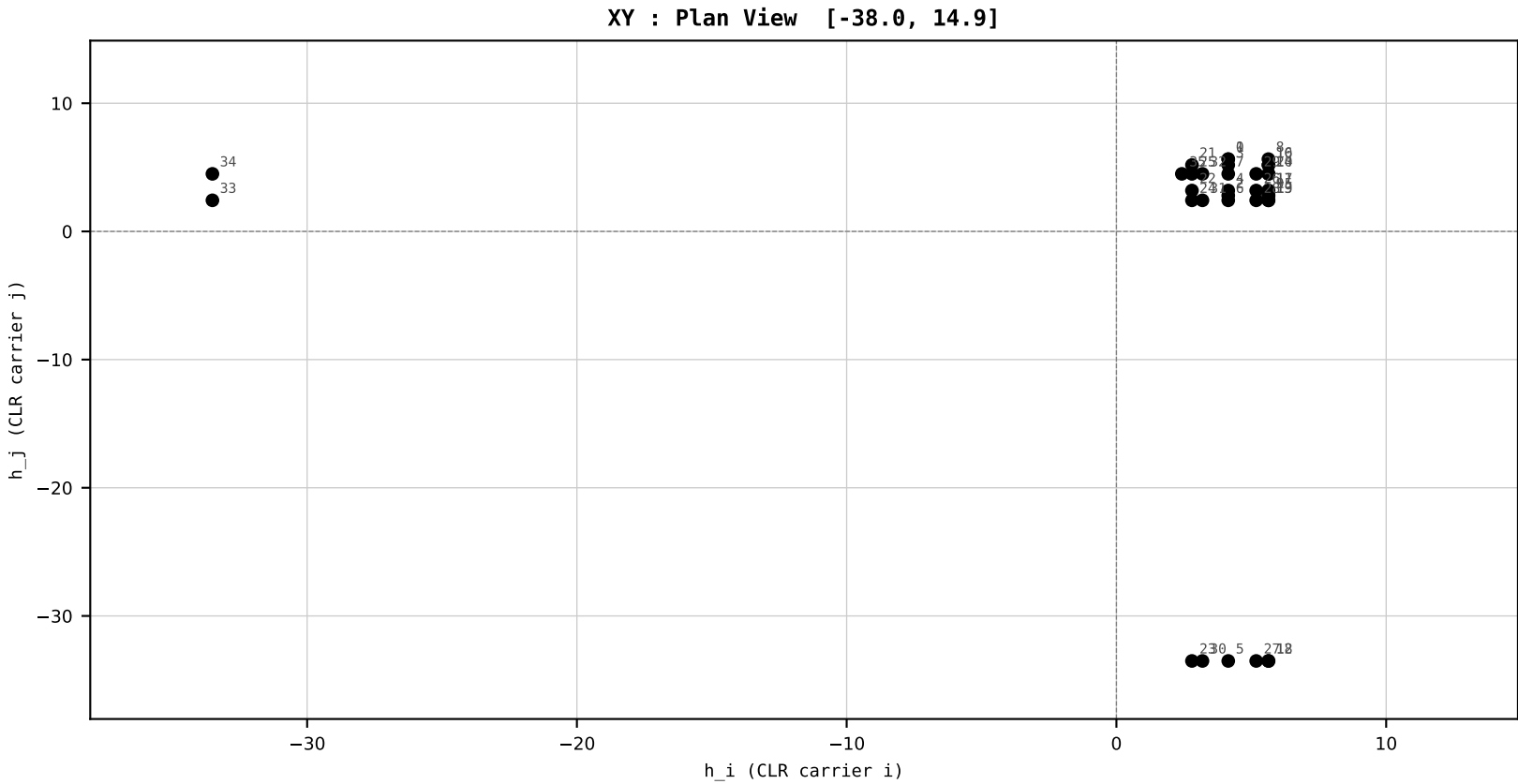
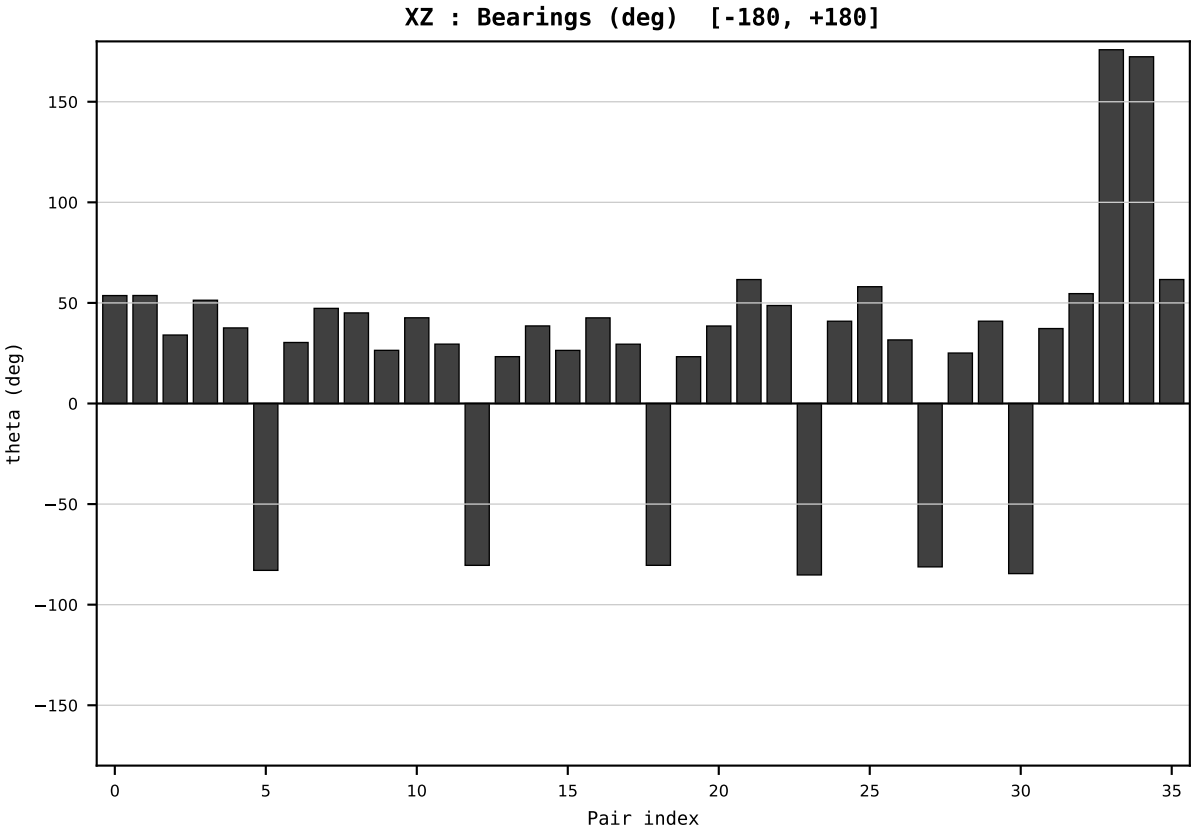
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.245653
Ring = Hs-2
E_metric= 914.5648
kappa_HS= 100889999999999984.00
omega = 1.2190 deg
d_A = 0.760916
Helm = Solar
Helm d = +0.590262
DR = 39.153 HLR
DR ratio= 1008900000067498000.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

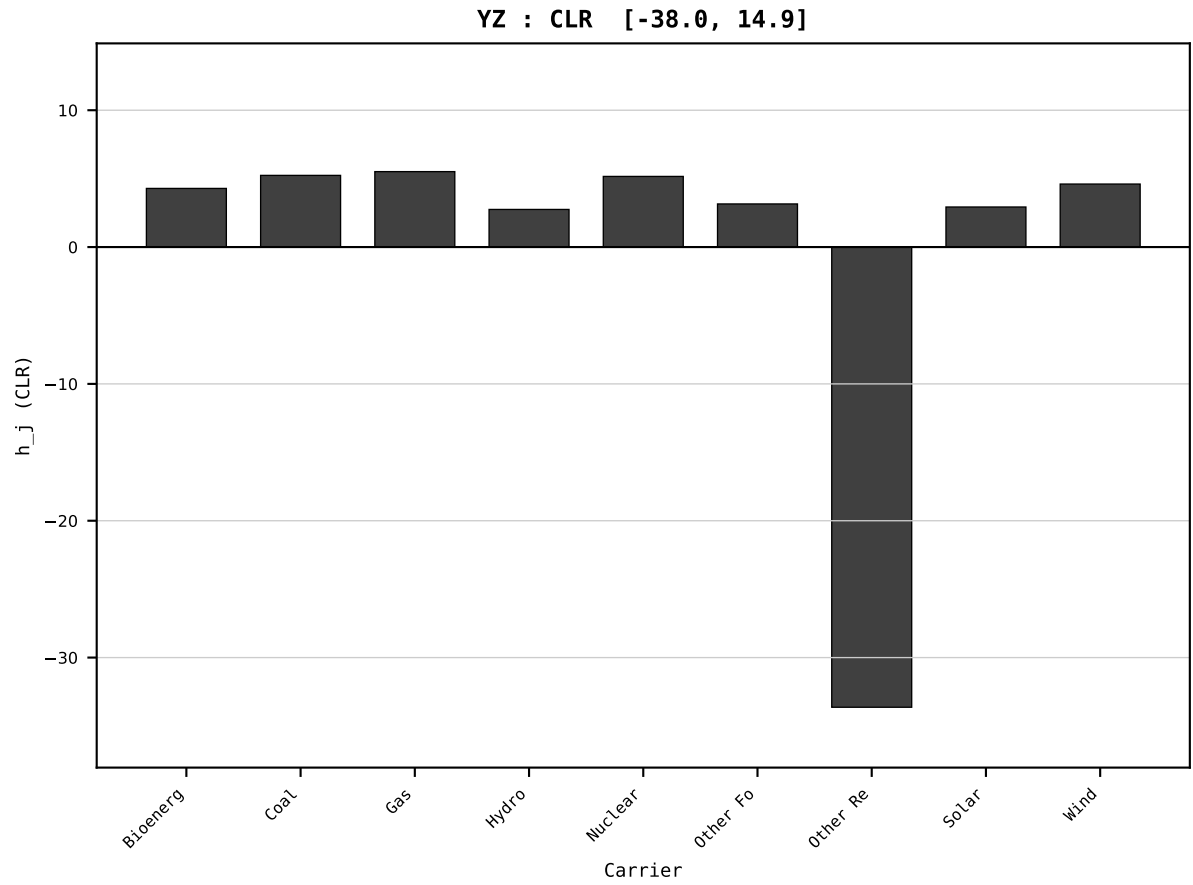
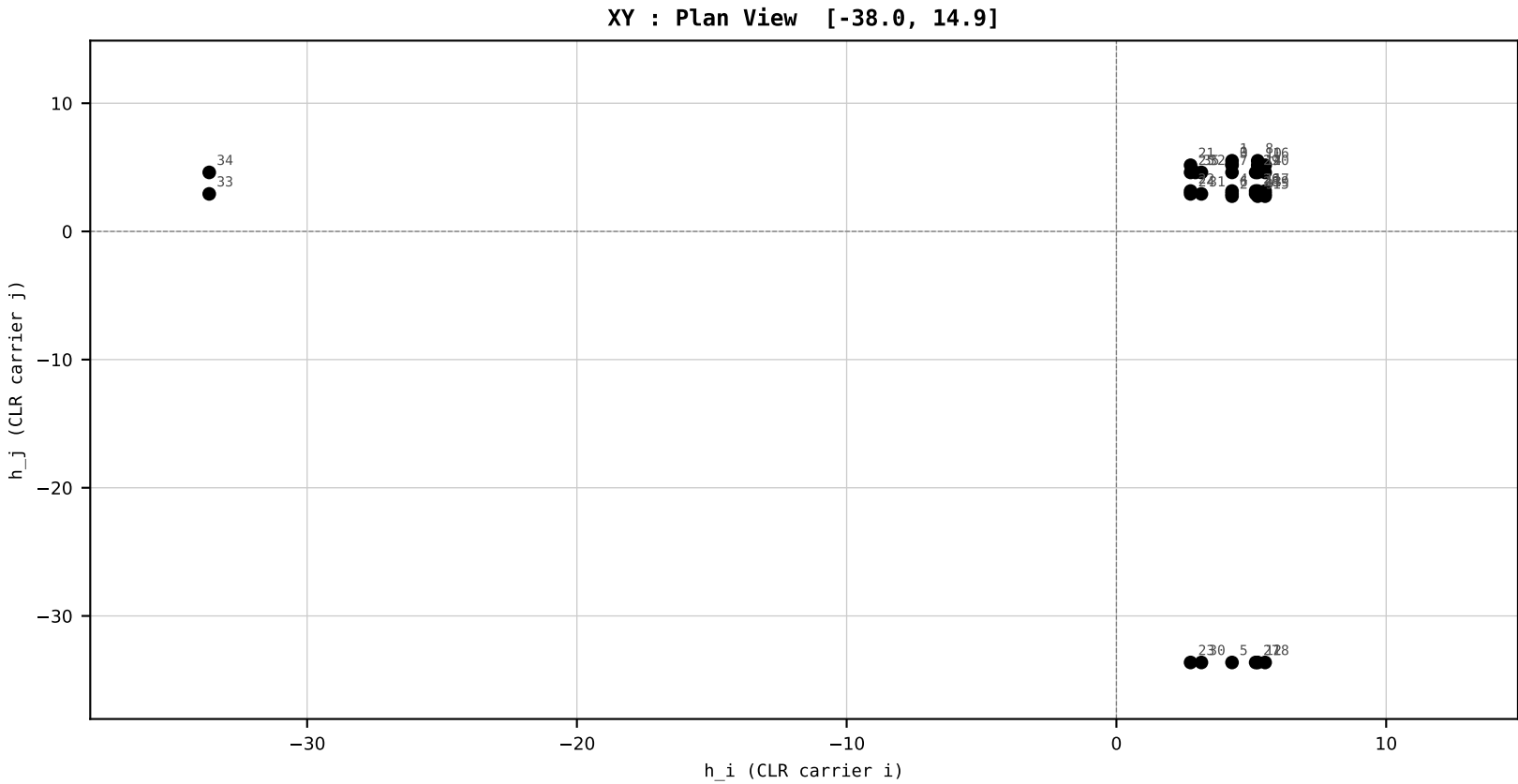
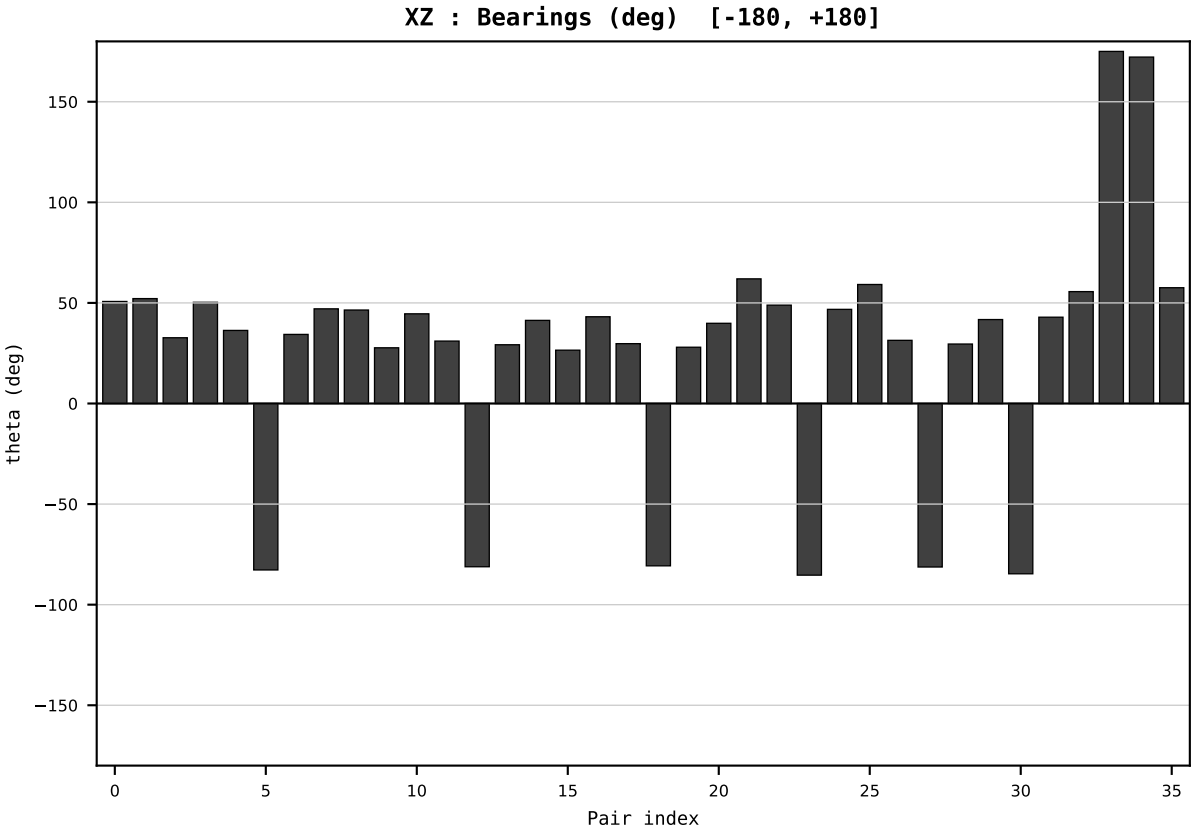
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.206119
Ring = Hs-2
E_metric= 919.4045
kappa_HS= 9987999999999984.00
omega = 1.0971 deg
d_A = 0.690200
Helm = Solar
Helm d = +0.501757
DR = 39.143 HLR
DR ratio= 99879999967813408.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

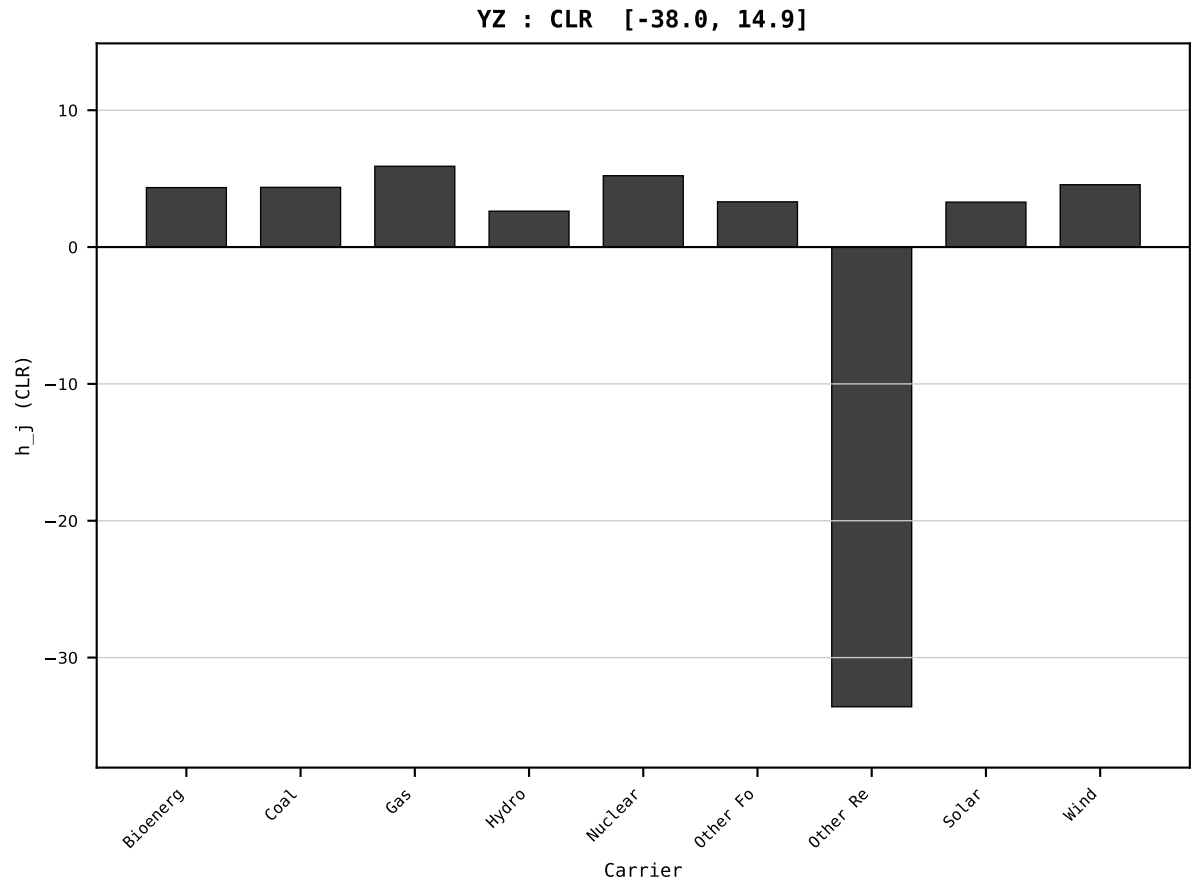
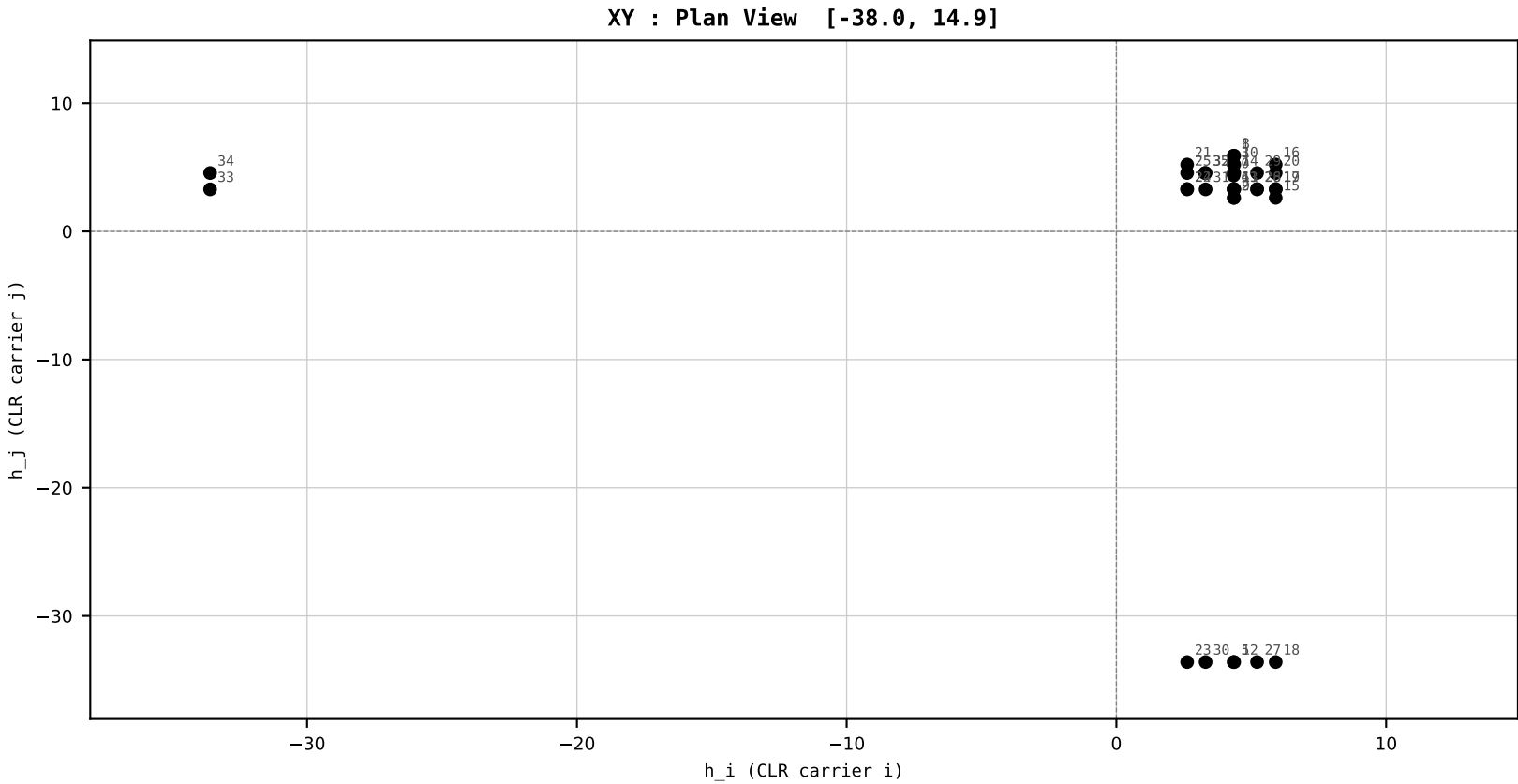
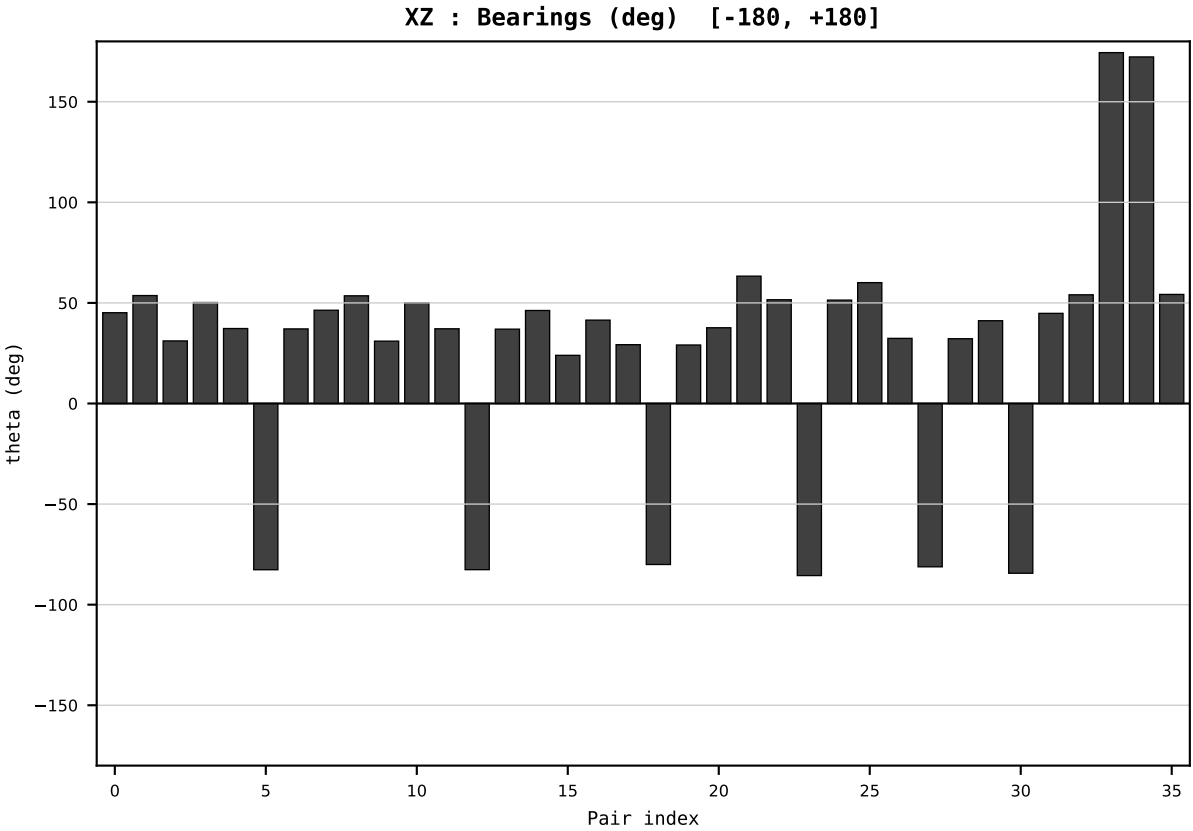
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.249877
Ring = Hs-2
E_metric= 916.7901
kappa_HS= 143129999999999984.00
omega = 1.6738 deg
d_A = 1.045760
Helm = Coal
Helm d = -0.873359
DR = 39.503 HLR
DR ratio= 143129999555321712.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

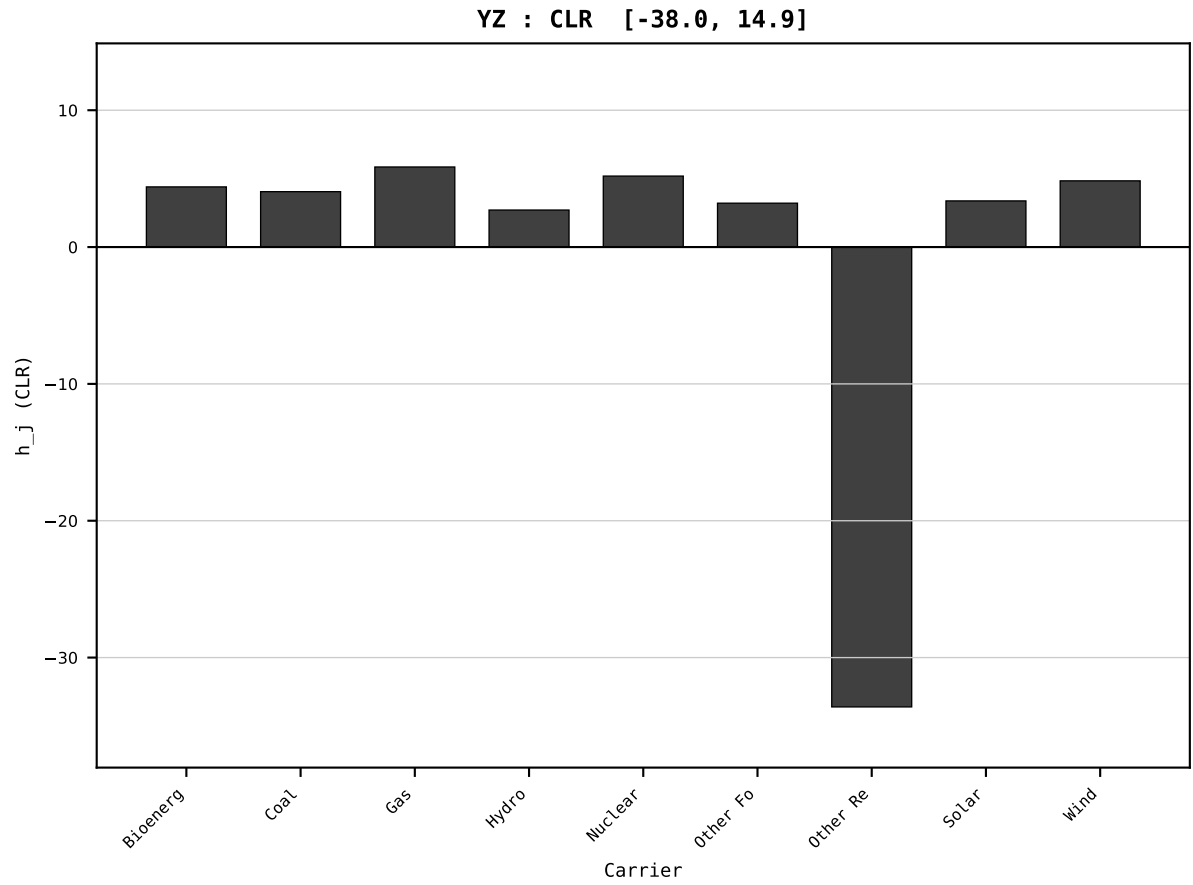
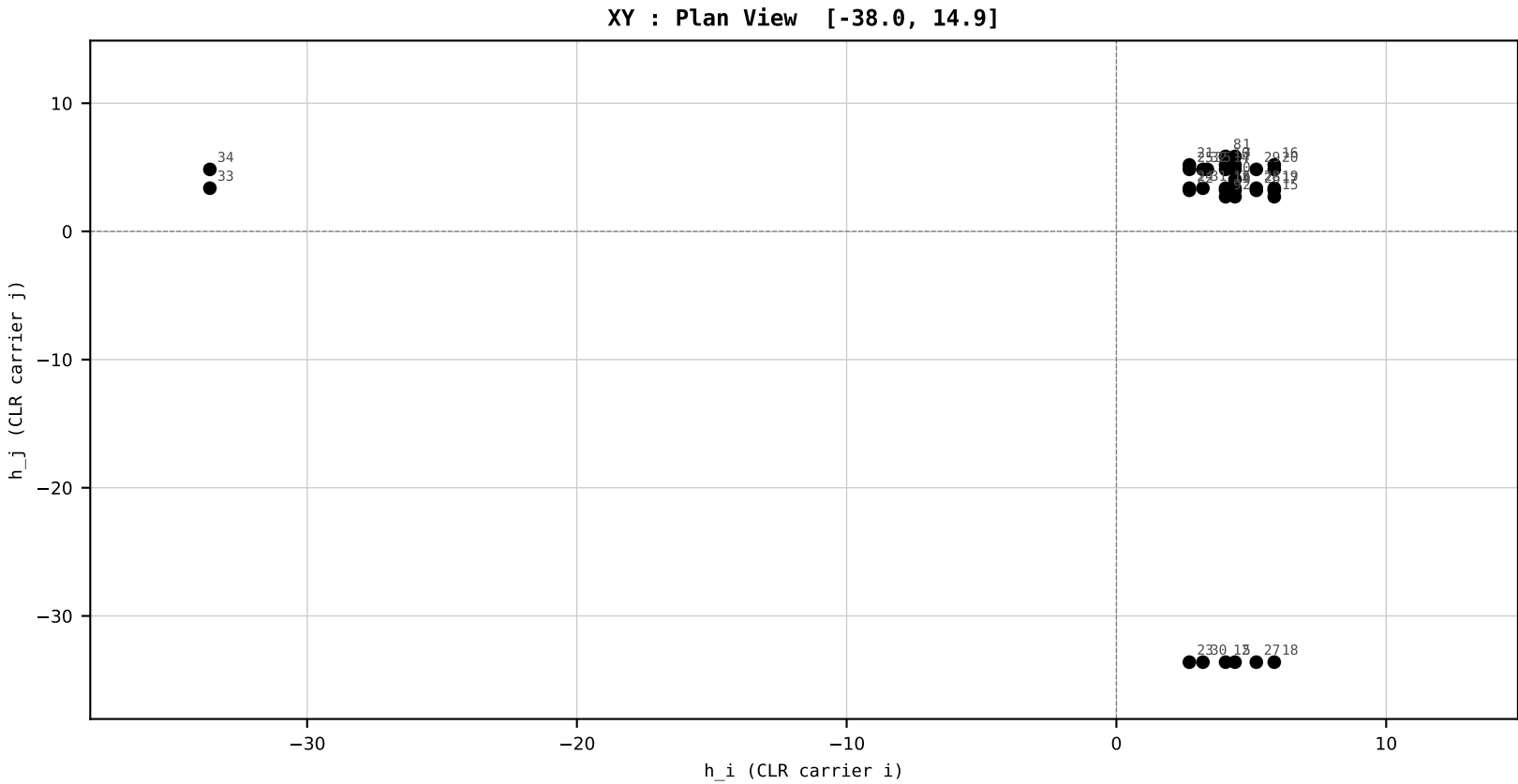
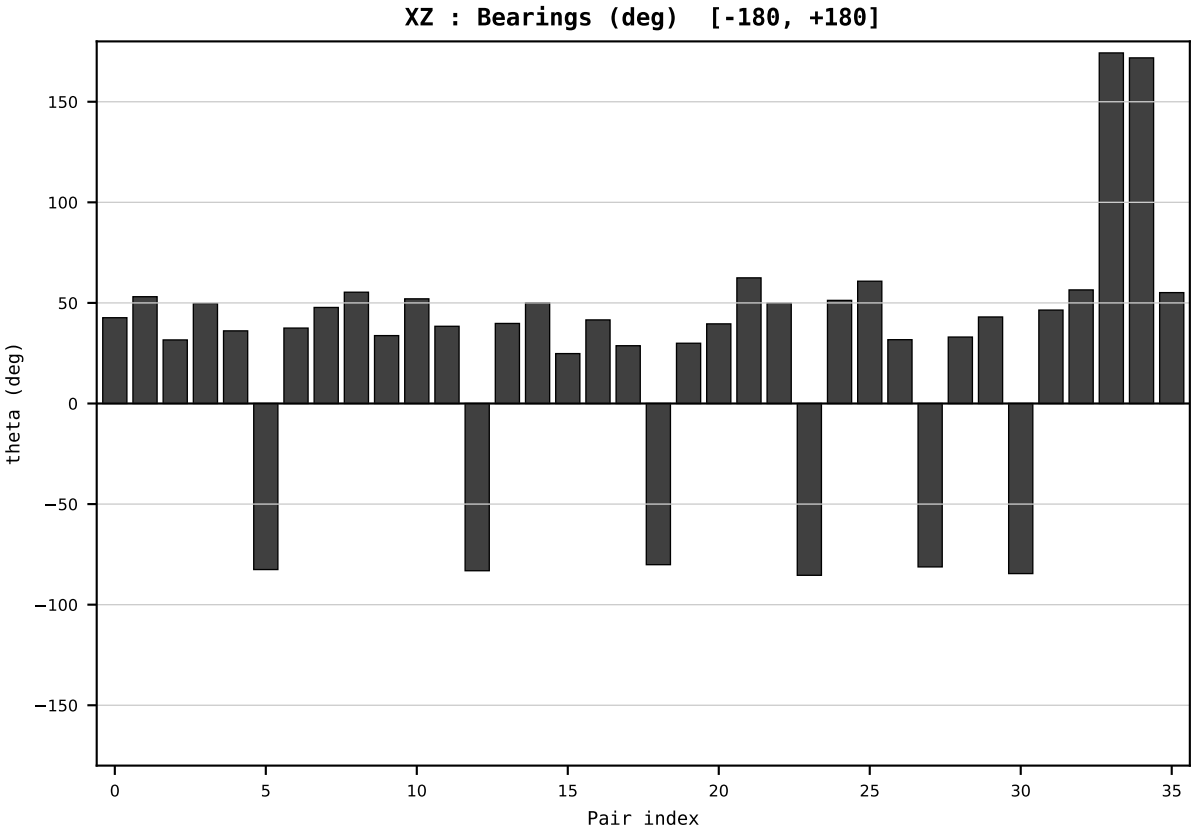
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.242414
Ring = Hs-2
E_metric= 917.2914
kappa_HS= 13674999999999984.00
omega = 0.7349 deg
d_A = 0.458615
Helm = Coal
Helm d = -0.316282
DR = 39.457 HLR
DR ratio= 136749999150403536.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

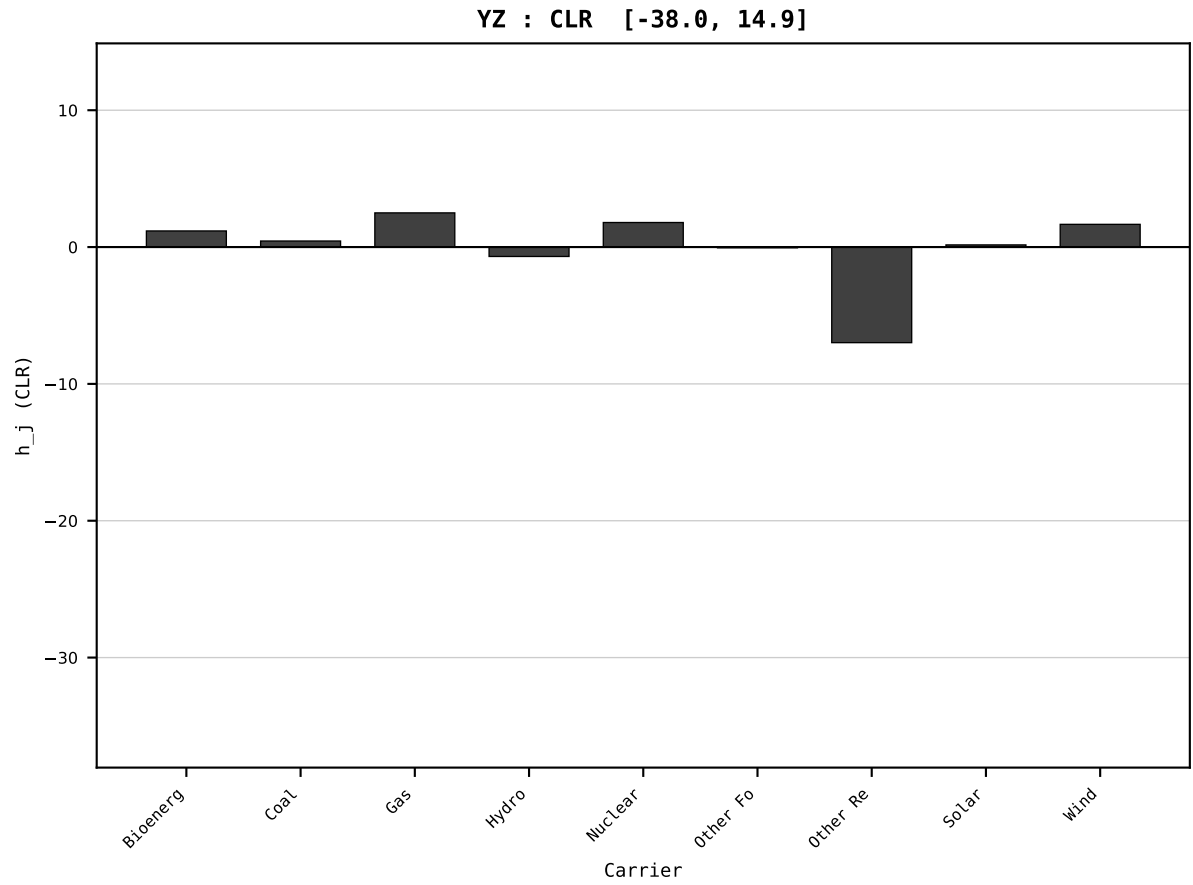
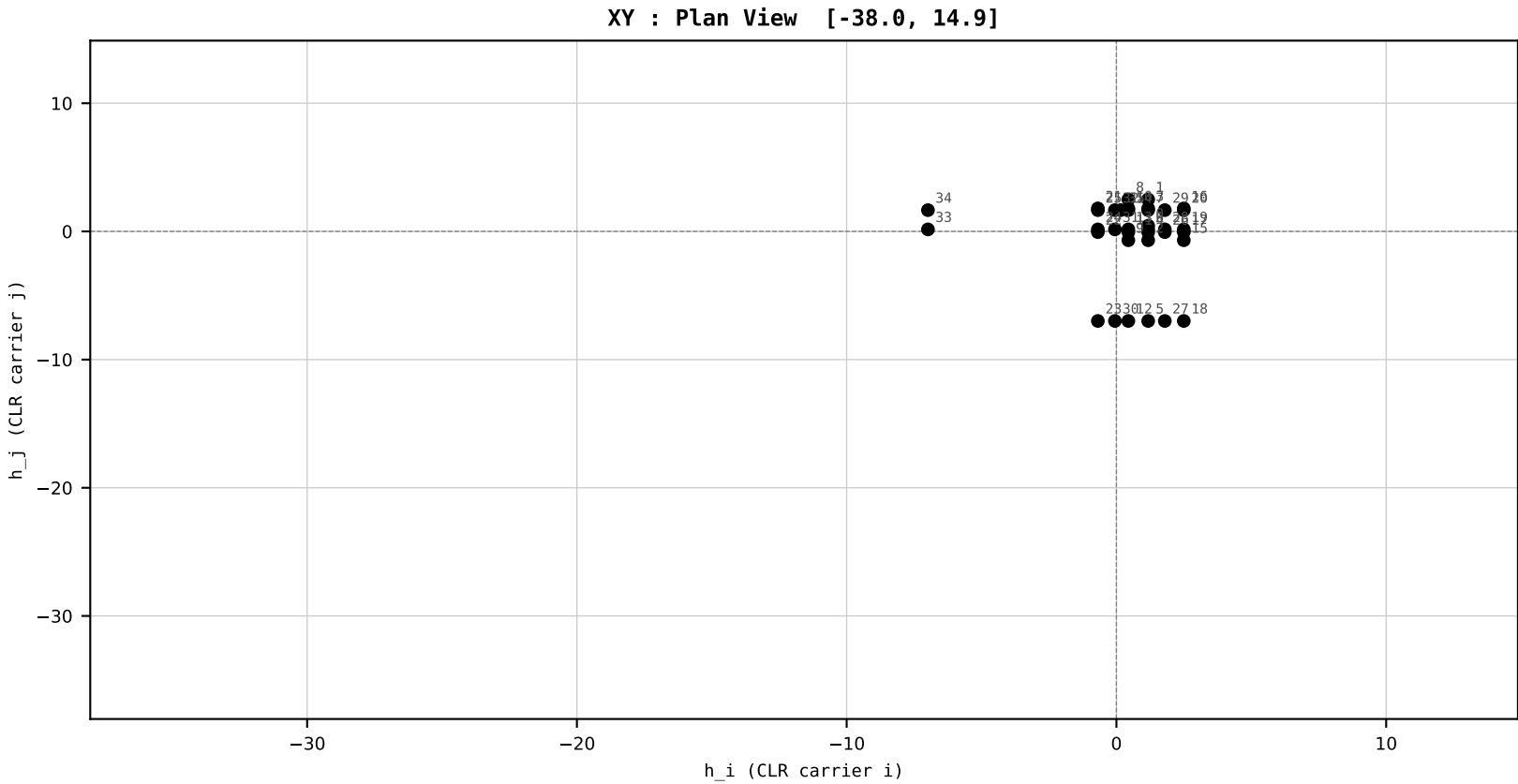
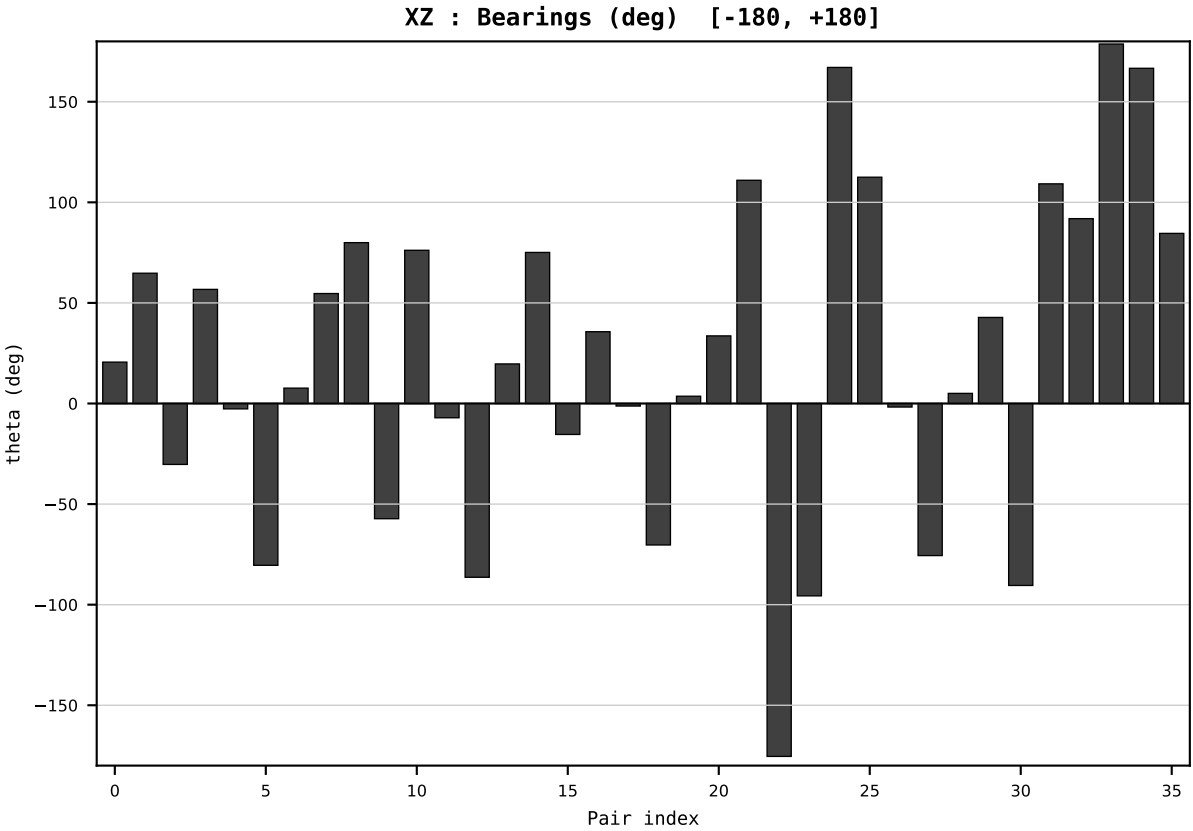
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.238427
Ring = Hs-2
E_metric= 63.1052
kappa_HS= 13149.00
omega = 16.6103 deg
d_A = 28.236007
Helm = Other Renewables
Helm d = +26.618855
DR = 9.484 HLR
DR ratio= 13149.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

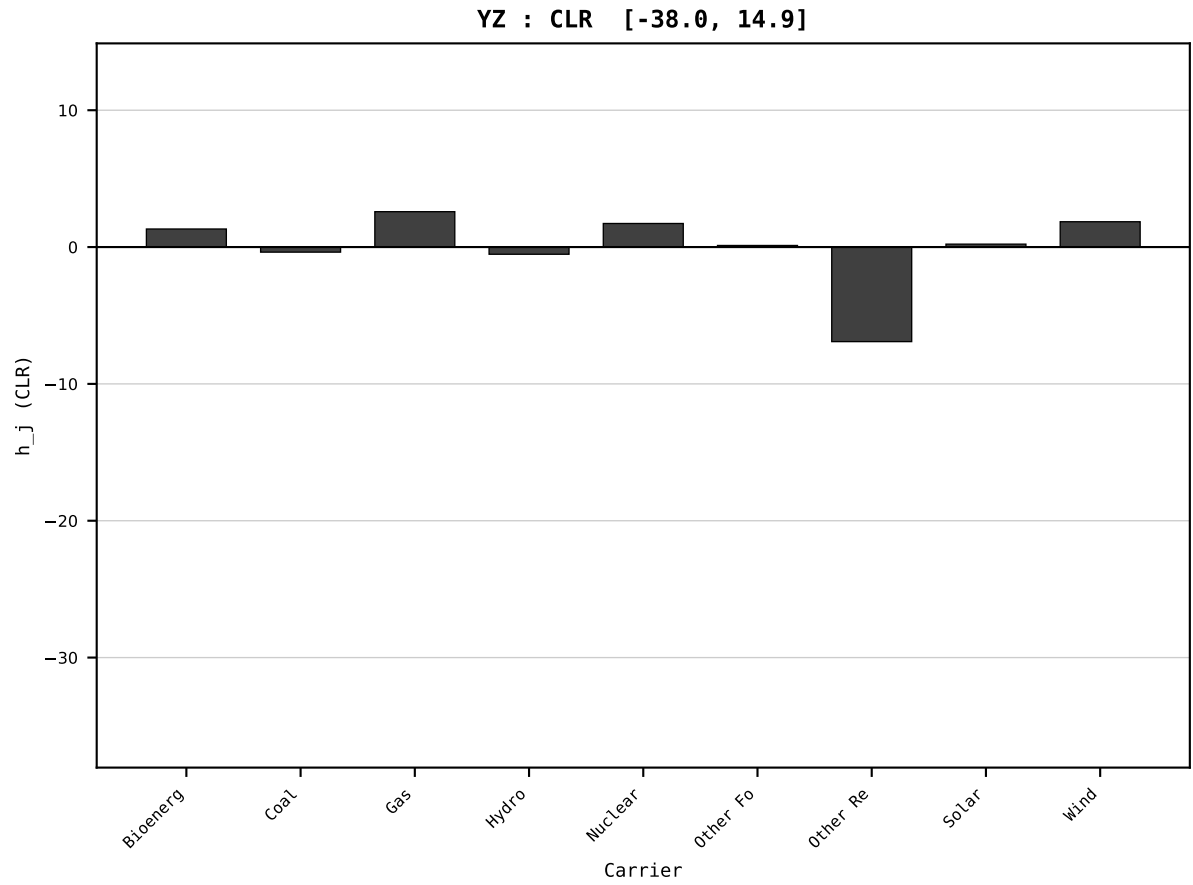
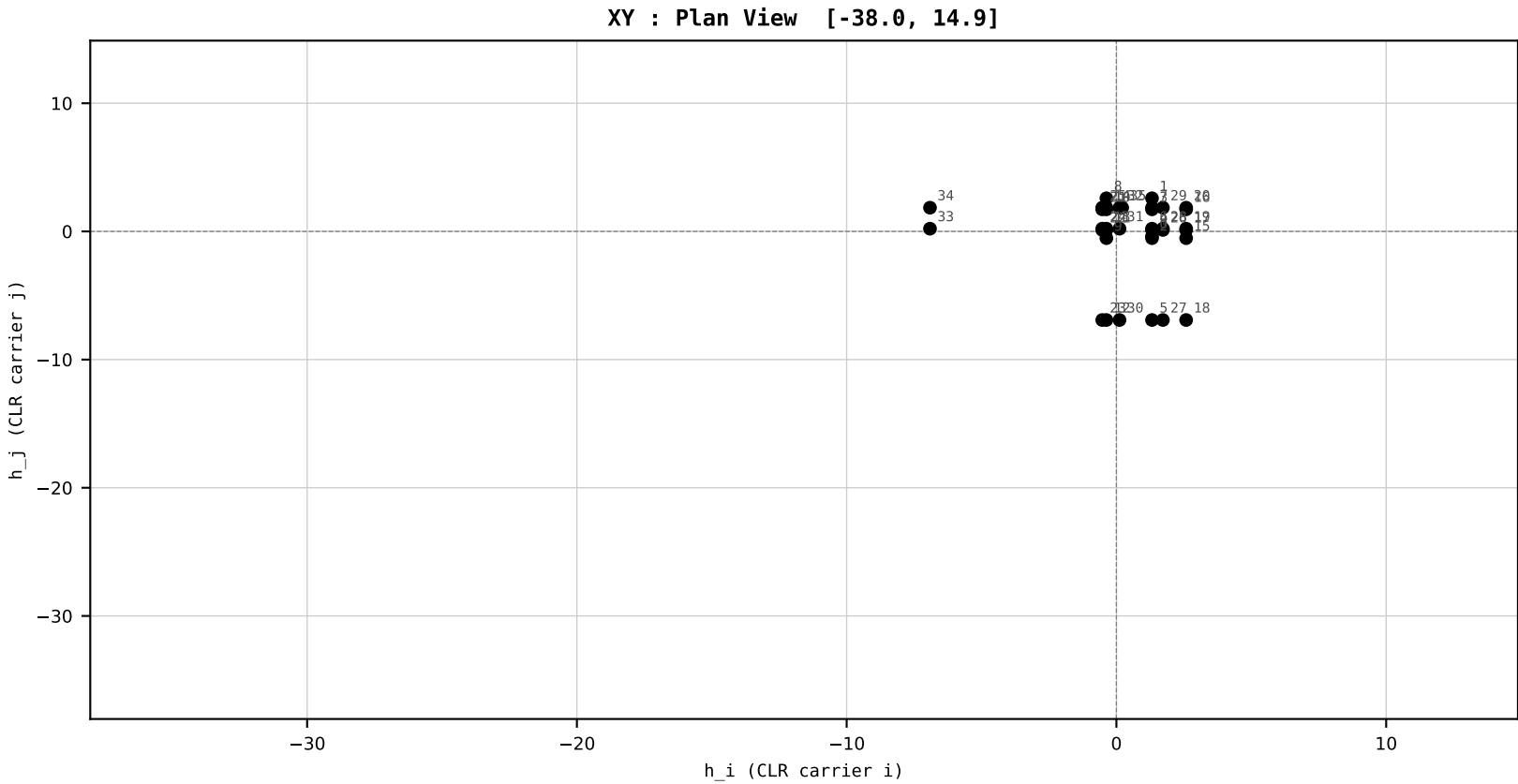
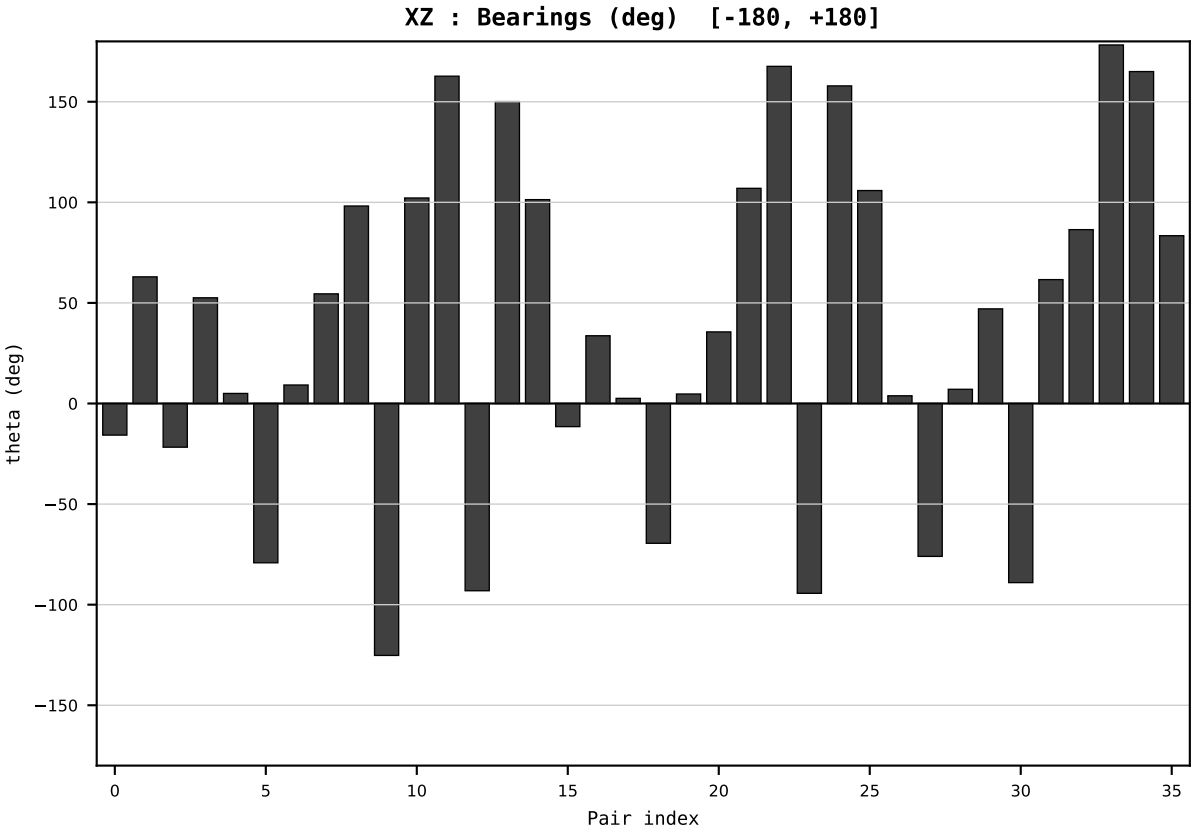
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.257979
Ring = Hs-2
E_metric= 63.0521
kappa_HS= 13309.00
omega = 6.4343 deg
d_A = 0.891439
Helm = Coal
Helm d = -0.813158
DR = 9.496 HLR
DR ratio= 13309.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

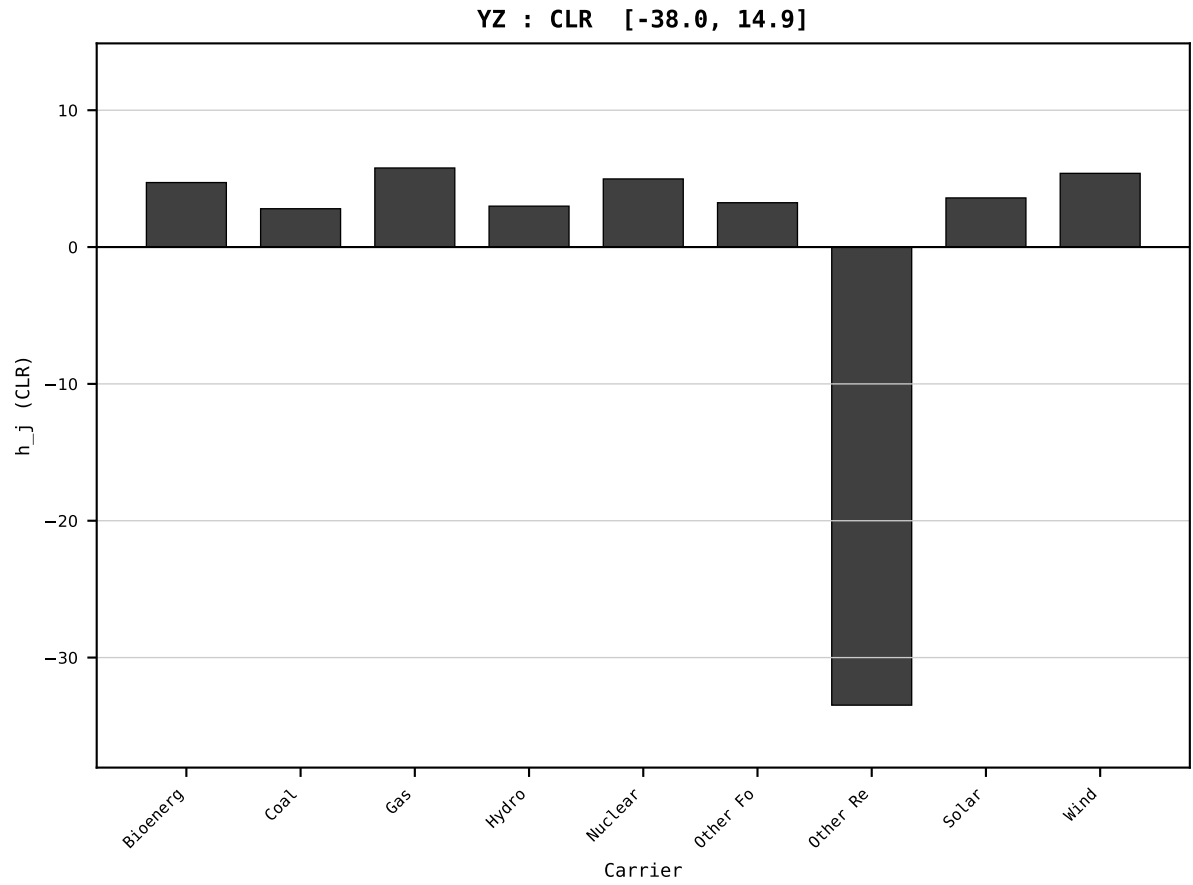
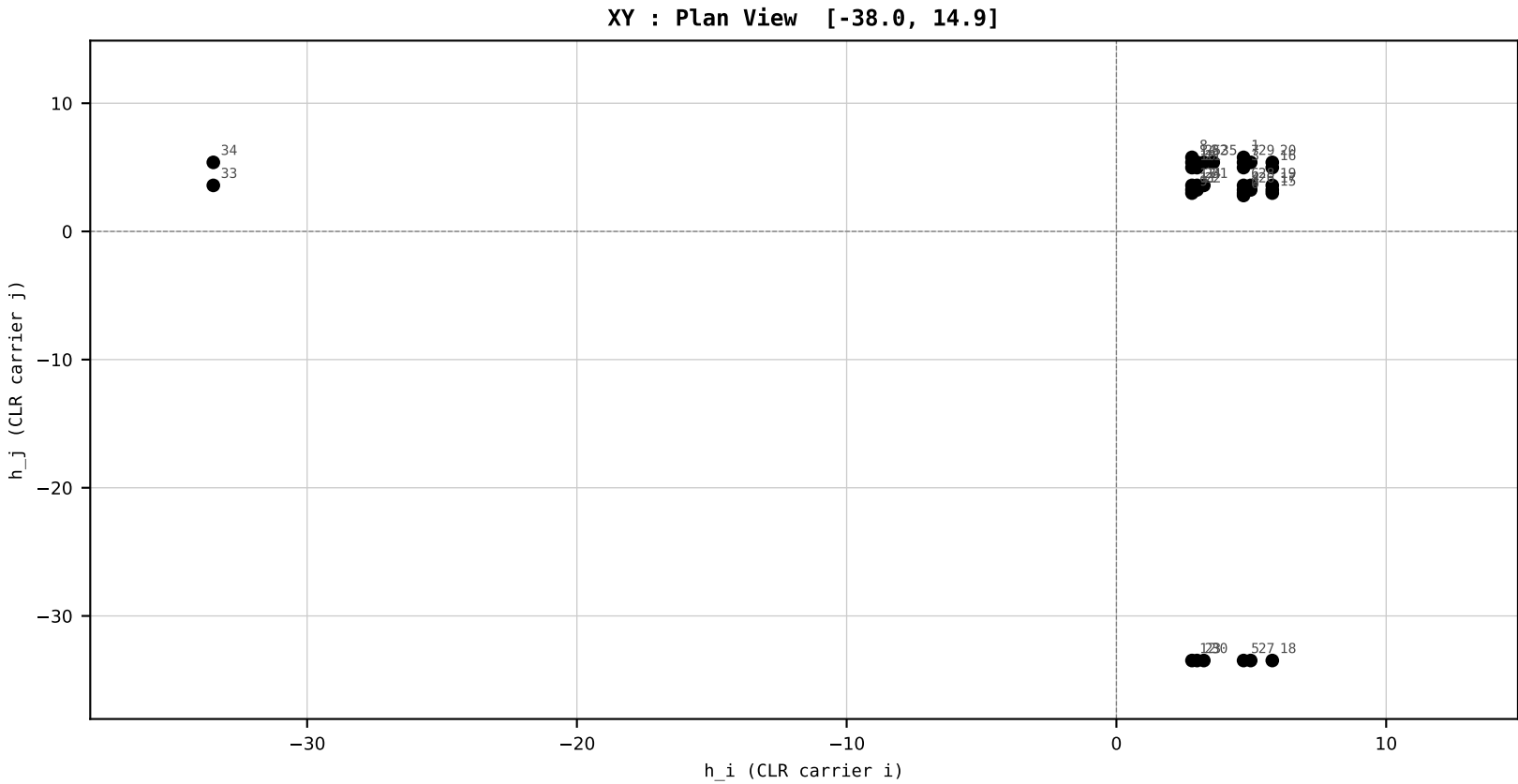
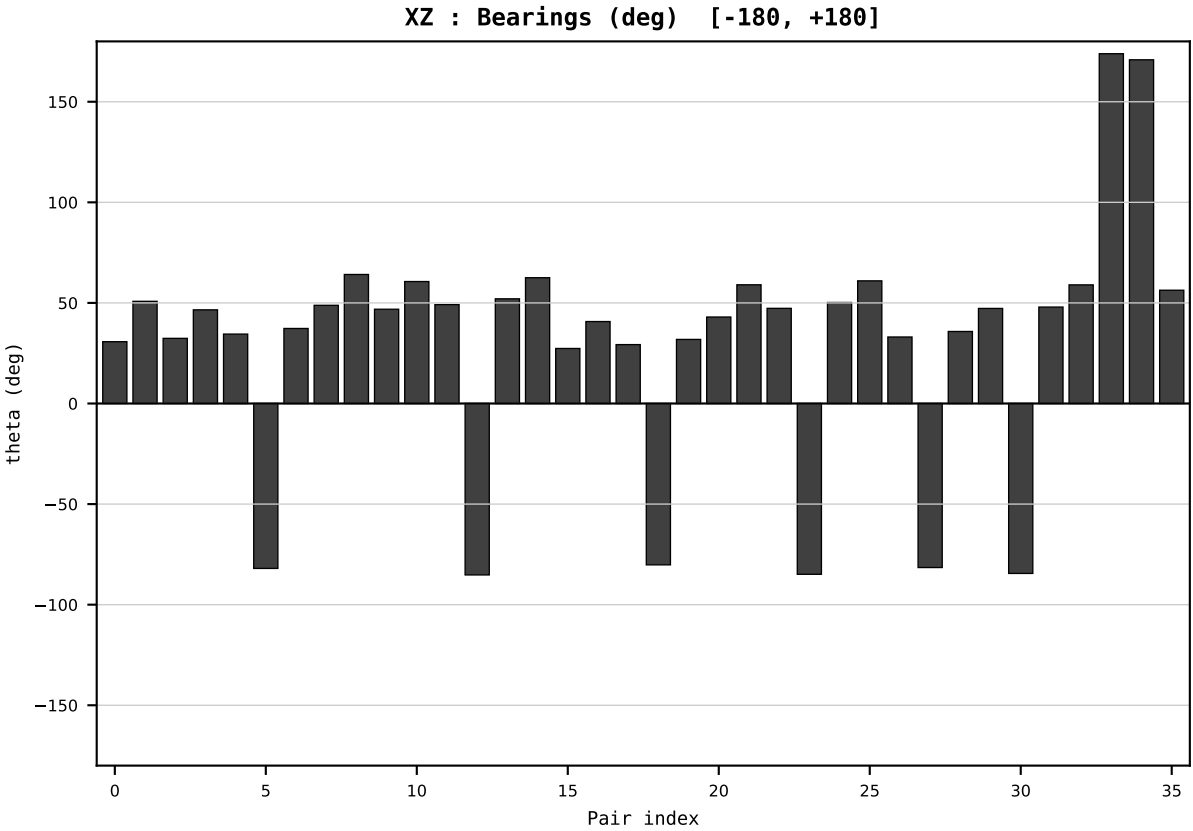
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.246847
Ring = Hs-2
E_metric= 915.4463
kappa_HS= 11188999999999984.00
omega = 17.7152 deg
d_A = 28.181681
Helm = Other Renewables
Helm d = -26.567038
DR = 39.256 HLR
DR ratio= 111889999918284832.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

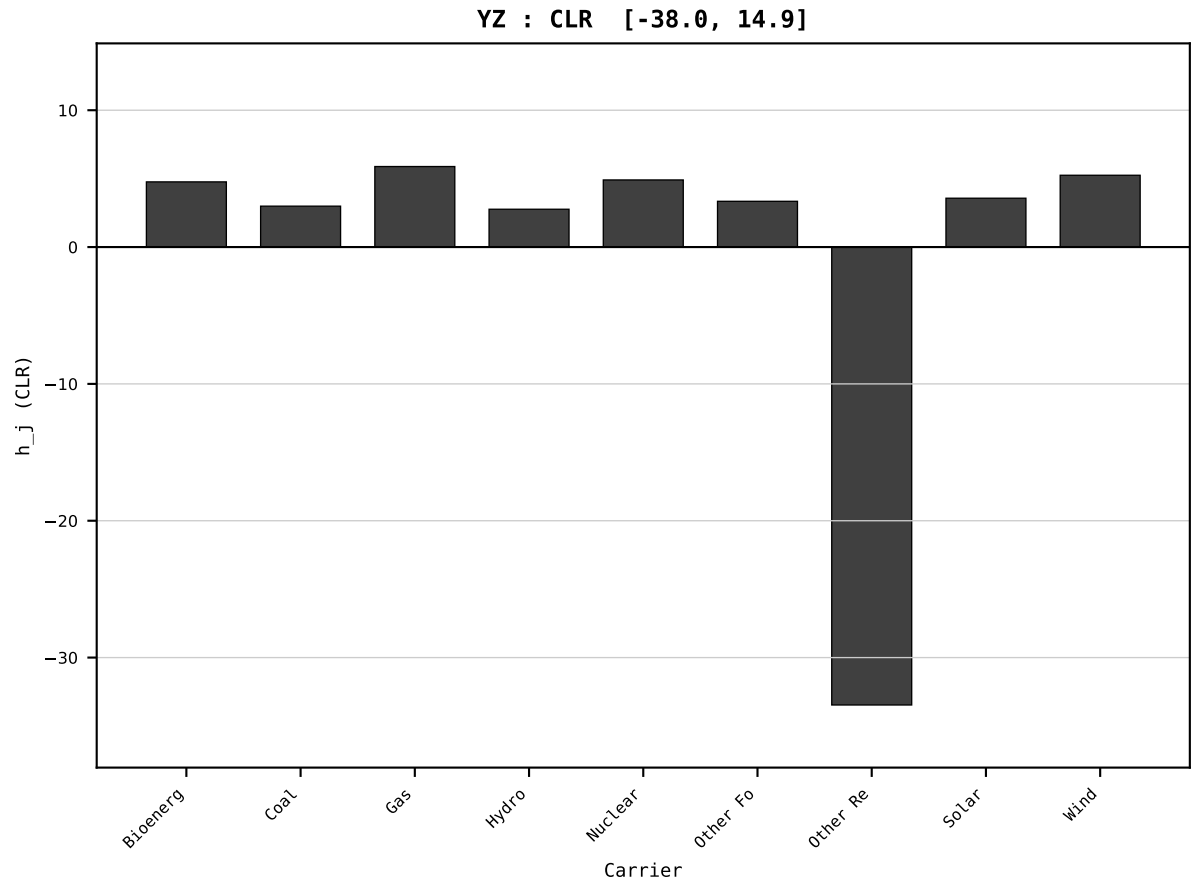
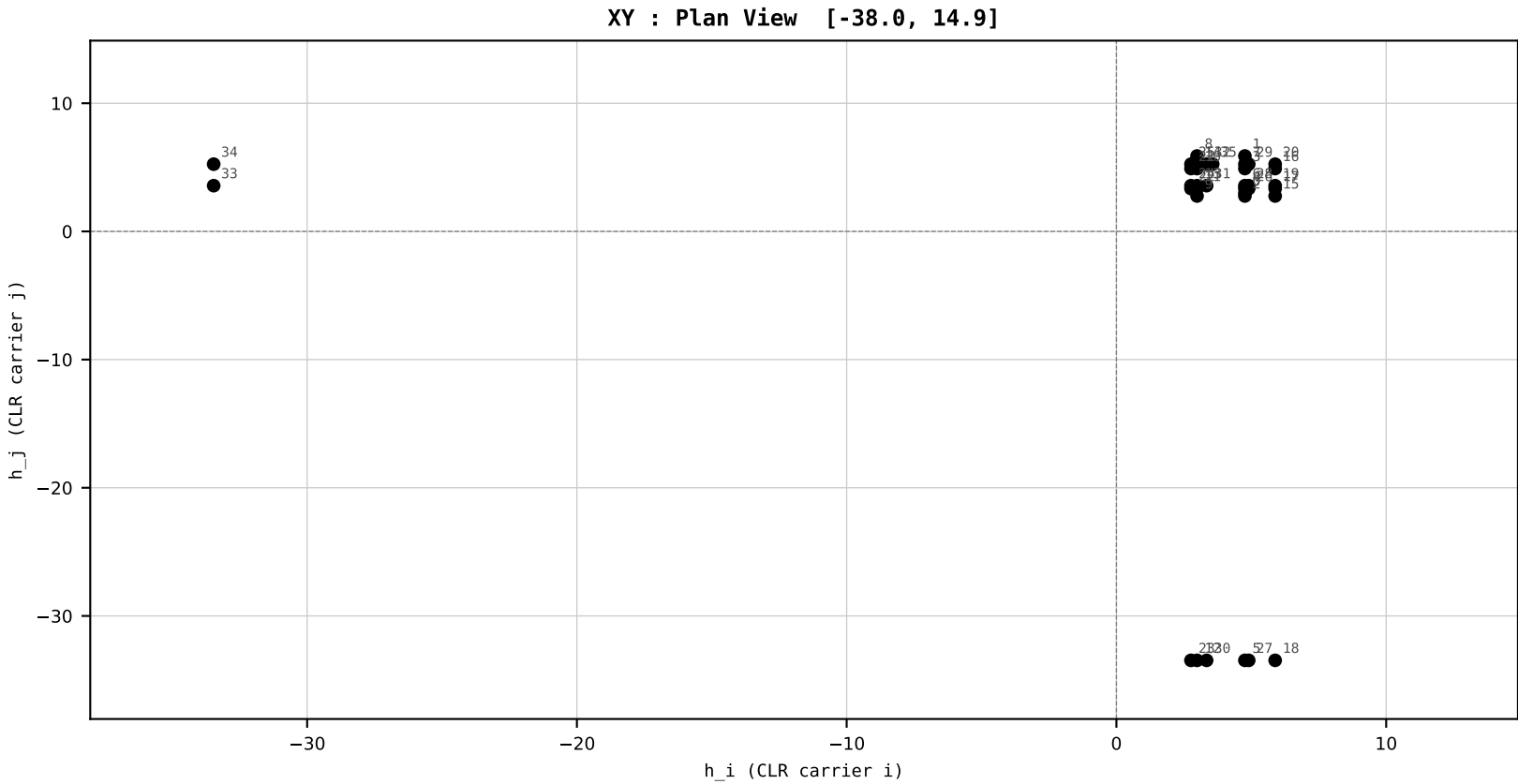
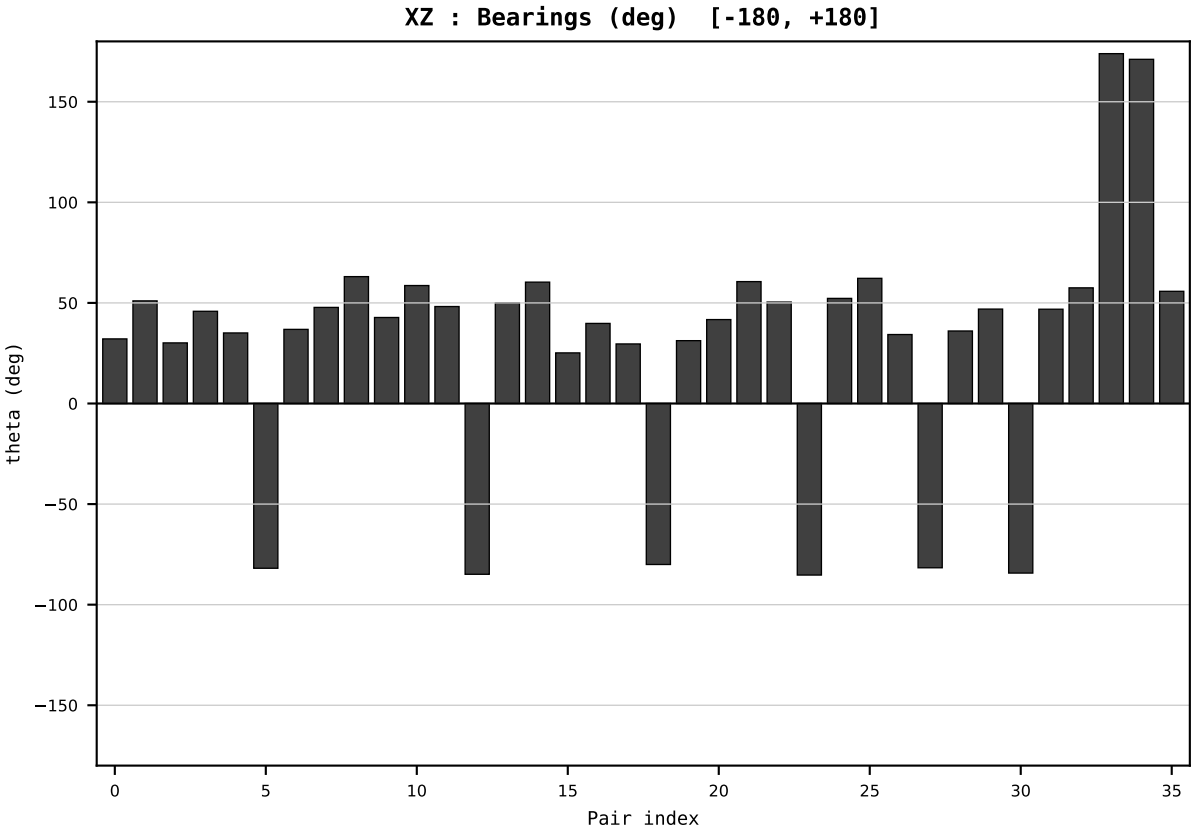
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.255549
Ring = Hs-2
E_metric= 914.9922
kappa_HS= 12284000000000000.00
omega = 0.5927 deg
d_A = 0.368914
Helm = Hydro
Helm d = -0.225695
DR = 39.350 HLR
DR ratio= 122839999940630944.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

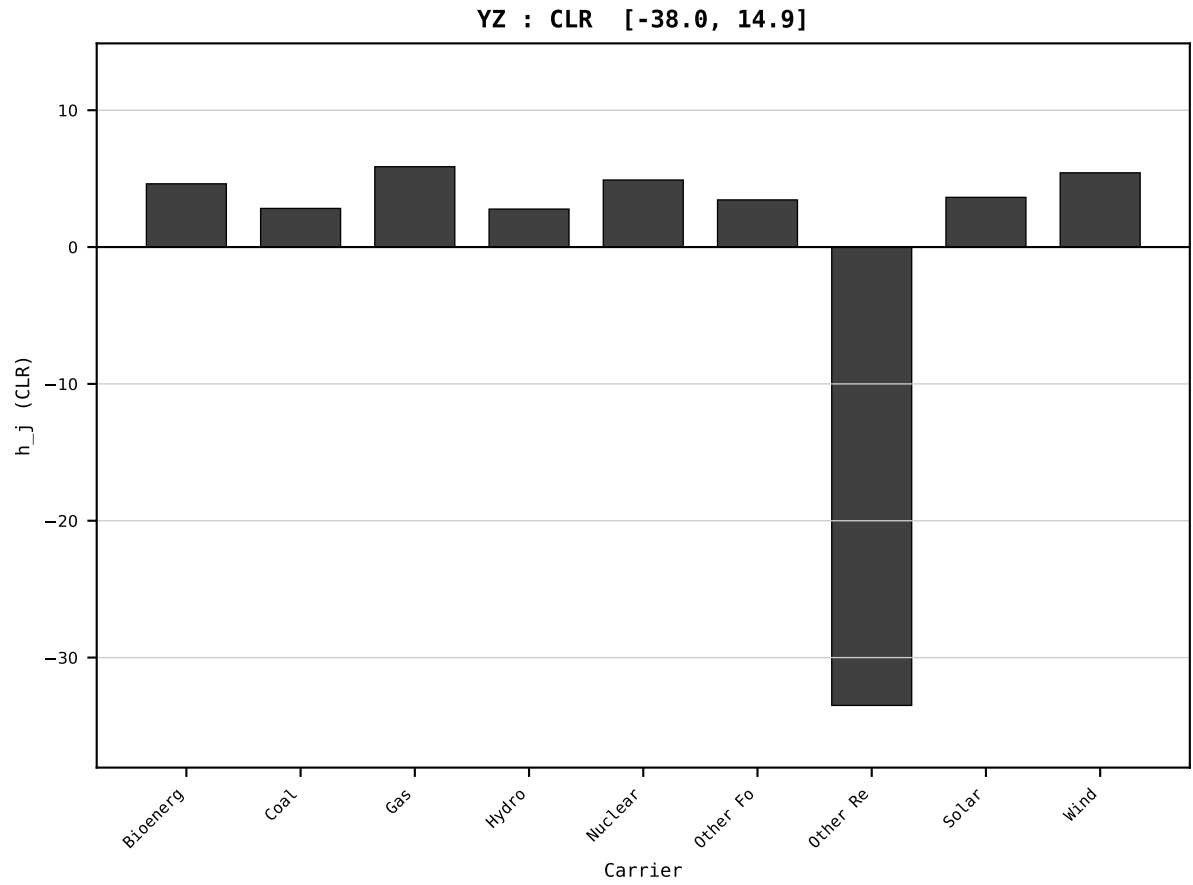
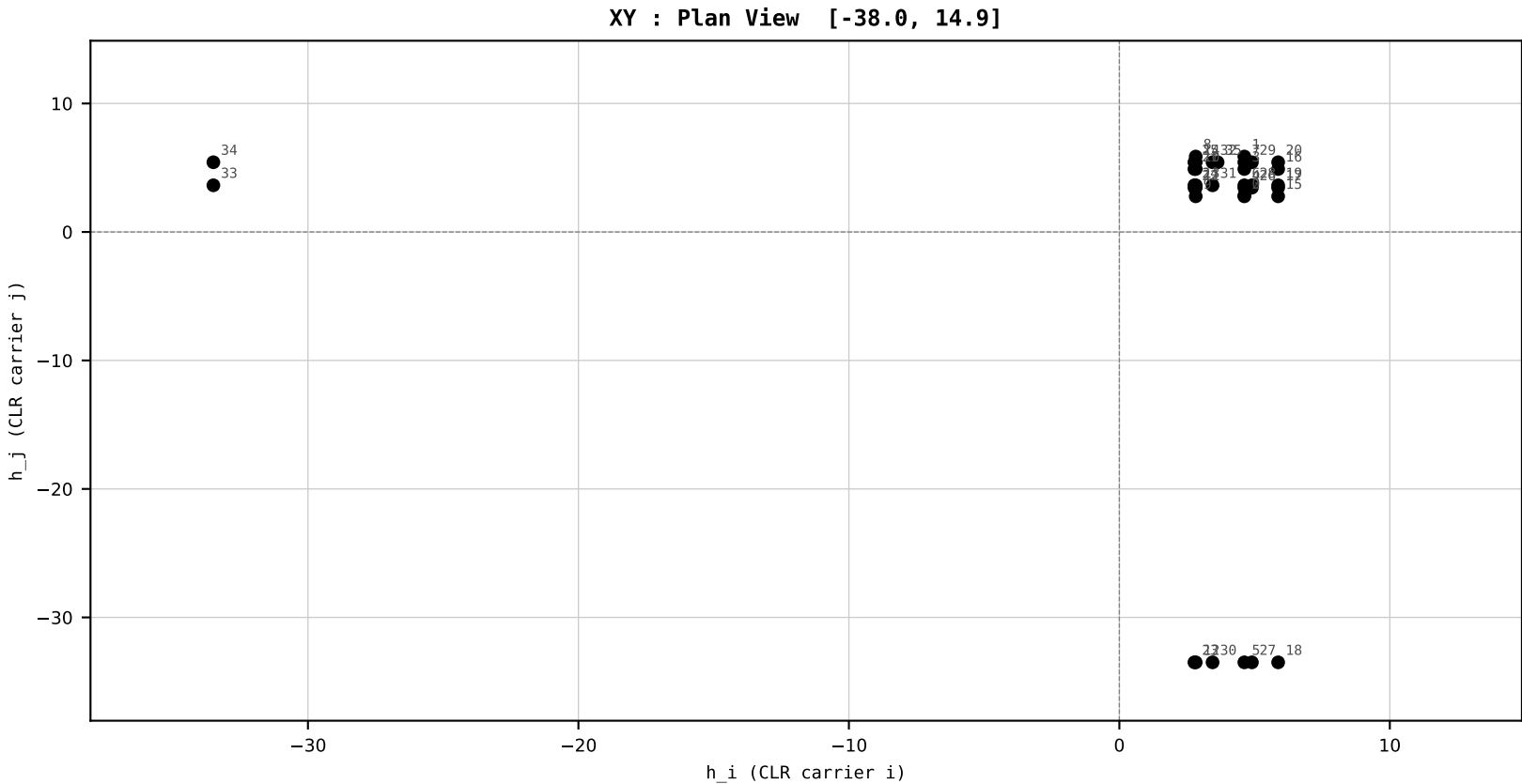
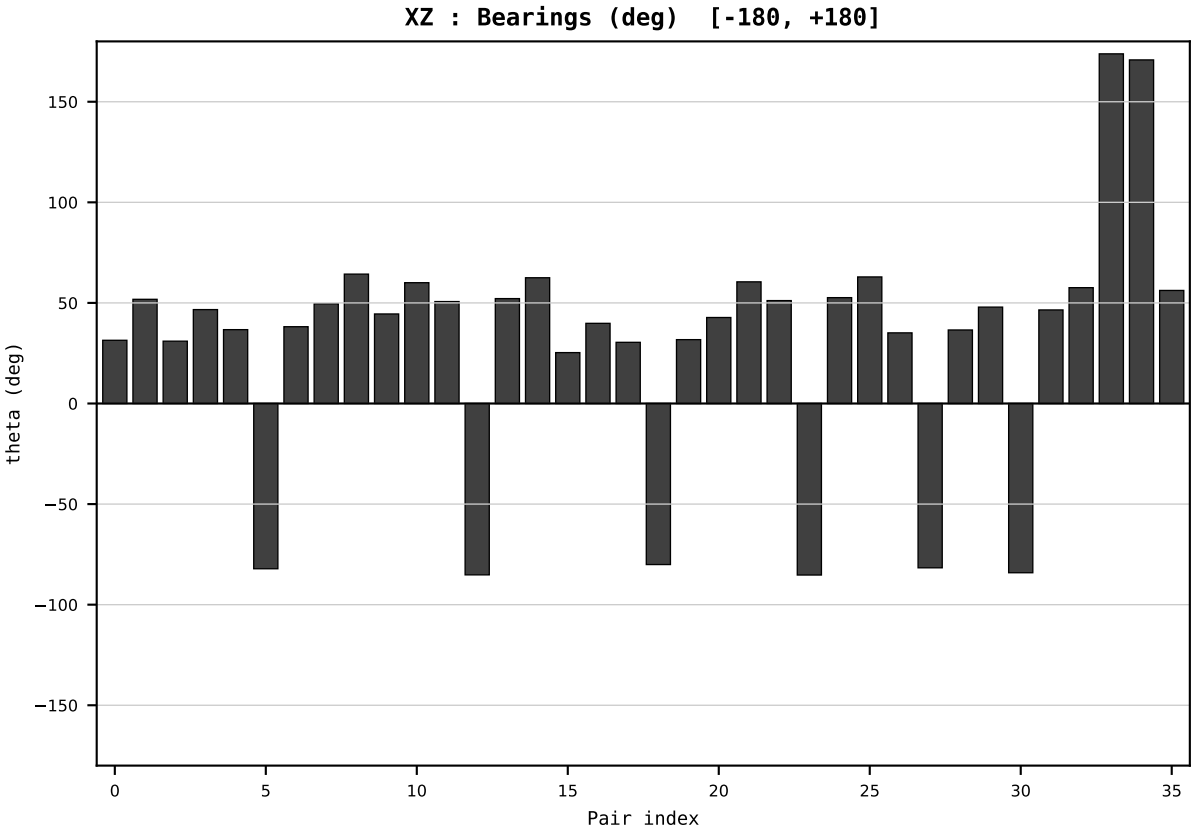
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.259322
Ring = Hs-2
E_metric= 914.4047
kappa_HS= 12510999999999984.00
omega = 0.4928 deg
d_A = 0.309051
Helm = Wind
Helm d = +0.179503
DR = 39.368 HLR
DR ratio= 125110000595524400.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



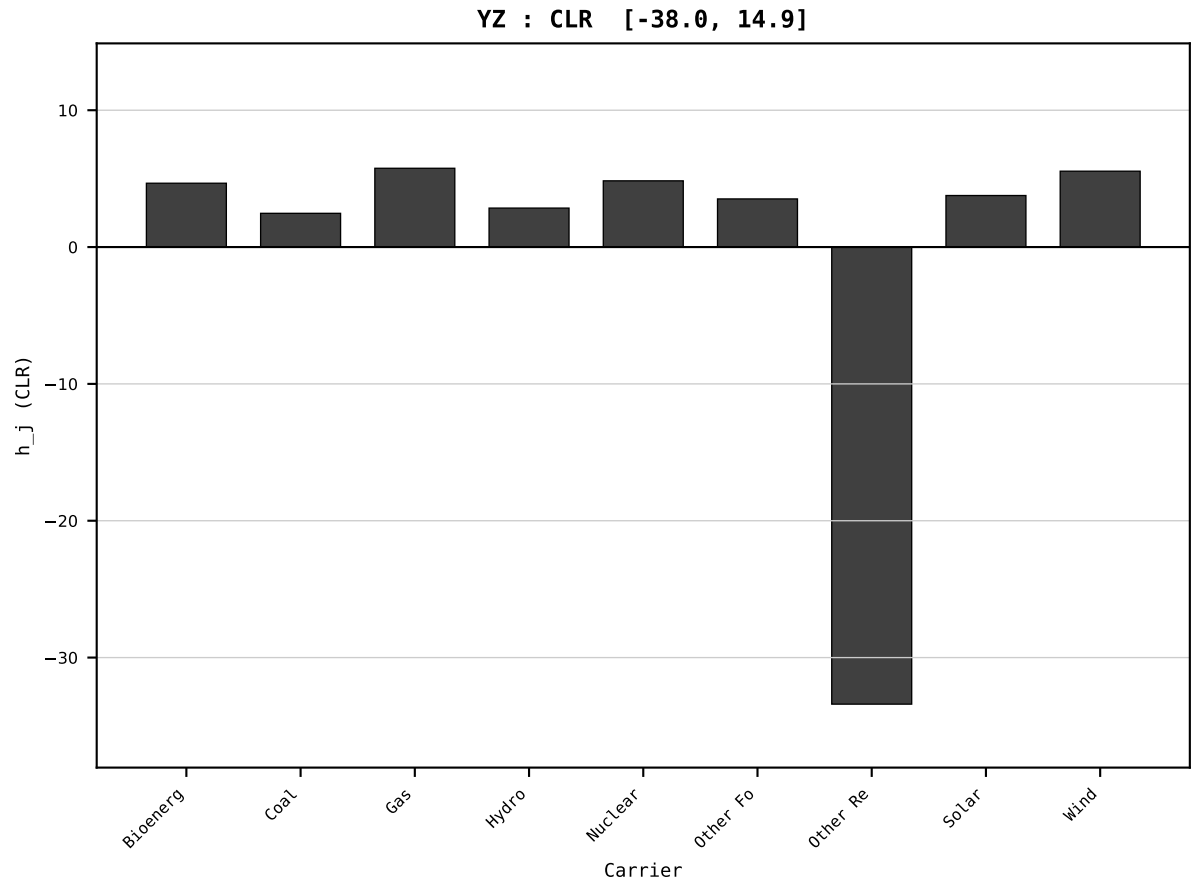
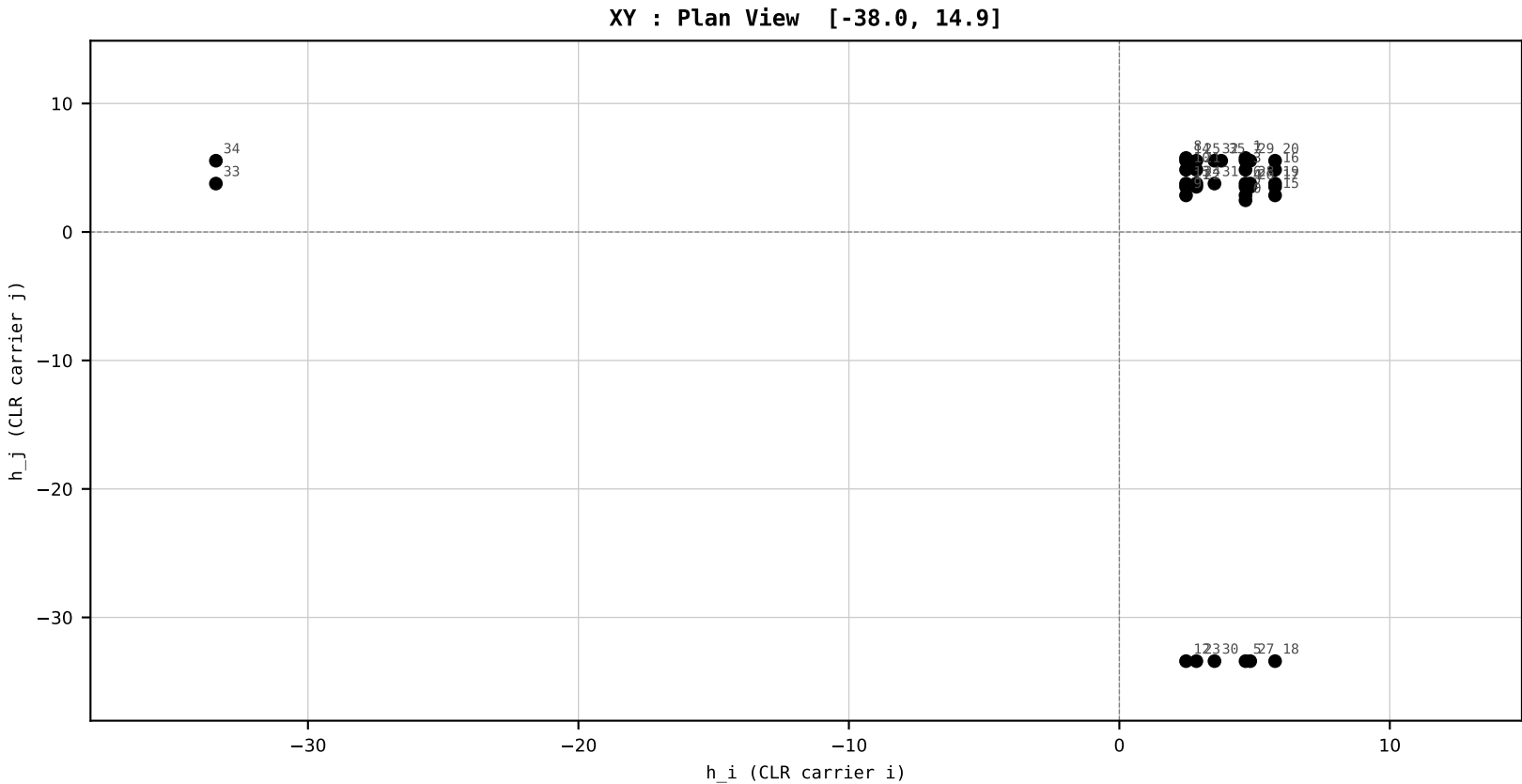
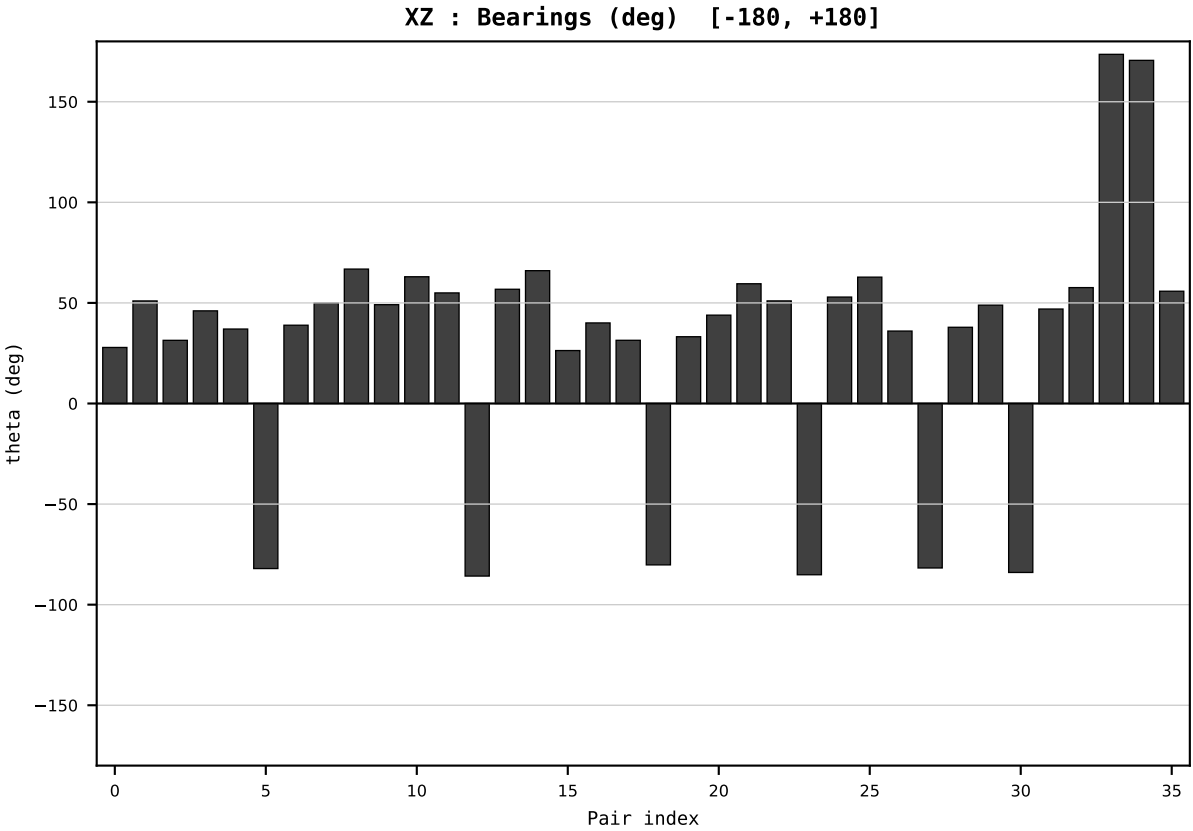
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.250656
Ring = Hs-2
E_metric= 914.8761
kappa_HS= 10165999999999984.00
omega = 0.6998 deg
d_A = 0.444546
Helm = Coal
Helm d = -0.358081
DR = 39.160 HLR
DR ratio= 101660000312153280.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

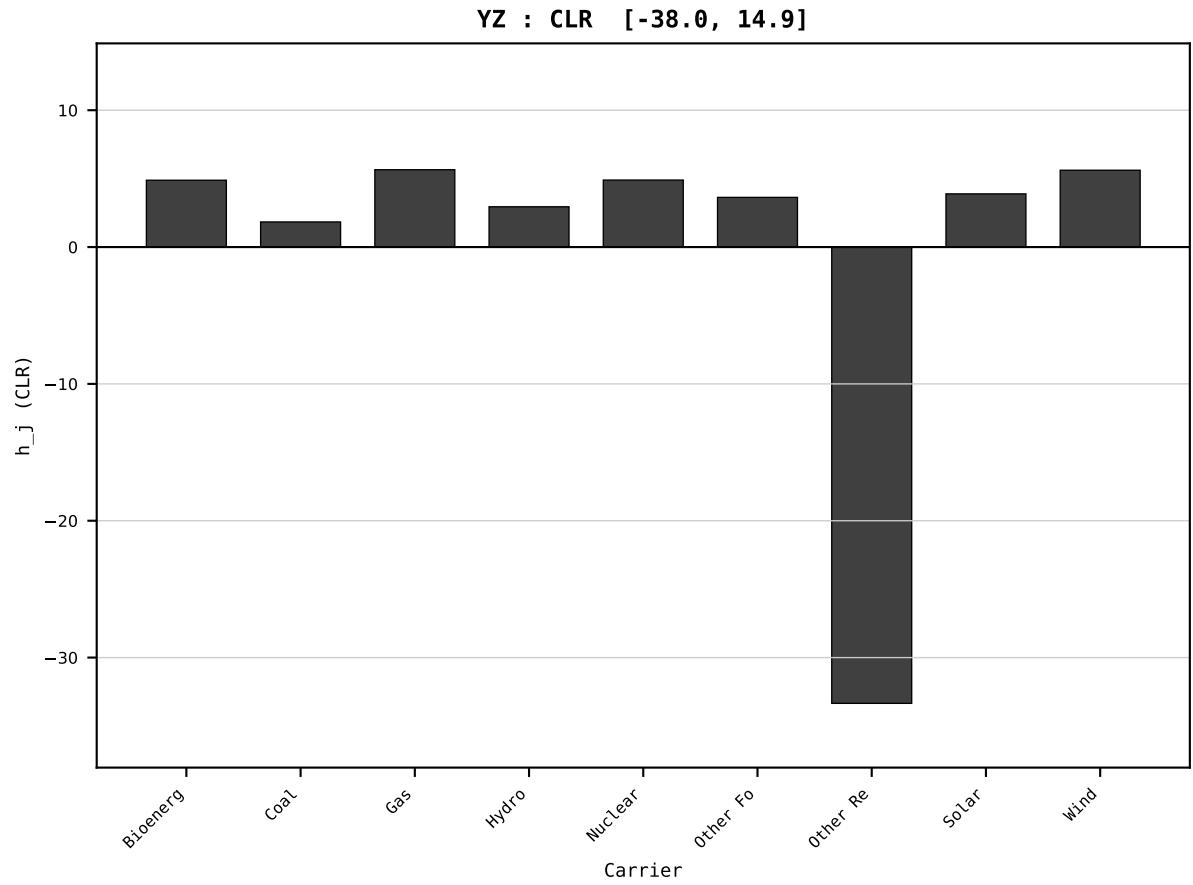
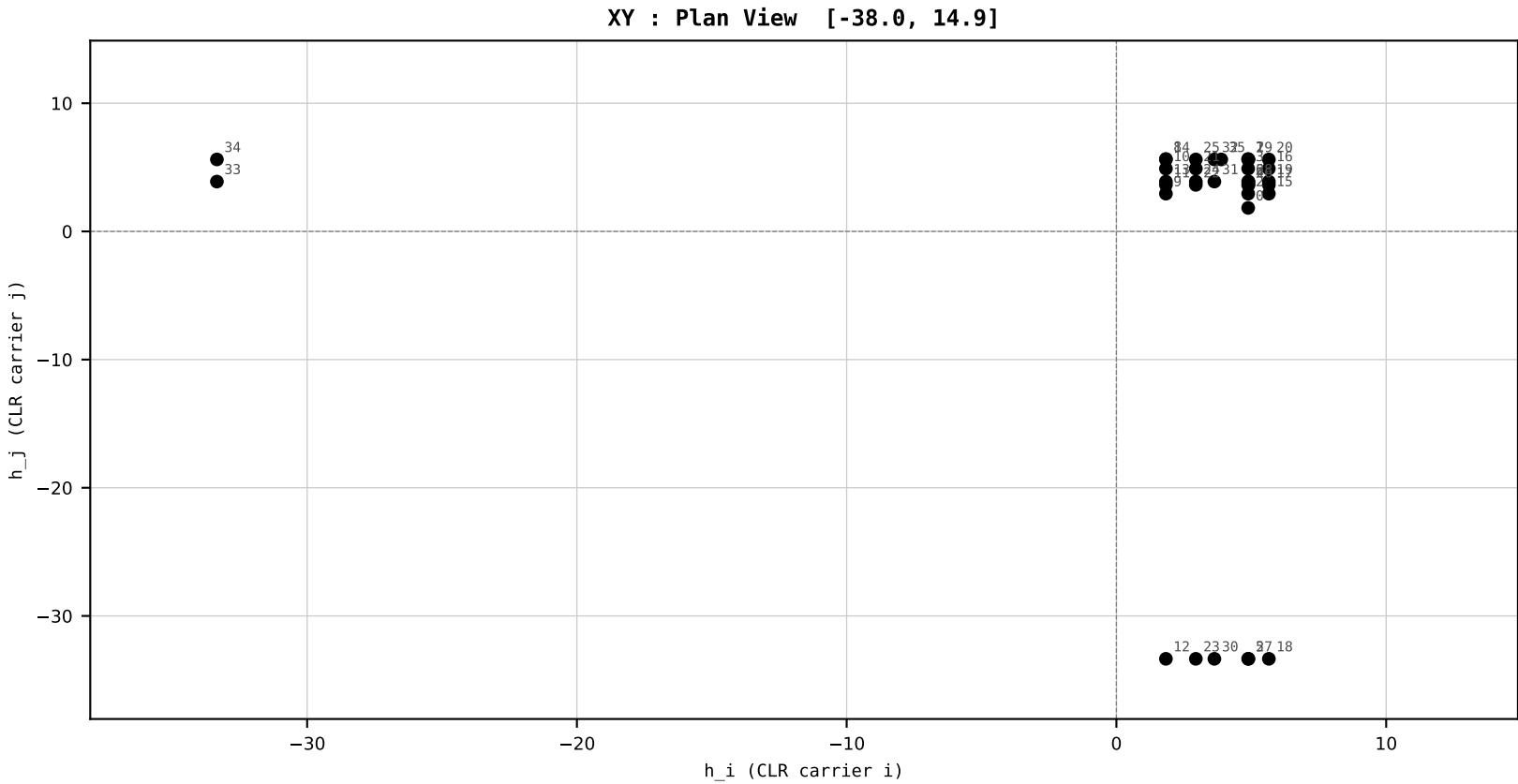
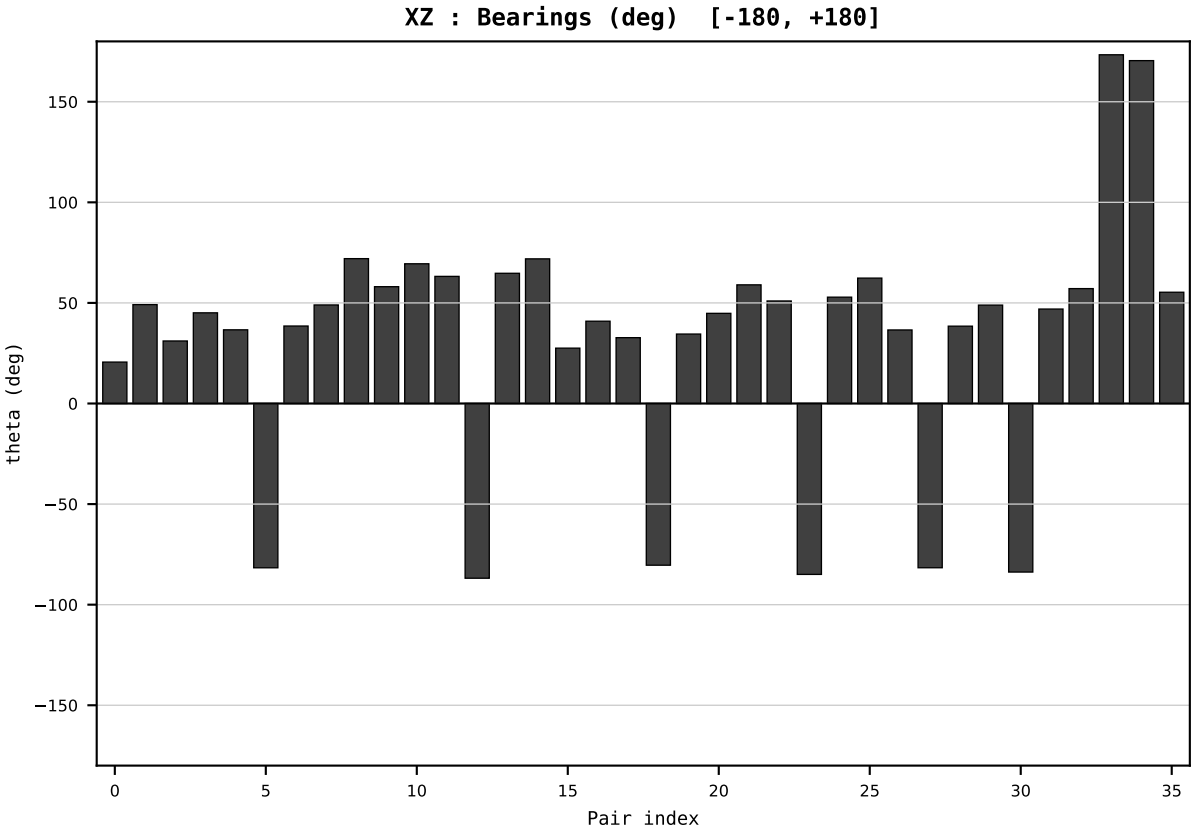
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.239203
Ring = Hs-2
E_metric= 915.1572
kappa_HS= 8630000000000000.00
omega = 1.1445 deg
d_A = 0.710901
Helm = Coal
Helm d = -0.630382
DR = 38.997 HLR
DR ratio= 86300000604094160.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

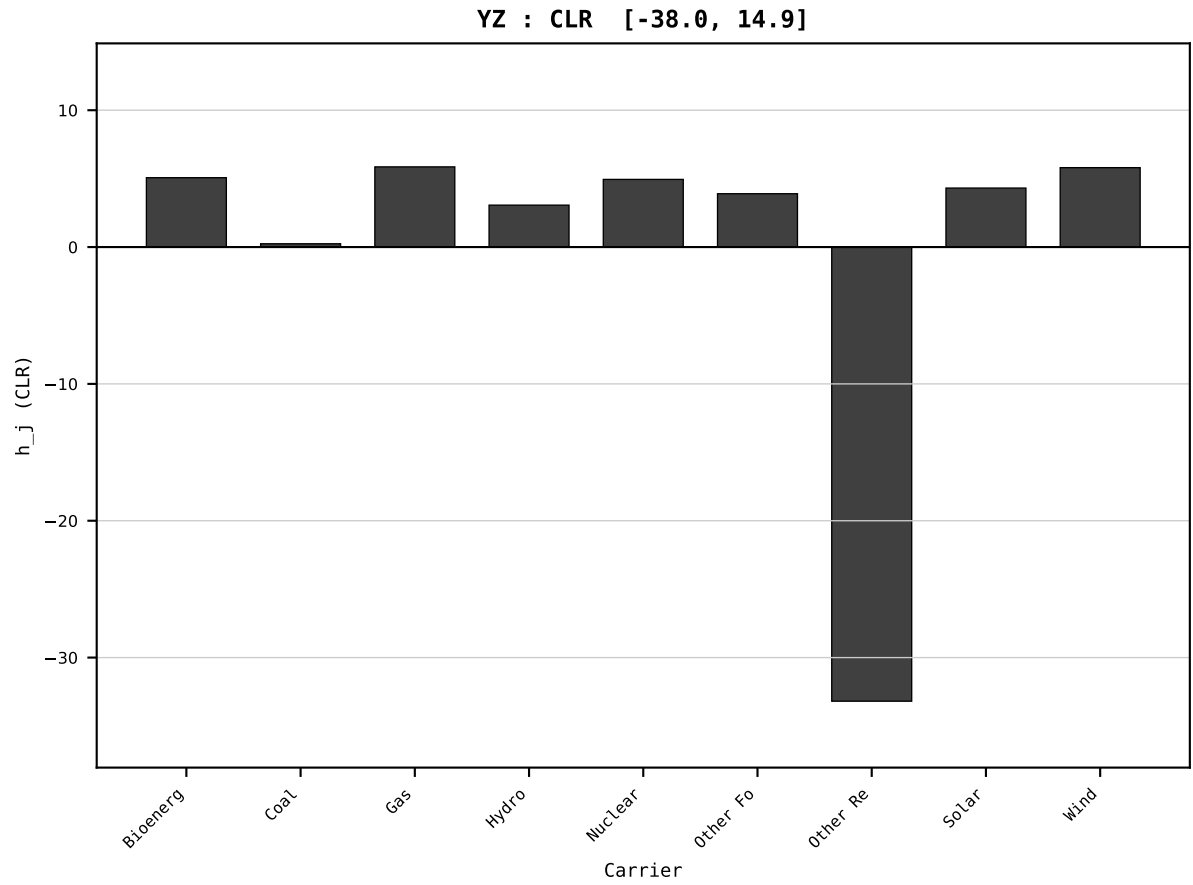
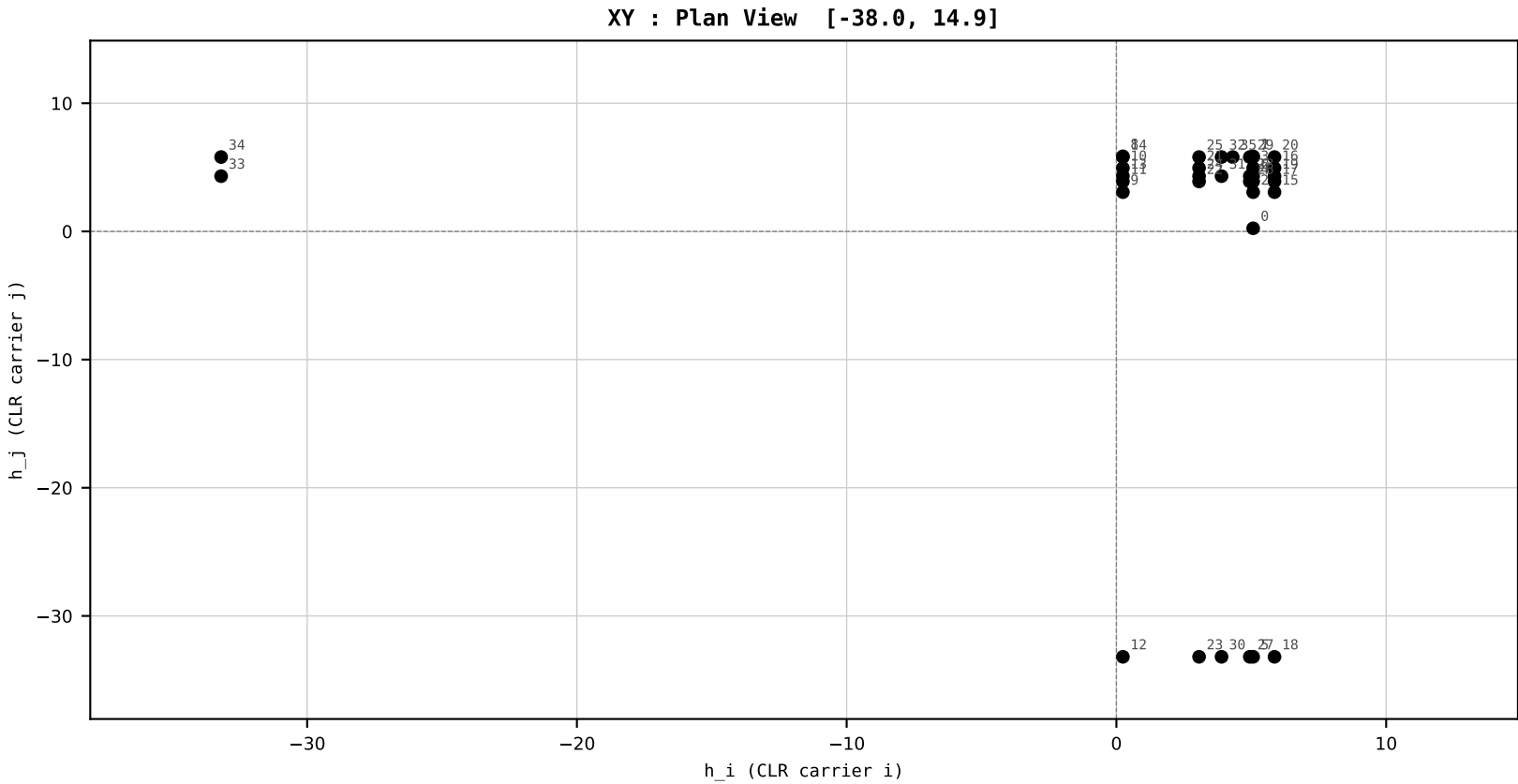
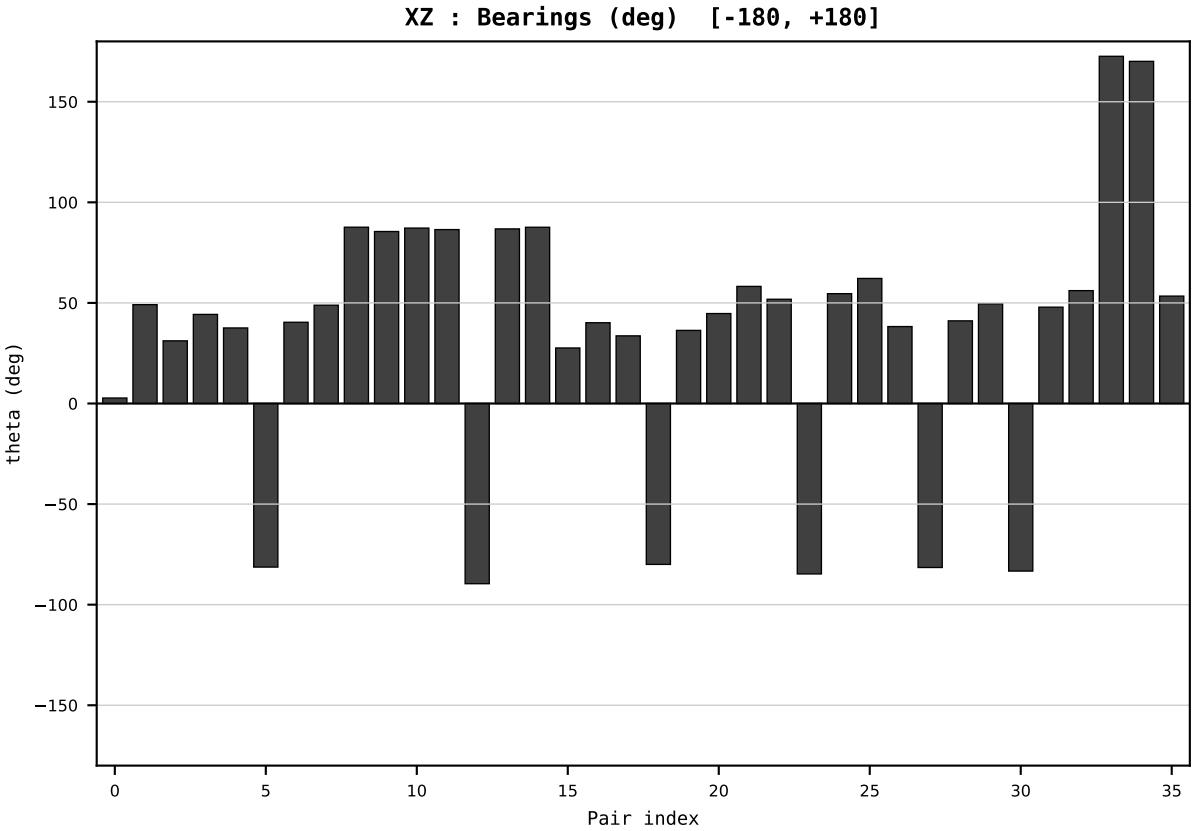
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

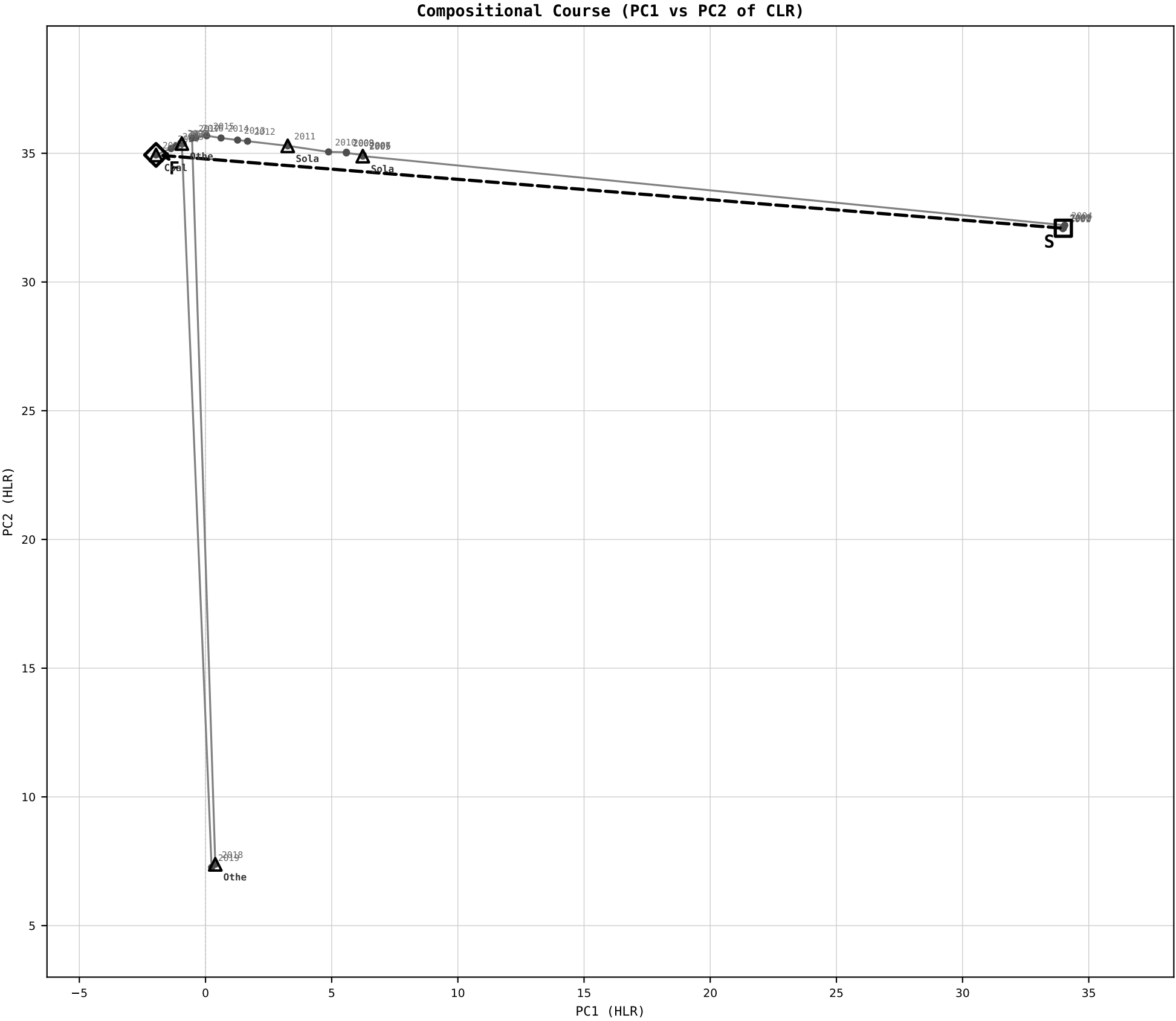
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.245474
Ring = Hs-2
E_metric= 914.6317
kappa_HS= 9090999999999984.00
omega = 2.7663 deg
d_A = 1.715852
Helm = Coal
Helm d = -1.593740
DR = 39.049 HLR
DR ratio= 909099999905049104.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:

$Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$

$\rightarrow$ PC projection

Unit: HLR (Higgins Log-Ratio)

Net distance = 36.2591 HLR

Path length = 100.3591 HLR

Directness = 0.3613

DR start = 39.537 HLR

ratio = 148079999529708608.0x

DR final = 39.049 HLR

ratio = 90909999905049104.0x

E_metric start = 2195.0806

E_metric final = 1262.8551

PC1 loadings:

Solar -0.931

Coal +0.202

Nuclear +0.151

PC2 loadings:

Other Re -0.941

Coal +0.160

Other Fo +0.123

Top displacement events:

2018 Other Re d=28.236

2020 Other Re d=28.182

2005 Solar d=28.133

2011 Solar d=1.727

2025 Coal d=1.716

The instrument reads.

The expert decides.

The loop stays open.

View B

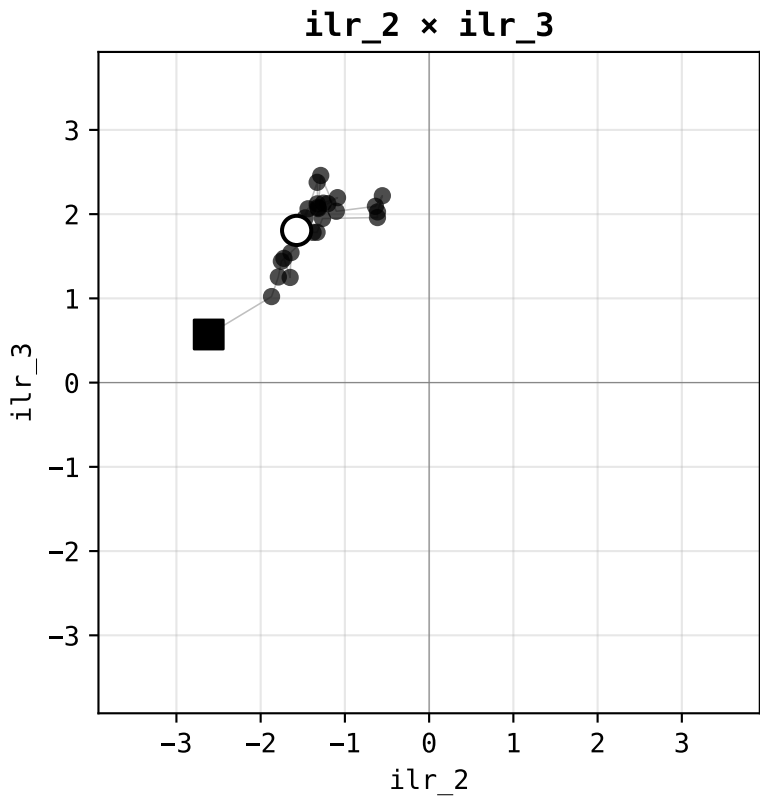
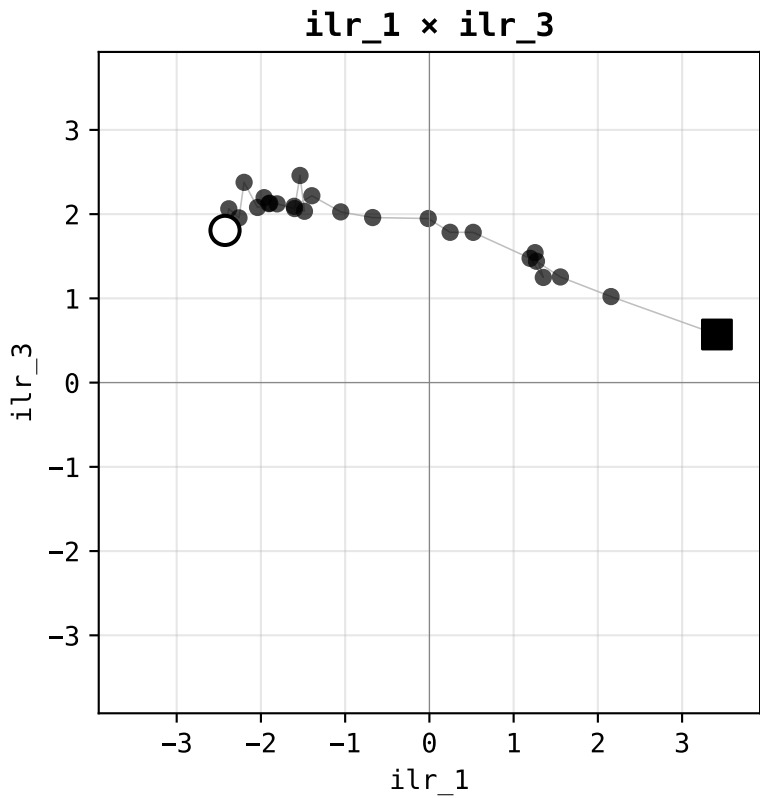
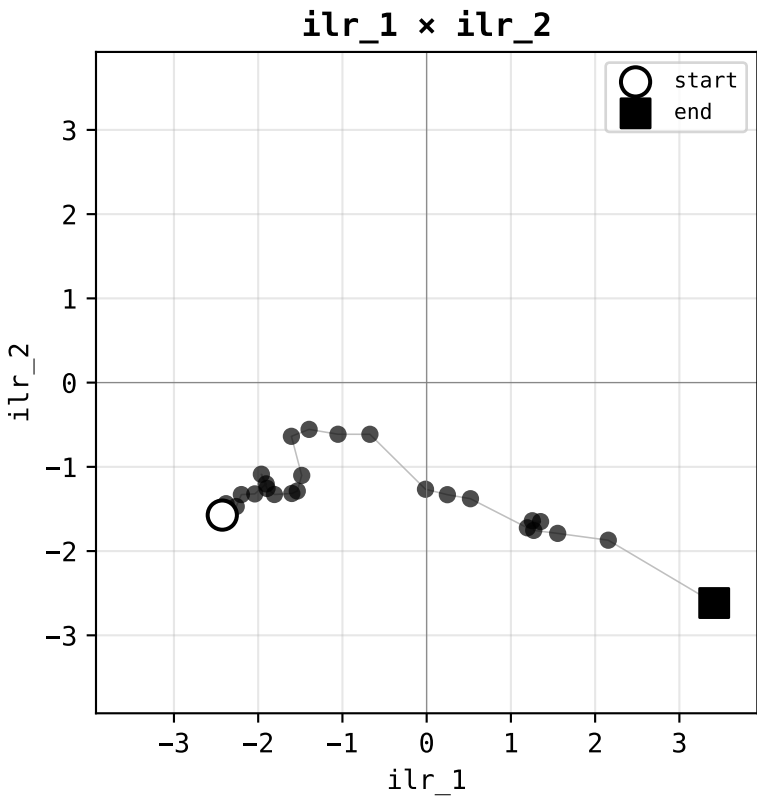
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ GBR United Kingdom

ILR-Helmert Orthogonal Triplet – GBR United Kingdom · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=26 years · D=9 carriers



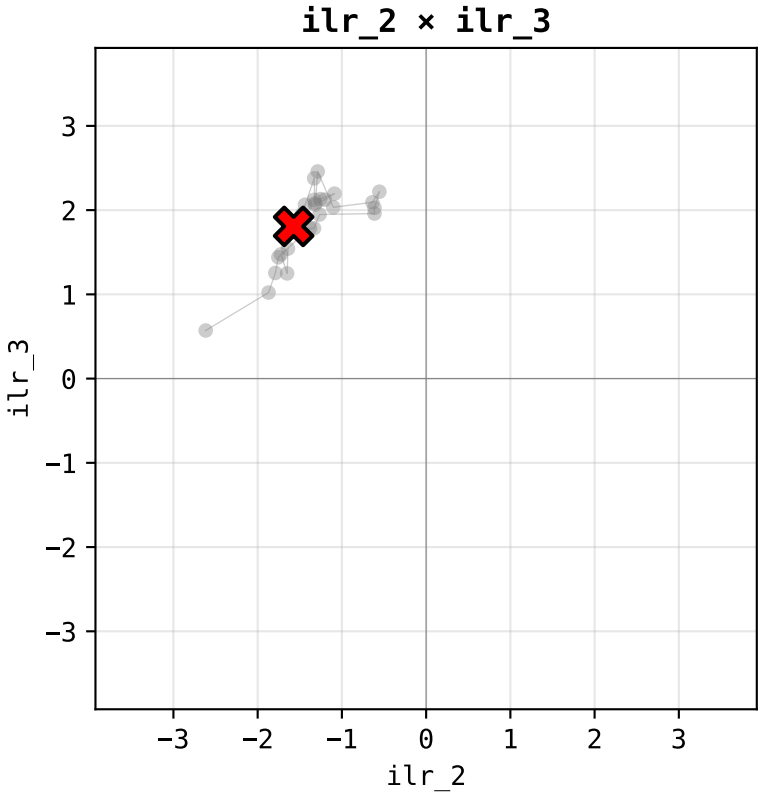
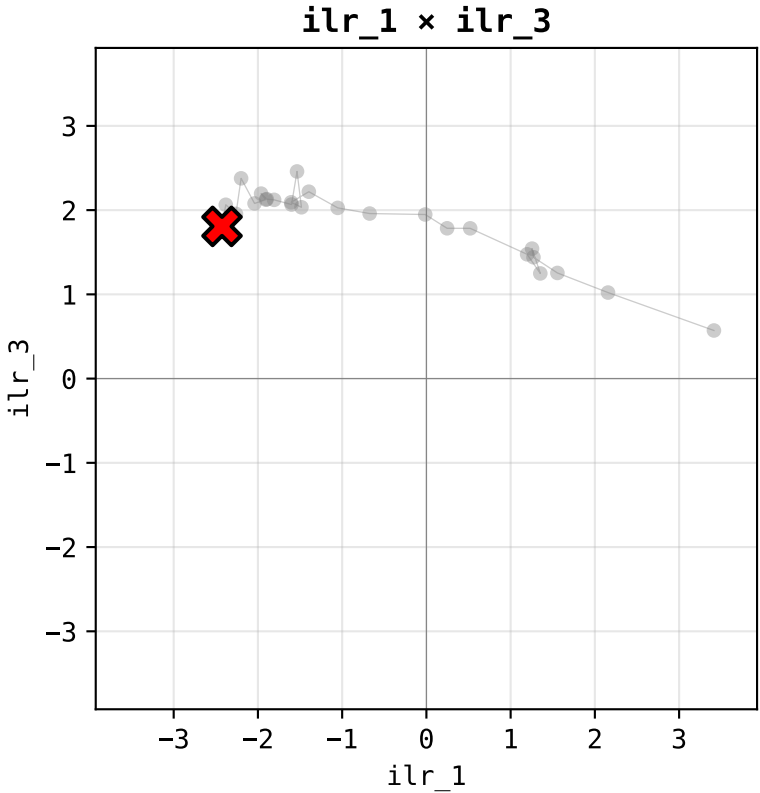
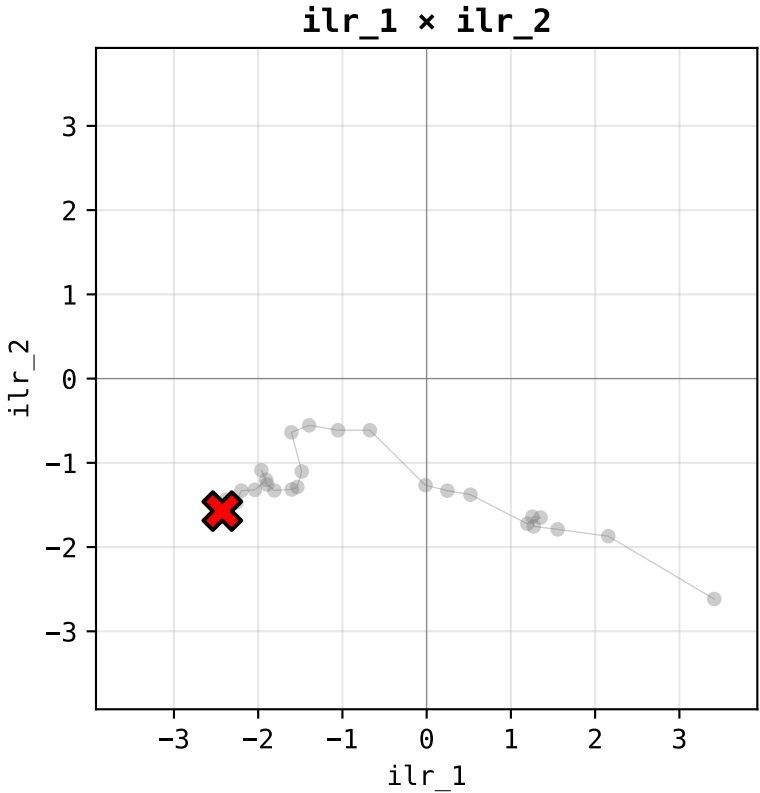
Helmert basis loadings (which carriers each ILR axis contrasts):

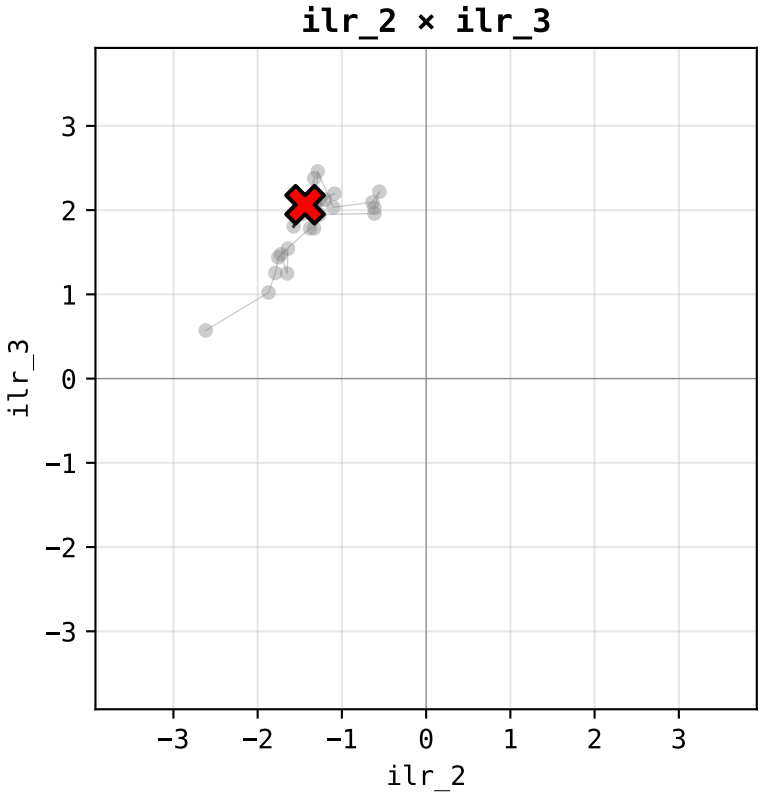
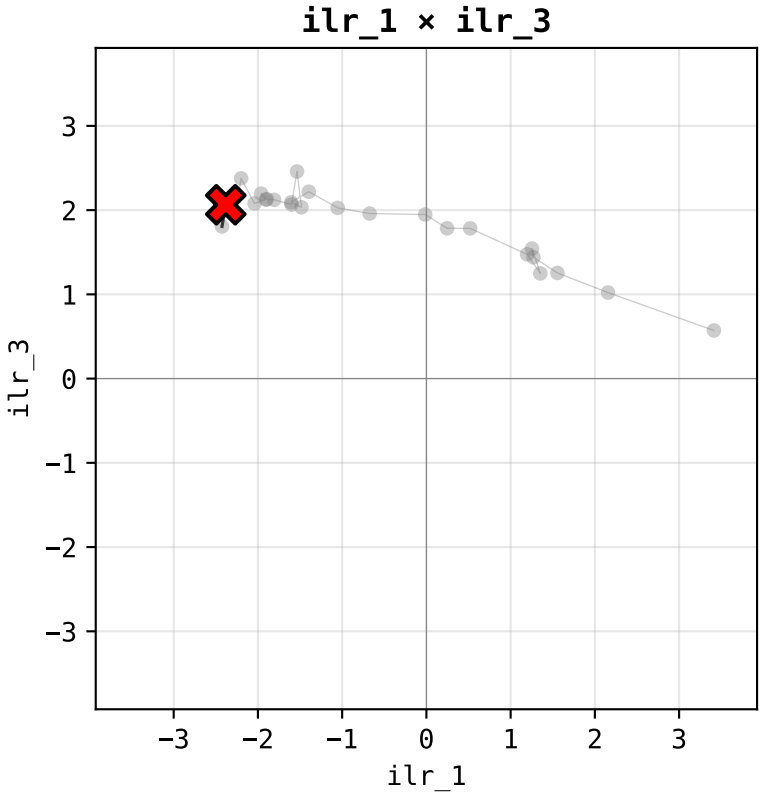
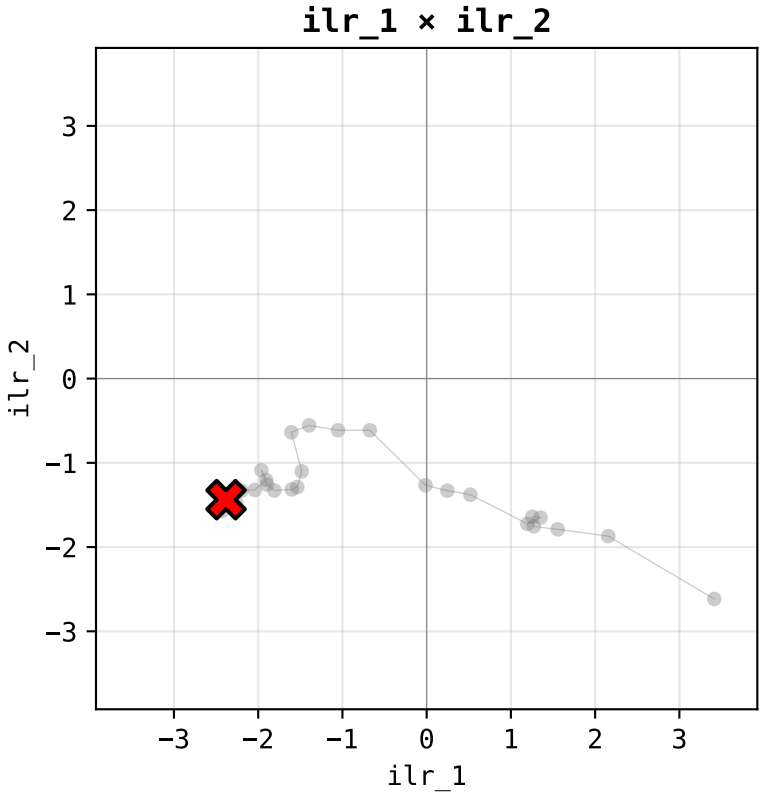
ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

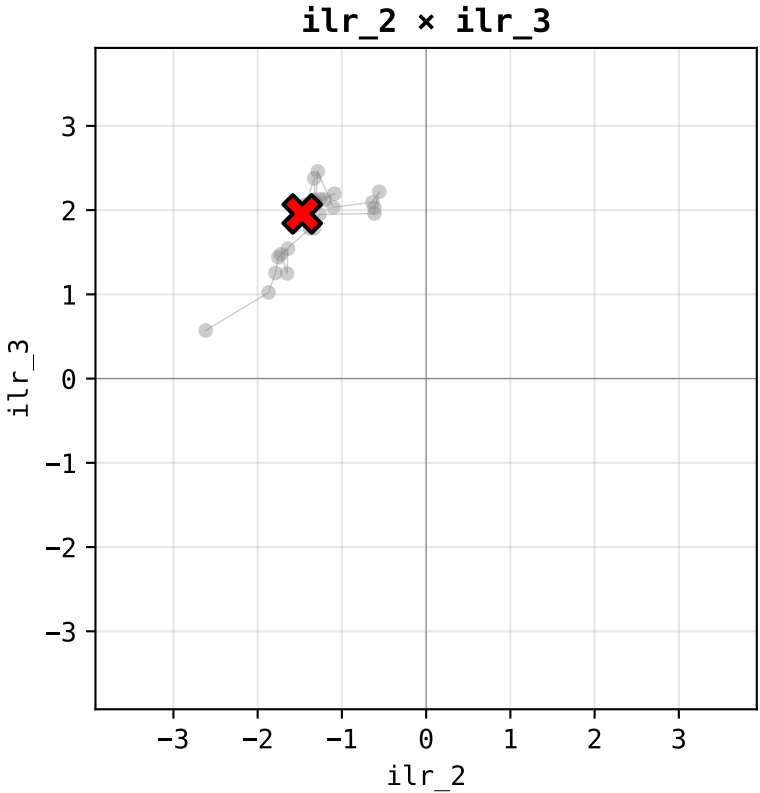
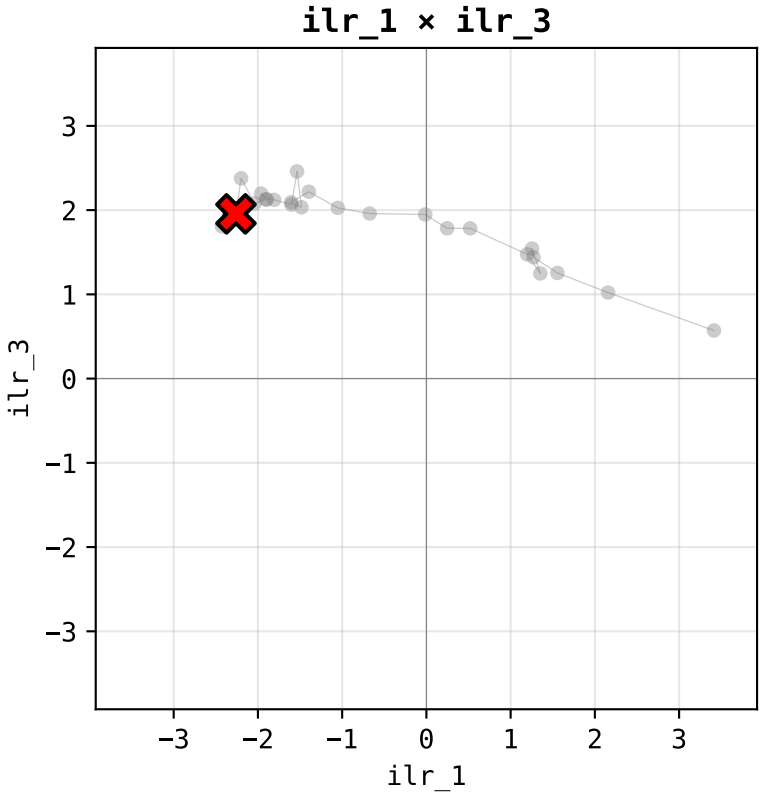
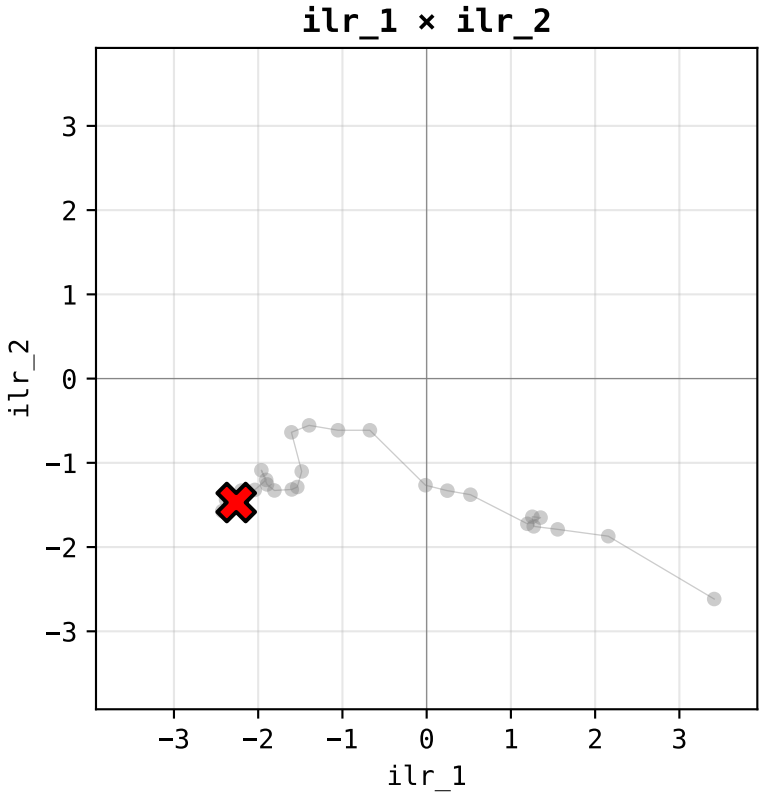
Trajectory:
start year: 2000
end year: 2025
N readings: 26

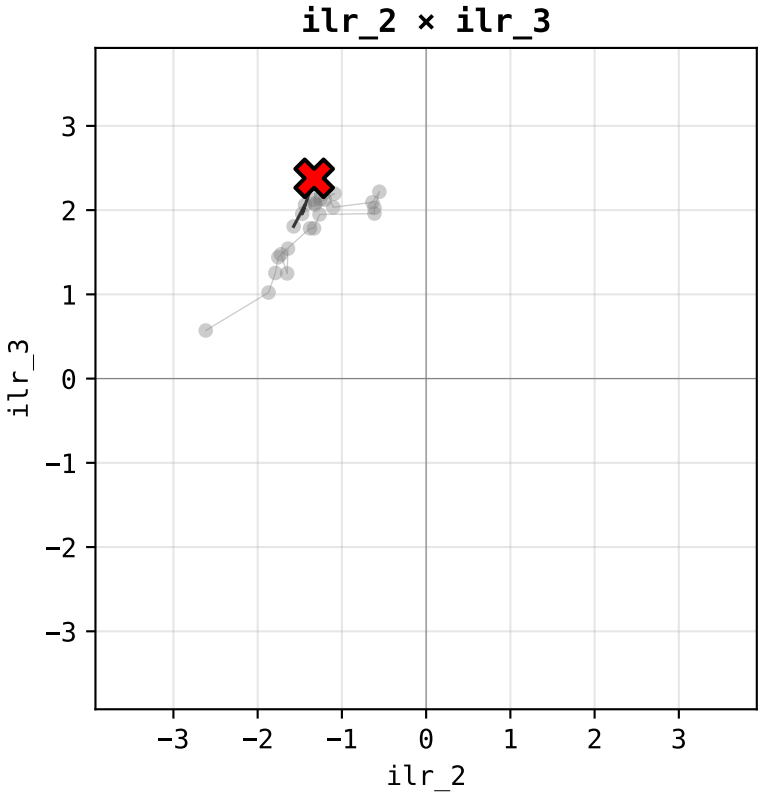
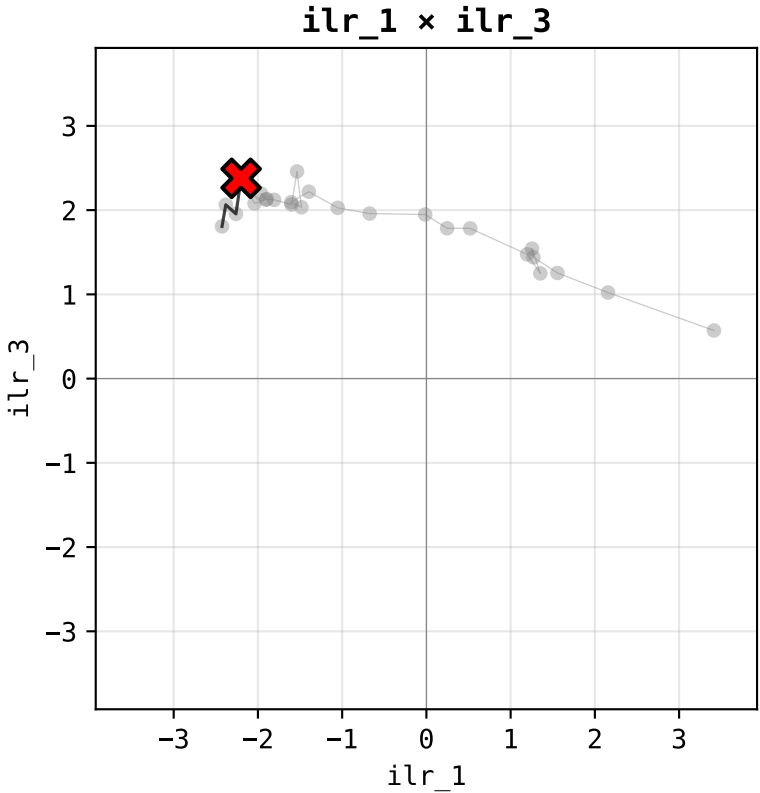
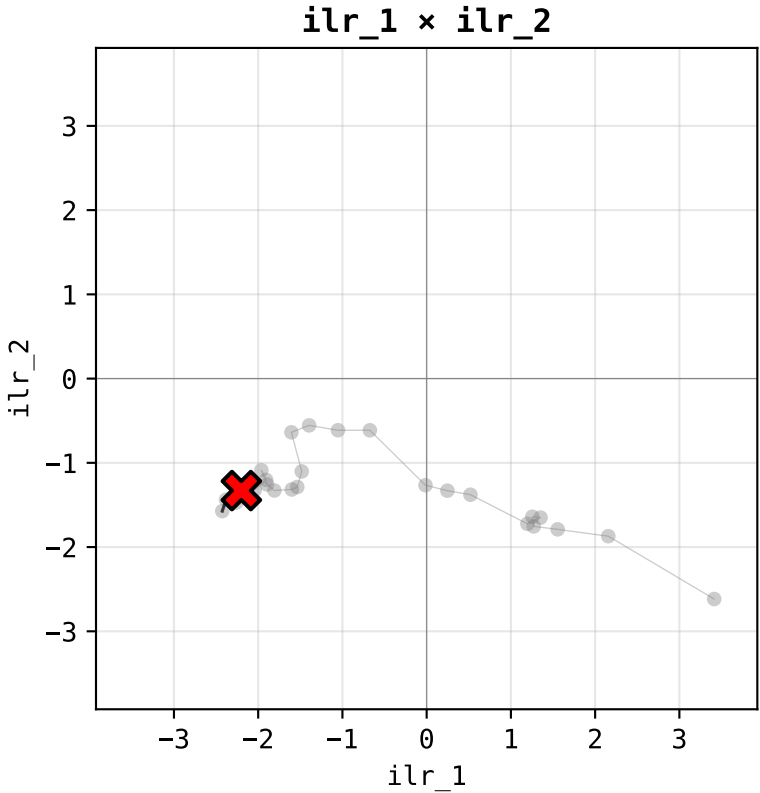
Display:
fixed scale ±3.93 ILR
Order 1 (first principles)
per Output Doctrine v1.0

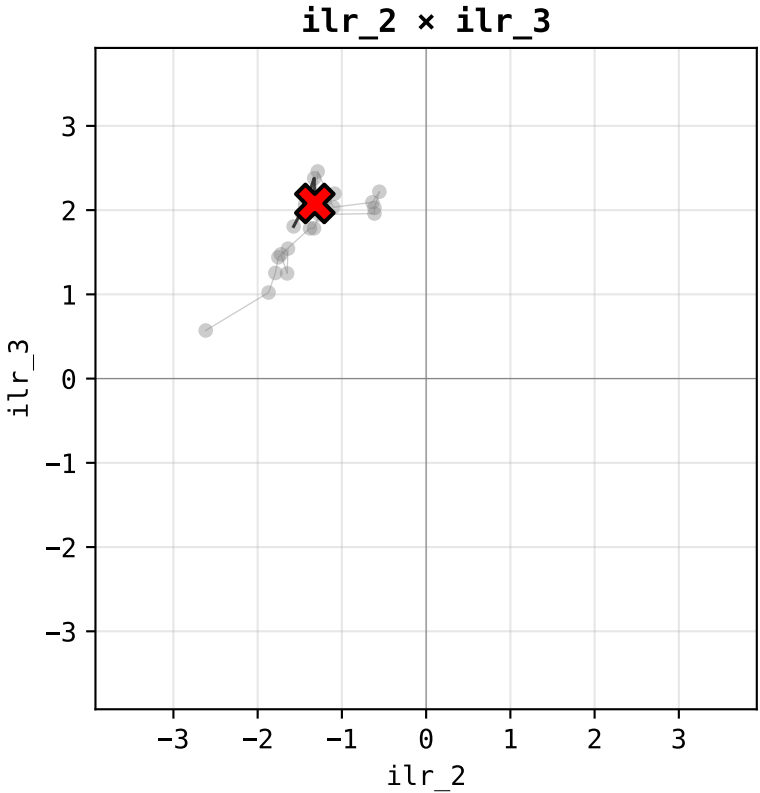
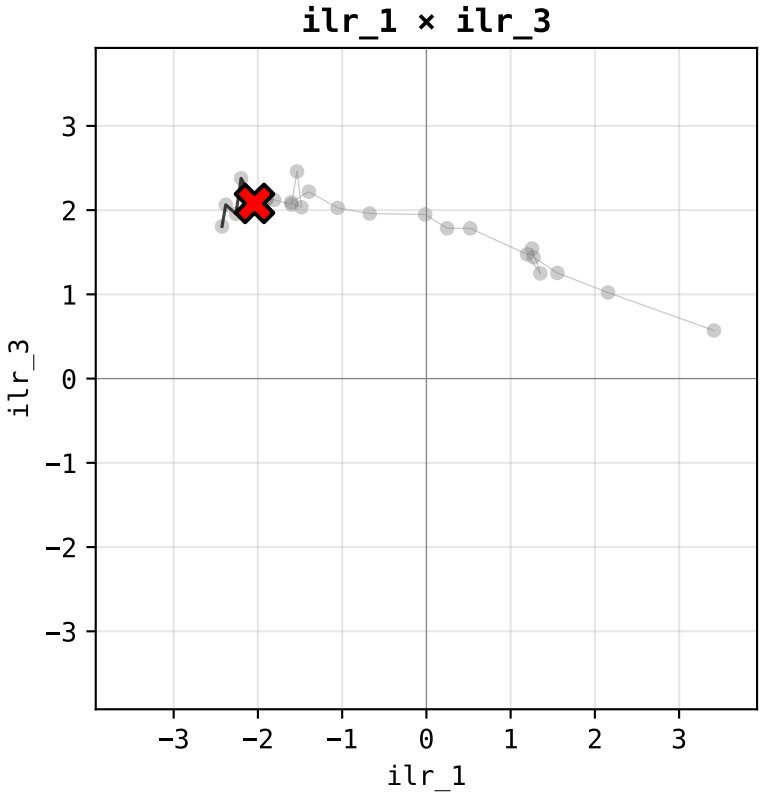
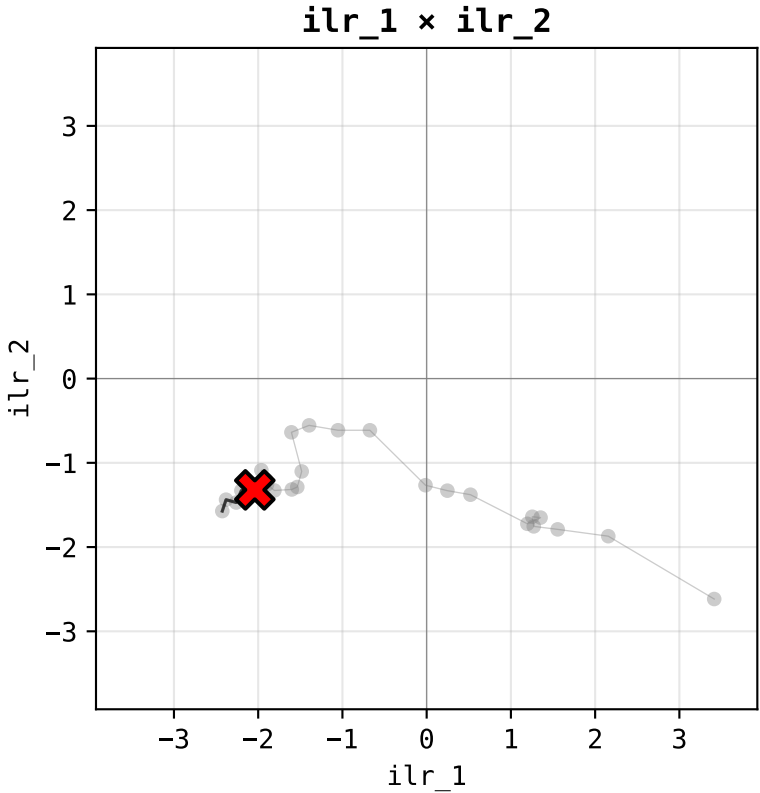
Symbology:
○ start · ■ end
· intermediate years

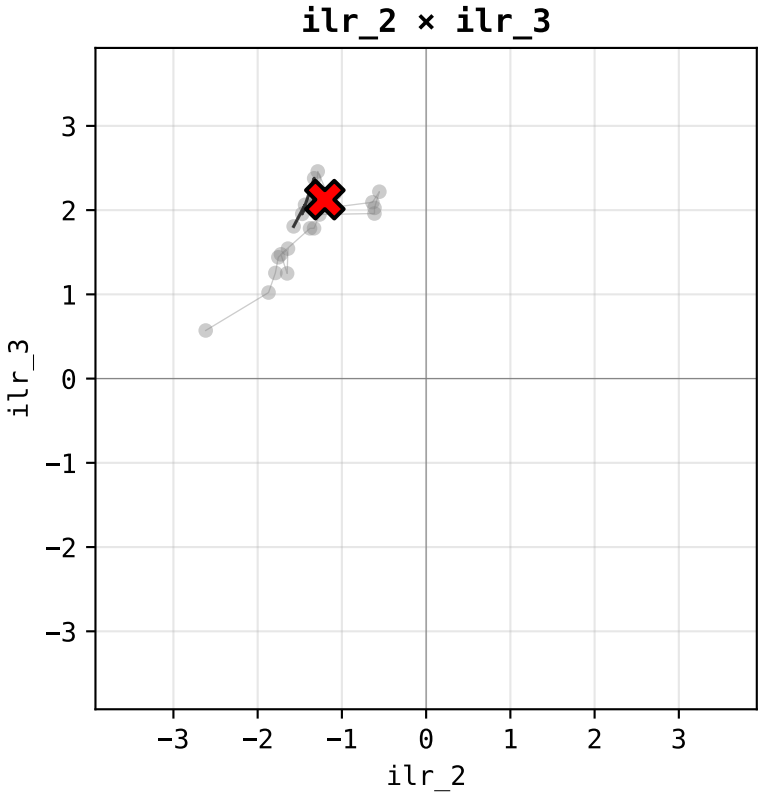
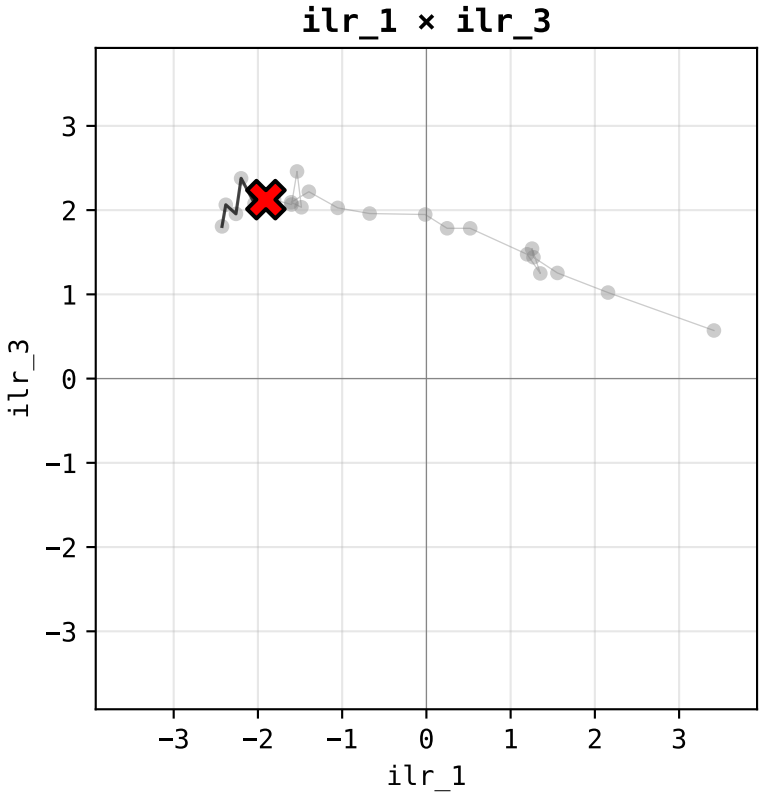
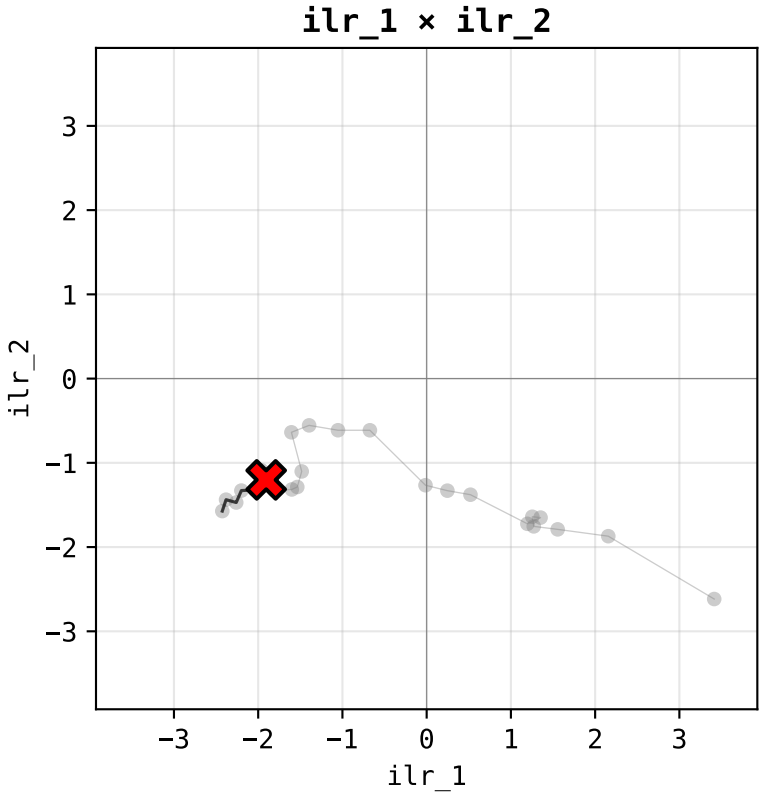


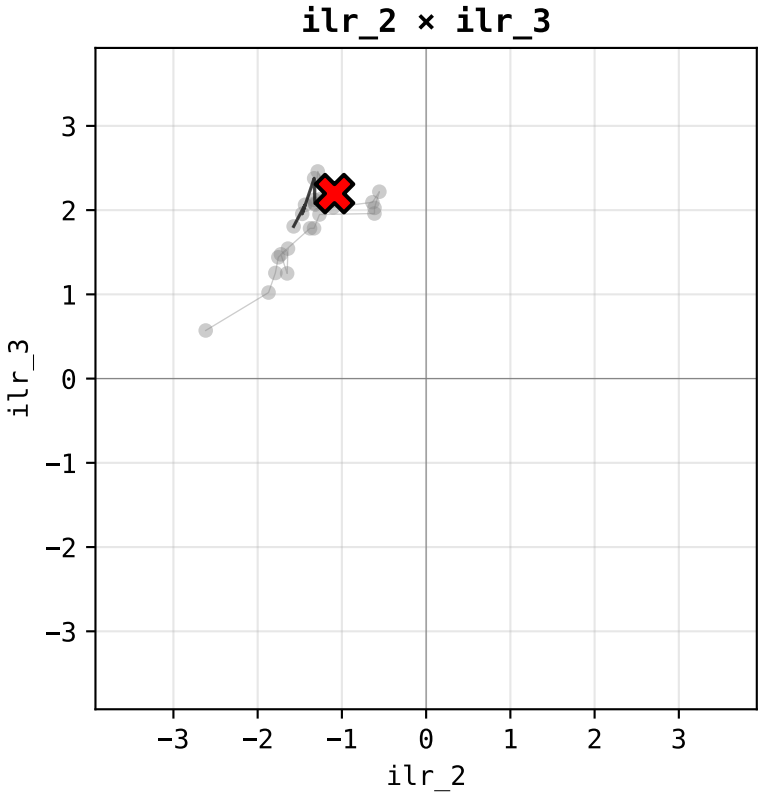
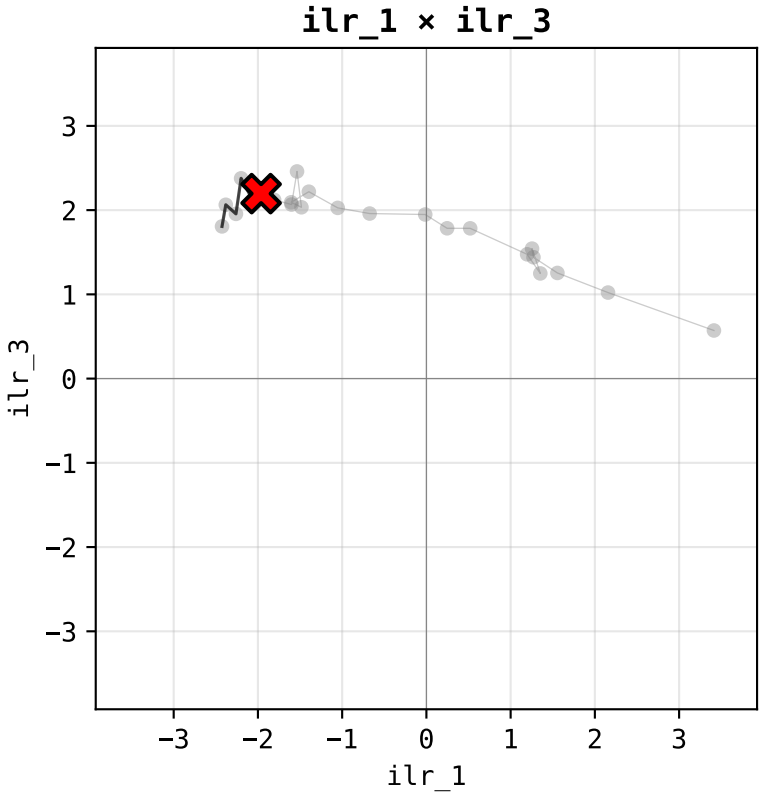
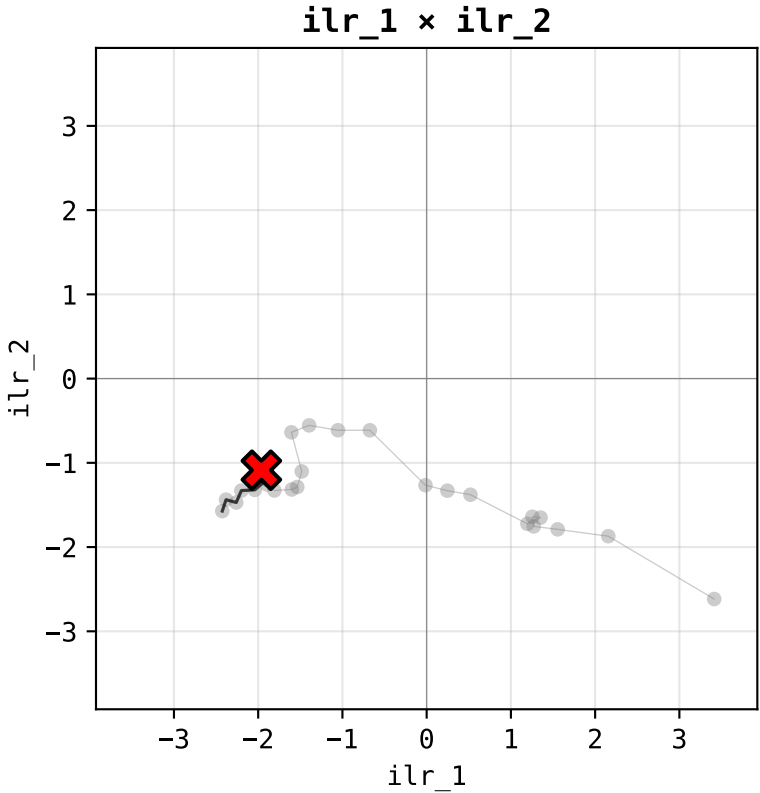


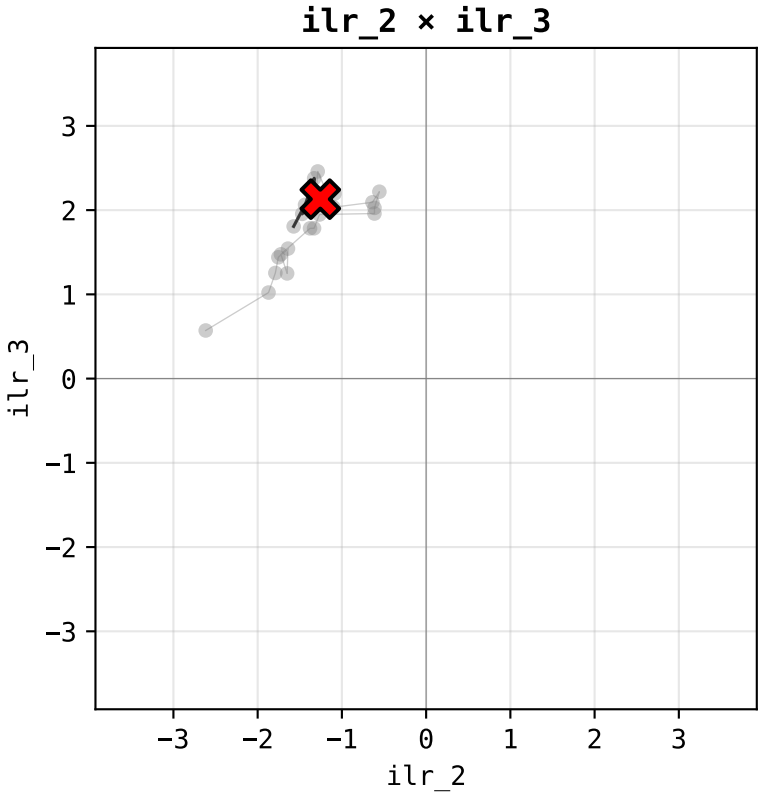
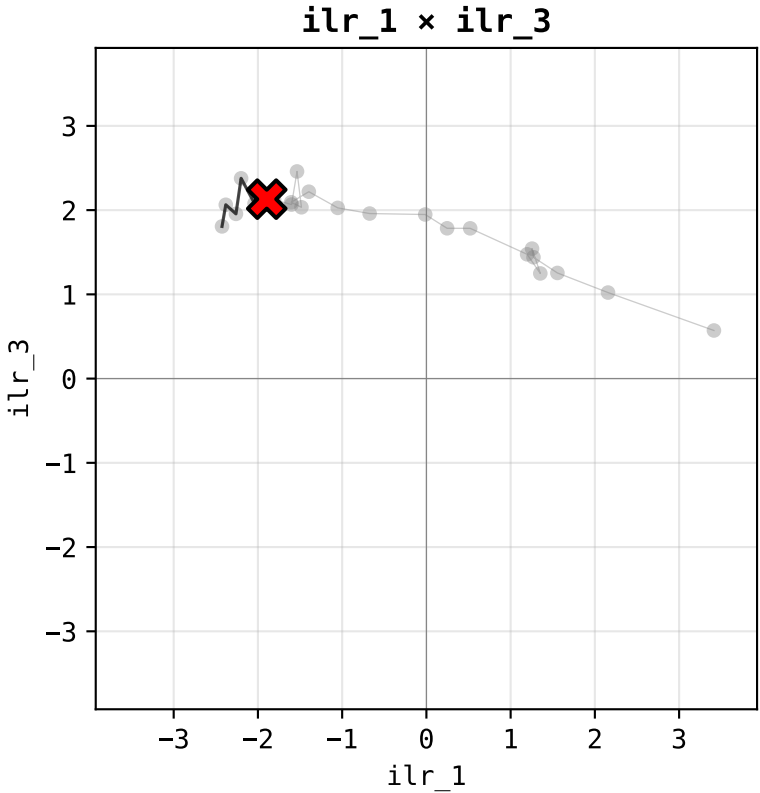
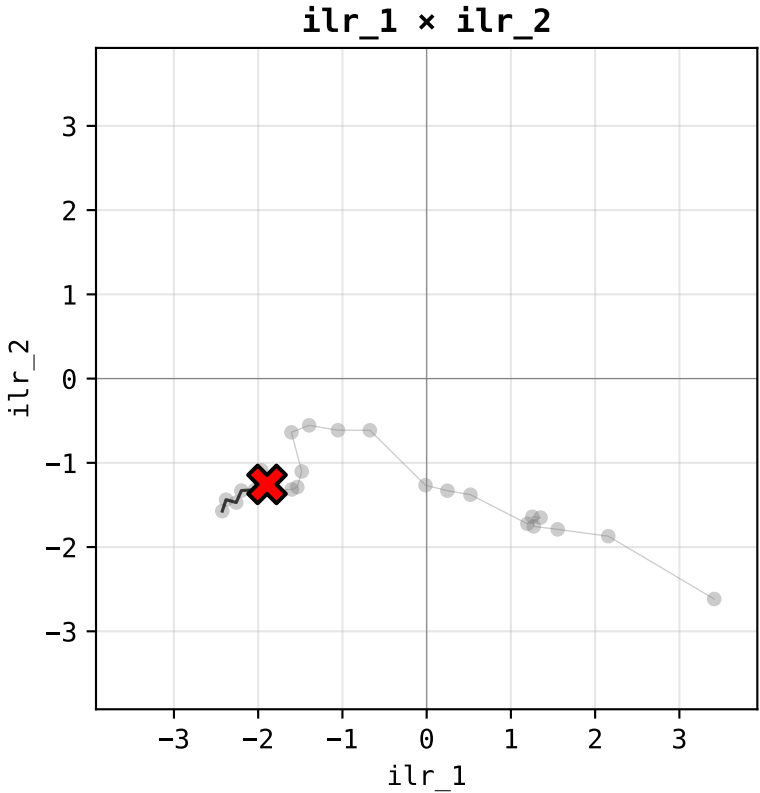


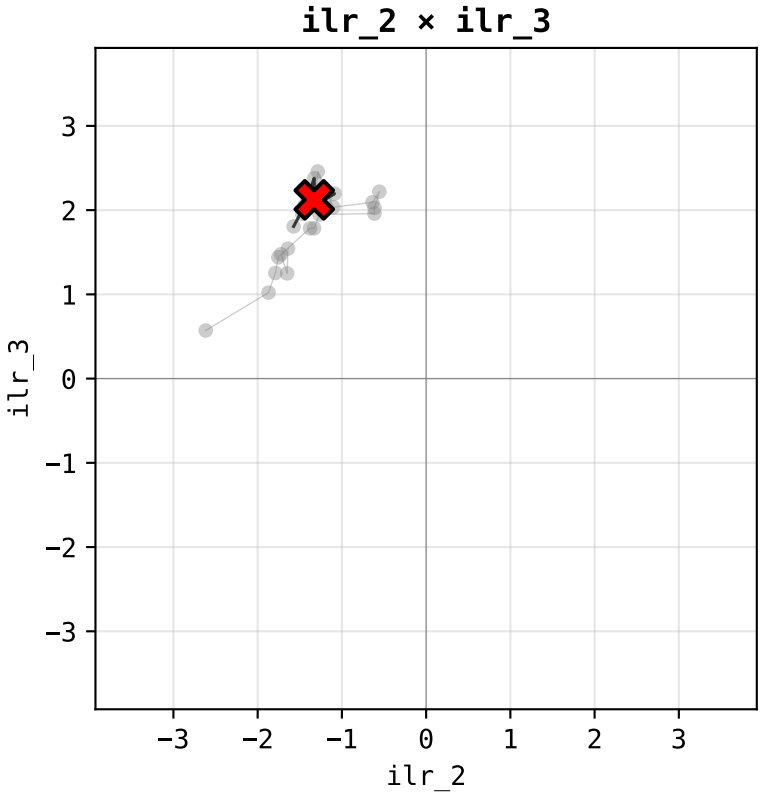
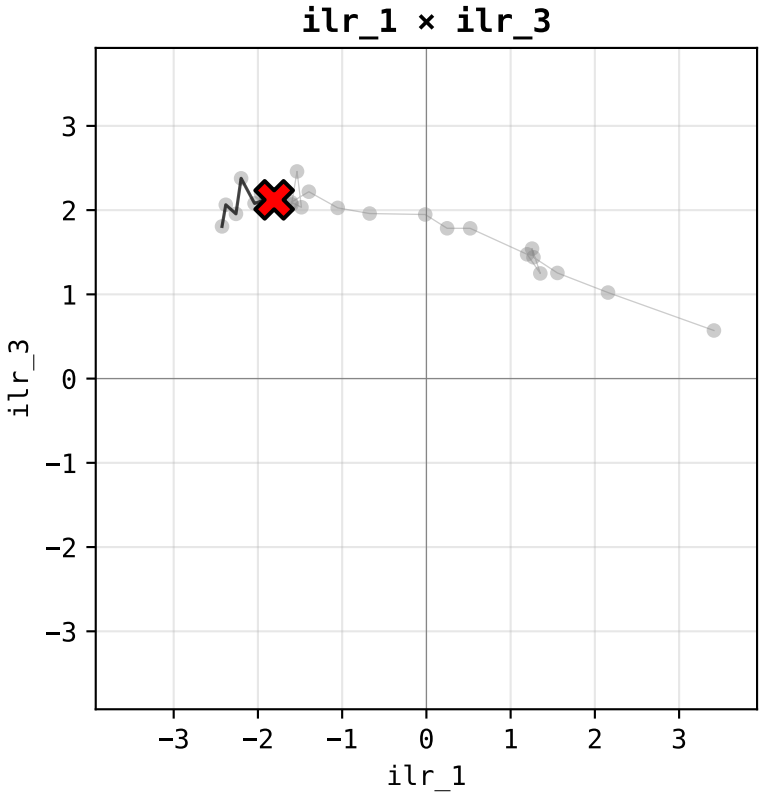
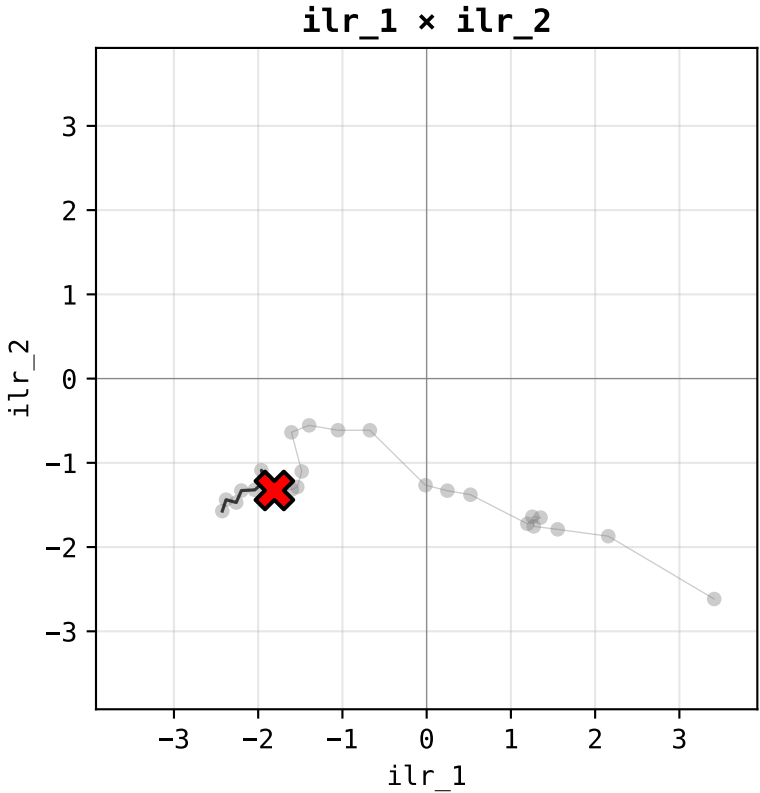




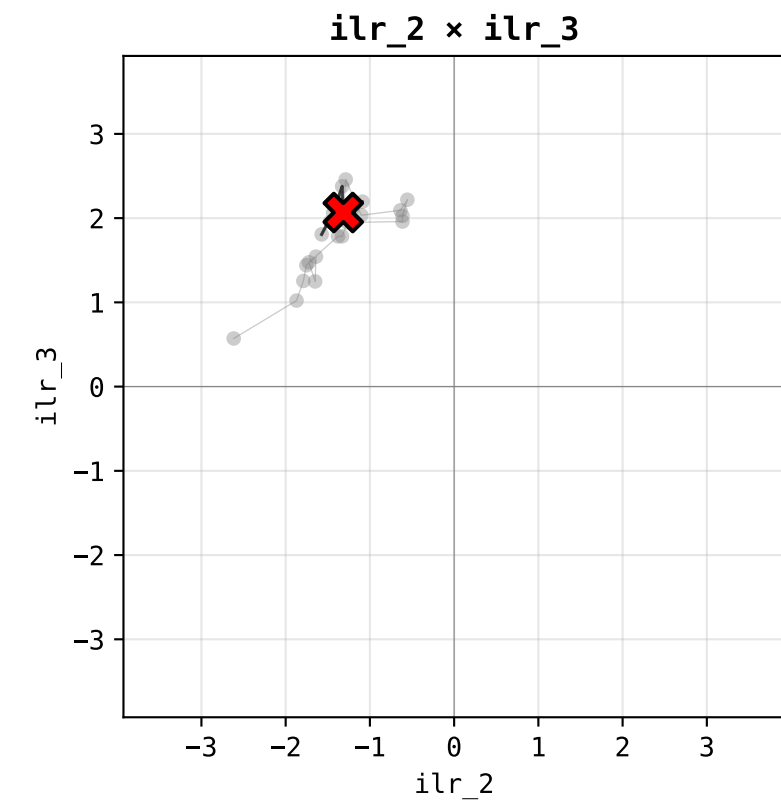
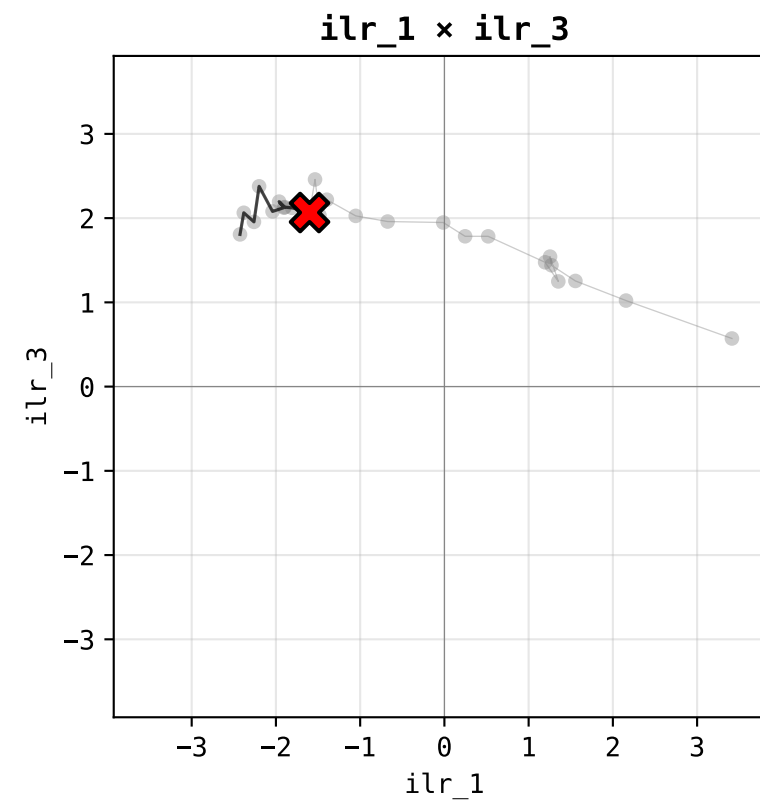
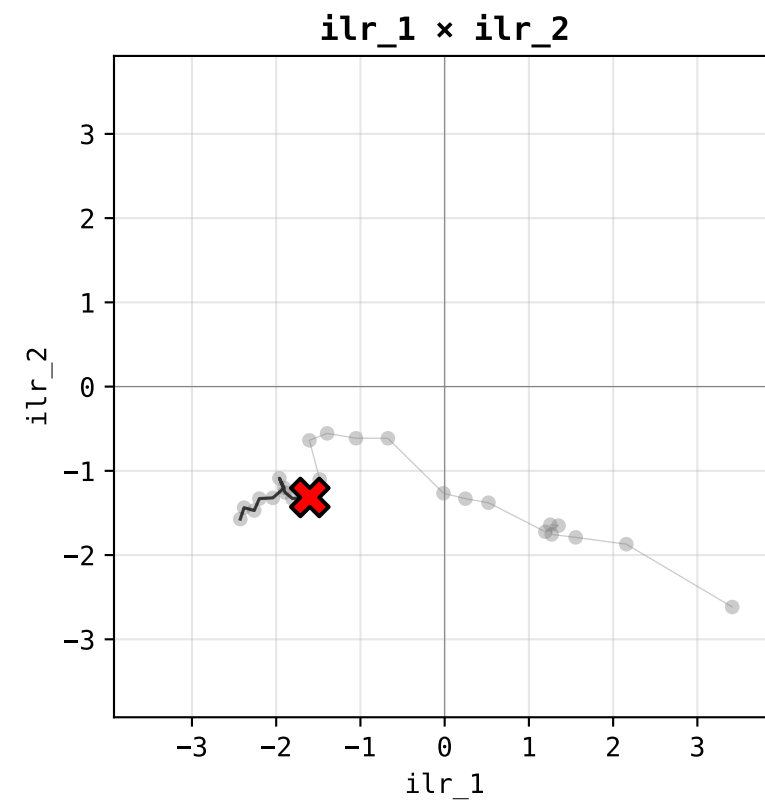






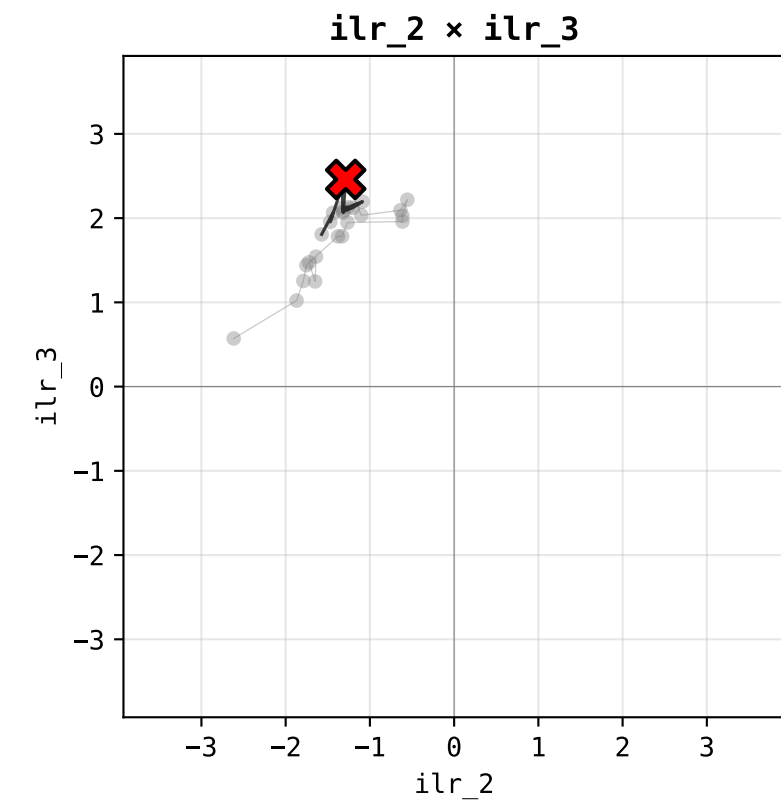
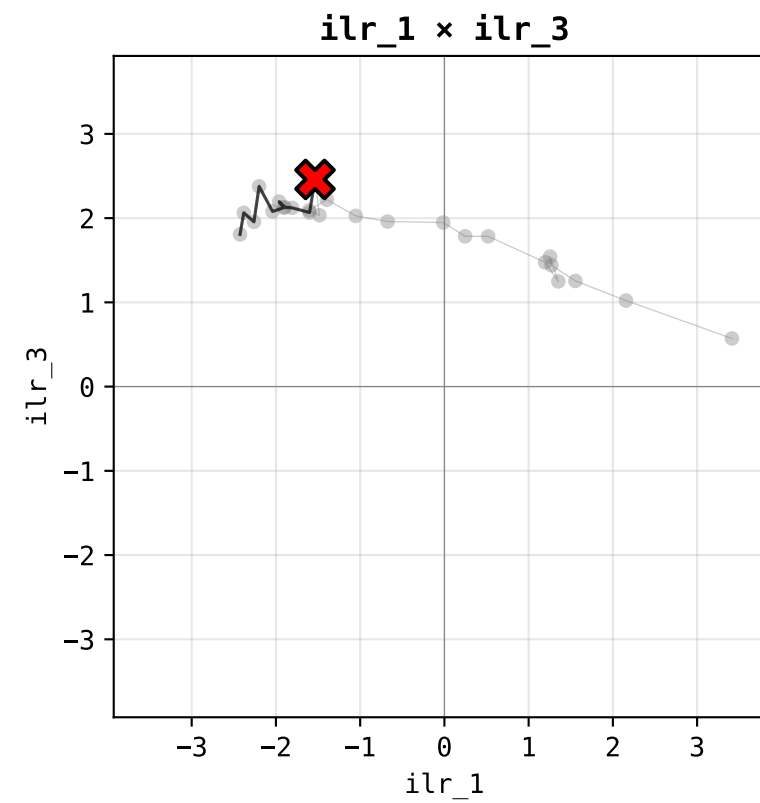
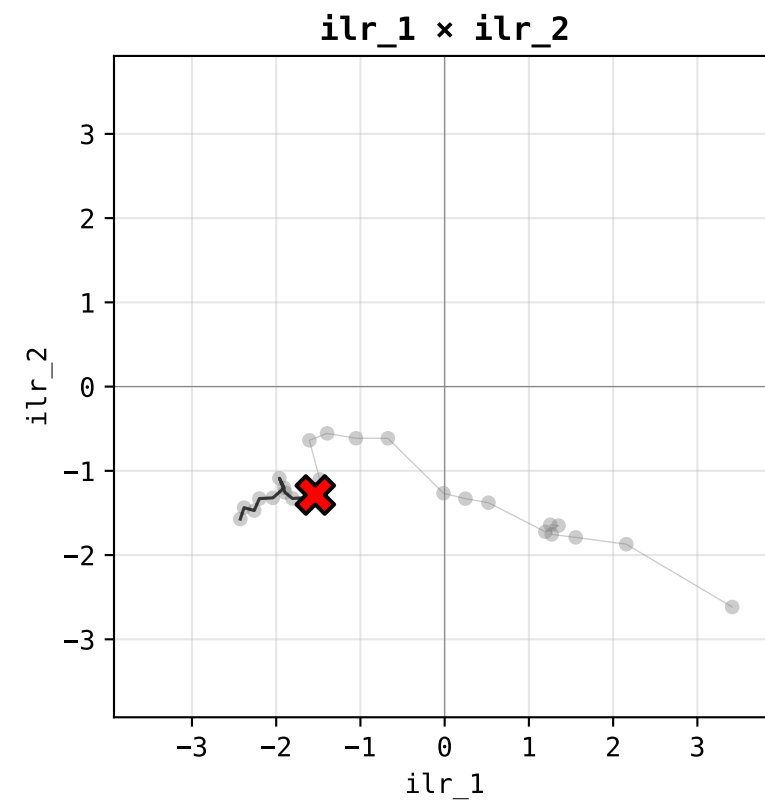


ILR Triplet – GBR United Kingdom · year 2009 (10/26)

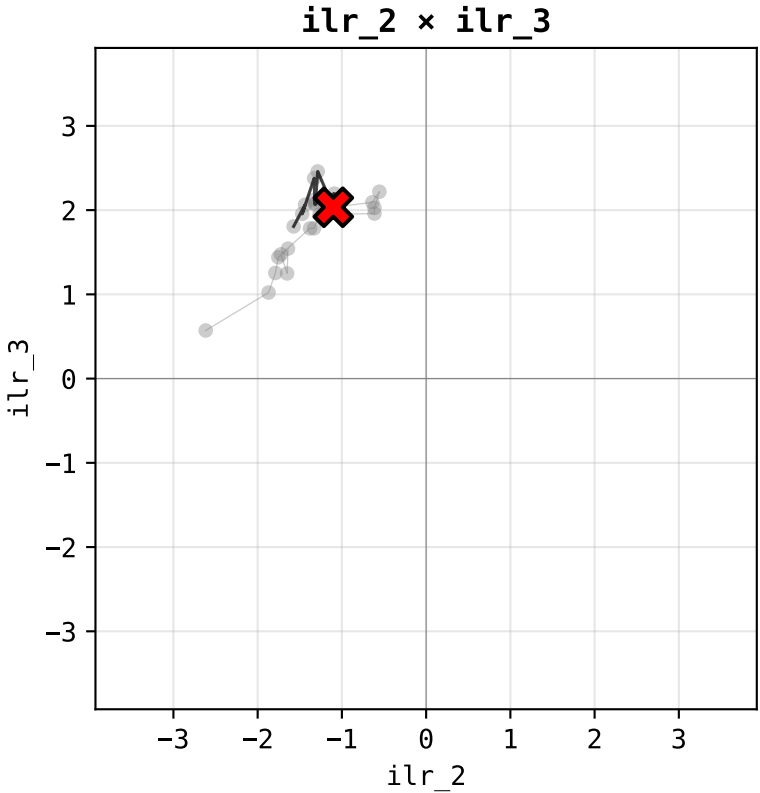
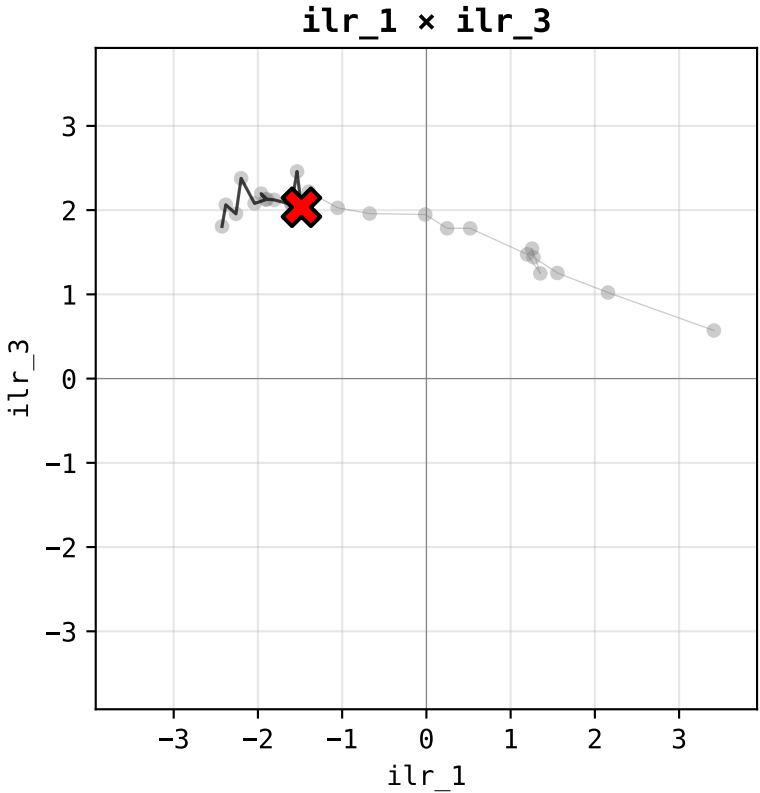
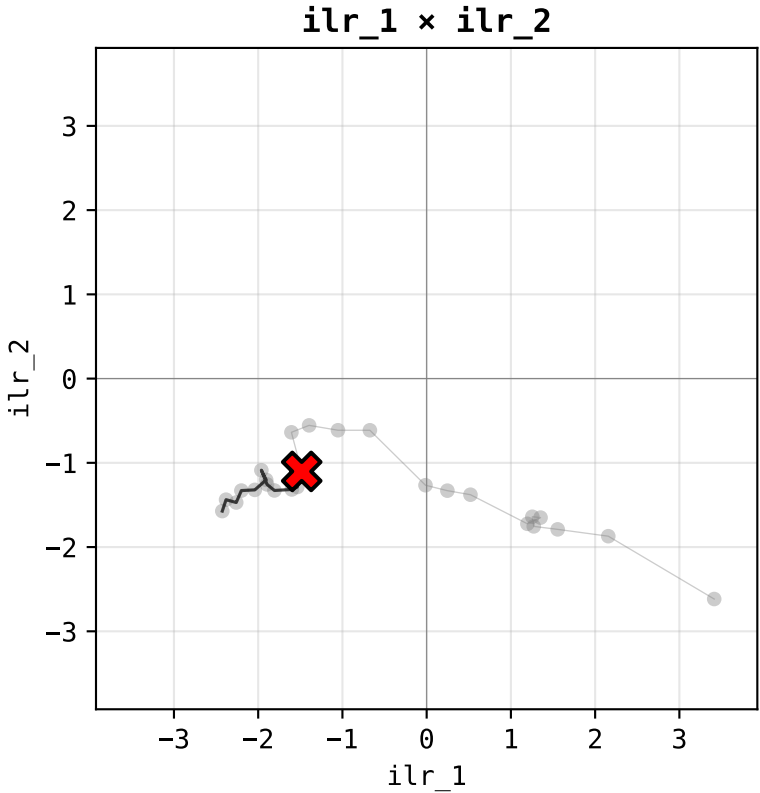


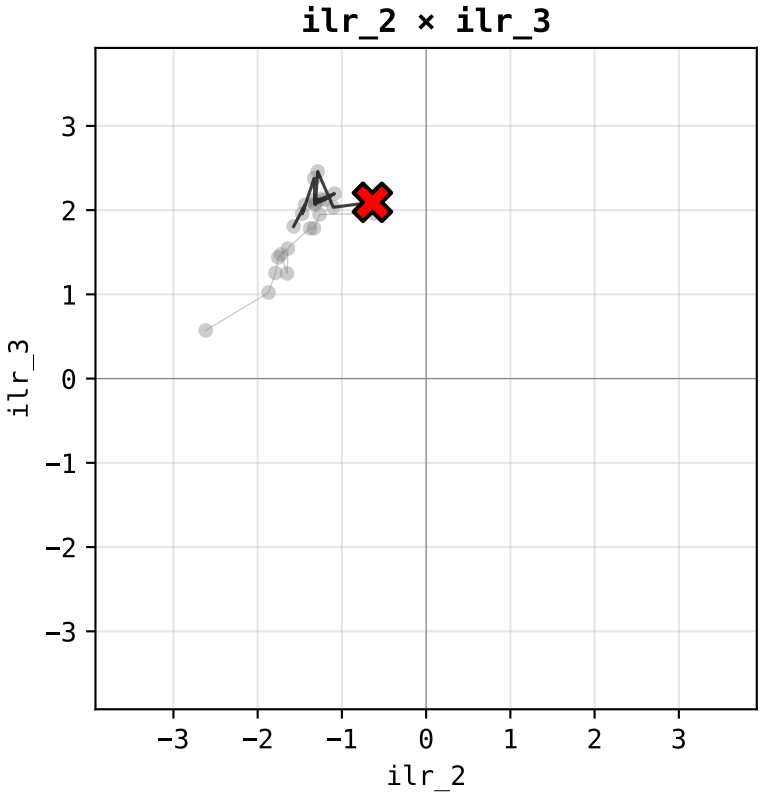
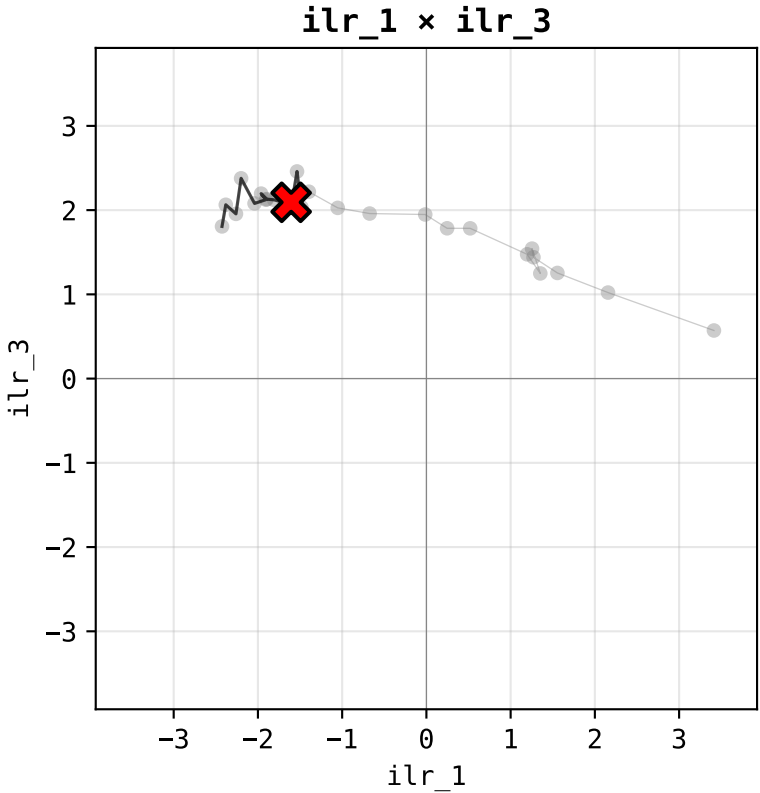
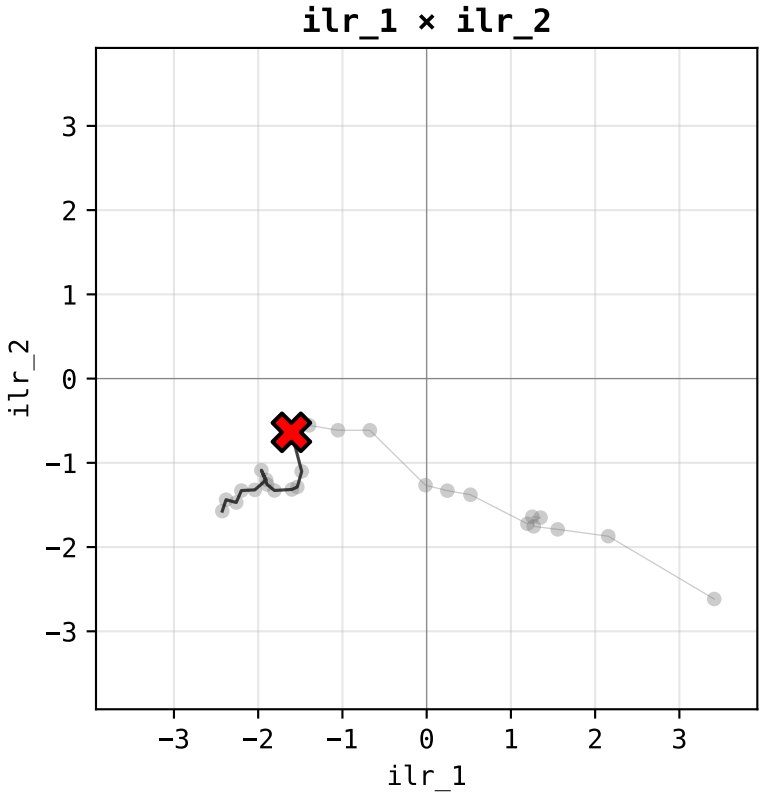
X = current year (2009) · black = past trajectory · gray = future · ilr range ±3.93

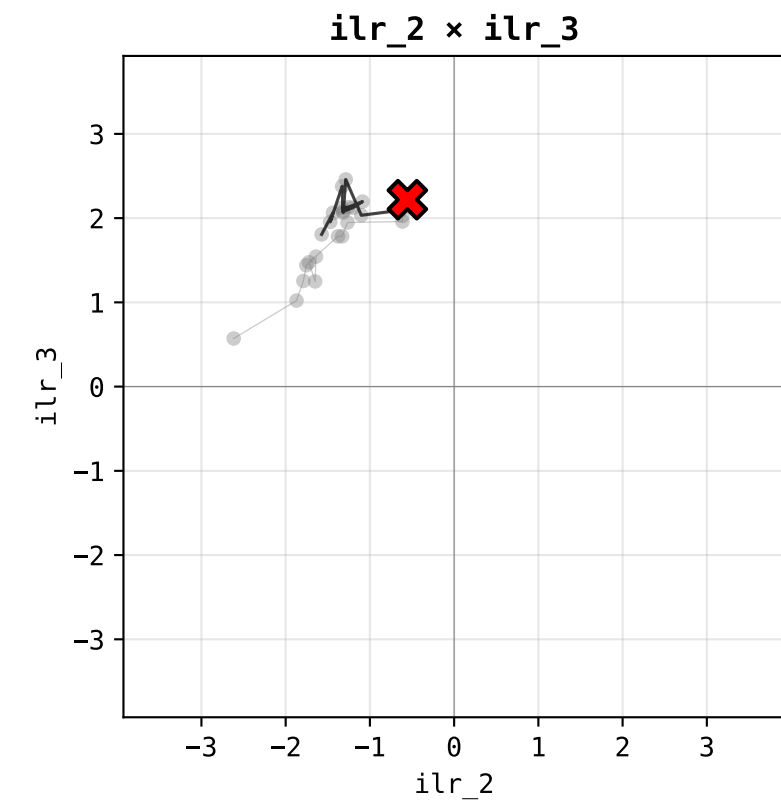
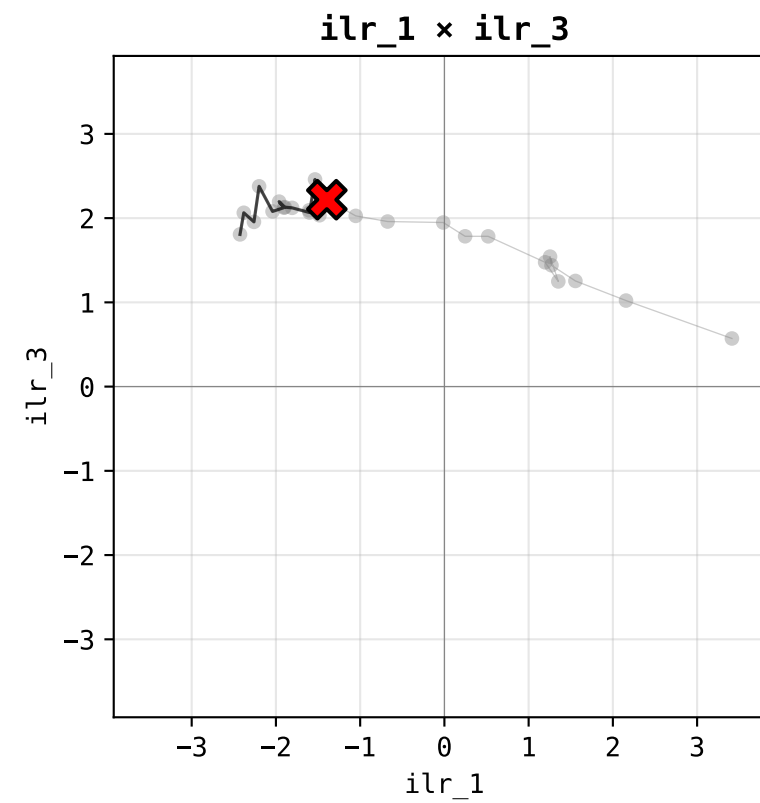
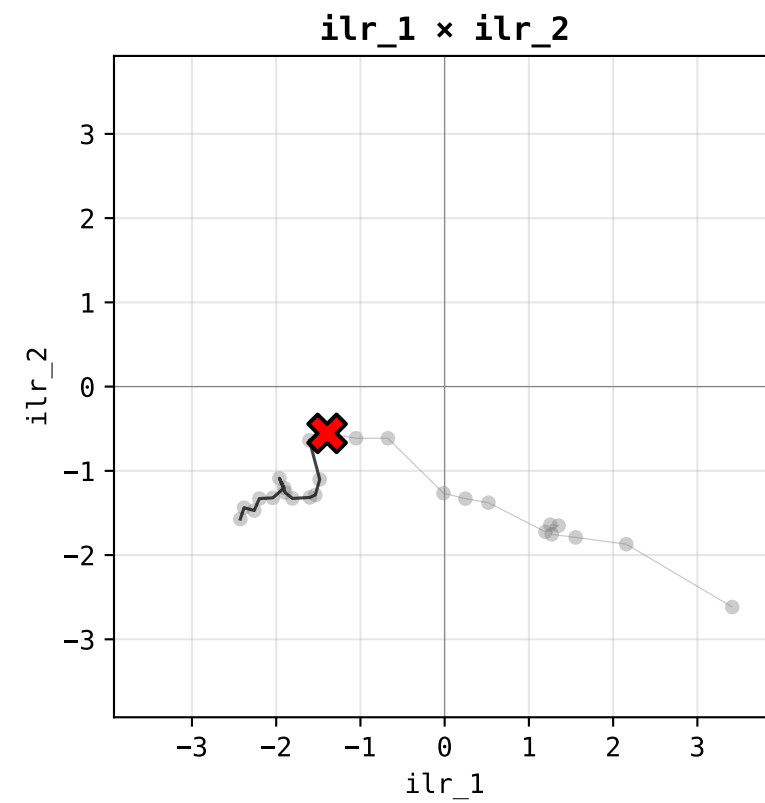
ILR Triplet – GBR United Kingdom · year 2010 (11/26)

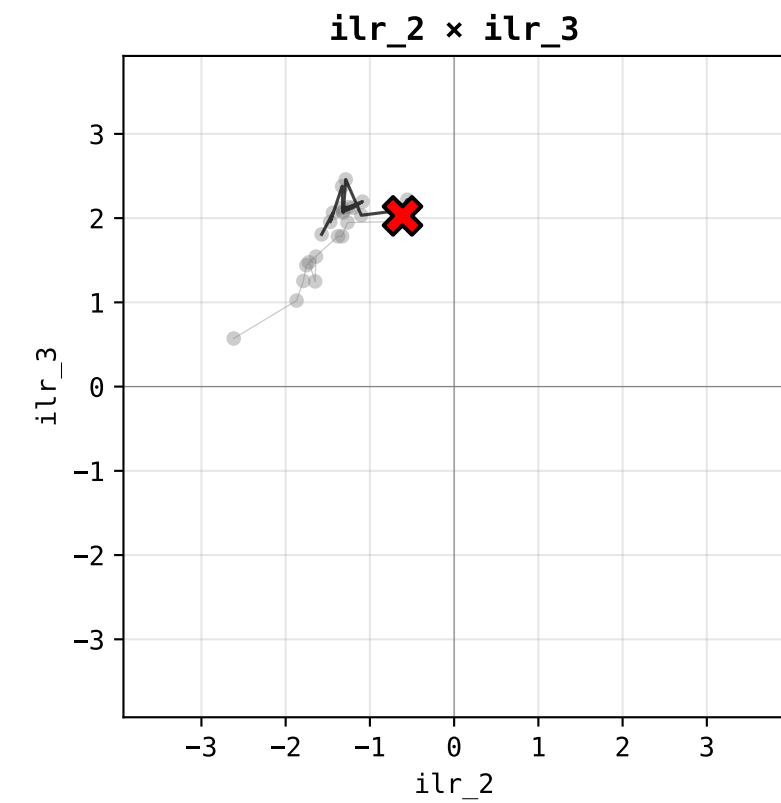
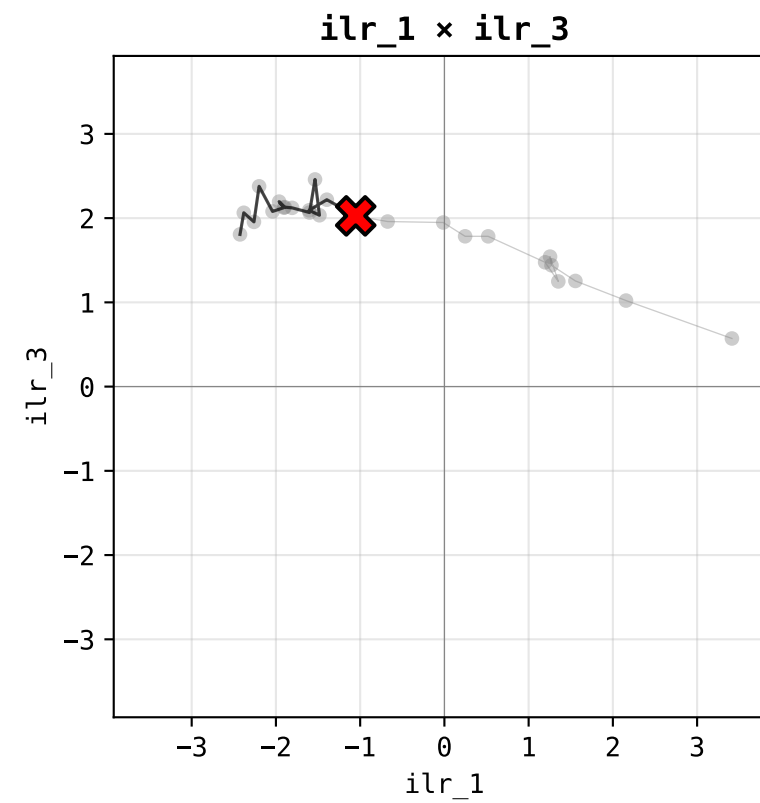
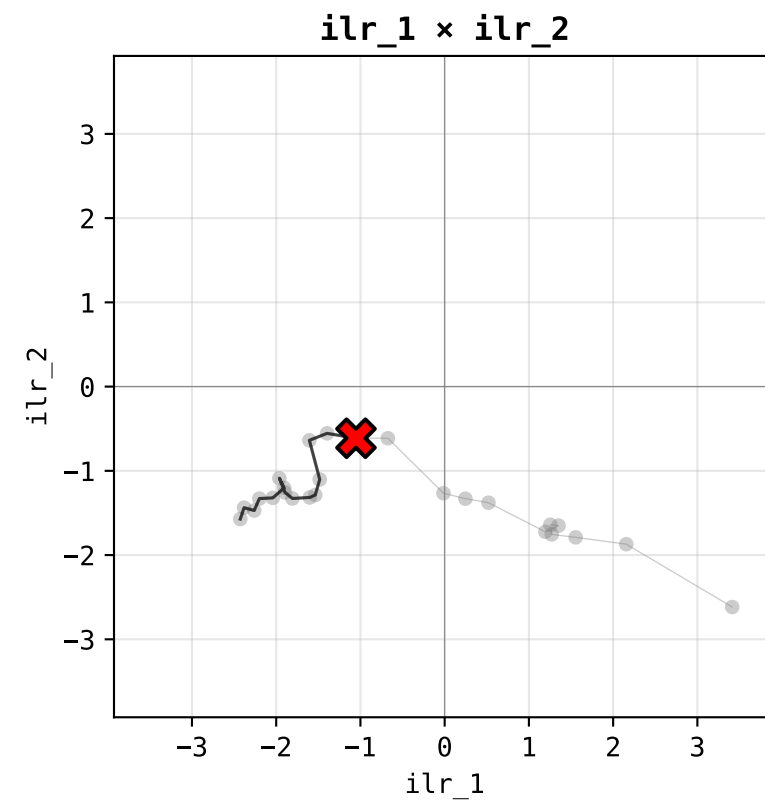


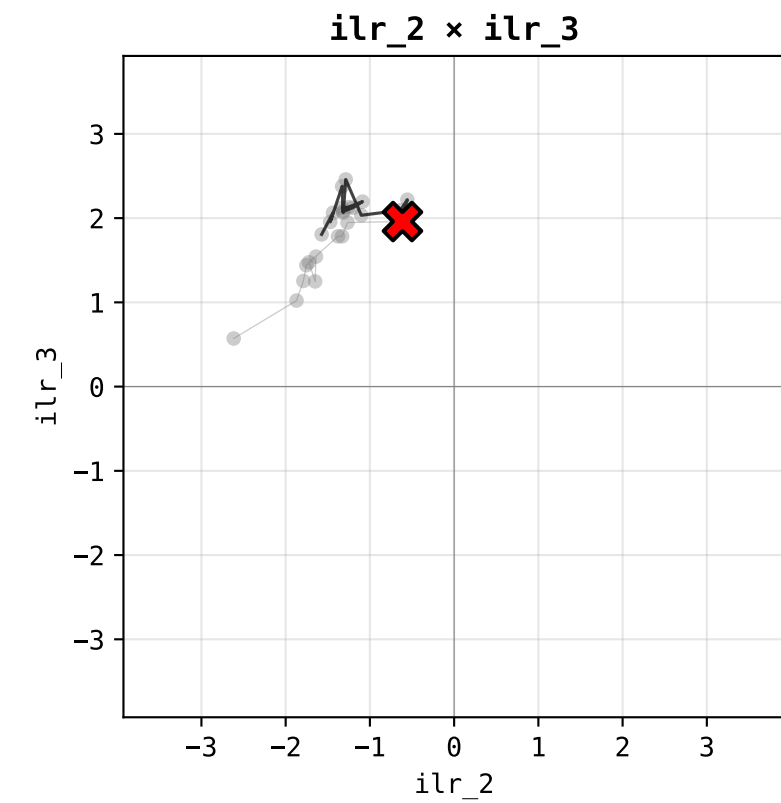
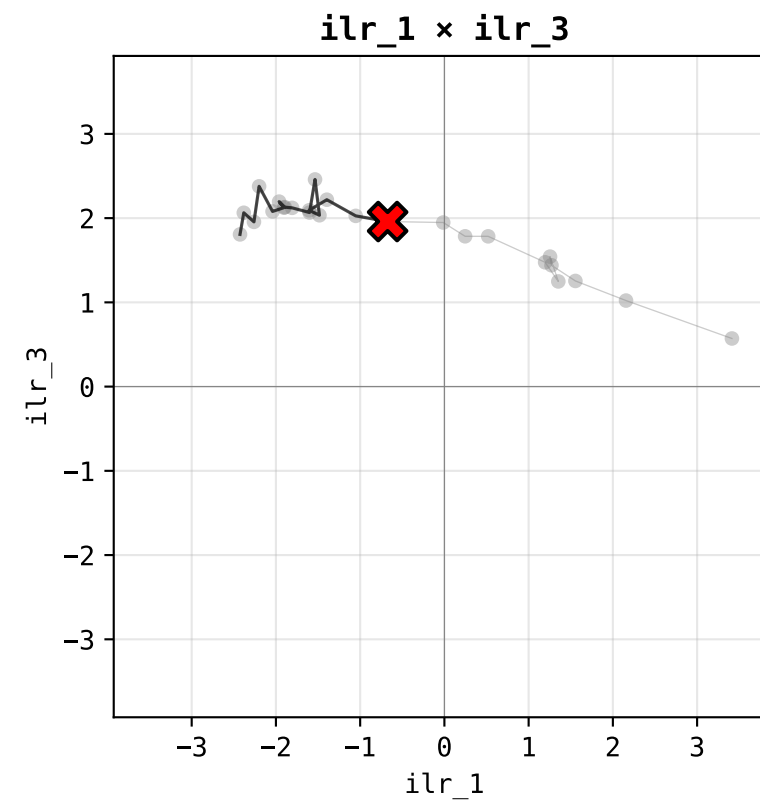
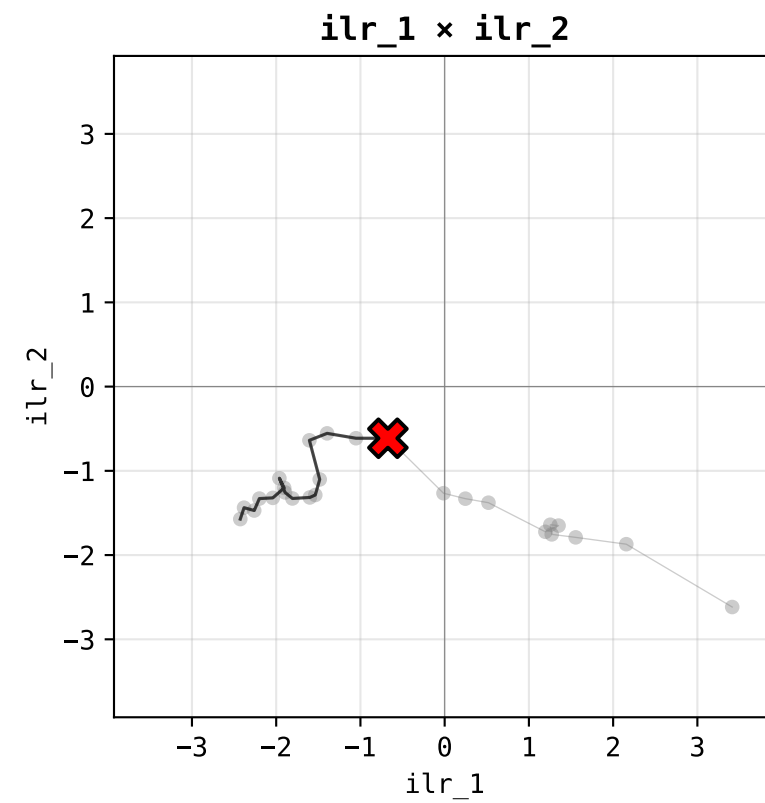
X = current year (2010) · black = past trajectory · gray = future · ilr range ±3.93

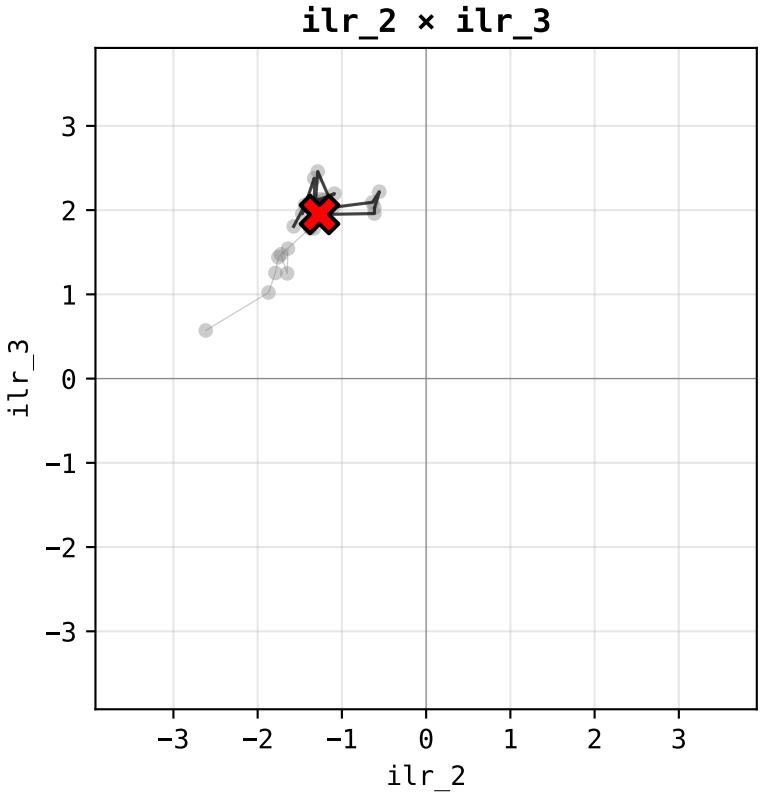
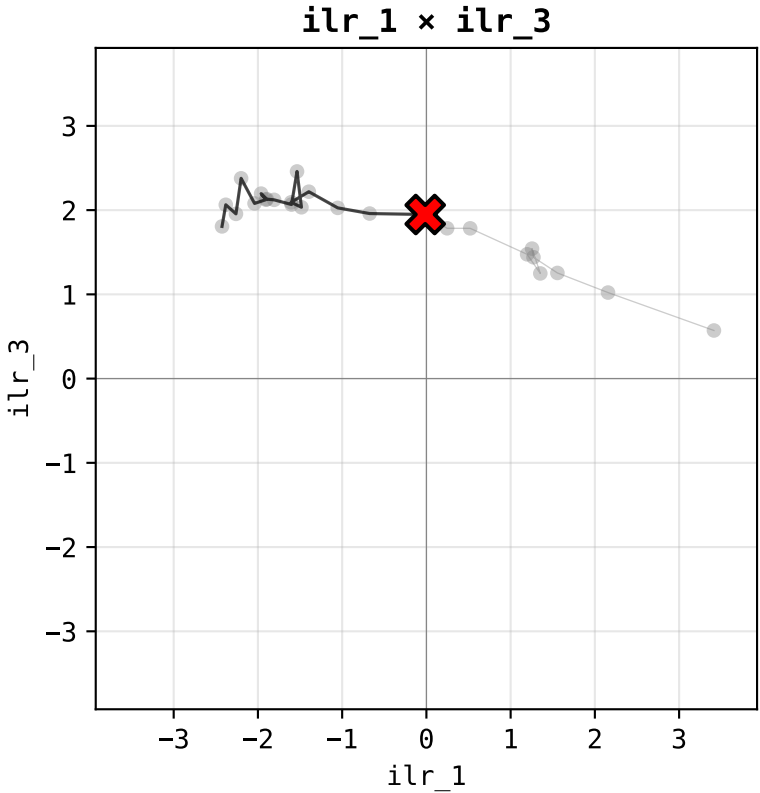
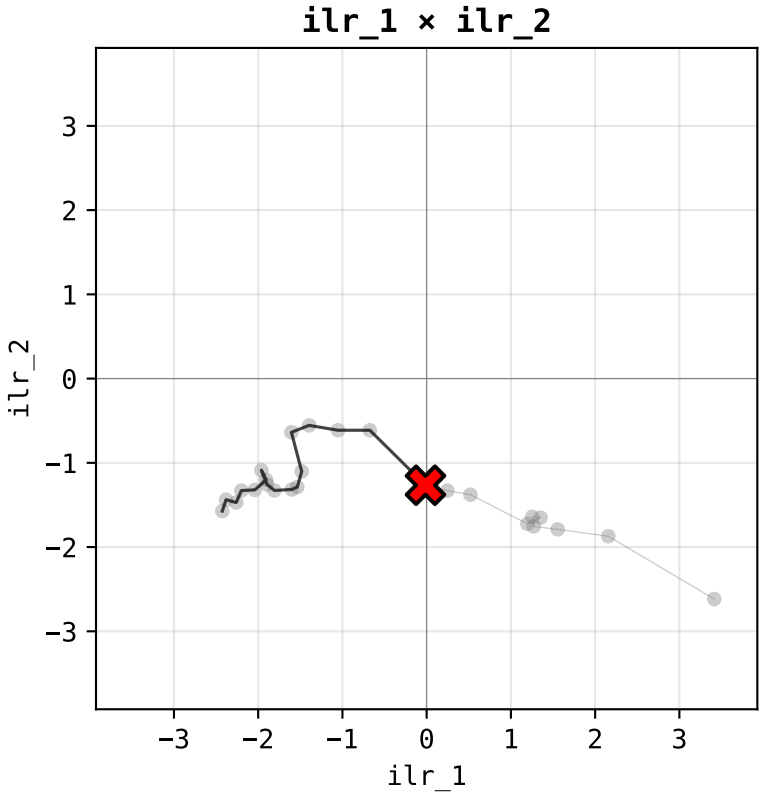


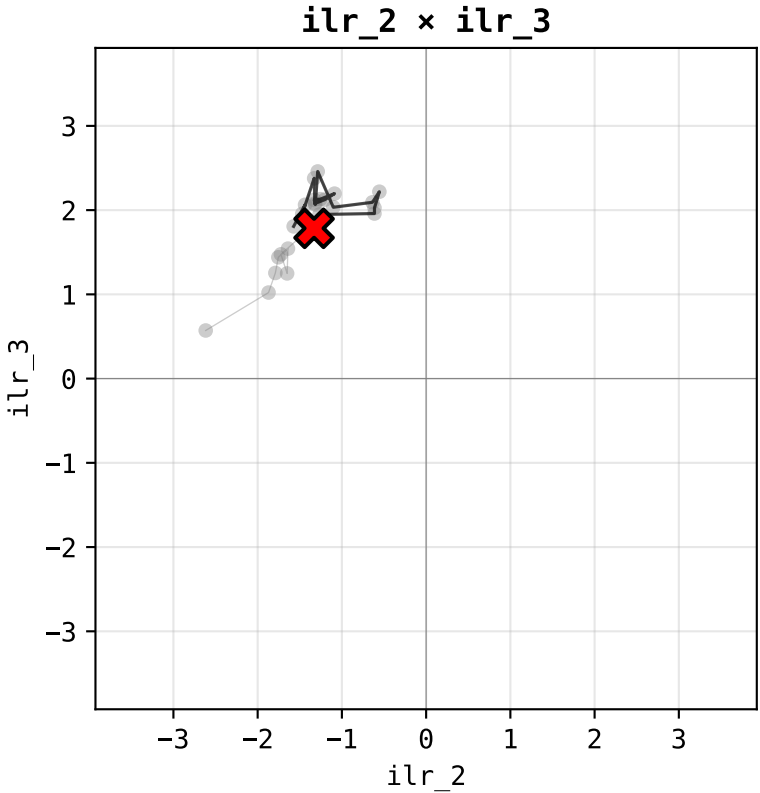
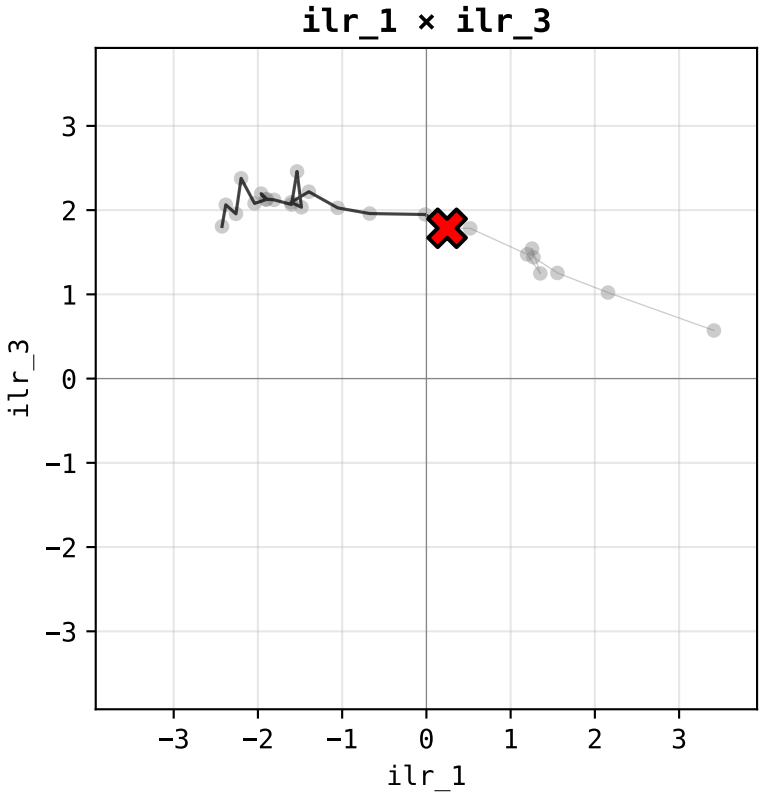
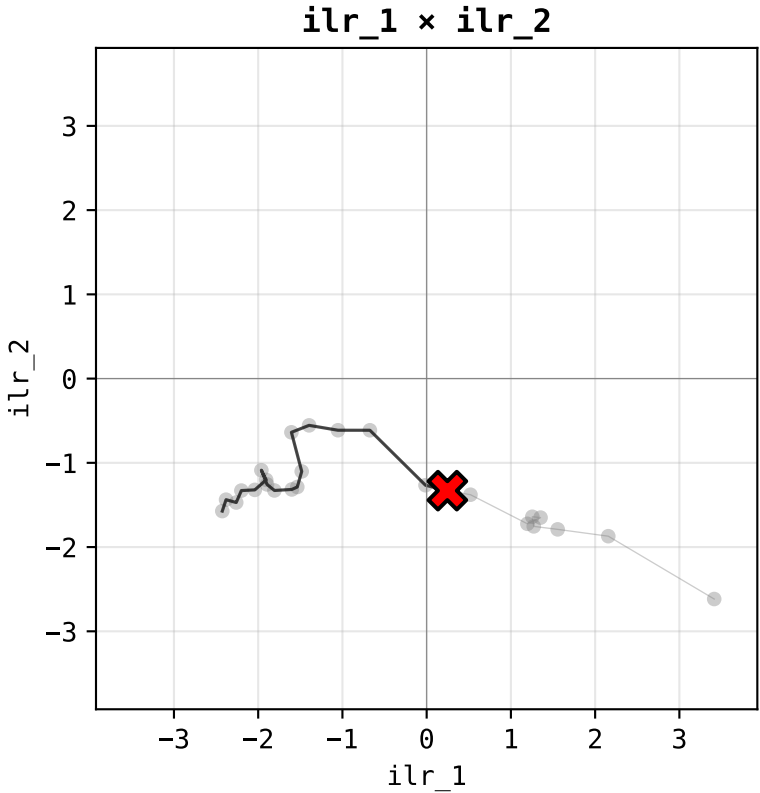


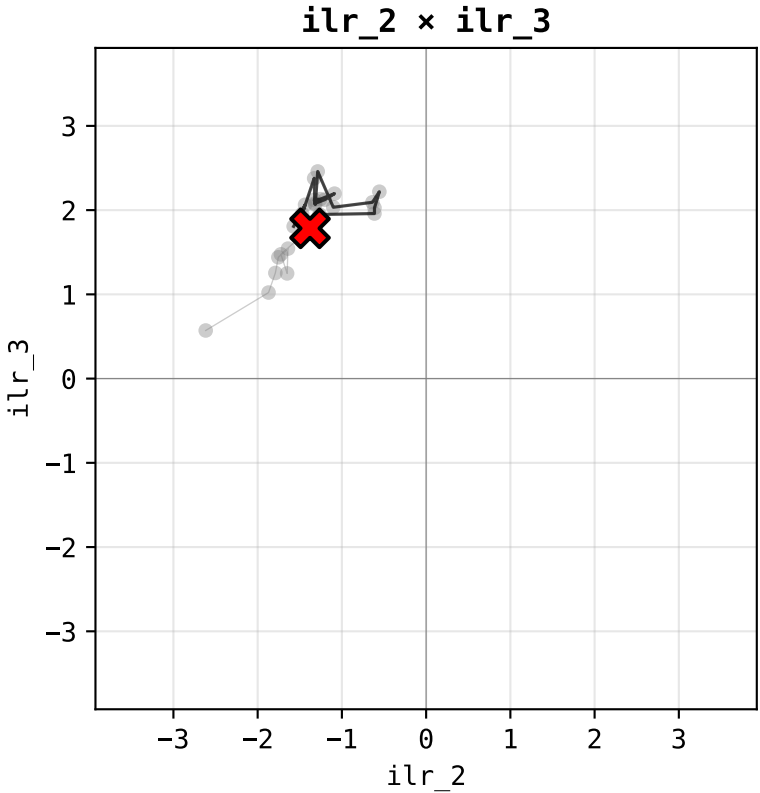
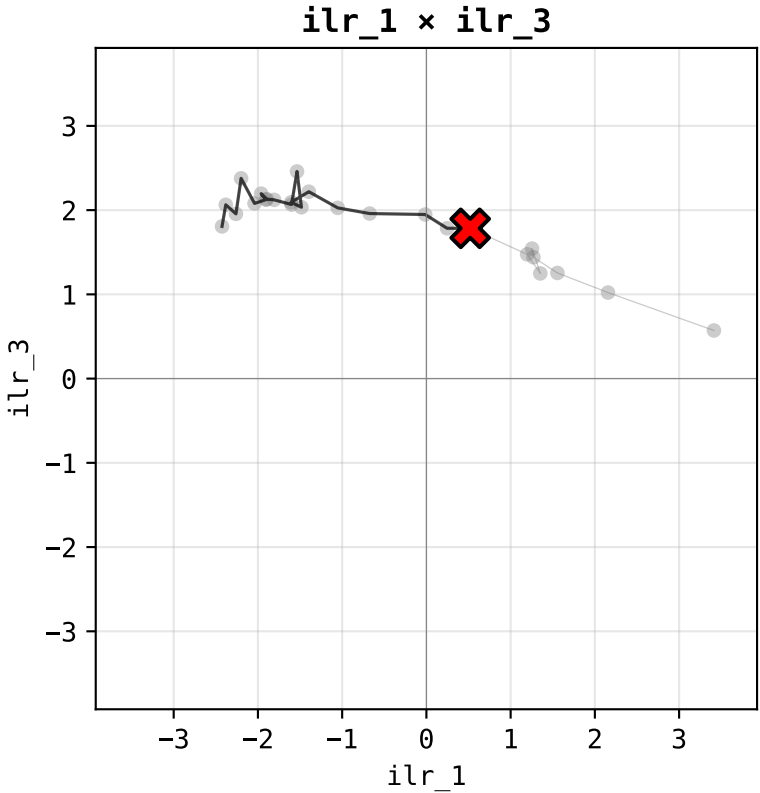
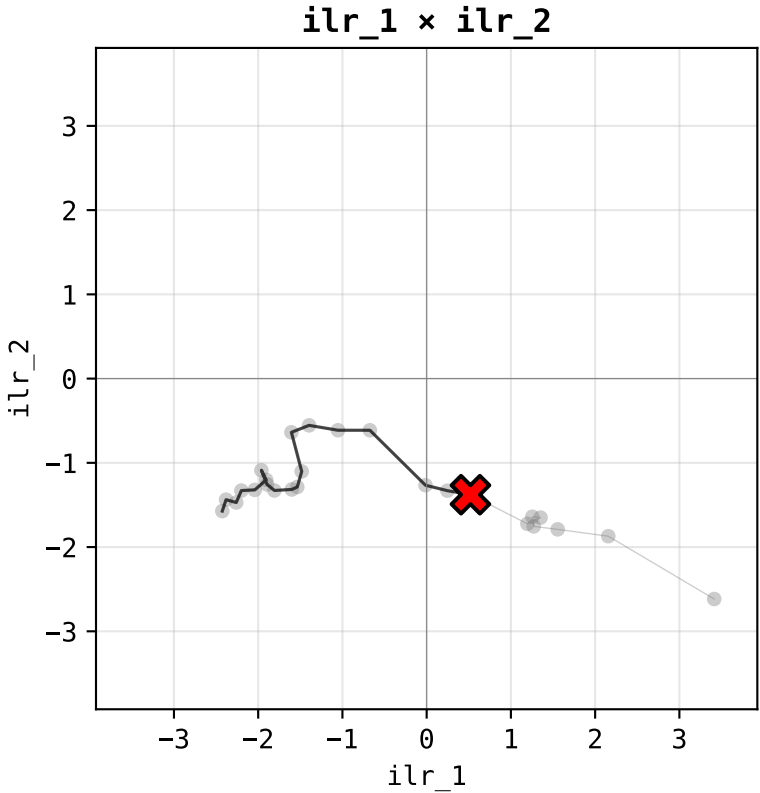


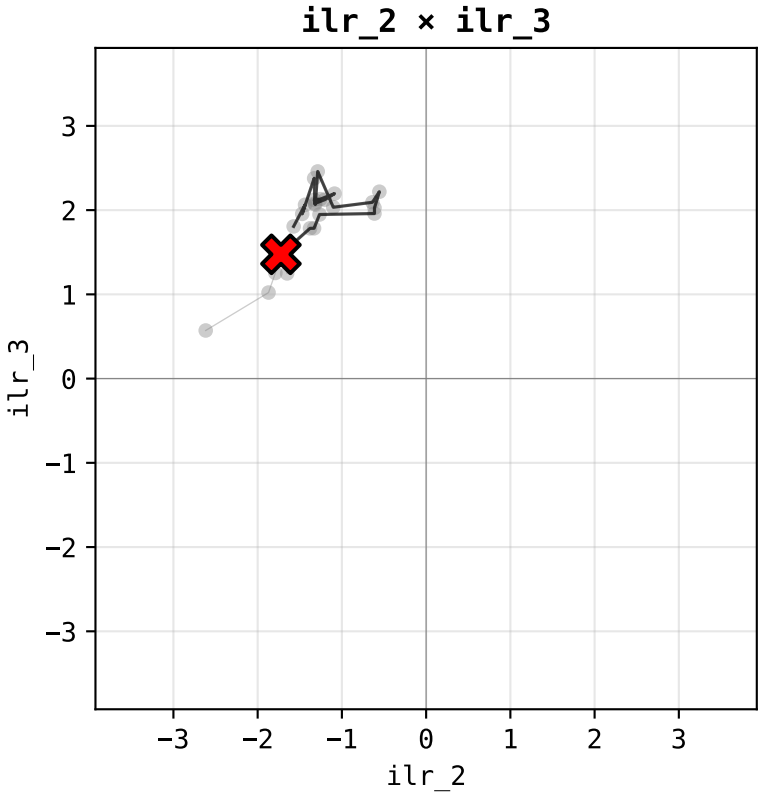
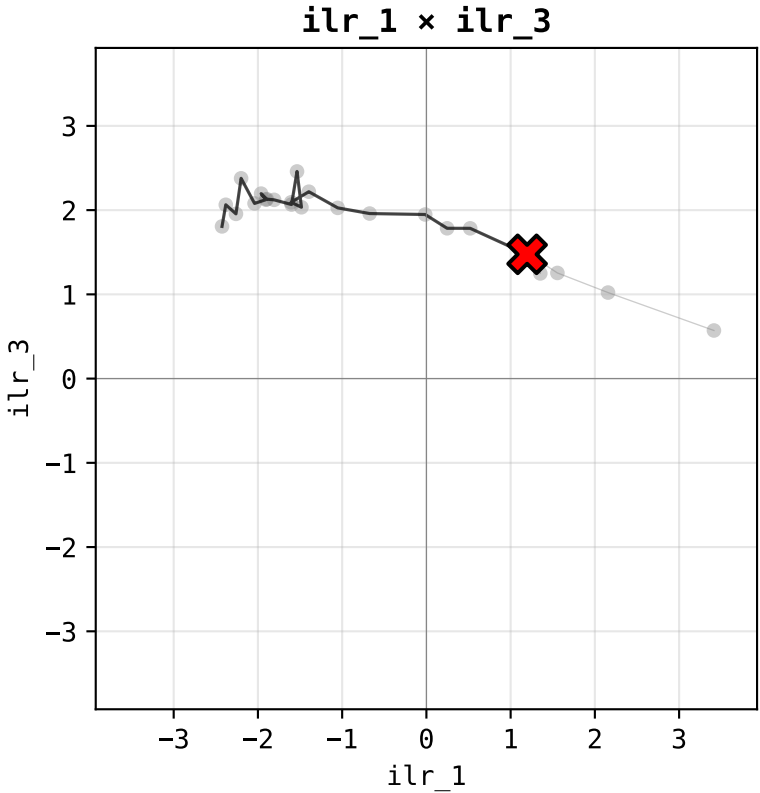
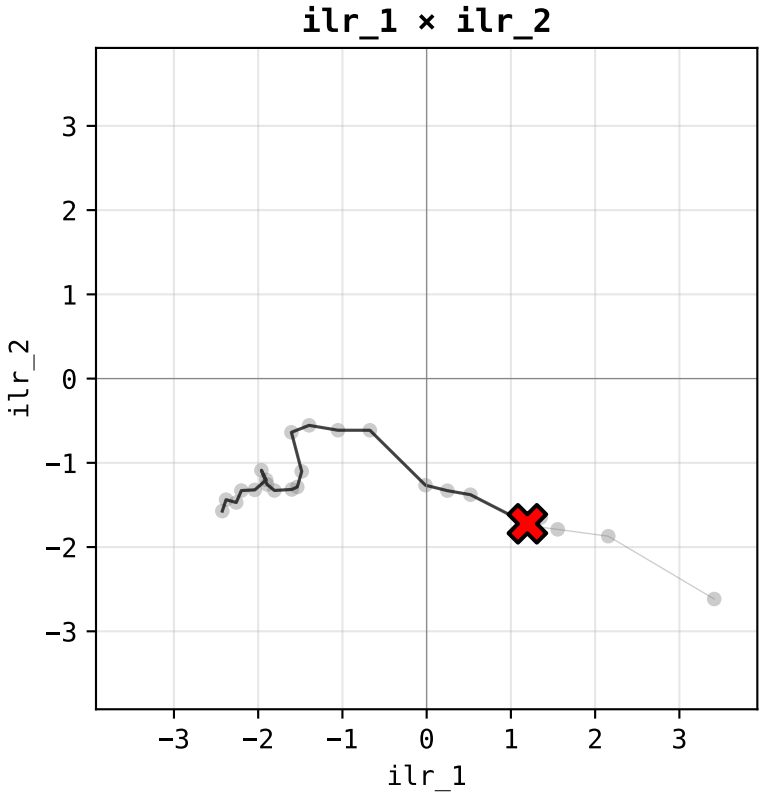


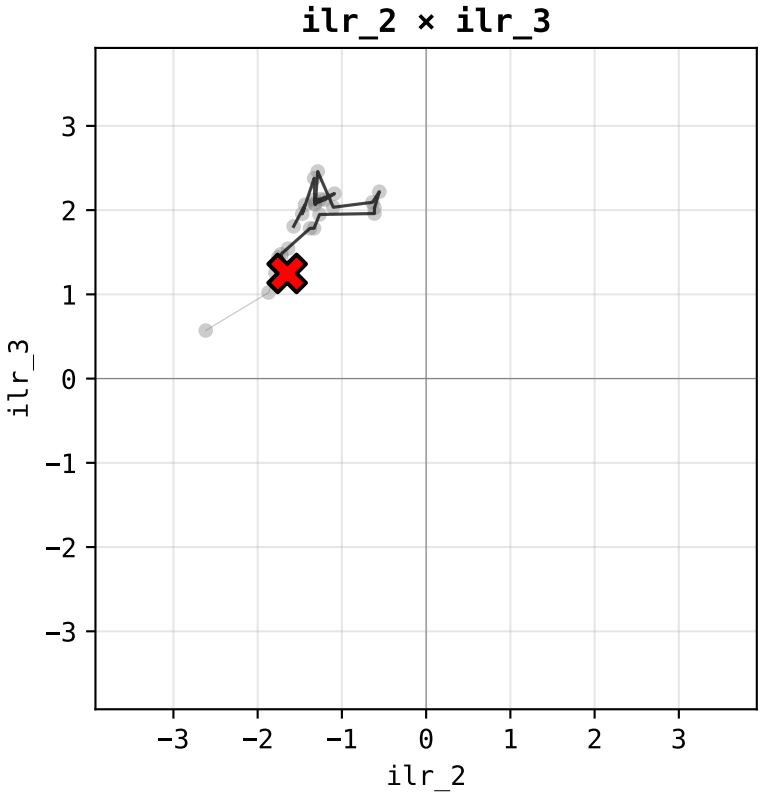
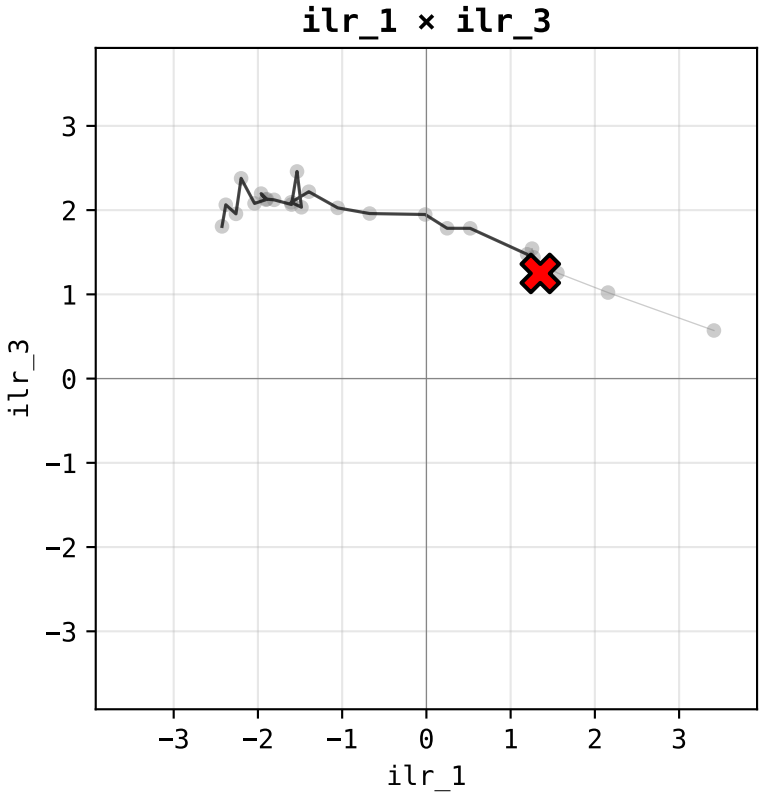
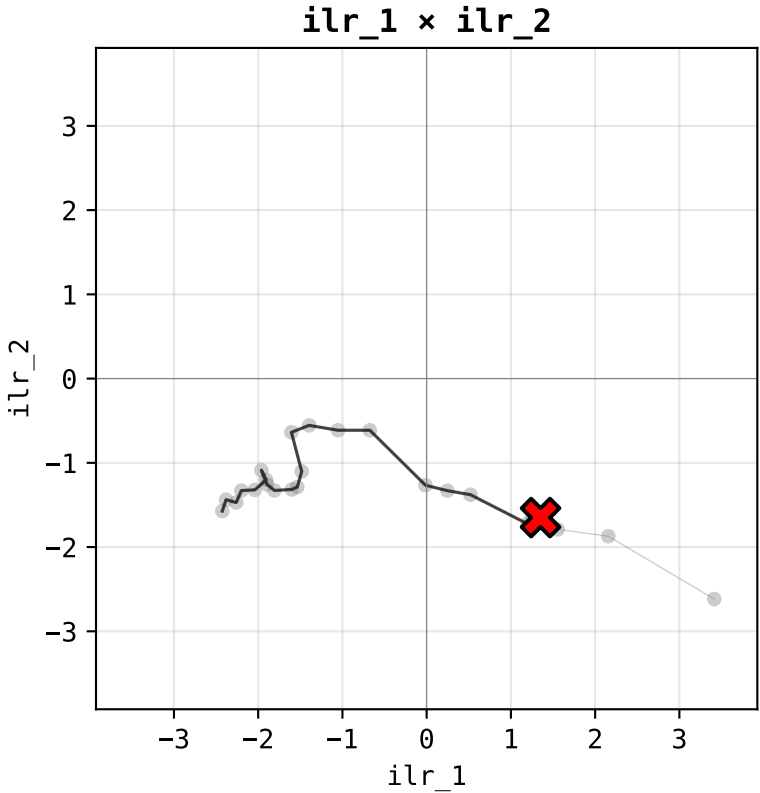


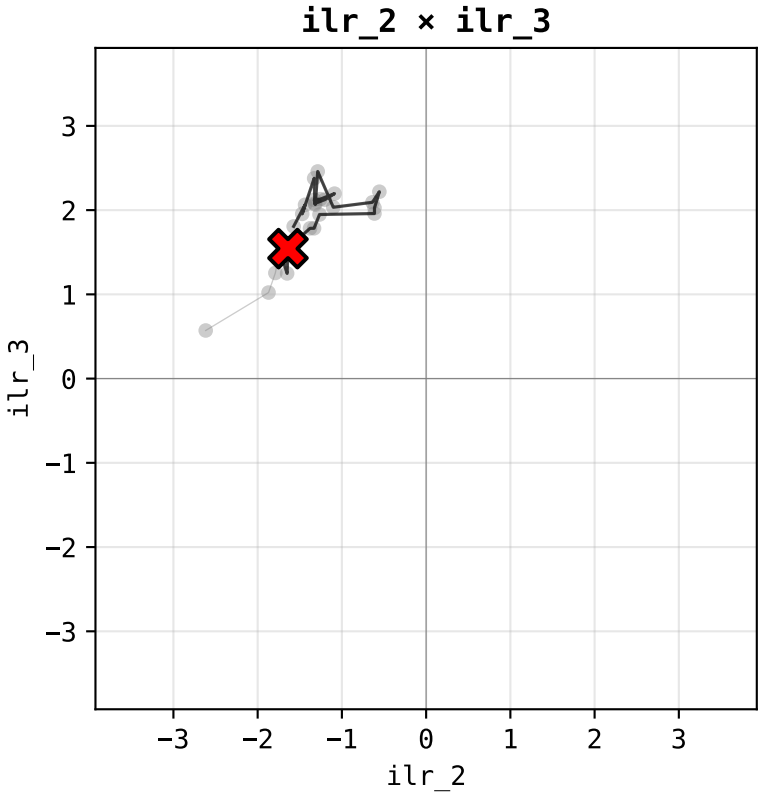
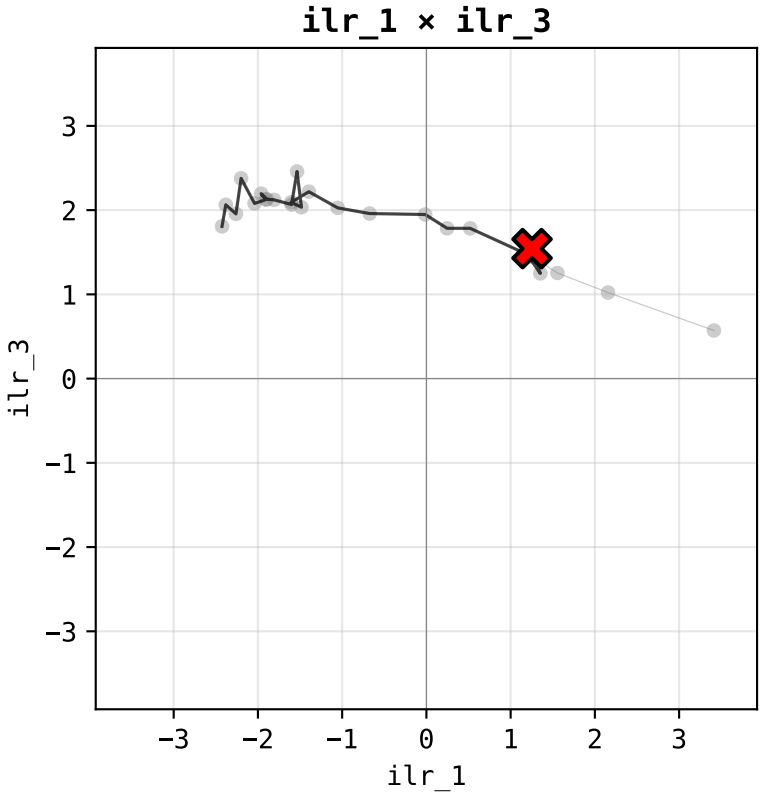
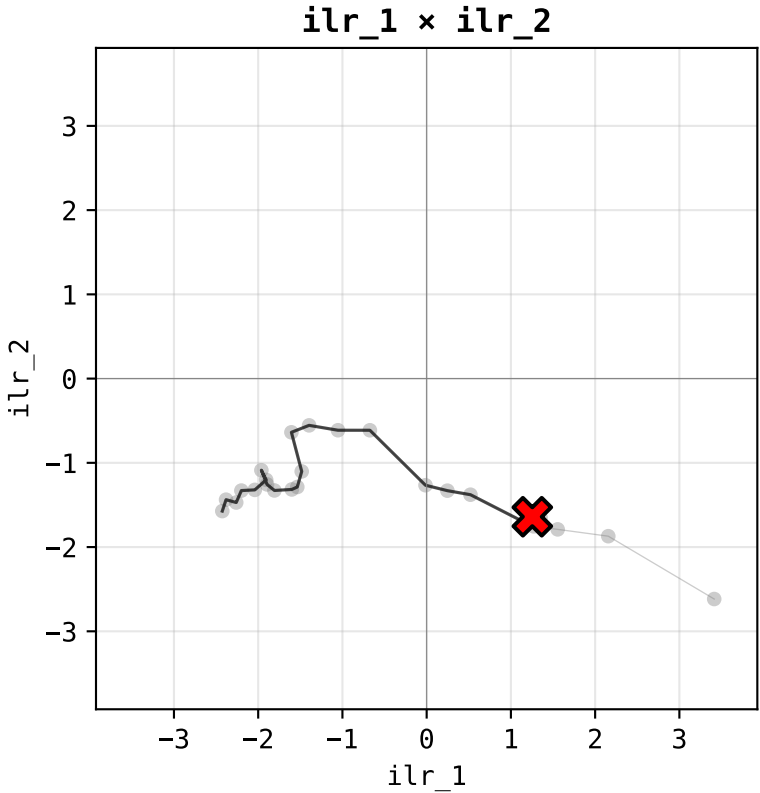


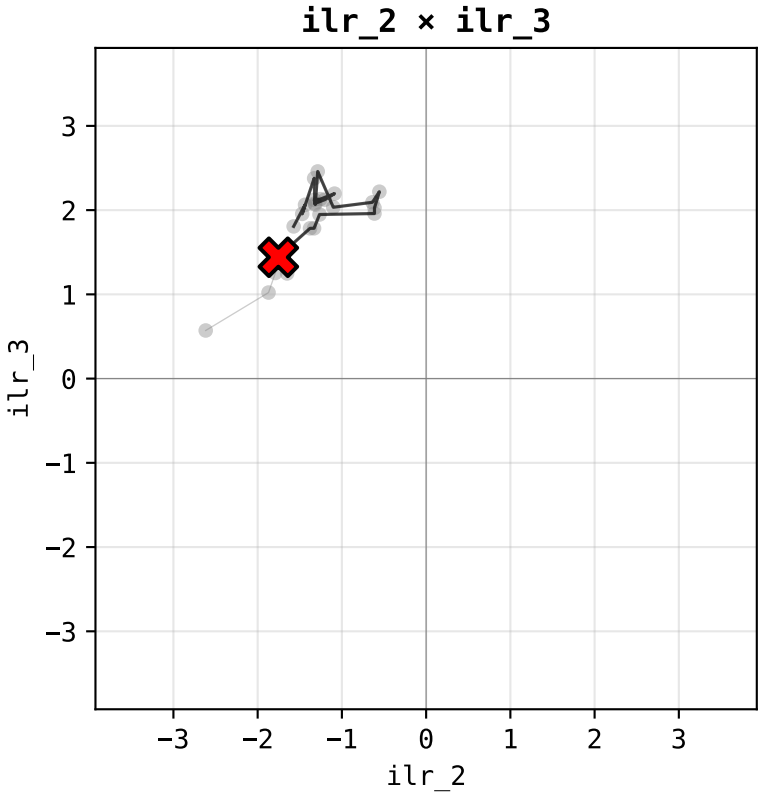
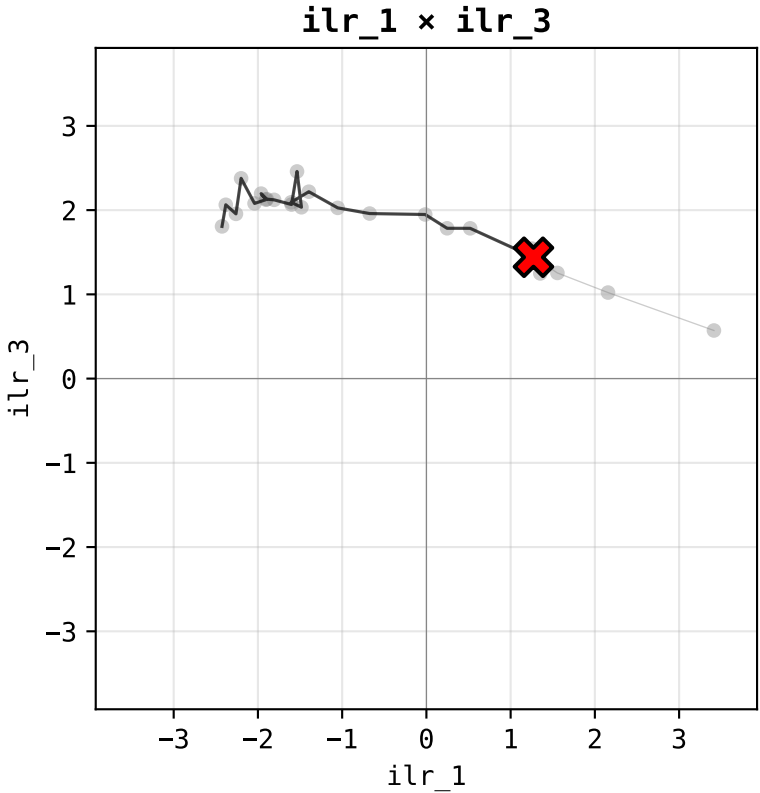
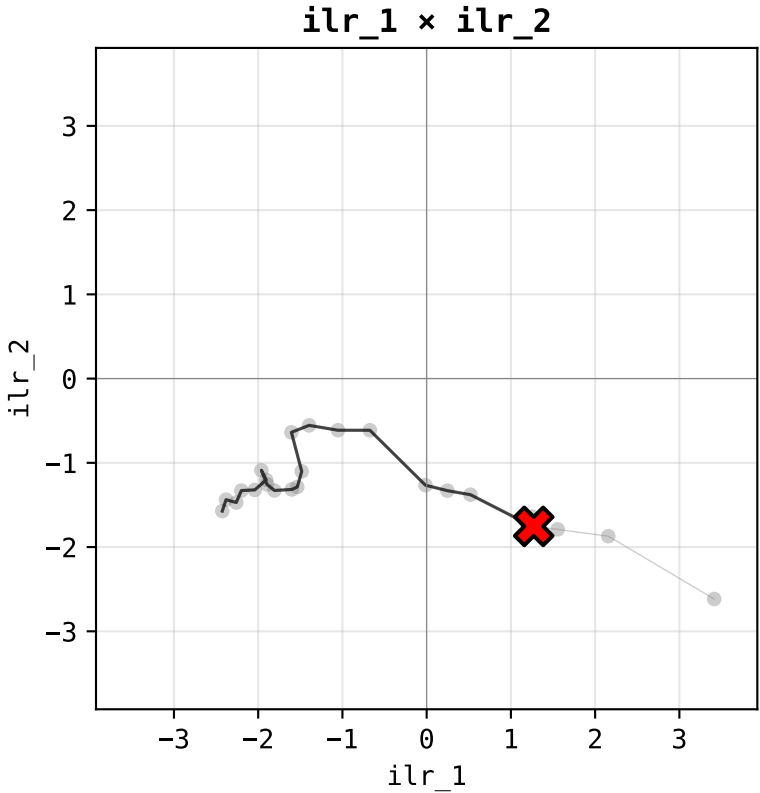


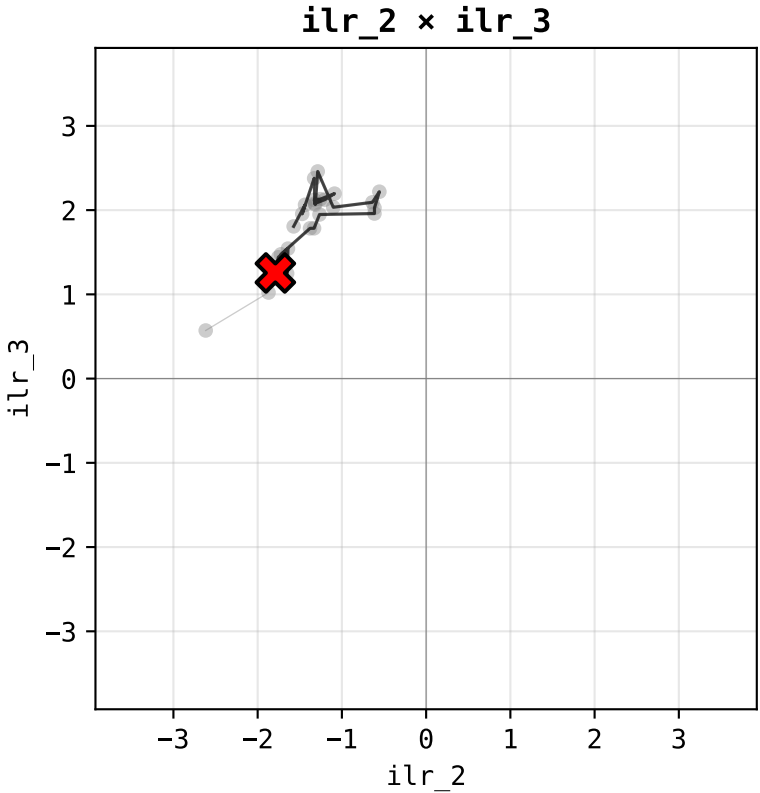
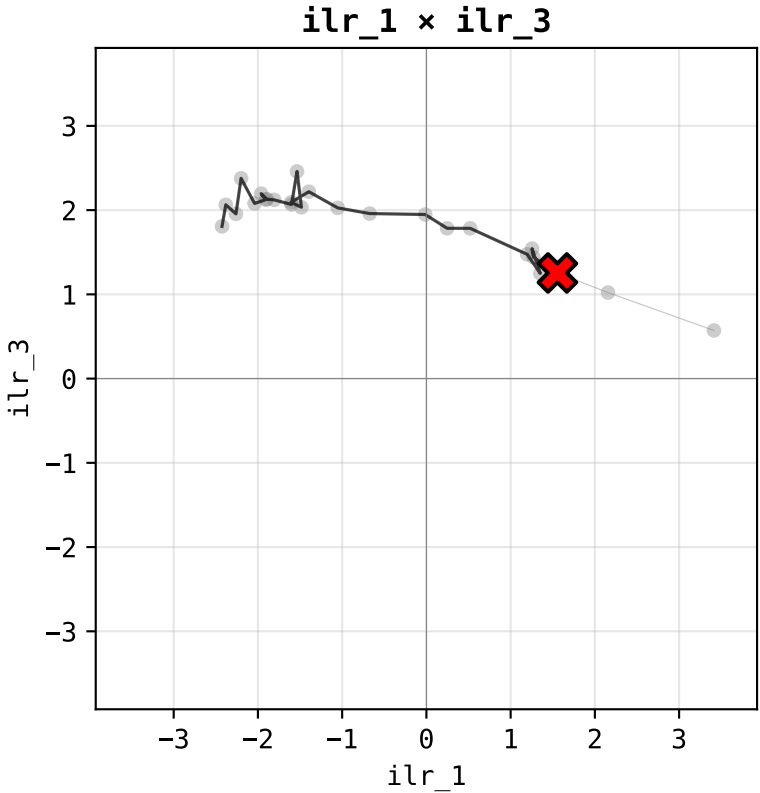
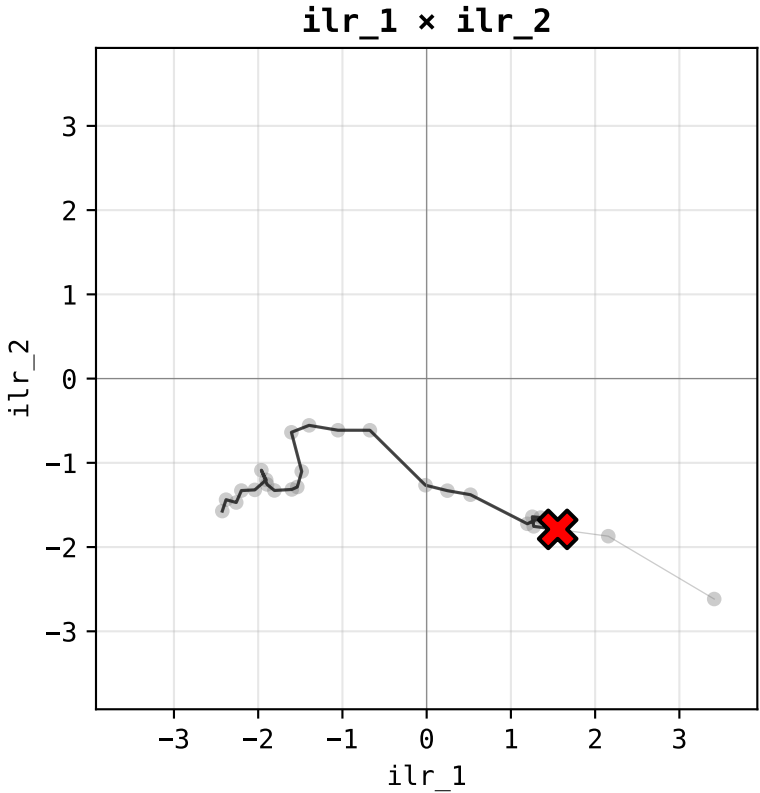




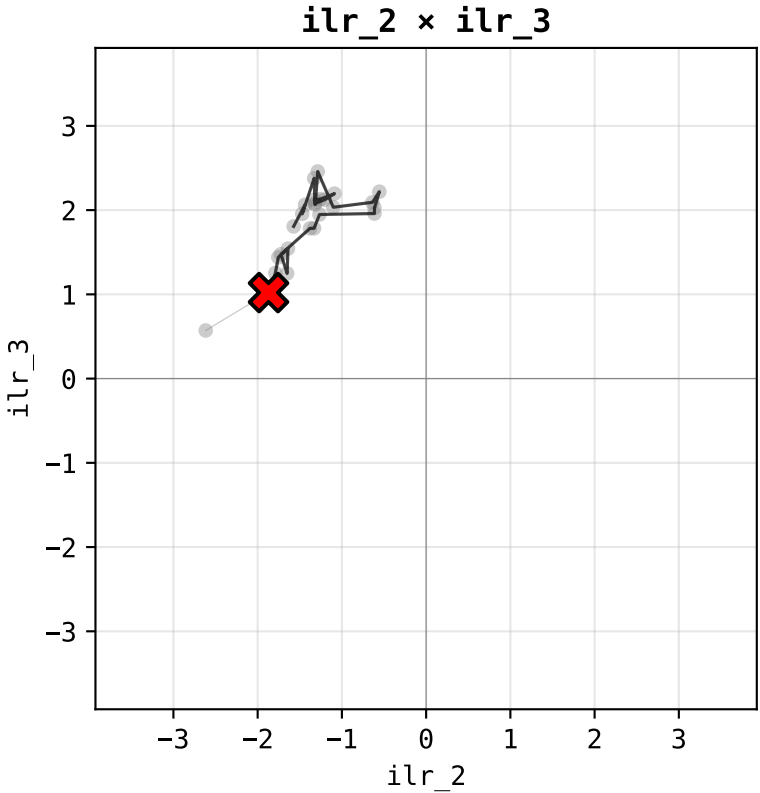
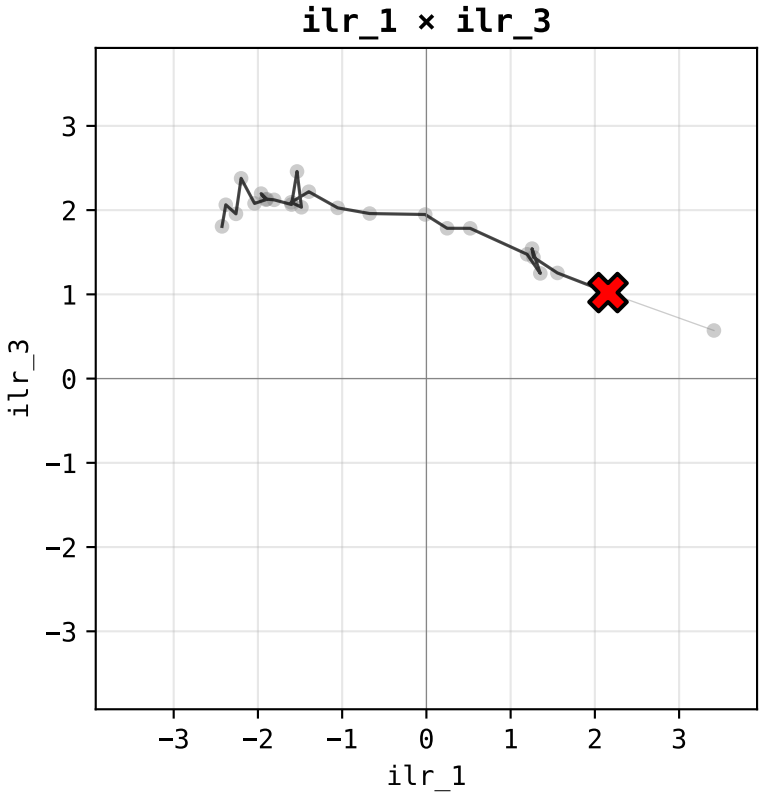
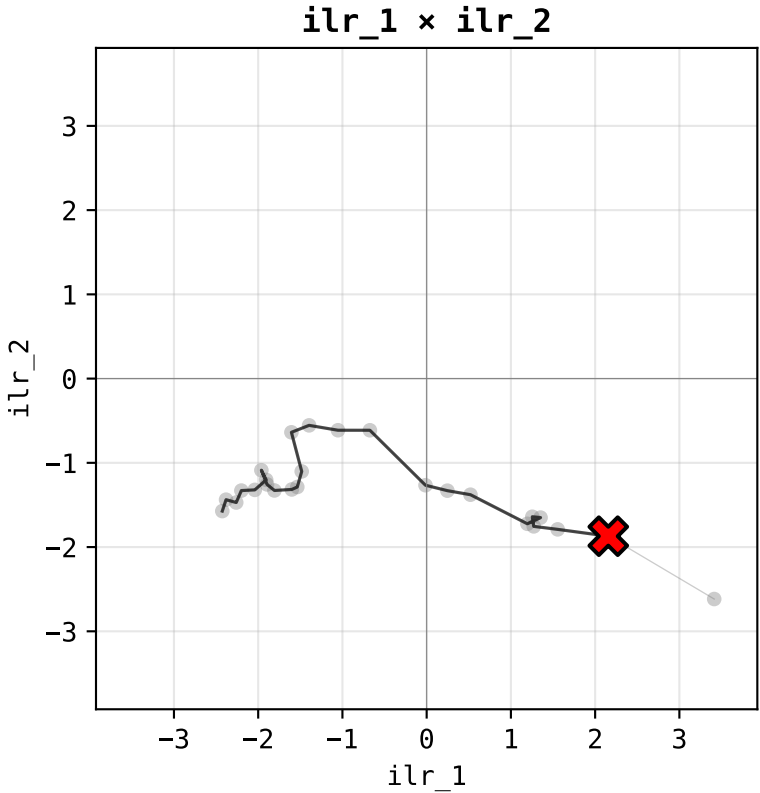




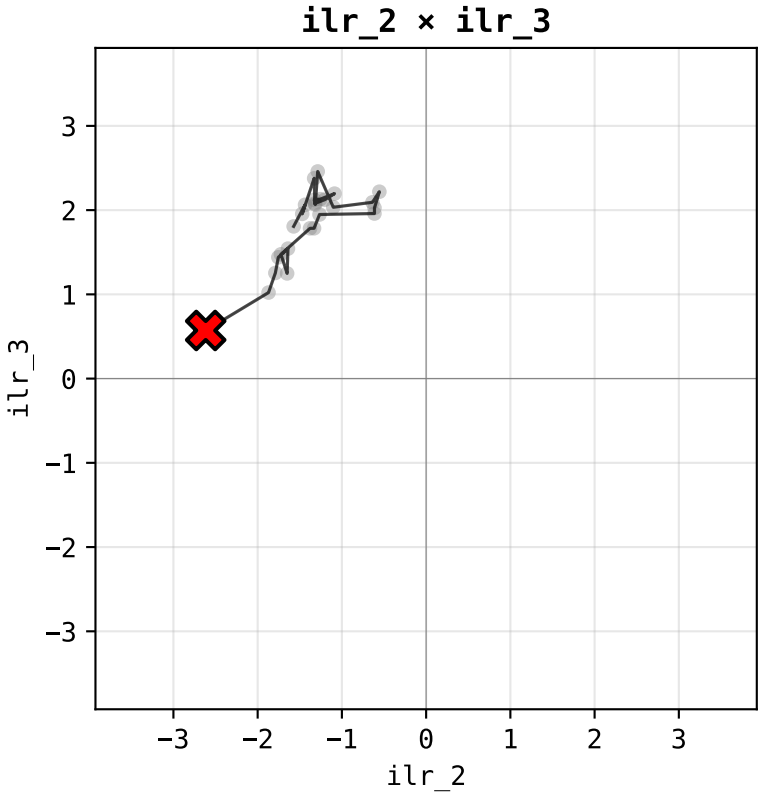
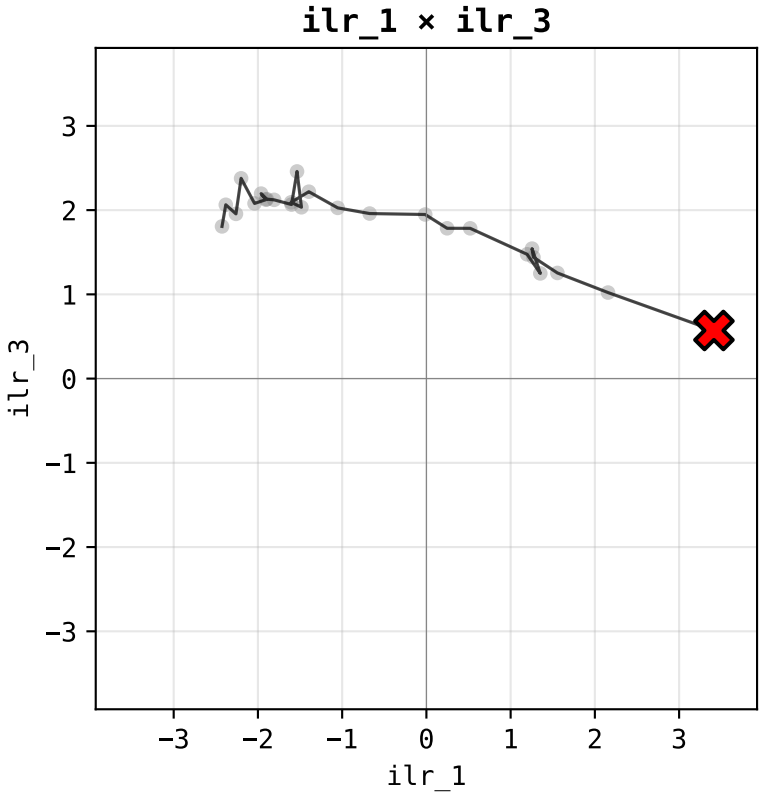
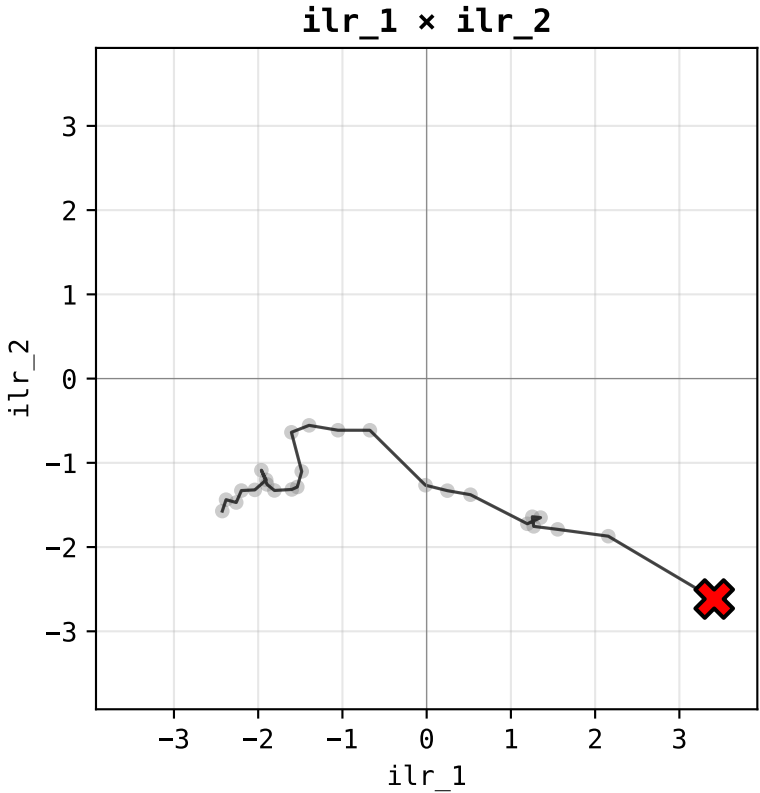




ILR Triplet – GBR United Kingdom · year 2024 (25/26)



X = current year (2024) · black = past trajectory · gray = future · ilr range ± 3.93



View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ IND India

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

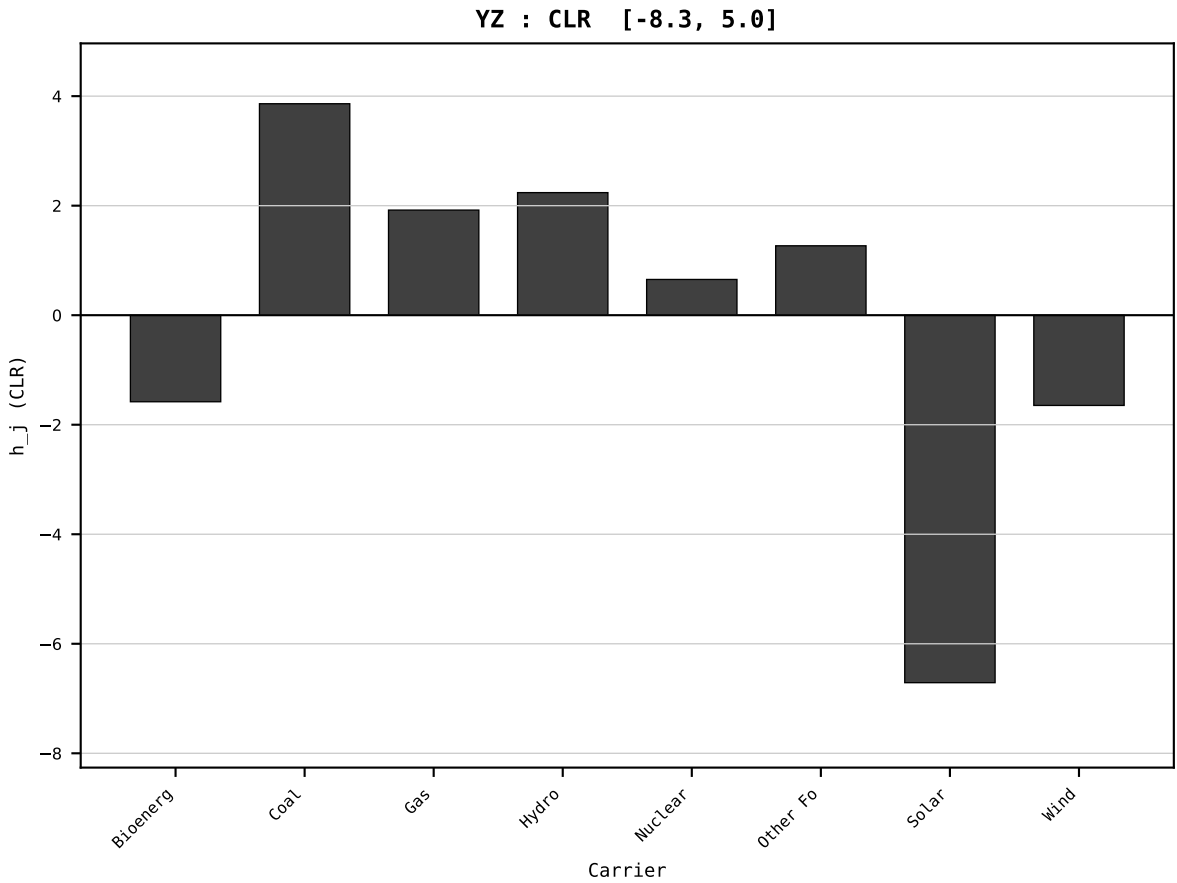
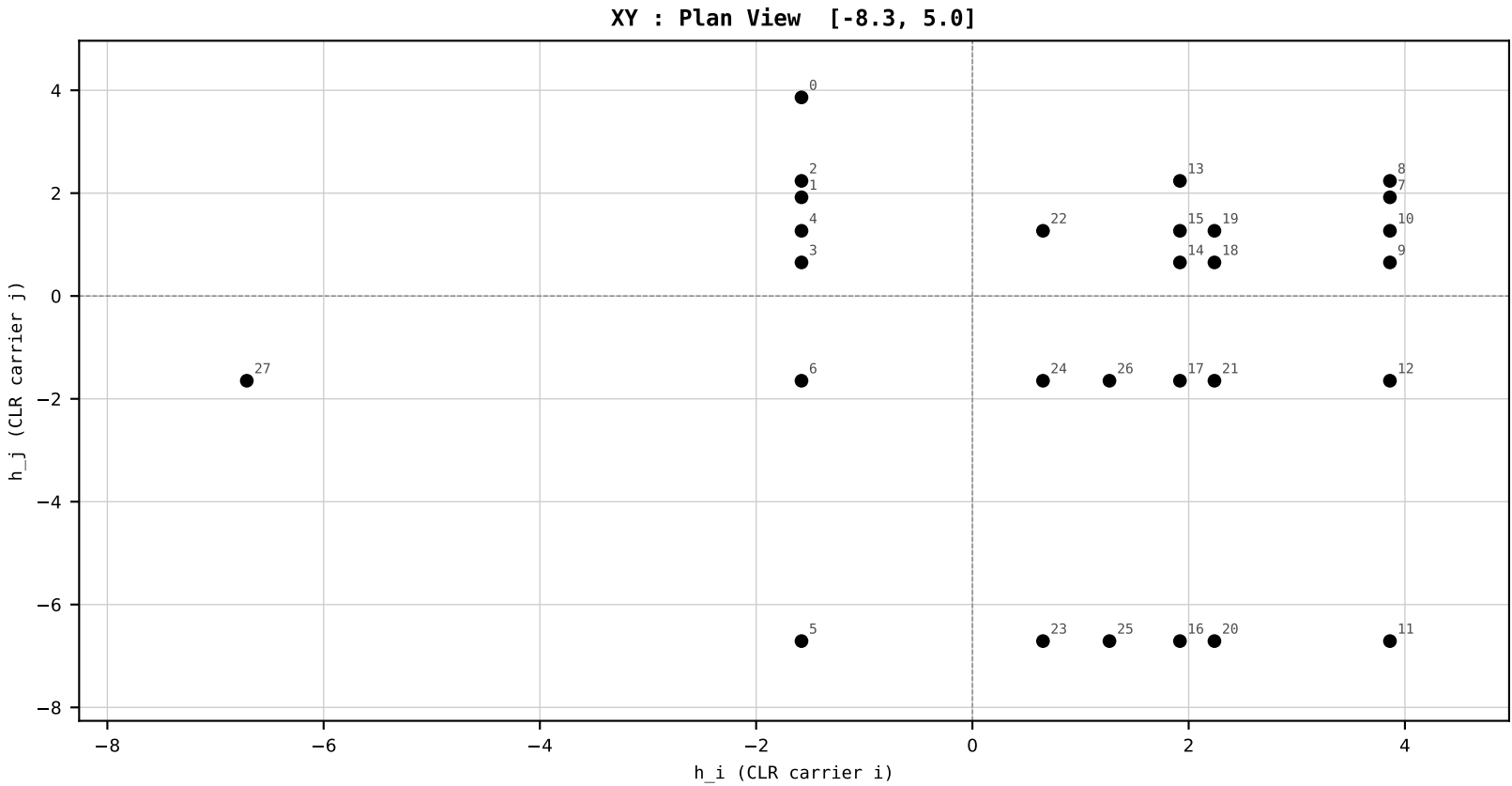
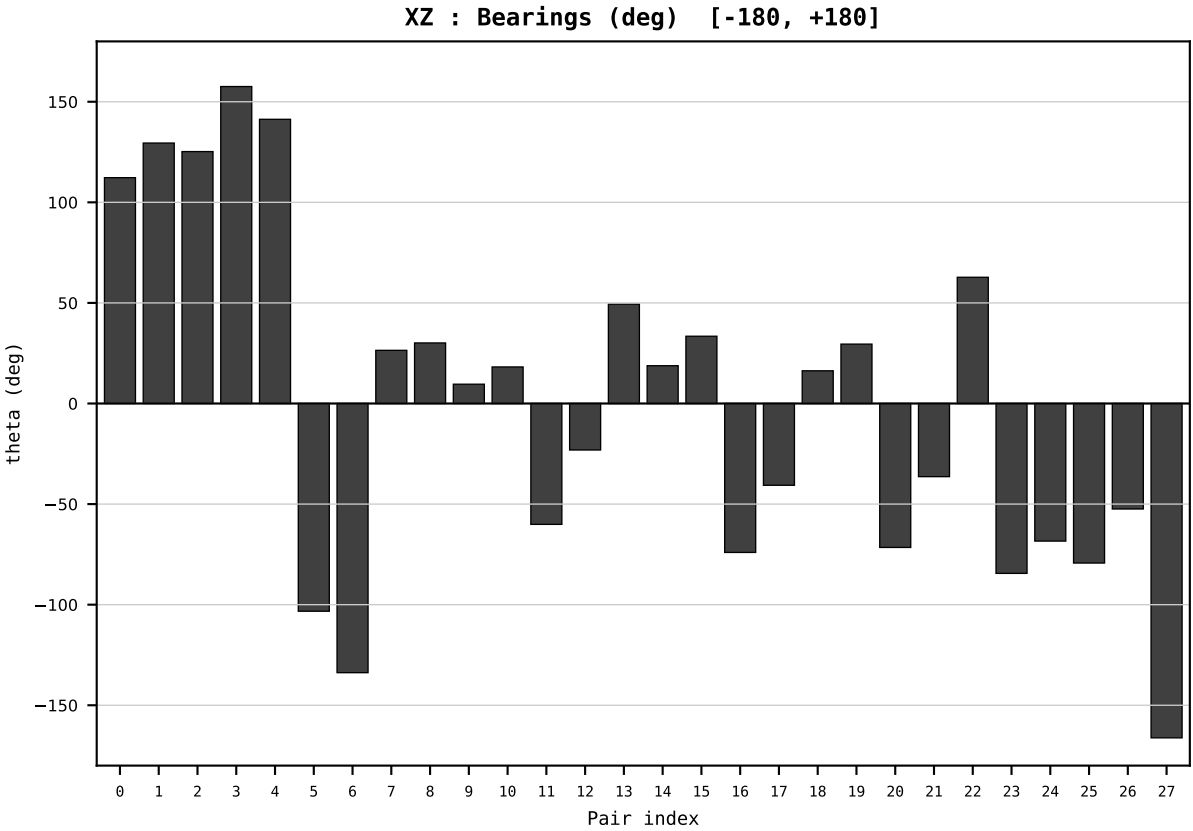
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.498576
Ring = Hs-3
E_metric= 75.8817
kappa_HS= 39023.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.572 HLR
DR ratio= 39023.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

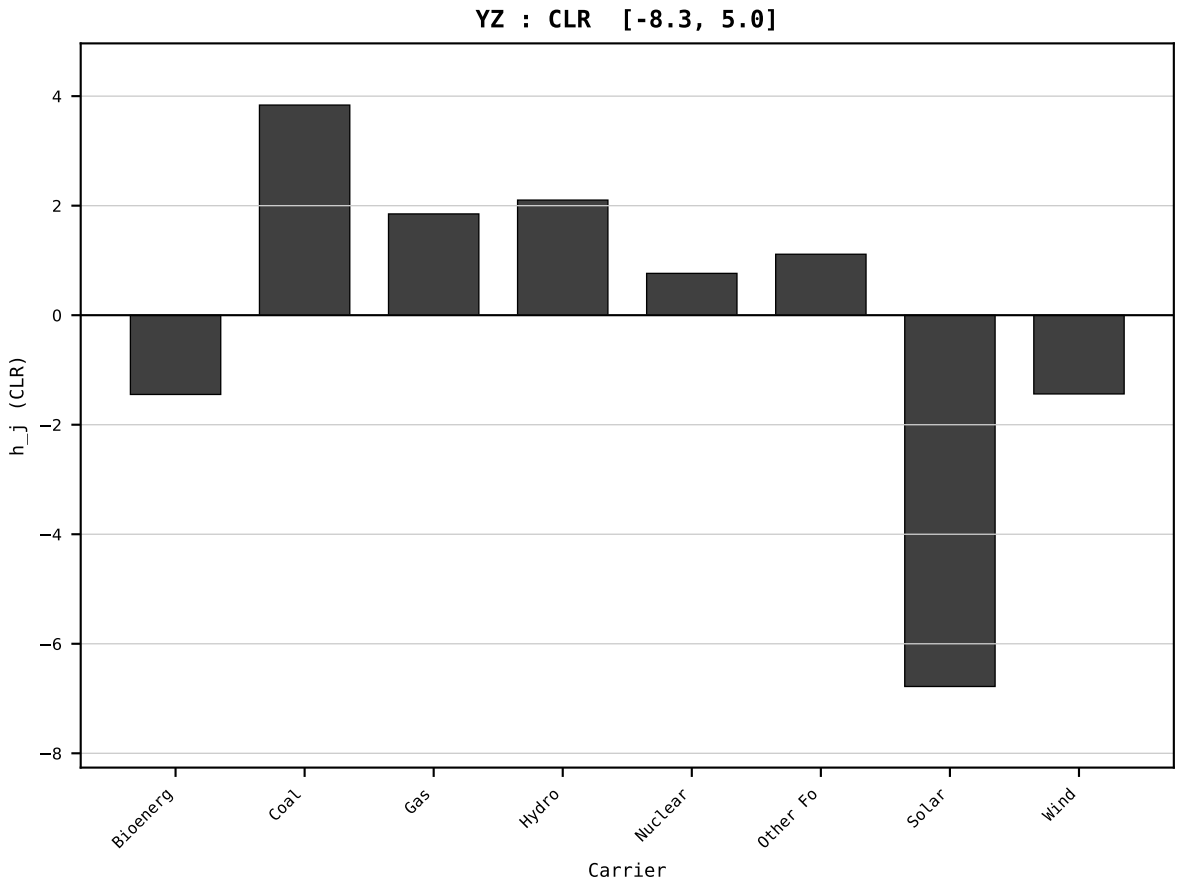
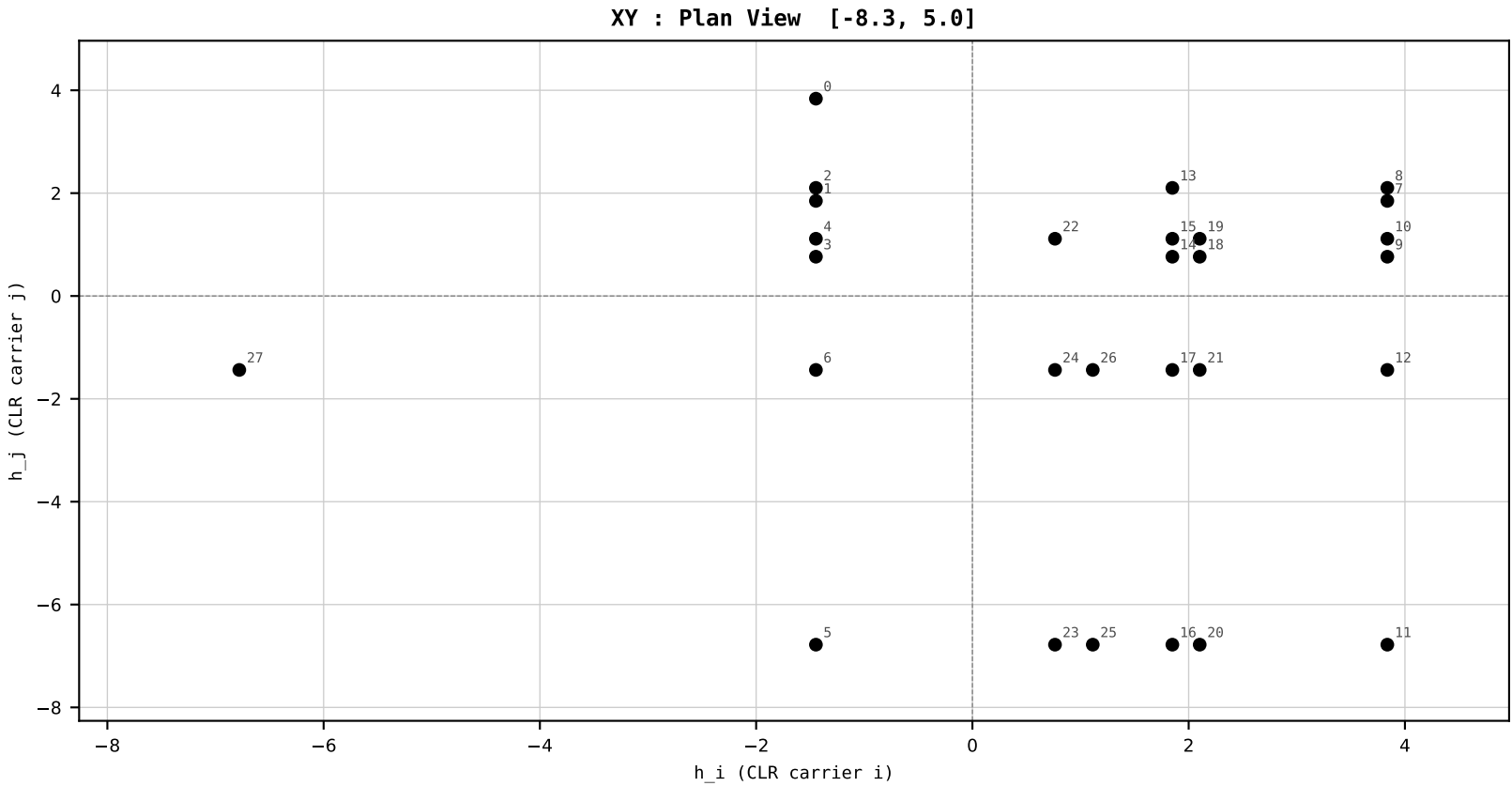
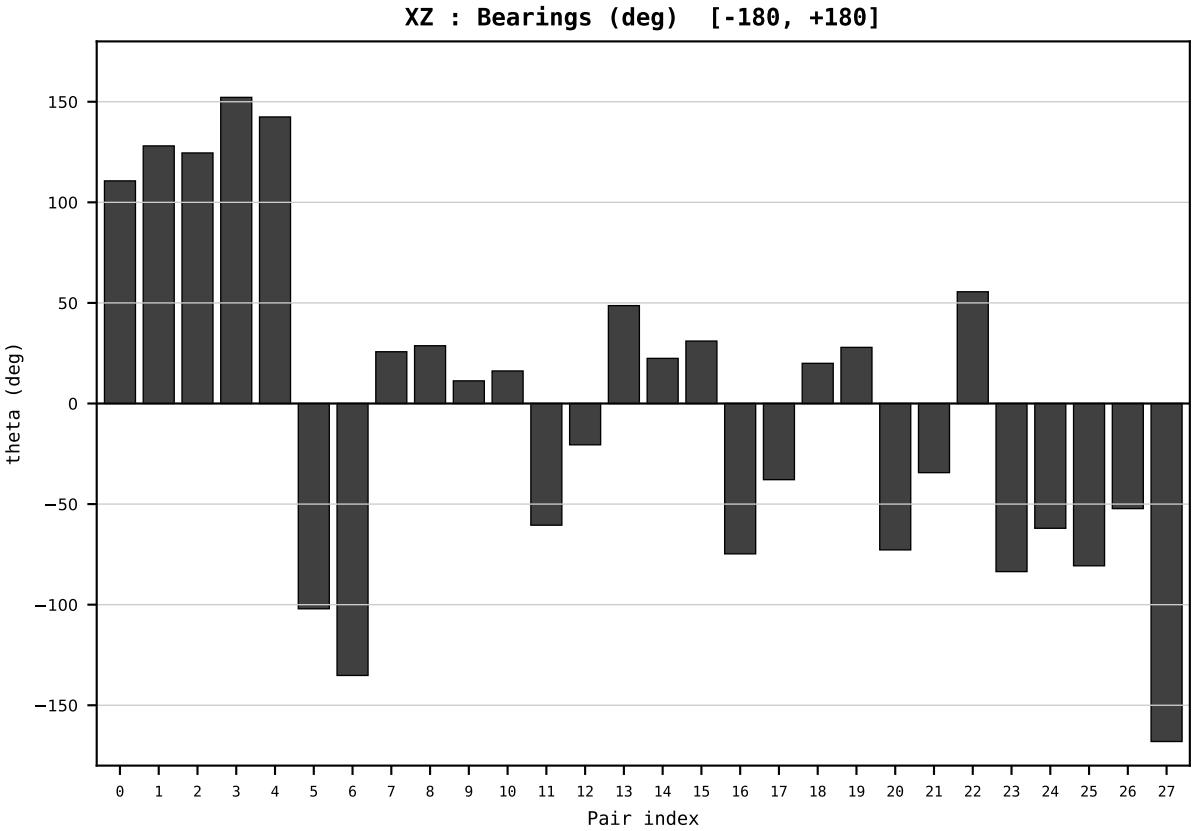
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.506780
Ring = Hs-4
E_metric= 74.5149
kappa_HS= 40829.00
omega = 2.2964 deg
d_A = 0.356346
Helm = Wind
Helm d = +0.210203
DR = 10.617 HLR
DR ratio= 40829.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

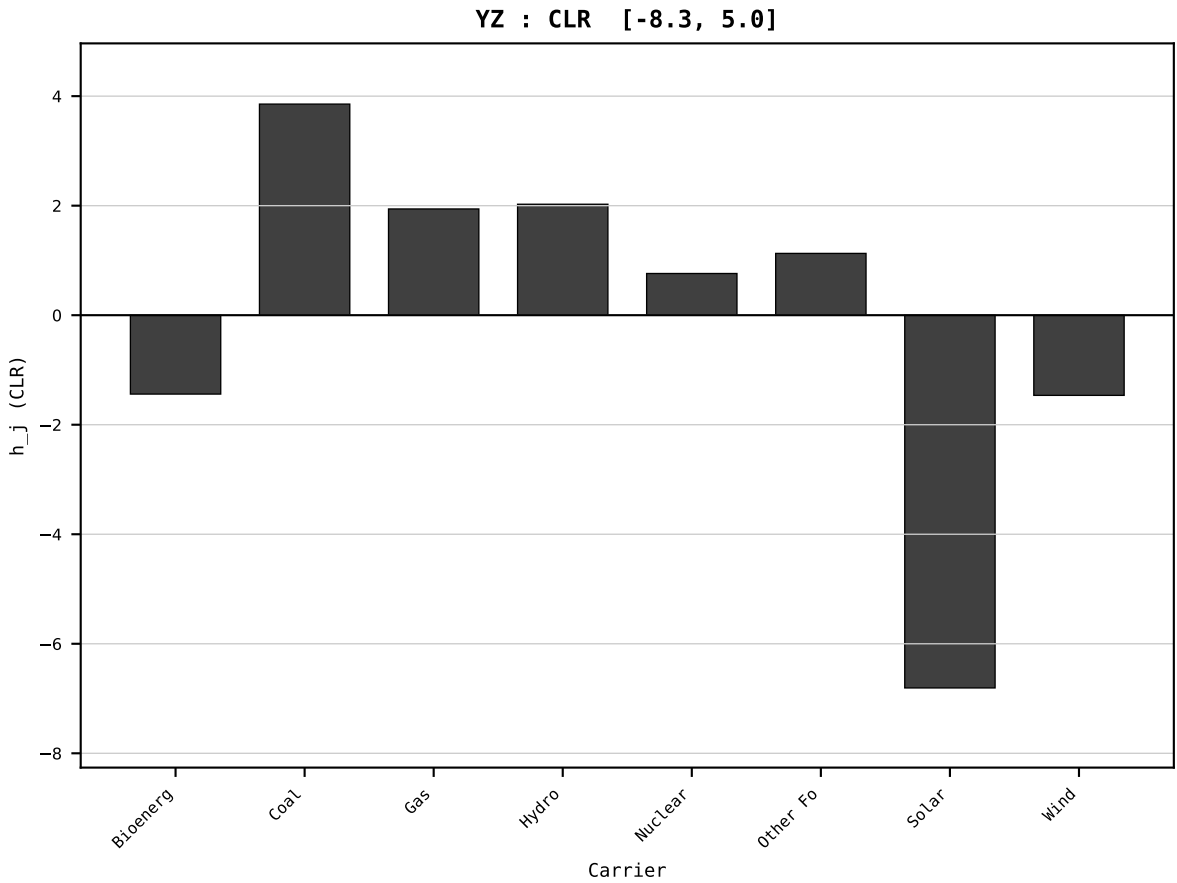
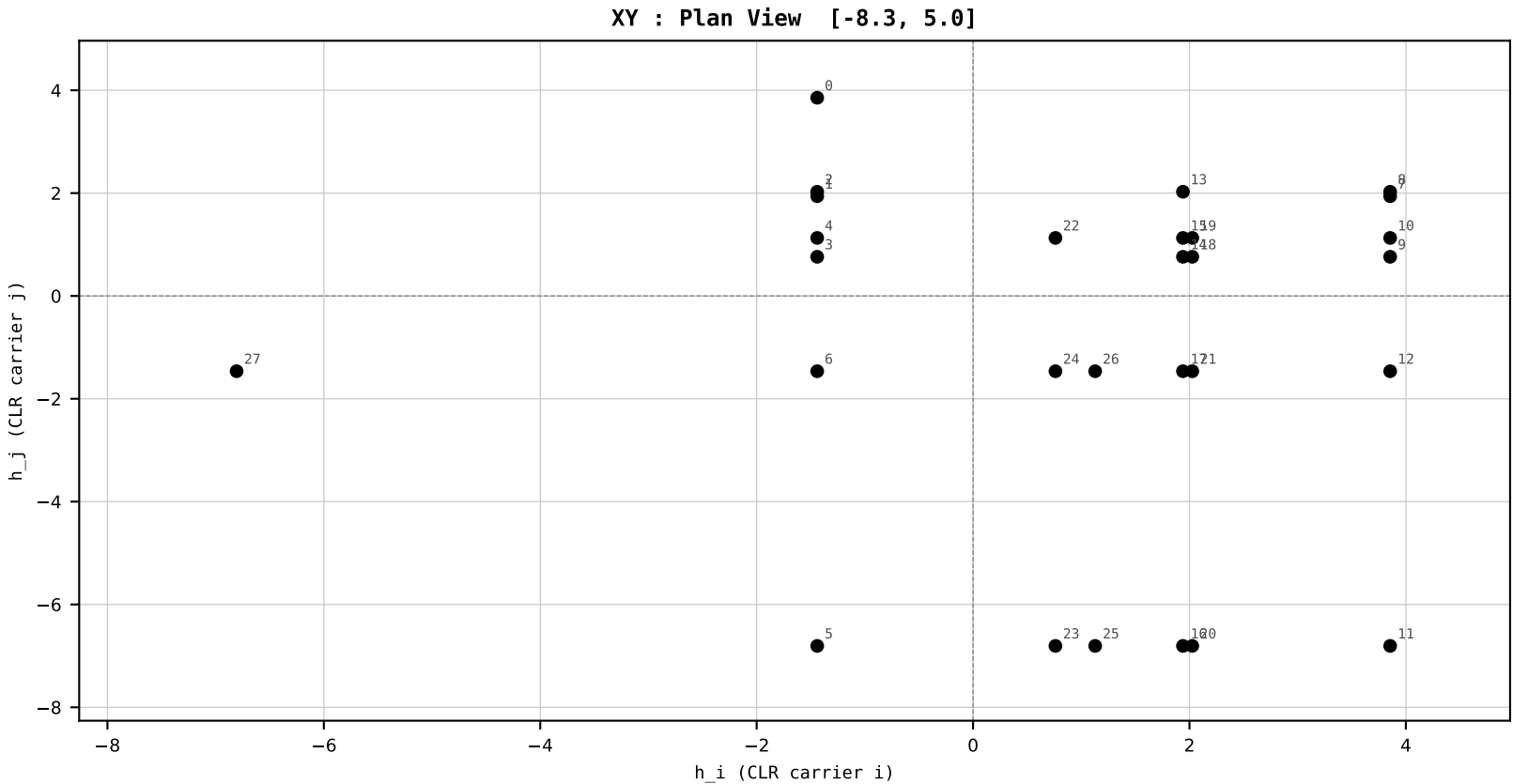
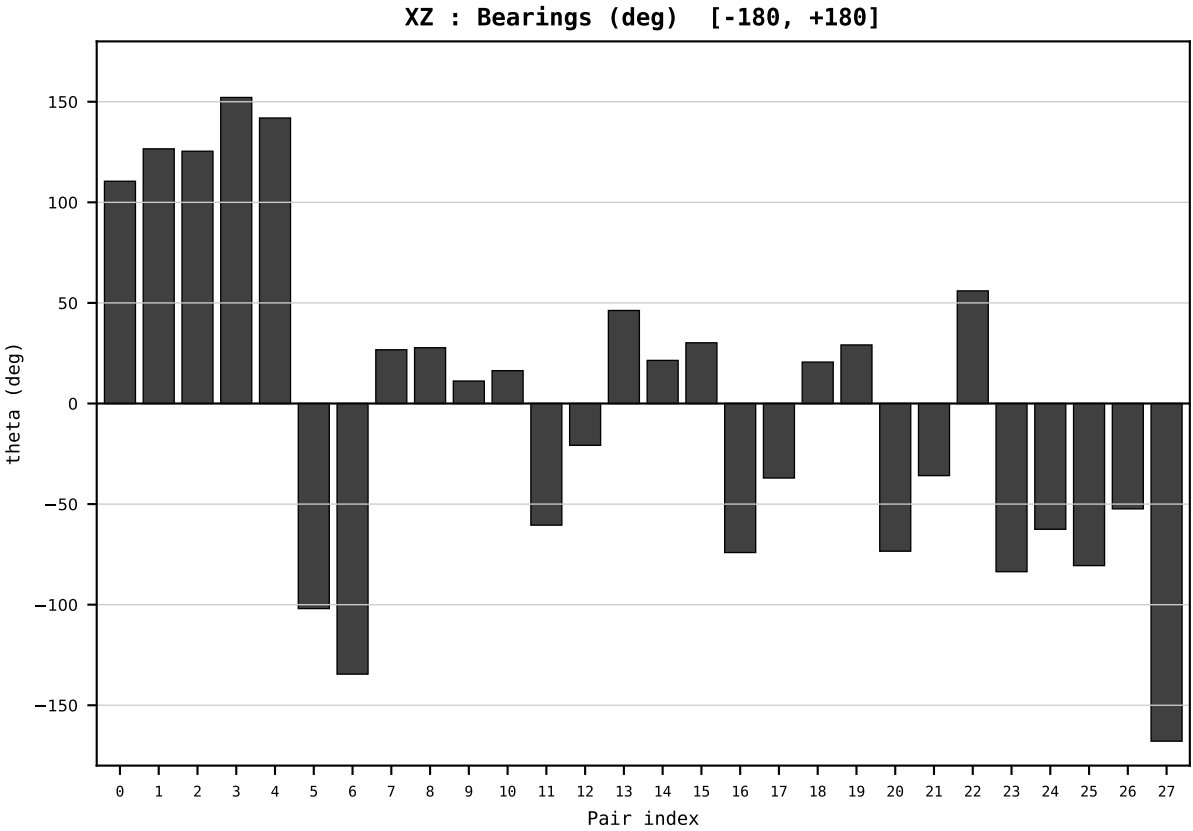
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.509498
Ring = Hs-4
E_metric= 75.1276
kappa_HS= 42670.00
omega = 0.8041 deg
d_A = 0.126449
Helm = Gas
Helm d = +0.090790
DR = 10.661 HLR
DR ratio= 42670.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

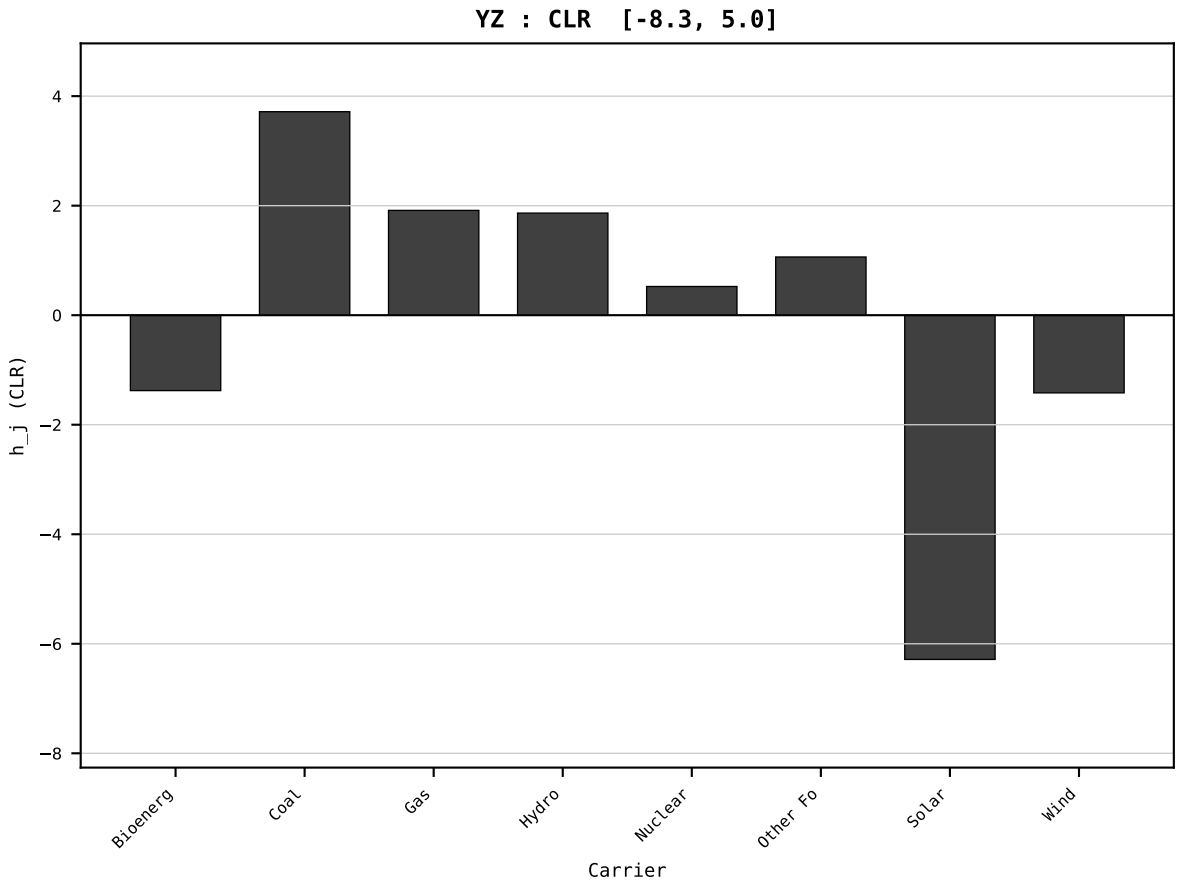
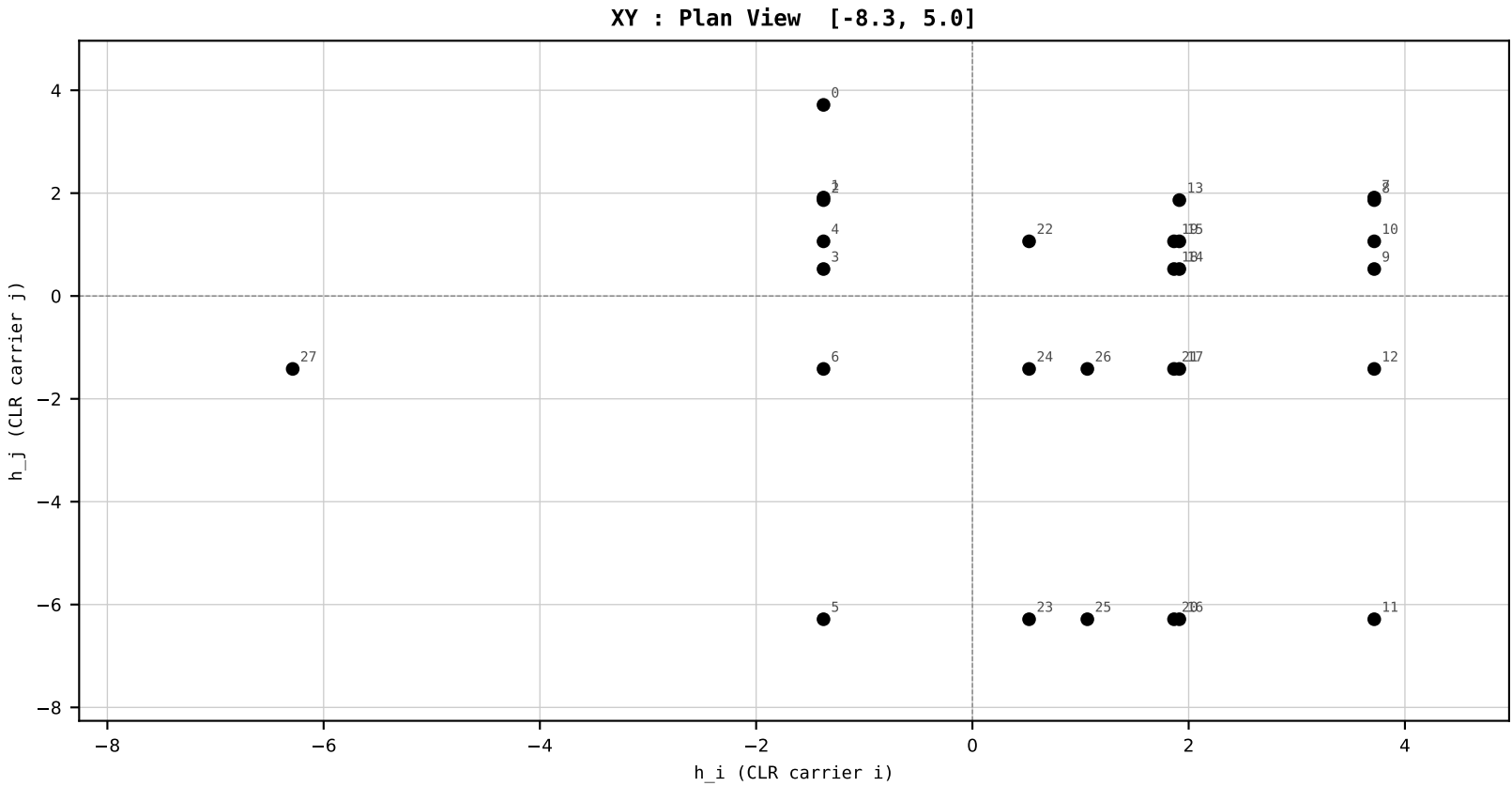
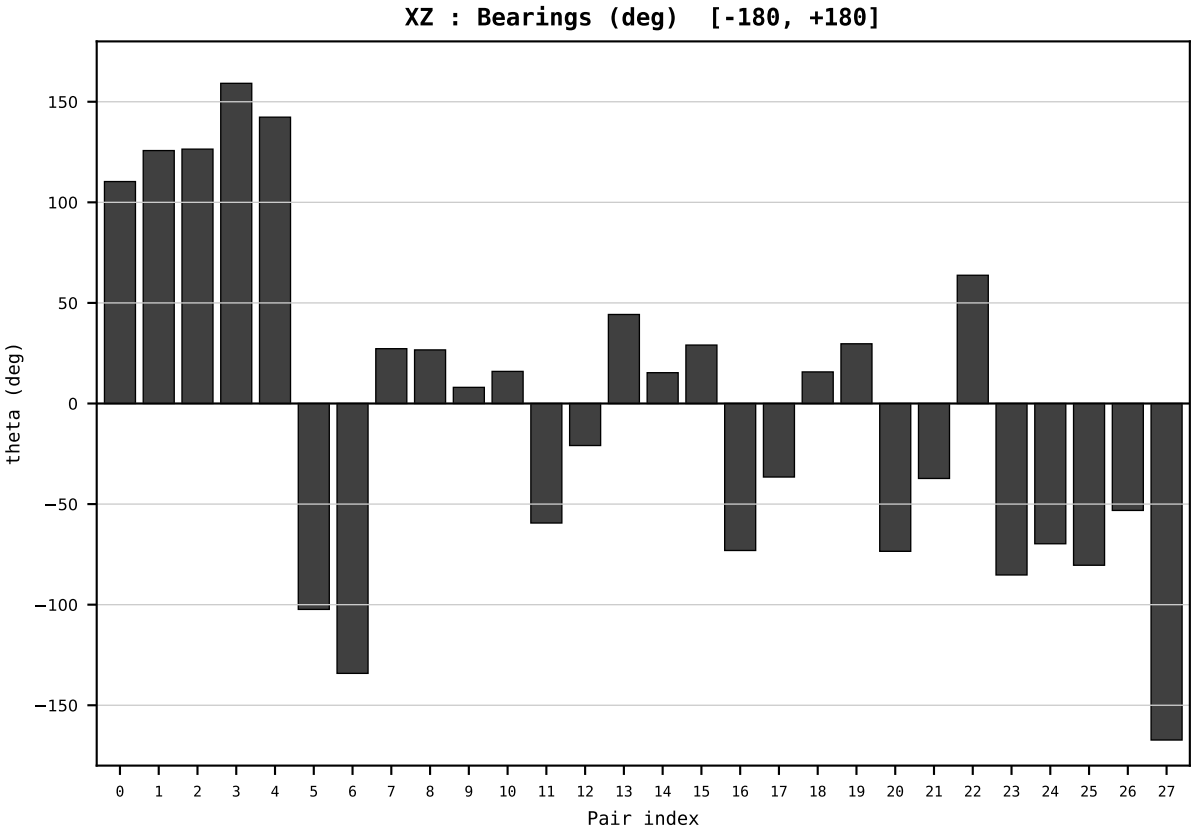
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.500822
Ring = Hs-4
E_metric= 65.7748
kappa_HS= 22073.50
omega = 1.8448 deg
d_A = 0.619370
Helm = Solar
Helm d = +0.520423
DR = 10.002 HLR
DR ratio= 22073.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

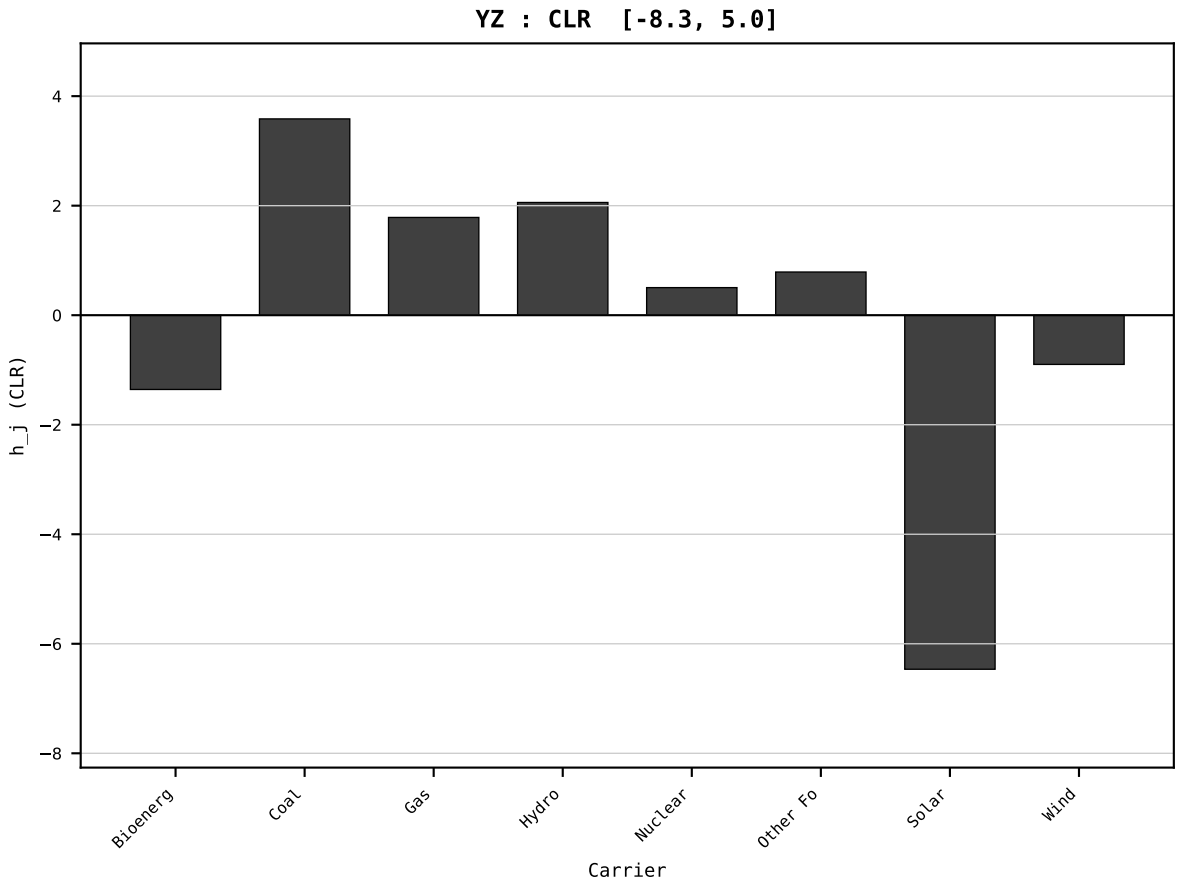
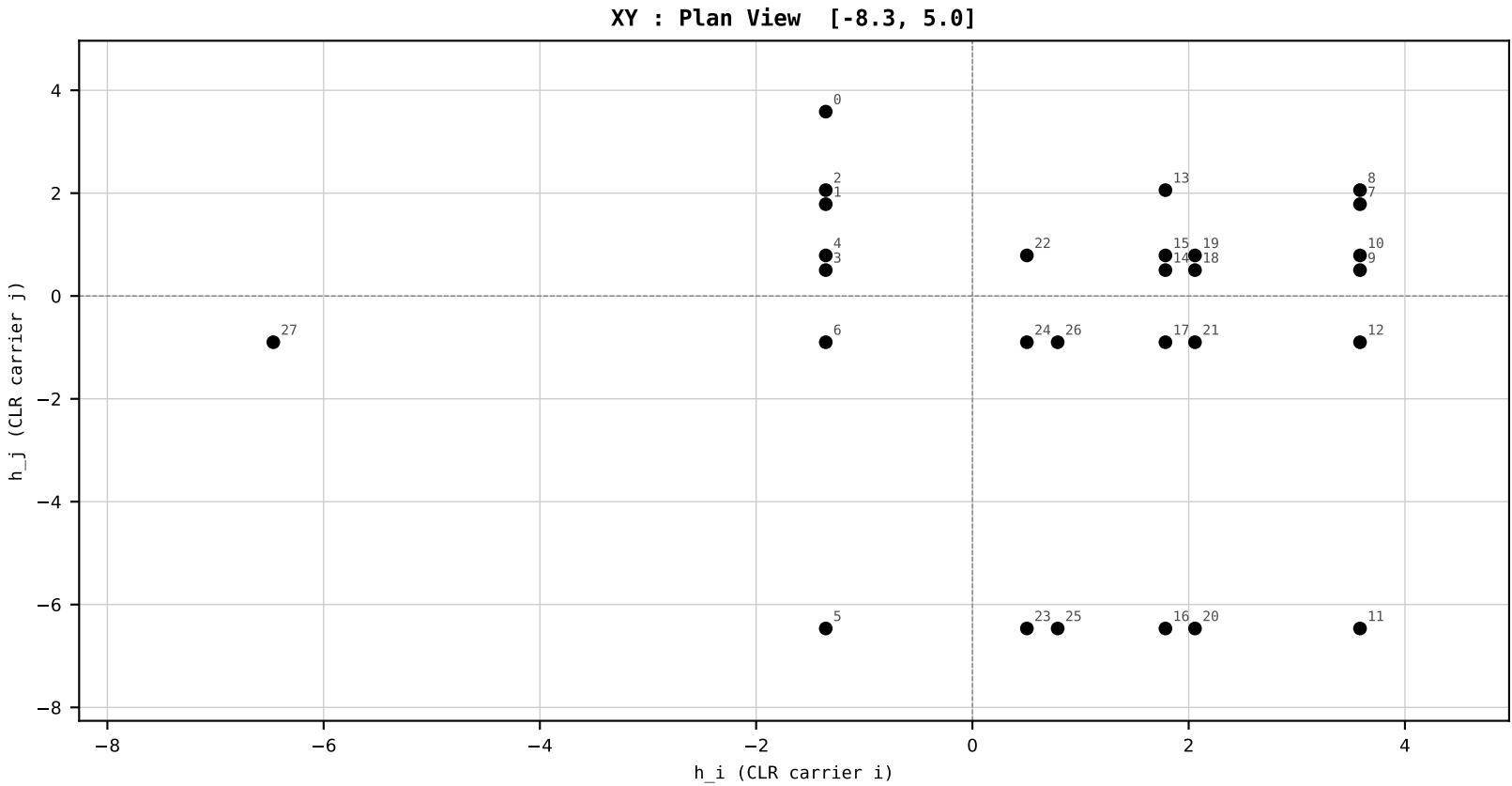
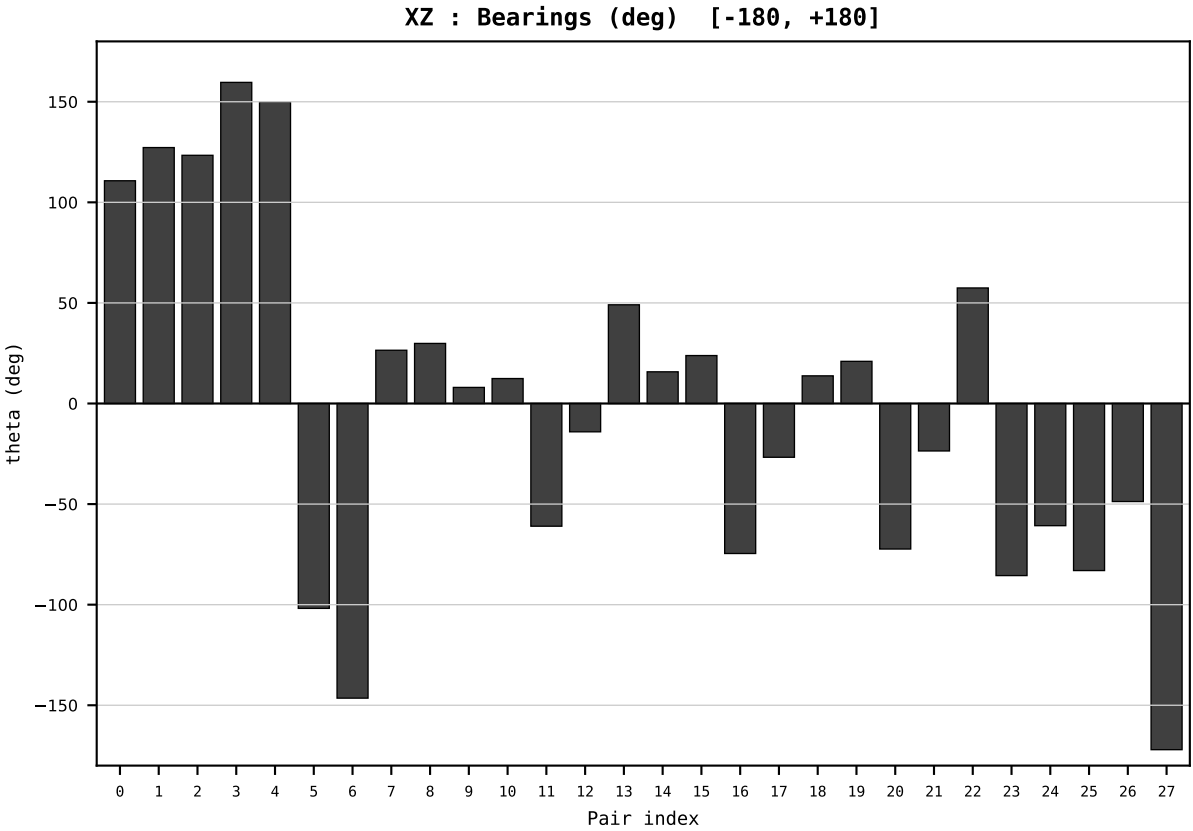
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.474628
Ring = Hs-3
E_metric= 65.5944
kappa_HS= 23150.50
omega = 4.7416 deg
d_A = 0.670616
Helm = Wind
Helm d = +0.519664
DR = 10.050 HLR
DR ratio= 23150.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

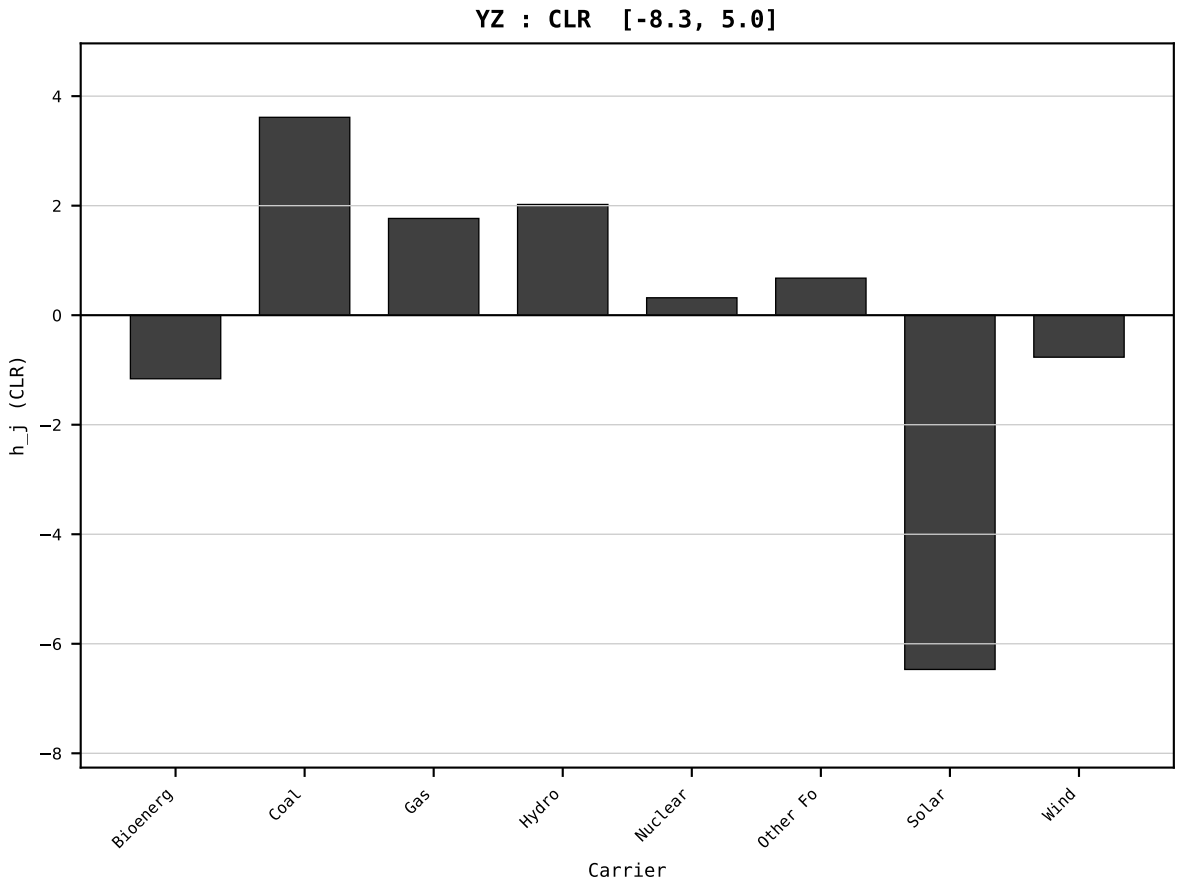
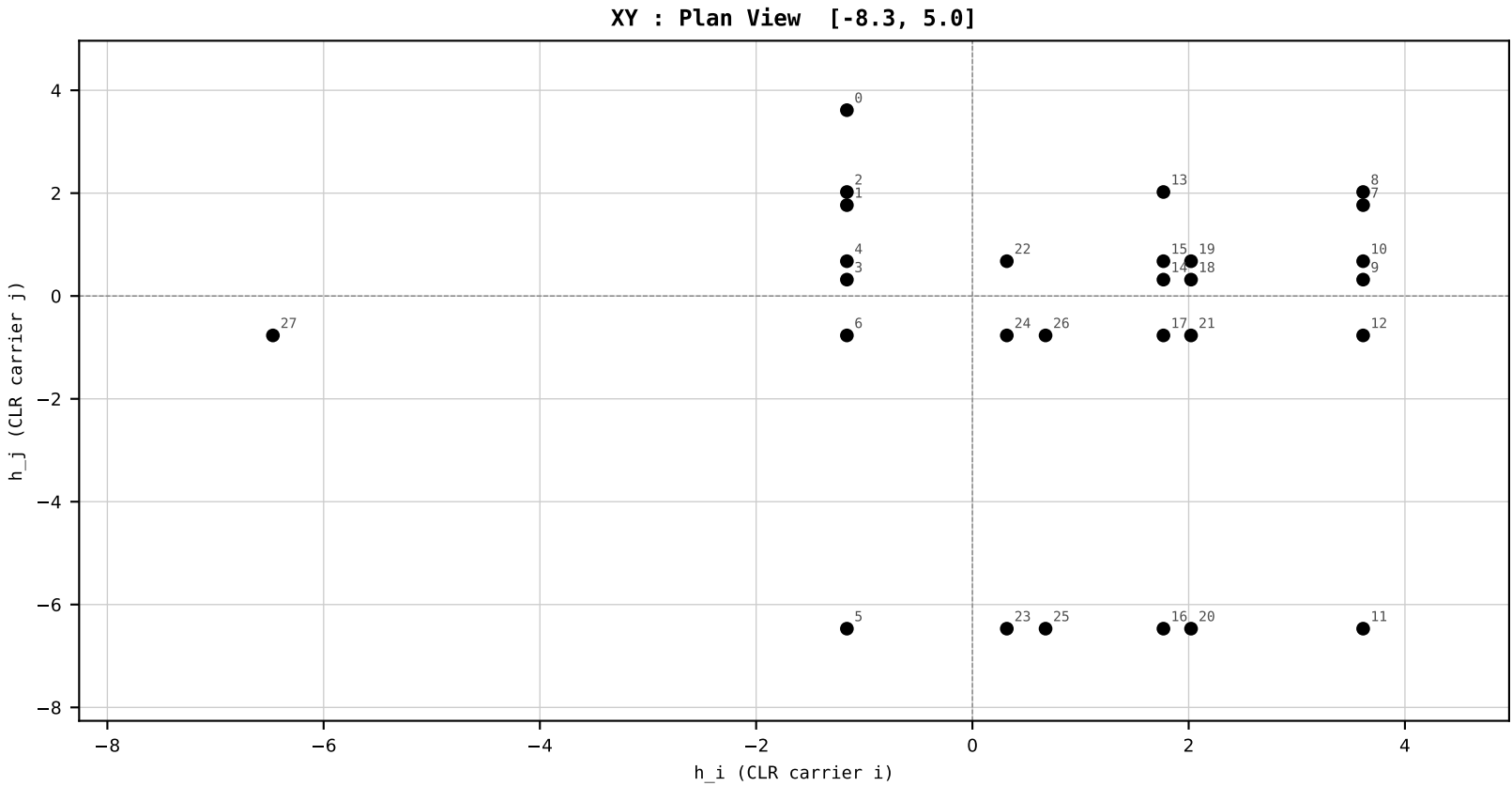
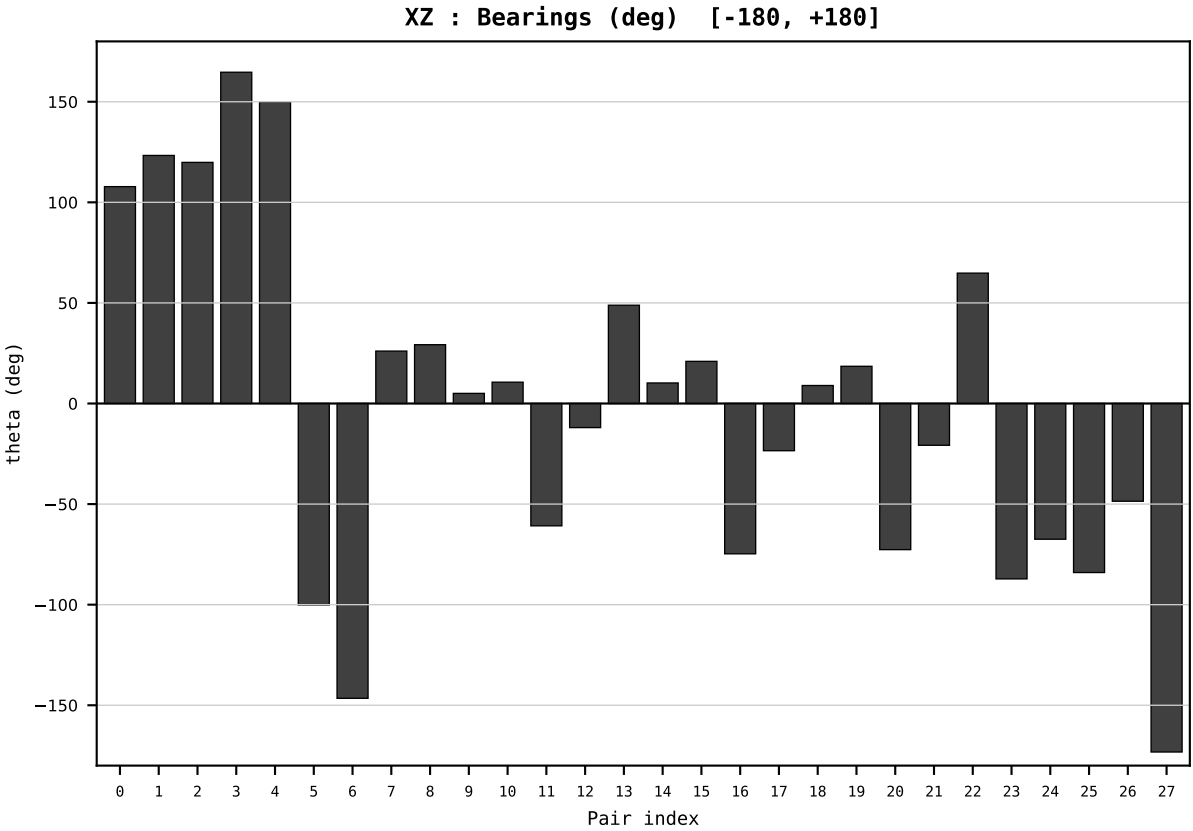
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.491003
Ring = Hs-3
E_metric= 64.6110
kappa_HS= 23924.00
omega = 2.2606 deg
d_A = 0.324101
Helm = Bioenergy
Helm d = +0.195431
DR = 10.083 HLR
DR ratio= 23924.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

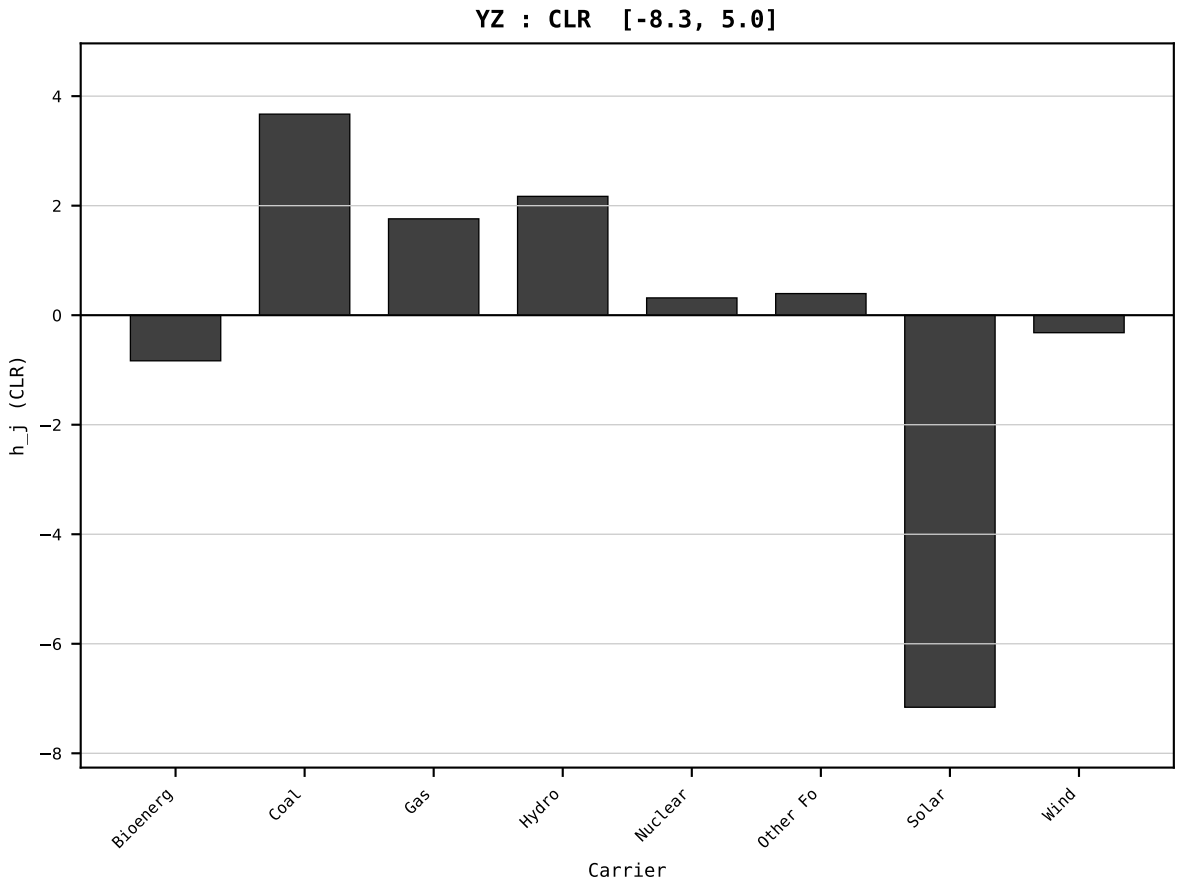
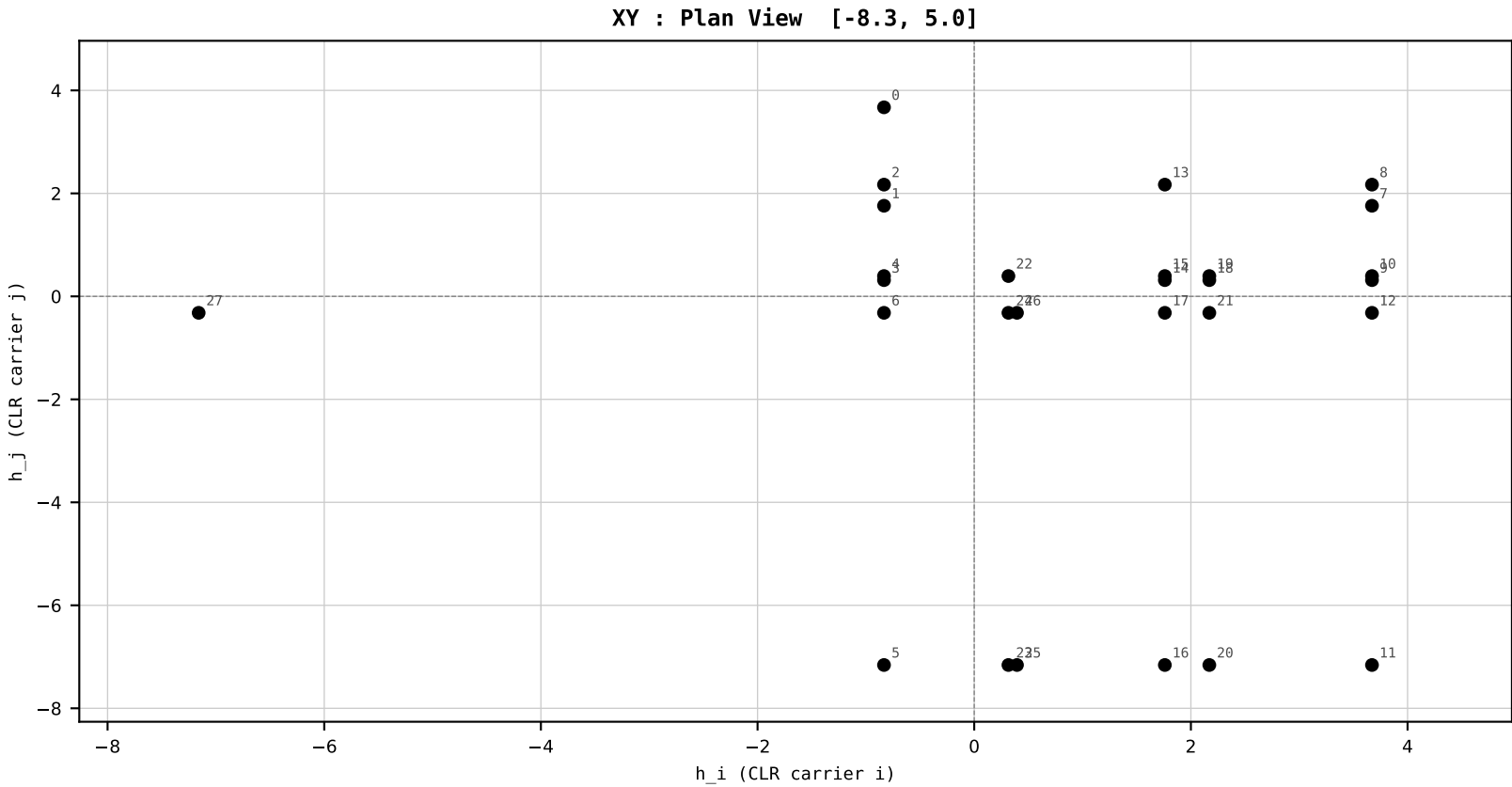
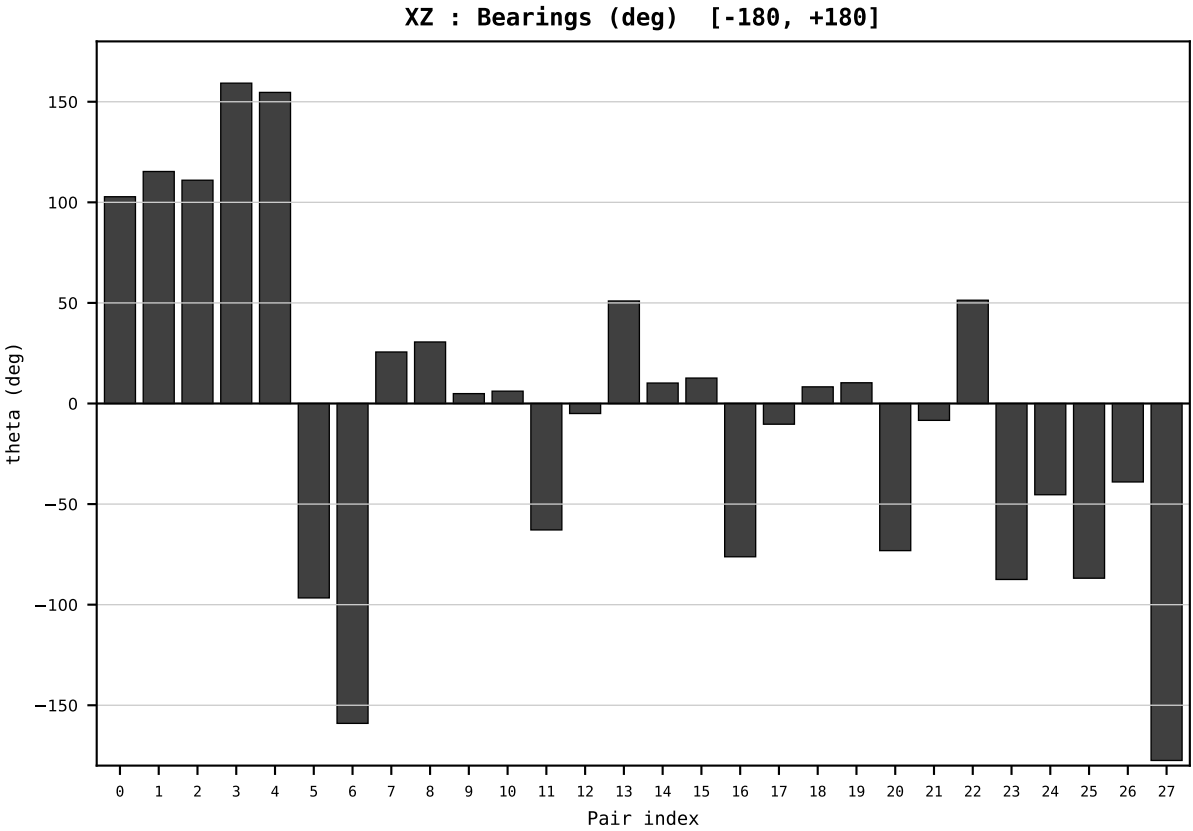
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.493296
Ring = Hs-3
E_metric= 73.5896
kappa_HS= 50546.00
omega = 5.3311 deg
d_A = 0.942605
Helm = Solar
Helm d = -0.689998
DR = 10.831 HLR
DR ratio= 50546.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

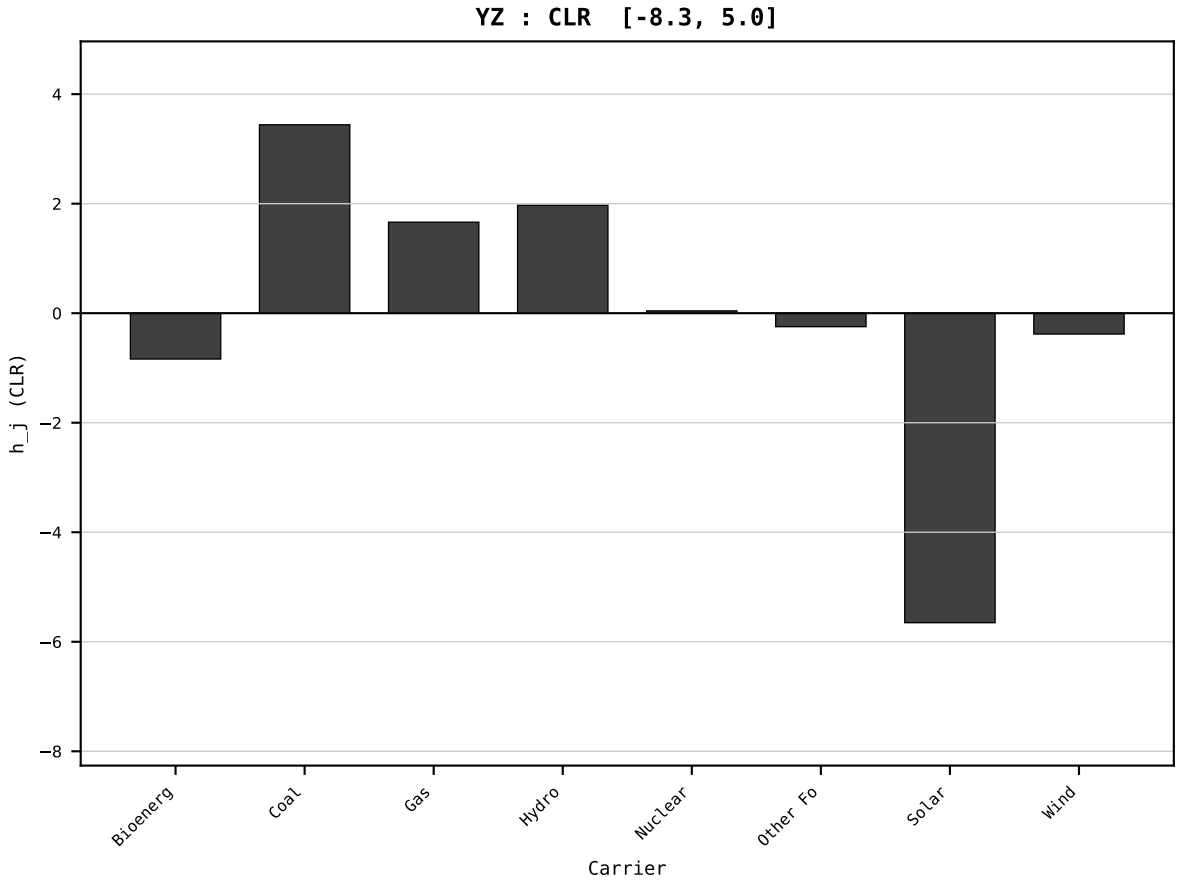
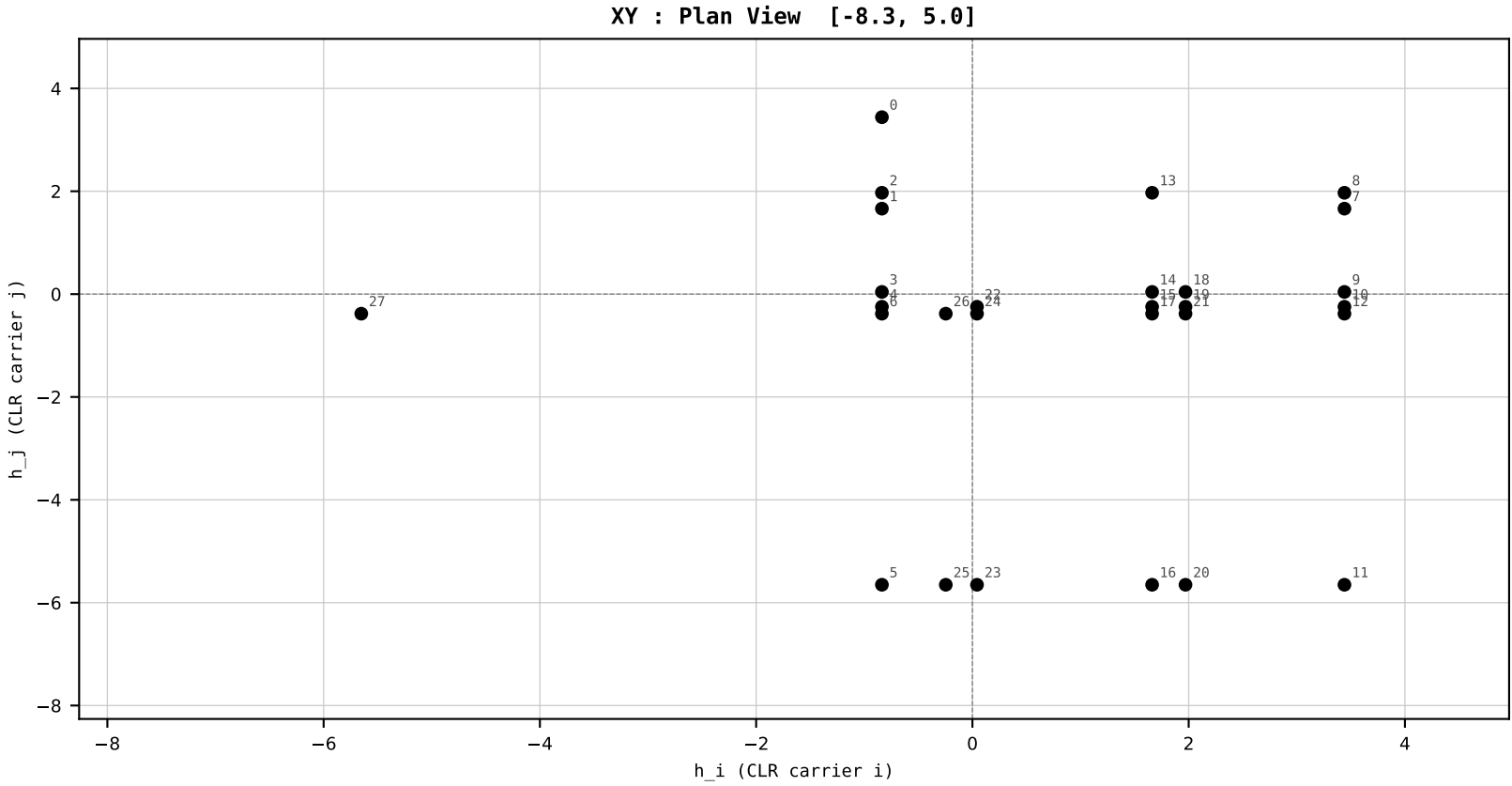
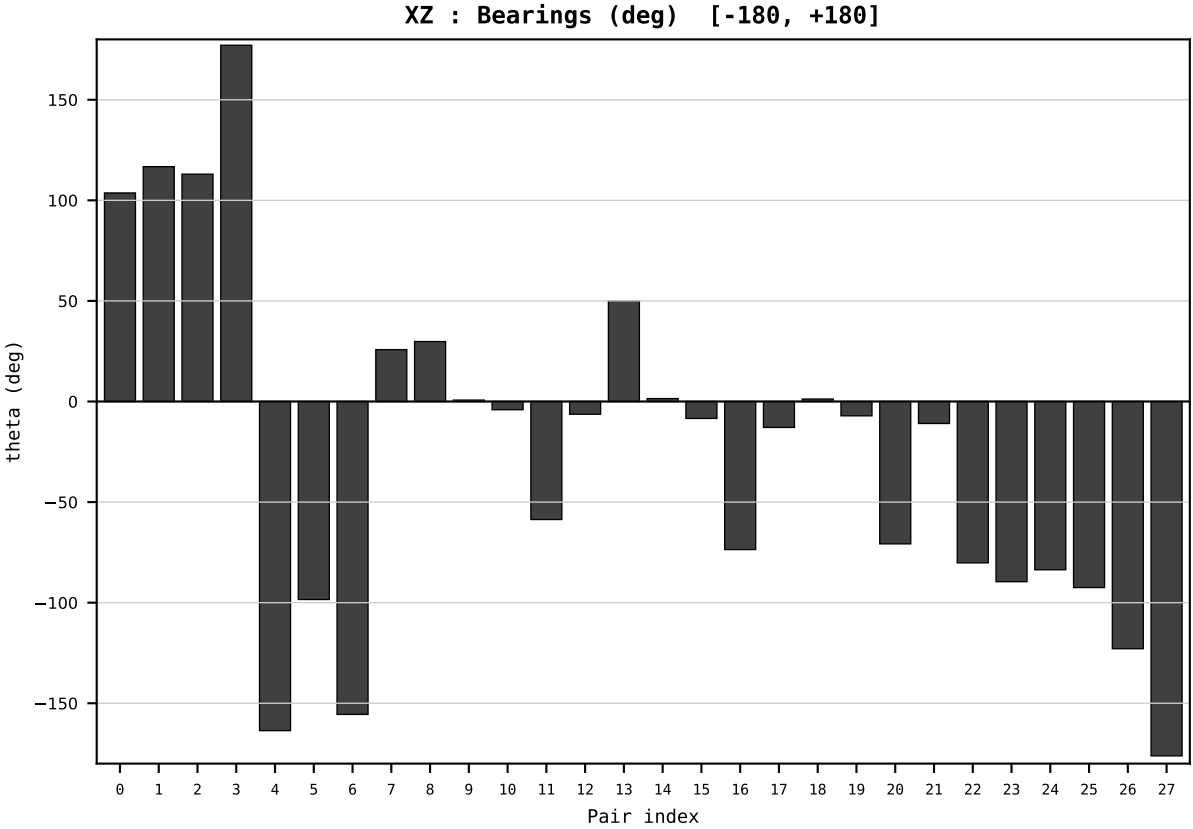
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.488879
Ring = Hs-3
E_metric= 51.3339
kappa_HS= 8887.83
omega = 6.8016 deg
d_A = 1.692216
Helm = Solar
Helm d = +1.507238
DR = 9.092 HLR
DR ratio= 8887.8x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

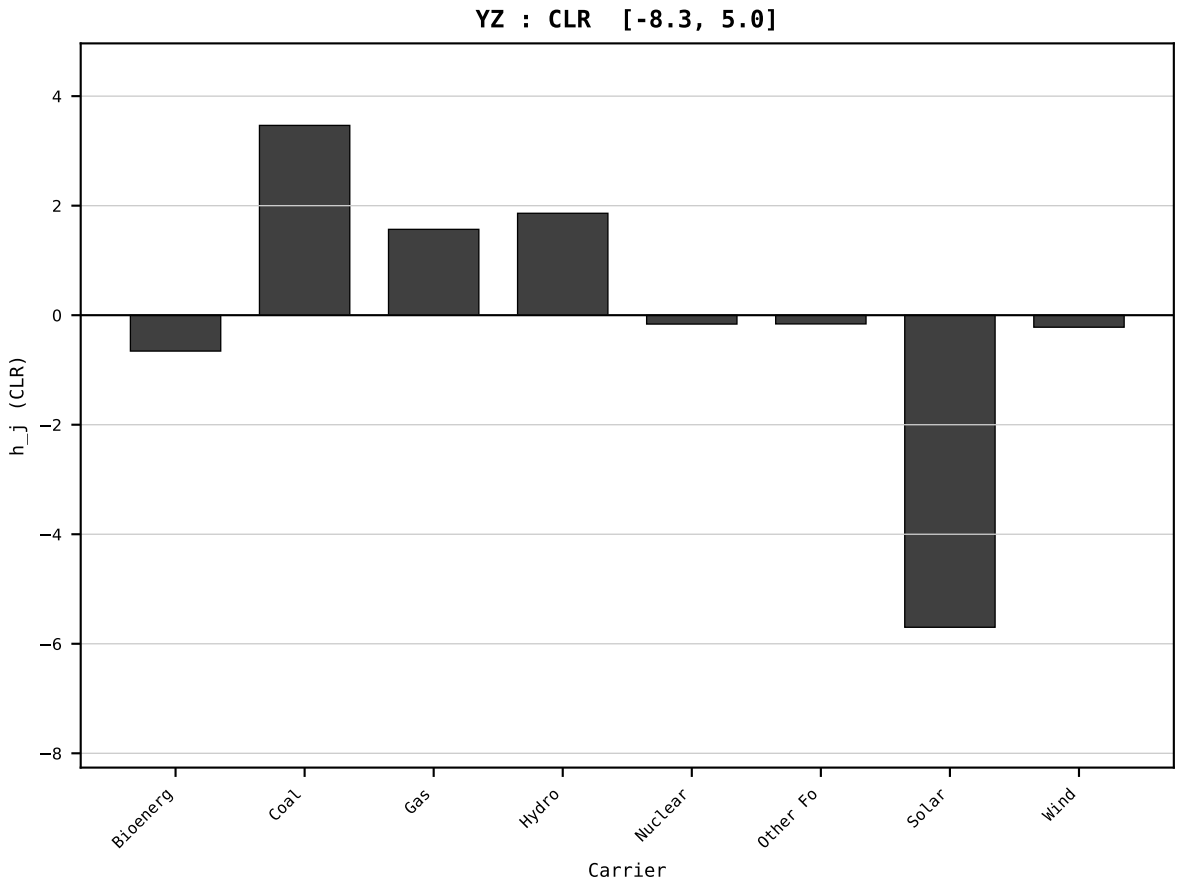
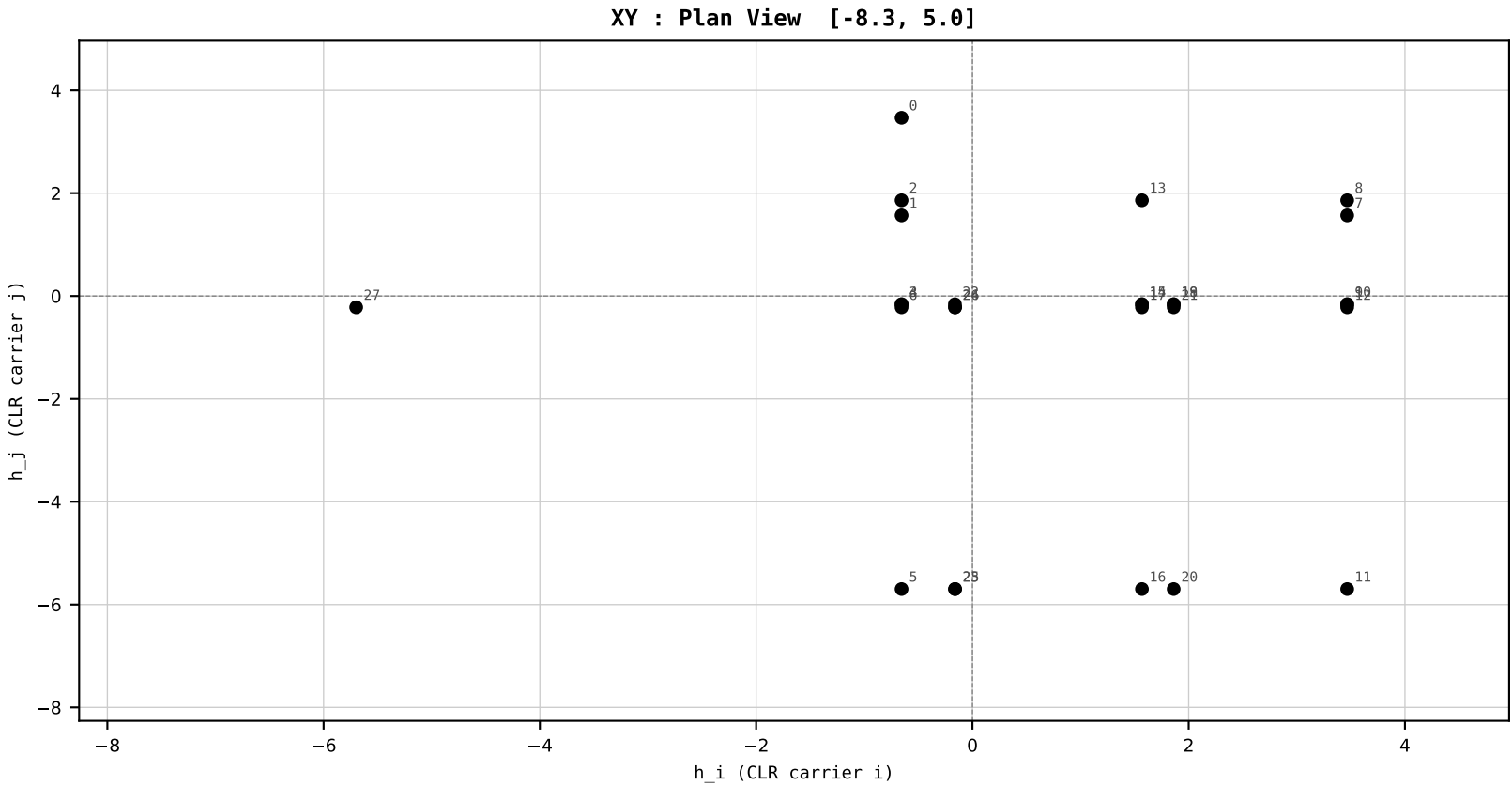
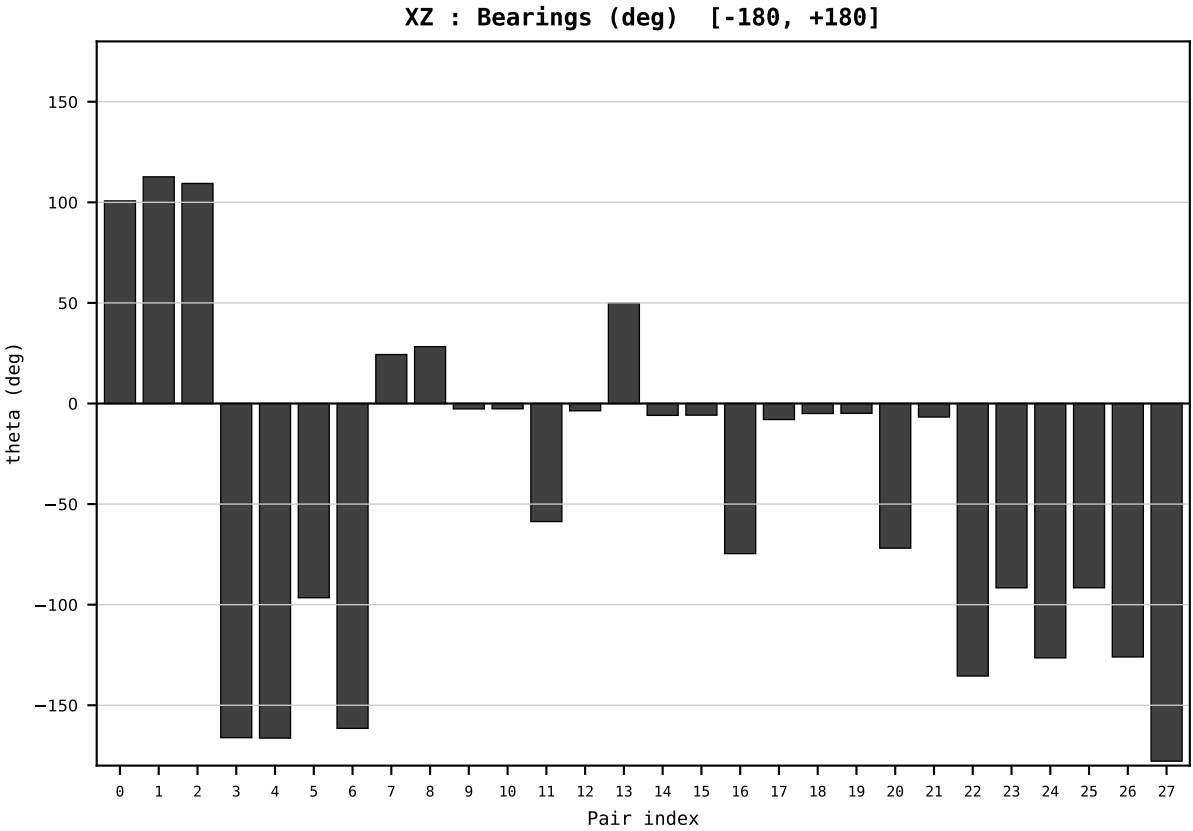
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.503376
Ring = Hs-4
E_metric= 50.9375
kappa_HS= 9552.00
omega = 2.9028 deg
d_A = 0.363314
Helm = Nuclear
Helm d = -0.204435
DR = 9.165 HLR
DR ratio= 9552.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=8 N=26 pairs=28

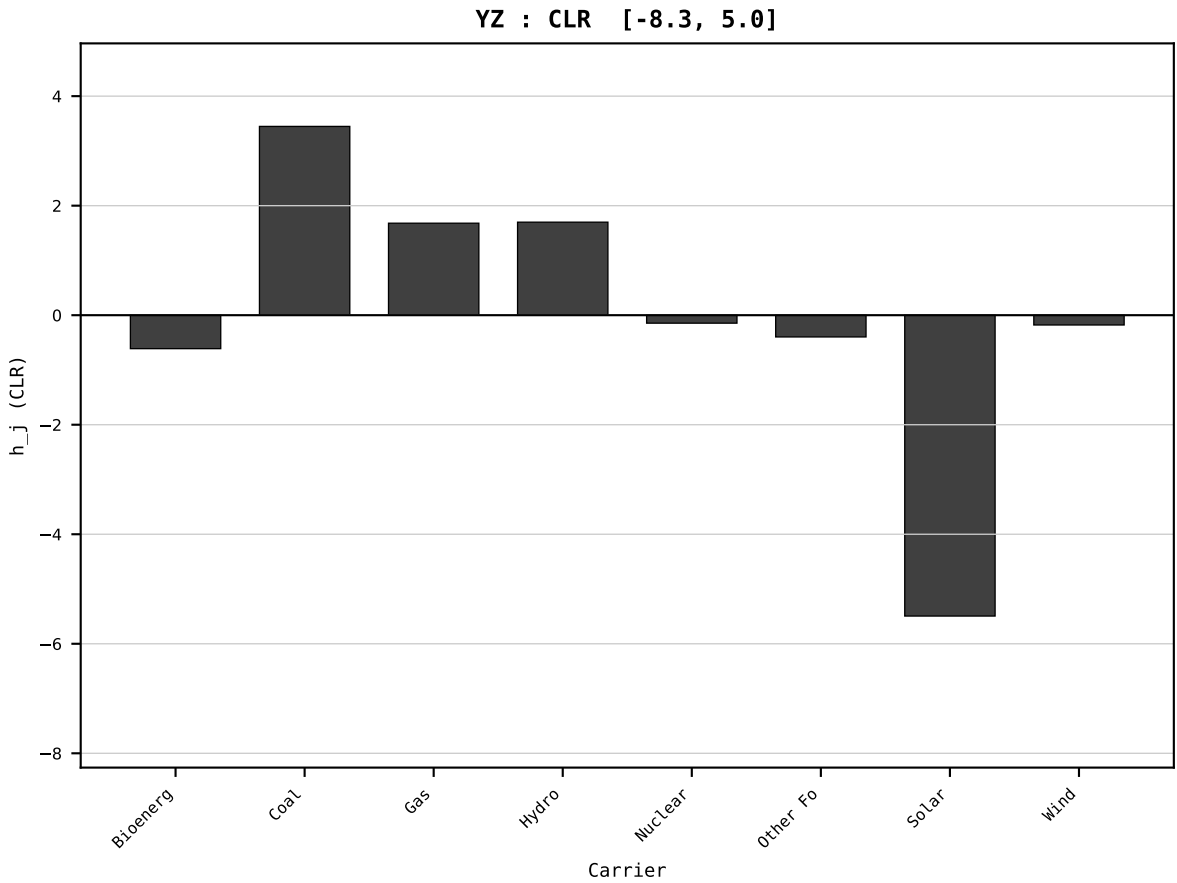
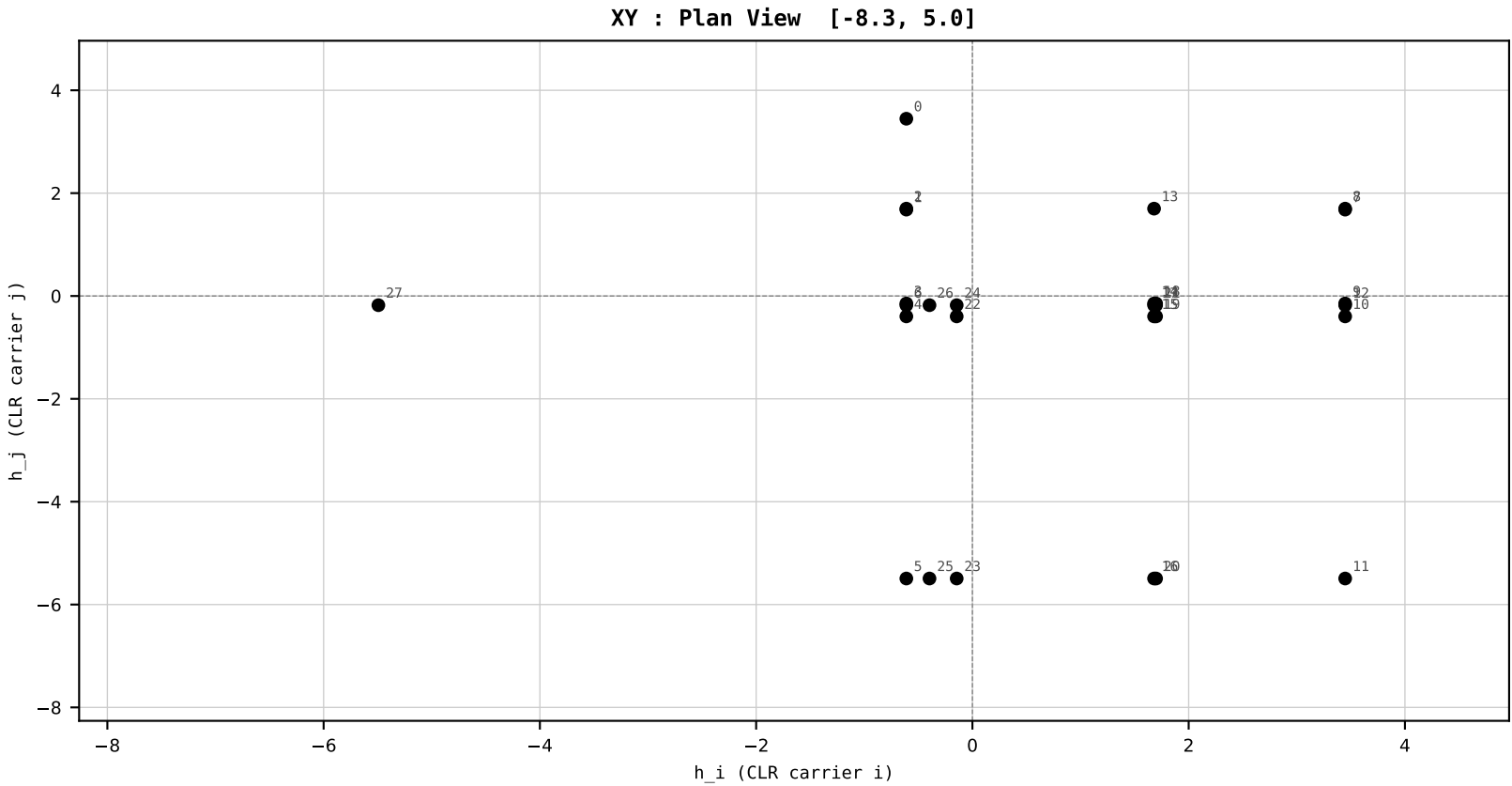
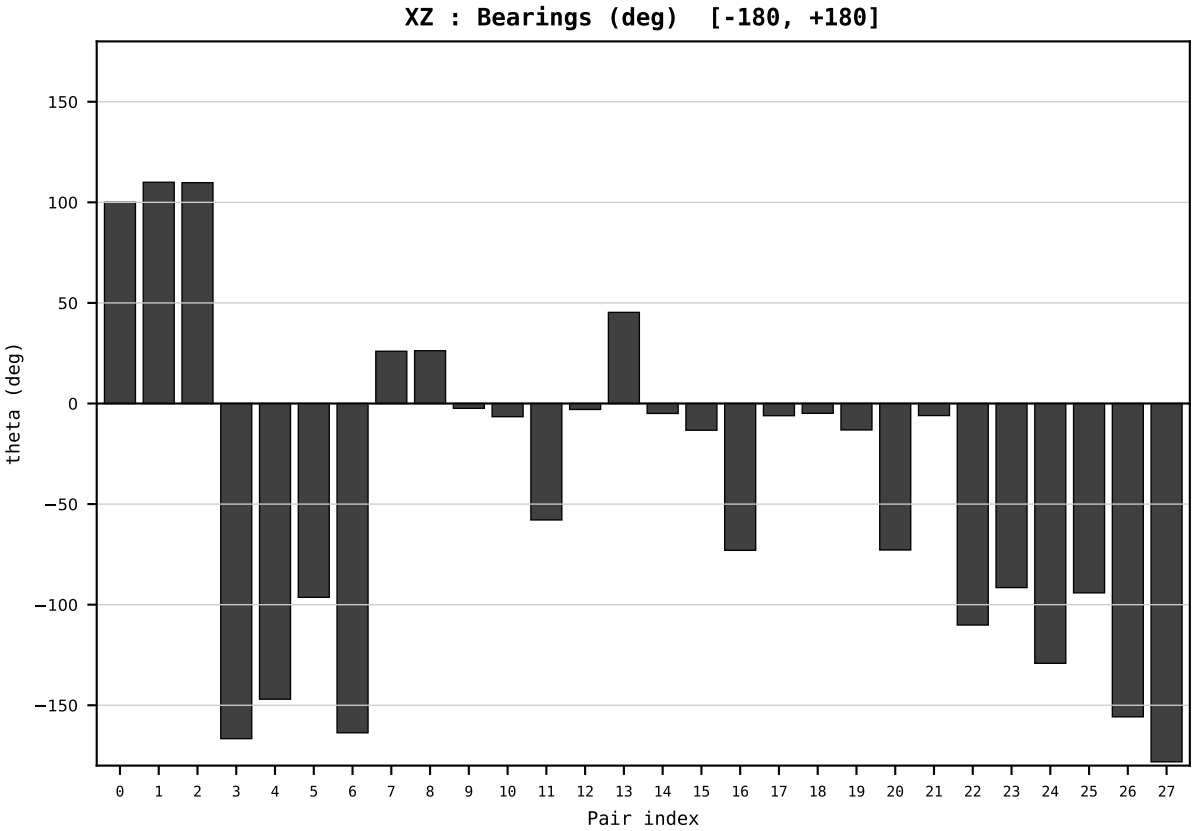
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.505927
Ring = Hs-4
E_metric= 48.3562
kappa_HS= 7638.88
omega = 2.6745 deg
d_A = 0.376406
Helm = Other Fossil
Helm d = -0.237572
DR = 8.941 HLR
DR ratio= 7638.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

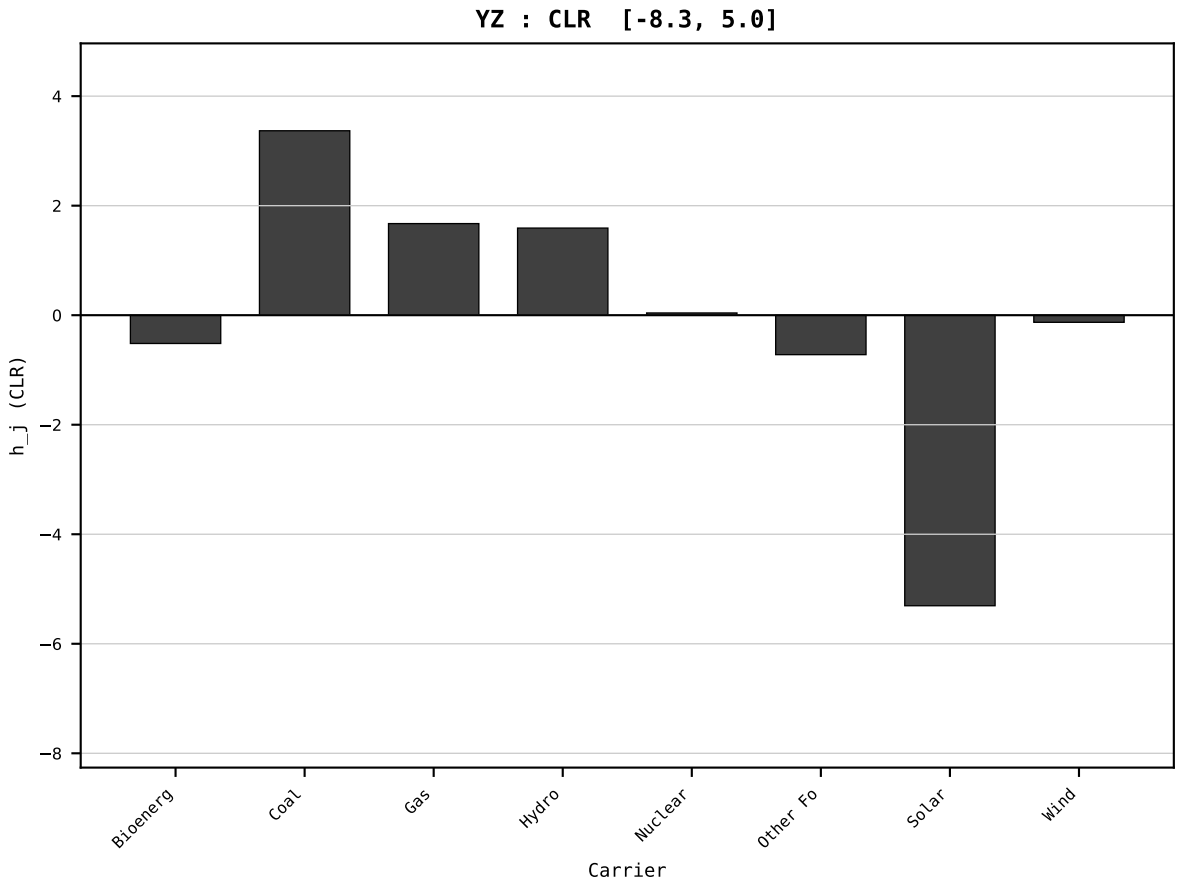
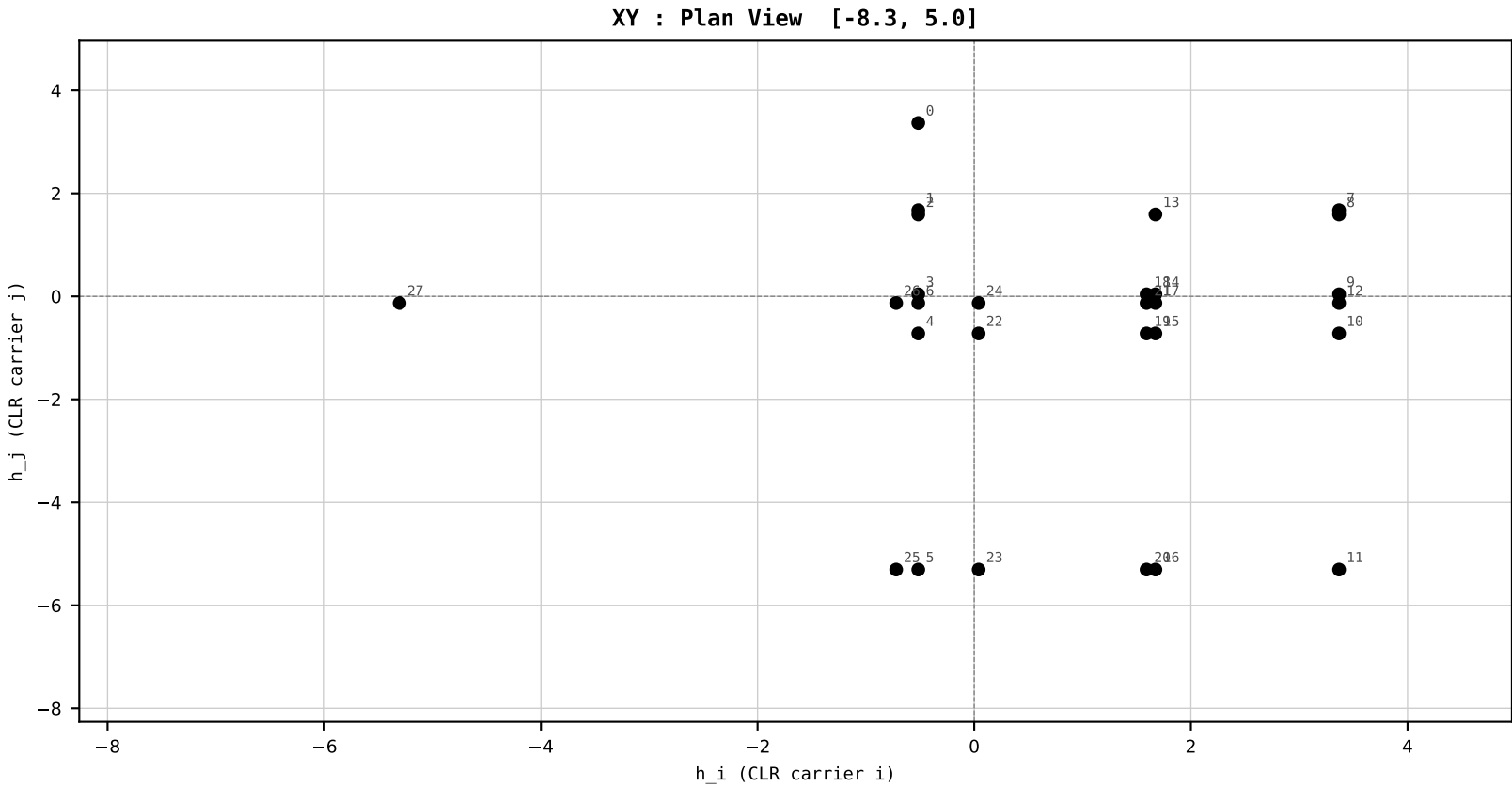
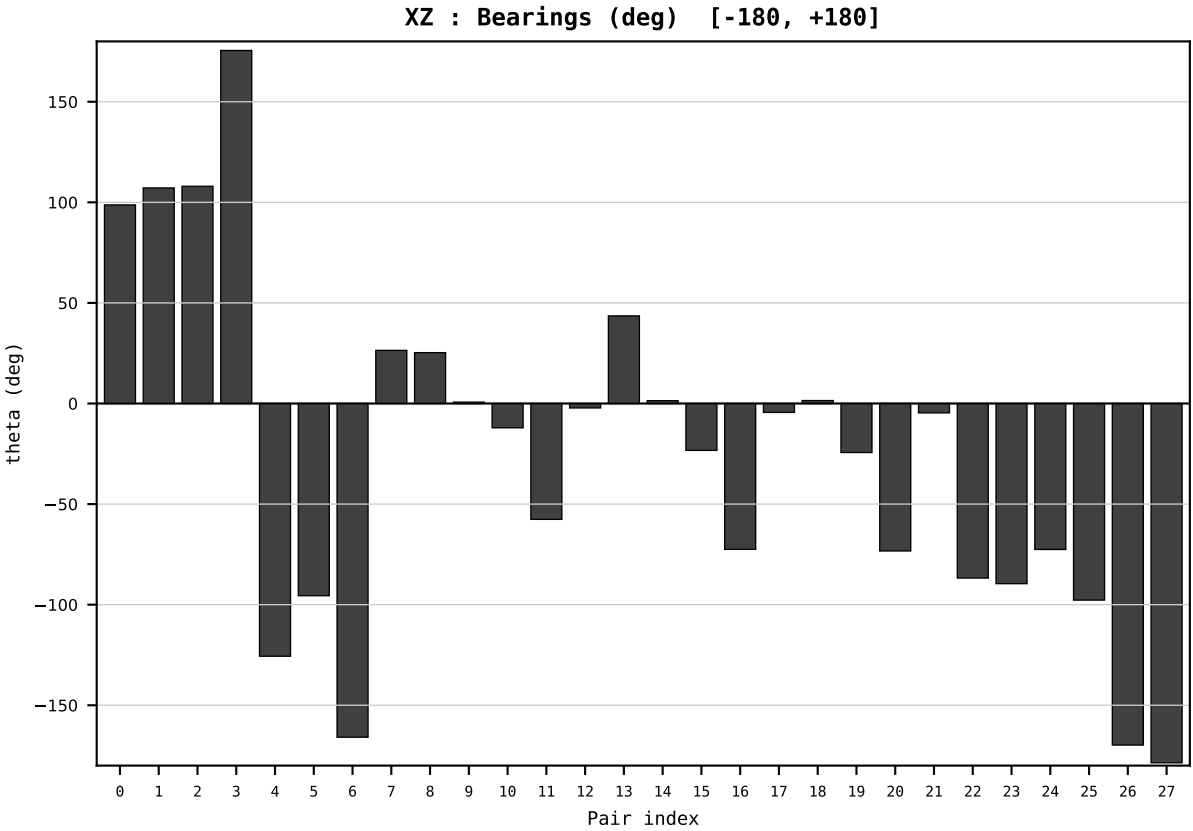
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.493325
Ring = Hs-3
E_metric= 45.6238
kappa_HS= 5845.18
omega = 3.3901 deg
d_A = 0.451796
Helm = Other Fossil
Helm d = -0.323553
DR = 8.673 HLR
DR ratio= 5845.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

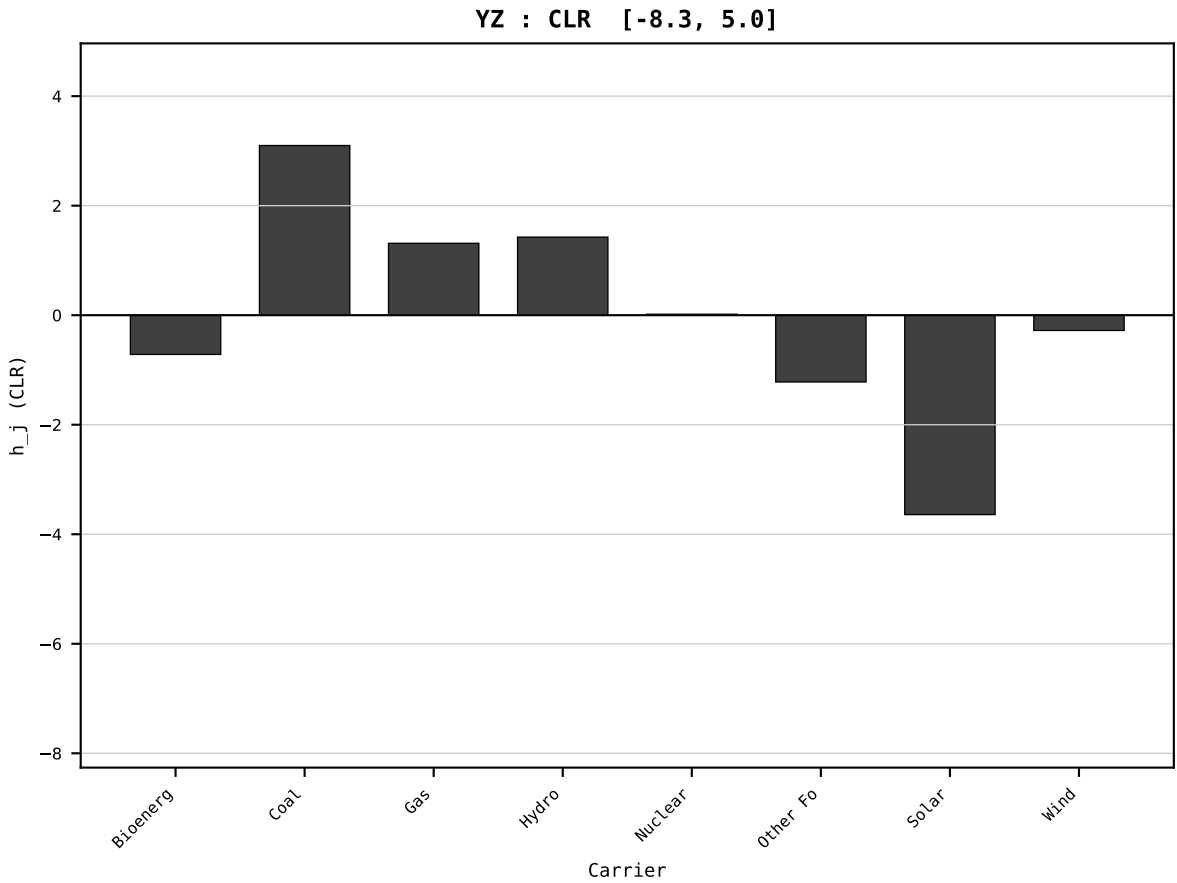
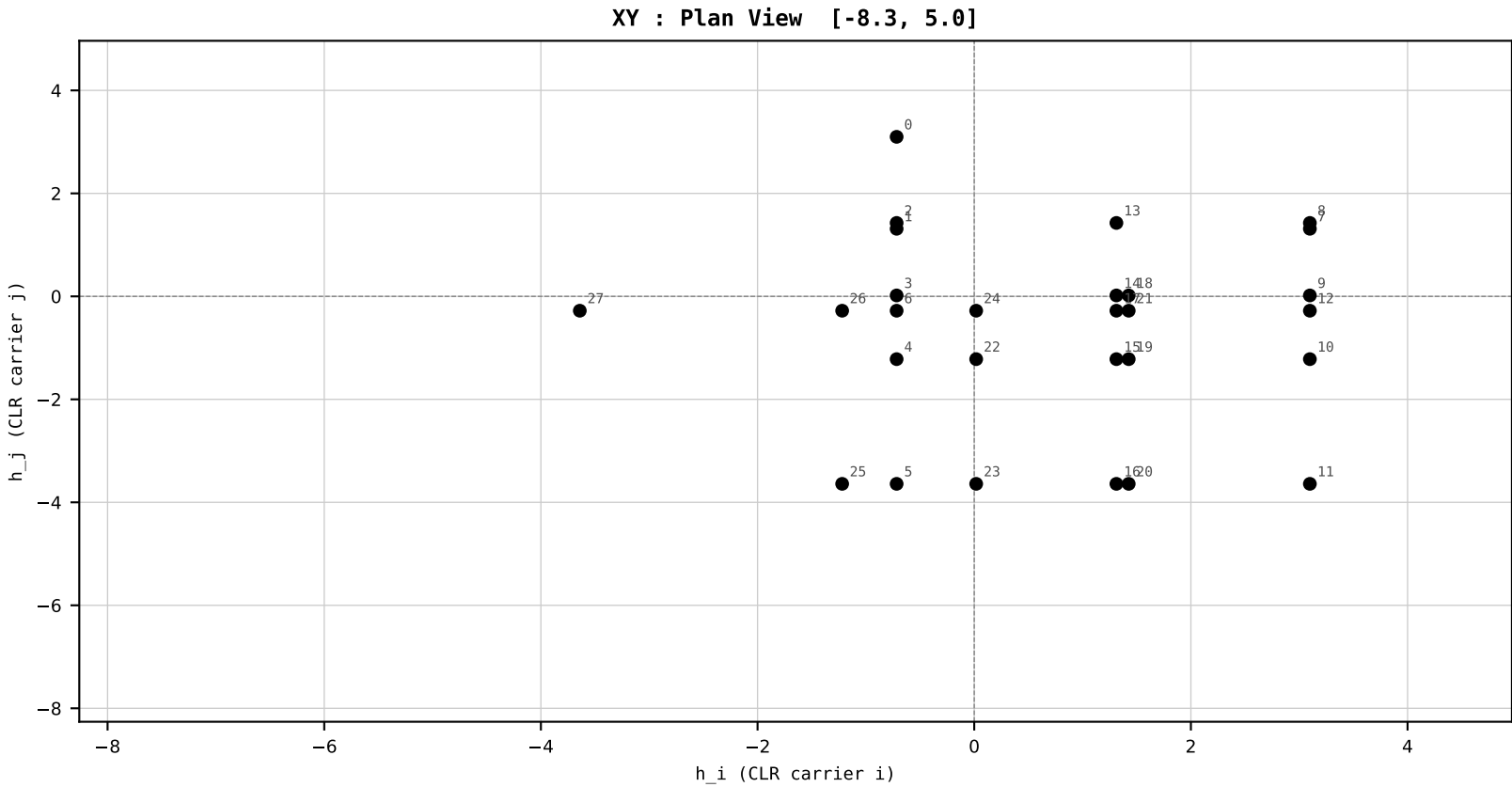
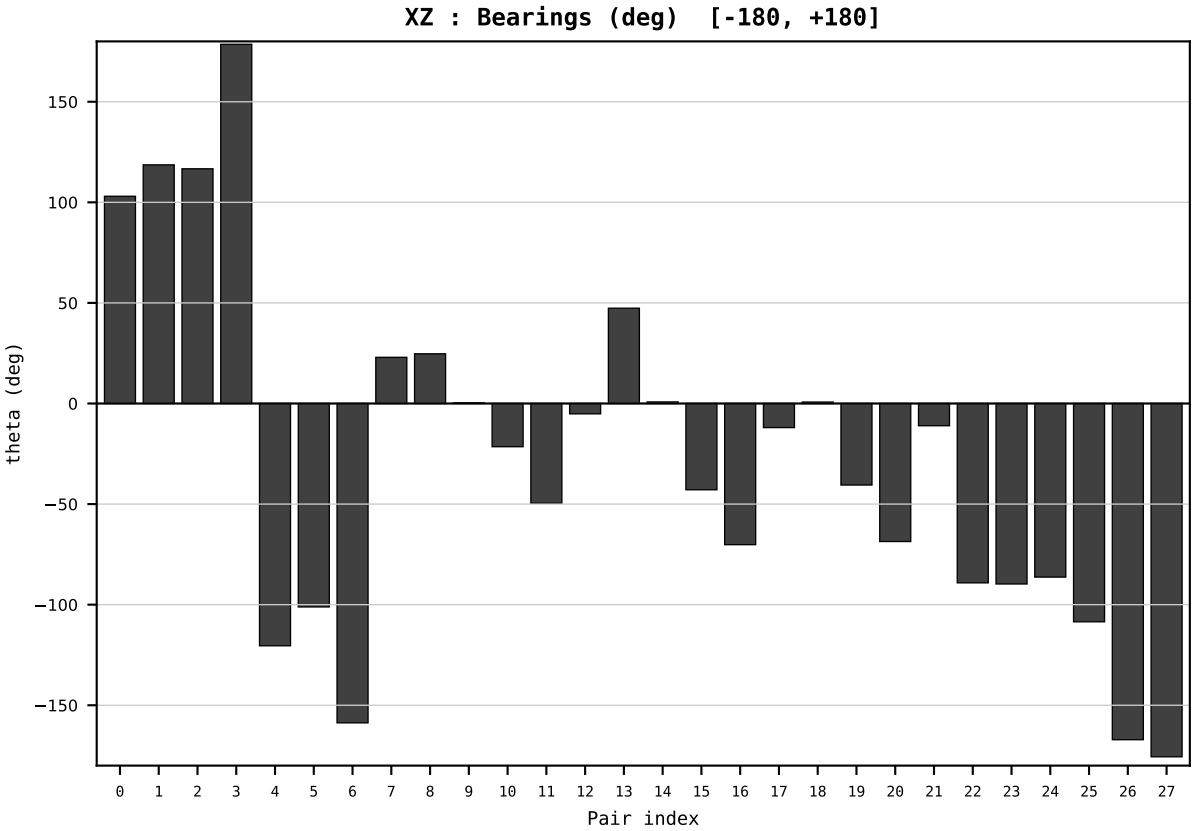
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.480839
Ring = Hs-3
E_metric= 28.6888
kappa_HS= 844.88
omega = 11.1115 deg
d_A = 1.819832
Helm = Solar
Helm d = +1.664605
DR = 6.739 HLR
DR ratio= 844.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

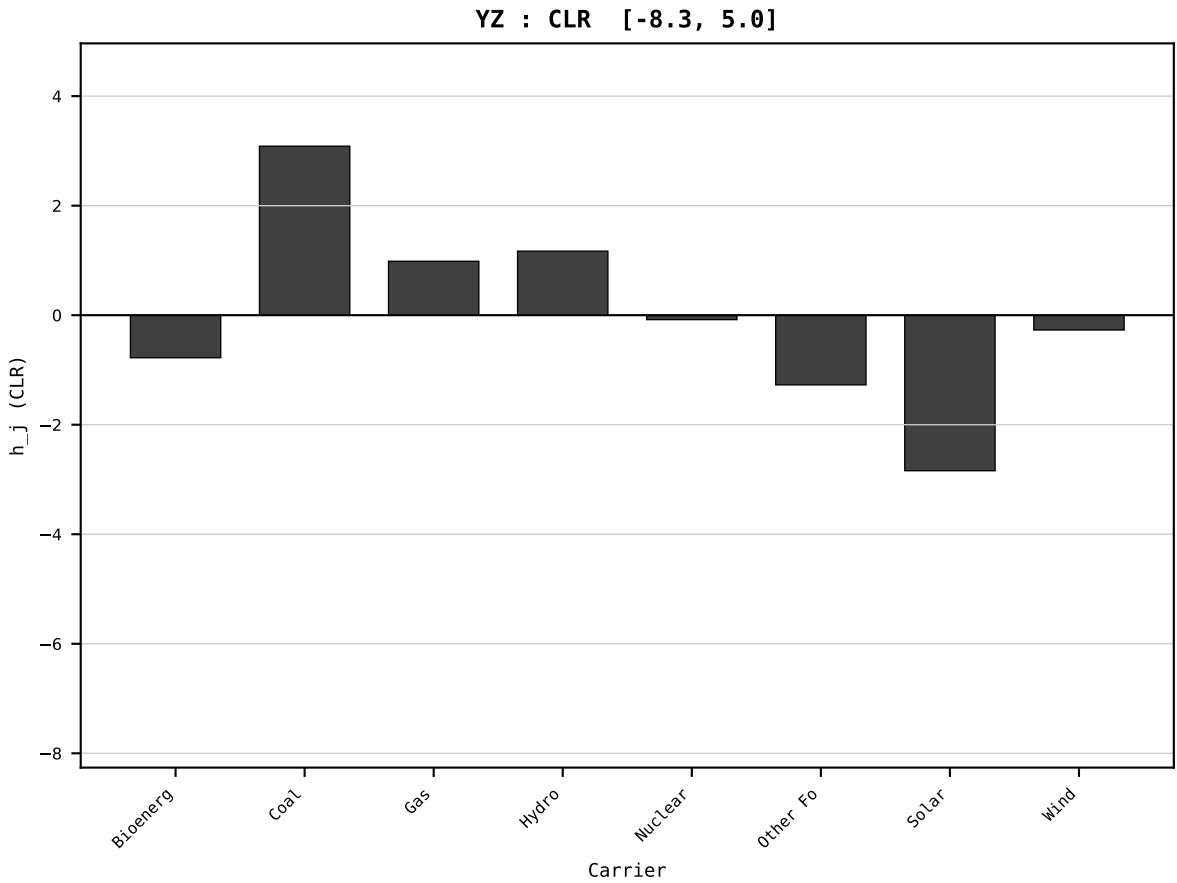
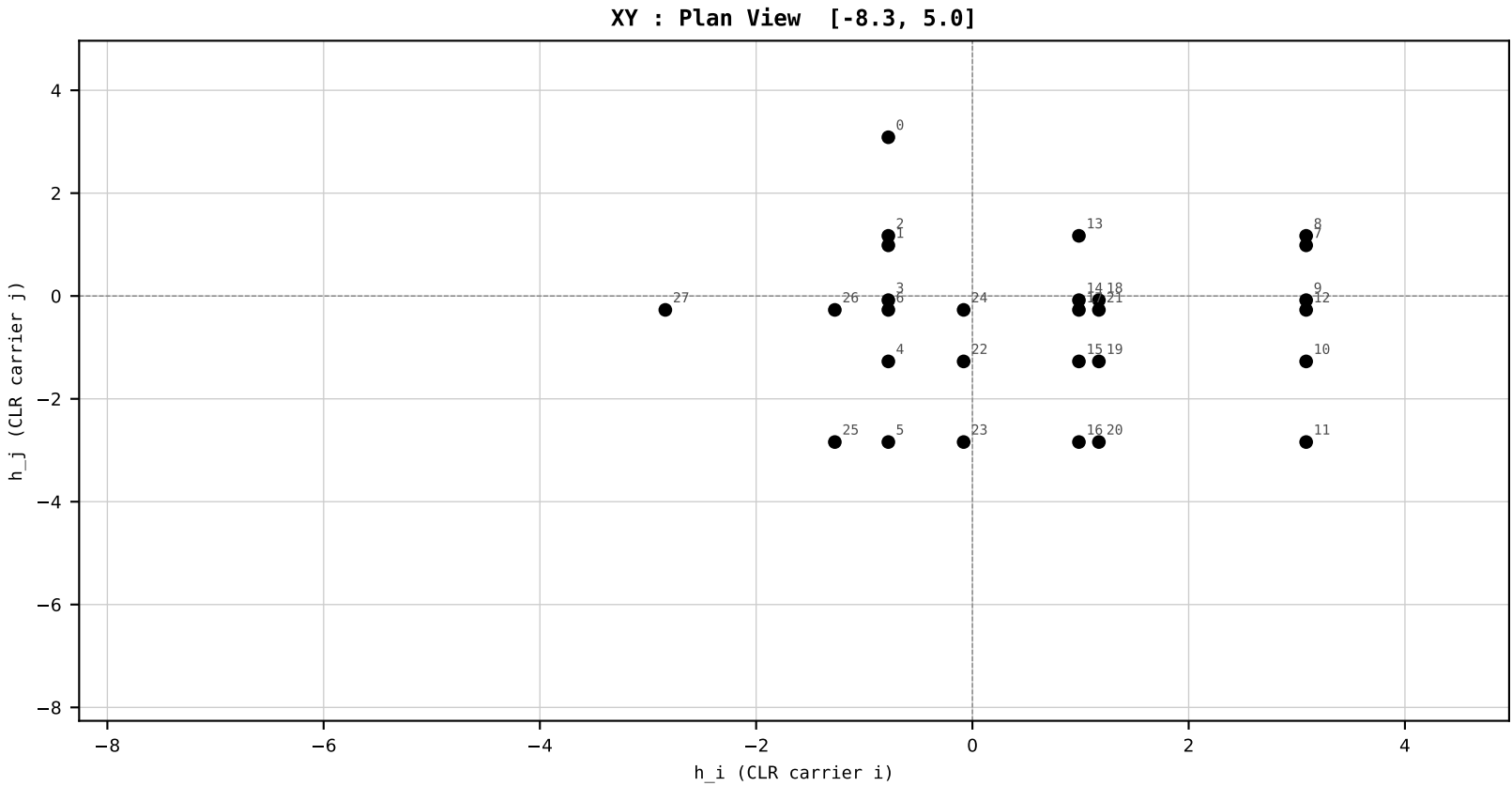
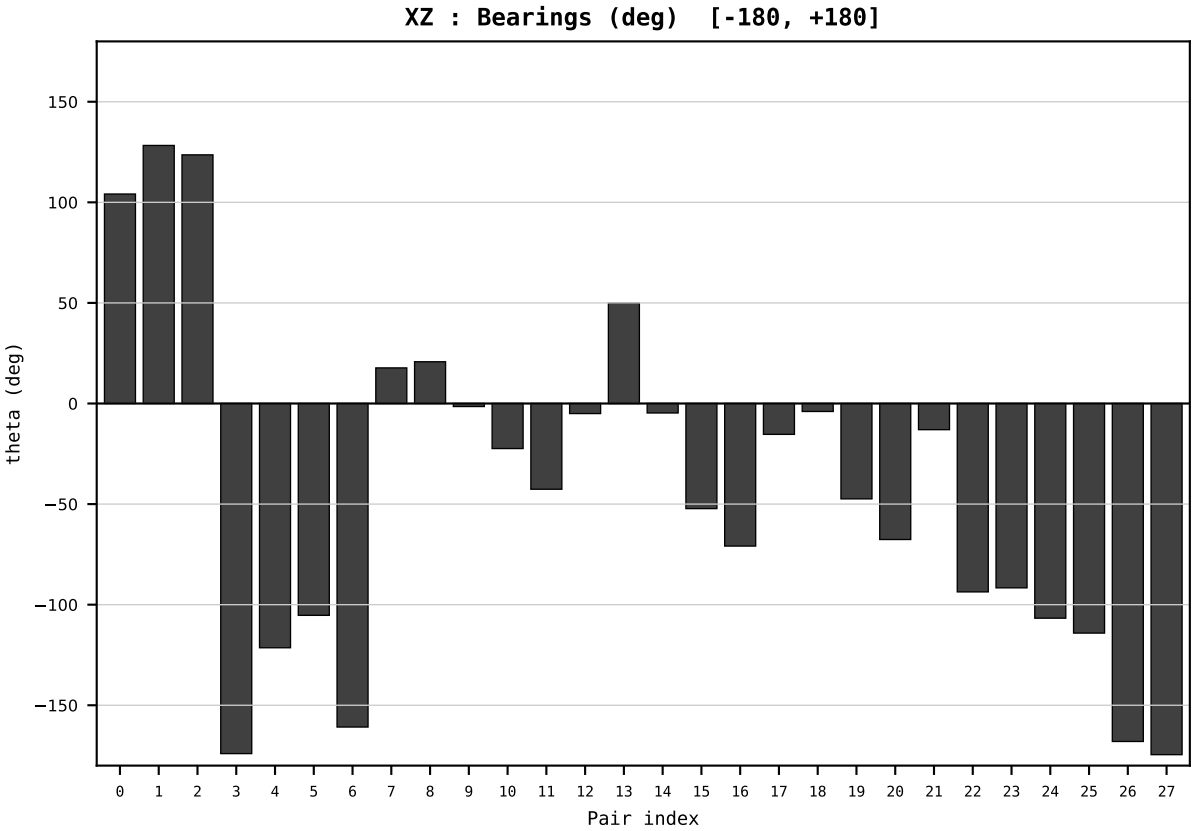
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.516572
Ring = Hs-4
E_metric= 22.2333
kappa_HS= 374.82
omega = 7.3940 deg
d_A = 0.911514
Helm = Solar
Helm d = +0.800856
DR = 5.926 HLR
DR ratio= 374.8x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=8 N=26 pairs=28

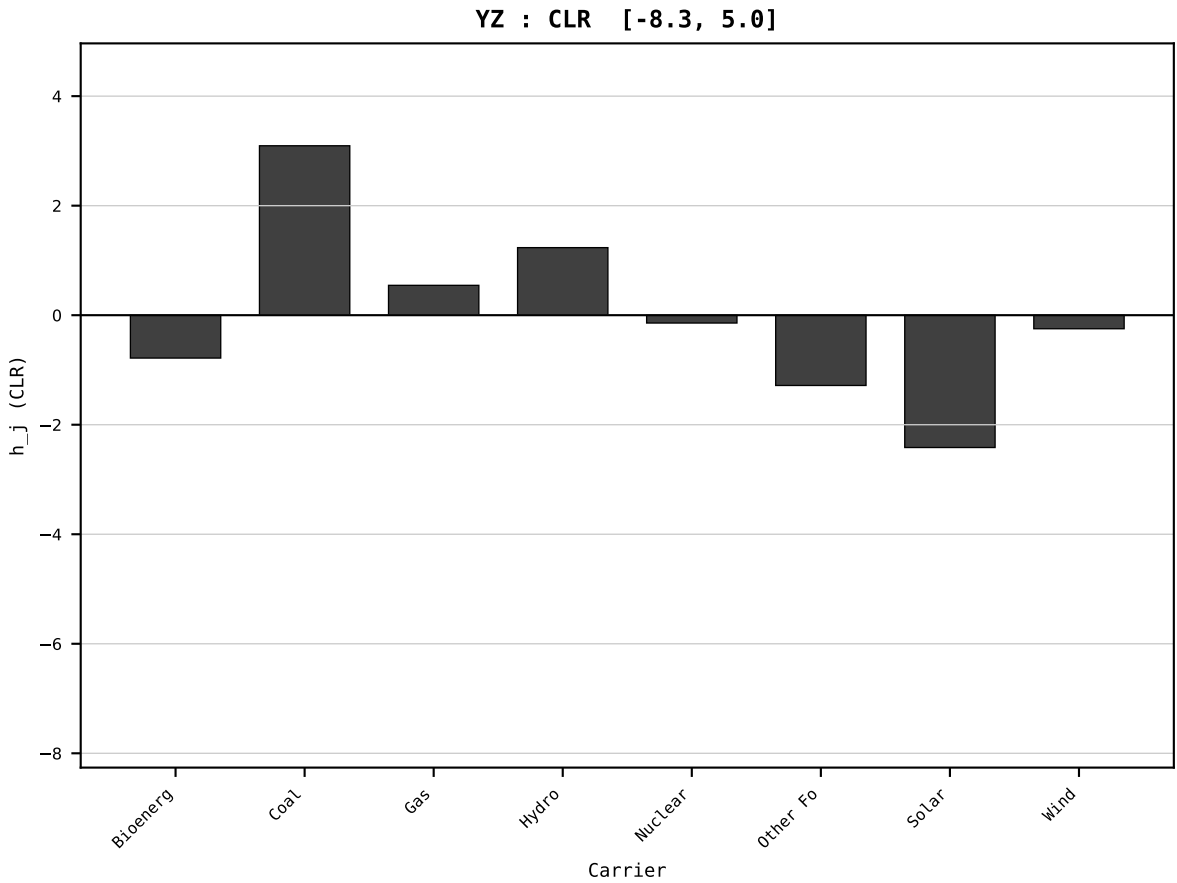
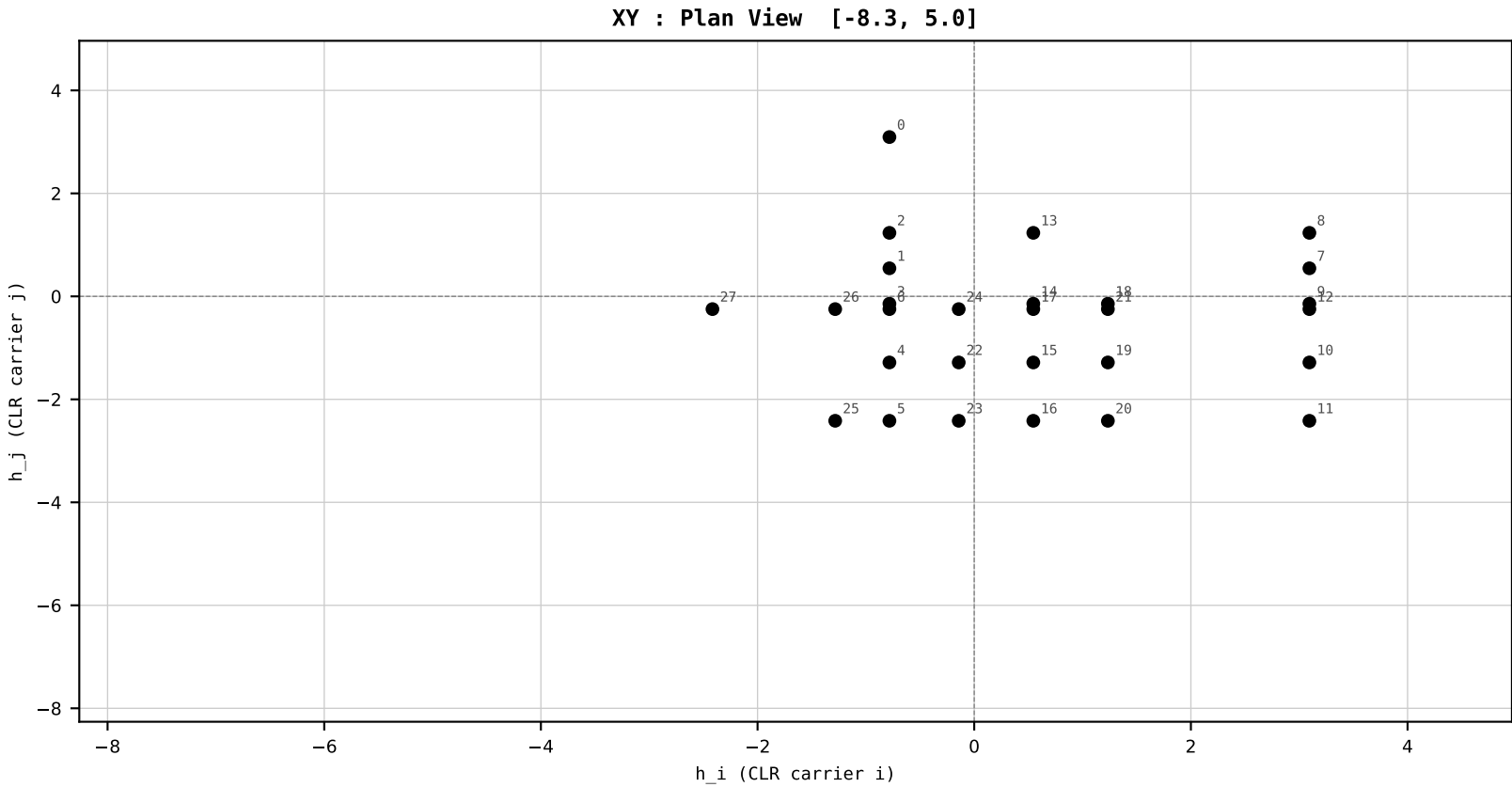
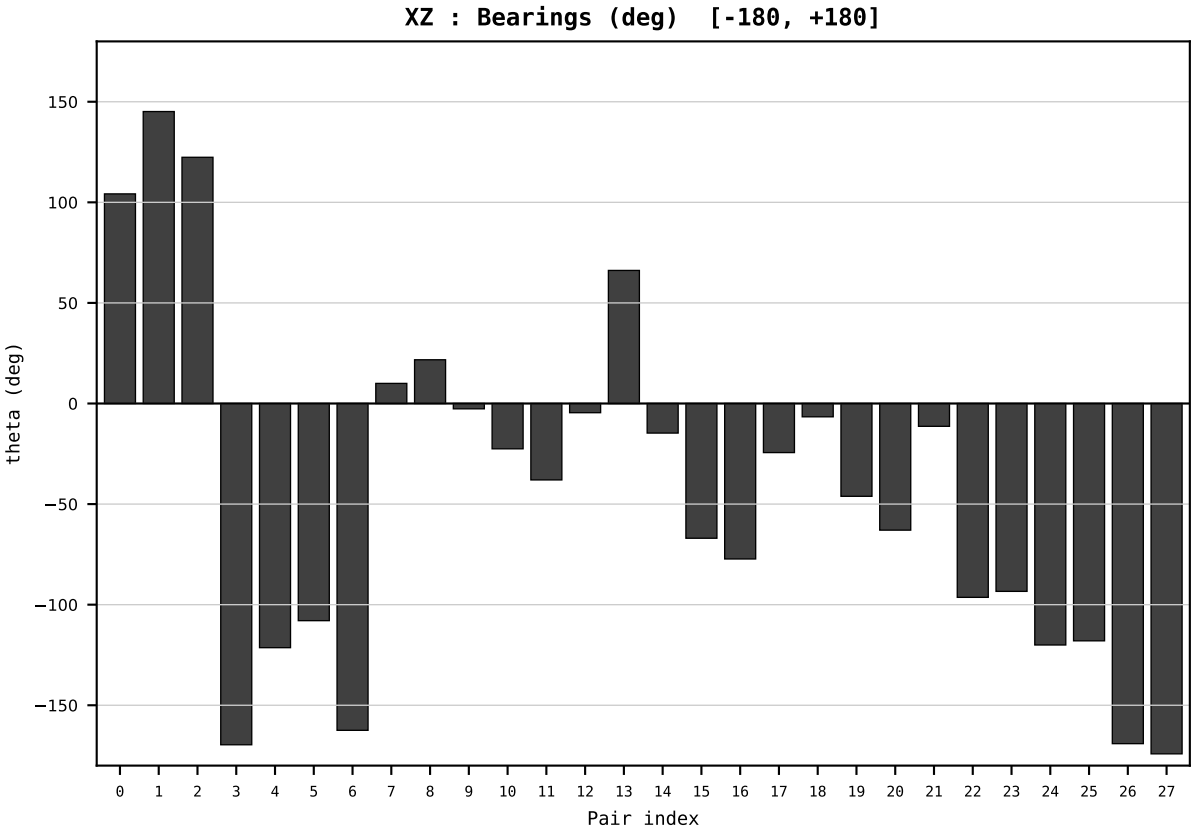
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.538501
Ring = Hs-4
E_metric= 19.5721
kappa_HS= 247.22
omega = 6.8358 deg
d_A = 0.617545
Helm = Gas
Helm d = -0.439615
DR = 5.510 HLR
DR ratio= 247.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

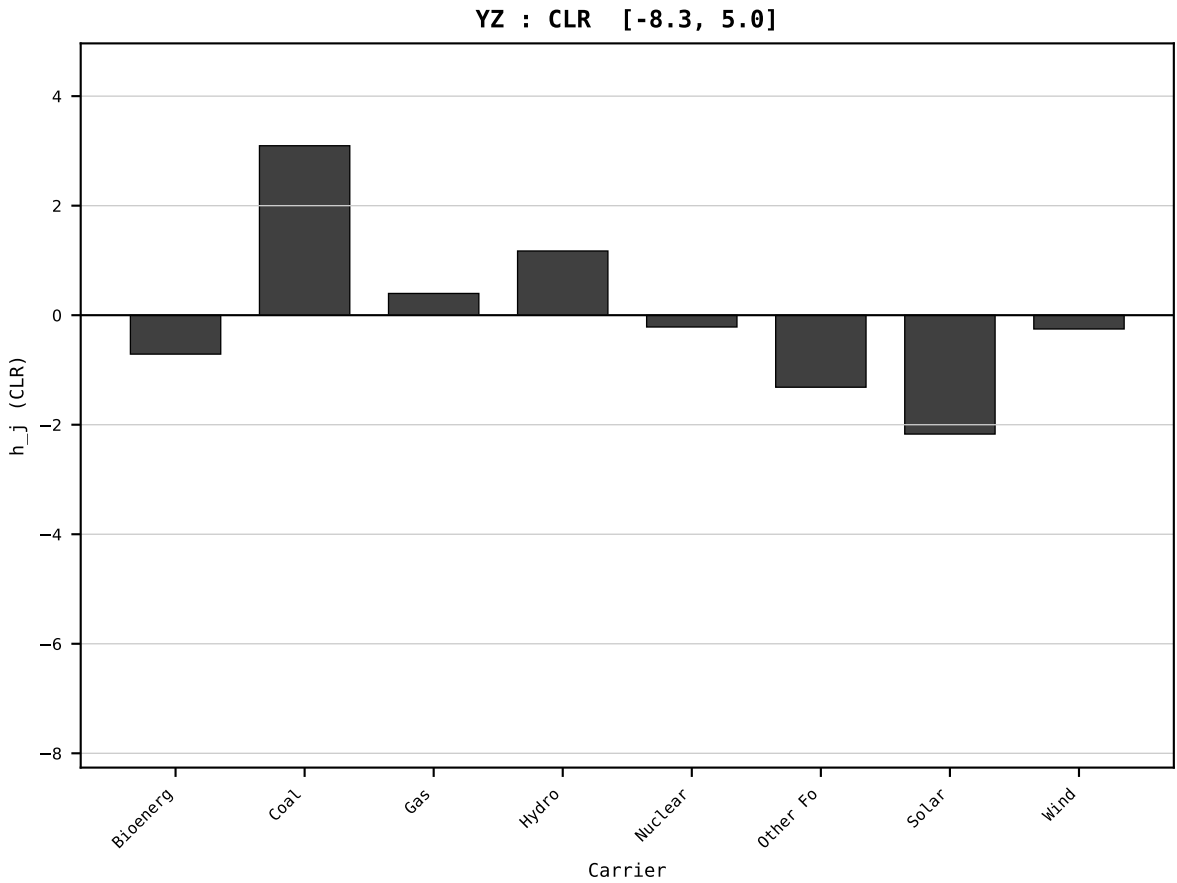
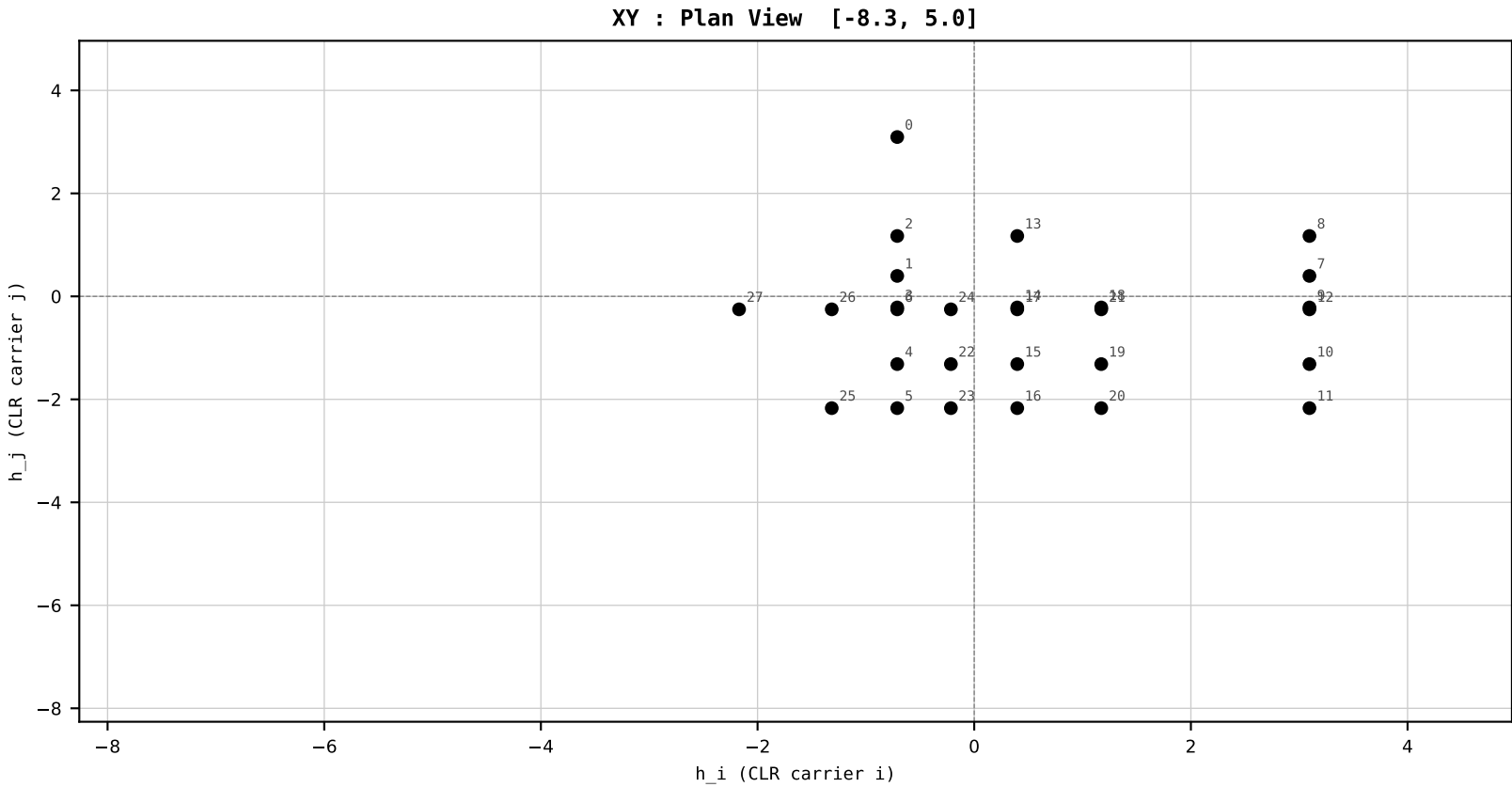
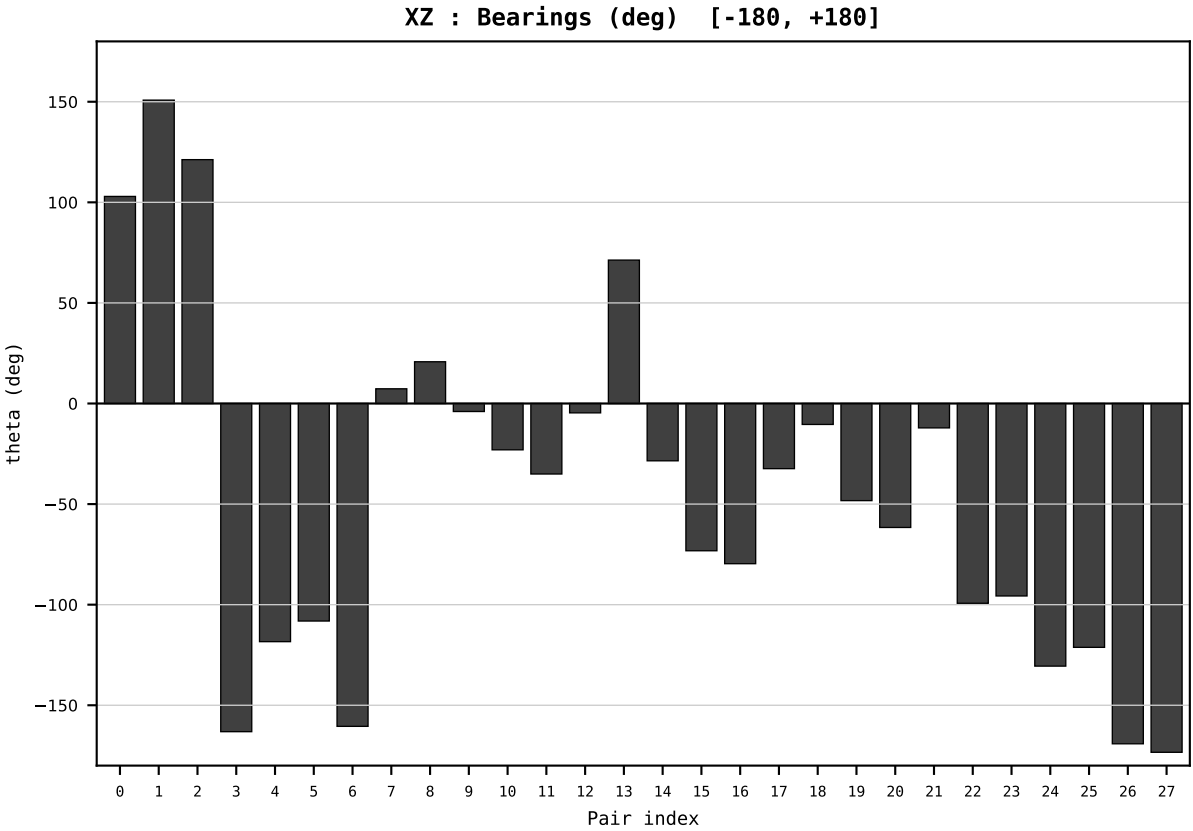
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.549920
Ring = Hs-4
E_metric= 18.1649
kappa_HS= 193.47
omega = 3.5245 deg
d_A = 0.312363
Helm = Solar
Helm d = +0.245715
DR = 5.265 HLR
DR ratio= 193.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

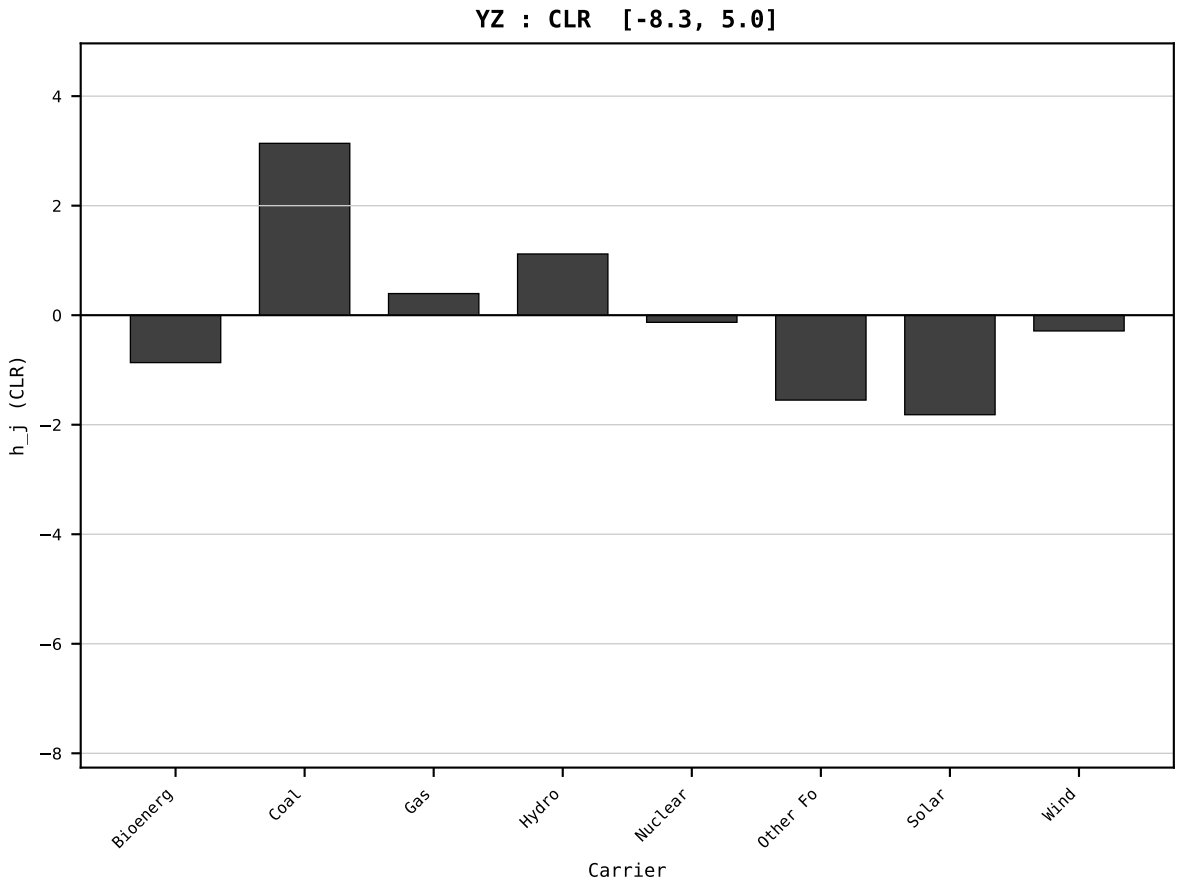
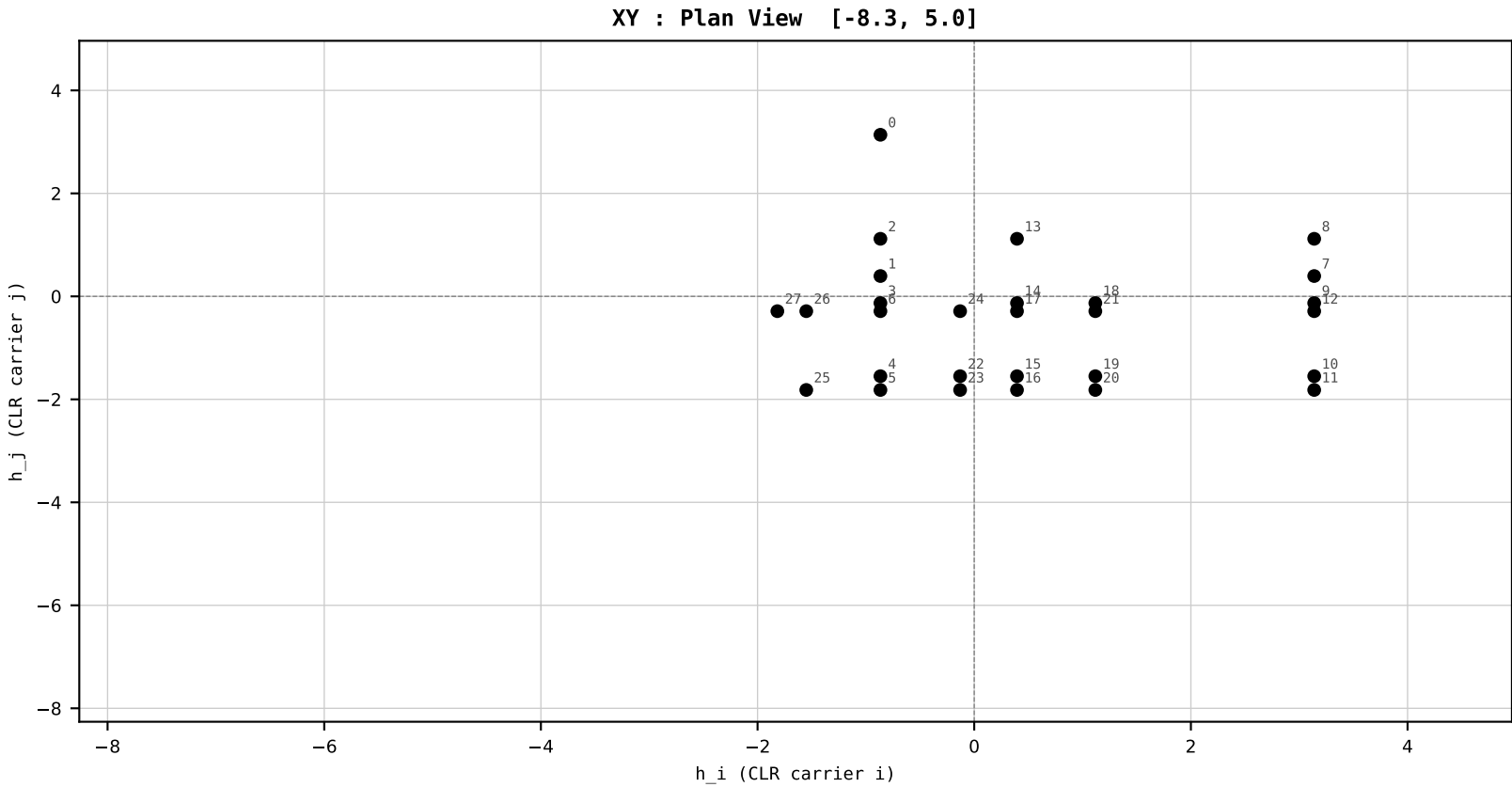
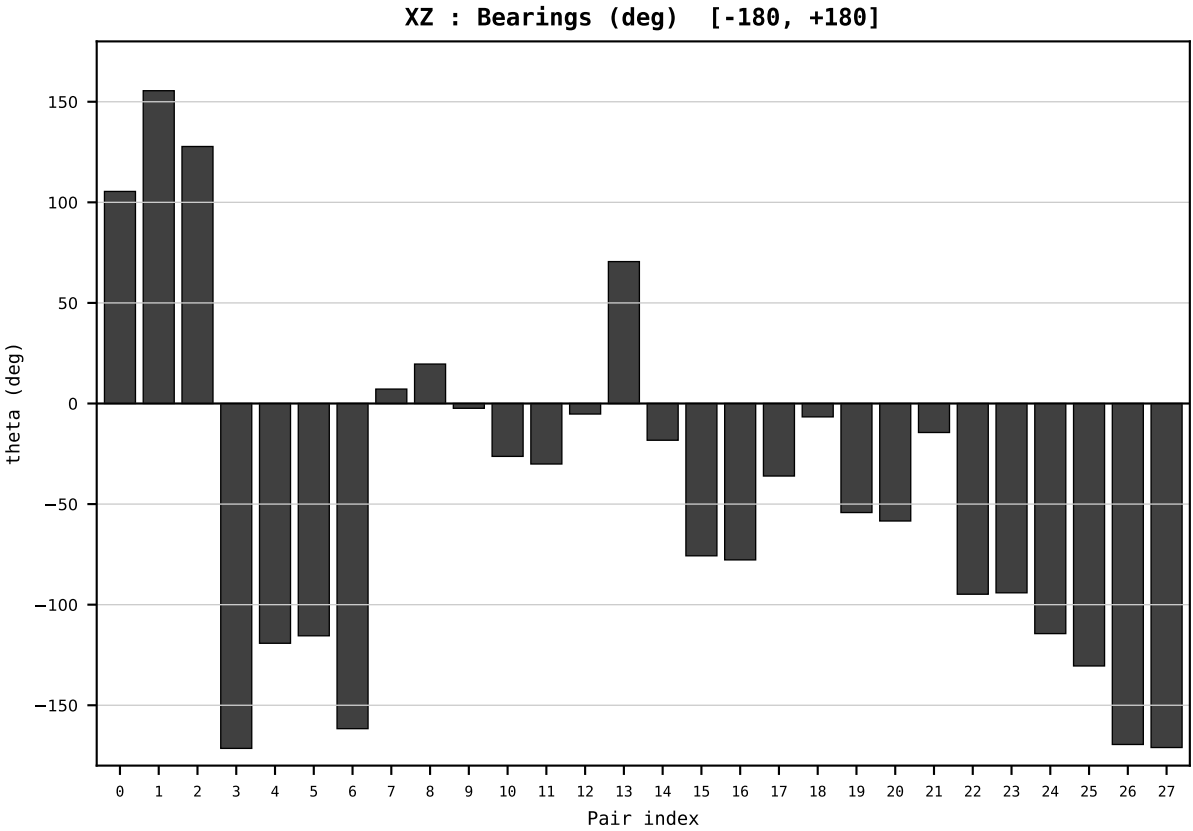
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.566019
Ring = Hs-4
E_metric= 17.8098
kappa_HS= 141.97
omega = 6.2793 deg
d_A = 0.466443
Helm = Solar
Helm d = +0.353359
DR = 4.956 HLR
DR ratio= 142.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=8 N=26 pairs=28

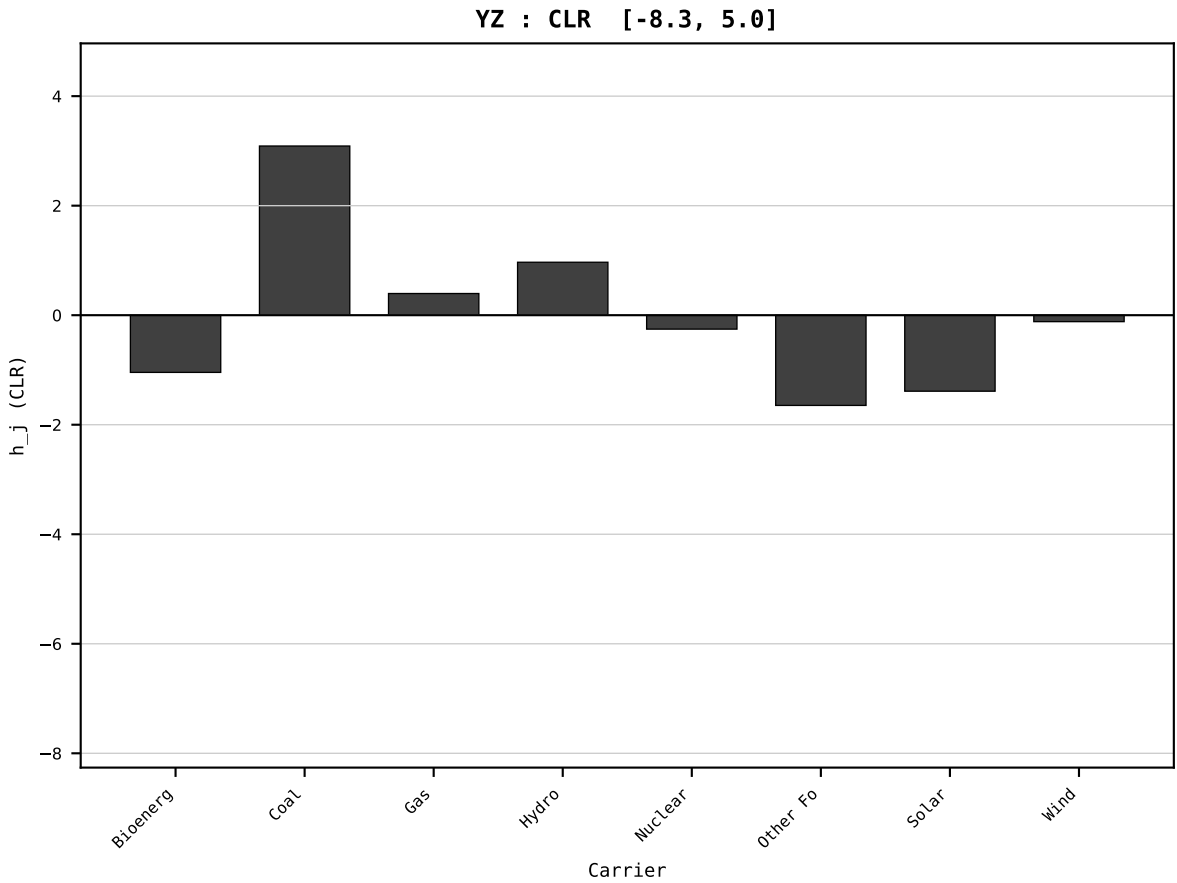
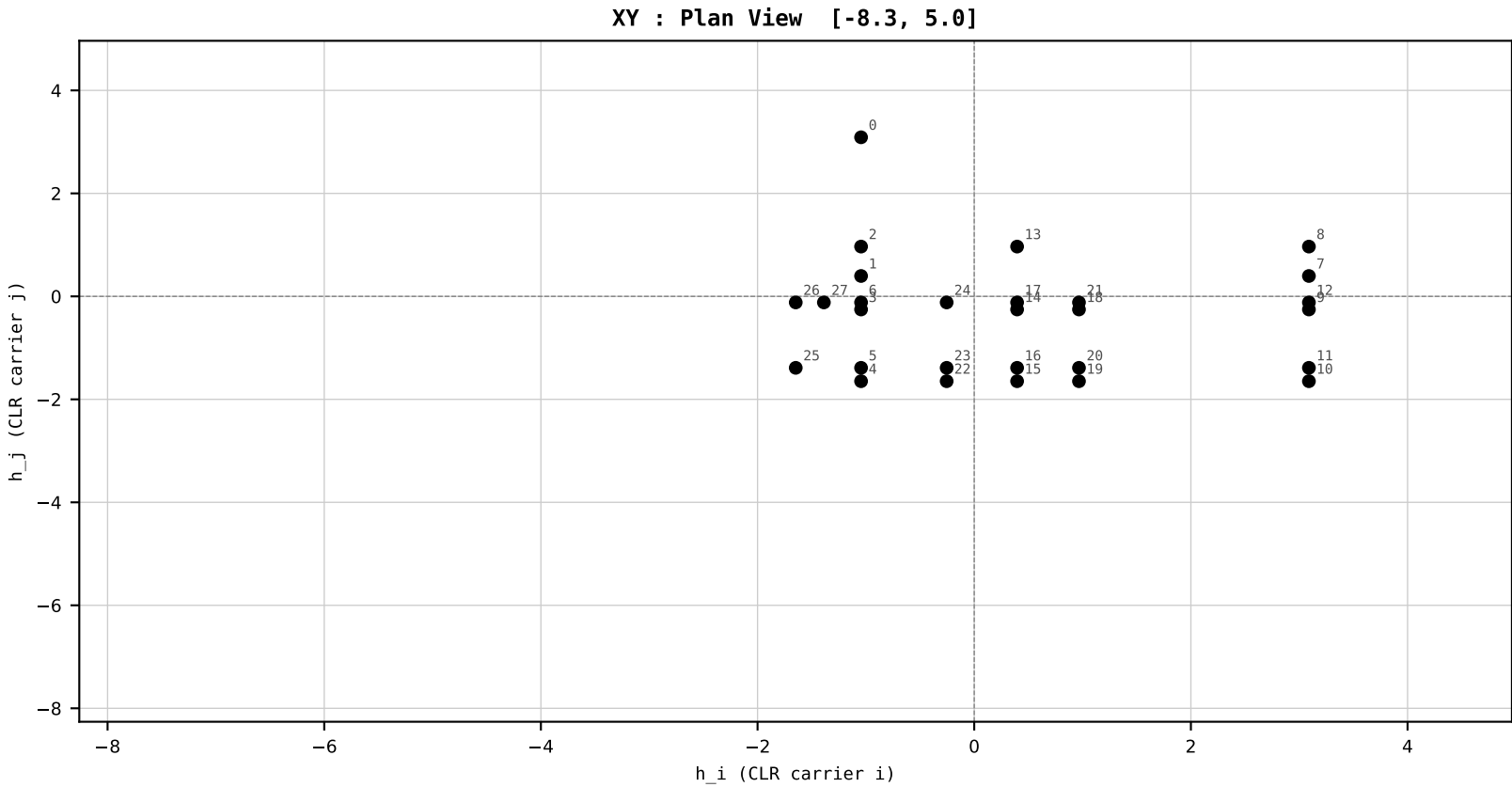
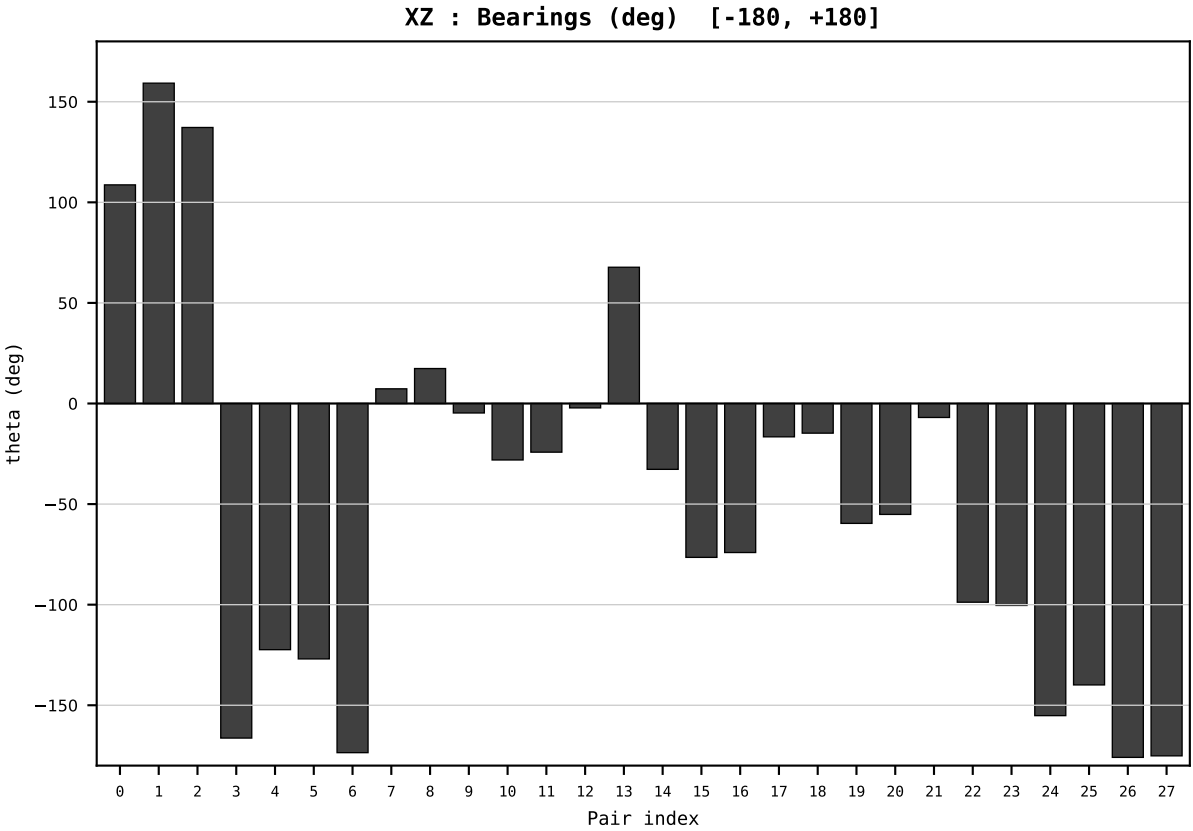
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.562151
Ring = Hs-4
E_metric= 16.4465
kappa_HS= 114.09
omega = 7.1743 deg
d_A = 0.543254
Helm = Solar
Helm d = +0.429368
DR = 4.737 HLR
DR ratio= 114.1x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

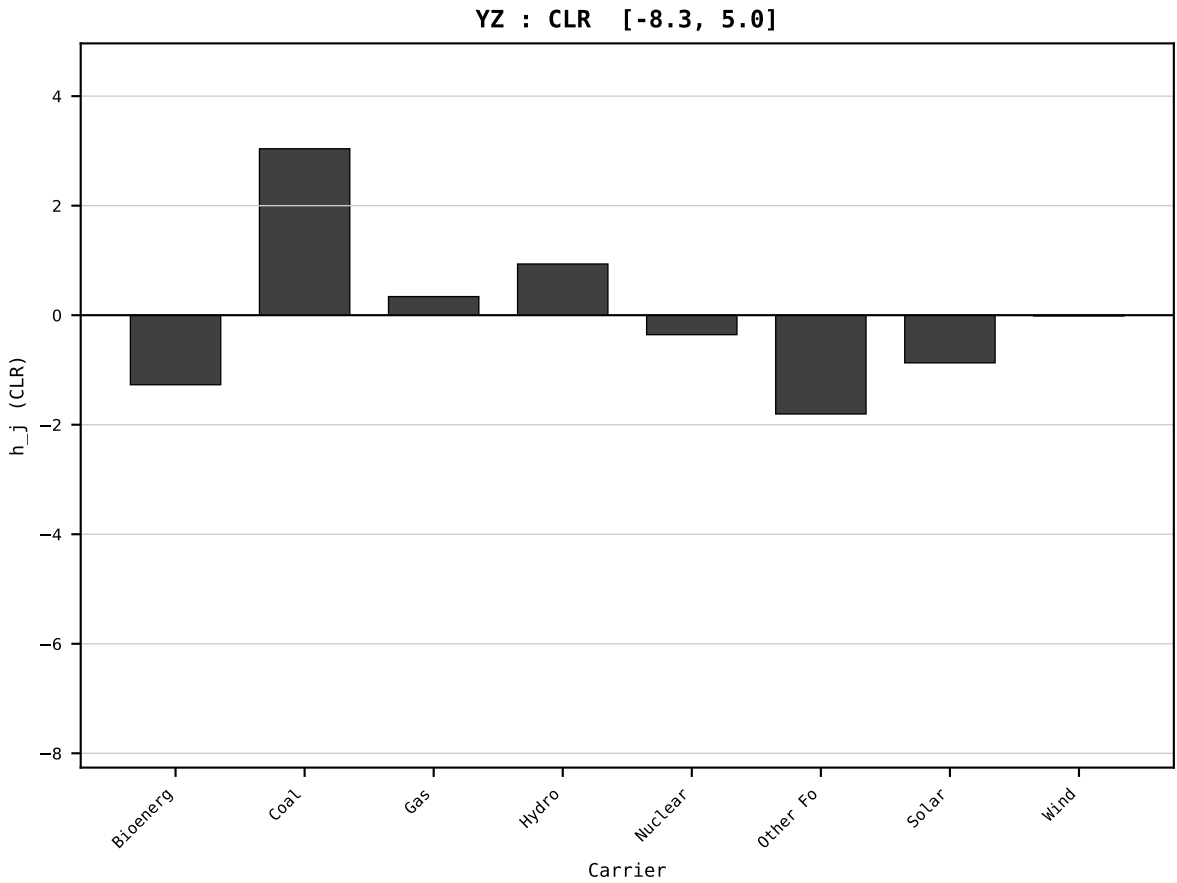
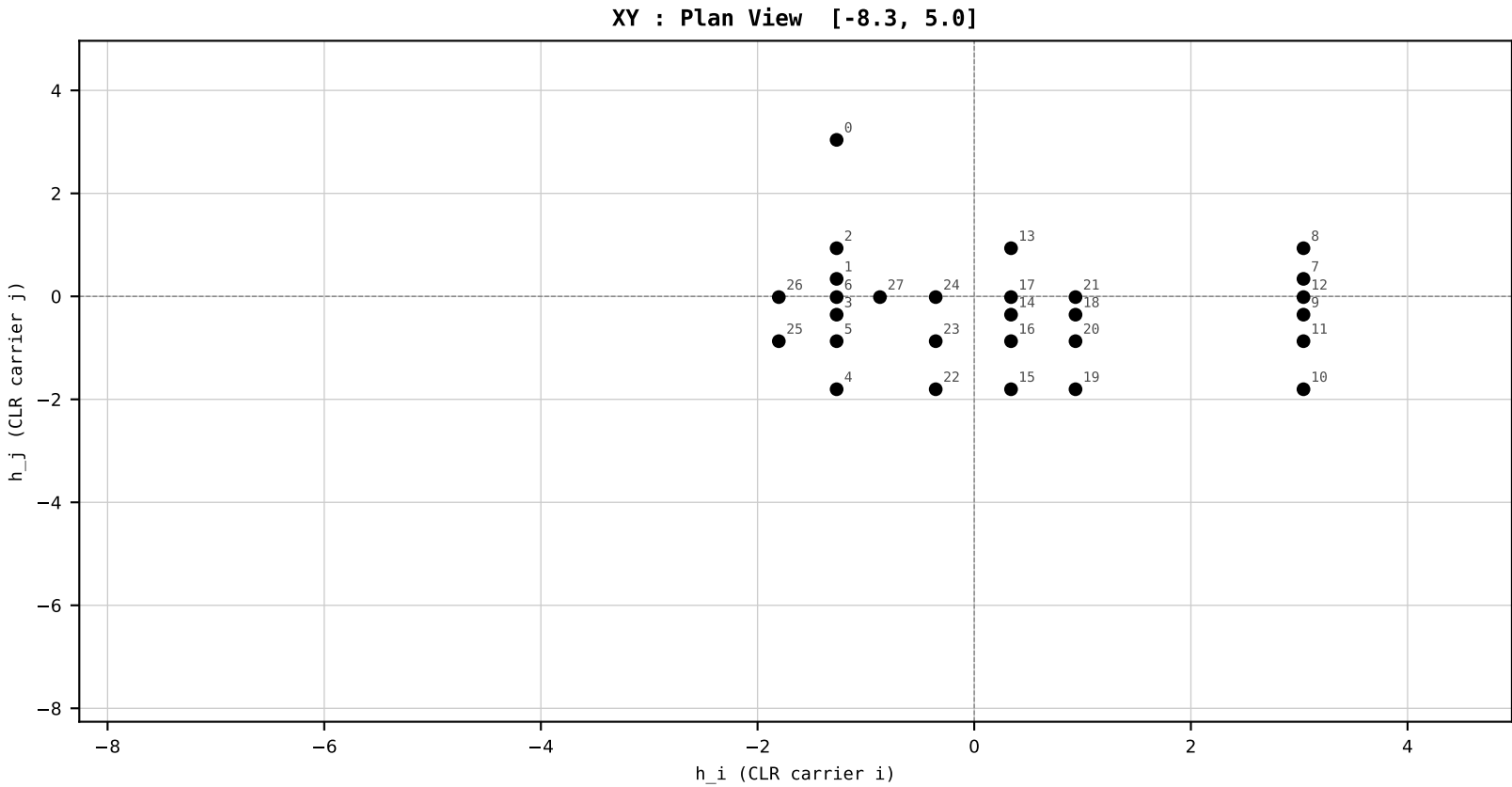
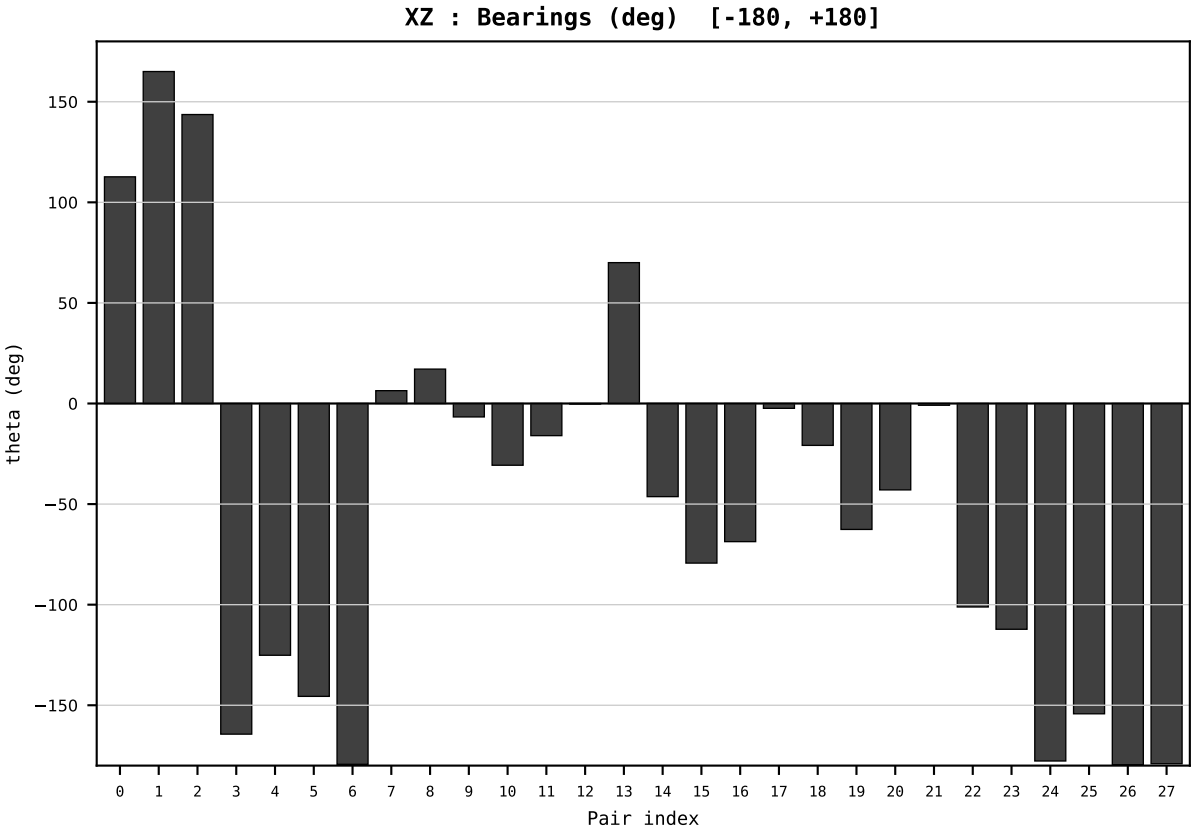
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.550408
Ring = Hs-4
E_metric= 15.9784
kappa_HS= 126.88
omega = 8.6362 deg
d_A = 0.609086
Helm = Solar
Helm d = +0.517691
DR = 4.843 HLR
DR ratio= 126.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

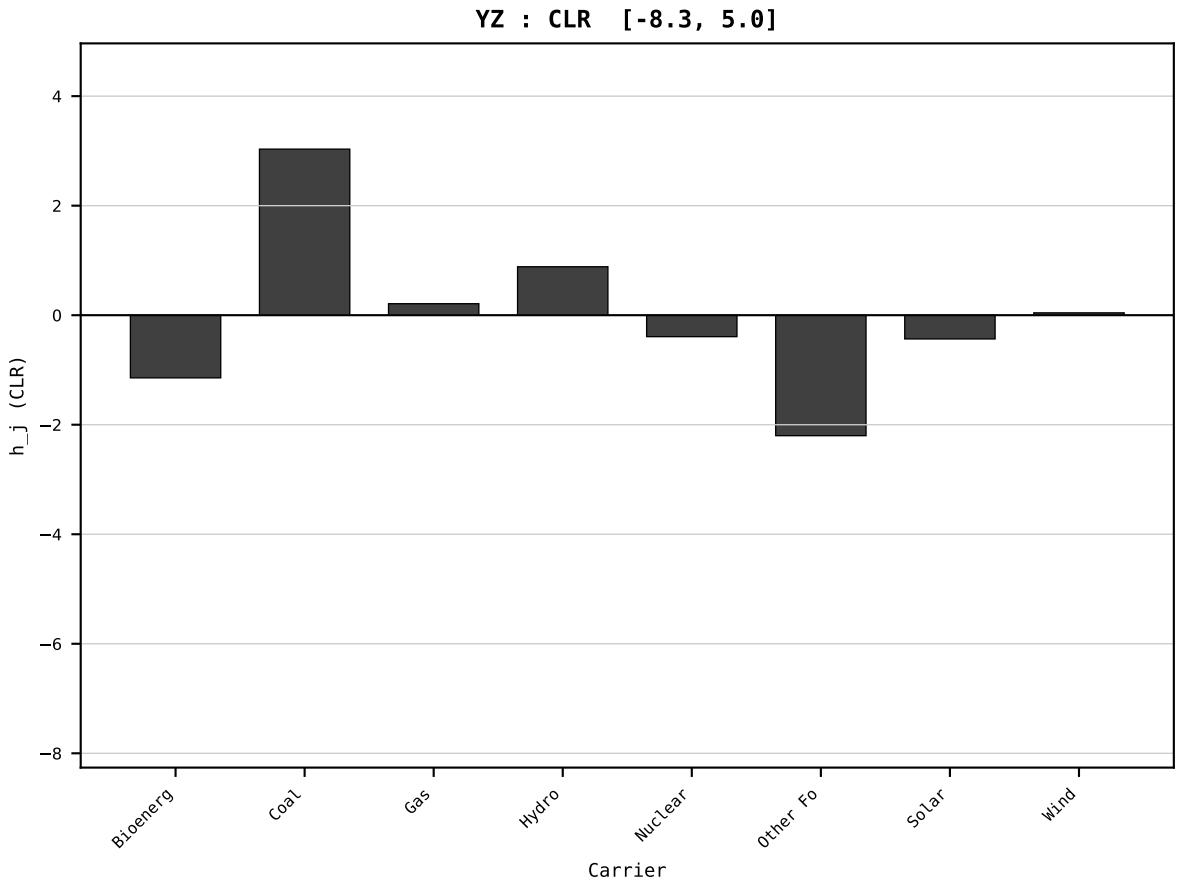
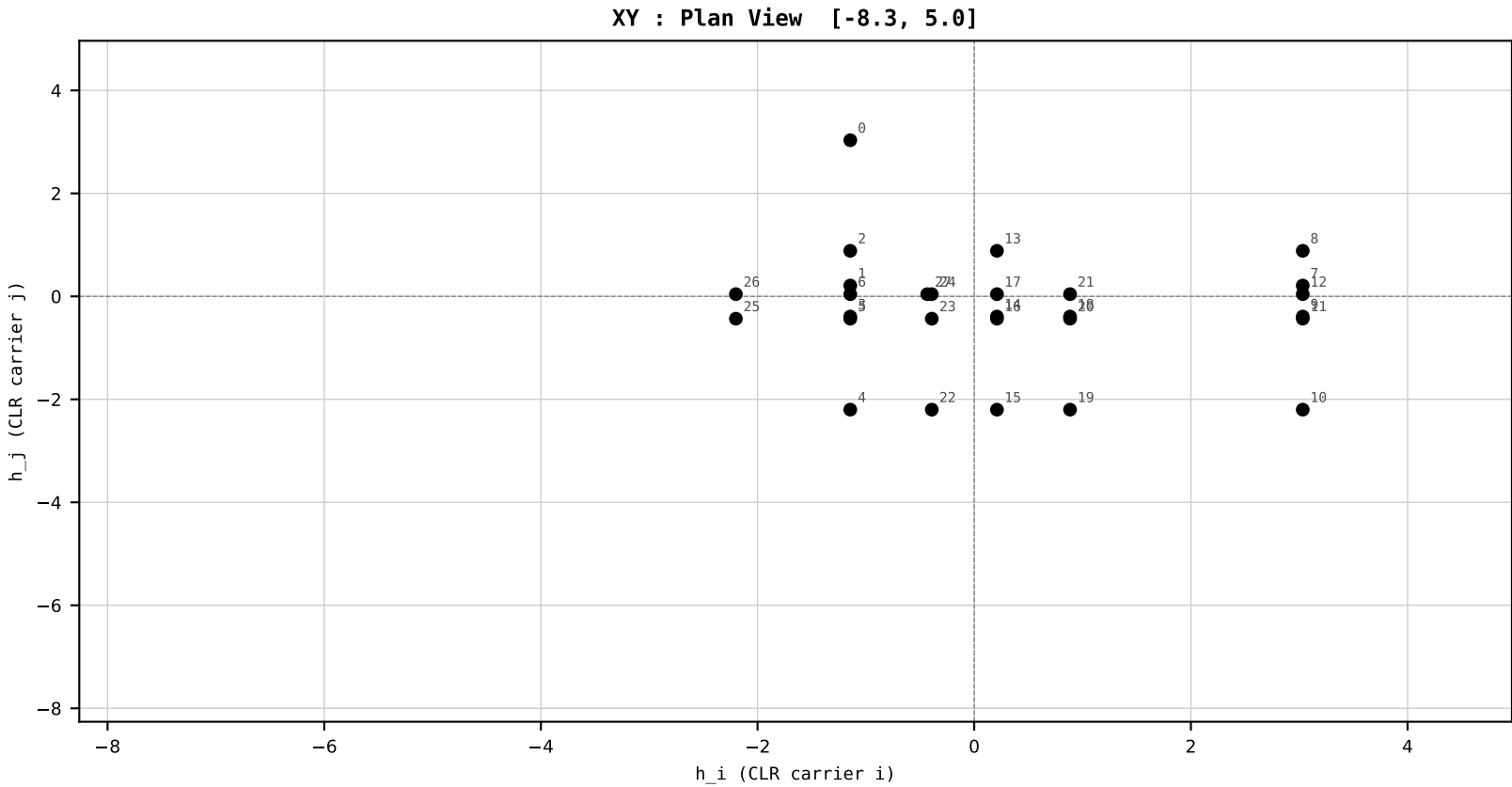
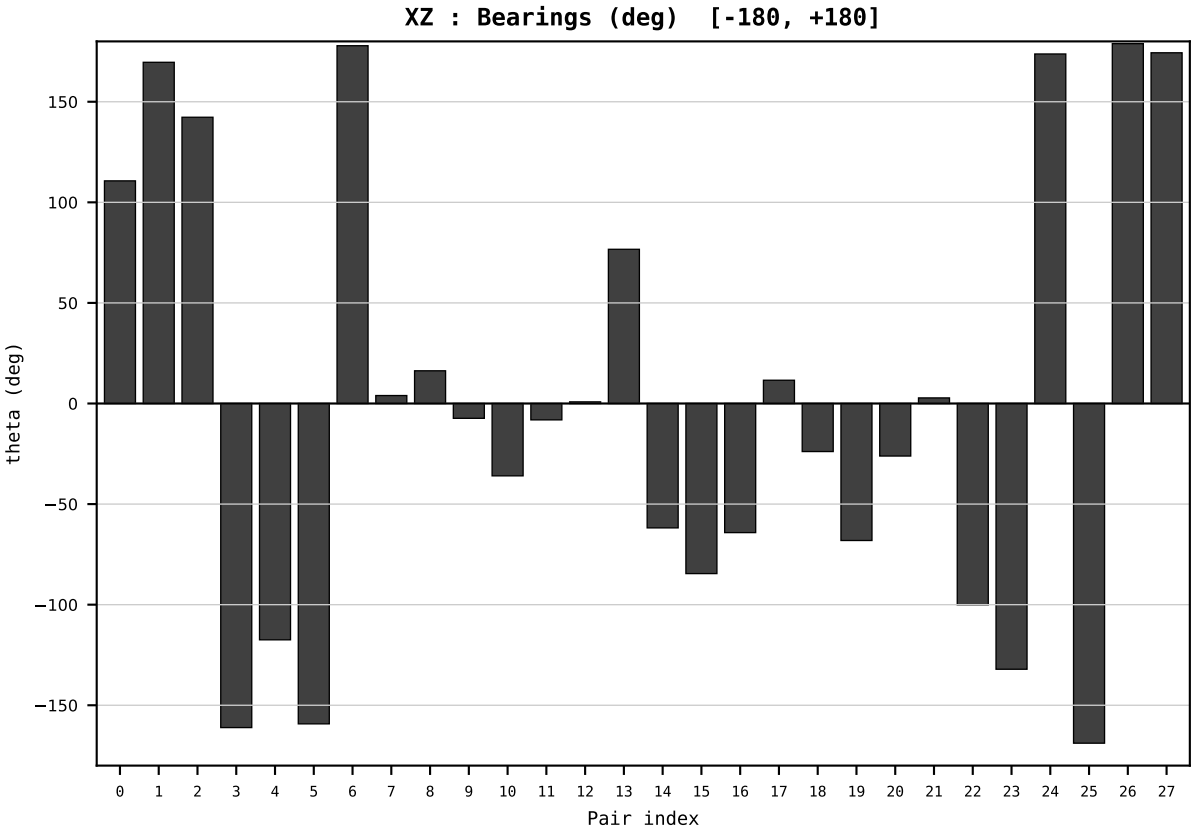
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.546713
Ring = Hs-4
E_metric= 16.5164
kappa_HS= 187.32
omega = 8.8095 deg
d_A = 0.622688
Helm = Solar
Helm d = +0.436690
DR = 5.233 HLR
DR ratio= 187.3x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



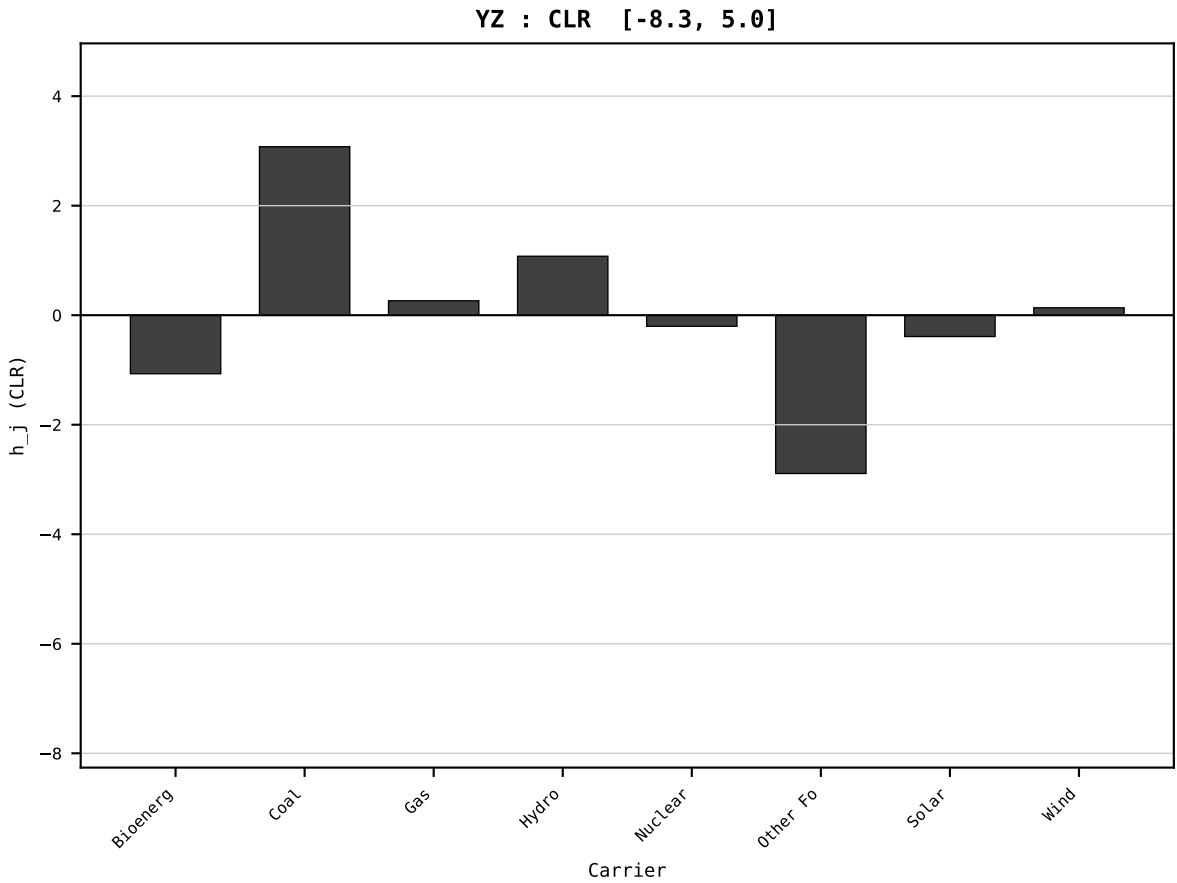
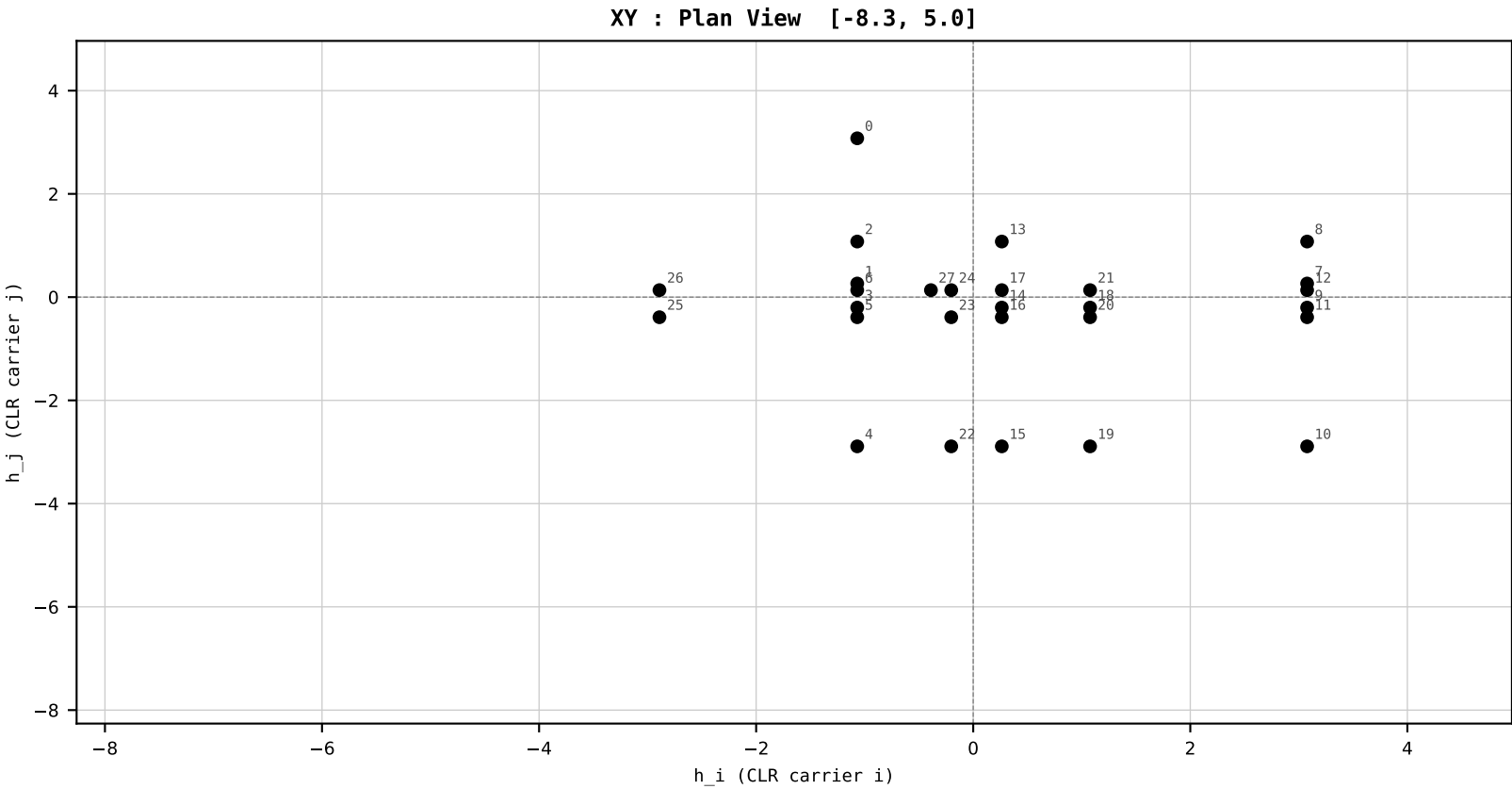
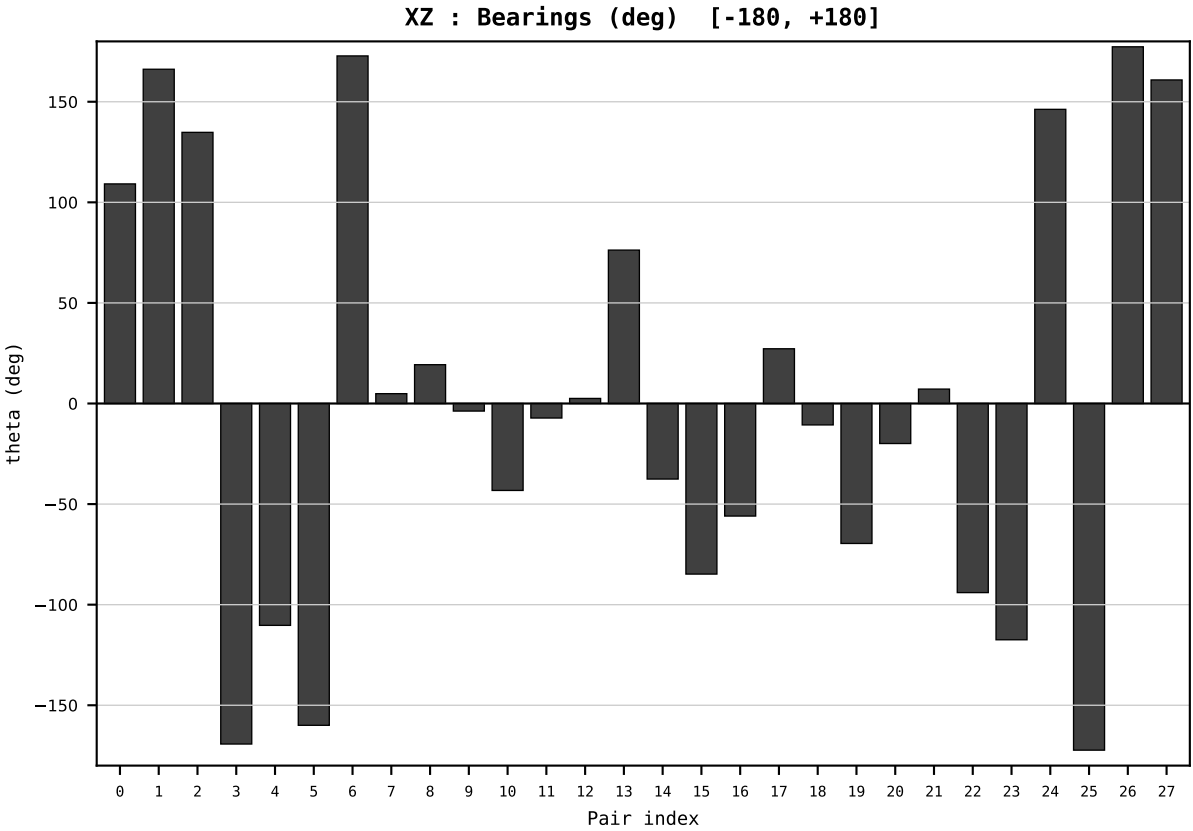
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.534043
Ring = Hs-4
E_metric= 20.4039
kappa_HS= 390.48
omega = 8.0922 deg
d_A = 0.755525
Helm = Other Fossil
Helm d = -0.690711
DR = 5.967 HLR
DR ratio= 390.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

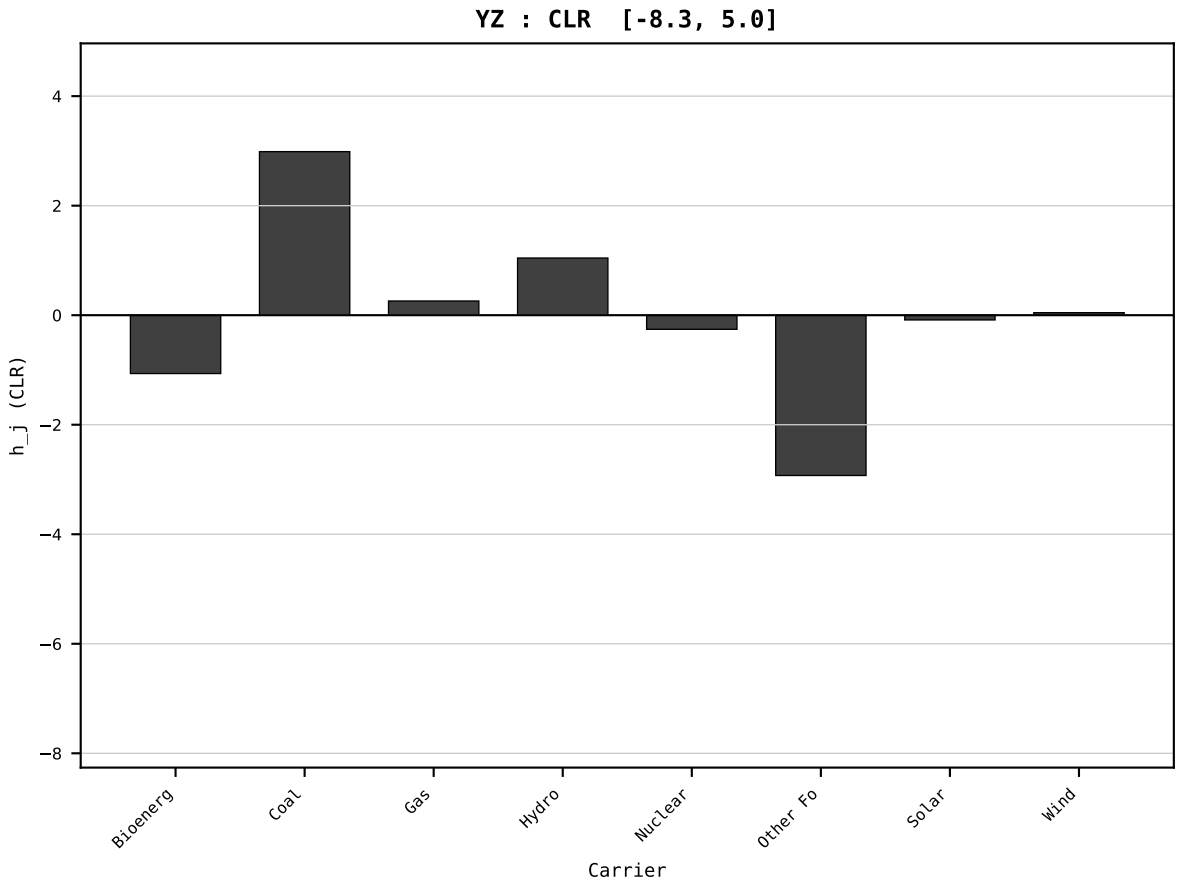
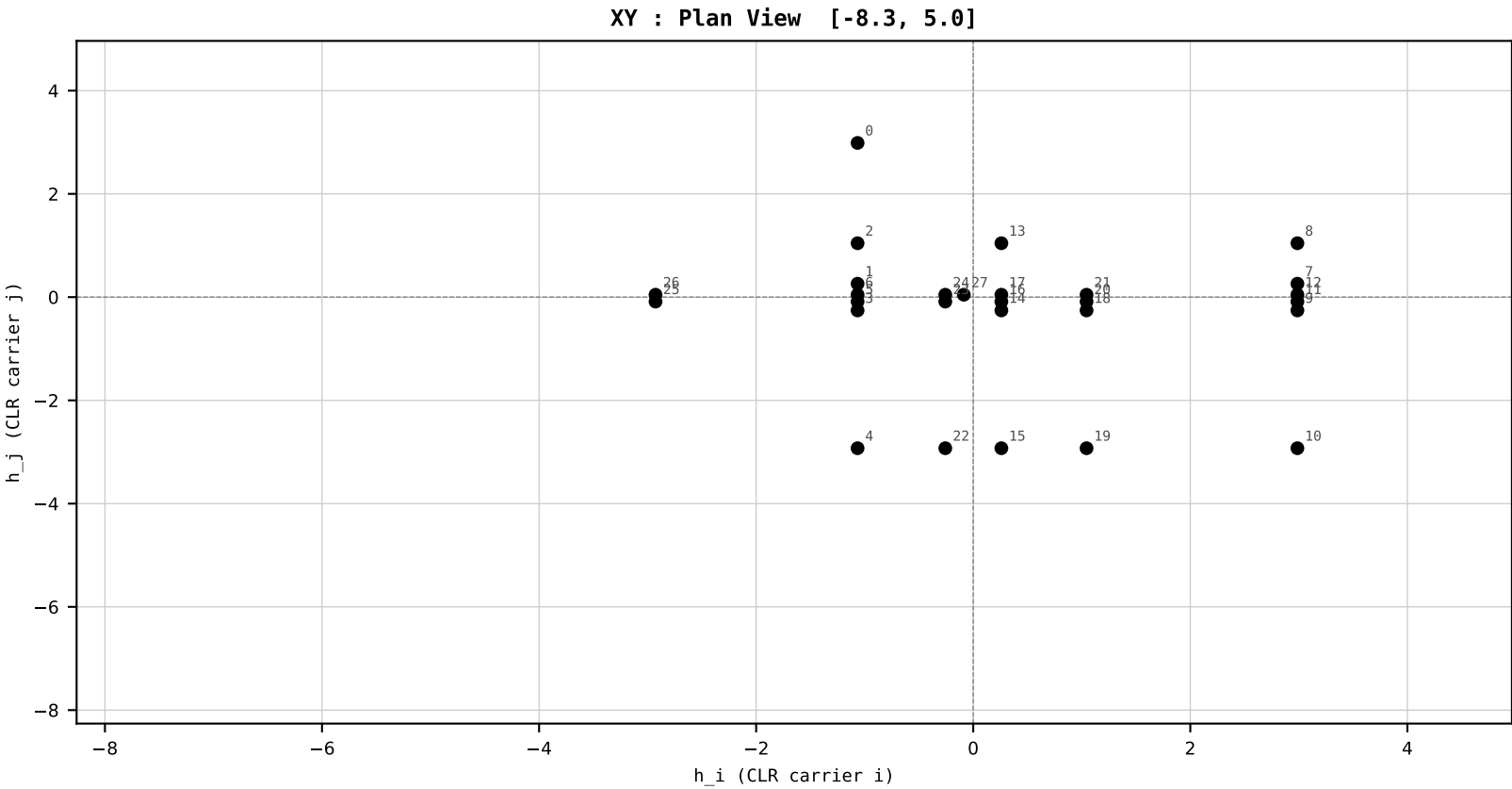
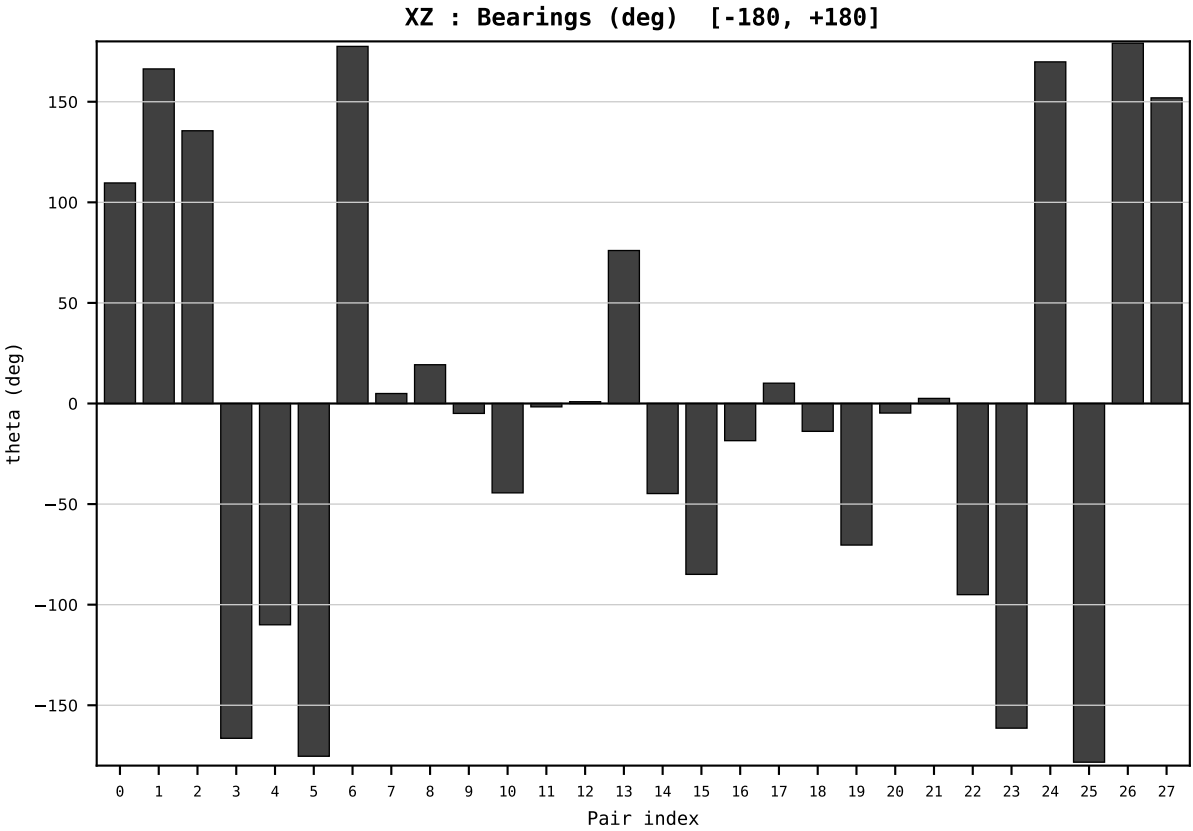
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.509401
Ring = Hs-4
E_metric= 19.8580
kappa_HS= 370.16
omega = 4.2261 deg
d_A = 0.336395
Helm = Solar
Helm d = +0.302967
DR = 5.914 HLR
DR ratio= 370.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

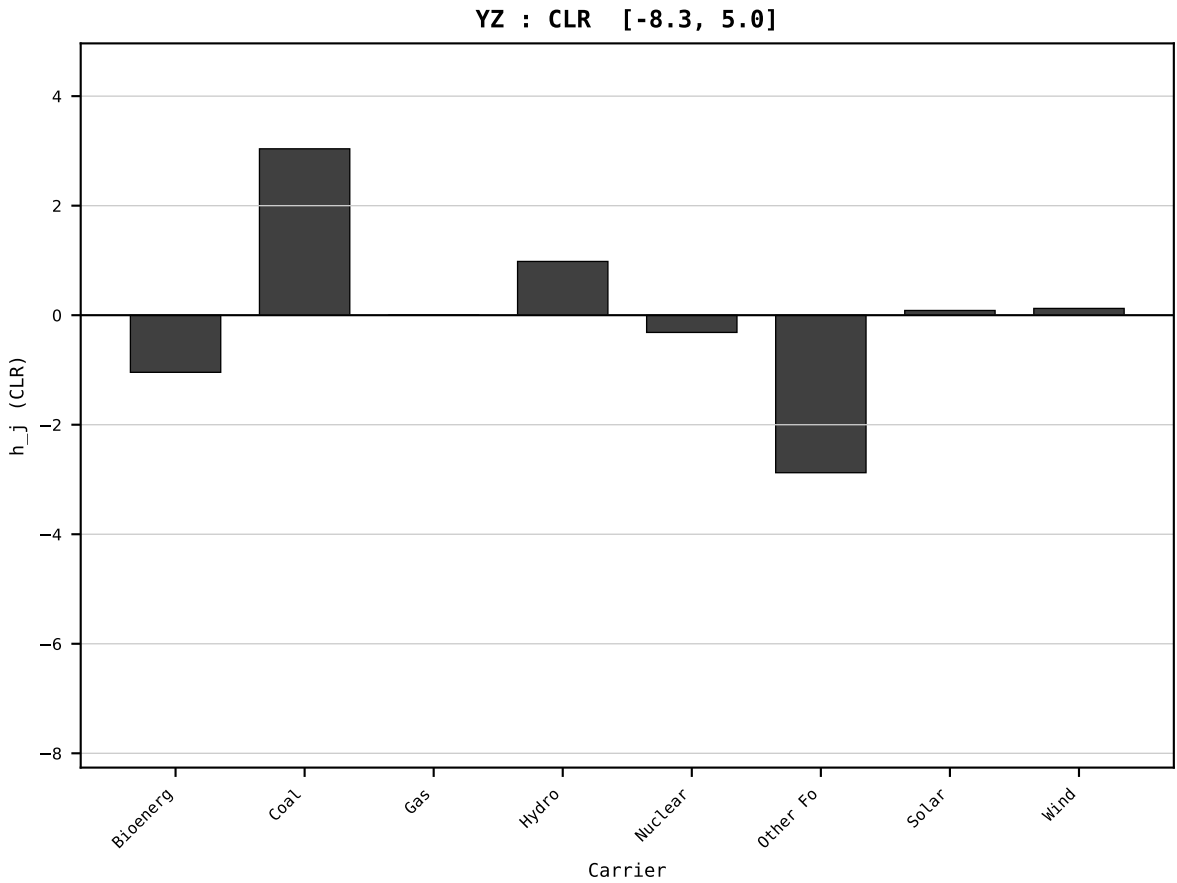
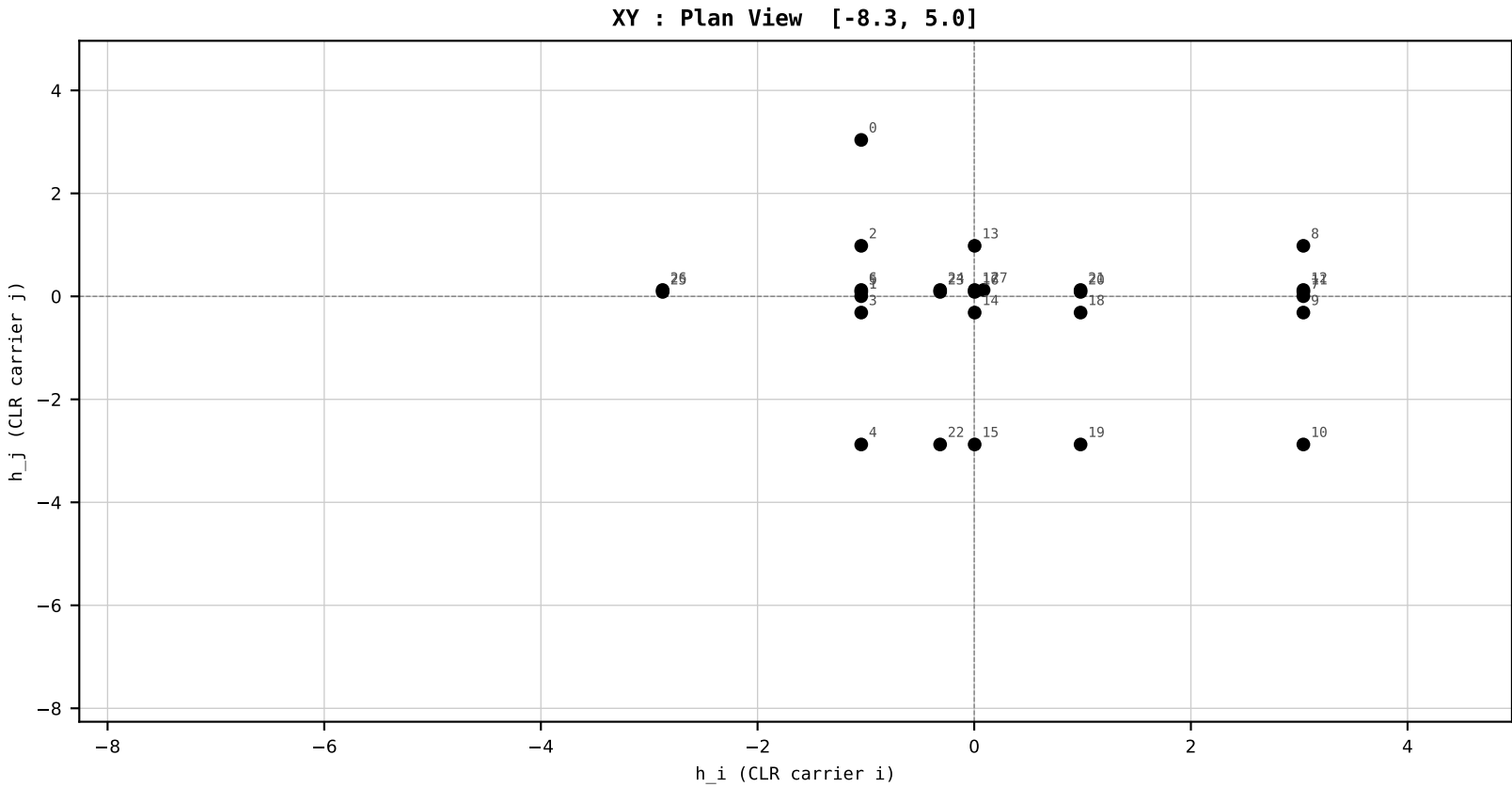
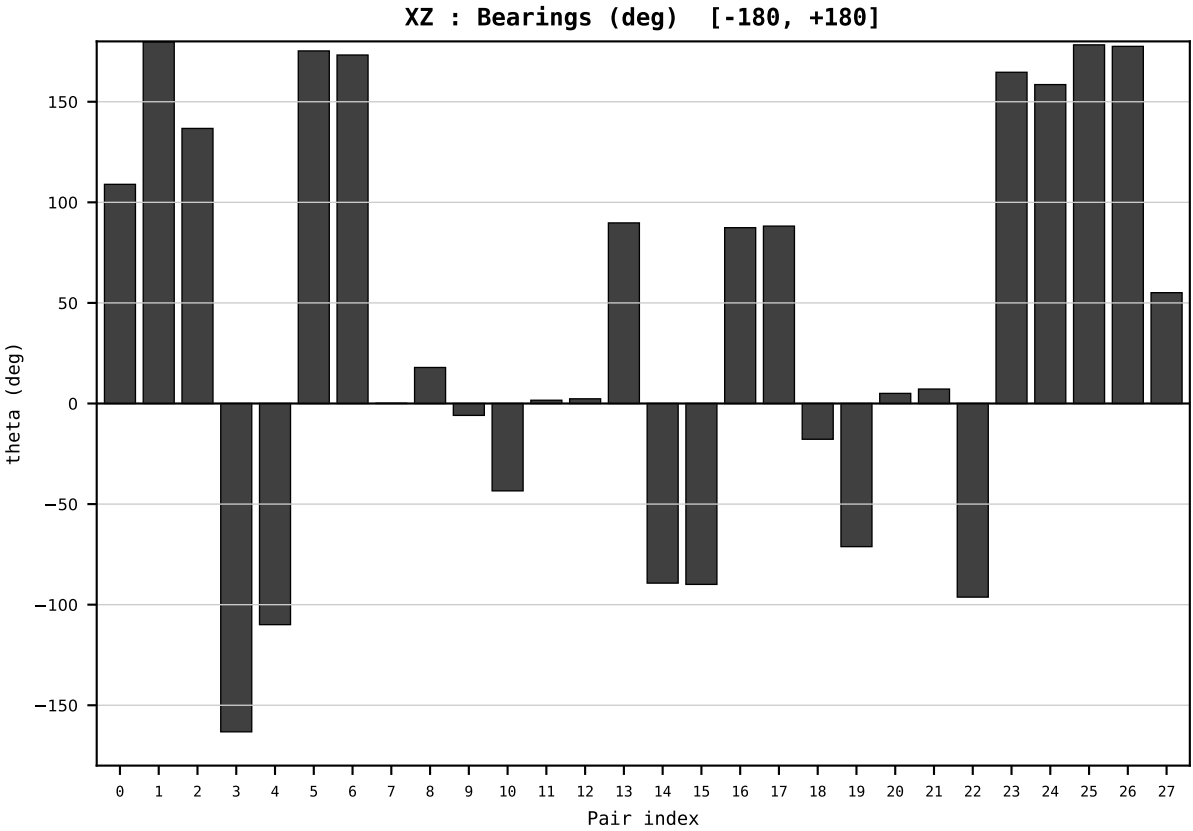
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.528191
Ring = Hs-4
E_metric= 19.6773
kappa_HS= 370.37
omega = 4.3477 deg
d_A = 0.337902
Helm = Gas
Helm d = -0.255484
DR = 5.915 HLR
DR ratio= 370.4x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

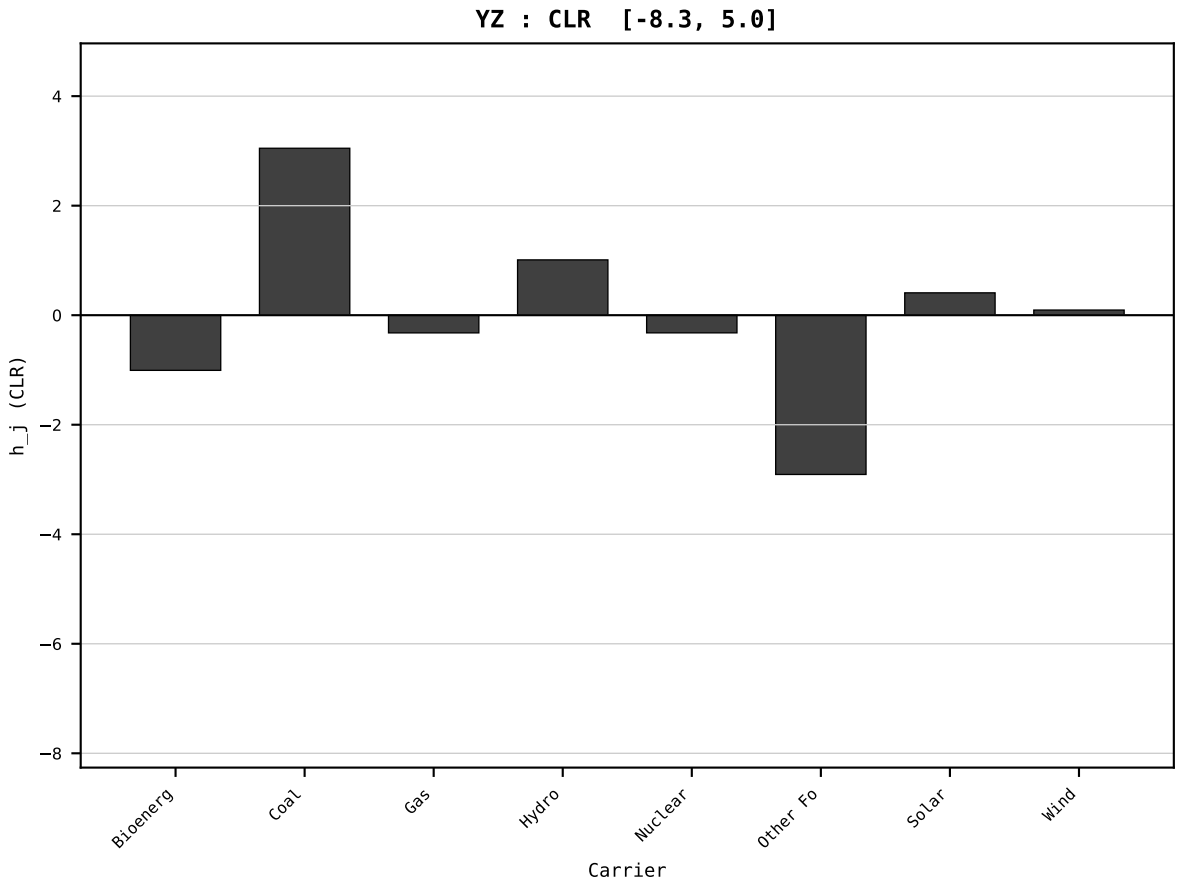
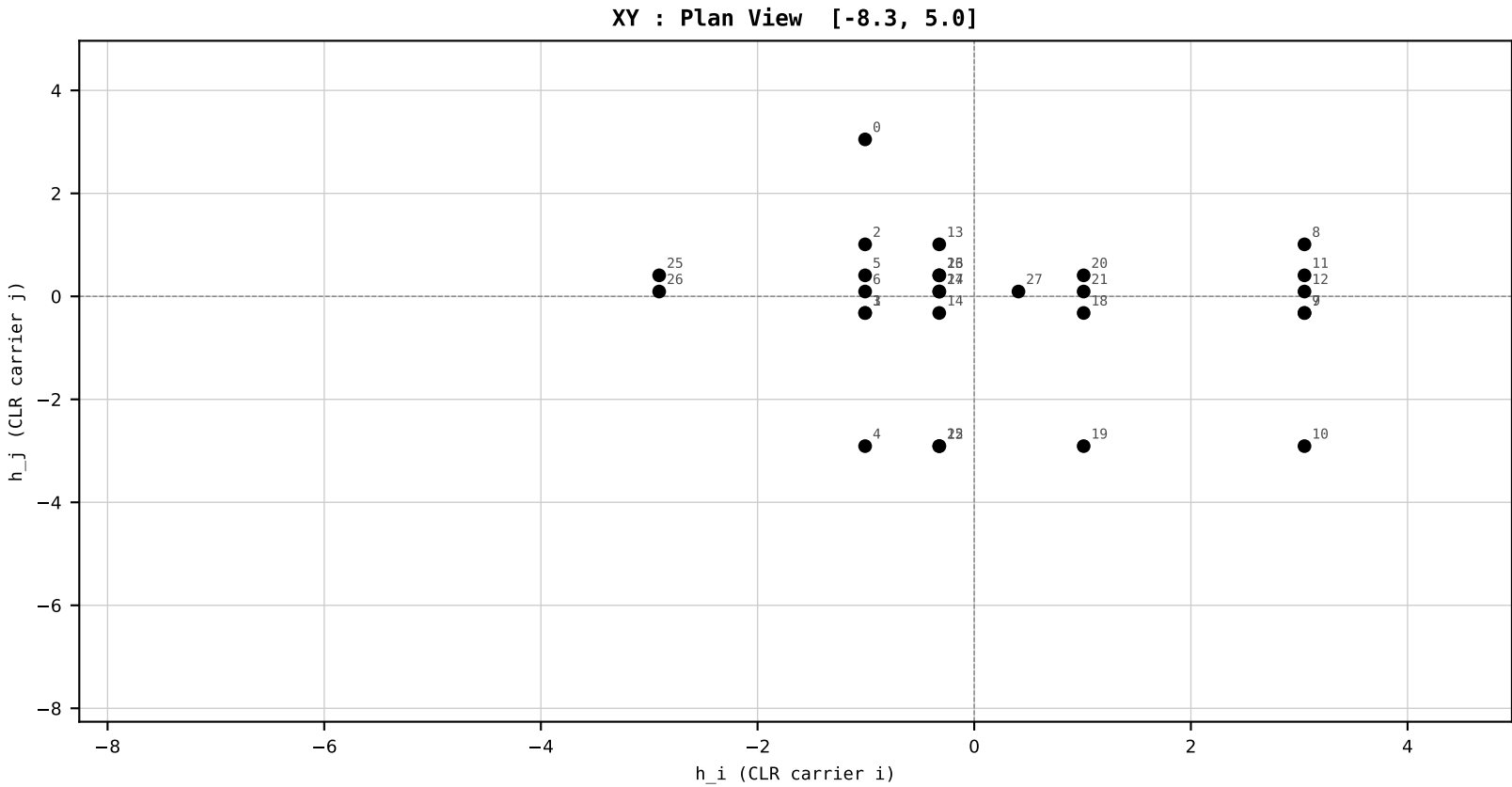
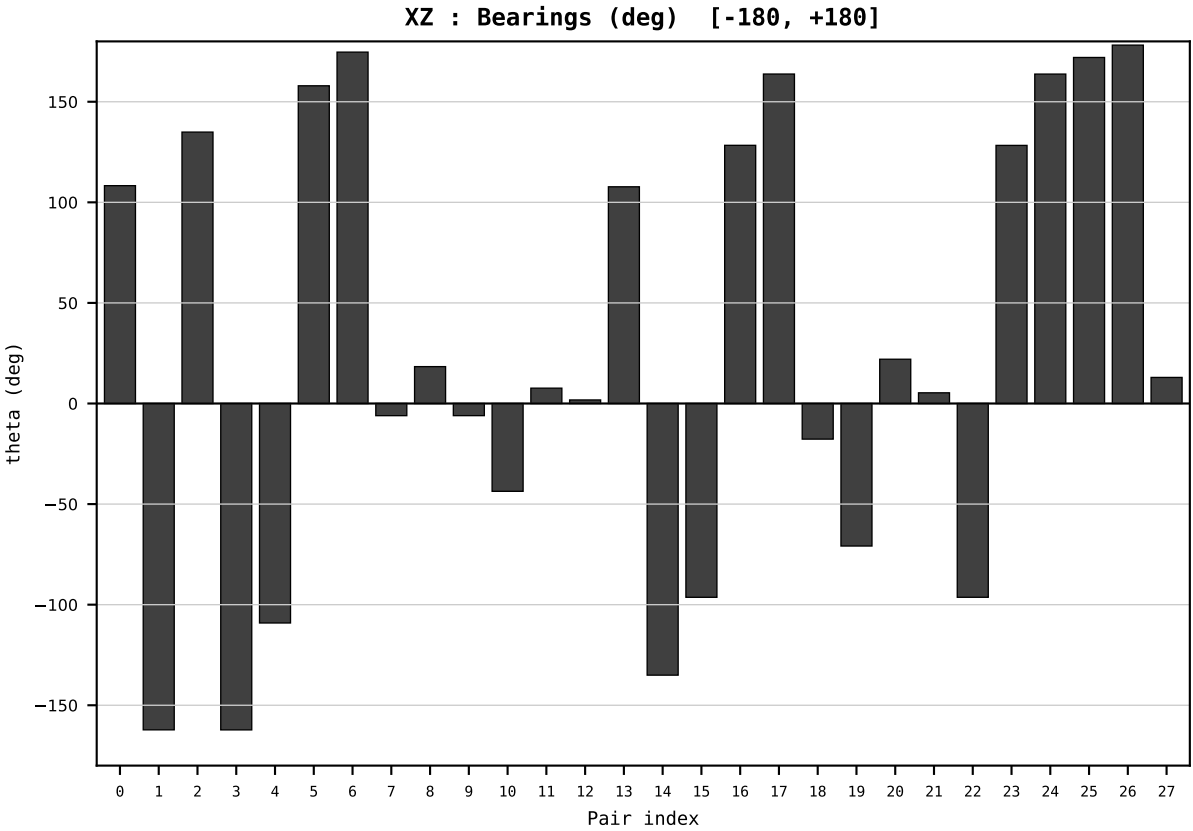
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.527016
Ring = Hs-4
E_metric= 20.1694
kappa_HS= 386.33
omega = 5.9073 deg
d_A = 0.463267
Helm = Gas
Helm d = -0.326832
DR = 5.957 HLR
DR ratio= 386.3x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



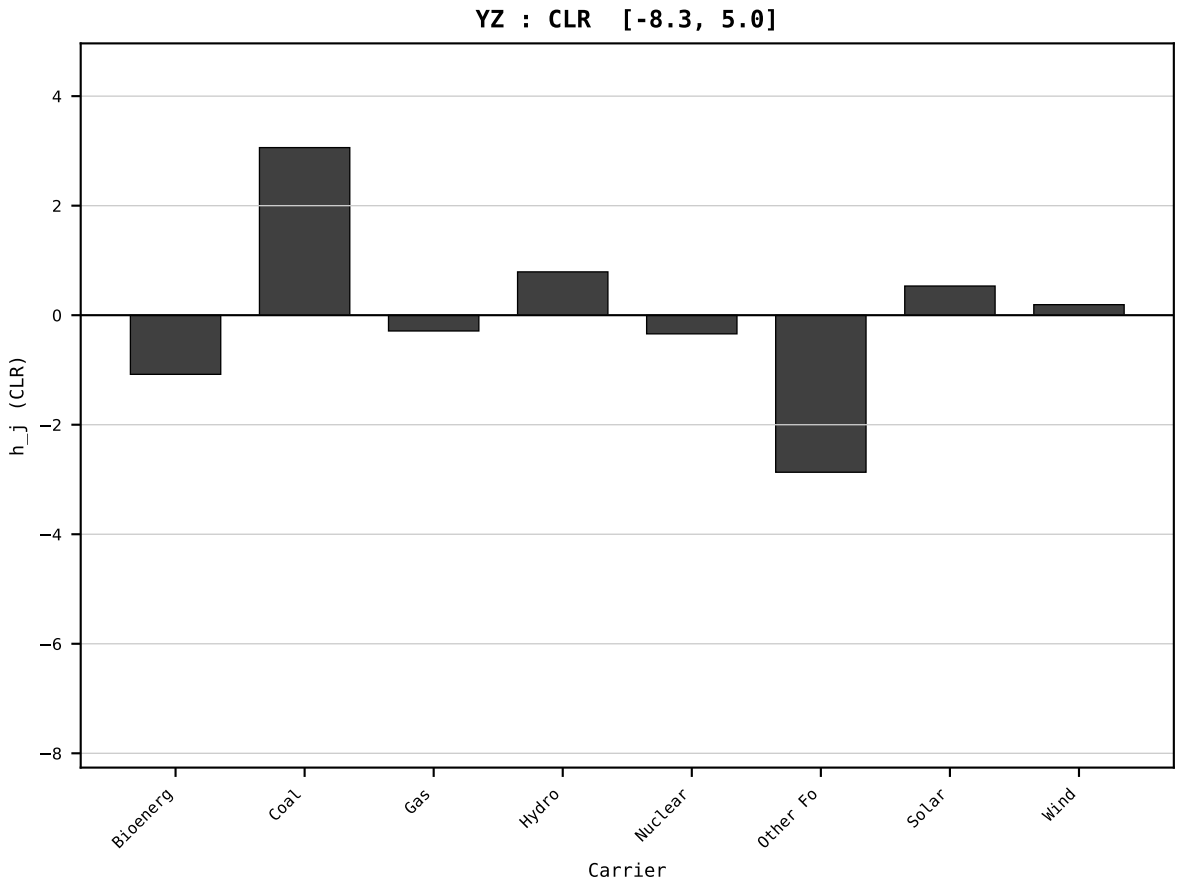
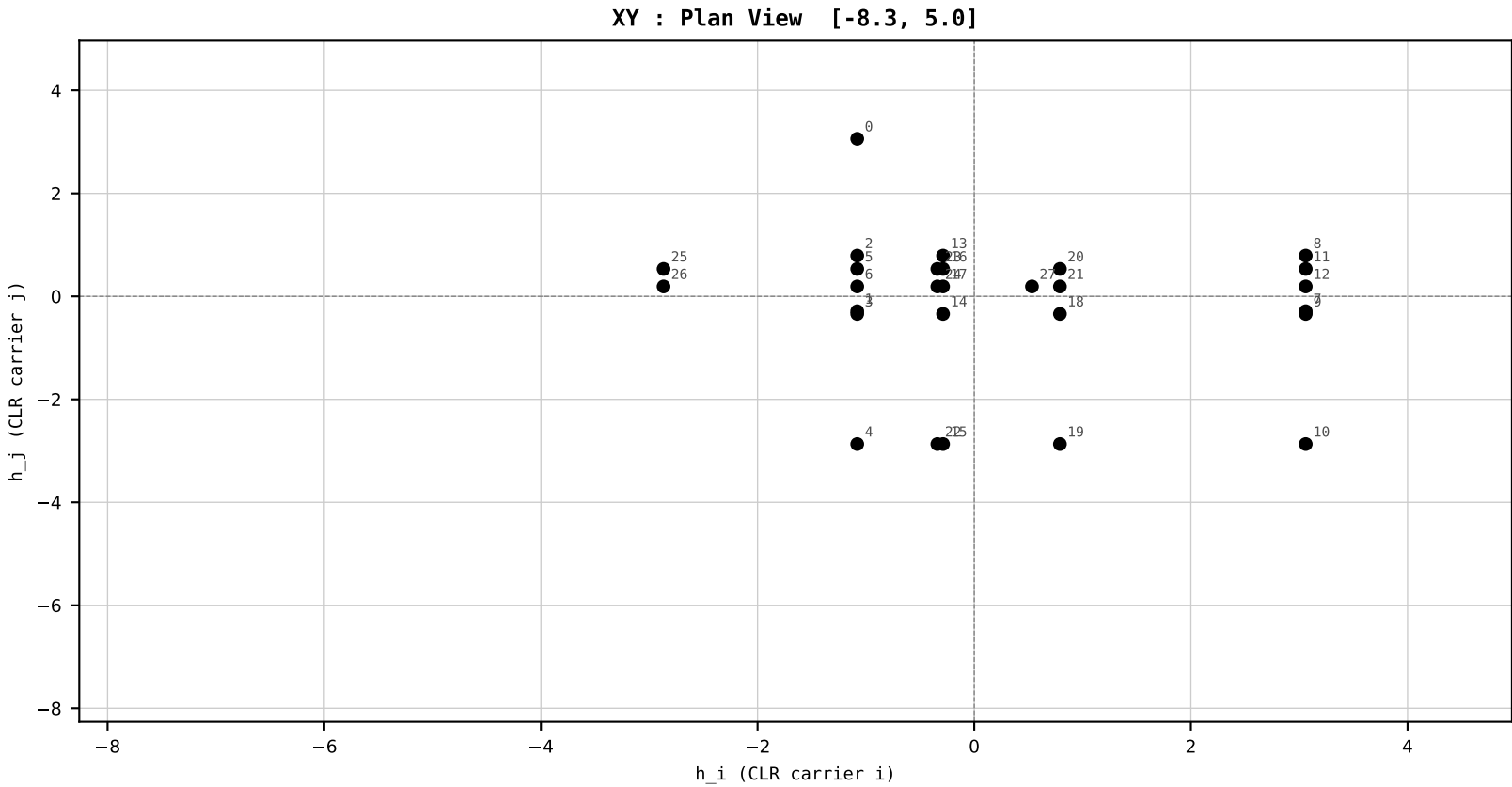
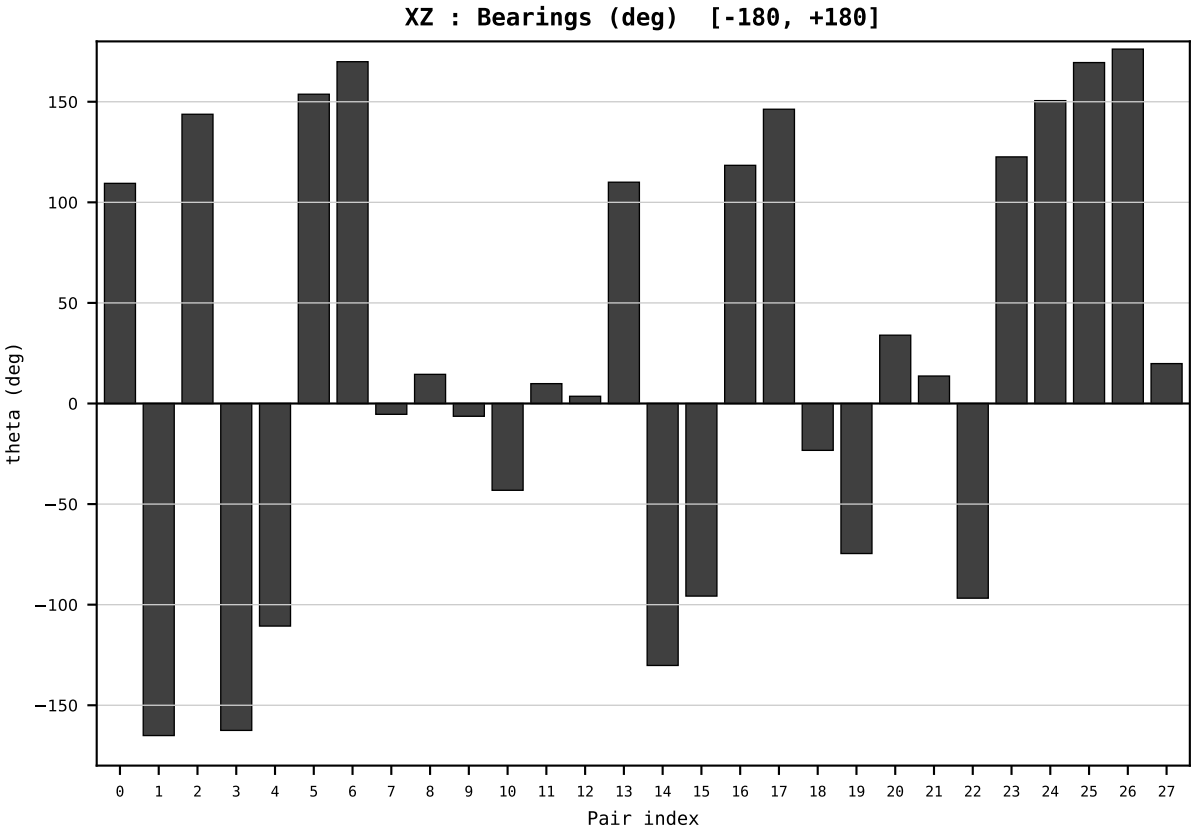
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.533376
Ring = Hs-4
E_metric= 19.9012
kappa_HS= 375.44
omega = 3.6403 deg
d_A = 0.285909
Helm = Hydro
Helm d = -0.219097
DR = 5.928 HLR
DR ratio= 375.4x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

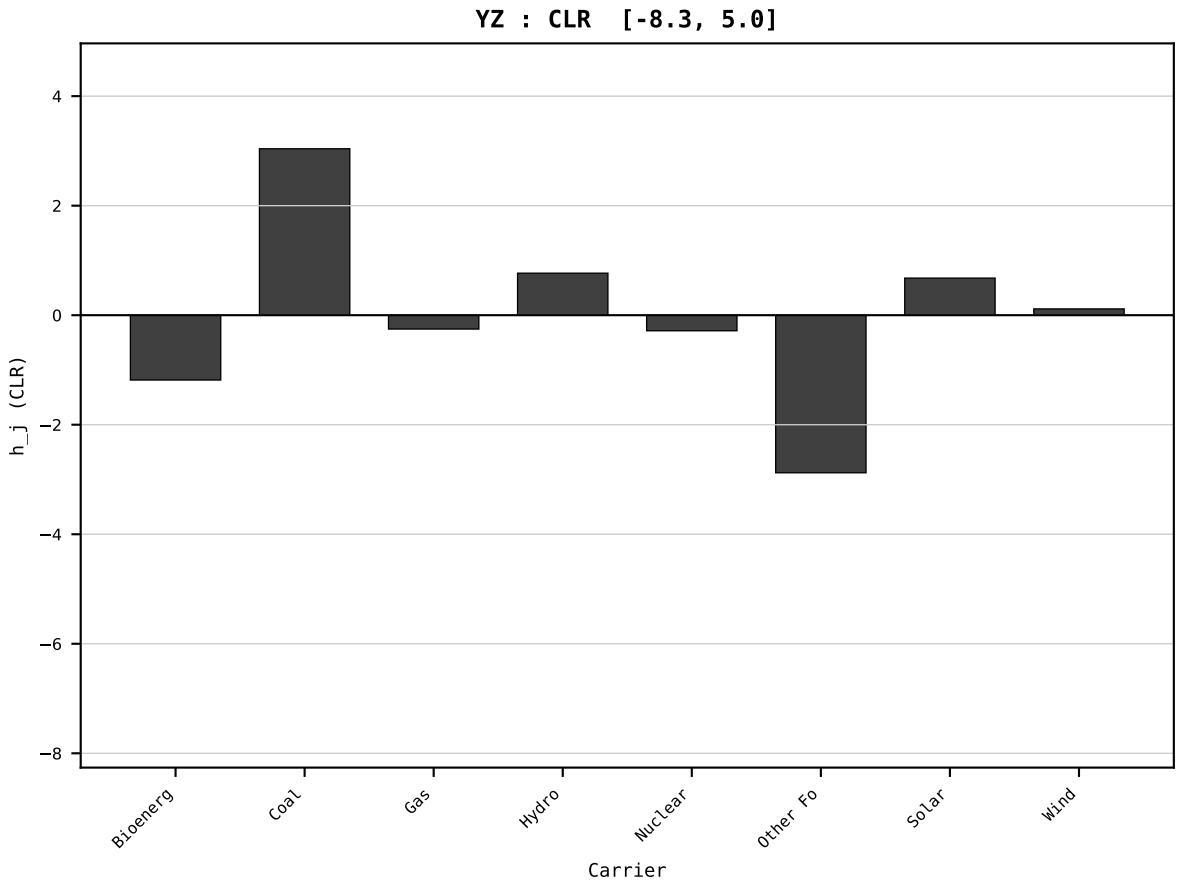
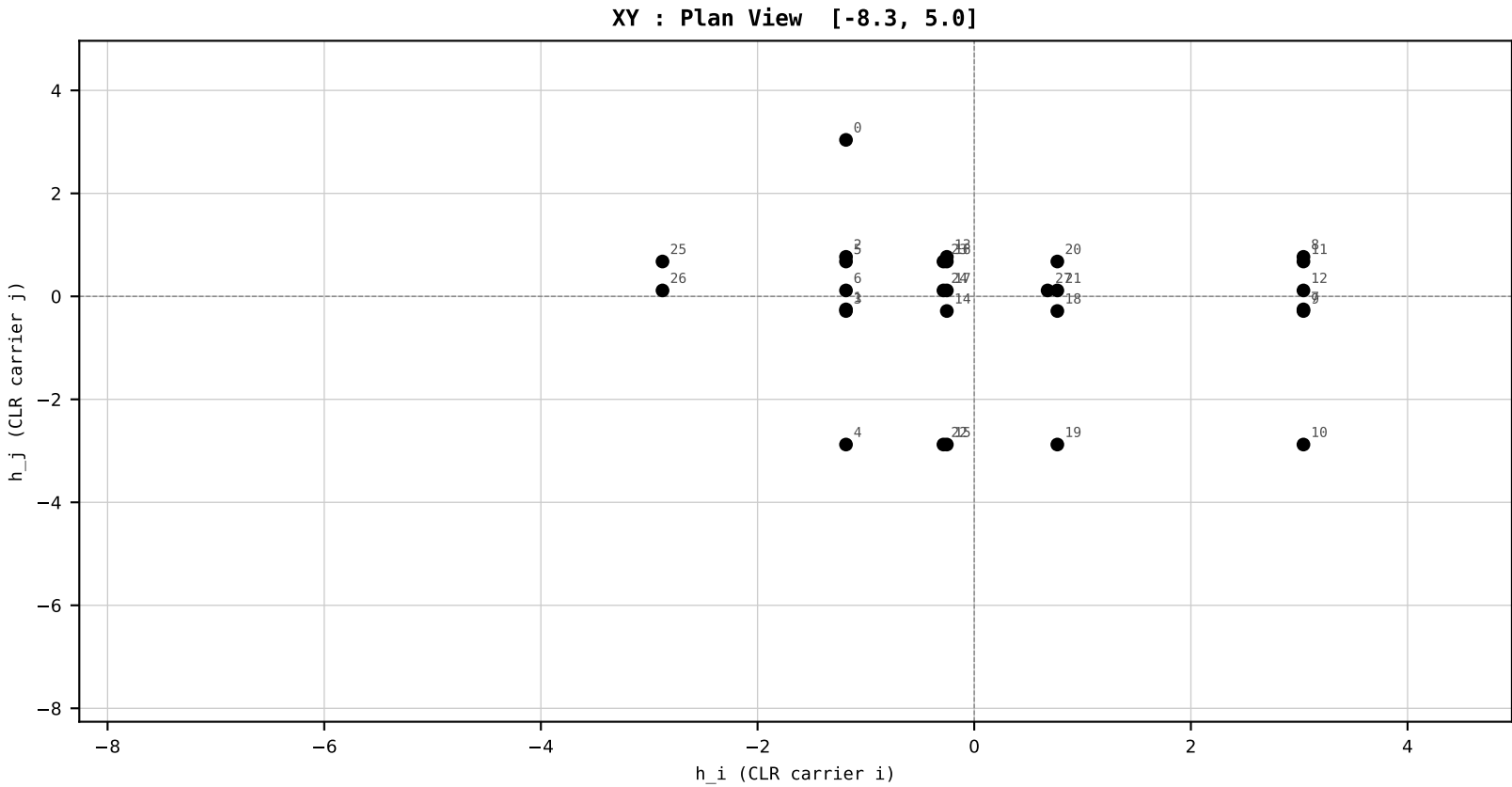
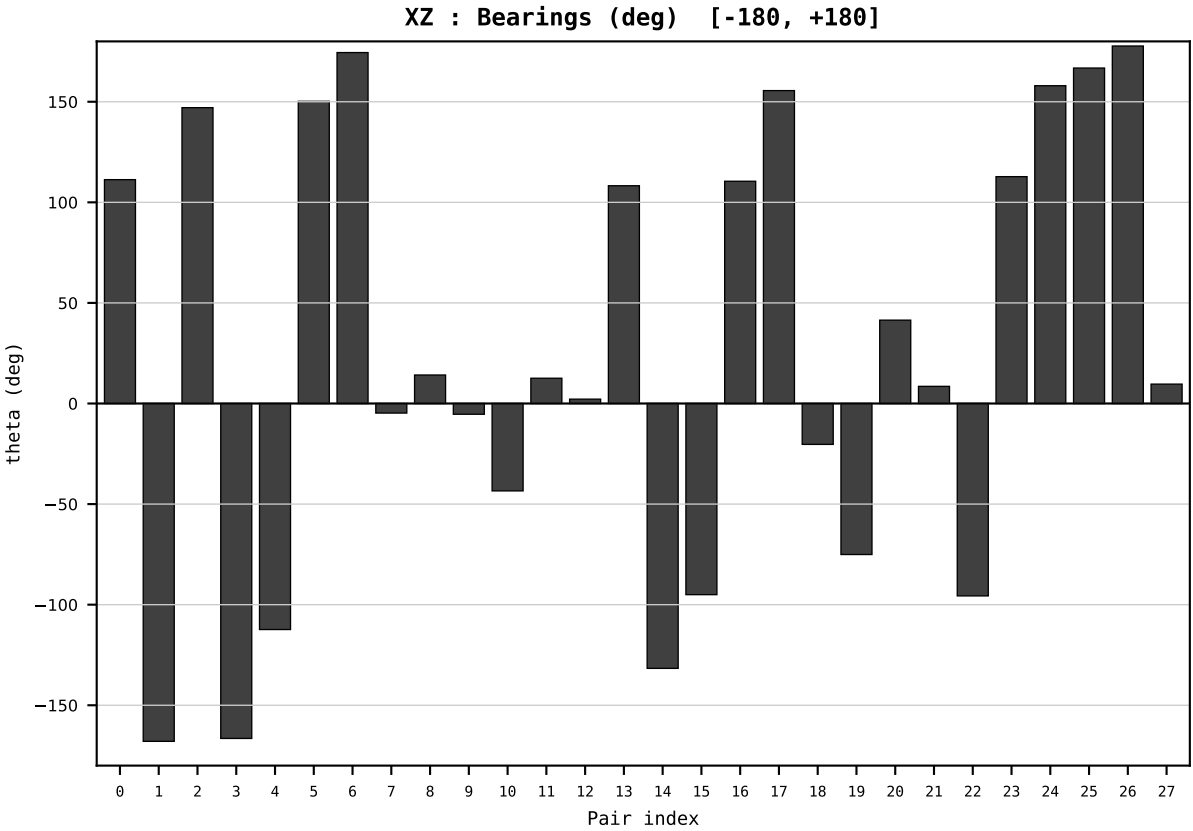
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.523895
Ring = Hs-4
E_metric= 20.1222
kappa_HS= 371.12
omega = 2.6415 deg
d_A = 0.207695
Helm = Solar
Helm d = +0.144532
DR = 5.917 HLR
DR ratio= 371.1x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

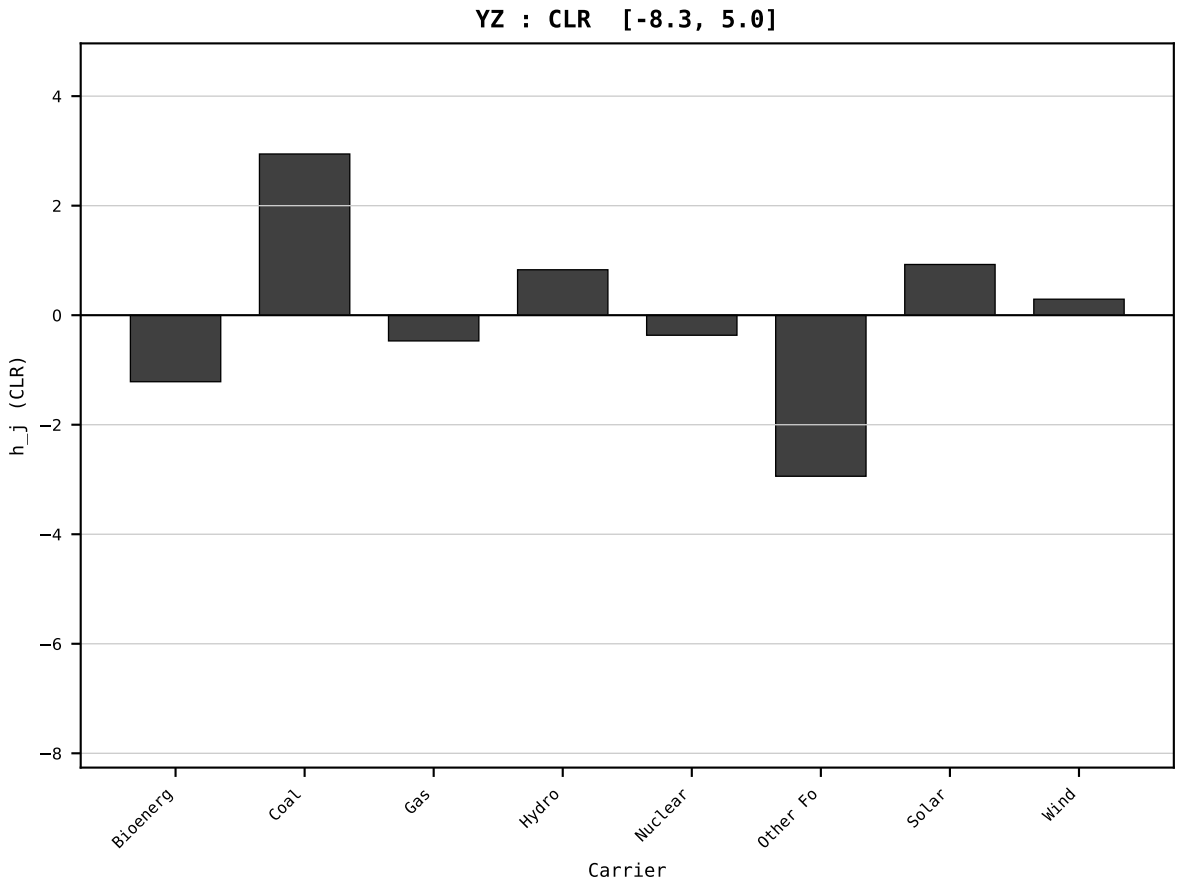
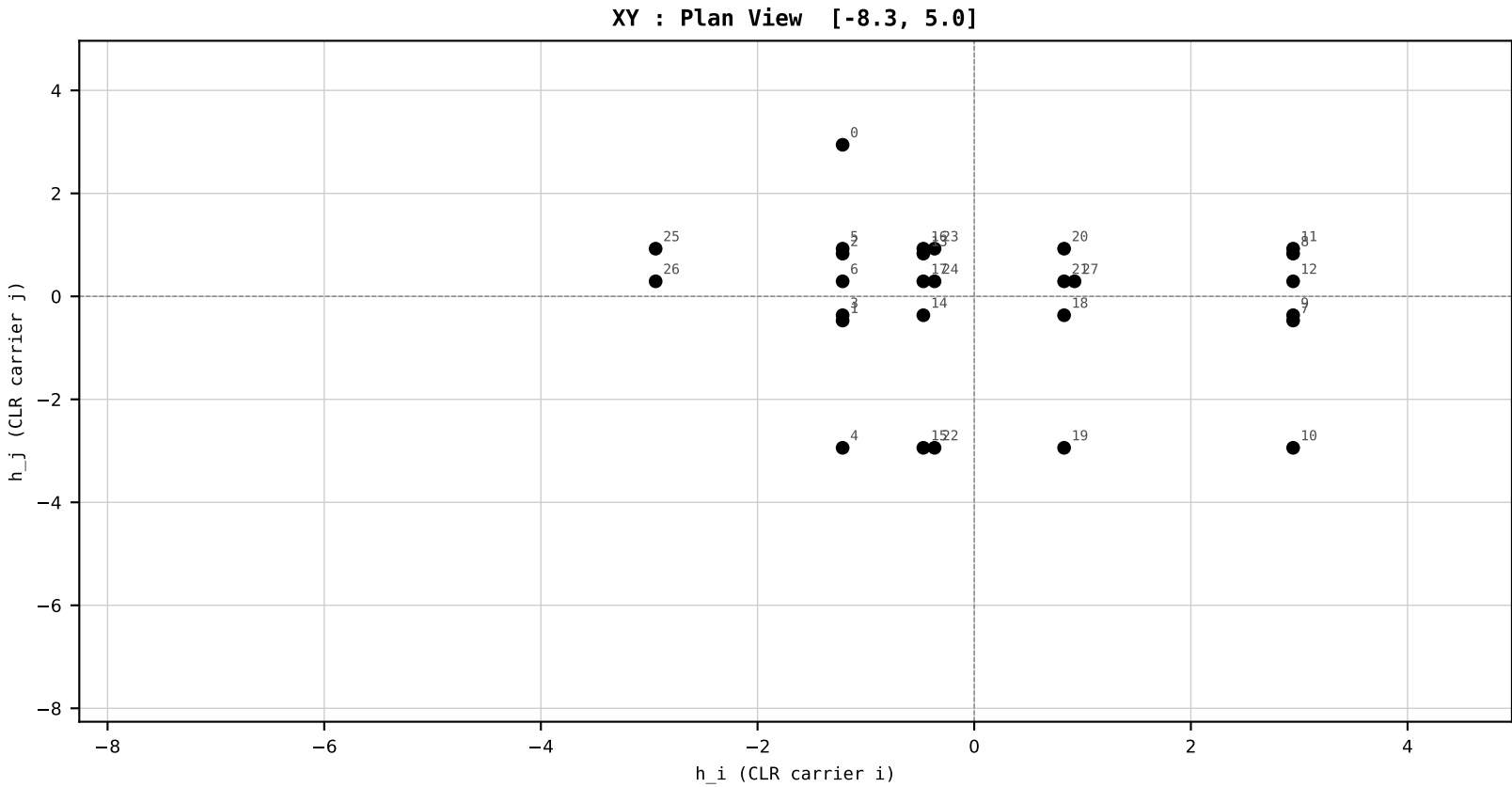
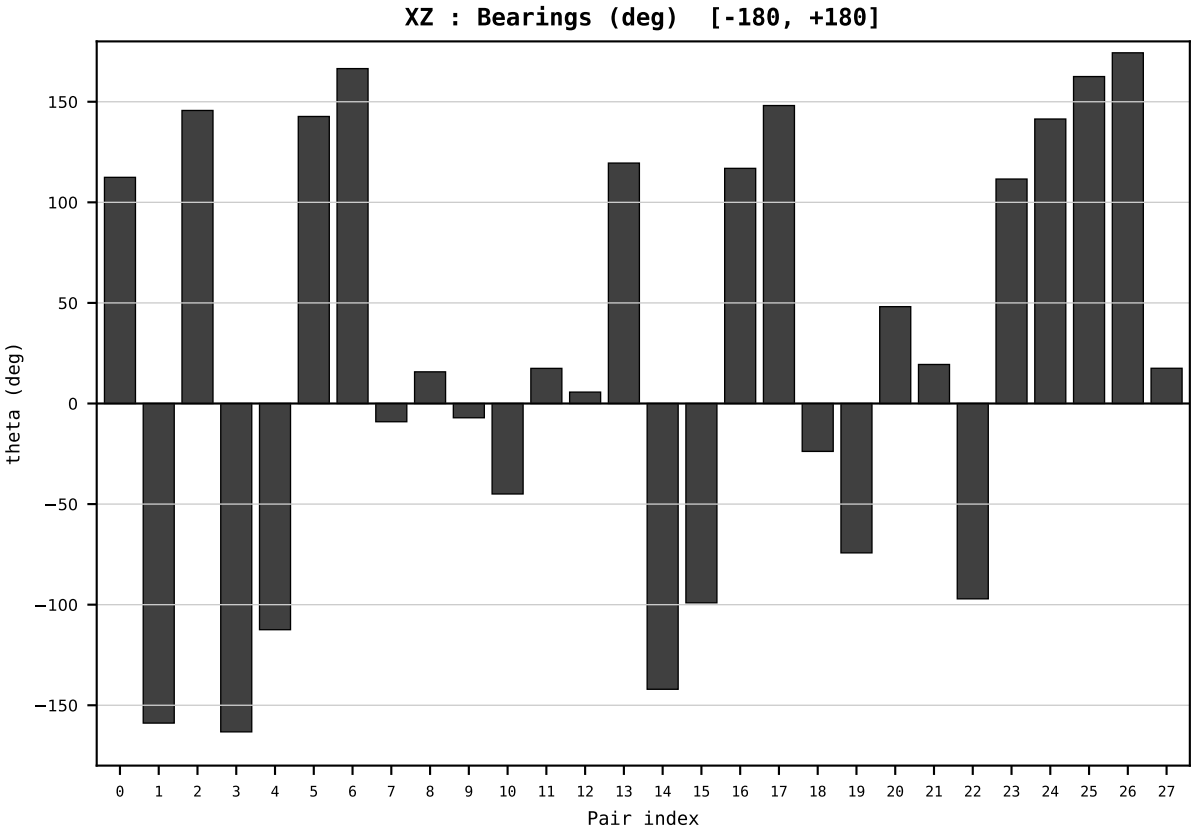
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

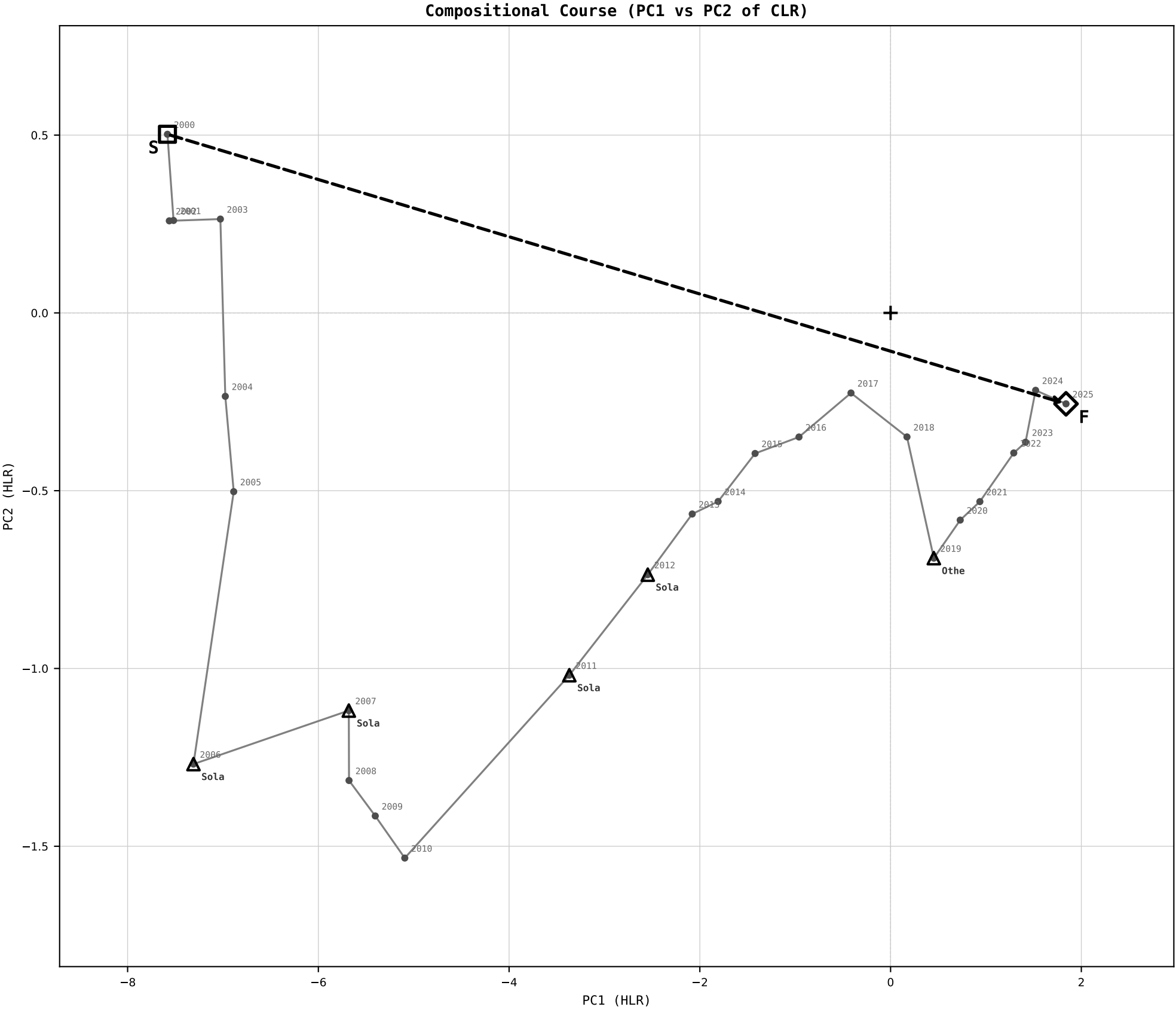
Dataset : ember_IND_India_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.484928
Ring = Hs-3
E_metric= 20.7772
kappa_HS= 359.55
omega = 5.0650 deg
d_A = 0.406114
Helm = Solar
Helm d = +0.248797
DR = 5.885 HLR
DR ratio= 359.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 9.4603 HLR
Path length = 14.6188 HLR
Directness = 0.6471

DR start = 10.572 HLR
ratio = 39023.0x
DR final = 5.885 HLR
ratio = 359.5x

E_{metric} start = 75.8817
E_{metric} final = 20.7772

PC1 loadings:
Solar +0.841
Other Fo -0.425
Gas -0.241

PC2 loadings:
Wind -0.574
Bioenerg -0.487
Other Fo +0.458

Top displacement events:
2011 Solar d=1.820
2007 Solar d=1.692
2006 Solar d=0.943
2012 Solar d=0.912
2019 Other Fo d=0.756

The instrument reads.
The expert decides.
The loop stays open.

View B

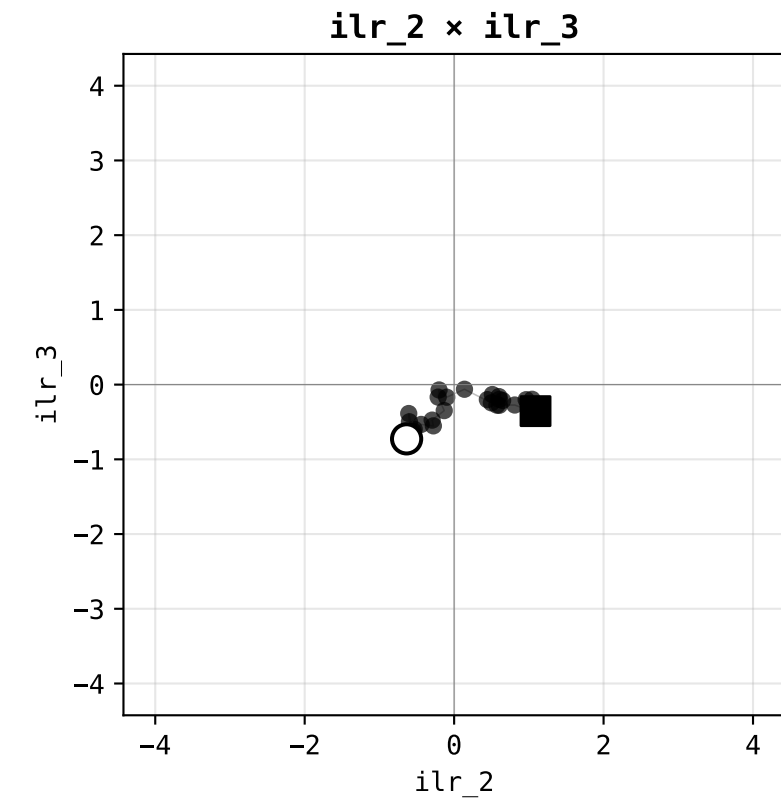
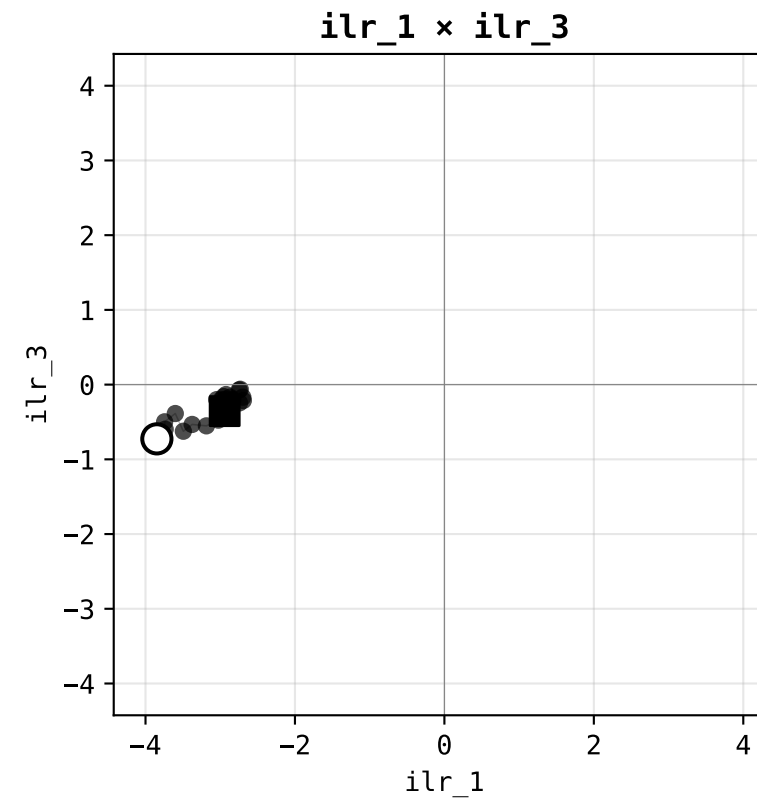
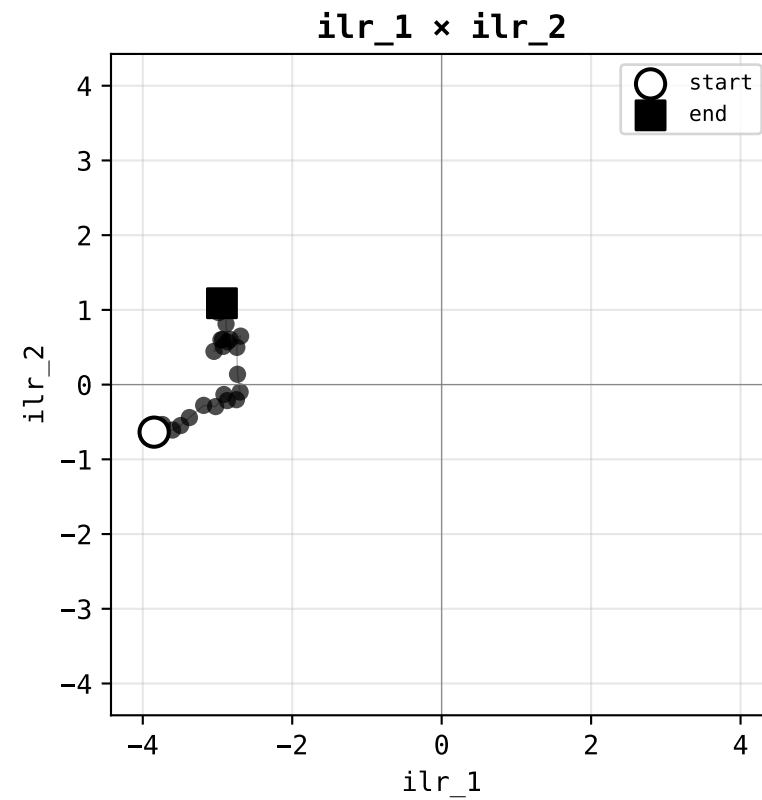
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ IND India

ILR-Helmert Orthogonal Triplet – IND India · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · $N=26$ years · $D=8$ carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:

start year: 2000
end year: 2025
N readings: 26

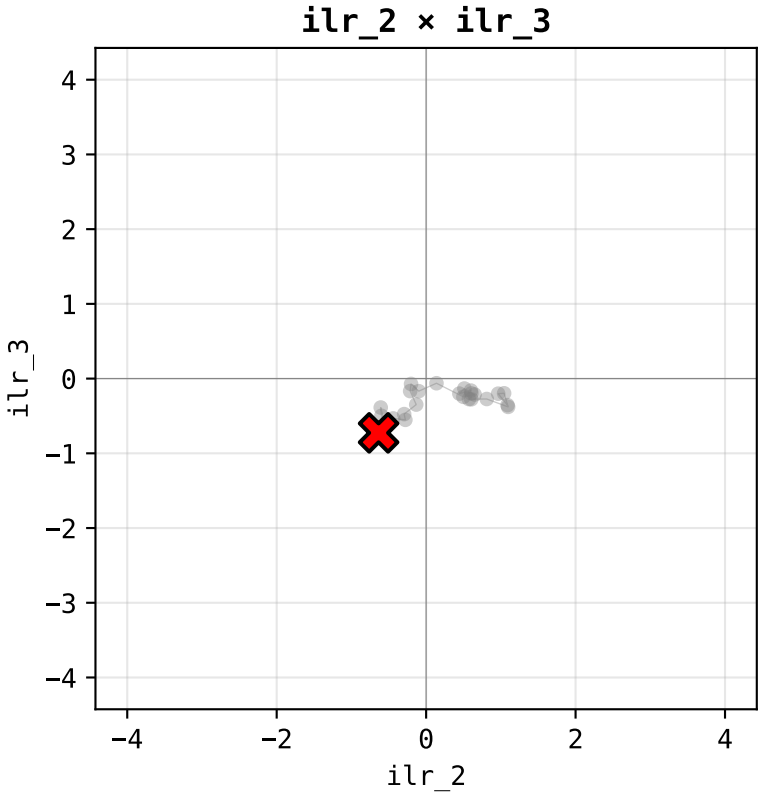
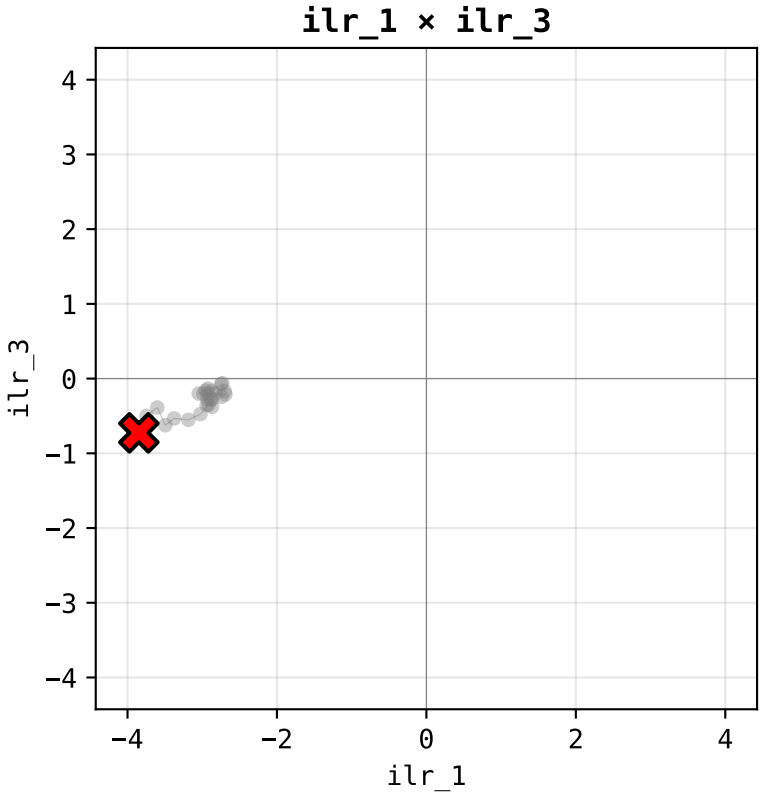
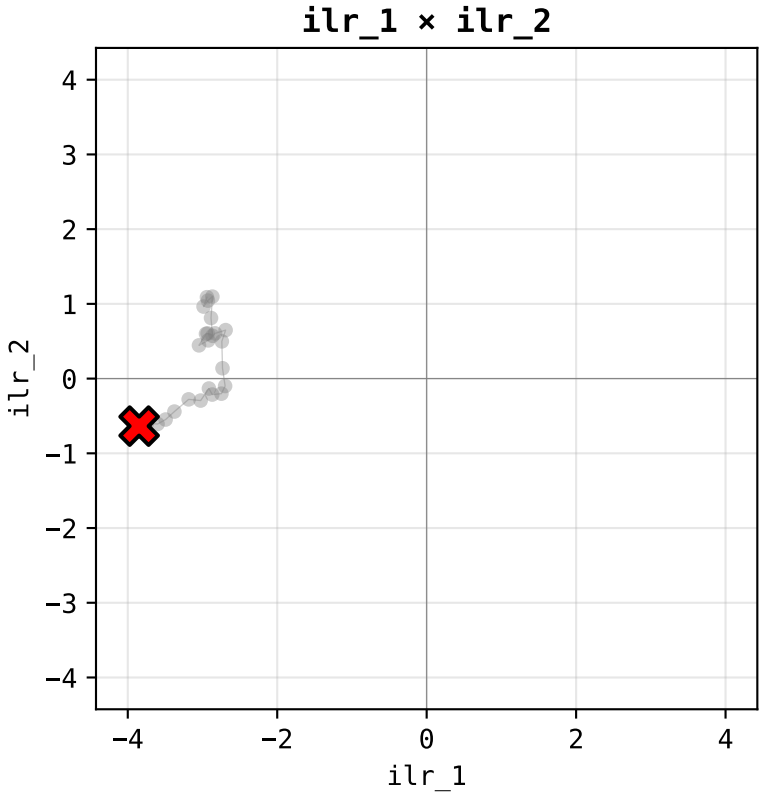
Display:

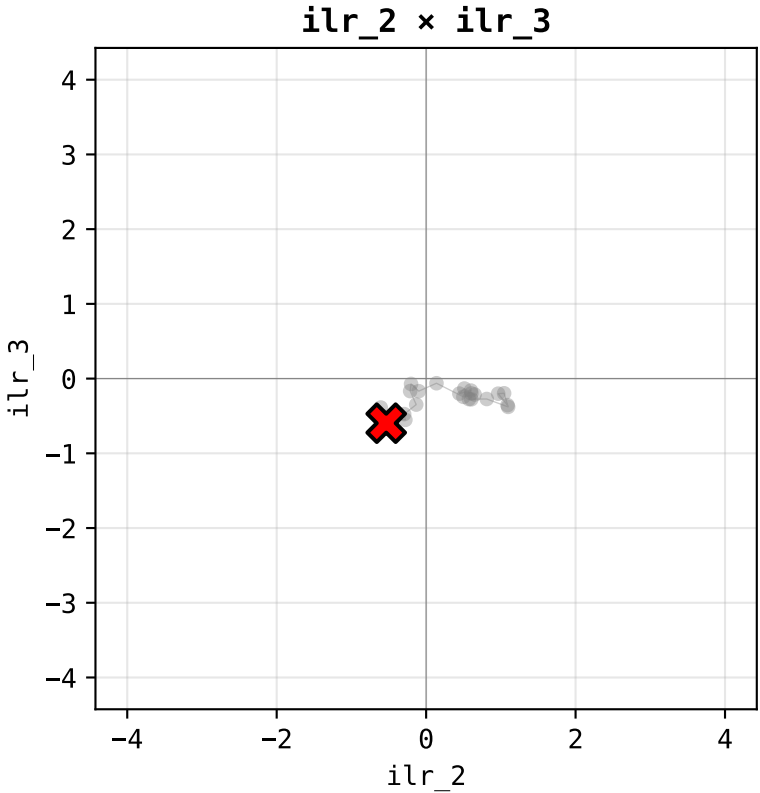
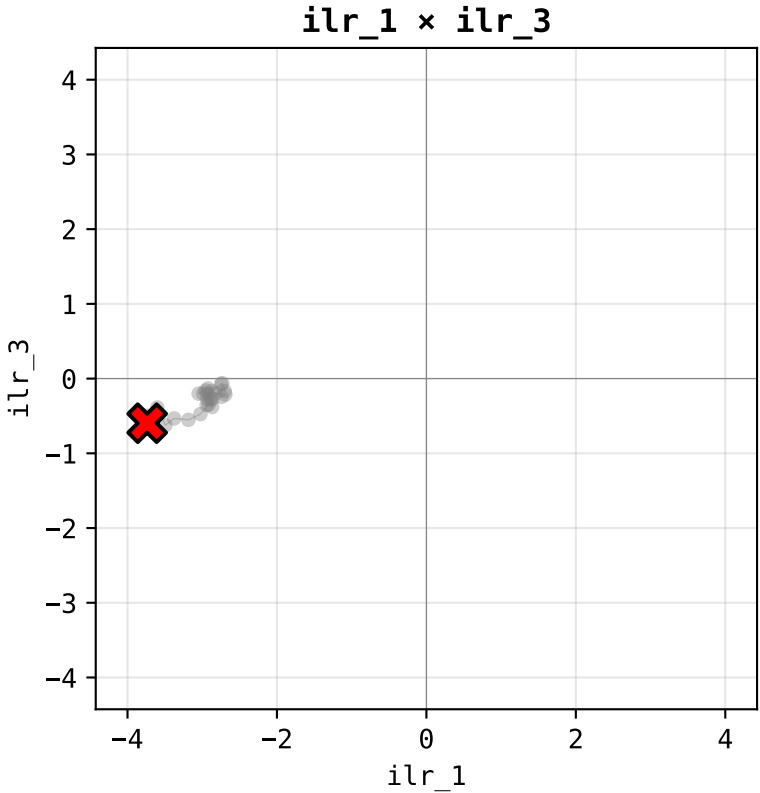
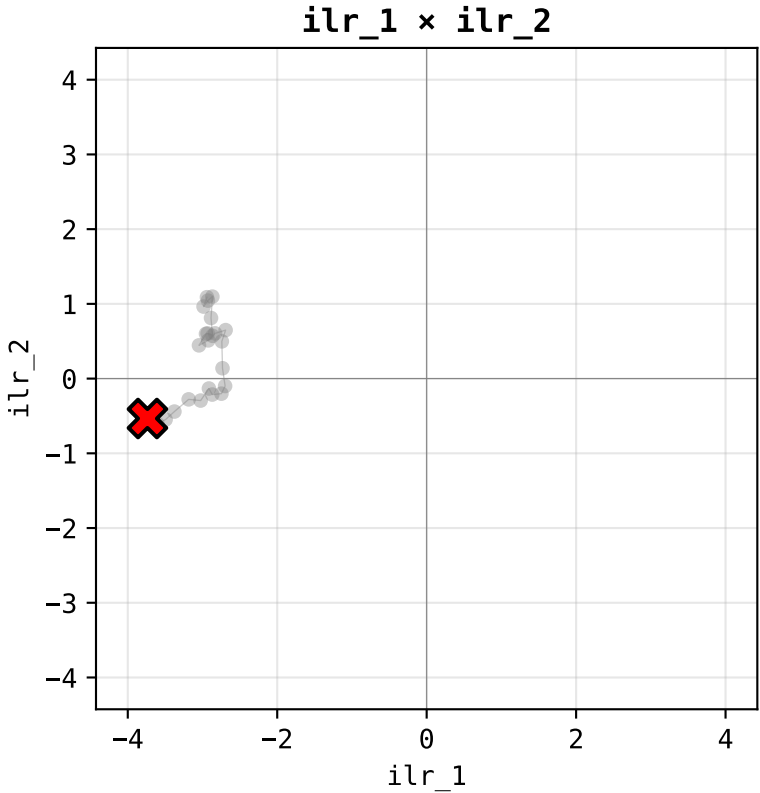
fixed scale ± 4.43 ILR
Order 1 (first principles)
per Output Doctrine v1.0

Symbology:

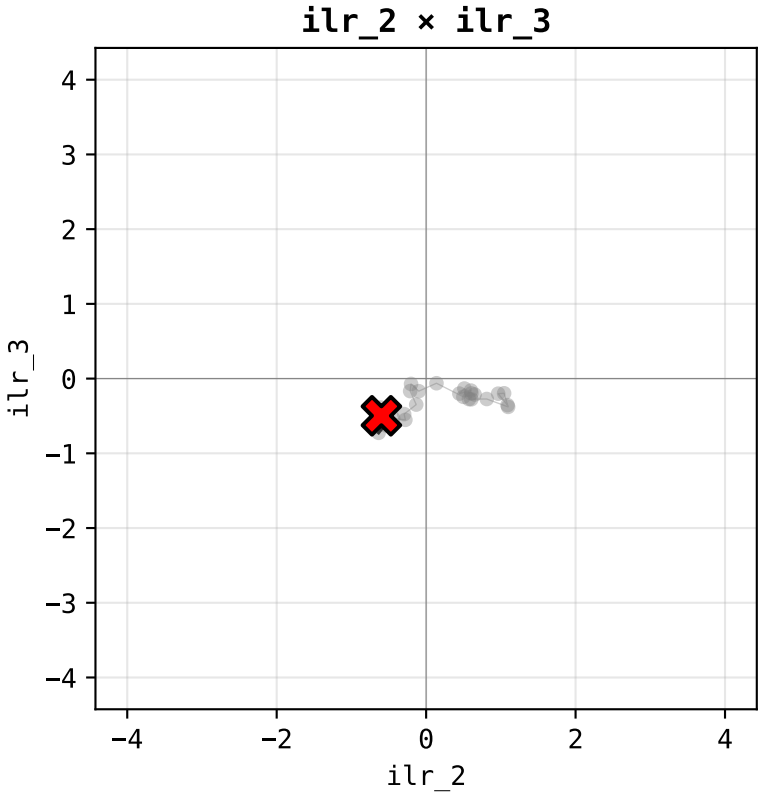
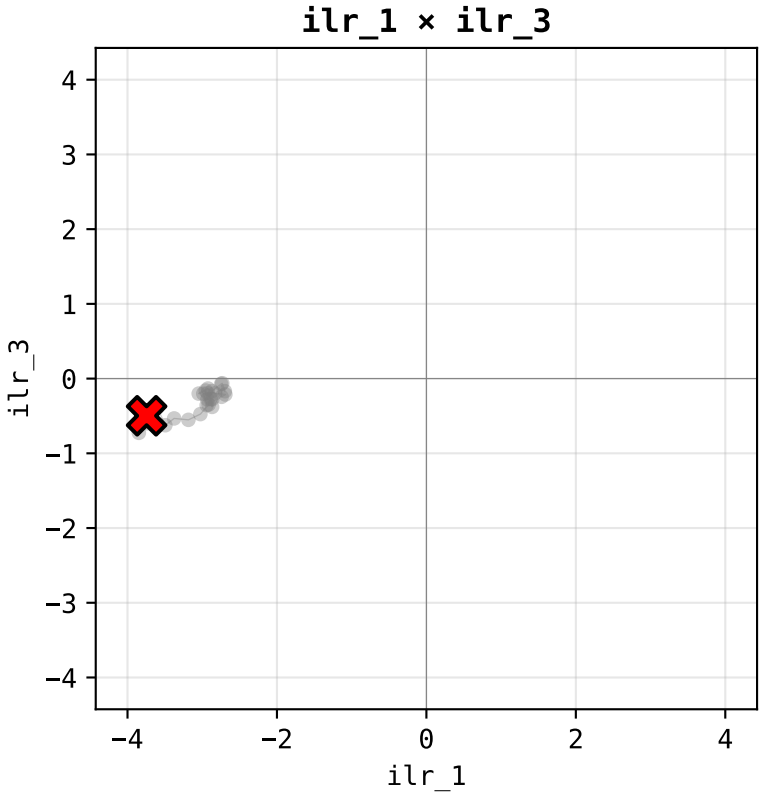
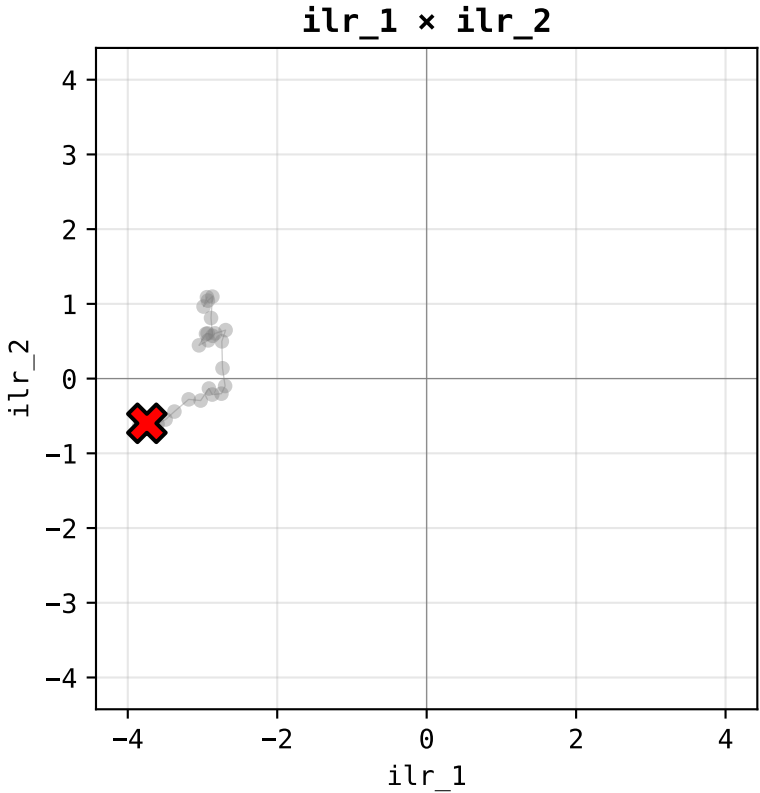
○ start · ■ end
· intermediate years

ILR Triplet – IND India · year 2000 (1/26)

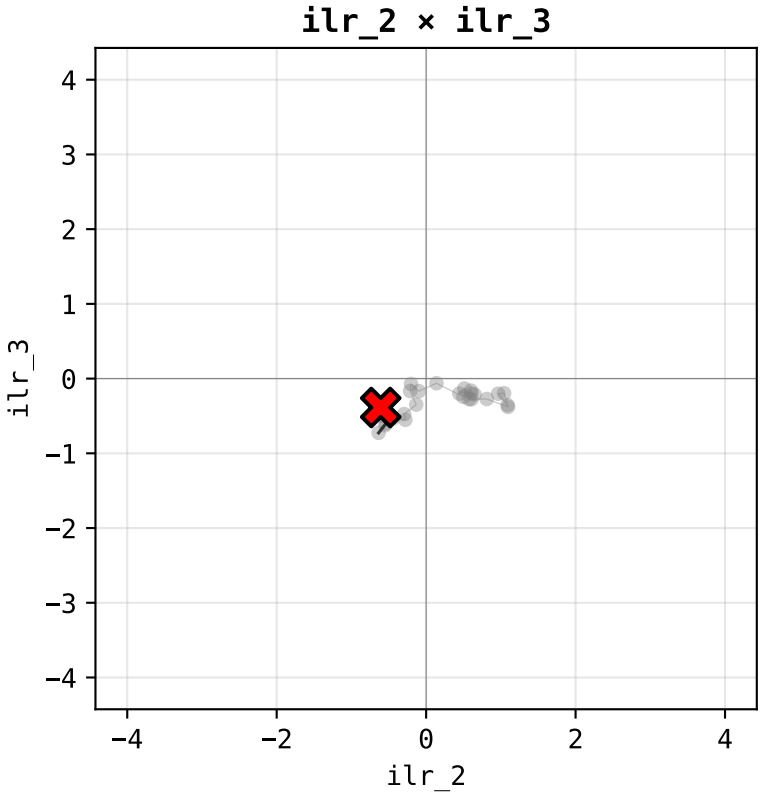
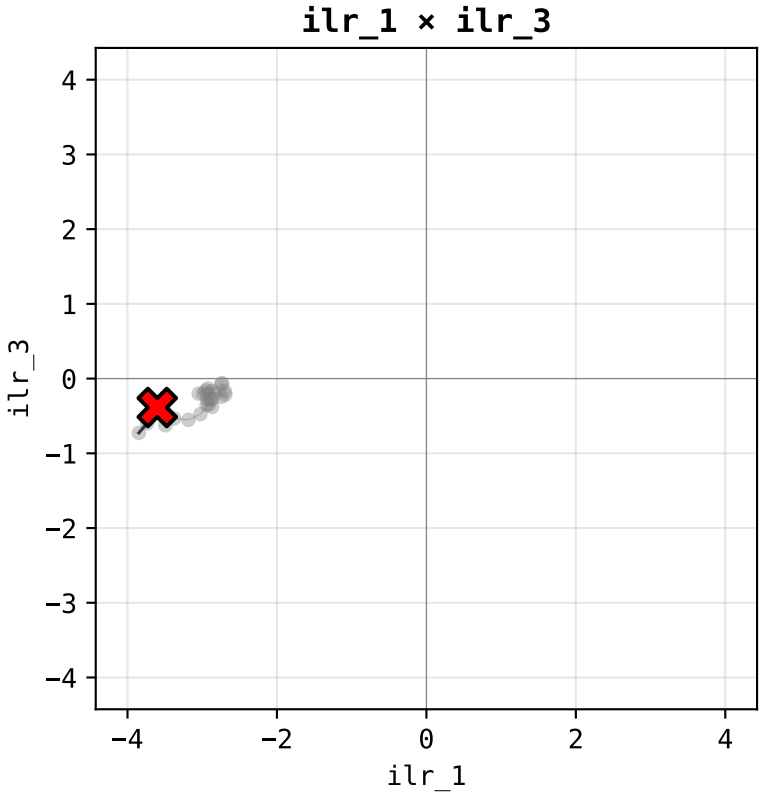
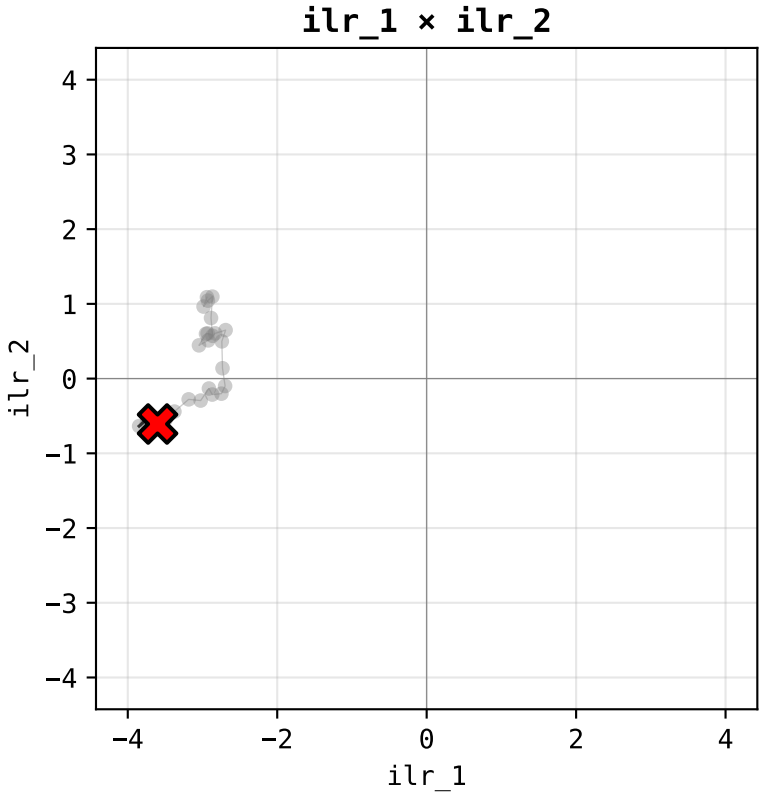




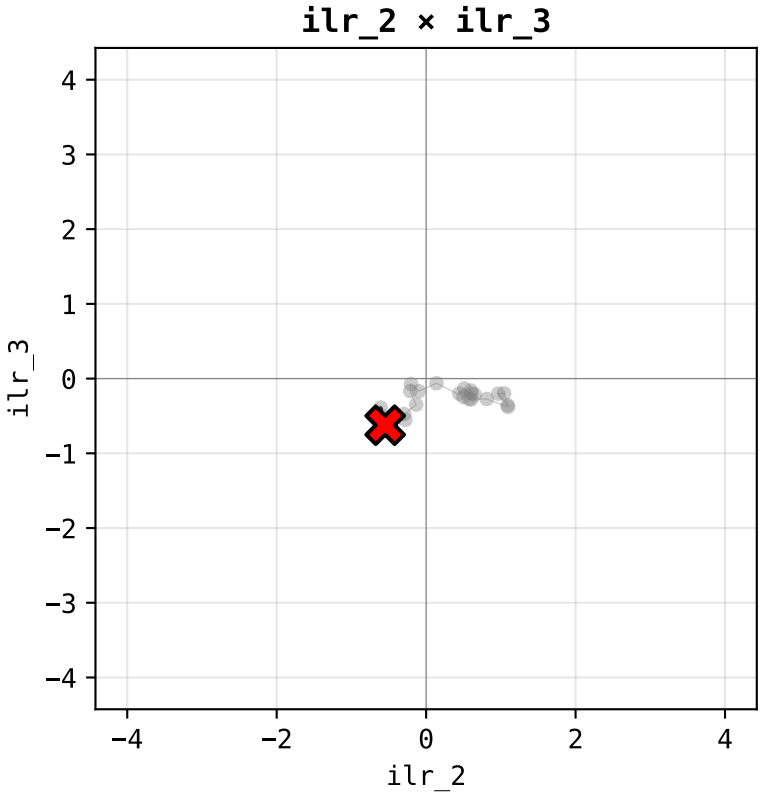
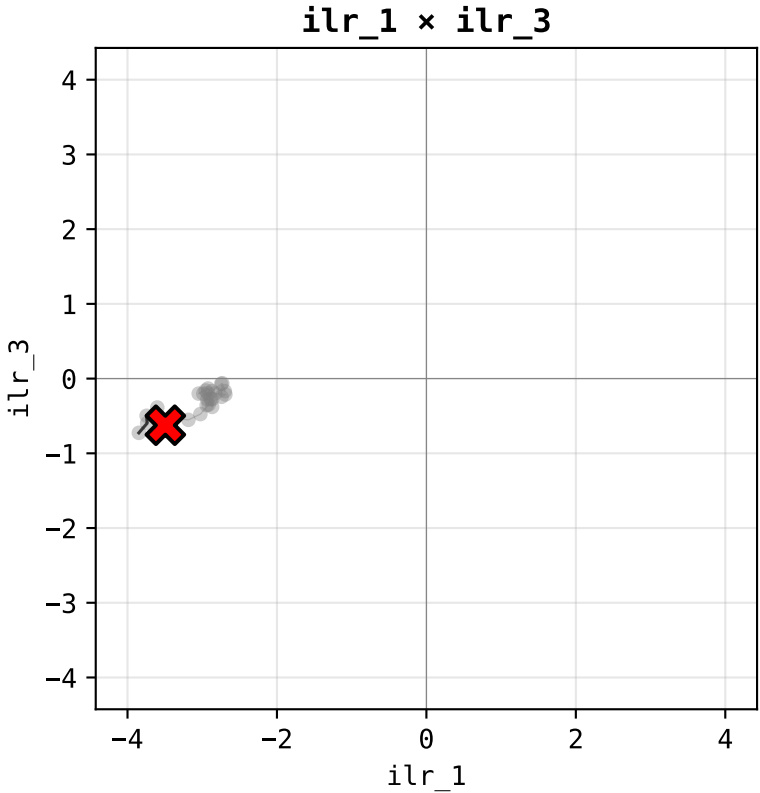
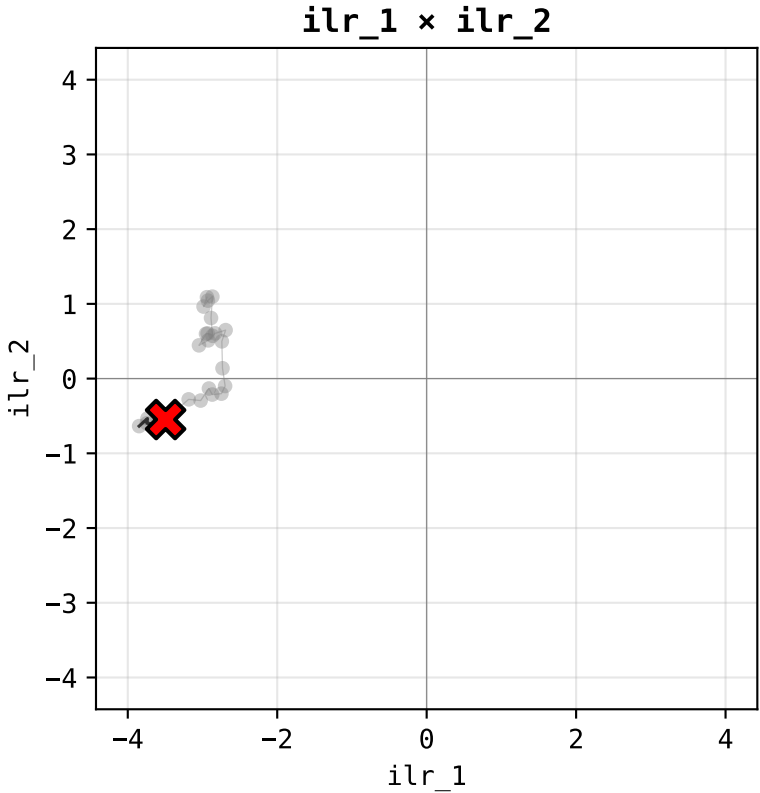
ILR Triplet – IND India · year 2002 (3/26)



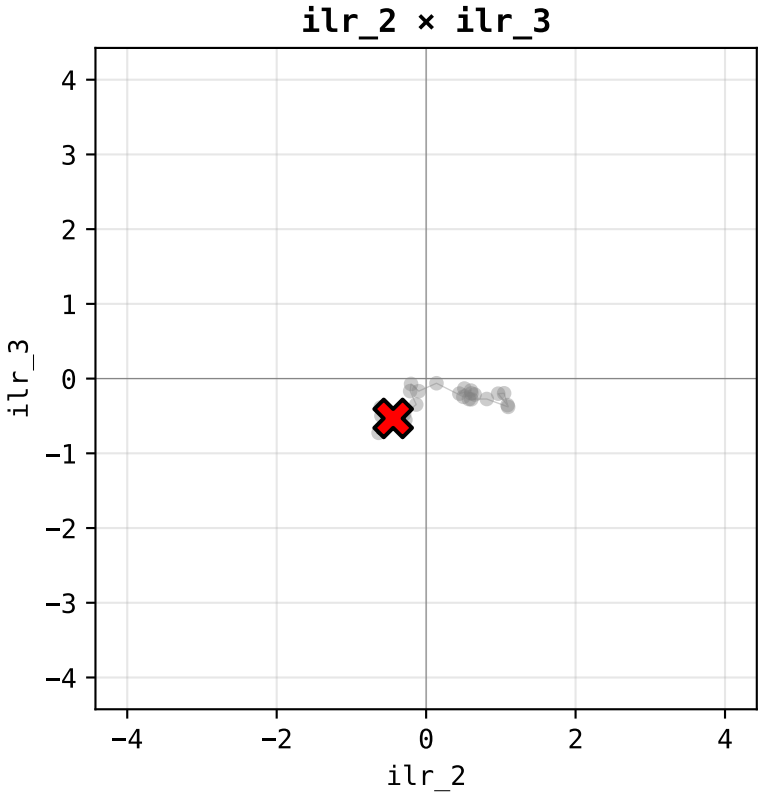
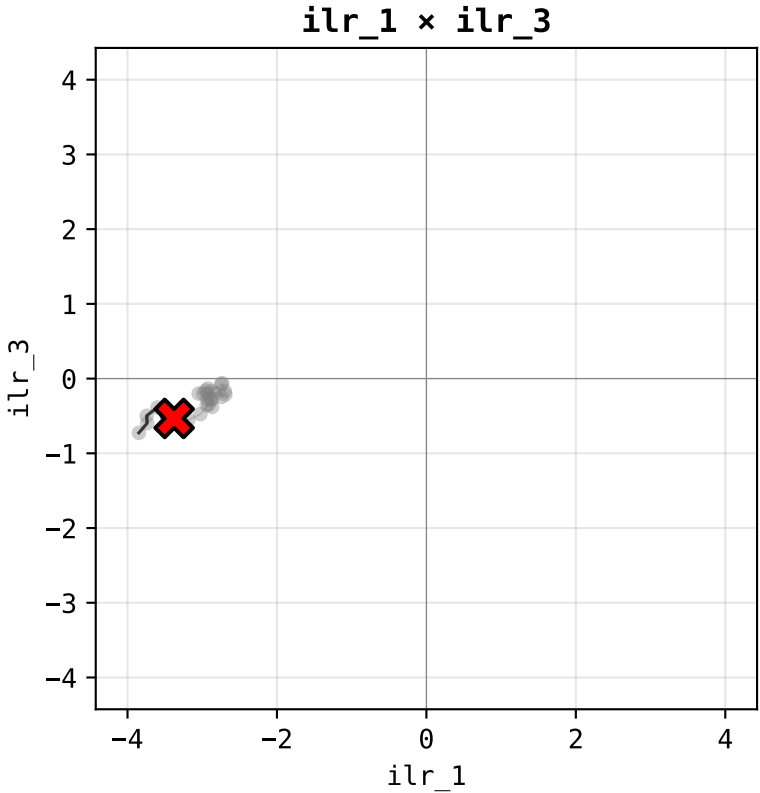
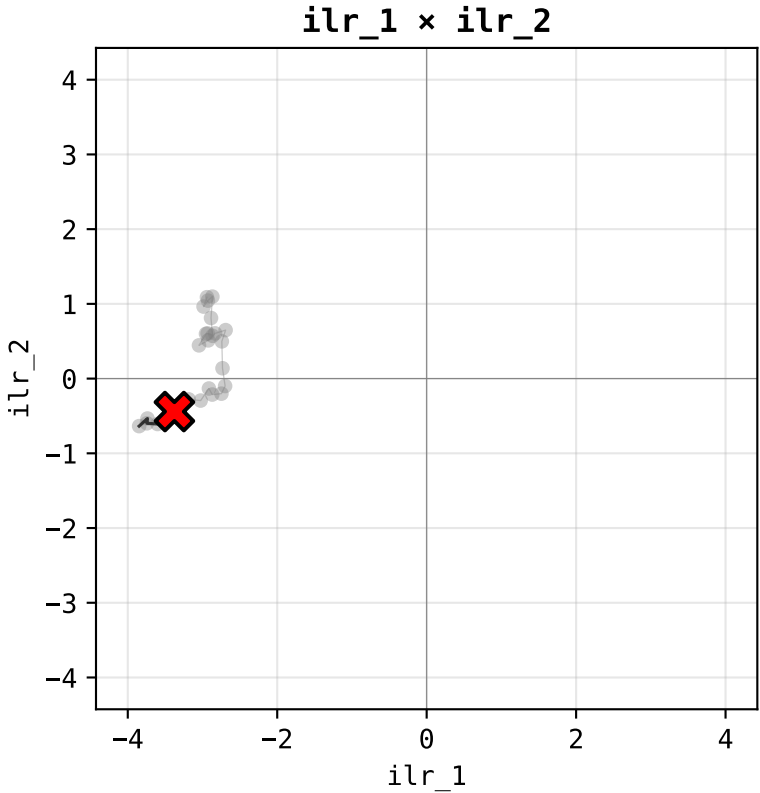
ILR Triplet – IND India · year 2003 (4/26)



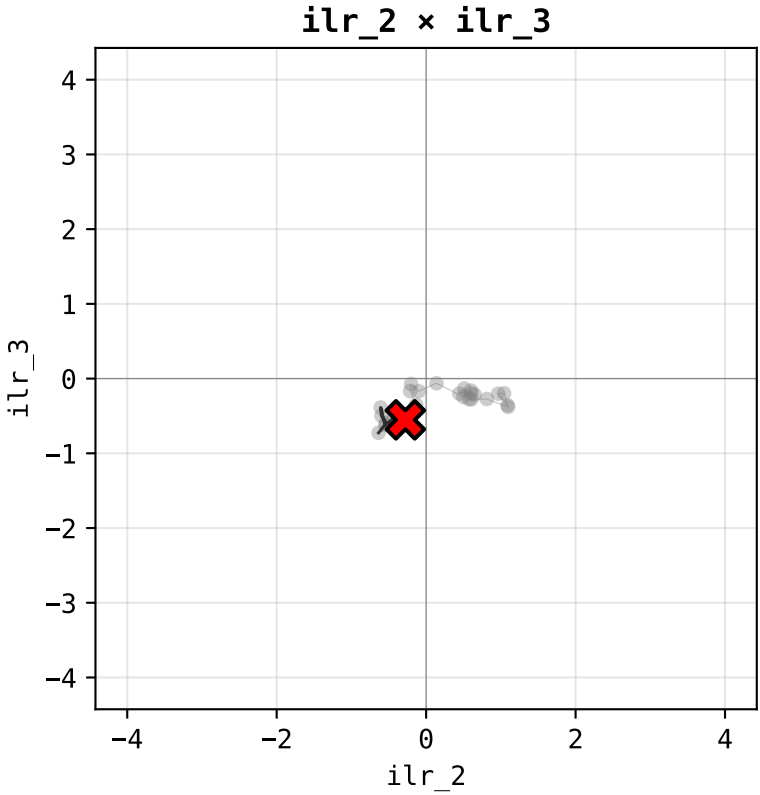
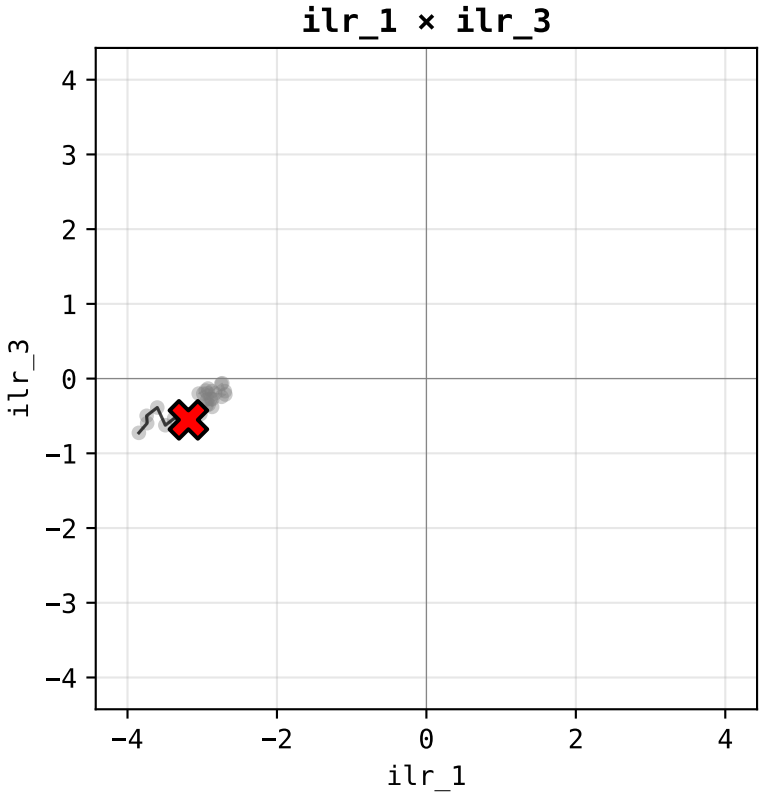
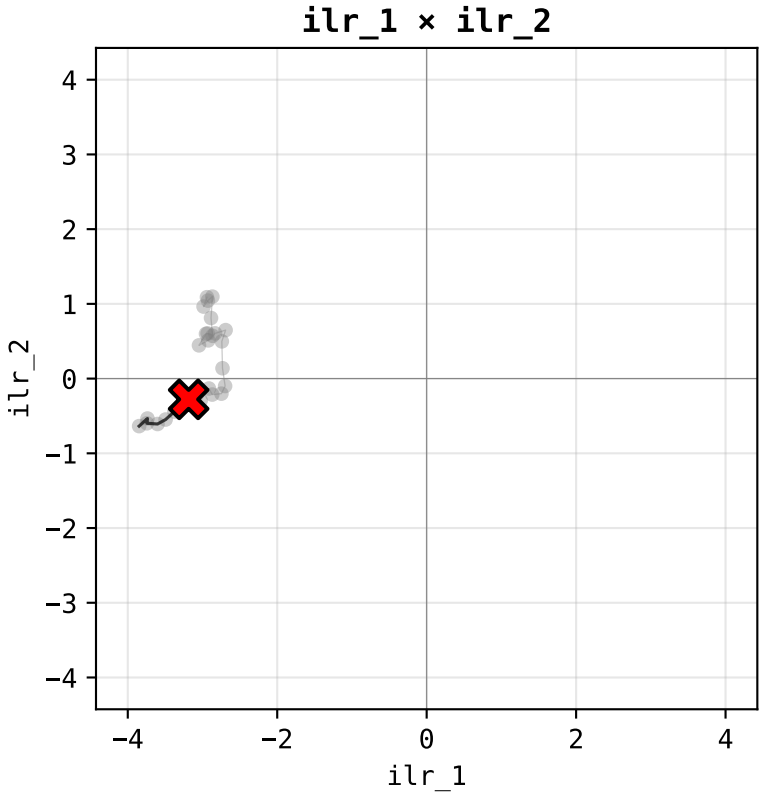
ILR Triplet – IND India · year 2004 (5/26)



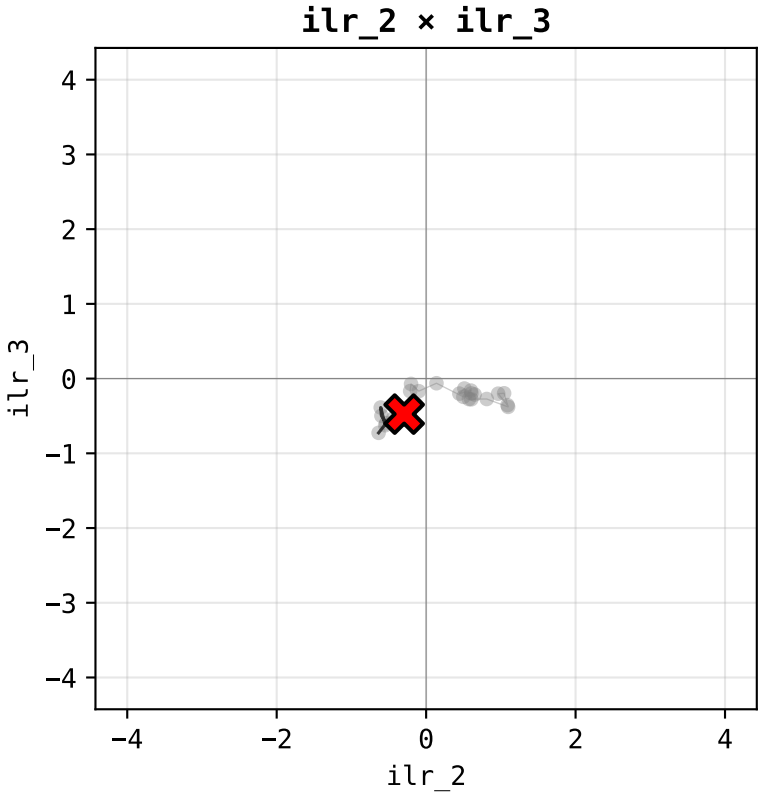
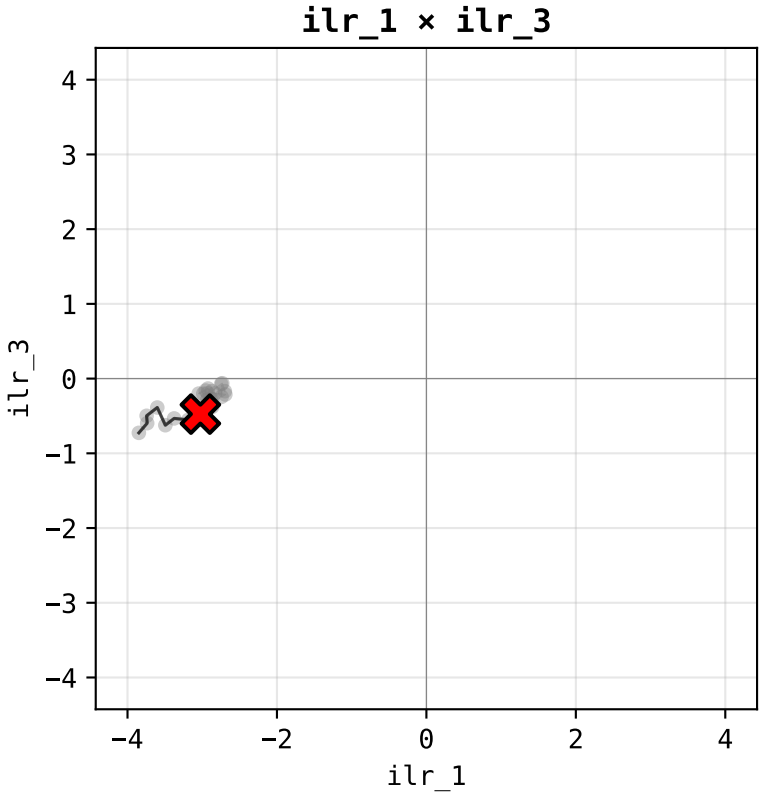
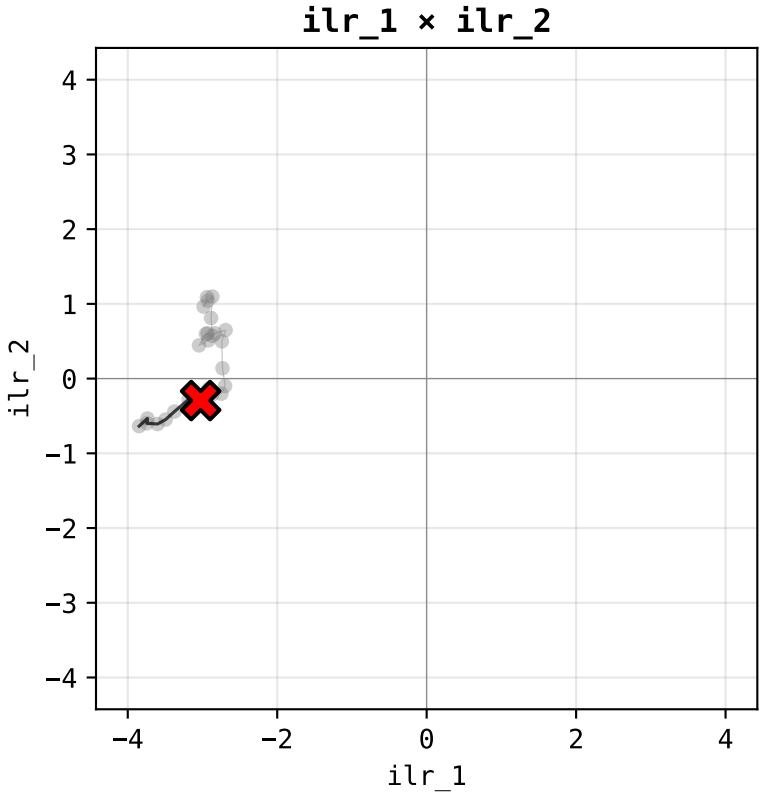
ILR Triplet – IND India · year 2005 (6/26)



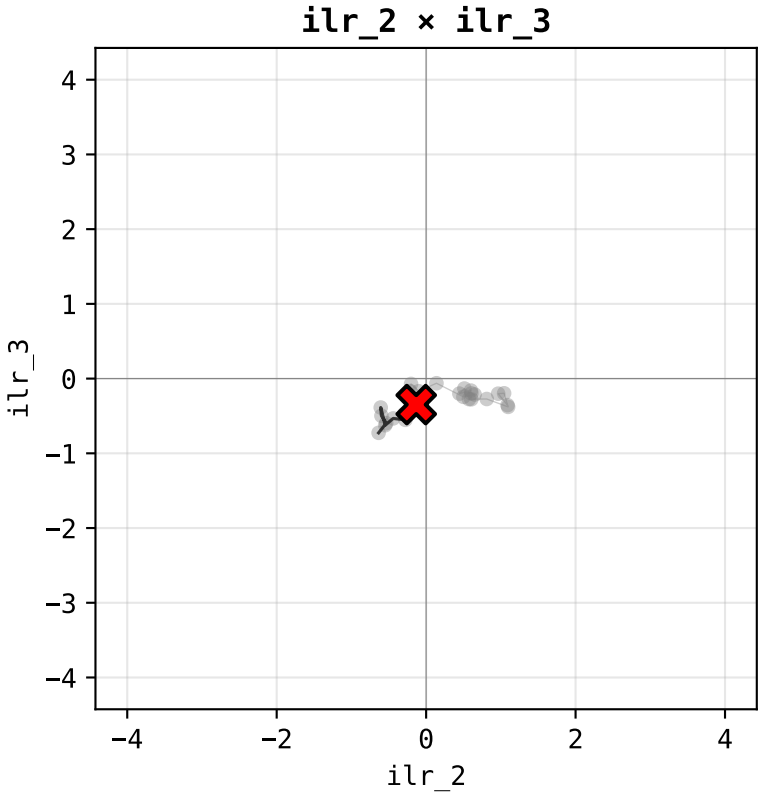
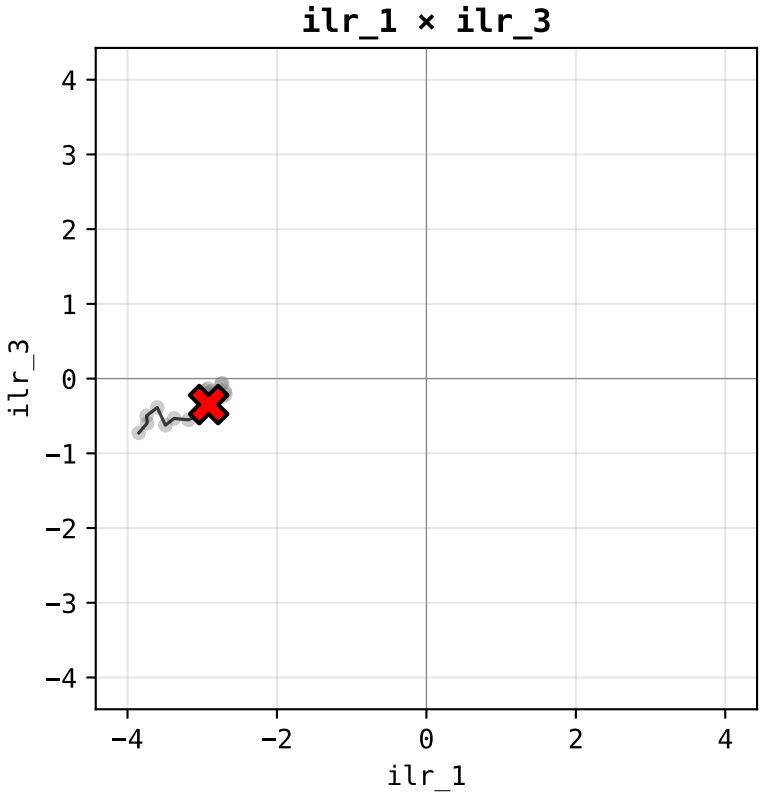
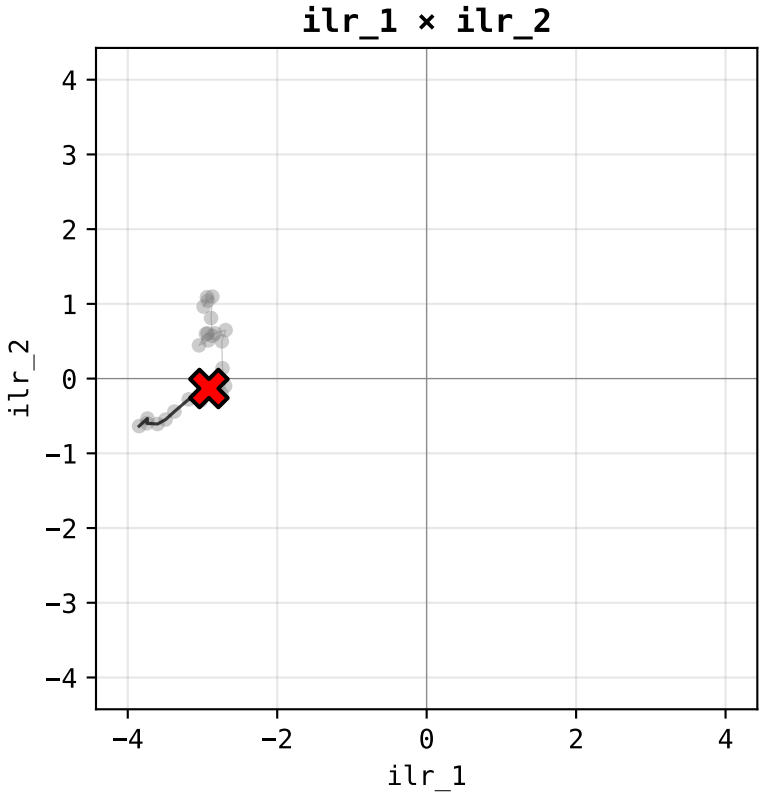
ILR Triplet – IND India · year 2006 (7/26)



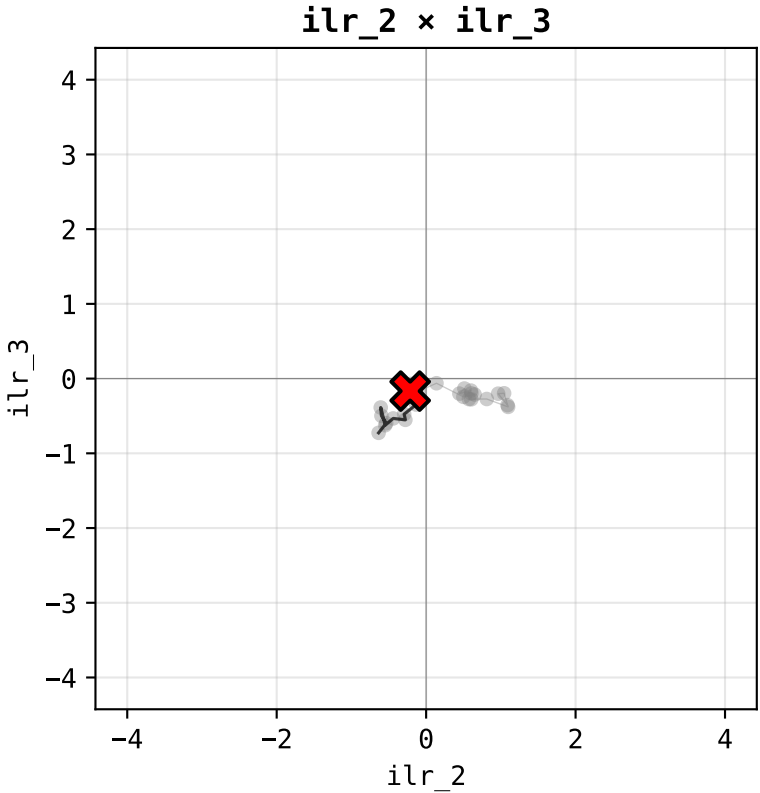
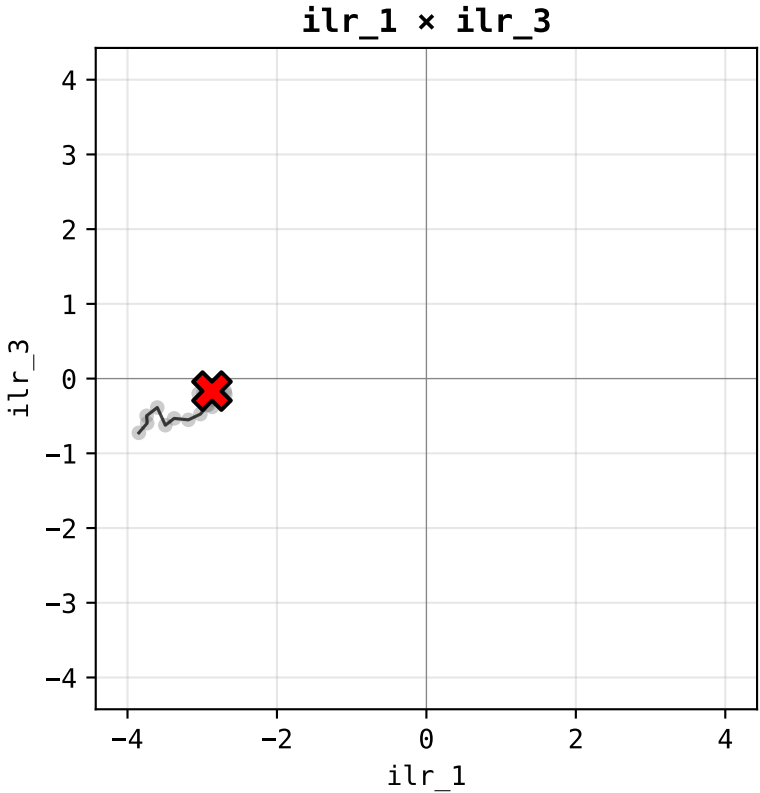
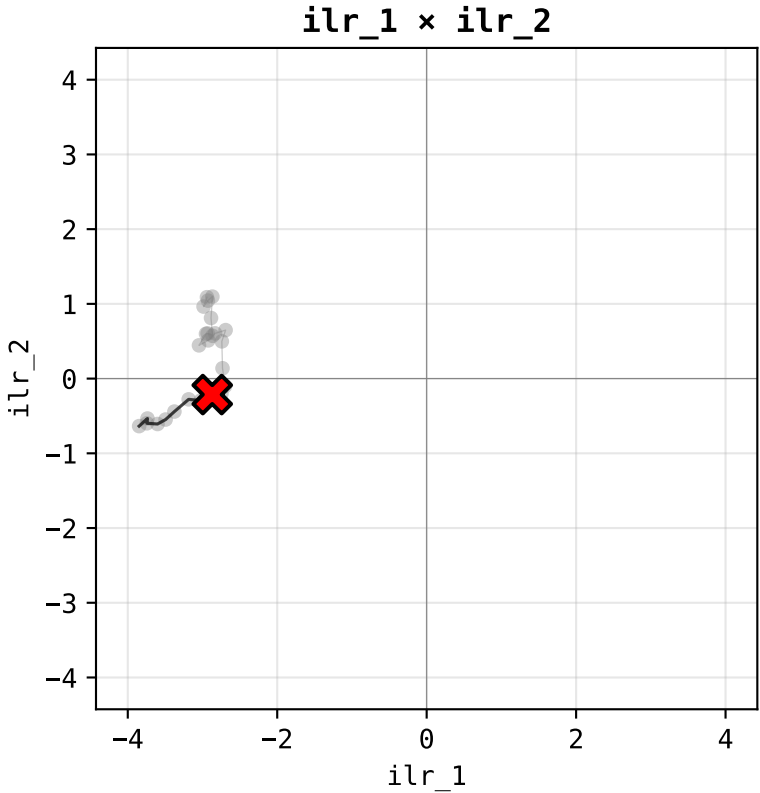
ILR Triplet – IND India · year 2007 (8/26)



ILR Triplet – IND India · year 2008 (9/26)

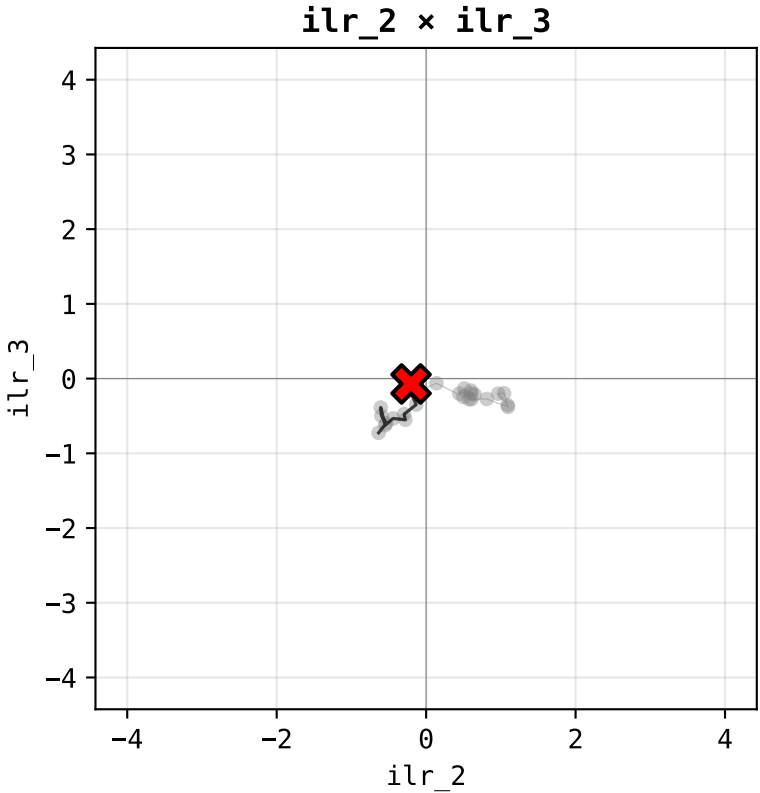
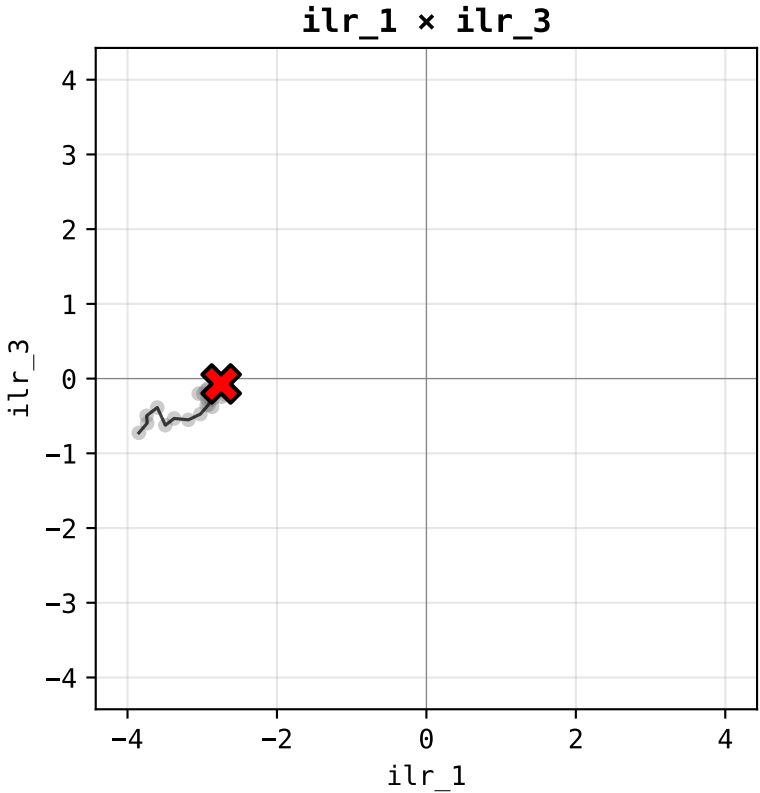
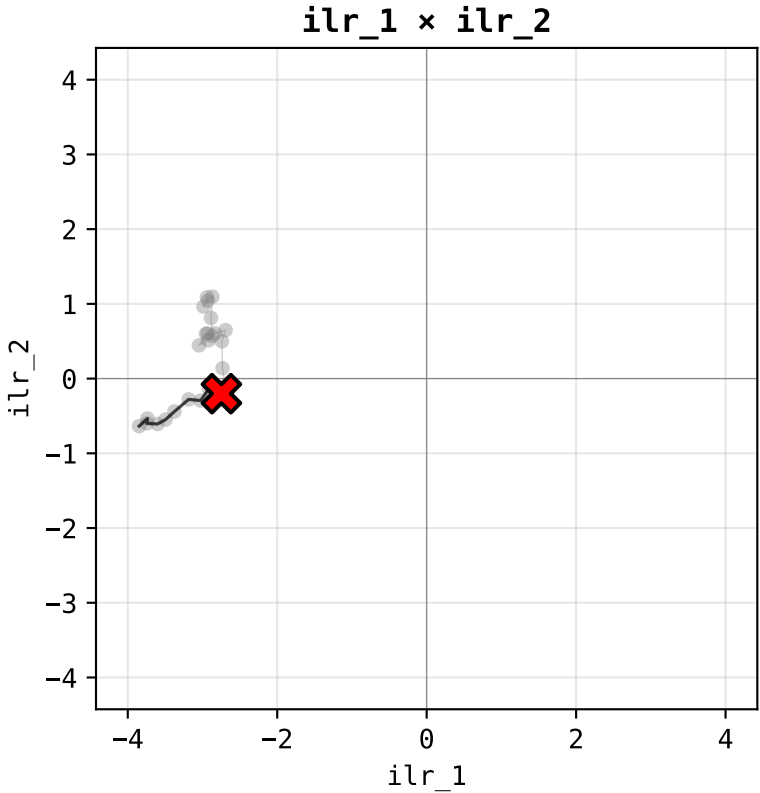


ILR Triplet – IND India · year 2009 (10/26)

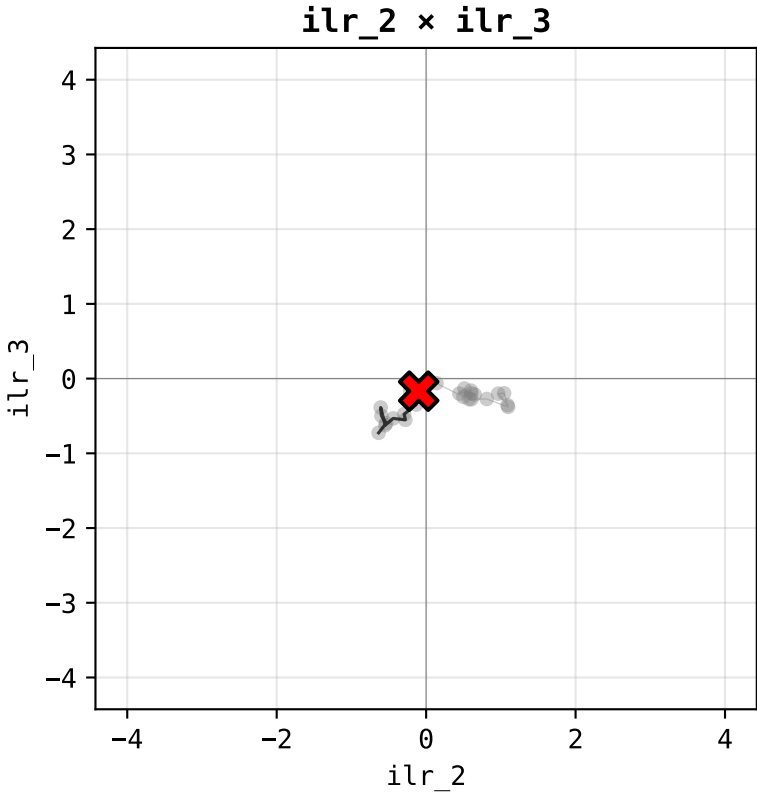
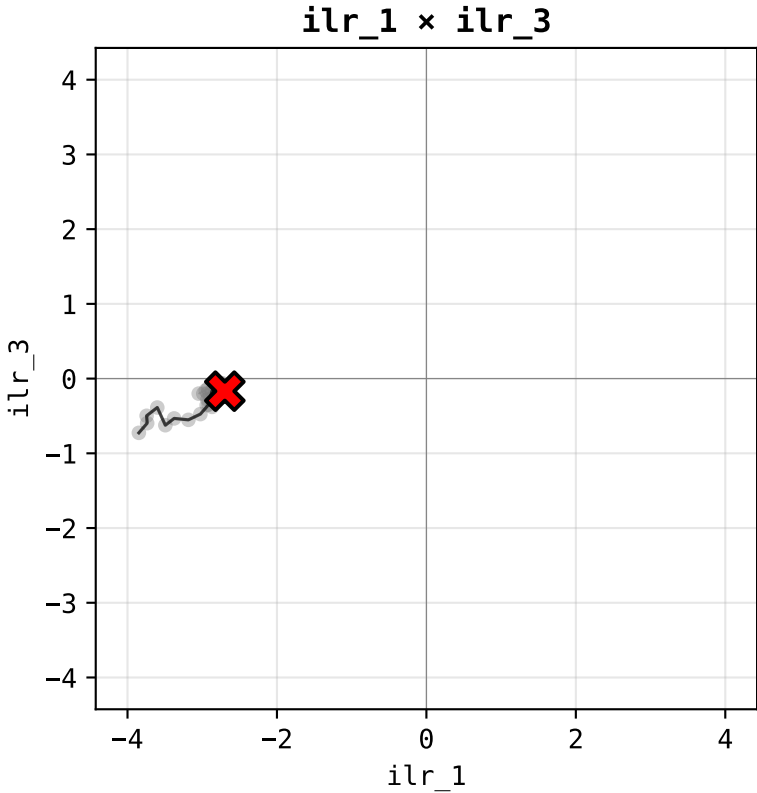
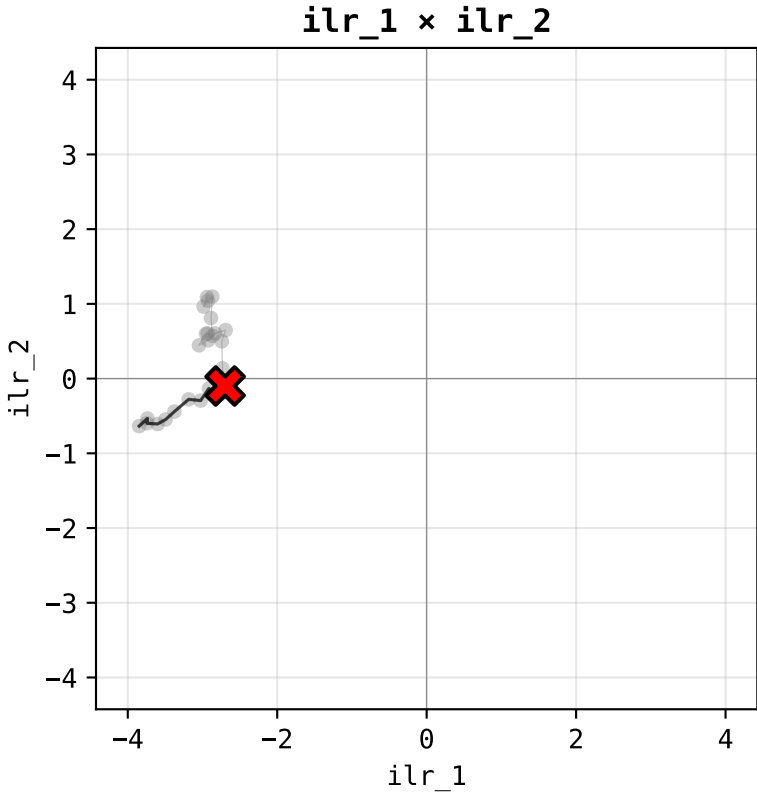


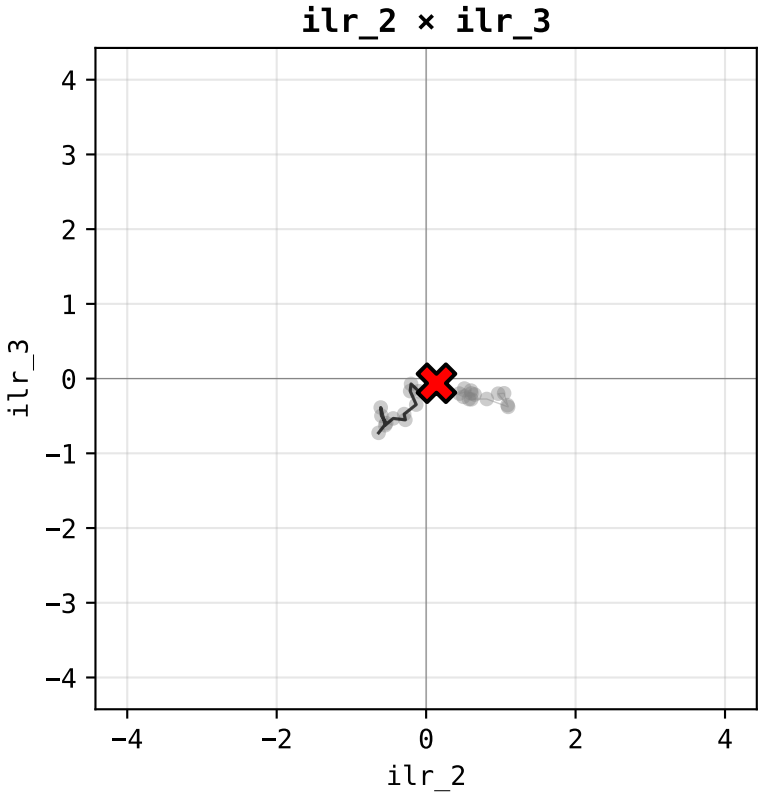
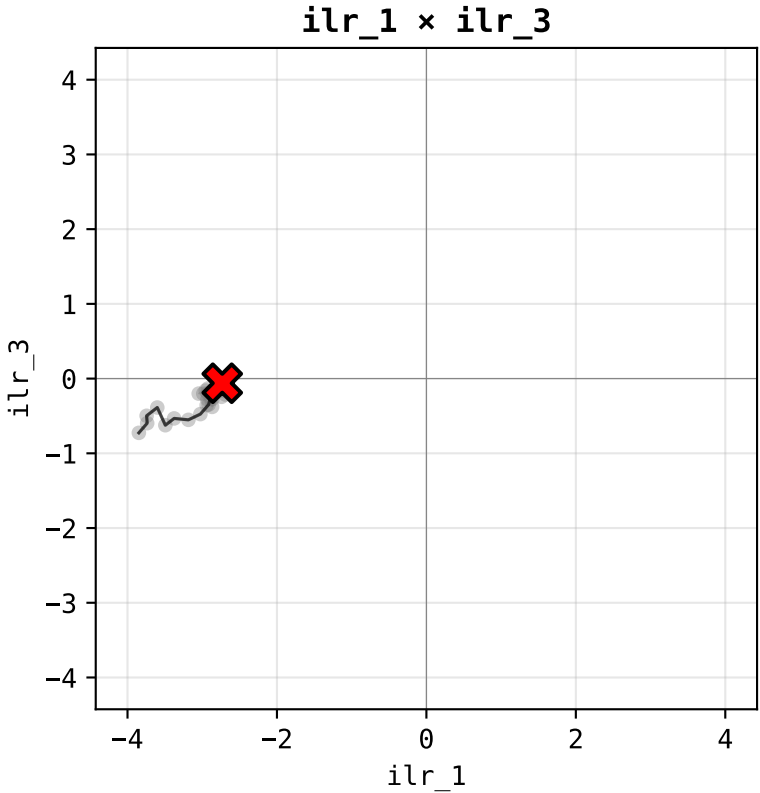
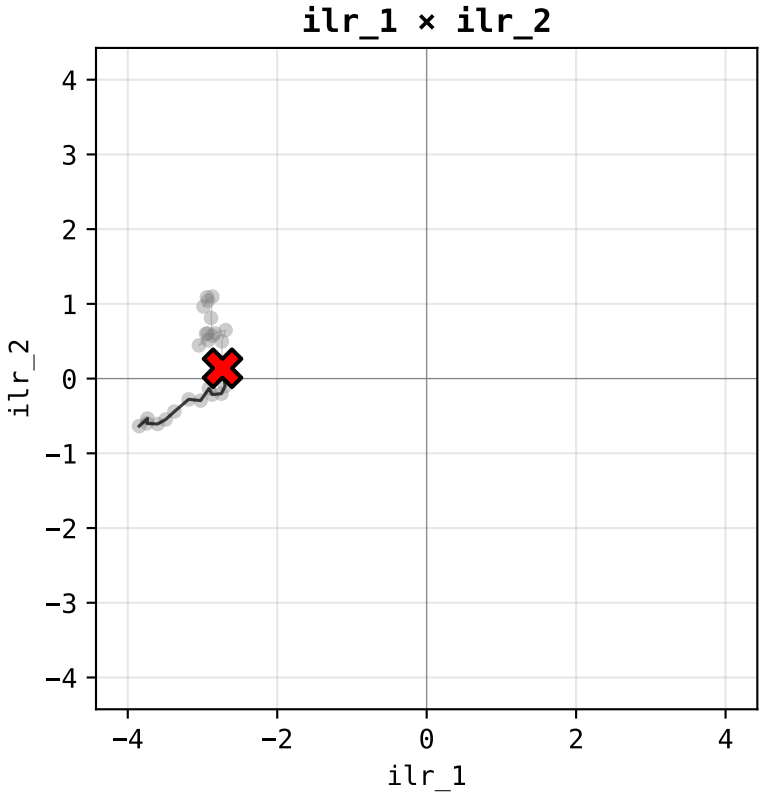
X = current year (2009) · black = past trajectory · gray = future · ilr range ±4.43

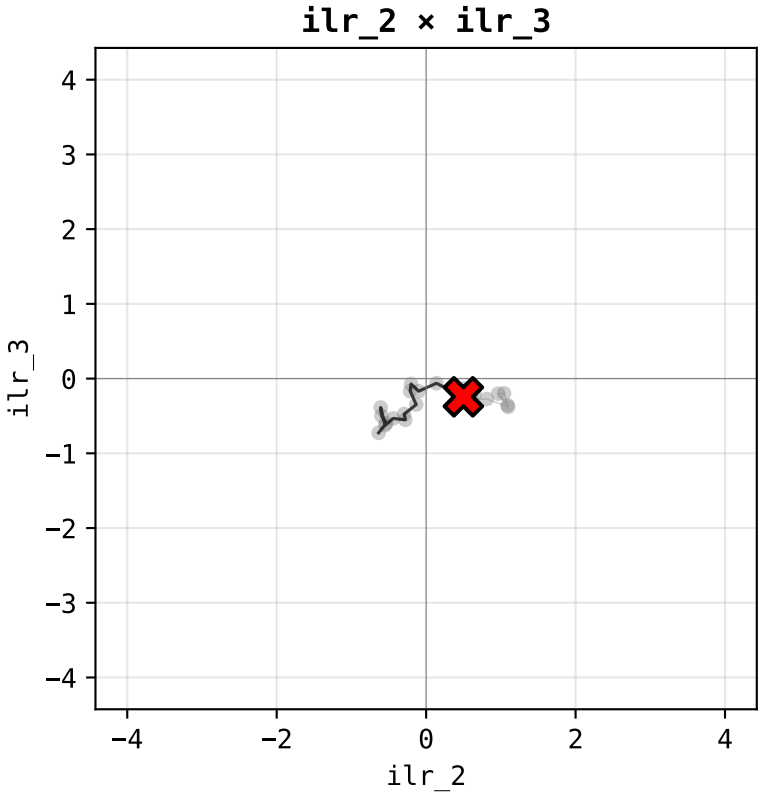
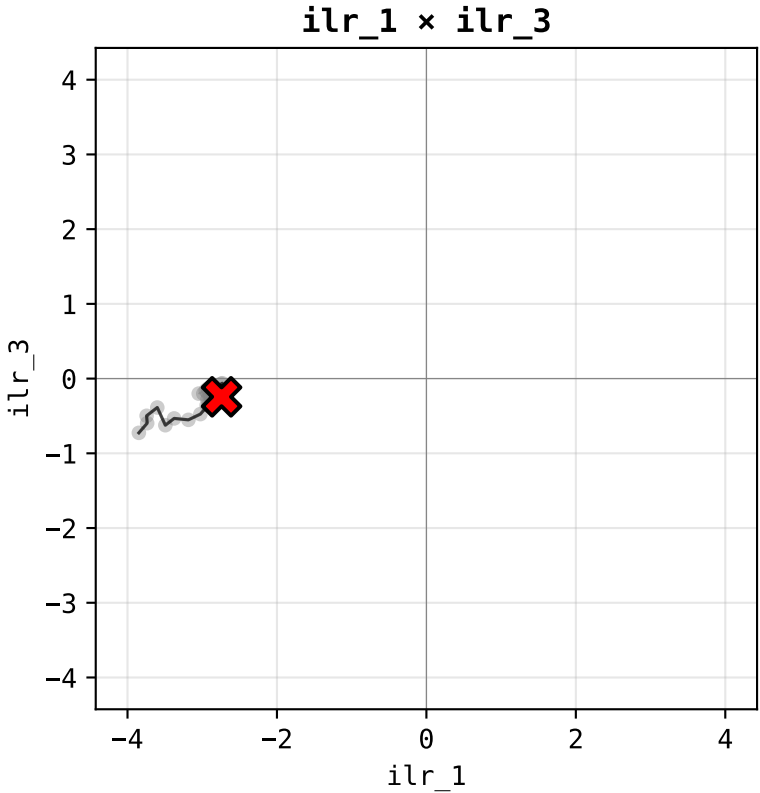
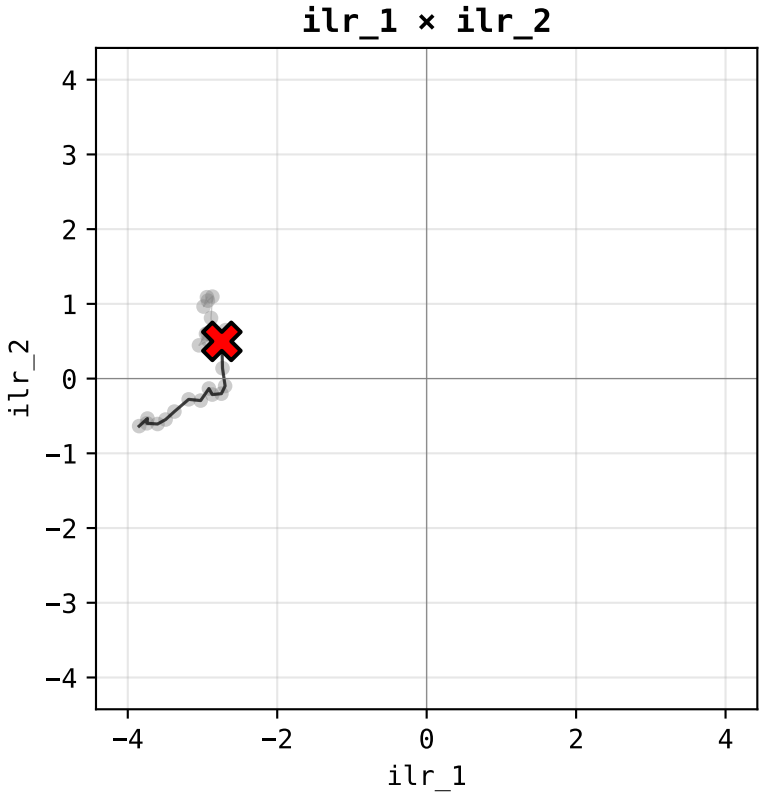
ILR Triplet – IND India · year 2010 (11/26)

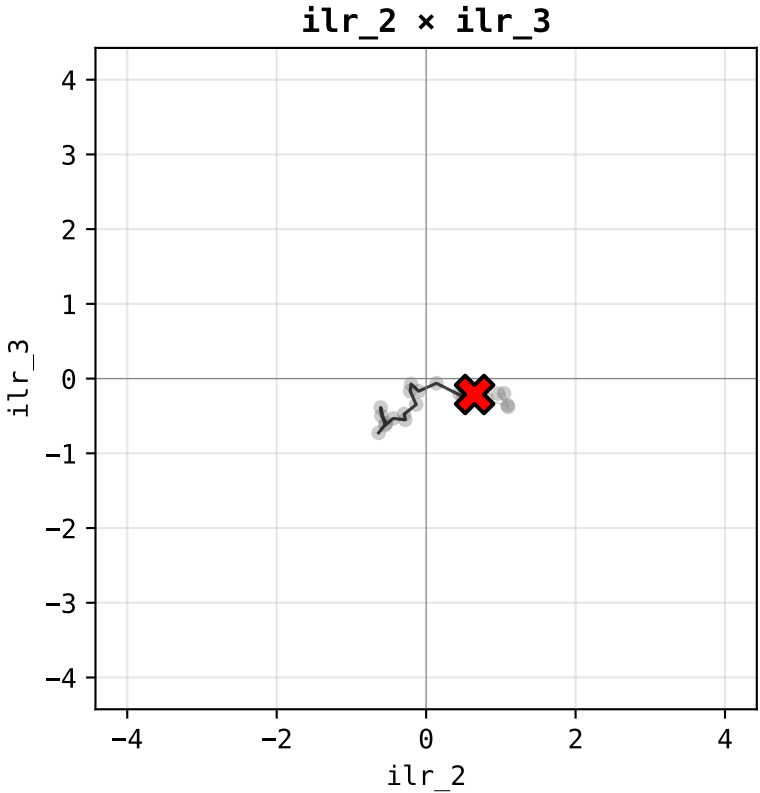
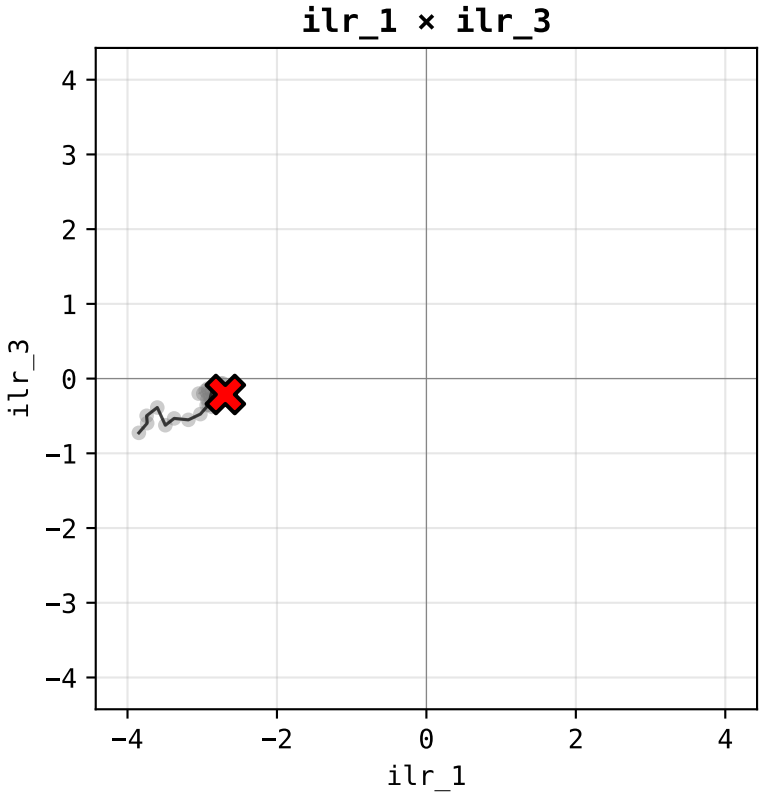
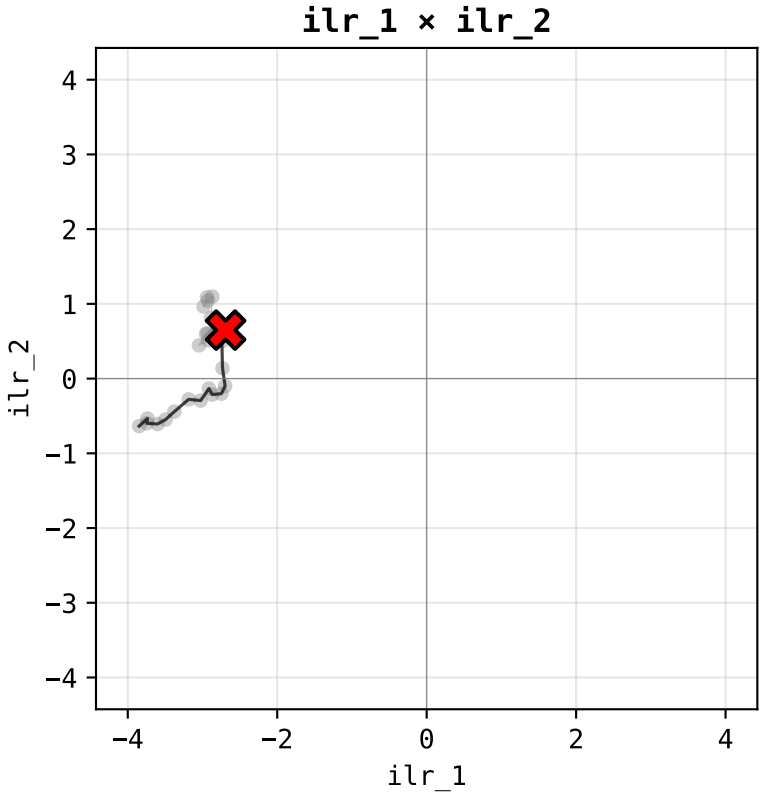


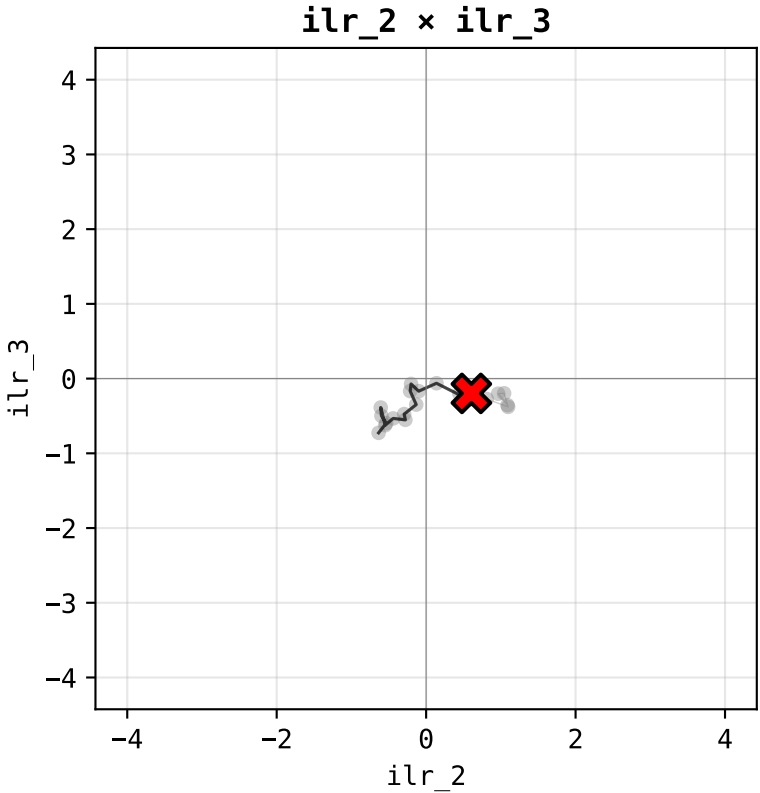
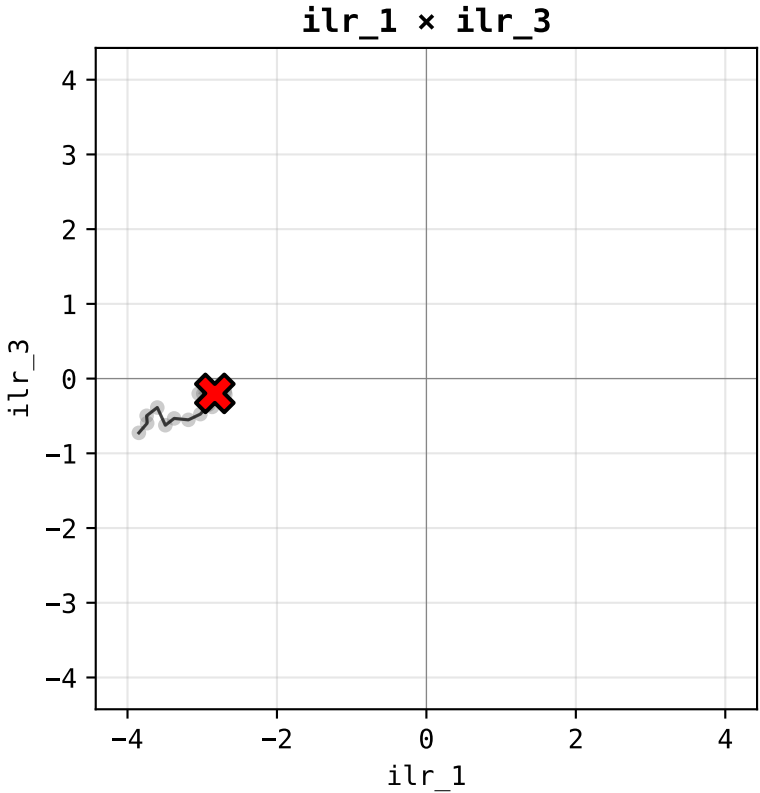
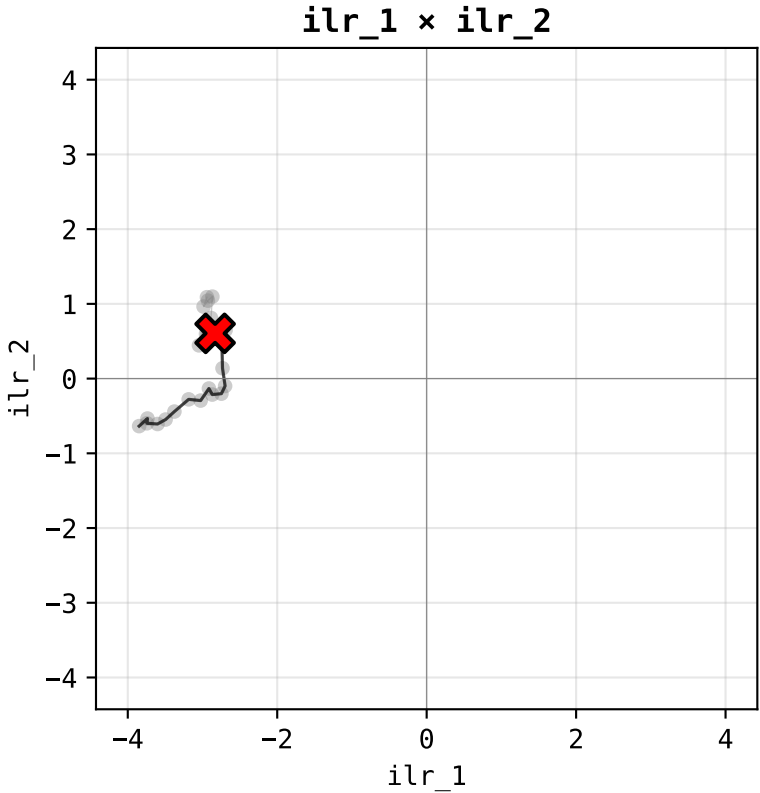
ILR Triplet – IND India · year 2011 (12/26)

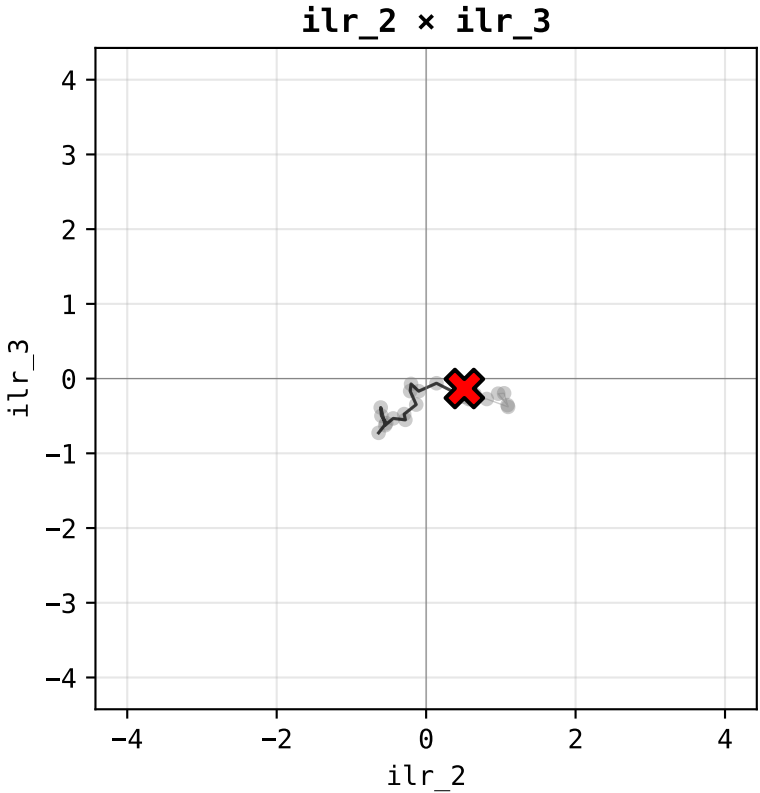
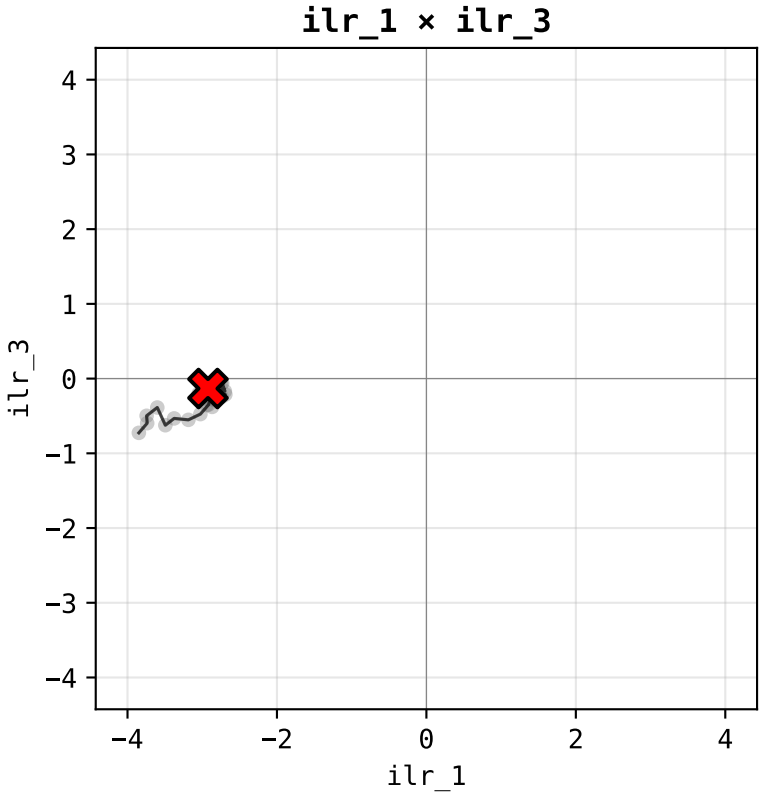
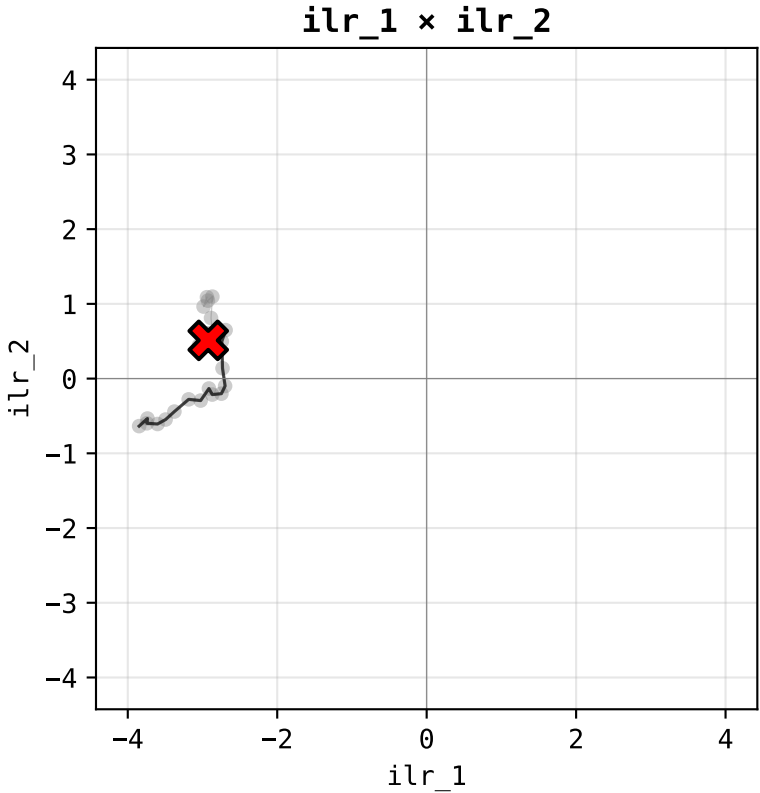


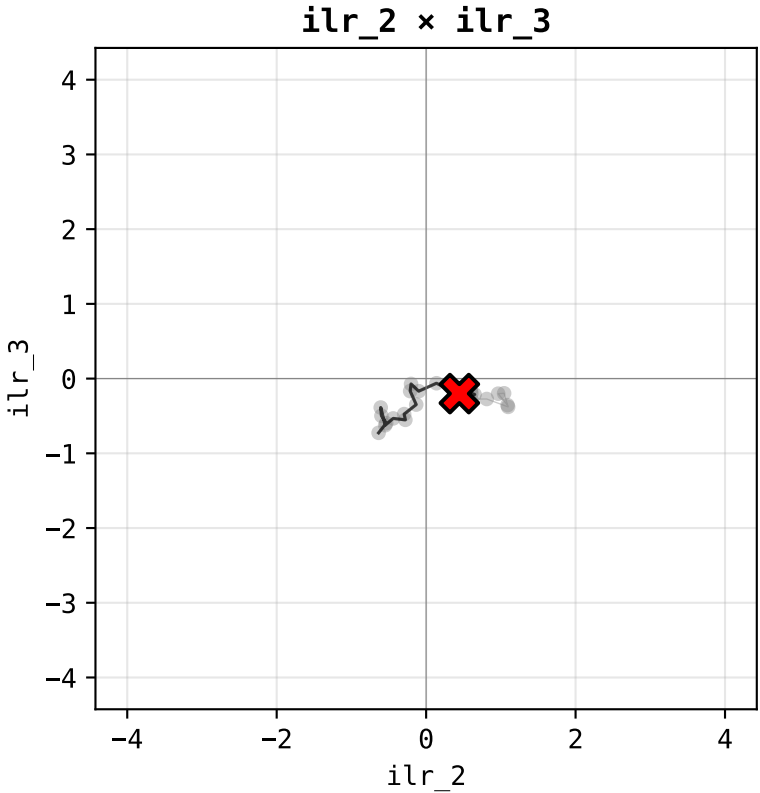
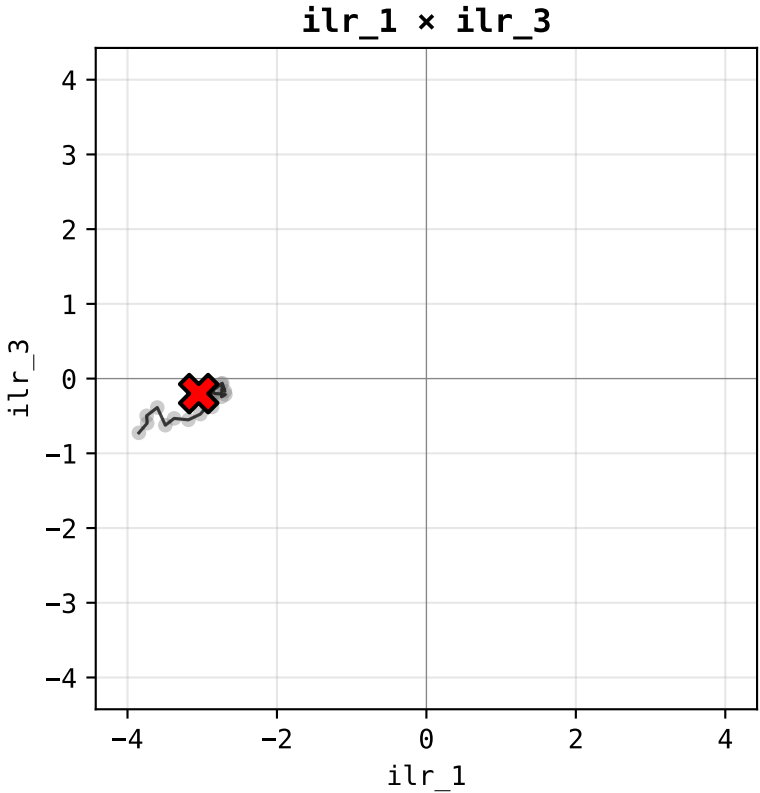
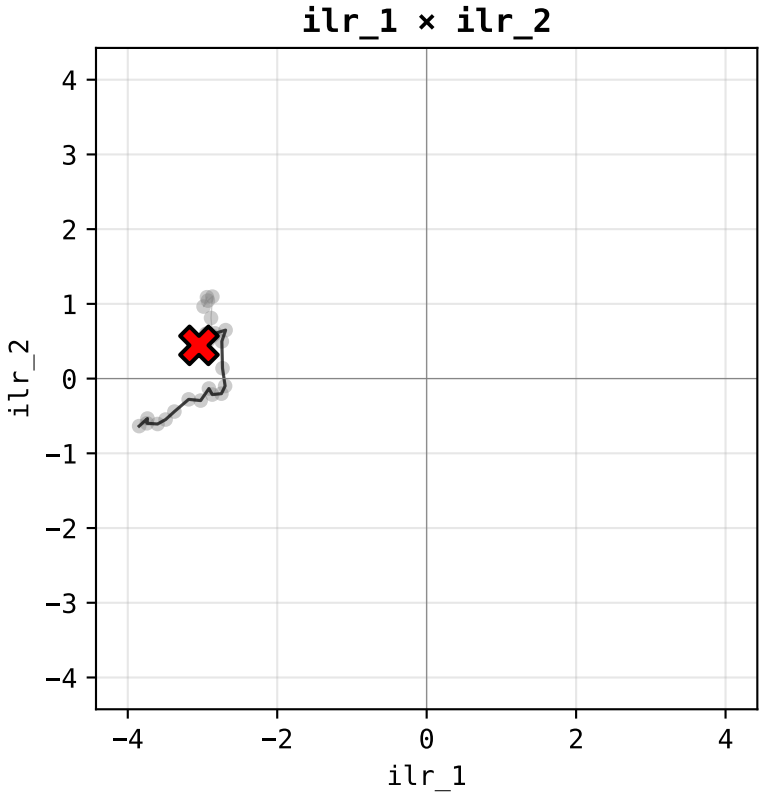




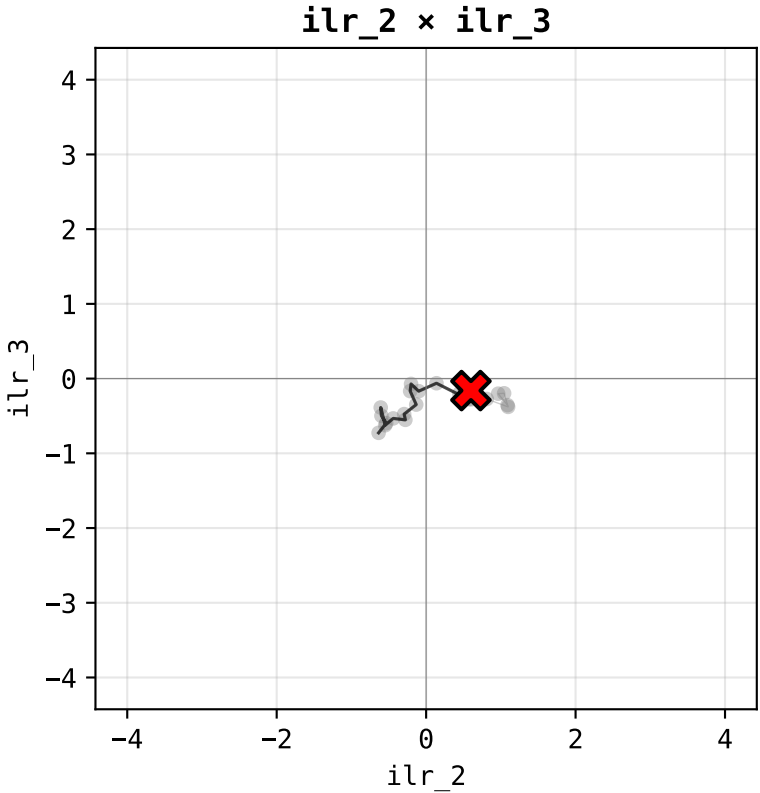
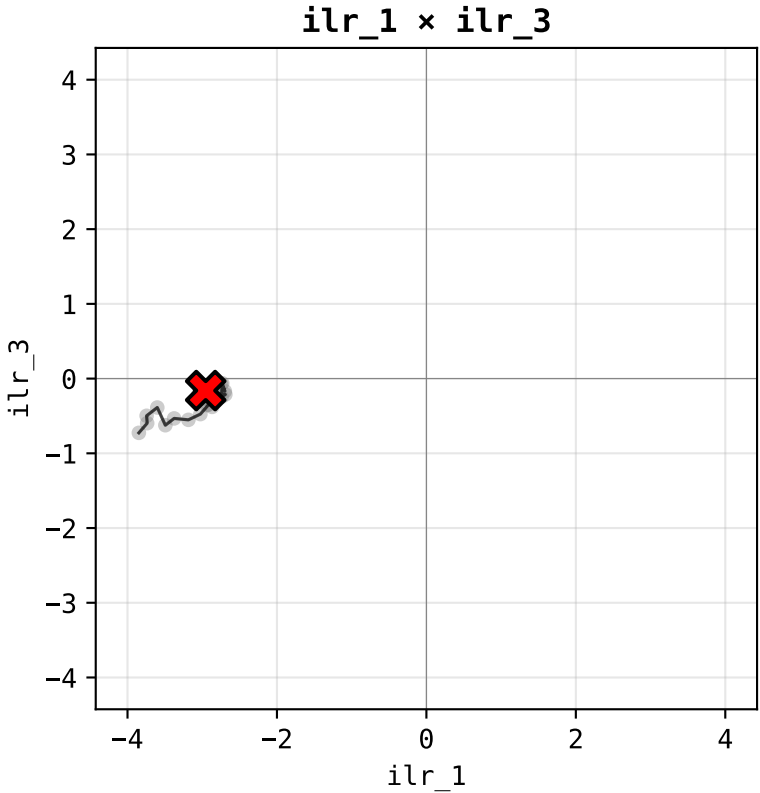
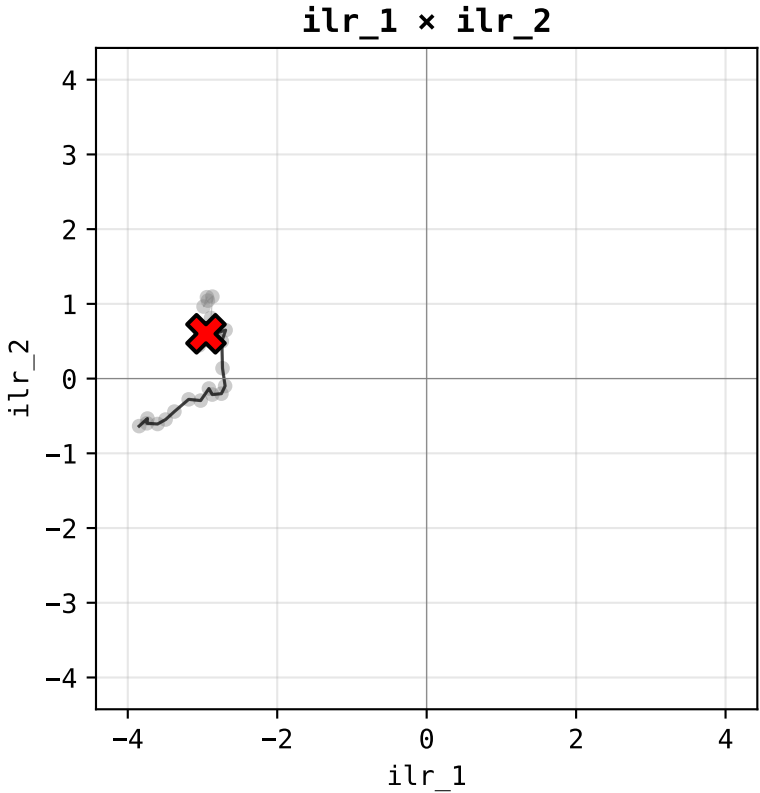


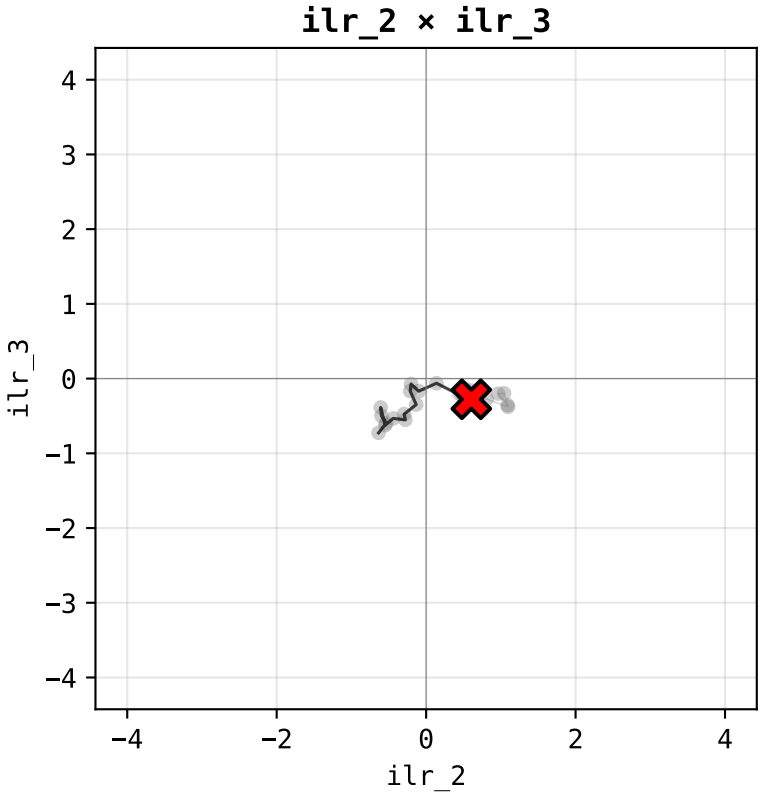
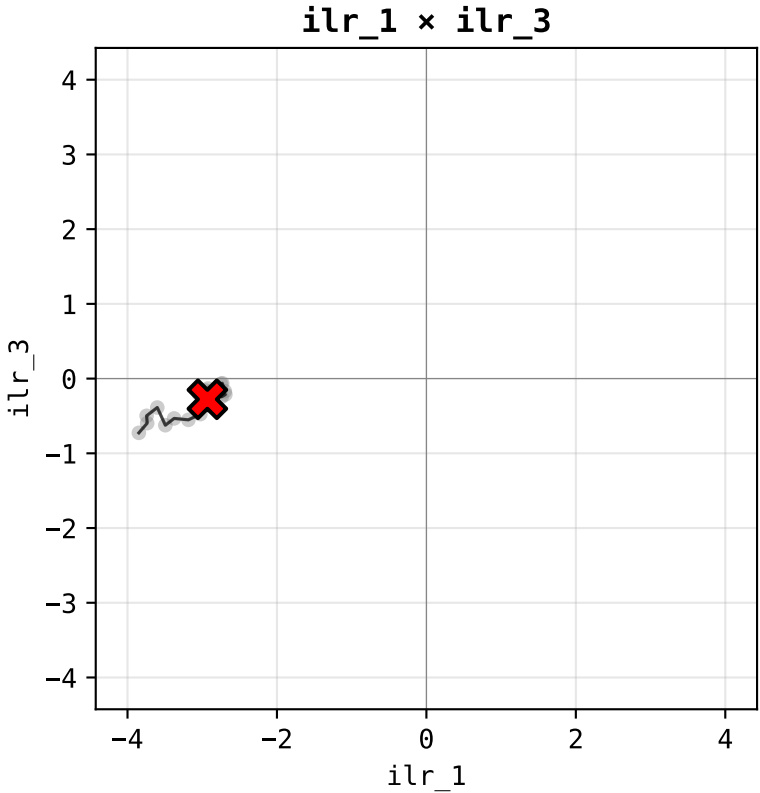
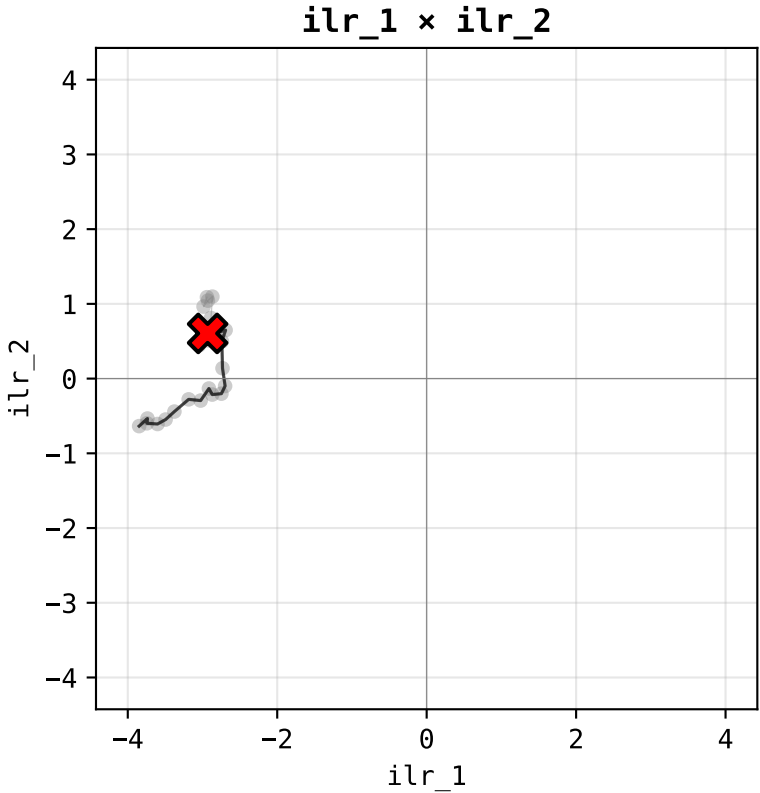


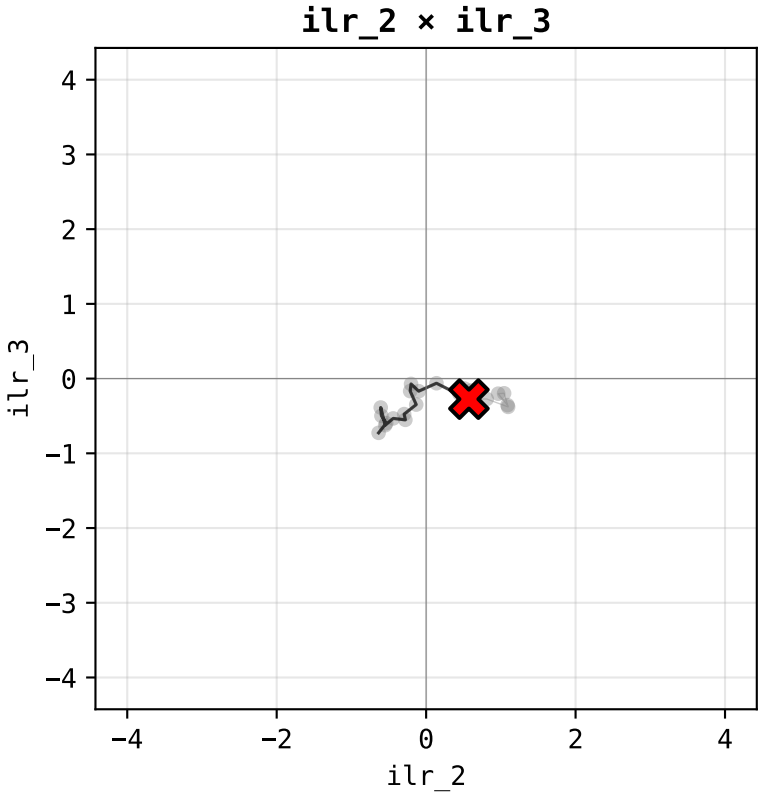
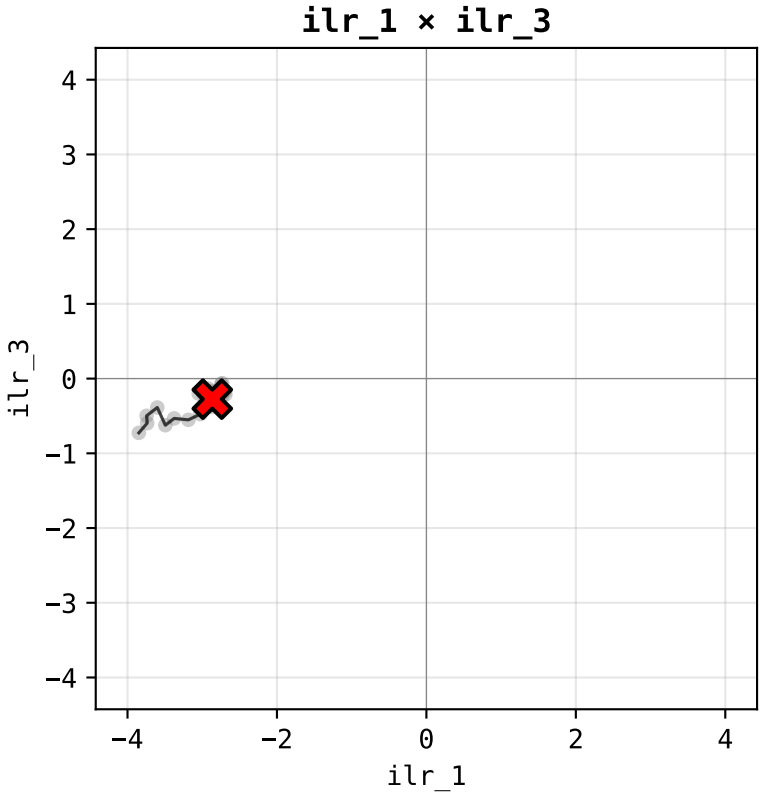
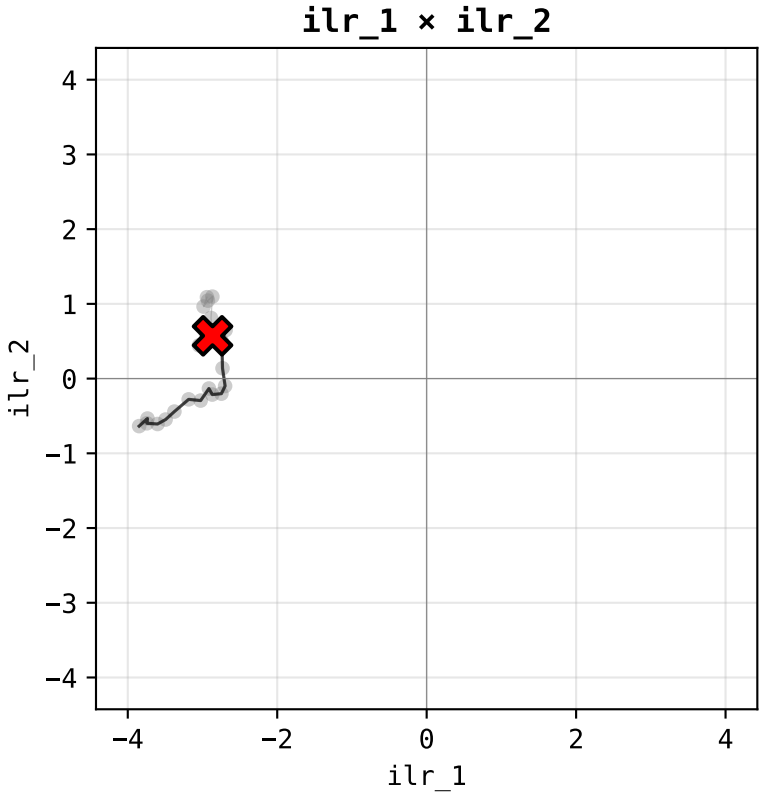


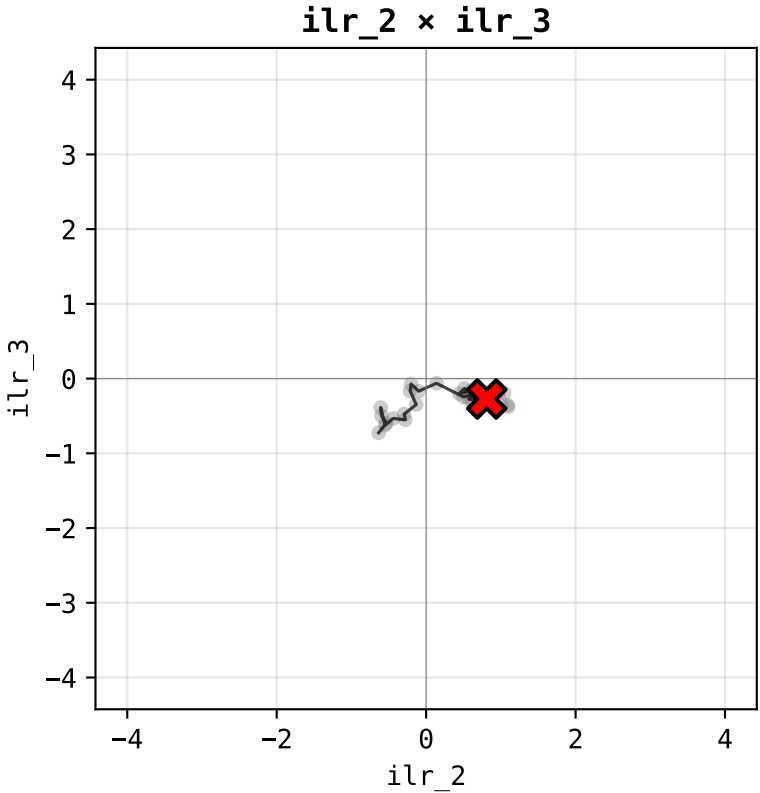
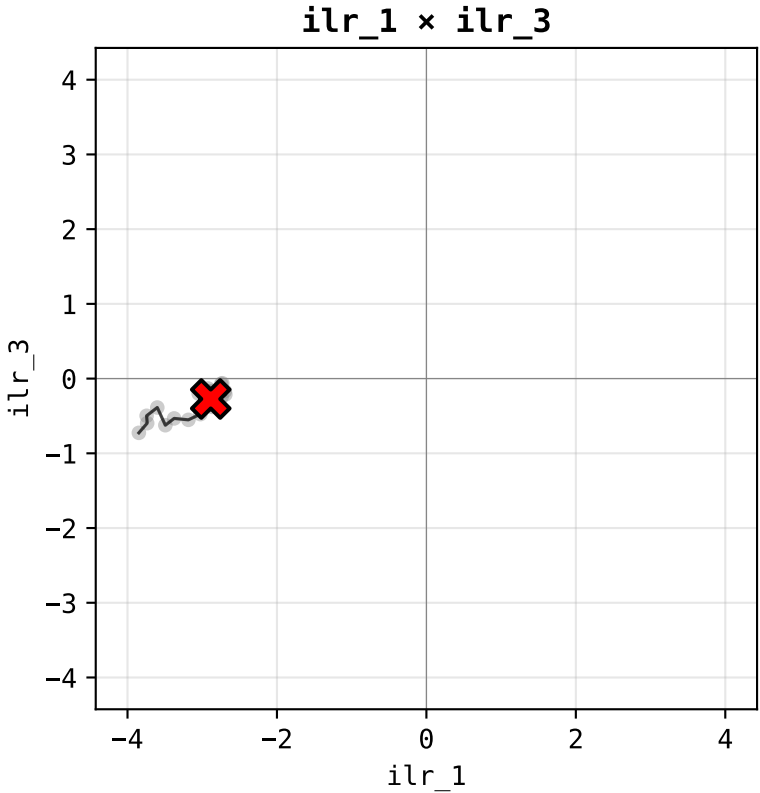
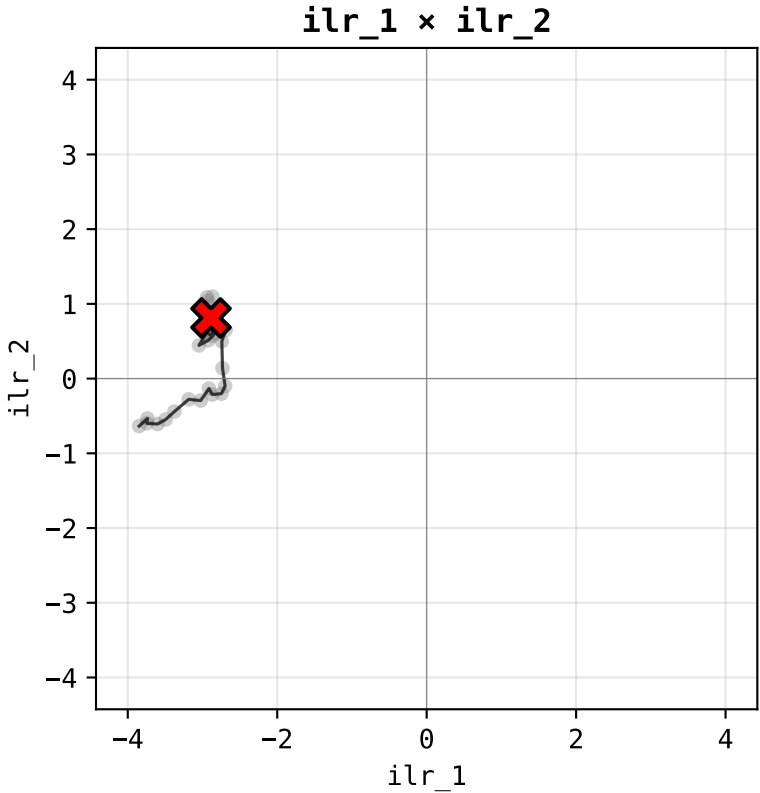


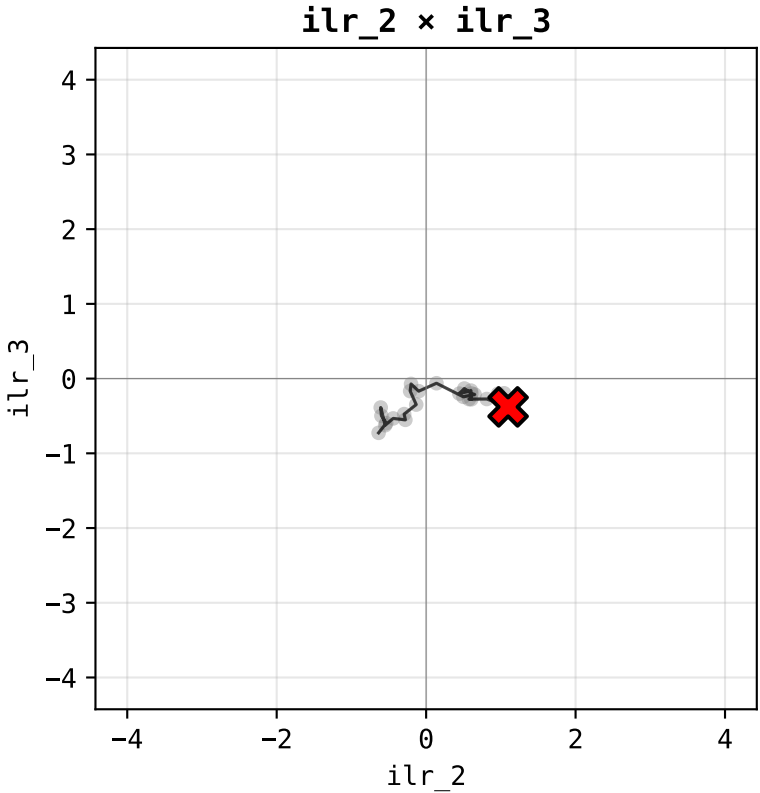
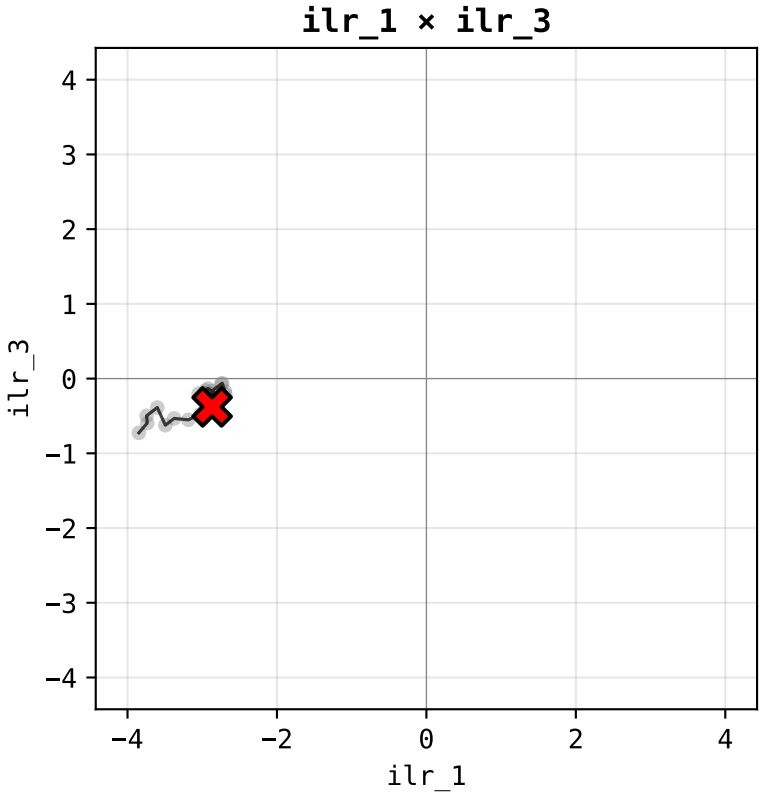
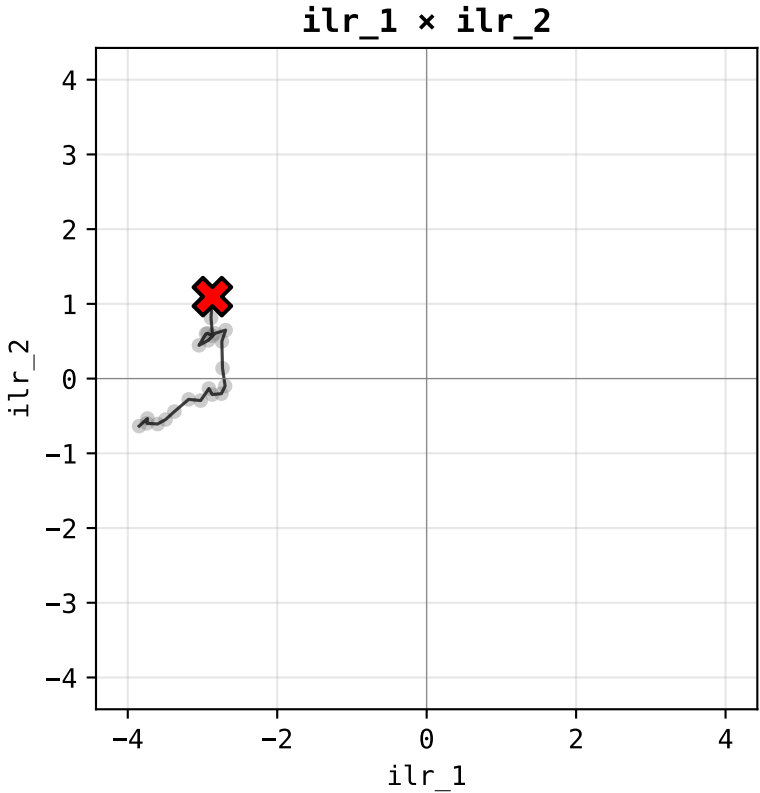
ILR Triplet – IND India · year 2018 (19/26)

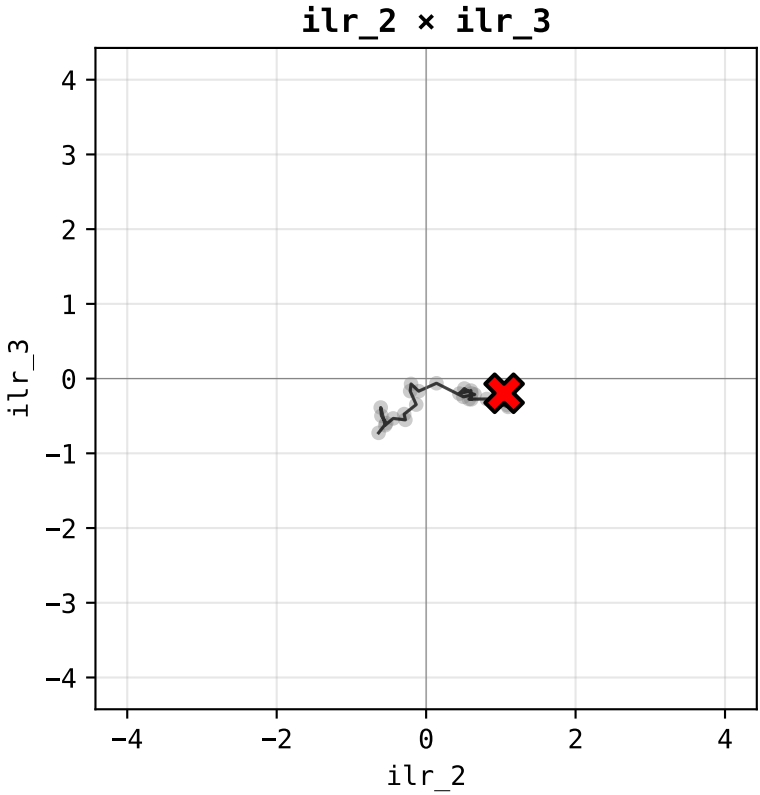
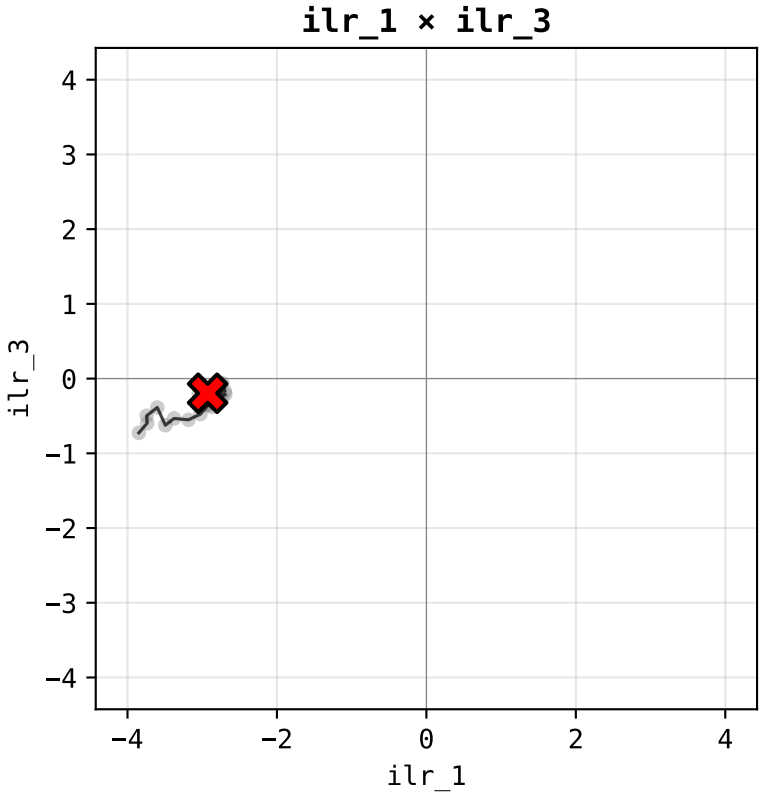
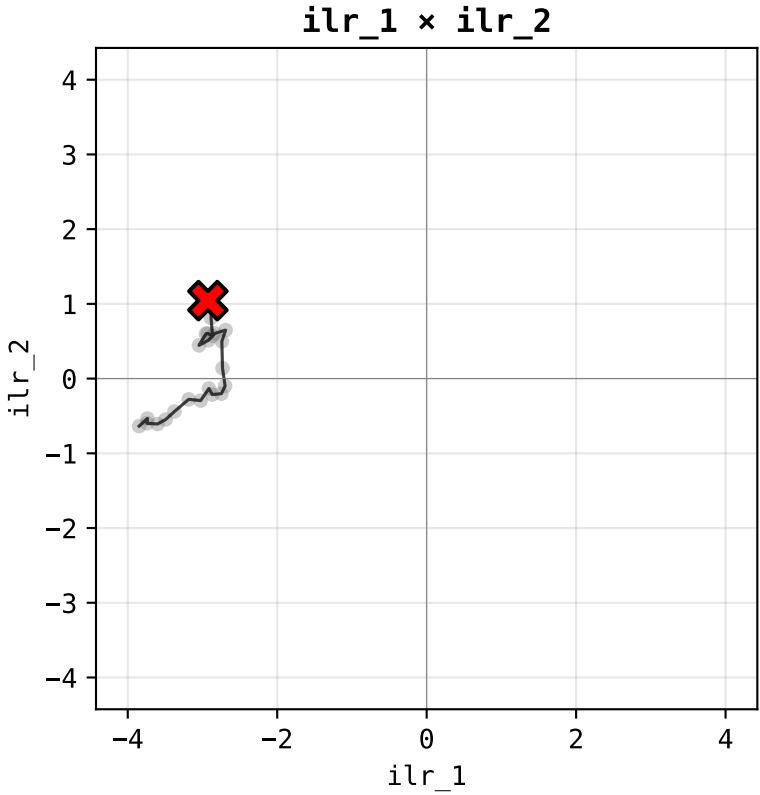


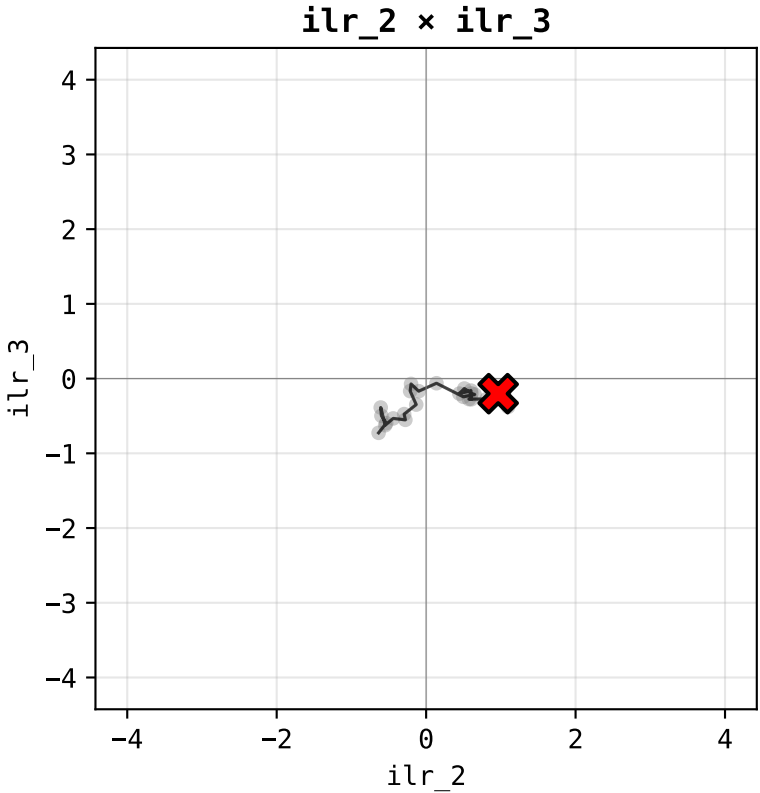
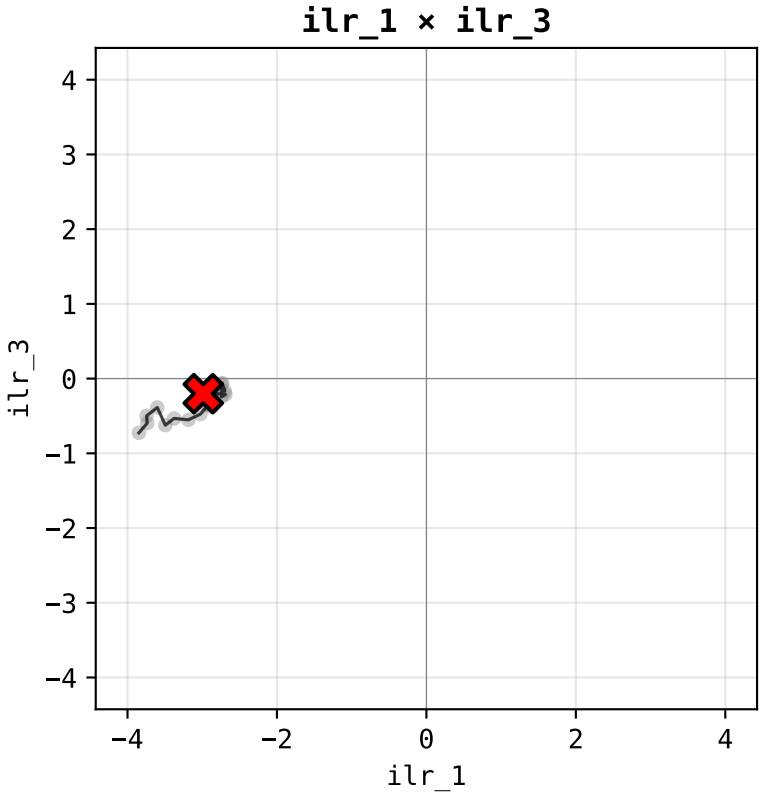
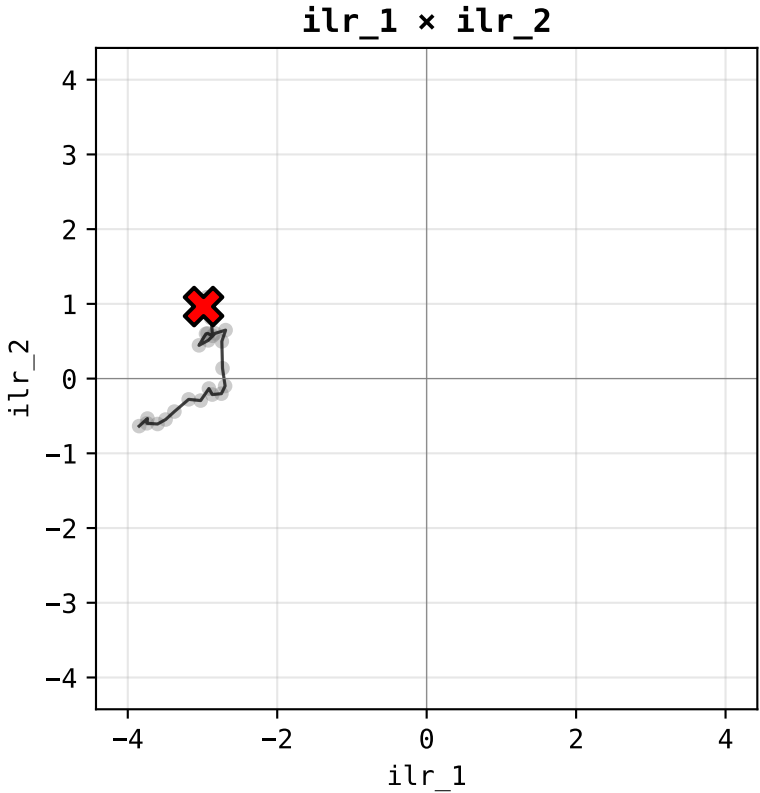


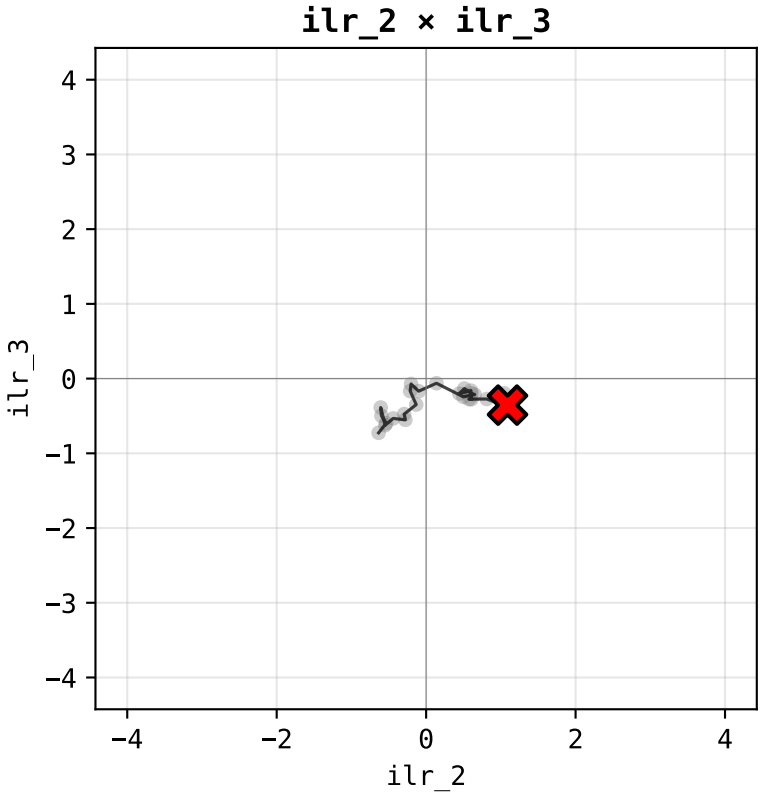
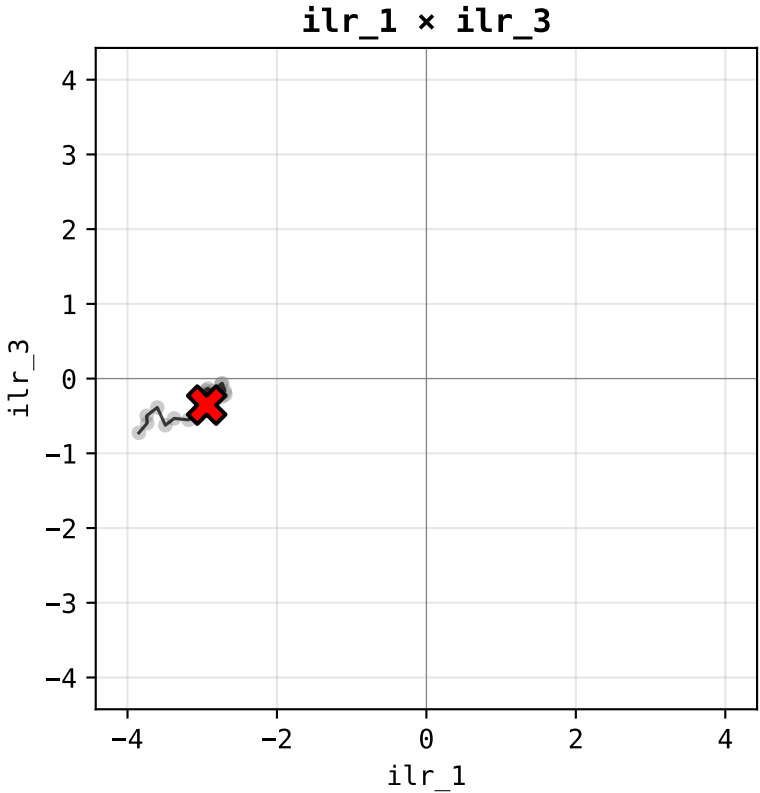
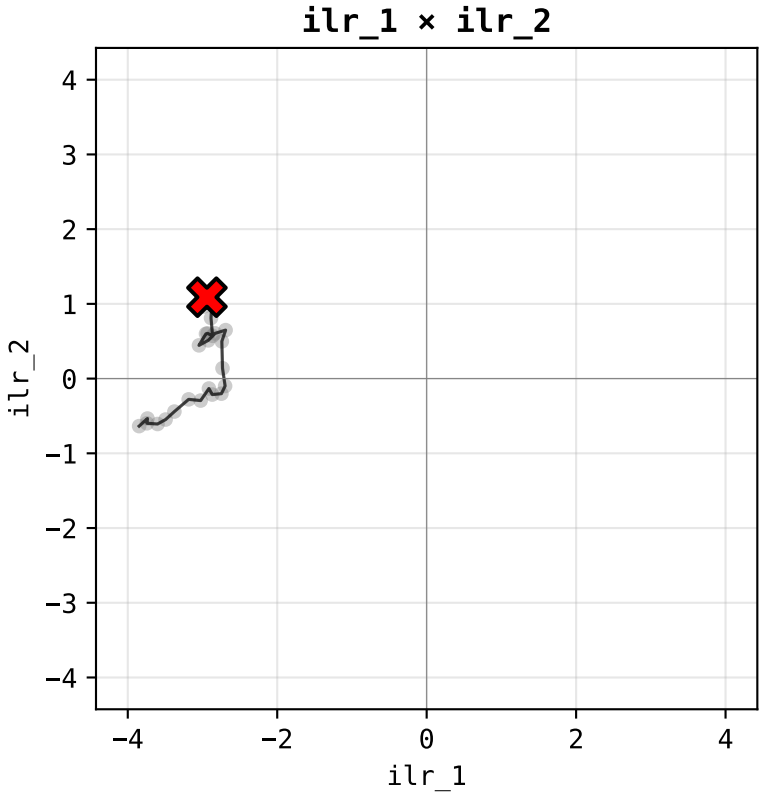












View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ JPN Japan

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

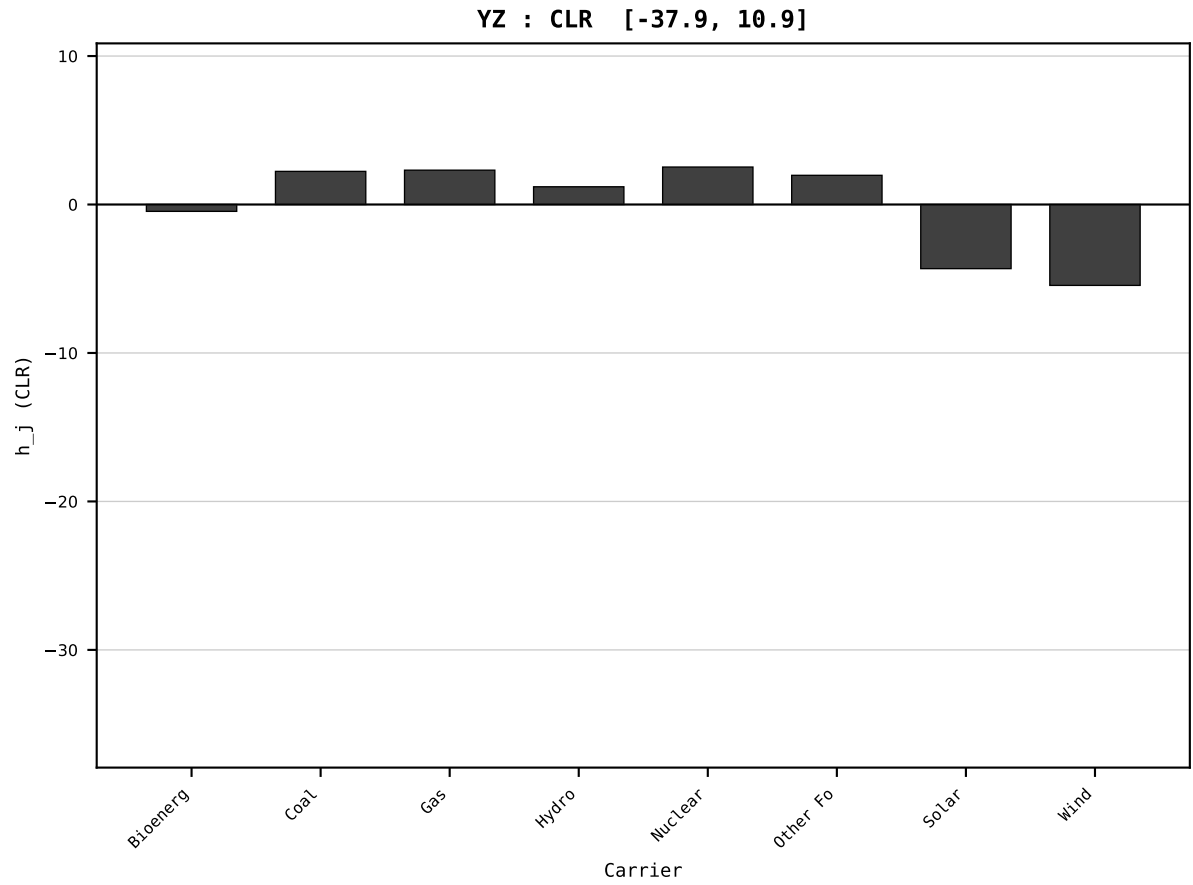
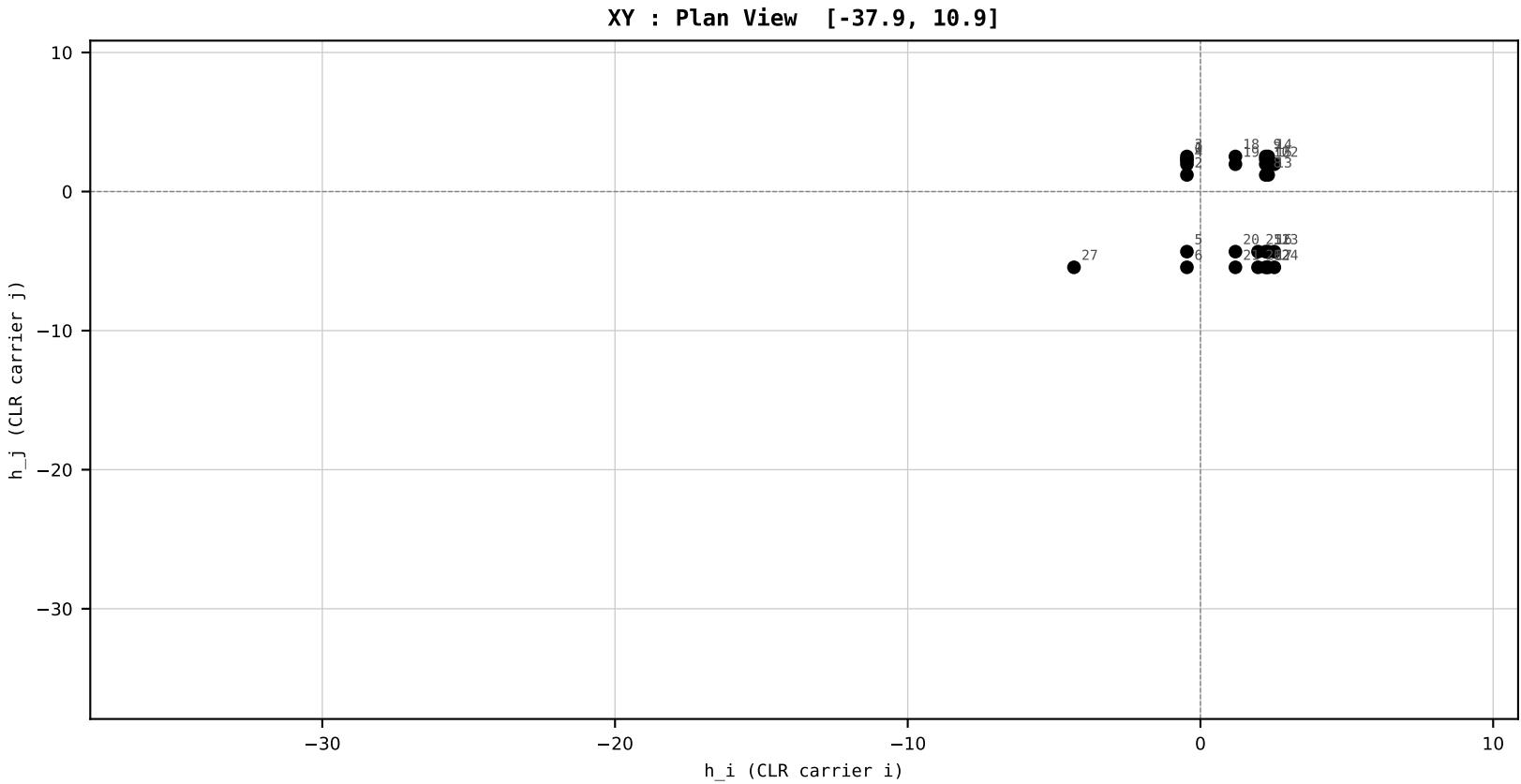
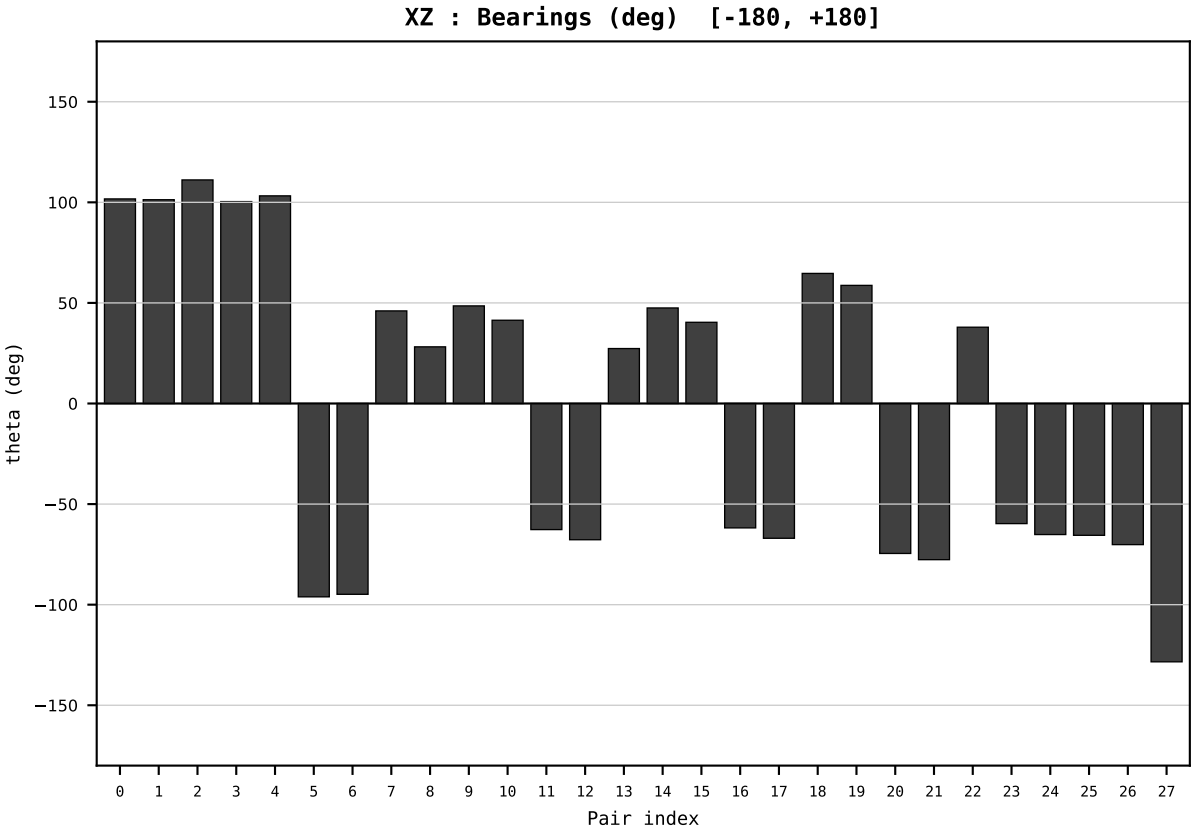
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.234683
Ring = Hs-2
E_metric= 70.5673
kappa_HS= 2901.09
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 7.973 HLR
DR ratio= 2901.1x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

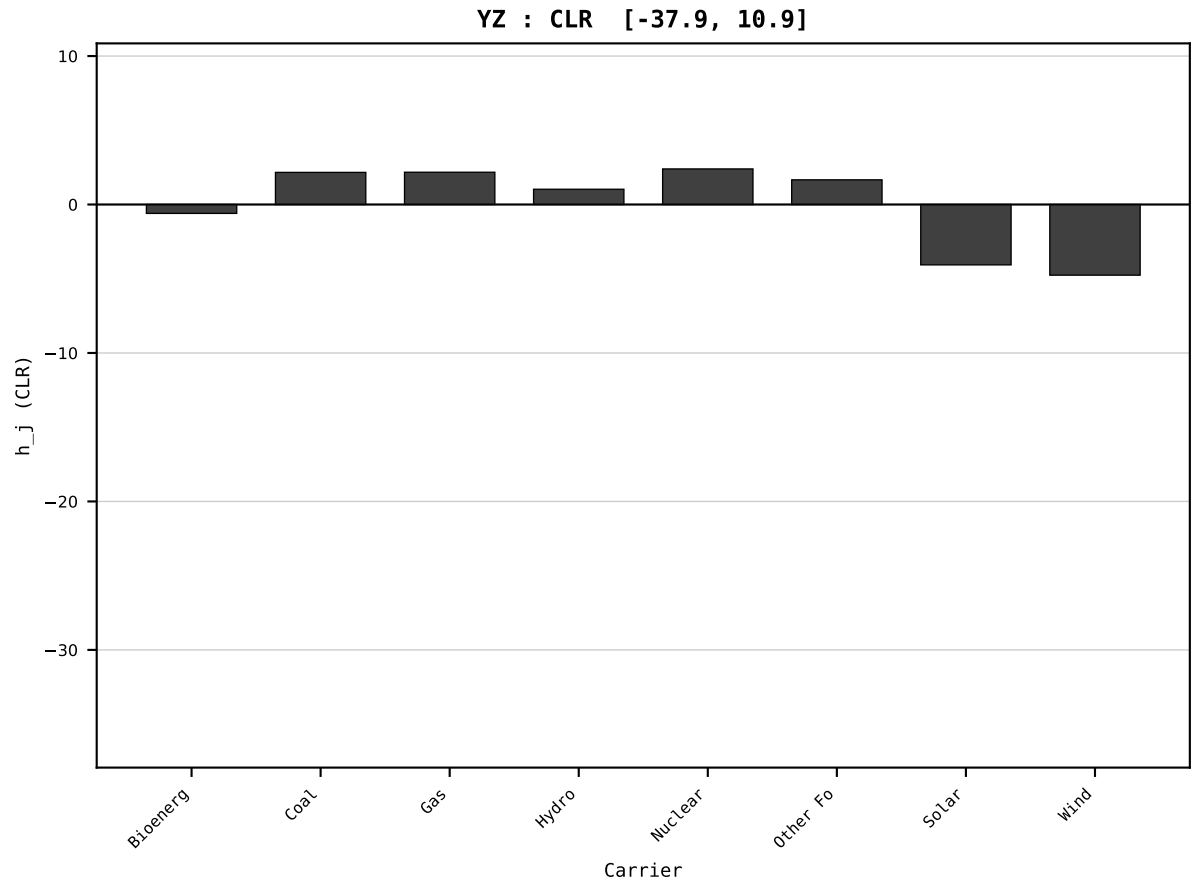
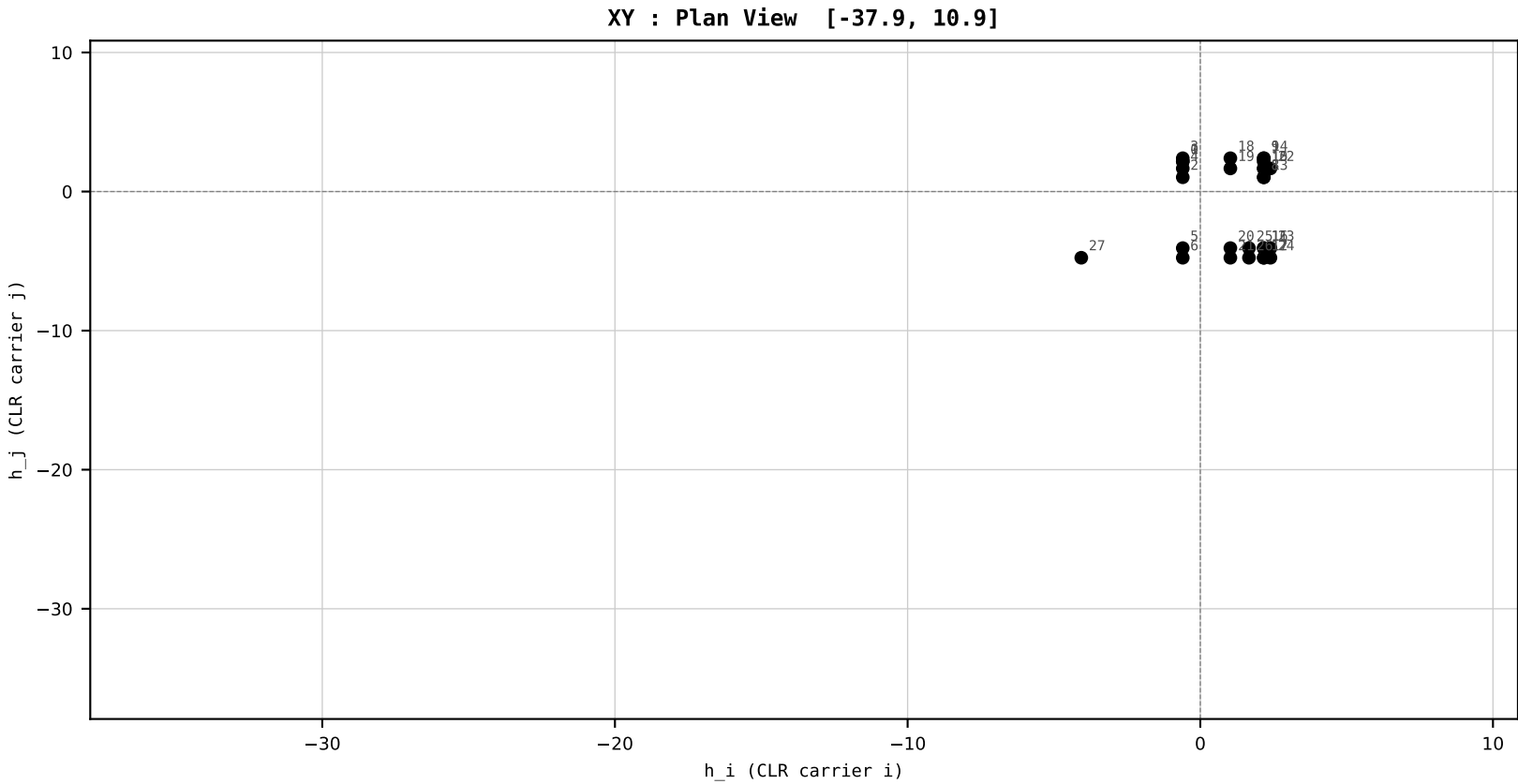
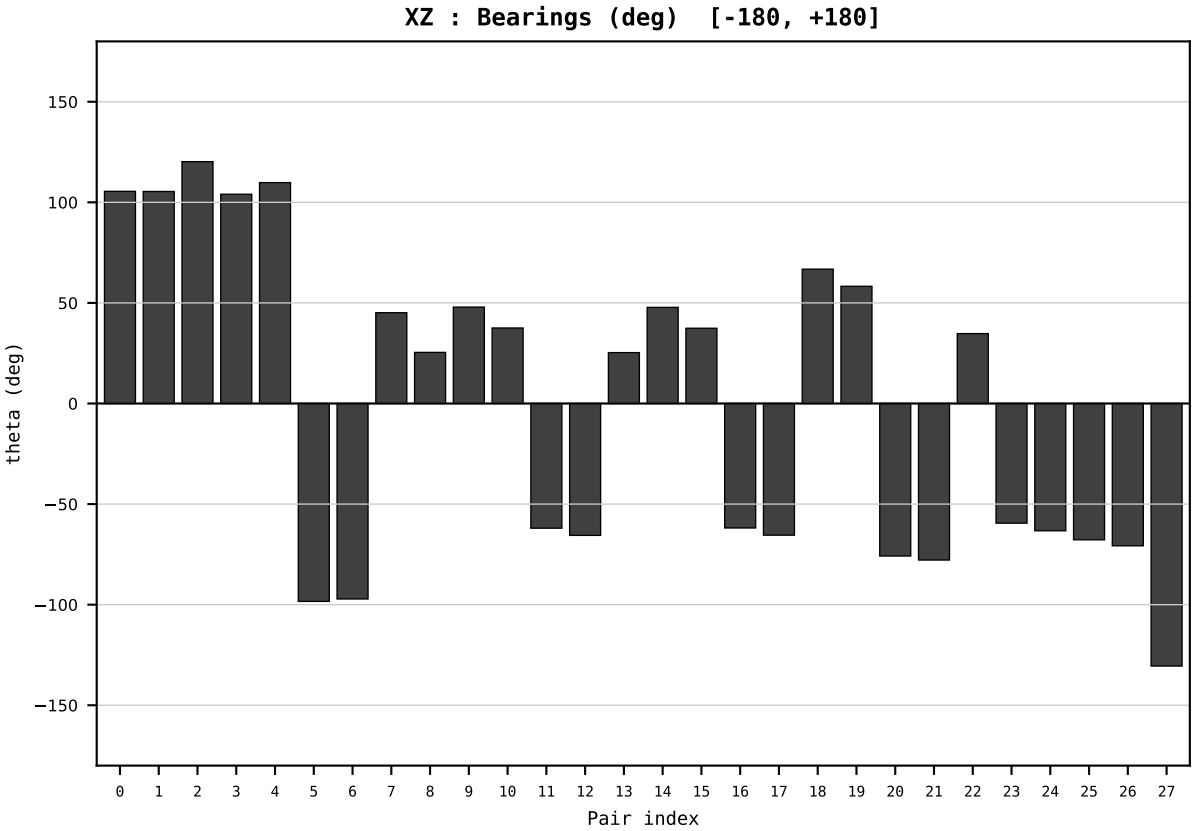
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.239561
Ring = Hs-2
E_metric= 58.5208
kappa_HS= 1282.16
omega = 2.8189 deg
d_A = 0.847846
Helm = Wind
Helm d = +0.689167
DR = 7.156 HLR
DR ratio= 1282.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

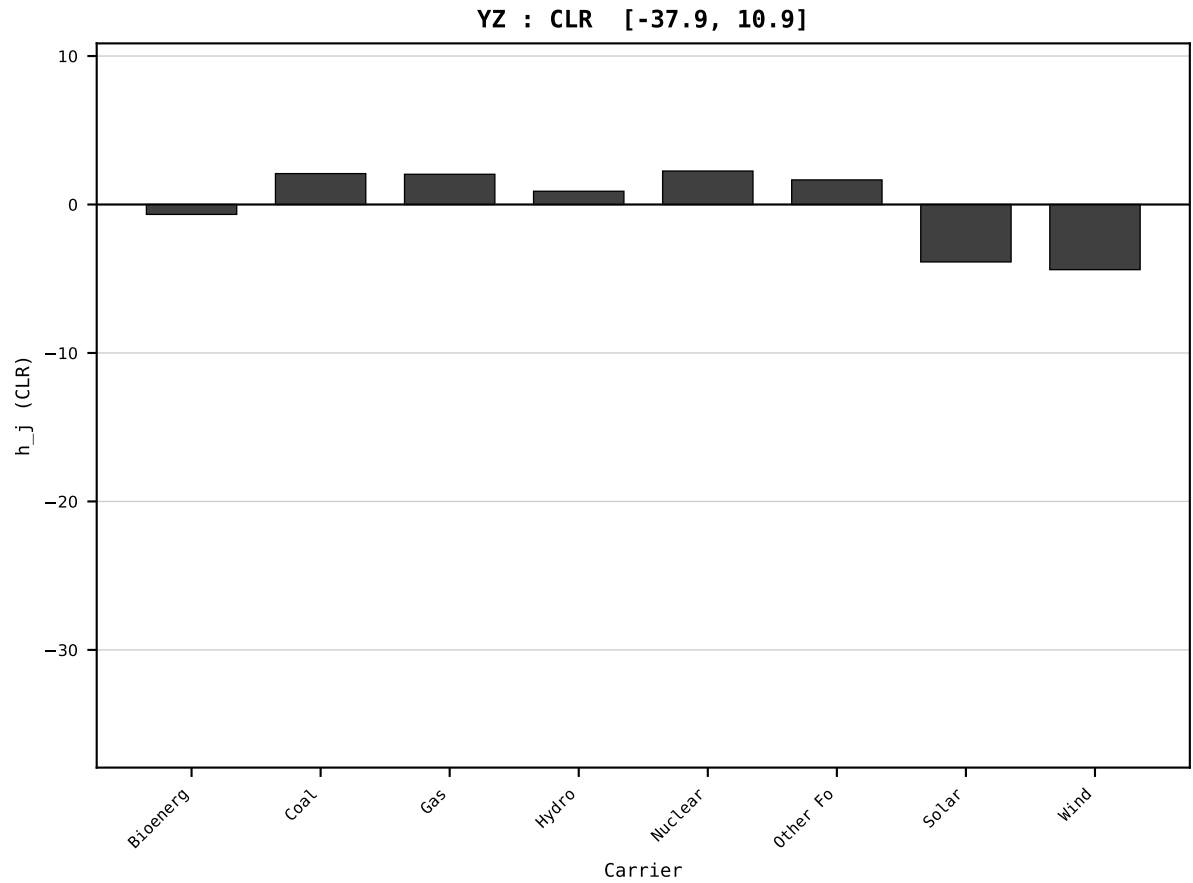
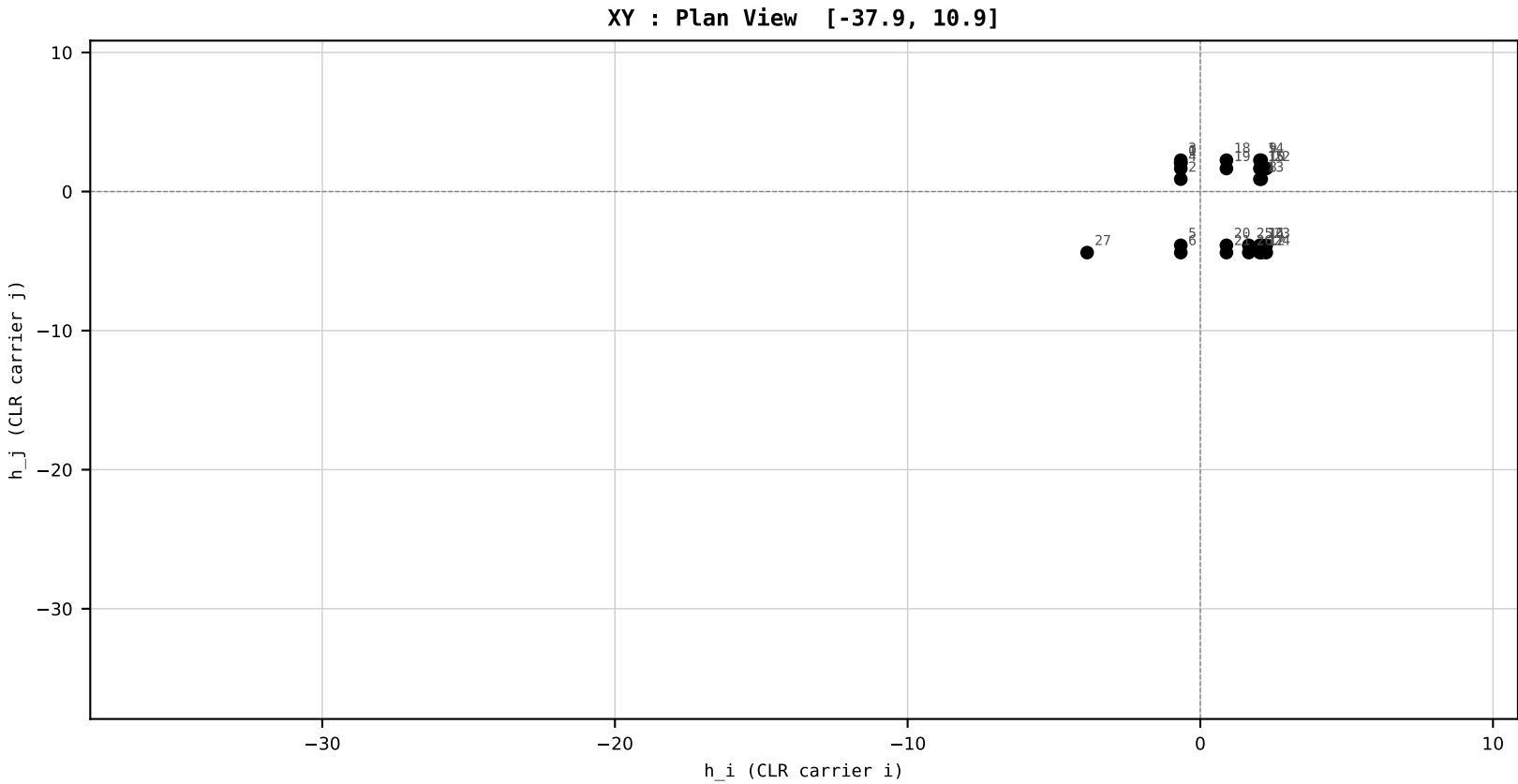
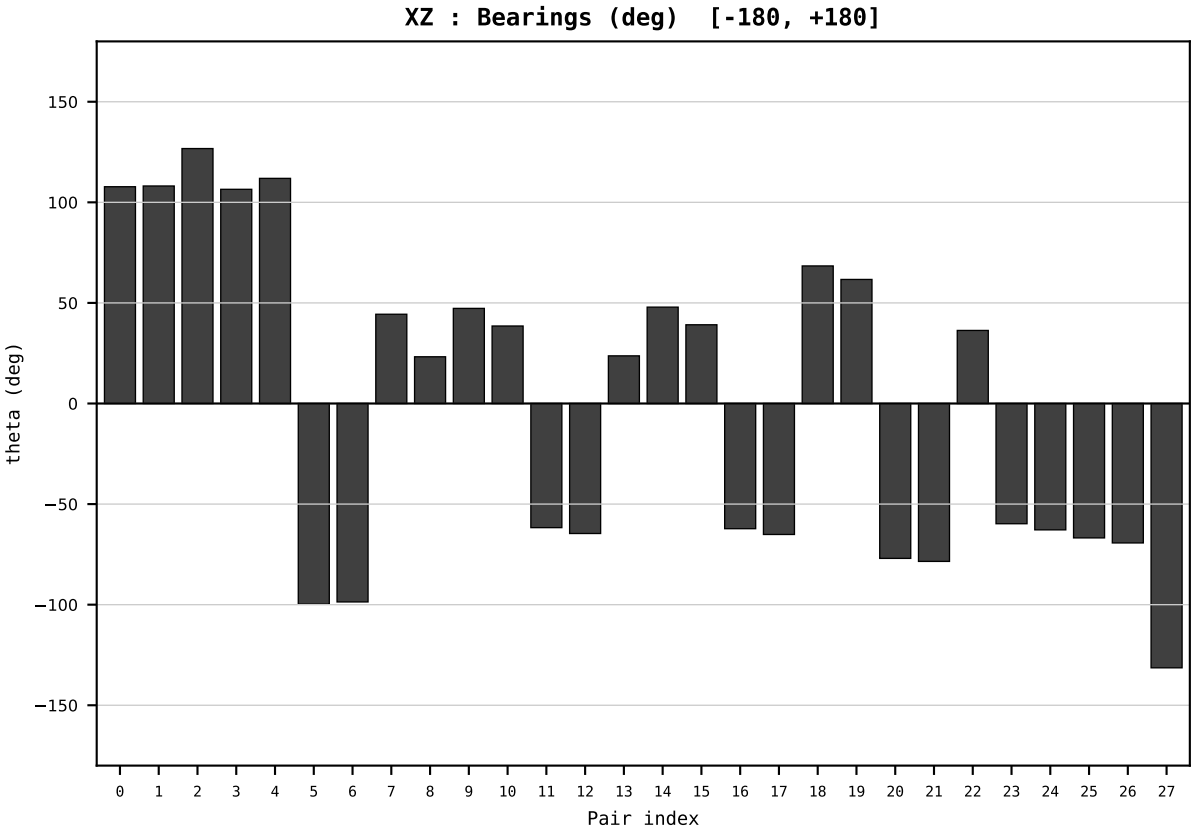
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.234830
Ring = Hs-2
E_metric= 51.7554
kappa_HS= 766.49
omega = 1.5248 deg
d_A = 0.496689
Helm = Wind
Helm d = +0.371738
DR = 6.642 HLR
DR ratio= 766.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

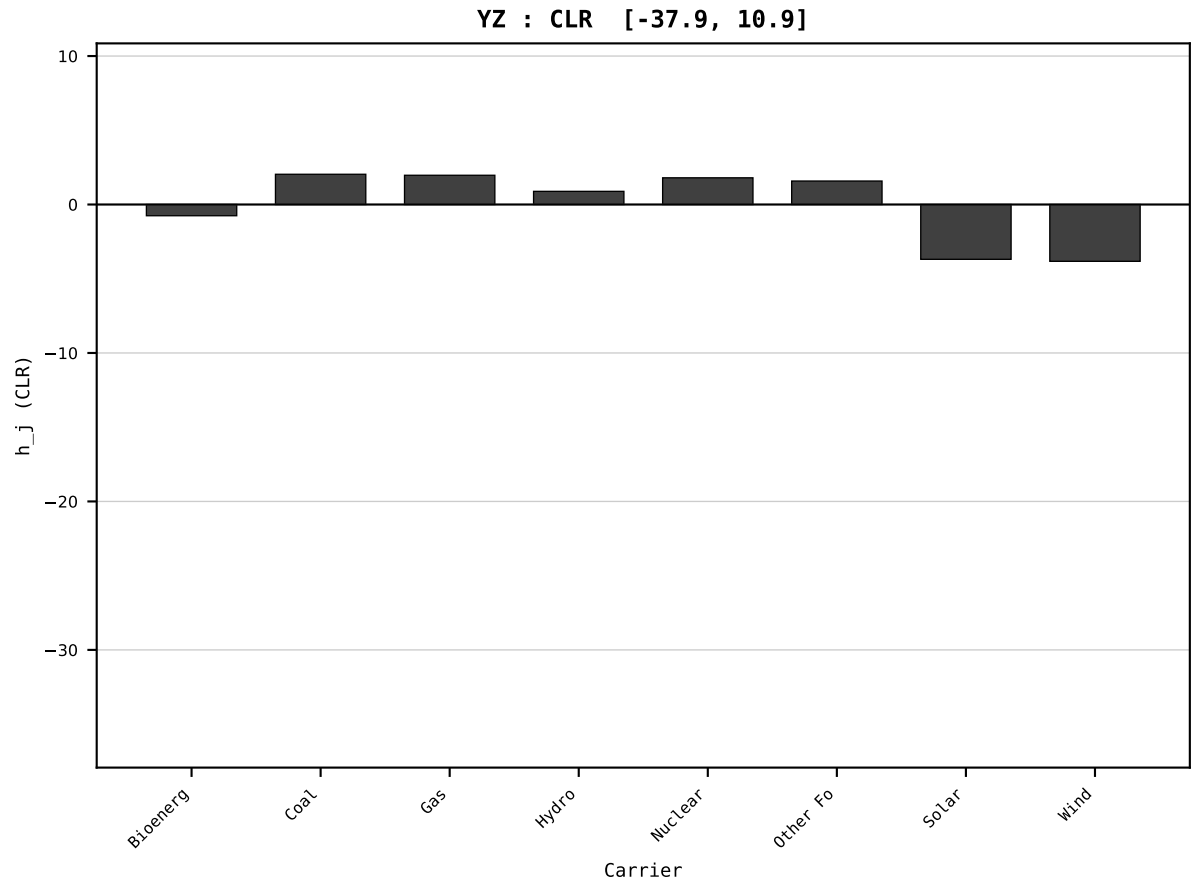
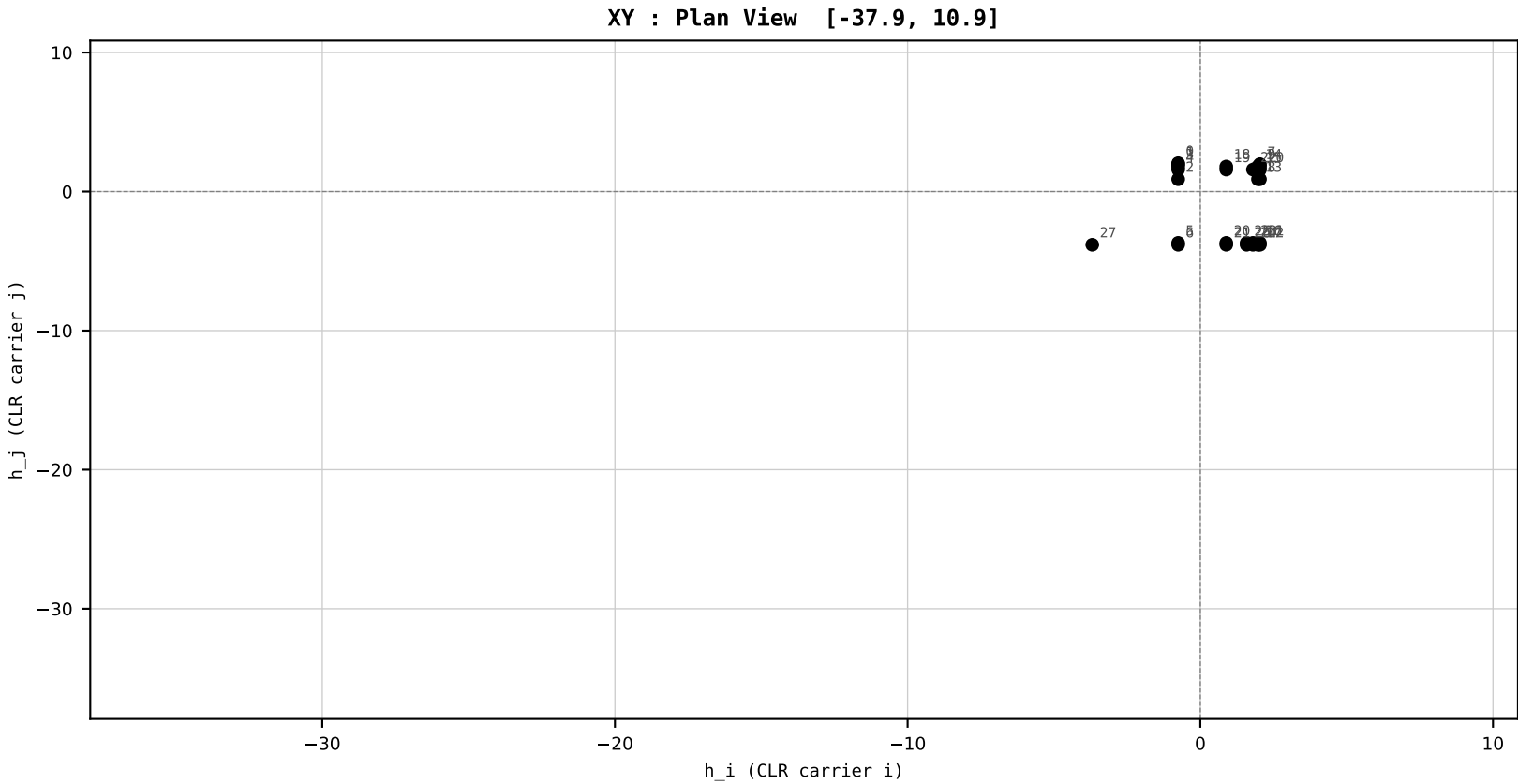
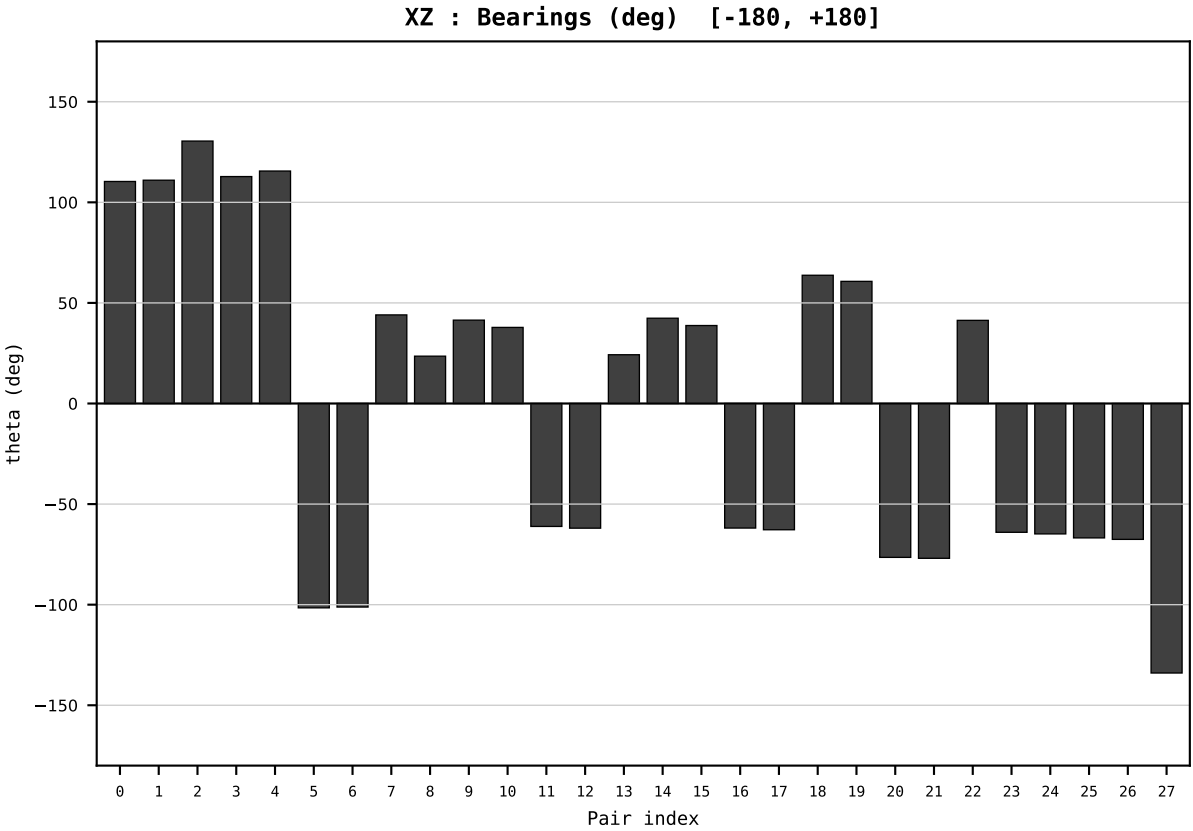
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.222849
Ring = Hs-2
E_metric= 43.3901
kappa_HS= 351.88
omega = 3.7738 deg
d_A = 0.757601
Helm = Wind
Helm d = +0.561548
DR = 5.863 HLR
DR ratio= 351.9x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

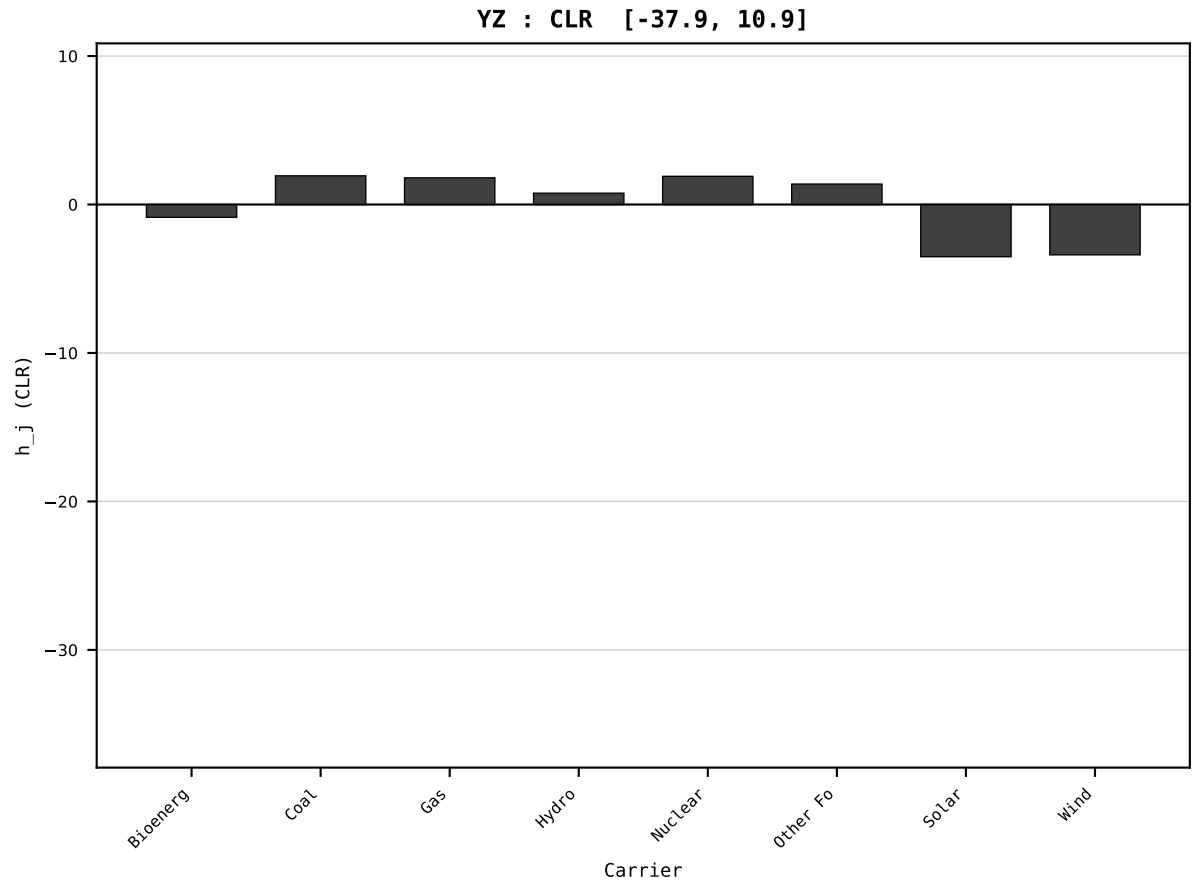
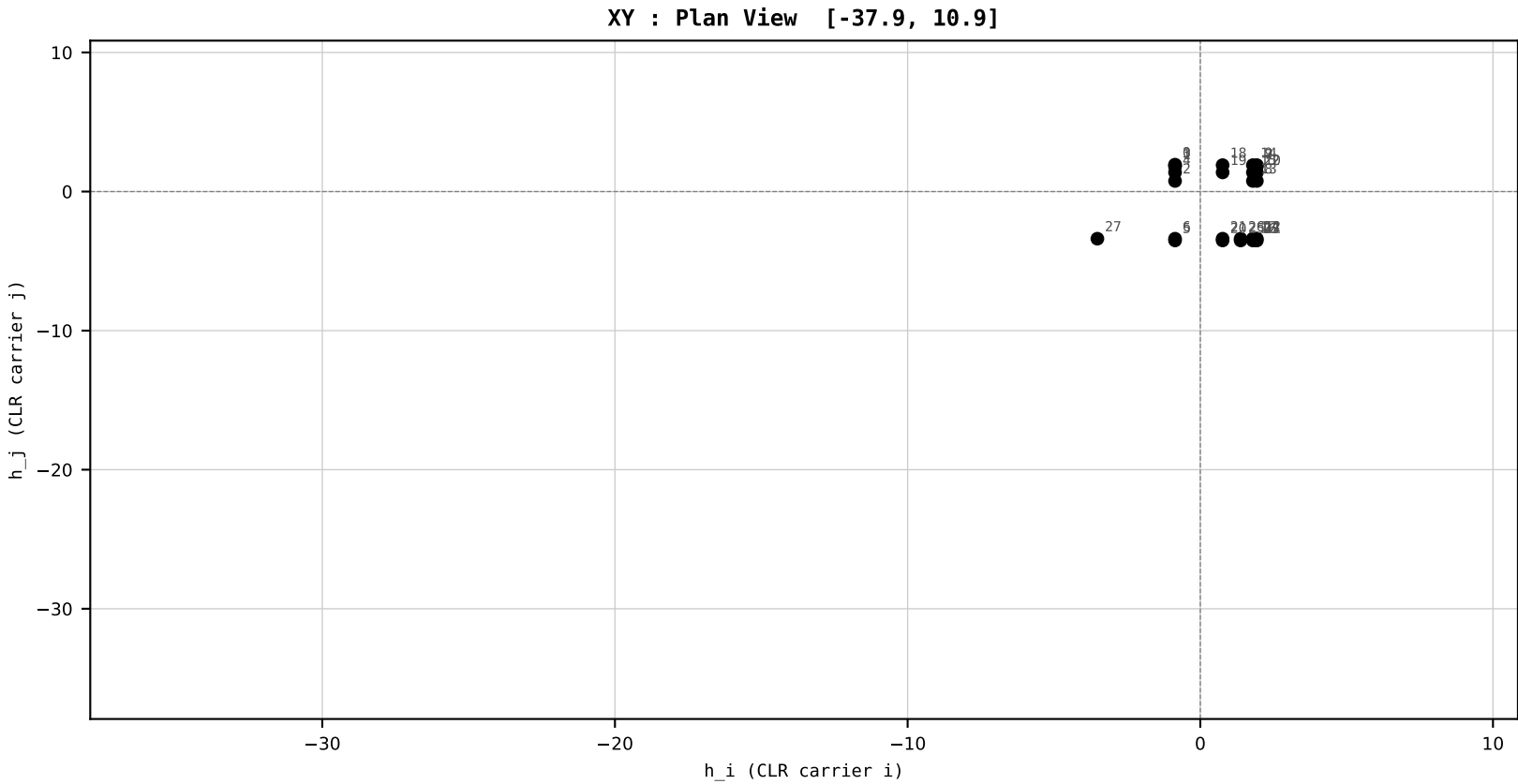
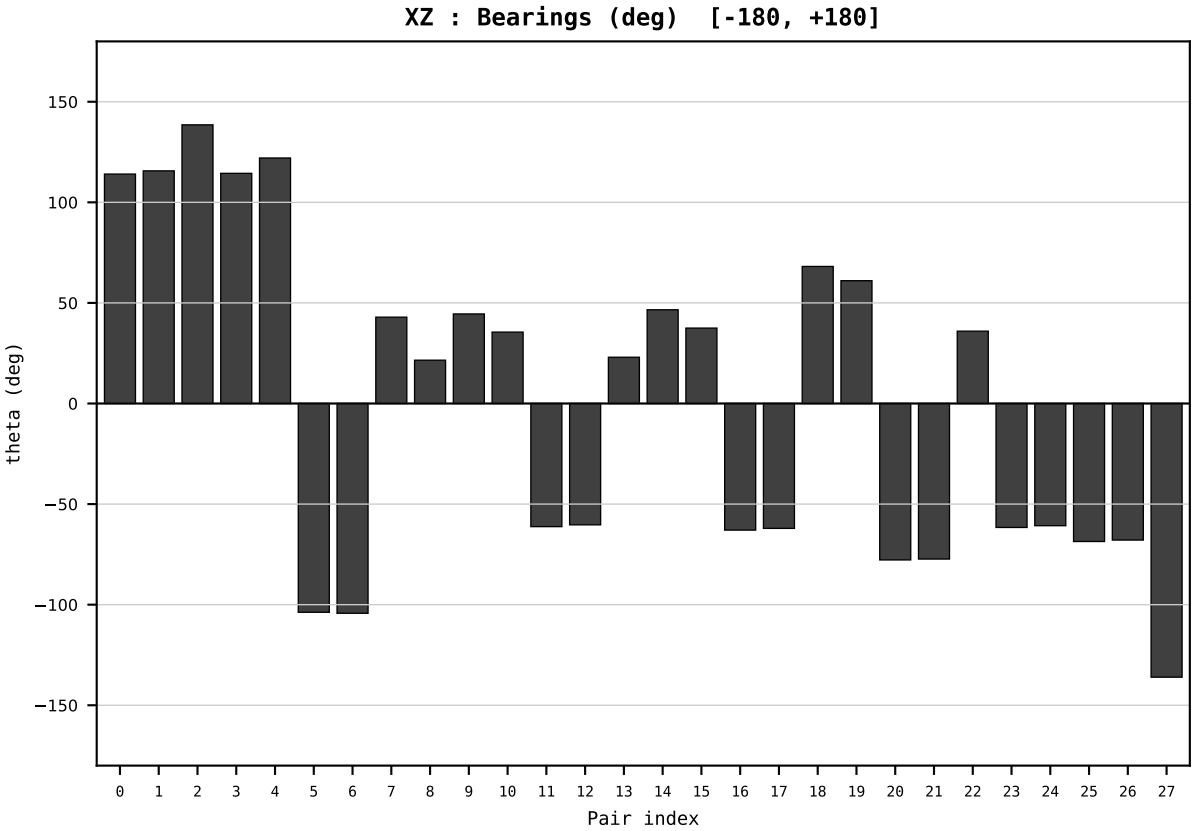
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.225250
Ring = Hs-2
E_metric= 37.6631
kappa_HS= 232.59
omega = 3.3324 deg
d_A = 0.582491
Helm = Wind
Helm d = +0.435321
DR = 5.449 HLR
DR ratio= 232.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

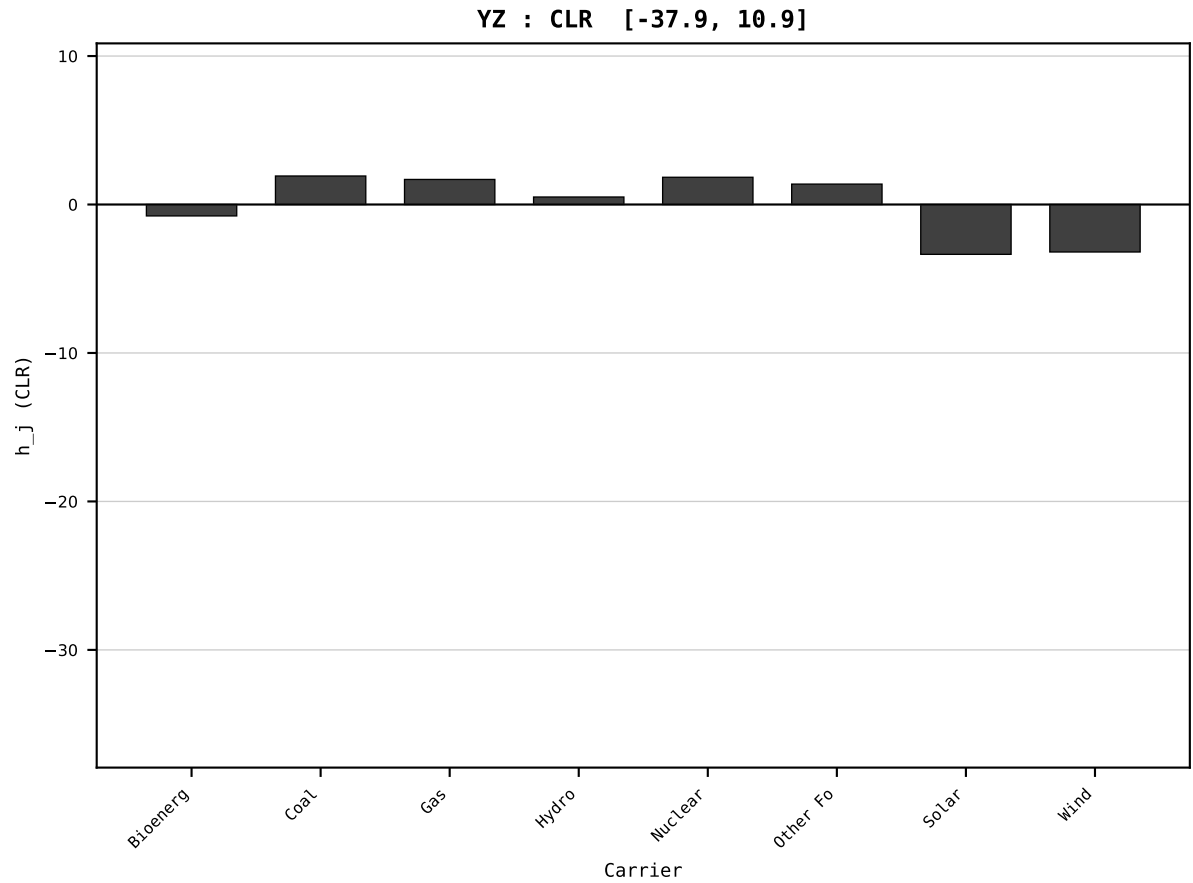
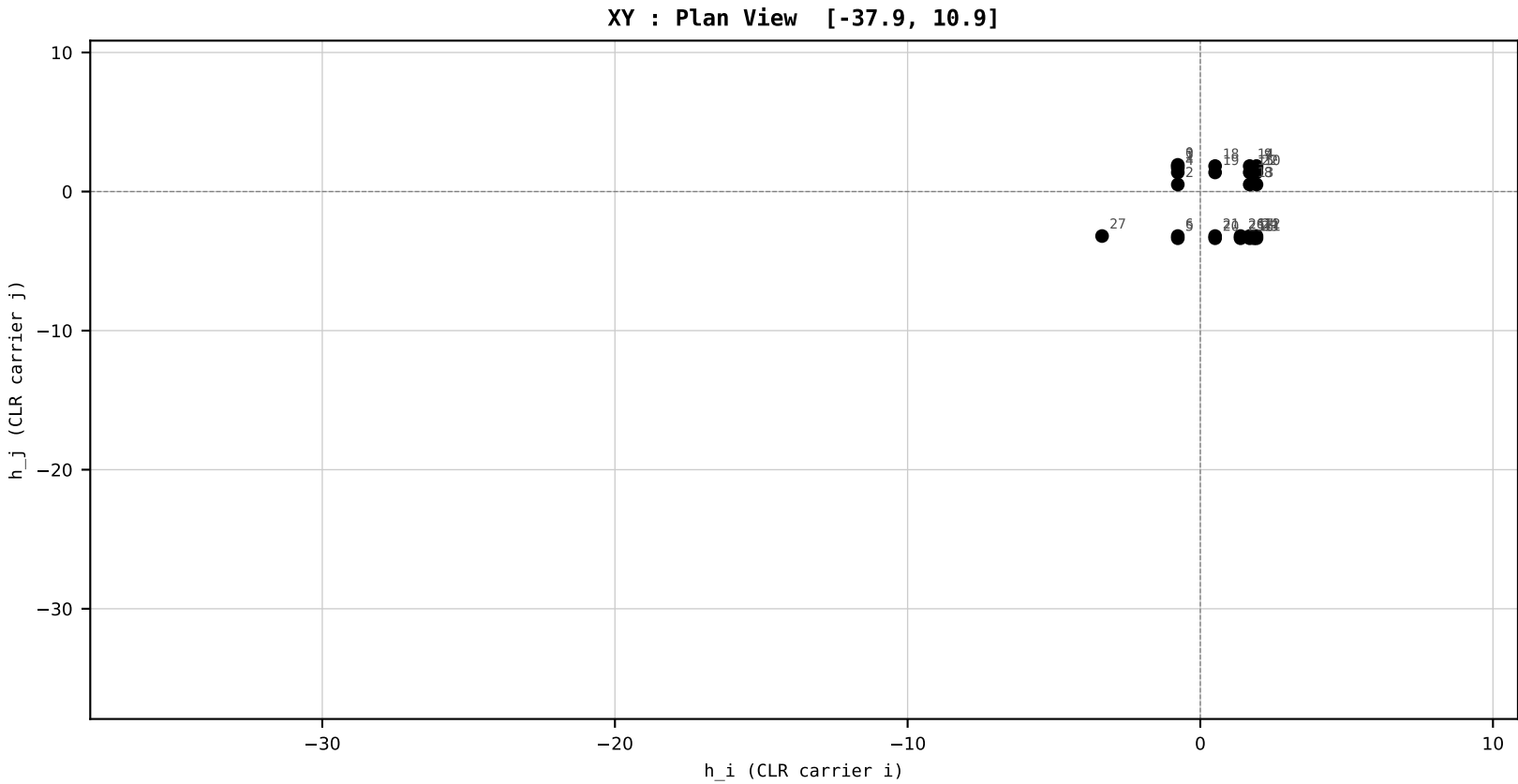
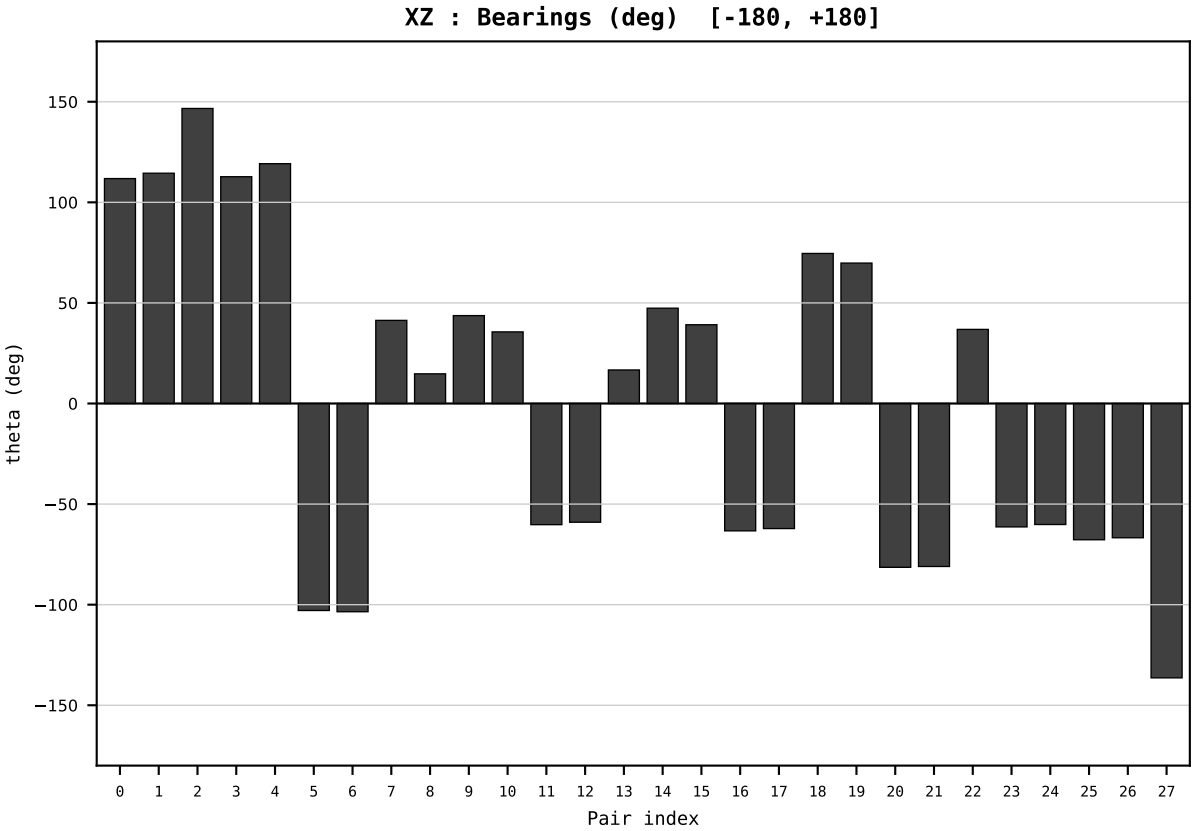
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227556
Ring = Hs-2
E_metric= 34.1389
kappa_HS= 195.83
omega = 2.4787 deg
d_A = 0.391966
Helm = Hydro
Helm d = -0.256970
DR = 5.277 HLR
DR ratio= 195.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

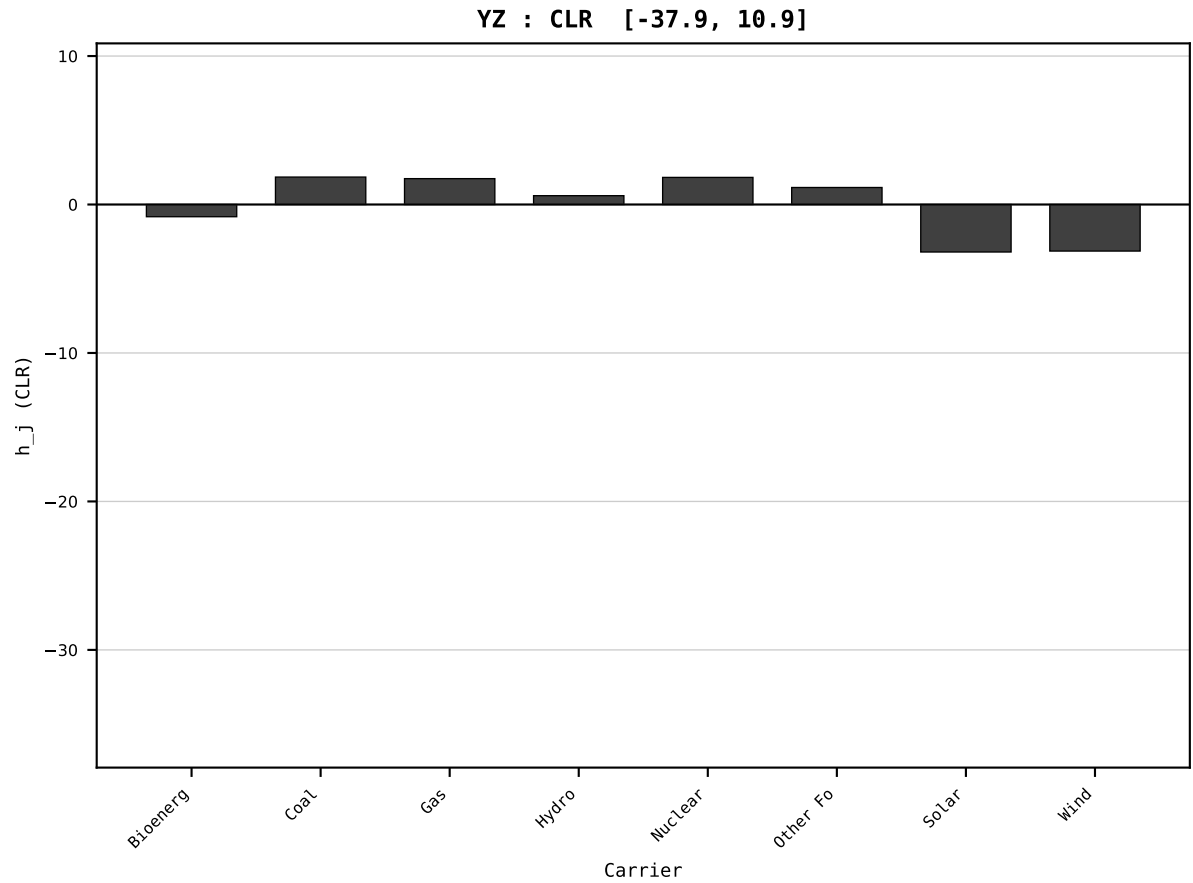
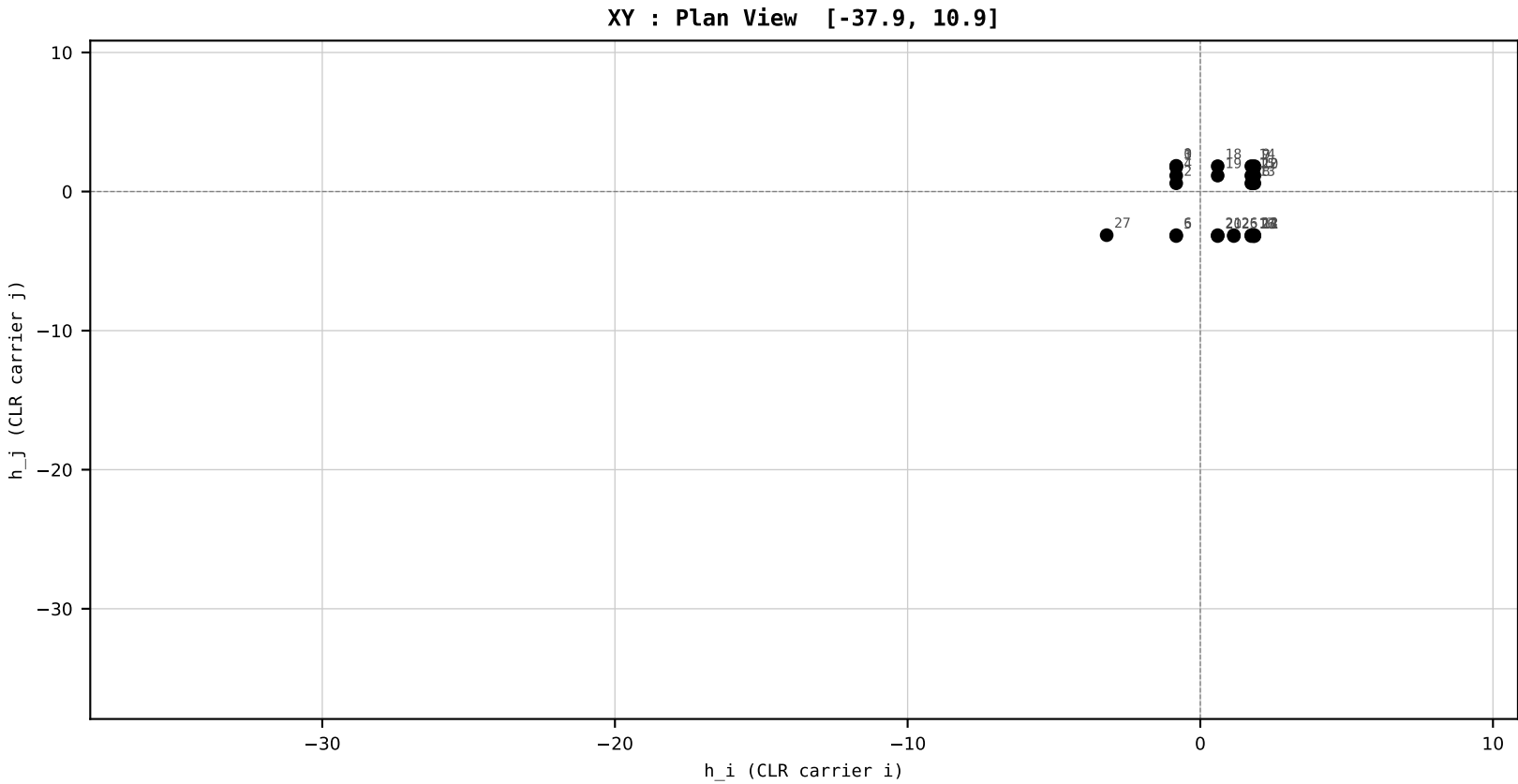
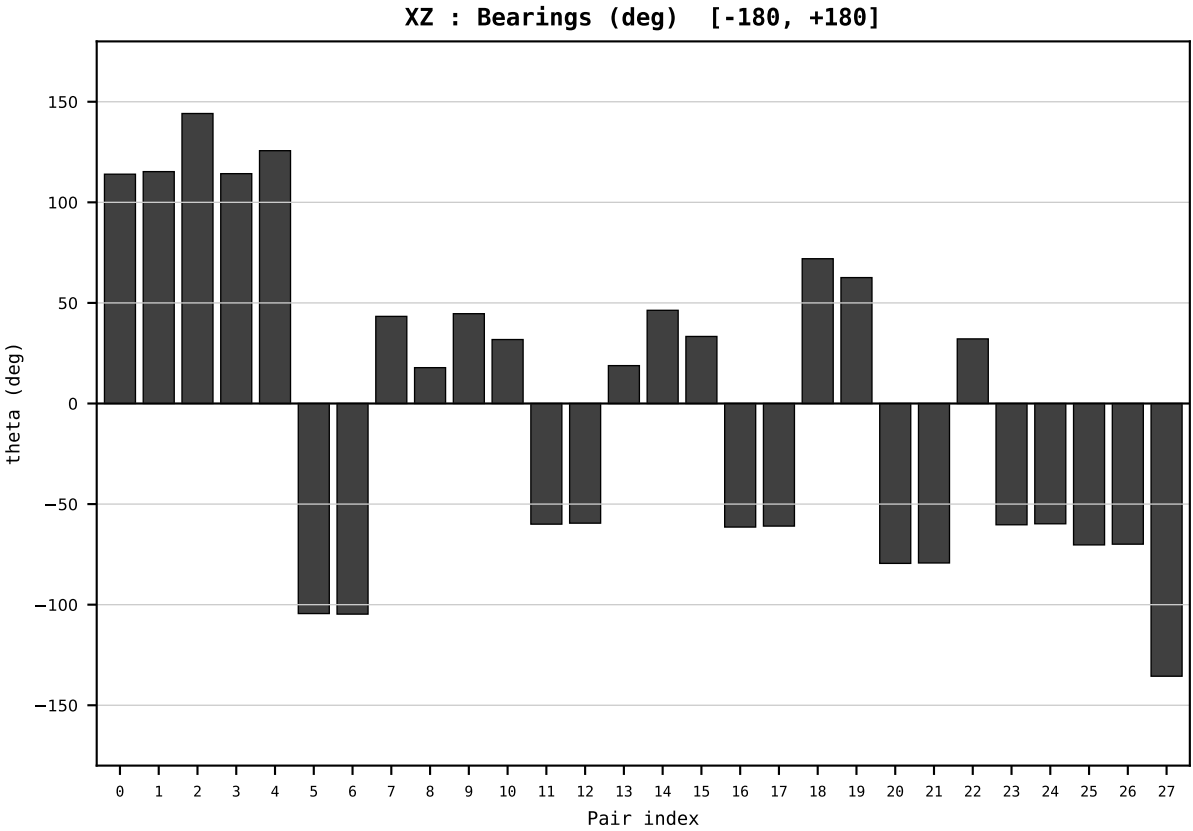
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227477
Ring = Hs-2
E_metric= 32.2009
kappa_HS= 155.67
omega = 2.6642 deg
d_A = 0.316211
Helm = Other Fossil
Helm d = -0.228493
DR = 5.048 HLR
DR ratio= 155.7x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

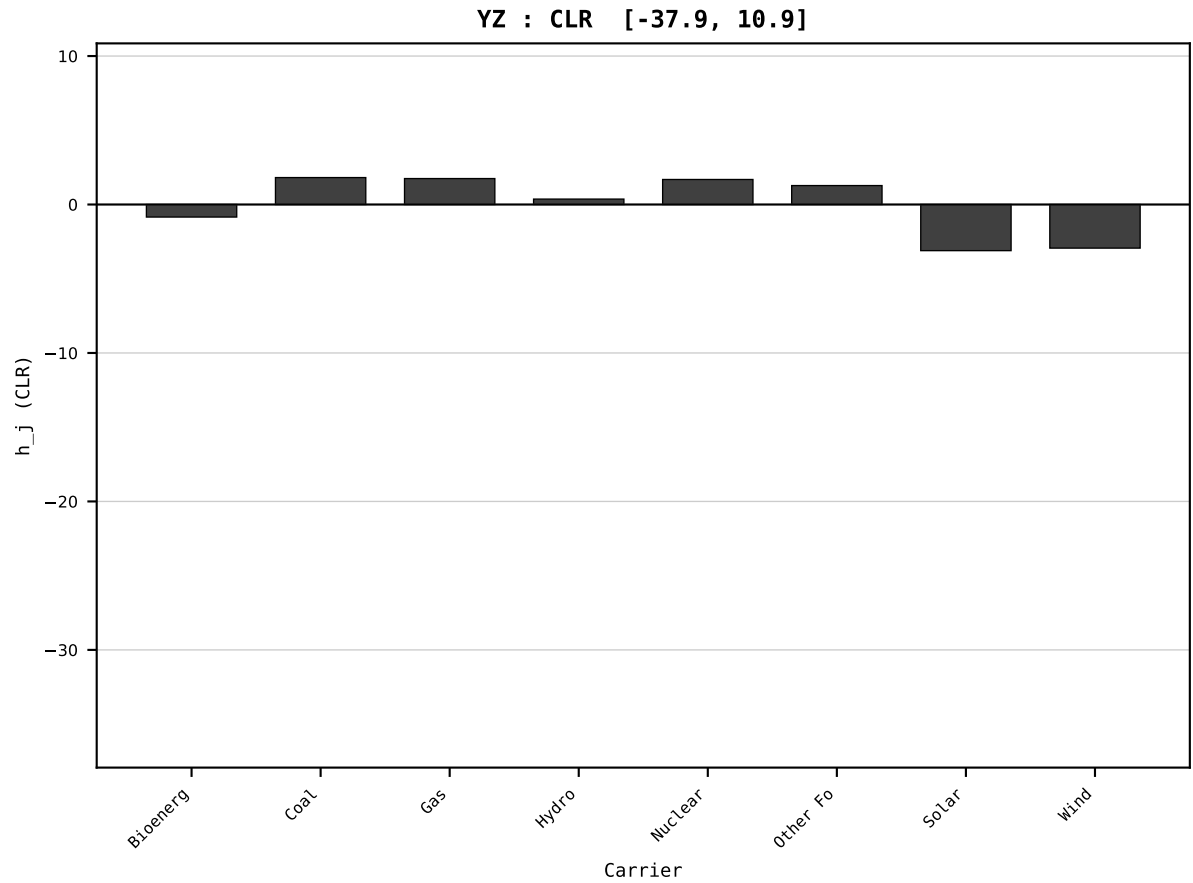
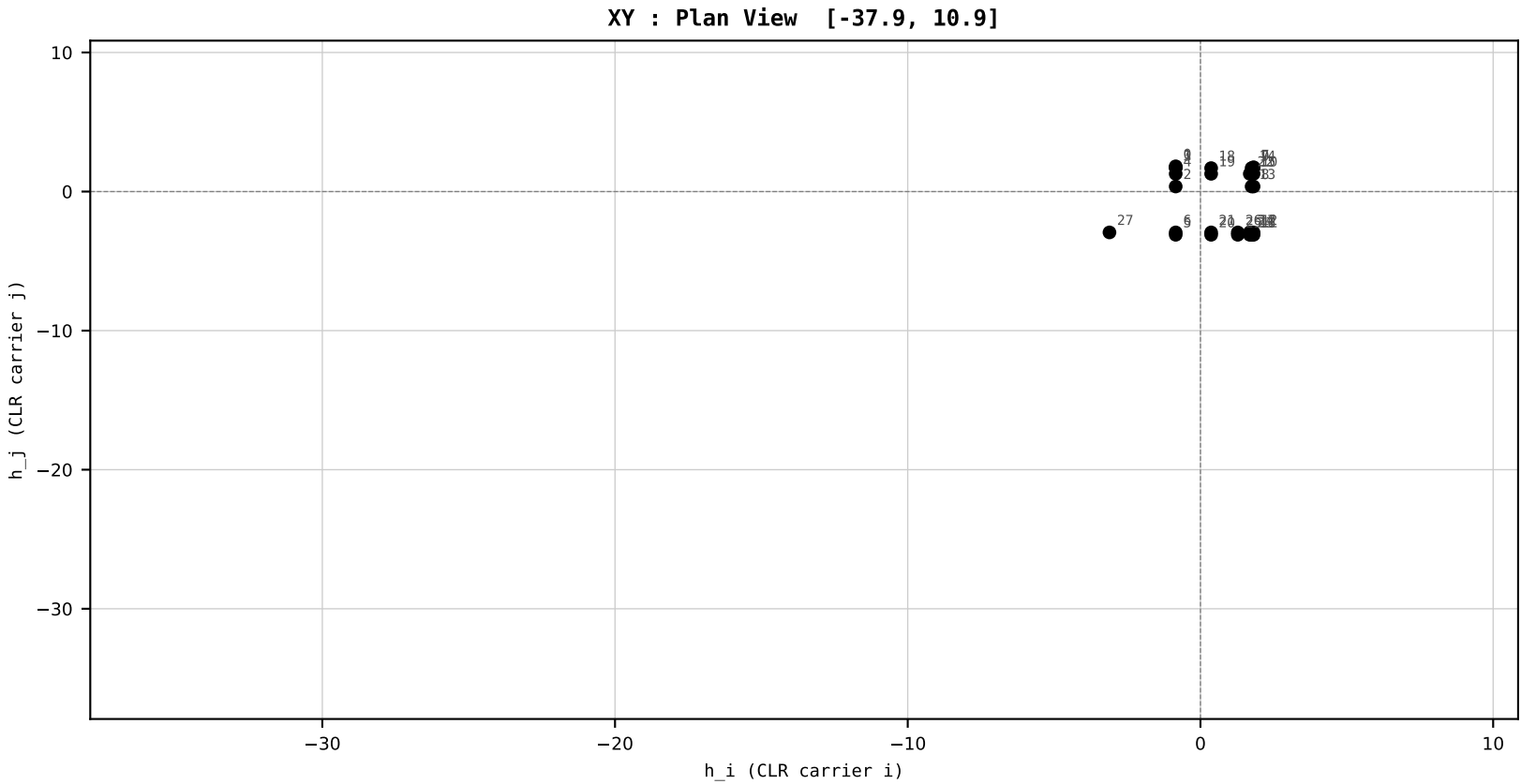
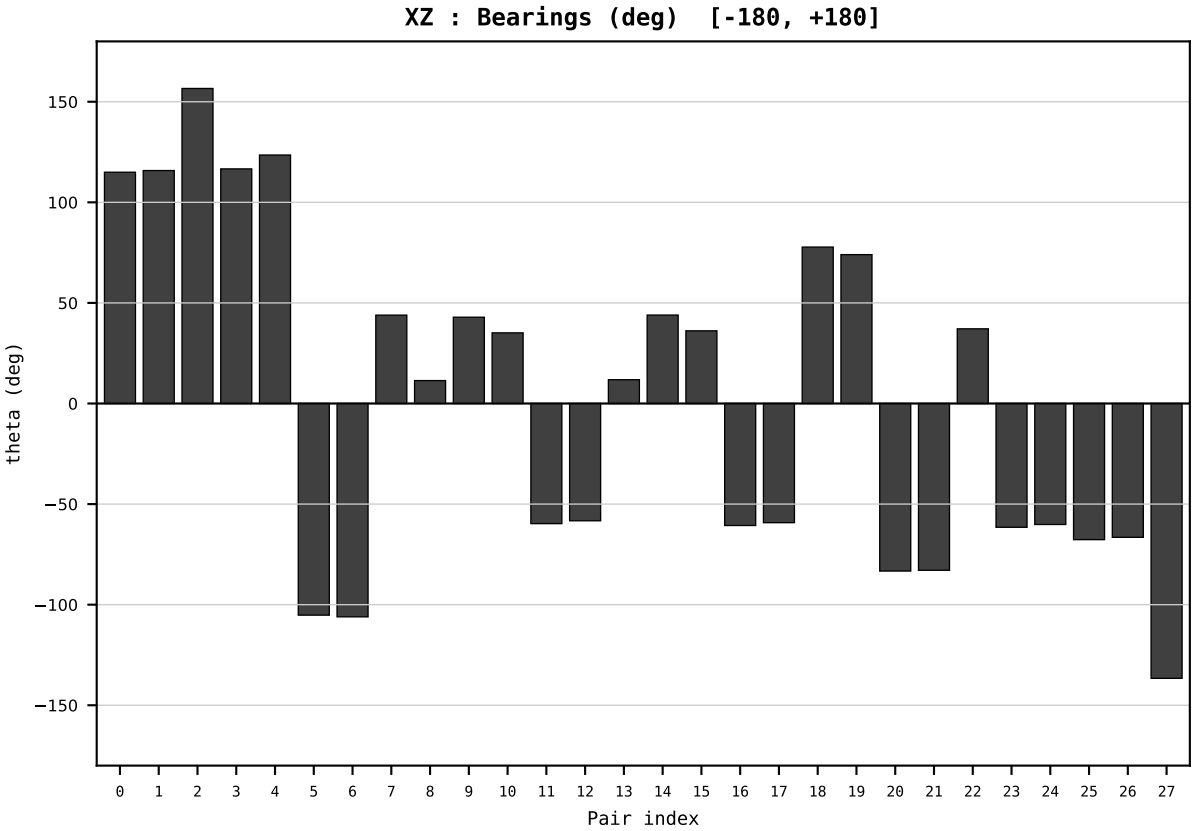
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227024
Ring = Hs-2
E_metric= 29.9546
kappa_HS= 137.36
omega = 3.2122 deg
d_A = 0.371740
Helm = Hydro
Helm d = -0.228639
DR = 4.923 HLR
DR ratio= 137.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



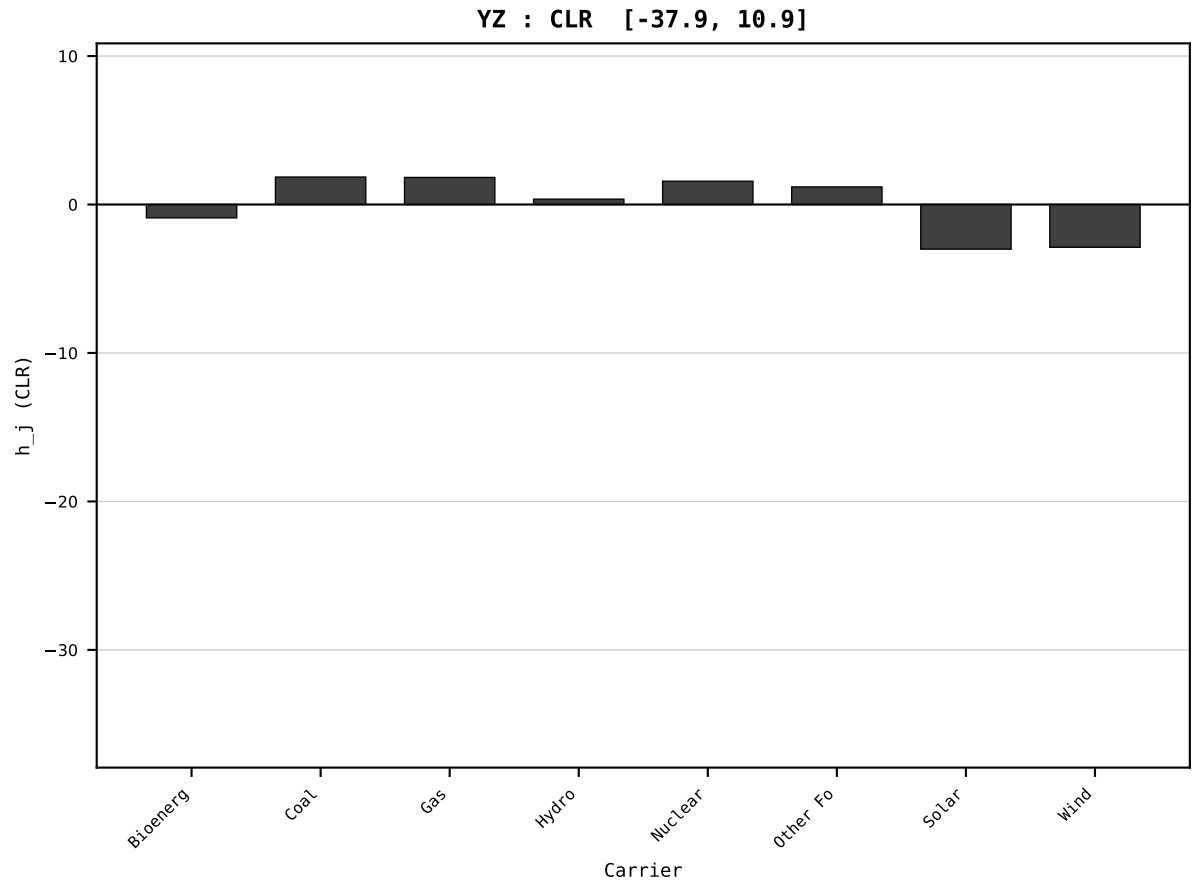
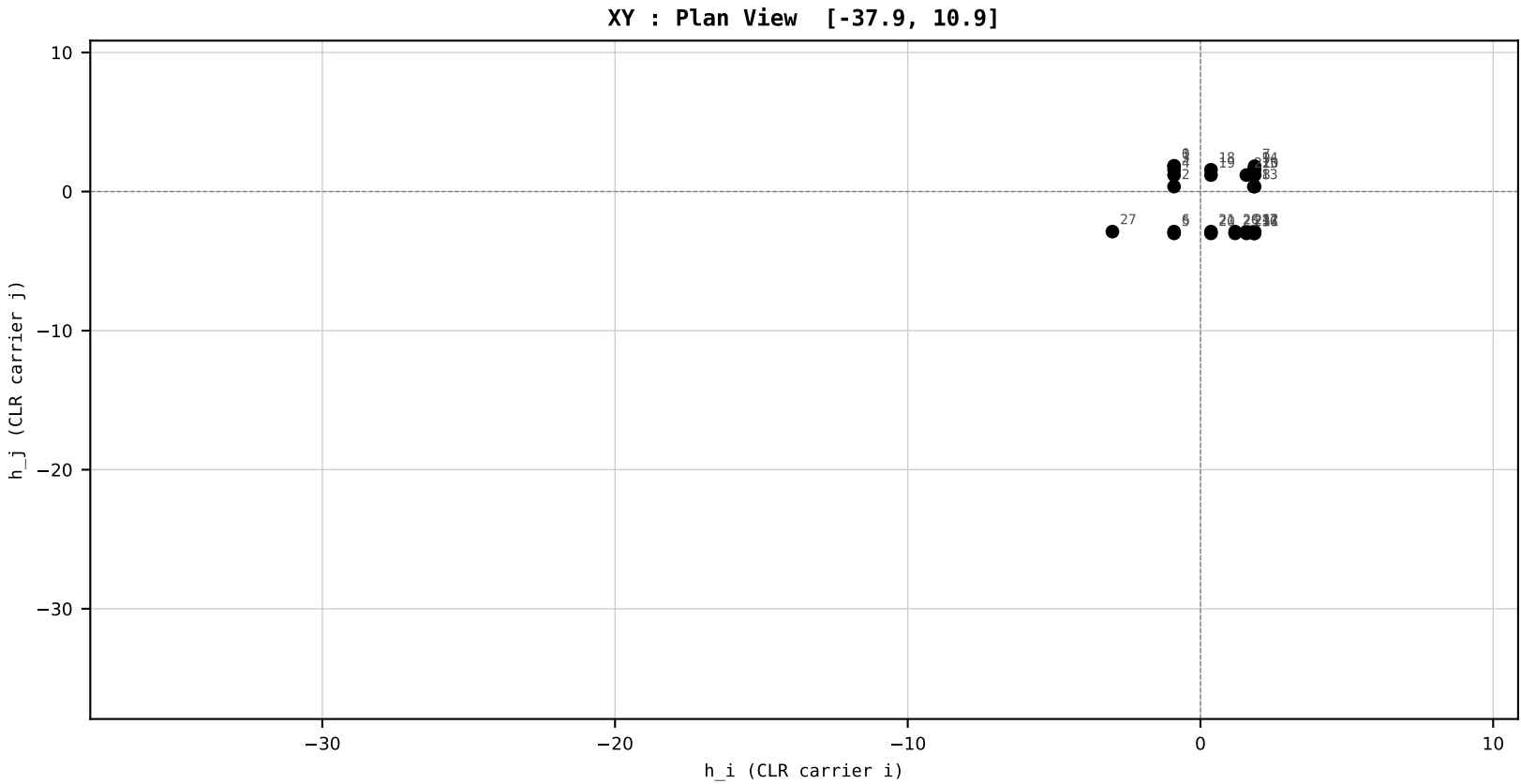
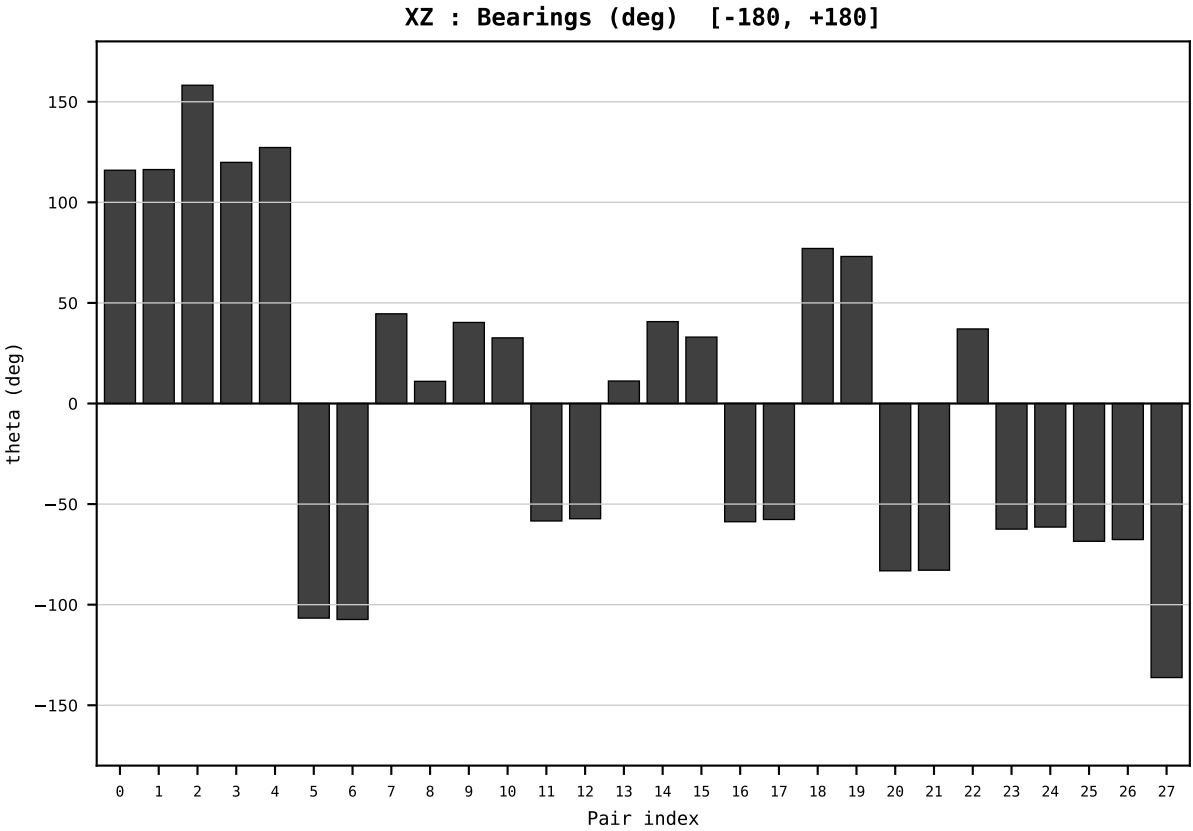
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.232256
Ring = Hs-2
E_metric= 28.9064
kappa_HS= 128.84
omega = 2.0079 deg
d_A = 0.213233
Helm = Nuclear
Helm d = -0.116881
DR = 4.859 HLR
DR ratio= 128.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



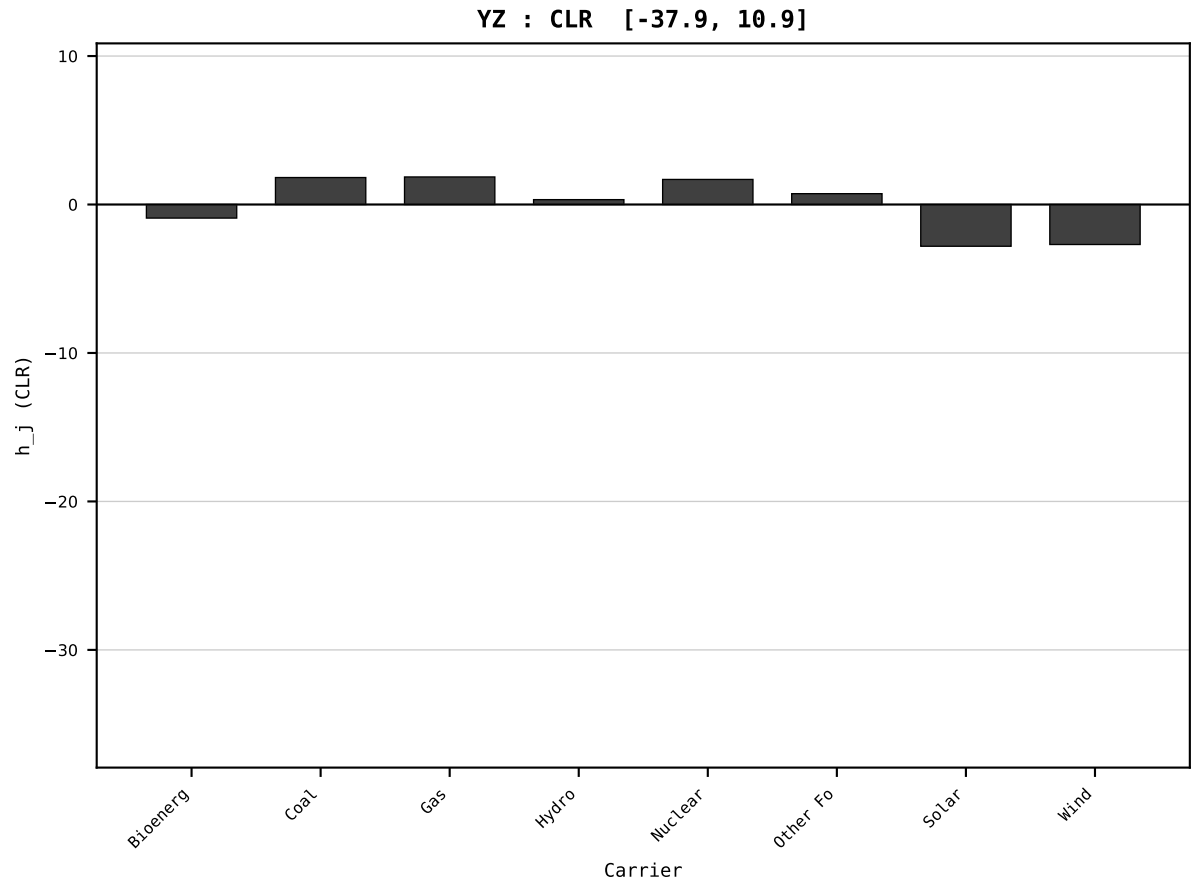
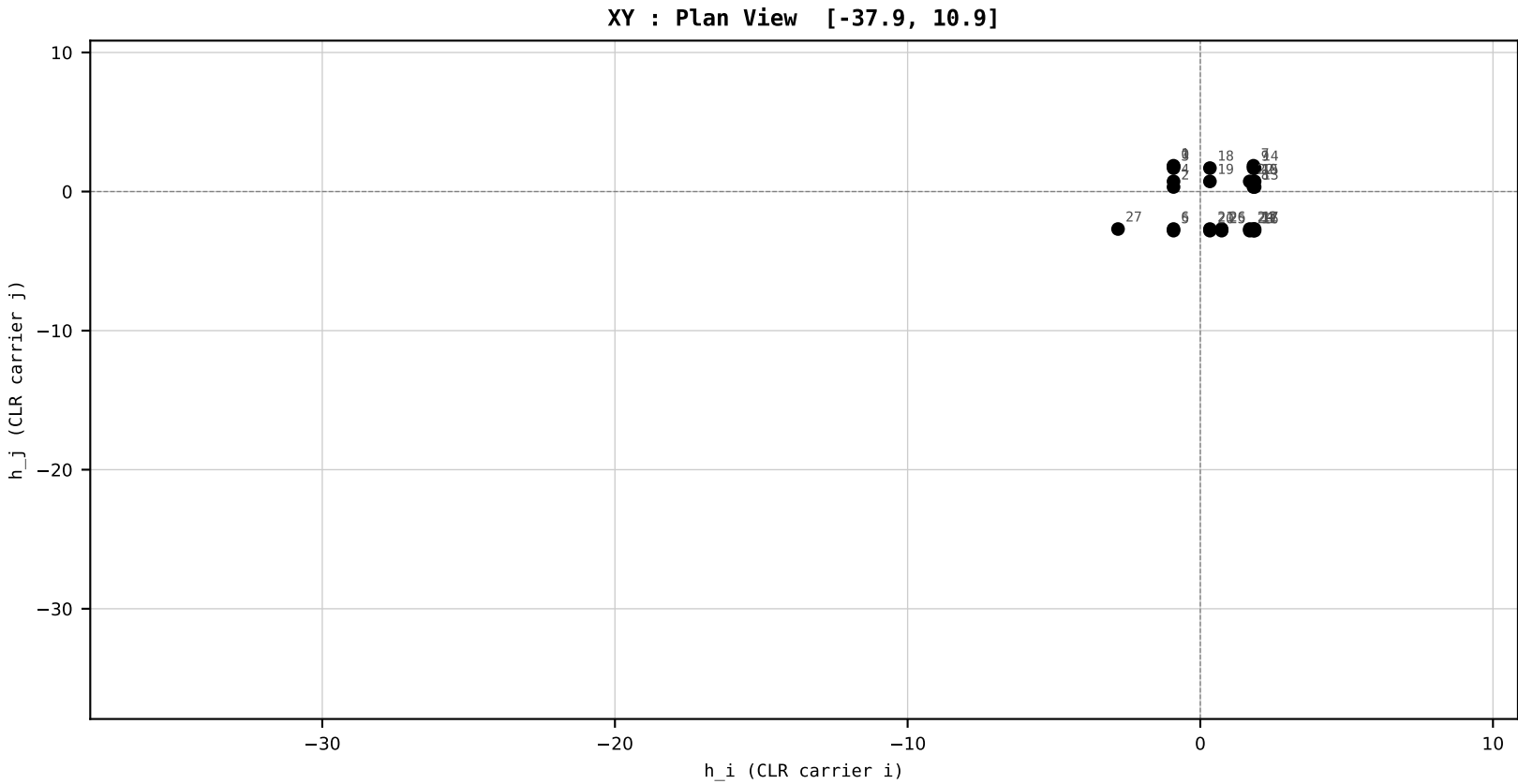
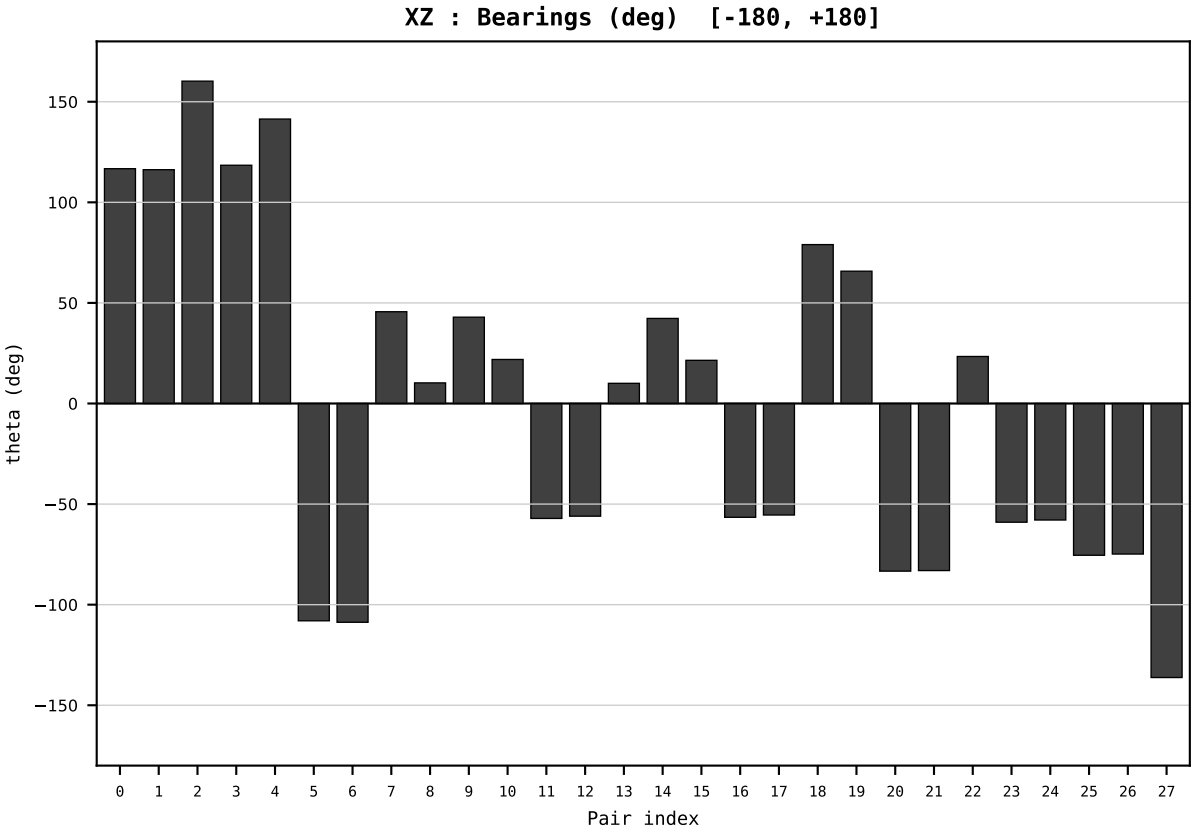
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.247095
Ring = Hs-2
E_metric= 26.2382
kappa_HS= 106.47
omega = 5.2774 deg
d_A = 0.545961
Helm = Other Fossil
Helm d = -0.454982
DR = 4.668 HLR
DR ratio= 106.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

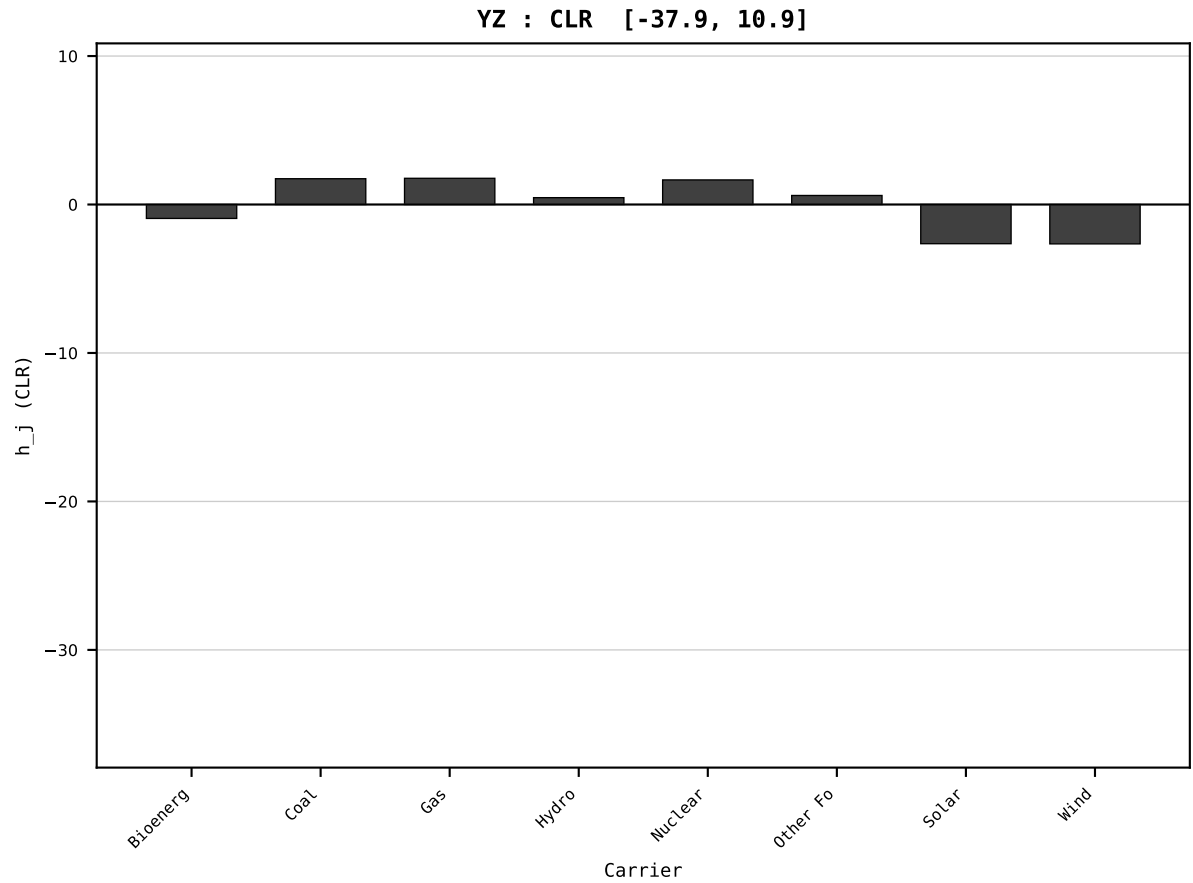
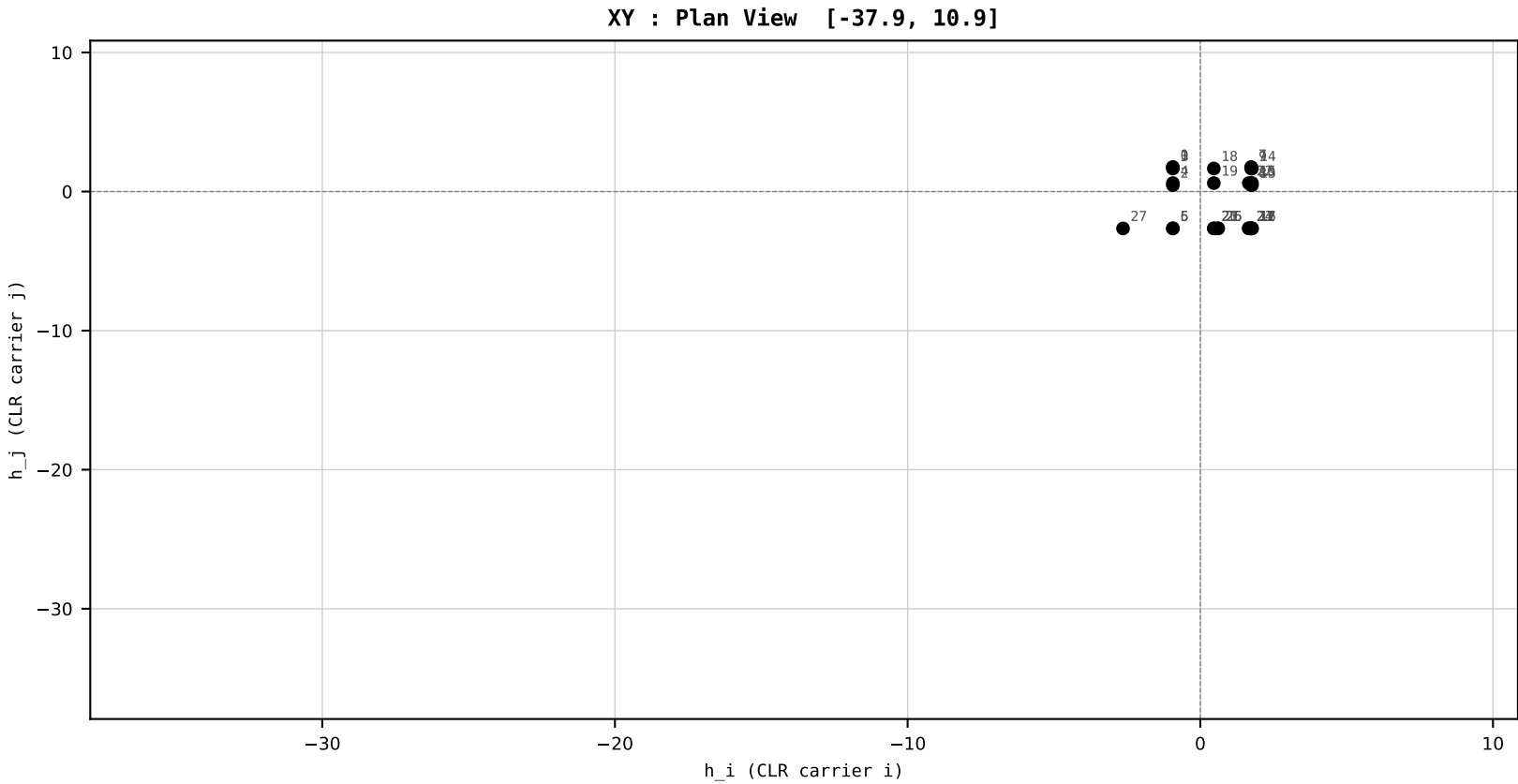
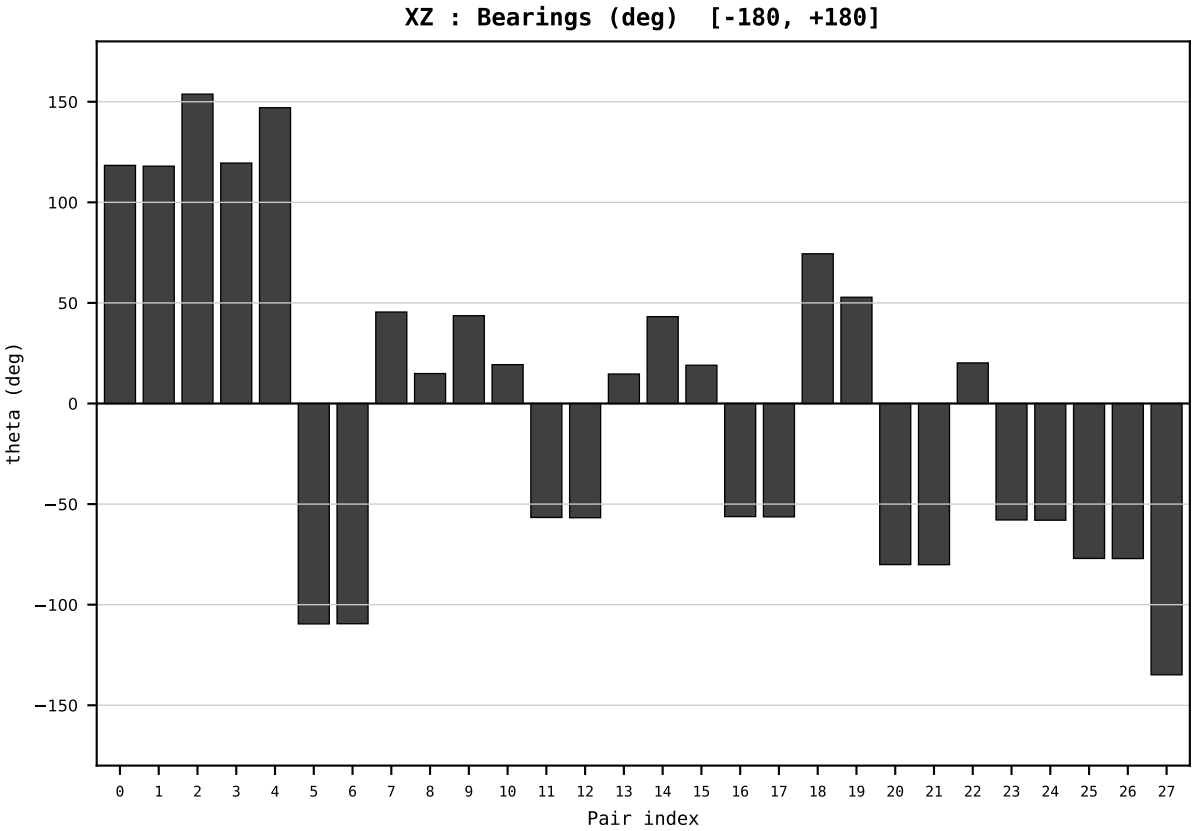
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.237884
Ring = Hs-2
E_metric= 24.3502
kappa_HS= 82.97
omega = 2.4032 deg
d_A = 0.282318
Helm = Solar
Helm d = +0.171611
DR = 4.419 HLR
DR ratio= 83.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

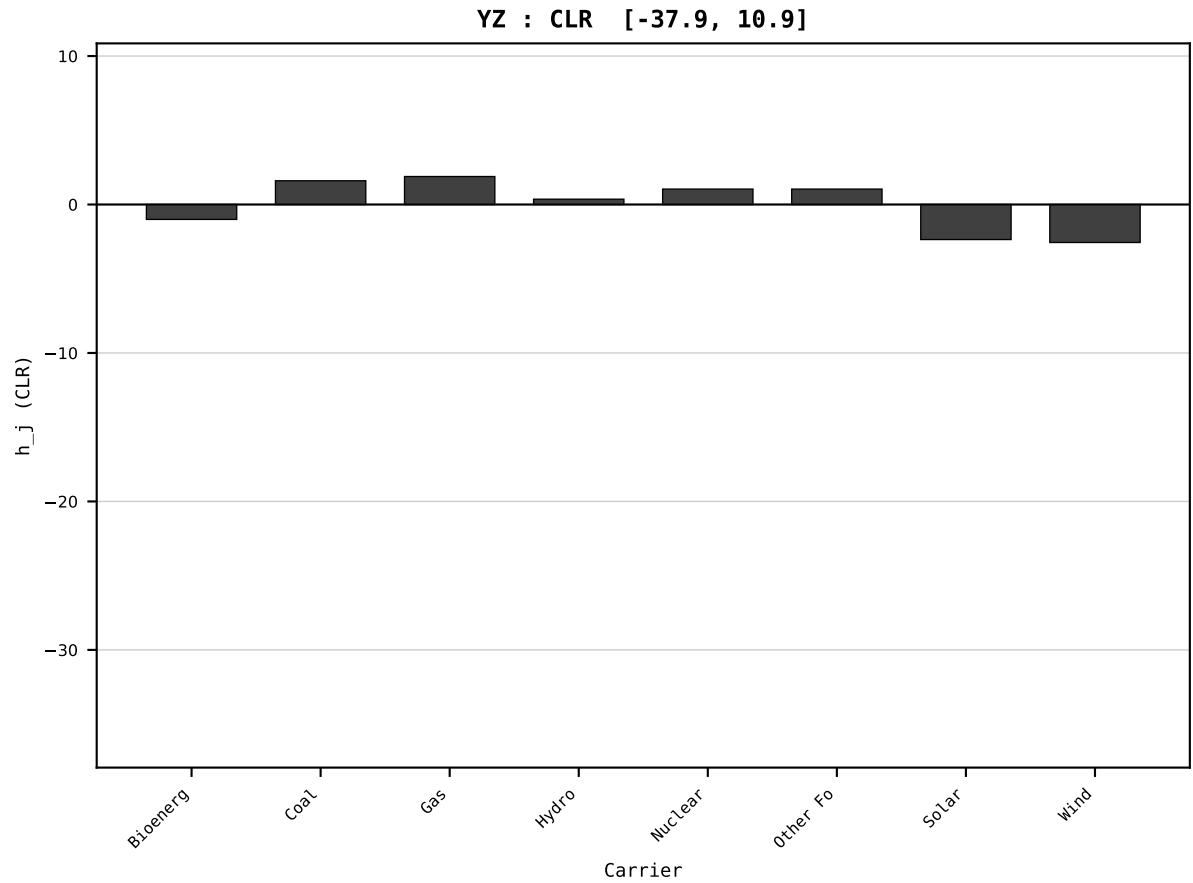
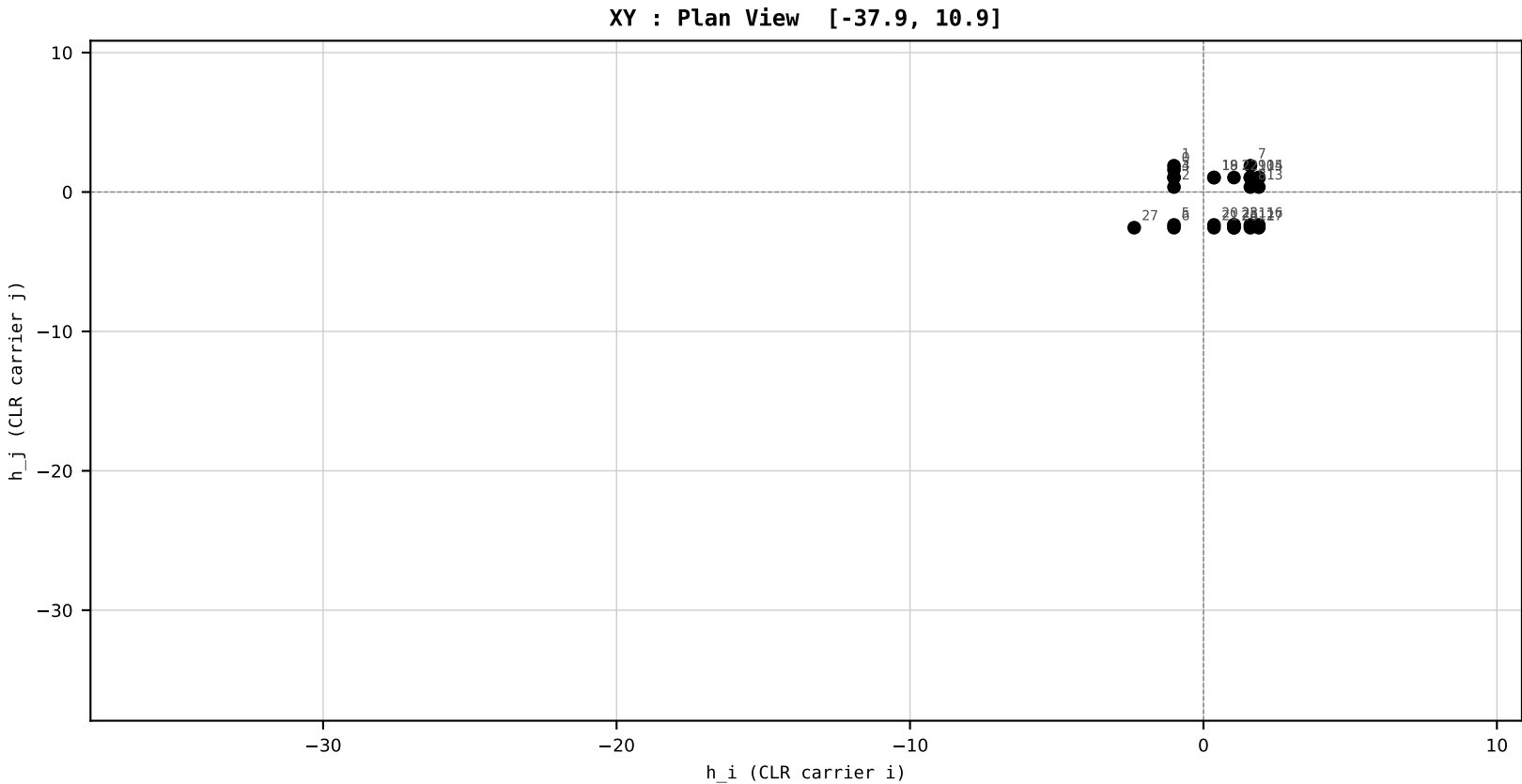
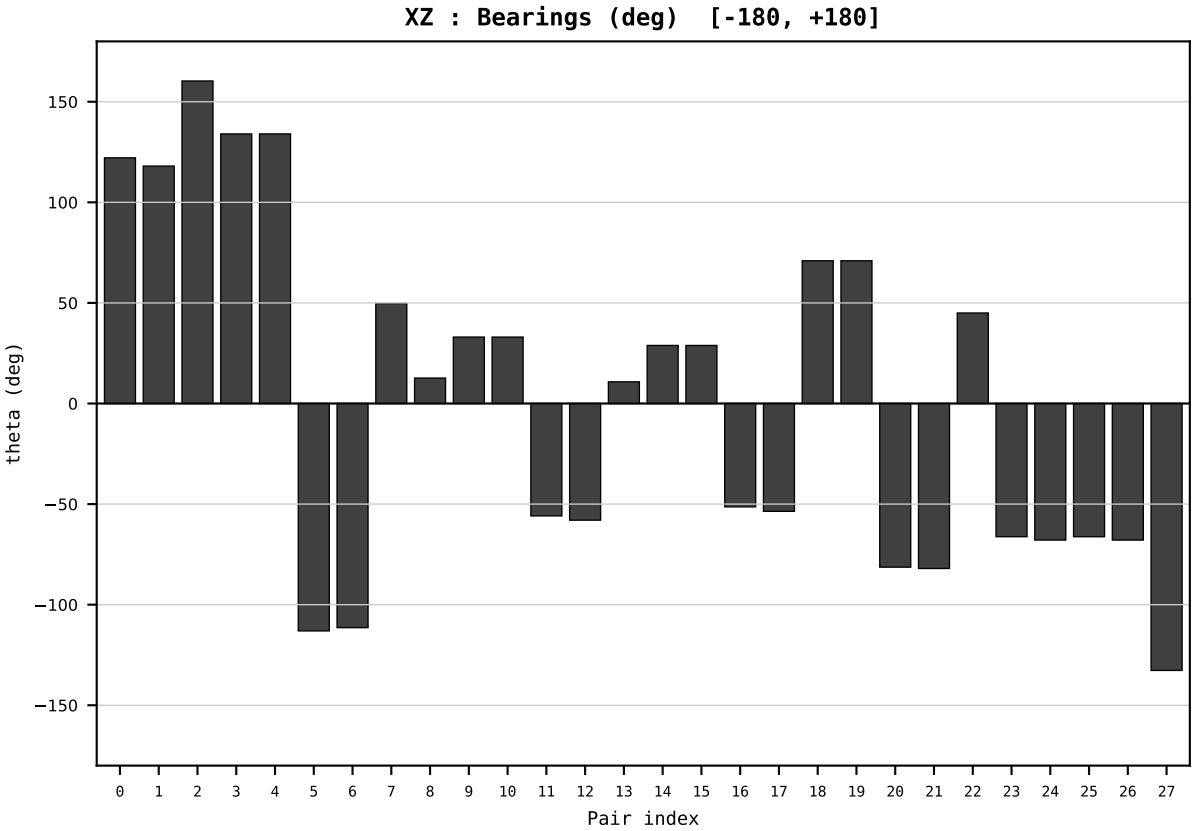
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.230931
Ring = Hs-2
E_metric= 21.5271
kappa_HS= 85.16
omega = 9.4013 deg
d_A = 0.837843
Helm = Nuclear
Helm d = -0.617218
DR = 4.445 HLR
DR ratio= 85.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



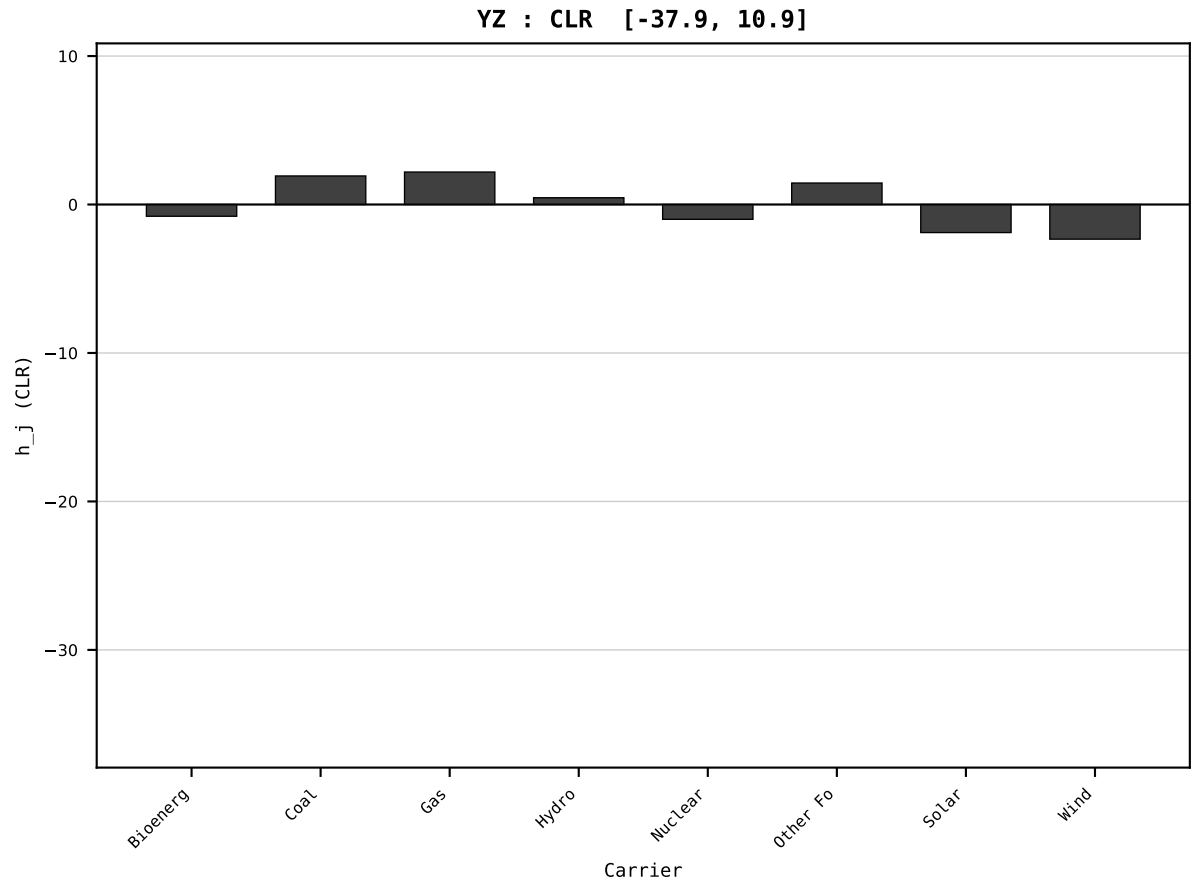
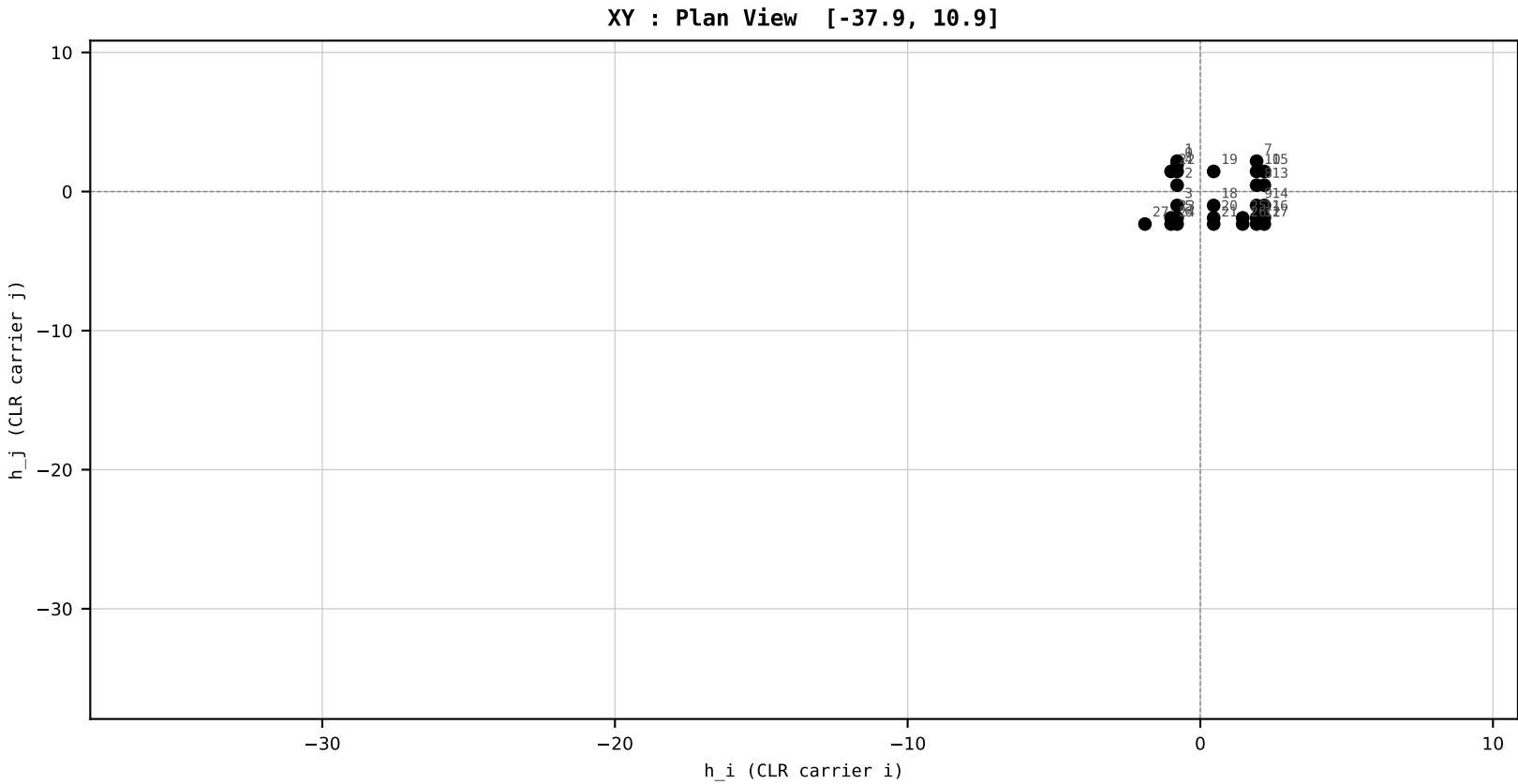
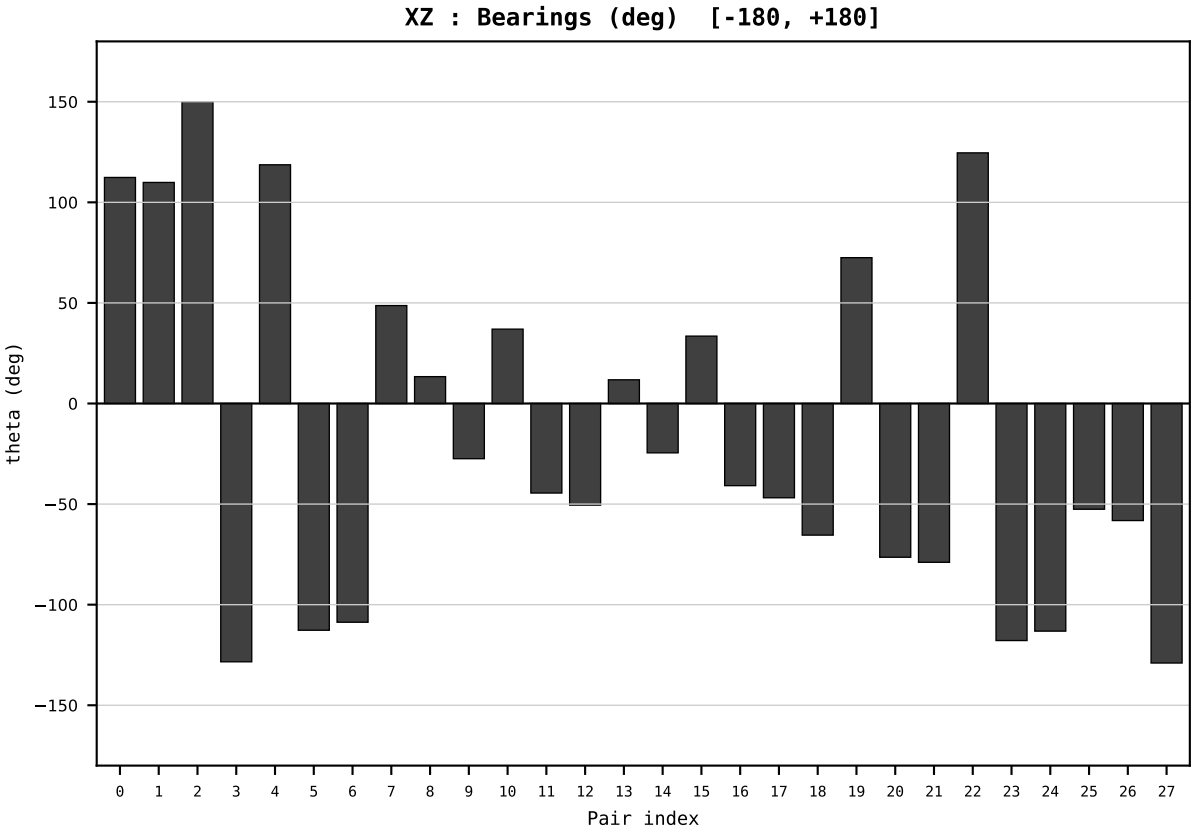
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.312291
Ring = Hs-3
E_metric= 21.4426
kappa_HS= 92.07
omega = 27.4642 deg
d_A = 2.200635
Helm = Nuclear
Helm d = -2.038262
DR = 4.523 HLR
DR ratio= 92.1x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



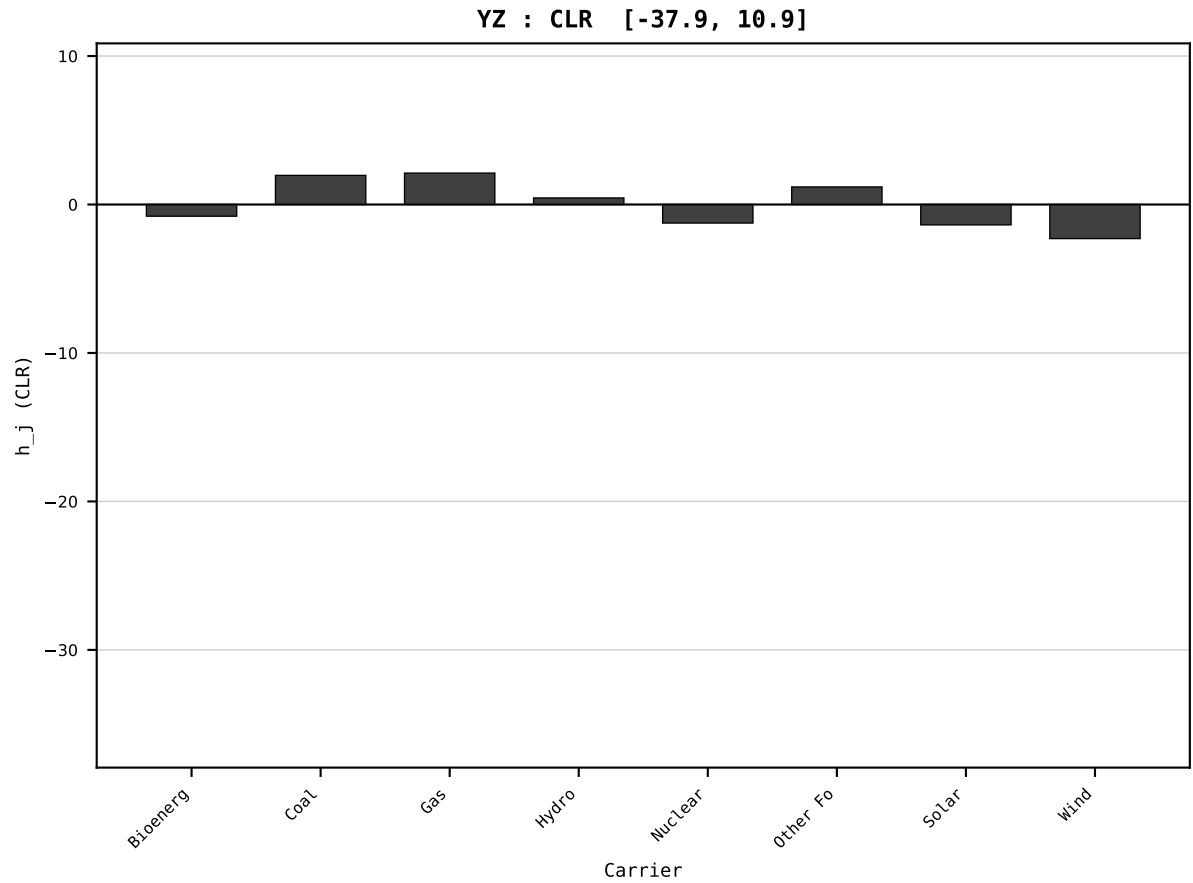
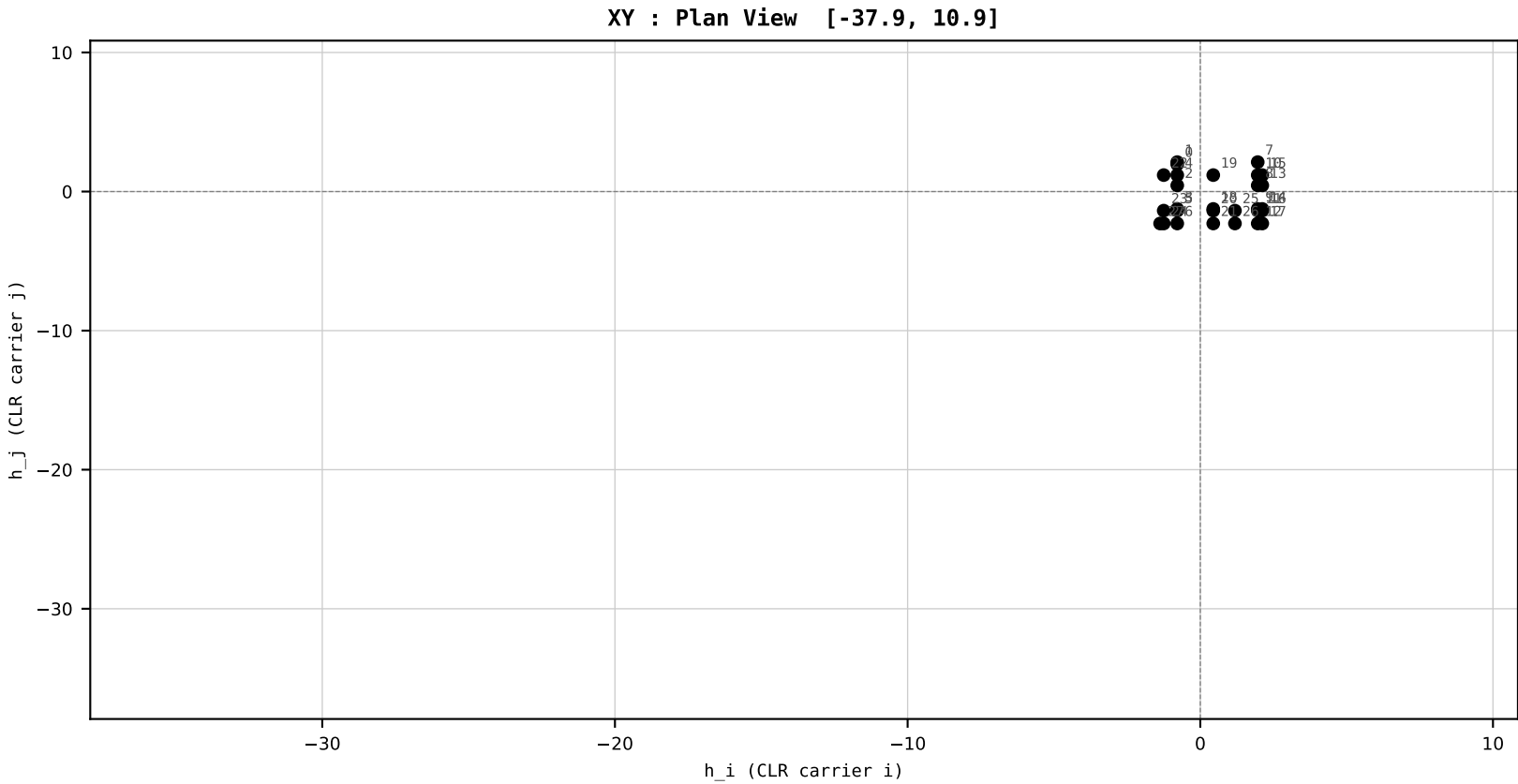
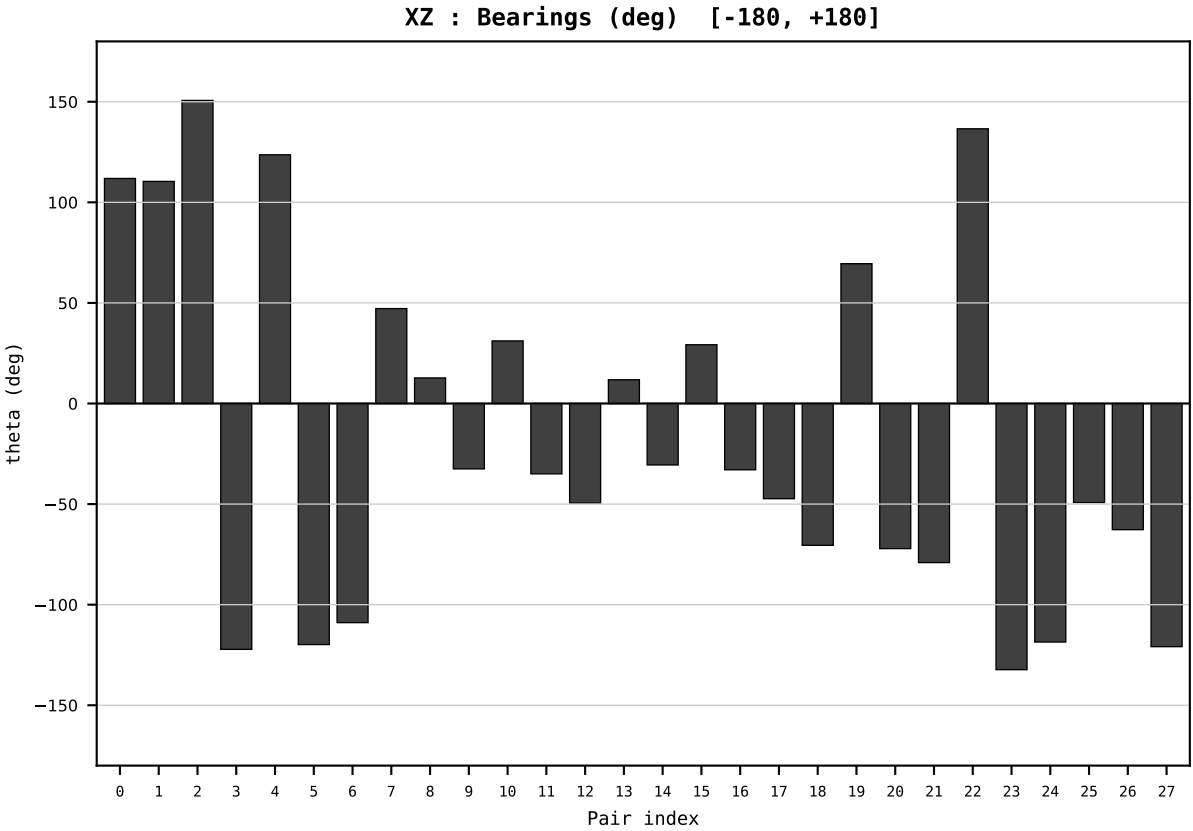
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.312583
Ring = Hs-3
E_metric= 19.2716
kappa_HS= 82.57
omega = 7.5396 deg
d_A = 0.639857
Helm = Solar
Helm d = +0.518125
DR = 4.414 HLR
DR ratio= 82.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

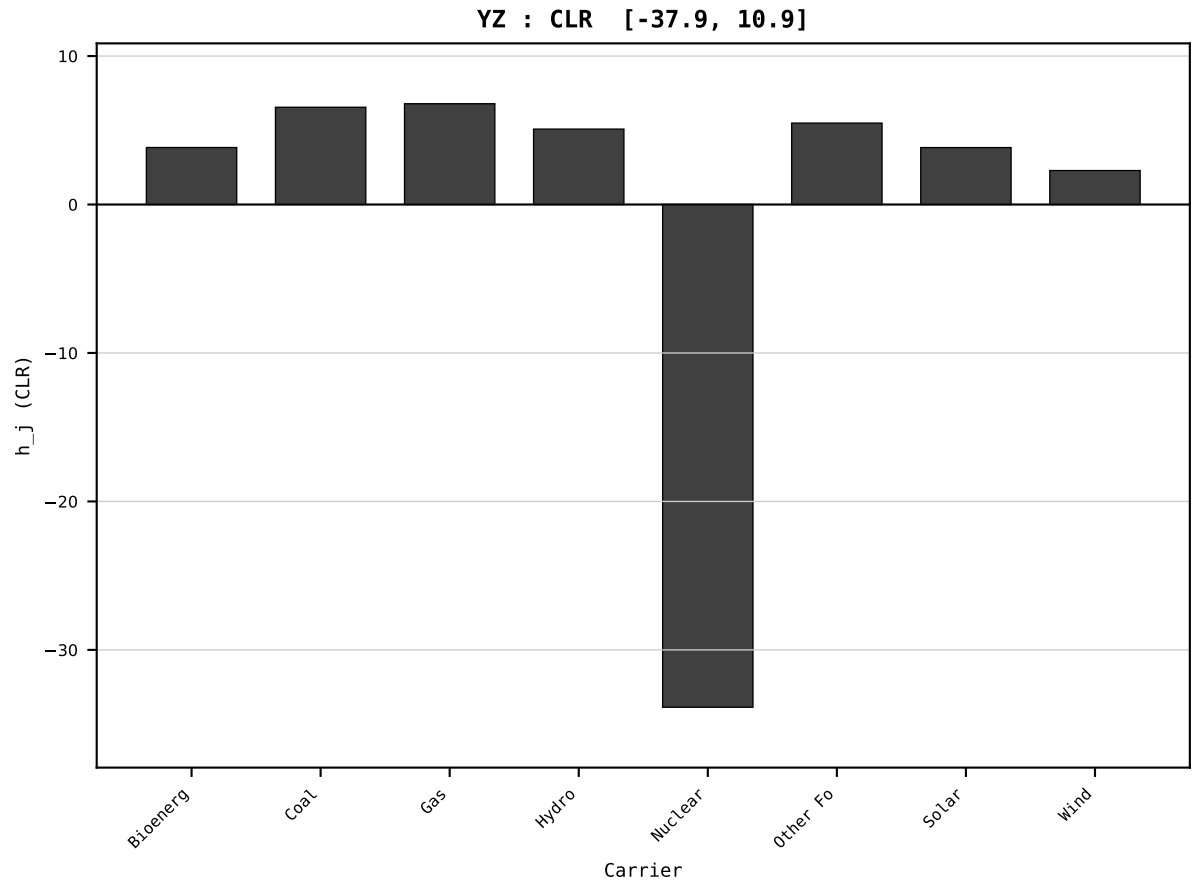
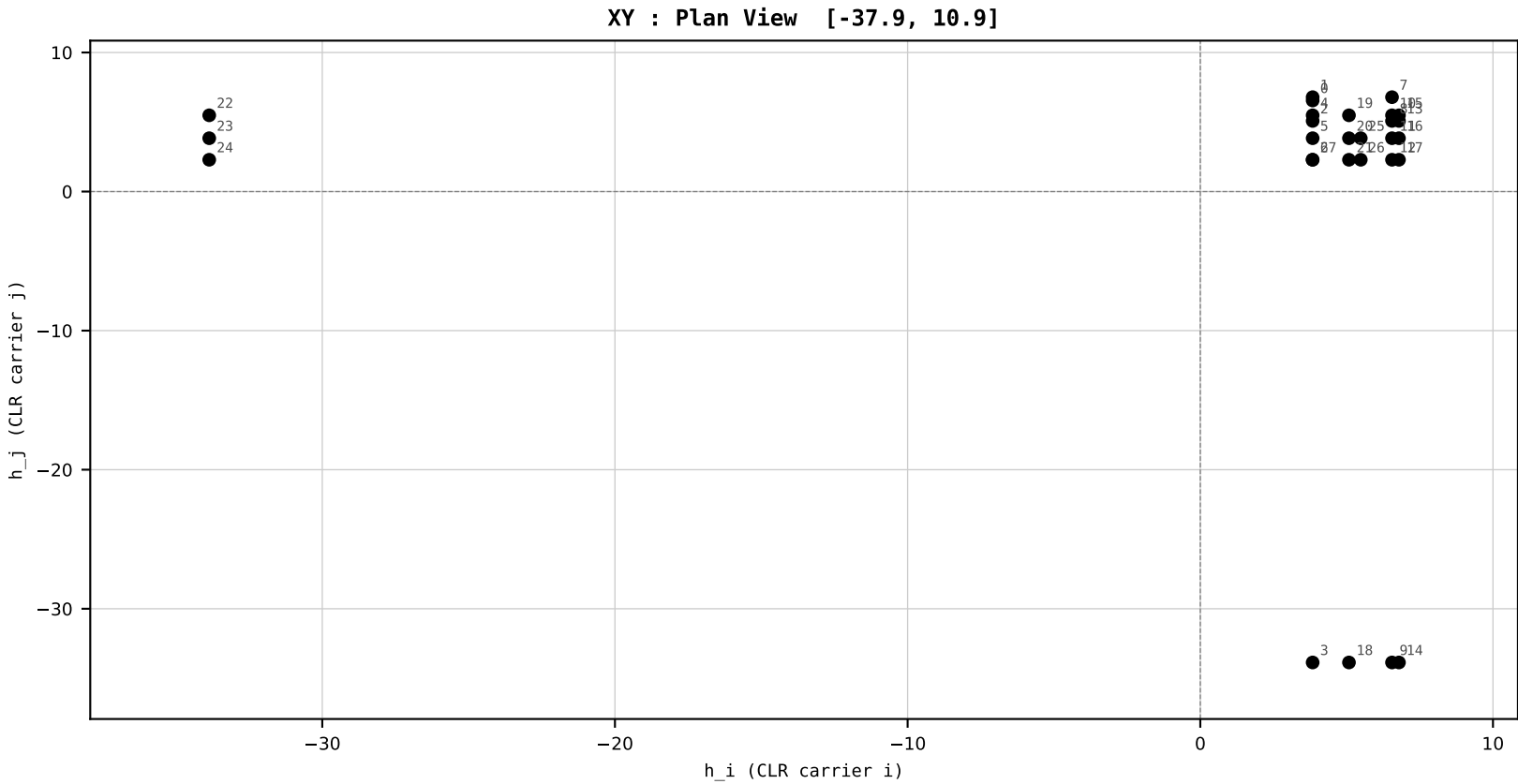
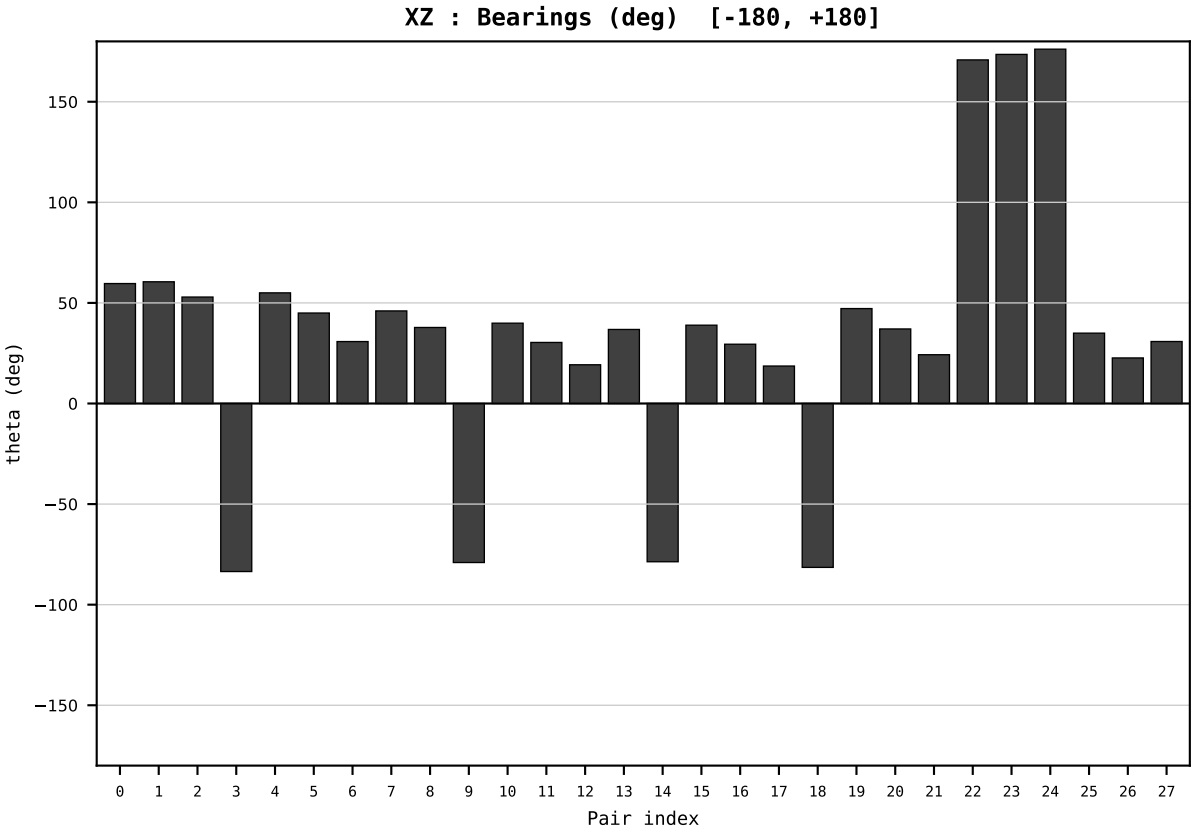
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.340926
Ring = Hs-3
E_metric= 896.7320
kappa_HS= 45087999999999936.00
omega = 66.1070 deg
d_A = 34.870573
Helm = Nuclear
Helm d = -32.612482
DR = 40.650 HLR
DR ratio= 450879998367363840.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



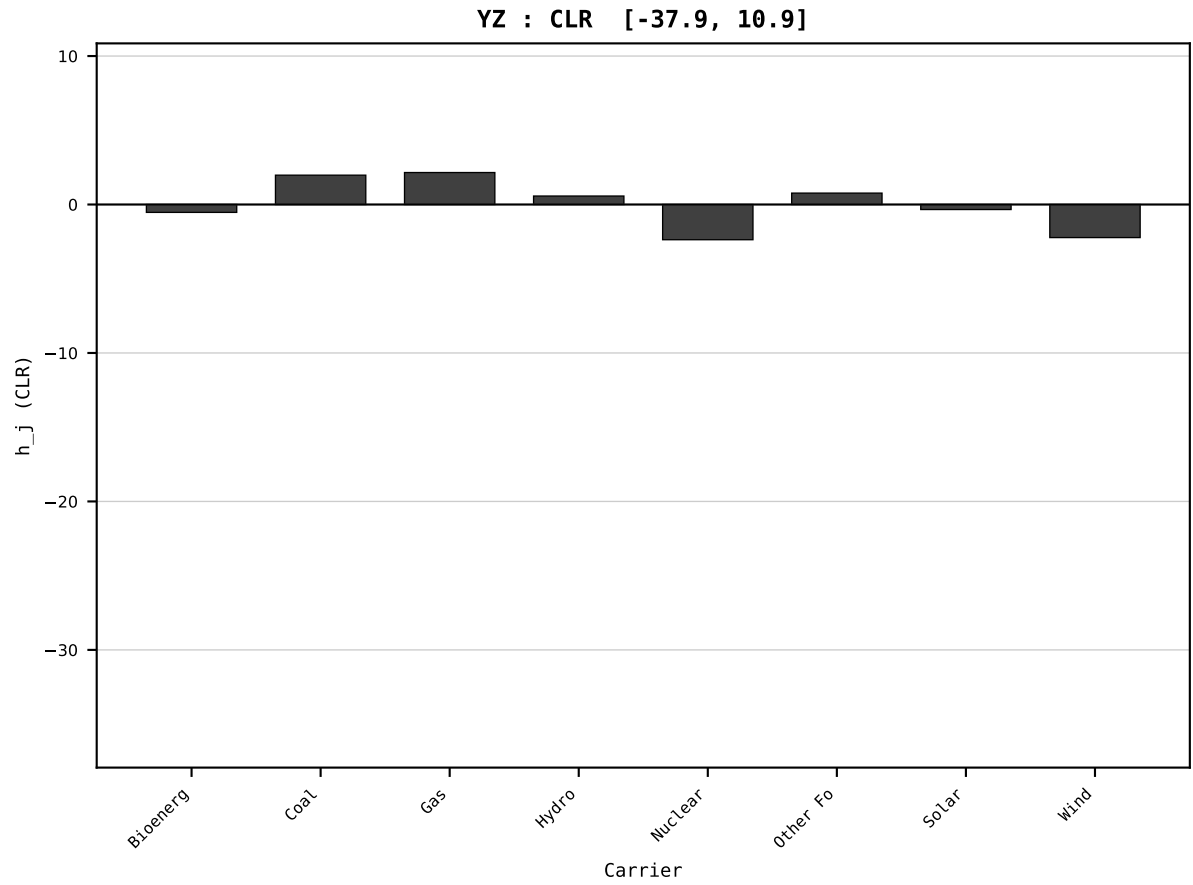
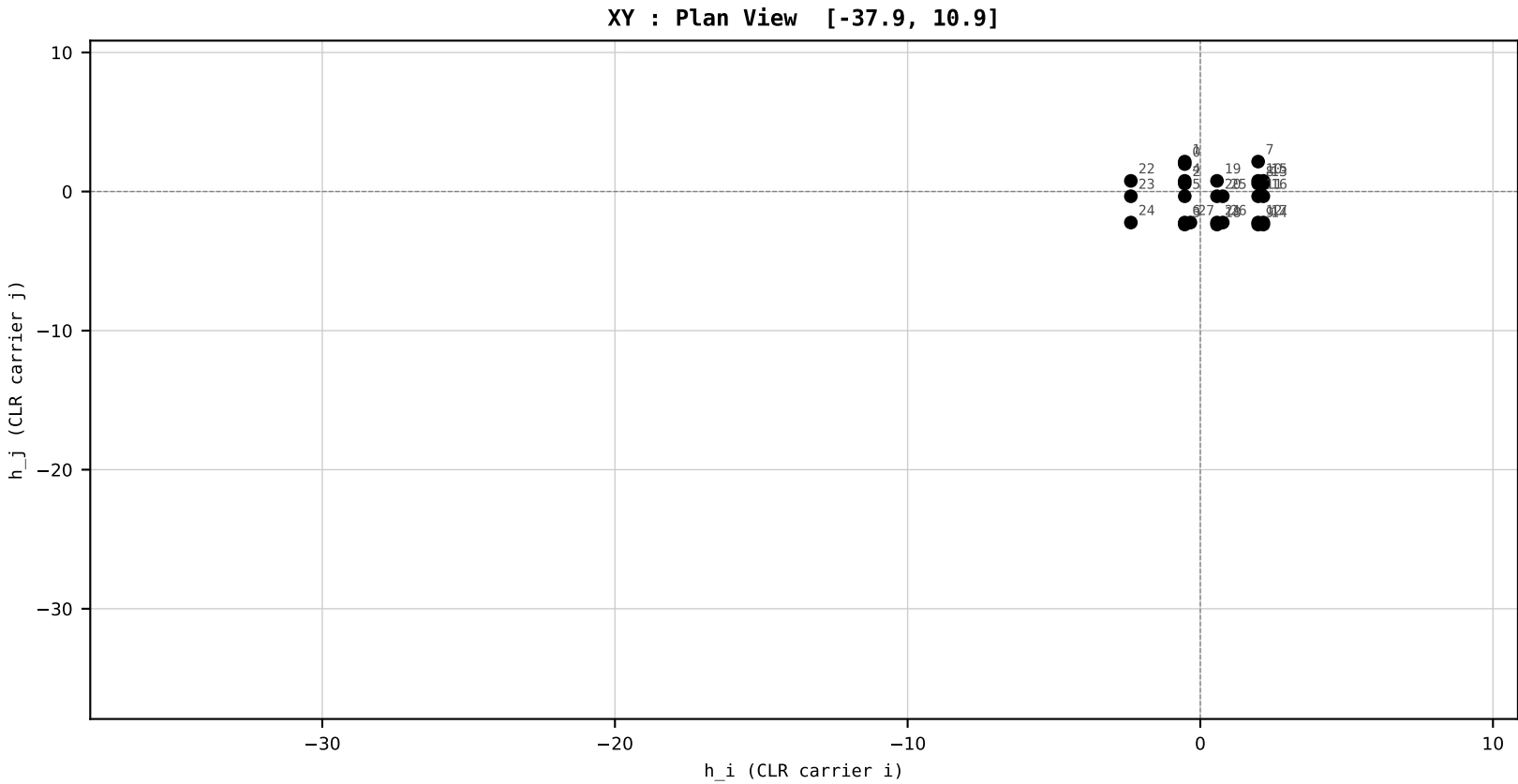
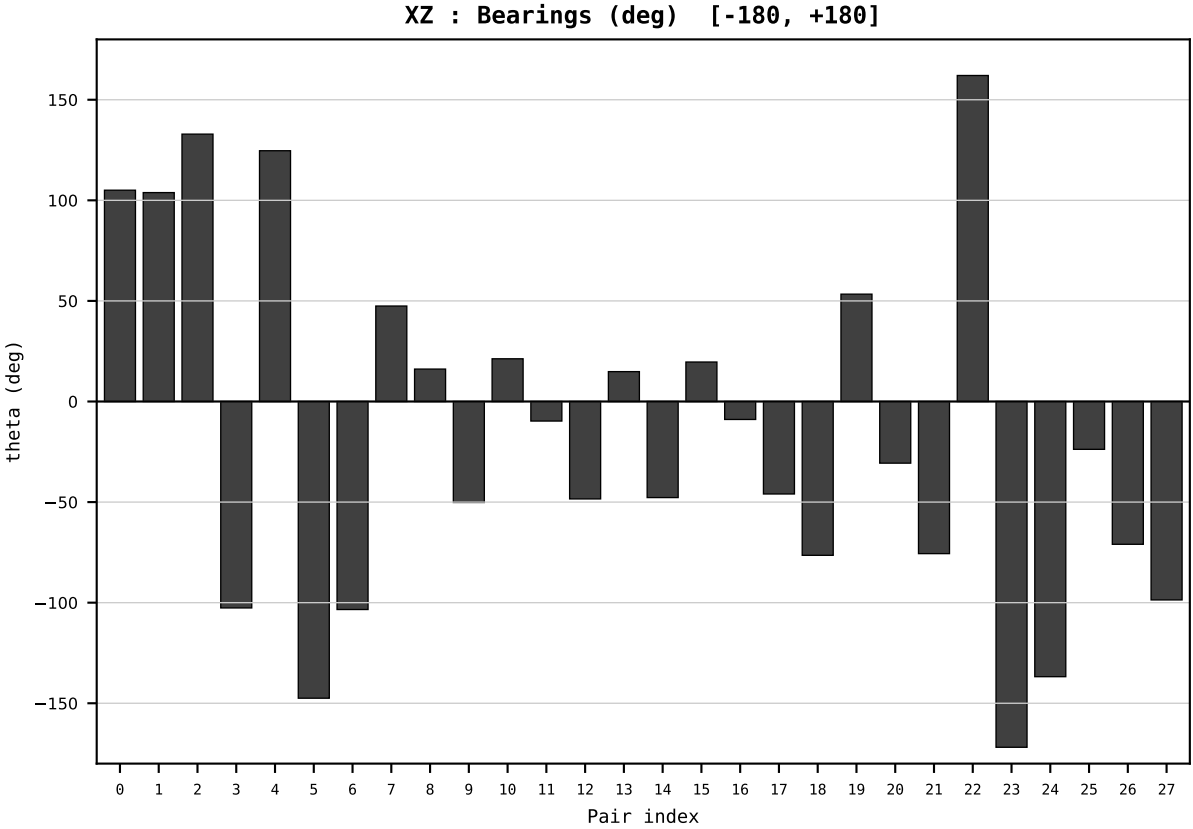
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.309676
Ring = Hs-3
E_metric= 20.4479
kappa_HS= 92.37
omega = 49.6746 deg
d_A = 33.667556
Helm = Nuclear
Helm d = +31.490431
DR = 4.526 HLR
DR ratio= 92.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=8 N=26 pairs=28

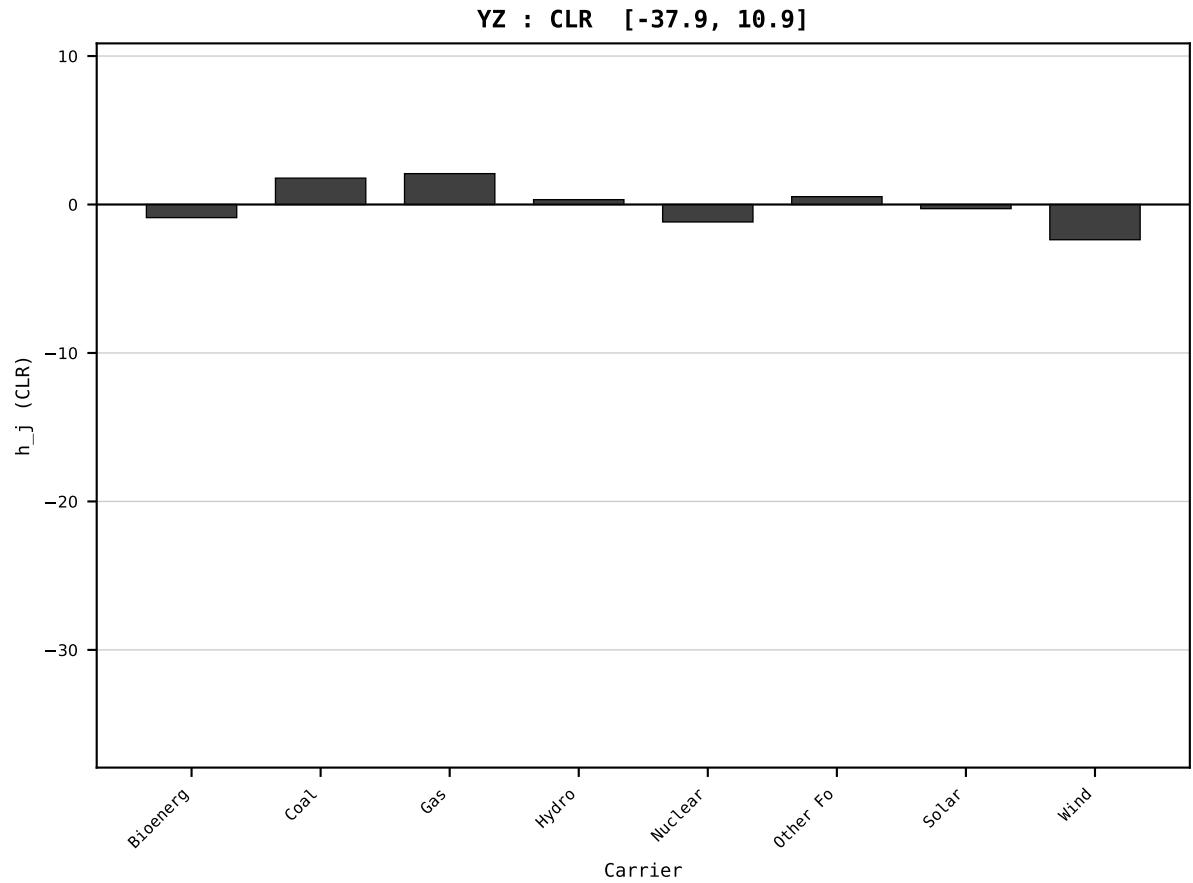
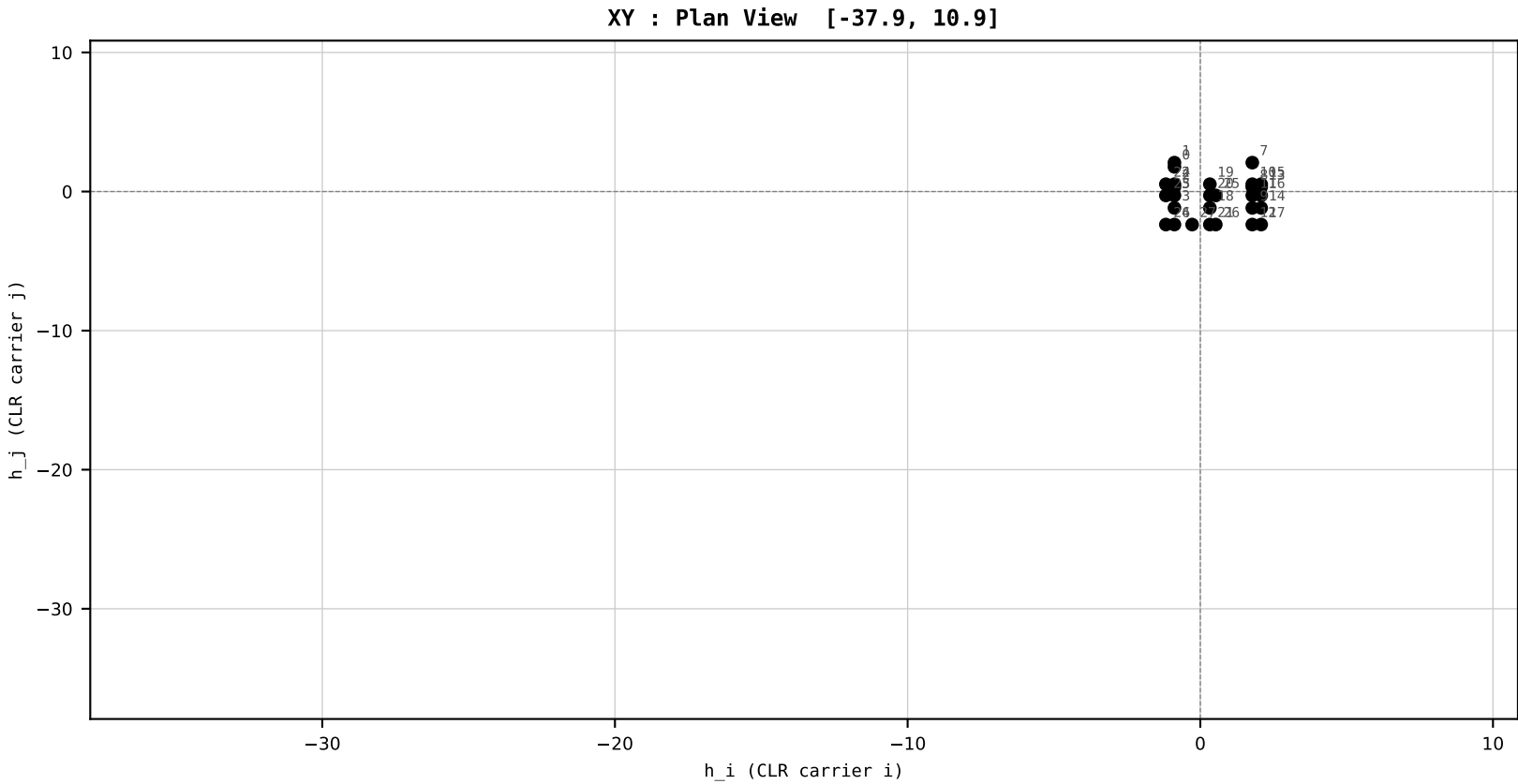
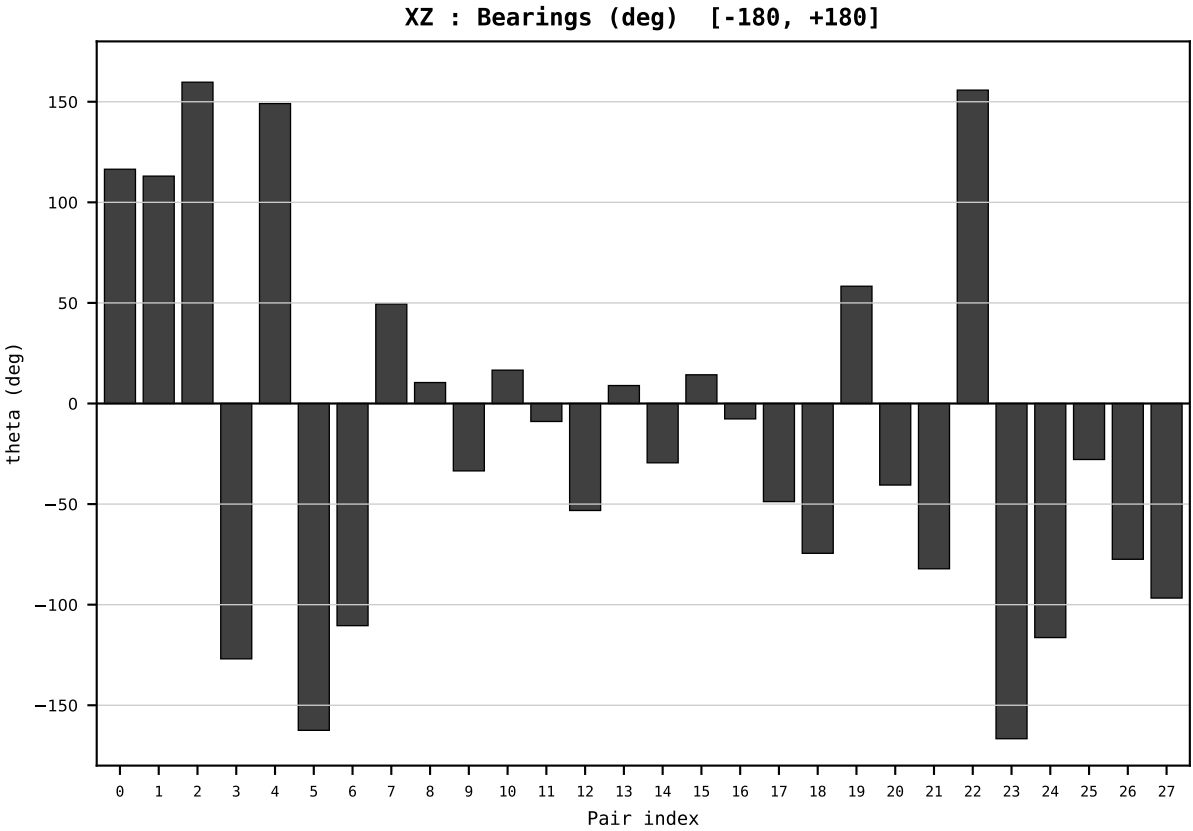
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.303033
Ring = Hs-3
E_metric= 15.7404
kappa_HS= 85.85
omega = 16.2591 deg
d_A = 1.320048
Helm = Nuclear
Helm d = +1.196283
DR = 4.453 HLR
DR ratio= 85.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

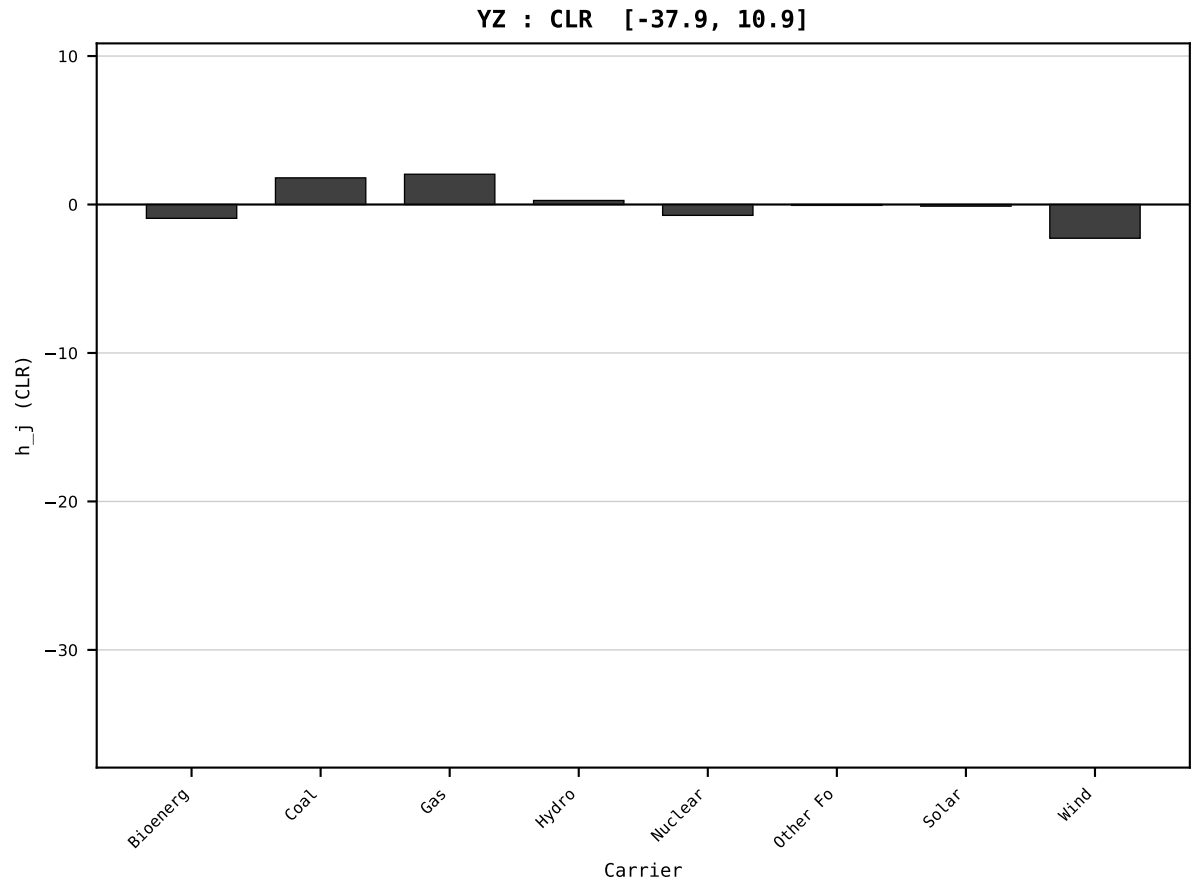
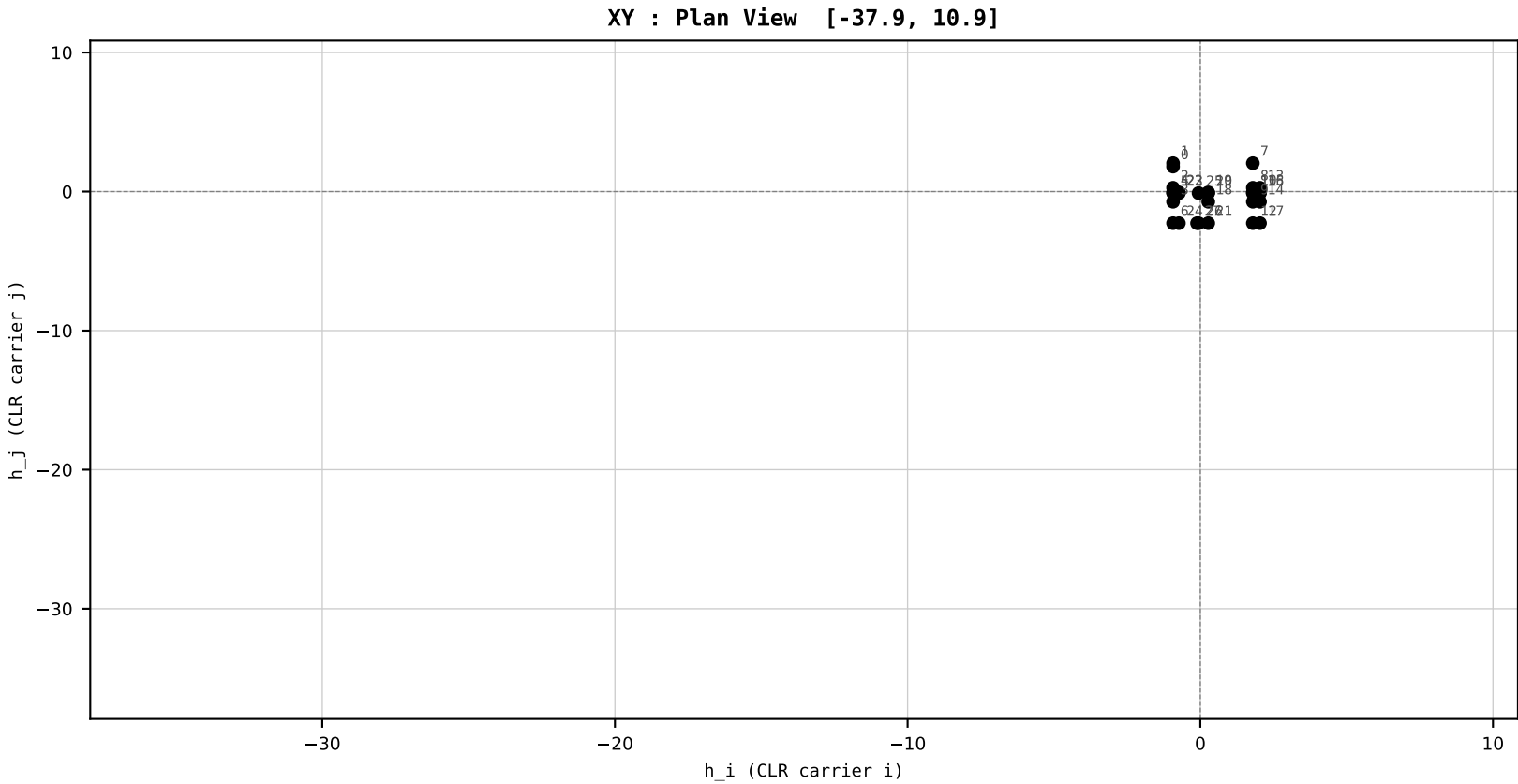
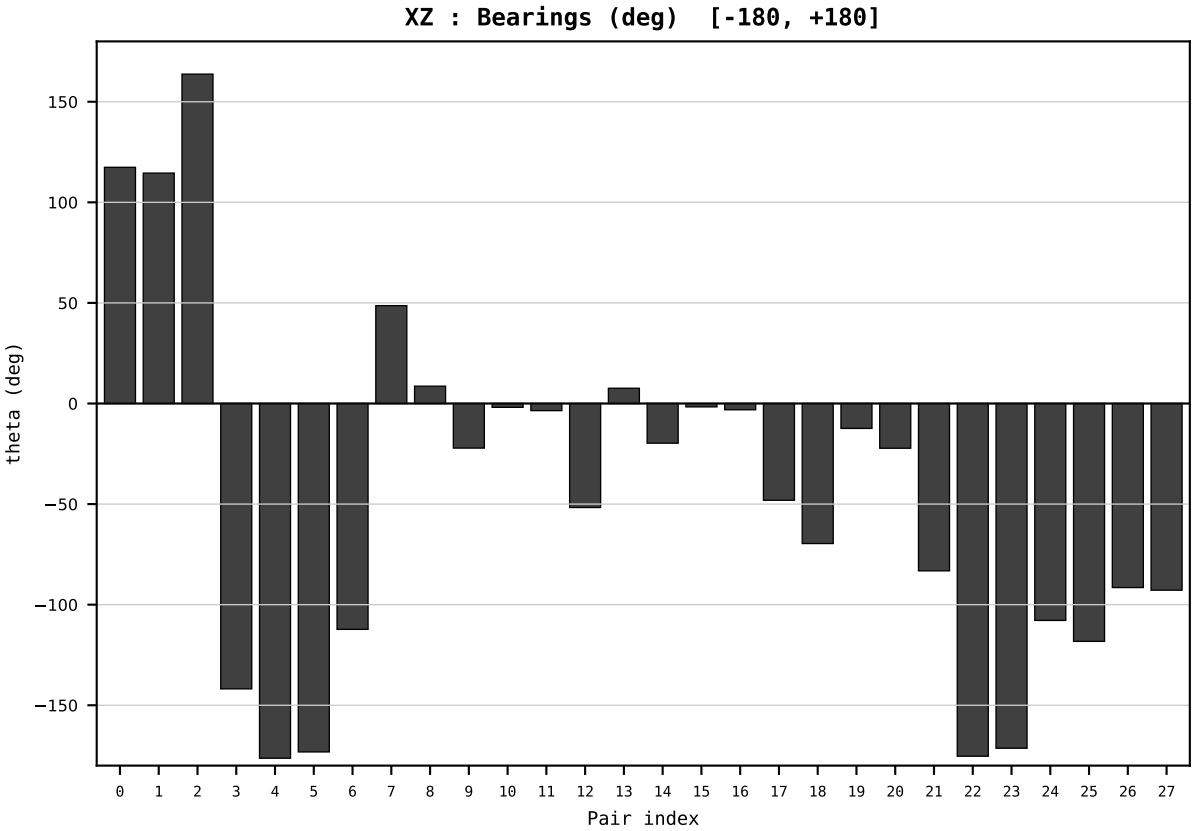
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.308283
Ring = Hs-3
E_metric= 14.0385
kappa_HS= 74.62
omega = 10.9468 deg
d_A = 0.767882
Helm = Other Fossil
Helm d = -0.588595
DR = 4.312 HLR
DR ratio= 74.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

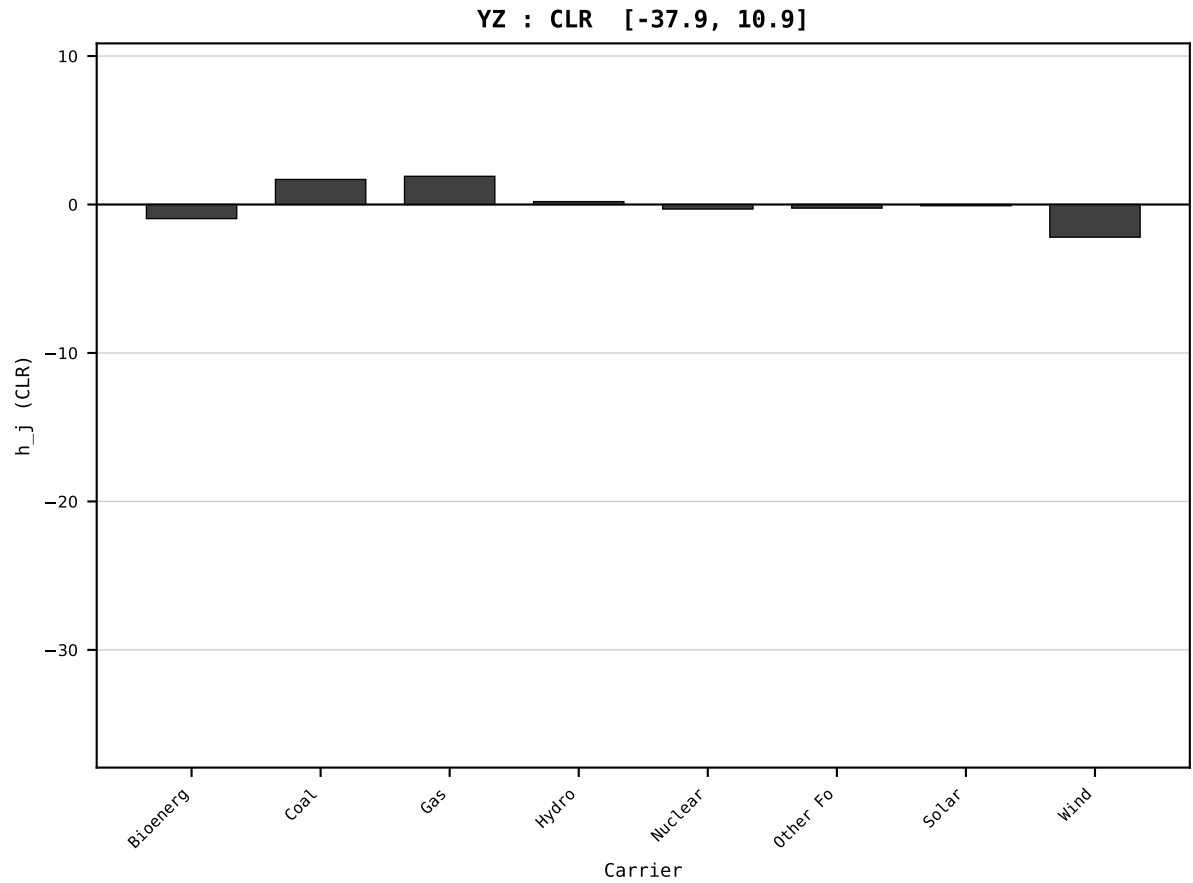
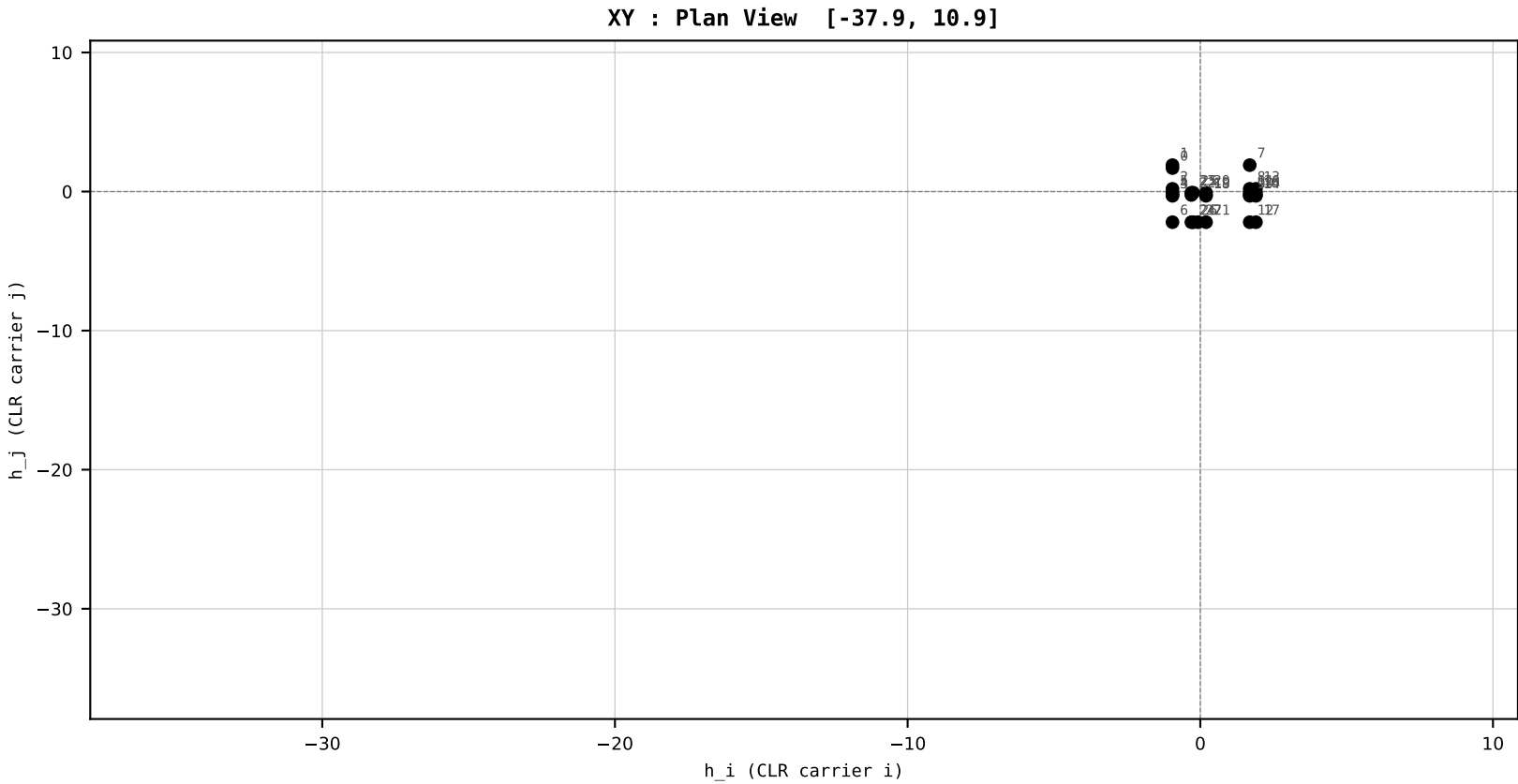
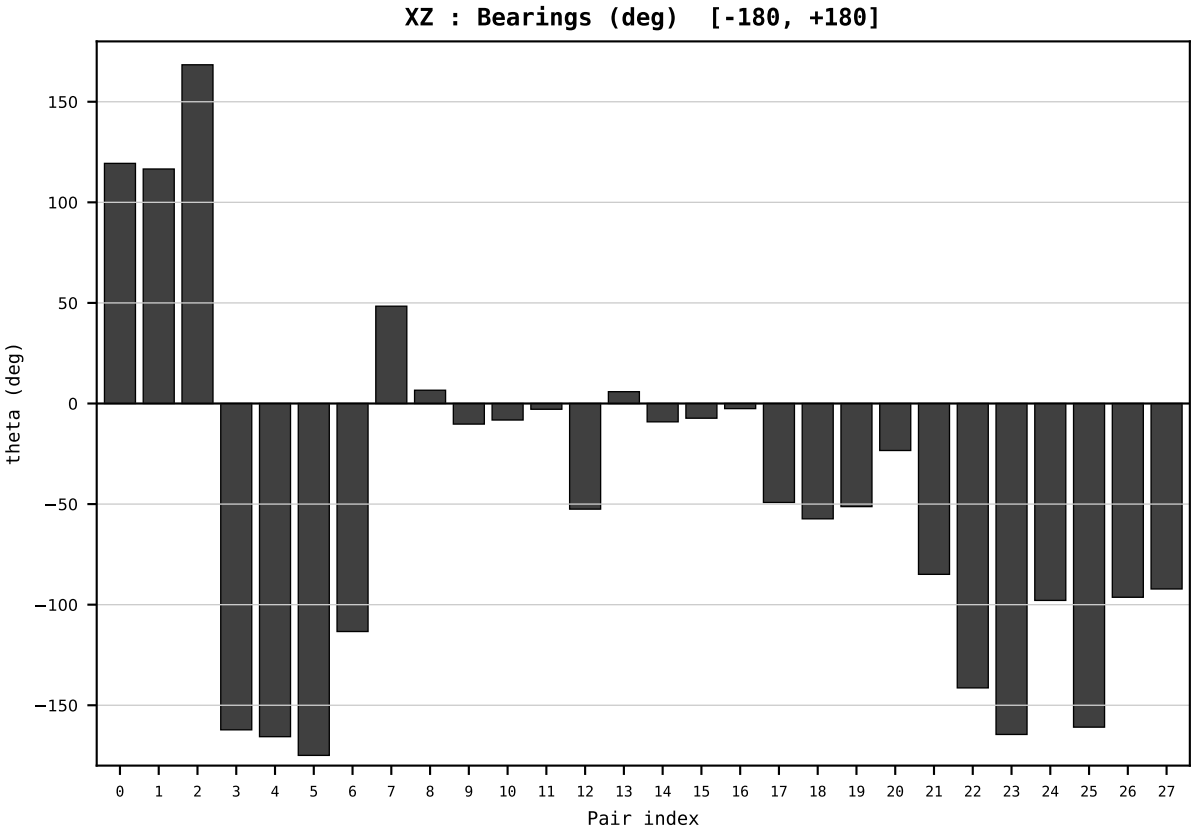
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.279970
Ring = Hs-2
E_metric= 12.4125
kappa_HS= 60.46
omega = 7.1966 deg
d_A = 0.507945
Helm = Nuclear
Helm d = +0.426213
DR = 4.102 HLR
DR ratio= 60.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



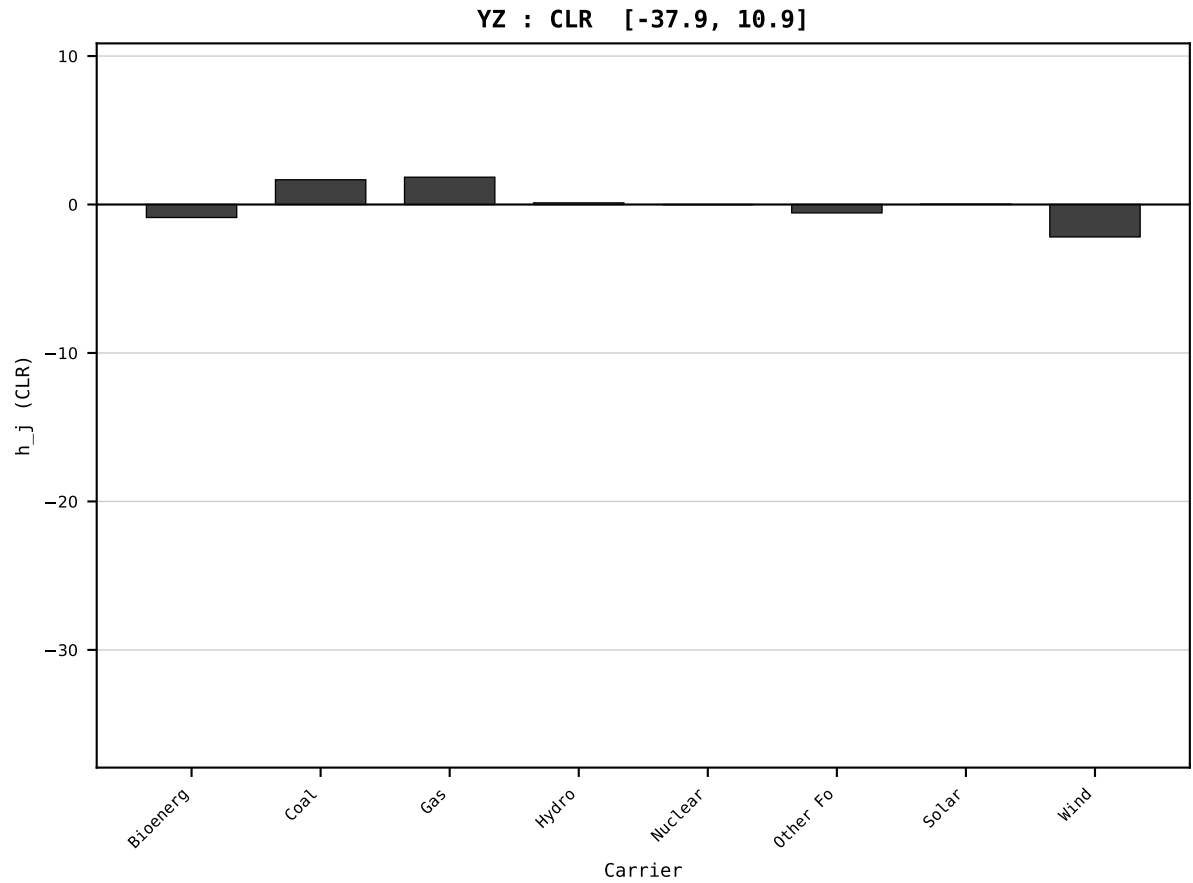
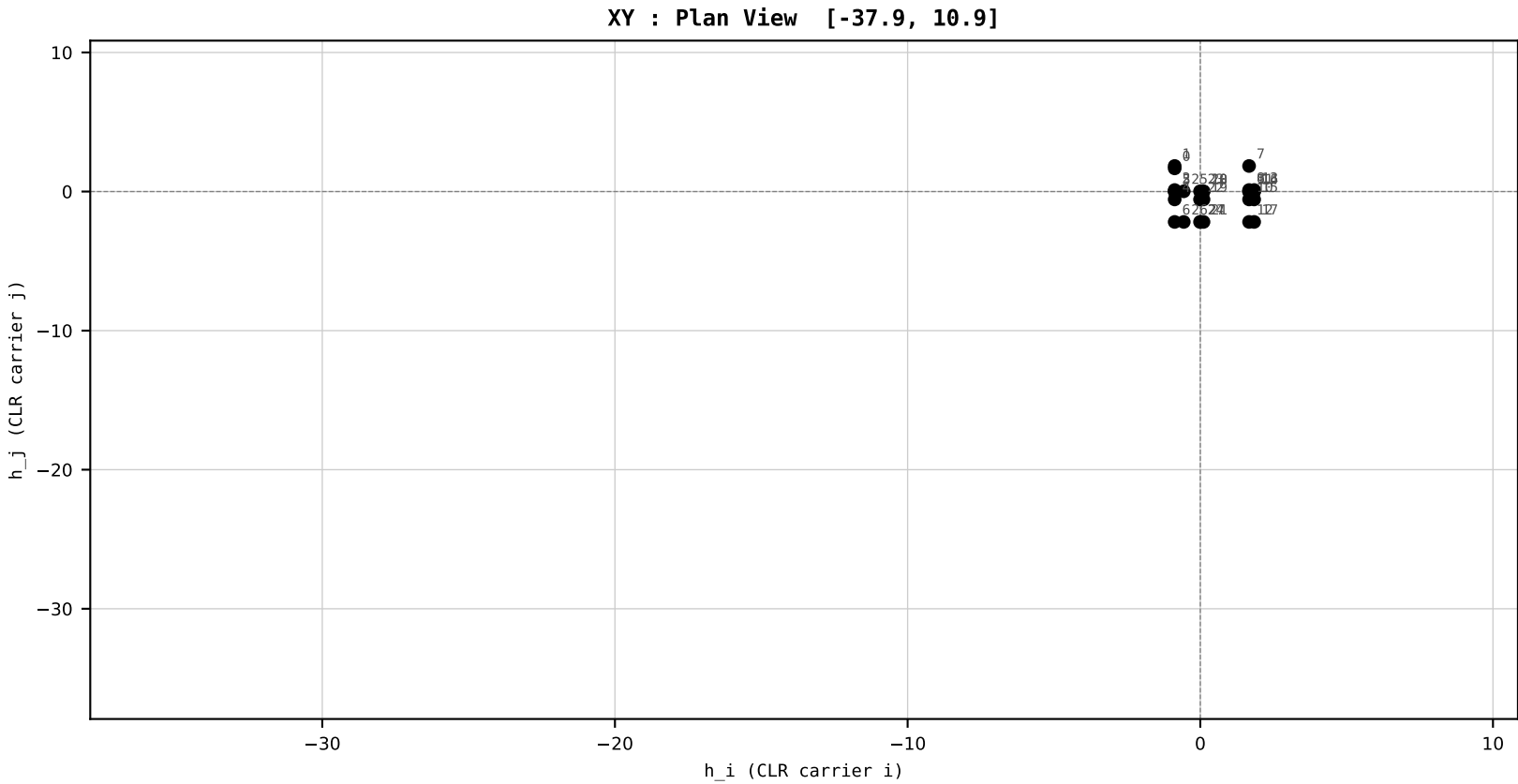
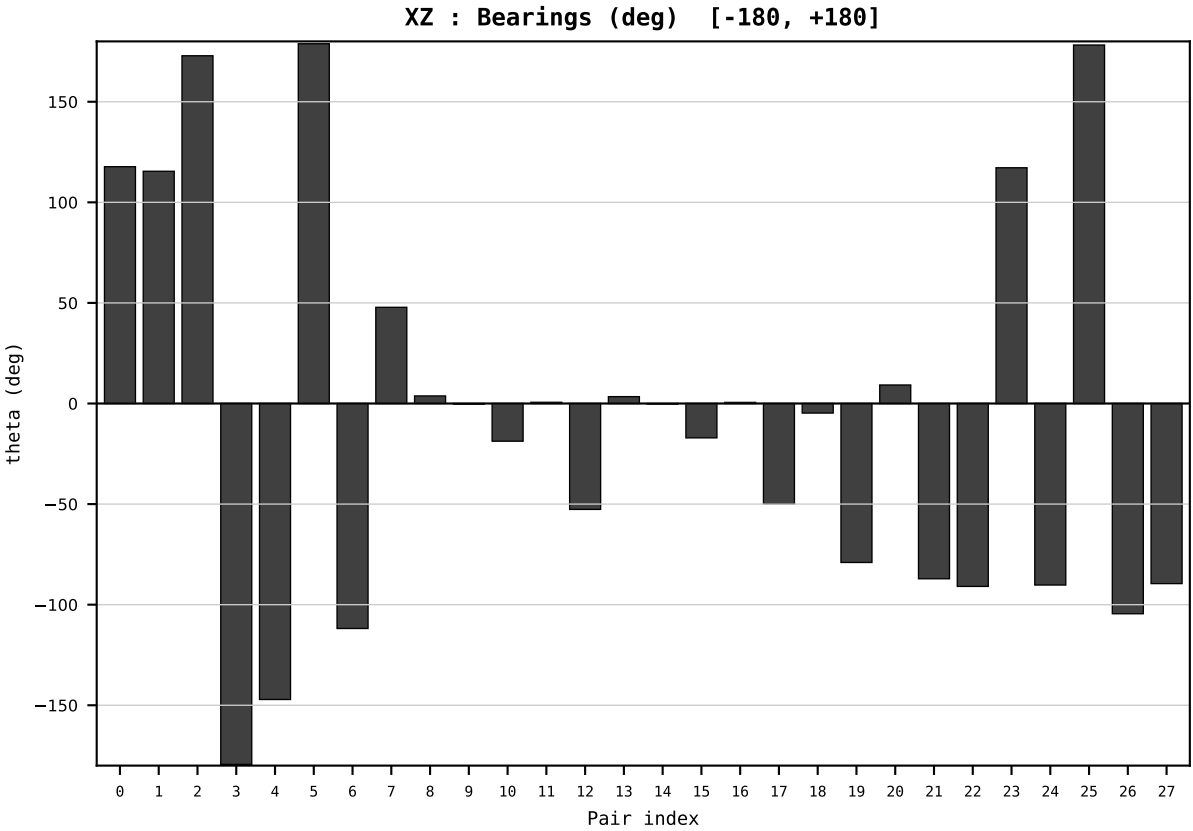
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.268630
Ring = Hs-2
E_metric= 12.0301
kappa_HS= 55.89
omega = 7.6236 deg
d_A = 0.467986
Helm = Other Fossil
Helm d = -0.321852
DR = 4.023 HLR
DR ratio= 55.9x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

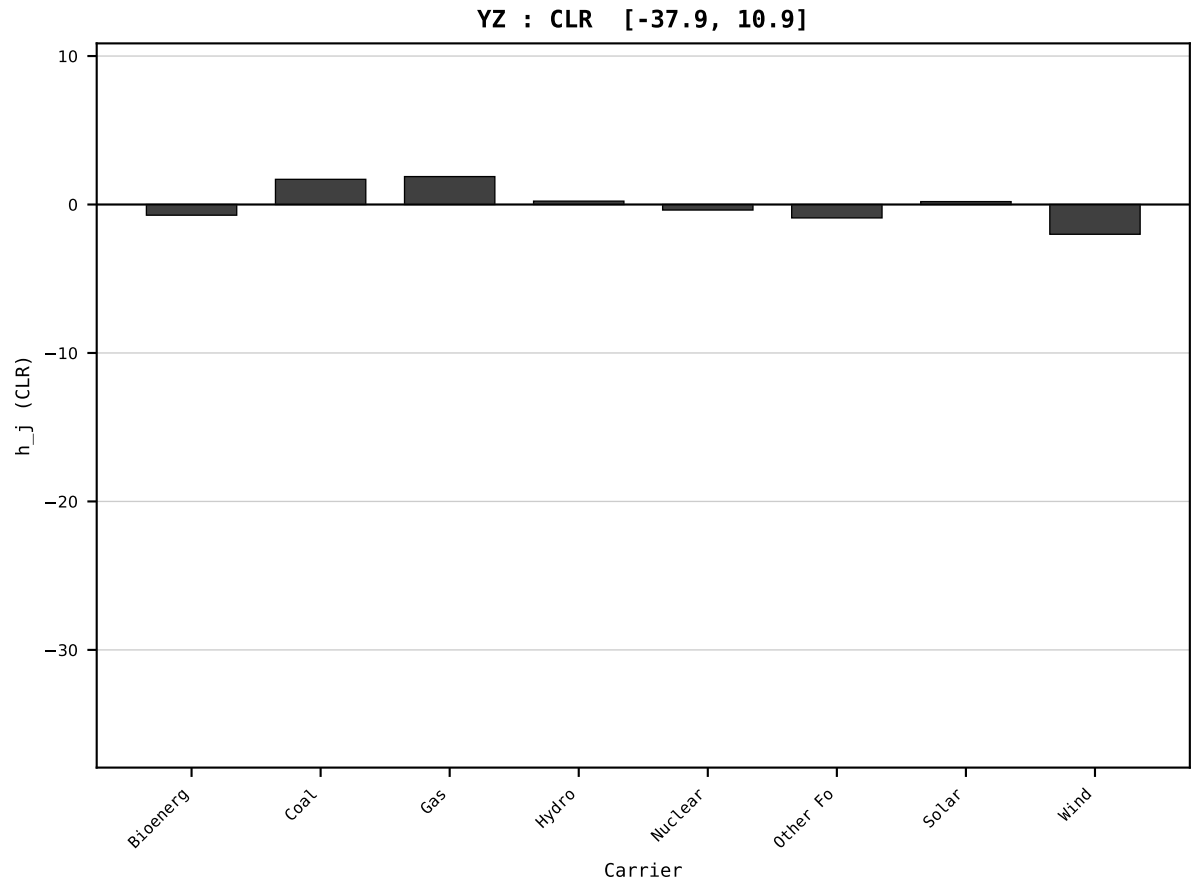
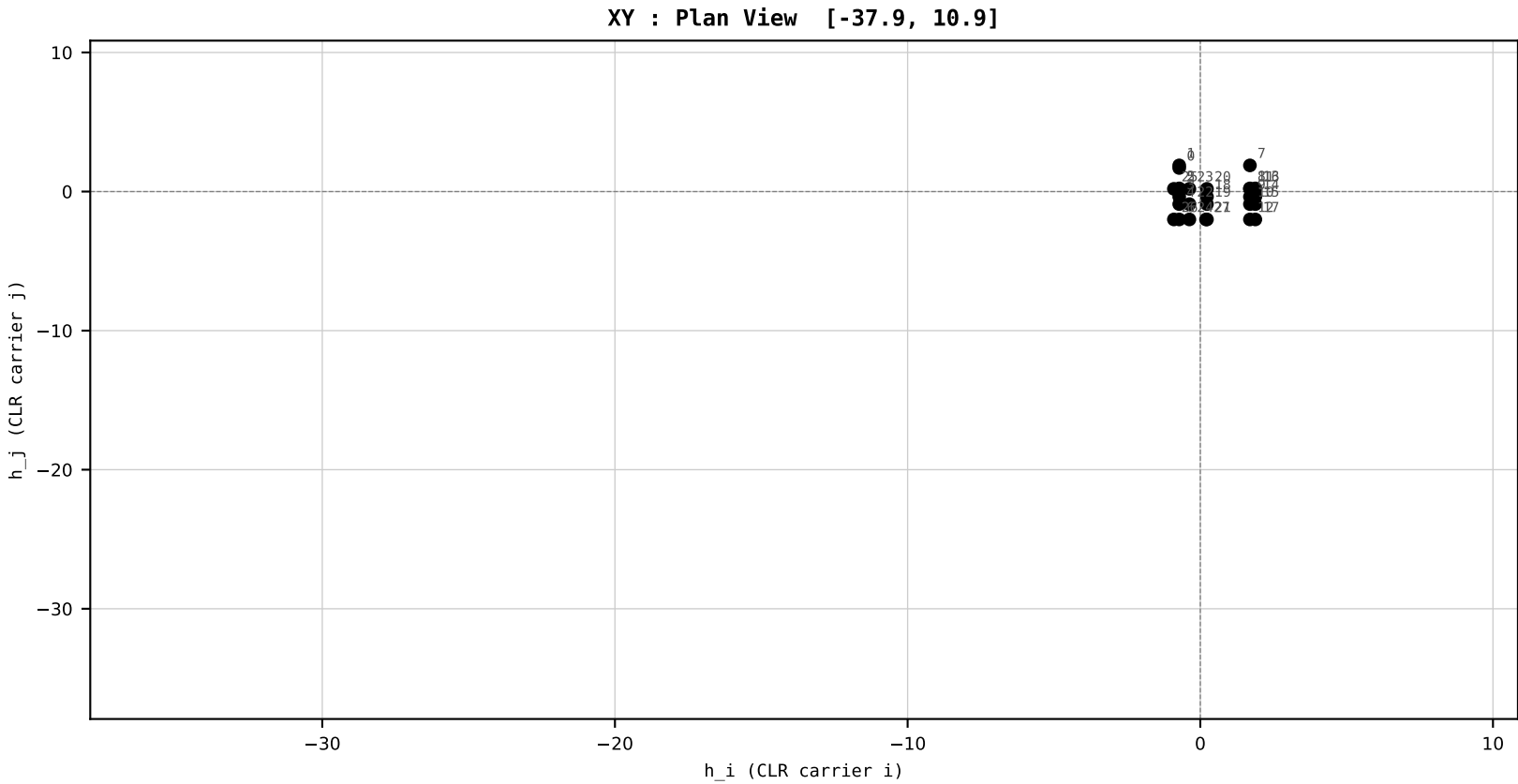
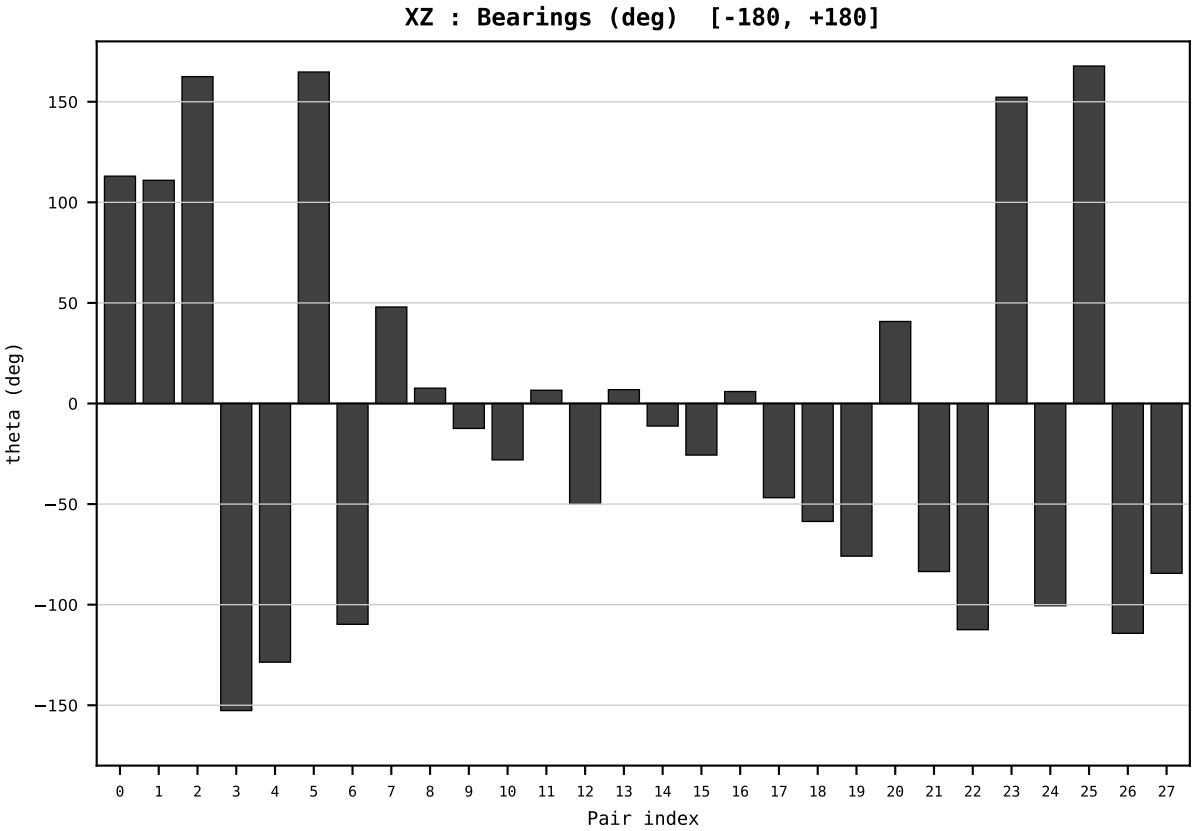
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.280146
Ring = Hs-2
E_metric= 11.9940
kappa_HS= 48.68
omega = 9.7926 deg
d_A = 0.591660
Helm = Nuclear
Helm d = -0.363464
DR = 3.885 HLR
DR ratio= 48.7x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

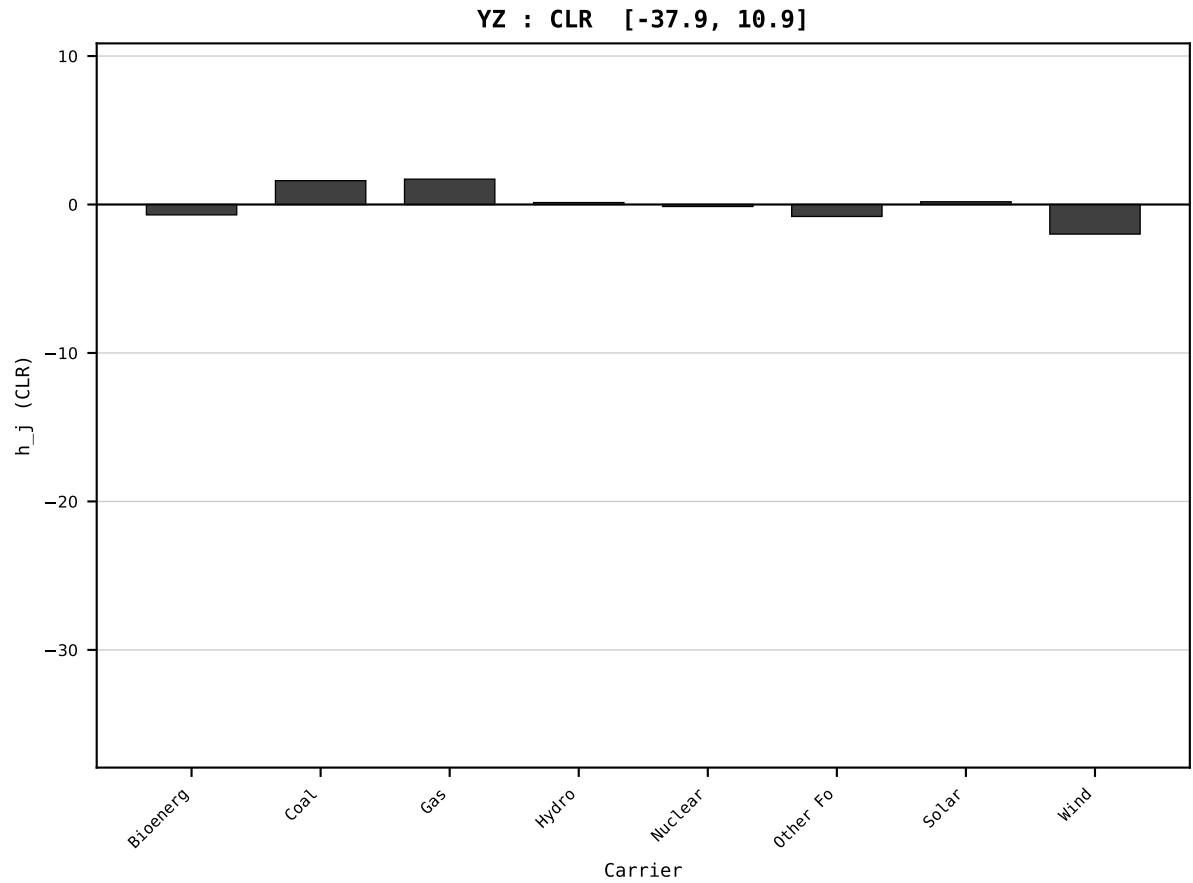
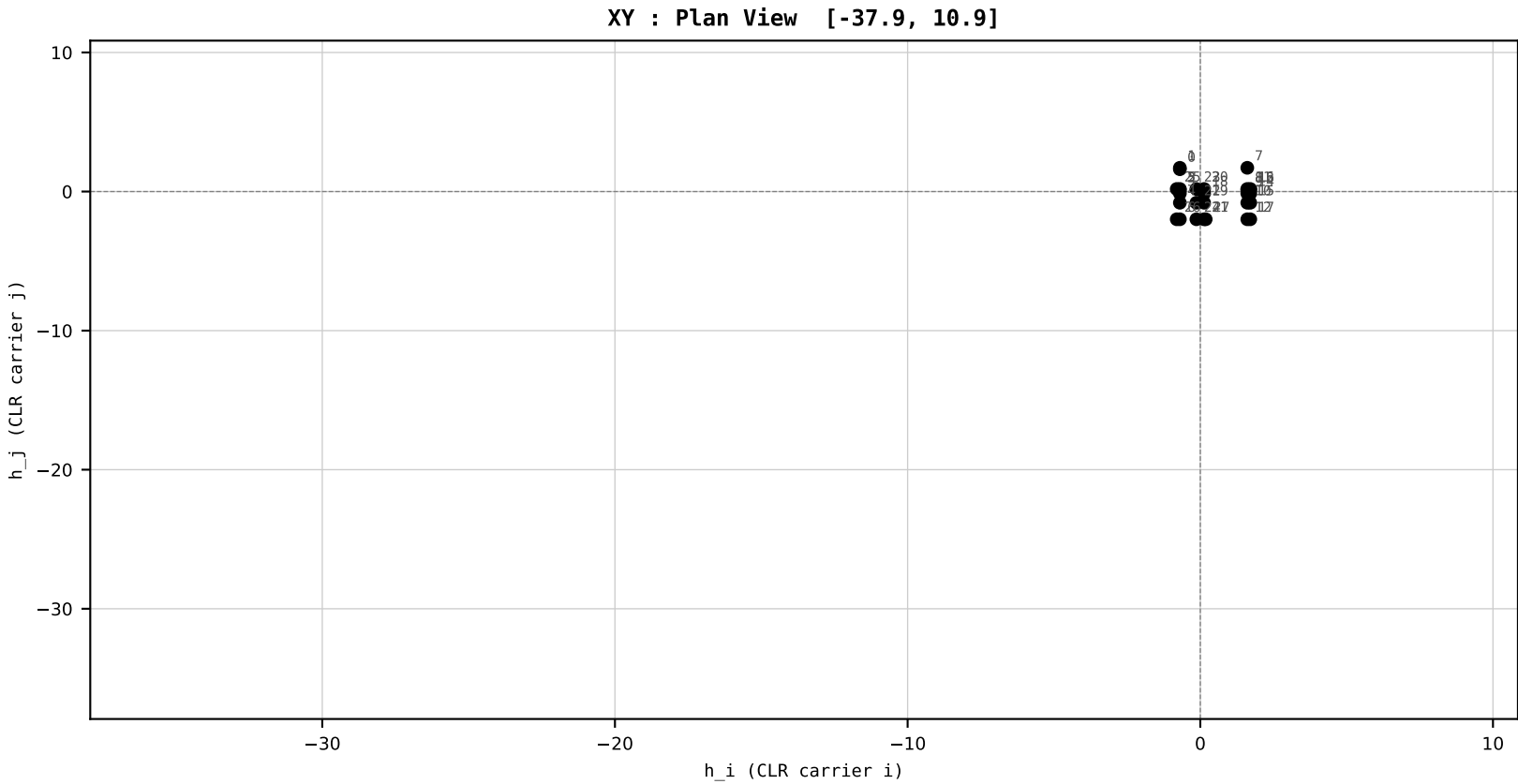
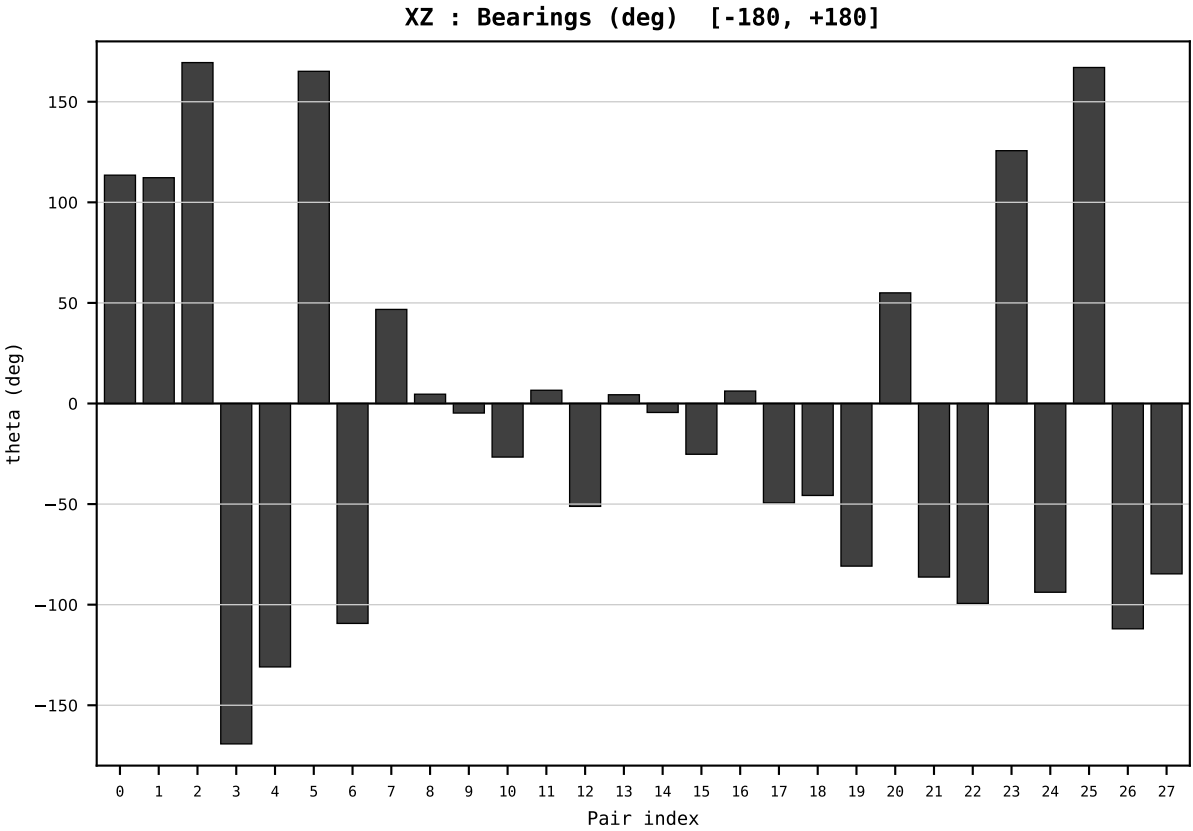
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.246834
Ring = Hs-2
E_metric= 10.6662
kappa_HS= 40.44
omega = 4.7105 deg
d_A = 0.339626
Helm = Nuclear
Helm d = +0.239503
DR = 3.700 HLR
DR ratio= 40.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



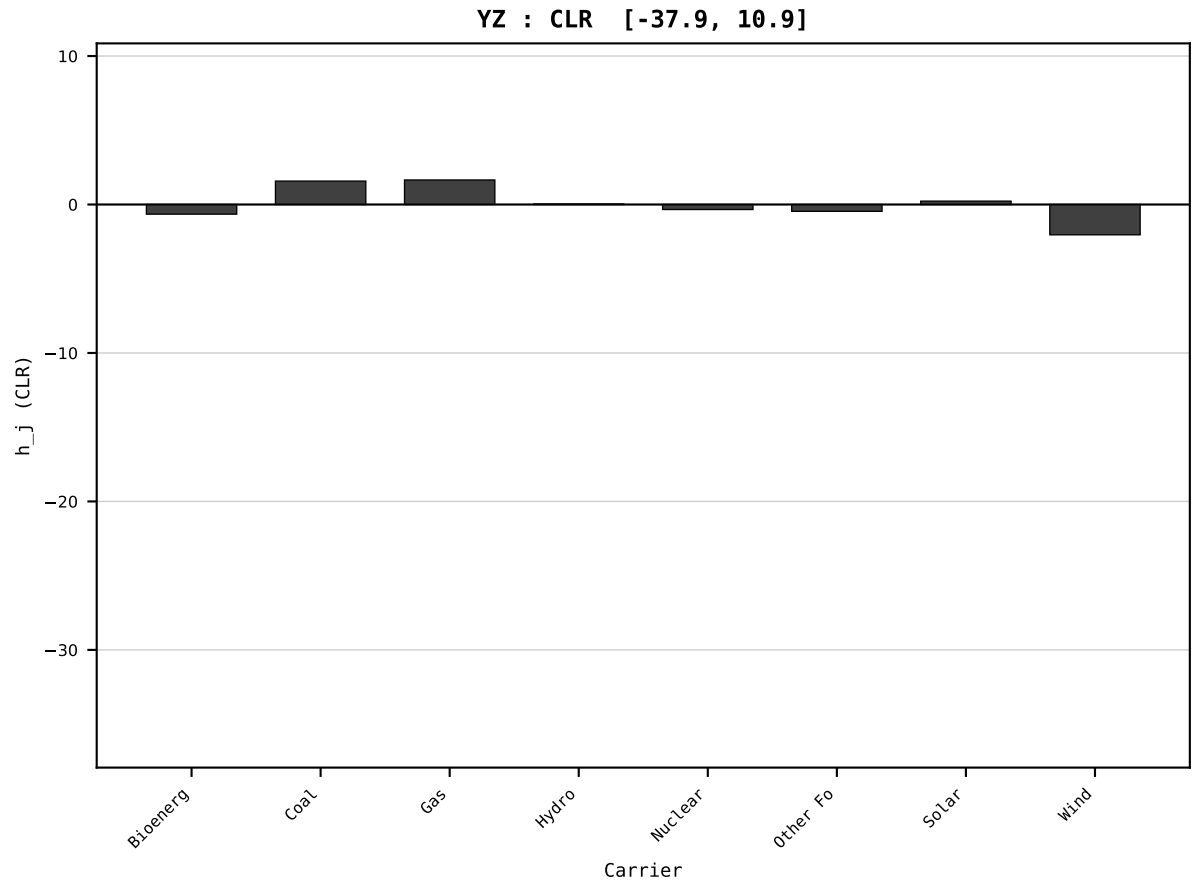
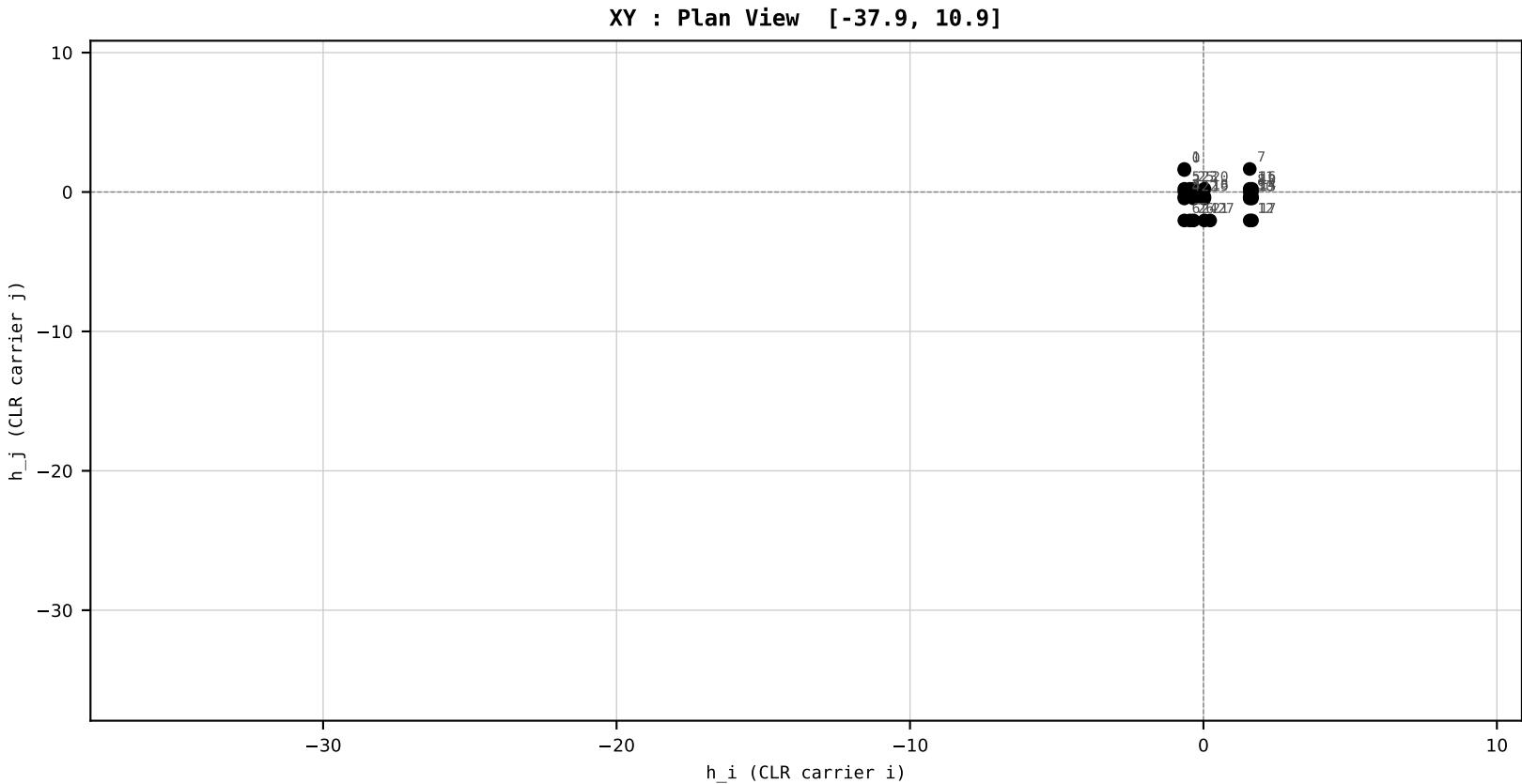
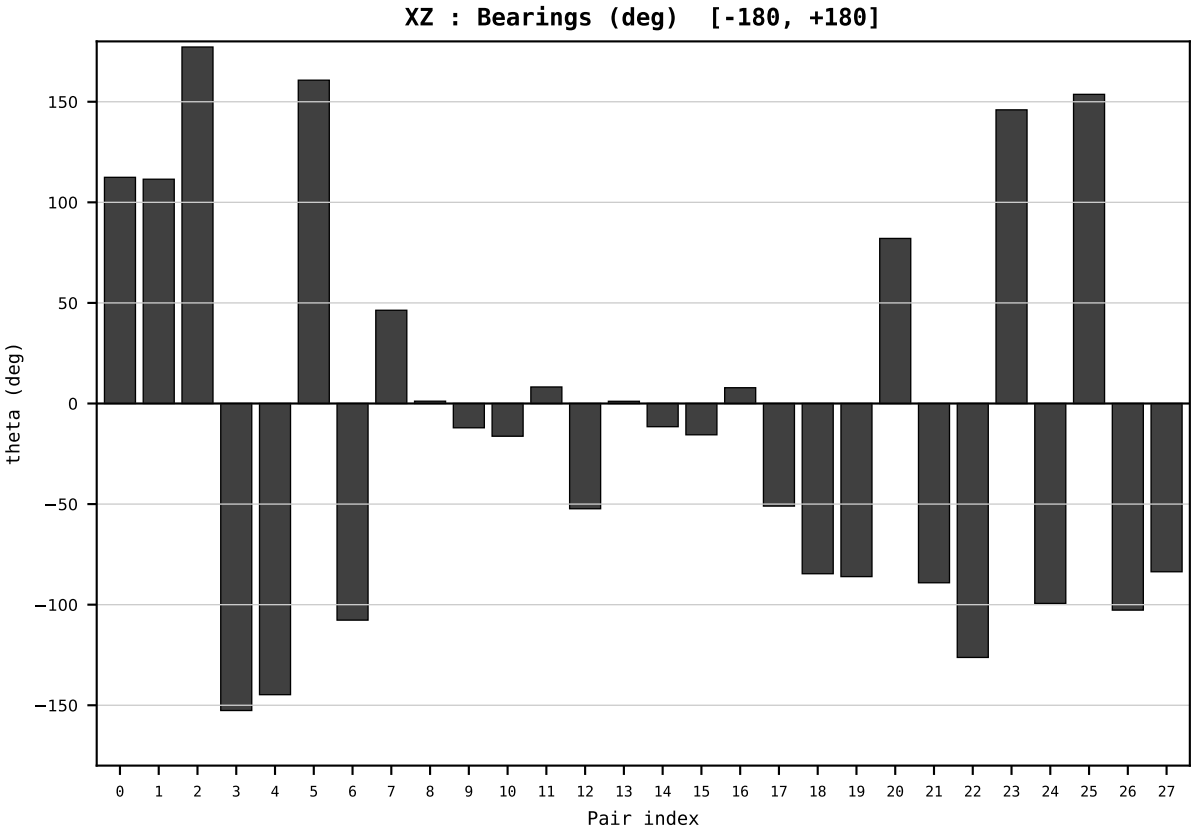
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.236919
Ring = Hs-2
E_metric= 10.1872
kappa_HS= 40.22
omega = 7.4261 deg
d_A = 0.424692
Helm = Other Fossil
Helm d = +0.344772
DR = 3.694 HLR
DR ratio= 40.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

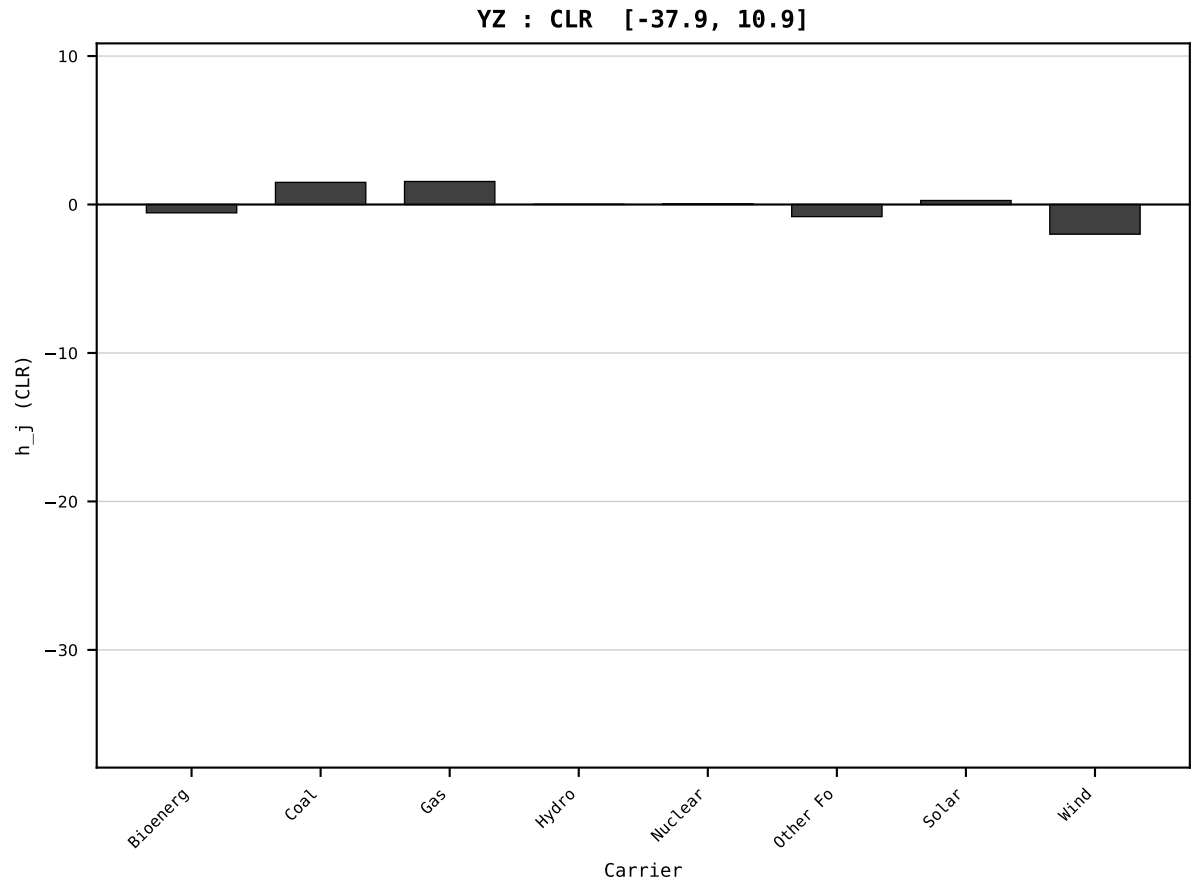
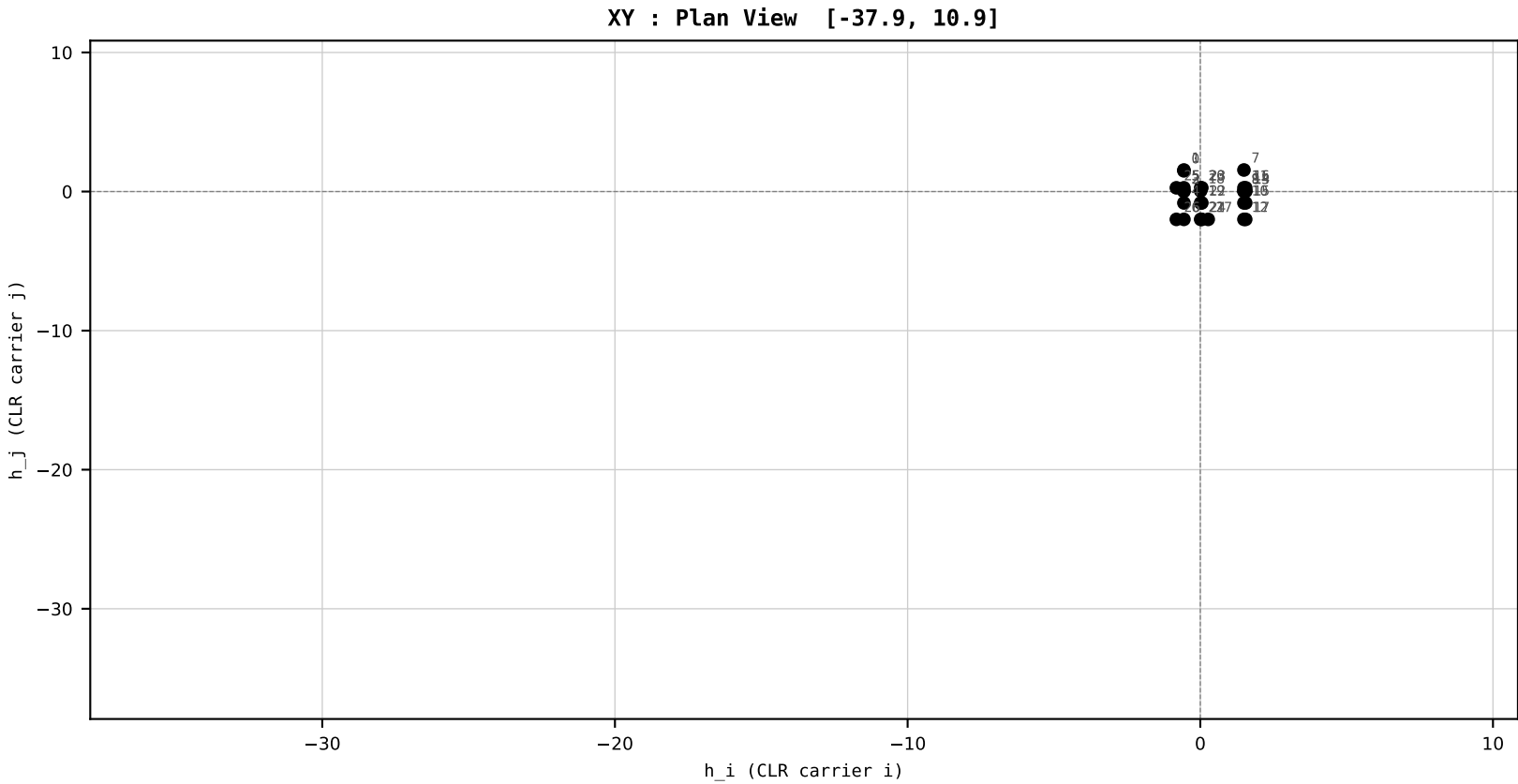
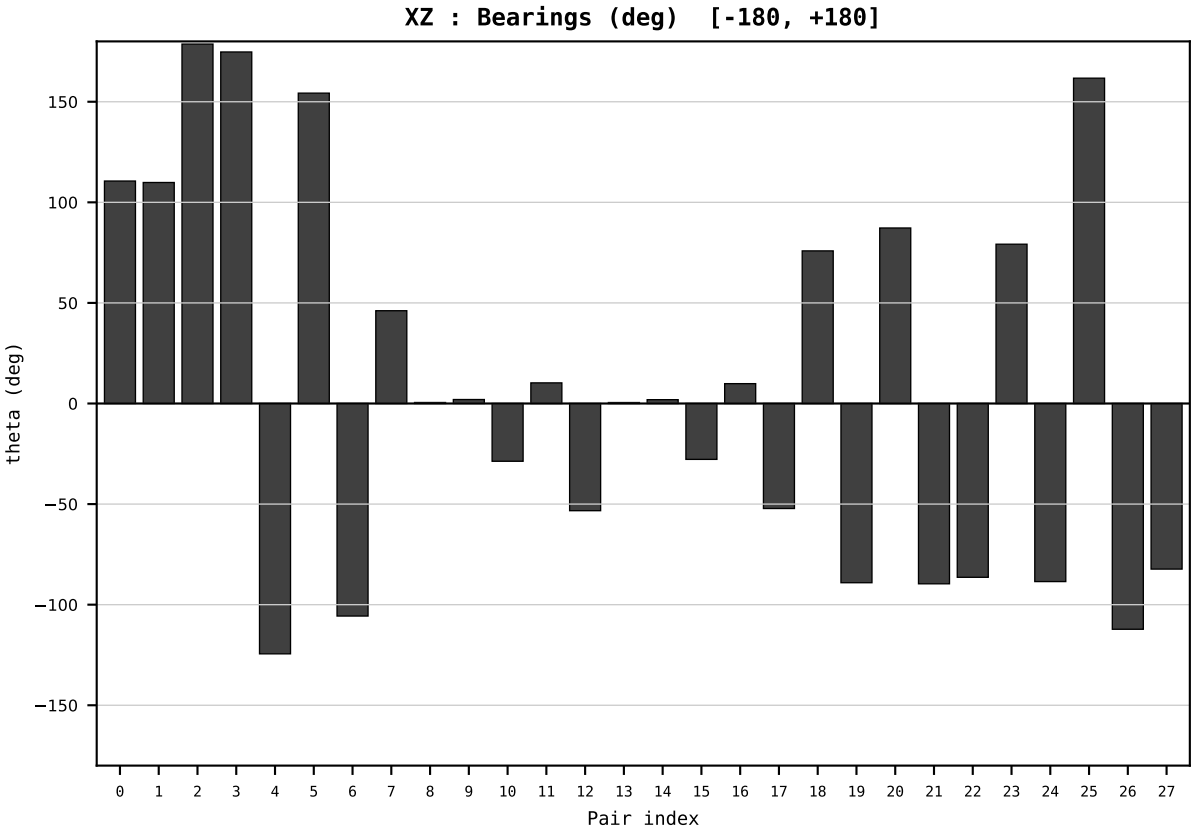
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.213214
Ring = Hs-2
E_metric= 9.6860
kappa_HS= 34.84
omega = 10.0007 deg
d_A = 0.555145
Helm = Nuclear
Helm d = +0.388708
DR = 3.551 HLR
DR ratio= 34.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



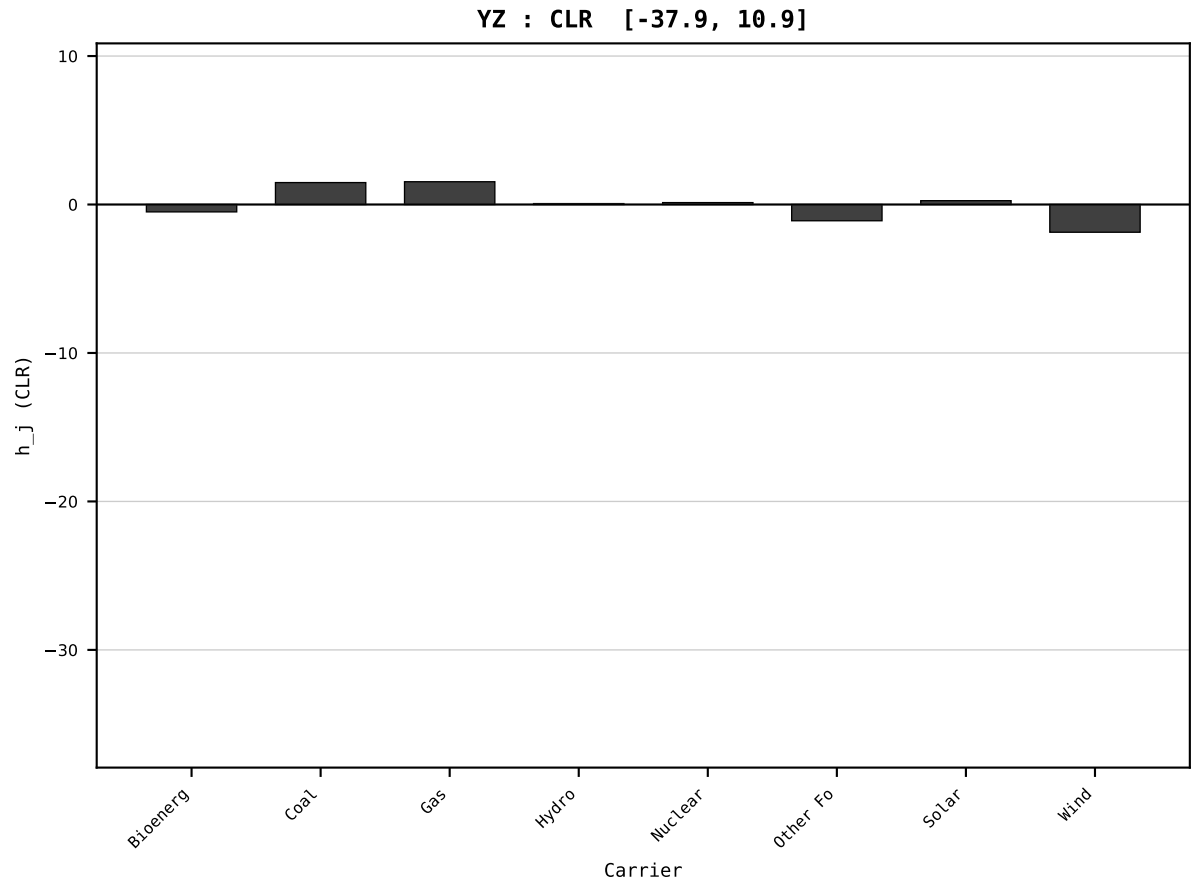
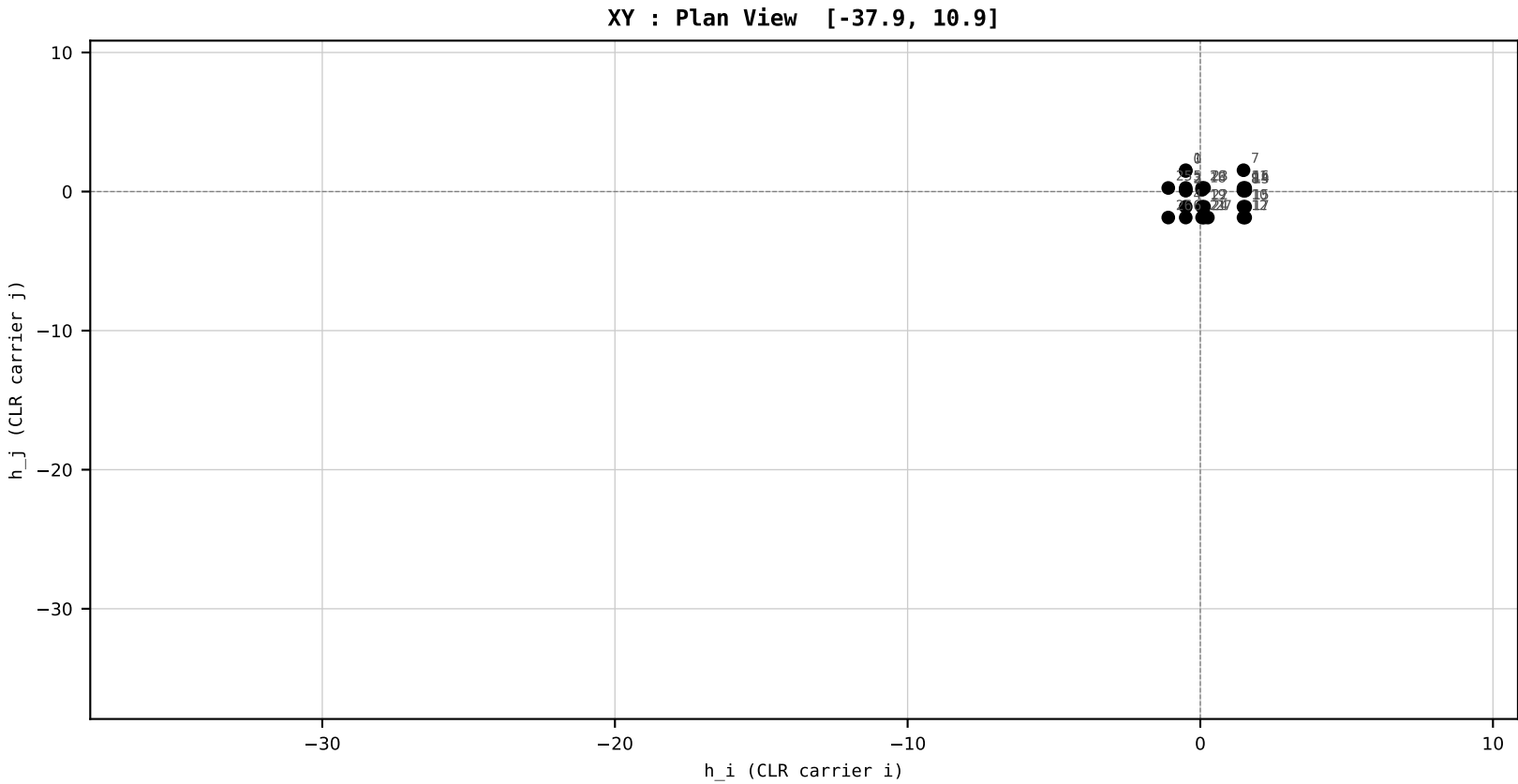
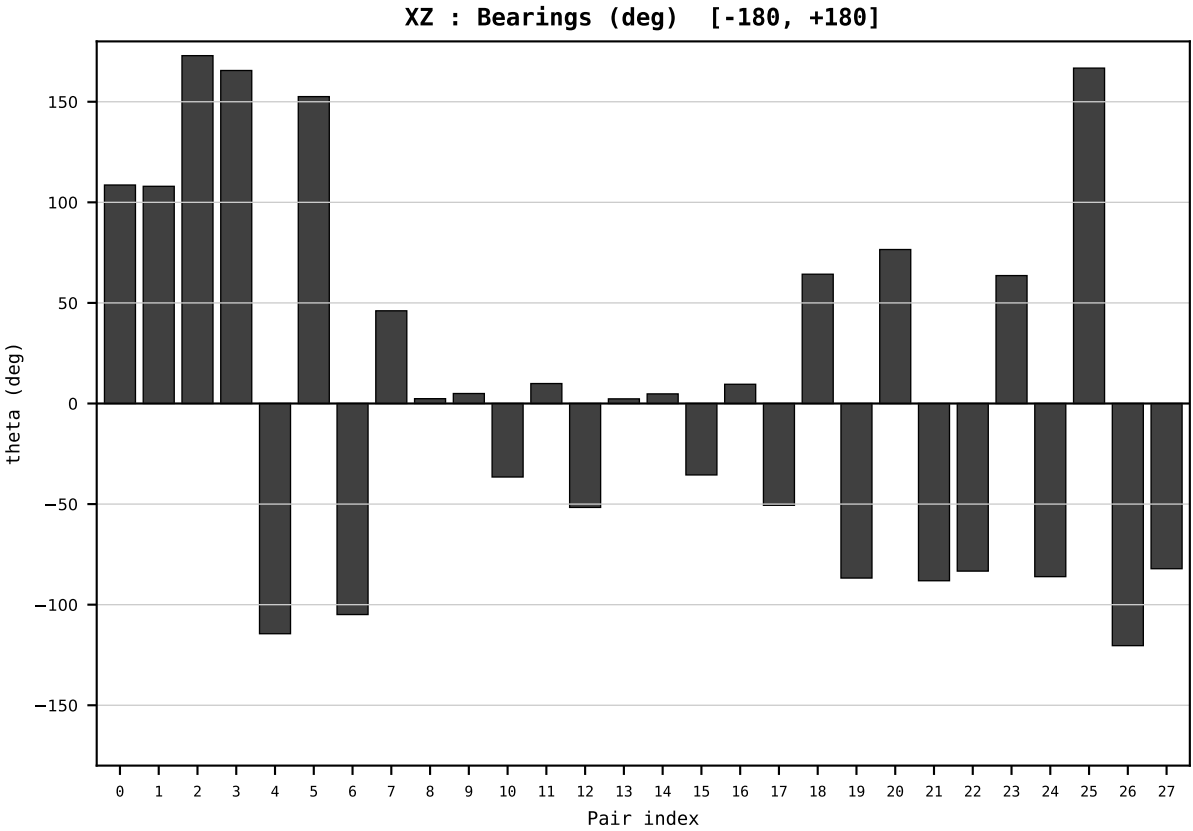
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.209872
Ring = Hs-2
E_metric= 9.5540
kappa_HS= 30.00
omega = 6.0418 deg
d_A = 0.327601
Helm = Other Fossil
Helm d = -0.277449
DR = 3.401 HLR
DR ratio= 30.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



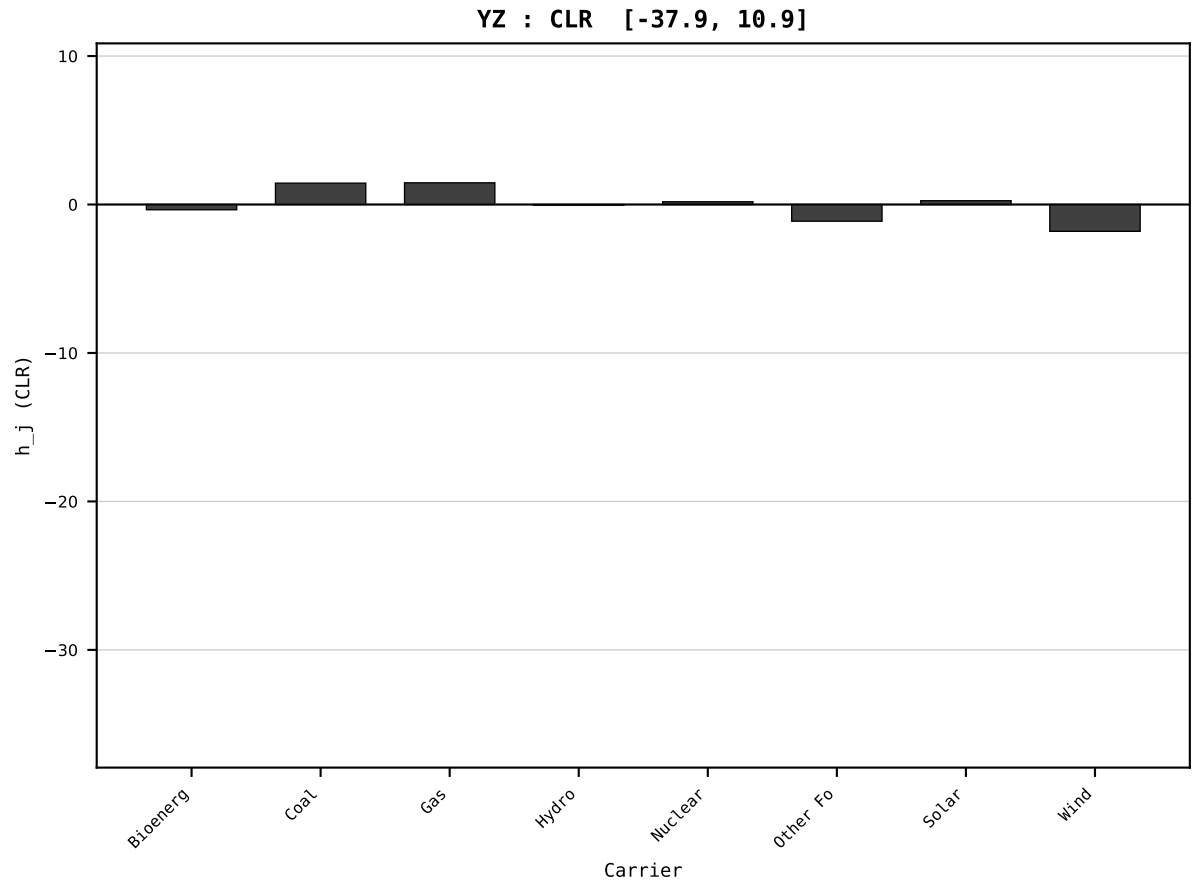
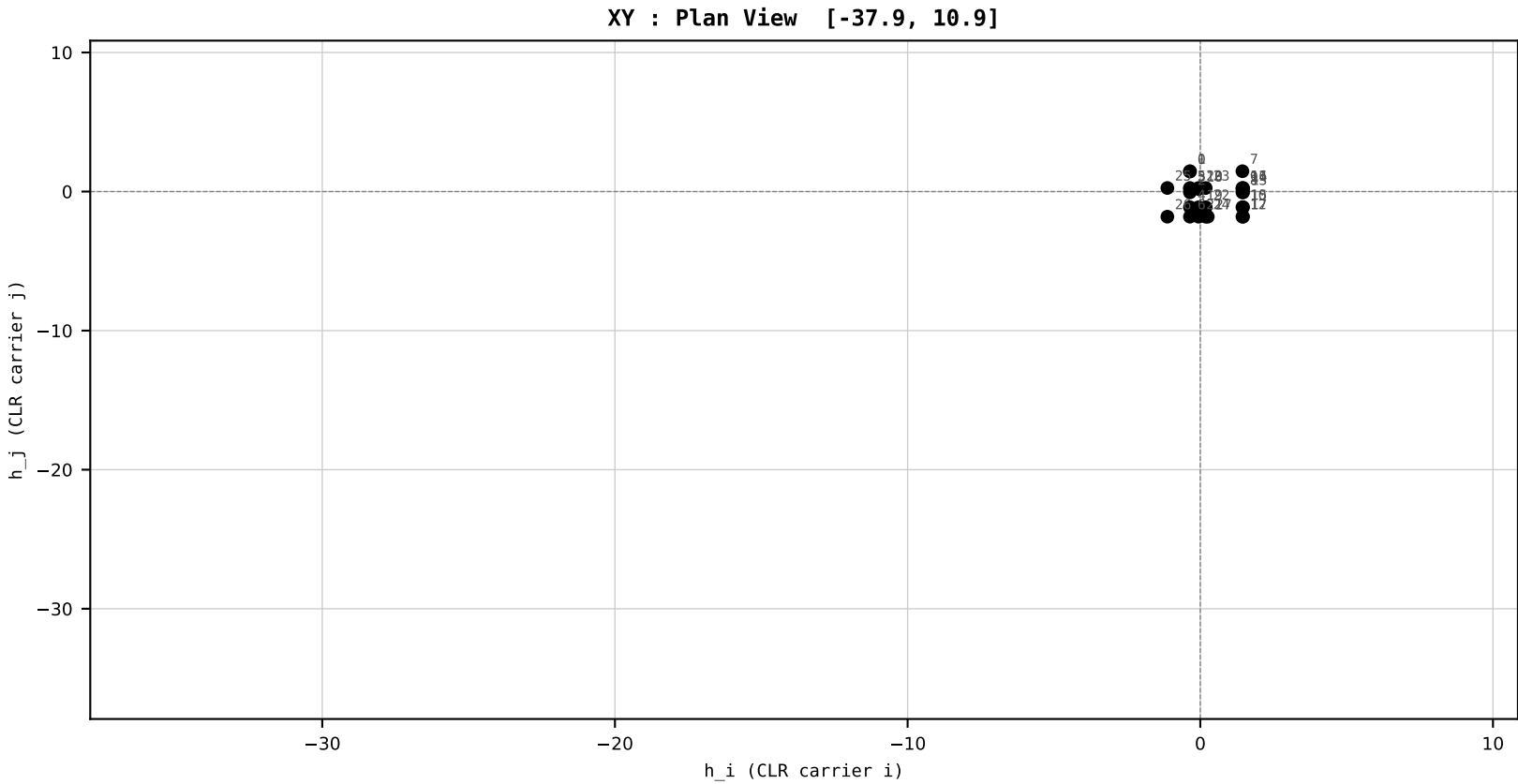
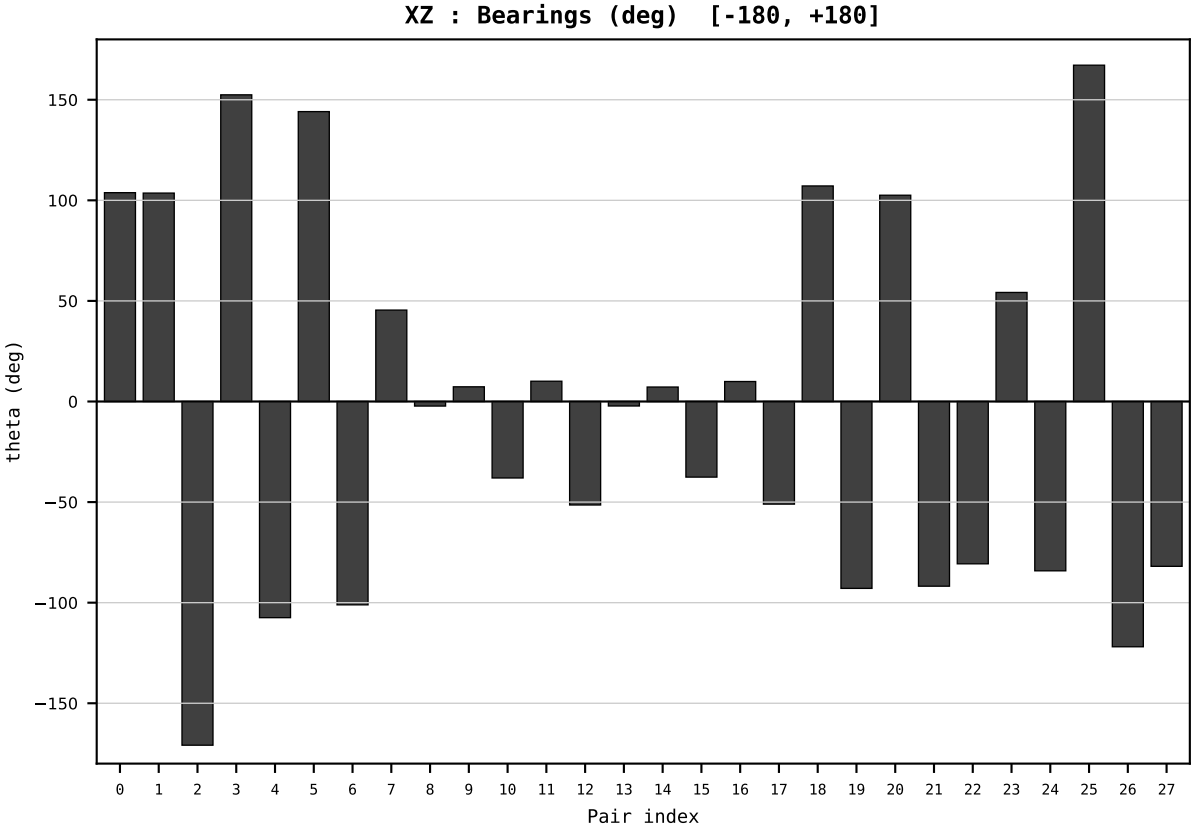
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

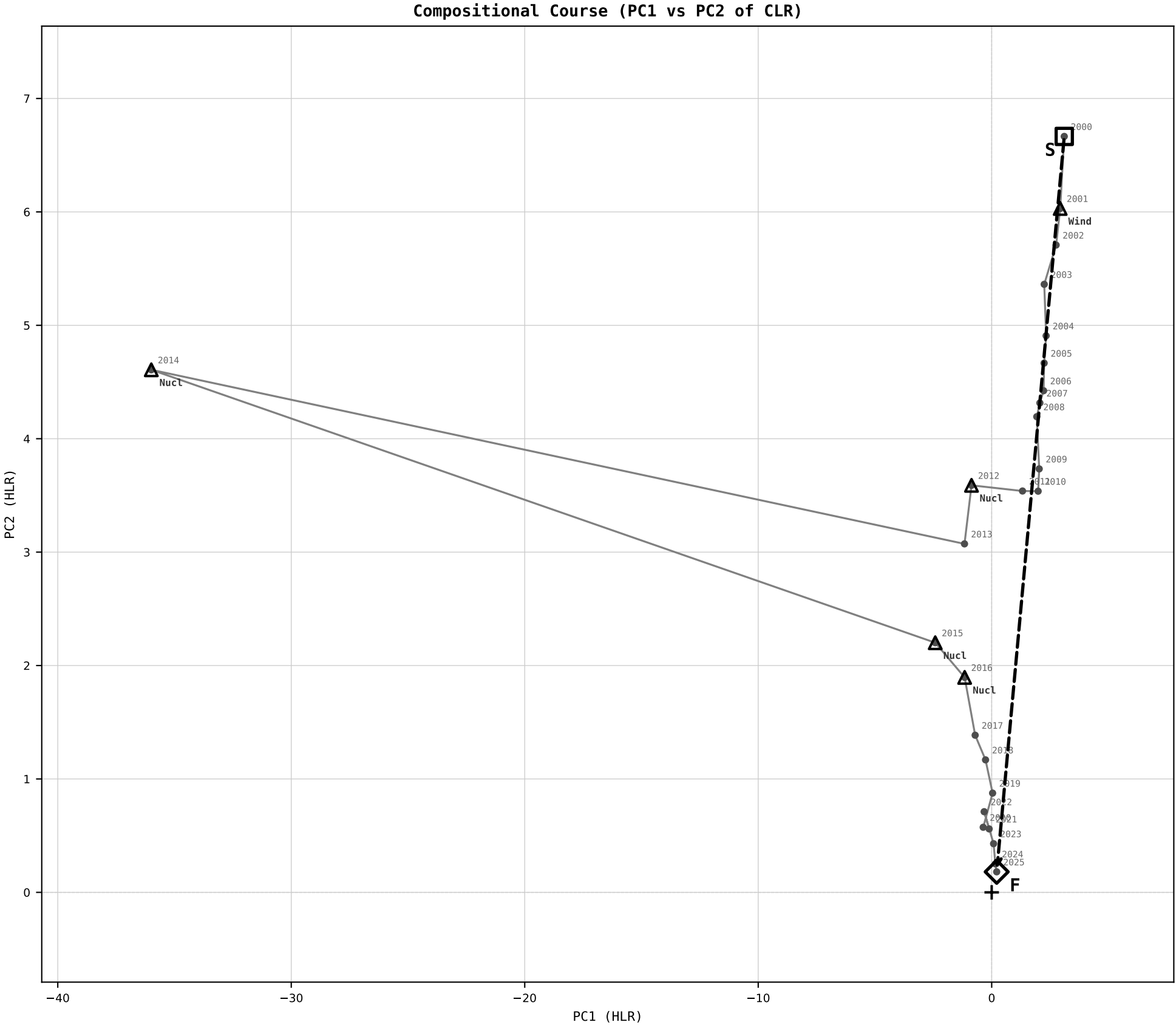
Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.197825
Ring = Hs-2
E_metric= 8.9676
kappa_HS= 26.27
omega = 3.7598 deg
d_A = 0.221651
Helm = Bioenergy
Helm d = +0.143563
DR = 3.268 HLR
DR ratio= 26.3x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 7.2233 HLR
Path length = 82.5468 HLR
Directness = 0.0875

DR start = 7.973 HLR
ratio = 2901.1x
DR final = 3.268 HLR
ratio = 26.3x

E_{metric} start = 70.5673
E_{metric} final = 8.9676

PC1 loadings:
Nuclear +0.934
Solar -0.184
Wind -0.152

PC2 loadings:
Solar -0.716
Other Fo +0.535
Wind -0.326

Top displacement events:
2014 Nuclear d=34.871
2015 Nuclear d=33.668
2012 Nuclear d=2.201
2016 Nuclear d=1.320
2001 Wind d=0.848

The instrument reads.
The expert decides.
The loop stays open.

View B

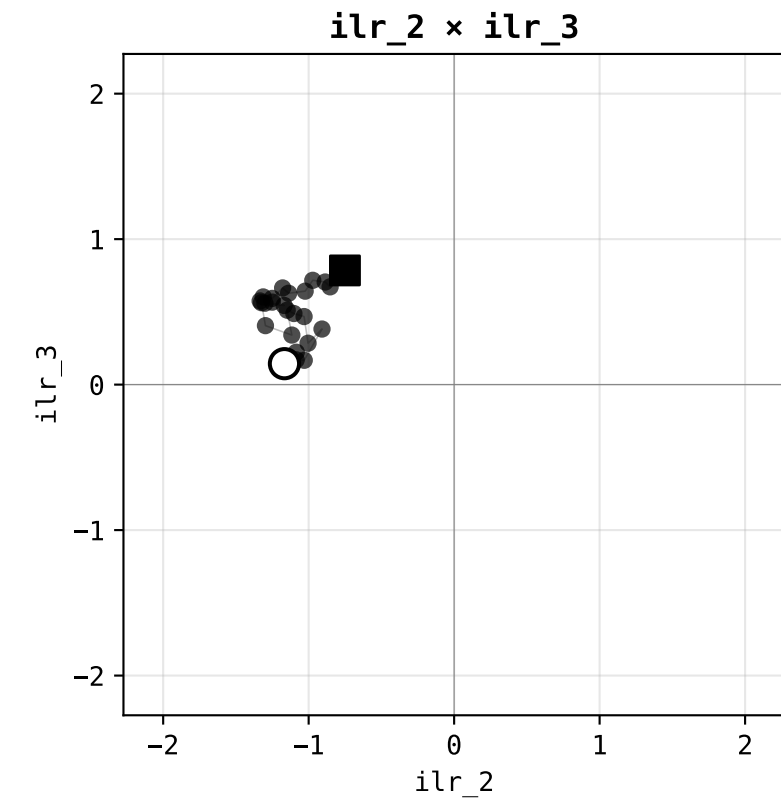
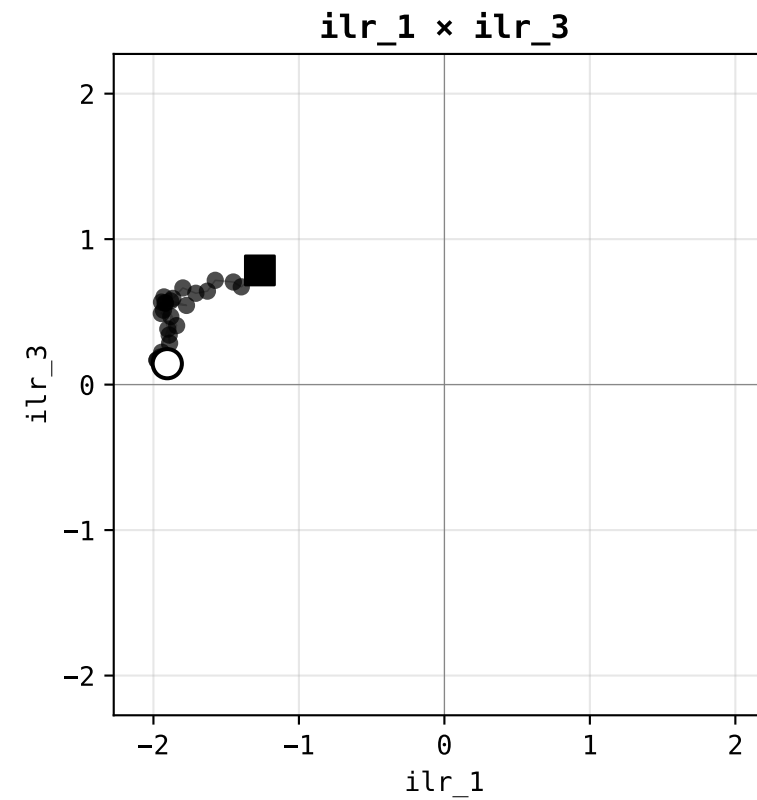
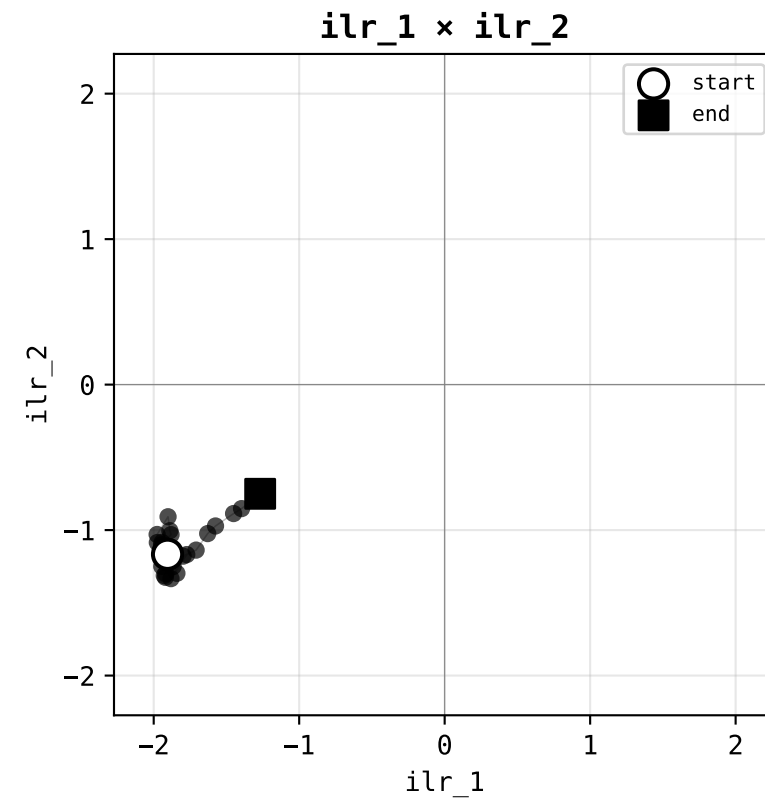
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ JPN Japan

ILR-Helmert Orthogonal Triplet – JPN Japan · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · $N=26$ years · $D=8$ carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:

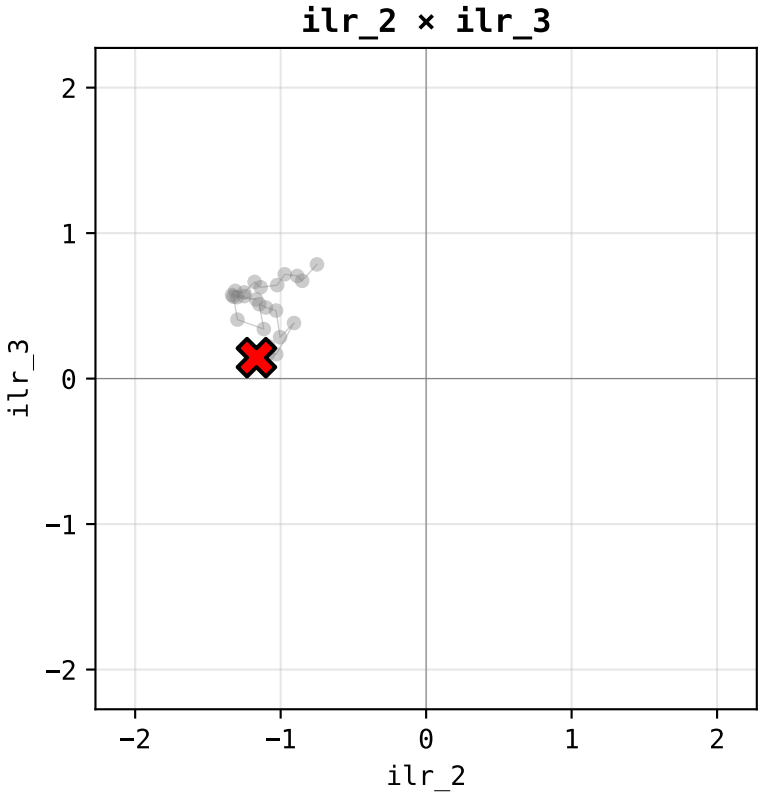
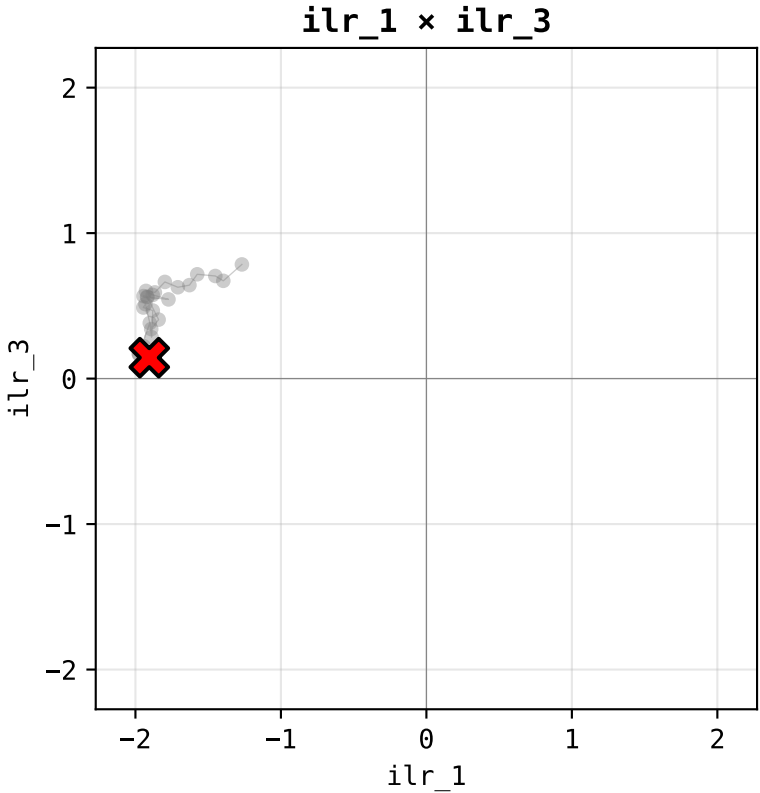
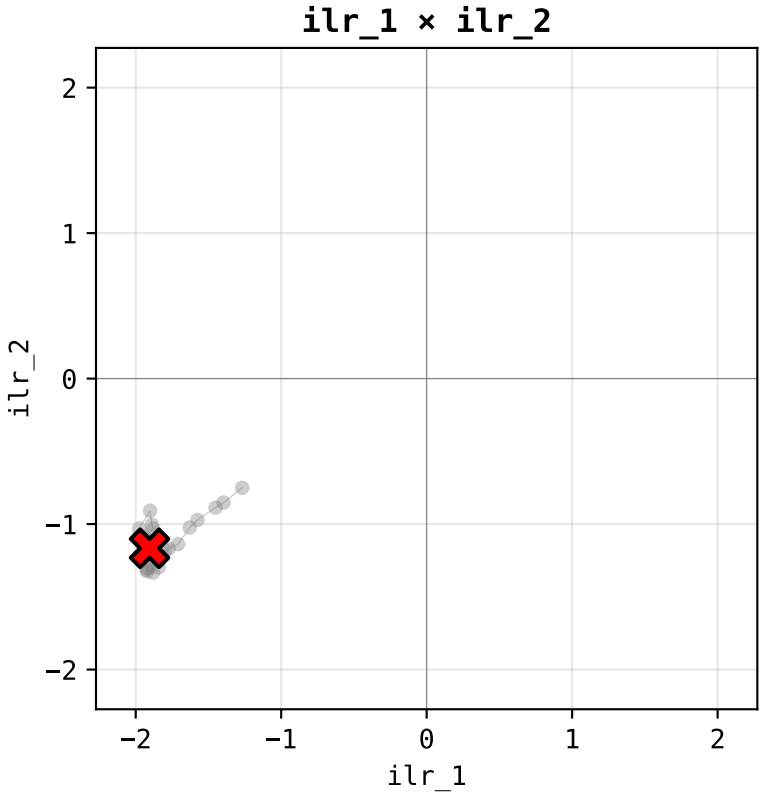
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end year: 2025
N readings: 26

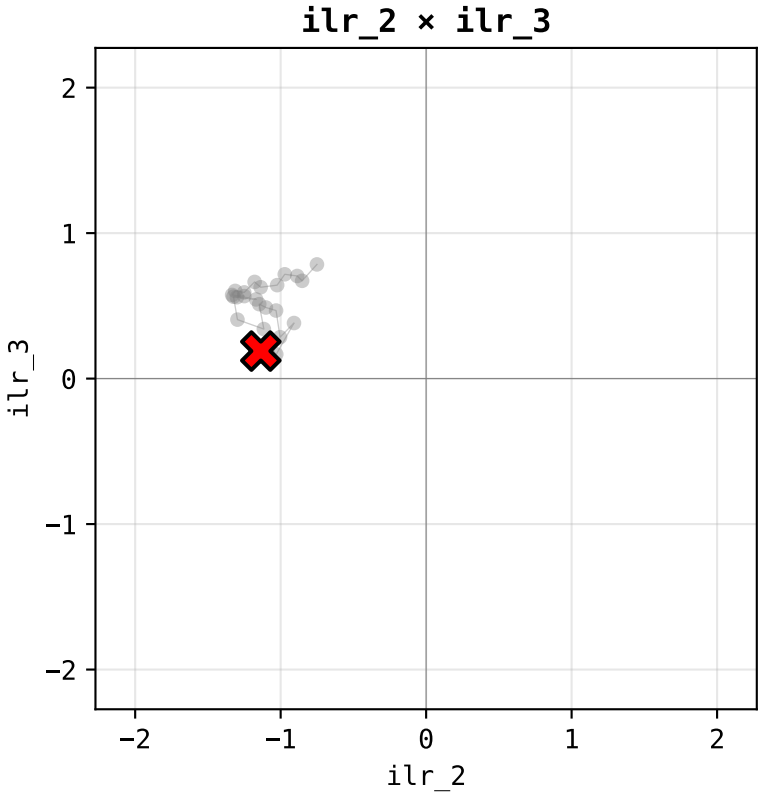
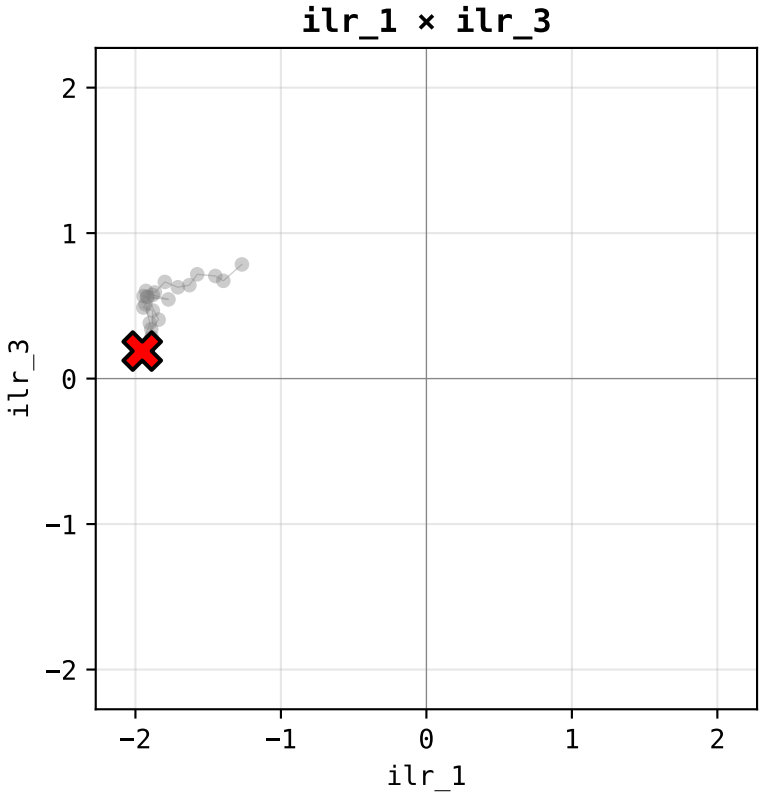
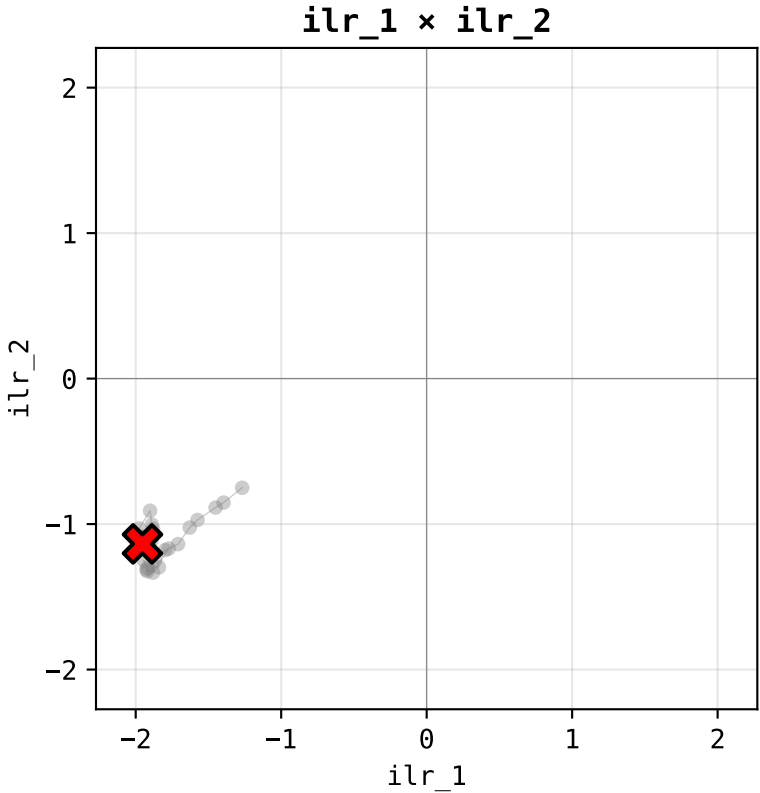
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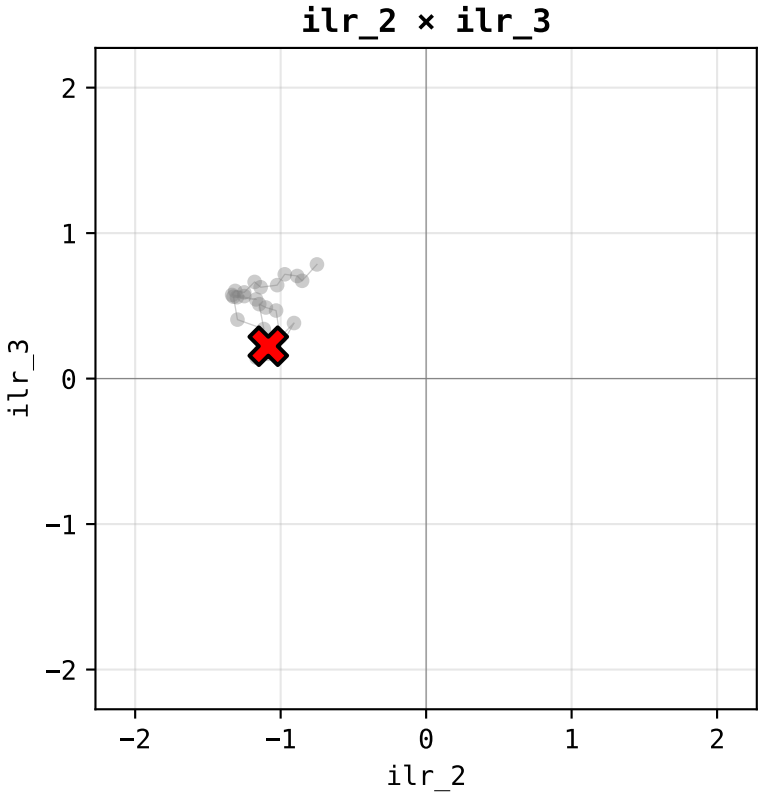
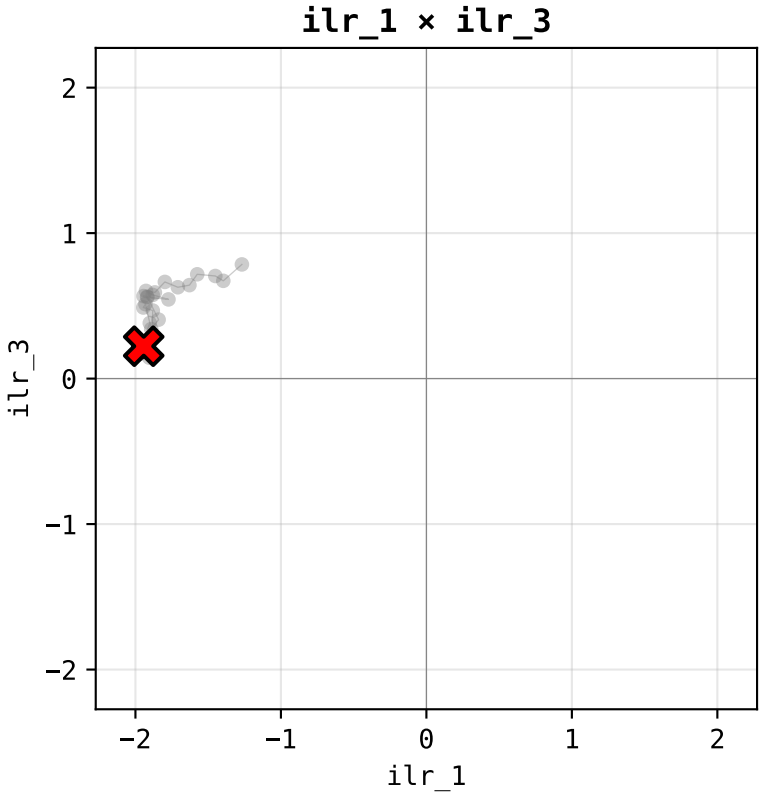
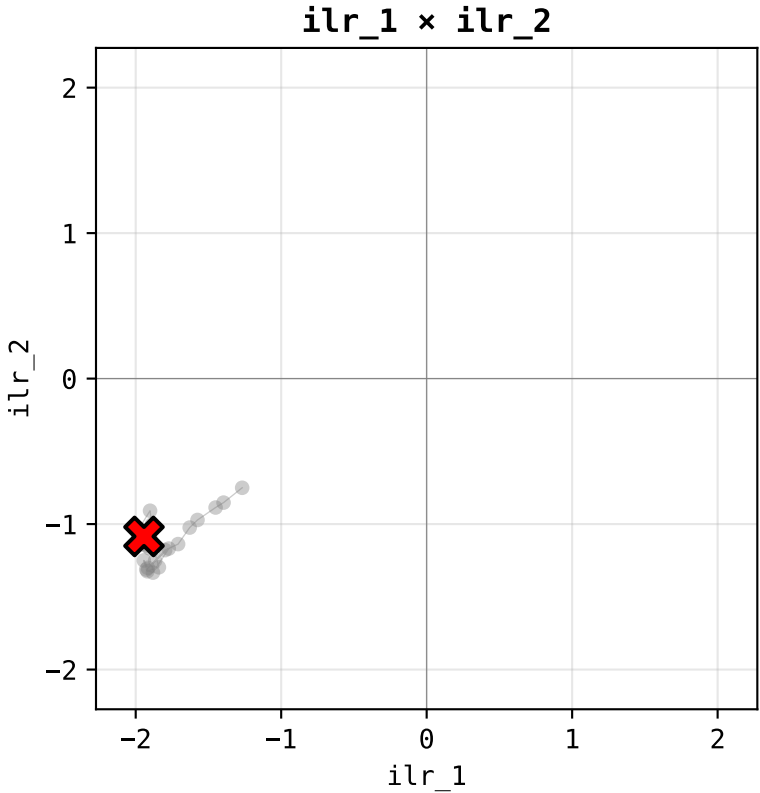
fixed scale ± 2.27 ILR
Order 1 (first principles)
per Output Doctrine v1.0

Symbology:

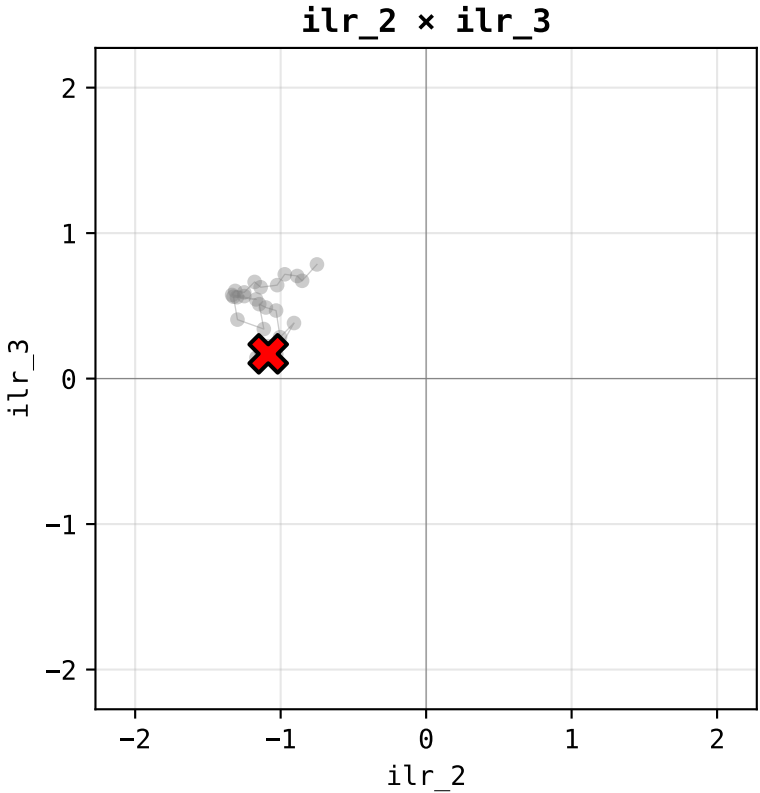
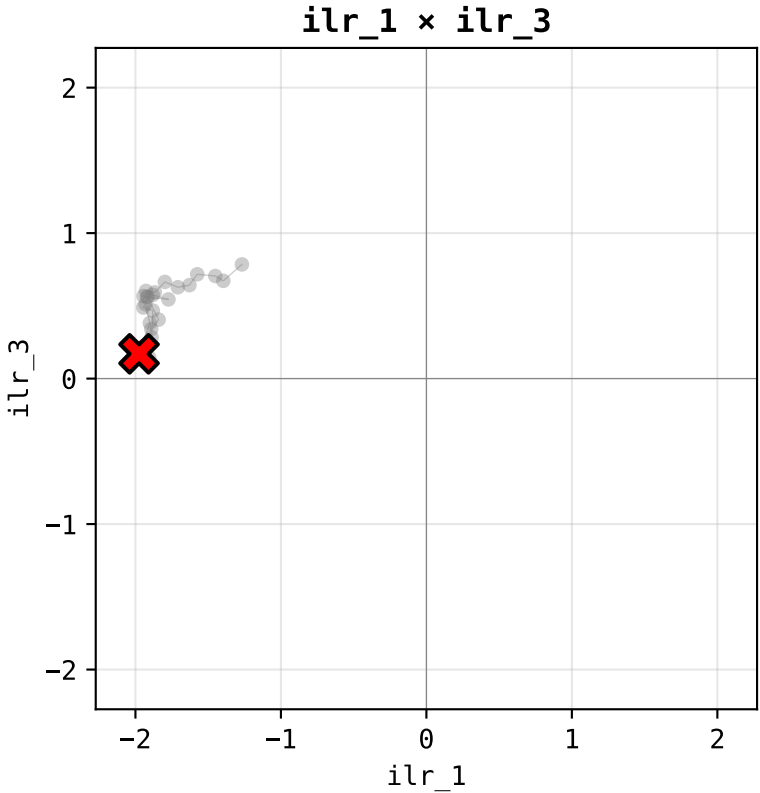
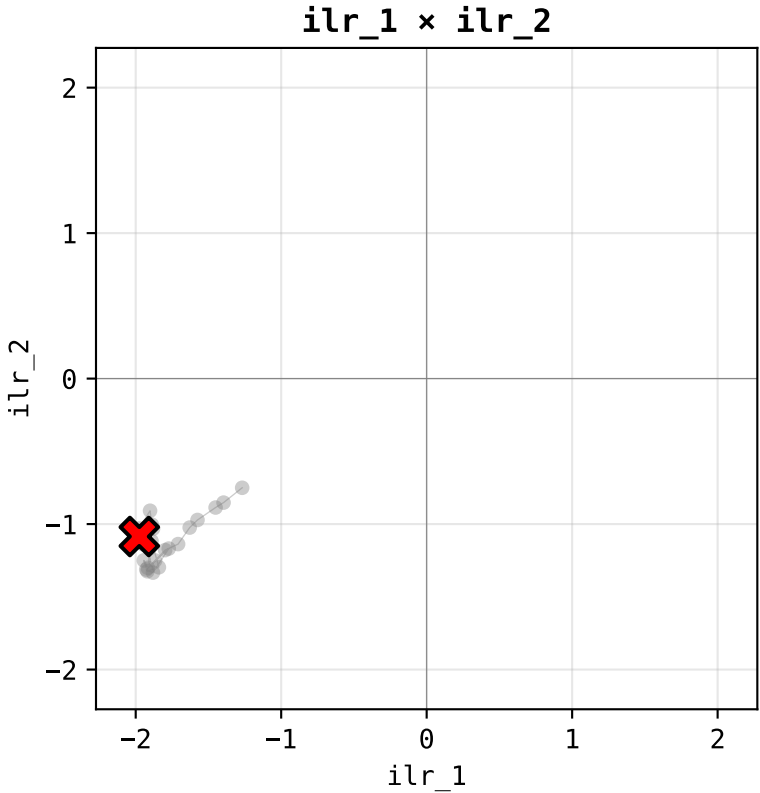
○ start · ■ end
· intermediate years

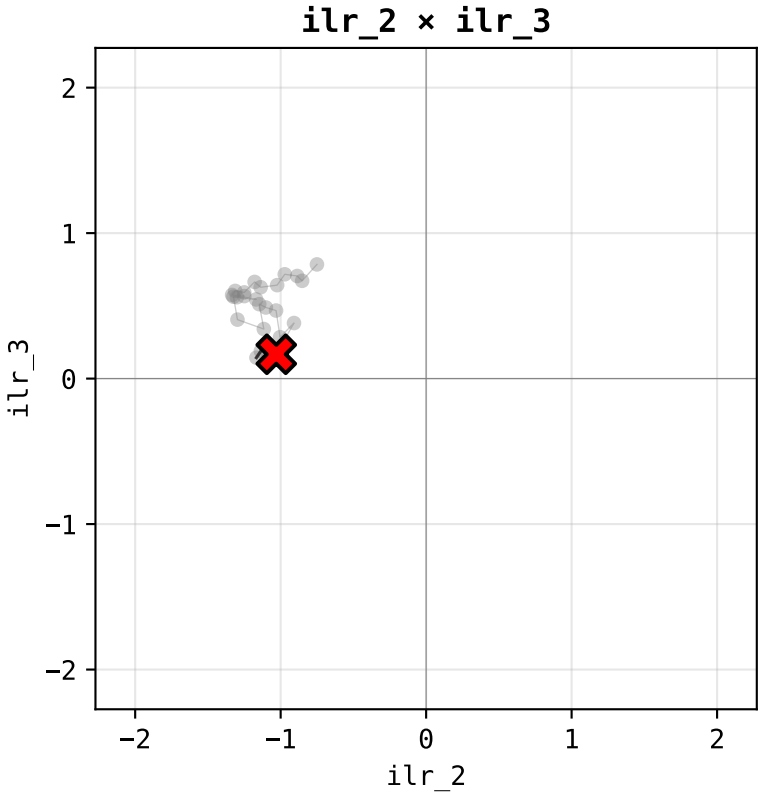
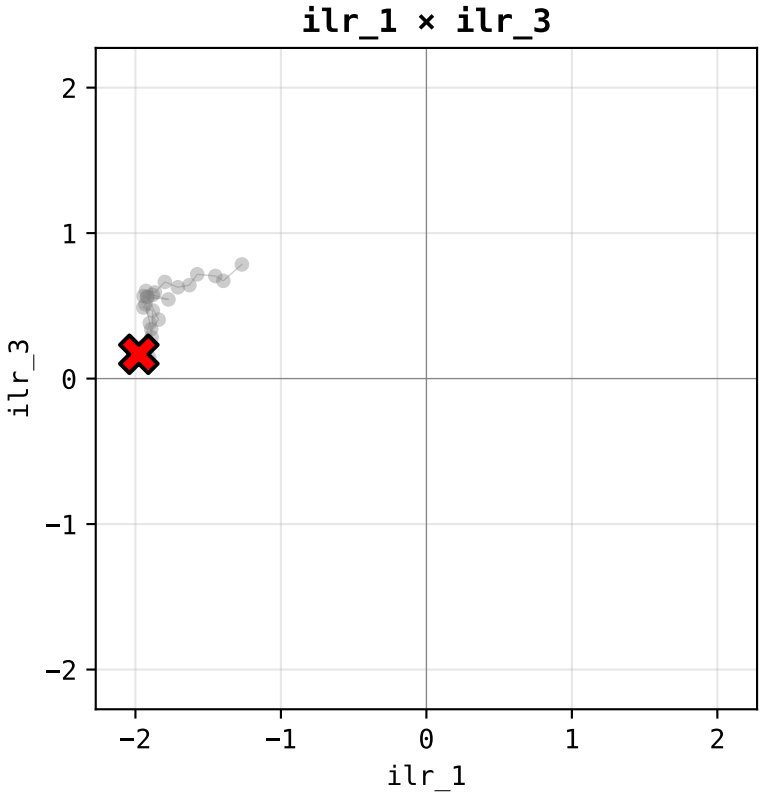
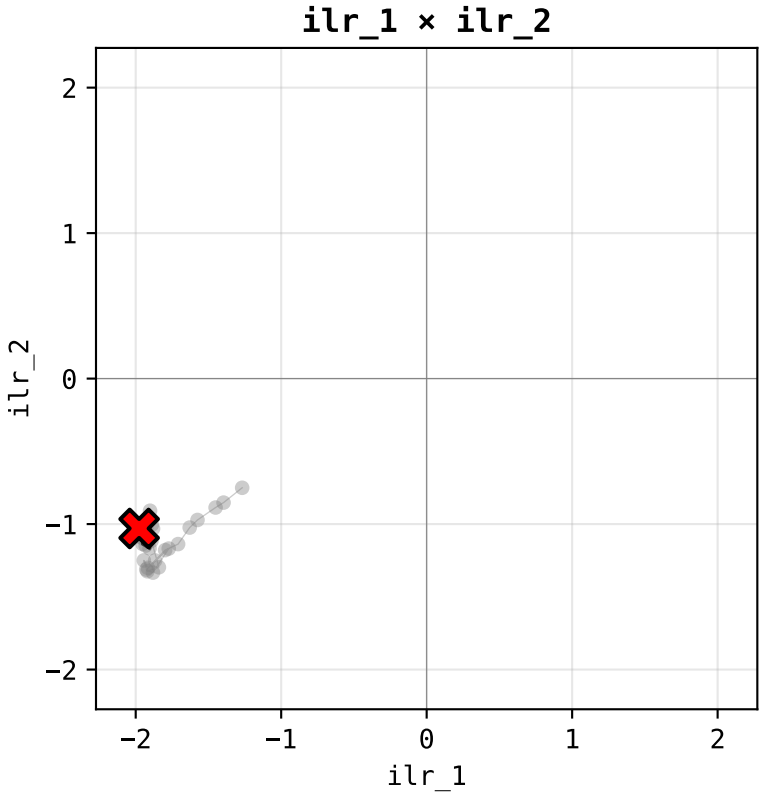


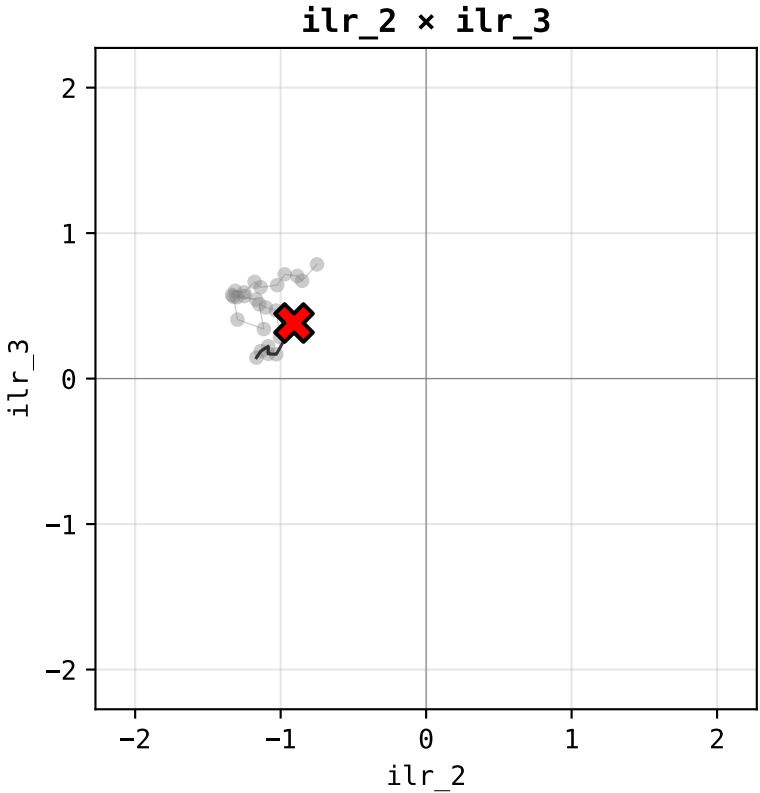
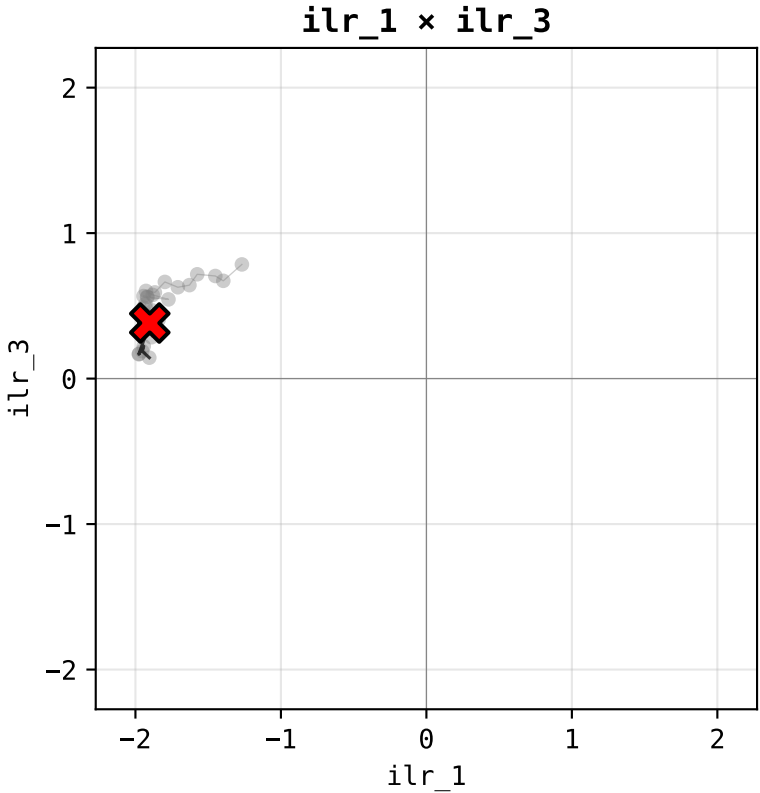
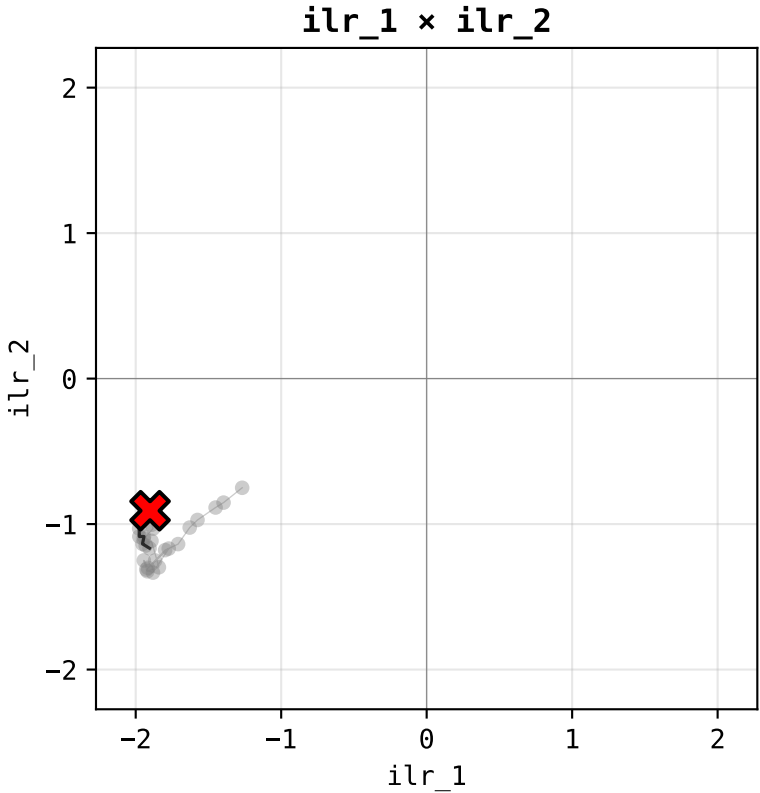




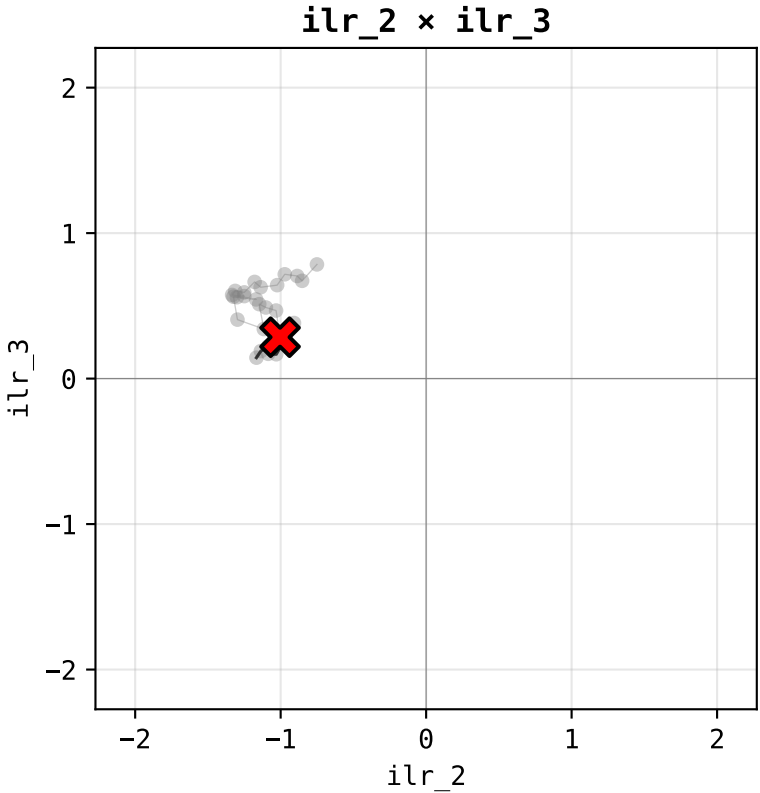
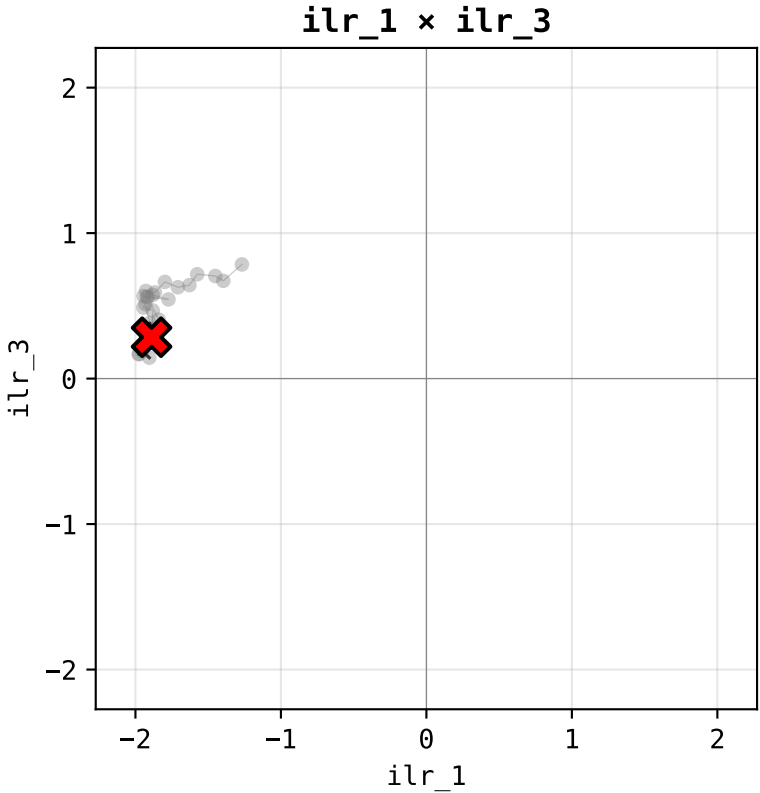
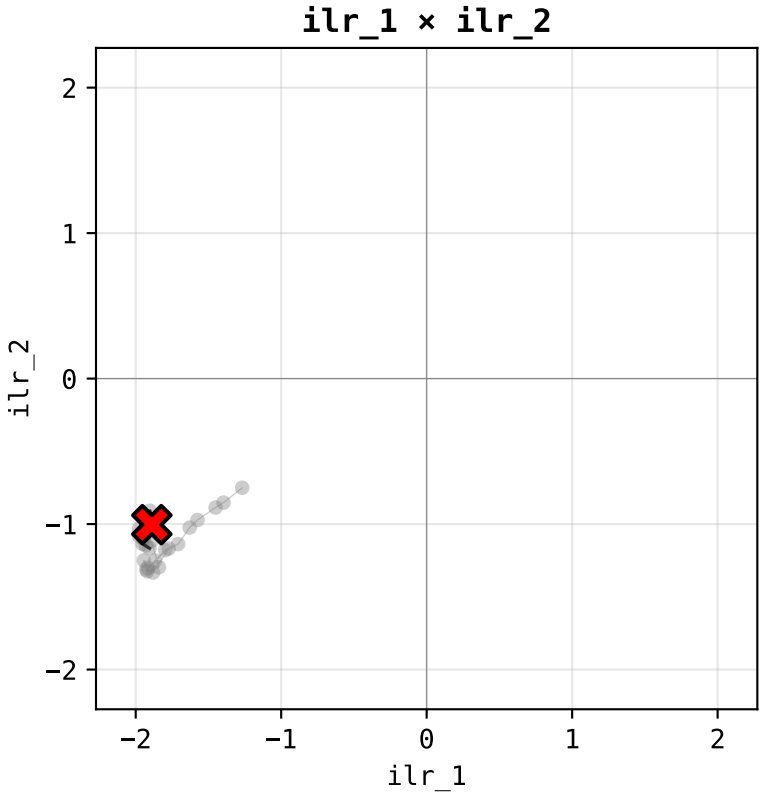
ILR Triplet – JPN Japan · year 2003 (4/26)



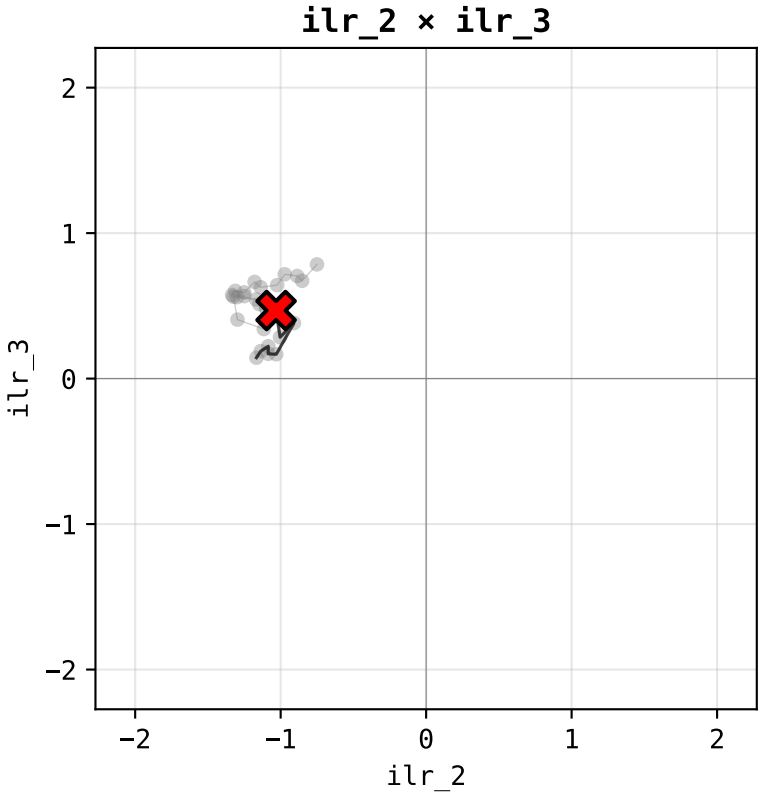
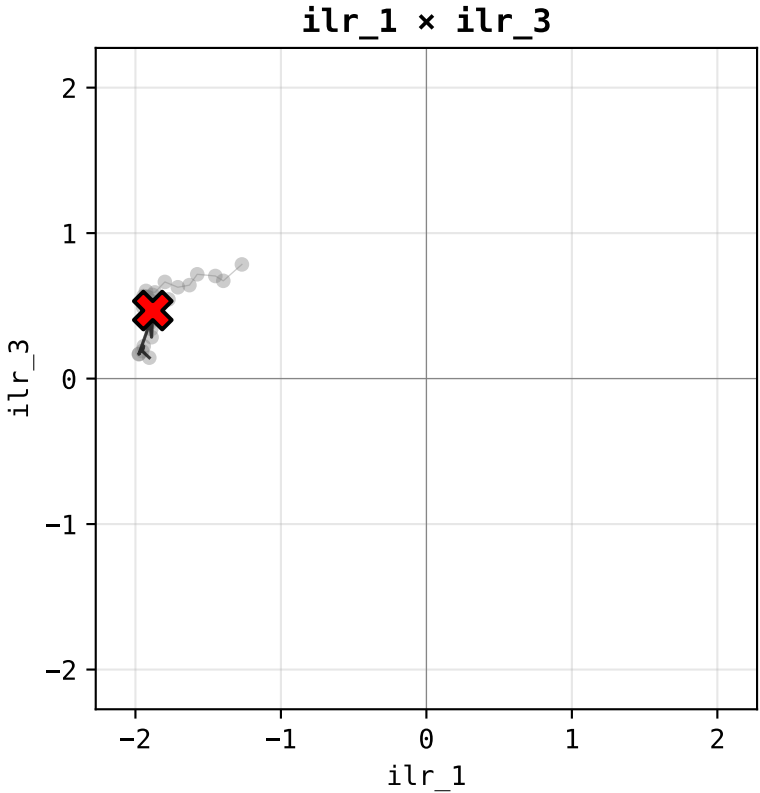
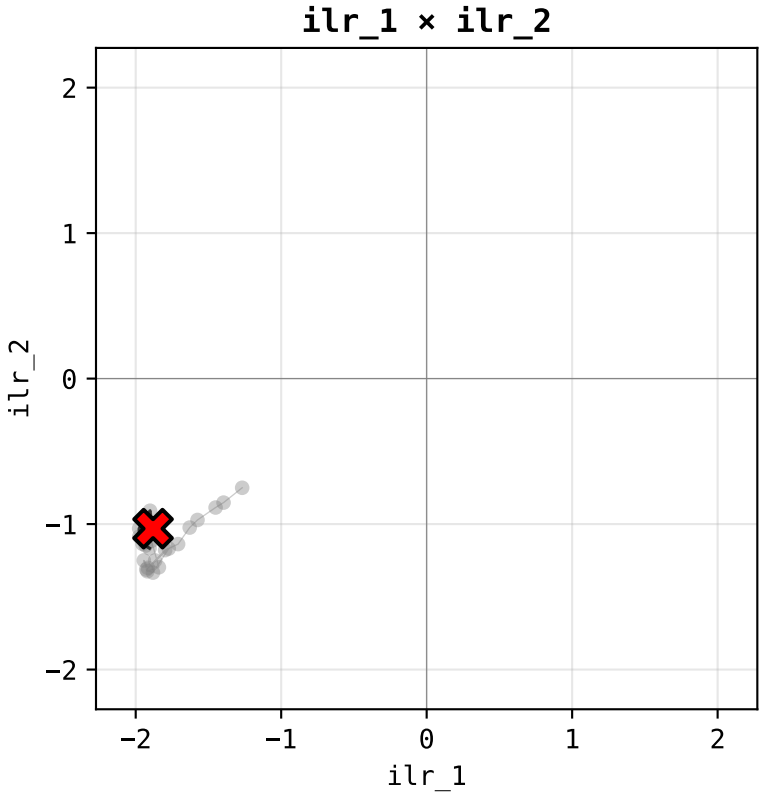




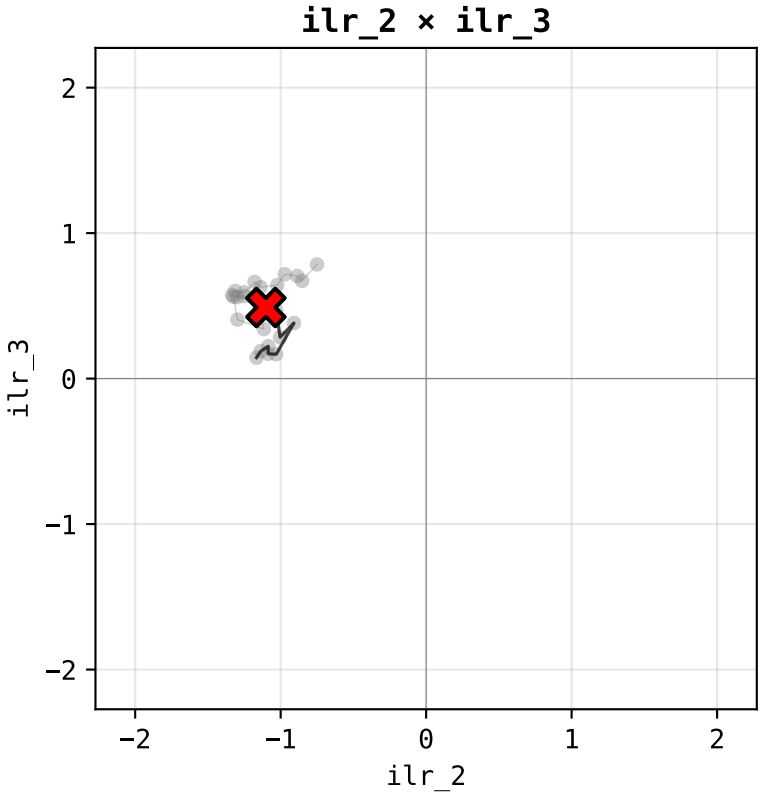
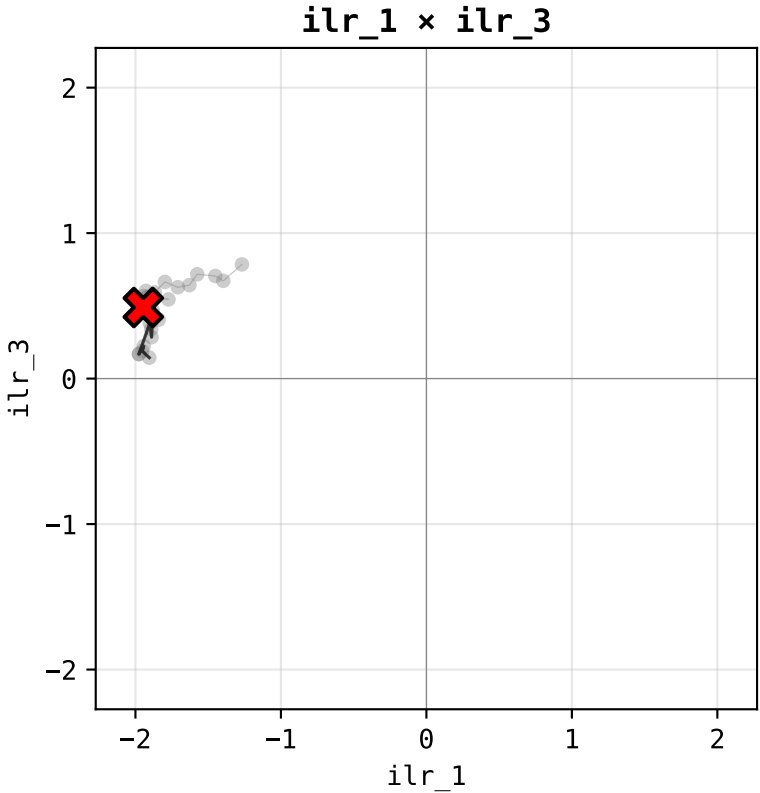
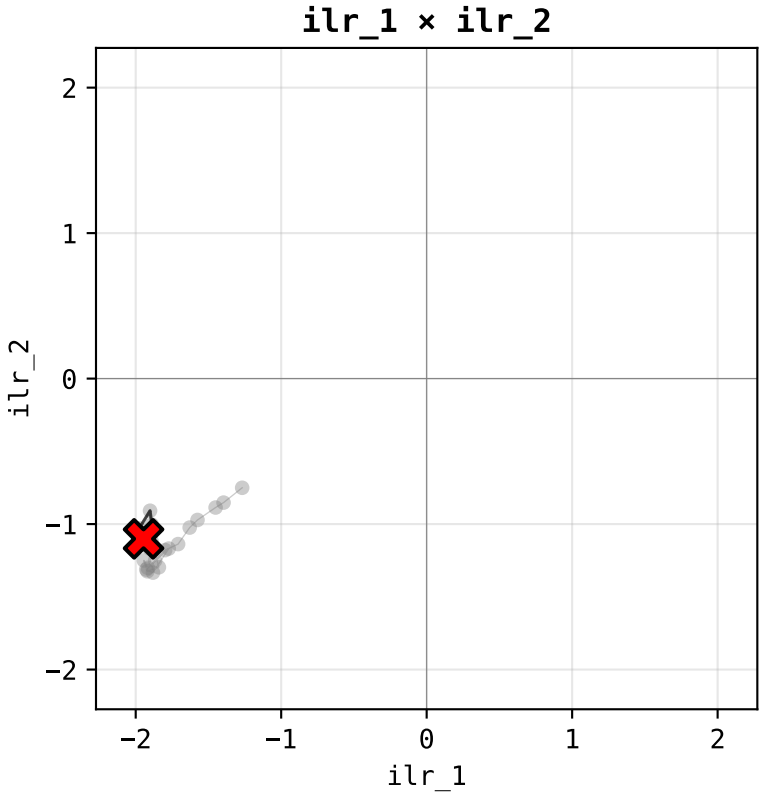
ILR Triplet – JPN Japan · year 2006 (7/26)



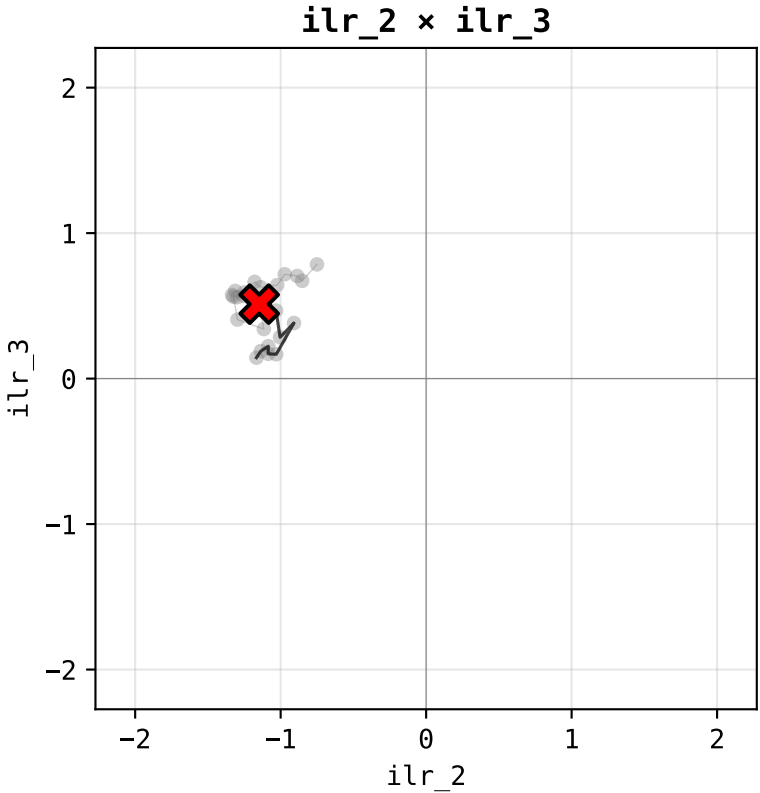
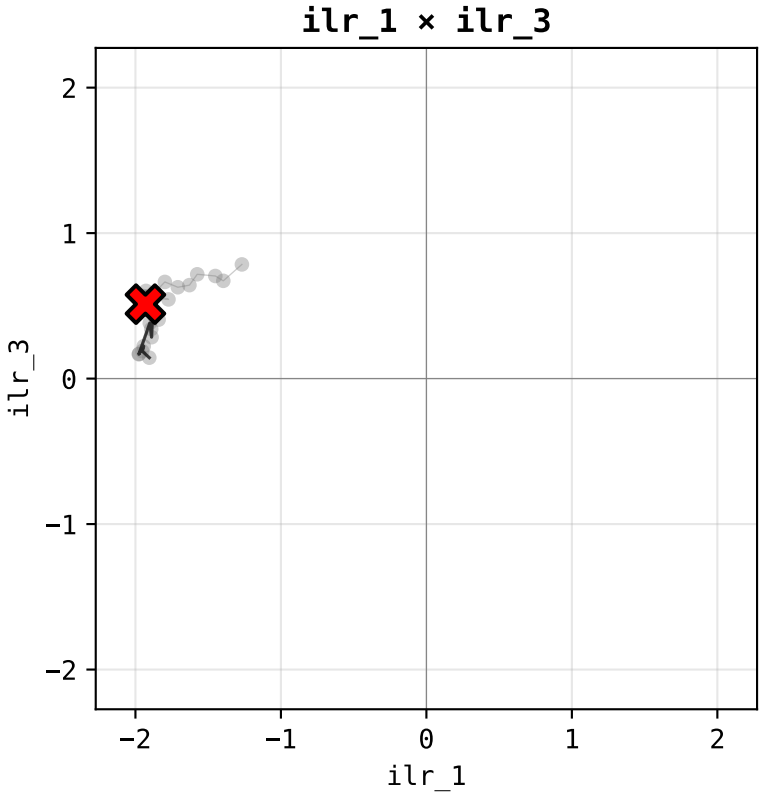
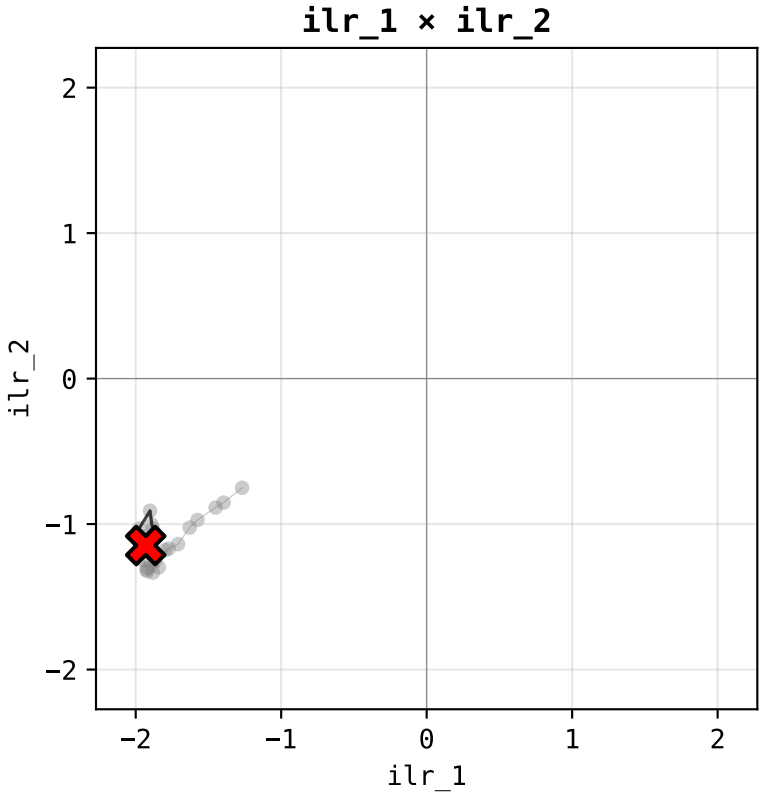
ILR Triplet – JPN Japan · year 2007 (8/26)



ILR Triplet – JPN Japan · year 2008 (9/26)

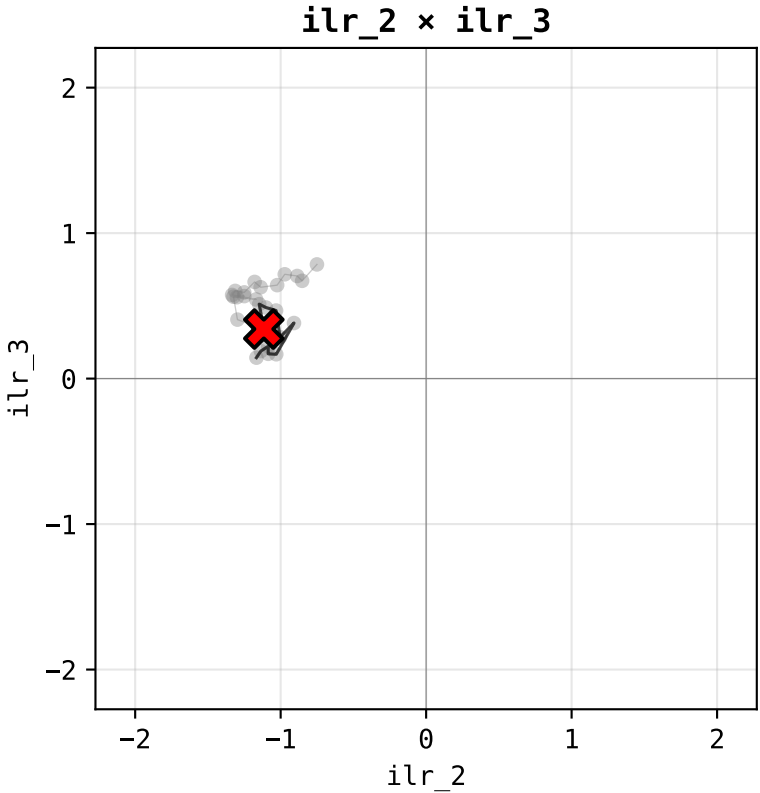
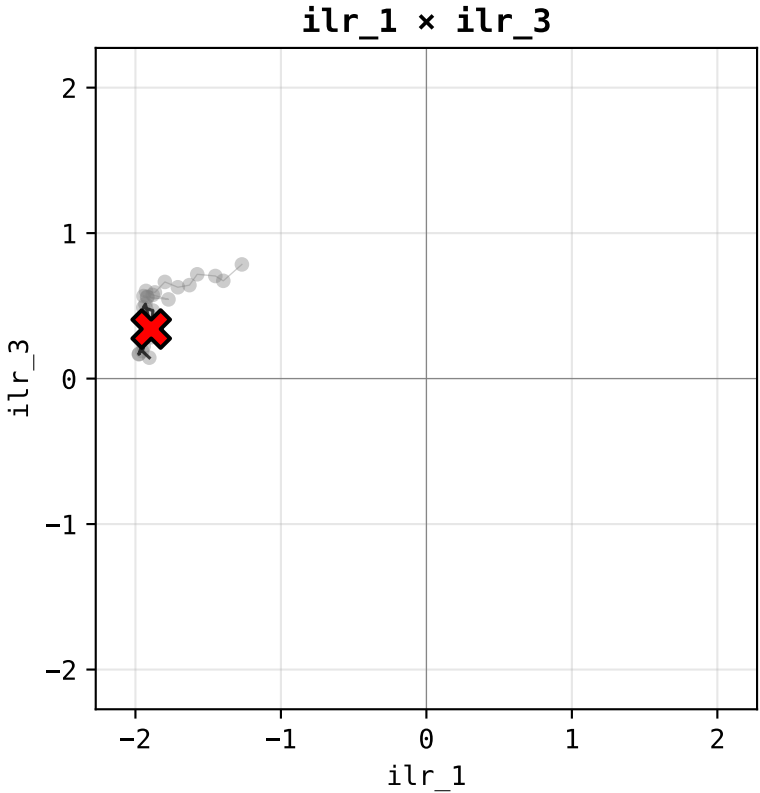
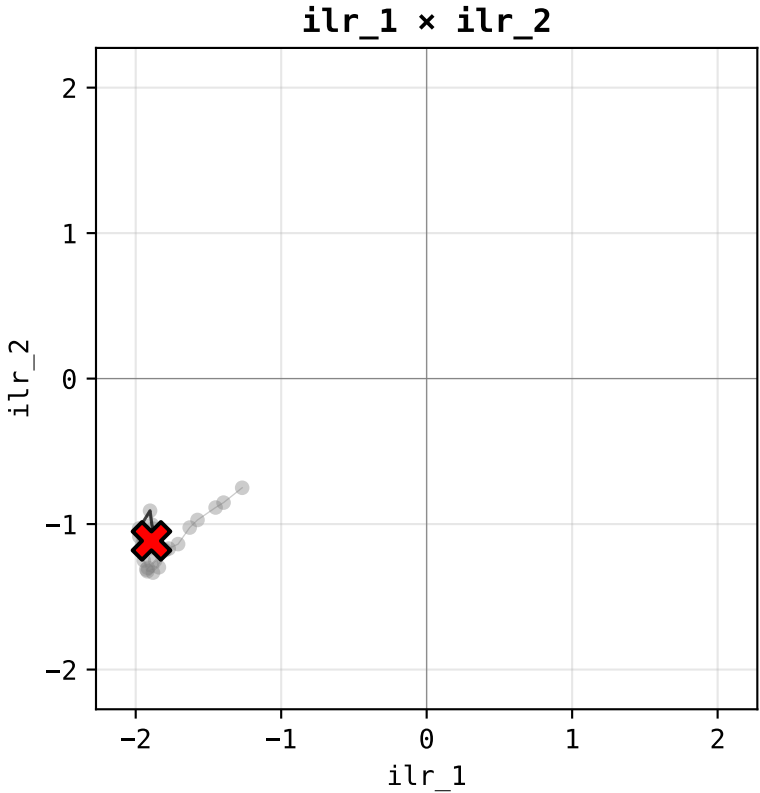


ILR Triplet – JPN Japan · year 2009 (10/26)



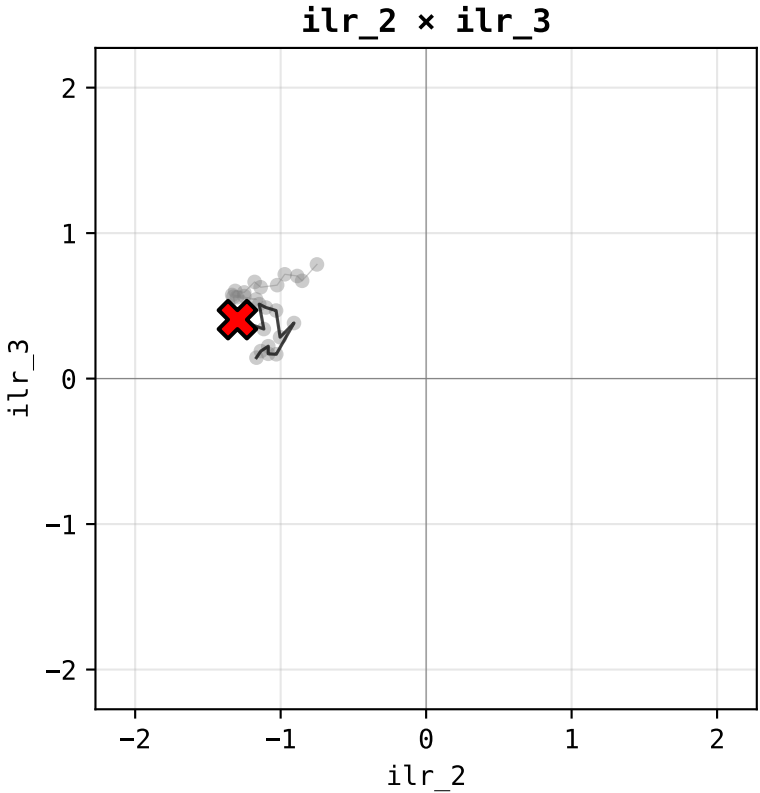
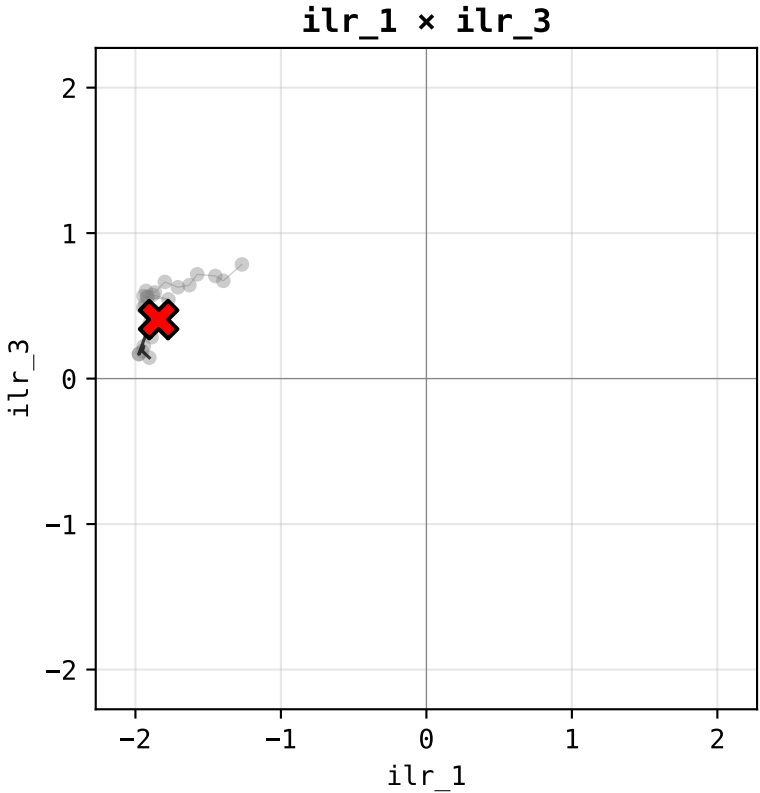
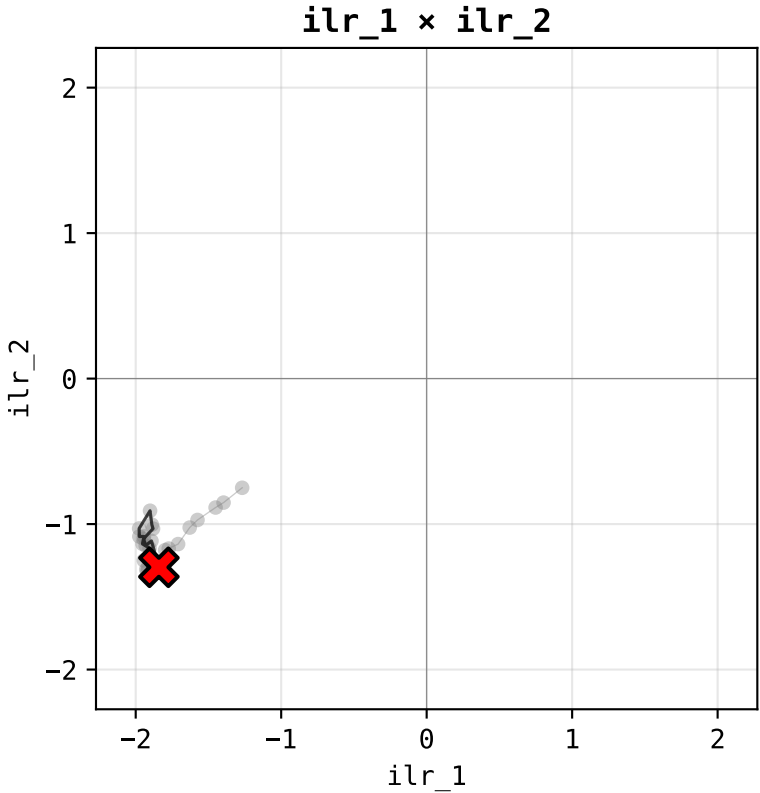
X = current year (2009) · black = past trajectory · gray = future · ilr range ±2.27

ILR Triplet – JPN Japan · year 2010 (11/26)

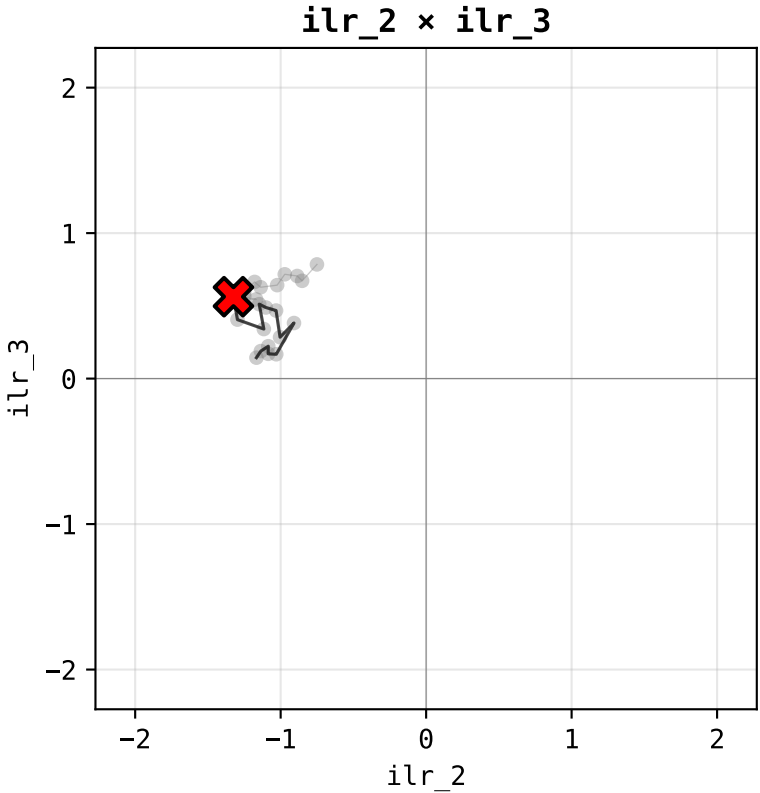
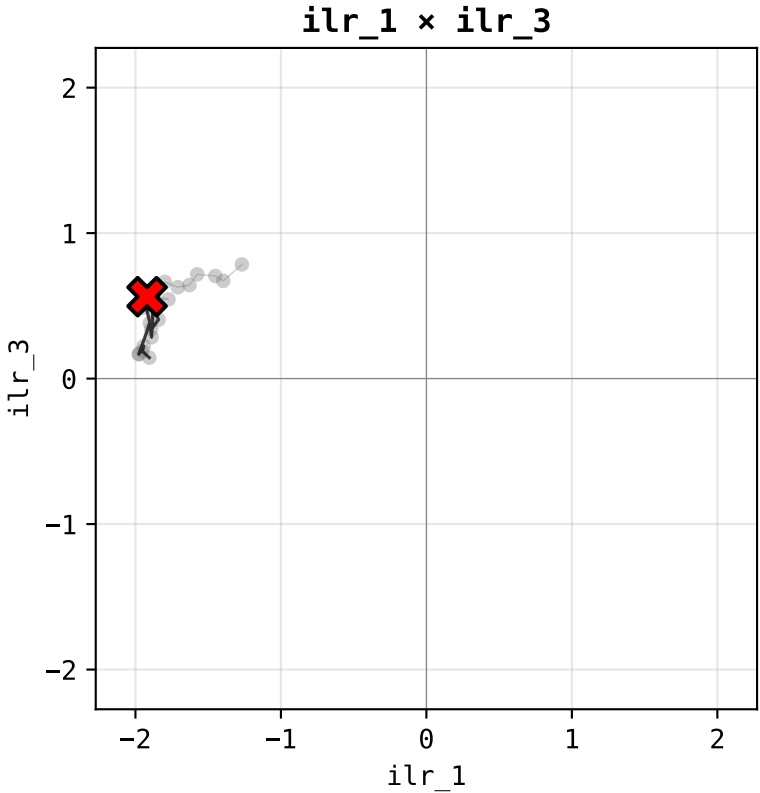
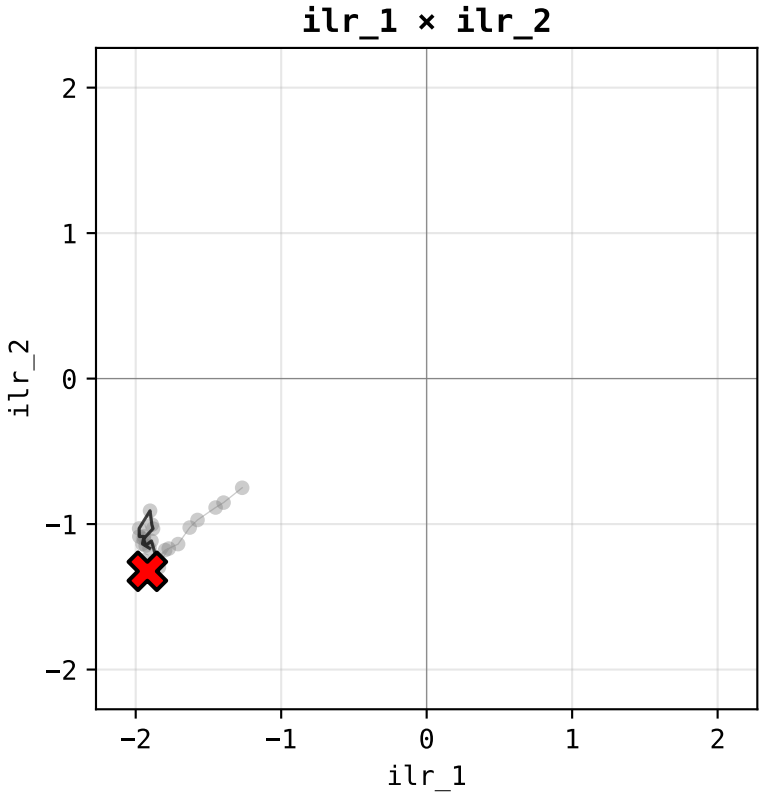


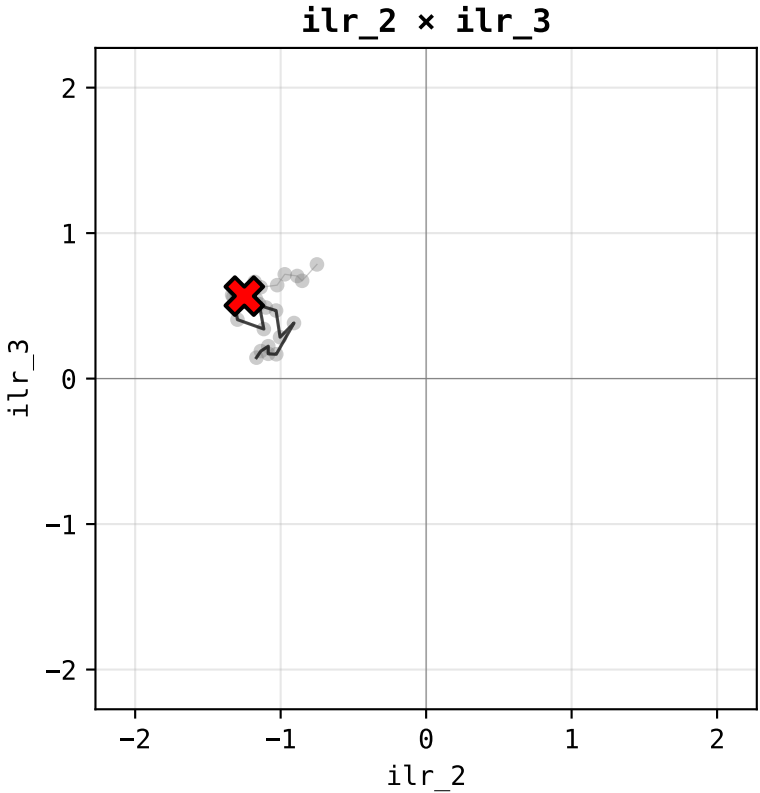
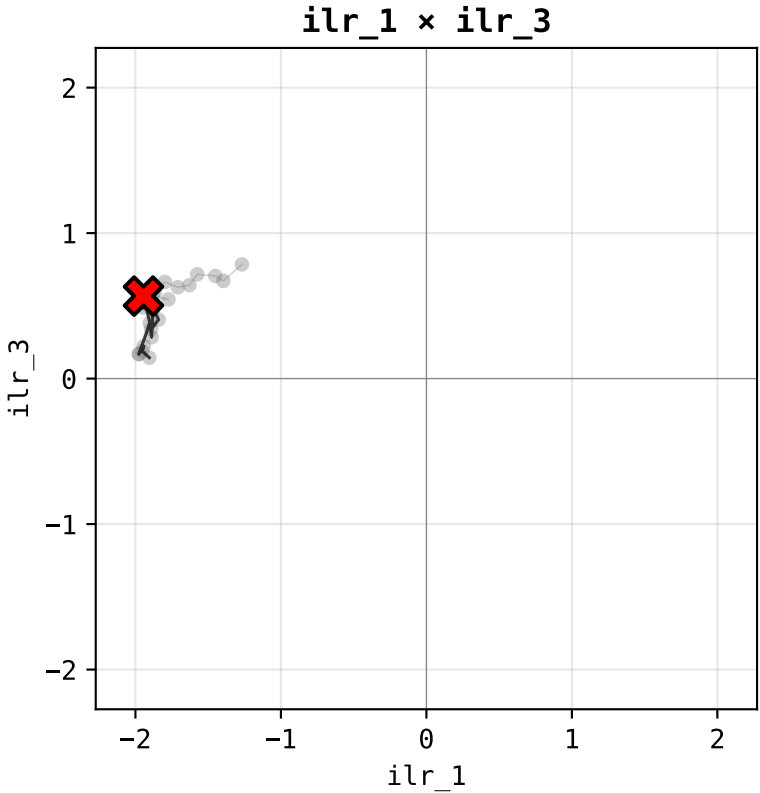
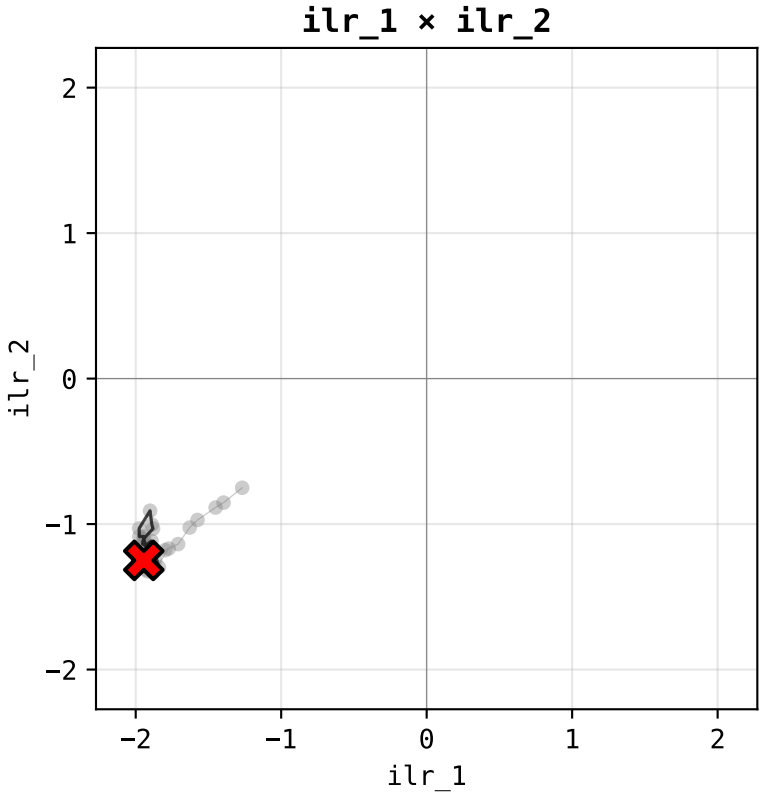
X = current year (2010) · black = past trajectory · gray = future · ilr range ±2.27

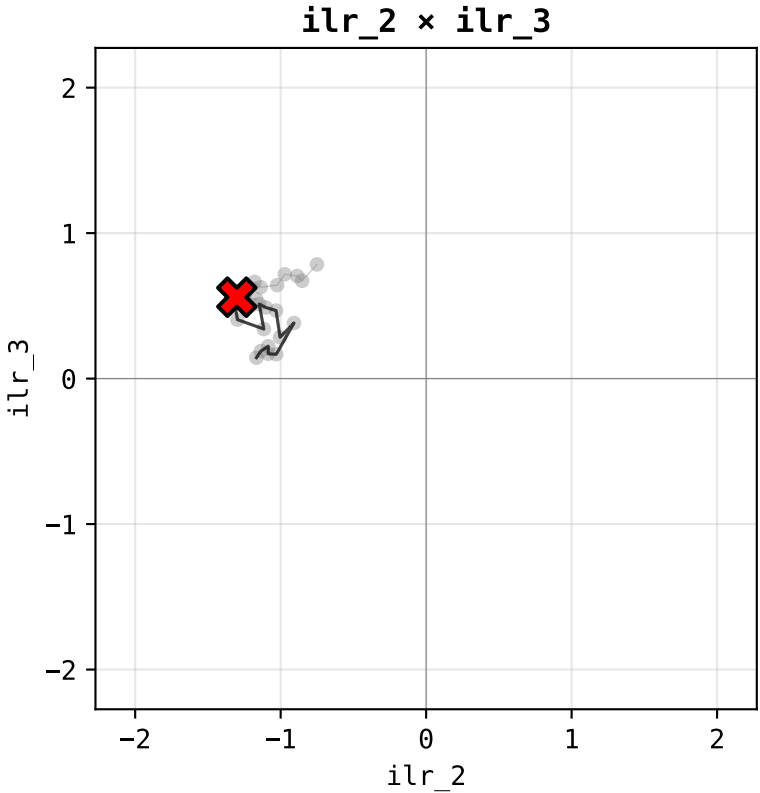
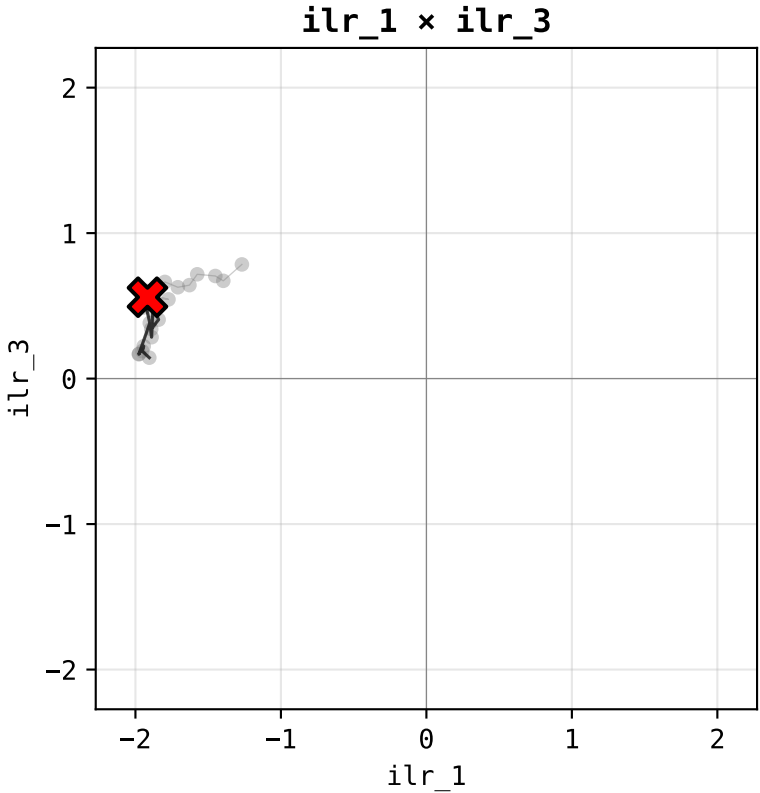
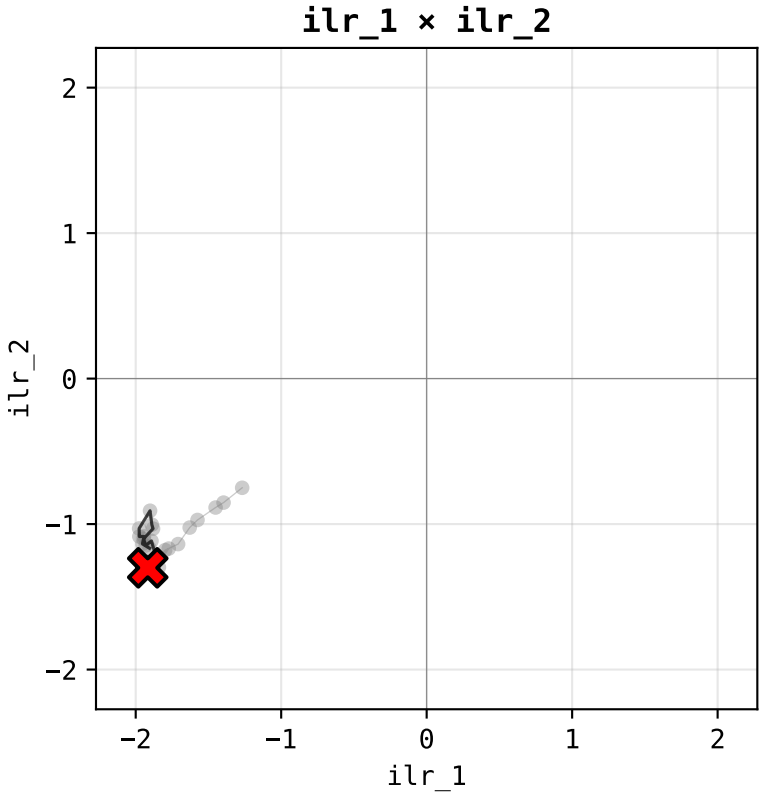
ILR Triplet – JPN Japan · year 2011 (12/26)

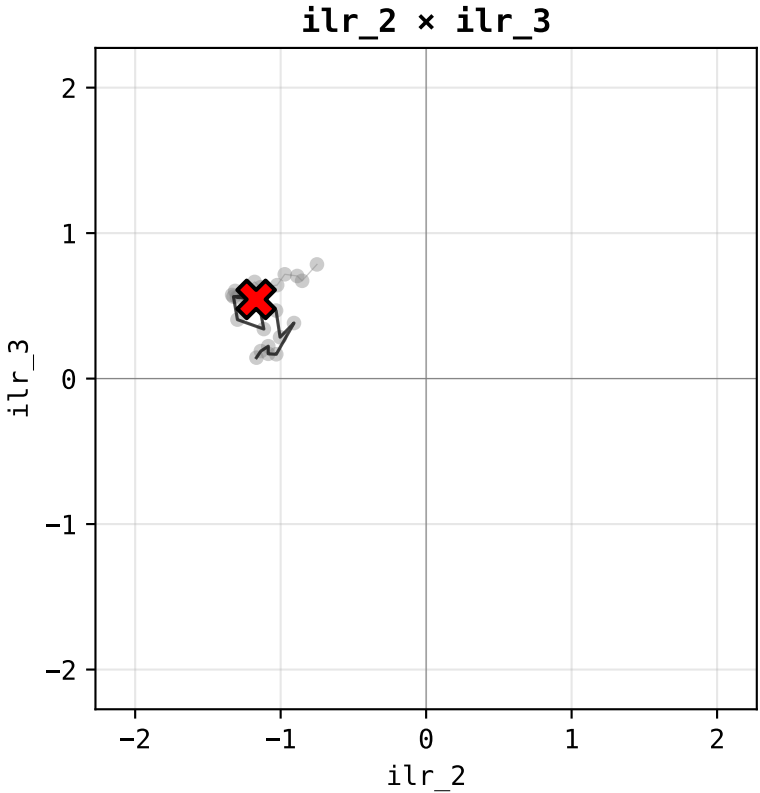
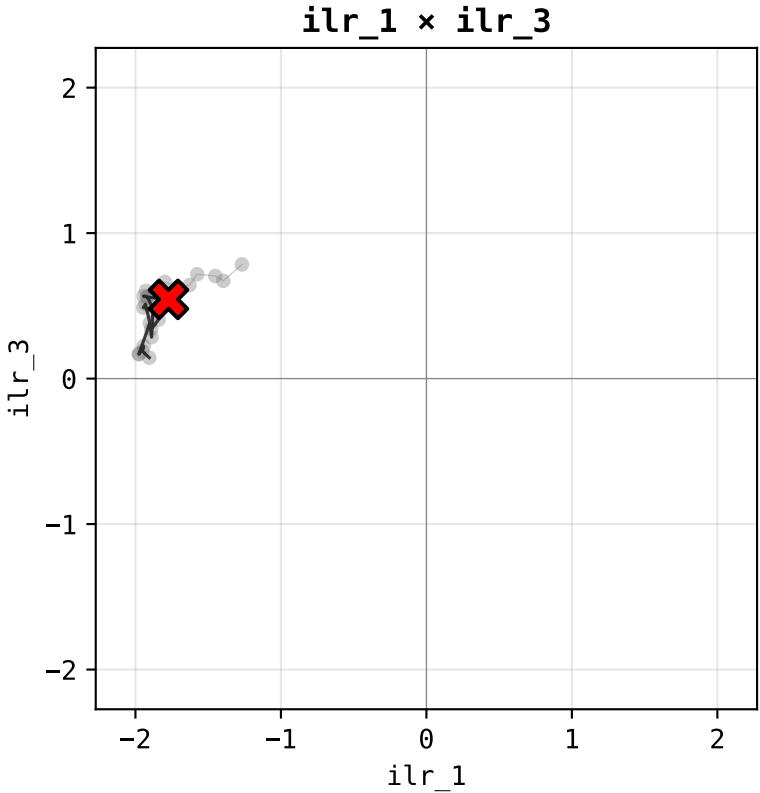
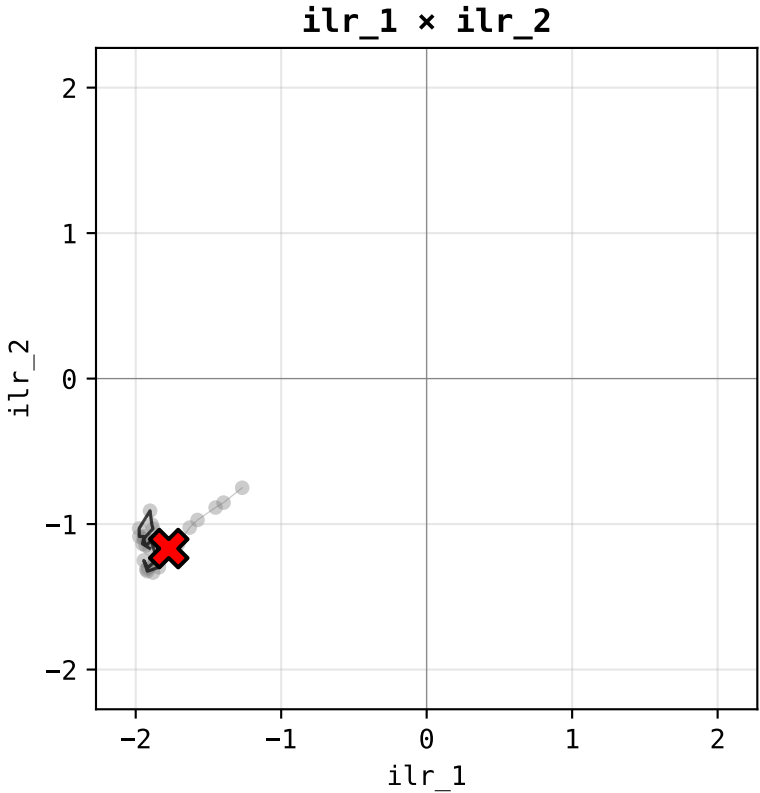


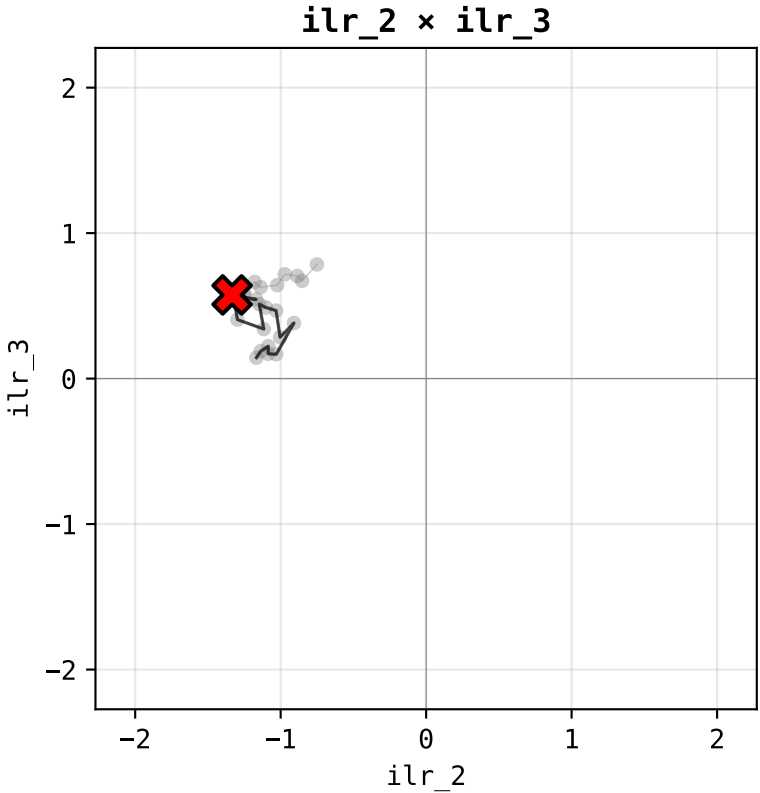
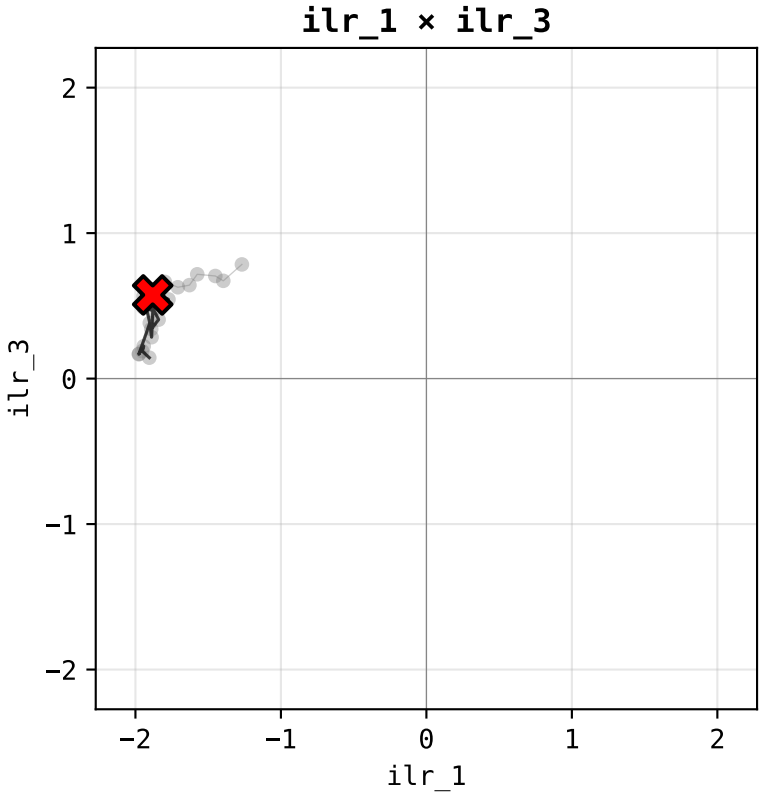
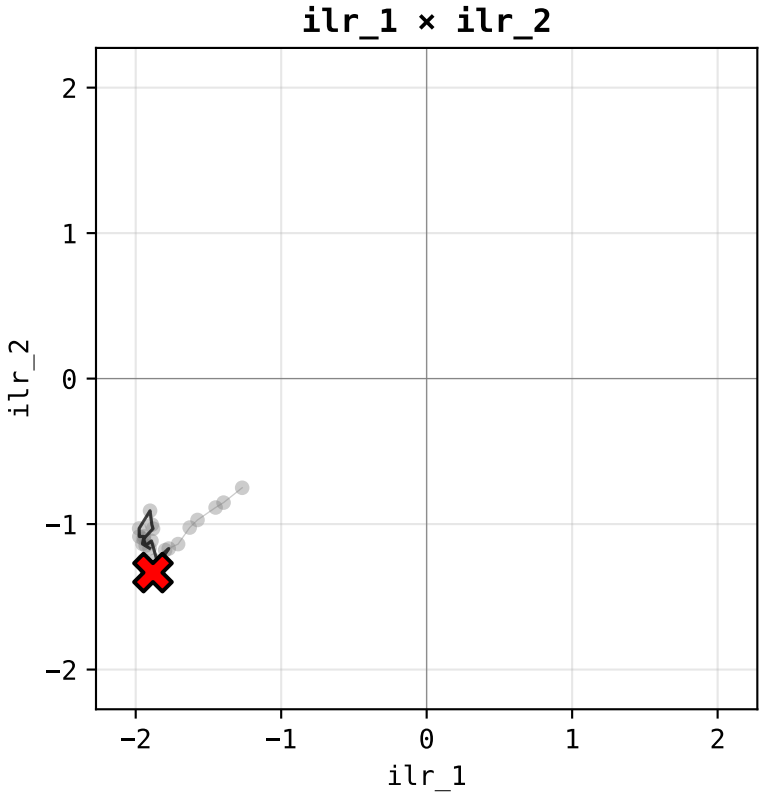
X = current year (2011) · black = past trajectory · gray = future · ilr range ±2.27

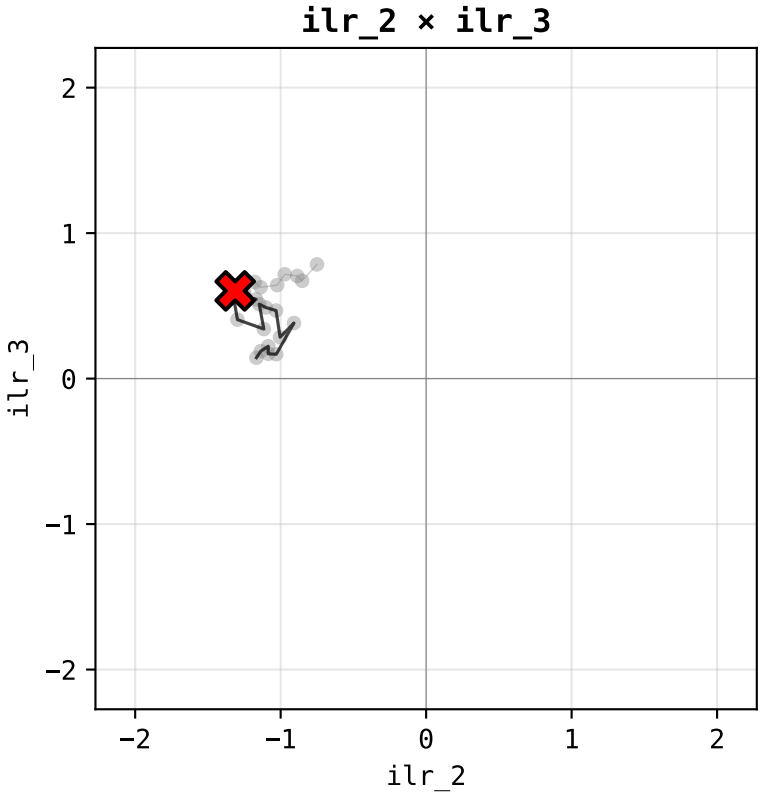
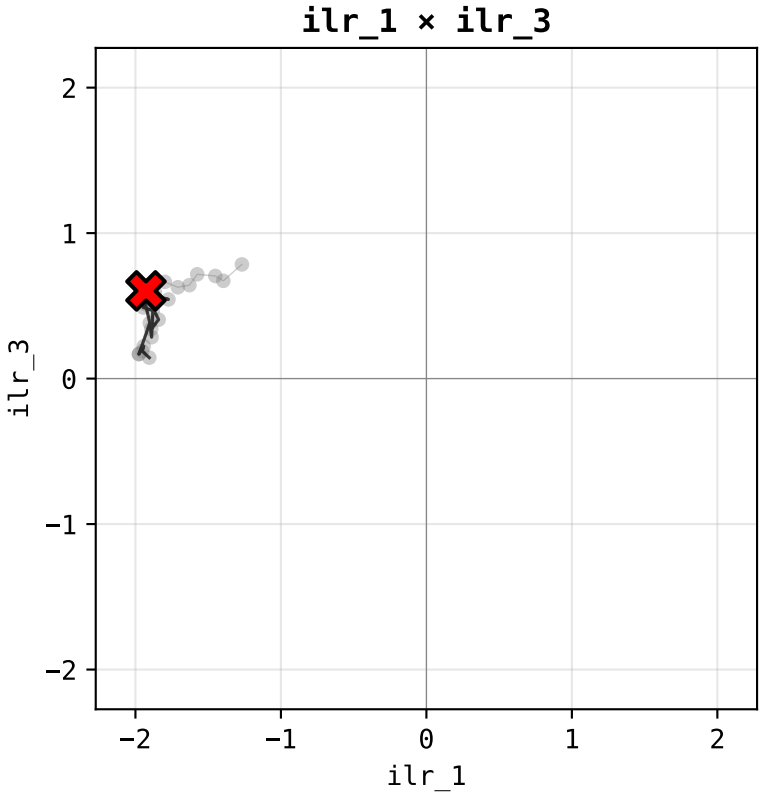
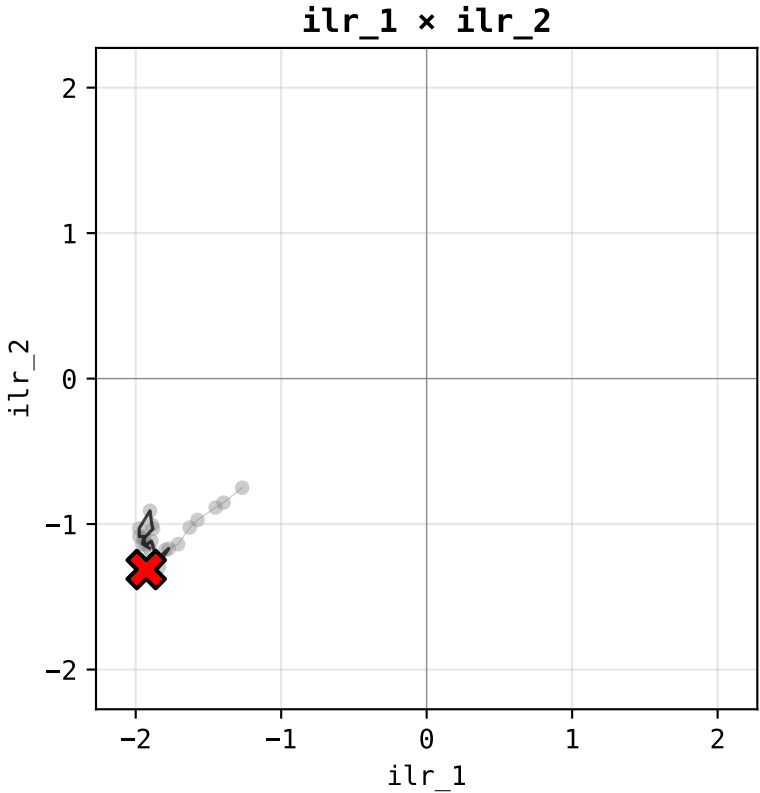


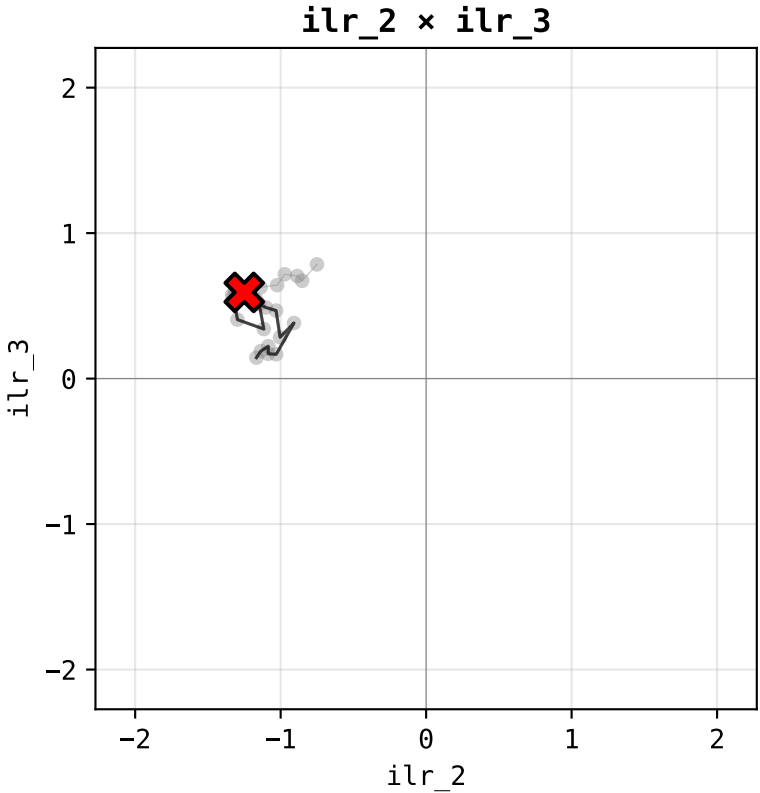
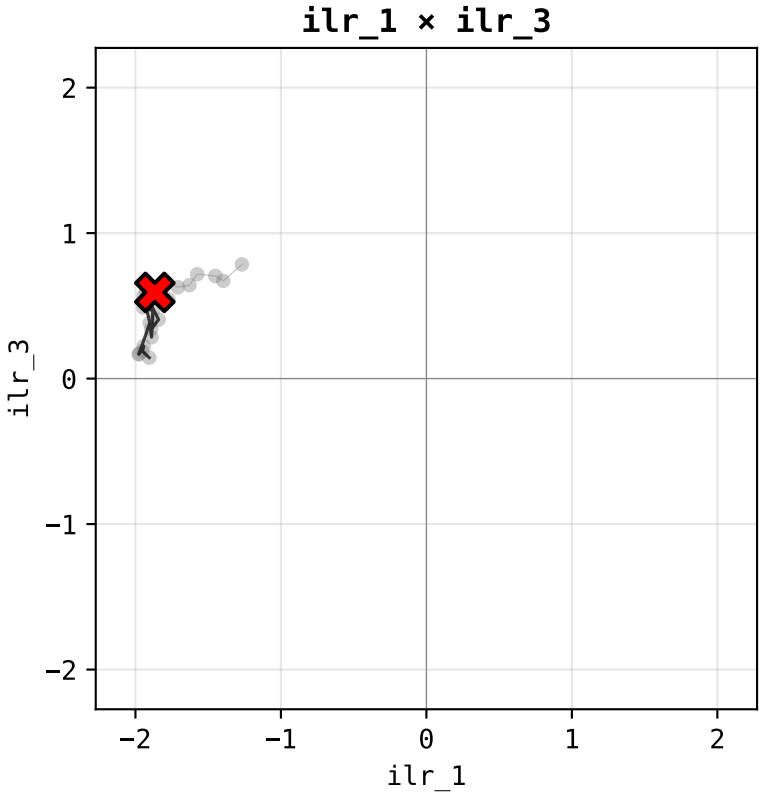
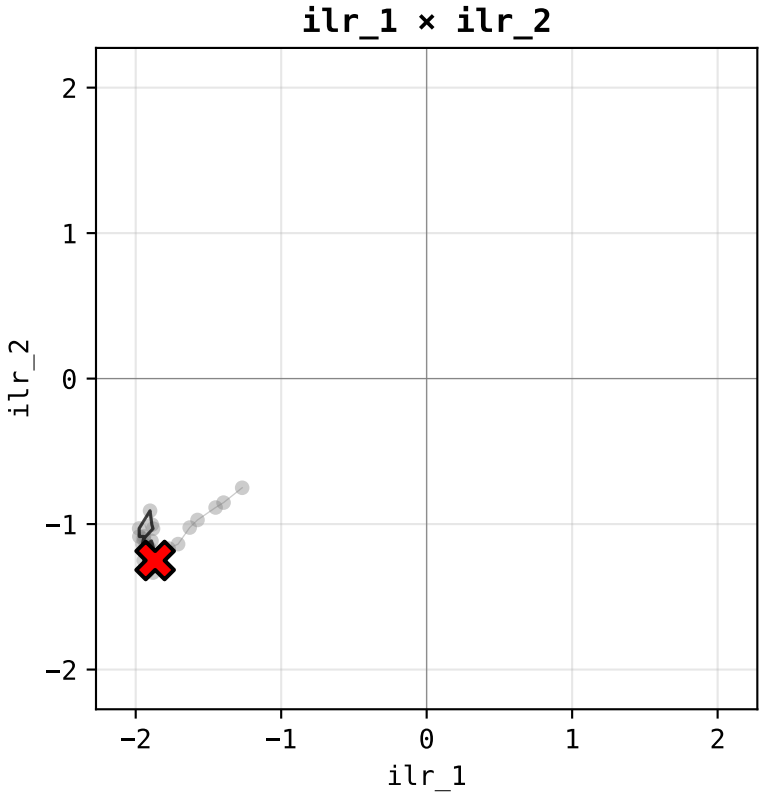


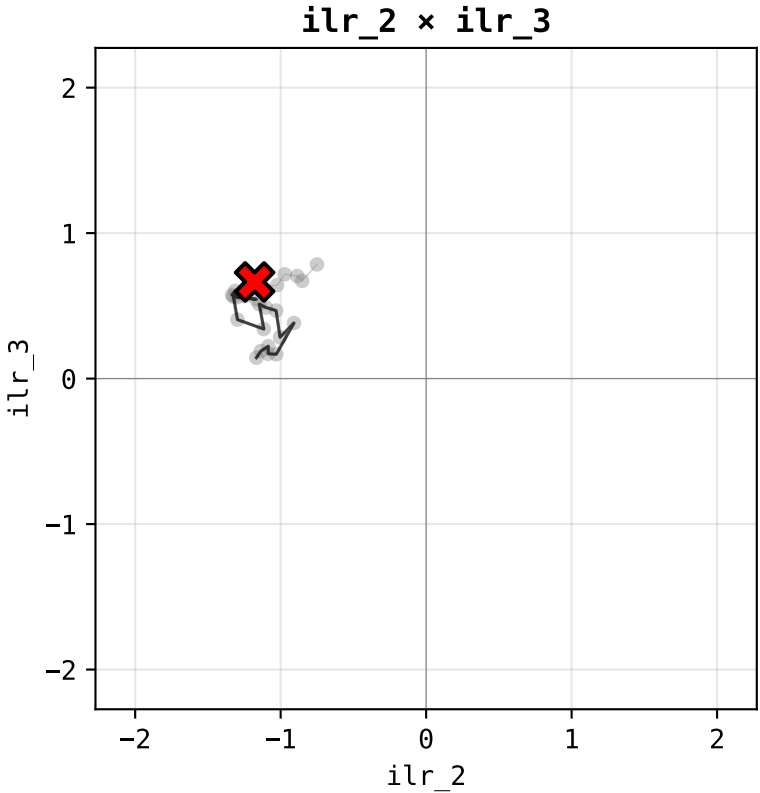
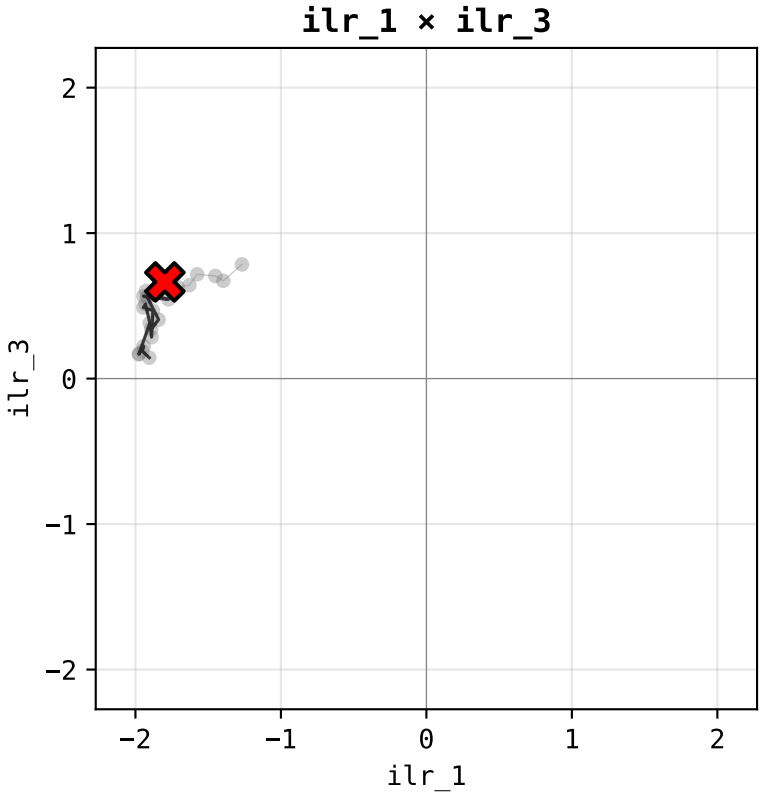
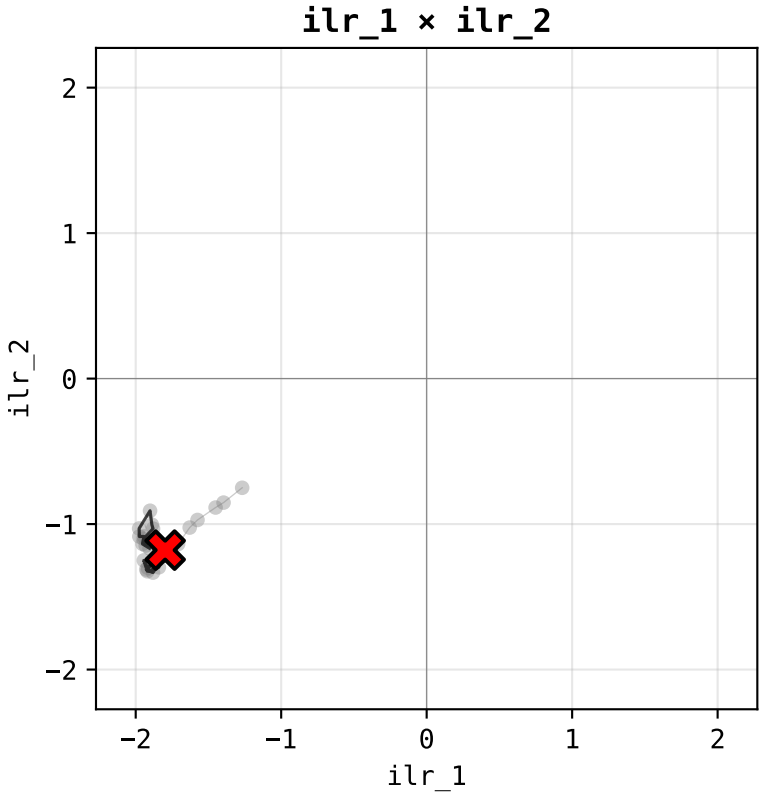


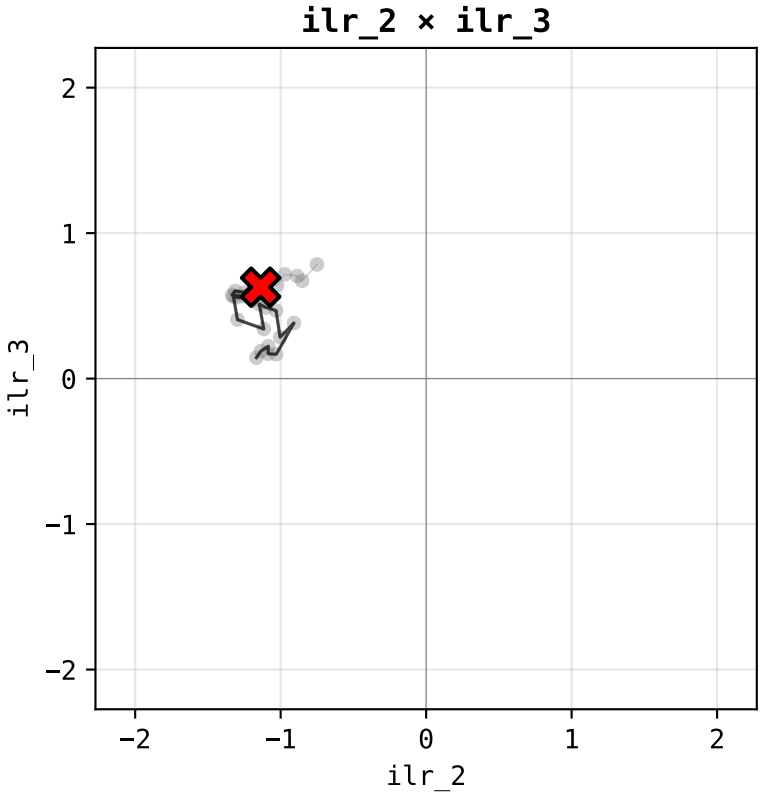
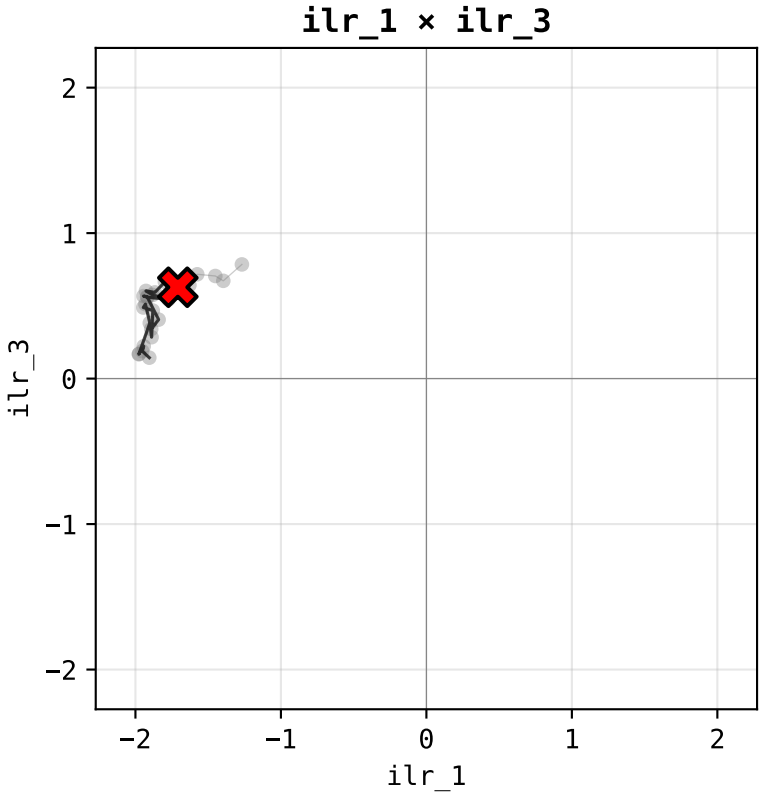
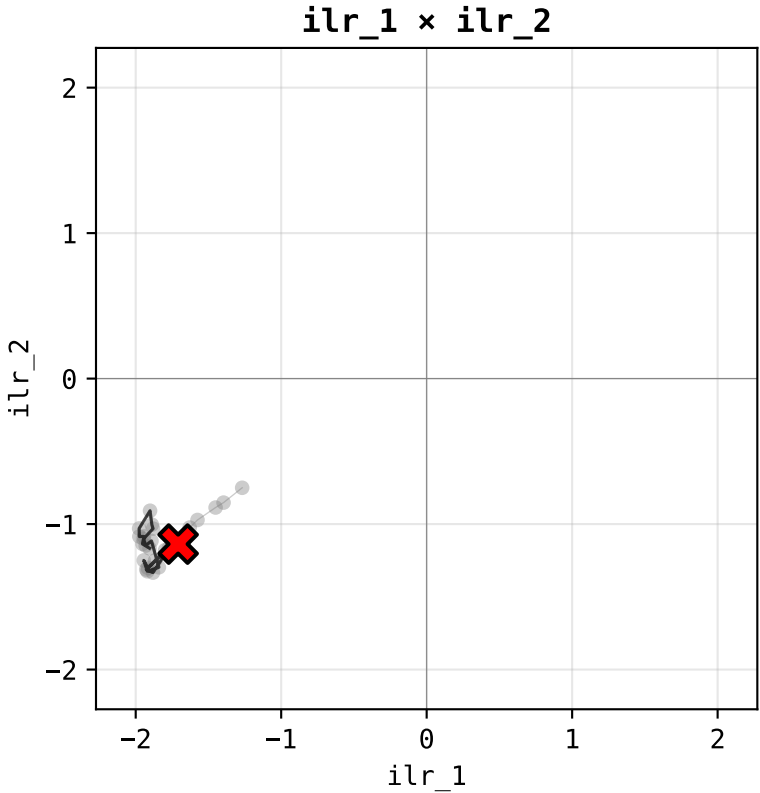


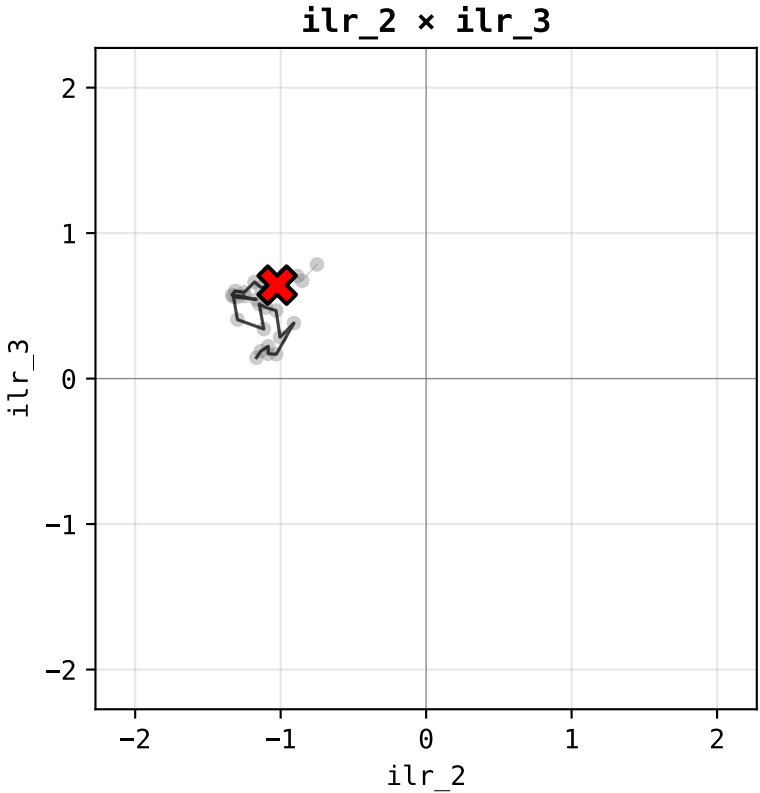
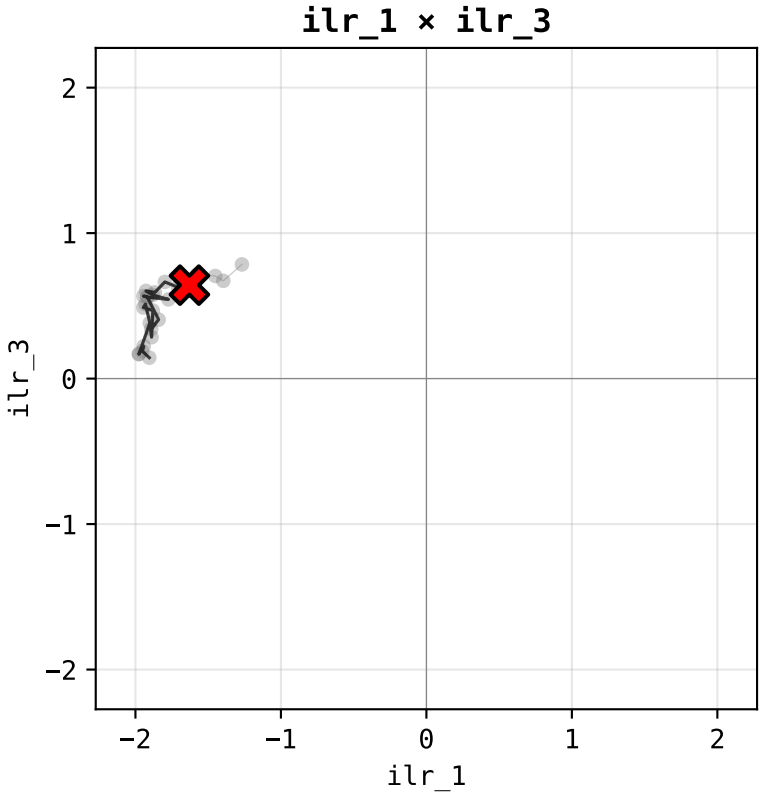
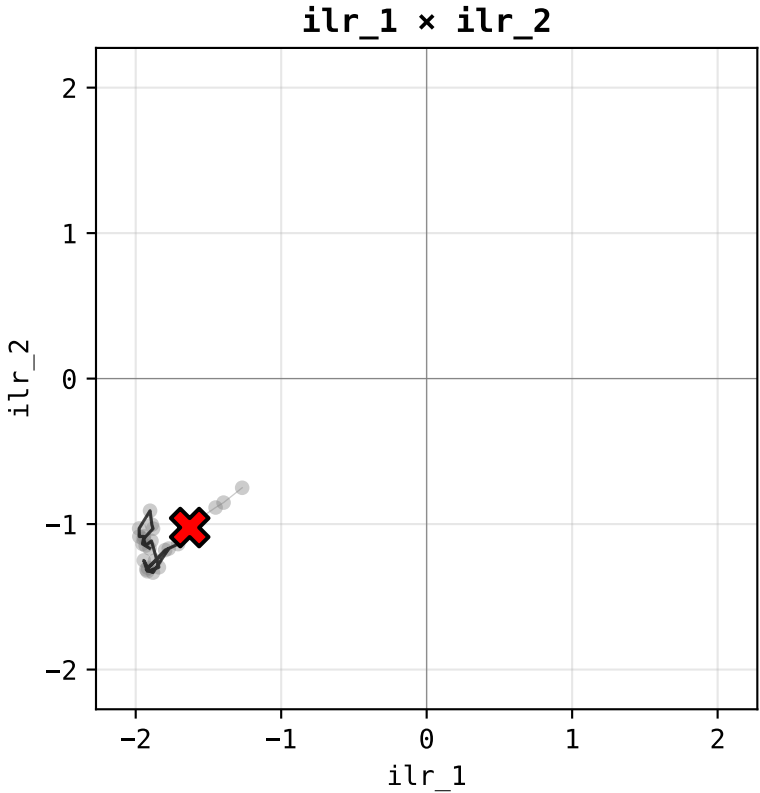


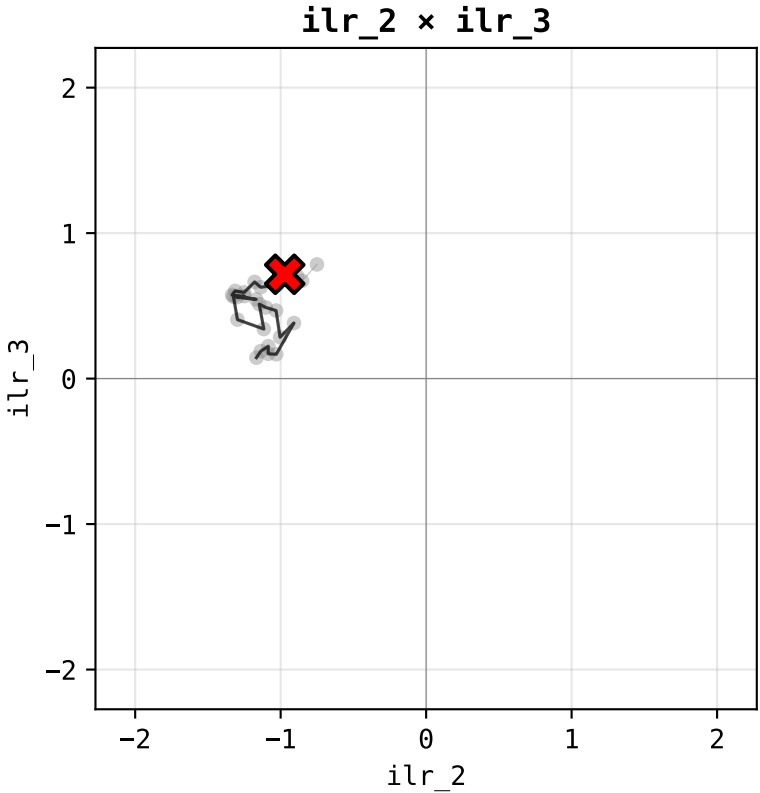
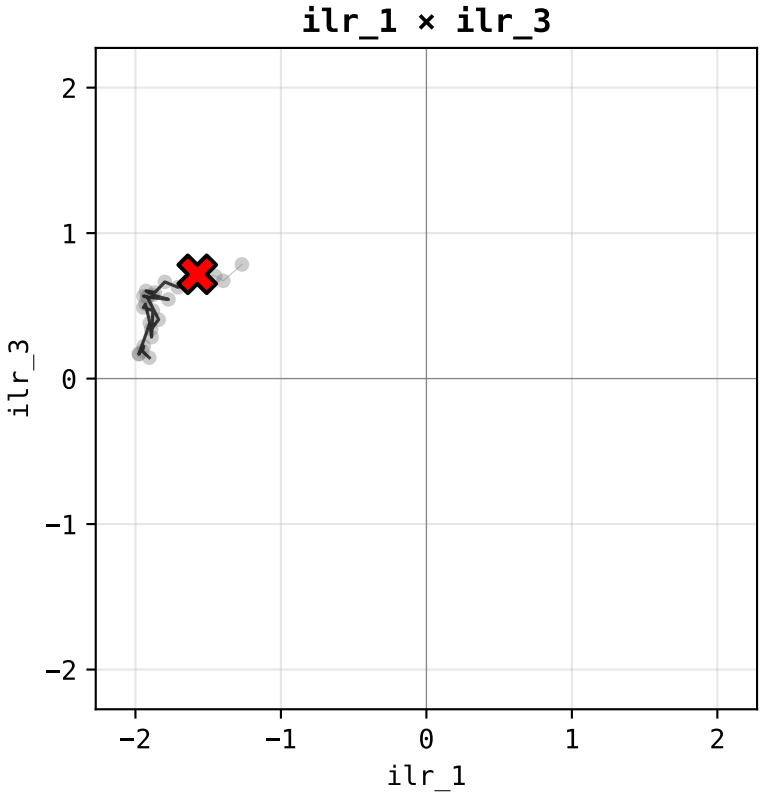
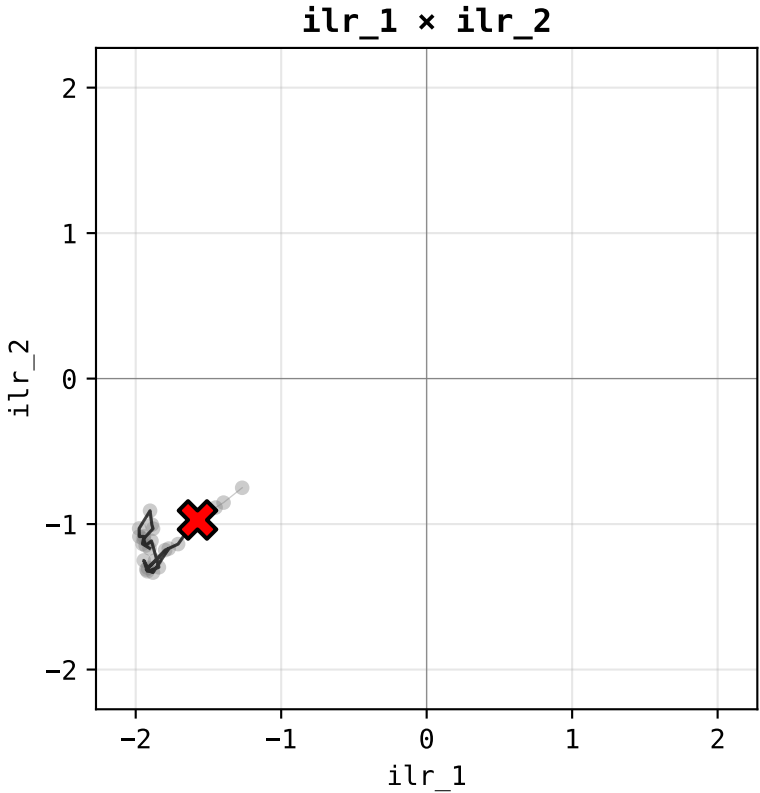


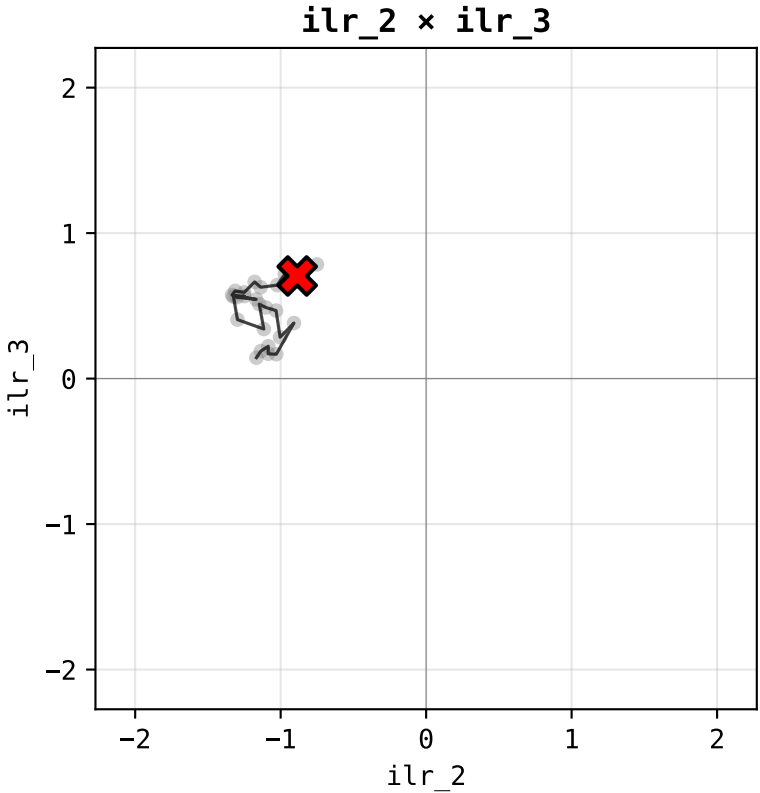
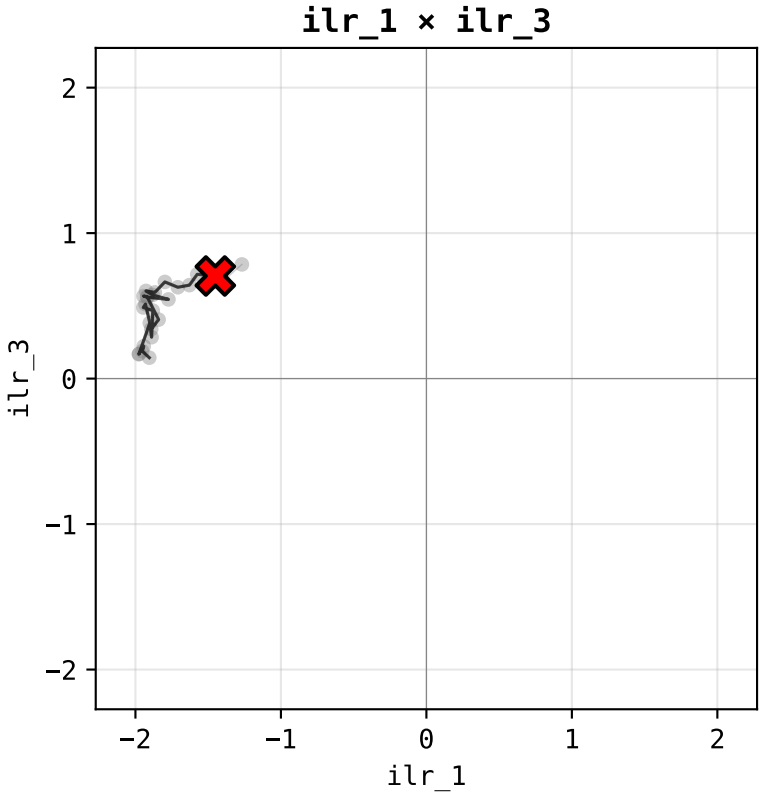
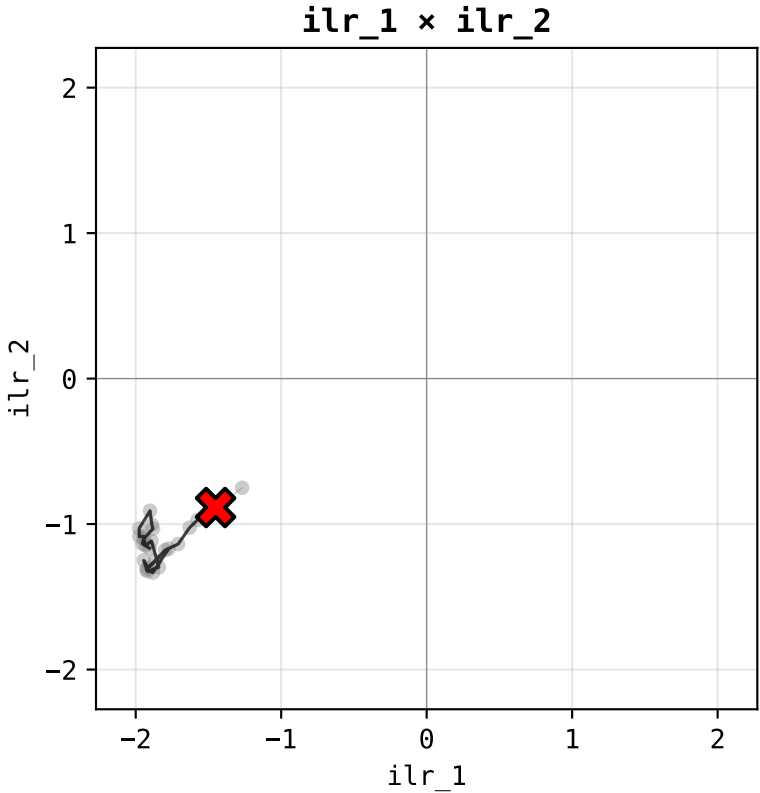


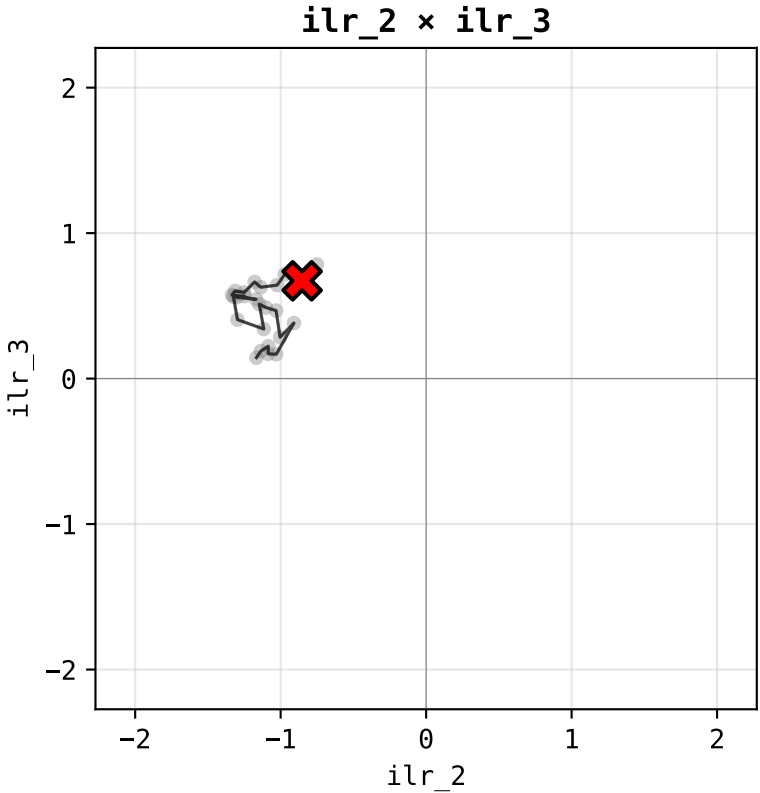
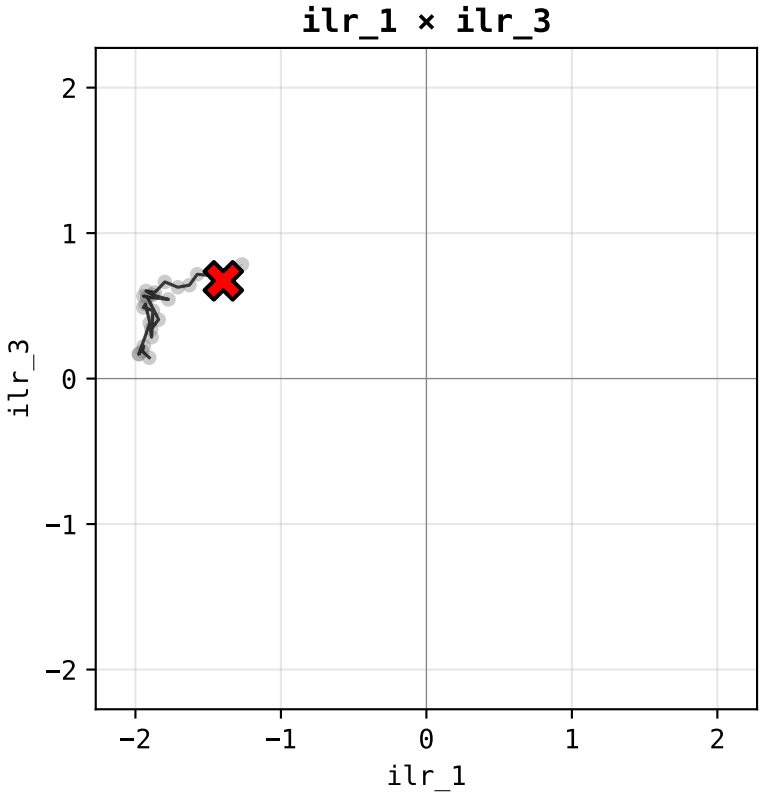
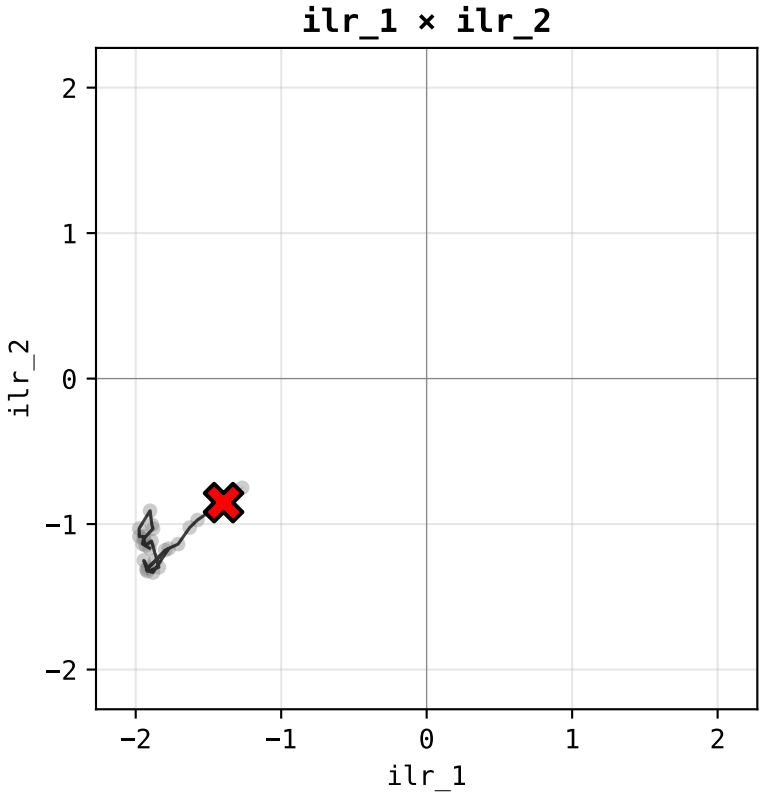


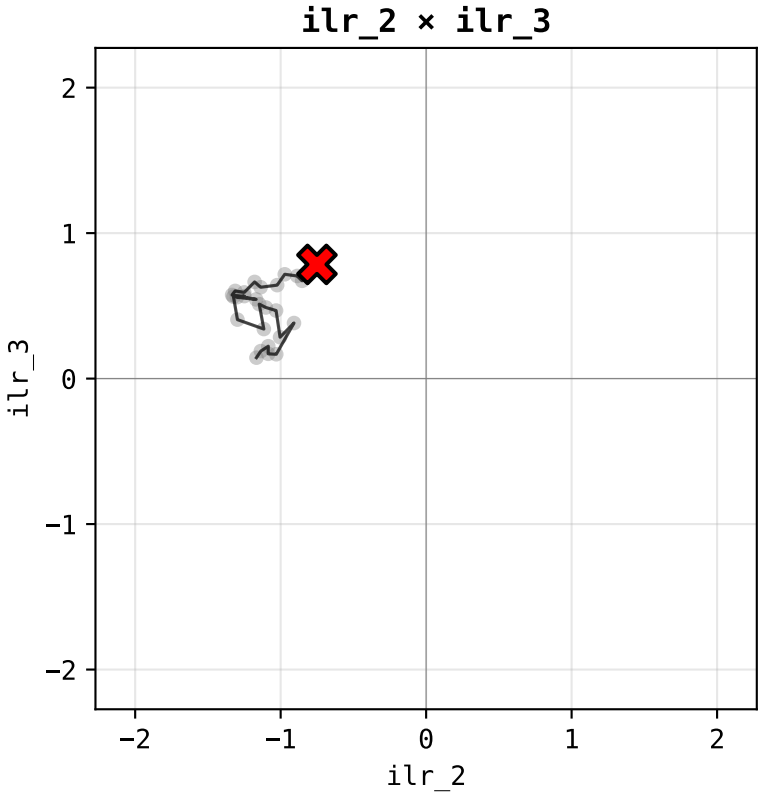
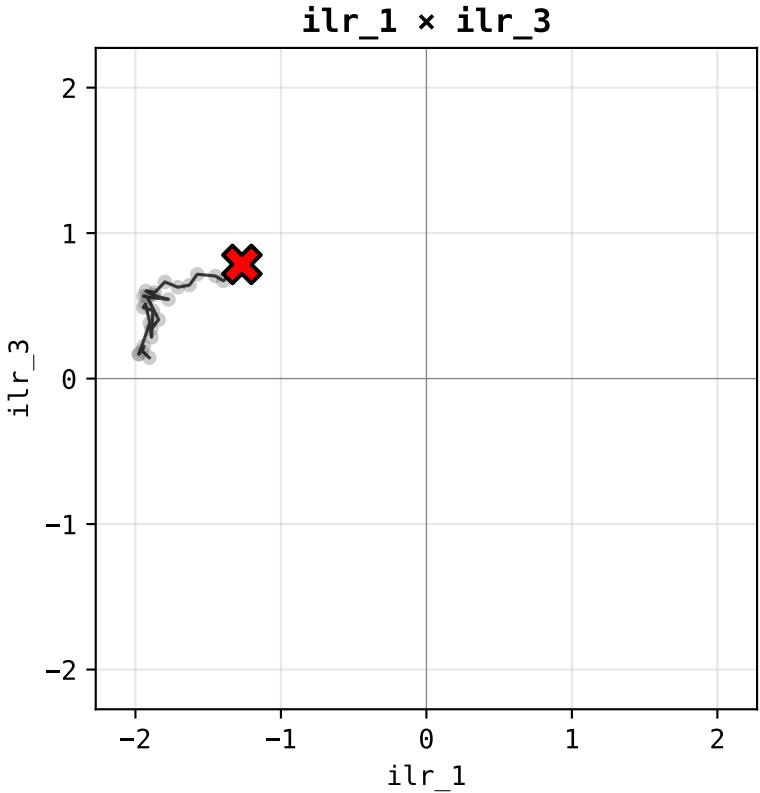
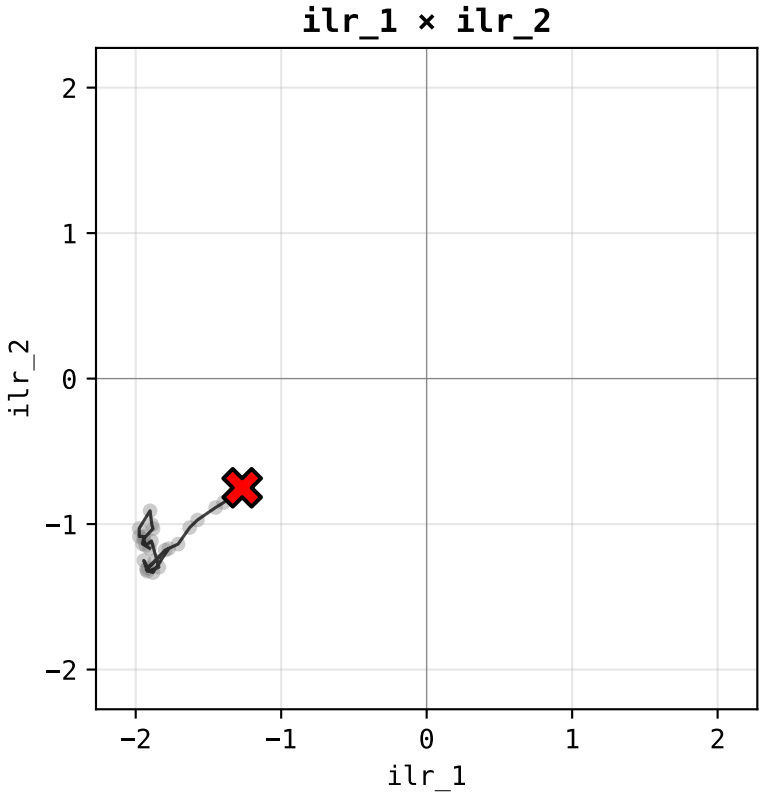












View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ USA United States

SECTION PLATE t=0 | Year 2001 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

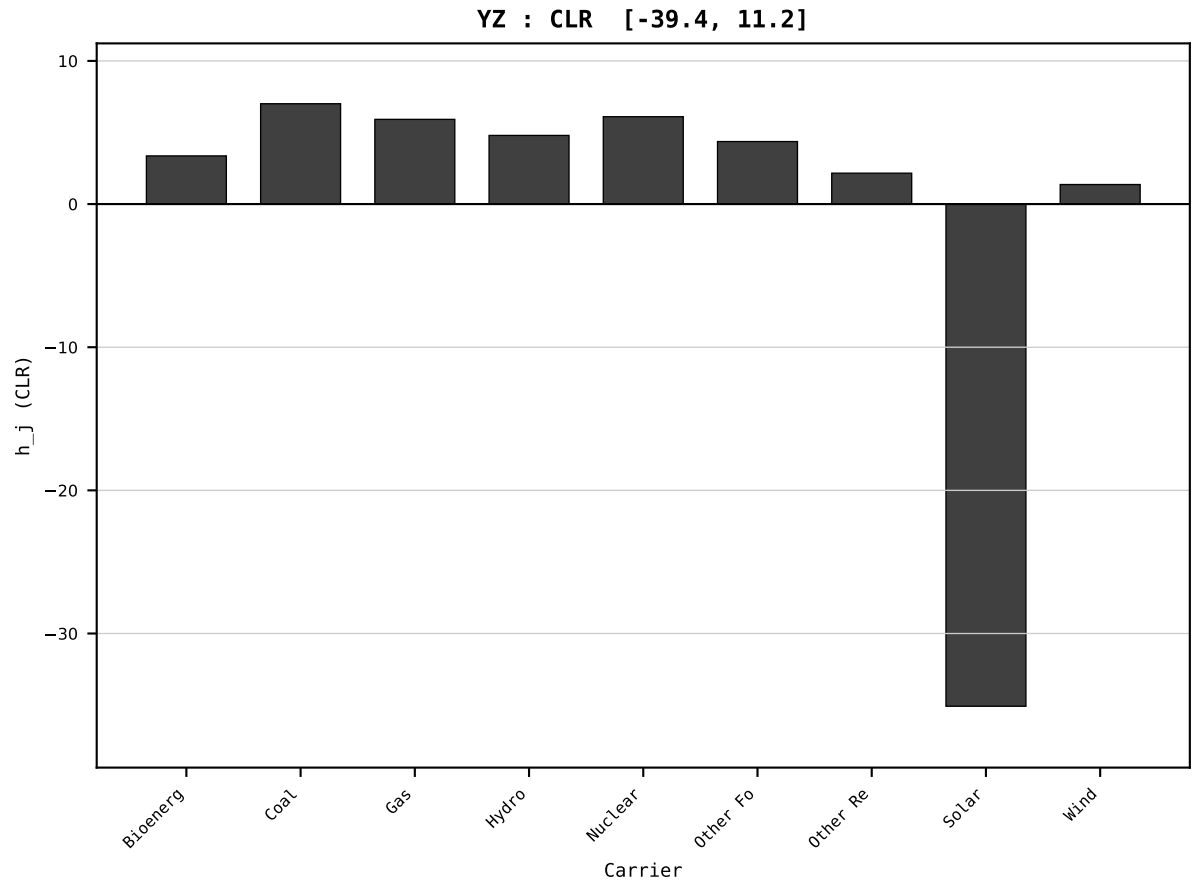
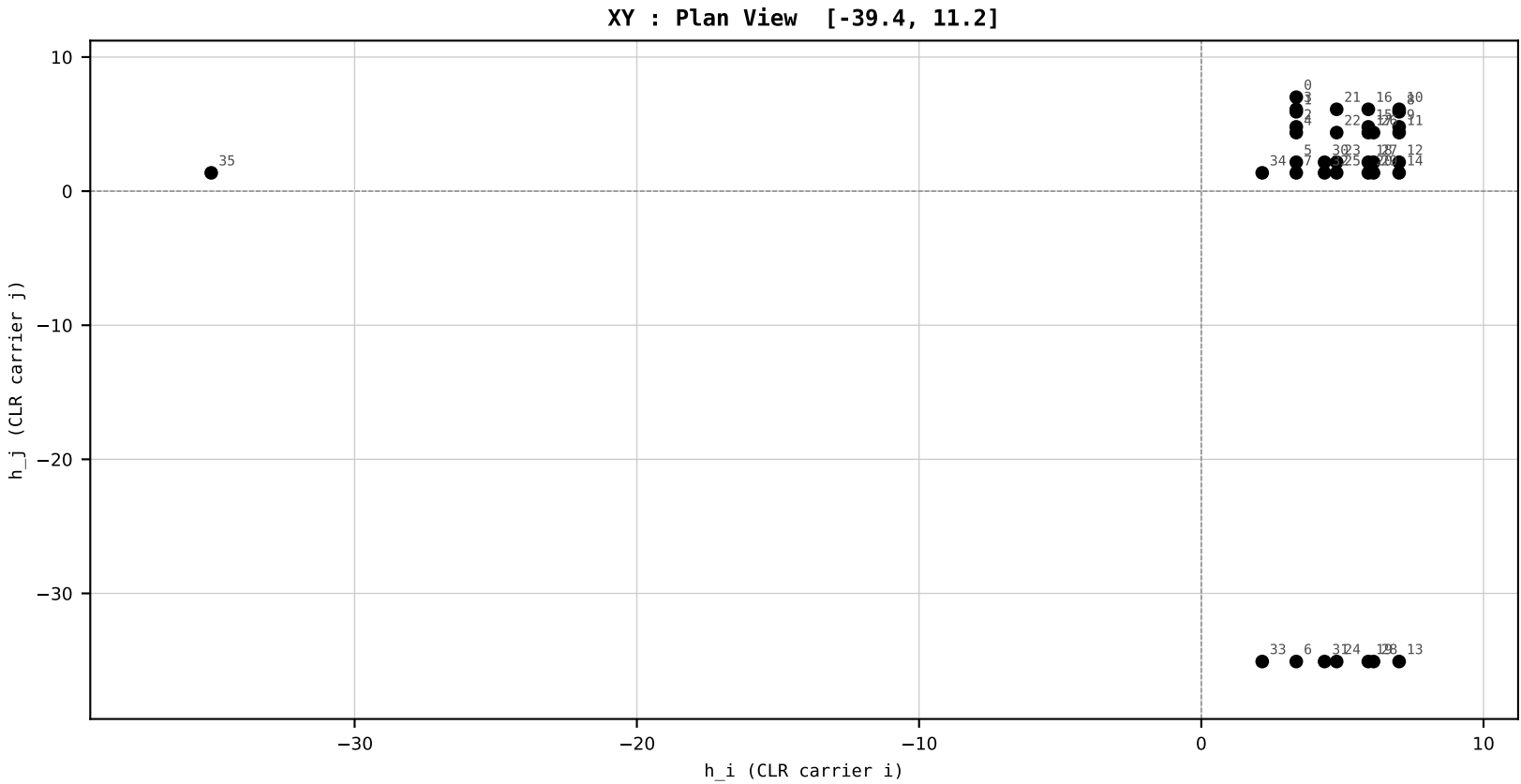
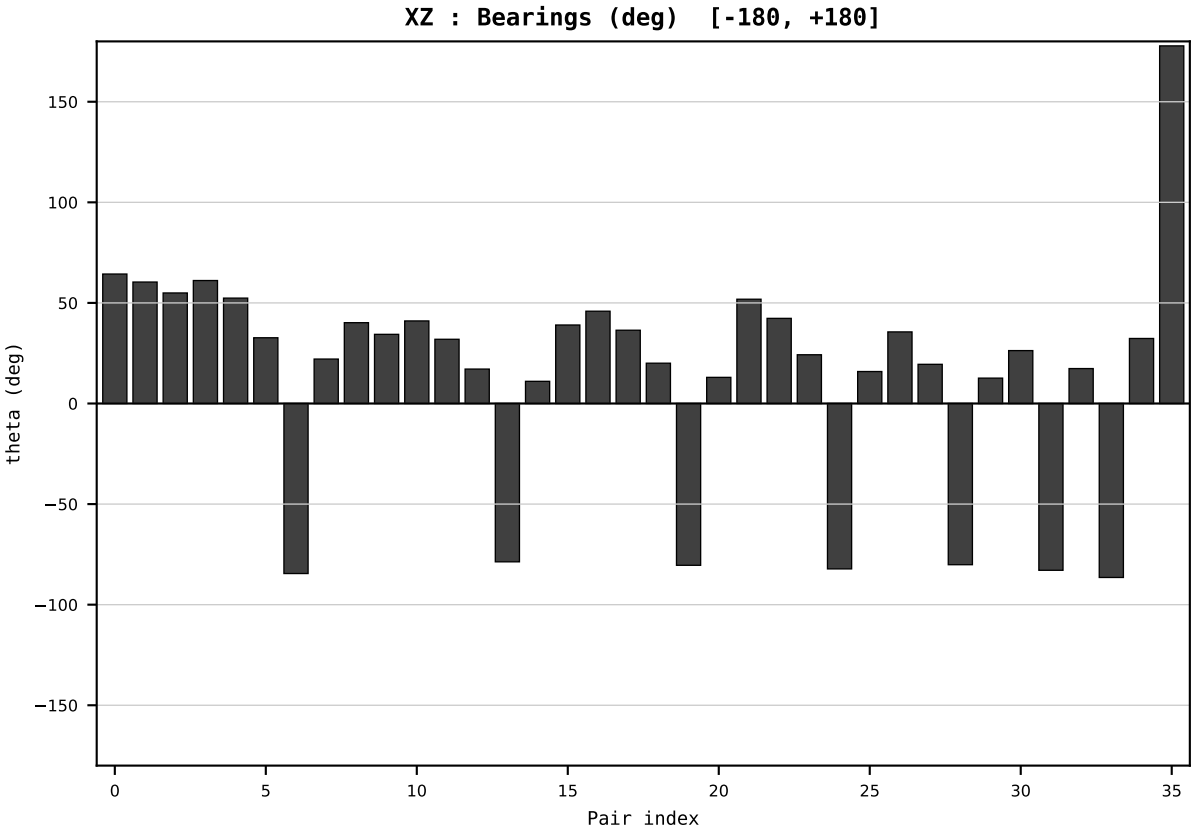
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2001
t : 0 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.388371
Ring = Hs-3
E_metric= 895.1337
kappa_HS= 190397000000000000.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 42.090 HLR
DR ratio= 1903969992697630976.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2002 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

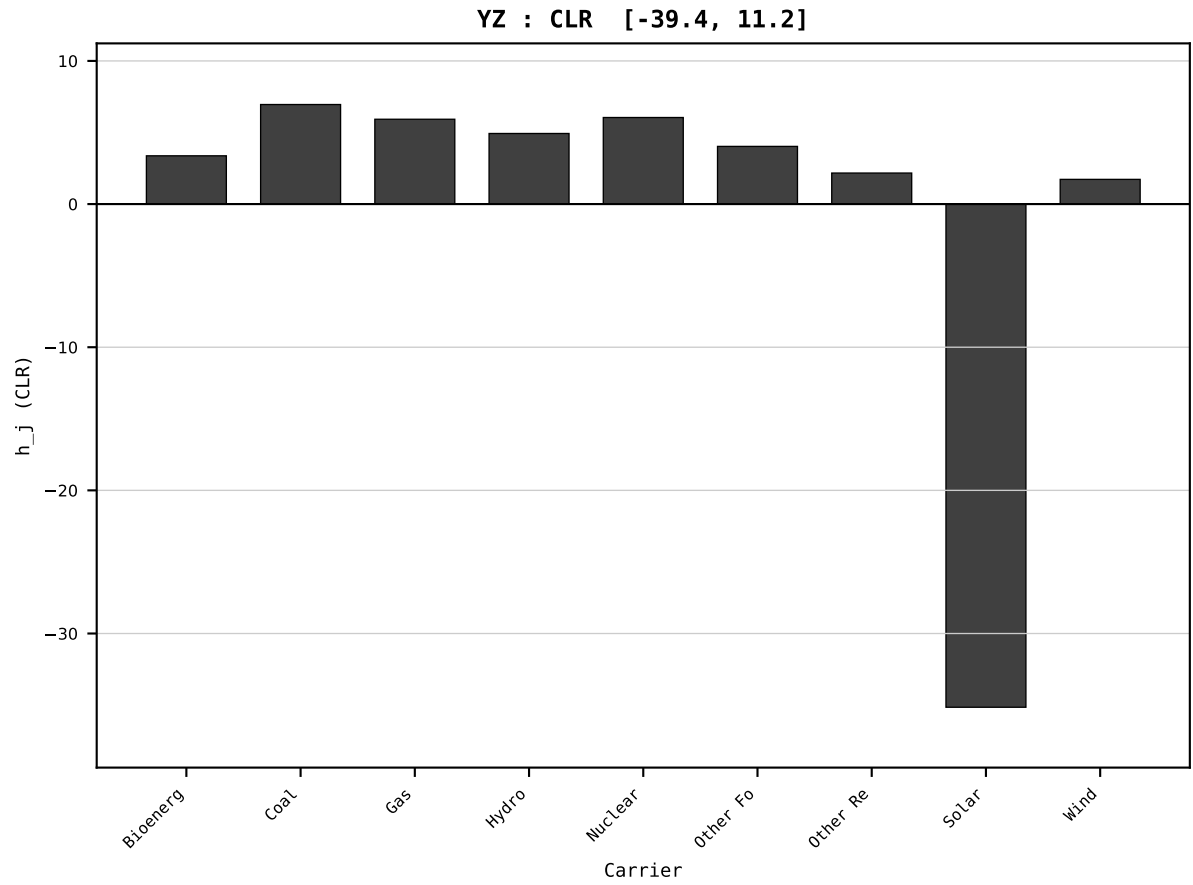
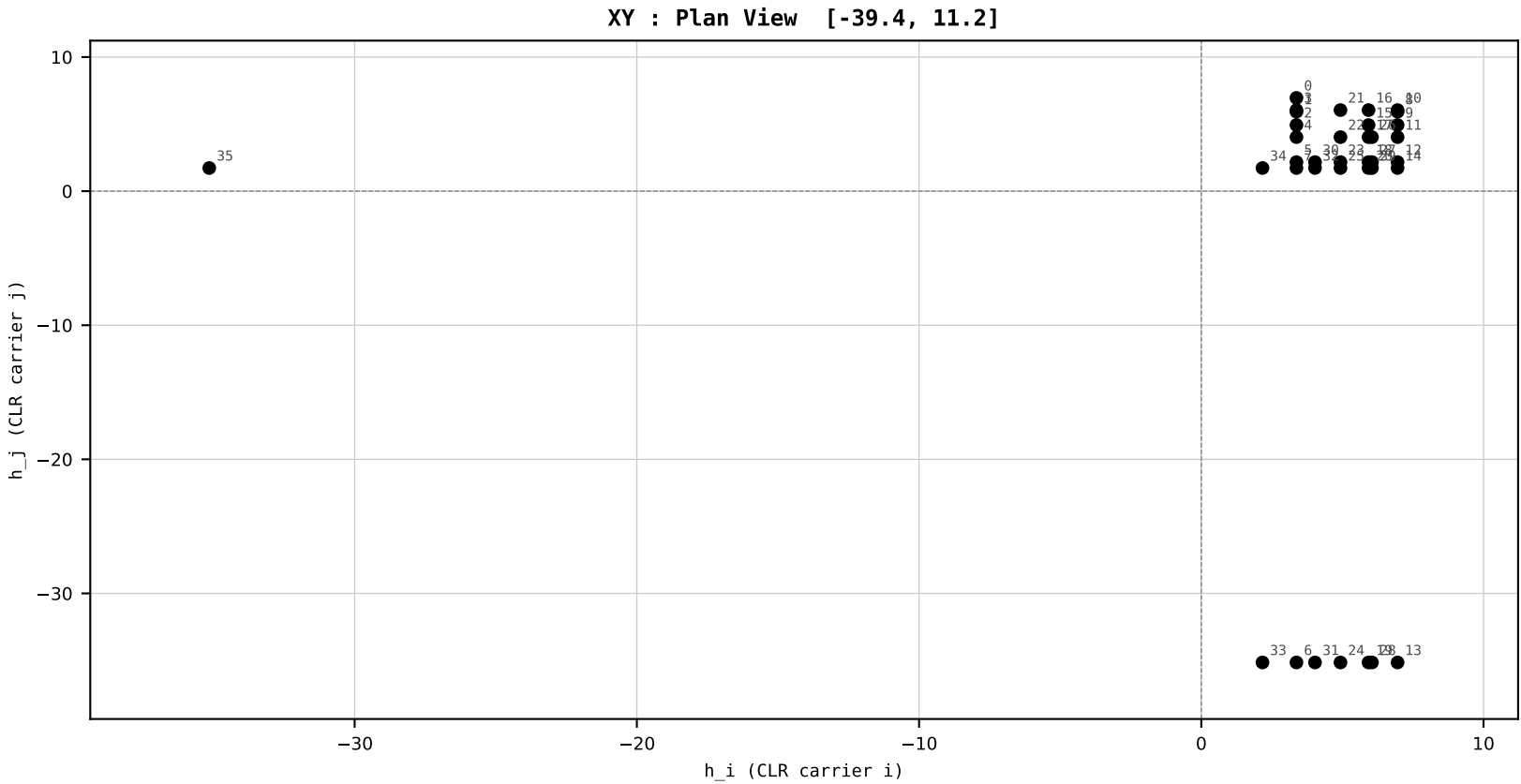
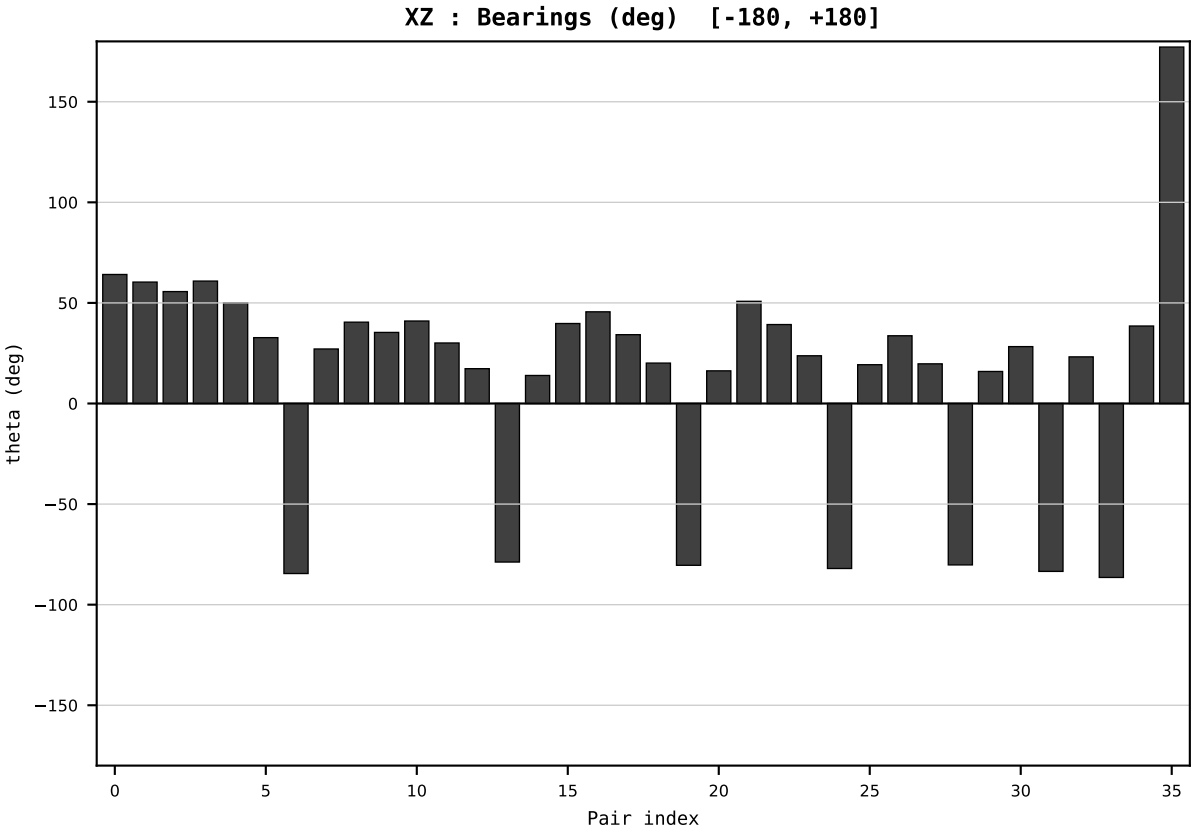
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2002
t : 1 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.383719
Ring = Hs-3
E_metric= 895.5128
kappa_HS= 193315000000000000.00
omega = 0.7941 deg
d_A = 0.522906
Helm = Wind
Helm d = +0.359790
DR = 42.106 HLR
DR ratio= 1933149998657515008.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2003 | D=9 N=25 pairs=36

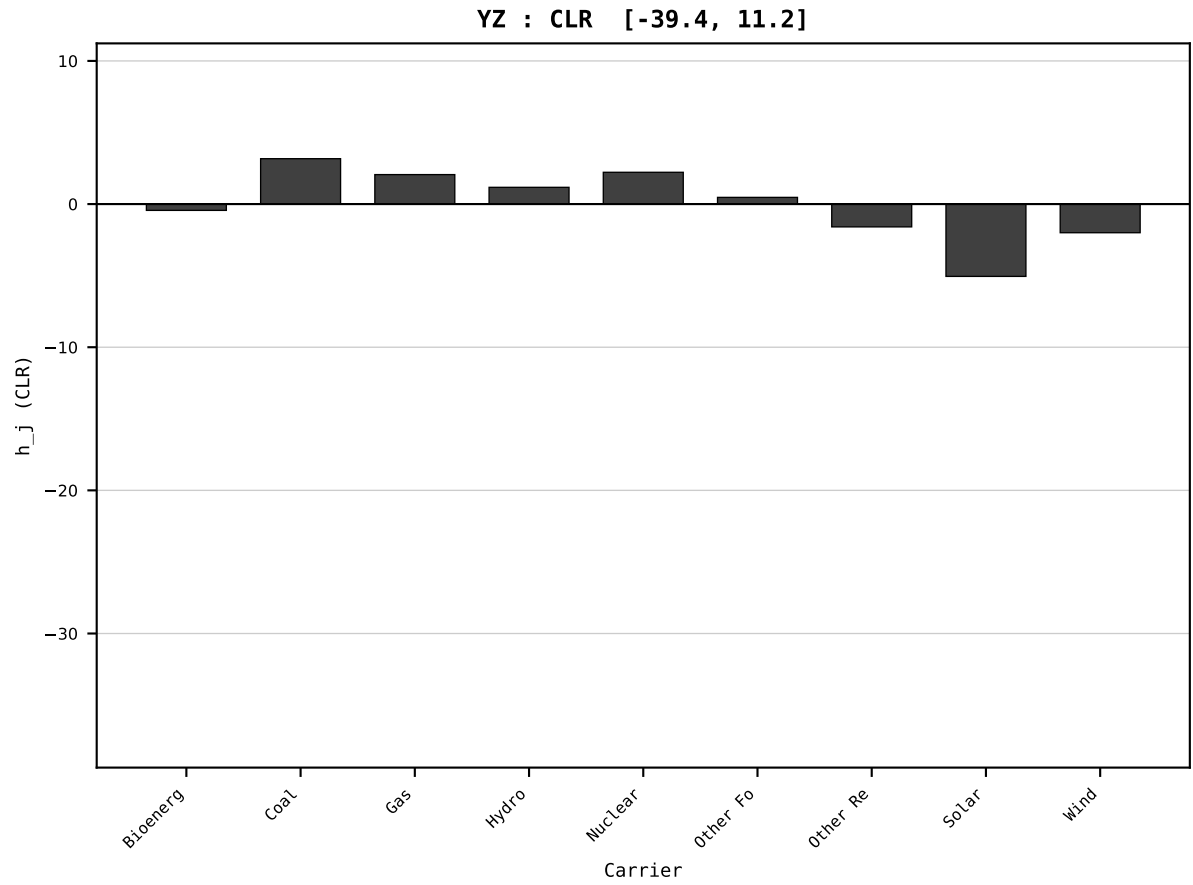
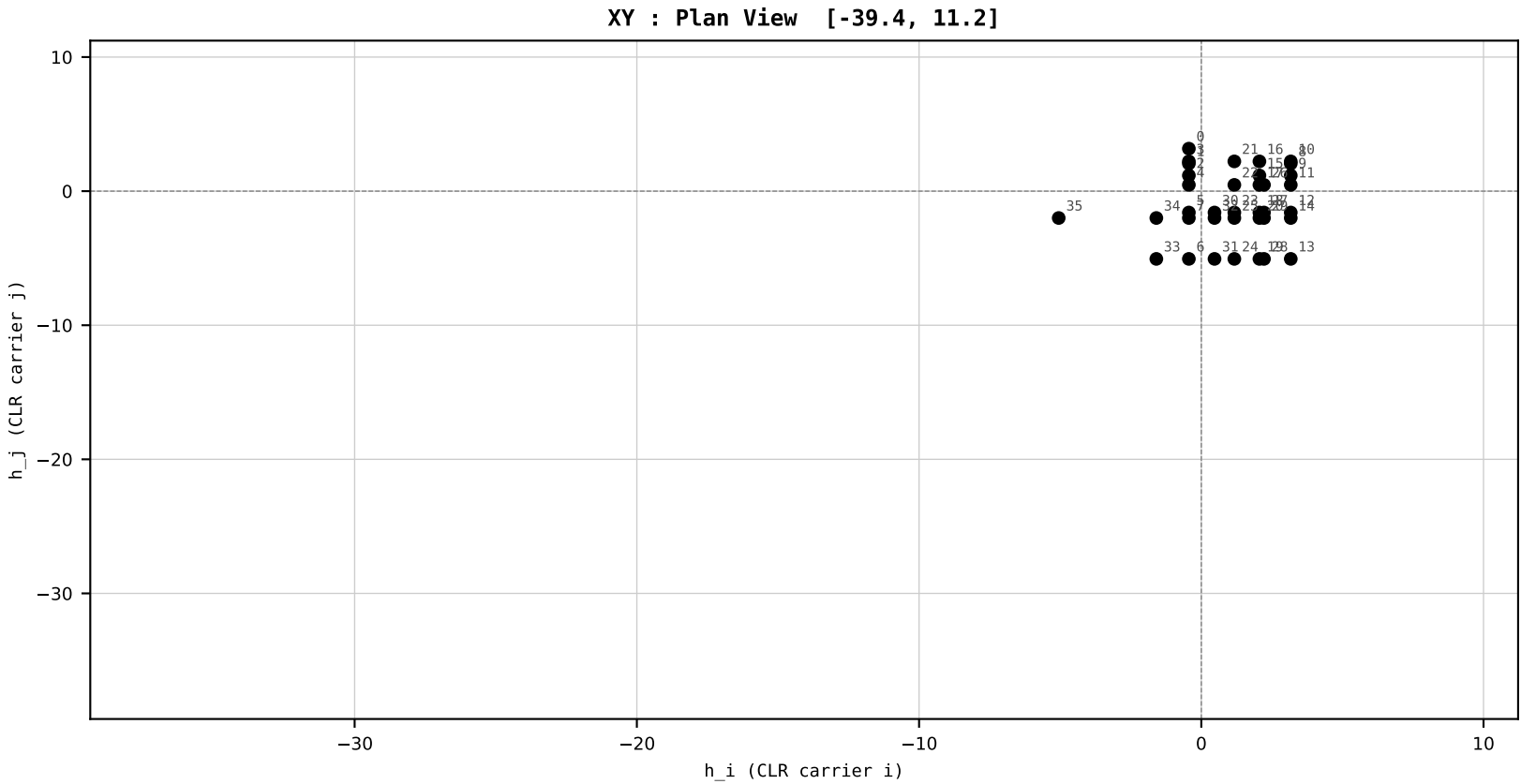
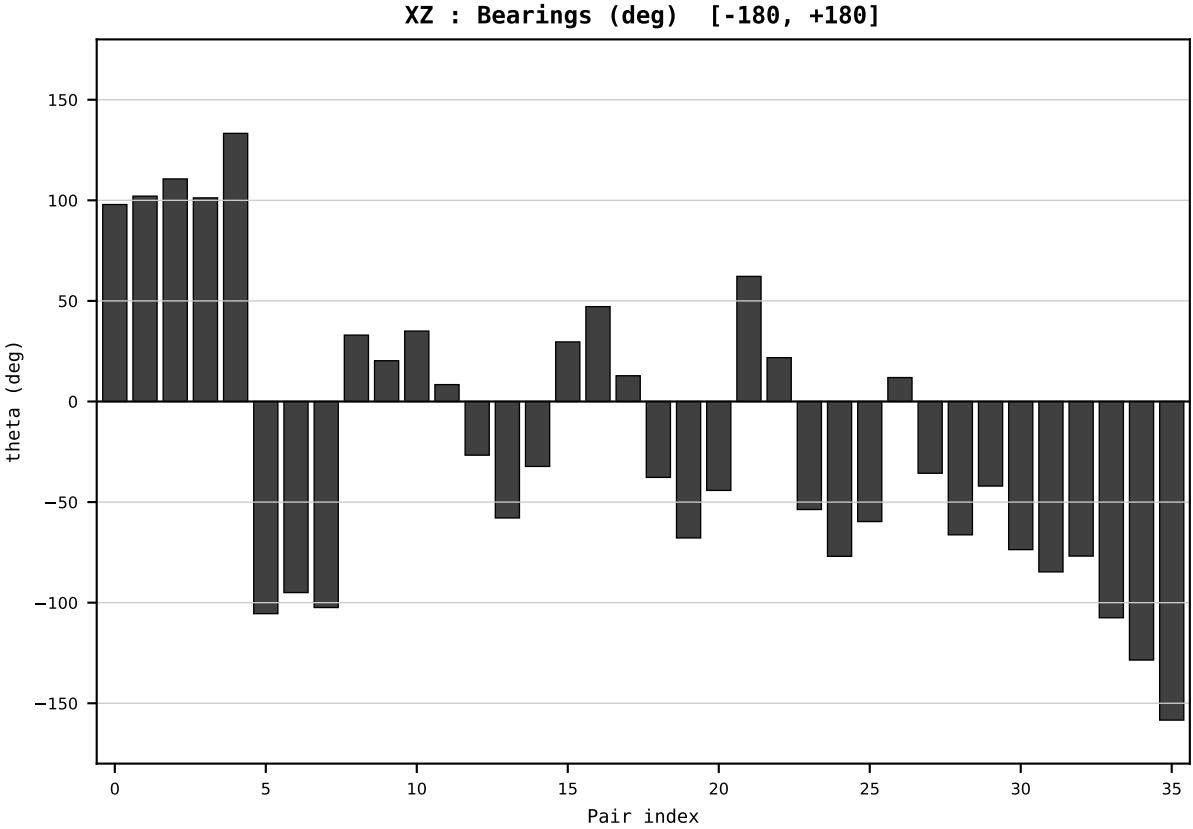
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2003
t : 2 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.379089
Ring = Hs-3
E_metric= 53.0775
kappa_HS= 3724.00
omega = 34.9884 deg
d_A = 31.925473
Helm = Solar
Helm d = +30.098741
DR = 8.223 HLR
DR ratio= 3724.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2004 | D=9 N=25 pairs=36

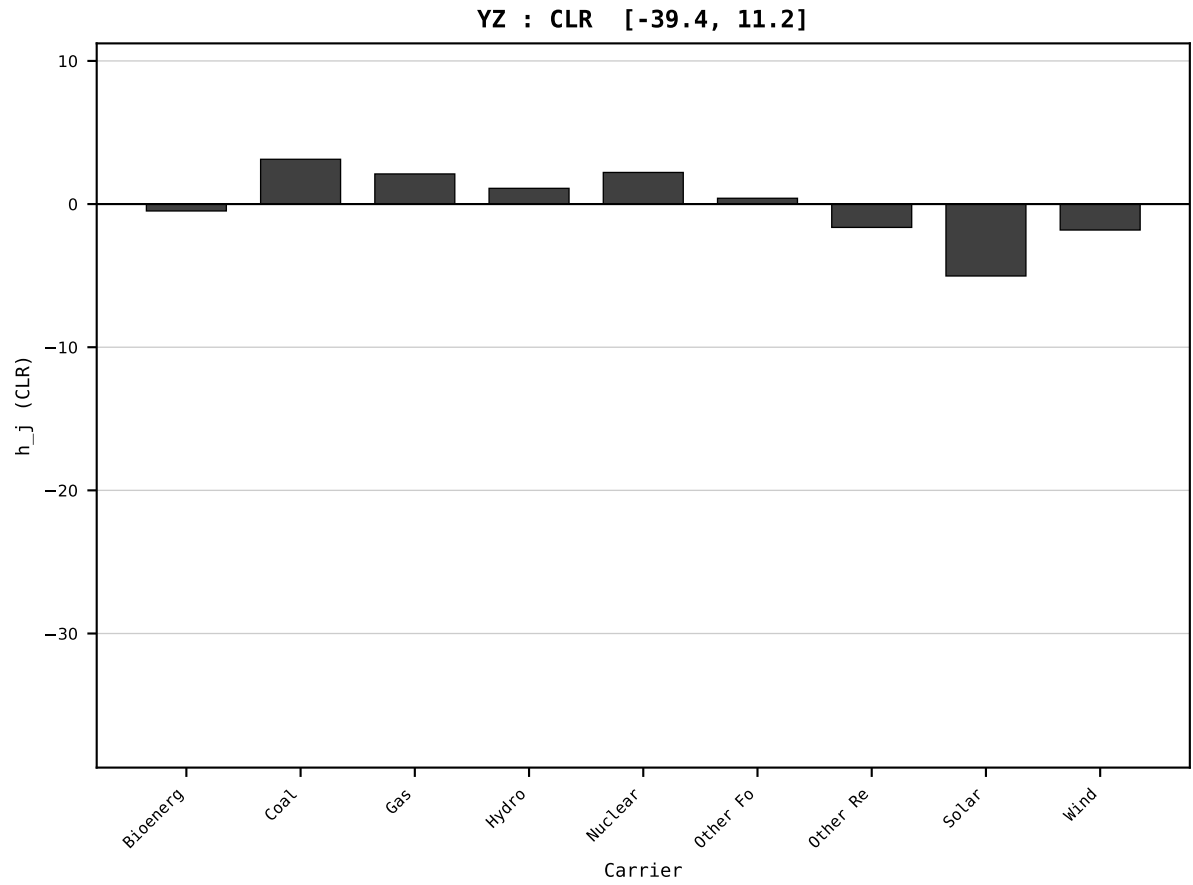
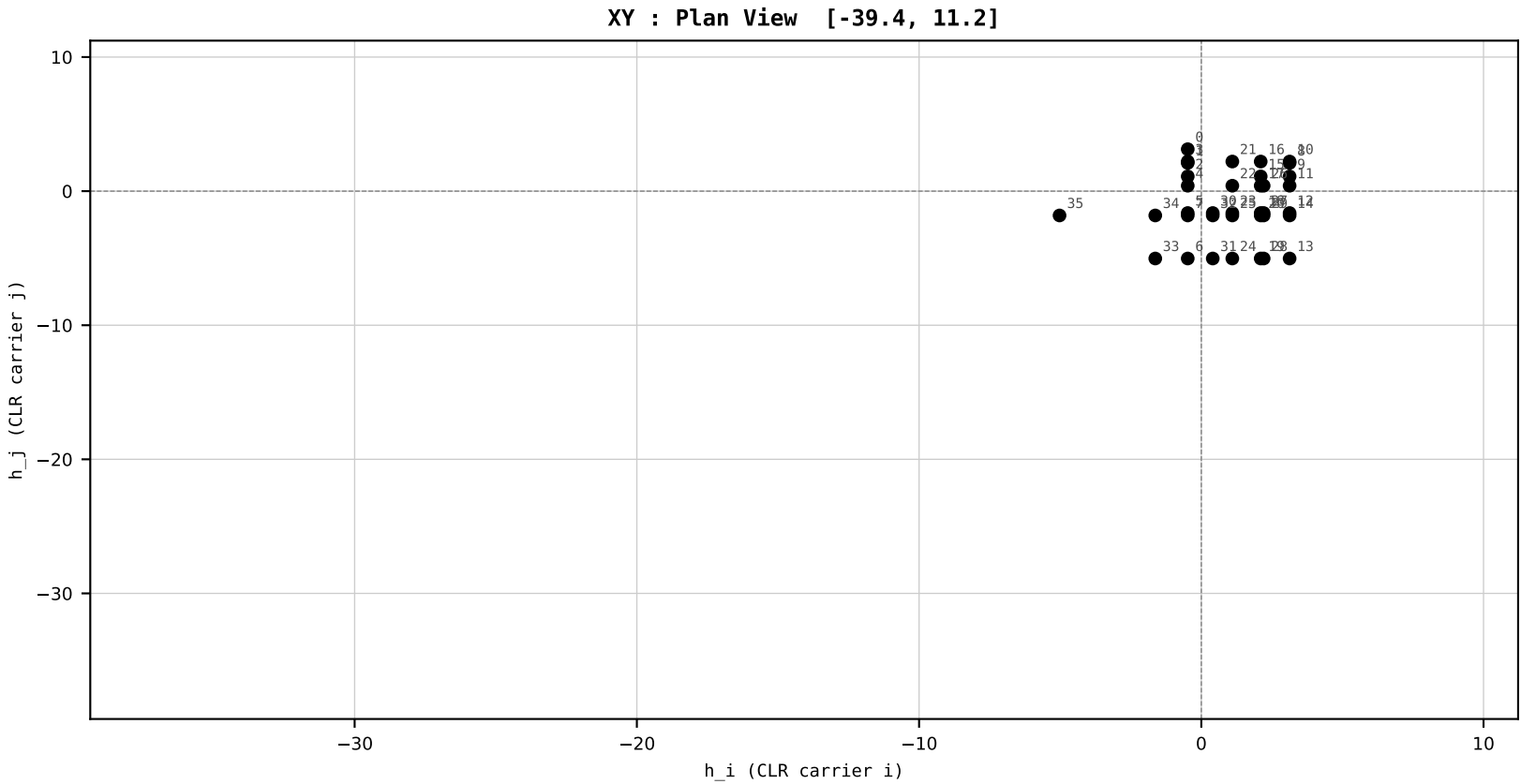
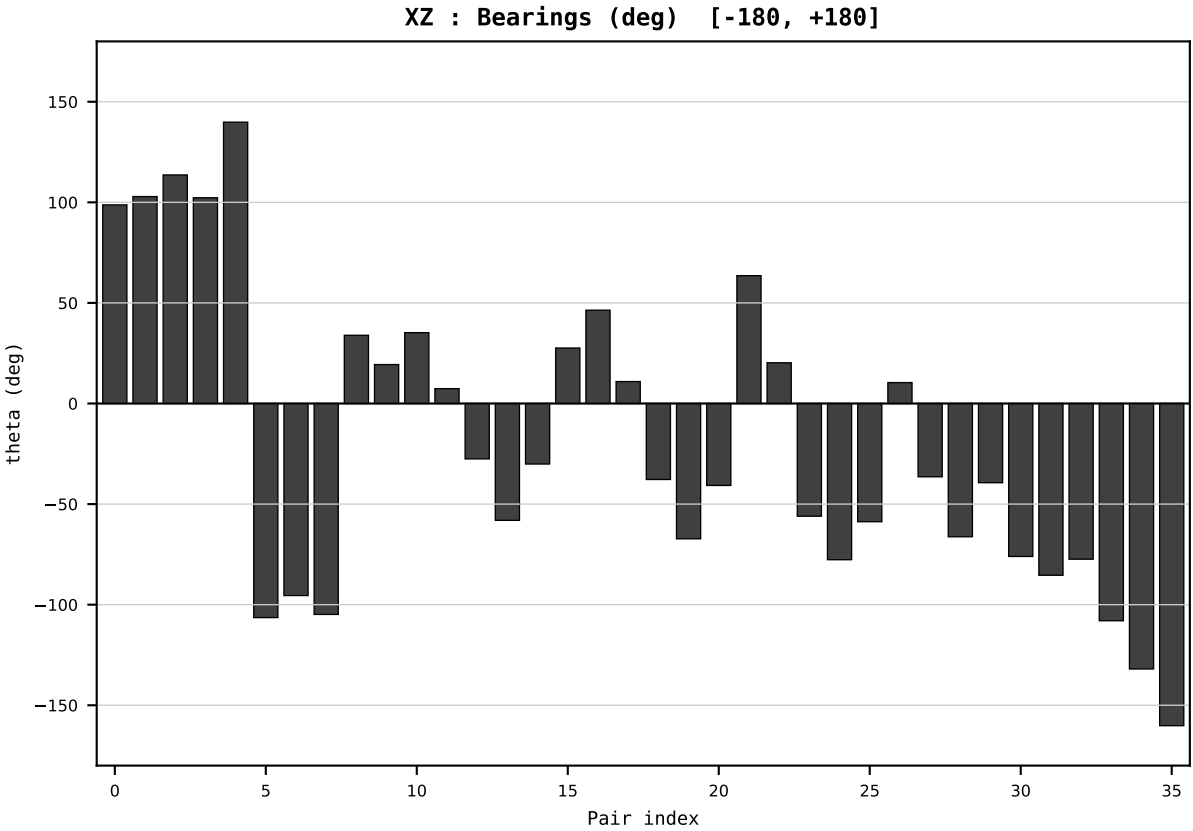
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2004
t : 3 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.376148
Ring = Hs-3
E_metric= 51.8804
kappa_HS= 3470.70
omega = 1.6887 deg
d_A = 0.228934
Helm = Wind
Helm d = +0.188988
DR = 8.152 HLR
DR ratio= 3470.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2005 | D=9 N=25 pairs=36

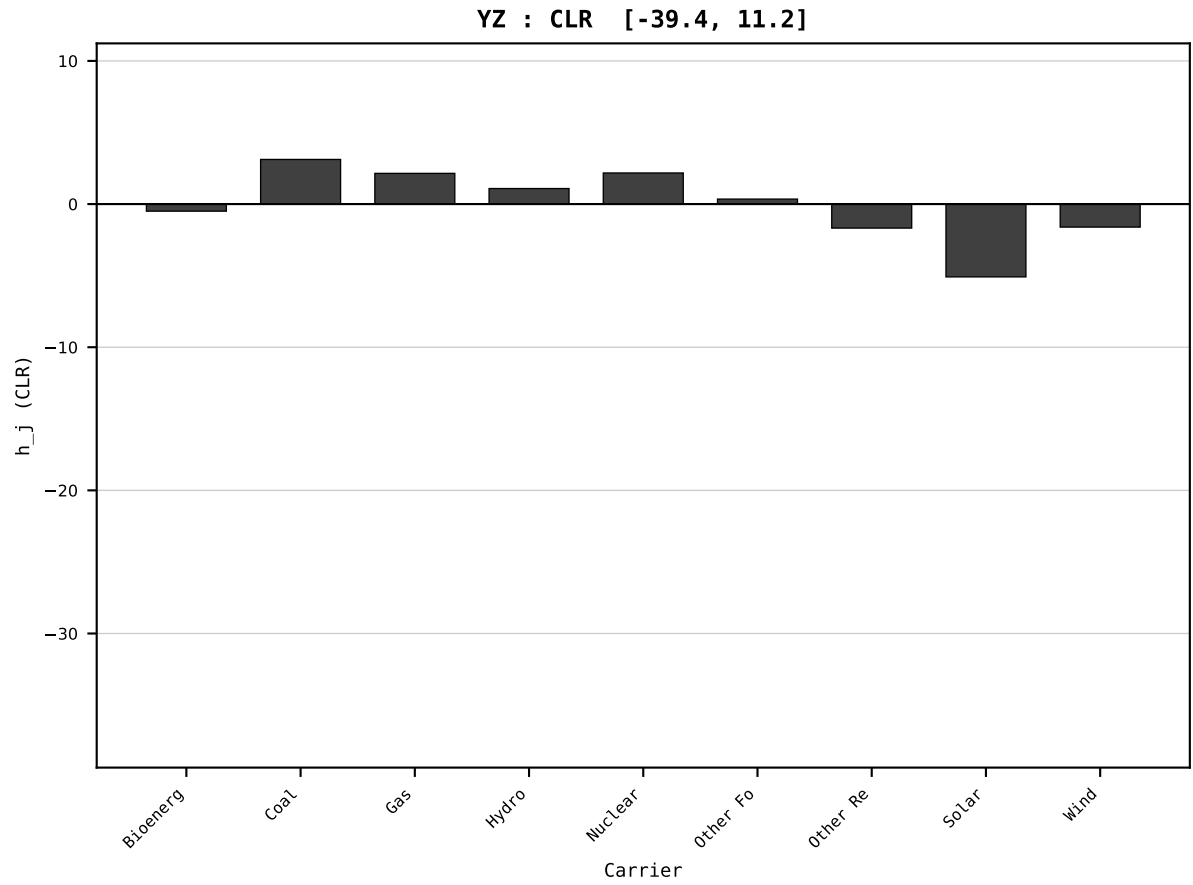
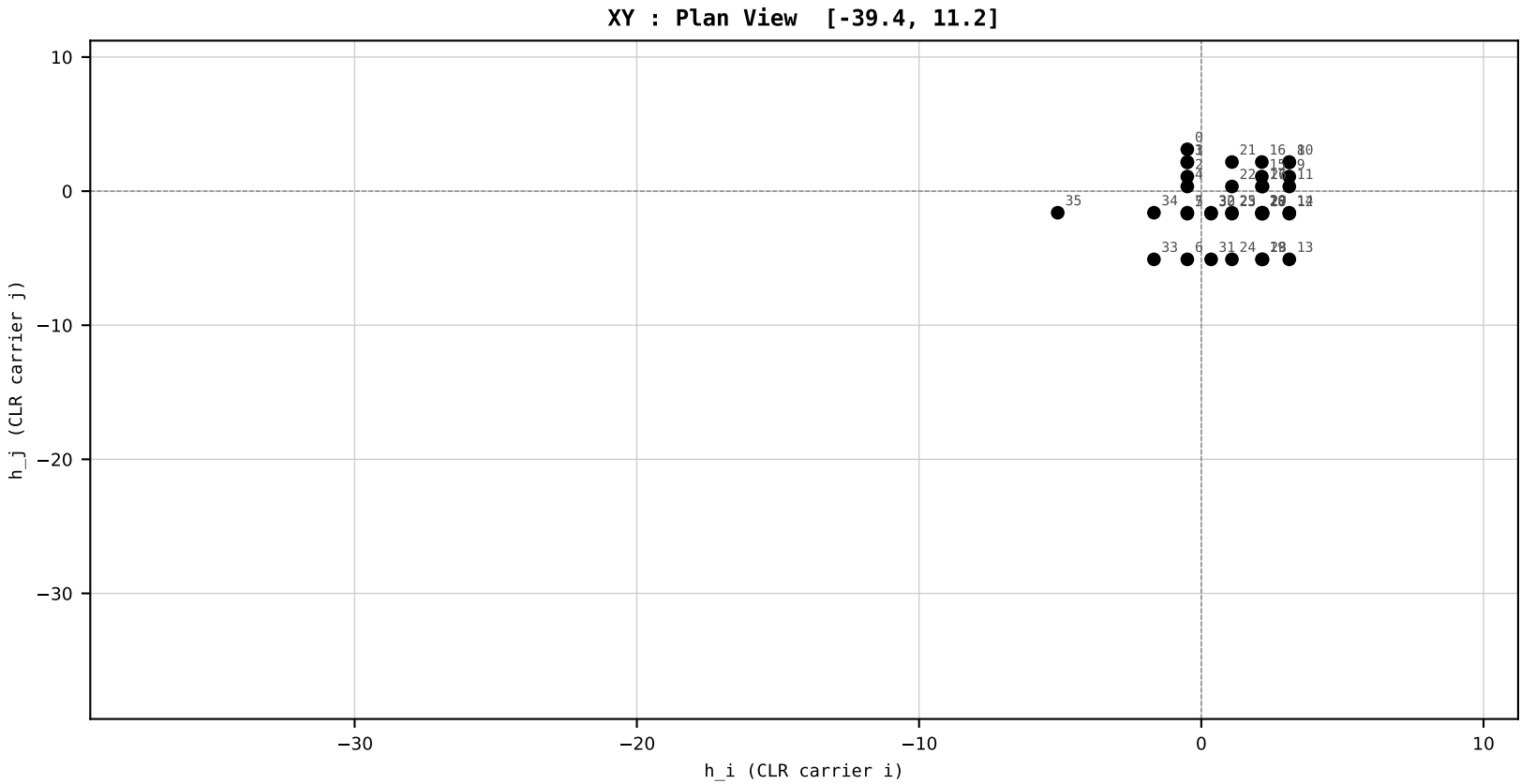
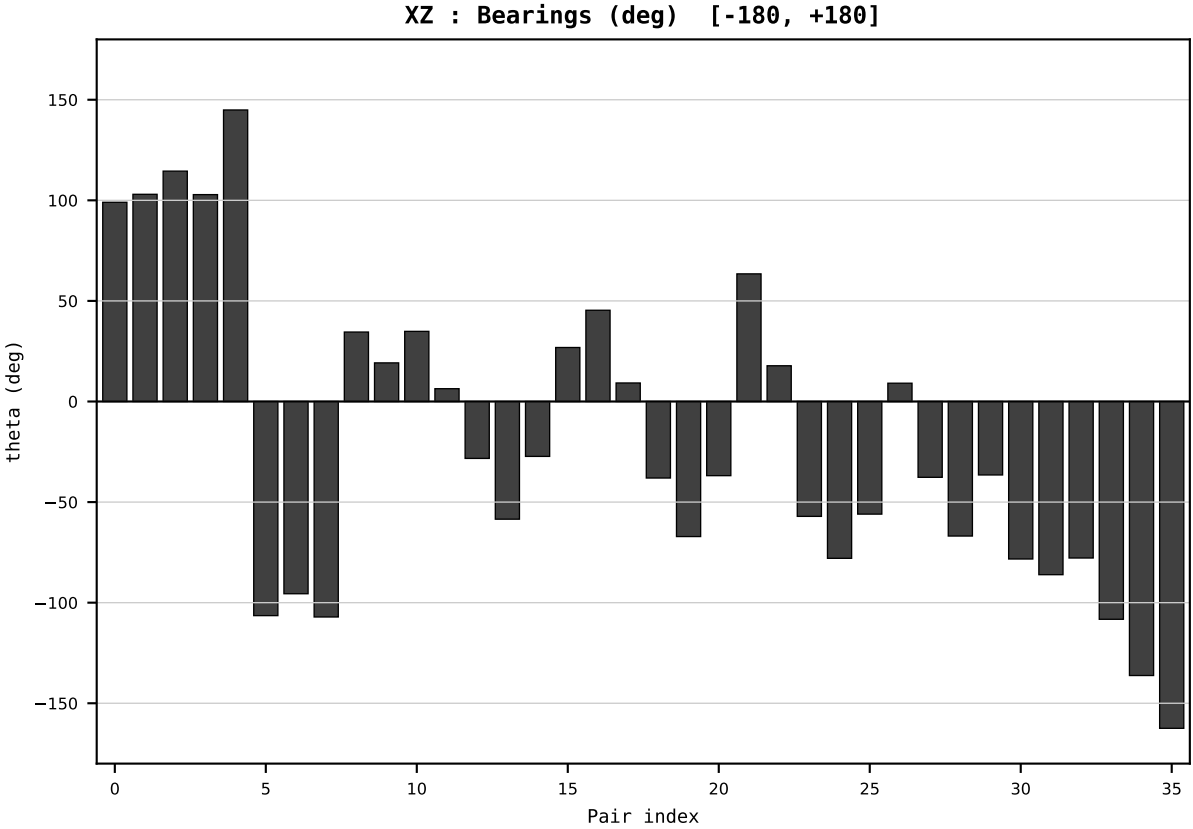
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2005
t : 4 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.375606
Ring = Hs-3
E_metric= 51.8713
kappa_HS= 3659.76
omega = 1.8643 deg
d_A = 0.234341
Helm = Wind
Helm d = +0.204228
DR = 8.205 HLR
DR ratio= 3659.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2006 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

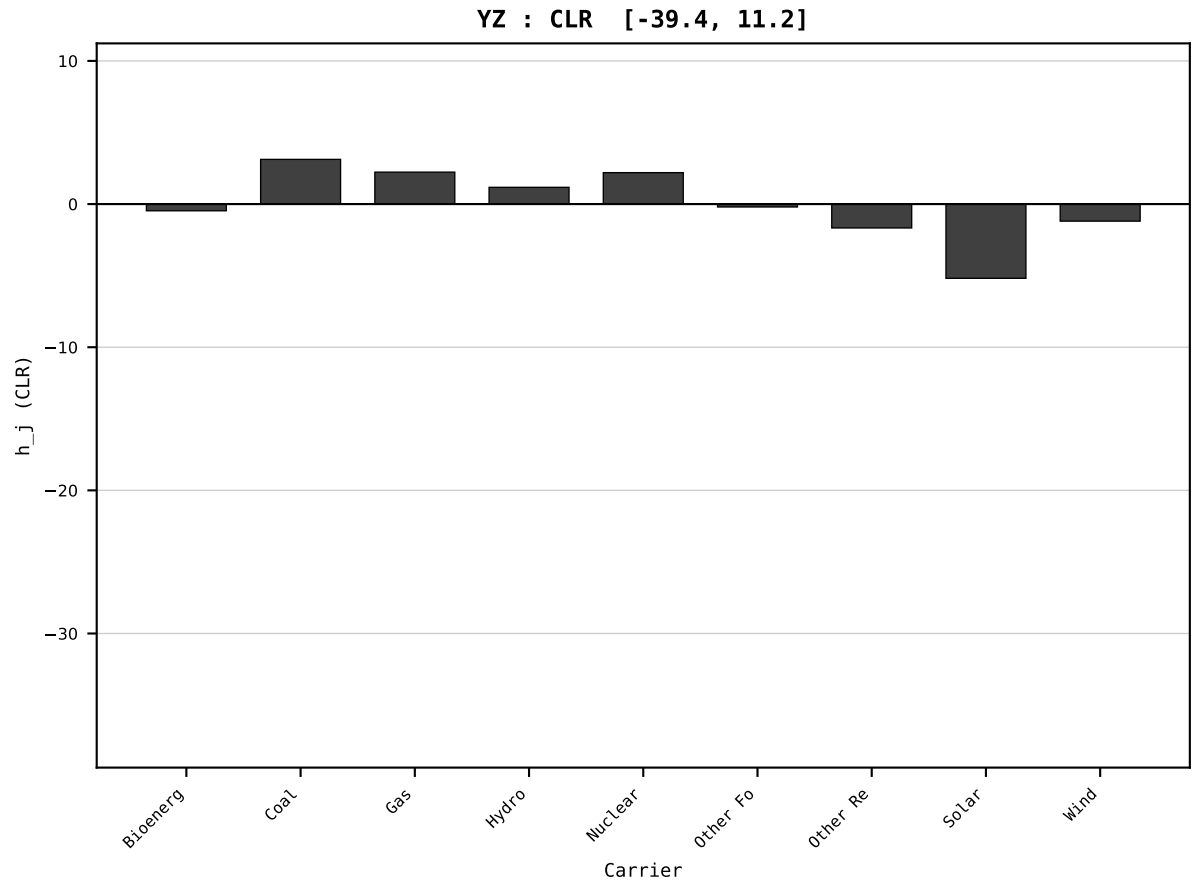
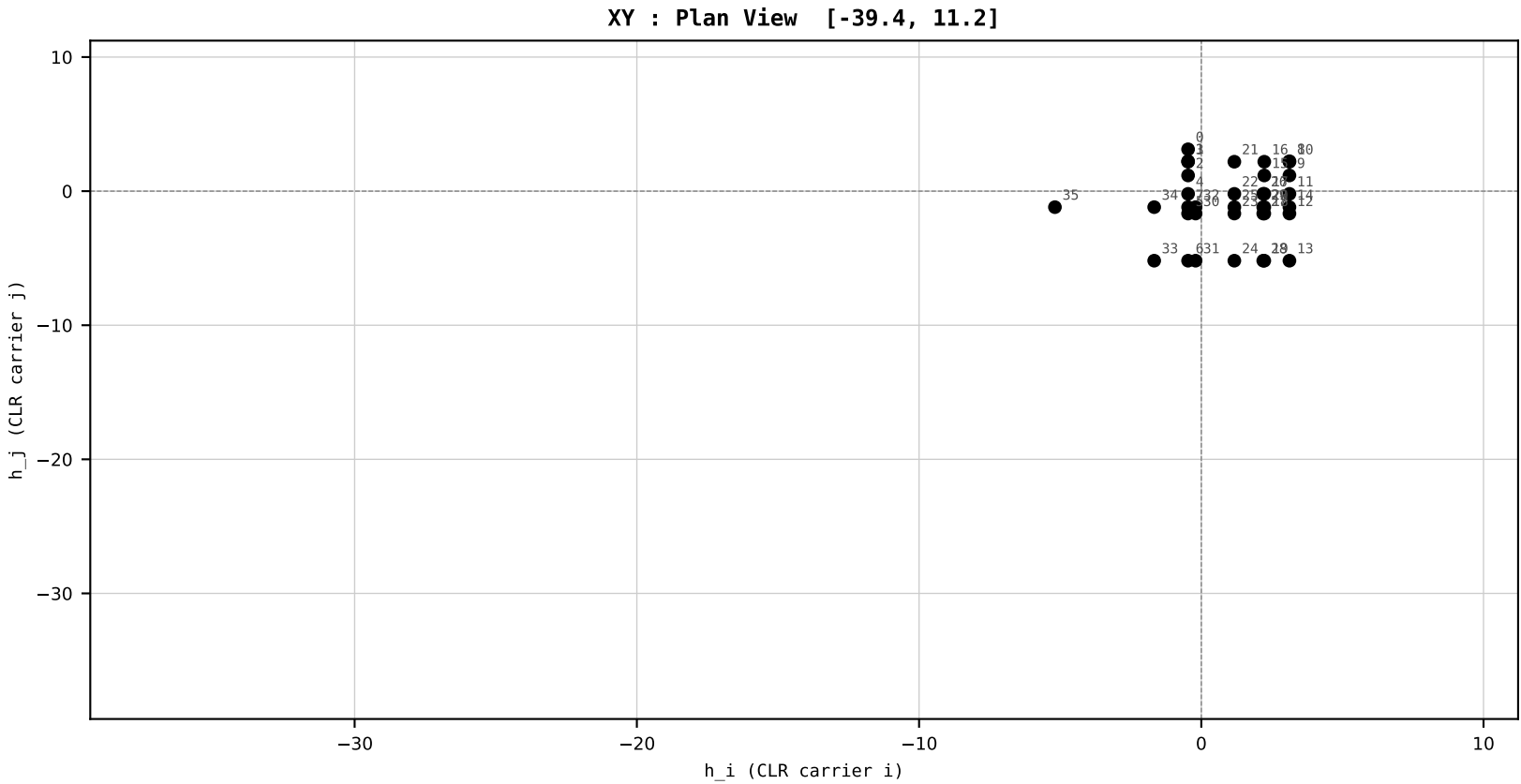
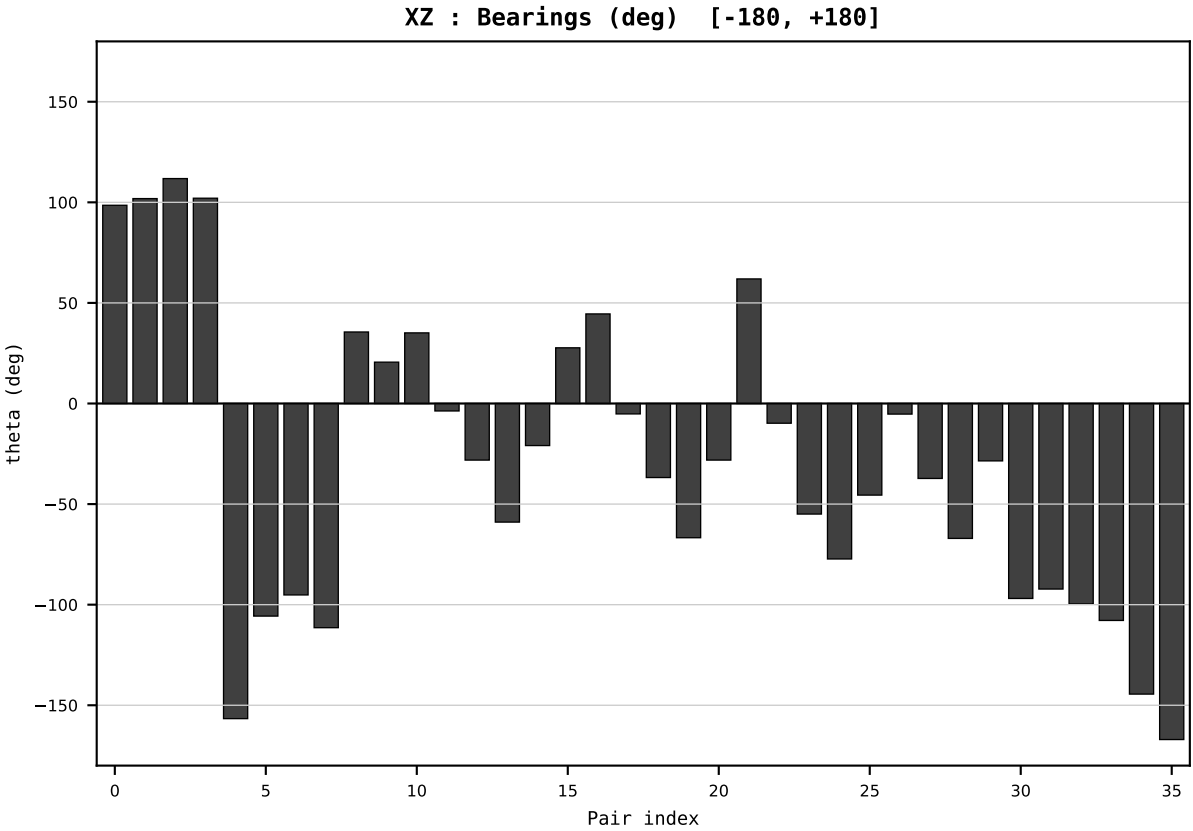
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2006
t : 5 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.379864
Ring = Hs-3
E_metric= 52.2936
kappa_HS= 4062.29
omega = 5.6182 deg
d_A = 0.707976
Helm = Other Fossil
Helm d = -0.550093
DR = 8.310 HLR
DR ratio= 4062.3x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2007 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

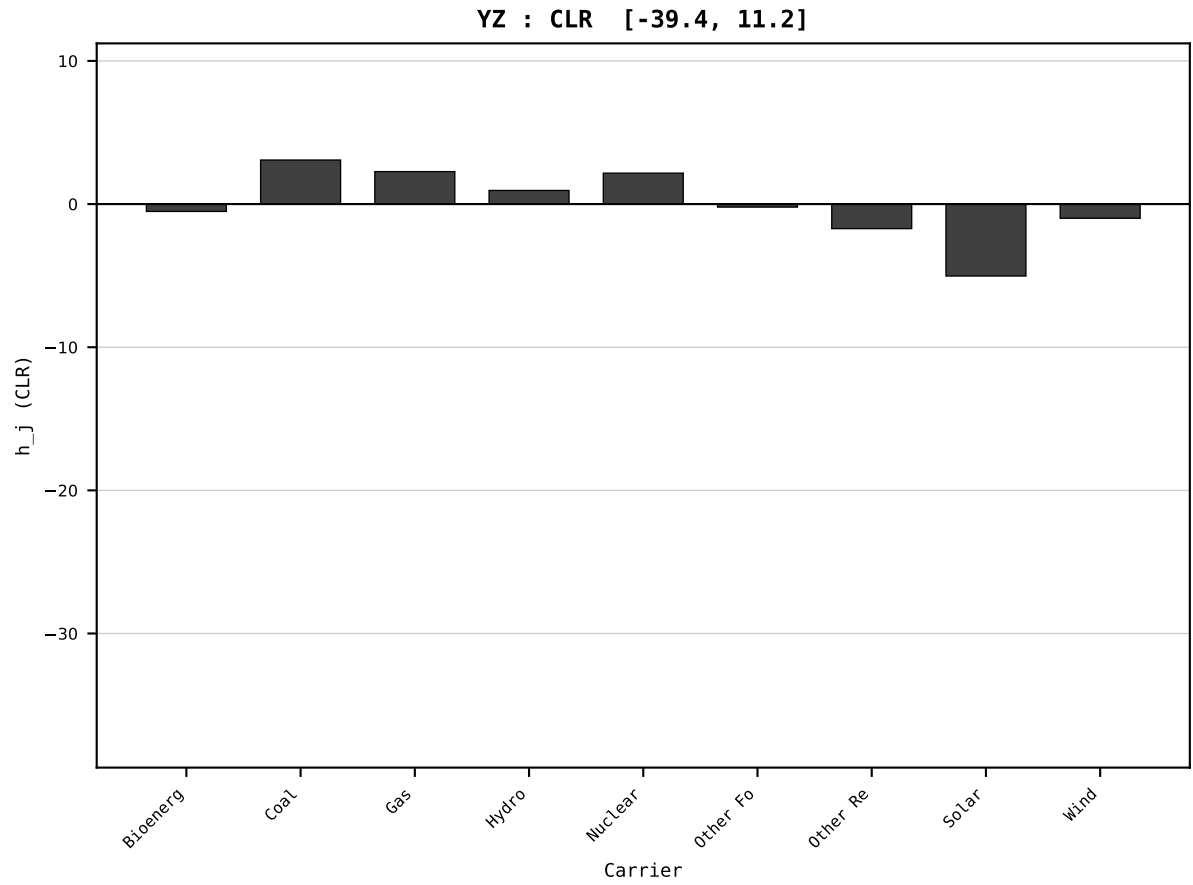
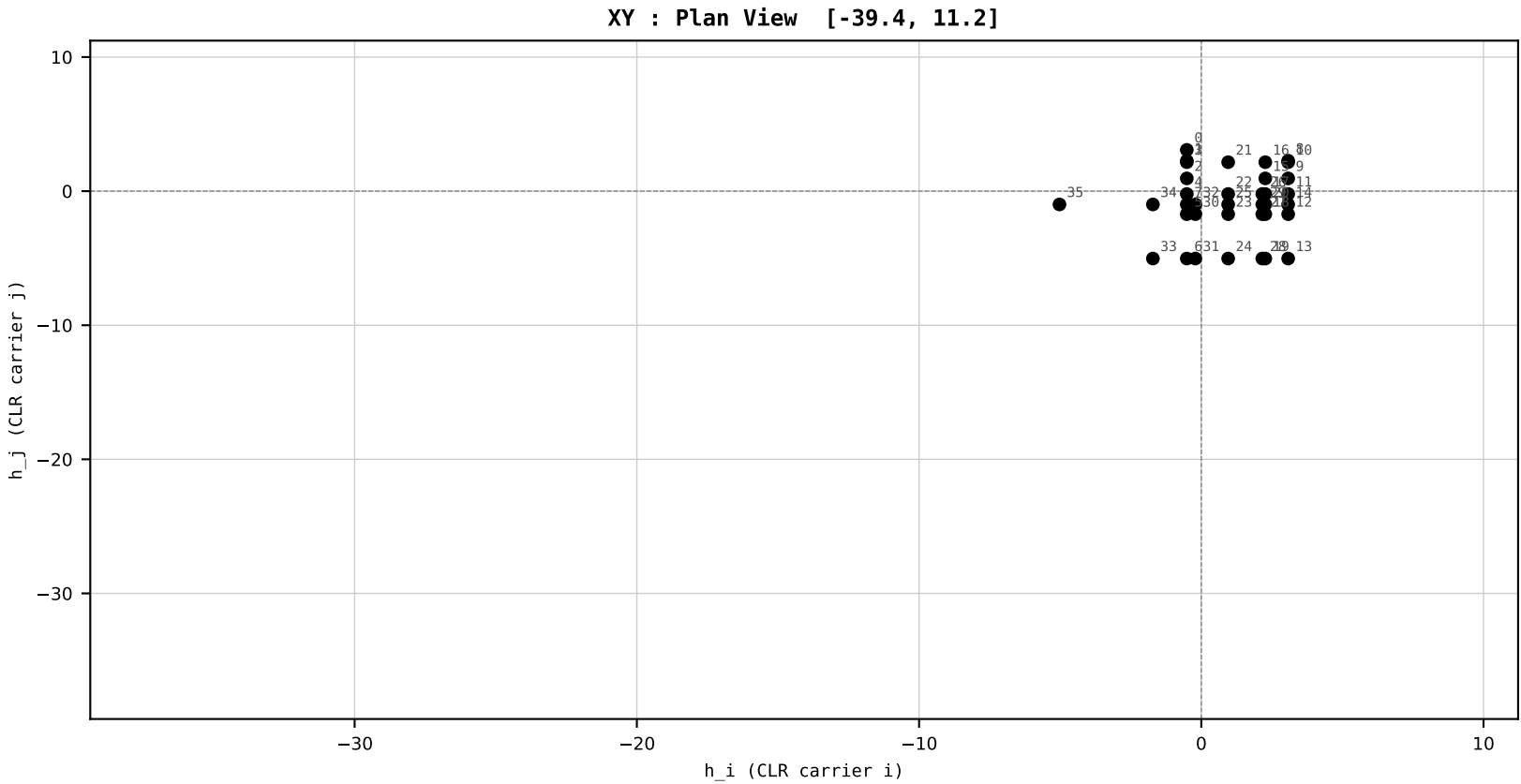
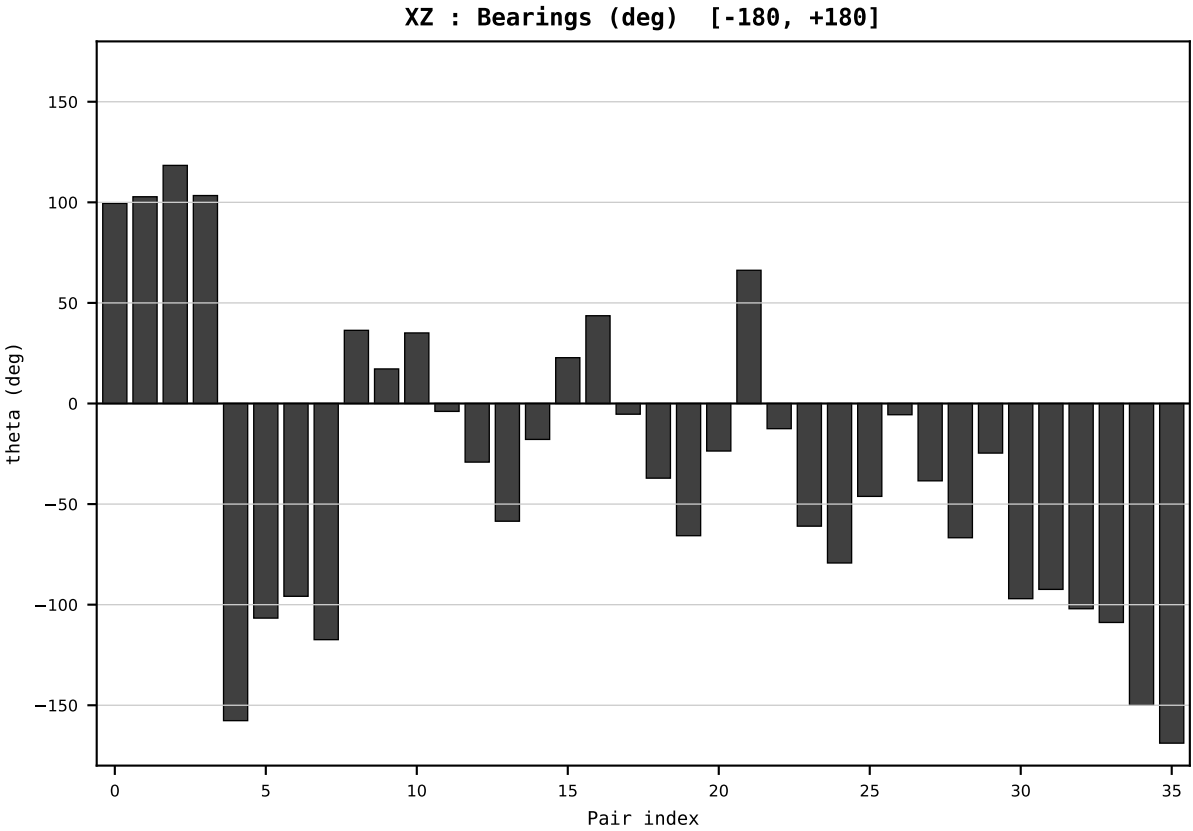
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2007
t : 6 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.381244
Ring = Hs-3
E_metric= 49.6813
kappa_HS= 3305.62
omega = 2.4037 deg
d_A = 0.350949
Helm = Hydro
Helm d = -0.219022
DR = 8.103 HLR
DR ratio= 3305.6x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2008 | D=9 N=25 pairs=36

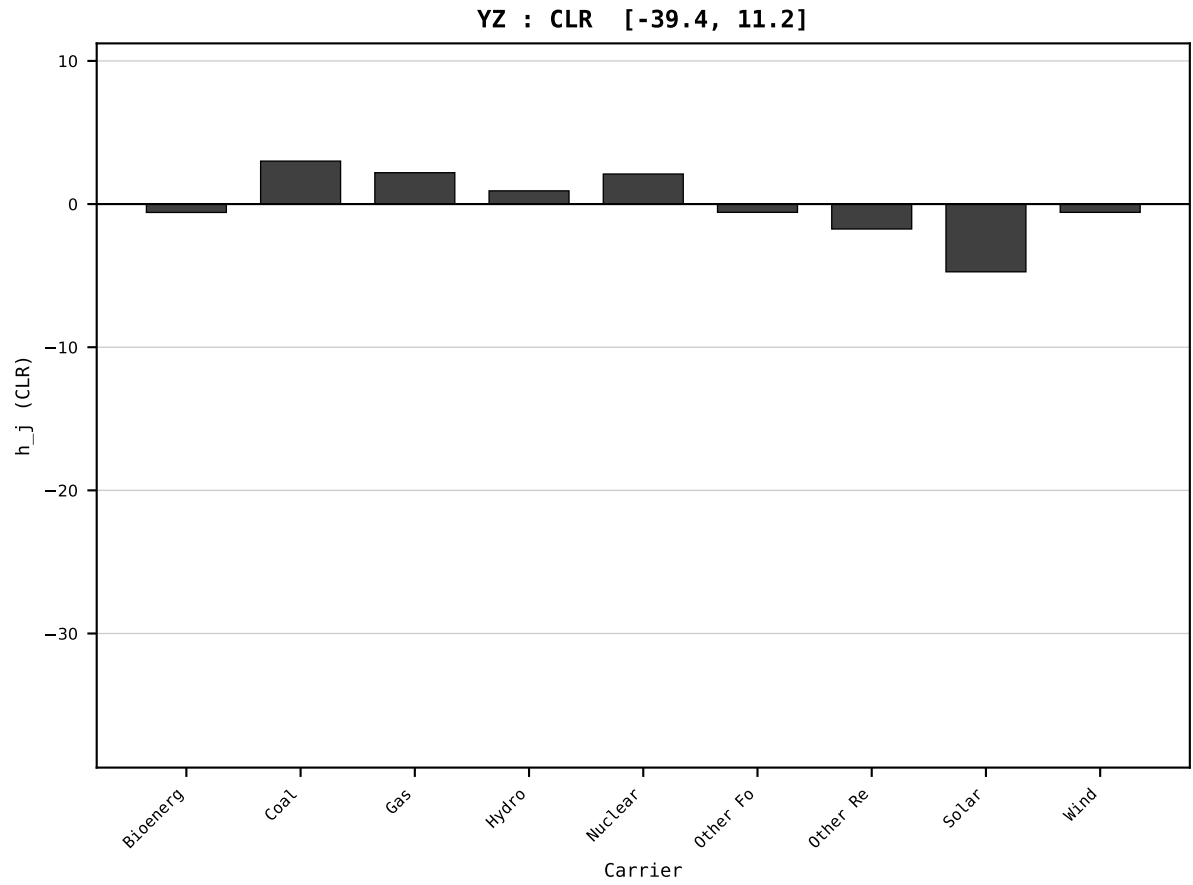
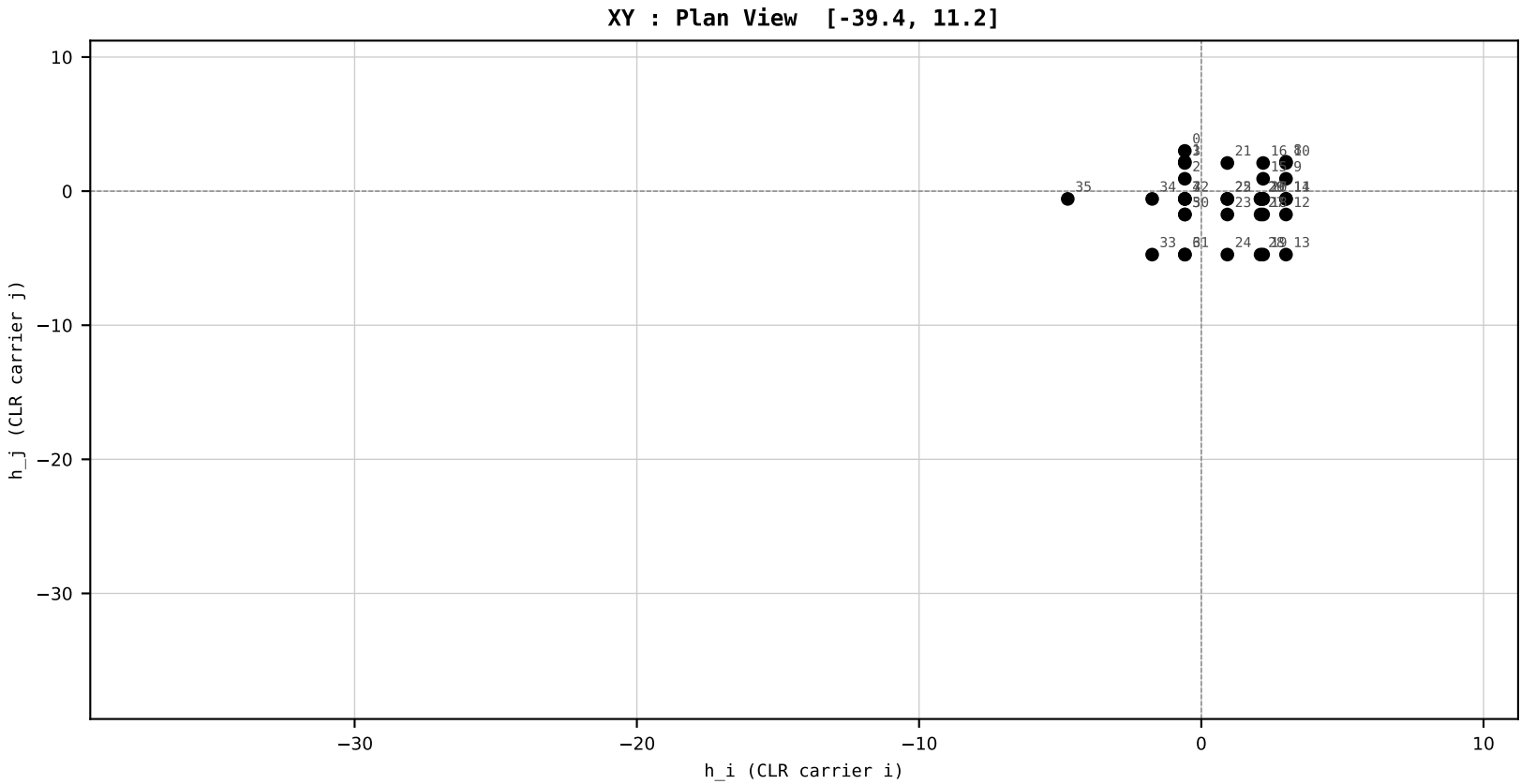
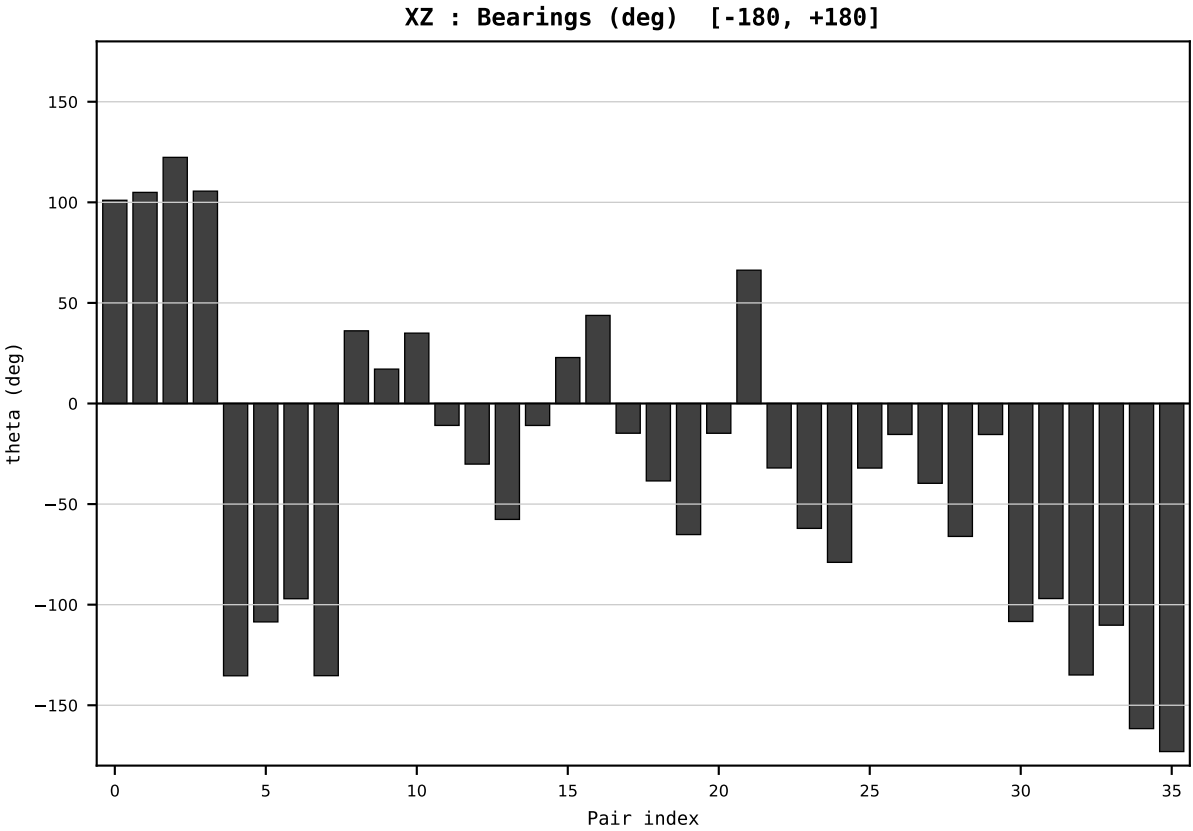
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2008
t : 7 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.376532
Ring = Hs-3
E_metric= 45.5003
kappa_HS= 2282.54
omega = 4.7104 deg
d_A = 0.642683
Helm = Wind
Helm d = +0.412825
DR = 7.733 HLR
DR ratio= 2282.5x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2009 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

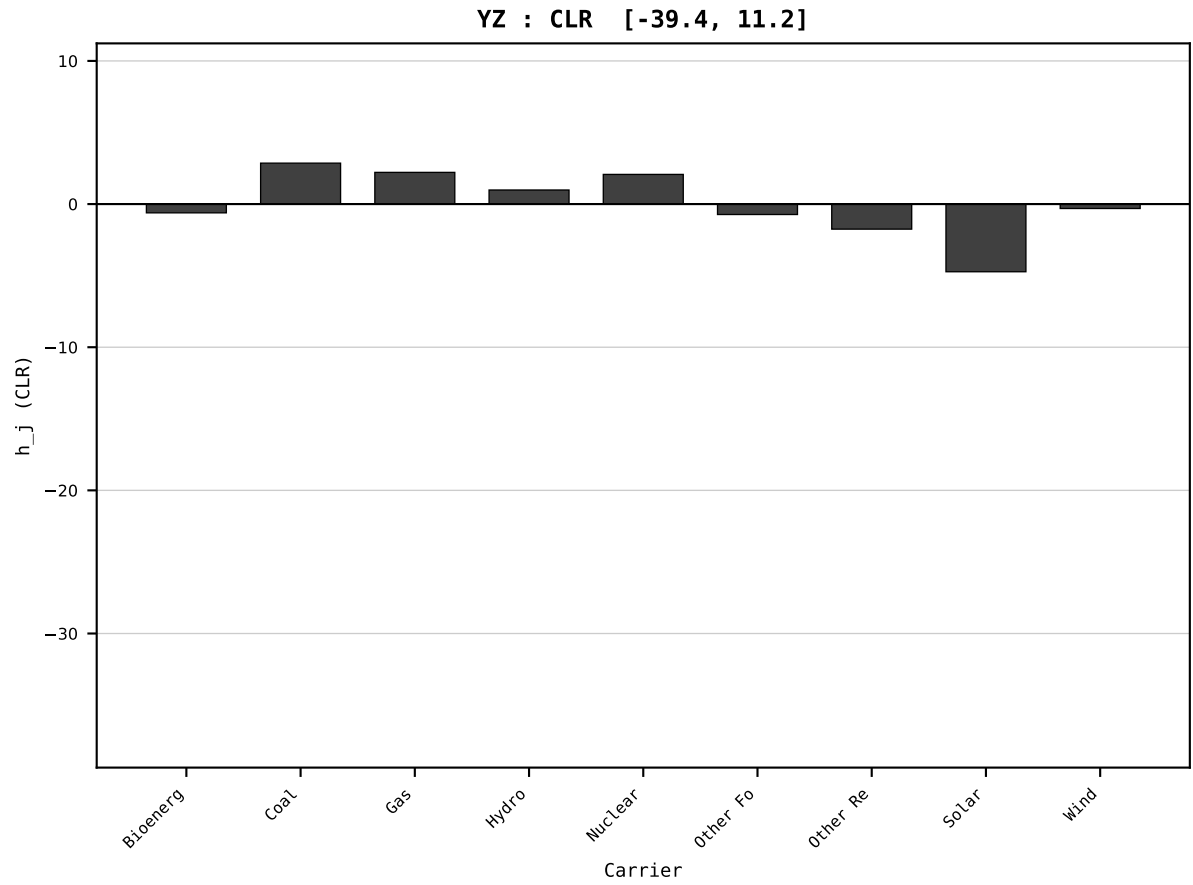
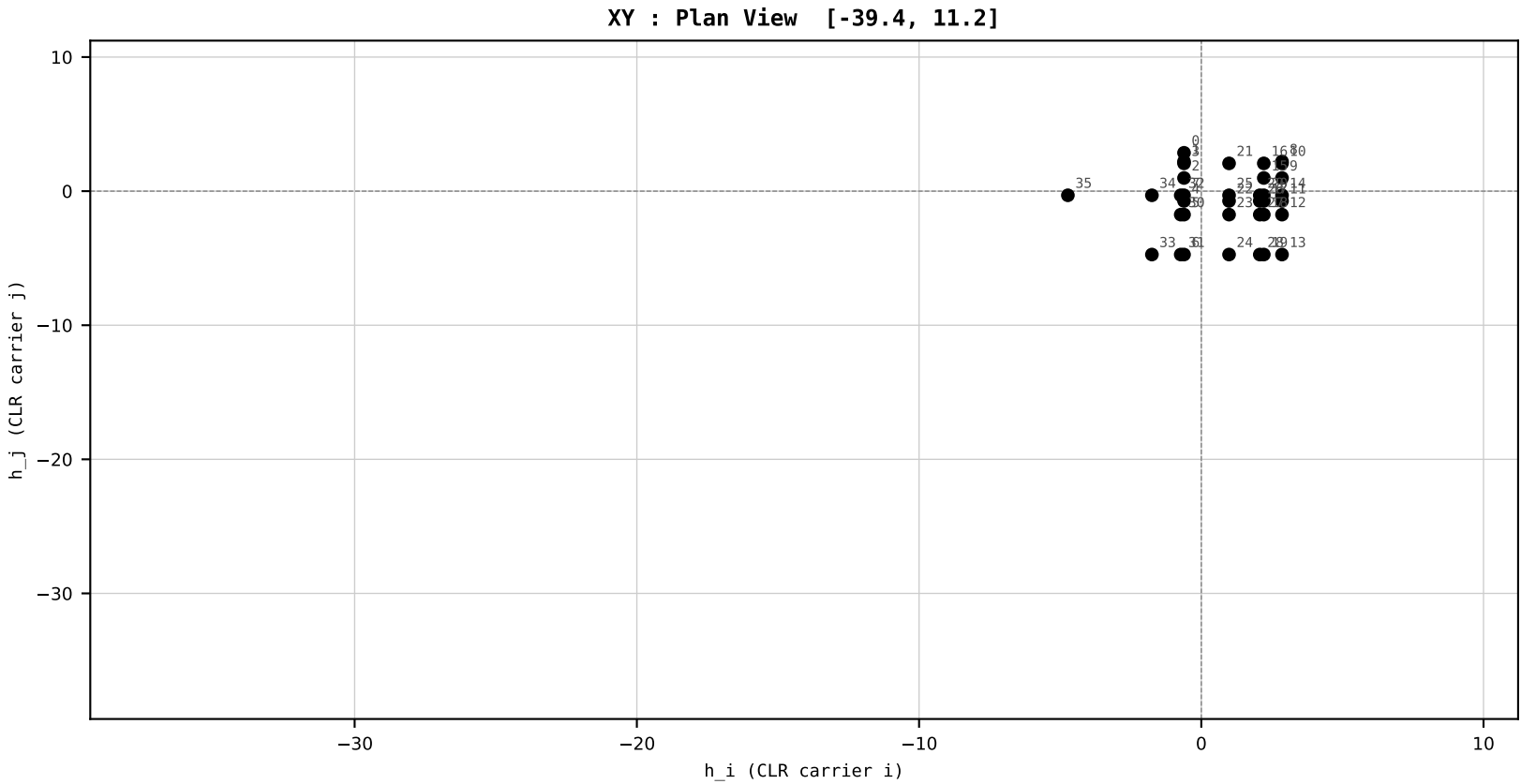
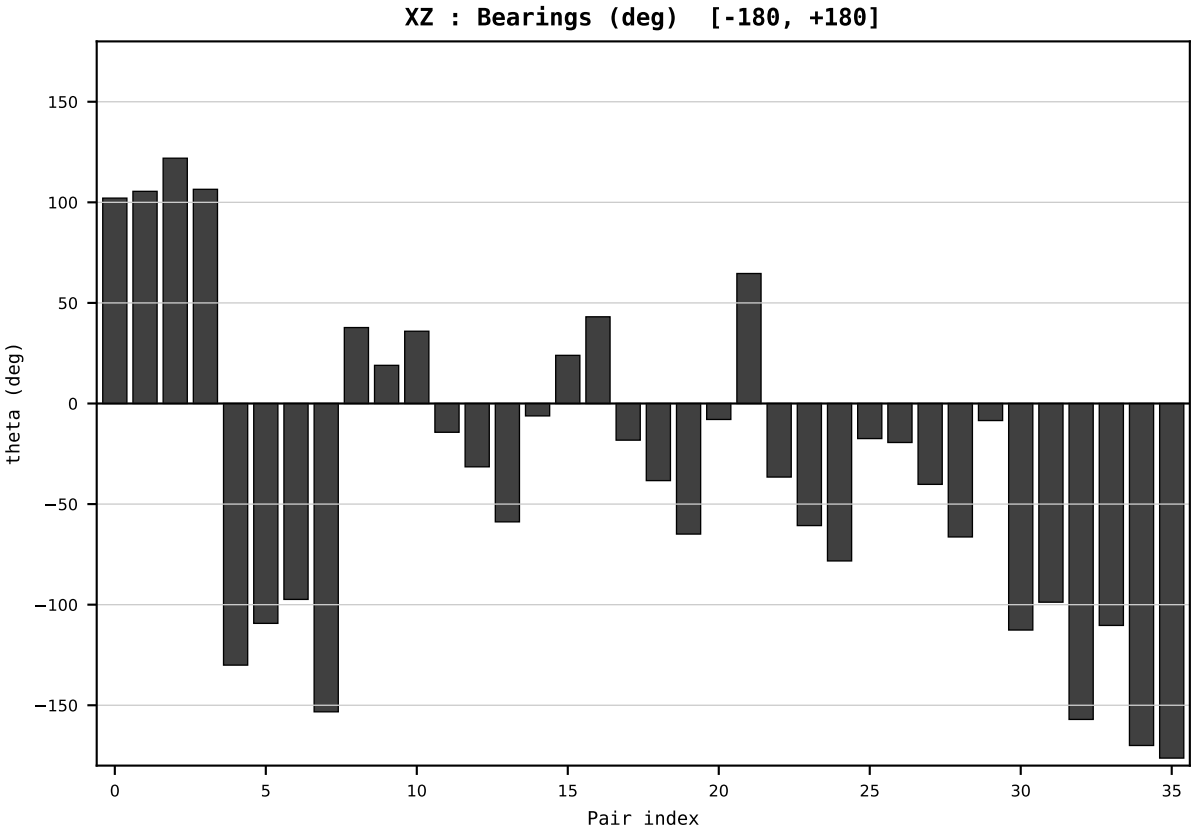
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2009
t : 8 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.353477
Ring = Hs-3
E_metric= 44.7578
kappa_HS= 1972.93
omega = 2.9378 deg
d_A = 0.348811
Helm = Wind
Helm d = +0.270064
DR = 7.587 HLR
DR ratio= 1972.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=9 | Year 2010 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

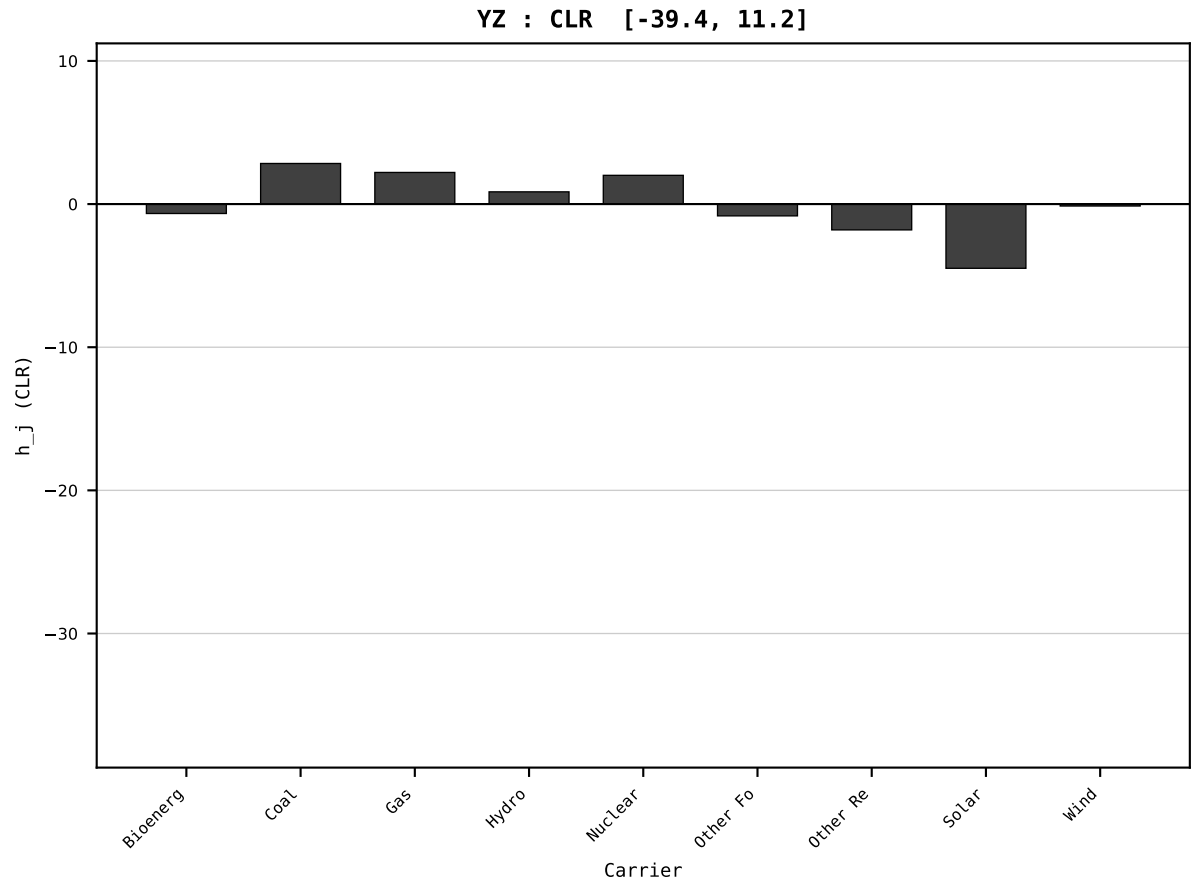
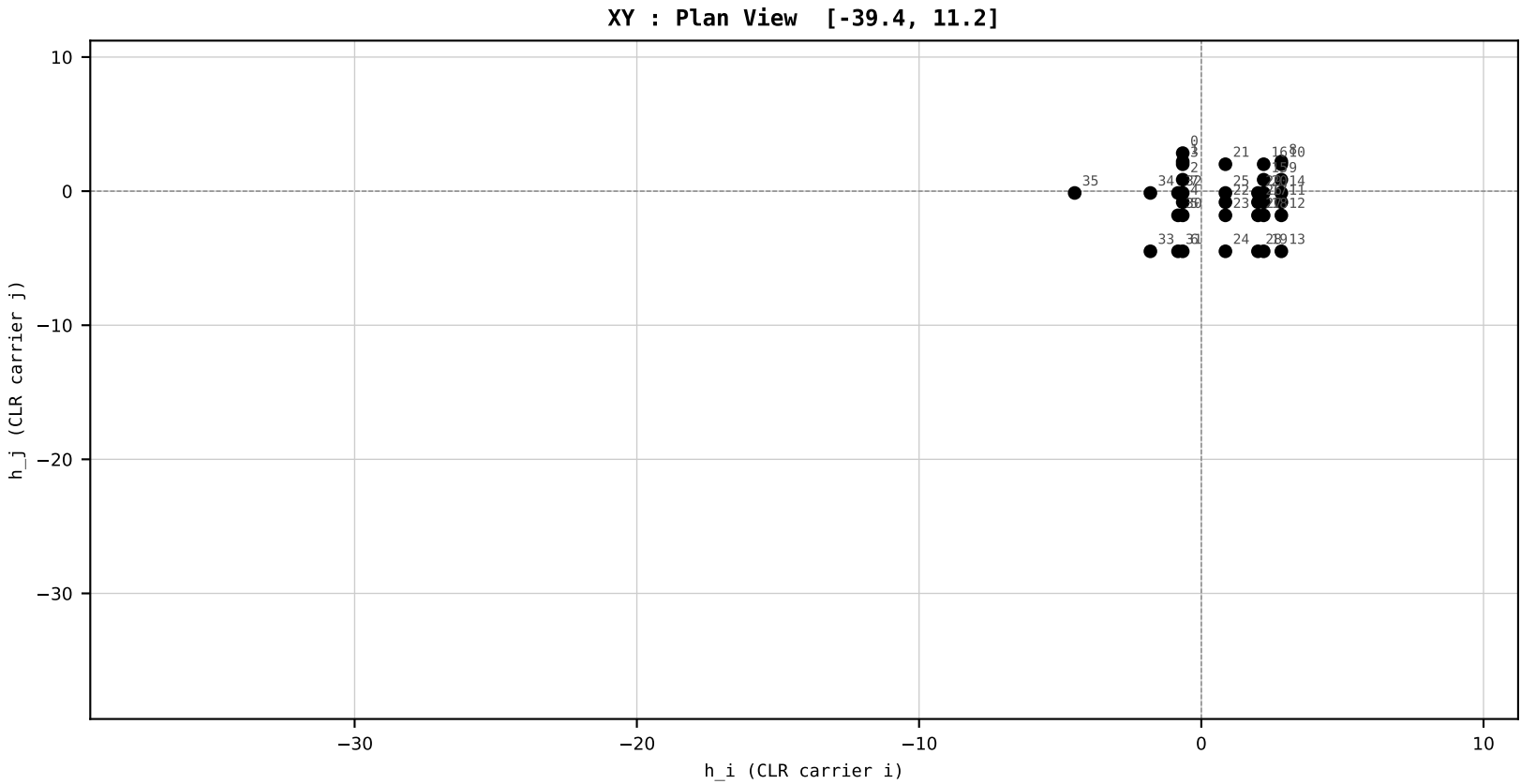
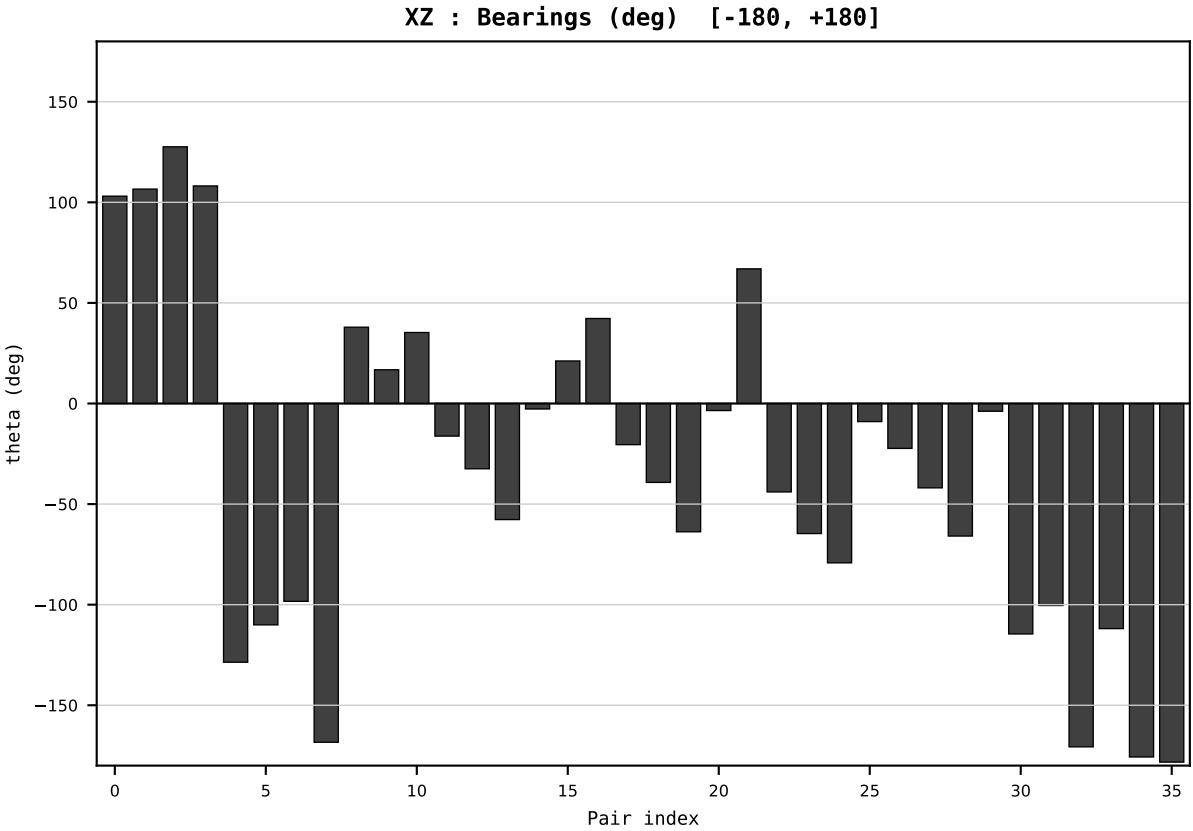
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2010
t : 9 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.355235
Ring = Hs-3
E_metric= 42.2037
kappa_HS= 1514.16
omega = 2.5467 deg
d_A = 0.351238
Helm = Solar
Helm d = +0.241100
DR = 7.323 HLR
DR ratio= 1514.2x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2011 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

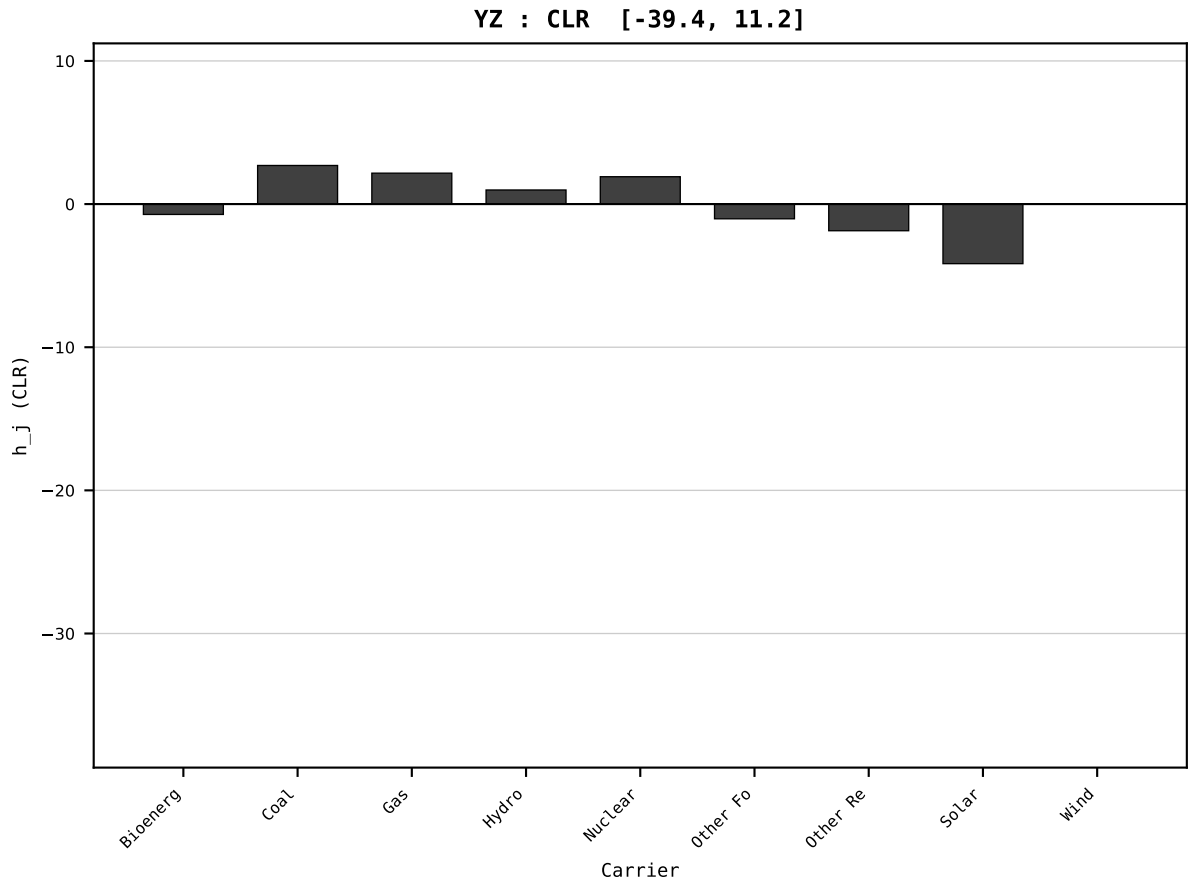
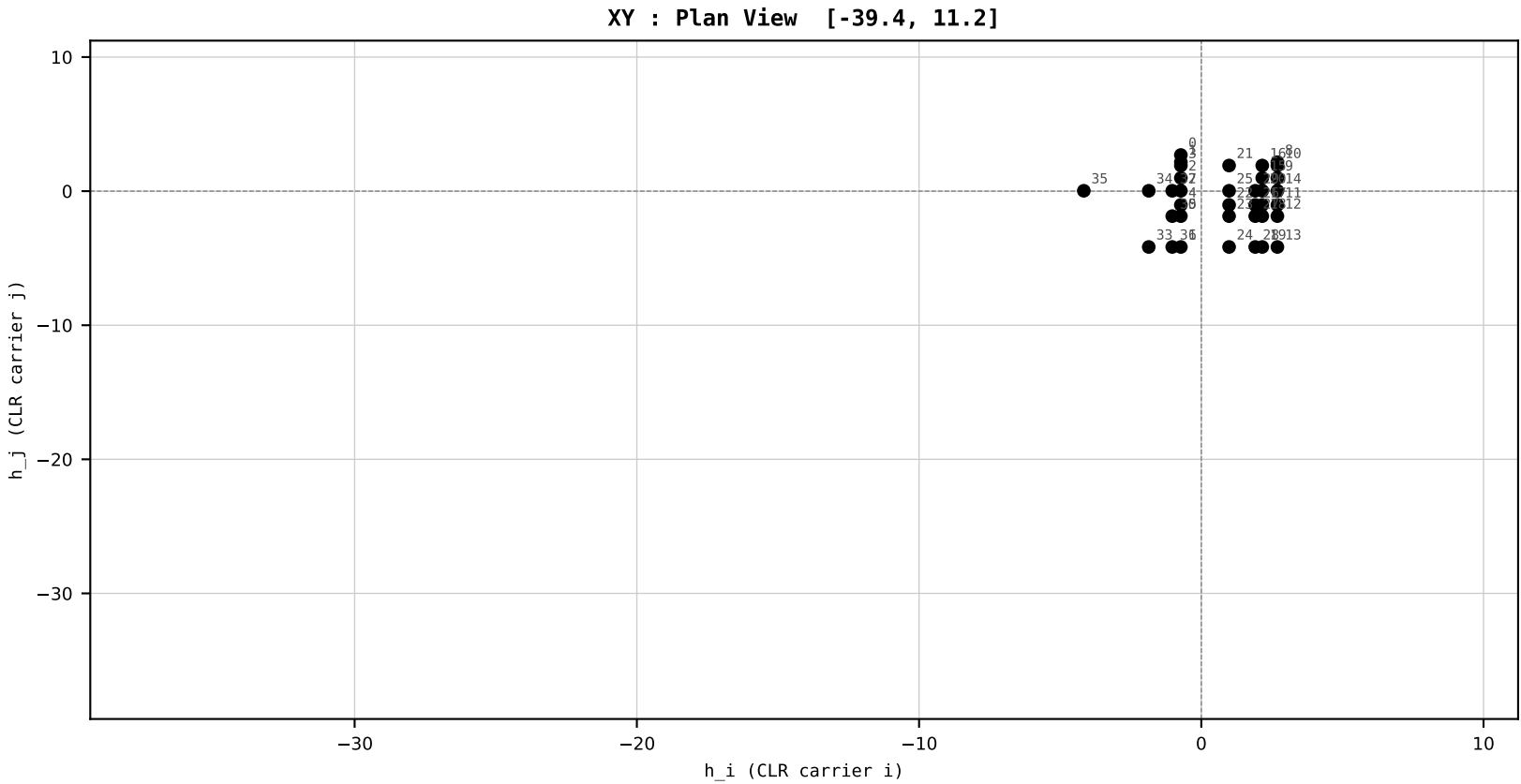
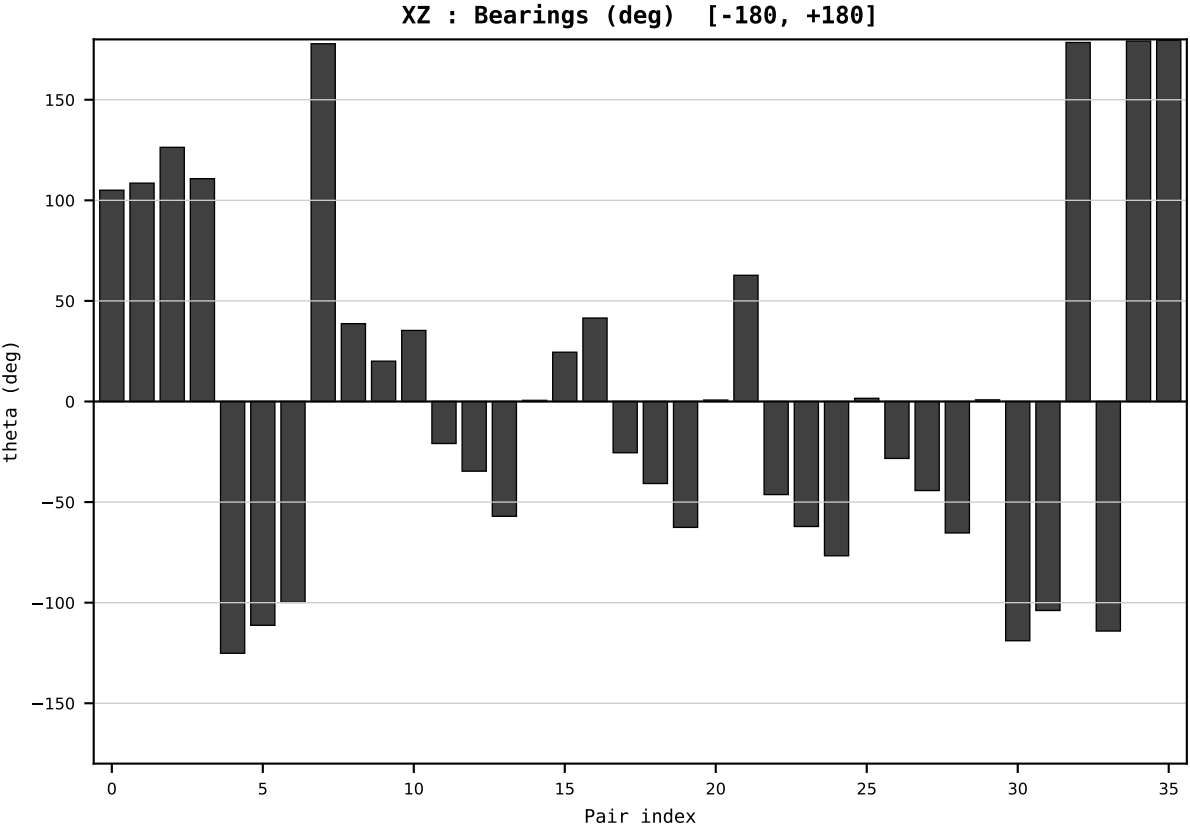
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2011
t : 10 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.334969
Ring = Hs-3
E_metric= 38.9355
kappa_HS= 952.43
omega = 3.6391 deg
d_A = 0.478871
Helm = Solar
Helm d = +0.323844
DR = 6.859 HLR
DR ratio= 952.4x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2012 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

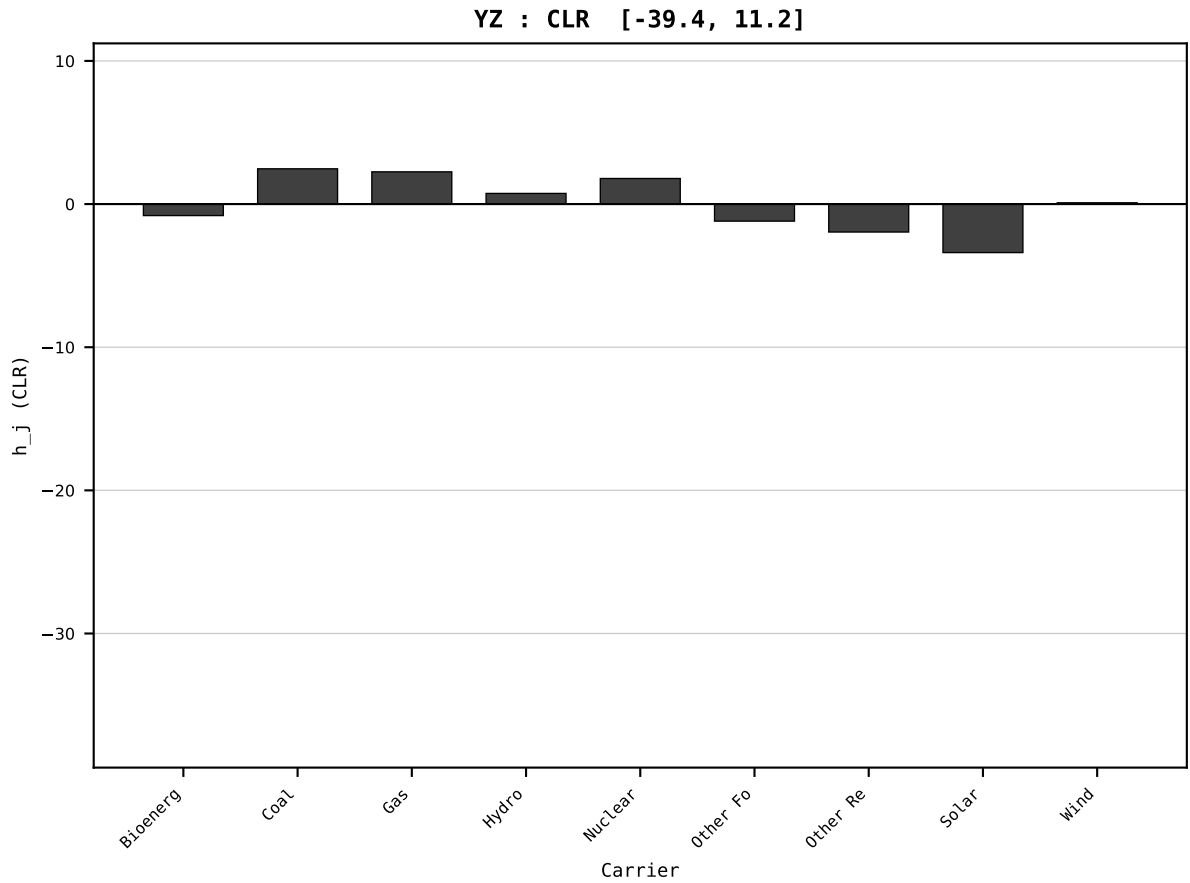
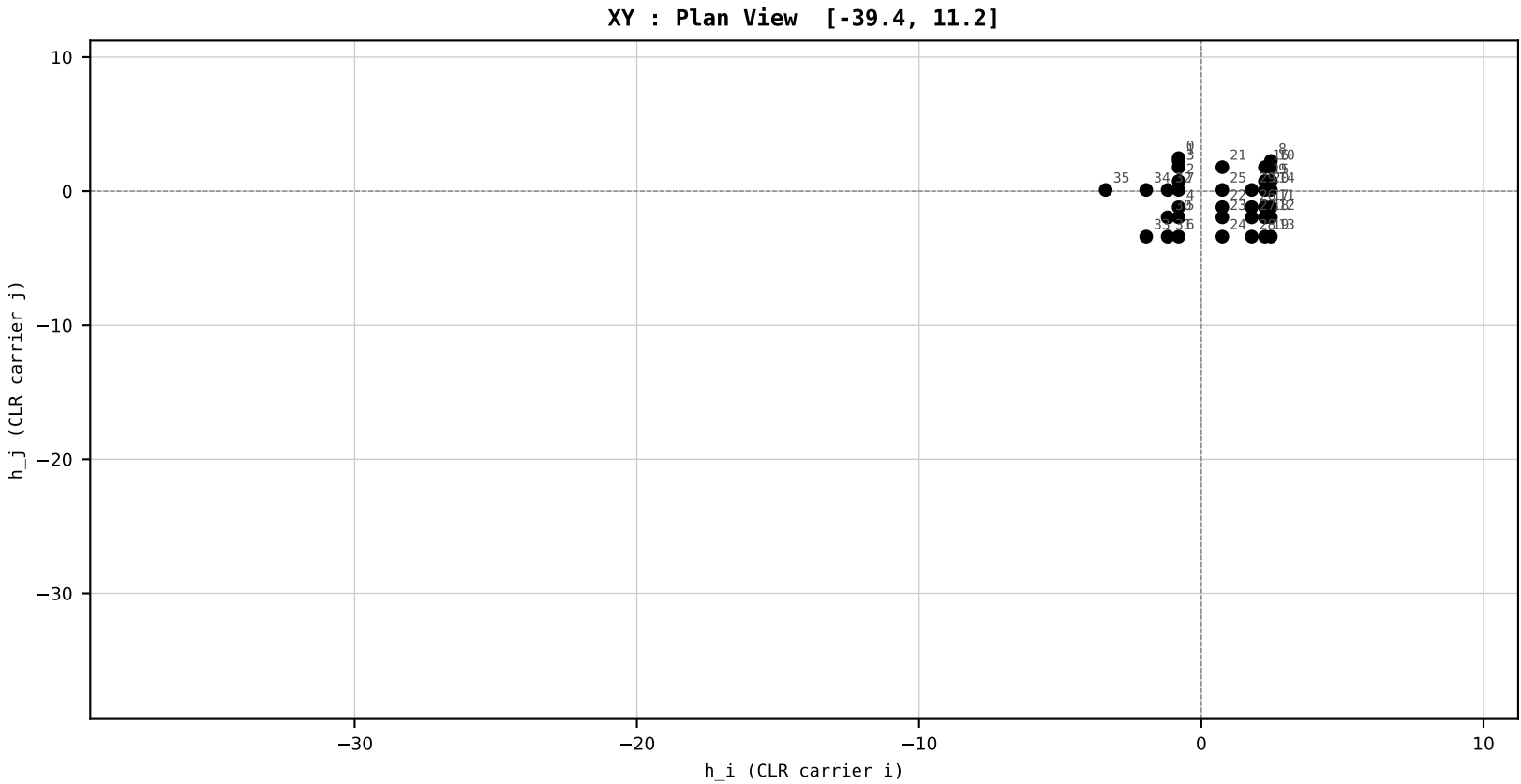
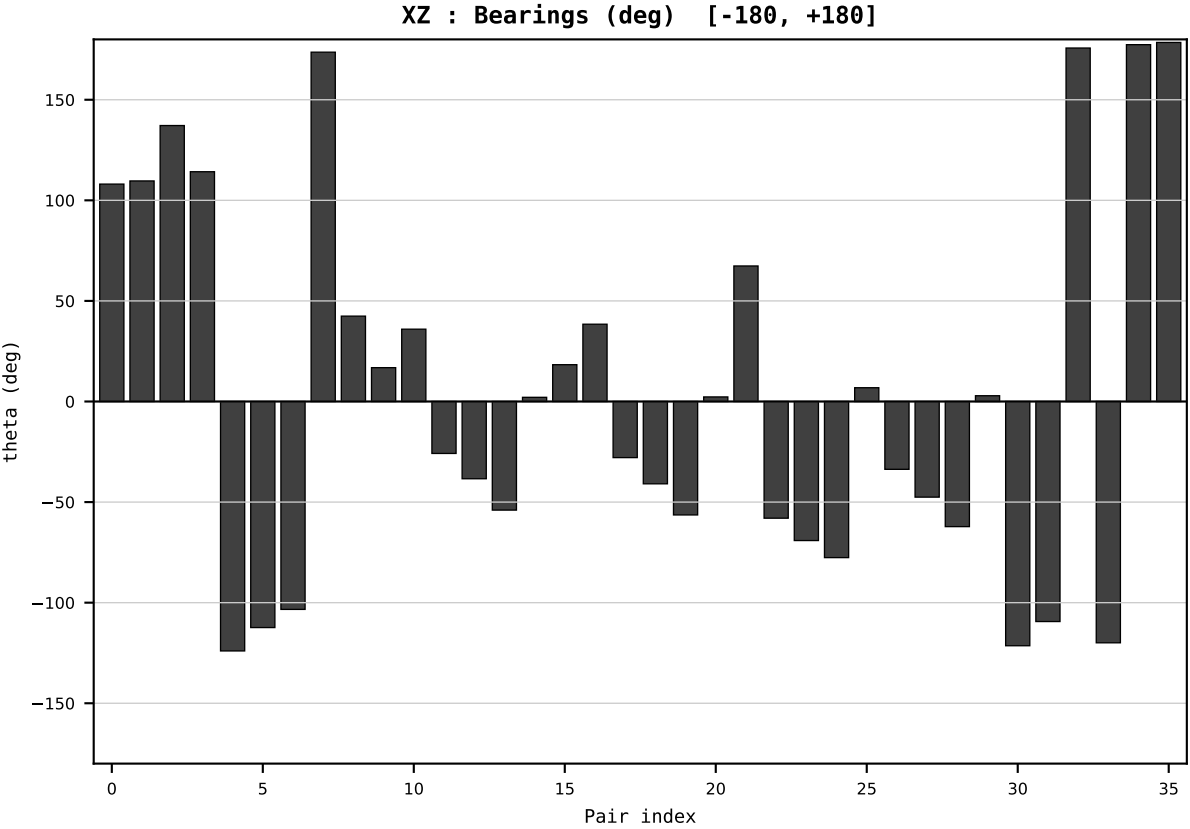
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2012
t : 11 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.325964
Ring = Hs-3
E_metric= 32.2877
kappa_HS= 348.86
omega = 6.5734 deg
d_A = 0.881530
Helm = Solar
Helm d = +0.772628
DR = 5.855 HLR
DR ratio= 348.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2013 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

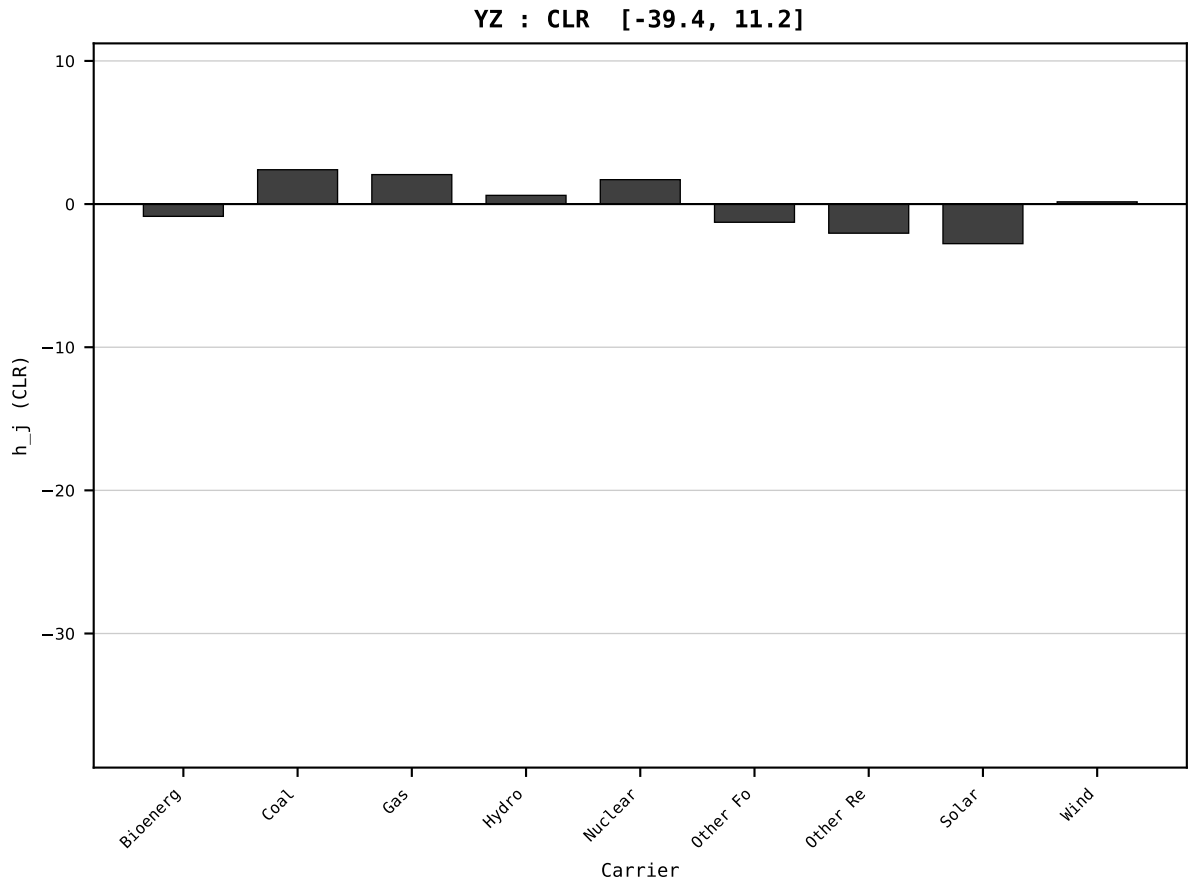
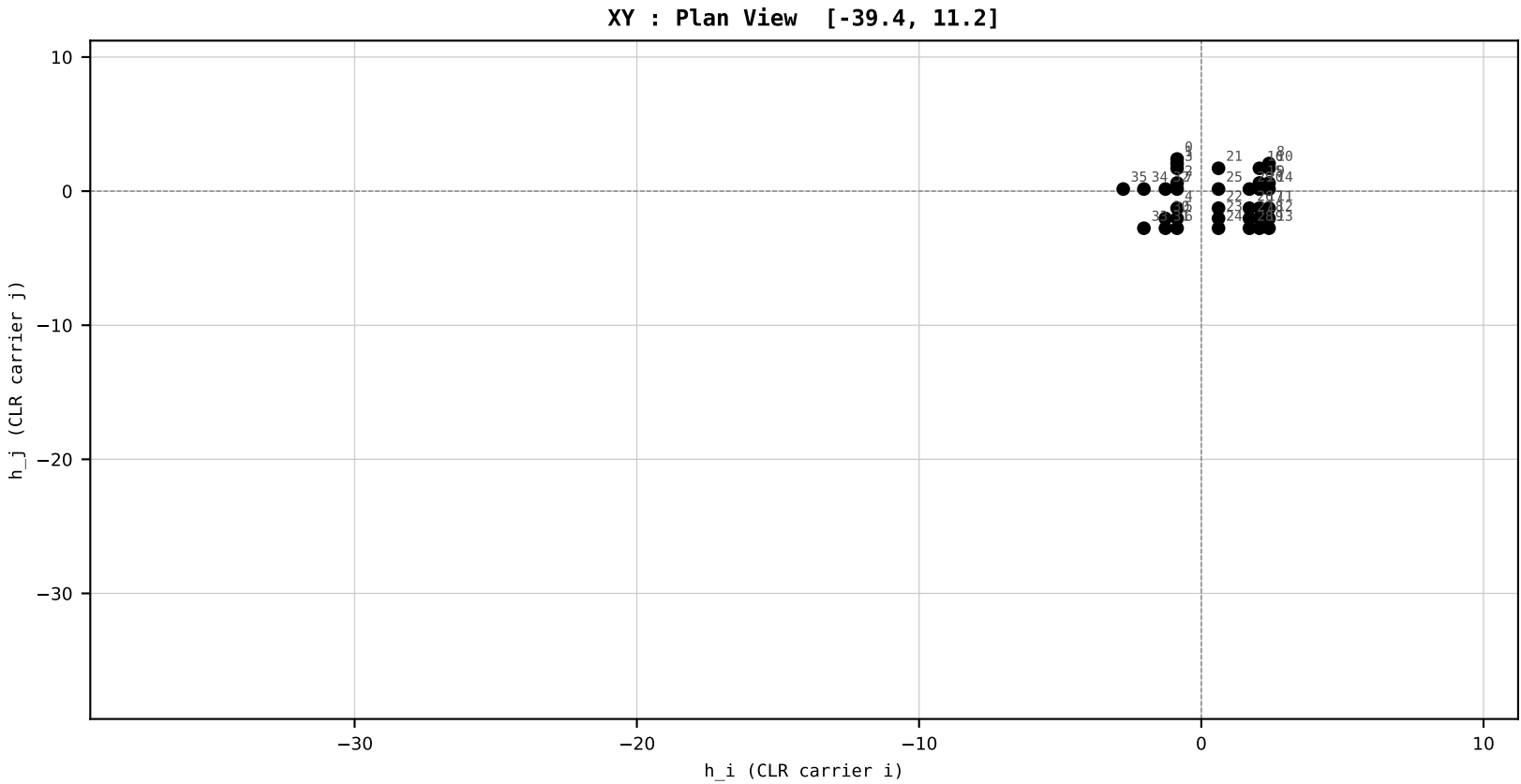
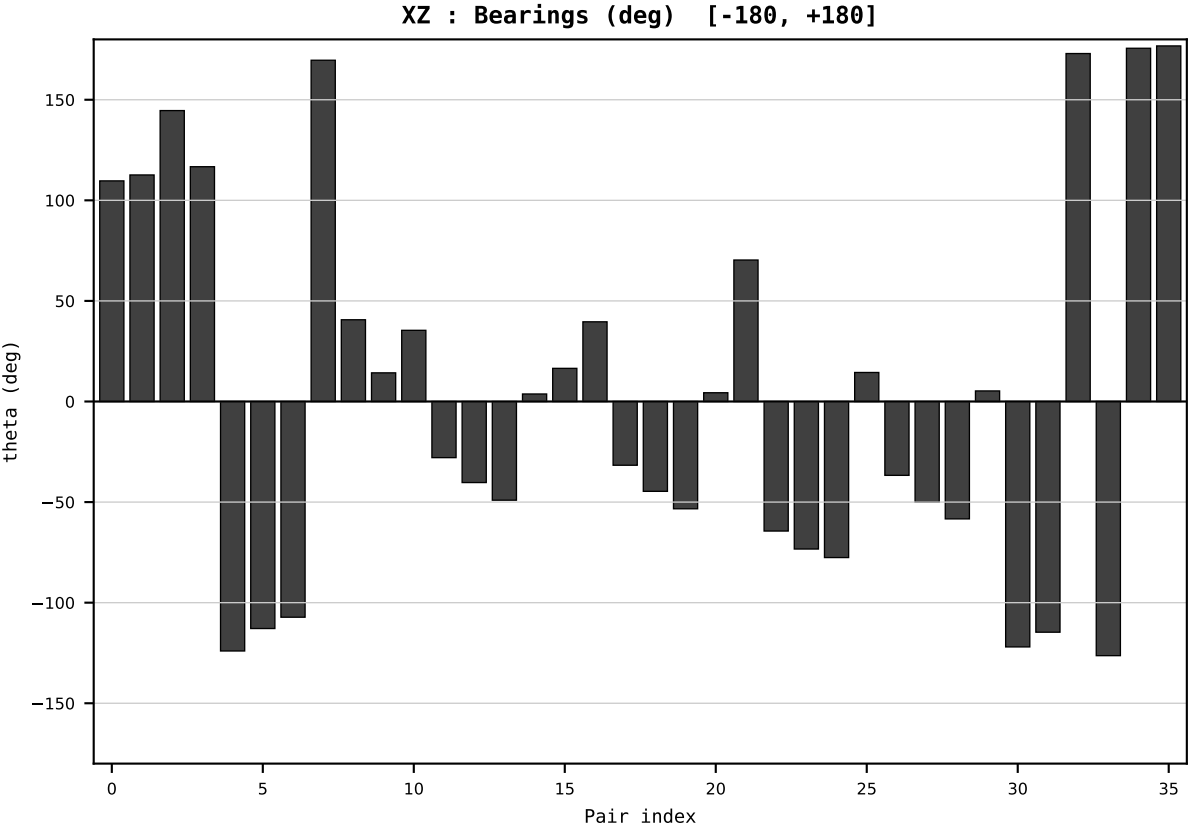
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2013
t : 12 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.318102
Ring = Hs-3
E_metric= 27.4388
kappa_HS= 175.10
omega = 5.5612 deg
d_A = 0.690898
Helm = Solar
Helm d = +0.624351
DR = 5.165 HLR
DR ratio= 175.1x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



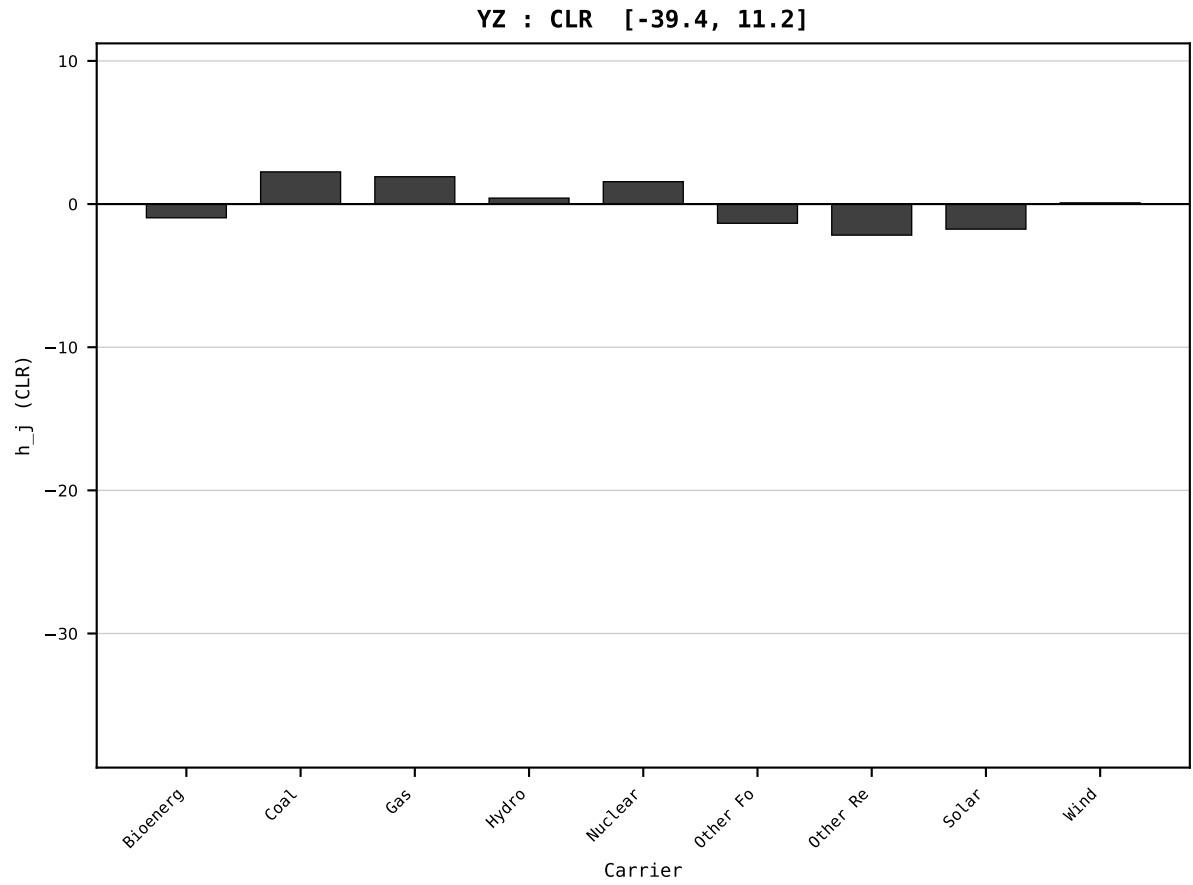
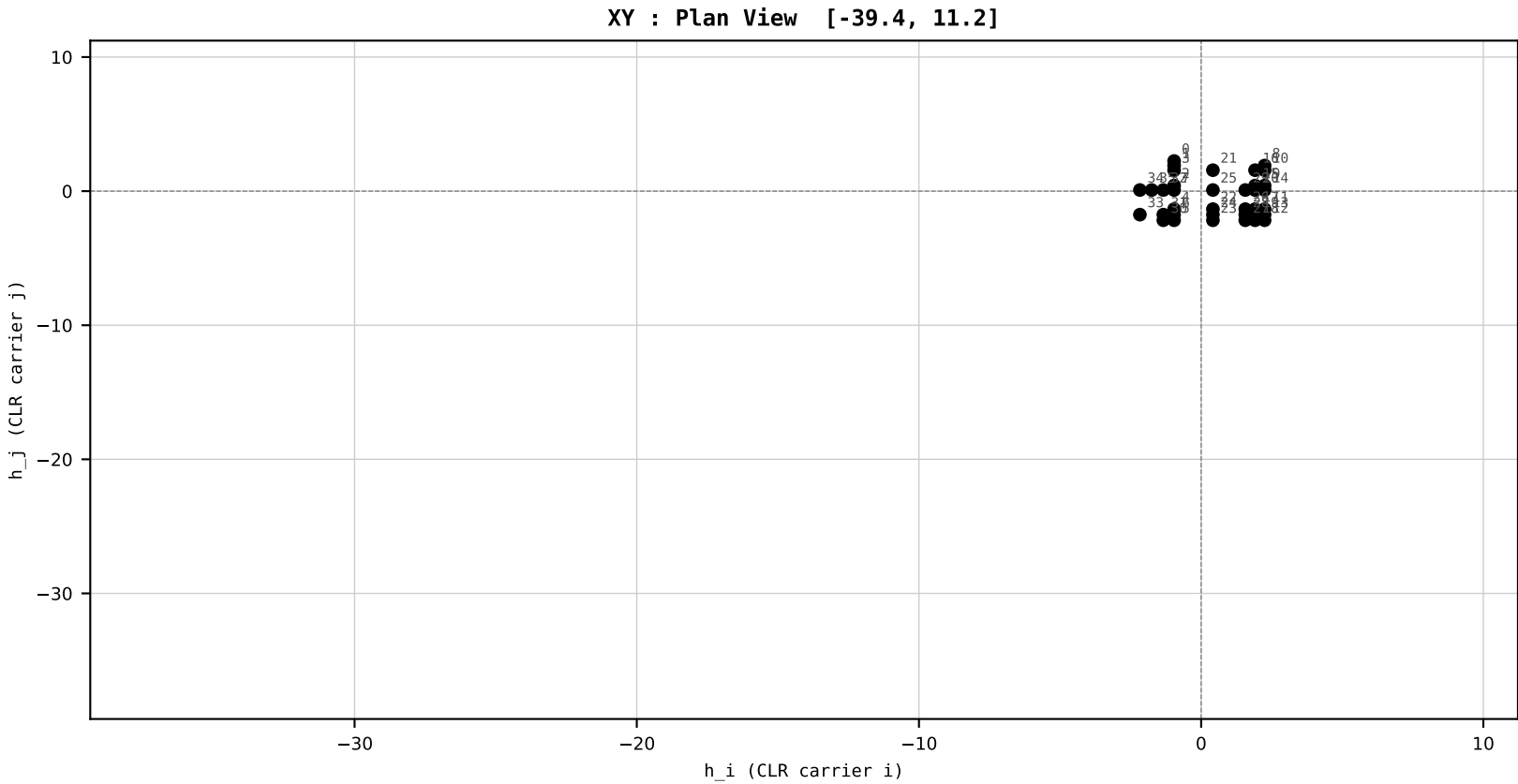
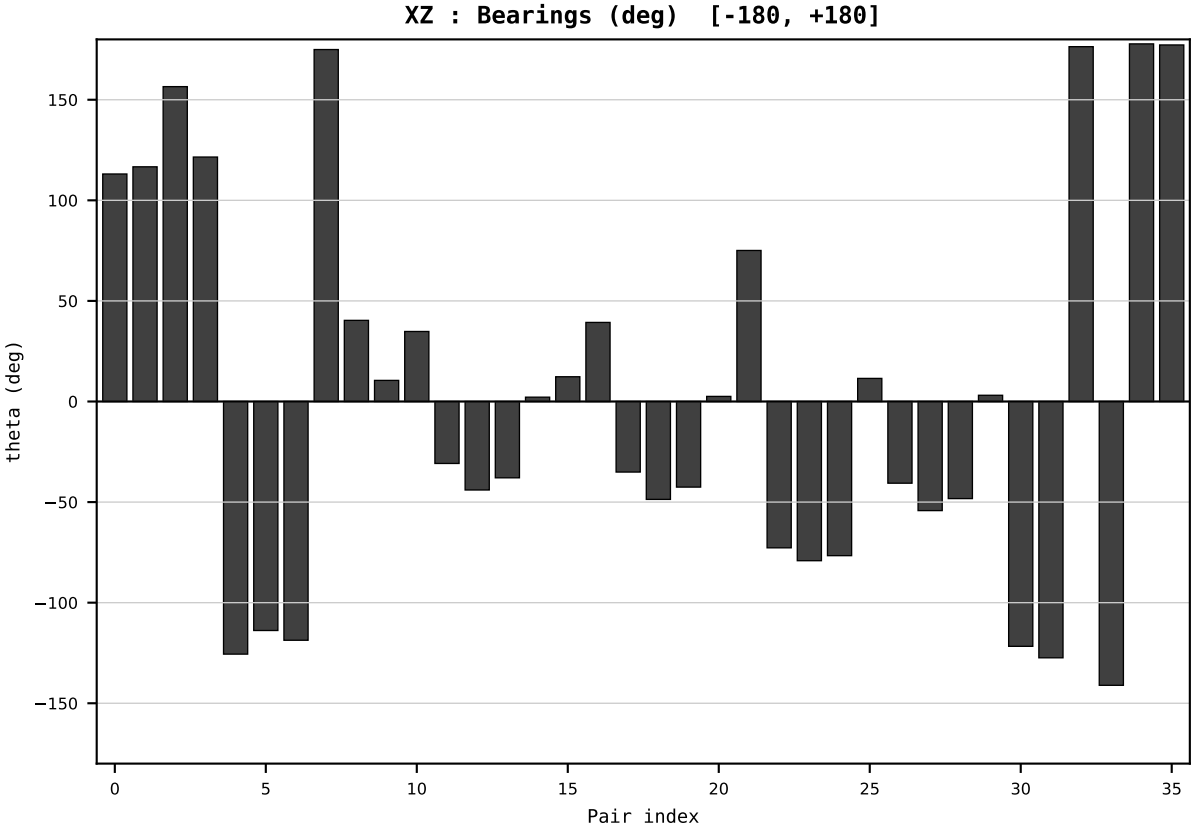
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2014
t : 13 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.306116
Ring = Hs-3
E_metric= 21.8268
kappa_HS= 83.03
omega = 10.6716 deg
d_A = 1.080374
Helm = Solar
Helm d = +1.013049
DR = 4.419 HLR
DR ratio= 83.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2015 | D=9 N=25 pairs=36

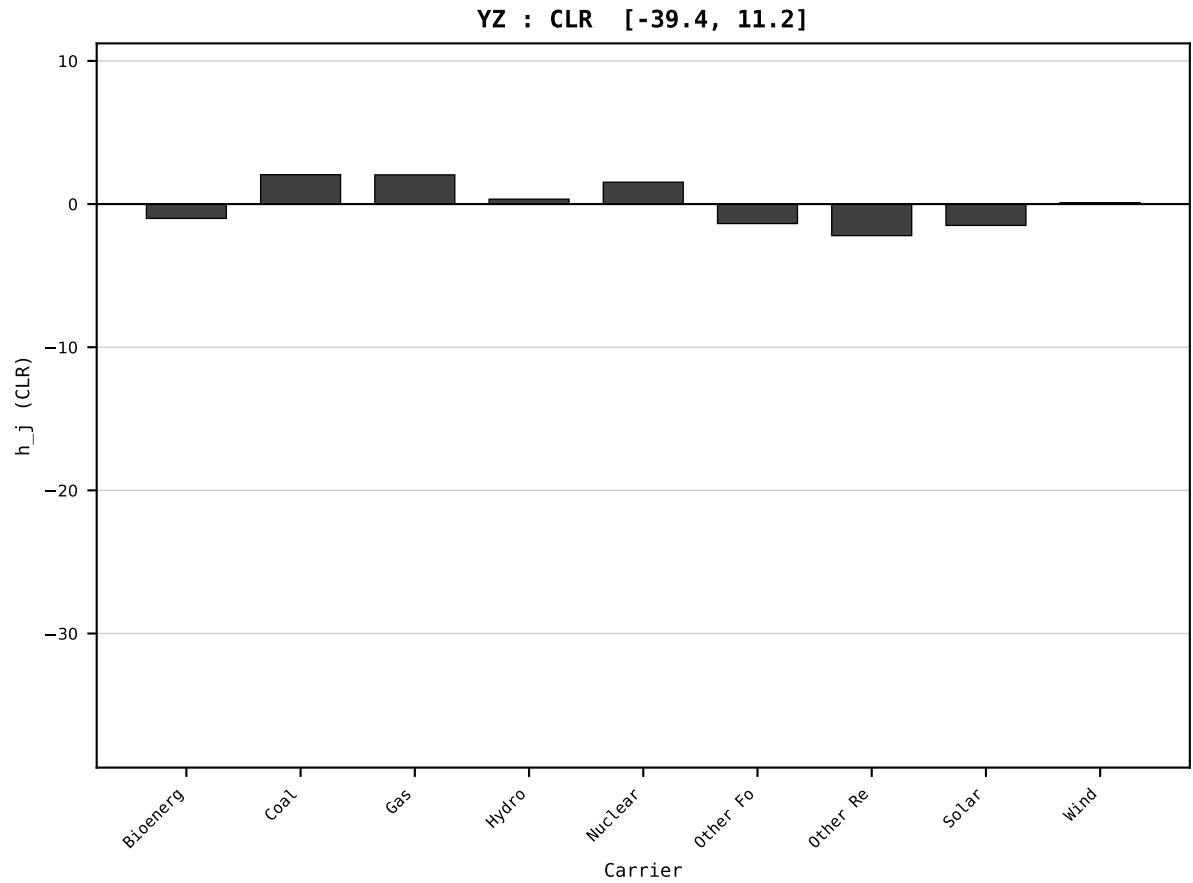
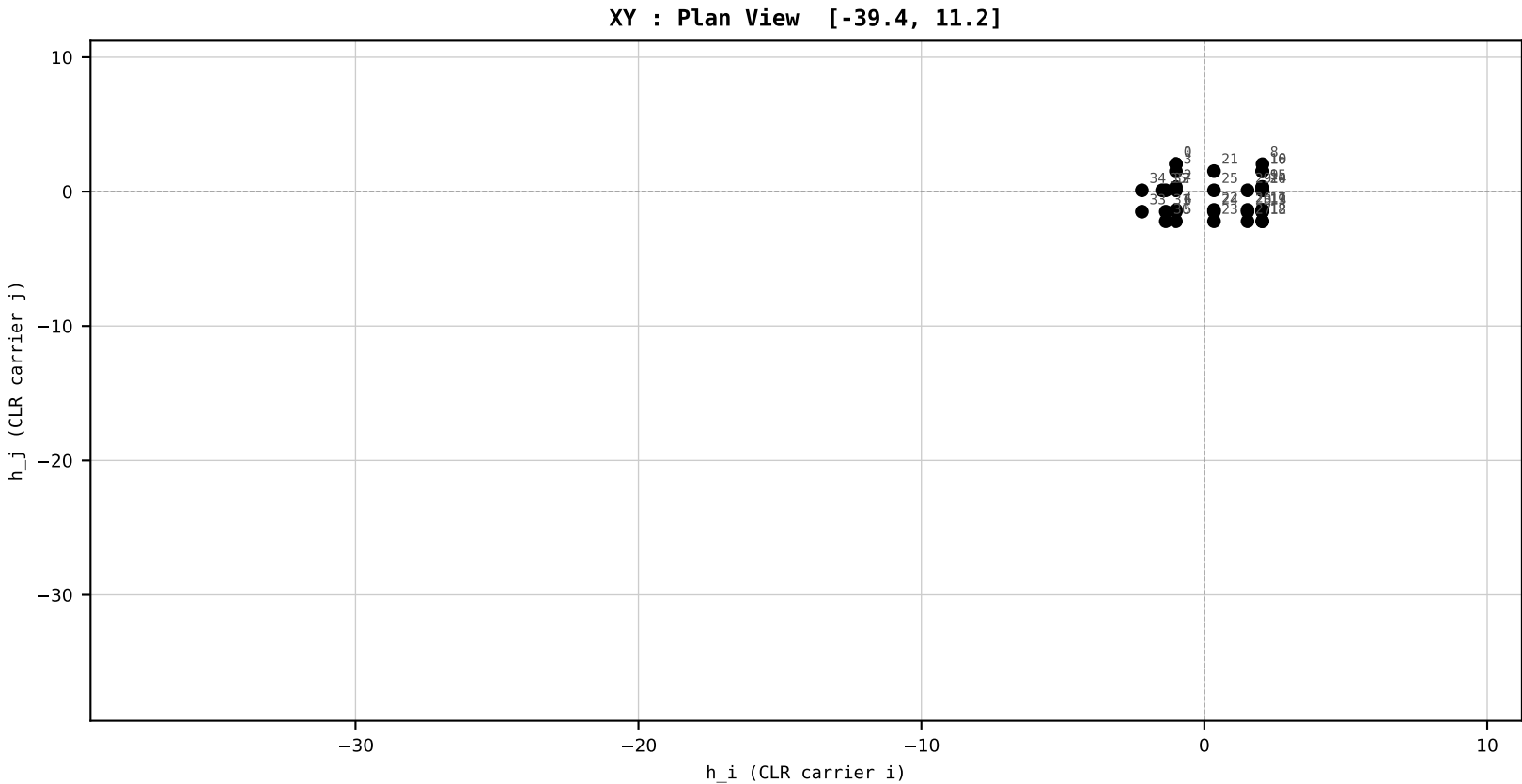
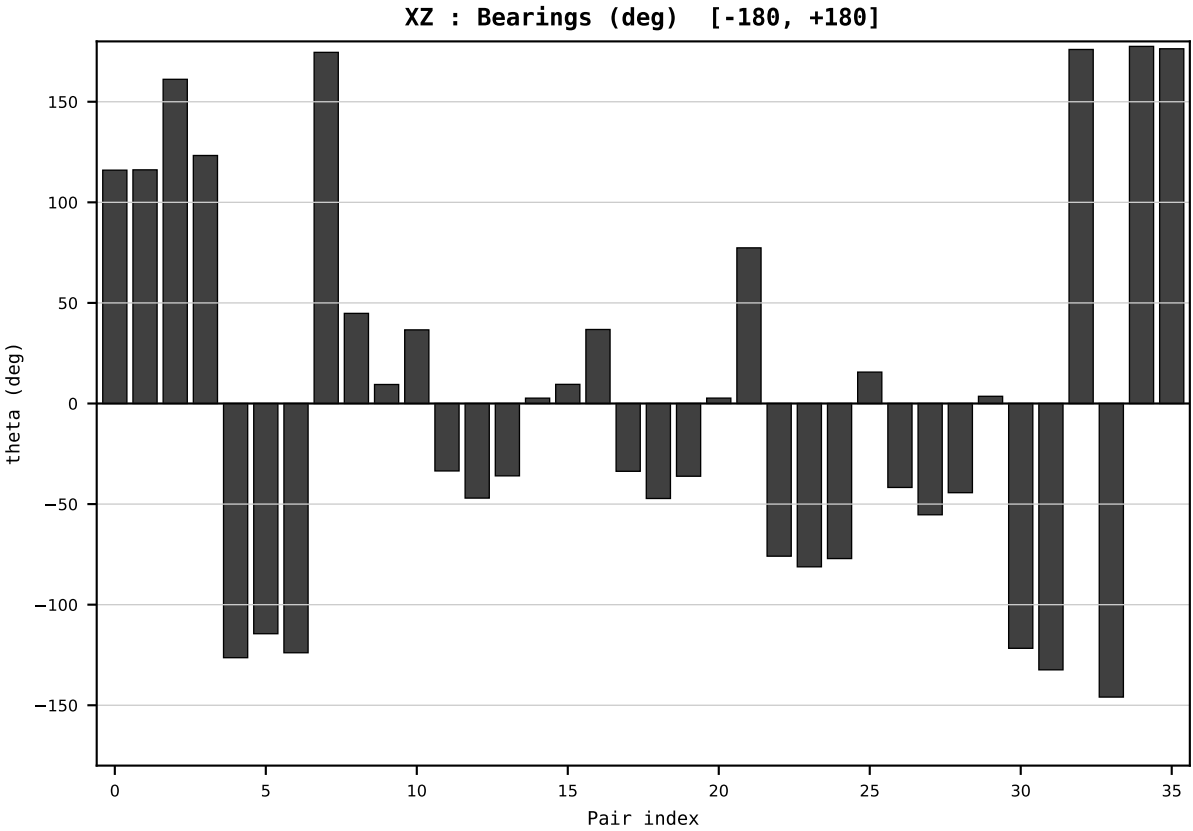
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2015
t : 14 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.296660
Ring = Hs-2
E_metric= 20.7750
kappa_HS= 70.73
omega = 4.3215 deg
d_A = 0.366154
Helm = Solar
Helm d = +0.262026
DR = 4.259 HLR
DR ratio= 70.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2016 | D=9 N=25 pairs=36

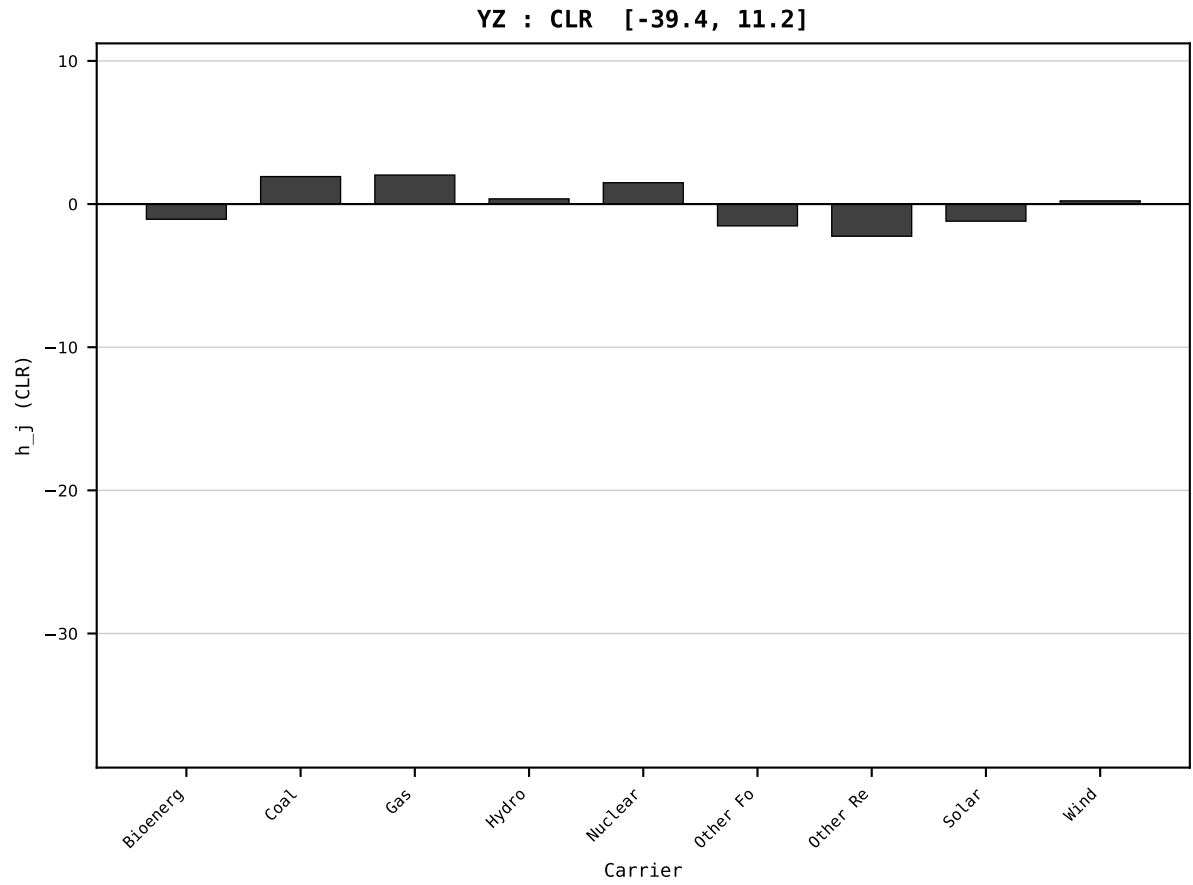
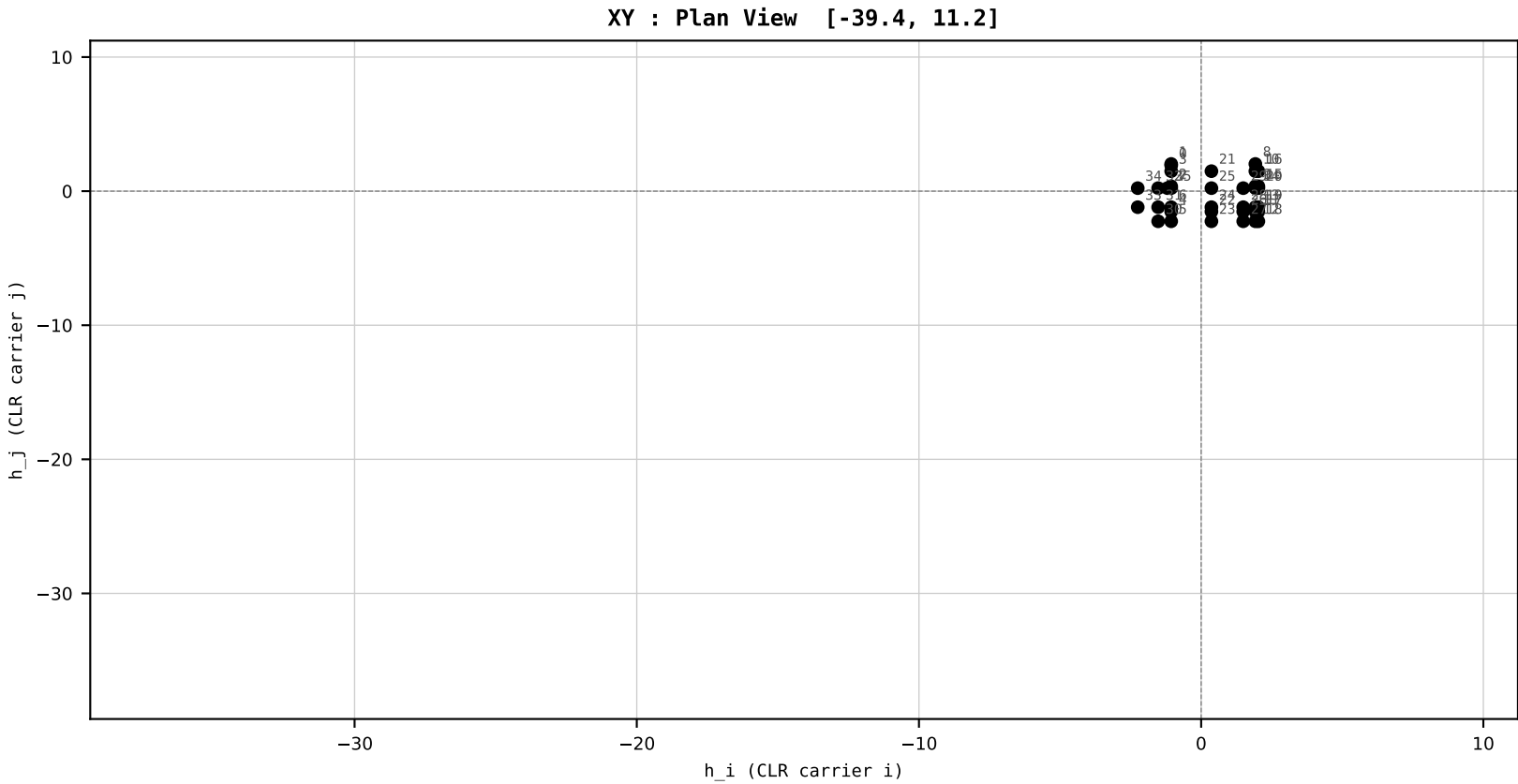
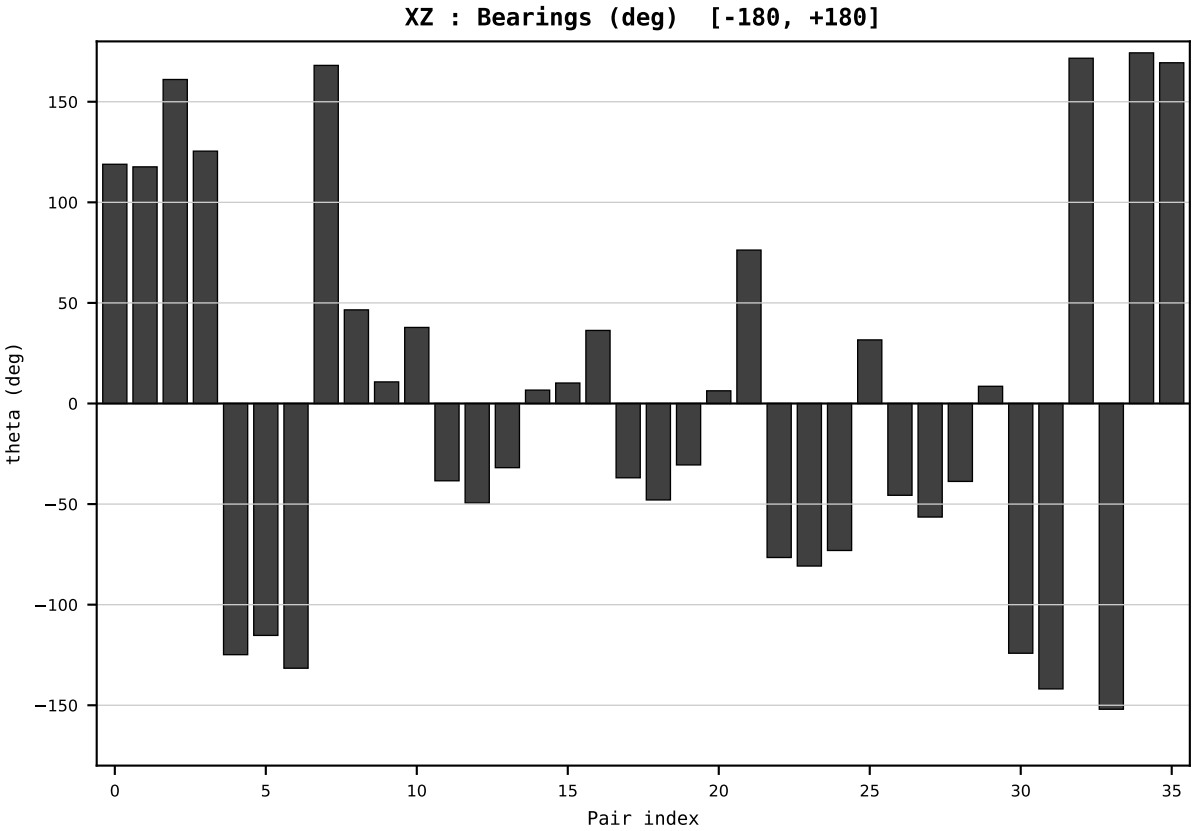
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2016
t : 15 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.282403
Ring = Hs-2
E_metric= 20.1405
kappa_HS= 71.90
omega = 4.9089 deg
d_A = 0.393670
Helm = Solar
Helm d = +0.294744
DR = 4.275 HLR
DR ratio= 71.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2017 | D=9 N=25 pairs=36

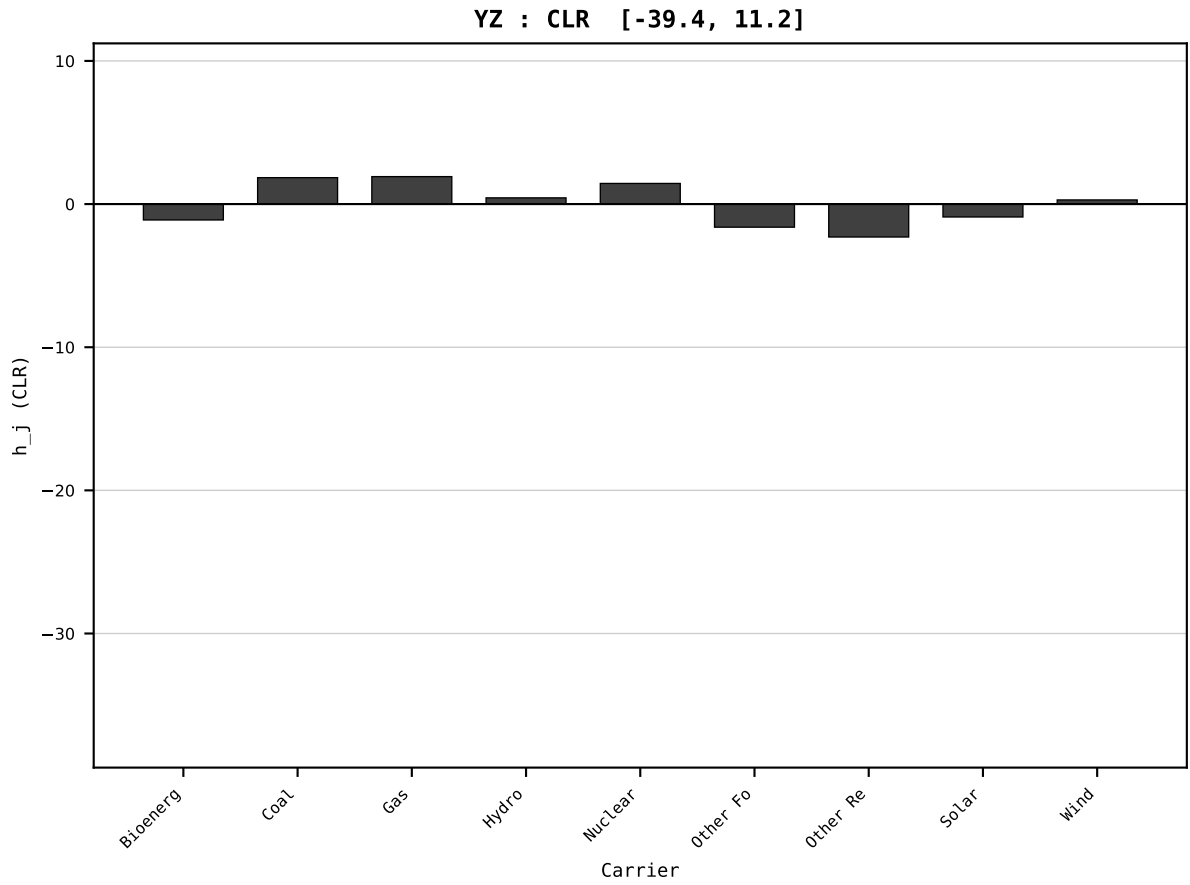
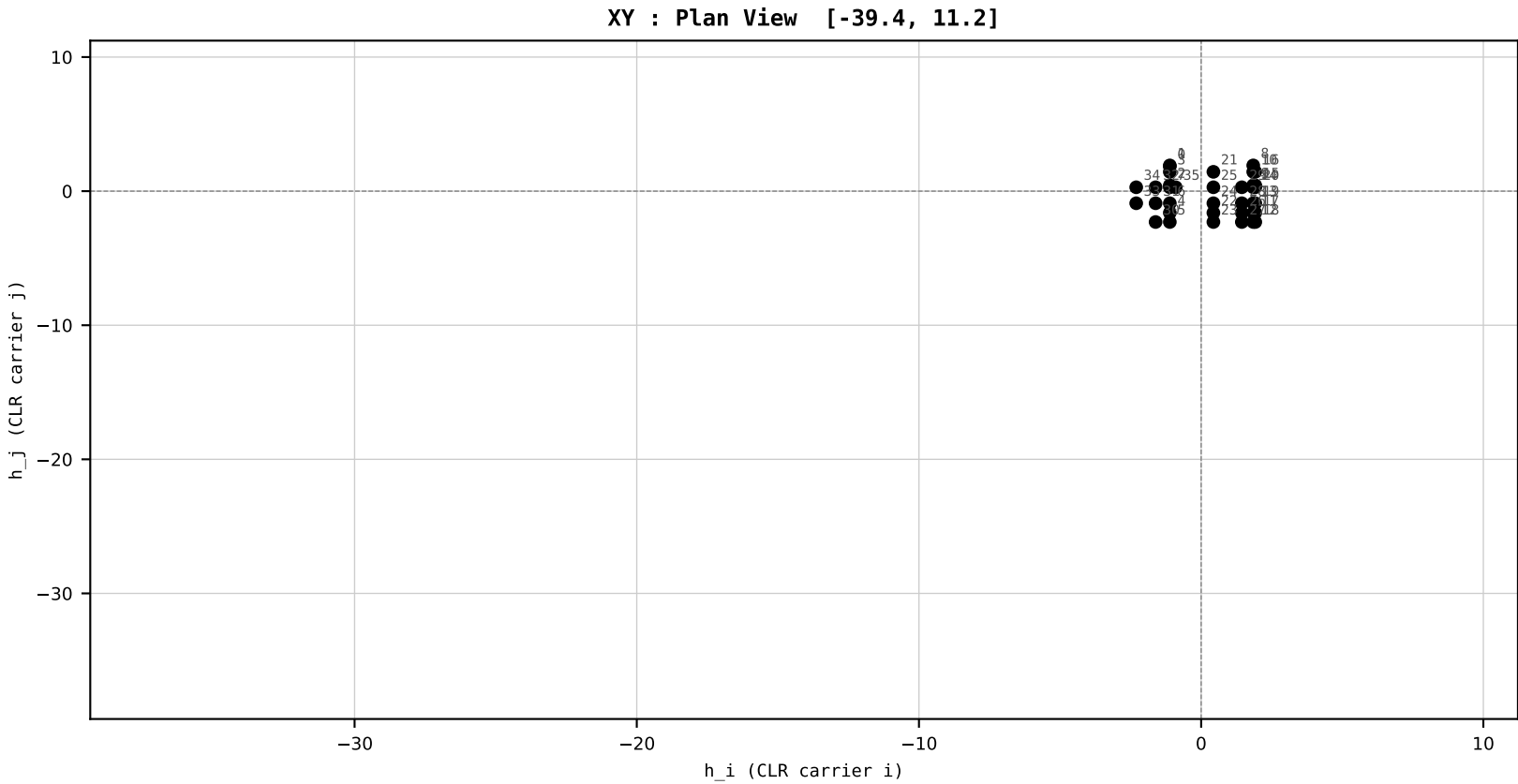
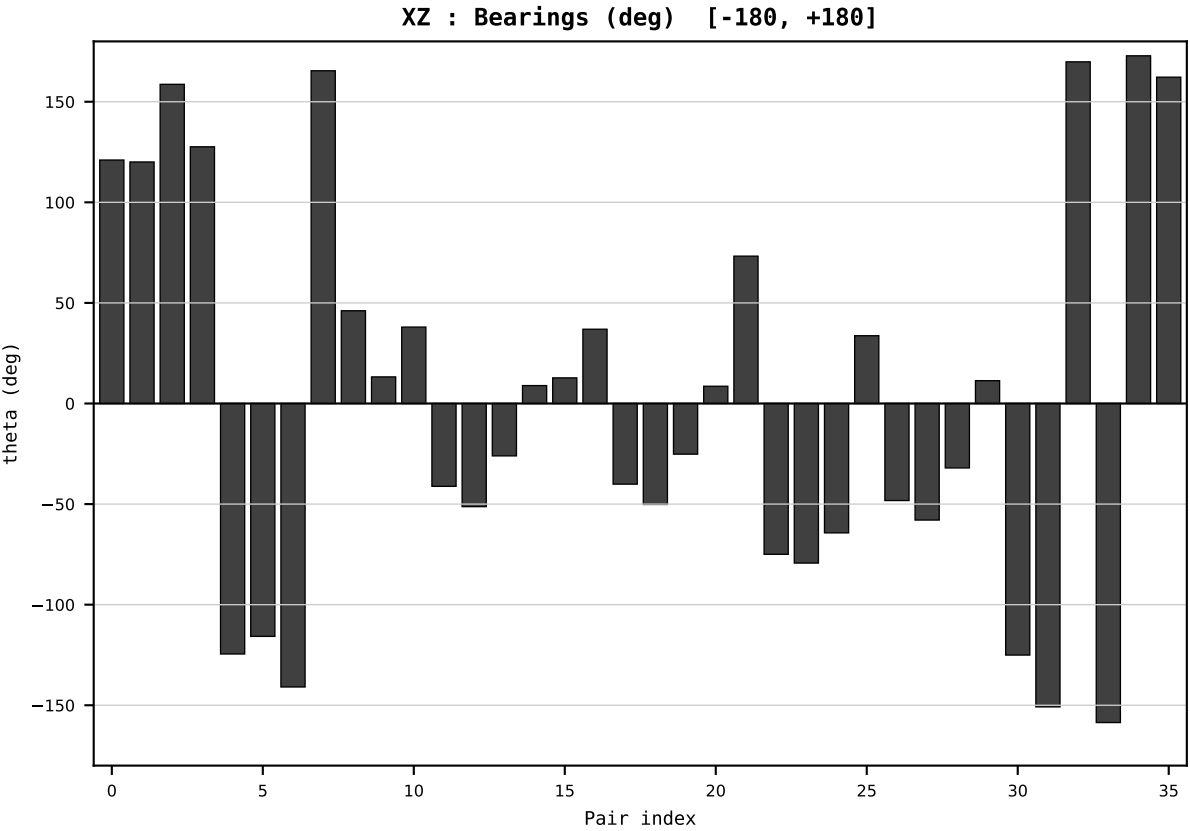
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2017
t : 16 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.263028
Ring = Hs-2
E_metric= 19.3839
kappa_HS= 67.98
omega = 4.5042 deg
d_A = 0.359563
Helm = Solar
Helm d = +0.294082
DR = 4.219 HLR
DR ratio= 68.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

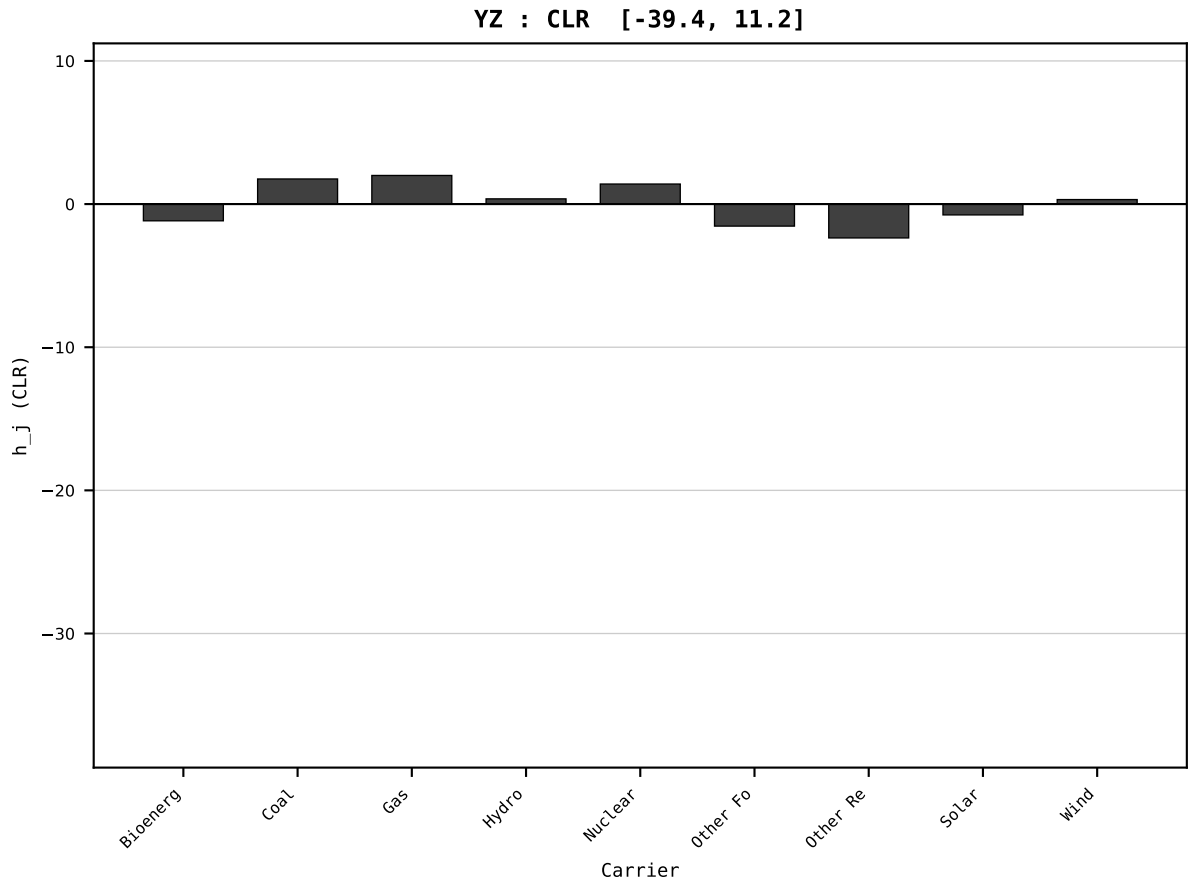
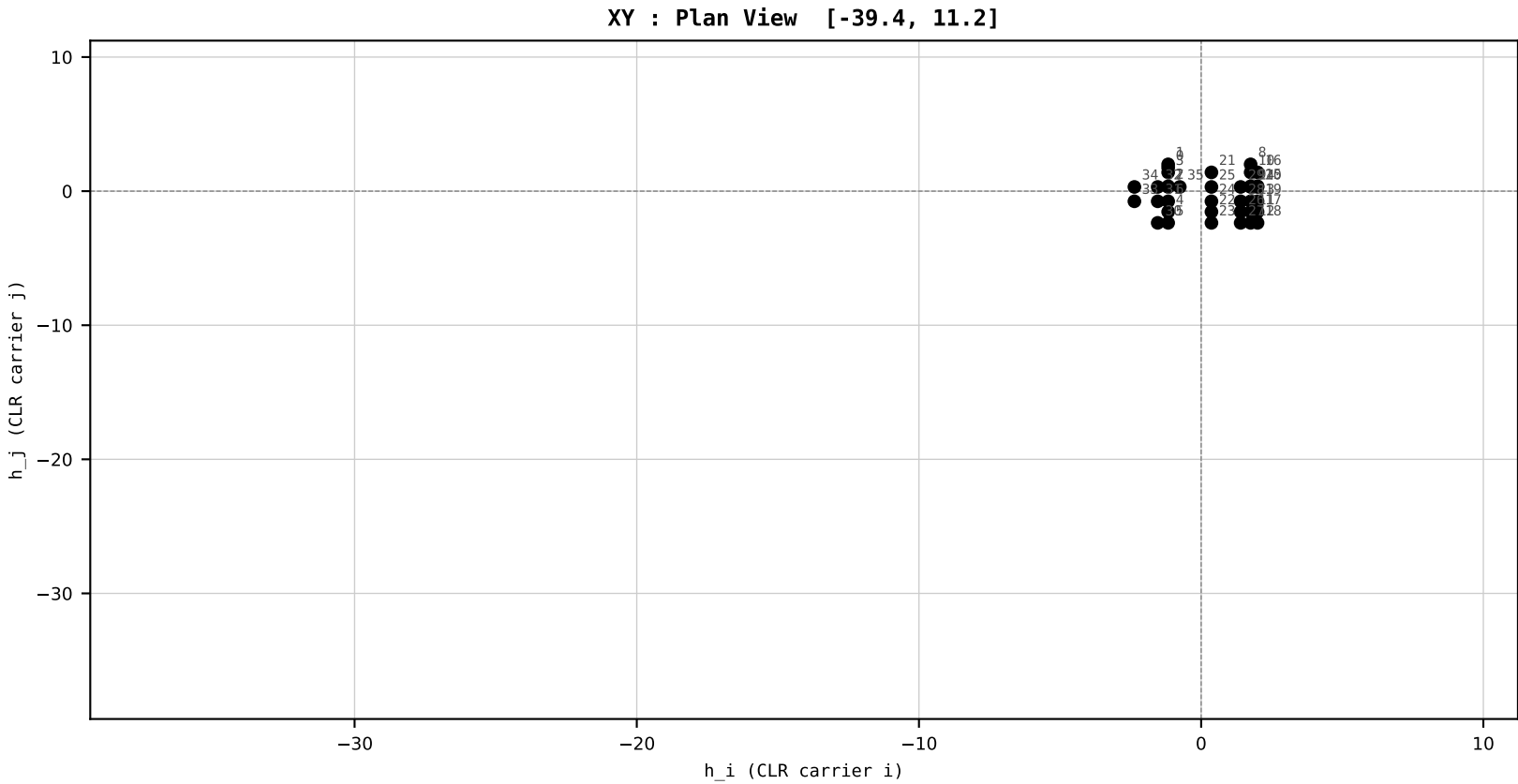
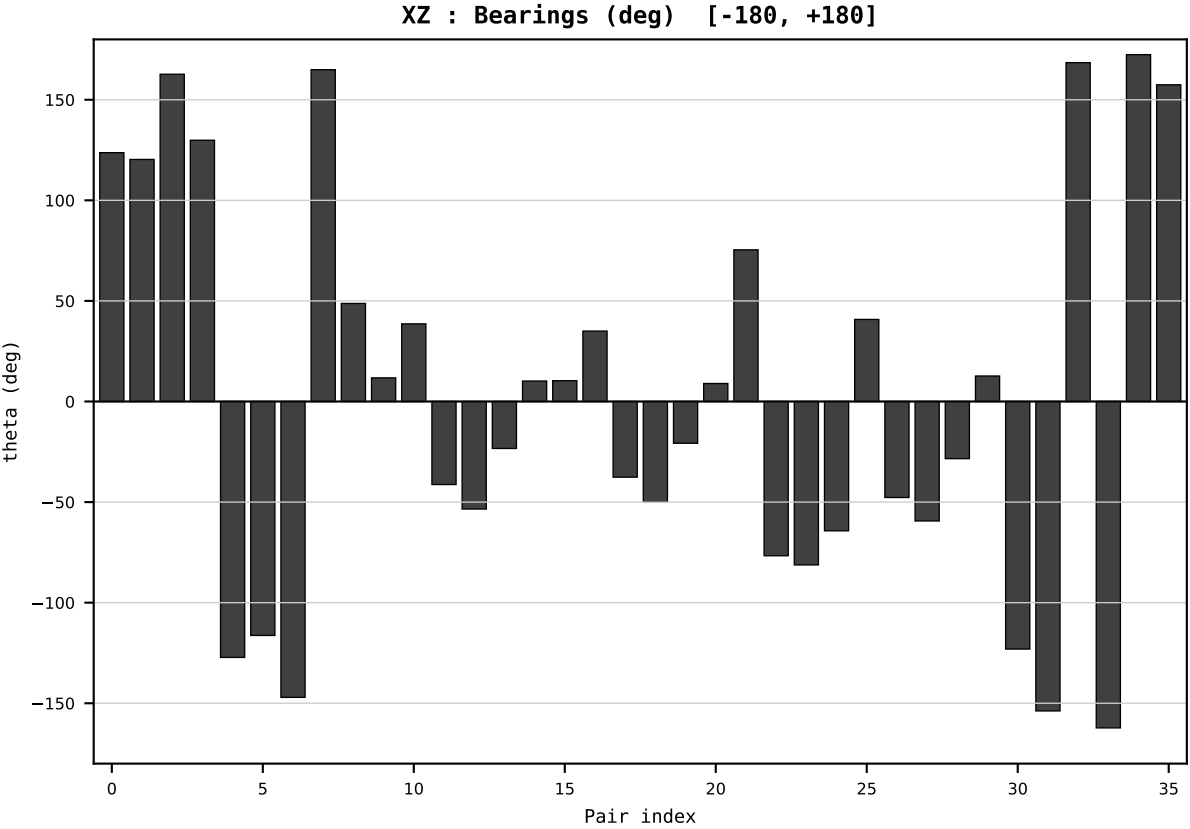
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2018
t : 17 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.264078
Ring = Hs-2
E_metric= 19.1683
kappa_HS= 78.61
omega = 3.0814 deg
d_A = 0.237367
Helm = Solar
Helm d = +0.144753
DR = 4.364 HLR
DR ratio= 78.6x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2019 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

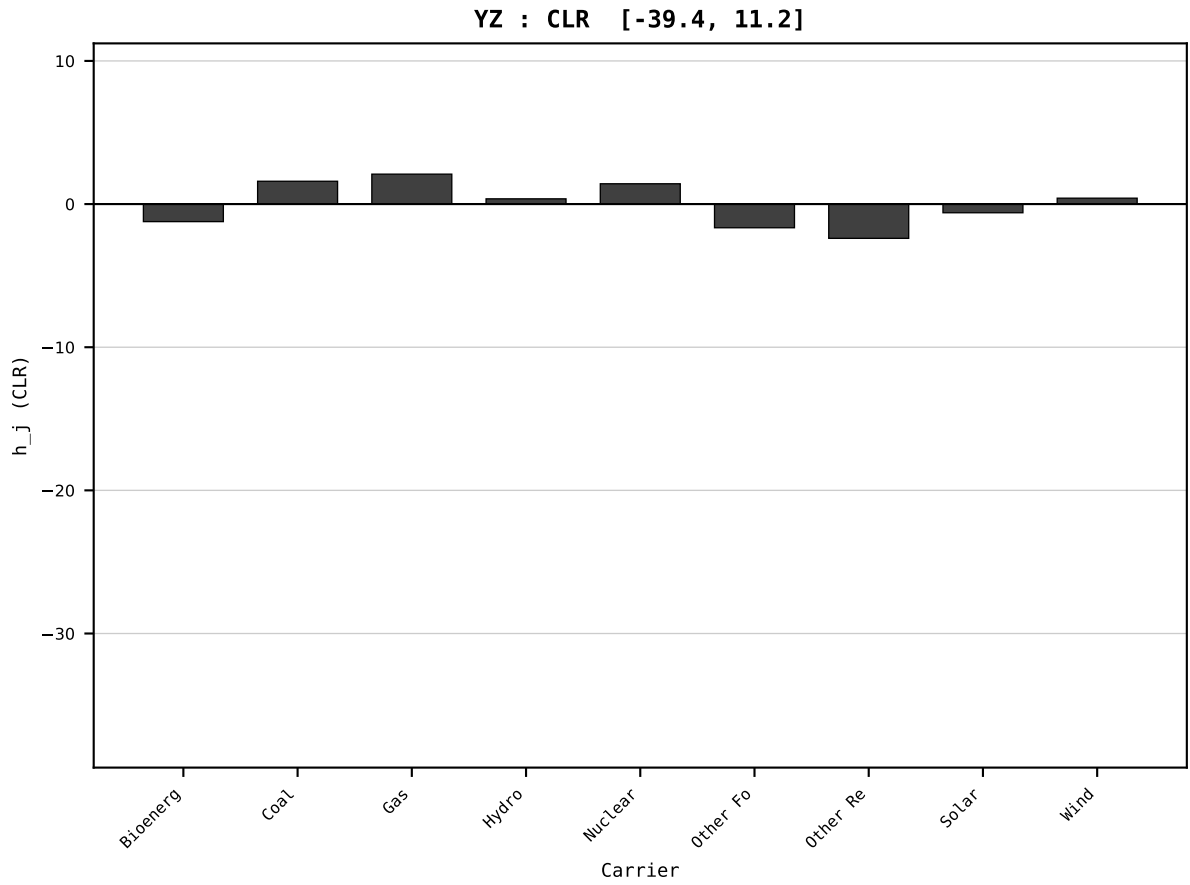
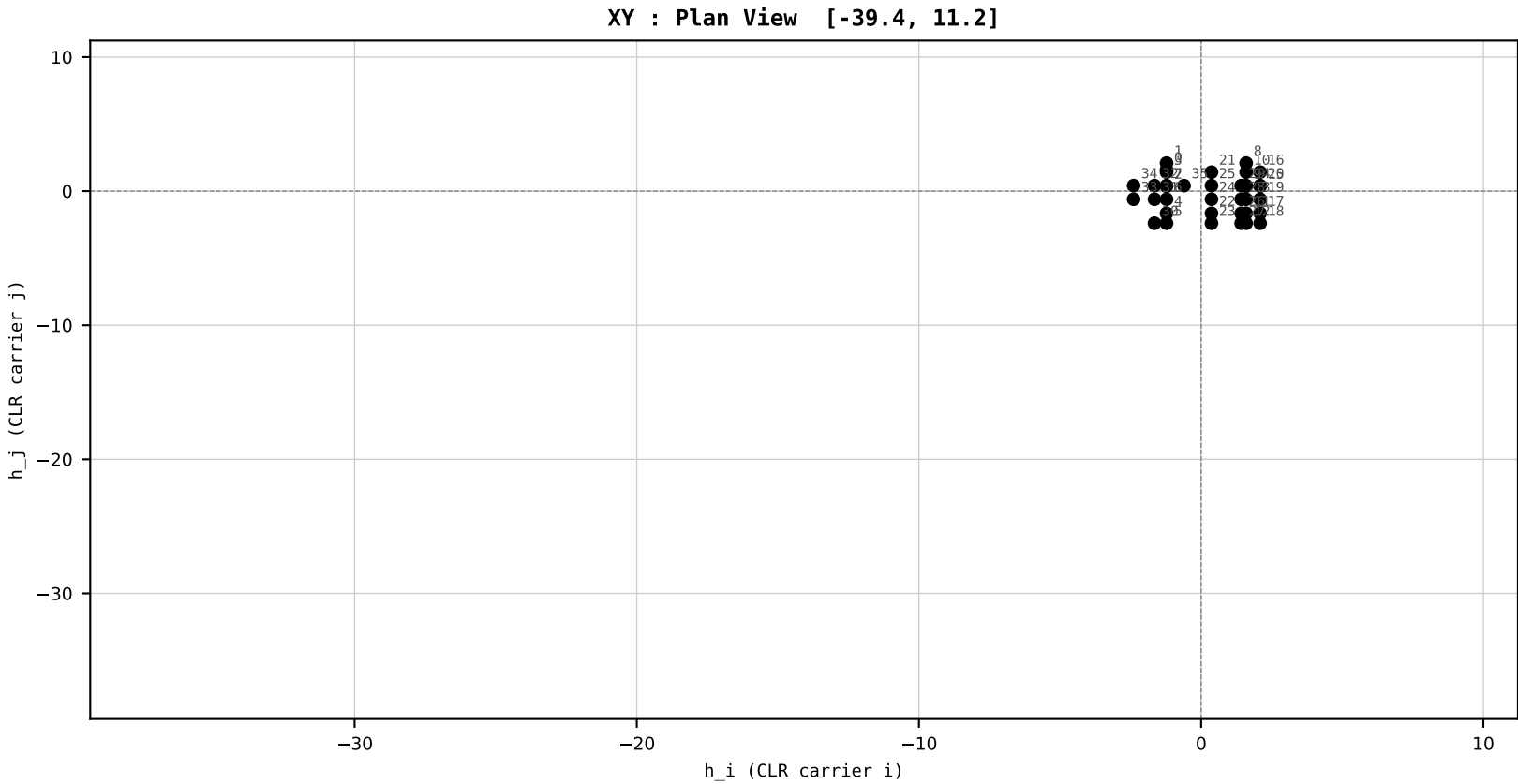
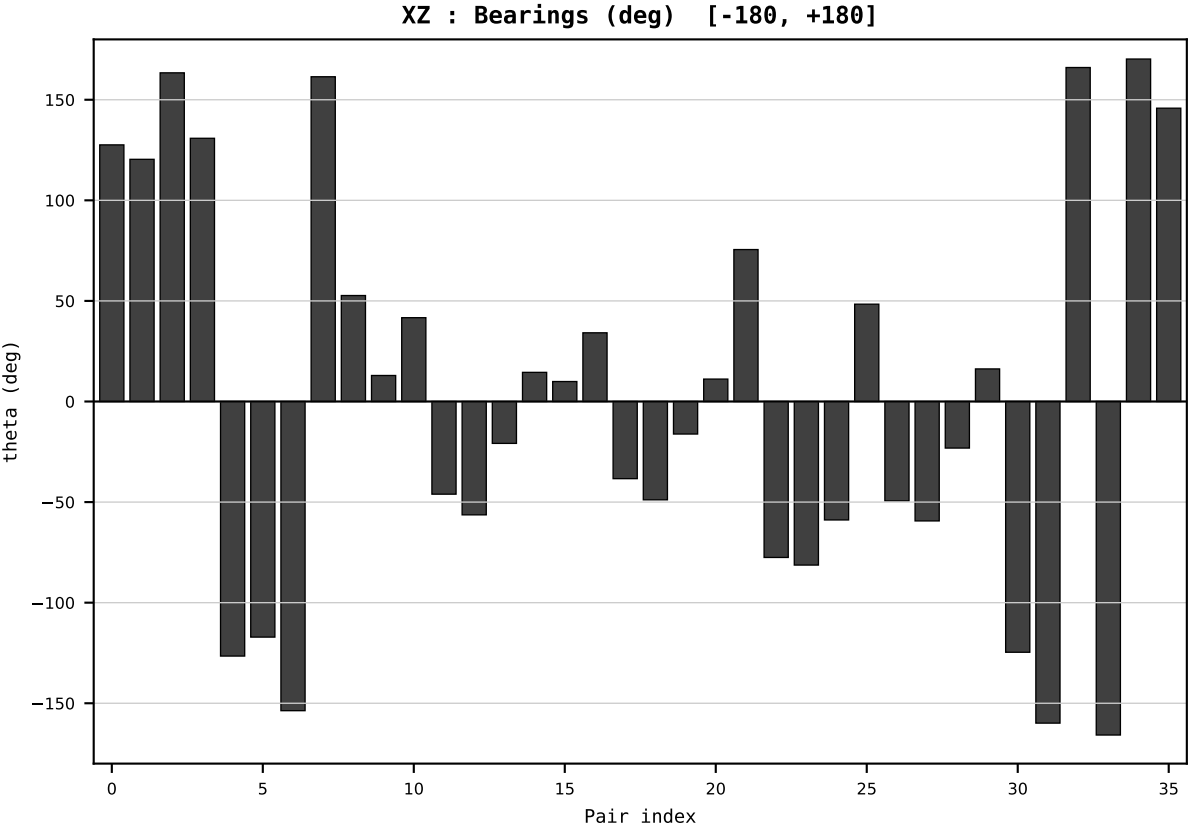
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2019
t : 18 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.264929
Ring = Hs-2
E_metric= 19.5686
kappa_HS= 88.69
omega = 3.7204 deg
d_A = 0.289310
Helm = Coal
Helm d = -0.159435
DR = 4.485 HLR
DR ratio= 88.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2020 | D=9 N=25 pairs=36

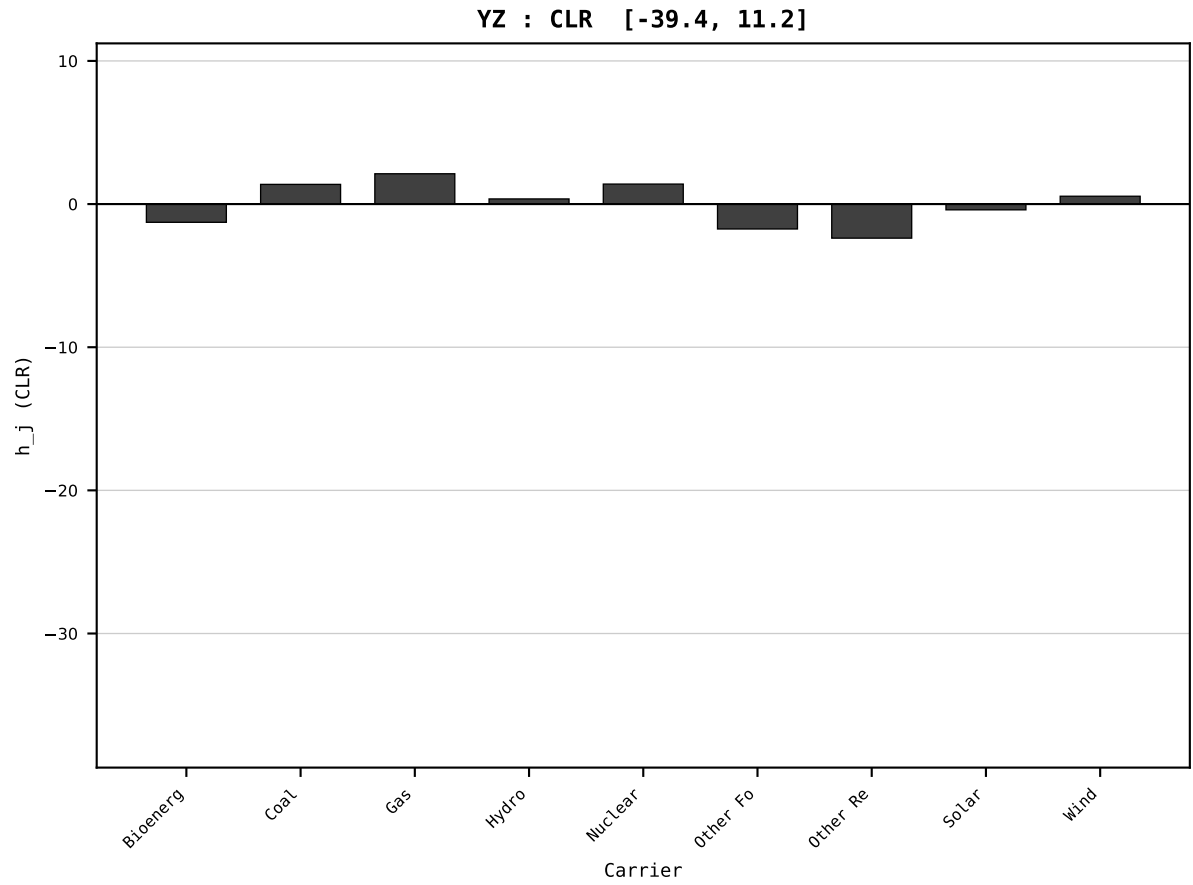
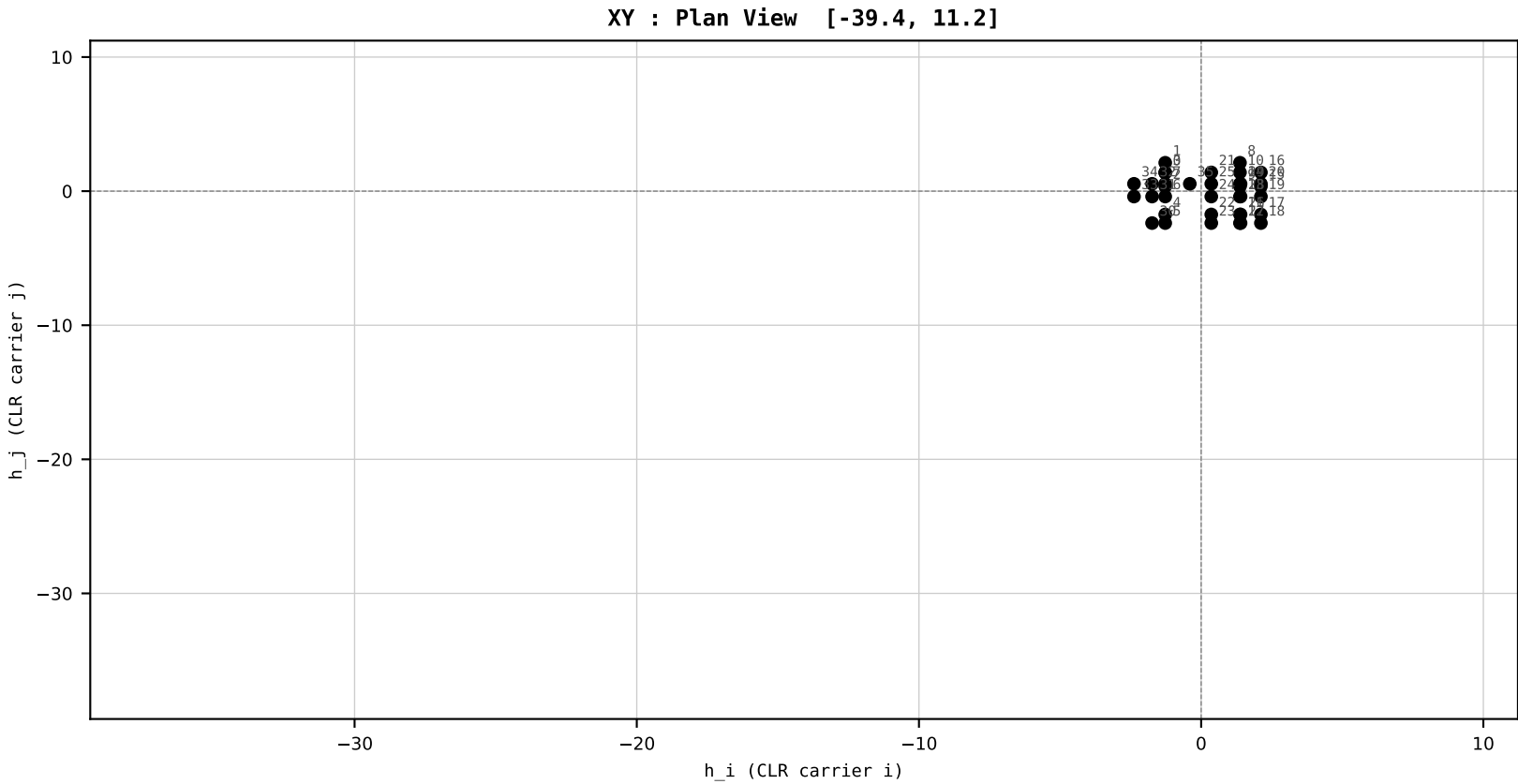
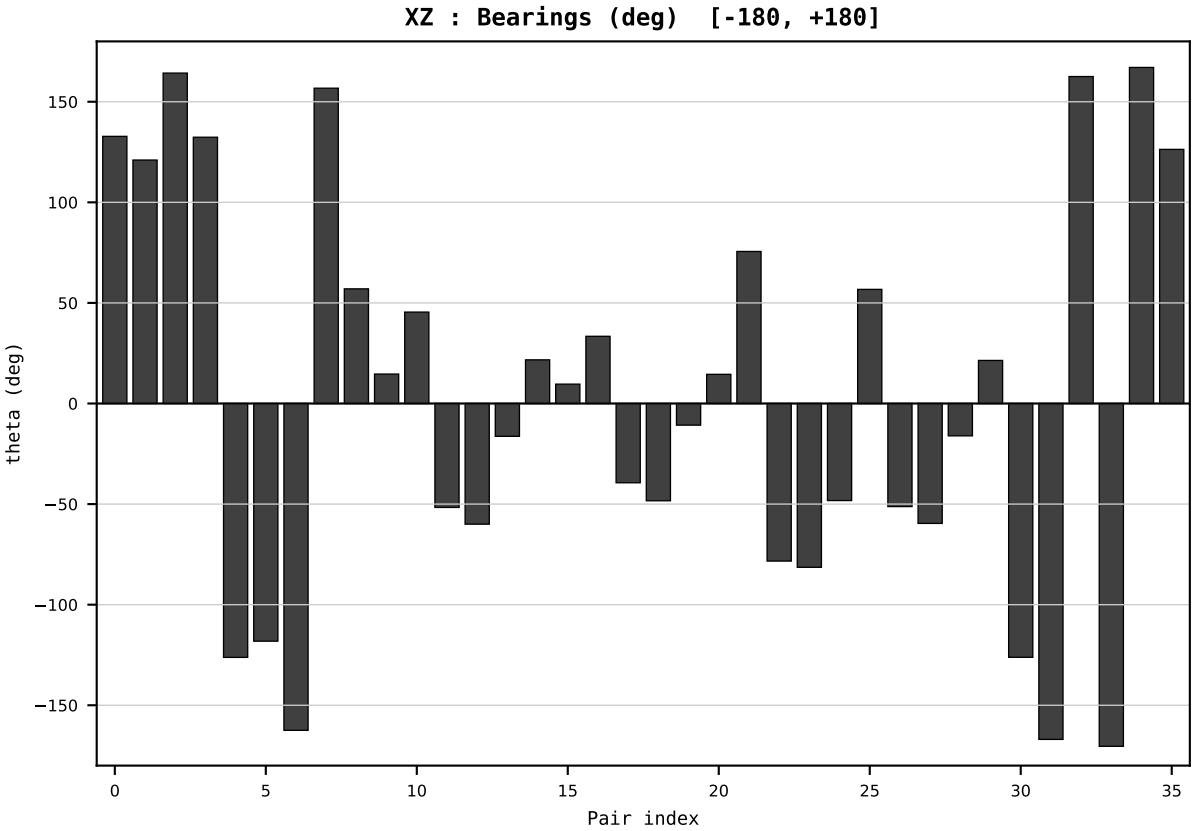
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2020
t : 19 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.259091
Ring = Hs-2
E_metric= 19.2298
kappa_HS= 89.83
omega = 4.4528 deg
d_A = 0.344356
Helm = Coal
Helm d = -0.218685
DR = 4.498 HLR
DR ratio= 89.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2021 | D=9 N=25 pairs=36

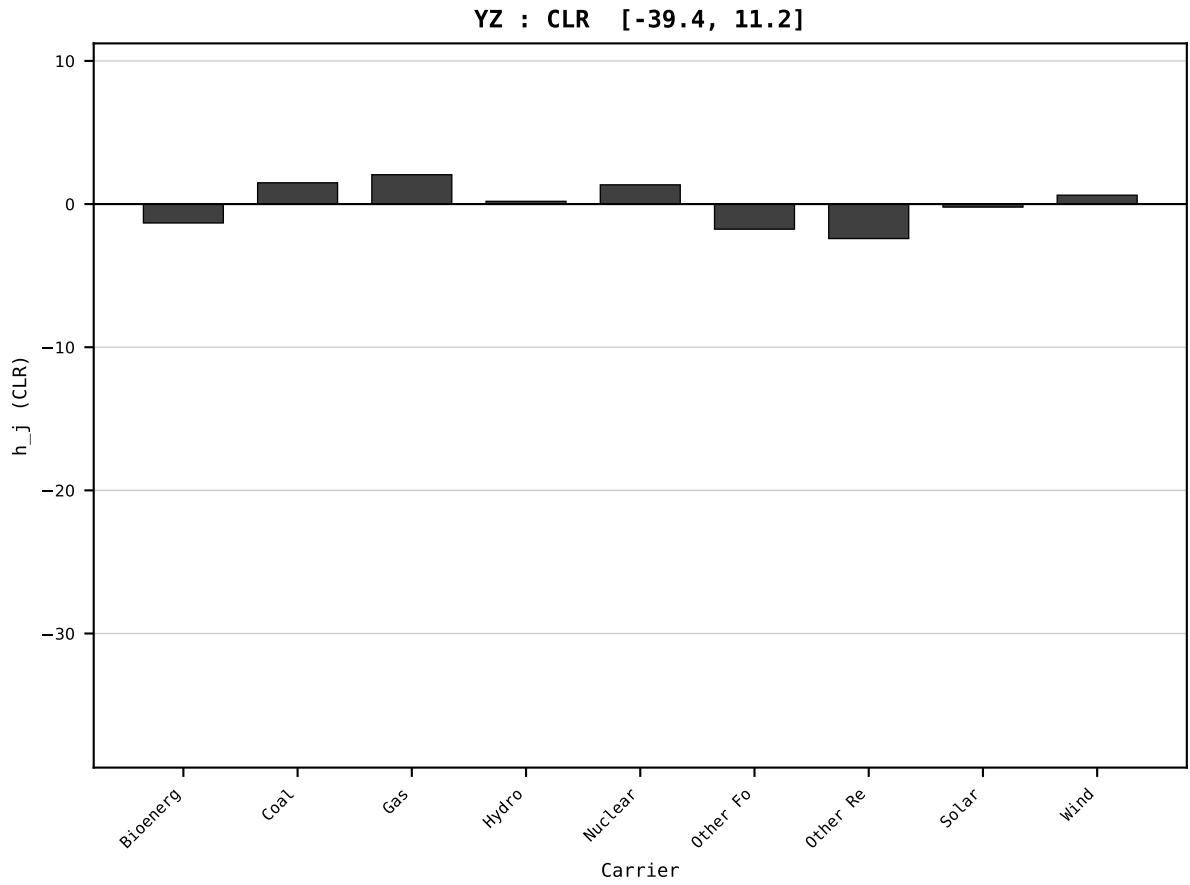
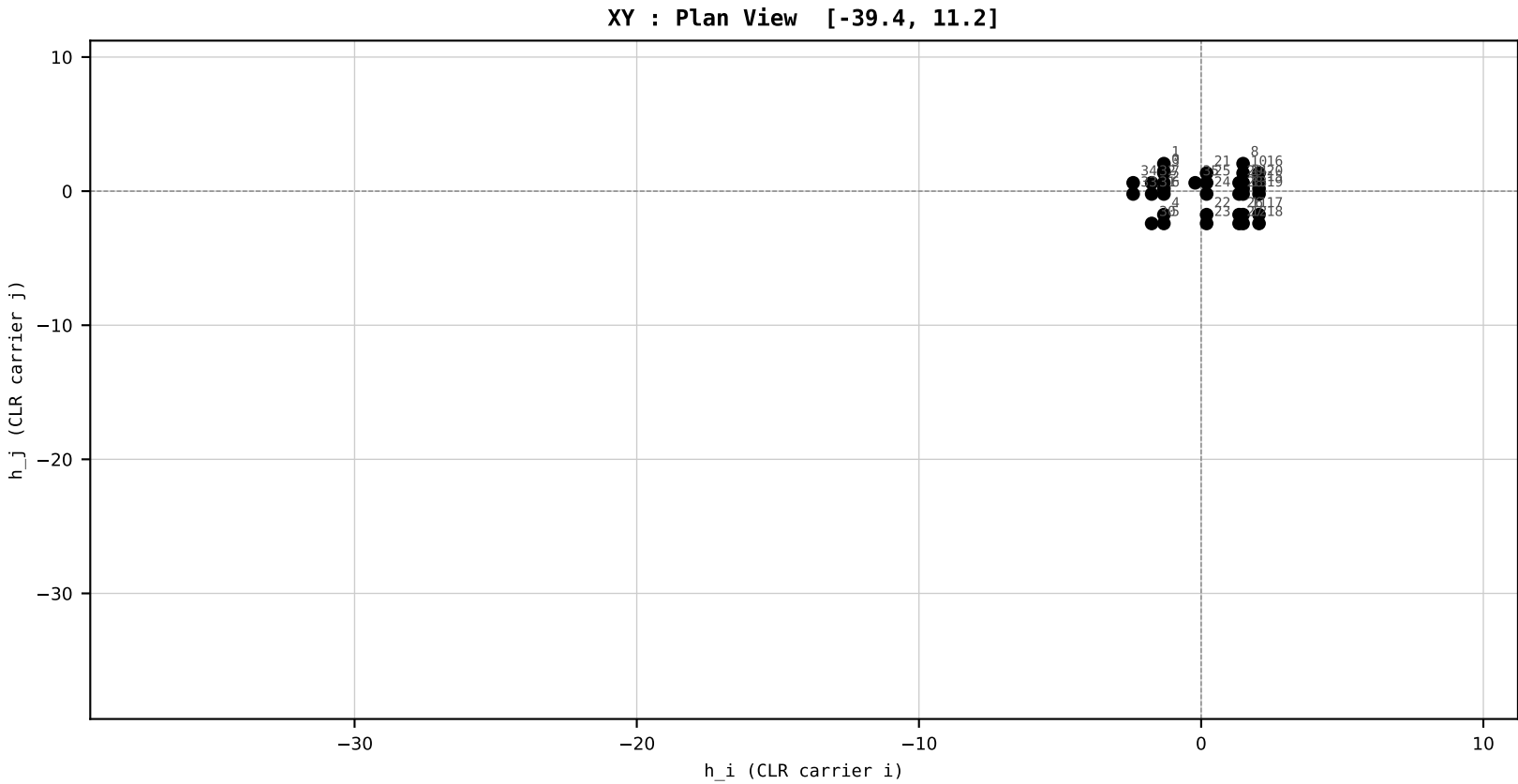
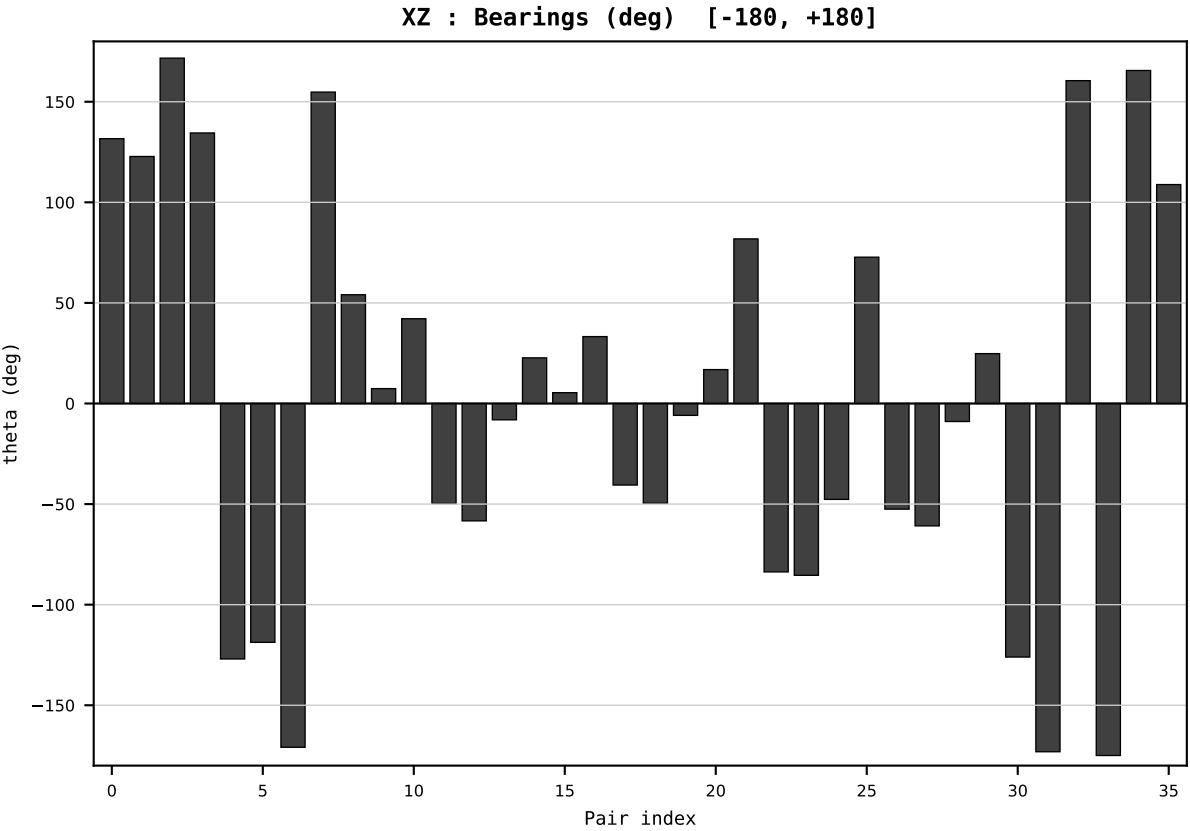
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2021
t : 20 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.250253
Ring = Hs-2
E_metric= 19.3124
kappa_HS= 86.48
omega = 3.9579 deg
d_A = 0.303328
Helm = Solar
Helm d = +0.190528
DR = 4.460 HLR
DR ratio= 86.5x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2022 | D=9 N=25 pairs=36

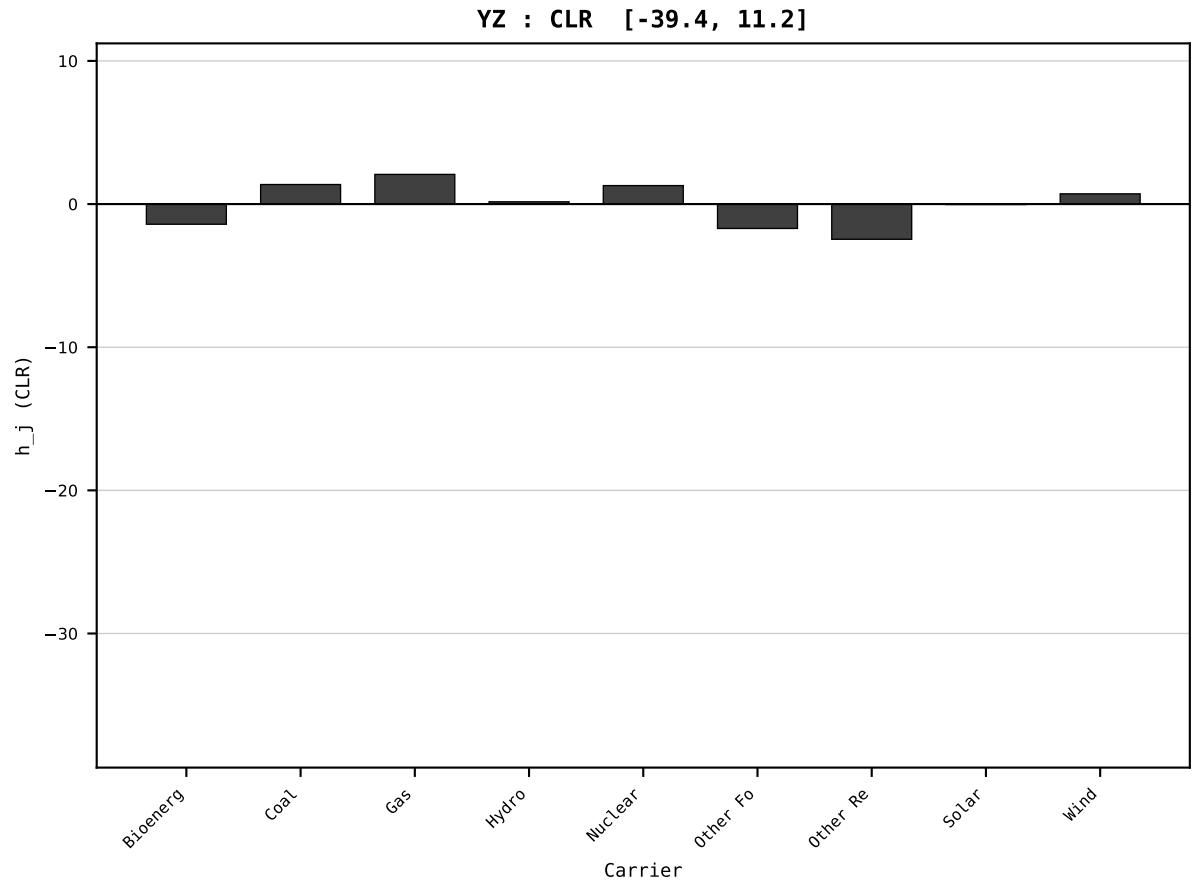
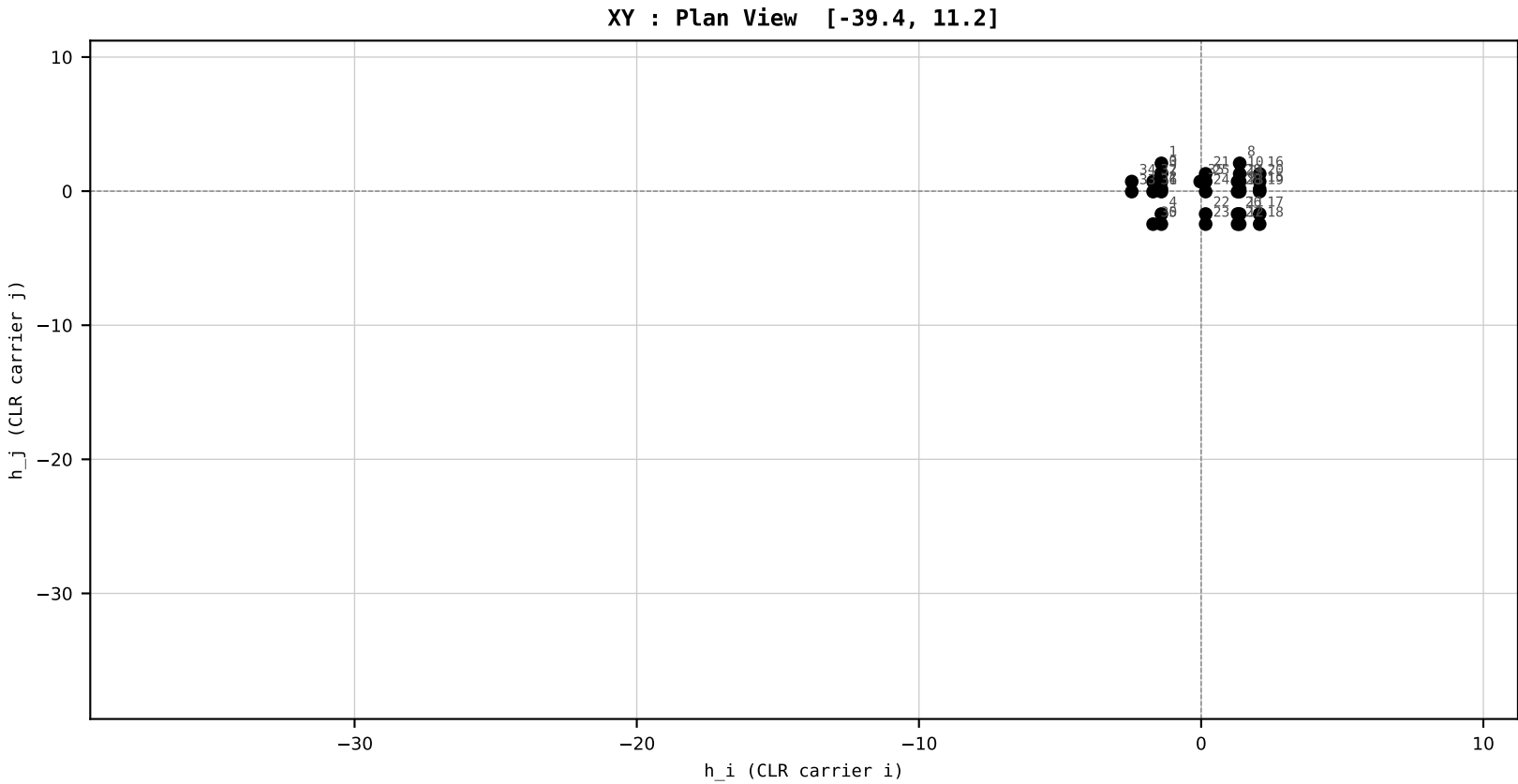
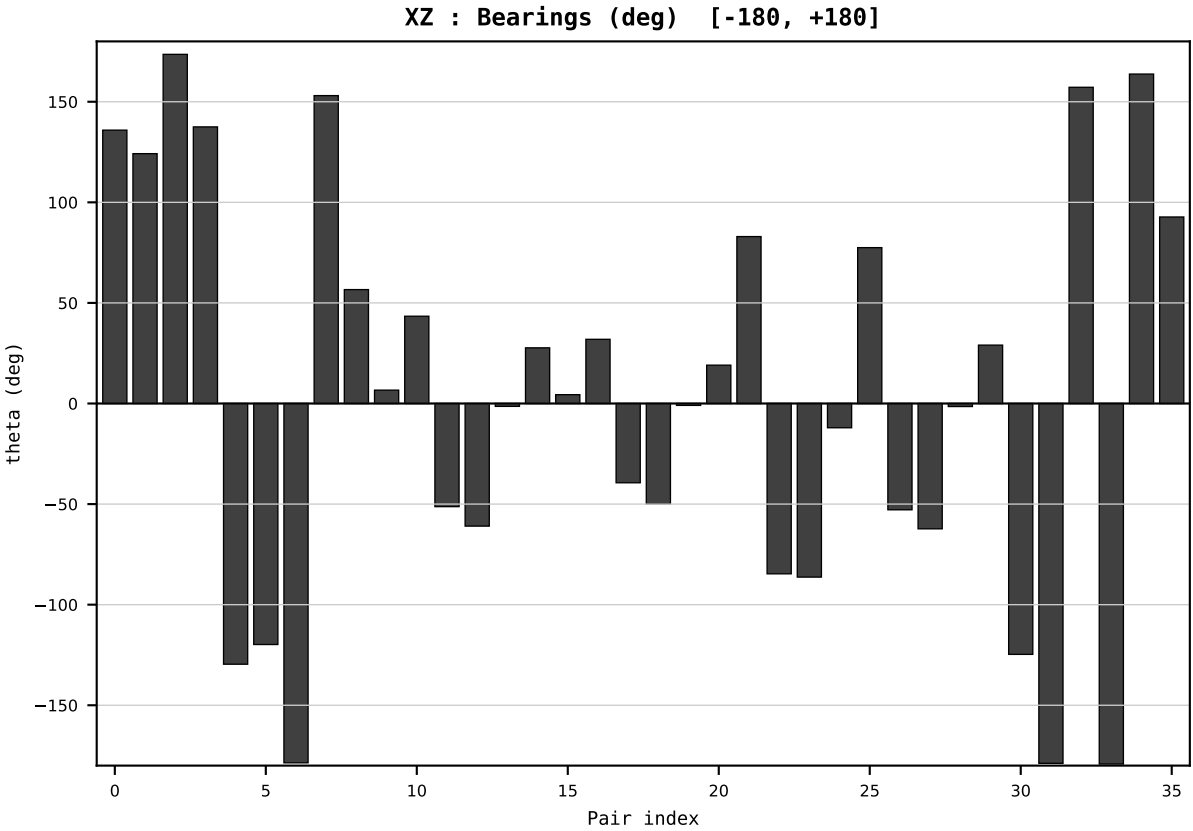
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2022
t : 21 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.246637
Ring = Hs-2
E_metric= 19.3055
kappa_HS= 93.00
omega = 3.5094 deg
d_A = 0.269105
Helm = Solar
Helm d = +0.177922
DR = 4.533 HLR
DR ratio= 93.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2023 | D=9 N=25 pairs=36

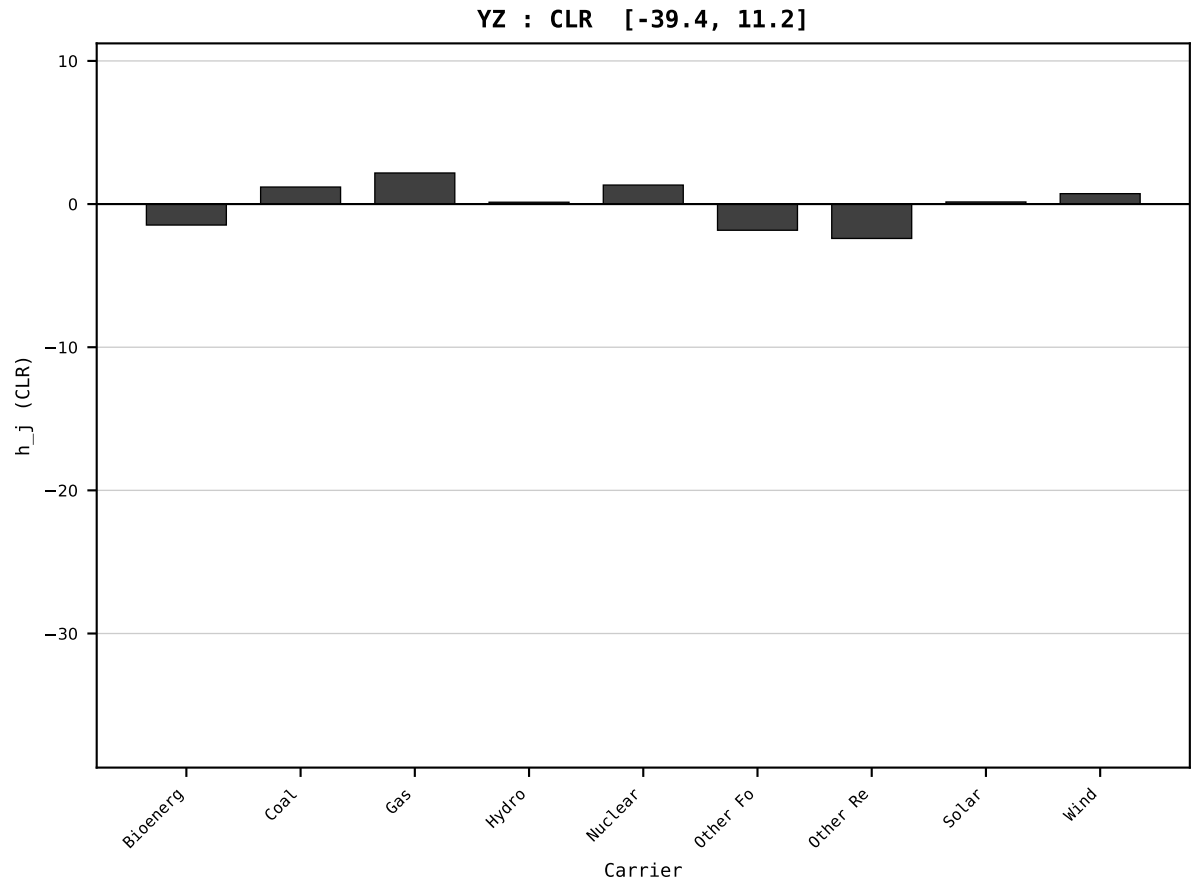
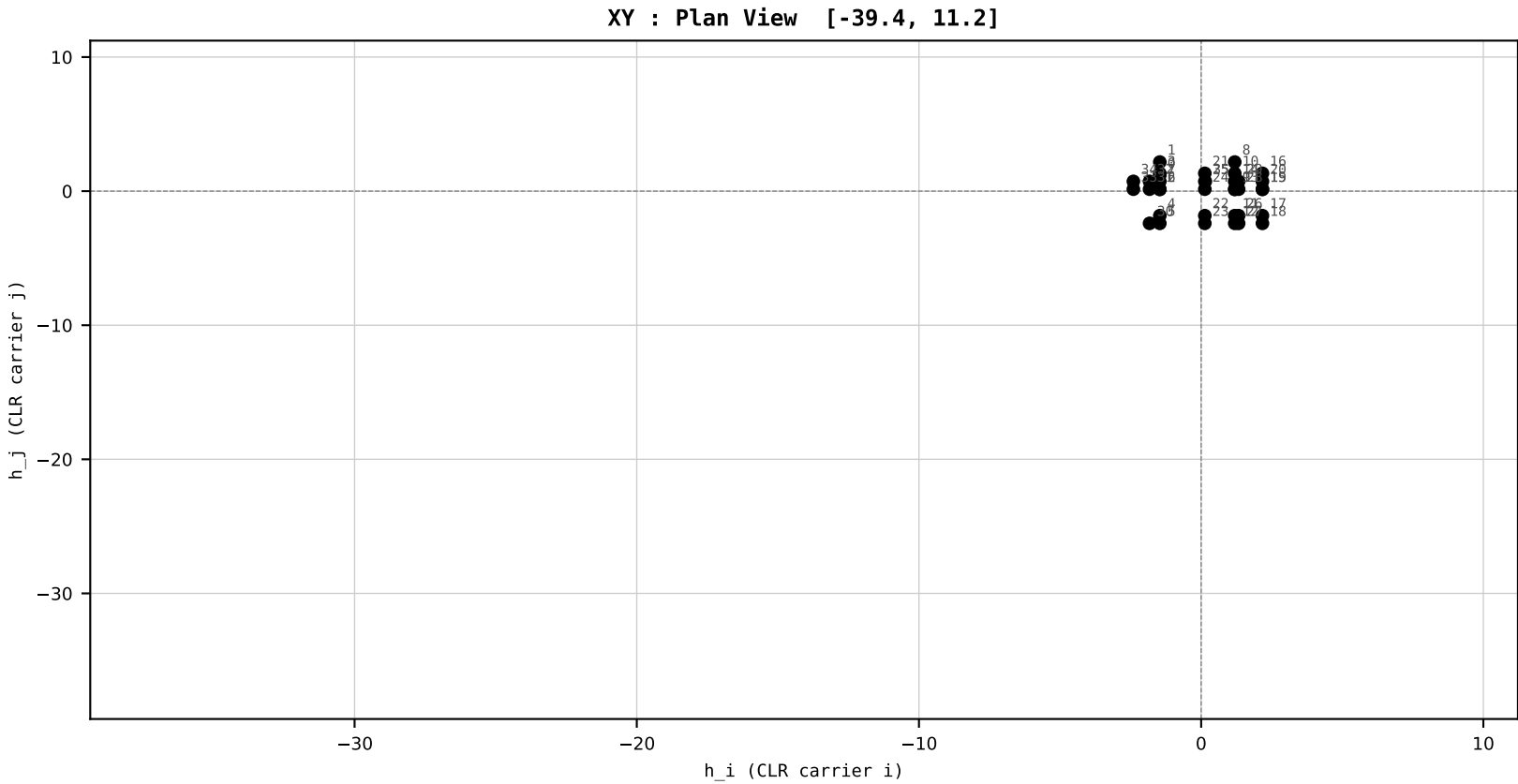
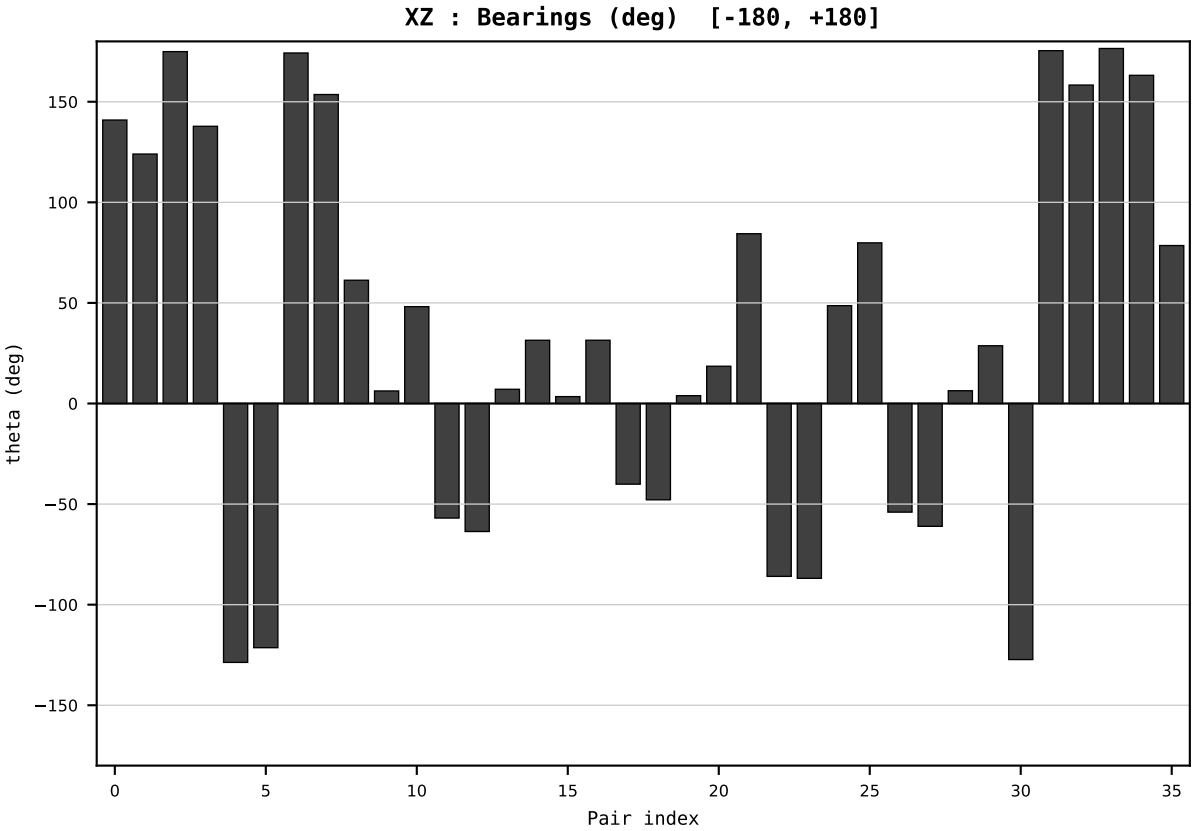
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2023
t : 22 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.258277
Ring = Hs-2
E_metric= 19.7208
kappa_HS= 96.83
omega = 4.0175 deg
d_A = 0.313215
Helm = Solar
Helm d = +0.181686
DR = 4.573 HLR
DR ratio= 96.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=23 | Year 2024 | D=9 N=25 pairs=36

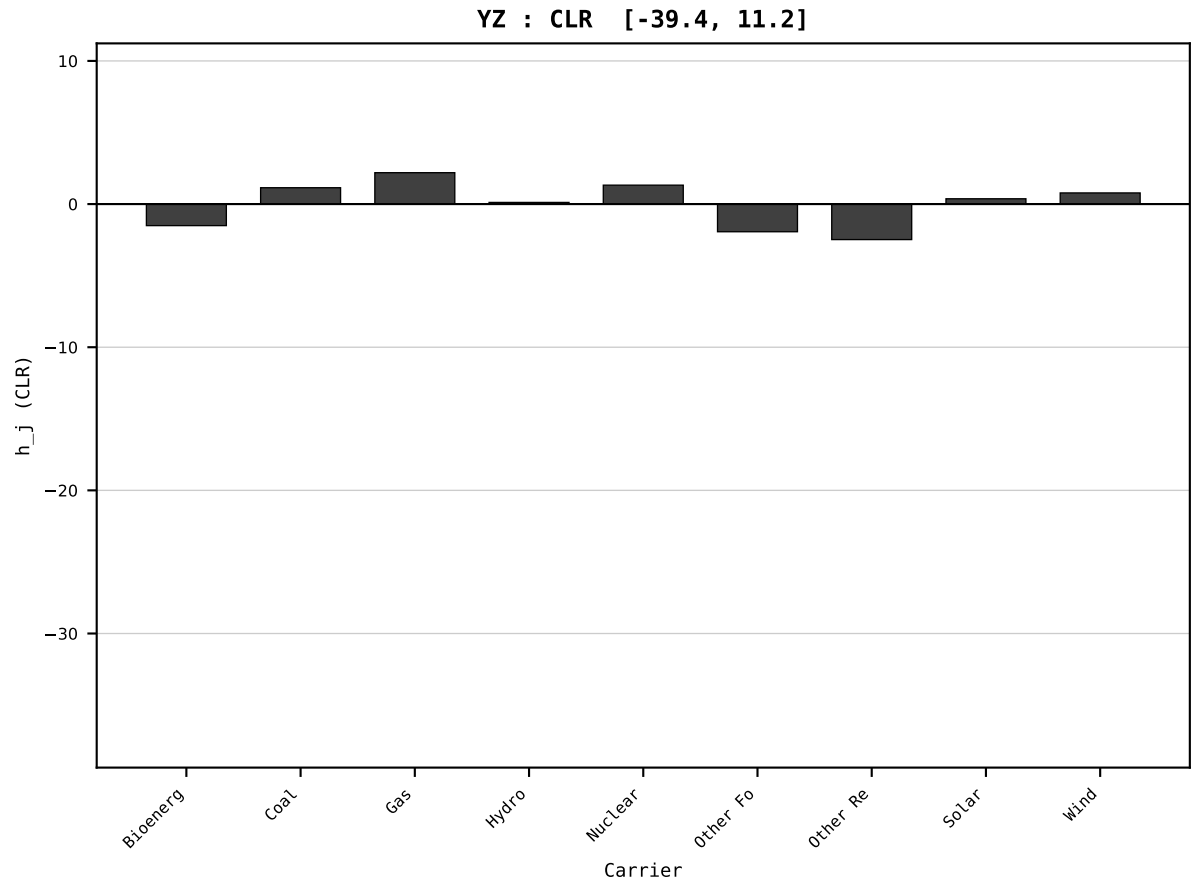
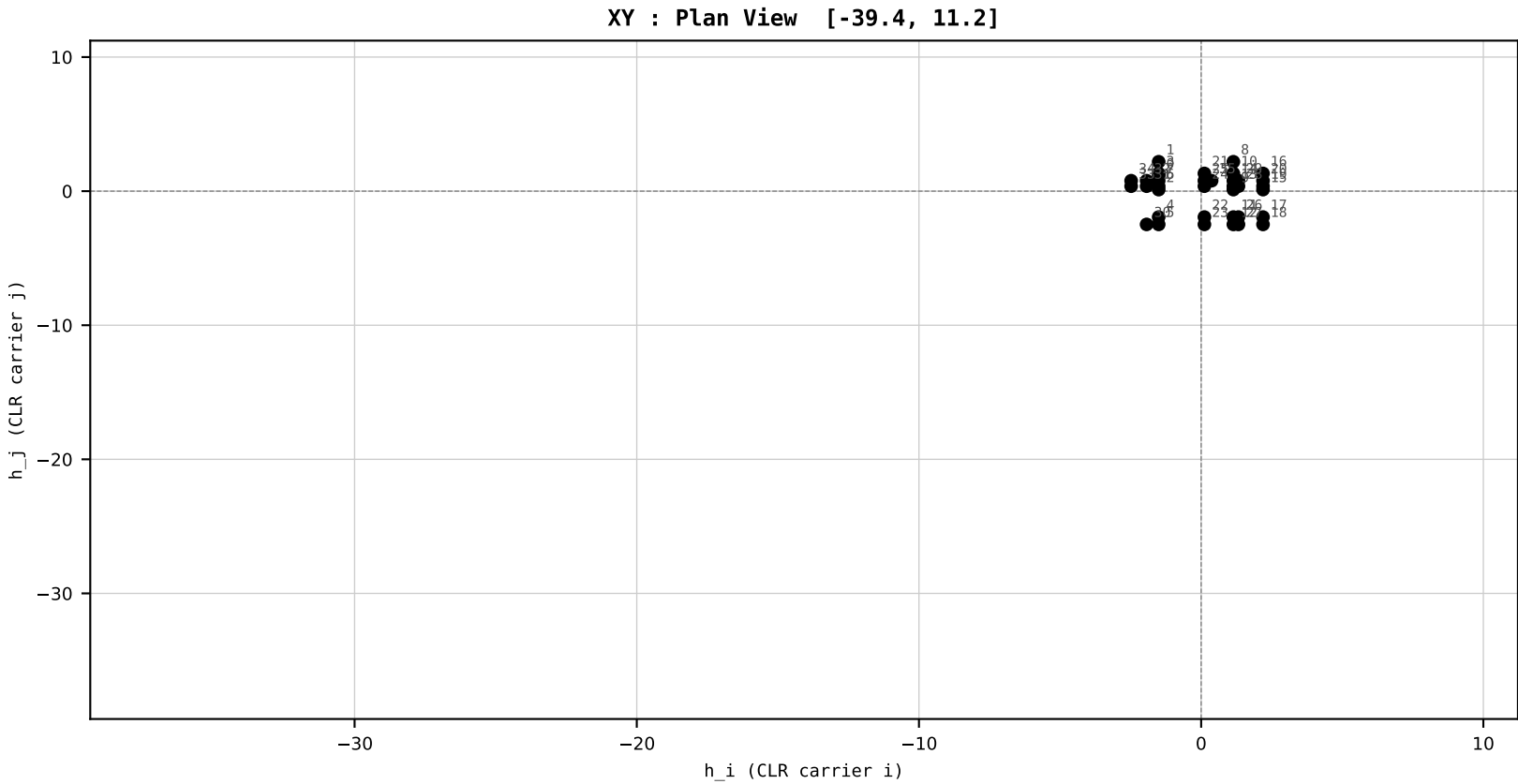
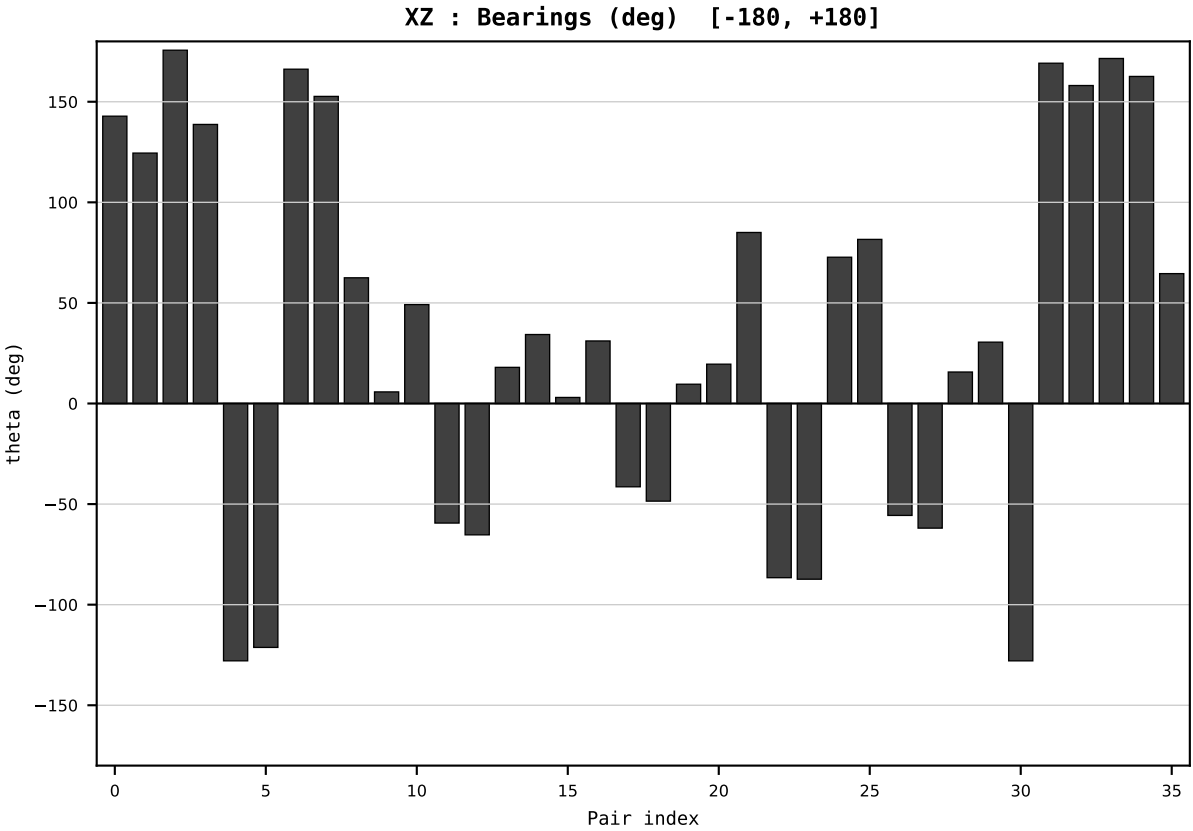
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2024
t : 23 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.256233
Ring = Hs-2
E_metric= 20.7697
kappa_HS= 107.02
omega = 3.1307 deg
d_A = 0.272026
Helm = Solar
Helm d = +0.222573
DR = 4.673 HLR
DR ratio= 107.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



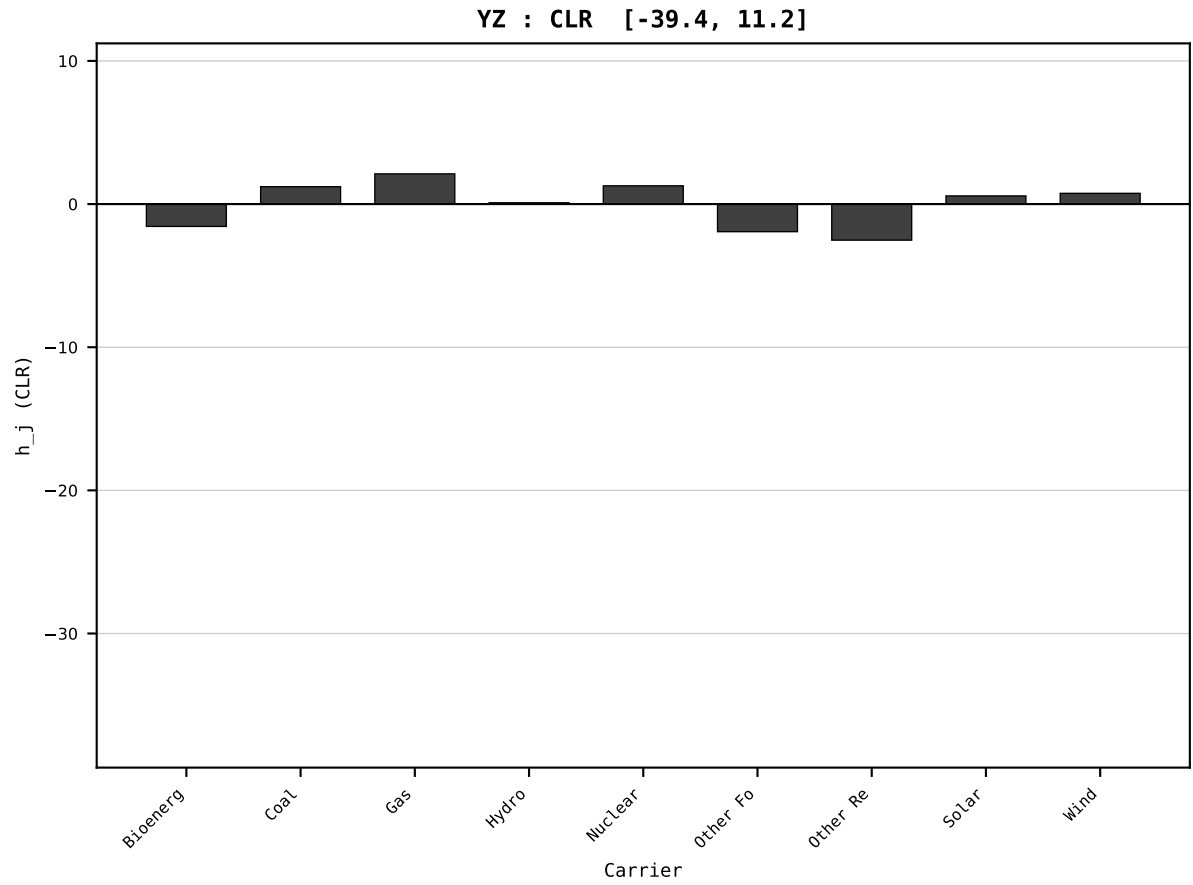
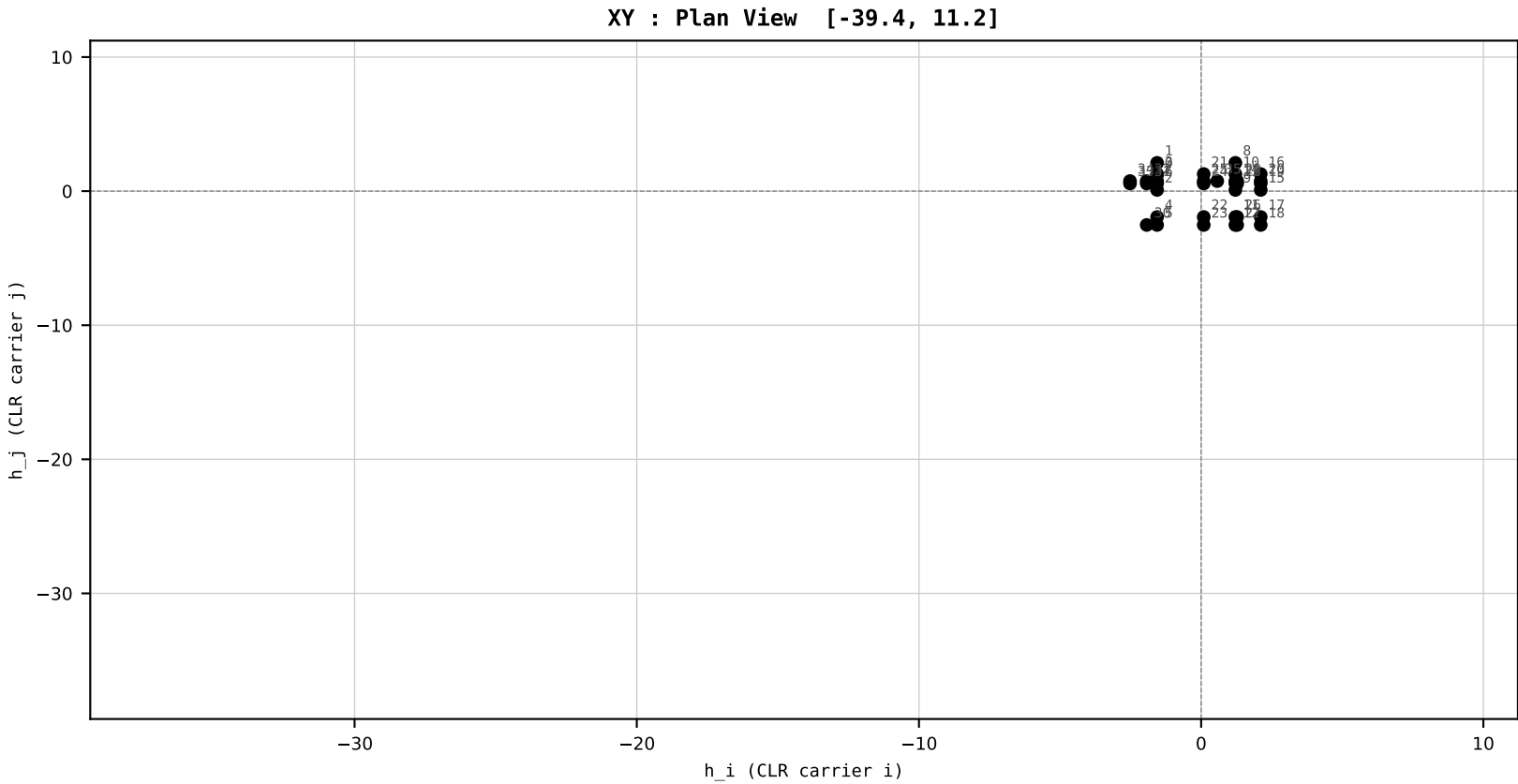
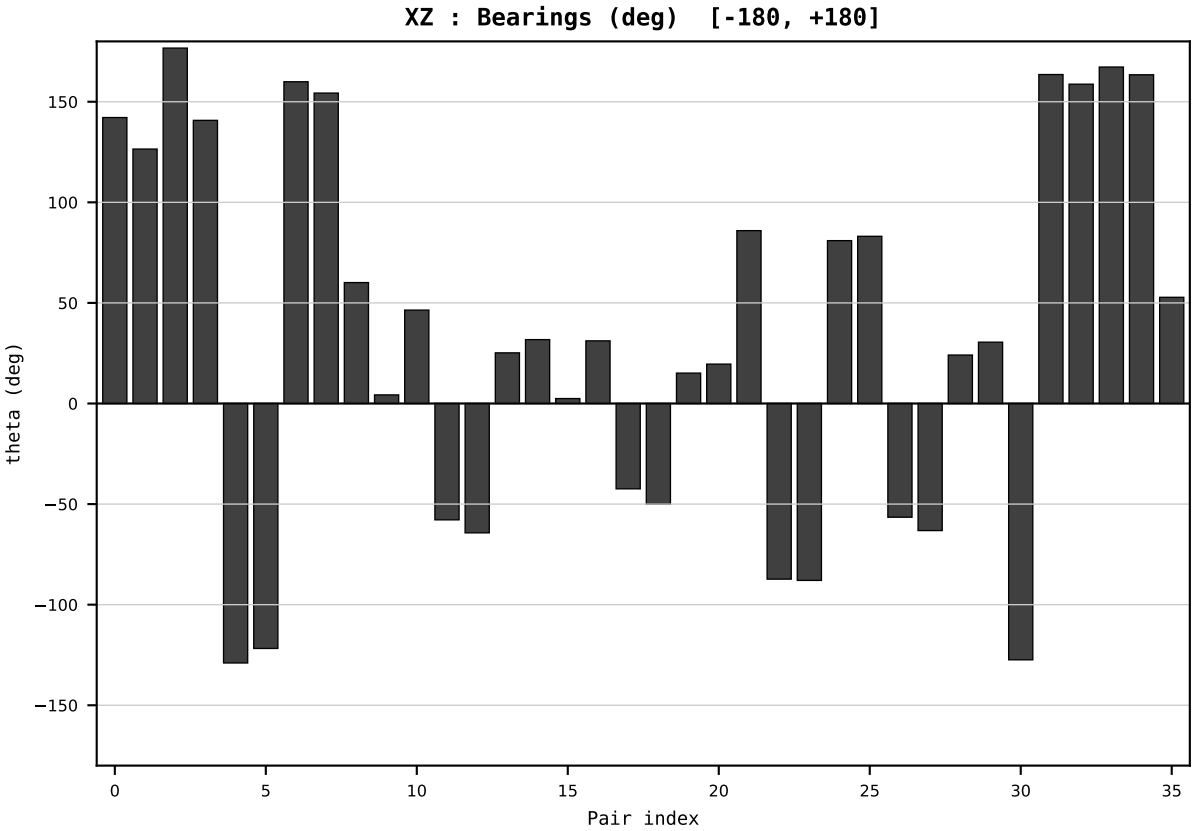
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

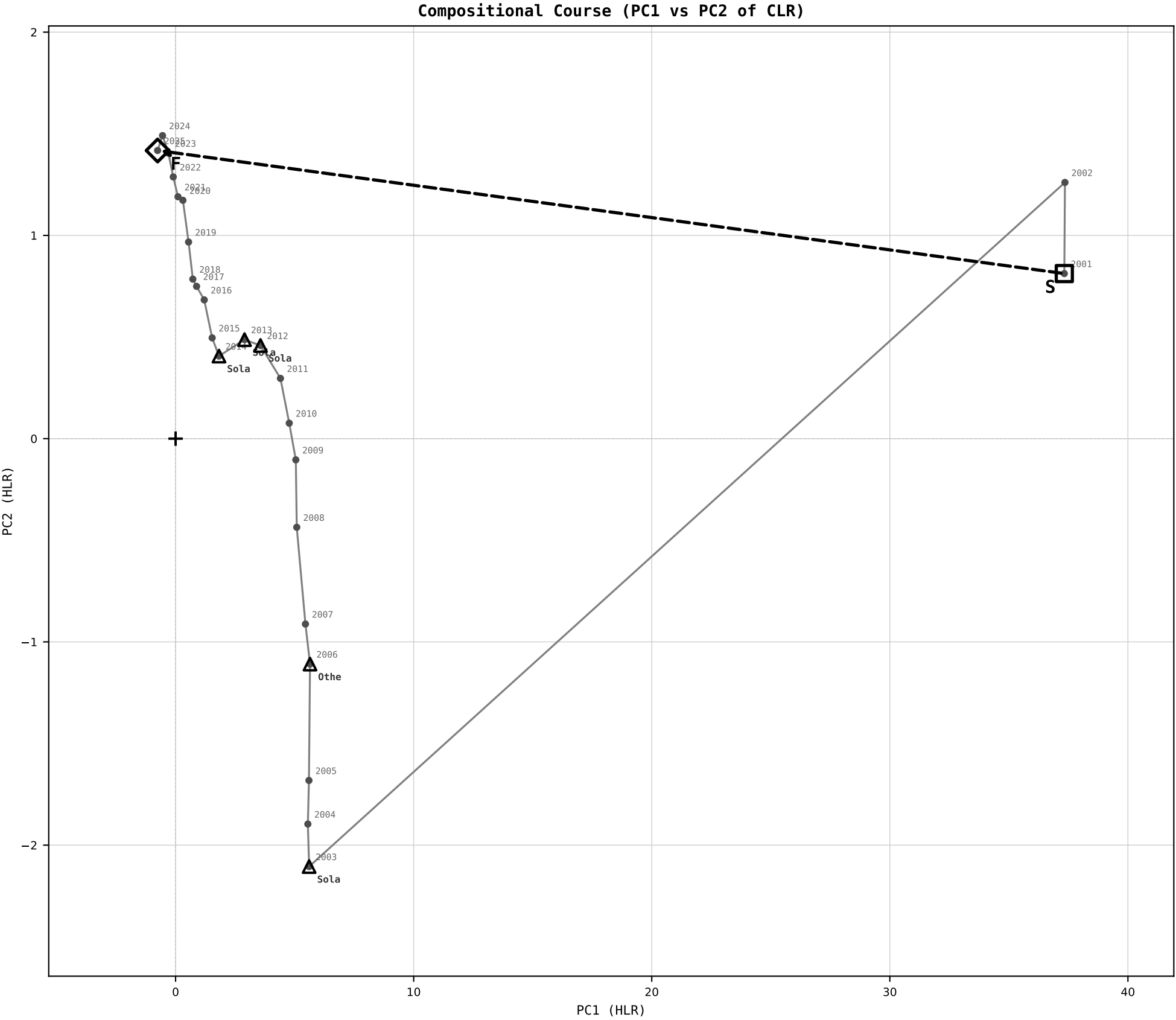
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2025
t : 24 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.240009
Ring = Hs-2
E_metric= 20.9704
kappa_HS= 102.78
omega = 3.0581 deg
d_A = 0.244792
Helm = Solar
Helm d = +0.200013
DR = 4.633 HLR
DR ratio= 102.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





View B

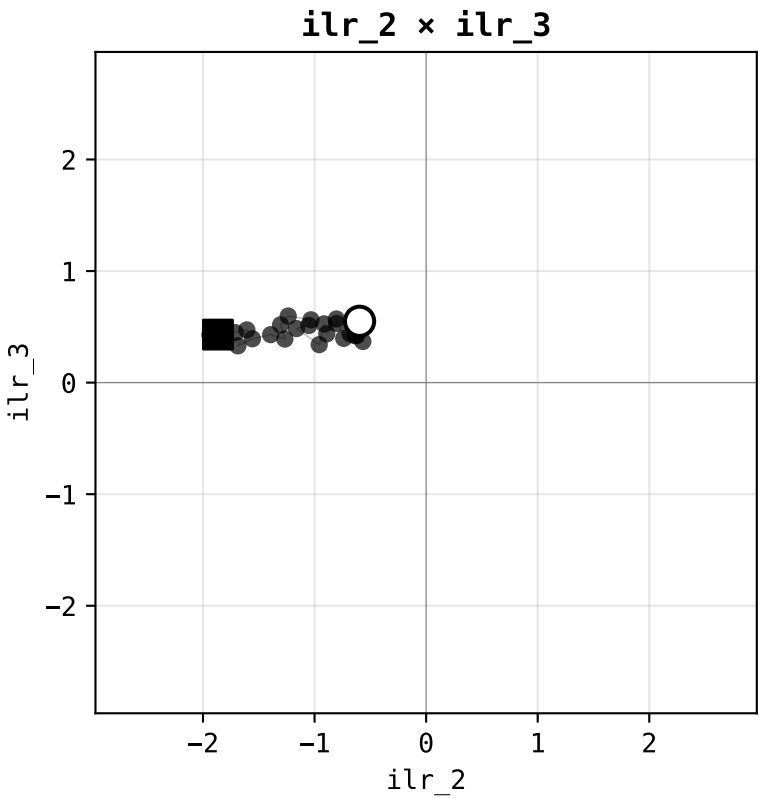
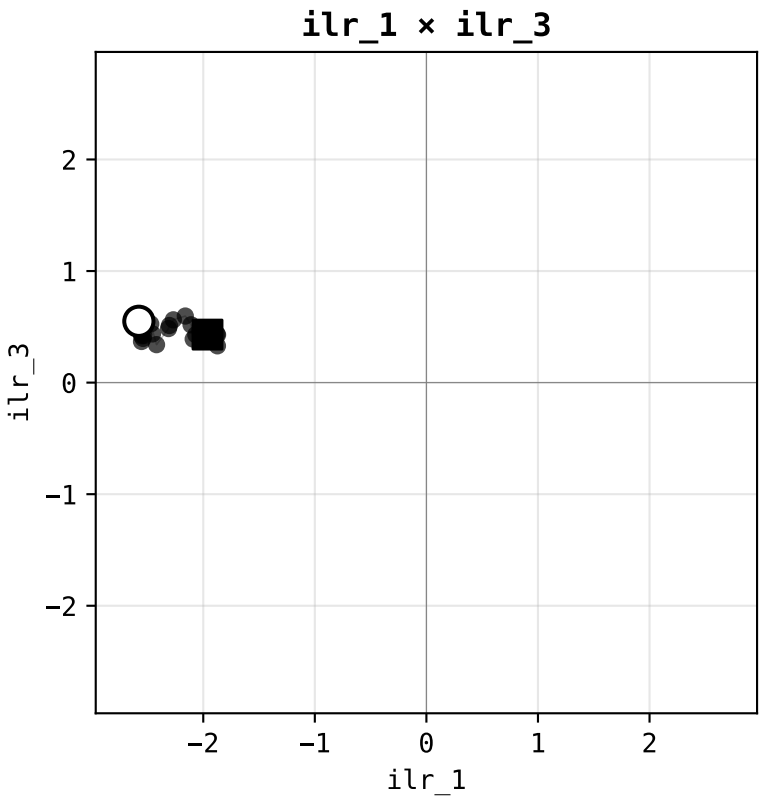
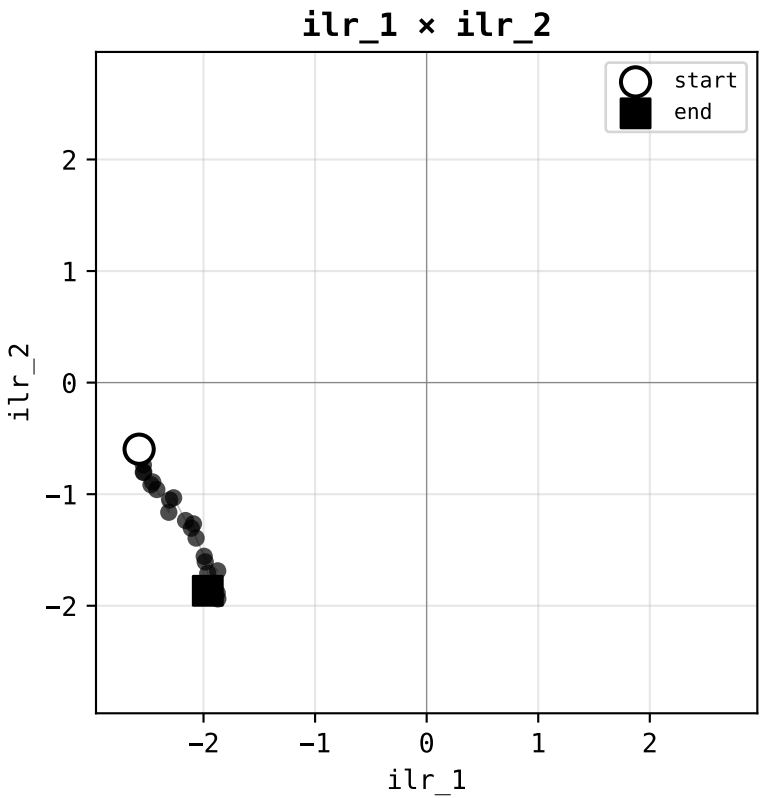
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ USA United States

ILR-Helmert Orthogonal Triplet – USA United States · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=25 years · D=9 carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

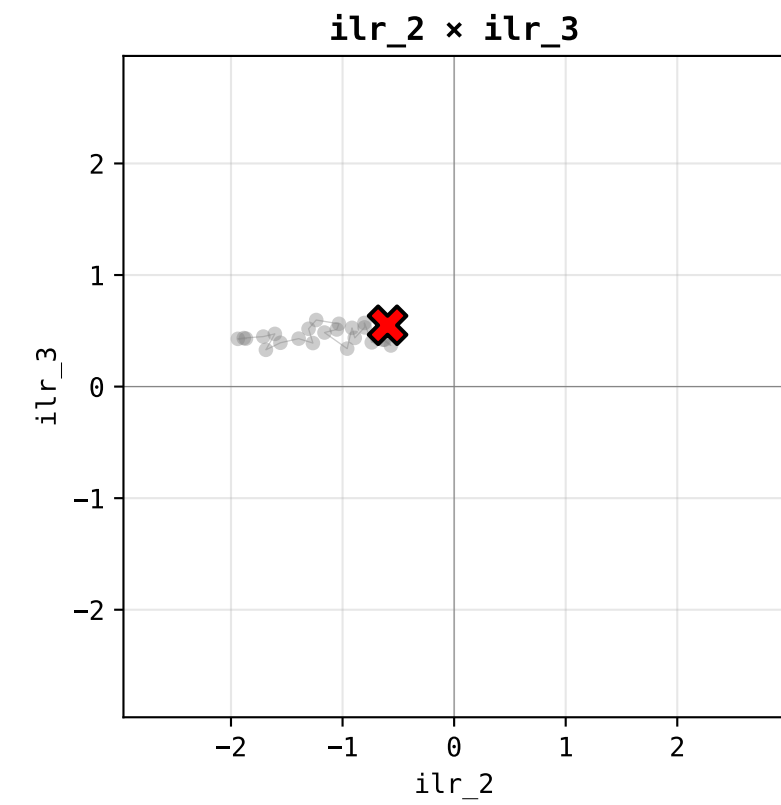
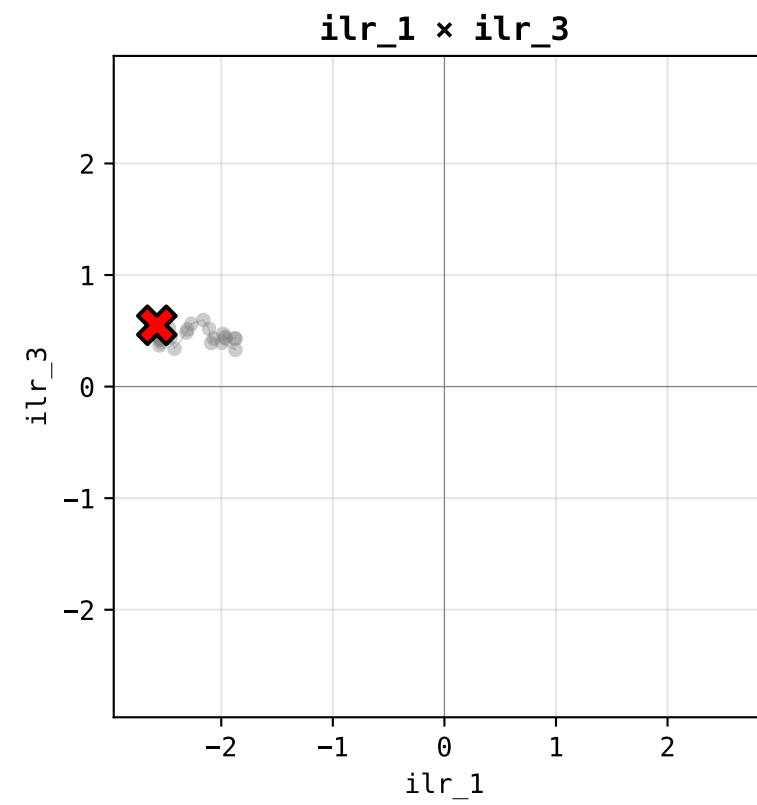
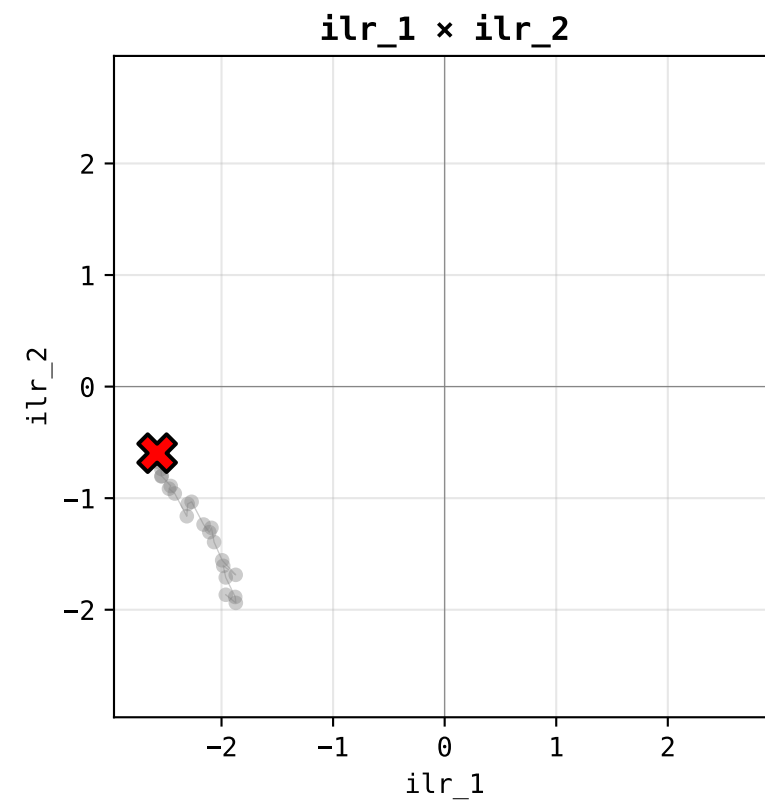
ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

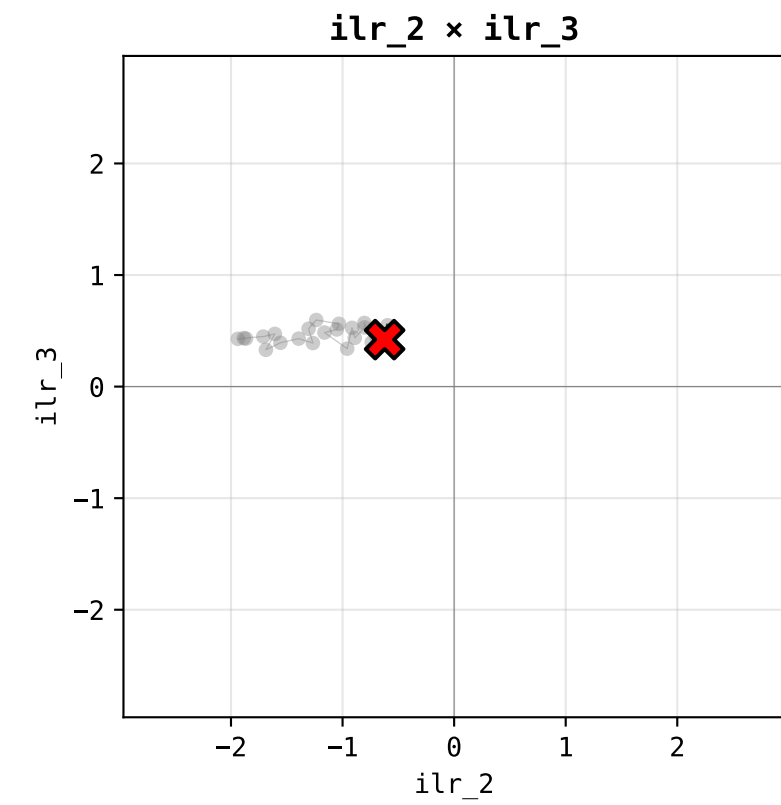
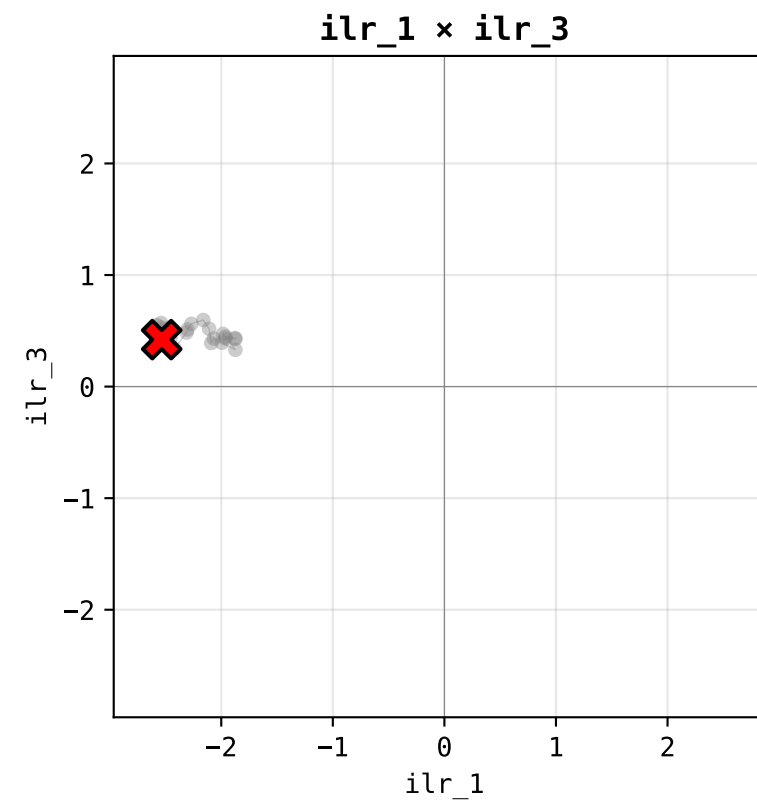
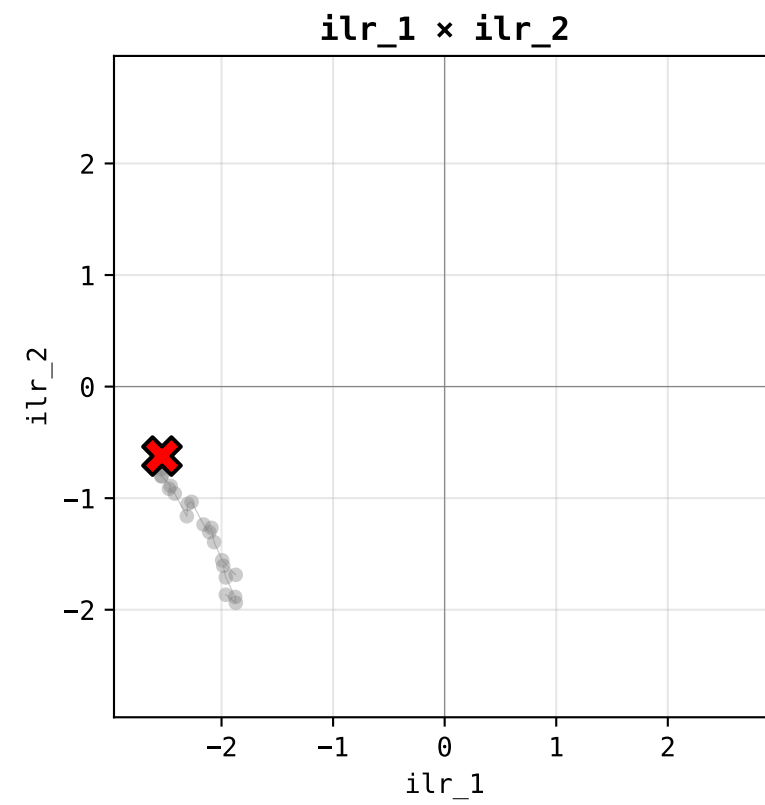
Trajectory:
start year: 2001
end year: 2025
N readings: 25

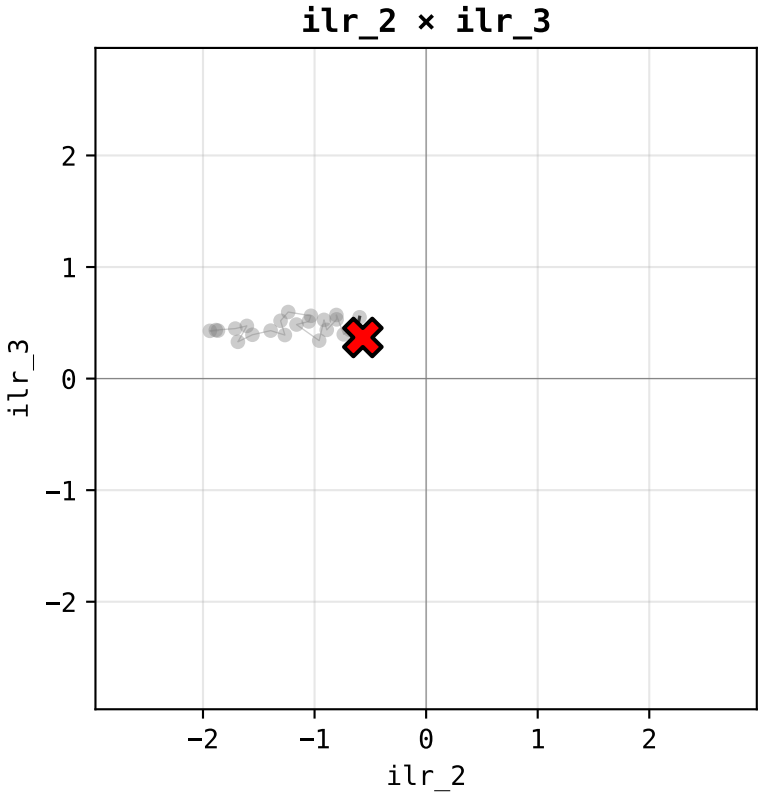
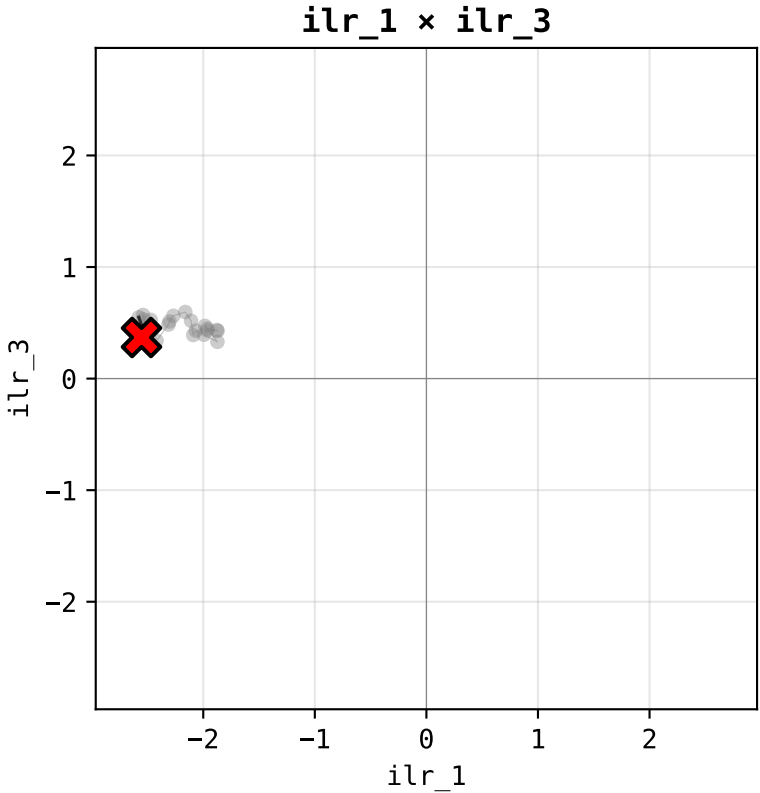
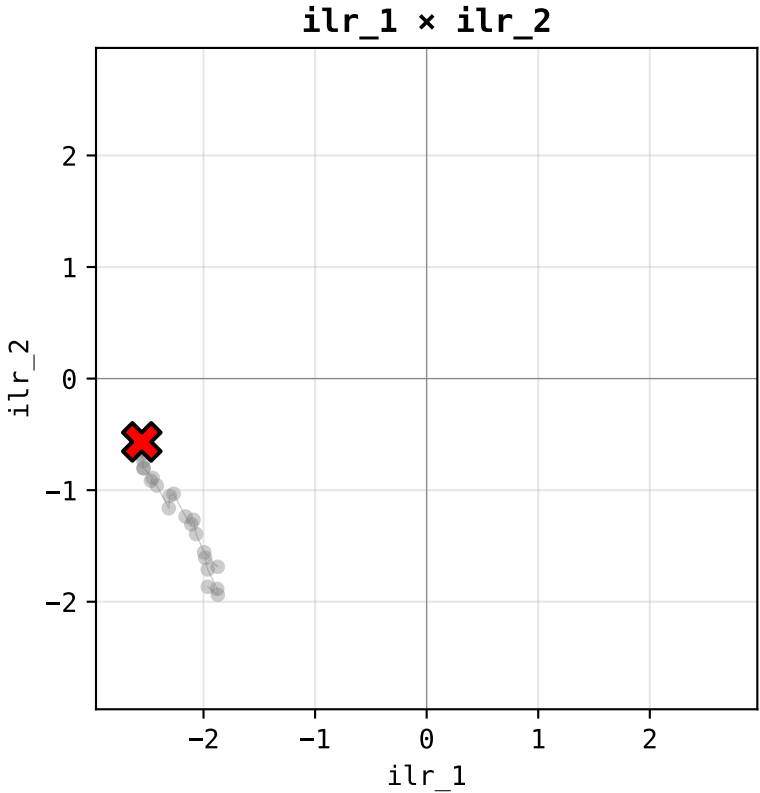
Display:
fixed scale ±2.96 ILR
Order 1 (first principles)
per Output Doctrine v1.0

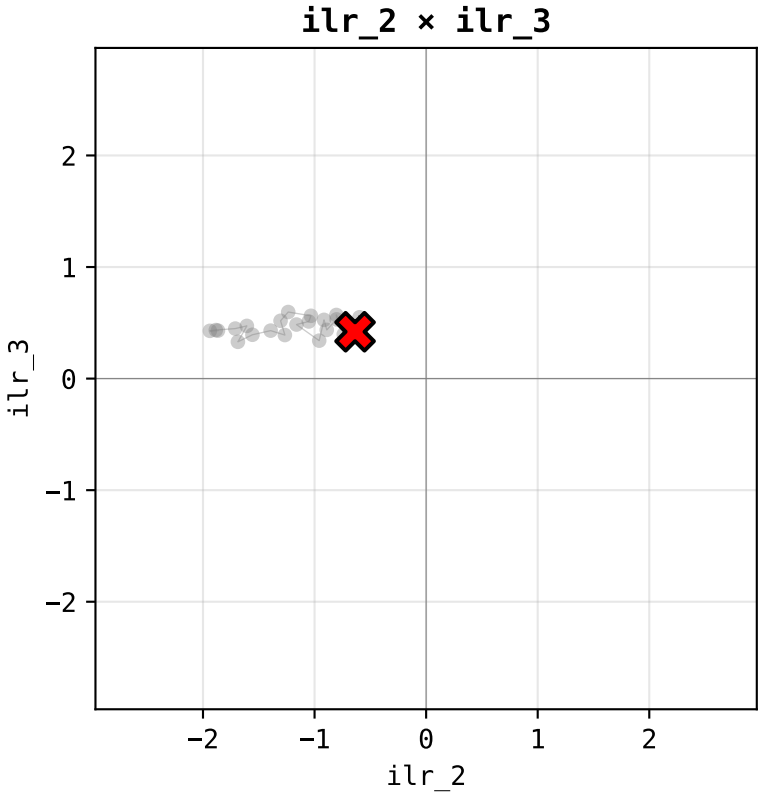
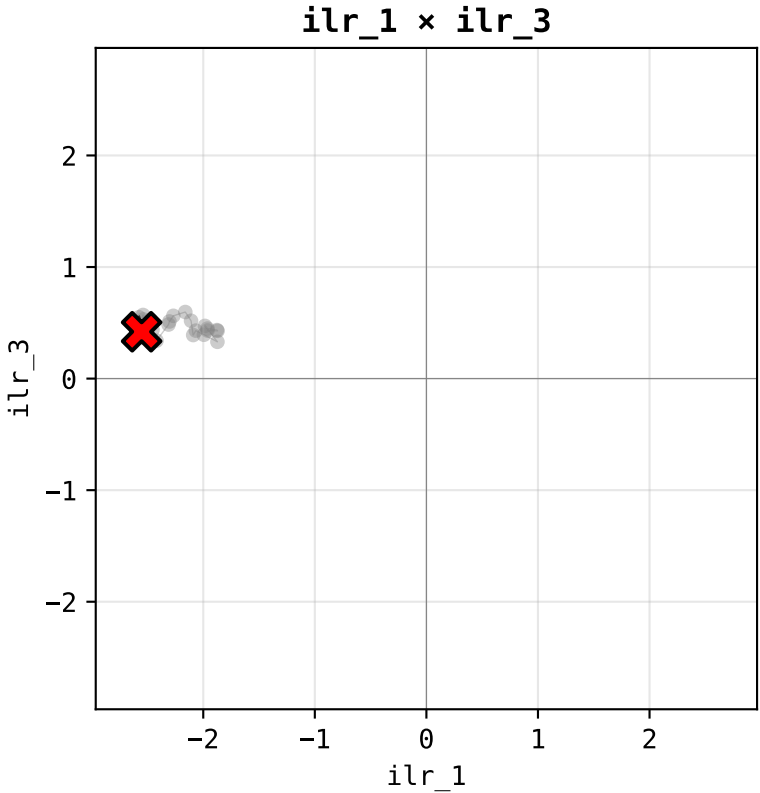
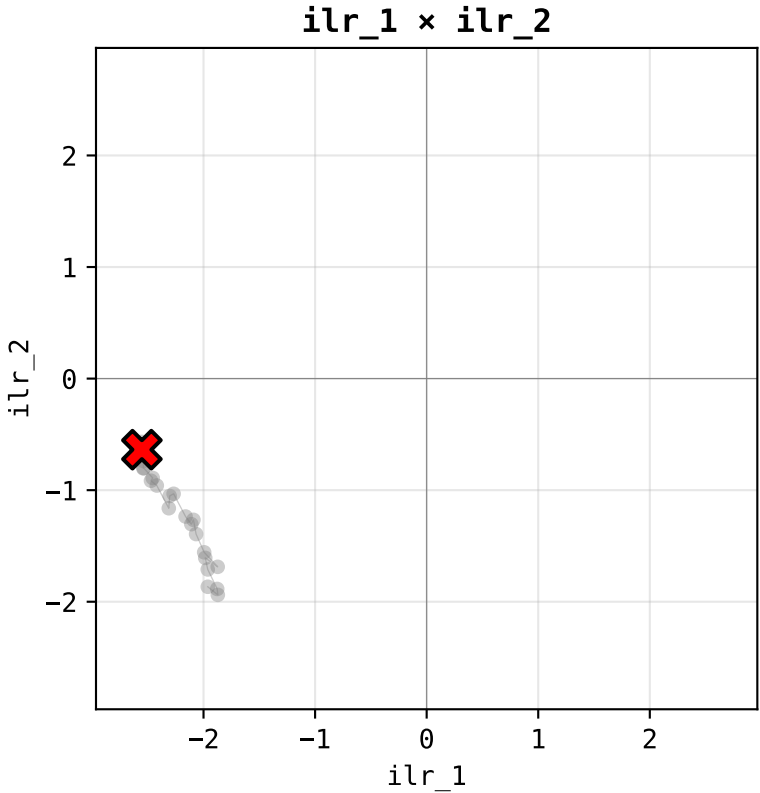
Symbology:
○ start · ■ end
· intermediate years

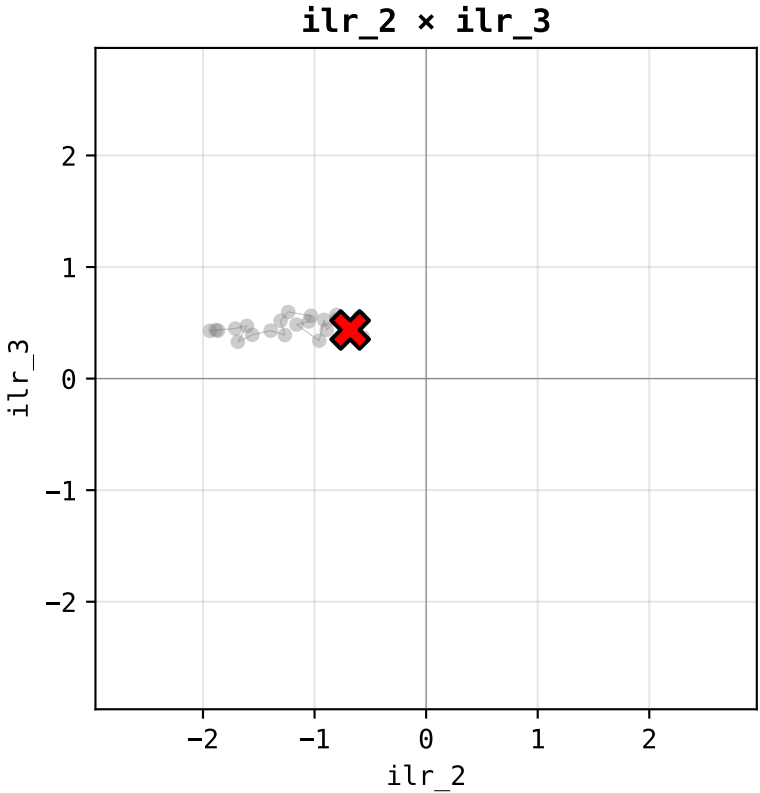
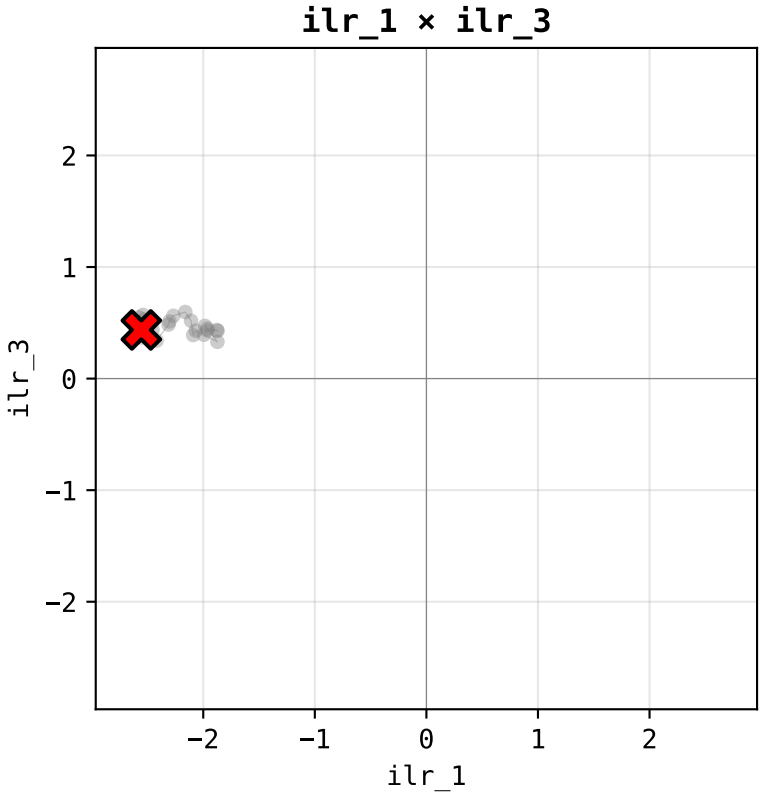
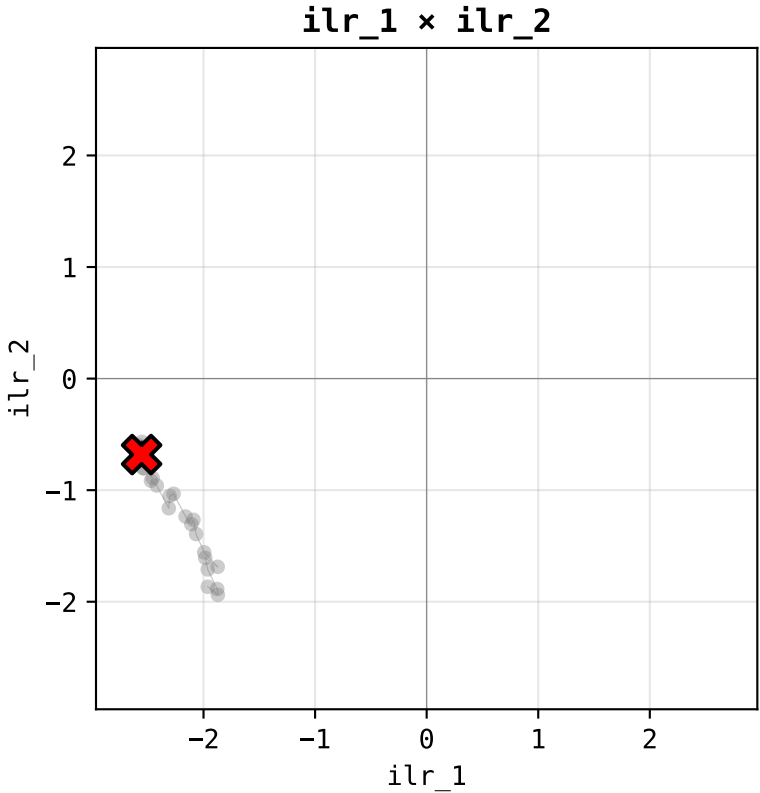
ILR Triplet – USA United States · year 2001 (1/25)



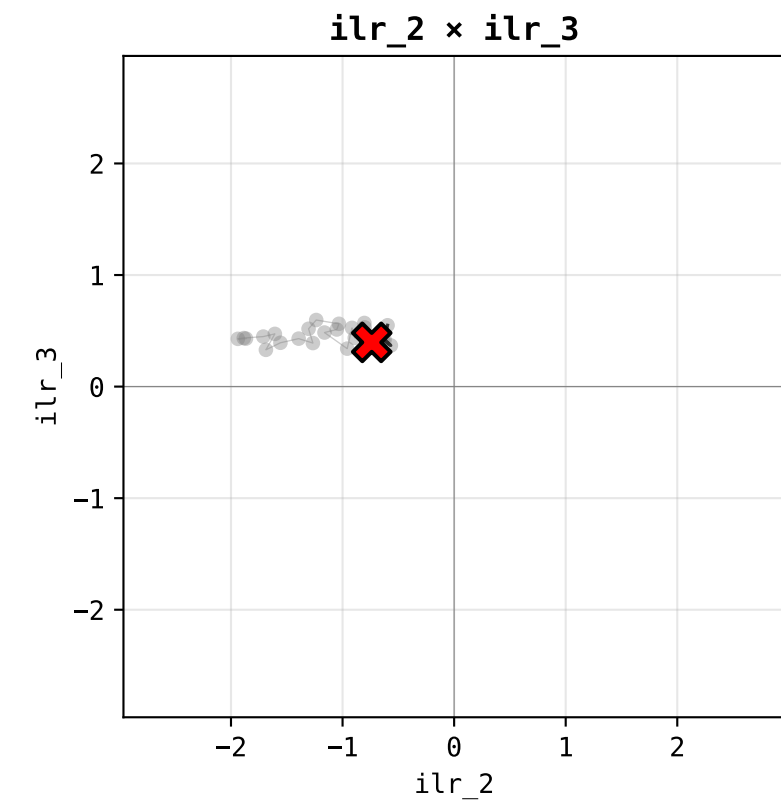
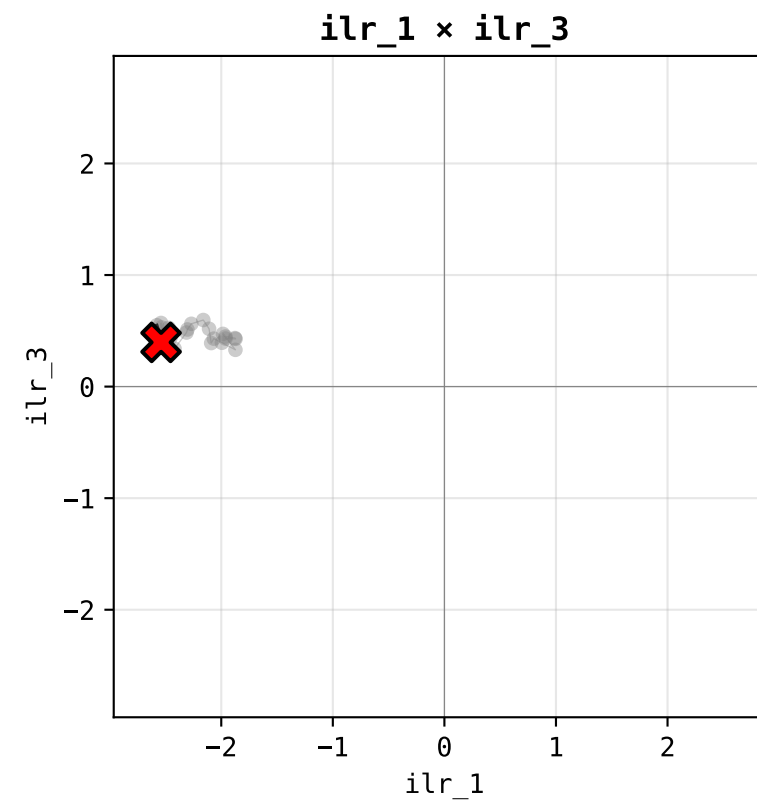
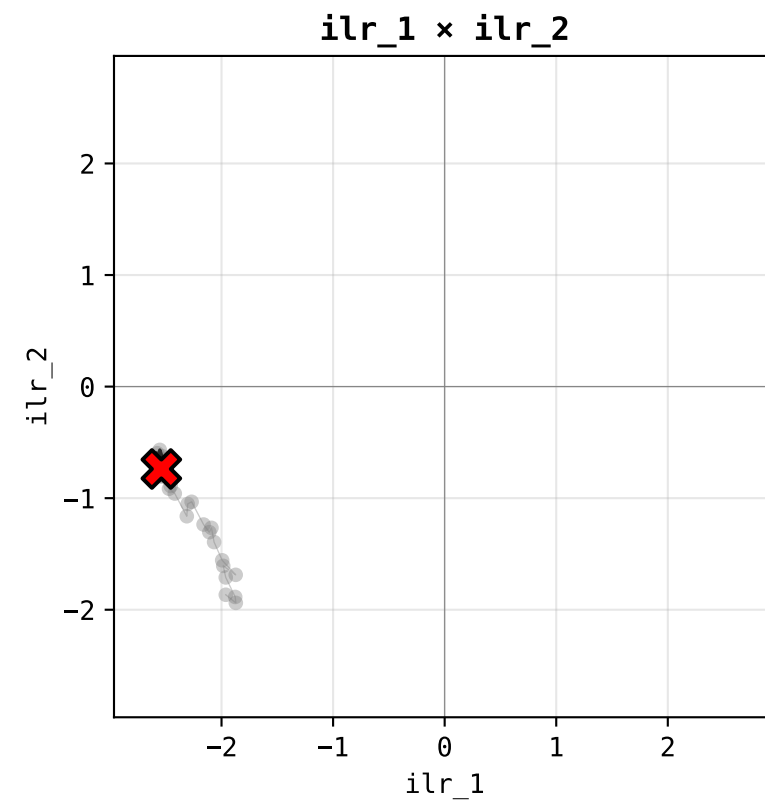




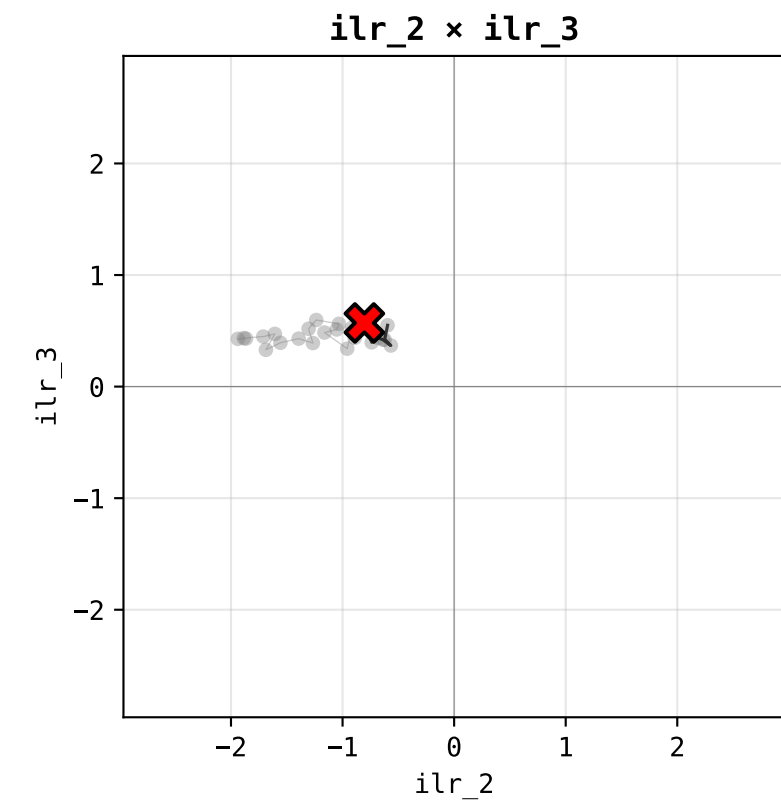
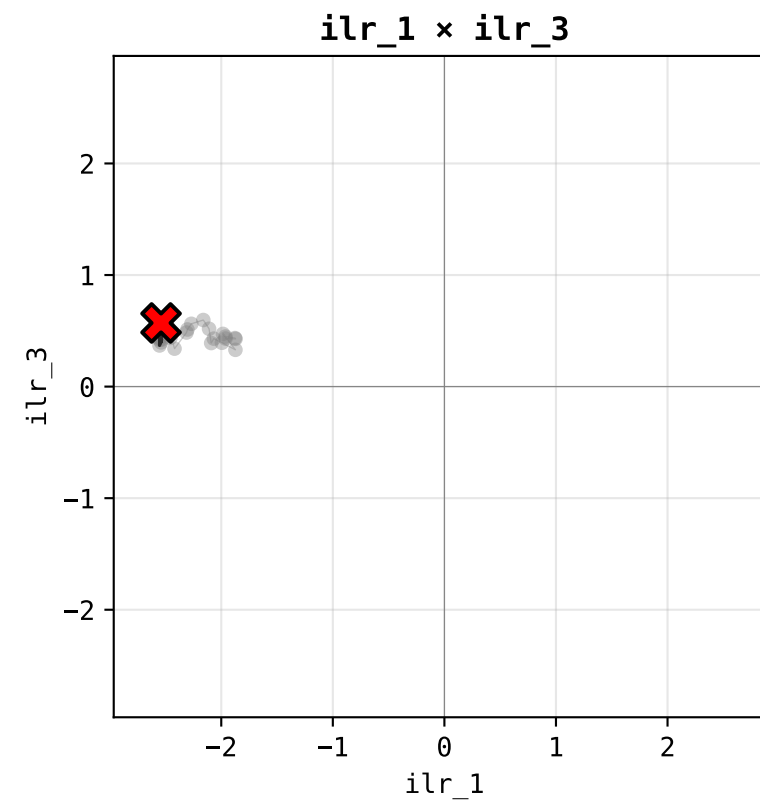
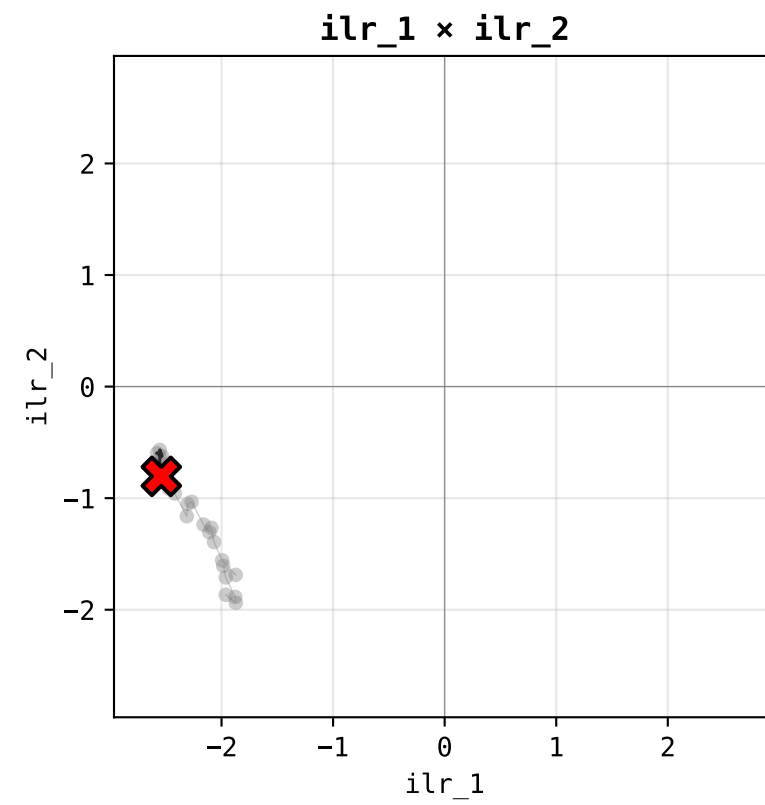




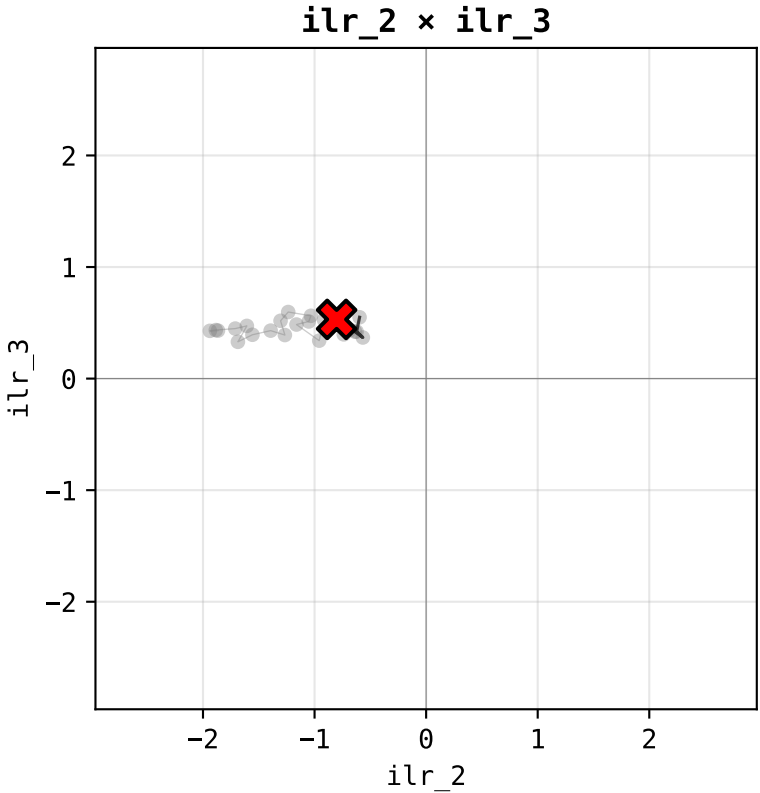
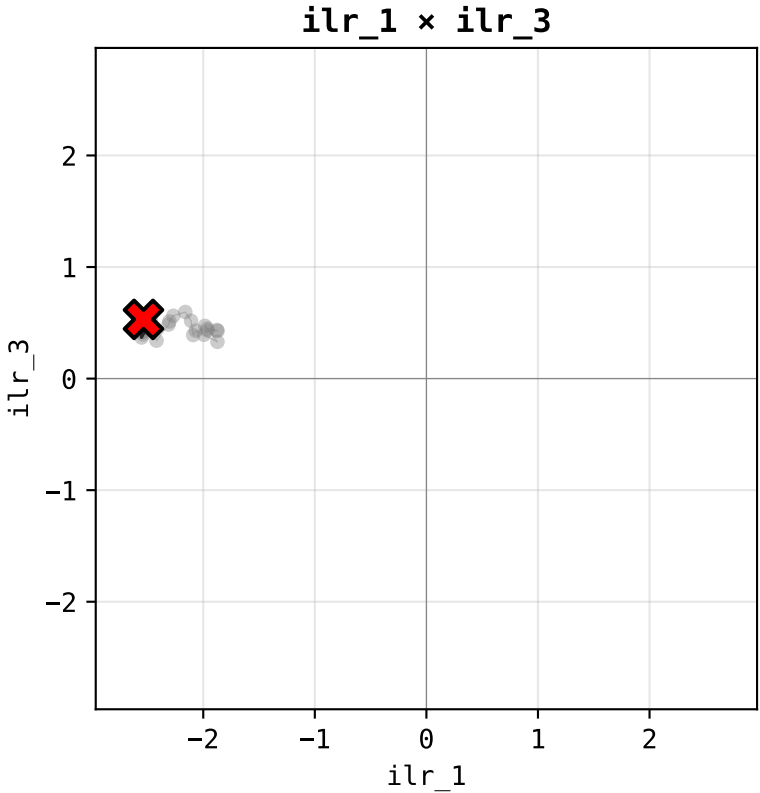
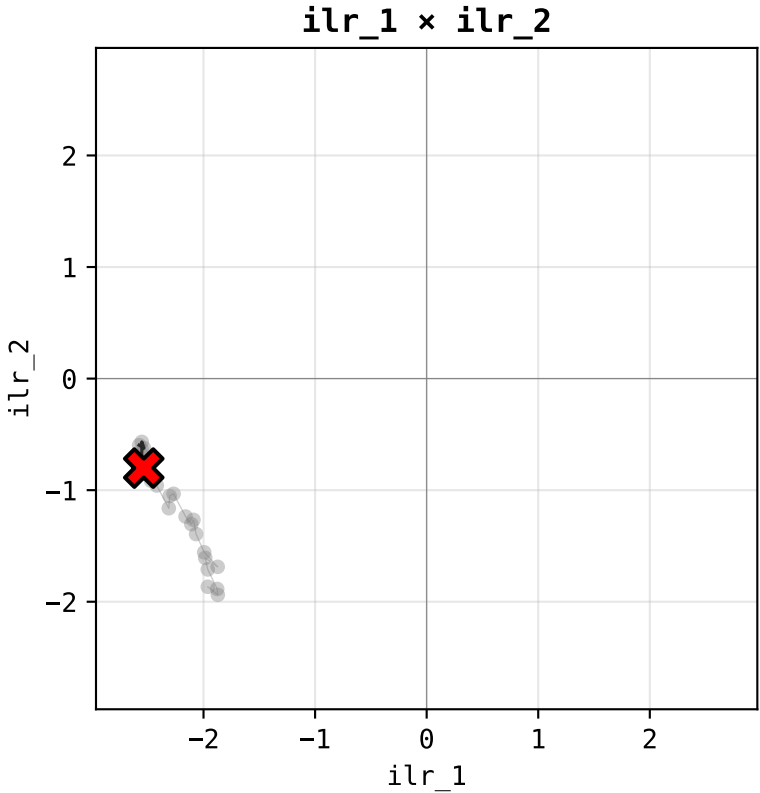
ILR Triplet – USA United States · year 2006 (6/25)



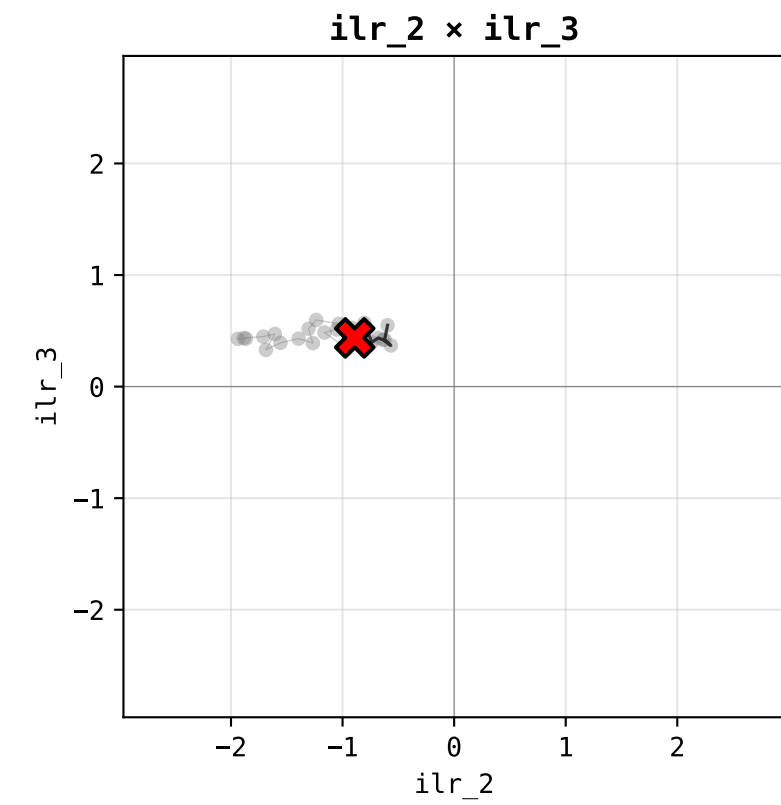
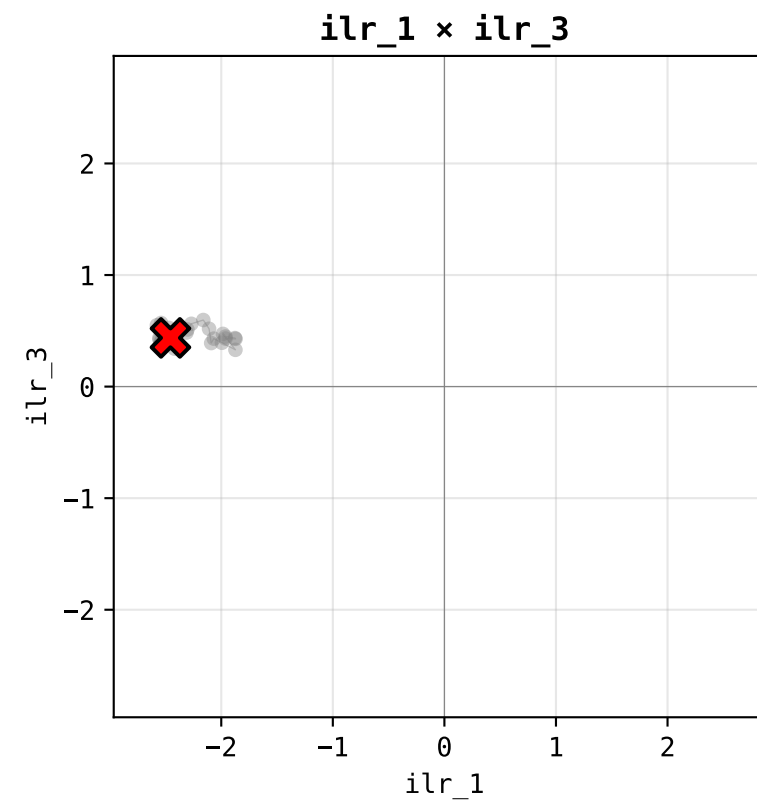
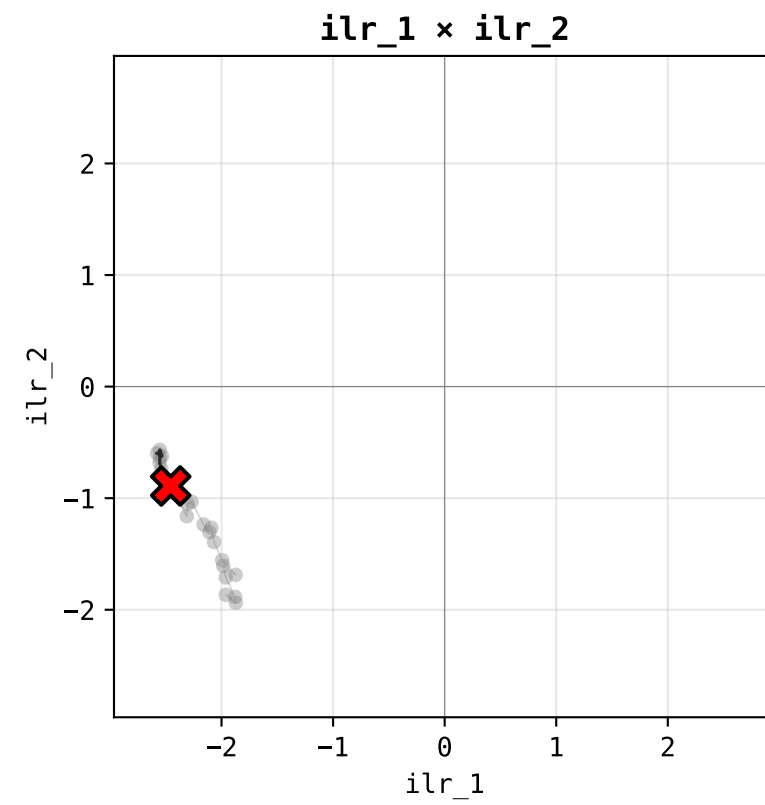
ILR Triplet – USA United States · year 2007 (7/25)



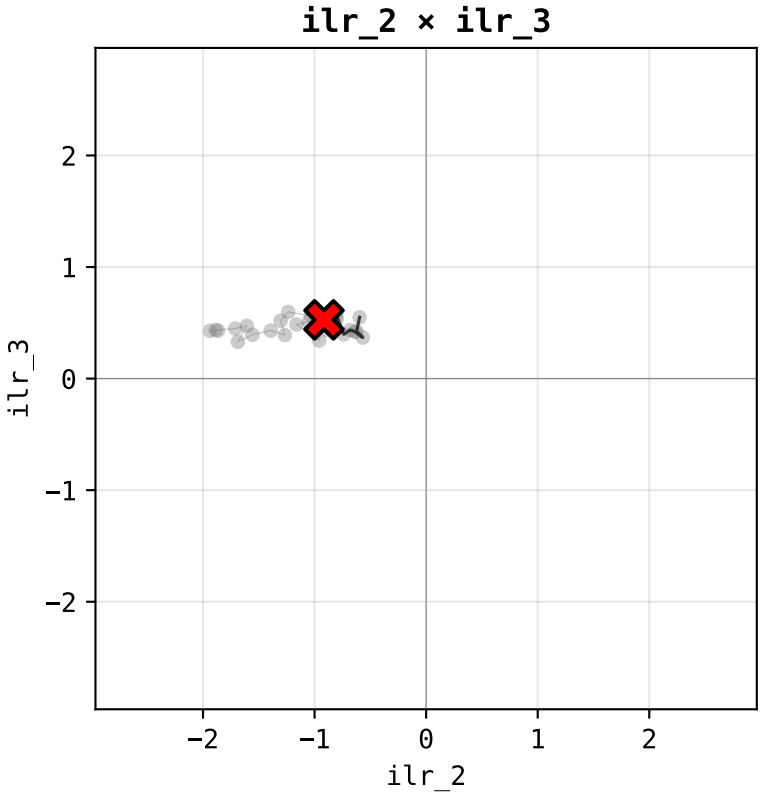
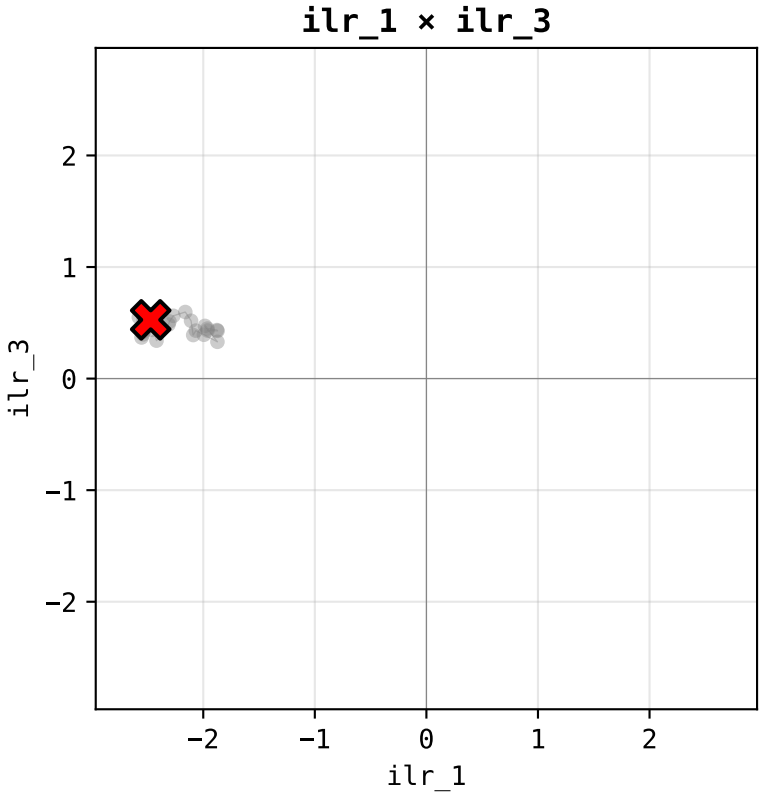
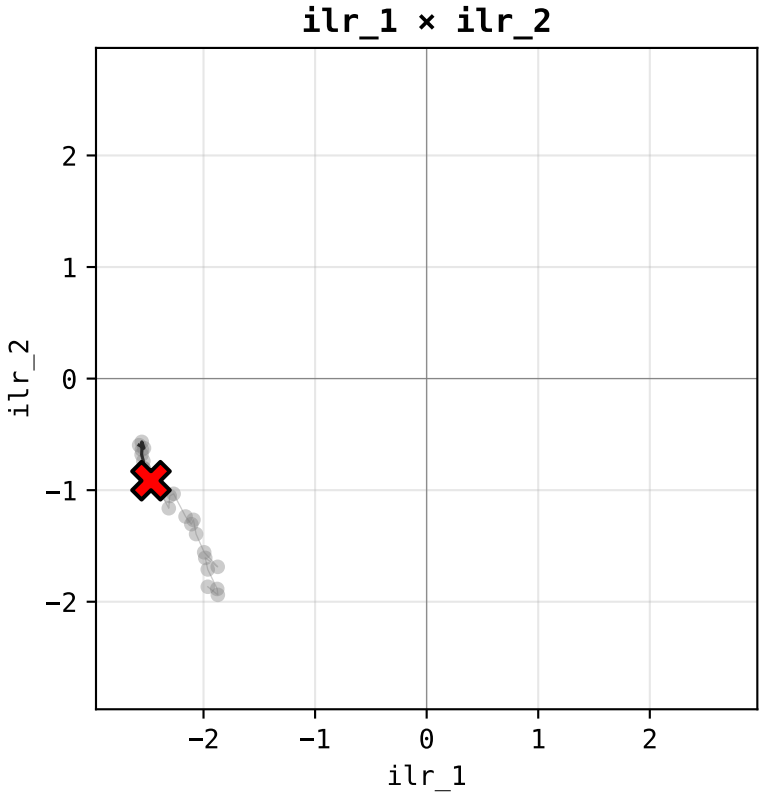
X = current year (2007) · black = past trajectory · gray = future · ilr range ±2.96

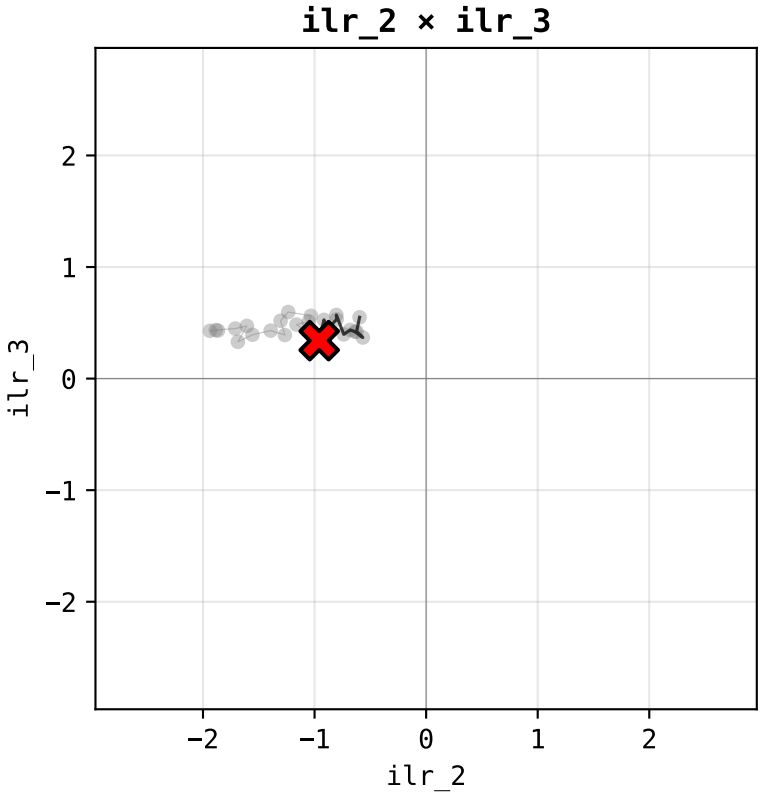
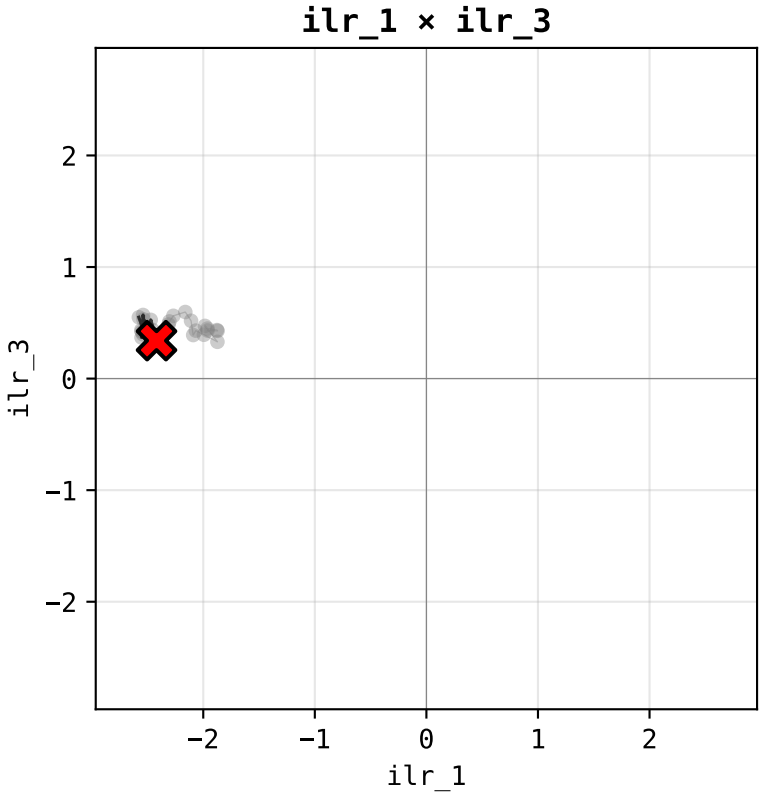
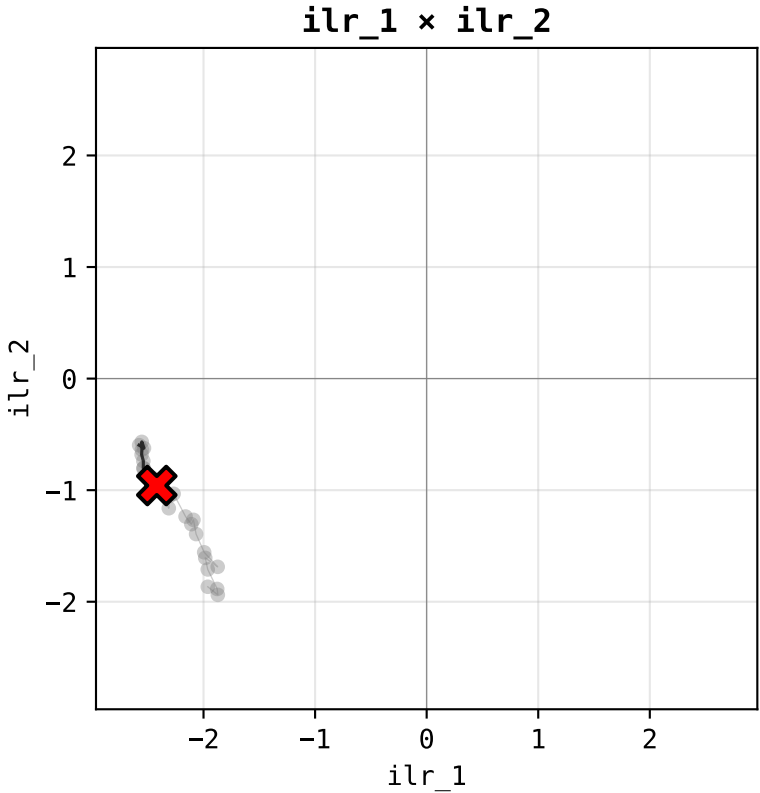


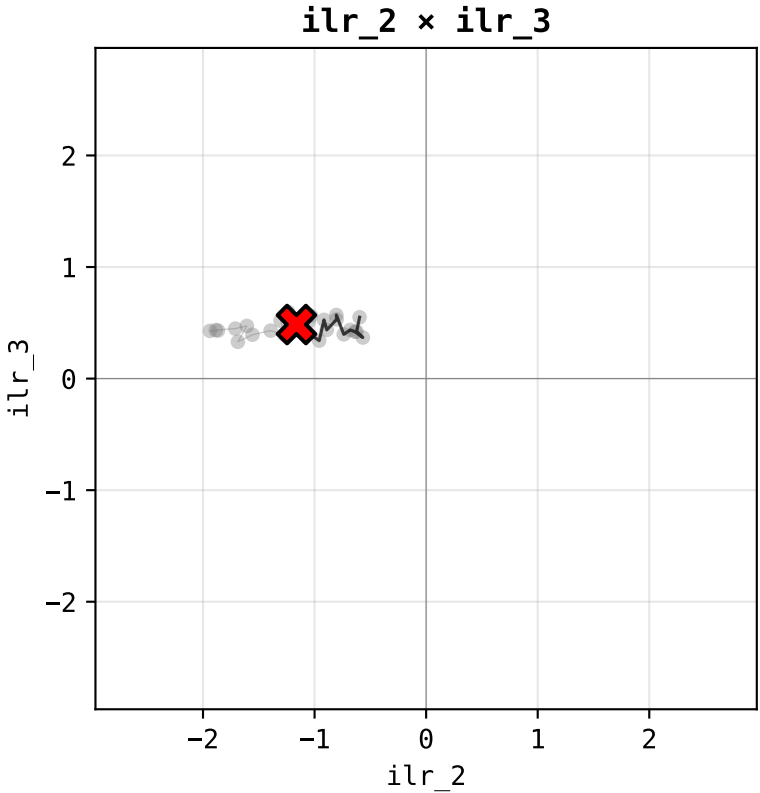
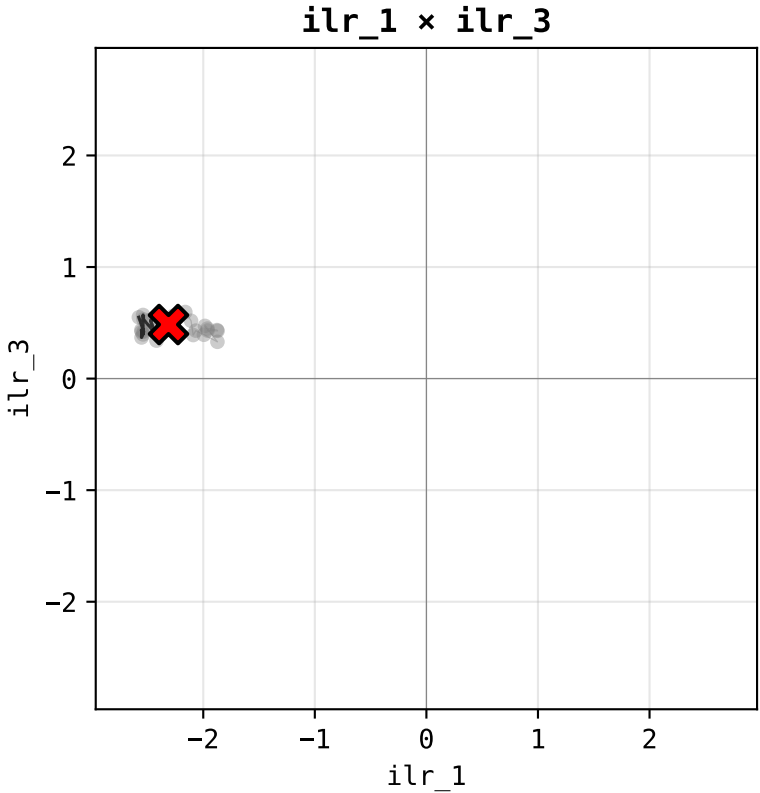
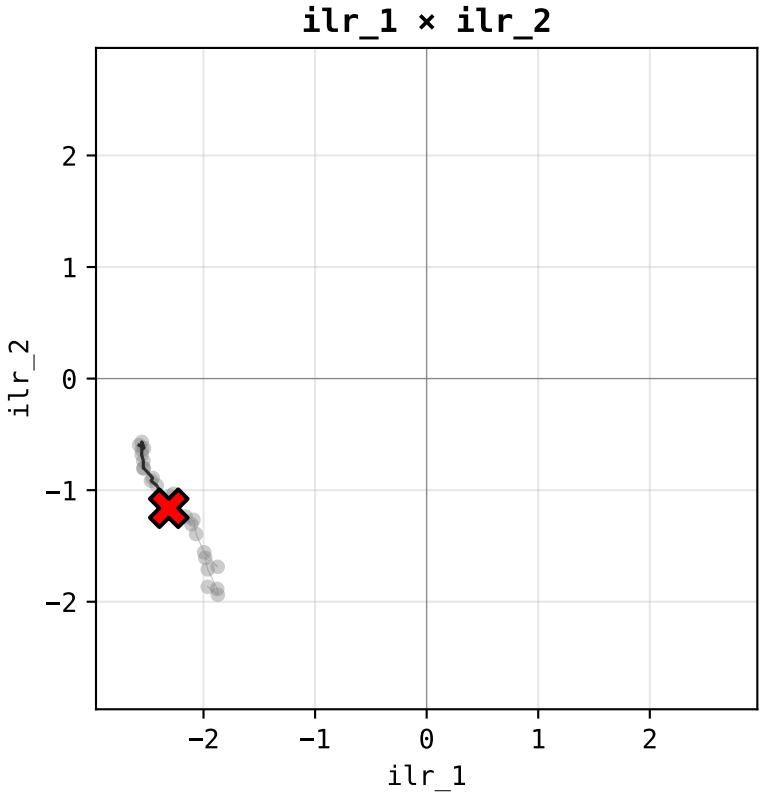
ILR Triplet – USA United States · year 2009 (9/25)

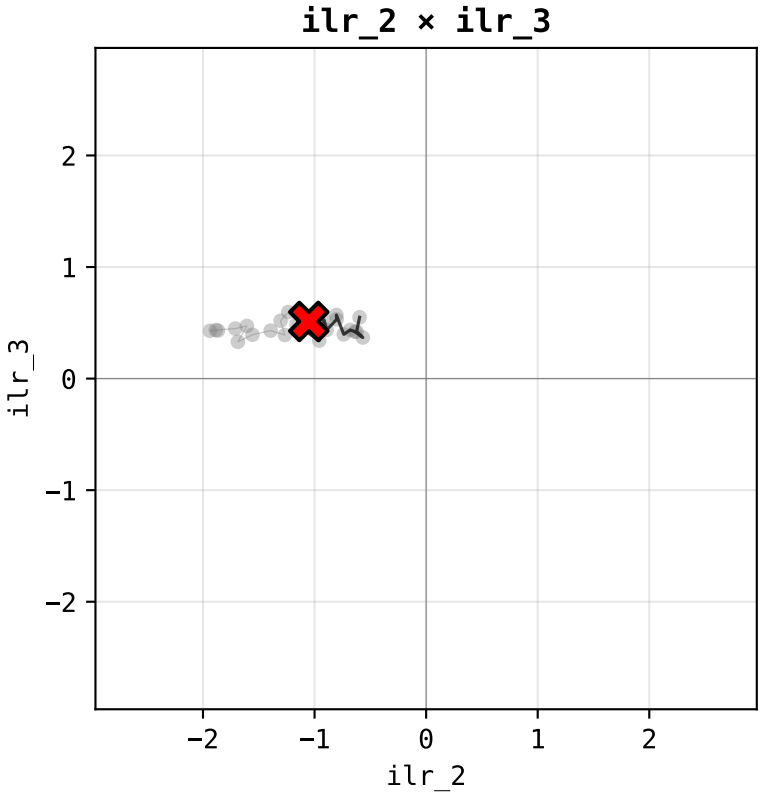
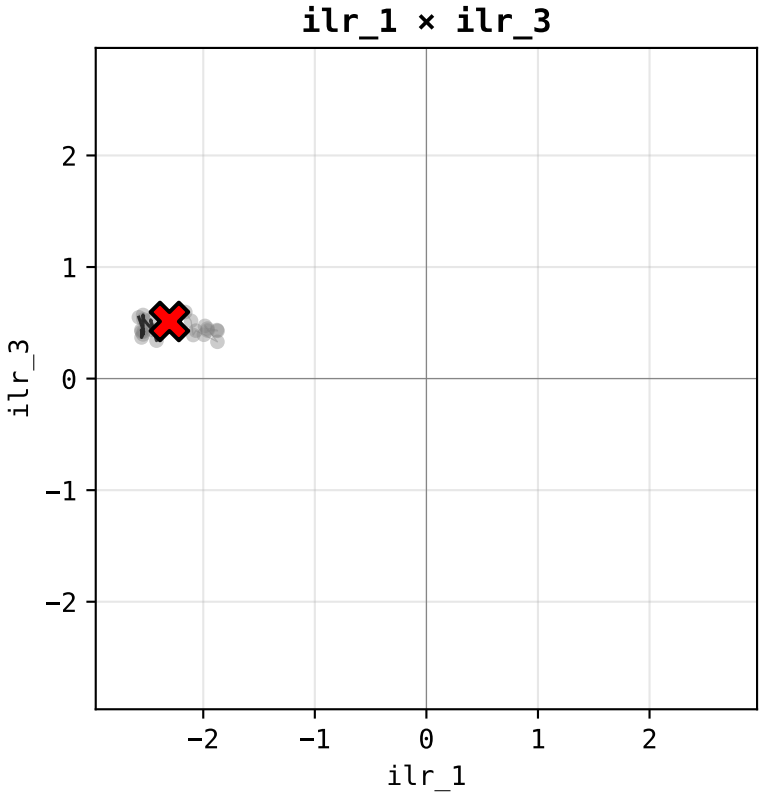
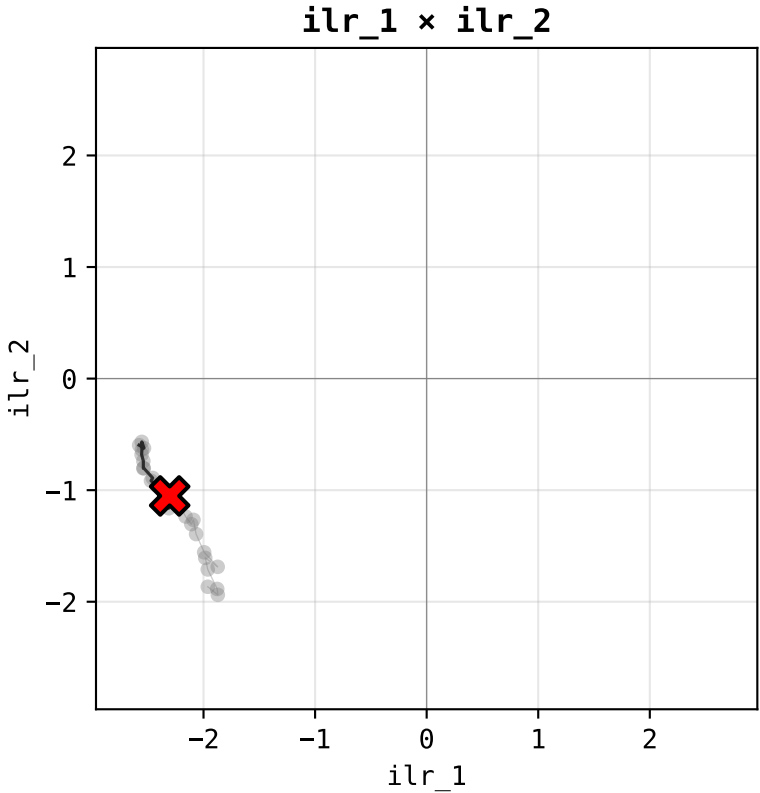


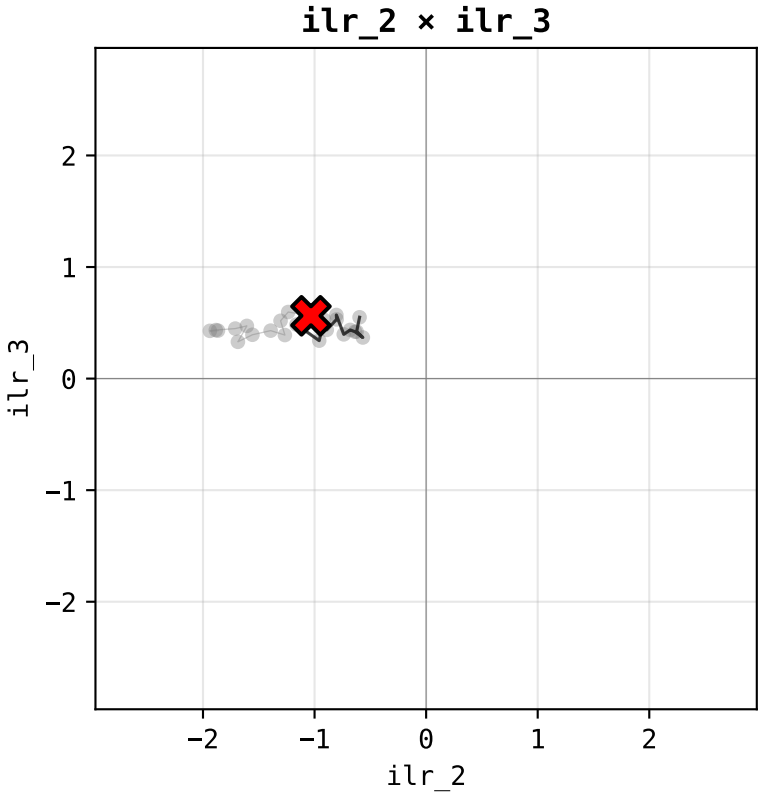
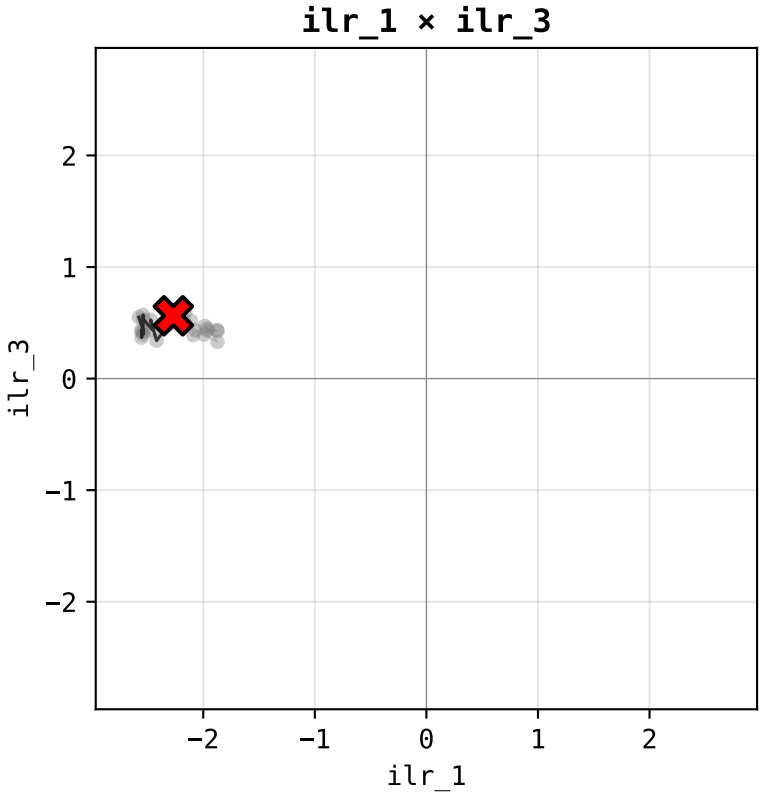
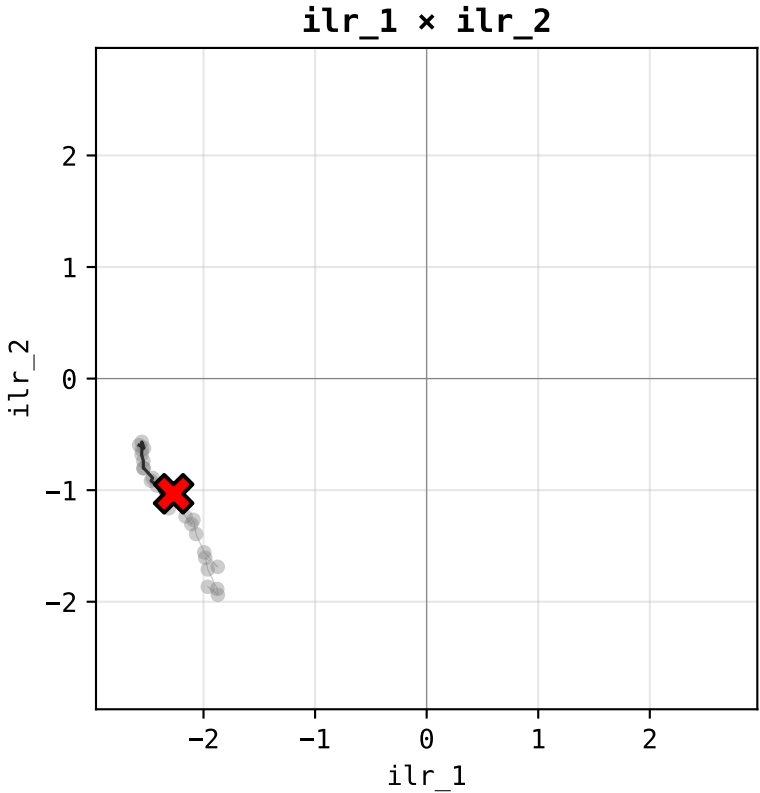
X = current year (2009) · black = past trajectory · gray = future · ilr range ±2.96

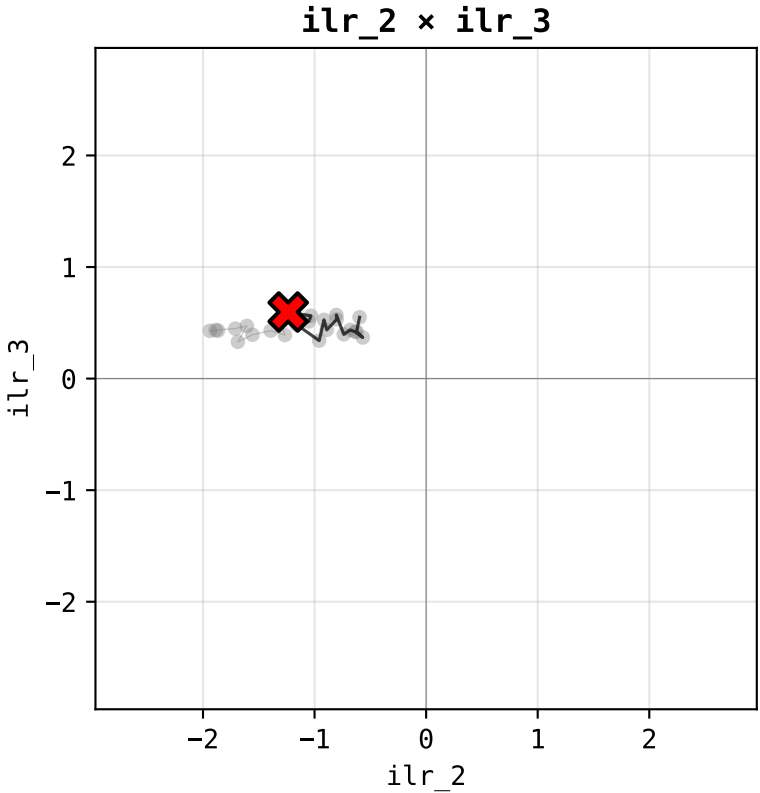
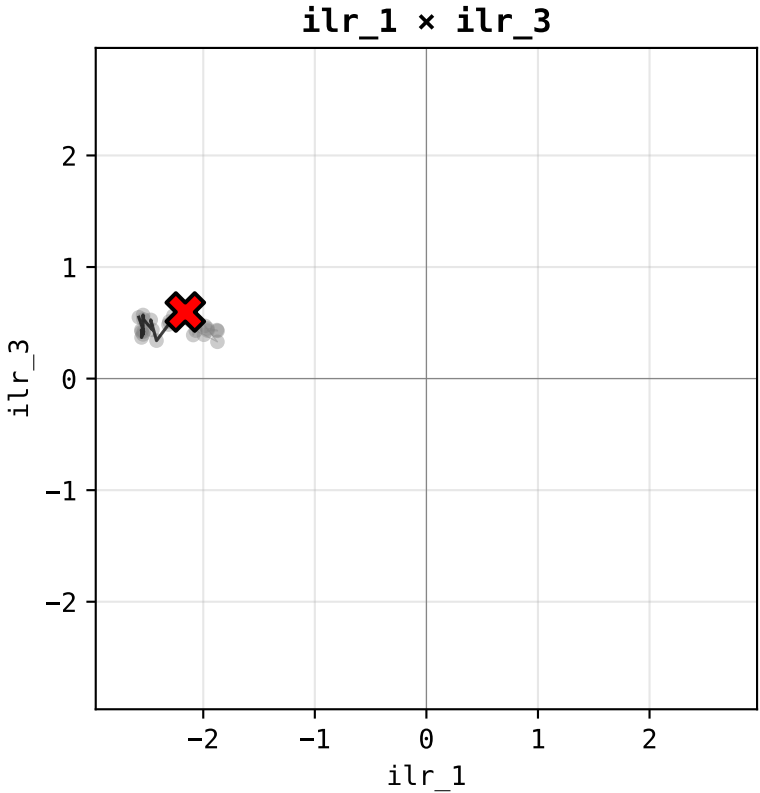
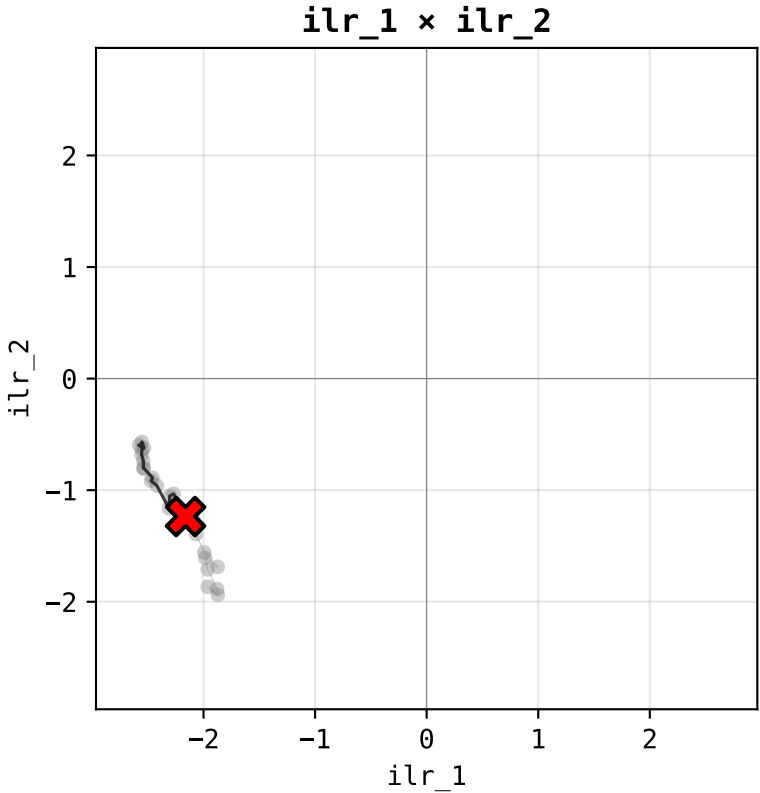


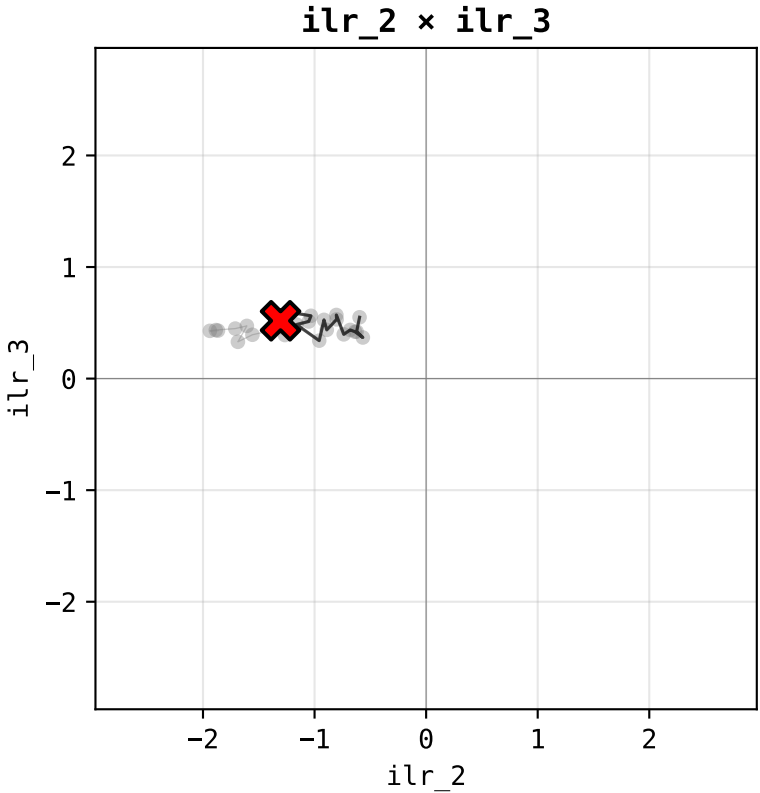
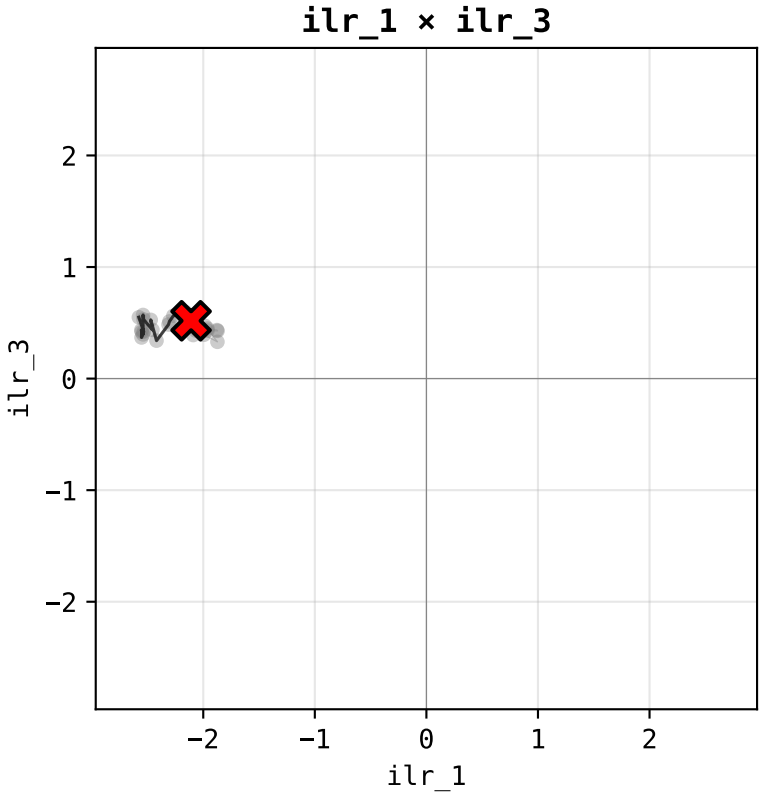
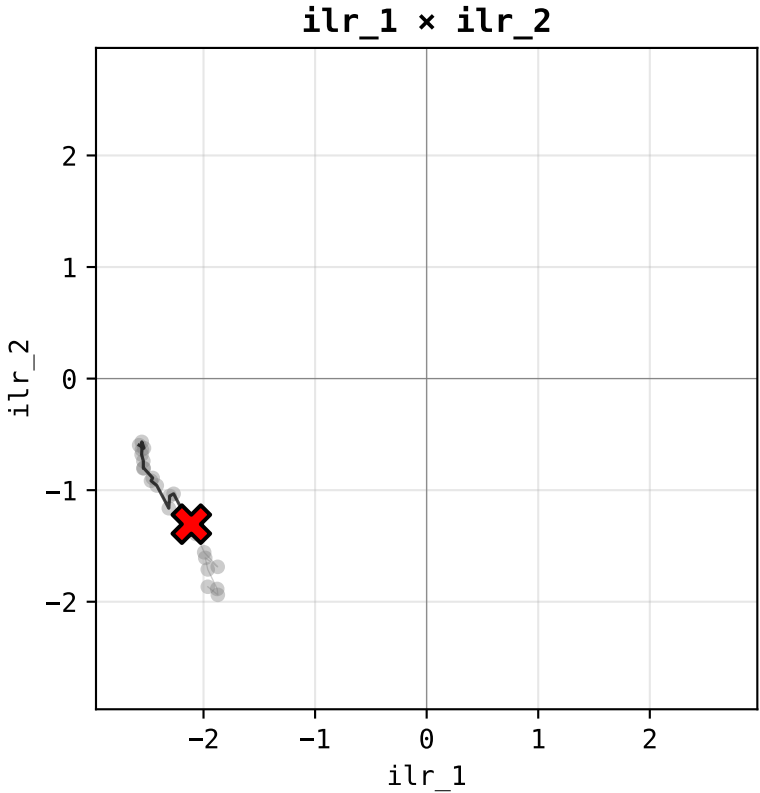


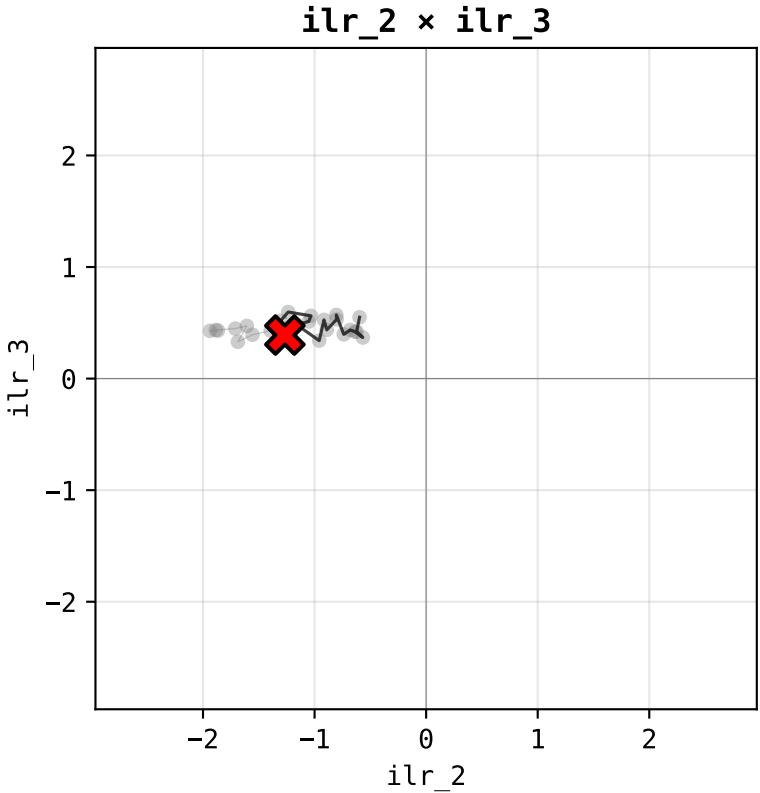
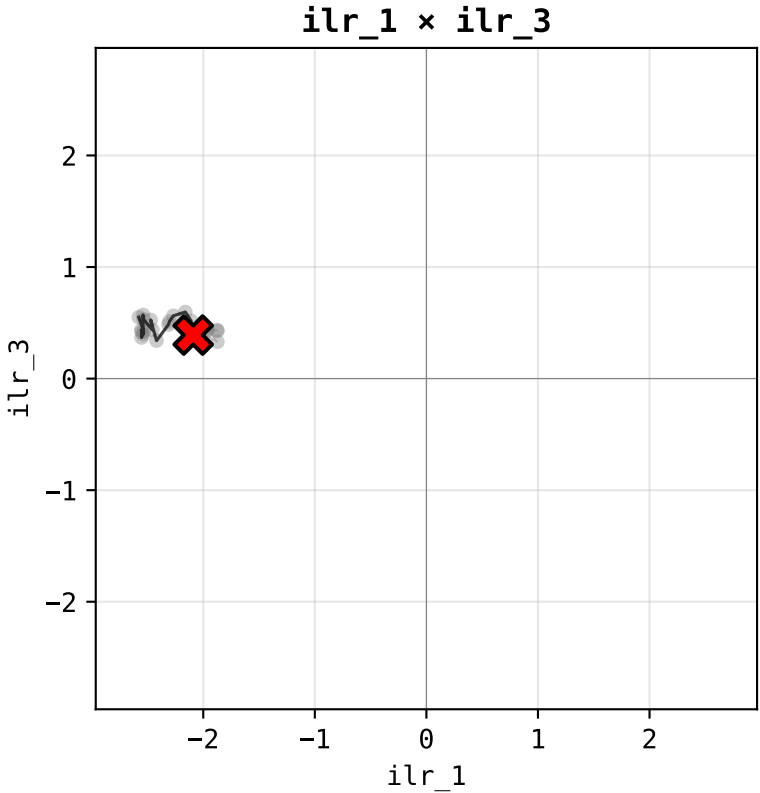
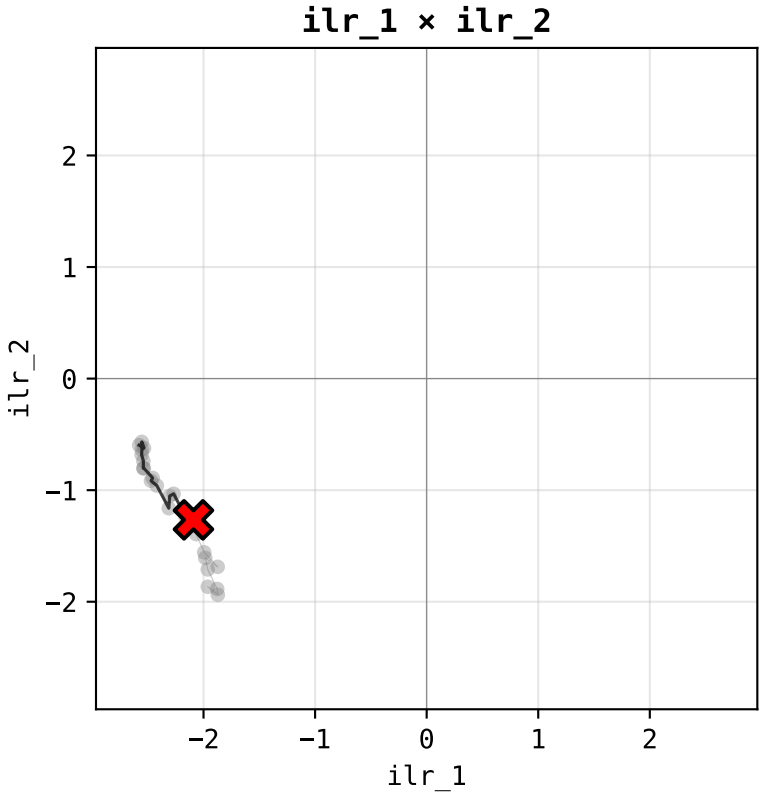


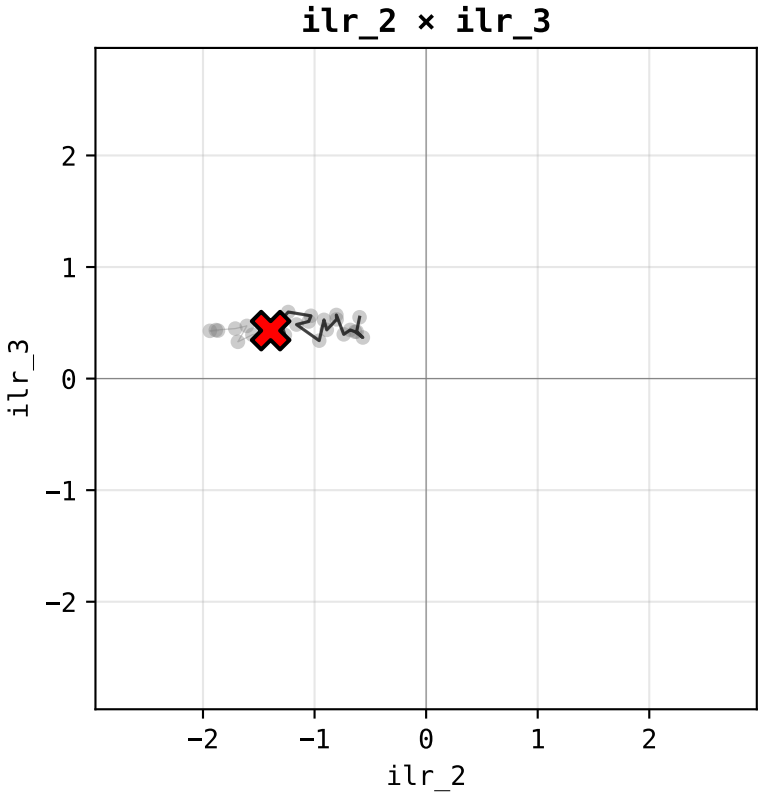
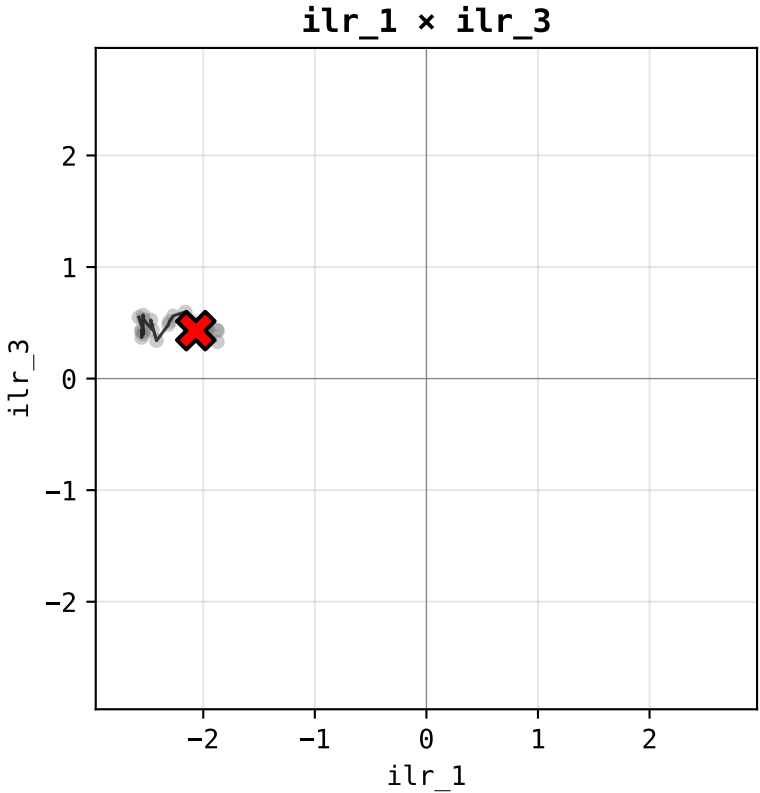
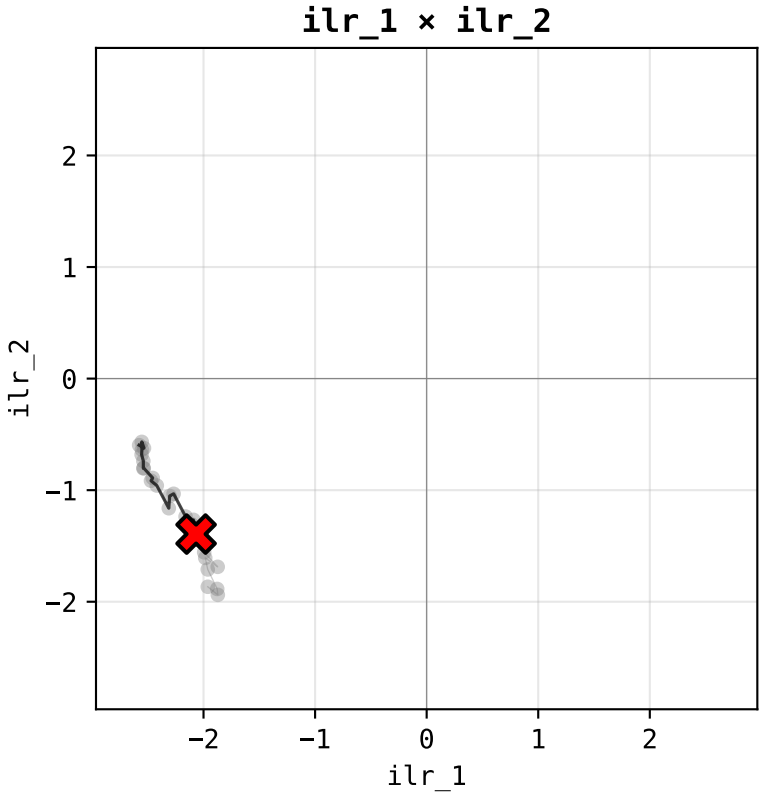


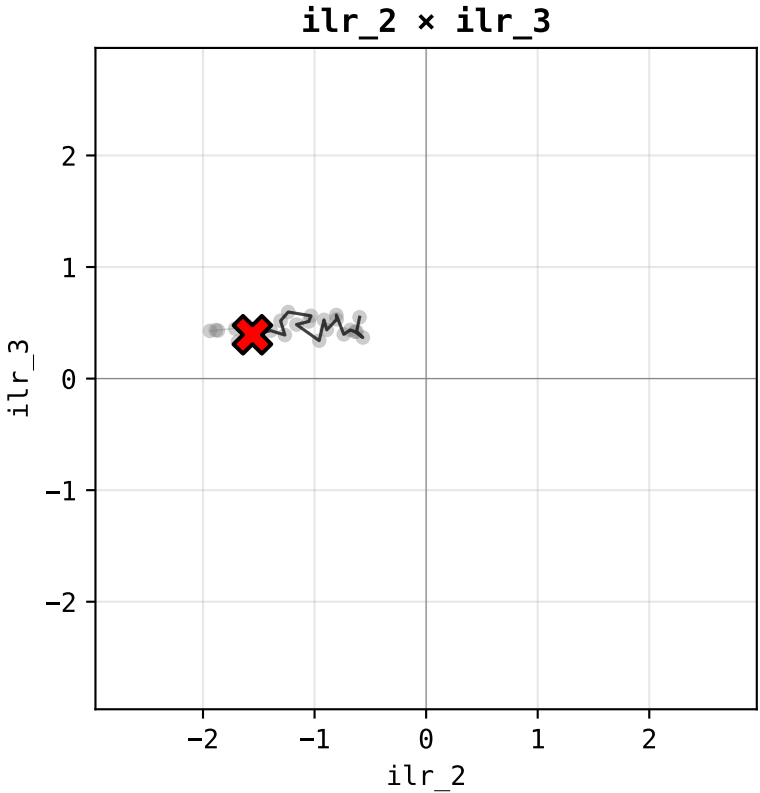
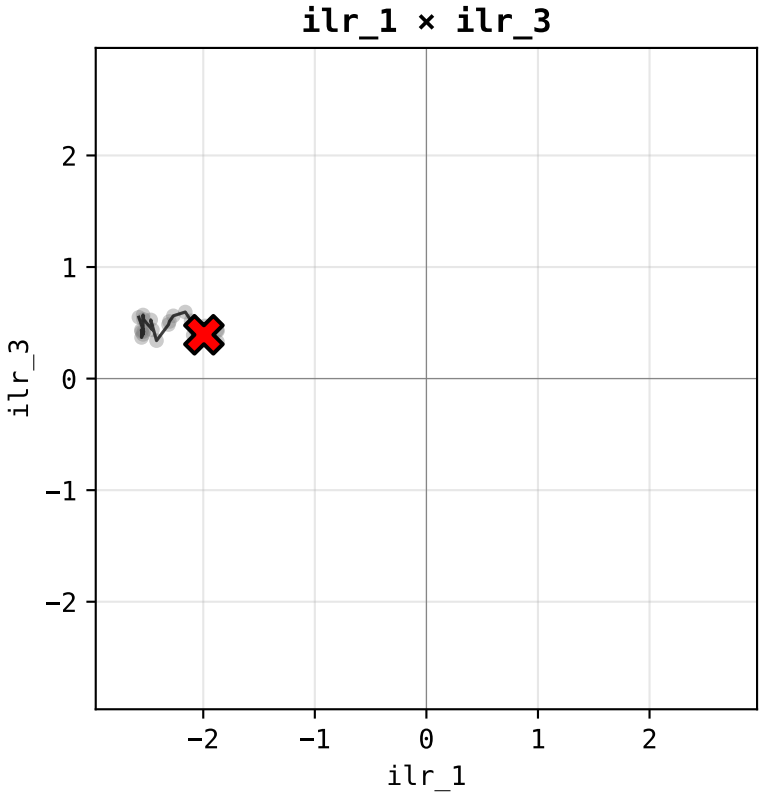
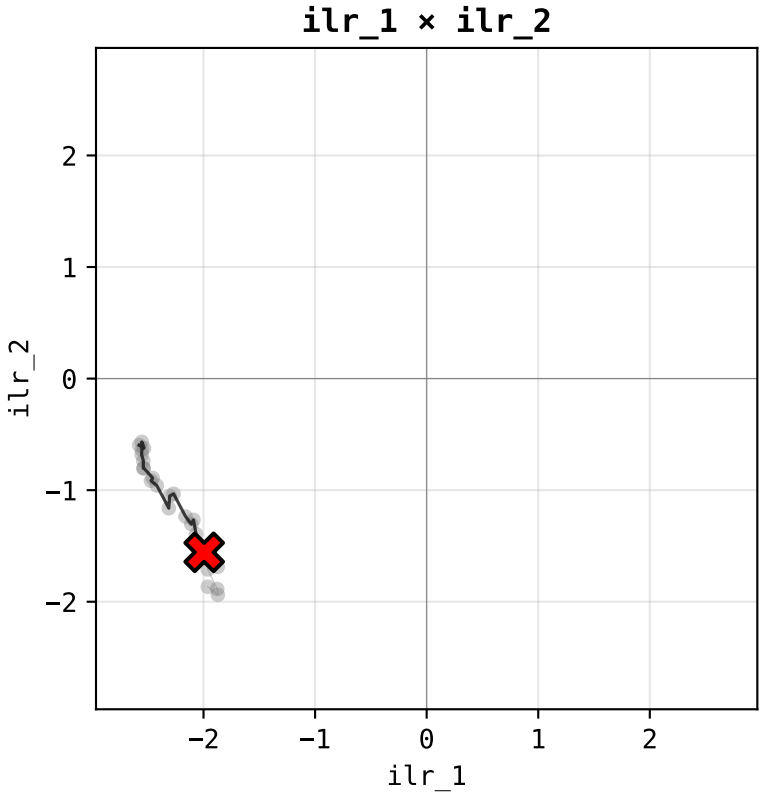


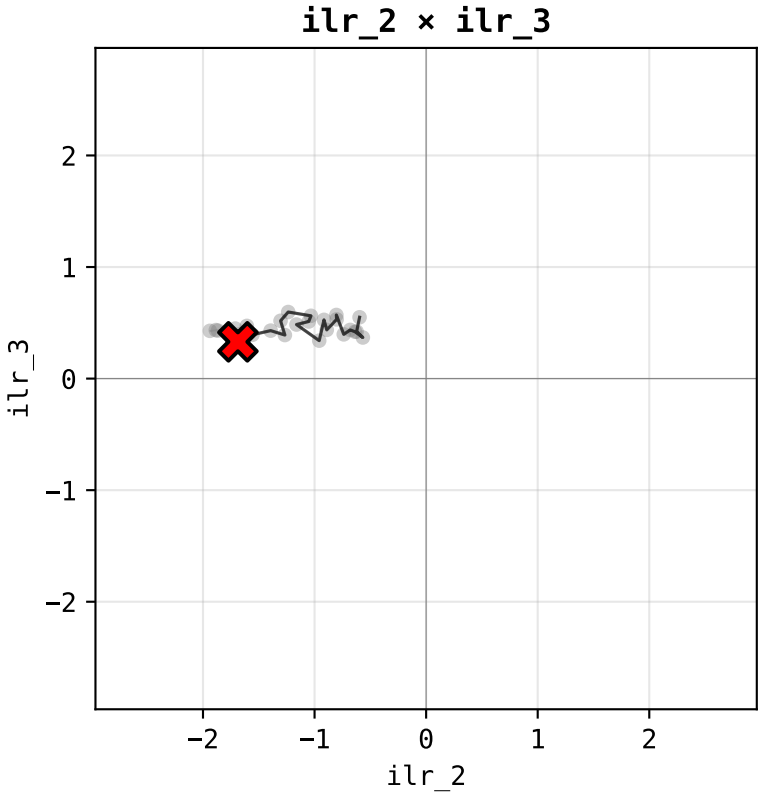
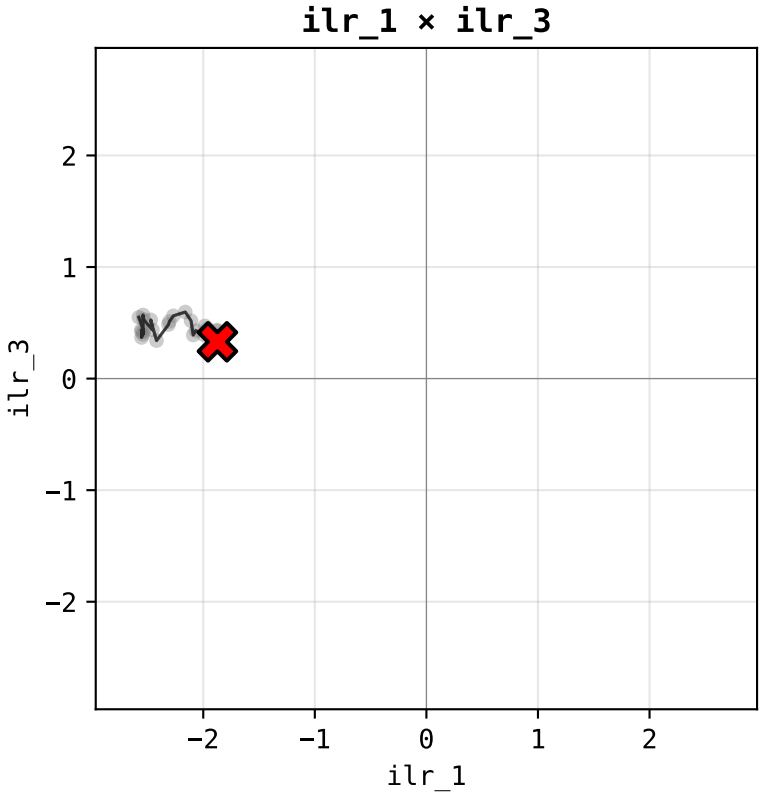
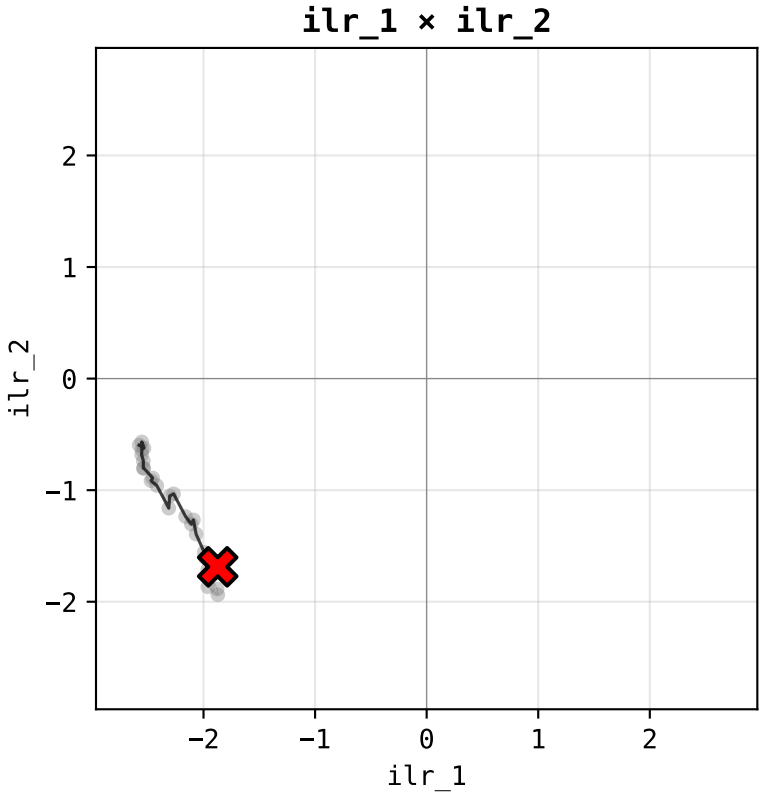


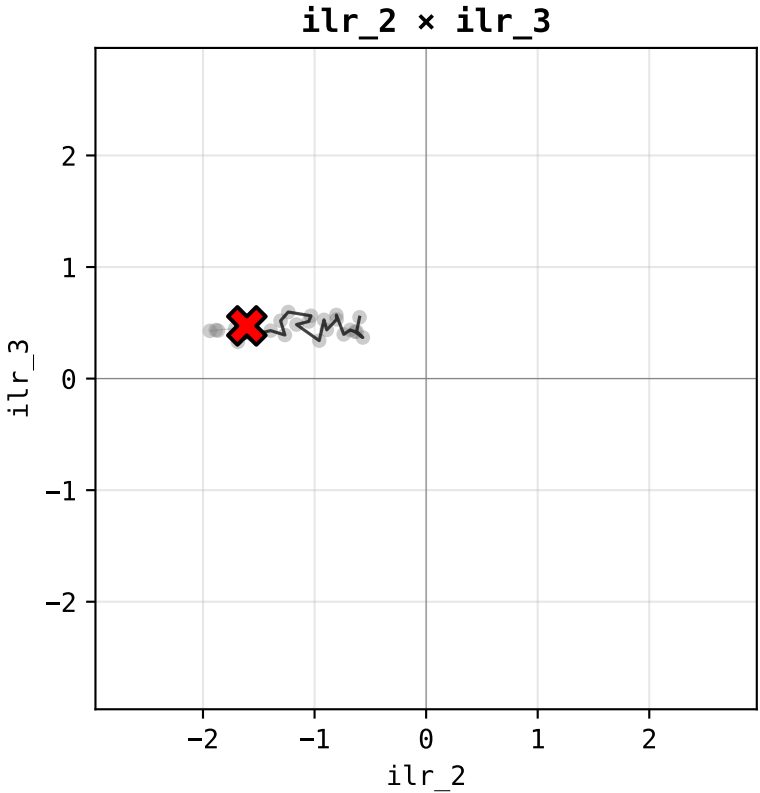
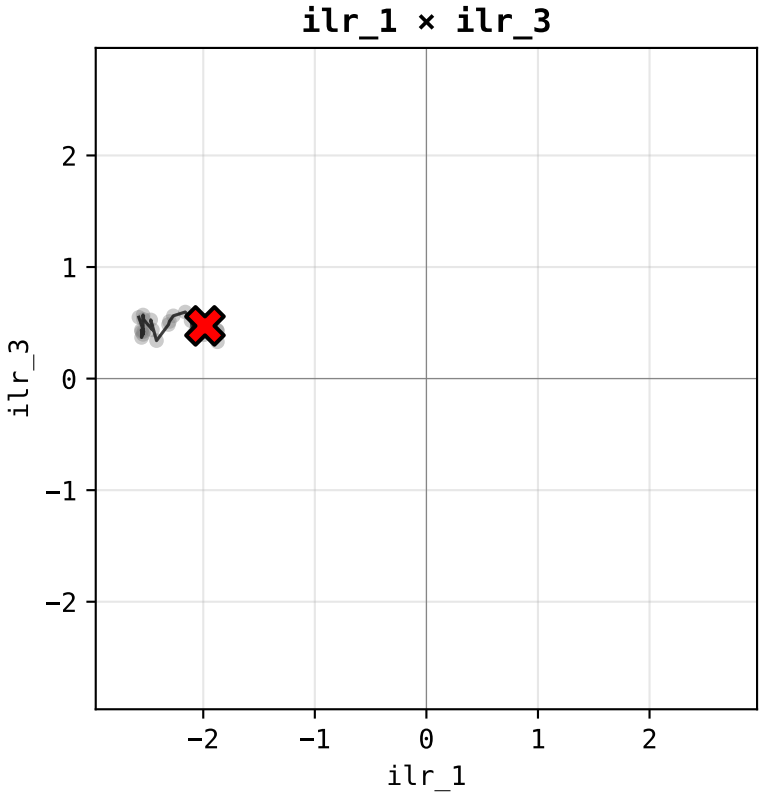
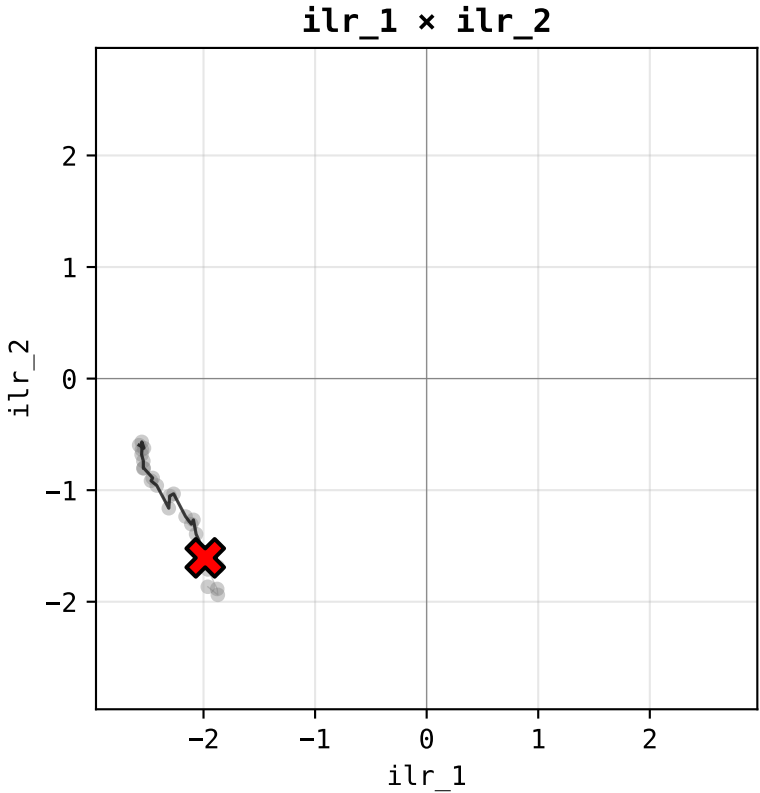


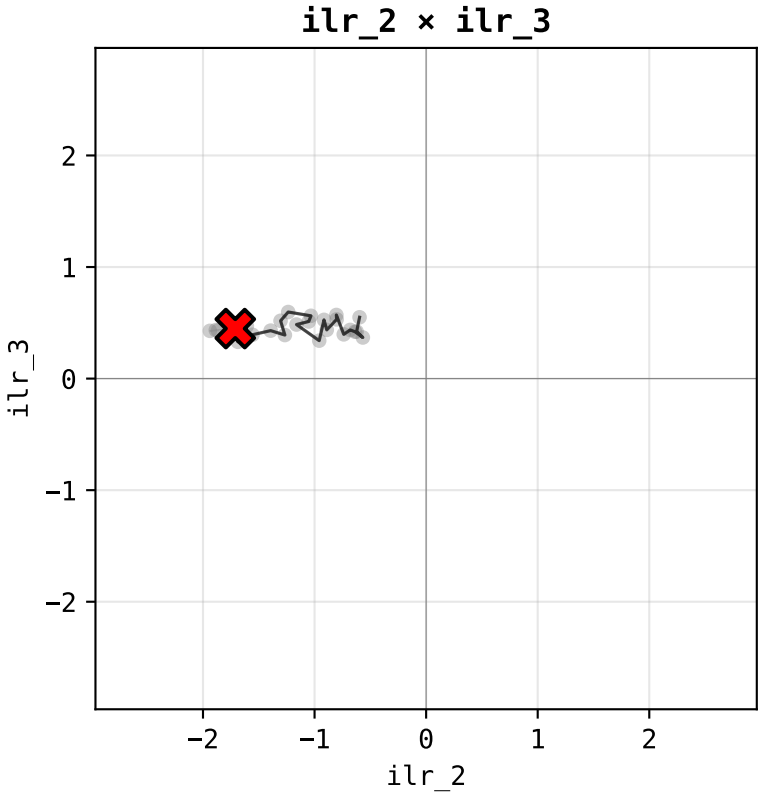
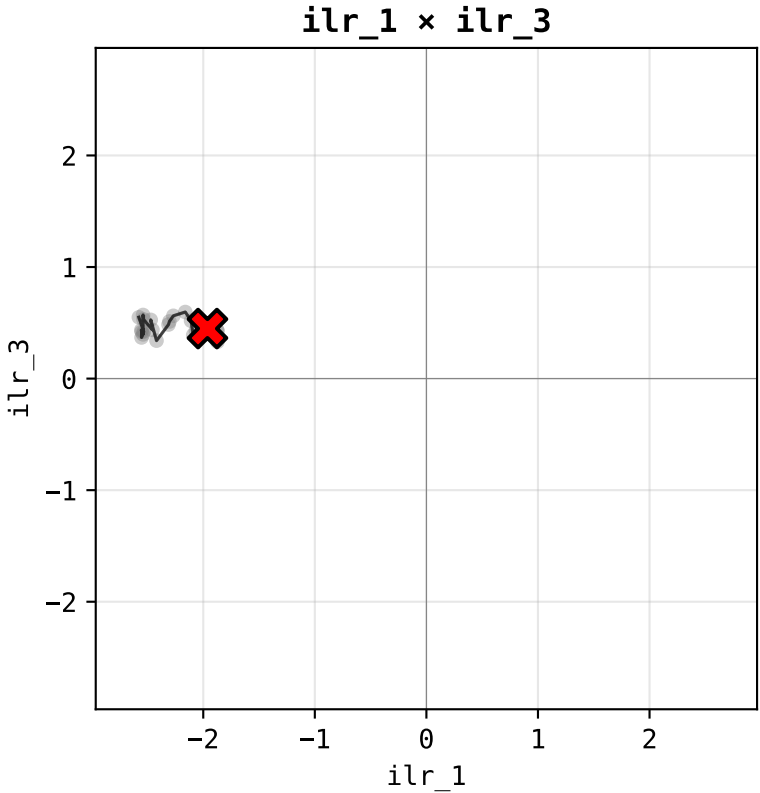
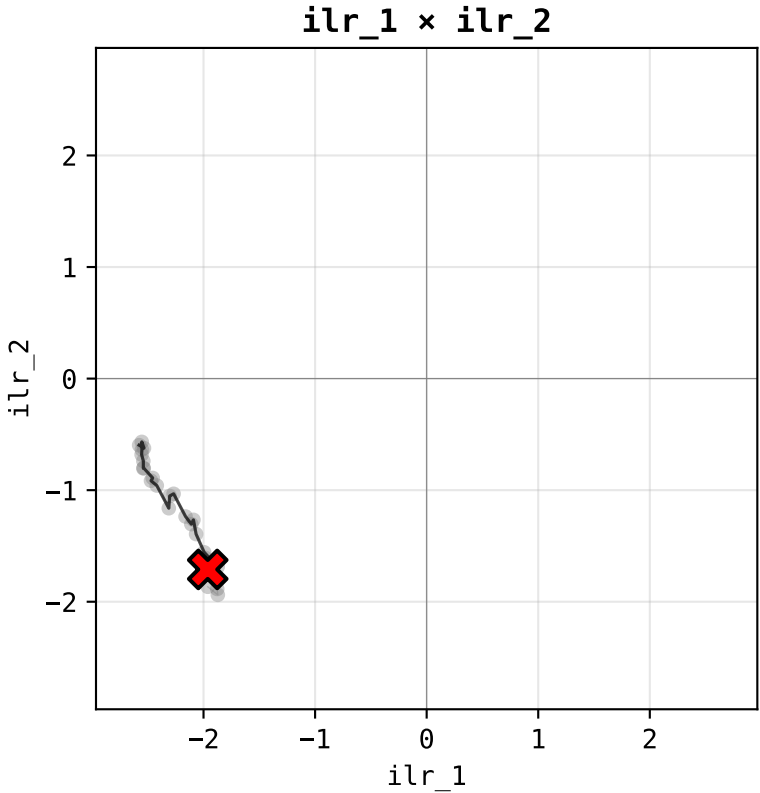


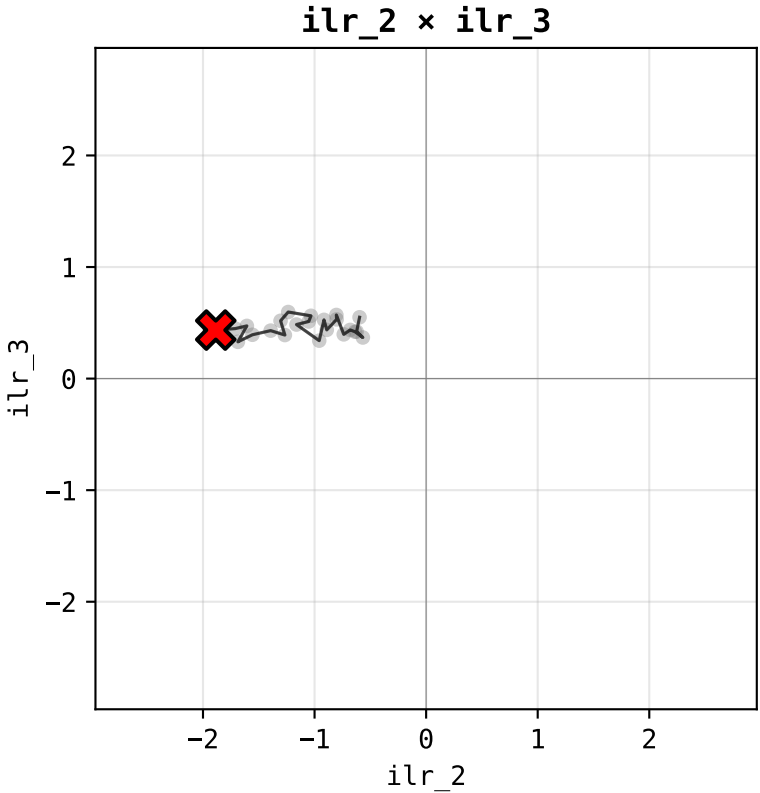
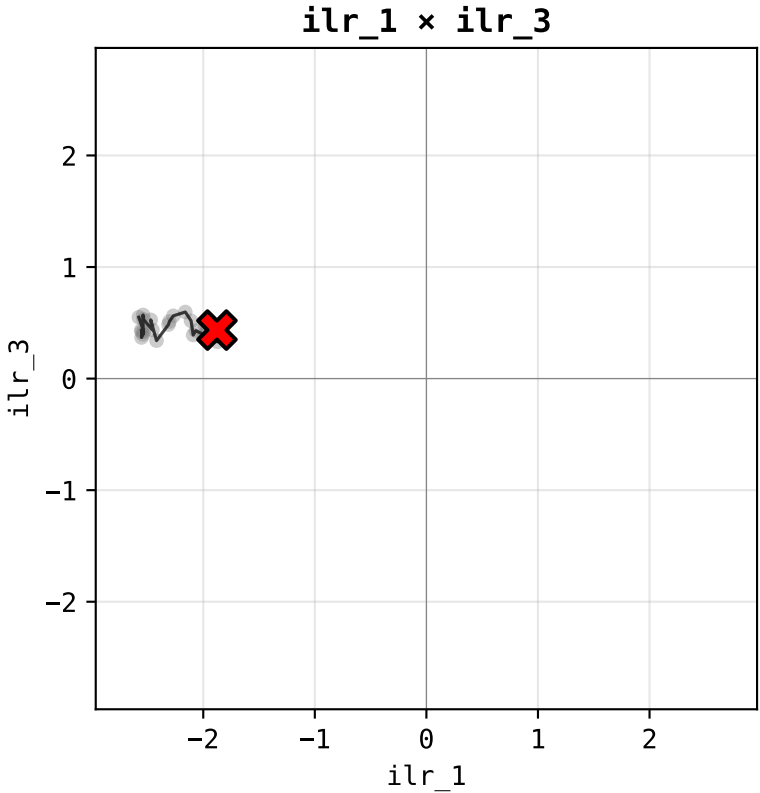
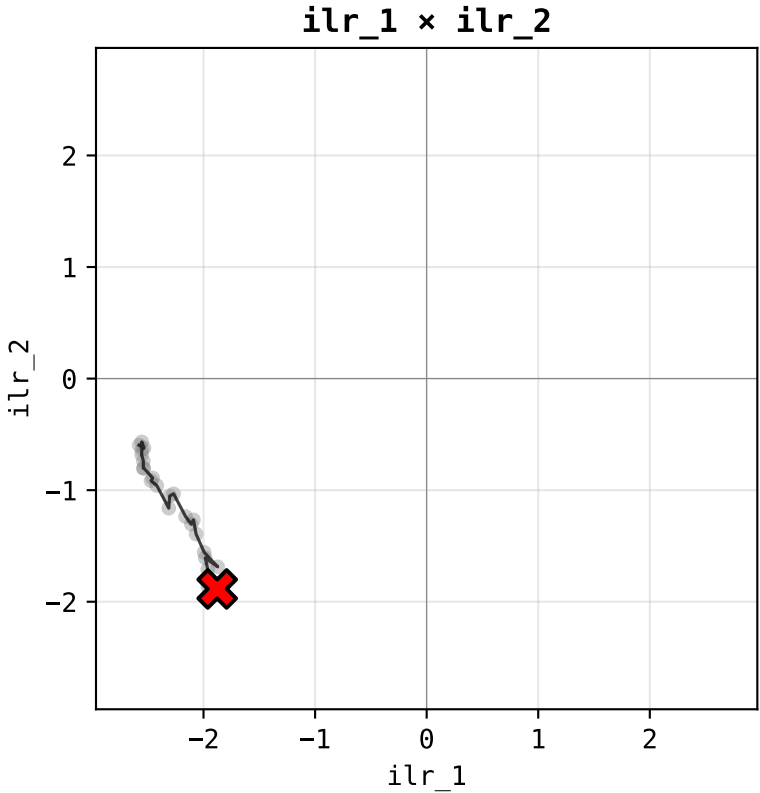


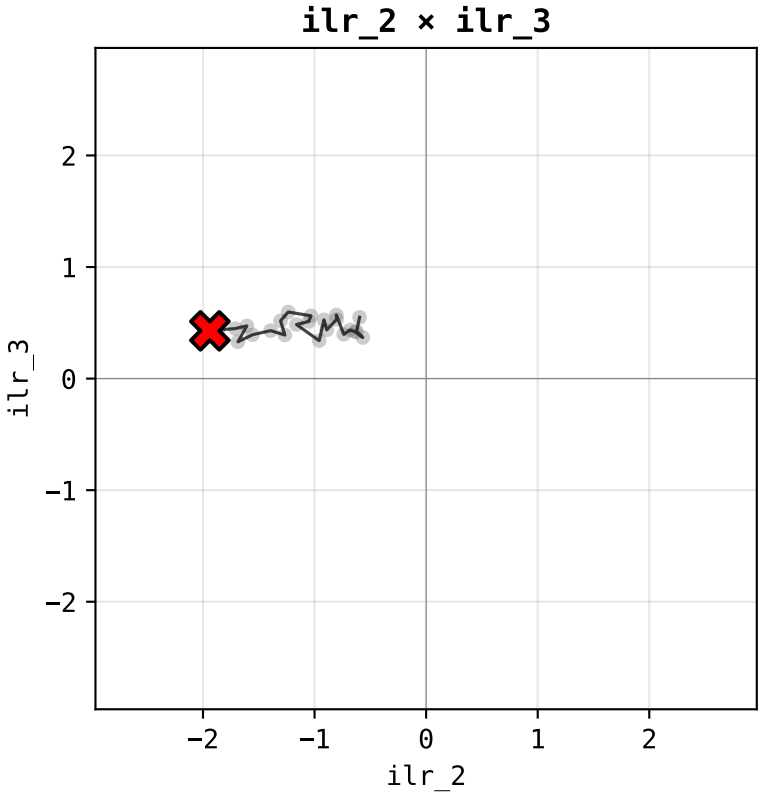
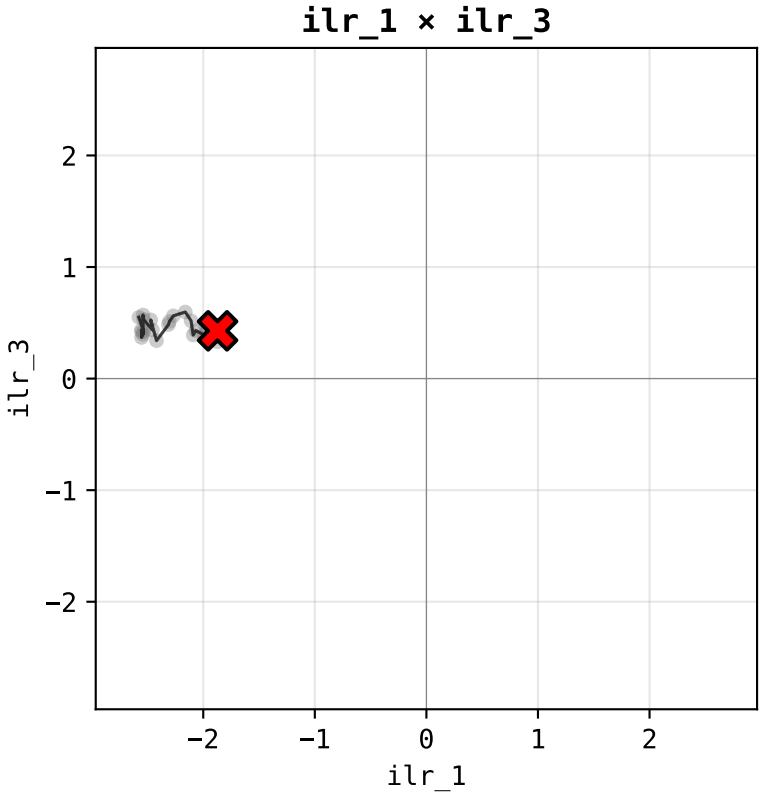
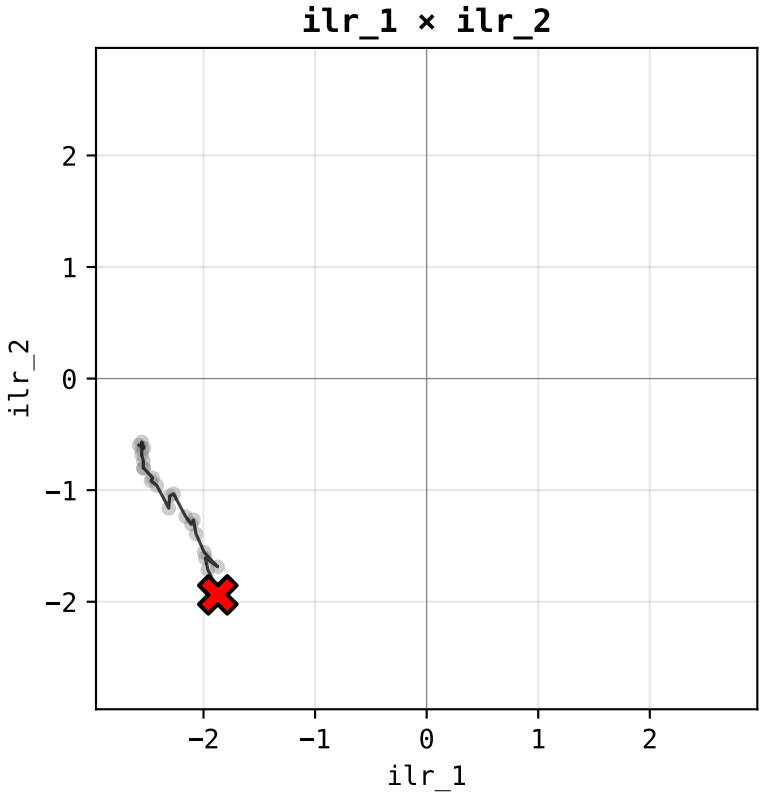


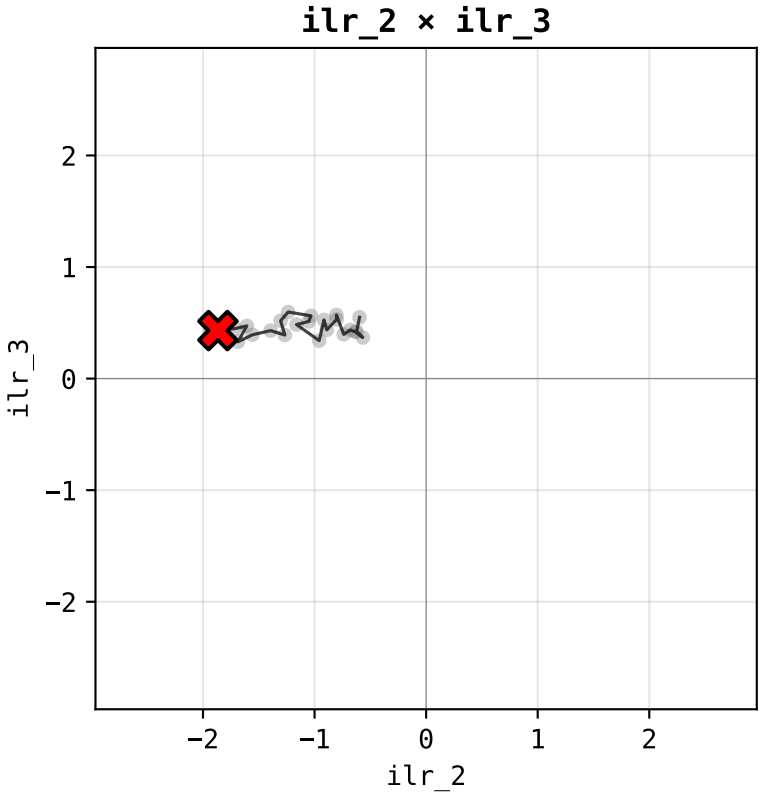
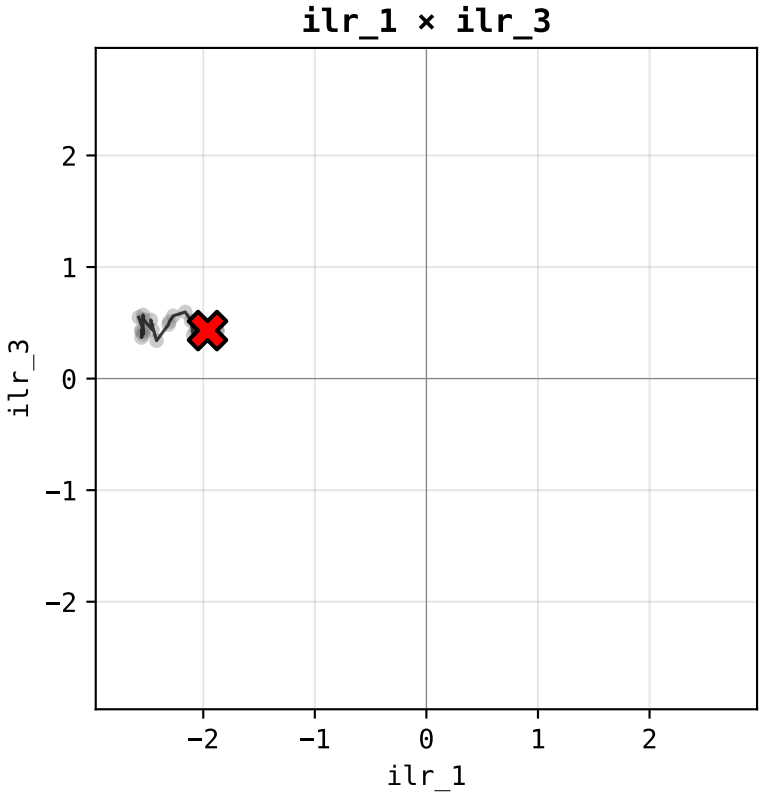
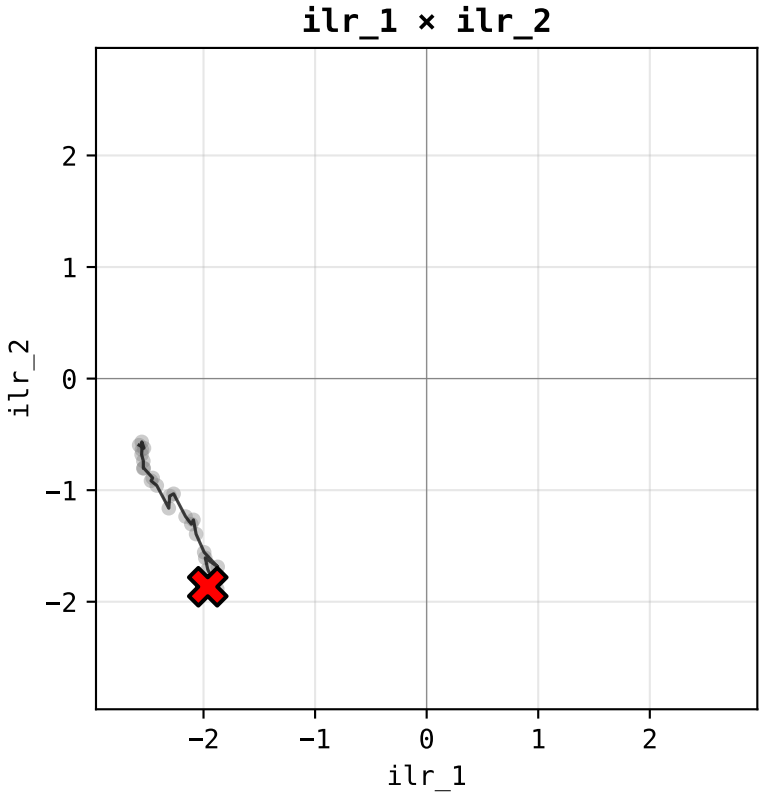












View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ WLD World

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

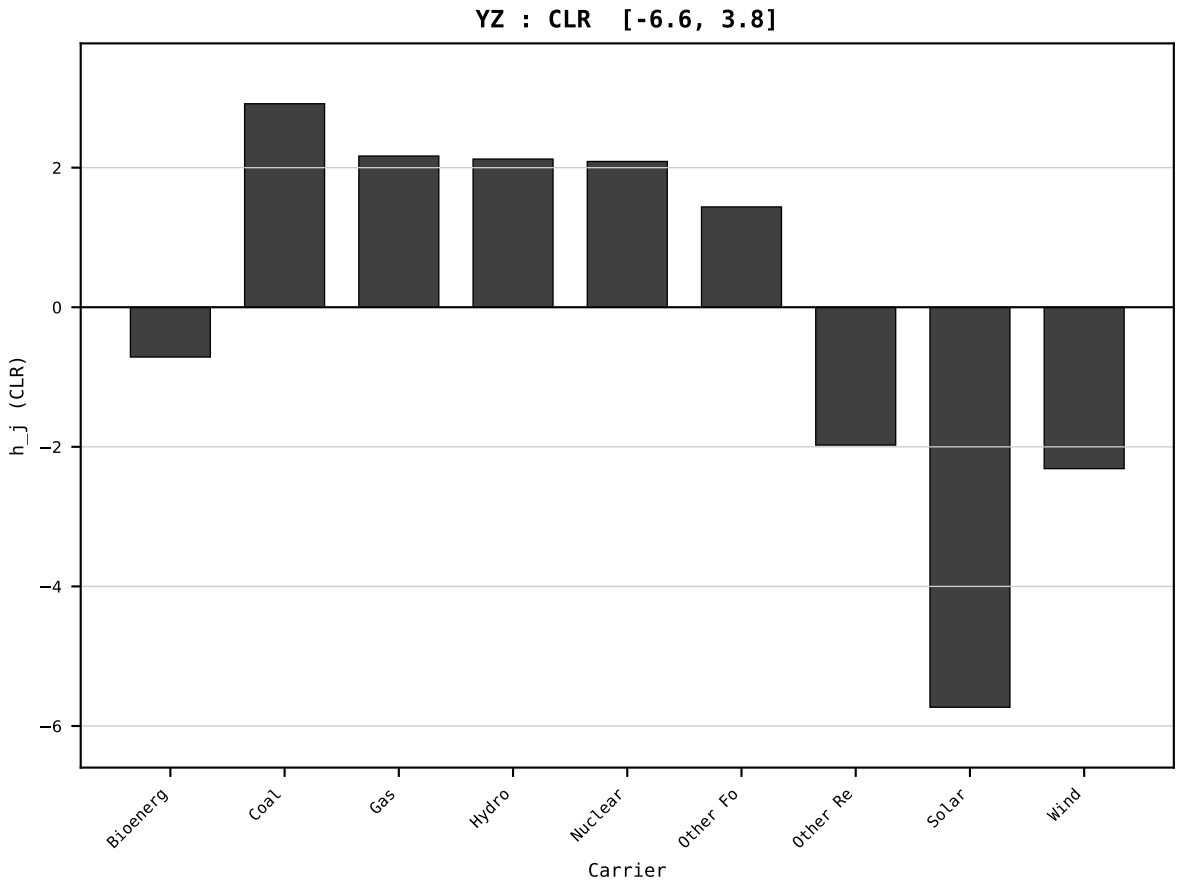
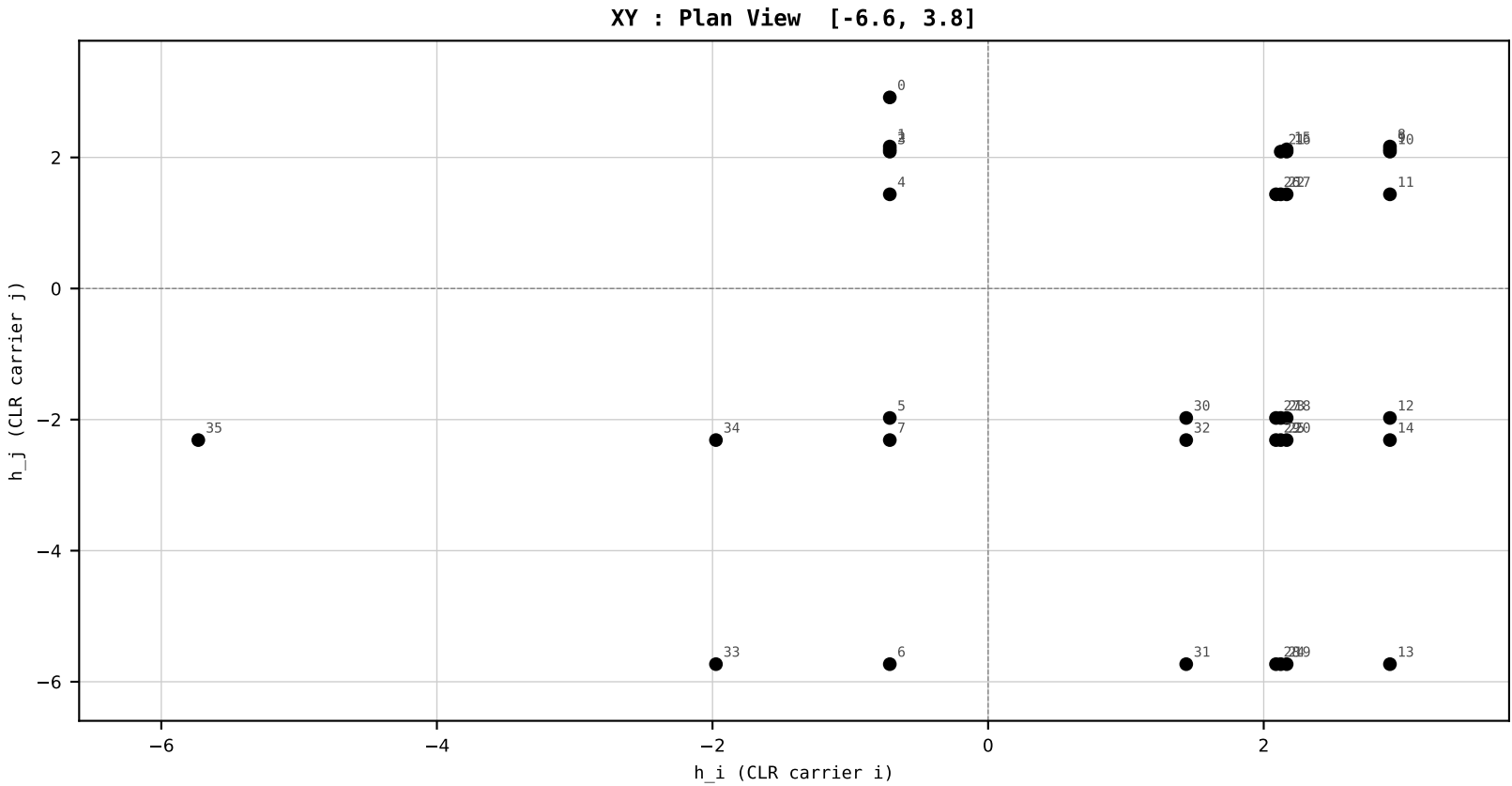
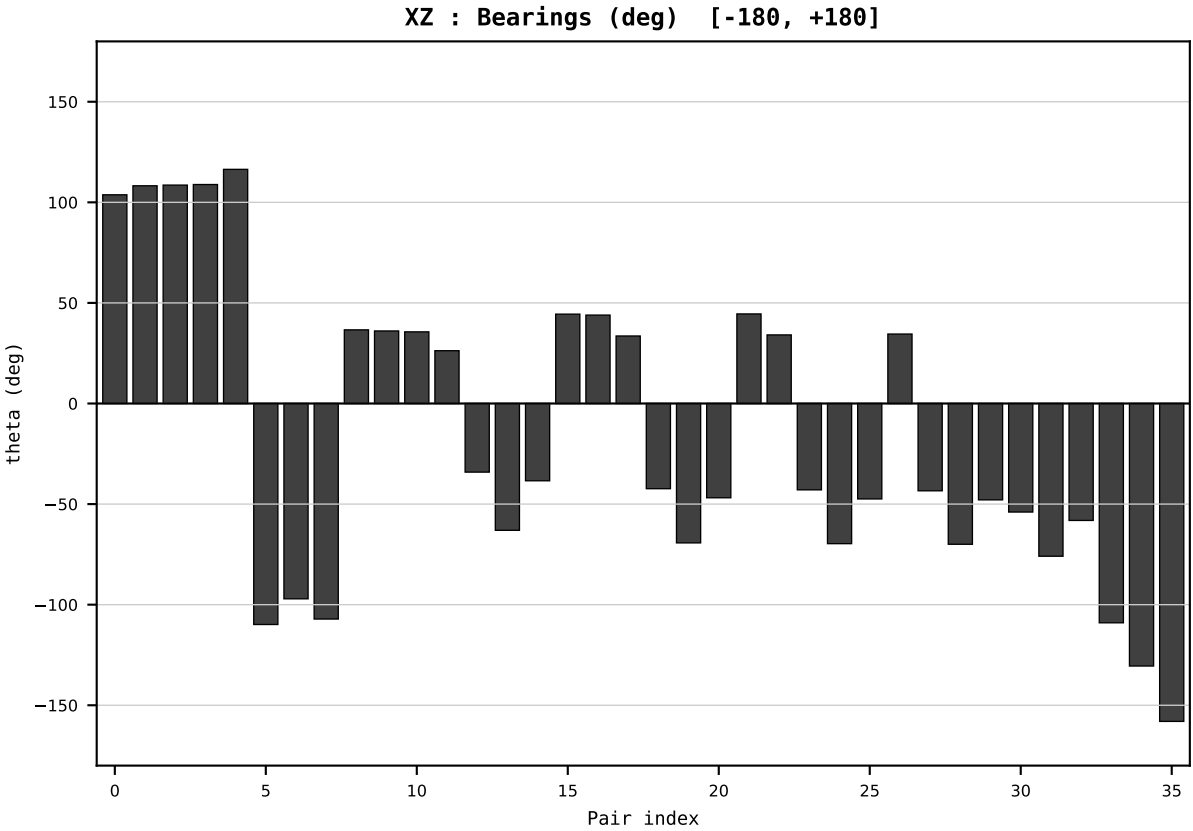
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287446
Ring = Hs-2
E_metric= 66.7408
kappa_HS= 5694.91
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 8.647 HLR
DR ratio= 5694.9x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

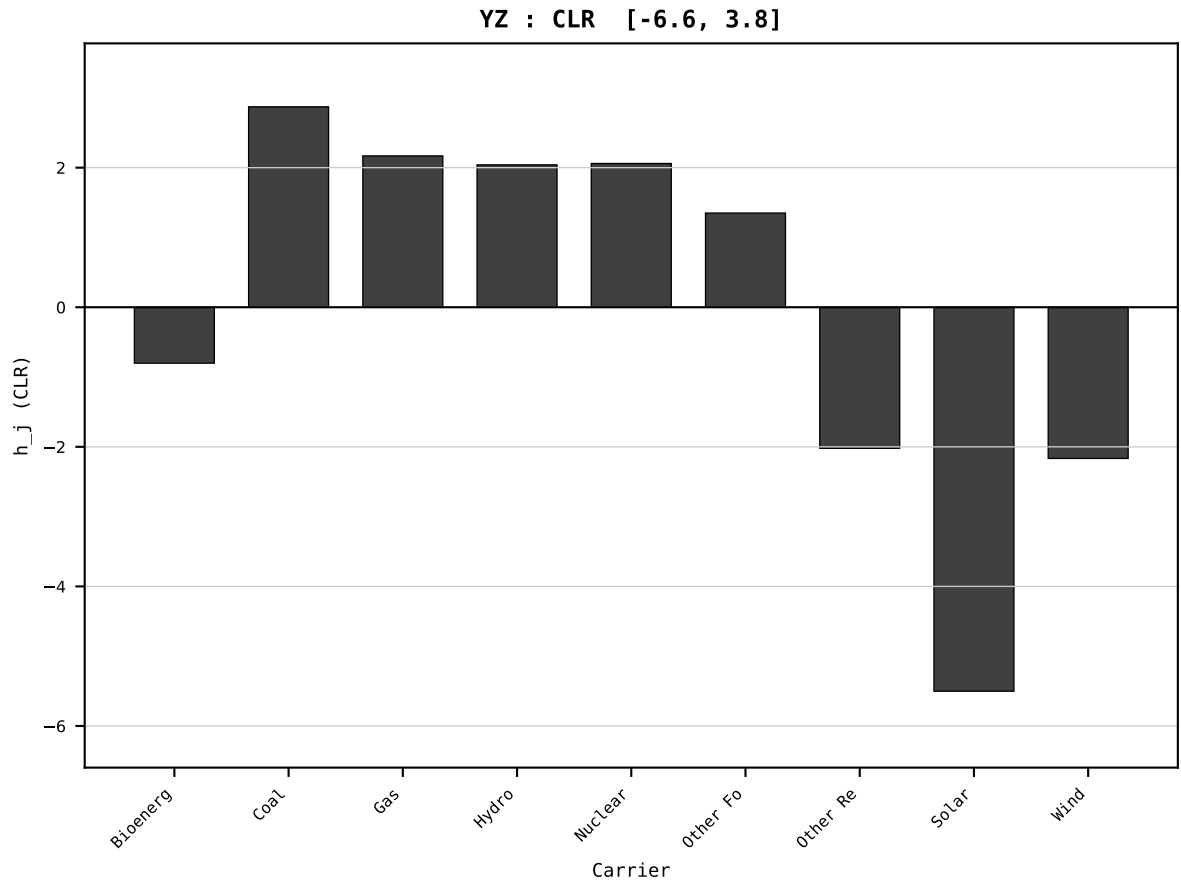
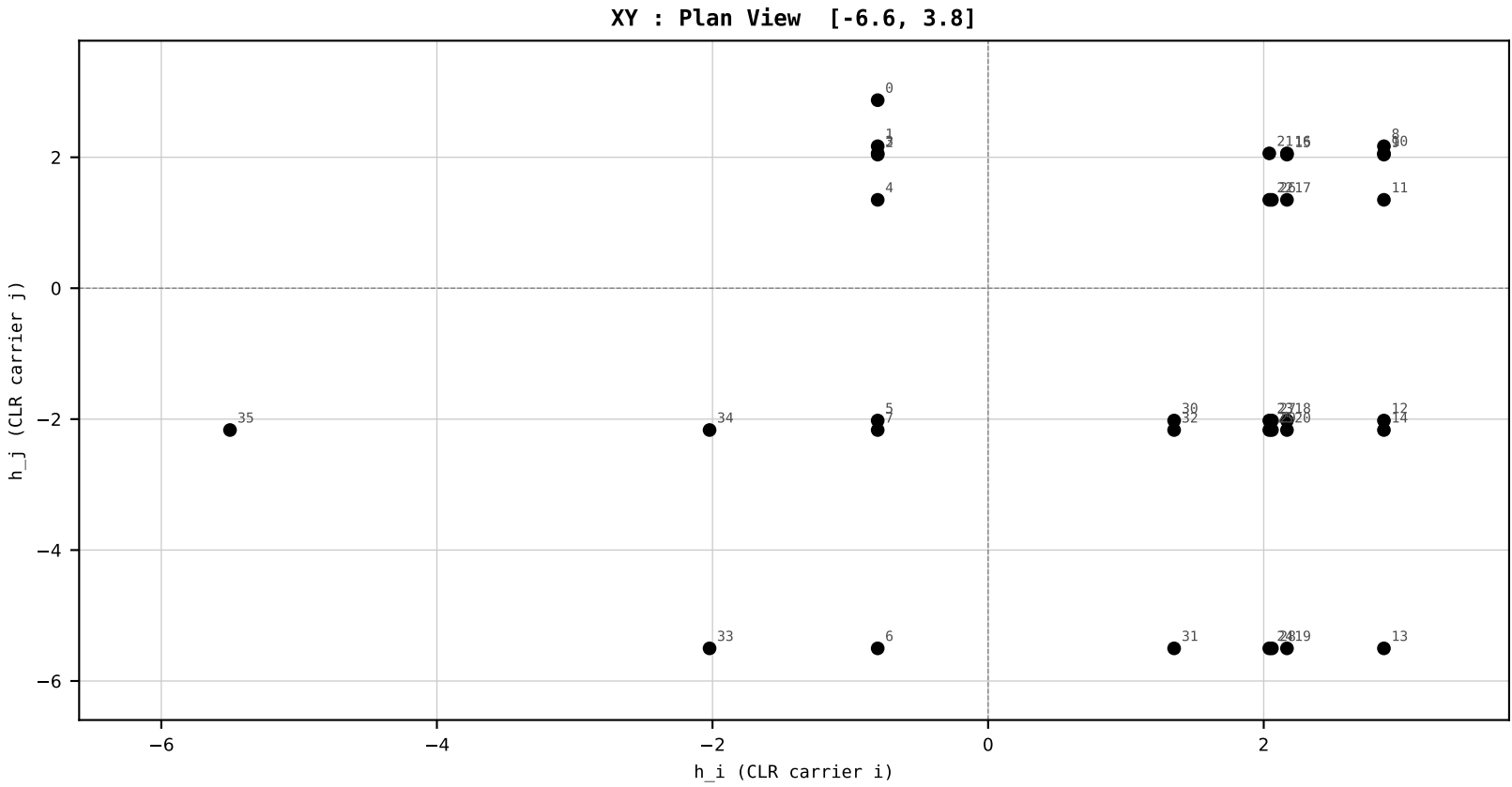
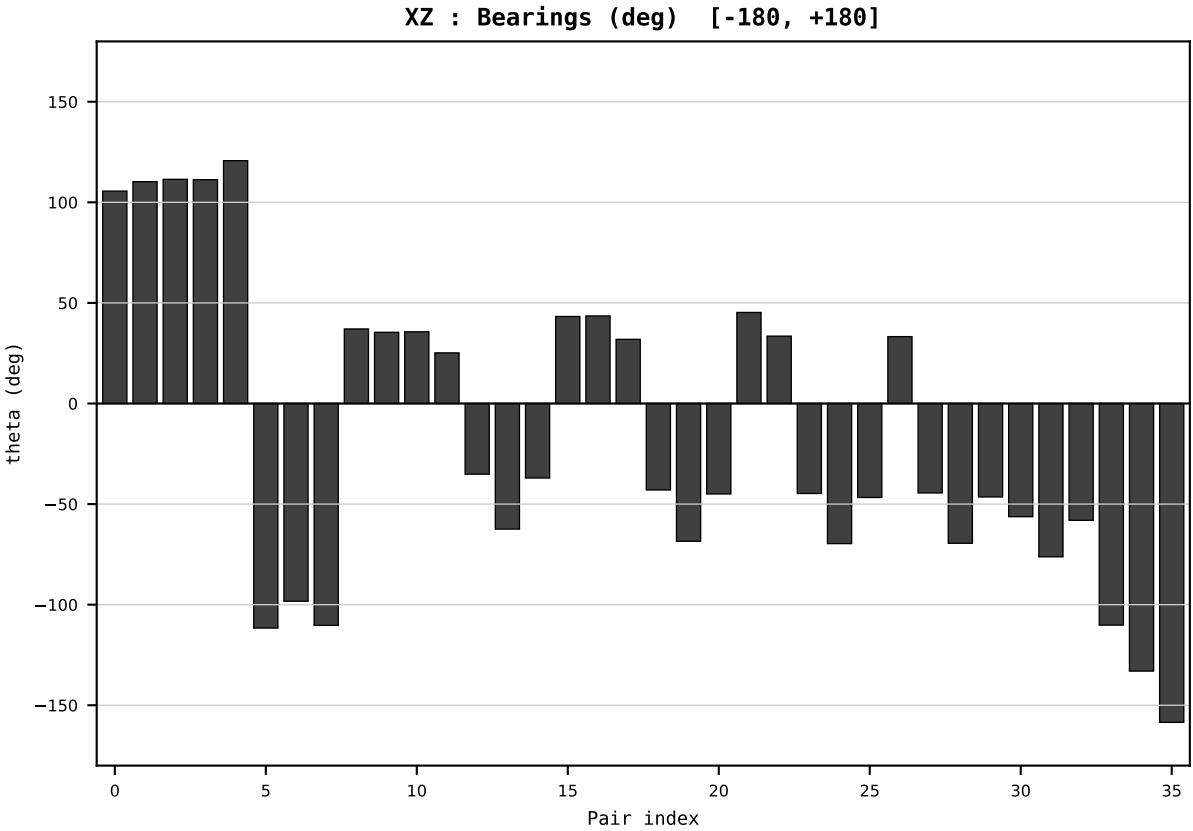
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.288557
Ring = Hs-2
E_metric= 62.8650
kappa_HS= 4331.38
omega = 1.4848 deg
d_A = 0.318532
Helm = Solar
Helm d = +0.229979
DR = 8.374 HLR
DR ratio= 4331.4x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

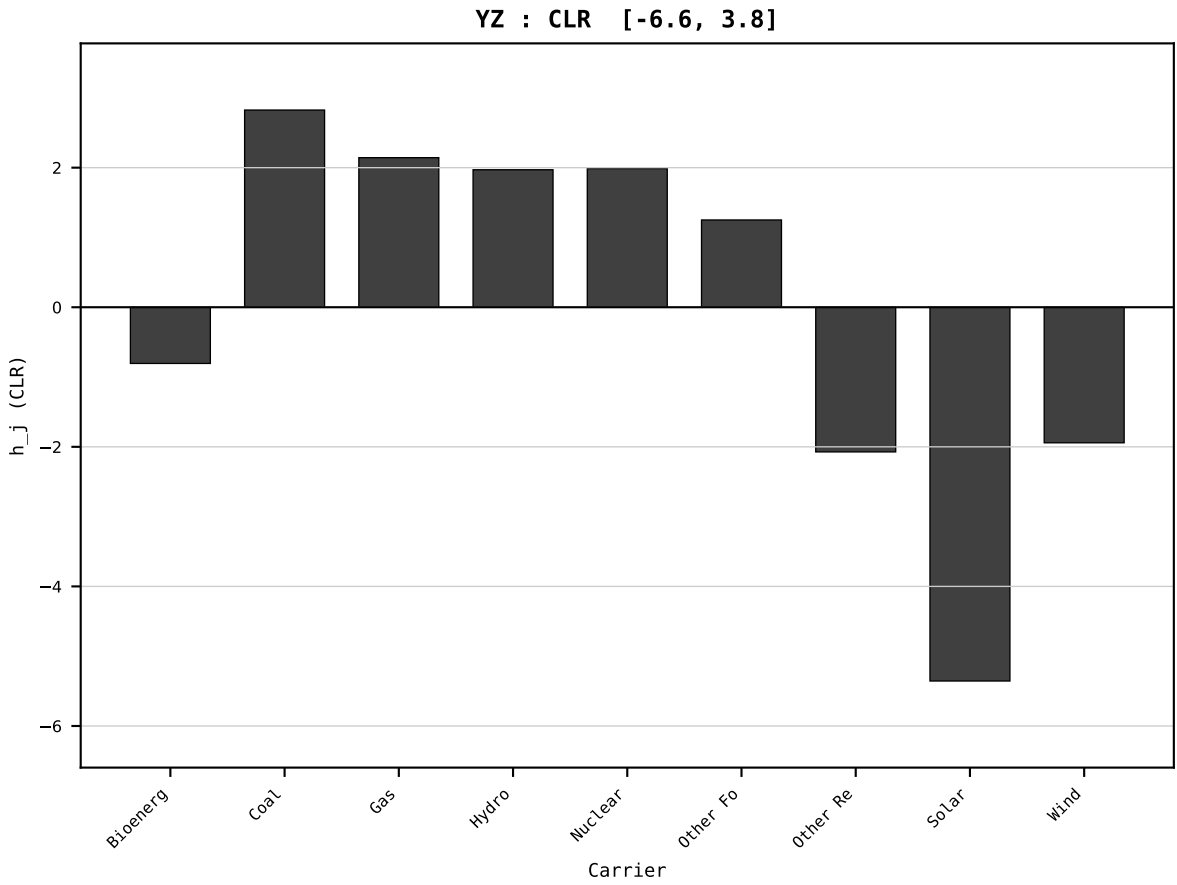
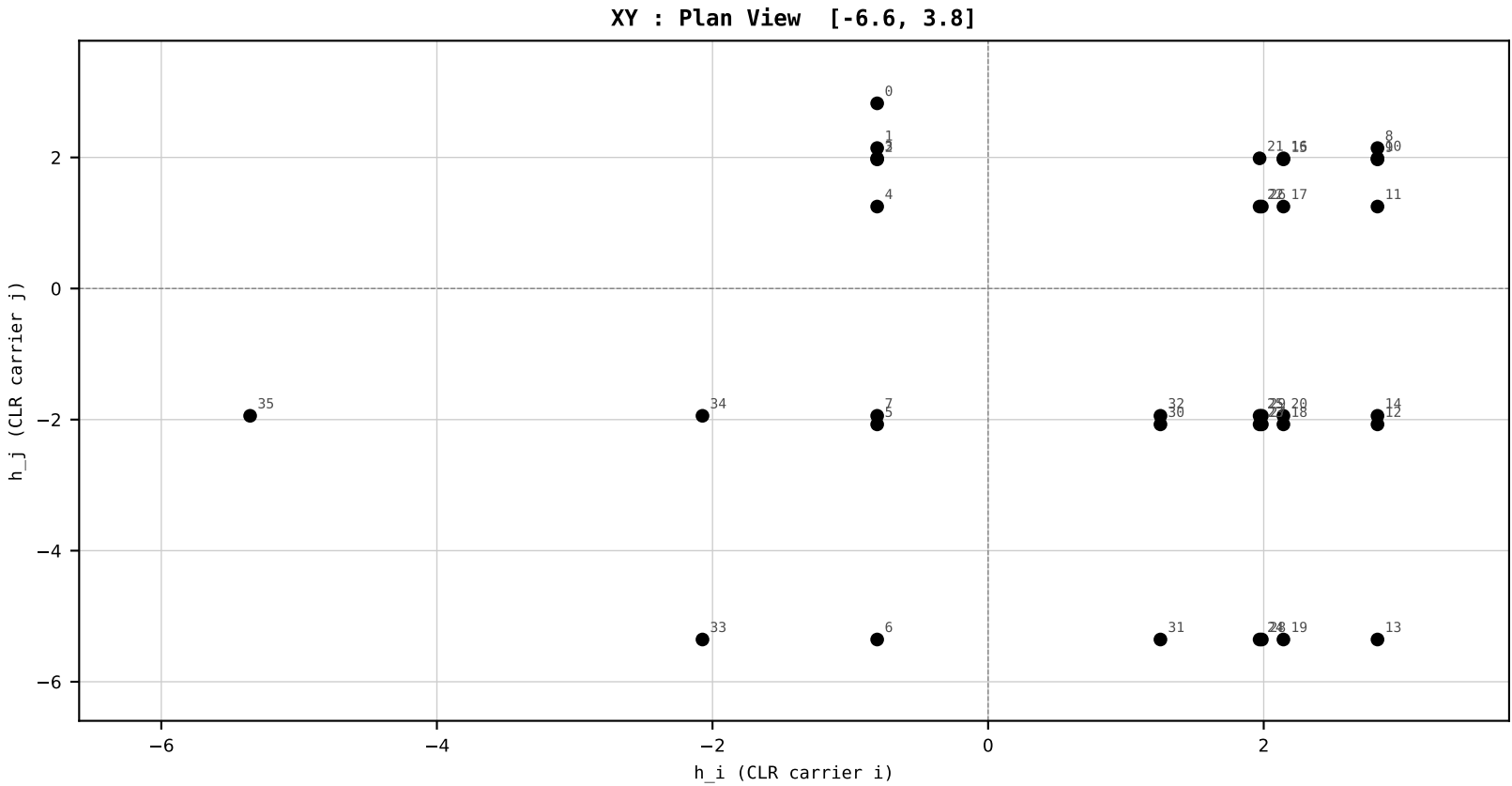
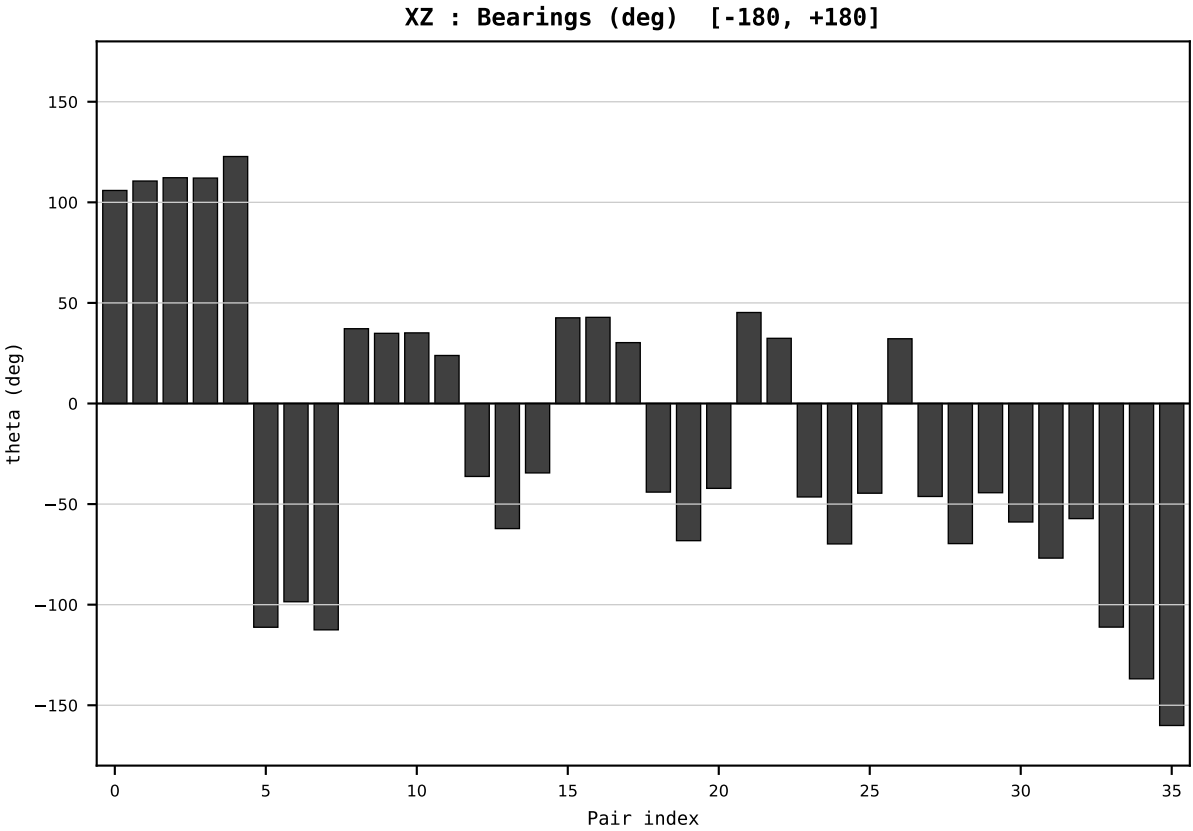
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.288986
Ring = Hs-2
E_metric= 59.3639
kappa_HS= 3572.27
omega = 1.5895 deg
d_A = 0.311719
Helm = Wind
Helm d = +0.224097
DR = 8.181 HLR
DR ratio= 3572.3x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

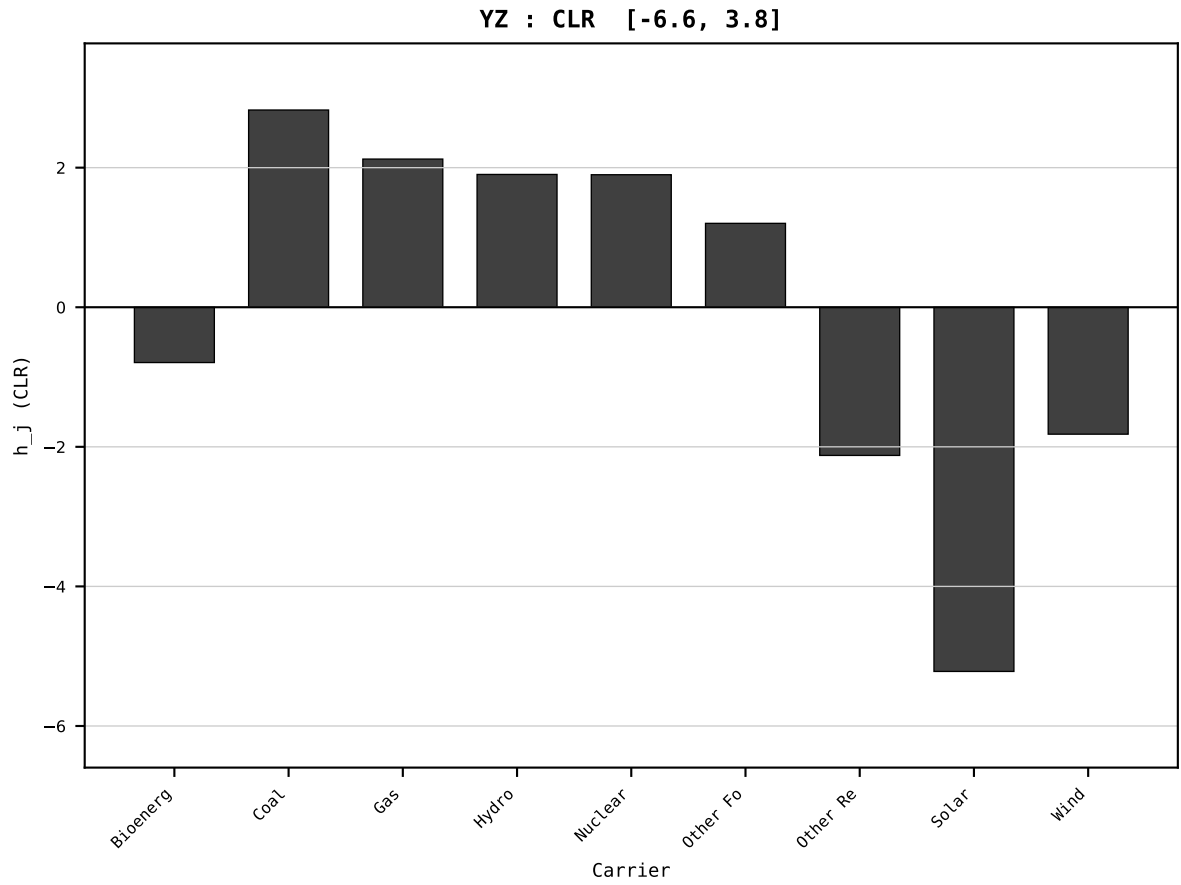
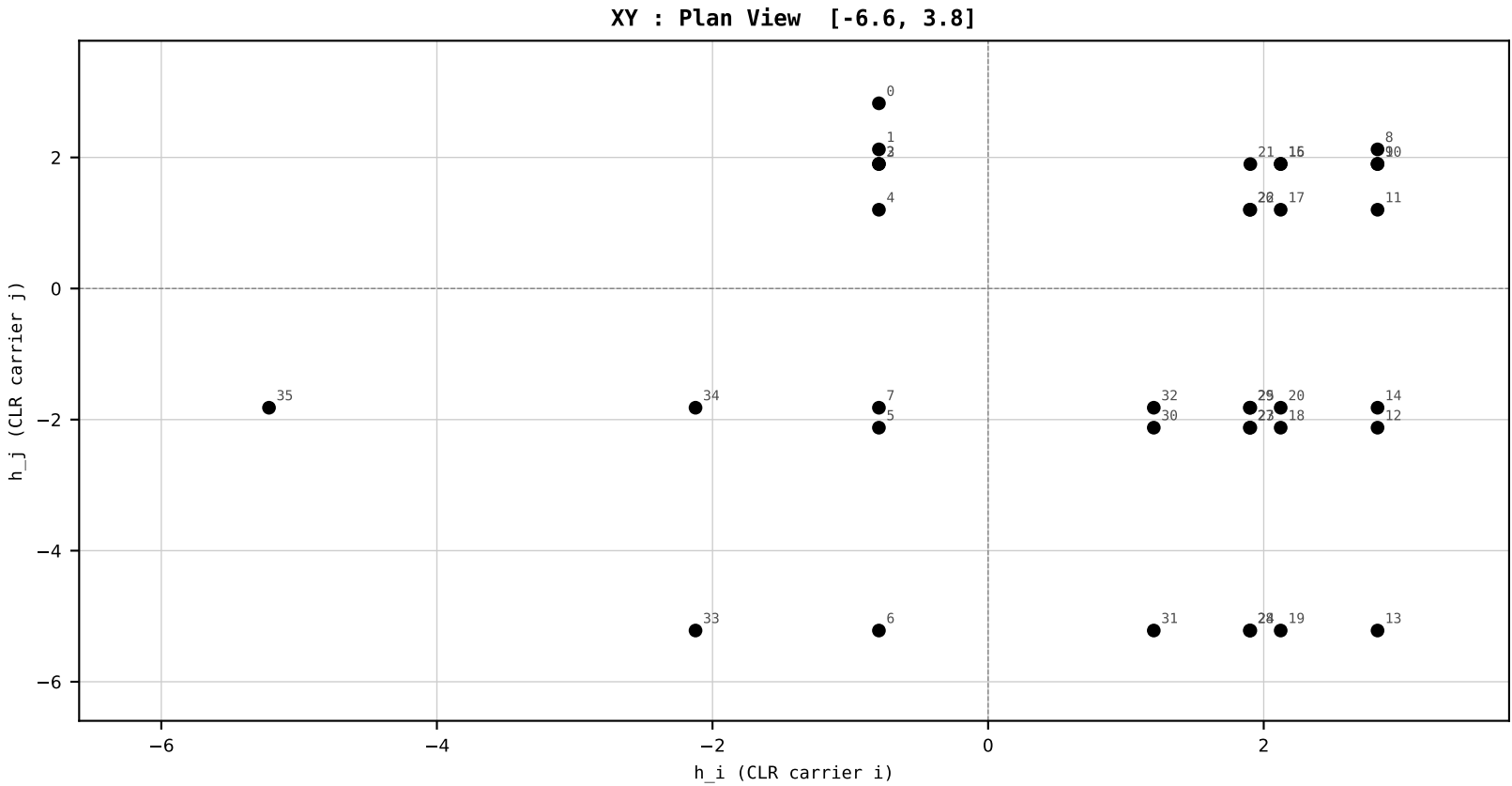
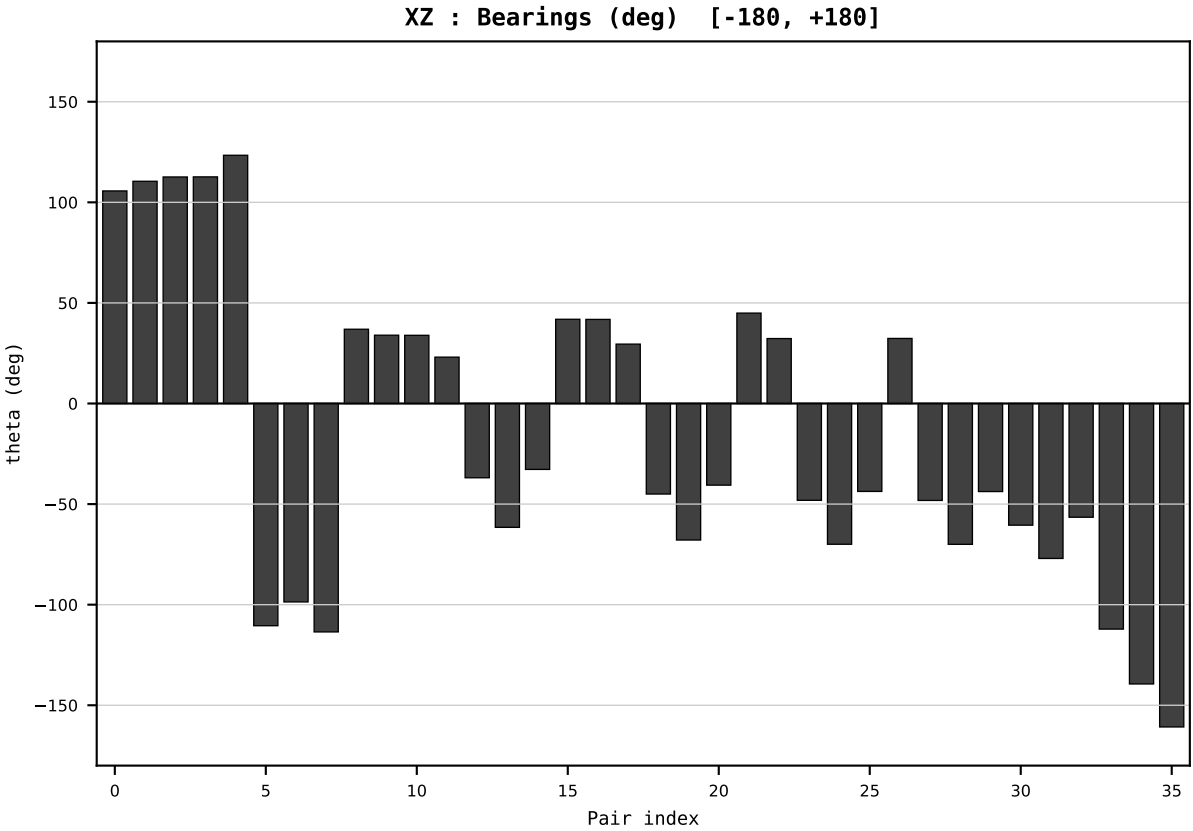
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.293186
Ring = Hs-2
E_metric= 56.8394
kappa_HS= 3116.09
omega = 1.1711 deg
d_A = 0.227352
Helm = Solar
Helm d = +0.137038
DR = 8.044 HLR
DR ratio= 3116.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

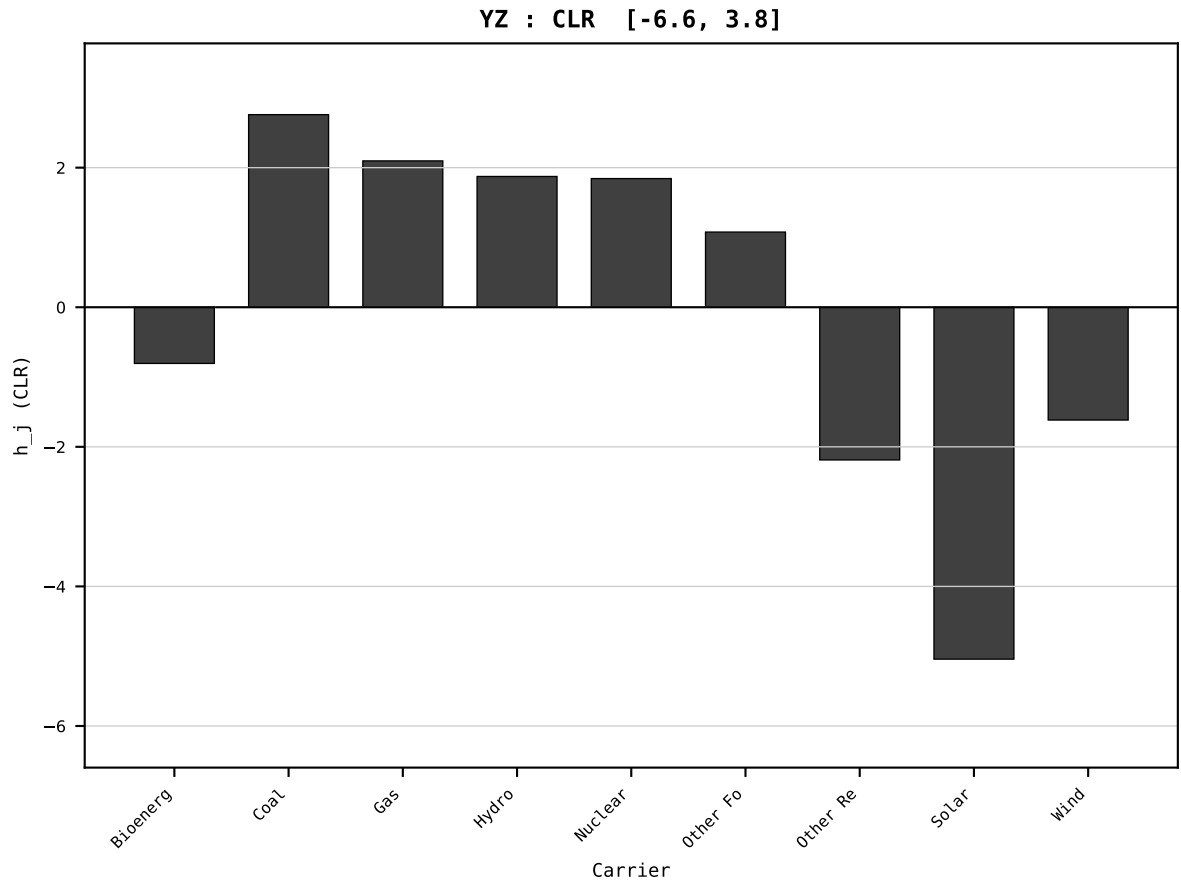
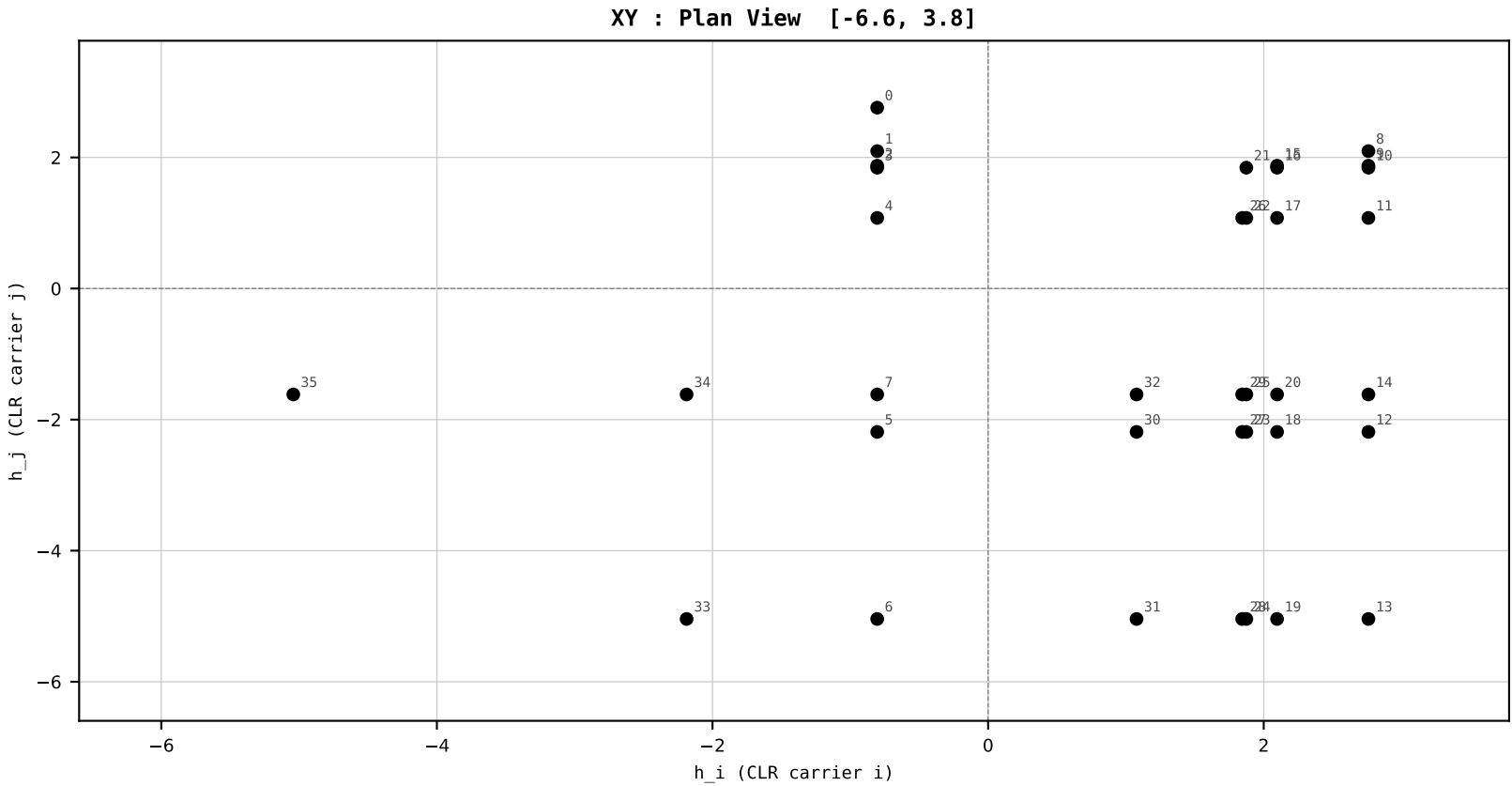
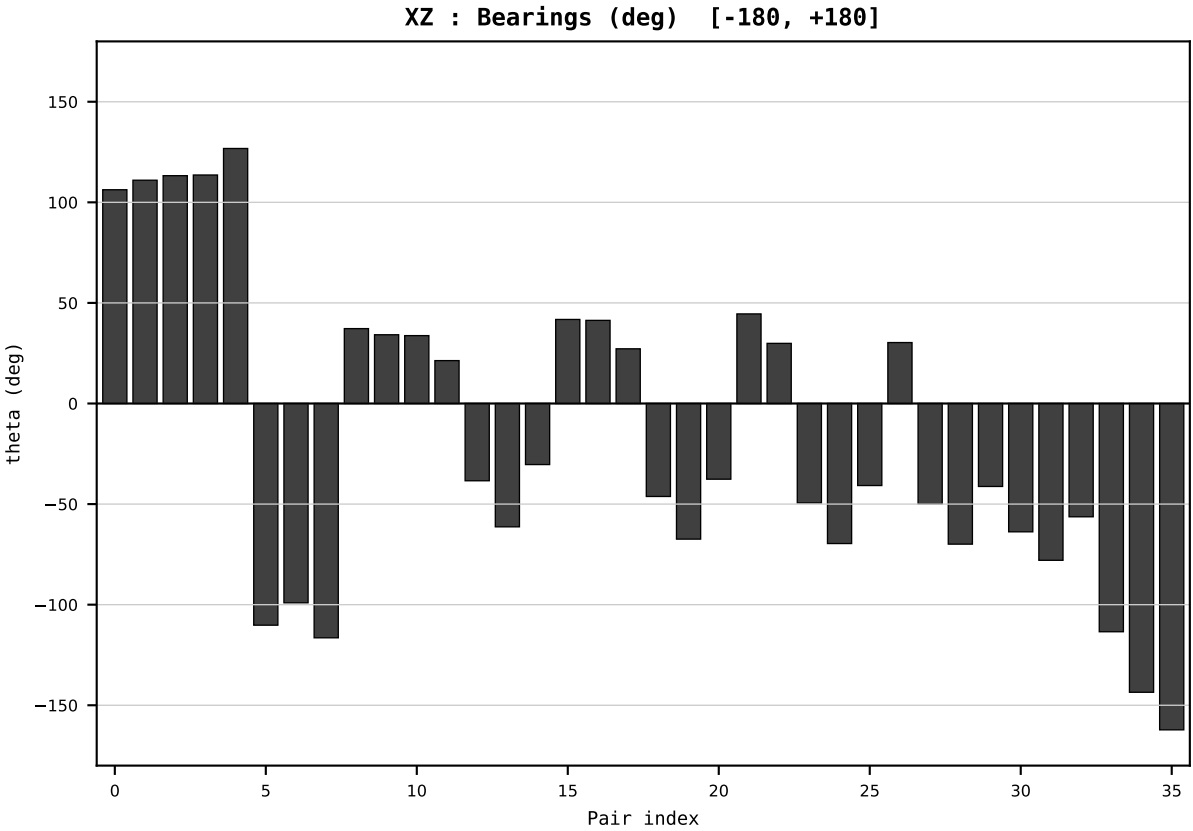
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.291066
Ring = Hs-2
E_metric= 53.5533
kappa_HS= 2445.07
omega = 1.7599 deg
d_A = 0.317755
Helm = Wind
Helm d = +0.202466
DR = 7.802 HLR
DR ratio= 2445.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

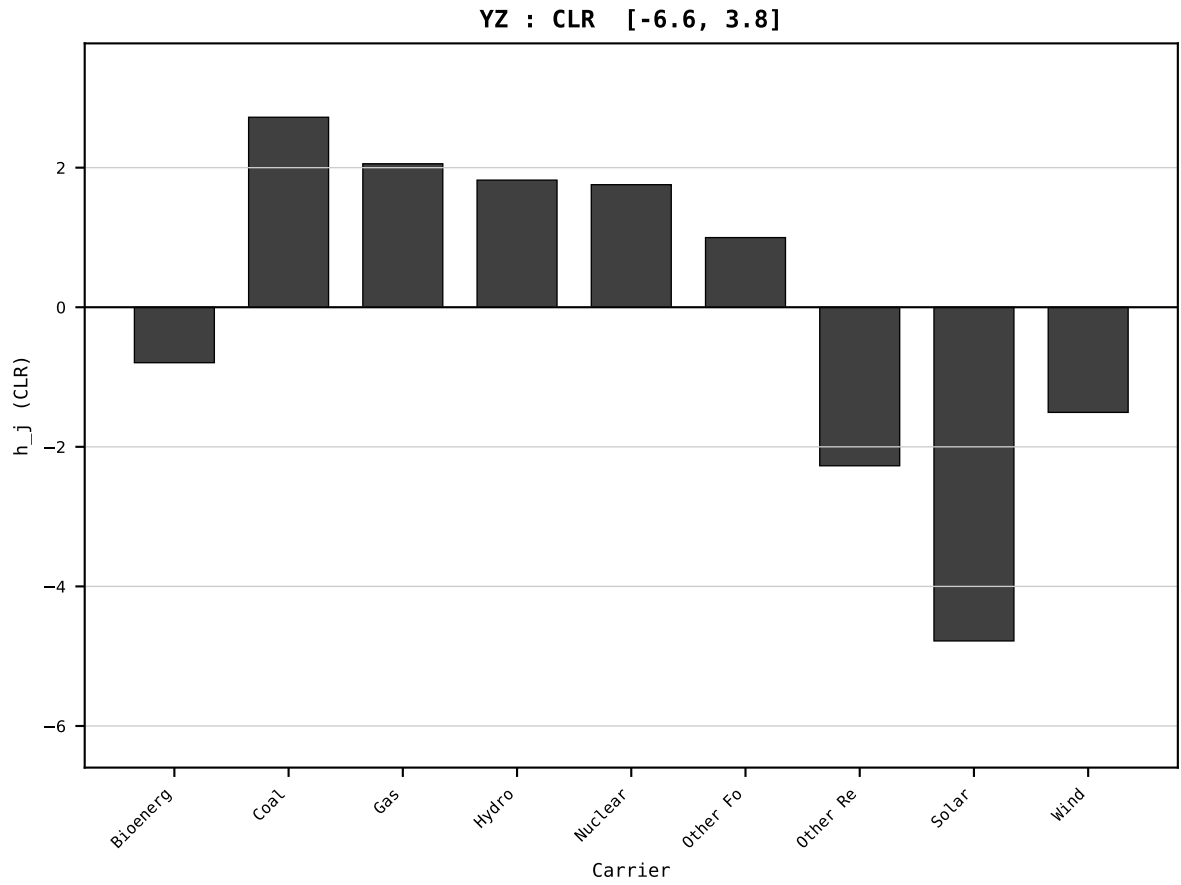
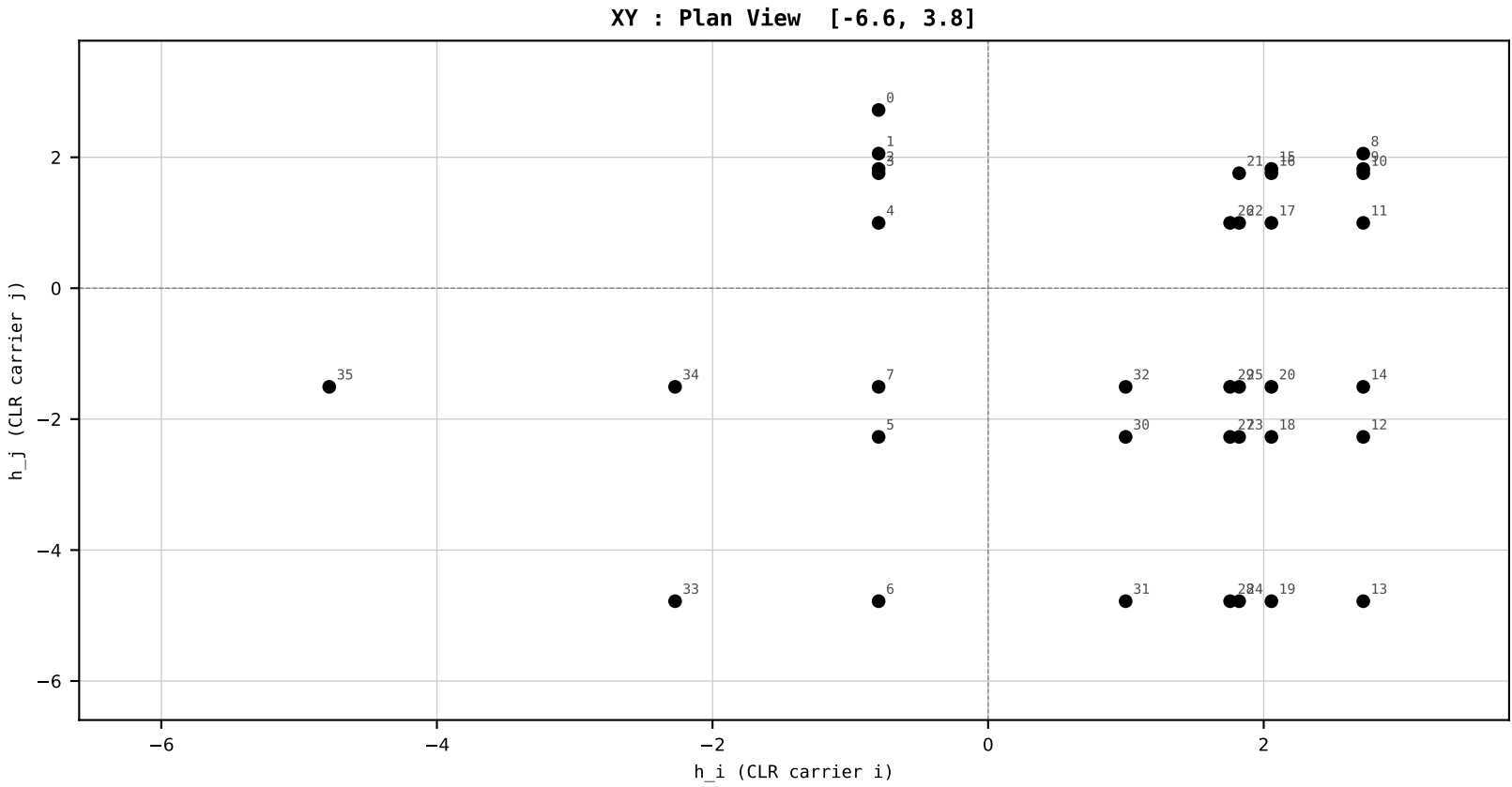
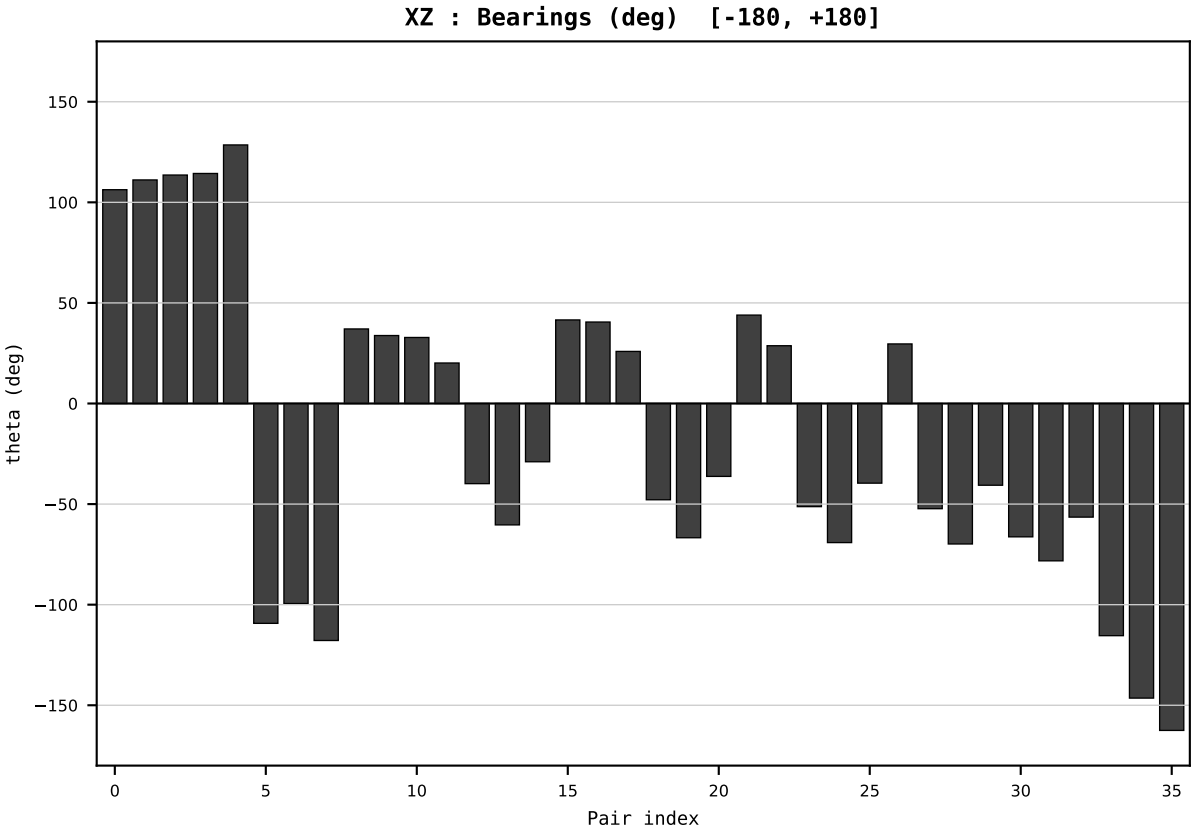
Dataset : ember_WLD_World_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.292003
Ring = Hs-2
E_metric= 49.9572
kappa_HS= 1814.48
omega = 1.6789 deg
d_A = 0.326946
Helm = Solar
Helm d = +0.260914
DR = 7.504 HLR
DR ratio= 1814.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

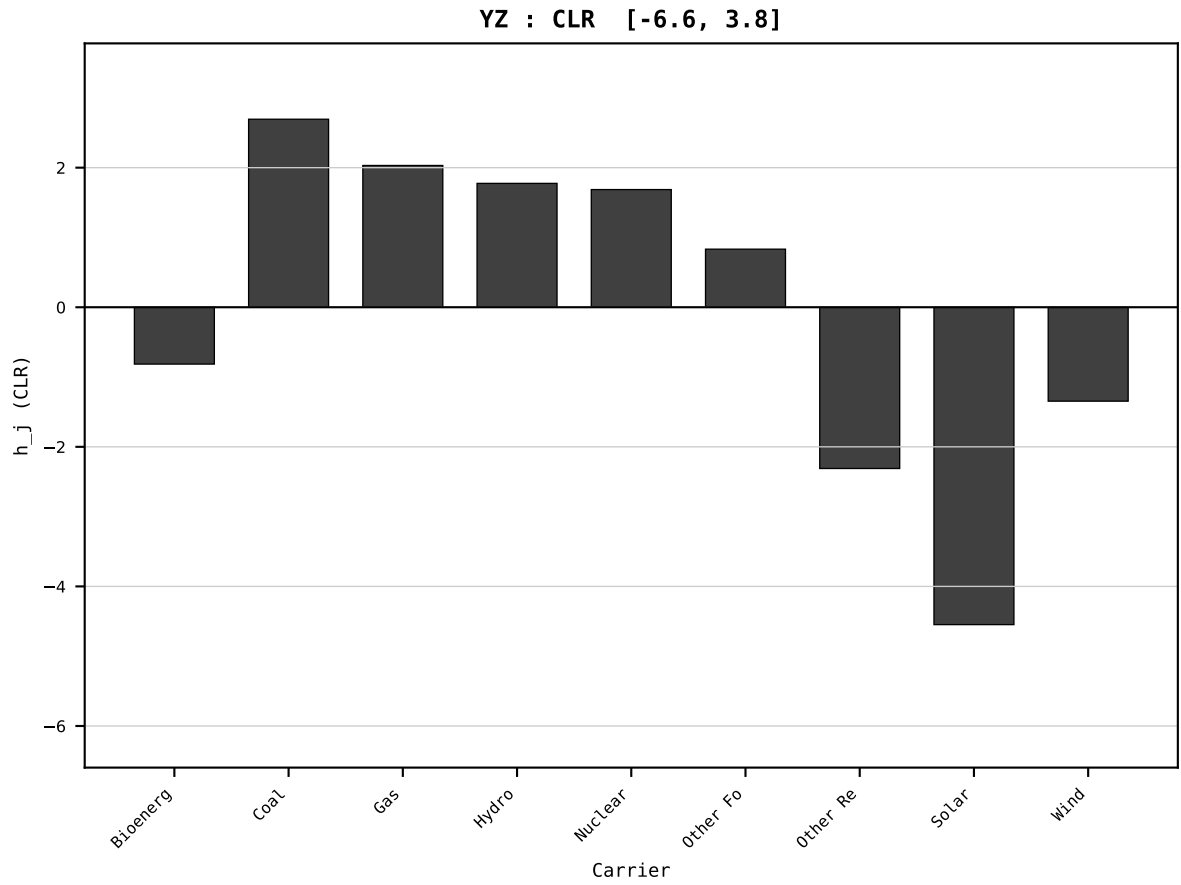
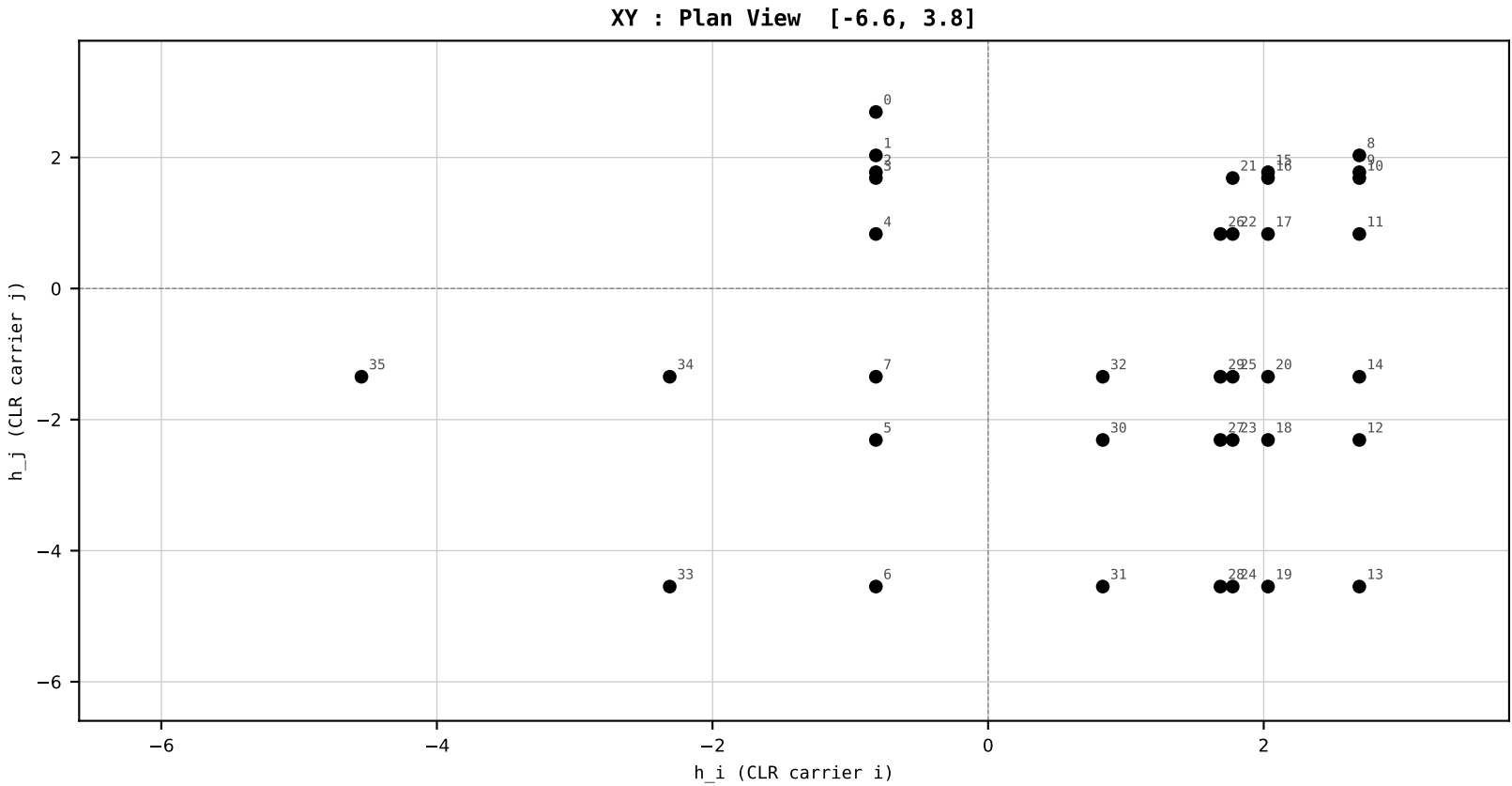
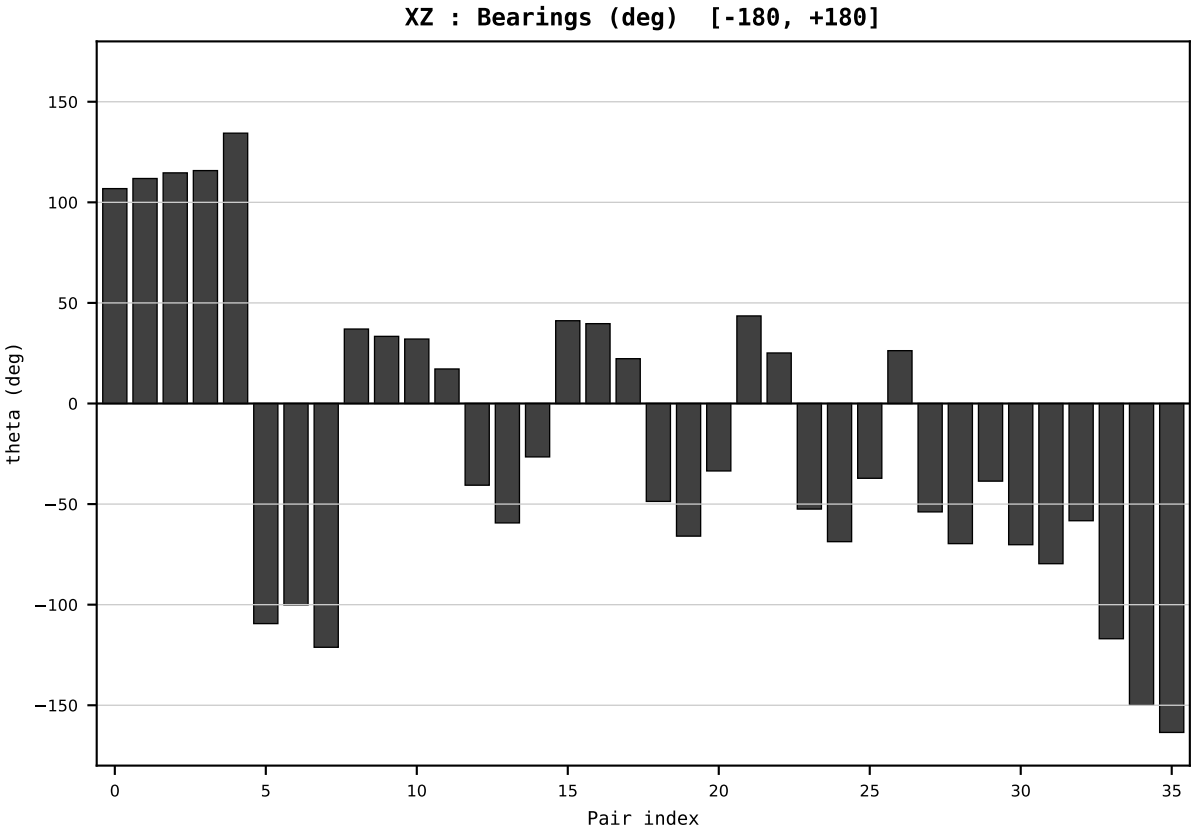
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.295981
Ring = Hs-2
E_metric= 46.5605
kappa_HS= 1396.12
omega = 1.9988 deg
d_A = 0.344208
Helm = Solar
Helm d = +0.233681
DR = 7.241 HLR
DR ratio= 1396.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

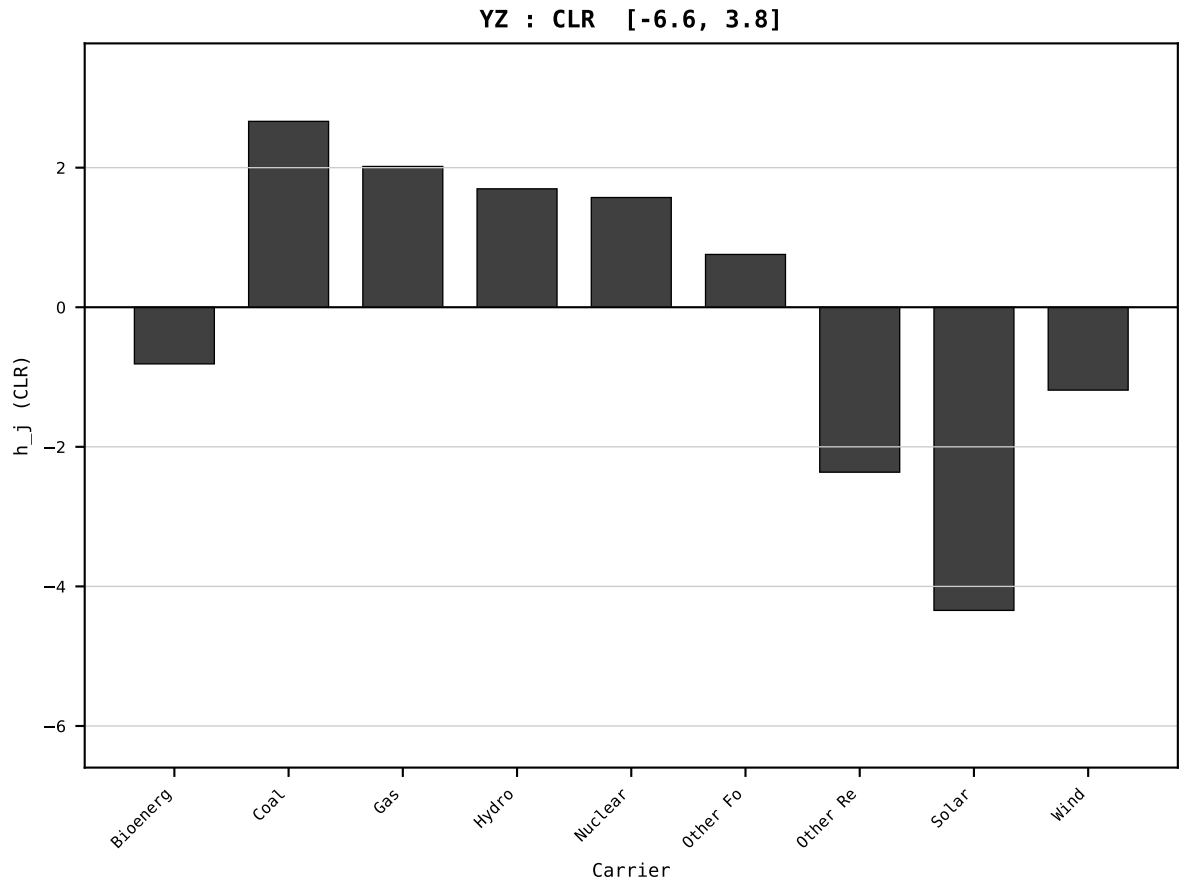
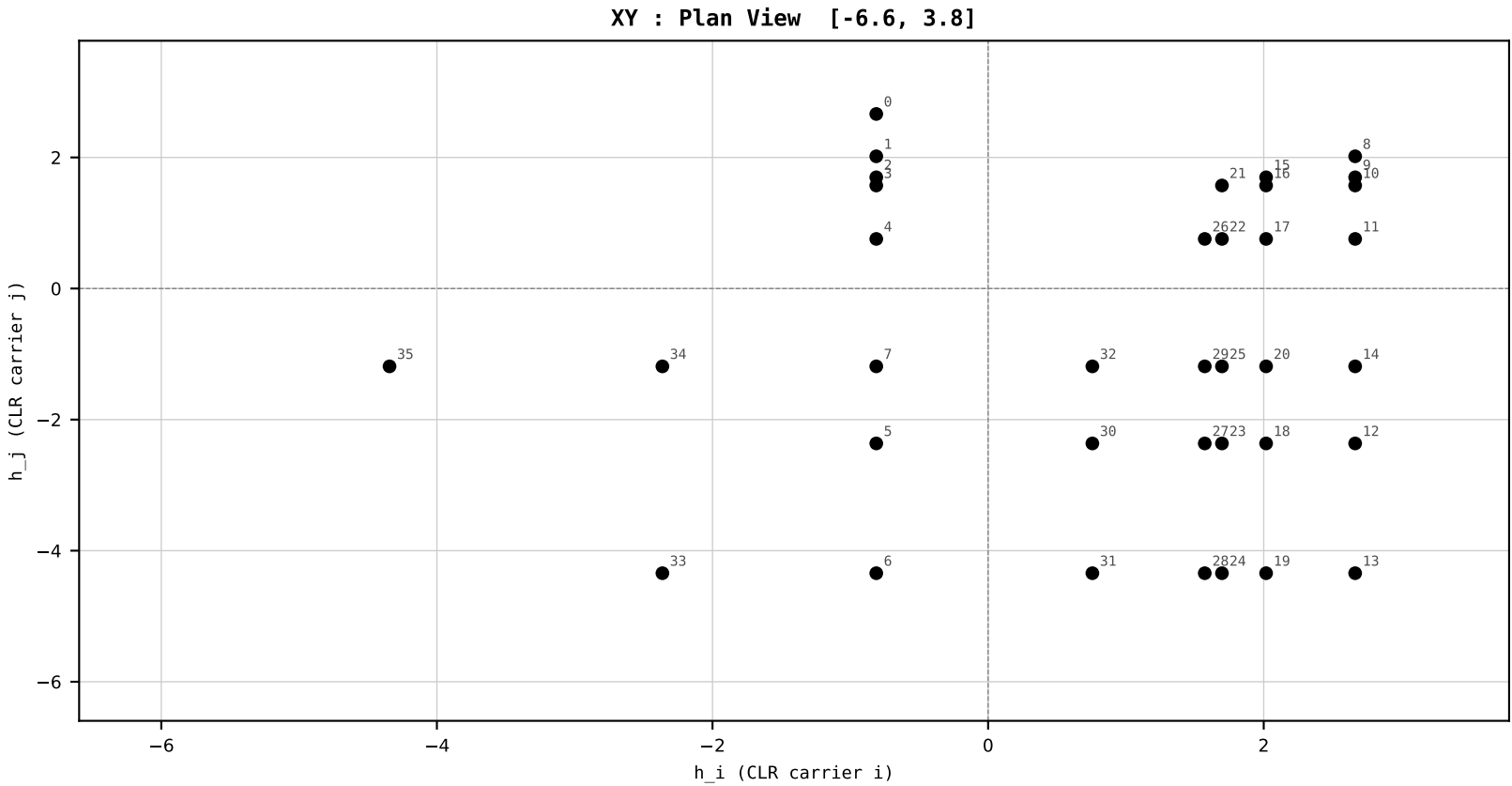
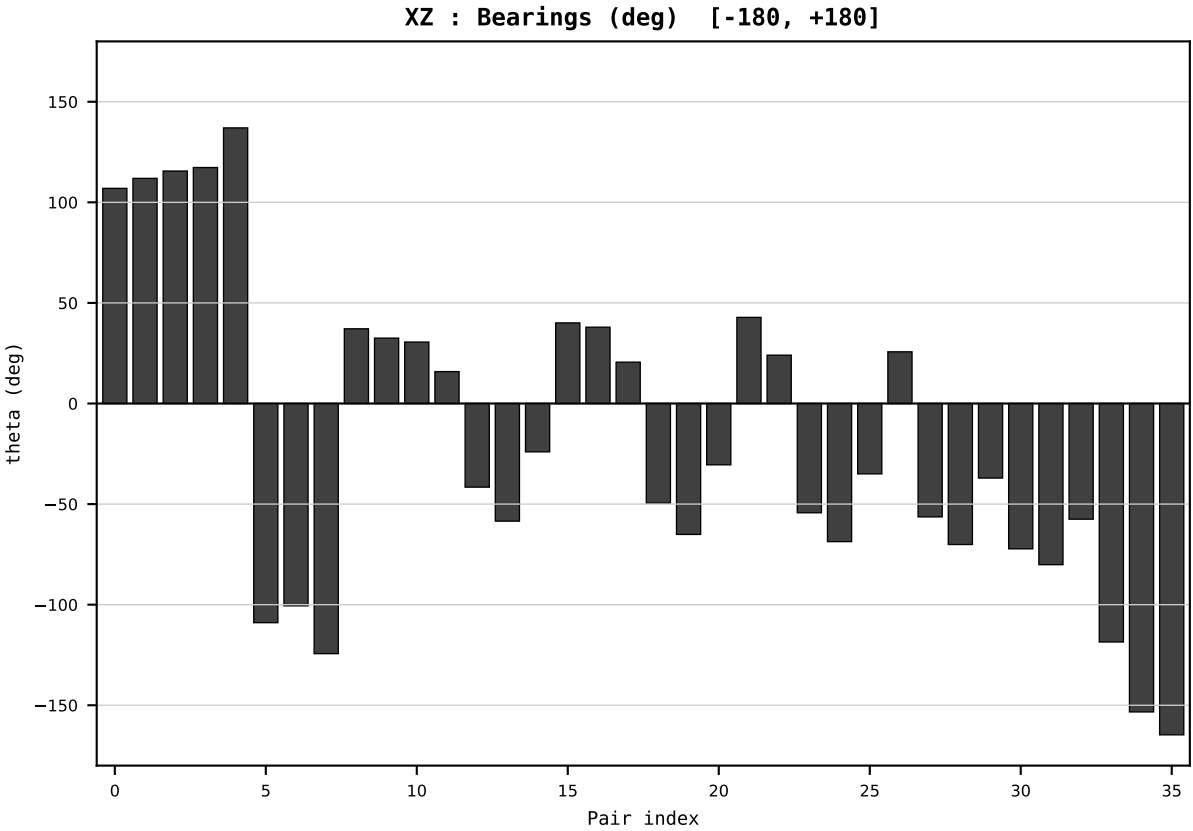
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.297889
Ring = Hs-2
E_metric= 43.6095
kappa_HS= 1104.50
omega = 1.8487 deg
d_A = 0.308559
Helm = Solar
Helm d = +0.203824
DR = 7.007 HLR
DR ratio= 1104.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



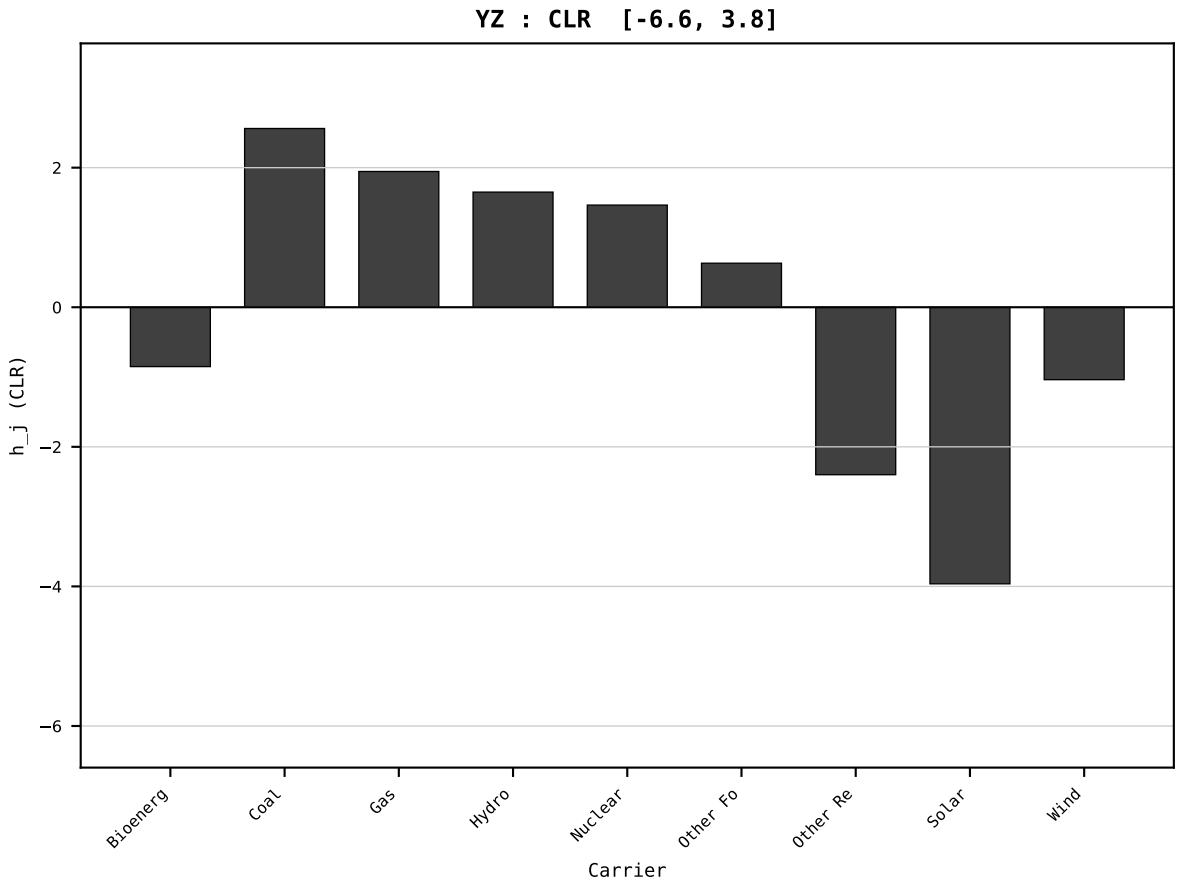
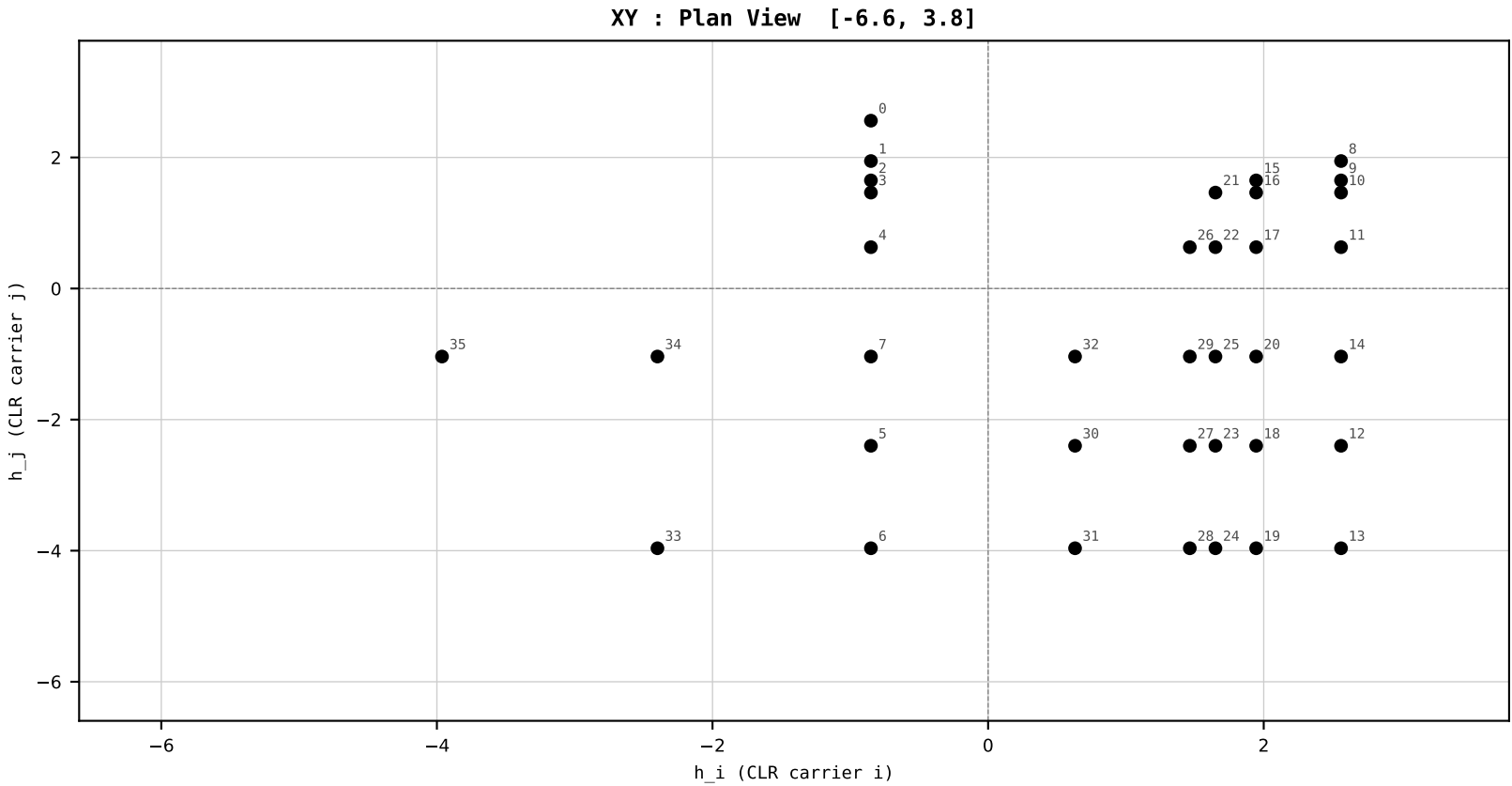
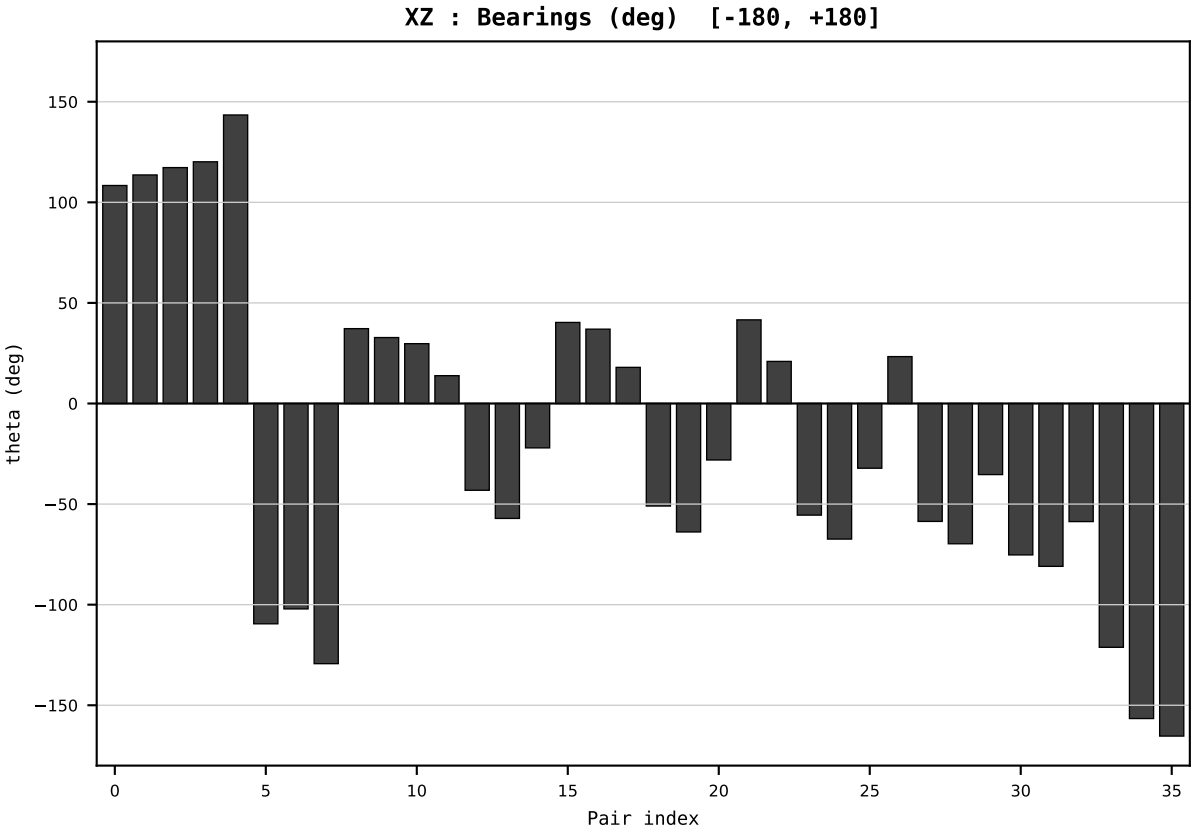
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.292239
Ring = Hs-2
E_metric= 38.8679
kappa_HS= 681.37
omega = 2.5134 deg
d_A = 0.464351
Helm = Solar
Helm d = +0.380700
DR = 6.524 HLR
DR ratio= 681.4x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



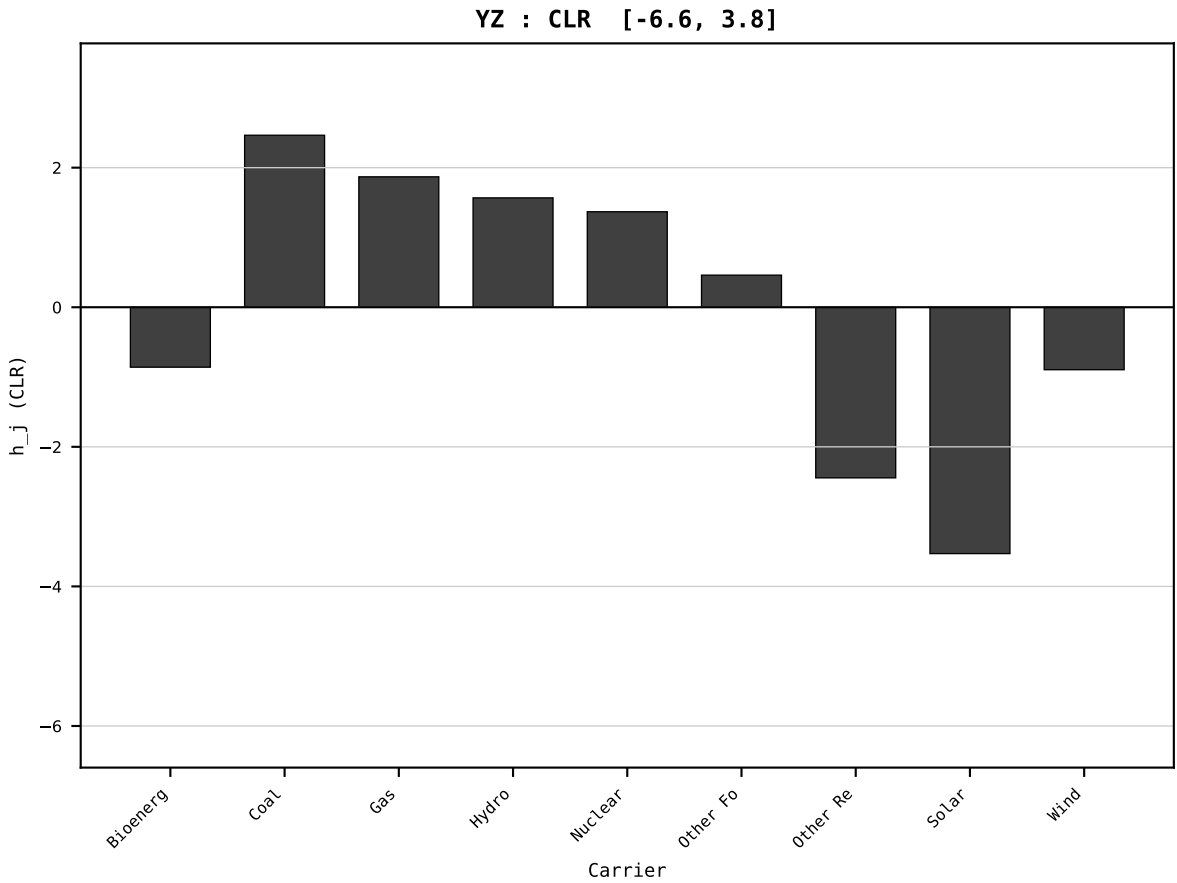
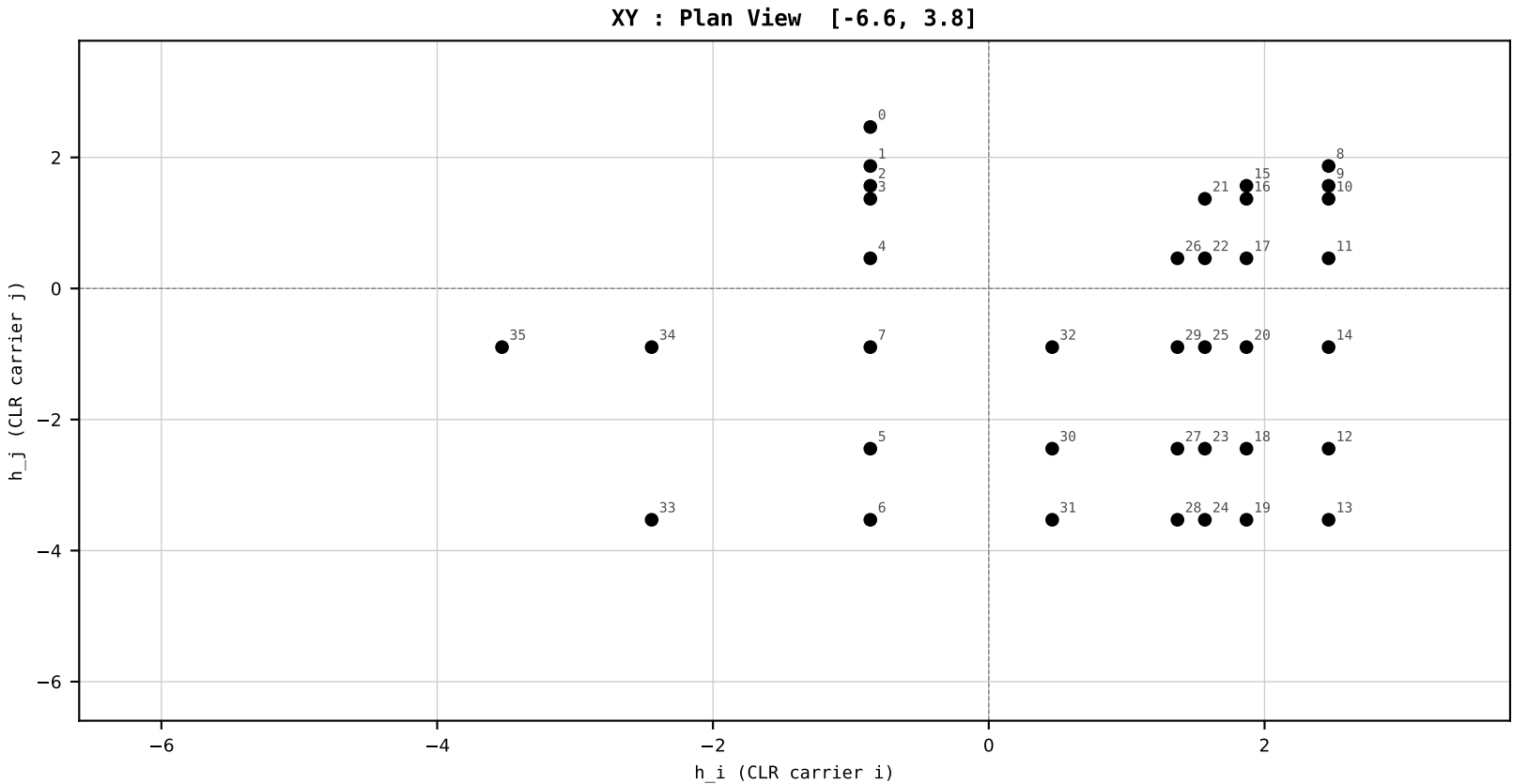
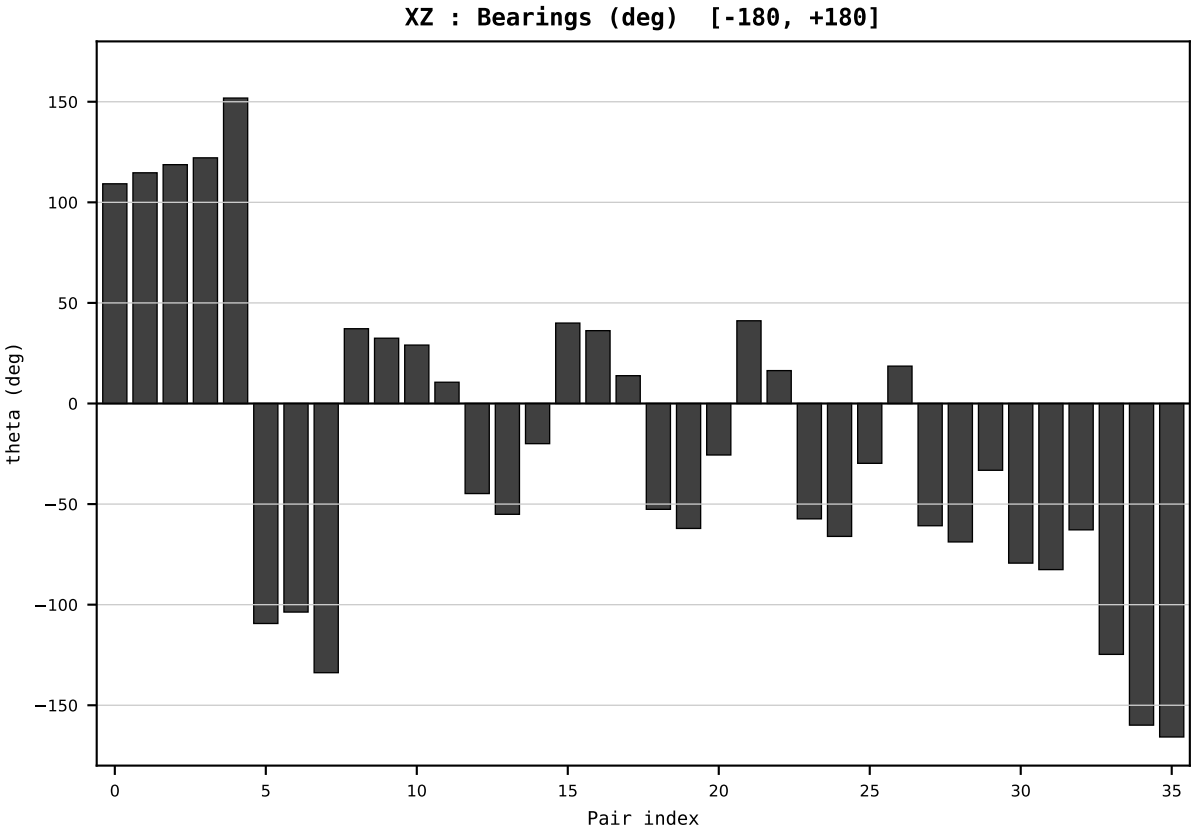
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287865
Ring = Hs-2
E_metric= 34.0820
kappa_HS= 401.15
omega = 3.1960 deg
d_A = 0.519977
Helm = Solar
Helm d = +0.433127
DR = 5.994 HLR
DR ratio= 401.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

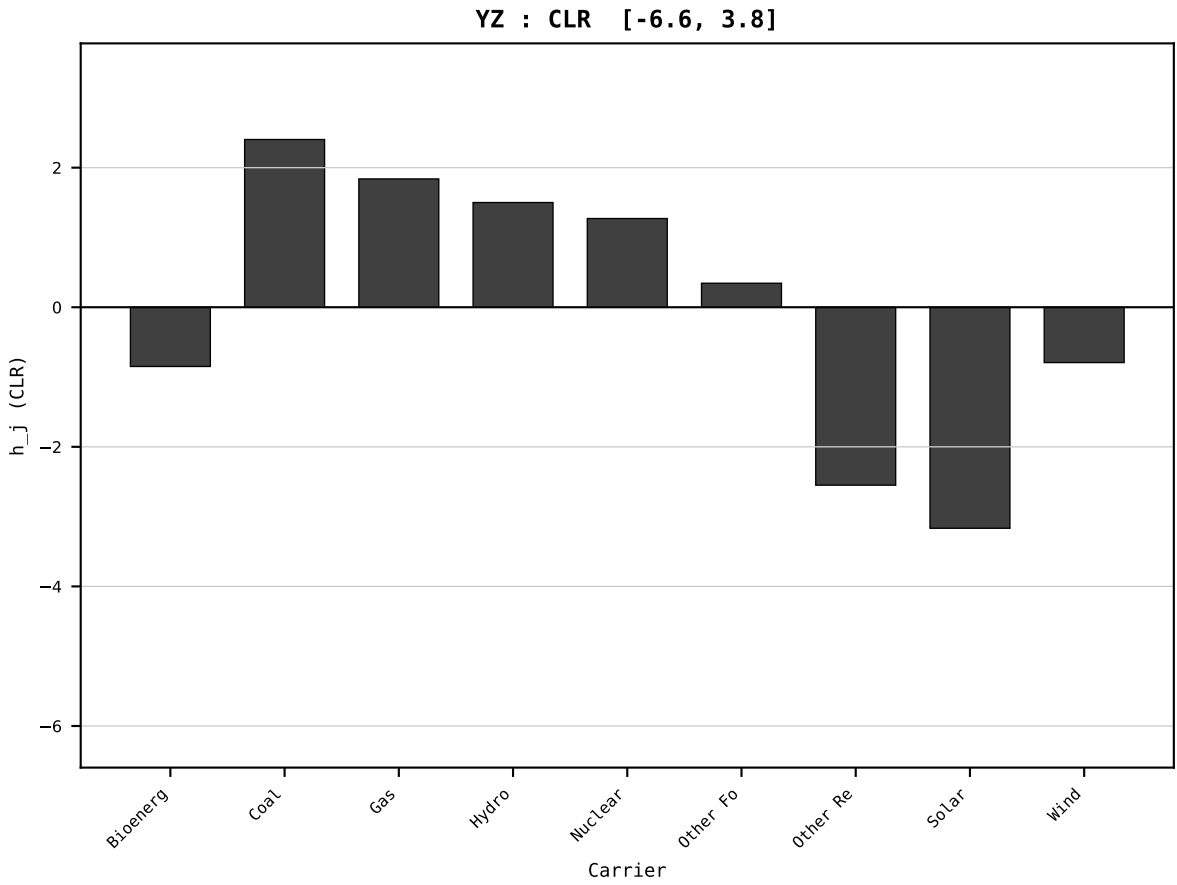
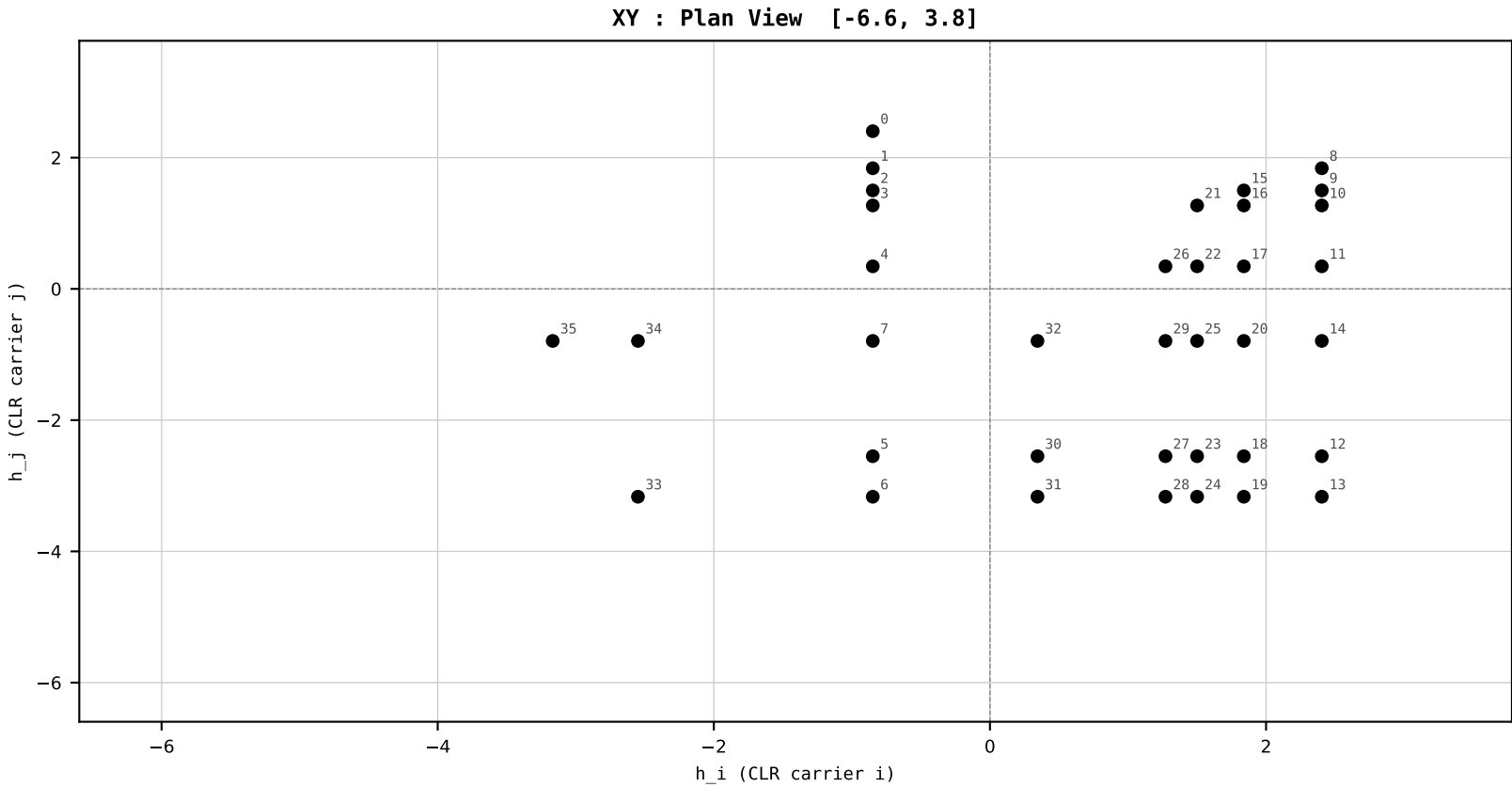
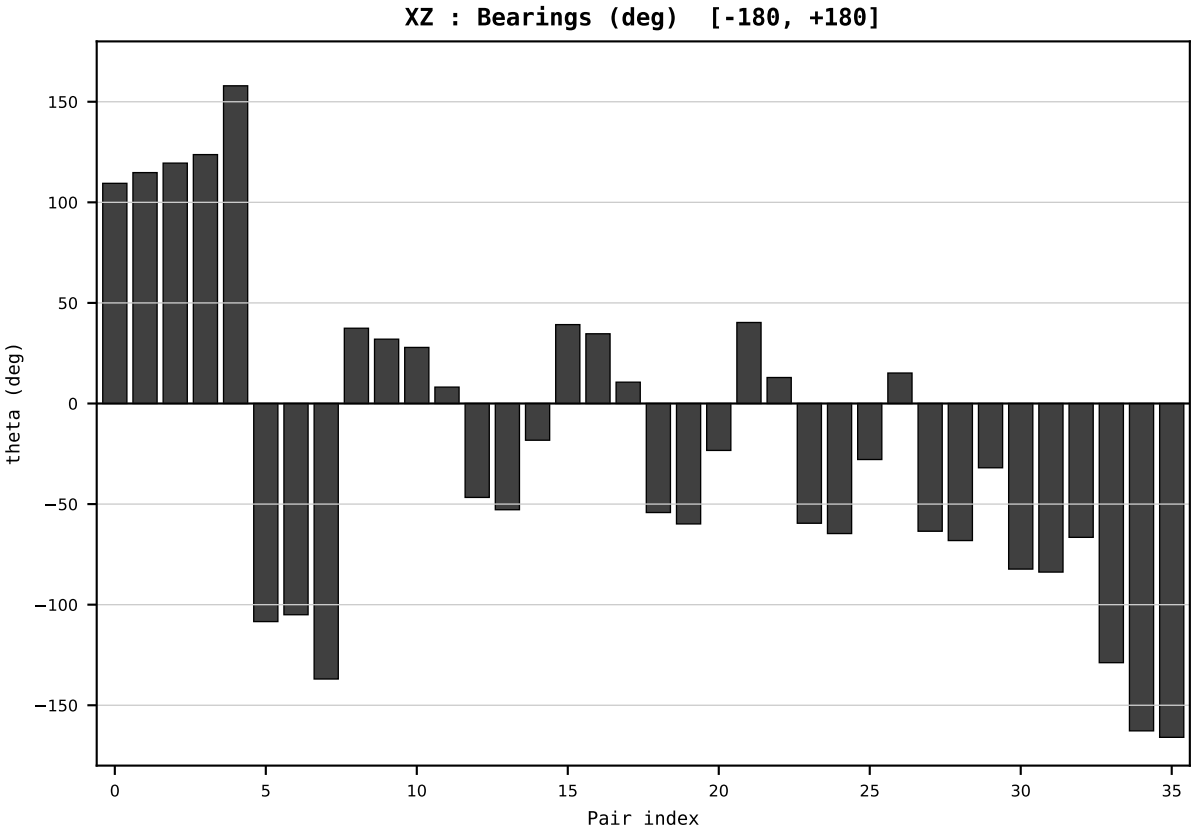
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.285810
Ring = Hs-2
E_metric= 31.0237
kappa_HS= 262.59
omega = 3.3815 deg
d_A = 0.430229
Helm = Solar
Helm d = +0.362834
DR = 5.571 HLR
DR ratio= 262.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

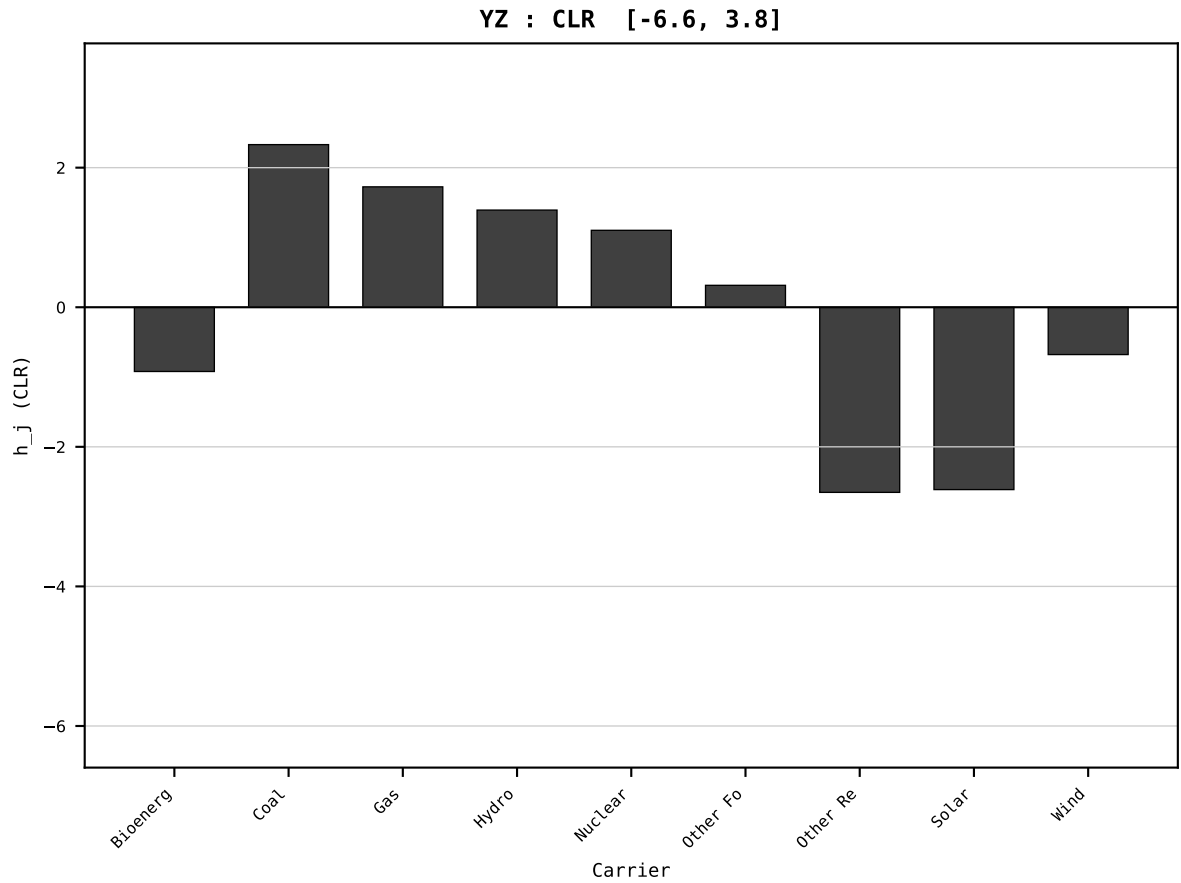
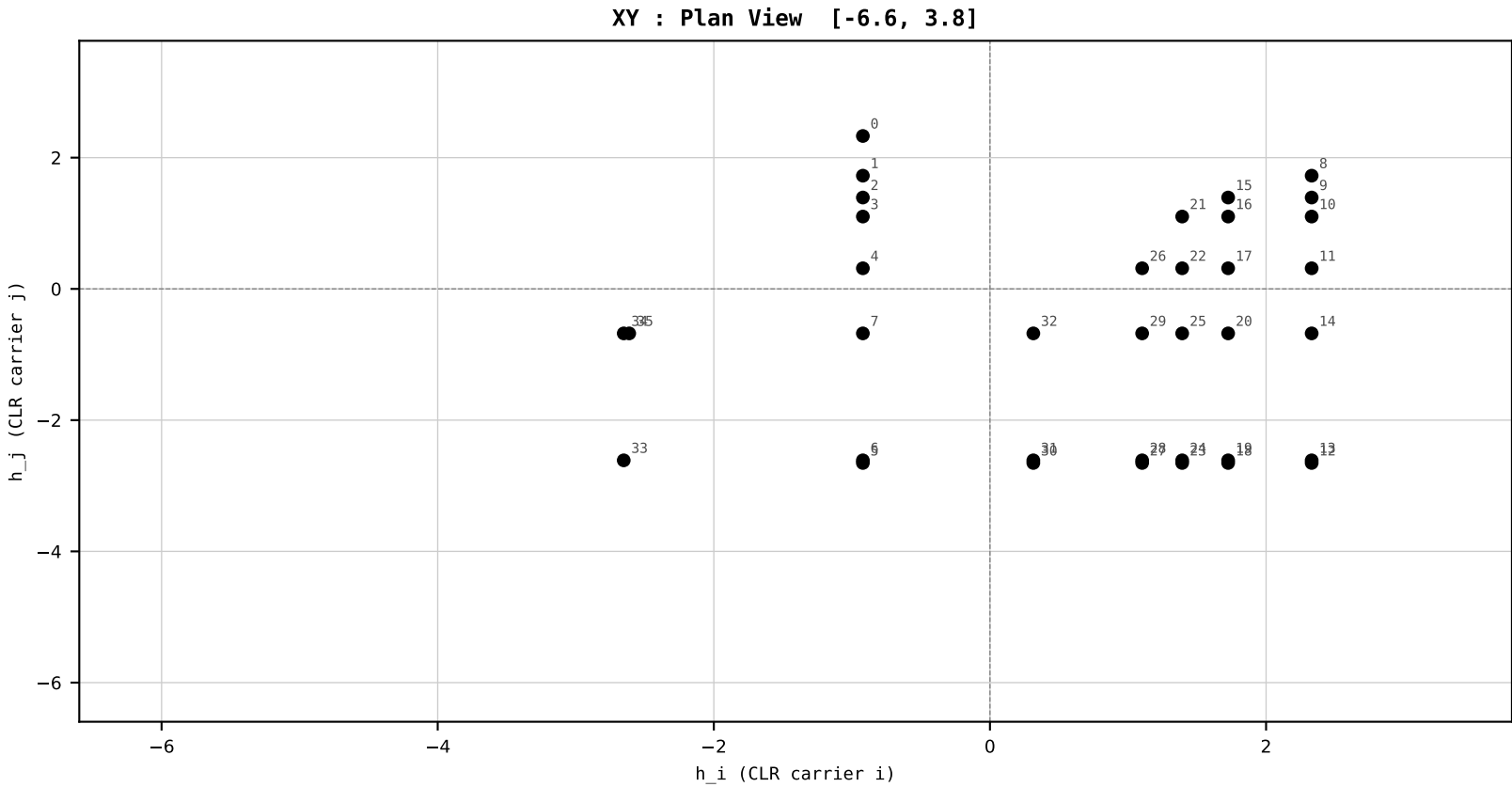
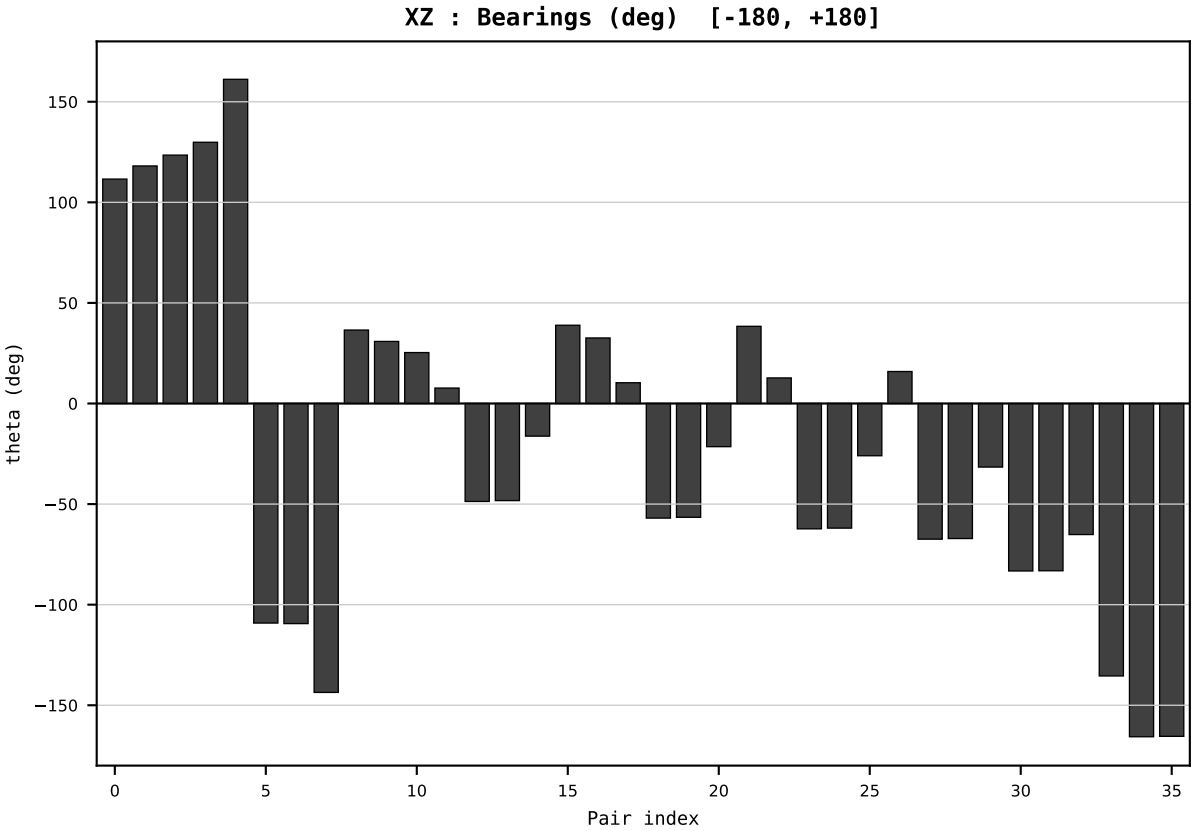
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.281092
Ring = Hs-2
E_metric= 26.8292
kappa_HS= 145.89
omega = 5.2622 deg
d_A = 0.628840
Helm = Solar
Helm d = +0.554151
DR = 4.983 HLR
DR ratio= 145.9x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

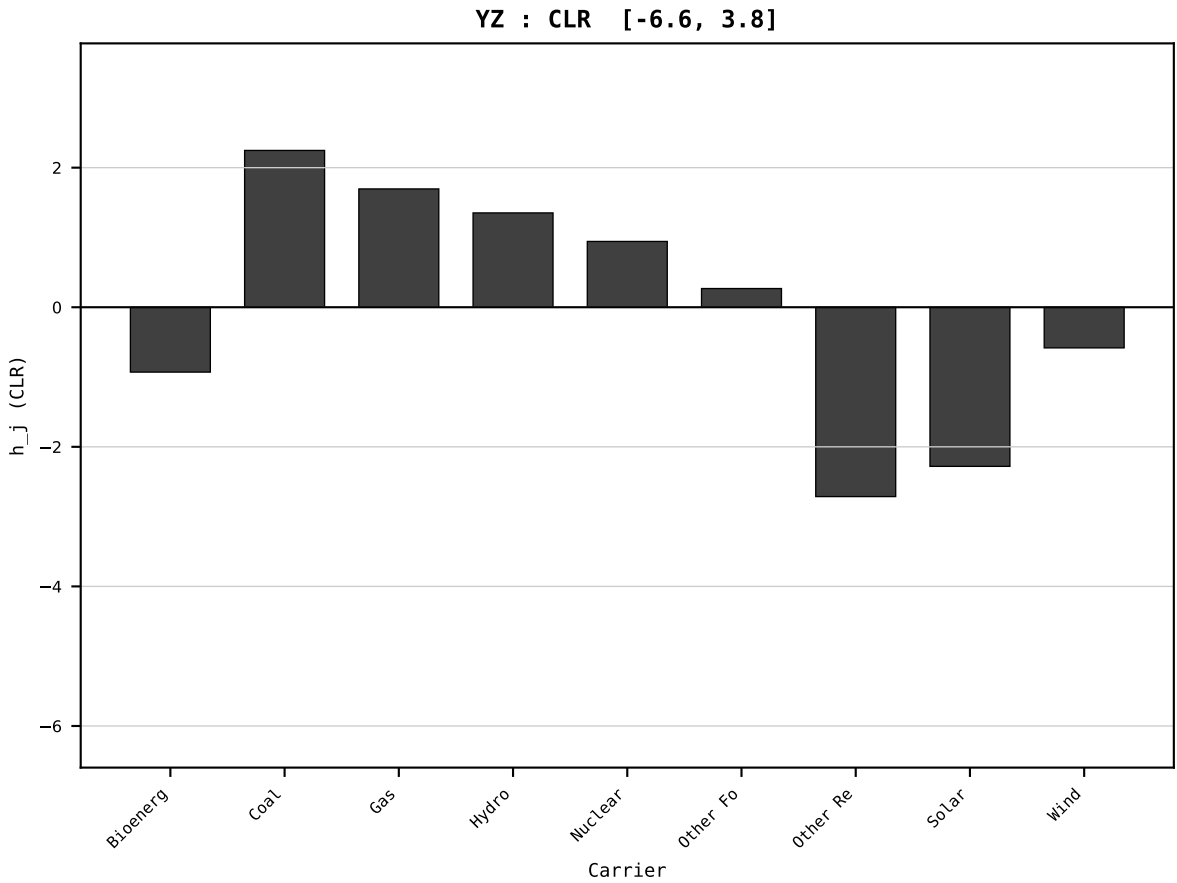
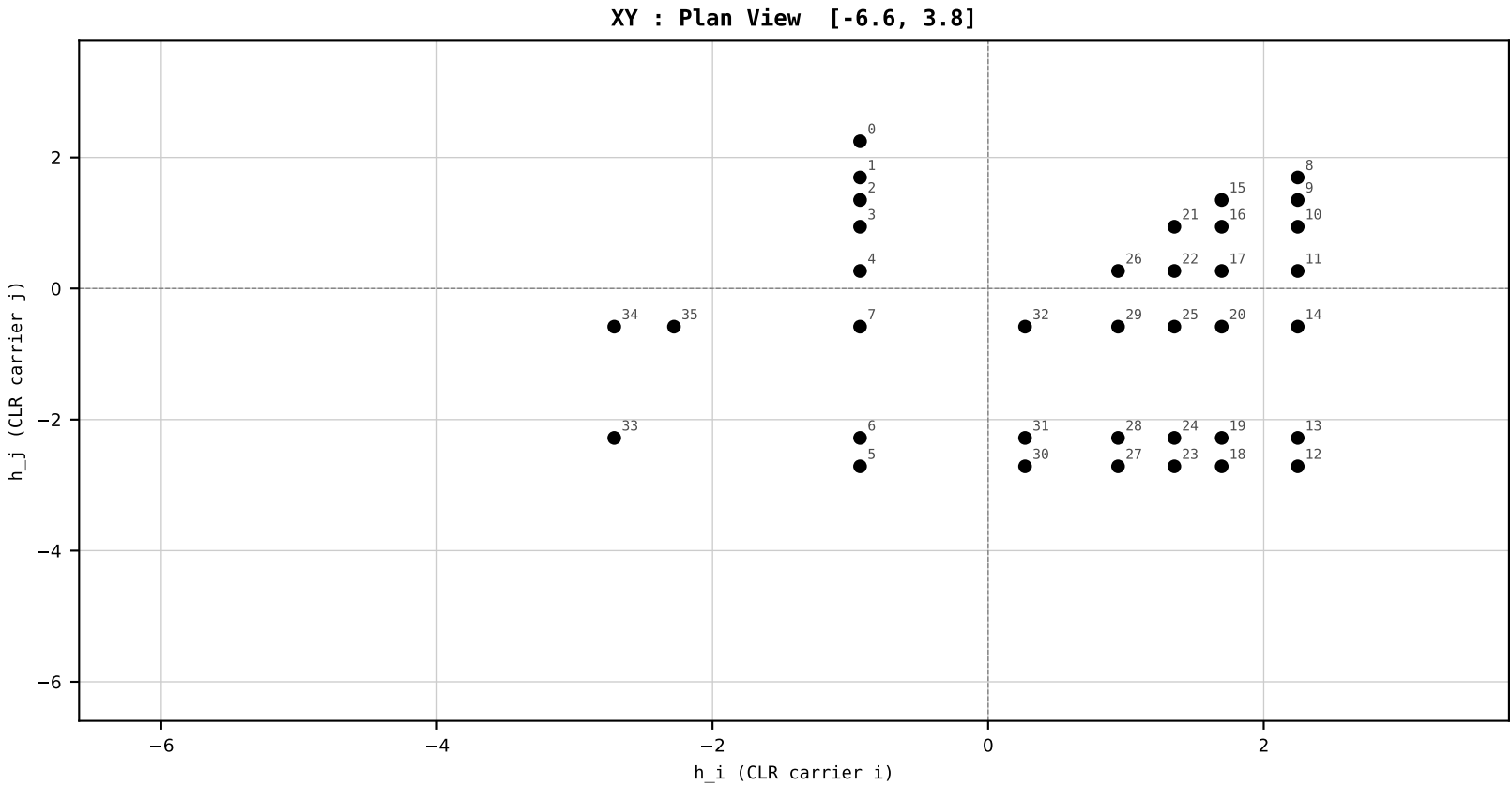
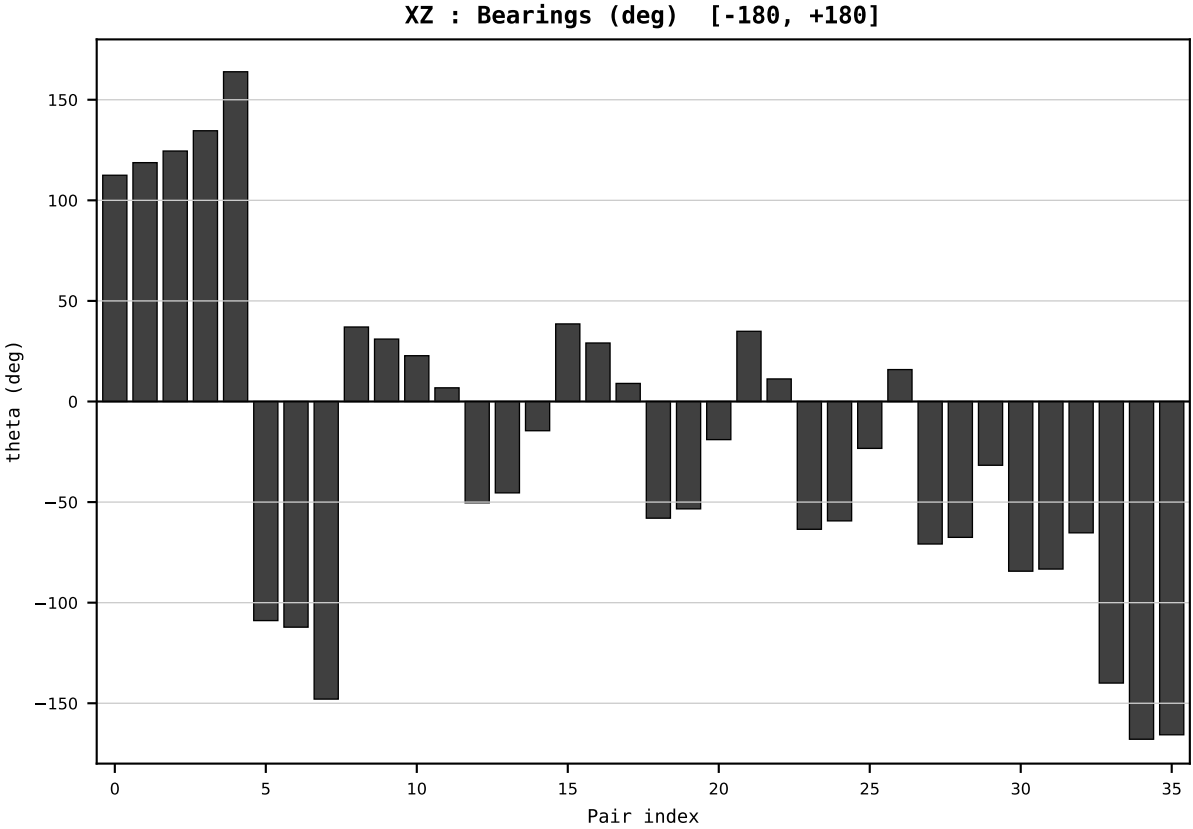
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.273763
Ring = Hs-2
E_metric= 24.4711
kappa_HS= 142.59
omega = 3.6979 deg
d_A = 0.401150
Helm = Solar
Helm d = +0.333042
DR = 4.960 HLR
DR ratio= 142.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



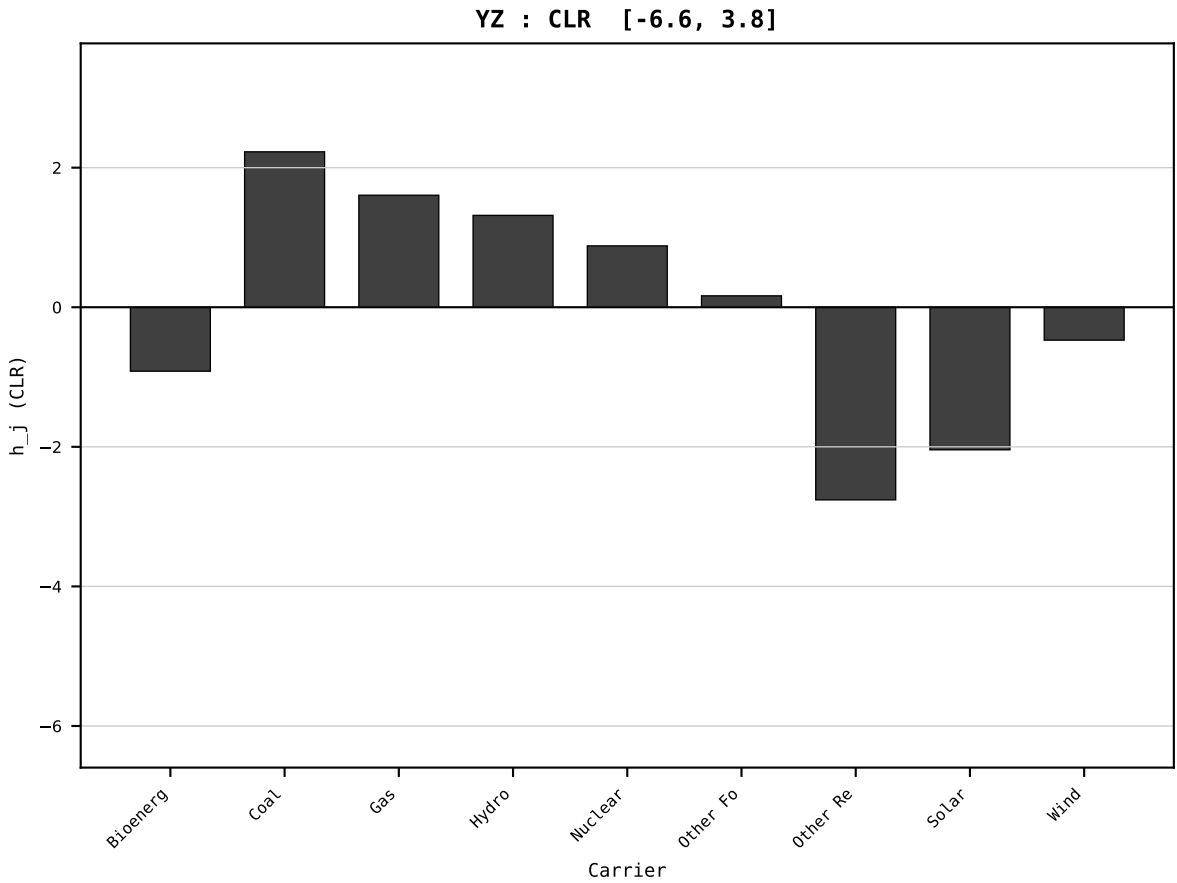
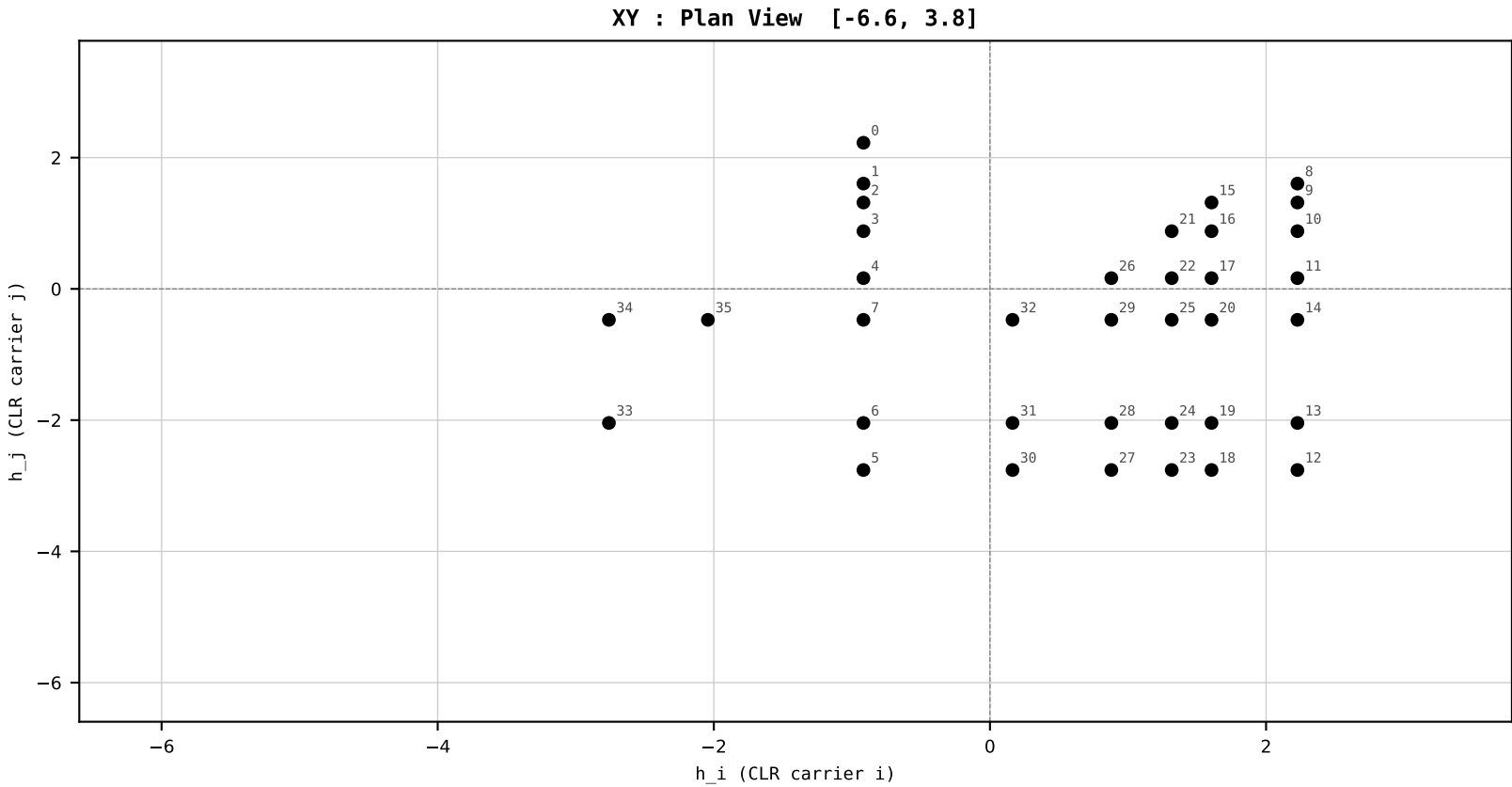
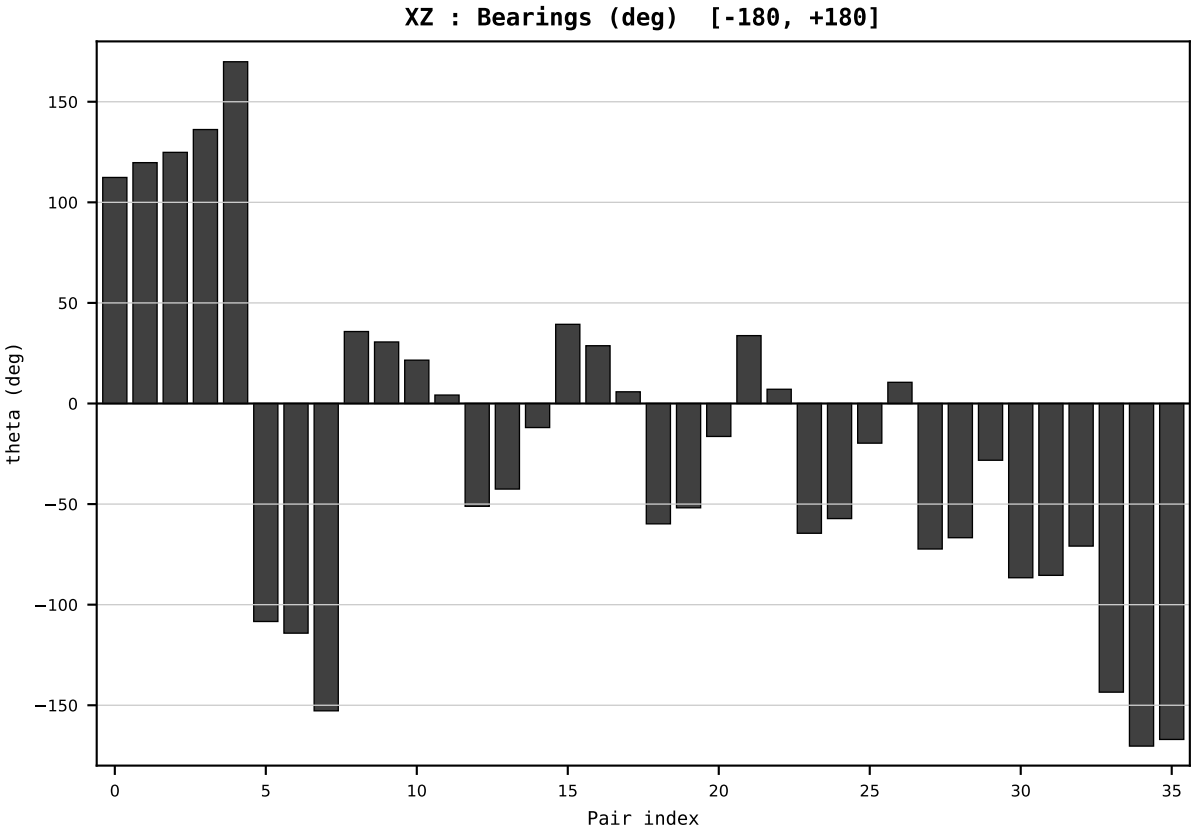
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.271626
Ring = Hs-2
E_metric= 22.9224
kappa_HS= 146.52
omega = 3.1180 deg
d_A = 0.308917
Helm = Solar
Helm d = +0.236796
DR = 4.987 HLR
DR ratio= 146.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



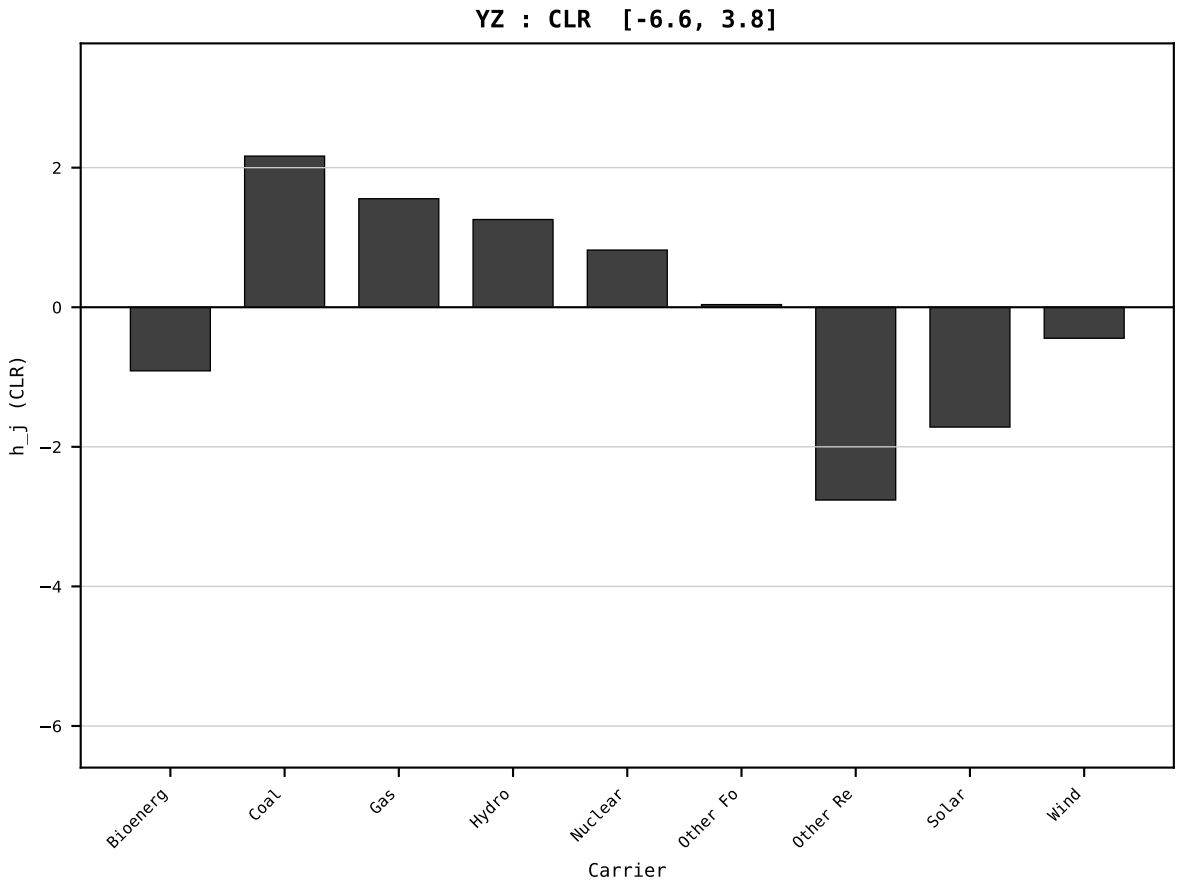
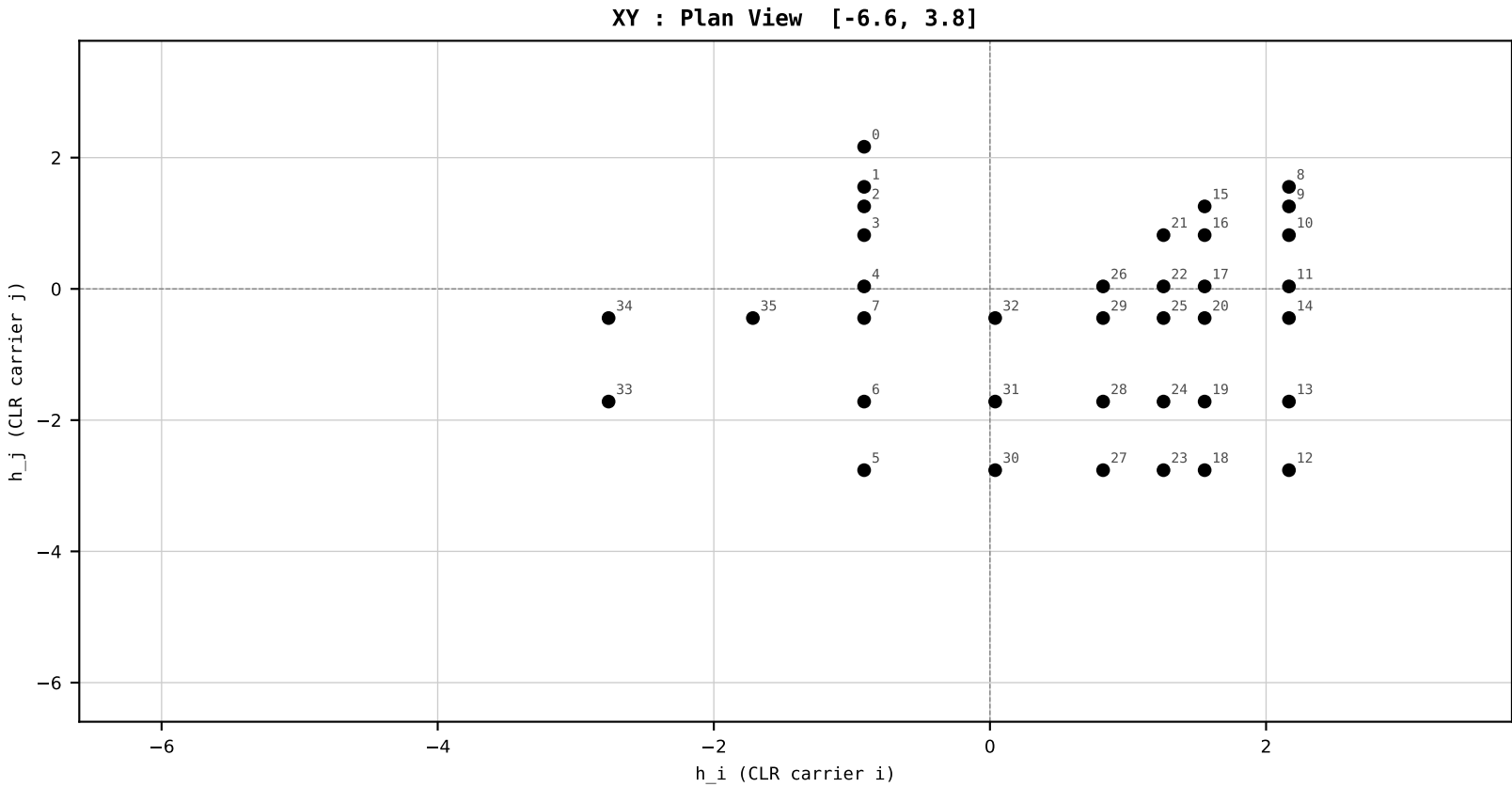
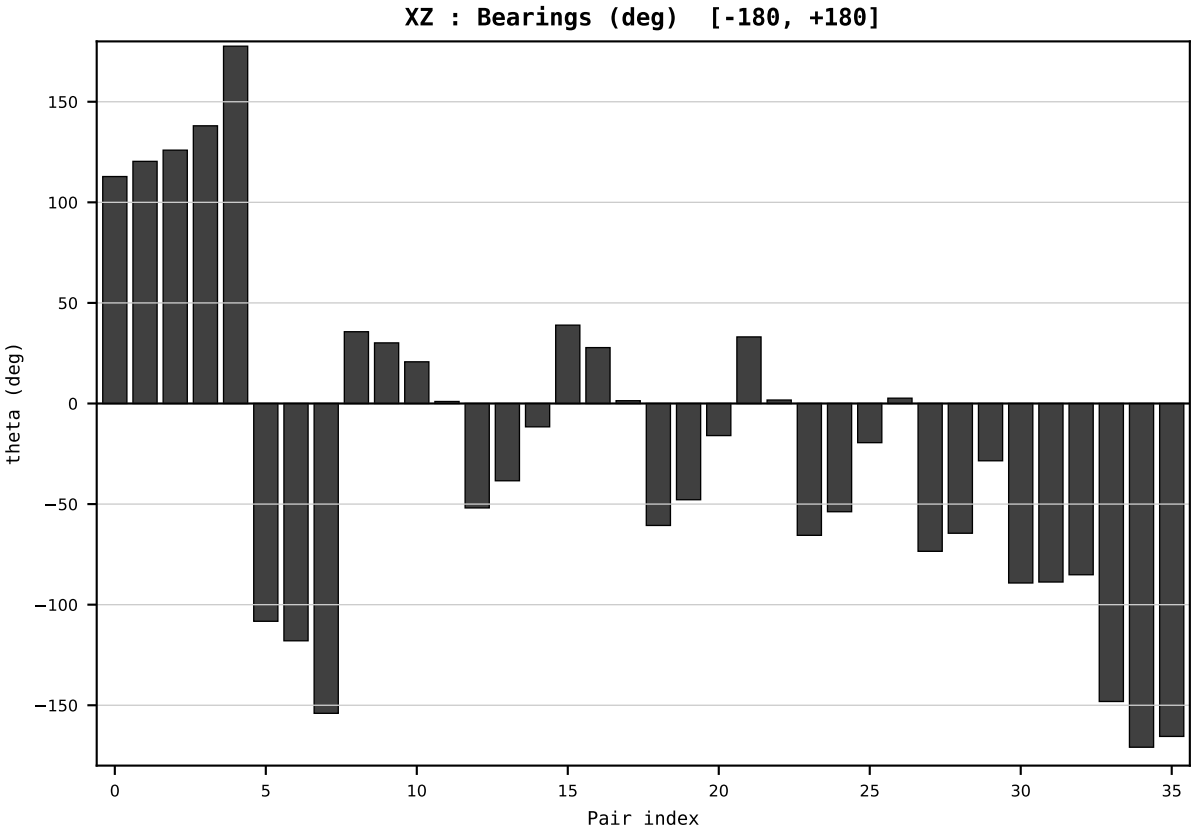
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.265704
Ring = Hs-2
E_metric= 20.9690
kappa_HS= 138.13
omega = 3.7243 deg
d_A = 0.368906
Helm = Solar
Helm d = +0.325856
DR = 4.928 HLR
DR ratio= 138.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



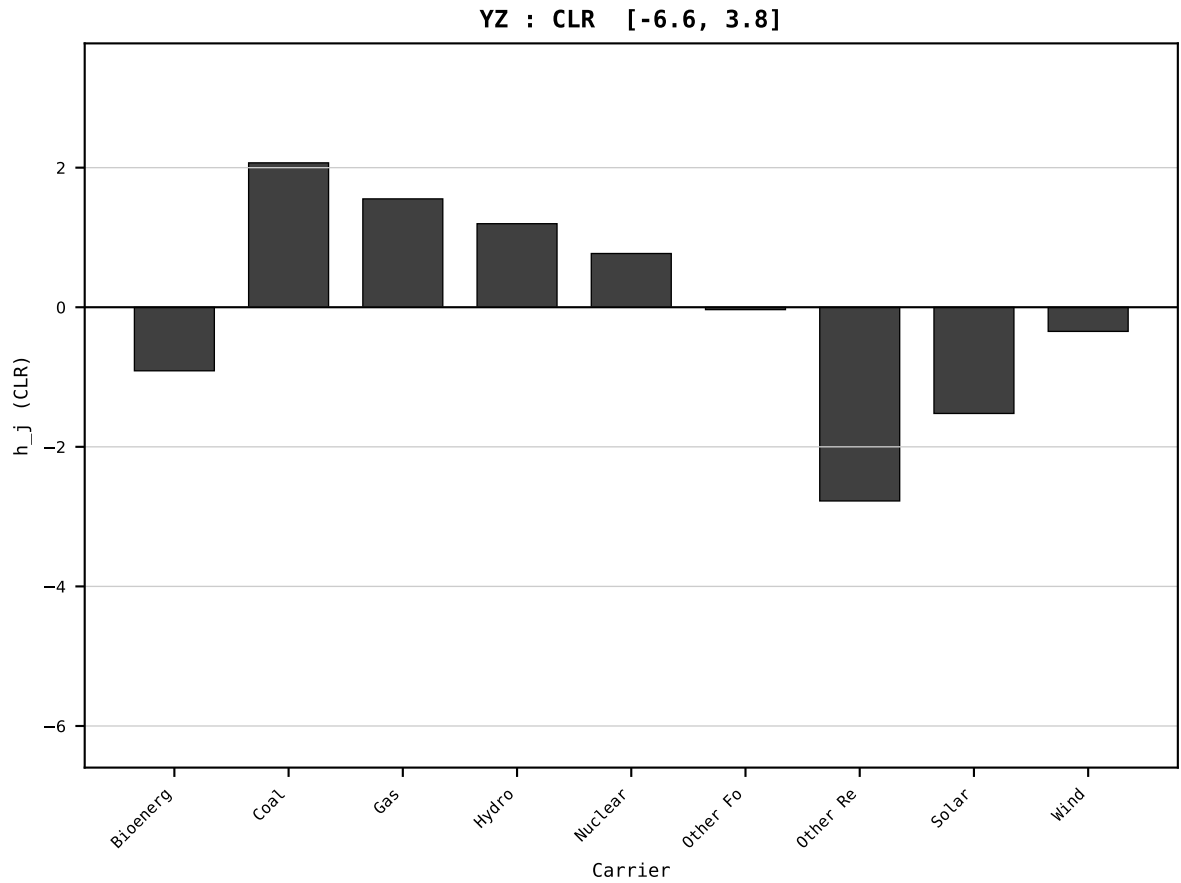
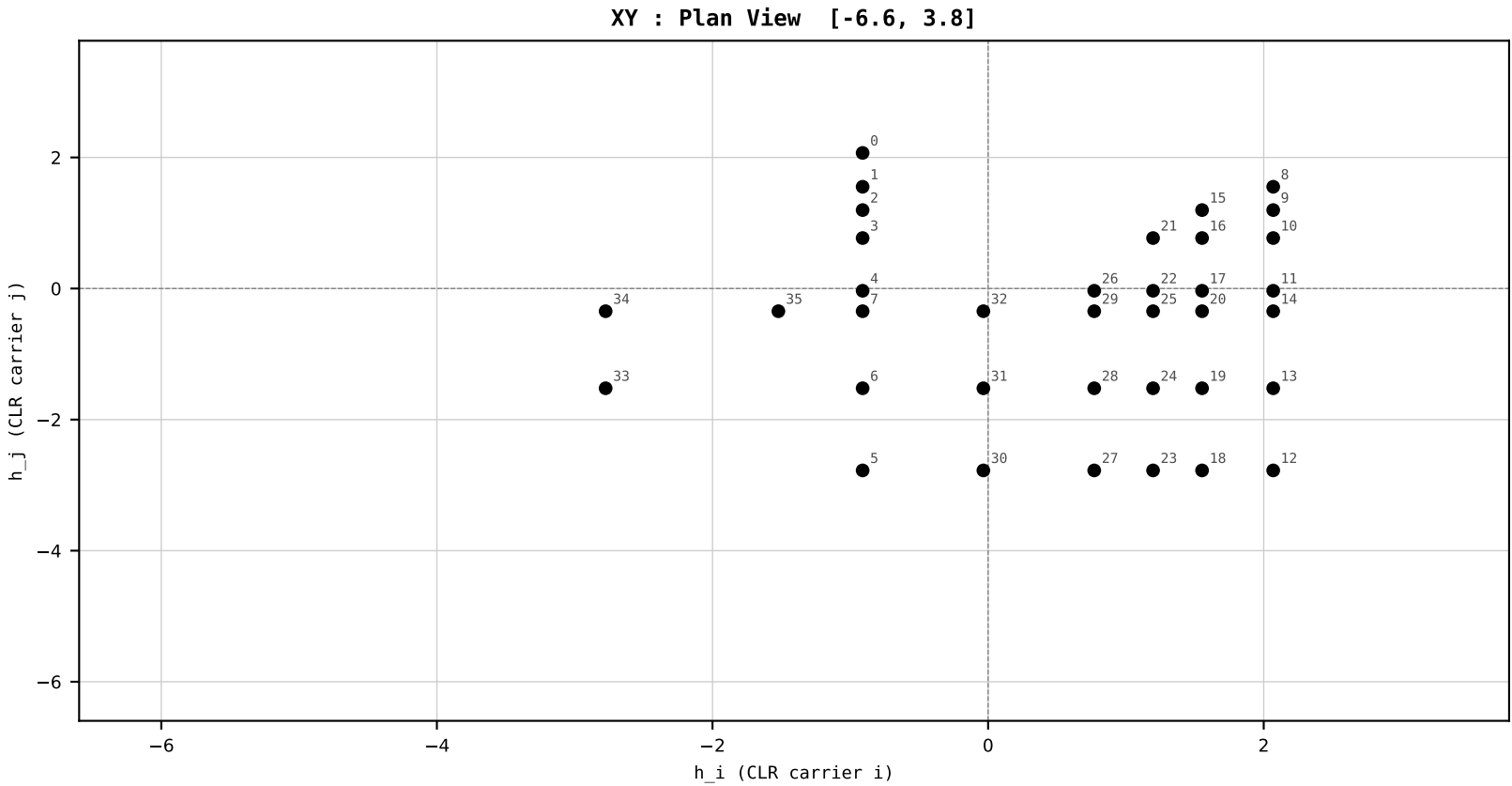
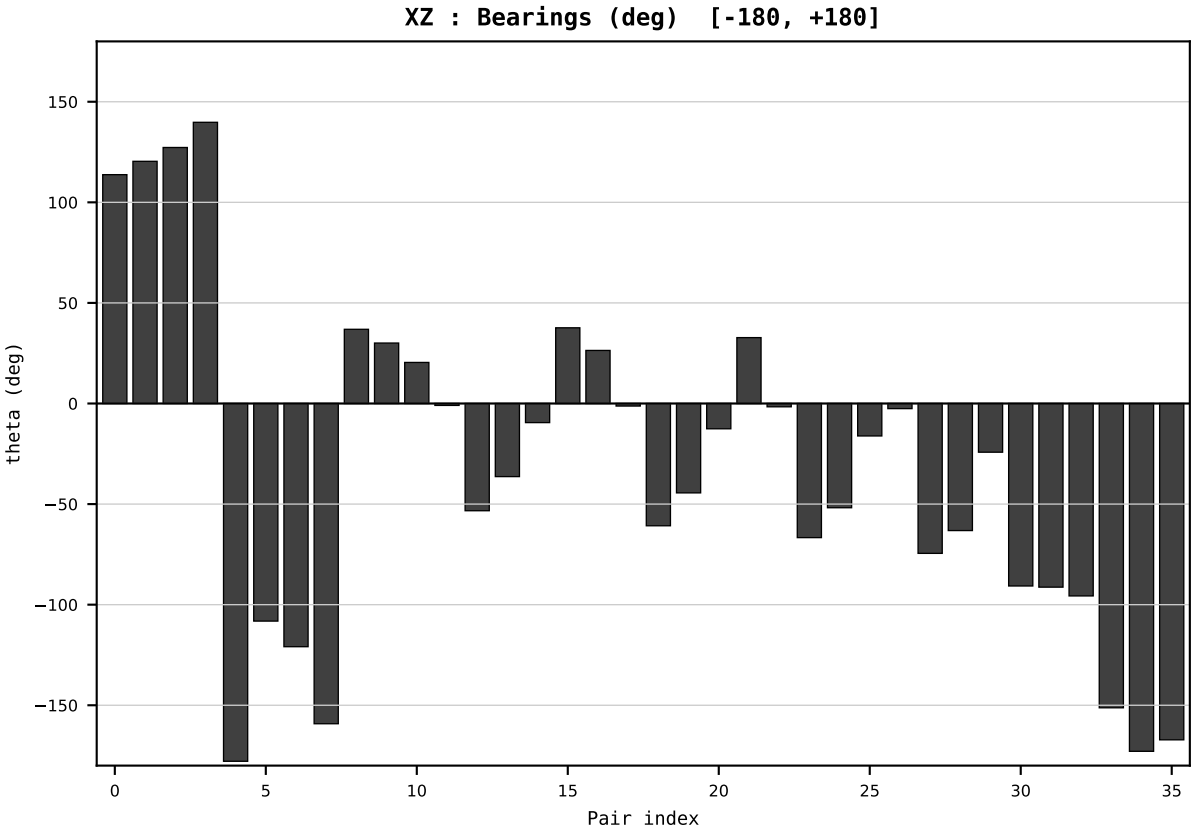
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.252906
Ring = Hs-2
E_metric= 19.6868
kappa_HS= 127.03
omega = 2.7958 deg
d_A = 0.261905
Helm = Solar
Helm d = +0.195252
DR = 4.844 HLR
DR ratio= 127.0x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



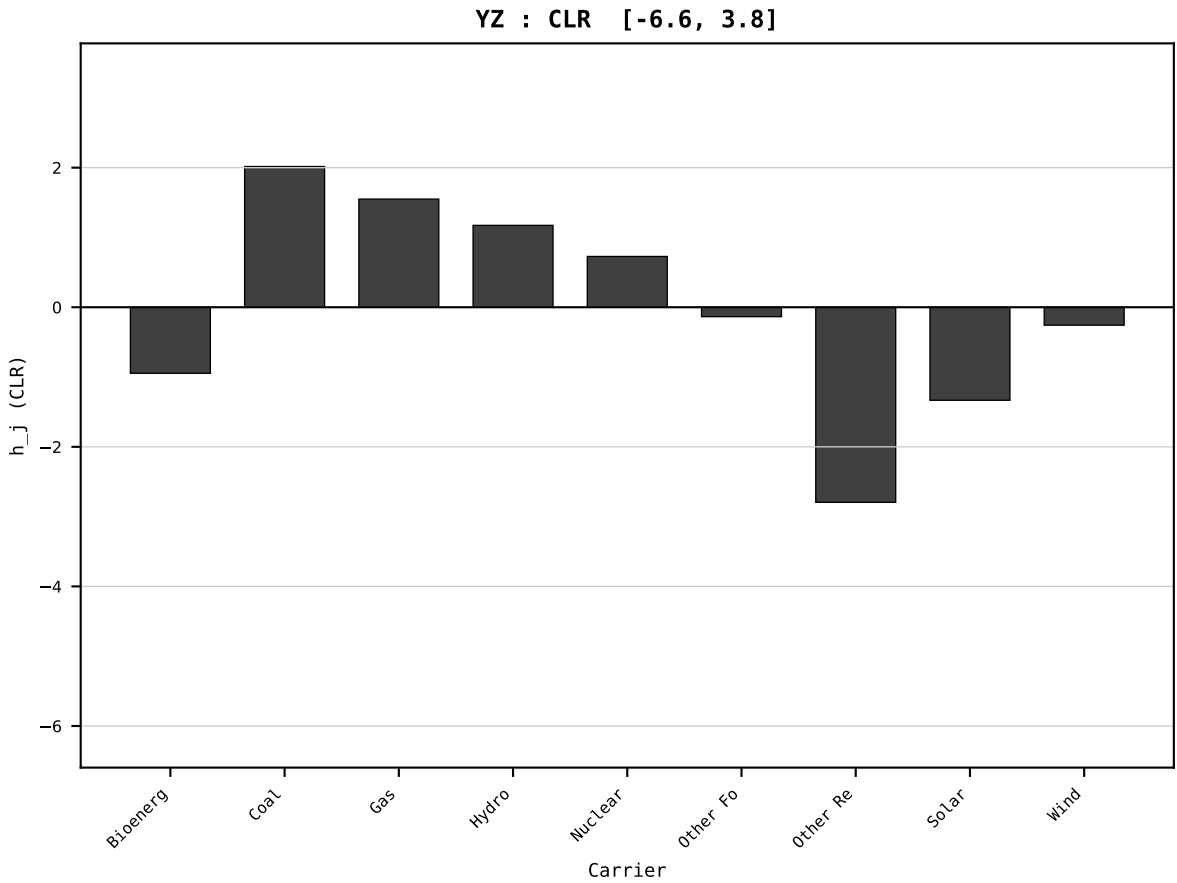
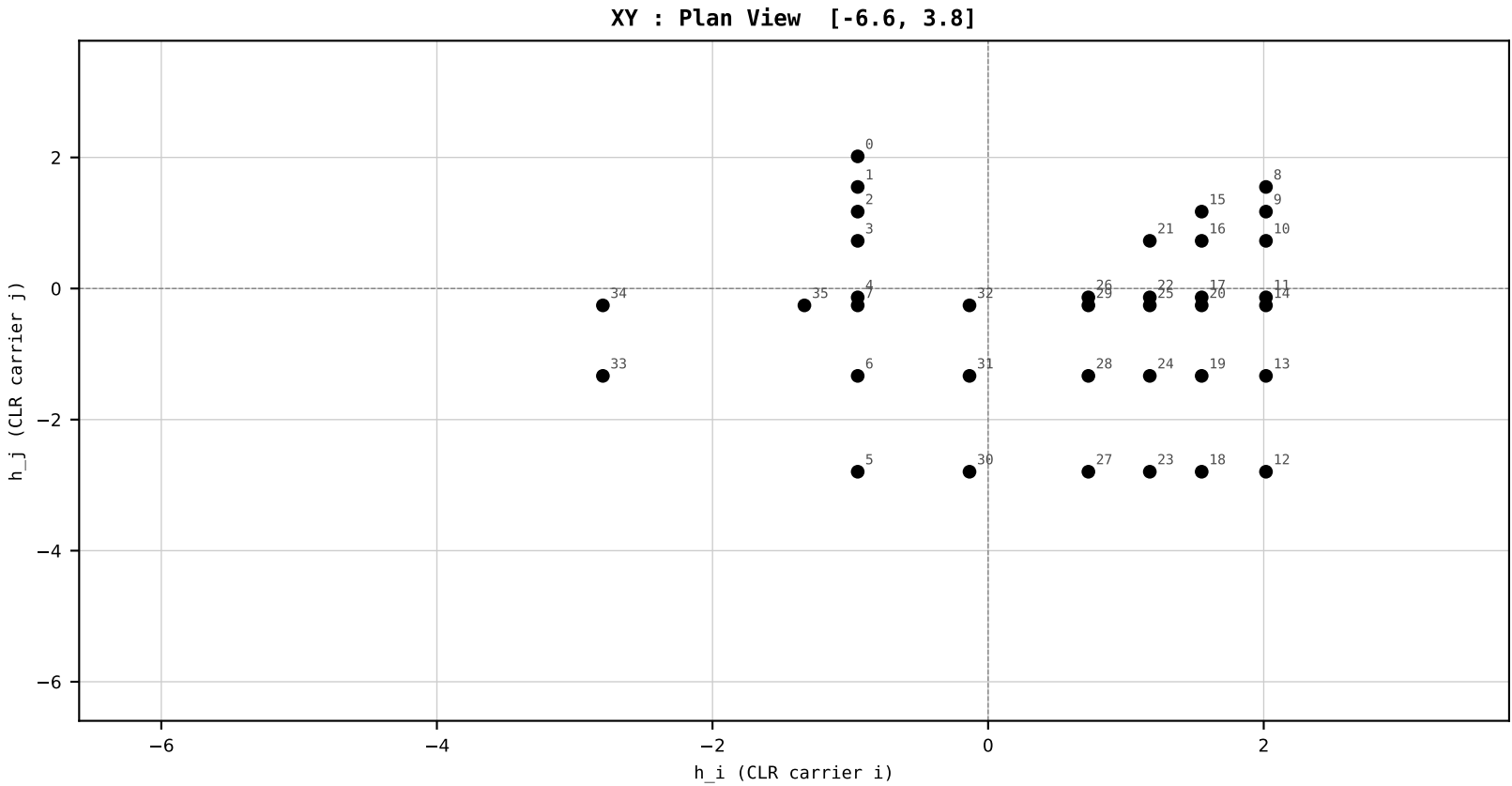
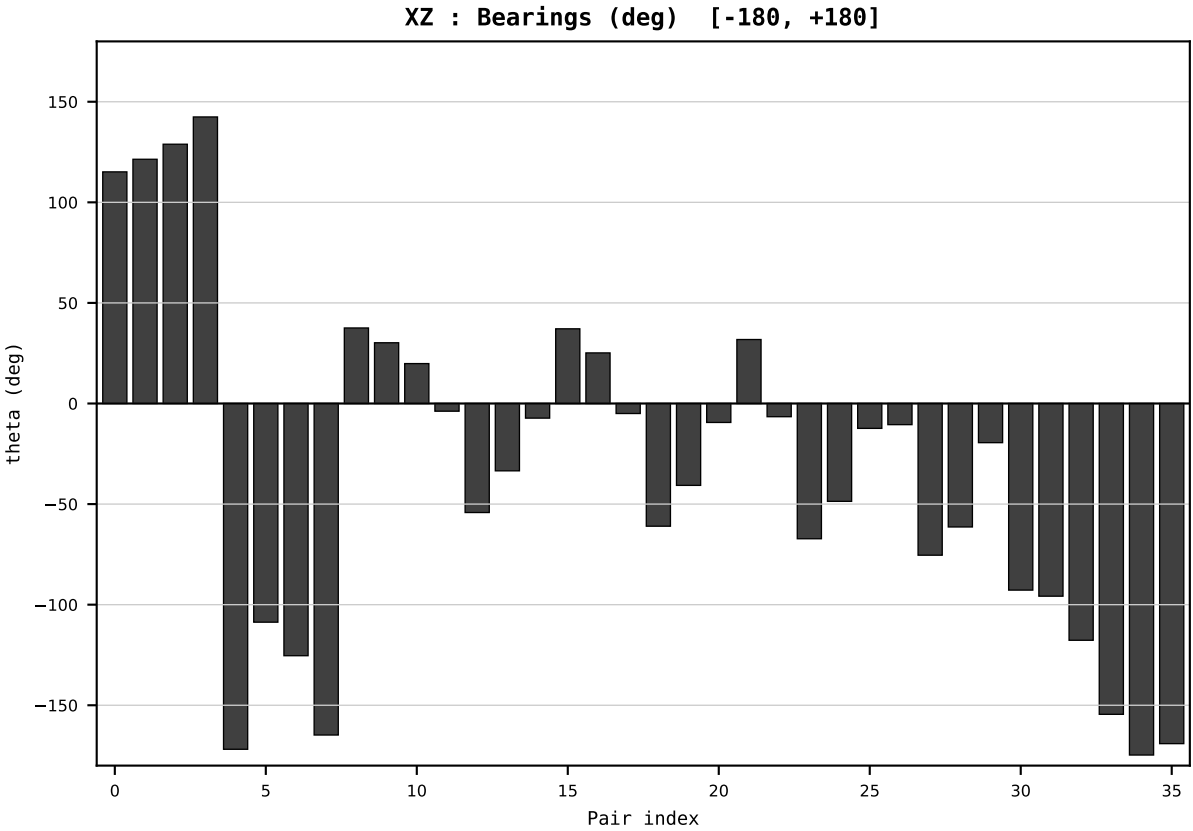
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.246638
Ring = Hs-2
E_metric= 18.9446
kappa_HS= 122.95
omega = 3.0133 deg
d_A = 0.246037
Helm = Solar
Helm d = +0.188869
DR = 4.812 HLR
DR ratio= 122.9x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

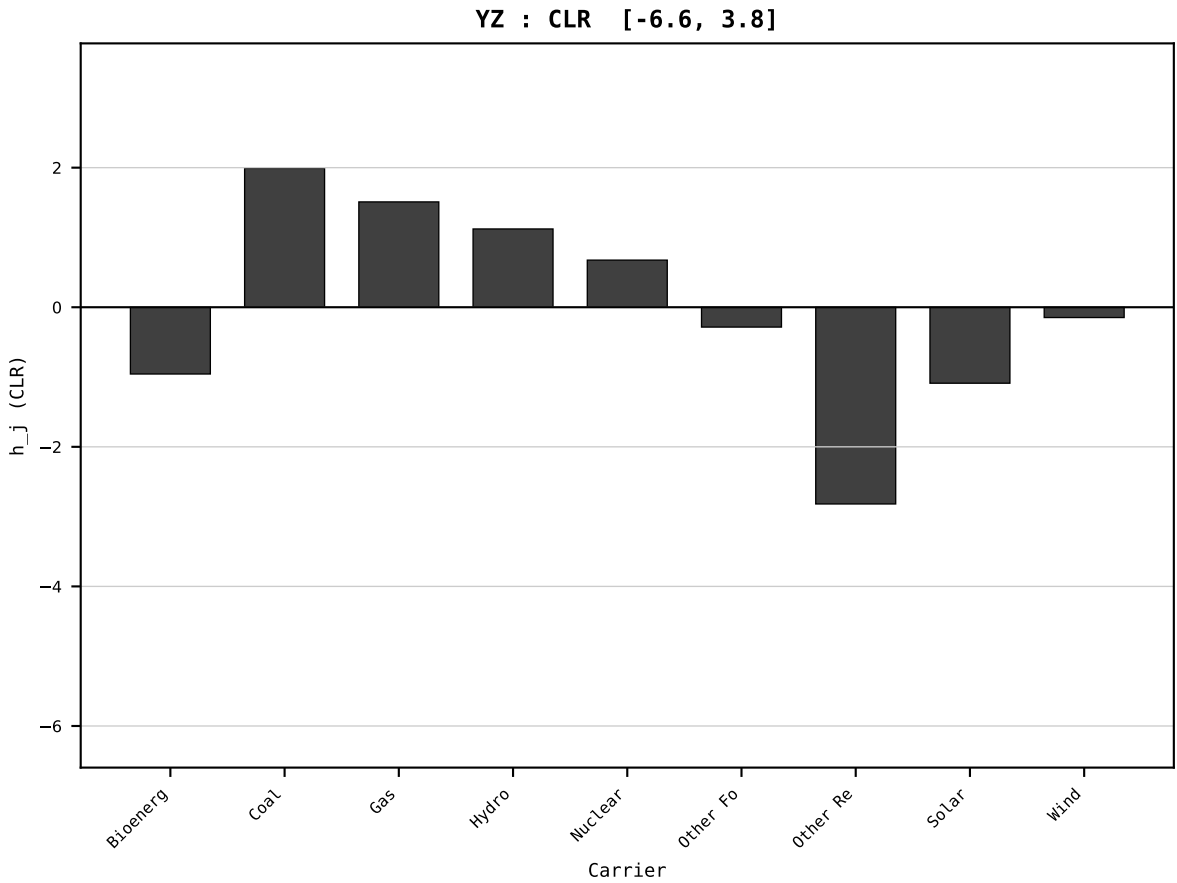
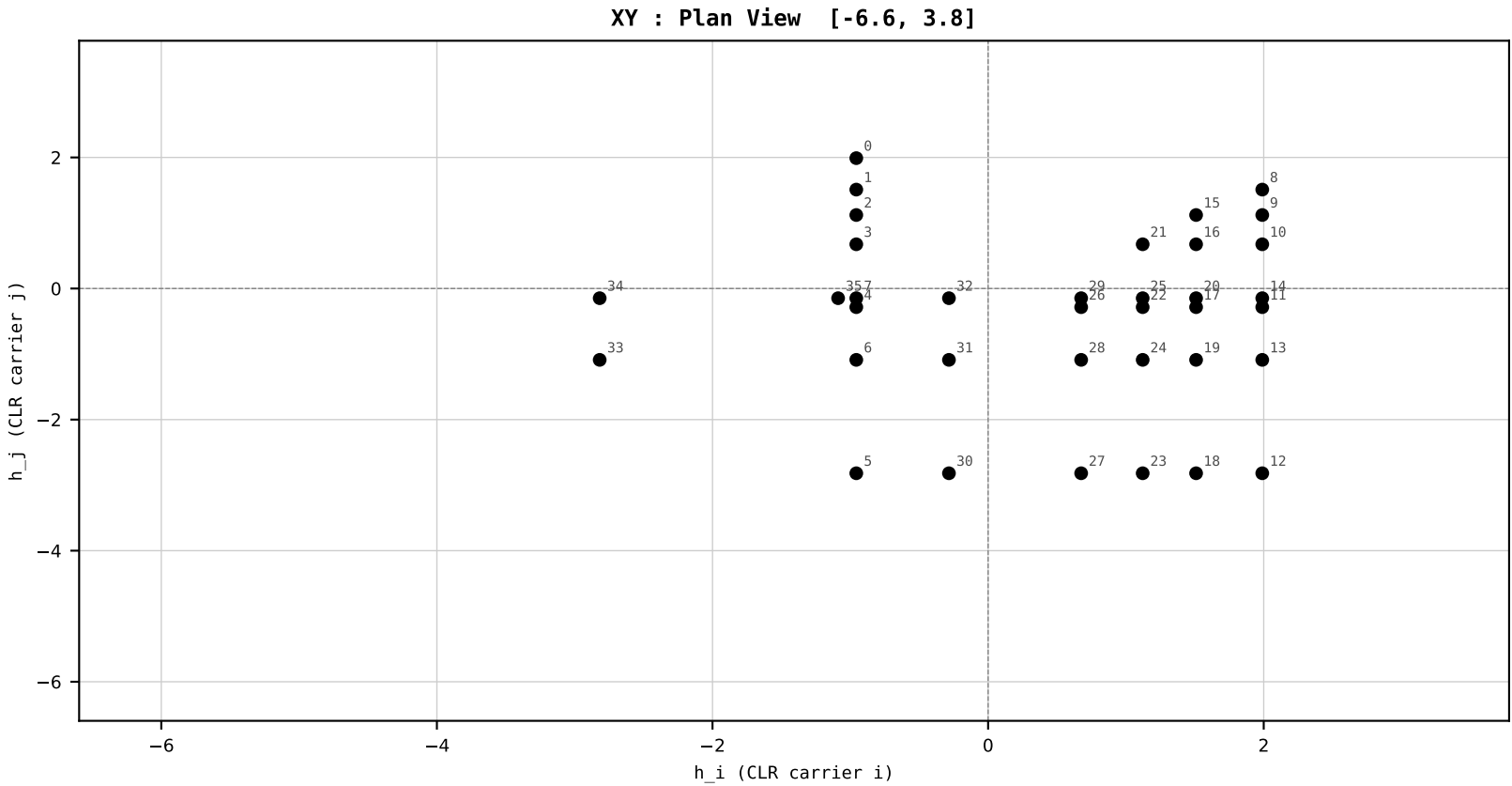
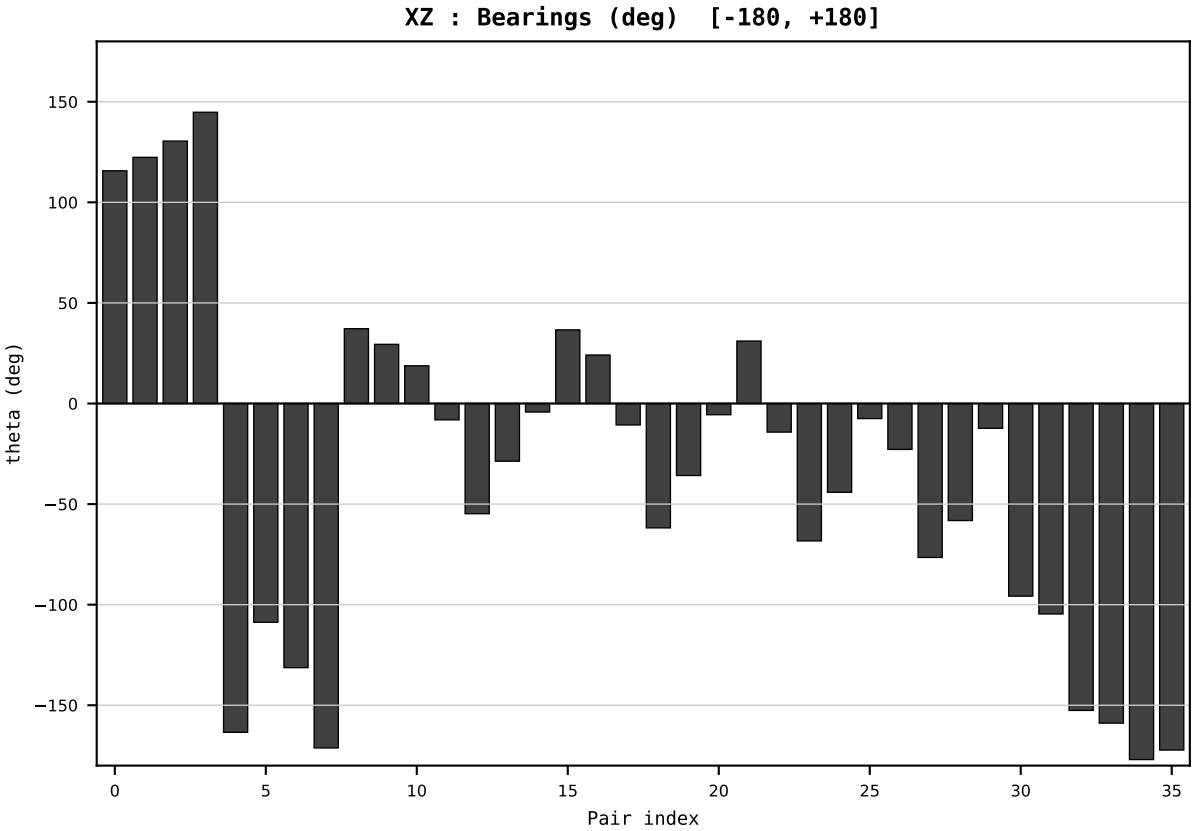
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.241826
Ring = Hs-2
E_metric= 18.0970
kappa_HS= 122.49
omega = 4.0572 deg
d_A = 0.320167
Helm = Solar
Helm d = +0.244504
DR = 4.808 HLR
DR ratio= 122.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



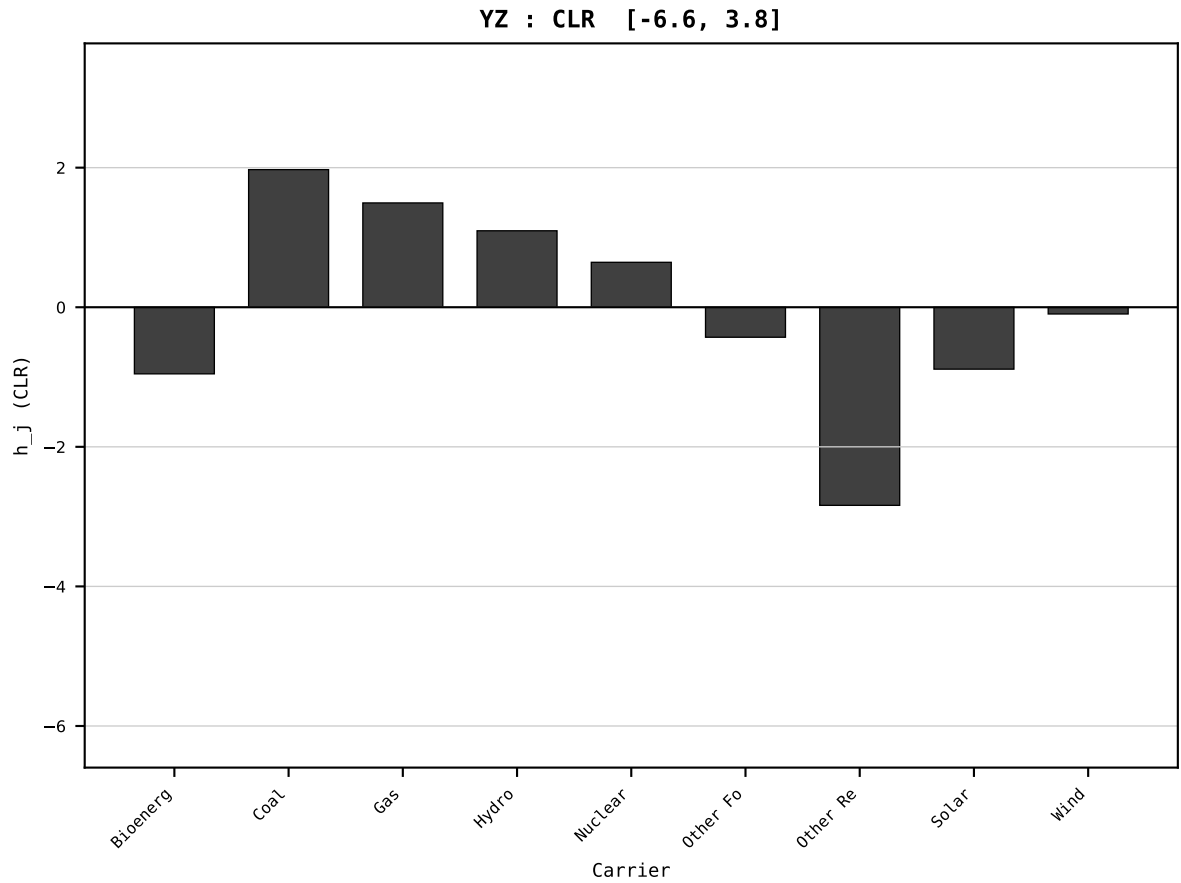
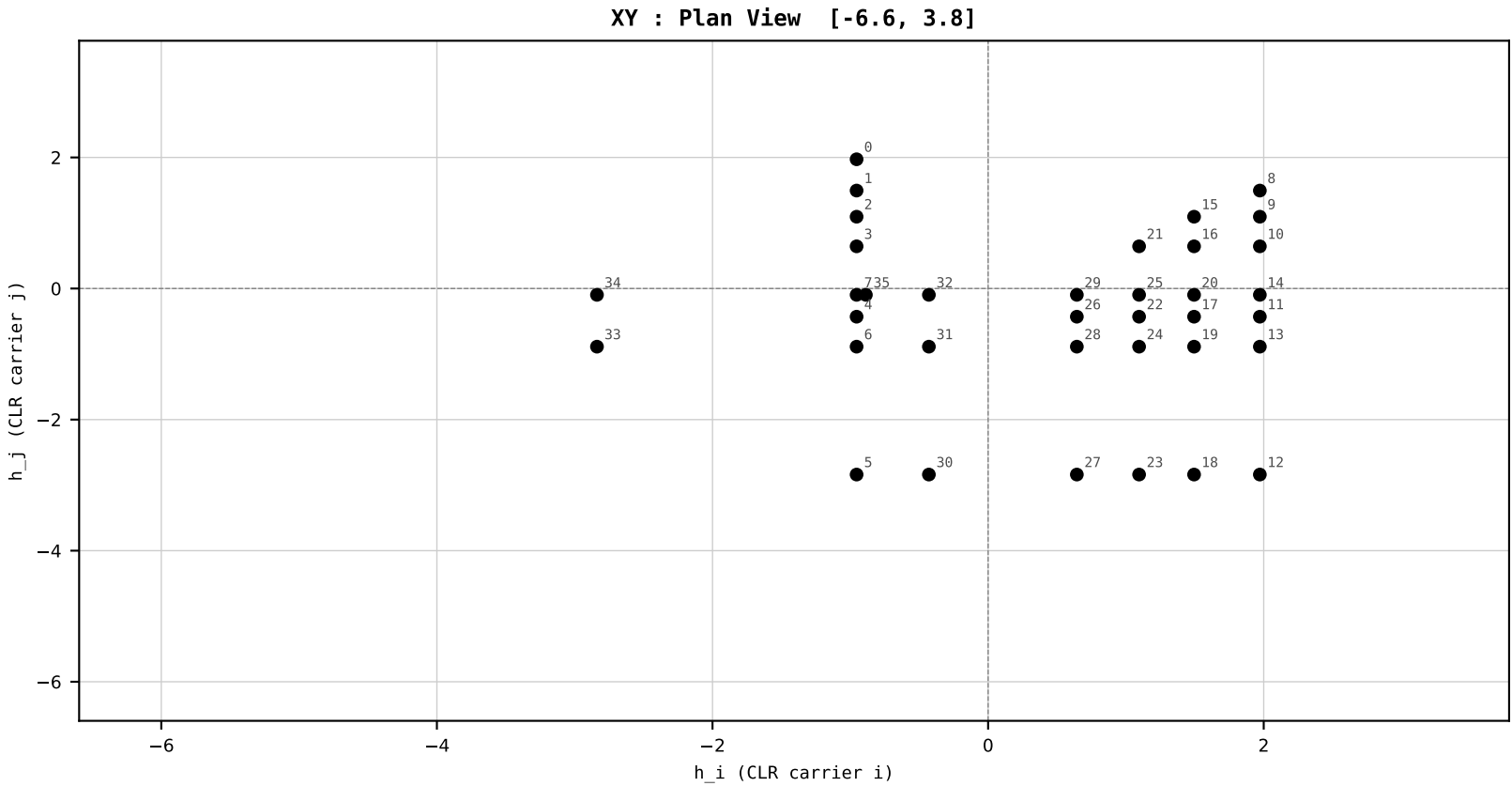
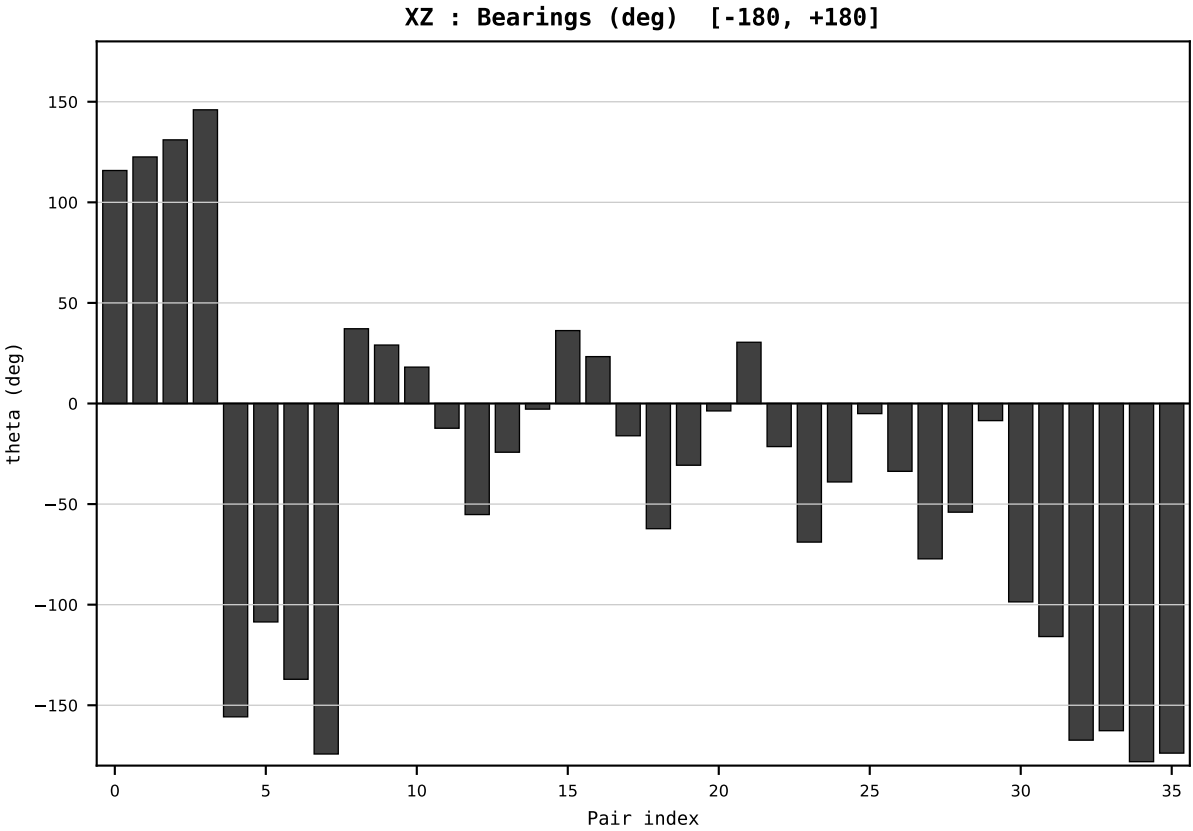
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.239063
Ring = Hs-2
E_metric= 17.6853
kappa_HS= 122.77
omega = 3.4435 deg
d_A = 0.258782
Helm = Solar
Helm d = +0.201485
DR = 4.810 HLR
DR ratio= 122.8x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



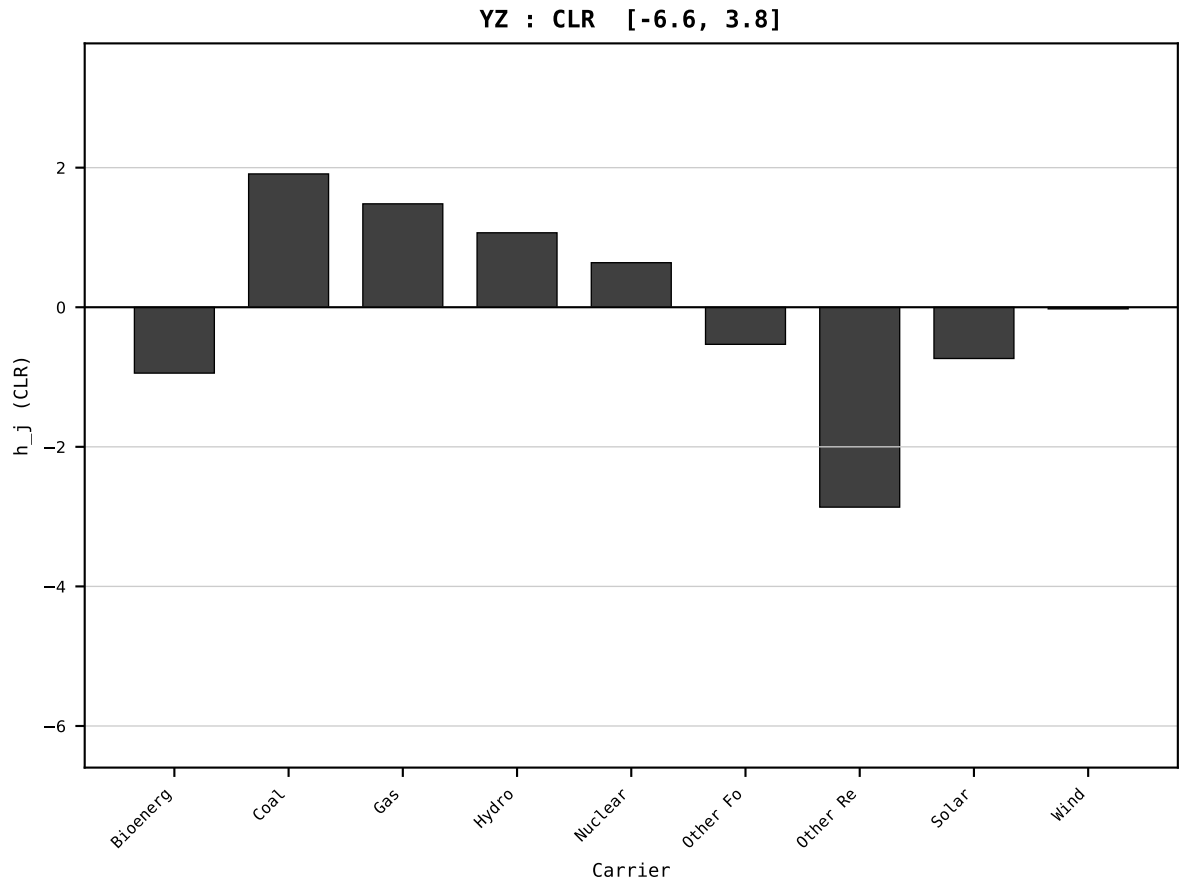
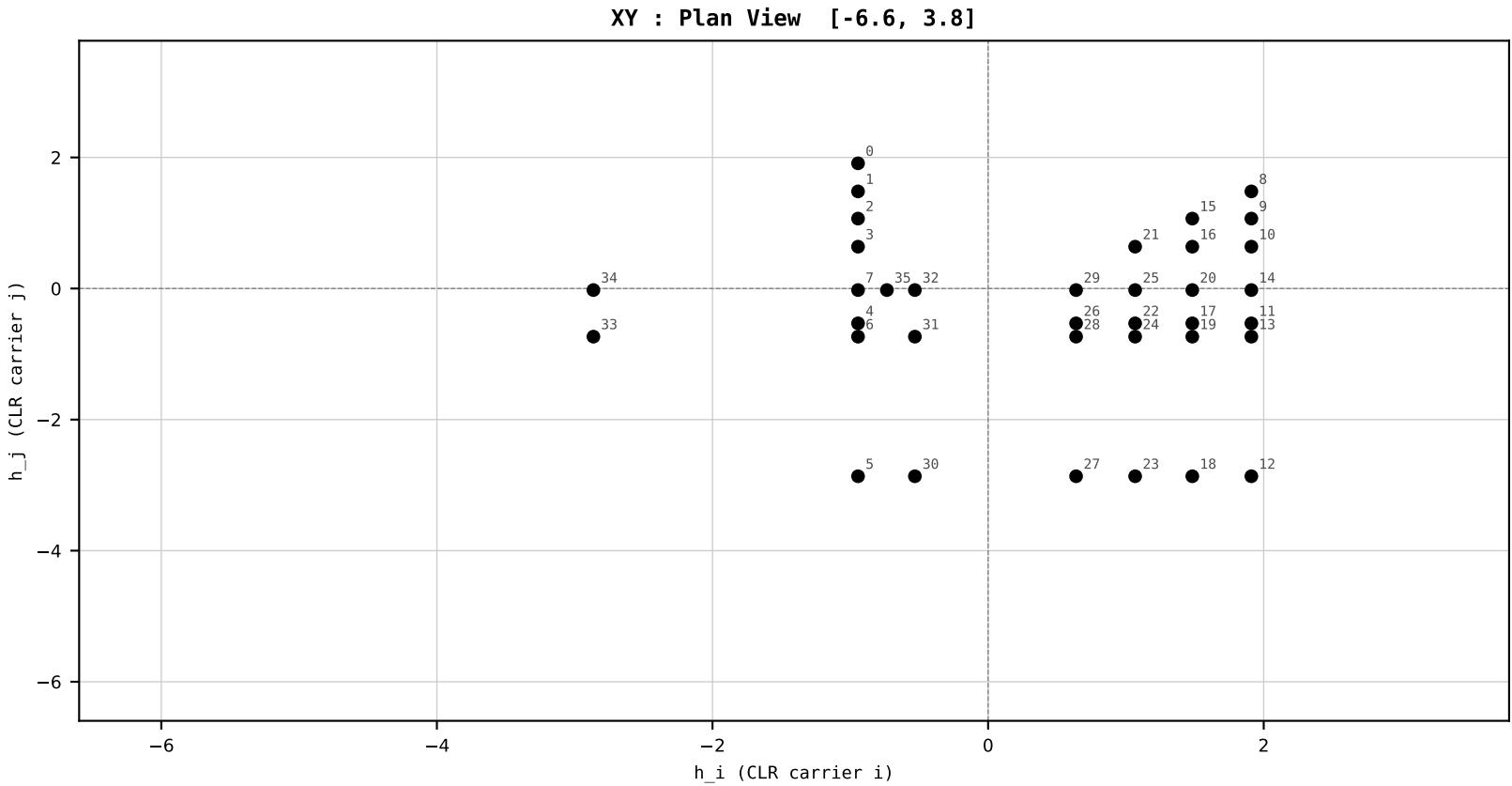
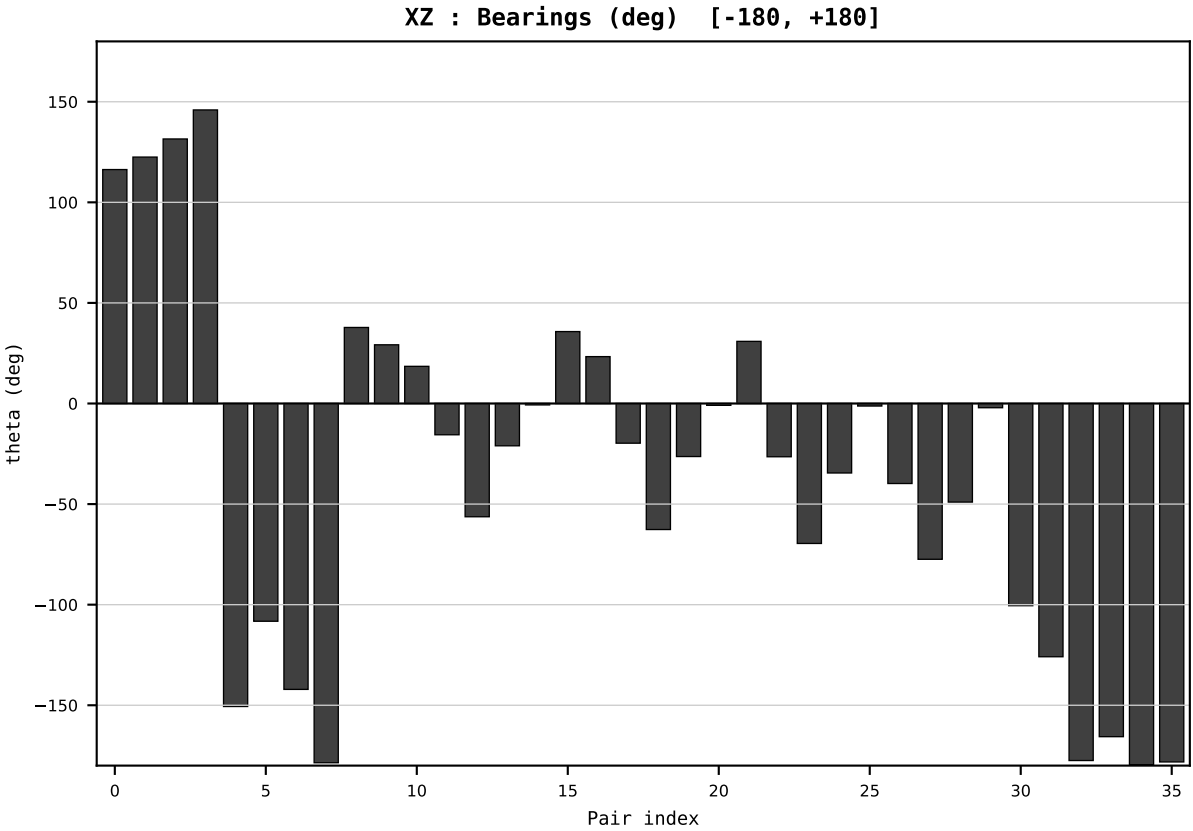
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.228939
Ring = Hs-2
E_metric= 17.3079
kappa_HS= 118.46
omega = 2.8245 deg
d_A = 0.211059
Helm = Solar
Helm d = +0.152684
DR = 4.775 HLR
DR ratio= 118.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

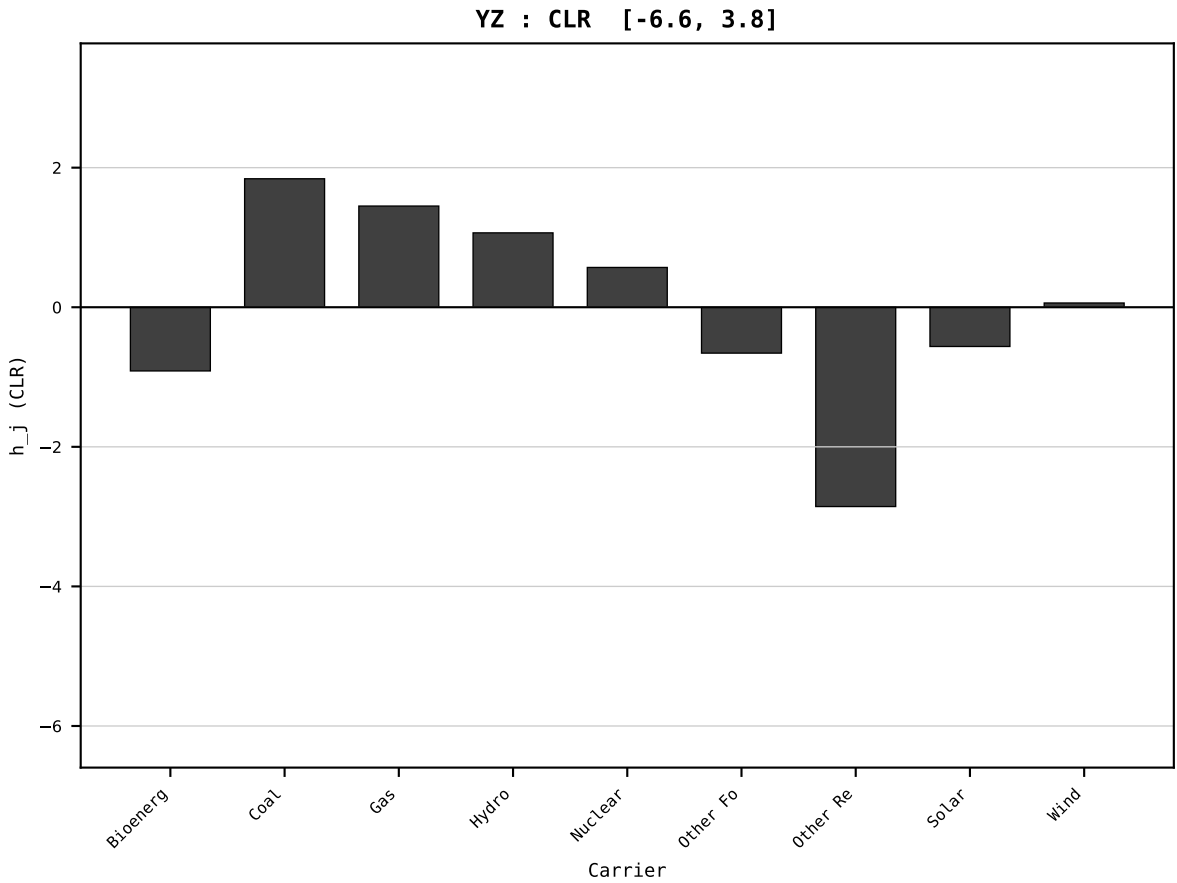
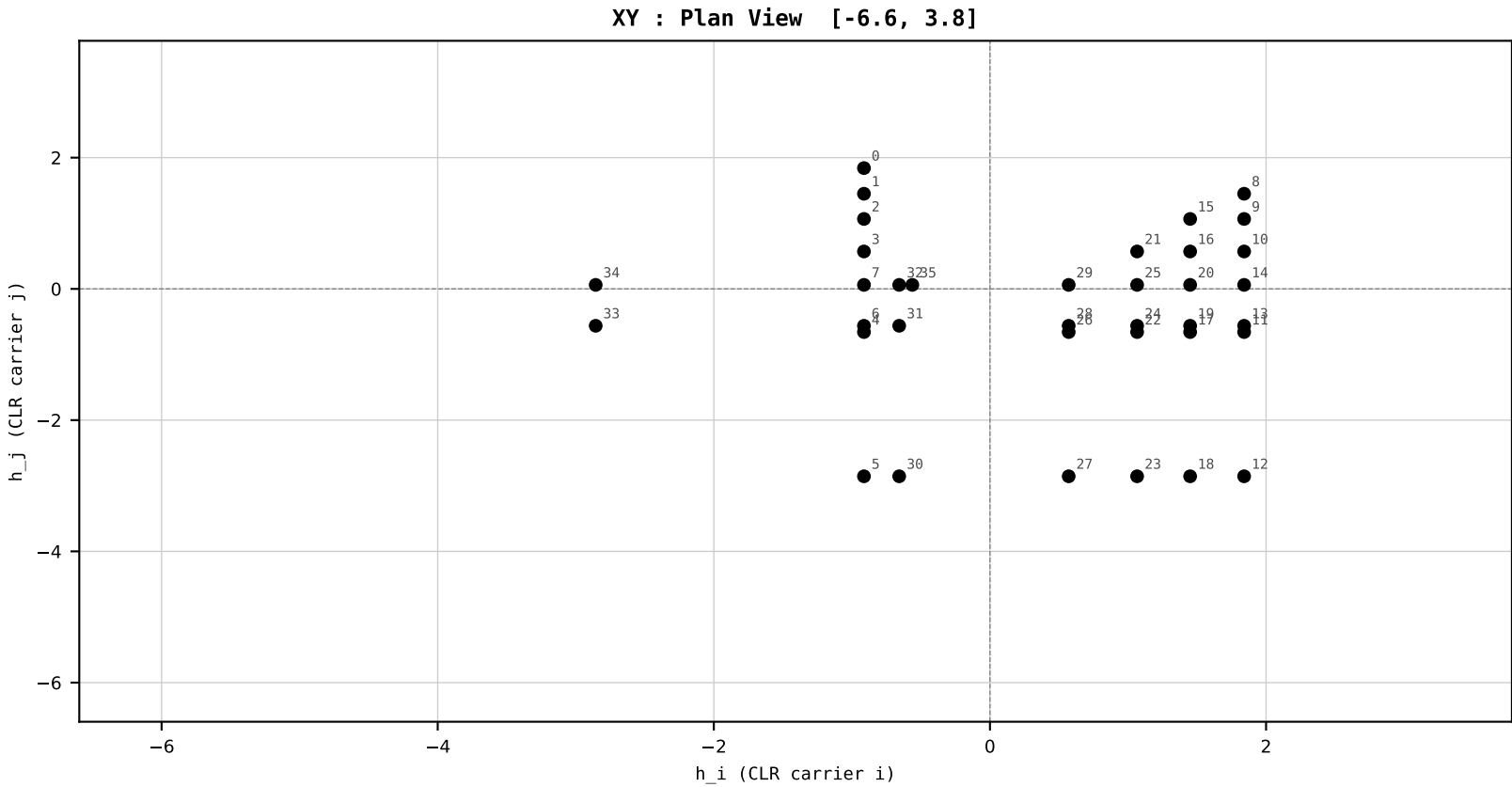
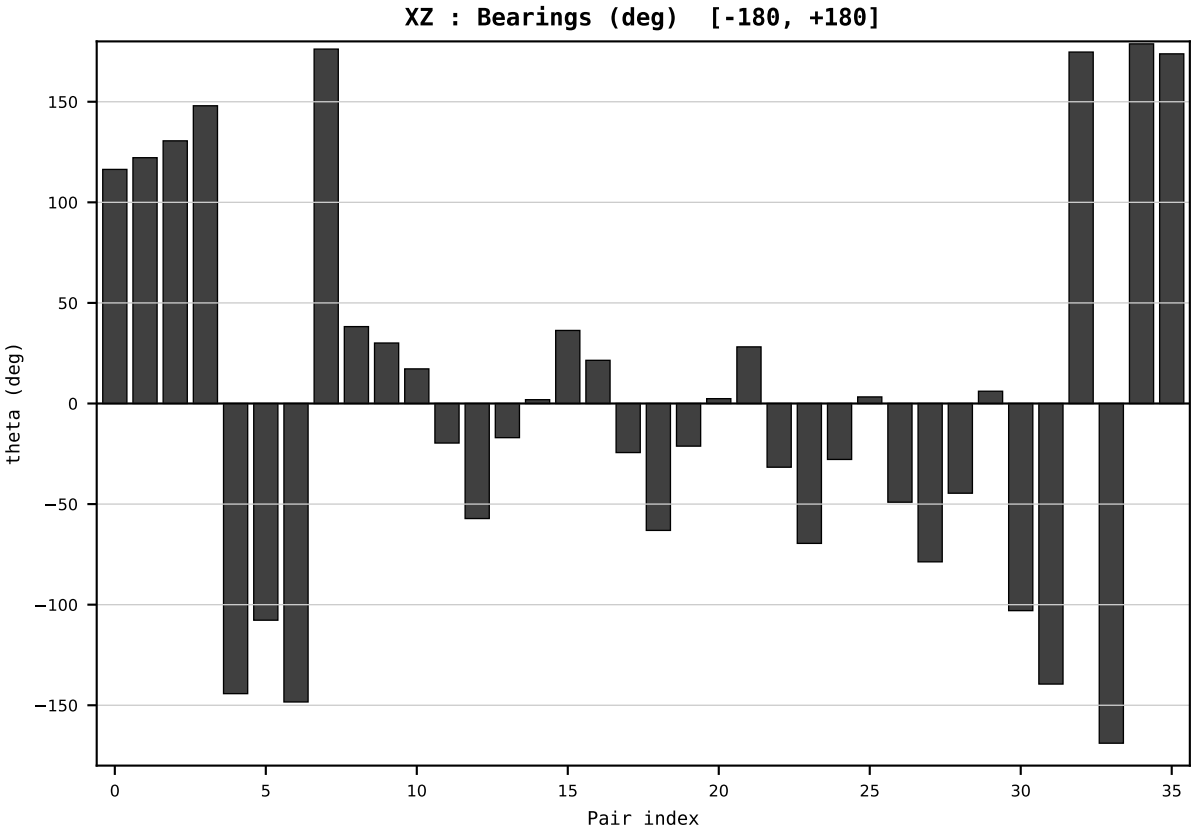
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.217796
Ring = Hs-2
E_metric= 16.6924
kappa_HS= 109.58
omega = 3.3646 deg
d_A = 0.253314
Helm = Solar
Helm d = +0.172148
DR = 4.697 HLR
DR ratio= 109.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

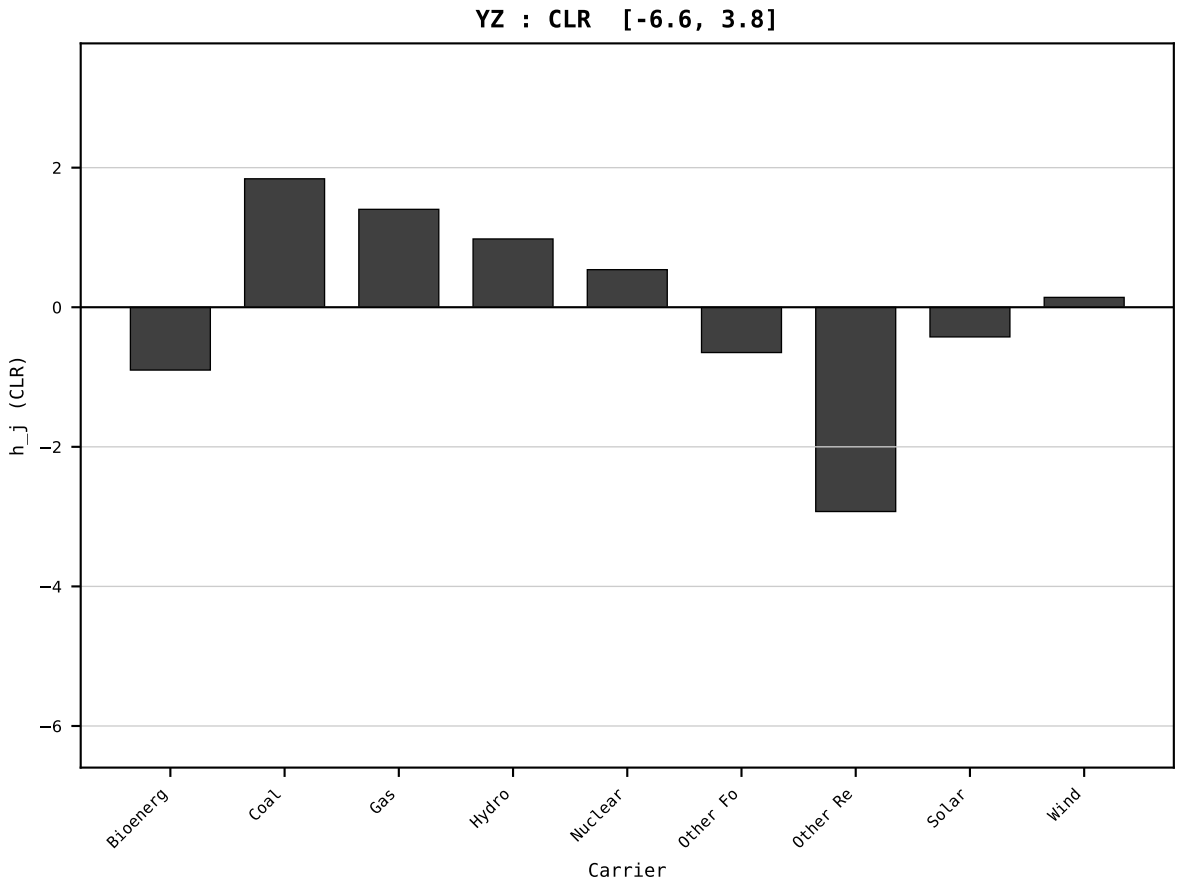
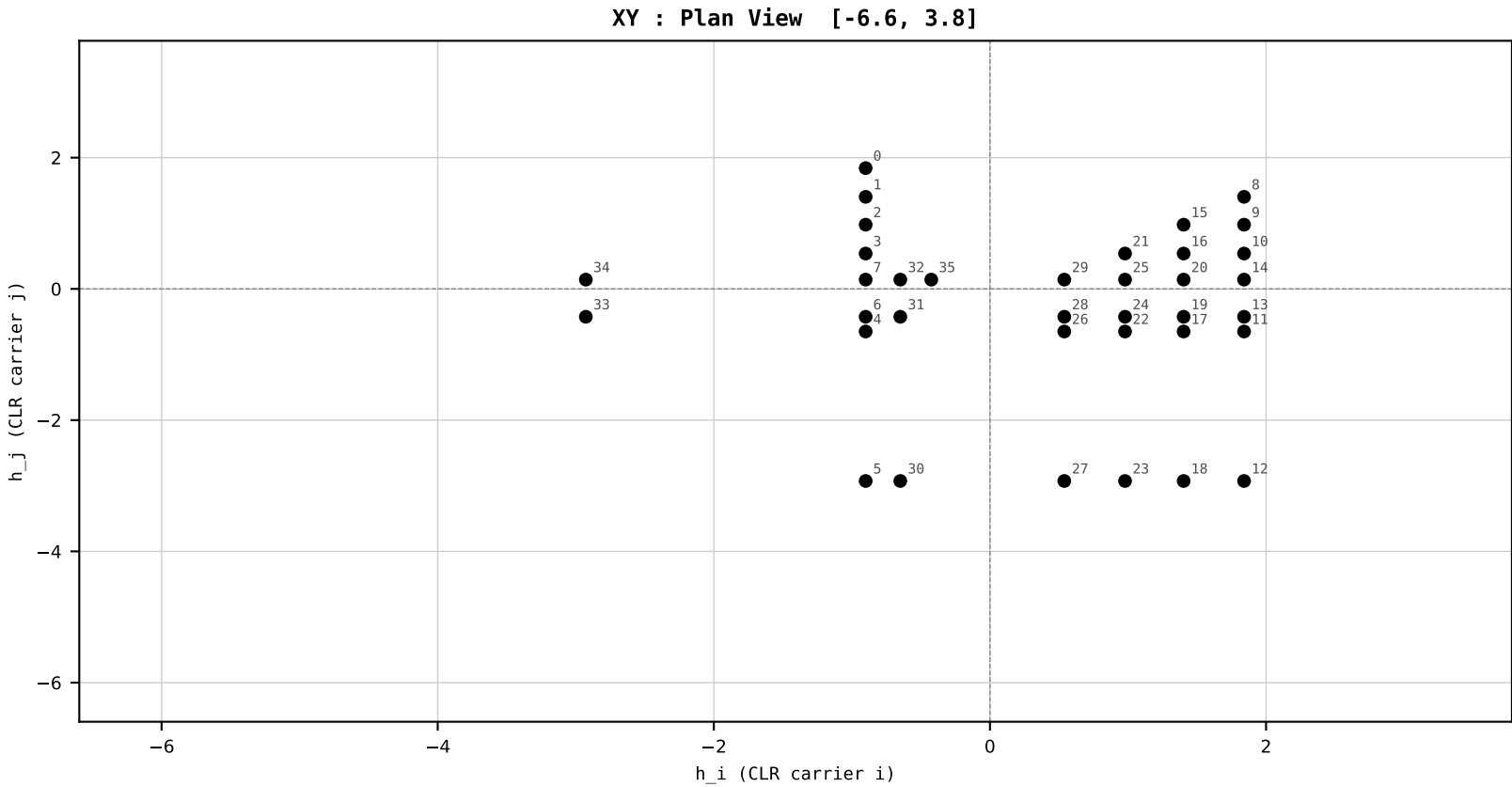
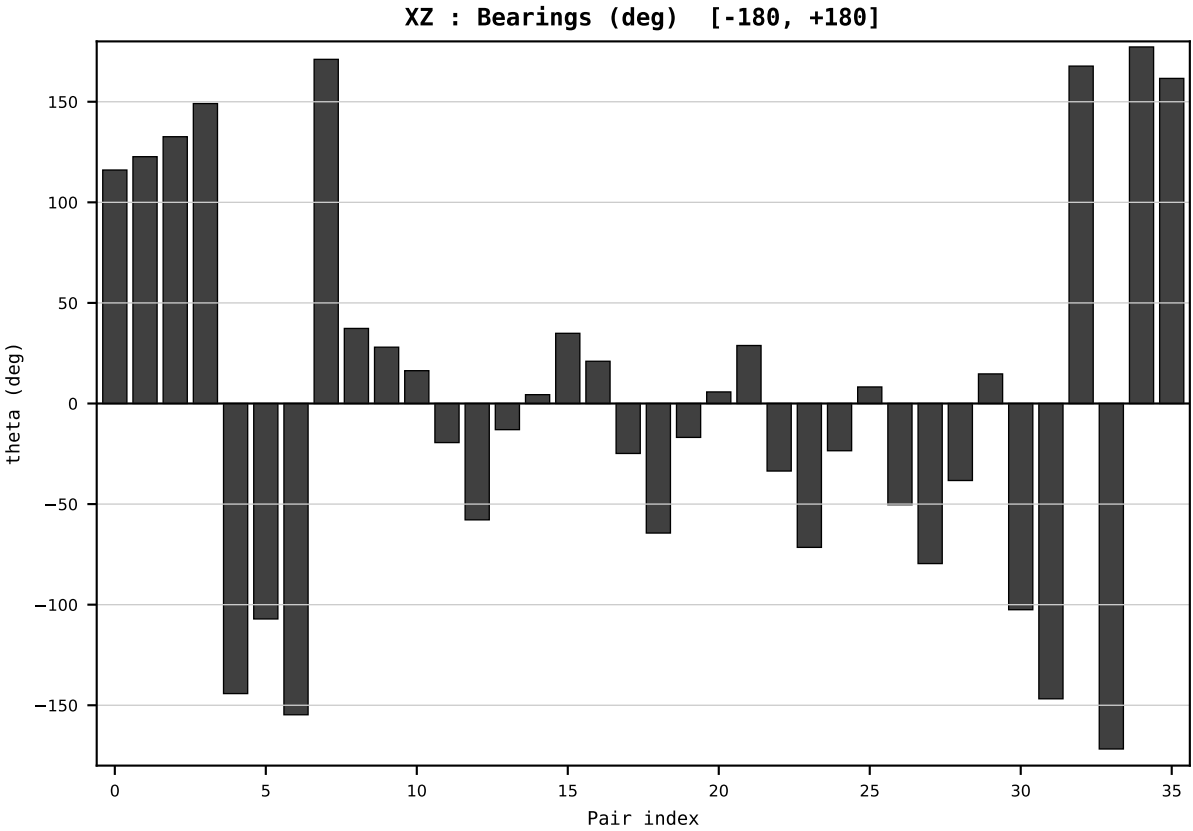
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.211988
Ring = Hs-2
E_metric= 16.6038
kappa_HS= 117.65
omega = 2.8560 deg
d_A = 0.203654
Helm = Solar
Helm d = +0.137219
DR = 4.768 HLR
DR ratio= 117.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



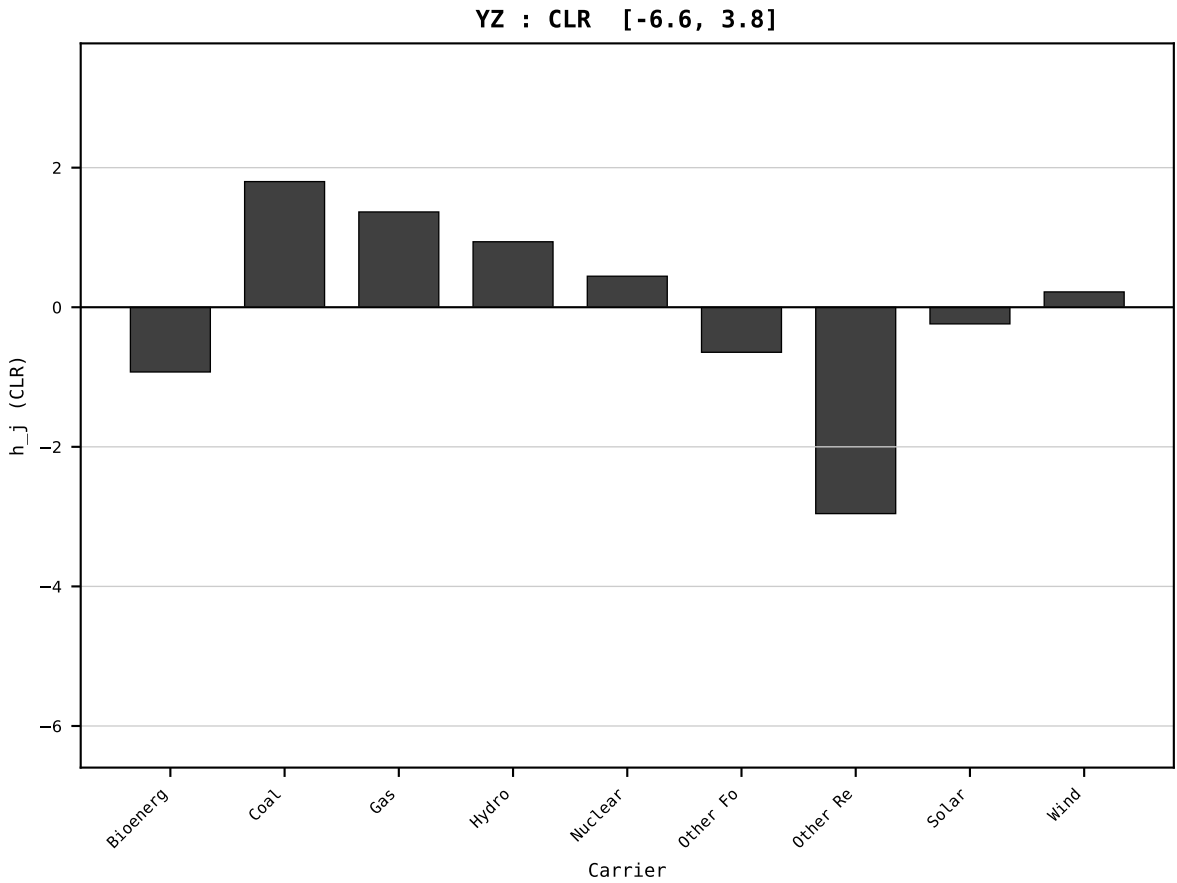
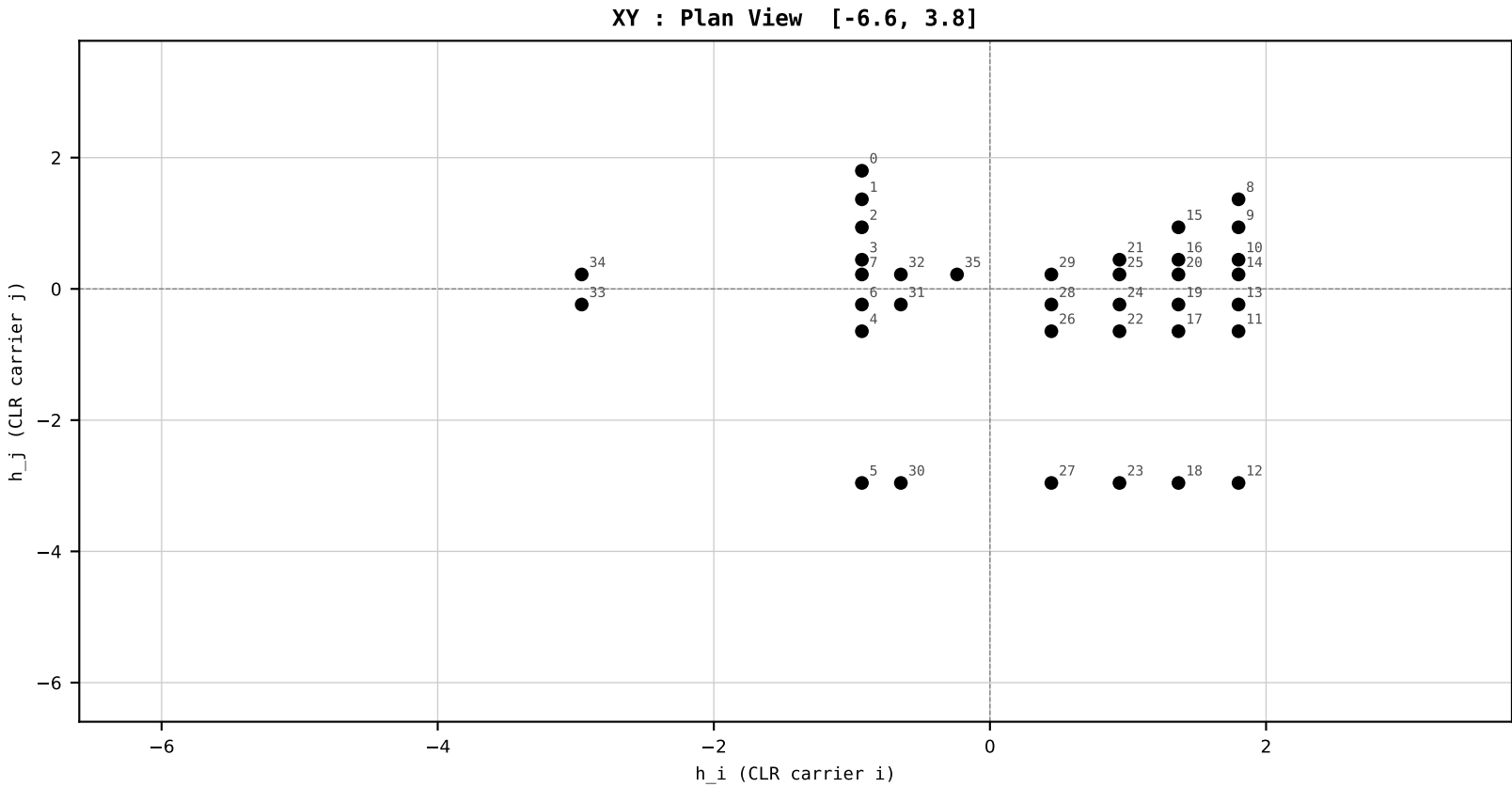
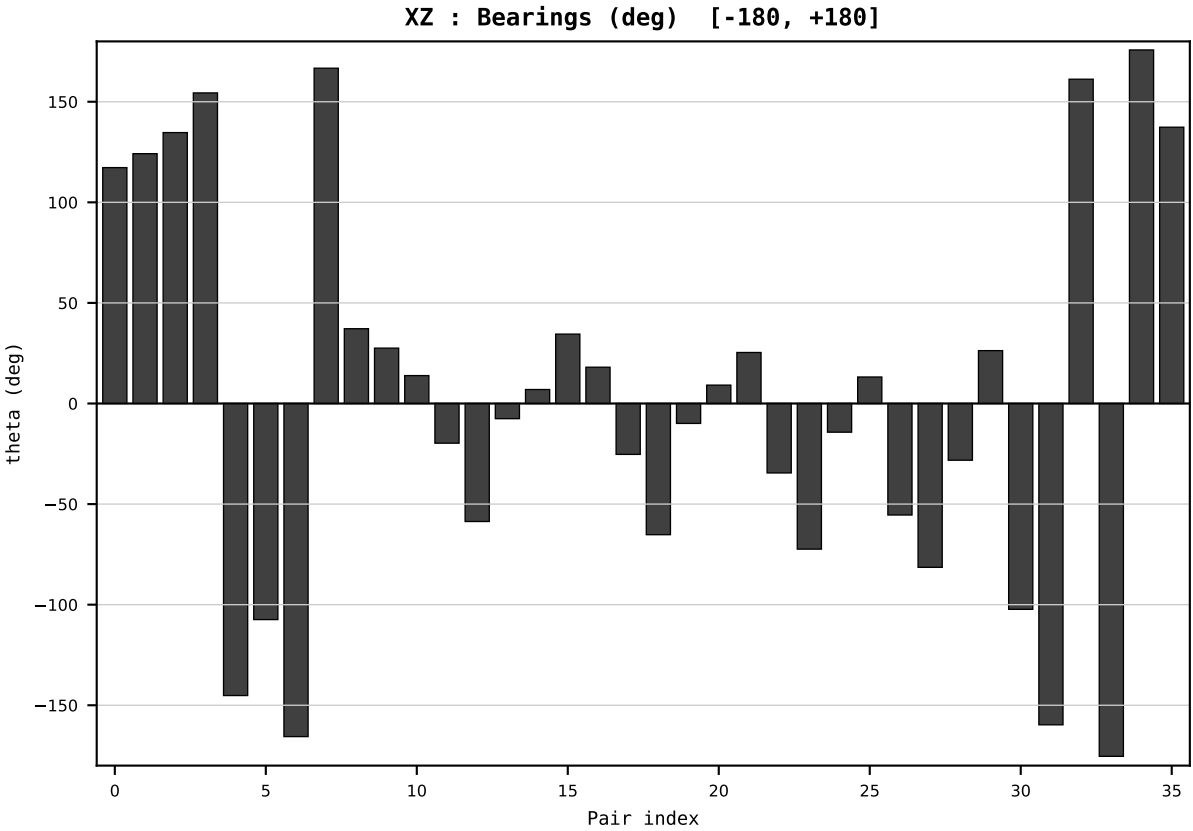
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.202475
Ring = Hs-2
E_metric= 16.3100
kappa_HS= 116.45
omega = 3.3044 deg
d_A = 0.236707
Helm = Solar
Helm d = +0.186591
DR = 4.757 HLR
DR ratio= 116.4x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=9 N=26 pairs=36

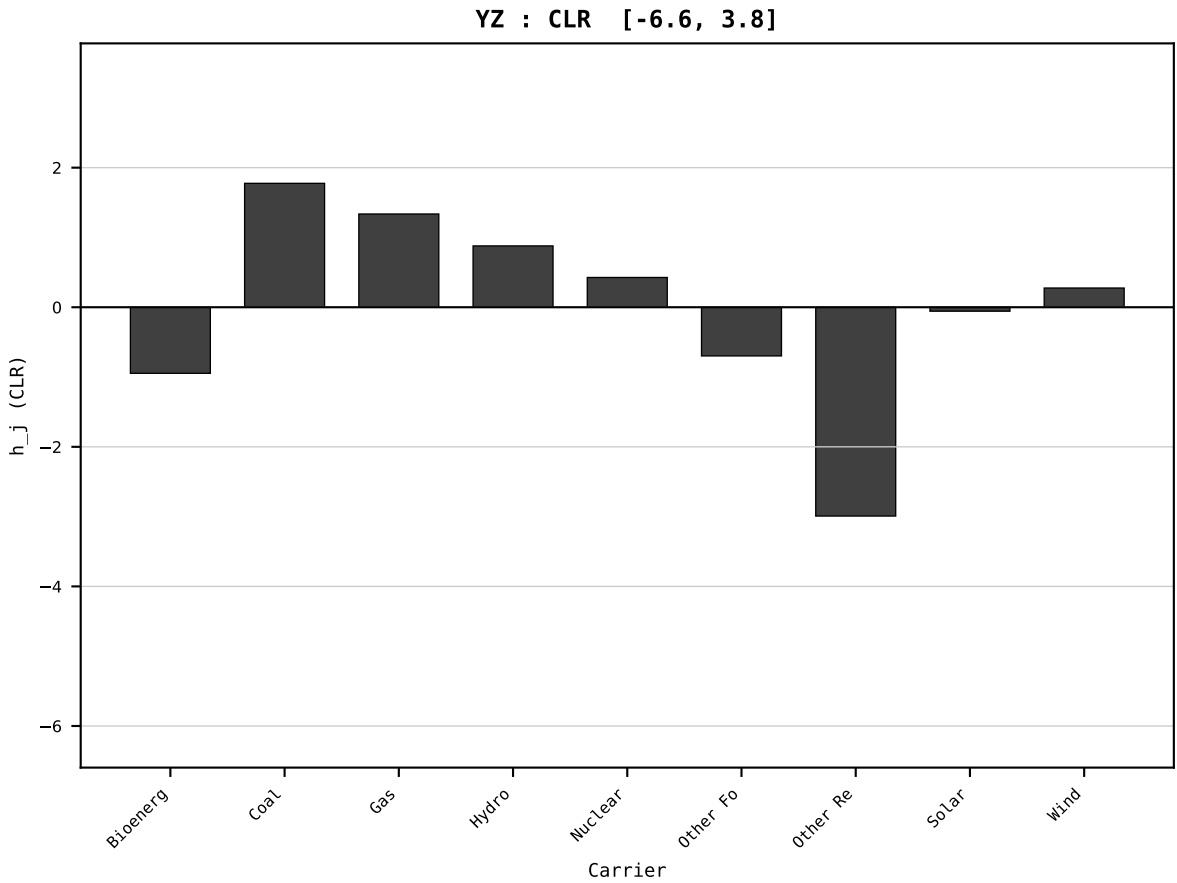
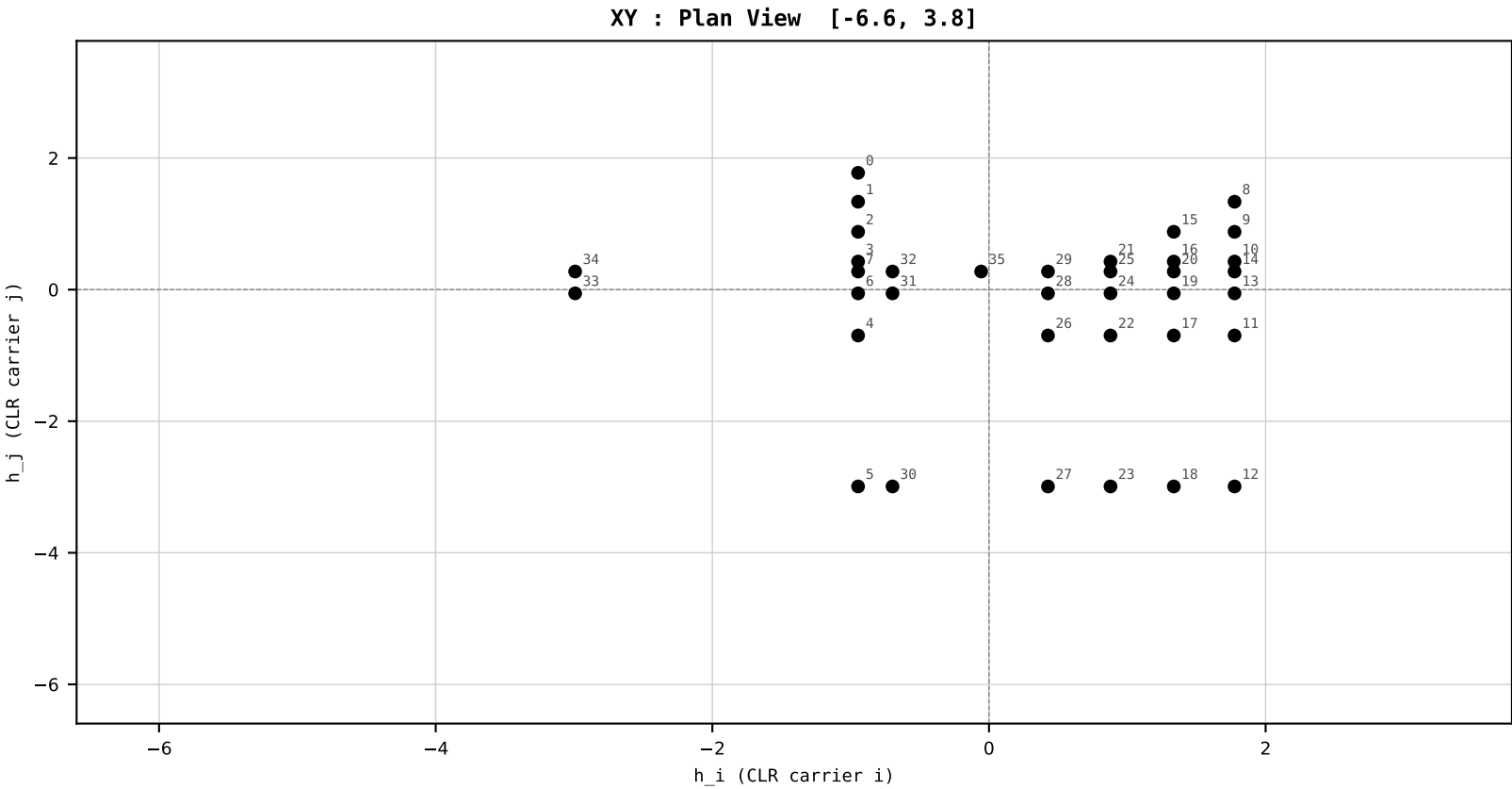
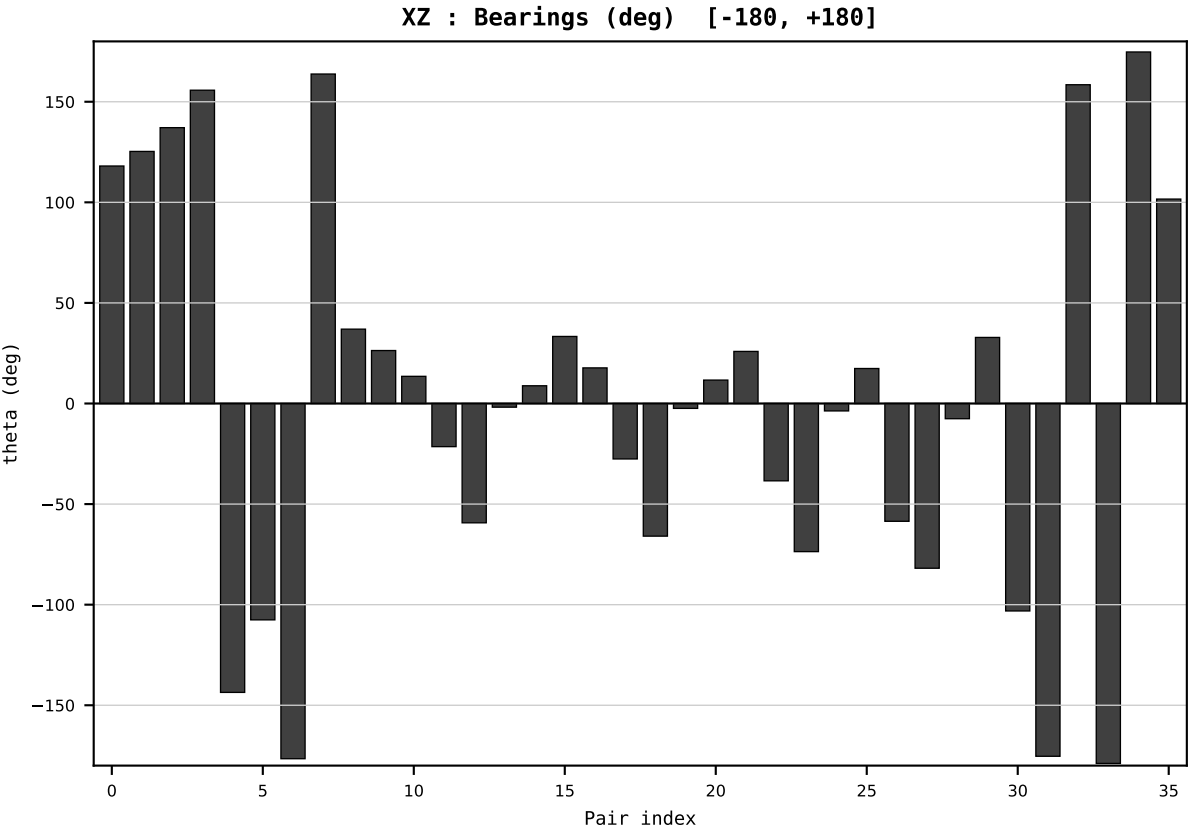
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.195016
Ring = Hs-2
E_metric= 16.3032
kappa_HS= 117.60
omega = 3.0360 deg
d_A = 0.213954
Helm = Solar
Helm d = +0.181807
DR = 4.767 HLR
DR ratio= 117.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

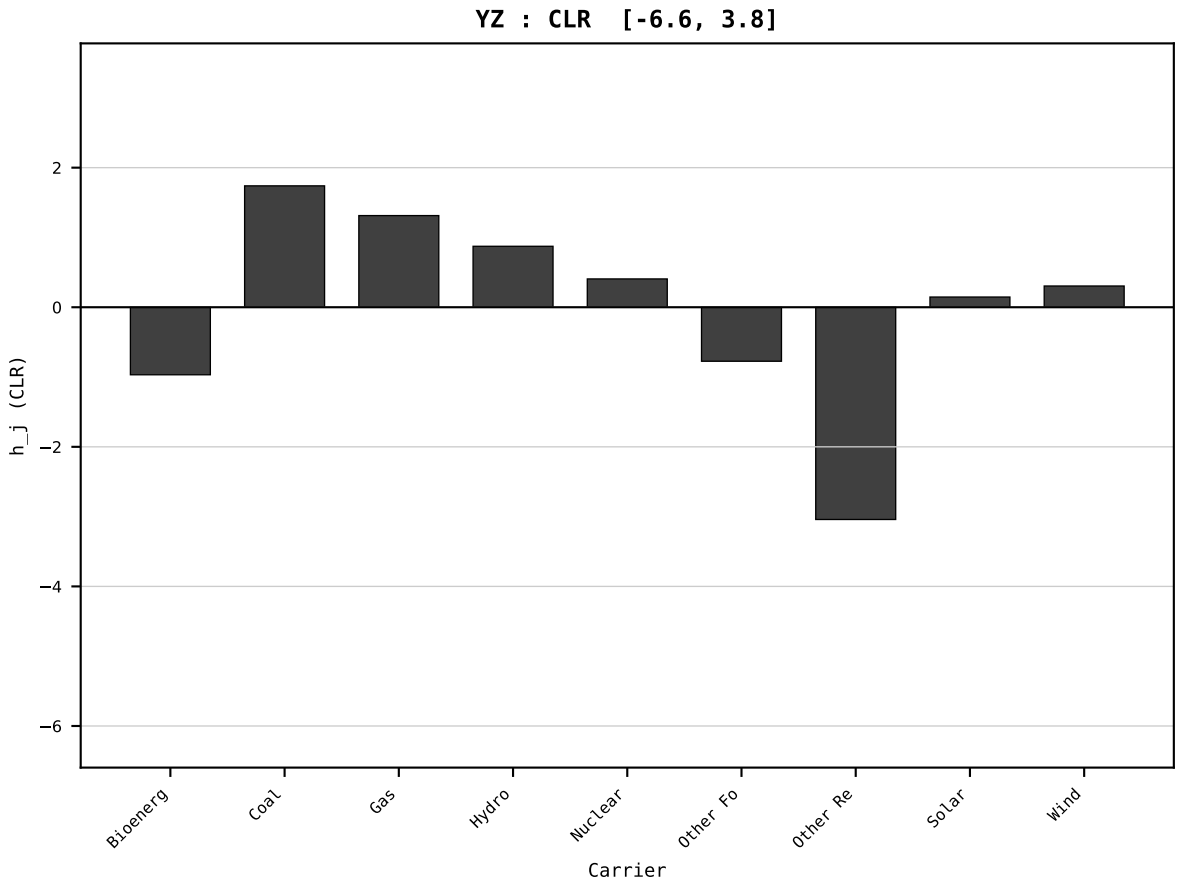
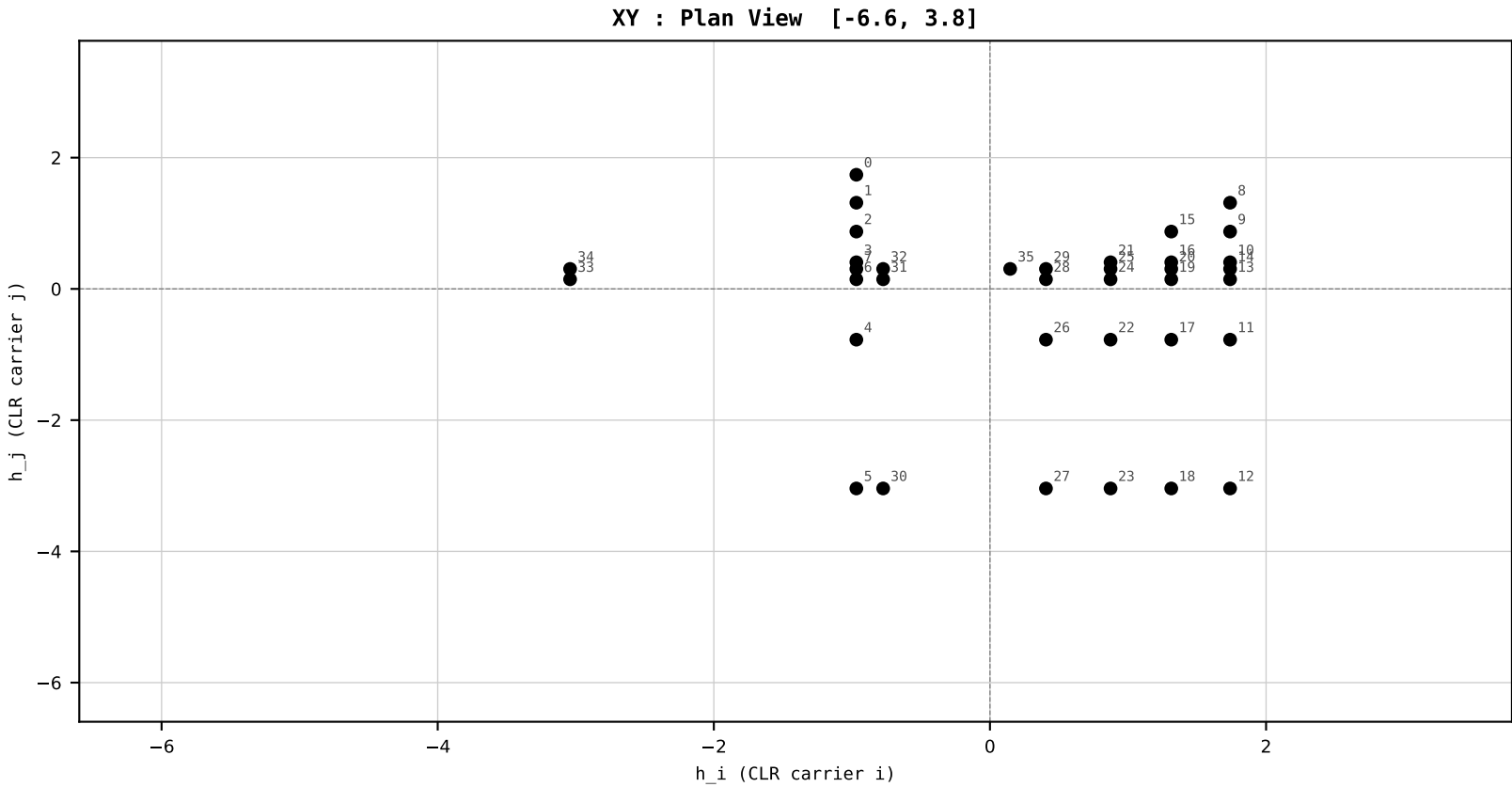
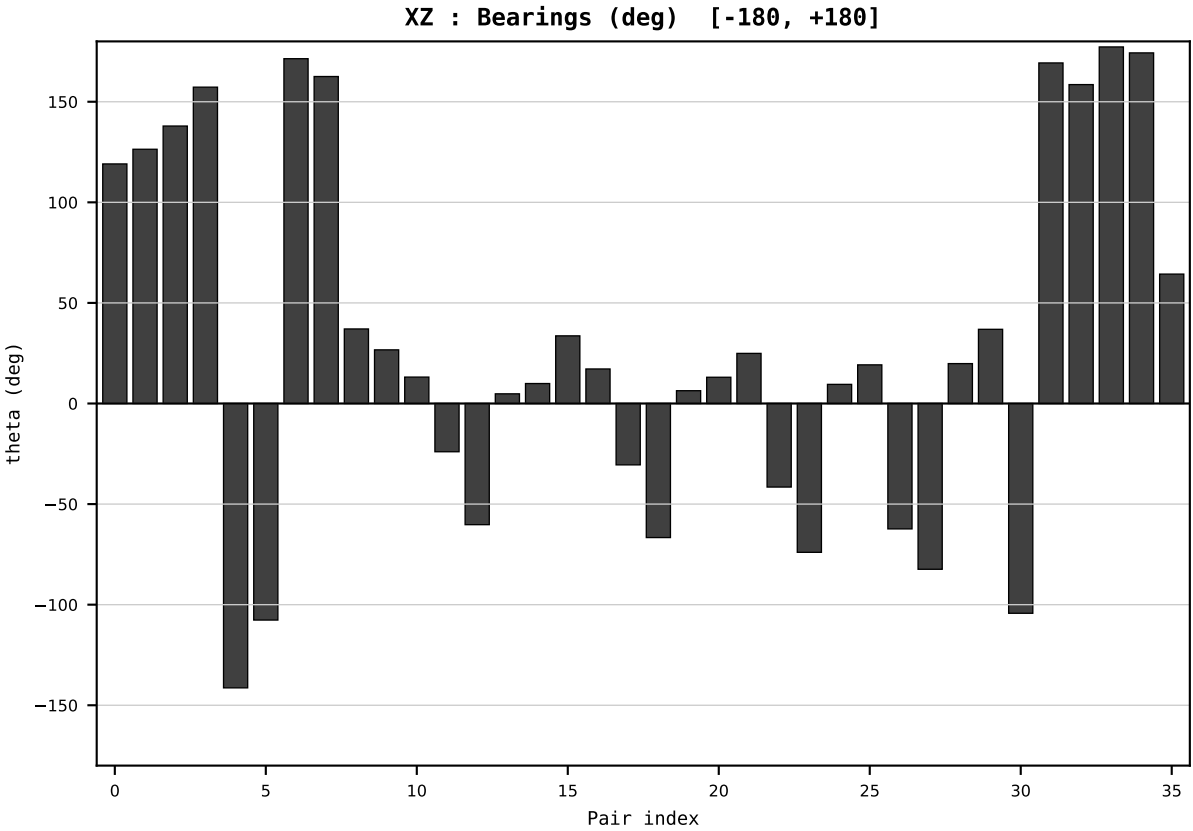
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.186834
Ring = Hs-2
E_metric= 16.5752
kappa_HS= 119.16
omega = 3.2199 deg
d_A = 0.230278
Helm = Solar
Helm d = +0.202924
DR = 4.781 HLR
DR ratio= 119.2x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

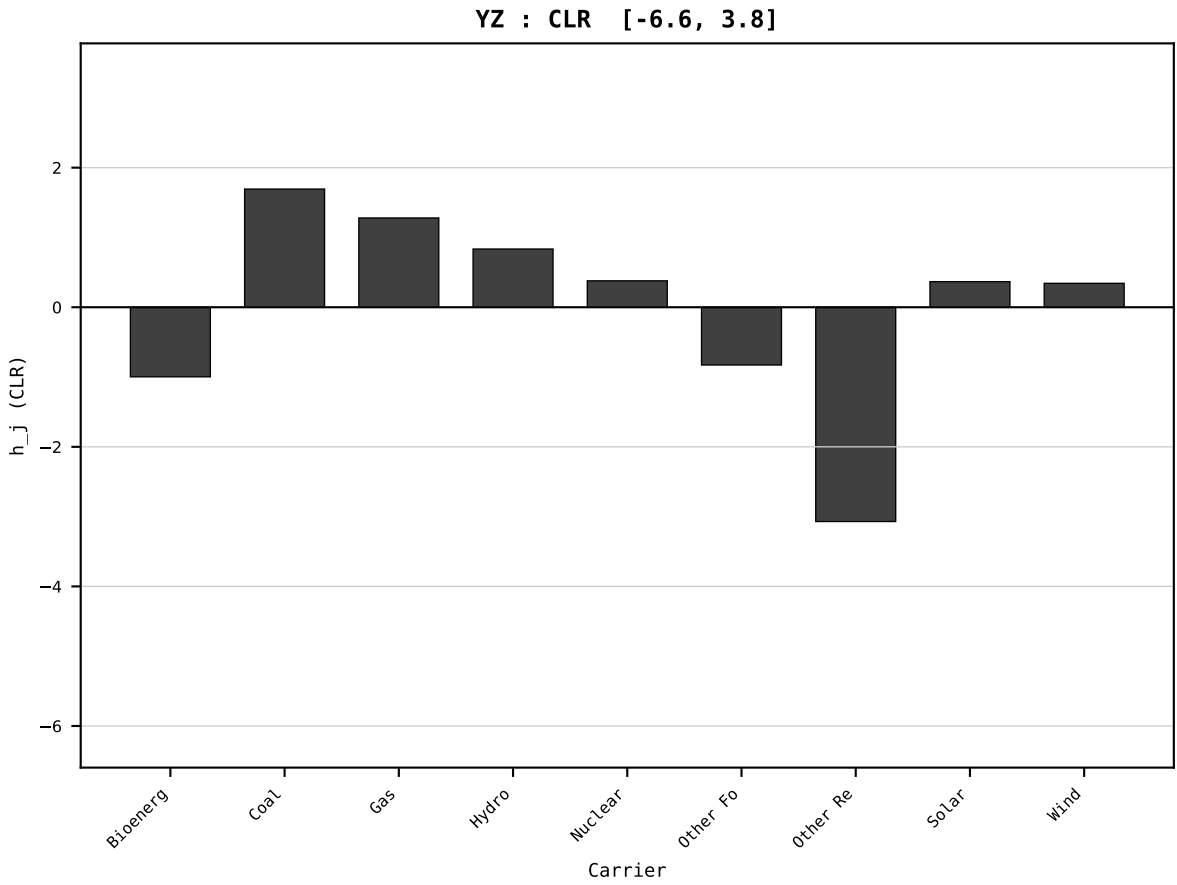
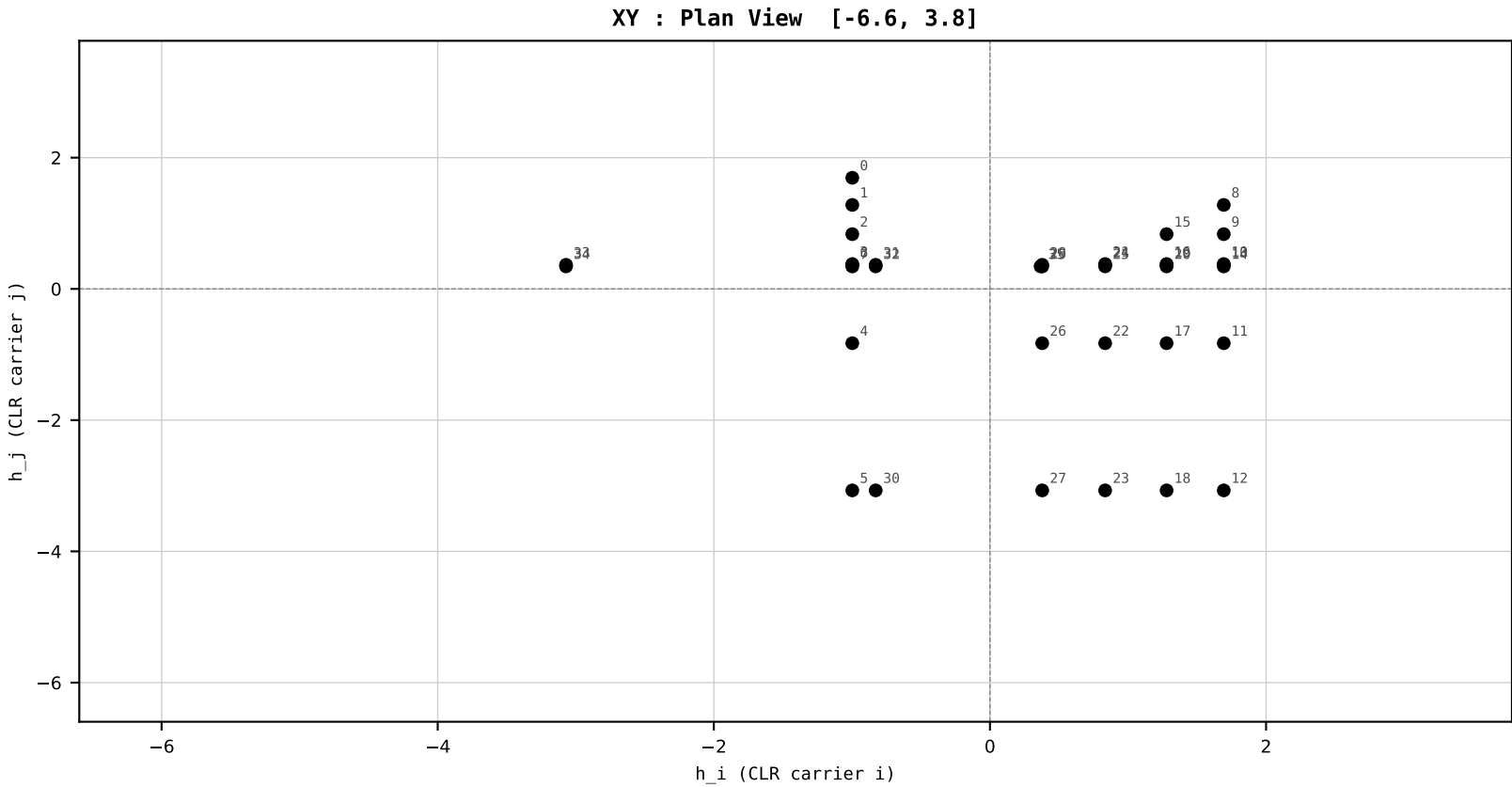
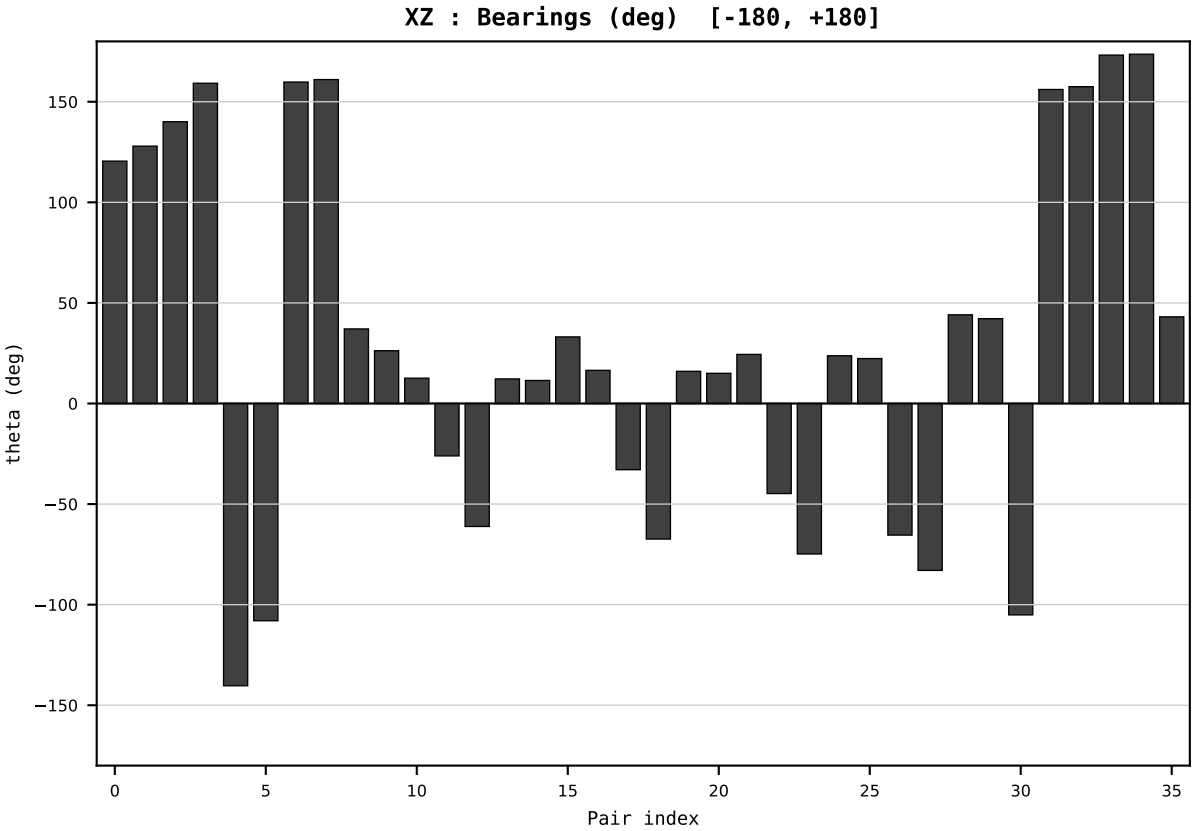
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

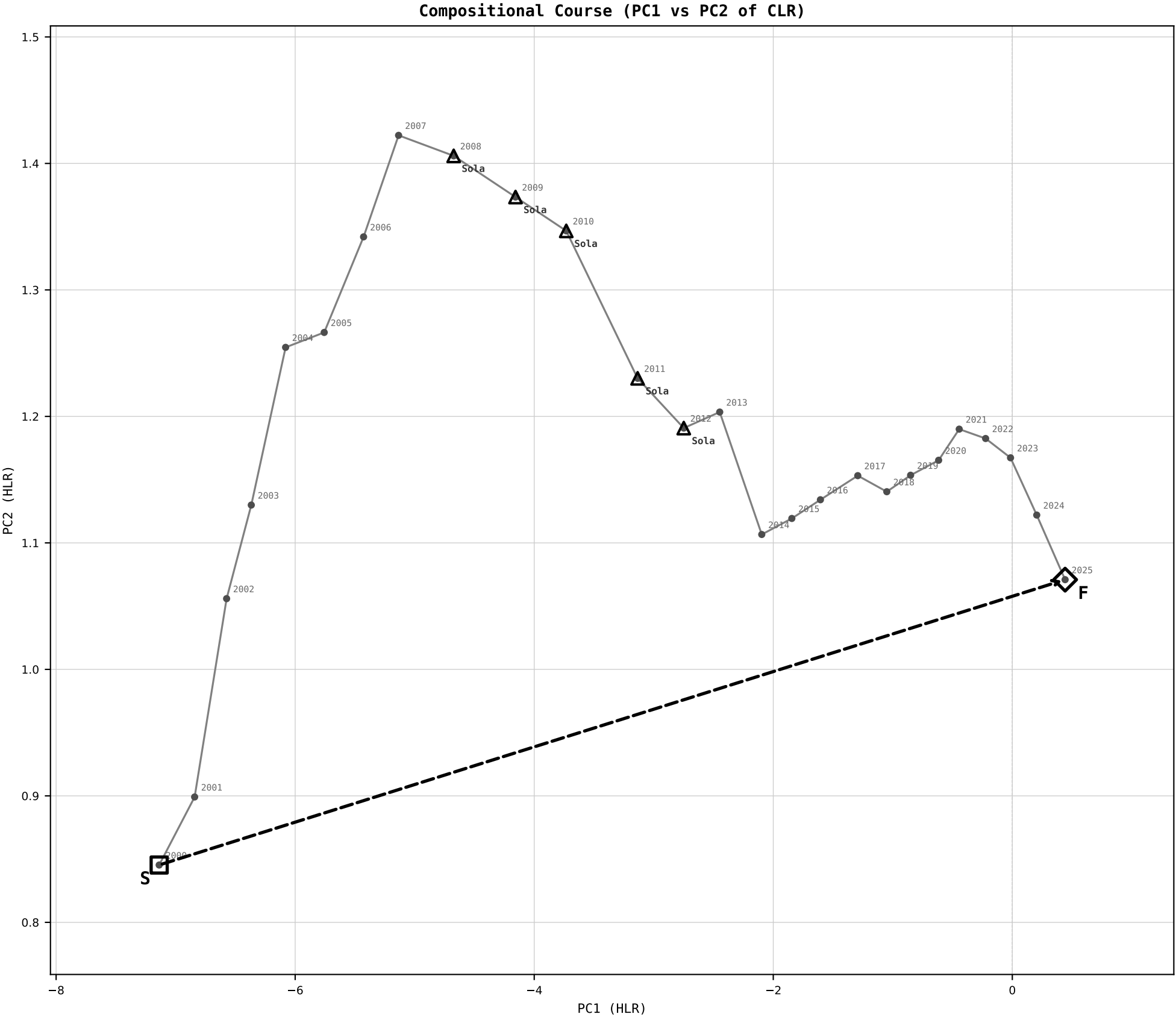
Dataset : ember_WLD_World_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.177094
Ring = Hs-2
E_metric= 16.7040
kappa_HS= 117.22
omega = 3.4412 deg
d_A = 0.245463
Helm = Solar
Helm d = +0.220443
DR = 4.764 HLR
DR ratio= 117.2x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow$ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 7.5841 HLR
Path length = 7.9588 HLR
Directness = 0.9529

DR start = 8.647 HLR
ratio = 5694.9x
DR final = 4.764 HLR
ratio = 117.2x

E_metric start = 66.7408
E_metric final = 16.7040

PC1 loadings:
Solar +0.816
Wind +0.324
Other Fo -0.296

PC2 loadings:
Wind +0.827
Solar -0.443
Other Fo -0.193

Top displacement events:
2011 Solar d=0.629
2009 Solar d=0.520
2008 Solar d=0.464
2010 Solar d=0.430
2012 Solar d=0.401

The instrument reads.
The expert decides.
The loop stays open.

View B

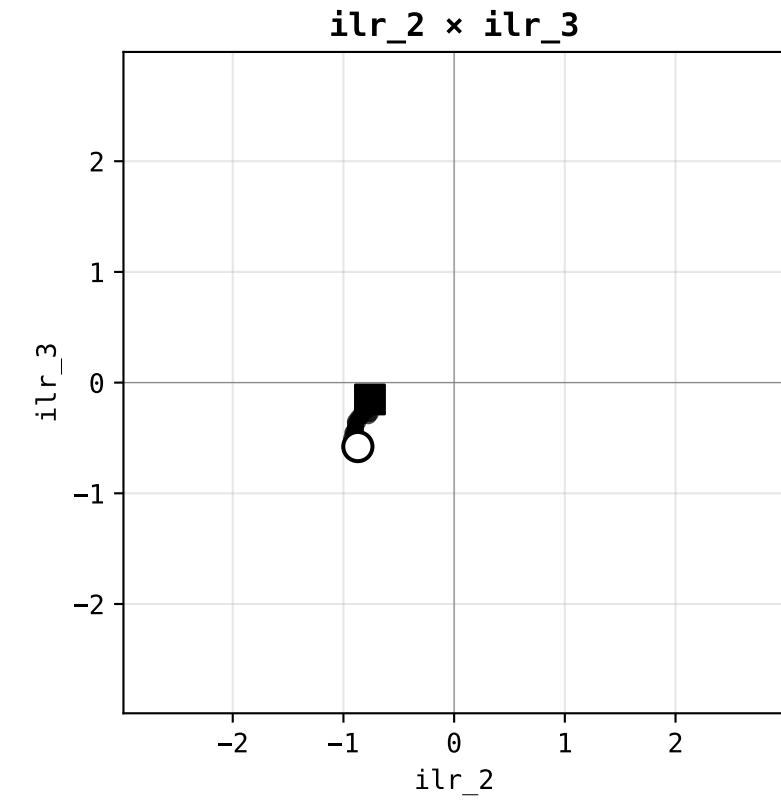
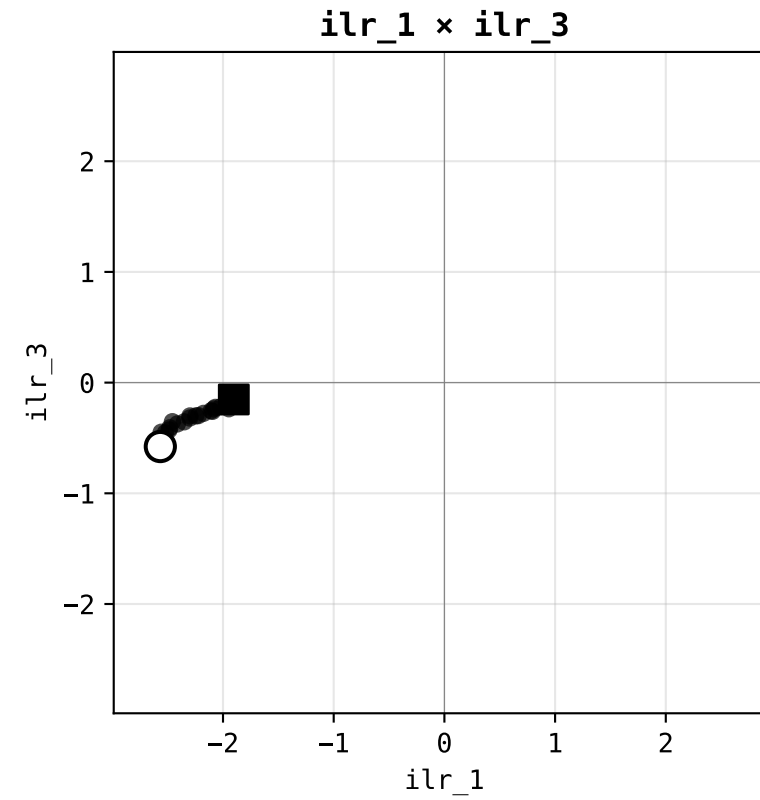
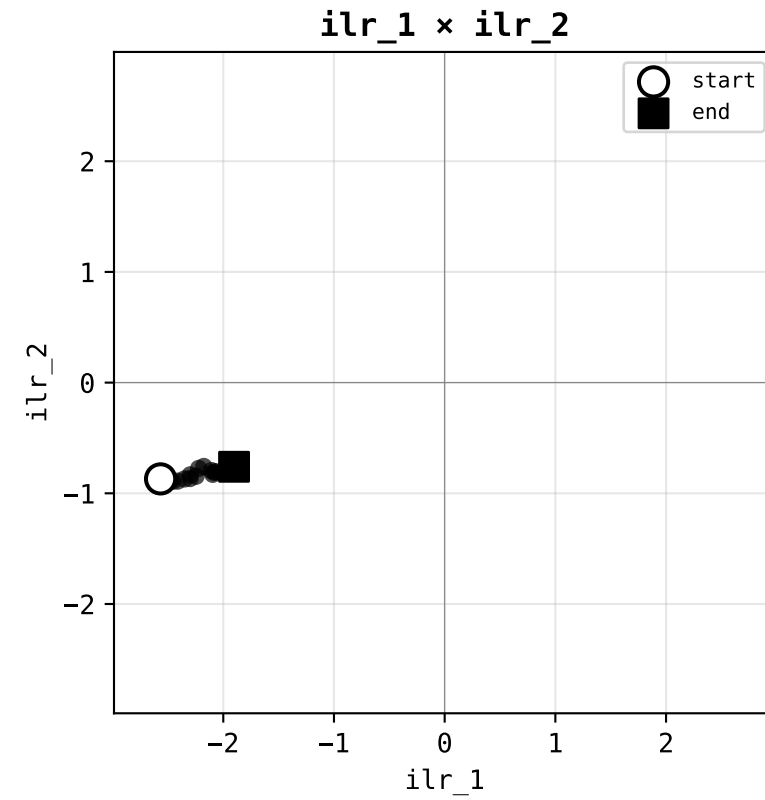
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ WLD World

ILR-Helmert Orthogonal Triplet – WLD World · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · $N=26$ years · $D=9$ carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:

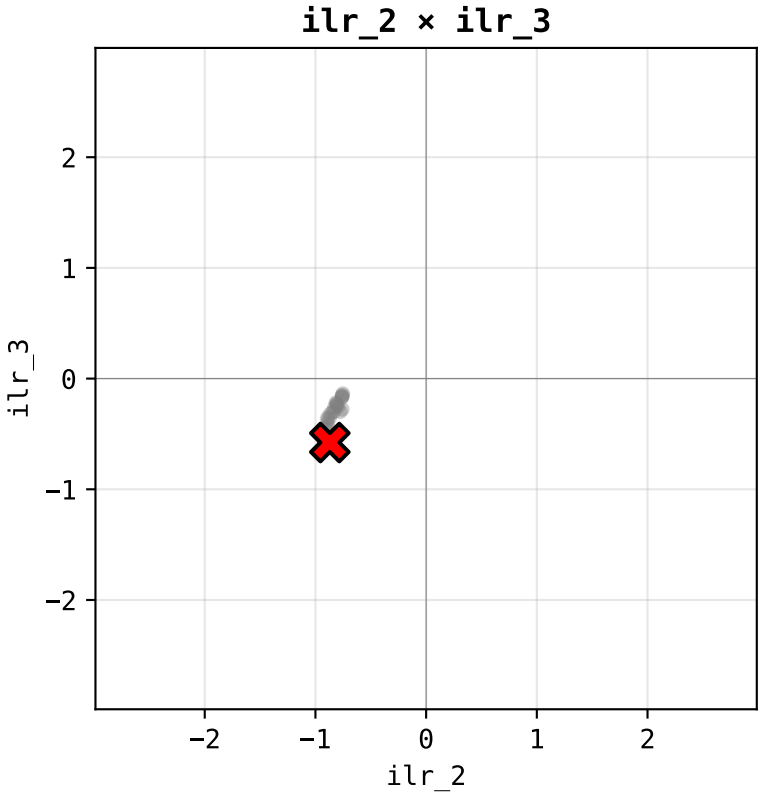
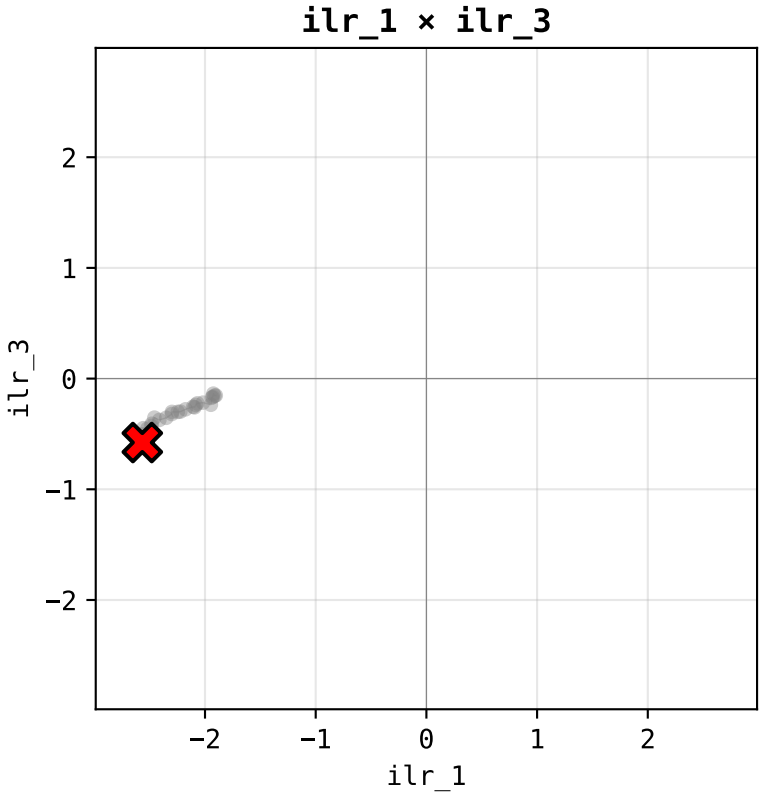
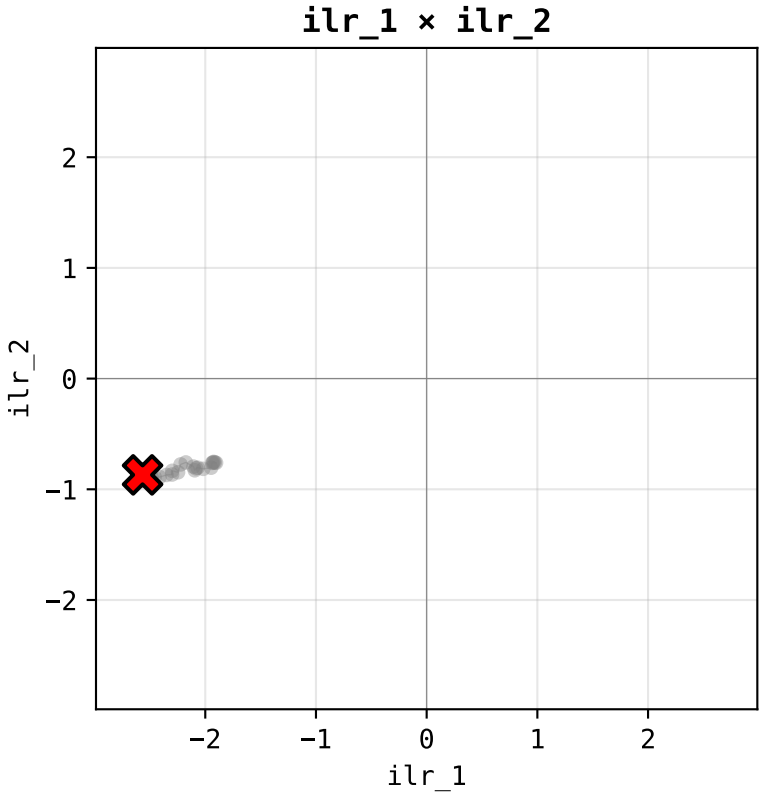
start year: 2000
end year: 2025
N readings: 26

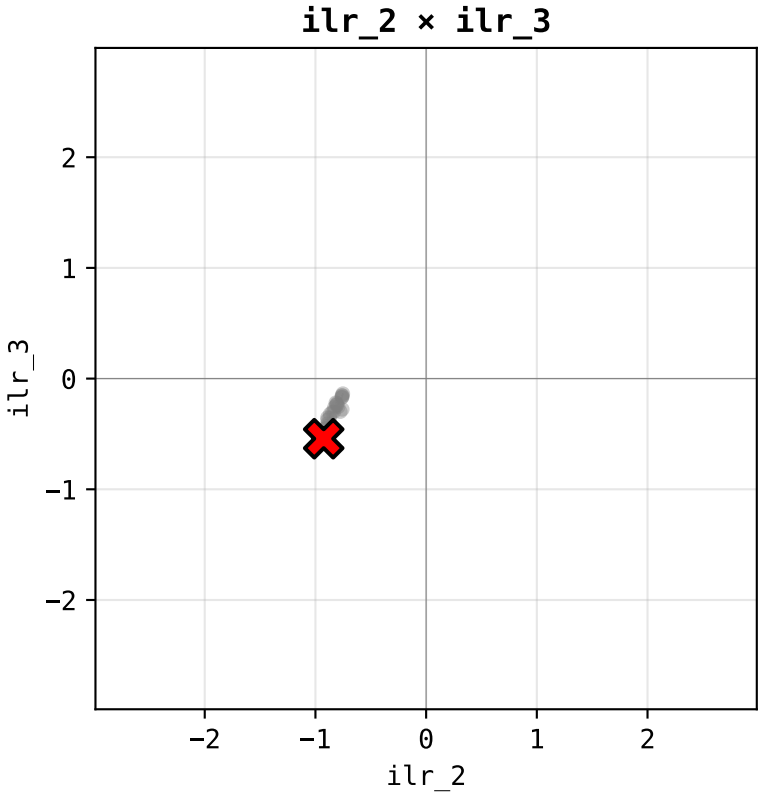
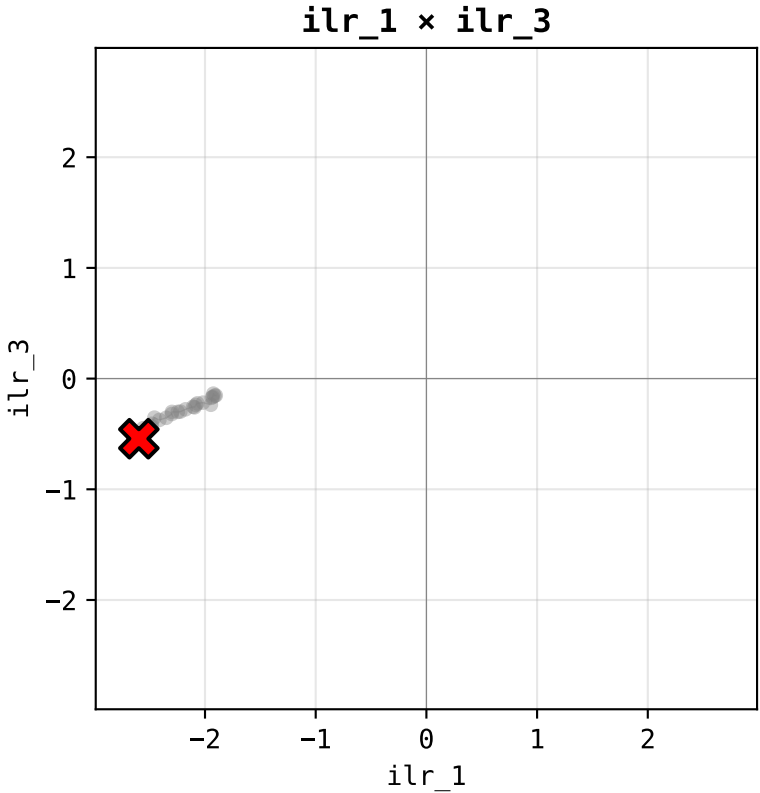
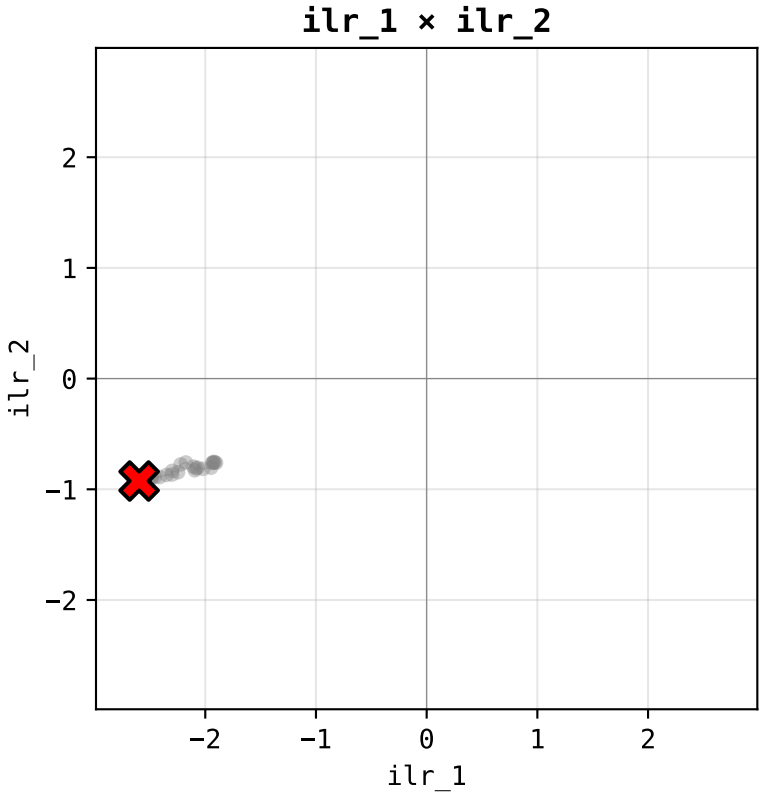
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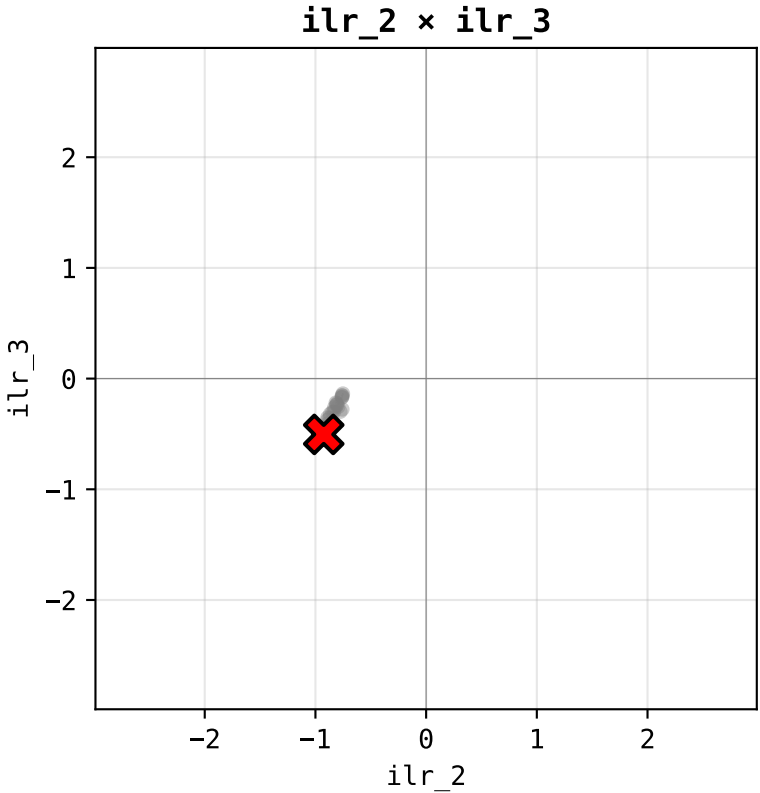
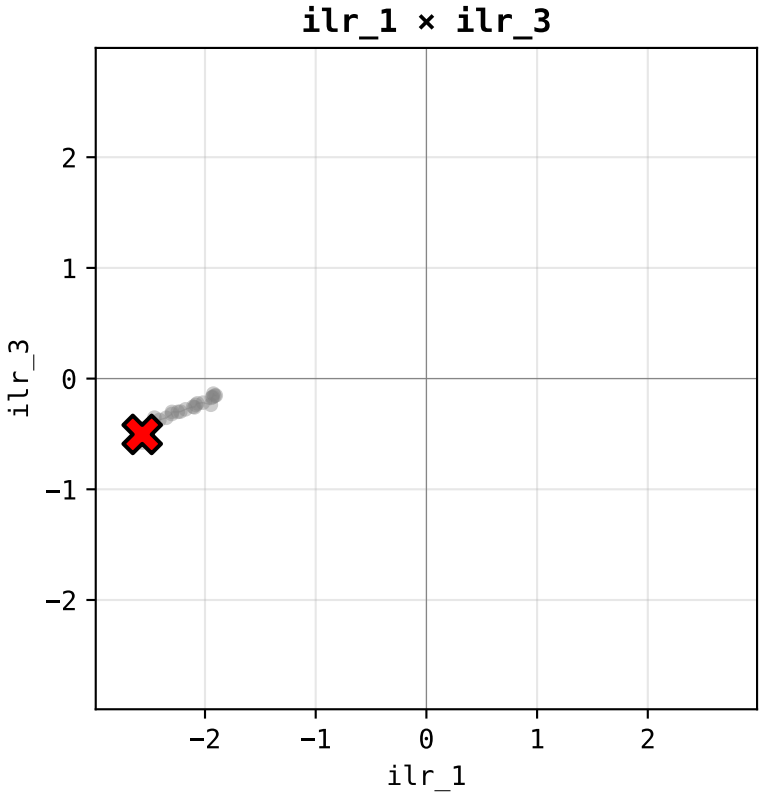
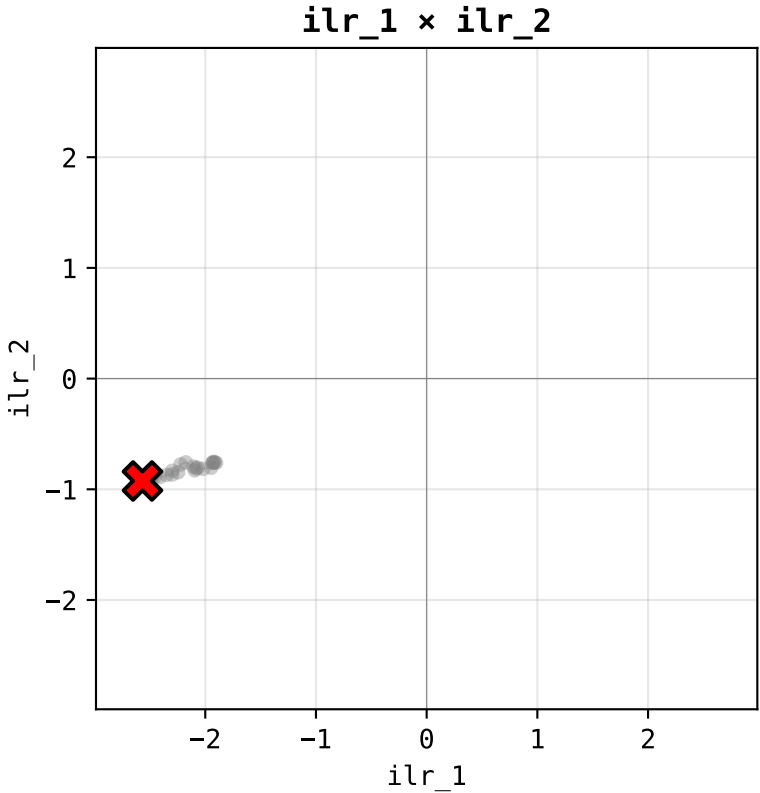
fixed scale ± 2.99 ILR
Order 1 (first principles)
per Output Doctrine v1.0

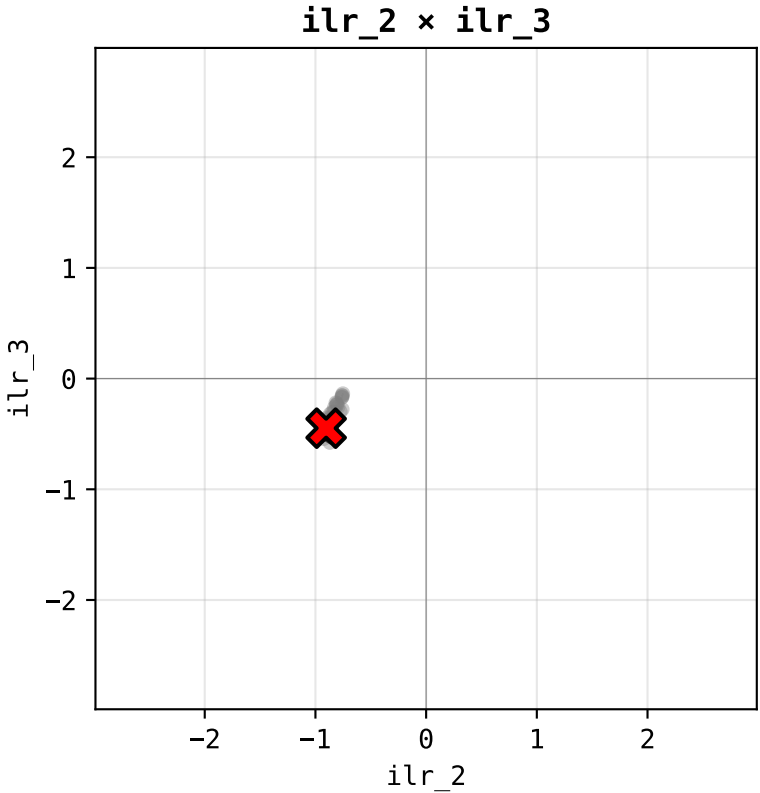
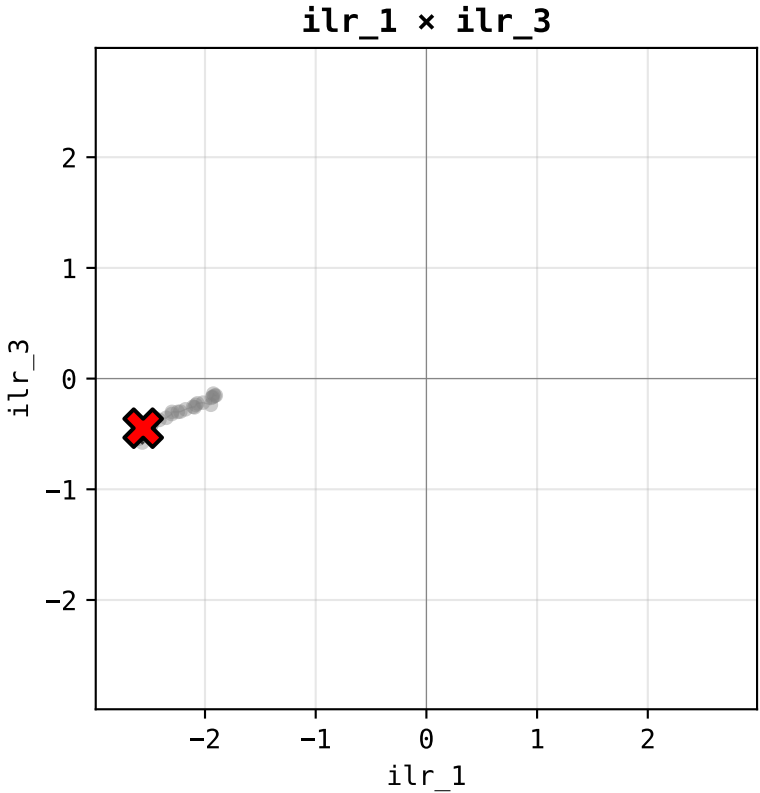
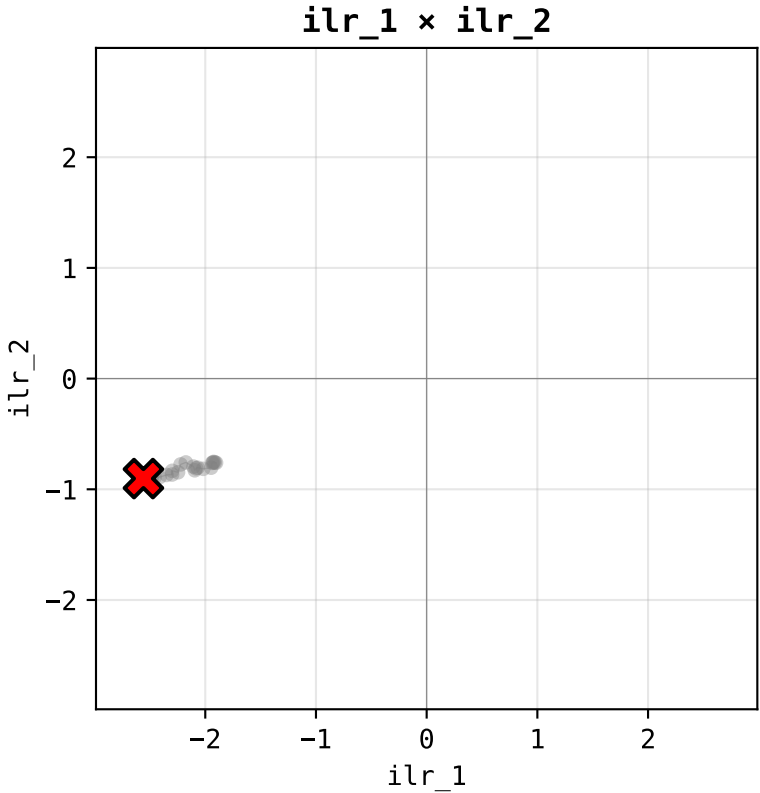
Symbology:

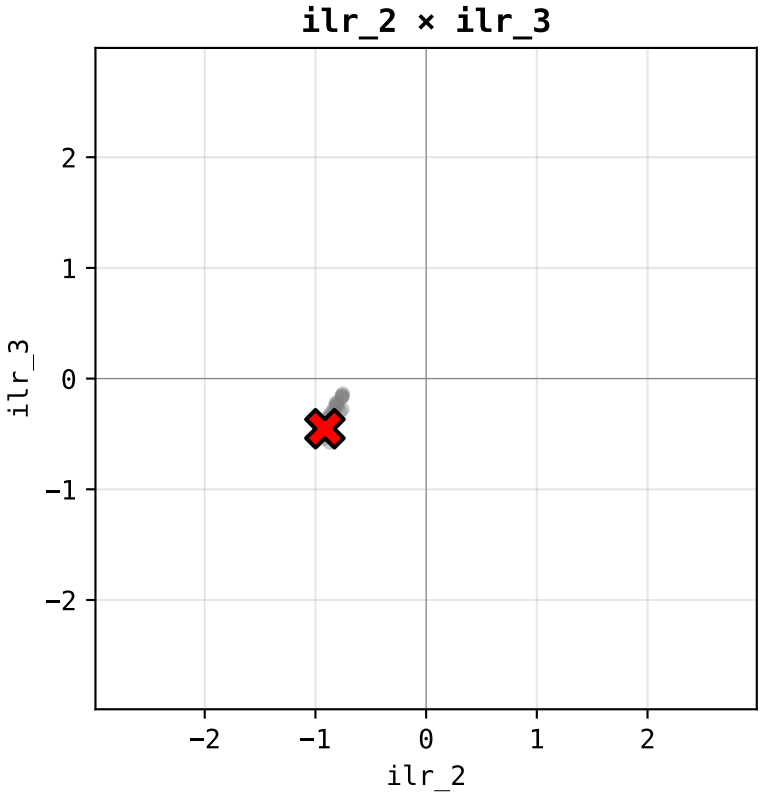
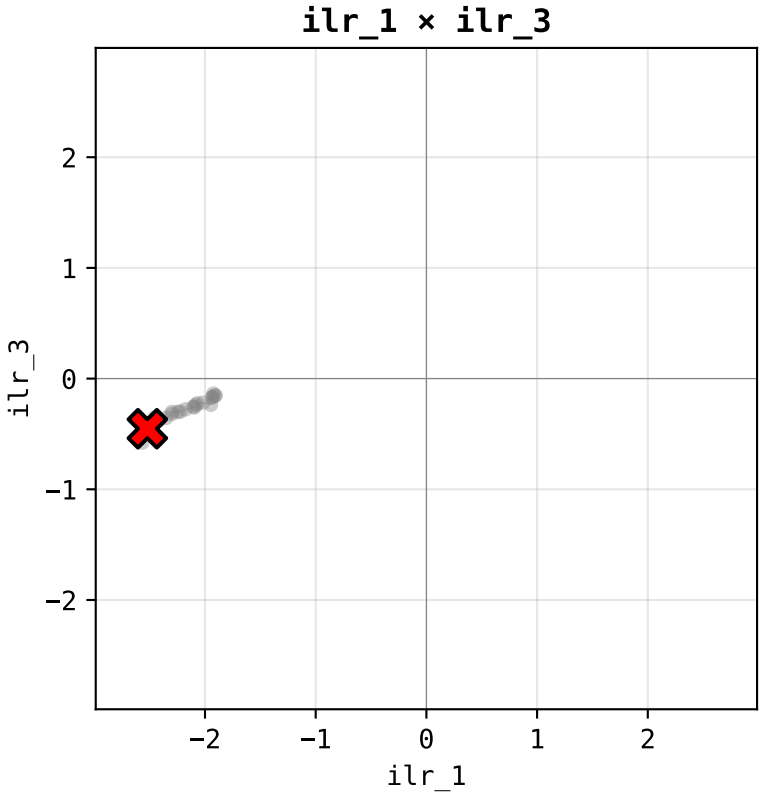
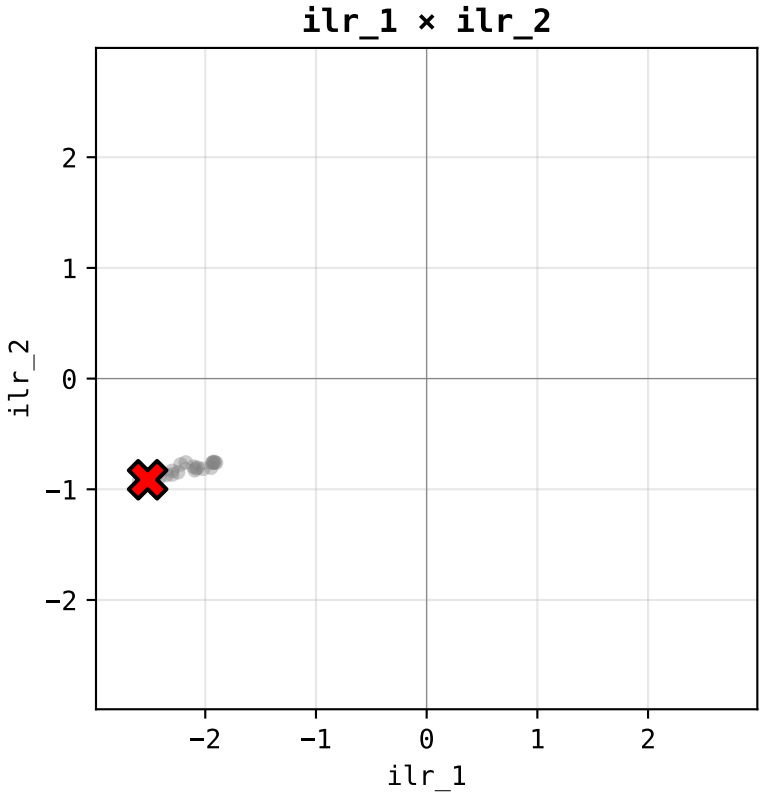
○ start · ■ end
· intermediate years

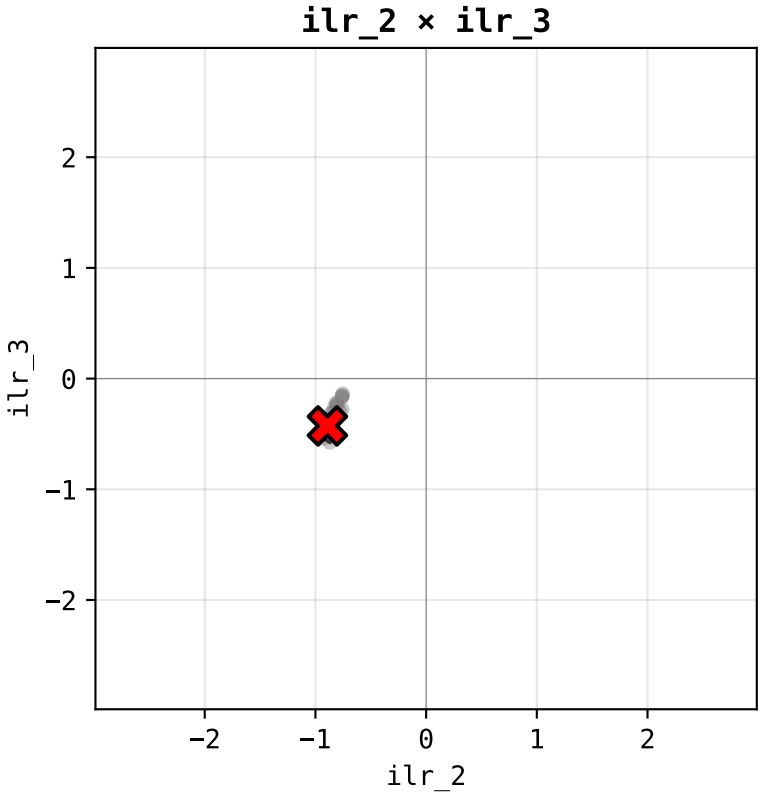
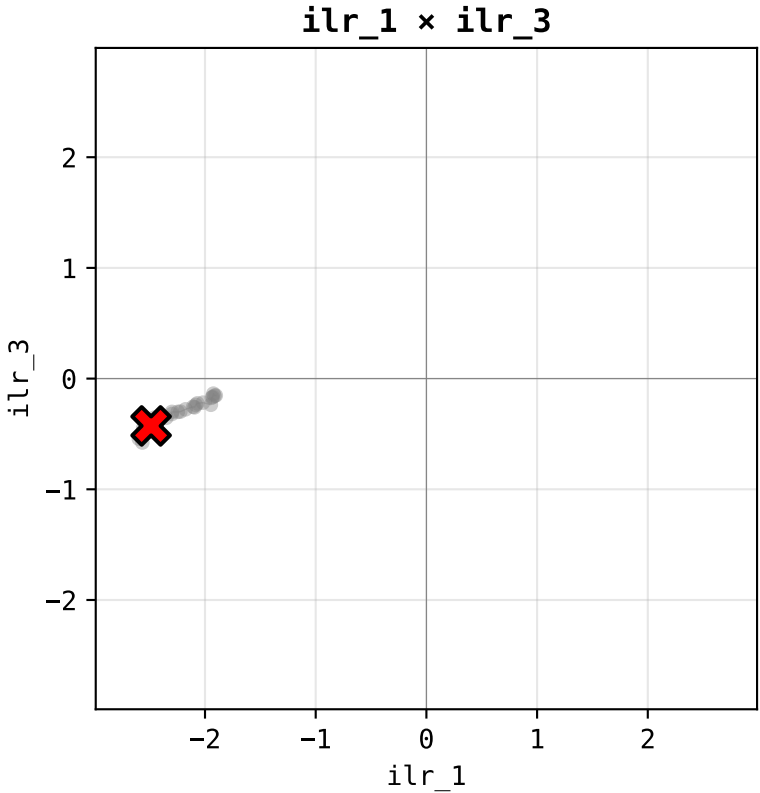
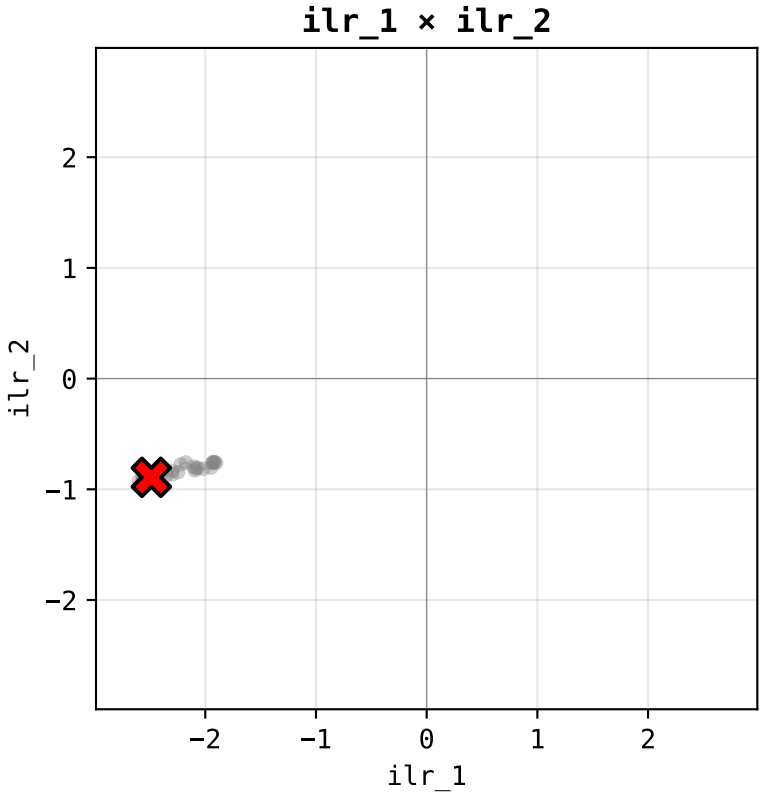




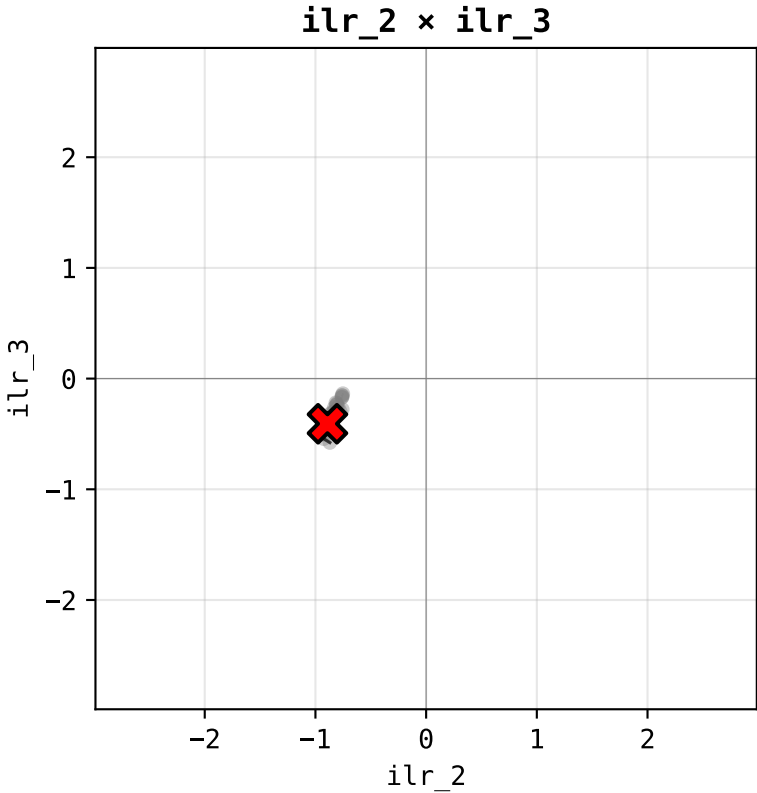
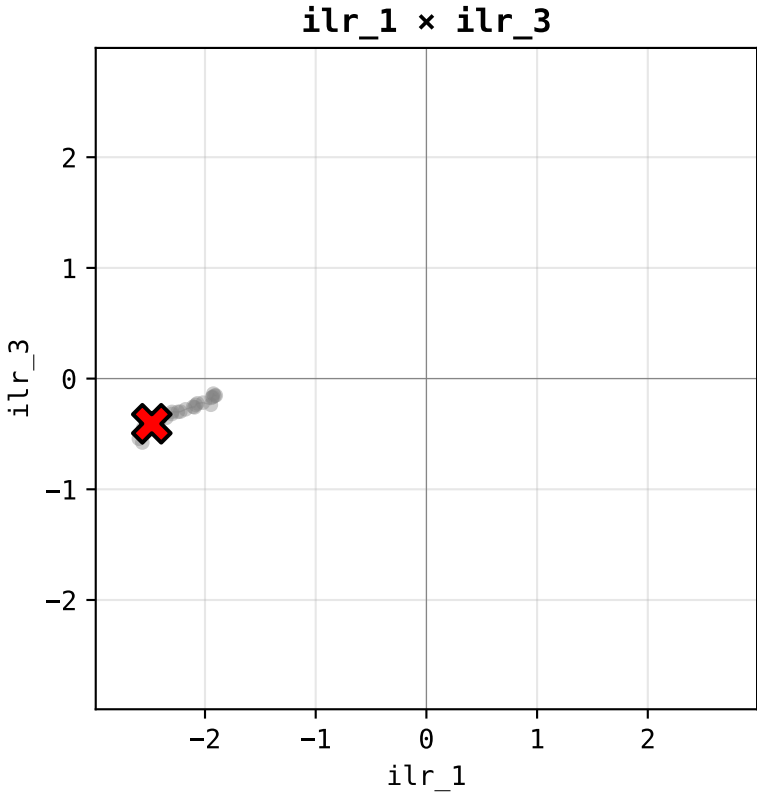
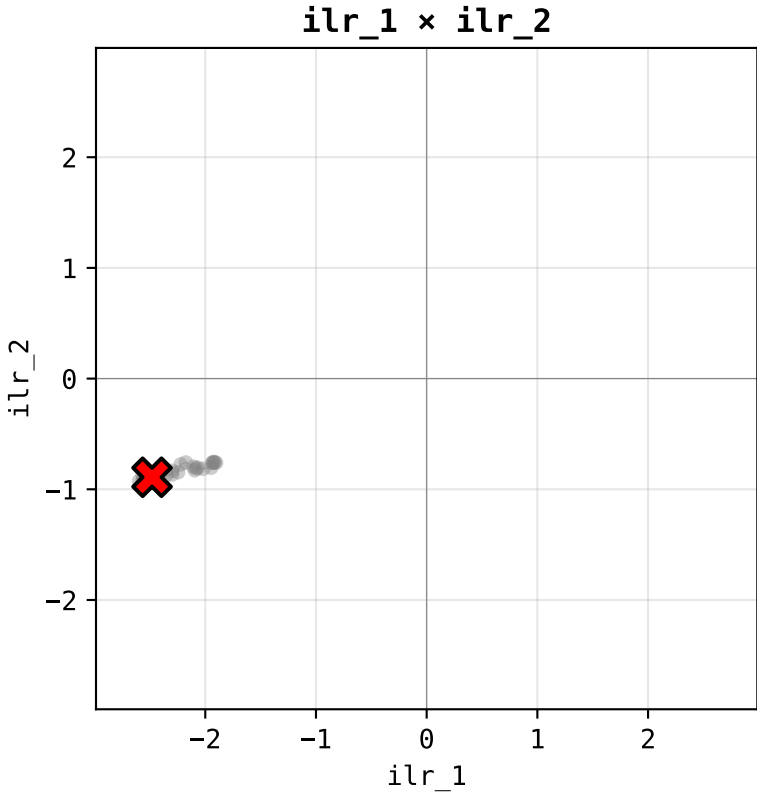


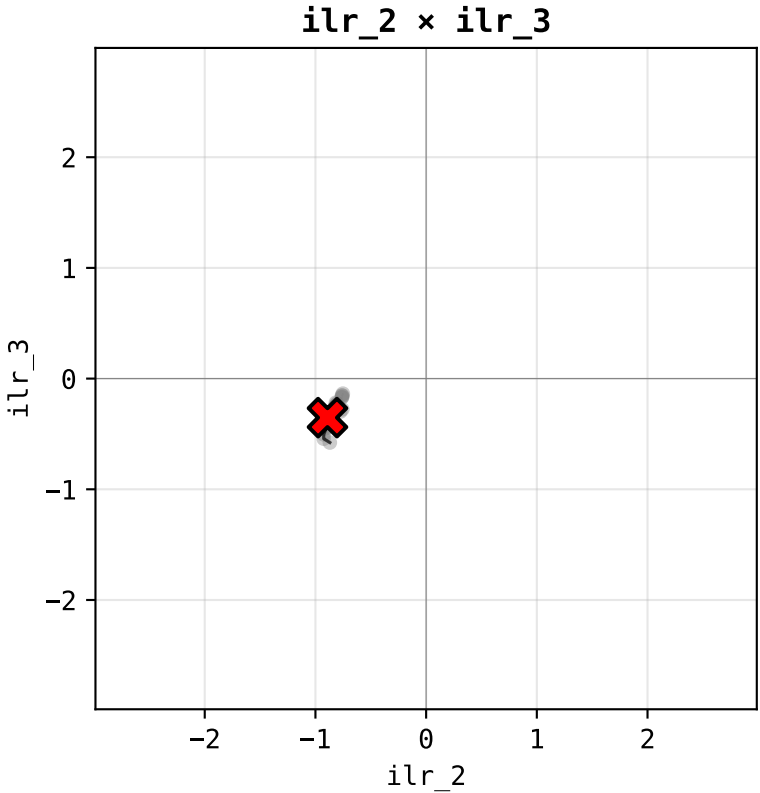
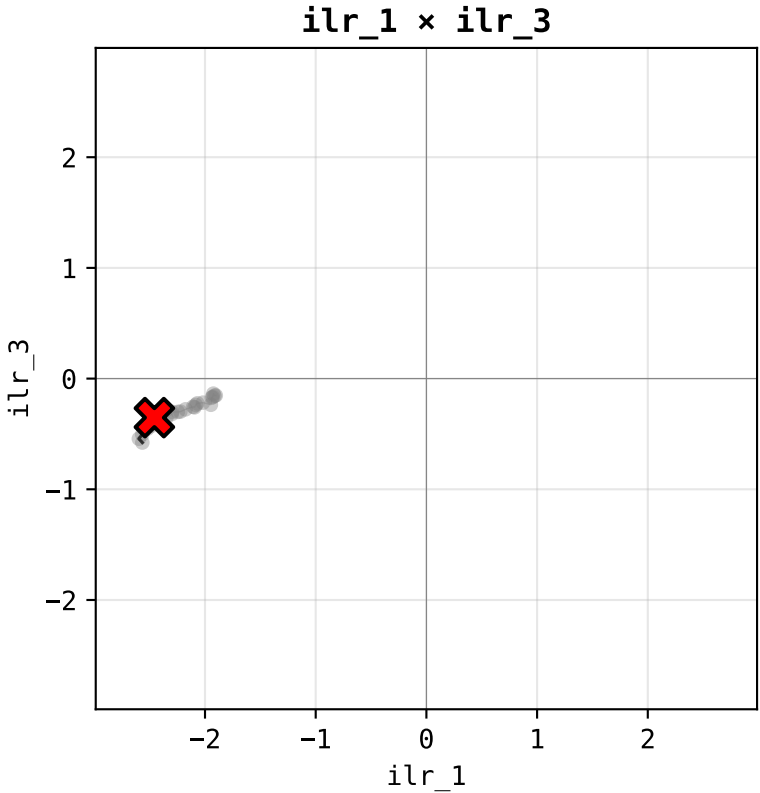
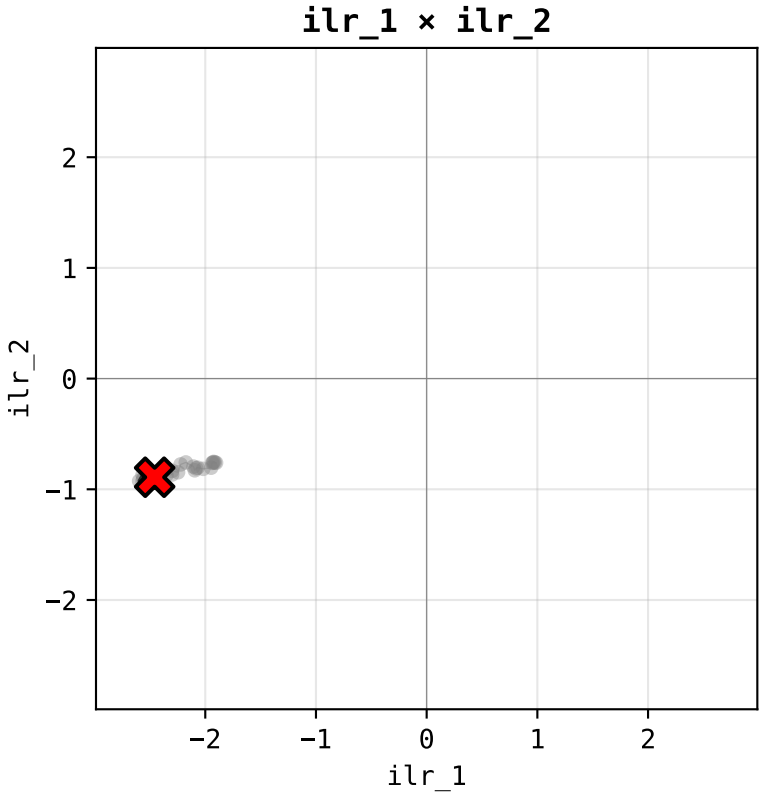




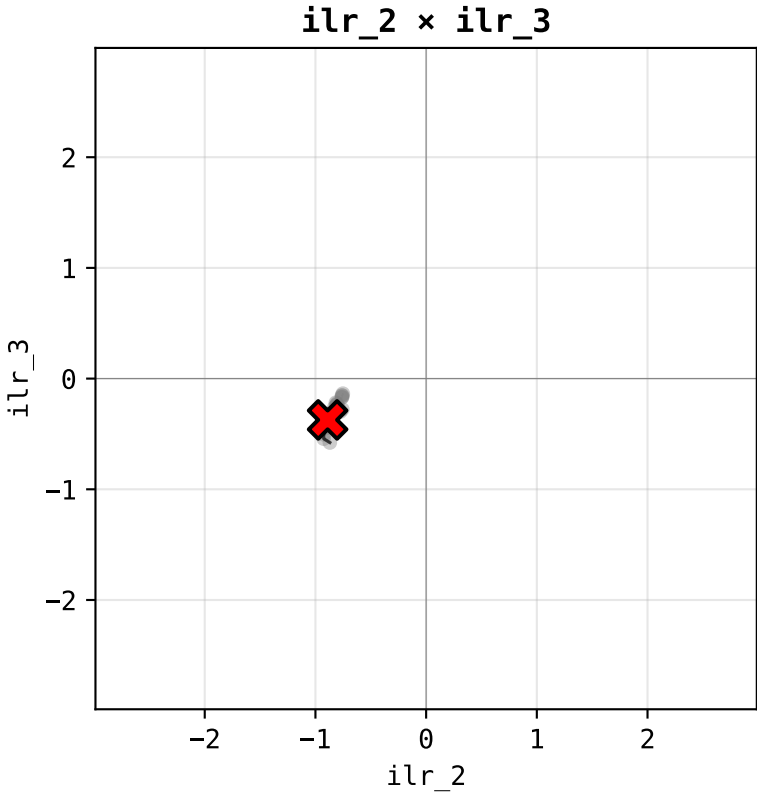
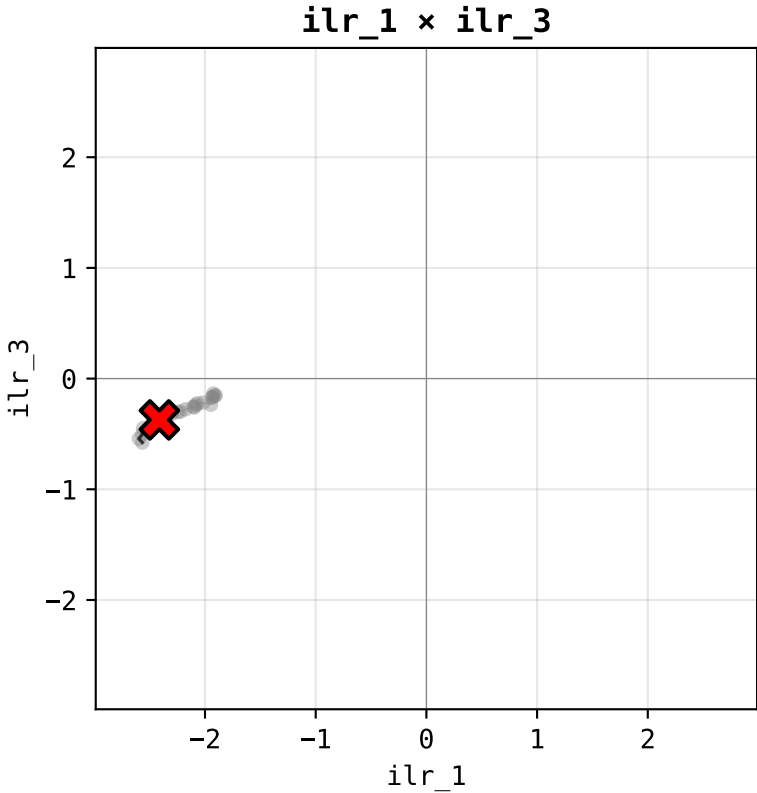
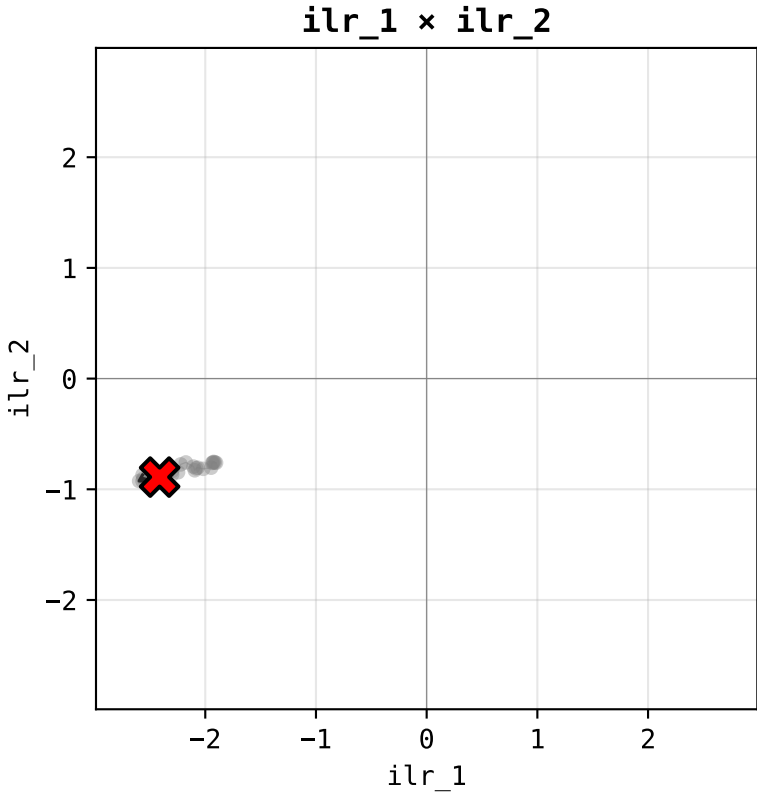


ILR Triplet – WLD World · year 2006 (7/26)

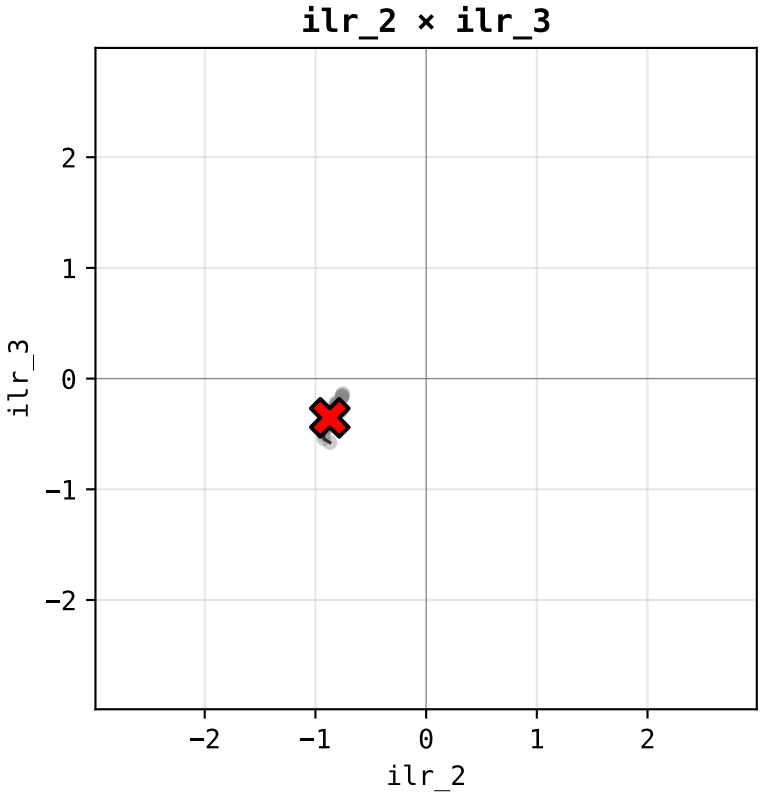
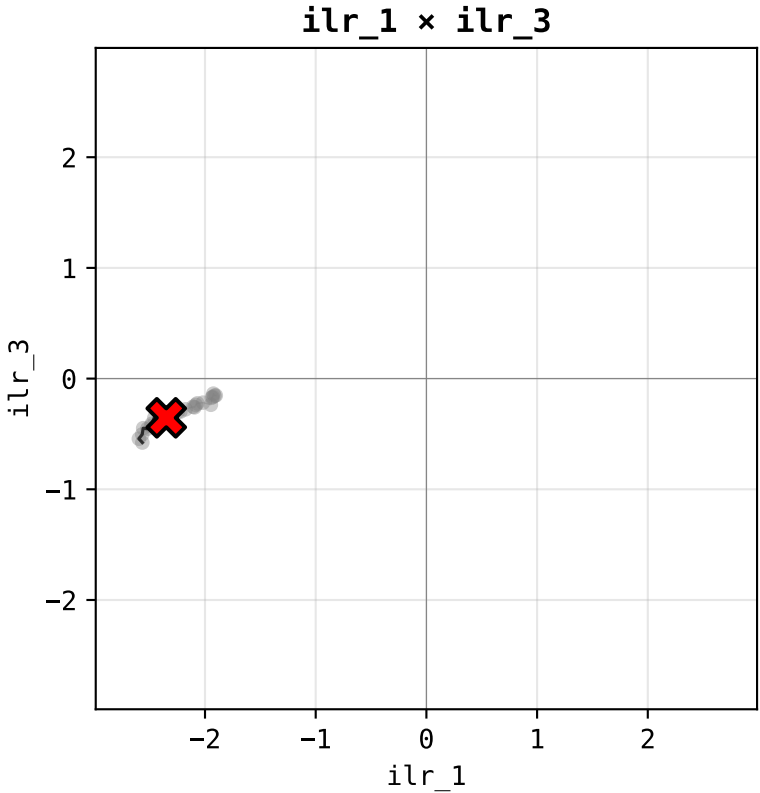
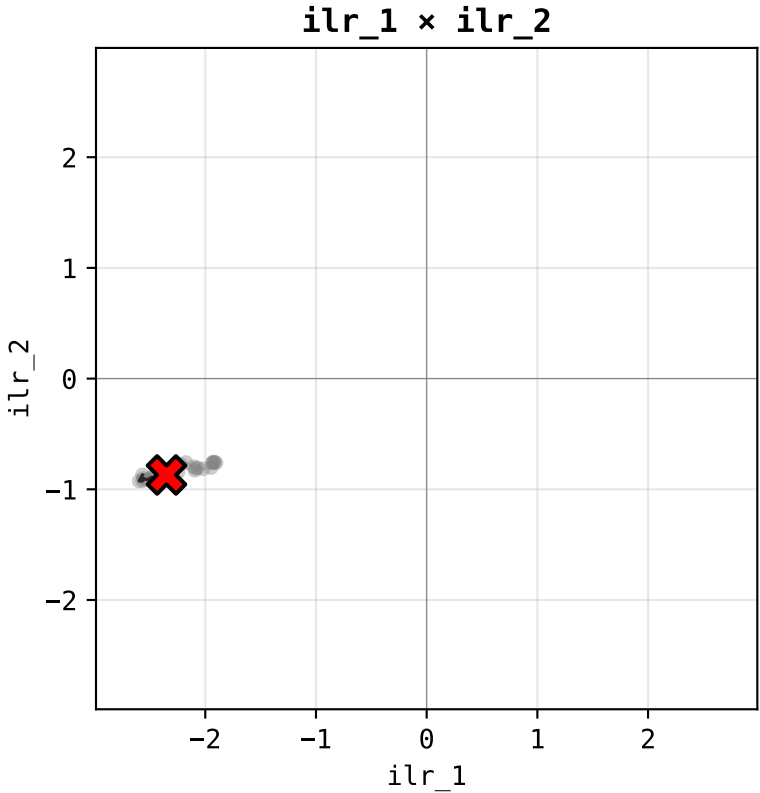




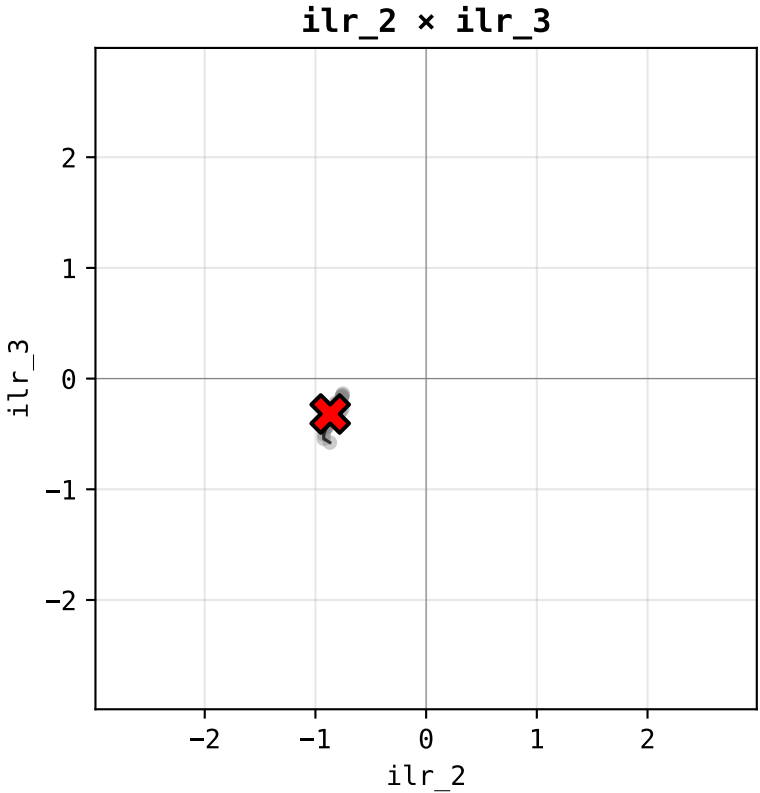
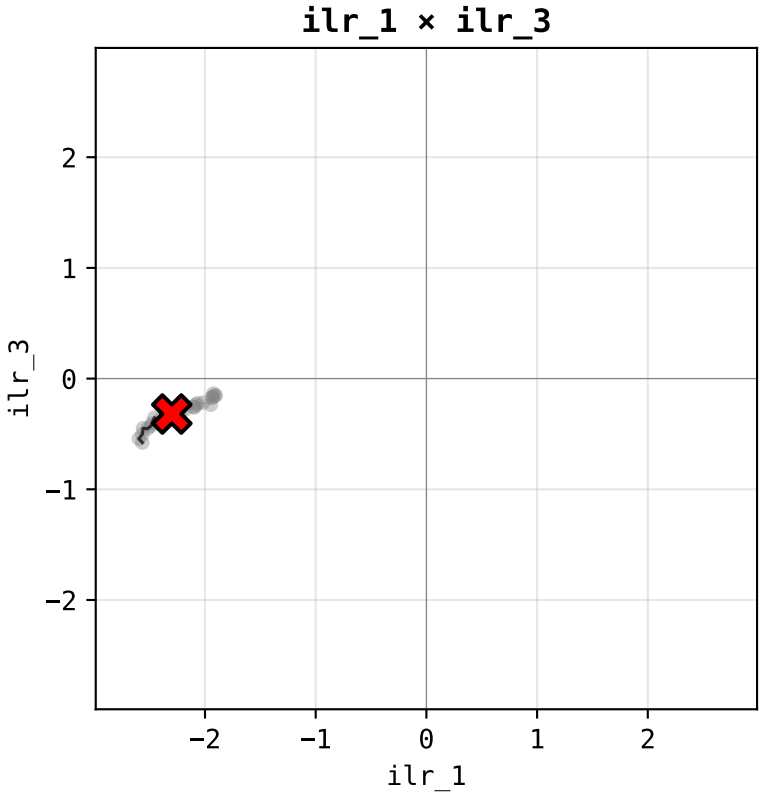
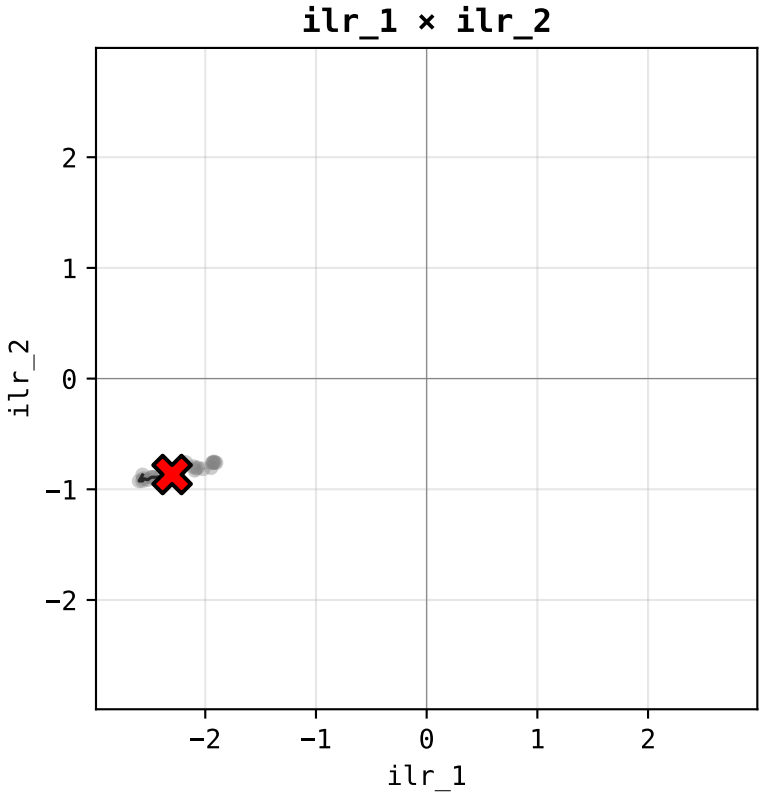
ILR Triplet – WLD World · year 2008 (9/26)



ILR Triplet – WLD World · year 2009 (10/26)



ILR Triplet – WLD World · year 2010 (11/26)



ILR Triplet – WLD World · year 2011 (12/26)

